



SEVENTH EDITION

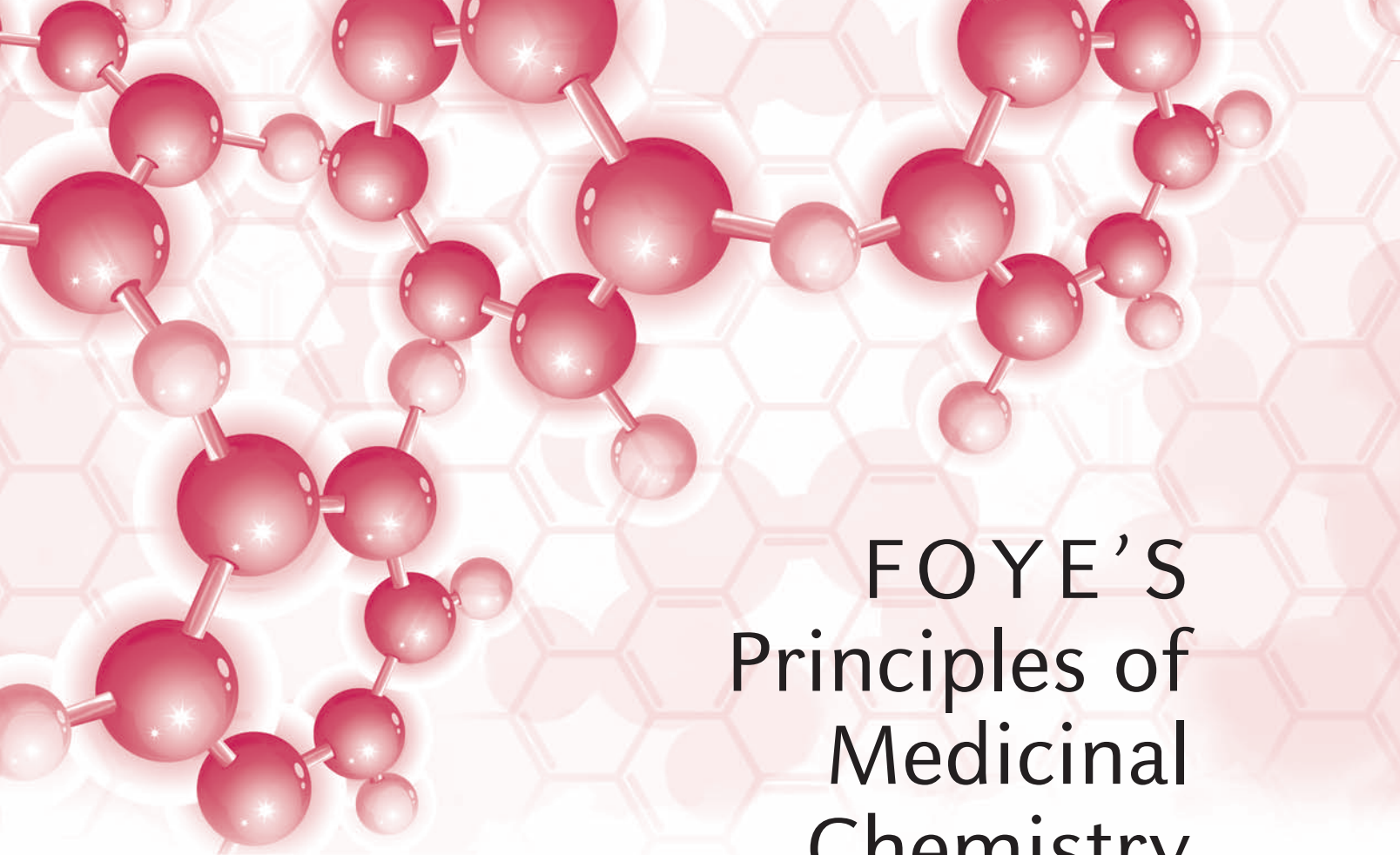
Foye's Principles of Medicinal Chemistry

Thomas L. Lemke | David A. Williams

Victoria F. Roche | S. William Zito

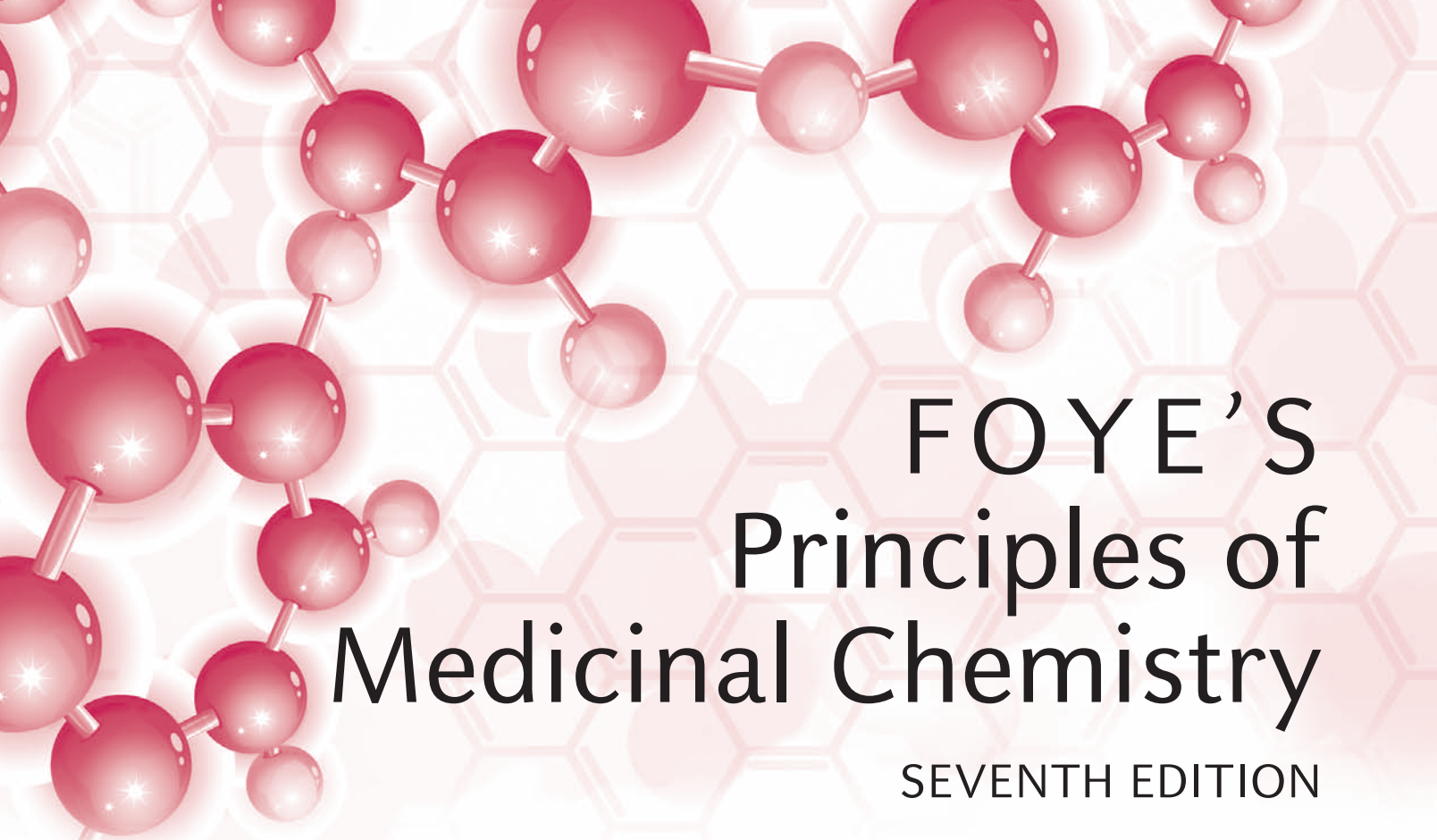


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FOYE'S Principles of Medicinal Chemistry

SEVENTH EDITION



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Seventh Edition

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This textbook is dedicated to our students and to our academic colleagues who mentor these students in the principles and applications of medicinal chemistry. The challenge for the student is to master the chemical, pharmacological, pharmaceutical and therapeutic aspects of the drug and utilize the knowledge of medicinal chemistry to effectively communicate with prescribing clinicians, nurses and other members of the health care team, as well as in discussing drug therapy with patients.

*Thomas L. Lemke
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Preface

As defined by IUPAC, medicinal chemistry is a chemistry-based discipline, involving aspects of the biological, medical and pharmaceutical sciences. It is concerned with the invention, discovery, design, identification and preparation of biologically active compounds, the study of their metabolism, the interpretation of their mode of action at the molecular level and the construction of structure-activity relationships (SAR), which is the relationship between chemical structure and pharmacological activity for a series of compounds.

As we look back 38 years to the first edition of Foye's *Principles of Medicinal Chemistry* and nearly 63 years to the first edition of Wilson and Gisvold's textbook, *Organic Chemistry in Pharmacy* (later renamed *Textbook of Organic Medicinal and Pharmaceutical Chemistry*), we can examine how the teaching of medicinal chemistry has evolved over the last half of the 20th century. Sixty years ago the approach to teaching drug classification was based on chemical functional groups; in the 1970s it was the relationship between chemical structure and pharmacological activity for a series of compounds, and today medicinal chemistry involves the integration of these principles with pharmacology, pharmacapeutics, and therapeutics into a single multi-semester course called pharmacodynamics, pharmacotherapeutics, or another similar name. Drug discovery and development will always maintain its role in traditional drug therapy, but its application to pharmacogenomics may well become the treatment modality of the future. In drug discovery, toxicogenomics is used to improve the safety of drugs mandated by U.S. Food and Drug administration by studying the adverse/toxic effects of drugs in order to draw conclusions on the toxic and safety risk to patients. The scope of knowledge in organic chemistry, biochemistry, pharmacology, and therapeutics allows students to make generalizations connecting the physicochemical properties of small organic molecules and peptides to the receptor and biochemical properties of living systems.

Creating new drugs to combat disease is a complex process. The shape of a drug must be right to allow it to bind to a specific disease-related protein (i.e., receptor) and to work effectively. This shape is determined by the core framework of the molecule and the relative orientation of functional groups in three dimensional space. As a consequence, these generalizations, validated by repetitive examples, emerge in time as principles of drug discovery and drug mechanisms, principles that describe the structural relationships between diverse organic molecules and the biomolecular functions that predict their mechanisms toward controlling diseases.

Medicinal chemistry is central to modern drug discovery and development. For most of the 20th century, the majority of drugs were discovered either by identifying the active ingredient in traditional natural remedies, by rational drug design, or by serendipity. As we have moved into the 21st century, drug discovery has focused on drug targets and high-throughput screening of drug hits and computer-assessed drug design to fill its drug pipeline. Medicinal chemistry has advanced during the past several decades from not only synthesizing new compounds but to understanding the molecular basis of a disease and its control, identifying biomolecular targets implicated as disease-causing, and ultimately inventing specific compounds (called "hits") that block the biomolecules from progressing to an illness or stop the disease in its tracks. Medicinal chemists use structure-activity relationships to improve the "hits" into "lead candidates" by optimizing their selectivity against the specific target, reducing drug activity against non-targets, and ensuring appropriate pharmacokinetic properties involving drug distribution and clearance.

These are tough times for the drug industry, as companies are looking at diminishing pipelines of potential new drugs, growing competition from generic versions of their drugs and increasing pressure from regulatory agencies to ensure that products are both safe and more effective than existing drugs. With the completion of sequencing of the human genome there are now greater challenges facing the drug industry for applications of new technologies in discovery and development. The number of drug targets once considered to be less than 500, has doubled and is expected to increase tenfold. Diseases that were once thought to be caused by a single pathology are now known to have differing etiologies requiring highly specific medications. In order to maintain its pipeline of new drugs, the drug industry is integrating biopharmaceuticals, such as therapeutic antibodies (e.g., in the treatment of arthritis), along with small-molecule drugs. As the drug industry undergoes reform, drug companies are developing collaborations with academia for new sources of drug molecules.

The editors of this textbook are all medicinal chemists, and our approaches to editing this seventh edition of Foye's *Principles of Medicinal Chemistry* are influenced by our respective academic backgrounds. We believe that our collaboration on this textbook represents a melding of our perspectives that will provide new dimensions of appreciation and understanding for all students. In

addition we recognize the benefits of medicinal chemistry can only be valuable if the science can be translated into improving the quality of life of our patients. As a result it is essential that the student apply the chemistry of the drugs to their patients and we have attempted to bridge the gap between the science of drugs and the real life situations through the use of scenarios and case studies. Finally in editing this multi-authored book we have tried to promote a consistent style in the organization of the respective chapters.

ORGANIZATIONAL PHILOSOPHY

The organizational approach taken in this textbook builds from the principles of drug discovery, physicochemical properties of drug molecules, and ADMET (absorption-distribution-metabolism-excretion-toxicity) to their integration into therapeutic substances with application to patient care. Our challenge has been to provide a comprehensive description of drug discovery and pharmacodynamic agents in an introductory textbook. To address the increasing emphasis in U.S. pharmacy schools on integrating medicinal chemistry with pharmacology and clinical pharmacy and the creation of one-semester principle courses, we organized the book into four parts: Part I: Principles of Drug Discovery; Part II: Drug Receptors Affecting Neurotransmission and Enzymes as Catalytic Receptors; Part III: Pharmacodynamic Agents (with further subdivision into drugs affecting different physiologic systems); and Part IV: Disease State Management. Parts I and II are designed for a course focused on principles of drug discovery and Parts II through IV are relevant to integrated courses in medicinal chemistry/pharmacodynamics/pharmacotherapeutics.

WHAT IS NEW IN THIS EDITION

The pharmacist sits at the interface between the health-care system and the patient. The pharmacist has the responsibility for improving the quality of life of the patient by assuring the appropriate use of pharmaceuticals. To do this appropriately, the pharmacist must bring together the basic sciences of chemistry, biology, biopharmaceutics and pharmacology with the clinical sciences. In an attempt to relate the importance of medicinal chemistry to the clinical sciences, each of the chapters in Part II, Pharmacodynamic Agents, through Part IV, Disease State Management, includes the following:

- **A clinical significance section:** At the beginning of most chapters, a practicing clinician has provided a statement of the clinical significance of medicinal chemistry to the particular therapeutic class of drugs.
- **A clinical scenario section:** At the beginning of the chapters in Part III and IV the clinician has provided a brief clinical scenario (mini-case) or real-life therapeutic problem related to the disease state under consideration. A solution to the case or problem appears at the end of the chapter along with the medicinal chemist's analysis of the solution.

The intent of this section is to pose a problem at the beginning of the chapter to stimulate the student's thinking as he/she reads through the chapter and then bring the learning "full circle" with the clinician's and chemist's solution to the case/problem revealed once the entire chapter has been read.

- **A case study:** Each of the above chapters ends with a case study (see the "Introduction to Medicinal Chemistry Case Studies" section of this preface). As with previous editions of Foye's Principles of Medicinal Chemistry these cases are meant help the student evaluate their comprehension of the therapeutically relevant chemistry presented in the chapter and apply their understanding in a standardized format to solving the posed problem. All cases presented in this text underwent review by a practicing pharmacist to ensure clinical accuracy and relevance to contemporary practice.

In addition, the reader will find at the beginning of most chapters a list of drugs (presented by generic or chemical names) discussed in that chapter. Additionally, at the beginning of each chapter, one will find a list of the commonly used abbreviations in the chapter.

Several new chapters appear in the seventh edition, including Chapter 5, Membrane Drug Transporters; Chapter 16, Anesthetics: General and Local Anesthetics; Chapter 19, CNS Stimulants and Drugs of Abuse; and Chapter 42, Obesity and Nutrition. Lastly, a second color has been added to this edition to help emphasize particular points in the chapters. In most figures where drug metabolism occurs the point of metabolism is highlighted in red with coloration of the functionality which has been changed.

STUDENT AND INSTRUCTOR RESOURCES

Student Resources

A Student Resource Center at <http://thePoint.lww.com/Lemke7e> includes the following materials:

- Full Text Online
- Additional Case Studies
- Answers to Additional Case Studies
- Practice Quiz Questions
- Drug Updates
- U.S. Drug Regulation: An Overview

Instructor Resources

We understand the demand on an instructor's time. To facilitate and support your educational efforts, you will have access to Instructor Resources upon adoption of *Foye's Principles of Medicinal Chemistry, 7th edition*. An Instructor's Resource Center at <http://thePoint.lww.com/Lemke7e> includes the following:

- Full Text Online
- Image Bank
- Answers to In-Text Case Studies
- Angel/Blackboard/WebCT Course Cartridges
- U.S. Drug Regulation: An Overview

ACKNOWLEDGEMENTS

We are indebted to our talented and conscientious contributors, for without them this book would not exist. This includes chapter authors, clinicians who wrote both the clinical significance sections and scenarios, and to Victoria Roche and Sandy Zito for creation of the exciting and educational case studies. We also thank our respective academic institutions for the use of institutional resources and for the freedom to exercise the creative juices needed to bring new ideas to a textbook in medicinal chemistry.

We are grateful for the many people at Lippincott Williams & Wilkins who were there to answer questions, make corrections, and support us through their encouraging words. Many of those who shepherded this book through the complex process of publication worked behind the scene and are not known to us, but we specifically acknowledge Andrea M. Klingler and Paula Williams (Product Managers), and David Troy (Acquisitions Editor) for their kind and gentle prodding.

Finally, we want to acknowledge our respective spouses, Pat and Gail, who were supportive of this time-consuming labor of love. Untold hours were spent away from the family sitting in front of our computers in order to bring this project to fruition.

Thomas L. Lemke, PhD
David A. Williams, PhD

INTRODUCTION TO MEDICINAL CHEMISTRY CASE STUDIES

We are pleased to share our newest medicinal chemistry case studies with student and faculty users of *Foye's Principles of Medicinal Chemistry*. One case study is provided at the end of most chapters. This preface is written to explain their scope and purpose, and to help those who are unfamiliar with our technique of illustrating the therapeutic relevance of chemistry get the most out of the exercise.

Like the more familiar therapeutic case studies, medicinal chemistry case studies are clinical scenarios that present a patient in need of a pharmacist's expert intervention. The learner, most commonly in the role of the pharmacist, evaluates the patient's clinical and personal situation and makes a drug product selection from a limited number of therapeutic choices. However, in a medicinal chemistry case study, only the structures of the potential therapeutic candidates are given. To make their

professional recommendation, students must conduct a thorough analysis of *key structure activity relationships (SAR)* in order to predict such things as relative potency, receptor selectivity, duration of action and potential for adverse reactions, and then apply the knowledge gained to meet the patient's therapeutic needs.

The therapeutic choices we offer in each case have been purposefully selected to allow students to review the therapeutically relevant chemistry of different classes of drugs used to treat a particular disease. We recognize that this approach might occasionally omit some compounds viewed by practitioners as drugs of choice within a class or the formulary entry at their practice sites. Faculty employing the cases as in-class or take-home assignments might alter the structural choices provided to meet their teaching and learning goals, and this is certainly acceptable. Regardless of how they are used, students working thoughtfully and scientifically through the cases will not only master chemical concepts and principles and reinforce basic SAR, but also learn how to actively use their unique knowledge of drug chemistry when thinking critically about patient care. This skill will be invaluable when, as practitioners, they are faced with a full gamut of therapeutic options to analyze in order to ensure the best therapeutic outcomes for their patients.

In short, here's what we hope students will gain by working our cases.

- Mastery of the important concepts needed to be successful in the medicinal chemistry component of the pharmacy curriculum;
- An ability to identify the relevance of drug chemistry to pharmacological action and therapeutic utility, and to discriminate between therapeutic options based on that understanding;
- An enhanced ability to think critically and scientifically about drug use;
- A commitment to caring about the impact of professional decisions on patients' quality of life;
- The ability to demonstrate the unique role of the pharmacist as the chemist of the health care team.

We hope you find these case studies both challenging and enjoyable, and we encourage you to use them as a springboard to more in-depth discussions with your faculty and/or colleagues about the role of chemistry in rational therapeutic decision-making.

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History and Evolution of Medicinal Chemistry

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The unprecedented increase in human life expectancy, which has almost doubled in a hundred years, is mainly due to drugs and to those who discovered them (1).

The history of all fields of science is comprised of the ideas, knowledge, and available tools that have advanced contemporary knowledge. The spectacular advances in medicinal chemistry over the years are no exception. Alfred Burger (1) stated that "...the great advances of medicinal chemistry have been achieved by two types of investigators: those with the genius of prophetic logic, who have opened a new field by interpreting correctly a few well-placed experiments, whether they pertained to the design or the mechanism of action of drugs; and those who have varied patiently the chemical structures of physiologically active compounds until a useful drug could be evolved as a tool in medicine." To place the development of medicinal chemical research into its proper perspective, one needs to examine the evolution of the ideas and concepts that have led to our present knowledge.

Drugs of Antiquity

The oldest records of the use of therapeutic plants and minerals are derived from the ancient civilizations of the Chinese, the Hindus, the Mayans of Central America, and the Mediterranean peoples of antiquity. The Emperor Shen Nung (2735 BC) compiled what may be called a pharmacopeia including *ch'ang shang*, an antimalarial alkaloid, and *ma huang*, from which ephedrine was isolated. Chaulmoogra fruit was known to the indigenous

American Indians, and the ipecacuanha root containing emetine was used in Brazil for the treatment of dysentery and is still used for the treatment of amebiasis. The early explorers found that the South American Indians also chewed coca leaves (containing cocaine) and used mushrooms (containing methylated tryptamine) as hallucinogens. In ancient Greek apothecary shops, herbs such as opium, squill, and *Hyoscyamus*, viper toxin, and metallic drugs such as copper and zinc ores, iron sulfate, and cadmium oxide could be found.

The Middle Ages

The basic studies of chemistry and physics shifted from the Greco-Roman to the Arabian alchemists between the 13th and 16th centuries. Paracelsus (1493–1541) glorified antimony and its salts in elixirs as cure-alls in the belief that chemicals could cure disease.

The 19th Century: Age of Innovation and Chemistry

The 19th century saw a great expansion in the knowledge of chemistry, which greatly extended the herbal pharmacopeia that had previously been established. Building on the work of Antoine Lavoisier, chemists throughout Europe refined and extended the techniques of chemical analysis. The synthesis of acetic acid by Adolph Kolbe in 1845 and of methane by Pierre Berthelot in 1856 set the stage for organic chemistry. Pharmacognosy, the science that deals with medicinal products of plant, animal, or mineral origin in their crude state, was replaced by physiologic chemistry. The emphasis was shifted from finding

new medicaments from the vast world of plants to finding the active ingredients that accounted for their pharmacologic properties. The isolation of morphine by Friedrich Sertürner in 1803, the isolation of emetine from ipecacuanha by Pierre-Joseph Pelletier in 1816, and his purification of caffeine, quinine, and colchicine in 1820 all contributed to the increased use of “pure” substances as therapeutic agents. In the 19th century, digitalis was used by the English physician and botanist, William Withering, for the treatment of edema. Albert Niemann isolated cocaine in 1860, and in 1864, he isolated the active ingredient, physostigmine, from the Calabar bean. As a result of these discoveries and the progress made in organic chemistry, the pharmaceutical industry came into being at the end of the 19th century (2).

The 20th Century and the Pharmaceutical Industry

Diseases of protozoal and spirochetal origin responded to synthetic chemotherapeutic agents. Interest in synthetic chemicals that could inhibit the rapid reproduction of pathogenic bacteria and enable the host organism to cope with invasive bacteria was dramatically increased when the red dyestuff 2,4-diaminoazobenzene-4'-sulfonamide (Prontosil) reported by Gerhard Domagk dramatically cured dangerous systemic gram-positive bacterial infections in man and animals. The observation by Woods and Fildes in 1940 that the bacteriostatic action of sulfonamide-like drugs is antagonized by *p*-aminobenzoic acid is one of the early examples in which a balance of stimulatory and inhibitory properties depends on the structural analogies of chemicals.

That, together with the discovery of penicillin by Alexander Fleming in 1929 and its subsequent examination by Howard Florey and Ernst Chain in 1941, led to a water-soluble powder of much higher antibacterial potency and lower toxicity than that of previously known synthetic chemotherapeutic agents. With the discovery of a variety of highly potent anti-infective agents, a significant change was introduced into medical practice.

DEVELOPMENTS LEADING TO VARIOUS MEDICINAL CLASSES OF DRUGS

Psychopharmacologic Agents and the Era of Brain Research

Psychiatrists have been using agents active in the central nervous system for hundreds of years. Stimulants and depressants were used to modify the mood and mental states of psychiatric patients. Amphetamine, sedatives, and hypnotics were used to stimulate or depress the mental states of patients. Was it the synthesis of chlorpromazine by Paul Charpentier that caused a revolution in the treatment of schizophrenia? Who really discovered chlorpromazine? Was it Charpentier, who first synthesized the molecule in 1950 at Rhone-Poulenc's research laboratory; Simone Courvoisier, who reported distinctive effects on animal behavior; Henri Laborit, a French

military surgeon who first noticed distinctive psychotropic effects in man; or Pierre Deniker and Jean Delay, French psychiatrists who clearly outlined what has now become its accepted use in psychiatry and without whose endorsement and prestige Rhone-Poulenc might never have developed it further as an antipsychotic? Because of the bitter disputes over the discovery of chlorpromazine, no Nobel Prize was ever awarded for what has been the single most important breakthrough in psychiatric treatment (Fig. 1).

The discovery of the antidepressant effects of the antitubercular drug iproniazid (isopropyl congener of isoniazid), which has monoamine oxidase (MAO)-inhibiting activity, led to a series of MAO inhibitor antidepressants including phenelzine (Nardil) and tranylcypromine (Parnate), which are still used clinically. Soon after, the first dibenzazepine (tricyclic) antidepressant imipramine was introduced by Ciba-Geigy Corporation in 1957 a series of tricyclic compounds synthesized initially as structural analogs of phenothiazines, were developed. The tricyclic antidepressants are not antipsychotic, but instead elevate mood by blocking the transport inactivation of monoamine neurotransmitters including norepinephrine and serotonin. In the late 1980s, a series of selective serotonin reuptake or transport inhibitors (SSRIs) were developed, starting with *R,S*-zimeidine from Astra Pharmaceutica (which proved to be toxic) and then *R,S*-fluoxetine (Prozac) from Eli Lilly and Company, the first commercially successful SSRI and the first psychotropic agent to attain an annual market above \$1 billion.

The anxiolytic agents, including a large series of benzodiazepines (including chlordiazepoxide [Librium] and diazepam [Valium] and the carbamate meprobamate [Miltown]), are examples of the serendipitous discovery of new drugs based on random screening of newly synthesized chemicals (Fig. 1). The discovery of these drugs was based on observations of effects on the behavior of animals used in screening bioassays. In 1946, Frank M. Berger observed unusual and characteristic paralysis and relaxation of voluntary muscles in laboratory animals for different series of compounds. At this point, the treatment of ambulatory anxious patients with meprobamate and psychotic patients with one of the aminoalkylphenothiazine drugs was possible.

There was a need for drugs of greater selectivity in the treatment of anxiety because of the side effects often

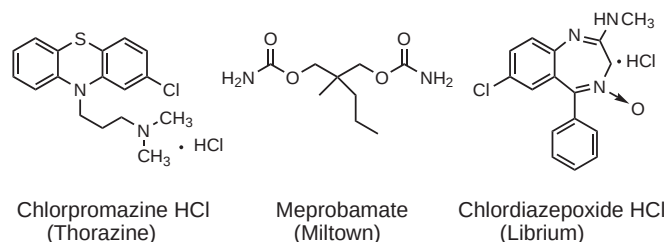


FIGURE 1 Psychopharmacologic agents.

encountered with phenothiazines. Leo Sternback, a chemist working in the research laboratory of Hoffman-La Roche in New Jersey, decided to reinvestigate a relatively unexplored class of compounds that he had studied in the 1930s when he was a postdoctoral fellow at the University of Cracow in Poland. He synthesized about 40 compounds in this series, all of which were disappointing in pharmacologic tests, so the project was abandoned. In 1957, during a cleanup of the laboratory, one compound synthesized 2 years earlier had crystallized and was submitted for testing to L.O. Randall, a pharmacologist. Shortly thereafter, Randall reported that this compound was hypnotic and sedative and had antistrychnine effects similar to those of meprobamate. The compound was named chlordiazepoxide and marketed as Librium in 1960, just 3 years after the first pharmacologic observations by Randall. Structural modifications of benzodiazepine derivatives were undertaken, and a compound 5 to 10 times more potent than chlordiazepoxide was synthesized in 1959 and marketed as diazepam (Valium) in 1963. The synthesis of many other experimental analogs soon followed, and by 1983, about 35 benzodiazepine drugs were available for therapy (see Chapter 15). Benzodiazepines are used in the pharmacotherapy of anxiety and related emotional disorders and in the treatment of sleep disorders, status epilepticus, and other convulsive states. They are used as centrally acting muscle relaxants, for premedication, and as inducing agents in anesthesiology.

Endocrine Therapy and Steroids

The first pure hormone to be isolated from the endocrine gland was epinephrine, which led to further molecular modifications in the area of sympathomimetic amines. Subsequently, norepinephrine was also identified from sympathetic nerves. The development of chromatographic techniques allowed the isolation and characterization of a multitude of hormones from a single gland. In 1914, biochemist Edward Kendall isolated thyroxine from the thyroid gland. He subsequently won the Nobel Prize in Physiology or Medicine in 1950 for his discovery of the activity of cortisone. Two of the hormones of the thyroid gland, thyroxine (T_4) and liothyronine (T_3), have similar effects in the body regulating metabolism, whereas the two hormones from the posterior pituitary gland—vasopressin, which exerts pressor and antidiuretic activity, and oxytocin, which stimulates lactation and uterine motility—differ considerably both in their chemical structure and physiologic activity. (Fig. 2)

Less than 50 years after the discovery of oxytocin by Henry Dale in 1904, who found that an extract from the human pituitary gland contracted the uterus of a pregnant cat, the biochemist Vincent du Vigneud synthesized the cyclic peptide hormone. His work resulted in the Nobel Prize in Chemistry in 1955.

A major achievement in drug discovery and development was the discovery of insulin in 1921 from animal

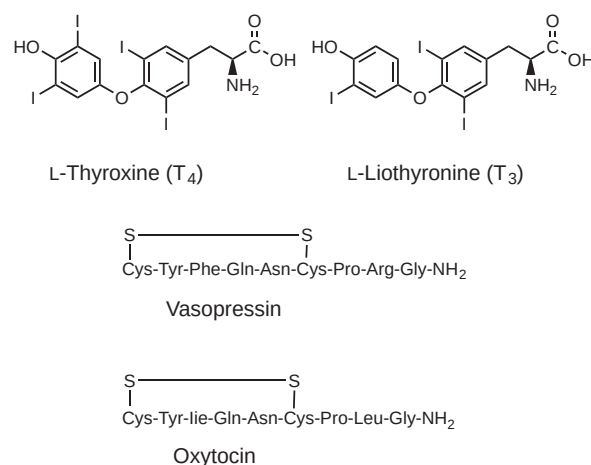


FIGURE 2 Hormones from the endocrine glands.

sources. Frederick G. Banting and Charles H. Best, working in the laboratory of John J.R. McLeod at the University of Toronto, isolated the peptide hormone and began testing it in dogs. By 1922, researchers, with the help of James B. Collip and the pharmaceutical industry, purified and produced animal-based insulin in large quantities. Insulin soon became a major product for Eli Lilly & Co. and Novo Nordisk, a Danish pharmaceutical company. In 1923, McLeod and Bunting were awarded the Nobel Prize in Medicine or Physiology, and after much controversy, they shared the prize with Collip and Best. For the next 60 years, cattle and pigs were the major sources of insulin. With the development of genetic engineering in the 1970s, new opportunities arose for making synthetic insulin that is chemically identical to human insulin. In 1978, the biotech company Genentech and the City of Hope National Medical Center produced human insulin in the laboratory using recombinant DNA technology. By 1982, Lilly's Humulin became the first genetically engineered drug approved by the U.S. Food and Drug Administration (FDA). At about the same time, Novo Nordisk began selling the first semisynthetic human insulin made by enzymatically converting porcine insulin. Novo Nordisk was also using recombinant technology to produce insulin. Recombinant insulin was a significant milestone in the development of genetically engineered drugs and combined the technologies of the biotech companies with the know-how and resources of the major pharmaceutical industries. Inhaled insulin was approved by the FDA in 2006. Many drugs are now available (see Chapter 27) to treat the more common type 2 diabetes in which insulin production needs to be increased. Insulin had been the only treatment for type 1 diabetes until 2005 when the FDA approved Amylin Pharmaceuticals' Symlin to control blood sugar levels in combination with the peptide hormone. The isolation and purification of several peptide hormones of the anterior pituitary and hypothalamic-releasing hormones now make it possible to produce synthetic peptide

agonists and antagonists that have important diagnostic and therapeutic applications.

Extensive and remarkable advances in the endocrine field have been made in the group of steroid hormones. The isolation and characterization of minute amounts of the active principles of the sex glands and from the adrenal cortex eventually led to their total synthesis. Male and female sex hormones are used in the treatment of a variety of disorders associated with sexual development and the sexual cycles of males and females, as well as in the selective therapy of malignant tumors of the breast and prostate gland. Synthetic modifications of the structure of the male and female hormones have furnished improved hormonal compounds such as the anabolic agents (see Chapter 40). Since early days, women have ingested every manner of substance as birth control agents. In the early 1930s, Russell Marker found that, for hundreds of years, Mexican women had been eating wild yams of the *Dioscorea* genus for contraception, with apparent success. Marker determined that diosgenin is abundant in yams and has a structure similar to progesterone. Marker was able to convert diosgenin into progesterone, a substance known to stop ovulation in rabbits. However, progesterone is destroyed by the digestive system when ingested. In 1950, Carl Djerassi, a chemist working at the Syntex Laboratories in Mexico City, synthesized norethindrone, the first orally active contraceptive steroid, by a subtle modification of the structure of progesterone. Gregory Pincus, a biologist working at the Worcester Foundation for Experimental Biology in Massachusetts studied Djerassi's new steroid together with its double bond isomer norethynodrel (Fig. 3).

By 1956, clinical studies led by John Rock, a gynecologist, showed that progesterone, in combination with norethindrone, was an effective oral contraceptive. G.D. Searle was the first on the market with Enovid, a combination of mestranol and norethynodrel. In 2005, it was estimated that 11 million American women and about

100 million women worldwide were using oral contraceptive pills. In 1993, the British weekly *The Economist* considered the pill to be one of the seven wonders of the modern world, bringing about major changes in the economic and social structure of women globally.

In the early 1930s, chemists recognized the similarity of a large number of natural products including the adrenocortical steroids such as hydrocortisone. The medicinal value of Kendall's Compound F and Reichstein's Compound M was quickly recognized. The 1950 Nobel Prize in Physiology or Medicine was awarded to Phillip S. Hench, Edward C. Kendall, and Tadeus Reichstein "...for their discovery relating to the hormones of the adrenal cortex, their structure and biological effects."

An interesting development in the study of glucocorticoids led in 1980 to the synthesis of the "abortion pill," RU-486, synthesized by Etienne-Emile Beaulieu, a consultant to the French pharmaceutical company, Rousel-Uclaf. Researchers at that time were investigating glucocorticoid antagonists for the treatment of breast cancer, glaucoma, and Cushing syndrome. In screening RU-486, researchers at Rousel-Uclaf found that it had both antiglucocorticoid activity as well as high affinity for progesterone receptors where it could be used for fertility control. RU-486, also known as mifepristone (Mifeprex), entered the French market in 1988, but sales were suspended by Rousel-Uclaf when antiabortion groups threatened to boycott the company. In 1994, the company donated the United States rights to the New York City-based Population Council, a nonprofit reproductive and population control research institution. Mifepristone is now administered in doctors' offices as a tablet in combination with misoprostol, a prostaglandin that causes uterine contractions to help expel the embryo. The combination of mifepristone and misoprostol is more than 90% effective. Plan B, also known as the "morning after pill," has been referred to as an emergency contraceptive. It contains levonorgestrel, the same progestin that is in "the pill," and should be taken within 3 days of unprotected sex and can reduce the risk of pregnancy by 89%.

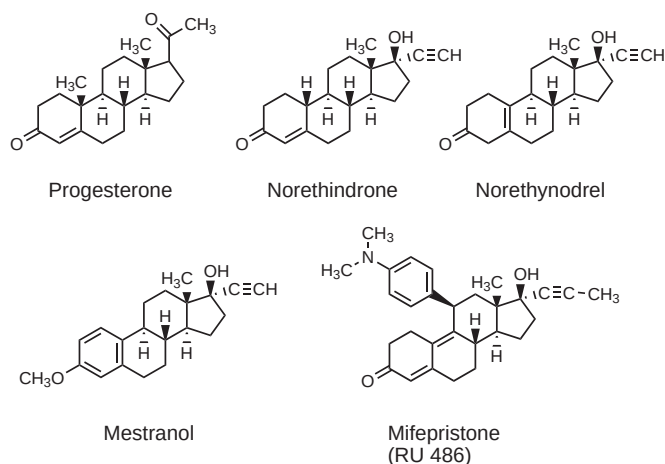


FIGURE 3 Steroidal agents.

Anesthetics and Analgesics

The first use of synthetic organic chemicals for the modulation of life processes occurred when nitrous oxide, ether, and chloroform were introduced in anesthesia during the 1840s. Horace Wells, a dentist in Hartford, Connecticut, administered nitrous oxide during a tooth extraction while Crawford Long, a Georgia physician, used ether as an anesthetic for excising a growth on a patient's neck. It was William Morton, a 27-year-old dentist, however, who gave the first successful public demonstration of surgical anesthesia on October 16, 1846, at the surgical amphitheater that is now called the Ether Dome at Massachusetts General Hospital. Morton attempted to patent his discovery but was unsuccessful, and he died penniless in 1868. Chloroform had also been used as an anesthetic at St. Bartholomew's Hospital in London. In

Paris, France, Pierre Fluorens tested both chloroform and ethyl chloride as anesthetics in animals.

The potent and euphoric properties of the extract of the opium poppy have been known for thousands of years. In the 16th century, the Swiss physician and alchemist, Paracelsus (1493–1541) popularized the use of opium in Europe. At that time, an alcoholic solution of opium, known as laudanum, was the method of administration. Morphine was first isolated in pure crystalline form from opium by the German apothecary, Fredrick W. Sertürner, in 1805 who named the compound “morphium” after Morpheus, the Greek god of dreams. It took another 120 years before the structure of morphine was elucidated by Sir Robert Robinson at the University of Oxford. The chemistry of morphine and the other opium alkaloids obtained from *Papaver somniferum* has fascinated and occupied chemists for over 200 years, resulting in many synthetic analgesics available today (see Chapter 20). (–)-Morphine was first synthesized by Marshall Gates at the University of Rochester in 1952. Although a number of highly effective stereoselective synthetic pathways have been developed, it is unlikely that a commercial process can compete with its isolation from the poppy. Diacetylmorphine, known as heroin, is highly addictive and induces tolerance. The illicit worldwide production of opium now exceeds the pharmaceutical production by almost 10-fold. In the United States, some 800,000 people are chemically addicted to heroin, and a growing number are becoming addicted to OxyContin, a synthetic opiate also known as oxycodone. Another synthetic opiate, methadone, relaxes the craving for heroin or morphine. A series of studies in the 1960s at Rockefeller University by Vincent Dole and his wife, Marie Nyswander, found that methadone could also be a viable maintenance treatment to keep addicts from heroin. It is estimated that there are about 250,000 addicts taking methadone in the United States. It has not been widely recognized in the United States that opiate addiction is a medical condition for which there is no known cure. More than 80% of United States heroin addicts still lack access to methadone treatment facilities, primarily due to the stigma against drug users and the medical distribution of methadone.

It has been only within the last 40 years that scientists have begun to understand the effects of opioid analgesics at the molecular level. Beckett and Casey at the University of London proposed in 1954 that opiate effects were receptor mediated, but it was not until the early 1970s that the stereospecific binding of opiates to specific receptors was demonstrated. The characterization and classification of three different types of opioid receptors, *mu*, *kappa*, and *delta*, by William Martin formed the basis of our current understanding of opioid pharmacology. The demonstration of stereospecific binding of radiolabeled ligands to opioid receptors led to the development of radioreceptor binding assays for each of the opioid receptor types, a technique that has been of major importance in the identification of selective opioids as well as many other

receptors. In 1973, Avram Goldstein, Solomon Snyder, Ernst Simon, and Lars Terenius independently described saturable, stereospecific binding sites for opiate drugs in the mammalian nervous system. Shortly thereafter, John Hughes and Hans Kosterlitz, working at the University of Aberdeen in Scotland, described the isolation from pig brains of two pentapeptides that exhibited morphine-like actions on the guinea pig ileum. At about the same time, Goldstein reported the presence of peptide-like substances in the pituitary gland showing opiate-like activity. Subsequent research revealed that there are three distinct families of opiate peptides: the enkephalins, the endorphins, and the dynorphins.

Hypnotics and Anticonvulsants

Since antiquity, alcoholic beverages and potions containing laudanum, an alcoholic extract of opium, and various other plant products have been used to induce sleep. Bromides were used in the middle of the 19th century as sedative-hypnotics, as were chloral hydrate, paraldehyde, urethane, and sulfenal. Joseph von Merring, on the assumption that a structure having a carbon atom carrying two ethyl groups would have hypnotic properties, investigated diethyl acetyl urea, which proved to be a potent hypnotic. Further investigations led to 5,5-diethylbarbituric acid, a compound synthesized 20 years earlier in 1864 by Adolph von Beyer. Phenobarbital (5-ethyl-5-phenylbarbituric acid) (Fig. 4) was synthesized by the Bayer Pharmaceutical Company and introduced to the market under the name Luminol. The compound was effective as a hypnotic, but also exhibited properties as an anticonvulsant. The success of phenobarbital led to the testing of more than 2,500 barbiturates, of which about 50 were used clinically, many of which are still in clinical use. Modification of the barbituric acid molecule also led to the development of the hydantoins. Phenytoin (also known as diphenylhydantoin or Dilantin) (Fig. 4) was first synthesized in 1908, but its anticonvulsant properties were not discovered until 1938. Because phenytoin was not a sedative at ordinary doses, it established that antiseizure drugs need not induce drowsiness and encouraged the search for drugs with selective antiseizure action.

Local Anesthetics

The local anesthetics can be traced back to the naturally occurring alkaloid cocaine isolated from *Erythroxylon coca*. A Viennese ophthalmologist, Carl Koller, had

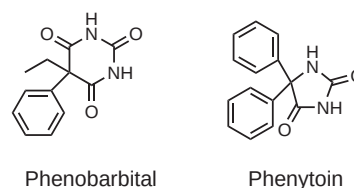


FIGURE 4 Examples of an early hypnotic and anticonvulsant.

experimented with several hypnotics and analgesics for use as a local anesthetic in the eye. His friend, Sigmund Freud, suggested that they attempt to establish how the South American Indians allayed fatigue by chewing leaves of the coca bush. Cocaine had been isolated from the plant by the Swedish chemist Albert Niemann at Gothenburg University in 1860. Koller found that cocaine numbed the tongue, and thus, he discovered a local anesthetic. He quickly realized that cocaine was an effective, nonirritating anesthetic for the eye, leading to the widespread use of cocaine in both Europe and the United States. (Carl Koller's nickname among Viennese medical students was "Coca Koller"). Richard Willstätter in Munich determined the structure of both cocaine and atropine in 1898 and succeeded in synthesizing cocaine 3 years later. Although today cocaine is of greater historic than medicinal importance and is widely abused, few developments in the chemistry of local anesthetics can disclaim a structural relationship to cocaine (Fig. 5). Benzocaine, procaine, tetracaine, and lidocaine all can be considered structural analogs of cocaine, a classic example of how structural modification of a natural product can lead to useful therapeutic agents.

Drugs Affecting Renal and Cardiovascular Function

Included in this category are drugs used in the treatment of myocardial ischemia, congestive heart failure, various arrhythmias, and hypercholesterolemia. Only two examples of drug development will be highlighted. Use of the cardiac drug digoxin dates back to the folk remedy foxglove attributed to William Withering who, in 1775, discovered that the foxglove plant, *Digitalis purpurea*, was beneficial to those suffering from abnormal fluid buildup. The active principles of digitalis were isolated in 1841 by E. Humolle and T. Quevenne in Paris. They consisted mainly of digitoxin. The other glycosides of digitalis were subsequently isolated in 1869 by Claude A. Nativelle and in 1875 by Oswald Schmiedberg. The correct structure of digitoxin was established more than 50 years later by Adolf Windaus at Gothenburg University. In 1929, Sydney Smith at Burroughs Wellcome isolated and separated a new glycoside from *D. purpurea*, known

as digoxin. This is now the most widely used cardiac glycoside. Today, dried foxglove leaves are processed to yield digoxin much like the procedure used by Withering. It takes about 1,000 kg of dried foxglove leaves to make 1 kg of pure digitalis.

It is the group of drugs used in the therapy of hypercholesterolemia that has received the greatest success and financial reward for the pharmaceutical industry during the last two decades. Cholesterol-lowering drugs, known as statins, are one of the cornerstones in the prevention of both primary and secondary heart diseases. Drugs such as Merck's lovastatin (Mevacor) and Pfizer's atorvastatin (Lipitor) are a huge success (Fig. 6). In 2004, Lipitor was the world's top selling drug, with sales of more than \$10 billion. As a class, cholesterol- and triglyceride-lowering drugs were the world's top selling category, with sales exceeding \$30 billion. The discovery of the statins can be credited to Akira Endo, a research scientist at Sankyo Pharmaceuticals in Japan (3). Endo's 1973 discovery of the first anticholesterol drug has almost been relegated to obscurity. The story of his research and the discovery of lovastatin are not typical but often escape attention. When Endo joined Sankyo after his university studies to investigate food ingredients, he searched for a fungus that produced an enzyme to make fruit juice less pulpy. The search was a success, and Endo's next assignment was to find a drug which would block the enzyme hydroxymethylglutaryl-coenzyme A (HMG-CoA) a key enzyme essential to the production of cholesterol. With Endo's interest and background, he searched for fungi that would block this enzyme. In 1973, after testing 6,000 fungal broths Endo found a substance made by the mold *Penicillium citrinum* that was a potent inhibitor on the enzyme needed to make cholesterol; it was named compactin (mevastatin) (Fig. 6). However, the substance did not work in rats but did work in hens and dogs. Endo's bosses were unenthusiastic about his discovery and discouraged further research with this compound. With the collaboration of Akira Yamamoto, a physician treating patients with extremely high cholesterol due to a genetic defect, Endo prepared samples of his drug, and it was administered to an 18-year-old

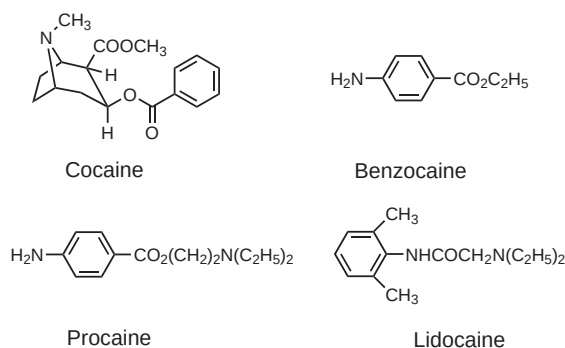


FIGURE 5 Synthetic local anesthetics development based on the structure of cocaine.

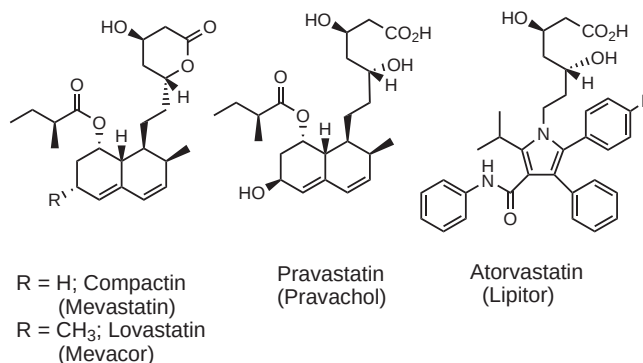


FIGURE 6 The first statins.

woman by Yamamoto. Further testing in nine patients led to an average of 27% lowering of cholesterol. In 1978, using a different fungus, Merck discovered a substance that was nearly identical to Endo's; this one was named lovastatin (Mevacor). Merck held the patent rights in the United States and, in 1987, started marketing it as Mevacor, the first FDA-approved statin. Sankyo eventually gave up compactin and pursued another statin that they licensed to Bristol-Myers Squibb Co., which was sold as Pravachol. In 1985, Michael S. Brown and Joseph Goldstein won the Nobel Prize in Physiology or Medicine for their work in cholesterol metabolism. It was only in January of 2006 that Endo received the Japan Prize, considered by many to be equivalent to the Nobel Prize. There is no doubt that millions of people whose lives have been and will be extended through statin therapy owe it to Akira Endo.

Anticancer Agents

Sulfur mustard gas was used as an offensive weapon by the Germans during World War I, and the related nitrogen mustards were manufactured by both sides in World War II. Later, investigations showed that the toxic gases had destroyed the blood's white cells, which subsequently led to the discovery of drugs used in leukemia therapy. These compounds, although effective antitumor agents, were very toxic. 6-Mercaptopurine (Fig. 7) was really the first effective leukemia drug developed by George Hitchings and his technician, Gertrude Elion, who, working together at Burroughs Wellcome Research Laboratories, shared the Nobel Prize in 1988. By a process now termed "rational drug design," Hitchings hypothesized that it might be possible to use antagonists to stop bacterial or tumor cell growth by interfering with nucleic acid biosynthesis in a similar way that sulfonamides blocked cell growth.

Unlike many cancer drugs available today, cisplatin is an inorganic molecule with a simple structure (Fig. 7). Cisplatin interferes with the growth of cancer cells by binding to DNA and interfering with the cells' repair mechanism and eventually causes cell death. It is used to treat

many types of cancer, primarily testicular, ovarian, bladder, lung, and stomach cancers. Cisplatin is now the gold standard against which new medicines are compared. It was first synthesized in 1845, and its structure was elucidated by Alfred Werner in 1893. It was not until the early 1960s when Barnett Rosenberg, a professor of biophysics and chemistry at Michigan State University, observed the compound's effect in cell division, which prompted him to test cisplatin against tumors in mice. The compound was found to be effective and entered clinical trials in 1971. There is an important lesson to be learned from Rosenberg's development of cisplatin. As a biophysicist and chemist, Rosenberg realized that when he was confronted with interesting results for which he could not find explanations, he enlisted the help and expertise of researchers in microbiology, inorganic chemistry, molecular biology, biochemistry, biophysics, physiology, and pharmacology. Such a multidisciplinary approach is the key to the discovery of modern medicines today. Although cisplatin is still an effective drug, researchers have found second-generation compounds such as carboplatin that have less toxicity and fewer side effects.

A third compound in the class of anticancer agents is paclitaxel (Taxol), discovered in 1963 by Monroe E. Wall and Masukh C. Wani at Research Triangle Park in North Carolina (Fig. 7). Taxol was isolated from extracts of the bark of the Pacific yew tree, *Taxus brevifolia*. The extracts showed potent anticancer activity, and by 1967, Wall and his coworkers had isolated the active ingredients; in 1971, they established the structure of the compound. Susan Horwitz, working at the Albert Einstein College of Medicine in New York, studied the mechanism of how Taxol kills cancer cells. She discovered that Taxol works by stimulating growth of microtubules and stabilizing the cell structures so that the killer cells are unable to divide and multiply. It was not until 1993 that Taxol was brought to the market by Bristol-Myers Squibb and soon became an effective drug for treating ovarian, breast, and certain forms of lung cancers. The product became a huge commercial success, with annual sales of approximately \$1.6 billion in 2000.

Old Drugs as Targets for New Drugs

Cannabis is used throughout the world for diverse purposes and has a long history characterized by usefulness, euphoria or evil, depending on one's point of view. To the agriculturist cannabis is a fiber crop; to the physician of a century ago it was a valuable medicine; to the physician of today it is an enigma; to the user, a euphoriant; to the police, a menace; to the trafficker, a source of profitable danger; to the convict or parolee and his family, a source of sorrow (4).

The plant, *Cannabis sativa*, the source of marijuana, has a long history in folk medicine, where it has been used for ills such as menstrual pain and the muscle spasms that affect multiple sclerosis sufferers. As in so many other areas of drug research, progress was achieved in the understanding of the pharmacology and biogenesis of a naturally occurring drug only when the chemistry had been well established and the researcher had at his

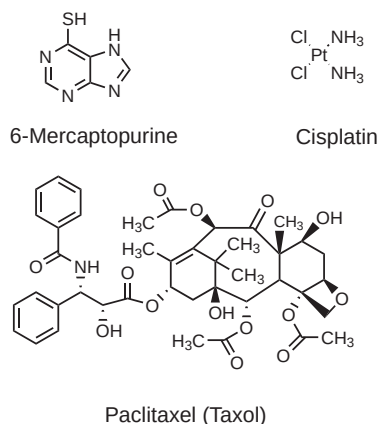


FIGURE 7 Anticancer drugs.

disposal pure compounds of known composition and stereochemistry. Cannabis is no exception in this respect, with the last 60 years producing the necessary know-how in the chemistry of the cannabis constituents so that chemists could devise practical and novel synthetic schemes to provide the pharmacologists with pure substances. The isolation and determination of the structure of tetrahydrocannabinol (Δ^9 -THC), the principal active ingredient, were performed in 1964 by Rafael Mechoulam at Hebrew University in Israel. Although cannabis and some of its structural analogs have been and are still used in medicine, in the last few years, research has focused on the endocannabinoids and their receptors as targets for drug development. It was shown that THC exerts its effects by binding to receptors that are targets of naturally occurring molecules termed endocannabinoids that have been involved in controlling learning, memory, appetite, metabolism, blood pressure, emotions such as fear and anxiety, inflammation, bone growth, and cancer. It is no surprise, then, that drug researchers are focusing on developing compounds that either act as agonists or antagonists of the endocannabinoids. In 1990, Lisa Matsuda and Tom Bonner at the National Institutes of Health cloned a THC receptor now called CB₁ from a rat brain. Shortly thereafter, Mechoulam and his coworkers identified the first of these endogenous cannabinoids called anandamide and, a few years later, identified 2-arachidonylglycerol (2-AG). In 1993, the second cannabinoid receptor, CB₂, was cloned by Muna Abu-Shaar at the Medical Research Council in Cambridge, United Kingdom. The drug rimonabant was an endocannabinoid antagonist developed by the French pharmaceutical company Sanofi-Aventis, and although it was approved initially for promoting weight loss, it has subsequently been removed from the market. The drug binds to CB₁ but not CB₂ receptors, resulting in the weight loss effect. Efforts to develop other endocannabinoids as therapeutic agents are in full swing in many laboratories and include preclinical testing for epilepsy, pain, anxiety, and diarrhea. Thus, a new series of drugs is being developed that are not centered on marijuana itself, but inspired by its active ingredient Δ^9 -THC, mimicking the endogenous substances acting in the brain or the periphery.

Molecular Imaging

The clinician now has at his or her disposal a variety of diagnostic tools to help obtain information about the pathophysiologic status of internal organs. The most widely used methods for noninvasive imaging are scintigraphy, radiography (x-ray and computed tomography [CT]), ultrasonography, positron emission tomography (PET), single photon emission computed tomography (SPECT), and magnetic resonance imaging (MRI). Chemists continue to make important contributions to the preparation of radiopharmaceuticals and contrast agents. These optical, nuclear, and magnetic methods are increasingly being empowered by new types of imaging agents. The effectiveness of new and old drugs to treat disease and to monitor

the response to therapies is now being routinely used in the drug discovery process.

The expanded use of the cyclotron in the late 1930s and the nuclear reactor in the early 1940s made available a variety of radionuclides for potential applications in medicine. The field of nuclear medicine was founded with reactor-produced radioiodine for the diagnosis of thyroid dysfunction. Soon other radioactive tracers, such as ¹⁸F, ¹²³I, ¹³¹I, ^{99m}Tc, and ¹¹C, became available. This, together with more sensitive radiation detection instruments and cameras, made it possible to study many organs of the body such as the liver, kidney, lung, and brain. The diagnostic value of these noninvasive techniques served to establish nuclear medicine and radiopharmaceutical chemistry as distinct specialties. A radiopharmaceutical is defined as any pharmaceutical that contains a radionuclide (5).

Historically, radioiodine has a special place in nuclear medicine. In 1938, Hertz, Roberts, and Evans first demonstrated the uptake of ¹²⁸I by the thyroid gland. ¹³¹I, with a longer half-life ($t_{1/2}$: 8 days), became available later and is now widely used. Although iodine has 24 known isotopes, ¹²³I, ¹³¹I, and ¹²⁵I are the only iodine isotopes currently used in medicine. At present, the most widely used PET radiopharmaceutical is the glucose analog ¹⁸F-FDG (2-fluoro-2-deoxy-D-glucose; ¹⁸F $t_{1/2}$ = 1.8 hrs), which is routinely used for functional studies of brain, heart, and tumor growth. The process is derived from the earlier animal studies quantifying regional glucose metabolism with [¹⁴C]-2-deoxyglucose, which passes through the blood-brain barrier by the same carrier-facilitated transport system used for glucose. With the advancement in the development of highly selective PET and SPECT ligands, the potential of the noninvasive imaging procedures will achieve wider application both in pharmacologic research and diagnosis of CNS disorders.

The Next Wave in Drug Discovery: Genomics

Imatinib (Gleevec) was discovered through the combined use of high-throughput screening and medicinal chemistry that resulted in the successful treatment of chronic myeloid leukemia. Through rational molecular modifications based on an understanding of the structure of logical alternative tyrosine kinase targets, improved activity against the platelet-derived growth factor receptor (PDGFR), epidermal growth factor receptor (EGFR) and vascular endothelial growth factor receptor (VEGFR) have been obtained. As a result of the success of imatinib, scientists are modifying their drug discovery and development strategies to one that considers the patient's genes, without abandoning the more traditional drugs. It has been known for many years that genetics plays an important role in an individual's well-being. Attention is now being paid to manipulating the proteins that are produced in response to malfunctioning genes by inhibiting the out-of-control tyrosine kinase enzymes in the body that play such an important role in cell signaling events in growth and cell division. Using the human genome, scientists with knowledge of

the sequencing of DNA and genes of various species have shown that some cancers are caused by genetic errors that direct the biosynthesis of dysfunctional proteins. Because proteins carry out the instructions from the genes located on the DNA, dysfunctional proteins such as the kinases deliver the wrong message to the cells, making them cancerous. The emphasis is now to inhibit the proteins in order to slow the progression of the cancerous growth.

An emphasis in the pharmaceutical industry and in academia is to develop drug formulations that guarantee that therapies will reach specific targets in the body. Vaccines based on a proprietary plasmid DNA that will activate skeletal muscles to manufacture desired proteins and antigens are being developed. Plasmid DNA vaccine technology represents a fundamentally new means of treatment that is of great importance for the future of drug targeting. There is currently an increase in the number of products coming out of biotechnology companies. Biotechnology drug discovery and drug development tools are used to create the more traditional small molecules. The promise of pharmacogenetics lies in the potential to identify sources of interindividual variability in drug responses that affect drug delivery and safety. Recent success stories in oncology demonstrate that the field of pharmacogenetics has progressed substantially. The knowledge created through pharmacogenetic trials can contribute to the development of patient-specific medicines as well as to improved decision making along the research and development value chain (6).

Combinatorial Chemistry and High-Throughput Screening

No discussion of the history and evolution of medicinal chemistry would be complete without briefly mentioning combinatorial chemistry and high-throughput screening. Combinatorial chemistry is one of the new technologies developed by academics and researchers in the pharmaceutical and biotechnology industries to reduce the time and cost associated with producing effective, marketable, and competitive new drugs. Chemists use combinatorial chemistry to create large populations of molecules that can be screened efficiently, generally using high-throughput screening. Thus, instead of synthesizing a single compound, combinatorial chemistry exploits automation and miniaturization to synthesize large libraries of compounds. Combinatorial organic synthesis is not random, but systematic and repetitive, using sets of chemical “building blocks” to form a diverse set of molecular entities.

Random screening has been a source of new drugs for several decades. Many of the drugs currently on the market were developed from leads identified through screening of natural products or compounds synthesized in the laboratory. However, in the late 1970s and 1980s, screening fell out of favor in the industry. Using traditional methods, the number of novel selective leads generated did not make this approach cost effective. The last 25 years have seen an enormous advance in the

understanding of critical cellular processes, leading to a more rationally designed approach in drug discovery. The availability of cloned genes for use in high-throughput screening to identify new molecules has led to a reexamination of the screening process. Targets are now often recombinant proteins (i.e., receptors) produced from cloned genes that are heterologously expressed in a number of ways. Combinatorial libraries complement the enormous numbers of synthetic libraries available from new and old synthetic programs. The development and use of robotics and automation have made it possible to screen large numbers of compounds in a short period of time. It should also be emphasized that computerized data systems and the analysis of the data have facilitated the handling of the information being generated, leading to the identification of new leads.

SUMMARY

It is fair to say that more than 50% of the drugs in use today had their origin in a plant, animal, or mineral that had been used as a cure for alleviating disease occurring in man. Examples of a number of discoveries of important drugs in use today are recounted as “case studies” in the drug discovery process and are described in more detail in the following chapters. The discoveries briefly described are in large measure due to the increased sophistication brought to bear in the isolation, identification, structure determination, and synthesis of the active ingredients of the drugs used empirically hundreds of years ago.

The emergence of the pharmaceutical industry took place in conjunction with the advances in organic/medicinal/pharmaceutical chemistry, pharmacology, bacteriology, biochemistry, and medicine as distinct fields of science in the late 19th century. Current research efforts are now focused not only on discovering new biologically active compounds using ever increasingly sophisticated technology, but also on gaining a better understanding of how and where drugs exert their effects at the molecular level. One should not underestimate, however, that the discoveries in the 20th and 21st centuries and earlier represent an amazing amount of insight, determination, and luck by researchers in chemistry, pharmacology, biology, and medicine. We owe gratitude and admiration to those earlier scientists who had the imagination and inspiration to develop drugs to cure so many illnesses.

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Part **I**

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Chapter 1

Drug Discovery from Natural Products

A. DOUGLAS KINGHORN

Abbreviations

CAM, complementary and alternative medicine

CNS, central nervous system

COPD, chronic obstructive pulmonary disease

DSHEA, Dietary Supplement Health and Education Act

GLP-1, glucagon-like peptide-1

HMG-CoA, 5-hydroxy-3-methylglutaryl-coenzyme A

HPLC, high-performance liquid chromatography

HTS, high-throughput screening

LC, liquid chromatography

M6G, morphine-6-glucuronide

MOA, memorandum of agreement

MS, mass spectrometry

NCE, new chemical entity

NMR, nuclear magnetic resonance

PVP, polyvinylpyrrolidone

SCE, single chemical entity

SPE, solid-phase extraction

THC, tetrahydrocannabinol

UNCLOS, United Nations Convention on the Law of the Sea

INTRODUCTION

“Pharmacognosy” is one of the oldest established pharmaceutical sciences, and the term has been used for nearly two centuries. Initially, this term referred to the investigation of medicinal substances of plant, animal, or mineral origin in their crude or unprepared state, used in the form of teas, tinctures, poultices, and other types of formulation (1–4). However, by the middle of the 20th century, the chemical components of such crude drugs began to be studied in more detail. Today, the subject of pharmacognosy is highly interdisciplinary, and incorporates aspects of analytical chemistry, biochemistry, biosynthesis, biotechnology, ecology, ethnobotany, microbiology, molecular biology, organic chemistry, and taxonomy, among others (5). The term “pharmacognosy” is defined on the Web site of the American Society of Pharmacognosy (www.phcog.org) as “the study of the

physical, chemical, biochemical, and biological properties of drugs, drug substances, or potential drugs or drug substances of natural origin, as well as the search for new drugs from natural sources.”

There seems little doubt that humans have used natural drugs since before the advent of written history. In addition to their use as drugs, the constituents of plants have afforded poisons for darts and arrows used in hunting and euphorants with psychoactive properties used in rituals. The actual documentation of drugs derived from natural products in the Western world appears to date as far back to the Sumerians and Akkadians in the third century BCE, as well as the Egyptian *Ebers Papyrus* (about 1600 BCE). Other important contributions on the uses of drugs of natural origin were documented by Dioscorides (*De Materia Medica*) and Pliny the Elder in the first century CE and by Galen in the second century. Written records also exist from about the same time period on plants

used in both Chinese traditional medicine and Ayurvedic medicine. Then, beginning about 500 years ago, information on medicinal plants began to be documented in herbals. In turn, the laboratory study of natural product drugs commenced approximately 200 years ago, with the purification of morphine from opium. This corresponds with the beginnings of organic chemistry as a scientific discipline. Additional drugs isolated from plant sources included atropine, caffeine, cocaine, nicotine, quinine, and strychnine in the 19th century, and then digoxin, reserpine, paclitaxel, vincristine, and chemical precursors of the steroid hormones in the 20th century. Even as we enter the second decade of the 21st century, approximately three quarters of the world's population are reliant on primary health care from systems of traditional medicine, including the use of herbs. A more profound understanding of the chemical and biologic aspects of plants used in the traditional medicine of countries such as the People's Republic of China, India, Indonesia, and Japan has occurred in recent years, in addition to the medicinal plants used in Latin America and Africa. Many important scientific observations germane to natural product drug discovery have been made as a result (1–4).

By the mid-20th century, therapeutically useful alkaloids had been purified and derivatized from the ergot fungus, as uterotonic and sympatholytic agents. Then, the penicillins were isolated along with further major structural classes of effective and potent antibacterials from terrestrial microbes, and these and later antibiotics revolutionized the treatment of infectious diseases. Of the types of organisms producing natural products, terrestrial microorganisms have been found to afford the largest number of compounds currently used as drugs for a wide range of human diseases, and these include antifungal agents, the “statin” cholesterol-lowering agents, immunosuppressive agents, and several anticancer agents (6,7).

At present, there remains much interest also in the discovery and development of drugs from marine animals and plants. However, to date, marine organisms have had a relatively brief history in serving as sources of drugs, with only a few examples approved for therapeutic use thus far. Although the oceans occupy 70% of the surface of the earth, an intense effort to investigate the chemical structures and biologic activities of the marine fauna and flora has only been ongoing for about 40 years (8).

The term “natural product” is generally taken to mean a compound that has no known primary biochemical role in the producing organism. Such low molecular weight organic molecules may also be referred to as “secondary metabolites” and tend to be biosynthesized by the producing organism in a biologically active chiral form to increase the chances of survival, such as by repelling predators or serving as insect pollination attractants, in the case of plants (9). There have been a number of studies to investigate the physicochemical parameters of natural products in recent years, and it has been concluded that “libraries” or collections of these substances

tend to afford a higher degree of “drug-likeness,” when compared with compounds in either synthetic or combinatorial “libraries” (10,11). This characteristic might well be expected, since natural products are produced by living systems, where they are subject to transport and diffusion at the cellular level. Small-molecule natural products are capable of modulating protein–protein interactions and can thus affect cellular processes that may be modified in disease states. When compared to synthetic compounds, natural products tend to have more protonated amine and free hydroxy functionalities and more single bonds, with a greater number of fused rings containing more chiral centers. Natural products also differ from synthetic products in the average number of halogen, nitrogen, oxygen, and sulfur atoms, in addition to their steric complexity (12,13). It is considered that natural products and synthetic compounds occupy different regions of “chemical space,” and hence, they each tend to contribute to overall chemical diversity required in a drug discovery program (13). Fewer than 20% of the ring systems produced among natural products are represented in currently used drugs (10). Naturally occurring substances may serve either as drugs in their native or unmodified form or as “lead” compounds (prototype bioactive molecules) for subsequent semisynthetic or totally synthetic modification, for example, to improve biologic efficacy or to enhance solubility (1–4,6,8,10,11).

In the present era of efficient drug design by chemical synthesis aided by computational and combinatorial techniques, and with other new drugs obtained increasingly by biotechnologic processes, it might be expected that traditional natural products no longer have a significant role to play in this regard. Indeed, in the past two decades, there has been a decreased emphasis on the screening of natural products for new drugs by pharmaceutical companies, with greater reliance placed on screening large libraries of synthetic compounds (10,11,14,15). However, in a major review article, Newman and Cragg from the U.S. National Cancer Institute pointed out that for the period from 1981 to 2006, about 28% of the new chemical entities (NCEs) in Western medicine were either natural products per se or semisynthetic derivatives of natural products. Thus, of a total of 1,184 NCEs for all disease conditions introduced into therapy in North America, Western Europe, and Japan over the 25.5-year period covered, 5% were unmodified natural products and 23% were semisynthetic agents based on natural product lead compounds. An additional 14% of the synthetic compounds were designed based on knowledge of a natural product “pharmacophore” (the region of the molecule containing the essential organic functional groups that directly interact with the receptor active site and, therefore, confers the biologic activity of interest) (16). The launch of new natural product drugs in Western countries and Japan has continued in the first decade of 21st century, and such compounds introduced to the market recently have been documented (14,16–18).

Thus, it is generally recognized that the secondary metabolites of organisms afford a source of small organic molecules of outstanding chemical diversity that are highly relevant to the contemporary drug discovery process. Potent and selective leads are obtained from more exotic organisms than before, as collection efforts venture into increasingly inhospitable locales throughout the world, such as deep caves in terrestrial areas and thermal vents on the ocean floor. On occasion, a natural lead compound may help elucidate a new mechanism of interaction with a biologic target for a disease state under investigation. Natural products may serve to provide molecular inspiration in certain therapeutic areas for which there are only a limited number of synthetic lead compounds. A valuable approach is the large-scale screening of libraries of partially purified extracts from organisms (11). However, there is a widespread perception that the resupply of the source organism of a secondary metabolite of interest may prove problematic and will consequently hinder the timely, more detailed, biologic evaluation of a compound available perhaps only in milligram quantities initially. In addition, natural product extracts have been regarded as incompatible with the modern rapid screening techniques used in the pharmaceutical industry, and some believe that the successful market development of a natural product-derived drug is too time consuming (10,11,14,15). A further consideration of the factors involved in the discovery of drugs from natural products will be presented in the next section of this chapter. This will be followed by examples of natural products currently used in various therapeutic categories, as well as a few selected representatives with present clinical use or future potential in this regard.

NATURAL PRODUCTS AND DRUG DISCOVERY

Collection of Source Organisms

There are at least five recognized approaches to the choice of plants and other organisms for the laboratory investigation of their biologic components, namely, random screening; selection of specific taxonomic groups, such as families or genera; a chemotaxonomic approach where restricted classes of secondary metabolites such as alkaloids are sought; an information-managed approach, involving the target collection of species selected by database surveillance; and selection by an ethnomedical approach (e.g., by investigating remedies being used in traditional medicine by “shamans” or medicine men or women) (19). In fact, if plant-derived natural products are taken specifically, it has been estimated that of 122 drugs of this type used worldwide from a total of 94 species, 72% can be traced to the original ethnobotanical uses that have been documented for their plant of origin (19). The need for increased natural products discovery research involving ethnobotany should be regarded as urgent, due to the accelerating loss in developing countries of indigenous cultures and languages, inclusive of

knowledge of traditional medical practice (20). However, it is common for a given medicinal plant to be used ethnomedically in more than one disease context, which may sometimes obscure its therapeutic utility for a specific disease condition. Another manner in which drugs have been developed from terrestrial plants and fungi is through following up on observations of the causes of livestock poisoning, leading to new drugs and molecular tools for biomedical investigation (21). When the origin of plants with demonstrated inhibitory effects in experimental tumor systems was considered at the U.S. National Cancer Institute, medicinal or poisonous plants with uses as either anthelmintics or arrow and homicidal poisons were three to four times more likely to be active in this regard than species screened at random (22).

Although some shallow water marine specimens may be collected simply by wading or snorkeling down to 20 feet below the water surface, scuba diving permits the collection of organisms to depths of 120 feet. Deep-water collections of marine animals and plants have been made by dredging and trawling and through the use of manned and unmanned submersible vessels. Collection strategies for specimens from the ocean must take into account marine macroorganism–microorganism associations that may be involved in the biosynthesis of a particular secondary metabolite of interest (8). Thus, there seems to be a complex interplay between many marine host invertebrate animals and symbiotic microbes that inhabit them, and it has been realized that several bioactive compounds previously thought to be of animal origin may be produced by their associated microorganisms instead (23).

The process of collecting or surveying a large set of flora (or fauna) for the purpose of the biologic evaluation and isolation of lead compounds is called “biodiversity prospecting” (24). Many natural products collection programs are focused on tropical rain forests, in order to take advantage of the inherent biologic diversity (or “biodiversity”) evident there, with the hope of harnessing as broad a profile of chemical classes as possible among the secondary metabolites produced by the species to be obtained. To exemplify this, there may be more tree species in a relatively small area of a tropical rainforest than in the whole of the temperate regions of North America. A generally accepted explanation for the high biodiversity of secondary metabolites in humid forests in the tropics is that these molecules are biosynthesized (a process of chemical synthesis by the host organism) for ecologic roles, in response to a continuous growing season under elevated temperatures, high humidity, and great competition due to the high species density present. Maximal biodiversity in the marine environment is found on the fringes of the ocean or sea bordering land, where there is intense competition among sessile (nonmoving) organisms, such as algae, corals, sponges, and some other invertebrate animals, for attachment space (25).

Great concern should be expressed about the continuing erosion of tropical rain forest species, which

is accelerating as the 21st century develops (26). Approximately 25 “hot spots” of especially high biodiversity have been proposed that represent 44% of all vascular plant species and 35% of all species of vertebrates in about 1.4% of the earth’s surface (27). At present, many of the endemic (or native) species to these biodiversity “hot spot” areas have been reported to be undergoing massive habitat loss and are threatened with extinction, especially in tropical regions (26,27).

After the United Nations Convention on Biological Diversity, passed in Rio de Janeiro in 1992, biologic or genetic materials are owned by the country of origin (24,28). A major current-day component of being able to gain access to the genetic resources of a given country for the purposes of drug discovery and other scientific study is the formulation of a memorandum of agreement (MOA), which itemizes access, prior informed consent (involving human subjects in cases where ethnomedical knowledge is divulged), intellectual property related to drug discovery, and the equitable sharing of financial benefits that may accrue from the project, such as patent royalties and licensing fees (24,28). When access to marine organisms is desired, the United Nations Convention on the Law of the Sea (UNCLOS) must also be considered (29).

Once a formal “benefit sharing” agreement is on hand, the organism collection process can begin. It is usual to initially collect 0.3 to 1 kg of each dried plant sample and about 1 kg wet weight of a marine organism for preliminary screening studies (30). In the case of a large plant (tree or shrub), it is typical to collect up to about four different organs or plant parts, since it is known that the secondary metabolite composition may vary considerably between the leaves, where photosynthesis occurs, and storage or translocation organs such as the roots and bark (31). There is increasing evidence that considerable variation in the profile of secondary metabolites occurs in the same plant organ when collected from different habitats, depending on local environmental conditions, and thus it may be worth reinvestigating even well-studied species in drug discovery projects. Taxa endemic (native) to a particular country or region are generally of higher priority than the collection of pandemic weeds. It is very important never to remove all quantities of a desired species at the site of collection, in order to conserve the native germplasm encountered. Also, rare or endangered species should not be collected; a listing of the latter is maintained by the *Red List of Threatened Species* of the International Union for Conservation of Nature and Natural Resources (www.redlist.org), covering terrestrial, marine, and freshwater organisms.

A crucial aspect of the organism collection process is to deposit voucher specimens representative of the species collected in a central repository such as a herbarium or a museum, so that this material can be accessed by other scientists, in case of need. It is advisable to deposit specimens in more than one repository, including regional and national institutions in the country in

which the organisms were collected. Collaboration with general and specialist taxonomists is very important, because without an accurate identification of a source organism, the value of subsequent isolation, structure elucidation, and biologic evaluation studies will be greatly reduced (31).

Organisms for natural products drug discovery work may be classified into the following kingdoms: Eubacteria (bacteria, cyanobacteria [or “blue-green algae”]), Archaea (halobacterians, methanogens), Protoctista (e.g., protozoa, diatoms, “algae” [including red algae, green algae]), Plantae (land plants [including mosses and liverworts, ferns, and seed plants]), Fungi (e.g., molds, yeasts, mushrooms), and Animalia (mesozoa [wormlike invertebrate marine parasites], sponges, jellyfish, corals, flatworms, roundworms, sea urchins, mollusks [snails, squid], segmented worms, arthropods [crabs, spiders, insects], fish, amphibians, birds, mammals) (24). Of these, the largest numbers of organisms are found for arthropods, inclusive of insects (~950,000 species), with only a relatively small proportion (5%) of the estimated 1.5 million fungi in the world having been identified. At present, with 300,000 to 500,000 known species, plants are the second largest group of classified organisms, representing about 15% of our biodiversity. Of the 28 major animal phyla, 26 are found in the sea, with eight of these exclusively so. There have been more than 200,000 species of invertebrate animals and algal species found in the sea (24). A basic premise inherent in natural products drug discovery work is that the greater the degree of phylogenetic (taxonomic) diversity of the organisms sampled, the greater the resultant chemical diversity that is evident.

Interest in investigating plants as sources of new biologically active molecules remains strong, in part because of a need to better understand the efficacy of herbal components of traditional systems of medicine (32). In the last decade, many new natural product molecules have been isolated from fungal sources (6,7). An area of investigation of great potential expansion in the future will be on other microbes, particularly of actinomycetes and cyanobacteria of marine origin, especially if techniques can continue to be developed for their isolation and culturing in the laboratory (33). Because as many as 99% of known microorganisms are not able to be cultivated under laboratory conditions, the technique of “genome mining” isolates their DNA and enables new secondary metabolite biochemical pathways to be exploited, leading to the possibility of producing new natural products (34). The endophytic fungi that reside in the tissue of living plants have been found to produce an array of biologically interesting new compounds and are worthy of more intensive investigation (35). It is of interest to note that in a survey of the origin of 30,000 structurally assigned lead compounds of natural origin, the compounds were derived from animals (13%), bacteria (33%), fungi (26%), and plants (27%) (12). For the year 2008, it was reported that 24 animal-, 25 bacterial-, 7 fungal-, and

108 plant-derived natural products were undergoing at least phase I clinical trials leading to drug development (36). Therefore, while natural product researchers tend to specialize in the major types of organism on which they work, it is reasonable to expect that the future investigation of all of their major groups mentioned earlier will provide dividends in terms of affording new prototype biologically active compounds of use in drug discovery.

Preparation of Initial Extracts and Preliminary Biologic Screening

Although different laboratories tend to adopt different procedures for initial extraction of the source organisms being investigated, it is typical to extract initially terrestrial plants with a polar solvent like methanol or ethanol, and then subject this to a defatting (lipid-removing) partition with a nonpolar solvent like hexane or petroleum ether, and then partition the residue between a semipolar organic solvent, such as chloroform or dichloromethane, and a polar aqueous solvent (31). Marine and aquatic organisms are commonly extracted fresh into methanol or a mixture of methanol–dichloromethane (30). A peculiarity of working on plant extracts is the need to remove a class of compounds known as “vegetable tannins” or “plant polyphenols” before subsequent biologic evaluation because these compounds act as interfering substances in enzyme inhibition assays, as a result of precipitating proteins in a nonspecific manner. Several methods to remove plant polyphenols have been proposed, such as passage over polyvinylpyrrolidone (PVP) and polyamide, on which they are retained. Alternatively, partial removal of these interfering substances may be effected by washing the final semipolar organic layer with an aqueous sodium chloride solution (31). However, it should be pointed out that there remains an active interest in pursuing purified and structurally characterized vegetable tannins for their potential medicinal value (37). Caution also needs to be expressed in regard to common saturated and unsaturated fatty acids that might be present in natural product extracts, because these may interfere with various enzyme inhibition and receptor binding assays. Fatty acids and other lipids may largely be removed from more polar natural product extracts, using the defatting solvent partition stage mentioned earlier (38).

Drug discovery from organisms is a “biology-driven” process, and as such, biologic activity evaluation is at the heart of the drug discovery process from crude extracts prepared from organisms. So-called high-throughput screening (HTS) assays have become widely used for affording new leads. In this process, large numbers of crude extracts from organisms can be simultaneously evaluated in a cell-based or non-cell-based format, usually using multiwell microtiter plates (39). Cell-based *in vitro* bioassays allow for a considerable degree of biologic relevance, and manipulation may take place so that a selected cell line may involve a genetically altered organism (40) or incorporate a reporter gene (41). In

noncellular (cell-free) assays, natural products extracts and their purified constituents may be investigated for their effects on enzyme activity (42) or on receptor binding (43). Other homogenous and separation-based assays suitable for the screening of natural products have been reviewed (44). For maximum efficiency and speed, HTS may be automated through the use of robotics and may be rendered as a more effective process through miniaturization.

Methods for Compound Purification and Structure Elucidation and Identification

Bioassay-directed fractionation is the process of isolating pure active constituents from some type of biomass (e.g., plants, microbes, marine invertebrates) using a decision tree that is dictated solely by bioactivity. A variety of chromatographic separation techniques are available for these purposes, including those based on adsorption on sorbents, such as silica gel, alumina, Sephadex, and more specialized solid phases, and methods involving partition chromatography inclusive of counter-current chromatography (45). Recent improvements have been made in column technology, automation of high-performance liquid chromatography (HPLC; a technique often used for final compound purification), and compatibility with HTS methodology (46). Routine structure elucidation is performed using combinations of spectroscopic procedures, with particular emphasis on ^1H - and ^{13}C -nuclear magnetic resonance (NMR) spectroscopy and mass spectrometry (MS). Considerable progress has been made in the development of cryogenic and capillary NMR probe technology, for the determination of structures of sub-milligram amounts of natural products (47). In addition, the automated processing of spectroscopic data for the structure elucidation of natural products is a practical proposition (48). Another significant advance is the use of “hyphenated” analytical techniques for the rapid structure determination of natural products without the need for a separate isolation step, such as liquid chromatography (LC)-NMR and LC-NMR-MS (11,46). The inclusion of an online solid-phase extraction (SPE) cartridge is advantageous in the identification of natural product molecules in crude extracts using LC-NMR, coupled with MS and circular dichroism spectroscopy (49).

Dereplication is a process of determining whether an observed biologic effect of an extract or specimen is due to a known substance. This is applied in natural product drug discovery programs in an attempt to avoid the re-isolation of compounds of previously determined structure. A step like this is essential to prioritize the resources available to a research program, so that the costly stage of bioassay-directed fractionation on a promising lead crude extract can be devoted to the discovery of biologically active agents representing new chemotypes (46,50). This has been particularly necessary for many years in studies on anti-infective agents from actinomycetes and bacteria and is also routinely applied to extracts from marine invertebrates and higher plants. Methods for

dereplication must be sensitive, rapid, and reproducible, and the analytical methods used generally contain a mass spectrometric component (50). For example, the eluant (effluent) from an HPLC separation of a crude natural product extract may be split into two portions, so that the major part is plated out into a microtiter plate, with the wells then evaluated in an *in vitro* bioassay of interest. The fractions from the minor portion of the column eluant are introduced to a mass spectrometer, and the molecular weights of compounds present in active fractions can be determined. This information may then be introduced into an appropriate natural products database, and tentative identities of the active compounds present in the active wells can be determined (50).

Metabolomics is a recently developed approach in which the entire or “global” profile of secondary metabolites in a system (cell, tissue, or organism) is catalogued under a given set of conditions. Secondary metabolites may be investigated by a detection step such as MS after a separation step such as gas chromatography, HPLC, or capillary electrophoresis (51). This type of technology has particular utility in systematic biology, genomics research, and biotechnology and should have value in future natural products drug discovery (51,52).

Compound Development

A major challenge in the overall natural products drug discovery process is to obtain larger amounts of a biologically active compound of interest for additional laboratory investigation and potential preclinical development. One strategy that can be adopted when a plant-derived active compound is of interest is to obtain a recollection of the species of origin. To maximize the likelihood that the recollected sample will contain the bioactive compound of previous interest, the plant recollection should be carried out in the same location as the initial collection, on the same plant part, at the same time of the year (31). Some success has been met with the production of terrestrial plant metabolites via plant tissue culture (53). For microbes of terrestrial origin, compound scale up usually may be carried out through cultivation and large-scale fermentation (6,7).

Although evaluation of crude extracts of organisms is not routinely performed in animal models because of limitations of either test material or other project resources, it is of great value to test *in vitro*–active natural products in a pertinent *in vivo* method to obtain a preliminary indication of the worthiness of a lead compound for preclinical development. There are also a variety of “secondary discriminator” bioassays that provide an assessment of whether or not a given *in vitro*–active compound is likely to be active *in vivo*, and these require quite small amounts of test material. For example, the *in vivo* hollow fiber assay was developed at the U.S. National Cancer Institute for the preliminary evaluation of potential anticancer agents and uses confluent cells of a tumor model of interest deposited in polyvinylidene fluoride fibers that are implanted in nude mice (31,54). It is also

important for pure bioactive compounds to be evaluated mechanistically for their effects on a particular biologic target, such as on a given stage of the life cycle of a pathogenic organism or cancer cell. Needless to say, a pure natural product of novel structure with *in vitro* and *in vivo* activity against a particular biologic target relevant to human disease acting through a previously unknown mechanism of action is of great value in the drug discovery process.

Once a bioactive natural product lead is obtained in gram quantities, it is treated in the same manner as a synthetic drug lead and is thus subjected to pharmaceutical development, leading to preclinical and clinical trials. This includes lead optimization via medicinal chemistry, combinatorial chemistry, and computational chemistry, as well as formulation, pharmacokinetics, and drug metabolism studies, as described elsewhere in this volume. Often, a lead natural product is obtained from its organism of origin along with several naturally occurring structural analogs, permitting a preliminary structure–activity relationship study to be conducted. This information may be supplemented with data obtained by microbial biotransformation or the production of semisynthetic analogs, to allow researchers to glean some initial information about the pharmacophoric site(s) of the naturally occurring molecule (10,11). Combinatorial biosynthesis is a contemporary approach with the ability to produce new natural product analogs, or so-called “unnatural” natural products, and these may be used to afford new drug candidates. This methodology involves the engineering of biosynthetic gene clusters in microorganisms and has been applied to the generation of polyketides, peptides, terpenoids, and other compounds. New advances in the biochemical and protein engineering aspects of this technique have led to a greater applicability than previously possible (55).

SELECTED EXAMPLES OF NATURAL PRODUCT–DERIVED DRUGS

In the following sections, examples are provided of both naturally occurring substances and synthetically modified compounds based on natural products with drug use. It is evident that many of the examples shown reflect considerable structural complexity and that the compounds introduced to the market have been obtained from organisms of very wide diversity. More detailed treatises with many more examples of natural product drugs are available (e.g., see references 1–4). Several recent reviews have summarized natural product drugs introduced to the market in recent years and substances on which clinical trials are being conducted (16–18,36).

Drugs for Cardiovascular and Metabolic Diseases

There is a close relationship between natural product drugs and the treatment of cardiovascular and

metabolic diseases. The powdered leaves of *Digitalis purpurea* have been used in Western medicine for more than 200 years, with the major active constituent being the cardiac (steroidal) glycoside digitoxin, which is still used now for the treatment of congestive heart failure and atrial fibrillation. A more widely used drug used today is digoxin, a constituent of *Digitalis lanata*, which has a rapid action and is more rapidly eliminated from the body than digitoxin. Deslanoside (deacetyl-lanatoside C) is a hydrolysis product of the *D. lanata* constituent lanatoside C and is used for rapid digitalization (1–4). The “statin” drugs used for lowering blood cholesterol levels are based on the lead compound mevastatin (formerly known as compactin), produced by cultures of *Penicillium citrinum*, and were discovered using a 5-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase assay. Because hypercholesterolemia is regarded as one of the major risk factors for coronary heart disease, several semisynthetic and synthetic compounds modeled on the mevastatin structure (inclusive of the dihydroxycarboxylic acid side chain) now have extremely wide therapeutic use, including atorvastatin, fluvastatin, pravastatin, and simvastatin. Lovastatin is a natural product drug of this type, isolated from *Penicillium brevecompactum* and other organisms (3). There is also a past history of the successful production of cardiovascular agents from a terrestrial vertebrate, namely, the angiotensin-converting enzyme inhibitors captopril and enalapril, which were derived from tetrotide, a nonapeptide isolated from the pit viper, *Bothrops jararaca* (56).

Two further new drugs derived from an invertebrate and a vertebrate source, respectively, are bivalirudin and exenatide. Bivalirudin is a specific and reversible direct thrombin inhibitor that is administered by injection and is used to reduce the incidence of blood clotting in patients undergoing coronary angioplasty. This compound is a synthetic, 20-amino acid peptide and was modeled on hirudin, a substance present in the saliva of the leech, *Haementeria officinalis* (57,58). Exenatide is a synthetic version of a 39-amino acid peptide (exenatide-4), produced by a lizard native to the southwest United States and northern Mexico, called the Gila monster, *Heloderma suspectum*, and acts in the same manner as glucagon-like peptide-1 (GLP-1), a naturally occurring hormone. This drug is also administered by injection and enables improved glycemic control in patients with type 2 diabetes (18,59).

Central and Peripheral Nervous System Drugs

A comprehensive review has appeared on natural products (mostly of experimental value) that affect the central nervous system (CNS), inclusive of potential analgesics, antipsychotics, anti-Alzheimer disease agents, antitussives, anxiolytics, and muscle relaxants, among other categories. The authors point out that apart from the extensive past literature on plants and their constituents as hallucinogenic agents, this area of research inquiry on natural products is not well developed but is likely to be productive in the future (60). Natural products also have the potential to treat drug abuse (61).

The morphinan isoquinoline alkaloid, (–)-morphine, is the most abundant and important constituent of the dried latex (milky exudate) of *Papaver somniferum* (opium poppy) and the prototype of the synthetic opioid analgesics, being selective for μ -opioid receptors (Fig. 1.1). This compound may be considered the paramount natural product lead compound, with many thousands of analogs synthesized in an attempt to obtain derivatives with strong analgesic potency but without any addictive tendencies (1–4). One derivative now in late clinical trials as a pain treatment is morphine-6-glucuronide (M6G), the major active metabolite of morphine, with fewer side effects than the parent compound (18,62). The pyridine alkaloid epibatidine, isolated from a dendrobatid frog (*Epipedobates tricolor*) found in Ecuador, activates nicotinic receptors and has a 200-fold more potent analgesic activity than morphine. The drug potential of epibatidine is limited by its concomitant toxicity, but it is an important lead compound for the development of future new analgesic agents with less addictive liability than the opiate analgesics (63). A nonopioid analgesic for the amelioration of chronic pain has been introduced to the market recently, namely, ziconotide, which is a synthetic version of the peptide, ω -conotoxin MVIIA. The conotoxin class is produced by the cone snail, *Conus magus*, and these compounds are peptides with 24- to 27-amino acid residues. Ziconotide selectively binds to N-type voltage-sensitive neuronal channels, causing a blockage of neurotransmission and a potent analgesic effect (18,64). This is one of the first examples of a new natural product drug from a marine source.

(–)- Δ^9 -trans-Tetrahydrocannabinol (tetrahydrocannabinol [THC]) is the major psychoactive (euphoriant) constituent of marijuana (*Cannabis sativa*). The synthetic form of THC (dronabinol) was approved more than 25 years ago to treat nausea and vomiting associated with cancer chemotherapy and has been used for a lesser amount

D-Phe-L-Pro-L-Arg-L-Pro-Gly-Gly-Gly-L-Asn-Gly-L-Asp-L-Phe-L-Glu-L-Glu-L-Ile-L-Pro-L-Glu-L-Glu-L-Tyr-L-Leu

Bivalirudin

L-His-Gly-L-Glu-Gly-L-Thr-L-Phe-L-Thr-L-Ser-L-Asp-L-Leu-L-Ser-L-Lys-L-Gln-L-Met-L-Glu-L-Glu-L-Glu-L-Ala-L-Val-L-Arg-L-Leu-L-Phe-L-Ile-L-Glu-L-Trp-L-Leu-L-Lys-L-Asn-Gly-Gly-L-Pro-L-Ser-L-Ser-Gly-L-Ala-L-Pro-L-Pro-L-Pro-L-Ser-NH₂

Exenatide

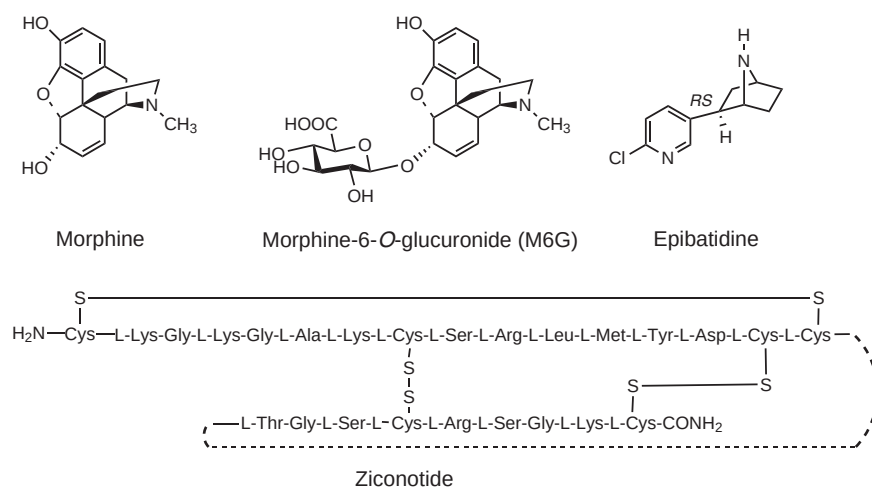
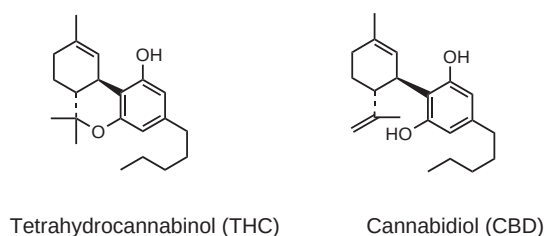
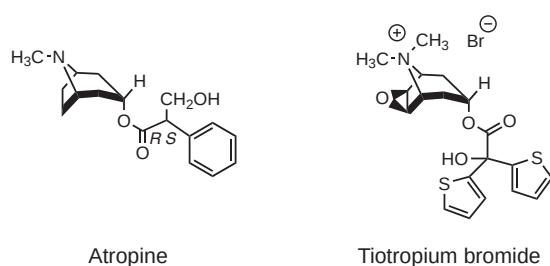


FIGURE 1.1 Analgesic compounds of natural origin or derived from naturally occurring analgesics.

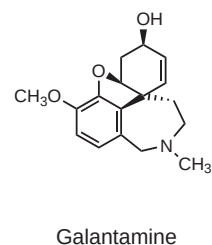
of time to treat appetite loss in HIV/AIDS patients (3). More recently, an approximately 1:1 mixture of THC and the structurally related marijuana constituent cannabidiol (CBD) has been approved in Canada and the United Kingdom for the alleviation of neuropathic pain and spasticity for multiple sclerosis patients and is administered in low doses as a buccal spray (18,65). There is considerable interest in using cannabinoid derivatives based on THC for medicinal purposes, but it is necessary to minimize the CNS effects of these compounds.



Another important natural product lead compound is the tropane alkaloid ester atropine [(±)-hyoscyamine], from the plant *Atropa belladonna* (deadly nightshade). Atropine has served as a prototype molecule for several anticholinergic and antispasmodic drugs. One recently introduced example of an anticholinergic compound modeled on atropine is tiotropium bromide, which is used for the maintenance treatment of bronchospasm associated with chronic obstructive pulmonary disease (COPD) (66).



In the category of anti-Alzheimer disease agents, galantamine hydrobromide is a selective acetylcholinesterase inhibitor that slows down neurologic degeneration by inhibiting this enzyme and by interacting with the nicotinic receptor (67). Galantamine (also known as “galanthamine”) is classified as an Amaryllidaceae alkaloid and has been obtained from several species in this family. Because commercial synthesis is not economical, it is obtained from the bulbs of *Leucojum aestivum* (snowflake) and *Galanthus* species (snowdrop) (1–4). There is some evidence that there is an ethnomedical basis for the current use of galantamine (68).



Anti-infective Agents

Since the introduction of penicillin G (benzylpenicillin) to chemotherapy as an antibacterial agent in the 1940s, natural products have contributed greatly to the field of anti-infective agents. In addition to the penicillins, other classes of antibacterials that have been developed from natural product sources are the aminoglycosides, cephalosporins, glycopeptides, macrolides, rifamycins, and tetracyclines. Antifungals, such as griseofulvin and the polyenes, and avermectins, such as the antiparasitic drug ivermectin, are also of microbial origin (1–4). Of the approximately 90 drugs in this category that were introduced in Western countries, including Japan, in the period from 1981 to 2002, almost 80% can be related to a microbial origin (16). Despite this, relatively few major

pharmaceutical companies are currently working on the discovery of new anti-infective agents from natural sources, due to possible bacterial resistance against new agents and concerns regarding regulation (17). Higher plants have also afforded important anti-infective agents, perhaps most significantly the quinoline alkaloid quinine, obtained from the bark of several *Cinchona* species found in South America, including *Cinchona ledgeriana* and *Cinchona succirubra*. Quinine continues to be used for the treatment of multidrug-resistant malaria and was the template molecule for the synthetic antimalarials chloroquine, primaquine, and mefloquine (1–4).

The following examples have been chosen to represent an array of different structural types of antibacterial agents recently introduced into therapy (Fig. 1.2) (6,14,17,18). Meropenem is a carbapenem (a group of β -lactam antibiotics in which the sulfur atom in the thiazolidine ring is replaced by a carbon) and is based on thienamycin (Fig. 1.2), isolated from *Streptomyces cattleya*. It is a broad-spectrum antibacterial that was introduced into therapy in the last decade as a stable analog of the initially discovered thienamycin (69). Tigecycline (Fig. 1.2) is member of the glycylcycline class of tetracycline antibacterials and is the 9-*tert*-butylglycylamido derivative of minocycline, a semisynthetic derivative of chlorotetracycline from *Streptomyces aureofaciens*. This is a broad-spectrum antibiotic with activity against methicillin-resistant *Staphylococcus aureus* (70). Daptomycin (Fig. 1.2) is the prototype member of the cyclic lipopeptide class of antibiotics and, although isolated initially from

Streptomyces roseosporus, is produced by semisynthesis. This compound binds to bacterial cell membranes, disrupting the membrane potential, and blocks the synthesis of DNA, RNA, and proteins. Daptomycin is bactericidal against gram-positive organisms including vancomycin-resistant *Enterococcus faecalis* and *Enterococcus faecium* and is approved for the treatment of complicated skin and dermal infections (71). Telithromycin (Fig. 1.2) is a semisynthetic derivative of the 14-membered macrolide erythromycin A from *Saccharopolyspora erythraea* and is a macrolide of the ketolide class that lacks a cladinose sugar but has an extended alkyl-aryl unit attached to a cyclic carbamate unit. It binds to domains II and V of the 23S rRNA unit of the bacterial 50S ribosomal unit, leading to inhibition of the ribosome assembly and protein synthesis. This macrolide antibiotic is used to treat bacteria that infect the lungs and sinuses, including community-acquired pneumonia due to *Streptococcus pneumoniae* (72).

Natural products have been a fruitful source of antifungal agents in the past, with the echinocandins being a new group of lipopeptides introduced recently (73). Of these, three compounds are now approved drugs, including the acetate of caspofungin, which is a semisynthetic derivative of pneumocandin B₀, a fermentation product of *Glarea lozoyensis*. Caspofungin inhibits the synthesis of the fungal cell wall β (1,3)-D-glucan, by noncompetitive inhibition of the enzyme β (1,3)-D-glucan synthase, producing both a fungistatic and a fungicidal effect (73). The compound is administered by slow intravenous infusion and is useful in treating infections by *Candida* species (74).

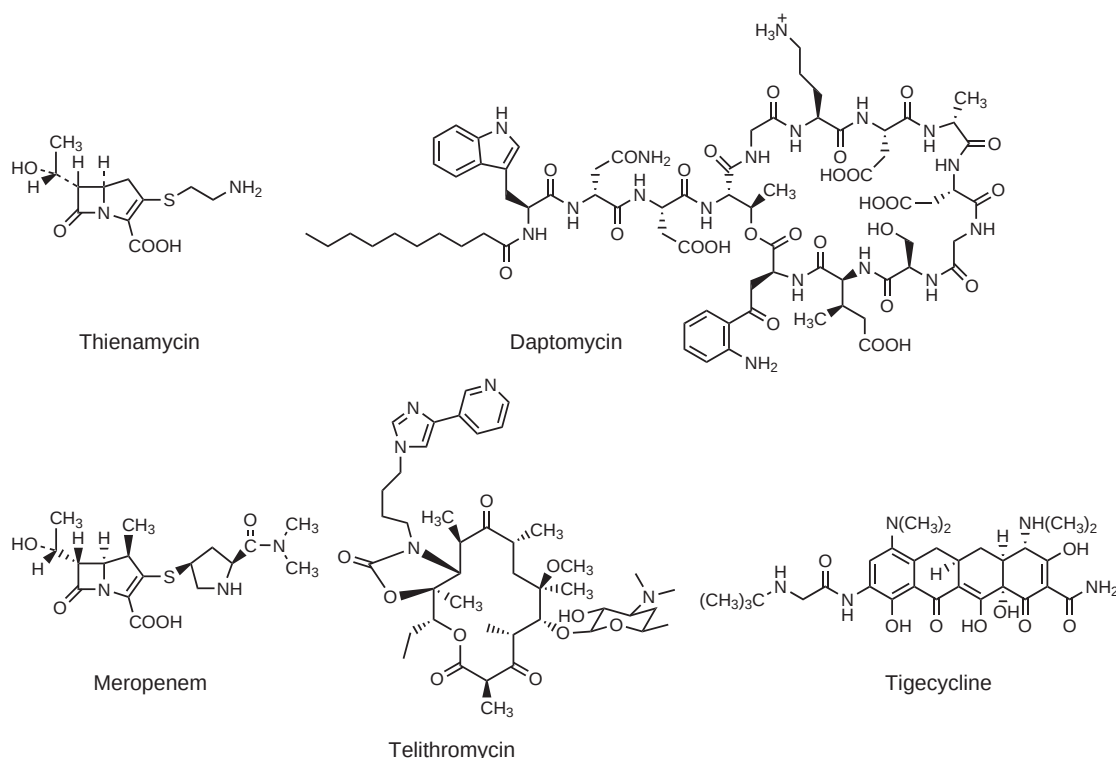
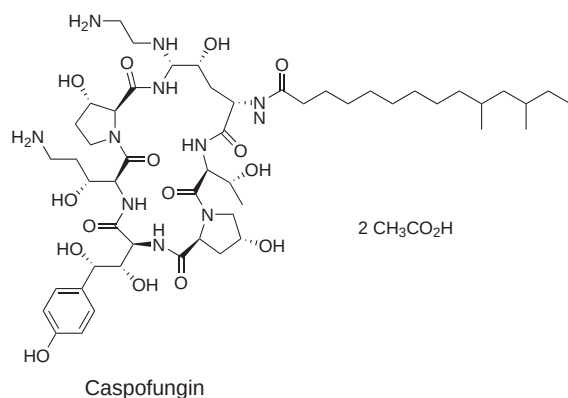


FIGURE 1.2 Examples of Natural and Semisynthetic Anti-infective Agents.



Malaria remains a parasitic scourge that is still extending in incidence. In 1972, the active principle from *Artemisia annua*, a plant used for centuries in Chinese traditional medicine to treat fevers and malaria, was established as a novel antimalarial chemotype. This compound, artemisinin (*qinghaosu* in Chinese), is a sesquiterpene lactone with an endoperoxide group that is essential for activity, and it reacts with the iron in haem in the malarial parasite, *Plasmodium falciparum* (Fig. 1.3). Because this compound is poorly soluble in water, a number of derivatives have been produced with improved formulation, including arteether and artemether. Although animal experiments have suggested that artemisinin derivatives are neurotoxic, this may not be the case in malaria patients (1–4). Artemisinin-based combination treatments such as coartemether (artemether and lume-fantrine) are now widely used for treating drug-resistant *P. falciparum* malaria (75). Coartemether is also known as artemisinin combination therapy and is registered in a large number of countries. A second ether derivative of artemisinin has also been developed, namely, arteether, and is registered in the Netherlands (76).

There are now about 30 approved drugs or drug combinations used to treat HIV/AIDS infections, with most of these being targeted toward the viral enzymes reverse transcriptase or protease. Bevirimat is a semisynthetic 3',3'-dimethylsuccinyl derivative of the oleanane-type triterpenoid betulinic acid, which is found widely in the plant kingdom, including several species used in traditional Chinese medicine. This compound is now undergoing clinical trials as a potential HIV maturation inhibitor (77,78).

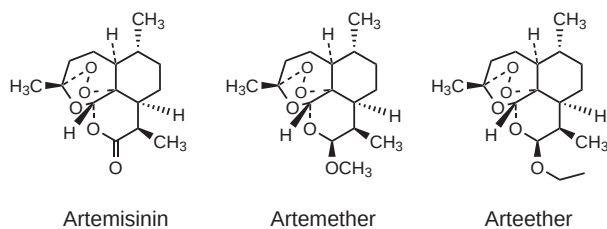
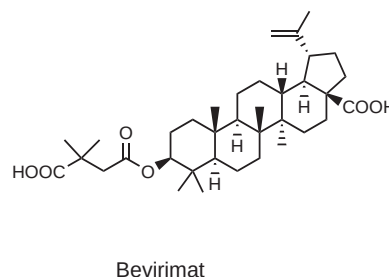


FIGURE 1.3 Artemisinin and two derivatives used for the treatment of malaria.



Anticancer Agents

For several decades, natural products have served as a useful group of structurally diverse cancer chemotherapeutic agents, and many of our most important anticancer agents are of microbial or plant origin. Thus, the antitumor antibiotics include the anthracyclines (daunorubicin, doxorubicin, epirubicin, idarubicin, and valrubicin), bleomycin, dactinomycin (actinomycin D), mitomycin C, and mitoxantrone. Four main classes of plant-derived antitumor agents are used: vinca (*Catharanthus*) bisindole alkaloids (vinblastine, vincristine, and vinorelbine); the semisynthetic epipodophyllotoxin derivatives (etoposide, teniposide, and etoposide phosphate); the taxanes (paclitaxel and docetaxel); and the camptothecin analogs (irinotecan and topotecan) (Fig. 1.4) (1–4,79).

The parent compounds paclitaxel (originally called “taxol”) and camptothecin were both discovered in the laboratory of the late Monroe E. Wall and of Mansukh Wani at Research Triangle Institute in North Carolina (Fig. 1.4). Like some other natural product drugs, several years elapsed from the initial discovery of these substances until their ultimate clinical approval in either a chemically unmodified or modified form. One of the factors that served to delay the introduction of paclitaxel to the market was the need for the large-scale acquisition of this compound from a source other than from the bark of its original plant of origin, the Pacific yew (*Taxus brevifolia*), because this would involve destroying this slow-growing tree. Paclitaxel and its semisynthetic analog docetaxel may be produced by partial synthesis. To enable this, the diterpenoid “building block,” 10-deacetylbaaccatin III, is used as a starting material, which can be isolated from the needles of the ornamental yew, *Taxus baccata*, a renewable botanical resource that can be cultivated in greenhouses (80). A major pharmaceutical company now manufactures paclitaxel by plant tissue culture. The initial source plant of camptothecin, *Camptotheca acuminata*, is a rare species found in regions south of the Yangtze region of the People’s Republic of China. Today, camptothecin is not only produced commercially from cultivated *C. acuminata* trees in mainland China, but also from the roots of *Nothapodytes nimmoniana* (formerly known as both *Nothapodytes foetida* and *Mappia foetida*), which is found in the southern regions of the Indian subcontinent (81). It is of interest to note that these two antineoplastic agents are particularly important not only because of the clinical effectiveness of their derivatives as cancer chemotherapeutic agents, having a significant proportion of the market share (80), but

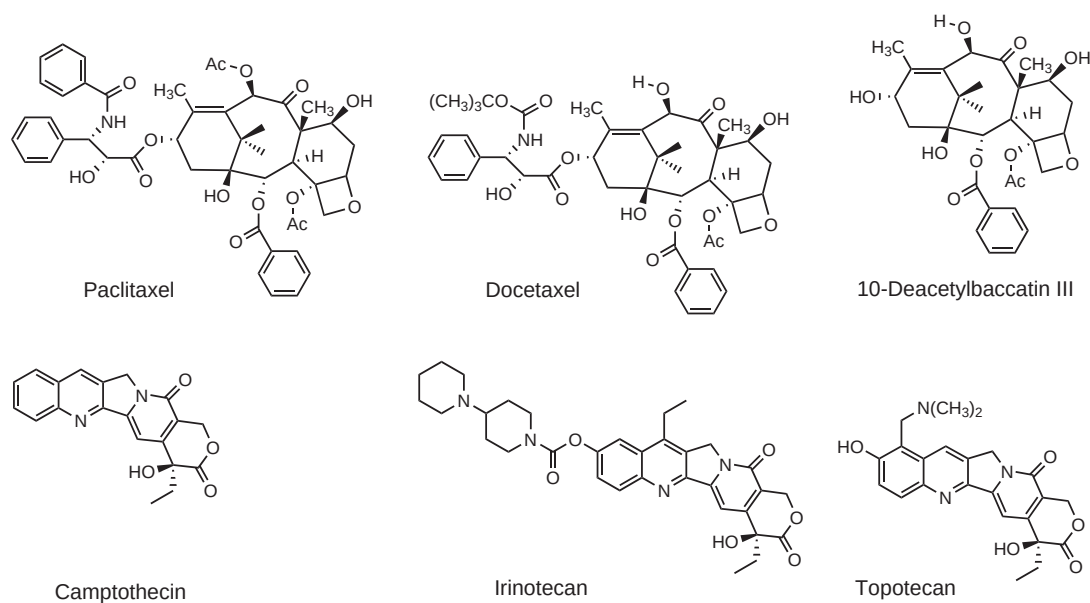


FIGURE 1.4 Lead anticancer compounds paclitaxel, camptothecin, and their respective derivatives.

also because they are prominent lead compounds for synthetic optimization. There are several taxanes and camptothecin derivatives in clinical trial (17,18). Interestingly, endophytic fungi have been reported to produce paclitaxel (82) and camptothecin (83), so it may be possible in the future to produce these important compounds by fermentation rather than by cultivation or other existing methods. Paclitaxel and camptothecin were each found to exhibit a unique mechanism of action for the inhibition of cancer cell growth, with paclitaxel shown to promote the

polymerization of tubulin and the stabilization of microtubules and with camptothecin demonstrated as the first inhibitor of the enzyme DNA topoisomerase I (84).

Several other natural product molecules or their derivatives have been introduced to therapy recently (Fig. 1.5) (17,18,85). Ixabepilone, a semisynthetic derivative of epothilone B, is now marketed in the United States for the treatment of locally advanced and metastatic breast cancer (86). The epothilones are derived from the terrestrial myxobacterium *Sorangium cellulosum*

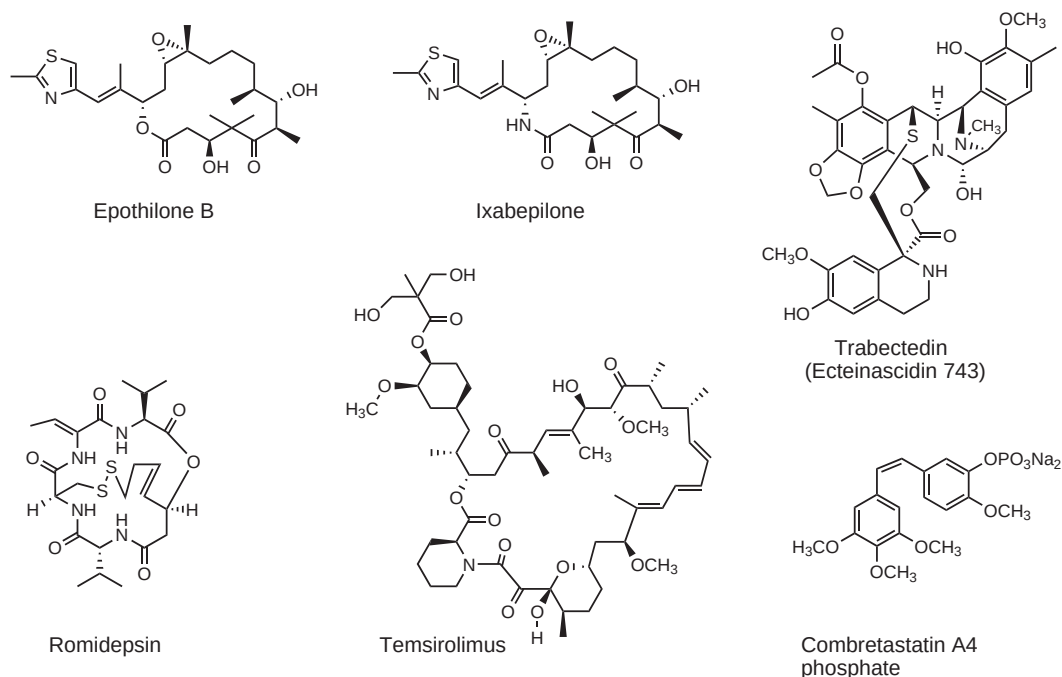


FIGURE 1.5 Potential cancer chemotherapeutic agents from marine, bacterial, plant, and fungal origin.

and have a similar type of action on tubulin as paclitaxel (87). Trabectedin (ecteinascidin-743 or ET-743) is an isoquinoline alkaloid obtained originally from the marine tunicate, *Ecteinascidia turbinata*, but now produced by partial synthesis from a microbial metabolite, cyanosafracin B, of *Pseudomonas fluorescens*. Trabectedin is an alkylating agent that binds to the minor groove of DNA and blocks cells in the G₂-M phase; it is used in Europe as second-line therapy for patients with soft tissue sarcoma (88,89). Romidepsin is a depsipeptide isolated from the soil bacterium, *Chromobacterium violaceum*, in 1994 (90). This compound is an inhibitor of histone deacetylase and was approved in the United States for the treatment of T-cell lymphoma (91). Temsirolimus is a semisynthetic ester derivative of sirolimus (rapamycin), with the latter compound isolated some time ago from *Streptococcus hygroscopicus* (92,93). Recently, temsirolimus was approved in the United States for treating advanced renal cell carcinoma, and mechanistically, this compound is an inhibitor of the mammalian target of rapamycin kinase (92,93). A promising new anticancer agent still in advanced clinical trials is combretastatin A4 phosphate, a water-soluble prodrug of combretastatin A4 from the South African plant, *Combretum caffrum* (94). Combretastatin A4 phosphate binds to tubulin and also affects tumor blood flow as a vascular disrupting agent (94,95).

Cancer chemoprevention is regarded as the use of synthetic or natural agents to inhibit, delay, or reverse the process of carcinogenesis through intervention before the appearance of invasive disease. This relatively new approach toward the management of cancer has involved gaining a better understanding of the mechanism of action of cancer chemopreventive agents (96). Among the natural products that have been studied for this purpose, there has been a renewed interest in the effects of the phytochemical components of the diet, and some of these compounds have been found to block cancer initiation (blocking agents) or reverse tumor promotion and/or progression (suppressing agents) (97). Members of many different structural types of plant secondary metabolites have been linked with potential cancer chemopreventive activity (97,98). Approximately 35 foods of plant origin have been found recently to produce cancer chemopreventive agents, such as curcumin from turmeric, epigallocatechin 3-*O*-gallate from green tea, *trans*-resveratrol from grapes and certain red wines, and *d*-sulforaphane from broccoli (Fig. 1.6) (97,98).

Immunomodulators

The fungal-derived cyclic peptide cyclosporin (cyclosporine A) was found some years ago to be an immunosuppressive agent in organ and tissue transplant surgery. Another compound with this same type of use and that also acts by the inhibition of T-cell activation is the macrolide tacrolimus (FK-506) from *Streptomyces tsukubaensis* (3).

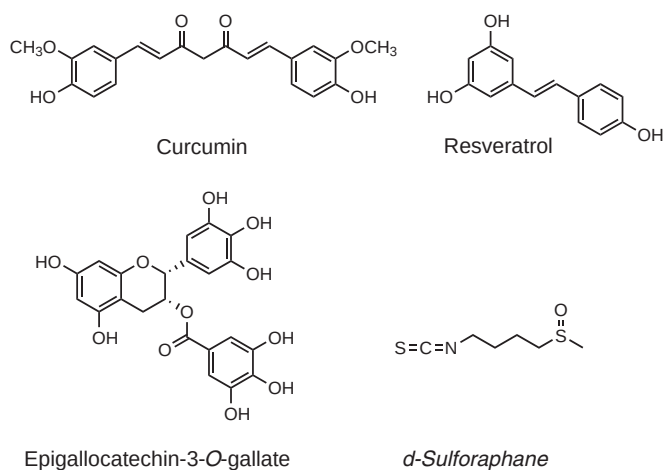


FIGURE 1.6 Naturally occurring cancer chemopreventive agents.

Two additional natural product-derived immunosuppressants are clinically available as mycophenolate sodium and everolimus (Fig. 1.7) (17,18). The active principle of both mycophenolate sodium and an earlier introduced form mycophenolate mofetil (a morpholinoethyl derivative) is mycophenolic acid, obtained from several *Penicillium* species. This compound is a reversible inhibitor of inosine monophosphate dehydrogenase, which is involved in guanosine nucleotide synthesis (17,99). Everolimus is an orally active semisynthetic 40-*O*-(2-hydroxyethyl) derivative of rapamycin (also known as sirolimus) and was originally obtained from *Streptomyces hygroscopicus*. Everolimus is a proliferation inhibitor that blocks growth factor-mediated transduction signals and prevents organ rejection through a different mechanism than mycophenolate mofetil (17,100).

BOTANICAL DIETARY SUPPLEMENTS

The use of phytomedicines (herbal remedies) as prescription products has been well established in Germany and several other countries in Western Europe for approximately 30 years. About 80% of physicians in Germany prescribe phytomedicines through the orthodox health

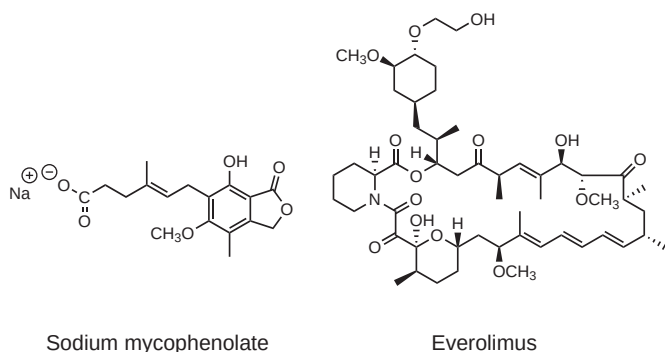


FIGURE 1.7 Naturally occurring immunosuppressants.

care system. Over the last 15 years, there has been a large influx of botanical products into community pharmacy practice and health food stores in the United States as a result of the passage of the Dietary Supplement Health and Education Act (DSHEA) in 1994. Such products are regulated by the U.S. Food and Drug Administration as foods rather than drugs and must adhere to requirements regarding product labeling and acceptable health claims (101). Currently, among the most popular botanical products used in the United States are those containing black cohosh, cranberry, echinacea, evening primrose, garlic, ginkgo, ginger, ginseng, green tea, milk thistle, saw palmetto, soy, St. John's wort, and valerian. These are purchased as either the crude powdered form in compressed tablets or capsules or as galenical preparations, such as extracts or tinctures, and are frequently ingested in the form of a tea (101). In addition to the United States, a parallel increased interest in herbal remedies has occurred in all countries in Western Europe, Canada, and Australia, in part because of a greater awareness of complementary and alternative medicine (CAM). Many clinical trials on these products have been conducted in Europe, and some are occurring in the United States under the sponsorship of the National Institutes of Health.

The recent widespread introduction of a large number of botanical dietary supplements has opened a new door in terms of research inquiry for natural product scientists in the United States. Not all of these products have a well-documented efficacy, however. Three important needs in the scientific investigation of herbal remedies are the characterization of active principles (where these are not known), the development of rigorous and validated analytical methods for quality control procedures, and the determination of their potential toxicity and interactions with prescription medications (102). Unlike compounds approved as single chemical entity (SCE) drugs, it is accepted that combinations of plant secondary metabolites may be responsible for the physiologic effects of herbal medicines. For example, both the terpene lactone (e.g., ginkgolide B; Fig. 1.8) and the flavonoid glycoside constituents of ginkgo (*Ginkgo biloba*) leaves are regarded as being necessary for alleviation of the symptoms of peripheral vascular disease for which this phytomedicine is used in Europe (101). Moreover, an acetone-soluble extract of *G. biloba* containing standardized amounts of flavone glycosides (24%) and terpene lactones (6%) has been used in many clinical trials on this herb (101). If the "active principles" of an herbal remedy are known or can be discovered, these substances can act as reference standards, and their specified concentration levels can be quantified in chemical quality control procedures, which are predominantly performed by HPLC. A number of official monographs for the standardization of botanical dietary supplements have been developed over the last decade in the United States (103). Other scientific challenges on herbal remedies are to establish more completely their dissolution, bioavailability, and shelf life. For

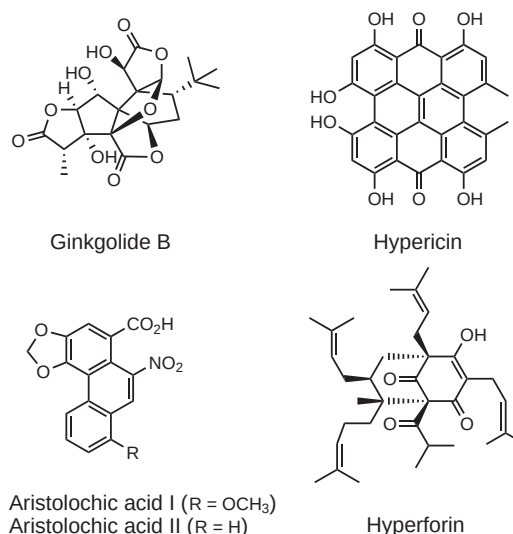


FIGURE 1.8 Chemicals found in various botanical dietary supplements.

example, it has been found that co-effectors such as certain procyanidins present in St. John's wort (*Hypericum perforatum*) can increase the bioavailability of hypericin (Fig. 1.8), one of the constituents of this plant known to exert antidepressant activity (104). These herbal products should be free of adulteration (the deliberate addition of nonauthentic plant material or of biologically active or inactive compounds), as well as free of other additives such as herbicides, pesticides, heavy metals, solvent residues, and microbial and biologic contaminants (101,102).

Unfortunately, many herbal remedies may pose toxicity risks or may be involved in harmful drug interactions. A drastic example of toxicity was caused by the herbal Chinese medicinal plant, *Aristolochia fangchi*, which was substituted in error for another Chinese plant in a weight-reducing regimen taken by a number of women in Belgium several decades ago. Years later, this was linked to the generation of severe renal disease characterized as interstitial fibrosis with atrophy of the tubules, as well as the development of tumors. These toxic symptoms, also known as "Chinese herb nephropathy," were attributed to the presence of the phenanthrene derivatives aristolochic acids I and II (Fig. 1.8) produced by *A. fangchi*, which have been found experimentally to intercalate with DNA (105). The presence of high levels of the phloroglucinol derivative hyperforin (Fig. 1.8) in St. John's wort (*Hypericum perforatum*) products has been found to induce cytochrome P450 enzymes (in particular CYP3A4), leading to decreased plasma concentrations of prescription drugs that may be coadministered, such as alprazolam, cyclosporin, digoxin, indinavir, irinotecan, simvastatin, and warfarin, as well as oral contraceptives (106).

In 2006, the first example of a new class of natural product prescription drugs was approved by the U.S. Food and Drug Administration, namely, a mixture of

sin catechins present in a standardized extract of the leaves of green tea (*Camellia sinensis*). This product is approved for the topical treatment of external genital and perianal warts. Stringent criteria must be followed in the manufacture and quality control of this product, and it was subjected to rigorous clinical trials (107).

FUTURE PROSPECTS

The beginning of the second decade of the 21st century seems to be an opportune time for renewed efforts to be made regarding the discovery of new secondary metabolite prototype, biologically active compounds from animals, fungi, microorganisms, and plants of terrestrial and marine origin. Although many pharmaceutical companies have reduced their investment in natural products research, in favor of screening libraries of synthetic compounds and combinatorial chemistry, this has coincided with disappointing numbers of SCE drugs being introduced in recent years (11,14–16). Fortunately, many smaller “biotech” companies have actively taken up the challenge of contemporary natural products drug discovery from organisms. There continues to be a steady stream of new natural product–derived drugs introduced into therapy for the treatment of many common human diseases (e.g., cancer, cardiovascular diseases, neurologic conditions) (17,18). However, there is ample potential for much greater utilization of natural product–derived compounds in the treatment or prophylaxis of such major worldwide scourges as HIV/AIDS, tuberculosis, hepatitis C, and tropical diseases (e.g., lymphatic filariasis, leishmaniasis, schistosomiasis). The search for such agents should be enhanced by the availability of extensive libraries of taxonomically authenticated crude extracts of terrestrial and marine origin, as well as of pure secondary metabolites from microorganisms, plants, and animals. In addition, this will be facilitated by recently developed techniques such as biocatalysis, combinatorial biosynthesis, combinatorial and computational chemistry, metabolic engineering, and tissue culture. The high “drug-like” quality of natural product molecules stands as a constant, and it only remains for natural product chemists and biologists to investigate these substances in the most technically ingenious and expedient ways possible.

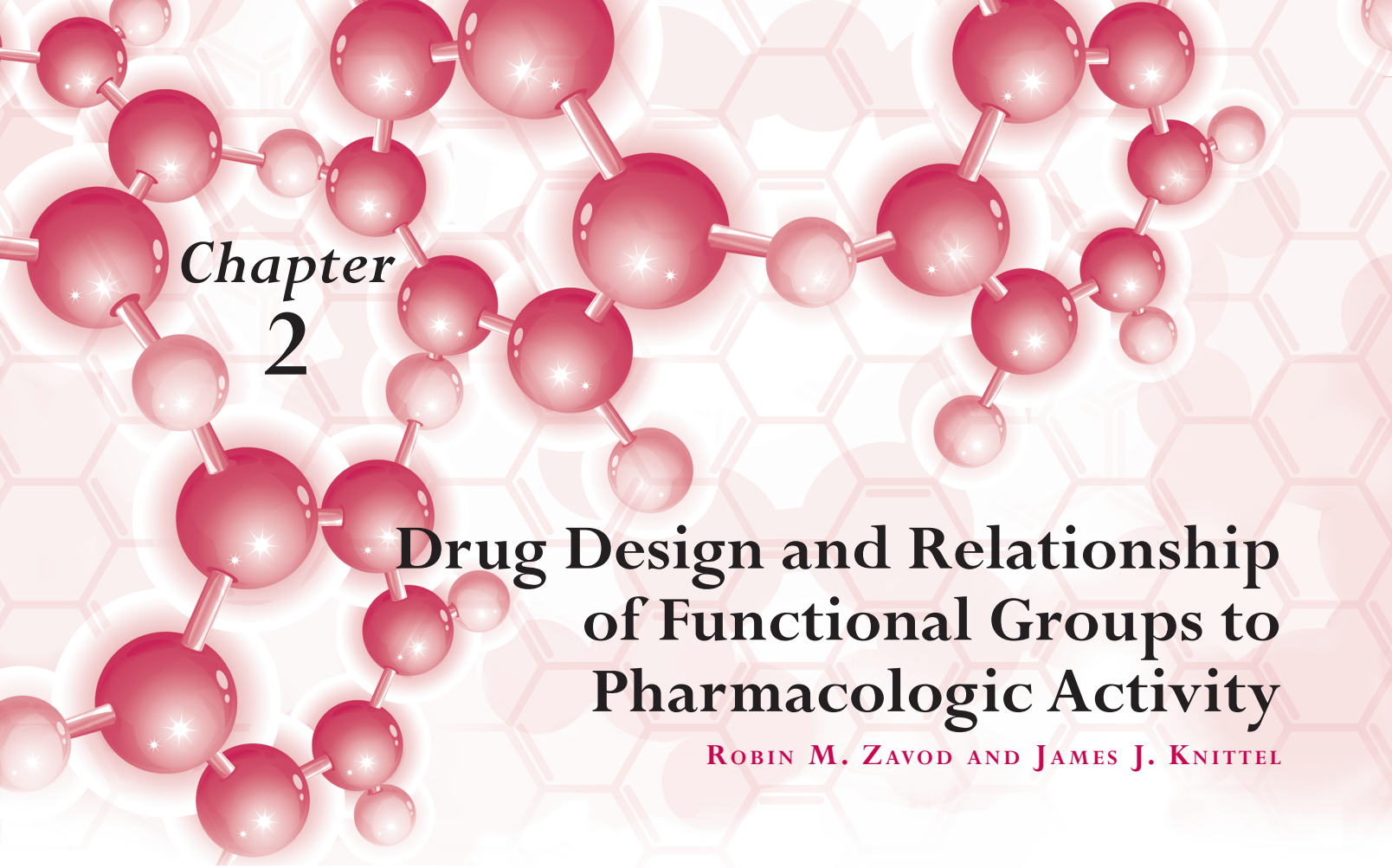
It should not be thought that after approximately 200 years of investigation, the prospects of finding new drugs of natural origin are nearing exhaustion; there is still a large scope for success in this type of endeavor. For example, if plants are taken specifically, less than 20% have been evaluated chemically or biologically. The urgency to perform this type of work cannot be understated in view of the increasing erosion of natural resources that will accelerate as the 21st century progresses.

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Chapter 2

Drug Design and Relationship of Functional Groups to Pharmacologic Activity

ROBIN M. ZAVOD AND JAMES J. KNITTEL

Abbreviations

HCl, hydrochloric acid
IV, intravenous
MW, molecular weight
NaOH, sodium hydroxide

PABA, *p*-aminobenzoic acid
QSAR, quantitative structure–activity relationship

SAR, structure–activity relationship
USP, U.S. Pharmacopeia

Medicinal chemistry is an interdisciplinary science at the intersection of organic chemistry, biochemistry (bioorganic chemistry), computational chemistry, pharmacology, pharmacognosy, molecular biology, and physical chemistry. This branch of chemistry is involved with the identification, design, synthesis, and development of new drugs that are safe and suitable for therapeutic use in humans and pets. It also includes the study of marketed drugs, their biologic properties, and their quantitative structure–activity relationships (QSARs).

Medicinal chemistry studies how chemical structure influences biologic activity. As such, it is necessary to understand not only the mechanism by which a drug exerts its effect, but also how the molecular and physicochemical properties of the molecule influence the drug's pharmacokinetics (absorption, distribution, metabolism, toxicity, and elimination) and pharmacodynamics (what the drug does to the body). The term “physicochemical properties” refers to how the functional groups present within a molecule influence its acid–base properties, water solubility, partition coefficient, crystal structure, stereochemistry, and ability to interact with biologic systems, such as enzyme active sites (Chapter 8) and receptor sites

(Chapter 3). To design better medicinal agents, the relative contribution that each functional group makes to the overall physicochemical properties of the molecule must be evaluated. Studies of this type involve modification of the molecule in a systematic fashion followed by a determination of how these changes affect biologic activity. Such studies are referred to as structure–activity relationships (SARs)—that is, the relationship of how structural features of the molecule contribute to, or take away from, the desired biologic activity.

Because of the foundational nature of the content of this chapter, there are numerous case studies presented throughout the chapter (as boxes), as well as at the end. In addition, a list of study questions at the conclusion of—and unique to—this chapter provides further self-study material related to the subject of medicinal chemistry/drug design.

INTRODUCTION

Chemical compounds, usually derived from plants and other natural sources, have been used by humans for thousands of years to alleviate pain, diarrhea, infection,

and various other maladies. Until the 19th century, these “remedies” were primarily crude preparations of plant material of unknown constitution. The revolution in synthetic organic chemistry during the 19th century produced a concerted effort toward identification of the structures of the active constituents of these naturally derived medicinals and synthesis of what were hoped to be more efficacious agents. By determining the molecular structures of the active components of these complex mixtures, it was thought that a better understanding of how these components worked could be elucidated.

RELATIONSHIP BETWEEN MOLECULAR STRUCTURE AND BIOLOGIC ACTIVITY

Early studies of the relationship between chemical structure and biologic activity were conducted by Crum-Brown and Fraser (1) in 1869. They demonstrated that many compounds containing tertiary amine groups exhibited activity as muscle relaxants when converted to quaternary ammonium compounds. Molecules with widely differing pharmacologic properties, such as strychnine (a convulsant), morphine (an analgesic), nicotine (a deterrent, insecticide), and atropine (an anticholinergic), could be converted to muscle relaxants with properties similar to those of tubocurarine when methylated (Fig. 2.1). Crum-Brown and Fraser therefore concluded that muscle relaxant activity required the presence of a quaternary ammonium group within the structure. This initial hypothesis was later disproven by the discovery of the natural neurotransmitter and activator of muscle contraction, acetylcholine (Fig. 2.2). Even though Crum-Brown and Fraser’s initial hypothesis that related chemical structure with action as a muscle relaxant was incorrect, it demonstrated the concept that molecular structure influences the biologic activity of chemical entities and that alterations in structure produce changes in biologic action.

With the discovery by Crum-Brown and Fraser that quaternary ammonium groups could produce molecules with muscle relaxant properties, scientists began to look for other functional groups that produce specific biologic responses. At the time, it was thought that specific chemical groups, or nuclei (rings), were responsible for specific biologic effects. This led to the postulate, that took some time to disprove, that “one chemical group gives one biological action” (2). Even after the discovery of acetylcholine by Loewi and Navrati (3), which effectively dispensed with Crum-Brown and Fraser’s concept of all quaternary ammonium compounds being muscle relaxants, this was still considered to be dogma and took a long time to refute.

SELECTIVITY OF DRUG ACTION AND DRUG RECEPTORS

Although the structures of many drugs or xenobiotics, or at least their functional group composition, were known at the start of the 20th century, how these compounds

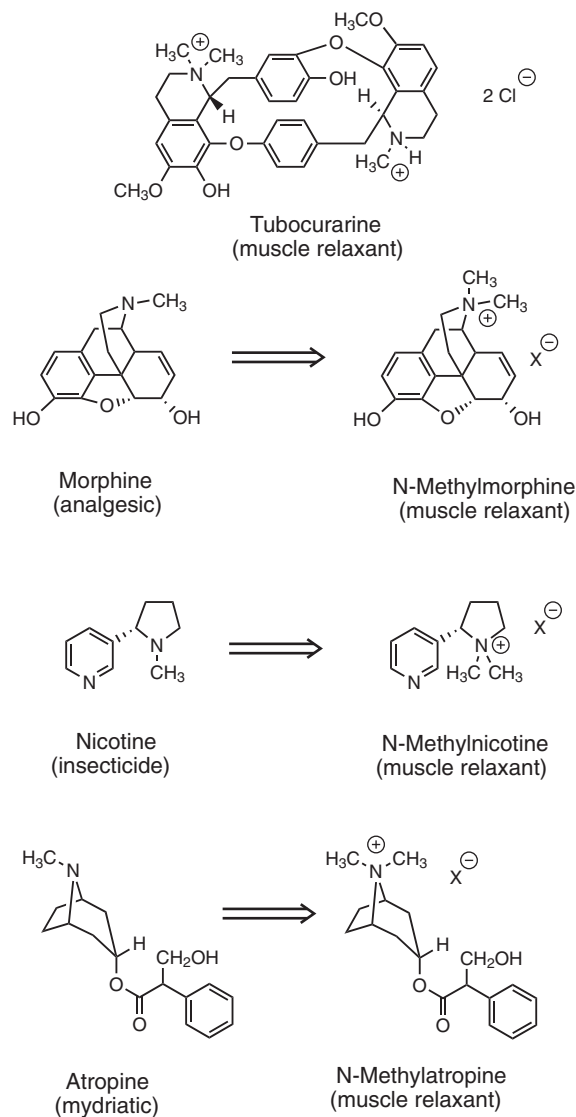


FIGURE 2.1 Effects of methylation on biologic activity.

exerted their effects was still a mystery. Using his observations with regard to the staining behavior of microorganisms, Ehrlich (4) developed the concept of drug receptors. He postulated that certain “side chains” on the surfaces of cells were “complementary” to the dyes (or drug) and suggested that the two could therefore interact with one another. In the case of antimicrobial compounds, interaction of the chemical with the cell surface “side chains” produced a toxic effect. This concept was the first description of what later became known as the receptor hypothesis for explaining the biologic action of chemical entities. Ehrlich also discussed selectivity

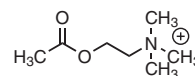


FIGURE 2.2 Acetylcholine, a neurotransmitter and muscle relaxant.

of drug action via the concept of a “magic bullet.” He suggested that this selectivity permitted eradication of disease states without significant harm coming to the organism being treated (i.e., the patient). This was later modified by Albert (5) and today is referred to as “selective toxicity.” An example of poor selectivity was demonstrated when Ehrlich developed organic arsenicals that were toxic to trypanosomes as a result of their irreversible reaction with thiol groups (-SH) on vital proteins. The formation of As-S bonds resulted in death to the target organism. Unfortunately these compounds were toxic not only to the target organism, but also to the host once certain blood levels of arsenic were achieved.

The “paradox” that resulted after the discovery of acetylcholine—how one chemical group can produce two different biologic effects (i.e., muscle relaxation and muscle contraction)—was explained by Ing (6) using the actions of acetylcholine and tubocurarine as his examples (see also Chapter 9). Ing hypothesized that both acetylcholine and tubocurarine act at the same receptor, but that one molecule fits to the receptor in a more complementary manner and “activates” it, causing muscle contraction. (Ing did not elaborate just how this activation occurred.) The blocking effect of the larger molecule, tubocurarine, could be explained by its occupation of part of the receptor, thereby preventing acetylcholine, the smaller molecule, from interacting with the receptor. With both molecules, the quaternary ammonium functional group is a common structural feature and interacts with the same region of the receptor. If one closely examines the structures of other molecules with opposing effects on the same pharmacologic system, this appears to be a common theme: Molecules that block the effects of natural neurotransmitters, such as norepinephrine, histamine, dopamine, or serotonin for example are called antagonists and, are usually larger in size than the native compound, which is not the case for antagonists of peptide neurotransmitters and hormones such as cholecystokinin, melanocortin, or substance P. Antagonists to these peptide molecules are usually smaller in size. However, regardless of the type of neurotransmitter (biogenic amine or peptide), both agonists and antagonists share common structural features with the neurotransmitter that they influence. This provides support to the concept that the structure of a molecule, its composition and arrangement of functional groups, determines the type of pharmacologic effect that it possesses (i.e., SAR). For example, compounds that are muscle relaxants that act via the cholinergic nervous system possess a quaternary ammonium or protonated tertiary ammonium group and are larger than acetylcholine (compare acetylcholine in Fig. 2.2 with tubocurarine in Fig. 2.1).

SARs are the underlying principle of medicinal chemistry. Similar molecules exert similar biologic actions in a qualitative sense. A corollary to this is that structural elements (functional groups) within a molecule most often contribute in an additive manner to the physicochemical properties of a molecule and, therefore, to its biologic

action. One need only peruse the structures of drug molecules in a particular pharmacologic class to become convinced (e.g., histamine H_1 antagonists, histamine H_2 antagonists, β -adrenergic antagonists). In the quest for better medicinal agents (drugs), it must be determined which functional groups within a specific structure are important for its pharmacologic activity and how these groups can be modified to produce more potent, more selective, and safer compounds.

An example of how different functional groups can yield chemical entities with similar physicochemical properties is demonstrated by the sulfanilamide antibiotics. In Figure 2.3, the structures of sulfanilamide and *p*-aminobenzoic acid (PABA) are shown. In 1940, Woods (7) demonstrated that PABA reverses the antibacterial action of sulfanilamide (and other sulfonamide-based antibacterials) and that both PABA and sulfanilamide have similar steric and electronic properties. Both molecules contain acidic functional groups, with PABA containing an aromatic carboxylic acid and sulfanilamide an aromatic sulfonamide. When ionized at physiologic pH, both compounds have a similar electronic configuration, and the distance between the ionized acid and the weakly basic amino group is also very similar. It should be no surprise that sulfanilamide acts as an antagonist to PABA metabolism in bacteria.

Biologic Targets for Drug Action

In order for drug molecules to exhibit their pharmacologic activity, they must interact with a biologic target, typically a receptor, enzyme, nucleic acid, or excitable membrane or other biopolymer. These interactions occur between the functional groups found in the drug molecule and those found within each biologic target. The relative fit of each drug molecule with its target is a function of a number of physicochemical properties including acid-base chemistry and related ionization, functional group shape and size, and three-dimensional spatial orientation. The quality of this “fit” has a direct impact on the biologic response produced. In this chapter, functional group characteristics are discussed as a means to better understand overall drug molecule absorption, distribution, metabolism, and excretion, as well as potential interaction with a biologic target.

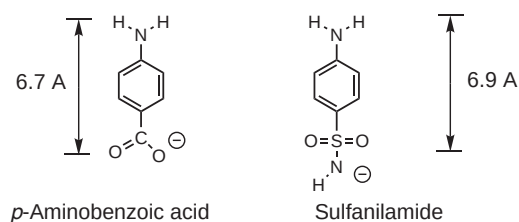
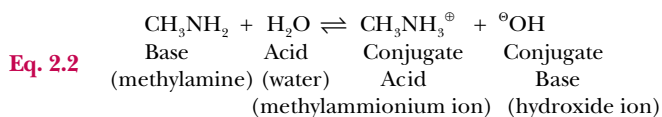
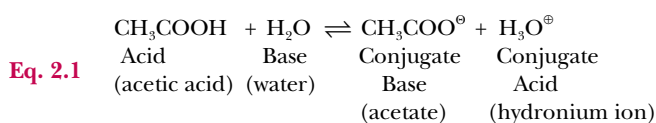


FIGURE 2.3 Ionized forms of *p*-aminobenzoic acid (PABA) and sulfanilamide, with comparison of the distance between amine and ionized acids of each compound. Note how closely sulfanilamide resembles PABA.

PHYSICOCHEMICAL PROPERTIES OF DRUGS

Acid–Base Properties

The human body is 70 to 75% water, which amounts to approximately 51 to 55 L of water for a 160-lb (73-kg) individual. For an average drug molecule with a molecular weight of 200 g/mol and a dose of 20 mg, this leads to a solution concentration of approximately 2×10^{-6} M (2 μ M). When considering the solution behavior of a drug within the body, we are dealing with a dilute solution, for which the Brønsted-Lowry (8) acid–base theory is most appropriate to explain and predict acid–base behavior. This is a very important concept in medicinal chemistry, because the acid–base properties of drug molecules have a direct effect on absorption, excretion, and compatibility with other drugs in solution. According to the Brønsted-Lowry Theory, an “acid” is any substance capable of yielding a proton (H^+), and a “base” is any substance capable of accepting a proton. When an acid gives up a proton to a base, it is converted to its “conjugate base.” Similarly, when a base accepts a proton, it is converted to its “conjugate acid” (Eqs. 2.1 and 2.2):



Note that when an acidic functional group loses its proton (often referred to as having undergone “dissociation”), it is left with an extra electron and becomes negatively charged. This is the “ionized” form of the acid. The ability of the ionized functional group to participate in an ion-dipole interaction with water (see the Water Solubility of Drugs section) enhances its water solubility. Many functional groups behave as acids (Table 2.1). The ability to recognize these functional groups and their relative acid strengths helps to predict absorption, distribution, excretion, and potential incompatibilities between drugs.

When a basic functional group is converted to the corresponding conjugate acid, it too becomes ionized. In this instance, however, the functional group becomes positively charged due to the extra proton. Most drugs that contain basic functional groups contain primary, secondary, and tertiary amines or imino amines, such as guanidines and amidines. Other functional groups that are basic are shown in Table 2.2. As with the acidic groups, it is important to become familiar with these functional groups and their relative strengths.

Functional groups that cannot give up or accept a proton are considered to be “neutral” (or “nonelectrolytes”)

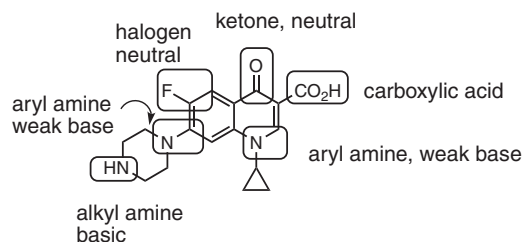


FIGURE 2.4 Chemical structure of ciprofloxacin showing the various organic functional groups.

with respect to their acid–base properties. Common neutral functional groups are shown in Table 2.3. Quaternary ammonium compounds are neither acidic nor basic and are not electrically neutral. Additional information about the acid–base properties of the functional groups listed in Tables 2.1 through 2.3 can be found in Gennaro (9) and Lemke (10).

A molecule can contain multiple functional groups with acid–base properties and, therefore, can possess both acidic and basic character. For example, ciprofloxacin (Fig. 2.4), a fluoroquinolone antibacterial agent, contains a secondary alkylamine, two tertiary arylamines (aniline-like amines), and a carboxylic acid. The two arylamines are weakly basic and, therefore, do not contribute significantly to the acid–base properties of ciprofloxacin under physiologic conditions. Depending on the pH of the physiologic environment, this molecule will either accept a proton (secondary alkylamine), donate a proton (carboxylic acid), or both. Thus, it is described as amphoteric (both acidic and basic) in nature. Figure 2.5 shows the acid–base behavior of ciprofloxacin in two different environments. Note that at a given pH (e.g., pH 1.0 to 3.5), only one of the functional groups (the alkylamine) is significantly ionized. To be able to make this prediction, an appreciation for the relative acid–base strength of both the acidic and basic functional groups is required. Thus, one needs to know which acidic or basic functional group within a molecule containing multiple functional groups is the strongest and which acidic or basic functional group is the weakest. The concept of pK_a not only describes relative acid–base strength of functional groups,

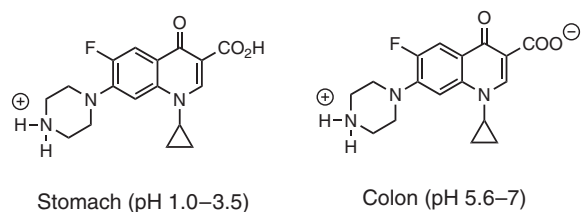
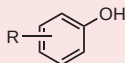
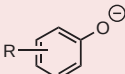
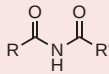
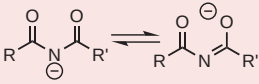
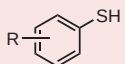
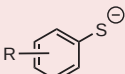
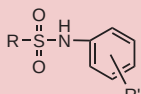
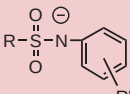
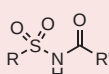
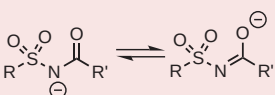
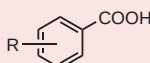
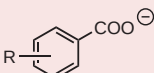


FIGURE 2.5 Predominate forms of ciprofloxacin at two different locations within the gastrointestinal tract.

TABLE 2.1 Common Acidic Organic Functional Groups and Their Ionized (Conjugate Base) Forms

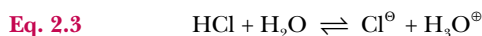
Acids (pKa)			Conjugate Base
Phenol (9-11)			Phenolate
Sulfonamide (9-10)			Sulfonamidate
Imide (9-10)			Imidate
Alkylthiol (10-11)	R-SH		Thiolate
Thiophenol (9-10)			Thiophenolate
N-Arylsulfonamide (6-7)			N-Arylsulfonamidate
Sulfonimide (5-6)			Sulfonimide
Alkylcarboxylic acid (5-6)			Alkylcarboxylate
Arylcarboxylic acid (4-5)			Arylcarboxylate
Sulfonic acid (0-1)			Sulfonate

Acid strength increases as one moves down the table.

but also allows one to calculate, for a given pH, the relative percentages of the ionized and un-ionized forms of the drug. As stated earlier, this helps to predict relative water solubility, absorption, and excretion for a given compound.

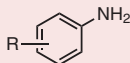
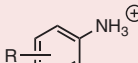
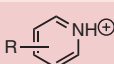
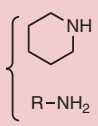
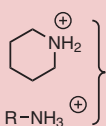
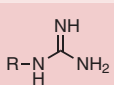
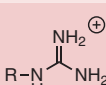
Relative Acid Strength (pK_a)

Strong acids and bases completely donate (dissociate) or accept a proton in aqueous solution to produce their respective conjugate bases and acids. For example, mineral acids, such as hydrochloric acid (HCl), or bases, such as sodium hydroxide (NaOH), undergo 100% dissociation in water, with the equilibrium between the ionized and un-ionized forms shifted completely to the right (ionized), as shown in Equations 2.3 and 2.4:



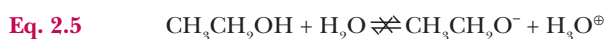
Acids and bases of intermediate or weak strength, however, incompletely donate (dissociate) or accept a proton, and the equilibrium between the ionized and un-ionized forms lies somewhere in the middle, such that all possible species can exist at any given time. Note that in Equations 2.3 and 2.4, water acts as a base in one instance and as an acid in the other. Water is therefore *amphoteric*—that is, it can act as an acid or a base, depending on the prevailing pH of the solution. From a physiologic perspective, drug molecules are always present as a dilute aqueous solution. The strongest base

TABLE 2.2 Common Basic Organic Functional Groups and Their Ionized (Conjugate Acid) Forms

Base			Conjugate Acid (pKa)
Arylamine			Arylammonium (5-6)
Aromatic amine			Aromatic ammonium (4-5)
Imine	$R-\overset{\text{H}}{\underset{\text{H}}{\text{C}}}=\text{NH}$	$R-\overset{\oplus}{\underset{\text{H}}{\text{C}}}=\text{NH}$	Iminium (3-4)
Alkylamines			Alkylammonium (2° 10-11) (1° 9-10)
Amidine			Amidinium (10-11)
Guanidine			Guanidinium (12-13)

that is present is OH^- , and the strongest acid is H_3O^+ . This is known as the “leveling effect” of water. Thus, some functional groups that have acidic or basic character do not behave as such under physiologic conditions in aqueous solution. For example, alkyl alcohols, such as ethyl alcohol, are not sufficiently acidic to become significantly ionized in an aqueous solution at a physiologically pH. Water is not sufficiently basic to remove the proton from ethyl alcohol to form the ethoxide ion

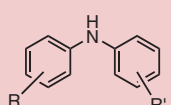
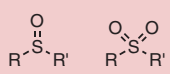
(Eq. 2.5). Therefore, under physiologic conditions, alcohols are neutral with respect to acid–base properties:



Predicting the Degree of Ionization of a Molecule

By knowing if there are acidic and/or basic functional groups present in a molecule, one can predict

TABLE 2.3 Common Organic Functional Groups That are Considered Neutral Under Physiologic Conditions

$R-\text{CH}_2-\text{OH}$	$R-\text{O}-R'$		
Alkyl alcohol	Ether	Ester	Sulfonic acid ester
		$R-\text{C}\equiv\text{N}$	
Amide	Diarylamine	Nitrile	Quaternary ammonium
		$R-\text{S}-R'$	
Amine oxide	Ketone & Aldehyde	Thioether	Sulfoxide Sulfone

whether a molecule is going to be predominantly ionized or un-ionized at a given pH. To be able to quantitatively predict the degree of ionization of a molecule, the pK_a values of each of the acidic and basic functional groups present and the pH of the environment in which the molecule will be located must be known. The magnitude of the pK_a value is a measure of relative acid or base strength, and the Henderson-Hasselbalch equation (Eq. 2.6) can be used to calculate the percent ionization of a compound at a given pH (this equation was used to calculate the major forms of ciprofloxacin in Fig. 2.5):

Eq. 2.6
$$pK_a = pH + \log \frac{[\text{acid form}]}{[\text{base form}]}$$

The key to understanding the use of the Henderson-Hasselbalch equation for calculating percent ionization is to realize that this equation relates a constant, pK_a , to the ratio of the acidic form of a functional group to its conjugate base form (and conversely, the conjugate acid form to its base). Because pK_a is a constant for any given functional group, the ratio of acid to conjugate base (or conjugate acid to base) will determine the pH of the solution. A sample calculation is shown in Figure 2.6 for the sedative hypnotic amobarbital.

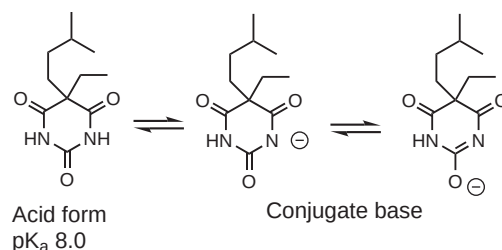
When dealing with a basic functional group, one must recognize the conjugate acid represents the ionized form of the functional group. Figure 2.7 shows the calculated percent ionization for the decongestant phenylpropanolamine. It is very important to understand that for a base, the pK_a refers to the conjugate acid or ionized form of the compound. To thoroughly comprehend this relationship, calculate the percent ionization of an acidic functional group and a basic functional group at different pH values and carefully observe the trend.

Water Solubility of Drugs

The solubility of a drug molecule in water greatly affects the routes of administration that are available, as well as its absorption, distribution, and elimination. Two key concepts to keep in mind when considering the water (or fat) solubility of a molecule are the potential for hydrogen bond formation and ionization of one or more functional groups within the molecule.

Hydrogen Bonds

Each functional group capable of donating or accepting a hydrogen bond contributes to the overall water solubility of the compound and increases the hydrophilic (water-loving) nature of the molecule. Conversely, functional groups that cannot form hydrogen bonds do not enhance hydrophilicity and will contribute to the hydrophobic (water-fearing) nature of the molecule. Hydrogen bonds are a special case of what are usually referred to as dipole-dipole interactions. A permanent dipole occurs



Question: At a pH of 7.4, what is the percent ionization of amobarbital?

Answer: $8.0 = 7.4 + \log \frac{[\text{acid}]}{[\text{base}]}$

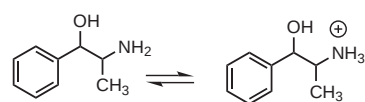
$$0.6 = \log \frac{[\text{acid}]}{[\text{base}]}$$

$$10^{0.6} = \frac{[\text{acid}]}{[\text{base}]} = \frac{3.98}{1}$$

$$\% \text{ acid form} = \frac{3.98 \times 100}{4.98} = 79.9\%$$

FIGURE 2.6 Calculation of percent ionization of amobarbital. Calculation indicates that 80% of the molecules are in the acid (or protonated) form, leaving 20% in the conjugate base (ionized) form.

as a result of an unequal sharing of electrons between the two atoms within a covalent bond. This unequal sharing of electrons only occurs when these two atoms have significantly different electronegativities. When a permanent dipole is present, a partial charge is associated



Question: What is the % ionization of phenylpropanolamine at pH 7.4?

Answer: $9.4 = 7.4 + \log \frac{[\text{acid}]}{[\text{base}]}$

$$2.0 = \log \frac{[\text{acid}]}{[\text{base}]}$$

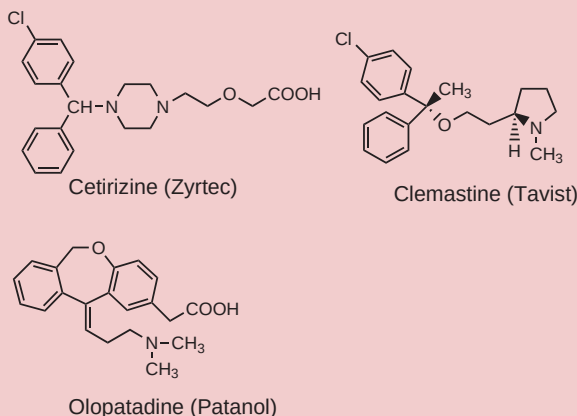
$$10^2 = \frac{[\text{acid}]}{[\text{base}]} = \frac{100}{1}$$

$$\% \text{ ionization} = \frac{100 \times 100}{101} = 99\%$$

FIGURE 2.7 Calculation of percent ionization of phenylpropanolamine. Calculation indicates that 99% of the molecules are in the acid form, which is the same as the percent ionization.

ABSORPTION/ACID-BASE CASE

A long-distance truck driver comes into the pharmacy complaining of seasonal allergies. He asks you to recommend an agent that will act as an antihistamine but that will not cause drowsiness. He regularly takes TUMS for indigestion due to the bad food that he eats while on the road.

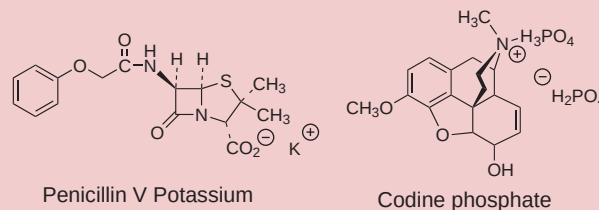


1. Identify the functional groups present in Zyrtec and Tavist, and evaluate the effect of each functional group on the ability of the drug to cross lipophilic membranes (e.g., blood–brain barrier). Based on your assessment of each agent's ability to cross the blood–brain barrier (and, therefore, potentially cause drowsiness), provide a rationale for whether the truck driver should be taking Zyrtec or Tavist.
2. Patanol is sold as an aqueous solution of the hydrochloride salt. Modify the structure present in the box to show the appropriate salt form of this agent. This agent is applied to the eye to relieve itching associated with allergies. Describe why this agent is soluble in water and what properties make it able to be absorbed into the membranes that surround the eye.
3. Consider the structural features of Zyrtec and Tavist. In which compartment (stomach [pH 1] or intestine [pH 6 to 7]) will each of these two drugs be best absorbed?
4. TUMS neutralizes stomach acid to pH 3.5. Based on your answer to question 3, determine whether the truck driver will get the full antihistaminergic effect if he takes his antihistamine at the same time that he takes his TUMS. Provide a rationale for your answer.

with each of these atoms along a single bond (one atom has a partial negative charge, and one atom has a partial positive charge). The atom with a partial negative charge has higher electron density than the other atom. When two functional groups that contain one or more permanent dipoles approach one another, they align such that the negative end of one dipole is electrostatically attracted to the positive end of the other. When the positive end of the dipole is a hydrogen atom, this interaction is referred to as a “hydrogen bond” (or H-bond).

ACID-BASE CHEMISTRY/COMPATIBILITY CASES

The intravenous (IV) technician in the hospital pharmacy gets an order for a patient that includes the two drugs drawn below. She is unsure if she can mix the two drugs together in the same IV bag and is not certain how water soluble the agents are.



1. Penicillin V potassium is drawn in its salt form, whereas codeine phosphate is not. Modify the structure above to show the salt form of codeine phosphate. Determine the acid–base character of the functional groups in the two molecules drawn above as well as the salt form of codeine phosphate.
2. As originally drawn above, which of these two agents is more water soluble? Provide a rationale for your selection that includes appropriate structural properties. Is the salt form of codeine phosphate more or less water soluble than the free base form of the drug? Provide a rationale for your answer based on the structural properties of the salt form of codeine phosphate.
3. What is the chemical consequence of mixing aqueous solutions of each drug in the same IV bag? Provide a rationale that includes an acid–base assessment.

Thus, for a hydrogen bonding interaction to occur, at least one functional group must contain a dipole with an electropositive hydrogen. The hydrogen atom must be covalently bound to an electronegative atom, such as oxygen (O), nitrogen (N), sulfur (S), or selenium (Se). Of these four elements, only oxygen and nitrogen atoms contribute significantly to the dipole, and we will therefore concern ourselves only with the hydrogen-bonding capability (specifically as hydrogen bond donors) of functional groups that contain a bond between oxygen and hydrogen atoms (e.g., alcohols) and functional groups that contain a bond between nitrogen and hydrogen atoms (e.g., primary and secondary amines and amides) (e.g., NH and CONH groups).

Even though the energy associated with each hydrogen bond is small (1 to 10 kcal/mol/bond), it is the additive nature of multiple hydrogen bonds that contributes to the overall water solubility of a given drug molecule. This type of interaction is also important in the interaction between a drug and its biologic target (e.g., receptor). Figure 2.8 shows several types of hydrogen bonding interactions that can occur with a couple of functional groups and water. As a general rule, the more hydrogen

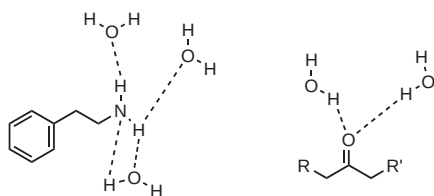


FIGURE 2.8 Examples of hydrogen bonding between water and hypothetical drug molecules.

bonds that are possible between a drug molecule and water, the greater the water solubility of the molecule. Table 2.4 lists several common functional groups and the number of hydrogen bonds in which they can potentially participate. Note that this table does not take into account the possibility of intramolecular hydrogen bond formation. Each intramolecular hydrogen bond decreases water solubility (and increases lipid solubility) because there is one less interaction possible with water.

Ionization

In addition to the hydrogen-bonding capacity of a molecule, another type of interaction plays an important role in determining water solubility: the ion–dipole interaction. This type of interaction can occur with organic salts. Ion–dipole interactions occur between either a cation and the partially negatively charged atom found in a permanent dipole (e.g., the oxygen atom in water) or an anion and

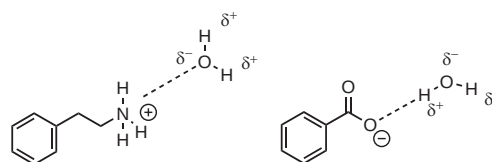


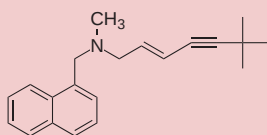
FIGURE 2.9 Examples of ion–dipole interactions.

the partially positively charged atom found in a permanent dipole (e.g., the hydrogen atoms in water) (Fig. 2.9).

Organic salts are composed of a drug molecule in its ionized form and an oppositely charged counterion. For example, the salt of a carboxylic acid is composed of the carboxylate anion (ionized form of the functional group) and a positively charged ion (e.g., Na^+) and the salt of a secondary amine is composed of the ammonium cation (ionized form of the functional group) and a negatively charged ion; e.g., Cl^-). Not all organic salts are very water soluble. To associate with enough water molecules to become soluble, the salt must be highly dissociable; in other words, the cation and anion must be able to separate and interact independently with water molecules. Highly dissociable salts are those formed from strong acids with strong bases (e.g., sodium chloride), weak acids with strong bases (e.g., sodium phenobarbital), or strong acids with weak bases (e.g., atropine sulfate). Examples of strong acids (strong acids are 100% ionized in water [i.e., no ionization constants or $\text{p}K_a$ values of <1]) include the hydrohalic (hydrochloric, hydrobromic, and hydrofluoric), sulfuric, nitric, and perchloric acids. All other acids (e.g., phosphoric, tartaric, acetic, and other organic acids, and phenols) are partially ionized with $\text{p}K_a$ values from 1 to 14 and are, therefore, considered to be moderate or weak acids. Hydroxides of sodium, potassium, and

ABSORPTION/BINDING INTERACTIONS CASE

A 24-year-old man comes into the pharmacy and asks you to recommend a treatment for the itching and burning he has recently noticed on both feet. He indicates that he would prefer a cream rather than a spray or a powder. Your recommendation to this patient is to use Lamisil (terbinafine), a very effective topical antifungal agent that is sold over the counter.



Terbinafine (Lamisil)

1. Identify the structural characteristics and the corresponding properties that make terbinafine an agent that can be used topically.
2. The biologic target of drug action for terbinafine is squalene epoxidase. Consider each of the structural features of this antifungal agent and describe the type of interactions that the drug will have with the target for drug action. Which amino acids are likely to be present in the active site of this enzyme?

TABLE 2.4 Common Organic Functional Groups and Their Hydrogen-Bonding Potential

Functional Groups	Number of Potential H-bonds
$\text{R}-\text{OH}$	3
$\text{R}-\text{C}(=\text{O})-\text{R}'$	2
$\text{R}-\text{NH}_2$	3
$\text{R}-\text{NH}-\text{R}'$	2
$\text{R}-\text{N}(\text{R}')-\text{R}''$	1
$\text{R}-\text{C}(=\text{O})-\text{O}-\text{R}'$	2

calcium are strong bases because they are 100% ionized, whereas other bases, such as amines, are of moderate or weak strength. The salt formed by a carboxylic acid with an alkylamine is the salt of a weak acid and weak base, respectively. This salt does not dissociate appreciably and cannot significantly contribute to the overall water solubility of a given drug molecule. In general, low molecular weight salts are water soluble, and high molecular weight salts are water insoluble. Examples of common organic salts used in pharmaceutical preparations are provided in Figure 2.10.

The extent to which ionized molecules are soluble in water is also dependent on the presence of intramolecular ionic interactions. Molecules with ionizable functional groups of opposite charges have the potential to interact with each other rather than with water molecules. When this occurs, these molecules often become water insoluble. A classic example is the amino acid tyrosine (Fig. 2.11). Tyrosine contains three very polar functional groups, two of which are ionizable (the alkylamine and carboxylic acid) depending on the pH of the environment.

The phenolic hydroxyl group is also ionizable (pK_a 9 to 10); however, it does not contribute significantly to the ionization of tyrosine under pharmaceutically or physiologically relevant conditions (<1% ionized at pH 7). Because of the presence of three very polar functional groups (two of them being ionizable), one would expect tyrosine to be very soluble in water, yet its solubility is

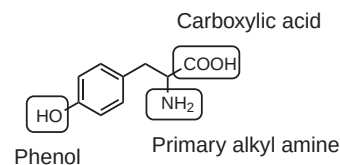


FIGURE 2.11 Organic functional groups in tyrosine (see text for pK_a values).

only 0.45 g/1,000 mL. The basic alkylamine (pK_a 9.1 for the conjugate acid) and the carboxylic acid (pK_a 2.2) are both ionized at physiologic pH, and a zwitterionic molecule results. These two charged groups are sufficiently close that a strong ion-ion interaction occurs, thereby keeping each group from participating in ion-dipole interactions with surrounding water molecules. This lack of interaction between the ions and the dipoles found in water results in a molecule that is very water insoluble (Fig. 2.12). Not all zwitterions or multiply charged molecules demonstrate this behavior; only those that contain ionized functional groups close enough for an ionic interaction to occur will be poorly soluble. Generally, the greater the separation between charges, the more highly water soluble one anticipates the molecule will be. This is only true, however, up to a certain number of carbon atoms. This will be discussed in more detail later.

Predicting Water Solubility: The Empirical Approach

Lemke (10) developed an empiric approach to predicting the water solubility of molecules based on the carbon-solubilizing potential of several functional groups. In his approach, if the solubilizing potential of the functional groups exceeds the total number of carbon atoms present, then the molecule is considered to be water soluble. Otherwise, it is considered to be water insoluble. Participation in intramolecular hydrogen bonding or ionic interactions decreases the solubilizing potential of a given functional group. It is difficult to quantitate how much such interactions will decrease a molecule's overall water solubility.

Table 2.5 shows the water-solubilizing potential for several functional groups common to many drugs. Because most drug molecules contain more than one functional group (i.e., are polyfunctional), the second column in the table will be of more utility. To demonstrate Lemke's method, consider the structure of anileridine. Anileridine (Fig. 2.13) is an opioid analgesic that contains three functional groups that contribute to water

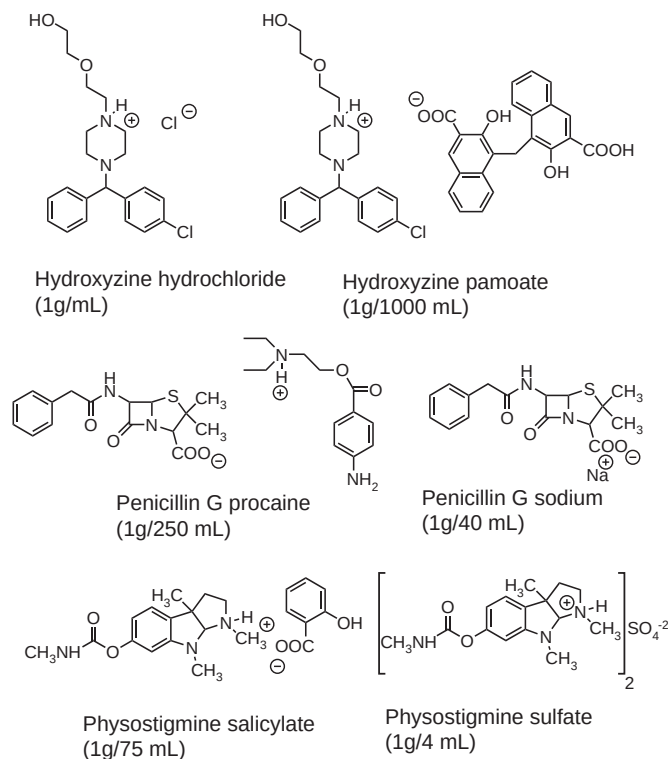


FIGURE 2.10 Water solubilities of different salt forms of selective drugs.

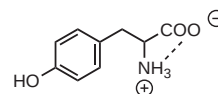
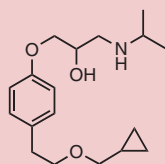


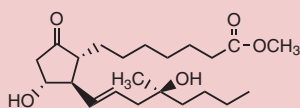
FIGURE 2.12 Zwitterionic form of tyrosine showing ion-ion bond.

BINDING INTERACTIONS CASE

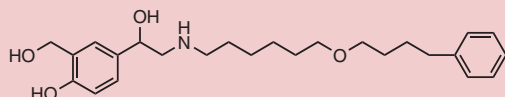
Each of these drug molecules interacts with a different biologic target and elicits a unique pharmacologic response. For each of the three molecules, list the types of interactions that are possible with a biologic target. For each type of interaction, provide one example of an amino acid that could participate in that interaction.



Betaxolol (Betoptic)



Misoprostol (Cytotec)



Salmeterol (Serevent)

Example: Binding interaction: Van der Waals
Amino acid: Leucine

solubility: an aromatic amine (very weak base), a tertiary alkylamine (weak base), and an ester (neutral). There are a total of 22 carbon atoms in the molecule and a solubilizing potential from the three functional groups of nine carbon atoms. Since the solubilizing potential of the functional groups is less than the total number of carbons that are present, it is predicted that anileridine is insoluble in water. This is, indeed, the case: The solubility of anileridine is reported in the U.S. Pharmacopeia (USP) as 1 g/10,000 mL, or 0.01%. Now consider the

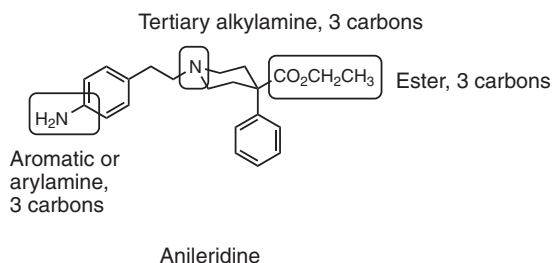


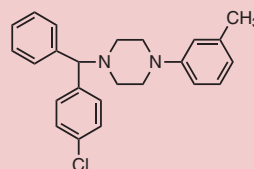
FIGURE 2.13 Identification of functional groups in anileridine.

WATER/LIPID SOLUBILITY CASE

When you look at any drug molecule, there are a number of functional groups present that contribute to the properties of that drug molecule. Identify the types of functional groups in each molecule and to which physical properties (water/lipid solubility) each contributes.

1. Structural feature

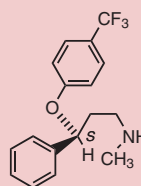
Physical property



Meclizine (Antivert)

2. Structural feature

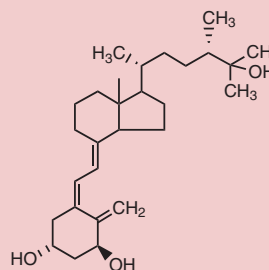
Physical property



Fluoxetine (Prozac)

3. Structural feature

Physical property



1,25-Dihydroxy Vit D₂

hydrochloride salt of anileridine. Not only do the three functional groups contribute a solubilizing potential of nine carbon atoms, the positive charge of the alkylammonium contributes also to its water solubility. Lemke (10) estimates that each ionized functional group (cationic or anionic) found within a drug molecule contributes a solubilizing potential of 20 to 30 carbon atoms. Thus, the solubilizing potential for all of the functional groups in anileridine hydrochloride is 29 to 39 carbon atoms, which is more than the total number of carbon atoms in the molecule. This salt should therefore be soluble in water, and it is (to the extent of 0.2 g/mL, or 20%).

INTERACTIONS/SOLUBILITY CASE

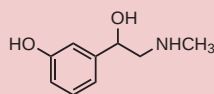
J.K. presents a prescription for her 6-month-old daughter for Donatussin Drops. She wants to know if this medication will have an effect on her daughter's alertness.

Components of Donatussin:

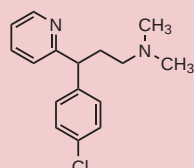
Phenylephrine (decongestant)

Chlorpheniramine (antihistamine)

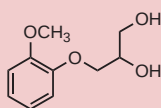
Guaifenesin (expectorant)



Phenylephrine



Chlorpheniramine



Guaifenesin

1. Identify the structural features/functional groups of phenylephrine and guaifenesin that contribute to improved water solubility (medication given as drops). List the type(s) of interactions that these groups have with water, and draw an example of these interactions (with appropriate labels).
2. Evaluate each of the three molecules, and determine if each molecule contains any functional groups that will allow the drug to cross the blood-brain barrier and have an effect on this child's alertness (create a list of relevant functional groups for each molecule). Based on your evaluation, which agent is likely to have the most significant effect? Identify what property is necessary for these agents to cross this biologic membrane.
3. Identify the binding interactions that chlorpheniramine and guaifenesin could have with their respective targets for drug action. Be sure to identify which functional groups will participate in each of these binding interactions.

Problem 6, found at the end of the chapter, provides an additional opportunity to use this approach to predict water solubility. Solubility data for these drug molecules can be found in the USP. In most instances, discrepancies between approximate and actual water solubilities can be rationalized by careful inspection of the chemical structure.

Predicting Water Solubility: Analytical/Quantitative Approach

Another method for predicting water solubility involves calculation of an approximate logP, or log of the partition coefficient for a molecule. This approach is based on an approximation method developed by Cates (11) and discussed in Lemke (10). In this approach, one sums

TABLE 2.5 Water-solubilizing Potential of Organic Functional Groups in a Mono- or Polyfunctional Molecule

Functional Group	Monofunction Molecule	Polyfunctional Molecule
Alcohol	5 to 6 carbons	3 to 4 carbons
Phenol	6 to 7 carbons	3 to 4 carbons
Ether	4 to 5 carbons	2 carbons
Aldehyde	4 to 5 carbons	2 carbons
Ketone	5 to 6 carbons	2 carbons
Amine	6 to 7 carbons	3 carbons
Carboxylic acid	5 to 6 carbons	3 carbons
Ester	6 carbons	3 carbons
Amide	6 carbons	2 to 3 carbons
Urea, carbonate, carbamate		2 carbons

Water solubility is defined as greater than 1% solubility (9).

the hydrophobic or hydrophilic properties of each functional group present in the molecule. Before we can calculate logP values, a brief explanation of the concept of partition coefficient is necessary.

Eq. 2.7
$$P = C_{\text{oct}} / C_{\text{water}}$$

In its simplest form, the partition coefficient, *P*, refers to the ratio of drug concentration in octanol (C_{oct}) to that in water (C_{water}) (Eq. 2.7). Octanol is used to mimic the amphiphilic nature of lipid, because it has a polar head group (primary alcohol) and a long hydrocarbon chain, or tail, similar to the fatty acid tail that makes up part of a lipid membrane. Because *P* is logarithmically related to free energy (12), *P* is generally expressed as log*P* and is, therefore, the sum of the hydrophobic and hydrophilic characteristics of the functional groups that make up the structure of the molecule. Thus, log*P* is a measure of the lipid/water solubility characteristics of the entire molecule. Because each functional group contained within the molecule contributes to the overall hydrophilic/hydrophobic character of the molecule, a hydrophilic/hydrophobic value (the hydrophobic substituent constant, π) can be assigned to each functional group. Equation 2.8 defines this relationship:

Eq. 2.8
$$\text{Log}P = \Sigma \pi \text{ (fragments)}$$

When calculating log*P* from hydrophobic substituent constants, the sum is usually referred to as log*P*_{calc} or Clog*P* [for software sources to calculate Clog*P*, see (16)] to distinguish it from an experimentally determined value (Mlog*P* or log*P*_{meas}). Over the years, extensive tables

of π values have been compiled for organic functional groups and molecular fragments (12–15). Table 2.6 is a highly abbreviated summary of π values from Lemke (10), based largely on the manuscript by Cates (11). Using the values in this table, a fairly reasonable estimate for the water solubility of many organic compounds (shown as logP) can be determined.

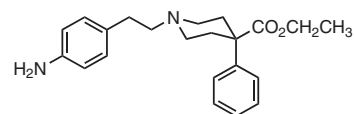
Again we will consider the structure of the opioid analgesic anileridine to demonstrate the calculation of logP (Fig. 2.13). This compound has a total of 22 carbon atoms, some aliphatic and some aromatic. We need to distinguish between the aliphatic and aromatic carbon atoms because the delocalized π orbitals for the sp^2 hybridized aromatic carbon atoms make them more polar than aliphatic carbons. The compound also contains one tertiary alkylamine, one aromatic or aryl amine, and one ester. Evaluation of esters and amides requires that the oxygen, nitrogen, and ester/amide carbon atoms are counted in this π value. The remaining aliphatic carbons are then counted. Figure 2.14 summarizes the logP calculation for anileridine. The calculation gives a ClogP value for anileridine of +4.8. Water solubility as defined by the USP is solubility greater than 3.3%, which equates to an approximate logP of +0.5. LogP values less than +0.5 are therefore considered to be water soluble, and those greater than +0.5 are considered to be water insoluble. According to our calculation, anileridine would be predicted to be insoluble in water. This calculation agrees with the more empiric procedure discussed earlier.

Other sample calculations are shown in Figure 2.15, and several problems are provided at the end of this chapter. In Figure 2.15, MlogP values (when available) and ClogP values (16) are included for comparison purposes (see Appendix A for additional ClogP values). Even though the π values from Table 2.6 are not as extensive as those in the computer program, there is good general agreement with most of these compounds with respect to their solubility (or insolubility) in water. In addition, other programs besides ClogP are available to predict logP values; some of these programs are available on the Internet. One must keep in mind that due to the assumptions made in these programs, they cannot produce results that are in total agreement with measured values or other prediction programs. ClogP values calculated from ACDLogP (16) are generally more accurate. Other programs for calculating logP values, such as Molinspiration (17) and Interactive Analysis (18), use different methods and assumptions and, therefore, do not always agree with ClogP predictions or experimentally determined values. This is not to say that the latter two programs do not give accurate results. Often, one or all of the programs will have reasonable agreement with measured values, but greater disagreement tends to occur as the number of functional groups in the molecule that participate as hydrogen-bond acceptor and/or hydrogen-bond donor groups increases. This increases the likelihood that intramolecular interactions will occur—something that is not always taken into account with these programs.

TABLE 2.6 Hydrophilic-lipophilic Values (π V) for Organic Fragments (10)

Functional Group	π value (aliphatic)	π value (aromatic)
H		0.00
Alkane	0.50	0.56 (CH_3); 1.02 (CH_2CH_3)
Alkene		0.82
C_6H_5 (phenyl)	2.15	1.96
Br, Cl, F, I	0.60; 0.39; −0.17; 1.00	0.86; 0.71; 0.14; 1.12
NO_2	−0.85	−0.28
NH_2 (primary amine)	−1.19	−1.23
NHR (secondary amine)	−0.67	0.47
NR_2 (tertiary amine)	−0.30	0.18
$-\text{NHC}=\text{OR}$ (amide)	−0.97	
SC_6H_5	2.32	
OH	−1.12	−0.67
OCH_3		−0.02
$-\text{OC}=\text{OR}$ (ester)	−0.27	−0.64
CHO (aldehyde)		−0.65
$\text{C}=\text{OCH}_3$ (ketone)		−0.55
CO_2H		−0.32
SO_2NH_2 (sulfonamide)		−1.82

The ability to predict the percent ionization or water solubility of a molecule should not be viewed as an exercise in arithmetic, but rather as a way to understand the solution behavior of molecules, especially as it relates to admixtures and the pharmacokinetic differences among molecules. The ionization state of a molecule not only influences its water solubility, but also its ability to traverse



Fragments	π
1 primary aromatic or aryl amine	−1.23
1 tertiary alkylamine	−0.30
9 aliphatic carbons	+4.5
2 phenyl rings	+4.30
1 ester	−0.27
logP	+7.0

FIGURE 2.14 Calculation of logP for anileridine.

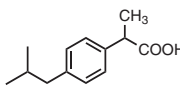
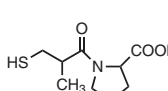
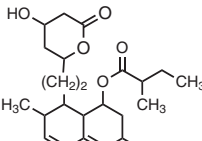
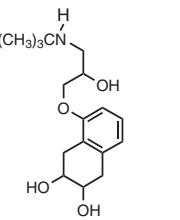
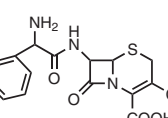
	<table><tr><th>Fragments</th><th>π</th></tr><tr><td>6 carbons</td><td>+3.0</td></tr><tr><td>1 phenyl</td><td>+2.15</td></tr><tr><td>1 carboxyl</td><td>-0.32</td></tr><tr><td>logP</td><td>+4.83</td></tr><tr><td>MlogP</td><td>+3.5; ClogP +3.68</td></tr></table>	Fragments	π	6 carbons	+3.0	1 phenyl	+2.15	1 carboxyl	-0.32	logP	+4.83	MlogP	+3.5; ClogP +3.68		<table><tr><th>Fragments</th><th>π</th></tr><tr><td>7 carbons</td><td>+3.5</td></tr><tr><td>1 sulfur</td><td>0.0</td></tr><tr><td>1 carboxyl</td><td>-0.32</td></tr><tr><td>1 amide</td><td>-0.97</td></tr><tr><td>logP</td><td>+2.21</td></tr><tr><td>ClogP</td><td>+1.16</td></tr></table>	Fragments	π	7 carbons	+3.5	1 sulfur	0.0	1 carboxyl	-0.32	1 amide	-0.97	logP	+2.21	ClogP	+1.16		<table><tr><th>Fragments</th><th>π</th></tr><tr><td>22 carbons</td><td>+11.0</td></tr><tr><td>1 alcohol</td><td>-1.12</td></tr><tr><td>2 carboxyls</td><td>-0.64</td></tr><tr><td>logP</td><td>+ 9.24</td></tr><tr><td>MlogP</td><td>+4.26; ClogP +4.08</td></tr></table>	Fragments	π	22 carbons	+11.0	1 alcohol	-1.12	2 carboxyls	-0.64	logP	+ 9.24	MlogP	+4.26; ClogP +4.08
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FIGURE 2.15 ClogP calculations for selected compounds.

membranes and, therefore, its ability to be absorbed. The distribution of the drug and its ability to bind to proteins other than its target are also greatly influenced by the ionization state and the hydrophilic/hydrophobic nature of the molecule.

STEREOCHEMISTRY AND DRUG ACTION

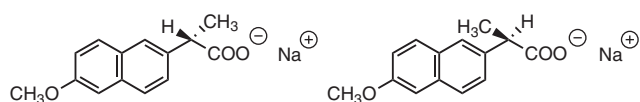
Stereoisomers are molecules that contain the same number and kinds of atoms, the same arrangement of bonds, but different three-dimensional structures; in other words, they only differ in the three-dimensional arrangement of atoms in space. There are two types of stereoisomers: enantiomers and diastereomers. Enantiomers are pairs of molecules for which the three-dimensional arrangement of atoms represents nonsuperimposable mirror images. Diastereomers represent all of the other stereoisomeric compounds that are not enantiomers. Thus, the term “diastereomer” includes compounds that contain double bonds (geometric isomers) and ring systems. Unlike enantiomers, diastereomers exhibit different physicochemical properties, including, but not limited to, melting point, boiling point, solubility, and chromatographic behavior. These differences in physicochemical properties allow the separation of individual diastereomers from mixtures with the use of standard chemical separation techniques, such as column chromatography or crystallization. Enantiomers cannot be separated using such techniques unless a chiral environment is provided or if they are first converted to diastereomers (e.g., salt formation with another enantiomer). Examples of enantiomers and diastereomers are provided in Figure 2.16.

The physicochemical properties of a drug molecule are dependent not only on what functional groups are present in the molecule but also on the spatial arrangement of these groups. This becomes an especially

important factor when the environment that a molecule is in is asymmetric, such as the human body. Proteins and other biologic targets are asymmetric in nature. How a particular drug molecule interacts with these macromolecules is determined by the three-dimensional orientation of the functional groups present. If critical functional groups in the drug molecule do not occupy the proper spatial region, then productive interactions with the biologic target will not be possible. As a result, it is possible that the desired pharmacologic activity will not be achieved. If, however, the functional groups within a drug molecule are located in the proper three-dimensional orientation, then the drug can participate in multiple key interactions with its biologic target. It is important to understand not only which functional groups contribute to the pharmacologic activity of a drug, but also the importance of the three-dimensional nature of these functional groups in predicting drug potency and potential side effects.

Approximately one in every four drugs currently on the market is some type of isomeric mixture. For many of these drugs, the biologic activity may only reside in one isomer (or at least predominate in one isomer). The majority of these isomeric mixtures are termed “racemic mixtures” (or “racemates”). A racemic mixture is comprised of equal amounts of both possible drug enantiomers. As mentioned earlier in this chapter, when enantiomers are introduced into an asymmetric, or chiral, environment, such as the human body, they display different physicochemical properties. This can lead to significant differences in their pharmacokinetic and pharmacodynamic behavior, resulting in adverse side effects or toxicity. For example, the individual isomers in a racemic mixture can exhibit significant differences in absorption (especially active transport), serum protein binding, and metabolism. As it relates to drug metabolism, it is certainly possible that only one of the

ENANTIOMERS



S-(+)-naproxen sodium

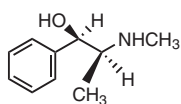
R-(-)-naproxen sodium



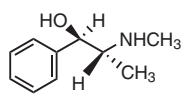
Levorphanol (anagetic)

Dextrophan (antitussive)

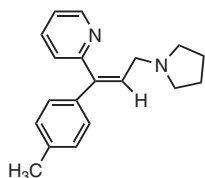
DIASTEREOMERS



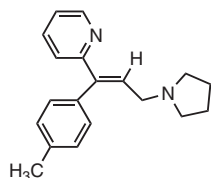
1R, 2S-(-)-Ephedrine



1R, 2R-(-)-Pseudoephedrine



Z-triprolidine (inactive)



E-triprolidine (active)

FIGURE 2.16 Examples of stereoisomers.

isomers can be converted into a toxic substance or can influence the metabolism of another drug (Chapter 4). Since stereochemistry can have a profound effect on both the pharmacokinetic and pharmacodynamic properties of a drug, it is important to review the foundational concepts

Stereochemical Definitions**Designation of Absolute Configuration**

At first, enantiomers were distinguished by their ability to rotate the plane of polarized light. Isomers that rotate the plane of polarized light to the right, or in a clockwise direction, were designated as dextrorotatory, indicated by a (+)- sign before the chemical name [e.g., (+)-amphetamine or dextroamphetamine]. The opposite designation, levorotatory or (–)–, was assigned to molecules that rotate the plane of polarized light to the left, or in a counterclockwise direction. The letters *d*- and *l*- were formerly used to indicate (+)- and (–)-, respectively. A racemate (racemic mixture)—that is, a 1:1 mixture of enantiomers—is indicated by placement of a (±)- before the compound name. This nomenclature is based on a

QUESTIONS WE CAN NOW ANSWER ABOUT ANY DRUG MOLECULE

Based on your knowledge of acid–base chemistry, from where will this drug primarily be absorbed?
What is the solubility of the drug in the stomach, plasma, or an aqueous IV?
What are the possible interactions that the drug could have with its respective target for drug action?
What is the compatibility of the drug if mixed with other drugs?
How should this drug be delivered? Is it stable in stomach acid?

physical property of the molecule and does not describe the absolute configuration or three-dimensional arrangement of atoms around the chiral center.

In the late 19th century, Fisher and Rosanoff developed a system of nomenclature based on the structure of glyceraldehyde (Fig. 2.17). Since there were no methods at that time to determine the absolute three-dimensional arrangement of atoms in space, the two isomers of glyceraldehyde were arbitrarily assigned the designation of D-(+)- and L-(–). It was not until the 1950s that the absolute configurations of these molecules were determined (Fisher had fortuitously guessed correctly). The configurations of other molecules were then assigned based on their relationship to D- or L-glyceraldehyde via synthetic methods or chemical degradation. Thus, via chemical degradation, it was possible to determine that (+)-glucose, (–)-2-deoxyribose, and (–)-fructose had the same terminal configuration as D-(+)-glyceraldehyde and, therefore, were assigned the D-absolute configuration. Amino acid configurations were assigned based on their relationship to D-(+)- and L-(–)-serine (Fig. 2.17). Unfortunately, this system becomes very cumbersome with molecules that contain more than one chiral center.

LEARNING THE LINGO: DRUG MOLECULE EVALUATION*Analysis of Individual Functional Groups:*

- Name of functional group
- Shape of functional group
- Hydrophobic vs. hydrophilic character
- Polar vs. nonpolar character
- Acidic vs. basic (pK_a) character
- Binding interactions
- Chemical/enzymatic stability

Analysis of the Whole Drug Molecule:

- Looking for functional group balance: water solubility and absorption
- Ionization issues: effect on solubility and absorption
- Drug combinations: acid–base interactions
- Drug interactions with biologic target: good fit or not?
- Stability and bioavailability: route of administration

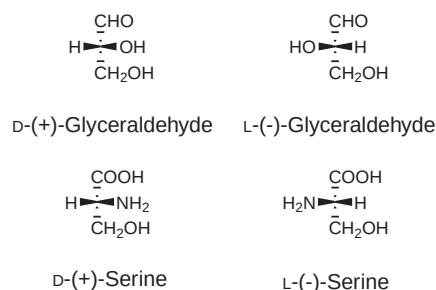
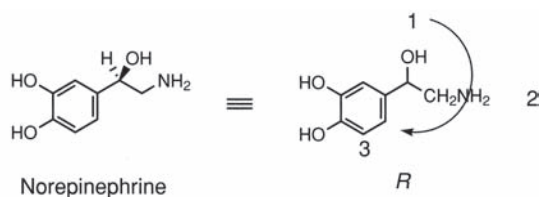


FIGURE 2.17 Relationship of optical isomers of serine to D- and L-glyceraldehyde.

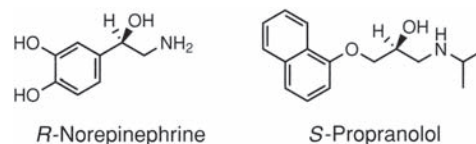
In 1956, a new system of stereochemical nomenclature was introduced by Cahn et al. (19) and is known as the Sequence Rule or CIP system. With this system, atoms attached to a chiral center are ranked based on their atomic number. Highest priority is given to the atom with the highest atomic number, and subsequent atoms are ranked accordingly, from highest to lowest. When a decision cannot be made in the assignment of priority—for example, two atoms with the same atomic number attached to the chiral center—this evaluation extends to the next atom until a priority can be established. When the molecule is then viewed from the side opposite to the lowest-priority atom, the priority sequence from highest to lowest can then be determined. If the priority sequence proceeds to the right, or in a clockwise direction, the chiral center is designated with an *R*-absolute configuration. The designation is *S* when the priority sequence proceeds to the left, or in a counterclockwise direction. An example of this is seen in the neurotransmitter norepinephrine.



Degradation studies demonstrate that (–)-norepinephrine is related to D-(–)-mandelic acid; therefore, it was given the D-designation using the Fisher system. With the CIP system, norepinephrine is assigned the *R*-absolute configuration.

It should be noted that the CIP nomenclature system uses a set of arbitrary rules and, therefore, should be viewed as a system that tracks absolute configuration only. In many instances, two molecules can have different absolute configurations as designated by the CIP system, but the same relative orientation of the functional groups relevant for biologic activity. An example of this is demonstrated when the absolute configuration of the nonselective β -adrenergic antagonist propranolol is compared to norepinephrine. Because of the presence of the ether oxygen atom, the priority sequence of the functional groups about the chiral

center results in the assignment of the *S*-absolute configuration for the more active enantiomer of propranolol. Close inspection of both *R*-norepinephrine and *S*-propranolol, however, shows that the hydroxy group, basic amine, and aromatic rings of both compounds occupy the same regions in three-dimensional space.



Stereochemistry and Biologic Activity

Easson-Stedman Hypothesis

In 1886, Piutti (20) reported different physiologic actions for the enantiomers of asparagine, with (+)-asparagine having a sweet taste and (–)-asparagine a bland one. This was one of the earliest observations that enantiomers can exhibit differences in biologic action. In 1933, Easson and Stedman (21) reasoned that differences in biologic activity between enantiomers resulted from selective reactivity of one enantiomer with its receptor. They postulated that such interactions require a minimum of a three-point fit to the receptor. This is demonstrated in Figure 2.18 for two hypothetical enantiomers. In Figure 2.18, the letters A, B, and C represent hypothetical functional groups that can interact with complementary sites on the hypothetical receptor surface, represented by A', B', and C'. Only

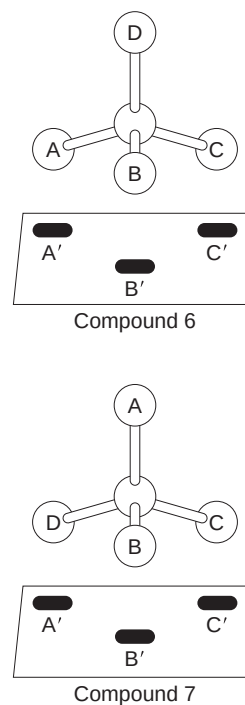


FIGURE 2.18 Optical isomers. Only in compound 6 do the functional groups A, B, and C align with the corresponding sites of binding on the asymmetric surface.

one enantiomer is capable of attaining the correct special orientation to enable all three functional groups to interact with their respective sites on the receptor surface. The inability of the other enantiomer to achieve the same number of interactions with the hypothetical receptor surface explains its reduced biologic activity. The Easson-Stedman Hypothesis states that the more potent enantiomer must be involved in a minimum of three intermolecular interactions with the surface of the biologic target and that the less potent enantiomer only interacts with two sites. This can be illustrated by looking at the differences in vasopressor activity of *R*(-)-epinephrine, *S*(+)-epinephrine, and the achiral *N*-methyldopamine (Fig. 2.19). With *R*(-)-epinephrine, the three points of interaction with the receptor site are the substituted aromatic ring, β -hydroxyl group, and the protonated secondary ammonium group. All three functional groups interact with their complementary sites on the receptor surface, resulting in receptor stimulation (in this case). With *S*(+)-epinephrine, only two interactions are possible (the protonated secondary ammonium and the substituted aromatic ring). The β -hydroxyl group is located in the wrong place in space and, therefore, cannot interact properly with the receptor. *N*-methyldopamine can achieve the same interactions with the receptor as *S*(+)-epinephrine; therefore, it is not surprising that its vasopressor response is the same as that of *S*(+)-epinephrine and less than that of *R*(-)-epinephrine.

Not all stereoselectivity seen with enantiomers can be attributed to differences in the ability of the drug molecule to interact with its biologic target. Differences in biologic activity can also result from differences in the ability of each enantiomer to reach the biologic target. Because the biologic system encountered by the drug is asymmetric, each enantiomer can experience selective penetration into membranes, metabolism, absorption at sites of loss (e.g., adipose tissue), and/or excretion. Figure 2.20 shows various phases that enantiomers can encounter before reaching the biologic target. An enantiomer cannot encounter stereoselective environments at each of these points; however, enantioselectivity at any

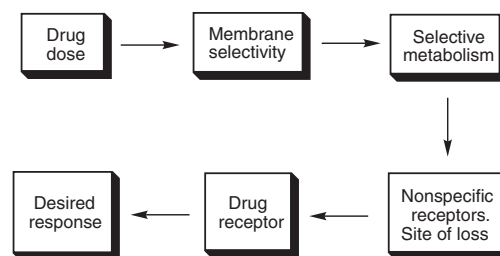


FIGURE 2.20 Selective phases to which optical isomers can be subjected before biologic response.

point can provide enough of an influence to cause one enantiomer to produce a significantly better pharmacologic effect than the other. Conversely, such processes can also contribute to untoward effects of a particular enantiomer. Differences in pharmacologic action among stereoisomers provides an excellent example of how not all pharmacologic effects of a drug are necessarily beneficial to the patient. Although there is no regulatory prohibition on the development of racemic agents, it is reasonable that single enantiomer drugs will become the overwhelming therapeutic choice in the future.

Diastereomers

As mentioned earlier, diastereomers are molecules that are nonsuperimposable, non-mirror images. This type of isomer can result from the presence of more than one chiral center in the molecule, double bonds, or ring systems. These isomers have different physicochemical properties, and as a result, it is possible that they can have differences in biologic activity.

Molecules that contain more than one chiral center probably are the most common type of drug-based diastereomers. Classic examples are the diastereomers ephedrine and pseudoephedrine (Fig. 2.21). When a molecule contains two chiral centers, there can be as

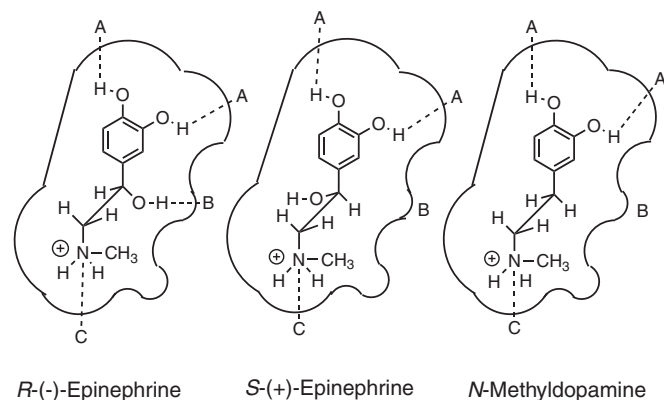


FIGURE 2.19 Drug receptor interaction of *R*(-)-epinephrine, *S*(+)-epinephrine, and *N*-methyldopamine.

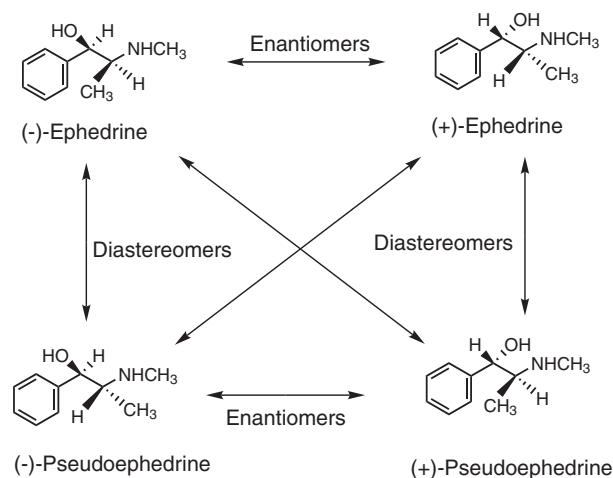


FIGURE 2.21 Relationship between the diastereomers of ephedrine and pseudoephedrine.

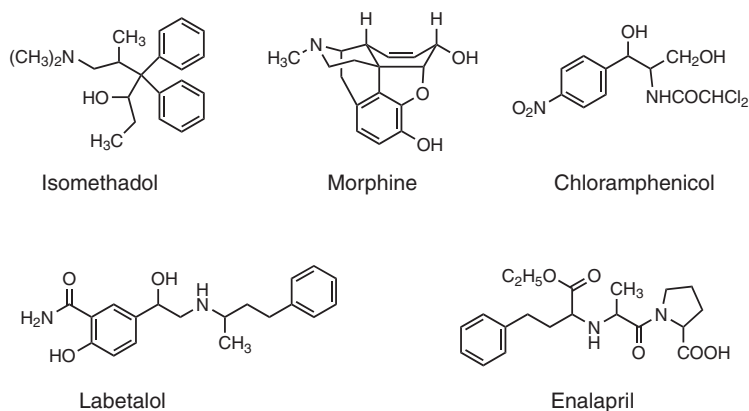
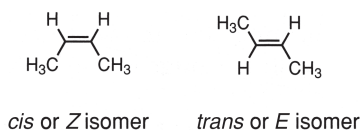


FIGURE 2.22 Examples of chiral drugs with two or more asymmetric centers.

many as four possible stereoisomers consisting of two sets of enantiomeric pairs. When considering an enantiomeric pair of molecules, there is inversion of both chiral centers. In diastereomers, there is inversion of only one chiral center. (Problem 9 at the end of this chapter helps to illustrate this point.) Figure 2.22 shows several examples that contain two or more chiral centers and, therefore, are diastereomeric (see Problem 10 at the end of this chapter).

Restricted bond rotation caused by carbon–carbon double bonds (alkenes or olefins) and similar systems, such as imines ($C=N$), can produce stereoisomers. These are also referred to as geometric isomers, although they more properly are classified as diastereomers. In this situation, substituents can be oriented on the same side or on opposite sides of the double bond. The alkene 2-butene is a simple example.



With 2-butene, it is readily apparent that the methyl groups can be on the same or on opposite sides of the double bond. When they are on the same side, the molecule is defined as the *cis*- or *Z*-isomer (from the German *zusammen*, meaning “together”); when they are on opposite sides, the designation is *trans*- or *E*- (from the German *entgegen*, meaning “opposite”). With simple compounds, such as 2-butene, it is easy to determine which groups in the molecule are *cis* or *trans* to one another. This becomes more difficult to determine, however, with more complex structures, where it is less obvious which substituents should be referred to when naming the compound. In 1968, Blackwood et al. (22) proposed a system for the assignment of “absolute” configuration with respect to double bonds. Using the CIP sequence rules, each of the two substituents attached to the carbon atoms comprising the double bond are

assigned a priority of 1 or 2, depending on the atomic number of the atom attached to the double bond. When two substituents of higher priority are on the same side of the double bond, this isomer is given the designation of *cis* or *Z*. When the substituents are on opposite sides, the designation is *trans* or *E*. The histamine H_1 -receptor antagonist triprolidine (Fig. 2.23) is a good example for demonstrating how this nomenclature system works. The *E*-isomer of triprolidine is more active both in vitro and in vivo, indicating that the distance between the pyridine and pyrrolidine rings is critical for binding to the receptor.

Diastereomers (as well as enantiomers) can also be found in cyclic molecules. For example, the cyclic alkane 1,2-dimethylcyclohexane can exist as *cis/trans*-diastereomers, and the *trans* isomer can also exist as an enantiomeric pair. In Figure 2.24, each of the *trans*-enantiomorphs is depicted in the two possible chair conformations for the cyclohexane ring. Since cyclohexane rings can exhibit significant conformational freedom, this allows for the possibility of conformational isomers. Isomers of this type will be discussed in the next section. When two or more rings share a common bond (e.g., decalin), rotation around the bonds is even more restricted. This prevents ring “flipping” from occurring, and as a result, diastereomers and enantiomers are generated.

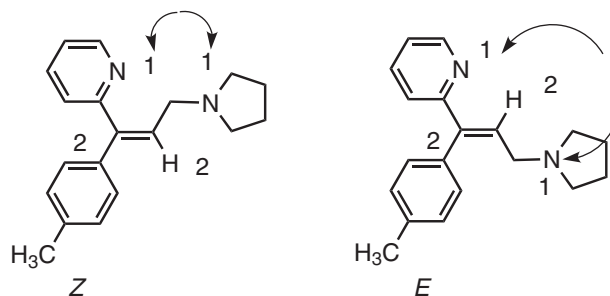


FIGURE 2.23 Geometric isomers of triprolidine.

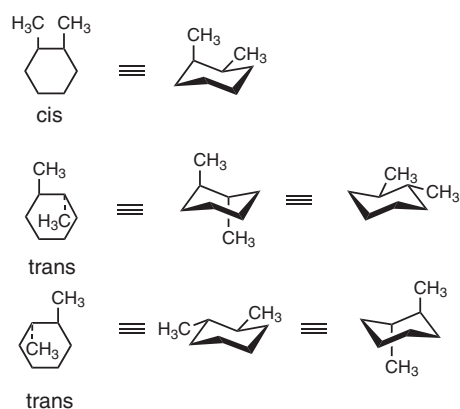
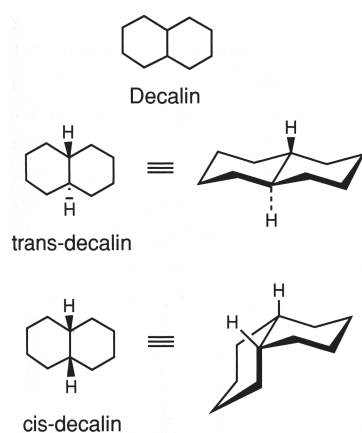


FIGURE 2.24 Diastereomers of 1,2-dimethylcyclohexane.

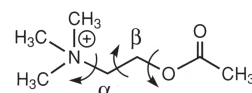


In the case of decalin, a two-ring system, the rings are fused together via a common bond in either the *trans* or *cis* configuration as shown. Steroids, a class of medically important compounds that consist of four fused rings (three cyclohexanes and one cyclopentane), exhibit significantly different biologic activity when the first two cyclohexane rings are fused into different configurations, referred to as the 5α - or 5β -isomers (Fig. 2.25). The α -designation indicates that the hydrogen atom in the 5-position is below the “plane” of the ring system; the β -designation refers to

the hydrogen atom being above this plane. What appears to be a very minor change in orientation for the substituent results in a very drastic change in the three-dimensional shape of the molecule and in its biologic activity. Figure 2.25 shows the diastereomers 5α -cholestane and 5β -cholestane as examples. The chemistry and pharmacology of steroids will be discussed in more detail in Chapters 28 (Adrenocorticoids), 40 (Men’s Health: Androgens), and 41 (Women’s Health: Estrogens and Progestins).

Conformational Isomerism

Conformational isomerism takes place via rotation about one or more single bonds. Such bond rotation results in nonidentical spatial arrangement of atoms in a molecule. This type of isomerism does not require much energy because no bonds are broken. In the conversion of one enantiomer into another (or diastereomer) bonds are broken, which requires significantly more energy. The neurotransmitter acetylcholine can be used to demonstrate the concept of conformational isomers.



Acetylcholine

Each single bond within the acetylcholine molecule is capable of undergoing rotation, and at room temperature, such rotations readily occur. Rotation around the $\text{C}\alpha\text{--C}\beta$ bond of acetylcholine was shown by Kemp and Pitzer (23) in 1936 not to be free but, rather, to have an energy barrier, which is sufficiently low that at room temperature acetylcholine exists in many interconvertible conformations (see Chapter 9). Rotation around the central $\text{C}\alpha\text{--C}\beta$ bond produces the greatest spatial rearrangement of atoms compared to rotation around any other bond. Since the atoms at the end of some of the bonds within acetylcholine are identical, rotation about several of these bonds produces redundant structures. When viewed along the $\text{C}\alpha\text{--C}\beta$ bond, acetylcholine can be depicted in

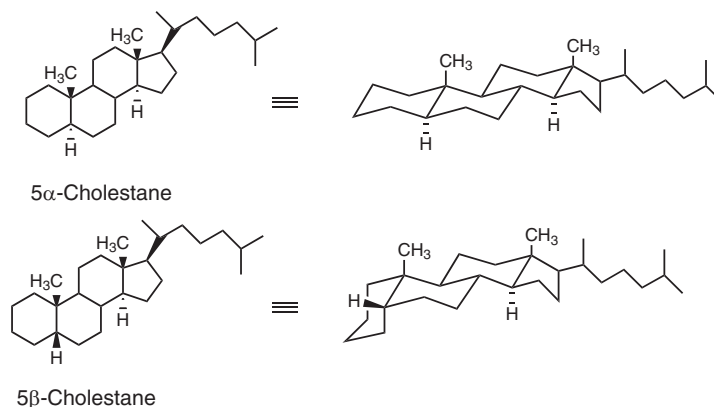


FIGURE 2.25 The 5α and 5β conformations of the steroid nucleus cholestane.

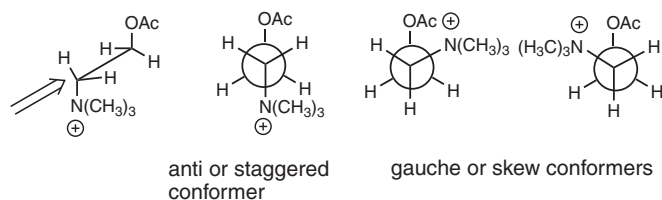


FIGURE 2.26 Anti and gauche conformations of acetylcholine.

the sawhorse or Newman projections, as shown in Figure 2.26. When the ester and trimethylammonium groups are 180° apart, the molecule is said to be in the anti, or staggered, conformation (or conformer or rotamer). This conformation allows maximum separation of the functional groups and is the most stable conformation energetically. It is possible that other conformations are more stable if factors other than steric interactions are considered (e.g., intramolecular hydrogen bonds). Rotation of one end of the $C\alpha-C\beta$ bond by 120° or 240° results in the two gauche, or skew, conformations shown in Figure 2.26. These are less stable than the anti conformer, although some studies suggest that an electrostatic attraction between the electron-poor trimethylammonium and electron-rich ester oxygen atom stabilizes this conformation. Rotation by 60° , 180° , and 240° produces the least stable conformations in which all of the atoms overlap, what are referred to as eclipsed conformations.

DRUG DESIGN: DISCOVERY AND STRUCTURAL MODIFICATION OF LEAD COMPOUNDS

Process of Drug Discovery

The process of drug discovery begins with the identification of new, previously undiscovered, biologically active compounds, often called “hits,” which are typically found by screening many compounds for the desired biologic properties. We will next explore the various approaches used to identify “hits” and to convert these “hits” into “lead” compounds and, subsequently, into drug candidates suitable for clinical trials. Sources of “hits” can originate from natural sources, such as plants, animals, or fungi; from synthetic chemical libraries, such as those created through combinatorial chemistry or historic chemical compound collections; from chemical and biologic intuition due to years of chemical–biologic training; from targeted/rational drug design; or from computational modeling of a target site such as an enzyme. Chemical or functional group modifications of the “hits” are performed in order to improve the pharmacologic, toxicologic, physiochemical, and pharmacokinetic properties of a “hit” compound into a “lead” compound. The lead compound to be optimized should be of a known chemical structure and possess a known mechanism of action, including knowledge of its functional groups (pharmacophoric groups) that are recognized by the receptor/active site and are responsible for that molecule’s affinity at the targeted receptor site. “Lead optimization” is the process whereby modifications

of the functional groups of the lead compound are carried out in order to improve its recognition, affinity, and binding geometries of the pharmacophoric groups for the targeted site (a receptor or enzyme); its pharmacokinetics; or its reactivity and stability toward metabolic degradation. The final step of the drug discovery process involves rendering the lead compound into a drug candidate that is safe and suitable for use in human clinical trials, including the preparation of a suitable drug formulation.

Natural Product Screening

Perhaps the most difficult aspect of drug discovery is that of lead discovery. Until the late 19th century, the development of new chemical entities for medicinal purposes was achieved primarily through the use of natural products, generally derived from plant sources (see Chapter 1). As the colonial powers of Europe discovered new lands in the Western Hemisphere and colonized Asia, the Europeans learned from the indigenous peoples of the newly discovered lands of remedies for many ailments derived from plants. Salicylic acid was isolated from the bark of willow trees after learning that Native Americans brewed the bark to treat inflammatory ailments. Structural optimization of this lead compound (salicylic acid) by the Bayer Corporation of Germany resulted in acetylsalicylic acid, or aspirin, the first nonsteroidal anti-inflammatory agent. South American natives used a tea obtained by brewing Cinchona bark to treat chills and fever. Further study in Europe led to the isolation of quinine and quinidine, which subsequently were used to treat malaria and cardiac arrhythmias, respectively. Following “leads” from folklore medicine, chemists of the late 19th and early 20th centuries began to seek new medicinals from plant sources and to assay them for many types of pharmacologic actions. This approach to drug discovery is often referred to as “natural product screening.” Before the mid-1970s, this was one of the major approaches to obtaining new chemical entities as “leads” for new drugs. Unfortunately, this approach fell out of favor and was replaced with the rational approaches to drug design developed during that period (see next section). Heightened awareness of the fragility of ecosystems, especially the rainforests, has fueled a resurgence of screening products from plants before they become extinct. A new field of pharmacology, called “ethnopharmacology,” which is the discipline of identifying potential natural product sources with medicinal properties based on native lore, has emerged as a result.

Compounds isolated from natural sources are usually tested in one or more bioassays for the ailment(s) that the plant material has been purported to treat. Interestingly, the treatment of different ailments can require different methods of preparation (e.g., brewing, chewing, or direct application to wounds) or different parts of the same plant (e.g., roots, stem, leaves, flowers, or sap). As it turns out, each method of administration or part of the plant used can produce one or more different chemical compounds that are necessary to generate the desired outcome.

Drug Discovery via Random Screening of Synthetic Organic Compounds

The random screening of synthetic organic compounds approach to the discovery of new chemical entities for a particular biologic action began in the 1930s, after the discovery of the sulfonamide class of antibacterials. All compounds available to the investigator (natural products, synthetic molecules), regardless of structure, were tested in the pharmacologic assays available at the time. This random screening approach was also applied in the 1960s and 1970s in an effort to find agents that were effective against cancer. Some groups did not limit their assays to identify a particular type of biologic activity but, rather, tested compounds in a wide variety of assays. This large-scale screening approach of drug “leads” is referred to as high-throughput screening, which involves the simultaneous bioassay of thousands of compounds in hundreds to thousands of bioassays. These types of bioassays became possible with the advent of computer-controlled robotic systems for the assays and combinatorial chemistry techniques for the synthesis of large numbers of compounds in small (milligram) quantities. This type of random screening eventually gave way to targeted dedicated screening and rational design techniques.

Drug Discovery from Targeted Dedicated Screening and Rational Drug Design

Rational drug design is a more focused approach that uses greater knowledge (structural information) about the drug receptor (targets) or one of its natural ligands as a basis to design, identify, or create drug “leads.” Testing is usually done with one or two models (e.g., specific receptor systems or enzymes) based on the therapeutic target. The drug design component often involves molecular modeling and the use of quantitative structure–activity relationships (QSARs) to better define the physicochemical properties and the pharmacophoric groups that are essential for biologic activity. The development of QSARs relies on the ability to examine multiple relationships between physical properties and biologic activities. In classic QSAR (e.g., Hansch-type analysis), an equation defines biologic activity as a linear free-energy relationship between physicochemical and/or structural properties. It permits evaluation of the nature of interaction forces between a drug and its biological target, as well as the ability to predict activity in molecules. These approaches are better for the development of a lead compound into a drug candidate than for the discovery of a lead compound.

Drug Discovery via Drug Metabolism Studies

New drug entities have been “discovered” as drug leads through investigation of the metabolism of drug molecules that already are clinical candidates or, in some instances, are already on the market. In this method,

metabolites of known drug entities are isolated and assayed for biologic activity using either the same target system or broader screen target systems. The broader screening systems are more useful if the metabolite under evaluation is a chemical structure that was radically altered from the parent molecule through some unusual metabolic rearrangement reaction. In most cases, the metabolite is not radically different from the parent molecule and, therefore, would be expected to exhibit similar pharmacologic effects. One advantage of evaluating this type of drug candidate is that a metabolite can possess better pharmacokinetic properties, such as a longer duration of action, better oral absorption, or less toxicity with fewer side effects (e.g., terfenadine and its antihistaminic hydroxylated metabolite, fexofenadine). As it turns out, the sulfonamide antibacterial agents were discovered in this way. The azo dye Prontosil was found to have only antibacterial action in vivo. It was soon discovered that this compound required metabolic activation via reduction of the diazo group to produce the active metabolite 4-aminobenzene sulfonamide (Fig. 2.27). The sulfonamide mimics the physicochemical properties of PABA, a crucial component in microbial metabolism. It is no surprise that the sulfonamide acts as a competitive inhibitor of the enzyme for which PABA is a substrate.

Drug Discovery from the Observation of Side Effects

An astute clinician or pharmacologist can detect a side effect in a patient or animal model that could lead, on further development, to a new therapeutic use for a particular chemical entity. Discovery of new lead compounds via exploitation of side-effect profiles of existing agents is discussed below.

One of the more interesting drug development scenarios is that of the phenothiazine antipsychotics (see Chapter 14). Molecules with this type of biologic activity can be traced back to the first histamine H_1 -receptor antagonists developed in the 1930s. In 1937, Bovet and Staub (24) were the first to recognize that it should be possible to antagonize the effects of histamine and, thereby, treat allergic reactions. They tested compounds that were known to act on the autonomic nervous system and, eventually, discovered that benzodioxanes (Fig. 2.28) significantly antagonized the effects of histamine. During an attempt

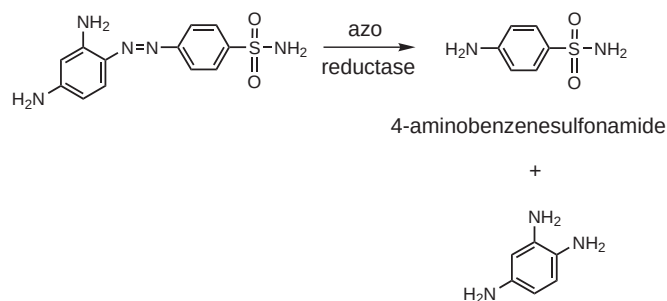


FIGURE 2.27 Metabolic conversion of prontosil to 4-aminobenzenesulfonamide.

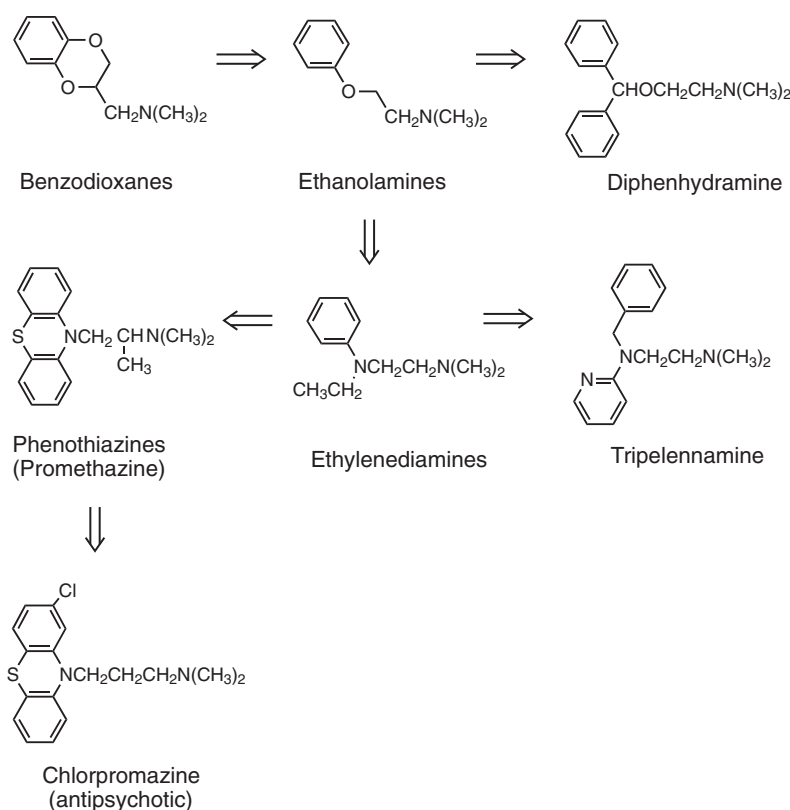


FIGURE 2.28 Development of phenothiazine-type antipsychotic drugs.

to improve the antihistaminergic action of the benzodioxanes, it was discovered that phenyl substituted ethanolamines also demonstrated significant antihistaminergic activity. Further development of this class generated two different classes of antihistamines, the diphenhydramine class of antihistamines represented by diphenhydramine (Fig. 2.28) and the ethylenediamine class, represented by tripelennamine (Fig. 2.28) (see also Chapter 32).

Incorporation of the aromatic rings of the ethylenediamines into the rigid and planar tricyclic phenothiazine structure produced molecules (e.g., promethazine) with good antihistaminergic action and relatively strong sedative properties (see also Chapter 32). At first, these compounds were found to be useful as antihistamines, but their very strong sedative properties led to their use as potentiating agents for anesthesia (25). Further development to increase the sedative properties of the phenothiazines resulted in the development of chlorpromazine in 1950 (26).

Chlorpromazine was found to produce a tendency for sleep, but unlike the antihistamine phenothiazines, it also produced a disinterest in patients with regard to their surroundings (i.e., tranquilizing effects). In patients with psychiatric disorders, an ameliorative effect on the psychosis and a relief of anxiety and agitation were noted. These observations suggested that chlorpromazine had potential for the treatment of psychiatric disorders. Thus, what started out as an attempt to improve antihistaminergic activity ultimately resulted in an entirely new class of chemical entities useful in the treatment of an unrelated disorder (27).

Another example of how new chemical entities can be derived from biologically unrelated molecules is illustrated by the development of the potassium channel agonist diazoxide (Fig. 2.29). This molecule was developed as a result of the observation that the thiazide diuretics, such as chlorothiazide, not only exhibited diuretic activity, due to inhibition of sodium absorption in the distal convoluted tubule, but also demonstrated a direct effect on the renal vasculature. Structural modification to enhance this direct effect led to the development of diazoxide and related potassium channel agonists for the treatment of hypertension (see Chapter 24).

Refinement of the Lead Structure

Determination of the Pharmacophore

Once a “hit” compound has been discovered for a particular therapeutic use, the next step is to identify the pharmacophoric groups. The pharmacophore of a drug molecule

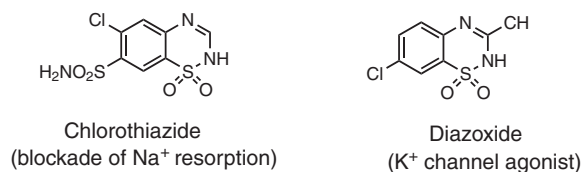


FIGURE 2.29 Structural similarity of chlorothiazide (a diuretic) and diazoxide (an antihypertensive that acts via opening of K^+ channels).

is that portion of the molecule that contains the essential functional group(s) that directly bind with the active site of the biologic target to produce the desired biologic activity. Because drug–target interactions can be very specific (think of a lock [receptor] and key [drug] relationship), the pharmacophore can constitute a small portion of the molecule. In many cases, a very structurally complex molecule can be “stripped down” to a simpler structure with retention of the pharmacophoric groups while maintaining the desired biologic action. An example of this is the opioid analgesic morphine, a tetracyclic compound with five chiral centers. Not only would structure simplification possibly provide molecules with fewer side effects, but a reduction in the number of chiral centers would greatly simplify the synthesis of morphine derivatives. Figure 2.30 shows how the morphine structure has been simplified in the search for molecules with fewer deleterious side effects, such as respiratory depression and addiction potential. Within each class are analogues that are less potent, equipotent, and many times more potent than morphine. As shown in the figure, the pharmacophore of morphine consists of a tertiary alkylamine that is at least four atoms away from an aromatic ring. A more detailed discussion of the chemistry and pharmacology of morphine can be found in Chapter 20.

Alterations in Alkyl Chains: Chain Length, Branching, and Rings

An increase or decrease in the length of an alkyl chain (homologation), branching, and alteration of ring size can have a profound effect on the potency and

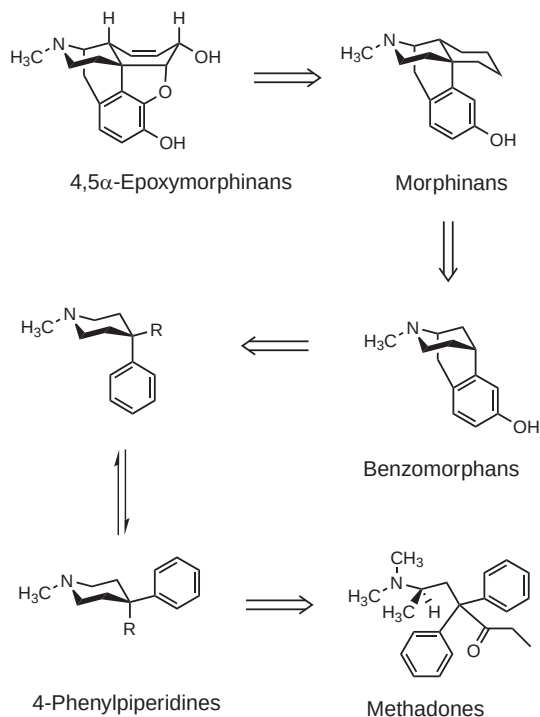
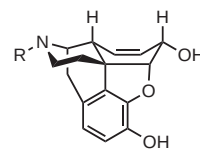


FIGURE 2.30 Morphine pharmacophore and its relationship to analgesic derivatives.

pharmacologic activity of the molecule. A change in the length of an alkyl chain by one CH_2 unit or branch alters the lipophilic character of the molecule and, therefore, its properties of absorption, distribution, and excretion. If the alkyl chain is directly involved in an interaction with the biologic target, then this type of alteration can influence the quality of those interactions. Molecules that are conformationally flexible can become less flexible if branching is introduced at a key position of an alkyl chain or the alkyl chain is incorporated into a ring equivalent. Changes in conformation can alter the spatial relationship between the pharmacophoric (functional) groups in the molecule and thereby influence interactions with the biologic target. Small structural changes are important to consider in the design of structural analogues.

An example that demonstrates how an increase in hydrocarbon chain length has significant effects not only on potency, but also on drug action (agonist vs. antagonist) is provided by a series of *N*-alkyl morphine analogues (Fig. 2.31). In this series, homologation of $\text{R}=\text{CH}_3$ (morphine) to $\text{R}=\text{CH}_2\text{CH}_2\text{CH}_3$ (*N*-propylnormorphine) produces a pronounced decrease in agonist activity and an increase in antagonist activity. When further homologated by one methylene unit $\text{R}=\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ (*N*-butylnormorphine), the resulting analog is totally devoid of agonist or antagonist activity (i.e., the compound is inactive). Additional increases in chain length ($\text{R}=\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{R}=\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$) produce compounds with increasing potency as agonists. When R is β -phenylethyl, the compound is a full agonist, with a potency approximately 14-fold that of morphine (28,29).

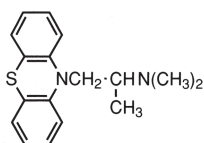
Branching of alkyl chains can also produce drastic changes in potency and pharmacologic activity. If the mechanism of action is closely related to the lipophilicity of the molecule, then hydrocarbon chain branching will result in a less lipophilic compound and a significant alteration in



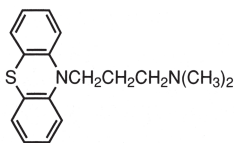
R	Pharmacological activity
$-\text{CH}_3$	Analgesic (morphine)
$-\text{CH}_2\text{CH}_3$	Opioid agonist activity decreased
$-\text{CH}_2\text{CH}_2\text{CH}_3$	Opioid antagonist activity increased
$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	Inactive as opioid agonist or antagonist
$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	} Opioid antagonist activity increased
$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	
$-\text{CH}_2\text{CH}_2$ 	14X potency of morphine

FIGURE 2.31 Effect of alkyl chain length on activity of morphine.

biologic effect. This decrease in lipophilicity is a result of the alkyl chain becoming more compact and causes less disruption of the hydrogen-bonding network of water. If the hydrocarbon chain is directly involved in interactions with its biologic target, then branching can produce major changes in pharmacologic activity. For example, consider the phenothiazines promethazine and promazine:



Promethazine



Promazine

The primary pharmacologic activity of promethazine is that of an antihistamine, whereas promazine is an antipsychotic. The only difference between the two molecules is the alkylamine side chain. In the case of promethazine, there is an isopropylamine side chain, whereas promazine contains an *n*-propylamine. In this case, simple modification of one carbon atom from a branched to a linear hydrocarbon radically alters the pharmacologic activity.

Positional isomers of aromatic ring substituents can also possess different pharmacologic properties. Substituents on aromatic rings can alter the electron distribution throughout the ring, which, in turn, can influence how the ring interacts with the biologic target. Aromatic ring substituents can also influence the conformation of the flexible portion of a molecule, especially if the substituents are located ortho to the same carbon attached to the flexible side chain. Ring substituents influence the conformations of adjacent substituents via steric interactions and can significantly alter interactions with the biologic target.

Functional Group Modification: Bioisosterism

Bioisosterism

When a lead compound is first discovered, it often lacks the required potency and pharmacokinetic properties suitable for making it a viable clinical candidate. These can include undesirable side effects, physicochemical properties, other factors that affect oral bioavailability (see also Chapter 3), and adverse metabolic or excretion properties. These undesirable properties are often the result of the presence of specific pharmacophoric (functional) groups in the molecule. Successful modification of the compound to reduce or eliminate these undesirable features without losing the desired biologic activity is the goal. Replacement or modification of specific pharmacophoric (functional) groups with other groups having similar properties is known as “isosteric replacement” or “bioisosteric replacement.”

In 1919, Langmuir (30,31) first developed the concept of chemical isosterism to describe the similarities in physical properties among atoms, functional groups, radicals,

TABLE 2.7 Comparison of Physical Properties of N₂O and CO₂

Property	N ₂ O	CO ₂
Viscosity at 20°C	148 × 10 ⁻⁶	148 × 10 ⁻⁶
Density of liquid at 10°C	0.856	0.858
Refractive index of liquid, D line 16°C	1.193	1.190
Dielectric constant of liquid at 0°C	1.593	1.582
Solubility in alcohol at 15°C	3.250	3.130

and molecules. The similarities among atoms described by Langmuir resulted primarily from the fact that these atoms contained the same number of valence electrons and came from the same columns within the periodic table. This concept of isosterism was limited to elements in adjacent rows and columns, inorganic molecules, ions, and small organic molecules, such as diazomethane and ketene. Table 2.7 shows a comparison of the physical properties of N₂O and CO₂ to illustrate Langmuir’s concept.

To account for similarities between functional groups with the same number of valence electrons but different numbers of atoms, Grimm (32) developed his hydride displacement law. This is not a “law” in the strict sense but, rather, more an illustration of similar physical properties among closely related functional groups. Table 2.8 presents an example of Grimm’s hydride displacement. Descending diagonally from left to right in the table, hydrogen atoms are progressively added to maintain the same number of valence electrons for each group of atoms within a column (thus the term “hydride”). Within each column, the groups are considered to be “pseudoatoms” with respect to each another. Thus, NH₂ is considered to be isosteric to OH, and so on. This early view of isosterism did not consider the actual location, motion, and resonance of electrons within the orbitals of these functional group replacements. Careful observation of this table reveals that some groups do share similar physicochemical properties, but others have very different properties, despite having the same number of valence electrons. For example, OH and NH₂ share similar hydrogen-bonding properties and, therefore, should be interchangeable if that is the only important criterion. The NH₂ group is basic, whereas the OH is neutral. Hence, at physiologic pH, the NH₂ group exists in

TABLE 2.8 Grimm’s Hydride Displacement “Law”

C	N	O	F	Ne
	CH	NH ₂	OH	FH
		CH ₂	NH ₂	OH ₂
			CH ₃	NH ₃

its protonated or conjugate acid form and the molecule becomes positively charged. If OH is being replaced by NH_2 , the additional positive charge could have a significant effect on the overall physicochemical properties of the molecule in which it is being introduced. The difference in physicochemical properties of the CH_3 group relative to the OH and NH_2 groups is even greater. In addition to acid-base character, this “law” fails to take into account other important physicochemical parameters, such as electronegativity, polarizability, bond angles, size, shape of molecular orbitals, electron density, and partition coefficients, all of which contribute significantly to the overall physicochemical properties of a molecule.

Instead of considering only partial structures, Hinsberg (33) applied the concept of isosterism to entire molecules. He developed the concept of “ring equivalents”—that is, functional groups that can be exchanged for one another in aromatic ring systems without drastic changes in physicochemical properties relative to the parent structure. Benzene, thiophene, and pyridine illustrate this concept (Fig. 2.32). A $-\text{CH}=\text{CH}-$ group in benzene is replaced by the divalent sulfur, $-\text{S}-$, in thiophene, and a $-\text{CH}=\text{CH}-$ is replaced by the trivalent $-\text{N}=\text{N}-$ to give pyridine. The physical properties of benzene and thiophene are very similar. For example, the boiling point of benzene is 81.1°C , and that of thiophene is 84.4°C (at 760 mm Hg). Pyridine, however, deviates, with a boiling point of 115 to 116°C . Hinsberg therefore concluded that divalent sulfur ($-\text{S}-$ or thioether) must resemble $-\text{C}=\text{C}-$ in shape, and these groups were considered to be isosteric. Note that hydrogen atoms are ignored in this comparison. Today, this isosteric relationship is seen in many drugs e.g., H_1 -receptor antagonists (Fig. 2.32).

It is difficult to relate biologic properties to physicochemical properties of individual atoms, functional groups, or entire molecules, because many physicochemical parameters are involved simultaneously and, therefore, are difficult to quantitate. Simple relationships as described earlier often do not hold up across the many types of biologic systems seen with medicinal agents. That is, what can work as an isosteric replacement in one biologic system may not work in another. Because of this, it was necessary to introduce the term “bioisosterism” to describe functional groups related in structure that have similar biologic effects. Friedman (34) introduced the term bioisosterism and defined it as follows: “Bioisosteres are (functional) groups or molecules that have chemical and physical similarities producing broadly similar biological properties.”

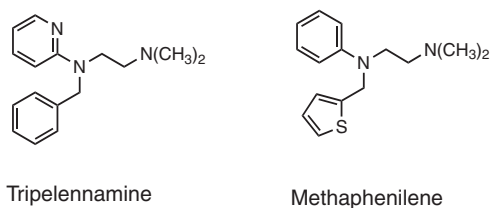


FIGURE 2.32 Isosteric substitution of thiophene for benzene and benzene for pyridine.

Burger (35) expanded this definition to take into account biochemical views of biological activity: “Bioisosteres are compounds or groups that possess near equal molecular shapes and volumes, approximately the same distribution of electrons, and which exhibit similar physical properties such as hydrophobicity. Bioisosteric compounds affect the same biochemically associated systems as agonist or antagonists and thereby produce biological properties that are related to each other.”

Classical and Nonclassical Bioisosteres

Bioisosteric groups can be subdivided into two categories: classical and nonclassical bioisosteres. Functional groups that satisfy the original conditions of Langmuir and Grimm are referred to as classical bioisosteres. Nonclassical bioisosteres do not obey steric and electronic definitions of classical bioisosteres and do not necessarily have the same number of atoms as the functional group that they replace. A wider set of compounds and functional groups are encompassed by nonclassical bioisosteres that produce, at the molecular level, qualitatively similar agonist or antagonist action. In animals, many hormones and neurotransmitters with very similar structures and biologic actions can be classified as bioisosteres. An example is the insulins isolated from various mammalian species. Even though these insulins can differ by several amino acid residues, they still produce the same biologic effects. (If this did not occur, the use of insulin to treat diabetes would have had to wait another 60 years for recombinant DNA technology to allow production of human insulin.)

TABLE 2.9 Classical Bioisosteres (Groups Within the Row Can Replace Each Other)

Monovalent bioisosteres

F, H
OH, NH_2
F, OH, NH_2 or CH_3 for H
SH, OH
Cl, Br, CF_3

Divalent bioisosteres:

$-\text{C}=\text{S}$, $-\text{C}=\text{O}$, $-\text{C}=\text{NH}$, $-\text{C}=\text{C}-$

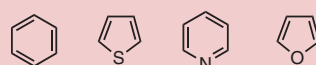
Trivalent atoms or groups:

$-\text{C}=\text{N}-$, $-\text{N}=\text{N}-$
 $-\text{P}=\text{N}-$, $-\text{As}=\text{N}-$

Tetrasubstituted atoms:

$-\text{N}^+(\text{R})_4-$, $-\text{C}^+(\text{R})_4-$, $-\text{P}^+(\text{R})_4-$, $-\text{As}^+(\text{R})_4-$

Ring equivalents:

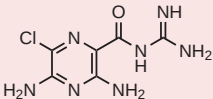
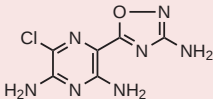
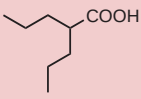
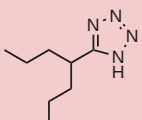
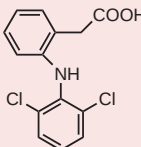
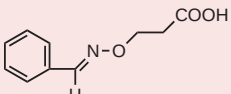
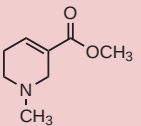
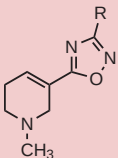
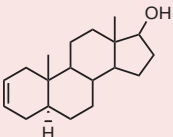
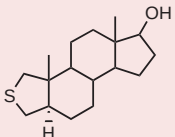
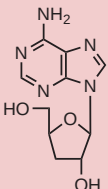
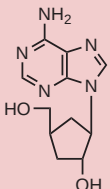
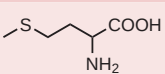
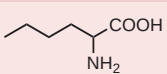


What can be a successful bioisosteric replacement for a given molecule that interacts with a particular biologic target quite has often no effect or abolishes biologic activity in another. Thus, the use of bioisosteric replacement (classical or nonclassical) in the design of new chemical entities (drug discovery) is highly dependent on the biologic system under investigation. No hard-and-fast rules exist to determine which bioisosteric replacement is going to work with a given molecule, although as the following tables and examples demonstrate, some generalizations are possible. Each category of bioisostere can be further subdivided as shown below, and examples are provided in Tables 2.9 and 2.10:

I. Classical bioisosteres

- A. Monovalent atoms and groups
- B. Divalent atoms and groups

TABLE 2.10 Nonclassical Bioisosteric Replacements

Compounds	Bioisosteric Replacement	References
		40
		41
		42
		43
		44
		45
		46

C. Trivalent atoms and groups

D. Tetrasubstituted atoms

E. Ring equivalents

II. Nonclassical bioisosteres

A. Exchangeable groups

B. Rings versus noncyclic structure

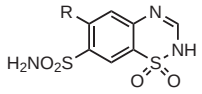
CLASSICAL BIOISOSTERES Substitution of hydrogen with fluorine is a common monovalent isosteric replacement. Sterically, hydrogen and fluorine are quite similar, with their van der Waals' radii measuring 1.2 and 1.35 Å, respectively. Because fluorine is the most electronegative element in the periodic table, any differences in biologic activity resulting from replacement of hydrogen with fluorine can be attributed to this property.

A classic example of hydrogen replacement by fluorine is development of the antineoplastic agent 5-fluorouracil from uracil. Another example is shown in Figure 2.33, in which the chlorine of chlorothiazide has been replaced with bromine or a trifluoromethyl group. For each of the substitutions, the electronic (σ , where σ^+ is electron withdrawing and σ^- is electron donating) and hydrophobic (π) properties of each group are maintained relatively constant, but the size of each group varies significantly, as indicated by the Taft steric parameter (E_s).

Figure 2.34 shows an example of classical isosteric substitution of an amino group for a hydroxyl group in folic acid. The amino group is capable of mimicking the tautomeric forms of folic acid and providing the appropriate hydrogen bonds to the enzyme active site.

A tetravalent bioisosteric replacement study was done by Grisar et al. (36) with a series of α -tocopherol analogues (Fig. 2.35). α -Tocopherol has been shown to scavenge lipoperoxyl and superoxide radicals and to accumulate in heart tissue. This is thought to be part of its mechanism of action to reduce cardiac damage resulting from myocardial infarction. All of the bioisosteric analogues were found to produce similar biologic activity.

NONCLASSICAL BIOISOSTERES As mentioned earlier, nonclassical bioisosteres are replacements of functional groups



R =	Cl	Br	CF ₃
σ	+0.23	+0.23	+0.54
π	+0.71	+0.86	+0.88
E_s	-0.97	-1.16	-2.40

FIGURE 2.33 Isosteric replacement of chlorine in thiazide diuretics. Comparison of physicochemical properties of the substituents.

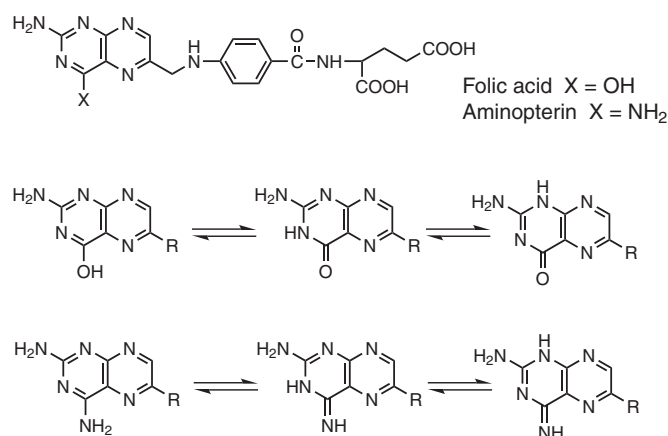


FIGURE 2.34 Isosteric replacement of OH by NH₂ in folic acid and possible tautomers of folic acid and aminopterin.

not defined by classical definitions. Some of these groups, however, mimic spatial arrangements, electronic properties, or some other physicochemical property of the molecule or functional group critical for biologic activity. One example is the use of a double bond to position essential functional groups into a particular spatial configuration critical for activity. This is shown in Figure 2.36 with the naturally occurring hormone estradiol and the synthetic analogue diethylstilbestrol. The *trans* isomer of diethylstilbestrol has approximately the same potency as estradiol, whereas the *cis* isomer is only one fourteenth as active. In the *trans* configuration, the phenolic hydroxy groups mimic the correct orientation of the phenol and alcohol in estradiol (37,38). This is not possible with the *cis* isomer, and more flexible analogues (Fig. 2.36) have little or no activity (39,40).

Another example of a nonclassical replacement is that of a sulfonamide group for a phenol in catecholamines (Fig. 2.37). With this example, steric factors appear to have less influence on receptor binding than acidity and hydrogen-bonding potential of the functional group on the aromatic ring. Both the phenolic hydroxyl of isoproterenol and the acidic proton of the arylsulfonamide have nearly the same pK_a (~10) (41) and are thereby capable of losing a proton and interacting with the biologic target as anions (Fig. 2.37). Because the replacement is not susceptible to metabolism by catechol *O*-methyltransferase, it has also the added advantage of increasing the duration of action and making the

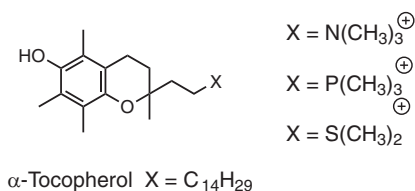


FIGURE 2.35 Tetravalent bioisosteres of α -tocopherol.

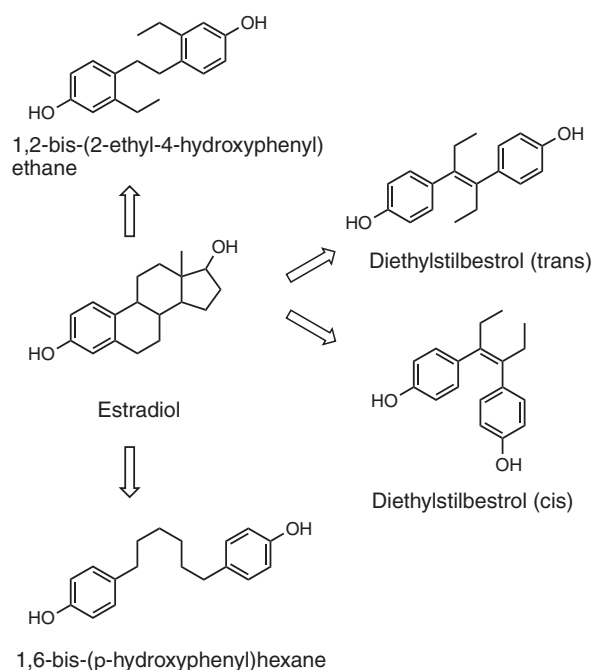


FIGURE 2.36 Noncyclic analogs of estradiol.

compound orally active. Other examples of successful bioisosteric replacements are shown in Table 2.10, and a more detailed description of the role of bioisosterism can be found in the review by Patani and LaVoie (42).

PEPTIDE AND PROTEIN DRUGS

Not all drugs are small molecules as described thus far. Some very important therapeutic agents are peptidic in nature (e.g., insulin, calcitonin) and, due to their physical chemical properties, generally cannot be delivered orally and must be administered parenterally. Peptides and proteins are very similar in that they are made up of units, or residues, of amino acids that are linked by amide bonds, also referred to as peptide bonds. There is no definitive number of amino acid residues that delineates a peptide from a protein. However, the term peptide refers generally to molecules that contain up to

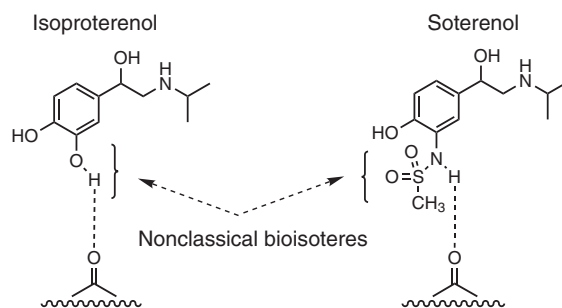


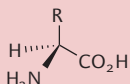
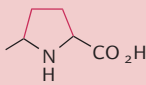
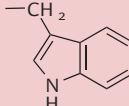
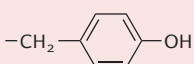
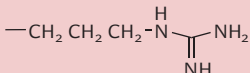
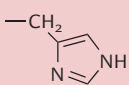
FIGURE 2.37 Bioisosteric replacement of *m*-OH of isoproterenol with a sulfonamido group and similar hydrogen-bonding capacity to a possible drug receptor.

50 amino acids. Molecules composed of more than 50 residues are generally referred to as proteins.

There are 20 naturally occurring amino acids that serve as the building blocks for both peptide and protein drugs (Table 2.11). Each amino acid contains a common

functional group “backbone” that includes a basic amine attached to the α -carbon of an acidic carboxylic acid. The α -carbon for each amino acid is substituted with a unique side chain. The amino acid side chains contribute significantly to the physical chemical properties of

TABLE 2.11 The Twenty Natural Occurring Amino Acids

Name		General Structure		
	3 Letter code	Single Letter code	Structure R =	pKa of side chain
Glycine	Gly	G	-H	none
Alanine	Ala	A	-CH ₃	none
Valine	Val	V	-CH(CH ₃) ₂	none
Isoleucine	Ile	I	-CH(CH ₃)CH ₂ CH ₃	none
Leucine	Leu	L	-CH ₂ CH(CH ₃) ₂	none
Proline	Pro	P	 (side chain highlighted)	none
Phenylalanine	Phe	F	-CH ₂ -C ₆ H ₅	none
Tryptophan	Trp	W		none
Methionine	Met	M	-CH ₂ CH ₂ SCH ₃	none
Cysteine	Cys	C	-CH ₂ SH	(acidic) ~8
Serine	Ser	S	-CH ₂ OH	none
Threonine	Thr	T	-CH(CH ₃)OH	none
Tyrosine	Tyr	Y		(acidic) ~10
Arginine	Arg	R		(basic) ~12.5
Lysine	Lys	K	-CH ₂ CH ₂ CH ₂ NH ₂	(basic) ~10.5
Histidine	His	H		(basic) ~6
Asparagine	Asn	N	-CH ₂ CONH ₂	none
Glutamine	Gln	Q	-CH ₂ CH ₂ CONH ₂	none
Aspartic Acid	Asp	D	-CH ₂ COOH	(acidic) ~3.8
Glutamic Acid	Glu	E	-CH ₂ CH ₂ COOH	(acidic) ~4

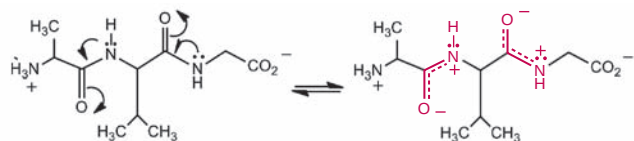


FIGURE 2.38 A tripeptide, Ala-Val-Gly, indicating the planarity of the peptide bonds caused by the restricted rotation around the amide bond.

the peptide that is formed from a unique sequence of amino acids.

As previously mentioned, the amino acid residues are linked by amide bonds, as shown in Figure 2.38. Each carboxylic acid forms an amide bond with the amine group of the next amino acid in the sequence. As with other amide bonds, conjugation between the lone pair of electrons on the nitrogen atom and the adjacent carbonyl group results in the amide bond having partial double-bond character due to a resonance structure as shown in Figure 2.38. This property has two major consequences: 1) the amide bond is therefore co-planar; and 2) there is restricted rotation around the C–N bond (Fig. 2.38). Because of this restricted rotation, there are two conformations possible, *cis* and *trans* (Fig 2.39). The *trans*-conformation is lower in energy due to fewer steric interactions (similar to that found with a carbon–carbon double bond) and is favored. When proline is one of the amino acid residues, the *cis*-conformation can be favored as a result of the amine group being part of a pyrrolidine ring. For this reason, the presence of proline in peptides and proteins is associated with a “kink” or bend in the overall conformation of the peptide chain.

Physical Chemical Properties of Peptides

Since the α -amine and α -carboxylic acid of each amino acid are involved in the peptide backbone (except at each terminus of the chain), the basic and acidic nature of these functional groups does not contribute to the overall physical chemical properties of the molecule. The functional groups found within the amino acid side chains are what are integral to the physicochemical properties of the peptide or protein and represent important points of interaction with the corresponding biologic target. Examination of Table 2.11 shows that the functional groups found within the amino acid side chains can be basic (e.g., amine, guanidine, imidazole), acidic (e.g., carboxylic acid, phenol, thiol), neutral (e.g., thioether, amide), or hydrocarbon (e.g., alkyl, aromatic rings) in nature. All of the functional groups found in amino acid side chains were discussed previously as components of small-molecule drugs. The primary differences in physical chemical properties between peptides and small molecules are due to their large size (molecular weight [MW]) and, as a result, the sheer number of different side chains

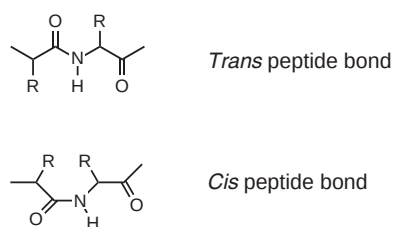


FIGURE 2.39 *Cis/trans* peptide bond configuration.

(i.e., functional groups) present in a given structure. As might be expected, the types and number of functional groups present in the side chains dictate how much more or less polar the peptide is compared to a small-molecule drug. It is certainly plausible that peptide-based drugs will not have optimal logP values for passive absorption across membranes and, given their large MW, will not readily cross membranes (see also Chapter 3).

Metabolism/Degradation of Peptide and Protein Drugs

Peptides and proteins are metabolized extensively by enzymes in the gastrointestinal tract, blood, interstitial fluid, vascular bed, and cell membranes, which results in very poor oral absorption and a short half-life for these molecules. The primary route of metabolism of peptides and proteins involves hydrolysis of the peptide bonds that link the amino acids by enzymes called peptidases. Some of these peptidases exhibit specificity for certain amino acid sequences or have specificity for either the amino or carboxyl terminus of the peptide (exopeptidase). For example, carboxypeptidases cleave off one C-terminal residue, dipeptidyl carboxypeptidases cleave dipeptides from the C terminus, aminopeptidases cleave off one N-terminal residue, and amidases (endopeptidases) cleave internal peptide bonds. There are also peptidase subclasses that exhibit specificity for certain amino acid sequences within the peptide or at either of the termini. Some peptidases (e.g., dipeptidyl peptidase IV) have been found to catalyze the degradation of naturally occurring peptides (e.g., GLP-1) (Fig. 2.40) (see also Chapter 37).

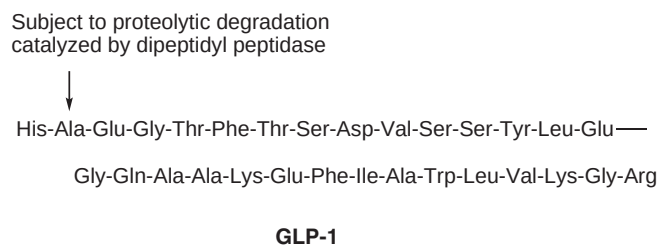


FIGURE 2.40 GLP-1 degradation by dipeptidyl peptidase IV.

SUMMARY

Medicinal chemistry involves the discovery of new chemical entities and the systematic study of the SARs of these compounds for disease state management. Such studies provide the basis for development of better and therapeutically safer medicinal agents from lead compounds found from natural sources, random screening, systematic screening, and focused rational design. Drug design goals include increasing the potency and duration of action of newly discovered compounds and decreasing adverse side effects.

For the pharmacist, it is also important to understand how the physicochemical properties influence the pharmacokinetic properties of the medicinal agents being dispensed. Such knowledge will help the pharmacist not only to better understand the clinical properties of these compounds but also to anticipate the properties of newly marketed agents. An understanding of the chemical properties of the molecule will allow the pharmacist to anticipate formulation problems (especially IV admixtures), as well as potential adverse interactions with other drugs as the result of serum protein binding and metabolism.

PROBLEMS

The following problems are provided for additional study:

1. Calculate the percent ionization of amobarbital at pH 2.0, 5.5, and 8.0. What trend is seen?
2. Calculate the percent ionization of phenylpropanolamine at pH 2.0, 5.5, and 8.0. Compare these results with those obtained in Problem 1.
3. Calculate the percent ionization of sulfacetamide in the stomach, duodenum, and ileum. Draw the structure of the predominate form of the drug in each tissue.
4. Referring to Figure 2.15, redraw each compound in its ionized form.
5. For the organic functional groups listed in Table 2.4, name each functional group, and redraw them, showing all potential hydrogen bonds with water.
6. Using the empiric method of Lemke, predict the water solubility for each of the following molecules (*Note*: Water solubility is defined as >1% solubility):

Aspirin	Carphenazine maleate
Chlordiazepoxide	Codeine

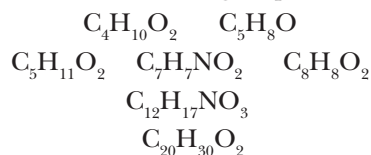
Codeine phosphate Cyproheptadine hydrochloride

Haloperidol Phenytoin

7. Calculate the logP value for each of the following:

Aspirin	Carphenazine
Codeine	Cyproheptadine
Haloperidol	Chlordiazepoxide
Phenytoin	

8. Using the Merck Index or other source, find the chemical structures for the following empirical formulae. List as many physicochemical properties as possible for each compound, and compare them within each group of isomers:



9. Using the Cahn-Ingold-Prelog rules, assign the absolute configuration to each chiral center of ephedrine and pseudoephedrine (Fig. 2.21).
10. For the compounds shown in Figure 2.22, indicate, using an*, where the chiral centers are in each molecule.
11. Draw each possible stereoisomer for chloramphenicol and enalapril. Assign the absolute stereochemistry to each chiral center.
12. I. Draw the Newman projection along the $\text{CH}_3\text{--N}$ bond of acetylcholine in the staggered conformation. Rotate the bond 120° and 240° . Are these rotamers conformational isomers? Explain why or why not.
II. Repeat the above exercise with the N1--C2 bond of acetylcholine.
13. Draw the three most stable rotamers of norepinephrine. Of these rotamers, is there the possibility of an intramolecular interaction that would stabilize what normally would be considered an unstable rotamer? Explain.

TABLE 2.12 Drug Structures not Included in Text for Problems at End of Chapter 2

Problem #	Drug Name	Drug structure
3	Sulfacetamide	
6 + 7	Aspirin	
6 + 7	Chlordiazepoxide	
6 + 7	Carphenazine maleate	
6 + 7	Codeine	
6 + 7	Cyproheptadine	
6 + 7	Phenytoin	
6 + 7	Haloperidol	
11	Chloramphenicol	

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Chapter 3

Physicochemical and Biopharmaceutical Properties of Drug Substances and Pharmacokinetics

SUNIL S. JAMBHEKAR

Abbreviations

ADME , absorption, distribution, metabolism, and excretion	f , bioavailability	pK_a , the negative logarithm of a dissociation constant, K_a
AP , absorption potential	F , absolute bioavailability	pK_w , negative logarithm of dissociation constant of water
ATP , adenosine 5'-triphosphate	FDA , U.S. Food and Drug Administration	PSA , polar surface area
ANDA , Abbreviated New Drug Application	F_{uni} , fraction of un-ionized form	PSAd , dynamic polar surface area
AUC , area under the plasma concentration–time curve	HCl , hydrochloric acid	P_w , permeability of a drug in the gut wall
$(AUC)_\infty$, total area under the plasma concentration–time curve	IM , intramuscular	Q , blood flow rate
AUMC , area under the first-moment curve	IV , intravenous	SC , subcutaneous
Cl , clearance	K , elimination rate constant	SITT , small intestinal transit time
C_o , concentration of drug in the organic or oil phase	K_m , the concentration of drug at half the V_{max}	SIV , small intestinal volume
$(C_p)_{max}$, plasma peak concentration	MAD , maximum absorbable dose	S_o , intrinsic solubility
C_s , solubility	MAT , mean absorption time	$t_{1/2}$, half-life
dC , concentration gradient	MDCK , Madin-Darby canine kidney	t_{max} , peak time
ER , extraction ratio	MRT , mean residence time	V , volume of distribution
	NDA , New Drug Application	V_1 , volume of the luminal content
	P or $\log P$, partition coefficient	V_{max} , maximum theoretical transfer rate
	P_{aq} , aqueous permeability	
	P-gp , P-glycoprotein	

Throughout its history, the pharmacy profession has been concerned primarily with the manner in which drugs produce their pharmacologic effects and the dosage forms through which drugs are administered. Since the early 20th century, efforts have been directed to determining, understanding, and providing rational explanations of drug effects on biologic systems, but we have been limited

by our ability to correlate the observed physiologic events with a reasonable hypothesis or concept. Pharmacists, at one time, were closely involved in formulating a prescription written by a physician for a patient. Today, most of the formulating is done by the pharmaceutical manufacturer.

Early descriptions of drug action were confined to their reference as tonic or toxic effects. This approach

was followed by the concept of receptor theory, which for decades remained primarily an operational concept that was useful for discussing the new actions of drugs on a molecular level (1). Research in receptor theories, however, has provided evidence that the drug receptors do exist as distinct entities, and limited success has been attained in the characterization of receptors (2,3).

An extension of the receptor theory of drug action is an increased emphasis on the importance of physicochemical properties of the drug and the relationship of such properties to the pharmacologic responses. Because these properties have an important role in determining biologic action of pharmaceuticals, it is appropriate to refer to these properties as biopharmaceutical properties of drug substances. Examples of such properties include solubility, partition coefficients, diffusivity, degree of ionization, and polymorphism, which in turn are determined by the chemical structure and stereochemistry of drug substances.

A consideration of these biopharmaceutical properties is fundamental to discussing several important aspects of the overall effects. For a given chemical entity (drug), there often will be a difference in physiologic availability and, presumably, in clinical responses, primarily because drug molecules must cross various biologic membranes and interact with intercellular and intracellular fluids before reaching the elusive region termed the “site of action.” Under these conditions, the biopharmaceutical properties of the drug must contribute favorably to facilitate absorption and distribution processes to augment the drug concentration at various active sites. Furthermore, equally important is the fact that these biopharmaceutical properties of a drug must ensure a specific orientation on the receptor surface so that a sequence of events is initiated that leads to the observed pharmacologic effects. Drug molecules that are deficient in the required biopharmaceutical properties can display generally marginal pharmacologic action or be totally ineffective.

Biopharmaceutics is the study of the influence of formulation factors on the therapeutic activity of a drug product or dosage forms. It involves the study of the relationship between some of the physicochemical properties of a drug and the biologic effects observed after the administration of a drug via various dosage forms or drug delivery systems. Almost any alteration in a drug delivery system is likely to alter the drug delivery rate and the amount of the drug delivered to the desired place in the body. This includes the chemical nature of the drug (e.g., ester, salts, complexes), the particle size and surface area of the drug, the type of dosage forms (e.g., solution, suspension, capsule, tablet), and the excipients and processes used in the manufacturing of the drug delivery systems.

Drugs, via drug delivery systems, are most often administered to human subjects by the oral route. Compared to other routes of drug administration, especially the intravenous route, the oral route is unusually complex with respect to the physicochemical conditions existing at the absorption site. Therefore, before we discuss how

the biopharmaceutical properties of a drug in a dosage form can affect the availability and action of that drug, it is prudent to review gastrointestinal physiology.

GASTROINTESTINAL PHYSIOLOGY

Figure 3.1 schematically represents the gastrointestinal tract and some of the problems encountered in consideration of drug absorption from the site following administration of a drug via dosage forms (4). The stomach is divided into two main parts: the body of the stomach and the pylorus. Histologically, these parts correspond to the pepsin- and hydrochloric acid (HCl)-secreting area and the mucus-secreting area, respectively, of the gastric mucosa. In the human, the stomach contents are usually in the pH range of 1.0 to 3.5, with pH 1.0 to 2.5 being the most common range. Furthermore, there is a diurnal cycle of gastric acidity in humans. During the night, stomach contents are, as a rule, more acidic (pH, ~1.3), and during the day, because of food consumption, the pH is less acidic. The recovery of stomach acidity, however, occurs quite rapidly. The presence of protein, being amphoteric in nature, acts as an excellent buffer, and as digestion proceeds, the liberated amino acids enormously increase the neutralizing capacity of the stomach.

The small intestine is divided anatomically into three sections: the duodenum, the jejunum, and the ileum. All three areas are involved in the digestion and absorption of food. The available absorbing area is increased by surface folds in the intestinal lining. The surface of these folds possesses villi and microvilli, which are tiny, finger-like projections that protrude from the epithelial lining of the intestinal wall as shown in Figure 3.2. The pH of the duodenal contents in the human is usually in the range of 5 to 7. There is a gradual decrease in acidity along the length of the gastrointestinal tract, with the ultimate pH being 7 to 8 in the lower ileum. It has been estimated that approximately 8 L of fluid enter the upper intestine per day, with approximately 7 L of this arising from digestive juices and fluids and approximately 1 L arising from oral intake. Over the entire length of the large and small intestine and the stomach is the brush border, which consists of a uniform coating (thickness, 3 mm) of mucopolysaccharide. This coating layer serves as a mechanical barrier to bacteria or food particles.

When a dosage form containing a drug or drug molecules moves from the stomach through the pylorus into the duodenum, the dosage form encounters a rapidly changing environment with respect to pH. Furthermore, digestive juices secreted into the small bowel contain many enzymes not found in the gastric juices. Digestion and absorption of foodstuff occur simultaneously in the small intestine. Intestinal digestion is the terminal phase of preparing foodstuff for absorption and consists of two processes: completion of the hydrolysis of large molecules to smaller ones, which are absorbed, and converting the finished product of hydrolysis into an aqueous solution or emulsion.

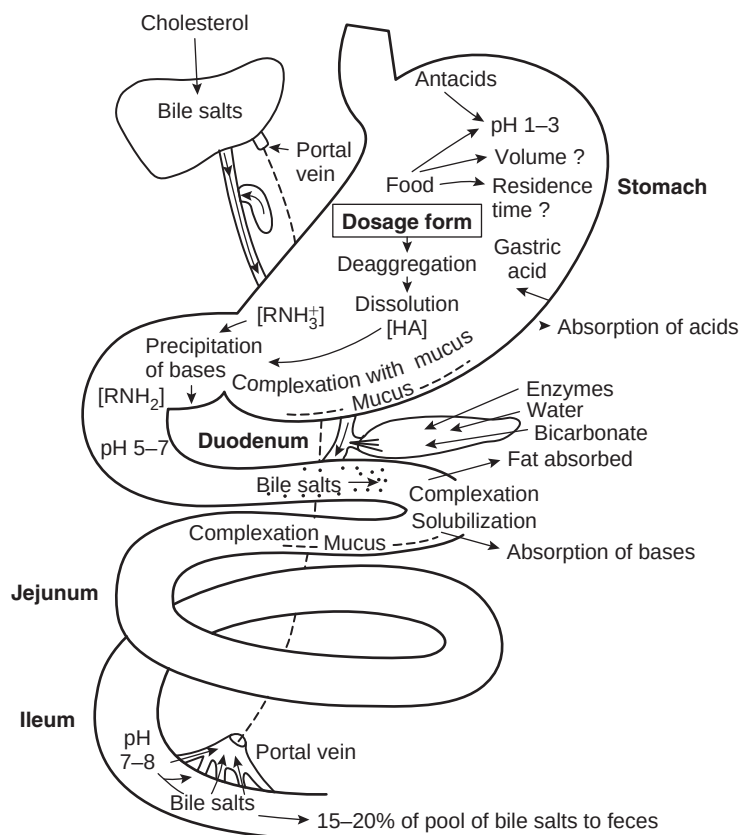


FIGURE 3.1 Processes occurring along with drug absorption when drug molecules travel down the gastrointestinal tract and the factors that affect drug absorption. (From Florence AT, Attwood D. *Physicochemical Principles of Pharmacy*, 2nd Ed. New York: Chapman and Hall, 1988, with permission.) (21).

Drug absorption, whether from the gastrointestinal tract or from other sites, requires the passage of the drug in a molecular form across the barrier membrane. Most drugs are presented to the body as solid or semisolid dosage forms, and the drug particles must first be released from these dosage forms. These drug particles must dissolve, and if they possess the desirable biopharmaceutical properties, they will pass from a region of high concentration to a region of low concentration across the membrane into the blood or general circulation (Fig. 3.3). Knowledge of biologic membrane structure

and its general properties is pivotal in understanding absorption processes and the role of the biopharmaceutical properties of drug substances.

Biologic Membrane

The prevalent view of the gastrointestinal membrane is that it consists of a bimolecular lipid layer covered on each side by protein, with the lipid molecule oriented perpendicular to the cell surface as shown in Figure 3.4. The lipid layer is interrupted by small, water-filled pores with a radius of approximately 4 Å. A molecule with a radius of 4 Å or less can easily pass through these water-filled pores. Thus, membranes have a specialized transport system to assist the passage of water-soluble material and ions through the lipid interior, a process sometimes termed "convective absorption." The rate of permeation of such

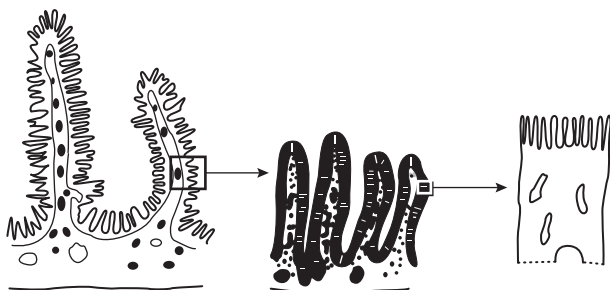


FIGURE 3.2 The epithelium of the small intestine at different levels of magnification. From left to right: the intestinal villi and microvilli that constitute the brush border.

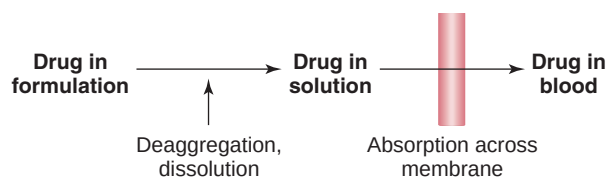


FIGURE 3.3 Sequence of events in drug absorption from formulations of solid dosage forms.

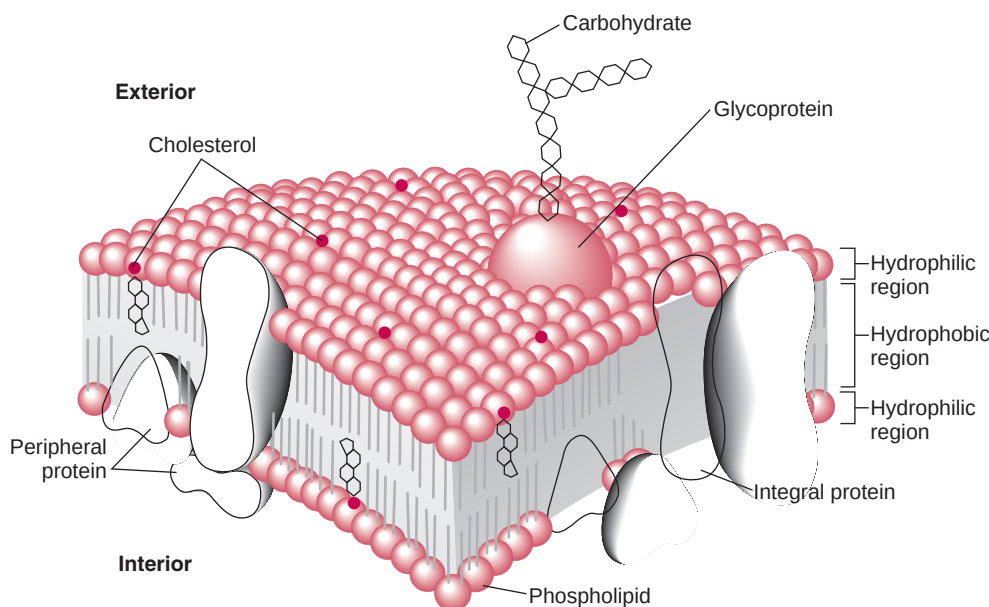


FIGURE 3.4 Basic structure of an animal cell membrane. (From Smith C, Marks A, Lieberman M, eds. *Basic Medical Biochemistry*. Baltimore: Lippincott Williams & Wilkins, 2004:159–163, with permission.)

small molecules through the pore is affected not only by the relative sizes of the holes and the molecules, but also by the interaction between permeating molecules and the membrane. When permeation through the membrane occurs, the permeating substance is considered to have transferred from solution in the luminal aqueous phase to the lipid membrane phase, then to the aqueous phase on the other side of the membrane. Biologic membranes differ from a polymeric membrane in that they are composed of small amphipathic molecules, phospholipids, and cholesterol. The protein layer associated with membranes is hydrophobic in nature. Therefore, biologic membranes have a hydrophilic exterior and a hydrophobic interior. Cholesterol is a major component of most mammalian biologic membranes, and its removal renders the membrane highly permeable. Cholesterol complexes with phospholipids, and its presence reduces the permeability of the membrane to water, cations, glycerides, and glucose. The shape of the cholesterol molecule allows it to fit closely with the hydrocarbon chains of unsaturated fatty acids in the bilayer. It is the general opinion that the cholesterol makes the membrane more rigid. The flexibility of the biologic membrane to reform and adapt to a changing environment is its important feature. The details of membrane structure are still widely debated, and a more recent membrane model is shown in Figure 3.4.

In addition to biopharmaceutical factors, several physiologic factors can also affect the rate and extent of gastrointestinal absorption. These factors are as follows: properties of epithelial cells, segmental activity of the bowel, degree of vascularity, effective absorbing surface area per unit length of gut, surface and interfacial tensions, electrolyte content and their concentration in

luminal fluid, enzymatic activity in the luminal contents, and gastric emptying rate of the drug from stomach.

Mechanisms of Drug Absorption

Drug transfer is often viewed as the movement of a drug molecule across a series of membranes and spaces (Fig. 3.5), which, in aggregate, serve as a macroscopic membrane. The cells and interstitial spaces lying between the gastric lumen and the capillary blood or structure between the sinusoidal space and the bile canaliculi are examples. Each of the cellular membranes and spaces can impede drug transport to varying degrees; therefore, any one of them is a rate-limiting step to the overall process of drug transport. This complexity of structure makes quantitative prediction of drug transport difficult. A qualitative description of the processes of drug transport across functional membranes follows.

Passive Diffusion

The transfer of most drugs across a biologic membrane occurs by passive diffusion, a natural tendency for molecules to move from a higher concentration to a lower concentration. This movement of drug molecules is caused by the kinetic energy of the molecules. The rate of diffusion depends on the magnitude of the concentration gradient (dC) across the membrane and is represented by the following equation:

$$3.1 \quad -\frac{dC}{dt} = K \cdot dC = K(C_{\text{abs}} - C_b)$$

where $-dC/dt$ is the rate of diffusion across a membrane; K is a complex proportionality constant that includes the

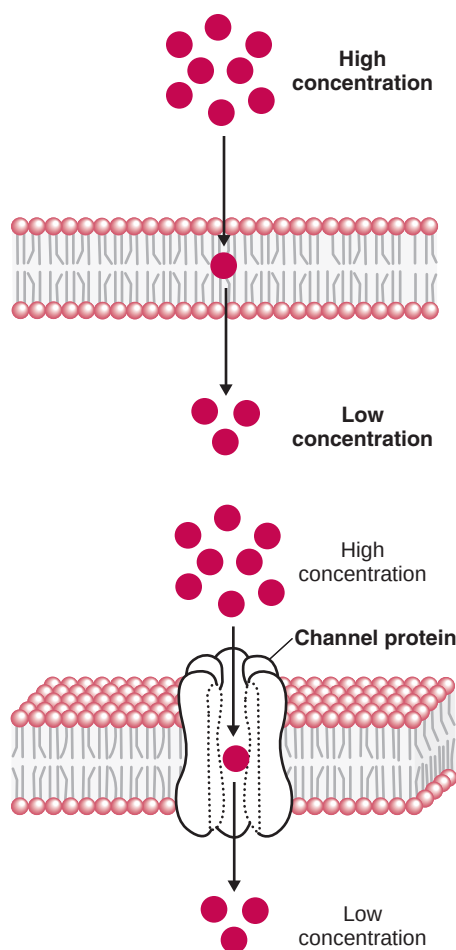


FIGURE 3.5 (A) Simple diffusion. (B). Membrane channels. (From Smith C, Marks A, Lieberman M, eds. Basic Medical Biochemistry. Baltimore: Lippincott Williams & Wilkins, 2004:159–163, with permission.)

area of membrane, the thickness of the membrane, the partition coefficient of the drug molecule between the lipophilic membrane and the aqueous phase on each side of the membrane, and the diffusion coefficient of the drug; C_{abs} is the drug concentration at the absorption site; and C_b is the drug concentration in the blood.

The gastrointestinal absorption of a drug from an aqueous solution requires transfer from the lumen to the gut wall followed by penetration of the epithelial membrane by a drug molecule to the capillaries of the systemic circulation. On entering the blood, the drug distributes itself rapidly in the blood. Because of the volume differences at absorption and distribution sites, the drug concentration in blood (C_b) will be much lower than the concentration at the absorption site (C_{abs}). This concentration gradient is maintained throughout the absorption process—that is, $(C_{abs} - C_b)$. As a result, the concentration gradient (dC in Eq. 3.1), is approximately equal to C_{abs} , so Equation 3.1 is written as:

$$3.2 \quad -\frac{dC}{dt} = K \cdot C_1$$



FIGURE 3.6 Effect of drug concentration on the rate of absorption when passive diffusion is operative.

Because absorption by passive diffusion is a first-order process, the rate of absorption (dC/dt in Eq. 3.2) is directly proportional to the concentration at the site of absorption (C_1). The greater the concentration of drug at the absorption site, the faster is the rate of absorption (Fig. 3.6). The percentage of dose absorbed at any time, however, remains unchanged.

A major source of variation is membrane permeability, which depends on the lipophilicity of the drug molecule. This is often characterized by its partition between oil and water. The lipid solubility of a drug, therefore, is a very important physicochemical property governing the rate of transfer through a variety of biologic membrane barriers. Figure 3.7 illustrates the role of partition coefficients in the

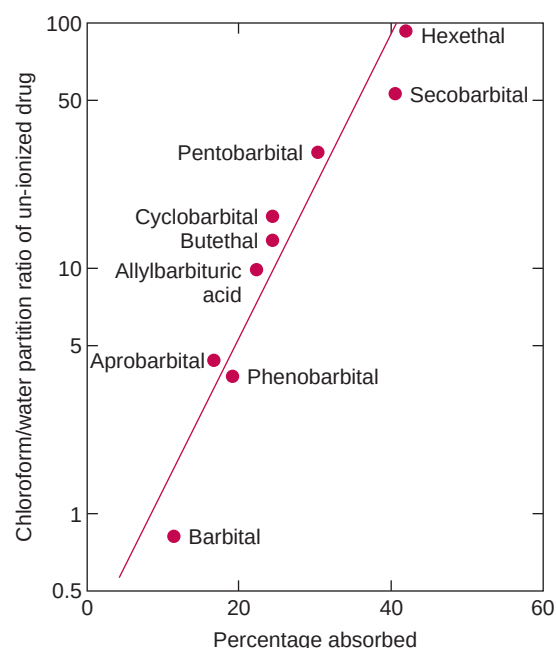


FIGURE 3.7 Comparison between colonic absorption of barbiturates in the rat and lipid-to-water partition coefficient of the un-ionized form of the barbiturates. (From Schanker LS. Absorption of drugs from the colon. J Pharmacol Exp Ther 1959;126:283–294, with permission.) (6).

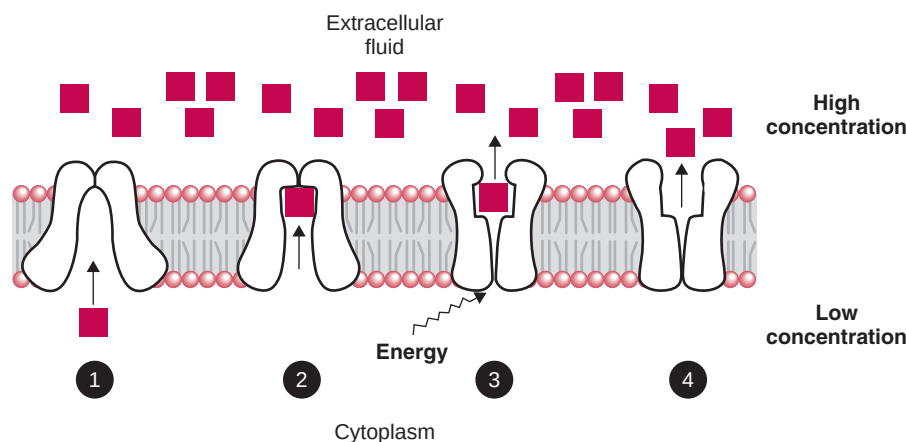


FIGURE 3.8 Active transport. (From Smith C, Marks A, Lieberman M, eds. Basic Medical Biochemistry. Baltimore: Lippincott Williams & Wilkins, 2004:159–163, with permission.)

drug absorption process from the colon and that a good correlation exists between the percentage of drug absorption and the partition coefficient of an un-ionized drug.

Carrier-Mediated or Active Transport

Although most drugs are absorbed from the gastrointestinal tract by passive diffusion, some drugs of therapeutic interest and some chemicals of nutritional value, such as amino acids, dipeptides and tripeptides, glucose, and folic acid, are absorbed by the action of transporter proteins (i.e., a carrier-mediated transport mechanism) (Fig. 3.8). In this type of transport, membranes have a specialized role. The usual requirement for active transport is structural similarities between the drug and the substrate normally transported across the membrane. Active transport differs from passive diffusion in the following ways: 1) The transport of the drug occurs against a concentration gradient; 2) the transport mechanism can become saturated at high drug concentration; and 3) a specificity for a certain molecular structure can promote competition in the presence of a similarly structured compound. This, in turn, can decrease the absorption of a drug. Active or facilitated absorption of a drug is usually explained by assuming that transporter proteins (i.e., carriers in membranes) are responsible for shuttling these solutes in mucosal or serosal direction. The number of apparent carriers in membranes, however, is limited. Therefore, the rate of transfer or absorption is described by the following equation:

$$3.3 \quad \text{Absorption rate} = \frac{dC}{dt} = V_{\max} \cdot C / K_m + C$$

where C is the solute concentration at the absorption site, and $V_{\max}$ (the maximum theoretical transfer rate) and K_m (the concentration of drug at half the $V_{\max}$) are constants. Low doses or concentrations, when $K_m \gg C$, reduce Equation 3.3 to:

$$3.4 \quad \frac{dC}{dt} = \frac{V_{\max}}{K_m} \cdot C = K_m \cdot C$$

Equation 3.4 indicates that apparent first-order kinetics is observed. Under these conditions, there are sufficient numbers of carriers available so that a constant proportion of solute molecules presented to the membrane are transported across the membranes. As the solute concentration increases, the number of free carriers is reduced, and the proportion of solute molecules transferred across the membrane is reduced until a maximum saturation is reached, when $C \gg K_m$:

$$3.5 \quad \text{Absorption rate} = \frac{dC}{dt} = V_{\max}$$

Equation 3.5 indicates that a further increase in solute concentration will not result in any further increase in the rate of absorption (Fig. 3.9). The capacity-limited characteristics of carrier-mediated processes suggest that the bioavailability of drugs absorbed in this manner should decrease nonlinearly with increasing doses.

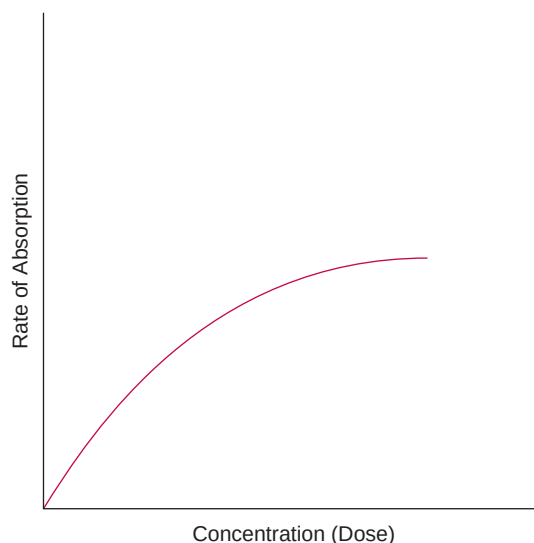


FIGURE 3.9 Relationship between drug concentration and rate of absorption when an active transport process is operative.

Therefore, the use of a large, single oral dose of these drugs is irrational, and if larger daily doses are necessary, one should use divided doses. Examples of substances that are actively transported include amino acids, methyldopa, fluorouracil, penicillamine, and levodopa.

Convective Absorption

The absorption of small molecules (molecular radii less than $\sim 4 \text{ \AA}$) through water-filled pores of biologic membrane is referred to as convective absorption. The rate of absorption because of this mechanism is equated to the product of a sieving coefficient, the rate of fluid or water absorption, and the concentration of solute in the luminal content. The sieving coefficient is indirectly related to the relative sizes of the pores and the molecules.

ION-PAIR ABSORPTION In 1967, Higuchi suggested that highly ionized compounds, such as quaternary ammonium compounds, can possibly be absorbed by an ion-pair mechanism (5). In vitro, a relatively large organic anion can combine with a relatively large cation to form an ion pair of neutral properties, which will then cross a water-organic solvent interface and transfer to an organic phase.

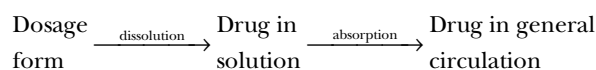
PHYSICOCHEMICAL FACTORS AFFECTING DRUG ABSORPTION

The pH-Partition Hypothesis on Drug Absorption

Drug absorption is influenced by many physiologic factors. Additionally, it also depends on many physicochemical properties of the drug itself. Some investigators (6–12) concluded from their research that most drugs are absorbed from the gastrointestinal tract by a process of passive diffusion of the un-ionized moiety across a lipid membrane. Furthermore, the dissociation constant, lipid solubility, and pH of the fluid at the absorption site determine the extent of absorption from a solution. The interrelationship among these parameters is known as the pH-partition theory. This theory provides a basic framework for the understanding of drug absorption from the gastrointestinal tract and drug transport across the biologic membrane. The principle points of this theory are as follows:

1. The gastrointestinal and other biologic membranes act like lipid barriers.
2. The un-ionized form of the acidic or basic drug is preferentially absorbed.
3. Most drugs are absorbed by passive diffusion.
4. The rate of drug absorption and amount of drug absorbed are related to the drug's oil-water partition coefficient (i.e., the more lipophilic the drug, the faster is its absorption).
5. Weak acidic and neutral drugs are absorbed from the stomach, but basic drugs are not.

When a drug is administered intravenously, it is immediately available to body fluids for distribution to the site of action. However, all extravascular routes, such as oral, intramuscular, sublingual, buccal, subcutaneous, dermal, rectal, and nasal routes, can influence the overall therapeutic activity of the drug, primarily because of its dissolution rate, a step that is necessary for a drug to be available in a solution form. When a drug is administered orally in a dosage form such as a tablet, capsule, or suspension, the rate of absorption across the biologic membrane frequently is controlled by the slowest step in the following sequence:



In many instances, the slowest step, or the rate-limiting step, in the sequence is the dissolution of the drug. When dissolution is the controlling step, any factors that affect the rate of dissolution must also influence the rate of absorption. This, in turn, affects the extent and duration of action. Several factors can influence the dissolution rate of drug from solid dosage forms and, therefore, the therapeutic activity. These factors include solubility of a drug, particle size and surface area of drug particles, crystalline and salt form of a drug, and rate of disintegration.

The absorption rate of drugs is also affected by interaction or formation of complexes in the gastrointestinal tract. Generally, such complex formation reduces the concentration of free drug at the absorption site. Because the complexed drug is absorbed either slowly or not at all, the net effect is the reduction of concentration of drug at the absorption site and slower rate of absorption.

Ionization and pH at Absorption Site

The fraction of the drug existing in its un-ionized form in a solution is a function of both the dissociation constant of a drug and the pH of the solution at the absorption site. The dissociation constant, for both weak acids and bases, is often expressed as the pK_a (the negative logarithm of a dissociation constant, K_a). The Henderson-Hasselbalch equation (Eq. 3.6) for the ionization of a weak acid, HA, is presented below.

$$3.6 \quad \text{pH} - \text{p}K_a = \log \frac{[\text{Ionized}]}{[\text{Un-ionized}]}$$

Assuming that α is the fraction of ionized species and that $1 - \alpha$ is the fraction remaining as the un-ionized form, Equation 3.6 is written as

$$3.7 \quad \text{pH} - \text{p}K_a = \log \frac{\alpha}{1 - \alpha}$$

or

$$3.8 \quad \frac{\alpha}{1 - \alpha} = \text{antilog} (\text{pH} - \text{p}K_a)$$

From Equation 3.8, the fraction or percentage of the absorbable and nonabsorbable forms of a weak acid can be calculated if the pH condition at the site of administration is known. Analogously, the dissociation or basicity constant for a weak base is derived as follows:

3.9 $pK_a + pK_b = pK_w$

where $\text{p}K_{\text{w}}$ is the negative logarithm of dissociation constant of water. Although the dissociation constant of a weak base is described by the term K_{b} , it is conventionally expressed in terms of K_{a} , because of the relationship expressed in Equation 3.9.

$$\text{3.10} \quad \text{p}K_a - \text{pH} = \log \frac{[\text{Ionized}]}{[\text{Un-ionized}]}$$

Again, assuming that α is the fraction of ionized species and that $1 - \alpha$ is the fraction of un-ionized species, Equation 3.10 becomes

$$\text{3.11} \quad \text{p}K_{\text{a}} - \text{pH} = \log \frac{\alpha}{1 - \alpha}$$

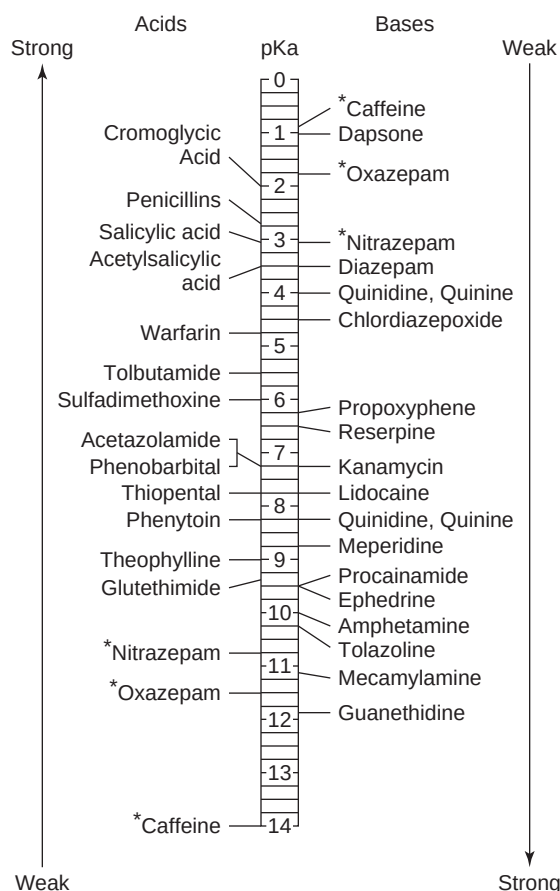


FIGURE 3.10 The pK_a values of certain acidic and basic drugs. Drugs denoted with an asterisk are amphoteric (13). (From Rowland M, Tozer T. Clinical Pharmacokinetics: Concepts and Application, 2nd Ed. Philadelphia: Lea and Febiger, 1989, with permission.)

or

$$\text{3.12} \quad \frac{\alpha}{1-\alpha} = \text{antilog} (\text{p}K_a - \text{pH})$$

From Equation 3.12, one can calculate the fraction or percentage of absorbable and nonabsorbable form of a weak base if the pH condition at the site of drug absorption is known. Figure 3.10 shows the pK_a values of several drugs and the relative acid or base strength of these compounds.

The relationship between pH and pK_a and the extent of ionization is given by Equations 3.8 and 3.12 for weak acids and weak bases, respectively. Accordingly, most weak acidic drugs are predominantly in the un-ionized form at lower pH of the gastric fluid and, therefore, are absorbed from the stomach as well as from the intestine. Some very weak acidic drugs, such as phenytoin and many barbiturates, the pK_a values of which are greater than 8.0, are essentially un-ionized at all pH values. Therefore, for these weak acidic drugs, transport is more rapid and independent of pH, provided that the un-ionized form is lipophilic or nonpolar. Furthermore, it is important to note that the fraction un-ionized changes dramatically only for weak acids with pK_a values between 3 and 7. Therefore, for the weak acids, a change in the rate of transport with pH is expected, as shown in Figure 3.11 (13). Although the transport of weak acids with pK_a values less than 3.0 should theoretically depend on pH, the fraction un-ionized is so low that transport across the gut membrane is slow even under the most acidic conditions.

Most weak bases are poorly absorbed, if at all, in the stomach, because they are present largely in the ionized form at pH 1 to 2. Codeine, a weak base with a pK_a of approximately 8, will have about 1 in every 1 million

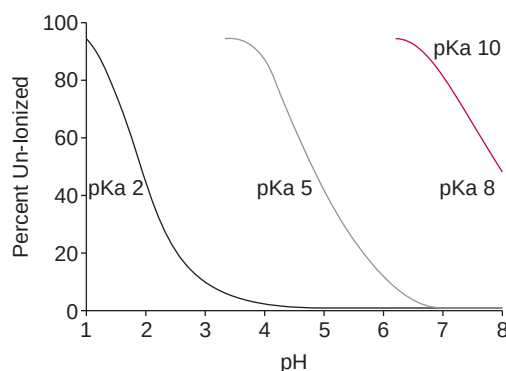


FIGURE 3.11 For very weak acids, pK_a values greater than 8.0 are predominantly un-ionized at all pH values between 1.0 and 8.0. Profound changes in the un-ionized fraction occur with pH for an acid with a pK_a value that lies within the range of 2.0 to 8.0. Although the fraction un-ionized of even strong acids increases with hydrogen ion concentration, the absolute value remains low at most pH values shown. (From Rowland M, Tozer T. Clinical Pharmacokinetics: Concepts and Application, 2nd Ed. Philadelphia: Lea and Febiger, 1989, with permission.) (13).

molecules in its un-ionized form at gastric pH 1.0. Weakly basic drugs with a pK_a of less than 4, such as dapsone, diazepam, and chlordiazepoxide, are essentially un-ionized through the intestine. Strong bases, which are those with pK_a values between 5 and 11, show pH-dependent absorption. Stronger bases, such as guanethidine ($pK_a > 11$) are ionized throughout the gastrointestinal tract and tend to be poorly absorbed.

The evidence of the importance of dissociation in drug absorption is found in the result of studies in which pH at the absorption site is changed (Tables 3.1 and 3.2). Table 3.2 clearly shows the decreased absorption of a weak acid at pH 8.0 compared to pH 1.0 (13). However, an increase to pH 8.0 promotes the absorption of a weak base with practically nothing absorbed at pH 1.0. The data in Table 3.2 also permit a comparison of intestinal absorption of acidic and basic drugs from buffered solutions ranging from pH 4.0 to 8.0 (14). These results are in agreement with the pH-partition hypothesis.

The pH-partition theory provides a basic framework for the understanding of drug absorption and, sometimes, is an oversimplification of a more complex process. For example, experimentally observed pH-absorption curves are less steep (Fig. 3.12) than those expected theoretically and are shifted to higher pH values for bases and lower pH values for acids. This deviation, observed experimentally, has been attributed by several investigators to factors such as limited absorption of ionized species of drugs, the presence of an unstirred diffusion layer adjacent to the cell membrane, and a difference between luminal pH and cell membrane surface pH.

Lipid Solubility

PARTITION COEFFICIENT Some drugs are poorly absorbed after oral administration even though they are available

TABLE 3.1 Comparison of Gastric Absorption of Acids and Bases at pH 1 and 8 in the Rat (13,44)			
	pK_a	% Absorbed at pH 1	% Absorbed at pH 8
Acids			
5-Sulfosalicylic acid	<2.0	0	0
5-Nitrosalicylic acid	2.3	52	16
Salicylic acid	3.0	61	13
Thiopental	7.6	46	34
Bases			
Aniline	4.6	6	56
<i>p</i> -Toluidine	5.3	0	47
Quinine	8.4	0	18
Dextromethorphan	9.2	0	16

TABLE 3.2 Comparison of Intestinal Absorption of Acids and Bases in the Rat at Several pH Values (13,44)					
% Absorbed from Rat Intestine					
	pK_a	pH 4	pH 5	pH 7	pH 8
Acids					
5-Nitrosalicylic acid	2.3	40	27	0	0
Salicylic acid	3.0	64	35	30	10
Acetylsalicylic acid	3.5	41	27	—	—
Benzoic acid	4.2	62	36	35	5
Bases					
Aniline	4.6	40	48	58	61
Aminopyrine	5.0	21	35	48	52
<i>p</i> -Toluidine	5.3	30	42	65	64
Quinine	8.4	9	11	41	54

predominantly in the un-ionized form in the gastrointestinal tract. This is attributed to the low lipid solubility of the un-ionized molecule. A guide to lipid solubility or lipophilic nature of a drug is provided by a property called the partition coefficient (P). Therefore, this parameter influences the transport and absorption processes of drugs, and it is one of the most widely used properties in quantitative structure–activity relationships (15) (see also Chapter 2).

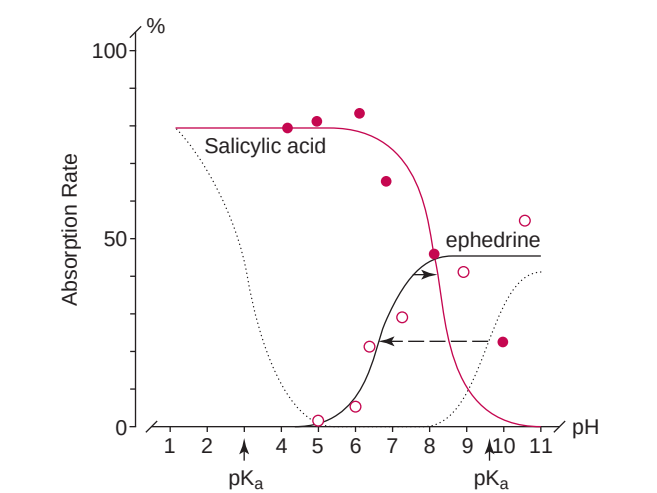


FIGURE 3.12 Relationship between absorption rates of salicylic acid and ephedrine and bulk phase pH in the rat small intestine in vivo. Dashed lines represent curves predicted by the pH-partition theory in the absence of an unstirred layer. (From Winne D. The influence of unstirred layers on intestinal absorption in intestinal permeation. In: Kramer M, Lauterbach F, eds. Workshop Conference Hoechst, Vol 4. Amsterdam: Excerpta Medica, 1977:58–64, with permission.) (16).

The movement of molecules from one phase to another is called partitioning. Drugs partition themselves between the aqueous phase and lipophilic membrane. Preservative emulsions partition between the water and oil phases; antibiotics partition from body fluids to microorganisms; and drug and other adjuvants can partition into the plastic and rubber stoppers of containers. Therefore, it is important that this process is understood.

If two immiscible phases are placed adjacent to each other, with one containing a solute soluble in both phases, the solute will distribute itself between two immiscible phases until equilibrium is attained; therefore, no further transfer of solute occurs. If we consider an aqueous (w) and an organic (o) phase, we write according to theory.

If the solute under consideration forms an ideal solution in either phases or in solvent, the activity coefficient can be replaced by the concentration term to yield Equation 3.13:

$$P = \frac{C_o}{C_w} \quad 3.13$$

Equation 3.13 is used conventionally to calculate the partition coefficient of a drug. In Equation 3.13, C_o , the concentration of drug in the organic or oil phase, is divided by the concentration in the aqueous phase. The greater the value of P , the higher is the lipid solubility of the solute. It has been demonstrated for several systems that the partition coefficient can be approximated by the solubility of the solute in the organic phase divided by the solubility in the aqueous phase. Therefore, the partition coefficient is a measure of the relative affinities of the solute for an aqueous or nonaqueous or oil phase. The effect of lipid solubility and, hence, the partition coefficient on the absorption of a series of barbituric acid derivatives is shown in Table 3.3. The term partition coefficient is more commonly expressed exponentially as log P .

TABLE 3.3 Comparison of Barbiturate Absorption in Rat Colon and Partition Coefficient (Chloroform/Water) of Undissociated Drug (5,45)

Barbiturate	Partition Coefficient	% Absorbed
Barbital	0.7	12
Apobarbital	4.9	17
Phenobarbital	4.8	20
Allylbarbital	10.5	23
Butethal	11.7	24
Cyclobarbital	13.9	24
Pentobarbital	28.0	30
Secobarbital	50.7	40
Hexethal	>100	44

It must be clearly understood that even though drugs with greater lipophilicity and, therefore, partition coefficient are better absorbed, it is imperative that drugs exhibit some degree of aqueous solubility. This is essential, because the availability of the drug molecule in a solution form is a prerequisite for drug absorption and the biologic fluids at the site of absorption are aqueous in nature. Therefore, from a practical viewpoint, drugs must exhibit a balance between hydrophilicity and lipophilicity. This factor is always taken into account while a chemical modification is being considered as a way of improving the efficacy of a therapeutic agent.

Examples of polar or hydrophilic molecules that are poorly absorbed following oral administration and, therefore, must be administered parenterally include gentamicin, ceftriaxone, and streptokinase. Lipid-soluble drugs with favorable partition coefficients generally are well absorbed after oral administration. Often, the selection of a compound with higher partition coefficient from a series of research compounds provides improved pharmacologic activity. Occasionally, the structure of an existing drug is modified to develop a similar pharmacologic activity with improved absorption. Chlortetracycline, which differs from tetracycline by the substitution of a chlorine at C-7, substitution of an n-hexyl (hexethyl) for a phenyl ring in phenobarbital, or replacement of the 2-carbonyl of pentobarbital with a 2-thio group (thiopental) are examples of enhanced lipophilicity (Fig. 3.13).

It is important to note that even a minor molecular modification of a drug can also promote the risk of altering the efficacy and safety profile of a drug. For this reason, medicinal chemists prefer the development of a lipid-soluble prodrug of a drug with poor oral absorption characteristics.

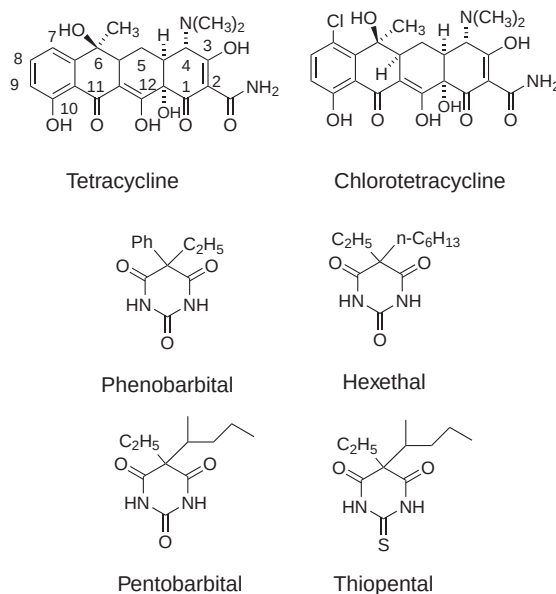


FIGURE 3.13 Drug pairs in which chemical modification enhances lipophilicity.

Estimation of Drug Absorption

When different chemical entities are being investigated for their potential as drug candidates, dosage form performance is one of the possible contributing factors to poor bioavailability. Historically, the concept of bioavailability is closely, if not solely, associated with dosage form performance. Because poor bioavailability in particular is increasingly an issue in the drug discovery and development process, application of the bioavailability principles and techniques has been extended to include animal studies in the selection of potential drug candidates for their full development (17,18).

As the drug travels down the gastrointestinal tract following its oral administration, part of the dose cannot be available for absorption for a number of reasons. These include its chemical degradation, physical inactivation because of binding and complexation with substances in the intestinal tract, incomplete dissolution of the dosage form, microbial metabolism, insufficient contact time in the gastrointestinal tract, poor solubility and poor permeability across the gastrointestinal mucosa, and metabolism within the gut wall. Of the absorbed dose, some of the drug is metabolized in transit during its first passage through the gut wall and the liver. Unchanged drug that reaches the hepatic portal vein can be extracted by the liver via metabolism or biliary excretion. Thus, the bioavailability (f) of an orally administered dose of a drug comprises the individual fractions that survive several barriers encountered by the drug during its first passage from gut lumen to the sampling site, and it is described (19), in general, by the following relationship:

$$3.14 \quad f = F_a F_g F_b$$

where F_a , F_g , and F_b are the fractions of intact drug absorbed (F_a) that escape irreversible elimination as the drug passes sequentially from the gastrointestinal tract across the gut wall (F_g) and traverses the liver (F_b) into systemic circulation. Thus, bioavailability of a drug is equal to or less than the fraction absorbed, depending on the extent of metabolism and loss during the absorption process. Therefore, poor blood levels of a drug can be a consequence of poor absorption or of good absorption accompanied by extensive metabolism.

There appear to be several common misconceptions (20) regarding the nature of absorption. Among these misconceptions are that intestinal absorption, permeability, fraction of drug absorbed, and in some cases, even bioavailability are equivalent properties and, consequently, are used interchangeably. Another common misconception is that absorption and permeability are discrete fundamental properties of a drug molecule and can be predicted solely from its chemical structure. In reality, however, drug absorption is quite a complex process dependent on drug properties such as solubility and permeability, formulation factors, and physiologic variables such as regional permeability differences, pH, luminal and mucosal enzymology, and intestinal motility, among others.

Publication of the so-called “Rule of Five” (21) has generated widespread interest regarding applying calculated physicochemical properties in the drug discovery process to separate out poor drug candidates before better drug candidates go into clinical trials. According to the Rule of Five, poor intestinal absorption is associated with and attributed to the molecule possessing any two of the following properties: molecular weight greater than 750 daltons, number of hydrogen bond donors greater than 5, number of hydrogen bond acceptors greater than 10, and calculated log P greater than 5. These guidelines have been proven to be very useful for approximate predictions of intestinal drug absorption. The critical role of lipid solubility in drug absorption is a major guiding principle in the drug discovery and development process. Because the lipid solubility of a drug molecule is the sum of the individual partition coefficients for each of its functional groups (see Chapter 2), the prediction of lipid solubility (calculated CLog P ; see Appendix A for tables of CLog P values) can be estimated.

A recent examination (22) of the relationship of molecular surface properties with biologic performance of a molecule has been revealing. Most notably, it has been demonstrated that polar surface area (PSA) of a drug molecule has a strong correlation predicting drug transport from human intestine and across the drug membrane. The PSA is defined as the sum of the Van der Waals surface areas for the polar atoms, oxygens, nitrogens, and attached hydrogen atom (or the number of H-bond donors and H-bond acceptors). (See www.molinspiration.com for calculating the PSA.) The PSA is a major determinant for oral absorption and brain penetration of drugs that are transported by the transcellular route (movement across cell membranes). This property should be considered in the early phase of drug screening. Another related parameter, dynamic PSA (PSAd), has surfaced (23) as a parameter of value in predicting membrane permeability and oral absorption in humans. Interpolation of the sigmoidal plot for 20 selected compounds suggests that when the PSAd is greater than 140 Å², incomplete absorption (< 10%) results, and when the PSAd value is less than 60 Å², drug absorption will be in excess of 90%.

A drug's absorption, as reflected in its bioavailability, is a fairly complex process, and although it is related to the drug structure, it is related in a complex manner. Failure to appreciate and understand these complexities, in an attempt to build models, can provide a prediction of marginal and low confidence. Both fraction absorbed and bioavailability are measures of the extent of absorption. Permeability, on the other hand, is related to the rate of absorption (20):

$$3.15 \quad J = P_e \cdot SA \cdot dC$$

where J is the absorptive flux and is equal to the permeability (P_e) of intestinal mucosa to the drug, the surface area available (SA), and the drug concentration gradient (dC) across the mucosa. Factors that can influence

permeability include structural characteristics of a drug, including size, shape, solubility, charge, and surface area.

It has been argued (24), on a theoretical basis, that a fundamental relationship exists between the rate measured as a permeability coefficient and the extent of absorption. This has led to the greater interest and increasing use of the *in vitro* permeability model to serve as an experimental surrogate for predicting oral absorption potential of drug candidates in drug discovery programs. Additionally, although in some instances (24,25) it has been possible to directly correlate absorption with permeability, more often poor correlation exists (20,26), as illustrated in Figure 3.14. At times, good absorption is observed for poorly permeable compounds. These poorly permeable but well-absorbed compounds exhibit high aqueous solubility, generally exceeding 2.5 mg/mL (20,22). This suggests that aqueous solubility can help to compensate for the poor *in vitro* permeability observed.

Estimating the extent of oral drug absorption and variation in drug absorption, therefore, can be of great value in the selection of a potential therapeutic agent and in identifying ways to optimize the oral drug delivery in patients. This can be facilitated by developing the predictive oral drug delivery models. In turn, these models permit the estimation of drug absorption without performing *in vivo* studies in humans and impart better understanding of the rate-limiting processes affecting drug absorption, which can assist in developing strategies for the development of oral drug delivery. There are three physical barriers to drug absorption: the dissolution resistance, the aqueous boundary layer resistance, and the membrane resistance (27–29).

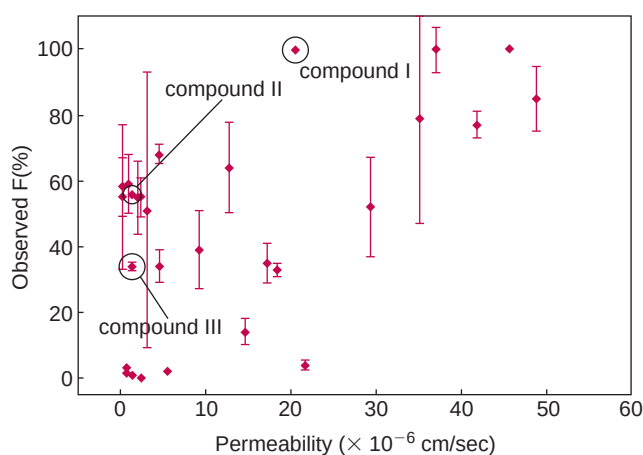


FIGURE 3.14 Poor correlation between oral bioavailability and permeability as measured using Caco-2 cells for three compounds. (From Burton P, Goodwin J, Vidamas T, et al. Predicting drug absorption: how nature made it a difficult problem. *J Pharm Exp Ther* 2002;303:889–895; and from Hilgers AR, Smith DP, Biermacher JJ, et al. Predicting oral absorption of drugs: a case study with novel class of antimicrobial agents. *Pharm Res* 2003;20:1149–1155, with permission.) (19).

Although physicochemical properties, such as solubility and permeability, and other properties, such as metabolic stability and toxicity, are important individually, the interrelationship of these properties is what eventually determines the *in vivo* performance of a drug. In particular, the role of solubility is dependent on the potency, which will determine the dose. In other words, low solubility is problematic for a high-dose drug; however, it can be more acceptable for a low-dose drug.

Measurement of Permeability

Although a variety of models (subcellular fraction, cell monolayer model, isolated intestinal tissue, and intestinal perfusion) are available to predict the permeability of a drug, the cell monolayer model and rat intestinal perfusion techniques are the most commonly used techniques.

Cell Monolayer

These models consist of cells grown on permeable inserts. Transport of compounds across the cell monolayer is used to quantitate the permeability of a new chemical entity in a rapid manner. One of the most popular cell lines is Caco-2, derived from human colon adenocarcinoma cells. The monolayer exhibits ion conductance and possesses transepithelial electrical resistance indicative of fully formed tight junctions that restrict the paracellular transport of a chemical entity. Although Caco-2 cells are the most commonly used cells, Madin-Darby canine kidney (MDCK) cells are becoming more widespread in use, in part because of the shorter culture time (4 to 7 days vs. 21 to 30 days for Caco-2 cells) needed for their use in permeability experiments.

Excellent correlation for permeability coefficient between MDCK and Caco-2 cells was observed for 55 compounds with known human intestinal absorption. Regardless of the type of cells used in determining permeability measurement, establishing the correlation between the permeability coefficient and the fraction of drug absorbed *in vivo* validates this approach.

Several clones of HT 29 cells have been used (30) to study different aspect of intestinal drug absorption. An enterocytic HT 29 clone, HT 29-18-C1, was proposed as a model to study intestinal permeability. The limitation of the cells is that these cells grow very slowly, and a large number of cultures failed to develop acceptable barrier characteristics.

Biopharmaceutical Drug Classification

It is clear from the discussion thus far that the physicochemical properties, such as drug solubility and drug permeability, have a critical role in the drug absorption process. The following biopharmaceutical drug classification system (30–32) has been developed to optimize the development of an oral dosage form taking into consideration two rate-limiting factors, drug permeability and drug dissolution, the latter of which is related to drug solubility.

CLASS I DRUGS (HIGH SOLUBILITY AND HIGH PERMEABILITY) Class I drugs provide both rapid dissolution and high membrane permeation. This class includes small-molecule hydrophilic drugs that are not ionized in the gastrointestinal tract. Examples include acetaminophen, valproic acid, ketoprofen, disopyramide, verapamil, propranolol, fluconazole, and metoprolol. Class I drugs are well absorbed and are affected by a limited set of interactions that alter drug absorption. Because gastric emptying frequently will control the rate of absorption for this class of drugs, interactions that delay gastric emptying will delay drug absorption. This is important for class I analgesic drugs, for which a rapid rate of absorption and quick rise in the plasma level to within the therapeutic range is needed to alleviate pain quickly.

CLASS II DRUGS (LOW SOLUBILITY AND HIGH PERMEABILITY) For immediate-release formulations of many poorly water-soluble drugs, the dissolution rate limits drug absorption. Along with this limitation, a greater impact on drug absorption will be observed with high oral doses. For example, the antifungal drug griseofulvin and the cardiac glycoside drug digoxin are both poorly water soluble and possess similar dissolution profiles, which limit the rate of drug absorption. The extent of griseofulvin absorption, however, is incomplete for a typical dose of 500 mg, whereas a normal, 0.25-mg oral dose of digoxin usually provides a fairly complete absorption. Other examples are diazepam and nifedipine.

Any interactions that increase drug solubility and dissolution rate in the gastrointestinal tract will exert a positive effect on the gastrointestinal absorption of this class of drugs. The absorption of this class of drugs is often enhanced in proportion to the fat content of the coadministered meal. This is attributed to the increased gastrointestinal fluid volume from a coadministered meal, stimulated gastrointestinal secretions, and biliary solubilization effects that increase the dissolution rate. Furthermore, increased gastric residence time as a function of the caloric density permits greater time for drug dissolution.

CLASS III DRUGS (HIGH SOLUBILITY AND LOW PERMEABILITY) For drugs possessing high water solubility, the intestinal membrane permeation rate is often the rate-limiting step in drug absorption from immediate-release dosage forms. Many drugs in this class (e.g., acyclovir and chloramphenicol) also show region-dependent absorption with better absorption in the upper small intestine. Therefore, any interactions that compromise upper intestinal absorption can result in a significant decrease in oral bioavailability. Consequently, these drugs show a sharp decrease in absorption with a coadministered meal that is independent of fat content. Meals tend to decrease the absorption of some drugs in this category as a result of simple physical barrier that compromises the availability of drug molecules to the upper intestinal membrane.

CLASS IV DRUGS (LOW SOLUBILITY AND LOW PERMEABILITY) Poor aqueous solubility cannot necessarily impart high lipophilicity and, therefore, high membrane permeation for a drug. Class IV drugs possess both low solubility and low permeability, both of which are undesirable for good drug absorption. Examples include furosemide and paclitaxel. Drugs in this class, however, still are administered orally if the plasma concentrations obtained are sufficient to produce the desired therapeutic effect and the drugs do not possess a narrow therapeutic index.

Role of Transporters in Drug Absorption

The oral route of drug administration remains the most popular and convenient route of administration, despite its many shortcomings and challenges. Although the advantages associated with oral administration far outweigh the limitations, a major limitation for oral absorption relates to the interactions of drugs with intestinal membrane transporters and metabolizing enzymes (33). The rapidly growing awareness of transporters affecting the rate and extent of intestinal drug absorption has attracted attention in drug discovery and development (Chapter 5). Intestinal membrane transporters affecting the rate of oral absorption are the influx peptide transporters (PepT1, PHTs, and HPT-1), bile salt transporter, phosphate transporter, nucleoside transporters, organic cation/anion transporters (OATP and OCTP), and fatty acid transporters. Transporters affecting drug efflux into the intestinal lumen include P-glycoprotein (P-gp), MRP2, BCRP, and MRP3. The primary intestinal enzyme affecting the absorption of drugs is CYP3A4, as well as the phase II enzymes, glutathione transferase, glucuronyltransferases, and sulfotransferases. Thus, inhibition of these membrane transporters and/or metabolizing enzymes and modulation of the expression of these membrane transporters and/or metabolizing enzymes are key factors affecting the rate and extent of drug absorption. Drug molecules recognized by OATP-B include bile acids, bilirubin and bilirubin glucuronides, estrogen and androgen sulfate conjugates, digoxin, pravastatin, fexofenadine, thyroid hormones, and other lipophilic organic anions. The PepT1 will transport peptide-like drugs, such as β -lactam antibiotics (penicillins and cephalosporins), angiotensin-converting enzyme inhibitors, rennin inhibitors, thrombin inhibitors, and di-/tripeptide prodrugs of antivirals (valacyclovir).

Efflux Transporters

More recently (33–37), the role of efflux transporters in influencing the permeability and the overall bioavailability of drugs has emerged and gained considerable attention. Among these transporters is P-gp, which is expressed on the luminal surface of normal intestinal mucosa. Unlike absorptive transporters that increase the uptake of a substrate from intestinal lumen, P-gp impedes uptake by returning the portion of drug entering the mucosa back to the lumen in a concentration-dependent

manner. Two types of P-gp have been observed in mammals: drug-transporting P-gp and phospholipid-transporting P-gp.

The localization suggests that P-gp functionally can protect the body against toxic xenobiotics by excreting these compounds into bile, urine, and the intestinal lumen and by preventing their accumulation in brain and testes. Thus, P-gp can have a significant role in drug absorption and disposition in human and animals. An increasing number of drugs have been shown to be substrate for P-gp, including HIV protease inhibitors and verapamil, which is also an inhibitor of P-gp and, thus, can increase the intestinal permeability of other drugs.

P-gp is a cell membrane-associated protein that transports a variety of substances. It has been studied extensively as a mediator of multidrug resistance in cancer, but only recently has the role of P-gp expressed in normal tissue as a determinant of drug pharmacokinetics and pharmacodynamics been investigated.

P-gp is a 170 kDa protein product of the *MDR1* gene. It is a dimer consisting of 1,280 amino acids, with 12 trans-membrane segments and 2 adenosine 5'-triphosphate (ATP)-binding domains. P-gp requires binding of ATP to both ATP-binding domains for the transport function. A proposed mechanism by which P-gp secretes substrates is illustrated in Figure 3.15.

Unlike most other transport proteins that recognize a few structurally similar substrates, P-gp recognizes a broad range of pharmacologically and structurally diverse compounds. In general, P-gp substrates are large, lipophilic compounds that tend to be cationic at

physiologic pH. An evaluation of 100 structurally diverse compounds revealed that P-gp substrates have a relatively high number of electron-donating groups (i.e., O, N, S, F, Cl, or groups with a π -electron orbital of an unsaturated system).

In recent years, the role of drug transporters in the intestinal epithelium as major determinants of drug absorption has been recognized, and P-gp and other transporters have been implicated in modulating the absorption and/or intestinal elimination of drugs. Reduction in the small intestinal transit time (SITT) of a drug can decrease the peak plasma concentration and area under the plasma concentration–time curve, the rate and extent of absorption, and, therefore, the bioavailability of a drug. Another potential consequence of increased transit has been proposed for digoxin and, possibly, other drugs that are substrate for P-gp in the small intestine. Because intestinal permeability of such compounds can depend on the relative activity of P-gp in the intestine, factors affecting this activity can also affect absorption. One determinant is drug concentration, which will influence the degree of saturation of the transporters. Another consideration is the specific activity of the transporters with the intestine itself. Evidence suggests that P-gp is not homogeneously distributed throughout the intestinal tract but, rather, increases in abundance from the proximal to the distal small intestine. Therefore, drugs that can be substrate for P-gp but that are partly permeable can be well absorbed in the duodenum and proximal jejunum, which have little P-gp. Drugs that inhibit P-gp can alter the absorption, disposition, and elimination of coadministered drugs and can enhance bioavailability or cause unwanted drug–drug interactions.

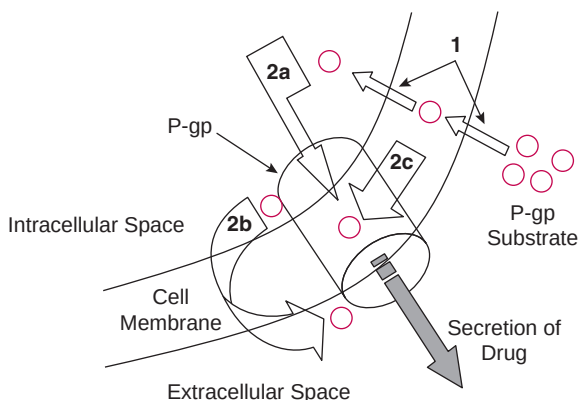


FIGURE 3.15 Proposed mechanism by which P-glycoprotein (P-gp) secretes substrates. (1) Passive drug uptake across cell membrane. (2a) Formation of hydrophobic channel (pore) between the intracellular and extracellular space. (2b) Flippase activity, whereby the drug is flipped from the inner leaflet to the outer leaflet of the cell membrane. (2c) “Vacuum cleaner model,” in which the drug interacts with P-gp in the lipid bilayer and is subsequently secreted back into the extracellular space. (From Matheney C, Lamb M, Brouwer K, et al. Pharmacokinetics and pharmacodynamic implications of P-glycoprotein modulation. *Rev Ther* 2001;21:778–796, with permission.)

Prodrugs

Prodrugs are bioreversible derivatives of drug molecules that undergo an enzymatic and/or chemical transformation in vivo to release the active parent drug, which can then exert the desired pharmacologic effect. In both drug discovery and development, prodrugs have become an established tool for improving physiochemical, biopharmaceutical, or pharmacokinetic properties of pharmacologically active agents. The rationale behind the use of a prodrug is generally to optimize absorption, distribution, metabolism, and excretion (ADME) processes. Prodrugs are usually designed to improve oral bioavailability due to poor absorption from the gastrointestinal tract. Prodrugs are now an established concept to overcome barriers to a drug’s usefulness. About 5% to 7% of drugs approved worldwide can be classified as prodrugs, and the implementation of a prodrug approach in the early stages of drug discovery is a growing trend. A few examples of noteworthy blockbuster prodrugs are omeprazole, simvastatin, lovastatin, enalapril, clopidogrel, valacyclovir, acyclovir, and oseltamivir (Fig. 3.16).

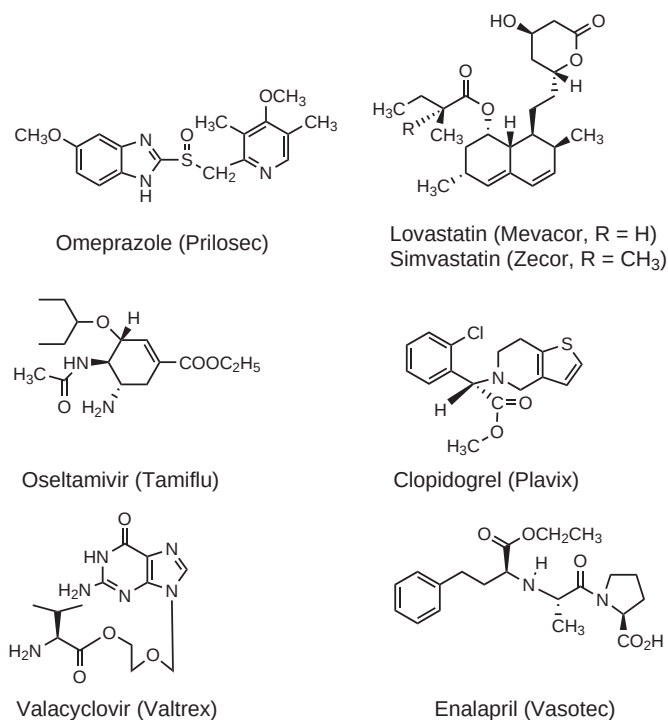


FIGURE 3.16 Commercially available blockbuster prodrugs and their structures.

The term “prodrug” or “proagent” was first introduced by Albert (38) to signify pharmacologically inactive chemical derivatives that could be used to alter the physicochemical properties of drugs, in a temporary manner, to increase their usefulness and/or to decrease associated toxicity. Since Albert discussed the concept of prodrugs in the late 1950s, such compounds have also been called “latentiated drugs,” “bioreversible derivatives,” and “congeners,” but “prodrug” is now the most commonly accepted term (39–41). Usually, “prodrug” implies a metabolizable covalent link between a drug and a chemical moiety, although some authors also use it to characterize some forms of salts of the active drug molecule. Approximately 49% of all marketed prodrugs are activated by hydrolysis, and 23% are bioprecursors (i.e., lacking a promoiety) activated by a biosynthetic reaction (42).

The prodrug approach gained attention as a technique for improving drug therapy in the early 1970s. Numerous prodrugs have been designed and developed since then to overcome pharmaceutical and pharmacokinetic barriers in clinical drug application such as low oral drug absorption, lack of site specificity, chemical instability, toxicity, and poor patient acceptance (e.g., bad taste, odor, pain at injection site) (42).

Prodrugs provide a rationale and opportunities to reach target physicochemical, pharmacokinetic, and pharmacodynamic properties. They are designed to overcome barriers such as insufficient chemical stability, poor solubility, unacceptable taste or odor, irritation or pain, insufficient oral absorption, inadequate blood–brain

barrier permeability, marked presystemic metabolism, and toxicity (39).

Strategic Considerations

Beaumont et al. (43), in a recent review on the design of ester prodrugs, conclude with the recommendations that the prodrug strategy should only be considered as a last resort to improve the oral bioavailability of important therapeutic agents. It can be considered in parallel with classical analoging as soon as a problem becomes apparent.

One prodrug strategy that is often applied to improve the solubility of drug candidates is to introduce a promoiety containing phosphate, sulfate, or succinate groups, which can ionize in relevant biologic media. Strategies to improve the permeability by prodrug formulation can be used to develop prodrugs that have increased passive diffusional permeability properties across the enterocytic membrane compared with the parent compound or to develop prodrugs as substrates for absorptive intestinal membrane transporters such as the peptide transporter PepT1.

Regardless of the strategy used, however, a prerequisite is that the drug candidates contain functional group(s) such as carboxylic acid, alcohol, thiol, or amine functionalities that can be applied in the formation of bioreversible linkages between the promoiety and the drug candidate. The most frequently applied prodrug linkage is the ester linkage; however, many other linkages can also be used.

Thus, drug candidates that are alcohols, amines, thiols, or carboxylic acids are formulated as the corresponding ester, amide, thioester, and ester prodrugs, respectively. Conventional ester linkages are, in general, rapidly cleaved in the gastrointestinal fluids and thus characterized by very short half-lives *in vivo* due to esterase-catalyzed hydrolysis. In cases where the prodrug linkage is hydrolyzed in gastrointestinal fluids, the prodrug formulation does not necessarily increase the oral bioavailability of the parent drug candidate.

Type of Prodrugs

Hard Prodrug: A hard prodrug is a biologically active compound with a high lipid solubility or high water solubility having a long biologic half-life. Examples include cocaine and heroin.

Soft Prodrug: A soft drug is a biologically active compound that is biotransformed *in vivo* in a rapid and predictable manner into nontoxic moieties. Examples include insulin and adrenaline.

Carrier-Linked Prodrug: A carrier-linked prodrug is a compound that contains an active drug linked to a carrier group that can be removed enzymatically. These prodrugs are generally esters or amides, and such a prodrug would have greatly modified lipophilicity due to the attached carrier. The active drug is realized by hydrolytic cleavage either chemically or enzymatically. The prodrug should be biologically

inactive. The chemical linkage between the parent drug and its promoiety must be bioreversible; the prodrug should be sufficiently stable to allow its formulation into an appropriate dosage form. The carrier-linked prodrug is subdivided into a bipartite prodrug that is comprised of one carrier attached to drug. A tripartite prodrug is carrier connected to a linker that is connected to a drug.

Bioprecursors: Bioprecursors are inert molecules obtained by chemical modification of the active drug but do not contain a carrier. Such a moiety has almost the same lipophilicity as the parent drug and is bioactivated generally by enzymatic redox metabolism.

Mutual Prodrug: Two, usually synergistic, drugs are attached to each other. A bipartite or tripartite prodrug is one in which the carrier is a synergistic drug with the drug to which it is linked.

Major Objectives of Prodrug Design

IMPROVED BIOAVAILABILITY Aqueous solubility, drug instability, passive intestinal absorption, targeted active absorption, formulation problems, taste palatability, and metabolic switching are well-established factors limiting bioavailability that need to be circumvented.

IMPROVED AQUEOUS SOLUBILITY Inadequate aqueous solubility is an important factor limiting parenteral, percutaneous, and oral bioavailability. In such cases, a prodrug strategy can bring great pharmaceutical and pharmacokinetic benefit. Charged promoieties (e.g., esters such as phosphates, hemisuccinates, aminoacyl conjugates, dimethylamino acetates) and neutral promoieties (e.g., polyethylene glycols) can be used (e.g., celecoxib and valdecoxib). One of the potential problems in this approach is that solubilizing groups can sometimes generate toxic effects.

IMPROVED PASSIVE INTESTINAL ABSORPTION Providing enhanced lipophilicity for increased passive intestinal absorption is the most frequent rationale when adopting a prodrug strategy (approximately 49% of all marked prodrugs are activated by hydrolysis).

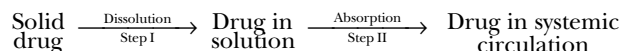
Other pharmaceutical applications of prodrugs include protection against fast metabolism (slow-release prodrugs), improvement of taste, improvement of odor, change of physical form for preparation of solid dosage forms (4), reduction of gastrointestinal irritation, and reduction of pain on injection.

Because prodrugs are designed to improve the permeability and oral absorption of the parent drug, they are more lipid soluble than the parent drug and should be rapidly converted to the parent compound during absorption from the gut wall, liver, or site of action. Examples of prodrugs include pivampicillin, the pivalate ester prodrug of ampicillin that is more lipid soluble and, therefore, more efficiently absorbed than the parent compound (35); valacyclovir, an L-valyl

ester prodrug of acyclovir; lisdexamfetamine, an L-lysinyamide prodrug of amphetamine that is slowly hydrolyzed to amphetamine; and methyldopa, a prodrug of methyl dopamine that is decarboxylated in the brain to methyl dopamine. The recognition that di- and tripeptides are transported from the intestine by their PepT1 transporter has led to the development of prodrugs designed as di- or tripeptide analogues (36). For example, the dipeptidyl analogue of methyldopa increased the intestinal absorption of methyldopa by more than 20-fold.

FACTORS AFFECTING THE ABSORPTION OF DRUGS FROM SOLID DOSAGE FORMS AND SUSPENSIONS

When a drug is administered orally via tablet, capsule, or suspension, the rate of absorption is often controlled by how fast the drug particles dissolve in the fluid at the site of administration. Hence, the dissolution rate is often the rate-limiting (slowest) step in the following sequence:



If the dissolution of the drug is slow or controlling the rate of absorption (Step I), then dissolution is the rate-determining step. Factors controlling dissolution, such as solubility, ionization, or surface area, will then control the overall dissolution process. Figure 3.17 describes the absorption of aspirin from solution and from two different types of tablets.

It is clear from Figure 3.17 that aspirin absorption is more rapid from solution than from tablet formulations. This rapid absorption of aspirin is an indication that the rate of absorption is dissolution rate limited. A general

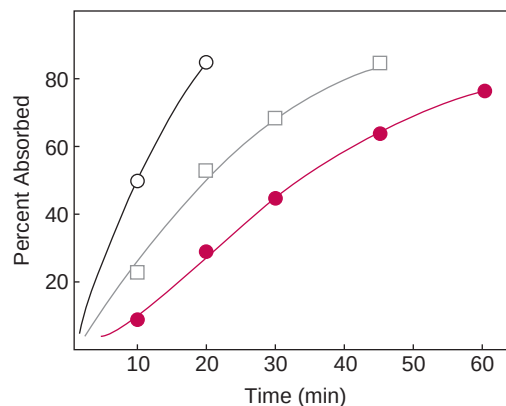


FIGURE 3.17 Absorption of aspirin after oral administration of a 650-mg dose in solution (o), in buffered tablets (□), or in regular tablets (●). (From Kwan KC. Oral bioavailability and first-pass effects. *Drug Metab Drug Dispos* 1997;25:1329–1336, with permission.) (18).

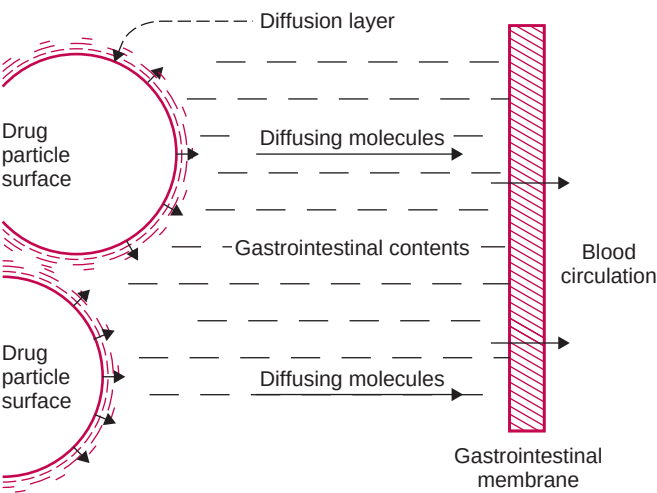


FIGURE 3.18 Dissolution from a solid surface.

relationship describing the dissolution of a drug was first reported by Noyes and Whitney (44). The equation derived by those authors is as follows:

3.16
$$\frac{dc}{dt} = KS(C_s - C_t)$$

where dc/dt is the dissolution rate, K is a constant, S is the surface area of the dissolution solid, C_s is the equilibrium solubility of drug in the solvent, and C is the concentration of drug in the solvent at time t .

The constant K in Equation 3.16 has been shown to be equal to D/h , where D is the coefficient of the dissolving material of the drug and h is the thickness of the diffusion layer surrounding the dissolving solid particles. This diffusion layer is a thin, stationary film of a solution adjacent to the surface of a solid particle (Fig. 3.17) and is saturated with drug (4); in other words, the drug concentration in the diffusion layer is equal to C_s , the equilibrium solubility. The term $(C_s - C_t)$ in Equation 3.16 represents the concentration gradient for the drug between the diffusion layer and the bulk solution. If dissolution is the rate-limiting step in the absorption process, the term C_t in Equation 3.16 is negligible

compared to C_s . Under this condition, Equation 3.16 is reduced to:

3.17
$$\frac{dc}{dt} = \frac{DSC_s}{h}$$

Equation 3.17 describes a diffusion-controlled dissolution process (4), which can be visualized as shown in Figure 3.18; when solid drug particles are introduced to the fluids at the absorption sites, the drug promptly saturates the diffusion layer. This is followed by the diffusion of drug molecules from the diffusion layer into the bulk solution, which is instantly replaced in the diffusion layer by molecules from the solid crystal or particle. This is a continuous process. Although it oversimplifies the dynamics of the dissolution process, Equation 3.17 is a qualitatively useful equation and clearly indicates the effects of some important factors on the dissolution and, therefore, the absorption rate of drugs. When dissolution is the rate-limiting factor in the absorption, then bioavailability is affected. These factors are listed in Table 3.4.

The Noyes-Whitney equations (Eqs. 3.16 and 3.17) demonstrate that the equilibrium solubility (C_s) is one of the major factors determining the rate of dissolution. Changes in the characteristics of solvents, such as pH, affecting the solubility of the drug, affect its dissolution rate. Similarly, the use of a different salt or other physicochemical form of a drug, which exhibits a solubility different from the parent drug, usually affects the dissolution rate. Increasing the surface area of a drug exposed to the dissolution medium, by reducing the particle size, usually increases the dissolution rate. In the discussion to follow, some of the more important factors affecting dissolution and, therefore, absorption are presented in greater detail.

Dissolution

pH and Solubility of Weak Acids and Bases

Solubility is another factor determining the rate of dissolution. As solubility increases, so does the dissolution rate. One way of increasing solubility is to use salts. Salts of weak acids and weak bases generally have much higher aqueous solubility than the free acid or base; therefore, if the drug is given as a salt, the solubility can be increased,

TABLE 3.4 The Effect of Changing Parameters from the Dissolution Equation on the Rate of Solution (4)		
Equation Parameter	Comments	Effect on Rate of Solution
D (diffusion coefficient of drug)	May be decreased in the presence of substances that increase viscosity of the medium	(–)
A (area exposed to solvent)	Increased by micronization and in “amorphous drugs”	(+)
δ (thickness of diffusion layer)	Decreased by increased agitation in gut or flask	(+)
C_s (solubility in diffusion layer)	That of weak electrolytes altered by change in pH by use of appropriate drug salt or buffer ingredient	(–) (+)
C (concentration in bulk)	Decreased by intake of fluid in stomach, by removal of drug by partition or absorption	(+)

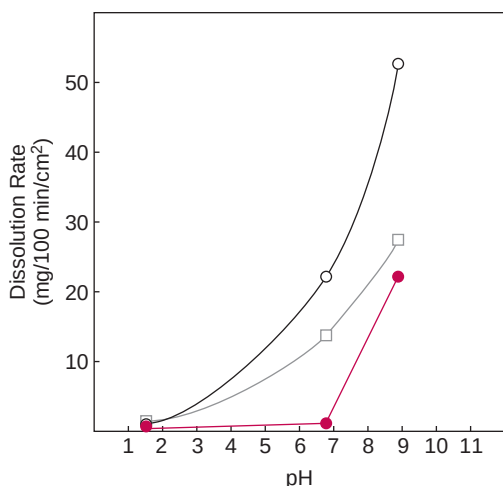


FIGURE 3.19 The pH-dependent dissolution of salicylic acid (o), benzoic acid (□), and phenobarbital (●). (From Gibaldi M. *Biopharmaceutics and Clinical Pharmacokinetics*, 4th Ed. Philadelphia: Lea and Febiger, 1991, with permission.) (45).

and we should have improved dissolution (Fig. 3.19). This factor can lead to quite different peak plasma concentrations after oral administration.

The solubility of weak acids and bases is a function of the pH of the medium. Therefore, differences in the dissolution rate are expected to occur in different regions of the gastrointestinal tract. The solubility of weak acid is obtained by

$$3.18 \quad C_s = [\text{HA}] + [\text{A}^-]$$

where $[\text{HA}]$ is the intrinsic solubility of the un-ionized acid (i.e., C_o) and $[\text{A}^-]$ is the concentration of its anion, which can be expressed in terms of its dissociation constant, K_a , and C_o ; that is,

$$3.19 \quad C_s = C_o + \frac{K_a C_o}{[\text{H}^+]}$$

In a similar manner, the solubility of a weak base is obtained by

$$3.20 \quad C_s = C_o + \frac{C_o [\text{H}^+]}{K_a}$$

By substituting Equations 3.18 and 3.19 into Equation 3.20 for the term C_s , the following dissolution rate equations are obtained:

For weak acids:

$$3.21 \quad \frac{dc}{dt} = \frac{K'(C_o + K_a C_o)}{[\text{H}^+]}$$

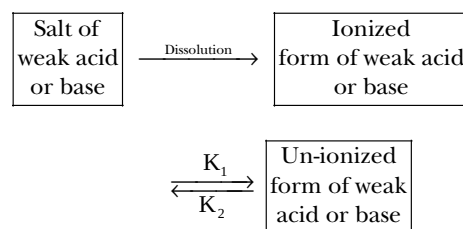
or

$$3.22 \quad \frac{dc}{dt} = \frac{K' C_o (1 + K_a)}{[\text{H}^+]}$$

and for a weak base:

$$3.23 \quad \frac{dc}{dt} = \frac{K' C_o (1 + [\text{H}^+])}{K_a}$$

Equations 3.21 through 3.23 show that K' is equal to DS/h . Equations 3.22 and 3.23 clearly suggest that the dissolution rate of weak bases decreases with increasing pH. Hence, the dissolution rate of weak bases is optimum in gastric fluid, but for weak acids, it is at a minimum. Furthermore, the dissolution rate of weak acids increases as the solid drug particles move to the more alkaline regions of the gastrointestinal tract. Figure 3.18 illustrates the dissolution rates of weak acids as a function of pH (45). The absorption of a salt of weak acid or base can be explained by using the following figure:



where K_1 and K_2 represent the rate constants associated with the formation of un-ionized and ionized species of a compound, respectively. The ratio of these two rate constants represents the dissociation constant of a compound. The absorption of the un-ionized species of a molecule disturbs the equilibrium of the process. To regain the equilibrium, some of the ionized species, therefore, are converted into un-ionized species, which are then absorbed through the membrane. This process, being a continuous one, permits the absorption of the un-ionized species to take place. Therefore, a drug molecule will eventually be absorbed.

The relatively poor dissolution of weak acids at the pH of gastric fluid further diminishes the importance of the stomach as a drug absorption site. Although gastric absorption of weak acids can occur from solution, it is unlikely that much of the drug dissolves and is absorbed during the short residence time as a solid dosage form in the stomach. A study by Ogata et al. (46) proposed that the critical value of solubility that separates acidic drugs from the absorption sites (stomach or intestine) is approximately 30 mg/mL in 0.1 N HCl when 1 g of drug is administered orally. Those authors found that if the solubility of a drug is less than 3 mg/mL, practically no absorption occurs in the stomach. Changes in the gastric pH also alter the solubility of certain drugs and can affect the dissolution and absorption rates. A patient with achlorhydria has a higher gastric pH and absorbs aspirin more rapidly than a normal subject. On the other hand, similar differences were not observed with respect to the absorption rates of acetaminophen, a much weaker acid, the solubility of which would be unaffected by changes in pH (47).

The relationships between dissolution rate and hydrogen ion concentration, described in Equations 3.22 and 3.23, are approximations and tend to overpredict the dissolution rate of both weak acids in the small intestine and weak bases in the stomach. In reality, the hydrogen ion concentration of the bulk is not equal to the hydrogen ion concentration of the diffusion layer.

Salts

The objective of drug discovery is to identify substances that are highly active in biologic systems. Screening for potential drug candidates is performed in solution at micromolar and nanomolar concentration levels.

Drug substances can be administered by various routes, including oral, which is the most frequently chosen route; parenteral by several modes of injection (intramuscular [IM], intravenous [IV], subcutaneous [SC], and others); intranasal inhalation; and topical to the various areas of the body surface, to name just a few more important routes. Various pharmaceutical dosage forms are formulated to administer drug substances by these routes. Each dosage form requires that a typical set of relevant parameters of the drug substance lie within a certain range, beyond which the development of said dosage form becomes increasingly difficult and eventually impossible. Also, in a mutual interdependence, the selection of dosage form and the applicable technology depend on the physical chemical properties profile of the chosen salt.

Approximately two thirds of the drug substances are weakly acidic or basic entities, and therefore, salt formation provides a significant opportunity to alter the physicochemical properties of drugs in the solid state without altering chemical integrity. The decision whether to pursue a salt form is usually made early in the development process on the basis of known properties of the uncharged molecules. Salt formation will often be considered when the drug has one or more unfavorable properties, including poor dissolution, low aqueous solubility ($< 10 \mu\text{g/mL}$), poor chemical stability, hygroscopicity, low melting point ($< 80^\circ\text{C}$), poor crystallization (forms oil or amorphous solid on crystallization attempts), or displays multiple polymorphs. A significant number of counterions are suitable for pharmaceutical salt selection studies of weakly acidic or basic drugs, and therefore, there is a broad range of opportunity to obtain different salts with improved and desired physical chemical properties.

The selection of suitable counterions for inclusion in a salt selection search is dependent on safety considerations. For example, chloride and sodium ions are regarded as safe to administer and well tolerated. Counterion selection often uses pK_a values of the acid and base involved to estimate the likelihood of successful salt formation.

Based on the pK_a values, a series of salt formers can then be selected from those frequently used pharmaceutically (Table 3.5), whereas others must be rejected. From experience, to form a stable salt, the pK_a values of an acid–base pair should differ by at least two pK_a units

TABLE 3.5 Most Frequently Used Acids and Bases as Salt Formers

Acids (Anions)	Bases (Cations)
Hydrobromic	Sodium
Hydrochloric	Potassium
Hydroiodic	Zinc
Sulfuric	Choline
Nitric	Calcium
Methanesulfonic	Magnesium
Maleic	Diethylamine
Phosphoric	Tromethamine
Fumaric	Silver
Citric	Ammonium
Succinic	
Acetic	
Tartaric	

(i.e., the pK_a of the acid should be at least two units lower than that of the base). This corresponds to a situation in which both compounds, brought together in water, are ionized to a degree of at least 90%. Strong mineral acids such as HCl ($pK_a \approx -6$) or H_2SO_4 ($pK_a = -3$) can form solid salts with the weak base atazanavir, having a pK_a as low as 4.25, whereas attempts to isolate a salt with either acetic acid ($pK_a = 4.76$) or benzoic acid ($pK_a = 4.19$) would fail with such a weak base.

The final decision on the selection of a salt of a drug substance is based on data collected from analytical and physical characterization of a certain number of prepared salts. The data are evaluated primarily against the requirements set by biopharmaceutical and therapeutic considerations for the use of the drug, pharmaceutical and chemical–technologic aspects in view of the development and manufacturing of dosage forms, and the synthesis and isolation of the drug substance. The most frequently encountered issues while selecting a suitable salt are discussed below.

For injectable solutions, high solubility of a drug under the conditions close to physiologic pH of 7.4 is essential. For a small-volume injectable such as IM and SC, the solubility should be as high as possible to accommodate the dose to be administered in 0.5 to 2 mL for SC or up to 5 mL for IM administration. The solubility requirement for IV injections is less stringent because volumes up to 20 mL can be administered. With lower solubility drugs, one has to resort to infusions of volumes of up to 1,000 mL. Injectable solutions of drugs require particularly high chemical stability. Ideally, a drug substance must withstand heat sterilization in solution and subsequent storage

for up to 5 years. For those drug substances lacking such optimum stability, it is possible to circumvent heat stress by sterile filtration. Naturally, for injectables, solid-state drug properties are of minor importance as long as they do not hamper processing or the dissolution of a lyophilizate.

For solid dosage forms, the most critical step, after swallowing a unit dose, is the release of the drug substance. Solubility and, therefore, dissolution can control or limit, respectively, this important process. Therefore, a solubility that is reasonably high in relation to the drug dose is desirable.

In selecting a salt form for a product that is applied topically, it is a prerequisite that it (salt of the drug) must not cause local irritation, whether the target of treatment is the skin surface, dermal tissue, or underlying organs (e.g., joints, muscles, tendons) or even if systemic effects are intended. Weakly acidic to neutral pH values (~5 to 7.5) are tolerated best, and deviations from this range should be avoided. Choosing the right salt form of a drug can offer the following advantages: Annoying polymorphism problems can be circumvented; high hygroscopicity resulting in deliquescence can be avoided; amorphous material can be turned to a crystalline salt; taste and smell problems can be minimized; the melting point can be raised to improve mechanical properties (e.g., for milling), to the extent that liquid bases or acids are turned into solids; and local irritation can be avoided (e.g., for inhalation).

With drug candidates exhibiting absorption difficulties, investigations can even include in vivo studies in animals (e.g., rat, dog) with experimental formulations to identify the most suitable salt form or solid-state form. Sometimes such a search can address the state of distribution of the active substance in an experimental formulation (e.g., micronized, nano-sized, amorphous, suspension, emulsion, microemulsion), and so the borderlines between chemical technical operations and pharmaceutical formulation and processing techniques can become quite diffuse.

Solubility and Dissolution Rate

Chemical stability includes potential interaction between drug entity and counterions and stability in the presence of pharmaceutical excipients (drug/excipient compatibility); stability of the morphic state in bulk form; stability in solid and suspension dosage forms; hydrate formation and stability during storage and processing; chemical stability of hydrated versus anhydrous forms (influence of released hydrate water during storage); and uptake of water of hydration by anhydrides in solid dosage forms with consequences for mechanical properties.

Regarding molecular weight, large counterions can surmount the tolerable drug amount to be packed into a solid dose unit; in contrast, for extremely low-dose drugs, a larger molecular weight can improve handling and content uniformity.

The dissolution rate of a particular salt is usually different from that of a parent compound. Sodium or

TABLE 3.6 Dissolution Rate of Weak Acids and Their Sodium Salts (44)

Compound	pK_a	Dissolution Rate (mg/100 min/cm ²)		
		0.1 N HCl pH 1.5	0.1 M Phosphate pH 6.8	0.1 M Borate pH 9.0
Benzoic acid	4.2	2.1	14	28
Sodium salt		980	1,770	1,600
Phenobarbital	7.4	0.24	1.2	22
Sodium salt		~200	820	1,430
Salicylic acid	3.0	1.7	27	53
Sodium salt		1,870	2,500	2,420
Sulfathiazole	7.3	<0.1	~0.5	8.5
Sodium salt		550	810	1,300

potassium salts of weak acids dissolve more rapidly than the free acid. The same is true with HCl or other salts of weak bases. Table 3.6 illustrates the dissolution rate differences between some weak acids and their sodium salts (45). The differences in the dissolution rates of salt and parent compound are explained by taking into consideration the pH of the diffusion layer. At a given pH, regardless of salt or free acids/bases, a drug will have a fixed solubility. The classical dissolution equation predicts a slower dissolution of a salt of a drug, and the concept of a diffusion layer becomes useful.

For sodium or potassium salts of weak acids, the pH of the solution in a diffusion layer is greater than the pH of the diffusion layer for the corresponding weak acid. However, the pH of the solution in the diffusion layer for hydrochloride salts of weak bases is always smaller than the diffusion layer of the corresponding free base. Therefore, effective solubility and dissolution rate of soluble salts on drug absorption are available in the literature. The potassium salt of penicillin V yields a higher peak plasma concentration of antibiotic than the corresponding free acid (48). Sodium salts of barbiturates are reported by Anderson (49) to provide a rapid onset of sedation. Some salts have a lower solubility and dissolution rate than their parent compounds. Examples include aluminum salts of weak acids and pamoate salts of weak bases. In these particular examples, insoluble films of either weak acids or pamoic acid appear to form in the dissolving solids and further retard the dissolution rate.

Surface Area and Particle Size

The surface area per gram (or per dose) of a solid drug is changed by altering the particle size. For example, a cube that is 1 cm on each side has a surface area of 6 cm². If this cube is broken into cubes with sides of 0.1 cm, the total

surface area is 60 cm^2 . If the particles are broken up by grinding, then irregular shapes with even larger surface areas are created. Generally, as the surface area increases, the drug will dissolve more rapidly. Therefore, many poorly soluble and slowly dissolving drugs currently are marketed in a micronized or microcrystalline form. The problems of low water solubility and particle size were not fully appreciated, but they have resulted in reducing the therapeutic dose of some drugs without sacrificing therapeutic efficacy. For example, since the original marketing of spironolactone, its dose has been reduced from 500 to 25 mg as a result of a reformulation that includes micronization. The bioavailability of digoxin increased from 40% to approximately 80% to 97% by reducing the particle size from 100 to approximately 10 nm. A similar result has been obtained for griseofulvin.

Polymorphism

Many pharmaceutical solids can exist in two or more crystalline forms called polymorphs (50,51). Polymorphism is the ability of the same drug molecule to crystallize into more than one different crystal structure that has a different arrangement and/or conformation of molecules in the crystal lattice. However, once they are in the solution phase, polymorphs share a common form. The different arrangements of atoms within the crystal unit cell can have a profound effect on physical and chemical properties of the final crystallized compound and on the final drug product. Amorphous solids consist of disordered arrangements of molecules that do not possess a distinguishable crystal lattice. Solvates are crystalline solid adducts containing either stoichiometric or nonstoichiometric amounts of a solvent incorporated within the crystal structure (50,51). If the incorporated solvent is water, the solvates are commonly known as hydrates. Polymorphs and/or solvates of a pharmaceutical solid or pharmaceutical excipients (e.g., lactose) can have different physical and chemical properties, such as melting point, chemical reactivity, apparent solubility, and dissolution rate. These properties of a drug substance can affect the intended shelf life (stability), rate of dissolution, and bioavailability/bioequivalence of the drug product. A metastable (amorphous) pharmaceutical solid form can change crystalline structure or solvate/desolvate in response to changes in environmental conditions or shelf-life storage. When such differences in physical properties are sufficiently large, bioavailability is altered, and it is often difficult to formulate a bioequivalent drug product using a different polymorph. The Biopharmaceutics Classification System criteria of high solubility and rapid dissolution should be considered in product development decisions when polymorphism exists. Drug substances with different physical form include warfarin sodium, famotidine, and ranitidine, and those with solvation or hydration state include terazosin hydrochloride, ampicillin, and cefadroxil.

Some drugs that exist as polymorphs can have different solubility properties and, thus, different dissolution

characteristics. Chloramphenicol palmitate and ritonavir provide good examples of how polymorphism can influence drug dissolution and, thus, drug bioavailability. Chloramphenicol palmitate is a broad-spectrum antibiotic known to crystallize in at least three polymorphic forms and one amorphous (metastable) form, B. The most stable form, known as form A, is the polymorph that is marketed, whereas the metastable form B is approximately eight times more soluble than form A, thus providing an eightfold difference in bioavailability. This large difference in bioavailability creates the danger of fatal dosages when the unwanted polymorph is unwittingly administered because of alterations in process and/or storage conditions. The HIV protease inhibitor ritonavir (Norvir) was withdrawn from the market because an undesirable polymorph of ritonavir had been produced during its shelf life. Ritonavir was found to exist in only one monoclinic form during development and early manufacturing. This form, called "form I," was not sufficiently bioavailable in the solid state by the oral route, requiring the initial product (Norvir) to be formulated as a capsule filled with a hydroalcoholic solution containing the dissolved drug. Two years after the product launch, several lots of Norvir capsules started failing dissolution specifications. Evaluation of the failed lots revealed that a second crystal form of ritonavir, "form II," had precipitated from the formulation during its shelf life. "Form II" was 50% less soluble compared to "form I," resulting in failure of batches in the dissolution test, affecting its bioavailability and causing the eventual withdrawal of the product from the market. Substantial time and effort went into identifying and correcting the problem. To ensure a continuous supply of this life-saving drug, a liquid formulation had to be introduced in the market until the issue of the polymorphic form was resolved. Hence, an inadvertent production of the "wrong" polymorph at the crystallization stage or any transformations of one dosage form to another during processing (e.g., drying, milling, granulation, compression, spray drying, or freeze-drying), storage, and scale-up can result in pharmaceutical dosage forms that are either ineffective or toxic. This highlights that identification of different solid forms of a drug substance and determination of their physical and chemical properties, thermodynamic stabilities, and conditions or kinetics of interconversion are essential for ensuring reproducible behavior of drug products.

Estimate of Dose Absorbed

Johnson and Swindell (52) proposed a simple predictive model that relates the aqueous solubility and absorption rate constant (K_a) to determine the maximum absorbable dose (MAD):

3.24

$$\text{MAD} = K_a C_s V t$$

where K_a is the first-order absorption rate constant; $C_s V t$ is a constant, and t is the time.

Hilgers et al. (26) argued that Equation 3.24, proposed by Johnson and Swindell (52), assumes that the drug absorption occurs under highly unrealistic conditions, such as the intestine is exposed to a saturated solution of a drug of interest for a time equal to the normal SITT. For interrelating permeability and solubility of a drug to estimate the absorption potential (30) as MAD by using the predictive model, Hilgers et al. (26) proposed a modified equation:

$$3.25 \quad \text{MAD} = S \cdot K_a \cdot \text{SIV} \cdot \text{SITT}$$

where S is solubility and SIV is the small intestinal volume.

Briefly, the MAD calculates the total mass of drug that could theoretically be absorbed if a saturated solution of a compound with solubility S in the SIV were absorbed with a first-order absorption rate constant (K_a) for a time equivalent to SITT.

The absorption rate constant (K_a) was determined from Caco-2 permeability value with the assumption that the value of K_a and rat ileum permeability values are approximately equal and, consequently, that the estimated rat permeability (P_e) was converted to an absorption rate constant (K_a) from the following relationship:

$$3.26 \quad K_a = P_e (A/V)$$

where V is the volume of the intestinal lumen and A is the surface area. For the rats used in this study (26), A/V was equal to 10 cm^{-1} .

Using Equations 3.24 and 3.25 for estimating the oral absorption of oxazolidinone antibiotics data obtained in rat and comparing the predicted MAD to the actually administered dose, Burton et al. (19) and Hilgers et al. (26) reported the relationship illustrated in Figure 3.20.

It is clear from Figure 3.20 that better prediction of MAD is achieved when solubility, permeability, and dose are taken into consideration. Despite this, drug permeability is an important determinant of drug absorption; therefore, it is informative to explore mechanisms contributing to permeability in the light of structure-based models for absorption prediction.

The relationship between the estimate of MAD and the fraction absorbed is hyperbolic (28), analogous to the relationship between drug permeability and fraction absorbed. Incorporating the solubility term in Equation 3.24 yields a more realistic value of estimate of MAD for poorly to moderately soluble compounds. Therefore, the concept of MAD serves to combine two major determinants of oral drug absorption (i.e., intestinal permeability and solubility). In turn, the value of MAD can provide a practical guideline in the early drug discovery program.

If the dose of a potential drug candidate is projected to be greater than the calculated estimated dose absorbed, this serves a cautionary notice that the following should be examined: the potential for solubilization in the gastrointestinal tract, and the formulation variables that could ultimately result in an increased solubility or drug

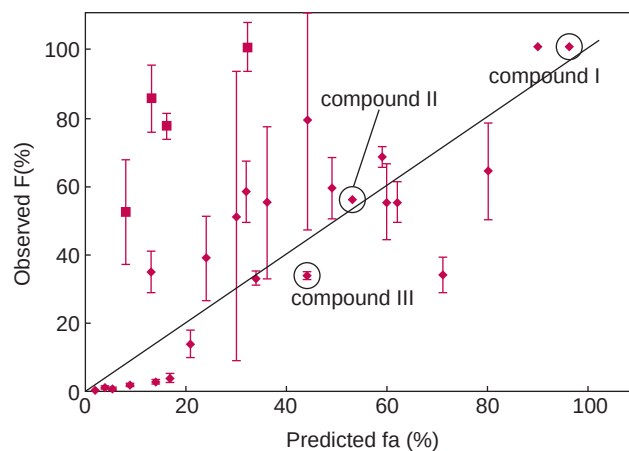


FIGURE 3.20 A better correlation between observed bioavailability and predicted bioavailability when both aqueous solubility and permeability parameters of drugs are used to predict the fraction absorbed for three compounds. (From Burton P, Goodwin J, Vidamas T, et al. Predicting drug absorption: how nature made it a difficult problem. *J Pharm Exp Ther* 2002;303:889–895; and adapted from Hilgers AR, Smith DP, Biermacher JJ, et al. Predicting oral absorption of drugs: a case study with novel class of antimicrobial agents. *Pharm Res* 2003;20:1149–1155, with permission.)

dissolution during the gastrointestinal transit to increase the fraction absorbed.

To reach the systemic circulation, a drug must move from the intestinal lumen through an unstirred water layer and mucous coat adjacent to the epithelial cell structure. Movement across the epithelial layers takes place by two independent routes, transcellular flux (i.e., movement across the cells) and paracellular flux (i.e., movement between adjacent epithelial cells). The solute molecules then encounter a basement membrane, interstitial space, and mesenteric capillary wall to access the mesenteric circulation. Any and all of these micro-environments can be considered a resistance to solute molecule movement, each with an associated permeability coefficient. Therefore, the overall process consists of a number of resistances (i.e., reciprocal of permeability) in series. Furthermore, the influence of drug structure with permeability in these different domains is different. For example, permeability in an unstirred water layer is inversely related to solute size, whereas paracellular permeability is a function of both size and charge. Furthermore, cations exhibit greater permeability than neutral species, which in turn manifest greater permeability than anions.

With respect to transcellular permeability, the relationship of solute structure with permeability depends on the mechanism. Historically, a passive diffusion pathway is assumed for most solutes. Nevertheless, a great number of solutes are identified as being associated with active absorption and secretory processes in intestinal epithelial cells. Additionally, although active transport involves specific interactions between a solute and transporter, passive diffusion is dependent on solute partitioning into the cellular

plasma membrane and the diffusion coefficient within the membrane. Both processes, however, are influenced by the physicochemical and structural characteristics of the drug. Factors influencing plasma membrane partitioning are solute size, lipophilicity, hydrogen-bonding potential, and charge characteristics, whereas the diffusion coefficient is dependent on size or total molecular surface area. In general, a non-PSA favors partitioning.

For solutes that exhibit marginal (or a lack of) membrane affinity, permeability is low, resulting primarily from paracellular diffusion of the solute. As the propensity of the solute to partition into cell membrane increases, so does the permeability, as a result of the significant increase in surface area of the transcellular pathway relative to the paracellular route. This increase in permeability will approach a plateau, the so-called “aqueous boundary layer–limited situation,” in which diffusion across the cell is very rapid relative to diffusion of the solute through the unstirred water/mucous layer.

In the case of ionizable solutes, permeability is also pH dependent. A neutral, uncharged species is capable of transcellular diffusion, whereas a charged species is restricted to the paracellular pathway. Thus, the observed permeability of such molecules is dependent on the relative concentrations of charged and neutral species.

Dressman et al. (53) developed an equation to determine the absorption potential (AP) of a drug by taking into consideration several of its physicochemical properties, including intrinsic solubility (S_o) of the un-ionized species of a drug, fraction of un-ionized form (F_{uni}) of a drug at a specific pH, volume of the luminal content (V_l), permeability of a drug in the gut wall (P_w), and the aqueous permeability (P_{aq}) of a drug. The equation is:

$$3.27 \quad F_{abs} = \left(\frac{P_w}{P_{aq}} \right) (F_{uni}) \left(\frac{S_o V_l}{X_o} \right)$$

where F_{abs} is the fraction absorbed and X_o is the administered dose. With the assumption that, in many cases, the

permeability ratio (i.e., P_w/P_{aq}) of a drug is proportional to its membrane-water coefficient (P), which can be correlated to the 1-octanol–water partition coefficient (P), Equation 3.27 was simplified to:

$$3.28 \quad AP = \log \left(P \cdot F_{uni} \cdot \frac{S_o V_l}{X_o} \right)$$

where AP is a dimensionless parameter.

Using Equation 3.28 and selecting drugs that represent a wide range of absorption characteristics, from poorly absorbed compounds to those that are virtually completely absorbed, the utility of the AP parameter as a predictor of the fraction absorbed was assessed (53). The physicochemical properties, such as the partition coefficient, solubility, fraction available in un-ionized form, dissociation constant, fraction of dose absorbed, and calculated dimensionless parameter AP for drugs selected in this study (53), are reported in Table 3.7. The observed correlation between the fractions absorbed and the AP is illustrated graphically in Figure 3.21.

From Table 3.7 and Figure 3.21, it is quite apparent that for the compounds selected in this study (53), the dimensionless parameter AP manifests a strong correlation to the fraction absorbed. Negative AP values correspond to poor drug absorption. For the range of AP values between 0 and 1, an increase in AP value correlates with an increase in fraction absorbed, whereas AP values greater than 1 indicate complete drug absorption.

SUMMARY

At one time, it was common to assume that the biologic response to a drug was simply a function of the intrinsic pharmacologic activity of the drug molecule. Today, when assessing the potency of most drugs, consideration is given to plasma drug concentration–response rather than dose–response relationships. The concentration of

TABLE 3.7 Physicochemical Properties and Calculated Absorption Potential (AP) for Representative Drugs (53)

Drug	P	S (mg/mL)	Dose (mg)	pK_a	F_{non} (pH 6.5)	AP	%ABS (range)
Acyclovir	0.018	1.3	200	9.5	1	−1.5	17 (12–23)
Chlorothiazide	0.54	0.4	250	6.7	0.6	−0.89	25 (10–40)
Griseofulvin	151	0.015	250	—	1	0.36	43 (35–51)
Hydrochlorothiazide	0.85	0.6	25	8.8	0.95	0.7	67 (50–90)
Phenytoin	295	0.014	100	9.2	0.99	1.0	90 (80–100)
Prednisone	26	0.235	20	—	1	1.9	99
Digoxin	56	0.024	0.25	—	1	3.13	>90

ABS, % absorbed.

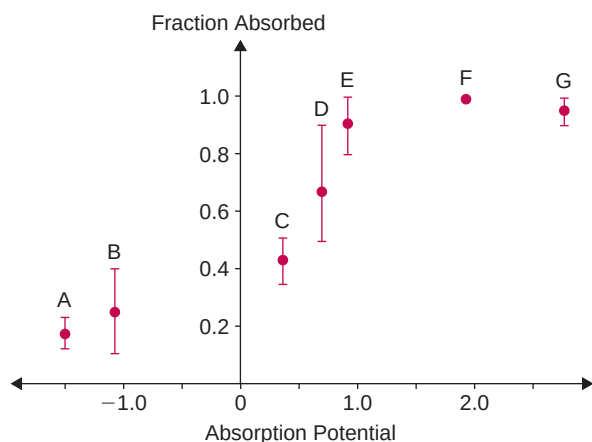


FIGURE 3.21 The relationship between absorption potential (AP) and fraction absorbed for seven representative drugs (see Table 3.7). A, acyclovir; B, chlorothiazide; C, micronized griseofulvin; D, hydrochlorothiazide; E, phenytoin; F, prednisolone; G, digoxin (Lanoxicap). (From Dressman JB, Amidon GL, Fleisher D. Absorption potential: estimating the fraction absorbed for orally administered compounds. *J Pharm Sci* 1985;74:588–589, with permission.)

a drug in the plasma is dependent on the rate and extent of absorption, which in turn is influenced by the physicochemical properties of drug substances. Drug absorption can markedly affect the onset and intensity of a biologic response to a drug. Clinically significant differences in the absorption of closely related drugs, such as lincomycin and clindamycin, penicillin and pivampicillin, or secobarbital and sodium secobarbital, are invariably the result of significant differences in their physicochemical properties.

Dissolution is simply a process by which a solid substance goes into solution. The determination of dissolution rates of pharmaceutical substances from dosage forms does not predict their bioavailability or their in vivo performance; rather, it indicates the potential availability of drug substance for absorption. Therefore, it is essential for pharmacists and pharmaceutical scientists to know and understand the importance of dissolution and its potential influence on the rate and extent of absorption and availability for drugs.

Factors affecting the dissolution rate of a drug from a dosage form can be related to the physicochemical properties of a drug, formulation of a dosage form, and dissolution apparatus and test parameters. Additionally, a brief introduction of the role of intestinal transporters and metabolizing enzymes (CYP3A4) in drug absorption was provided. The role of intestinal permeability in drug absorption process, techniques of measuring intestinal permeability, and methods that permit the estimation of fraction of dose absorbed by considering the physicochemical properties of a drug also were discussed. Additionally, the biopharmaceutical drug classification based on the two physicochemical properties of drugs,

solubility and permeability, and the implications of these properties on drug absorption were covered.

PHARMACOKINETICS

Introduction

The events following drug administration are divided into two phases: a pharmacokinetic phase, in which the ability to adjust a dose, alter the dosage form, and alter the frequency and route of administration are related to the drug concentration–time relationship in the body; and a pharmacodynamic phase, in which the drug concentration at the sites of action is related to the magnitude of effects produced. Once both of these phases have been defined for a drug, a dosage regimen for a drug can be established to achieve the optimum therapeutic goals in individual patients and to predict what can happen when a dosage regimen is changed.

The sites into which drugs are routinely administered are broadly classified as intravascular and extravascular. Intravascular administration refers to the placement of a drug directly into blood, either IV or intra-arterially. Because the drug is placed directly into blood, it is imperative that a drug administered intravascularly be given as a solution. The extravascular routes of administration include oral, IM, sublingual, buccal, SC, dermal, rectal, and nasal routes. To enter the blood, a drug administered extravascularly must be absorbed from the site of administration. In addition, if a drug is orally administered through solid dosage forms, such as tablets or capsules, then the drug must first dissolve at the site of administration. Therefore, the dissolution of a drug is essential before absorption occurs. However, no such absorption step is required when a drug is administered IV.

Pharmacokinetics is the scientific discipline that deals with the mathematical description of biologic processes affecting drugs and affected by drugs. In addition to signifying the relationship of ADME processes to the intensity and time course of pharmacologic effects of drugs, pharmacokinetics describes the time course of a drug's ADME processes, which take place following the administration of a drug. Therefore, it is necessary to describe and analyze these processes and their effects in relation to their rates, rate constants, or time course. A qualitative description of these processes is quite insufficient and seldom leads to adequate and accurate characterizations of the effects of drugs on the body and effects of the body on the drugs. Pharmacokinetics is a quantitative study whose purposes are:

1. To develop mathematical expressions that permit one to describe the temporal changes of the drug concentration;
2. To determine constraints that describes ADME processes succinctly;
3. To make predictions and extrapolations based on the mathematical expressions; and

4. To help establish a dosage regimen that, in turn, will result in improved drug utilization in patients.

At a fundamental level, pharmacokinetics is a useful tool for pharmacists and physicians when optimizing the dosage regimen of drugs for individuals who can differ in their therapeutic response and their ability to absorb and eliminate drugs. Adjustment of dosage regimen to account for individual differences and disease states is, in essence, an exercise in clinical pharmacokinetics and clinical pharmacy practice.

A basic tenet of pharmacokinetics is that the magnitudes of both the desired response and toxicity are functions of drug concentration in the blood. Furthermore, it is not only the efficacy of a drug at the site of action that determines the intensity and duration of its pharmacologic or therapeutic effects, but also the amount of a drug and the rate at which the drug gets to the site of action. The vital processes of the body can delay the transport of drug molecules across membranes, convert drug molecules into metabolites, and remove them from the body as metabolites and/or the unchanged form. In turn, this can result in therapeutic failure, as a result of drug concentrations being too low, or unacceptable toxicity, as a result of the drug concentration being too high. Between these concentrations limits lies a region associated with therapeutic success. This region is regarded as a therapeutic range, or “therapeutic window.” Each drug can possess its own therapeutic window, and because the drug concentration is rarely measured at the site of drug action, the drug concentration is measured at alternative and more accessible sites, such as plasma or serum and urine.

Figure 3.22 illustrates the concentration or therapeutic window for a drug. The terms “minimum toxic concentration” and “minimum effective concentration” describe the limits of the therapeutic range for a drug. If the administered dose of a drug produces the plasma concentration within this range, the drug will

likely produce its therapeutic effect. The term “onset of action” is defined as the time at which the drug enters the therapeutic range (i.e., above the minimum effect concentration), and “termination of action” is defined as when the plasma concentration of a drug falls below the therapeutic range. The time span between the termination and the onset of action is described as the duration of action.

It is clear from Figure 3.22 and the previous definitions that an optimum dosage regimen might be defined as one that maintains the plasma concentration of a drug within the therapeutic range. Furthermore, it can be obvious from the previous discussion that the success of a drug in providing the desired drug concentration depends on factors such as how rapidly the drug reaches the general circulation from the site of administration, particularly following the oral and other extravascular routes; whether the drug is reaching the general circulation in sufficient amounts to provide plasma concentration within the therapeutic range; and the pharmacokinetic properties of a drug.

The purpose of this section is to provide students with a brief overview and the functional understanding of basic pharmacokinetics and its application to how physicochemical properties of drug molecules affect pharmacokinetic properties. Emphasis is placed on how to carry out pharmacokinetic analysis of the data and on how to use the pharmacokinetic parameters for predictive purposes. The mathematical equations presented have been chosen because of their general utility for predicting the plasma concentrations following administration of a drug by intravascular and extravascular routes. Furthermore, this discussion attempts to review and illustrate how the chemical modification of a drug through molecular modifications can alter selected pharmacokinetic parameters of drugs and, therefore, possibly the pharmacologic response in the drug discovery and development process.

Table 3.8 describes the dimensions of various pharmacokinetics parameters and provides examples of the corresponding typical dimensions reported in the literature.

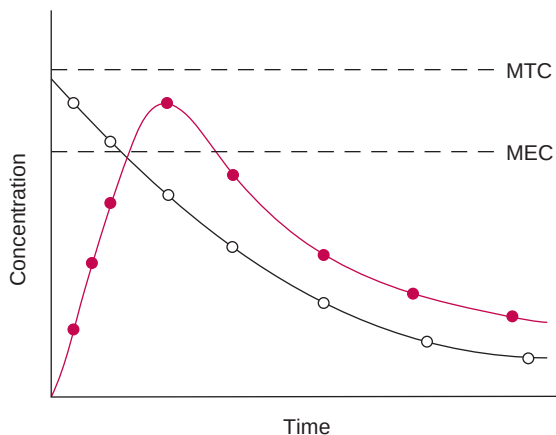


FIGURE 3.22 Typical plasma concentration versus time profile following administration of a dose of a drug by intravascular (o) and extravascular routes (●).

Compartmental Concepts

The most commonly used approach to pharmacokinetic characterization of a drug is to depict the body as a system of compartments, even though these compartments often do not have any apparent physiologic reality. These frequently used compartmental models are illustrated in Figure 3.23.

The one-compartment model considers the body as a single homogenous unit (central compartment). This simplest model is particularly useful for pharmacokinetic analysis of plasma concentration versus time for drugs that are very rapidly distributed in the body. The two-compartment model consists of a central component, which includes the plasma and other highly perfused organs, connected to a peripheral or tissue compartment.

TABLE 3.8 Dimensions of Measurement and Typical Dimensions for Important Pharmacokinetics Parameters

Pharmacokinetics Parameter	Dimension of Measurement	Examples of Typical Dimensions
Mass of drug (e.g., mass of drug in blood, mass of drug at the site of administration, mass of drug in urine, and mass of metabolite in urine)	Amount	mg, μg , etc.
Plasma and/or serum concentration	Amount per unit volume	$\text{mg}\cdot\text{L}^{-1}$, $\mu\text{g}\cdot\text{mL}^{-1}$, mg/L, $\mu\text{g}/\text{mL}$, etc.
Elimination half-life	Time	hr, min, etc.
All first-order rate constants (e.g., elimination, distribution, disposition, absorption, and intercompartmental transfer rate constants)	Reciprocal of time	hr^{-1} , min^{-1} , etc.
All the apparent volumes of drug distribution (e.g., V for one compartment model, and V_c and V_b for two-compartment model)	Volume or volume per unit weight of a subject	L, mL, etc. or $\text{L}\cdot\text{kg}^{-1}$, $\text{mL}\cdot\text{kg}^{-1}$, L/kg, mL/kg, etc.
Zero-order rate constant	Amount, concentration, or percentage per unit time	$\text{mg}\cdot\text{hr}^{-1}$, $\mu\text{g}\cdot\text{mL}\cdot\text{hr}^{-1}$, $\%\cdot\text{hr}^{-1}$, mg/hr, $\mu\text{g}/\text{hr}$, $\%\text{hr}$, etc.
All rates (zero as well as first order; e.g., absorption rate, elimination rate, and excretion rate)	Amount per time or concentration per time	$\text{mg}\cdot\text{hr}^{-1}$, $\mu\text{g}\cdot\text{hr}^{-1}$, $\mu\text{g}\cdot\text{mL}^{-1}\cdot\text{hr}^{-1}$, mg/hr, $\mu\text{g}/\text{hr}$, $\mu\text{g}/\text{mL}/\text{hr}$, etc.
All clearances (e.g., systemic, renal, metabolic, and creatinine clearance)	Volume per unit time or volume per unit time per unit weight	$\text{mL}\cdot\text{hr}^{-1}$, $\text{L}\cdot\text{hr}^{-1}$, $\text{mL}\cdot\text{hr}^{-1}\cdot\text{kg}^{-1}$, $\text{L}\cdot\text{hr}^{-1}\cdot\text{kg}^{-1}$, mL/hr, L/hr, mL/hr/kg, etc.
Peak time	Time	hr, min, etc.
Dose administered (e.g., single intravenous or oral dose, loading dose, and maintenance dose)	Amount or amount per unit weight	mg, μg , $\text{mg}\cdot\text{kg}^{-1}$, mg/kg etc.
Peak concentration (e.g., for a single dose or multiple doses) etc.	Mass per unit volume	$\text{mg}\cdot\text{L}^{-1}$, $\text{ng}\cdot\text{mL}^{-1}$, mg/L, ng/mL
Area under the plasma concentration–time curve (AUC)	Mass per volume $\times$ time	$\text{mg}\cdot\text{L}^{-1}\cdot\text{hr}$, $\text{ng}\cdot\text{mL}^{-1}\cdot\text{hr}$, mg/L/hr, ng/mL/hr, etc.
Dosing interval	Time	hr, min., etc.
Absolute or relative bioavailability	Dimensionless	
Blood flow rate	Volume per unit time	$\text{L}\cdot\text{hr}^{-1}$, $\text{mL}\cdot\text{hr}^{-1}$, L/hr, mL/hr, etc.
Extraction ratios	Dimensionless	
Infusion rate	Mass per unit time or mass per unit time per unit weight	$\text{mg}\cdot\text{kg}^{-1}$, $\mu\text{g}\cdot\text{kg}^{-1}$, $\mu\text{g}\cdot\text{kg}^{-1}\cdot\text{hr}^{-1}$, mg/kg, $\mu\text{g}/\text{kg}$, $\mu\text{g}/\text{kg}/\text{hr}$, etc.

Each compartment can be considered to include a group of tissues, fluids, or parts of organs. A somewhat more complex model, illustrated in Figure 3.24, is the three-compartment model, which consists of a central

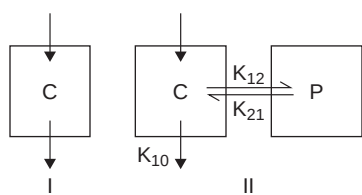


FIGURE 3.23 Schematic representation of the one-compartment (I) and two-compartment (II) models commonly used in pharmacokinetics. Arrows represent transfer of a drug because of the first-order process. C, central compartment (plasma, highly perfused organs); P, peripheral (tissue) compartment.

compartment connected to more than one peripheral compartment that differ in their relative accessibility to a drug. This model is chosen if the available data warrant such a model.

The selection of a model depends greatly on the site and tissue being sampled, the frequency of sampling collection, and the ultimate goals of the study. The general operating rule in selecting a model for pharmacokinetic analysis of plasma concentration versus time data is to postulate the minimum number of compartments necessary to accurately describe the pharmacokinetics of a given drug. An approach to selecting the minimum number of compartments should be performed, unless the experimental evidence dictates that such limited selection can lead to errors in estimating the pharmacokinetic parameters of drugs and in the use of equations to predict blood levels of drugs.

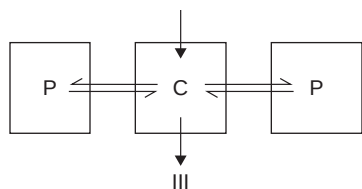


FIGURE 3.24 Schematic representation of a three-compartment (III) model commonly used in pharmacokinetics. Arrows represent transfer of a drug because of the first-order process. C, central compartment; P, peripheral compartment.

Linear and Nonlinear Pharmacokinetics

Linear Pharmacokinetics

Many processes in pharmacokinetics can be accurately described by a first-order process. This means that the rate of drug metabolism, the rate of transfer of a drug between compartments, and the rate of absorption and elimination of drugs from the body are directly proportional to the size of the dose administered. It is also true that passive diffusion is responsible for the transfer of a drug in the body and that a directly proportional relationship exists between the administered dose and the resulting drug concentration in the body. This dose proportionality is often used as an indicator of linear pharmacokinetics. It is important, however, to recognize that the pharmacokinetic parameters, such as elimination half-life and elimination rate constant, are independent of the size of the administered dose. Therefore, linear pharmacokinetics is regarded as dose-independent kinetics.

Nonlinear Pharmacokinetics

The rate of elimination of drugs (e.g., ethanol, salicylate, phenytoin) by metabolism and by other transfer processes, such as protein carrier systems, is not removed from the body by a first-order process, which means that the elimination rate is not proportional to the concentration of drug or to the dose administered, but follows zero-order kinetics—that is, the rate of change of drug concentration is independent of the drug concentration. A constant amount of a drug, rather than the constant percentage of the remaining amount of drug, is eliminated per unit time (i.e., mg/min or $\mu\text{g/mL/min}$).

The most frequently reported reason for the use of nonlinear kinetics is that metabolism and transfer processes require protein carrier systems. These systems are specific with respect to substrates that have finite capacities. The kinetics of these processes are described by the Michaelis-Menten equation (Eqs. 3.3–3.5).

Nonlinear, or dose-dependent, elimination kinetics can also be the result of effects other than the limited capacity of metabolism or elimination processes. If a drug is partly reabsorbed from the renal tubules by a recycling process with limited capacity, then the elimination of large doses proceeds relatively more rapidly than the elimination of smaller doses. Similarly, lesser binding of

drugs to plasma constituents or tissues at higher dosing can result in relatively more rapid drug elimination than is observed at lower drug concentrations.

Evidence suggests that some drug metabolites can inhibit their own formation. This process of product inhibition can also cause dose-dependent effects, with large doses being eliminated relatively more slowly than small doses. The rate of decline of a drug concentration in the postdistribution phase, at any given level of a drug in the body, will be independent of the dose in the case of simple Michaelis-Menten kinetics. In cases of product inhibition, this rate tends to decrease with increasing doses.

Intravascular Administration

IV BOLUS ADMINISTRATION AND ONE-COMPARTMENT MODEL After the administration by IV injection, if a drug distributes very rapidly in the body, this confers on the body the characteristics of a one-compartment model, and if the drug elimination from the body can be described by a first-order process, then a plot of the logarithm of plasma drug concentration as a function of time yields a straight line, as shown in Figure 3.25.

The equation responsible for describing plasma drug concentration against time is as follows:

$$3.29 \quad C_p = (C_p)_0 e^{-Kt}$$

where C_p is the plasma drug concentration at time t ; $(C_p)_0$ is the initial plasma drug concentration (i.e., $t = 0$) immediately after the injection; K is the first-order

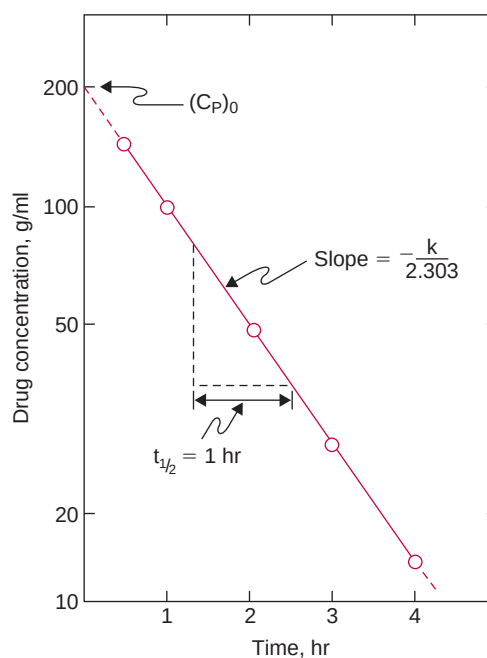


FIGURE 3.25 Schematic representation of plasma concentration of drug in the body as a function of time following rapid intravenous injection.

elimination rate constant; and t represents time. Equation 3.29 is also written as follows:

$$3.30 \quad \ln C_p = \ln (C_p)_0 - Kt$$

or

$$3.31 \quad \log C_p = \log (C_p)_0 - \frac{Kt}{2.303}$$

The initial plasma drug concentration, $(C_p)_0$, can be obtained by extrapolation of the line (Fig. 3.24) to $t = 0$ or the y-intercept of plasma drug concentration versus time plot. Figure 3.24 shows that a plot of the log of plasma drug concentration versus time will be linear under the stated condition.

Three primary factors determine the plasma concentration of the administered drug: 1) the route of administration; 2) the uptake of drug by body tissues; and 3) the elimination of the drug from the body. In the case of IV administration, because the drug is introduced directly into the blood, there is no delay as a result of the absence of the drug absorption process. The drug plasma level, however, depends on the size of the dose, and the maximum plasma drug concentration occurs immediately after completion of the dose administration.

Elimination Rate Constant and Half-Life

The half-life of a drug is a major factor in determining the dosing frequency, which is a function of the drug's clearance (Cl) and apparent volume of distribution (V). The relationship of a drug's half-life to the prediction of human pharmacokinetics and dosing regimen is shown in Figure 3.26 (54–56). Drugs with short half-lives are more likely to require frequent administration, whereas those with long half-lives tend to require dosing once daily. The two pharmacokinetic parameters that determine the half-life of a drug are clearance and volume of distribution. Dosing regimen is also linked to other factors, such as the drug pharmacodynamics and the drug concentrations associated with side effects versus those minimally required for efficacy.

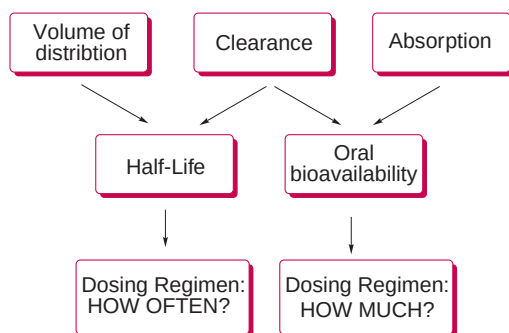


FIGURE 3.26 The relationship of volume of distribution to the prediction of human kinetics.

The elimination rate constant (K) can be determined from the slope of the straight line as follows:

$$3.32 \quad (\text{slope}) (2.303) = -K$$

However, it is much easier to determine the elimination rate constant by making use of the following relationship:

$$3.33 \quad K = \frac{0.693}{t_{1/2}}$$

where $t_{1/2}$ is the time required for any drug concentration to decrease by half (i.e., 50%) and is also known as the biologic or elimination half-life. The elimination half-life is a pharmacokinetic property of a drug, and it is independent of the size of the administered dose when the administered drug exhibits the characteristics of a first-order process.

The elimination half-life ($t_{1/2}$) and the elimination rate constant (K) of a drug also have an important role in determining the plasma concentration of a drug at a given time. For instance, a drug with a short elimination half-life will be eliminated from the body much more quickly than a drug with a longer elimination half-life. These two parameters of a drug, therefore, become important in maintaining the desired drug blood levels in the body. In essence, these two parameters provide a quantifiable index of the presence of a drug in the body.

The process of drug elimination includes metabolism and excretion, which begin almost immediately when blood circulation distributes some of the drug to organs capable of metabolizing the drug or excreting it from the body. Among the organs of drug elimination, the liver is the principal site of metabolism, and the kidney is primarily responsible for the excretion of unmetabolized drugs and their metabolites. Other organs, however, can also participate in the elimination of selected drugs.

The consequence of metabolism of the drug to metabolite depends on the pharmacologic activity of the individual metabolite(s). Metabolites can be active or completely inactive. Active metabolites can be more or less potent than the parent drug, and they can exhibit similar or dissimilar action. The kinetics of distribution and elimination of metabolites can differ from those of the parent drug, because each metabolite differs from the parent drug in its physicochemical properties as a result of functional group additions or changes.

The elimination rate constant (K) is the sum of the individual rate constants that characterize the elimination of a drug from the body and the form of metabolite or unchanged drug. Thus:

$$3.34 \quad K = K_u + K_m$$

where K_u and K_m represent the first-order rate constants associated with excretion and the form of metabolite or unchanged drug, respectively, removed from the blood.

Apparent Volume of Distribution

Plasma or serum samples, collected immediately following the administration of equal doses (i.e., X_0) of two different drugs, can exhibit large differences in the plasma drug concentration. This is because drug distribution in the body is largely a function of the drug's physicochemical properties and, therefore, of the chemical structure of a drug. The sole purpose of this parameter (V or V_d) is to relate the amount (mg) and the concentration (L^{-1}) of drug in the body at a given time. Therefore, it is important to recognize that knowledge of this parameter is essential for the determination of the dose of a drug required to attain the desired initial plasma concentration. The relationship of V to the prediction of human pharmacokinetics and dosing regimen is shown in Figure 3.26. Because it is not a true physical volume but, rather, a mathematical term that describes the behavior of a drug in the body with regard to its degree of partitioning between the plasma compartment and the remainder of the body, V is called "apparent." Drug partitioning into tissues represents a complex combination of physicochemical (lipophilicity, pK_a , plasma protein binding) and physiologic parameters. Drugs that are equally bound to plasma proteins can yield different V values, because the drug with the greater tissue binding will yield the larger V . However, drugs with equal tissue binding can differ in V , with the compounds having the greater plasma protein binding yielding the smaller volume of distribution.

This relationship is shown in Equation 3.35:

$$(X)_t = V(C_p)_t \quad (3.35)$$

where $(X)_t$ and $(C_p)_t$ are the amount of the drug and its plasma concentration, respectively, at a given time. V is determined by rearranging Equation 3.35 as follows:

$$V = \frac{(X)_0}{(C_p)_t} = \frac{\text{Dose } (X_0)}{(C_p)_0} \quad (3.36)$$

where $(X)_0$ is the administered dose of the drug and $(C_p)_0$ is its initial plasma concentration.

V of a drug is a property of the drug rather than of the biologic system and describes the extent to which a particular drug is distributed in the body tissues. The magnitude of V as a rule does not correspond to plasma, extracellular, or total body volume space but can vary from a few liters (7 to 10 L) to several hundred liters (≥ 200 L) in a 70-kg subject. The higher the value of V , the greater is the extent to which the drug is distributed in the body tissues, organs, or both. Furthermore, because body tissues, biologic membranes, and organs are lipophilic in nature, the value of V is correlated with the lipophilicity of a drug, which in turn is influenced by the drug's chemical structure, pK_a , and protein binding. The more lipophilic the drug, the greater is the value of V and the smaller is the initial plasma concentration (assuming the administered doses of drugs are identical). Conversely, if the drug is hydrophilic, the drug will penetrate to a lesser extent into tissue; consequently, the aqueous plasma concentration will be

higher and V will be smaller. Therefore, the value of V is influenced by the lipophilicity of the drug molecule.

In most situations, V is independent of the drug concentration, because doubling the amount of a drug in the body usually results in doubling of its plasma concentration (linear pharmacokinetics). V is constant for a drug and remains independent of the dose administered and disease state. Other factors that can influence the blood composition, total body fluid, and the permeability characteristic of tissue, can bring about changes in the value of V . Furthermore, because V reflects the extent to which a drug will penetrate into tissue, alteration in the permeability characteristics of tissue will alter V . In addition, V of a drug can vary in infants, adults, and the geriatric population.

Many acidic drugs, including salicylates, sulfonamides, penicillins, and anticoagulants, are either highly bound to plasma proteins or are too water soluble to enter intracellular fluid and cannot penetrate tissues in a significant amount. Therefore, these drugs have low volumes of distribution and low tissue-to-plasma concentration ratios. A given dose of these drugs will yield relatively higher plasma concentrations. It can be implied that analytical problems in the determination of drug concentration are minimized or do not exist. Basic drugs, including tricyclic antidepressants and antihistamines, are extensively bound to extracellular tissues and are absorbed by adipose tissues. The V for these drugs is large—often larger than the total body space. For example, the value of V for amphetamine is approximately 200 L (3 L/kg). The relatively small doses and V for amphetamine produce a low plasma concentration, which makes its quantitative detection in plasma a difficult task.

Theoretical limits for V could be as low as 5 to 7 L (equivalent to the volume of the body fluid if the drug totally fails to penetrate the tissues or is extremely hydrophilic) or as high 200 L or greater. Because of differences in the magnitude of the V of a drug, a given dose of a drug with a relatively high V will provide low initial drug concentrations, and vice versa (Eq. 3.35).

Clearance

One of the most important pharmacokinetic properties of a drug is clearance (Cl). In pharmacokinetic terms, clearance refers to the hypothetical volume of distribution from which the drug is entirely removed or cleared in unit time (mL/min or $mL \cdot min^{-1}$). In other words, Cl is an index of drug elimination. Cl is a function both of the intrinsic ability of certain organs, such as kidneys and liver, to excrete or metabolize a drug and of the blood flow rate to these organs. The concept of Cl is illustrated by assuming elimination in a single organ as depicted in Figure 3.27.

Under the conditions described in Figure 4.27, the venous concentration of drug (C_v) will always be less than the arterial concentration (C_a) because of the drug being eliminated or excreted during the passage of blood through the organ. The rate at which the drug enters the organ is the product of blood flow rate (Q) and the arterial concentration. The rate at which the drug leaves the

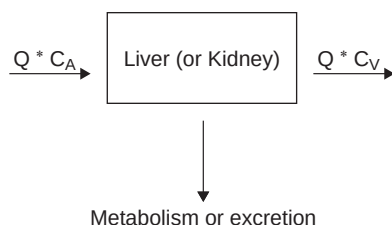


FIGURE 3.27 Schematic representation of drug elimination by the liver (or kidney). $Q \cdot C_A$ is the rate at which the drug enters the liver (or kidney), and $Q \cdot C_v$ is the rate at which the drug leaves the liver (or kidney). The venous drug concentration (C_v) is less than the arterial drug concentration (C_A).

organ, however, is equal to the product of blood flow rate and venous concentration:

$$3.37 \quad \text{The rate in} = Q \cdot C_A$$

$$3.38 \quad \text{The rate out} = Q \cdot C_v$$

The difference between the rate in and rate out is the rate of elimination of a drug by the organ:

$$3.39 \quad \text{rate of elimination} = Q(C_A - C_v)$$

The dimensionless ratio of elimination rate (K) to the rate at which a drug enters the organ (QC_A) is defined as the extraction ratio (ER) and is obtained as follows:

$$3.40 \quad \text{ER} = \frac{Q(C_A - C_v)}{QC_A} = \frac{(C_A - C_v)}{C_A}$$

ER of a drug ranges from 0 to 1 depending on how well the organ eliminates or excretes the drug from the blood flowing through it. If an organ does not eliminate the drug, then $C_A = C_v$, and the ER is 0 (low extraction ratio). If, however, the organ avidly removes the drug so that $C_v \approx 0$, then ER is 1 (high extraction ratio). If liver is the organ responsible for metabolizing the drug, then the extraction ratio is described by using the notation, E_H .

Using the extraction ratio number (i.e., ER or E_H), the drugs have been classified as having a low (ER < 0.3), intermediate (ER = 0.3 to 0.7), or high (ER > 0.7) ER. Table 3.9 lists the representative drugs with their hepatic or extraction ratios. The influence of blood flow and intrinsic clearance of an organ on the clearance of a drug is determined by the ER of the drug.

Cl can also be viewed as a proportionality constant relating the elimination rate of a drug to its plasma concentrations at a given time and is expressed as:

$$3.41 \quad (Cl) = \frac{\text{rate of elimination}}{\bar{C}_p}$$

where $\bar{C}_p$ is the average plasma concentration of a drug at a time that corresponds to the rate of elimination.

It follows from an earlier equation (Eq. 3.41) that:

$$3.42 \quad (Cl) = Q \cdot \text{ER}$$

where Q and ER have been previously defined and, because the drug elimination follows a first-order process, Cl is independent of the drug concentrations or the dose administered.

The total-body clearance of a drug from the blood is equal to the ratio of the overall elimination rate to drug concentration (Eq. 3.41), where the overall elimination rate is comprised of the sum of the elimination processes occurring in all organs and the removal of a drug in all its forms. Therefore, the overall clearance, $(Cl)_s$, represents the renal clearance (i.e., unchanged form of a drug) and

TABLE 3.9 Hepatic and Renal Extraction Ratios of Selected Drugs and Metabolites

Low (< 0.3)	Intermediate (0.3–0.7)	High (> 0.7)
Hepatic extraction ^a		
Carbamazepine	Aspirin	Alprenolol
Diazepam	Quinidine	Arabinosyl-cytosine
Digitoxin	Codeine	Desipramine
Indomethacin	Nortriptyline	Doxepin
Phenobarbital		Isoproterenol
Phenytoin		Lidocaine
Procainamide		Meperidine
Salicylic Acid		Morphine
Theophylline		Nitroglycerin
Tolbutamide		Pentazocine
Valproic Acid		Propoxyphene
Warfarin		Propranolol
Renal extraction ^a		
Atenolol	Cimetidine	(Many) Glucuronides
Cefazolin	Cephalothin	Hippurates
Chlorpropamide	Procainamide	(Some) Penicillins
Digoxin	(Some) Penicillins	(Many) Sulfates
Furosemide		
Gentamicin		
Lithium		
Phenobarbital		
Sulfisoxazole		
Tetracycline		

^aAt least 30% of the drug is eliminated by this route.

the metabolic clearance (i.e., removal of a drug as metabolic by kidney). It is also useful to keep in mind that the Cl can be expressed as the product of the apparent volume of distribution (V) and the elimination rate constant (K) for drugs that exhibit characteristics of a one-compartment model. Thus:

$$3.43 \quad (Cl)_s = VK$$

Hepatic Clearance

Although metabolism can take place in many organs, the liver frequently has the greater metabolic capacity and, therefore, has been the most thoroughly studied. The most direct quantitative measure of the liver's ability to eliminate a drug is hepatic clearance, $(Cl)_H$, which includes biliary excretion clearance and hepatic metabolic clearance:

$$3.44 \quad (Cl)_H = Q_H E_H$$

where Q_H is the sum of the hepatic portal and hepatic arterial blood flow rates, the values of which are 1,050 and 300 mL/min, respectively.

Under conditions of normal body functions, the pharmacokinetic behavior of most drugs can be established within reasonable limits, and optimal dosage regimens can be designed using the observed values of the pharmacokinetic parameters of the drug. However, when renal function is compromised as a result of acute or chronic renal diseases or the patient's age, drugs that are predominantly eliminated through the kidneys are likely to be retained in the body for a longer duration and accumulate to the extent of providing toxic drug levels with repeated dosing. If the drug is converted to a metabolite, the accumulation of active metabolite can also lead to a toxic effect, and although most metabolites are inactive, their accumulation with repeated dosing can produce toxic reactions by displacement of the parent drug from plasma protein and by inhibiting further drug metabolism.

Renal failure can result from a variety of pathologic conditions. If renal impairment is rapid in onset and short in duration, then renal failure is described as acute. The primary cause of acute renal failure can be prerenal (i.e., acute congestive heart failure or shock), intrarenal (i.e., acute tubular necrosis), or postrenal (i.e., hypercalcemia). Although renal failure is generally reversible, complete restoration of renal function can take 6 to 12 months.

Chronic renal failure is almost always caused by intrinsic renal diseases and is characterized by slow, progressive development. Unlike the acute condition, chronic renal impairment is generally irreversible. The degree or loss of kidney functional capacity in the chronic condition is best described in terms of the intact "nephron" hypothesis, in which the diseased kidney is comprised of nephrons that are essentially nonfunctional because of pathologic conditions along with normal nephrons. Progressive

renal impairment is the result of an increasing fraction of nonfunctional nephrons.

The prolonged and progressive nature of chronic renal failure is of particular concern in older patients, who can require a variety of medications, both for their renal condition and for other unrelated conditions. The inability of these patients to adequately excrete drugs and drug metabolites and the influence of their uremic conditions on the functions of other physiologic systems require careful adjustments of drug dosage to obtain accurate and adequate blood levels without increased toxicity.

Drugs are cleared by kidneys as a result of passive filtration through the glomeruli or by active transport in the kidney tubule. Once in the nephrons, drugs (and their unconjugated metabolites) can also be reabsorbed into the circulation. The glomerular filtration rate is measured using any compound that is filtered by glomeruli and not reabsorbed. Although exogenous compounds, such as urea and inulin, are used for this purpose, the relative ease of using endogenous creatinine has made this the method of universal choice. In principle, the following equation determines the relationship between the creatinine clearance, $(Cl)_{cr}$, the serum creatinine concentration, $(\bar{C}_s)_{cr}$ and creatinine excretion rate, $(dX_p/dt)_T \cdot cr$:

$$3.45 \quad (Cl)_{cr} = \frac{(dX_p/dt)_T \cdot cr}{(\bar{C}_s)_{cr}}$$

Serum creatinine concentration is constant unless there is a change in the rate of production of creatinine in the body or creatinine clearance. The creatinine clearance in normal kidneys is approximately 110 to 130 mL/min. This value declines with progressive renal impairment, and it drops to zero with severe renal impairment. Creatinine clearance values of 20 to 30 mL/min signify moderate renal impairment; values of less than 10 mL/min signify severe renal impairment. Creatinine is poorly secreted and not subject to tubular reabsorption; therefore, its clearance is a useful measure of the glomerular filtration rate. Although creatinine clearance tells us about only one aspect of renal function (i.e., filtration), it is an excellent indicator for assessing the severity of renal impairment.

The extent to which decreased renal function influences drug elimination is a function of the percentage of circulating drug being cleared by the kidneys. From the literature, the influence of renal impairment on the elimination half-life of a drug will clearly be a direct function of the percentage of the drug cleared through the kidneys. If the elimination half-life of a drug that is cleared essentially unchanged via the kidneys is plotted against the endogenous creatinine clearance (Fig. 3.28), the result will be a hyperbola.

IV Bolus Administration (Two-Compartment Model)

After the administration of a drug IV, a finite amount of time has to pass before distribution equilibrium is attained

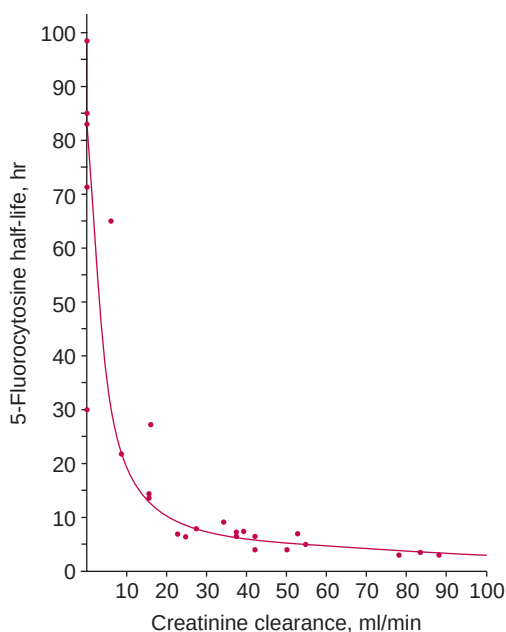


FIGURE 3.28 Curvilinear relationship between the elimination half-life of 5-fluorocytosine and renal function (creatinine clearance). (45).

in the body. During this distribution phase, the plasma drug concentration will decline more rapidly than during the postdistribution phase, as shown in Figure 3.29. There are three possible types of two-compartment models, which differ between whether the elimination of the drug occurs from the central compartment, the peripheral compartment, or both. The three types of two-compartment models are, mathematically, indistinguishable on the basis of available concentration data. The type of two-compartment model illustrated in Figure 3.23 is most

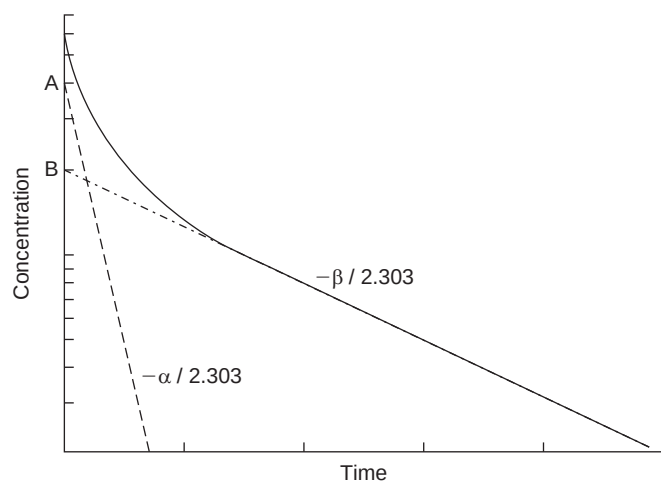


FIGURE 3.29 Semilogarithmic plot of drug concentration in the plasma against time following administration of a rapid intravenous injection when the body can be represented as a two-compartment open model. The dashed line is obtained by “feathering” the curve.

often used to describe the pharmacokinetics of drugs. In this model, it is assumed that drug elimination from a two-compartment model occurs exclusively from the central compartment, because the sites of metabolism and excretion (i.e., liver and kidney) are well perfused with blood and, therefore, presumably rapidly accessible to drug in the systemic circulation. Whether this distribution phase is apparent depends on the early collection of blood samples. A distribution phase can last for only a few minutes or for several hours.

A semilogarithmic plot of plasma drug concentration as a function of time (Fig. 3.29) after rapid IV injection of a drug can often be resolved into two linear components, which can be done graphically by using the residual, or “feathering,” method, as shown in Figure 3.29, in which the slopes of rapid and slow disposition phases will permit the determination of α and β , respectively, in Equation 3.46. The intercepts on the concentration axis are designated A and B. The entire plasma concentration–time curve can be described by the following equation:

$$3.46 \quad C_p = Ae^{-\alpha t} + Be^{-\beta t}$$

where α and β are the first-order distribution and disposition rate constants, respectively. A biexponential decline in the plasma drug concentration justifies, mathematically, the representation of the body as a two-compartment model.

Extravascular Route of Administration

When a drug is administered by extravascular routes (such as orally), absorption is requisite for a drug to reach the general circulation. Absorption is defined as a process whereby a drug proceeds from the site of administration to the site of measurement within the body, generally blood, plasma, or serum. Figure 3.30 represents the passage of a drug through the gastrointestinal tract into the general circulation.

When a drug is orally administered, there are several possible sites for drug loss. One such site is the gastrointestinal lumen, where drug decomposition can occur. If it is assumed that the drug survives destruction in the gut lumen and is metabolized by enzymes as it passes through the membrane of the gastrointestinal tract, then even though the drug leaves the site of administration, it is considered not to be systemically absorbed. Indeed, loss at any site in the gastrointestinal tract before reaching the site of measurement can contribute to a decrease in the systemic absorption of the drug. The requirement for an orally administered drug to pass through the gastrointestinal tract makes the extent of absorption not always complete. The loss of a drug as it passes for the first time through gastrointestinal membrane and the lining, during absorption, is known as the first-pass effect. Figure 3.31 represents the time course of a drug and metabolite at each site in the body.

The rate or the change in the amount of drug in the body (dX/dt) following administration of a drug by an

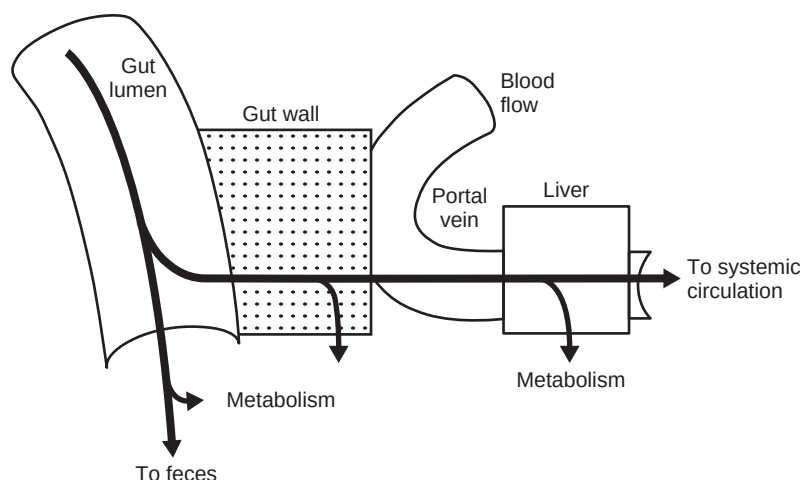


FIGURE 3.30 After oral administration, a drug must pass sequentially through the gut lumen, the gut wall, and then the liver before reaching the general circulation. Metabolism can occur in the lumen before absorption, in the gut wall during absorption, or in the liver after absorption and before reaching the systemic circulation. (From Rowland M, Tozer T. *Clinical Pharmacokinetics: Concepts and Application*, 2nd Ed. Philadelphia: Lea and Febiger, 1989, with permission.)

extravascular route is a function of both the absorption rate ($K_a X_a$) and the elimination rate (KX):

$$3.47 \quad \frac{dX}{dt} = K_a X_a - KX$$

where $K_a X_a$ is the first-order absorption rate, KX is the first-order elimination rate, and K_a and K are the first-order absorption and elimination rate constants, respectively.

When the absorption rate is greater than the elimination rate (i.e., $K_a X_a > KX$), the amount of drug in the body and plasma drug concentration increase with time. Conversely, when the amount of drug remaining at the absorption site (X_a) is sufficiently small, the elimination

rate exceeds the absorption rate (i.e., $KX > K_a X_a$); therefore, the amount of drug in the body and the drug concentration in the plasma decrease with time.

The peak plasma concentration [$(C_p)_{\max}$] after drug administration occurs at the moment when the absorption rate equals the elimination rate (i.e., $K_a X_a = KX$). The faster the drug is absorbed, the higher the $(C_p)_{\max}$ and the shorter the time required following administration of a dose to observe the $(C_p)_{\max}$. Integration of Equation 3.47 from $t = 0$ to $t = t^*$ and converting the amount to the concentration results in the following equation:

$$3.48 \quad C_p = \frac{K_a F(X_a)_0}{V(K_a - K)} (e^{-Kt} - e^{-K_a t})$$

where $(X_a)_0$ is the administered dose and F is the fraction of the administered dose that is absorbed and available to reach the general circulation. Equation 3.48 is often used to determine plasma concentration after administration of a drug by an extravascular route when the administered drug manifests the characteristics of a one-compartment model.

The absorption rate constant (K_a) of a drug frequently is larger than the elimination rate constant (K). Under such a condition, at some time after drug administration, the value of the term e^{-Kt} in Equation 3.48 approaches zero, indicating that no more drug is available for absorption, and Equation 3.48 simplifies to:

$$3.49 \quad C_p = \frac{K_a F(X_a)_0}{V(K_a - K)} (e^{-Kt})$$

$$3.50 \quad C_p = \text{Intercept} (e^{-Kt})$$

$$3.51 \quad \log C_p = \log \text{Intercept} - \frac{Kt}{2.303}$$

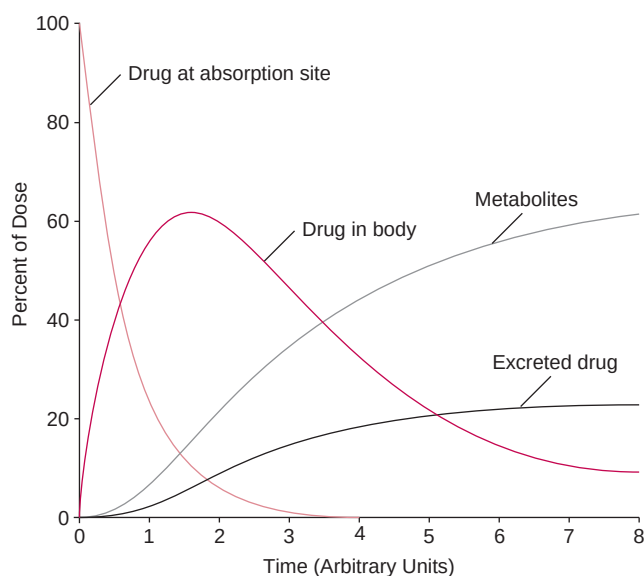


FIGURE 3.31 Time course of a drug at each site following an extravascular administration.

When the absorption is complete, the term $X_a K_a$ disappears from Equation 3.47 and the equation is reduced to:

$$3.52 \quad -\frac{dX}{dt} = KX$$

During the postabsorption phase, the decline in the plasma concentration with time follows first-order kinetics. A typical plot of plasma concentration versus time is shown in Figure 3.32, where the intercept of the extrapolated line (I^*) is a complex function of K_a and K , respectively, as well as the dose or amount absorbed, $F(X_a)_0$, and the apparent volume of distribution (V). It is, however, incorrect to assume that the intercept approximates the ratio of dose over the V unless the drug is rapidly and completely absorbed, which rarely occurs.

Importance of Absorption Rate

The influence of absorption on the drug concentration time profile is shown in Figure 3.33. The administration of an equal dose of a drug in three different dosage forms or by three different extravascular routes or three different formulations results in threefold the drug concentration in the plasma. The faster the drug is absorbed (i.e., $K_a \gg K$), the greater is $(C_p)_{\max}$ and the shorter the time required to achieve peak plasma drug concentration.

Many drugs do not exhibit demonstrable pharmacologic effects or do not elicit a desired degree of pharmacologic response unless a minimum concentration is reached at the site of an action and, therefore, a minimum therapeutic concentration in the plasma. Thus, the absorption rate of a drug can affect the clinical response if it fails to yield the minimum effective concentration. As evident in

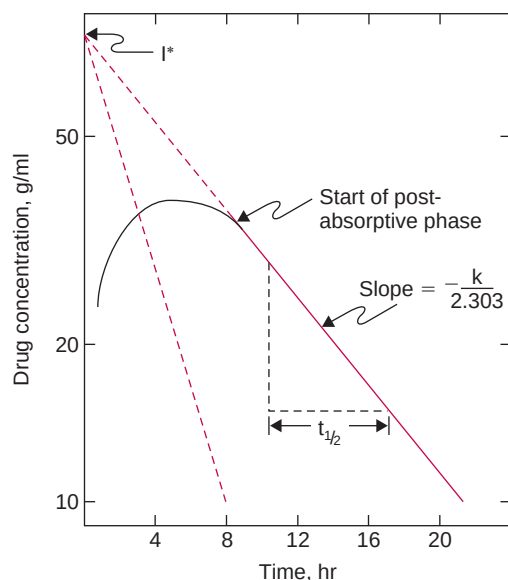


FIGURE 3.32 Typical semilogarithmic plot of drug concentration versus time profile in plasma following the administration of a drug by an extravascular route. The dashed line represents the “feathered line” used to obtain the absorption rate constant (K_a).

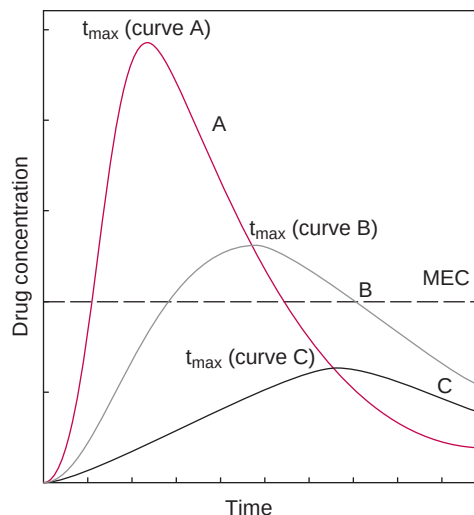


FIGURE 3.33 A plasma concentration versus time profile illustrating the influence of absorption rate constant (K_a) on the rate of absorption as reflected in the peak time ($t_{\max}$) and peak plasma concentration $(C_p)_{\max}$.

Figure 3.33, the more rapid the absorption of a drug is, the faster its onset of response (i.e., curve A). When the drug is absorbed rather slowly (curve C), the minimum effective concentration is just barely attained. The intensity of maximum pharmacologic effects is a function of the drug concentration. The data presented in Figure 3.33 suggest that the administered dose of a drug in curve A can produce a more intense response than observed in curves B and C.

The peak plasma drug concentration is always lower following administration of a drug by the extravascular route compared with its initial plasma concentration following administration of an identical dose by IV solution. In the former, at peak time, some drugs can still remain at the absorption site and some has been eliminated, whereas the entire dose is in the body immediately following the IV administration.

The delay between drug administration and a drug reaching the general circulation can be of particular importance when a rapid onset of effect is desired. This delay is termed “lag time,” and it is anywhere between a few minutes to many hours. Lag time generally is attributed to the slow and poor absorption of the drug, either because of slow disintegration and dissolution of the drug from the dosage form or because of slow removal of the coating material from coated tablets.

Determination of Peak Time

The determination of peak time ($t_{\max}$) can be achieved by using the following equation:

$$3.53 \quad t_{\max} = \frac{\ln (K_a / K)}{(K_a - K)}$$

where K_a and K are the first-order absorption and elimination rate constants, respectively. Equation 3.53 shows

that the $t_{\max}$ is a function only of the relative magnitude of the absorption and elimination rate constants. As the rate of absorption decreases (i.e., smaller K_a value), the $t_{\max}$ will be higher, as shown in Figure 3.33, progressing from curve C to curve A.

The rate of drug absorption varies when the extravascular route is changed, when the formulation of a drug is changed, or when the dosage form is changed. These changes will be reflected in different $t_{\max}$ for the same dose of a drug. However, $t_{\max}$ will be unaffected by a mere change in the size of the administered dose. In many disease states, the impairment in the renal function can affect the elimination rate constant, thereby producing a change in the $t_{\max}$.

Determination of Peak Plasma Concentrations

The maximum peak plasma drug concentration, $(C_p)_{\max}$, in the body occurs at time $t_{\max}$, which is described by substituting $t_{\max}$ for time t in Equation 3.48:

$$3.54 \quad (C_p)_{\max} = \frac{K_a F(X_a)_0}{V(K_a - K)} (e^{-K t_{\max}} - e^{-K_a t_{\max}})$$

Equation 3.54 is further simplified into:

$$3.55 \quad (C_p)_{\max} = \text{Intercept} (e^{-K t_{\max}} - e^{-K_a t_{\max}})$$

where the intercept of the plasma drug concentration versus time is equal to $K_a F(X_a)_0 / V(K_a - K)$, as previously described in Equations 3.49 through 3.51.

Peak Time and Peak Plasma Concentration

The most important property of any oral dosage form, intended for a systemic therapeutic effect, is the ability of the oral dosage form to deliver the active ingredient to the bloodstream in an amount and at a rate sufficient to produce the desired therapeutic effect. This property of a dosage form is described as bioavailability.

Bioavailability, essentially, captures two features: how fast the drug enters the systemic circulation (rate of absorption) and how much of the administered dose of a drug enters the systemic circulation (extent of absorption). Assuming that the therapeutic effect of a drug is directly related to the drug concentration in a patient's blood, these two properties of an oral dosage form are, in principle, important in correlating the therapeutic response to a drug's dose. The onset of response is directly linked to the rate of absorption, whereas the time-dependent extent of response is connected to the extent of adsorption.

The bioavailability of a drug, following the administration of an oral dose, can vary because of either patient-related or dosage form-related factors. Patient-related factors can include the nature and timing of meals, age, disease, and gastrointestinal physiology. The dosage form- or drug-related factors can include the chemical form of the drug (such as salt vs. free drug), its physical properties (such as crystal structure and particle size),

and an array of formulation (excipients) and manufacturing (processing) variables.

Therefore, the bioavailability of a drug is of clinical, academic, and regulatory interest. The regulatory interest includes agencies such as the U.S. Food and Drug Administration (FDA) and its global partners, which approve safe and effective drug products. Applications from manufacturers seeking regulatory approval of a new drug (New Drug Application [NDA]) must provide exhaustive information about the drug's pharmacokinetics (ADME). The more pertinent interests in the bioavailability of a drug relate to questions about the absolute extent of absorption (absolute bioavailability), the importance of product formulation changes during the drug development process, the comparability of different oral dosage forms, and whether the products can be administered with meals.

Manufacturers seeking regulatory approval of generic drug products by filing an Abbreviated New Drug Application (ANDA) must provide detailed bioavailability evidence showing comparative performance of their product against the innovator's branded product with respect to the rate of absorption and the extent of absorption. To declare product equivalence, while assessing and evaluating the bioequivalence of two or more chemically or pharmaceutically equivalent drug products, both the rate and extent of absorption of the products are expected to be the same within the statistical tolerance of $\pm 80\%$. In the United States, the FDA has the responsibility to determine how the rate and extent are to be assessed and set standards for what is meant by "significant differences," "same," and "similar." The underlying principle is that generic products should be bioequivalent for the drug or its active metabolite or both to be considered by the FDA for therapeutic equivalence. Therefore, the measurements of the rate and extent of absorption are indicators of therapeutic outcomes or, at least, markers to assess drug product performance.

The extent of drug absorption is generally evaluated by measuring the area under the plasma concentration-time curve (AUC), a well-accepted criterion to assess the extent of absorption. In linear pharmacokinetics, all observed plasma drug concentrations rise and decline in proportion to changes in the extent of absorption. The rate of absorption, however, is assessed by measuring peak time ($t_{\max}$) and peak plasma concentration $[(C_p)_{\max}]$. The rate of absorption is turning out to be the nemesis of bioequivalence testing because there is disagreement among scientists on how best to assess the rate of absorption. $(C_p)_{\max}$ and the time of its occurrence ($t_{\max}$) are considered as universal measures of absorption rate. $(C_p)_{\max}$ depends on the extent of absorption and, for this reason, is often not a reliable measure of rate. Moreover, $(C_p)_{\max}$ is relatively insensitive to the changes in the absorption rate. The measure of $t_{\max}$ has been found to be a relatively sensitive measure of absorption rate. It is, however, a discrete measure that depends on the frequency of the blood sampling in the absorption and elimination phases. Because

rate is not a single number and varies with time, it is often difficult to measure absorption rate precisely.

With limitations and debates on assessing the rate of absorption accurately, $(C_p)_{\max}$ and $t_{\max}$ have an important role characterizing the absorption of a drug following its oral administration. Collectively, both parameters can be used to draw some inferences from the AUC data following the oral administration of a drug. As previously stated, both $t_{\max}$ and $(C_p)_{\max}$ are used by the FDA to assess the comparative bioavailability and bioequivalence for a drug product. $t_{\max}$ can be used to assess the onset of action following oral administration of an identical dose of drug via a different dosage form, different route of administration, or different formulation of the same dosage form. $t_{\max}$ is a reflection of the onset of action of a drug. The shorter the $t_{\max}$, the quicker the onset of action of a drug is likely to be. Differences in onset of action and $t_{\max}$ are routinely observed when the same dose of a drug is administered by different routes or different dosage forms to a subject.

$(C_p)_{\max}$ also has an important role in assessing comparative bioavailability or bioequivalence. Assuming that plasma concentration is a reflection of the pharmacologic effects of a drug, the higher the drug $(C_p)_{\max}$ (within the therapeutic range), the better will be the pharmacologic effect or the intensity of the pharmacologic effect. Therefore, $(C_p)_{\max}$ is also used to determine the route of drug administration and the selection of a formulation.

Bioavailability

The bioavailability of a drug is defined as the rate and extent to which the administered dose of a drug reaches the general circulation. As a rule, rapid and complete absorption of a drug is desirable if it is used for pain, allergy response, insomnia, and other conditions for which a quick onset of action is desired. As previously indicated in Figure 3.33, the more rapid the absorption is, the shorter the onset of action and the greater the intensity of a pharmacologic response. Bioavailability determines the amount of administered dose that reaches the circulation, which is also related to rate of drug clearance, as shown in Figure 3.26.

The efficacy of a single dose is a function of both the rate and the extent of absorption. Thus, for two dosage forms or two oral routes to be comparable with regard to the bioavailability following the administration of a drug, the absorption rate of a drug and the extent to which a drug reaches the general circulation from each dosage form or oral route of administration must be comparable.

The useful estimate of relative absorption rates of a drug from different products, through different routes of administration or different conditions (i.e., with or without food or in the presence of other drugs, etc.), can be made by comparing the magnitude of time of occurrence of $t_{\max}$, $(C_p)_{\max}$, and total area under the plasma concentration–time curve [AUC_0^∞]. The peak time and peak plasma concentration can be determined by using Equations 3.53 and 3.54 or 3.55, respectively, and the

extent of absorption can be determined as described in the following section.

Estimating the Extent of Absorption

The extent of absorption is estimated by determining the total area under the plasma drug concentration–time curve, AUC_0^∞ , or the total amount of an unchanged drug excreted in urine, $(X_u)_\infty$, following the administration of a drug. AUC_0^∞ can be estimated by several methods, such as by using a planimeter, which is an instrument for measuring the area of a planar irregular figure, and by using the cut-and-weight method, which weighs the paper of plasma concentration–time curve. The weight is converted to weight per unit area. The most common method, however, is the application of the trapezoidal rule and equation, when possible. In a single-dose study, we cannot determine the area under the plasma concentration–time curve by using the trapezoidal rule alone. In this case, a widely used practice is to determine the area under the plasma concentration–time curve from $t = 0$ to $t = t^*$ (the last sampling time) by means of trapezoidal rule and estimate the remaining area by using the following equation:

$$3.56 \quad (AUC)_{t^*}^\infty = C_p^* / K$$

where $(AUC)_{t^*}^\infty$ is the area under the plasma concentration–time curve from the last sampling time to time ∞ , C_p^* is the last observed plasma concentration, and K is the first-order elimination rate constant. This $(AUC)_{t^*}^\infty$ will be added to the area under the curve obtained from $t = 0$ to $t = t^*$ to calculate the total area under the plasma concentration–time curve:

$$3.57 \quad (AUC)_0^\infty = (AUC)_0^{t^*} + (AUC)_{t^*}^\infty$$

When an IV administration of a drug exhibits the characteristics of a one-compartment model, the total area under the plasma concentration–time curve is estimated by the following equation:

$$3.58 \quad (AUC)_0^\infty = \frac{\text{dose}}{VK}$$

where VK is the systemic clearance of a drug.

Following the administration of a drug by an IV injection, if it is necessary to use a two-compartment model, the area under the plasma concentration–time curve from $t = 0$ to $t = t^*$ (the last sampling time) is estimated by using trapezoidal rules, as previously mentioned. The area under the plasma concentration–time curve from $t = t^*$ to $t = \infty$ can then be computed using the following equation:

$$3.59 \quad (AUC)_{t^*}^\infty = C_p^* / \beta$$

where C_p^* is the last observed plasma concentration and β is the first-order disposition rate constant.

When a drug is administered by an oral route, one can use the following equation to determine AUC_0^∞ :

$$3.60 \quad (AUC)_0^\infty = \frac{F(\text{dose})}{VK}$$

If it is desired to assess the relative extent of drug absorption from a product, AUC_0^∞ of the product is compared to the AUC_0^∞ obtained for a reference drug standard. The reference standard can be an IV injection, an orally administered aqueous or water-miscible solution, or another product accepted as a standard.

When it is desired to assess the absolute bioavailability (F), the reference drug standard becomes an IV injection, and when it is desired to judge the bioequivalence, the reference standard is an innovator product. If the AUC_0^∞ values are identical following the administration of equal doses of a test product and the reference IV solution, we conclude that the test product is completely absorbed and not subject to presystemic metabolism.

Frequently, however, the standard is an innovator product or another established product. If the AUC_0^∞ values are identical following the administration of equal doses of the test and reference products, we conclude that the test product is completely bioavailable relative to the standard. It is essential to use the term “relative to the standard” because we do not know if the standard is completely absorbed or available. Additionally, when two products produce comparable $(C_p)_{\max}$ and $t_{\max}$ and the reference standard is an innovator product, then the products are judged to be bioequivalent.

By using the ratio of AUC_0^∞ for extravascular to IV routes, one can determine the F of a drug from a test product as follows:

$$3.61 \quad F = \frac{(AUC)_0^\infty \text{ oral}}{(AUC)_0^\infty \text{ IV solution}}$$

where F is the absolute bioavailability of a drug or the fraction of the administered dose that reaches the general circulation following the administration of equal dose of a drug. If the administered doses of a drug are different than the AUC_0^∞ , then estimates can be scaled approximately to permit comparison under identical conditions or equivalent doses—assuming, of course, that AUC_0^∞ is directly proportional to the administered dose. The relative bioavailability (F_{rel}) of a drug from a test product is determined by using the following expression:

$$3.62 \quad F_{\text{rel}} = \frac{(AUC)_0^\infty \text{ test product}}{(AUC)_0^\infty \text{ reference standard}}$$

Equation 3.62 assumes that the doses administered from each product are identical, and if not, the AUC_0^∞ values should be scaled for the dose differences.

Therefore, the bioavailability of a drug following its extravascular administration can be expressed as:

$$3.63 \quad F = \frac{(X_u)_\infty \text{ extravascular}}{(X_u)_\infty \text{ intravascular}}$$

To determine the assessment of relative bioavailability (F_{rel}), Equation 3.63 becomes:

$$3.64 \quad F_{\text{ref}} = \frac{(X_u)_\infty \text{ test product}}{(X_u)_\infty \text{ standard product}}$$

In addition, Equations 3.63 and 3.64 are applicable under the condition that the administered doses are identical. The utility of these equations depends on how much of the drug is eliminated by urinary excretion, the sensitivity of the analytical procedure, and the variability in urinary output of the drug. Many drugs are extensively metabolized, with little, if any, drug appearing in an unchanged form in the urine. In such cases, the bioavailability is estimated from the plasma concentration–time data.

Presystemic or First-Pass Metabolism

After oral administration, a drug must pass sequentially from the gastrointestinal lumen, through the gut wall, and through the liver before reaching the general circulation (Fig. 3.30). Because the gut wall and liver are the sites of drug metabolism, a fraction of the amount of drug absorbed can be eliminated or metabolized before reaching the general circulation. Therefore, an oral dose of a drug can be completely absorbed yet incompletely available to reach the general circulation because of presystemic or first-pass effect (metabolism) in the gut wall or liver. If such is the case, it will be reflected in the values of AUC_0^∞ for the administered dose.

Criteria have been developed to identify and quantify the extent of presystemic metabolism and to indicate when it is occurring. The determination of presystemic metabolism requires only that the systemic availability of a drug is less than the fraction of the dose absorbed. The fraction absorbed can be determined from the urinary excretion of a drug and metabolite after oral administration of a drug relative to that after IV administration. Many drugs undergoing presystemic metabolism in humans have been identified on the basis of this type of information. Differentiation between the gut wall and the liver as the site of presystemic metabolism is more difficult in humans, although relatively easy in animals.

The liver is the most important site of presystemic elimination because of high levels of drug-metabolizing enzymes, its ability to rapidly metabolize different types of drugs, and its unique anatomic location. The following are selected examples of drugs that are subject to considerable hepatic first-pass metabolism: the β -blockers propranolol and metoprolol; the analgesics propoxyphene, meperidine, and pentazocine; the antidepressants imipramine and nortriptyline; and the antiarrhythmic lidocaine.

Hepatic presystemic metabolism is most easily understood when liver is the sole organ of drug elimination. Under these conditions, the clearance of the drug, as determined following IV administration of the drug, is equal to:

$$3.65 \quad (Cl)_H = \frac{\text{dose}}{(AUC)_0^\infty}$$

Hepatic clearance, however, is also equal to:

$$3.66 \quad (Cl)_H = Q_H \cdot E_H$$

where $(Cl)_H$ is the hepatic clearance, Q_H is the hepatic blood flow rate, and E_H is the dimensionless hepatic extraction ratio of the drug. Hepatic blood flow rate (Q_H) has a mean range from approximately 1.1 to 1.8 L/min, with an average of approximately 1.5 L/min. The hepatic extraction ratio (E_H) of a drug can range from 0 to 1, depending on the liver's ability to metabolize the drug. The maximum hepatic clearance of a drug, excluding hepatic metabolism, is equal to hepatic blood flow; this occurs when $E_H = 1.0$. The fraction of a drug eliminated from portal blood (Fig. 3.30) during the absorption phase is given by the hepatic extraction ratio (E_H); the remainder of the drug (i.e., $1 - E_H$) escapes into the systemic circulation and then is cleared from the circulation by the liver according to Equation 3.66.

If the fraction of the oral dose is absorbed and then subjected to hepatic presystemic metabolism, the AUC_0^∞ following oral administration of a drug is given by:

$$3.67 \quad (AUC)_0^\infty = \frac{F(X_a)_0(1 - E_H)}{Q_H \cdot E_H}$$

Because $Q_H \cdot E_H$ is equal to hepatic clearance (Eq. 3.66), which under these conditions is given by the ratio of an IV dose to an area under the concentration–time curve, AUC_0^∞ IV, Equation 3.67 is rewritten as:

$$3.68 \quad \frac{(AUC)_0^\infty \text{ oral}}{(AUC)_0^\infty \text{ IV}} = \frac{F(X_a)_0(1 - E_H)}{(X_a)_0 \text{ IV}}$$

The ratio of AUC_0^∞ after oral and IV administration of equal doses of drugs is the systemic availability (i.e., the fraction absorbed). If it is assumed that the drug is completely absorbed (i.e., $F = 1$), then Equation 3.68 reduces to:

$$3.69 \quad F = (1 - E_H)$$

Equation 3.69 shows that the systemic availability of the drug depends on the E_H of the drug, and those drugs with low E_H , such as antipyrine, tolbutamide, and warfarin, undergo little presystemic metabolism.

An estimate of E_H is made from determination of the clearance of a drug following IV administration and comparing this value to the mean value of liver blood flow according to Equation 3.66, when rearranged:

$$3.70 \quad E_H = \frac{(Cl)_H}{Q_H}$$

The IV clearance of propranolol is approximately 1.05 L/min. Assuming that the average liver blood flow is approximately 1.5 L/min, we can determine that the hepatic extraction for propranolol (E_H) is 0.7 and that the fraction absorbed (F) is 0.3. This means that even though propranolol is well absorbed, only 30% of the oral dose is available for systemic circulation.

This type of information, in conjunction with the value of the fraction absorbed (F), has been used to substantiate the predominantly hepatic presystemic elimination of several drugs, including propranolol, lidocaine, and pentazocine. The plasma concentrations for pentazocine, following the oral administration of a 100-mg dose and IV administration of a 30-mg dose, are shown in Figure 3.34, which shows that even though the IV dose is smaller, this route of administration provides higher plasma concentration than an oral dose. The systemic availability (F) of pentazocine after oral administration was reported to be 11% to 32%, with a mean of 18%. This low systemic availability is consistent with its high hepatic clearance.

Repetitive Drug Administration (Multiple Dosing)

If a fixed IV dose of a drug is administered repeatedly at a constant time interval (τ), the plasma concentration of a drug at any time can be calculated using the following expression:

$$3.71 \quad (C_p)_t = \frac{X_0}{V} \left(\frac{1 - e^{-nKt}}{1 - e^{-Kt}} \right) e^{-Kt}$$

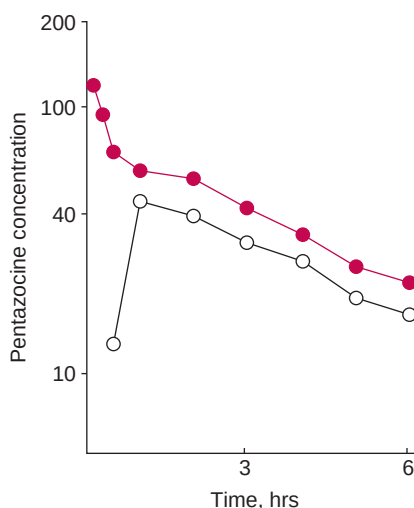


FIGURE 3.34 Pentazocine concentration in plasma (ng/mL) after administration of 100 mg orally (○) or 30 mg intravenously (●). (From Ehrnebo M, Boreus L, Lonroth U. Bioavailability and first-pass metabolism of oral pentazocine in man. *Clin Pharmacol Ther* 1977;22:888–892, with permission.) (57).

where n is the number of doses that have been administered, t is the time between $t = 0$ and $t = \tau$, τ is the dosing interval, X_0 is the dose administered, V is the apparent volume of distribution of the drug, and K is the elimination rate constant. At the plateau, Equation 3.71 reduces to:

$$(C_p)_\infty = \frac{X_0}{V} \left(\frac{e^{-Kt}}{1 - e^{-K\tau}} \right) \quad 3.72$$

where $(C_p)_\infty$ is the steady-state plasma concentration.

The maximum plasma concentration of a drug (Fig. 3.35) at steady-state, $(C_p)_\infty \text{ max}$, and its minimum plasma concentration at steady-state, $(C_p)_\infty \text{ min}$, is determined by setting $t = 0$ and $t = \infty$, respectively. Equation 3.71 then becomes:

$$(C_p)_\infty \text{ max} = \frac{X_0}{V} \left(\frac{1}{1 - e^{-K\tau}} \right) \quad 3.73$$

and

$$(C_p)_\infty \text{ min} = \frac{X_0}{V} \left(\frac{1}{1 - e^{-K\tau}} \right) e^{-K\tau} \quad 3.74$$

When drugs are administered as repetitive doses (multiple doses), it is often of practical use to determine the “average” plasma concentration at the plateau or steady-state, $(\bar{C}_p)_{ss}$ average. This is obtained by:

$$(\bar{C}_p)_{ss} \text{ average} = \frac{X_0}{VK\tau} \quad 3.75$$

where τ is the dosing interval, X_0 is the administered dose, and VK is its systemic clearance. Equation 3.75 clearly indicates that by knowing the V and the K , obtained from the administration of a single IV bolus dose, the average plasma concentration of a drug can be predicted for the IV bolus administration of a fixed dose (X_0) at a constant dosing interval (τ). Equation 3.75 also clearly indicates that only the size of the dose (X_0) and the dosing interval (τ)

can be adjusted to obtain the desired average steady-state plasma drug concentration.

It is important to recognize that the average steady-state plasma drug concentration, $(\bar{C}_p)_{ss}$ average, is neither the arithmetic nor geometric mean of $(C_p)_\infty \text{ max}$ and $(C_p)_\infty \text{ min}$ but, rather, the ratio of the AUC_0^∞ curve during the dosing interval (τ) at the plateau over the dosing interval (τ).

When the drug is administered by the oral route (Fig. 3.36), the mathematical expressions are more complex than the analogous equations for IV administration:

$$(C_p)_\infty = \frac{K_a F(X_a)_0}{V(K_a - K)} \left(\frac{e^{-Kt}}{1 - e^{-K\tau}} - \frac{e^{-K_a t}}{1 - e^{-K_a \tau}} \right) \quad 3.76$$

where $(X_a)_0$ is the dose administered; F is the fraction absorbed; K_a and K are the first-order absorption and the elimination rate constants, respectively; V is the apparent volume of distribution; and t and τ are time and dosing intervals, respectively. Following the administration of each successive dose in the postabsorption period, Equation 3.76 reduces to:

$$(C_p)_\infty \text{ min} = \frac{K_a F(X_a)_0}{V(K_a - K)} \left(\frac{e^{-K\tau}}{1 - e^{-K\tau}} \right) \quad 3.77$$

The average steady-state plasma concentration of a drug when administered by an extravascular route is obtained using the following equation:

$$(C_p)_{ss} \text{ average} = \frac{F(X_a)_0}{VK\tau} \quad 3.78$$

Or by substituting Equation 3.60 into Equation 3.78:

$$(C_p)_{ss} \text{ average} = \frac{(AUC)_0^\infty}{\tau} \quad 3.79$$

where F is the fraction absorbed or absolute bioavailability of a drug. Taking the ratio of Equation 3.78 over Equation 3.75 after the attainment of the steady-state condition permits one to determine the bioavailability of a drug; of course, this assumes that the administered doses are identical.

Repeated administration of a fixed dose at a constant dosing interval (τ) produces a gradual increase of drug levels in the body until the steady-state condition is attained. This increase is the result of drug accumulation factor (τ) because of the sequential dosing of the drug. Therefore, predicting the degree of accumulation of a drug under defined conditions becomes important. Multiplying each side of Equation 3.75 by V and dividing by the administered dose, Equation 3.80 is obtained:

$$\frac{X_{ss} \text{ average}}{X_0} = \frac{1}{K\tau} = \frac{1.44t_{1/2}}{\tau} \quad 3.80$$

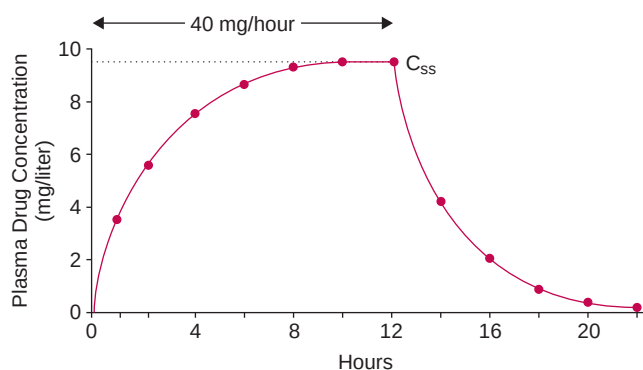


FIGURE 3.35 A typical plasma concentration versus time profile for a drug administered intravenously as a fixed dose (X_0) at a fixed dosing interval (τ).

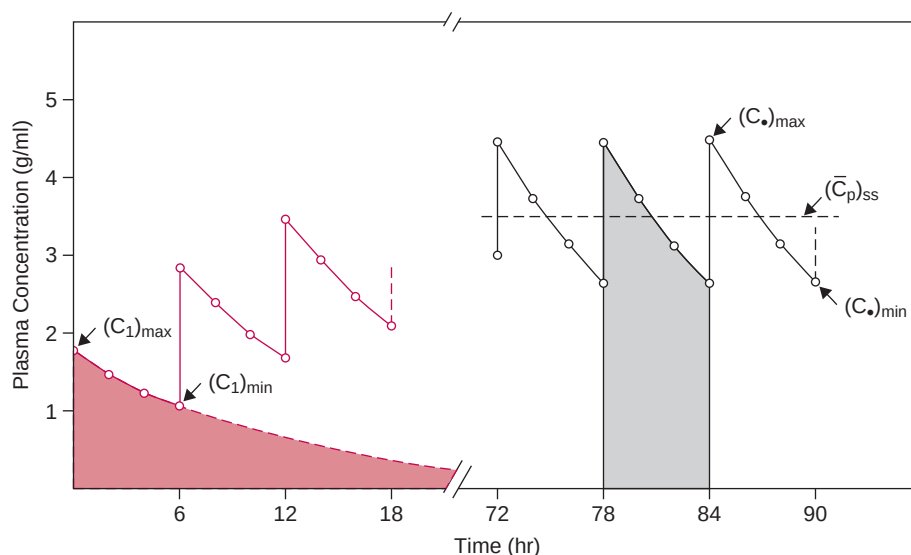


FIGURE 3.36 A typical plasma concentration versus time profile for a drug administered orally as a fixed dose $(X_a)_0$ at a fixed dosing interval (τ) .

Where $X_{ss} \text{ average} / X_0 = R = \text{drug accumulation factor}$ and $X_{ss} \text{ average}$ is the “average” amount of drug in the body at the steady-state condition.

The ratio of the average amount of a drug at its steady-state and the administered dose is defined as drug accumulation (R). Equation 3.80 describes that the magnitude of drug accumulation is a function of the elimination half-life of a drug and the chosen dosing interval. For example, if a drug with an elimination half-life of 12 hours (i.e., $K = 0.0577 \text{ h}^{-1}$) is administered every 6 hours (τ), the ratio of $X_{ss} \text{ average}$ over dose is 2.9. This means that repeated administration of a fixed dose of a drug in the body is approximately 2.9-fold the amount administered in a single dose. It is also clear from Equation 3.80 that the drug accumulation ratio (R) is directly proportional to the elimination half-life of the drug ($t_{1/2}$) and inversely proportional to the dosing interval (τ); however, R is independent of the size of the administered dose.

Because considerable time can elapse before a steady-state condition is attained as a result of repeated drug administration, it is often desirable to administer a large dose initially (i.e., loading dose) to achieve the desired drug levels immediately. Equation 3.29, which describes the time course of drug concentration after a single IV bolus dose, is written as:

$$(C_p)_1 \min = \frac{X_0}{V} (e^{-K\tau}) \quad 3.81$$

where $(C_p)_1 \min$ is the drug concentration immediately before administration of the second dose of the same size as the first one (i.e., the minimum concentration occurs at $t = \tau$ following administration of the first dose). The minimum steady-state plasma concentration, $(C_p)_\infty \min$,

is given by Equation 4.74. Thus, the ratio of $(C_p)_\infty \min$ to $(C_p)_1 \min$ (i.e., Eqs. 3.74 and 3.81) is another way to measure the drug accumulation (R), which can be calculated by means of the following expression:

$$R = \frac{(C_p)_\infty \min}{(C_p)_1 \min} = \frac{1}{1 - e^{-K\tau}} \quad 3.82$$

This ratio of minimum drug concentrations, numerically, is not equal to the ratio of the “average” dose of a drug at steady-state and the dose administered (Eq. 3.80).

If one wished to administer a loading dose (D_L) that produces the minimum concentration equal to $(C_p)_\infty \min$:

$$(C_p)_1 \min = (C_p)_\infty \min = \frac{D_L}{V} e^{-K\tau} \quad 3.83$$

Dividing Equation 3.74 by Equation 3.83 will result in:

$$1 = \frac{D}{D_L (1 - e^{-K\tau})} \quad 3.84$$

Equation 3.84, on rearrangement, yields an expression to determine the loading dose (D_L):

$$\frac{D_L}{D} = \frac{1}{(1 - e^{-K\tau})} \quad 3.85$$

In Equations 3.84 and 3.85, D_L is the loading dose and D is the maintenance dose. Equation 3.85 permits the calculation of loading dose for the chosen maintenance dose and dosing interval (τ) and is applicable for the administration of a drug not only by an IV bolus but also by the extravascular route. When a drug is administered

by the extravascular route, however, it is essential that each maintenance dose be administered following the complete absorption of a drug from the previous dose. Conversely, Equation 3.85 also permits the determination of the maintenance dose required to maintain the minimum drug level produced by the administration of the initial dose for any chosen dosing interval.

Plasma Protein Binding in Pharmacokinetics

Drug binding to plasma proteins affects drug distribution and elimination as well as the pharmacologic effect of a drug. The high molecular weight of plasma proteins restricts their passage across capillaries, and their low lipid solubility prevents them from crossing the cell membrane. Analogously, binding of drugs to plasma protein restricts their passage across cell membranes. Only that fraction of the drug concentration that is freely circulating or unbound can penetrate the cell membrane and be subject to glomerular filtration. Hepatic metabolism of most drugs is also limited by the availability of free fraction of drug in the blood. Because the interaction of drugs with plasma protein is a rapidly reversible process, one can view the plasma protein-binding phenomenon as being temporary storage of a drug, subject to instant recall.

Drug binding to plasma protein can be attributed to ionic, Van der Waals, and hydrogen bonding. The most important contribution to drug binding in the plasma is made by albumin, which comprises approximately 50% of the total plasma protein. In healthy subjects, albumin concentration in the plasma is approximately 4 g/100 mL. During pregnancy and other diseases, however, low levels of plasma protein are observed. Although albumin binds a wide variety of drug molecules, it has a particularly important role in the binding of weak acidic and neutral drugs.

α_1 -Acid glycoprotein is another important binding protein with an affinity for basic drugs. α_1 -Acid glycoprotein is a low molecular weight protein. It is an acute-phase reactant, and its concentration in plasma rises in inflammation, malignant diseases, and stress. Conversely, its plasma concentration falls in hepatic diseases and nephrotic syndrome. The average concentration of α_1 -acid glycoprotein is approximately 40 to 100 mg/100 mL. Other plasma proteins have a limited role in drug binding. The drug-protein interactions can be described by applying the law of mass action:



where D_F and D_B represent the free and bound drug, respectively, and K_1 and K_2 are the association and dissociation rate constants, respectively. Thus:

$$3.87 \quad K = \frac{K_1}{K_2} = \frac{[D_B]}{[D_F][\text{free sites}]}$$

3.88

$$K = \frac{K_1}{K_2} = \frac{[D_B]}{[D_F][nP - D_B]}$$

where K is the equilibrium association constant, K_1 and K_2 are binding rate constants, n is the number of available binding sites per mole of protein, and $[D_F]$, $[D_B]$, and P are the molar concentration of free drug, bound drug, and protein, respectively.

The binding rate constants K_1 and K_2 appear to be large, because the equilibrium is established almost immediately. The value of the equilibrium constant (K) varies from zero, at which essentially no drug is bound, to greater than 10^6 , at which almost all drug is bound to the protein. The fraction of drug in the plasma that is free or unbound (f_p) is then obtained as follows:

3.89

$$f_p = \frac{[D_F]}{[D_F] + [D_B]}$$

3.90

$$f_p = \frac{[D_F]}{[D_T]}$$

where $[D_T]$ is the total drug concentration in the plasma. In most cases, for a given amount of drug in the body, the greater the binding of drug to plasma protein, the larger is the total drug concentration of drug in the plasma. Changes in drug binding as a rule affect the blood level of total drug and have a role in pharmacokinetic variability.

The free fraction of drug in the plasma (f_p) depends on the magnitude of the equilibrium constant (K), the total drug concentration, $[D_T]$, and the protein concentration, $[P]$. In theory, there are limited numbers of binding sites on the protein. As the plasma drug concentration increases, the number of available free sites decreases, and therefore, the fraction of available free drug increases. In reality, however, the fraction of unbound drug in the plasma for most drugs, when administered in therapeutic doses, is essentially constant over the entire drug concentration range. Concentration-dependent changes in the fraction of free drug in the plasma are most likely to occur with drugs exhibiting a high association constant ($K = 10^5$ to 10^6) administered in large doses.

The relationship between bioavailability and area under the plasma concentration-time curve is nonlinear and absorption rate dependent when the plasma protein binding of a drug is concentration dependent. Two drug products from which a drug is equally well absorbed will produce different values for the area under the plasma concentration-time curve if a difference exists in the absorption rate. Generally, such a comparison will overestimate the extent of drug absorption of the more slowly absorbed product.

The clearance of many drugs from the blood is directly proportional to the free drug fraction in the plasma (f_p). The steady-state concentration of these drugs is inversely proportional to the free fraction in the plasma. However,

clearance of some drugs is largely independent of plasma protein binding. The direction and magnitude of the effect of plasma protein binding on the elimination half-life of a drug depends on the size of the drug's V and whether the drug exhibits restrictive clearance (i.e., has an intrinsic clearance less than the liver blood flow). The half-life of a restrictively cleared drug with relatively small V (i.e., $V < 0.25$ L/kg) can show a small decrease in elimination half-life when there is decrease in plasma protein binding (i.e., an increase in the free fraction in plasma). Conversely, the half-life of a nonrestrictively cleared drug with a small apparent volume of distribution can show a small increase in half-life when the free fraction is increased. Drugs with a large value for V (i.e., $V > 0.5$ L/kg) either will be essentially independent of the changes in plasma protein binding (restrictive clearance) or will show an increase in half-life that is directly proportional to the increase in free fraction (nonrestrictive clearance).

The classical methods of studying protein binding of drugs include equilibrium dialysis and ultrafiltration. Ultrafiltration provides quick measurements but is not as accurate as the equilibrium dialysis method. Detailed discussions for studying protein binding can be found in textbooks listed in the Suggested Readings.

STATISTICAL MOMENT ANALYSIS

Statistical moment analysis is a noncompartmental method, based on statistical moment theory, for calculation of the absorption, distribution, and elimination parameters of a drug. This approach to estimating pharmacokinetic parameters has gained considerable attention in recent years.

The zero moment in the drug plasma concentration–time curve is the total area under the plasma concentration–time curve from $t = 0$ to $t = \infty$, AUC_0^∞ . Estimates of the area under this curve are useful in calculating bioavailability as well as drug clearance, which is the ratio of dose over area under the concentration–time curve for an IV dose.

The first moment of the plasma concentration–time profile is the total area under the concentration–time curve resulting from plot of the product of plasma concentration and time (i.e., $C_p t$) versus time, as illustrated in Table 3.10 and Figure 3.37 (45).

Column 2 of Table 3.10 shows the concentration versus time data obtained following a 1-hour constant-rate infusion, and column 3 also includes concentration–time values for the product of concentration $\times$ time. These values are plotted against time in Figure 3.34. The area under the curve for the concentration–time plot is obtained by using the trapezoidal rule. The total area under the curve for the product of concentration $\times$ time is termed the “area under the first-moment curve” (AUMC).

The ratio of the AUMC over the area under the concentration–time curve for any drug, according to the theory, is the assessment of the mean residence time

TABLE 3.10 Drug Concentration and Drug Concentration–Time Data During and After a 1-Hour, Constant-Rate Infusion (45)

Time (hours)	Concentration ($\mu\text{g/mL}$)	Concentration–Time ($\mu\text{g/mL/h}$)
0.5	3.2	1.6
1.0	5.9	5.9
2.0	4.2	8.4
3.0	3.0	9.0
4.0	2.1	8.4
5.0	1.5	7.5
6.0	1.1	6.6
8.0	0.5	4.0

(MRT). The MRT provides a quantitative estimate of the persistence of a drug in the body, and like the half-life of a drug, MRT or persistence is a function of distribution and elimination of a drug. Comparison of MRTs following administration of a drug as an IV bolus or via any other extravascular route provides information regarding the mean absorption time (MAT).

One of the most useful properties of statistical moment analysis is that it permits estimation of the apparent volume of distribution that is independent of drug elimination (i.e., regardless of the model chosen to describe the concentration–time data).

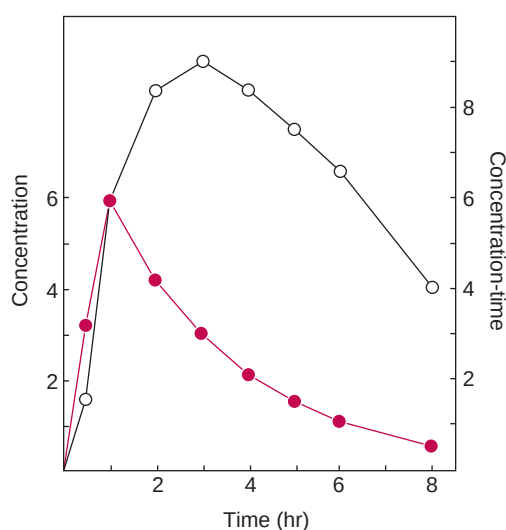


FIGURE 3.37 Plots of drug concentration ($\mu\text{g/mL}$; ●) and drug concentration $\times$ time ($\mu\text{g/mL/h}$; ○) versus time during and after 1 hour of a constant-rate infusion. The area under the concentration versus time plot to infinity is AUC; the area under the concentration $\times$ time versus time plot to infinity is AUMC. (45).

Mean Residence Time (MRT)

The MRT of a drug after administration of a single dose is provided by the following equation:

$$3.91 \quad \text{MRT} = \frac{\text{AUMC}}{\text{AUC}}$$

The MRT for a drug administered IV provides a useful estimate of the persistence time in the body. Therefore, in this sense, it is related to the half-life of a drug. When applied to a drug that distributes rapidly (i.e., one-compartment model), it has been shown that:

$$3.92 \quad \text{MRT} = \frac{1}{K}$$

where K is the elimination rate constant.

Because the half-life of a drug is equal to $0.693/K$, half-life is a measurement of the time required to eliminate 50% of the administered dose. The MRT, however, indicates the time required to eliminate 63.2% of the administered dose.

When a drug is administered by an extravascular route, statistical moment analysis theory can also be used for estimating the rate of absorption. This approach, however, requires the calculation of MRT for IV as well as extravascular routes, because the method is based on the differences in MRT for different modes of administration. In general:

$$3.93 \quad \text{MAT} = (\text{MRT})_{\text{EV}} - (\text{MRT})_{\text{IV}}$$

where $(\text{MRT})_{\text{EV}}$ is the MRT following administration of a drug by an extravascular route and $(\text{MRT})_{\text{IV}}$ is the MRT for the IV bolus dose. When the administered drug follows the first-order process:

$$3.94 \quad (\text{MRT})_{\text{EV}} = \frac{1}{K_a}$$

where K_a is the first-order absorption rate constant. Under these conditions:

$$3.95 \quad K_a = \frac{1}{\text{MAT}}$$

and the absorption half-life is obtained by $0.693/\text{MAT}$.

The statistical moment theory offers an attractive alternative for the evaluation and estimation of the absorption data, and even in the absence of intravenous data, this method permits the ranking of several dosage forms, with respect to drug release and absorption, from the available MRT values.

SUMMARY

From this discussion, the efficacy of a drug is not determined by its pharmacodynamic characteristics alone, but

efficacy also depends, to a large extent, on the pharmacokinetic parameters of the drug, because ADME processes control the rate and extent to which an administered dose of a drug reaches its site of action.

Because of the high degree of structural variability of drugs, multiplicity of kinetics, and metabolite kinetics, the task of establishing a clear correlation between the structured chemistry of substituents and their pharmacokinetic properties appears somewhat daunting. The pharmacokinetic fate of a drug molecule, however, appears to be a consequence of its physicochemical properties and, therefore, can, to some extent, be predicted from its chemical structure.

Although the drug in the formulation has received considerable attention, many of the alterations in the formulation can be considered as chemical changes. Most of what has been reported applies primarily to the gastrointestinal absorption of drugs and can be viewed as attempts to:

1. Maximize the rate of absorption by increasing the rate of dissolution (i.e., micronization, salt of acid or bases, amorphous form and metastable polymorph, etc.)
2. Decrease the loss of a drug because of its degradation in the stomach (i.e., acid, insoluble esters or salt, and chemically stable derivatives of a drug)
3. Extend the duration of action by reducing the rate of a drug's release from a dosage form (e.g., timed release, depot-forming injectable, macro-crystals, and slowly dissolving salts)
4. Decrease the loss of a drug by reducing the complex formation

These examples for enhancing gastrointestinal absorption represent the response to a particular problem with the parent compound and, therefore, can be viewed as "corrective" research. It is of considerable interest to see this aspect of research become "predictive and preventive," in which the pharmacokinetic parameters of drugs are required in the early phases of drug discovery to optimize the effectiveness of drugs.

An immediate problem facing those who would consider optimizing all factors of a drug is physically locating the receptor site and defining the ideal time course for the drug-receptor interaction and sustained effects. An ideal drug molecule should reach the site of action, arrive rapidly in sufficient quantity, remain at the site of action for sufficient time, be excluded from other sites, and be removed from the site, when appropriate. Although such an ideal drug molecule rarely exists, alternate approaches are chosen to optimize the effectiveness of a drug. Furthermore, if a correlation exists between a biologic response and the plasma blood levels of a drug, then the pharmacokinetic parameters (ADME) have an important role in influencing the biologic response, because these parameters influence the magnitude of the blood level of a drug in the body. The task of examining the examples of drugs illustrating the connection between biologic

response and pharmacokinetics study is not an easy one, but the results do convey the important facts that:

1. Pharmacokinetic parameters influence the biologic responses, which are critical in drug design.
2. Pharmacokinetic parameters can be modified by subtle structural changes, which in turn can influence the desired blood level.

The ultimate goal is to design a drug molecule that exhibits the desired pharmacologic effect as a result of the proper balance of ADME processes. Figure 3.37 illustrates how modification of a parent structure can influence the availability of a drug to the receptor site (58).

The following are some of the processes shown in Figure 3.38 that are altered by changing a substituent group on the drug molecule:

- I. Supply and loss
 - A) Rate of transfer from the dosage form
 - B) Binding of a drug in the depot
 - C) Stability of a drug in the depot
- II. Distribution in the body
 - A) Binding of a drug in the central and peripheral compartments
 - B) Apparent volume of distribution
 - C) Transfer of a drug to the receptor sites
- III. Drug–receptor interaction

Consider the following well-known example for the design of a urinary tract anti-infective. The site of infection is the urinary tract. The example selected is the prodrug methenamine. In acidic pH, methenamine is converted to formaldehyde, which acts as an antibacterial agent (Fig. 3.39). Tablets of methenamine are often enteric coated to prevent conversion to formaldehyde in the stomach. Methenamine is cleared intact from the kidney into the urine, where it is hydrolyzed to formaldehyde if the pH is less than 6.5. The rate of hydrolysis is controlled by the urinary pH.

The influence of structural effects on pharmacokinetic parameters can be illustrated using the following examples. The steady-state levels of the antibiotic carbenicillin

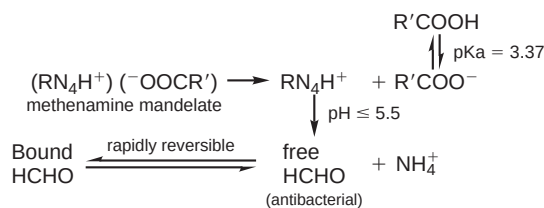


FIGURE 3.39 Conversion of methenamine to formaldehyde in acidic pH.

are twice those of ampicillin. These higher blood levels of carbenicillin following IV administration have been attributed to its efficacy in the treatment of relatively resistant infections, such as *Pseudomonas* species. The reason for these differences in the higher steady-state plasma concentration is the larger apparent volume of distribution for ampicillin, because the elimination rate constants are similar. If all the factors were equal, one can argue that an increased value for the apparent volume of distribution is a clinical advantage, because bacteria germinate more frequently in the tissue than in the blood. The effectiveness of an antibiotic depends on its penetration into tissues, particularly inflamed tissue. Thus, if plasma protein binding is equal for both antibiotics, the antibiotic with a larger volume of distribution would appear to be reaching the site of action with better efficacy, but this is by no means unequivocal. Therefore, the spectrum of research activity in the area of antibiotics would imply that the following goals for molecular modifications are generally pursued:

1. Increased tissue distribution
2. Longer half-life to maintain a higher blood level and decrease the frequency of dose administration
3. Decreased binding capacity to foods and plasma protein

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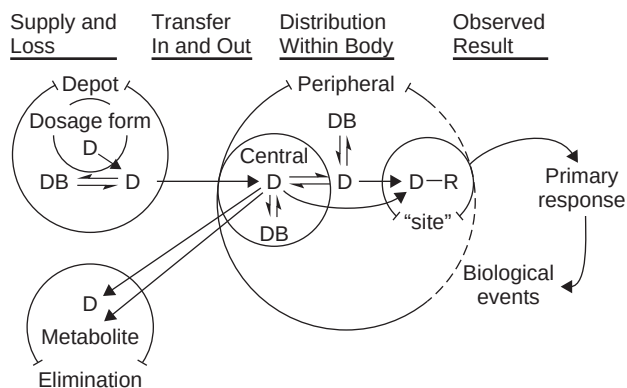
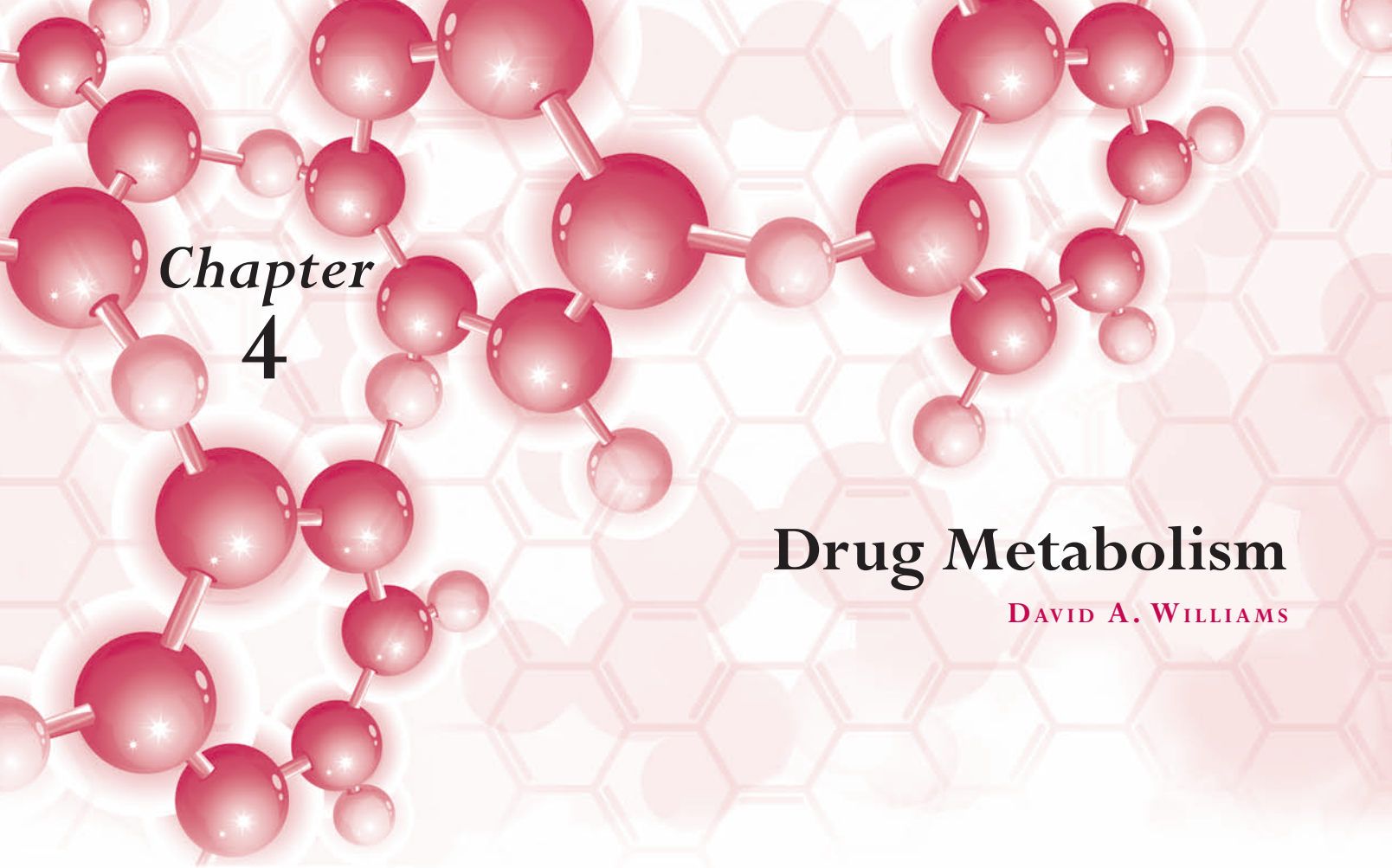


FIGURE 3.38 How some of the processes can be altered by changing a substituent group on a drug molecule.

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Chapter 4

Drug Metabolism

DAVID A. WILLIAMS

Abbreviations

ADH, alcohol dehydrogenases
AhR, aromatic hydrocarbon receptor
ALDH, aldehyde dehydrogenases
ALT, alanine aminotransferase
AST, aspartate aminotransferase
Ah, aromatic hormone
AKR, aldo-keto reductase
ATP, adenosine triphosphate
CAR, constitutive androstane receptor
CoA, coenzyme A
COMT, catechol-*O*-methyltransferase
COX, cyclooxygenase
DHEA, dehydroepiandrosterone
DIH, drug-induced hepatotoxicity
EHC, enterohepatic cycle
EM, fast or extensive metabolizers
ER, endoplasmic reticulum
FAD, flavin dinucleotide
FADH₂, reduced form of flavin dinucleotide
FDA, U.S. Food and Drug Administration
FMN, flavin mononucleotide
FMNH₂, reduced form of flavin mononucleotide
FMO, flavin-containing monooxygenase

GSH, glutathione
GSSG, oxidized glutathione
GST, glutathione *S*-transferases
HOCl, hypochlorous acid
HSAB, hard-soft acid-bases
IDR, idiosyncratic drug reaction
kDa, unit of molecular weight
LDL, low-density lipoprotein
MAO, monoamine oxidase
MAOI, monoamine oxidase inhibitors
MDMA, 4-methylenedioxy-methamphetamine
MIC, metabolite-intermediate complex
MPO, myeloperoxidase
MPP⁺, 1-methyl-4-phenylpyridinium ion
MPTP, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine
NAD⁺, nicotinamide adenine dinucleotide
NADH, the reduced form of nicotinamide adenine dinucleotide
NADP⁺, nicotinamide adenine dinucleotide phosphate
NADPH, the reduced form of nicotinamide adenine dinucleotide phosphate

NAT, *N*-acetyltransferase
NSAID, nonsteroidal anti-inflammatory drug
OATP, organic anion transporter polypeptide
P450, cytochrome P450 monooxygenase
PAH, polycyclic aromatic hydrocarbon
PAPS, 3'-phosphoadenosine-5'-phosphosulfate
PGG₂, prostaglandin G₂
PGI₂, prostacyclin
P-gp, P-glycoprotein
PM, poor metabolizers
PON, paraoxonase
PPAR, peroxisome proliferator-activated receptor
PXR, pregnane X receptor
ROS, reactive oxygen species
SULT, sulfotransferase
TPMT, thiopurine *S*-methyltransferase
UGT, UDP-glucuronosyl transferase
UDP, uridine diphosphate
UDPGA, UDP-glucuronic acid
UM, ultrarapid metabolizers

SCENARIO

Mark D. Watanabe, PharmD, PhD, BCPP

Y.H., a 32-year-old female with a history of chronic schizophrenia, presented for a routine follow-up visit to the behavioral health clinic. Her psychotic symptoms had been well controlled on risperidone 3 mg daily. At her previous appointment 1 month ago, Y.H. had reported being depressed, so her psychiatrist initiated a trial of paroxetine 20 mg daily as treatment. As the pharmacist was interviewing Y.H. for a medication history, he

noticed that there was a resting tremor in her left hand. Y.H. acknowledged that it had first appeared within the last few weeks. She also stated that she was just starting to feel less depressed.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

What is a poison?

All substances are poisons;

There is none that is not a poison.

The right dose differentiates a poison from a drug.

—Paracelsus (1493–1541)

Humans are exposed throughout their lives to a large variety of drugs and nonessential exogenous (foreign) compounds (collectively termed “xenobiotics”) that can pose health hazards. Drugs taken for therapeutic purposes as well as occupational or private exposures to vapors of volatile chemicals or solvents pose possible health risks; smoking and drinking lead to the absorption of large amounts of substances with potentially adverse health effects. Furthermore, ingestion of natural toxins in vegetables and fruits, pesticide residues in food, as well as carcinogenic pyrolysis products from fats and protein formed during the charbroiling of meat have to be considered as additional health risks. Most of these xenobiotics undergo enzymatic biotransformations by xenobiotic-metabolizing enzymes in the liver and extrahepatic tissues and are eliminated by excretion as hydrophilic metabolites. In some cases, especially during oxidative metabolism, numerous chemical procarcinogens form reactive metabolites capable of covalent binding to biopolymers, such as proteins or nucleic acids—critical components that can lead to mutagenicity, cytotoxicity, and carcinogenicity. Therefore, insight into the biotransformation and bioactivation of xenobiotics becomes an indisputable prerequisite to assess drug safety and estimate risks associated with chemicals and drugs.

Detoxication and toxic effects of drugs and other xenobiotics have been studied extensively in various mammalian species. Frequently, differences in sensitivity to these toxic effects were observed and can now be attributed to genetic differences between species in the isoenzyme/isoforms of cytochrome P450 monooxygenases (P450). The level of expression of the P450 enzymes is regulated by genetics and by a variety of endogenous factors such as hormones, gender, age, and disease, as well as the presence of environmental factors, such as inducing agents. Drugs were developed

and prescribed under the old paradigm that “one dose fits all,” which largely ignored the fact that humans (both adults and children) are genetically and metabolically different, resulting in variable responses to drugs.

Drugs can no longer be regarded as chemically stable entities that elicit the desired pharmacologic responses, after which they are excreted from the body. Drugs undergo a variety of chemical changes in humans brought about by enzymes of the liver, intestine, kidney, lung, and other tissues, with subsequent alterations in the nature of their pharmacologic activities, durations of activity, and toxicities. Thus, the pharmacologic and toxicologic activities of a drug (or xenobiotic) are, in many ways, consequences of its metabolism.

Drug therapy is becoming more oriented toward control of metabolic, genetic, and environmental illnesses (e.g., cardiovascular disease, mental illness, cancer, and diabetes) than toward short-term therapies. In most cases, drug therapies last for months or even years, and the problem of drug toxicities resulting from long-term therapy has become increasingly important.

The practice of simultaneous prescriptions of several drugs has become common. Thus, an awareness of possible drug–drug interactions is essential to avoid catastrophic synergistic effects and chemical, enzymic, and pharmacokinetic interactions that can produce toxic side effects.

The study of xenobiotic metabolism has developed rapidly during the past few decades (1–9). These studies have been fundamental in the assessment of drug efficacy and safety and in the design of dosage regimens; in the development of food additives and the assessment of potential hazards of contaminants; in the evaluation of toxic chemicals; and in the development of pesticides and herbicides and the assessment of their metabolic fates in insects, other animals, and plants. The metabolism of drugs and other xenobiotics is fundamental to many toxic processes, such as carcinogenesis, teratogenesis, and tissue necrosis. Often, the same enzymes involved in drug metabolism also carry out the regulation and metabolism of endogenous substances

such as steroids, vitamin D, prostaglandins, and lipids. Consequently, the inhibition and induction of these enzymes by drugs and xenobiotics can have a profound effect on the normal processes of intermediary metabolism, such as tissue growth and development, hematopoiesis, calcification, and lipid metabolism.

Familiarity with the mechanisms of drug metabolism can often aid the prediction of the consequences of drug–drug interactions, drug–food interactions, and herbal drug–drug interactions, and help to explain patients' adverse responses to drug regimens. Incorporating pharmacogenomics into the selection of drug regimens will change the way in which drugs are prescribed for patients. Selection based on a patient's individual genetic makeup could eliminate previously unpredictable responses to drug treatment caused by genetic polymorphisms that effect metabolism, clearance, and tolerance. Pharmacogenomic testing to determine a patient's phenotype (i.e., poor metabolizer) and, thus, his or her ability to metabolize drugs will become common in the future. Such knowledge will assure improved selection of proper drug regimens and doses before therapy begins.

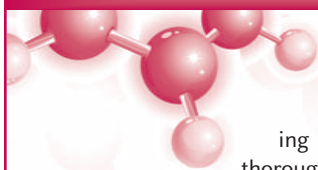
The increased knowledge of drug metabolism, fed by the need for better safety evaluations of drugs and chemicals, has resulted in a proliferation of publications (e.g., *Drug Metabolism Reviews*, *Drug Metabolism and Disposition*, and *Xenobiotica*) and a series of monographs that present the current state of knowledge of foreign compound metabolism from biochemical and pharmacologic viewpoints (3–9).

PATHWAYS OF METABOLISM

Drugs, plant toxins, food additives, environmental chemicals, insecticides, and other chemicals foreign to the body undergo enzymic transformations that results frequently in the loss of pharmacologic activity. The term “detoxication” describes the result of such metabolic changes. Although drug metabolism leads as a rule to detoxication, the processes of oxidation, reduction, glucuronidation, sulfonation (sulfo-conjugation), and other enzyme-catalyzed reactions can lead to the formation of metabolites having therapeutic or toxic effects. This process is often referred to as “bioactivation.” One of the earliest discoveries of an example of bioactivation was the reduction of Prontosil to the antibacterial agent sulfanilamide. Other examples of drug metabolism that lead to therapeutically active drugs include the hydroxylation of acetanilide to acetaminophen, the *N*-demethylation of the antidepressant imipramine to form desipramine, and conversion of the anxiolytic diazepam to form desmethyldiazepam. The insecticide parathion is desulfurized by both insects and mammals to form paraoxon.

Most drugs and other xenobiotics are metabolized by enzymes normally associated with the metabolism of endogenous constituents (e.g., steroids and biogenic amines). The liver is the primary site of drug metabolism, although other xenobiotic-metabolizing enzymes are found in nervous tissue, kidney, lung, plasma, and the gastrointestinal tract (digestive secretions, bacterial flora, and the intestinal wall).

CLINICAL SIGNIFICANCE



The basic principles of drug metabolism may inform a wide variety of clinical decisions regarding pharmacotherapy. For example, a thorough understanding enables a careful assessment for drug–drug interactions in particular patient cases. Drugs, as chemical entities, can be substrates, inhibitors, or inducers of metabolic enzymes. The interplay of these roles potentially influences serum drug concentrations in ways that may directly affect the desired outcome (i.e., decreases in levels that may prevent therapeutic efficacy or increases that may enhance risks of toxicity). There are certainly many more theoretical pharmacokinetic interactions that invoke these mechanisms than are actually seen in clinical practice. Through careful observation and analysis of unexpected and possibly concentration-dependent events, one could more readily identify and document which interactions are of greater clinical significance by virtue of their actual occurrence in patients. The clinician would then be poised to recommend appropriate dosage adjustments or medication changes based on the actual outcomes of these interactions.

In addition, the biotransformation of drugs may produce reactive metabolites which, through basic chemical reactions, can interact with the components of cellular membranes and proteins in a manner that disrupts normal structure and function. A working knowledge of those functional groups within drug molecules that may be more susceptible to reactive metabolite formation could help explain toxic sequelae when they emerge during a medication trial. This could be useful information whenever alternative therapeutic agents within a given chemical class are being considered.

Ongoing discoveries from studies in the pharmacogenetics field are expanding the drug metabolism literature in directions that hint at the future prospect of truly individualized drug regimens. The need to keep abreast of these new developments is both compelling and exciting, and their application builds upon the principles presented in this chapter.

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Although hepatic metabolism continues to be the most important route of metabolism for xenobiotics and drugs, other biotransformation pathways have a significant role in the metabolism of these substances. Among the more active extrahepatic tissues capable of metabolizing drugs are the intestinal mucosa, kidney, and lung (see the discussion of extrahepatic metabolism). The ability of the liver and extrahepatic tissues to metabolize substances to either pharmacologically inactive or bioactive metabolites before they achieve systemic blood levels is termed “first-pass metabolism” or the “presystemic first-pass effect.” Other metabolite reactions occurring in the gastrointestinal tract are associated with bacteria and other microflora in the tract. The bacterial flora can affect metabolism through: 1) production of toxic metabolites, 2) formation of carcinogens from inactive precursors, 3) detoxication, 4) exhibition of species differences in drug metabolism, 5) exhibition of individual differences in drug metabolism, 6) production of pharmacologically active metabolites from inactive precursors, and 7) production of metabolites not formed by animal tissues.

Phase 1 Reactions

The pathways of xenobiotic metabolism are divided into two major categories. Phase 1 reactions (biotransformations) include oxidation, hydroxylation, reduction, and hydrolysis. In each of these enzymatic reactions, a new functional group is introduced into the substrate molecule, an existing functional group is modified, or a functional group or acceptor site for Phase 2 transfer reactions is exposed, thus making the xenobiotic more polar and, therefore, more readily excreted.

Phase 2 Reactions

Phase 2 reactions (conjugation) are enzymatic syntheses whereby a functional group, such as alcohol, phenol, or amine, is masked by the addition of a new group, such as acetyl, sulfate, glucuronic acid, or certain amino acids, which usually increases the polarity of the drug or xenobiotic. Most substances undergo both Phase 1 and Phase 2 reactions sequentially.

Those xenobiotics that are resistant to metabolizing enzymes or are already hydrophilic are excreted largely unchanged. This basic pattern of xenobiotic metabolism is common to all animal species, including humans, but species can differ in details of the reaction and enzyme control.

FACTORS AFFECTING METABOLISM

As indicated earlier, drug therapy is becoming oriented more toward controlling metabolic, genetic, and environmental illnesses rather than short-term therapy associated with infectious diseases. In most cases, drug therapy lasts for months or even years, and the

problems of drug–drug interactions and chronic toxicity from long-term drug therapy have become more serious. Therefore, a greater knowledge of drug metabolism is essential. Several factors influencing xenobiotic metabolism include:

1. *Genetic factors.* Individual differences in drug effectiveness (drug sensitivity or drug resistance), drug–drug interactions, and drug toxicity can depend on racial and ethnic characteristics with the population frequencies of the many polymorphic genes and the expression of the metabolizing enzymes. Pharmacogenetics focuses primarily on genetic polymorphisms (mutations) responsible for interindividual differences in drug metabolism and disposition (10). Genotype–phenotype correlation studies have validated that inherited mutations result in two or more distinct phenotypes causing very different responses following drug administration. The genes encoding for CYP2A6, CYP2C9, CYP2C19, and CYP2D6 are functionally polymorphic; therefore, at least 30% of P450-dependent metabolism is performed by polymorphic enzymes. For example, mutations in the *CYP2D6* gene result in poor, intermediate, or ultrarapid metabolizers of more than 30 cardiovascular and central nervous system drugs. Thus, each of these phenotypic subgroups experiences different responses to drugs extensively metabolized by the CYP2D6 pathway ranging from severe toxicity to complete lack of efficacy. For example, ethnic specificity has been observed with the sensitivity of the Japanese and Chinese to ethanol as compared to Caucasians, CYP2C19 polymorphism (affects ~20% of Asians and ~3% of Caucasians) and the variable metabolism of omeprazole (proton pump inhibitor) and antiseizure drugs, and the polymorphic paraoxonase-catalyzed hydrolysis of the neurotoxic organophosphates and lipid peroxides (atherosclerosis) (see the discussion of genetic polymorphism).
Incorporating pharmacogenomics, the study of heritable traits affecting patient response to drug treatment, into drug therapy will alter the way in which drug regimens are chosen for patients based on their individual genetic makeup (10), thus eliminating the unpredictable response of drug treatment as a result of genetic polymorphisms that affect metabolism, clearance, and tolerance. Understanding how individuals are genetically predisposed to differences in metabolism risk could facilitate development of new classes of drugs that are metabolized by nonpolymorphic P450 enzymes.
2. *Physiologic factors.* Age is a factor as both very young and old have impaired metabolism. Hormones

(including those induced by stress), sex differences, pregnancy, changes in intestinal microflora, diseases (especially those involving the liver), and nutritional status can also influence drug and xenobiotic metabolism.

Because the liver is the principal site for xenobiotic and drug metabolism, liver disease can modify the pharmacokinetics of drugs metabolized by the liver (11–13). Several factors identified as major determinants of the metabolism of a drug in the diseased liver are the nature and extent of liver damage, hepatic blood flow, the drug involved, the dosage regimen, and the degree of participation of the liver in the pharmacokinetics of the drug. Liver disease affects the elimination half-life of some drugs but not of others, although all undergo hepatic biotransformation (Table 4.1). Some results have shown that the capacity for drug metabolism is impaired in chronic liver disease, which could lead to drug overdosage. Consequently, as a result of the unpredictability of drug effects in the presence of liver disorders, drug therapy in these circumstances is complex, and more than usual caution is needed (13).

Substances influencing drug and xenobiotic metabolism (other than enzyme inducers) include lipids, proteins, vitamins, and metals. Dietary lipid and protein deficiencies diminish microsomal drug-metabolizing activity. Protein deficiency leads to reduced hepatic microsomal protein and lipid metabolism; oxidative metabolism is decreased due to an alteration in endoplasmic reticulum (ER) membrane permeability affecting electron transfer. In terms of toxicity, protein deficiency would increase the toxicity of drugs and xenobiotics by reducing their oxidative microsomal metabolism and clearance from the body.

3. *Pharmacodynamic factors.* Dose, frequency, and route of administration, plus tissue distribution and protein binding of a drug, affect its metabolism.
4. *Environmental factors.* Competition of ingested environmental substances with other drugs and xenobiotics for metabolizing enzymes, and poisoning of enzymes by toxic chemicals such as carbon monoxide or pesticide synergists alter metabolism. Induction of enzyme expression (in which the number of enzyme molecules is increased, while the activity is constant) by other drugs and xenobiotics is another consideration.

Such factors can change not only the kinetics of an enzyme reaction but also the whole pattern of metabolism, thereby altering bioavailability, pharmacokinetics, pharmacologic activity, or toxicity of a xenobiotic. Species differences

TABLE 4.1 The Effect of Liver Disease in Humans on the Elimination Half-Life of Various Drugs^a

Difference Reported	No Difference Reported
Acetaminophen	Chlorpromazine
Amylbarbital	Dicoumarol
Carbenicillin	Phenytoin
Chloramphenicol	Phenylbutazone
Clindamycin	Salicylic acid
Diazepam	Tolbutamide
Hexobarbital	
Isoniazid	
Lidocaine	
Meperidine	
Meprobamate	
Pentobarbital	
Phenobarbital	
Prednisone	
Rifamycin	
Tolbutamide	
Theophylline	

^aClearance is disputable but can be increased.

of responses to xenobiotics must be considered in the extrapolation of pharmacologic and toxicologic data from animal experiments to predict effects in humans. The primary factors in these species differences probably are the rates and patterns of drug and xenobiotic metabolism.

DRUG BIOTRANSFORMATION PATHWAY (PHASE 1)

Human Hepatic Cytochrome P₄₅₀ Enzyme System

Oxidation is probably the most common reaction in xenobiotic metabolism. This reaction is catalyzed by a group of membrane-bound monooxygenases found in the smooth ER of the liver and other extrahepatic tissues, termed the “cytochrome P₄₅₀ monooxygenase enzyme system” (14) (hereafter, the abbreviation P₄₅₀ will be used for this enzyme system). Additionally, P₄₅₀ has been called a mixed-function oxidase or microsomal hydroxylase. The tissue homogenate fraction containing the smooth ER is called the microsomal fraction. P₄₅₀ functions as a multicomponent electron-transport system, responsible for the oxidative metabolism of a variety of endogenous substrates (e.g., steroids, fatty acids, prostaglandins, and bile acids) and exogenous substances (xenobiotics) including drugs, carcinogens, insecticides, plant toxins, environmental pollutants, and other foreign chemicals. Central to the functioning of this unique superfamily of heme proteins is an iron protoporphyrin. The iron protoporphyrin is coordinated to the sulfur of cysteine and has the ability to form a complex with carbon monoxide, resulting in a complex that has its primary absorption maximum at 450 nm (thus the title of these metabolizing P₄₅₀ enzymes). P₄₅₀ has an absolute requirement

for NADPH (the reduced form of nicotinamide adenine dinucleotide phosphate) and molecular oxygen (dioxygen). The rate at which various compounds are metabolized by this system depends on the animal species, strain, nutritional status, type of tissue, age, and pretreatment of animals. The variety of reactions catalyzed by P450 (Table 4.2) include the oxidation of alkanes and aromatic compounds; the epoxidation of alkenes, polycyclic hydrocarbons, and halogenated benzenes; the dealkylation of secondary and tertiary amines and ethers; the deamination of primary amines; the conversion of amines to *N*-oxides, hydroxylamine, and nitroso derivatives; and the dehalogenation of halogenated hydrocarbons. It also catalyzes the oxidative cleavage of organic thiophosphate esters, the sulfoxidation of some thioethers, the conversion of phosphothionates to phosphate derivatives, and the reduction of azo and nitro compounds to primary aromatic amines.

The most important function of P450 is to “activate” molecular oxygen (dioxygen), permitting the incorporation of one atom of oxygen into an organic substrate molecule concomitant with the reduction of the other

atom of oxygen to water. The introduction of a hydroxyl group into the hydrophobic substrate molecule provides a site for subsequent conjugation with hydrophilic compounds (Phase 2), thereby increasing the aqueous solubility of the product for transport and excretion from the organism. This enzyme system not only catalyzes xenobiotic transformations in ways that typically lead to detoxication but also, in some cases, in ways that lead to products having greater cytotoxic, mutagenic, or carcinogenic properties. A nonheme, microsomal flavo-protein monooxygenase is responsible for the oxidation of certain nitrogen- and sulfur-containing organic compounds.

Components of P450

P450 consists of at least two protein components: a heme protein called P450 and a flavoprotein called NADPH-P450 reductase, containing both flavin mononucleotide (FMN) and flavin dinucleotide (FAD). P450 is the substrate- and oxygen-binding site of the enzyme system, whereas the reductase serves as an electron carrier, shuttling electrons from NADPH to P450. A third component essential for electron transport

TABLE 4.2 Hydroxylation Mechanisms Catalyzed by Cytochrome P450

Aromatic hydroxylation
$\text{CH}_3\text{CO}-\overset{\text{H}}{\text{N}}-\text{C}_6\text{H}_5 \xrightarrow{[\text{OH}]} \text{CH}_3\text{CO}-\overset{\text{H}}{\text{N}}-\text{C}_6\text{H}_4-\text{OH}$
Aliphatic hydroxylation
$\text{R}-\text{CH}_3 \xrightarrow{[\text{OH}]} \text{R}-\text{CH}_2-\text{OH}$
Deamination
$\text{R}-\text{CH}(\text{NH}_2)-\text{CH}_3 \xrightarrow{[\text{OH}]} (\text{R}-\text{C}(\text{OH})(\text{NH}_2)-\text{CH}_3) \longrightarrow \text{R}-\text{CO}-\text{CH}_3 + \text{NH}_3$
O-Dealkylation
$\text{R}-\text{O}-\text{CH}_3 \xrightarrow{[\text{OH}]} (\text{R}-\text{O}-\text{CH}_2-\text{OH}) \longrightarrow \text{R}-\text{OH} + \text{CH}_2\text{O}$
N-Dealkylation
$\text{R}-\text{N}(\text{CH}_3)_2 \xrightarrow{[\text{OH}]} (\text{R}-\text{N}(\text{CH}_2\text{OH})(\text{CH}_3)) \longrightarrow \text{R}-\overset{\text{H}}{\text{N}}-\text{CH}_3 + \text{CH}_2\text{O}$
$\text{R}-\text{NH}-\text{CH}_3 \xrightarrow{[\text{OH}]} (\text{R}-\text{NH}-\text{CH}_2\text{OH}) \longrightarrow \text{R}-\text{NH}_2 + \text{CH}_2\text{O}$
N-Oxidation
$(\text{CH}_3)_3-\text{N} \xrightarrow{[\text{OH}]} ((\text{CH}_3)_3-\text{NOH}) \longrightarrow (\text{CH}_3)_3-\text{NO} + \text{H}^\oplus$
Sulfoxidation
$\text{R}-\text{S}-\text{R}' \xrightarrow{[\text{OH}]} (\text{R}-\underset{\text{OH}}{\text{S}}-\text{R}') \longrightarrow \text{R}-\underset{\text{O}}{\text{S}}-\text{R}' + \text{H}^\oplus$

from NADPH to P450 is a phospholipid, phosphatidylcholine, which facilitates the transfer of electrons from NADPH-P450 reductase to P450 (14). Although the phospholipid does not function in the system as an electron carrier, it has great influence on the P450 monooxygenase system. The phospholipid constitutes approximately one-third of the hepatic ER and contributes to a negatively charged environment at neutral pH.

Of the three components involved in microsomal oxidative xenobiotic metabolism, P450 is important due to its vital role in oxygen activation and substrate binding. P450 is an integral membrane protein deeply imbedded in the membrane matrix. The environment surrounding the enzyme is negatively charged at neutral pH by the phospholipids. The electron components of P450 are located on the cytoplasmic side of the ER and the hydrophobic active site toward the lumen of the ER (15). The active site of P450 consists of a hydrophobic substrate-binding domain in which is imbedded an iron protoporphyrin (heme) prosthetic group. This group is exactly like that of hemoglobin, peroxidase, and the b-type cytochromes. The iron in the iron protoporphyrin is coordinated with four nitrogens via a tetradentate link to the porphyrin ring. X-ray studies reveal that in the ferric state, the two nonporphyrin ligands are water and cysteine (Fig. 4.1). The cysteine thiolate ligand (proximal) is present in all states of the enzyme and is absolutely essential for the formation of the reactive oxenoid intermediate. The sixth (distal) coordination position is occupied by an easily exchangeable ligand, most likely water, which is labile and readily exchanged for stronger ligands such as cyanide, amines, imidazoles, and pyridines. The ferrous form loses the water ligand completely, leaving the sixth position open for binding ligands such as oxygen and carbon monoxide.

The vast array of xenobiotics presents a unique challenge to the human body, to metabolize these many lipophilic foreign compounds; this makes it impractical to

have one enzyme to metabolize each compound or each class of compounds. Therefore, whereas most cellular functions are in general very specific, oxidation of xenobiotics necessitates P450s with diverse substrate specificities and regioselectivities (multiple sites of oxidation). Several types of P450 enzymes can be found in a single animal species. Humans have 57 genes and more than 59 pseudogenes divided among 18 families of P450 genes and 43 subfamilies, each coding for a different version of the enzyme (isoform), so that together the P450s can metabolize almost any lipophilic compound to which the human is exposed.

Classification of the P450 Multigene Family

Nebert et al. (16) and Gonzalez and Gelboin (17) classified the P450 supergene family on the basis of structural (evolutionary) relationships. P450 monooxygenases resulting from this supergene family have been subdivided into families that possess greater than 40% amino acid homology with subfamilies of greater than 55% homology (16,17). The P450s are named with the root symbol CYP followed by an Arabic numeral designating the family member (e.g., CYP1, CYP2, or CYP3), a letter denoting the subfamily (e.g., CYP1A, CYP2C, CYP2D, or CYP2E), and another Arabic numeral representing the individual gene. Names of genes are written in italics. The nomenclature system is based solely on sequence similarity among the P450s and does not indicate the properties or functions of individual P450s. Of the more than 17 P450 isoforms that have been identified so far, the primary isoforms responsible for drug metabolism in the liver are presented in Figure 4.2 (18). It is evident that the CYP3A and CYP2C families are the isoforms most involved in the metabolism of clinically relevant drugs, whereas the CYP1A2 isoform is predominantly involved in the bioactivation of environmental substances.

In all probability P450s evolved for the regulation of endogenous substances, such as metabolism of cholesterol to maintain membrane integrity, and for steroid

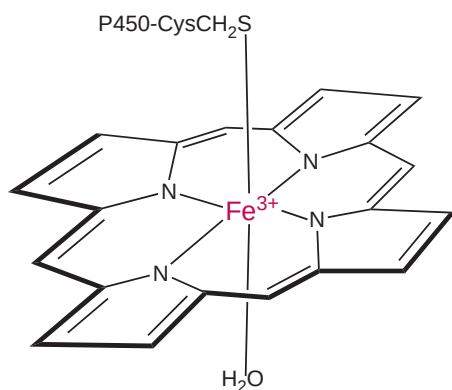


FIGURE 4.1 Ferric heme thiolate catalytic center of P450. The porphyrin side chains are deleted for clarity.

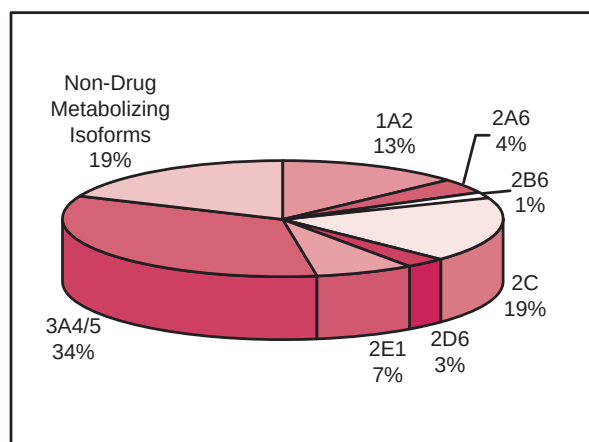


FIGURE 4.2 Total human P450 isoforms expressed in the liver that metabolize drugs.

biosynthesis and metabolism, rather than for metabolizing foreign compounds. P450s are either located in the inner mitochondrial membrane and involved in highly specific steroid hydroxylations or are bound to the ER of the cell and have broad substrate specificity. In evolutionary terms, P450s evolved from a common ancestor, and only more recently (during the last 100 million years), P450 genes have taken on the role of producing enzymes to metabolize a vast array of lipophilic foreign compounds. The xenobiotic P450 genes probably emerged from the steroidogenic P450s to enhance animal survival by synthesizing new P450s to metabolize plant toxins concentrated in the food chain. It is not surprising that animals and humans possess a vast array of P450 enzymes capable of handling a multitude of xenobiotics. Interindividual variation in the expression of xenobiotic P450 genes (genetic polymorphism) or in their inducibility can be associated with differences such as individual susceptibilities to cigarette smoke carcinogenesis. Certain P450 isoforms that clearly exhibit genetic polymorphisms are known to metabolize and as a rule inactivate therapeutic agents. The extent of P450 polymorphism in humans is being investigated to determine the risks of protection against cancers. Food mutagens are typically carcinogens in tissues, but they are activated by CYP1A2 and CYP3A in the liver. Specific forms of P450 in hepatic microsomes are regulated by hormones (e.g., the CYP3A subfamily) and are induced or inhibited by drugs, food toxins, or other environmental xenobiotics (see section on induction and inhibition of P450 isoforms). Identification of a specific P450 isoform as the primary form responsible for metabolism of a specific drug in humans permits reconciliation of the drug's toxicity or other pharmacologic effects.

Substrate Specificity

No evidence exists that the active oxygenating species differ among P450s, suggesting that the substrate specificities, substrate affinities, regioselectivities, and rates of reaction are probably consequences of topographic features of the active site of apoproteins (14,15,19–21). Because a primary function of these enzymes is the metabolism of hydrophobic substrates, it is likely that hydrophobic forces are important in the binding of many substrates to the apoproteins. Nonspecific binding is consistent with the multiple substrate orientations at the active site necessary for the broad regioselectivities observed. A specific binding requirement would decrease the diversity of substrates. Some P450 isoforms have constrained binding sites, and thus, they metabolize small organic molecules (e.g., CYP2E1); CYP1A1/2 isoforms have planar binding sites and metabolize only aromatic planar compounds (i.e., polycyclic aromatic hydrocarbons [PAHs]); CYP2D6 exhibits high affinities for specific apoprotein interactions (hydrogen bonds, ion-pair formation) for specific substrates such as lipophilic amines; and CYP3A4 has broader affinities

for a variety of lipophilic substrates (with molecular weights of 200 to 1,200 daltons). If P450 isoforms are tightly membrane bound, substrate access to the active sites is limited to compounds that can diffuse through the membranes, whereas different P450 isoforms that are less tightly bound will metabolize hydrophilic compounds.

In the past, the P450s were often referred to as having broad and overlapping specificities, but it became apparent that the broad substrate specificity could be attributed to multiple isoenzymic forms of P450. The phenotype of an individual with respect to the forms and amounts of individual P450s expressed in the liver can determine the rate and pathway of the metabolic clearance of a compound (see the discussion of genetic polymorphism). Significant differences exist between humans and other animal species with respect to the catalytic activities and regulation of expression of hepatic drug-metabolizing P450s. These differences often make it difficult to extrapolate results of P450-mediated metabolism studies performed experimentally in animal species to humans. Caution is warranted in the extrapolation of rodent data to humans, as some isoforms are similar between species (e.g., CYP1A and CYP3A subfamilies) whereas others are different (e.g., CYP2A, CYP2B, CYP2C, and CYP2D subfamilies).

The unique and diverse characteristics of the P450 ensure that predicting the metabolism of xenobiotics will be difficult. To date, no crystal structure for a mammalian membrane-bound P450 isoform has been described.

Other P450 Isoforms

A large number of other P450 isoforms catalyzing the oxidation of steroids, bile acids, fat-soluble vitamins, and other endogenous substances are shown in Table 4.3. Included are the following: CYP4, an ω -hydroxylase that hydroxylates the ω -methyl group of C20 to C22 fatty acids; CYP5, which causes platelet aggregation through synthesis of thromboxane A_2 ; CYP7A, which catalyzes the rate-determining step in the biosynthesis of bile acids from cholesterol; CYP7B, a brain-specific form of 7 α -hydroxylase that produces several neurosteroids; CYP8A, which regulates hemostasis through prostacyclin (PGI_2) synthesis and opposes the action of CYP5; CYP8B, which catalyzes bile acid biosynthesis; CYP11A1, which has steroid 20 α -hydroxylase and steroid 22-hydroxylase activity, the first steps in mitochondrial steroid biosynthesis that oxidatively cleave the 17-side chain of cholesterol to pregnenolone so that defects in this enzyme lead to a lack of glucocorticoids, feminization, and hypertension; CYP11B1, which is found in the inner mitochondrial membrane of the adrenal cortex and catalyzes the 11 β -hydroxylation of 11-deoxycortisol to hydrocortisone or 11-deoxycorticosterone to corticosterone; CYP11B2, which hydroxylates corticosterone at the 18-position to aldosterone; CYP17A1,

TABLE 4.3 Other Cytochromes with Indicated Specificities and Actions

CYP450s	Activity	Location	Action
CYP4	ω -hydroxylases		ω -Hydroxylation of fatty acids, leukotrienes, eicosanic acids
CYP5	thromboxane A ₂ synthase		Arachidonic acid to thromboxane A ₂ effecting platelet aggregation
CYP7A	7 α -hydroxylase		Cholesterol to bile acids formation
CYP7B	7 α -hydroxylase	Brain	Neurosteroids formation: 7 α -hydroxydehydroepiandrosterone and 7 α -hydroxypregnenolone
CYP8A	prostacyclin synthase		Synthesis of prostaglandin I ₂ regulating hemostasis
CYP8B	12 α -hydroxylase		Catalyzes bile acid biosynthesis
CYP11A1	steroid 20 α -/22-hydroxylase	Adrenal mitochondria	Catalyze mitochondrial steroid biosynthesis via 17-side chain cleavage of cholesterol to pregnenolone -lack of leads to feminization and hypertension
CYP11B1	11 β - hydroxylase	Adrenal cortex	Catalyzes 11-deoxycortisol to hydrocortisone or 11-deoxycorticosterone to corticosterone
CYP11B2	18-hydrolyase	Adrenal zona glomerulosa	Hydroxylates corticosterone at the 18-position leading to aldosterone
CYP17A1	17 α -hydroxylase and 17,20-lyase	ER of adrenal cortex	Production of testosterone and estrogen- lack of affects sexual development
CYP19A	Aromatase	ER of gonads, brain, adipose tissue	Aromatization of ring A of testosterone leading to estrogen-lack of causes estrogen deficiency
CYP21A1	C21 steroid hydroxylase	Adrenal cortex	17-hydroxyprogesterone to cortisol
CYP24	24-hydroxylase	Mitochondria	Catalyzes the degradation/inactivation of vitamin D metabolites
CYP26A1	<i>trans</i> -retinoic acid hydroxylase		Terminates retinoic acid signal
CYP26B1	retinoic acid hydroxylase		Catalyzes the hydroxylation of <i>cis</i> -retinoic acids
CYP26C	retinoic acid hydroxylase		Unknown
CYP27A1	27-hydroxylase		Catalyzes the oxidation of the cholesterol in bile acid biosynthesis
CYP27B1	vitamin D ₃ 1- α -hydroxylase	Mitochondria	Activates vitamin D ₃
CYP27C1			Unknown
CYP39			Catalyzes the 7-hydroxylation of 24-hydroxy cholesterol
CYP46	cholesterol 24-hydroxylase		Unknown
CYP51	lanosterol 14 α -demethylase		Converts lanosterol into cholesterol

which exhibits 17 α -hydroxylase and 17,20-lyase activities (two enzymes in one), which are required for production of testosterone and estrogen (lack of this enzyme affects sexual development at puberty); and CYP19A, an aromatase, which is found in the ER of gonads, brain, and adipose tissue and is a key steroidogenic enzyme that catalyzes ring A of testosterone aromatization to estrogen (lack of this enzyme causes estrogen deficiency and failure of females to develop at puberty). Aromatase inhibitors target estrogen-sensitive breast

cancers, and a number of environmental contaminants can act as aromatase inhibitors, thereby disrupting endocrine function in humans and wildlife through suppression of circulating estrogen levels. CYP21A1 has C21 steroid hydroxylase activity, and a lack of this enzyme prevents cortisol synthesis, thereby diverting excess 17-hydroxyprogesterone into overproduction of testosterone biosynthesis. CYP24 (mitochondrial 25-hydroxyvitamin D₃ 24-hydroxylase) catalyzes the degradation/inactivation of vitamin D metabolites; CYP26A1

is involved in terminating the retinoic acid signal and thus turning off a developmental switch. CYP26B1 catalyzes the hydroxylation of *cis*-retinoic acids, which are not recognized by the CYP26A1. CYP26C, a retinoic acid hydroxylase, has an unknown function. CYP27A1, a 27-hydroxylase, catalyzes the oxidation of the cholesterol 17-side chain as the first step in bile acid biosynthesis to the feedback inhibitors cholic acid and chenodeoxycholic acid and 25-hydroxyvitamin D₃. CYP27B1 (mitochondrial vitamin D₃ 1- α -hydroxylase) activates vitamin D₃. CYP27C1 has an unknown function. CYP39 catalyzes the 7-hydroxylation of 24-hydroxy cholesterol with unknown function. CYP46 (cholesterol 24-hydroxylase) has an unknown function. CYP51 (lanosterol 14 α -demethylase) converts lanosterol into cholesterol, which is the enzyme inhibited by ketoconazole.

Cytochrome P450 Isoforms Metabolizing Drugs/Xenobiotics (18–20)

Figure 4.3 shows the participation (%) of hepatic P450 isoforms in the metabolism of drugs and xenobiotics (19–21). P450s are the most important enzymes responsible for Phase 1 drug metabolism. The polymorphic nature of the P450s influences individual drug responses and drug–drug interactions and induces adverse drug reactions. The crystal structures of eight mammalian P450s (CYP1A2, CYP2B4, CYP2C5, CYP2A6, CYP2D6, CYP2C8, CYP2C9, and CYP3A4) have been published. Outstanding is the fact that approximately two-thirds of all drugs are metabolized by one isoform, CYP3A4, increasing the potential for drug–drug interactions. When two drugs are metabolized by the same isoform, only one drug can serve as a substrate at a time, increasing the likelihood of a drug–drug interaction, especially if one drug has a low therapeutic threshold.

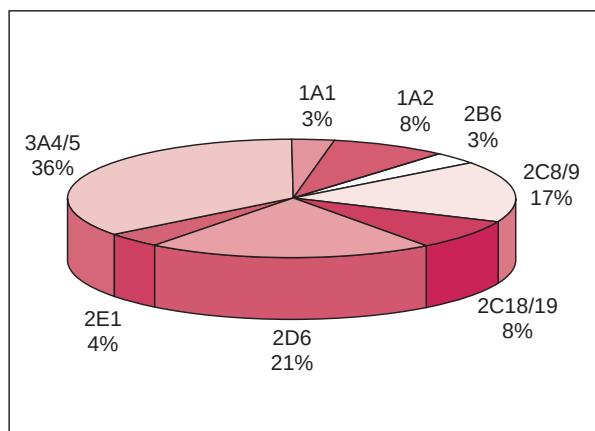


FIGURE 4.3 Percentage of clinically important drugs metabolized by human P450 isoforms.

Family 1

Family 1 consists of three subfamilies (CYP1A1, CYP1A2, and CYP1B1), three genes, and one pseudogene. The human CYP1A subfamily has an integral role in the metabolism of estrogens and two important classes of environmental carcinogens, polycyclic aromatic hydrocarbons (PAHs) and arylamines (Tables 4.4 and 4.5) (22). The PAHs are commonly present in the environment from industrial combustion processes and tobacco products. Several potent carcinogenic arylamines result from the pyrolysis of amino acids in cooked meats and cause colon cancer in rats. Environmental and genetic factors can alter the expression of this subfamily of enzymes.

CYP1A1 The CYP1A1 subfamily (also called aromatic hydrocarbon hydroxylase) is expressed primarily in extrahepatic tissues, small intestine, placenta, skin, and lung as well as in the liver in response to the presence of

TABLE 4.4 Some Substrates and Reaction Type for Human Subfamily CYP1A2^a

Acetaminophen (imino quinone)
Amitriptyline (<i>N</i> -demethylation)
Caffeine (<i>N</i> ¹ - and <i>N</i> ³ -demethylation)
Chlordiazepoxide
Cinacalcet
Clomipramine (<i>N</i> -demethylation)
Clopidogrel
Clozapine
Cyclobenzaprine
Desipramine (<i>N</i> -demethylation)
Diazepam
Duloxetine
Erlotinib
Estradiol (2- and 4-hydroxylation)
Fluoroquinolones (3'-hydroxylation of piperazine ring)
Flutamide
Fluvoxamine
Haloperidol
Imipramine (<i>N</i> -demethylation)
Levobupivacaine
Mexiletine
Mirtazapine
Naproxen
Nortriptyline
Olanzapine
Ondansetron
Propafenone
Propranolol
Ramelteon
Riluzole
Ropinirole
Ropivacaine
Tacrine
Theophylline
Tizanidine
Verapamil
R-warfarin
Zileuton
Zolmitriptan

^aDrugs in bold italic have been reported to cause drug–drug interactions.

TABLE 4.5 Some Procarcinogens and Other Toxins Activated by Human Cytochrome P450s

CYP1A1	CYP1A2	CYP2E1	CYP3A4
Benzo[a]pyrene and other polycyclic aromatic hydrocarbons	4-Aminobiphenyl 2-Naphthylamine 2-Aminofluorene 2-Acetylaminofluorene 2-Aminoanthracene Heteropolycyclic amines (2-aminoquinolines) Aflatoxin B1 Ipomeanol	Benzene Styrene Acrylonitrile Vinyl bromide Trichloroethylene Carbon tetrachloride Chloroform Methylene chloride N-nitrosodimethylamine 1,2-Dichloropropane Ethyl carbamate	Aflatoxin B1 Aflatoxin G1 Estradiol 6-Aminochrysene Polycyclic hydrocarbon Dihydrodiols

CYP1A1 inducers such as PAHs (i.e., in cigarette smoke and the carcinogen 3-methylcholanthrene), α -naphthoflavone (a noncarcinogenic inducer related to dietary flavones), and indole-3-carbinol (found in Brussels sprouts and related vegetables). CYP1A1 metabolizes a range of PAHs, including a large number of procarcinogens and promutagens (Table 4.5). Diethylstilbestrol and 2- and 4-hydroxyestradiol (catecholestrogens) are oxidized by CYP1A1 to their quinone analogues, which are normally reduced to inactive metabolites (23). In the absence of a detoxifying reduction step, however, the quinones can accumulate and initiate carcinogenic processes or cell death by covalently damaging DNA or cellular proteins. Interindividual variations in the inducible expression of CYP1A1 might be related to genetic differences in aromatic hormone (Ah) receptor expression, which could explain differences in individual susceptibilities to cigarette smoke-induced lung cancer. Therefore, genetic factors appear to be important in the expression of the *CYP1A1* gene in humans and its involvement in human carcinogenesis. Women who smoke are at greater risk than men to develop lung cancer (adenocarcinoma) and chronic obstructive pulmonary diseases. The mechanism for induction of the *CYP1A1* gene begins with binding of the inducing agents to a cytosolic receptor protein, the Ah receptor, which is then translocated to the nucleus where it binds to the DNA of the *CYP1A1* gene, thus enhancing the rate of transcription. The presence of the Ah receptor in hepatic and intestinal tissues can have implications beyond xenobiotic metabolism and can have a role in the induction of other genes for the control of cellular growth and differentiation. On the other hand, CYP1A1 can metabolize procarcinogens to hydroxylated inactive compounds that are not mutagenic. The question of how the bowel protects itself from ingested compounds known to be activated by CYP1A1 (i.e., PAH) remains unanswered (22).

CYP1A2 The CYP1A2 subfamily (also known as phenacetin *O*-deethylase, caffeine demethylase, or anti-pyrene *N*-demethylase) is one of the major P450s in human liver that catalyzes the oxidation (and in some

cases the bioactivation) of arylamines, nitrosamines, and aromatic hydrocarbons and the bioactivation of promutagens and procarcinogens, caffeine, clozapine, tacrine, theophylline, estrogens, and other substances (Tables 4.4 and 4.5). CYP1A2 is expressed in the liver (13% to 40%), intestine, and stomach and is induced by smoking, PAHs, and isosafrole (a noncarcinogenic dietary compound). CYP1A2 is primarily responsible for activation of the carcinogen aflatoxin B1 in ordinary human exposure and of the pneumotoxin ipomeanol. CYP1A2 is subject to reversible and/or irreversible inhibition by a number of drugs, natural substances, and other compounds. *CYP1A2* gene expresses a 515-residue protein with a molecular mass of 58.3 kDa. The CYP1A2 structure exhibits a relatively compact, planar active site cavity that is highly adapted for the size and shape of its substrates. A large interindividual variability in the expression and activity of CYP1A2 has been observed, which is largely caused by genetic, epigenetic, and environmental factors (e.g., smoking). CYP1A2 is primarily regulated by the aromatic hydrocarbon receptor (AhR). Induction or inhibition of CYP1A2 may provide partial explanation for some clinical drug interactions. More than 15 variant alleles of the *CYP1A2* gene have been identified, and some of them have been associated with altered drug clearance and response and disease susceptibility. Evidence for polymorphism of this isoform has been reported, and it is likely that low CYP1A2 activity will be associated with altered susceptibility to the bioactivation of procarcinogens, promutagens, and other xenobiotics known to be substrates for this enzyme. The expression of the *CYP1A2* gene in the stomach becomes an important issue for gastric carcinogenesis induced by smoking and by the metabolic activation of the procarcinogenic arylamines to mutagens (22). Clinical studies have suggested that the *N*-demethylation of imipramine is greater in smokers than in nonsmokers.

Family 2

The human CYP2A family comprises 16 subfamilies (CYP2A6, CYP2A7, CYP2A13, CYP2B6, CYP2C8, CYP2C9,

CYP2C18, CYP2C19, CYP2D6, CYP2E1, CYP2F1, CYP2J2, CYP2R1, CYP2S1, CYP2U1, CYP2W1), 16 genes, and 16 pseudogenes. CYP2A7 is nonfunctional. CYP2A13 is highly active in the metabolic activation of a major tobacco-specific carcinogen, 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone, in the human respiratory tract, with a catalytic efficiency much greater than that of other CYP2A. Expressed CYP2A13 is more active than CYP2A6 in the metabolic activation of hexamethylphosphoramide, *N,N*-dimethylaniline, 2'-methoxyacetophenone, and *N*-nitrosomethylphenylamine, but is much less active than CYP2A6 in coumarin 7-hydroxylation.

CYP2A6 CYP2A6 has a low level of hepatic expression and represents only approximately 4% of the total of hepatic P450 isoforms (Fig. 4.2). It is also expressed at low levels in lung and nasal epithelium and exhibits high interindividual variability (polymorphism). Lung CYP2A6 bioactivates many tobacco smoke-specific carcinogens, procarcinogens, and nitrosamines. Mutant alleles of this isoenzyme have been associated with an elevated risk for small-cell lung cancer. Hepatically, CYP2A6 is the major enzyme catalyzing the 7-hydroxylation of coumarin (coumarin 7-hydroxylase) and C5 oxidation of nicotine to cotinine (nicotine C-poxidase) (Table 4.6). Other substrates include the hydroxylation of aflatoxin B1, naproxen, tacrine, clozapine, mexiletine, and cyclobenzaprine. In *in vitro* studies with human microsomes and expressed human CYP2A6, selegiline and its desmethyl metabolite are mechanism-based CYP2A6 inhibitors of nicotine metabolism. Thus, inhibition of nicotine metabolism by selegiline could increase plasma nicotine *in vivo*. There is overlapping substrate specificity with CYP2A13. CYP2A6 exhibits polymorphism, with an incidence of 2% in the Caucasian population. This population is characterized as poor metabolizers. Smokers with a defective *CYP2A6* gene smoke fewer cigarettes, implicating a genetic factor in nicotine dependence.

CYP2A13 CYP2A13 is expressed highest in nasal epithelium, followed by lung and trachea, with high interindividual variability. CYP2A13 activates tobacco smoke-specific carcinogens, procarcinogens, and nitrosamines into DNA-altering compounds that cause lung cancer. Individuals with a high level of CYP2A13 expression could likely have an increased risk of developing tobacco smoking-related lung cancers. Although CYP2A13 and CYP2A6 are 93.5% identical in amino acid sequences, CYP2A13 exhibits higher affinity for metabolizing tobacco smoke carcinogens than does CYP2A6. A comparison of the crystal structures of CYP2A13 and CYP2A6 has shown them to be very similar, and like CYP2A6, the CYP2A13 active site cavity is small and highly hydrophobic with a cluster of phenylalanine residues in the active site. Amino acid differences

TABLE 4.6 Some Substrates for CYP2A6

Acetaminophen	Ifosfamide
Aminopyrine	Methoxsalen
Caffeine	Nicotine
Cinnarizine	Omeprazole
Cisapride	Phenacetin
Clomethiazole	Pilocarpine
Cyclophosphamide	Promazine
Disulfiram	Propofol
Efavirenz	Tamoxifen
Etodolac	Tretinoin
Fenchone	Verbenone
Flunarizine	Zidovudine
Halothane	

Adapted from Di YM, Chow VD, Yang LP, et al. Structure, function, regulation and polymorphism of human cytochrome CYP2A6. *Curr Drug Metab*. 2009;10:754–780.

between CYP2A6 and CYP2A13 at key positions could be the cause of the significant variations observed in ligand binding and catalysis.

CYP2B6 The human CYP2B6 subfamily has not been as fully characterized at the molecular level as have other members of the human P450 family. This family represents less than 1% of the total of hepatic P450 isoforms, and its level of expression mainly in the liver is low and may be co-regulated with CYP3A4. CYP2B6 has been thought to have an insignificant role in human drug metabolism. This enzyme exhibits polymorphic and ethnic differences in CYP2B6 expression and therefore could be important in drug–drug interactions. CYP2B6 can metabolize approximately 8% of clinically used drugs including cyclophosphamide, ifosfamide, tamoxifen, ketamine, artemisinin, nevirapine, efavirenz, bupropion, sibutramine, nicotine, sertraline, and propofol (Table 4.7). CYP2B6 also metabolizes arachidonic acid, lauric acid, 17 β -estradiol, estrone, ethinylestradiol, and testosterone and can also bioactivate several procarcinogens and toxicants. The crystal structure of CYP2B6 has not been resolved, whereas several pharmacophore and homology models of human CYP2B6 have been reported. CYP2B6 is closely regulated by the androstane receptor, which can activate CYP2B6 expression upon ligand binding. Pregnane X receptor and glucocorticoid receptor also have a role in the regulation of CYP2B6. Induction of CYP2B6 may partially explain some clinical drug interactions observed. For example, coadministered carbamazepine decreases the blood levels of bupropion. There is a wide interindividual variability in the expression and activity of CYP2B6. Such a large variability is probably a result of the effects of genetic polymorphisms and exposure to drugs that are inducers or inhibitors of CYP2B6. CYP2B6 substrates are by and large small nonplanar molecules, neutral or weakly basic, highly lipophilic with one or two hydrogen-bond acceptors, and frequently active in the CNS. Phenobarbital appears to induce its formation.

TABLE 4.7 Some Substrates for Human Subfamily CYP2B6

Amitriptyline	Diclofenac	Meperidine	Sevoflurane
Artemisinin	Diphenhydramine	Methadone	Sibutramine
Bupropion	Disulfiram	Methoxyflurane	S-mephenytoin
Carbamazepine	Efavirenz	Mianserin	S-mephobarbital
Carbaryl	Ethylbenzene	Nevirapine	Tamoxifen
Cinnarizine	Fenchone	Nicotine	Temazepam
Cisapride	Fluoxetine	Nor-LAAM	Thiotepa
Clomethiazole	Haloperidol	Prasugrel	Tramadol
Clopidogrel	Ifosfamide	Promazine	Tretinoin
Cyclophosphamide	Imipramine	Propofol	Valproic acid
Desmethylelegiline	Ketamine	Selegiline	Verapamil
Diazepam	L-Acetyl- α -methadol	Sertraline	

Adapted from Mo SL, Liu YH, Duan W, et al. Substrate specificity, regulation, and polymorphism of human cytochrome CYP2B6. *Curr Drug Metab.* 2009;10:730–53.

CYP2C The CYP2C subfamily is the most complex family, consisting of three subfamilies (CYP2C8, CYP2C9, and CYP2C19), metabolizing approximately 25% of the clinically important drugs (Fig. 4.3), including *S*-warfarin, *S*-mephenytoin, and tolbutamide (Table 4.8). The CYP2C subfamily represents approximately 20% of the total of P450 isoforms in the liver (Fig. 4.2). CYP2C8 is expressed primarily in extrahepatic tissues (kidney, adrenal, brain, uterus, breast, ovary, and intestine) and metabolizes the tricyclic antidepressants, diazepam and verapamil. Its level of expression is less than that of the other CYP2C9 and CYP2C19 subfamilies.

CYP2C9 metabolizes approximately 15% of clinical drugs, including the methyl hydroxylation of tolbutamide and glyburide; 4'-hydroxylation of diclofenac, celecoxib, and phenytoin; oxidation of the hydroxymethyl group of losartan to a carboxylic acid; and 7-hydroxylation of *S*-warfarin and many nonsteroidal anti-inflammatory drugs (NSAIDs) (Table 4.8). CYP2C9 is highly polymorphic. Some natural and herbal compounds are also metabolized by CYP2C9, probably leading to the formation of toxic metabolites. CYP2C9 also metabolizes endogenous compounds such as steroids, melatonin, retinoids, and arachidonic acid. Many CYP2C9 substrates are weak acids, but CYP2C9 has also the capacity to metabolize neutral, highly lipophilic compounds. Ligand-based and homology models of CYP2C9 have been reported, which have provided insight into how its substrates are bound to the active site of CYP2C9. The resolved crystal structure of CYP2C9 has confirmed the importance in substrate specificity and ligand orientation. For example, CYP2C9 is activated by dapsone and its analogues and *R*-lansoprazole in a stereospecific and substrate-dependent manner, probably through binding to the active site and inducing positive cooperativity. CYP2C9 is subject to induction by rifampin, phenobarbital, and dexamethasone, indicating the involvement of pregnane X receptor, the androstane receptor, and the glucocorticoid receptor in the regulation of CYP2C9. A number of compounds inhibit CYP2C9, which could

provide an explanation for some clinically important drug interactions. Tienilic acid, suprofen, and silybin are mechanism-based inhibitors of CYP2C9. Given the importance of CYP2C9 in drug metabolism and the presence of polymorphisms, it is important to identify drugs as potential substrates, inducer, or inhibitors of CYP2C9 in drug–drug interactions.

CYP2C19 is found primarily in the liver and intestine. The expression of CYP2C19 in the liver is less than that of CYP2C9 and also exhibits polymorphism (differences in the DNA sequence for the *CYP2C* gene), which changes the enzyme's ability to metabolize these substrates (i.e., poor metabolizer phenotype). Because of this genetic difference in expressing CYP2C isoforms, it is important to be aware of a person's race when prescribing drugs that are metabolized differently by different racial populations (see the section concerning genetic polymorphism). CYP2C19 (*S*-mephenytoin hydroxylase) is the isoform associated with the 4'-hydroxylation of *S*-mephenytoin. The CYP2C subfamily is apparently not inducible in humans.

CYP2D6 CYP2D6 polymorphism is, perhaps, the most studied P450 (see the section on genetic polymorphism). This enzyme is responsible for at least 30 different drug oxidations, representing approximately 21% of clinically important drugs (Fig. 4.3). CYP2D6 is only 3% expressed in the liver and minimally expressed in the intestine, and it does not appear to be inducible (Fig. 4.2). Because there might not be any other way to clear drugs metabolized by CYP2D6 from the system, poor metabolizers of CYP2D6 substrates can be at severe risk for adverse drug reactions or drug overdoses. The metabolism of debrisoquine by CYP2D6 is one of the most studied examples of metabolic polymorphism, with the molecular basis for defective metabolism well understood (Table 4.9) (see the section on genetic polymorphism). This isoform metabolizes a wide variety of lipophilic amines and is probably the only P450 for which a charged or ion-pair interaction is important for substrate binding. It appears

TABLE 4.8 Some Substrates and Reaction Type for Human Subfamily CYP2C^a

CYP2C8	CYP2C9	CYP2C9 (con't)	CYP2C19
Amiodarone	Amitriptyline	Losartan (hydroxymethyl to carboxylic)	Amitriptyline
Amodiaquine	Bosentan	Mefenamic acid	Carisoprodol
Benzphetamine	Buprenorphine	Meloxicam	Cilostazol
Carbamazepine	Carvedilol	(R)-Mephenytoin	Citalopram
Cerivastatin	Celecoxib	Mestranol	Clomipramine
Docetaxel	Chlorpheniramine	Montelukast	Clotidogrel
Fluvastatin	Chloramphenicol	Naproxen	Cyclophosphamide
Isotretinoin	Clomipramine	Nateglinide	Desipramine
Paclitaxel	Chlorpropamide	Omeprazole	Diazepam (N-demethylation)
Phenytoin	Clotidogrel	Phenobarbital	Escitalopram
Pioglitazone	Dapsone	Phenytoin (4'-hydroxylation)	Esomeprazole
Repaglinide	Desogestrel	Phenylbutazone (4-hydroxylation)	Formoterol
Retinol	Diclofenac (4'-hydroxylation)	Piroxicam	Hexobarbital
Rosiglitazone	Diazepam	Prasugrel	Imipramine (N-demethylation)
Tolbutamide	Diphenhydramine	Ramelteon	Indomethacin
Torsemide	Donepezil	Rosiglitazone	Lansoprazole
Verapamil	Dorzolamide	Rosuvastatin	Loratidine (N-decarboethoxylation)
Zopiclone	Dronabinol	Sildenafil	(S)-Mephenytoin (4'-hydroxylation)
	Eletriptan	Sulfamethoxazole	(R)-Mephenytoin (N-demethylation)
	Etodolac	Sulfinpyrazone (aromatic hydroxylation)	(R)-Mephobarbital
	Fluoxetine	Suprofen	Moclobemide
	Flurbiprofen (4'-hydroxylation)	Tamoxifen	Nelfinavir
	Fluvastatin	Tenoxicam	Nilutamide
	Formoterol	Terbinafine	Omeprazole (hydroxylation)
	Glibenclamide	Tienilic acid (thiophene ring hydroxylation)	Pantoprazole
	Gliclazide	Tolbutamide (p-methylhydroxylation)	Pentamidine
	Glyburide	Tolterodine	Phenobarbital
	Glimepiride	Torsemide	Phenytoin (ring hydroxylation)
	Glipizide	Δ^1 -THC (7-hydroxylation)	Progesterone
	Glyburide	Testosterone (16 α -hydroxylation)	Proguanil (cyclization)
	Halothane	Trimethadione	Propranolol (side chain hydroxylation)
	Hexobarbital (3'-hydroxylation)	Valdecocix	Rabeprazole
	Ibuprofen (i-butylmethyl-hydroxylation)	Vardenafil	Teniposide
	Imipramine	Valproic acid	Thioridazine
	Indomethacin	Valsartan	Progesterone
	Irbesartan	Voriconazole	Tolbutamide
	Irinotecan	S-Warfarin (7'-hydroxylation)	Voriconazole
	Ketamine	Zafirlukast	R-Warfarin
	Lomoxicam	Zileuton	
		Zopiclone	

^aDrugs in bold italic have been reported to cause drug-drug interactions.

also to preferentially catalyze the hydroxylation of a single enantiomer (stereoselectivity) in the presence of enantiomeric mixtures. Quinidine is an inhibitor of CYP2D6, and concurrent administration with CYP2D6 substrates results in increased blood levels and toxicities for these substrates. If the pharmacologic action of the CYP2D6 substrate depends on the formation of active metabolites, quinidine inhibition results in a lack of a therapeutic response. The interaction of two substrates for CYP2D6 can prompt a number of clinical responses. For example, when substrates have unequal affinities for CYP2D6, the first-pass hepatic metabolism of a substrate (drug) with high affinity for CYP2D6 will inhibit binding of a second substrate that has a lower affinity for this enzyme. The result of this will be a rapid absorption of the second unmetabolized substrate, leading to a higher plasma concentration and to the increased potential for adverse reaction or toxicity.

CYP2E1 Few drugs are metabolized by CYP2E1; however, this isoform has a major role in the metabolism of numerous halogenated hydrocarbons (including volatile general anesthetics) and of a range of low molecular weight organic compounds including dimethylformamide, acetonitrile, acetone, ethanol, and benzene, as well as in the activation of acetaminophen to its reactive metabolite, *N*-acetyl-*p*-benzoquinoneimine (Table 4.10) (24,25). CYP2E1 is of most interest as a result of the oxidation of ethanol into its reactive products, acetaldehyde and 1-hydroxyethyl radicals, and its ability to activate small molecular weight products into electrophilic reactive metabolites, which result in toxicity and carcinogenicity. This enzyme then oxidizes acetaldehyde produced from ethanol to acetic acid. This isoform is only 7% expressed in the liver and is also expressed in the kidney, intestine, and lung. CYP2E1 is inducible under diverse pathophysiologic conditions including

TABLE 4.9 Some Substrates and Reaction Type for Human CYP2D6 Isoform^a

Alprenolol (4-hydroxylation)	Dexfenfluramine	Lidocaine (3-hydroxylation)	Pindolol
Amitriptyline (10-hydroxylation)	Dextromethorphan (O-demethylation)	Maprotiline	Promethazine (ring hydroxylation, S-oxidation, N-demethylation)
Amphetamine	Diphenhydramine (N-demethylation, ring hydroxylation, cleavage ether bond)	Meperidine	
Aripiprazole	Dolasetron (hydroxylation of indole ring)	Methadone	Propafenone (4-hydroxylation)
Atomoxetine	Donepezil	Methamphetamine	Propoxyphene
Bifuralol (1'-hydroxylation)	Doxepin	Methoxyamphetamine (4-hydroxylation, N-demethylation)	Propranolol (4'-hydroxylation)
Bisoprolol	Duloxetine	Metoclopramide	Quetiapine
Captopril	Encainide (N-demethylation, O-demethylation)	Metoprolol (O-demethylation)	Quinidine (hydroxylation)
Carvedilol	Fenfluramine	Mexiletine (4-hydroxylation and methyl hydroxylation)	Ranolazine
Cevimeline	Fluphenazine	Minaprine	Risperidone
Chlorpheniramine (N-demethylation, ring hydroxylation, deamination)	Fentanyl	Mirtazapine	Ritonavir
Chlorpromazine	Flecainide (O-dealkylation)	Morphine	Sertraline
Chlorpropamide	Fluoxetine (N-dealkylation)	Nebivolol	S-Metoprolol
Cinacalcet	Fluvoxamine	Nortriptyline (10-hydroxylation)	Sparteine (N-oxidation)
Clemastine	Formoterol	Olanzapine	Tamoxifen
Clomipramine (hydroxylation)	Galantamine	Ondansetron (hydroxylation of indole ring)	Thioridazine (aromatic hydroxylation)
Clozapine (aromatic hydroxylation)	Guanoxan (6- and 7-hydroxylation)	Oxycodone (O-demethylation)	Timolol (O-dealkylation)
Codeine (O-demethylation)	Haloperidol	Paroxetine	Tolterodine (2-hydroxylation)
Cyclobenzaprine	Hydrocodone	Perhexiline (4'-hydroxylation)	Tramadol
Darifenacin	Hydroxyzine (ring hydroxylation)	Perphenazine (aromatic hydroxylation)	Trazodone
Debrisoquine (4-hydroxylation)	Imipramine (2-hydroxylation)	Phenacetin	Tripelennamine
Desipramine	Indoramin (6-hydroxylation)	Phenformin (4-hydroxylation)	Tropisetron (hydroxylation of indole ring)
			Venlafaxine

^aDrugs in bold italic have been reported to cause drug–drug interactions.

diabetes, obesity, fasting, cancer, alcoholic liver disease, and nonalcoholic liver disease, as well as by the chemical substances ethanol, isoniazid, 4-methylpyrazole, and other chemicals (see Table 4.14). Mitochondrial CYP2E1 induces the formation of reactive oxygen species (ROS), which is associated with oxidative stress and lipid peroxidation and augments ethanol-induced hepatocellular injury. CYP2E1 is also known as the microsomal ethanol-oxidizing system, benzene hydroxylase, or aniline hydroxylase. CYP2E1 is induced in alcoholics, and there is a polymorphism associated with this isoform, which is most common in Chinese people. This isoform also appears to be involved in smoking-induced cancer (c.f., CYP1A2). Most of the compounds that induce CYP2E1 are also substrates for this enzyme. The induction of this enzyme in humans can cause enhanced susceptibility to

the toxicity and carcinogenesis of CYP2E1 substrates. Some evidence shows interindividual variation in the in vitro liver expression of this isoform. Ketogenic diets (increased serum ketone levels), including those deficient in carbohydrates or high in fat, are known to enhance the metabolism of halogenated hydrocarbons in rats (25). The mechanism of induction appears to be a combination of an increase in CYP2E1 transcription, mRNA translation efficiency, and stabilization of CYP2E1 against proteolytic degradation. The induction of CYP2E1 resulting from ketosis (i.e., due to starvation, a high-fat diet, uncontrolled diabetes, or obesity) or from exposure to alcoholic beverages or other xenobiotics can be detrimental to individuals simultaneously exposed to halogenated hydrocarbons (increased hepatotoxicity as a result of exposure to halothane or chloroform). Chronic

TABLE 4.10 Some Substrates and Reaction Type for Human CYP2E1 Isoform^a

Acetaminophen (<i>p</i> -benzoquinone imine)
Styrene (epoxidation)
Theophylline (C-8 oxidation)
<i>Disulfiram</i>
Halogenated Hydrocarbons
Dehalogenation of chloroform, methylene chloride
Volatile Anesthetics (fluorinated hydrocarbons)
Enflurane, halothane, methoxyflurane, sevoflurane, desflurane
Miscellaneous Organic Solvents
Ethanol (to acetaldehyde)
Glycerin
Dimethylformamide (<i>N</i> -demethylation)
Acetone
Diethylether
Benzene (hydroxylation)
Aniline (hydroxylation)
Acetonitrile (hydroxylation to cyanohydrin)
Pyridine (hydroxylation)

^aDrugs in bold italics have been reported to cause drug–drug interactions.

alcohol intake is known to enhance the hepatotoxicity of halogenated hydrocarbons. Testosterone appears to regulate CYP2E1 levels in the kidney, and pituitary growth hormone regulates hepatic levels of CYP2E1. Kidney damage in response to halocarbon exposure was greater in male rats than in female rats. This finding could have implications for sexual differences in the nephrotoxicity of CYP2E1 substrates in humans.

Family 3

The human CYP3 family comprises four subfamilies CYP3A4, CYP3A5, CYP3A7, CYP3A43, four genes, and two pseudogenes.

CYP3A4 The CYP3A4 subfamily includes the most abundantly expressed P450s in the human liver and intestine (extrahepatic metabolism). Although CYP3A4 is responsible for approximately two-thirds of CYP3A-mediated drug metabolism, the other minor isoforms (CYP3A5, CYP3A7, and CYP3A43) also contribute. CYP3A5 is the best studied of the minor CYP3A isoforms, with 85% sequence identity with CYP3A4. Approximately 20% of human livers express CYP3A5. The expression of CYP3A5 shows ethnic differences, with the wild-type CYP3A5*1 allele more common in Africans than in Caucasians or Asians. In individuals who express CYP3A5, 17% to 50% of the total hepatic CYP3A is this isoform. Additionally, CYP3A5 is also expressed in a range of extrahepatic tissues (kidney, small intestine, lung, adrenal glands) and is inducible via the pregnane X receptor. Both CYP3A4 and CYP3A5 exhibit significant overlap in substrate specificity but can differ in catalytic activity and regioselectivity. Results from a comparison of CYP3A4 and CYP3A5 enzyme kinetics indicate that CYP3A5 has different enzymic characteristics than CYP3A4 in some CYP3A-catalyzed

reactions. The enzyme kinetics of CYP3A5 suggest faster substrate turnover than that observed with CYP3A4.

Approximately one-third of the total P450 in the liver and two-thirds in the intestine is CYP3A4. This isoform is responsible for the metabolism of testosterone and more than one-third of the clinically important drugs. CYP3A4 is expressed in the intestine, lung, placenta, kidney, uterus, and brain and is glucocorticoid-inducible. CYP3A7 is predominantly expressed in fetal liver (~50% of total fetal P450 enzymes) but is also found in some adult livers and extrahepatically. CYP3A7 has a specific role in the hydroxylation of retinoic acid, 16 α -hydroxylation of steroids, and hydroxylation of allylic and benzylic carbons, and therefore, it is relevant in both normal development and in carcinogenesis. The most recently discovered CYP3A isoform is CYP3A3. In addition to a low level of liver expression, it is expressed in prostate and testis. Its substrate specificity is currently unclear. Polymorphisms that predict the absence of active enzyme have been identified.

The CYP3A4 subfamily metabolizes approximately 50% to 60% of clinically important drugs (Table 4.11) and is inhibited by a number of xenobiotics (Table 4.12). CYP3A4 also appears to activate aflatoxin B1 and, possibly, to metabolize benzo[a]pyrene. The interindividual differences reported for metabolism of nifedipine, cyclosporine, triazolam, and midazolam are probably related to changes in induction and not to polymorphism. Binding of CYP3A is predominantly lipophilic (14). Drugs known to be substrates for CYP3A4 have low and variable oral bioavailabilities that might be explained by prehepatic metabolism by a combination of intestinal CYP3A4 and P-glycoprotein in the enterocytes of the intestinal wall (see the section on oral bioavailability). Therefore, it is the expression and function of CYP3A4 that governs the rate and extent of metabolism of the substrates for the CYP3A subfamily. The induction of the CYP3A subfamily by phenobarbital in humans could ultimately be responsible for many of the well-documented interactions between barbiturates and other drugs (19,20).

Family 4

The human CYP4 family consists of 6 subfamilies and 13 CYP4 proteins, of which the CYP4A, CYP4B, and CYP4F subfamilies have been most studied. The CYP4A subfamily (fatty acid hydroxylases) catalyzes the hydroxylation of the terminal ω -carbon and, to a lesser extent, the ω -1 position of saturated and unsaturated fatty acids, as well as the ω -hydroxylation of various prostaglandins. The *CYP4A1*, *CYP4A2*, and *CYP4A3* genes are expressed mostly in liver and kidney, and their expression is induced by a class of chemicals known as peroxisome proliferators, which includes the hypolipidemic drug, clofibrate. Induction of CYP4A expression by clofibrate is due to transcriptional activation, mediated possibly via a peroxisome proliferator-activated receptor (PPAR). *CYP4A* gene expression is hormonally regulated. There is a close association between CYP4A1 induction, peroxisome proliferation, and induction of the peroxisomal fatty acid metabolizing

TABLE 4.11 Substrates for Human CYP3A4 Isoform^a

<i>Alfentanil</i>	Citopram	Efavirenz	Lansoprazole	Prasugrel	<i>Prednisone</i>	<i>Zileuton</i>
Alfuzosin	Clarithromycin	<i>Eplerenone</i>	Letrozole	Propranolol	<i>Prednisolone</i>	<i>Ziprasidone</i>
Almotriptan	Clindamycin	<i>Ergotamine</i>	Lercanidipine	<i>Quetiapine</i>	Dexamethasone	Zolpidem
<i>Alprazolam</i>	Clomipramine	<i>Erlotinib</i>	<i>Lidocaine</i>	<i>Quinidine</i>	<i>Sunitinib</i>	<i>Zonisamide</i>
Amitriptyline	Clonazepam	<i>Erythromycin</i>	Lopinavir	Quinine	<i>Tacrolimus</i>	
Amiodarone	<i>Clopidogrel</i>	Esomeprazole	Loratidine	Rabeprazole	<i>Tadalafil</i>	
<i>Amlodipine</i>	Clozapine	<i>Eszopiclone</i>	<i>Lovastatin</i>	<i>Ramelteon</i>	<i>Tamoxifen</i>	
Amprenavir	Cocaine	<i>Ethinylestradiol</i>	<i>Methadone</i>	<i>Ranolazine</i>	<i>Telithromycin</i>	
Aprepitant	Codeine	<i>Ethosuximide</i>	<i>Midazolam</i>	Repaglinide	Temazepam	
Aripiprazole	<i>Colchicine</i>	Etonogestrel	<i>Mifepristone</i>	Rifampin	Δ^1 -THC	
Astemizole	Cyclophosphamide	<i>Etoposide</i>	<i>Mirtazepine</i>	<i>Rifabutin</i> - related	Theophylline	
<i>Atazanavir</i>	<i>Cyclosporine</i>	Exemestane	<i>Modafinil</i>	compounds	<i>Tiagabine</i>	
<i>Atorvastatin</i>	Dapsone	<i>Felodipine</i>	<i>Mometasone</i>	<i>Ritonavir</i>	Tinidazole	
Azole antifungals	<i>Darifenacin</i>	<i>Fentanyl</i>	<i>Montelukast</i>	Salmeterol	<i>Tipranavir</i>	
Bepiridil	Delavirdine	Fexofenadine	Nateglinide	<i>Saquinavir</i>	<i>Tolterodine</i>	
Bexarotene	Desogestrel	Finasteride	<i>Nelfinavir</i>	<i>Sertraline</i>	Toremifene	
Bromocriptine	Dextromethorphan	Flutamide	Nevirapine	Sibutramine	Tramadol	
Budesonide	Diazepam	<i>Fluticasone</i>	<i>Nicardipine</i>	<i>Sildenafil</i>	<i>Trazodone</i>	
Buprenorphine	<i>Dihydroergotamine</i>	Galantamine	<i>Nifedipine</i>	<i>Simvastatin</i>	<i>Triazolam</i>	
Buspirone	<i>Diisopyramide</i>	Gleevec	<i>Nisoldipine</i>	<i>Sirolimus</i>	Trimetrexate	
Cafegot	Diltiazem	Haloperidol	Nitrendipine	<i>Solifenacin</i>	Valdecoxib	
Caffeine	Docetaxel	Hydrocodone	Norethindrone	Sorafenib	Valproic acid	
Cannabinoids	Dofetilide	Imatinib	<i>Omeprazole</i>	Steroids	<i>Vardenafil</i>	
<i>Carbamazepine</i>	Dolasetron	Imipramine	<i>Ondanestron</i>	Testosterone	<i>Verapamil</i>	
Cerivastatin	Domperidone	<i>Indinavir</i>	<i>Oral contraceptives/</i>	Progesterone	<i>Vinblastine</i>	
Cevimeline	Donepezil	Irinotecan	<i>progestins</i>	<i>Estradiol</i>	<i>Vincristine</i>	
Chlorpheniramine	Doxorubicin	<i>Isradipine</i>	Oxybutynin	<i>17α-Ethinylestradiol</i>	<i>Voriconazole</i>	
Cilostazol	Dronabinol	<i>Itraconazole</i>	<i>Paclitaxel</i>	<i>Norethisterone</i>	<i>R-Warfarin</i>	
<i>Cinacalcet</i>	Duasteride	<i>Keotconazole</i>	Pantoprazole	Hydrocortisone	Zaleplon	
			Pioglitazone	Methylprednisolone		

^aDrugs in bold italic have been reported to cause drug–drug interactions.

system. The CYP4A subfamily is involved in the metabolism of eicosanoic acids (e.g., arachidonic acid), leading to the formation of physiologically important metabolites involved in such processes as blood flow in the kidney, and brain. CYP4B1 is a major bioactivating factor catalyzing the formation of reactive metabolites, such as the pneumotoxin, 4-ipomeanol. The CYP4F subfamily consists of seven proteins and are ω -hydroxylases, which function as eicosanoid regulators. This family is responsible for inactivating the vascular effects of leukotrienes, ω -3 fatty acids, and tocopherols (vitamin E analogs).

Clearly, no one animal model or combination of animal models represents the metabolic capabilities of humans. By having a complete understanding of the factors (e.g., inducers, inhibitors, and effects of disease states) that alter the expression and activity of an enzyme responsible for the metabolism of a particular compound and by a determination of responsible isoforms and patient phenotyping, it might be possible to predict drug–drug interactions and metabolic clearance.

An alphabetical listing of the clinically important drugs and the CYP450 isoforms catalyzing their oxidative metabolism is presented in Table 4.13.

Catalytic Cycle of Cytochrome P450: Steps of the Catalytic Cycle

The many variant P450 isoforms that have been isolated show a remarkable uniformity for the catalytic

mechanism (14,21,26,27). The current view illustrating the cyclic mechanism for reduction and oxygenation of P450 with stepwise interactions with substrate molecules, electron donors, and oxygen is shown in Figure 4.4 and can be summarized as follows (26,27):

Step A. The Fe^{+3} -P450 complex binds reversibly to a molecule of the substrate (RH), resulting in a complex analogous to an enzyme–substrate complex. Binding of the substrate facilitates the first one-electron reduction step.

Step B. The substrate complex of Fe^{+3} -P450 undergoes reduction to a Fe^{+2} -P450 substrate complex by an electron originating from NADPH and transferred by the flavoprotein (NADPH-P450 reductase) from the FMN ₂/FADH complex.

Step C. The reduced Fe^{+2} -P450 complex readily binds dioxygen as the sixth ligand of Fe^{+2} to form a dioxygen- Fe^{+2} -P450 complex.

Step D. The dioxygen- Fe^{+2} -P450 complex rearranges by resonance, as a result of the strong electronegativity of oxygen, to an Fe^{+3} -P450-superoxide substrate complex.

Step E. The Fe^{+3} -P450-superoxide substrate complex undergoes further reduction by accepting the rapid delivery of a second electron from the flavoprotein (or possibly cytochrome b_5) to form the equivalent of a two-electron-reduced peroxy- Fe^{+3} -P450 complex. If the electron is not delivered rapidly, the

TABLE 4.12 Cytochrome P450 Inhibitors^a

CYP1A2	CYP2B6	CYP2C8	CYP2C19	CYP2C9	CYP2D6	CYP2E1	CYP3A4/5/7
Amiodarone Atazanavir Cimetidine Ciprofloxacin Citalopram Clarithromycin Diltiazem Enoxacin Erythromycin Ethinyl Estradiol Fluoroquinolones Fluvoxamine Interferon Ketoconazole Isoniazid Methoxsalen Mibefradil	Amiodarone Amlodipine Azelastrine Citalopram Citalopram Desipramine Disulfiram Doxorubicin Ethinyl estradiol Fluoxetine Fluvoxamine Isoflurane Ketoconazole Mestranol Methimazole Miconazole Nelfinavir Orphenadrine Paroxetine Sertraline Sorafenib Tamoxifen Ticlopidine Thiotepa Venlafaxine	Anastrozole Gemfibrozil Glitazones Montelukast Nicardipine Sulfapyrazone Trimethoprim	Cimetidine Citalopram Delavirdine Efavirenz Felbamate Fluconazole Fluoxetine Fluvestatin Fluvoxamine Indomethacin Isoniazid Ketoconazole Lansoprazole Modafinil Omeprazole Oxcarbazepine Probenicid Ticlopidine Topiramate	Amiodarone Atazanavir Cimetidine Clodogrel Cotrimoxazole Delavirdine Disulfiram Efavirenz Fenofibrate Fluconazole Fluorouracil Fluoxetine Fluvastatin Fluvoxamine Gemfibrozil Imatinib Isoniazid Itraconazole Ketoconazole Leflunomide Lovastatin Methoxsalen Metronidazole Mexiletine Modafinil Nalidixic acid Norethindrone Norfloxacin Omeprazole Oral Contraceptives Paroxetine Phenylbutazone Probenicid Sertraline Sulfamethoxazole Sulfaphenazole Sulfonamides Tacrine Teniposide Ticlopidine Tipranavir Troleandomycin Voriconazole Zafirlukast Zileuton	Amiodarone Bupropion Celecoxib Chloroquine Chlorpheniramine Chlorpromazine Cimetidine Cinacalcet Citalopram Clemastine Clomipramine Cocaine Darifenacin Desipramine Diphenhydramine Doxepin Doxorubicin Duloxetine Escitalopram Fluoxetine Fluphenazine Halofantrine Haloperidol Hydroxychloroquine Hydroxyzine Imatinib Levomethoprine Methadone Metoclopramide Mibefradil Midodrine Moclobemide Norfluoquine Paroxetine Perphenazine Propafenone Propoxyphene Propranolol Quinacrine Quinidine Ranitidine Ranolazine Ritonavir Sertraline Terbinafine Thioridazine Ticlopidine Tipranavir Triptennamine	Diethyl dithio- carbamate Disulfiram	Amiodarone Ampranavir Aprepitant Atazanavir Cimetidine Ciprofloxacin Clarithromycin Cyclosporine Danazol Delavirdine Diethyl dithio-carbamate Diltiazem Efavirenz Erythromycin Ethinyl Estradiol Fluconazole Fluoxetine Fluvoxamine Gestodene Imatinib Indinavir Isoniazid Itraconazole Ketoconazole Methylprednisolone Metronidazole Mibefradil Miconazole Mifepristone Nelfinavir Nicardipine Nifedipine Norethindrone Norfluoquine Norfluoquine Oxiconazole Prednisone Quinine Ranolazine Ritonavir Roxithromycin Saquinavir Sertraline Telithromycin Troleandomycin Verapamil Voriconazole Zafirlukast Zileuton Zolpidem

^aP450 isoform inhibitors presented in bold italics have been associated with drug–drug interactions of clinical relevance or with drug–drug interaction warnings that can require dosage adjustment. Data from Stockley's *Drug Interactions, Their Mechanisms, Clinical Importance and Management*. London and Chicago: Pharmaceutical Press, 2010.

TABLE 4.13 Substrates for the P450 Isoforms Catalyzing Their Metabolism

Acetaminophen	1A2, 2E1, 3A4, 2C9	Efavirenz	2B6
Albendazole	3A4, 1A2	Eletriptan	2C9
Alfentanil	3A4	Enalapril	3A4
Alprazolam	3A4	Encainide	2D6
Amiodarone	3A4, 2C9	Enflurane	2E1
Amitriptyline	1A2, 2C9, 2D6, 3A4, 2C19	Ergot alkaloids	3A4
Amlodipine	3A4	Erythromycin	3A4
Amphetamine	2D6	Esomeprazole	2C19
Aripipraole	2D6, 3A4	Estradiol	1A2
Anastrozole	3A4	Estrogens, oral	3A4
Astemizole	3A4	Ethanol	2E1
Atomoxetine	2D6	Ethinyl estradiol	3A4
Atorvastatin	3A4	Ethosuximide	3A4
Bepidil	3A4	Etodolac	2C9
Bisoprolol	2D6	Etoposide	3A4
Bosentan	2C9	Felodipine	3A4
Bupropion	2C9, 2B6, 2A6, 1A2	Fenfluramine	2D6
Busulfan	3A4	Fentanyl	2D6, 3A4
Caffeine	1A2	Fexofenadine	3A4
Cannabinoids	3A4	Finasteride	3A4
Carbamazepine	2C8, 3A4, 2B6	Flecainide	2D6
Carisoprodol	2C19	Fluconazole	3A4
Carvedilol	2C9, 2D6	Flurbiprofen	2C9
Celecoxib	2C9	Fluoxetine	2C9, 2D6, 2B6
Cerivastatin	3A4	Fluvastatin	2C8, 2C9
Cevimeline	2D6	Fluphenazine	2D6
Chlordiazepoxide	1A2	Flutamide	1A2, 3A4
Chloroquine	3A4	Fluvoxamine	1A2, 2D6
Chlorpromazine	2D6, 3A4	Formoterol	2C9, 2C19, 2D6
Chlorzoxazone	2E1	Galantamine	2D6
Cilostazol	2C19	Glimepiride	2C9
Cimetidine	3A4	Glipizide	2C9
Cisapride	3A4	Glyburide	2C9, 3A4
Citalopram	2C19, 3A4	Granisetron	3A4
Clarithromycin	3A4	Halofantrine	3A4
Clindamycin	3A4	Haloperidol	1A2, 2D6
Clomipramine	1A2, 2C9, 2C19, 2D6, 3A4	Halothane	2E1, 2C9
Clopidogrel	2C19, 2C9, 1A2, 3A4/5	Hexobarbital	2C19, 2C9
Clonazepam	3A4	Hydrocodone	2D6, 3A4
Clozapine	1A2, 2D6, 2C19, 3A4	Hydrocortisone	2D6, 3A4
Cocaine	3A4	Ibuprofen	2C9
Codeine	2D6, 3A4	Ifosfamide	2B6, 3A4
Cyclobenzaprine	1A2, 2A6, 2D6, 3A4	Imipramine	1A2, 2C19, 2C9, 2D6, 3A4
Cyclophosphamide	2B6, 2C19, 3A4	Indinavir	2D6, 3A4
Cyclosporine	3A4	Indomethacin	2C9, 2C19
Dapsone	2C9, 3A4	Irbesartan	2C9
Delavirdine	3A4	Isoflurane	2E1, 2B6
Desipramine	1A2, 2C19, 2D6	Isoniazid	2E1
Desogestrel	2C9	Isotretinoin(retinoids)	1A2, 2C8, 3A4
Dexamethasone	3A4	Isradipine	3A4
Dexfenfluramine	2D6	Itraconazole	3A4
Dextromethorphan	2D6, 3A4	Ketamine	2B6, 2C9
Diazepam	1A2, 2C19, 2C9, 3A4	Ketoconazole	3A4
Diclofenac	2C8, 2C9	Labetalol	2D6
Diltiazem	3A4	Lansoprazole	2C19, 3A4
Diphenhydramine	2C9	Lidocaine	2D6, 3A4
Disopyramide	3A4	Leflunomide	1A2, 2C9, 2C19
Divalproex sodium	2C19	Losartan	2C9, 3A4
Docetaxel	2C8, 3A4	Lovastatin	3A4
Dolasetron	2D6, 3A4	Maprotiline	2D6
Donepezil	2D6, 3A4, 2C9	Meclobemide	2C19
Dorazolamide	2D6	Mefenamic acid	2C9
Doxepin	3A4	Mefloquine	3A4
Doxorubicin	2C9	Meloxicam	2C9
Dronabinol	2C9	Meperidine	2D6, 2B6

TABLE 4.13 Substrates for the P450 Isoforms Catalyzing Their Metabolism (Continued)

Mephenytoin	2C19	Rabeprazole	2C19
Mephobarbital	2C9, 2B6	Rapaglinide	2C8
Mestranol	2C9	Retinoic acid	2C8
Methadone	1A2, 2D6, 2B6	Rifabutin	3A4
L- α -Methadol	2B6	Rifampin	3A4
Methamphetamine	2D6	Riluzole	1A2
Metoprolol	2D6	Risperidone	2D6, 3A4
Mexiletine	1A6, 2D6, 2A6	Ritonavir	2A6, 2C19/9, 2D6, 2E1, 3A4
Mibefradil	3A4	Ropinirole	1A2
Miconazole	3A4	Ropivacaine	1A2, 2D6
Midazolam	3A4	Rosiglitazone	2C8, 2C9
Mirtazapine	1A2, 2D6, 3A4	Rosuvastatin	2C8, 2C9
Modafinil	3A4	Salmeterol	3A4
Montelukast	2C9	Saquinavir	3A4
Morphine	2D6	Selegiline	2D6, 2B6, 2A6
Naproxen	1A2, 2C18, 2C9, 2A6	Sertindole	2D6
Nateglinide	2C9	Sertraline	2D6, 2C19, 3A4, 2B6
Navelbine	3A4	Sevoflurane	2E1
Nefazodone	3A4	Sibutramine	2B6
Nelfinavir	2B6, 3A4	Sildenafil	2C9, 3A4
Nevirapine	3A4	Simvastatin	3A4
Nicardipine	3A4	Sufentanil	3A4
Nicotine	2A6, 2B6, 2A13	Sulfamethoxazole	2C9
Nifedipine	3A4/5	Suprofen	2C9
Nilutamide	2C19	Tacrine	1A2, 2A6
Nimodipine	3A4	Tacrolimus	3A4
Nisoldipine	3A4	Tamoxifen	1A2, 2A6, 2B6, 2D6, 2E1, 3A4
Nitrendipine	3A4	Terfenadine	3A4
Nortriptyline	1A2, 2D6	Testosterone	3A4
Olanzapine	1A2, 2D6, 3A4	Δ^9 -THC	2C9
Omeprazole	2C19, 2C9, 3A4	Theophylline	1A2, 2E1, 3A4
Ondansetron	1A2, 2D6, 2E1, 3A4	Thiabendazole	1A2
Oral contraceptives	3A4	Thioridazine	2C19, 2D6
Oxycodone	2D6	Tiagabine	3A4
Paclitaxel	2C8, 3A4	Timolol	2D6
Pantoprazole	2C19	Tolbutamide	2C8, 2C9, 2C19
Paroxetine	2D6	Tolteridine	2D6, 2C9
Perphenazine	2D6	Torsemide	2C9
Phenacetin	1A2	Tramadol	2D6
Phenformin	2D6	Trazodone	2D6
Phenol	2E1	Valproic acid	2C9, 2A6, 2B6
Phenobarbital	2B6, 2C9	Valdecocib	2C9
Phenytoin	3A4	Vardenafil	3A4
Pimozide	2D6, 3A4	Venlafaxine	2D6
Pindolol	2C8, 3A4	Verapamil	1A2, 3A4, 2C8
Pioglitazone	2C18, 2C9	Vinblastine	3A4
Piroxicam	2C18, 2C9	Vinca alkaloids	3A4
Prasugrel	3A4, 2B6, 2C9	Vincristine	3A4
Pravastatin	3A4	Voriconazole	2C9
Praziquantel	2B6, 3A4	Warfarin	2C18, 2C9
Prednisone	3A4	R-warfarin	1A2
Progesterone	3A4, 2C19	S-warfarin	2C9, 2C18
Proguanil	2C18, 2C19	Yohimbine	2D6
Propafenone	1A2, 2D6, 3A4	Zafirlukast	2C9
Propofol	2B6, 2C9	Zaleplon	3A4
Propoxyphene	2D6	Zileuton	1A2, 2C9, 3A4
Propranolol	1A2, 2C18, 2C19, 2D6	Zolpidem	3A4
Quetiapine	3A4	Zoplicone	3A4
Quinidine	3A4		
Quinine	3A4		

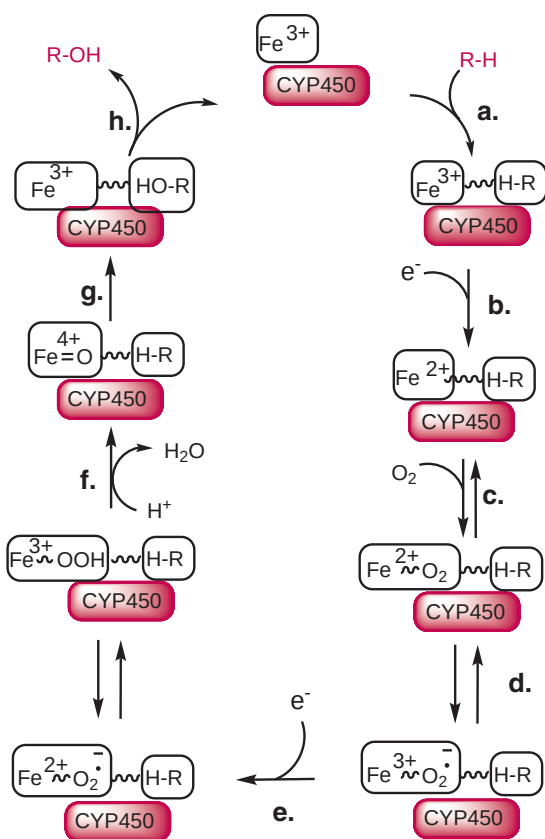


FIGURE 4.4 Cyclic mechanism for P₄₅₀. The substrate is RH, and the valence state of the heme iron in P₄₅₀ is indicated.

complex dissociates and is aborted (uncoupled) from subsequent substrate hydroxylation at this step by xenobiotics, which can cause release of the superoxide anion, which decomposes to hydrogen peroxide and dioxygen, with regeneration of the starting point of the cycle, the Fe^{3+} -P₄₅₀ complex (step A).

Step F. The peroxy- Fe^{3+} -P₄₅₀ complex undergoes heterolytic cleavage of peroxide anion to form water and a highly electrophilic perferryl oxenoid intermediate ($\text{Fe}^{3+}=\text{O}$) or a ferryl ($\text{Fe}^{4+}=\text{O}$) oxygen-cysteine-porphyrin resonance-stabilized complex (the more favorable complex). The ferryl oxenoid species represents the catalytically active oxygenation species.

Step G. Abstraction of a hydrogen atom from the substrate by the ferryl oxenoid species produces a carbon-centered radical-ferryl-hydroxide complex; radical addition to a π -bond; or electron abstraction from a heteroatom to form a heteroatom-centered radical-cation-ferryl intermediate.

Step H. Subsequent radical recombination (oxygen rebound) or electron transfer (deprotonation) yields the hydroxylated product and regeneration of the Fe^{3+} -P₄₅₀ complex (step A).

Until the final step, the oxidizable substrate has been an inactive spectator in the chemical events of oxygen

activation. None of the preceding oxygenated intermediates has been sufficiently reactive to abstract hydrogen from the substrate. The ferryl-iron oxenoid complex (step G), however, is a competent hydrogen abstractor, even for relatively inert terminal methyl groups on hydrocarbon chains. Thus, this is evidence that the oxidant exhibits regioselectivity in its choice of hydrogen atoms, balancing stability of the resulting carbon radical with stereochemical constraints (Fig. 4.5). Because the inert aliphatic region of the substrate has been converted to a highly reactive radical, the process is described as substrate activation. Various studies have shown that hydroxylation or alkene formation proceeds not by a direct one-step insertion of an oxygen atom but rather by a two-step two-electron process that involves radical or cationic substrate intermediates with subsequent hydroxyl radical recombination (oxygen rebound) to form products (Fig. 4.6) (28).

Despite considerable experimental evidence, the proposed mechanism and intermediates of the monooxygenation of unsaturated substrates (alkenes, alkynes, and aromatics) remain controversial (19,20,29–31). The proposed mechanism for the oxidation of π -bonds in alkenes is a stepwise sequence of one-electron transfers between the radical complex and the ferryl oxygen intermediate [$\text{Fe}^{4+}=\text{O}$], leading to alkene oxidation (Fig. 4.6). After the initial formation of an unsaturated P₄₅₀ π -complex, the one-electron transfer either yields a radical σ -complex, which can either collapse to an arene or to an alkene epoxide (steps A or D, Fig. 4.6); undergoes a 1,2-group migration to form a carbonyl product (steps A and B, Fig. 4.6); or produces a vinyl hydroxylated product (step C, Fig. 4.6)

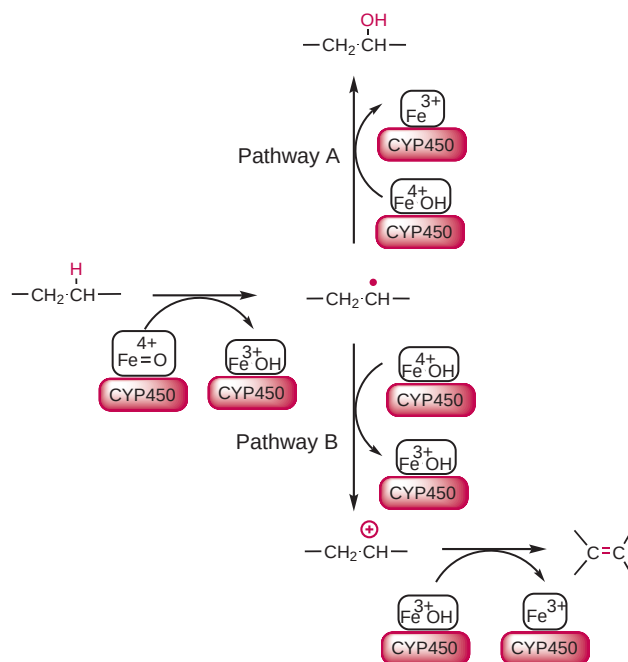


FIGURE 4.5 Proposed mechanisms for the hydroxylation and dehydrogenation of alkanes.

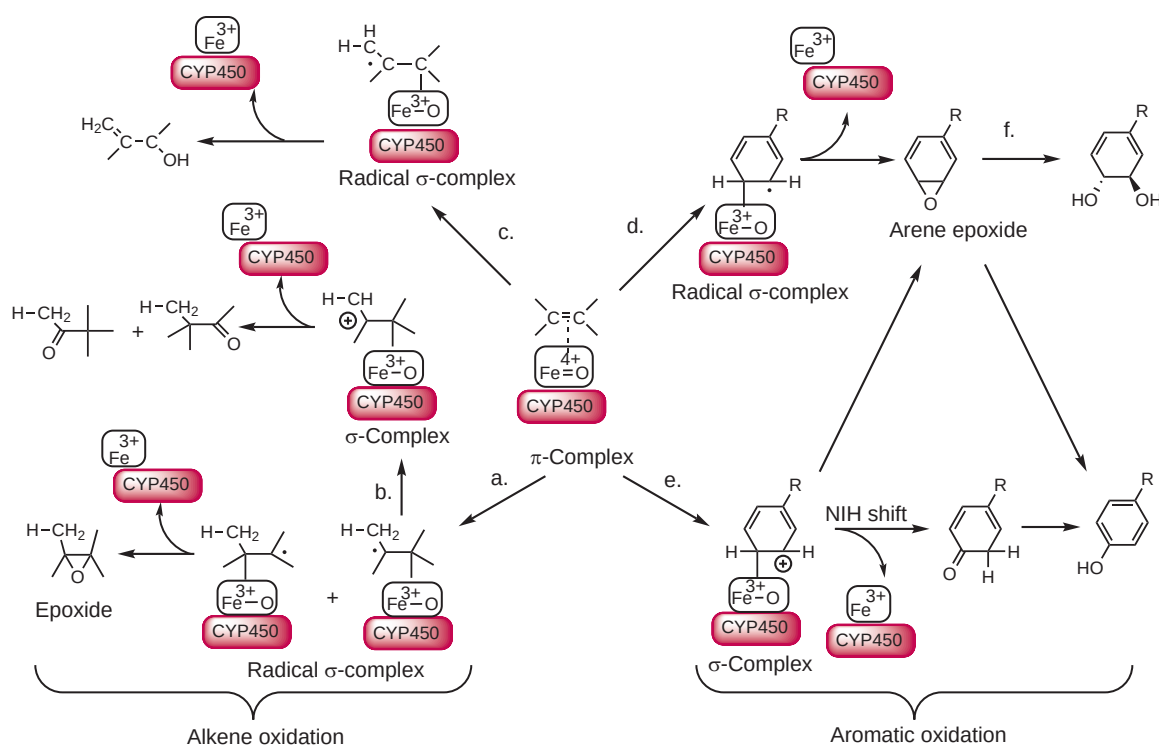


FIGURE 4.6 Proposed mechanisms for the oxidation of alkene and aromatic compounds.

or a σ -complex, which can break down to a phenol (step E, Fig. 4.6). The presence of a hydroxyl radical in the porphyrin ring allows some substrate radicals to covalently bond by *N*-alkylation of a pyrrole nitrogen rather than recombination with $(\text{Fe}-\text{OH})^{+3}$ radical. This deviation from the normal course of reaction explains the suicide inhibition exhibited by some xenobiotics, such as the oral contraceptives, erythromycin, and paroxetine (32,33).

In the case of aromatic oxidations (Fig. 4.6), following the initial formation of an arene P450 π -complex, one-electron transfer yields either a π -complex or a radical σ -complex. The radical σ -complex can collapse to the arene epoxide (Fig. 4.6, step D), or the π -complex can proceed to a σ -complex followed by an NIH shift (1,2-group migration) to a phenolic product (Fig. 4.6, step E). Arene oxides are highly unstable entities and rearrange (NIH shift) nonenzymatically to phenols or hydrolyze enzymatically with epoxide hydrolase to 1,2-dihydrodiols (*trans* configuration) (Fig. 4.6, step F), which subsequently are dehydrogenated to 1,2-diphenols. The oxidation of aromatic compounds can be highly specific to individual P450 isoforms, suggesting that substrate binding and orientation at the active site can dominate the mechanism of oxidative catalysis.

The exact mechanism of oxidative *N*, *O*, and *S*-dealkylations is unclear. There are two competing mechanisms: single electron transfer or hydrogen atom abstraction (Fig. 4.7). Heteroatom-containing substrates most commonly undergo hydroxylation adjacent (α) to the heteroatom, as compared to other positions. Reactions of this

type include *N*, *O*, and *S*-dealkylations, as well as dehydrohalogenations and oxidative deamination (dealkylation) reactions. The single electron transfer pathway is abstraction of an electron from the heteroatom to produce a radical cation, followed by the loss of the α -proton from the more labile α -carbon to generate a carbon radical that can recombine with the ferric-bound hydroxyl radical intermediate (oxygen rebound) to generate an unstable geminal hydroxy heteroatom-substituted intermediate (e.g., carbinolamine, halohydrin, hemiacetal, hemiketal, or hemithioketal) that breaks down, releasing the heteroatom and

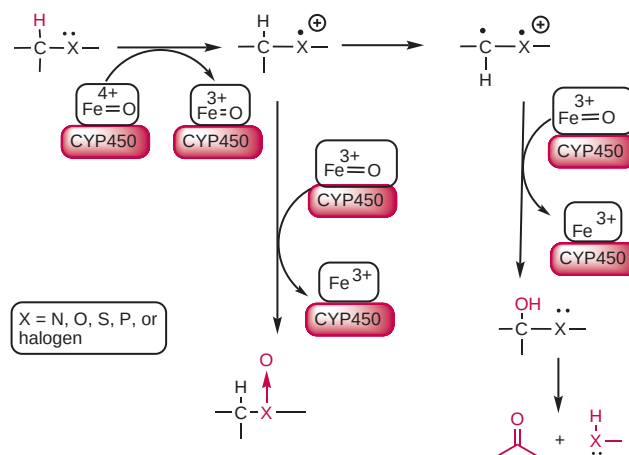


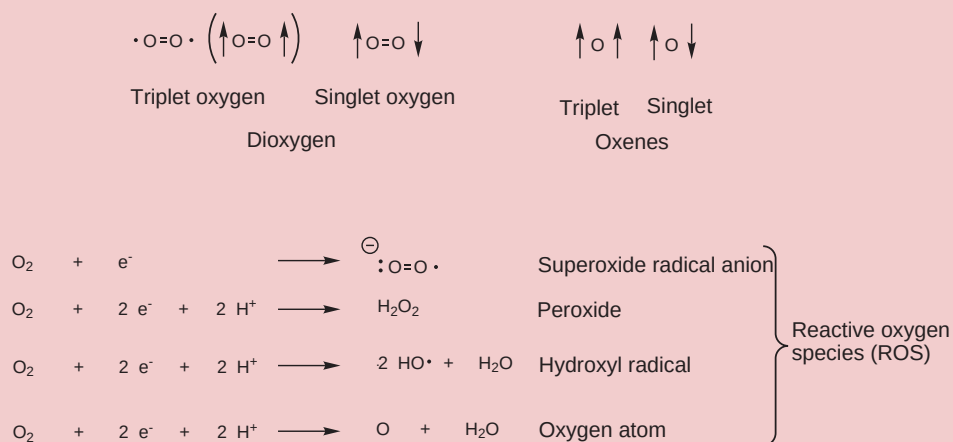
FIGURE 4.7 Proposed mechanism for heteroatom-compound oxidation, dealkylation and dehalogenation.

OXYGEN ACTIVATION

Elemental oxygen (dioxygen) is a relatively unreactive form of oxygen that exists as an unpaired diradical in the triplet form. Alternatively, singlet oxygen is a form of dioxygen in which the diradical electrons are paired. In the singlet form, oxygen is too reactive for biologic systems to handle. Free oxygen atoms (oxenes), which are formed by splitting dioxygen, are highly reactive but are not known to exist in biochemical processes. The solution to the problem of a reactive form of oxygen lies in the suggestion that the reduction of dioxygen occurs to a reactive oxygen species (ROS), such as superoxide radical anion, peroxide, hydroxyl radical, or oxygen atom.

Any of these ROS could oxidize an organic substrate with the net insertion of an oxygen atom. In each case, reductive reactions

are required for activation of dioxygen to one of the ROS from electrons supplied by NADPH. The generation of a carbon-centered radical and a hydroxyl radical with triplet oxygen atom has been found to be relevant to a number of enzymatic and chemical reactions involving oxenoids (oxygen rebound mechanism) (28). The function of P450 monooxygenases is mostly hydroxylation of a substrate. A reactive radical-like iron oxenoid intermediate is generated that is reactive enough to split aliphatic C-H bonds, add to bonds α to heteroatoms, or remove single electrons from heteroatoms to produce radical-cations. Although the mechanisms of P450 are not fully understood, the reactive oxygen intermediate has not been isolated or even spectroscopically observed.



forming a carbonyl compound (29–31). The resultant carbon radical is stabilized by the heteroatom. This pathway is feasible for nitrogen, which is easily oxidized and electron rich. However, if the recombination reaction with the ferric-bound hydroxyl radical intermediate (oxygen rebound) is faster than the deprotonation of the α -proton from the adjacent α -carbon, the oxidation will simply result in oxidation of the heteroatom, as in the formation of *N*-oxides or oxidation of sulfides to sulfoxides and/or sulfones.

Xenobiotics containing heteroatoms (N, S, P, and halogens) are frequently metabolized by heteroatom oxidation to their corresponding heteroatom oxides (tertiary amines to *N*-oxides, sulfides to sulfoxides and/or sulfones, or phosphines to phosphine oxides). Heteroatom oxidation can also be attributed to a microsomal flavin-containing monooxygenase. As is the case with heteroatom α -hydroxylation, one-electron oxidation of the heteroatom occurs as the first step to form the heteroatom ferric-hydroxide radical intermediate, which collapses to generate the heteroatom oxide. This reaction is favored by the absence of α -hydrogens and by the stability of the heteroatom radical-cation (29–31).

All known oxidative reactions catalyzed by P450 monooxygenases can be described in the context of a mechanistic scheme that involves the ability of a high-valent iron oxenoid species to bring about the stepwise one-electron oxidation through the abstraction of hydrogen atoms, abstraction of electrons from heteroatoms, or addition to π -bonds. A series of radical recombination reactions completes the oxidation process.

Induction and Inhibition of Cytochrome P450 Isoforms

Induction

Many of the enzymes involved in drug metabolism can be upregulated by exposure to drugs and environmental chemicals or by other coingested/inhaled compounds leading to increased rates of metabolism, thereby altering their pharmacologic and toxicologic effects (34–36). Prolonged administration of a drug or xenobiotic can enhance the metabolism of itself and a wide variety of other compounds. This phenomenon is known as enzyme induction, a dose-dependent phenomenon.

Drugs and xenobiotics exert this effect by inducing transcription of P450 mRNA and synthesis of xenobiotic-metabolizing enzymes in the smooth ER of the liver and other extrahepatic tissues (34,35). Enzyme induction is an adaptive response associated with increases in liver weight, induction of gene expression, and morphologic changes in hepatocytes. Induction is the process whereby the rate of enzyme synthesis is increased relative to the rate of enzyme synthesis in the uninduced organism. In many older studies of mammalian systems, the term “induction” was inferred from the increase in enzyme activity, but the amount of enzyme protein had not been determined. Enzyme induction is important to interpret the results of chronic toxicities, mutagenicities, or carcinogenesis and to explain certain unexpected drug–drug interactions in patients. Enzyme inducers trigger pathways involving the constitutive androstane receptor (CAR), PPAR, AhR, and pregnane X receptor (PXR) (34). CAR is a member of the nuclear receptor superfamily and, along with PXR, upregulates the expression of P450 proteins responsible for the metabolism and excretion of these xenobiotics, and thus is a key regulator of xenobiotic CYP450 metabolism.

Many drugs and xenobiotics stimulate the upregulation of P450 isoforms, as shown in Table 4.14. These stimulators have nothing in common so far as their pharmacologic activities or chemical structures are concerned, but they are all metabolized by one or more of the P450 isoforms. Most are lipid soluble at physiologic pH. PAHs in cigarette smoke, xanthines and flavones in foods, halogenated hydrocarbons in insecticides, polychlorinated biphenyls, and food additives are but a few of the environmental chemicals that alter the activities of P450 enzymes (37).

Enzyme induction, which can last as long as 1 to 3 weeks as drug blood levels decrease, can alter the pharmacokinetics and pharmacodynamics of a drug, with clinical

implications for the therapeutic actions of the drug as well as increased potential for drug–drug interactions. As a result of induction, a drug can be metabolized more rapidly to metabolites that are more potent, more toxic, or less active than the parent drug. Induction can also enhance the activation of procarcinogens or promutagens. Not all inducing agents enhance their own metabolism; for example, phenytoin induces CYP3A4 but is hydroxylated by CYP2C9. Some of the more common enzyme inducers of P450 subfamilies, which can also be substrates for the same P450 isoforms, include phenobarbital (CYP2B6, CYP2C, and CYP3A4), rifampicin (CYP3A4), and cigarette smoke (CYP1A1/2) (Table 4.14). The broad range of drugs metabolized by these CYP450 subfamilies (Table 4.13) and that are also affected by these enzyme inducers raises the issue of clinically significant drug–drug interactions and their clinical implications. Examples of a clinically significant P450 drug–drug interaction and an herbal drug–drug interaction are rifampin and oral contraceptives and St. John’s wort and oral contraceptives, respectively. Both rifampin and St. John’s wort induce the expression of CYP3A4, thereby reducing the serum levels of the oral contraceptive due to increased oxidative metabolism of the oral contraceptives by CYP3A4 to form less active metabolites, thereby increasing the risk for pregnancy. Drugs poorly metabolized by P450 enzymes are less affected by enzyme induction. Inducers of P450 isoforms also stimulate oxidative metabolism and synthesis of endogenous substances: for example, the hydroxylation of androgens, estrogens, progestational steroids (synthetic oral contraceptives), glucocorticoids, vitamin D, and bilirubin, which decreases their biologic activity. These enzyme inducers might also be implicated in deficiencies associated with these steroids. For example, the induction of C-2 hydroxylation of estradiol and synthetic estrogens by phenobarbital,

TABLE 4.14 Drugs That Induce the Expression of P450 Isoforms^a

Amprenavir	3A4	4-Methylpyrazole	2E1
Aprepitant	2C9	Modafinil	1A2, 2B6, 3A4
Barbiturates	3A4, 2C9, 2C19, 2B6	Nevirapine	3A4, 2B6
Carbamazepine	1A2, 3A4, 2C8, 2C9, 2D6, 2B6	Norethindrone	2C19
Charbroiled meats	1A1/2,	Omeprazole	1A1/2, 3A4
Cigarette smoke	1A1/2	Oxcarbazepine	3A4
Clotrimazole	1A1/2, 3A4	Phenobarbital	3A4, 2C, 2B6, 2D6, 1A2
Ethanol	2D6, 2E1	Phenytoin	3A4, 1A2, 2B6, 2C8, 2C19, 2D6
Efavirenz	3A4	Primidone	3A4, 2C9, 1A2, 2B6, 2D6
Erythromycin	3A4	Psoralen	1A1/2
Glucocorticoids	3A4, 2A6	Ethosuximide	3A4
(Dexamethasone, prednisone)	2C19	Polycyclic aromatic hydrocarbons	1A1/2
Griseofulvin	3A4	Rifampin	2C8, 2C9, 2C19, 2D6, 3A4, 2B6
Isoniazid	2E1	Rifampicin	2C8
Lansoprazole	1A1/2, 3A4	Rifabutin	2C8, 3A4
Mephenytoin	2B6	Rifapentine	3A4
		Ritonavir	2D6, 3A4
		St. John’s Wort	1A2, 2C9, 3A4
		Troglitazone	3A4
		Topiramate	3A4

^aDrugs in bold italic have been reported to cause drug–drug interactions.

dexamethasone, or cigarette smoking in women results in the increased formation of the principal and less active metabolite of these estrogenic substances, reducing their effectiveness (38). Thus, cigarette smoking in premenopausal women could result in an estrogen deficiency, increasing the risks of osteoporosis and early menopause. Postmenopausal women who smoke and take estrogen replacement therapy can lose the effectiveness of the estrogen.

In addition to enhancing metabolism of other drugs, many compounds, when chronically administered, stimulate their own metabolism, thereby decreasing their therapeutic activities and producing a state of apparent tolerance. This self-induction can explain some of the changes in drug toxicity observed during prolonged treatments. The duration of the sedative action of phenobarbital, for example, becomes shorter with repeated doses and can be explained in part on the basis of increased metabolism.

The time course of induction varies with different inducing agents and with different P450 isoforms. CYP1A induction always involves AhR. Increased transcription of P450 mRNA has been detected as early as 1 hour after administration of phenobarbital, with maximum induction occurring after 48 to 72 hours. After administration of PAH such as 3-methylcholanthrene and benzo[a]pyrene, maximum induction of the CYP1A subfamily is reached within 24 hours. Less potent inducers of hepatic drug metabolism can take as long as 6 to 10 days to produce maximum induction (34,35). Exposures to a variety of different xenobiotics can increase the hepatic content of specific isoforms of P450 (34–36). Therefore, the process of enzyme induction involves the adaptive increase in the content of specific enzymes in response to the enzyme-inducing agent. Other inducible metabolizing enzymes include uridine diphosphate (UDP)–glucuronosyl transferase and glutathione transferase.

Specific Inducers

PHENOBARBITAL AND RIFAMPIN Phenobarbital and rifampin are probably the enzyme inducers that have been studied most extensively. These drugs could alter the pharmacokinetics and pharmacodynamics of many concurrently administered drugs listed in Tables 4.8 (CYP2C) and 4.11 (CYP3A4), which raises the issue of clinically significant drug–drug interactions.

CIGARETTE SMOKE Cigarette smoke has been shown to increase the hydrocarbon-inducible isoforms CYP1A1 and CYP1A2 in the lungs, liver, small intestine, and placenta of cigarette smokers. A decrease in the pharmacologic action and stimulation of the CYP1A1/2 metabolism of several drugs is the end result. Cigarette smoking has been reported to lower blood levels of drugs metabolized by CYP1A1/2, including theophylline, imipramine, estradiol, pentazocine, and propoxyphene; to decrease the urinary excretion of nicotine; and to decrease drowsiness caused by chlorpromazine, diazepam, and chlordiazepoxide.

DIETARY SUBSTANCES A diet containing Brussels sprouts, cabbage, and cauliflower (cruciferous vegetables) was found to stimulate P450 activity in rat intestine (39). It was subsequently determined that indole-3-carbinol in the plants is released from the breakdown of glucobrassicin, a thioglucoside also called a glucosinolate, during gastrointestinal digestion, which was responsible for the enzyme induction. Other examples of chemicals found naturally in foods that are released during digestion that enhance metabolism in animals are flavones, saffrole, eucalyptol, xanthines, β -ionone, and organic peroxides. Volatile oils in soft woods (e.g., cedar) have been shown to be enzyme inducers.

ALCOHOL Sober alcoholics show an increase in CYP2E1 enzyme activity, leading to more rapid clearance of drugs and xenobiotics that are substrates for this isoform from the body. As discussed previously, hepatic CYP2E1 oxidizes ethanol, and chronic ethanol intake increases the activity of CYP2E1 through enzyme induction (24). When intoxicated, alcoholics are more susceptible to the actions of various drugs due to inhibition of drug metabolism as a result of an excessive quantity of alcohol in the liver and an additive or synergistic effect in the central nervous system. The basis for this inhibition is unknown. Furthermore, moderate ethanol consumption reduces the clearance of some drugs, presumably due to competition between ethanol and the other drugs for hepatic biotransformation. The changes in drug metabolism in alcoholics can also be attributed to other factors, such as malnutrition, other drugs, and the trace chemicals that are sources of flavors and odors of particular alcoholic beverages. Heavy drinkers metabolize phenobarbital, tolbutamide, and phenytoin more rapidly than nonalcoholics, which can be clinically important when this leads to problems in adjusting drug therapies for alcoholics.

Inhibition

Another method to alter in vivo effects of xenobiotics metabolized by P450s is the use of inhibitors (Table 4.12). The P450 inhibitors can be divided into three categories according to their mechanisms of action: reversible inhibition, metabolite intermediate complexation of P450, or mechanism-based inactivation of P450 (36,40,41). The polysubstrate nature of P450 is responsible for the large number of documented interactions associated with the inhibition of drug oxidation and drug biotransformation.

REVERSIBLE INHIBITION Reversible inhibition of P450 is the result of reversible interactions at the heme–iron active center of P450, the lipophilic sites on the apoprotein, or both. The interaction occurs before the oxidation steps of the catalytic cycle, and the effects dissipate quickly when the inhibitor is discontinued. The most effective reversible inhibitors are those that interact strongly with both the apoprotein and the heme iron. It is widely accepted that inhibition has an important impact on the oxidative metabolism and pharmacokinetics of drugs

that have a metabolism that cosegregates with that of an inhibitor (Tables 4.4 and 4.8 to 4.11) (40,41). Drugs interacting reversibly with P450 include the fluoroquinolone antimicrobials, cimetidine, the azole antifungals, quinidine (specific for CYP2D isoforms), and diltiazem. Cimetidine is the only H_2 antagonist that inhibits P450 by interacting directly with the P450 heme-iron through one of its imidazole ring nitrogen atoms. Cimetidine is not a universal inhibitor of P450 oxidative metabolism, but it does bind differentially to several P450 isoforms (Table 4.12). Cimetidine inhibits the oxidation of theophylline (CYP1A2), chlordiazepoxide (CYP3A4), diazepam (CYP2C), propranolol (CYP2C18/19 and CYP2D6), warfarin (CYP2C9), and antipyrine (CYP1A2), but not that of ibuprofen (CYP2C9), tolbutamide (CYP2C8/9), mexiletine (CYP2D6), the 6-hydroxylation of steroids (CYP3A4/5), and carbamazepine (CYP3A4/5) (36). The imidazole-based azole antifungals are potent inhibitors of CYP3A4 and of the P450-mediated biosynthesis of endogenous steroid hormones. The azole antifungals exert their fungistatic effects through inhibition of fungal P450, inhibiting the oxidative biosynthesis of lanosterol to ergosterol, thereby affecting the integrity and permeability of the fungal membranes.

P450 COMPLEXATION INHIBITION Noninhibitory alkylamine drugs have the ability to undergo P450-mediated oxidation to nitrosoalkane metabolites (Fig. 4.8), which have high affinities to form stable complexes with the reduced (ferrous) heme-P450 intermediates of the CYP2B, CYP2C, and CYP3A subfamilies. This process is termed “metabolite intermediate complexation” (40,41). Thus the P450 isoform is unavailable for further oxidation, and synthesis of the new enzyme is required to restore P450 activity. The process relies on at least one cycle of the P450 catalytic cycle to generate the required heme intermediate. The macrolide antibiotics troleandomycin, erythromycin, and clarithromycin, as well as their analogues, are selective inhibitors of CYP3A4 that are capable of inducing expression of hepatic and extrahepatic CYP3A4 mRNA and of induction of their own biotransformation into nitrosoalkane metabolites. The clinical significance of this inhibition with CYP3A4 is the long-lived impairment of metabolism of a large number of coadministered substrates of this isoform, and the potential for drug-drug interactions and time-dependent nonlinearities in their pharmacokinetics upon long-term administration

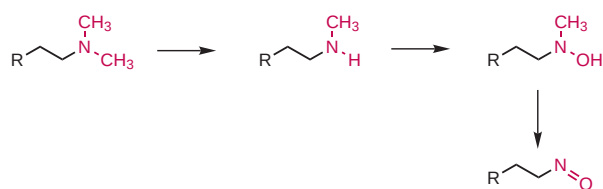


FIGURE 4.8 Sequence of oxidation of dialkylamine to nitroso metabolite intermediate.

(Tables 4.11 and 4.12). For macrolides to be so metabolized, they must possess an unhindered dimethylamino sugar, and the compound itself must be lipophilic. Other alkylamine-based drugs that demonstrate this type of inhibition include orphenadrine (antiparkinson drug), the antiprogesterin mifepristone (CYP3A4), and SKF525A (the original P450 inhibitor). Methylenedioxyphenyl compounds (i.e., the insecticide synergist piperonyl butoxide and the flavoring agent isosafrole) generate metabolite intermediates that form stable complexes with both the ferric and ferrous state of P450.

MECHANISM-BASED INHIBITION Certain drugs that are not inhibitors of P450 contain functional groups that, when oxidized by P450, generate metabolites that irreversibly bind to the enzyme. This process is termed “mechanism-based inhibition” (“suicide inhibition”) and requires at least one catalytic P450 cycle either during or subsequent to the oxygen-transfer step, when the drug is activated to the inhibitory species. Alkenes and alkynes were the first functionalities found to inactivate P450 by generation of a radical intermediate that alkylates the heme structure (see the section on alkene and alkyne hydroxylation) (32,33,40,41). Iron is lost from the heme, and abnormal *N*-alkylated porphyrins are produced. Drugs that are mechanism-based inhibitors of P450 include the 17 α -acetylenic estrogen, 17 α -ethinyl estradiol, the 17 α -acetylenic progestin, norethindrone (norethisterone), and their radical intermediates that *N*-alkylate the heme of CYP3A4 (see Fig. 4.11 and later discussion); chloramphenicol and its oxidative dechlorinated acyl metabolite that alkylates P450 apoprotein; cyclophosphamide and its generated acrolein and phosphoramidate mustard that alkylates CYP2B6 apoprotein; spironolactone and its 7-thio metabolite that alkylates heme; 8-methoxypsoralen (a furocoumarin) and its epoxide metabolite that alkylates the P450 apoprotein of CYP2A6; nicotine (CYP1A6 and CYP1A13 isoforms); isoniazid that alkylates CYP1A2, CYP2A6, CYP2C19, and CYP3A4 isoforms; 21-halosteroids; halo-carbons; and secobarbital. The selectivity of P450 isoform destruction by several of these inhibitors indicates involvement of this isoform in its bioactivation of such drugs.

Oxidations Catalyzed by Cytochrome P450 Isoforms

ALIPHATIC AND ALICYCLIC HYDROXYLATIONS The accepted mechanism of hydroxylation of alkane C-H bonds is shown in Figure 4.6 and has been reviewed in detail elsewhere (29–31). The principal metabolic pathway of the methyl group is oxidation to the hydroxymethyl derivative followed by P450 oxidation to the carboxylic acid (e.g., tolbutamide) (Fig. 4.9). On the other hand, some methyl groups are oxidized only to the hydroxymethyl derivative, without further oxidation to the acid. Where there are several equivalent methyl groups, as a rule only one methyl group is oxidized. In the case of aromatic methyl groups, the para methyl is the most vulnerable because it is less sterically hindered.

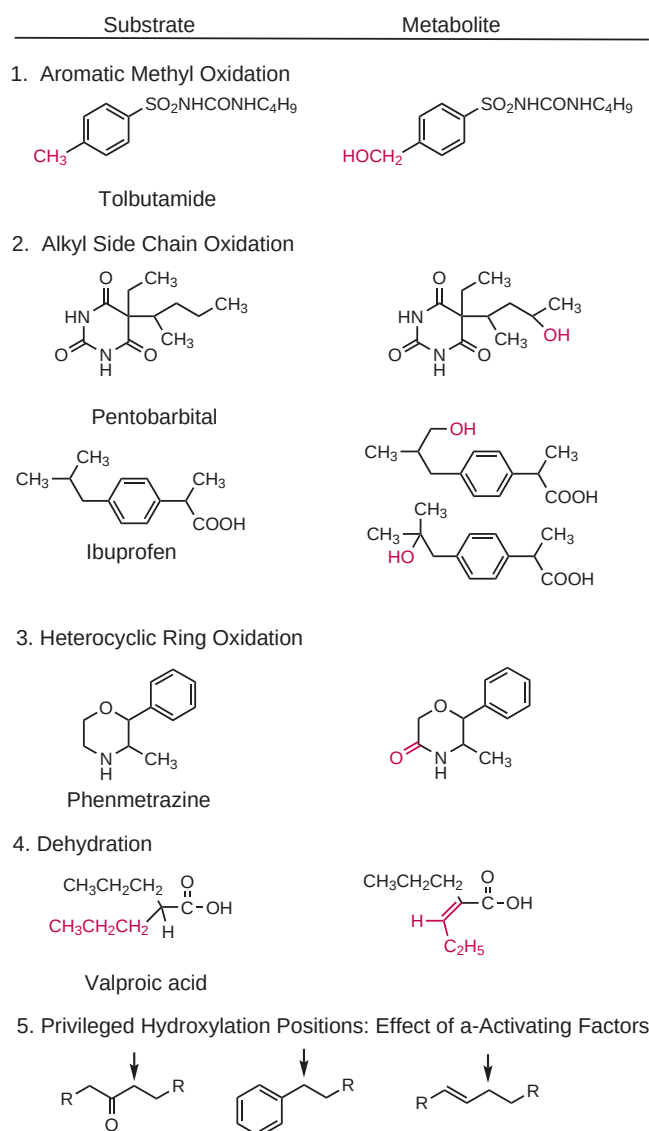


FIGURE 4.9 Examples of oxidative metabolism of aliphatic and alicyclic hydrocarbons catalyzed by CY P450.

Alkyl side chains are often hydroxylated on the terminal or penultimate carbon atoms (e.g., pentobarbital) (Fig. 4.9). The isopropyl group is an interesting side chain that is hydroxylated at the secondary carbon and at either of the equivalent methyl groups (e.g., ibuprofen) (Fig. 4.9). Hydroxylation of alkyl side chains attached to an aromatic ring does not follow the general rules for alkyl side chains, because the aromatic ring activates the α -position of hydroxylation (Fig. 4.9). In general, oxidation occurs preferentially on the benzylic methylene group and, to a lesser extent, at other positions on the side chain.

The methylene groups of an alicycle are readily hydroxylated, usually at the least hindered position, or at an activated position—for example, α to a carbonyl (cyclohexanone), α to a double bond (cyclohexene), or α to a phenyl ring (tetralin). The products of hydroxylation often show regioselectivity and stereoisomerism. Nonaromatic heterocycles

normally undergo oxidation at the α -carbon adjacent to the heteroatom (e.g., phenmetrazine) (Fig. 4.9).

In addition to hydroxylation reactions, P450s can catalyze the dehydrogenation of an alkane to an alkene (olefin). The reaction is thought to involve formation of a carbon radical and electron transfer to the ferryl complex of P450, which produces a carbocation that is deprotonated to a dehydrogenated product alkene (Fig. 4.5) (29–31). An example of the ability of P450 to function as both a dehydrogenase and a monooxygenase has been demonstrated with the antiseizure drug valproic acid. Whereas the major metabolic products in humans are formed by β -oxidation and acyl glucuronidation, several alkenes are also formed, including (*E*)-2-ene isomer (Fig. 4.9) (42). Presumably, the CYP3A subfamily catalyzes these reactions. The factors that determine whether P450 catalyzes hydroxylation (oxygen rebound/recombination) or dehydrogenation (electron transfer) remain unknown, but hydroxylation is usually favored. In some instances, the product of dehydrogenation can be the primary product (i.e., 6,7-dehydrogenation of testosterone).

ALKENE AND ALKYNE HYDROXYLATION The oxidation of alkenes yields primarily epoxides and a series of products derived from 1,2-migration (see previous discussion) (Fig. 4.6). The stereochemical configuration of the alkene is retained during epoxidation. The epoxides can differ in reactivity. Those that are highly reactive either undergo pH-catalyzed hydrolyses to excretable vicinal dihydrodiols or covalent reactions (alkylation) with macromolecules such as proteins or nucleic acids, which lead to tissue necrosis or carcinogenicity. Moreover, the ubiquitous epoxide hydrolase can catalyze the rapid hydrolysis of epoxides to nontoxic vicinal dihydrodiols. Several drugs (carbamazepine, cyproheptadine, and protriptyline), however, were found to form stable epoxides at the 10,11-position during biotransformation (Fig. 4.10). The fact that these epoxides could be detected in the urine indicates that these oxides are not particularly reactive and should not readily react covalently with macromolecules.

The epoxidation of terminal alkenes is accompanied by the mechanism-based (“suicide”) *N*-alkylation of the heme–porphyrin ring. If the π -complex attaches to the alkene at the internal carbon, the terminal carbon of the double bond can irreversibly *N*-alkylate the pyrrole nitrogen of the porphyrin ring (32,33). The heme adduct formation is observed mostly with monosubstituted, unconjugated alkenes (i.e., 17 α -ethylenic steroids and 4-ene metabolite of valproic acid).

In addition to the formation of epoxides, heme adducts, and hydroxylated products, carbonyl products are also created. These latter products result from the migration of atoms to adjacent carbons (i.e., 1,2-group migration). For example, during the P450-catalyzed oxidation of trichloroethylene, a 1,2-shift of chloride occurs to yield chloral (Fig. 4.10).

Like the alkenes, alkynes (acetylenes) are more readily oxidized, but usually faster. Depending on which of the

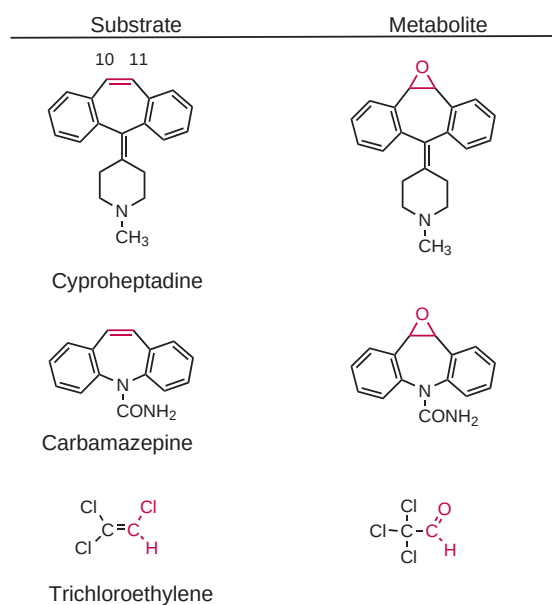


FIGURE 4.10 Examples of oxidative metabolism of alkenes catalyzed by P450.

two alkyne carbons are attacked, different products are obtained (32,33). If attachment of P450 occurs on the terminal alkyne carbon, a hydrogen atom migrates, forming a ketene intermediate that readily hydrolyzes with water to form an acid or that alkylates nucleophilic protein side chains (i.e., lysinyl or cysteinyl) to form a protein adduct (Fig. 4.11). The effect of attaching the ferryl oxygen at the internal alkenyl carbon is *N*-alkylation of a pyrrole nitrogen in the porphyrin ring by the terminal acetylene carbon, with the formation of a keto heme adduct (Fig. 4.11). The latter mechanism has been proposed for the irreversible inactivation of CYP3A4 with 17 α -alkenyl steroids (i.e., 17 α -ethinyl estradiol).

AROMATIC HYDROXYLATION The metabolic oxidation of aromatic carbon atoms by P450 depends on both the isoform catalyzing the oxidation and the oxidation potential of the aromatic compound. The products are typically phenolic products, and the position of hydroxylation can be activated/deactivated by the type of substituents on the

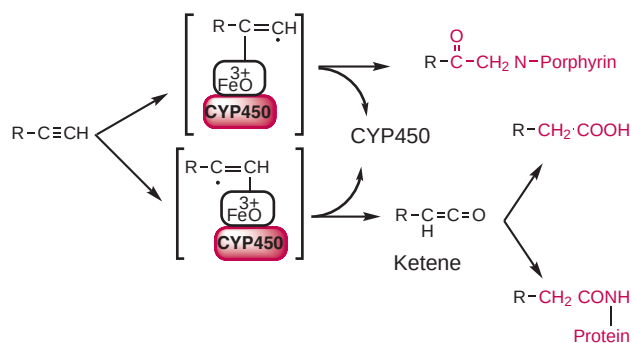


FIGURE 4.11 Alkyne oxidation catalyzed by P450.

ring, according to the theories of aromatic electrophilic substitution (Fig. 4.12). For example, electron-donating substituents enhance *o*- and *p*-hydroxylation, whereas electron-withdrawing substituents reduce or prevent *o*- or *p*-hydroxylation. Moreover, steric factors must also be considered, since oxidation occurs mostly at the least hindered position. For monosubstituted benzene compounds, *p*-hydroxylation frequently predominates, with some ortho product also formed (Fig. 4.12). When more than one phenyl ring is present, only one ring is typically hydroxylated (e.g., phenytoin).

Traditionally, hydroxylation of aromatic compounds by P450 has been thought to be mediated by an arene oxide (epoxide) intermediate, followed by the “NIH shift,” as discussed previously (29–31) (Fig. 4.6). The formation of phenols and the isolation of urinary dihydrodiols, catechols, and glutathione conjugates (mercapturic acid derivatives) implicate arene oxides as intermediates in the metabolism of both benzene and substituted benzenes. Arene oxides are also prone to conjugate with glutathione to form premercapturic acids (see section on glutathione conjugation).

The CYP1A2 and CYP3A subfamilies are important contributors to 2- and 4-hydroxylation of estradiol, and CYP3A4 is an important contributor to 2-hydroxylation of the synthetic estrogens (e.g., 17 α -ethinyl estradiol) (38). The principal metabolite (as much as 50%) for estradiol is 2-hydroxyestradiol, with 4-hydroxy and 16 α -hydroxyestradiol as minor metabolites (Fig. 4.12). The 2-hydroxy metabolite of both estradiol and ethinyl estradiol have limited or no estrogenic activity, whereas the C-4 and

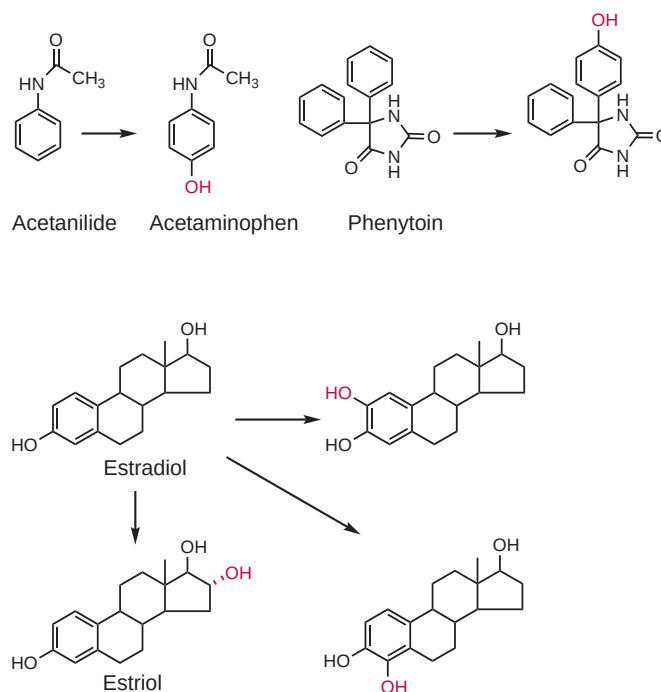


FIGURE 4.12 Examples of oxidative metabolism of aryl compounds catalyzed by P450.

C-16 α -hydroxy metabolites have a potency similar to that of estradiol. In humans, 16 α -hydroxyestradiol is the primary estrogen metabolite both in pregnancy and in breast cancer. The metabolites 16 α -hydroxyestrone and 4-hydroxyestrone can be carcinogenic in specific cells since they are capable of damaging cellular proteins and DNA after their activation to quinone intermediates.

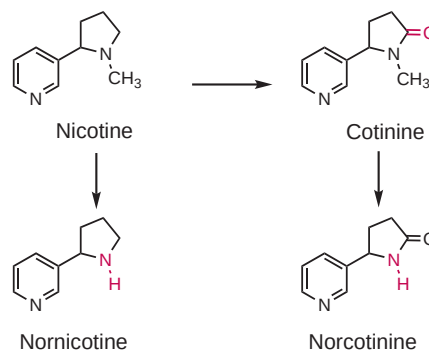
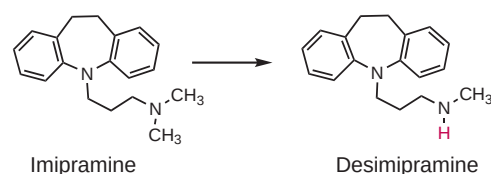
Xenobiotic-metabolizing enzymes not only detoxify xenobiotics but also cause the formation of active intermediates (bioactivation), which in certain circumstances can elicit a diversity of toxicities, including mutagenesis, carcinogenesis, and hepatic necrosis (37). In addition to glutathione, some nucleophiles, such as other sulfhydryl compounds (most effective), alcohols, and phosphates, can react with arene oxides. Many of these nucleophiles are found in proteins and nucleic acids. The covalent binding of these bioactive epoxides to intracellular macromolecules provides a molecular basis of these toxic effects (see the discussion of toxicity due to oxidative metabolism).

N-DEALKYLATION, OXIDATIVE DEAMINATION, AND N-OXIDATION

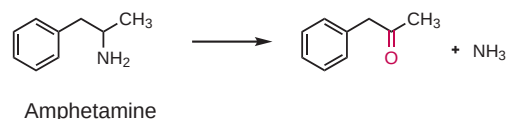
N-dealkylation Dealkylation of secondary and tertiary amines to yield primary and secondary amines, respectively, is one of the most important and frequently encountered reactions in drug metabolism. The proposed mechanism of oxidative *N*-dealkylation involves α -hydrogen abstraction or an electron abstraction from the nitrogen by the ferryl oxygen and has been previously discussed (29–31) (Fig. 4.7).

Typical *N*-substituents removed by oxidative dealkylation are methyl, ethyl, *n*-propyl, isopropyl, *n*-butyl, allyl, and benzyl. Dealkylation occurs mostly with a smaller alkyl group. Substituents that are more resistant to dealkylation include *t*-butyl (no α -hydrogen) and cyclopropylmethyl. In general, tertiary amines are dealkylated to secondary amines faster than secondary amines are dealkylated to primary amines. This rate difference has been correlated with lipid solubility. Appreciable amounts of secondary and primary amines therefore accumulate as metabolites that are more polar than the parent amine, thus slowing the rate at which the amines diffuse across membranes and reducing their accessibility to receptors. Frequently, these amine metabolites contribute to the pharmacologic activity of the parent substance (e.g., imipramine) (Fig. 4.13), or they produce unwanted side effects, such as hypertension, which result from the *N*-dealkylation of *N*-isopropylmethoxamine to methoxamine. The design of an analogous drug without these unwanted drug metabolites can be achieved by proper choice of replacement substituents, such as substitution of the *N*-isopropyl group in *N*-isopropylmethoxamine with a *tert*-butyl (*N*-*tert*-butylmethoxamine or butoxamine). *N*-dealkylation of substituted amides and aromatic amines occurs in a similar manner. *N*-substituted nonaromatic nitrogen heterocycles undergo oxidation on the α -carbon to a lactam (nicotine to cotinine) as well as *N*-dealkylation (nicotine to nornicotine, cotinine, and norcotinine) (Fig. 4.13).

Dealkylation:



Deamination:



N-Oxidation

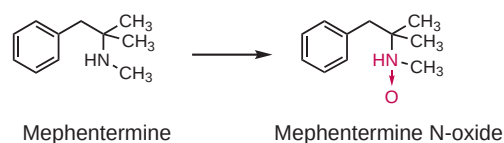


FIGURE 4.13 Examples of *N*-dealkylation, oxidative deamination, and *N*-oxidation reactions catalyzed by P450.

Oxidative Deamination The mechanism of oxidative deamination follows a pathway similar to that of *N*-dealkylation. Initially, oxidation to the iminium ion occurs, followed by decomposition to the carbonyl metabolite and ammonia. Oxidative deamination can occur with α -substituted amines, exemplified by amphetamine (Fig. 4.13). Disubstitution of the α -carbon inhibits deamination (e.g., phentermine). Some secondary and tertiary amines, as well as amines substituted with bulky groups, can undergo deamination directly, without *N*-dealkylation (e.g., fenfluramine). Apparently, this behavior is associated with increased lipid solubility.

N-oxidation In general, *N*-oxygenation of amines forms stable *N*-oxides with tertiary amines and amides and hydroxylamines with primary and secondary amines, when no α -protons are available (e.g., mephentermine and arylamines) (Fig. 4.13). Tertiary amines having one or more hydrogens on the adjacent carbon dealkylate via the *N*-oxide. Rearrangement of the *N*-oxide to a carbinolamine, which subsequently collapses, produces the secondary amine. Amine metabolites can be *N*-conjugated, increasing their excretion.

O- AND S-DEALKYLATION Oxidative *O*-dealkylation of ethers is a common metabolic reaction with a mechanism of dealkylation analogous to that of *N*-dealkylation: oxidation of the α -carbon and subsequent decomposition of the unstable hemiacetal to an alcohol (or phenol) and a carbonyl product (29–31). *S*-Dealkylation of thioethers to hemithioacetals is a minor pathway in comparison to the major direct P450 oxidation of sulfur to sulfoxides and/or sulfones. The majority of ether groups in drug molecules are aromatic ethers (e.g., codeine, prazosin, and verapamil). For example, codeine is *O*-demethylated to morphine (Fig. 4.14). The rate of *O*-dealkylation is a function of chain length (i.e., increasing chain length or branching reduces the rate of dealkylation). Steric factors and ring substituents influence the rate of dealkylation but are complicated by electronic effects. Some drug molecules contain more than one ether group, in which case only one ether is dealkylated. The methylenedioxy group undergoes variable rates of dealkylation to the 1,2-diphenolic metabolite. Metabolism of such a group is also capable of forming a stable complex with and inhibiting P450.

By analogy to *O*- and *N*-dealkylations, esters and amides can also be P450 oxidized by attaching the α -carbon to the ether oxygen of the ester (or α -carbon to the amide nitrogen). This reaction has not been usually recognized because of it is obscured by the competing, more common hydrolysis of esters and amides.

Aliphatic and aromatic methyl thioethers undergo *S*-dealkylation to thiols and carbonyl compounds. For example, 6-methylthiopurine is demethylated to give the active anticancer drug 6-mercaptapurine (Fig. 4.14). Other thioethers are oxidized to sulfoxides or sulfones (see *N*- and *S*-oxidations).

DEHALOGENATION Many halogenated hydrocarbons, such as insecticides, pesticides, general anesthetics, plasticizers, flame retardants, and commercial solvents, undergo a variety of different dehalogenation biotransformations (24,25). Because of our potential exposure to these halogenated compounds as drugs and environmental pollutants in air, soil, water, or food, it is important to understand the interactions between metabolism and toxicity. Some halogenated hydrocarbons form glutathione or mercapturic acid conjugates, whereas others undergo dehydrohalogenation and reductive dehalogenation catalyzed by

CYP2E1. In many cases, reactive intermediates, including radicals, anions, and cations, are produced that can react with a variety of tissue molecules.

Halogenated hydrocarbons differ in their chemical reactivity as a result of the electron-withdrawing properties of the halogens on adjacent carbon atoms, resulting in the α -carbon developing electrophilic character. The halogen atoms have also the ability to stabilize α -carbon cations, free radicals, carbanions, and carbenes.

Oxidative dehydrohalogenation is a common metabolic pathway of many halogenated hydrocarbons (25,29–31). P450-catalyzed oxidation generates the transient gem-halohydrin (analogous to alkane hydroxylation) that can eliminate the hydrohalic acid to form carbonyl derivatives (aldehydes, ketones, acyl halides, and carbonyl halides) (Fig. 4.7). This reaction requires the presence of at least one halogen and one α -hydrogen. gem-Trihalogenated hydrocarbons are more readily oxidized than are the gem-dihalogenated and monohalogenated compounds. The acyl and carbonyl halides formed are reactive metabolites that can react either with water to form carboxylic acids or nonenzymatically with tissue molecules (with a potential for eliciting increased toxicity). Chloramphenicol (RNHCOCHCl_2) is biotransformed into an acyl halide (RNHCOCOCl) that selectively acylates the apoprotein of P450 (40,41).

An excellent example of oxidative dehydrohalogenation leading to significant hepatotoxicity and nephrotoxicity is seen with the fluorinated inhalation anesthetics (Fig. 4.15). The toxicity of halothane and the fluranes is related to their metabolism to either an acid chloride (or fluoride) or a trifluoroacetate intermediate. The CYP2E1 has been identified as the isoform catalyzing the biotransformation of the fluranes (25,43,44). The hydroxylated intermediate decomposes spontaneously to reactive intermediates, an acid chloride (or fluoride) or trifluoroacetate, that can either react with water to form halide anions and a fluorinated carboxylic acid or bind covalently to tissue proteins to produce an acylated protein. The acylated protein becomes a “hapten,” stimulating an immune response and a hypersensitivity reaction. Halothane has received the most attention as a result of its ability to cause “halothane-associated” hepatitis. This immunologic reaction occurs after repeated exposure in surgical patients to trifluoroacetylate protein. The patient is sensitized to future exposures of the volatile anesthetic. After subsequent exposure to a fluorinated anesthetic, the antigenic trifluoroacetylate protein stimulates an immune response, producing halothane-like hepatitis. Because of the common metabolic pathway involving CYP2E1 for enflurane, isoflurane, desflurane, and methoxyflurane, halothane-exposed patients who have halothane hepatitis can show cross-sensitization to one of the other fluranes, triggering an idiosyncratic hepatic necrosis. The formation of antigenic protein is related to the amount of CYP2E1-catalyzed metabolism of each agent: halothane (20% to 40%) > enflurane (2% to 8%) > isoflurane (0.2% to 1.0%) > desflurane (<0.1%). Enough fluoride ion is generated from oxidative dehalogenation during flurane anesthesia to produce subclinical nephrotoxicity. Interestingly, female rats metabolize halothane more slowly than do male

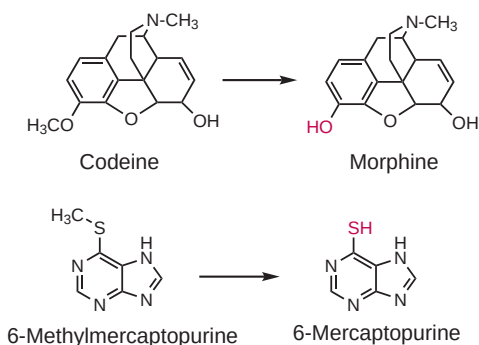


FIGURE 4.14 Examples of *O*- and *S*-dealkylations catalyzed by P450.

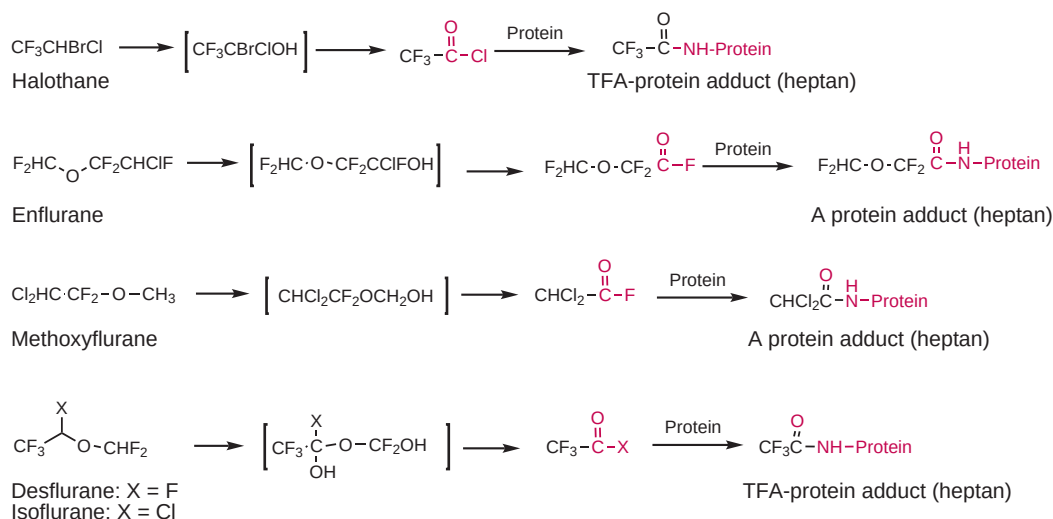


FIGURE 4.15 CYP2E1-catalyzed metabolism of fluorinated volatile anesthetics to antigenic proteins.

rats and are less susceptible to hepatotoxicity than males are. For patients with preexisting liver dysfunction, isoflurane or desflurane can be a better choice of general anesthetic.

In today's environment, most humans have been exposed to many CYP2E1-inducing agents (including recreational, industrial, and agricultural chemicals and alcohol), having an unknown effect on hepatic toxicity from volatile anesthetics. Enhanced activity of CYP2E1 has been observed in obesity, isoniazid therapy, ketogenic diets, and alcoholism.

AZO AND NITRO REDUCTION In addition to the oxidative systems, liver microsomes also contain enzyme systems that catalyze the reduction of azo and nitro compounds to primary amines. A number of azo compounds, such as Prontosil and sulfasalazine (Fig. 4.16), are converted to aromatic primary amines by azoreductase, an NADPH-dependent enzyme system in the liver microsomes. Evidence exists for the participation of P450 in some reductions. Nitro compounds (e.g., chloramphenicol and nitrobenzene) are reduced to aromatic primary amines by a nitroreductase, presumably through nitrosamine and hydroxylamine intermediates. These reductases are

not solely responsible of reduction of azo and nitro compounds; reduction by the bacterial flora in the anaerobic environment of the intestine can also occur.

Oxidations Catalyzed by Flavin Monooxygenase

Flavin-containing monooxygenase (FMO) oxidizes drugs, xenobiotics, and environmental chemicals containing a "soft nucleophile," usually nitrogen or sulfur. FMO, unlike P450, is a monooxygenase, utilizing the two-electron reducing equivalents of NADPH (hydride anion) to reduce one atom of molecular oxygen to water while the other oxygen atom is used to oxidize the substrate. Because oxygen activation occurs before substrate addition, any compound binding to 4 α -hydroperoxyflavin, the enzyme-bound monooxygenating FMO intermediate, is a potential substrate. The products formed from FMO-catalyzed oxidation are consistent with a two-electron oxidation of the heteroatom. Unlike P450, FMO does not catalyze epoxidation reactions or hydroxylation at unactivated carbon atoms of xenobiotics. FMO and P450 also exhibit similar tissue and cellular location, molecular weight, and substrate specificity, and exist as multiple enzymes under developmental control.

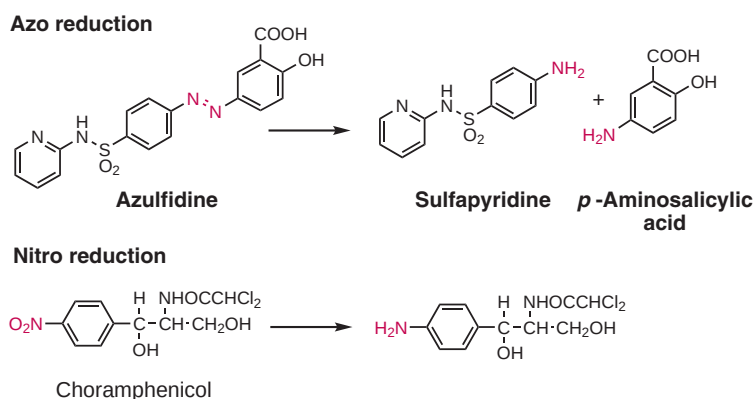


FIGURE 4.16 P450-catalyzed reduction of azo and nitro compounds.

Humans express five different FMOs (FMO1, FMO2, FMO3, FMO4, and FMO5) in a tissue-specific manner, with different substrate specificities. The human FMO functional gene family consists of five families each expressing a single protein unlike P450. Three of the five expressed human FMO genes, *FMO1*, *FMO2*, and *FMO3*, exhibit genetic polymorphisms. FMO3 is the prominent form in adult human liver that is likely to be associated with the bulk of FMO-mediated metabolism and contributes to the disease known as trimethylaminuria. FMO2 is largely found in the lung. Because of their low expression, less is known about the substrate specificity for FMO4 and FMO5. Thus, these enzymes do not contribute significantly to human drug metabolism. FMO does not require a reductase to transfer electrons from NADPH. The catalytic cycles of FMO and P450 are very different. Another distinction is the lack of induction of FMOs by xenobiotics. In general, P450 is the major contributor to oxidative xenobiotic metabolism. However, FMO activity

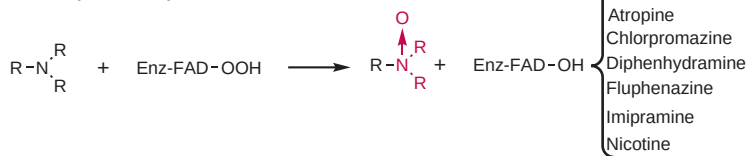
may be of significance in a number of cases and should not be overlooked. FMO and P450 have overlapping substrate specificities, but often yield different metabolites with potentially diverse toxicologic/pharmacologic consequences. The physiologic functions of FMO are poorly understood (45). Unlike P450, which exhibits interindividual variation in expression to both genetic and environmental factors, FMO is not regulated by environmental factors, and interindividual variability in FMO enzyme expression would be predominately genetic in origin.

N- and S-Oxidations

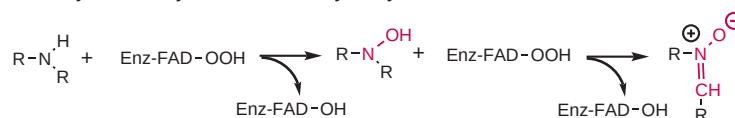
FMO constitutes an alternative biotransformation pathway for converting *N*- and *S*-containing lipophilic drugs and xenobiotics to more polar materials that are more efficiently excreted in the urine. Typically, FMO catalyzes oxygenation of the *N*- and *S*-heteroatoms ("soft nucleophiles") (Fig. 4.17) but not heteroatom dealkylation

Nitrogen compounds:

Tert-acyclic and cyclic amines to *N*-oxides

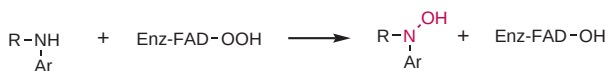


Sec-acyclic and cyclic amines to hydroxylamines and nitrones

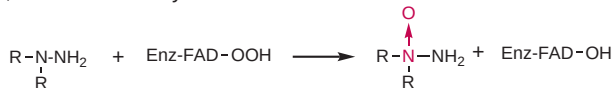


Drugs: Desmethylinipramine; Desmethyltrifluoperazine

N-alkyl and *N,N*-dialkylaryl amines to hydroxylamines

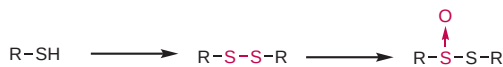


1,1-Disubstituted hydrazines



Sulfur compounds:

Thiols and disulfides



Thioethers/Sulfides

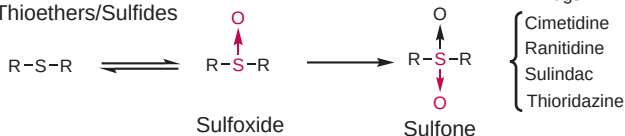


FIGURE 4.17 Examples of flavin monooxygenase (FMO) oxidations.

reactions. Normally, FMO is not inducible by phenobarbital, nor is it affected by P450 inhibitors. With few exceptions, however, xenobiotic substrates of FMO are also substrates of the isoforms of P450, producing similar oxidation products. Which monooxygenase is responsible for the oxidation can be readily determined, because FMO is thermally labile in the absence of NADPH, whereas P450 is stable.

Of the many nitrogen functional groups in xenobiotics, only secondary and tertiary acyclic, cyclic, and arylamines as well as hydroxylamines and hydrazines are oxidized by FMO and excreted in the urine (Fig. 4.17). The tertiary amines form stable amine oxides (*N*-oxides), which are highly polarized, hydrophilic, and less basic compounds that are reduced back to the parent tertiary amine in the gastrointestinal tract. On the other hand, secondary amines are sequentially oxidized to hydroxylamines, nitrones, and a complex mixture of products. Secondary *N*-alkylarylamines can be *N*-oxygenated to reactive *N*-hydroxylated metabolites, which are responsible for the toxic, mutagenic, and carcinogenic activity of these aromatic amines. For example, the chemically unstable hydroxylamine intermediates of aromatic amines degrade into bladder carcinogens (see the discussion of this type of toxic mechanism under glucuronic acid conjugation), and the hydroxamic acid intermediates of *N*-arylacetamides are bioactivated into liver carcinogens. Hepatic FMO, however, will not catalyze the oxidation of primary alkyl- or arylamines, except for the carcinogenic *N*-hydroxylated derivatives of 2-aminofluorene, 2-aminoanthracene, and other amino PAHs.

S-Oxidation occurs almost exclusively by FMO (Fig. 4.17). Sulfides are oxidized to sulfoxides and sulfones; thiols are oxidized to disulfides and thiocarbamates; and mercaptopyrimidines and mercaptoimidazoles (i.e., the antithyroid drug methimazole) are oxidized via sulfinates

(RSOH) to sulfinates (R-SO₂-OH), all of which are eliminated in the urine. FMO *S*-oxygenates a number of sulfur-containing substrates and, in some cases, with great stereoselectivity. The hepatotoxicity of thioureas (e.g., the antimycobacterial drug ethionamide) is the result of metabolic bioactivation of thioureas due to FMO3-dependent *S*-oxidation to produce toxic sulfinic acid metabolites. Metabolic bioactivation of ethionamide by bacterial FMOs may be its antimycobacterial mechanism of action.

Primary aromatic amines and amides, aromatic heterocyclic amines and imines, and the aliphatic primary amine phentermine are *N*-oxidized by P450 to hydroxylamines. P450 oxidizes carbon disulfide to carbon dioxide and hydrogen sulfide and the antipsychotic phenothiazines to sulfoxides.

Catalytic Cycle for Flavin Monooxygenase

The major steps in the catalytic cycle for FMO are shown in Figure 4.18 (45,46). Like most of the other monooxygenases, FMO requires NADPH and oxygen as cosubstrates to catalyze the oxidation of the xenobiotic substrate. Unlike P450, however, the xenobiotic being oxidized does not need to be bound to the 4 α -hydroperoxyflavin intermediate (FAD-OOH) for oxygen activation to occur. Apparently, FMO is present within the cell in its enzyme-bound activated hydroperoxide (Enz-FAD-OOH) state ready to oxidize any suitable lipophilic substrate that binds to it. The FMO uses a nonradical, nucleophilic displacement type of mechanism binding dioxygen with a reduced flavin. The reactive oxygen intermediate is a reactive derivative of hydrogen peroxide, flavin-4 α -hydroperoxide (Fig. 4.18, insert), which is reactive enough to successfully attack a lone electron pair on a heteroatom, such as nitrogen or sulfur, but not reactive enough to attack a typical C-H bond. These studies suggest the xenobiotic substrate interacts with

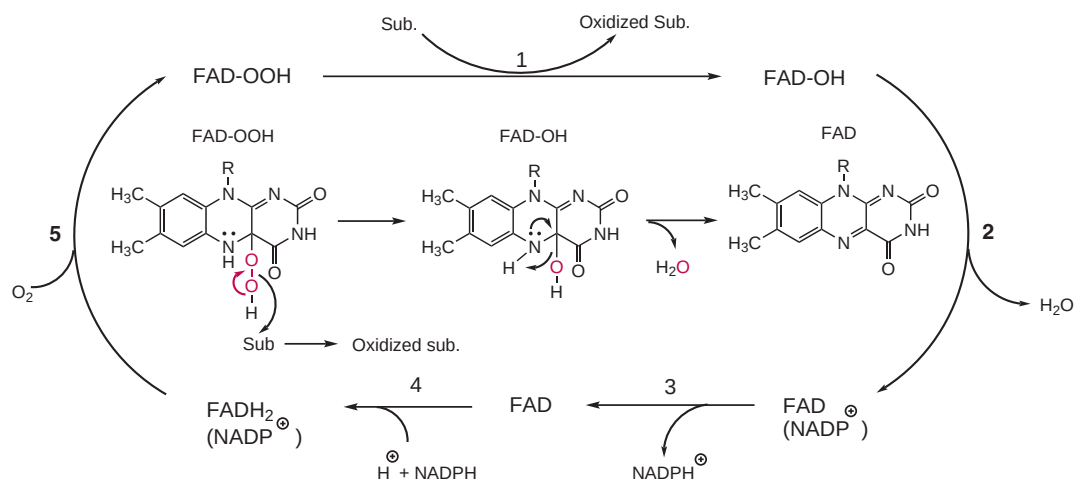


FIGURE 4.18 Flavin monooxygenase (FMO) catalytic cycle. Oxygenated substrate is formed by nucleophilic attack of a substrate (Sub.) by the terminal oxygen of the enzyme-bound hydroperoxyflavin (FAD-OOH), followed by heterolytic cleavage of the peroxide (1). The release of H₂O (2) or of NADP⁺ (3) is rate-limiting for reactions catalyzed by liver FMO. Reduction of flavin by NADPH (4) and addition of oxygen (5) complete the cycle by regeneration of the oxygenated FAD-OOH.

the 4 α -hydroperoxyflavin form of FMO and is oxidized by oxygen transfer from Enz-FAD-OOH to form the oxidized product. Neither the substrate nor the oxidized substrate is essential for any other steps in the cycle. Steps 2 to 5 simply regenerate the oxygenating agent Enz-FAD-OOH from Enz-FAD-OH, NADPH, oxygen, and a proton. Any compound readily crossing cell membranes by passive diffusion and penetrating to the FMO-bound hydroperoxyflavin intermediate is a potential substrate, thus explaining the broad substrate specificity exhibited for FMO. The fact that the xenobiotic substrate is not required for activation of the FMO-hydroperoxyflavin state distinguishes FMO from P450 monooxygenases, in which substrate binding initiates the P450 catalytic cycle and activation of oxygen to the ferryl oxygenating agent. It is not unusual for FMO oxidation products to undergo reduction to the parent xenobiotic, which can enter into repeated redox reactions (termed “metabolic cycling”).

Results of substrate specificity studies suggest that the number of ionic groups on endogenous substrate is an important factor enabling FMO to distinguish between xenobiotic and endogenous substrates, preventing the indiscriminate oxidation of physiologically important amine and sulfur compounds (47). Without exception, FMO readily catalyzes the oxidation of uncharged amines or sulfur compounds (in equilibrium with its respective monocation or monoanion; for sulfur compounds, the charge is on sulfur atom). The FMO will not catalyze the oxidation of dianions (e.g., thiamine pyrophosphate), dications (e.g., polyamines), dipolar ions (e.g., amino acids and peptides), or other polyionic compounds with one or more anionic groups (i.e., COO⁻) distal to the heteroatom (e.g., coenzyme A).

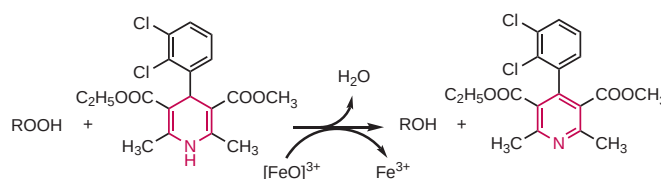
Unlike the P450 system, only three isoforms of hepatic FMO have been characterized in the adult human liver (48,49): FMO1, the major form in fetal tissue; FMO2; and FMO3. The availability of different forms of FMO can be of clinical importance in the pharmacologic and toxicologic properties of FMO-dependent drug oxidations.

The substrate specificity for FMO3 is distinct from that of FMO1. FMO3 *N*-oxygenates primary, secondary, and tertiary amines, whereas FMO1 is only efficient at *N*-oxygenating tertiary amines. FMO3 is sensitive to steric features of the substrate, and aliphatic amines with linkages between the nitrogen atom and a large aromatic group such as a phenothiazine of at least five carbons are *N*-oxygenated significantly more efficiently than those substrates with two or three carbons. Amines with smaller aromatic substituents, such as phenethylamines, are often efficiently *N*-oxygenated by FMO3. Interindividual variation in FMO3-dependent metabolism of drugs, xenobiotics, and environmental chemicals is therefore more likely to be a result of genetic and not environmental effects. Certain mutations of the *FMO3* gene have been associated with deficient *N*-oxygenation of trimethylamine, which results in an inherited genetic condition called trimethylaminuria. Allelic variations of FMO3 might cause abnormal metabolism of drugs, which has clinical implications

for human drug metabolism. For example, (*S*)-nicotine *N*-1'-oxide formation is a highly stereoselective product of FMO3 for adult humans who smoke cigarettes. Thus, *FMO3* genetic polymorphism and poor metabolism phenotype in certain human populations could contribute to adverse drug reactions or exaggerated clinical response to certain medications. *FMO3* may be another example of an environmental gene that participates in a protective mechanism to help shield humans from potentially toxic exposure to chemicals.

Peroxidases and Other Monooxygenases

Peroxidases are hemoproteins and are closely related enzymes to P450 monooxygenase (50,51). The differences between peroxidase and P450 is that the cysteine axial ligand of P450 has been replaced by a histidine residue as well as the inclusion of polar amino acids within proximity of the heme-active site. These polar amino acid differences in the peroxidase heme-active site allow the peroxidase to rapidly catalyze the reduction of hydroperoxides to alcohol (water in the case of hydrogen peroxide) and the simultaneous reoxidation of peroxidase. The normal course of peroxide (ROOH)-catalyzed oxidation involves the formation of the ferryl-oxo intermediate, analogous to the ferryl complex in P450. Thus, the catalytic cycle of peroxidases is simpler than that of P450. Unlike P450, which is capable of oxidizing almost any type of substrate, peroxidase-catalyzed oxidation of drugs and xenobiotics is limited to electron-rich substrates that are easily oxidized (e.g., heteroatom oxygenation, aromatization [oxidation] of 1,4-dihydropyridines [calcium channel blockers] and arylamines).



Cyclooxygenase (COX; also known as prostaglandin synthase or prostaglandin endoperoxide synthetase) is a widely distributed heme-peroxidase enzyme responsible for the formation of important biologic mediators called prostanoids, including prostaglandins, prostacyclin, and thromboxane from arachidonic acid (see also Chapter 31). COX catalyzes the heme-peroxidation of arachidonic acid by means of hydrogen atom abstraction from arachidonic acid by a tyrosine radical generated by a ferryl-oxo-heme intermediate to hydroperoxy endoperoxide prostaglandin G₂ (PGG₂), which is then reduced to prostaglandin H₂, regenerating the COX. Two oxygen molecules then react with the arachidonic acid radical to produce PGG₂. During the reduction of PGG₂ to prostaglandin H₂, drug substrates can be oxidized. Three COX isoenzymes are known: COX-1, COX-2, and COX-3. Inhibitors of COX are the NSAIDs.

Myeloperoxidase (MPO) is a heme-peroxidase enzyme most abundantly present in neutrophil granulocytes (a subtype of white blood cells). MPO differs from P450 and other peroxidases in that the substrates are hydrogen peroxide and chloride anion. The chloride anion is oxidized to hypochlorous acid (HOCl) in the granulocyte, which is cytotoxic and very effective in killing bacteria, virus, and other pathogens—the natural function of neutrophils. HOCl is also used to disinfect drinking water. HOCl can also oxidize drugs to reactive intermediates (e.g., nitrenium ions), which can be responsible for causing agranulocytosis. Examples of HOCl-generated reactive intermediates associated with agranulocytosis are the atypical antipsychotic clozapine, the antimalarial amodiaquine, the antipyretic aminopyrine, and the phosphodiesterase cardiotonic vesnarinone.

Other peroxidases include horseradish peroxidase (found in plants), lactoperoxidase found in breast milk, and thyroid peroxidase found in the thyroid gland to produce thyroid hormones from iodide.

Other monooxygenases catalyzing oxidation reactions similar to P450 include dopamine β -monooxygenase, a mammalian copper-containing enzyme catalyzing carbon hydroxylation, epoxidation, S-oxygenation, and N-dealkylation reactions, and nonheme iron-containing enzymes from bacteria and plants.

Nonmicrosomal Oxidations

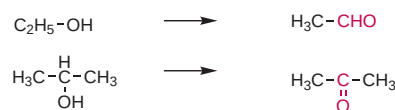
In addition to the microsomal monooxygenases, other oxidases and dehydrogenases that catalyze oxidation reactions are present in the mitochondrial and soluble fractions of tissue homogenates.

Oxidation of Alcohols

Alcohol dehydrogenases (ADH) are a family of nicotinamide adenine dinucleotide (NAD)-specific enzyme dehydrogenase enzymes that occur in many organisms and are responsible for catalyzing oxidation of primary and secondary alcohols to aldehydes and ketones, respectively, with reduction of NAD^+ to NADH. ADHs can also catalyze the reverse reaction, reducing NADH to NAD^+ . In humans and many other animals, genetic evidence suggests that early on in evolution, ADHs were effective for metabolizing endogenous and exogenous formaldehyde and metabolizing naturally occurring toxic alcohols contained in foods or produced by bacteria in the digestive tract. Another evolutionary purpose may be metabolism of the endogenous alcohol vitamin A (retinol) to generate the hormone retinoic acid. In yeast, plants, and many bacteria, some ADHs catalyze the opposite reaction as part of fermentation to ensure a constant energy supply of NAD^+ for the host organism.

In humans, ADH exists as a dimer with a mass of 80 kDa and is encoded by at least seven different genes. There are five classes (I to V) of ADHs, but the hepatic form that is primarily present in humans is class I. Class I consists of A, B, and C subunits that are encoded by the

genes ADH1A, ADH1B, and ADH1C. The human genes that encode class II, III, IV, and V ADHs are ADH4, ADH5, ADH7, and ADH6, respectively. The large number of ADH classes gives the body the capability to metabolize/detoxify a variety of primary and secondary alcohols. Human ADH is present at high levels in the liver and the lining of the stomach and exhibits a broad specificity for alcohols. Most primary alcohols are readily oxidized to their corresponding aldehydes. Some secondary alcohols are oxidized to the ketones, whereas other secondary and tertiary alcohols are excreted either unchanged or as their conjugate metabolite. Some secondary alcohols also show mixed activity due to steric factors and a lack of substrate affinity for the enzyme. The mechanism of dehydrogenation involves abstraction by the enzyme and transfer of a hydride anion from the alcohol to NAD^+ , reducing NAD^+ to NADH, and subsequent formation of an aldehyde. ADH contains a two Zn^{+2} ions in each subunit. One of those Zn ions is located at the catalytic site and binds the hydroxyl group of the alcohol in place for dehydrogenation to occur. ADH is located in the soluble fraction of tissue homogenates.



Oxidation by ADH is the principal pathway for ethanol metabolism, but the CYP2E1 isoform has also a significant role in ethanol metabolism and tolerance. Two-thirds of ingested ethanol is oxidized by ADH and the remainder by CYP2E1; during intoxication, however, ethanol induces the expression of CYP2E1. The induction of CYP2E1 contributes to the activation of other drugs and xenobiotics, increasing the vulnerability of heavy drinkers to anesthetic drugs, over-the-counter analgesics, prescription drugs, and chemical carcinogens. In turn, the excessive amounts of acetaldehyde generated cause hepatotoxicity, lipid peroxidation of membranes, formation of protein adducts, and other hepatocellular changes.

The toxicity of methanol and ethylene glycol in humans has long been recognized, but frequent reports of such toxicity are not surprising given the number of consumer products containing methanol and ethylene glycol (automotive antifreeze). Methanol (wood alcohol or methyl alcohol) is commonly used as a solvent in organic synthetic procedures and is available to consumers in a variety of products, including solid fuels (Sterno), paint removers, motor fuels, antifreeze, and alcoholic beverages (an adulterant in wines or an unintentional ingredient). Oral methanol toxicity in humans is characterized by its rapid absorption from the gut, followed by a latent period of many hours before metabolic acidosis (lowered blood pH and bicarbonate levels) and ocular toxicity are evident. The metabolic acidosis and

blindness result from the excessive accumulation of formic acid and the inability of the hepatic tetrahydrofolate pathway to oxidize formate to carbon dioxide. The rate of elimination of methanol from the blood is relatively slow compared to the rapid rate for ethanol, accounting for its long latency period. Its half-life ranges from 2 to 3 hours at low blood concentration to 27 hours at high blood concentration. Evidence supports the singular role of liver ADH in the metabolism of methanol to formaldehyde, although it is oxidized slowly by ADH (approximately one-sixth the rate of ethanol). The demonstration that methanol is a substrate for ADH provides a rational basis for the use of ethanol in the treatment of methanol toxicity. Ethanol depresses the rate of methanol oxidation by acting as a competitive substrate for ADH, reducing the formation of formaldehyde. On the other hand, formaldehyde is not customarily detected in the blood as a result of its rapid metabolism by ADH to formate. Although human exposure to methanol vapor is less prevalent, methanol is rapidly absorbed through the skin or by inhalation, and depending on the severity of exposure, this can result in methanol poisoning. Ethylene glycol is oxidized to hydroxyacetaldehyde and glyoxal and, subsequently, to oxalate by ADH. When eliminated into the urine, oxalate forms calcium oxalate crystals that can block the kidney tubules. 4-Methylpyrazole (fomepizole) is an alcohol dehydrogenase inhibitor that is used as an antidote for the treatment of methanol or ethylene glycol poisoning. 1,4-Butanediol is a solvent that has become popular as a date-rape drug and drug of abuse due to its metabolism to γ -hydroxybutyrate (see Chapter 13), which binds to the γ -hydroxybutyrate receptor, which in turn produces central nervous system sedation with amnesia.

ADH also functions as a reductase when it catalyzes the reduction of an aldehyde or ketone to an alcohol. In addition, other NADP- or NAD-dependent dehydrogenases in the cytosol are capable of reducing a variety of ketones. Ketones are stable to further oxidation and, consequently, yield reduction products as major metabolites. Examples of reduction include the sedative-hypnotic chloral hydrate to trichloroethanol, the opioid antagonist naltrexone to 6- β -hydroxynaltrexol, the opioid analgesic methadone to α -methadol, the antipsychotic haloperidol to hydroxyhaloperidol, and the antiemetic dolasetron to dihydrodolasetron. These alcohol metabolites are all pharmacologically active.

Aldehyde Dehydrogenase

Aldehyde dehydrogenases (ALHDs) (not to be confused with aldehyde oxidase) are a family of polymorphic NAD⁺-specific enzymes that catalyze the NAD⁺-dependent oxidation (dehydrogenation) of aldehydes to carboxylic acids, to be subsequently metabolized by the body's muscle and heart. There are three mammalian classes of these enzymes: class 1 (low K_m , cytosolic, ALDH1), class 2

(low K_m , mitochondrial, ALDH2), and class 3 (high K_m , such as those expressed in tumors, stomach, and cornea). In all three classes, constitutive (an enzyme whose activity is constant and active) and inducible forms exist. ALDH1 and ALDH2 are the most important enzymes for aldehyde oxidation, and both are tetrameric enzymes composed of 5-kDa subunits. These enzymes are found in many tissues of the body but are at the highest concentration in the liver. ALDH1 and ADH are responsible for the metabolism of retinol to retinoic acid. ALDH2 has a major role in the hepatic detoxication of acetaldehyde and other aldehydes produced from oxidation of ethanol and other primary alcohols and oxidation of endogenous aldehydes, such as those produced by the oxidation of biogenic amines (norepinephrine, serotonin, dopamine). These aldehydes can be toxic, and health problems arise when the aldehyde cannot be cleared. The accumulation of unmetabolized acetaldehyde in the blood has been shown to be a carcinogen in lab animals. For example, when high levels of acetaldehyde occur in the blood, facial flushing, light headedness, palpitations, nausea, and general "hang-over" symptoms occur. These symptoms are indicative of a disease known as "Asian flush" or "Oriental flushing syndrome," due to a mutant form of ALDH2, termed *ALDH2*2*, where a lysine residue replaces a glutamate in the active site. These individuals have about 8% of the wild-type (normal) *ADH2* allele. The mutant allele shows a higher K_m for NAD⁺ than the wild-type allele. This mutation is common in Japan, where 41% of a nonalcoholic control group were ALDH2 deficient and where only 2% to 5% of an alcoholic group were ALDH2 deficient. In Taiwan, the numbers are similar, with 30% of the control group showing the deficiency and 6% of alcoholics displaying the deficiency. The deficiency is manifested by slow acetaldehyde removal and low alcohol tolerance, which perhaps leads to a lower frequency of alcoholism. These symptoms are also the same as those observed in people who drink ethanol while being treated with the drug disulfiram (Antabuse), which is why it is used to treat alcoholism. These patients show higher blood levels of acetaldehyde and become violently ill upon consumption of even small amounts of alcohol. Several other drugs (e.g., metronidazole) cause a similar reaction known as "disulfiram-like reaction."



Aldo-Keto Reductases

Aldo-keto reductases (AKRs) are a superfamily of enzymes that are typically monomeric, cytosolic, NAD(P)H-dependent oxidoreductases that share high sequence identity (<40%) and similar three-dimensional structures and are involved in the formation of hyperosmotic sugars, which can contribute to diabetic complications. They also metabolize reactive aldehydes, prostaglandins, steroid hormones, chemical carcinogens, and drugs.

AKRs detoxify endogenous toxicants (e.g., cytotoxic/mutagenic bifunctional electrophiles derived from the decomposition of lipid hydroperoxides), detoxify nicotine-derived nitrosamino ketone carcinogens in tobacco, and activate polycyclic aromatic trans-dihydrodiols to yield reactive ortho-quinones. AKRs also detoxify mycotoxin metabolites (e.g., aflatoxin dialdehyde), protecting against hepatocellular carcinomas. AKRs could have a central role in the cellular response to osmotic, electrophilic, and oxidative stress.

Molybdenum Hydroxylases

Molybdenum hydroxylases are non-P450 enzymes capable of catalyzing the oxidation of drugs. The molybdenum hydroxylases, which include aldehyde oxidase, xanthine oxidase, and xanthine dehydrogenase, are more commonly found in the cytosol of mammalian liver and carry out the oxidation and detoxification of a number of structurally different azaheterocycles (52). The efficient oxidation of endogenous purine nucleosides suggests that their metabolism and detoxification might be an important physiologic role of the molybdenum hydroxylases. Among the azaheterocycles metabolized are derivatives of pyridine, quinoline, pyrimidine, purine, quinazoline, and pteridines. These hydroxylases, as a rule, oxidize the α -carbon to the nitrogen of the azaheterocycle to oxo metabolites (also known as lactams). The molybdenum hydroxylases are cytosolic homodimeric enzymes (two identical monomers) with a common electron-transfer system in each enzyme subunit: a molybdopterin (a pyranopteridine ligand) cofactor that binds the single molybdenum ion, two iron-sulfur centers, and FAD molecule. The molybdenum hydroxylases catalyze their reactions differently than P450 and other hydroxylase enzymes, requiring water rather than molecular oxygen as the source of the oxygen atom incorporated into the metabolite and with the concomitant reduction of molecular oxygen to superoxide (53,54). The active sites possess a catalytically labile $\text{Mo}^{5+}\text{-OH}$ (or, possibly, $\text{Mo}^{6+}\text{-OH}_2$) group that is transferred to the substrate during the course of the hydroxylation reaction.

ALDEHYDE OXIDASE Aldehyde oxidase is a molybdenum cofactor-specific enzyme that generates carboxylic acids and hydrogen peroxide from aldehydes in the presence of oxygen. The enzyme is a homodimer and requires FAD in addition to molybdenum as cofactors. In addition to metabolizing some aldehydes, aldehyde oxidase also oxidizes a variety of azaheterocycles but not thia- or oxaheterocycles. Of the various purine nucleosides metabolized by aldehyde oxidase, the 2-hydroxy and 2-amino derivatives are more efficiently metabolized, and for the N^9 -substituents, the typical order of preference of the acyclic nucleosides is as follows: 9-[(hydroxy-alkyloxy)methyl]-purines > 2'-arabinofuranosyl > H. The kinetic rate constants for purine analogues reveal that the pyrimidine portion of the purine ring system is more important for substrate affinity

than the imidazole portion. Aldehyde oxidase is inhibited by potassium cyanide and menadione (synthetic vitamin K).

Aldehyde oxidase metabolizes an assortment of azaheterocycles including the short-acting sedative-hypnotic drug zaleplon (a pyrazolo[1,5 α] pyrimidine derivative) to its 5-oxo metabolite; the anticancer drug thioguanine to 8-oxothioguanine; the α_2 -adrenergic agonist brimonidine (a pyrimidine derivative) to its 2-oxo-, 3-oxo-, and 2,3-dioxo- metabolites; quinine and quinidine to their 2-quinolone metabolites; the proantiviral drug famciclovir (a purine derivative) to its active 6-oxo metabolite (penciclovir); O^6 -benzylguanine to its 8-oxo metabolite (also formed primarily from CYP3A4); the metabolism of the anticancer drug DACA (an acridine-4-carboxamide derivative) to its 9-acridone metabolite; and the antiseizure drug zonisamide (a 1,2-benzisoxazole derivative) primarily by reductive cleavage of the 1,2-benzisoxazole ring to 2-sulfamoylacetylphenol. Although the azaheterocycles thiazole and oxazole are not metabolized by aldehyde oxidase, their carbocyclic analogues, benzothiazole, benzoxazole and 1,2-benzisoxazole, are metabolized. On the other hand, the heterocycles, benzothiophene and benzofuran, which do not contain a nitrogen atom, are not metabolized by or inhibited by aldehyde oxidase. The hepatotoxic and neurotoxic 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) is metabolized by aldehyde oxidase to its nontoxic MP-2-pyridone metabolite (MPTP lactam). Both aldehyde oxidase and xanthine oxidase contribute to the first-pass hepatic metabolism of orally administered methotrexate (a 2,4-diaminopteridine) to its 7-hydroxymethotrexate metabolite.

Potent inhibitors of aldehyde oxidase include the selective estrogen receptor modulator raloxifene, tamoxifen, estradiol, and ethinyl estradiol. Other classes of drugs that demonstrate inhibition of aldehyde oxidase included phenothiazines, tricyclic antidepressants, tricyclic atypical antipsychotic agents, dihydropyridine calcium channel blockers, loratadine, cyclobenzaprine, amodiaquine, maprotiline, ondansetron, propafenone, domperidone, quinacrine, ketoconazole, verapamil, tacrine, and salmeterol. Thus, the possibility of clinical drug interactions mediated by inhibition of this aldehyde oxidase could be observed between these drugs and drugs for which aldehyde oxidase is the primary route of metabolism.

XANTHINE OXIDASE AND XANTHINE DEHYDROGENASE Xanthine oxidase and xanthine dehydrogenase represent different forms of the same gene product and are involved in the oxidative metabolism of purines with the formation of the uric acids. Xanthine dehydrogenase can be converted to xanthine oxidase by reversible sulfhydryl oxidation; xanthine dehydrogenase and xanthine oxidase are interconvertible.



These two enzyme forms and their reactions often are referred to as xanthine oxidoreductase. Xanthine oxidase is the rate-limiting enzyme in human purine catabolism of hypoxanthine to uric acid and can also generate ROS. Xanthine oxidase is a cytosolic 370-kDa molybdo-flavoenzyme that requires a molybdopterin cofactor, two iron-sulfur centers, and FAD. Both xanthine oxidase and xanthine dehydrogenase have important roles in the metabolism of a number of purine anticancer drugs to their active and inactive metabolites. Although xanthine oxidase is strongly inhibited by the antigout drug allopurinol, aldehyde oxidase also oxidizes it to oxypurinol. Only xanthine dehydrogenase requires NAD^+ as an electron acceptor for the oxidation of azaheterocycles. 6-Mercaptopurine is metabolized by xanthine oxidase to 6-mercaptopuric acid. In humans, xanthine oxidase is normally found in the liver and not free in the blood. Xanthinuria is a rare genetic disorder where the lack of xanthine oxidase leads to high concentration of xanthine in blood and can cause health problems such as renal failure. There is no specific treatment for this disorder except to avoid foods high in purine and to maintain a high fluid intake.

Oxidative Deamination of Amines

Monoamine oxidase (MAO) and diamine oxidase catalyze oxidative deamination of amines to the aldehydes in the presence of oxygen. The aldehyde products can be metabolized further to the corresponding alcohol or acid by aldehyde oxidase or dehydrogenase.

Monoamine Oxidase

MAO is a mitochondrial membrane flavin-containing enzyme that catalyzes the oxidative deamination of monoamines where oxygen is used to remove an amine group from a monoamine substrate, resulting in the formation of the corresponding aldehyde and ammonia according to the following equation:

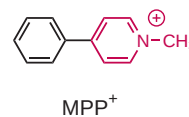
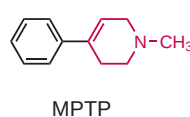


Substrates for this enzyme include monoamines and secondary and tertiary amines in which the amine substrates are methyl groups. The amine must be attached to an unsubstituted methylene group, and compounds having substitution at the α -carbon atom are poor substrates for MAO (e.g., aniline, amphetamine, and ephedrine) but are oxidized by the P450 enzymes rather than by MAO (Fig. 4.13). For secondary and tertiary amines, alkyl groups larger than a methyl and branched alkyl groups (i.e., isopropyl, *t*-butyl, or β -phenylisopropyl) inhibit MAO oxidation, but such substrates can function as reversible inhibitors of MAO. Nonselective irreversible inhibitors of MAO include hydrazides (phenelzine) and tranylcypromine and the MAO-B selective inhibitors pargiline and selegiline. MAO is important in regulating the metabolic degradation of catecholamines and serotonin in neural tissues, and hepatic MAO has a crucial defensive

role in inactivating circulating monoamines or those that originated in the gastrointestinal tract and were absorbed into the systemic circulation (e.g., tyramine).

There are two types of MAO: MAO-A and MAO-B. They show dissimilar substrate preferences and different sensitivities to inhibitors. MAO-A is found mainly in peripheral adrenergic nerve terminals and shows substrate preference for 5-hydroxytryptamine, norepinephrine, and epinephrine, and is also found in the liver, gastrointestinal tract, and placenta. MAO-B is found principally in platelets and shows selectivity for nonphenolic, lipophilic β -phenethylamines. Common substrates to both MAO-A and MAO-B are dopamine, tyramine, and other monophenolic phenylethylamines.

A contaminant in the synthesis of reversed esters of meperidine, MPTP, was discovered to be a highly selective neurotoxin for dopaminergic cells, producing parkinsonism (see also Chapter 13) (47). The neurotoxicity of MPTP is associated with cellular destruction in the substantia nigra along with severe reductions in the concentration of dopamine, norepinephrine, and serotonin. The neurotoxic action for MPTP involves a sequence of events beginning with the metabolic activation of MPTP to the toxic metabolite MPP⁺ (1-methyl-4-phenylpyridinium ion) by MAO-B, specific uptake and accumulation of MPP⁺ in the nigrostriatal dopaminergic neurons, and ending with the inhibition of oxidative phosphorylation (of NADH dehydrogenase in complex I). This inhibition results in mitochondrial injury, depriving the sensitive nigrostriatal cells of oxidative phosphorylation with their eventual cell death (neurotoxic actions of MPP⁺). The MAO-B inhibitors (e.g., deprenyl) blocked this biotransformation.



Because of the vital role that MAOs have in the inactivation of neurotransmitters, MAO dysfunction (too much or too little MAO activity) is thought to be responsible for a number of neurologic disorders. For example, unusually high or low levels of MAOs in the body have been associated with depression, schizophrenia, substance abuse, attention deficit disorder, and migraines. Monoamine oxidase inhibitors (MAOIs) are one of the major classes of drugs prescribed for the treatment of depression, although they are last-line treatment due to risk of the drugs interacting with diet or other drugs. Excessive levels of catecholamines (epinephrine, norepinephrine, and dopamine) may lead to a hypertensive crisis, and excessive levels of serotonin may lead to serotonin syndrome.

Diamines, such as $\text{H}_2\text{N-(CH}_2)_n\text{-NH}_2$, in which n is less than six, are not attacked and show little affinity for MAO. If the intermolecular distance between the amine groups is increased, the rate of oxidation by MAO increases.

Evidently, the second amine group interferes with attachment of the amine to the enzyme.

Diamine Oxidase

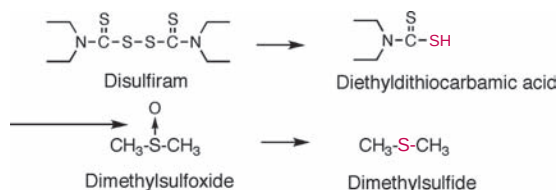
Diamine oxidase, a flavin-containing enzyme, catalyzes the aerobic oxidation of diamines and histamine in much the same way that MAO attacks monoamines, forming aldehydes. This enzyme is inhibited by carbonyl-blocking reagents and produces hydrogen peroxide, supporting the role of pyridoxal phosphate and the flavin prosthetic groups in the catalytic action of the enzyme. Diamine oxidase is found in the cytosolic fraction, kidneys, intestines, liver, lung, and nervous tissue. It limits the biologic effects of histamine and the polymethylene amines putrescine and cadaverine. It also attacks monoamines, but at a higher substrate concentration.



Plasma amine oxidases are in blood plasma of mammals and include spermine oxidase, which deaminates spermine and other polyamines.

Miscellaneous Reductions

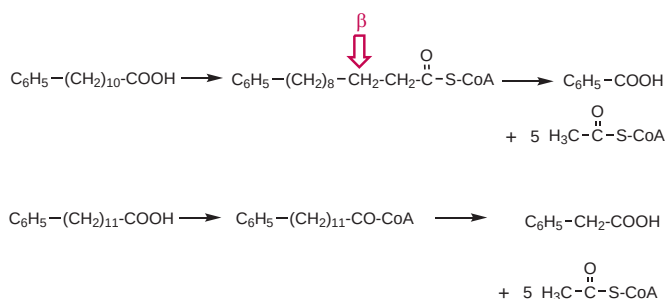
Disulfides (e.g., disulfiram), sulfoxides (e.g., dimethylsulfoxide), *N*-oxides, double bonds such as those in progestational steroids, and dehydroxylation of aromatic and aliphatic hydroxyl derivatives are examples of reductions occurring in microsomal or nonmicrosomal (cytosol enzymes) fractions.



Various studies regarding the biotransformation of xenobiotic ketones have established that ketone reduction is an important metabolic pathway in mammalian tissue. Because carbonyl compounds are lipophilic and can be retained in tissues, their reduction to the hydrophilic alcohols and subsequent conjugation are critical to their elimination. Although ketone reductases (carbonyl reductases) can be closely related to the alcohol dehydrogenases, they have distinctly different properties and use NADPH as the cofactor. Carbonyl reductase participates in arachidonic acid metabolism. The metabolism of xenobiotic ketones to free alcohols or conjugated alcohols has been demonstrated for aromatic, aliphatic, alicyclic, and unsaturated ketones (e.g., naltrexone, naloxone, hydromorphone, and daunorubicin). The carbonyl reductases are distinguished by the stereospecificity of their alcohol metabolites.

β-Oxidation

Alkyl carboxylic acids, as their coenzyme A (CoA) thioesters, are metabolized by oxidation at the β -carbon to the carboxylic carbon (β -oxidation). This pathway involves the oxidative cleavage of two carbon units at a time (as acetate), beginning at the carboxyl terminus and continuing until no more acetate units can be removed. The reaction is terminated when a branch (e.g., valproic acid) or aromatic group is encountered. The metabolism of even and odd phenylalkyl acids can serve as an example:



Hydrolysis

In general, esters and amides are hydrolyzed by enzymes in the blood, liver microsomes, intestine, kidneys, and other tissues. Esters and certain amides are rapidly hydrolyzed by a group of enzymes termed “carboxylesterases.” The more lipophilic the amide, the more favorable it is as a substrate for this enzyme. In most cases, the hydrolysis of an ester or amide bond in a toxic substance results in bioinactivation to hydrophilic metabolites that are readily excreted. Some of these metabolites can yield conjugated metabolites (i.e., glucuronides). Amides are very common in food as proteins and peptides we eat. Not surprisingly, there are a large number of proteolytic enzymes in the gastrointestinal tract called amino endopeptidases and amino exopeptidases that hydrolyze ingested proteins into amino acids.

Carboxylesterases include cholinesterase (pseudocholinesterase), arylcarboxylesterases, liver microsomal carboxylesterases, and other unclassified liver carboxylesterases. Cholinesterase hydrolyzes choline-like esters (e.g., acetylcholine, succinylcholine) and procaine as well as acetylsalicylic acid. Genetic variant forms of cholinesterase have been identified in human serum (e.g., succinylcholine toxicity when administered as a ganglionic blocker for muscle relaxation). Meperidine is hydrolyzed only by liver microsomal carboxylesterases (Fig. 4.19). Diphenoxylate is hydrolyzed to its active metabolite diphenoxylate acid within 1 hour (Fig. 4.19). Presumably, the lack of a central pharmacologic action of diphenoxylate is attributed to the formation of a zwitterionic diphenoxylate acid, which is readily eliminated in the urine.

A distinct type of esterase is the enzyme serum paraoxonase (PON), which appears to act as an important guardian against the neurotoxicity of organophosphates and functions as an antioxidant by preventing the oxidation

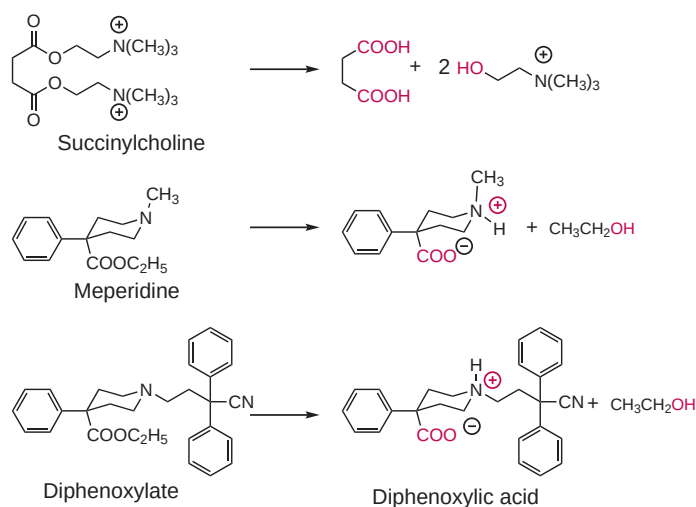


FIGURE 4.19 Examples of hydrolysis reactions.

of low-density lipoprotein (LDL), lowering the risk of developing coronary artery disease and atherosclerosis (55). The PON1 (A-esterase) is similar to an arylesterase in that it catalyzes the hydrolysis of phenyl acetate and other aryl esters. Without PON1, the organophosphate is free to react with and irreversibly inhibit acetylcholinesterase (see Chapter 9). Additionally, PON1 exhibits a substrate-dependent polymorphism. Individuals who are susceptible to the toxic effects of organophosphates such as paraoxon and chlorpyrifos (Dursban) are deficient in this isoenzyme. PON2 is an intracellular protein found in a variety of cell that protects cells against oxidative damage, and PON3 is similar to PON1 in activity but differs from it in substrate specificity.

Esters that are sterically hindered are hydrolyzed more slowly and can appear unchanged in the urine. For example, approximately 50% of a dose of atropine appears unchanged in the urine of humans, and the remainder of atropine appears to consist of unhydrolyzed biotransformed products.

As a rule, amides are more stable to esterase hydrolysis than are esters, and it is not surprising to find amides of drugs excreted largely unchanged. This fact has been exploited in developing the antiarrhythmic drug procainamide. Procaine is not useful as a result of its rapid esterase hydrolysis, but 60% of a dose of procainamide was recovered unchanged from the urine of humans, with the remainder being mostly *N*-acetylprocainamide. On the other hand, the deacylated metabolite of indomethacin (a tertiary amide) is one of the major metabolites detected in human urine. Amide hydrolysis of phthalylsulfathiazole and succinylsulfathiazole by bacterial enzymes in the colon releases the antibacterial agent sulfathiazole.

SUMMARY

Phase 1 metabolic transformations introduce new and polar functional groups into the molecule, which can produce one or more of the following changes:

1. Decreased pharmacologic activity (deactivation)
2. Increased pharmacologic activity (activation)
3. Increased toxicity (carcinogenesis, mutagenesis, cytotoxicity)
4. Altered pharmacologic activity

Drugs exhibiting increased activity or activity different from the parent drug usually undergo further metabolism and conjugation resulting in deactivation and excretion of the inactive conjugates.

DRUG CONJUGATION PATHWAYS (PHASE 2)

Conjugation reactions represent probably the most important xenobiotic biotransformation reaction (7,56–58). Xenobiotics are as a rule lipophilic, well absorbed from the blood, but excreted slowly in the urine. Only after conjugation (Phase 2) reactions have added an ionic hydrophilic moiety, such as glucuronic acid, sulfate ester, or glycine, to the xenobiotic is water solubility increased and lipid solubility decreased enough to make urinary elimination possible. The major proportion of the administered drug dose is excreted as conjugates into the urine and bile. Conjugation reactions can be preceded by Phase 1 reactions. For xenobiotics with a functional group available for conjugation, conjugation can be its fate.

The major conjugation reactions (glucuronidation and sulfonation) were traditionally thought to terminate pharmacologic activity by transforming the parent drug or Phase 1 metabolites into readily excreted ionic polar products (7,56). Moreover, these terminal metabolites would have no significant pharmacologic activity (i.e., poor cellular diffusion and affinity for the active drug's receptor). This long-established view has changed with the discoveries that morphine 6-glucuronide has more analgesic activity than morphine in humans and that minoxidil sulfate is the active metabolite for the antihypertensive minoxidil. For most xenobiotics, conjugation

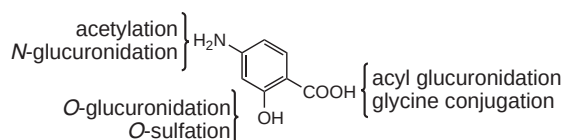


FIGURE 4.20 Sequential conjugation pathways for *p*-aminosalicylic acid.

is a detoxification mechanism. However, some compounds could form reactive intermediates that have been implicated in carcinogenesis, allergic reactions, and tissue damage.

Sequential conjugation of the same drug substance produces multiple conjugated products, for example *p*-aminosalicylic acid in Figure 4.20. Thus, the drug or xenobiotic can be a substrate for more than one metabolizing enzyme. Different conjugation pathways could compete for the same functional group with the outcome being an array of metabolites excreted in the urine or feces. Factors determining the outcome of this interplay include availability of cosubstrates, enzyme kinetics ($V_{\max}$), affinity of the drug/xenobiotic (K_m) for the metabolizing enzyme, and type of tissues (e.g., hepatic, kidney, lung). When a cosubstrate is low or depleted, the competing conjugation reactions can take control. The reactivity of the functional group is also determined by all subsequent conjugation reactions. For example, major conjugation reactions are sulfonation, ether glucuronidation, and methylation for the phenolic hydroxyl groups; acetylation, sulfonation, and glucuronidation for amine groups; and amino acid conjugation, CoA conjugation, and ester glucuronidation for carboxyl groups.

Conjugation enzymes can show stereospecificity toward enantiomers when a racemic drug is administered. The metabolite pattern of the same drug when administered orally can be different when administered intravenously as a result of presystemic intestinal conjugation.

Glucuronic Acid Conjugation

Glucuronide formation is probably the major and most common route for Phase 2 metabolism to water-soluble metabolites, which accounts for the major share of the conjugated metabolites found in the urine and bile (7,56). The significance of glucuronidation lies in the available supply of glucuronic acid in the liver and in the many functional groups forming glucuronide conjugates (e.g., phenols, alcohols, carboxylic acids, and amines).

Mechanism of Glucuronide Conjugation

Glucuronide conjugation involves the direct condensation of the xenobiotic (or its Phase 1 product) with the activated form of glucuronic acid, UDP-glucuronic acid (UDPGA). The overall scheme of the glucuronide conjugation reaction is shown in Figure 4.21. The

reaction between UDPGA and acceptor compound is catalyzed by a family of UDP-glucuronosyl transferases (UGT), a multigene family of isozymes located along the ER of liver, epithelial cells of the intestine, and other extrahepatic tissues. Its unique location in the ER along with the P450 isoforms has important physiologic effects in the neutralization of reactive metabolites generated by the P450 isoforms and in controlling the levels of reactive metabolites present in these tissues (Fig. 4.22). This cartoon depicts the spatial orientation and the interrelationship of the ER membrane-bound enzymes such as P450, UGTs, and membrane-bound transporters (57). The transporters carry the UDPGA and xenobiotics (D) from the cytosol into the ER lumen and then transport the glucuronide metabolite from the ER lumen into the cytosol. The presence of the active site for UGT toward the ER lumen catalyzes the reaction between the substrate and UDPGA. Glucuronidation is a low-affinity and high-capacity Phase 2 reaction. The resultant glucuronide has the β -configuration about carbon 1 of glucuronic acid. With the attachment of the hydrophilic carbohydrate moiety containing an easily ionizable carboxyl group ($\text{p}K_a = 3\text{--}4$), a lipid-soluble substance is converted into a conjugate that is reabsorbed poorly by the renal tubules from the urine and excreted more readily into the urine or, in some cases, into the bile. Endogenous substances conjugated with glucuronic acid include catecholamines, steroids, bilirubin, and thyroxine. Not all glucuronides are excreted by the kidneys. Some are excreted into the intestinal tract with bile (enterohepatic cycling), where β -glucuronidase in the intestinal flora hydrolyzes the C1-*O*-glucuronide to the aglycone (xenobiotic or their metabolites) for reabsorption into the portal circulation (see section on enterohepatic cycling).

UGT Families

UGTs have been classified into families according to similarities in amino acid sequences, analogous to the P450 family. The human UGT family is divided into two subfamilies, UGT1 (UGT1A1, 1A2, 1A3, 1A4, 1A5, 1A6, 1A7, 1A8, 1A9, and 1A10) and UGT2 (UGT2B4, 2B7, 2B10, 2B11, 2B15, 2B17, and 2B28) (7,58). Considerable overlap in substrate specificities exists between the two families. UGT1A1 isoform is primarily responsible for the glucuronidation of bilirubin, estradiol, and other estrogenic steroids; UGT1A3 and UGT1A4 catalyze the glucuronidation of drugs with tertiary amines to form quaternary glucuronides and hydroxylated xenobiotics; UGT1A6 exhibits limited substrate specificity for planar phenolic substances; UGT1A9 has a wide range of substrate specificity and can glucuronidate nonplanar phenols, plant substances (e.g., anthraquinones and flavones), steroids, other phenolic drugs, and C-glucuronidate phenylbutazone; and UGT1A10 glucuronidates mycophenolic acid an inhibitor of inosine monophosphate dehydrogenase. Human family 2 isoform UGT2B4 is homologous to UGT2B7 and catalyzes

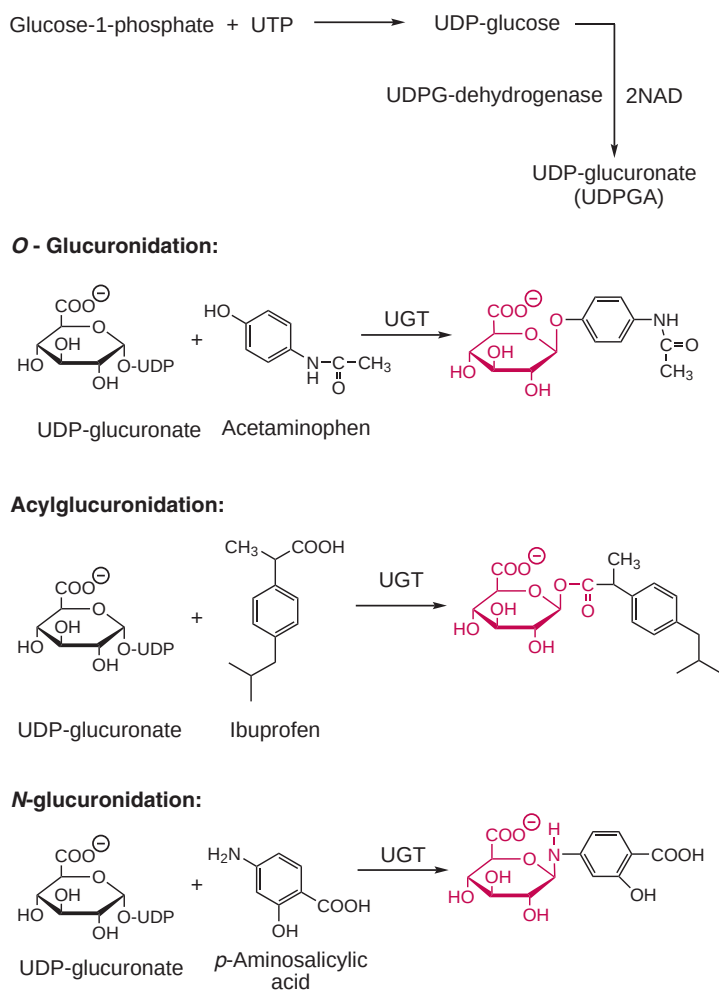


FIGURE 4.21 Glucuronidation pathway catalyzed by UDP-glucuronosyl transferases (UGTs).

the glucuronidation of the 6 α -hydroxyl group of bile acids; UGT2B7 glucuronidates the largest number of substrates including the 3- and 6-glucuronidation of morphine and 6-glucuronidation of codeine; UGT2B11 glucuronidates a wide range of planar phenols, bulky alcohols, and polyhydroxylated estradiol metabolites; and UGT2B15 catalyzes the glucuronidation of the 17 α -hydroxyl group of dihydrotestosterone and other steroidal compounds as well as phenolphthalein. UGT1A isoforms are inducible with 3-methylcholanthrene and cigarette smoking, and the UGT2B family is inducible by barbiturates. Approximately 40% of the glucuronides are produced by UGT2B7, 20% by UGT1A4, and 15% by UGT1A1.

UGT Distribution

The human liver has been established as the most important tissue for all routes of metabolism, including glucuronidation. Studies have shown that the rate of glucuronidation is not uniform in the different sections of the liver: UGT1A6 content was greatest in

the middle section of the liver but can also be found in the bile duct epithelium and in the endothelium of the hepatic artery and portal vein; UGT2B2 was uniformly distributed throughout the liver. The UGTs expressed in the intestine include UGT1A1 (bilirubin-glucuronidating isoform), UGT1A3, UGT1A4, UGT1A6, UGT1A8, UGT1A9, and UGT1A5. Substrate specificities of intestinal UGT isoforms are comparable to those in the liver. The UGT isoforms in the intestine can glucuronidate orally administered drugs, such as morphine, acetaminophen, α - and β -adrenergic agonists, and other phenolic phenethanolamines, as well as other dietary xenobiotics, reducing their oral bioavailability (first-pass metabolism). Although UGT isoforms are found in kidney, brain, nasal epithelia, and lung, they are not uniformly distributed, with UGT1A6 being the isoform that is ubiquitous in extrahepatic tissue.

O-, N-, and S-Glucuronides

The xenobiotics forming glucuronides with alcohols and phenols are ether O-glucuronides. Aromatic and some

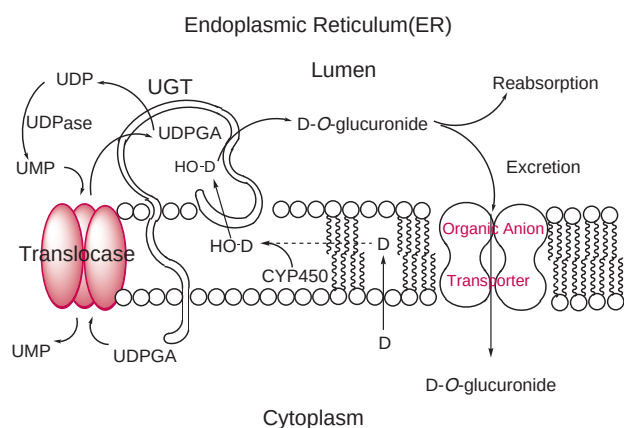


FIGURE 4.22 Proposed topologic model of UGT. A lipophilic drug (D) reaches the active site of P450 from the membrane or cytoplasm and is hydroxylated (HO-D). The hydroxylated metabolite is transferred to UGT, where glucuronidation occurs, followed by release into the lumen and excretion from the cell. The UDPGA is synthesized in the cytoplasm and transported via translocase. (Adapted from Oesch F. Metabolic transformation of clinically used drugs to epoxides: new perspectives in drug–drug interactions. *Biochem Pharmacol* 1976;25:1935–1937, with permission.)

aliphatic carboxylic acids form ester (acyl) glucuronides. Aromatic amines form *N*-glucuronides and sulfhydryl compounds form *S*-glucuronides, both of which are more labile to acid as compared with the *O*-glucuronides (Fig. 4.21). Some tertiary amines (e.g., tripeleminamine) have been reported to form quaternary ammonium *N*-glucuronides. Substances containing a 1,3-dicarbonyl structure (e.g., phenylbutazone) can undergo formation of *C*-glucuronides by direct conjugation without previous metabolism. The acidity of the methylene carbon of the 1,3-dicarbonyl group determines the degree of *C*-glucuronide formation.

Acyl Glucuronides

Drug–acyl glucuronides are reactive conjugates at physiologic pH. The acyl group of the *C*¹-acyl glucuronide can migrate via transesterification from the original *C*-1 position of the glucuronic acid to the *C*-2, *C*-3, or *C*-4 positions (Fig. 4.21). The resulting positional isomers are not hydrolyzed by β -glucuronidase, giving the appearance of a new unknown conjugate. However, under physiologic or weakly alkaline conditions, the *C*¹-acyl glucuronide can hydrolyze in the urine to produce the parent substance (aglycone) or undergo acyl migration to an acceptor macromolecule, forming an immunoreactive hapten. The pH-catalyzed migration of the acyl group from the drug *C*¹-*O*-acyl glucuronide to a protein or other cellular constituent occurs with the formation of a covalent bond to the protein (59). The acylated protein becomes a “hapten” and could stimulate an immune response against the drug, resulting in the expression of a hypersensitivity reaction or other forms of immunotoxicity. A high incidence of

anaphylactic reactions has been reported for several NSAIDs, for example, benoxaprofen, zomepirac, indoprofen, alclofenac, ticrynafen, and ibufenac, which have been removed from the market. All of these NSAIDs are metabolized by humans to acyl glucuronides. Similar reactions have been reported for other NSAIDs, including tolmetin, sulindac, ibuprofen, ketoprofen, and acetylsalicylic acid. The frequency of the immunotoxic response can be related to the stability of the acyl glucuronide, the chemical rate kinetics for the migration of the acyl group, and the concentration and stability/half-life of the antigenic protein. When the acyl glucuronide is the primary metabolite in patients with decreased renal function (i.e., elderly individuals) or when probenecid is coadministered, renal cycling of the unconjugated (aglycone) parent drug or metabolite is likely to occur, resulting in the plasma accumulation of the aglycone. The reduced elimination of the acyl glucuronide increases its hydrolysis to either the aglycone or the migration of the *C*¹-*O*-acyl group to an acceptor macromolecule.

Bioactivation and Toxic Glucuronides

As a rule, glucuronides are biologically and chemically less reactive than their parent molecules and are readily eliminated without interaction with intracellular substances. However, some glucuronide conjugates are more pharmacologically active than the parent drug (60). Morphine, for example, forms the 3-*O*- and 6-*O*-glucuronides in the intestine and in the liver. The 3-*O*-glucuronide is the primary glucuronide metabolite of morphine with a blood concentration 20-fold that of morphine. Pharmacologically, it is an opiate antagonist. On the other hand, 6-*O*-glucuronide, with a blood concentration twice that of morphine, is a more potent μ -receptor agonist and, whether administered orally or parentally, is 650-fold more analgesic than morphine in humans. Thus, the analgesic effects of morphine are the result of a complex interaction of the drug and its two glucuronide metabolites with the opiate receptor. The 6-*O*-glucuronide is transported into the brain via an anion-transport system.

Glucuronidation is also capable of promoting cellular injury (e.g., hepatotoxicity and carcinogenesis) by facilitating the formation of reactive electrophilic (electron-deficient) intermediates and their transport into target tissues (37). The induction of bladder carcinogenesis by aromatic amines can result from the *O*-glucuronidation of the *N*-hydroxylarylamines. These *O*-glucuronides become concentrated in the urine where they are either hydrolyzed by the acid pH of the urine to produce *N*-hydroxylaryl amines or eliminate water from the *N*-hydroxylarylamines under these acidic conditions to yield the formation of an electrophilic aryl nitrenium species (see Fig. 4.26, metabolite 5). This reactive species binds covalently with endogenous cellular constituents (e.g., nucleic acids and proteins), initiating carcinogenesis.

Sulfonation and glucuronidation occur side by side, often competing for the same substrate (most commonly phenols, i.e., acetaminophen). The balance between sulfonation and glucuronidation is influenced by such factors as animal species, dose, availability of cosubstrates, and inhibition and induction of the respective transferases.

Sulfonation (Sulfo-Conjugation)

Sulfonation (sulfo-conjugation) is an important conjugation reaction in the biotransformation of steroid hormones, catecholamine neurotransmitters, thyroxine, bile acids, phenolic drugs, and other xenobiotics (7,61,62). The major physiologic consequence of sulfonation of a drug or xenobiotic is its increased aqueous solubility and excretion, because the pK_a of the sulfate group is approximately 1 to 2. Sulfate esters are almost totally ionized in physiologic solution and possess a smaller volume of distribution than unconjugated steroids and drugs. For some drugs, sulfonation can result in their bioactivation to reactive electrophiles or therapeutically active conjugates (e.g., minoxidil sulfate). Cytosolic sulfotransferases are, in general, associated with the sulfonation of phenolic steroids, neurotransmitters, and xenobiotics. Membrane-bound sulfotransferases are localized in the Golgi apparatus of most cells and are responsible for the sulfonation of glycosaminoglycans, glycoproteins, and the tyrosinyl group of peptides and protein but are usually not associated with xenobiotic metabolism.

Mechanism of Sulfonation

Xenobiotics are sulfonated by the transfer of a sulfonic group ($-\text{SO}_3^-$) from 3'-phosphoadenosine-5'-phosphosulfate (PAPS) to the acceptor molecule, a cytosolic reaction catalyzed by a family of multigene sulfotransferases (Fig. 4.23); PAPS is formed enzymatically from adenosine triphosphate (ATP) and inorganic sulfate. Sulfonation is a reaction principally of phenols and, to a lesser extent, of alcohols to form highly ionic and hydrophilic sulfate esters ($\text{R-O-SO}_3\text{H}$). The availability of PAPS and its precursor inorganic sulfate determines the sulfonation reaction rate.

The total pool of sulfate is frequently limited and can be readily exhausted due to the large number of substrates. Sulfonation is a high-affinity and low-capacity Phase 2 reaction that works in conjunction with glucuronidation on overlapping substrates. Thus, sulfonation predominates at low substrate concentrations, and glucuronidation predominates at high substrate concentrations (62). Thus, when sulfonation becomes saturated from increasing doses of a drug, sulfonation becomes a less predominant pathway, and the rate of glucuronidation increases until it reaches saturation and the drug is either excreted unchanged or other metabolism pathways become activated. For example, at high doses of acetaminophen, glucuronidation more often than not predominates over that of sulfonation, which prevails at low doses. Other precursors for sulfate include L-methionine and L-cysteine. When PAPS, inorganic sulfate, or the sulfur amino acids are low or depleted or when a substrate for sulfonation is given in high doses, competing reactions with glucuronidation can take control. Additionally, *O*-methylation is a competing reaction for a catecholamine.

Sulfotransferase Family

In humans, sulfotransferases are divided into two families, SULT1 and SULT2 (61). SULT1A1, SULT1A2, and SULT1A3 catalyze the sulfonation of many phenolic drugs, catecholamine, hormones, aromatic amines, and other xenobiotics (62). SULT1A1/2 (formerly known as phenol sulfotransferase thermally stable) preferentially sulfonates small planar phenols in the micromolar concentration range, estradiol and synthetic estrogens, phytoestrogens, acetaminophen, the *N*-oxide of minoxidil, and *N*-hydroxyaromatic and heterocyclic amines. The crystal structure for SULT1A1 has provided insights into this enzyme's substrate specificity and catalytic function, including its role in the sulfonation of endogenous substrates such as estrogens. SULT1A3 (formerly known as phenol sulfotransferase thermally labile) selectively sulfonates the catecholamines dopamine, norepinephrine, and epinephrine, as well as the *N*-oxide of minoxidil and thyroid hormones, but not estrogenic steroids and other hydroxy

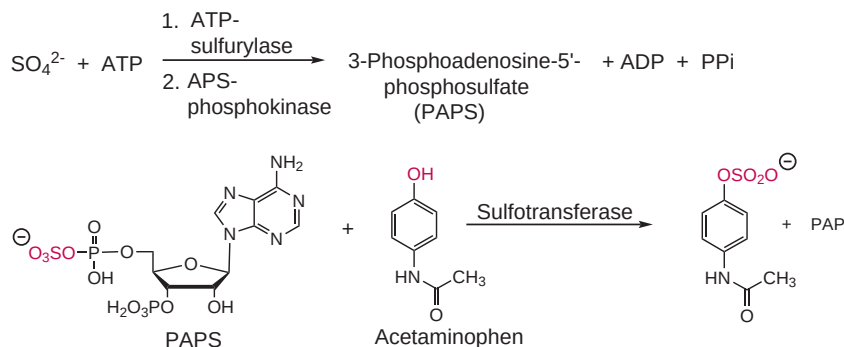


FIGURE 4.23 Sulfonation pathways.

steroids; SULT1B1 catalyzes the sulfonation of the thyroid hormones; SULT1C1 is involved with the bioactivation of procarcinogens via sulfonation; SULT1E1 (formerly known as estrogen sulfotransferase) preferentially sulfonates estradiol in the nanomolar range; SULT2A1 (formerly known as dehydroepiandrosterone [DHEA] sulfotransferase) conjugates DHEA, estradiol (micromolar range), the synthetic estrogens, and other estrogen metabolites; and SULT2B1 (formerly known as hydroxysteroid sulfotransferase) sulfonates DHEA and pregnenolone.

Sulfonation is an important reaction in the transport and metabolism of steroids. Sulfonation decreases the biologic activity of the steroid because the steroid sulfate esters are not capable of binding to their receptors. It provides for the transport of an inactive form of the steroid to its target tissue, where the active steroid is regenerated by sulfatases at the target tissue.

Sulfotransferase Distribution

The SULT1A families are abundantly expressed in the liver, small intestine, brain, kidneys, and platelets (61). For example, phenol is sulfonated by a sulfotransferase in the liver, kidneys, and intestines, whereas steroids are sulfonated only in the liver. The broad diversity of compounds sulfonated in human tissues results, in part, from the multi-isoforms of the cytosolic sulfotransferases and their overlapping substrate specificities. Sulfate esters are almost totally ionized and, therefore, are excreted mostly in the urine, but biliary elimination is common for steroids. When biliary sulfate esters are hydrolyzed in the intestine by sulfatases, the parent drug (or xenobiotic) or its metabolites can be reabsorbed into the portal circulation to be resulfonated for eventual elimination in the urine as a sulfate ester (enterohepatic cycling). The rate of sulfonation appears to be age dependent, decreasing with age. An important site of sulfonation, especially after oral administration, is the intestine. The result is a presystemic first-pass effect, decreasing drug bioavailability of several drugs for which the primary route of conjugation is sulfonation. Drugs such as isoproterenol, albuterol, steroid hormones, α -methyldopa, acetaminophen, and fenoldopam are sulfonated. Competition for intestinal sulfonation between coadministered substrates can influence their bioavailability with either an enhancement of or a decrease in therapeutic effects. An example would be coadministration of a 500-mg dose of acetaminophen and 0.03-mg dose of the oral contraceptive ethinyl estradiol.

Bioactivation and Toxicity

As with glucuronidation, sulfonation is a detoxication reaction, although sulfate esters have been reported to be pharmacologically active (e.g., minoxidil sulfate, dehydroepiandrosterone sulfate, and morphine 6-sulfate) or to be converted into unstable sulfate esters that form reactive intermediates implicated in carcinogenesis

and tissue damage (62). SULT1A1 can also sulfonate procarcinogens such as hydroxymethyl polycyclic aromatic hydrocarbons producing reactive intermediates capable of forming DNA adducts, potentially resulting in mutagenesis. Sulfonation of an alcohol generates a good leaving group and can be an activation process for alcohols to produce a reactive electrophilic species (37). However, like the *N*-glucuronides, *N*-sulfates are capable of promoting cytotoxicity by facilitating the formation of reactive electrophilic intermediates. Sulfonation of *N*-oxygenated aromatic amines is an activation process for some arylamines that can eliminate sulfate to produce an electrophilic species capable of reacting with proteins or DNA (e.g., 2-acetylaminofluorene). *N*-sulfonation of arylamines to arylsulfamic acids ($R-NHSO_3H$) is a minor pathway.

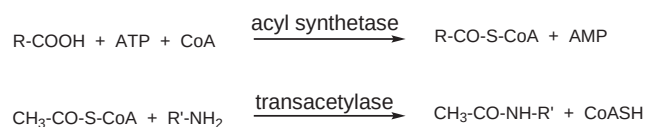
Stereoselectivity

SULT1A3 displays stereoselectivity in the sulfonation of chiral phenolic phenethanolamines. This isoform can be responsible, in part, for the enantiomer-specific metabolism observed for the β -adrenergic agonists. For example, the (+)-enantiomers of terbutaline and isoproterenol and the (–)-enantiomer of albuterol are selectively sulfonated.

Conjugation with Amino Acids

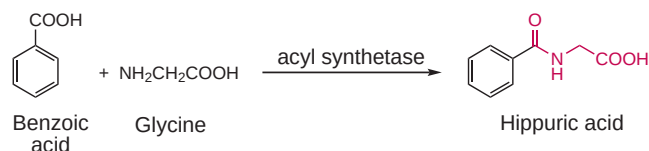
Conjugation with amino acids is an important metabolic route in the metabolism of drug or xenobiotic carboxylic acids before elimination (56). This pathway is low in the newborn and could contribute to the toxicity of administered benzoic acid derivatives (e.g., the bacterial preservative benzyl alcohol in intravenous fluids). Glycine, the most common amino acid, forms water-soluble ionic conjugates with aromatic, arylaliphatic, and heterocyclic carboxylic acids. These glycine conjugates are usually less toxic than their precursor acids and are readily excreted into the urine and bile. These reactions involve the formation of an amide or peptide bond between the xenobiotic carboxylic acid and the amino group of glycine. The xenobiotic is first activated to its CoA thioester before reacting with the amino group (Fig. 4.24). The formation of the xenobiotic acyl CoA thioester is of critical importance in intermediary metabolism of lipids as well as intermediate- and long-chain fatty acids.

The major metabolic biotransformations for xenobiotic carboxylic acids include conjugation with either glucuronic acid or glycine. The metabolic fate of these carboxylic acids depends on the size and type of substituents adjacent to the carboxyl group. Most unbranched aliphatic acids are completely oxidized by β -oxidation to acetic acids and do not typically form conjugates, although branched aliphatic and arylaliphatic acids are resistant to β -oxidation and form glycine or acyl glucuronide conjugates: substitution of the α -carbon favors glucuronidation rather than glycine conjugation. Benzoic and heterocyclic aromatic acids are mostly conjugated



Examples:

Glycine conjugation:



Acetylation:

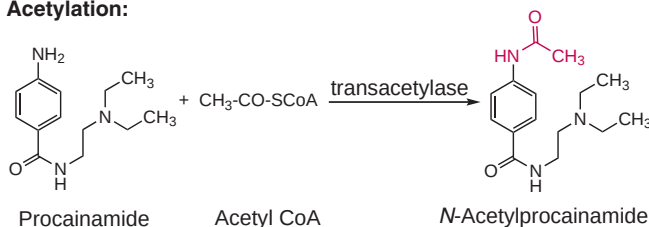


FIGURE 4.24 Amino acid conjugation pathways of carboxylic acids with glycine and acetylation pathways.

with glycine. Glycine conjugation is preferred for xenobiotic carboxylic acids at low doses, and glucuronidation is preferred at high doses with broad substrate selectivity. In humans, glutamine can also form a conjugate with phenylacetic acids and related arylacetic acids. Bile acids form conjugates with glycine and taurine by the action of enzymes in the microsomal fraction rather than in the mitochondria.

In contrast to the enhanced reactivity and toxicity of the various glucuronide, sulfate ester, acetyl, and glutathione conjugates, amino acid conjugates have not proven to be toxic. It has been proposed that amino acid conjugation is a detoxication pathway for reactive acyl CoA thioesters.

Conjugation with CoA

Several carboxylic acid-containing drugs (e.g., zomepirac and benzoxaprofen) have been implicated in rare but serious adverse reactions. These carboxylic acids were withdrawn from the market in the late 1980s as a result of unpredictable idiosyncratic reactions that could have been caused by carboxylic acid–protein adducts formed by reaction of their reactive acyl glucuronide or acyl CoA thioesters with endogenous proteins. Carboxylic acids can be bioactivated via two distinct pathways: UGT-catalyzed conjugation with glucuronic acid to acyl glucuronides, or acyl CoA synthetase-catalyzed formation of acyl CoA thioesters. The reactive CoA thioester intermediates of carboxylic acids are electrophilic and, therefore, can contribute to the acylation of target proteins (see Reactive Metabolites Resulting from Bioactivation and Fig. 4.30-1). Acyl CoA thioester serves as an obligatory

intermediate in the formation of glycine conjugates and carnitine esters, which are involved in mitochondrial acyl transfer or elimination of acyl groups into urine. Therefore, their appearance in metabolism studies and urine is of significance, because they serve as biomarkers for the formation of acyl CoA thioesters, which can provide the link between protein-reactive acyl CoA thioesters and the rare and unpredictable idiosyncratic drug reactions in humans.

A nontoxic reaction involving acyl CoA thioesters includes chiral inversions of the 2-arylpropionic acids (“profens”), a major group of NSAIDs that exist in two enantiomeric forms (59). The anti-inflammatory activity (inhibition of cyclooxygenase) for the NSAIDs resides with the *S*(+)-enantiomer. The intriguing aspect for the metabolism of the NSAIDs is their unidirectional chiral inversion from the *R*(–)- to the *S*(+)-enantiomer (Fig. 4.25). The NSAID acyl CoA thioester is the critical intermediate for this chiral inversion of the 2-arylpropionic acids, and the formation of the thioester is stereospecific for the pharmacologically inactive *R*-enantiomer (61). Racemic ibuprofen and related anti-inflammatory 2-arylpropionic acids (e.g., benoxaprofen, carprofen, cicliprofen, clidanac, fenoprofen, indoprofen, ketoprofen, loxoprofen, and naproxen) undergo in vivo metabolic inversion to the more active *S*-enantiomer via the formation, epimerization, and hydrolysis of their respective acyl CoA thioesters (63). The unidirectional *R*- to *S*-inversion of ibuprofen is attributed to the stereoselective thioester formation of *R*-ibuprofen CoA, not to the stereoselectivity of either the epimerization or hydrolysis steps (64). *S*(+)-ibuprofen does not form its CoA thioester in vivo. Because the formation of 2-arylpropionyl CoA thioester is analogous to the activation and metabolism of medium and long-chain fatty acids, it seems possible that conditions either elevating (e.g., diabetes or fasting)

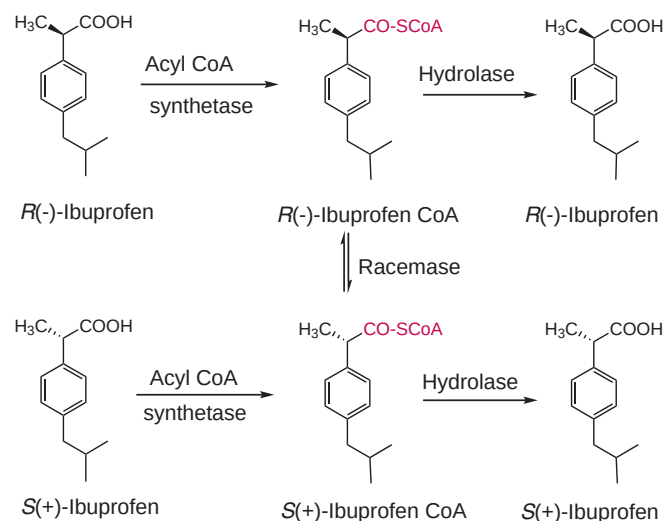


FIGURE 4.25 Coenzyme A (CoA) conjugation pathway: Stereospecific inversion of *R*(–) to *S*(+)-ibuprofen.

or depleting CoA can alter CoA thioester formation of the 2-arylpropionic acids and their in vivo metabolic inversion. Amino acid conjugation (i.e., CoA activation) is more sensitive to steric hindrance than is glucuronidation (e.g., arylacetic acids).

Acetylation

Acetylation is principally a reaction of amino groups involving the transfer of acetyl CoA to primary aliphatic and aromatic amines, amino acids, hydrazines, or sulfonamide groups (56). The liver is the primary site of acetylation, although extrahepatic sites have been identified. Sulfonamides, being difunctional, can form either N^1 or N^4 acetyl derivatives and in some instances, the diacetylated derivative has been identified. Secondary amines are not acetylated. Acetylation can produce conjugates that retain the pharmacologic activity of the parent drug (e.g., *N*-acetylprocainamide) (Fig. 4.24).

The existence of genetic polymorphism in the rate of acetylation has important consequences for drug therapy and tumorigenicity of xenobiotics. Acetylation polymorphism has been associated with differences in human drug toxicity between the two acetylator phenotypes, slow and fast acetylators. Slow acetylators are more prone to drug-induced toxicities and accumulate higher blood concentrations of the unacetylated drug (e.g., hydralazine and procainamide-induced lupus erythematosus, isoniazid-induced peripheral nerve damage, and sulfasalazine-induced hematologic disorders) than do fast acetylators. Fast acetylators eliminate the drug more rapidly by conversion to its relatively nontoxic *N*-acetyl metabolite. However, for some drug substances, fast acetylators can pose a greater risk of liver toxicity than slow acetylators because fast acetylators produce toxic metabolites more rapidly (e.g., isoniazid forms the hepatotoxic monoacetylhydrazine metabolite). Thus, differences in acetylator phenotype can influence adverse drug reactions.

The possibility arises that genetic differences in acetylating capacity can confer differences in susceptibility to chemical carcinogenicity from arylamines. The tumorigenic activity of arylamines (1 in Fig. 4.26) can be the result of a complex series of sequential metabolic reactions commencing with *N*-acetylation (2 in Fig. 4.26), subsequent oxidation to arylhydroxamic acids (3 in Fig. 4.26), and metabolic transformation to acetoxarylaminates by *N*, *O*-acetyltransferase (4 in Fig. 4.26). The acetoxarylamine can eliminate the acetox group to form the reactive arylnitrenium ion (5 in Fig. 4.26), which is capable of covalently binding to nucleic acids and proteins, thus increasing the risk for development of bladder and liver tumors (65). The rapid acetylator phenotype is expected to form the acetoxarylamine metabolite at a greater rate than the slow acetylator and, thereby, to present a greater risk for development of tumors compared with the slow acetylator.

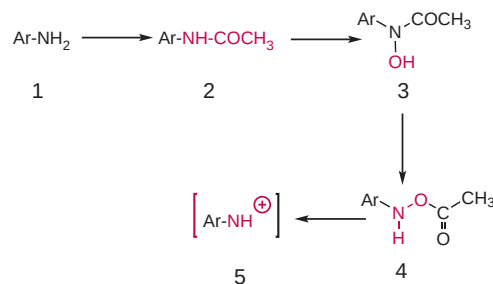


FIGURE 4.26 Bioactivation of acetylated arylamines.

Glutathione Conjugation and Mercapturic Acid Synthesis

Mercapturic acids are *S*-derivatives of *N*-acetylcysteine synthesized from glutathione (GSH) (53,54). The mercapturic acid pathway appears to have evolved as a protective mechanism against xenobiotic-induced hepatotoxicity or carcinogenicity, serving to detoxify a large number of noxious substances that we inhale or ingest or that are produced daily in the human body. Most xenobiotics that are metabolized to mercapturic acids first undergo conjugation with GSH catalyzed by the enzyme glutathione *S*-transferase (GST), a multigene isoenzyme family that is abundant in the soluble supernatant liver fractions. In humans, the GST family consists of cytosolic dimeric isoenzymes of 45 to 55 kDa size that have been assigned to at least eight classes: alpha (five members), kappa (one member), mu (six members), omega (two members), pi (one member), theta (two members), zeta (one member), and microsomal (three members). The principal drug substrates for the mu family are the nitrosourea and mustard-type anticancer drugs. The theta isoform metabolizes small organic molecules, such as solvents, halocarbons, and electrophilic compounds (e.g., α,β -unsaturated carbonyl compounds). The GSH conjugation reaction to mercapturic acid metabolites is depicted in Figure 4.27.

GST increases the ionization of the thiol group of GSH, increasing its nucleophilicity toward electrophiles and thereby increasing the rate of conjugation with these potentially harmful electrophiles, thereby protecting other vital nucleophilic centers in the cell, such as nucleic acids and proteins from these electrophiles. Glutathione is also capable of reacting non-enzymatically with nucleophilic sites on neighboring macromolecules (Fig. 4.27) (see also subsequent section on metabolic activation). Once conjugated with GSH, the electrophiles are, as a rule, excreted in the bile and urine.

A range of functional groups yields thioether conjugates of GSH as well as products other than thioethers (Fig. 4.27). The nucleophilic attack by GSH occurs on electrophilic carbons with leaving groups (e.g., halogen [alkyl, alkenyl, aryl, or aralkyl halides], sulfonates [alkylmethanesulfonates], and nitro [alkyl nitrates])

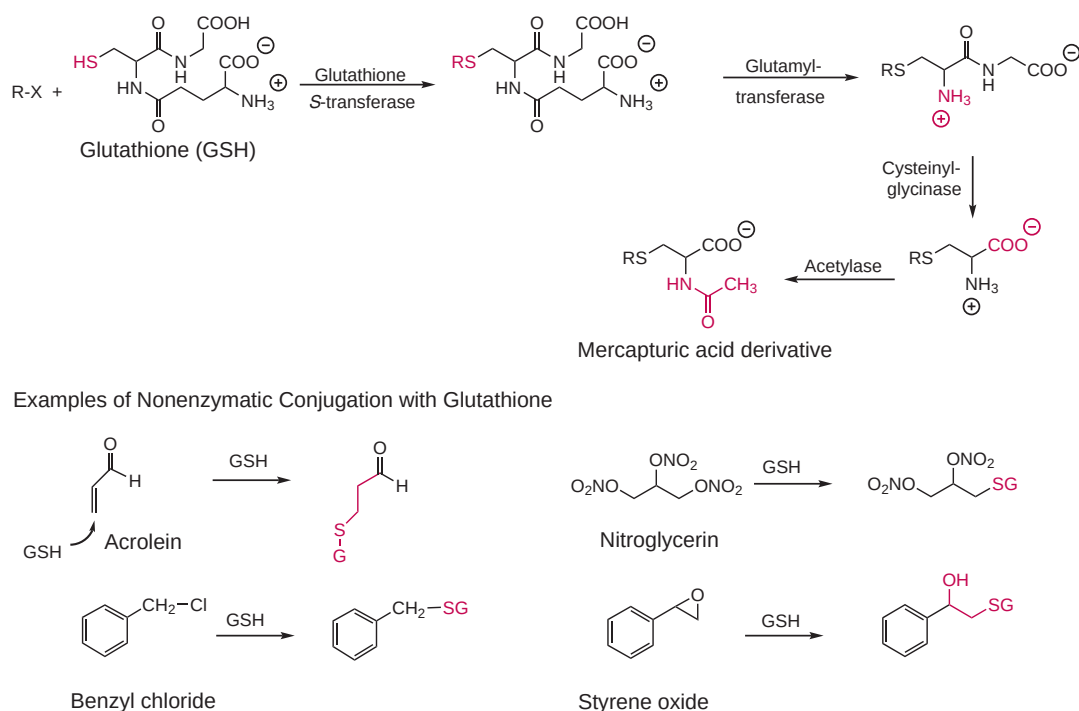


FIGURE 4.27 Glutathione and mercapturic acid conjugation pathways.

groups), ring opening of small ring ethers (epoxides and β -lactones, e.g., β -propiolactone), and the Michael-type addition to the activated β -carbon of an α,β -unsaturated carbonyl compound (e.g., acrolein) (see Reactive Metabolites Resulting from Bioactivation and Fig. 4.32A). Organic nitrate esters (e.g., the coronary vasodilator nitroglycerin) undergo a redox reaction that results in the concurrent oxidation of GSH to GSSG (through formation of the labile *S*-nitrate conjugation product) and reduction of the nitrate ester to an alcohol and inorganic nitrite. The lack of substrate specificity gives argument to the fact that glutathione transferase has undergone adaptive changes to accommodate the variety of xenobiotics to which it is exposed. The conjugation of an electrophilic compound with GSH is usually a reaction of detoxication, but in some cases, carcinogens have been activated through conjugation with GSH (29–31).

The enzymatic conjugation of GSH with epoxides provides a mechanism for protecting the liver from injury caused by certain bioactivated intermediates (see the subsequent Metabolic Bioactivation section). Not all epoxides are substrates for this enzyme, but the more chemically reactive epoxides appear to be better substrates. Important among the epoxides that are substrates for this enzyme are those produced from halobenzenes and PAHs through the action of a P450 monooxygenase. Epoxide formation represents bioactivation, because the epoxides are reactive and potentially toxic, whereas their GSH conjugates are inactive. Conjugation of GSH with the epoxides of aryl hydrocarbons eventually results in the formation of hydroxymercapturic

acids (premercapturic acids), which undergo acid-catalyzed dehydration to the mercapturic acids. The halobenzenes are typically conjugated in the para position.

Monohalogenated, gem-dihalogenated, and vicinal dihalogenated alkanes undergo glutathione transferase-catalyzed conjugation reactions to produce *S*-substituted glutathione derivatives that are metabolically transformed into the more stable and less toxic mercapturic acids. This common route of metabolism occurs through nucleophilic displacement of a halide ion by the thiolate anion of glutathione (see the discussion on glutathione conjugation). The mutagenicity of the 1,2-dihaloethanes (e.g., the pesticide and fumigant ethylene dibromide) has been attributed to GSH displacing bromide with the formation of the *S*-(2-haloethyl) glutathione, which subsequently rearranges to a reactive episulfonium ion electrophile that, in turn, alkylates DNA. Many of the halogenated hydrocarbons exhibiting nephrotoxicity undergo the formation of similar *S*-substituted cysteine derivatives.

A correlation exists between the hepatotoxicity of acetaminophen and levels of GSH in the liver. The probable mechanism of toxicity that has emerged from animal studies is that acetaminophen is CYP1A2 and CYP2E1 oxidized to the *N*-acetyl-*p*-benzoquinonimine intermediate, which conjugates with and depletes hepatic GSH levels (Fig. 4.28). This action allows the benzoquinonimine to bind covalently to tissue macromolecules. The mercapturic acid derivative of acetaminophen represents approximately 2% of the administered dose of acetaminophen. Thus, the possibility exists that those toxic metabolites

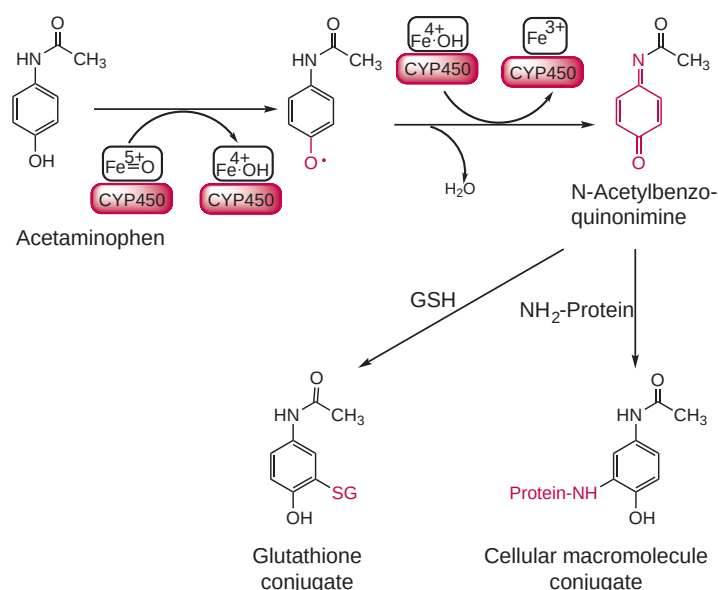


FIGURE 4.28 Proposed mechanism for the P450-catalyzed oxidation of acetaminophen to its *N*-acetyl-*p*-benzoquinoneimine intermediate, which can further react with either glutathione (GSH) or cellular macromolecules (NH₂-protein).

that are mostly detoxified by conjugating with GSH exhibit their hepatotoxicity (or, perhaps, carcinogenicity) because the liver has been depleted of GSH and is incapable of inactivating them. Pretreatment of animals with phenobarbital often hastens the depletion of GSH by increasing the formation of epoxides or other reactive intermediates.

Methylation

Methylation is a common biochemical reaction but appears to be of greater significance in the metabolism of endogenous compounds than for drugs and other xenobiotics (56). Methylation differs from other conjugation processes in that the *O*-methyl metabolites formed can, in some cases, have as great or greater pharmacologic activity and lipophilicity than the parent molecule (e.g., the conversion of norepinephrine to epinephrine). Methionine is involved in the methylation of endogenous and exogenous substrates, because it transfers its methyl group via the activated intermediate *S*-adenosylmethionine to the substrate under the influence of methyl transferases (Fig. 4.29). Methylation results principally in the formation of *O*-methylated, *N*-methylated, and *S*-methylated products.

O-Methylation

O-Methylation is catalyzed by the magnesium-dependent enzyme catechol-*O*-methyltransferase (COMT) transferring a methyl group primarily to the *m*- or 3-hydroxy group of the catechol moiety (3,4-dihydroxyphenyl moiety) of norepinephrine, epinephrine, or dopamine, or, less frequently, to the *p*- or 4-hydroxy group of these catecholamines (regioselectivity), as well as by their deaminated metabolites. COMT

does not methylate monophenols or other dihydroxy noncatechol phenols. The meta:para product ratio depends greatly on the type of substituent attached to the catechol ring. Substrates specific for COMT include the catecholamines norepinephrine, epinephrine, and dopamine; the catechol amino acids L-DOPA and α -methyl-DOPA; and 2- and 4-hydroxyestradiol (catechol-like) metabolites of estradiol. The enzyme is thought to function in the biologic inactivation of the adrenergic neurotransmitter norepinephrine as well as other endogenous and exogenous catechol-like substances. It is found in liver, kidneys, nervous tissue, and other tissues.

Hydroxyindole-*O*-methyltransferase, which *O*-methylates *N*-acetylserotonin, serotonin, and other hydroxyindoles, is found in the pineal gland and is involved in the formation of melatonin, a hormone associated with the dark-light diurnal cycle in humans. This enzyme differs from COMT in that it does not methylate catecholamines and has no requirement for magnesium iron.

N-Methylation

N-Methylation of various amines is among several conjugate pathways for metabolizing amines. Specific *N*-methyltransferases catalyze the transfer of active methyl groups from *S*-adenosylmethionine to the acceptor substance. Phenylethanolamine-*N*-methyltransferase methylates a number of endogenous and exogenous phenylethanolamines (e.g., normetanephrine, norepinephrine, and norephedrine) but does not methylate phenylethylamines. Histamine-*N*-methyltransferase methylates specifically histamine, producing the inactive metabolite *N*¹-methylhistamine. Amine-*N*-methyltransferase will *N*-methylate a variety of primary and secondary amines from a number of sources,

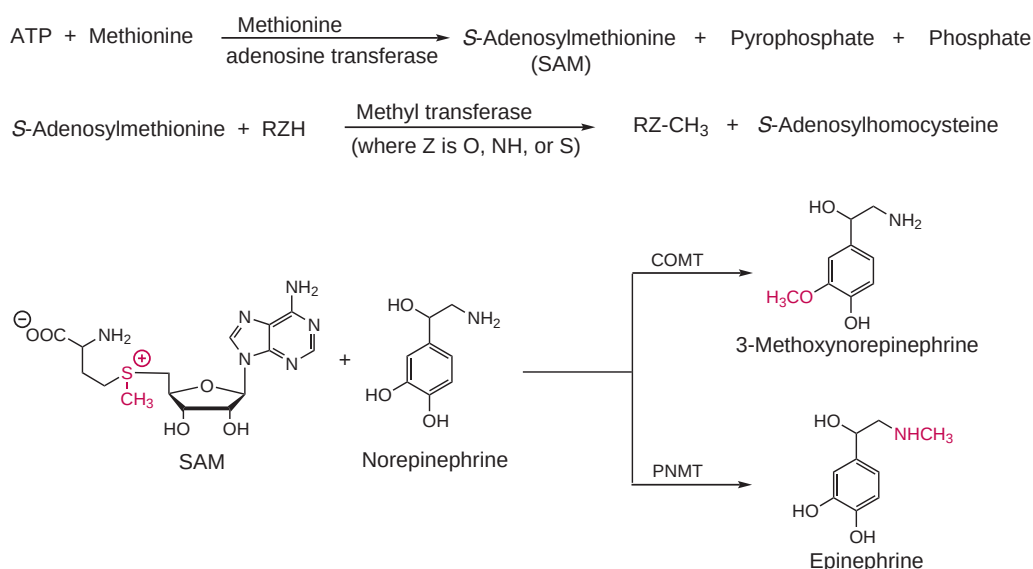


FIGURE 4.29 Methylation pathways.

including endogenous biogenic amines (serotonin, tryptamine, tyramine, and dopamine) and drugs (desmethylinipramine, amphetamine, and normorphine). Amine-*N*-methyl transferases seem to have a role in recycling *N*-demethylated drugs.

S-Methylation

Thiols are, as a rule, toxic, and the role of thiol *S*-methyl transferases is a nonoxidative detoxification pathway of these compounds. *S*-Methylation of sulfhydryl compounds also involves a microsomal enzyme requiring *S*-adenosylmethionine. Although a wide range of exogenous sulfhydryl compounds are *S*-methylated by this microsomal enzyme, none of the endogenous sulfhydryl compounds (e.g., cysteine and GSH) can function as substrates. Clearly, *S*-methylation represents a detoxication step for thiols. Dialkylthiocarbamates (e.g., disulfiram) and the antithyroid drugs (e.g., 6-propyl-2-thiouracil), mercaptans, and hydrogen sulfide (from thioglycosides as natural constituents of foods, mineral sulfides in water, fermented beverages, and bacterial digestion) are *S*-methylated. Other drugs undergoing *S*-methylation include captopril, thiopurine, azaprine, penicillamine, and 6-mercaptopurine.

Thiopurine methyltransferase or thiopurine *S*-methyltransferase (TPMT) catalyzes the *S*-methylation of thiopurine drugs, such as azathioprine, 6-mercaptopurine, and 6-thioguanine. Interindividual variations in sensitivity and toxicity to thiopurine are correlated with TPMT genetic polymorphisms. Defects in the TPMT gene lead to decreased methylation and decreased inactivation of the thiopurine drugs, leading to enhanced bone marrow toxicity that may cause myelosuppression, anemia, bleeding tendency, leukopenia, and infection. Approximately 5% of all thiopurine therapies will fail due to toxicity as a result of TPMT polymorphism.

Conjugation of Cyanide

The toxicity of hydrogen cyanide is the result of its ionization to cyanide ion in biologic tissues. Cyanide anion is a powerful metabolic inhibitor that arrests cellular respiration by inactivating cytochrome enzymes that are fundamental to the respiratory process as well as combining with hemoglobin to form cyanomethemoglobin, which is incapable of transporting oxygen to tissues. With the wide prevalence of cyanoglycosides in plant-food materials, the ability to detoxify cyanide is a vital function of the liver, erythrocytes, and other tissues. Rhodanase, a mitochondrial enzyme in liver and other tissues, catalyzes the formation of thiocyanate from cyanide rapidly in the presence of thiosulfate and colloidal sulfur, but cysteine and GSH are poor sulfur donors. The detoxification of cyanide depends on the availability of a physiologic pool of thiosulfate, the origin of which is not known. A possible source for thiosulfate is the transamination of cysteine to β -mercaptopyruvate and transfer of the mercapto group by a sulfur transferase to sulfite-producing thiosulfate. Depletion of the thiocyanate pool increases cyanide toxicity. However, in the presence of excess cyanide, minor pathways for cyanide metabolism can occur, including oxidation to cyanate (NCO^-), reaction with cobalamin (vitamin B_{12}) to form cyanocobalamin, and the formation of 2-iminothiazolidine-4-carboxylic acid from the nonenzymatic reaction between cysteine and cyanide (66).



ELIMINATION PATHWAYS

Most xenobiotics are lipid-soluble and are altered chemically by the metabolizing enzymes, usually into less toxic and more water-soluble substances, before being

excreted into the urine (or, in some cases, bile). The formation of conjugates with sulfonate, amino acids, and glucuronic acid is particularly effective in increasing the polarity of drug molecules. The principal route of excretion of drugs and their metabolites is in the urine. If drugs and other compounds foreign to the body are not metabolized in this manner, substances with a high lipid–water partition coefficient could be reabsorbed readily from the urine through the renal tubular membranes and into the plasma. Therefore, such substances would continue to be recirculated, and their pharmacologic or toxic effects would be prolonged. Very polar or highly ionized drug molecules often are excreted in the urine unchanged.

Urine

Tubular reabsorption is greatly reduced by conversion of a drug into a more hydrophilic substance with a lower partition coefficient. In general, the more resistant a drug is to the metabolizing enzymes, the greater the therapeutic action and the smaller the dose needed to achieve a particular therapeutic goal.

Urine is not the only route for excreting drugs and their metabolites from the animal body. Other routes include bile, saliva, lungs, sweat, and milk. The bile has been recognized as a major route of excretion for many endogenous and exogenous compounds.

Enterohepatic Cycling of Drugs

The liver is the principal organ for the metabolism and eventual elimination of xenobiotics from the human body in either the urine or the bile. When eliminated in the bile, steroid hormones, bile acids, drugs, and their respective conjugated metabolites are available for reabsorption from the duodenal–intestinal tract into the portal circulation, undergoing the process of enterohepatic cycling (EHC) (67). Nearly all drugs are excreted in the bile, but only a few are concentrated in the bile. For example, the bile salts are so efficiently concentrated in the bile and reabsorbed from the gastrointestinal tract that the entire body pool of bile acids/salts are recycled multiple times per day. Therefore, EHC is responsible for the conservation of bile acids, steroid hormones, thyroid hormones, and other endogenous substances. In humans, compounds excreted into the bile typically have a molecular weight greater than 500 daltons, whereas with rats, the critical molecular weight is 325 daltons. Consequently, biliary excretion is more common in rats than in humans. In humans, compounds with a molecular weight between 300 and 500 daltons are excreted in both urine and bile. Some compounds would not be expected to be excreted in the bile as a result of a molecular weight of less than 300 daltons and a relatively nonpolar structure. Compounds excreted into bile are, as a rule, hydrophilic substances that can be either charged (anionic; e.g., dyes) or uncharged (e.g., cardiac glycosides and steroid hormones). Biotransformation of

this type of compound by means of Phase 1 and Phase 2 reactions would produce a conjugated metabolite that is anionic and hydrophilic with a molecular weight greater than that of the parent compound. The conjugated metabolites are most often present as their glucuronide conjugates, because glucuronidation adds 176 daltons to the molecular weight of the parent compound. Unchanged drug in the bile is either excreted with the feces, metabolized by the bacterial flora in the intestinal tract, or reabsorbed into the portal circulation via EHC.

The bacterial intestinal flora is directly involved in EHC and the recycling of drugs through the portal circulation (see the discussion of extrahepatic metabolism). A conjugated drug and metabolites excreted via the bile can be hydrolyzed by enzymes of the bacterial flora, releasing the parent drug or its Phase 1 metabolite for reabsorption into the portal circulation (68). Among the numerous compounds metabolized in the enterohepatic circulation are the estrogenic and progestational steroids, digitoxin, indomethacin, diazepam, pentaerythritol tetranitrate, mercurials, arsenicals, and morphine. The oral ingestion of xenobiotics inhibiting the gut flora (i.e., nonabsorbable antibiotics) can affect the pharmacokinetics of the initial drug.

The impact of EHC on the pharmacokinetics and pharmacodynamics of a drug depends on the importance of biliary excretion of the drug relative to renal clearance and on the efficiency of gastrointestinal absorption. The EHC becomes dominant when biliary excretion is the major clearance mechanism for the drug. Because the majority of the bile is stored in the gallbladder and released on the ingestion of food, intermittent spikes in the plasma drug concentration are observed following reentry of the drug from the bile via EHC. From a pharmacodynamic point of view, the net effect of EHC is to increase the duration of a drug in the body and to prolong its pharmacologic action.

Chronic treatment with the enzyme inducer phenobarbital enhances the biliary excretion of drug molecules and their metabolites by increasing liver size, bile flow, and more efficient transport into the bile. This behavior is not shared by all inducers of the P450 monooxygenases. The route of administration can also influence excretion pathways. Direct administration into the portal circulation might be expected to result in more biliary excretion than could be expected via the systemic route.

DRUG METABOLISM AND AGE

Approximately 40% of the population is older than 65 years of age and is responsible for more than 50% of the national drug expenditures. People older than 85 years represent approximately 2% of the population; they are the most medicated and account for more than one-third of all prescription drugs dispensed. The average elderly patient in a health care facility could receive as many as 10 medications daily, which results in the potential

TABLE 4.15 Effect of Age on the Clearance of Some Drugs

No Change	Decrease
Acetaminophen*	Alprazolam
Aspirin	Amitriptyline
Diclofenac	Carbenoxolone
Digitoxin	Chlordiazepoxide
Diphenhydramine	Chlormethiazole
Ethanol	Clobazam
Flunitrazepam	Desmethyldiazepam
Heparin	Diazepam
Lormetazepam	Labetalol
Midazolam	Lidocaine
Nitrazepam	Lorazepam
Oxazepam	Morphine
Phenytoin*	Meperidine
Prazosin	Nifedipine and other dihydropyridines
Propylthiouracil	Norepinephrine
Temazepam	Nortriptyline
Thiopental*	Phenytoin
Tolbutamide*	Piroxicam
Warfarin	Propranolol
	Quinidine
	Quinine
	Theophylline
	Verapamil

*Drugs for which clearance is disputable but can be increased.

for a greater incidence of adverse drug reactions. The widespread use of medications in the elderly increases the potential for an increased incidence of drug-related interactions, which can be related to changes in drug *v* (Table 4.15). The interpretation of the age-related alteration in drug response must consider the contributions of absorption, distribution, metabolism, and excretion (69). Drug therapy in the elderly is expected to become one of the more significant problems for clinical medicine. It has been well documented that the metabolism of many drugs and their elimination is impaired in the elderly.

Metabolism in the Elderly

The decline in drug metabolism due to old age is associated with physiologic changes that have pharmacokinetic implications affecting the steady-state plasma concentrations and renal clearance for the parent drug and its metabolites (70,71). Those changes relevant to the bioavailability of drugs in the elderly are decreases in hepatic blood flow, glomerular filtration rate, hepatic P450 enzyme activity, plasma protein binding, and body mass. Because the rate of a drug's elimination from the blood through hepatic metabolism is determined by hepatic blood flow, protein binding, and intrinsic renal clearance, a reduction in hepatic blood flow can lead to an increase in drug bioavailability and decreased renal clearance, with the symptoms of drug overdose and toxicity as the outcome (e.g., warfarin in patients with congestive heart failure). Drugs for which renal elimination is dependent on hepatic blood flow

have a high extraction ratio and undergo extensive first-pass metabolism when administered orally. Available evidence suggests that age is associated with a reduction in first-pass metabolism of some, but certainly not all, drugs. Those orally administered drugs exhibiting a reduction in first-pass metabolism in the elderly include the dihydropyridine calcium antagonists, chlormethiazole, diazepam, lorazepam, chlordiazepoxide, alprazolam, propranolol, verapamil, labetalol, theophylline, morphine, amitriptyline, and nortriptyline. The bioavailability of drugs with low extraction ratios depends on the percentage of drug–protein binding and not on first-pass hepatic metabolism. Inasmuch as drug binding to plasma proteins is an important factor in the rate of drug metabolism, it appears not to be a significant factor in the elderly.

Age-related changes in drug metabolism are a complicated interplay between age-related physiologic changes, genetics, environmental influences (diet and nutritional status, smoking, and enzyme induction), concomitant diseases states, and drug intake. In most studies, the elderly appear just as responsive to drug-metabolizing enzyme activity (Phase 1 and Phase 2) as young individuals. All the common pathways of drug conjugation, including glucuronidation, sulfonation, and glycine conjugation, are variably affected by aging. Given the number of factors that determine the rate of drug metabolism, it is not surprising that the effects of aging on drug elimination by metabolism have yielded variable results even for the same drug. Therefore, the bioavailability of a drug in the elderly and the potential for drug toxicity is dependant largely on its extraction ratio and mode of administration. The fact that drug renal elimination can be altered in old age suggests that initial doses of metabolized drugs should be reduced in older patients and then modified according to the clinical response (70,71). A decrease in hepatic drug metabolism coupled with age-related alterations in clearance, volume of distribution, and receptor sensitivity can lead to prolonged plasma half-life and increased drug toxicity (Table 4.15).

Drug–Drug Interactions

Although drug–drug interactions constitute only a small proportion of adverse drug reactions in the elderly, they are important, because they are often predictable and, therefore, avoidable or manageable. Their frequency is related to the age of the patient, the number of drugs prescribed, the number of physicians involved in the patient's care, and the presence of increasing frailty. The most important mechanisms for drug–drug interactions are the inhibition or induction of drug metabolism. Interactions involving a loss of action of one of the drugs are at least as frequent as those involving an increased effect. Although only approximately 10% of potential interactions result in clinically significant adverse events, death or serious clinical consequences

are rare, but low-grade, clinical morbidity in the elderly can be much more common. Nonspecific complaints (e.g., confusion, lethargy, weakness, dizziness, incontinence, depression, and falling) should all prompt a closer look at the patient's drug list. A number of strategies can be adopted to decrease the risk of potential clinical problems. The number of drugs prescribed for each individual should be limited to as few as necessary. The use of drugs should be reviewed regularly, and unnecessary agents should be withdrawn if possible, with subsequent monitoring. Patients should be encouraged to engage in a "prescribing partnership" by alerting physicians, pharmacists, and other health care professionals to symptoms that occur when new drugs are introduced. Health care professionals should develop a strategy for monitoring their drug treatment, looking for the drug-drug interactions that have been encountered.

Those P450 substrates reported to cause drug-drug interactions are shown in Tables 4.4 through 4.14 in bold italics.

Fetal Metabolism

The ability of the human placenta to metabolize xenobiotics is well established. Drug-metabolizing enzymes expressed in the placenta include CYP1A1, CYP2E1, CYP3A4/5, CYP3A7, CYP4B1, epoxide hydrolase, alcohol dehydrogenase, glutathione transferase, UGTs, sulfotransferases, and *N*-acetyltransferase. A 1973 clinical study reported that women ingest an average of 10 drugs during pregnancy, not including general anesthetics, intravenous fluids, vitamins, iron, nicotine, cosmetic products, artificial sweeteners, or exposure to environmental contaminants. The majority of these drugs readily cross the placenta, thus exposing the fetus to a large number of xenobiotic agents. The knowledge regarding the effects of prenatal exposure to drugs, environmental pollutants (e.g., smoking), and other xenobiotics (e.g., ethanol) on the fetus has led to a decrease in the exposure to these substances during pregnancy. The human fetus is at special risk from these xenobiotics due to low or negligible levels of the P450 monooxygenase system. Placentas of tobacco smokers have shown a significant increase in the rate of placental CYP1A activity. Concern for this CYP1A activity is increasing, because this enzyme system is known to catalyze the formation of reactive metabolites capable of covalently binding to macromolecules producing permanent effects (e.g., teratogenicity, hepatotoxicity, or carcinogenicity) in the fetus and newborn. A more disturbing fact is that the conjugation enzymes (i.e., glucuronosyl transferases, epoxide hydrolase, glutathione transferase, and sulfotransferase), which are important for the formation of Phase 2 conjugates of these reactive metabolites, are also found in low to negligible levels in the fetus and newborn, increasing the exposure of these infants to these potentially toxic metabolites.

The placenta is not a barrier protecting the fetus from xenobiotics; because most every drug present in the maternal circulation will cross the placenta and reach the fetus. For some drugs, however, the placental efflux transport protein, P-glycoprotein (P-gp; discussed later and in Chapter 7), functions as a maternofetal barrier, pumping drugs and P-gp substrates out of the fetal circulation back into the maternal circulation (72) and protecting the fetus from exposure to potentially harmful teratogenic xenobiotics/drugs and endogenous substances that have been absorbed through the placenta. The P-gp inhibitors should be carefully evaluated for their potential to increase fetal susceptibility to drug or chemical-induced teratogenesis. On the other hand, selective inhibition of P-gp could be used clinically to improve pharmacotherapy of the unborn child. Depending on the pharmacologic activity of the parent substance or its metabolites, adult maternal drug metabolism can be viewed as complimentary yet contradictory. Because metabolites are usually more water soluble than the parent substance, drug metabolites when formed by maternal drug metabolism can be trapped and accumulate on the fetal side of the placenta. Such accumulation can result in drug-induced toxicities or developmental defects. However, the difference between fetal and adult metabolism can be used advantageously and constitutes the rationale for transplacental therapy (e.g., the administration of betamethasone several days before delivery can increase the production of surfactant in the fetal lung and prevent respiratory distress syndrome in the neonate).

Neonatal Metabolism

From the day of birth, the neonate is exposed to drugs and other foreign compounds persisting from pregnancy as well as those transferred via breast milk. Many of the drug-metabolizing enzymes operative in the neonate are developed during the first year of life. The routine use of therapeutic agents during labor and delivery, as well as during pregnancy, is widespread, and consideration must be given to the fact that potentially harmful metabolites can be generated by the newborn. Consequently, the use of drugs capable of forming reactive metabolic intermediates (e.g., isotretinoin [Accutane]) should be avoided during pregnancy, delivery, and the neonatal period. The activity of Phase 1 and Phase 2 drug-metabolizing enzymes is high at birth but decreases to normal levels with increasing age. Evidence suggests increased activity of drug-metabolizing enzymes in liver microsomes of neonates resulting from treatment of the mother during the pregnancy with enzyme inducers (e.g., phenobarbital).

GENETIC POLYMORPHISM

The reality of drug therapy is that many drugs do not work in all patients. By current estimates, the percentage of patients who will react favorably to a specific drug

ranges from 20% to 80%. Drugs have been developed and dosage regimens prescribed under the old paradigm that “one dose fits all,” which ignores largely the fact that humans are genetically different, resulting in interindividual differences in drug metabolism and disposition (10). It is widely accepted that genetic factors have an important impact on the oxidative metabolism and pharmacokinetics of drugs. Genotype–phenotype correlation studies (pharmacogenetics) have shown that inherited mutations in P450 genes (alleles) result in distinct phenotypic subgroups. For example, mutations in the *CYP2D6* gene result in poor (PM), intermediate (or extensive [EM]), and ultrarapid (UM) metabolizers of *CYP2D6* substrates (73) (Table 4.9). Each of these phenotypic subgroups experience different responses to drugs extensively metabolized by the *CYP2D6* pathway, ranging from severe toxicity to complete lack of efficacy. Genetic studies confirm that “one dose does not fit all,” leaving the question of why we would continue to develop and prescribe drugs under the old paradigm. As early as 1997, the U.S. Food and Drug Administration (FDA) recognized that identifying genetic polymorphisms might allow the safe dosing, marketing, and approval of drugs that would otherwise not be approved and advised pharmaceutical companies to incorporate the knowledge of genetic polymorphisms into drug development (see sidebar). Importantly, pharmacogenomic testing (the study of heritable traits affecting patient response to drug treatment) can significantly increase the likelihood of developing drug regimens that benefit most patients without severe adverse events.

Polymorphisms are expressed for a number of metabolizing enzymes, but the polymorphic P450 isoforms that are most important for drug metabolism include *CYP2A6*, *CYP2A13*, *CYP2B6*, *CYP2C9*, *CYP2C19*, and *CYP2D6*. These isoforms have been studied extensively, and despite their low abundance in the liver, they have been found to catalyze the metabolism of many drugs. Most gene variants are expressed into inactive, truncated proteins or fail to express any protein. There is, as yet, no clear information about *CYP1A1* and *CYP3A4/5* polymorphism. These polymorphic isoforms give rise to phenotypic subgroups in the population differing in their ability to perform clinically significant biotransformation reactions with obvious clinical ramifications (73). Metabolic polymorphism can have several consequences; for example, when enzymes that metabolize drugs used either therapeutically or socially are deficient, adverse or toxic drug reactions can occur in these individuals. The discovery of genetic polymorphism resulted from the observation of increased frequency of adverse effects or no drug effects after normal doses of drugs to some patients (e.g., hyper-central nervous system response from the administration of the antihistamine doxylamine or no analgesic response with codeine). Polymorphism is a difference in DNA sequence found

FDA ADVISORY ON GENETIC POLYMORPHISM

“When a genetic polymorphism affects an important metabolic route of elimination, large dosing adjustments can be necessary to achieve the safe and effective use of the drug...indeed in some cases understanding how to adjust the dose to avoid toxicity can allow the marketing of a drug that would have an unacceptable level of toxicity were its toxicity unpredictable and unpreventable.”

—FDA, *Guidance of Industry, Drug Metabolism/Drug Interaction Studies in the Drug Development Process: Studies in Vitro*, April 1997.

at 1% or greater in a population and expressed as an amino acid substitution in the protein sequence of an enzyme resulting in changes in its rate of activity ($V_{\max}$) or affinity (K_m). Thus, mutant DNA sequences can lead to interindividual differences in drug metabolism. Furthermore, the polymorphisms do not occur with equivalent frequency in all racial or ethnic groups. As a result of these differences, it is important to be aware of a person’s race and ethnicity when giving drugs that are metabolized differently by different populations (73–76). Because no other way exists to adequately clear these drugs from the body, PMs can be at greater risk for adverse drug reactions or toxic overdoses. The signs and symptoms of these overdoses are primarily extensions of the drug’s common adverse effects or pharmacologic effects (Table 4.16) (73). The level of adverse reactions or overdosage depends very much on the overall contribution of the mutant isoform to the drug’s metabolism. Perhaps the most interesting explanation for the various mutant isoforms is that they evolved as protective mechanisms against alkaloids and other common substances in the food chain for the different ethnicities. Although much effort has gone into finding polymorphisms of *CYP3A4* and *CYP1A2* genes, none has yet to be discovered.

Occasionally, one derives benefit from an unusual P450 phenotype. For example, cure rates for peptic ulcer treated with omeprazole are substantially greater in individuals with defective *CYP2C19* due to the sustained high plasma levels achieved.

CYP2A6

CYP2A6 is of particular importance, because it activates a number of procarcinogens to carcinogens and is the major isoform metabolizing the anticoagulant warfarin and nicotine (74). Approximately 15% of Asians express the *CYP2A6*4del* allele, and 2% of Caucasians express the other *CYP2AD6*2* allele. Both of these alleles express zero enzymes or a nonfunctional enzyme; these individuals are referred to as having the PM phenotype. A benefit from being a PM of *CYP2A6* substrates might be

TABLE 4.16 Impact of Human P450 Polymorphisms on Drug Treatment in Poor Metabolizers

Polymorphic Enzyme	Decreased Clearance	Adverse Effects (overdosage)	Reduced Activation of Prodrug
CYP2C9	S-Warfarin	Bleeding	Losartan
	Phenytoin	Ataxia	
	Losartan		
	Tolbutamide	Hypoglycemia	
	NSAIDs	GI bleeding?	
CYP2C19	Omeprazole		Proguanil
	Diazepam	Sedation	
CYP2D6	Tricyclic antidepressants	Cardiotoxicity	Tramadol
	SSRIs	Serotonin syndrome	
	Antiarrhythmic drugs	Arrhythmias	Ethylmorphine
	Perhexiline	Neuropathy	
	Haloperidol	Parkinsonism?	
	Perphenazine		
	Zuclopenthixol		
	S-Mianserin		
	Tolterodine		
	Nicotine		

the protection from some carcinogens and smoking as a result of the high plasma levels of nicotine achieved with fewer cigarettes.

CYP2B6

CYP2B6 is of particular interest due to its wide interindividual variability in the expression and enzyme activity (75). Such a large variability for an enzyme that is not highly expressed is probably a result of effects of genetic polymorphisms and exposure to drugs that are inducers or inhibitors of CYP2B6. Variant alleles resulting from splicing defects or gene deletions have been identified in individuals with altered metabolic activity or impaired enzyme function as PMs. At least 28 allelic variants and some subvariants of *CYP2B6* have been shown to have an effect on drug clearance and drug response. For example, HIV-infected African-American individuals with a defective variant of *CYP2B6* are PMs of efavirenz, with plasma levels approximately threefold higher than individuals with the normal *CYP2B6* gene. A different variant of *CYP2B6* is associated with approximately twofold greater plasma levels of nevirapine in HIV-infected patients. A benefit from being a PM of CYP2B6 antiviral drugs might be enhanced HIV protection as a result of the higher plasma levels of the antiviral drugs achieved with low dosages.

Smokers with another variant of *CYP2B6* may be more vulnerable to abstinence symptoms and relapse following treatment with bupropion as a smoking cessation agent.

CYP2C9 and CYP2C19

CYP2C9 and CYP2C19 metabolize approximately 15% of clinical drugs such as phenytoin, *S*-warfarin, tolbutamide, losartan, and many NSAIDs (76). Although CYP2C19 metabolizes fewer drugs than CYP2D6 does, the drugs that CYP2C19 does metabolize are clinically important (Table 4.8). A deficit of CYP2C19 is found in the PM phenotype, which is seen in 8% to 13% of Caucasians, 20% to 30% of the Asian population (11% to 23% of Japanese and 5% to 37% of Chinese), up to 20% of the black African-American population, 14% to 15% of Saudi Arabians and Ethiopians, and up to 70% of Pacific Islanders (76). The more common mutant allele in these individuals is *CYP2C19**2, which expresses an inactive enzyme. The large interindividual variability observed in the therapeutic response to the antiepileptic drug mephenytoin is attributed to CYP2C19 polymorphism, which catalyzes the *p*-hydroxylation of its *S*-stereoisomer (76). The *R*-enantiomer is *N*-demethylated by CYP2C8 with no difference in its metabolism between PMs and EMs.

CYP2C9 is highly polymorphic, with at least 33 variants of CYP2C9 being identified so far. *CYP2C9*2* is frequent among Caucasians, with approximately 1% of the population being homozygous carriers and 22% being heterozygous. The corresponding figures for the *CYP2C9*3* allele are 0.4% and 15%, respectively. Clinical studies have addressed the importance of the *CYP2C9*2* and **3* alleles as a determining factor for drug clearance and drug response. CYP2C9 polymorphisms are relevant for the efficacy and adverse effects of numerous NSAIDs, sulfonylurea antidiabetic drugs, and, most critically, oral anticoagulants belonging to the class of vitamin K epoxide reductase inhibitors (e.g., warfarin) (76). A deficiency of this isoform, however, is seen in 8% to 13% of Caucasians, 2% to 3% of African Americans, and 1% of Asians. Individuals with the PM phenotype who possess this deficient isoform variant are ineffective in clearing S-warfarin (so much so that they can be fully anticoagulated on just 0.5 mg of warfarin per day) and in clearing phenytoin, which has a potentially very toxic narrow therapeutic range. On the other hand, the prodrug losartan will be poorly activated and ineffective.

CYP2D6

CYP2D6 is of particular importance, because it metabolizes a wide range of commonly prescribed drugs, including antidepressants, antipsychotics, β -adrenergic blockers, and antiarrhythmics (Table 4.9). CYP2D6 deficiency is a clinically important genetic variation of drug metabolism characterized by three phenotypes: UM, EM, and PM. The PM phenotype is inherited as an autosomal recessive trait, with 5 of 30 of the known *CYP2D6* gene mutations leading to either zero expression or the expression of a nonfunctional enzyme (73). Approximately 12% to 20% of Caucasians express the *CYP2D6*4* allele, and 5% express the other *CYP2D6* alleles. Up to 34% of African Americans express the *CYP2D6*17* allele, and 5% express the other *CYP2D6* alleles. Up to 50% of Chinese express the *CYP2D6*10* allele, and 5% express the other *CYP2D6* alleles (these individuals are referred to as PM) (73,75). Conversely, the 20% to 30% of Saudi Arabians and Ethiopians who express the *CYP2D6*2XN* allele are known as UMs of CYP2D6 substrates, because they express excess enzyme as a result of having multiple copies of the gene (73). Inasmuch as CYP2D6 is not inducible, individuals of Ethiopian and Saudi Arabian descent have developed genetically a different strategy to cope with the (presumed) high load of alkaloids and other substances in their diet; thus, the high expression of CYP2D6 using multiple copies of the gene. Those individuals who are deficient in CYP2D6 will be predisposed to adverse effects or drug toxicity from antidepressants or neuroleptics caused by inadequate metabolism or long half-lives, but the metabolism of prodrugs in these patients will be ineffective due to lack of

activation (e.g., codeine, which must be metabolized by *O*-demethylation to morphine). Those with the UM phenotype will require a dose that is much higher than normal to attain therapeutic drug plasma concentrations (e.g., one patient required a daily dose of ~300 mg of nortriptyline to achieve therapeutic plasma levels) or a lower dose for prodrugs that require metabolic activation. Individuals with the PM phenotype are also characterized by loss of CYP2D6 stereoselectivity in hydroxylation reactions. It can be anticipated that large differences in steady-state concentration for CYP2D6 substrates will occur between individuals with the different phenotypes when they receive the same dose. Depending on the drug and reaction type, a 10- to 30-fold difference in blood concentrations can be observed in the PM-phenotype debrisoquine polymorphism (76).

Acetylation

Acetylation, a nonmicrosomal form of metabolism, also exhibits polymorphisms and was first demonstrated in the acetylation of isoniazid. Several forms of *N*-acetyl transferase (NAT1 and NAT2) occur in humans. NAT1 (arylamine *N*-acetyltransferase) is widely distributed and catalyzes the *N*-acetylation of acidic arylamines, such as *p*-aminobenzoic acid and *p*-aminosalicylic acid, and so its expression levels in the body have toxicologic importance with regard to drug toxicity and cancer risk, for example bladder cancer and colorectal cancer. NAT2 is found predominately in the liver. Some clinically used drugs undergoing NAT2-catalyzed *N*-acetylation include isoniazid, procainamide, hydralazine, phenelzine, dapsone, caffeine, some benzodiazepines, and the carcinogenic secondary *N*-alkylarylamines (2-aminofluorene, benzidine, and 4-aminobiphenyl). Both NAT1 and NAT2 have a cysteine group at the active site, which is acetylated by acetyl CoA. The *S*-acetyl group is then transferred to the substrate. α -Substituents to the amino group sterically block acetylation. Polymorphisms in NAT2 gene in human populations can be segregated into fast, intermediate, and slow acetylator phenotypes. Polymorphisms in NAT2 are also associated with higher incidences of cancer and drug toxicity. Intestinal *N*-acetyl transferase appears not to be polymorphic (i.e., 5-aminosalicylic acid). The proportion of the fast acetylation phenotype is approximately 30% to 45% in Caucasians, 89% to 90% in Asians, and 100% in Canadian Eskimos. Drug-induced systemic lupus erythematosus from chronic procainamide therapy is more likely to appear in slow acetylators.

The antituberculosis drugs isoniazid and rifampin in combination are known to cause drug-induced hepatotoxicity (DIH). A higher risk of DIH during antituberculosis treatment has been reported in India compared with the Western populations. The occurrence of DIH in India was approximately 19%, with a

higher prevalence of NAT2 slow acetylator genotypes in DIH (~70%) compared with non-DIH (~45%). The frequency of NAT2*5/*7 and NAT2*6/*7 genotypes was higher in DIH than non-DIH (20% vs. 7% and 20% vs. 5%, respectively). Thus, genotyping of the NAT2 gene could possibly identify the high-risk group for developing antituberculosis treatment-induced hepatitis prior to medication.

Other Polymorphic Metabolizing Enzymes

The polymorphism for CYP2E1 is expressed more in Chinese than in Caucasians. Those with the CYP2E1 PM phenotype exhibit tolerance to alcohol and less toxicity from halohydrocarbon solvents.

Individuals with a defective variant of *FMO3* are poor metabolizers of trimethylamine to its *N*-oxide metabolite, and therefore exhibit trimethylaminuria, in which affected individuals excrete diet-derived free trimethylamine in the urine. This disease has probably been around since 1400 BC, when a young woman was described as being rejected by society because she smelled like rotting fish. Even Shakespeare likely was referring to a person with trimethylaminuria in a scene from *The Tempest*, Act I, scene II, line 26: "What have we here? A man or a fish? Dead or alive? A fish! He smells like a fish; a very ancient and fishlike smell."

In human populations, serum PON1 exhibits a substrate-dependent polymorphism to the neurotoxic effects of organophosphates in those susceptible individuals who are deficient in PON1 (i.e., PM phenotype) (55). The PON1 catalyzes the hydrolysis of paraoxon, chlorpyrifos (Dursban), and other organophosphates.

Polymorphism has been associated with serum cholinesterases (particularly succinyl cholinesterase, causing skeletal muscle paralysis), alcohol dehydrogenases, aldehyde dehydrogenases, epoxide hydrolase, and xanthine oxidase (74). Approximately 50% of the Asian population lack aldehyde dehydrogenase, resulting in high levels of acetaldehyde following ethanol ingestion and causing nausea and flushing. People with genetic variants of cholinesterase respond abnormally to succinylcholine, procaine, and other related choline esters. The clinical consequence of reduced enzymic activity of cholinesterase is that succinylcholine and procaine are not hydrolyzed in the blood, resulting in prolongation of their respective pharmacologic activities.

A suggestion has been made that those with EM phenotypes can be more prone than those with PM phenotypes to develop cancers, because they are better able to activate procarcinogens. Such interindividual variations can have a major influence in determining the risk of cancer. The activity of a particular P450 isoform can be a rationale for predicting the individual risk from exposure to carcinogenic compounds.

Our increasing knowledge of genetic polymorphism has contributed a great deal to our understanding about interindividual variation in the metabolism of drugs, including how to change dose regimens accordingly to

minimize drug toxicity and improve therapeutic efficacy. In humans, drugs not subject to polymorphic metabolism also exhibit substantial interindividual variation in their disposition, which is attributed to a great extent to environmental factors (e.g., inducing agents, smoking, and alcohol ingestion).

For many of these drugs, clear gene-dose and gene-effect relationships have been observed in patients. In this regard, the normal and mutant alleles can be considered as useful biomarkers in monitoring drug response and adverse effects. Genetic testing of mutant alleles is expected to have a role in predicting drug clearance and conducting individualized pharmacotherapy. However, clinical studies with large samples are warranted to establish gene-dose and gene-effect relationships for those enzymes exhibiting polymorphisms and their substrate drugs.

ORAL BIOAVAILABILITY

Oral bioavailability (see Chapter 3) is the fraction of the total dose of a drug that reaches the systemic circulation. The low oral bioavailability for a drug can be the result of disintegration and dissolution properties of the drug formulation, solubility of the drug molecule in the gastrointestinal environment, membrane permeability, presystemic intestinal metabolism, hepatic first-pass metabolism, or susceptibility to membrane transporters, such as P-gp efflux. Other routes of administration (e.g., subcutaneous, intravenous, inhalation, and nasal) for susceptible drugs have been investigated in an attempt to overcome the pronounced presystemic metabolism. The extent of first-pass metabolism depends on the drug delivery system because a formulation can increase or decrease the rate of dissolution, the residence time of a drug in the gastrointestinal tract, and the dose. The more prolonged the residence time, the greater the efficiency of first-pass metabolism. The drug form and delivery system should yield optimal bioavailability and pharmacokinetic profiles, resulting in a reproducible clinical response.

Studies are being performed to determine the effect of presystemic and hepatic first-pass metabolism on the toxicity and carcinogenicity of xenobiotics. For a non-therapeutic toxic substance, the existence of a first-pass effect is desirable, because the liver can bioinactivate the substance, preventing its distribution to other parts of the body. On the other hand, first-pass metabolism can increase a drug's toxicity by biotransforming the toxicant to a more toxic metabolite, which can reenter the blood and exert its toxic effect.

Presystemic First-Pass Metabolism

Although hepatic metabolism continues to be the most important route of metabolism for xenobiotics, the ability of the liver and intestine to metabolize

substances to either pharmacologically inactive or bioactive metabolites before reaching systemic blood levels is called prehepatic or presystemic first-pass metabolism, which results in the low systemic availability for susceptible drugs. Sulfonation and glucuronidation are major pathways of presystemic intestinal first-pass metabolism in humans for acetaminophen, phenylephrine, terbutaline, albuterol, fenoterol, and isoproterenol.

The discovery that CYP3A4 is found in the mucosal enterocytes of the intestinal villi signifies its role as key determinant in the oral bioavailability of its numerous drug substrates (Table 4.11) (77). Drugs known to be substrates for CYP3A have a low and variable oral bioavailability that can be explained by presystemic first-pass metabolism by the small intestine P450 isoforms. The concentration of functional intestinal CYP3A is influenced by genetic disposition, induction, and inhibition, which to a great extent determines drug blood levels and therapeutic response. Xenobiotics, when ingested orally, can modify the activity of intestinal CYP3A enzymes by induction, inhibition, and stimulation. By modulation of the isoform pattern in the intestine, a xenobiotic could alter its own metabolism and that of others in a time- and dose-dependent manner. The concentration of CYP3A in the intestine is comparable to that of the liver. The oral administration of dexamethasone induces the formation of CYP3A and erythromycin inhibits it. The glucocorticoid inducibility of CYP3A4 can also be a factor in differences of metabolism between males and females. Studies have suggested that intestinal CYP3A4 C-2 hydroxylation of estradiol contributes to the oxidative metabolism of endogenous estrogens circulating with the enterohepatic recycling pool (38). Norethisterone has a low oral bioavailability of 42% due to oxidative first-pass metabolism (CYP3A), but levonorgestrel is completely available in women with no conjugated metabolites.

Several clinically relevant drug–drug interactions between orally coadministered drugs and CYP3A4 can be explained by a modification of drug metabolism at the P450 level. If a drug has high presystemic elimination (low bioavailability) and is metabolized primarily by CYP3A4, then coadministration with a CYP3A4 inhibitor can be expected to alter the drug's pharmacokinetics by reducing its metabolism, thus increasing its plasma concentration. Drugs and some foods (e.g., grapefruit juice) that are known inhibitors, inducers, or substrates for intestinal CYP3A4 can potentially interact with the metabolism of a coadministered drug, affecting its area under the curve and rate of clearance (Tables 4.12 and 4.14) (78). Inducers can reduce absorption and oral bioavailability, whereas these same factors are increased by inhibitors. For example, erythromycin can enhance the oral absorption of another drug by inhibiting its metabolism in the small intestine by CYP3A4. By virtue of being competitive substrates for CYP3A4, prednisone, prednisolone, and methylprednisolone

(but not dexamethasone) are competitive inhibitors of synthetic glucocorticoid metabolism. This is because a major metabolic pathway for synthetic glucocorticoids involves CYP3A4. In addition to coadministered drugs, metabolic interactions with exogenous CYP3A4 substrates secreted in the bile are possible. The poor oral bioavailability for cyclosporine is attributed to a combination of intestinal metabolism by CYP3A4 and efflux by P-gp (79).

Because the intestinal mucosa is enriched with glucuronosyltransferases, sulfotransferases, and glutathione transferases, presystemic first-pass metabolism for orally administered drugs susceptible to these conjugation reactions results in their low oral bioavailability (65). Presystemic metabolism often exceeds liver metabolism for some drugs. For example, more than 80% of intravenously administered albuterol is excreted unchanged in urine, with the balance as glucuronide conjugates, whereas when albuterol is administered orally, less than 5% is systemically absorbed as a result of intestinal sulfonation and glucuronidation. Presystemic metabolism is a major pathway in humans for most β -adrenergic agonists, such as glucuronides or sulfate esters for terbutaline, fenoterol, albuterol, and isoproterenol, morphine (3-*O*-glucuronide), acetaminophen (*O*-sulfate), and estradiol (3-*O*-sulfate). The bioavailability of orally administered estradiol or ethinyl estradiol in females is approximately 50%. Mestranol (3-methoxyethinyl estradiol) has greater bioavailability, however, because it is not significantly conjugated. Levodopa has a low oral bioavailability as a result of its metabolism by intestinal L-aromatic amino acid decarboxylase. The activity of this enzyme depends on the percentage bound of its cofactor pyridoxine (vitamin B₆). Tyramine, which occurs in fermented foods such as cheeses and red wines, ripe bananas, and yeast extracts, is metabolized by both MAO-A and MAO-B in the gut wall.

The extensive presystemic first-pass sulfonation of phenolic drugs, for example, can lead to increased bioavailability of other drugs by competing for the available sulfate pool, resulting in the possibility of drug toxicity (56). Concurrent oral administration of acetaminophen with ethinyl estradiol resulted in a 48% increase in ethinyl estradiol blood levels. Ascorbic acid, which is sulfonated, also increases the bioavailability of concurrently administered ethinyl estradiol. Sulfonation and glucuronidation occur side by side, often competing for the same substrate, and the balance between sulfonation and glucuronidation is influenced by several factors, such as species, doses, availability of cosubstrates, inhibition, and induction of the respective transferases.

First-Pass Metabolism

Several orally administered drugs are known to undergo liver first-pass metabolism during their transport to the systemic circulation from the gastrointestinal tract

TABLE 4.17 Examples of Drugs Exhibiting First-Pass Metabolism

Acetaminophen	Isoproterenol	Oxprenolol
Albuterol	Lidocaine	Pentazocine
Alprenolol	Meperidine	Propoxyphene
Aspirin	Methyltestosterone	Propranolol
Cyclosporin	Metoprolol	Salicylamide
Desmethylinipramine	Dihydropyridines	Terbutaline
Fluorouracil	(Nifedipine)	Verapamil
Hydrocortisone	Nortriptyline	
Imipramine	Organic nitrates	

(e.g., metoprolol). Thus, the liver can remove substances from the blood after their absorption from the gastrointestinal tract, thereby preventing distribution to other parts of the body. This effect can seriously impair the bioavailability of an orally administered drug, reducing the amount of the drug that reaches the systemic circulation and, ultimately, its receptor to produce its pharmacologic effect. Drugs subject to first-pass metabolism are included in Table 4.17.

P-Glycoprotein

Another factor that must be considered in the oral bioavailability of many CYP3A4 substrates is intestinal P-gp (80). Originally discovered as a transmembrane transporter protein associated with the resistance (elimination) of anticancer drugs, P-gp can also have

a role in how a drug is absorbed, distributed, metabolized, and eliminated from the body (see Chapter 5) (79–81). Considering its role as a transporter protein (efflux pump), it is logical that it should exhibit saturable (nonlinear) kinetics. P-gp exhibits a broad specificity for a large number of substrates, inhibitors, and inducers (Table 4.18). The common link between P-gp substrates is that most of the same compounds are also substrates for CYP3A4. The close physical location of P-gp and CYP3A4 in the endothelial cells of the intestinal mucosa allows these proteins to work in concert with each other to decrease drug plasma concentrations of CYP3A4 substrates, suggesting a complementary protective mechanism for these two proteins, forming a barrier to the intestinal absorption of CYP3A4 substrates. Hepatic and renal P-gp also appears to function in a complementary manner, promoting the elimination of substrates into the bile and urine, respectively. For example, if a drug is a substrate for intestinal P-gp, its oral absorption will be incomplete and this same drug will be actively transported by the renal tubules into the urine, enhancing its elimination. On the other hand, inhibiting P-gp would be expected to improve the oral bioavailability of P-gp substrates, but if the inhibitor is also a substrate for CYP3A4, increased metabolism (presystemic) would occur. Drugs with low oral bioavailability or high first-pass metabolism can be particularly susceptible to alterations in the transport kinetics of P-gp. Because P-gp exhibits saturation (nonlinear) kinetics, drugs

TABLE 4.18 Some Substrates, Inhibitors, and Inducers for P-Glycoprotein

Substrate		Inhibitors		Foods
Acetolol	Lidocaine*	Amiodarone*	Lovastatin*	Daidzein
Amiodarone*	Loperamide*	Amitriptyline*	Maprotiline	Genistein
Atorvastatin*	Methotrexate	Astemizole*	Mefloquine*	Grapefruit juice*
Celiprolol	Mibefradil	Atorvastatin*	Mibefradil*	Orange juice
Cimetidine*	Nadolol	Carvedilol	Midazolam*	Isoflavones
Ciprofloxacin	Nelfinavir*	Chlorpromazine*	Mifepristone*	
Colchicine	Nicardipine*	Clarithromycin*	Nelfinavir*	Inducers
Cyclosporin*	Ondansetron*	Cyclosporin*	Nicardipine*	Dexamethasone*
Daunorubicin*	Paclitaxel*	Desipramine	Nifedipine*	Prazosin
Debrisoquine	Pravastatin*	Dexverapamil*	Nitrendipine*	Progesterone*
Dexamethasone*	Quinidine*	Diltiazem*	Ofloxacin	Quercetin
DHEA	Quinolones	Dipyridamole	Prochlorperazine	Rifampin
Digoxin*	Ranitidine	Disulfiram	Progesterone*	St. John's Wort*
Diltiazem*	Rifampin*	Doxepin		
Docetaxel*	Ritonavir*	Erythromycin*	Propranolol	
Domperidone*	Saquinavir*	Flupentixol	Propafenone*	
Doxorubicin*	Tacrolimus*	Felodipine*	Quinidine*	
Enoxacin	Taxol*	Fluphenazine	Quinine*	
Erythromycin*	Teniposide*	Glibenclamide	Reserpine	
Estradiol*	Terfenadine*	Haloperidol*	Rifampin*	
Etoposide*	Timolol	Hydrocortisone*	Ritonavir*	
Fexofenadine*	Verapamil*	Imipramine*	Saquinavir*	
Hydrocortisone*	Vinblastine*	Itraconazole*	Tacrolimus*	
Idarubicin	Vincristine*	Ivermectin	Testosterone*	
Indinavir*	Vindesine*	Ketoconazole*	Tamoxifen*	
Ivermectin		Lidocaine*	Trimipramine	
			Verapamil*	

*CYP3A4 substrate, inhibitor, or inducer.

with low dosages can have their oral bioavailability enhanced by increasing its oral dosage, thus saturating the P-gp pump. As with CYP3A4, there is significant interindividual variation (4- to 10-fold) in the intestinal expression of P-gp, which could explain the variance observed in the pharmacokinetics for CYP3A4 substrates. The interactive nature of CYP3A4 and P-gp will be of importance in controlling and improving the oral bioavailability of CYP3A4 substrates and drug regimens. The presence of inhibitors of P-gp in grapefruit juice (e.g., 6,7-dihydroxybergamottin and other furanocoumarins) has confirmed that the inhibition of efflux transport of drugs and of drug metabolism by CYP3A4 could be an important cause of drug-grapefruit juice interaction (82).

In summary, oral bioavailability for xenobiotics is dependent on a combination of factors, including physical properties of the drugs and formulation and biologic factors such as metabolizing enzymes, membrane permeability, and the membrane efflux pump, P-gp.

EXTRAHEPATIC METABOLISM

Because the liver is the primary tissue for xenobiotic metabolism, it is not surprising that our understanding of mammalian P450 monooxygenase is based chiefly on hepatic studies. Although the tissue content of P450s is highest in the liver, P450 enzymes are ubiquitous, and their role in extrahepatic tissues remains unclear. P450 pattern in these tissues differs considerably from that in the human liver (83). In addition to liver tissue, P450 enzymes are found in lung, nasal epithelium, intestinal tract, kidney, and adrenal tissues and brain. It is possible that the expression of the polymorphic genes and induction of the isoforms in the extrahepatic tissues can affect the activity of the P450 isoforms in the metabolism of drugs, endogenous steroids, and xenobiotics. Therefore, characterization of P450, UGT, SULT, and other polymorphic drug-metabolizing enzymes in extrahepatic tissues is important to our overall understanding about the biologic importance of these isoform families to improved drug therapy, design of new drugs and dosages forms, toxicity, and carcinogenicity.

The mucosal surfaces of the gastrointestinal tract, the nasal passages, and the lungs are major portals of entry for xenobiotics into the body and, as such, are continuously exposed to a variety of orally ingested or inhaled airborne xenobiotics, including drugs, plant toxins, environmental pollutants, and other chemical substances. As a consequence of this exposure, these tissues represent a major target for necrosis, tumorigenesis, and other chemically induced toxicities. Many of these toxins and chemical carcinogens are relatively inert substances that must be bioactivated to exert their cytotoxicity and tumorigenicity. The epithelial cells of these tissues are capable of metabolizing a wide

variety of exogenous and endogenous substances, and these cells provide the principal and initial source of biotransformation for these xenobiotics during the absorptive phase. The consequences of such presystemic biotransformation is either a decrease in the amount of xenobiotics available for systemic absorption by facilitating the elimination of polar metabolites or toxification by activation to carcinogens, which can be one determinant of tissue susceptibility for the development of intestinal cancer. The risk of colon cancer can depend on dietary constituents that contain either procarcinogens or compounds modulating the response to carcinogens.

Intestinal Metabolism

Many of the clinically relevant aspects of P450 can, in fact, occur at the level of the intestinal mucosa and could account for differences among patients in dosing requirements. The intestinal mucosa is enriched especially with CYP3A4 isoform, glucuronosyl transferases, sulfotransferases, and GSTs, making it particularly important for orally administered drugs susceptible to oxidation (77), glucuronidation, or sulfonation conjugation pathways (56), or glutathione conjugation pathways. The highest concentrations of P450s occur in the duodenum, with gradual tapering into the ileum. In the human intestine, CYP2E, CYP3A, CYP2C8, CYP2C9, CYP2C19, and CYP2D6 have been identified. Therefore, intestinal P450 isoforms provide potential presystemic first-pass metabolism of ingested xenobiotics affecting their oral bioavailability (e.g., hydroxylation of naloxone) or bioactivation of carcinogens or mutagens. It is not surprising that dietary factors can affect the intestinal P450 isoforms. For example, a 2-day dietary exposure to cooked Brussels sprouts significantly decreased the 2 α -hydroxylation of testosterone yet induced CYP1A2 activity for PAH. An 8-oz glass of grapefruit juice inhibited the sulfoxidation metabolism of omeprazole (CYP3A4) but not its hydroxylation (CYP2C19), thus increasing its systemic blood concentration. These types of interactions between a drug and a dietary inhibitor could result in a clinically significant drug interaction.

Intestinal UGT isoforms can glucuronidate orally administered drugs, such as morphine, acetaminophen, α - and β -adrenergic agonists, and other phenolic phenethanolamines and other dietary xenobiotics, resulting in a reduction of their oral bioavailability (increasing first-pass metabolism), thus altering their pharmacokinetics and pharmacodynamics. The UGTs expressed in the intestine include UGT1A1 (bilirubin-glucuronidating isoform), UGT1A3, UGT1A4, UGT1A6, UGT1A8, UGT1A9, and UGT1A5. Substrate specificities of intestinal UGT isoforms are comparable to those in the liver. Glucuronidase hydrolysis of biliary glucuronide conjugates in the intestine can contribute to EHC of the parent drug.

Likewise, the sulfotransferases in the small intestine can sulfonate orally administered drugs and xenobiotics

for which the primary route of conjugation is sulfonation (e.g., isoproterenol, albuterol, steroid hormones, α -methyl dopa, acetaminophen, and fenoldopam), decreasing their oral bioavailability and, thus, altering their pharmacokinetics and pharmacodynamics. Competition for intestinal sulfonation between coadministered substrates can influence their bioavailability with either an enhancement or a decrease of therapeutic effects. Sulfatase hydrolysis of biliary sulfate esters in the intestine can contribute to EHC of the parent drug.

The occurrence of intestinal P450 enzymes and bacterial enzymes in the microflora allows the metabolism of relatively stable environmental pollutants and food-derived xenobiotics (i.e., plants contain a variety of protoxins, promutagens, and procarcinogens) into mutagens and carcinogens (68). For example, cruciferous vegetables (Brussels sprouts, cabbage, broccoli, cauliflower, and spinach) are all rich in indole compounds (e.g., indole 3-carbinol), which with regular and chronic ingestion are capable of inducing some intestinal P450s (CYP1A subfamily) and inhibiting others (CYP3A subfamily). It is likely that these vegetables would also alter the metabolism of food-derived mutagens (e.g., heterocyclic amines produced during charbroiling of meat are P450 *N*-hydroxylated and become carcinogenic in a manner similar to arylamines) and carcinogens.

The extent of a drug's metabolism in the small bowel and its role in clinically relevant drug interaction remain to be evaluated and must be taken into account during oral pharmacokinetics analysis of future drug interaction studies. Clinically significant interaction will not always occur when a drug is combined with other isoform subfamily substrates. Oral coadministration of a drug with drugs that interact with its metabolism need not be avoided. The blood concentration of the drug must be monitored closely, however, and the dose should be adapted to avoid adverse drug reactions.

Intestinal Microflora

When drugs are orally ingested or there is considerable biliary excretion of a drug or its metabolites into the gastrointestinal tract, such as with a parentally administered drug (EHC or recirculation), the intestinal bacterial microflora can have a role in the metabolism of these drugs. The microflora has an important role in the enterohepatic recirculation of xenobiotics via their conjugated metabolites (e.g., digoxin, the oral contraceptives norethisterone and ethinyl estradiol, and chloramphenicol) and endogenous substances (steroid hormones, bile acids, folic acid, and cholesterol), which reenter the gut via the bile (68). Compounds eliminated in the bile are conjugated with glucuronic acid, glycine, sulfate, and glutathione, and once secreted into the small intestine, the bacterial β -glucuronidase, sulfatase, nitroreductases, and various glycosidases catalyze the hydrolysis of the conjugates. The activity of orally administered conjugated

estrogens (e.g., Premarin) involves the hydrolysis of the sulfate esters by sulfatases, releasing estrogens to be reabsorbed from the intestine into the portal circulation. The clinical use of oral antibiotics (e.g., erythromycin, penicillin, clindamycin, and aminoglycosides) has a profound effect on the gut microflora and the enzymes responsible for the hydrolysis of drug conjugates undergoing EHC. Bacterial reduction includes nitro reduction of nitroimidazole, azo reduction of azides (sulfasalazine to 5-aminosalicylic acid and sulfapyridine), and reduction of the sulfoxide to its sulfide. The sulfoxide of sulindac is reduced by both gut microflora and hepatic P450s. Other ways in which bacterial flora can affect metabolism include the following: 1) production of toxic metabolites, 2) formation of carcinogens from inactive precursors, 3) detoxication, 4) exhibition of species differences in drug metabolism, 5) exhibition of individual differences in drug metabolism, 6) production of pharmacologically active metabolites from inactive precursors, and 7) production of metabolites not formed by animal tissues. In contrast to the predominantly hepatic oxidative and conjugative metabolism of the liver, gut microflora is largely degradative, hydrolytic, and reductive, with a potential for both metabolic activation and detoxication of xenobiotics.

Lung Metabolism

Some of the hepatic xenobiotic biotransformation pathways are also operative in the lung (84,85). Because of the differences in organ sizes, the total content of the pulmonary xenobiotic-metabolizing enzyme systems is usually lower than in the liver, creating the impression of a minor role for the lung in xenobiotic elimination. CYP2E1 is expressed in the lung to the greatest extent. The expression of other P450s, FMO, epoxide hydrolase, and the Phase 2 conjugation pathways are comparable to those in the liver. Thus, the lungs can have a significant role in the metabolic elimination or activation of small molecular weight inhaled xenobiotics. When drugs are injected intravenously, intramuscularly, or subcutaneously, or after skin absorption, the drug initially enters the pulmonary circulation, after which the lung becomes the organ of first-pass metabolism for the drug. The blood levels and therapeutic response of the drug are influenced by genetic disposition, induction, and inhibition of the pulmonary metabolizing enzymes. By modulation of the P450 isoform pattern in the lung, a xenobiotic could alter its own metabolism and that of others in a time- and dose-dependent manner. Because of its position in the circulation, the lung provides a second-pass metabolism for xenobiotics and their metabolites exiting from the liver, but it is also susceptible to the cytotoxicity or carcinogenicity of hepatic activated metabolites. Antihistamines, β -blockers, opioids, and tricyclic antidepressants are among the basic amines known to accumulate in the lungs as a result of their binding to surfactant phospholipids in lung tissue. The significance of this relationship to potential pneumotoxicity remains to be seen.

Nasal Metabolism

The nasal olfactory mucosa is recognized as a first line of defense for the lung against airborne environmental xenobiotics, because the mucosa is constantly exposed to the airborne external environment (86). Drug metabolism in the olfactory nasal epithelium represents a major metabolic pathway for protecting the brain against the entry of drugs, inhaled environmental pollutants, or other volatile chemicals into the CNS and brain. In some instances, these enzymes can biotransform drugs, environmental pollutants, or airborne xenobiotics into more reactive and potentially toxic metabolites, increasing the risk of carcinogenesis in the nasopharynx and lung (e.g., procarcinogens and nitrosamines in cigarette smoke). P450 activity is high in olfactory tissue, and olfactory-specific isoforms have been identified, including CYP1A2, CYP2A6, CYP2A13, CYP2E1, CYP3A4, and CYP2G1, as well as several isoforms of the flavin-dependent monooxygenases. Phase 2 enzymes found in the olfactory mucosa include UGTs, glutathione-S-transferases, carboxylesterases, aldehyde oxidases, and epoxide hydrolases. The most striking feature of the nasal epithelium is that P450 catalytic activity is higher than in any other extrahepatic tissue, including the liver. Nasal decongestants, essences, anesthetics, alcohols, nicotine, and cocaine have been shown to be metabolized in vitro by P450 enzymes from the nasal epithelium. Because the P450s in the nasal mucosa are active, first-pass metabolism should be considered when delivering susceptible drugs to the nasal tissues.

The nasal mucosa is also exposed to a wide range of odorants. Odorants, which are mostly small lipophilic molecules, enter the mucosa and reach the odorant receptors on sensory neurons by a transient process requiring signal termination, which could be provided by biotransformation of the odorant in the epithelial supporting cells. Thus metabolism of odorants could be involved in both the termination and initiation of olfactory stimuli.

Metabolism in Other Tissues

The isoforms of P450s and their regulation in the brain are of interest in defining the possible involvement of P450s in central nervous system toxicity and carcinogenicity. P450s in the kidney and adrenal tissues include isoforms primarily involved in the hydroxylation of steroids, arachidonic acid, and 25-hydroxycholecalciferol.

STEREOCHEMICAL ASPECTS OF DRUG METABOLISM

In addition to the physicochemical factors that affect xenobiotic metabolism, stereochemical factors have an important role in the biotransformation of drugs. This involvement is not unexpected, because the xenobiotic-metabolizing enzymes are also the same enzymes that metabolize certain endogenous substrates, which for the most part are chiral molecules. Most of these enzymes

show stereoselectivity but not stereospecificity; in other words, one stereoisomer enters into biotransformation pathways preferentially but not exclusively. Metabolic stereochemical reactions can be categorized as follows: substrate stereoselectivity, in which two enantiomers of a chiral substrate are metabolized at different rates; product stereoselectivity, in which a new chiral center is created in a symmetric molecule and one enantiomer is metabolized preferentially; and substrate-product stereoselectivity, in which a new chiral center of a chiral molecule is metabolized preferentially to one of two possible diastereomers (87). An example of substrate stereoselectivity is the preferred decarboxylation of *S*- α -methyldopa to *S*- α -methyldopamine, with almost no reaction for *R*- α -methyldopa. The reduction of ketones to stereoisomeric alcohols and the hydroxylation of enantiotropic protons or phenyl rings by monooxygenases are examples of product stereoselectivity. For example, phenytoin undergoes aromatic *p*-hydroxylation of only one of its two phenyl rings to create a chiral center at C-5 of the hydantoin ring, methadone is reduced preferentially to its α -diastereomeric alcohol, and naltrexone is reduced to its 6- β -alcohol. An example of substrate-product stereoselectivity is the reduction of the enantiomers of warfarin and the β -hydroxylation of *S*- α -methyldopamine to (1*R*,2*S*)- α -methylnorepinephrine, whereas *R*- α -methyldopamine is hydroxylated only to a negligible extent. In vivo studies of this type often can be confused by the further biotransformation of one stereoisomer, giving the false impression that only one stereoisomer was formed preferentially. Moreover, some compounds show stereoselective absorption, distribution, and excretion, which proves the importance of performing in vitro studies. Although studies regarding the stereoselective biotransformation of drug molecules are not yet extensive, those that have been done indicate that stereochemical factors have an important role in drug metabolism and, in some cases, could account for the differences in pharmacologic activity and duration of action between enantiomers (see the discussion of chiral inversion of the NSAIDs).

METABOLIC BIOACTIVATION: ROLE IN HEPATOTOXICITY, IDIOSYNCRATIC REACTIONS, AND CHEMICAL CARCINOGENESIS

Drug-Induced Hepatotoxicity

Drug-induced hepatotoxicity is the leading cause of hepatic injury, accounting for approximately half of all cases of acute liver failure in the United States (88,89). Recent studies have shown that drug-induced hepatotoxicity represents a larger percentage of adverse drug reactions than reported previously and that the incidence and severity of drug-induced liver injury is underestimated among the general population.

Acetaminophen overdose is the leading cause for calls to Poison Control Centers (>100,000 calls/year) and accounts

for more than 56,000 emergency room visits, 2,600 hospitalizations, and an estimated 458 deaths from acute liver failure each year. Among the listed drugs in Table 4.19, acetaminophen is the most frequent hepatotoxic agent and can cause extensive hepatic necrosis with as little as 10 to 12 g (30 to 40 tablets). Chronic alcohol intake enhances acetaminophen hepatotoxicity by more than five times as compared to acute alcohol intake, yet acetaminophen is heavily marketed for its safety as compared to nonsteroidal analgesics. United States drug manufacturers continue to market and promote extra-strength acetaminophen products (500 to 750 mg/tablet) and a variety of extra-strength acetaminophen–drug combination products. Self-poisoning with acetaminophen (paracetamol) is also a common cause of hepatotoxicity in the Western world. To reduce the number of acetaminophen poisonings in the United Kingdom, over-the-counter (nonprescription) sales of acetaminophen are limited to 16 tablets per packet.

Drug-induced hepatic damage is also the most frequent reason that new therapeutic agents are not approved by the FDA (e.g., ximelagatran in 2004) and the most common adverse drug reaction leading to withdrawal of a drug from the market (Table 4.19). Hepatotoxicity almost always involves metabolism with P450 enzymes rather than Phase 2 enzymes. More than 600 drugs, chemicals, and herbal remedies can cause hepatotoxicity, of which more than 30 drugs have either been withdrawn from the United States market as a result of hepatotoxicity or have carried a black box warning for hepatotoxicity since 1990. Table 4.19 includes some of the more common drugs that have exhibited drug-induced

SOME DRUGS EXHIBITING DRUG-INDUCED HEPATOTOXICITY NOT APPROVED FOR USE IN UNITED STATES

Drugs with reactive metabolites that were used in other countries but never approved in the United States include: alpidem, amineptine, amodiaquine, cinchophen, dihydralazine, dilevalol, ebrotidine, glafenine, ibufenac, isoxicam, niperotidine, perhexiline, pirofen, and tilbroquinol.

hepatotoxicity ranging from severe, requiring the drug's regulatory withdrawal from the market (**bold** in Table 4.19); moderate to severe, requiring black box warning restrictions (*italics* in Table 4.19); or mild to moderate, requiring frequent liver function monitoring.

However, Watkins et al. (90) reported in the *Journal of the American Medical Association* that one-third of 106 patients taking a maximum daily acetaminophen dose of 4 g for 8 days, either alone or in combination with hydrocodone, exhibited a threefold increase in liver enzymes associated with acetaminophen-induced liver injury. This threefold increase in transaminase levels is a signal for potential liver safety concerns in those individuals who are at risk of acetaminophen-induced liver toxicity. Drug-induced injury is most common and includes hepatic necrosis and steatosis, which can affect significant portions of the liver (88). Drugs reported to cause hepatocellular necrosis include acetaminophen, methyl dopa, valproic acid, trazodone,

TABLE 4.19 Some Drugs Causing Hepatic Injury^a

Acarbose	Fenofibrate	Methotrexate	Rosiglitazone
Acetaminophen	<i>Fluconazole</i>	Methsuximide	Rosuvastatin
Allopurinol	<i>Flutamide</i>	Methyldopa	<i>Saquinavir</i>
Amiodarone	Fluvastatin	Nabumetone	Simvastatin
Amprenavir	Gemfibrozil	Naproxen	Sulindac
Anagrelide	Gemtuzumab	Nefazodone (2005)	<i>Tacrine</i>
<i>Atomoxetine</i>	Griseofulvin	<i>Nevirapine</i>	Tamoxifen
Atorvastatin	<i>Halothane</i>	Niacin (SR)	Tasosartan (1998)
Azathioprine	Imatinib	Nitrofurantoin	<i>Terbinafine</i>
Bicalutamide	Indinavir	Olanzapine	Testosterone
Bosentan	<i>Infliximab</i>	Oxaprozin	Thioguanine
Bromfenac (1998)	<i>Interferon-β_{1a}</i>	Peg-interferon-α _{2a}	Tizanidine
Carbamazepine	<i>Interferon-β_{1b}</i>	Pemoline (2005)	<i>Tolcapone</i>
Celecoxib	Isoflurane	Pentamidine	Troglitazone (2000)
Dapsone	<i>Isoniazid</i>	Pioglitazone	<i>Trovafoxacin</i>
Deferasirox	Fenofibrate	Piroxicam	<i>Valproic acid</i>
<i>Diclofenac</i>	Isotretinoin	Pravastatin	Voriconazole
Disulfiram	Itraconazole	<i>Pyrazinamide</i>	Ximelagatran (2004)
<i>Duloxetine</i>	<i>Ketoconazole</i>	Ribavirin	<i>Zileuton</i>
Efavirenz	<i>Ketorolac</i>	Rifabutin	<i>Zifirlukast</i>
Ethotoin	Lamivudine	<i>Rifampin</i>	
Ethosuximide	<i>Leflunomide</i>	Riluzole	
<i>Felbamate</i>	Lovastatin	Ritonavir	

^aDrugs in **bold** have exhibited severe drug-induced hepatotoxicity and were withdrawn either voluntarily or by a regulatory agency (year given). Drugs in *italics* have exhibited moderate–severe drug-induced hepatotoxicity requiring a black box warning restricting their use. The other drugs have exhibited mild to moderate drug-induced hepatotoxicity that can need frequent liver transaminase testing for those at risk (see Table 4.20).

nefazodone, venlafaxine, and lovastatin. Drug-induced liver damage occurs after a prolonged period of drug administration.

The most commonly used indicators of hepatotoxicity (i.e., liver injury) are increased levels of the liver transaminases, aspartate aminotransferase (AST) and alanine aminotransferase (ALT) (88,89). Drug-induced hepatotoxicity can develop rapidly, often before abnormal laboratory tests are noticed, which are characterized by rapid elevations in ALT and AST of 8 to 500 times the upper normal limit, with variable elevations in bilirubin. Drugs causing acute liver injury (hepatocellular necrosis) exhibit elevations in hepatic transaminases ranging from 50 to 100 times higher than the normal level. On the other hand, the elevations of ALT and AST in alcoholic liver disease are two to three times higher than normal. Some hepatotoxins, however, do not elevate transaminases, whereas nonhepatic toxins can elevate ALT.

Most drug-induced hepatotoxicity is of an idiosyncratic nature, occurring in a small percentage of patients (1 in 5,000) who ingest the drug (88,89). These reactions tend to be of two distinct types: 1) hypersensitivity reactions that are immune mediated, occurring within the first 4 to 6 weeks, and are associated with fever, rash, eosinophilia, and a hepatitis-like picture (e.g., phenytoin, sulindac, and allopurinol); and 2) metabolic idiosyncratic reactions that tend to occur at almost any time during the first year of treatment (e.g., troglitazone and isoniazid). The incidence of overt idiosyncratic liver diseases varies

with the drug, ranging from approximately 1 in 100 with isoniazid to 1 in 1,000 with phenytoin, to 1 in 10,000 or more with sulindac and troglitazone, and 1 in 100,000 with diclofenac. To detect a single case of drug-induced hepatotoxicity with 95% confidence requires the number of patients studied to be threefold the incidence of the reaction. For one adverse drug reaction in 10,000 patients, at least 30,000 patients need to be evaluated. Thus, many drugs are approved before liver toxicity is observed. It is the responsibility of postmarketing surveillance and monitoring of liver transaminases to identify potential cases of liver-adverse drug reactions.

Risk factors (Table 4.20) for drug-induced liver injury, such as age, gender, genetic predisposition, multiple drugs or dietary supplements, and degree of alcohol consumption, appear to increase the susceptibility to drug-induced hepatotoxicity (88,89). Patients with mild to moderate chronic liver disease do not appear to be at increased risk for idiosyncratic hepatic injury from drugs. However, the drugs in Table 4.19 should be used with caution in these patients, because such patients can have altered metabolism of these drugs and, therefore, can be at increased risk for liver injury. The coadministration of drugs in Table 4.19 with enzyme inducers, such as phenobarbital, phenytoin, ethanol, and/or cigarette smoke, can induce hepatic enzymes, resulting in the enhancement of hepatotoxicity.

Most hepatic adverse effects associated with drugs occur in adults rather than children. Drug-induced liver injury occurs at a higher rate in patients older than 50

TABLE 4.20 Risk Factors for Drug-Induced Liver Injury

Race	Some drugs exhibit different toxicities based on race as a result of individual P450 polymorphism. For example, blacks and Hispanics can be more susceptible to isoniazid (INH) toxicity.
Age	Elderly persons are at increased risk of hepatic injury due to decreased clearance, drug-to-drug interactions, reduced hepatic blood flow, variation in drug binding, and lower hepatic volume. In addition, poor diet, infections, and multiple hospitalizations are important reasons for drug-induced hepatotoxicity. Hepatic drug reactions are rare in children (e.g., acetaminophen, halogenated general anesthetics).
Gender	Although the reasons are unknown, hepatic drug reactions are more common in females. Females are more susceptible to acetaminophen, halothane, nitrofurantoin, diclofenac, and sulindac.
Alcohol	Alcoholics are susceptible to drug toxicity, because alcohol induces liver injury and cirrhotic changes that alter drug metabolism. Alcohol causes depletion of glutathione (hepatoprotective) stores, making the person more susceptible to toxicity by drugs (e.g., acetaminophen, statins).
Liver disease	Patients with chronic liver disease are not uniformly at increased risk of hepatic injury. Although the total P450 level is reduced, some patients can be affected more than others. The modification of doses in persons with liver disease should be based on knowledge of the specific P450 isoform involved in the metabolism. Patients with HIV infection who are coinfecting with hepatitis B or C virus are at increased risk for hepatotoxic effects. Similarly, patients with cirrhosis are at increased risk to hepatotoxic drugs (e.g., methotrexate, methylidopa, valproic acid).
Genetic factors	Genetic (polymorphic) differences in the formation of P450 isoforms (2C family and 2D6) can result in abnormal reactions to drugs, including idiosyncratic reactions (see Table 4.18).
Other comorbidities	Patients with AIDS, renal disease, or diabetes mellitus; persons who are malnourished; and persons who are fasting can be susceptible to drug reactions as a result of low glutathione stores.
Pharmacokinetics	Long-acting drugs can cause more injury than shorter-acting drugs, as well as sustained-release drug product formulations.
Drug adulterants	Contaminants are often found in noncertified herbal supplements (e.g., hepatitis C).

years, and drug-associated jaundice occurs also more frequently in the geriatric population (88,89). This age-related risk can be the result of increased frequency of drug exposure, multidrug therapy, and age-related changes in drug metabolism.

For reasons that are unclear, drug-induced liver injury affects females more than males: Females accounted for approximately 79% of all reactions to acetaminophen and 73% of all idiosyncratic drug-induced reactions (88). Females exhibit increased risk of hepatic injury from drugs such as atorvastatin, nitrofurantoin, methyldopa, and diclofenac.

Genetic factors as a result of enzyme polymorphism in affected individuals can decrease the ability to metabolize or eliminate drugs, thus increasing their duration of action and the drug exposure and/or decreasing the ability to modulate the immune response to drugs or metabolites. Chronic ingestion of alcohol can also predispose many patients to increased hepatotoxicity from drugs by lowering the store of glutathione (a detoxifying mechanism), which prevents trapping of the toxic metabolites as mercapturate conjugates that are excreted in the urine.

Other factors include the effect of drug formulation (sustained vs. rapid release; increased exposure) on pharmacokinetics, for example, elimination half-life of the drug or adulterants (e.g., enzyme inducers) in dietary supplements.

Drug-induced hepatotoxicity can be categorized as an intrinsic (predictable) or idiosyncratic (unpredictable) drug reaction (89). Most drugs involved in hepatotoxicity belong to the idiosyncratic group. Intrinsic hepatotoxins produce liver injury in a dose-related manner when the toxic amount of drug is ingested without bioactivation, such as these toxins found in the *Amanita* mushroom. Fortunately, few drugs are intrinsic hepatotoxins. Idiosyncratic hepatotoxicity is the result of the toxic effects of a drug's metabolites.

The common trigger for both mild and severe forms of hepatotoxicity is bioactivation of relatively inert functional groups to reactive electrophilic intermediates, which is considered to be an obligatory event in the etiology of many drug-induced idiosyncratic hepatotoxicities (91,92). A great deal of evidence now shows that reactive metabolites are formed from drugs known to cause idiosyncratic hepatotoxicity, but how these toxic species initiate and propagate tissue damage remains poorly understood. However, the relationship between bioactivation and the occurrence of hepatic injury is not simple. For example, many drugs at therapeutic doses undergo bioactivation in the liver but are not hepatotoxic. The tight coupling of bioactivation with bioinactivation pathways can be one reason for the lack of hepatotoxicity with these drugs. Examples of bioinactivation (detoxification) pathways include glutathione conjugation of quinones by GSTs and hydration of arene oxides to dihydrodiols by epoxide hydrolases. When reactive metabolites are poor substrates for

such detoxifying enzymes, they can escape bioinactivation and, thereby, damage proteins and nucleic acids, prompting hepatotoxicity.

Most drugs, however, are not directly chemically reactive but, through the normal process of drug metabolism, can form electrophilic, chemically reactive metabolites (90–92). Formation of chemically reactive metabolites is mainly catalyzed by P450 enzymes, but products of Phase 2 metabolism (e.g., acylglucuronides, acyl CoA thioesters, or *N*-sulfates) can also lead to toxicity. However, if P450 drug bioactivation is closely coupled with Phase 2 bioinactivation (e.g., glutathione conjugation to mercapturates), then the net chemical process is one of detoxification if the final product is rapidly cleared. Toxicity can accrue when accumulation occurs of a chemically reactive metabolite that, if not detoxified, can lead to covalent modification of biologic macromolecules. The identity of the target macromolecule and the functional consequence of its modification will dictate the resulting toxicologic response. P450 enzymes are present in many organs, mainly the liver but also the kidney and lung, and thus can bioactivate chemicals to cause organ-specific toxicity. Evidence for the formation of reactive metabolites was found for 5 of the 6 drugs that have been withdrawn from the market since 1995 and for 8 of the 15 drugs that have black box warnings. Evidence for reactive metabolite formation has been found for acetaminophen, bromfenac, diclofenac, clozapine, and troglitazone (91–93). Acetaminophen is the most studied hepatotoxin. The current hypothesis of how reactive metabolites lead to liver injury suggests that hepatic (target) proteins can be modified by reactive metabolites. Much more important can be the identification of the target proteins modified by these toxic metabolites and how this reaction alters the function of the target proteins. Additionally, it is important to note that the toxicity of reactive metabolites can also be mediated by noncovalent binding mechanisms, which can have profound effects on normal liver physiology. Technologic developments in the wake of the genomic revolution now provide unprecedented power to characterize and quantify covalent modification of individual target proteins and their functional consequences (93). Such information should dramatically improve our understanding of drug-induced hepatotoxic reactions. Moreover, covalent binding per se does not necessarily lead to drug hepatotoxicity. The regioisomer of acetaminophen, 3-hydroxyacetanilide, becomes covalently bound to hepatic proteins in rodents without inducing hepatotoxicity (94).

Therefore, it is necessary to identify targets for these reactive metabolites (i.e., covalently modified macromolecules) that are critical to the toxicologic process. Hard electrophiles react as a rule with hard nucleophiles, such as the basic groups in DNA and lysine ω -amino residues in proteins. Soft electrophiles react with soft nucleophiles, which include cysteine residues in proteins and in

HARD/SOFT ACIDS/BASES

The nonenzymatic reaction of an electrophilic metabolite with a nucleophilic molecule occurs usually via a substitution or addition mechanism involving the donation of an electron pair by the nucleophile to an acceptor molecule, an electrophile, with subsequent formation of a covalent bond and an adduct product (Coles B. Effects of modifying structure on electrophilic reactions with biological nucleophiles. *Drug Metab Rev* 1984–1985;15:1307–1334). The most accepted concept classifies electrophiles and nucleophiles according to Pearson's "hard-soft acid-bases" (HSAB) model. Thus, hard electrophiles have a formal positive charge at the electrophilic center, and the valence electrons are not easily delocalized or polarized (e.g., acylium ions, carbocations, nitrenium ions; see Figs. 4.30 and 4.33), whereas soft electrophiles have a partial positive charge density and valence electrons that are delocalized (polarized), such as activated double bonds of α,β -unsaturated carbonyl compounds as shown in Fig. 4.32. Hard nucleophiles have high electronegativity (oxygen and nitrogen groups) and low polarization of valence electrons, whereas soft nucleophiles have low electronegativity and are more polarizable. The softest biologic nucleophilic sites are cysteine thiol groups on proteins and GSH. The primary and secondary amino groups of lysine and histidine or the hydroxyl groups of serine or threonine on proteins are hard nucleophiles, whereas the hardest nucleophiles are the oxygen atoms of purines and pyrimidines on DNA and RNA.

Based on the HSAB theory, the reaction rates and selectivities of electrophiles and nucleophiles are dependent upon comparable states of "hardness." Specifically a soft electrophile such as quinoneimine will react predominantly with a soft nucleophile such as the thiol group of cysteine. A hard electrophile such as the acylium ion formed from acyl glucuronide will react with hard nucleophiles such as the hydroxyl group of serine. This preferential nonenzymatic reactivity is due primarily to the high-energy transition state that acts as a barrier to the reaction of a hard electrophile with a soft nucleophile such as the delocalized double bonds of *p*-benzoquinoneimines, *p*-benzoquinonemethides,

and other α,β -unsaturated carbonyl intermediates that react by Michael-type addition of the nucleophile to the polarized (partial positive charge) located on the β -carbon.

Adduct formation is dependent not only on the physiochemical nature of the electrophilic but also on the microenvironment of the nucleophilic center, which can vary significantly even among centers of the same elemental type (e.g., sulfur or amino groups). Thus, nucleophilic reactivity among free sulfhydryl groups on proteins can be diverse, and consequently, soft electrophilic metabolites will form adducts with the more reactive thiol groups on a given protein or free thiols on glutathione. This diversity in nucleophilic reactivity is a function of both steric and electronic factors mediated primarily by the tertiary structure of a protein. For example, adjacent acidic and basic amino acids on a protein significantly influence the reactivity of the target nucleophilic group. Depending on the physiochemical nature of the electrophile, the resulting electrophilic metabolite can produce toxicity by reacting with either soft thiol nucleophilic sites on proteins and free thiols such as GSH or harder nucleophilic centers on DNA and RNA to produce adducts.

For example, metabolic epoxidation of an aromatic ring produces an epoxide, a relatively hard electrophilic metabolite that will form a ring-opened hydroxyl adduct primarily with hard nucleophilic centers on guanine and adenine on DNA and not with soft thiol nucleophiles. On the other hand, a soft electrophile, such as the *N*-acetyl *p*-benzoquinone imine metabolite of acetaminophen, will form adducts with thiol groups on proteins and free thiols on GSH, but not with hard nucleophilic groups on lysine or serine on proteins or with those on guanine and adenine on DNA. Glutathione offers little protection against carcinogens, most of which are hard nucleophiles. These examples show that the bioactivated metabolite can exhibit distinct electrophilic characteristics and different nucleophilic target molecules. Thus, soft electrophiles are associated with organ-specific toxicities (e.g., hepatotoxicity, renal toxicity), and hard electrophiles are associated with carcinogenicity.

glutathione. Unfortunately, no simple rules predict the target macromolecules for a particular chemically reactive metabolite or the biologic consequences of a particular modification. Furthermore, noncovalent interactions have also a role, because covalent binding of hepatotoxins is not indiscriminate with respect to proteins. Even within a single protein, there can be selective modification of an amino acid side chain found repeatedly in the primary structure. Thus, the microenvironment (e.g., pK_a and hydrophobicity) of the amino acid in the tertiary structure appears to be the crucial determinant of selective binding and, therefore, the impact of covalent binding on protein function. In turn, the extent of binding and the biochemical role of the protein will determine the toxicologic insult of drug bioactivation. The resulting pathologic consequences will be a balance between the

rates of protein damage and the rates of protein replacement and cellular repair.

Drug-Induced Idiosyncratic Reactions

Idiosyncratic drug reactions (IDR; type B adverse drug reactions) occur in from 1 in 1,000 to 1 in 50,000 patients, are not predictable from the known pharmacology or toxicology of the drug, are not produced experimentally in vitro and in vivo, and are dose independent. The occurrence of IDRs during late clinical trials or after a drug has been released can lead to severe restriction of its use and even its withdrawal. The IDRs do not, as a rule, result from the drug itself because most people can tolerate the drug, but rather from a unique set of patient characteristics, including gender, age, genetic predisposition, and a lack of drug-metabolizing enzymes,

that can increase the risk of these adverse drug reactions. Most IDRs are caused by hypersensitivity reactions and can result in hepatocellular injury. The hepatic injury occurs within 1 week to 12 months after initiation of drug therapy and often is accompanied by systemic characteristics of allergic drug reactions, such as rash and fever. Signs of hepatic injury reappear with subsequent administration of the same drug with only one or two doses. Hypersensitivity reactions can be severe and associated with fatal reactions as a multiorgan clinical syndrome characterized by the following: 1) fever; 2) rash; 3) gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea, or abdominal pain); 4) generalized malaise, fatigue, or achiness; and 5) respiratory symptoms (e.g., dyspnea, cough, or pharyngitis). Examples of drugs causing IDRs through a hypersensitivity mechanism include penicillin, methyldopa, chlorpromazine, erythromycin, azathioprine toxicity in thiopurine methyltransferase-deficient individuals, sulfonamide and acetaminophen hepatotoxicity in alcoholics and in UGT1A6-deficient cats, ivermectin neurotoxicity in Collie dogs deficient in P-gp, perhexilene hepatotoxicity in CYP2D6-deficient individuals, phenytoin toxicity in CYP2C9-deficient individuals, and valproic acid hepatotoxicity.

The clinical features of some cases of drug-induced idiosyncratic hepatotoxicity strongly suggest an involvement of the immune system (95). These clinical characteristics include the following: 1) concurrence of rash, fever, and eosinophilia; 2) delay of the initial reaction (1 to 8 weeks) or requirement of repeated exposure to the culprit drug; 3) rapid recurrence of toxicity on reexposure to the drug; and 4) presence of antibodies specific for native or drug-modified hepatic proteins. Our current understanding of drug-induced adaptive immune responses is largely based on the hapten hypothesis.

Idiosyncratic drug reactions that are connected with hepatotoxicity involve the formation of reactive metabolites (91,92). Such reactions are not predictable, but current bioanalytical technology has enabled the *in vivo* identification of the formation of reactive metabolites, as evidenced by the detection of biomarkers (i.e., mercapturate or cysteine adducts) in urine, drug-specific antibodies, or antibodies to P450 isoforms (93). As a result, some drugs known to cause hepatic injury continue to be used, because the drug's benefit outweighs its risk and no alternative efficacious drug exists. For example, isoniazid, a drug commonly used to treat tuberculosis, is implicated in approximately 15% to 20% of the individuals who show increased serum transaminases after receiving the drug as a single agent for tuberculosis prophylaxis. Of these individuals, an estimated 1 in 1,000 patients can develop severe hepatic necrosis. Additionally, NSAIDs, including cyclooxygenase-2 inhibitors, commonly are associated with idiosyncratic liver injury. Most of the idiosyncratic toxins listed in Table 4.21 that have been studied to date produce reactive metabolites.

Current hypotheses regarding IDRs suggest that metabolic activation of a drug to a reactive metabolite

is a necessary yet insufficient step in the generation of an idiosyncratic reaction (91,92). Evidence for this hypothesis comes from drugs that are associated with hepatotoxicity (Table 4.21) and the detection of drug metabolite-specific antibodies in affected patients.

For the other drugs that have been associated with idiosyncratic hepatotoxicity but that do not have black box warnings, either evidence for hepatotoxicity was not available or suitable studies had not been carried out. High doses increase the risk for an IDR (e.g., clozapine at 300 mg/d vs. olanzapine at 20 mg/d). Strong evidence also exists that T cells have an important role in immune-mediated IDR and can trigger cell death. In patients exhibiting IDR, pretreatment with immunosuppressants prevented the IDR (rash). The incidence of IDRs also appears to be lower in patients with low T-cell counts.

The hapten hypothesis proposes that the reactive metabolites of the drugs act as haptens and bind covalently to endogenous proteins to form immunogenic drug-protein adducts triggering either antibody or cytotoxic T-cell responses (92,96). The hapten hypothesis is supported by the detection of antibodies that recognize drug-modified hepatic proteins in the sera of drug-induced liver injury patients. For example, antibodies that recognize trifluoroacetate-altered hepatic proteins have been detected in the sera of patients with halothane-induced hepatitis. Such drug-specific antibodies or autoantibodies that recognize native liver proteins

TABLE 4.21 Some Examples of Idiosyncratic Toxins

Abacavir	Hypersensitivity
Acetaminophen	Hepatotoxicity
Amiodarone	Hepatotoxicity
Aromatic anticonvulsants	Hypersensitivity
Cefaclor	Hepatotoxicity
Clozapine	Agranulocytosis
Diclofenac	Hepatotoxicity
Felbamate	Aplastic anemia
Fibrates	Hepatotoxicity
Halothane	Hepatotoxicity
Indomethacin	Hepatotoxicity
Isoniazid	Hepatotoxicity
Levamisole	Hepatotoxicity
Nefazodone	Hepatotoxicity
Nevirapine	Agranulocytosis
Oral contraceptives	Hepatotoxicity
Paroxetine	Hepatotoxicity
Phenytoin	Hepatotoxicity
Statins	Hepatotoxicity
Sulfonamides	Stevens-Johnson syndrome
Tamoxifen	Hepatotoxicity
Tacrine	Hepatotoxicity
Tienilic acid	Hypersensitivity
Ticlopidine	Agranulocytosis
Troglitazone	Hepatotoxicity
Valproic acid	Hepatotoxicity
Vesnarinone	Agranulocytosis
Penicillinamine ^a	Hypersensitivity

^aDoes not produce reactive metabolites.

have also been found in patients with liver injury caused by other drugs, such as diclofenac. In patients who developed IDRs of the liver and other organs, drug-specific T cells have been detected, and in some cases, T-cell clones were generated. Most drugs are small molecules and are unlikely to form haptens. Electrophilic acylators (hard electrophiles) can react with the lysine ω -amino residues (hard nucleophiles) of the target protein or guanosine residues of DNA. How much chemical modification is required to trigger an IDR remains unknown. Halothane is the most studied molecule for supporting this hypothesis regarding IDRs (91). Therefore, it is not surprising that irreversible chemical modification of a protein, which has a profound effect on function, is a mechanism of idiosyncratic hepatotoxicity. However, it is important to note that a number of drugs (e.g., penicillins, aspirin, and omeprazole) rely on covalent binding to proteins for their efficacy; thus, prevention of their covalent binding through chemical modification of the compound can also, inadvertently, lead to loss of efficacy.

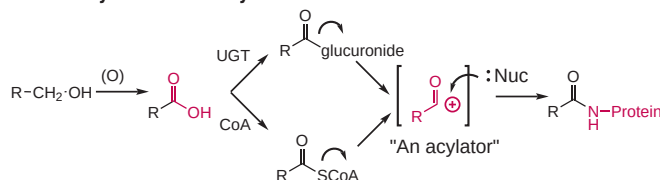
Drug-induced stress and/or damage of hepatocytes can trigger activation and inflammatory responses of the immune system within the liver (95,96). Evidence to support this idea has been obtained mainly from studies of liver injury induced by overdoses of acetaminophen, which is one of the few drugs that provide an experimental animal model of drug-induced liver injury. Evidence is growing that the initial benzoquinoneimine-induced hepatocyte damage can lead to activation of immune cells within the liver, thereby stimulating hepatic infiltration by inflammatory cells. Activated T cells of the immune system produce a range of inflammatory mediators, including cytokines, chemokines, and reactive oxygen and nitrogen species, that contribute to the progression of liver injury. On the other hand, the immune cells also represent a major source of hepatoprotective factors.

Reactive Metabolites Resulting from Bioactivation

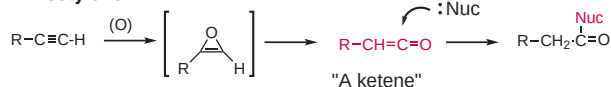
Electrophiles

The concept that small organic molecules can undergo bioactivation to electrophiles and free radicals and elicit toxicity by chemical modification of cellular macromolecules has its basis in chemical carcinogenicity and the pioneering work of Miller and Miller (97) and Gillette et al. (98–100). A number of different types of reactive metabolites exist; however, they can be broadly classified as either electrophiles (Fig. 4.30) or free radicals (Fig. 4.31) (93,101). These reactive metabolites are short-lived, with half-lives of usually less than 1 minute, and are not normally detectable in plasma or urine except as Phase 2 conjugates or other biomarkers. Electrophiles are reactive because they possess electron-deficient centers (polarization-activated double bonds or positive-charge acylators) (Figs. 4.30 and 4.32) and can form covalent bonds with electron-rich biologic

1. Carboxylic acids as acylators:

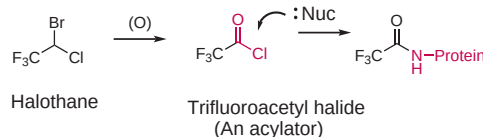


2. Acetylene:

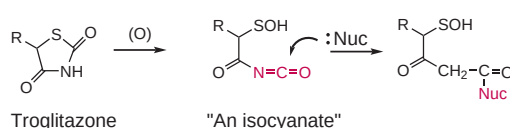


Example: Ethinylestradiol

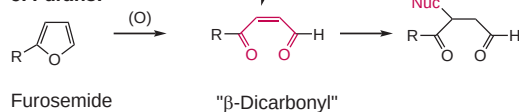
3. Halogenated hydrocarbon:



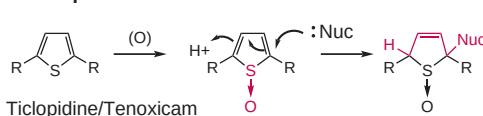
4. Thiazolidinedione:



5. Furans:



6. Thiophene:



7. Aromatics and olefins:

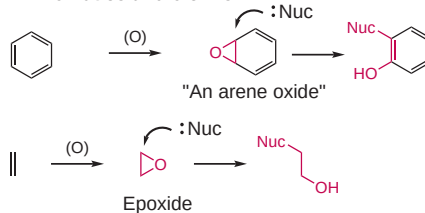
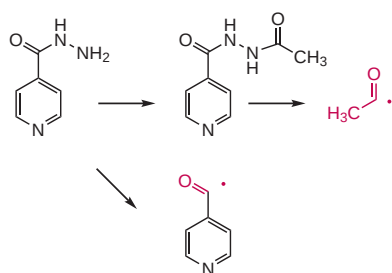


FIGURE 4.30 Some examples of electrophilic intermediates resulting from bioactivation. Nuc, nucleophiles.

nucleophiles. They are either soft electrophiles that react directly with soft nucleophiles (:Nuc), such as the thiol groups in either glutathione or cysteine residues within proteins, or hard electrophiles that react with hard nucleophiles, such as basic groups in DNA and lysine ω -amino residues in proteins, or are mediated by bioactivation enzymes, such as glutathione transferase

1. Isoniazid



2. Hydrazines



3. Sulfhydryl



Captopril

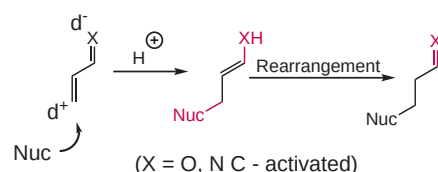
FIGURE 4.31 Drug bioactivation to free radicals.

or epoxide hydase. Softness and hardness are associated with the polarizability of the electrophilic/nucleophilic species (see Hard/Soft Acids/Bases). Activated double bonds are soft electrophilic intermediates as shown in Figure 4.32.

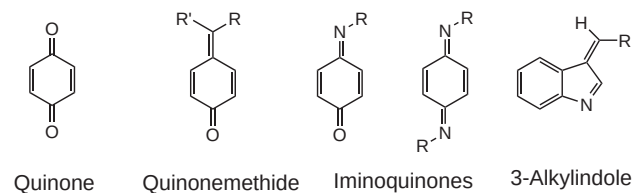
Examples of activated double bond electrophiles include α,β -unsaturated carbonyl compounds, quinones, quinoneimines, quinonemethides, and diiminoquinones, as shown in Figure 4.32B. These electrophilic intermediates are highly polarized and can react with nucleophiles in a 1,4-Michael-type addition at the more electrophilic or β -carbon of the activated double bond intermediate to the addition product (Fig. 4.32A). Specific examples of activated double bond electrophiles that have been proposed for the anticancer drug leflunamide, the food antioxidant butylated hydroxytoluene, acetaminophen, the antiandrogen flutamide, the anticonvulsant felbamate, and the cytotoxic cyclophosphamide are shown in Figure 4.32C. The bioinactivation pathways for these electrophilic intermediates can involve either direct addition, with or without transferases, depending on the degree of polarization and reactivity of the electrophilic intermediate (hard vs. soft electrophiles).

Other commonly found electrophilic intermediates for drug molecules in Figure 4.30 include the formation of ketenes from the bioactivation of acetylenic groups (e.g., ethinylestradiol) (Fig. 4.30-2); isocyanates from thiazolidinediones (e.g., the “glitazones”) (Fig. 4.30-4) (93); acylium ions from halogenated hydrocarbons (e.g., halothane) (Fig. 4.30-3) (93) and carboxylic acids; β -dicarbonyl from furans (e.g., furosemide) (Fig. 4.30-5) (93); activated thiophene-S-oxide from thiophenes, such as ticlopidine; tenoxicam, which cause the IDR of agranulocytosis (Fig. 4.30-6) (93); and epoxides and arene oxides from olefins and aromatic compounds

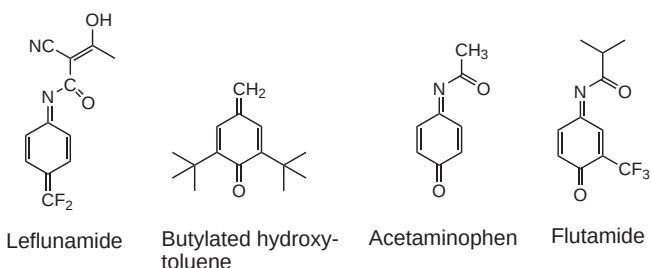
A. 1,4-Addition of a nucleophile in a Michael-type reaction:



B. Examples of activated double bonds:



C. Reactive intermediates from drugs that form activated double bonds:



R = phenyl - felbamate
R = H - acrolein from cyclophosphamide

FIGURE 4.32 Some examples of activated double bonds as electrophilic reactive intermediates.

(Fig. 4.30-7) (93). Drugs possessing structural features prone to metabolic epoxidations are abundant. Therefore the incidence of epoxide metabolites in mediating adverse biologic effects has aroused concern about clinically used drugs known to be metabolized to epoxides. Metabolically produced epoxides have been reported for allobarbitol, secobarbitol, protriptyline, carbamazepine, and cyproheptadine and are implicated with 8-methoxypsoralen and other furanocoumarins (6,7-dihydroxybergamottin in grapefruit juice), phenytoin, phenisuximide, phenobarbital, mephobarbital, lorazepam, and imipramine (81). The alarming biologic effects of some epoxides, however, do not imply that all epoxides have similar effects. Epoxides vary greatly in molecular geometry, stability, electrophilic reactivity, and relative activity as substrates for epoxide-transforming enzymes (e.g., epoxide hydrolase, glutathione S-transferase, and others).

Some carboxylic acid-containing drugs have been implicated in rare IDRs, which was the basis for the market withdrawal of the NSAIDs zomiperac and benzoxaprofen. These drugs (e.g., NSAIDs, fibrates, “statins,” and valproic acid) can be bioactivated to acyl glucuronides or acyl CoA thioesters (Fig. 4.30-1). These products are electrophilic acylators that can acylate target proteins if they escape inactivation by *S*-glutathione-thioester formation. A crucial factor is the concentration of acyl glucuronides in hepatocytes due to their transport by conjugate export pumps, where acylglucuronides can selectively acylate canalicular membrane proteins. Acyl CoA esters can be either rapidly hydrolyzed or further metabolized in hepatocytes. Evidence is accumulating that acyl glucuronides can alter cellular function by haptenation of peptides, target protein acylation or glycation, or direct stimulation of neutrophils and macrophages. The role of acyl CoA reactive metabolites is less clear. It should be noted that some noncarboxylic acid drugs can be biotransformed by oxidative metabolism in the liver to the respective carboxylic acids.

Free Radicals

P450 activates molecular dioxygen to generate ROS, such as singlet oxygen ($^1\text{O}_2$) or superoxide (see Oxygen Activation, p. 118). Reactive metabolites that possess unpaired electrons are free radicals (molecular species that contain an odd unpaired electron), which can react with molecular oxygen (ground state triplet) to generate intracellular oxidative stress damage (93,98–100). Free radicals abstract a hydrogen atom from other molecules rather than becoming covalently bound. Free radical reactions can be self-propagating by abstracting a hydrogen atom from the double bond of a lipid that initiates a chain reaction leading to lipid peroxidation, oxidative stress, or other types of modification of biologic molecules.

Some examples of free radicals generated by the bioactivation of drug molecules are shown in Figure 4.31. Isoniazid is acetylated to its major metabolite acetylisoniazid, which is hydrolyzed to acetylhydrazine and isonicotinic acid (Fig. 4.31-1). Acetylhydrazine is further metabolized by the CYP2E1 to an *N*-hydroxy intermediate that hydrates into an acetyl radical, which can then initiate the process that leads to hepatic necrosis. Other carbon-centered radicals are formed from hydrazines such as the antihypertensive hydralazine and thio-radicals from the angiotensin-converting enzyme inhibitor captopril (Fig. 4.31-2 and -3).

Bioinactivation Mechanisms

Several enzyme systems exist as cellular defense (detoxification) pathways against the chemically reactive metabolites generated by P450 metabolism (91,92,102,103). These include GST, epoxide hydrolase, and quinone reductase, as well as catalase, glutathione peroxidase, and superoxide dismutase, which detoxify the peroxide

and superoxide byproducts of metabolism. The efficiency of the bioinactivation process is dependent on the inherent chemical reactivity of the electrophilic intermediate, its affinity and selectivity of the reactive metabolite for the bioinactivation enzymes, the tissue expression of these enzymes, and the rapid upregulation of these enzymes and cofactors mediated by the cellular sensors of chemical stress. The reactive metabolites that can evade these defense systems can damage target proteins and nucleic acids by either oxidation or covalent modification.

The most abundant agents of cellular defense are thiols. Glutathione is a soft electrophile and, therefore, will only react noncatalytically with soft electrophiles, such as activated double bonds (Fig. 4.32). Glutathione conjugation to mercapturates is one of the most important defenses against hepatocellular injury. Glutathione protects cellular enzymes and membranes from toxic metabolites, and its inadequate stores can compromise efficient detoxification of the reactive metabolites. The subsequent inability to detoxify the reactive metabolites can result in hepatocellular injury. The rate-limiting factor for glutathione synthesis is the intracellular concentration of cysteine. *N*-acetylcysteine is often used as an alternative to glutathione to trap the iminoquinone intermediate in the treatment of acute acetaminophen toxicity. Glutathione has a protective role in the hepatic tissue injury produced by acetaminophen but not by furosemide.

The relationship between bioactivation, bioinactivation, and DNA adduct formation has been well established for a number of hepatocarcinogens. Glutathione conjugation of hard electrophiles becomes more efficient when catalyzed by GSTs, an important example being the detoxication of the hepatocarcinogen aflatoxin. Aflatoxin, a hepatocarcinogen and a hepatotoxin found in mold growing on peanuts, is converted into aflatoxin B1 epoxide in rodents, which is more readily detoxified by GST enzymes than by epoxide hydrolase. The balance between these transferase reactions explains the greater DNA damage in humans compared with rodents, because human forms of GST are less able to catalyze the conjugation of aflatoxin epoxide compared with the rodent forms. Transgenic knockout mice have been used to establish the role of bioactivation by P450 and bioinactivation by GSTs for a number of carcinogenic PAHs.

Substances that detoxify free radicals include the antioxidants vitamin C, vitamin E, and carotene, which scavenge free radicals, including reactive metabolites and ROS generated as a consequence of chemical stress.

Specific Examples

Some examples of bioactivation to hepatotoxic or IDR electrophilic intermediates are shown in Figure 4.33. Bioactivation can occur by both oxidation and

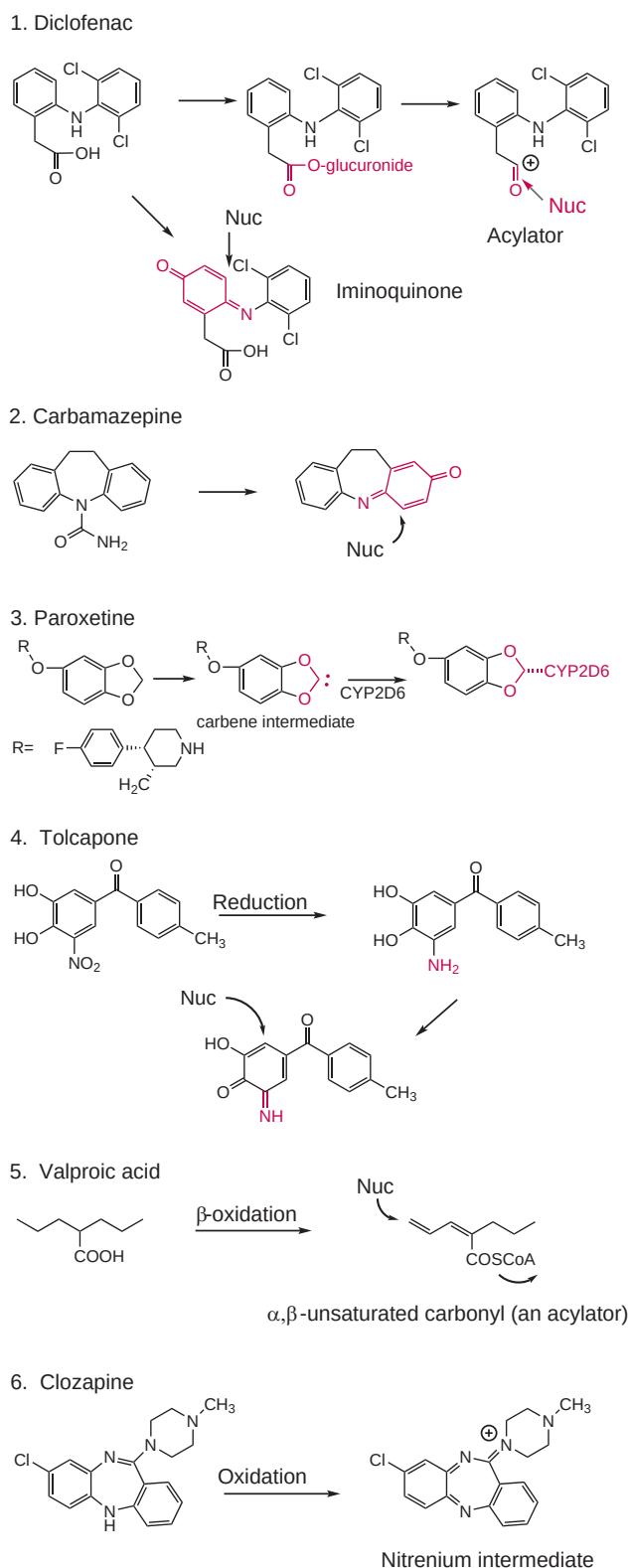


FIGURE 4.33 Some examples of drug bioactivation to their hepatotoxic intermediates.

conjugation reactions, such as those with diclofenac, which undergoes the formation of an acyl glucuronide and/or acyl CoA (acylator intermediates), or to produce

iminoquinones via formation of a phenol intermediate (Fig. 4.33-1) (92). The anticonvulsant carbamazepine is 2-hydroxylated, and the elimination of the amide group yields the reactive quinoneimine intermediate (Fig. 4.33-2); and the antidepressant paroxetine and other xenobiotics with the common methylenedioxyphenyl nucleus undergo methylene oxidation to a carbene intermediate (Fig. 4.33-3). In the COMT inhibitor used in the treatment of parkinsonism, tolcapone, the nitro group is first reduced to an amine and then oxidized to a α -quinoneimine (Fig. 4.33-4). The mitochondrial/hepatotoxicity of the anticonvulsant valproic acid results from the formation of an activated α,β -unsaturated CoA thioester via mitochondrial β -oxidation, most commonly associated with the oxidation of fatty acids (Fig. 4.33-5). The agranulocytosis resulting from the ingestion of the antipsychotic clozapine is bioactivated by its oxidation by hypochlorous acid in neutrophils to a nitrenium intermediate (Fig. 4.33-6) (93). The effect of structure modification for troglitazone that reduced its hepatotoxicity is shown in Figure 4.34. The p -dihydroxy elements of the chroman ring nucleus (outlined in red in Fig. 4.34) of troglitazone is bioactivated to an activated double bond (p -quinone) and has been replaced with a pyridine ring that is not bioactivated, although the thiazolidone ring can be bioactivated to an isocyanate (Fig. 4.30) (92).

The oxidation of acetaminophen to the chemically reactive N -acetyl- p -benzoquinoneimine (Fig. 4.32) is catalyzed by the isoforms CYP1A2 and CYP2E1, which can either react covalently with glutathione to form an inactive product or with cellular macromolecules, initiating the processes leading to hepatic necrosis (Fig. 4.28) (93). The usual route for acetaminophen metabolism is glucuronidation. If insufficient UDP-glucuronic acid is present, then bioactivation will dominate.

Furosemide, a frequently used diuretic drug, is reportedly a human hepatocarcinogen. The hepatic toxicity apparently results from metabolic activation of the furan ring to a β -dicarbonyl intermediate (Fig. 4.30-5) (93). Ticlopidine and tenoxicam, reported to cause agranulocytosis, do so via metabolic activation of the thiophene ring to an S -oxide (Fig. 4.30-6) (93). The agranulocytosis resulting from the ingestion of clozapine is via its bioactivation to a nitrenium ion intermediate (Fig. 4.33-6) (96–101).

Drug-Induced Chemical Carcinogenesis

The mechanism whereby xenobiotics are transformed into chemical carcinogens is usually accepted as bioactivation to reactive metabolites, which are responsible for initiating carcinogenicity (98–100). Many carcinogens elicit their cytotoxicity through a covalent linkage to DNA. This process can lead to mutations and, potentially, to cancer. Most chemical carcinogens of concern are chemically inert but require activation by the xenobiotic-metabolizing enzymes before they can undergo

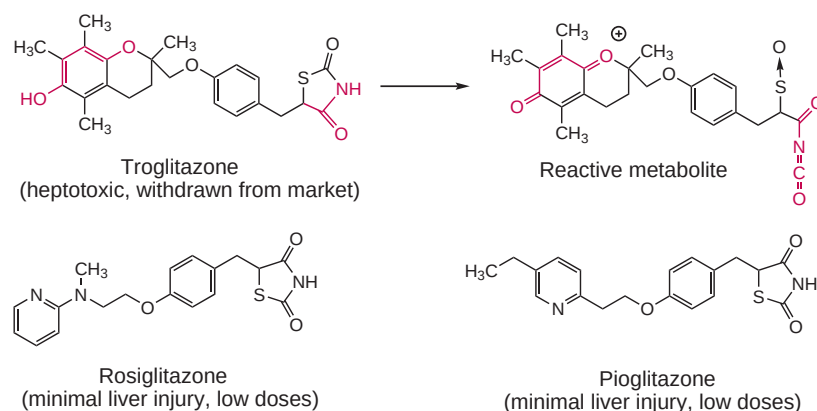


FIGURE 4.34 The effect of structure modification on the drug-induced hepatotoxicity of troglitazone.

reaction with DNA or proteins (cytotoxicity). There are many ways to bioactivate procarcinogens, promutagens, plant toxins, drugs, and other xenobiotics (37) (Fig. 4.35). Oxidative bioactivation reactions are by far the most studied and common. Conjugation reactions (Phase 2), however, are also capable of activating these xenobiotics to produce electrophiles, in which the conjugating derivative acts as a leaving group. These reactive metabolites are mostly electrophiles, such as epoxides, quinones, or free radicals formed by the P450 enzymes or FMO. The reactive metabolites tend to be oxygenated in sterically hindered positions, making them poor substrates for subsequent bioinactivation transferases, such as epoxide hydrolase and GST. Therefore, their principal fate is formation of covalent linkage to intracellular macromolecules, including enzyme proteins and DNA. Experimental studies indicate that the CYP1A subfamily can oxygenate aromatic hydrocarbons (e.g., PAHs) in sterically hindered positions to arene oxides. Activation by *N*-hydroxylation of polycyclic aromatic amines (e.g., aryl *N*-acetamides) appears to depend on either FMO

or P450 isoforms. The formation of chemically reactive metabolites is important, because they frequently cause a number of different toxicities, including tumorigenesis, mutagenesis, tissue necrosis, and hypersensitivity reactions.

Our understanding of these reactions was advanced by the studies of Miller and Miller (97) and Gillette et al. (98–100). They proposed that the proportion of the dose that binds covalently to critical macromolecules could depend on the quantity of the reactive metabolite that is formed.

A scheme illustrating the complexities of drug-induced chemical carcinogenesis is shown in Figure 4.35. Reactions that proceed via the open arrows eventually lead to neoplasia. Some carcinogens can form the “ultimate carcinogen” directly through P450 isoform bioactivation; others, like the PAHs (e.g., benzo[*a*]pyrene), appear to involve a multistep reaction sequence forming an epoxide, reduction to a diol by epoxide hydrolase, and perhaps, the formation of a second epoxide group on another part of the molecule. Other procarcinogens form the *N*-hydroxy intermediate that requires transferase-catalyzed conjugation (e.g., *O*-glucuronide and *O*-sulfate) to form the “ultimate carcinogen.” The quantity of “ultimate carcinogen” formed should relate directly to the proportion of the dose that binds or alkylates DNA.

The solid-arrow reaction sequences in Figure 4.35 are intended to show detoxification mechanisms, which involve several steps. First, the original chemical can form less active products (phenols, diols, mercapturic acids, and other conjugates). Second, the “ultimate carcinogen” can rearrange so as to be prevented from its reaction with DNA (or whatever the critical macromolecule is). Third, the covalently bound DNA can be repaired. Fourth, immunologic removal of the tumor cells can occur. Several mechanisms within this scheme could regulate the quantity of covalently bound carcinogen: 1) The activity of the rate-limiting enzyme, such as epoxide hydrolase, P450 isoform, or one of

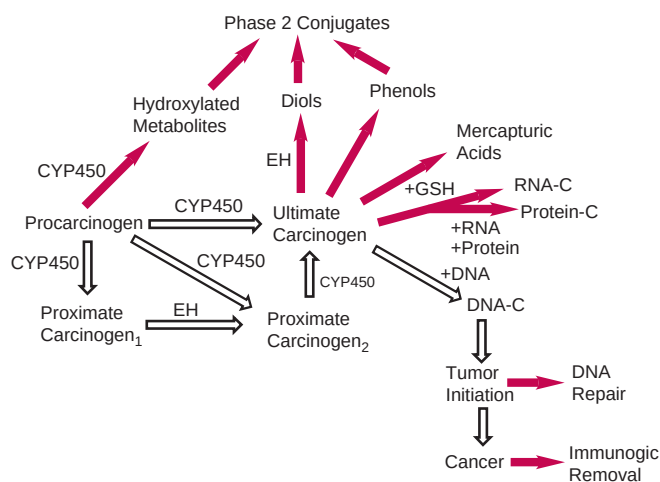


FIGURE 4.35 The bioactivation of procarcinogens and a proposed mechanism of chemical carcinogenesis.

the transferases, could be involved; 2) the availability of cosubstrates, such as glutathione, UDP-glucuronic acid, or PAPS, can be rate-limiting; 3) relative P450 activities for detoxification and activation must be considered; 4) availability of alternate reaction sites for the ultimate carcinogen (e.g., RNA and protein can be involved); 5) and possible specific transport mechanisms that deliver either the procarcinogen or its ultimate carcinogen to selected molecular or subcellular sites.

It is now well established that numerous organic compounds that are essentially nontoxic as long as their structure is preserved can be converted into cytotoxic, teratogenic, mutagenic, or carcinogenic compounds by normal biotransformation pathways in both animals and humans (37). The reactive electrophilic intermediate (hard acids) involves its reaction with cellular constituents (hard bases) forming either detoxified products or binding covalently with essential macromolecules, initiating processes that eventually lead to the toxic effect. A better understanding of the mechanisms underlying these reactions can lead to more rational approaches to the development of nontoxic therapeutic drugs. For the present, it seems that new advances in drug therapy cannot occur without some risk of causing structural tissue lesions. Special attention to risk factors is required for drugs that will be used for long periods in the same patient.

Some toxic chemicals exert their toxic action by lethal injury or biologic auto-oxidation (radical lipid peroxidation). Lethal injury involves the disruption of cellular energy metabolism by inhibition of oxidative phosphorylation or adenosine triphosphatase, resulting in disruption of subcellular organelles, cell death, and tissue necrosis. Because the early stages of lethal injury are reversible, complete recovery can occur. Auto-oxidation is the process whereby cellular components are irreversibly oxidized and damaged by free radicals or free radical-generating systems. This results in the oxidation and depletion of glutathione, various thiol enzymes, or lipid peroxidation, which in turn leads to the disruption of cellular membranes and to cell death, tissue necrosis, and death of the organism. When cell death does not occur, nonlethal changes, such as mutations and malignant transformations, are likely.

DRUG-DRUG INTERACTIONS

Drug-drug interactions represent a common clinical problem, which has been compounded by the introduction of many new drugs and the expanded use of herbal medicines. Between 1999 and 2005, approximately 100 drug-drug interactions were reported, of which approximately 50% involved P450 inhibition. Drug-drug interactions occur when the efficacy or toxicity of a medication is changed by coadministration

of another substance, drug, food (e.g., grapefruit), or herbal product (103,104). Pharmacokinetic interactions often occur as a result of a change in drug metabolism. For example, CYP3A4 oxidizes more than 60% of the clinically used drugs with a broad spectrum of structural features, and its location in the small intestine and liver permits an effect on both presystemic and systemic drug disposition. Some drug-drug interactions with CYP3A4 substrates/inhibitors can also involve inhibition of P-gp. Other clinically important drug-drug interactions resulting from coadministration of CYP3A4 substrates or inhibitors include rhabdomyolysis with the coadministration of some 3-hydroxy-3-methylglutaryl-CoA reductase inhibitors ("statins"), symptomatic hypotension with some dihydropyridine calcium antagonists, excessive sedation from benzodiazepine or nonbenzodiazepine hypnotics, ataxia with carbamazepine, and ergotism with ergot alkaloids.

The clinical importance of any drug-drug interaction depends on factors that are drug-, patient-, and administration-related. Drugs with low oral bioavailability or high first-pass metabolism are particularly susceptible to drug-drug interactions as a result of coadministration of inhibitors that alter absorption, distribution, and elimination. In general, a doubling or more in the plasma drug concentration has the potential for enhanced adverse or beneficial drug response. Less pronounced pharmacokinetic drug interactions can still be clinically important for drugs with a steep concentration-response relationship or narrow therapeutic index. In most cases, the extent of drug interaction varies markedly among individuals; this is likely to be dependent on interindividual differences in P450 content (polymorphism), preexisting medical conditions, and possibly, age. Drug-drug interactions can occur under single-dose conditions or only at steady-state. The pharmacodynamic consequences can or cannot closely follow pharmacokinetic changes. Drug-drug interactions can be most apparent when patients are stabilized on the affected drug and the P450 substrates or inhibitors are then added to the regimen (103). One reason for the increased incidence of drug-drug interactions is the practice of simultaneously prescribing several potent drugs as well as concurrently ingesting nonprescription products and herbal products.

Although drug-drug interactions constitute only a small proportion of adverse drug reactions, they have become an important issue in health care. Many of the drug-drug interactions can be explained by alterations in the metabolic enzymes in the liver and other extrahepatic tissues, and many of the major pharmacokinetic interactions between drugs are caused by hepatic P450 isoenzymes being affected by coadministration of other drugs. Some drugs act as potent enzyme inducers, whereas others are inhibitors. Drug-drug interactions

involving enzyme inhibition, however, are much more common. Understanding these mechanisms of enzyme inhibition or induction is extremely important to give appropriate multidrug therapies. In the future, individuals at greatest risk for drug–drug interactions and adverse events need to be identified. P450s have a dominant role in the metabolism and elimination of drugs from the body, and their substrates are shown in Tables 4.4 and 4.8 through 4.11. Drugs in bold italics have been associated with drug–drug interactions (103). Inhibitors of P450 are shown in Table 4.12. Pharmacokinetic interactions can arise when the biotransformation and elimination of a drug are impaired by coadministered drugs. Thus, drugs can compete for biotransformation by a common P450. Adverse drug reactions, including toxicity, can occur if elimination is dependent on a P450 that exhibits defective gene variants. Thus, the genetic makeup of the individual (see the section on genetic polymorphism) has a major influence on the duration of drug action as well as on drug efficacy and safety. P450 pharmacogenetics affects the tendency for certain drug–drug interactions to occur. Thus, the future safe use of drug combinations in patients can require genotyping and phenotyping of individuals before the commencement of therapy. Identification of subjects who metabolize drugs in a different fashion from the general population should minimize the impact of pharmacogenetic variation on drug pharmacokinetics.

Metabolism-Based Enzyme Inhibition

Many drug–drug interactions are the result of inhibition or induction of P450 enzymes. Metabolism-based enzyme inhibition involves mostly competition between two drugs for the enzyme-active site. Metabolic drug–drug interactions occur when drug A (or its metabolite) alters the pharmacokinetics of a coadministered drug B by inhibiting, activating, or inducing the activity of the enzymes that metabolize drug B. Inhibitory drug–drug interactions could result in serious adverse effects, including fatalities in patients receiving multiple medications. This process is competitive and begins with the first dose of the inhibitor, and the extent of inhibition correlates with the relative affinities for the enzymes and the metabolic half-lives of the drugs involved. However, mechanism-based (irreversible) inhibition results from a metabolite that binds irreversibly with a covalent bond to the enzyme, rendering the enzyme inactive. Enzyme-specific P450 inhibitors, including metabolism- and mechanism-based inhibitors, are metabolized by specific P450 isoforms and are by and large excluded from further consideration for new drug development. Not only is CYP3A4 the most abundant isoform in human liver, but it also metabolizes more than 60% of the drugs in clinical use, which renders CYP3A4 highly susceptible to both metabolism- (reversible) and mechanism-based

inhibition. The CYP3A subfamily is involved in many clinically significant drug–drug interactions, and therefore metabolism-based inhibition of CYP3A can cause clinically significant drug–drug interactions, such as those involving nonsedating antihistamines and cisapride, which can result in cardiac dysrhythmias. For example, inhibitors of CYP1A2 can increase the risk of toxicity from clozapine or theophylline. Inhibitors of CYP2C9 can increase the risk of toxicity from phenytoin, tolbutamide, and oral anticoagulants (e.g., warfarin). Inhibitors of CYP3A4 can increase the risk of toxicity from many drugs, including carbamazepine, cyclosporine, ergot alkaloids, lovastatin, protease inhibitors, rifabutin, simvastatin, tacrolimus, and vinca alkaloids. Inhibitors of CYP2D6 can increase risk of toxicity of many antidepressants, opiate analgesics, and psychotherapeutic agents. An excellent example of a metabolism-based inhibition drug–drug interaction that resulted in a life-threatening ventricular arrhythmia associated with QT prolongation (torsades de pointes) occurred when CYP3A4 substrates or inhibitors were coadministered with terfenadine, astemizole, cisapride, or pimozide. This potentially lethal drug interaction led to the withdrawal of terfenadine and cisapride from clinical use and to the approved marketing of fexofenadine, the active metabolite of terfenadine, which does not have this interaction. Examples of enzyme inducers include barbiturates, carbamazepine, glutethimide, griseofulvin, phenytoin, primidone, rifabutin, and rifampin. Some drugs, such as ritonavir, can act as either an enzyme inhibitor or an enzyme inducer, depending on the situation. Drugs metabolized by CYP3A4 or CYP2C9 are particularly susceptible to enzyme induction.

Mechanism-Based Enzyme Inhibition

Mechanism-based inhibition differs from reversible inhibition in that the inhibitors require enzymatic activation by the target enzyme prior to exerting their inhibitory effect. This initial activation step leads to the formation of an active inhibitor, often referred to as the metabolite-intermediate complex (MIC). The MIC can then exert its inhibitory effect by either forming a direct covalent link with the enzyme or forming a noncovalent tight binding complex. Mechanism-based inhibition is characterized by NADPH-, time-, and concentration-dependent enzyme inactivation, occurring when some drugs are converted by P450s to reactive metabolites (103). Mechanism-based inactivation of CYP3A4 by drugs can be the result of chemical modification of the heme, the apoprotein, or both, as a result of covalent binding of the modified heme to the protein. The clinical pharmacokinetic effect of a mechanism-based CYP3A4 inhibitor is a function of its enzyme kinetics (i.e., K_m and V_{max}) and the rate of synthesis of new or replacement enzyme. Predicting drug–drug interactions involving CYP3A4 inactivation is possible when

pharmacokinetic principles are followed. Such prediction can become difficult, however, because the clinical outcomes of CYP3A4 inactivation depend on many factors associated with the enzyme, the drugs, and the patients.

Some of the clinically important drugs that have been identified to be mechanism-based CYP3A4 inhibitors include antibacterials (e.g., clarithromycin, erythromycin, isoniazid), anticancer drugs (e.g., irinotecan, tamoxifen, raloxifene), antidepressants (e.g., fluoxetine, paroxetine), anti-HIV agents (e.g., ritonavir, delavirdine), antihypertensives (e.g., dihydralazine, verapamil), steroids and their receptor modulators (e.g., ethinyl estradiol, gestodene, raloxifene), dihydrotestosterone reductase inhibitors (e.g., finasteride), and some herbal constituents (e.g., bergamottin, glabridin). Drugs inactivating CYP3A4 often possess several common moieties such as a tertiary amine, furan ring, or an acetylene group. The chemical properties of a drug critical to CYP3A4 inactivation include formation of reactive metabolites by P450 isoenzymes, P450 inducers, and P-gp substrate and the occurrence of clinically significant pharmacokinetic interactions with coadministered drugs.

Compared to the more common metabolism-based (reversible) inhibition, mechanism-based inhibitors of CYP3A4 more frequently cause pharmacokinetic-pharmacodynamic drug-drug interactions and are frequently the cause of unfavorable drug-drug interactions, because the inactivated CYP3A4 must be replaced by newly synthesized CYP3A4 protein. The resultant drug interactions may lead to adverse drug effects, including some fatal events. Also, raloxifene, a drug approved for the treatment of osteoporosis and chemoprevention of breast cancer in postmenopausal women at high risk for invasive breast cancer, has demonstrated the ability to act as a mechanism-based inhibitor of CYP3A4 by forming adducts with the apoprotein. It has been established that 3'-hydroxyraloxifene is produced exclusively via CYP3A4-mediated oxygenation to produce a reactive diquinone methide excluding the alternative arene oxide pathway.

However, predicting drug-drug interactions involving CYP3A4 inactivation is difficult because the clinical outcomes depend on a number of factors that are associated with drugs and patients. The apparent pharmacokinetic effect of a mechanism-based inhibitor of CYP3A4 would be a function of its enzyme kinetics and the synthesis rate of new or replacement enzyme. Most CYP3A4 inhibitors are also P-gp substrates/inhibitors, confounding the *in vitro* to *in vivo* extrapolation. The clinical significance of CYP3A inhibition for drug safety and efficacy warrants closer understanding of the mechanisms for each inhibitor. Furthermore, such inactivation may be exploited for therapeutic gain in certain circumstances. Clinicians should have good knowledge about these CYP3A4 inhibitors and avoid their combination use.

Although CYP2D6 constitutes a relatively minor fraction of the total hepatic P450 content (Fig. 4.2), the contribution of this isoform is significant as a result of its role in the metabolism and clearance of many therapeutic agents that target the cardiovascular and central nervous system (Fig. 4.3) (Table 4.9). In addition, clinically significant polymorphisms in the *CYP2D6* gene have been identified in a variety of populations with altered metabolic activity as PMs or individuals with impaired enzyme function resulting from splicing defects or gene deletions. On the other hand, EMs or individuals with normal enzyme function are heterozygous or homozygous for the wild-type allele. *In vivo* clearance of CYP2D6 substrates in PMs is usually much lower than in EMs, leading to higher plasma concentrations and the potential for clinical toxicities with therapeutic doses. For example, paroxetine is a selective serotonin reuptake inhibitor that is both a substrate for and an inhibitor of CYP2D6 (also see Chapter 19). Paroxetine is metabolized by CYP2D6 via the formation of a carbene intermediate of the methylenedioxy group yielding an irreversible complex with CYP2D6 (Fig. 4.33) (105). Also, an *in vitro* study of 3,4-methylenedioxy-methamphetamine (MDMA, ecstasy) suggested that a typical recreational MDMA dose could inactivate most hepatic CYP2D6 within an hour, and the return to a basal level of CYP2D6 could take at least 10 days, impacting its pharmacokinetics.

Because of the pivotal role of P450 isoenzymes in drug metabolism, significant inactivation of these isoforms and particularly the major human hepatic and intestinal CYP3A4 and hepatic CYP2D6 could result in drug-drug interactions and adverse drug reactions. Compared with the reversible inhibition of CYP3A4 and CYP2D6, mechanism-based inhibitors of these isoenzymes more frequently cause pharmacokinetic/pharmacodynamic drug-drug interactions, as the inactivated isoenzyme has to be replaced by newly synthesized P450 protein. Pharmacokinetic interactions often occur as a result of a change in drug metabolism. For example, the macrolides increased the plasma concentrations of a number of therapeutic agents that are substrates of CYP3A4 (Table 4.11). Diltiazem has been shown to inhibit the metabolism of a variety of coadministered drugs including carbamazepine, quinidine, midazolam, and lovastatin (Table 4.11). Inhibition of CYP3A by ritonavir explains, in part, the remarkable elevation of blood concentrations and area under the plasma concentration-time curve of other concomitantly administered drugs that are extensively metabolized by CYP3A4 and have significant first-pass metabolism. These drugs include rifabutin (400%), clarithromycin (77%), ketoconazole, saquinavir (5,000%), amprenavir (210%), nelfinavir (152%), lopinavir (7,700%), and indinavir (380%) (103). Furthermore, such inactivation can be exploited for therapeutic gain in certain circumstances (e.g., the protease inhibitors ritonavir/indinavir).

Beneficial Drug–Drug Interactions

By understanding the unique functions and characteristics of these P450 isoenzymes, health care practitioners can better anticipate and manage drug–drug interactions. They can also predict or explain an individual's response to a particular therapeutic regimen. A beneficial drug interaction, for example, is the coadministration of a CYP3A4 inhibitor with cyclosporine. By inhibiting CYP3A4, the plasma concentrations of cyclosporine (a CYP3A4 substrate) are increased, which allows a reduction of the cyclosporine dosage, thereby improving clinical efficacy and reducing its cost. Similarly, certain HIV protease inhibitors, such as saquinavir, have a low oral bioavailability due to intestinal CYP3A4 metabolism. The oral bioavailability of saquinavir can be profoundly increased by the addition of a low dose of the mechanism-based CYP3A4 inhibitor, ritonavir. This concept of altering drug pharmacokinetics by adding a low, subtherapeutic dose of a mechanism-based CYP3A4 inhibitor (ritonavir) to increase the oral bioavailability of another protease inhibitor, lopinavir (CYP3A4 substrate), led to the marketing of Kaletra, a new drug combination of lopinavir and ritonavir.

Another beneficial mechanism-based inhibition is of the P2Y₁₂ receptor (a protein found on the surface of blood platelet cells and that is an important regulator in blood clotting) by the antiaggregating platelet prodrugs, clopidogrel and prasugrel. Clopidogrel is bioactivated to its reactive metabolite by CYP2C19, CYP1A2, and CYP3A4, whereas prasugrel is bioactivated by CYP3A4 and CYP2B6. The bioactivated metabolites can then react with the P2Y₁₂ receptor protein.

Grapefruit Juice–Drug Interactions

The discovery that grapefruit juice can markedly increase the oral bioavailability of CYP3A4 metabolized drugs was based on an unexpected observation from an

interaction study between the dihydropyridine calcium channel antagonist felodipine and ethanol in which grapefruit juice was used to mask the taste of the ethanol. Subsequent investigations confirmed that grapefruit juice significantly increased the oral bioavailability of felodipine by reducing presystemic felodipine metabolism through selective inhibition of CYP3A4 expression in the intestinal wall (106).

Grapefruit juice is a beverage often consumed at breakfast for its health benefits and to mask the taste of drugs or foods. Unlike other citrus fruit juices, however, grapefruit juice can significantly increase the oral bioavailability of drugs that are metabolized primarily by intestinal CYP3A4, causing an elevation in their serum concentrations (Table 4.22). Those drugs with high oral bioavailabilities (>60%), however, are all likely safe to take with grapefruit juice, because their high oral bioavailability leaves little room for elevation by grapefruit juice. The importance of the interaction appears to be influenced by individual patient susceptibility, type and amount of grapefruit juice, and administration-related factors.

Grapefruit juice can alter oral drug pharmacokinetics by different mechanisms. Irreversible inactivation of intestinal CYP3A4, which can persist up to 24 hours, is produced by grapefruit juice given as a single, normal, 200- to 300-mL drink or by whole fresh fruit segments (Table 4.22). As a result, presystemic metabolism is reduced, and oral drug bioavailability is increased. Enhanced oral drug bioavailability can occur up to 24 hours after juice consumption. Inhibition of P-gp is a possible mechanism that increases oral drug bioavailability by reducing intestinal and/or hepatic efflux transport. Inhibition of organic anion–transporting polypeptides by grapefruit juice and apple juice has been observed; intestinal uptake transport appeared to be decreased as oral drug bioavailability was reduced. Numerous medications used in the prevention or

TABLE 4.22 Some CYP3A4 Substrates and Interactions with Grapefruit Juice (78)

Drug	Interaction ^a	Drug	Interaction ^a	Drug	Interaction ^a
Calcium channel blocker		HIV protease inhibitors		CNS Drugs	
Amlodipine	Y	Indinavir	N?	Buspirone	Y
Felodipine	Y	Nelfinavir	N?	Carbamazepine	Y
Nifedipine	Y	Ritonavir	N?	Diazepam	Y
Nimodipine	Y	Saquinavir	Y	Midazolam	Y
Nisoldipine	Y	Macrolides		Triazolam	Y
Nitrendipine	Y	Clarithromycin	N	Immunosuppressants	
Pranidipine	Y	HMG-CoA reductase inhibitors		Cyclosporine	Y
Antiarrhythmics		Atorvastatin	Y	Tacrolimus	Y?
Diltiazem	N	Cerivastatin	Y?	Other	
Verapamil	N	Fluvastatin	N?	Methadone	Y
Antihistamines		Lovastatin	Y	Sildenafil	Y
Ebastine	Y?	Pravastatin	N?		
Loratidine	Y?	Simvastatin	Y		

^aY (yes) and N (no) indicate published evidence of the presence or absence of an interaction with grapefruit juice. Y? and N? indicate expected findings based on available data. Those drugs with Y or Y? should not be consumed with grapefruit juice in an unsupervised manner.

treatment of coronary artery disease and its complications have been observed or predicted to interact with grapefruit juice. Such drug–drug interactions can increase the risk of rhabdomyolysis when dyslipidemia is treated with the HMG-CoA reductase inhibitors (“statins”). Such drug–drug interactions could also cause excessive vasodilatation when hypertension is managed with the dihydropyridines felodipine, nicardipine, nifedipine, nisoldipine, or nitrendipine. An alternative agent could be amlodipine. The therapeutic effect of the angiotensin II type I receptor antagonist losartan can be reduced by grapefruit juice. Grapefruit juice interacting with the antidiabetic agent repaglinide can cause hypoglycemia, and interaction with the appetite suppressant sibutramine can cause elevated blood pressure and heart rate. In angina pectoris, administration of grapefruit juice could result in atrioventricular conduction disorders with verapamil or attenuated antiplatelet activity with clopidogrel. Grapefruit juice can enhance the drug toxicity for antiarrhythmic agents, such as amiodarone, quinidine, disopyramide, or propafenone and for the congestive heart failure drug carvedilol. Some drugs for the treatment of peripheral or central vascular disease also have the potential to interact with grapefruit juice. Interaction with sildenafil, tadalafil, or vardenafil for erectile dysfunction can cause serious systemic vasodilatation, especially when combined with a nitrate. In stroke, interaction with nimodipine can cause systemic hypotension.

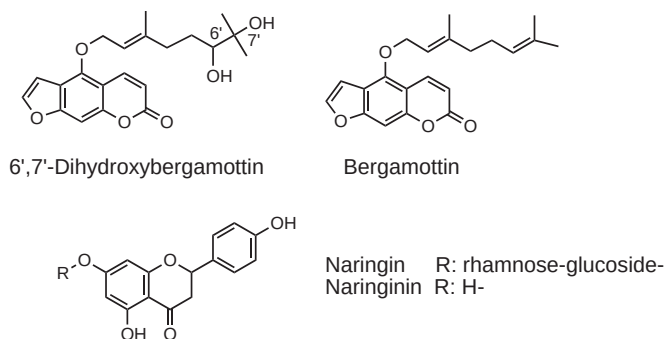
If a drug has low inherent oral bioavailability from presystemic metabolism by CYP3A4 or efflux transport by P-gp and the potential to produce serious overdose toxicity, avoidance of grapefruit juice entirely during pharmacotherapy appears mandatory. Although altered drug response is variable among individuals, the outcome is difficult to predict, and avoiding the combination will guarantee that toxicity is prevented. The elderly are at particular risk, because they often are prescribed medications and frequently consume grapefruit juice.

The mechanism by which grapefruit juice produces its effect is through inhibition of the enzymatic activity and a decrease in the intestinal expression of CYP3A4. The P-gp efflux pump also transports many CYP3A4 substrates; thus, the presence of inhibitors of P-gp in grapefruit juice (e.g., 6',7'-dihydroxybergamottin and other furanocoumarins) could be a related factor for drug–grapefruit juice interactions (82). Numerous studies have shown that grapefruit juice acts on intestinal CYP3A4, not at the hepatic level.

Does the quantity of juice matter? The majority of the presystemic CYP3A4 inhibition is obtained following ingestion of one glass of grapefruit juice; however, 24 hours after ingestion of a glass of grapefruit juice, 30% of its effect is still present (78). The reduction in intestinal CYP3A4 concentration is rapid: A 47% decrease occurred

in a healthy volunteer within 4 hours after consuming grapefruit juice. Daily ingestion of grapefruit juice results in a loss of CYP3A4 from the small intestinal epithelium. Consumption of very large quantities of grapefruit juice (six to eight glasses/day) can lead to inhibition of hepatic CYP3A4.

The active constituents found in grapefruit juice responsible for its effects on CYP3A4 include flavonoids (e.g., naringenin and naringin) and furanocoumarins (e.g., bergamottin and 6',7'-dihydroxybergamottin) (82). Of particular interest are the effects of naringin and 6',7'-dihydroxybergamottin on the activity of intestinal CYP3A4.



The majority of studies to date have used either freshly squeezed grapefruit juice, reconstituted frozen juice, commercial grapefruit juice, grapefruit segments, or grapefruit extract; all are capable of causing drug–drug interactions with CYP3A4 substrates (blended grapefruit juices have not yet been investigated). The active constituents in grapefruit juice are present not just in the juice but also in the pulp, peel, and core of the fruit and are responsible for its flavor. Bergamottin and 6',7'-dihydroxybergamottin are potent mechanism-based inhibitors of CYP3A4, and naringenin isomers are competitive inhibitors of CYP3A4 (82). Higher concentrations of 6',7'-dihydroxybergamottin and naringin are present in grapefruit segments. Thus, any therapeutic concern for a drug interaction with grapefruit juice should now be extended to include whole fruit and other products derived from the grapefruit peel. The difference in the *in vitro* CYP3A4 inhibition between grapefruit juice and orange juice is that orange juice contained no measurable amounts of 6',7'-dihydroxybergamottin.

If a patient has been taking medication with grapefruit juice for some time without ill effects, is it safe to continue to do so? Much of this unpredictability results from the inconsistency of the juice concentrations and the sporadic manner in which grapefruit juice is consumed, suggesting that this approach cannot be entirely safe (106). Given the unpredictability of the effect of grapefruit juice on the oral bioavailability of the drugs in Table 4.22, patients should be advised to avoid this combination, thus preventing the

onset of potential adverse effects. Each patient's situation should be considered, and advice should be based on consumption history and the specific medications involved. The benefits of increased and controlled drug bioavailability by grapefruit juice can, in the future, be achieved through either standardizing the constituents or coadministration of the isolated active ingredients. This would then lead to a safe, effective, and cost-saving means to enhance the absorption of many therapeutic agents.

P-Glycoprotein–Drug Interactions

From the earlier discussion regarding P-gp, it is obvious that P-gp-mediated transport has an important role in pharmacokinetic-mediated drug–drug interactions (79). Thus, inhibition of P-gp-mediated transport could dramatically increase the systemic bioavailability of an otherwise poorly absorbed drug. Similar consequences could be expected with a reduction in renal or biliary clearances (e.g., digoxin). Numerous investigations with drugs such as digoxin, etoposide, cyclosporine, vinblastine, paclitaxel, loperamide, domperidone, and ondansetron demonstrate that P-gp has an important role in determining the pharmacokinetics of substrate drugs (79). For example, if drug A is a substrate for both P-gp and *only* for CYP3A4 and if a second drug B is added that is an inhibitor for both P-gp and CYP3A4 (Table 4.18), then the plasma drug concentration for unmetabolized drug A will be elevated, with increased potential for drug–drug interactions as adverse effects or for causing a drug overdose. If drug A is a substrate for multiple P450 isoforms, however, then drug A will be metabolized by these other isoforms, with minimal effect on plasma drug concentrations. On the other hand, if the second drug B is only an inhibitor for P-gp, then drug A will be subject to CYP3A4 metabolism, thus decreasing the plasma concentration for drug A to subtherapeutic levels. The effect of P-gp inhibition is to increase the oral bioavailability so that the later actions of CYP3A4 inhibition will be increased. One of the best examples is the interaction between digoxin and quinidine. Quinidine blocks P-gp in the intestinal mucosa and in the proximal renal tubule; thus, digoxin elimination into the intestine and urine is inhibited, increasing the plasma digoxin concentration to toxic levels. Another example is loperamide, which is an opiate antidiarrheal normally kept out of the brain by the P-gp pump; however, inhibition of P-gp allows accumulation of loperamide in the brain, leading to respiratory depression. Increasingly, the relevant clinical data for drug–drug interactions can be found on the World Wide Web.

The components of grapefruit juice reportedly inhibit P-gp, and this can be one of the mechanisms for the increase in bioavailability of drugs that are substrates for P-gp (Table 4.18). Although fexofenadine is a P-gp substrate, rather than a decrease in its plasma levels when it

is coadministered with grapefruit juice, the blood levels are increased as a result of fexofenadine being a substrate of the organic anion transporter polypeptide (OATP) in the intestine. Studies have shown that apple and other fruit juices are more potent inhibitors of OATP than of P-gp.

Food–Drug Interactions

Drug pharmacokinetics can be altered by the fat content of food through changes in drug solubility as well as the nutritional status of a patient (107). The fact that grapefruit juice can increase the bioavailability of certain drugs by reducing their presystemic intestinal metabolism has led to renewed interest in the area of “food–drug interactions,” with particular interest regarding the effects of grapefruit constituents. Specific naturally occurring chemicals in food have been associated with drug–drug interactions. For example, severe hypertensive reactions have occurred when patients treated with antidepressant MAO inhibitors have ingested cheeses and other foods rich in the biogenic amine tyramine (see Chapter 18).

Drug–Dietary Supplement Interactions

The increasing use of dietary supplements presents a special challenge in health care; thus, there is an increasing need to predict and avoid these potential adverse drug–dietary supplement interactions. The present interest and widespread use of herbal remedies has created the possibility of interaction between them and over-the-counter or prescription drugs if they are used simultaneously. As herbal medicines become more popular, herbal hepatotoxicity is being increasingly recognized. Females appear to be predisposed to hepatotoxicity, and coadministered agents that induce P450 enzymes (e.g., St. John's wort) can also increase individual susceptibility to some dietary supplements. Currently, nearly one in five adults taking prescription medicines is also taking at least one dietary supplement. The mechanisms for drug–dietary supplement interactions are similar to those for drug–drug interactions affecting the pharmacokinetics of the respective drug. Little is known regarding the pharmacokinetic properties of many of the substances in dietary supplements. Therefore, the potential for drug–dietary supplement interactions has greatly increased.

St. John's Wort

A commonly reported drug–dietary supplement interaction is between St. John's wort and HIV protease inhibitors, leading to drug resistance and treatment failure. St. John's wort is a popular dietary supplement often used for depression. Of the two substances found in St. John's wort, hypericin and hyperforin, hyperforin appears to be the main constituent, with *in vitro* selective serotonin reuptake inhibitor activity (also see Chapter 18).

Hyperforin appears also to be the more potent inducer of CYP3A enzymes based on in vitro and in vivo studies.

The FDA has issued a statement that “concomitant use of St. John’s wort with protease inhibitors or non-nucleoside reverse transcriptase inhibitors is not recommended.” St. John’s wort appeared to have minimal effects on the CYP3A4 enzymes after acute administration; however, chronic administration (≥ 2 weeks) of St. John’s wort selectively induced CYP3A4, with a greater effect in the small intestine than in the liver. Administration of St. John’s wort for 8 weeks decreased the plasma levels of norethindrone, a low-dose oral contraceptive; reduced the half-life of ethinyl estradiol, consistent with increased CYP3A activity; increased breakthrough bleeding; and reduced contraceptive efficacy. Based on these in vivo and in vitro studies, the efficacy of drugs that are substrates for the CYP3A family or P-gp can be reduced with coadministration of St. John’s wort. St. John’s wort should be listed along with other known CYP3A inducers (e.g., rifampin and rifabutin) as possibly decreasing plasma levels of CYP3A substrates. The drug products Kaletra (lopinavir and ritonavir), Mifeprex (mifepristone, RU-486), Nuvaring (etonogestrel/ethinyl estradiol), Gleeevec (imatinib), Neoral (cyclosporine), Rapamune (sirolimus), and Prograf (tacrolimus) include information about drug interactions with St. John’s wort in their labeling. Thus, patients ingesting St. John’s wort products should be advised that St. John’s wort can have potentially dangerous drug–drug interactions with some prescription drugs and to consult a physician before taking St. John’s wort if currently taking anticoagulants, oral contraceptives, antidepressants, antiseizure medications, drugs to treat HIV or prevent transplant rejections, or any other prescription drug.

St. John’s wort also decreases the absorption of digoxin and fexofenadine, apparently by inducing P-gp in the intestinal and renal endothelium, increasing their elimination in the intestine and urine, respectively, and their plasma concentrations.

Echinacea

In vitro or in vivo chronic administration studies of echinacea, an herbal product used for the treatment of colds and viral infections, inhibited hepatic CYP1A2 and intestinal CYP3A activities and induced hepatic CYP3A. Based on these preliminary findings, the effect of echinacea on various CYP3A substrates can vary depending on the relative contribution to a given drug’s overall clearance by intestinal CYP3A versus hepatic CYP3A in the individual substrate’s clearance pathway.

Ginkgo Biloba

In vitro studies with ginkgo biloba, often used for memory improvement, exhibited induction of CYP2C19. The extent of induction appears to be CYP2C19 genotype dependent.

Kava

Reports of hepatotoxicity have been associated with the use of kava, a popular drug in Europe and North America. Hepatotoxicity was not observed when kava was prepared as a water infusion but was observed with solvent-extracted products available in stores and on the Internet. The three kava lactones (methysticin, desmethoxyyangonin, and yangonin; active principles) are potent inhibitors of CYP1A2, CYP2C9, CYP2C19, CYP2E1, and CYP3A4, with methysticin being the most potent enzyme inhibitor as well as the most cytotoxic. The potent inhibition of P450 enzymes suggests a high potential for drug–drug interactions with drugs and other herbs that are metabolized by the same P450 enzymes. Long-term use or use in individuals with liver disorders should be avoided, and liver function transaminases need to be checked frequently.

Other Dietary Supplements Exhibiting Drug-Induced Hepatotoxicity

DHEA and androstenedione are testosterone precursors that have been associated with hepatic toxicity and should be avoided in those with hepatic disease or coingestion with other potentially hepatotoxic products or enzyme inducers that might increase the risk of liver damage. Liver enzymes should be monitored once or twice a year. Boldo, an herbal medicine, can cause hepatotoxicity and exacerbate existing liver disease. Because chaparral, comfrey, germander, skullcap, and valerian root can cause acute and chronic liver injury, these products should be considered unsafe. Pennyroyal oil can cause acute hepatotoxic liver injury, which has been attributed to the bioactivation of the terpine, *R*-(+)-pulegone, resulting in depletion of hepatic glutathione. In some cases, unknown adulterants found in these herbal products can be responsible for the hepatotoxicity. Black cohosh, commonly used by women for menopausal symptoms, including hot flashes and sleep disorders, formed quinone metabolites in vitro, but no mercapturate conjugates were detected in urine samples from women who consumed multiple oral doses of up to 256 mg of a standardized black cohosh extract. At moderate doses of black cohosh, the risk of liver injury is minimal (108).

Silybum marianum (milk thistle) is used in the treatment of chronic or acute liver disease as well as in protecting the liver against toxicity (109). *Silybum* is cited as one of the oldest known herbal medicines. The active constituents of milk thistle are flavonolignans, which are known collectively as silymarin. The most remarkable use of silymarin is in the treatment of *Amanita* mushroom poisoning. *Amanita* mushrooms possess two extremely powerful hepatotoxins, amanitin and phalloidin (the median lethal dose of amanitin is 0.1 mg/kg body weight). Severe liver damage (and death) is avoided if silymarin is administered within 24 hours following ingestion of *Amanita*. It is also a hepatoprotective for chronic exposure to ethanol and acetaminophen toxicity.

Miscellaneous Drug–Drug Interactions

The ability of drugs and other foreign substances to stimulate (induction) metabolism of other drugs has already been discussed. Phenobarbital, for example, stimulates metabolism of a variety of drugs (e.g., phenytoin and coumarin anticoagulants). Stimulation of bishydroxycoumarin metabolism can create a problem in patients undergoing anticoagulant therapy. If phenobarbital administration is stopped, the rate of metabolism of the anticoagulant decreases, resulting in greater plasma concentrations of bishydroxycoumarin and enhanced anticoagulant activity, increasing the possibility of hemorrhage. Serious side effects have resulted from this type of interaction. These observations indicate that combined therapy of a potent drug (e.g., bishydroxycoumarin) and an inducer of drug metabolism (e.g., phenobarbital) can create a hazardous situation if the enzyme inducer is withdrawn and therapy with the potent drug is continued without an appropriate decrease in dose.

Some drugs are competitive inhibitors of nonmicrosomal metabolic pathways. Serious reactions have been reported in patients treated with an MAO inhibitor, such as tranylcypromine or iproniazid, because they are, as a rule, sensitive to a subsequent dose of a sympathomimetic amine (e.g., amphetamine) or a tricyclic antidepressant (e.g., amitriptyline), which is metabolized by MAO.

Allopurinol, a xanthine oxidase inhibitor used for the treatment of gout, inhibits metabolism of 6-mercaptopurine and other drugs metabolized by this enzyme. A serious drug interaction results from the concurrent use of allopurinol for gout and 6-mercaptopurine to block the immune response from a tissue transplant or as

antimetabolite in neoplastic diseases. In some cases, however, allopurinol is used in conjunction with 6-mercaptopurine to control the increase in uric acid elimination from 6-mercaptopurine metabolism. The patient should be supervised closely, because when given in large doses, allopurinol, an inhibitor of purine metabolism, can have serious effects on bone marrow.

GENDER DIFFERENCES IN DRUG METABOLISM

The role of gender as a contributor to variability in xenobiotic metabolism and IDRs, which are more common in women than in men, is not clear, but increasing numbers of reports show differences in metabolism between men and women, raising the intriguing possibility that endogenous sex hormones, hydrocortisone, or their synthetic equivalents can influence the activity of inducible CYP3A. For example, *N*-demethylation of erythromycin was significantly higher in females than males. Nevertheless, the *N*-demethylation was persistent throughout adulthood. In contrast, males exhibited unchanged *N*-demethylation values.

Gender-dependent differences of metabolic rates have been detected for some drugs. Side-chain oxidation of propranolol was 50% faster in males than in females, but no differences between genders were noted in aromatic ring hydroxylation. *N*-demethylation of meperidine was depressed during pregnancy and for women taking oral contraceptives. Other examples of drugs cleared by oxidative drug metabolism more rapidly in men than in women included chlordiazepoxide and lidocaine. Diazepam, prednisolone, caffeine, and acetaminophen are metabolized slightly faster by women than by men. No gender differences have been observed in the clearance of phenytoin, nitrazepam, and trazodone, which interestingly are not substrates for the CYP3A subfamily. Gender differences in the rate of glucuronidation have been noted.

More investigation is warranted, and future pharmacokinetic studies examining the alteration in drug metabolism in one gender need to be reexamined with respect to the other gender. Even in postmenopausal women, CYP3A function can be altered and influenced by the lack of estrogen or the presence of androgens.

MAJOR PATHWAYS OF METABOLISM

Table 4.12 contains an extensive list of commonly used drugs and the P450 isoforms that catalyze their metabolism. In addition, Phase 1 and Phase 2 metabolic pathways for some common drugs are listed in Table 4.23.

TABLE 4.23 Metabolic Pathways of Common Drugs

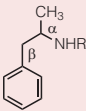
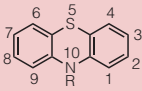
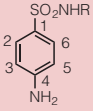
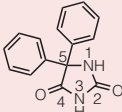
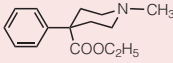
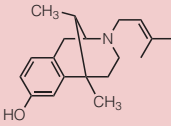
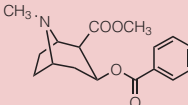
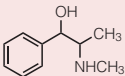
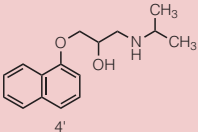
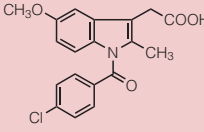
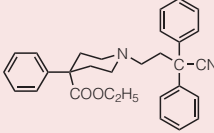
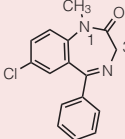
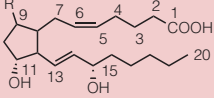
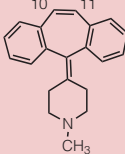
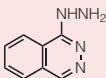
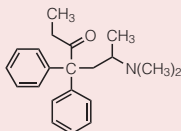
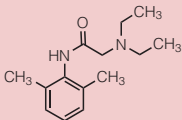
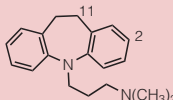
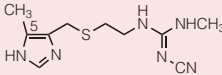
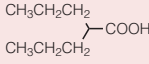
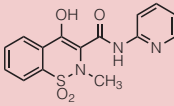
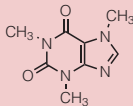
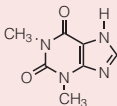
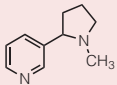
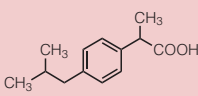
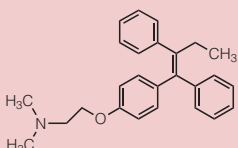
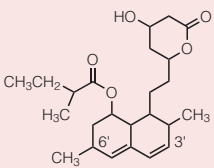
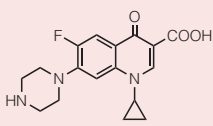
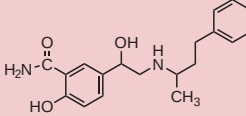
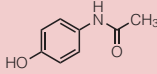
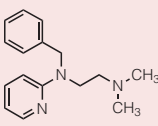
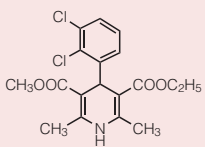
Drug	Pathway	Drug	Pathway
Amphetamines 	Deamination (followed by oxidation and reduction of the ketone formed) <i>N</i> -oxidation <i>N</i> -dealkylation Hydroxylation of the aromatic ring Hydroxylation of the β -carbon atom Conjugation with glucuronic acid of the acid and alcohol products from the ketone formed by deamination	Barbiturates 	Oxidation and complete removal of substituents at carbon 5 <i>N</i> -dealkylation at N^1 and N^3 Desulfuration at carbon 2 (thiobarbiturates) Scission of the barbiturate ring at the 1:6 bond to give substituted malonylureas
Phenothiazines 	<i>N</i> -dealkylation in the N^{10} side chain <i>N</i> -oxidation in the N^{10} side chain Oxidation of the heterocyclic S atom to sulfoxide or sulfone Hydroxylation of one or both aromatic rings Conjugation of phenolic metabolites with glucuronic acid or sulfate Scission of the N^{10} side chain	Sulfonamides 	Acetylation at the N^4 amino group Conjugation with glucuronic acid or sulfate at the N^4 amino group Acetylation or conjugation with glucuronic acid at the N^1 amino group Hydroxylation and conjugation in the heterocyclic ring, R
Phenytoin 	Hydroxylation of one aromatic ring Conjugation of phenolic products with glucuronic acid or sulfate Hydrolytic scission of the hydantoin ring at the bond between carbons 3 and 4 to give 5,5-diphenylhydantoic acid	Meperidine 	Hydrolysis of ester to acid <i>N</i> -dealkylation Hydroxylation of aromatic ring <i>N</i> -oxidation Both <i>N</i> -dealkylation and hydrolysis Conjugation of phenolic products
Pentazocine 	Hydroxylation of terminal methyl groups of the alkenyl side chain to give cis and trans (major) alcohols Oxidation of hydroxymethyl product of the alkenyl side chain to carboxylic acids Reduction of alkenyl side chain and oxidation of terminal methyl group	Cocaine 	Hydrolysis of methyl ester Hydrolysis of benzoate ester <i>N</i> -dealkylation Both hydrolysis and <i>N</i> -dealkylation
Phenmetrazine 	Oxidation to lactam Aromatic hydroxylation <i>N</i> -oxidation Conjugation of phenolic products	Ephedrine 	<i>N</i> -dealkylation Oxidative deamination Oxidation of deaminated product to benzoic acid Reduction of deaminated product to 1,2-diol
Propranolol 	Aromatic hydroxylation at C-4' <i>N</i> -dealkylation Oxidative deamination Oxidation of deaminated product to naphthoxylactic acid Conjugation with glucuronic acid <i>O</i> -dealkylation	Indomethacin 	<i>O</i> -demethylation <i>N</i> -deacylation of <i>p</i> -chlorobenzoyl group Both <i>O</i> -dealkylation and <i>N</i> -deacylation Conjugation of phenolic products with glucuronic acid Other conjugation products
Diphenoxylate 	Hydrolysis of ester to acid Hydroxylation of one aromatic ring attached to the <i>N</i> -alkyl side chain	Diazepam 	<i>N</i> -dealkylation at N^1 Hydroxylation at carbon 3 Conjugation with glucuronic acid Both <i>N</i> -dealkylation of N^1 and hydroxylation at carbon 3
Prostaglandins 	Reduction of double bonds at carbons 5 and 6 and 13 and 14 Oxidation of 15-hydroxyl to ketone β -Oxidation of carbons 3, 5, and 7 ω -Oxidation of carbon 20 to acid	Cyproheptadine 	<i>N</i> -dealkylation 10,11-Epoxy formation Both <i>N</i> -dealkylation and 10,11-epoxidation

TABLE 4.23 Metabolic Pathways of Common Drugs (Continued)

Drug	Pathway	Drug	Pathway
Hydralazine 	<i>N</i> -acetylation with cyclization to a methyl- <i>s</i> -triazolophthalazine <i>N</i> -formylation with cyclization to an <i>s</i> -triazolophthalazine Aromatic hydroxylation of benzene ring Oxidative loss of hydrazinyl group to 1-hydroxy Hydroxylation of methyl of methyl- <i>s</i> -triazolophthalazine Conjugation with glucuronic acid	Methadone 	Reduction of ketone to hydroxyl Aromatic hydroxylation of one aromatic ring <i>N</i> -dealkylation of alcohol product <i>N</i> -dealkylation with cyclization to pyrrolidine
Lidocaine 	<i>N</i> -dealkylation Oxidative cyclization to a 4-imidazolide <i>N</i> -oxidation of amide N Aromatic hydroxylation ortho to methyl Hydrolysis of amide	Imipramine 	<i>N</i> -dealkylation Hydroxylation at C-11 Aromatic hydroxylation (C-2) <i>N</i> -oxidation Both <i>N</i> -dealkylation and hydroxylation
Cimetidine 	<i>S</i> -oxidation Hydroxylation of 5-methyl	Valproic acid 	CoA thioester Dehydrogenation to (E) 2-ene Dehydrogenation to (E) 2,4-diene Dehydrogenation to 4-ene 3-Hydroxylation
Piroxicam 	Pyridine 3'-hydroxylation Hydrolysis of amide Decarboxylation	Caffeine 	<i>N</i> ³ -demethylation <i>N</i> ¹ -demethylation <i>N</i> ⁷ -demethylation to theophylline C-8 oxidation to uric acids Imidazole ring opened
Theophylline 	<i>N</i> ³ -demethylation <i>N</i> ¹ -demethylation C-8 oxidation to uric acids Imidazole ring opened 1-Me xanthine to 1-Me uric acid—xanthine oxidase	Nicotine 	Pyrrolidine 5'-hydroxylation to cotinine Pyrrolidine <i>N</i> -oxidation (FMO) <i>N</i> -demethylation (nornicotine and nortinine) Pyridine <i>N</i> -methylation 3'-Hydroxylation of cotinine
Ibuprofen 	CoA thioester and epimerization of R- to S+ enantiomer Methyl hydroxylation to CH ₂ OH CH ₂ OH to COOH Acylglucuronide	Tamoxifen 	<i>N</i> -demethylation 4'-Hydroxylation <i>N</i> -oxidation (FMO) 4'- <i>O</i> -sulfate 4'- <i>O</i> -glucuronide
Lovastatin 	6'-Hydroxylation 3'-Side chain hydroxylation 3'-Hydroxylation β-oxidation of lactone <i>O</i> -glucuronides	Ciprofloxacin 	Piperazine 3'-hydroxylation <i>N</i> -sulfonation
Labetalol 	<i>O</i> -sulfate (major) <i>O</i> -glucuronide	Acetaminophen 	<i>O</i> -glucuronide <i>O</i> -sulfate Oxidation to <i>N</i> -acetyl- <i>p</i> -benzoquinoneimine Conjugation of <i>N</i> -acetyl- <i>p</i> -benzoquinoneimine with glutathione
Tripelennamine 	<i>p</i> -Hydroxylation Benzylic C-hydroxylation <i>N</i> -depyridinylation <i>N</i> -debenzylation	Felodipine 	Aromatization Ester hydrolysis Methyl hydroxylation

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Mark D. Watanabe, PharmD, PhD, BCPP

Because both the risperidone and paroxetine appeared to demonstrate a therapeutic effect for Y.H., the pharmacist recommended that the dose of risperidone be decreased to 2 mg daily while continuing paroxetine 20 mg daily. At her next appointment the following month, Y.H. stated that her mood was “getting back on an even keel” and her left hand was no longer shaking.

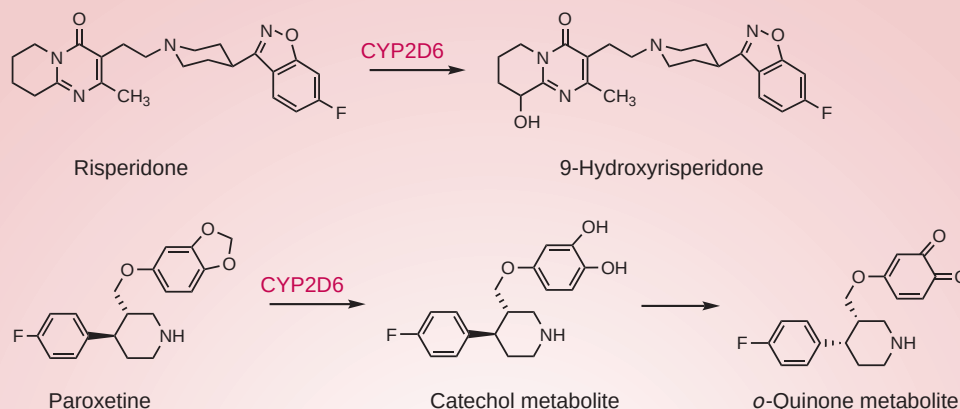
Chemical Analysis

Victoria Roche and S. William Zito

Risperidone is an antipsychotic agent that acts through the inhibition of dopamine D_2 and serotonin 5-HT_2 receptors. If serum levels exceed those associated with safe therapeutic use, adverse effects associated with excessive D_2 receptor antagonism,

including tremor, can commonly manifest. Like many psychotropic agents, risperidone is metabolized by CYP2D6. The active metabolite generated by this isoform is 9-hydroxyrisperidone. Paroxetine is an antidepressant that works by inhibiting the reuptake of 5-HT. It too is metabolized by CYP2D6, yielding an inactive catechol metabolite that can further oxidize to an orthoquinone.

Coadministration of paroxetine and risperidone would result in competition for limited quantities of CYP2D6. As risperidone serum levels increase, the side effects of tremor, dystonia, and akathisia could reasonably be anticipated. Maintaining these two CYP2D6 substrates in Y.H.’s mental health drug regimen demands a reduction in the dose of risperidone so that the metabolism-mediated increase in serum concentration keeps blood levels within the therapeutic range and out of the toxic range. The reduction in dose from 3 to 2 mg daily appears to have accomplished this goal.



CASE STUDY

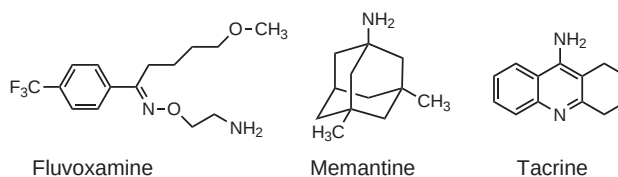
Victoria Roche and S. William Zito

NH is a 79-year-old widow who has been in a slow cognitive decline for the past 3 years. Always fiercely independent, her progressive mental impairment has prompted a serious depression that has been fairly well controlled with fluvoxamine, a serotonin reuptake transporter inhibitor antidepressant drug metabolized by CYP1A2 and 2D6. Recently, NH became lost and disoriented while driving to the local supermarket. In a panic, she caused a car accident that resulted in significant property damage but, thankfully, no personal injuries. In 3 months she will be moving into Brookside Manor, a new assisted living facility

known for its high-quality care of Alzheimer’s patients. NH is a pack-a-day smoker, but Brookside is totally smoke free. NH is dreading the smoking cessation program she will be required to complete before she moves in, but does understand (at times) that she has no choice.

From a cognitive impairment therapy standpoint, NH was doing well on the NMDA receptor antagonist memantine (Namenda), which is minimally metabolized by CYP isoforms and predominantly excreted unchanged. However, the most recent upheaval in her life and the pending change in her

living situation have prompted a worsening of her cognitive dysfunction symptoms. Her physician is now trying tacrine (Cognex), a centrally acting acetylcholinesterase inhibitor, in place of Namenda. After a week on this medication NH had initially complained of diarrhea and “feeling hot and sweaty” (symptoms of tacrine toxicity). Now that she’s quitting smoking, these symptoms have intensified, she’s experiencing urinary incontinence, and her hepatic transaminase enzymes are significantly elevated. As the consultant pharmacist for Brookside, how would you advise this patient and her current providers?



1. Conduct a thorough metabolic analysis of the therapeutic options in the case.
2. Apply the chemical understanding gained from the metabolic analysis to this patient's specific needs to make a therapeutic recommendation.

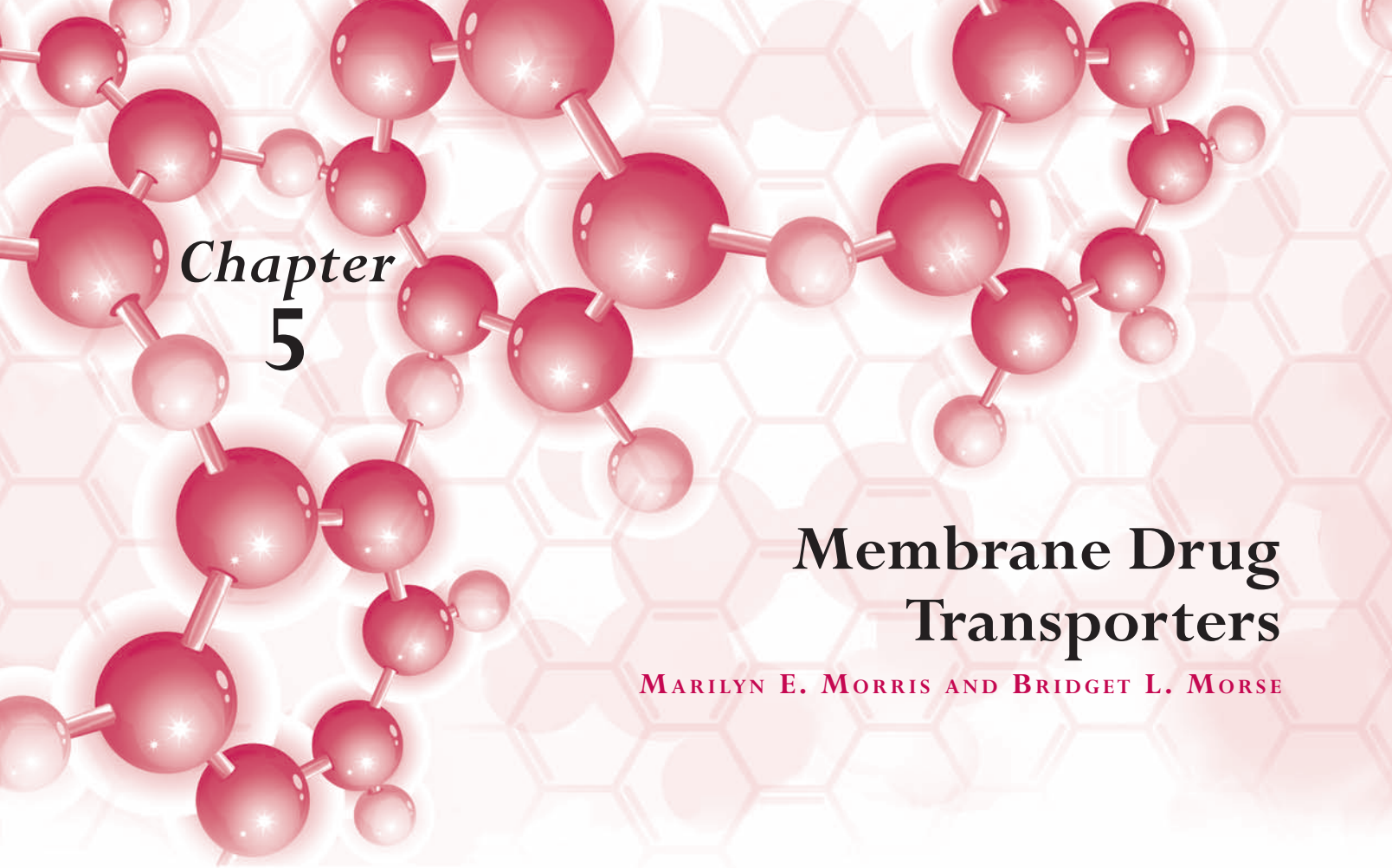
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Chapter 5

Membrane Drug Transporters

MARILYN E. MORRIS AND BRIDGET L. MORSE

Drugs Covered in This Chapter

- | | | |
|----------------|------------------------|----------------|
| • Acyclovir | • Gabapentin | • Probenecid |
| • Adefovir | • Gabapentin enacarbil | • Repaglinide |
| • Atorvastatin | • Gemfibrozil | • Rifampin |
| • Cerivastatin | • Lamivudine | • Rosuvastatin |
| • Cidofovir | • Methotrexate | • Simvastatin |
| • Cisplatin | • Nitrofurantoin | • Tenofovir |
| • Cyclosporine | • Olmesartan | • Topotecan |
| • Digoxin | • Oxaliplatin | • Valacyclovir |
| • Doxorubicin | • Paclitaxel | • Valsartan |
| • Fexofenadine | • Penicillin | |
| • Furosemide | • Pravastatin | |

Abbreviations

- | | | |
|---|---|--|
| ABC , ATP-binding cassette | GI , gastrointestinal | OATP , organic anion transporting polypeptide |
| ACE , angiotensin-converting enzyme | GHB , γ -hydroxybutyrate | OCT , organic cation transporter |
| ATP , adenosine triphosphate | HMG-CoA , 3-hydroxy-3-methyl-glutaryl-coenzyme A | OCTN , organic cation/carnitine transporter |
| AUC , area under the curve | LAT , L-type amino acid transporter | P-gp , P-glycoprotein |
| BBB , blood-brain barrier | MATE , multidrug and toxin extrusion transporter | PEPT , peptide transporter |
| BCRP , breast cancer resistance protein | MCT , monocarboxylate transporter | PHT , peptide/histidine transporter |
| BSEP , bile salt export pump | MDR , multidrug resistance | PXR , pregnane X receptor |
| CNT , concentrative nucleoside transporter | MRP , multidrug resistance-associated protein | SLC , solute carrier |
| ENT , equilibrative nucleoside transporter | OAT , organic anion transporter | SMCT , sodium-coupled monocarboxylate transporter |
| FXR , farnesoid X receptor | | SNP , single nucleotide polymorphism |

INTRODUCTION

The objectives of the chapter are:

- Emphasize the significance of transport in pharmacokinetics and pharmacodynamics
- Provide an overview of transport mechanisms and classification of drug transporters
- Summarize transporters relevant to drug disposition and their relevant drug substrates
- Provide evidence and mechanisms for transport interactions relevant to therapeutics
- Provide an understanding of mechanisms underlying interindividual variability in the pharmacokinetics of drug transporter substrates

Transporters have fundamental roles in cell homeostasis and physiologic function by facilitating the movement of molecules across biological membranes. Transporters are responsible for maintaining ionic and osmotic gradients necessary for normal cell activity. Transporters facilitate the oxygen binding and release in red blood cells, and they are necessary for the transport of nutrients to vital organs. The primary function of transporters is to transport endogenous substances, such as hormones, glucose, and amino acids; however, many of these transporters also transport xenobiotics. It is these drug transporters that are of importance when considering drug disposition and drug response.

Drug transporters are localized to barrier membranes of the body responsible for xenobiotic entry and exit. They are expressed in organs of absorption, such as the intestine, and clearance organs including the liver and kidney. Transporters are also expressed on membranes that separate particularly susceptible organs from the rest of the body, including the blood–brain and blood–placenta barriers, where they facilitate the influx of nutrients and efflux of potentially harmful xenobiotics. Because of their location on barrier membranes, transporters have an important role in drug pharmacokinetics and pharmacodynamics. Transporters can have a role in drug absorption and can facilitate or prevent drug entry into the body. Also, transporters have a role in drug distribution, as they facilitate the movement of drugs between the blood and peripheral tissues. The role of transporters in drug distribution can also affect drug response by allowing or preventing drug access to the site of action. One of the most interesting roles of drug transporters is their indirect effects on drug metabolism. Transporters can restrict or allow a drug's distribution into organs that contain drug-metabolizing enzymes, particularly the liver and the intestine. In this regard, there can be extensive transport-metabolism interplay, and transport or metabolism can be the rate-limiting process controlling

drug elimination. Transporters are also responsible for transport and removal of drug metabolites. Finally, transporters have a significant role in drug excretion, as they are present in the kidney and on the canalicular membrane of the liver where they can facilitate drug elimination into the urine or bile, respectively.

Because transporters can have a significant effect on drug absorption, distribution, and clearance, changes in the function of these transporters can significantly alter drug pharmacokinetics and pharmacodynamics. Factors leading to changes in transporter function include interactions with other drugs, disease states, and genetic variability in expression. Understanding transporter effects and the effects of changes in transporter function is important for effective therapeutic use of drugs that interact with transporters.

MECHANISMS OF MEMBRANE TRANSPORT

Membrane transport is essential for almost every drug to be therapeutically effective. This can be achieved by different mechanisms of transport (Fig. 5.1). Passive diffusion is the simplest way for a drug to pass through a membrane and depends only on the existence of a concentration gradient for a molecule across the membrane. Because of the lipophilicity of biologic membranes, diffusion is energetically unfavorable for drugs that are relatively hydrophilic, particularly for drugs that are predominantly ionized at physiologic pH, and these molecules require facilitated transport. Facilitated transport refers to any transport aided by a facilitating protein. Similar

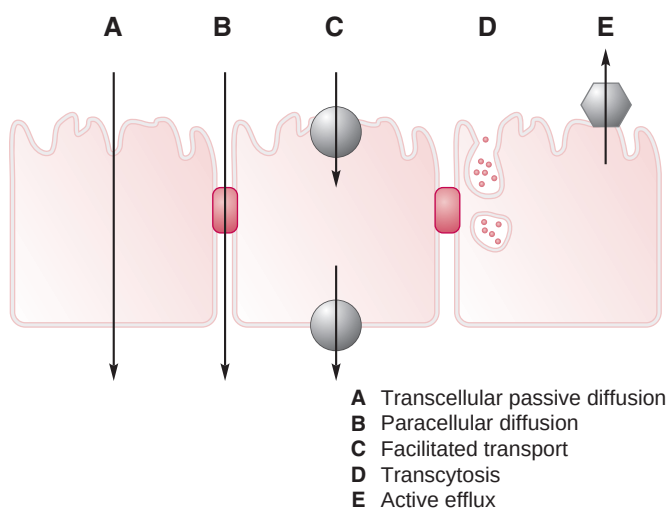


FIGURE 5.1 Mechanisms of membrane transport. Transport mechanisms most commonly used by therapeutic agents include passive diffusion and facilitated transport.

to diffusion, passive facilitated transport depends only on the existence of a concentration gradient and involves movement of molecules down this gradient. Active transport uses a separate energy source to move molecules against their concentration gradient. Drug transporters can use either passive or active transport mechanisms. Paracellular transport (transfer of substances between cells of an epithelial cell layer) and transcytosis (vesicular transfer of substances across the interior of a cell) are less commonly used mechanisms of membrane transport. Mannitol can pass through membranes via paracellular transport and can be used as a probe for this type of transport. Transcytosis is often receptor mediated and is a mechanism of transport for endogenous substances such as insulin and transferrin.

Molecules can use one or more of these mechanisms to cross biological membranes. Transporter substrates can be lipophilic and also capable of passive diffusion; therefore, drug transport can often be described with the following equation:

$$J = \frac{V_{\max} \times C}{K_m + C} + P \times C$$

where J is the rate of total drug transport, $V_{\max}$ is the maximum rate of the saturable transporter system, K_m reflects the affinity of a drug substrate for the transporter system, P is the rate of nonsaturable transport or passive diffusion, and C is the drug concentration. The importance of a transporter depends on to what extent transporter-mediated transport ($V_{\max}/K_m$) is responsible for membrane passage of a drug compared to other mechanisms like passive diffusion (P). Transporter-mediated transport becomes particularly important when the need is present to move a molecule against its concentration gradient, as this type of transport can only be achieved by active transporters and P due to passive diffusion is negligible.

When considering drug transport, it is important to consider the polarization of biological membranes, in that different membranes can have different properties relating to their function as a barrier, including the expression of different transporters. The membrane facing the systemic circulation is referred to as the basolateral membrane, or the sinusoidal membrane in the liver. The membrane facing the exterior, such as the gut lumen, is referred to as the apical membrane. The apical membrane is also called the brush border membrane in the kidney or intestine and the canalicular membrane in the liver. Transport is actually a two-step process in which molecules must be transported across the apical then basolateral membrane or vice versa. Drug transport is a concerted action between transporters expressed on both membranes.

CLASSIFICATION OF TRANSPORTERS

Facilitative Versus Active Transporters

Although all transporters are facilitating proteins, the terms facilitative and active transporters refer to transporters that transport substrates down and against their concentration gradients, respectively. Transporters that are classified as facilitative transporters only move molecules down their concentration gradient, without the use of a separate energy source. The transporters of this type should not be confused with channels. Channels also facilitate the movement of ions and hydrophilic molecules across membranes and down their concentration gradient; however, channels control transport or flow of substrates by gating mechanisms, whereas transporters bind to their substrates and undergo a conformational change to transport the substrate across a membrane.

Primary Versus Secondary Active Transporters

Most drug transporters are active transporters that use an energy source other than the drug's concentration gradient to transport substrates across membranes from a region of lower concentration to a region of higher concentration (i.e., against its concentration gradient). Active transporters are further classified as primary or secondary active transporters. Primary active transporters most commonly use adenosine triphosphate (ATP) as an energy source for substrate transport. Secondary active transporters use the concentration gradient of another substance, such as protons or sodium ions, but also other ionic endogenous substances as energy sources to drive transport. The concentration gradients that drive secondary active transport are generally created by primary active transporters; an example of interplay between the two types of transporters is depicted in Figure 5.2. As shown, primary active transport by the Na^+/K^+ ATPase results in a concentration gradient of Na^+ ions. This concentration gradient drives secondary active transport by the Na^+/H^+ exchanger, and the resulting proton gradient drives transport of the drug substrate against its concentration gradient. The Na^+/K^+ ATPase and Na^+/H^+ exchanger do not transport drug substrate, but are involved in drug transport by producing the sodium and proton gradients used as driving forces. Molecules used to drive secondary active transport are also substrates for the secondary active transporter and are simultaneously transported across the membrane. This simultaneous transport may be in the same direction, referred to as *symport*, or in opposite directions, referred to as *antiport*.

Influx Versus Efflux Transporters

Influx transporters transport substrates from extracellular spaces into cells. Efflux transporters transport

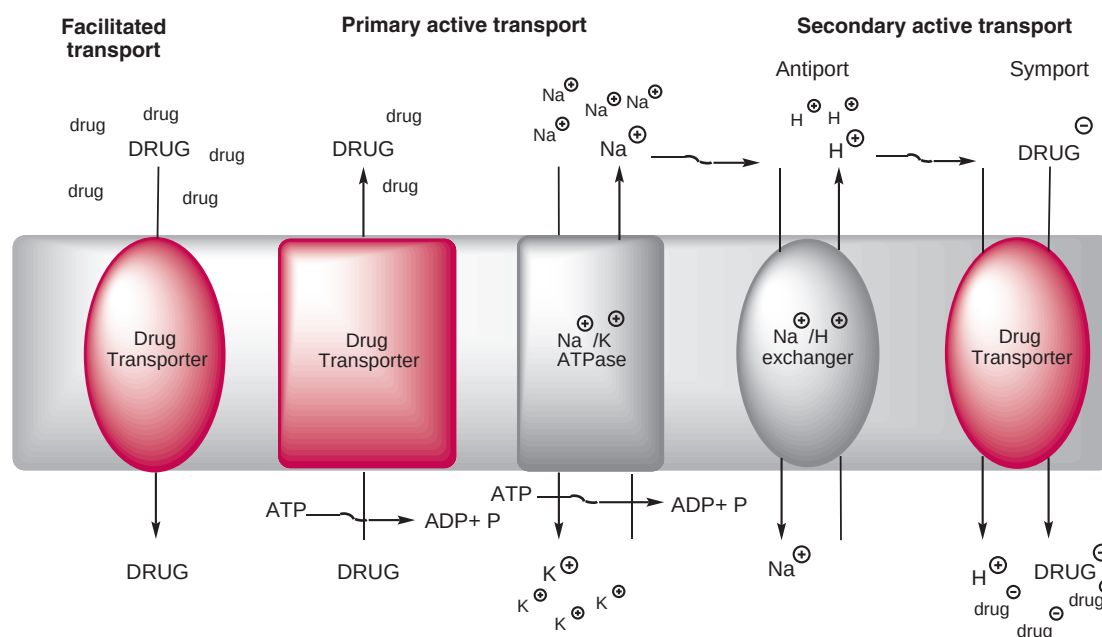


FIGURE 5.2 Types of drug transporters. Primary active transporters use ATP and have the ability to transport substrates against their concentration gradient. Secondary active transporters used gradients created by primary active transport to transport drug substrates. Facilitated transporters only transport substrates down their concentration gradient. Transporters that transport drug substrates are shown in red, whereas those shown in gray provide driving forces for drug transport.

substrates out of cells. Transporters are usually responsible for either drug influx or drug efflux, but in some cases facilitate both types of transport.

Secretory Versus Absorptive Transporters

Secretory transporters facilitate drug clearance and are responsible for transport of drugs from the blood, such as excretion into the urine. Absorptive transporters allow a drug access into the blood, such as those facilitating absorption in the gut or reabsorption in the kidney. A transporter can be both secretory and absorptive depending on its physiologic locations. The same transporter is often expressed at both the site responsible for drug absorption and that responsible for drug clearance; hence, the transporter would serve both secretory and absorptive functions. When considering transport across membranes of the brain and placenta, the terms secretion and absorption are applied differently. With respect to these sites, absorptive transporters allow access into the sites and out of the circulation, and secretory transporters efflux drugs out of these sites and back into the blood circulation.

ABC Versus SLC Transporters

Drug transporters have been classified into two families, namely the ABC (ATP-binding cassette) and SLC (solute carrier) families. Members of the ABC family are primary active transporters that use ATP as an energy

source. Most SLC transporters are secondary active transporters, using the concentration gradients of several different molecules as a driving force for transport. The SLC family also includes a handful of facilitative transporters. Many members of each family have a role in drug disposition, and these transporters are summarized in Table 5.1.

ABC Transporters

ABC transporters are efflux transporters, many of which are located on the apical side of biological membranes, facilitating drug secretion. ABC transporters are primary active transporters and use ATP as an energy source. The widespread physiologic expression of ABC transporters and their extensive range of substrates make it inevitable for these transporters to have effects on drug pharmacokinetics and elicit clinically significant drug interactions. Substrates and inhibitors for ABC transporters are given in Table 5.2.

MULTIDRUG RESISTANCE (MDR) PROTEINS (ABCB)

MDR1 (ABCB1) MDR1, more commonly known as P-glycoprotein (P-gp), is the most extensively researched of all the transporters. The structure of P-gp consists of two homologous halves each with six transmembrane-spanning domains and one ATP binding site (Fig. 5.3). Transport by P-gp has been discovered to be notably complicated, with at least two binding sites and three different proposed mechanisms by which it transports substrates (1).

TABLE 5.1 Drug Transporters

Transporter Family	Family Member	Gene Name	Location	Role in Drug Disposition
MDR	MDR1	<i>ABCB1</i>	Intestine, liver, kidneys, brain, heart, placenta	• Role in oral absorption, biliary clearance, renal secretion, and drug penetration of blood–brain barrier • Drug–drug interactions • Relevant polymorphisms • Role in multidrug resistance
MRP	MRP1	<i>ABCC1</i>	Liver, intestine, brain	• Role in multidrug resistance • Facilitates basolateral drug efflux • Role in oral absorption, biliary clearance, and renal secretion of drugs and drug conjugates • Facilitates sinusoidal efflux of drugs and drug conjugates • Facilitates renal secretion and sinusoidal efflux of drugs and drug conjugates
	MRP2	<i>ABCC2</i>	Intestine, liver, kidney	
	MRP3	<i>ABCC3</i>	Liver, brain	
	MRP4	<i>ABCC4</i>	Kidney, liver, brain	
BCRP	BCRP1	<i>ABCG2</i>	Intestine, liver, kidney, brain, heart, placenta	• Role in oral absorption, biliary secretion, and drug penetration of blood–brain barrier • Drug–drug interactions • Relevant polymorphisms • Role in multidrug resistance
OCT	OCT1	<i>SLC22A1</i>	Liver	• Facilitates sinusoidal uptake • Drug–drug interactions • Relevant polymorphisms • Facilitates renal secretion • Drug–drug interactions • Relevant polymorphisms
	OCT2	<i>SLC22A2</i>	Kidney	
OCTN	OCTN1	<i>SLC22A4</i>	Kidney, intestine	• Role in oral absorption and renal secretion • Role in oral absorption and renal secretion
	OCTN2	<i>SLC22A5</i>	Kidney, intestine	
OAT	OAT1	<i>SLC22A6</i>	Kidney	• Facilitates renal secretion • Drug–drug interactions • Facilitates sinusoidal uptake • Facilitates renal secretion • Drug–drug interactions • Facilitates renal reabsorption
	OAT2	<i>SLC22A7</i>	Liver	
	OAT3	<i>SLC22A8</i>	Kidney	
	OAT4	<i>SLC22A11</i>	Kidney	
OATP	OATP1A2	<i>SLCO1A2</i>	Brain, intestine	• Role in oral absorption and brain uptake • Facilitates sinusoidal uptake • Drug–drug interactions • Relevant polymorphisms • Facilitates sinusoidal uptake • Relevant polymorphisms • Role in oral absorption and sinusoidal uptake • Role in oral absorption and sinusoidal uptake • Facilitates renal secretion
	OATP1B1	<i>SLCO1B1</i>	Liver	
	OATP1B3	<i>SLCO1B3</i>	Liver	
	OATP2A1	<i>SLCO2A1</i>	Intestine, liver	
	OATP2B1	<i>SLCO2B1</i>	Intestine, liver	
MATE	OATP4C1	<i>SLCO4C1</i>	Kidney	
	MATE1	<i>SLC47A1</i>	Kidney, liver	• Role in renal and biliary secretion • Drug–drug interactions • Facilitates renal secretion
	MATE2-K	<i>SLC47A2</i>	Kidney	
MCT	MCT1	<i>SLC16A7</i>	Ubiquitous	• Role in oral absorption and renal reabsorption • Overexpression in tumor cells
PEPT	PEPT1	<i>SLC15A1</i>	Intestine, kidney	• Facilitates oral absorption • Drug–drug interactions • Facilitates renal reabsorption
	PEPT2	<i>SLC15A2</i>	Kidney	
CNT	CNT1	<i>SLC28A1</i>	Intestine, kidney, liver	• Role in absorption and disposition of nucleoside analogs • Role in absorption and disposition of nucleoside analogs • Role in distribution and renal reabsorption of nucleoside analogs
	CNT2	<i>SLC28A2</i>	Intestine, Kidney, liver	
	CNT3	<i>SLC28A3</i>	Kidney, brain, placenta	
ENT	ENT1	<i>SLC29A1</i>	Ubiquitous	• Role in absorption and disposition of nucleoside analogs • Role in absorption and disposition of nucleoside analogs
	ENT2	<i>SLC29A2</i>	Ubiquitous	

TABLE 5.2 Relevant Drug Substrates and Inhibitors of ABC Transporters

Transporter	Substrates	Inhibitors
MDR1	Amphotericin B, cyclosporine, daunorubicin, dexamethasone, digoxin, diltiazem, docetaxel, doxorubicin, erythromycin, etoposide, fexofenadine, hydrocortisone, ketoconazole, methadone, mitoxantrone, morphine, paclitaxel, phenytoin, rifampin, ritonavir, saquinavir, tacrolimus, verapamil, vinblastine, vincristine	Clarithromycin, cyclosporine, elacridar, quinidine, verapamil
MRP2	Ampicillin, ceftriaxone, cisplatin, daunomycin, doxorubicin, etoposide, fluorouracil, glucuronide conjugates, glutathione conjugates, irinotecan, methotrexate, mitoxantrone, olmesartan, pravastatin, ritonavir, rosuvastatin, saquinavir, SN-38-glucuronide, valsartan, vinblastine, vincristine	Cyclosporine, probenecid, efavirenz, emtricitabine
MRP3	Fexofenadine, glucuronide conjugates, methotrexate	Efavirenz, emtricitabine
MRP4	Adefovir, furosemide, methotrexate, tenofovir, topotecan	Diclofenac
BCRP	Ciprofloxacin, imatinib, irinotecan, methotrexate, mitoxantrone, nitrofurantoin, rosuvastatin, sorafenib, topotecan	Elacridar

P-gp is expressed in most tissues and is involved in drug transport in the intestine, liver, kidney, brain, and placenta, and in tumor cells. This is one reason P-gp has been and remains a transporter of clinical interest. The ability of P-gp to transport a wide range of substrates is another remarkable feature. These substrates span many therapeutic areas and drug classes, making it difficult to determine the structure–activity relationship for P-gp-mediated transport. Digoxin serves as a probe for P-gp in both in vitro and in vivo studies (2). Other substrates include many chemotherapeutic drugs and HIV protease inhibitors. One common feature of P-gp substrates is that they are generally lipophilic, commonly containing aromatic rings (Fig. 5.4). The relative lipophilicity of these substrates implies that most P-gp substrates are also capable of passive diffusion. This means that P-gp transport must be efficient to have an effect on the passage of its substrates across membranes, and P-gp transport does not always result in clinically relevant effects. P-gp substrates are rarely specific and are also often substrates of other transporters and drug-metabolizing enzymes,

particularly CYP3A4, making it difficult to attribute in vivo pharmacokinetic effects or interactions entirely to P-gp alone. Intestinal efflux by P-gp has been demonstrated to contribute to the low bioavailability of many compounds (3). Significant drug interactions have also been demonstrated to involve P-gp in the intestine, as well as in the kidneys (4,5). P-gp is also likely important in pregnancy in that it protects the fetus from xenobiotics by effluxing them back into the maternal blood circulation (6). In addition, as the original name “multidrug resistance protein” (MDR) implies, P-gp is also recognized as an important factor for chemotherapeutic drug resistance due to its overexpression in tumor cells, and P-gp inhibitors have recently gained interest as concurrent treatment with chemotherapy (7).

MULTIDRUG RESISTANCE–ASSOCIATED PROTEINS (ABCC)

MRP2 (ABCC2) Multidrug resistance–associated protein 2 (MRP2) is primarily localized to three apical membrane barriers, the liver canalicular membrane, the brush border membrane in the kidney, and the

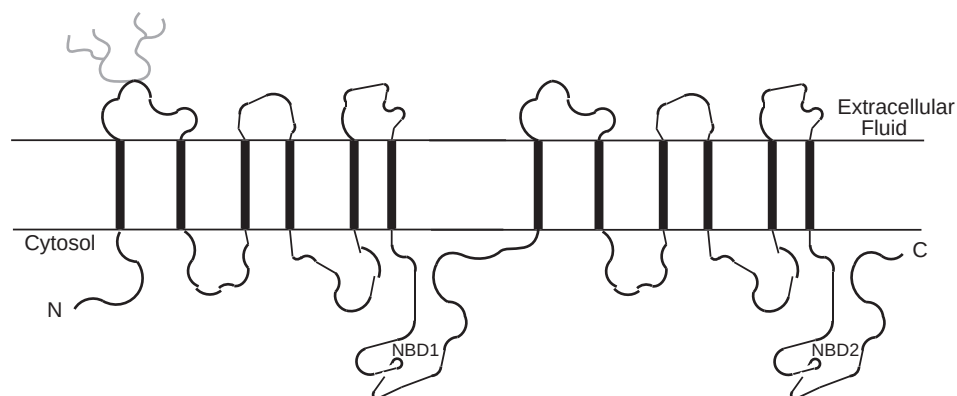


FIGURE 5.3 Structure of P-glycoprotein including 12 transmembrane-spanning domains and 2 nucleotide-binding domains (NBDs).

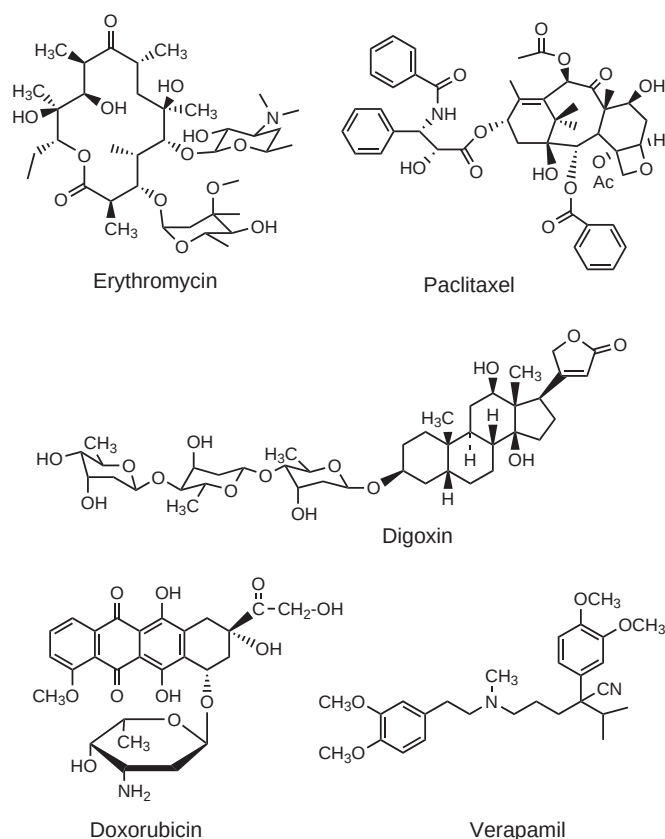


FIGURE 5.4 Examples of P-gp substrates.

apical membrane of the gut, where it serves to excrete substrates into the bile, urine, and back into the gut lumen, respectively. The major substrates of MRP2 are conjugates, including glucuronide, glutathione, and sulfate conjugates, and the primary function of MRP2 is the secretion of bile acid conjugates across the canalicular membrane into the bile. MRP2 substrates also include phase 2 drug metabolites and many unconjugated therapeutic drugs including vinca alkaloids. Glutathione symport is often required for transport of unconjugated substances (8). MRP2 can contribute to the renal secretion of some drug substrates; however, the main role of MRP2 in drug disposition appears to be hepatobiliary transport (1,9). Due to transport of drug conjugates, MRP2 has an important role in enterohepatic cycling. MRP2 transports conjugates into the bile, which empties into the gut lumen. Bacteria in the gut metabolize the conjugate back to the parent drug, which can then be reabsorbed back into the systemic blood circulation. In this way, MRP2 can be responsible for maintaining plasma concentrations of drugs that are not substrates, due to transport of their conjugated metabolites. P-gp is also expressed on the canalicular membrane, although it does not appear to have a role in biliary secretion of drug conjugates.

Other MRPs

In contrast to the apical expression of MRP2, MRPs 1, 3, 4, 5, and 6 are basolateral efflux transporters. MRP1 is another ABC transporter that can be overexpressed in cancer cells, resulting in resistance to chemotherapy (8). MRP3 has an important role in the transport of conjugated drug metabolites in the liver, particularly glucuronide conjugates. Located on the sinusoidal membrane, MRP3 serves to efflux metabolites out of hepatocytes into the plasma, resulting in conjugate excretion into the urine. MRP3 also transports conjugated bile acids and can be induced in cholestatic conditions (10). This also occurs with MRP4, which is located on the sinusoidal membrane as well. MRP4 is also expressed on the brush border membrane of the kidney and has a role in anionic drug secretion into the urine (9).

BREAST CANCER RESISTANCE PROTEIN

BCRP (ABCG2) Breast cancer resistance protein (BCRP) is actually a “half” transporter, in that it consists of only six transmembrane-spanning domains and one ATP binding site (Fig. 5.5). Like P-gp, BCRP expression is widespread. Its highest expression is in the placenta, and it is present in the intestine, liver, kidney, brain, heart, and ovaries (11). As its name implies, BCRP has also been attributed with causing chemotherapeutic drug resistance due to overexpression in tumor cells, and among its substrates are several chemotherapeutic agents. BCRP shares many substrates with P-gp and is commonly localized with P-gp, making it difficult to attribute drug transport or inhibition to BCRP. However, the use of BCRP inhibitors and knockout animals has demonstrated the importance of BCRP in drug disposition. Significant effects of BCRP transport have been demonstrated with drug absorption, with biliary clearance, and interestingly on drug transport in the mammary gland, mediating the transport of some drugs into breast milk (12–14). BCRP has a similar role to P-gp in pregnancy, as it is highly expressed at the

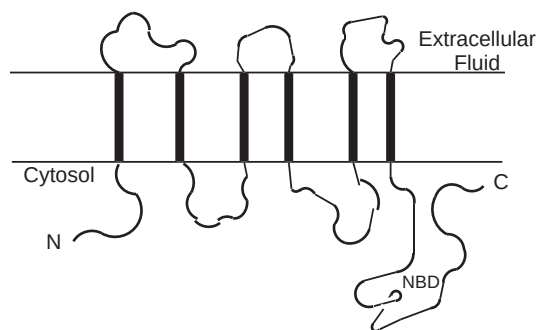


FIGURE 5.5 Structure of BCRP. BCRP is regarded as a half transporter due to its structure including only six transmembrane-spanning domains and one nucleotide-binding domain (NBD).

blood–placenta barrier, where it also serves a protective function to efflux potentially harmful xenobiotics (6).

SLC Transporters

Members of the SLC transporter family are secondary active transporters and, as such, use the concentration gradients of many other substances to transport drug substrates. The concentration gradients of these other substances generally facilitate drug influx, although in some physiologic locations, they can facilitate efflux, and some transporters can be capable of transport in both directions. Like the ABC transporters, SLC transporters are physiologically expressed throughout the body. They transport many different xenobiotics along with many endogenous substances, leading to effects on drug disposition and drug interactions. Substrates and inhibitors of SLC transporters are given in Table 5.3.

ORGANIC ANION TRANSPORTING POLYPEPTIDES (SLCO)

OATP1B1 (SLCO1B1) This transporter is one of the most clinically relevant transporters in that its substrates include drugs such as the 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors, or statins, which are widely prescribed. This transporter

is localized specifically on the sinusoidal membrane of the liver, where it serves to influx substrates into hepatocytes, where they can be metabolized or transported into bile. Organic anion transporter polypeptide (OATP) 1B1 and other OATPs are antiporters, using the high intracellular concentration of glutathione as a driving force. Although the name implies anion transport, OATPs are also capable of transporting cationic and neutral compounds, with OATP1B1 displaying the broadest substrate specificity overall. OATP1B1 transports many endogenous substrates including bilirubin and estrogen conjugates. Common drug substrates of OATP1B1 and other OATPs are given in Figure 5.6. Inhibitors of this transporter are also commonly used in the clinic for similar or concomitant disease states and include drugs like gemfibrozil. Rifampin functions as an inhibitor as well as being a substrate. This transporter has also gained attention due to the recognition that it is highly polymorphic, with a large number of single nucleotide polymorphisms (SNPs) having been identified in the *SLCO1B1* gene. Many of these SNPs have been demonstrated to translate into decreased hepatic uptake in vivo, resulting in decreased clearance of statins and other OATP1B1 substrates (15).

TABLE 5.3 Relevant Drug Substrates and Inhibitors of SLC Transporters

Transporter	Substrates	Inhibitors
OATP1B1	Atorvastatin, bosentan, caspofungin, cerivastatin, enalapril, fexofenadine, fluvastatin, methotrexate, olmesartan, pravastatin, repaglinide, rifampin, rosuvastatin, simvastatin, SN-38, valsartan	Atorvastatin, clarithromycin, cyclosporine, erythromycin, gemfibrozil, paclitaxel, rifampin, ritonavir, saquinavir, tacrolimus, telmisartan
OATP1B3	Digoxin, docetaxel, fexofenadine, olmesartan, paclitaxel, rifampin	Clarithromycin, cyclosporine, erythromycin, rifampin, ritonavir
OATP1A2	Digoxin, erythromycin, fexofenadine, imatinib, levofloxacin, methotrexate, pitavastatin, rocuronium, rosuvastatin, saquinavir	Grapefruit juice, rifampin, ritonavir, saquinavir, verapamil
OATP2B1	Atorvastatin, bosentan, fexofenadine, pravastatin, rosuvastatin	Cyclosporine, gemfibrozil, rifampin
OAT1	Acyclovir, adefovir, cidofovir, ciprofloxacin, lamivudine, methotrexate, penicillins, tenofovir, zidovudine	Probenecid
OAT3	Bumetanide, cefaclor, ceftizoxime, furosemide, NSAIDs, penicillins	Probenecid
OCT1	Metformin, lamivudine, oxaliplatin	Disopyramide, quinidine
OCT2	Cisplatin, lamivudine, metformin, oxaliplatin, procainamide	Cetirizine, cimetidine, quinidine
MATE1	Metformin, oxaliplatin	Cimetidine, quinidine, procainamide
MCT1	Atorvastatin, salicylic acid, pravastatin, valproic acid, γ -hydroxybutyric acid	Dietary flavonoids
PEPT1	Captopril, cefadroxil, cephalexin, enalapril, valacyclovir	Glycyl-proline, zinc
PEPT2	Captopril, cefadroxil, cephalexin, enalapril, valacyclovir	Fosinopril
CNT/ENT	Clofarabine, gemcitabine, ribavirin, other nucleoside analogs	Dipyridamole

NSAIDs, nonsteroidal anti-inflammatory drugs.

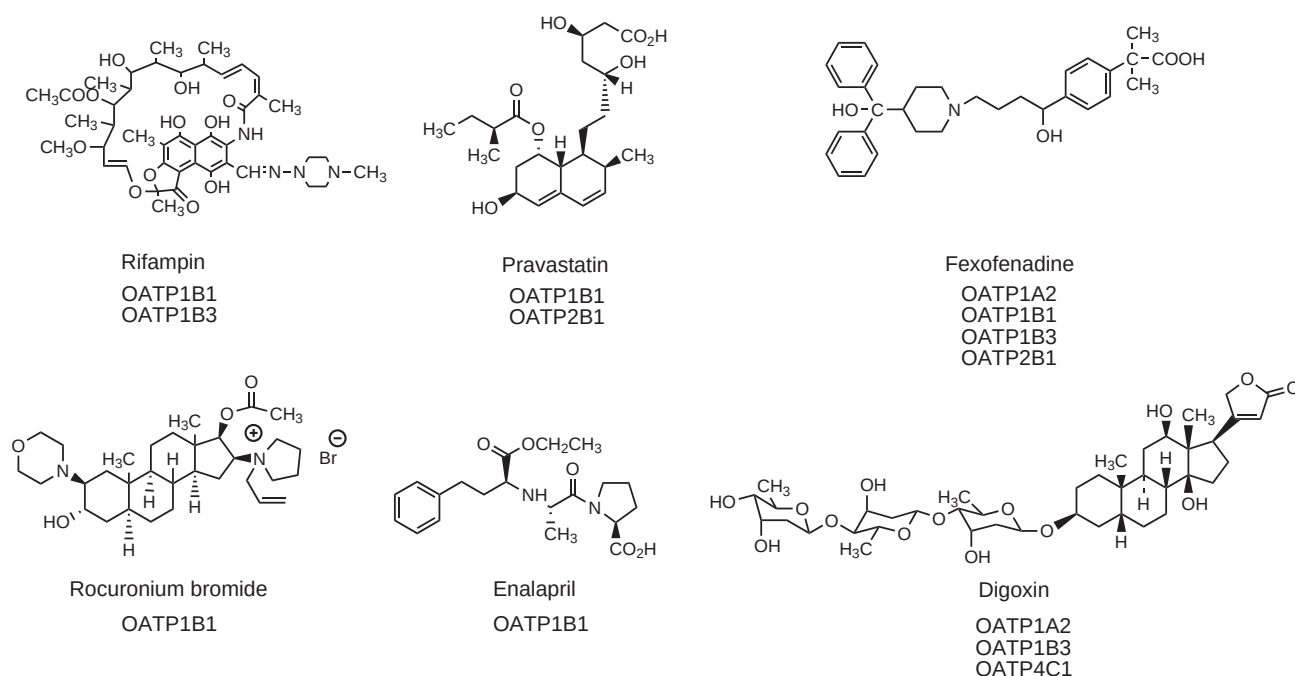


FIGURE 5.6 Common OATP substrates.

Other OATPs

OATP1B3 is expressed like OATP1B1 exclusively on the sinusoidal membrane of the liver. The substrates for OATP1B3 are generally shared by OATP1B1, with the exceptions of digoxin, docetaxel, and paclitaxel (15). OATP1B3 can act as a compensatory transport mechanism when transport by OATP1B1 is deficient due to inhibition or genetic polymorphisms. The other OATPs with identified drug substrates and effects on drug disposition include OATP1A2, 2B1, and 4C1. OATP1A2 is expressed in the intestine and expressed highly in the brain and is responsible for both oral absorption and blood–brain barrier transport of drugs and endogenous substances. OATP2B1 is also expressed on the sinusoidal membrane of the liver with OATP1B1 and 1B3; however, unlike these transporters, it is also expressed in other tissues including the kidney and intestine. OATP4C1 is expressed on the basolateral membrane of the kidney. Among its substrates are digoxin and the antidiabetic drug sitagliptin, and this transporter can have a role in the renal secretion of these drugs (16,17).

Organic Anion Transporters (SLC22A)

The organic anion transporters (OATs) are highly expressed in the kidney where they have a significant role in the renal clearance of many anionic drugs. The driving forces and polarized renal expression of OATs support renal secretion of drug substrates from the blood circulation into the urine. OAT1 is expressed exclusively in the kidney, on the basolateral membrane of renal tubule cells, along with OAT3. These

transporters facilitate anion transport into renal tubule cells through antiport with intracellular dicarboxylates like α -ketoglutarate. Located on the apical membrane, OAT4 can transport drugs from the tubule cells into the urine, but it is likely primarily responsible for renal reabsorption (18). OATs transport an expansive range of drugs, and the importance of OAT expression in the kidney has been demonstrated with numerous clinically relevant drug interactions. Coadministration of OAT inhibitors can lead to decreased renal secretion of OAT substrates including penicillin antibiotics and diuretic agents (Tables 5.3 and 5.4) (9). Along with renal transport, OATs also mediate hepatic anion transport. The primary location of OAT2 is on the liver sinusoidal membrane, where it facilitates anion uptake into hepatocytes. OATs are also present on the choroid plexus (a structure in the brain where cerebrospinal fluid is produced), where they can have a role in transfer of drugs between the blood and cerebrospinal fluid.

Organic Cation/Carnitine Transporters (SLC22)

Organic cation transporters (OCTs) are electrogenic uniporters, which use only the cation concentration as a driving force. They require no cosubstrate, resulting in unequal distribution of ionic charges across the membrane. OCT1 is expressed in the liver and OCT2 in the kidney, both on the basolateral membrane. Although they are named cation transporters, these transporters are also capable of transporting some anionic and neutral compounds. Drug interactions and drug toxicity have been attributed to OCT-mediated

TABLE 5.4 Clinically Relevant Transporter-Mediated Drug Interactions^a

Transporter	Substrate	Inhibitor(s)	Effect on Substrate Pharmacokinetics
MDR1	Digoxin	Clarithromycin (4), cyclosporine (95), ritonavir (5)	↑ plasma AUC, ↓ Cl_R and/or ↑ F
	Paclitaxel	Cyclosporine (43)	↑ F
BCRP	Topotecan	Elacridar (13)	↑ F
	Rosuvastatin	Atazanavir/ritonavir (44)	↑ plasma AUC and C_{max}
OATP1B1/1B3	Statins	Gemfibrozil (54), ^b cyclosporine (58), ^b lopinavir/ritonavir (96)	↑ plasma AUC, ↓ Cl
	Bosentan	Lopinavir/ritonavir (97)	↑ AUC and C_{max}
OAT1/3	Acyclovir, cidofovir, furosemide, penicillins	Probenecid (9,74)	↑ plasma AUC, ↓ Cl_R
OCT2/MATE1	Metformin	Cimetidine (19)	↑ plasma AUC, ↓ Cl_R

AUC, area under the curve; Cl , clearance; Cl_R , renal clearance; F, bioavailability; C_{max} , peak plasma concentration.

^a Interactions that have been demonstrated in humans to elicit clinically significant effects on substrate pharmacokinetics or pharmacodynamics in vivo.

^b Also involves enzyme inhibition.

transport, due to their substrate range that includes commonly prescribed therapeutic agents, such as metformin (Table 5.3). Coadministration of OCT inhibitors (e.g., cimetidine) and genetic variation in OCT2 have been demonstrated to increase plasma concentrations of the antidiabetic drug metformin due to its decreased renal secretion (Tables 5.4) (19,20). Renal transport of cisplatin by OCT2 has also been identified as a significant factor in cisplatin-induced nephrotoxicity (21). OCT3 transports monoamines and is primarily located in the brain and placenta, where its primary role can be to eliminate catecholamines from the fetal blood circulation (22). Decreased OCT3 transport has also been suggested to be related to cases of preeclampsia (23).

Organic cation/carnitine transporter (OCTN) 1 and OCTN2 are proton antiporters and are primarily expressed in the kidney, where they are localized on the apical membrane. Expression of human OCTN1/2 mRNA has also been detected in the intestine (24). These transporters share substrates with OCTs, and their apical expression allows for the concerted movement of these substrates across membranes. These

transporters are of particular importance in the placenta, as they are responsible for transporting carnitine to the fetus (22). Some antiepileptics, like valproate, inhibit OCTNs, and it has been suggested that inhibition of carnitine transport can be partly responsible for teratogenicity and other adverse effects associated with valproate use (25). Mutations in OCTN2 lead to systemic carnitine deficiency syndrome, causing early-onset cardiomyopathy, including congestive heart failure (26). The anticholinergic drugs ipratropium and tiotropium are transported primarily by OCTN2 and, to a lesser extent, by OCTN1, by bronchial epithelial cells. These findings are consistent with the pharmacologic activity of the drugs after administration via inhalation.

Multidrug and Toxin Extrusion Transporters (SLC47A)

Multidrug and toxin extrusion transporter (MATE) 1 and MATE2-K are cation transporters that use proton antiport as a driving force. MATE1 is located primarily on the apical membrane of the kidney and canalicular membrane of the liver, where it acts in concert with OCTs expressed on

TABLE 5.5 Nuclear Receptors Involved in Transporter Regulation

Nuclear Receptor or Transcription Factor	Agents That Act on Receptor	Transporters Affected
Pregnane X receptor (PXR)	Rifampin, cytokines	MDR1, MRP2, OATP1A2
Constitutive androstane receptor (CAR)	Phenobarbital	MDR1, MRP2, MRP4
Farnesoid X receptor (FXR)	Bile acids	MRP2, MRP3, MRP1, OATP1B3
Peroxisome proliferator-activated receptor γ (PPAR γ)	Thiazolidinediones	BCRP

the basolateral membranes of these organs to transport cationic substrates, including therapeutic agents. MATE2-K is kidney specific. The significance of MATE1 transport in metformin pharmacokinetics was characterized in MATE1 knockout mice, in which the renal clearance of metformin was decreased to less than 20% that of controls, indicating the importance of this transporter for metformin renal secretion (27). It has also been demonstrated that MATE1 has a role in the nephrotoxicity of platinum agents, where MATE1 secretion appears protective due to the prevention of renal cell accumulation (28,29). Many drug substrates and inhibitors of these transporters have been identified, and the therapeutic relevance of these transporters is expected to increase as their transport and effects on in vivo drug disposition are further characterized (Table 5.3) (30).

Monocarboxylate Transporters (SLC16A and SLC5A)

The first discovered monocarboxylate transporters (MCTs) were proton coupled, transporting monocarboxylates via symport. Because the transport of substrates by MCTs is driven by the proton gradient, these transporters can influx or efflux substrates depending on this gradient. The MCTs are ubiquitously expressed, and their role in the transport of endogenous substances, including lactate and pyruvate, has been extensively characterized (31). Many therapeutic agents have also been identified as substrates; however, the clinical relevance of MCT transport of these agents is unclear (Table 5.3). A relevant therapeutic aspect of MCT1 is its high expression in the intestine, making this transporter a target as a means for oral drug absorption (32). MCT1 has also been identified to be overexpressed in tumor cells, and inhibition of this transporter represents a possible therapeutic strategy for some cancers (33). The drug of abuse γ -hydroxybutyrate (GHB) has been identified as an MCT substrate, and the relevance of MCT transport of this drug has been established. Inhibition of the MCT-mediated renal reabsorption of GHB has been demonstrated to increase its renal and total clearance, making MCT inhibition a possible strategy for the treatment of GHB overdose (34).

Another subset of MCTs has been identified that use sodium for monocarboxylate symport and are referred to as sodium-coupled MCTs (SMCTs). Expression of SMCTs is also widely distributed, including the apical membranes of the intestine and kidney, where they likely act in conjunction with proton-coupled MCTs to transport monocarboxylates across membranes.

Peptide Transporters (SLC15A)

Peptide transporters (PEPTs) mediate the transport of di- and tripeptides via proton symport. PEPT1 has very high expression on the apical membrane of the intestine and, as such, is used for drug delivery. Transport mediated by PEPT1 facilitates sufficient oral absorption to allow oral administration for many therapeutic drugs and prodrugs structurally designed to target this transporter (Fig. 5.7 and

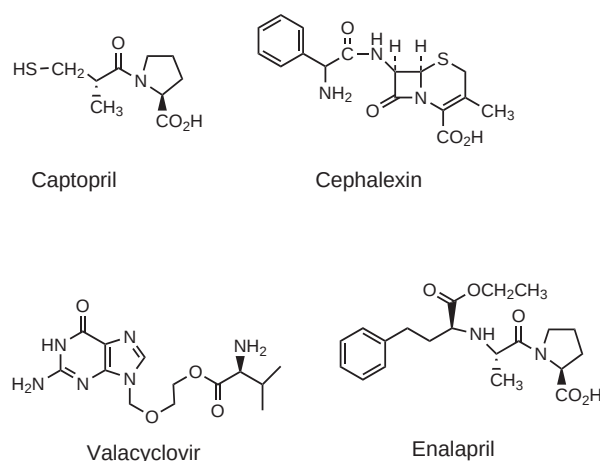


FIGURE 5.7 Drug substrates of peptide transporters.

Table 5.3) (32). PEPT1 is also present at lower levels on the brush border membrane of the kidney; however, PEPT2 is the major peptide transporter at this site. PEPT2 is also localized to the apical membrane where it serves mainly to reabsorb peptides from the urine in renal tubule cells. It also facilitates transport of drug substrates similar to PEPT1 including β -lactam antibiotics and angiotensin-converting enzyme (ACE) inhibitors (Table 5.3) (35). Two peptide/histidine transporters (PHTs) were recently identified, PHT1 and PHT2. Expression of these transporters has been demonstrated in the intestine, and they likely have a role in peptide absorption, along with PEPT1. Expression of mRNA for these transporters has also been identified in several other human tissues (36). The role of the PHTs in drug disposition has yet to be elucidated.

Nucleoside Transporters (SLC28 and SLC29)

Two nucleoside transporter groups exist, the concentrative nucleoside transporters (CNTs) and the equilibrative

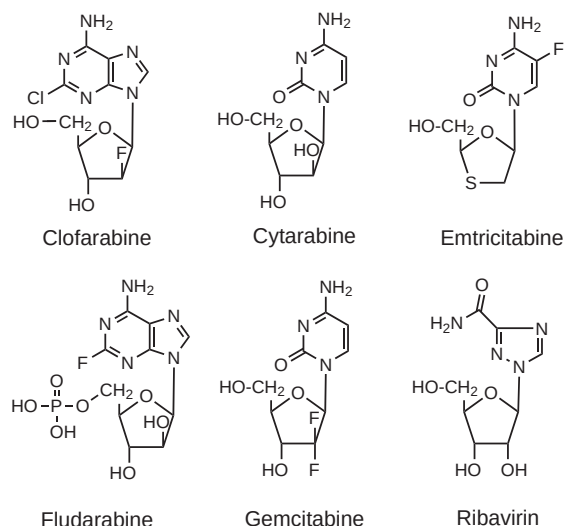


FIGURE 5.8 Drug substrates of nucleoside transporters.

EVALUATING TRANSPORTER EFFECTS IN VIVO

Many transporters share substrates with other transporters and with drug-metabolizing enzymes, making it difficult to attribute *in vivo* effects to a specific transporter. The use of probe substrates and specific inhibitors makes this identification easier, but these are not available for all transporters or may not be suitable for *in vivo* use. Knockout animals can be used to assess the effects of a specific transporter *in vivo*, whereas transfected cell lines or oocytes and specific silencing of a transporter using small interfering, double-stranded RNA can be useful to assess effects *in vitro*. When evaluating the effects of drug transport on bioavailability, it can be difficult to attribute effects to transport in the intestine alone, as changes in both bioavailability and clearance can affect plasma concentrations after oral administration. It is necessary to assess administration of substrates and inhibitors both orally and intravenously to determine if transport primarily affects substrate bioavailability or systemic clearance. Often both parameters are affected by changes in transport because the same transporters are responsible for both processes. Significant effects on absorption cannot always be demonstrated with efflux transporter substrates *in vivo*. This can be due to the very high luminal concentrations of substrate causing transporter saturation and to the passive membrane permeability of these compounds. A lack of effect can also be due to shared transporter substrates, allowing compensatory transport by one transporter when another is inhibited.

nucleoside transporters (ENTs). CNTs are present on the apical membrane of cells and are sodium dependent, whereas ENTs are facilitative and located primarily on basolateral membranes (37). These two groups of transporters work together to achieve vectorial transport of purines and pyrimidines. The expression of both groups of transporters is widespread, and although their primary function is to facilitate cellular uptake of nucleosides, they also transport many nucleoside analogs (Fig. 5.8), and these transporters are associated with toxicity and therapeutic response to these drugs (Table 5.3) (37,38).

EFFECT OF TRANSPORT ON DRUG PHARMACOKINETICS AND DRUG-DRUG INTERACTIONS

Intestinal Transport

The importance of transporter-mediated influx or efflux in the intestine depends primarily on two factors. The first is the contribution of transporter-mediated transport compared to the other mechanisms of drug passage across the gut wall. The second is drug concentration. Drug concentrations are generally much higher in the gut lumen compared to the systemic blood circulation due to a lower volume of distribution. This causes the concentration gradient driving diffusion to be very large but, more importantly, can lead to transporter saturation. This can lead to low

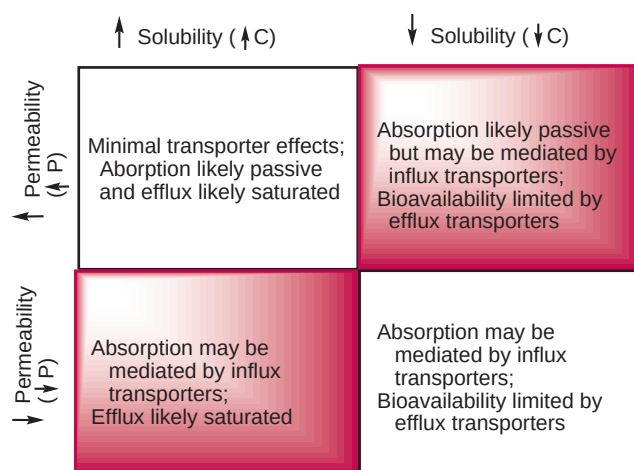


FIGURE 5.9 Role of transporters in drug absorption. High aqueous drug solubility will result in high luminal drug concentrations, allowing for possible transporter saturation. High permeability results in high nonsaturable transport, allowing these drugs to be orally absorbed without dependence on transporters. (Adapted from Shugarts S, Benet LZ. The role of transporters in the pharmacokinetics of orally administered drugs. *Pharm Res* 2009;26:2039–2054.)

bioavailability due to saturation of influx transporters and nonlinearity in drug absorption. Transporter saturation can also aid in drug absorption in that high drug concentrations in the enterocyte can saturate efflux transporters. High drug concentrations in the gut can also result in drug interactions affecting oral absorption. The U.S. Food and Drug Administration–recommended Biopharmaceutics Classification System names solubility and permeability as determining factors of oral absorption (39), factors which can be translated into effects on drug concentration (*C*) and nonsaturable transport (*P*) in the intestine. Considering these factors, the predicted effects of drug transporters on oral absorption are given in Figure 5.9 (40).

The polarized expression of relevant drug transporters in the intestine is illustrated in Figure 5.10. Clinical relevance of drug transport is determined not only by the presence of transporters on a biologic membrane, but also by their relative expression. The transporters with highest intestinal expression include BCRP, PEPT1, P-gp, and MCT1 (41). High expression of influx transporters, like PEPT1 and MCT1, allows the use of these high-capacity transporters as a drug delivery mechanism. High expression of efflux transporters makes these transporters important in limiting drug bioavailability.

Along with drug transporters, present in the enterocytes forming the gut wall are drug-metabolizing enzymes, including phase 1 and phase 2 enzymes. First-pass metabolism in the gut can be a reason for low oral bioavailability. Transporters often share substrates with drug-metabolizing enzymes, and the intestine is one site in which substantial transport–metabolism interplay has been recognized. In particular, it has been noted that P-gp shares many substrates with the enzyme CYP3A4, and it is proposed that P-gp can prevent oral absorption through drug efflux

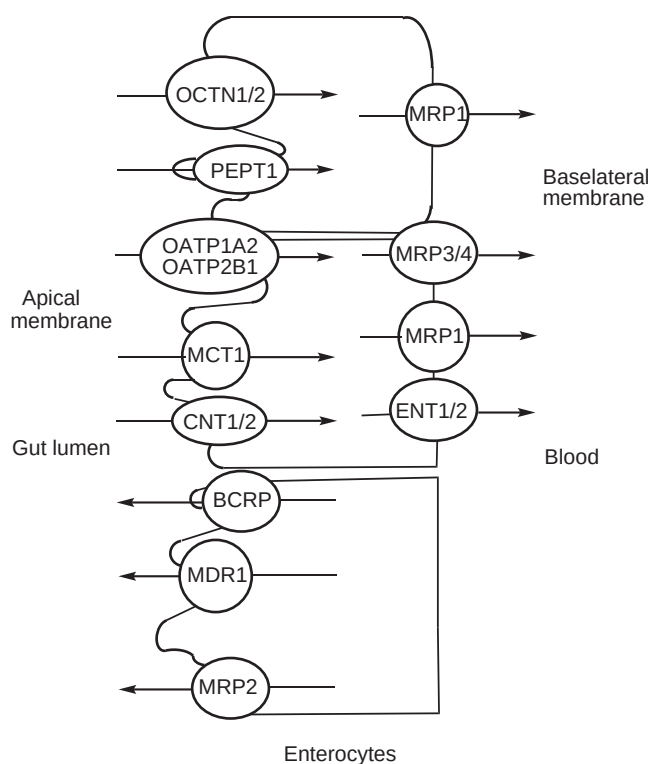


FIGURE 5.10 Polarized expression of drug transporters in the intestine.

and by facilitating drug metabolism by CYP3A4 (42). This is achieved by the cycling of substrates through the gut lumen and enterocytes, allowing enzymes such as CYP3A4

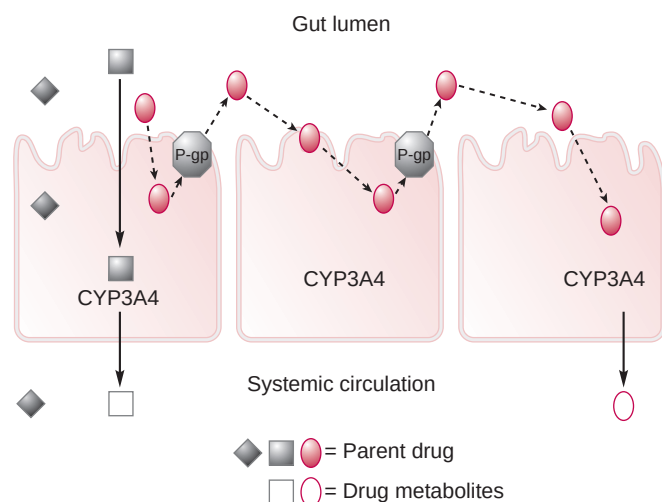


FIGURE 5.11 Transport-metabolism interplay in the intestine. Drugs may be orally absorbed without undergoing metabolic conversion. Drugs may undergo metabolism in the gut, and this may occur upon initial entry into the enterocyte or after cycling between the gut lumen and the enterocyte. P-gp facilitates CYP3A4 metabolism by allowing the enzyme multiple opportunities for drug metabolism, as demonstrated by the molecule shown in red. (Adapted from Benet LZ. The drug transporter-metabolism alliance: uncovering and defining the interplay. *Mol Pharm* 2009;6:1631–1643.)

multiple opportunities to metabolize their substrates. This interplay is depicted in Figure 5.11.

Efflux Transporters and Drug Bioavailability

Studies using specific P-gp inhibitors and Mdr1 knockout animals have demonstrated the importance of P-gp in the oral bioavailability of drugs including HIV protease inhibitors, β -receptor antagonists, morphine, and many other therapeutic agents (3). Many of the drug interactions in the intestine involve P-gp or other efflux transporters, usually involving inhibition of transport, resulting in increased bioavailability of the transporter substrate. Administration of P-gp inhibitors and inducers has demonstrated clinically significant effects on the bioavailability of P-gp substrates, including digoxin and paclitaxel (4,43).

Drug bioavailability can also be limited by BCRP. Administration of the P-gp/BCRP inhibitor elacridar increased the bioavailability of topotecan sixfold in Mdr1 knockout mice, and oral coadministration with elacridar has been shown to increase the bioavailability of topotecan in humans to almost 100% (12,13). Similarly, oral coadministration of the statin rosuvastatin, a BCRP substrate, and the antivirals atazanavir/ritonavir, known to inhibit BCRP, significantly increased rosuvastatin exposure in humans, most likely due to inhibition of BCRP-mediated intestinal efflux and biliary secretion (44).

Influx Transporters and Drug Bioavailability

Select influx transporters in the intestine can be used to facilitate oral absorption. The high expression of PEPT1

P-GP AND DIGOXIN: CLINICAL SIGNIFICANCE

Digoxin undergoes little metabolism in humans, and transport by P-gp is the primary determinant of digoxin pharmacokinetics. One of the first drug interactions noted with digoxin was that with cyclosporine, leading to decreased digoxin clearance and digoxin-associated arrhythmias (45). It was later discovered that cyclosporine and other therapeutic agents decrease digoxin clearance due to inhibition of P-gp-mediated renal secretion (46). Nonrenal clearance of digoxin is minimal; however, P-gp inhibitors have similarly been demonstrated to decrease digoxin biliary excretion (47). P-gp function also affects digoxin absorption, and concomitant oral administration of clarithromycin was demonstrated to increase digoxin bioavailability along with decreasing digoxin renal clearance (4). Because in vivo P-gp inhibition can affect both absorption and clearance, there exists substantial risk of elevated digoxin plasma concentrations and digoxin-associated toxicity with concomitant inhibitor administration.

P-gp induction can also affect digoxin pharmacokinetics. In a study of human volunteers, chronic rifampin administration resulted in a 3.5-fold increase in duodenal P-gp expression leading to a decrease in digoxin bioavailability (48). P-gp expression correlated well with digoxin area under the curve (AUC) in this study, emphasizing the significance of intestinal P-gp on digoxin exposure.

MCT₁ AND ORAL DRUG DELIVERY

The antiepileptic agent, gabapentin, has unfavorable oral pharmacokinetics, with low, dose-dependent bioavailability and poor rates of clinical response. In an effort to improve the oral delivery of gabapentin, a prodrug, gabapentin enacarbil, was designed to target absorption by MCT₁. In a pilot study comparing the pharmacokinetics of gabapentin to the prodrug, gabapentin displayed dose-dependent bioavailability ranging from 27% to 65%, whereas gabapentin enacarbil displayed bioavailability above 68% for all doses and dose-proportional oral exposure (52). Gabapentin enacarbil resulted in higher plasma concentrations of gabapentin at similar doses compared to gabapentin itself. This case emphasizes the utility of high-capacity transporters like MCT₁ for facilitating oral drug delivery and achieving clinically desirable drug pharmacokinetics.

has made it a target for oral drug delivery, and among its substrates are ACE inhibitors, β -lactam antibiotics, and the antiviral drug valacyclovir. Nonlinearity in the oral pharmacokinetics of valacyclovir can be attributed to saturable PEPT₁-mediated absorption (49). Other transporters involved in oral drug delivery include OATP1A2 and 2B1, OCTN1/2, and MCT₁ (32).

Influx transporters can also be responsible for drug interactions in the intestine, although fewer drug interactions have been noted than for efflux transporters. Plasma concentrations of fexofenadine, an OATP1A2 substrate, are decreased with concomitant ingestion of grapefruit juice (50). This is most likely due to transporter inhibition by the furanocoumarins in grapefruit juice, to which drug interactions with intestinal P-gp have also been attributed (51).

Hepatic Transport

As a clearance organ, the liver can eliminate drug through two pathways, metabolism and biliary excretion. Uptake

into the liver is necessary for both of these processes, and for hydrophilic or ionic drugs, this is often a transporter-mediated process. Active transporters are also necessary for biliary excretion, as diffusion of drugs into the bile is unlikely due to the low volume of the bile canaliculus resulting in high drug concentrations. The transporters facilitating both drug uptake and drug excretion in the liver are depicted in Figure 5.12.

Role of Transporters in Drug Metabolism and Drug Interactions at the Sinusoidal Membrane

As with the intestine, there exists transporter–metabolism interplay in the liver because transporters regulate the distribution of drug substrates into the liver and therefore their access to drug-metabolizing enzymes. Just as the significance of a transporter on a drug's membrane permeability is determined by the contribution of that transport process to overall transport, uptake by transporters needs to be a rate-limiting process for uptake to be a significant factor in drug elimination. Uptake by OATP1B1 located on the sinusoidal membrane appears to be the rate-limiting process for elimination of many of its substrates, and changes in OATP1B1 transport have been demonstrated to affect drug clearance. The relevance of OATP transport has been demonstrated with the HMG-CoA reductase inhibitors, several of which are OATP1B1 substrates. Drug interactions resulting in OATP1B1 inhibition and numerous polymorphisms in the *SLCO1B1* gene demonstrate decreased clearance of statins and other OATP substrates, including repaglinide, olmesartan, and the active metabolite of irinotecan, SN-38 (15,53). Polymorphisms in the other sinusoidal uptake transporters have also been identified, and effects on their drug substrates emphasize their role in hepatic uptake. Polymorphism in *SLCO1B3* has been correlated with increased myelosuppression associated with docetaxel, and polymorphisms in *SLC22A1* have been demonstrated to be related to decreased metformin clearance (53).

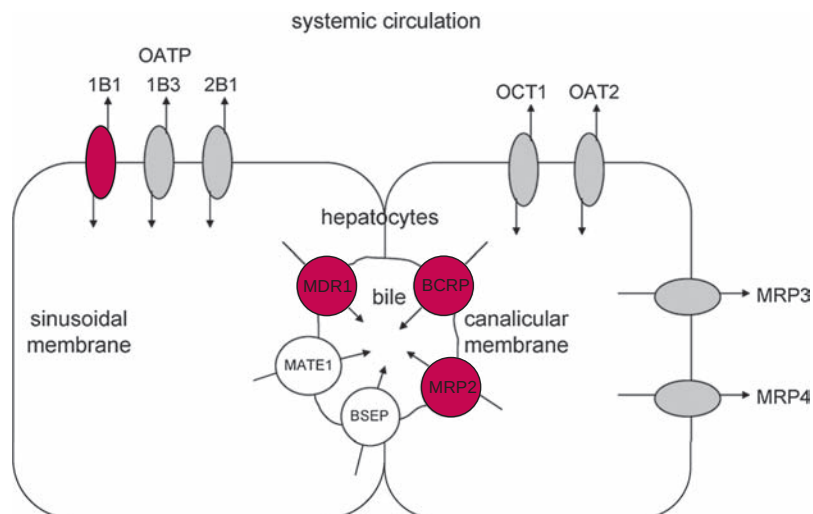


FIGURE 5.12 Drug transporter expression in hepatocytes. Transporters most relevant to drug disposition are shown in red.

OATP1B1 AND STATIN DISPOSITION

Although the metabolism and disposition of statin drugs can involve many drug-metabolizing enzymes and drug transporters, OATP1B1 has emerged as a protein of significance, with increasing clinical evidence supporting its involvement in the clearance, response, and toxicity of these drugs. One of the first interactions noted was the several-fold increase in cerivastatin plasma concentrations following gemfibrozil administration, resulting in severe cases of rhabdomyolysis (54). This drug interaction was determined to be due to inhibition of OATP1B1 and CYP2C8 by gemfibrozil (55). Gemfibrozil has been shown to increase the plasma AUC of other statins, which undergo little metabolism, indicating the significance of OATP-mediated transport (15,55,56). Similar effects on statin pharmacokinetics have also been demonstrated with other OATP1B1 inhibitors, including cyclosporine and rifampin (57–59). The effect of OATP1B1 polymorphisms on statin pharmacokinetics and response further emphasizes the significance of OATP-mediated uptake. Studies of subjects with the common SLCO1B1 c.521CC genotype have reported increases in plasma AUC from 65% with rosuvastatin to 221% with simvastatin compared to subjects carrying the wild-type allele (15). Polymorphisms in SLCO1B1 have also been correlated with increased rates of myopathy, along with decreased lipid-lowering effects (60,61). Alterations in the function of this transporter can significantly affect the safe and effective use of statins, and the use of OATP inhibitors with statin therapy can require dosage adjustment. SLCO1B1 genotyping may become useful for individualizing statin therapy.

Role of Transporters in Biliary Drug Excretion and the Effect of Drugs on Canalicular Membrane Transport

Some of the first drug interactions discovered to affect biliary clearance of drugs in humans involved the ABC transporter P-gp. It was demonstrated that quinidine and verapamil, known P-gp inhibitors, significantly decreased the biliary clearance of digoxin, resulting in increased digoxin plasma concentrations (47,62). P-gp efflux also has a role in the biliary excretion of doxorubicin, as demonstrated by an 80% to 90% decrease in doxorubicin biliary clearance in Mdr1 knockout mice (63,64). MRP2 also mediates biliary excretion of doxorubicin, along with that of other therapeutic agents including pravastatin and valsartan (Table 5.2) (64–66). MRP2 has a significant role in the biliary excretion of drug conjugates and in enterohepatic circulation, but it appears that BCRP has a more significant role in the excretion of sulfate conjugates, along with its role in secreting unconjugated drugs. Decreased biliary excretion of acetaminophen sulfate and other sulfate conjugates has been reported to be decreased in mice lacking Bcrp but not in those lacking Mdr2 (67,68). BCRP knockout or inhibition results in decreased biliary clearance of therapeutic agents including topotecan, nitrofurantoin, and ciprofloxacin (12,14,69). In addition, polymorphisms in the *ABCG2* gene have been shown to

correlate with increased response to rosuvastatin, which can be explained by increased concentrations in the liver, the site of action, due to decreased biliary excretion by BCRP (70). Drugs have also been demonstrated to interact with the bile salt export pump (BSEP), which is expressed at the canalicular membrane and is responsible for bile acid secretion. Drugs with significant hepatotoxicity, such as cyclosporine, rifampin, and troglitazone, have been demonstrated to inhibit BSEP, leading to liver injury presumably due to inhibition of bile acid transport (71,72). The primary metabolite of troglitazone, troglitazone sulfate, has been demonstrated to inhibit BSEP and likely contributes to troglitazone-induced hepatotoxicity (71).

Renal Transport

Renal clearance is mediated by three primary pathways: glomerular filtration, tubular secretion, and renal reabsorption. Small-molecule drugs that are not protein bound will be filtered at the glomerulus. Drugs that undergo glomerular filtration can have negligible renal clearance if they are sufficiently lipophilic to be passively reabsorbed from the tubular lumen back into systemic circulation. Drugs that are hydrophilic or ionized will usually be passively excreted into the urine unless they are actively reabsorbed by transporters. Drugs can have extensive renal clearance, sometimes much greater than the glomerular filtration rate, if they undergo renal secretion. Renal secretion of drugs from the blood into urine requires active transport because drug concentrations in the tubular lumen are generally higher than those in the blood. The following equation can be used to determine renal clearance considering the possibility of all three processes:

$$Cl_r = (GFR * f_{up} + Cl_{rs}) (1 - FR)$$

where *GFR* is the glomerular filtration rate; *f_{up}* is the fraction unbound in the plasma; *Cl_{rs}* is the clearance by renal secretion; and *FR* is the fraction of unchanged drug that is filtered and secreted in the urine, which is reabsorbed. Transporters are involved in both active secretion and active reabsorption. Secretion and reabsorption require transport of drug across both apical and basolateral membranes, and the polarized expression of organic anion and organic cation transporters on both membranes allows directional transport of ionized substrates. The transporters responsible for active secretion and active reabsorption and their polarized expression in the kidney are illustrated in Figure 5.13.

Renal Clearance of Anions

Tubular secretion of anions across the basolateral membrane is mediated primarily by OAT1 and OAT3. Substrates of these transporters include β-lactam antibiotics, nonsteroidal anti-inflammatory drugs, and many other anionic therapeutic agents (Table 5.3). Probenecid is a known inhibitor of OATs, and coadministration with OAT substrates leads to decreased renal clearance of

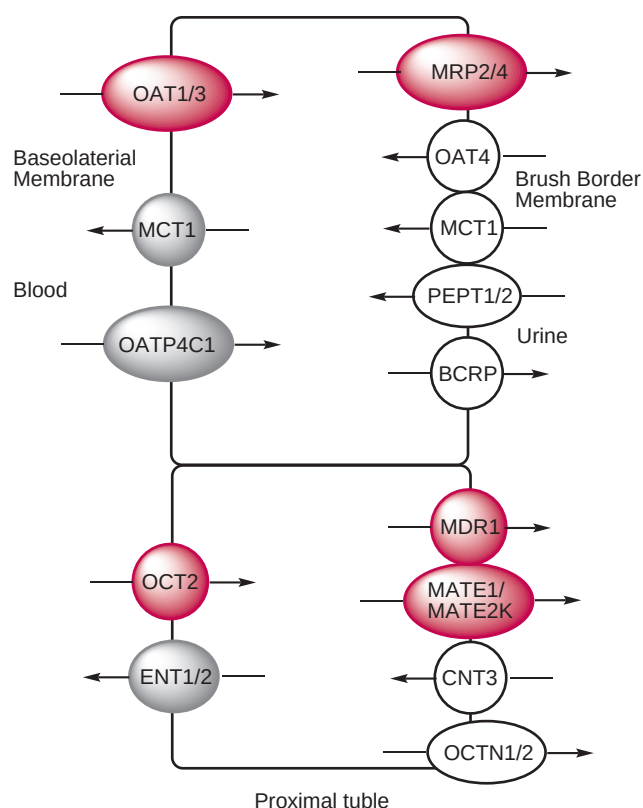


FIGURE 5.13 Drug transporter expression in the proximal renal tubule. Transporters most relevant to drug disposition are shown in red.

these substrates. One of the most highly recognized transporter interactions is that between probenecid and penicillin derivatives, and coadministration of probenecid has been demonstrated to decrease the renal clearance of numerous drugs of this class including piperacillin, nafcillin, ticarcillin, and others (9). Inhibition of OAT transport by probenecid also causes decreased renal clearance of diuretics like furosemide and results in diminished diuretic effect due to decreased tubular drug concentrations (73,74). OAT-mediated drug interactions affect renal secretion of many other OAT substrates including methotrexate, acyclovir, and zidovudine (9). Secretion of anions across the renal brush border membrane is primarily mediated by MRP2 and MRP4. Probenecid is also an MRP2 inhibitor, and some of its effects on renal clearance can be due to combined inhibition of OATs and MRP2. The role of MRP4 in renal anion secretion has recently been elucidated, and it can be of greater importance at this site than MRP2 because MRP4 expression was reported to be five times that of MRP2 in human renal cortex (75). MRP4 knockout mice display significant renal cell accumulation of the nucleotide phosphonates adefovir and tenofovir (76). Interestingly, cidofovir does not appear to be transported by MRP4, and this may be a reason for the significant nephrotoxicity associated with this drug. These drugs are also OAT substrates, and OAT inhibition can have therapeutic potential for preventing renal accumulation and toxicity with these agents (76).

RENAL CLEARANCE BY OCT2/MATE: CISPLATIN TOXICITY

The platinum agents are used clinically for treatment of a variety of cancers. Treatment with cisplatin has been limited by its adverse effects, including nephrotoxicity, ototoxicity, and neurotoxicity (78). Transport by OCT2 has been attributed with facilitating cisplatin toxicity in the kidney, demonstrated by decreased nephrotoxicity with coadministration of OCT2 inhibitors and in Oct2 knockout animals (79,80). Genetic polymorphism in OCT2 has also been correlated with decreased serum creatinine concentrations in patients undergoing cisplatin treatment (21). Oxaliplatin, another OCT2 substrate, is not associated with severe nephrotoxicity. Oxaliplatin is efficiently transported by MATE1 and MATE2-K, allowing for secretion of this drug from tubule cells into the urine (29). Cisplatin is a poor MATE substrate, and cisplatin has been demonstrated to accumulate in renal tubule cells (81). Carboplatin, another platinum-based agent, is a substrate for neither OCT2 or MATE transporters and, similarly to oxaliplatin, does not display nephrotoxicity (29). It is apparent that the effects of cisplatin in the kidney can be attributed to OCT2-mediated influx followed by inefficient apical efflux, resulting in renal cell accumulation and toxicity. Recently, OCT2 was also found to be present in the cochlea, and Oct2 knockout prevented cisplatin-induced ototoxicity, along with reducing nephrotoxicity (80). OCT2 inhibition represents a potentially clinically relevant therapy for the prevention of both toxicities during cisplatin therapy.

Renal Clearance of Cations

Renal secretion of cations involves transport across the basolateral membrane by OCT2 followed by transport across the brush border membrane by MATE1, OCTNs, or possibly P-gp. Metformin is a substrate of both OCT2 and MATE1, and inhibition or knockout of either of these transporters results in significant decreases in metformin renal clearance (19,27). Genetic variants of both OCT2 and MATE1 have also been correlated with decreased metformin clearance and/or increased glucose-lowering effects with metformin treatment (20,77). Platinum-based chemotherapeutic agents also use the OCT2/MATE transport system, and this transport is a factor in toxicity of these agents (23).

Blood–Brain Barrier Transport

The blood–brain barrier (BBB) serves an important protective function for the CNS and presents both structural and metabolic barriers for xenobiotic penetration into the brain. The endothelial cells of the BBB form tight junctions with minimal fenestrations, limiting BBB penetration to drug molecules that are capable of transcellular diffusion. For entry of polar or charged essential nutrients such as amino acids and glucose, transport processes are necessary. Transporters present on the BBB for this function have been exploited to facilitate brain penetration of agents like levodopa, one of the most commonly used

agents in Parkinson disease, which is transported into the brain by the L-type amino acid transporter (LAT) 1 (82). Influx transporters including LAT1 remain important for brain delivery of select drugs; however, attention has recently shifted to the efflux transporters expressed at the BBB, including P-gp, BCRP, and MRPs. Secretion of drug into the blood by these transporters has been recognized as another important barrier for drug penetration into the brain. Mdr1 knockout mice display substantially higher brain/plasma ratios for many P-gp substrates, with over 10-fold increases exhibited for digoxin, paclitaxel, and protease inhibitors compared to wild-type animals (83–85). Inhibition of P-gp has been used as a strategy to increase BBB penetration and has been demonstrated to be effective in animal models (43,84). The use of P-gp inhibitors in studies with human subjects shows similar but much more modest effects on brain penetration. This decreased effect is due in part to the lower inhibitor concentrations reached at doses used in human subjects, and evaluating the effects of these inhibitors at relevant concentrations is a priority for determining the utility of this strategy (85). Bcrp knockout animals have demonstrated the ability of this efflux transporter to limit brain penetration of drugs such as sorafenib (86). Many anticancer drugs are cosubstrates for P-gp and BCRP, and knockout of both transporters has demonstrated synergistic effects on brain accumulation of these drugs (86–88). The use of coinhibitors represents a potential strategy for improving treatment of brain tumors with these agents (43).

Regulation of Transporters and Interindividual Variability

Transcriptional regulation of transporters, effects mediated by nuclear receptors, can contribute to interindividual variability. The nuclear receptors that affect transporter expression are summarized in Table 5.5. Rifampin, an agonist of the nuclear receptor pregnane X (PXR), induces the expression of several enzymes and transporters including CYP3A4 and MDR1. During rifampin treatment, decreased drug exposure is observed with CYP3A4/MDR1 cosubstrates such as cyclosporine, calcium channel blockers, and chemotherapeutic agents, leading to a complete loss of response for some of these substrates (89,90). Interesting effects on transporter and enzyme regulation also occur in the presence of disease states. During cholestasis, hepatic transporter expression changes to facilitate bile acid secretion into the plasma by MRP3 and MRP4 and decrease secretion into the bile by MRP2 and BSEP (10). This change in transporter function has been attributed to the effect of bile acids on the farnesoid X receptor (FXR), causing upregulation of sinusoidal efflux transporters and downregulation of those at the canalicular membrane (91). Changes in enzyme and transporter expression have also been demonstrated in other disease states, including cancer, chronic renal failure, inflammation, and epilepsy, which display considerable interindividual variability (92–94). Current literature supports the

involvement of select nuclear receptors in the regulation of affected proteins in some of these conditions (93). Understanding regulatory effects with these diseases and the mechanisms by which enzyme and transporter functions are altered will aid in accurate dosing of these patient populations.

CONCLUSIONS

Transporters have important roles in the pharmacokinetics of their drug substrates. Transport can result in drug–drug interactions, nonlinear pharmacokinetics, and interindividual variability. Changes in transporter function can translate into drug toxicity, although transporter interactions also have the potential to be beneficial for drug therapy. Application of transport principles and understanding the effects of transport on drug substrate concentrations promote the safe and effective use of these agents. Further investigation of transporter effects *in vivo* is expected to reveal additional significant effects of drug transporters on drug disposition.

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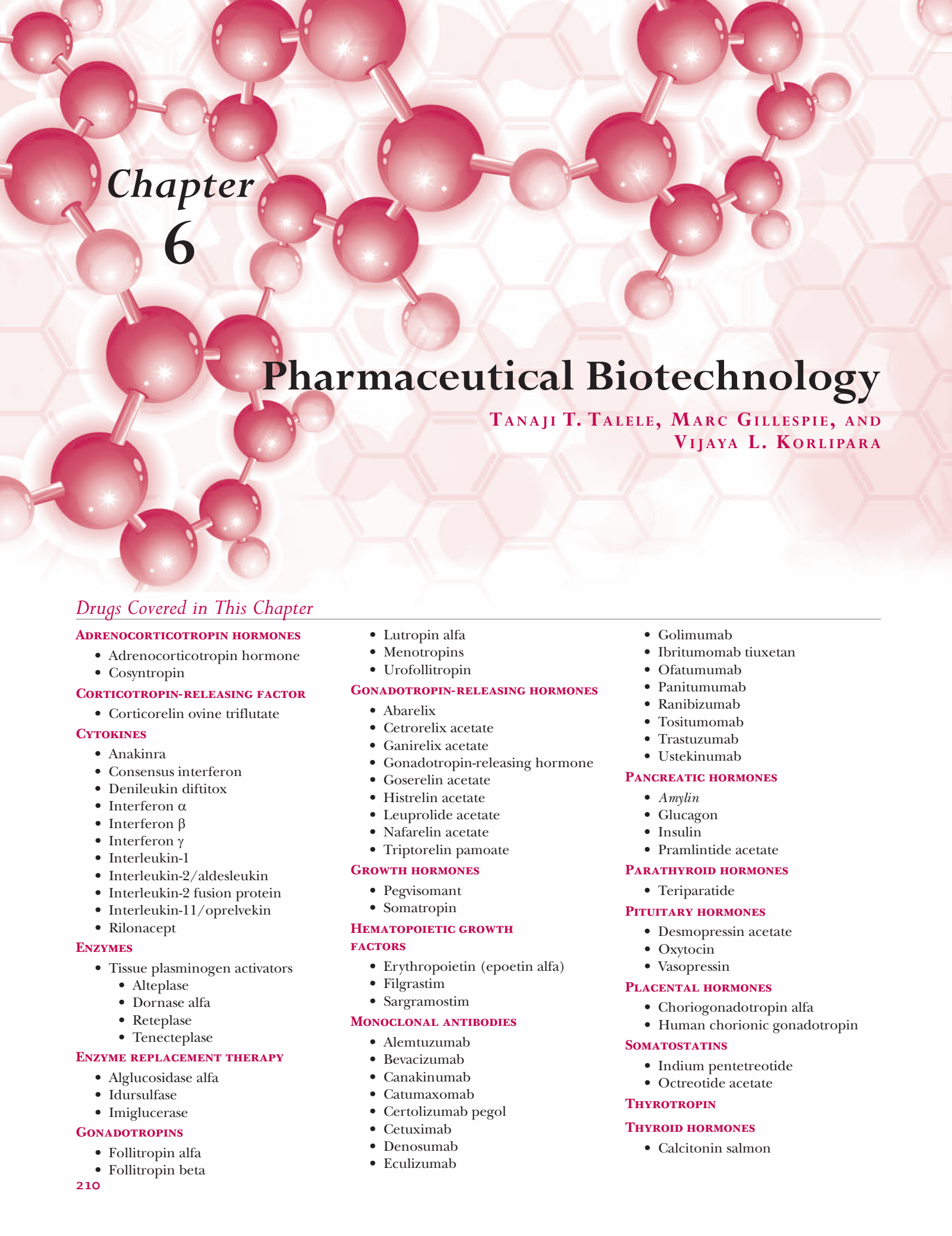
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Chapter 6

Pharmaceutical Biotechnology

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Drugs Covered in This Chapter

ADRENOCORTICOTROPIN HORMONES

- Adrenocorticotropin hormone
- Cosyntropin

CORTICOTROPIN-RELEASING FACTOR

- Corticorelin ovine triflutate

CYTOKINES

- Anakinra
- Consensus interferon
- Denileukin diftitox
- Interferon α
- Interferon β
- Interferon γ
- Interleukin-1
- Interleukin-2/aldesleukin
- Interleukin-2 fusion protein
- Interleukin-11/oprelvekin
- Rilonacept

ENZYMES

- Tissue plasminogen activators
 - Alteplase
 - Dornase alfa
 - Reteplase
 - Tenecteplase

ENZYME REPLACEMENT THERAPY

- Alglucosidase alfa
- Idursulfase
- Imiglucerase

GONADOTROPINS

- Follitropin alfa
- Follitropin beta

- Lutropin alfa
- Menotropins
- Urofollitropin

GONADOTROPIN-RELEASING HORMONES

- Abarelix
- Cetrorelix acetate
- Ganirelix acetate
- Gonadotropin-releasing hormone
- Goserelin acetate
- Histrelin acetate
- Leuprolide acetate
- Nafarelin acetate
- Triptorelin pamoate

GROWTH HORMONES

- Pegvisomant
- Somatropin

HEMATOPOIETIC GROWTH FACTORS

- Erythropoietin (epoetin alfa)
- Filgrastim
- Sargramostim

MONOCLONAL ANTIBODIES

- Alemtuzumab
- Bevacizumab
- Canakinumab
- Catumaxomab
- Certolizumab pegol
- Cetuximab
- Denosumab
- Eculizumab

- Golimumab
- Ibritumomab tiuxetan
- Ofatumumab
- Panitumumab
- Ranibizumab
- Tositumomab
- Trastuzumab
- Ustekinumab

PANCREATIC HORMONES

- Amylin
- Glucagon
- Insulin
- Pramlintide acetate

PARATHYROID HORMONES

- Teriparatide

PITUITARY HORMONES

- Desmopressin acetate
- Oxytocin
- Vasopressin

PLACENTAL HORMONES

- Choriogonadotropin alfa
- Human chorionic gonadotropin

SOMATOSTATINS

- Indium pentetreotide
- Octreotide acetate

THYROTROPIN

THYROID HORMONES

- Calcitonin salmon

• TUMOR NECROSIS FACTORS

- Abatacept
- Etanercept

OTHER GROWTH FACTORS

- Clotting factor VIII, factor IX, and anticoagulants

VACCINES

Abbreviations

ACTH, adrenocorticotropin hormone
ART, assisted reproductive therapy
ATP, adenosine triphosphate
bp, base pair
cDNA, complementary DNA
CDR, complementarity-determining region
CF, cystic fibrosis
CHO, Chinese hamster ovary
CLL, chronic lymphocytic leukemia
CRF, corticotropin-releasing factor
CSF, colony-stimulating factor
CT, calcitonin
CTCL, cutaneous T-cell lymphoma
CTLA-4, cytotoxic T-lymphocyte-associated antigen-4
CTP, cytidine triphosphate
ddNTP, 2',3'-dideoxynucleotide triphosphate
DM, diabetes mellitus
DMARD, disease-modifying antirheumatic drug
DTPA, diethylenetriaminepentaacetic acid
EGF, epidermal growth factor
EGFR, epidermal growth factor receptor
EpCAM, epithelial cell adhesion molecule
EPO, erythropoietin
ERT, enzyme replacement therapy
Fab, functional human antibody
FDA, U.S. Food and Drug Administration

FSH, follicle-stimulating hormone
GAG, glycosaminoglycan
G-CSF, granulocyte colony-stimulating factor
gDNA, genomic DNA
GH, growth hormone
GM-CSF, granulocyte-macrophage colony-stimulating factor
GnRH, gonadotropin-releasing hormone
GTP, guanosine triphosphate
HAMA, human anti-mouse antibody
hCG, human chorionic gonadotropin
hGH, human growth hormone
5-HM, 5-hydroxymethyl metabolite
IL, interleukin
IFN, interferon
IgG, immunoglobulin G
¹¹¹**In**, indium-111
IRR, infusion-related reaction
IVF, in vitro fertilization
LH, luteinizing hormone
MAb, monoclonal antibody
MAPK, mitogen-activated protein kinase
M-CSF, macrophage colony-stimulating factor
MPS, mucopolysaccharidosis
mRNA, messenger RNA
MS, multiple sclerosis
MTX, methotrexate
MW, molecular weight

NHL, non-Hodgkin lymphoma
oCRF, sheep corticotropin-releasing factor
4OH-TAM, 4-hydroxy tamoxifen
PCR, polymerase chain reaction
PDGF, platelet-derived growth factor
PEG, polyethylene glycol
PI3K, phosphatidylinositol 3-kinase
PTH, parathyroid hormone
RA, rheumatoid arthritis
RANK, receptor activator of nuclear factor-κB
RANKL, receptor activator of nuclear factor-κB ligand
rDNA, recombinant DNA
rh, recombinant human
rRNA, ribosomal RNA
SC, subcutaneous
SNP, single nucleotide polymorphism
TNF, tumor necrosis factor
t-PA, tissue-type plasminogen activator
tRNA, transfer RNA
UGT1A1, UDP-glucuronosyl transferase 1A1
UTP, uridine triphosphate
VEGF, vascular endothelial growth factor
VEGFR, vascular endothelial growth factor receptor
VIP, vasoactive intestinal peptide

INTRODUCTION

Pharmaceutical biotechnology generates basic scientific knowledge, useful therapeutic and diagnostic products, and promising methodologies for future research and clinical applications. The techniques of biotechnology lead to the development of novel therapeutics, improved methods of manufacturing pharmaceuticals, and significant contributions to our understanding of disease etiology, pathophysiology, and biochemistry. Genomics, transcriptomics, proteomics, pharmacogenomics, and metabolomics, the core approaches to pharmaceutical biotechnology, are making major contributions in three areas:

1. Identification of new genes
2. Identification of drug targets
3. Development of “personalized medicine”

Although many of the first biotechnology-derived therapeutics were initially used in acutely ill, hospitalized

patients, the products of today’s pharmaceutical biotechnology have an impact on chronic disease patient populations constituting much of ambulatory care practice.

PHARMACOGENETICS

The application of genetic information from the Human Genome Project to disease diagnosis and drug therapy is a major development that is bringing about a wealth of change. The main benefits of pharmacogenetic screening are a reduction in the health care costs arising from drug toxicity and lack of efficacy in drug therapy.

Mutations are the result of heritable, permanent changes in DNA base sequence. These usually come about by one of three mechanisms: substitution, addition, or deletion. The number of base pairs involved in a single gene mutation can vary from a single base pair change or single nucleotide polymorphism (SNP), to a large number of gross changes in the gene structure

CLINICAL SIGNIFICANCE



The biotechnology industry has grown explosively over the past 30 years, with U.S. health care biotech revenues of publicly traded biotech industry companies rising from \$8 billion in 1992 to \$56.6 billion in 2009 (1). Fifteen new biopharmaceutical products were approved by the U.S. Food and Drug Administration in 2009.

Improved manufacturing of pharmaceuticals was the first major contribution of biotechnology to pharmaceutical care in the 1980s. Biotechnology-produced human insulin, growth hormone, and erythropoietin—all replacements of highly specific, endogenous molecules—were major advances in therapy. Since the late 1980s, however, pharmaceutical biotechnology also has helped to identify new compounds with new mechanisms of action. Significant contributions to our understanding about the mechanism of disease at the molecular level continue to be made by biotechnology researchers and will translate into newer, better pharmaceuticals. The impact of biotechnology on pharmaceutical care is expected to increase exponentially as advances in technology continue to yield novel medicinal agents, such as the colony-stimulating factors, tissue-type plasminogen activator, new vaccines, DNase, fusion protein drugs, and specific monoclonal antibodies. Products of biotechnology are playing a critical role in the discovery and design, as well as in the production, of treatments for life-threatening diseases including genetic disorders, cancer, and AIDS.

Pharmaceutical care using biotechnology-derived products requires:

1. An understanding of how the handling and stability of biopharmaceuticals differ from those of other drugs that pharmacists dispense.
2. A preparation of the product for patient use, including reconstitution or compounding, if required.
3. Patient education regarding their disease, benefits of the prescribed biopharmaceutical, potential side effects or drug interactions to be aware of, and the techniques to self-administer the biotechnology drug.
4. Patient counseling about the reimbursement issues involving an expensive product.
5. Monitoring of the patient for compliance.

Key areas of biotechnology growth include pharmacogenomics/pharmacogenetics; recombinant-based therapies, such as vaccines delivered by novel administrative routes; genetic modification of the patient; and cloning of human tissues and organs. The field has matured sufficiently with the generic-like versions of original biotech drugs, called “biogenerics,” “biosimilars,” or “follow-on biologics” (2,3).

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involving insertion, deletion, or rearrangements of a very large number of base pairs.

Mutations result in a variety of changes to the gene product. If the base change occurs within the reading frame of the gene product, it may alter the amino acid sequence of the gene product (protein). SNPs have been defined as single base mutations that occur in 1% or more of the population (4). Many of these SNPs may be responsible for the production of a malfunctioning gene product that, in turn, is responsible for a serious disease. The well-used example is the A → T mutation in the β -globin gene, in which the GAG glutamic acid codon is changed to the GTG valine codon, resulting in abnormal aggregation of the hemoglobin molecules and causing sickle cell anemia. Another example is one form of β -thalassemia that results from a mutation of a CAG glutamine codon to the TAG stop codon early in the DNA sequence, resulting in premature termination of the translation process and total loss of the β subunit of the adult hemoglobin molecule.

In principle, one is able to manipulate the therapeutic outcome of a patient population by separating patients who will not respond normally to a particular drug therapy from those who will, based on an analysis of established genetic biomarkers related to the drug's metabolic

profile (Fig. 6.1). Patients who are homozygous for the genetic biomarkers indicating poor metabolism of the recommended therapeutic agent may, depending on the therapeutic window of the drug, exhibit a response resembling an overdose to the normal drug dose and require a reduced dose to achieve therapeutic levels (patient A in Fig. 6.2), whereas patients carrying the biomarkers indicating extensive metabolism may exhibit a lack of efficacy at normal doses and require an increased dose of the drug to achieve therapeutic activity (patient D in Fig. 6.2). The patients who are homozygous or heterozygous for the normal (wild-type) genetic biomarker (patients B and C, respectively, in Fig. 6.2) may require no alteration of the “normal” dose of the drug for adequate therapeutic outcome.

The application of genetic variation to drug therapy centered on the variations seen in a number of drug metabolism enzymes and much of the pharmacogenetic research has developed with the phase I (oxidative) and phase II (conjugative) metabolic enzymes. Currently, considerable data relate SNPs in the coding and regulatory regions of these genes to poor metabolism of particular drugs that have the potential to result in drug overdoses in patients carrying the genetic variations and given the “normal” dose of drug (Fig. 6.3).

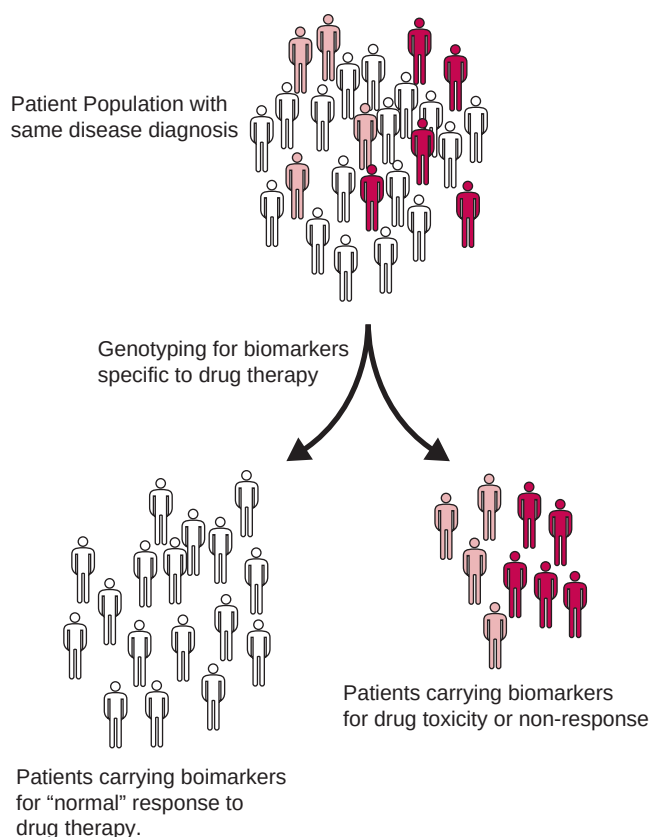


FIGURE 6.1 Manipulating therapeutic outcomes: manipulating therapeutic outcomes through pharmacogenetic analysis of a population with the same disease diagnosis to eliminate patients with genetic biomarkers indicating an adverse response to a particular drug therapy.

The action of a drug on the body is a complex interaction involving the metabolism of drugs, the transport of drugs throughout the body, and the interaction of the drug with its target. The parameters that describe this interaction are the pharmacokinetic (absorption, distribution, metabolism, and excretion) and pharmacodynamic (drug–target interaction) features of drug therapy. The biologic or pharmacodynamic targets of drug therapy are often candidate genes for biotechnology research (Fig. 6.4).

Transport proteins that move drugs through membrane barriers are critical in the absorption, distribution, and excretion of drugs and drug metabolites, whereas enzymes are critical in the metabolism of drugs in the body (Fig. 6.4). The pharmacodynamic pathway is represented most abundantly by the drug receptor pathway of drug response. This involves drug receptors as well as the numerous biochemical events that occur subsequent to drug–receptor interaction and before the measurement of the pharmacologic response. The complexity of the pharmacodynamic pathway can be simplified if one takes the modular biology approach and considers the many biochemical pathways between the drug–receptor interaction and the drug response to be linked modules of biochemical systems.

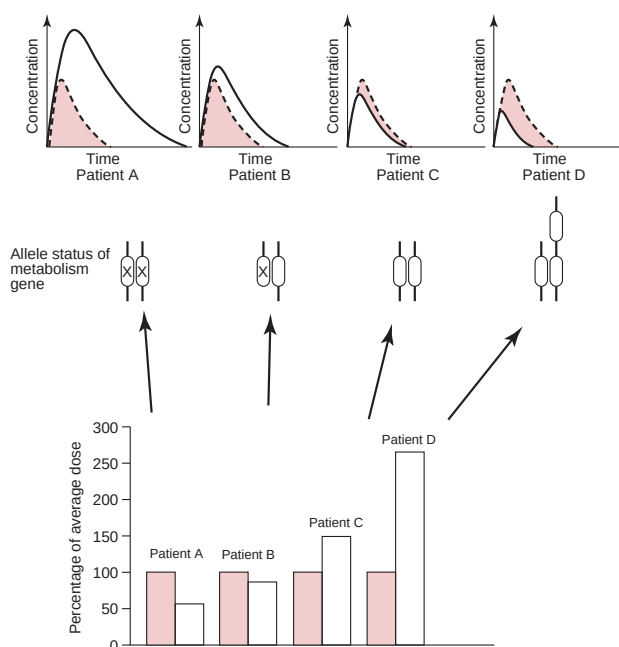


FIGURE 6.2 Dose–genotype relationship. A schematic representation of the potential benefit of adjusting drug dose to patient genotype. The theoretical "normal" drug concentration/time curve is depicted by the dashed line in the concentration/time curves of Patients A to D at the top of the figure, whereas the theoretical "genetic" drug concentration/time curve for each patient is expressed by the solid line. The allele status for each patient is shown below the theoretical plots. The oval containing the "X" indicates the presence of the detrimental allelic variation. The extra oval in Patient D indicates duplication of one of the normal alleles. The bar graph at the bottom of the figure indicates the normal doses (shaded) compared to the dose adjustments that should be made based on the patient's allele status (Kirchheiner J, Brockmoller J. Clinical consequences of cytochrome P450 2C9 polymorphisms. *Clin Pharmacol Ther* 2005;77:1–16.)

Thus, the candidate genes for the therapeutic drug pathway include the genes for drug metabolism, drug transport, and drug response (Table 6.1).

Genomics

Genomics is the study of the full complement of genetic information, both coding and noncoding, in an organism's genome. The main genomics technique is the use of SNPs as biomarkers to inform clinicians about subtypes of disease that require differential treatments and provide pharmacists with information for selection of the best therapeutic methodology to effectively manage the disease as well as provide an indication of the patients at risk of experiencing adverse reactions or those who will not respond to a given drug dose.

The general genotyping procedure for detecting known SNPs consists of polymerase chain reaction (PCR) amplification of the region of interest in the genome, discrimination of the alleles in that region, and detection of the discrimination products. Many genotyping

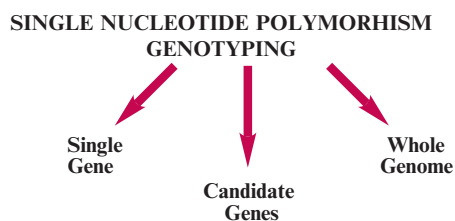


FIGURE 6.3 Single-nucleotide polymorphism (SNP) genotyping: approaches to genotyping for drug therapy. The single-gene approach has been used successfully in pharmacogenetic studies of a number of phase I and phase II metabolic enzymes. The candidate gene approach involves the selection of the genes for proteins that are demonstrated to be involved in a particular drug action, including metabolic enzymes, transport proteins, drug targets, and proteins involved in modular drug response pathways. The whole-genome approach uses SNPs evenly spaced throughout the genome.

technologies are available to choose from, and each has their advantages and disadvantages (Table 6.2). Discriminating between the alleles in the genomic region being examined is generally done by one of the five techniques illustrated in Figure 6.5.

The technique of choice for high-throughput genotyping is the minisequencing/primer extension technique augmented with the 5'-tag technology to permit the purification of the extended products via specific tags (Fig. 6.6). The genomic region of interest is PCR amplified, and the amplified DNA is exposed to allele-specific primers with 5'-tags that are complementary to oligonucleotides immobilized at specific locations in the wells of microtiter plates or on specific colored beads. Single base extension using DNA polymerase and fluorescently labeled 2',3'-dideoxynucleotide triphosphates (ddNTPs) in solution is used to detect the SNP. The resulting fluorescently labeled and tagged DNA fragments are captured by complementary tags

on microtiter plates or color-coded beads. Fluorescence resulting from laser excitation of the plates or beads is read by a detector and fed to a computer database for analysis and storage. Companies providing the single-base-extension tag-capture technology also provide a primer design service so that users can take advantage of multiplexing limited only by the capacity to design effective primers for the amplification and detection steps.

Transcriptomics

Transcriptomics is the technology behind the study of the full complement of mRNA transcripts in the cell's transcriptome and also is known as expression profiling. Methods of expression profiling are either "open" or "closed." Open systems do not require any advance knowledge of the sequence of the genome being examined; closed systems require some advance knowledge and usually involve the use of oligonucleotide or complementary DNA (cDNA) array hybridization technologies (the gene chip) and quantitative PCR (Fig. 6.7).

The cDNA arrays are prepared by spotting gene-specific PCR products, including full-length cDNAs, collections of partially sequenced cDNAs, or randomly chosen cDNAs from any library of interest onto a glass, silicon, gel, or bead matrix or a nylon or nitrocellulose membrane. The matrix (most often glass) is coated with polylysine, amino silanes, or amino-reactive silanes to assist in the attachment of the cDNA probes. The PCR products of the clones are purified and spotted onto the matrix by robots through contact printing or non-contact piezo or ink-jet devices. After cross-linking the probe to the matrix by ultraviolet irradiation, the probe is made single stranded by heat or alkali treatment. The mRNA target is prepared from both a test and reference sample by reverse transcription to produce cDNA, which is then labeled with fluorescent probes (one for the test and another for the reference). The fluorescent targets are pooled and hybridized to the probe array under very stringent conditions. Laser excitation of the hybridized samples on the matrix and comparison of the fluorescence intensity of the reference sample with the fluorescence intensity of the test sample using computer algorithms yield an emission characteristic of the increase or decrease of mRNA expression under test conditions.

Proteomics

Proteomics is the technology behind the study of the total protein complement of a genome, or the complete set of proteins expressed by a cell, tissue, or organism. The presence of an open reading frame in a DNA sequence indicates the presence of a gene, but it does not indicate gene transcription, RNA editing, translation, or posttranslational modification and the presence of isoforms. Analysis of the transcriptome does not indicate alteration of protein levels by proteolysis, recycling, and sequestration. Therefore, it is important to determine protein levels, protein expression, and protein-protein interactions directly.

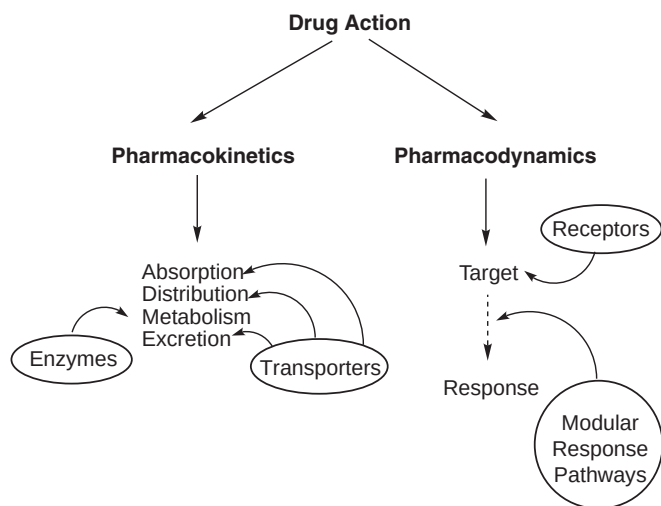


FIGURE 6.4 Schematic representation of the multigenic therapeutic drug pathway.

TABLE 6.1 Specific Examples of the Application of Allelic Variation in Drug Therapy

Protein	Action	Drug	Therapy	Efficacy/Toxicity
CYP450 2D6	Drug metabolism	Codeine	Pain	Efficacy
CYP450 2C9	Drug metabolism	Warfarin	Coagulation	Toxicity (hemorrhage)
Thiopurine S-methyltransferase	Drug metabolism	Mercaptopurine	Acute lymphoblastic leukemia	Toxicity (myelotoxic)
HER2/neu	Drug target	Herceptin	Breast cancer	Efficacy
bcr/abl	Drug target	Gleevec	Chronic myelogenous leukemia	Efficacy
EGFR	Drug target	Erbix	Colon cancer	Efficacy
Apolipoprotein E4	Marker	Tacrine	Alzheimer disease	Efficacy
Cholesteryl ester transferase	Marker	Statins	Atherosclerosis	Efficacy
ATP-binding cassette B1	Drug transport	Saquinavir	HIV	Efficacy
		Indinavir	Leukemia	
		Ritonavir		
		Daunorubicin		
		Etoposide		
UDP-glucuronosyl-transferase 1A1	Drug metabolism	Irinotecan	Cancer	Toxicity (myelotoxic)
N-Acetyltransferase 2	Drug metabolism	Isoniazid	Tuberculosis	Toxicity (hepatotoxic)
Pseudocholinesterase	Drug metabolism	Suxamethonium	Muscle relaxation during surgery	Toxicity (prolonged apnea)

Proteomics is divided into three main areas (5):

1. Microcharacterization for large-scale identification of proteins and posttranslational modifications (5).
2. Differential gel electrophoresis for comparison of protein levels (6).
3. Protein–protein interaction studies using protein chips (7), mass spectrometry isotope-coded affinity tag technology (8), and the yeast two-hybrid system (9).

Metabolomics

Metabolomics is the technology behind the measurement of metabolite concentrations, fluxes, and secretions in cells and tissues (metabolome). Metabolomics is at the cross-roads of genotype/phenotype interactions, where the interrelationship of metabolic pathways is considered to be the fundamental component of an organism's phenotype. Metabolomics is distinguished by the fact that metabolites are not directly linked to the genetic code but actually are products of a concerted action of many enzyme networks and other regulatory proteins (10). Metabolites also are more complex than nucleic acids, which are composed of four nucleotides, or proteins, which in turn are composed of 20 different amino acids. As a result, metabolites cannot be sequenced like genes and proteins so that their characterization requires that the structure of each metabolite be determined individually through description of elemental composition and stereochemical orientation of functional groups.

Metabolites are more than just products of enzyme-catalyzed reactions. They also are sensors and regulators of complex molecular interactions in the organism, and as a result, the composition of the metabolome can be altered by changes in the individual's environment. Therefore, a study of the metabolome is complicated not only by the uniqueness of the individual's genome but also by the uniqueness of the individual's environment. On the other side of the coin, however, one could consider the metabolome to be a very sensitive indicator of the individual's phenotype (11,12).

Metabolomics can be approached by using several different but related strategies:

1. Target analysis investigates the primary metabolic effect of a genetic variation, in which the analysis usually is limited to the substrate and/or product of the protein expressed by the altered gene.
2. Metabolic profiling limits the investigation to a number of predefined metabolites, usually in a specific metabolic pathway.
3. Metabolomics is a comprehensive analysis (both identification and quantification) of metabolites in a biologic system, which investigates the effect of multiple genetic variations on many different biochemical pathways.
4. Metabolic fingerprinting is a strategy to screen a large number of metabolites for biologic relevance in diagnostic or other analytical procedures in which it is important to rapidly distinguish

TABLE 6.2 Genotyping Technologies

Technique ^a	Mechanism of Allele Discrimination	Reaction Formate	Detection Mechanism ^b
Reverse dot blot	Hybridization	Solid support	Colorimetric
Microarray	Hybridization	Solid support	Fluorescence
DASH	Hybridization	Solid support	Fluorescence
Molecular beacons	Hybridization	Homogeneous reaction	FRET
5'-Nuclease cleavage (TaqMan)	Hybridization	Homogeneous reaction	FRET or FP
Cleavase (Invader Assay)	Hybridization	Homogeneous reaction	FRET, FP, MS
Biosensor microchip	Hybridization	Solid support	Electrochemical and fluorescence
AS-PCR	Primer extension	Homogeneous reaction	Fluorescence or FRET
AS-PE	Primer extension	Solid support	Fluorescence
APEX	Primer extension	Solid support	Fluorescence
FP-TDI	Primer extension	Homogeneous reaction	FRET or FP
MALDI-TOF MS	Primer extension	Homogeneous reaction	MS
Pyrosequencing	Primer extension	Homogeneous reaction or solid support	Luminescence
BioBeads and TagArrays	Primer extension	Homogeneous reaction and solid support capture	Fluorescence
OLA	Ligation	Homogeneous and solid support	Colorimetric and fluorescence
RCA	Ligation	Solid support	Fluorescence
Closed tube	Ligation	Homogeneous reaction	FRET
Microsphere/microarray ligation	Ligation	Homogeneous reaction	Fluorescence

^aDASH, dynamic allele-specific hybridization; AS-PCR, allele-specific polymerase chain reaction; AS-PE, allele-specific primer extension; APEX, arrayed primer extension; FP-TDI, fluorescence polarization template directed dye terminator incorporation; MALDI-TOF-MS, matrix-assisted laser desorption/ionization time-of-flight mass spectrometry; OLA, oligonucleotide ligation assay; RCA, rolling circle amplification.

^bFRET, fluorescence resonance energy transfer; FP, fluorescence polarization; MS, mass spectrometry.

between individuals in a population. Metabolic fingerprinting is the ultimate characterization of an individual's phenotype for disease diagnosis and drug therapy. Once the technologic problems of high-throughput metabolic analysis are developed in this area, it will be a major competitor of SNP analysis in pharmacogenomics.

The analytical technologies used in metabolomic investigations are nuclear magnetic resonance and mass spectrometry alone or in combination with liquid or gas chromatographic separation of metabolites. Other techniques include thin-layer chromatography, Fourier-transform infrared spectrometry, metabolite arrays, and Raman spectroscopy.

PHARMACEUTICAL BIOTECHNOLOGY METHODS

Pharmaceutical biotechnology is defined, at its most basic level, as the manipulation of nucleic acids in the production of therapeutic and diagnostic agents. In order to understand both DNA and RNA, the fundamental genetic material, it is important to understand the

basic process that moves information from DNA to RNA. This process starts with transcription.

Transcription

The genetic information in DNA is transcribed to the intermediate RNA molecule that moves to the cytoplasm, where it directs the synthesis of the gene product it encodes using ribosomes. RNA differs chemically from DNA in that the sugar molecule is ribose and thymine in DNA is replaced by uracil in RNA. Structurally, RNA contains both single-stranded runs and short, double helical regions.

There are several types of RNA molecules in the cell, three of which are highlighted here. Messenger RNA (mRNA) is transcribed from a particular DNA sequence. Transfer RNAs (tRNAs) are covalently linked to specific amino acids and carry the anticodon triplet that recognizes a complementary trinucleotide sequence of mRNA specific for the amino acid that it carries. The ribosome contains both ribosomal RNA (rRNA) and proteins.

Several differences exist in the mechanism of transcription between prokaryotes and eukaryotes. These differences are important when evaluating if recombinant

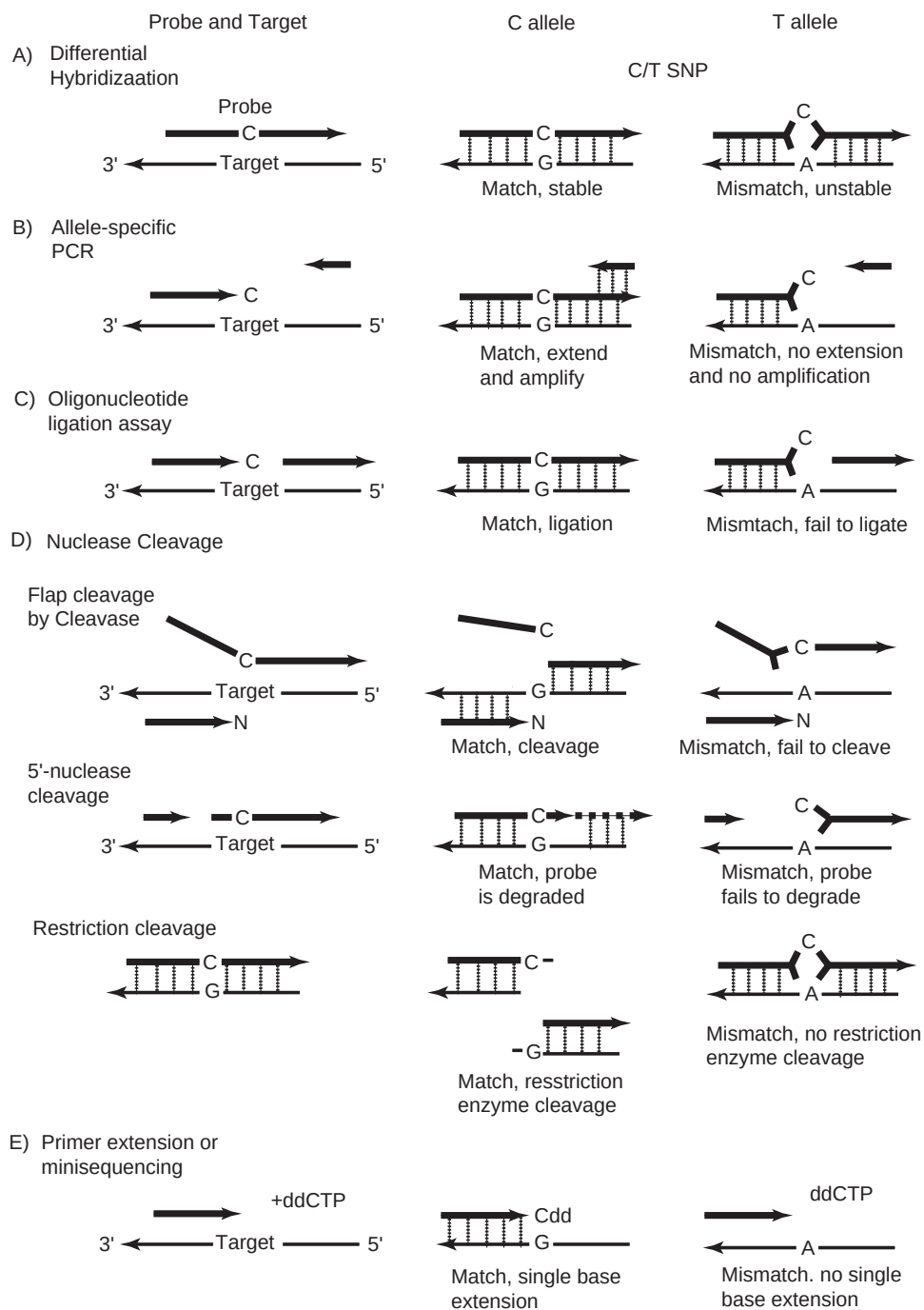


FIGURE 6.5 Allele discrimination strategies for the detection of the C allele of a C/T transition. Figure summarizes the five main biochemical reaction principles that underlie the single nucleotide polymorphism (SNP) genotyping technologies. Bold arrows indicate the probes; thinner arrows indicate the target. (From Syvanen AG. Accessing genetic variation: genotyping single-nucleotide polymorphisms. *Nat Rev Genet* 2001;2:930–942; Carlson CS, Newman TL, Nickerson DA. SNPing in the human genome. *Curr Opin Chem Biol* 2001;5:78–85, with permission.)

expression of a protein should be carried out in a eukaryotic or prokaryotic system. For our purposes, we describe transcription in a eukaryotic system.

Transcription is carried out by RNA polymerases, of which there are three types in the eukaryote. RNA pol I catalyzes the synthesis of rRNAs, RNA pol II is responsible for the synthesis of mRNA, and RNA pol III synthesizes

tRNA. All three polymerases are large enzymes containing 12 or more subunits.

Each eukaryotic RNA polymerase copies DNA from the 3' end, thus catalyzing mRNA formation in the 5'→3' direction and synthesizing RNA complementary to the antisense DNA template strand. The reaction requires the precursor nucleotides adenosine triphosphate

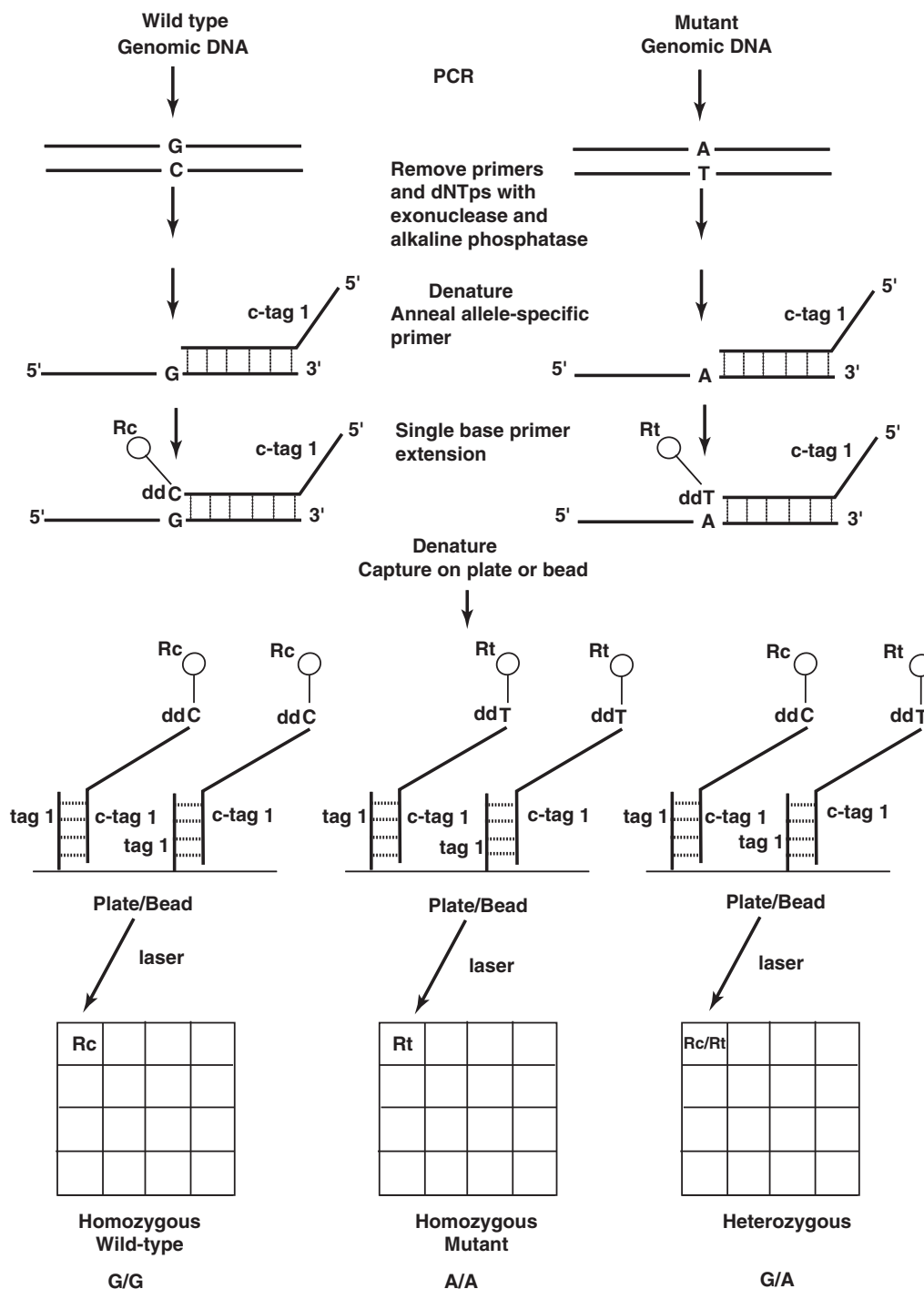


FIGURE 6.6. Tag-array and bead technology. Tag 1 is specific for the G/A mutation at a particular position in the gene under investigation. Other tags can be designed for other single nucleotide polymorphisms (SNPs) to permit multiplexing in the same well of the microtiter plate to produce an “array of arrays.” R, reporter fluorescent label.

(ATP), guanosine triphosphate (GTP), cytidine triphosphate (CTP), and uridine triphosphate (UTP) and does not require a primer for initiation of transcription. The five stages of eukaryotic transcription include initiation, elongation and termination, capping, polyadenylation, and splicing.

Initiation

Eukaryotes have different RNA polymerase-binding promoter sequences than prokaryotes. The TATA consensus sequence of the eukaryotic promoter region is located 25 to 35 base pairs (bp) upstream from the transcription start site (Fig. 6.8). The low activity of

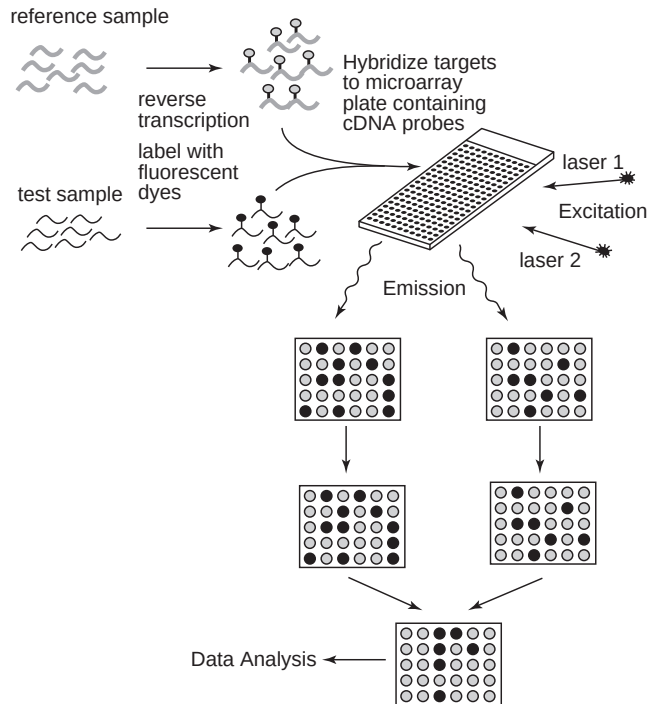


FIGURE 6.7 Schematic representation of a cDNA microarray chip. Probes of interest are obtained from DNA clones and printed on coated glass microscope slides. Analysis of mRNA expression is carried out by taking total mRNA from both a test and control sample, fluorescently labeling the samples with either Cy3- or Cy5-dUTP during a single round of reverse transcription, pooling the fluorescent targets, and hybridizing to the immobilized probes on the array under very stringent conditions. Laser excitation of the incorporated targets yields an emission with a characteristic spectrum, which is measured using a scanning confocal laser microscope and analyzed using a computer program. (From Duggan DJ, Bittner M, Chen Y, et al. Expression profiling using cDNA microarrays. *Nat Genet* 1999;21[Suppl 1]:10–14.)

basal promoters is greatly increased by the presence of upstream regulatory elements located 40 to 200 bp upstream of the promoter sequence and includes the SP1 box, the CCAAT box, and the hormone response elements. Transcription from many eukaryotic promoters

can be stimulated by control elements, called enhancers, located many thousands of base pairs away from the transcription start site.

Elongation and Termination

RNA polymerase moves along the DNA template until a terminator sequence is reached. The RNA molecule made from a protein-coding gene by RNA pol II in eukaryotes is called pre-mRNA. The pre-mRNA from a eukaryotic protein-coding gene is extensively processed within the nucleus before export to the cytoplasm.

Capping

At the end of polymerization, the 5' end of the pre-mRNA molecule is modified by addition of an *N*-7-methyl guanine molecule (Fig. 6.9).

Polyadenylation

The 3' end of the pre-mRNA is generated by nucleases catalyzed cleavage followed by the addition of a run, or tail, of 100 to 200 adenosine nucleotides, resulting in what is called the poly(A) tail. Cleavage and polyadenylation require specific sequences in the DNA and its pre-mRNA transcript that are part of the transcription termination signal. The poly(A) tail helps stabilize the mRNA molecule, reducing its sensitivity to 3'-nuclease activity.

RNA Splicing

Splicing is the precise excision of the intron sequences and joining of exons to produce a functional mRNA molecule.

Translation

Once the fully processed mRNA has been transported from the nucleus to the cytoplasm, protein synthesis occurs. The triplet genetic code carried on the mRNA is translated into a protein sequence by the ribosome. Amino acids are delivered to the ribosome by tRNAs that carry specific amino acids attached to their 3'-terminus based on the tRNA anticodon sequence. The anticodon complements the triplet codon sequence in mRNA, and

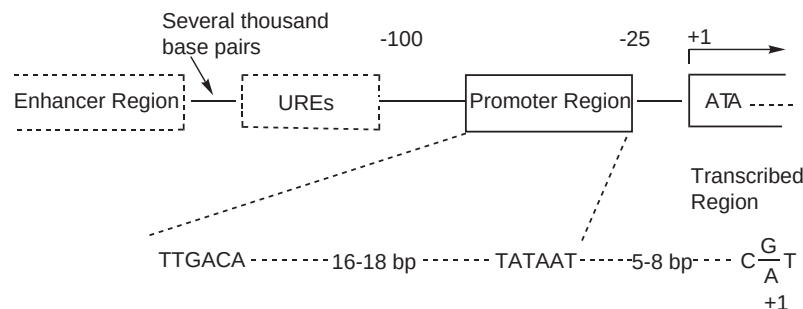


FIGURE 6.8 Schematic representation of the eukaryotic transcription unit showing the relationship between the promoter, upstream regulatory elements, and enhancer region. Note that transcription usually starts with a purine base in the 1 position. (Reproduced from Turner et al. *BIOS Instant Notes in Molecular Biology*. Garland Science/Taylor & Francis Books, Inc., 1997, with permission.)

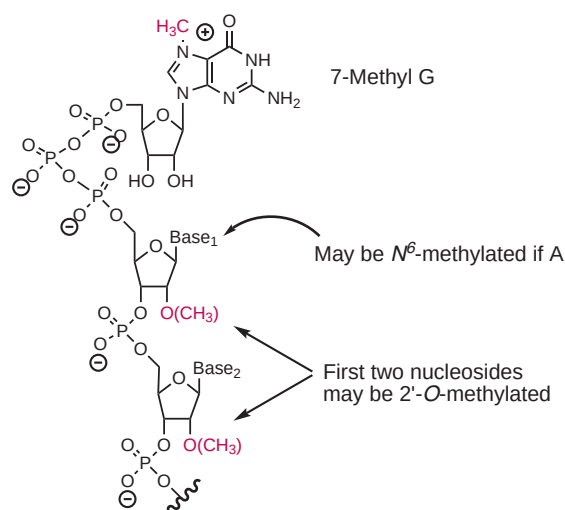


FIGURE 6.9 Chemical structure of the 5'-cap of mRNA. Cap 0 consists of the 7-methylguanosine triphosphate attached to the 5' end of the mRNA. Cap 1 consists of the 7-methylguanosine and 2'-O-methylation of the 5' base. Cap 2 consists of the 7-methylguanosine and the 2'-O-methylation of the first two bases in the sequence.

61 triplet sequences code for 20 amino acids (Table 6.3). Three codons are nonsense or stop codons that terminate translation. The code is degenerate with 18 of the 20 common amino acids coded for by more than one codon. Two amino acids, Met (AUG) and Trp (UGG), each have one unique codon. From a fixed start point on the mRNA (start codon, AUG), which establishes the open reading frame, each group of three bases in the coding region of the mRNA represents a codon that is recognized by a complementary triplet on the tRNA molecule.

There are four stages in the protein synthesis in both prokaryotes and eukaryotes:

1. **Initiation**—assembly of the ribosome on an mRNA molecule.
2. **Elongation**—repeated cycles of amino acid addition.
3. **Termination**—recognition of the stop codon, release of the new protein chain, and breakdown of the synthetic complex.
4. **Posttranslational modification**—usually includes protein cleavage by carboxy or amino peptidases, and chemical modification such as acetylation, sulfonylation, phosphorylation, hydroxylation, lipidation, and/or addition of polysaccharides.

Genes

A gene is the segment of genomic DNA (gDNA) involved in producing a polypeptide chain. The mRNA assembled from the gene includes regions preceding (the leader or 5'-untranslated region) and following (the trailer or 3'-untranslated region) the coding region as well as

TABLE 6.3 The Genetic Code

First Position	Second Position				Third Position
	U	C	A	G	
U	Phe UUU	Ser UCU	Tyr UAU	Cys UGU	U
	Phe UUC	Ser UCC	Tyr UAC	Cys UGC	C
	Leu UUA	Ser UCA	Stop UAA	Stop UGA	A
	Leu UUG	Ser UCG	Stop UAG	Trp UGG	G
C	Leu CUU	Pro CCU	His CAU	Arg CGU	U
	Leu CUC	Pro CCC	His CAC	Arg CGC	C
	Leu CUA	Pro CCA	Gln CAA	Arg CGA	A
	Leu CUG	Pro CCG	Gln CAG	Arg CGG	G
A	Ile AUU	Thr ACU	Asn AAU	Ser AGU	U
	Ile AUC	Thr ACC	Asn AAC	Ser AGC	C
	Ile AUA	Thr ACA	Lys AAA	Arg AGA	A
	Met AUG	Thr ACG	Lys AAG	Arg AGG	G
G	Val GUU	Ala GCU	Asp GAU	Gly GGU	U
	Val GUC	Ala GCC	Asp GAC	Gly GGC	C
	Val GUA	Ala GCA	Glu GAA	Gly GGA	A
	Val GUG	Ala GCG	Glu GAG	Gly GGG	G

intervening sequences such as introns that are removed in the processing of the pre-mRNA. With the discovery of other processes that contribute to the penultimate sequence of the mature mRNA, the definition of a gene is evolving (13).

Cloning and the Preparation of DNA Libraries

Two discoveries in the early 1970s provided breakthroughs in nucleic acid chemistry: the discovery of bacterial enzymes capable of cleaving nucleic acids at specific, palindromic (symmetrical) base sequences (Figs. 6.10 and 6.11) and the discovery of bacterial plasmids as vehicles (vectors) to amplify and carry the gene fragments produced by those restriction enzymes. Plasmids are small,

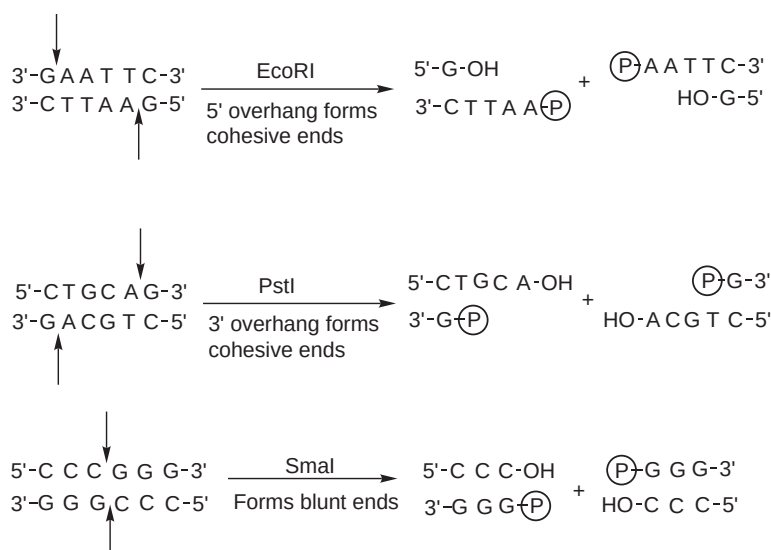


FIGURE 6.10 Actions of restriction endonucleases *EcoRI*, *PstI*, and *SmaI* at their recognition sequences. Note that *EcoRI* and *PstI* enzymes produce “sticky ends,” with overlapping sequences, whereas *SmaI* produces “blunt ends,” or nonoverlapping sequences.

circular, extrachromosomal nucleic acid molecules in bacteria that replicate independently within the bacterial cell. Restriction enzymes (and now PCR) are used to produce relatively small DNA fragments that are inserted into bacterial plasmid vectors, forming recombinant DNA (rDNA) molecules. The rDNA vectors are inserted into bacterial hosts, where the plasmid replicates with the bacteria, producing a large number of identical rDNA molecules known as clones, thus completing the process known as DNA cloning.

SEQUENCING DNA FRAGMENTS The two previous methods of nucleic acid sequencing, the Maxam and Gilbert method (chemical method) (14) and Sanger’s dideoxy

method (15), have been superseded by the far faster massive parallel sequencing methods. In short, high-throughput sequencing involves breaking up gDNA into fragments and placing the individual fragments onto specially designed microbeads where the many copies of each fragment have been amplified. The amplified fragments are then loaded into the small wells of a substrate. As the wells are loaded with sample, reagents are pumped across the plate. The addition of the reagents results in an enzymatic reaction between complementary bases in the DNA fragments, and a fluorescent signal is released and read by a computational sequence analyzer (16).

SYNTHESIZING OLIGONUCLEOTIDES The need for short oligonucleotides of known sequence has grown tremendously with the need for radiolabeled and fluorescently labeled probes to isolate and characterize nucleic acids. The phosphite triester (17) and the phosphotriester methods (18) are convenient solid-phase automated techniques for the synthesis of oligonucleotides (Fig. 6.12).

POLYMERASE CHAIN REACTION Working with small quantities of nucleic acids isolated from cell and tissue sources is difficult, and there is often a need to amplify these sequences. PCR is used to amplify fragments of DNA without the need for cloning (19). The process can amplify samples that contain as little as a single nucleotide fragment used as template. There is a requirement for the sequences flanking the boundaries of the fragment to be amplified so that oligonucleotide primers can be prepared.

PROTEIN SYNTHESIS THROUGH RECOMBINANT DNA Once the gene coding for the desired protein has been identified and isolated (Fig. 6.13), the genetic material is

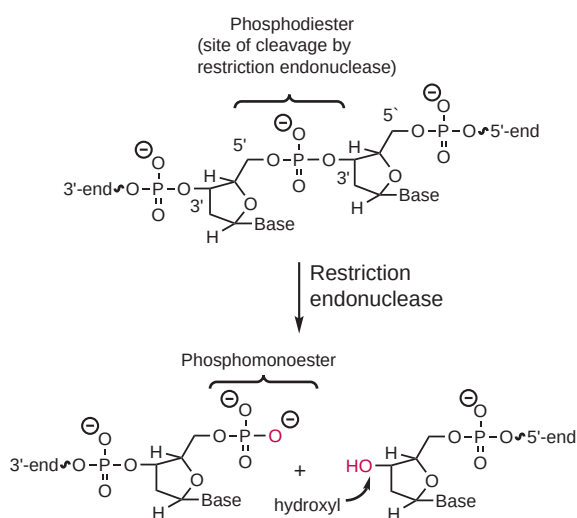


FIGURE 6.11 Cleavage by restriction endonuclease.

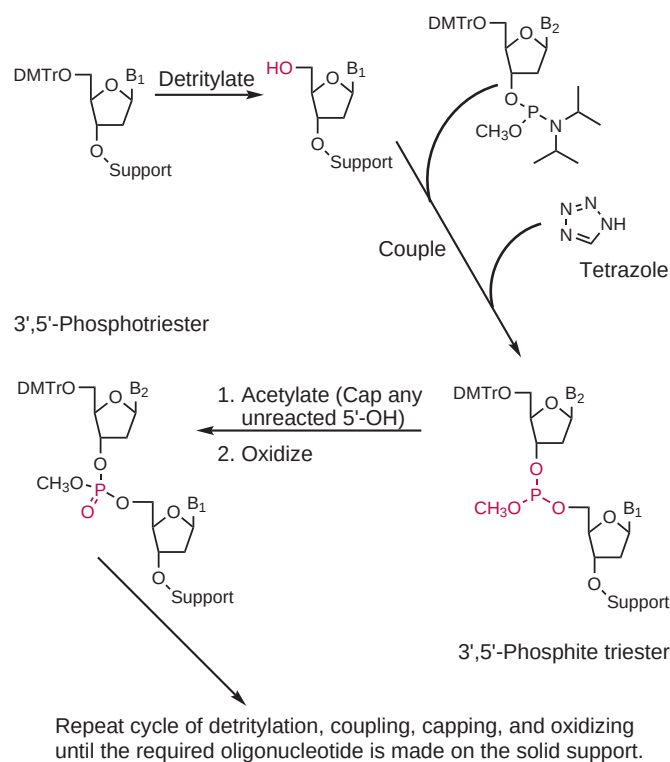


FIGURE 6.12 Solid-phase phosphate triester method of oligonucleotide synthesis.

introduced into cells on a vector capable of DNA replication and initiation of transcription. A cloning vector is a carrier molecule, the vehicle that is used to insert foreign DNA into a host cell. Typically, vectors are genetic elements that can be replicated in a host cell separately from that cell's chromosomes. Bacterial plasmids are circular DNA of only a few thousand base pairs that replicate freely within the cells and are ideal for carrying the gene into the host organism. DNA fragments coding for the desired protein can be cloned from gDNA or cDNA and inserted into the vector that carries the code to synthesize the protein in the host.

The vector containing the code for the target protein is then inserted into the host. Host cells are typically bacteria (e.g., *Escherichia coli*), eukaryotic yeast (e.g., *Saccharomyces cerevisiae* [baker's yeast]), or mammalian cell lines. The choice of host system is influenced primarily by the type of protein to be expressed and by the key differences among the various host cells (20). Overall protein yields generally are much lower in mammalian cells, but in some cases, this may be the only system that produces the specific protein of interest. It should be noted that recombinant proteins produced in gram-negative bacteria may contain endotoxins.

The host cells containing the vector are grown under selection to identify a clone that contains the desired gene and is able to produce the best protein (21). The selected cloned cells are used as inoculum first for a

small-scale cell culture/fermentation, which is then followed by larger fermentations in bioreactors. The medium is carefully controlled to enhance cell reproduction and protein synthesis. The host produces its natural proteins along with the desired protein, which may be secreted into the growth medium. The protein of interest can then be isolated from the fermentation, purified, and formulated to give a potential rDNA-produced pharmaceutical.

PROTEIN ISOLATION AND PURIFICATION The isolation and purification of the final protein product from the complex mixture of cells, cellular debris, medium nutrients, and other host metabolites is a challenging task (20,21). The structure, purity, potency, and stability of the recombinant protein must be considered. Often, sophisticated filtrations, phase separations, precipitation, and complex multiple-column chromatographic procedures are required to obtain the desired protein. Although isolation of the recombinant protein, produced in culture in relatively large amounts, generally is easier than isolating the native protein, ensuring the stability and retention of the bioactive three-dimensional structure (correct protein folding) of any biopharmaceutical is a more arduous task. In addition, recombinant proteins from bacterial hosts require removal of endotoxins, whereas viral particles may need to be removed from mammalian cell culture products (22). A discussion of these techniques is beyond the scope of this chapter; however, useful reviews on the extraction and purification (20,21,23) and the analysis and chromatography (24–28) of biotechnology products are available as a resource for further information.

GENERAL PROPERTIES OF BIOTECHNOLOGY-PRODUCED MEDICINAL AGENTS

rDNA and hybridoma technologies have made it possible to produce large quantities of highly pure, therapeutically useful proteins. The rDNA-derived proteins and monoclonal antibodies (MAbs) are not dissimilar to the other protein pharmaceuticals or biopharmaceuticals that pharmacists have dispensed in the past. As polymers of amino acids joined by peptide bonds, the properties of these proteins differ generally from those of small organic molecule pharmaceuticals.

Stability of Biotech Pharmaceuticals

The instability of proteins, including protein pharmaceuticals, can be separated into three distinct classes. Chemical instability results from bond formation or cleavage yielding a modification of the protein and a new chemical entity. Photoinstability of protein drugs upon exposure to light results in chemical and physical instability. Physical instability involves a change to the secondary

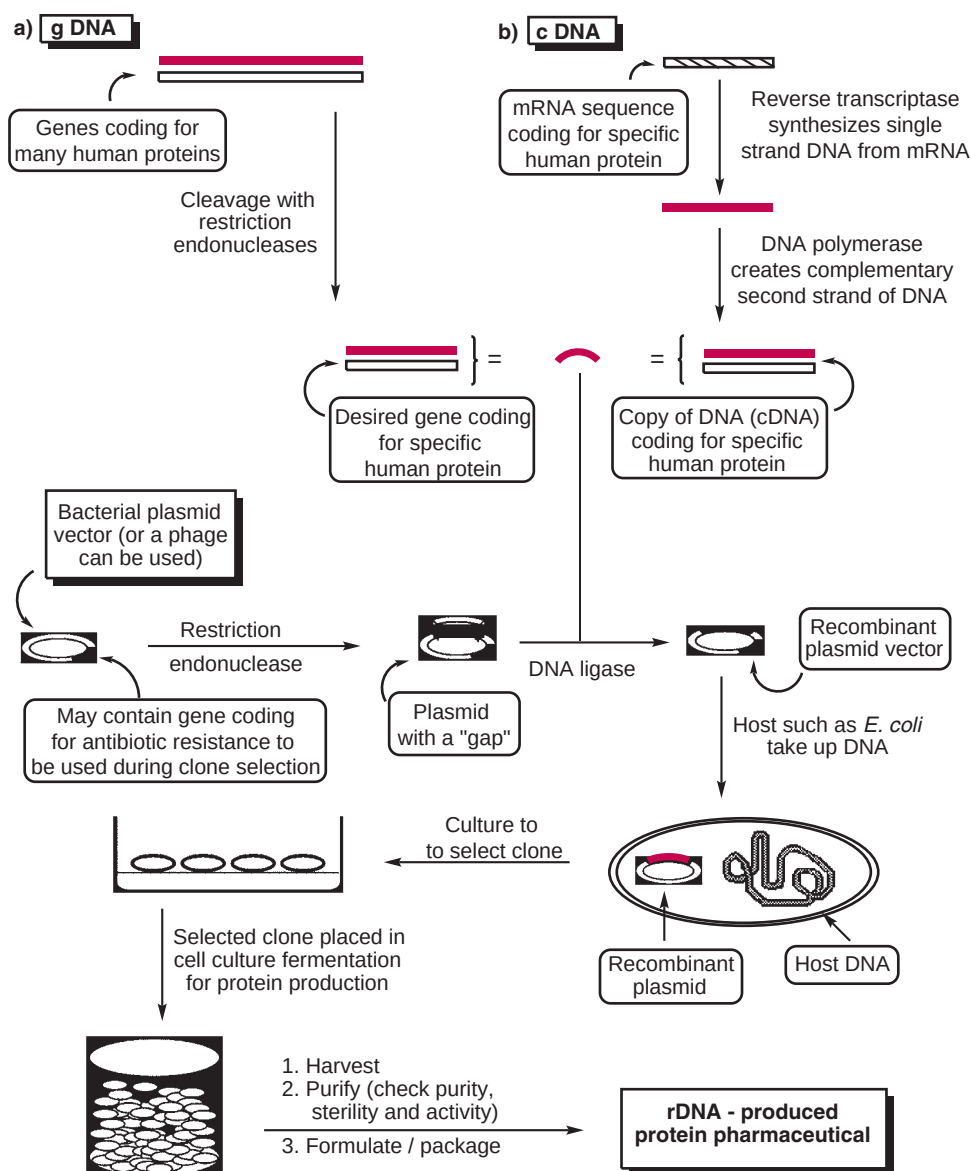


FIGURE 6.13 Summary of typical rDNA production of a protein from either (a) genomic DNA or (b) cDNA.

or higher-order structure of the protein rather than a covalent bond-breaking modification (29–36).

Chemical Instability

A variety of reactions give rise to the chemical instability of proteins, including hydrolysis, oxidation, racemization, β -elimination, and disulfide exchange (Fig. 6.14). Each of these changes may cause a loss of biologic activity. Proteolytic hydrolysis of peptide bonds results in fragmentation of the protein chain. It is well established that in dilute acids, aspartic acid residues in proteins are hydrolyzed at a rate at least 100-fold faster than that of other peptide bonds because of the mechanism of the reaction. An additional hydrolysis reaction is the deamidation of the neutral residue of asparagine and glutamine side-chain linkages, forming the ionizable

carboxylic acid residues aspartic acid and glutamic acid (Fig. 6.14a). This conversion may be considered primary sequence isomerization.

Oxidative degradative reactions can occur to the side chains of sulfur-containing methionine and cysteine residues and the aromatic amino acid residues histidine, tryptophan, and tyrosine in proteins during their isolation and storage. The weakly nucleophilic thioether group of methionine ($R-S-CH_3$) can be oxidized at low pH by hydrogen peroxide as well as by oxygen in the air to the sulfoxide ($R-SO-CH_3$) and the sulfone ($R-SO_2-CH_3$). The thiol (sulfhydryl, $R-SH$) group of cysteine can be successively oxidized to the corresponding sulfenic acid ($R-SOH$), disulfide ($R-SS-R$), sulfinic acid ($R-SO_2H$), and finally, sulfonic acid ($R-SO_3H$). A number of factors, including pH, influence the rate of this oxidation.

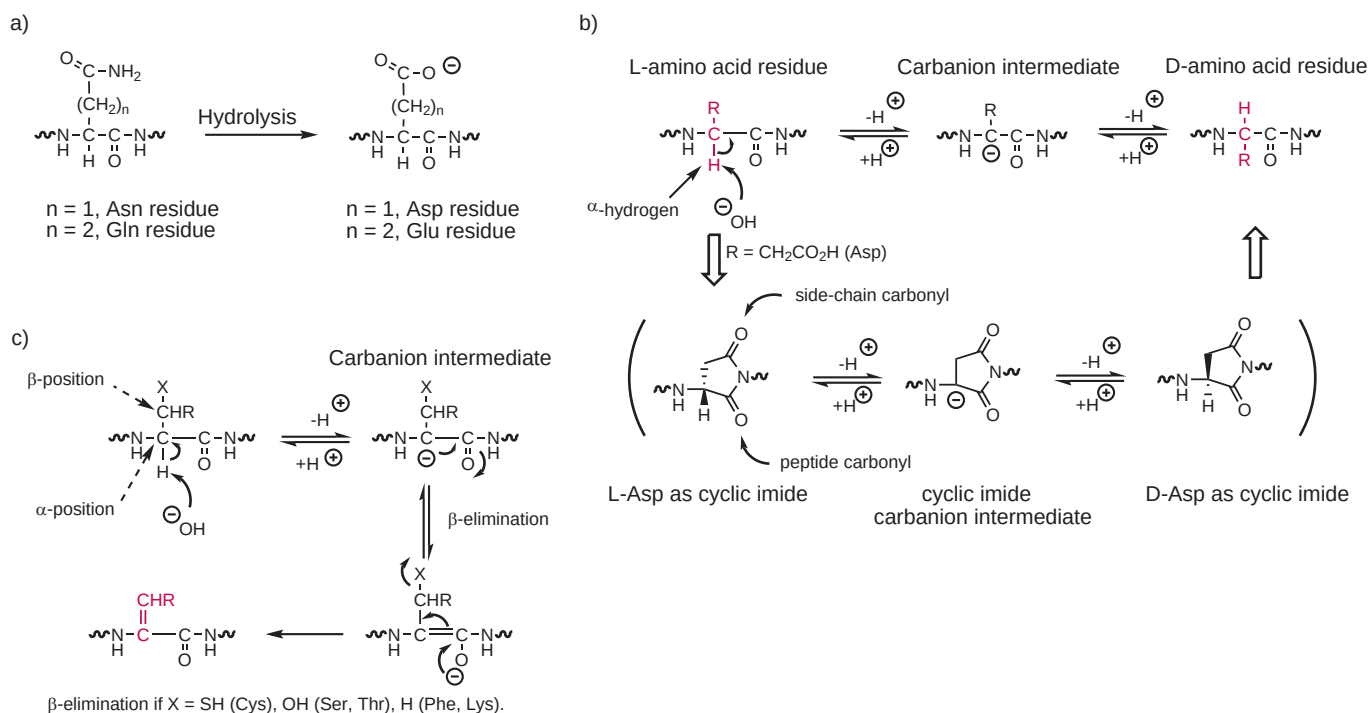


FIGURE 6.14 Chemical instability of protein biopharmaceuticals. (a) Hydrolysis. (b) Base-catalyzed racemization. (c) β -Elimination.

Oxidation of histidine, tryptophan, and tyrosine residues is believed to occur with a variety of oxidizing agents, resulting in the cleavage of the aromatic rings.

Base-catalyzed racemization reactions may occur in any of the amino acids except achiral glycine to yield residues in proteins with mixtures of L- and D-configurations. The α -methine hydrogen is removed to form a carbanion intermediate (Fig. 6.14b). The degree of stabilization of this intermediate controls the rate of this reaction. Racemization generally alters the proteins' physicochemical properties and biologic activity. Also, racemization generates nonmetabolizable D-configuration forms of the amino acids. Aspartate residues in proteins racemize at a 10^3 -fold faster rate than when free, in contrast to the 2- to 4-fold increase for the other residues. The facilitated rate of racemization for aspartic acid residues is believed to result from the formation of a stabilized cyclic imide.

Proteins containing cysteine, serine, threonine, phenylalanine, and lysine are prone to β -elimination reactions under alkaline conditions (Fig. 6.14c). The reaction proceeds through the same carbanion intermediate as racemization. The reaction is influenced by a number of additional factors, including temperature and the presence of metal ions.

The interrelationships of disulfide bonds and free sulfhydryl groups in proteins are important factors influencing the chemical and biologic properties of protein pharmaceuticals. Disulfide exchange can result in incorrect pairings and major changes in the higher-order structure (secondary and above) of proteins. The exchange may occur in neutral, acidic, and alkaline media.

Photoinstability

The exposure of proteins to light and the ensuing chemical and physical degradation have been studied extensively for many years (36). The peptide backbone, tryptophan, tyrosine, phenylalanine, and cysteine are the primary targets of photodegradation in proteins. Primary or type I photodegradation begins with absorption of light, resulting in excitation of an electron to higher energy singlet states. Absorption occurs through either the peptide backbone or by the amino acid side chains of tryptophan, tyrosine, phenylalanine, and cysteine. In aqueous solution, the absorption wavelengths are 180 to 230 nm for the peptide backbone, 280 to 305 nm for tryptophan, 260 to 290 nm for tyrosine, 240 to 270 nm for phenylalanine, and 250 to 300 nm for cysteine. Although tryptophan is present in relatively low abundance in proteins, it has the highest molar absorption coefficients and is therefore a major player in the photodegradation of protein drugs.

The major photolytic pathways of tryptophan in proteins are shown in Figure 6.15. Following absorption of light, the excited state tryptophan will either relax to the lowest energy singlet state followed by fluorescence, undergo intersystem crossing to the triplet state (3 tryptophan), or eject an electron with formation of a tryptophan radical cation that will rapidly deprotonate to form the neutral tryptophan radical. The tryptophan radical may extract a hydrogen from a nearby tyrosine, repairing the tryptophan and forming a tyrosine phenoxy radical; react with oxygen, if present, forming a peroxy radical on the Trp; or react with nearby amino acids.

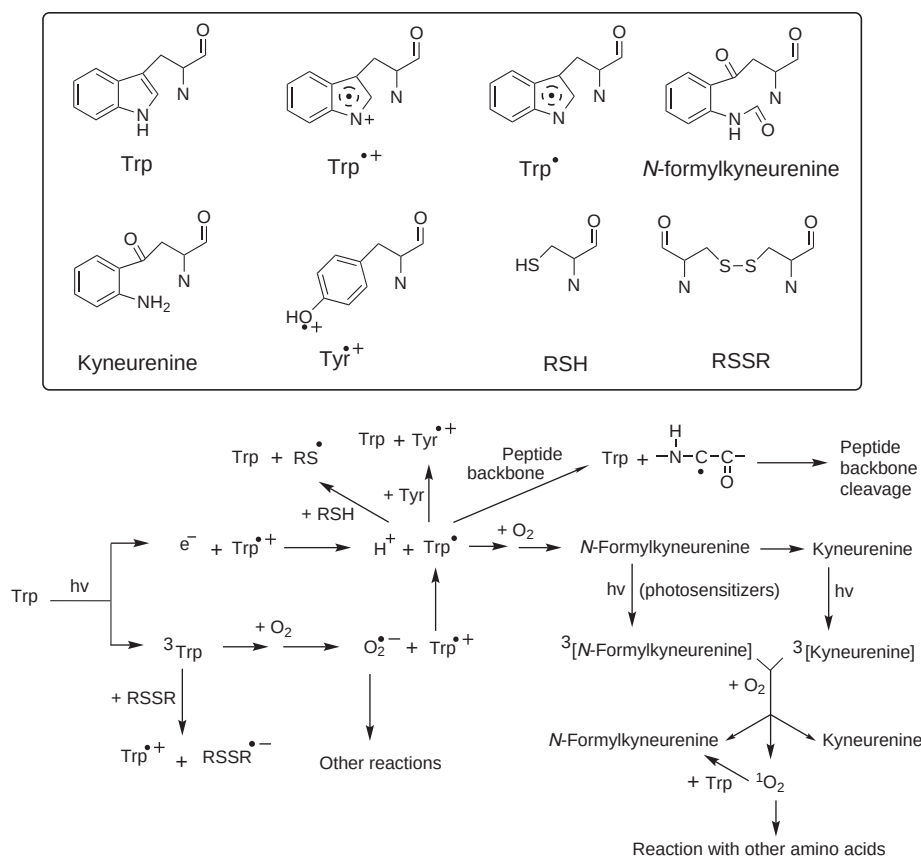


FIGURE 6.15 Structure of tryptophan (Trp) and common photolytic pathways.

The ejected electron can become solvated forming an e^-_{aq} or migrate along the peptide backbone and react with cystine residues, forming a disulfide radical anion as discussed below. Under anaerobic conditions, the tryptophan triplet state generally either relaxes back to the ground state with formation of light at 420 to 500 nm or electron transfers to a nearby cystine with subsequent reactions as discussed earlier. In the presence of oxygen, formation of the tryptophan-based peroxy radical can undergo further reaction to produce *N*-formylkyneurenine and kyneurenine. Interestingly, the *N*-formylkyneurenine and kyneurenine absorb light at longer wavelengths than Trp, thereby acting as photosensitizers to visible light, causing additional damage to the protein. Likewise, photodegradation of tyrosine, phenylalanine, cysteine, and histidine residues in proteins can occur.

Changes in the primary structure of the protein can result in changes in the secondary and tertiary structures impacting long-term stability, bioactivity, and immunogenicity of the peptide and protein drugs. Complete protection of the proteins from light is the only method to prevent photodegradation. This is achieved with primary and secondary packaging containers such as cardboard boxes that block incoming light to the protein. Addition of excipients such as methionine is also used to further reduce the protein aggregation as in the case

of darbepoetin alfa formulation. Methionine is known to react with peroxide to form methionine sulfoxide and likely reduces protein aggregation through its effect as a peroxide scavenger. Because photodegradation reactions occur through generation of singlet oxygen species, which are generated during the photolytic process by reaction of molecular dioxygen, some biopharmaceutical products, both liquid and lyophilized, are packaged in inert atmospheres.

Physical Instability

Generally not encountered in most small organic molecules, physical instability is a consequence of the polymeric nature of proteins. Proteins adopt secondary, tertiary, and quaternary structures, which influence their three-dimensional shape and, therefore, their biologic activity. Any change to the higher order structure of a protein may alter both. Physical instability includes denaturation, adsorption to surfaces, and noncovalent self-aggregation (soluble and precipitation). The most widely studied aspect of protein instability is denaturation. Noncovalent aggregation, however, is one of the primary mechanisms of protein degradation (29).

A protein, in principle, can be folded into a virtually infinite number of conformations. Denaturation occurs by disrupting the weaker noncovalent interactions that hold a protein together in its secondary and tertiary

structures. Temperature, pH, and the addition of organic solvents and solutes may cause denaturation. The process can be reversible or irreversible. In general, denaturation affects the protein by decreasing aqueous solubility, altering three-dimensional molecular shape, increasing susceptibility to enzymatic hydrolysis, and causing the loss of the native protein's biologic activity.

Handling and Storage of Biotechnology-Produced Products

The preparation and administration of protein drugs of synthetic, recombinant, or hybridoma origin are dissimilar to the nonprotein pharmaceuticals. Proteins generally have more limited shelf stability. The average shelf life for a biotechnology product is 12 to 18 months, compared with more than 36 months for a small-molecule weight drug. Although each individual biotechnology drug may be different, several generalizations can be made.

Proper storage of the lyophilized and the reconstituted drug is essential. Most of these drugs are expensive, so special care must be taken not to inactivate the therapeutic protein during storage and handling. The human proteins have limited chemical and physical stability, which is shortened on reconstitution. Expiration dating ranges from 2 hours to 30 days. The self-association of either native state or misfolded protein subunits may readily occur under certain conditions. This can lead to aggregation and precipitation and results in a loss of biologic activity. Self-association mechanisms depend on the conditions of formulation and may occur as a result of hydrophobic interactions.

Many of the biotechnology-produced drugs are stored refrigerated, but not frozen (2°C to 8°C). In general, temperature extremes must be avoided. One example is the rDNA-produced, blood clot-dissolving drug alteplase. A recombinant version of a naturally occurring human tissue-type plasminogen activator, lyophilized alteplase is stable at room temperature for several years if protected from light (37). Freezing or exposure to excessive heat decreases the physical stability of the protein. Anything that causes denaturation or self-aggregation, even though labile peptide bonds are not broken, may inactivate the protein. Some pharmacy facilities may need to increase cold storage capacity to accommodate biotech storage needs. If the patient must travel any distance home after receiving the medication, the pharmacist should help package the biotechnology product according to the manufacturer's directions. This may mean supplying a reusable cooler for the patient's use. Because the protein drug should not be frozen, the cooler should contain an ice pack rather than dry ice.

Some rDNA-derived pharmaceuticals, particularly the cytokines (e.g., the interferons, interleukin-2, and colony-stimulating factors), require human serum albumin in their formulation to prevent adhesion of the protein drug to the glass surface of the vial, which results in loss of protein (37–39). The amount of human serum albumin added varies with the biotech product. The

vials should not be shaken to prevent foaming of the albumin, which causes protein loss or inactivation of the biotechnology-derived proteins. Care must be exercised in reconstituting protein pharmaceuticals. The diluent used for reconstitution of biotechnology drugs varies with the product and is specified by the manufacturer. Diluents can include normal saline, bacteriostatic water, and 5% dextrose. Several reviews of biotechnology drugs written for pharmacists contain additional information on the subjects of handling and storage (40–45).

Pharmacokinetic Considerations of Biotechnology-Produced Proteins

The processes of absorption into, distribution within, metabolism by, and excretion from the body (i.e., ADME) of biotechnology-produced pharmaceuticals are important factors affecting the time course of their pharmacologic effect. To deliver quality pharmaceutical care with biotech products, a pharmacist must be able to apply pharmacokinetic principles to establish and maintain a nontoxic, therapeutic effect. The pharmacokinetics of protein and peptide drugs differ in some pharmacokinetic aspects from those of the small-molecule organic agents with which we are most familiar. Although a lengthy discussion of this topic is beyond the scope of this chapter, a brief overview of metabolism follows. Useful reviews also are available for further information (46,47).

The plasma half-life of most administered proteins and peptides is relatively short, because they are susceptible to a wide variety of metabolic reactions. The enzymes involved in peptide bond hydrolysis and, thus, the degradation of peptides and proteins are known as peptidases and can be found in the blood, in the vascular bed, in the interstitial fluid, on cell membranes, and within cells. These enzymes include carboxypeptidases (cleaves C-terminal residues), dipeptidyl carboxypeptidases (cleaves dipeptides from the C-terminus), aminopeptidases (cleaves N-terminal residues), and amidases (cleaves internal peptide bonds). For the most part, these enzymes are all specific for the natural amino acids of the L-configuration. Overall, the metabolic products of most proteins and peptides are not considered to be a safety issue. They generally are broken down into amino acids and reincorporated into new, endogenously biosynthesized proteins.

Metabolic oxidation reactions may occur to the side chains of sulfur-containing residues, similar to that observed for in vitro chemical instability. Methionine can be oxidized to the sulfoxide, whereas metabolic oxidation of cysteine residues forms a disulfide. Metabolic reductive cleavage of disulfide bridges in proteins may occur, yielding free sulfhydryl groups.

Biotechnology Drug Delivery

Protein-based pharmaceuticals, whether produced by biotechnology or isolated from traditional sources, present challenges to drug delivery because of the unique demands imposed by their physicochemical and biologic

properties. Although a detailed discussion of this topic is beyond the scope of this chapter, a brief overview follows. Useful reviews also are available for further information (31,48–52).

Delivery of large molecular weight, biotechnology-produced drugs into the body is difficult because of the poor absorption of these drugs, the acid lability of peptide bonds, and the rapid enzymatic degradation of these drugs in the body. In addition, protein pharmaceuticals are susceptible to physical instability, complex feedback control mechanisms, and peculiar dose–response relationships.

Given the limitations of today's technology, the strongly acidic environment of the stomach, the peptidases in the gastrointestinal tract, and the barrier to absorption presented by gastrointestinal mucosal cells preclude successful oral administration of most protein and peptide drugs. Therefore, administration of all the biotechnology-produced protein drugs and some natural or synthetic peptides currently is parenteral (by intravenous, subcutaneous, or intramuscular injections) to provide a better therapeutic profile. Manufacturers supply most of these drugs as sterile solutions without a preservative. In such cases, it is recommended that only one dose be prepared from each vial to prevent bacterial contamination.

Novel solutions to overcome delivery problems associated with biotechnology protein products and natural or synthetic peptide drugs are being explored. Oral drug delivery approaches in development for proteins and peptide drugs include conjugated systems (e.g., PEGylation with polyethylene glycol), amino acid substitution, liposomes, microspheres, erythrocytes as carriers, and viruses as drug carriers (30).

PEGylation is frequently used to improve the pharmacokinetic properties of biotechnology-produced drugs (53). This method involves the attachment of a flexible strand or strands of polyethylene glycol (PEG) to a protein. PEG is a simple, water-soluble, nontoxic polymer that is nonimmunogenic, is readily cleared from the body, and has been approved for human administration by mouth, injection, and topical application. The most common process used for PEGylation of protein is to activate the PEG with functional groups suitable for reaction with lysine and N-terminal amino groups. PEGylated proteins are less rapidly broken down by the body's enzymes than are unmodified proteins. PEG can extend the duration of action of proteins and peptides in the body from minutes, to hours, to days, depending on the PEG molecule used. By increasing the biologic half-life and improving the efficacy of proteins in the body, these modifications can reduce the frequency of injections a patient requires. They also reduce the rapidity and intensity of the body's immune reaction against molecules such as interferon. Several PEGylated protein products (e.g., pegvisomant, certolizumab pegol, peginterferon $\alpha 2a$) are currently marketed, and many are under development.

When metabolism studies indicate a predominant cleavage site, attempts can be made to replace that residue with another that retains the receptor-binding activity of the protein/peptide drug while yielding enhanced resistance to peptidase activity. Often, this can be accomplished by replacing the offending L-residue with its enantiomer, the D-amino acid, or another D-residue. Many peptidases are unable to cleave peptide bonds consisting of a "fraudulent" D-amino acid, and peptides containing such changes are designed as a means to increase their half-life. Also, the replacement of an L-amino acid with L-proline or N-methylation of the amide nitrogen offers the possibility of generating a peptide that is more resistant to enzymatic hydrolysis. The introduction of pseudopeptide bonds (54) and the design of retro-inverso peptides (55) are two examples of strategies that can afford peptidase resistance to peptide drugs.

Adverse Effects

An important consideration in the pharmaceutical care of a patient being administered a biotechnology-produced medicinal agent is the potential for adverse reactions. Many of the protein drugs are biotechnology-derived versions of endogenous human proteins normally present, on stimulus, in minute quantities near their specific site of action. Therefore, the same protein administered in much larger quantities may cause adverse effects not commonly observed at normal physiologic concentrations (46,47,56). Careful monitoring of patients administered biotechnology-produced drugs is critical for the health care team.

Immunogenicity

The immune system may respond to an antigen, such as a protein pharmaceutical, by triggering the production of antibodies. Biotechnology-derived proteins may possess a different set of antigenic determinants (i.e., regions of a protein recognized by an antibody) because of structural differences between the recombinant protein and the natural human protein (29,46,52). Factors that can contribute to this immunogenicity include lack of or incorrect glycosylation, amino acid modifications, and amino acid additions and deletions. A number of recombinant proteins produced with bacterial vectors contain an N-terminal methionine in addition to the natural human amino acid sequence. Bacterial vector-derived recombinant protein preparations also may contain small amounts of immunoreactive bacterial polypeptides as possible contaminants. Additionally, immunogenicity may result from proteins that are misfolded, denatured, or aggregated.

THERAPEUTIC PEPTIDE AND PROTEIN DRUGS

The following sections introduce a variety of peptide and protein hormones, both natural and synthetic, and their recombinant analogs that are commercially available for the treatment of various diseases or are used for

diagnostic purposes. These hormones are obtained synthetically, from natural sources, or via genetic engineering. They are listed, for the most part, according to their endocrine organ of origin. Space does not allow us to discuss other biologically significant peptides and proteins. Table 6.4, however, summarizes some of the interesting peptides and proteins used in medicine. Recombinant DNA technology provides a powerful tool for new pharmaceutical product development and production. Biotechnology-produced medicinal agents discussed in this chapter include hormones, enzymes, cytokines, hematopoietic growth factors, other growth factors, blood clotting factors and anticoagulants, vaccines, and monoclonal antibodies.

Production of Peptide and Protein Drugs

The production of many of important peptides and proteins, via synthetic procedures as well as through biotechnology, involves the pharmaceutical industry. Of necessity, peptides of low to moderate molecular weight, chemically modified peptides, and those containing pseudopeptide bonds or fraudulent amino acids will continue to be made by synthetic procedures rather than through biotechnology. Many of these hormones are available as chemical analogs, and these also

are discussed, particularly the chemical changes made and the implications of these changes to the resulting biologic action.

Peptide and Protein Hormones and Analogs

Hormones of Hypothalamic Origin

The hypothalamus, a relatively small organ that is located in the brain and is responsible for thermoregulation, among other functions, is the secretory source for a number of peptide hormones that are transported to the pituitary gland situated immediately below it. These hormones regulate the synthesis of other peptide hormones produced by the anterior pituitary (adenohypophysis) and are thus called releasing hormones, releasing factors, or inhibitory factors, as the case may be. The release of these hypothalamic hormones is regulated via cholinergic and dopaminergic stimuli from higher brain centers, and their synthesis and release are controlled by feedback mechanisms from their target organs.

Gonadotropin-Releasing Hormone

Gonadotropin-releasing hormone (GnRH) is a decapeptide (Fig. 6.16) that causes the release of the gonadotropins, luteinizing hormone (LH) and follicle-stimulating hormone (FSH), from the anterior pituitary gland, but

TABLE 6.4 Additional Significant Peptides and Proteins of Biologic Interest

Generic Name	Trade Name	General Use or Action
Exenatide	Byetta	39-Peptide known as an incretin mimetic; an agonist of glucagon-like peptide-1, used adjunctively in type 2 diabetes mellitus
Liraglutide	Victoza	Analog of human glucagon-like peptide-1 of rDNA origin used adjunctively in type 2 diabetes mellitus
Insulin-like growth factor-1 (human)		Protein of rDNA origin that has orphan drug status and is used in treating major burns requiring hospitalization
Secretin, synthetic human	Chirhostim	27-peptide used to evaluate exocrine pancreas function and in diagnosis of gastrinoma (Zollinger-Ellison syndrome)
Thymopentin	Timunox	Synthetic 5-peptide consisting of residues 32–36 of thymopoietin, an immunologically active polypeptide. It is an investigational drug used for its immunomodulating properties.
Thyrotropin alfa	Thyrogen	Recombinant form of human thyroid-stimulating hormone (rhTSH); used diagnostically for serum thyroglobulin testing
Carfilzomib		Multiple myeloma
Nesiritide	Natreacor	Acute decompensated congestive heart failure
Enfuvirtide	Fuzeon	HIV-1 infection
Ziconotide	Prialt	Chronic pain
Lanreotide	Somatuline	Acromegaly
Romiplostim	Nplate	Chronic idiopathic thrombocytopenic purpura
Degarelix	Firmagon	Advanced prostate cancer
Botulinum toxin A	Dysport	Inhibits release of acetylcholine, leading to its utility in diseases characterized by excessive efferent activity in motor nerves
Ecaltantide	Kalbitor	Acute attacks of hereditary angioedema
Tesamorelin acetate	Egrifta	Analog of growth hormone-releasing hormone for HIV-associated lipodystrophy

not in equal amounts. Therefore, GnRH is intimately involved in the control of both male and female reproduction. Medicinal chemists have capitalized on the relatively simple decapeptide structure of GnRH by preparing many analogs as potential fertility and antifertility agents, several of which are commercially available, especially those that are referred to as superagonists. It is known that GnRH can be degraded by enzymatic cleavage between Tyr 5–Gly 6 and Pro 9–Gly 10 (57). Structure–activity relationship studies of GnRH analogs have shown that when Gly 6 is replaced with certain D-amino acids, as well as with changes in the peptide C-terminus, they generally are less susceptible to proteolytic enzymes, resulting in a longer-lasting action. For that reason, they are referred to as superagonists. Furthermore, half-life of GnRH analogs is enhanced by substituting hydrophobic D-amino acids at position 6 (Fig. 6.16).

In physiologic doses, GnRH agonists are able to induce ovulation and spermatogenesis by increasing LH and FSH levels and the resulting sex steroid levels, as does the normal hormone. In larger pharmacologic (therapeutic) doses, however, GnRH agonists, especially the superagonists, block implantation of the fertilized egg, cause luteolysis of the corpus luteum, and can act as post-coital contraceptive agents (although not approved for this latter use). This “paradoxical” antifertility effect seen with the superagonists has been attributed to the fact that GnRH must be administered in a low-dose, pulsatile manner for it to be therapeutically effective as a fertility agent. Natural GnRH release from the hypothalamus occurs in a pulsatile manner. When GnRH or, especially, a superagonist is administered in pharmacologic doses each day, LH and FSH levels will initially rise but then begin to fall after a few days because of target tissue desensitization/downregulation of pituitary GnRH receptors. The continued use of these agents in a nonpulsatile manner will result in a drastic drop of the gonadal steroid levels to near castrate levels in both males and females, thereby giving rise to their use in such conditions as precocious puberty, endometriosis, and advanced metastatic breast and prostate carcinoma. Typically, however, the GnRH superagonists take approximately 2 weeks to finally desensitize the GnRH receptors, and during this time, there is a transient rise in LH and FSH levels, which often results in an initial “flare-up” of the original symptoms. The following discussion concerns the medicinal chemistry and medical use of the commercially available GnRH analogs (58).

SPECIFIC DRUGS

Leuprolide Acetate Leuprolide acetate, a synthetic nonapeptide analog of GnRH that possesses greater potency than the natural hormone, is a superagonist that is commercially available. Leuprolide acetate contains substitutions that hinder enzymatic degradation, D-Leu and NH-Et in place of glycine 6 and glycine 10 -NH₂, respectively (Fig. 6.16). Leuprolide acetate exhibits 15-fold higher potency than natural GnRH. When given continuously

and in therapeutic doses, leuprolide acetate inhibits LH and FSH secretion by desensitizing/downregulating the GnRH receptors, as discussed previously. After an initial stimulation, chronic administration of leuprolide acetate results in suppression of ovarian and testicular steroidogenesis. In premenopausal females, estrogens are reduced to postmenopausal levels; in males, testosterone is reduced to castrate levels.

Leuprolide acetate is administered by daily injections or as depot injections every month, every 3 months, every 4 months, or every 6 months as a palliative treatment in advanced prostatic carcinoma (as an alternative to orchiectomy). An implant version (Viadur) also is available for long-term palliative therapy; after implantation of the device into the upper arm, leuprolide acetate is continuously released over a 12-month period. Because dihydrotestosterone, a metabolite of testosterone, is able to stimulate the growth of prostate cancer, the ability of leuprolide acetate to bring testosterone to near castrate levels is why this drug finds use as a palliative in the advanced disease. The addition of a nonpeptidyl antiandrogen, such as flutamide or bicalutamide, to the leuprolide acetate regimen inhibits adrenal and testicular synthesized androgens from binding to or being taken up by target prostate cancer tissue. This combination therapy helps to control the initial flare-up, by blocking all sources of androgen, and is referred to as maximal androgen blockade.

Leuprolide acetate, in monthly and every-3-month depot formulations, is useful in treating women diagnosed with endometriosis, but not for longer than 6 months because of the chance of developing osteoporosis. Because estrogens stimulate the growth of endometrial tissue, the ability of this drug to drastically reduce estrogen levels suggests why leuprolide acetate is useful in treating endometriosis.

Central precocious puberty that is idiopathic, or gonadotropin dependent, can cause the development of secondary sexual characteristics in girls before the age of 8 years and in boys before the age of 9 years. In addition to the psychological and physiologic changes that occur because of entering puberty too early, there is the risk of the child failing to reach his or her full adult height. Therefore, leuprolide acetate’s ability to suppress LH and sex steroid levels (testosterone and estradiol) to prepubertal levels is the reason that leuprolide acetate is approved for treating children with this disease. Use of this drug in a child with precocious puberty will slow or stop that child’s secondary sexual development, slow linear growth and skeletal maturation, and in girls, bring about the cessation of menstruation.

Uterine leiomyomas (fibroids), which are benign neoplasms derived from smooth muscle, can cause, among other problems, excessive vaginal bleeding that may progress to anemia. Leuprolide acetate, concomitant with iron therapy, is used in treating the anemia that arises from uterine leiomyoma. The decrease in the formation of the steroid sex hormones reduces fibroid and uterine

volume, produces a relief in the clinical symptoms (pelvic pain), and stops the excessive vaginal bleeding, thus correcting the anemia.

Goserelin Acetate Goserelin acetate, like leuprolide acetate, is a synthetic superagonist nonapeptide analog of GnRH that possesses greater potency than the natural hormone. It contains D-Serine (Bu^t) and NH-NHCONH₂ in place of glycine 6 and glycine 10 -NH₂, respectively (Fig. 6.16). That is, the C-terminal modification simply has an NH substituting for the CH₂ of glycine, and like the C-terminal change in leuprolide acetate, this inhibits enzymatic degradation of the peptide by the postproline carboxyamide peptidase.

Goserelin acetate is available in the form of a small, solid pellet that is administered as subcutaneous (SC) implant for the palliative treatment of advanced, metastatic breast cancer in pre- and perimenopausal women or, similarly, as a palliative in advanced prostatic cancer. The rationale for this drug's use is, as described earlier, its ability as a superagonist to bring the levels of estradiol or testosterone to near castrate levels, thus slowing the progression of breast or prostate carcinoma, respectively. Additionally, goserelin acetate is approved for use in treating endometriosis for up to 6 months.

Goserelin acetate also is used in combination with the antiandrogen flutamide for shrinking prostate carcinoma before radiation therapy. This maximal androgen blockade combination is used when the prostate carcinoma has been staged as locally confined to the prostate gland, with one or both lobes as well as the seminal vesicles involved. The treatment should start 8 weeks before radiation treatment begins and be continued throughout the radiation therapy.

Furthermore, women who are to undergo hysterectomy for menorrhagia can benefit from previous treatment with goserelin acetate, because it is able to induce endometrial thinning. This thinning of the endometrium improves the operating environment by causing less intrauterine bleeding, increased postoperative

amenorrhea, and decreased dysmenorrhea following surgery, which is why goserelin acetate is approved for inducing endometrial thinning prior to a patient undergoing a hysterectomy for heavy menstrual bleeding.

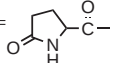
Nafarelin Acetate Nafarelin acetate, another synthetic superagonist decapeptide analog of GnRH, contains D-Nal(2) 6 [Nal = 3-(2-naphthyl)-Ala] in place of glycine 6, but the C-terminus, glycine 10 -NH₂, is identical with natural GnRH (Fig. 6.16). Nafarelin acetate is available as a 0.2% nasal spray for the relief of endometriosis. Estrogen, of course, is needed for the growth of endometrial tissue; thus, decreased estrogen leads to shrinkage of errant endometrial tissue. The observed side effects of nafarelin acetate are related to falling estrogen levels and include decreased libido, amenorrhea, hot flashes, and vaginal dryness. When used consistently, nafarelin acetate will inhibit ovulation and stop menstruation.

Nafarelin acetate also is used in children, male and female, for the treatment of central precocious puberty. By suppressing the release of LH, the estradiol or testosterone levels fall to prepubertal levels; early secondary sexual development is arrested, linear growth and skeletal maturation are slowed, and in girls, menstruation stops.

Histrelin Acetate Histrelin acetate, a superagonist analog of GnRH, contains D-histidine (N¹-Bzl) 6 in place of glycine 6, and the C-terminus is identical with leuprolide acetate—namely, NH-Et in place of glycine 10 -NH₂ (Fig. 6.16). This GnRH analog is commercially available in the form of an implantable device (SC in the upper arm) that slowly releases the drug over a 12-month period, resulting in decreased testicular steroidogenesis, for the palliative treatment of advanced prostate cancer.

Triptorelin Pamoate Triptorelin pamoate is another superagonist of GnRH, which like nafarelin acetate contains only a single amino acid substitution (D-tryptophan 6 for glycine 6) when compared to the natural hormone

endopeptidase cleavage		post-proline carboxyamide peptidase	
1	5	9	10
pGlu-His-Trp-Ser-Tyr-Gly-Leu-Arg-Pro-Gly-NH ₂			
_____ D-Leu ⁶ _____		Pro-NH-Et	
_____ D-Ser(Bu ^t) ⁶ _____		Pro-NH-NHCONH ₂	
_____ D-Nal(2) ⁶ _____		Gly-NH ₂	
_____ D-His(N ¹ -Bzl) ⁶ _____		Pro-NH-Et	
_____ D-Trp ⁶ _____		Gly-NH ₂	
Drug		Trade Name	
GnRH			
Leuprolide acetate		Eligard, Lupron, Viadur	
Goserelin acetate		Zoladex	
Nafarelin acetate		Synarel	
Histrelin acetate		Vantas	
Triptorelin pamoate		Trelstar LA and Depot	

FIGURE 6.16 GnRH-based drugs that are commercially available. Note that leuprolide, goserelin, nafarelin, histrelin, and triptorelin are all superagonists and contain a D-amino acid in place of Gly⁶, and that three of the five are missing the C-terminal Gly (the line indicates an identical sequence of amino acids). Note: pGlu = pyroglutamic acid =  Bu^t = *t*-butyl; D-Nal(2) = D-3-(2-naphthyl)alanine; N¹-Bzl = 1-benzyl-D-histamine.

(Fig. 6.16). In the treatment of advanced prostate cancer, it is important to reduce serum testosterone levels to very low levels, which can be achieved surgically by orchiectomy. When this surgical method is unacceptable to the patient, an alternative approach is “chemical castration,” which can be achieved by use of estrogen therapy, leuprolide, goserelin or histrelin acetates, and triptorelin pamoate. This product is available for intramuscular depot injection (monthly or every 3 months), wherein serum testosterone concentration drops to a level generally seen in surgically castrated men.

Ganirelix Acetate Ganirelix acetate is an analog of GnRH with substitutions at residues 1, 2, 3, 6, 8, and 10 (Fig. 6.17). It is not a superagonist but, rather, is a synthetic decapeptide with high antagonist activity and the first GnRH antagonist to be marketed. It is approved for the suppression of LH surges in women who are undergoing ovarian hyperstimulation fertility treatment; LH surges normally promote ovulation. The goal of this drug is to significantly reduce the number of medication days necessary to suppress the LH surge, thereby maintaining eggs in the ovaries. In vitro fertilization (IVF) treatment cycles were historically initiated by the administration of leuprolide acetate to suppress the premature release of LH. This inhibits ovulation so that the eggs remain available for retrieval by a fertility specialist. For this purpose, leuprolide acetate usually is injected for as many as 26 days. Clinical studies have shown that ganirelix acetate can shut down the LH surge in only 5 days of treatment, that the suppression of LH is more pronounced than that of FSH, and that the shorter treatment time minimizes unpleasant side effects, such as hot flashes and headaches.

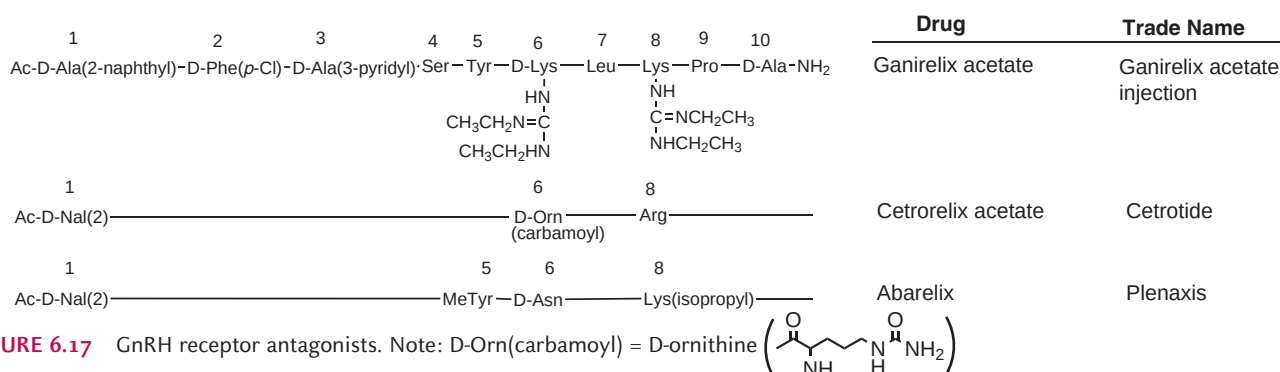
Cetrorelix Acetate Cetrorelix acetate is an analog of GnRH with amino acid substitutions at residues 1, 2, 3, 6, and 10 and differing from ganirelix at amino acids 1, 6, and 8 (Fig. 6.17). These substitutions are synthetic, non-DNA-directed amino acids and, like ganirelix acetate, impart GnRH antagonist activity to cetrorelix acetate. This drug also is marketed for use in women undergoing assisted reproductive therapy (ART) procedures, in which it is necessary to control their LH surge. This allows the follicles to develop to a size, as determined by

ultrasound, that increases the success of timed insemination and oocyte retrieval for IVF. Like ganirelix acetate, cetrorelix acetate has an advantage over GnRH agonists, such as leuprolide acetate, because it reduces the fertility therapy cycle to days rather than weeks.

Abarelix Abarelix is another GnRH receptor antagonist with substitutions or alterations of the natural hormone at seven residues; the amino acids at positions 4, 7, and 9 are identical with those in GnRH. Abarelix differs from ganirelix at amino acids 1, 5, 6, and 8 (Fig. 6.17). Unlike ganirelix and cetrorelix, however, abarelix is not used in women undergoing ART but, rather, as a palliative in men with advanced prostate cancer. Like the superagonists, abarelix affords a drop in testosterone to castrate levels. Abarelix does not, however, causes the initial rise in testosterone levels and resulting flare-up of symptoms often seen with the superagonists. Still, because of the risk of life-threatening allergic reactions associated with its use, this drug is specifically relegated to a select group of men with advanced symptomatic prostate cancer who are not candidates for other hormone therapies and who choose not to undergo orchiectomy. Abarelix is available only from health care providers who have participated in a special education program that allows them to identify allergic reactions associated with its use.

Somatostatins

Somatostatin is a cyclic 14-peptide that was first isolated by Guillemin in 1973 and is probably the most thoroughly investigated and most important of the inhibitory factors produced by the hypothalamus (Fig. 6.18). The principal activity of somatostatin is inhibition of the release of growth hormone (GH) from the anterior pituitary. Too much GH, as in pituitary tumors, causes acromegaly, a form of gigantism. However, too little GH leads to dwarfism. Somatostatin also has been found in the pancreas and the gastrointestinal tract, where it inhibits the secretion of both insulin and glucagon from the pancreas as well as the secretion of a variety of intestinal peptides (e.g., gastrin, secretin, pepsin, and renin). The short half-life of somatostatin, which is less than 3 minutes, has precluded its use as a therapeutic agent. Many derivatives of somatostatin have been prepared to increase its duration



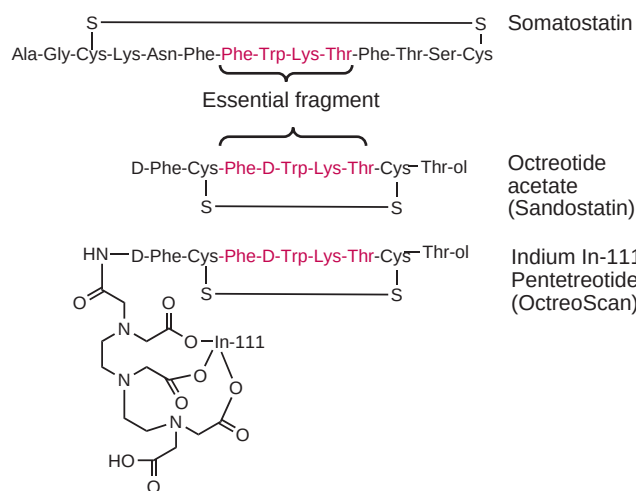


FIGURE 6.18 Somatostatin and its analog octreotide.

of action and/or to enhance its selectivity of action. The culmination of these structure–activity studies was the development of octreotide acetate.

SPECIFIC DRUGS

Octreotide Acetate Octreotide acetate, a long-acting octapeptide analog of somatostatin, has a half-life of approximately 100 minutes. A comparison of the primary structures of octreotide and somatostatin suggests little similarity, but studies suggested that the essential fragment for its activity was the tetrapeptide phenylalanine7-tryptophan8-lysine9-threonine10 (Fig. 6.18). These studies helped in the design of the potent drug now known as octreotide acetate (59). This drug suppresses the secretion of gastroenteropancreatic peptides, such as gastrin, vasoactive intestinal peptide (VIP), insulin, and glucagon, as well as pituitary GH. Furthermore, it is more potent than natural somatostatin in inhibiting the release of glucagon, insulin, and GH.

Octreotide acetate is used by SC injection in the palliative treatment of patients with metastatic carcinoid tumors, which are tumors of the endocrine system, gastrointestinal tract, and lung. It is also used in the palliative management of VIP-secreting tumors (VIPomas, usually pancreatic tumors). Patients with VIPomas suffer a profuse, watery diarrhea syndrome, and octreotide acetate is able to help by decreasing the release of damaging intestinal tumor cell secretions. Octreotide also helps to reduce hypokalemia by correcting electrolyte imbalances.

Octreotide acetate is able to decrease the secretion of GH from the pituitary and is used in treating patients with acromegaly who are unresponsive to previous pituitary radiation therapy or surgery. It is used in the treatment of acromegaly, because it reduces the blood levels of both GH and insulin-like growth factor-1. The long-acting repository form of octreotide acetate also is used in treating acromegaly, carcinoid tumors, and VIPomas, but in monthly depot injections.

Octreotide for intravenous injection is used in the treatment of acute bleeding from esophageal varices. Variceal bleeding occurs in about half the patients with cirrhosis of the liver and is responsible for about one-third of deaths in these patients. Octreotide is a potent vasoconstrictor that reduces portal and collateral blood flow by constricting visceral vessels, which leads to reduced portal blood pressure and decreases the bleeding.

Indium-111 Pentetreotide (OctreoScan) Somatostatin receptors have a broad distribution in normal tissue as well as in a variety of human malignancies (e.g., small-cell lung, brain, breast, pituitary, and endocrine pancreatic cancers). For this reason, octreotide, which binds to somatostatin receptors, was converted to a radionuclide-containing peptide by reacting the amino terminus with an active ester of diethylenetriaminepentaacetic acid (DTPA) to give DTPA-octreotide and then chelated to the radionuclide indium-111 (^{111}In) (Fig. 6.18). This radiopharmaceutical is used as a diagnostic agent for the early detection and localization of small tumors and their metastases in the body, especially tumors that originate from neuroendocrine cells.

Corticotropin-Releasing Factor

The primary function of corticotropin-releasing factor (CRF), a hypothalamic 41-peptide, is the regulation of the release of adrenocorticotropin hormone (ACTH) from the anterior pituitary gland, which then stimulates the release of hydrocortisone from the adrenal gland. Human CRF, which was previously commercially available, has been replaced by the sheep CRF (oCRF).

Corticotropin-Releasing Factor (Actrel) The ovine form of the hormone oCRF, which shares 83% homology with the human form, has a longer half-life and greater potency (60). Usually, ACTH deficiency in humans is associated with a pituitary disorder rather than with a deficiency of CRF, and oCRF can be used to distinguish between pituitary hypersecretion of ACTH (Cushing disease) and ectopic ACTH secretion, both of which are conditions that cause a hypersecretion of hydrocortisone. In the case of ectopic ACTH secretion, the administration of oCRF will elicit little to no response in the production of ACTH and hydrocortisone. Therefore, synthetic oCRF trifluoroacetate salt is marketed as a diagnostic agent for patients with ACTH-dependent Cushing syndrome.

Hormones Originating in the Anterior Lobe of the Pituitary Gland

The pituitary, lying just below the hypothalamus, is a small gland that can be divided into an anterior and a posterior lobe. This gland is responsible for the secretion of several important peptide hormones, two of which are released by the posterior lobe. The anterior pituitary peptide hormones control important functions such as growth, reproduction, metabolic and ion balance, as well as a number of other functions.

Growth Hormone

Human GH, a 191-amino acid protein with a molecular weight (MW) of 22 kDa is roughly spherical with a hydrophobic interior. GH is secreted by the anterior pituitary in response to the liberation of GH-releasing factor from the hypothalamus. This contrasts with somatostatin, which inhibits the release of GH. The primary function of GH in the body is to promote skeletal growth. When GH is absent during childhood, or if there is an inadequate supply, dwarfism results. Before 1985, children of short stature were sometimes treated with human GH (hGH) of pituitary origin, which was obtained from cadavers, but hGH of natural origin was discontinued by the U.S. Food and Drug Administration (FDA) when several young adults who had received hGH died. Their deaths were attributed to contaminated hGH, and the contaminant was later found to be an infective agent, known as a prion, that causes Creutzfeldt-Jakob disease, a rare and fatal neurodegenerative disease (61). Naturally occurring hGH has been replaced with material that is prepared by rDNA methodology (62).

SPECIFIC DRUGS

Somatropin (Genotropin, Humatrope, Norditropin, Nutropin, Saizen, Serostim, Zorbtive) Somatropin, which is an hGH that is prepared by rDNA procedures, contains exactly the same sequence of 191 amino acids as the natural hormone. Several of these products are indicated for the long-term treatment of children who fail to grow because of inadequate secretion of the body's normal endogenous GH and for growth problems associated with chronic renal failure (63). Even adults who are diagnosed with GH deficiency, which can arise as a result of pituitary or hypothalamic disease, surgery, radiation therapy, or other reasons, can benefit from hGH replacement therapy. Turner syndrome, a genetic disease in which there is a complete or partial absence of one of the two X chromosomes in females, causes short stature as one of its many symptoms. Girls suffering from Turner syndrome can benefit from the use of Humatrope, Norditropin, or Nutropin during their growth years. A long-acting dosage form of somatotropin is approved for use in children with Prader-Willi syndrome, a rare genetic disorder that causes short stature; an involuntary, continuous urge to eat that is life-long and may be life-threatening; low muscle tone; and cognitive disorders. The formulation was designed to reduce the frequency of injections to once or twice a month by encapsulating the agent in biodegradable microspheres.

The anabolic properties of hGH are the basis for the orphan drug uses of several of these recombinant products. hGH is used in AIDS-associated catabolism or weight loss, in cachexia resulting from AIDS, and as an anabolic agent in patients with severe burns. In addition, the anabolic property of recombinant hGH is the reason for the use of Zorbtive, along with specialized nutritional support, in the treatment of short bowel syndrome.

Pegvisomant (Somavert) This product contains 191 amino acid residues (of recombinant origin), the same number as in GH, but there are substitutions at residues 18, 21, 120, 167, 168, 171, 172, 174, and 179. This product is then covalently linked to several PEG molecules, and this PEGylated protein is a GH receptor antagonist. Thus, pegvisomant binds to the GH receptor and blocks endogenous GH from binding. The result is a blocking of the GH-stimulated overproduction of insulin-like growth factor-1 that contributes to the disabling symptoms and long-term health problems associated with acromegaly.

Gonadotropins

The gonadotropins, FSH and LH, are large glycoproteins released by the anterior pituitary on stimulation from GnRH produced by the hypothalamus. Both FSH and LH consist of two noncovalently associated α and β subunits. The α subunits contain 92 amino acids and are identical in both hormones, whereas the β subunit consists of 121 amino acids in LH and 111 amino acids in FSH, with the β subunits being dissimilar (64). Thus, the β subunit gives each hormone its specific function.

Both LH and FSH are referred to as gonadotropins, because they act on the male and female gonads, which results in the production of the sex steroids testosterone and estradiol, respectively. In the female, FSH and LH act in concert in regulating ovarian function: egg maturation, ovulation, and transformation of the ruptured follicle into the corpus luteum. In males, spermatogenesis is dependent on these two hormones. FSH is a 34 kDa glycoprotein (~14% carbohydrate) that, in females, facilitates the maturation of ovarian follicle cells and their secretion of estradiol, whereas in males, it stimulates the maturation of sperm in the testes. In females, LH promotes ovulation, formation of the corpus luteum, and progesterone secretion; in males, it enables the secretion of testosterone from the testes.

Male infertility is generally caused by the quality and/or quantity of the sperm produced. In the case of female infertility, however, it may be caused by several factors, such as the inability to produce an egg, to ovulate, for fertilization to occur, and for implantation of the fertilized ovum in the uterus. Several of the commercially available gonadotropins can help in enhancing both male and female fertility and are discussed below.

SPECIFIC DRUGS

Menotropins (Menopur, Repronex) Menotropins are natural products that are obtained from the urine of postmenopausal women and then biologically standardized (international units) for FSH and LH activities in an approximate ratio of 1:1. Menotropins are used in males with primary (hypothalamic) or secondary (pituitary) hypogonadism to stimulate spermatogenesis, provided they have been treated previously with human chorionic gonadotropin (hCG; a peptide hormone of placental origin that has activity very similar to LH; discussed below) to effect masculinization

(increased testosterone production). In females, menotropins and hCG are given sequentially for the purposes of inducing ovulation in women who are having difficulty ovulating as a result of either hypothalamic or pituitary hormonal dysfunction. The menotropins are given for 7 to 12 days, and after clinical evaluation (via ultrasound) indicates the presence of a mature follicle, a single dose of hCG is given to simulate the typical LH surge that normally triggers ovulation. Also, women use the combination of menotropins and hCG to promote the development of multiple follicles when they are participating in an IVF program requiring the recruitment of follicles.

Follitropins Follitropins are hormonal products that consist entirely of FSH and are used to stimulate ovarian follicle growth in women who do not have primary ovarian failure. In the absence of an adequate endogenous LH surge, however, hCG must be given following the use of follitropins to stimulate ovulation.

Urofollitropins (Bravelle) Urofollitropin, a natural product like the menotropins, is obtained from the urine of postmenopausal women and then highly purified so as to contain only FSH, reportedly with only minute amounts of LH. Urofollitropin is used for its ability to stimulate follicle development, such as in women undergoing drug-induced pituitary suppression (GnRH antagonist or superagonist) for purposes of IVF (i.e., multiple follicle development or egg donation). When the number and size of the ovarian follicles are correct, as determined by ultrasound, hCG is administered so as to effect ovulation, and the oocytes are retrieved for IVF.

This drug also is indicated for women who have infertility caused by polycystic ovary syndrome, which generally is observed clinically as enlarged, cystic ovaries containing relatively small follicles. These patients often develop hirsutism, their androgen and LH levels appear elevated while their FSH levels are low, and the early exposure to these improper hormone levels may be causing the follicular atresia (65).

Lately, it has become desirable to have a nonnatural source of pure FSH. It is believed that exposure to increased amounts of LH early in follicular development, as would be the case with menotropins, is detrimental to fertility. Also, urofollitropin is derived from menopausal urine, and the supply of this natural product is limited and, in addition, contains variable levels of urine-derived contaminant proteins (66). These problems have given rise to recombinant forms of FSH.

Follitropin alfa and follitropin beta are human FSH preparations produced by rDNA technology (67). The production of rhFSH in a Chinese hamster ovary (CHO) cell line has proved to be particularly challenging. Follitropin alfa was the first heterodimeric glycoprotein to be produced by rDNA technology. Before the product of rDNA origin was available for infertility treatment, FSH was isolated from urine at less than 5% purity.

Follitropin Alfa (Gonal-F, Gonal-F RFF, Gonal-F RFF Pen) Follitropin alfa is a human FSH preparation of rDNA origin. Because FSH is a glycoprotein, alterations in the carbohydrate side-chain attachments afford different isoforms, which leads to different pharmacokinetic and pharmacodynamic properties. Because it is of recombinant origin and not isolated from urine, it is free of any additional substances, such as urinary proteins and LH.

Like urofollitropin, it is marketed for enhancing the development of multiple follicles that can then be induced to ovulate, via hCG administration, so that the oocytes can be collected for IVF. It also is used in treating women who wish to become pregnant and are anovulatory because of polycystic ovary syndrome, in whom it can enhance follicle maturation before hCG administration for final ovulation.

Men with infertility also can benefit from therapy with follitropin alfa if their infertility is related to hypothalamic or pituitary hormonal dysfunction and not primary testicular failure, because it induces spermatogenesis. Just as with therapy using menotropins, pretreatment with hCG is performed for 3 months to achieve serum testosterone levels within the normal range, before hCG and follitropin alfa therapy.

Follitropin Beta (Follistim AQ) Follitropin beta is a human FSH preparation of rDNA origin that differs chemically from natural FSH and follitropin alfa only by slight variances in the composition of the carbohydrate side chains. In fact, the primary and tertiary structures of both follitropins alfa and beta are indistinguishable from those of human FSH of natural origin. Furthermore, bioassays and physicochemical studies indicate that follitropins alfa and beta are indistinguishable from each other. Therefore, follitropin beta is approved for the same medical uses as follitropin alfa.

Lutropin Alfa (Luveris) Lutropin alfa is the first pure human LH preparation and is of rDNA origin. According to physicochemical and biologic assays, it is indistinguishable from human LH of natural origin. It is indicated for use in infertile women who are undergoing ART, specifically those with severe LH deficiency. It is meant to be coadministered with follitropin alfa so as to stimulate follicular growth in these women. When the follicles are of correct size, as determined by ultrasound examination, hCG is administered to stimulate ovulation.

Anterior Pituitary Hormones

Adrenocorticotropin hormone (H.P. Acthar Gel) The anterior pituitary, under the influence of the hypothalamic hormone hCRF, releases ACTH, a single-chain peptide of 39 amino acids (also known as corticotropin). The sequence of 24-amino acid residues beginning from the amino terminus contains all the biologic activity of the parent. The remaining 15 C-terminal residues confer species specificity as well as enhance the stability of ACTH toward proteolytic cleavage. Amino acids 1 through 24, which are critical for ACTH activity, are identical in

humans, pigs, sheep, and beef, whereas the species differ only slightly from each other in the final 15 amino acids. The main action of ACTH on the adrenal cortex involves the release of the glucocorticoid hormone hydrocortisone and the mineralocorticoid hormone aldosterone.

Commercial ACTH is obtained from natural sources and is available in 16% gelatin (repository gel) to prolong its release after SC or intramuscular injection. ACTH has both anti-inflammatory and immunosuppressant properties, which contributes to its use in the treatment of acute exacerbations of multiple sclerosis.

Cosyntropin (Cortrosyn) Cosyntropin is a synthetic polypeptide that consists of amino acids 1 through 24 of human ACTH, but that has the full biologic activity of its parent. Because it is of synthetic origin, it is less allergenic than ACTH of natural origin. Cosyntropin is used as a diagnostic agent in the screening of patients suspected of having adrenocortical insufficiency. Normally, parenteral administration of cosyntropin acts rapidly on the adrenal cortex to effect release of plasma hydrocortisone; when performing the test, the hydrocortisone levels are compared to a control blood sample taken earlier.

Thyrotropin (Thyrogen)

Thyroid-stimulating hormone (thyrotropin) is a 28 to 30 kDa glycoprotein secreted by the anterior lobe of the pituitary gland that is necessary for the growth and function of the thyroid. A recombinant thyrotropin α useful for the detection and treatment of cancer was approved by the FDA in 1998.

Hormones Released from the Posterior Lobe of the Pituitary Gland

As previously discussed, the pituitary gland is responsible for the secretion of several peptide hormones, only two of which are released by the posterior lobe. In fact, these two hormones, oxytocin and vasopressin, are synthesized in neurons originating in the hypothalamus and are transported to the posterior pituitary for storage until release is required.

Oxytocin Oxytocin is a cyclic nonapeptide that, like somatostatin, contains a ring that encompasses a disulfide bridge (Fig. 6.19). Oxytocin has uterotonic action, contracting the muscles of the uterus during gestation, and plays an important role in milk ejection. Exogenous oxytocin most commonly is used for induction of labor, wherein it improves uterine contractions to achieve early vaginal delivery for fetal or maternal reasons (e.g., preeclampsia, Rh factor problems, pregnancy that has exceeded 42 weeks). It also finds use following delivery of the placenta, because it promotes contraction and vasoconstriction and helps to control postpartum bleeding.

Vasopressin Human vasopressin, or Arg-vasopressin, is chemically very similar to oxytocin and therefore sometimes is referred to as [phenylalanine 3, arginine 8]

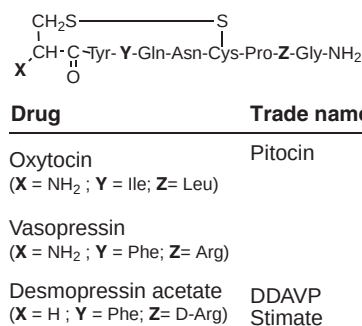


FIGURE 6.19 Structural relationship between oxytocin, vasopressin, and its analog.

oxytocin (Fig. 6.19). The physiologic role of vasopressin is the regulation of water reabsorption in the renal tubules, and thus, it is often referred to as the antidiuretic hormone. In high doses, vasopressin promotes the contraction of arterioles and capillaries, resulting in an increase in blood pressure, thus the name vasopressin. An inadequate output of pituitary antidiuretic hormone can cause diabetes insipidus, which is characterized by the chronic excretion of large amounts of pale urine and results in dehydration and extreme thirst.

Desmopressin Acetate Desmopressin, as its acetate salt, is a synthetic analog of vasopressin in which the N-terminal cysteine is devoid of its α -amino function (1-deamino) and where arginine 8 is present as its D-isomer (D-arginine 8), thus the commercial acronym DDAVP (Fig. 6.19). The presence of D-Arg and the absence of the N-terminal amine in the desmopressin structure have increased its half-life such that it is available for oral, parenteral, or nasal use. It is used by all three of these routes of administration to prevent or control polydipsia (excessive thirst), polyuria, and dehydration of patients with diabetes insipidus caused by a deficiency of vasopressin. It also has been approved for the treatment of nocturnal enuresis (bed-wetting), which is believed to be caused by an absence of the normal nighttime rise in vasopressin levels.

Desmopressin is known to cause an increase in both plasma factor VIII (antihemophilic factor) and plasminogen activator. Therefore, it is approved by the FDA for use, parenterally and nasally, in reducing spontaneous or trauma-induced bleeding episodes in patients with hemophilia A and type I von Willebrand disease, provided that their plasma factor VIII activity is greater than 5%. Stimate, the nasal spray used in treating patients with hemophilia A and type I von Willebrand disease, is 15-fold the concentration of DDAVP nasal spray; the latter is used in treating diabetes insipidus.

Hormones of Placental Origin

If, after ovulation occurs in females, the liberated ovum is fertilized and then implants in the endometrium, the

resulting placenta that forms between mother and fetus begins to release a hormone, hCG, the function of which is to maintain and prolong the life of the ovarian corpus luteum. The corpus luteum is important for the continued production of progesterone. Progesterone is especially important because it prepares the uterus for pregnancy and helps in the maintenance of the placenta. hCG begins to appear in the maternal bloodstream and urine shortly after conception and implantation of the fertilized ovum. As a result of this early release of hCG, its detection in the urine forms the basis for the home pregnancy kits that have become so popular in the early prediction of pregnancy.

Human Chorionic Gonadotropin (hCG, Pregnyl) Placental hCG is a complex protein that consists of an α and β subunit. The α subunit consists of 92 amino acids that are identical in sequence with those found in both LH and FSH, whereas the β subunit contains 145 amino acids and is responsible for its biologic specificity. The biologic actions of hCG closely resemble those of LH, but the former has a longer half-life than the latter and has minimal FSH activity.

Like LH, hCG stimulates the production of testosterone by the testes; therefore, it is used in treating male hypogonadism and prepubertal cryptorchidism in young males (age, 4 to 9 years), in whom it stimulates testicular descent. In treating infertility caused by pituitary dysfunction, hCG, in combination with menotropins (as discussed previously) or clomiphene, can induce ovulation and pregnancy in anovulatory females. This hCG, which is of natural origin, is purified from the urine of pregnant women.

Choriogonadotropin Alfa (Ovidrel) Choriogonadotropin alfa is obtained by rDNA technology and is biologically and chemically identical to hCG of natural origin. It is used, like hCG of natural origin, for inducing ovulation in women with anovulatory infertility. Following proper pretreatment with a GnRH antagonist or superagonist to desensitize the pituitary, women participating in ART are treated with a follicle-stimulating agent (e.g., menotropins) to effect the final maturation of the follicles within the ovaries. Ultrasonograms are used to determine proper follicle maturation before the administration of choriogonadotropin alfa to induce ovulation. A distinct advantage of this product is that it can be self-administered by the patient via SC injection.

Hormone of Parathyroid Origin

The parathyroid glands (four) exist as two pairs, one pair of which is embedded on the back surface of each lobe (two) of the thyroid gland. These very small glands are responsible for the secretion of parathyroid hormone (PTH), the action of which is the regulation of both calcium and phosphate metabolism within bone and kidney. In humans, the Ca^{2+} concentration is carefully regulated, and when it falls below a normal level, the parathyroid

glands secrete PTH, an 84-amino acid, single-chain protein (see Chapter 30 and Fig. 30.2). Depending on whether PTH is administered intermittently or continuously, it can either stimulate bone formation or breakdown (resorption), respectively.

Teriparatide (Forteo) Teriparatide, a polypeptide prepared by rDNA techniques, consists of the first 34 amino acids from the N-terminal end of PTH (see Chapter 30 and Fig. 30.2). It has been shown to contain all the structural requirements for the full biologic activity of PTH. When teriparatide is administered daily by SC injection, it stimulates osteoblastic activity at the expense of osteoclastic activity, and this enhances bone formation. This is the basis for teriparatide's use in treating high-risk patients in danger of bone fracture resulting from osteoporosis, men with primary or hypogonadal osteoporosis, and women with postmenopausal osteoporosis.

Hormone Secreted by the Parafollicular C Cells of the Thyroid Gland

The majority of the thyroid gland contains follicular cells responsible for the production of the thyroid hormones. A second population of endocrine cells within the thyroid known as C (clear) cells, or parafollicular cells, produce the hormone calcitonin (CT), which has an opposing action to that of PTH in that it decreases the Ca^{2+} concentration in body fluids. It accomplishes this by inhibiting the activity of osteoclasts (i.e., decreasing Ca^{2+} release from bone by inhibiting bone resorption). The actual biosynthesis and release of CT is regulated by the concentration of Ca^{2+} in plasma (i.e., when it is high, CT secretion increases).

Salmon Calcitonin (Fortical, Miacalcin) Calcitonin is a single-chain polypeptide consisting of 32 amino acids (Fig. 6.20). Calcitonins as obtained from different species are identical at seven of the first nine amino acids, contain glycine at position 28, and all terminate with proline- NH_2 . The C-terminal proline amide (proline- NH_2) is very important for the biologic function of CT, as is the disulfide bridge between cysteine amino acids at positions 1 and 7. In contrast, the amino acids from 10 through 27 can be varied and seem to influence CT's potency as well as its duration of action. Salmon CT differs from human CT at 16 amino acids.

Only salmon CT is commercially available for medical use, because on a weight basis, it is approximately 45-fold more potent than human CT. Salmon CT, in parenteral form, is approved for treating Paget disease of bone (generally seen in older persons; involves increased bone resorption and softening of bones), postmenopausal osteoporosis, and hypercalcemia of malignancy (multiple myeloma or advanced breast carcinoma). Salmon CT also is available in a nasal spray formulation, which is used exclusively in the treatment of postmenopausal osteoporosis.

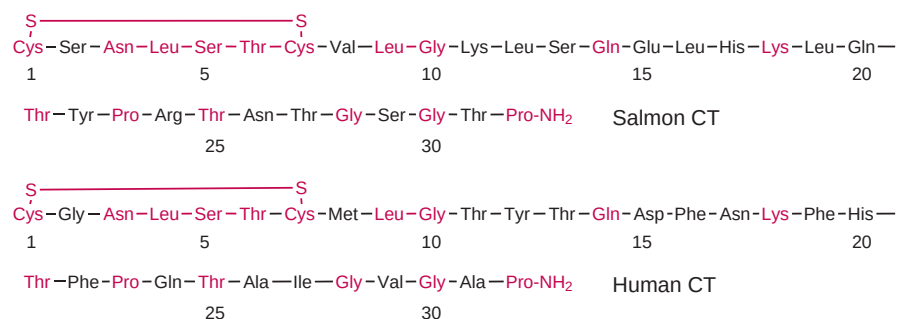


FIGURE 6.20 Primary structures of salmon and human calcitonin (CT). Similarities are highlighted in red.

Hormones of Endocrine Pancreatic Origin

The exocrine pancreas consists mostly (~99%) of gland cells known as pancreatic acini, which are responsible for secreting several digestive enzymes. The endocrine pancreas, or the remaining 1% of the gland, consists of a group of cells known as pancreatic islets or islets of Langerhans. Each of these islets consists of four distinct cell types, designated as α , β , γ , and δ cells. The α cells secrete glucagon, the β cells insulin and amylin, the δ cells a peptide that is identical with somatostatin of hypothalamic origin, and the γ cells pancreatic polypeptide, of which little concerning its physiologic action is known. Insulin, glucagon, and somatostatin are essential in regulating carbohydrate, lipid, and amino acid metabolism. Insulin is responsible for promoting the storage of glucose as glycogen and effecting hypoglycemia, whereas glucagon mobilizes glucose from its glycogen stores and causes hyperglycemia. The primary action of somatostatin of hypothalamic origin is to inhibit the release of GH from the pituitary, but pancreatic somatostatin suppresses the production of both insulin and glucagon. Amylin, which is cosecreted with insulin from the β cells, has physiologic actions that include slowing of gastric emptying, suppression of postprandial glucagon secretion, reduction of food intake, and inhibition in the secretion of both stomach acid and pancreatic digestive enzymes (see Chapter 27).

SPECIFIC DRUGS

Insulin Insulin has anabolic properties that include the stimulation of both skeletal muscle and liver cells to incorporate glucose and convert it to glycogen, to synthesize proteins from amino acids in the blood, and to act on fat cells to enhance their uptake of glucose and the synthesis of fat. In short, insulin encourages anabolism rather than catabolism, because it promotes the synthesis of glycogen, proteins, and lipids. A deficiency of insulin, which characterizes the disease diabetes mellitus (DM), causes extreme changes in the entire metabolic pattern of individuals with DM. Patients with DM often demonstrate elevated blood glucose levels, excess glucose in the urine, and failure to properly use carbohydrate and lipids. Untreated DM can be fatal. Even when treated, however, there can be numerous circulatory and renal complications, and some metabolic abnormalities may lead to blindness (diabetic retinopathy).

The human insulin molecule, consisting of 51 amino acids, has the structural characteristics of a large protein, yet is only the size of a polypeptide. Two disulfide bonds (cysteine [Cys] A7 to Cys B7 and Cys A20 to Cys B19) link two polypeptide chains, with the A-chain consisting of 21 amino acids and the B-chain consisting of 30 amino acids. An additional disulfide loop is found in the A-chain between Cys A6 and Cys A11 (Fig. 6.21).

The primary sequences of insulin from several species are known, and porcine insulin is the closest to that

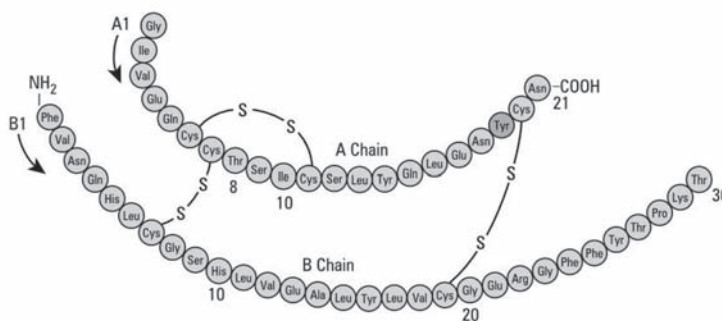


FIGURE 6.21 Primary structure of human insulin chains A and B, including the interchain disulfide bonds A7-B7 and A20-B19 and intrachain disulfide bond A6-A11.

of humans. Their A-chains are identical, and they differ only in their B-chains, with alanine 30 (porcine) in place of threonine 30 (human). Human and bovine insulin differ in each chain, with alanine 8 and valine 10 in the A-chain (bovine) and alanine 30 in the B-chain (bovine). Nearly all human insulin is produced by rDNA technology, and insulin obtained from porcine sources is being phased out. Recombinant insulin approved in 1982 for the treatment of insulin-dependent diabetes, is the first FDA-approved rDNA drug. The biotechnology solution has several advantages over insulin derived from animal sources:

1. It should have potentially fewer serious immune reactions.
2. It is pyrogen-free.
3. It is not contaminated with other peptide hormones, such as glucagon, somatostatin, and proinsulin, as found in isolated products.
4. It can be produced in larger amounts.

The first successful attempt to tailor a protein hormone for therapy by rDNA techniques has yielded interesting insulin analogs (68–70).

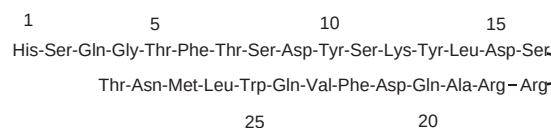
Human insulin of rDNA origin is available commercially as Humulin, Novolin, and several analogs. The Humulin and Novolin products are produced using genetically modified strains of two different microorganisms. Humulin is prepared using recombinant *E. coli* bacteria. The pharmaceutical preparation is reported to contain less than 4 ppm of immunoreactive bacterial polypeptides that act as possible contaminants. Baker's yeast (*S. cerevisiae*) serves as the recombinant organism for the production of Novolin. Before 1986, Humulin was produced by chemically joining together the separately rDNA-derived A-chain and B-chain. Today, the product is prepared by enzymatically cleaving the connecting peptide in recombinant proinsulin.

Studies in animals, healthy adults, and patients with type 1 DM have shown human insulin to have pharmacologic effects identical to those of purified porcine insulin. A comparable pharmacokinetic profile also has been shown. Human insulin, however, administered intramuscularly or intravenously, may have a slightly faster onset and slightly shorter duration of action when compared with purified porcine insulin in patients with diabetes. The usual precautions concerning toxic potentials observed with insulin of animal origin should be followed with rDNA human insulin. As would be expected, the recombinant product has been shown to be less immunogenic than animal insulins.

Insulin remains the only treatment option for type 1 diabetes and is still widely used to treat patients with type 2 diabetes who do not respond adequately to other pharmacotherapies. Recombinant DNA technology has led to the development of insulin analogs that have greater utility in certain situations and may more closely resemble the normal diurnal pattern of insulin secretion (Table 27.6). The newly engineered analogs

have specific amino acid sequence modifications that improve absorption properties and biologic profiles (71). Insulin lispro (Humalog) has a more rapid onset and shorter duration of action than regular human insulin. Unlike regular insulin that must be injected 30 to 60 minutes before a meal, recombinant insulin lispro is effective when injected 15 minutes before a meal. The analog differs from natural human insulin, because the B-chain amino acids B28 proline and B29 lysine are exchanged. Insulin aspart (Novolog), which is homologous with human insulin except for the single amino acid substitution of aspartic acid for proline at B28, is effective when injected 5 to 10 minutes before a meal. Insulin glargine (Lantus) is the newest rDNA-derived human insulin analog. An ultra-long-acting agent, insulin glargine differs from human insulin in that the amino acid asparagine at residue A21 is replaced by glycine, and two arginines are added to the C-terminus of the B chain. When administered subcutaneously, insulin glargine has a duration of action up to 24 to 48 hours. This change in action profile resulted from structural modifications that enhanced the product's basicity, thus causing the product to precipitate at neutral pH postinjection and, therefore, increasing its duration of action.

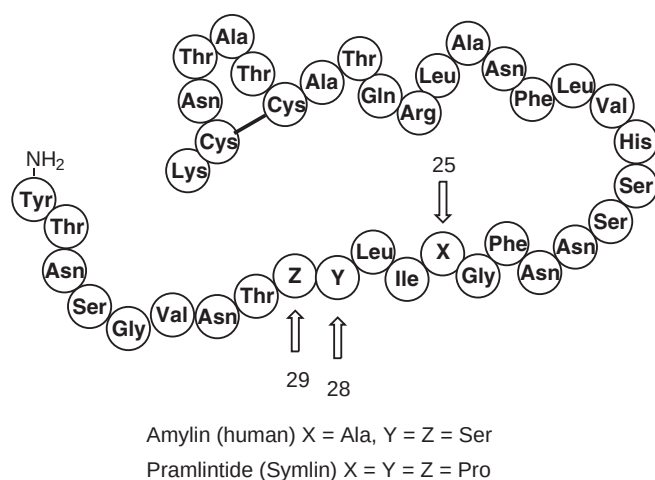
Glucagon (GlucaGen)



Glucagon, a 29-amino acid, straight-chain polypeptide of α -cell pancreatic origin, triggers liver glycogenolysis and gluconeogenesis, thereby elevating glucose levels. The principal action of glucagon is the liver-mediated release into the blood of abnormally high concentrations of glucose, which causes hyperglycemia. This means that glucagon has an effect on blood glucose levels that is opposite to what occurs with insulin.

Glucagon of rDNA origin is now available (69). Replacing the bovine product with the rDNA-derived drug would eliminate the risk of acquiring bovine spongiform encephalopathy from glucagon therapy. Human glucagon of rDNA origin is marketed for the treatment of severe hypoglycemic reactions in patients with diabetes, as can occur when there is an overdose of insulin. In patients with type 1 diabetes, the increase in glucose resulting from glucagon administration may not be sufficient, and supplemental carbohydrates may need to be administered quickly, especially in children.

AMYLIN Amylin is a 37-amino acid peptide that is structurally similar to CT. Amylin works together with insulin to regulate glucose concentrations after a meal. When in solution, amylin is viscous, unstable, and tends to aggregate; therefore, it cannot be used parenterally and is not commercially available.



Pramlintide acetate (Symlin) Pramlintide acetate is a synthetic analog of amylin (a 37-amino acid peptide) with proline substitutions at amino acids 25, 28, and 29. These substitutions change its physical properties such that it is commercially available for SC injection. When pramlintide is used in combination with insulin, it slows gastric emptying, lowers blood glucose levels after meals, and affords a feeling of fullness that leads to decreased caloric intake and the potential for weight loss. Pramlintide has been approved for use in adults with type 1 or type 2 diabetes as an adjunct along with insulin.

ENZYMES

Tissue-Type Plasminogen Activator

The fibrinolytic system is activated in response to the presence of an intracellular thrombus or clot. The process of clot dissolution is initiated by the conversion of plasminogen to plasmin. Plasminogen activation is catalyzed by two endogenous highly specific serine proteases, urokinase-type plasminogen activator and tissue-type plasminogen activator (t-PA).

The mature human t-PA is a glycoprotein consisting of a single chain of 527 amino acids. Its MW is approximately 70 kDa. Human t-PA contains 35 cysteines assigned to 17 disulfide bonds (Fig. 26.24). A serine protease domain of approximately 260 amino acids is located at the carboxy-terminal end of this protein. A fibronectin “finger” domain, two kringle domains, and an epidermal growth factor domain also are present. The t-PA protease domain is approximately 35% to 40% homologous with typical serine protease, such as bovine trypsin and chymotrypsin.

Mammalian cells produce two t-PA variants of N-linked glycosylation, type 1 (at asparagines 117, 184, and 448) and type 2 (only at asparagines 117 and 448). The rate of fibrin-dependent plasminogen activation is two- to threefold faster for type 2 compared with type 1. The cDNA obtained from a human melanoma cell line was expressed in CHO cells to achieve glycosylation and a protein identical to the natural protein. Protein engineering

studies have produced variant t-PA molecules with modified pharmacokinetics, affinity for fibrin, catalytic activity, and side effects.

Three rDNA thrombolytic agents are approved in the United States (20,72–74) (Fig. 26.24). The first is alteplase, an enzyme equivalent to human t-PA. It is indicated for the treatment of acute myocardial infarction (administered as a bolus), acute massive pulmonary embolism (administered by intravenous infusion), and ischemic stroke. It is the first fibrin-selective thrombolytic agent preferentially activating fibrinogen bound to fibrin. Thus, the thrombolytic effect is localized to a blood clot and avoids systemic activation of fibrinogen, preventing bleeding elsewhere in the body. Plasma t-PA concentrations are proportional to the rate of infusion. Alteplase is rapidly cleared from circulating plasma, with 50% cleared within 5 minutes after termination of infusion. The mechanisms for clearance of t-PA from the blood are poorly understood. Detectable levels of antibody against alteplase have been found in patients receiving the drug, although 12 days to 10 months later, antibody determinations have been negative.

The second, reteplase, a recombinant, is a nonglycosylated deletion mutation of human t-PA containing 355 of 527 amino acids of native t-PA. The drug is indicated for acute myocardial infarction and is given as a 10 U + 10 U double bolus.

The most recent addition to the marketed rDNA-derived t-PAs is tenecteplase. This recombinant protein contains three modifications from natural human t-PA. In the kringle 1 domain of natural t-PA, threonine 103 is replaced by arginine; the kringle 1 domain asparagine 117 is replaced by glutamine; and in the protease domain, four amino acids (lysine, histidine, and two arginines) are replaced by four alanines. The drug is indicated for acute myocardial infarction. Bleeding at the injection site is similar to that with alteplase, but there is a reduction in the noncerebral bleeding complications. It is administered as a single, 5-second bolus.

DNase—Dornase Alfa (Pulmozyme) According to the Cystic Fibrosis Foundation, cystic fibrosis (CF) is the most common fatal genetic disorder, afflicting approximately 30,000 patients, most of whom die before the age of 30 years. They develop thick mucous secretions and suffer from severe, frequent lung infections. Studies during the 1950s and 1960s determined that CF-related secretions in the lungs contained large amounts of DNA. Mucous-thickening DNA release resulted from an inflammatory response and ensuing white blood cell death. The enzyme DNase I specifically cleaves extracellular DNA, such as that found in the mucous secretion of CF patients, and has no effect on the DNA of intact cells. The FDA has approved a recombinant human DNase (75).

The enzyme DNase I is a glycoprotein containing 260 amino acids with an approximate MW of 37 kDa.

The recombinant protein is expressed by genetically engineered CHO cells encoding for the native enzyme, although DNase I was not purified or sequenced from human sources at the time. A degenerate sequence, based on the sequence of bovine DNase (263 amino acids), was used to synthesize probes and screen a human pancreatic DNA library. The primary amino acid sequence of rhDNase is identical to native human DNase I.

The only FDA-approved DNase product, dornase alfa (inhalation solution), has been developed as a therapeutic agent for the management of CF. It is administered by nebulizer aerosol delivery systems.

Dornase alfa is indicated for daily administration in conjunction with standard CF therapies to reduce the frequency of respiratory infections requiring parenteral antibiotics to improve pulmonary function. The breakdown of DNA in infected sputum results in improved airflow in the lung and reduced risk of bacterial infection. Although effective for the management of the respiratory symptoms of CF, dornase alfa is not a replacement for antibiotics, bronchodilators, and daily physical therapy. This product also finds application in treating chronic bronchitis.

Enzyme Replacement Therapy

Enzyme replacement therapy (ERT) is available for many of the lysosomal storage diseases, including Gaucher disease, Fabry disease, Pompe disease, mucopolysaccharidosis (MPS) type I (Hurler, Scheie, and Hurler/Scheie syndromes), MPS II (Hunter syndrome), and MPS VI (Maroteaux-Lamy syndrome) (76). The most frequent drug-related adverse events of ERT are infusion-related reactions. Antibodies formed against infused enzymes are usually of the immunoglobulin G (IgG) serotype. Signs and symptoms of infusion-related reactions (IRRs) can include cutaneous reactions, pyrexia, headache, and hypertension. Precise relationship between antibody formation and occurrence of IRRs is unclear. Not all IRRs are antibody mediated, and not all patients who develop antibodies develop IRRs (76).

Imiglucerase (Cerezyme) Type 1 Gaucher disease, the most common form, is an inherited disorder. Fewer than 1 in 40,000 people in the general population have Gaucher disease. Patients with the disease lack the normal form of the enzyme glucocerebrosidase. They cannot break down glucocerebroside, leading to accumulation of glucocerebroside within the lysosomes. This leads to the poor functioning of macrophages and an accumulation of “Gaucher cells” in the spleen, liver, and bone marrow. Imiglucerase is an analog of glucocerebrosidase produced by rDNA technology using mammalian cell culture system. The drug is a monomeric glycoprotein consisting of 497 amino acids (approximate MW = 60 kDa), with four Nlinked glycosylation sites, and differs from the human placental glucocerebrosidase

by one amino acid (histidine 495 replaces arginine). Imiglucerase carries out the normal function of the missing enzyme and has been shown to be safe in long-term safety studies (77).

Alglucosidase Alfa (Myozyme, Lumizyme) Pompe disease, a rare genetic disorder affecting nearly 1 in every 40,000 births, is caused by the mutation in a gene that makes the α -glucosidase enzyme (78). It was the first disease to be identified as a lysosomal storage disorder. α -Glucosidase enzyme is responsible for the breakdown of glycogen to glucose. The lack of degradation results in the accumulation of glycogen in lysosomes in all tissues, most notably in skeletal and cardiac muscles. The disease is characterized as early onset or late onset Pompe disease based on the age of onset of disease. Severe α -glucosidase deficiency manifests during infancy with rapidly progressing muscle weakness, hypotonia, and cardiomyopathy and eventually lethal muscle disorder. As with other lysosomal storage disorders, ERT is the patient's only hope. Alglucosidase alfa was developed and launched in 2006 for the treatment of infants and children with Pompe disease (79). Another alglucosidase alfa drug (Lumizyme) was approved by the FDA in 2010 for patients age 8 years and older with late-onset (noninfantile) Pompe disease. Alglucosidase alfa is a recombinant human enzyme produced by transfected CHO cells as a 110 kDa precursor that targets lysosomes via the mannose-6-phosphate receptor. Following endocytosis, the enzyme is transformed to its mature 76 kDa form that restores glycogen processing and reverses accumulation. Alglucosidase alfa therapy has been shown to improve cardiac and muscle function.

Idursulfase (Elaprase) MPS II (Hunter syndrome), a disease with an incidence of approximately 1 in 162,000 births, is caused by the deficiency of the lysosomal enzyme iduronate-2-sulfatase. This enzyme is responsible for a crucial step in degradation of two glycosaminoglycans (GAGs), dermatan sulfate and heparin sulfate, in the lysosomes of various cells. Iduronate-2-sulfatase deficiency causes an accumulation of GAGs in tissue, and the patient's only recourse is ERT. The clinical manifestations of this deficiency are short stature, joint stiffness, harsh facial features, hepatosplenomegaly, and progressive mental retardation. Recombinant iduronate-2-sulfatase has been available in the United States since 2006 and is produced from HT-1080 cells for proper translational attachment of N-linked oligosaccharides and the crucial mannose-6-phosphate groups as the targeting phosphate into lysosomes (80). In addition to being fully glycosylated with eight mannose-6-phosphate groups, the enzyme possesses sialylated moieties that improve its stability in circulation. For full activity, cysteine 59 must undergo modification to formylglycine. Idursulfase has been designated as an orphan product by the FDA.

Cytokines

Cytokines communicate in a dynamic cellular network during an immune/inflammatory response to an antigen (75,81–83). Lymphokine and monokine are the terms used for a cytokine derived from lymphocytes and macrophages, respectively. Chemokines are a group of at least 25 structurally homologous, low molecular weight cytokines that stimulate leukocyte movement and regulate the migration of leukocytes from the blood to tissues. Cytokines, usually released and targeted to produce a localized effect, regulate the growth, differentiation, and activation of the hematopoietic cells responsible for the maintenance of the immune response. A wide array of glycoproteins, including interferons, interleukins, hematopoietic growth factors, and tumor necrosis factors (TNFs), are cytokines. Cytokines can only act on target cells that express receptors for that cytokine. There are five families of cytokine receptor proteins: class I cytokine receptors, class II cytokine receptors, TNF receptors, chemokine receptors, and immunoglobulin superfamily receptors (Table 6.5) (20,84–90).

INTERFERONS The interferons (20,85) are a family of cytokines discovered in the late 1950s with broad-spectrum antiviral and potential anticancer activity, making them biologic response modifiers. Biotherapy (the therapeutic use of any substance of biologic origin) of cancer is different than standard chemotherapy. That is, biotherapeutic agents belong to a group of compounds that enhance normal immune interactions (therefore, they also are immunomodulators) with cells in a specific or nonspecific fashion. Chemotherapeutics interact directly with the cancer cells themselves. Three types of naturally occurring interferons have been found in small quantities: leukocyte interferon (IFN- α), produced by lymphocytes and macrophages; fibroblast interferon (IFN- β), produced by fibroblasts, epithelial cells, and macrophages; and immune interferon (IFN- γ), synthesized by CD4⁺, CD8⁺, and natural killer lymphocytes. Both IFN- α and IFN- β , also known as type I interferons, exhibit approximately 30% primary sequence homology but

no structural similarity to IFN- γ , a type II interferon. All three are glycoproteins. Previously only available in low yields by chemical synthesis or isolation, several rDNA interferon pharmaceuticals now have been marketed in the United States, including three IFN- α products, two IFN- β agents, and an IFN- γ drug.

Interferon- α At least 24 different human genes producing 16 distinct mature IFN- α molecules with slight structural variations are known (20,85,91). Human IFN- α proteins generally are composed of either 165 or 166 amino acids. The two primary subtypes, IFN- α 2a (Roferon-A) and IFN- α 2b (Intron A), both contain 165 amino acids, differing only at position 23, with IFN- α 2a containing a lysine residue and IFN- α 2b an arginine residue at this position. They have MWs of approximately 19 kDa. Although cultures of genetically modified *E. coli* produce two recombinant FDA-approved IFN- α products, IFN- α 2a and IFN- α 2b, their method of purification differs. Purification of IFN- α 2a includes affinity chromatography using a murine MAb, whereas that of IFN- α 2b does not.

Interferon- α possesses complex antiviral, antineoplastic, and immunomodulating activities. Although the precise mechanism of action of IFN- α is not known, it is believed to interact with cell surface receptors to produce the biologic effects. The actions appear to result from a complex cascade of biologic modulation and pharmacologic effects that include the modulation of host immune responses; cellular antiproliferative effects; cell differentiation, transcription, and translation processes; and reduction of oncogene expression. Interferon- α is filtered through the glomeruli in the kidney and undergoes rapid proteolytic degradation during tubular reabsorption.

Consensus Interferon (IFN Alfacon-1) Hepatitis C infection results in a chronic disease state in 50% to 70% of cases and is now the most important known cause of chronic liver disease (92). Following the acute phase, as many as 80% of patients may progress to the chronic phase of the infectious disease. An estimated 20% of patients with a chronic form of the disease progress to cirrhosis. The only agents shown to be effective in the treatment of hepatitis C are the interferons. A unique recombinant molecule, a consensus IFN, known as IFN alfacon-1, has been approved for the treatment of chronic hepatitis C infection. It is a 19.5 kDa recombinant protein produced in *E. coli* and contains 166 amino acids in a relationship in which each amino acid position in the molecule contains the most commonly occurring amino acid among all the natural IFN- α subtypes. IFN alfacon-1 exhibits 5- to 10-fold higher biologic activity when compared to either IFN- α 2a or IFN- α 2b.

Interferon- β Normally produced by fibroblasts, human IFN- β (20,85,91) was first cloned and expressed in 1980; however, its instability made it unsuitable for clinical use.

TABLE 6.5 The Five Families of Cytokine Receptors and Some Ligands

Receptor Families	Ligands
Class I cytokine receptors	IL-2, IL-7, IL-9, IL-11, IL-13, IL-15, GM-CSF, G-CSF
Class II cytokine receptors	INF- α , INF- β , INF- γ
Tumor necrosis factor receptors	TNF- α , TNF- β , CD30, CD40, FAS
Chemokine receptors	IL-8, RANTES, MIP-1, PF-4, MCAF
Immunoglobulin superfamily receptors	IL-1, M-CSF

The more stable recombinant IFN- β 1b (Betaseron), a 165-amino acid analog of human IFN- β , differs from the native protein with a serine residue substituted for cysteine at position 17. The highly purified rDNA technology-derived product has a MW of 18.5 kDa. It is produced in a recombinant *E. coli*. Approved in 1993 by the FDA, IFN- β 1b is indicated for the treatment of patients with exacerbating-relapsing multiple sclerosis (MS). The National MS Society says that 250,000 to 300,000 Americans have this disease, with more than 60% of patients falling into the exacerbating-relapsing category. A vial of recombinant IFN- β 1b contains 0.3 mg of protein with dextrose and human albumin as stabilizers. The exact mechanism of action of IFN- β 1b is not known. Its immunomodulating effects, however, may benefit patients with MS by decreasing the levels of endogenous IFN- γ . Levels of IFN- γ are believed to rise before and during acute attacks in patients with MS.

Whereas IFN- β 1b of rDNA origin was the first to the market, IFN- β 1a (Avonex) also is now available. IFN- β 1a is produced in mammalian cells and has the same amino acid sequence and carbohydrate side chain as natural IFN- β . Recombinant IFN- β 1a is administered to patients once weekly by intramuscular injection. This differs from the SC administration every other day of rDNA-produced IFN- β 1b.

Interferon- γ Human IFN- γ (20,85,91,93) is a single-chain glycoprotein with a MW of approximately 15.5 kDa. The cytokine mainly exists as a noncovalent dimer of differentially glycosylated chains in solution in vivo. Glycosylation does not appear to be necessary for biologic activity. The 140-amino acid IFN- γ 1b (Actimmune) is produced by fermentation of a recombinant *E. coli*. IFN- γ 1b was approved in 1990 by the FDA.

IFN- γ 1b possesses biologic activity identical to the natural human IFN- γ derived from lymphoid cells. Although all the IFNs share certain biologic effects, IFN- γ differs distinctly from IFN- α and IFN- β by its potent capacity to activate phagocytes involved in host defense. These activating effects include the ability to enhance the production of toxic oxygen metabolites within phagocytes, resulting in a more efficient killing of various microorganisms. This activity is the basis for the use of IFN- γ 1b in the management of chronic granulomatous disease. Chronic granulomatous disease is a group of rare X-linked or autosomal genetic disorders of the phagocytic oxygen metabolite-generating system, leaving patients susceptible to severe infections. The drug extends the time that patients spend without being hospitalized for infectious episodes. Investigational applications of IFN- γ include the treatment of renal cell carcinoma, small-cell lung cancer, infectious disease, trauma, atopic dermatitis, asthma, allergies, rheumatoid arthritis, and venereal warts.

INTERLEUKINS Interleukins are cytokines involved in immune cell communication. Synthesized by monocytes, macrophages, and lymphocytes, interleukins serve

as soluble messengers between leukocytes. Currently, at least 18 interleukins have been observed. One of the most studied cytokines is interleukin (IL)-2, originally called T-cell growth factor because of its ability to stimulate growth of T lymphocytes.

Interleukin-1 IL-1 is a major inflammatory mediator and exists in two forms: IL-1 α and IL-1 β . Each form is a product of two separate genes, but is related to each other structurally at a three-dimensional level. IL-1 β is a systemic, hormone-like mediator intended to be released from cells, whereas IL-1 α is primarily a regulator of intracellular events and mediator of local inflammation. The recombinant protein anakinra, which is an IL-1 receptor antagonist, and the fusion protein rilonacept, which blocks the excessive IL-1 β signaling are commercially available and described below (94,95).

Anakinra (Kineret) Rheumatoid arthritis (RA) is a chronic, inflammatory disease affecting synovial joints. Patients with persistent, active disease are traditionally treated with disease-modifying antirheumatic drugs (DMARDs). IL-1 receptor antagonist is an endogenous cytokine that blocks the binding of proinflammatory cytokine IL-1 to its receptor, thereby balancing the cartilage destruction and bone resorption mediated by IL-1. Anakinra is the first recombinant human IL-1 receptor antagonist and differs from the native human protein in that it is not glycosylated and has an additional N-terminal methionine (rmetHuIL-1 receptor antagonist). Anakinra, a 17.3 kDa recombinant protein expressed in *E. coli*, was approved in 2001 for treating the signs and symptoms and the joint-destructive components of RA (94–96). However, the challenge for anakinra for occupying the large number of IL-1 receptors is formidable, as these receptors are expressed on all cells except red blood cells; moreover, anakinra is rapidly excreted by the kidney, and blood levels are low after 24 hours. IL-1 receptors are also readily generated each day, necessitating a daily SC injection.

Rilonacept (Arcalyst) Cryopyrin-associated periodic syndromes are inherited disorders caused by mutations in the nucleotide-binding domain, leucine-rich family, pyrin domain containing-3 (*NLRP-3*) gene, which encodes the protein cryopyrin. Cryopyrin regulates the protease caspase-1 and controls activation of IL-1 β . Mutations in the *NLRP-3* gene can cause an overactive inflammasome, resulting in excessive levels of activated IL-1 β , which causes inflammation, joint pain, rash or skin lesions, fever and chills, eye redness or pain, and fatigue. Rilonacept is a 252 kDa recombinant fusion protein produced by the CHO cells and approved by the FDA in 2008 for the long-term treatment of familial cold autoinflammatory syndrome and Muckle-Wells syndrome, two cryopyrin-associated periodic syndromes that are extremely rare, affecting approximately 300 people in the United States. Rilonacept's mechanism of

intervention involves blockade of excessive IL-1 β signaling (94,95,97). Rilonacept combines the extracellular binding domains of the human IL-1 receptor component and IL-1 receptor accessory protein in a single chain, with two of these chains joined to the fragment crystallizable (Fc) portion of human IgG creating a dimeric molecule. Rilonacept serves as an effective soluble IL-1 β sink or trap since the coupled receptor components bind IL-1 β with higher affinity than either individual receptor. It exerts its effects through multicomponent receptor system and is more effective than drugs that target only one of the components such as the IL-1 receptor antagonist anakinra. With its weekly subcutaneous dosing regimen, it may have better patient compliance than anakinra. Concomitant administration of other IL-1 blockers and anti-TNF- α agents has been associated with an increased risk of serious infection and neutropenia. Because IL-1 blockade may interfere with the immune response to infections, patients should not initiate rilonacept with an existing infection. Patients with chronic inflammation tend to have suppressed formation of CYP450 enzymes. Treatment with rilonacept is expected to normalize CYP450 distribution, so plasma levels of coadministered drugs that are CYP450 substrates with narrow therapeutic indexes should be carefully monitored for potential dose modification.

Interleukin-2 Human IL-2 is a 133-amino acid, 15.4 kDa protein that is O-glycosylated at a threonine in position 3. An intramolecular disulfide bond between cysteine 58 and cysteine 105 is essential for biologic activity. A recombinant version of IL-2 is marketed as aldesleukin (Proleukin) (20,85,91,98–101). Aldesleukin differs from the native protein by the absence of glycosylation, a lack of the N-terminal alanine residue at position 1 (132 amino acids), and the replacement of cysteine with serine at position 125 of the primary sequence. Sequence changes were accomplished by site-directed mutagenesis to the IL-2 gene before cloning and expression. Aldesleukin exists as noncovalent microaggregates with an average size of 27 recombinant IL-2 molecules. The recombinant drug does possess the biologic activity of the native protein.

Aldesleukin is used in cancer biotherapy as a biologic response modifier for the treatment of metastatic renal cell carcinoma and metastatic melanoma. Side effects are the major dose-limiting factor, because aldesleukin is an extremely toxic drug. The manufacturer's labeling should be consulted for full details. Careful patient selection and thorough patient monitoring are essential. The incidence of nonneutralizing anti-IL-2 antibodies in patients treated on an every-8-hour regimen is quite high (76% in one clinical study).

Interleukin-2 Fusion Protein Using ligation chemistry approaches during the preparation of recombinant proteins, researchers have created biologically active molecules that combine the activities of two individual

proteins into “fusion molecules.” These fusion technologies hold promise for developing custom molecules expressing a wide variety of dual activities.

Denileukin Difitox (Ontak). The FDA approved the fusion protein denileukin difitox in 1999 for the treatment of patients with persistent or recurrent cutaneous T-cell lymphoma (CTCL) whose malignant cells express the CD25 component of the IL-2 receptor (102–105). CTCL is a general term for a group of low-grade, non-Hodgkin lymphomas (NHLs) affecting approximately 1,000 new patients per year. Malignant T cells manifest initially in the skin. Over time, there is systemic involvement. For many patients, CTCL is a persistent, disfiguring, and debilitating disease that requires multiple treatments. Malignant CTCL cells express one or more of the components of the IL-2 receptor. Thus, the IL-2 receptor may be a homing device to attract a “killer drug.” The incidence of CTCL in the United States is approximately 1,500 cases per year. Men are twice as likely as women to have the disease.

Denileukin difitox is a rDNA-derived, cytotoxic IL-2 fusion protein that is composed of two different components: the first 389 amino acids of diphtheria toxin fused to amino acids 2 to 133 of human IL-2. Thus, the fusion protein binds to IL-2 receptors on neoplastic T cells and gets internalized via receptor-mediated endocytosis, bringing diphtheria toxin directly to kill the CTCL target. Interestingly, although denileukin difitox was originally thought to act primarily by binding to CD25, now that the components of IL-2 receptor are better delineated, it appears that clinical activity of denileukin difitox is associated with the presence of CD122.

Interleukin-11 IL-11 is a thrombopoietic growth factor that directly stimulates the proliferation of hematopoietic stem cells and megakaryocyte progenitor cells (106). This induces megakaryocyte maturation, resulting in increased platelet production. IL-11 is a 178-amino acid glycosylated cytokine produced by bone marrow stromal cells. Primary osteoblasts and mature osteoclasts express mRNAs for an IL-11 receptor (IL-11R α). Thus, bone-forming cells and bone-resorbing cells are potential targets of IL-11. In 1997, the FDA approved an rDNA-derived version of IL-11 produced in *E. coli*. Oprelvekin (Neumega) contains only 177 amino acids, lacking the amino terminal proline of the native IL-11. Produced in *E. coli*, the cytokine analog is nonglycosylated. Oprelvekin has potent thrombopoietic activity in animal models of compromised hematopoiesis. It is indicated for the prevention of thrombocytopenia following myelosuppressive chemotherapy. Pharmacists should monitor for possible fluid retention and electrolyte states when it is used with chronic diuretic therapy.

TUMOR NECROSIS FACTOR TNFs, a family of cytokines produced mainly by activated mononuclear phagocytes, have both beneficial and potentially harmful effects, mediating

cytotoxic and inflammatory reactions (20). The TNFs are endogenous pyrogens capable of inducing chills, fever, and other flu-like symptoms. TNF- α , also called cachectin (and commonly referred to as TNF), and TNF- β , also called lymphotoxin, both bind to the same receptor and induce similar biologic activities. Biologic effects of TNF- α include selective toxicity against a range of tumor cells, mediation of septic shock, activation of elements of the immune system in response to gram-negative bacteria, and induction/regulation of inflammation. The TNF- α of rDNA origin has been studied extensively, but it has not been developed into a useful drug.

Etanercept is a rDNA-produced fusion protein that binds specifically to TNF and blocks its interaction with cell surface TNF receptors (107,108). It is indicated for the treatment of moderate to severe active RA in adults and for juvenile RA in patients who have had an inadequate response to one or more DMARDs. It is a genetically engineered protein that includes two components. The extracellular, ligand-binding portion (p75) of the human TNF receptor is linked as a fusion protein to the Fc portion of the human IgG1 antibody. Each etanercept molecule binds specifically to two TNF molecules found in the synovial fluid of patients with RA, blocking the interaction of TNF with the TNF receptor. The drug inhibits both TNF- α and TNF- β . The Fc portion of the fusion protein helps to clear the etanercept–TNF complex from the body.

Abatacept (Orencia) With the recognition that T cells play a central role in the pathogenesis of RA, abatacept has been developed as a novel, rational approach to interfere with the upstream effector of the inflammation to meet the needs of patients who fail to respond adequately to existing therapy using traditional DMARDs such as methotrexate (MTX) or TNF-blocking agents. Abatacept is a soluble 92 kDa human fusion protein approved for the treatment of RA as a new class of DMARDs (109). Structurally, abatacept consists of a fusion between the extracellular domain of human cytotoxic T-lymphocyte-associated antigen-4 (CTLA-4) linked to the modified Fc portion of human IgG1. Abatacept selectively modulates the CD80/CD86:CD28 costimulatory signal required for full T-cell activation. By targeting the activation of T cells, an upstream event in the immune cascade that underlies RA, abatacept has the potential to impact multiple downstream aspects of RA immunogenesis, such as the production of cytokines, autoantibodies, and inflammatory proteins. The immunoglobulin portion of the abatacept serves as a handle to facilitate purification of the protein that is produced by rDNA technology in a mammalian cell expression system. It also enhances the solubility and serum half-life of the fusion protein. Serum half-life of 14.7 days is independent of dose, and rate of elimination remains constant, supporting the lack of antidrug antibody generation. Abatacept was found to be well tolerated for up to 7 years of exposure in patients with RA who had an inadequate response to MTX or other traditional DMARDs or who failed to respond to treatment with

anti-TNF agents (110). The potential for increased risk of infection is an important safety concern in patients with RA receiving biologic therapies. During abatacept treatment, live vaccinations should be avoided because the drug may diminish the effectiveness of some immunizations, and anti-TNF agents should be avoided due to risk of infections.

HEMATOPOIETIC GROWTH FACTORS Hematopoiesis is the complex series of events involved in the formation, proliferation, differentiation, and activation of red blood cells, white blood cells, and platelets. Hematopoietic growth factors are cytokines that regulate these events (20,111–114). Investigators have identified and cloned at least 20 factors, including IL-3 (or multi-colony-stimulating factor [CSF]), IL-4, IL-5, IL-6, IL-7, erythropoietin (EPO), granulocyte-macrophage colony-stimulating factor (GM-CSF), granulocyte colony-stimulating factor (G-CSF), macrophage colony-stimulating factor (M-CSF), and stem cell factor. Figure 6.22 summarizes the elaborate hematopoietic cascade. All blood cells originate within the bone marrow from a single class of pluripotent stem cells. In response to various external and internal stimuli, regulated by hematopoietic growth factors, stem cells give rise to additional new stem cells (self-renewal) and differentiate into mature, specialized blood cells.

Erythropoietin–Epoetin Alfa (Epogen, Procrit) EPO (20,111,115–119), a glycoprotein with a MW of 30 to 34 kDa produced by the kidney, stimulates the division and differentiation of erythroid progenitors in the bone marrow, increasing the production of red blood cells. Epoetin alfa (sometimes called rHuEPO- α), a recombinant EPO prepared from cultures of genetically engineered mammalian CHO cells, consists of the identical 165–amino acid sequence of endogenous EPO. The MW is approximately 30.4 kDa. The protein contains two disulfide bonds (linking cysteine 7 with 161 and 29 with 33) and four sites of glycosylation (one O-site and three N-sites); the disulfide bonds and glycosylation are necessary for the hormone's biologic activity. Deglycosylated natural EPO and bacterial-derived EPO (without glycosylation) have greatly decreased in vivo activity, although in vitro activity is largely conserved. The sugars may play a role in thermal stability or the prevention of aggregation in vivo.

The marketed products are formulated as a sterile, colorless, preservative-free liquid for intravenous or SC administration. Epoetin alfa is indicated for the treatment of various anemias. Epoetin alfa represents a major scientific advance in the treatment of patients with chronic renal failure, serving as a replacement therapy for inadequate production of endogenous EPO by failing kidneys. Epoetin alfa may decrease the need for infusions in dialysis patients. By several mechanisms related to elevating the erythroid progenitor cell pool, epoetin alfa increases the production of red blood cells.

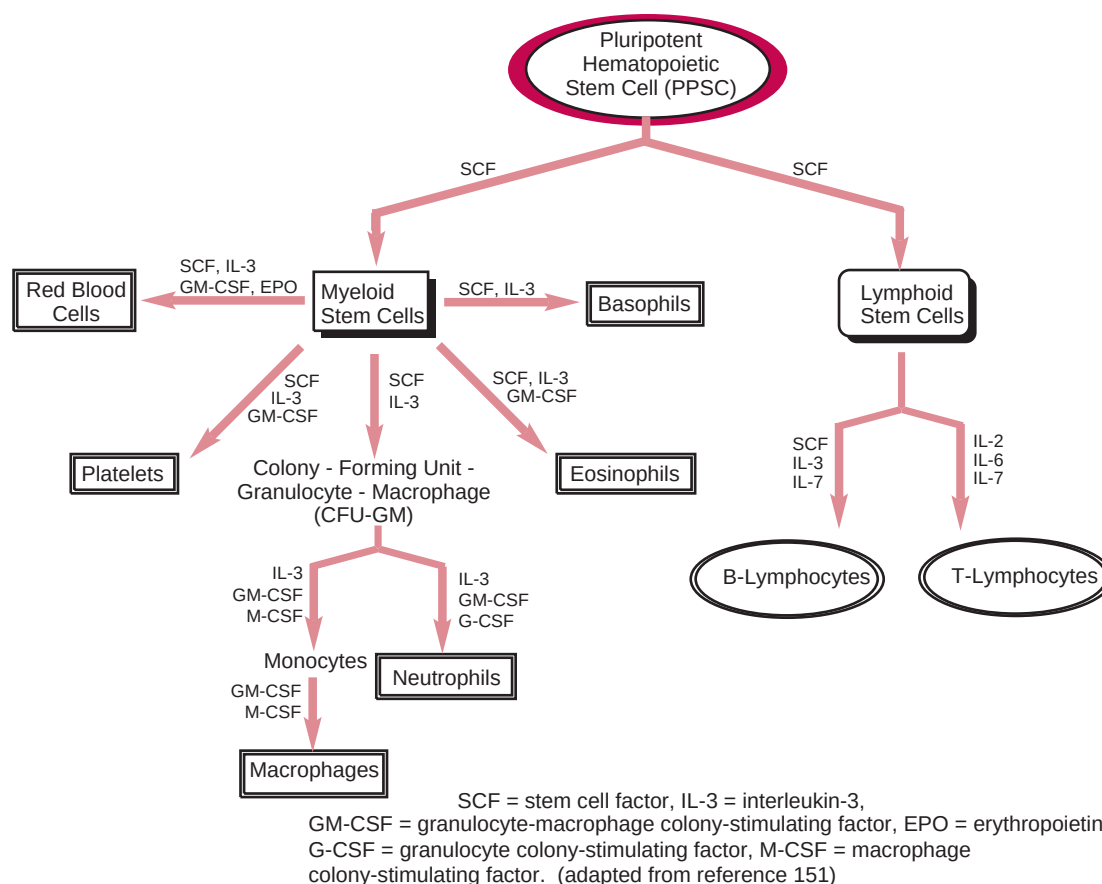


FIGURE 6.22 Schematic overview of hematopoiesis.

The manufacturer's full prescribing information should be consulted for dosing regimens, because the dose is titrated individually to maintain the patient's target hematocrit. The circulating half-life is 4 to 13 hours in patients with chronic renal failure. Peak serum levels are achieved within 5 to 24 hours following subcutaneous administration.

COLONY-STIMULATING FACTORS The CSFs are glycoprotein cytokines that promote progenitor proliferation, differentiation, and some functional activation. The name "colony-stimulating factor" results from the fact that these proteins often are assayed by their ability to stimulate the formation of cell colonies in bone marrow cultures. The names added to "CSF" reflect the types of cell colonies that arise in these assays.

Granulocyte Colony-Stimulating Factor—Filgrastim (Neuprogen). Recombinant DNA-derived G-CSF (20,111,120–124), or filgrastim, was approved by the FDA in 1991 to decrease the incidence of infection in patients with nonmyeloid malignancies who are receiving myelosuppressive anticancer drugs. Filgrastim is a 175-amino acid, single-chain protein with a MW of 18.8 kDa. Filgrastim, produced by a recombinant bacteria, differs from the endogenous human protein by the addition of

a methionine at the N-terminus (recombinant methionyl G-CSF is sometimes called r-metHuG-CSF) and the lack of glycosylation. Glycosylation, however, does not appear to be necessary for the biologic activity.

It is administered by intravenous infusion or by SC injection or infusion. The drug is rapidly absorbed, with peak serum concentrations in 4 to 5 hours. Elimination half-life is approximately 3.5 hours.

Filgrastim is lineage selective for the neutrophil lineage type of white blood cells, whereas GM-CSF is multilineage, stimulating progenitors of neutrophils, monocytes, basophils, and eosinophils. The drug reduces the period of neutropenia, the number of infections, and the number of days the patient is on antibiotics. Filgrastim generally is well tolerated, with medullary bone pain being the most frequently encountered side effect.

Granulocyte-Macrophage Colony-Stimulating Factor—Sargramostim (Leukine). GM-CSF (20,111,120–124) has been produced by rDNA technology in the yeast *S. cerevisiae*. Sargramostim is a glycoprotein of 127 amino acids, differing from the endogenous human GM-CSF by substituting leucine at position 23. Also, the glycosylation pattern may differ from that of the native protein.

Cellular division, maturation, and activation are induced through the binding of GM-CSF to specific receptors

expressed on the surface of target cells. On 2-hour intravenous infusion, the alpha half-life is 12 to 17 minutes, followed by a slower decrease (beta half-life) of 2 hours.

OTHER GROWTH FACTORS Growth factors are cytokines responsible for regulating cell proliferation, differentiation, and function (20). They act as intercellular signals. Each cell type's response is specific for each particular growth factor and differs from growth factor to growth factor. Platelet-derived growth factor (PDGF) is an endogenous growth-promoting protein that is released from cells involved in the healing process and is evident at the cell proliferation stage of a healing open wound. A recombinant human (rH) PDGF B homodimer has been produced from genetically engineered *S. cerevisiae* cells (125). Becaplermin is the B-chain of the PDGF B protein. Thus, becaplermin is also referred to as rHPDGF-BB. The 25 kDa protein is formulated into a gel that mimics natural PDGF when applied to diabetic foot ulcers.

Clotting Factor VIII, Factor IX, and Anticoagulants

Antihemophilic factor, or factor VIII, is required for the transformation of prothrombin (factor II) to thrombin by the intrinsic clotting pathway (20,74,126). Hemophilia A, a lifelong bleeding disorder, results from a deficiency of factor VIII. Conventional biotherapy for the treatment of hemophilia A includes protein concentrates from human plasma collected by transfusion services or commercial organizations. Therefore, the concentrates may contain other native human proteins and microorganisms, such as viruses (e.g., HIV and hepatitis), derived from infected blood. Four versions of recombinant factor VIII (antihemophilic factor), highly purified, microorganism-free proteins are now available. All four therapeutic proteins are produced by the insertion of cDNA encoding for the entire factor VIII protein into mammalian cells. The mature, heavily glycosylated protein is composed of 2,332 amino acids (1 to 2 million daltons) and contains sulfate groups. Stability of the large protein is a concern. The products have proved to be safe and effective for reducing bleeding time in patients. There is the possibility, however, of induction of inhibitors in previously untreated patients.

Hemophilia B results when a patient is deficient in specific clotting factor IX (20). It affects males primarily and makes up approximately 15% of all hemophilia cases. A recombinant human factor IX is now available.

Surgeons have used medicinal leeches (*Hirudo medicinalis*) for years to prevent thrombosis in fine vessels of reattached digits. Hirudin is the potent, specific thrombin inhibitor isolated from the leech. Lepirudin is a rDNA-derived (recombinant yeast) polypeptide that differs from the natural polypeptide, having a terminal leucine instead of isoleucine and missing the sulfate group at Tyr 63 (127).

Vaccines

There are two types of immunization: active immunization and passive immunization. Active immunization is the induction of an immune response either through exposure

to an infectious agent or by deliberate immunization with a vaccine (vaccination) made from the microorganism or its products to develop protective immunity. Passive immunization involves the transfer of products produced by an immune animal or human (preformed antibody or sensitized lymphoid cells) to a previously nonimmune recipient host, usually by injection. Sufficient active immunity may take days, several weeks, or even months to induce (possibly including booster vaccinations), but it generally is long lasting (even lifelong) through the clonal selection of genetically specific immunologic memory B and T lymphocytes. Passive immunity, although often providing effective protection against some infection, is relatively brief, lasting only until the injected immunoglobulin or lymphoid cells have disappeared (a few weeks or months). Thus, vaccines enable the body to resist infection by diseases. In response to an injection of vaccine, the immune system makes antibodies, which recognize surface antigens found in the vaccine. If the subject is later exposed to a virulent form of the virus, the immune system is primed and ready to eliminate it. Many viral vaccines are produced from the antigens isolated from pooled human plasma of virus carriers. Vaccinations are among the most cost-effective and widely used public health interventions. Although generally safe, the minimal risk of vaccine-produced infections can be eliminated by administration of highly purified vaccine antigens of recombinant origin (20,127–129). The different types of vaccines are described below, and the vaccines currently available in the United States are listed in Table 6.6.

Live, Attenuated Vaccines

To make a live, attenuated vaccine, the disease-causing organism is grown under special laboratory conditions that cause it to lose its virulence, or disease-causing properties.

Inactivated Vaccines

Inactivated vaccines are produced by killing the disease-causing microorganism with chemicals or heat.

Subunit Vaccines

Sometimes vaccines developed from antigenic fragments are able to evoke an immune response, often with fewer side effects than might be caused by a vaccine made from the whole organism.

Toxoid or Inactivated Vaccines

A toxoid is an inactivated toxin, the harmful substance produced by a microbe. Many of the microbes that infect people are not themselves harmful. It is the powerful toxins they produce that can cause illness. To inactivate such powerful toxins, vaccine manufacturers treat them by chemical means (formalin solution) and irradiation to completely cripple any disease-causing ability.

Conjugate Vaccines

The bacteria that cause some diseases, such as pneumococcal pneumonia and certain types of meningitis, have special outer coats. These coats disguise antigens so that

TABLE 6.6 Some FDA-Approved Vaccines

Generic Name	Trade Name	Vaccine Type	Protection Against
Anthrax	BioThrax	Inactivated	Anthrax
Chicken pox	Varivax	Live attenuated	Chicken pox
Diphtheria	Pediarix	Toxoid	Diphtheria
<i>Haemophilus influenza</i> type b (Hib)	ActHib, PedvaxHib	Conjugate	<i>H. influenza</i>
Hepatitis A	Havrix, Vaqta	Inactivated	Hepatitis A
Hepatitis B	Engerix-B, Recombivax	Recombinant subunit	Hepatitis B
HepA-HepB	Twinrix	Inactivated-recombinant	Hepatitis A and B
Herpes zoster	Zostavax	Live attenuated	Herpes zoster
Human papilloma virus	Gardasil, Cervarix	Recombinant	Cervical cancer
Influenza	Fluarix, Fluvirin, Fluzone, Flulaval, Afluria, Agriflu, FluMist	Inactivated Live attenuated	Influenza
Japanese encephalitis	Ixiaro, JE-Vax	Inactivated	Japanese encephalitis
Measles	Attenuvax	Live attenuated	Measles
Meningococcal conjugate	Menactra, Menomune, Menveo	Inactivated	Meningococcal
Mumps	MumpsVax	Live attenuated	Mumps
Pertussis	Boostrix	Subunit	Pertussis
Pneumococcal	Prevnar, PCV13, Pneumovax23	Conjugate	Pneumococcal
Polio	Ipol	Inactivated	Polio
Sipuleucel-T	Provenge	Autologous	Prostate cancer
Rabies	Imovax, Rabavert	Inactivated	Rabies
Rotavirus	Rotarix, RotaTeq	Live attenuated	Rotavirus
Rubella	Meruvax	Live attenuated	Measles
Tetanus toxoid (TT)		Toxoid	Tetanus
Typhoid	Typherix, Vivotif Berna, Typhim Vi	Inactivated	Typhoid
Varicella	Varivax	Live attenuated	Varicella
Vaccinia (smallpox)	ACAM2000	Live attenuated	Smallpox
Yellow fever	YF-Vax	Live attenuated	Yellow fever

the immature immune systems of infants and younger children are unable to recognize these harmful bacteria. In a conjugate vaccine, proteins or toxins from a second type of organism, one that an immature immune system can recognize, are linked to the outer coats of the disease-causing bacteria. This enables a young immune system to respond and defend against the disease agent.

DNA Vaccines or Naked Vaccine

Genes encoding antigens of an infectious organism are expressed by the host's own cells. Genes are inserted into a bacterial plasmid under the control of a mammalian promoter. The chimeric plasmid is either directly

injected into muscle or the DNA is conjugated to a solid matrix such as gold particles.

Recombinant Vector Vaccines

A vaccine vector, or carrier, is a weakened virus or bacterium into which harmless genetic material from another disease-causing organism can be inserted.

MONOCLONAL ANTIBODIES

Introduction to Antibodies

The cell-mediated branch of the immune system includes the antibody-secreting B cells or plasma cells (81–83).

Antibodies or immunoglobulins are soluble proteins that are produced in response to an antigenic stimulus. As part of the normal immune system, each B cell produces as many as 100 million antibody proteins (polyclonal antibodies) directed against bacteria, viruses, and other foreign invaders. Antibodies act by binding to a particular antigen, thereby “tagging” it for removal or destruction by other immune system components.

The production of antigen-“neutralizing” antibodies or immunoglobulins and the detection of a sufficient antibody titer are important concepts for an understanding of vaccinations and exposure to antigens. The humoral response to an antigen involves the creation of memory B cells and the transformation of B lymphocyte into plasma cells that serve as factories for the production of secreted antibodies. Approximately 4 days after initial contact with an antigen (immunization), immunoglobulin M antibodies (one of five types of immunoglobulin structure) appear and then peak approximately 4 hours later. Approximately 7 days after exposure, IgG, the major class of circulating immunoglobulin, appears. The antibodies bind to the antigen and effect additional immune system-mediated events, “neutralizing” the antigen and leading to its elimination. The concentration of an immunoglobulin specific for a given antigen at a given time is referred to as the antibody titer and may be a measure of the effectiveness of the initial antigen exposure/vaccination to elicit immunologic memory.

Antibody Structure

Antibodies are glycoproteins. The simplest structure of an immunoglobulin molecule consists of two identical long peptide chains (the heavy chains) and two identical short polypeptide chains (the light chains) interconnected by several disulfide bonds. The selectivity of any immunoglobulin for a particular antigen is determined by its structure and, specifically, by the variable or

antigen-binding regions (Fig. 6.23). Enzymatic digestion of the antibody with papain yields the functional human antibody (Fab) fragment, which contains the antigen-binding sites, and the Fc fragment, which specifies the other biologic activities of the molecule.

Hybridoma Technology

MAbs are ultrasensitive, hybrid immune system-derived proteins designed to recognize specific antigens. Nobel Laureates Kohler and Milstein first reported MAbs in 1975 (130). MAbs have been used in laboratory diagnostics, site-directed drugs, and home test kits (85,128,131–134). The B lymphocyte produces a wide range of structurally diverse antibody proteins with varying degrees of specificity in response to a single antigen stimulus. Because of their structural diversity, these antibodies would be called polyclonal antibodies. MAbs are homogeneous hybrid proteins produced by a selected, single clone of an engineered B lymphocyte. They are designed to recognize specific sites or epitopes on antigens.

Hybridoma technology (the technology used to produce MAbs) consists of combining or fusing two different cell lines: a myeloma cell (generally from a mouse) and a plasma spleen cell (B lymphocyte) capable of producing an antibody that recognizes a specific antigen (Fig. 6.24). The resulting fused cell, or hybridoma, possesses some of the characteristics of both original cells: the myeloma cell’s ability to survive and reproduce in culture (immortality), and the plasma spleen cell’s ability to produce antibodies to a specific antigen.

Monoclonal antibodies are more attractive than polyclonal antibodies for diagnostic and therapeutic applications because of their increased specificity of antigen recognition. Thus, they can serve as target-directed “homing devices” to find and attach to the targeted antigen. Developments in hybridoma technology have led to highly specific diagnostic agents for home use in

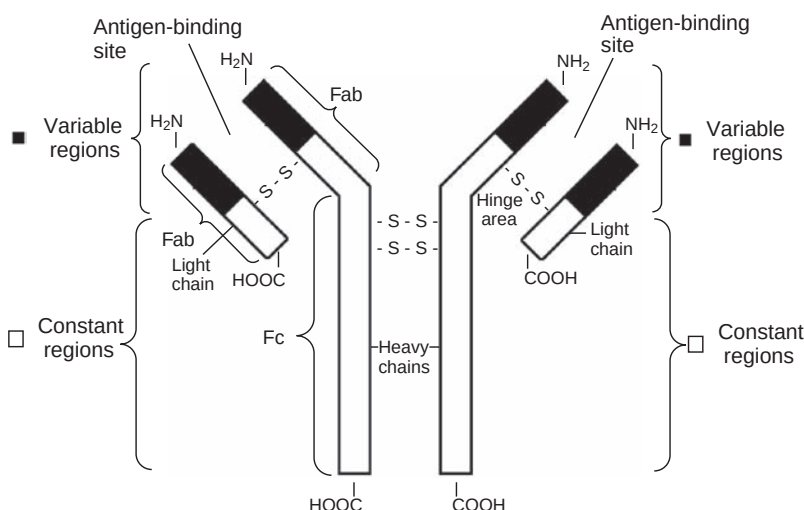


FIGURE 6.23 Schematic model of an antibody molecule.

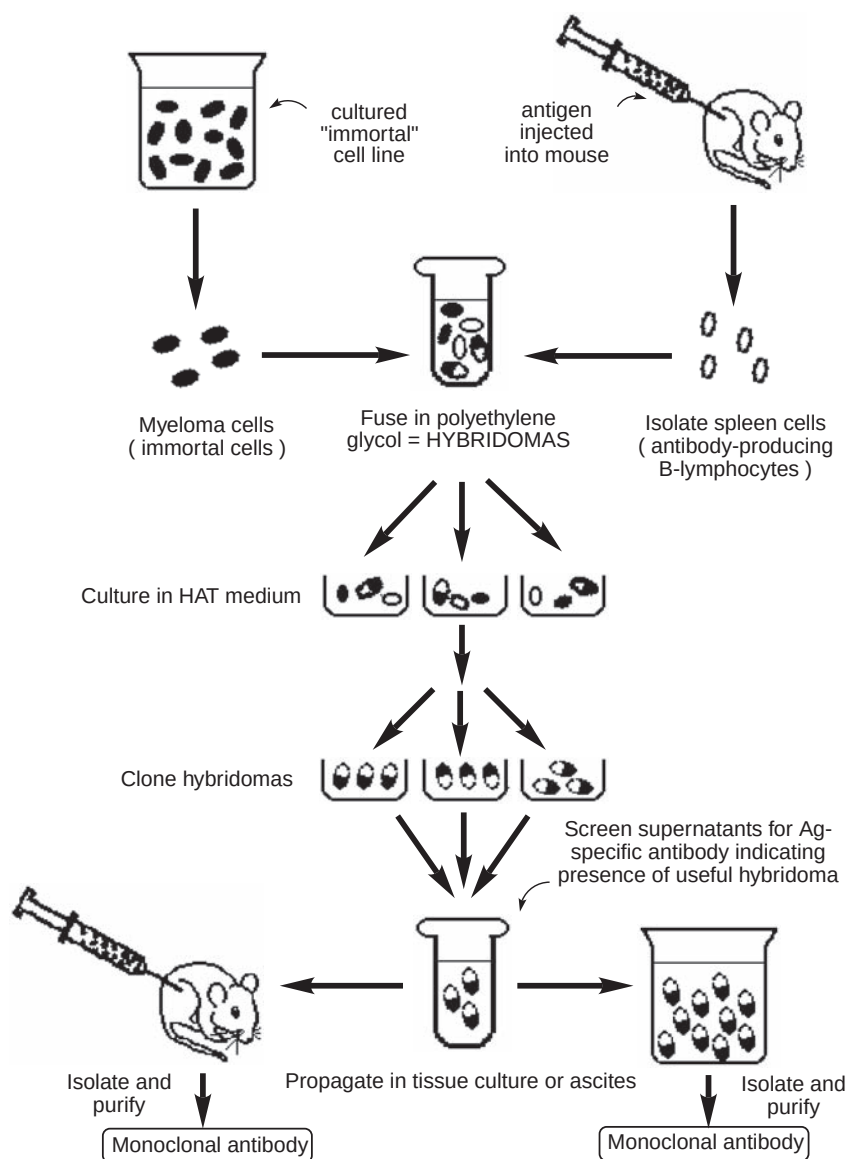


FIGURE 6.24 Outline of hybridoma creation and monoclonal antibody (MAB) production.

pregnancy testing and ovulation prediction kits; laboratory use in detection of colorectal cancer, ovarian cancer, and others; and design of site-directed therapeutic agents, such as trastuzumab to combat metastatic breast cancer and abciximab as an adjunct for the prevention of cardiac ischemic complications.

Monoclonal Antibody Immunogenicity

Nearly 25 years after the pioneering work of Kohler and Milstein, MABs began to realize their therapeutic potential. Until recently, most MABs were murine proteins based on their production. Initial clinical trials of murine MABs showed that these mouse proteins were highly immunogenic in patients after just a single dose (135). Human patients formed antibodies to combat the foreign MAB that was administered. The human anti-mouse antibody response is known as HAMA. So, far from being

the “magic bullets” that were proposed, immunogenic murine MABs were useless in chronic therapy. Thus, a different approach was needed to eliminate the unwanted immune response (HAMA) in patients.

The variable regions of an immunoglobulin must be of a specific chemical structure with the ability to bind to the antigen they “recognize.” The part within the variable region that forms the intermolecular interactions with the antigen is the complementarity-determining region (CDR). The variable domains of antibody light and heavy chains each contain three CDRs.

It was determined that immune responses against the mouse-produced MABs were directed against both the variable and the constant regions of the antibody. Human and murine antibodies are very homologous in chemical structure. Thus, an MAB should be engineered to decrease the immunogenicity of an MAB by replacing

the mouse constant regions of an IgG with human constant regions, making the antibody less mouse-like (131,136–139). In practice, what generally occurs is the variable heavy-chain and variable light-chain domains (CDRs) of human immunoglobulins are replaced with those of a murine antibody, which possesses the requisite antigen specificity. This “chimeric” MAb will retain its ability to recognize the antigen (a property of the murine MAb) and retain the many effector functions of an immunoglobulin (both murine and human), but be much less immunogenic (a property of a human immunoglobulin). A chimeric MAb, containing approximately 70% human sequence, will have a longer half-life than its murine counterpart in a human patient. Therapeutic MAbs that are chimeric include catumaxomab, cetuximab, and infliximab.

The discovery of the conserved structure of antibodies, particular IgGs, across many species suggested the possibility of chimeric antibodies, and the realization that the homology extended to the antigen-binding site facilitated the engineering of humanized immunoglobulins. Advances in phage display technology and the production of transgenic animals has led to the production of humanized or fully human MAbs (119,129–132). Functional human antibody fragments (e.g., Fab fragments) can be displayed on the surface of bacteriophages. A bacteriophage, also called a phage, is a virus that infects bacteria. The expression of these human antibody fragments on the phage surface has facilitated efficient screening of large numbers of phage clones (phage display) for antigen-binding specificity (133). Once a fragment with the requisite antigen specificity is selected, it can be isolated and engineered into a humanized MAb (replacing up to 95% of the murine protein sequence) or a fully human MAb (100% human sequence). Transgenic strains of mice have been genetically engineered to possess most or all of the essential human antibody genes. Thus, on immunization with a foreign antigen, the transgenic mice will develop humanized or fully human antibodies in response. Both of these techniques, although very complex and expensive, have yielded FDA-approved, humanized antibody pharmaceuticals, such as alemtuzumab, bevacizumab, eculizumab, and trastuzumab. The half-life of humanized antibodies is dramatically enhanced (from hours to weeks), and immunogenicity is drastically reduced.

Monoclonal Antibody Diagnostic Agents

Several ultrasensitive diagnostic MAb-based products have enjoyed great success; these include a variety of imaging agents for the detection of blood clots and cancer cells. A monoclonal Fab fragment, technetium-99m–arcitumomab (CEA-Scan), can detect the presence and indicate the location of recurrent and metastatic colorectal cancer. Colorectal cancer and ovarian cancer can be detected with satumomab pendetide (OncoScint CR/OV). Capromab pendetide (ProstaScint) is used for

detection, staging, and follow-up of patients with prostate adenocarcinoma. Small-cell lung cancer can be detected with nofetumomab (Verluma). The first imaging MAb for myocardial infarction is imicromab pentetate (MyoScint).

Monoclonal Antibody–Based, In-Home Diagnostic Kits

The strong trend toward self-care, coupled with a heightened awareness by the public of available technology and an emphasis on preventive medicine, has increased the use of in-home diagnostics (140–142). MAb specifically minimizes the possibilities of interference from other substances that might yield false-positive test results. The antigen being selectively detected by MAb-based pregnancy test kits is human chorionic gonadotropin, the hormone produced if fertilization occurs and that continues to increase in concentration during the pregnancy. Table 6.7 lists some examples of MAb-containing in-home pregnancy test kits as well as some examples of MAb-based, in-home ovulation prediction kits.

Monoclonal Antibody Therapeutic Agents

Hybridoma technology and advanced antibody engineering has led to the design of an increasing number of site-directed therapeutic agents for the treatment and prevention of transplant rejection, therapy in rheumatoid arthritis, treatment of NHL, and other indications. These products are examples of murine (e.g., tositumomab), chimeric (e.g., cetuximab), human (e.g., panitumumab), and humanized (e.g., trastuzumab) MAbs, and they represent significant advances in pharmacotherapy. Murine MAbs are derived from mice. Chimeric MAbs contain 65% to 90% human protein that is fused with the murine antibody variable region, which allows for functional complement activation and antibody-dependent cell-mediated cytotoxicity in humans. Variations of chimeric MAbs, such as partially humanized and de-immunized MAbs, are 95% human protein and are composed of a few critical residues involved in the antigen binding site from the murine antibody or modified murine variable domains containing nonimmunogenic amino acid sequences, respectively. As described earlier, to prevent any HAMA response, fully humanized MAbs containing only human protein sequences have been developed from mice that have had human immunoglobulin genes placed in their genome. Therapeutic MAbs (Table 6.8) can be divided into three main classes based on their mechanism of action: 1) MAbs as directed targeted therapy: these MAbs either block or stimulate a particular cell membrane molecule (e.g., growth factor signal receptor) or ligand (vascular endothelial growth factor [VEGF]) and thereby inhibit tumor growth or activate effector cells; 2) cytotoxicity by chaperoning cytotoxic molecules (immunconjugates): these MAbs are conjugated to various cytotoxic molecules/atoms including chemotherapy or radioisotopes such as

TABLE 6.7 Some MAb-Based In-Home Test Kits^a

Manufacturer	Product Distributor	Positive End Point
Pregnancy		
Answer Plus	Carter Products	Plus in test window = +
Answer Quick & Simple	Carter Products	Plus in test window = +
Clear Blue Easy	Unipath	Blue line in large window = +
Clear Blue Easy One Min.	Unipath	Blue line in large window = +
Conceive	Quidel	Pink to purple test line = +
1 Step E.P.T.	Warner Lambert	Pink color in test and control = +
Clear Blue Easy	Unipath	Blue line in large window = +
Clear Blue Easy One Min.	Unipath	Blue line in large window = +
Conceive	Quidel	Pink to purple test line = +
1 Step E.P.T.	Warner Lambert	Pink color in test and control = +
Clear Blue Easy	Unipath	Blue line in large window = +
Clear Blue Easy One Min.	Unipath	Blue line in large window = +
Ovulation		
Answer Quick & Simple	Carter Products	Purple stick line darker than reference = +
Conceive 1 Step	Quidel	Pink to purple test line darker than reference = +
First Response 1 Step	Carter Products	Purple test line darker than reference = +
ClearPlan Easy	Unipath	Blue test line in large window similar or darker than line in small window = +
Ovukit Self Test	Quidel	White to shades of blue compared to LH surge guide = +
OvuQuick	Quidel	Test spot appears darker than reference = +
Q Test	Quidel	Purple test line darker than reference = +

^aInformation from Pray W. Nonprescription Product Therapeutics, 2nd Ed. Philadelphia: Lippincott Williams & Wilkins, 2006:778–780; Quattrocchi E, Hove I.

Ovulation & pregnancy home testing products. U.S. Pharmacist 1998;23:54–63; and Rosenthal WM, Briggs GC. Home testing and monitoring devices.

In: Allen LV Jr, Berardi RR, DeSimone EM II, et al., eds. Handbook of Nonprescription Drugs. Washington, DC: American Pharmaceutical Association, 2000:917–942.

yttrium-90, which is in clinical use, cellular toxins such as diphtheria toxin, or biologic agents such as IFN; and 3) modulating an immunologic mechanism: in this case, MABs exert their cytotoxic effects by antibody-dependent cell-mediated cytotoxicity or complement-dependent cytotoxicity.

SPECIFIC DRUGS

Catumaxomab (Removab) Catumaxomab is a 180 kDa, chimeric murine IgG2a/rat IgG2b anti-EpCAM/anti-CD3, trifunctional MAB produced in the rat/mouse quadroma cell line (143). The human epithelial cell adhesion molecule (EpCAM) protein is overexpressed on a variety of carcinomas, and its inhibition has emerged as a viable approach for the immunotherapy of cancer. In addition to targeting EpCAM, catumaxomab has anti-CD3 arm that targets the T-cell antigen CD3; binding to the CD3 receptor activates T cells to release cytotoxic cytokines and to promote T-cell-mediated lysis. The

trifunctionality comes from an intact Fc region that selectively binds to FCγ receptor–positive accessory cells, such as macrophages, dendritic cells, and natural killer cells that promote phagocytosis and antibody-dependent cell cytotoxicity. Catumaxomab was generated using a hybrid hybridoma or quadroma, where two hybridoma cells producing different MABs are fused together to form a new hybrid cell that contains sets of genes encoding the two parent antibodies. Catumaxomab is used for the intraperitoneal treatment of malignant ascites in patients with EpCAM-positive carcinomas where standard therapy is not available or no longer feasible.

Trastuzumab (Herceptin) One of the major advances in breast cancer treatment in the last decade has been the development of MABs targeting growth factor receptors. Advances in the understanding of tumor pathobiology and molecular biology have facilitated the development of these targeted therapies, in particular in breast cancer.

TABLE 6.8 FDA-Approved MAb Therapeutic Agents

Generic Name	Trade Name	MAb Type	Indication
Abciximab	ReoPro	Chimeric	Adjunct to percutaneous transluminal coronary angioplasty for prevention of acute cardiac complications
Adalimumab	Humira	Fully human	Rheumatoid arthritis
Alemtuzumab	Campath	Humanized	B-cell chronic lymphocytic leukemia
Basiliximab	Simulect	Chimeric	Prevention of transplant rejection
Belimumab	Benlysta	Fully human	Systemic lupus erythematosus plus standard therapy
Bevacizumab	Avastin	Humanized	Metastatic colorectal cancer
Canakinumab	Ilaris	Fully human	Cryopyrin-associated periodic syndromes and Muckle-Wells syndrome
Catumaxomab	Removab	Chimeric	Malignant ascites
Certolizumab pegol	Cimzia	Humanized	Crohn disease
Cetuximab	Erbix	Chimeric	Colorectal cancer
Daclizumab	Zenapax	Humanized	Prevention of transplant rejection
Denosumab	Prolia	Fully human	Postmenopausal osteoporosis in women
Eculizumab	Soliris	Fully human	Paroxysmal nocturnal hemoglobinuria
Efalizumab	Raptiva	Humanized	Psoriasis
Etanercept	Embril	Fusion MAb	Rheumatoid and psoriatic arthritis and ankylosing spondylitis
Gemtuzumab	Mylotarg	Fusion MAb	Treatment of CD33-positive acute myeloid leukemia
Golimumab	Simponi	Fully human	Rheumatoid and psoriatic arthritis and ankylosing spondylitis
Ibritumomab tiuxetan	Zevalin	Murine	Non-Hodgkin lymphoma
Infliximab	Remicade	Chimeric	Crohn disease
Ipilimumab	Yervoy	Fully human	Late-stage melanoma
Muromonab-CD3	Orthoclone-OKT3	Murine	Reversal of transplant rejection
Natalizumab	Tysabri	Humanized	Multiple sclerosis
Ofatumumab	Arzerra	Fully human	Chronic lymphocytic leukemia
Omalizumab	Xolair	Humanized	Asthma
Palivizumab	Synagis	Humanized	Respiratory syncytial virus prophylaxis
Panitumumab	Vectibix	Fully human	First- or second-line treatment of metastatic colorectal cancer
Ranibizumab	Lucentis	Fully human	Neovascular age-related macular degeneration
Rituximab	Rituxan	Chimeric	Treatment of B-cell non-Hodgkin lymphoma
Tositumomab	Bexxar	Murine	Non-Hodgkin lymphoma
Trastuzumab	Herceptin	Humanized	Refractory breast cancer
Ustekinumab	Stelara	Fully human	Moderate to severe psoriasis

HER2 (or erbB2) is a transmembrane tyrosine kinase receptor that belongs to the family of the epidermal growth factor receptor (EGFR). HER2 protein overexpression is observed in 25% to 30% of primary breast cancers. Activated erbBs stimulate many intracellular

signaling pathways, mainly the mitogen-activated protein kinase (MAPK) and the phosphatidylinositol 3-kinase (PI3K)-Akt pathways. In 1998, trastuzumab was approved by the FDA for clinical use largely based on a randomized clinical trial that compared chemotherapy with

chemotherapy plus trastuzumab as a front-line treatment for patients with erbB2-overexpressing metastatic breast cancer (144).

Trastuzumab is a recombinant humanized anti-erbB2 MAb that binds the extracellular domain of the receptor and blocks intracellular signaling. Both cytostatic and cytotoxic mechanisms of action of trastuzumab were identified in preclinical studies. In vitro, downregulation of erbB2 disrupts receptor dimerization and signaling through the downstream PI3K cascade. It effectively prevents cell proliferation of the erbB2-overexpressing breast cancer SK-BR-3 cell line.

Trastuzumab, administered as single-agent, first-line therapy in women with erbB2-overexpressing metastatic breast cancer and in women with erbB2-overexpressing metastatic breast cancer that has progressed after chemotherapy, produces durable objective responses with a median duration of response of 9.1 months (144).

Bevacizumab (Avastin) Angiogenesis, the process of developing new blood vessels from existing ones, is critical for tumor cell growth, survival, invasion, and metastasis. Numerous growth factors work in the tumor microenvironment to promote angiogenesis. Of these, VEGF may be among the most important factors implicated in angiogenesis, based on its specificity as an endothelial cell mitogen and the ability of many tumor types to produce it in physiologically relevant quantities. Oncogenic activation, loss of tumor suppressor factors, and tumor hypoxia lead to upregulation of VEGF production. There are four isoforms of the VEGF (A, B, C, and D) and three types of VEGF receptors (VEGFRs). Once produced, VEGF-A binds to two tyrosine kinase receptors, termed VEGFR-1 and VEGFR-2, which are predominantly located on the surface of vascular endothelial cells. VEGFR-2 appears to be the more important of the two receptors for mediating the angiogenic effects of VEGF, whereas VEGFR-1 may be a decoy receptor that modulates the amount of VEGF available for binding to VEGFR-2. VEGF and its receptors have emerged as anticancer targets based on their central and specific role in angiogenesis. In principle, VEGF-targeted therapy may inhibit tumor growth by blocking new vessel growth. Bevacizumab was approved in 2004 for the treatment of metastatic colorectal cancer in combination with fluorouracil-based chemotherapy (145).

Bevacizumab, a 149 kDa recombinant humanized monoclonal IgG1 antibody (93% human, 7% murine sequences) directed against VEGF-A, is believed to globally prevent the binding of all VEGF isoforms to all VEGFRs. Bevacizumab is composed of two identical light chains (214 amino acids) and two heavy chains (453 amino acids) and is produced in a CHO cell expression system (145).

Panitumumab (Vectibix) EGFR, a transmembrane cell surface glycoprotein belonging to the subfamily of type I tyrosine kinase receptors, is overexpressed in certain human cancers including colon and rectum cancers. The

binding of ligands such as EGF and transforming growth factor- α to EGFR triggers autophosphorylation and internalization of EGFR, thereby activating various signaling pathways involved in proliferation, angiogenesis, inhibition of apoptosis, and metastasis.

Panitumumab is a 147 kDa recombinant, fully human IgG2 anti-EGFR MAb produced in genetically engineered mammalian CHO cells. Panitumumab was discovered using Abgenix's XenoMouse technology and was approved by the FDA in 2006 (146). Proposed mechanisms explaining the antitumor activity of panitumumab include downregulation of EGFR expression resulting from receptor internalization, induction of apoptosis via inhibition of EGFR signaling pathways and induction of cell cycle arrest, induction of autophagy, and inhibition of angiogenesis. Panitumumab is used in combination with chemotherapy for the first- or second-line treatment of metastatic colorectal cancer or as monotherapy for the treatment of chemotherapy-refractory metastatic colorectal cancer in patients with wild-type rather than mutant *K-ras* tumors. *K-ras* protein is a membrane bound G protein and is activated by receptor tyrosine kinases and is found to be mutated in 30% to 40% of colorectal cancer patients (147,148). Panitumumab has an acceptable tolerability profile when administered as monotherapy or in combination with chemotherapy. It is associated with skin-related toxicities, which is a characteristic of EGFR inhibitors.

Cetuximab (Erbix) As discussed previously, EGFR overexpression is detected in many human cancers, including colorectal cancer. Cetuximab was approved in 2004 for the second-line treatment of colorectal cancer either alone or in combination with chemotherapy (149). Cetuximab, a recombinant human/mouse chimeric MAb with an approximate MW of 152 kDa, is produced in mammalian murine myeloma cell culture. It binds specifically to the extracellular domain of the human EGFR on normal and tumor cells and competitively inhibits the binding of the EGF and other ligands, such as transforming growth factor- α . Binding of cetuximab to the EGFR blocks phosphorylation and activation of receptor-associated kinases, resulting in inhibition of cell growth, induction of apoptosis, decreased matrix metalloproteinase, and VEGF reduction. Cetuximab is eliminated by binding to EGFRs in various tissues, followed by internalization of the antibody-EGFR complex.

Ofatumumab (Arzerra) Chronic lymphocytic leukemia (CLL) is the most common adult leukemia and one of the most common malignant lymphoid diseases. Based on 2007 worldwide estimates, leukemia accounted for more than 330,000 new cases and more than 245,000 deaths (150). CLL cells are malignant B cells that have low surface expression levels of CD20 molecules. B cells normally protect the body from invading pathogens by developing into plasma cells, which make antibodies.

These antibodies directly inactivate pathogens or attach to pathogens to prepare them for destruction by other white blood cells. Ofatumumab received accelerated approval in 2009 for the treatment of patients with CLL refractory to fludarabine and alemtuzumab.

Ofatumumab is a fully human 149 kDa IgG1κ anti-CD20 MAb generated via transgenic mouse and hybridoma technology and is produced in a recombinant murine cell line (NS0) using standard mammalian cell cultivation and purification technologies (151). Ofatumumab binds specifically to both the small and large extracellular loops of the CD20 molecule. Ofatumumab is highly potent in lysing B cells, which is presumably due to its binding to the membrane-proximal, small extracellular loop of the target CD20 protein and its slow release from the target molecule (152). As with other human monoclonal antibodies, ofatumumab induces both antibody-dependent cell-mediated cytotoxicity and complement-dependent cytotoxicity.

Alemtuzumab (Campath-1H) Alemtuzumab was introduced in 2001 for the treatment of patients with B-cell CLL who had been treated with alkylating agents and failed fludarabine therapy. Alemtuzumab is an rDNA-derived humanized IgG1κ MAb specific for the cell surface glycoprotein CD52 expressed on normal and malignant human peripheral-blood B and T lymphocytes as well as natural killer cells, monocytes, macrophages, and tissues of the male reproductive system. It is produced in CHO cells and has an approximate MW of 150 kDa. The mechanism of action is not completely understood but involves a number of effects, including complement-mediated cell lysis, antibody-dependent cellular toxicity, and the induction of apoptosis (153,154).

Ibritumomab Tiuxetan (Zevalin) Radioimmunotherapy is an innovative form of cancer therapy, combining a MAb against a specific target antigen with a source of radiation such as a radioisotope. Technical advances have made it possible to link radionuclides such as yttrium-90 (^{90}Y) to MAbs specifically to target radiation to lymphoma cells. Yttrium-90 is a beta-emitting radionuclide that delivers 90% of its radiation (2.3 MeV) over a mean path length of 5 mm and has a half-life of 64 hours. These characteristics are particularly advantageous for treating bulky, poorly vascularized tumors and tumors with heterogeneous antigen expression. Ibritumomab tiuxetan therapeutic regimen became the first radioimmunotherapy approved by the FDA in 2002 for the treatment of NHL, the fifth leading type of cancer in the United States, with approximately 300,000 people affected by this cancer at present.

Ibritumomab tiuxetan, a 148 kDa radioimmunoconjugate, is a short-course therapy that uses immunobiologic and radiolytic mechanisms of action to destroy both dividing and nondividing tumor cells. Ibritumomab is a murine IgG1κ MAb produced in CHO cells. It is directed against the CD20 antigen found

on the surface of malignant B lymphocytes in patients with B-cell NHL as well as on normal mature B lymphocytes. Tiuxetan, [*N*-[2-bis(carboxymethyl)amino]-3-(*p*-isothiocyanatophenyl)propyl]-[*N*-[bis(carboxymethyl)amino-2-(methyl)-ethyl]glycine] forms a stable covalent thiourea-type linkage with the lysine and arginine residues of the antibody and can chelate a radionuclide via its carboxyl groups (Fig. 6.25). Ibritumomab tiuxetan can chelate indium-111 (^{111}In) for imaging or ^{90}Y for therapy. Thus, the antibody specifically targets radiation to CD20-positive cells while sparing normal non-lymphoid cells. Ibritumomab tiuxetan is indicated for the treatment of patients with relapsed or refractory, low-grade, follicular or transformed B-cell NHL, including patients with rituximab-refractory follicular NHL. After ^{90}Y Zevalin enters the bloodstream, the MAb ibritumomab recognizes and attaches to the CD20 antigen, allowing beta radiation emitted by the ^{90}Y isotope to penetrate and damage the B cell as well as neighboring cells (155).

Tositumomab (Bexxar) Tositumomab was introduced in 2003 as a novel radioimmunotherapeutic antibody for the treatment of B-cell NHL. Therapeutic regimen of the drug is composed of tositumomab and iodine-131 (^{131}I)-tositumomab. Tositumomab is a murine IgG2a lambda monoclonal antibody directed against the CD20 antigen, which is found on the surface of normal and malignant B lymphocytes. Tositumomab is produced in an antibiotic-free culture of mammalian cells and is composed of two murine gamma 2a heavy chains of 451 amino acids each and two lambda light chains of 220 amino acids each. The approximate MW of tositumomab is 150 kDa. ^{131}I -tositumomab is a radio-iodinated derivative of tositumomab that has been covalently linked to ^{131}I through tyrosine residues.

Tositumomab actions include induction of apoptosis, complement-dependent cytotoxicity, and antibody-dependent cellular cytotoxicity. Additionally, cell death is associated with ionizing radiation from the radioisotope. ^{131}I has a half-life of 8 days and emits β and γ rays. The antitumor

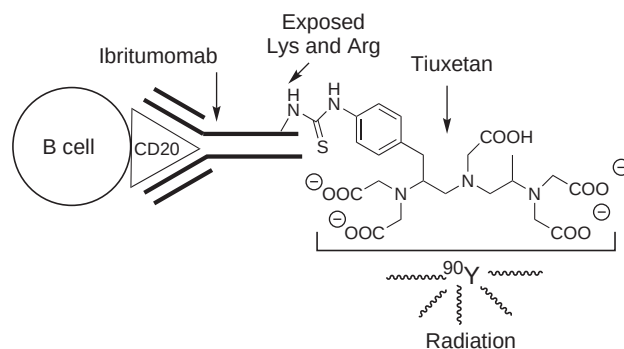


FIGURE 6.25 Structure of ibritumomab tiuxetan radioimmunoconjugate monoclonal antibody (MAb) and its binding to the target CD20 on the surface of the B cell.

activity results from the β -particles. Thus the mechanisms of action of this radioimmunotherapeutic drug involve combining tumor-targeting ability of the cytotoxic MAb with patient-specific delivery of radiation dose directly to the tumor cells. The therapeutic regimen of the radioimmunotherapeutic antibody is indicated for the treatment of patients with CD20 antigen-expressing relapsed or refractory, low-grade, follicular, or transformed NHL, including patients with rituximab-refractory NHL (155).

Storage of the radioimmunotherapeutic drug requires special considerations. The lead pot containing ^{131}I -tositumomab must be stored in a freezer at a temperature of -20°C or below until it is removed for thawing prior to administration to the patient. Thawed dosimetric and therapeutic doses of ^{131}I -tositumomab are stable for up to 8 hours at 2°C to 8°C or at room temperature. Any unused portion must be discarded according to federal and state laws.

Canakinumab (Ilaris) Canakinumab was approved in 2009 for the treatment of cryopyrin-associated periodic syndromes known as familial cold autoinflammatory syndrome and Muckle-Wells syndrome in patients 4 years of age or older. The two syndromes are serious inherited autoimmune inflammatory disorders that are believed to result from cryopyrin-activated overproduction of IL-1 β . Canakinumab is a 150 kDa, fully human, IgG1 anti-IL-1 β MAb produced in mouse hybridoma Sp2/0-Ag14 cell line by rDNA technology (156). Compared to other drugs targeting IL-1 β , such as anakinra and rilonacept, canakinumab possesses a less frequent dosing regimen and instigates fewer injection site reactions. Because IL-1 β suppression can hinder the immune response, the labeling recommends that patients receive all recommended vaccinations before starting canakinumab therapy. No live vaccines should be administered during treatment with canakinumab, and patients should be tested for latent tuberculosis before receiving canakinumab. Similar to rilonacept, the labeling warns against the concomitant administration of canakinumab and TNF inhibitors, and therapeutic monitoring is recommended with concurrent administration of drugs that interact with the CYP450 system.

Certolizumab Pegol (Cimzia) Crohn disease is an immunologically mediated chronic relapsing inflammatory bowel disease of the gastrointestinal tract that can potentially affect the entire gastrointestinal tract, but most commonly occurs in the terminal ileum and colon. Active Crohn disease is associated with excessive levels of proinflammatory cytokines including TNF- α . It is growing in incidence in both developing and developed countries. The incidence rate in North America is among the highest with 1 to 15 cases per 100,000 person-years. Certolizumab pegol is a TNF- α blocker approved for the treatment of moderate to severe Crohn disease (157,158). It is specifically indicated for reducing signs and symptoms and maintaining clinical response in adult patients

who have had an inadequate response to conventional therapy. Certolizumab pegol is a 91 kDa humanized antibody fragment Fab' that is manufactured in *E. coli*, purified, and conjugated to PEG. The addition of PEG moiety significantly enhances the plasma half-life of the antibody, allowing for less frequent dosing. Certolizumab is the third anti-TNF- α biologic to be marketed for Crohn disease behind infliximab and adalimumab and has higher affinity than both of these drugs for TNF- α . Unlike the full-length antibodies infliximab and adalimumab, certolizumab pegol does not contain an Fc region and, therefore, does not induce complement activation, antibody-dependent cellular cytotoxicity, or apoptosis in vitro. As with other TNF- α inhibitors, the drug is associated with an increased risk of opportunistic infections and malignancy, and it carries a warning of the risk of serious infections such as histoplasmosis.

Golimumab (Simponi) The proinflammatory cytokine TNF- α has been implicated as the primary mediator of articular inflammation in diseases such as RA, psoriatic arthritis, and ankylosing spondylitis. Targeting TNF- α has been a successful strategy in the intervention of a range of immunoinflammatory disorders. Golimumab is a 150 kDa, fully human, IgG1 anti-TNF MAb produced in murine Sp2/0 cell line. It was approved by the FDA in 2009 (159). Drugs previously introduced include infliximab (chimeric MAb), etanercept (fusion protein comprised of the ligand-binding segment of the soluble TNF receptor) in the late 1990s, and adalimumab (fully human) and certolizumab pegol (PEGylated humanized) in more recent years. Golimumab binds to both soluble and transmembrane forms of TNF- α , is the first once-monthly SC agent to enter the market, and is currently approved for the treatment of RA and psoriatic arthritis in combination with MTX and for treatment of ankylosing spondylitis. The drug is administered as an SC injection once a month. Live vaccines should not be administered while being treated with golimumab. The drug carries a boxed warning regarding the risk of serious infections including tuberculosis and invasive fungal infections, such as histoplasmosis. Furthermore, patients are also warned that lymphoma and other malignancies, some fatal, have been reported in children and adolescent patients treated with TNF blockers, of which golimumab is a member.

Ustekinumab (Stelara) Psoriasis is a chronic inflammatory disease that affects approximately 2% to 3% of the world population. The most common form of the disease is plaque psoriasis. Conventional treatment options for psoriasis include topical corticosteroids, phototherapy, and systemic drugs, but all of these drugs have limitations. Ustekinumab is a fully human 149 kDa IgG1 κ MAb developed in transgenic mouse and expressed in unspecified recombinant cell line. Ustekinumab inhibits IL-12 and IL-23 signaling by blocking p40 binding to IL-12/23 receptors. It was approved by the FDA in 2009 for the

treatment of moderate to severe psoriasis in patients who cannot tolerate other treatment modalities with ease or in whom these modalities have failed (160). As a human IgG1κ MAb, ustekinumab is expected to be degraded into small peptides and amino acids via catabolic pathways in the same manner as endogenous IgG. The mean half-life of ustekinumab ranges from 15 to 46 days across all psoriasis studies following intravenous and SC administration.

Eculizumab (Soliris) Paroxysmal nocturnal hemoglobinuria is a clonal hematopoietic stem cell disorder that is characterized by the production of abnormal red blood cells. Eculizumab, a fully humanized anti-CD5 MAb with an approximate MW of 148 kDa, is used for the treatment of patients with paroxysmal nocturnal hemoglobinuria to reduce hemolysis (161,162). It is the first therapy to be introduced for this rare and life-threatening form of hemolytic anemia. Eculizumab is effective in controlling serum hemolytic activity. Eculizumab has been granted orphan drug status from both the FDA and European regulatory agencies. The drug carries a black box warning for the potential increased risk of meningococcal infections and requires patients to receive meningococcal vaccine at least 2 weeks prior to receiving the first dose of eculizumab.

Ranibizumab (Lucentis) Ranibizumab is a recombinant, fully humanized IgG1 MAb fragment produced by the *E. coli* expression system. It neutralizes all active forms of VEGF-A and is indicated for the treatment of neovascular age-related macular degeneration (163,164). The full-length anti-VEGF-A MAb, bevacizumab, is approved for the treatment of colorectal cancer. The Fab domain of ranibizumab differs from the Fab domain of bevacizumab by six amino acids. The smaller size of ranibizumab (~48 kDa) is expected to facilitate retinal penetration and hence is more suitable for intraocular use. The binding of ranibizumab to VEGF-A prevents the interaction with its receptors (VEGFR-1 and VEGFR-2) on the surface of endothelial cells, thereby reducing endothelial cell proliferation, vascular leakage, and new blood vessel formation. Ranibizumab is administered once a month by intravitreal injection.

Denosumab (Prolia) Osteoporosis is a chronic, debilitating disease in which the bones become porous and break easily. Approximately 10 million people in the United States are estimated to have osteoporosis, and almost 34 million Americans are estimated to have low bone mass, placing them at increased risk for osteoporosis. Receptor activator of nuclear factor-κB ligand (RANKL) is a protein expressed by osteoblasts and bone-lining cells that binds to receptor activator of nuclear factor-κB (RANK) receptors on osteoclast and osteoclast precursors. The RANKL/RANK complex stimulates osteoclast precursors to mature to osteoclasts and increases osteoclast activity on bone resorption. After menopause, as estrogen levels drop, RANKL levels increase. Denosumab, a 147 kDa fully human MAb produced in genetically engineered

mammalian CHO cells, is specific for RANKL and was approved by the FDA in 2010 for the treatment of postmenopausal osteoporosis in women (also see Chapter 30). By inhibiting RANKL, denosumab inhibits osteoclast function and bone resorption. Denosumab has been shown to increase bone mineral density and decrease fracture risk in postmenopausal women with osteoporosis (165).

PHARMACOGENOMICS AND PERSONALIZED MEDICINE

Personalized medicine is defined as the utilization of molecular biomarkers from an individual's genome, transcriptome, proteome, and metabolome under the influence of the individual's envirome in the assessment of predisposition to disease, screening and early diagnosis of disease, assessment of prognosis, pharmacogenomic prediction of therapeutic drug efficacy and risk of toxicity, and monitoring of illness until the final therapeutic outcome is determined (Fig. 6.26) (166).

The concept of personalized medicine was anticipated by Sir William Osler (1849–1919), a well-known Canadian physician during his time. He recognized that “variability is the law of life, and as no two faces are the same, so no two bodies are alike, and no two individuals react alike and behave alike under the abnormal conditions we know as disease.” Personalized medicine has rapidly advanced the prediction of disease incidence as well as the prevention of incorrect drug prescription based on a person's clinical, genetic, and environmental information. The goal of personalized medicine is optimizing the medical care and outcomes for each patient (167).

Pharmacogenomics uses genomic tools to understand the genotype effects of relevant genes on the behavior of a drug, as well as the effects of a drug on gene expression. The best examples of successful pharmacogenomic applications are presented below.

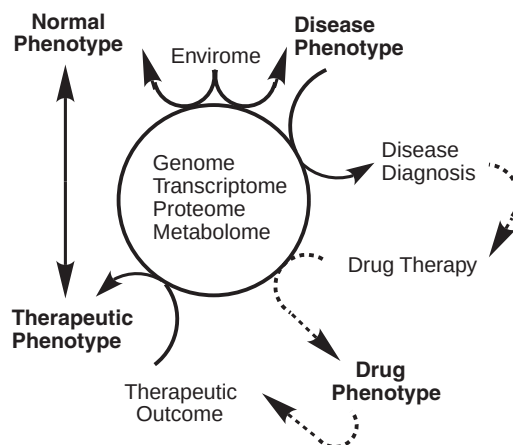


FIGURE 6.26 Schematic representation of the concept of molecular profiling for personalized medicine. Dashed arrows indicate the critical components of pharmacogenomics: disease diagnosis, drug therapy, drug phenotype, and therapeutic outcome.

The oral anticoagulant warfarin is prescribed for the long-term treatment and prevention of thromboembolic events. An investigation of the pharmacokinetic and pharmacodynamic drug properties of warfarin indicated the additive involvement of two genes when determining the dosage. One of these genes encodes CYP2C9, which is responsible for the metabolic clearance (~80%) of the pharmacologically potent *S*-enantiomer of warfarin. There are three allele types, CYP2C9*1, *2, and *3, and both CYP2C9*2 and *3 cause a reduction in warfarin clearance. A 10-fold difference in warfarin clearance was observed between groups of individuals having the genotype of the highest metabolizer (CYP2C9*1 homozygote) and lowest metabolizer (CYP2C9*3 homozygote) (168).

Trastuzumab is a MAb that specifically targets breast cancers overexpressing the *HER2/neu* gene and thus is marketed solely for the subset of breast cancer patients (~10%) overexpressing the *HER2/neu* gene. Because trastuzumab was developed for marker-positive individuals who comprise a rather low proportion of breast cancer patients, trastuzumab therapy may be one of the best examples of a genomic technology paving the way for personalized medical treatment (169).

Tamoxifen is a prodrug that is metabolized by members of the CYP450 family into two active metabolites: 4-hydroxy tamoxifen (4OH-TAM) and 4-hydroxyl-*N*-desmethyltamoxifen (endoxifen) (Fig. 6.27). CYP3A4/5 is responsible for the conversion of tamoxifen into *N*-desmethyltamoxifen, which is then converted into its active metabolite, endoxifen, by CYP2D6. CYP2D6 is also responsible for the conversion of tamoxifen into 4OH-TAM. In a recent study of steady-state levels of tamoxifen and active tamoxifen metabolites, there was interpatient variability for all three metabolites. A recent study investigating the CYP2D6*4 allele (inactive enzyme), a poor metabolizer that is common in Caucasians, in patients being treated with tamoxifen found that individuals who were homozygous for CYP2D6*4 had significantly

lower endoxifen levels than patients who had the wild-type gene. This study clearly indicates that genotyping of patients with impaired CYP2D6 function may be beneficial in a clinical setting to determine which patients will derive the most benefit from tamoxifen therapy (170).

Tolterodine [(*R*)-*N,N*-diisopropyl-3-(2-hydroxy-5-methylphenyl)-phenylpropanamine] is a new antimuscarinic drug for the treatment of urinary urge incontinence and other symptoms associated with an overactive bladder. Two different oxidative metabolic pathways, hydroxylation and *N*-dealkylation, have been identified in humans (Fig. 6.28). Hydroxylation to the pharmacologically active 5-hydroxymethyl metabolite (5-HM) is catalyzed by CYP2D6, whereas the *N*-dealkylation pathway is catalyzed by CYP3A4. Further oxidation of 5-HM catalyzed by alcohol and aldehyde dehydrogenases yields the carboxylic acid of tolterodine and its *N*-dealkylated form, along with *N*-dealkylated 5-HM. As described previously, CYP2D6 is subject to genetic polymorphism, with important implications for drugs that are metabolized by this enzyme such as tolterodine. Clinical studies have demonstrated that individuals with reduced CYP2D6-mediated metabolism represent a high-risk group in the population, with a propensity to develop adverse drug effects. In fast metabolizers, the mean systemic clearance of tolterodine was found to be 44 L/hour, yielding a half-life of 2 to 3 hours. In contrast, poor metabolizers have a five-fold lower clearance and a mean half-life of 9 hours, which results in a sevenfold higher maximum serum concentration of tolterodine at steady-state (171).

Irinotecan (also known as CPT-11 or Camptosar) is an approved topoisomerase I inhibitor used to treat patients with metastatic colon cancer. Acute and delayed diarrhea and neutropenia often occur after treatment with irinotecan. Diarrhea following treatment with irinotecan occurs due to the excretion of an active metabolite (SN-38, 10-hydroxy-7-ethylcamptothecin) initially into the bile and subsequently

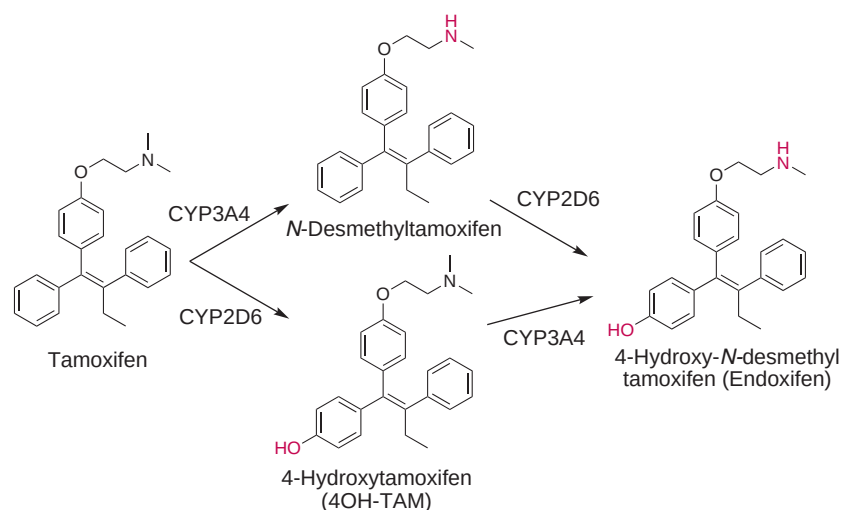


FIGURE 6.27 Tamoxifen metabolic pathway.

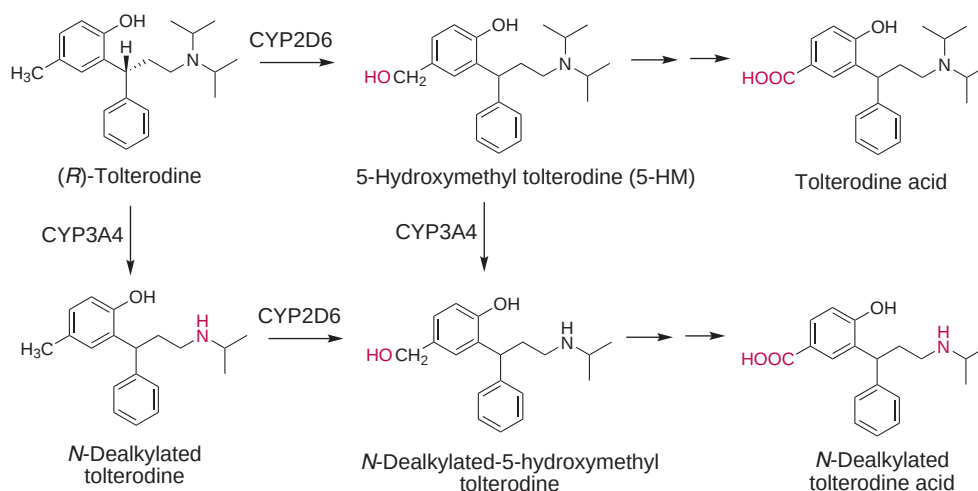


FIGURE 6.28 Tolterodine metabolic pathway.

the colon (Fig. 6.29; for more details, see Chapter 37 and Fig. 37.18). Irinotecan treatment is associated with an increased frequency of severe and potentially life-threatening toxicity among patients with genetic polymorphisms that markedly reduce glucuronidation of SN-38. Patients with decreased capacity to glucuronidate SN-38 (e.g., patients homozygous for specific UDP-glucuronosyl transferase 1A1 [UGT1A1] genotypes such as UGT1A1*28 [TA7]) are at increased risk for severe neutropenia after treatment with irinotecan or SN-38 than patients with wild-type sequence (172).

GENE THERAPY

The premise of gene therapy is that genes can (173) be used as pharmaceutical products to cause in vivo

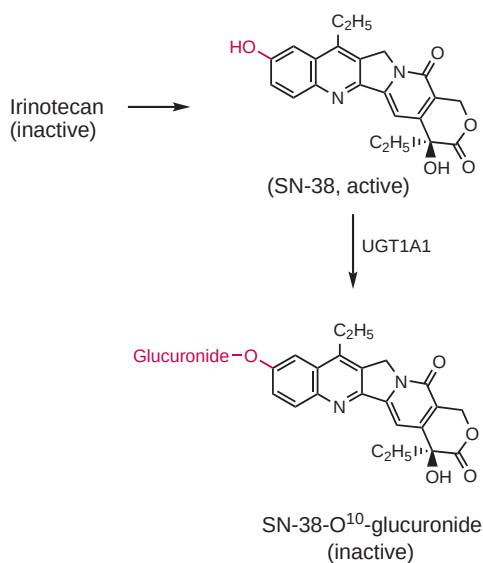


FIGURE 6.29 Metabolic activation of irinotecan.

production of therapeutic proteins. There is considerable confidence that gene therapy, as a therapeutic paradigm, will provide a number of novel pharmaceutical products, diagnostics, and therapeutic approaches in the years to come (173). A major problem in gene therapy is similar to the problem encountered in all forms of drug therapy—the assurance of drug efficacy through efficient delivery of the therapeutic agent to its biologic target in a fully functional form. Gene therapy is unique in the sense that it is the product of gene expression, the protein, and not the gene itself that is the therapeutic agent. Hence, we must not only deliver the gene to its proper target but also assure that when the gene reaches its target, it will arrive in a form that will produce the therapeutic agent in such a form that it, too, will be assured of reaching its specified target (174–176). A number of gene therapy products are under various phases of clinical investigation for different disease conditions including a severe combined immunodeficiency disease due to adenosine deaminase deficiency (CD34⁺ cells transduced with adenosine deaminase gene), glioblastoma (Toca511, a retroviral replicating vector), age-related macular degeneration (AAV2-sFLT01), Parkinson disease (CERE-120: adeno-associated virus delivery of neurturin), advanced osteocarcinoma (Her2 chimeric antigen receptor expressing T cells), CLL (autologous T lymphocytes engrafted with a chimeric antigen receptor targeting the kappa light chain), and Alzheimer disease (CERE-110: adeno-associated virus delivery of nerve growth factor). The steps (administration, delivery, and expression) involved in the delivery of the therapeutic gene are shown in Figure 6.30 (177). Because gene therapy has yet to reach the market, a detailed discussion on gene therapy and the ethical challenges it presents is beyond the scope of this chapter.

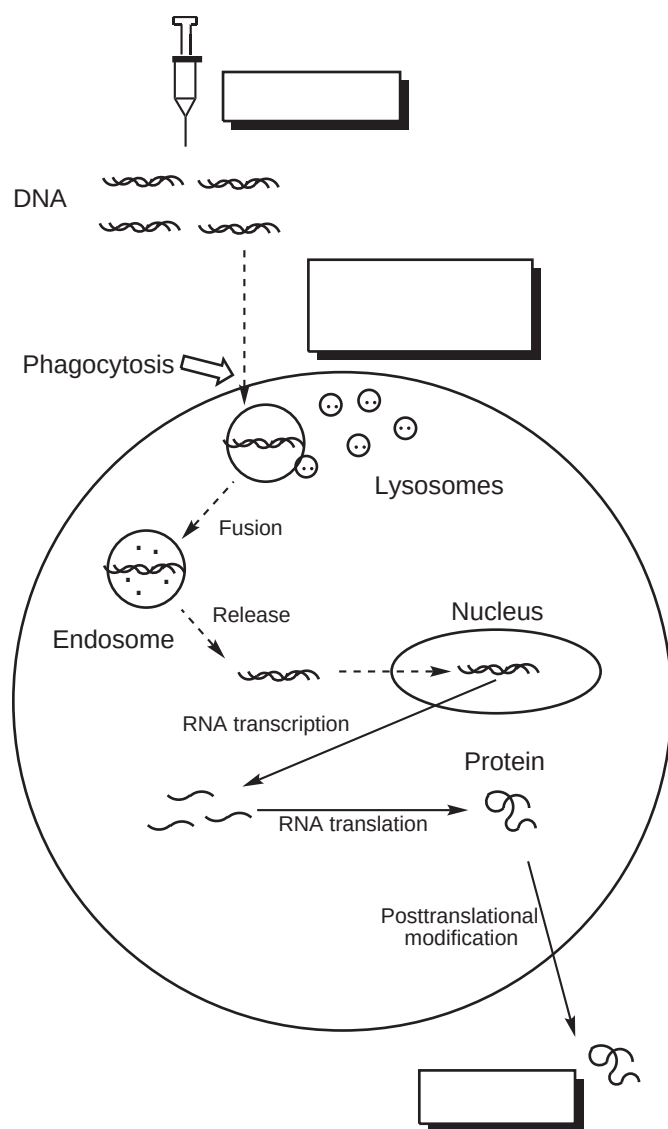


FIGURE 6.30 Steps in gene therapy. Gene therapy involves the steps of administration, delivery, and expression. (Note: dashed arrow represents uptake and transport).

SUMMARY

Completion of the Human Genome Project in 2003 resulted in the elucidation of the entire sequence of the 3 billion base pairs in the human genome, estimated to contain some 25,000 genes. The full impact that this scientific advance will have on our lives has yet to be

determined, but the social, legal, ethical, and economic issues are certain to be extensive and complex. The manipulation and analysis of the genomic information obtained from the Human Genome Project is paving the path for transformative biomedical developments. DNA-based tests are among the first commercial medical applications. Gene tests can be used to diagnose and confirm disease, provide prognostic information about the course of disease, and predict the risk of future disease in healthy individuals.

Knowledge of genes involved in diseases, disease pathways, and drug-response sites will result in the discovery of novel therapeutic targets beyond the 500 or so that were known prior to the completion of the Human Genome Project, providing a tremendous opportunity for the discovery of drugs that work through previously unexplored mechanisms.

Discovery of genomic biomarkers can help determine the type of drug therapy. For example, one of the most commonly used predictive biomarkers in cancer is immunohistochemical staining for the presence of the estrogen receptor in breast cancer. Only estrogen receptor-positive breast tumors are likely to respond to antihormonal therapy. Identification of biomarkers for various diseases can help in the selection of patients most likely to benefit from a potential drug, which in turn will speed the design of clinical trials that are more efficient and have the potential to reduce drug approval time and associated costs. Predicting toxicity is another important outcome of the availability of the genomic information. For example, patients who have mutations in the thiopurine S-methyltransferase gene will metabolize the chemotherapeutic mercaptopurine drugs at a reduced rate and can be overdosed on their treatment. Similarly, an SNP in the coding region of genes expressing drug-metabolizing CYP450 enzymes results in either poor (overdose) or fast (subtherapeutic dose) metabolism of drugs.

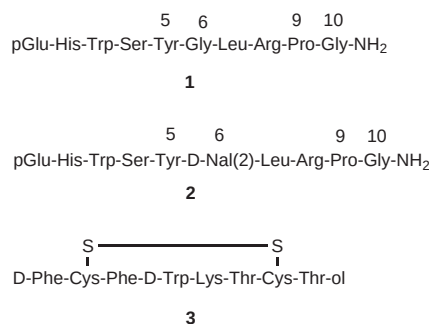
The future will see an altered form of health care using a genetic information infrastructure to contain costs and predict outcomes, create advanced personalized therapies, and develop a predict-and-manage paradigm of health care. The recent years have witnessed the introduction of a steady stream of biotechnology-derived drugs. The trend is expected to continue for the foreseeable future. The expiration of patents on biotechnology-derived drugs will introduce an era of more affordable biosimilars or follow-on biologics.

CASE STUDY

S. William Zito and Victoria Roche

TS is a 24-year-old white woman who is 8 months postpartum. She complains of excruciating back pain in the sacral area as well as left lower quadrant pain. The pain worsens about 1.5 weeks before her periods and continues during her menses. She gets only mild relief with nonsteroidal anti-inflammatory drugs. Since the delivery TS has been taking oral contraceptive pills; the pills regulate her cycles but do not seem to alleviate the pain. TS recounts that during her pregnancy she was pain-free for the first time in 5 years. The menstrual pain began when she was approximately 19 years old. On further questioning she notes occasional painful intercourse, especially just before menses. A pelvic examination reveals a retroverted uterus, which is tender to palpation. Cervix and vaginal mucosa are unremarkable. The Papanicolaou, *Neisseria gonorrhea*, and chlamydia tests were negative.

Because oral contraceptive pills do not alleviate the symptoms, TS's gynecologist suspects endometriosis and wants to prescribe a gonadotropin-releasing hormone superagonist to help control the symptoms while awaiting laparoscopy and biopsy for diagnostic confirmation. Which of the following peptides do you recommend?



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 7

Receptors as Targets for Drug Discovery

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Abbreviations

ATP, adenosine triphosphate

cAMP, cyclic adenosine
monophosphate

cGMP, cyclic guanosine
monophosphate

DAG, diacylglycerol

DAT, dopamine transporter

ERK, extracellular signal-regulated
kinase

GDP, guanosine diphosphate

GPCR, G protein–coupled receptors

Grb2, growth factor receptor-bound
protein 2

GSK-3, glycogen synthase kinase 3

GTP, guanosine triphosphate

HTS, high-throughput screening

IP₃, inositol 1,4,5-triphosphate

JAK, Janus kinase

LGIC, ligand-gated ion channel

MAPK, mitogen-activated protein
kinase

MEK, MAPK kinase

PLC, phospholipase C

STAT, signal transducer and activator
of transcription

INTRODUCTION

The human body is an example of an exquisitely designed, extremely complex machine that functions day-in and day-out to allow for survival of the organism in response to a never-ending onslaught of external challenges. When one considers the enormous variety of environmental stressors to which the body is continually subjected, it is not surprising to anticipate the existence of a multitude of checks and balances associated with its physiologic and biochemical systems. These systems, including endocrine, nervous, and enzymatic, typically function in concert to adapt to changing environmental conditions. Such systems are designed to respond quickly (i.e., within milliseconds) and for a short time; others are designed to act more slowly but usually have significantly longer durations ranging from minutes to hours to days

and, in some case, even months to years. Together, these systems support the organism's survival. Malfunctioning of the control of such systems often leads to disease and, potentially, the eventual demise of the individual.

The use of specific chemical compounds to treat disease dates back to early humans. Many primitive cultures used plants and other natural sources in an attempt to mitigate the influences of evil spirits and other factors rooted in superstition, which were believed to be the foundations of such illnesses. Over the centuries, a number of serendipitous observations involving the ability of largely botanical preparations to alter disease processes laid the foundation for the modern, more systematic approach to the discovery of medicinals for therapeutic use. The collaboration of chemical and biological scientists continues this quest for the “magic bullet” to treat those diseases that challenge the individual's well-being.

HISTORICAL PERSPECTIVES

For years, it had been known that some drugs were capable of producing their effects by acting at specific sites within the body. Claude Bernard was the first to demonstrate this in the mid-1800s, with his classical experiments involving curare (1). He showed that this neuromuscular blocking agent, which was used as an arrow poison by the South American natives, was capable of preventing skeletal muscle contraction following nerve stimulation, but was without effect when the muscle was stimulated directly. This work demonstrated for the first time a localized site of action for a drug and, most importantly, suggested that a gap, or synapse, existed between the nerve and the muscle. From these findings, he postulated that some chemical substance normally communicated the information between the nerve and the target tissue—in this case, the muscle. These findings established the foundations for what is known today as “chemical neurotransmission,” a process frequently disrupted by diseases and, likewise, the target of many therapeutic agents.

Investigations by J.N. Langley (2) in the early 1900s established the initial foundations for the interaction of drugs with specific cellular components, later to be identified and termed “receptors.” Before this time, many leading experts believed that most drugs acted nonselectively on virtually all the cells in the body to produce their biologic responses, with a response resulting from their general physical characteristics (e.g., lipid solubility) and not related to specific structural features of the compound. Langley noted that the natural product pilocarpine, which act to mimic the parasympathetic division of the autonomic nervous system, were very selective and also extremely potent. Additionally, the natural product atropine was capable of blocking, in a rather selective fashion, the effects of pilocarpine and parasympathetic nervous system stimulation. Importantly, he concluded that these two compounds interacted with the same component of the cell.

Paul Ehrlich (3), a noted microbiologist during the late 19th and early 20th centuries, is credited with coining the term “receptive substance,” or “receptor.” His observations that various organic compounds appeared to produce their antimicrobial effects with a high degree of selectivity led him to speculate that drugs produced their effects by binding to such a receptive substance. The interaction or binding of the drug with the receptor was analogous to a “lock” (the receptor) and a “key” (the drug), which gave rise to the lock and key fit theory for drug receptors. Thus, certain organic compounds would properly fit into the receptor and activate it, leading to a high degree of specificity. Although such a situation might be considered to be ideal for drug therapy, few drugs actually interact only with their intended receptors. The frequency of side effects is not associated with a simple extension of their desired pharmacologic actions; instead, drug molecules can also bind with other receptors or nonreceptor entities on or within cells to produce a host of other—and often undesirable—effects.

Some drugs produce their desired therapeutic effects without interaction with a specific receptor. For instance, osmotic diuretics produce their pharmacologic effects simply by creating an osmotic gradient in the renal tubules and, thereby, promoting the elimination of water in the urine. This is purely the result of a physical characteristic of the drug. Similarly, antacids produce their beneficial effects by chemically neutralizing the hydrochloric acid found in the stomach. No absorption of the drug is even required for its effects to be realized. Often, the failure of a drug to be absorbed and, thus, only act locally at the desired biologic site constitutes a tremendous advantage regarding the safety of that compound. Unfortunately, from a practical standpoint, most pharmacologic agents require absorption following administration to reach the intended target; thus, side effects typically are a serious consideration.

More sophisticated mechanisms can also be involved in the nonreceptor actions of therapeutic agents. For instance, the antineoplastic agent mechlorethamine, a nitrogen mustard, produces its beneficial (pharmacologic) and adverse (toxicologic) effects via interaction with many cellular components in both cancerous and normal cells. Via its conversion to a highly reactive electrophilic ethyleniminium ion intermediate, this agent reacts with nucleophilic cellular components, such as amino, hydroxyl, sulfhydryl, phosphate, carboxyl, and imidazole groups. In particular, by alkylating the N-8 position of guanine in DNA, this agent produces miscoding (cytosine normally base pairs with guanine in DNA; however, thymine now substitutes for cytosine) and the eventual death of the cell (4). When one realizes that all replicating cells contain an N-8 nitrogen in guanine in their DNA, it is easy to see why mechlorethamine produces nonselective destruction of cells throughout the body. Thus, no specific receptor is involved in the actions of this class of pharmacologic agent.

DRUG DISCOVERY

The discovery of pharmacologic agents by modern pharmaceutical companies and universities often involves the use of receptor–ligand binding techniques. Following the synthesis of a series of new chemically related compounds, which can constitute hundreds to thousands of compounds, the determination of the desired biologic activity was once a rather daunting task. Before the advent of receptor–ligand binding techniques, the initial screening of these compounds involved injecting each agent individually into experimental animals or incubating each agent with isolated tissues (e.g., intestine, heart, and skeletal muscle), which are techniques that require a large investment of resources, including personnel, time, animals, and money. Today, receptor–ligand binding techniques, such as high-throughput screening (HTS), are used to narrow large numbers of compounds down to those that display the greatest

affinity for a receptor, thereby significantly decreasing the time and cost associated with identifying “lead” compounds (5). Initial *in silico* analytical techniques use molecular modeling software to “virtually screen” trillions of candidate structures found in “chemical libraries” for the potential to bind to target proteins or other molecules of interest. Once the structural features of candidate molecules are identified, HTS techniques are employed that combine computer-controlled robotics, detectors, and data processing to identify compounds that interact with target molecules or biochemical pathways. Typically, HTS uses microtiter plates that contain from 96 to 3,456 wells that hold volumes of liquid as small as nanoliters (Fig. 7.1). The liquid in the wells contains substrates or cells of interest to which the chemical entities are added. Automated analytical machines then detect reactions and identify “hits” that represent compounds that will undergo further testing. The most current HTS processes can screen 100 million reactions in 10 hours. “Hits” are confirmed by retesting initially using identical assay conditions. Once confirmed, dose-response curves are generated using a range of concentrations of the compound. Next, the compounds undergo further testing using alternative assays and experimental conditions. The original “hit” compound can be structurally modified to optimize activity, potency, and stability using approaches such as combinatorial chemistry to create a “lead” or drug candidate. Converting a “hit” to a “lead” is the most important and difficult part of drug discovery. However, a danger associated with this initial screening approach is the failure to recognize potentially useful compounds that might require biotransformation before exerting a biologic effect, such as a prodrug. It should be remembered that ligand binding based on the affinity of a drug for its receptor does not differentiate agonists from antagonists. Despite these potential pitfalls associated with receptor–ligand binding techniques, modern drug discovery relies heavily on these approaches.



FIGURE 7.1 A pipetting robot system for fully automated high-throughput screening arrays.

Another contemporary approach to drug development is “rational drug design,” which is a process for discovering new drug entities through detailed knowledge of the structure of the biologic target. For example, the elucidation of the three-dimensional crystal structure of the bacterial leucine transporter (Fig. 7.2) provided a template to serve as a model for analogous mammalian monoamine transporters such as the dopamine transporter (DAT) (6). With knowledge of the amino acid sequences for both the leucine transporter and DAT, it was possible using computer modeling and site-directed mutagenesis techniques to discover chemical bonding interactions between key amino acids in the binding site of DAT and substrates such as dopamine and amphetamine. With this information, it becomes possible to design “leads” for ligands that could serve as either substrates or inhibitors of DAT. Drugs developed through this process can then be effective in treating a number of diseases such as Parkinson disease, attention-deficit disorders, and drug addiction.

AFFINITY: THE ROLE OF CHEMICAL BONDING

During the early 1900s, A.V. Hill used the natural products nicotine and curare in isolated muscle preparations and noted the effects of temperature in his experiments (1). He concluded that the ability of a drug to produce an effect must result from specific chemical interactions between the drug and specific sites. He also noted that the effects of many drugs were reversible, because washing the isolated tissue often restored the sensitivity of the tissue to nerve stimulation. These studies set the foundation for our understanding of the chemical interactions between drugs and receptors.

When a drug interacts with a receptor, multiple chemical interactive forces, both weak and strong, between two molecules (a drug and its receptor) are believed to be responsible for the initial interaction. Such forces having a role in ligand–receptor binding include covalent, ionic, and hydrogen bonds and hydrophobic interactions as shown in Fig. 7.3. Compounds attracted to a receptor macromolecule are said to have affinity for that receptor and can be classified as agonists or antagonists. Compounds with affinity for a receptor are referred to as ligands. An agonist is a compound that has affinity for its receptor and thus is capable of producing a positive biologic response as a result of its interaction with its receptor (7). The ability to produce a response is termed “efficacy” or “intrinsic activity.” On the other hand, antagonists are compounds with “affinity” for a receptor but do not activate the receptor to produce a biologic response. Antagonists are said to have affinity, but lack intrinsic activity. The affinity of a compound for a receptor is dependent on its correct three-dimensional characteristics, such as its size, stereochemical orientation of its functional groups, and its physical and electrochemical properties (e.g., ionic and dipole interactions).

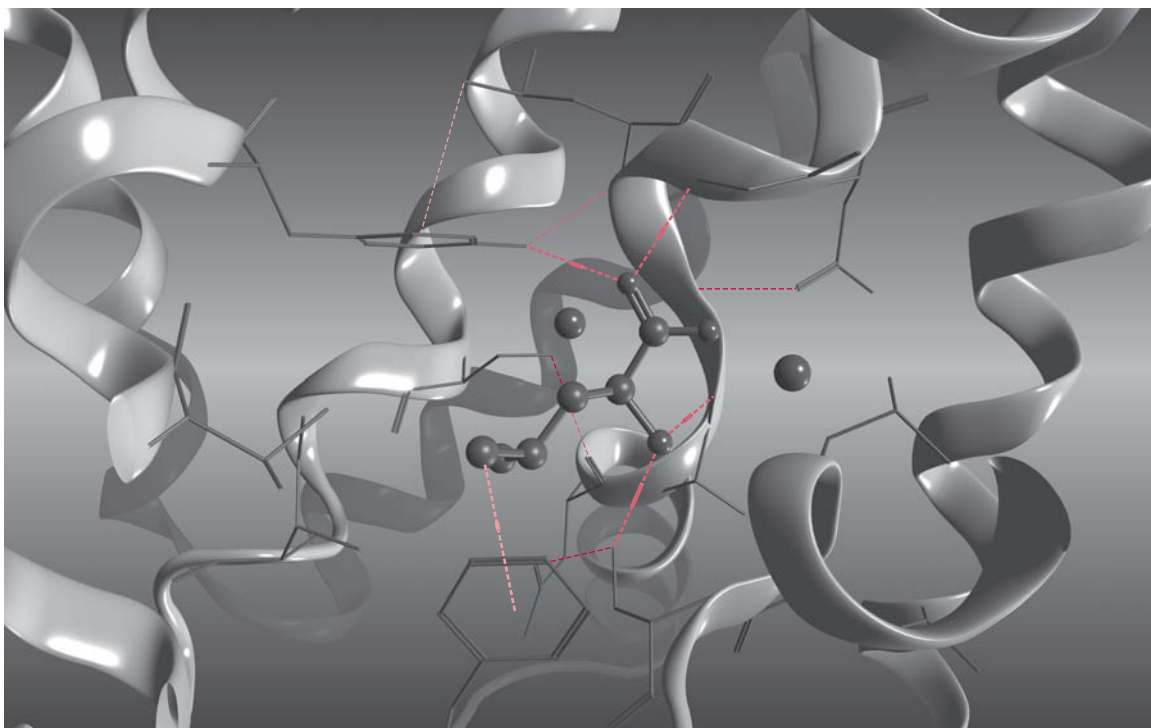


FIGURE 7.2 Molecular recognition of the bacterial leucine transporter (LeuT, pdb:3GWW) with co-crystallized leucine substrate (shown as ball-and-stick). Side-chain residues within 4.5 Å (line rendering) of the leucine molecule are shown, along with two Na⁺ ions (black spheres), which are required for transport. Hydrogen bonding and hydrophobic interactions between LeuT and substrate are shown as dotted lines, respectively. Figure rendered using Molecular Operating Environment (MOE) software (CCG, Montreal, Quebec, Canada). (From Zhou Z, Zhen J, Karpowich NK, et al. Antidepressant specificity of serotonin transporter suggested by three LeuT-SSRI structures. *Nat Struct Mol Biol* 2009;16:652–657.)

Assuming that a compound has been distributed to the general vicinity of a receptor, based on its physical characteristics, the binding of the drug to the receptor will then initially depend on the types of chemical bonds that can be established between the drug and its receptor. The overall strengths of these bonds will vary (Fig. 7.3) and will determine the degree of affinity between the drug and its receptor.

Covalent Bond

The strongest of bonds involved in drug–receptor interactions is the covalent bond, in which two atoms, one from the ligand and one from the receptor, share a pair of electrons to form a covalent bond. Because of the significant strength of the covalent bond (50 to 150 kcal/mol), covalent bonding often produces a situation in which the ligand is irreversibly bound by the receptor and, thus, leads to the receptor's eventual destruction via endocytosis and chemical destruction. Full recovery of cellular function therefore requires the synthesis of new receptors.

An example of an irreversible covalent bond formation between drug and its receptor involves the long-lasting blockade of α -adrenoceptors by phenoxybenzamine (a haloalkylamine) (see Chapter 10). Phenoxybenzamine

under physiologic pH forms a highly reactive carbo-nium ion intermediate that covalently links by reaction (alkylation) with amino, sulfhydryl, or carboxyl groups to the α -adrenoceptor. The receptor is thus rendered irreversibly nonfunctional and, eventually, destroyed. The synthesis of a new receptor requires a number of days, thus accounting for the extremely prolonged duration of the inhibition associated with this agent. This property of phenoxybenzamine to irreversibly bind the α -adrenoceptor by a covalent bond is critical for the

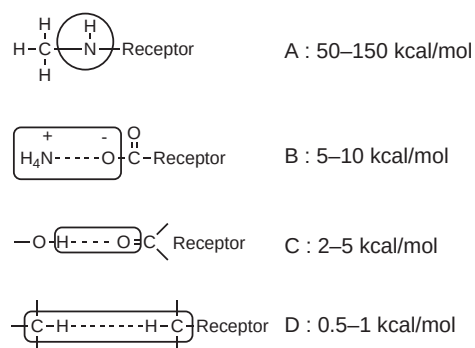


FIGURE 7.3 Various drug–receptor bonds. (A) Covalent. (B) Ionic. (C) Hydrogen. (D) Hydrophobic.

demonstration of spare receptors (8,9). Because other receptors and cellular components also contain molecular groups that are capable of reacting with the bioactivated (reactive) phenoxybenzamine intermediate, it is not surprising to find that receptors that mediate the actions of other neurotransmitters (such as acetylcholine, serotonin, or histamine) are also subject to alkylation and inhibition, demonstrating the lack of selectivity of phenoxybenzamine.

Another important example of a class of compounds that produces its effects via a covalent bond to its receptor is the organophosphate acetylcholinesterase inhibitors (see also Chapter 9). Examples of such agents include the insecticides parathion and malathion and the nerve-gas agents sarin, soman, and tabun. These type of inhibitors are capable of reacting (phosphorylating or forming a covalent phosphate bond) with the active site of acetylcholinesterase, which normally is responsible for metabolizing acetylcholine, the neurotransmitter found at the neuromuscular junction and within many sites of the autonomic and central nervous systems. Reaction of the enzyme with its normal substrate acetylcholine leads to a readily hydrolyzable acetylated enzyme, which rapidly regenerates to the active enzyme. Covalent bonding by the organophosphates results in phosphorylation of a serine residue within the active site of the enzyme, which is extremely stable and essentially irreversible. Recovery of enzymatic function in the tissue requires the synthesis of new enzyme molecules.

Ionic Bond

When two ions of opposite charge are attracted to each other through electrostatic forces, an ionic bond is formed. The strength of this type of bond varies between 5 and 10 kcal/mol, and it decreases proportionally to the square of the distance between the two atoms. The ability of a drug to bind to a receptor via ionic interactions therefore increases significantly as the drug molecule diffuses closer to the receptor. Additionally, the strength associated with the ionic bond is strong enough to support an initial transient interaction between the receptor and the drug, but unlike the covalent bond, the ionic bond is not so strong as to prevent dissociation of the drug–receptor complex.

The tendency of an atom to participate in ionic bonding is determined by its degree of electronegativity. Hydrogen, as a standard, has an electronegativity value of 2.2; fluorine is 4.2, chlorine 2.9 and nitrogen 3.1 (Linus Pauling units). Fluorine and chlorine atoms, as well as hydroxyl, sulfhydryl, and carboxyl groups, form strong ionic bonds because of a stronger attraction for electrons compared with that of hydrogen. On the other hand, alkyl groups do not participate in ionic bonds because of a weaker tendency to attract electrons compared with that of hydrogen.

Hydrogen Bond

A hydrogen bond (or hydrogen bonding) is a strong electrostatic dipole–dipole interaction between a hydrogen

atom and an electronegative atom, such as oxygen, nitrogen, or fluorine. The hydrogen bond is extremely strong because oxygen, nitrogen, and fluorine are extremely good at attracting the relative positive charge of hydrogen, resulting in an extreme dipole situation. This type of bond can occur between molecules (intermolecular hydrogen bonds) or within the same molecule (intramolecular hydrogen bonds). At 2 to 5 kcal/mol, a single hydrogen bond is stronger than a van der Waals interaction, but weaker than covalent or ionic bonds, and thus would not be expected to support a drug–receptor interaction alone. However, when multiple intermolecular hydrogen bonds are formed between drugs and receptors, as typically is the case, a substantial amount of stability is conferred on the drug–receptor interaction, an essential requirement for drug–receptor interactions. For example, a water molecule behaves as an electronic dipole and can easily form intermolecular hydrogen bonds with other water molecules, which gives water its high boiling point of 100°C. Intramolecular hydrogen bonding is partly responsible for the secondary, tertiary, and quaternary structures of proteins and nucleic acids.

Hydrophobic Interactions

Hydrophobic interactions (hydrophobic effect; fear of water) are intermolecular interactions or dispersion forces that occur between nonpolar organic molecules and contribute to the binding forces that attract a ligand to its receptor, other than ionic, covalent, or hydrogen bonds. These interactions are often referred to as van der Waals forces or London forces, which require two nonpolar molecules to come in close range to one another, or between groups within the same molecule. Dispersion forces (or London dispersion forces) are induced dipole–dipole electrostatic interactions between atoms/molecules at close distances and occur over a large surface area (i.e., at the interface of the ligand and binding site) and thus contribute to receptor binding. London forces are weaker than van der Waals forces. These forces tend to align the atoms/molecules in order to increase their interaction, thereby reducing their potential energy. Theorists have suggested that for these forces to operate, a momentary dipolar structure needs to exist to allow such association. This induced dipolar interaction results from a temporary imbalance of charge distribution between or within molecules. These forces are very weak (0.5 to 1 kcal/mol) and decrease proportionally to the seventh power of the interatomic distance. The hydrophobic effect is the ability of polar water molecules to exclude (repel) nonpolar hydrocarbon-like molecules.

AFFINITY: THE ROLE OF CONFORMATION

Most therapeutically useful drugs bind only transiently and reversibly to their intended receptor. The combination of a variety of bonds, including ionic, hydrogen, and van der Waals attractive forces, collectively

contributes to the strength of drug binding to the receptor. The functional groups (critical portion of the structure) of the drug that bind to the receptor are termed “pharmacophores” or “pharmacophoric groups.” Once the drug has bound, a biologic response may result (e.g., especially if the drug is an agonist). Either following or during the process of binding to the receptor, a conformational change can occur in the receptor that initiates the activation of the biologic response and increases the affinity/binding between the drug and its receptor. This conformational change in the receptor can also contribute to the dissociation of the drug–receptor complex. This simple explanation of the interaction between a drug and its receptor producing a biologic response is commonly referred to as the “occupancy theory,” which predicts that the biologic response is directly related to the number of receptors bound by an agonist.

Another theory of drug–receptor interactions is the “rate theory,” which suggests that the number of drug–receptor interactions per unit time determines the intensity of the response. Thus, drugs that associate with and then rapidly dissociate from the receptor, thus allowing other drug molecules to subsequently interact with the receptor, would be expected to produce the most robust responses. The “induced-fit theory” suggests that as the drug (agonist) approaches the inactive state of the receptor (conformation A), the agonist induces a specific conformational change (perturbation) in the inactive receptor, which allows for effective drug–receptor binding or formation of the drug–receptor complex (conformation B) and the desired biologic response (Fig. 7.4). According to this theory, the receptor exists in an inactive state/configuration (conformation A) without the proper conformation to form the drug–receptor complex. Following dissociation of the drug–receptor complex (conformation B), the receptor then reverts to its original or inactive state (conformation A). According to this theory, an antagonist, on the other hand, induces a nonspecific conformational change in the receptor (conformation C), which fails to produce the desired drug–receptor complex; thus, no biologic response is elicited. Combining the induced-fit and rate theories yields the “macromolecular perturbation theory,” which suggests that two types of conformational changes exist and that the rate of their existence determines the observed biologic response. This theory can partially account for the activity of partial agonists. Finally, the “activation-aggregation theory” indicates that receptors are always in a dynamic equilibrium between active and inactive states. Agonists function by shifting the equilibrium toward the activated state, whereas antagonists prevent the activated state. This theory can account for the activity of inverse agonists, which produce neither a typical agonist response nor an antagonist response (i.e., blocking the receptor) but, rather, produce biologic responses opposite to those of the agonist.

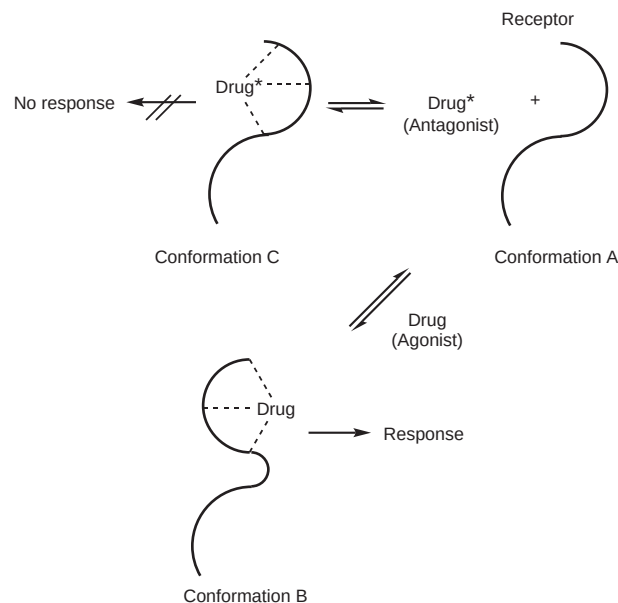


FIGURE 7.4 Diagrammatic representation of drug-induced fit theory, in which an agonist (Drug) or antagonist (Drug*) interacts with three different conformations of the receptor.

AFFINITY: THE ROLE OF STEREOCHEMISTRY

Very specific three-dimensional requirements must be satisfied for a compound to effectively act as an agonist. To elegantly demonstrate the specificity of a drug for its receptor, the unique three-dimensional characteristics of chiral compounds can be used as an example. As early as 1901, Pasteur (10) noted the significance of asymmetric compounds in biologic systems. Since that time, much has been learned from chiral compounds regarding three-dimensional binding requirements of receptors. For instance, although the individual enantiomers (i.e., nonsuperimposable mirror images) of norephedrine (2-amino-3-phenyl-1-propanol) (Fig. 7.5) have identical molecular weights, melting points, lipid solubility, and empirical formulae, these compounds have significantly different α -adrenoceptor agonistic activities. The 1*R*,2*S* enantiomer (levo) is approximately 100-fold more potent than the 1*S*,2*R* enantiomer (dextro) both in vivo and

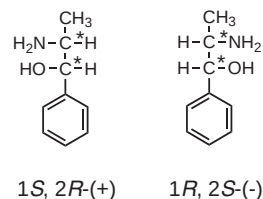


FIGURE 7.5 Projection formulae of 2-amino-3-phenyl-1-propanol stereoisomers. (From Maher TJ, Johnson DA. Review of chirality and its importance in pharmacology. *Drug Dev Res* 1991;24:149–156, with permission.)

in vitro (11,12). (Because there are two chiral centers, there is also another set of stereoisomers called diastereomers, 1*S*,2*S* and 1*R*,2*R*, that have an even different pharmacologic profile.) Thus, the greater efficacy of the 1*R*,2*S* enantiomer most likely is dependent on its ability to bind and activate the receptor as a result of its preferential fit into the receptor.

Labetalol, an adrenoceptor-blocking agent structurally related to epinephrine, has two asymmetric centers, and therefore, four diastereomers exist (Fig. 7.6). The formulation available for use as a mixed α - and β -adrenoceptor blocker contains equal amounts of each diastereomer. The *R,R*-isomer accounts for much of the β -adrenoceptor blocking activity, whereas the *S,R*-isomer has the greatest effect on α -adrenoceptors. The *S,S*-isomer has some α -adrenoceptor blocking activity but no activity at β -adrenoceptors. The *R,S*-isomer is essentially devoid of activity at both α - and β -adrenoceptors.

Many synthetically prepared therapeutic agents are a mixture of two enantiomers (racemates), with one enantiomer, termed the “eutomer,” being largely responsible for the desired pharmacologic effect (13). The other enantiomer, termed the “distomer,” can be inactive or even contribute more significantly to the toxicity of the therapeutic agent. For example, the bronchodilator albuterol is available as equimolar mixtures of *R*- and *S*-enantiomers. *R*-Albuterol binds to the β_2 -adrenergic receptor with high affinity to produce bronchodilation, whereas its *S*-enantiomer exhibits weak binding to the β_2 -adrenergic receptor and induces airway bronchoconstriction (toxicity).

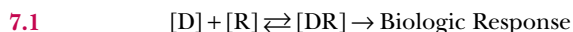
Thus, knowledge of chirality and stereochemistry of drug action has a significant role in drug development due to the advances gained in receptor theory as well as in therapeutics. It has been generally accepted in drug discovery that the development of enantiomerically pure drugs could provide safer and more efficacious alternatives to the racemate, when the beneficial effects are attributable to receptor selectivity by one of the enantiomers. An understanding of the role of chirality in the properties of the drug molecule is important to the rational process of developing chiral drugs as well as the

clinical use of chiral drugs. The pharmacokinetic properties of a racemate can be very different from the single enantiomer, with differences in bioavailability as a result of stereoselective metabolism, transport, or elimination. Stereochemical aspects of drug action have been known for many decades; however, only within the last couple of decades has emphasis been placed on stereochemistry of drug disposition.

DOSE-RESPONSE RELATIONSHIPS

A.J. Clark is generally given credit for being the first to apply the law of mass action principles to the concept of drug–receptor interactions, thus providing further evidence for the dose–effect phenomenon (14). This concept, as applied by Clark, states that the greater the number of agonist molecules at the site of the receptors, the greater the response (i.e., a direct relationship); however, these principles of the law of mass action in drug–receptor interactions have been questioned. The law of mass action applies to compounds dissolved in fluids that are allowed to freely diffuse. Since much is known about the anchoring of most receptors to, or within, cell membranes where drug–receptor interactions are thought to occur, this environment would actually constitute a solid–liquid interface. Thus, the law of mass action as applied to compounds dissolved in fluids where they are allowed to freely diffuse might not be completely applicable.

Equation 7.1 illustrates the interaction of a drug ([D]) with a receptor ([R]), which results in a drug–receptor complex ([DR]) and a biologic response. The interaction between most therapeutically useful drugs and their receptors is generally reversible:



After administration of a drug, one can monitor the biologic responses produced. Plotting the dose or concentration of the drug versus the effect produced (% response) yields a rectangular hyperbolic function, as illustrated in Figure 7.7A. This type of function is mathematically difficult from which to accurately extrapolate quantitative information due to the constantly changing slope of the curve. When the effect produced is plotted against the log of the drug concentration or the dose administered, however, a sigmoidal function results, as illustrated in Figure 7.7B. This function possesses a relatively linear portion of the curve about its central point, thereby making quantitative extrapolations more accurate.

Dose–response curves are typically plotted to determine both quantitative and qualitative parameters of potency and efficacy. Potency is inversely related to the dose required to produce a given response (typically half-maximum), and efficacy is the ability of a drug to produce a full response (100% maximum). In Figure 7.8, drug X is equally efficacious to drug Y, but drug X is

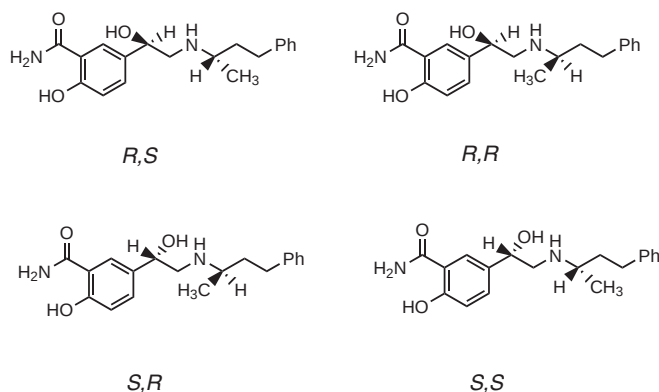


FIGURE 7.6 Diastereomers of labetalol.

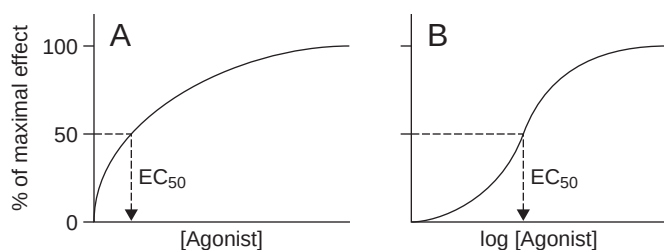


FIGURE 7.7 Plot of (A) dose or concentration and (B) log of the dose or concentration of a drug versus the effect produced.

more potent than drug Y. That is to say, both drug X and drug Y produce a 100% response, but drug X reaches that response at a lower dose. Visual inspection of such dose-response curves allows easy qualitative interpretations (e.g., in a series of curves). Those curves positioned to the left are more potent than those positioned to the right, as illustrated in Figure 7.8, drug Z is more potent than drug Y, and drug Z is equipotent to drug X. However, comparisons of efficacy are visually apparent, because the greater the maximum response (i.e., efficacy), the higher the maximum point on the dose-response curve. Thus, in Figure 7.8, drug X and drug Y are of equal efficacy, and drug X and drug Y are of greater efficacy than drug Z.

The potency of an agonist is decreased in situations where the agonist must compete with an antagonist for binding at the same site (competitive antagonism). The potency of the agonist is decreased as the concentration of antagonist is increased. Graphically, this is demonstrated as a parallel shift to the right in the dose-response curve with no decrease in maximal response (Fig. 7.9A). An antagonist can also inhibit activation of a receptor by binding to a separate allosteric site (noncompetitive antagonism). In this case, by binding to the allosteric site, the antagonist prevents the conformational change in receptor tertiary structure to the active state even with agonist binding to the primary site. Graphically, in the presence of a noncompetitive binding, the dose-response curve is shifted to the right, and the maximal response is reduced (Fig. 7.9B).

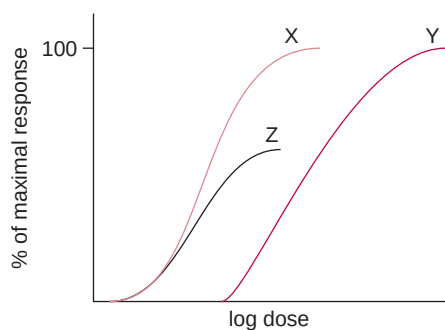


FIGURE 7.8 Dose-response relationship.

PRESYNAPTIC AND POSTSYNAPTIC RECEPTOR LOCATIONS

When an action potential arrives at the nerve cell's axon, a depolarization-induced exocytosis of the neurotransmitter from its storage sites in the presynaptic terminal occurs. Through this process, the action potential continues the flow of information to the target site, the postsynaptic cell. The neurotransmitter diffuses across the extracellular fluid-filled space known as the synapse to interact with its postsynaptic receptors. The released neurotransmitter, however, is also capable of interacting with presynaptic receptors located on the neurons that just released the neurotransmitter. The function of these receptors typically involves the regulation of nerve transmission, and they are termed "autoreceptors," because the neurotransmitters that activate them function to control their own release.

An exquisite example of both receptor locations and the action of autoreceptors in the control of neurotransmission is observed in norepinephrine-containing postganglionic neurons of the sympathetic nervous system (Fig. 7.10) (15). Norepinephrine, which is capable of stimulating both α - and β -adrenoceptors, is initially released from the presynaptic neuron and is present at low concentration in the synapse. Low concentrations of norepinephrine are capable of preferentially stimulating β -adrenoceptors located presynaptically, which function to increase the release of more neurotransmitter and, thereby, magnify the intended response. The epinephrine released from the adrenal medulla during sympathetic stimulation is also thought to have an important role in facilitating neuronal norepinephrine release. This is an example of a positive-feedback system, which allows a rapid rise in the concentration of the neurotransmitter and, thus, the intended signal. Following this initial period of robust norepinephrine release, very high norepinephrine concentrations result in the synaptic cleft, which is then capable of stimulating other presynaptic autoreceptors, this time terminating the additional release of neurotransmitter. This negative-feedback system allows the signal to be terminated very quickly.

Together, the presynaptic facilitatory β -adrenoceptor-mediated mechanism and the presynaptic inhibitory α -adrenoceptor-mediated mechanism allow a rapid, robust, and well-controlled signal to be delivered. If one were to design a system that was to respond quickly to stressors, such as the sympathetic nervous system is believed to be designed to do, then a system that turns on rapidly and can be terminated quickly would be ideal and, presumably, an evolutionary advantage. Many other neurotransmitter autoreceptors have been identified, such as in the serotonergic, dopaminergic, and histaminergic transmitter systems. Examples exist whereby a neurotransmitter can interact with a presynaptic receptor to influence the release of a

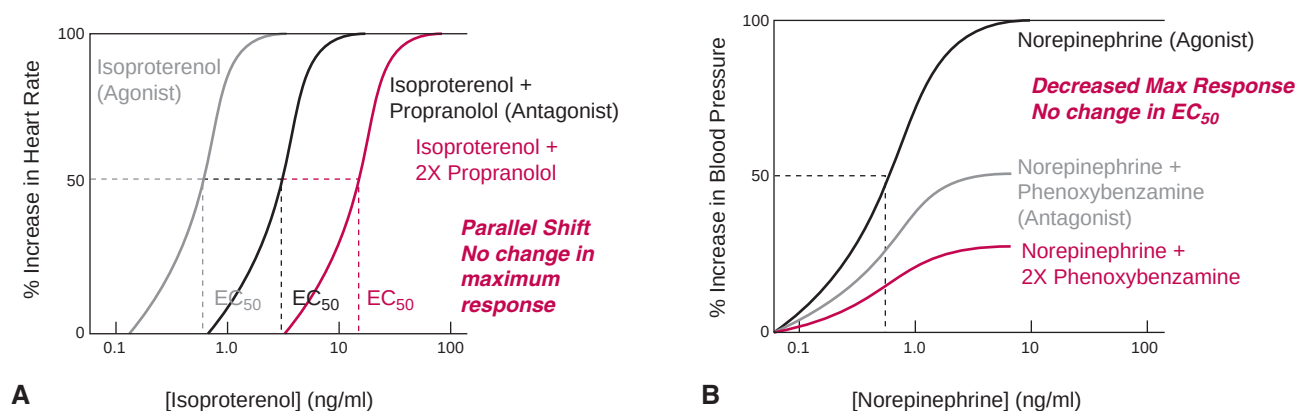


FIGURE 7.9 Dose-response curves for the effect of the agonist isoproterenol on heart rate in: (A) the presence of two concentrations of the competitive antagonist propranolol. Note the rightward shift of the dose-response curve with no change in maximum effect; and dose-response curves for the effect of the agonist norepinephrine on blood pressure (B) in the presence of two concentrations of the noncompetitive antagonist phenoxybenzamine. Note the decrease in maximum response and the absence of a rightward shift of the dose-response curves. (From Clarkson CW. Basic principles of pharmacology. Available at: http://tmedweb.tulane.edu/pharmwiki/doku.php/drug_receptor_theory.)

different neurotransmitter. For instance, norepinephrine released from neurons in the gastrointestinal tract can function to decrease acetylcholine release. This is termed a heteroreceptor.

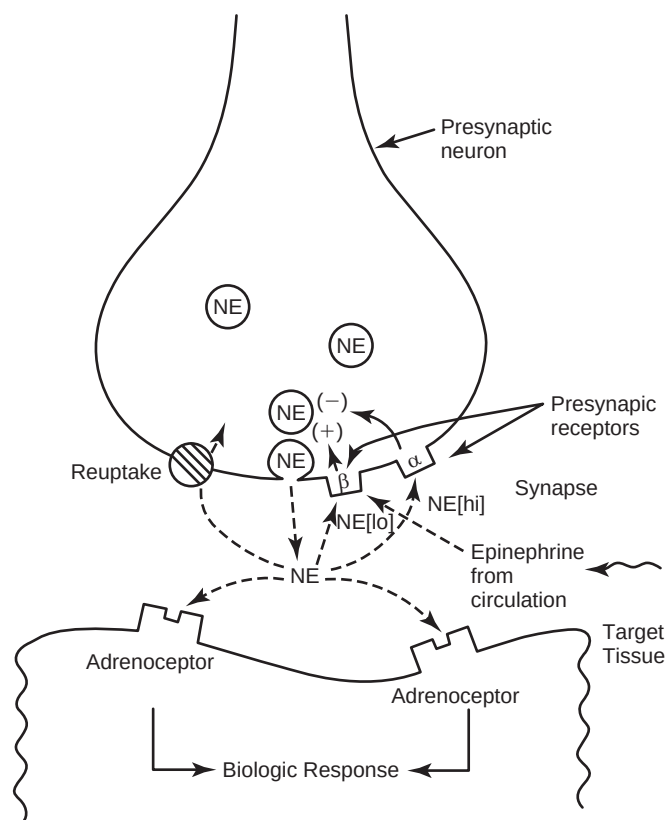


FIGURE 7.10 Autoreceptor control of neurotransmission as observed in norepinephrine-containing postganglionic neurons of the sympathetic nervous system. NE, norepinephrine; NE[lo], low norepinephrine concentration; NE[hi], high norepinephrine concentration; α, α-adrenoceptor; β, β-adrenoceptor; (-), inhibit; (+), stimulate.

DRUG RECEPTORS AND THE BIOLOGIC RESPONSE

There are four major families of receptors that drugs, which mimic, modify, or antagonize endogenous neurotransmitters, hormones, or autoids, are capable of interacting with in the body (Fig. 7.11). Some receptors allow a rapid response to a released neurotransmitter/hormone or an administered drug. These responses are generally important for the immediate response to a significant homeostatic challenge to the individual. Both ion channel and G protein-coupled receptors tend to be rapid responders, with some catalytic receptors also characterized this way. These responses tend to be short-lived. On the other hand, many catalytic and just about all cytoplasmic/nuclear receptors are likely to respond much more slowly, on the order of hours to days, and these responses are much longer in duration than the rapid-responding receptors.

Signal Transduction

Signal transduction is a communication process by which a cell converts an extracellular signal or stimulus by transmitting this signal across the cell membrane to the interior of the cell. Proteins on the cell's extracellular surface function as receptors for specific molecules, ligands, or agonists (first messengers). The binding of the ligand to the receptor initiates an interlinked series of processes, termed "signal transduction," that involve a sequence of biochemical reactions inside the cell, which are carried out by enzymes, proteins, and ions (especially calcium) that are linked through second messengers, such as cyclic adenosine monophosphate (cAMP) or inositol 1,4,5-trisphosphate (IP₃). This signal is relayed via a second messenger that results in specific cellular responses or changes in gene expression in the nucleus as shown in Figure 7.12. Such processes take place in as little as a millisecond or as long as a few

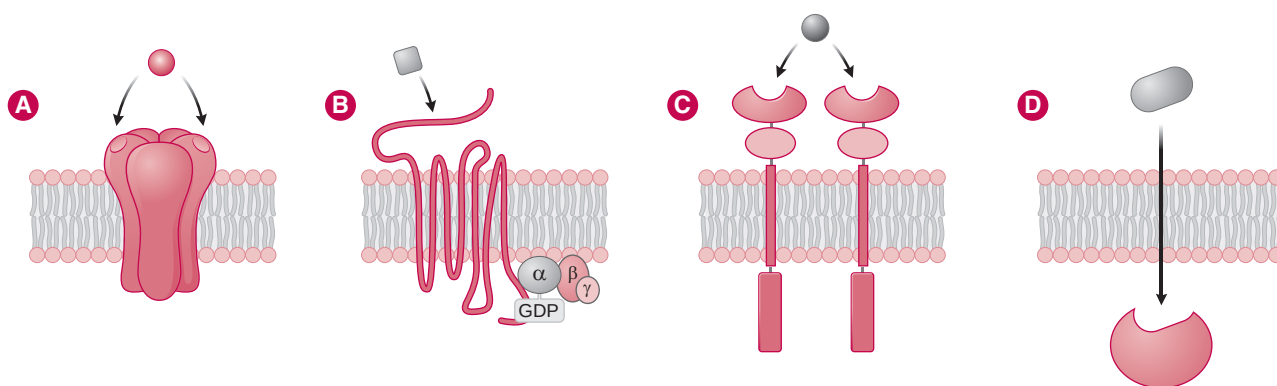


FIGURE 7.11 Major classes of drug receptors. (A) Transmembrane ligand-gated ion channel receptor. (B) Transmembrane G protein-coupled receptor (GPCR). (C) Transmembrane catalytic receptor or enzyme-coupled receptors. (D) Intracellular cytoplasmic/nuclear receptor. (From Simon JB, Golan DE, Tashjian A, et al., eds. Chapter 1: drug-receptor interactions. In: Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy. Baltimore: Lippincott Williams & Wilkins, 2008:3–16, with permission.)

seconds. Slower processes are rarely referred to as signal transduction.

The process of signal transduction serves several critical roles. First, it enables extracellular molecules to affect cellular function without entering the intracellular environment. This “long distance” communication is accomplished by the binding of a ligand (agonist) to the receptor protein and stabilizing the receptor structure in an active conformation. The active receptor conformation can then facilitate the flow of ions through a membrane channel by the removal of steric or electronic hindrances (opening of channel gates). For those receptors that produce signals via activation of intracellular metabolic pathways, the active state or conformation of the receptors changes the intracellular molecular environment in such a way as to activate, directly or indirectly, intracellular regulatory enzymes. Second, several different signals can affect one another by facilitating or inhibiting the activation of regulatory enzymatic proteins via common or opposing metabolic pathways. Thus, signal transduction mechanisms can interact in such a way as to yield an integrated response to multiple stimuli. Third, via the activation of enzymes and the production of second messengers (e.g., cAMP, diacylglycerol [DAG], and IP_3), an initially weak signal can be amplified many times, and its duration prolonged, to produce a robust cellular response. This amplification occurs through several mechanisms. The kinetic time frame for enzyme activation and the presence of key metabolites can be much longer than the time of receptor activation itself. Thus, a brief activation of a small number of receptors can result in a magnified response by the cell. Amplification can also occur via a molecular “cascade,” in which one initial signal can trigger a multitude of intracellular reactions that lead to an enhanced cellular response. Outcomes of signal transduction can include one or more of the following: 1) a change in cell membrane polarity in electrically

excitable tissues, such as nerves and muscles, which then results in the facilitation or inhibition of an action potential, thus affecting the excitability of the tissue; 2) the activation of cytosolic metabolic cascades, resulting in alterations of cellular morphology or function; and 3) gene activation, leading to the synthesis of new proteins that can then modify cellular structures and physiology.

Transmembrane Ion Channels

Ligand-Gated Ion Channels

The most rapid cellular responses to receptor activation are mediated via ligand-gated ion channels (LGICs) (Fig. 7.11A). The main component of this signal transduction pathway is a plasma membrane-spanning protein composed of multiple peptide subunits, each of which contains four membrane-spanning domains. The nicotinic acetylcholine receptor is, perhaps, the best characterized LGIC. The nicotinic receptor is composed of five distinct subunits, two α and, depending on the receptor subtype, various combinations of additional β , γ , and δ subunits. The binding of an acetylcholine molecule to the binding site on each of two α subunits induces a conformational change in the receptor, opening a sodium-selective ion channel through the center of the protein (16). The result is depolarization of the surrounding plasma membrane. Other neurotransmitter-activated LGICs include γ -aminobutyric acid, glycine, glutamate, and some serotonin receptors (17). These receptors share a similar structural conformation and function to the nicotinic receptor, except for the specificity of the ligand-binding site and selectivity of the channel for particular ions. The primary reason for the rapidity (milliseconds) of the cellular response with LGICs is that the transduction of the signal requires the activation of a single molecule. Therefore, this transduction mechanism is especially suited for physiologic processes necessitating

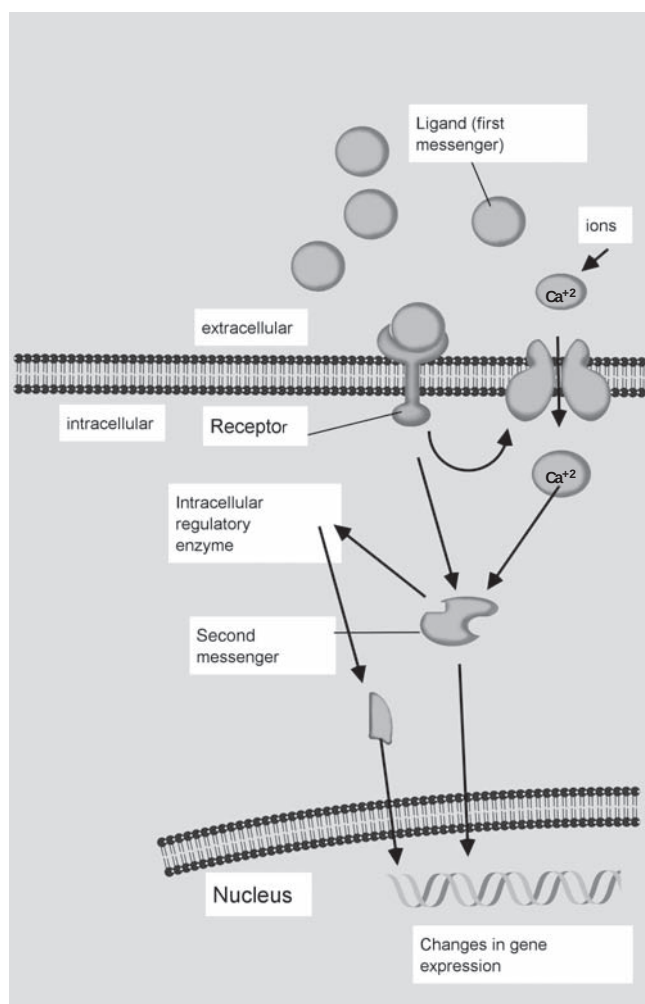


FIGURE 7.12 Overview of signal transduction in cellular regulation and gene expression via second messengers.

an immediate response, such as the stimulation of nerves and muscles.

Voltage-Gated and Second Messenger-Gated Channels

Other ion channels are controlled by either voltage changes or second messenger molecules. An example of a voltage-gated channel includes the sodium channels responsible for impulse conduction in sensory nerve fibers that transmit information about pain and temperature. Following administration, the local anesthetic lidocaine enters the nerve cell via diffusion in its unprotonated form. Once inside the nerve cell, lidocaine is protonated and, in this charged form, is capable of blocking the sodium channel from the intracellular side. Some second messenger molecules (e.g., cAMP and IP₃) generated following the activation of G protein-coupled receptors can influence the degree of channel opening or closing. The most common channels influenced by these second messengers include those for calcium and potassium.

Transmembrane G Protein-Coupled Receptors

The G protein-coupled receptors (GPCRs) are a class of large membrane-bound proteins that share a well-conserved structure and transduce their signal via the activation of an intracellular guanine nucleotide-binding protein (G protein). This family of proteins has seven hydrophobic (heptahelical) domains that span the plasma membrane; therefore, it is sometimes referred to as having a serpentine structure (Fig. 7.11B). The extracellular region of the protein is composed of the amino terminus and several loops, which comprise the ligand-binding site. Smaller ligands tend to bind deep within the extracellular loops, close to the plasma membrane, whereas larger molecules have binding sites that are more superficial. The carboxy end of the receptor is located in the area of the protein that protrudes into the cytoplasm. The intracellular side of the receptor also includes the binding site for the G protein, which usually binds to the third loop between the sixth and seventh transmembrane regions of the protein. Close to the carboxy terminus are serine and threonine residues, which are targets for adenosine triphosphate (ATP)-dependent phosphorylation. Following prolonged activation, phosphorylation of these residues is hypothesized to occur via a negative-feedback regulatory metabolic pathway, which facilitates the binding of modulating molecules that subsequently impair the coupling of G proteins to the receptor. The result is receptor desensitization.

More than 100 different GPCRs bind to a variety of ligands encompassing biogenic amines, such as acetylcholine, norepinephrine, and serotonin; amino acid neurotransmitters, such as glutamate and glycine; and peptide hormones, such as angiotensin II and somatostatin. There are multiple GPCR types for a single ligand. The result is the possibility that a single ligand can activate a variety of transduction pathways and produce a multiplicity of cellular responses. Thus, a receptor is defined not just by which ligand binds to it but also by how the signal is transduced and the resultant physiologic response. As an example, at least nine different adrenergic receptor subtypes exist (18). Norepinephrine can bind to the β_1 receptor, which is coupled to a G protein (designated G_s). Following receptor stimulation of G_s, there is activation of the enzyme adenylyl cyclase, thus leading ultimately to an increase in heart rate and force of contraction. Norepinephrine binding to α_1 -receptors, on the other hand, results in the binding to a different G protein (G_q), which activates the production of the second messengers IP₃ and DAG, which then initiate a cascade of intracellular events leading to smooth muscle contraction. Therefore, a single ligand can induce a wide range of responses as a consequence of coupling to different G proteins. Which G protein is activated depends on factors such as the presence and availability of individual G proteins within a particular cell type, kinetic issues (e.g., the binding affinity

of the G protein for the receptor protein), and finally, the affinity of the activated G protein subunits for signal transduction enzymes.

G Proteins

G proteins are heterotrimeric in structure with the subunits (in decreasing size) designated as α , β , and γ . At least 13 types of G proteins have been identified, which are divided among four families, G_s , G_i , G_q , and G_{12} . Individual G proteins transduce the receptor activation signal via one of a number of second messenger systems discussed below. The best-understood second messenger systems associated with each G protein family are summarized in Table 7.1.

The characteristics of the α subunit are what determine the designation of the G protein. Receptor activation leads to a conformational change in the associated G protein, triggering the release of bound guanosine diphosphate (GDP) from the α subunit, which is then replaced by a molecule of guanosine triphosphate (GTP). With the binding of GTP, the α -subunit GTP complex dissociates from the $\alpha\beta$ subunits and binds to a particular target enzyme, resulting in its activation or inhibition. Within a short period of time, the α subunit catalyzes the dephosphorylation of the associated GTP molecule to GDP, resulting in the reassociation of the α subunit with the $\alpha\beta$ subunits and, thus, the return of the G protein to the inactivated state (Fig. 7.13). Variations on this scheme include the activation of proteins such as G protein-gated ion channels by dissociated $\alpha\beta$ subunits and the ability of receptor proteins to activate more than a single G protein. The simultaneous activation of more than one type of GPCR results in the initiation of multiple signals, which can then interact with one another (a phenomenon commonly referred to as cross-talk). This interaction can be of several types: If both receptors use a common signal transduction pathway, the activation can result in an additive response by the cell. Conversely, if simultaneous receptor activation triggers opposing signal transduction pathways, the outcome will be an attenuated cellular response. Other types of interactions can include the desensitization or activation of other receptor proteins or second messenger pathways. The final

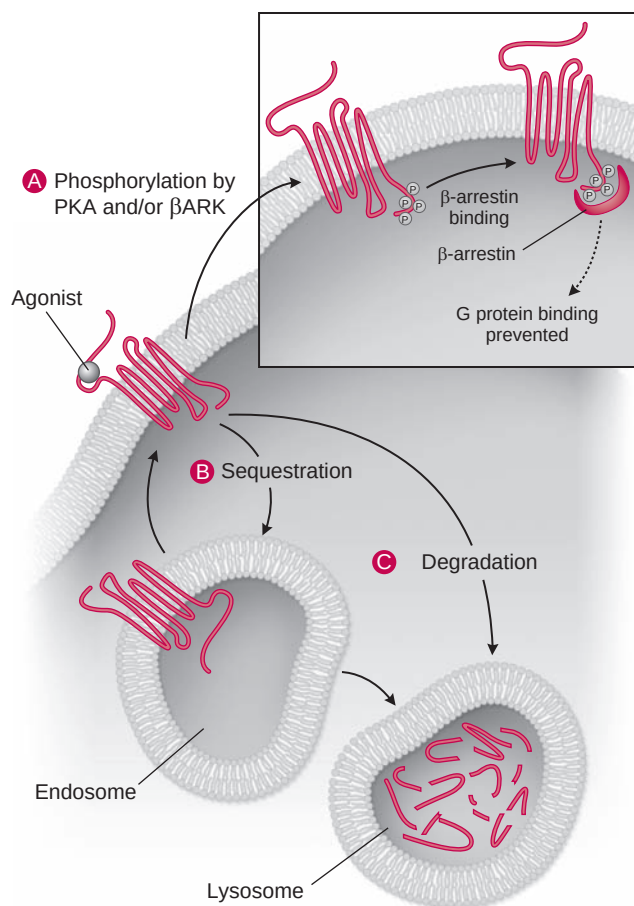


FIGURE 7.13 β -Adrenoceptor regulation. Following repeated stimulation of the β -adrenoceptor by an agonist, phosphorylation of amino acids on the intracellular C-terminal domain by protein kinase A (PKA) and/or β -adrenergic receptor kinase (β ARK) can (A) prevent subsequent G protein binding, (B) enhance removal of the receptor from the membrane via sequestration into endosomes, or (C) enhance degradation via lysosomal internalization and cleavage. (From Simon JB, Golan DE, Tashjian A, et al., eds. Chapter 1: drug-receptor interactions. In: Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy. Baltimore: Lippincott Williams & Wilkins, 2008:16, with permission.)

outcome of the activation of multiple signals is an integrated response by the cell.

Second Messenger Pathways

As previously discussed, in response to receptor activation, G proteins activate plasma membrane-bound enzymes, which then trigger a metabolic cascade that results in a cellular response (19). The products of these enzymatic actions are termed “second messengers,” because they mobilize other enzymatic and structural proteins, which then produce the cellular response. The enzymes catalyzing the synthesis of second messengers fall generally into two categories: those that convert the purine triphosphates ATP and GTP into their respective cyclic monophosphates (e.g., cAMP, cyclic guanosine

TABLE 7.1 G-Protein Transducers and Second Messengers

G Protein Transducer Family	Second Messenger System
G_s	Stimulates adenylyl cyclase activity and Ca^{2+} channels
G_i	Inhibits adenylyl cyclase activity and activates K^+ channels
G_q	Stimulates phospholipase C activity
G_{12}	Modulates sodium/hydrogen ion exchanger

monophosphate [cGMP]), and enzymes that synthesize second messengers from plasma membrane phospholipids (e.g., DAG and IP_3). The most thoroughly studied second messenger system is controlled by a family of 10 plasma membrane-bound isoenzymes of adenylyl cyclase*, which catalyze the conversion of ATP to cAMP (Fig. 7.14). Adenylyl cyclase is activated by the G_s family of G proteins and inhibited by the G_i family. Following synthesis, cAMP activates cAMP-dependent protein kinases by triggering the dissociation of regulatory subunits from catalytic subunits. The catalytic subunits then activate other target proteins via phosphorylation, which then trigger the cellular response. The magnitude of the cellular response is proportional to the concentration of cAMP. Degradation of cAMP occurs via phosphodiesterases or by reducing cAMP concentration via active transport out of the cell. The result is termination of the signal.

A similar, although less ubiquitous, second messenger pathway is associated with guanylyl cyclase**. Guanylyl cyclase is activated in response to catalytic receptors selective for ligands including atrial natriuretic factor and nitric oxide. When stimulated, guanylyl cyclase then catalyzes the synthesis of cGMP from GTP. cGMP subsequently activates cGMP-dependent protein kinases, which then activate other proteins. The actions of cGMP are terminated by enzymatic degradation of the second messenger or the dephosphorylation of substrates. One effect of this second messenger pathway is relaxation of smooth muscle via the dephosphorylation of myosin light chains.

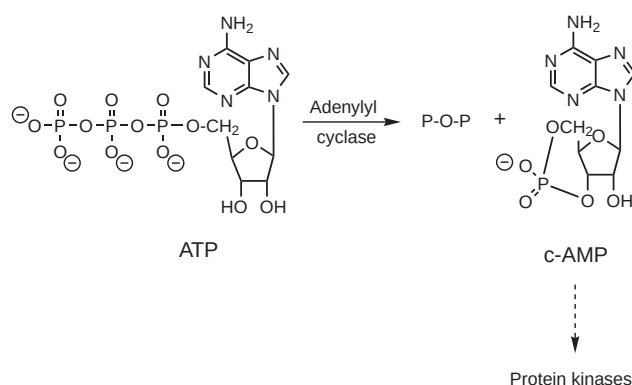


FIGURE 7.14 Conversion of adenosine triphosphate (ATP) to cyclic adenosine monophosphate (cAMP) catalyzed by adenylyl cyclase.

*This enzyme is known by two names, *adenylate cyclase* (EC 4.6.1.1), its official name from the International Union of Biochemistry and Molecular Biology Nomenclature Committee, or its alternative name, adenylyl cyclase.

**This enzyme is known by two names, *guanylate cyclase* (EC 4.6.1.2), its official name from the International Union of Biochemistry and Molecular Biology Nomenclature Committee, or its alternative name, guanylyl cyclase.

The generation of second messengers from plasma membrane phospholipids is mediated primarily by G protein activation of phospholipase C (PLC) (20). There are three families of PLC, designated PLC- β , PLC- γ , and PLC- δ . PLC- β can be activated by the α subunit of the G_q family of G proteins or the $\beta\gamma$ subunits of other G proteins. PLC- γ is activated via tyrosine kinase receptors, but the mechanism for PLC- δ is not yet understood. On activation, PLC hydrolyzes phosphatidyl inositol-4,5-bisphosphate to DAG and IP_3 (Fig. 7.15). The water-soluble IP_3 diffuses into the cytoplasm, where it triggers the release of calcium from intracellular stores. Intracellular calcium then binds to the protein calmodulin and also to protein kinase C, both of which then stimulate, via protein phosphorylation, a broad range of enzymes and other proteins, including specific kinases. The other product of PLC, DAG, is lipid soluble and remains in the plasma membrane, where it facilitates the activation of protein kinase C by calcium. The signal is terminated via inactivation of IP_3 by dephosphorylation, whereas DAG is inactivated by phosphorylation to phosphatidic acid or deacetylation to fatty acids. The

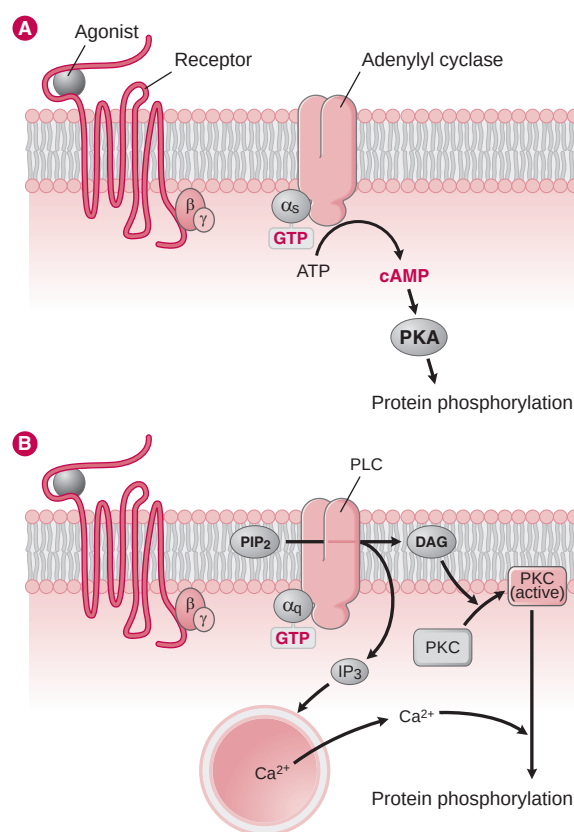


FIGURE 7.15 Second messenger mechanisms for cyclic adenosine monophosphate (cAMP) and inositol trisphosphate. (From Simon JB, Golan DE, Tashjian A, et al., eds. Chapter 1: drug–receptor interactions. In: Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy. Baltimore: Lippincott Williams & Wilkins, 2008:3–16, with permission.)

concentration of intracellular calcium is reduced by sequestration within cytoplasmic organelles or transport out of the cell. Activation of phospholipase D hydrolyzes phosphatidylcholine to phosphatidic acid, which can then be metabolized to DAG via phosphatidate phosphohydrolase. This pathway prolongs the duration of elevated levels of DAG. Phospholipase A_2 is activated by increased concentrations of intracellular calcium and metabolizes phosphatidylcholine to arachidonic acid. Arachidonic acid then functions as a substrate for the synthesis of autocoids, including prostaglandins, thromboxane A_2 , and leukotrienes.

Frizzled is another family of GPCR proteins that regulate functions such as embryonic development, the formation of neural synapses, and cell polarity via a signal transduction cascade composed of protein kinases such as glycogen synthase kinase 3 (GSK-3), phosphatases, and proteolytic enzymes. A target molecule for this cascade is β -catenin, which promotes specific gene expression.

With the development of screening assays of cDNA libraries, a number of GPCRs have been identified for

which there is no known endogenous ligand. These receptors are termed “orphan receptors.” When the endogenous ligand is identified, the receptor is said to be “adopted.” Strategies for identifying the ligand of orphan receptors include: 1) expression of the GPCR in a recombinant assay system; 2) screening candidate ligands against the receptor; 3) detecting active ligands by activation of signal transduction cascades; and 4) further testing of the ligand against other GPCRs to determine selectivity. An example of an adopted orphan GPCR is Axor 35, which was ultimately characterized as the histamine H_4 receptor (Fig. 7.16) (21).

Transmembrane Catalytic Receptors

Catalytic receptors are a class of plasma membrane-bound receptors characterized by a monomer with a ligand-binding site in the extracellular domain, a single membrane-spanning domain, and an intracellular domain with enzymatic activity (Fig. 7.11C). This family of receptors is activated predominately by peptide hormones, such as insulin, epidermal growth factor,

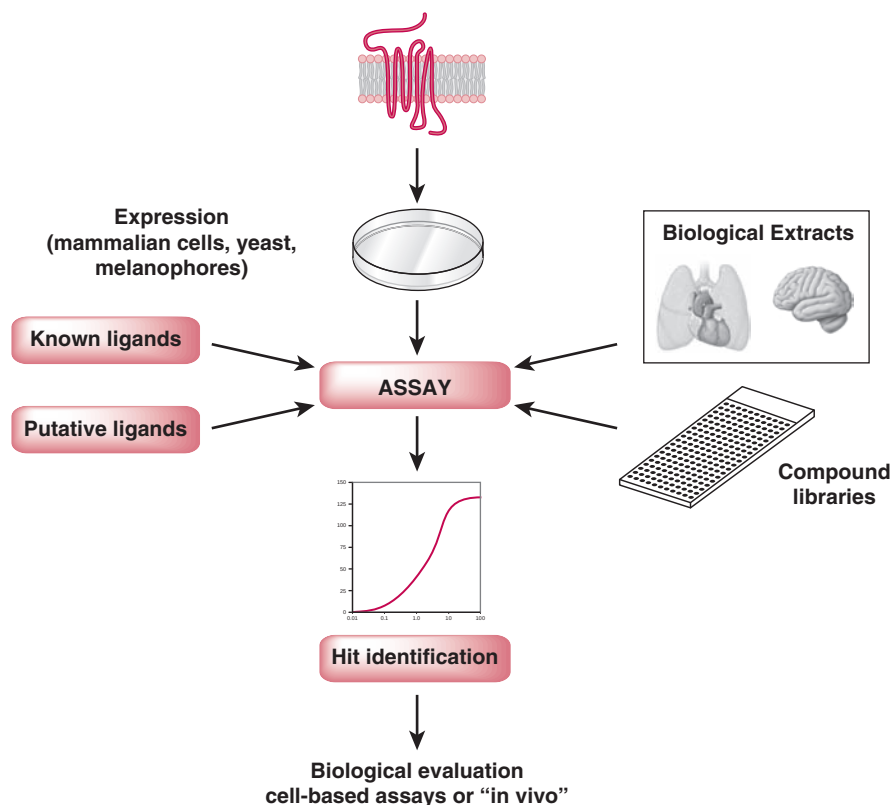


FIGURE 7.16 Strategy for the identification of ligands for orphan GPCRs. Orphan GPCRs are expressed in a recombinant expression system, such as mammalian cells, yeast, or *Xenopus* melanophores. Following expression, it is usual to generate an assay amenable to the screening of candidate ligands in microtiter plate formats. The identification of an activating ligand is detected according to the activation of an intracellular signaling cascade. An activating ligand is identified according to its ability to cause a concentration-dependent increase in the activity of a signaling cascade. Once identified, the ligand can be further characterized against other GPCRs to determine its activity and selectivity profile prior to being used in cell-based, tissue, and, in some cases, whole-animal experiments in order to study the physiologic role of the newly liganded receptor.

platelet-derived growth factor, and atrial natriuretic factor. The catalytic portion of the receptor functions as a protein kinase, targeting primarily tyrosine residues; however, the receptor for atrial natriuretic factor, rather than having kinase activity, activates guanylyl cyclase and metabolizes GTP to cGMP (22). Receptor activation occurs by ligand binding, which then triggers dimerization of receptor proteins via the cross-phosphorylation of tyrosine residues. The dimeric protein is the active form of the catalytic receptor. One consequence of activation via dimerization is that the intracellular signal can be maintained even after the ligand has dissociated from the binding site. The phosphorylation of intracellular proteins by this receptor type results in effects such as the opening of ion channels, changes in cytoplasmic function, or initiation of genomic expression. Other catalytic receptors, activated by transforming growth factor- β , use serine/threonine kinases in regulating cellular growth and differentiation. Major signal transduction pathways activated by receptor tyrosine

kinases include the mitogen-activated protein kinase (MAPK) cascade.

The MAPK Cascade

The MAPK cascade is comprised of a series of proteins that activate enzymatic substrates via phosphorylation (Fig. 7.17). Once a tyrosine hydroxylase receptor is activated by a ligand (most commonly hormones that affect gene expression), an adaptor protein, growth factor receptor-bound protein 2 (Grb2), binds to the receptor and the complex then activates a small GTPase called Ras. Ras then binds to and activates the serine/threonine-specific protein kinase c-Raf. c-Raf then phosphorylates a MAPK kinase (MEK), which phosphorylates MAPK (MAPKs are also referred to as extracellular signal-regulated kinases [ERKs]). Activated MAPK can then activate transcription factors that bind to DNA sequences that control the transfer of genetic information from DNA to messenger RNA. Ultimately, activation of catalytic receptors results in the stimulation of gene

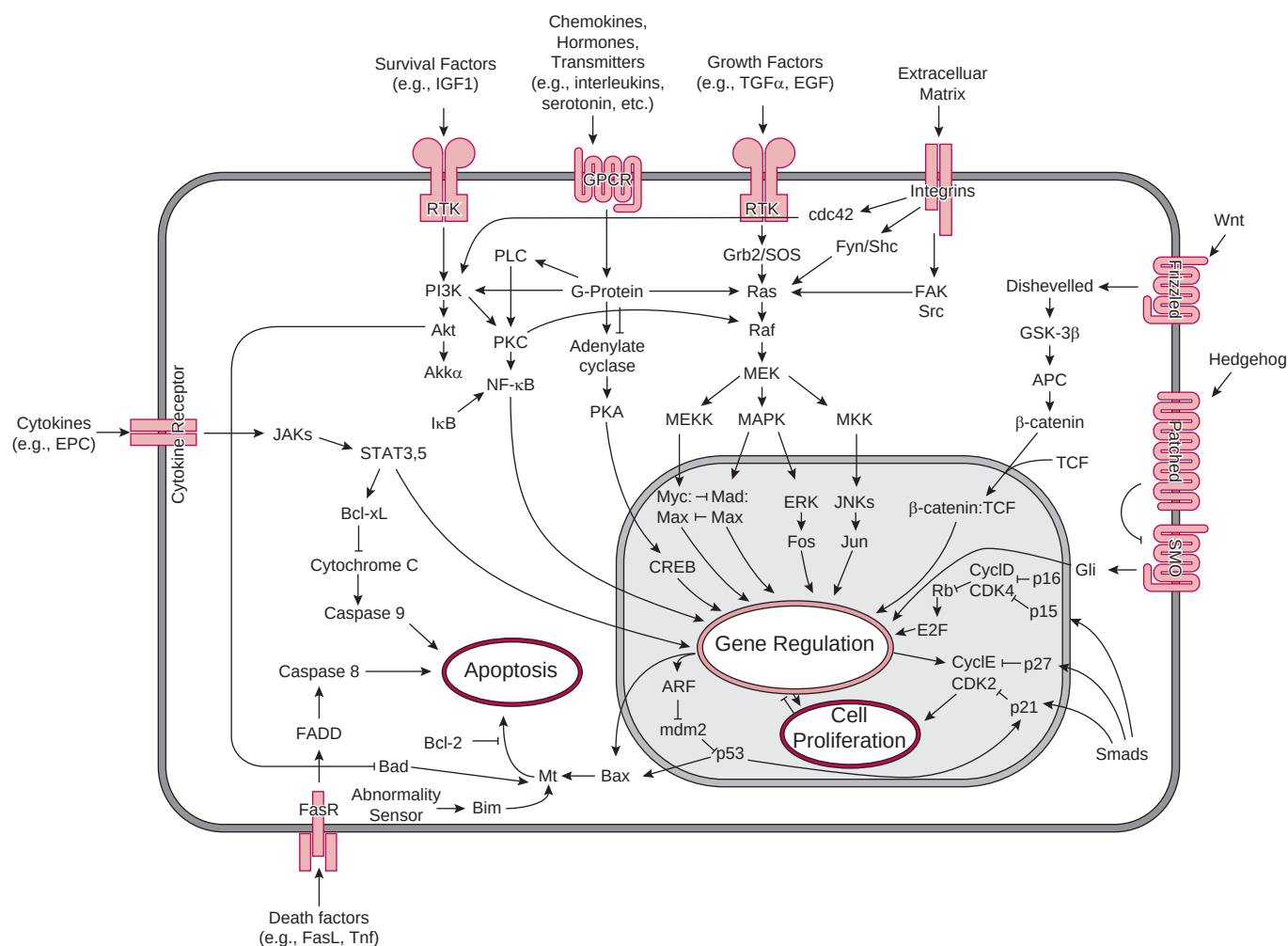


FIGURE 7.17 Signal transduction cascades that regulate gene expression and apoptosis. (From Wikipedia. Mitogen-activated protein kinase. Available at: http://en.wikipedia.org/wiki/Mitogen-activated_protein_kinase.)

expression, cell growth and division, differentiation, and apoptosis.

Enzyme-Coupled Receptors

Enzyme-coupled receptors are similar in their function to catalytic receptors, except rather than inherent catalytic activity, enzyme-coupled receptors bind to separate enzymatic proteins (Fig. 7.11C). This class of receptor binds cytokines, including growth hormone, erythropoietin, and interferon. Like catalytic receptors, enzyme-coupled receptors are activated via dimerization following ligand binding. Kinase activity is accomplished by separate, noncovalently bound protein kinases of the Janus kinase (JAK) family. After dimerization, JAKs are activated and phosphorylate receptor tyrosine residues. The phosphorylated receptor then binds other molecules, termed “signal transducers and activators of transcription” (STATs), which are phosphorylated by the JAKs and, subsequently, dissociate into the cytoplasm. The STATs then translocate to the nucleus, where they initiate gene transcription.

Intracellular Cytoplasmic/Nuclear Receptors

Cytoplasmic/nuclear receptors differ from those described earlier in that they are not associated with the plasma membrane but, rather, are located within the cytoplasm or are bound to the surface of the nucleus (Fig. 7.11D). These receptors are composed of a single polypeptide with three functional domains. The amino terminal contains a binding site for a modulator protein, heat shock protein-90, which is associated with the receptor in the absence of agonist. In the middle of the receptor peptide is a binding site for DNA, and the carboxy terminus contains the ligand-binding site. Cytoplasmic/nuclear receptors are activated by lipid-soluble ligands, which passively diffuse through the plasma membrane (23). Agonists include nitric oxide, steroid hormones, and vitamin D. Ligand binding activates the receptor by inducing the dissociation of heat shock protein-90. The receptor then translocates to the nucleus and binds to a DNA response element, which then initiates translation of the target gene. The response to this type of signal transduction is relatively slow, requiring 30 minutes to several hours following protein binding. Moreover, the duration of the response can last long after the concentration of the ligand has fallen to zero. The duration of the response is related to turnover rate of the synthesized protein; however, it is also affected by a ligand with extremely high binding affinity, possibly resulting in prolonged receptor activation.

RECEPTOR SUBTYPES

Careful examination of the effects of a series of sympathomimetics by Ahlquist (24) led him to postulate the existence of at least two types of adrenoceptors, which he termed α and β . Realizing that adrenoceptor agonists were capable of causing either relaxation or contraction

of isolated smooth muscles, he noted that although a compound like norepinephrine had potent excitatory actions but weak inhibitory actions, another catecholamine, isoproterenol, had potent inhibitory actions but weak excitatory actions. When a series of related compounds were tested for potency in various tissues, it was demonstrated that for the α -adrenoceptor, the order of potency was epinephrine $\geq$ norepinephrine $\ggg$ isoproterenol, and for the β -adrenoceptor, the order of potency was isoproterenol $>$ epinephrine $\geq$ norepinephrine. Following the findings of Ahlquist, others used specific antagonists that had become available to further support this designation of receptor subtypes. With the development of highly selective antagonists and procedures for cloning and determining the amino acid sequence of proteins, the classification of receptors into subtypes has expanded at a tremendous rate (25). For example, the α -adrenoceptor noted earlier can be subclassed as α_{1A} , α_{1B} , α_{1C} , α_{1D} , α_{2A} , α_{2B} , and α_{2C} based on cloning experiments (Table 7.2).

The therapeutic significance of such distinctions is not yet known, however, due to a lack of selective agonists or antagonists. Some therapeutic distinction can be made between α_1 and α_2 adrenoceptors in a general way, in that α_1 vasoconstriction antagonism by prazosin and central nervous system α_2 -adrenoceptor stimulation by clonidine are useful in the treatment of hypertension. Similarly, β -adrenoceptor antagonists are available that antagonize β_1 -adrenoceptors with some selectivity. Thus, the use of metoprolol, a selective β_1 -antagonist, is effective and relatively safe in hypertensive patients with compromised airway function, whereas the use of a nonselective (β_1 and β_2) antagonist, such as propranolol, would be clearly contraindicated in such a patient.

A summary of the most important receptor subtypes from a therapeutic standpoint is presented in Table 7.3. The reader should realize that this is a simplification of what is currently known about the various receptor subtypes. For instance, within the general category of serotonin receptors, at least 13 subtypes can be identified from cloning experiments. The lack of selective agonists and antagonists to characterize the pharmacology of each of these subtypes, however, has hindered our understanding of their individual functions and importance from a therapeutic standpoint. Our ability to eventually develop drugs that selectively manipulate such receptor subtypes has enormous therapeutic implications.

SPARE RECEPTORS

Biologic systems often have built-in safety factors to enhance the efficiency of receptor–stimulus coupling and, thereby, assure the desired neurotransmission. In many tissues containing α -adrenoceptors, only a small percentage of the available receptors need to be occupied to produce a maximum response. This depends on the particular tissue being studied and the agonist used.

TABLE 7.2 Adrenoreceptor Families

Receptor Type	Subtype	Transduction Mechanism	Tissue Function
α_1	1A	Activates $G_{q/11}$	Smooth muscle and myocardial contraction
α_1	1B	Activates $G_{q/11}$	Smooth muscle contraction
α_1	1D	Activates $G_{q/11}$	Smooth muscle contraction
α_2	2A	Activates $G_{i/o}$	Hypotension, sedation, analgesia, anesthesia
α_2	2B	Activates $G_{i/o}$	Vasoconstriction
α_2	2C	Activates $G_{i/o}$	Not established
β_1	—	Activates G_s	Increases heart rate and force of contraction
β_2	—	Activates G_s	Smooth muscle relaxation
β_3	—	Activates or inhibits adenylyl cyclase	Lipolysis, cardioinhibition

Therefore, 100% occupancy of the available receptors is not always required, because spare receptors (or a receptor reserve) are present. Studies using phenoxybenzamine, which alkylates the α -adrenoceptor and, therefore, irreversibly inactivates the receptor, indicate that only 5% to 10% of the available receptors need to be activated to elicit a maximum response to a full, or strong, agonist, such as norepinephrine or phenylephrine (8,9,26). To obtain a maximum response to a partial agonist like ephedrine, however, nearly 100% of the receptors need to be occupied. The explanation behind this difference can involve a less than ideal receptor–drug interaction for partial agonists. A partial agonist can function as an antagonist if it interferes with the ability of a full agonist to bind to its receptor and produce a response. In the absence of a full agonist, however, the partial agonist only displays agonistic activity.

DYNAMIC NATURE OF RECEPTORS

As is characteristic of most individual components of living systems, receptors are not static but, rather, are constantly in a state of dynamic adaptation. One could envision these protein molecules floating within the fluid mosaic of the biologic membrane awaiting interaction with normal physiologic signals. The function of such receptors, once stimulated, involves attempts to respond to perturbations of the normal physiology of the cell or organism. The role in maintaining homeostasis within the organism requires constant adaptation of receptor number and/or sensitivity in response to the changing environment in the vicinity of the receptor. One approach to controlling receptor activation is by regulating the concentration of neurotransmitter at the receptor binding site. This is accomplished by the modulation of release via activation of presynaptic receptors, as described earlier. Altering the rate of synthesis,

degradation, or efficiency of the enzymes that create or degrade neurotransmitter molecules is also used to regulate neurotransmitter concentrations at the receptor.

A second mechanism for modulating the cellular response to receptor activation is to alter receptor number and/or sensitivity. The process is best understood for GPCRs but has also been characterized for ion channel receptors, such as the nicotinic receptor. The alteration in the availability or functional capacity of a given receptor constitutes an adaptive mechanism whereby the cell or organism is protected from agonist overload. For example, the chronic administration of a β -adrenoceptor agonist, such as isoproterenol, is known to produce a desensitization of the β -adrenoceptors in the heart (27). During the period of overstimulation, the receptor becomes desensitized to further activation via phosphorylation by G protein–coupled kinases, such as protein kinase A or β -adrenoceptor kinase, at serine and threonine residues on the C-terminal domain that interferes with G_s binding (Fig. 7.13). Desensitization can also occur at other receptors with analogous phosphorylation sites (heterologous desensitization). A more prolonged or powerful overstimulation typically results in a decrease in receptor number and is termed “down-regulation.” In these GPCRs, downregulation initiated by phosphorylation of amino acids near the intracellular C-terminal domain results in binding to β -arrestins that facilitate their internalization via a clathrin-dependent pathway. Following internalization, the receptor can either be recycled back into the plasma membrane via endosomes or degraded by lysosomes. Reduced receptor numbers can also be accomplished via changes in the transcription and/or translation of genes that code for the receptor. Changes in receptor number or efficiency of receptors that lead to diminished responses and, thus, diminished efficacy of a drug with repeated or long-term use are examples of pharmacodynamic tolerance. As a

TABLE 7.3 Survey of Receptor Subtypes

Receptor Class	Subtype	Selective Selective Agonist	Antagonist	Effector	Cloned
Adrenoceptors	α_1	Phenylephrine	Prazosin	IP ₃ /DAG	Yes
	α_2	Clonidine	Yohimbine	↓cAMP	Yes
	β_1	Dobutamine	Atenolol	↑cAMP	Yes
	β_2	Terbutaline	Butoxamine	↑cAMP	Yes
Dopamine receptors	D ₁	Fenoldopam	Dihydroxidine	↑cAMP	Yes
	D ₂	Bromocriptine	(-)Sulpiride	↓cAMP	Yes
Excitatory amino acid receptors	NMDA	NMDA	D-AP5	↓Na ⁺ /Ca ²⁺	Yes
	AMPA	AMPA	CNQX	↑Na ⁺	Yes
	Kainate	Kainate	?	↑Na ⁺ /K ⁺	Yes
GABA receptors	GABA _A	Muscimol	Bicuculline	↑Cl ⁻	Yes
	GABA _B	Baclofen	Saclofen	↓cAMP	Yes
Histamine receptors	H ₁	2-(<i>m</i> -Fluorophenyl) Histamine	Mepyramine	IP ₃ /DAG	Yes
	H ₂	Dimapril	Ranitidine	↑cAMP	Yes
Muscarinic receptors	M ₁	Oxotremorine	Pirenzepine	IP ₃ /DAG	Yes
	M ₂	Oxotremorine	AF-DX116	↓cAMP	Yes
Nicotinic receptors	N _{muscle}	?	Decamethonium	↑Na ⁺ /Ca ²⁺	Yes
	N _{neuronal}	?	Hexamethonium	↑Na ⁺ /Ca ²⁺	Yes
Opioid receptors	Mu	Sufentanil	CTAP	↓cAMP	Yes
	Delta	[D-Ala ²] deltorphin	Naltrindole	↓cAMP	Yes
	Kappa	Dynorphin	Nor-binaltorphimine	↓cAMP	Yes
Serotonin receptors (5-HT)	5-HT _{1A}	8-OH-DPAT	Spiperone	↓cAMP	Yes
	5-HT _{2A}	α-Methyl-5-HT	Ketanserin	IP ₃ /DAG	Yes

?, no known selective compounds available; cloned, receptor subtype has been cloned and the amino acid structure is known. Chemical abbreviations used: AF-DX116, 11-[[2-[(diethylamino)methyl]-1-piperidinyl]acetyl]-5-11-dihydro-6H-pyrido[2,3-b] [1,4]benzodiazepine-6-one; AMPA, D,L-α-amino-3-hydroxy-5-methyl-4-isoxalone propionic acid; cAMP, cyclic adenosine 3',5'-monophosphate; CNQX, 6-cyano-7-nitroquinoxaline-2,3-dione; CTAP, D-Phe-Cys-Tyr-DTrp-Arg-Thr-Pen-Thr-NH₂; DAG, diacyl glycerol; D-AP5, D-amino-5-phosphonopentanoate; GABA, γ-aminobutyric acid; 5-HT, 5-hydroxytryptamine, serotonin; IP₃, inositol 1,4,5-triphosphate; NMDA, N-methyl-D-aspartate; 8-OH-DPAT, 8-hydroxy-2-(di-n-propylamino)tetralin.

general principle, the body will always attempt to maintain homeostasis, whether perturbed by environmental challenges, disease processes, or even the administration of drugs. Actually, the body sees the administration of drugs as a perturbation of homeostasis and, usually, attempts to overcome the effects of the drug by invoking receptor adaptations. Often, however, with appropriate dosing schedules, drugs can be used with little observed receptor adaptation such that the desired pharmacologic effect continues to be observed.

In a similar fashion to the example described earlier, chronic administration of the β-adrenoceptor antagonist propranolol leads to a state of receptor supersensitivity or upregulation. The cells within the tissue, such as the heart, sense an alteration in the normal rate of basal β-adrenoceptor stimulation and, thus, respond by either increasing the number or the affinity of the receptors for their natural agonists, norepinephrine and epinephrine. An enhanced efficiency of the interaction between the receptor and its transducing systems can also account for

a portion of the observed supersensitivity. The knowledge that such a receptor adaptation occurs has paramount practical therapeutic implications, because abrupt withdrawal of this class of agents can precipitate acute myocardial infarction. Thus, this practice should be scrupulously avoided.

Some pathophysiologic states are characterized by perturbations in receptor dynamics. Prinzmetal angina is thought to be characterized by an imbalance between vasodilatory β_2 -adrenoceptor function and vasoconstrictor α_1 -adrenoceptor function. In this disease state, the excessive alpha vasoconstriction of coronary arteries leads to myocardial ischemia and pain. The inadvertent use of a β -adrenoceptor antagonist, which is safely employed in typical angina pectoris to prevent β -adrenoceptor vasodilation, can leave unopposed alpha vasoconstrictor inputs and actually precipitate anginal pain. Thus, an understanding of the role that receptors have in physiology, pathophysiology, and pharmacology is essential for optimal therapeutic interventions.

FUTURE DIRECTIONS

Our understanding about the nature and role of receptors has increased tremendously since the early work of Langley and Ehrlich. Today, with the advances made in the field of molecular biology, it is possible to clone individual receptor subtypes (Table 7.3) and determine their function in cell culture. By modifying the amino acid structure at those sites believed to be involved in

agonist binding, a better appreciation for the interaction of drugs currently available—and the rational design of those awaiting discovery—can be realized. Additionally, as we begin to be able to determine the structure of receptor subtypes through cloning techniques, we hopefully will better understand those disease processes that result from, or lead to, receptor adaptations or dysfunction.

Promising therapeutic approaches that exploit ligand receptor interactions include gene therapy and pharmacogenomics. Gene therapy uses knowledge about genetic defects in receptors, enzymes, and other proteins to treat inherited diseases. The approach of gene therapy is to administer genetically engineered viruses as vectors that contain normal genes for proteins that can function in place of defective gene products. One target of gene therapy research is cystic fibrosis, a genetic disorder that affects the protein transmembrane conductance regulator, a cAMP-regulated chloride channel. The goal of gene therapy in this case is to introduce functional transmembrane conductance regulator genes into affected tissues and ultimately decrease the thick mucus that is the primary pathologic manifestation of the disease. Pharmacogenomics is based on drug therapy targeting particular isoforms of receptors and enzymes expressed as a consequence of the individual genotype of the patient. The expectation of the pharmacogenomic approach is to optimize drug therapy by selecting drugs and doses that have the best potency and efficacy for the expressed protein.

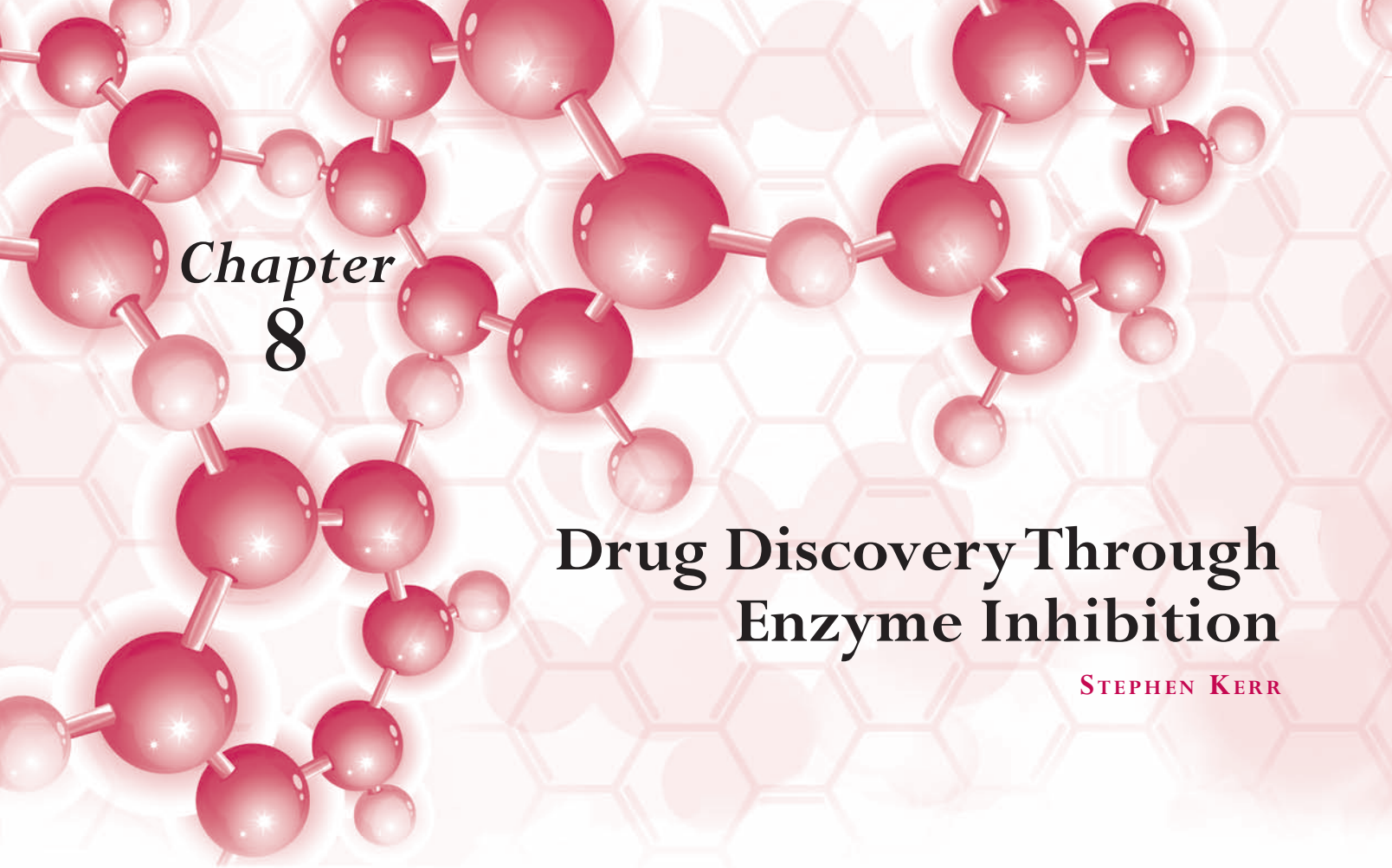
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Chapter 8

Drug Discovery Through Enzyme Inhibition

STEPHEN KERR

Abbreviations

ACE, angiotensin-converting enzyme
AChE, acetylcholinesterase
AMP, adenosine monophosphate
ANP, atrial natriuretic peptide
Ara-C, cytosine arabinoside
ATP, adenosine triphosphate
AZT, azidothymidine
cAMP, cyclic adenosine monophosphate
cGMP, cyclic guanosine monophosphate
CML, chronic myelogenous leukemia
dC, deoxycytidine
ddC, 2',3'-dideoxycytidine
DHFR, dihydrofolate reductase

dNTP, deoxynucleotide triphosphate
EC, Enzyme Commission
FdUMP, 5-fluorodeoxyuridine monophosphate
5-FU, 5-fluorouracil
GABA-T, γ -aminobutyric acid transaminase
GTP, guanosine triphosphate
 k_{cat} , catalytic or first-order rate constant
 K_d , enzyme–substrate dissociation constant
 K_i , inhibition constant
 K_m , Michaelis-Menten constant or apparent substrate–enzyme dissociation constant

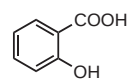
M, moles/L
mTOR, mammalian target of rapamycin
PAP, 2-phenylamionopyrimidine
PDE, phosphodiesterase
PLP, pyridoxal phosphate
RT, reverse transcriptase
SAR, structure–activity relationship
3-TC, 3-thiacytidine
TKI, tyrosine kinase inhibitor
TLCK, tosyl-lysyl-chloromethyl-ketone
TPCK, tosyl-phenylalanyl-chloromethyl-ketone
 v , velocity or rate of enzyme reaction
 V_{max} , maximum velocity or rate of enzyme reaction

OVERVIEW OF ENZYMES AS CATALYTIC RECEPTORS

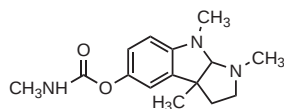
A Perspective

The concept of using small molecules that specifically target one or more enzymatic systems in the body is not new. Historically, compounds that were extracted from natural products have been used as medicinal agents (see Chapter 1). Subsequently, they have been shown to

have their therapeutic effect by targeting certain systemic enzymes (1). A classic example is the bark of the willow tree, known since ancient days to have antipyretic and analgesic effects. Its active ingredient, salicin, a glycoside, is metabolized in vivo to salicylic acid, which is a known inhibitor of cyclooxygenase, a key enzyme in the formation of prostaglandins, which are mediators of pain and fever. Similarly, physostigmine, isolated from the West African Calabar bean, was used as a treatment for glaucoma in the mid-1800s.

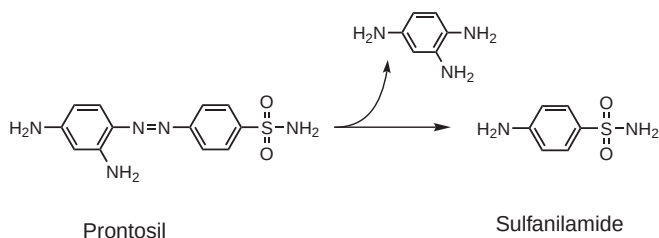


Salicylic acid



Physostigmine

Physostigmine's mechanism of action was only later determined to be inhibition of acetylcholinesterase (1). Inhibition of acetylcholinesterase in the eye leads to improved drainage and, thus, a decrease in the intraocular pressure giving relief to glaucoma patients. It was only in the 20th century, with the concept of the “magic bullet” having selective toxicity, introduced by Ehrlich as a rational approach to chemotherapy (1), that the concept of rational design and discovery of enzyme inhibitors followed. The discovery in 1935 of the antibacterial activity of the azo dye prontosil by Domagk (2), and the subsequent explanation in 1940 by Woods (3) of its metabolic reduction to sulfanilamide, an antimetabolite of *p*-aminobenzoic acid, finally paved the way for the rational design of enzyme inhibitors (Fig. 8.1). *p*-Aminobenzoic acid is an essential metabolite used in the bacterial synthesis of folic acid. Sulfanilamide, by its structural resemblance to *p*-aminobenzoic acid, competes for and selectively inhibits the bacterial enzyme dihydropteroate synthase (Fig. 8.2). In the absence of dihydropteroic acid, the bacteria are unable to synthesize tetrahydrofolic acid, an essential cofactor in one-carbon transfers that is involved in the *de novo* synthesis of purines and in the synthesis of thymidylate. This concept of designing drugs as antimetabolites, or structural analogs of essential metabolites, became the hallmark for the development of enzyme inhibitors. This was especially important in cancer therapy during the early days of rational drug design (4). As mechanisms of enzymes became better understood, the inhibitor design strategy grew more sophisticated, resulting in more potent and selective inhibitors being developed. The present-day focus on drug design through enzyme inhibition makes use of the antimetabolite theory as well as detailed kinetic and mechanistic information about the enzymatic pathways. These strategies use sophisticated assays, enzyme crystal structures and active site environments, site-directed mutagenesis experiments of catalytic groups of enzymes, and molecular docking experiments employing computers equipped with modeling or docking software. However, it must be mentioned that in the drug design



Prontosil

Sulfanilamide

FIGURE 8.1 Metabolic reduction of prontosil to sulfanilamide.

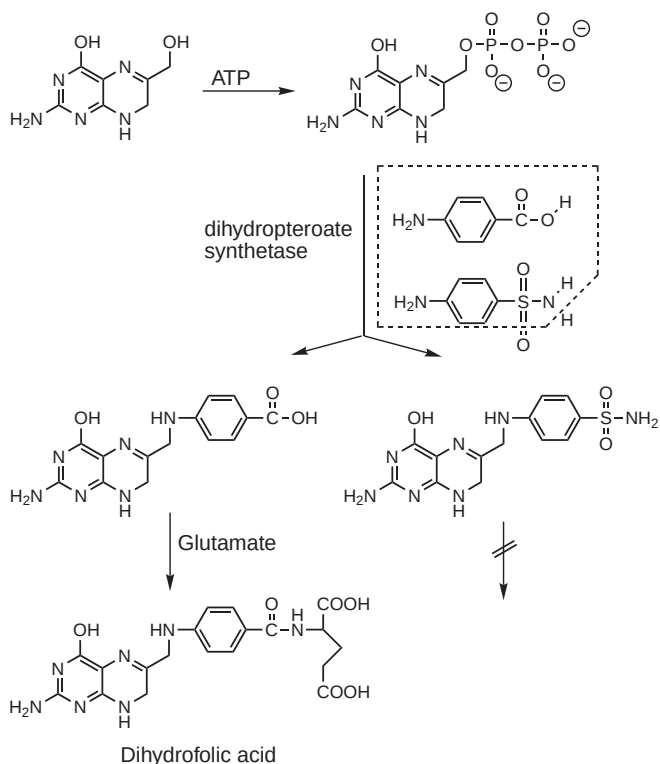


FIGURE 8.2 Metabolic pathway leading to dihydrofolic acid and its inhibition by sulfanilamide. This figure shows the structural resemblance to *p*-aminobenzoic acid.

process, designing a potent inhibitor of an enzyme is only the first step in the long and difficult process of drug development. Other factors, including pharmacokinetic profile of the inhibitor, toxicities and side effects, and animal and preclinical studies, must all be satisfactorily completed or addressed before the inhibitor can even enter clinical studies as a new drug candidate. Hence, even though an enormous amount of data exist regarding enzyme inhibitors, only a selected few turn out to be marketable drugs (Table 8.1). In the following sections, an overview of enzymes as catalytic proteins and general concepts of enzyme inhibitors and their rational design into drugs will be discussed with selected examples.

Overview

Enzymes are specialized proteins that function as catalysts to increase the rate of biochemical reactions. By interacting with substrates (reactant molecules upon which an enzyme acts), enzymes catalyze chemical reactions involved in the biosynthesis of many cellular products. Enzymes derive their name from Greek, in which the term *enzyme* means “in yeast” and was mainly used to distinguish the whole microorganism, such as yeast (“organized ferments”), from that of extracts of the whole microorganisms (“unorganized ferments”). The implication was that enzymes were unorganized ferments in yeast. Although the vast majority of enzymes are proteins, certain nucleic acids (RNAs) also possess

TABLE 8.1 A Partial Listing of Enzyme Inhibitor Drugs Presently Used as Drugs

Inhibitor (Drug)	Enzyme Inhibited	Use	Organism
Caspofungin	1,3- β -Glucan synthase	Antifungal	Fungal
Trilostane	3 (or 17) β -Hydroxysteroid dehydrogenase	Breast cancer	Human
Sildenafil	3',5'-Cyclic GMP phosphodiesterase	Erectile dysfunction	Human
Theophylline	3',5'-Cyclic nucleotide phosphodiesterase	Asthma	Human
Nitisinone	4-Hydroxyphenylpyruvate dioxygenase	Tyrosinemia	Human
Finasteride	Steroid 5 α -reductase	Benign prostatic hyperplasia	Human
Pyridostigmine	Acetylcholinesterase	Myasthenia gravis	Human
Pentostatin	Adenosine deaminase	Cancer	Human
Cycloserine	Alanine racemase	Tuberculosis	Bacterial
Fomepizole	Alcohol dehydrogenase	Alcoholism	Human
Disulfiram	Aldehyde dehydrogenase	Alcoholism	Human
Acarbose	α -Amylase	Diabetes	Human
Miglitol	α -Glucosidase	Diabetes	Human
Ethambutol	Arabinosyltransferase	Tuberculosis	Bacterial
Zileuton	Arachidonate 5-lipoxygenase	Inflammation	Human
Carbidopa	Aromatic L-amino acid decarboxylase	Parkinson disease	Human
Clavulanic acid	β -Lactamase	In combination with penicillins	Bacterial
Acetazolamide	Carbonate dehydratase (carbonic anhydrase)	Glaucoma	Human
Entacapone	Catechol O-methyltransferase	Parkinson disease	Human
Miglustat	Ceramide glucosyltransferase	Gaucher disease	Human
Methotrexate	Dihydrofolate reductase	Cancer	Human
Trimethoprim	Dihydrofolate reductase	Antibacterial	Bacterial
Sulfamethoxazole	Dihydropteroate synthase	Antibacterial	Bacterial
Topotecan	DNA topoisomerase	Cancer	Human
Ciprofloxacin	DNA gyrase	Antibacterial	Bacterial
Acyclovir	DNA-directed DNA polymerase	Antiviral (anti-HSV)	Viral
Rifampin	DNA-directed RNA polymerase	Antibacterial	Bacterial
Bacitracin	Dolichyl phosphatase	Antibacterial	Bacterial
Isoniazid	Fatty acid enoyl reductase	Tuberculosis	Bacterial
Oseltamivir	Viral neuraminidase	Anti-influenza	Viral
Fondaparinux	Factor Xa	Thrombosis	Human
Alendronate	Farnesyl-diphosphate farnesyltransferase	Osteoporosis	Human
Pyrazinamide	Mycobacterial fatty acid synthase	Tuberculosis	Bacterial
Valproic acid	Histone acetyltransferase	Seizures	Human
Nelfinavir	HIV protease	AIDS (anti-HIV)	Viral
Esomeprazole	H ⁺ /K ⁺ -ATPase	Gastroesophageal reflux disease	Human

(Continued)

TABLE 8.1 A Partial Listing of Enzyme Inhibitor Drugs Presently Used as Drugs (Continued)

Inhibitor (Drug)	Enzyme Inhibited	Use	Organism
Atorvastatin	HMG-CoA reductase	Hyperlipidemia	Human
Mycophenolate	IMP dehydrogenase	Immune suppression	Human
Propylthiouracil	Iodide peroxidase	Hyperthyroid	Human
Cilastatin	Renal dehydropeptidase	In combination with imipenem	Human
Eflornithine	Ornithine decarboxylase	Trypanosomes	Parasitic
Allopurinol	Xanthine oxidase	Gout	Human
Captopril	Peptidyl-dipeptidase A (angiotensin-converting enzyme)	Hypertension	Human
Pemetrexed	Phosphoribosylglycinamide formyltransferase	Cancer	Human
Aprotinin	Plasma kallikrein	Thrombosis	Human
Aminocaproic acid	Plasmin	Thrombosis	Human
Etodolac	Prostaglandin-endoperoxide synthase (cyclooxygenase)	Inflammation	Human
Bortezomib	Proteasome endopeptidase complex	Myeloma	Human
Imatinib	Protein-tyrosine kinase	Cancer	Human
Gemcitabine	Ribonucleoside-diphosphate reductase	Cancer	Human
Ribavirin	IMP dehydrogenase	Anti-viral (broad spectrum)	Viral
Azidothymidine	HIV reverse transcriptase	AIDS (anti-HIV)	Viral
Lamivudine	Reverse transcriptase	AIDS, hepatitis B	Viral
Penicillin	Serine-type D-Ala—Ala carboxypeptidase	Antibiotic	Bacterial
Digitoxin	Na ⁺ /K ⁺ -ATPase	Congestive heart failure	Human
Terbinafine	Squalene monooxygenase	Antifungal	Fungal
Itraconazole	Sterol 14 α -demethylase	Antifungal	Fungal
Lepirudin	Thrombin	Thrombosis	Human
Floxuridine	Thymidylate synthase	Cancer	Human
Orlistat	Triacylglycerol lipase	Obesity	Human
Metyrosine	Tyrosine 3-monooxygenase	Pheochromocytoma	Human
Fosfomycin	UDP-N-acetylglucosamine 1-carboxyvinyltransferase	Antibacterial	Bacterial
Aminoglutethimide	Monooxygenase	Breast cancer	Human
Acetohydroxamic acid	Urease	Gastritis	Bacterial
Dicumarol	Vitamin K-epoxide reductase	Thrombosis	Human

(Adapted from Robertson J. Mechanistic basis of enzyme-targeted drugs. *Biochemistry* 2005;44:5561–5771, with permission.)

enzymatic activity (i.e., ribozymes). Enzymes are the most efficient catalysts known in nature, because they have the ability to increase reaction rates by enormous factors. Like all catalysts, enzymes have the ability to lower the activation energy of reactions, and the tremendous catalytic power they possess results from their inherent ability to provide stabilization to the reacting molecules at their activated complex states. A preliminary enquiry shows that enzymatic rates for

reactions in aqueous solutions are limited by the diffusion rate constant so that the second-order rate constant of an enzyme ($k_{\text{cat}}/K_{\text{m}}$) is approximately 10^9 M s^{-1} (diffusion rate constant of water) (5). This implies that every collision of a reactant molecule (substrate) with the enzyme leads to product formation. Because for many enzymes, K_{m} (the Michaelis-Menten or apparent substrate–enzyme dissociation constant) values are in the micromolar or submicromolar range (10^{-4} M),

one can compute the k_{cat} (the catalytic or first-order rate constant or the number of molecules of substrate converted to product per unit time per enzyme) value to be approximately 10^5 . Estimates (6) for uncatalyzed reactions in water have ranged from 10^{-1} to 10^{-20} ; thus, the rate enhancements for enzyme-catalyzed reactions over the noncatalyzed reaction (also referred to as the proficiency of an enzyme) can range from 10^6 - to over 10^{25} -fold—truly remarkable proficiencies. Moreover, enzymes display great specificity toward particular chemical bonds (bond specificity; e.g., peptidases for peptide bonds) or functional groups (group specificity; e.g., esterases for esters), or they display absolute specificity toward a single molecule (e.g., carbonic anhydrase, which catalyzes the hydration of carbon dioxide). Furthermore, this specificity and catalytic proficiency are carried out, in most cases, at ambient temperatures and normal pressures in aqueous solutions. It is no wonder that enzymes have intrigued scientists for centuries.

Enzymes have been classified on the type of reaction catalyzed, and six major classes (families) of enzymes, numbered from 1 to 6, have been assigned by the Enzyme Commission (EC) of the International Union of Biochemistry and Molecular Biology (7). These classes are as follows: 1) oxidoreductases (e.g., dehydrogenases); 2) transferases (group transfer enzymes; e.g., kinases); 3) hydrolases (hydrolytic reactions; e.g., esterases); 4) lyases (formation or removal of double bonds; e.g., hydratase—addition of water across a double bond); 5) isomerases (e.g., mutarotation of glucose by mutases); and 6) ligases (joining of two substrates at the expense of energy, also referred to as synthetases). All discovered enzymes are identified by the prefix EC followed by an Arabic numeral based on the major class of reaction catalyzed, as indicated earlier. Furthermore, this is followed by a series of three more Arabic numerals, which indicate the subclass (functionality), sub-sub class (specific bond type), and serial number of the enzyme in that class, respectively. For example, the enzyme acetylcholinesterase has been given the following assignment: EC 3.1.1.7. As can be seen in this example, the numeral 3 indicates that this enzyme belongs to the family of hydrolases, the first 1 indicates that the nature of the bond being hydrolyzed is an ester, the second 1 indicates that specific ester bond is a carboxylic acid ester, and the last number is the serial number of this enzyme in this sub-subclass.

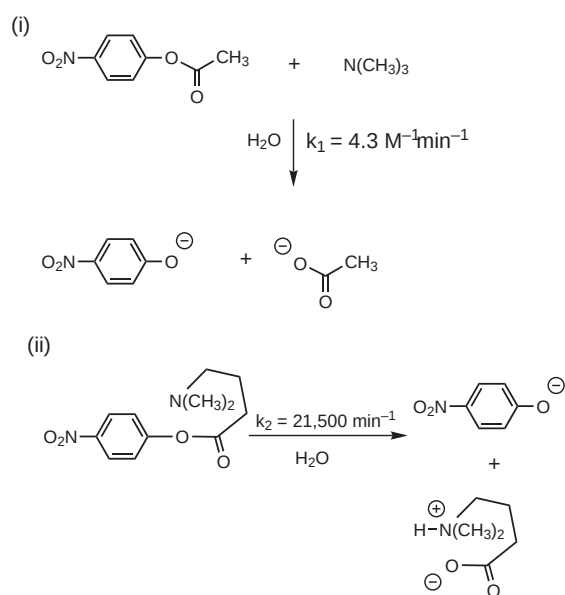
Many enzymes require cofactors, which enable them to carry out the catalysis. These small molecules (including ions) are intimately bound to the enzyme and are essential to the functioning of the enzyme. As macromolecular proteins, enzymes have the inherent ability to bind substrates. In effecting the transformation of these substrates to products (catalyzing the chemical reaction), enzymes use all the necessary tools in the chemical bonding arsenal to hold these substrates extremely tight in the active site as the reaction occurs. Because all chemical reactions require

covalent bond breakage and formation, the substrate must go through a transition state, or an “activated complex,” which is a destabilizing event, because bonds are being polarized and there is development of a partial charge. It is the inherent capability of the enzyme to greatly stabilize such activated complexes that give them their role in nature and allows their phenomenal catalytic power.

Pauling (8) first proposed the stabilization theory of the activated complex by enzymes. He concluded that the active site of the enzyme is complementary to the structure of the activated complex so that the binding of the enzyme to the activated complex is extremely tight. The ability to stabilize such complexes and correspondingly reduce the activation energy of the reaction and, thus, enhance the reaction rate is caused by many factors, both noncovalent as well as covalent (9). Noncovalent influences include entropic effects, such as proximity and orientation; restricted motion, where enzymes, by their involvement, have the inherent ability to bring reacting molecules closer together and in the correct orientation for bonds to form; desolvation effects, to strip solvent (water) molecules from the reactants; transition state electrostatic stabilization, to stabilize the partial charges being developed in the activated complex; induced-fit effects, in which the flexibility of the enzyme can accommodate the substrate, intermediate, and/or product; strain and distortion effects, to increase the reactivity of compounds; and many other such ancillary effects. Moreover, covalent effects also play a vital role in the enzyme’s catalytic role. Covalent effects imply a covalent bond being made by the enzyme (or its cofactor) to the substrate being transformed. These covalent effects would include general acid-base catalysis, in which amino acid side-chain groups partake in the overall biochemical reaction through proton donation (general acid) or proton abstraction (general base), as well as nucleophilic and electrophilic catalysis, in which there is bond formation with an amino acid side-chain group (or cofactor) to the substrate.

It should be mentioned that “covalent catalysis” has traditionally implied a group transfer reaction between one substrate and another being facilitated through an enzyme-group intermediate (e.g., the enzyme sucrose phosphorylase transfers a glucose moiety from sucrose to phosphate giving the products glucose-1-phosphate and fructose through an enzyme-glucosyl intermediate). Many amino acid side-chain groups (i.e., acidic and basic amino acids, e.g., aspartate, glutamate, histidine, lysine, arginine, tyrosine, and cysteine) as well as nucleophilic amino acid groups, such as cysteine, allow such covalent effects. Recent analysis of enzyme rate enhancements has postulated that noncovalent effects allow an increase of as much as 11-fold over the noncatalyzed reaction, whereas for those enzymes with rate enhancements greater than 11-fold over noncatalyzed reactions, covalent catalysis in the transition state accounts for this exceptional increase in rate enhancement (10).

Noncovalent effects like proximity and orientation may be explained on the basis that enzymes have the inherent ability to affect the order of reactions by the “effective concentration” principle—for example, to change a second-order reaction to a first-order one by bringing the reacting molecules closer to each other so that there is a lower amount of entropic loss (restriction of molecular motion) for the reacting substrates. In the example shown in Figure 8.3, which is taken from physical-organic chemical studies (11), one can consider the hydrolysis of a nitrophenyl ester by an amine in two scenarios: reaction (i), with two individual molecules reacting (i.e., amine and ester), versus reaction (ii), with a single molecule having an “in-built” amine on the ester molecule. If the reaction rate constants are compared [one needs to keep in mind that we are comparing a first-order rate constant for (ii), with a second-order rate constant for (i), which may not be a true comparison, because mechanisms are likely to be different], however, one finds that the first-order rate reaction [reaction (ii)] has a rate enhancement over reaction (i) by a factor of approximately 5,000 M attributed to the fact that reaction (i) needs to give up more degrees of freedom (entropy) [as compared to reaction (ii)] to react (i.e., form productive collision complexes, leading to product formation). This proximity effect indicates that by using such a system (bringing the reacting centers closer to each other), one can effectively increase the concentration of the reactants by this factor, resulting in a faster reaction. This enhancement factor of 5,000 M (i.e., effectively changing a second-order reaction to that of a first-order one) has been termed the “effective



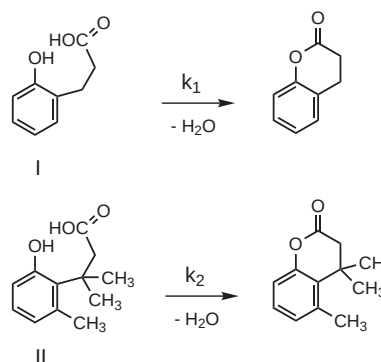
Effective concentration = $k_2/k_1 = 5,000 \text{ M}$ (approx.)

FIGURE 8.3 Concept of proximity and effective concentration.

concentration,” because this is an unrealistic increase in the concentration of the reactants, which gives rise to the higher rate constant (i.e., it is impossible to make a solution of 5,000 M of the reactants).

The next example, which is shown in Figure 8.4, illustrates that besides the proximity factor, one also can use “orientation” effects to bring about an increase in the reaction rate constant (12). For example, as shown in Figure 8.4, one may use alkyl groups to sterically hinder rotation about single bonds, effectively freezing the molecule in a particular conformation to provide maximum orbital overlap for bonding. Thus, using a “dimethyl lock” system in molecule II, which restricts rotation, ensures that lactonization for molecule II is faster by a factor of 4×10^4 than molecule I. Both of these effects, proximity and orientation, are thought to be part of an enzyme’s arsenal of tools in allowing enzymes to lower the activation energy for reactions. Other noncovalent effects, such as desolvation of the reactant molecules, can also be effectively achieved by enzymes. Enzymes have the capacity, by lining their exterior and/or interior surfaces with appropriately situated hydrophobic amino acids (e.g., phenylalanine, valine, tryptophan, leucine), to effectively strip away water molecules from substrates as they enter into the active site of the enzyme through such channels. Thus, no further expense of energy is required to desolvate the substrate before the reaction.

As previously mentioned, enzymes also use covalent chemistry as a means to effect catalysis. Indeed, it has been postulated that covalent chemistry plays a far greater role in enzyme catalysis than previously thought (10). Nucleophilic catalysis by hydrolytic enzymes, such as the serine proteases or esterases, are classic examples of covalent effects in enzyme catalysis. In such systems, for example, as in the case of the serine protease chymotrypsin, which hydrolyzes peptide bonds containing aromatic amino acids (e.g., phenylalanine and tyrosine), the covalent effects occur at the level of general acid, general base, and nucleophilic covalent catalysis.



Relative rate, $k_2/k_1 = 4 \times 10^4$

FIGURE 8.4 Concept of orientation.

Figure 8.5 illustrates an accepted mechanism for such enzymes. It should be noted that all serine proteases contain a “catalytic triad” (a set of three amino acid groups in the active site that function together and are directly involved in catalysis), designated as Ser-195, His-57, and Asp-102 (numerals represent the amino acid position in the protein primary structure), which are present in the active sites of these proteases. The catalysis is effected by making the hydroxyl group of the Ser more nucleophilic for attack at the carbonyl center of the peptide bond. One should recall that, in general, hydroxyl groups have pK_a values in the range of greater than 14 and, as such, are not acidic and unable to ionize at physiologic pH. However, because of the catalytic triad of amino acids, the proton from the Ser-OH group is transferred to the Asp moiety through the His group in a “charge relay system” such that the Ser-OH group is made into the highly nucleophilic alkoxide ion. This is achieved by the Asp group acting as a general base to pick up the proton from His, which can also abstract a proton from the Ser-OH group (see I in Fig. 8.5). Thus, His behaves as a tautomeric catalyst in this enzyme (i.e., acts as both a general acid and a general base) and, in essence, relays the proton from Ser to Asp. The Ser (as its alkoxide) is now a much more powerful nucleophile and can attack the peptidyl carbonyl group to generate a “tetrahedral oxyanion intermediate” (see II in Fig. 8.5), which collapses to liberate a new amino terminus of the peptide and the acylated Ser enzyme (see III in Fig. 8.5). The next part of the reaction involves a water molecule (which also is made more nucleophilic by a similar mechanism; see III in Fig. 8.5) that goes on to hydrolyze, through a tetrahedral intermediate (see IV in Fig. 8.5), the Ser-acyl bond

to liberate the new carboxyl terminus of the peptide ($R_1\text{COOH}$) and the free enzyme, which can be recycled for another round of catalysis.

Knowledge of the mechanisms and interactions, both noncovalent and covalent, that allow enzymes to function as such efficient catalysts provides the medicinal chemist with insights to design molecules that achieve selective inhibition of the enzyme. Such knowledge also paves the way to the design and discovery of drugs.

GENERAL CONCEPTS OF ENZYME INHIBITION

The body is composed of thousands of different enzymes, many of them acting in concert to maintain homeostasis. Although disease states may arise from the malfunctioning of a particular enzyme, or the introduction of a foreign enzyme through infection by microorganisms, inhibiting a specific enzyme to alleviate a disease state is a challenging process. Most bodily functions occur through a cascade of enzymatic systems, and it becomes extremely difficult to design a drug molecule that can selectively inhibit an enzyme and result in a therapeutic benefit. However, to address this problem, the basic mechanism of enzyme action needs to be understood. Once knowledge of a particular enzymatic pathway is determined and the mechanism and kinetics are worked out, the challenge is then to design a suitable inhibitor that is selectively used by the enzyme causing its inhibition.

As outlined earlier, enzymes (E) represent the best-known biochemical catalysts, because they are uniquely designed to carry out specific biochemical reactions in a highly efficient manner (5). They initially act by binding a substrate (S) to form an enzyme–substrate complex [E·S],

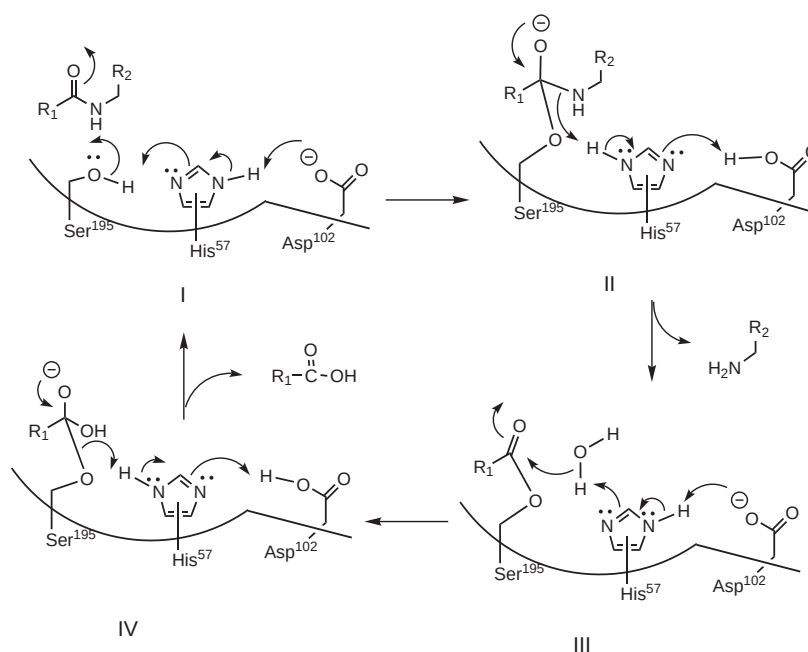


FIGURE 8.5 Acid, base, and nucleophilic covalent catalysis by chymotrypsin.

which undergoes specific chemistry (catalysis) to give the enzyme–product complex $[E \cdot P]$, followed by dissociation of product (P) and liberating the free enzyme (E). Equation 8.8.1 represents a simplified version of this scenario:

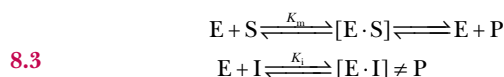


where K_d is the enzyme–substrate dissociation constant and k_{cat} represents the rate constant for the catalytic step (chemical modification step or slowest step in the overall pathway). If the binding step of $E + S$ to form $[E \cdot S]$ is relatively fast as compared to the catalytic step and one assumes steady-state conditions, then K_m [the substrate concentration at half-maximum velocity ($V_{\text{max}}/2$)], may be equated to the K_d , as shown in Equation 8.2: where v = velocity/rate of the reaction

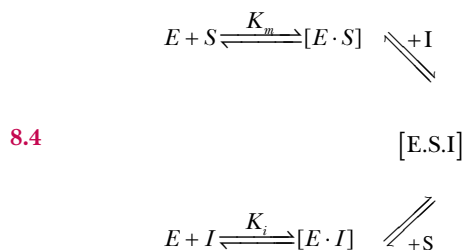
$$8.2 \quad \begin{aligned} \text{Michaelis-Menten equation: } v &= \frac{V_{\text{max}}[S]}{K_m + [S]} \\ \text{Lineweaver-Burk equation: } \frac{1}{v} &= \frac{K_m}{V_{\text{max}}} \cdot \frac{1}{[S]} + \frac{1}{V_{\text{max}}} \end{aligned}$$

The rate of the reaction can then be derived in terms of K_m and V_{max} (or $k_{\text{cat}} = V_{\text{max}}/[E]$). From the knowledge of the dissociation constant ($K_{m(d)}$) and the rate constant for catalysis (k_{cat}), it is then possible to compare inhibitors and the dissociation constant for the inhibitors, K_i , in relation to the natural substrates and the effect on the catalytic rates. These kinetic parameters, k_{cat} and $K_{m(d)}$ (or K_i) can then give an indication as to the affinity (K_i versus $K_{m(d)}$) and specificity (k_{cat}/K_i or $k_{\text{cat}}/K_{m(d)}$) of the inhibitor for a particular enzyme. Equations 8.3 and 8.4 represent the general scheme of reversible inhibition, and Figure 8.6 illustrates graphically and mathematically the relationship of the velocity of the enzyme reaction to the substrate $[S]$ and inhibitor $[I]$ concentration as well as the kinetic parameters K_m , K_i , and V_{max} (or $k_{\text{cat}} = V_{\text{max}}/[E]$):

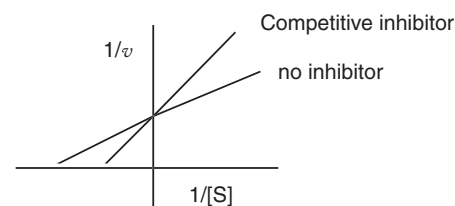
Competitive Inhibition



Noncompetitive Inhibition

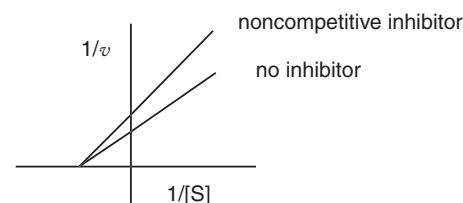


Inhibition of enzymes may be broadly classified under two categories—reversible and irreversible inhibitors—as shown in Equation 8.5:



$$v = \frac{V_{\text{max}}}{1 + (K_m/[S]) (1 + [I]/K_i)}$$

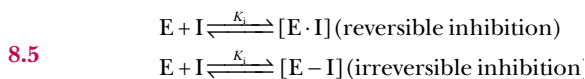
(K_m increases, V_{max} unchanged)



$$v = \frac{V_{\text{max}} / (1 + [I]/K_i)}{1 + (K_m/[S])}$$

(K_m unchanged, V_{max} decreases)

FIGURE 8.6 Graphic representation of competitive and noncompetitive enzyme inhibition.



In the presence of inhibitor, the enzyme–substrate complex, $[E \cdot S]$, is replaced by the enzyme–inhibitor complex, $[E \cdot I]$, which may block or retard the formation of product. In the presence of a reversible inhibitor, the enzyme is tied up and the reaction retarded or stopped; however, the enzyme can be subsequently regenerated from the enzyme–inhibitor complex, $[E \cdot I]$, to react again with substrate and produce product (see Eqs. 8.3 and 8.4). On the other hand, irreversible inhibition implies that the enzyme cannot be regenerated, and the only way for catalysis to proceed would be if new molecules of the enzyme are generated from gene transcription and translation. Irreversible inhibition is commonly associated with covalent bond formation between inhibitor and enzyme $[E - I]$, which *cannot* be easily broken and is often defined as a time-dependent loss of enzyme activity. Reversible inhibition, on the other hand, does not necessarily imply noncovalent bond formation. In many instances, reversible inhibition can occur through covalent bond formation, but these bonds can be hydrolyzed to regenerate free enzyme and inhibitor. Thus, for a reversible enzyme inhibitor, there is no time-dependent loss of activity, and enzyme activity can always be recovered. There are instances when reversible inhibition tends to look kinetically like irreversible inhibition. This scenario results whenever there is a tight binding of a reversible inhibitor to the enzyme; consequently, the

dissociation of the enzyme from this enzyme–inhibitor complex is extremely slow. Kinetically, it is extremely difficult to distinguish this type of inhibition from an irreversible inhibitor, because over time, the enzyme does tend to look like it loses its activity and, for all practical purposes, the enzyme behaves as if it were irreversibly tied up. To differentiate between tight-binding reversible and irreversible inhibitors, one can dialyze the enzyme–inhibitor complex. In case of the reversible inhibitor, on dialysis, the inhibitor will be removed from the enzyme, resulting in recovery of the enzyme activity; however, this does not occur with the irreversible inhibitor.

Reversible Enzyme Inhibition

Reversible enzyme inhibition may be classified under two main headings, competitive and noncompetitive, with both following Michaelis-Menten kinetics. Competitive inhibition, by definition, requires that the inhibitor competes with the substrate for binding to the enzyme at the active site, and this binding is mutually exclusive. That is, if the inhibitor binds to the enzyme, the substrate will not be able to bind, and vice versa. However, competitive inhibition also suggests that the inhibition can be reversed in the presence of saturating amounts of substrate, because in this case, all enzyme active sites will be occupied by substrate-displacing inhibitor. In contrast, noncompetitive inhibition implies independent binding (i.e., both inhibitor and substrate may bind to the enzyme at different sites). Because binding of the inhibitor to the enzyme is at a site other than the active site, noncompetitive inhibition cannot be reversed by increasing the concentration of substrate. Graphing the kinetics of inhibition as shown in Figure 8.6, the Lineweaver-Burk plot of $1/V$ versus $1/[S]$ shows distinguishing characteristics between the two types of inhibition. In competitive inhibition, there is no change in the $V_{\max}$ of the reaction (the intercept on the y-axis remains constant in the presence of inhibitor). However, the slope of the curve ($K_m/V_{\max}$) is different with the inhibitor present, and K_m changes because of the presence of the competitive inhibitor (Fig. 8.6, competitive inhibition). In the case of noncompetitive inhibition, only the $V_{\max}$ of the reaction decreases, while the K_m remains unchanged (intercept on the x-axis unchanged with inhibitor; Fig. 8.6, noncompetitive inhibition).

Most of the rationally designed and clinically useful reversible inhibitors are competitive inhibitors. Table 8.1 lists several currently approved drugs that act as enzyme inhibitors. The majority of enzyme inhibitors generally bear some structural resemblance to the natural substrate of the enzyme. The design of such inhibitors would thus seem to be a logical and rational task that is uniquely suited to the medicinal chemist who can use the principles of bioisosteric modification of natural enzyme substrates and metabolites, or modification of “lead” structures and structure–activity relationships (SARs), to create selective and potent inhibitors (see Chapter 2). However, there are pitfalls in this endeavor because even

the most rationally designed drug must still overcome transport and other cellular barriers before exerting its effects. In the case of the noncompetitive inhibitors, the design is not as straightforward. These inhibitors can have widely differing structures, which in many instances bear no resemblance to the natural substrate. In general, inhibitors of the noncompetitive type have been obtained primarily through random screening of chemically novel molecules followed by further synthetic manipulation of the pharmacophore to optimize their inhibitory effects.

Examples of Reversible Inhibitors

The design of enzyme inhibitors has included random screening of synthetic chemical agents, natural products, and combinatorial libraries followed by molecular optimization or SARs of so-called “lead” structures as well as bioisosteric analogs of the enzyme substrates themselves. Drugs (e.g., finasteride) also have been developed for one indication but, based on observed side effects, have led to other uses.

The rational approach in the design of enzyme inhibitors is greatly aided if the enzymatic reaction is characterized in terms of its kinetic mechanism. Such a characterization would include the knowledge of the kinetic parameters (rate constants and dissociation constants) of individual steps in the overall reaction pathway as well as the characterization of (any) intermediates involved in these individual steps. Examples of such “rational” inhibitors include both reversible and irreversible inhibitors of enzymes.

ANTIMETABOLITES Antimetabolites are agents that interfere with the functioning of an essential metabolite and that most often are designed as structural analogs of the natural metabolite. As described earlier, the mechanism of action of sulfanilamide is that of a competitive inhibitor of *p*-aminobenzoic acid. However, in the case of the sulfanilamide, the mechanism was only determined after the bacterial inhibitory action was noted. This is often the case when a drug is discovered to have a certain therapeutic effect and, later, this effect is “rationalized” as being caused by an enzyme-inhibitory action (1). Other classic examples of a competitive inhibitor acting as an antimetabolite include a number of nucleoside analogs used as antiviral and anticancer agents. These agents bear

USE OF FINASTERIDE

Finasteride (Proscar) an inhibitor of steroid 5α -reductase, an enzyme involved in the catalytic reduction of testosterone to dihydrotestosterone, was originally developed as an agent to treat prostate hyperplasia. In addition to this original use, finasteride is also indicated as an agent (Propecia) to stimulate hair growth for treatment of male pattern baldness. This benefit was recognized as a useful side effect during clinical trials of finasteride as an antiprostata agent.

structural resemblance to natural nucleosides, which in their triphosphate form are substrates for nucleic acid polymerases involved in the synthesis of nucleic acids. Nucleic acid polymerases catalyze the condensation of the free 3'-hydroxy end of a nucleic acid with an incoming 5'-triphosphate derivative of a nucleoside (deoxynucleotide triphosphate [dNTP]), resulting in a 3',5'-phosphodiester linkage. Hence, nucleoside analogs, to compete with the natural substrate in the synthesis of nucleic acid, must be converted intracellularly to their mono-, di-, and finally, triphosphate derivative before exerting their inhibitory effects on nucleic acid synthesis. Certain drug design strategies incorporate a "masked" phosphate group on the nucleoside such that once absorbed, they enter into the systemic circulation as the monophosphate (13). The majority of these analogs are designed such that they lack the 3'-hydroxy group and are dideoxy derivatives of the natural substrate. Thus, these analogs ensure that once they are incorporated into a nucleic acid, further extension of the nucleic acid is prevented because of the lack of a 3'-hydroxy group.

Inhibition of HIV-reverse Transcriptase

Azidothymidine (AZT). The advent of AIDS stimulated a great interest in designing inhibitors against the essential viral polymerase—HIV-reverse transcriptase (HIV-RT). AZT is a potent inhibitor of HIV-RT (14), the retroviral polymerase that catalyzes the formation of proviral DNA from viral RNA. Structurally, AZT is similar to the natural nucleoside thymidine but has an azide group ($-N_3$) rather than a hydroxy group ($-OH$) at the 3'-position of the sugar deoxyribose (Fig. 8.7). AZT is activated intracellularly to its triphosphate and competes with thymidine triphosphate for uptake by HIV-RT into DNA (15). Once incorporated, further chain extension of the DNA is prevented, because there is no 3'-hydroxyl group to continue the DNA synthesis. In this fashion, AZT is an effective chain terminator of viral DNA synthesis.

Dideoxycytidine (Zalcitabine) and 3-thiacytidine (Lamivudine). 2',3'-Dideoxycytidine (ddC) is another antiretroviral agent used against HIV-RT. In this case, ddC resembles the natural metabolite, deoxycytidine (dC), and as in the case of AZT, it is a 3'-deoxy analog of dC, where the 3'-OH group of dC is replaced by a hydrogen atom. Similarly, 3-thiacytidine (3-TC) is another anti-HIV agent that resembles dC. However, in this example, rather than replacing the 3'-hydroxyl functionality as in ddC, the 3'-carbon position of the sugar has been substituted by a sulfur atom.

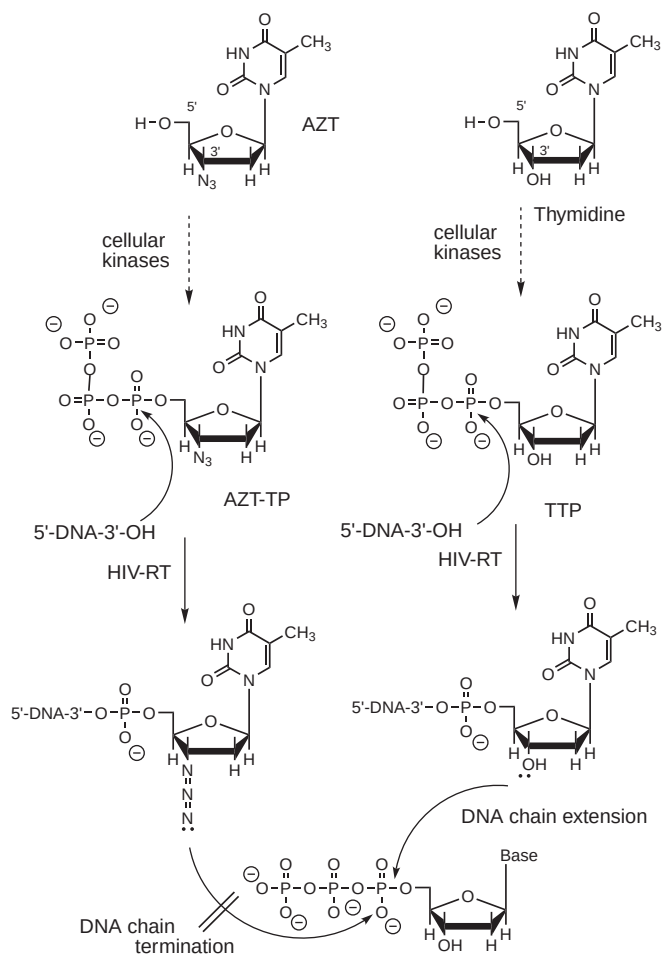
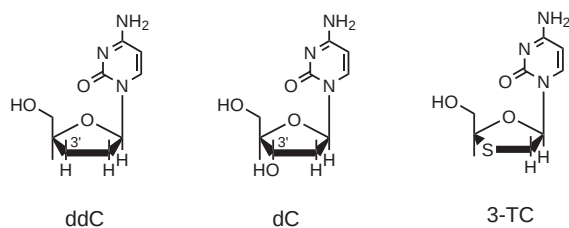
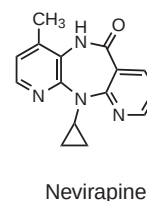


FIGURE 8.7 Activation, incorporation, and chain-terminating action of azidothymidine (AZT), a thymidine analog, as a reversible inhibitor of human immunodeficiency virus–reverse transcriptase (HIV-RT).

Nevirapine. An example of a potent noncompetitive inhibitor of HIV-RT is the drug nevirapine, a benzodiazepine analog (16), which is extremely tight-binding to the enzyme, having a K_i in the nanomolar range.



As can be seen in the structure of nevirapine, the drug bears no resemblance to any of the natural nucleotide substrates and was discovered through a random screening program. X-ray crystallographic studies of HIV-RT complexed with nevirapine have shown it binding in a hydrophobic pocket at a site adjacent to and slightly overlapping the nucleotide-binding site of HIV-RT (17). Kinetic studies with the enzyme have revealed an

HISTORIC DEVELOPMENT OF AZT

Interestingly, AZT was originally synthesized as an anticancer agent to inhibit cellular DNA synthesis, but it was found to be too toxic. Subsequently, in the mid-1980s, during a random screening of nucleoside agents for potential inhibitory effects against HIV-RT, AZT was found to have selectivity for the HIV-RT (14). As such, its effects on host cell polymerases result in its dose-limiting bone marrow toxicity.

extremely slow binding rate for the drug; however, once bound, the polymerization rate for the reaction is effectively reduced (18).

However, a drawback for nevirapine is that the virus can develop resistance very rapidly, through mutation of the amino acid groups in the binding pocket (19). Thus, its usefulness is limited to combination therapy with other antiretroviral agents rather than single-drug therapy.

Reversible Inhibitors Used in Cancer Therapy. The design of several anticancer agents have been based on the antimetabolite theory. Because cancer results in overproliferation and uncontrolled cell growth, antimetabolite drugs designed against cancer have been based on inhibiting DNA synthesis in the cell (see Chapter 37). Thus, these drugs have been targeted against those enzymes, including the nucleic acid polymerases, thymidylate synthase and dihydrofolate reductase (DHFR), which have a role in DNA synthesis. Examples of drugs that have been designed against nucleic acid polymerases include cytosine arabinoside (Ara-C) and 5-fluorouracil (5-FU) (Fig. 8.8). Ara-C is first converted to its triphosphate, and as such, it functions as an antimetabolite of deoxycytidine triphosphate to inhibit DNA polymerase. As can be seen from its structure, Ara-C is the arabino isomer of cytidine, where the 2'-hydroxyl group in Ara-C is in the arabino configuration rather than the ribo configuration of cytidine. Because of this stereochemical change in the placement of the 2'-hydroxyl group, Ara-C resembles more dC rather than cytidine. In this way, Ara-C inhibits DNA polymerases by competing with dC. 5-FU is an analog of the pyrimidine base uracil, where the hydrogen at the 5-position in uracil has been substituted by the bioisosteric fluorine (F) atom. This makes 5-FU look very similar to uracil. 5-FU, after conversion to 5-fluorodeoxyuridine monophosphate (FdUMP), becomes an inhibitor of thymidylate synthase, the enzyme involved in the de novo synthesis of thymidylate. In this case, FdUMP is an antimetabolite of deoxyuridine monophosphate.

Methotrexate is a potent inhibitor of DHFR, the enzyme responsible for the reduction of folic acid to dihydro- and tetrahydrofolic acid, precursors to one-carbon donation in purine and pyrimidine de novo synthesis. Methotrexate is an analog of folic acid where the 4-hydroxyl group (–OH) on the pteridine ring of

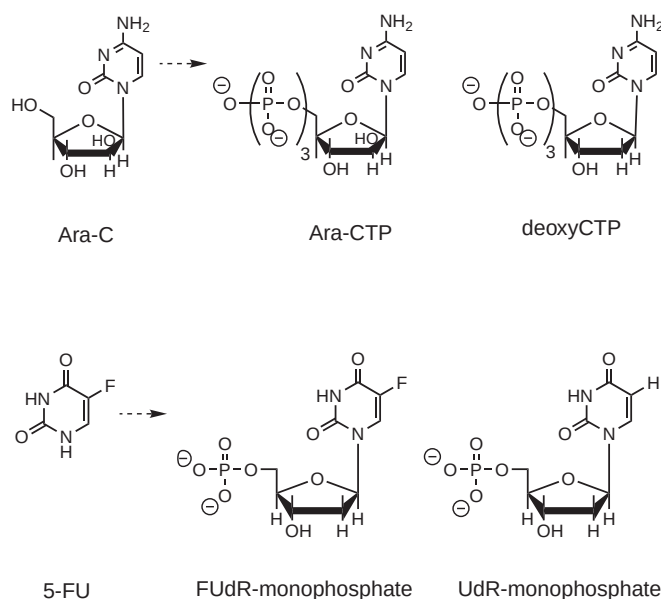


FIGURE 8.8 Structures of pyrimidine antimetabolites used in cancer chemotherapy.

folic acid has been replaced by an amino (–NH₂) group and the nitrogen atom at the 10-position is methylated (Fig. 8.9). These substitutions led to methotrexate having an affinity for DHFR orders of magnitude greater than that for the natural metabolite, folic acid, and allows it to be an extremely potent inhibitor of DHFR.

INHIBITION OF ACETYLCHOLINESTERASE Acetylcholinesterase (AChE) is the enzyme that catalyzes the catabolism of the neurotransmitter acetylcholine to acetate and choline (see Chapter 9). Thus, inhibition of AChE would lead to increased concentrations of acetylcholine in both of the cholinergic muscarinic and nicotinic synapses resulting in a prolonged cholinergic action. Inhibitors of AChE have found use in cases of myasthenia gravis, glaucoma, and Alzheimer disease. To appreciate the design of these inhibitors, it is useful to first understand the mechanism

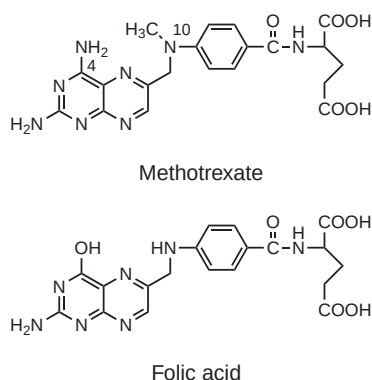


FIGURE 8.9 Methotrexate, the antimetabolite of folic acid.

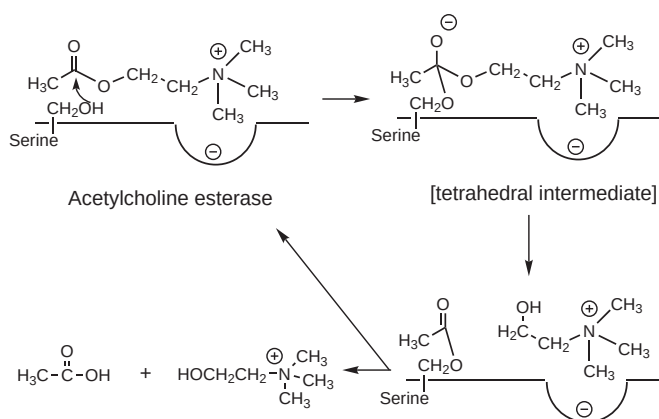


FIGURE 8.10 Mechanism of hydrolysis of acetylcholine by acetylcholine esterase.

of action of AChE. AChE has an anionic active site that can bind the positively charged quaternary ammonium group of the choline functionality and an active esteratic site that contains a nucleophilic $\text{Ser}-\text{CH}_2\text{OH}$ group that is involved in the hydrolysis of the ester bond as shown in Figure 8.10. The hydrolysis mechanism involves attack of the nucleophilic $\text{Ser}-\text{CH}_2\text{OH}$ group on the ester carbonyl group of acetylcholine to form a tetrahedral intermediate that collapses, releasing choline and forming an acetylated $\text{Ser}-\text{CH}_2\text{O}-\text{AChE}$ complex, which subsequently hydrolyzes with water to release AChE, acetic acid (as acetate ion), and choline.

Physostigmine has been used in the treatment of glaucoma. It is an alkaloid with a carbamate moiety that resembles the ester linkage of acetylcholine. Being an alkaloid, it is protonated at physiologic pH and, thus, can bind to the anionic site of AChE. Similar to the mechanism for the hydrolysis of acetylcholine by AChE, the $\text{Ser}-\text{CH}_2\text{OH}$ group of AChE can attack the carbamate carbonyl group of physostigmine, and in the process, the $\text{Ser}-\text{CH}_2\text{OH}$ group is carbamylated as shown in Figure 8.11. This carbamyl $\text{Ser}-\text{CH}_2\text{O}-\text{AChE}$ intermediate is more stable to

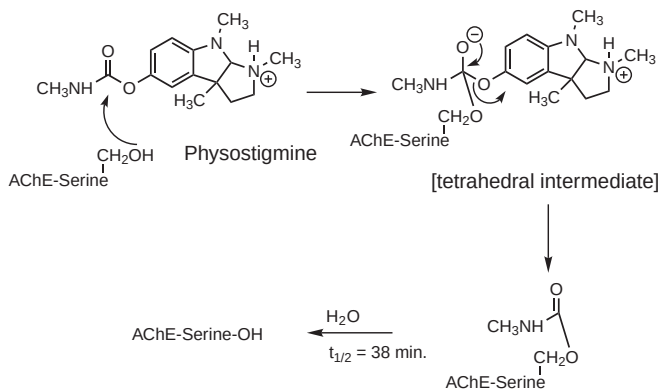


FIGURE 8.11 Mechanism of inhibition of acetylcholine esterase by physostigmine.

enzymatic hydrolysis than is acetylated $\text{Ser}-\text{CH}_2\text{O}-\text{AChE}$, and its subsequent hydrolysis by water occurs extremely slowly. The carbamylated AChE intermediate is slowly regenerated with a half-life of 38 minutes—more than 10 million-fold (10^7) slower than that for AChE's natural substrate, acetylcholine. This is an example of a reversible inhibitor involved in covalent bond formation with the enzyme that ultimately gets hydrolyzed.

INHIBITORS OF ANGIOTENSIN-CONVERTING ENZYME Angiotensin-converting enzyme (ACE) is a carboxypeptidase having a zinc ion as a cofactor and is involved in the renin-angiotensin cascade of blood pressure control (20). The design of the antihypertensive drug captopril, a clinically important and potent reversible inhibitor of ACE, is an example of one of the early successes of a rationally designed enzyme inhibitor (21) (see Chapter 23). The design of captopril was based on several factors. These included knowledge that ACE was similar in its enzymatic mechanism to carboxypeptidase A, except that ACE cleaved off a dipeptide whereas carboxypeptidase A cleaved single amino acid groups from the carboxyl end of the protein. The discovery of benzy succinic acid as a potent inhibitor of carboxypeptidase A and studies of a potent pentapeptide inhibitor of ACE, $\text{BPP}_{5\alpha}$ (Glu-Lys-Trp-Ala-Pro), from the venom of the Brazilian viper (*Bothrops jararaca*) showed that the N-terminal peptide fragments, including tetra-, tri-, and dipeptide fragments (Ala-Pro) of $\text{BPP}_{5\alpha}$, retained some inhibitory activity. Benzy succinic acid has been described as a bi-product inhibitor of carboxypeptidase A, wherein its design was based on the combination of the products of the peptidase reaction (i.e., the two peptide fragments, one with a free carboxyl end that coordinates the zinc ion of carboxypeptidase A and the other with a free amino terminus) (Fig. 8.12) (22). In the case of benzy succinic acid, the amino ($-\text{NH}_2$) functionality is replaced by the bioisosteric methylene ($-\text{CH}_2-$) group. Using the above concepts, it was rationalized that succinyl amino acids could behave similarly as bi-product inhibitors of ACE. Starting with a succinyl-proline moiety, the SAR developmental effort finally resulted in captopril with the substitution of the stronger zinc coordinating mercapto functionality in place of the carboxylic acid moiety of succinic acid and a stereospecific R methyl group on the succinyl moiety to represent the methyl group on the natural L-Ala group in L-Ala-Pro (the dipeptide fragment that had previously shown inhibitory activity). Captopril soon became clinically successful as an antihypertensive agent and, especially in combination with diuretics, in controlling hypertension.

Following on the heels of captopril was another bi-product ACE inhibitor, enalaprilat (Fig. 8.12). Enalaprilat incorporated a phenylethyl moiety with the S-configuration and made use of a hydrophobic binding pocket in ACE that was overlooked during the design of captopril (23). Recalling that the tripeptide fragment of $\text{BPP}_{5\alpha}$ (L-Trp-L-Ala-L-Pro) contained the aromatic tryptophan (Trp) group and

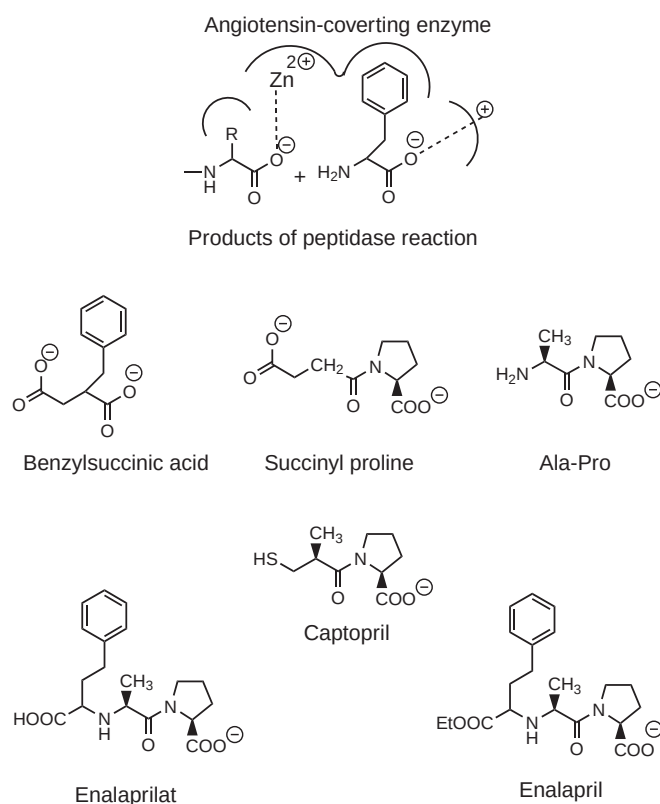


FIGURE 8.12 Angiotensin-converting enzyme (ACE) inhibitors and efforts that led to their development.

showed weak inhibitory properties suggested the benefit of an aromatic binding site. Substituting the tryptophan moiety with a phenyl group allowed the design of enalaprilat, which retained the carboxylic group as the coordinating ligand for zinc, which resulted in a 20-fold increase in antihypertensive potency over captopril. However, the diacid enalaprilat as its dianion was poorly absorbed from the gastrointestinal tract; thus, a prodrug monoethyl ester of enalaprilat, called enalapril, was developed. Enalapril had superior pharmacokinetics to enalaprilat and was rapidly metabolized to the active drug.

Transition-State Analogs

Transition-state analogs are compounds that resemble the substrate portion of the hypothetical transition state of an enzymatic reaction. All chemical reactions progressing from substrate to product must cross an energy barrier and proceed through a transition state or activated high-energy complex. This energy barrier is described as the activation energy. In the case of enzyme-catalyzed reactions, it is accepted that the enzyme reduces this energy barrier as compared to the nonenzyme-catalyzed reaction. Factors contributing to this reduced energy barrier are several and include stabilization of the transition state and intermediate forms of the reaction during the course of transition from substrate to product as well as conformational effects of distortion of the substrate

while traversing toward the product (5). In 1948, Pauling (8) initially suggested that compounds resembling the transition state of an enzyme-catalyzed reaction would be effective inhibitors of the enzyme, because the substrate transition state should have the greatest affinity for the enzyme. Later, Wolfenden (24) proposed that thermodynamically, it is possible to relate the hypothetical equilibrium dissociation constants between substrate and its transition state of an enzyme-catalyzed reaction with that of the nonenzyme-catalyzed one. Using such an analysis, he showed that the ratio of the hypothetical transition-state dissociation constants of the nonenzyme-catalyzed reaction to that of the enzyme-catalyzed one is equal to the ratio of the first-order rate constants of formation of transition state for enzyme-catalyzed reaction to noncatalyzed reaction. Because the ratio of the enzyme-catalyzed rate constant to that of the noncatalyzed one ranges from 10^7 to 10^{10} , it follows that the substrate transition state would bind the enzyme 10^7 - to 10^{10} -fold more tightly than the substrate itself (25). Hence, transition-state analogs that resemble the substrate would be extremely tight-binding compounds. To design a transition-state inhibitor, knowledge of the enzyme chemistry and its mechanism is a basic requirement. However, it must be understood that these substrate transition states are, by nature, unstable transient species existing for no longer than a few picoseconds. Nevertheless, experimental evidence has shown that even crudely designed transition-state inhibitors resembling the substrate are extremely potent inhibitors (25).

TRANSITION-STATE INHIBITOR OF ADENOSINE DEAMINASE Adenosine deaminase hydrolyzes adenosine (or deoxyadenosine) to inosine (or deoxyinosine) and is important for purine metabolism (Fig. 8.13). High levels of adenosine are toxic to B cells of the immune system and can result in an immunocompromised state. People who also lack the gene for adenosine deaminase have the genetic condition of severe combined immunodeficiency and are extremely susceptible to opportunistic infections. Many cancer and antiviral agents are also degraded by this enzyme; hence, there is a role for the development of inhibitors of this enzyme (26). The mechanism proposed for adenosine deaminase is nucleophilic attack of water at the 6-position of adenosine to form a tetrahedral intermediate (Fig. 8.13). The transition state presumably resembles this intermediate.

During the course of the deaminase reaction, the hybridization of the 6-carbon changes from an sp^2 -hybridized state to an sp^3 state. Subsequently, there is a loss of ammonia to form the product inosine. To develop a transition-state inhibitor for this enzyme, one would have to factor in this change in the hybridization of the substrate molecule; thus, molecules having an sp^3 -hybridized carbon at this position and resembling the substrate would potentially be candidates for transition-state inhibitors. The compound 1,6-dihydro-6-hydroxymethylpurine has such geometry, and its potent

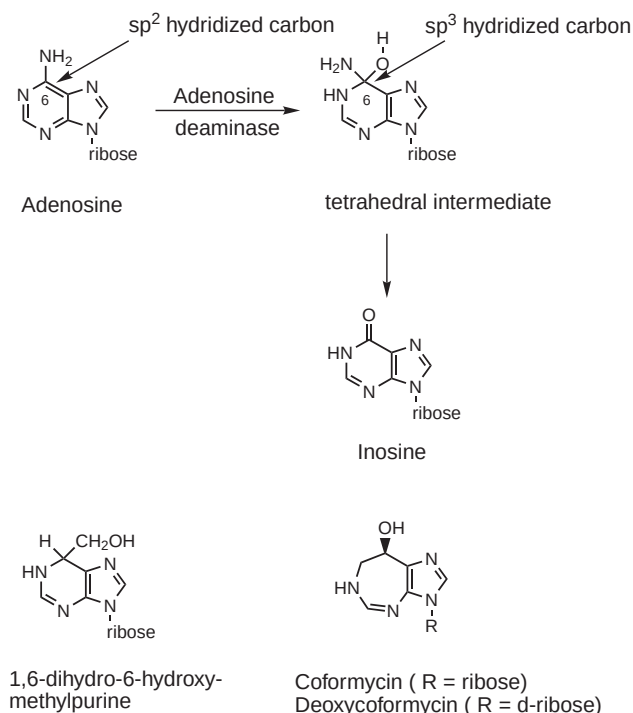


FIGURE 8.13 Mechanism and transition-state inhibitors of adenosine deaminase.

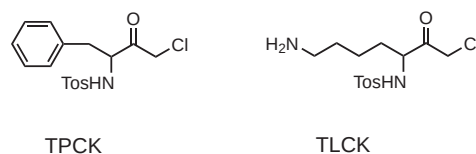
inhibitory properties of adenosine deaminase ($K_i < 1 \mu\text{M}$, as compared to a K_m for adenosine of 31 M) have been rationalized as being transition-state inhibitors (27). Two compounds that nature has provided, coformycin and its deoxyribose analog, deoxycoformycin (Fig. 8.13), are extremely potent inhibitors of adenosine deaminase ($K_i = 2 \text{ pM}$). Both of these compounds contain a seven-member ring structure, which through its flexibility is presumed to resemble the hypothesized distorted sp^2 – sp^3 transition state that forms during the addition of water to adenosine (28).

Irreversible Enzyme Inhibition

As previously described, irreversible enzyme inhibition is defined as “time-dependent inactivation of the enzyme,” which implies that the enzyme has been permanently modified, because it can no longer carry out its catalytic function. This modification is the result of a covalent bond being formed between the inhibitor and an amino acid group in the protein. Furthermore, the covalent bond is extremely stable and, for all practical purposes, is not hydrolyzed to give back the enzyme in its original state or structure. In most examples of irreversible inhibition, a new enzyme must be generated through gene transcription and translation for the enzyme to continue its normal catalytic action. Basically, there are two types of irreversible enzyme inhibitors, the affinity labels or active site-directed irreversible inhibitors and the mechanism-based irreversible enzyme inactivators.

Affinity Labels and Active Site-Directed Irreversible Inhibitors

The affinity labels are those chemical entities that are inherently reactive and can target any nucleophilic group in the enzyme, especially those residing in and around the catalytic center of the protein. These agents generally resemble the substrate so that they can bind in the active site of the enzyme. In most examples, these agents also contain an electrophilic group, which includes groups such as halo-methyl ketones ($\text{X-CH}_2\text{C=O}$, where X = halide), sulfonyl fluorides (SO_2F), nitrogen mustards [$(\text{ClCH}_2\text{CH}_2)_2\text{NH}$], diazoketones (COCHN_2), and other such reactive groups, that can “label,” or alkylate, a nucleophilic amino acid group in the enzyme. They generally tend to be indiscriminate in their action and have little therapeutic value, because they are nonselective and, thus, inherently toxic. They have been mainly used as biochemical tools to probe active sites of enzyme to differentiate the types of amino acid groups both in and around the catalytic center of an enzyme. The classic example of an affinity label is TPCK (tosyl-phenylalanyl-chloromethyl-ketone), an irreversible inhibitor of the serine protease chymotrypsin (29).



Because TPCK resembles the amino acid phenylalanine, it can bind to the active site of the chymotrypsin, the selectivity of which is for such hydrophobic amino acid groups (Phe and Tyr). During the course of normal peptide hydrolysis, the reactive chloromethyl-ketone labels the nucleophilic histidine group present as part of the catalytic triad (Ser-His-Asp) in the active site of the protease (Fig. 8.14). Another similarly designed affinity

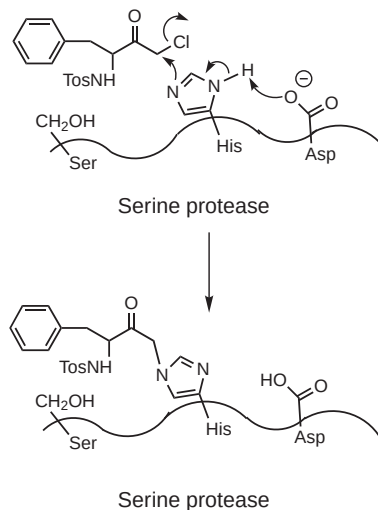


FIGURE 8.14 Mechanism of affinity label of serine protease with TPCK (tosyl-phenylalanyl-chloromethyl-ketone).

label is TLCK (tosyl-lysyl-chloromethyl-ketone), the specificity of which is for the protease trypsin. Trypsin cleaves peptide bonds adjacent to the basic amino acids, lysine and arginine. It was found that TLCK was a specific inhibitor of trypsin but had no activity for chymotrypsin. On the other hand, TPCK, although extremely specific for chymotrypsin, showed no activity for trypsin.

Because of the inherent reactivity and nonselectivity of these affinity labels and their limited utility in drug therapy, the late B.R. Baker (30) extended this concept to design inhibitors that would have greater selectivity and specificity and, thus, be potential drug candidates. He designed several analogs, termed active site-directed irreversible inhibitors, targeted toward thymidylate synthase, a key enzyme involved in the *de novo* metabolism of thymidylate. These analogs contained a substrate-binding region linked to a reactive group, such as a halomethyl ketone, by a tether whose chain length could be manipulated. The substrate portion of the analog ensures both affinity and rapid binding to the enzyme active site. Once the substrate is bound, areas in and around the binding site and on the surface of the enzyme could be probed for nucleophilic amino acid groups. By manipulating the length of the tether, ideal inhibitors could then be designed such that any suitably located, sufficiently nucleophilic amino acid group on the surface of the enzyme could potentially be alkylated by the halomethyl ketone (Fig. 8.15). Once alkylated, the tether “bridges” the active site with the labeled amino acid group, thus “tying up” and preventing further catalysis by the enzyme.

Mechanism-Based Irreversible Enzyme Inhibitors

OVERVIEW The mechanism-based irreversible inhibitors have also been termed “suicide substrates,” “ k_{cat}

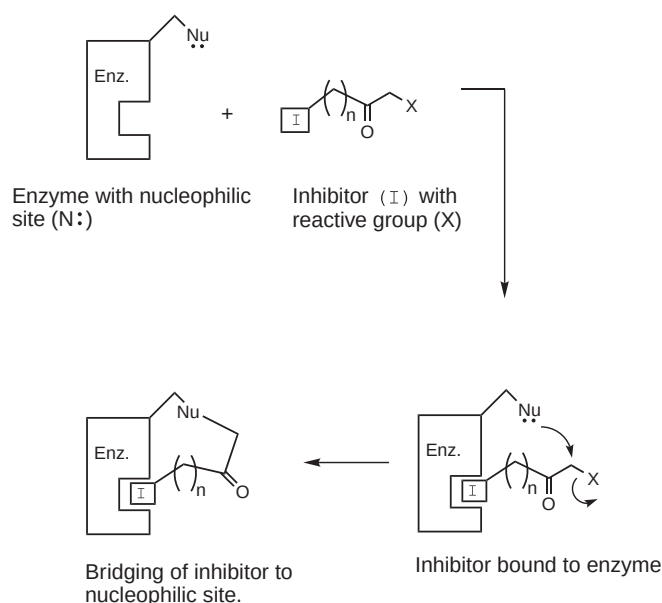
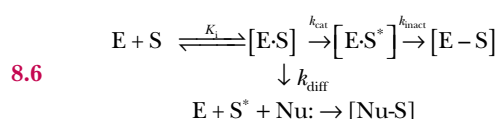


FIGURE 8.15 Model showing Baker's (30) active site-directed irreversible inhibitors.

inhibitors,” “Trojan horse inhibitors,” or “latent alkylating agents.” These inhibitors are inherently unreactive but, on normal catalytic processing by the enzyme, are activated into highly reactive moieties (31–33). These reactive groups can irreversibly alkylate a nucleophilic amino acid group or cofactor in the enzyme and, in essence, cause the enzyme's death (“suicide”). Basically, these inhibitors have a latent reactive group that only becomes apparent after binding and acted on by the normal catalytic machinery of the enzyme. This type of inhibitor design differs from the active site-directed inhibitors in that these inhibitors have one more level of built-in selectivity. The kinetic scheme for such inhibition is shown in Equation 8.6, in which enzyme, E, binds with substrate (inhibitor), S, to give an [E–S] complex with dissociation constant of K_i :



Next, the [E–S] complex is converted into a highly activated complex [E–S*] by the catalytic machinery (k_{cat}) of the enzyme, which can then go on to alkylate the enzyme, [E–S]. Note that it is possible for the reactive species [S*] to diffuse (k_{diff} dissociate) from the enzyme and react with some other target (nucleophilic species)—that is, the system is “leaky.” However, if this happens, the inhibitor cannot be classified as a “true” suicide substrate, because specificity is lost.

Several requirements need to be met by these inhibitors in order for them to be classified as suicide substrates. These include the following: 1) inactivation should be time dependent (reaction is irreversible); 2) kinetics should be first order; 3) the enzyme should show saturation kinetics; 4) the substrate should be able to protect the enzyme; and 5) stoichiometry of the reaction should be 1:1 (one active site to one inhibitor).

EXAMPLES OF MECHANISM-BASED IRREVERSIBLE INHIBITORS During the past three decades, besides the rational design of hundreds of molecules that have been synthesized and tested as mechanism-based irreversible inhibitors, it has also come to light that nature itself has known about this mechanistic mode of enzyme inhibition and provided us with several extremely potent mechanism-based irreversible inhibitors. Below are a few selected examples to demonstrate the mode of action of these inhibitors.

Halo Enol Lactones Halo enol lactones are examples of irreversible inhibitors for serine proteases. These analogs were developed by Katzenellenbogen and coworkers at the University of Illinois (34). On normal catalytic processing by the Ser-CH₂-OH group, the analogs produce a reactive halo-methyl ketone, which subsequently alkylates a nearby nucleophilic group on the enzyme (Fig. 8.16). Other inhibitors for

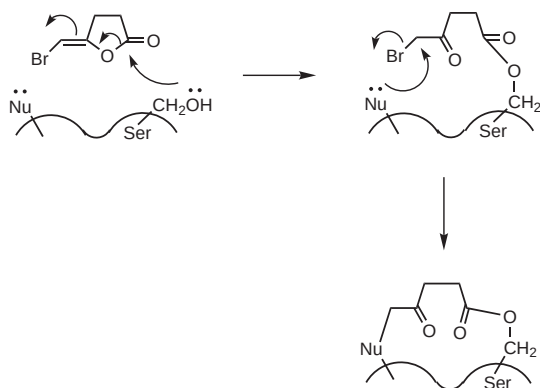


FIGURE 8.16 Mechanism-based inhibition of serine proteases by Katzenellenbogen's halo enol lactone.

the serine proteases have been designed by various researchers (32).

Clavulanic Acid Clavulanic acid is a potent inhibitor of bacterial β -lactamase (35). This enzyme is a serine protease and can hydrolyze β -lactams, such as the penicillin antibiotics. It is the principal enzyme responsible for penicillin-resistant bacteria. Clavulanic acid is a β -lactam and, when given in combination with penicillin, is preferentially taken up by β -lactamase and hydrolyzed. However, during the process of hydrolysis, the clavulanic acid undergoes a cleavage, leading to the formation of a "Michael acceptor," which subsequently alkylates a nucleophilic group on β -lactamase, causing irreversible inhibition as shown in Figure 8.17. Such combinations of a β -lactamase inhibitor and a penicillin have resulted in clinically useful agents, for example clavulanic acid plus amoxicillin (Augmentin).

Gabaculin Gabaculin, a naturally occurring neurotoxin, is a potent mechanism-based inhibitor of the enzyme γ -aminobutyric acid transaminase (GABA-T) with an interesting mechanism of action (36). GABA-T is a pyridoxal phosphate (PLP)-dependent enzyme

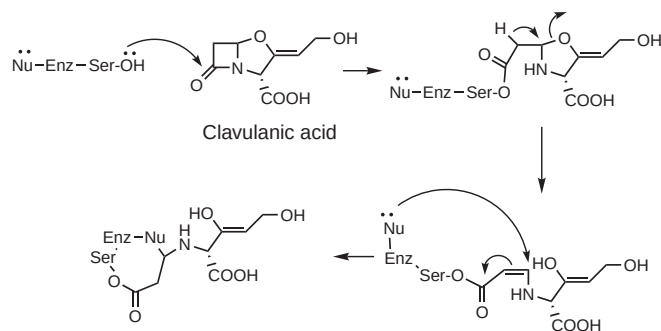


FIGURE 8.17 Mechanism-based inhibition of β -lactamases by clavulanic acid.

(PLP is a cofactor) involved in the catabolism (transamination) of the excitatory neurotransmitter, GABA, to succinate semialdehyde and pyridoxamine as shown in Figure 8.18. As part of the normal catalytic mechanism of PLP-dependent enzymes, the *m*-amino group of gabaculin initially reacts with the aldehyde group of PLP to form an adduct, which subsequently undergoes an aromatization reaction, resulting in an extremely stable covalent bond with PLP. Hence, in this case, rather than an enzymatic nucleophilic group being alkylated, the cofactor is unavailable, resulting in the inhibition.

Finasteride Finasteride is a clinically useful agent in the treatment of prostate hyperplasia and male pattern baldness. It is a potent inhibitor that targets human steroid- 5α -reductase, the enzyme responsible for the reduction of testosterone to more potent dihydrotestosterone (Fig. 8.19) (see Chapter 40). The inhibitory action of finasteride has been attributed both to its similarity in structure to testosterone, which allows the drug to bind to the enzyme and be reduced to dihydrofinasteride in place of testosterone, as well as to its ability to act as a mechanism-based inhibitor, during which it makes the cofactor NADPH unavailable by forming a covalent NADP-dihydrofinasteride adduct as shown in Figure 8.19 (see Chapter 40). This adduct very slowly releases dihydrofinasteride with a half-life of 1 month (37).

Recent Inhibitors of Clinically Important Enzymes

Cyclic Nucleotide Phosphodiesterase Inhibitors

Phosphodiesterases (PDEs) belong to a large family of enzymes that hydrolyze phosphate ester bonds, linking two hydroxyl groups with phosphoric acid. In nature, there are numerous biomolecules containing such phosphodiester bonds including glycerol phosphates (phospholipids), sugar phosphates, inositol phosphates, nucleic acids, and nucleotides including the cyclic nucleotides. Although, all of these phosphodiester bonds are essentially hydrolyzed by a PDE, the specific name of phosphodiesterase (PDE) has been applied to enzymes that hydrolyze the cyclic nucleotide monophosphates, cyclic adenosine monophosphate (cAMP) and cyclic guanosine monophosphate (cGMP), both of which are involved in signal transduction and second messenger cell signaling (Fig. 8.20) (see Chapter 3). Enzymes that hydrolyze the phosphodiester bonds other than the cyclic nucleotides are known by other names, such as nucleases (DNases or RNases, for nucleic acids) and phospholipases for glycerol phosphates.

The PDEs hydrolyze the 3'-5' phosphodiester bond of its primary substrates, cAMP or cGMP, leading to the formation of AMP (5'-adenosine monophosphate) and GMP (5'-guanosine monophosphate), respectively. The PDEs can bind either cyclic nucleotide with selectivity

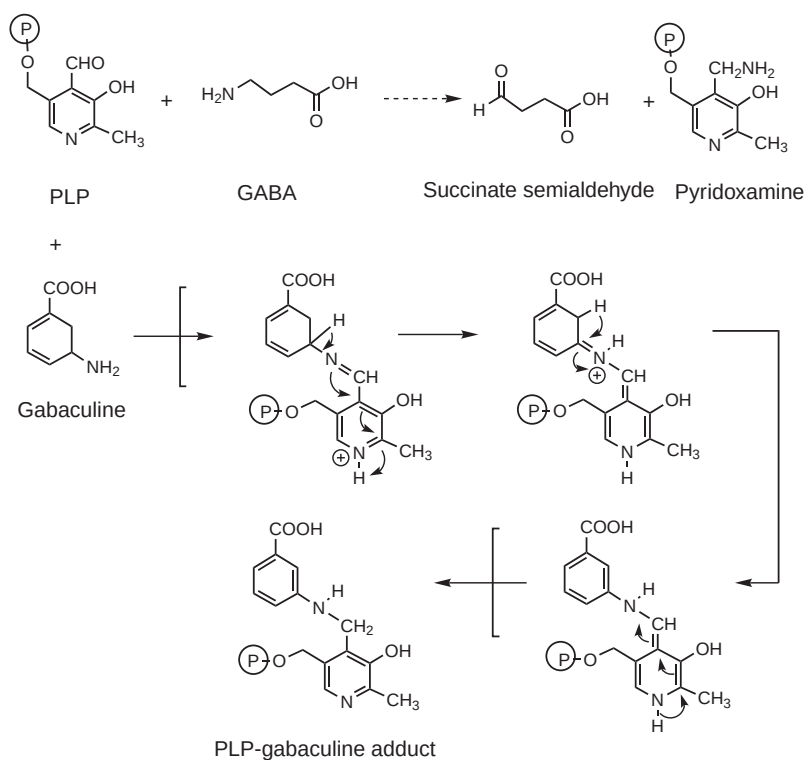


FIGURE 8.18 Pyridoxal phosphate-dependent γ -aminobutyric acid transaminase (GABA-T) reaction and the mechanism of suicide inhibition by gabaculine.

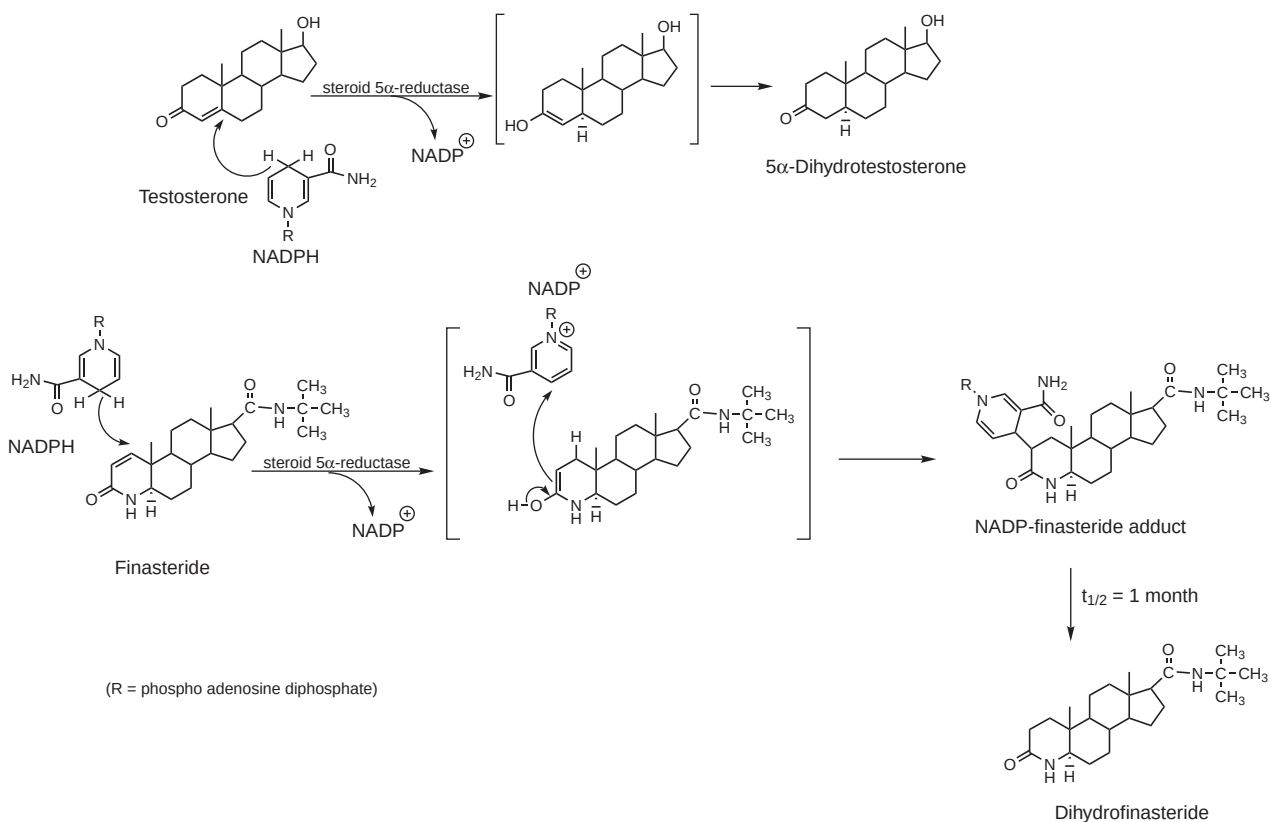


FIGURE 8.19 Mechanism of steroid reductase reduction on finasteride and testosterone and the structure of hypothesized NADPH-dihydrofinasteride adduct (37).

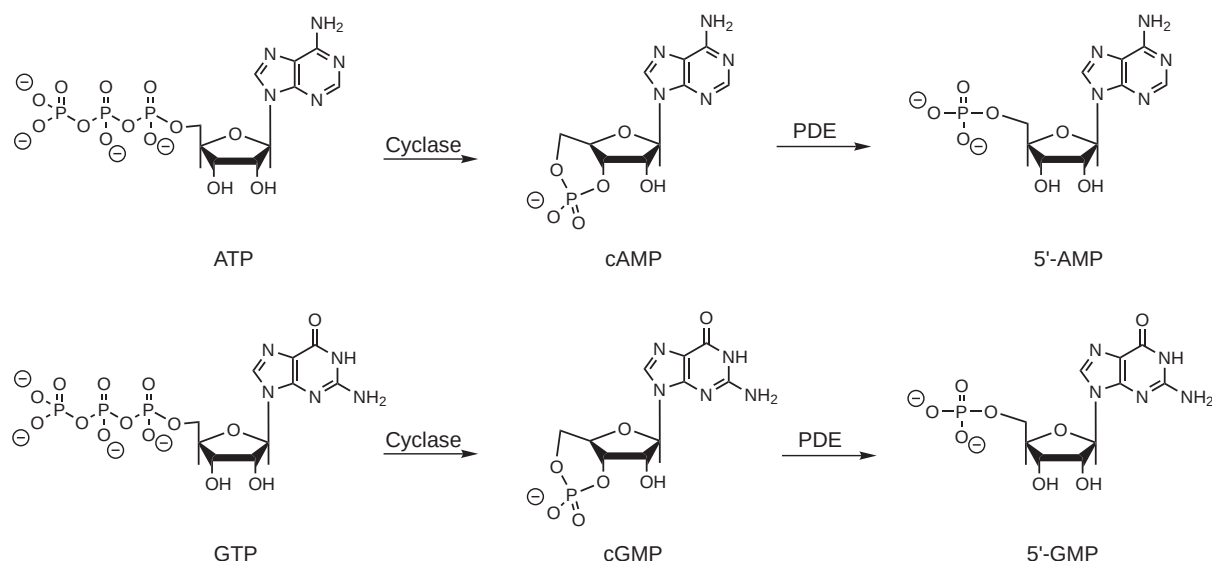


FIGURE 8.20 Enzymatic interconversions of cyclic and linear nucleotides.

depending on the particular isoenzyme. cAMP and cGMP are formed from their precursor nucleotide triphosphates, adenosine triphosphate (ATP) and guanosine triphosphate (GTP), respectively (Fig. 8.20). There are many different isoforms of the PDEs present in cells, which allows for great diversity in substrate specificity and cell regulation. This diversity further governs specific roles for each of these PDEs in various cellular locales and physiologic and pathologic conditions. Currently, the PDEs have been classified into 11 different families based on gene products (amino acid homology) and are estimated to comprise over 100 different mRNAs from 21 different genes due to alternative splicing and transcription start sites (38,39). The nomenclature of these PDE families is based on the species of origin, the gene, and the variant discovered. For example, HsPDE2A3 signifies that the origin is from homo sapiens (Hs), PDE denotes that it is a cyclic nucleotide phosphodiesterase, the 2A indicates that it is from the PDE gene 2 family A, and 3 indicates that it is the third variant that was reported in the gene database. The different PDEs have been shown to regulate various cellular activities, and hence because of this diverse functional ability, one could develop drugs to control these PDEs in a selective manner (Table 8.2) (40–42). Structural studies of several of these PDEs have shown that they all contain a “catalytic domain” with a consensus sequence denoting a metal ion binding site (a phosphorylase sequence, containing a signature recognition sequence for all PDEs) of two His and two Asps that bind $\text{Zn}^{2+}/\text{Mg}^{2+}$; a “glutamine switch,” which accounts for the substrate specificity (cAMP or cGMP) where an invariant glutamine amino acid moiety can rotate to either H-bond with cAMP or cGMP or is constrained from binding to either cAMP or cGMP, thus favoring binding to the other nucleotide; and a

“regulatory domain” based on the specific PDE family (e.g., Ca^{2+} -calmodulin binding site for PDE1) as well as an allosteric binding site for cGMP (38,39).

X-ray crystallographic studies with inhibitors bound with the enzyme have indicated that there are several binding modes for inhibition and the active site architecture is often unique for different PDEs. Inhibitors can bind to the enzyme through H-bonds with amino acid groups that make up the active site for cAMP/cGMP binding, with groups that line the channel leading to the active site, as well as by H-bonding with water to groups that bind to the active site metal ions. Moreover, because of the order of magnitude differences in cellular concentrations of the cyclic nucleotides (as compared to its precursor, nucleotide triphosphate molecule, <1 to $10\ \mu\text{M}$ for cAMP/cGMP vs. mM concentrations for ATP/GTP) (38), these PDEs become attractive targets to design drugs. (Since competitive inhibition for a natural substrate whose cellular concentration is $1\ \mu\text{M}$ is achieved much more easily than if the substrate concentration was $1\ \text{mM}$). Structural knowledge of the different binding modes for inhibitors with the PDEs has given rise to the rapid development of selective PDE inhibitors to allow for specific interactions along regulatory pathways involving such PDEs. While inhibitors of PDEs have been known for some time (e.g., caffeine and theophylline have been in use as therapeutic agents for many decades as non-selective inhibitors of PDE), the development of drugs to selectively inhibit a particular PDE is a more recent discovery. Moreover, selective PDE inhibitors would also have a better safety profile, as one would expect a decreased amount of side/adverse effects. For example, theophylline, a nonselective PDE inhibitor, is known to have a narrow therapeutic index because of its interaction with multiple PDEs.

TABLE 8.2 Specificity and Potency of PDE Inhibitors

PDE	Specificity	Inhibitor	IC ₅₀ ^a (nM)	PDE	Specificity	Inhibitor	IC ₅₀ ^a (nM)
PDE1B	Duel	Rolipram	>200,000	PDE6	cGMP	Sildenafil	50 ^b
		Cilomilast	87,000			Vardenafil	11 ^c
		Roflumilast	>200,000			Tadalafil	2,000 ^b
		Sildenafil	1,500	PDE7B	cAMP	Rolipram	>2,000,000
		Vardenafil	300			Cilomilast	44,000
PDE2A	Duel	Tadalafil	50,000			Roflumilast	>200,000
		Rolipram	>200,000	PDE8A	cAMP	Sildenafil	78,000
		Cilomilast	160,000			Vardenafil	1,900
		Roflumilast	>200,000			Tadalafil	74,000
		Sildenafil	35,000			Rolipram	>200,000
PDE3B	cAMP > cGMP	Vardenafil	3,100			Cilomilast	7,000
		Tadalafil	130,000			Roflumilast	>200,000
		Rolipram	>200,000	PDE9A	cGMP	Sildenafil	>200,000
		Cilomilast	87,000			Vardenafil	57,000
		Roflumilast	>200,000			Tadalafil	>200,000
PDE4B	cAMP	Sildenafil	15,000			Rolipram	>200,000
		Vardenafil	580			Cilomilast	>200,000
		Tadalafil	280,000			Roflumilast	>200,000
		Rolipram	570	PDE10A	cAMP<cGMP	Sildenafil	5,600
		Cilomilast	25			Vardenafil	680
PDE4D	cAMP	Roflumilast	0.84			Tadalafil	150,000
		Sildenafil	20,000			Rolipram	140,000
		Vardenafil	3,800			Cilomilast	73,000
		Tadalafil	9,200			Roflumilast	>200,000
		Rolipram	1,100	PDE11A	Duel	Sildenafil	6,800
PDE5A	cGMP	Cilomilast	11			Vardenafil	880
		Roflumilast	0.68			Tadalafil	19,000
		Sildenafil	14,000			Rolipram	>200,000
		Vardenafil	3,900			Cilomilast	21,000
		Tadalafil	19,000			Roflumilast	25,000
PDE5A	cGMP	Rolipram	>200,000	PDE5A	cGMP	Sildenafil	6,100
		Cilomilast	53,000			Vardenafil	240
		Roflumilast	17,000			Tadalafil	10
		Sildenafil	2.2				
		Vardenafil	1.0				
		Tadalafil	1.2				

^a Data are from Sutherland and Rall (40) unless stated otherwise. The numbers shown in the table are the 50% inhibition concentration (IC₅₀). The cAMP and cGMP concentrations used were far below the K_m of all the PDEs assayed except for PDE9A, in which case it was close to the K_m . The IC₅₀s obtained are good approximations of the inhibition constant K_i . Selectivities of an inhibitor may be determined by taking ratios of the numbers given in the table.

^b K_i values are from Card et al. (41).

^c Value from Hatzelmann and Schudt (42).

PDE5 SELECTIVE INHIBITORS The impetus for the development of PDE inhibitors arose from the knowledge that vasodilation (for treatment of high blood pressure) could be achieved through the stimulation of atrial natriuretic peptide (ANP), an endogenous peptide that allows for renal excretion of sodium/water. Furthermore it was also known that ANP stimulated the synthesis of cGMP through its activation of guanyl cyclase. Thus, inhibition of cGMP hydrolysis was a rational target for the development of PDE inhibitors as vasodilators. Since ANP worked in the kidney, the idea was to target the specific PDE in the kidney. The lead compound for the development of these PDE inhibitors was a xanthine derivative, zaprinast (Fig. 8.21), previously shown to demonstrate weak PDE inhibition (43). Analysis of the heterocyclic ring of zaprinast and comparison with the purine heterocycle of cGMP led to the development of a number of derivatives displaying PDE5 inhibitory activity, including the design of one containing a pyrazolopyrimidinone, which ultimately resulted in the development of the potent PDE5 inhibitor sildenafil (44). During clinical trials of sildenafil as an antihypertensive and vasodilator, it was discovered to have a (beneficial) side effect regarding erectile dysfunction. This pharmacologic effect was recognized, appreciated, and commercialized into a new line of therapeutic agents to treat erectile dysfunction. Inhibition of cGMP hydrolysis by PDE5 in the corpus cavernosum in the penis results in elevated levels of cGMP, leading to increased smooth muscle relaxation and increased blood flow and maintenance of an erection. Presently, PDE5 selective inhibitors include the drugs sildenafil (Viagra), vardenafil (Levitra), and tadalafil (Cialis) (Fig. 8.21) (see Chapter 40). They have been some of the most widely marketed and commercially successful drugs in recent years to treat erectile

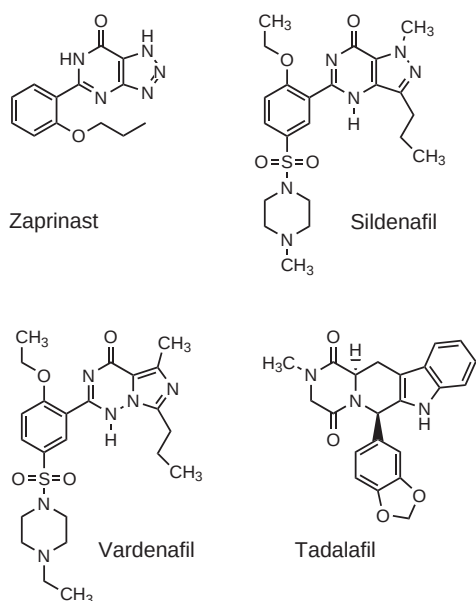


FIGURE 8.21 Inhibitors of PDE5.

dysfunction in males and also more recently have been approved to treat pulmonary hypertension (see Chapter 24). These drugs have high selectivity to inhibit PDE5 over the other PDE classes. PDE5 is a PDE that specifically hydrolyzes cGMP to GMP at low substrate levels and also has a high affinity binding site for cGMP on its regulatory domain. PDE5, originally isolated from platelets, was later found to be a regulator in vascular smooth muscle contraction present in the lungs and brains. The drugs are all reversible inhibitors of PDE5 where they bind in the active site of PDE5, the heterocyclic nucleus of the inhibitors binding in the space reserved for the guanosine ring (of cGMP). A side effect is hypotension, and also, because the active site structures of PDE5 and PDE6 are similar in architecture, some cross inhibition of PDE6 enzymes has been observed. PDE6 enzymes regulate the phototransduction cascade where PDE6 rapidly reduces the steady-state concentration of cGMP in response to light stimuli. Thus, a side effect of PDE5 inhibitors involves decreased vision in some patient populations. The drug tadalafil, however, is approximately 1,000-fold more selective for PDE5 than for PDE6. Several other selective competitive PDE inhibitors are approved for use, and these include the PDE4 inhibitors cilomilast and roflumilast for treatment of chronic obstructive pulmonary disease (Fig. 8.22) as well as the PDE3 inhibitors dipyridamole and cilostazol, as antiplatelet drugs (Fig. 8.23), and inamrinone and milrinone, as positive inotropic agents (Fig. 8.24).

Protein Kinase Inhibitors

Protein kinases belong to the family of group transfer phosphorylating enzymes that transfer a phosphate group from ATP onto amino acid groups of proteins. It is estimated that there are over 500 of these kinases encoded by the human genome, and this reaction, when coupled with a phosphatase (i.e., reversible phosphorylation of proteins), offers cells a precise regulatory mechanism to control their differentiation, maturation,

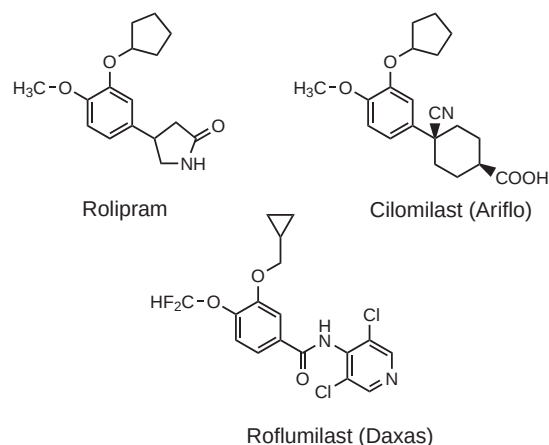


FIGURE 8.22 Competitive inhibitors of PDE4.

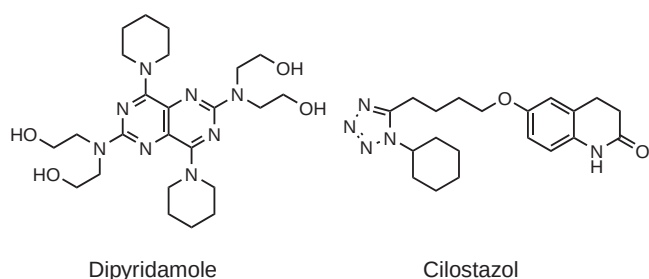


FIGURE 8.23 Antiplatelet inhibitors of PDE₃.

proliferation, apoptosis, and other cellular functions (45). The substrate amino acid groups on proteins that are phosphorylated belong largely to the hydroxyl-bearing amino acids, such as serine, threonine, and tyrosine. Therefore, these kinases are referred to as either serine/threonine kinases or tyrosine kinases. Because these proteins are involved in regulatory functions of the cell, it becomes apparent that mutations or aberrations of expression of these proteins can lead to dysregulation of cellular functions, giving rise to tumors and cancers. Indeed, research has shown that of the 518 kinase genes present in the human genome, 244 map to disease loci and cancer (45). Moreover, targeting these enzymes with inhibitors would be a way to selectively target cancers without the noxious side effects seen with conventional anticancer drugs such as the alkylating agents or antitumor antibiotics (see Chapter 37). Research over the last decade for kinase inhibitors has provided a rationale to develop targeted anticancer therapy where such kinases that manifest themselves in certain cancers have been specifically targeted for inhibition, resulting in dramatic decreases of cancer cells and greater survival times for the patient (46). Examples of such targeted anticancer drugs have involved primarily the development of the tyrosine kinase inhibitors (TKIs), several of which are U.S. Food and Drug Administration approved (see Chapter 37); however, more recently, inhibitors of serine/threonine kinases are also being developed and are in clinical trials.

TYROSINE KINASE INHIBITORS The tyrosine kinases are a group of enzymes responsible for signal transduction and intracellular signaling functions, many of which are involved in cell differentiation and proliferation.

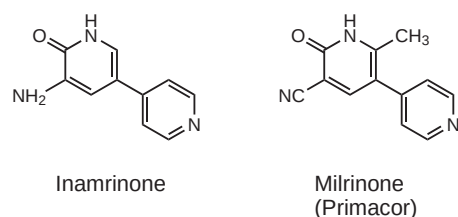


FIGURE 8.24 PDE₃ inhibitors with cardiac and pulmonary effects.

They can be divided into two major types depending on where they act in the cell: receptor tyrosine kinases, membrane-spanning proteins having an extracellular ligand binding domain and an intracellular catalytic (kinase) domain involved in the transduction of extracellular signals from the membrane to the cytoplasm, or the non-receptor tyrosine kinases involved in cytosolic signaling events. Inhibitors for both of these kinases have been developed and have shown excellent and selective activity in cancers manifested by aberrant expression of these enzymes. The development of a TKI to selectively treat chronic myelogenous leukemia (CML) was the impetus leading to the large number of presently available TKIs. CML in the majority of patients is due to a reciprocal translocation of chromosomes 9 and 22, resulting in a fusion of the *abl* (Abelson leukemia virus) gene of chromosome 9 to the *bcr* (break point cluster) gene of chromosome 22, leading to the *bcr-abl* fusion gene (the Philadelphia chromosome). Although the *abl* gene normally produces a non-receptor tyrosine kinase whose activity is highly regulated, the *bcr-abl* fusion gene produces a tyrosine kinase that is constitutively active and whose activity is required for the transformation of cells to become malignant (47). The knowledge of this direct correlation between expression of the abnormal fusion protein and CML allowed for the development of specific inhibitors for this enzyme and other such dysregulated kinases that are overexpressed in many cancers. Using a high-throughput screening program to develop inhibitors for receptor tyrosine kinases as a possible treatment for such cancers, 2-phenylaminopyrimidine (PAP) became a lead compound. Further SAR optimization and refinement of this lead led to imatinib (Gleevec) (Fig. 8.25), the first targeted drug for treatment of CML (48,49). Note that the SAR to imatinib included addition of a pyridine, a methyl, and a benzamide to enhance the potency of the basic PAP nucleus. The piperazinyl functionality helped to increase water solubility, allowing for better “drug-like” properties. Imatinib also proved to be inhibitory in many other cancers with overexpressed kinases, such as gastrointestinal stromal tumors (which overexpress *c-kit*), myelodysplastic diseases associated with platelet-derived growth factor receptor, and Philadelphia chromosome-positive adult lymphoblastic leukemia (50). X-ray

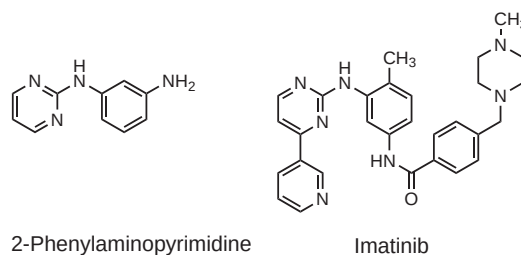


FIGURE 8.25 Structures of 2-phenylaminopyrimidine and imatinib.

crystallographic studies with imatinib co-crystallized with the tyrosine kinase expressed from *abl* showed that imatinib binds to the ATP binding site of the enzyme in its inactive conformation (off state) and this binding prevented the kinase from achieving its productive binding conformation with ATP (51,52). These kinases all have an “activation loop,” which contains a tyrosine phenolic group (Tyr 393), the major phosphorylated group, that allows switching the kinase from inactive to active forms and allows for ATP binding. Studies showed that with the enzyme-bound imatinib, the phenolic group of Tyr 393 was not phosphorylated and the conformation of this activation loop in the nonphosphorylated enzyme changed to that resembling substrate (ATP) being bound to the kinase. In this way, the altered geometry brought about by imatinib binding to the active site of the enzyme prevented the enzyme from binding its true substrate, ATP. Resistance develops to imatinib due to mutations in the hydrophobic pocket that prevent access of imatinib to the enzyme in the off state, thus allowing for the kinase to bind ATP and cancer to progress. TKIs have also been developed to bind the kinase in its “on state” (active conformation), where the drugs can access hydrophobic regions in the ATP binding pocket, and to bind both “on” and “off” forms (dual-mode inhibitors that are more potent than imatinib) (52).

Since the introduction of imatinib, many other TKI drugs have been developed for related tyrosine kinases such as epidermal growth factor receptor, platelet-derived growth factor receptor, and vascular endothelial growth factor receptor. All of these inhibitors make use of a heterocyclic functionality (resembling the adenine of ATP) and bind in the putative binding site of ATP (a highly conserved nucleotide binding region in the catalytic domain of the kinase). The inhibitors make use of differences in the variable region of the enzyme surrounding the ATP binding pocket, which allows for specific binding interactions with the various groups present

on the individual inhibitors. Resistance to these inhibitors manifests due to mutations to these variable regions on the enzyme as well as to cellular efflux pumps being activated. Examples of some of these inhibitors such as dasatinib (for imatinib resistance), sunitinib (inhibitor of vascular endothelial growth factor receptor), and gefitinib (inhibitor of epidermal growth factor receptor) are shown in Figure 8.26.

SERINE/THREONINE KINASE INHIBITORS The serine/threonine kinases are a family of enzymes that phosphorylate the hydroxyl groups of serine and threonine present on enzymes. One of the most studied of these enzymes is protein kinase C, an intracellular kinase that is activated by Ca^{2+} ions or diacylglycerol, leading to various signal transduction events in the cell through its activation of the mitogen-activated protein kinase family (53). Examples of other serine/threonine kinases include mammalian target of rapamycin (mTOR), a protein involved in the activation of immune B and T cells, and the Aurora kinases, which are involved in cell cycle regulation during mitosis. Inhibitors of mTOR include rapamycin (sirolimus) a macrolide originally isolated from a microbe on Easter Island and used as an immunosuppressant drug. The Aurora kinase inhibitors encompass inhibitors for Aurora A and Aurora B kinases, both of which are overexpressed in cancer cells leading to dysregulation during mitosis and chromosomal destabilization (54). Generally, mitosis is a highly regulated process with multiple checkpoints to regulate chromosomal segregation. One of these regulatory mechanisms involves phosphorylation of proteins by serine/threonine kinases (mitotic kinases). The Aurora kinases are one such mitotic kinase where they interact with many proteins, including tumor suppressors and activators from the mitotic entry stage all the way to cytokinesis. It is not surprising then that they are overexpressed in many tumors (breast, colon, gastric, ovarian, and pancreatic) and thus have become an

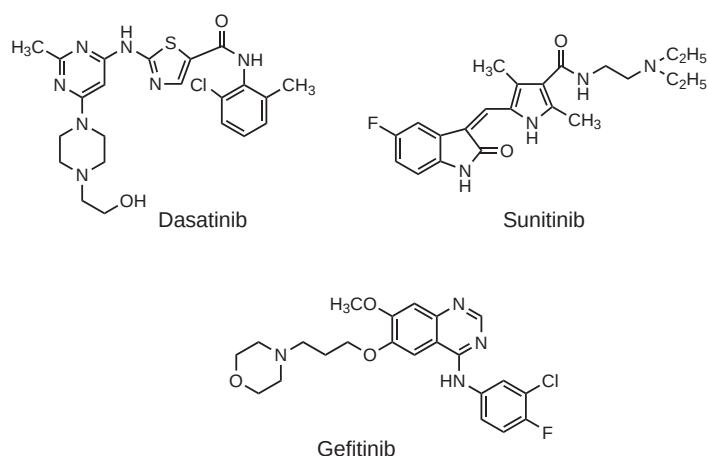


FIGURE 8.26 Structures of the tyrosine kinase inhibitors, dasatinib, sunitinib, and gefitinib.

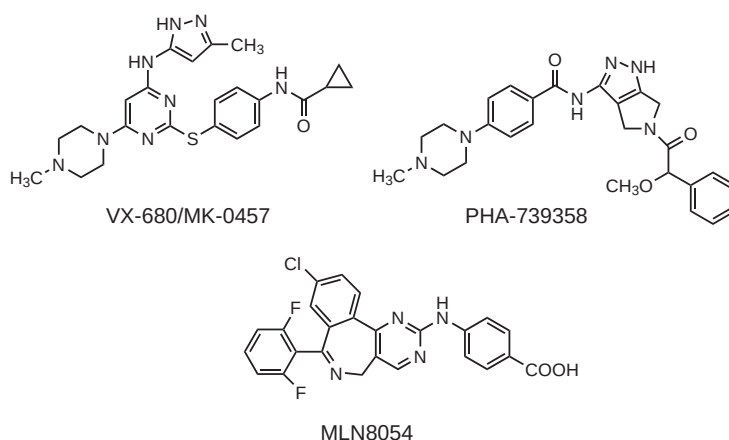


FIGURE 8.27 Structures of the Aurora kinase inhibitors undergoing clinical trials.

attractive target for anticancer drug development (55). Examples of some of these Aurora kinase inhibitors in clinical trials include VK680/MK-0457, PHA-739358, and MLN8054 (Fig. 8.27).

SUMMARY

This chapter has attempted to give the reader an overview of enzyme catalysis and the various ways in which enzymes can act as catalytic proteins. Based on these properties and mechanisms of enzyme action, the essentials of drug design and discovery through enzyme inhibition with a few choice examples have been outlined. The reader is referred to the Suggested Reading material for more detailed explanations and insights regarding the rationale and design strategies of enzyme inhibitors. In conclusion, drug design by enzyme inhibition is a continually developing enterprise. There will always be the need to discover more selective and more potent inhibitors in an effort to increase the therapeutic benefit to patients. This chapter has tried to give a brief insight into this fascinating area of medicinal chemistry and the various types of enzyme inhibitors that can be rationally designed.

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Part
II

**DRUG RECEPTORS
AFFECTING
NEUROTRANSMISSION
AND ENZYMES AS
CATALYTIC RECEPTORS**

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Chapter 9

Drugs Affecting Cholinergic Neurotransmission

E. KIM FIFER

Drugs Covered in This Chapter

ACETYLCHOLINE MIMETICS—MUSCARINIC AGONIST

- Methacholine chloride
- Carbachol chloride
- Bethanechol chloride
- Pilocarpine
- Cevimeline

ACETYLCHOLINESTERASE INHIBITORS (ANTICHOLINESTERASES)

- Physostigmine
- Neostigmine
- Pyridostigmine
- Carbaryl
- Edrophonium
- Echothiophate iodide

ACETYLCHOLINESTERASE INHIBITORS FOR THE TREATMENT OF ALZHEIMER DISEASE

- Tacrine
- Donepezil
- Rivastigmine
- Galantamine

INSECTICIDAL ACETYLCHOLINESTERASE INHIBITORS

- Parathion
- Malathion
- Paraoxin
- Schradan
- Dichlofenthion
- Chlorpyrifos

CHOLINESTERASE REACTIVATOR—TREATMENT OF ACETYLCHOLINESTERASE INHIBITOR POISONING

- Pralidoxime chloride

ACETYLCHOLINE ANTAGONISTS—MUSCARINIC ANTAGONISTS

- Atropine
- Scopolamine
- Homatropine
- Tiotropium
- Trosipium
- Ipratropium
- Glycopyrrolate
- Procyclidine

- Trihexyphenidyl
- Tolterodine
- Orphenadrine
- Benztropine
- Flavoxate
- Propantheline
- Oxybutynin
- Solifenacin
- Darifenacin
- Fesoterodine

NICOTINIC ANTAGONISTS—NEUROMUSCULAR BLOCKING AGENTS

- Decamethonium bromide
- Succinylcholine chloride
- d-Tubocurarine
- Metocurine
- Pancuronium bromide
- Vecuronium bromide
- Pipecuronium bromide
- Rocuronium bromide
- Atracurium besylate
- Mivacurium chloride
- Doxacurium chloride

Abbreviations

acetyl-S-CoA, acetyl coenzyme A
AChE, acetylcholinesterase
AChEI, acetylcholinesterase inhibitor
AD, Alzheimer disease
ATP, adenosine triphosphate
CNS, central nervous system
CoA, coenzyme A
GABA, γ -aminobutyric acid

GDP, guanosine diphosphate
GPCR, G protein–coupled receptor
GTP, guanosine triphosphate
IP₃, inositol-1,4,5-triphosphate
M, moles/L
mAChR, muscarinic acetylcholinergic receptor
nAChR, nicotinic acetylcholinergic receptor

NM, Neuromuscular (somatic) nAChR
NN, Neuronal Ganglionic nAChR
NMR, nuclear magnetic resonance
2-PAM, 2-pyridine aldoxime methyl chloride
SAR, structure–activity relationship

SCENARIO

Kathryn Neill, PharmD

LT is a 62-year-old man who calls the pharmacy after taking his new medication inappropriately. He brought in a new prescription for tiotropium (Spiriva) 2 days ago. A 1-month supply of capsules was dispensed with the Handihaler inhalation device. LT was counseled regarding proper use and demonstrated good understanding and technique. LT says this morning he “just popped the

Spiriva in my mouth with my blood pressure pill and swallowed it.” He wants to know if he should take ipecac or try to vomit and whether to take another dose with the inhaler for his lungs.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

No other mammalian system or chemical neurotransmitter has been studied as exhaustively as the parasympathetic nervous system and acetylcholine. Acetylcholine functions as the neurotransmitter for many different neurons (Fig. 9.1) (1). In the autonomic nervous system, it is released by pre- and postganglionic fibers of the parasympathetic division, preganglionic fibers of the sympathetic division, and a few postganglionic fibers of the sympathetic division (e.g., sweat and salivary glands). It is also released by neurons of the somatic (voluntary) nervous system and by some neurons in the CNS. Neurons that release acetylcholine are referred to as cholinergic, as are the receptors on which these neurons synapse. These receptors are further classified as either muscarinic or nicotinic, depending on their ability to bind the naturally occurring alkaloids muscarine or nicotine, respectively.

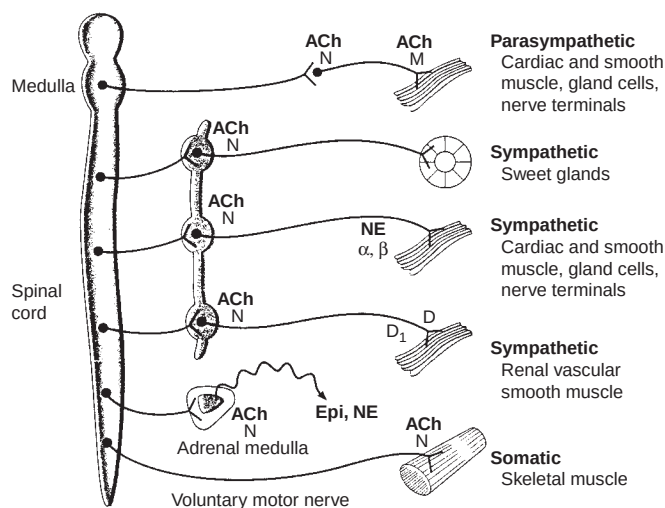
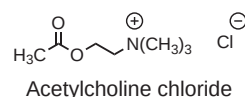


FIGURE 9.1 Schematic representation of autonomic and somatic motor nerves. The sites of action of acetylcholine (ACh), norepinephrine (NE), epinephrine (Epi), and dopamine (D) are indicated. Cholinergic receptors are designated as nicotinic (N) or muscarinic (M). (From Katzung BG. Introduction to autonomic pharmacology. In: Katzung BG, Masters SB, Trevor AJ, eds. Basic and Clinical Pharmacology, 11th Ed. New York: McGraw-Hill, 2009:78, with permission.)



Parasympathetic nerve impulses stimulate contraction of smooth muscle in the gastrointestinal tract and urinary tract, contraction of the ciliary muscle of the eye, and relaxation of smooth muscle of blood vessels, and also decrease heart contractility and rate.

Chemical compounds that cause stimulation of the parasympathetic nervous system are called cholinomimetic or, more specifically, parasympathomimetic agents. Cholinomimetic agents are 1) agonists that act directly on cholinergic receptors or 2) function as inhibitors of acetylcholinesterase (AChE), the enzyme responsible for hydrolysis of acetylcholine. Those compounds that possess affinity for cholinergic receptors but exhibit no intrinsic activity are called cholinergic antagonists, cholinolytic, or parasympatholytic agents. This chapter is devoted to the discussion of cholinergic agonists and antagonists and to the biochemistry of cholinergic neurotransmission.

Studies of the parasympathetic nervous system and cholinergic agents led to the concept of neurochemical transmission and were instrumental in the development of early drug receptor hypotheses and our understanding of the stereochemical influence on drug action. An excellent summary of this history is presented by Holmstedt (2).

In 1914, Dale defined two subdivisions of the parasympathetic nervous system when he observed that ethers and esters (including acetylcholine) of choline produced effects similar to those of either muscarine (muscarinic effects) or nicotine (nicotinic effects) (Fig 9.2) (2). The initial experiments were performed using an ergot extract contaminated with acetylcholine, although Dale was unaware of this contamination. Ewins, a chemist who collaborated with Dale, isolated acetylcholine from the ergot extract and subsequently synthesized acetylcholine, thus allowing Dale to show that the unexpected muscarinic effects observed with the ergot preparation were the result of acetylcholine. He proposed the term “parasympathomimetic” to describe the ability of acetylcholine to produce the same effects as electrical stimulation of parasympathetic nerves, and he suggested that

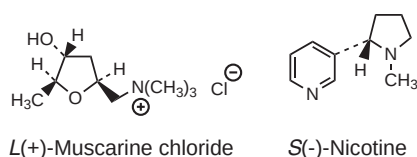


FIGURE 9.2 Structures of L-(+)-muscarine chloride and (-)-nicotine.

acetylcholine was a chemical neurotransmitter in the parasympathetic nervous system. Dale also observed that the action of acetylcholine was short-lived, and he proposed that tissues contained an esterase that hydrolyzed acetylcholine. In 1921, Loewi (3) demonstrated that a chemical compound mediated impulses between nerves; he referred to the chemical substance in his preparation as *vagusstoff*. In 1926, Loewi and Navratil (4) provided experimental evidence suggesting that *vagusstoff* was acetylcholine.

These classic studies are the foundation of our current understanding of the role of acetylcholine in cholinergic nerve transmission and our recognition of muscarinic acetylcholinergic receptors (mAChR) and nicotinic acetylcholinergic receptors (nAChRs). These classic studies provided the stimulus for subsequent studies of acetylcholine biochemistry, synthesis of new organic compounds (e.g., cholinergic and anticholinergic drugs), and purification of cholinergic receptors.

The concept that mAChRs and nAChRs can explain the different physiologic responses produced by acetylcholine was derived from this early research of Dale and Loewi. Although it is now recognized that there are multiple subclasses of both mAChRs and nAChRs, the general classification of these two types of cholinergic receptors continues to effectively explain the different physiologic responses produced by acetylcholine.

Because of the important role of acetylcholine as a chemical neurotransmitter in the autonomic nervous system, an imbalance in parasympathetic tone can lead to serious consequences. Conceptually, deficiencies in acetylcholine could be treated by administering the neurotransmitter itself, but acetylcholine is a poor therapeutic agent. Its actions are nonselective, producing effects at all cholinergic receptor sites, which could result in serious consequences for the patient. Because acetylcholine is a quaternary ammonium salt, it is poorly absorbed across biologic membranes, resulting in poor bioavailability regardless of the route of administration. Furthermore, its ester functional group is rapidly hydrolyzed in the acidic conditions of the gastrointestinal tract and by esterases in plasma.

Muscarinic cholinergic agents are used postsurgically to reestablish smooth muscle tone of the gastrointestinal and urinary tracts in order to relieve abdominal distention and urinary retention. These agents are also used to

CLINICAL SIGNIFICANCE

Agents affecting cholinergic neurotransmission are some of the most widely studied agents to date.

They are also one of the most intriguing classes of study in that the clinical utility of

these compounds runs the gamut from the life-saving potential of atropine given to patients undergoing cardiac life support to the life-threatening capacity of chemical warfare agents, such as sarin. Advancements in our understanding of muscarinic and nicotinic receptor activity and compounds that modulate these effects have led to decreased morbidity and mortality and enhanced quality of life for millions of individuals throughout the world. In addition, those agents employed as insecticides or pesticides have had tremendous economic impact. Agents affecting cholinergic neurotransmission are used to treat a variety of clinical conditions, including impaired or excessive gastric motility/secretion, glaucoma, bradycardia, Alzheimer's disease, Parkinson's disease, and myasthenia gravis. Nicotinic antagonists are used to facilitate surgical procedures by reducing the amount of anesthetic or sedative agent required, thus reducing risk to the patient. A thorough understanding of the structure–activity relationships of these compounds has led to an ability to enhance their desired pharmacodynamic effects while minimizing unwanted or harmful adverse effects.

For the clinician, the impact of the application of this knowledge is multifaceted. Examples include the synthesis of

newer chemical compounds used to treat Alzheimer's disease that provide greater affinity for acetylcholinesterase in the brain than in the periphery, decrease the frequency of dosing required, and alleviate the risk of hepatotoxicity associated with the first Alzheimer's agent, tacrine. These advances have increased the utility of these agents and given hope to countless individuals and families affected by this devastating disease. Modifications to neuromuscular blocking agents have resulted in differences in onset and length of activity, reduction in adverse effects (e.g., hypotension), and alternate routes of elimination, which increase their utility for patients with certain comorbid conditions (e.g., cardiac disease or renal dysfunction). Finally, it is important for the clinician to recognize the capacity of certain chemical configurations to be allergenic or more prone to producing adverse effects so the best agent for a particular patient can be selected.

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treat some forms of glaucoma by enhancing the outflow of aqueous humor, thereby reducing intraocular pressure. Cholinomimetic compounds with central nervous system (CNS) activity are used for the treatment of cognitive disorders (e.g., Alzheimer disease). Those that exhibit nicotinic effects are commonly used to treat myasthenia gravis.

Cholinergic muscarinic antagonists (anticholinergic drugs) are sometimes referred to as antispasmodics due to their ability to reduce smooth muscle spasms resulting from overstimulation of the gastrointestinal smooth muscles. Newer drugs in this class have found use in the treatment of overactive bladder.

Many synthetic cholinergic agonists have been designed using structure–activity relationships (SARs) based on the structure of acetylcholine. To design cholinergic agents that are selective for specific cholinergic receptors, it is necessary to have a thorough understanding of acetylcholine neurochemistry as well as the chemical nature and role of cholinergic receptors.

CHOLINERGIC RECEPTORS

History

Knowledge of the structure and function of cholinergic receptors has increased substantially since the concept of distinct mAChRs and nAChRs was first postulated. Early efforts to describe these receptors were hindered in that receptors were only a concept. Indeed, the existence of receptors was not established until 1973, when Pert and Snyder (5) provided demonstrable evidence for the existence of opiate receptors.

Early attempts to characterize cholinergic receptors were based on SAR and stereochemical studies of cholinergic agonists and antagonists. This led to synthesis of agonists and antagonists with exceptionally high affinity and selectivity for cholinergic receptors as well as to synthesis of radiolabeled cholinergic ligands with high specific radioactivity. Parallel advances in biochemistry, molecular pharmacology, and molecular biology made possible purification and sequencing of small quantities of protein, measurement of ligand binding to cell membranes and subcellular components, and cloning and sequencing of genes. This led to isolation, purification, and amino acid sequencing of one of the nAChRs—the first acetylcholine receptor and the first neurotransmitter receptor to be fully characterized (6,7). Subsequently, mAChRs were isolated, purified, and sequenced using these techniques.

Current research in pharmacology and molecular biology indicates that multiple mAChR and nAChR subtypes exist (8,9). However, the traditional classification of mAChRs and nAChRs adequately describes the actions of most cholinergic medicinal agents and is used throughout this chapter. Furthermore, most of the current therapeutic agents acting at mAChRs exhibit little selectivity for the receptor subtypes, with the exception of a few recently introduced anticholinergic agents for treatment of overactive bladder.

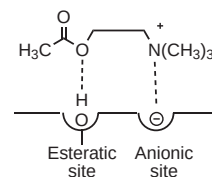


FIGURE 9.3 Original representation of the mAChR.

Muscarinic Receptors

The SAR regarding affinity and efficacy of cholinergic agonists provided the basis for early models of mAChR structure. An early model of the mAChR, depicted in Figure 9.3, illustrates the importance of muscarinic agonists having an ester functional group and a quaternary ammonium group separated by two carbons. This model depicts ionic bonding between the positively charged quaternary nitrogen of acetylcholine and a negative charge at the anionic site of the receptor. The negative charge was suggested to result from a carboxylate ion from the free carboxyl group of a dicarboxylic amino acid (e.g., aspartate or glutamate) at the binding site on the receptor protein. This model also involved a hydrogen bond between the ester oxygen of acetylcholine and a hydroxyl group contributed by the esteratic site of the receptor.

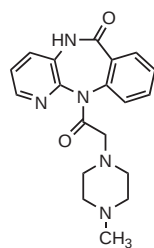
Although this early mAChR model accounted for two important SAR requirements for muscarinic agonists, it failed to explain the following: 1) at least two of the alkyl groups bonded to the quaternary nitrogen must be methyl groups; 2) the known stereochemical requirements for agonist binding to the receptor; and 3) the fact that all potent cholinergic agonists have only five atoms between the quaternary nitrogen and the terminal hydrogen atom. This last point is known as Ing's "Rule of Five" (10).

Subsequent models of the mAChR have depicted the receptor as a binding site on a protein molecule and explained more completely the structural and stereochemical requirements for cholinergic agonist activity. Interestingly, it was once proposed that the mAChR and AChE were the same entity, but this proposal was dispelled by experiments demonstrating that interaction of cholinergic ligands with the mAChR did not lead to hydrolysis of the ligand. None of these models, however, completely explained the diverse pharmacologic effects produced by all muscarinic agonists and antagonists.

Subsequent developments suggested that mAChR stimulation results in at least two important events: 1) inhibition of adenyl cyclase*, and 2) activation of phospholipase C and subsequent biosynthesis of second messengers. Both involve a guanosine triphosphate (GTP)-dependent mechanism. Two other important developments that helped characterize mAChRs were the synthesis of radiolabeled muscarinic ligands and the utilization of molecular biology techniques.

*This enzyme is known by two names, *adenylate cyclase* (EC 4.6.1.1), its official name from the International Union of Biochemistry and Molecular Biology Nomenclature Committee, or its alternative name, *adenyl cyclase*.

Heterogeneity in the mAChR population was first suggested in the late 1970s during pharmacologic studies using the muscarinic antagonist pirenzepine. At the time, pirenzepine was the only muscarinic antagonist to block gastric acid secretion at concentrations that did not block the effects of muscarinic agonists. This observation initiated research that ultimately led to discovery of mAChR subtypes, designated as M_1 , M_2 , and M_3 based on their pharmacologic responses to various ligands. Rapid advances in molecular biology led to cloning of cDNAs that encoded for five mAChRs, designated as m1 through m5; m1, m2, and m3 correspond to the respective M_1 through M_3 receptors identified by their pharmacologic specificity. The International Union of Pharmacology Committee on Receptor Nomenclature and Drug Classification has recommended that the uppercase nomenclature M_1 through M_5 be used to designate both pharmacologic and molecular subtypes (8).



Pirenzepine

All mAChR subtypes (M_1 to M_5) are found in the CNS, whereas other tissues can contain more than one. These receptors are summarized in Table 9.1 (11). As more mAChR subtypes have been discovered, it has become apparent that there is a lack of known antagonists exhibiting “very high subtype selectivity” and that there “are no muscarinic agonists with high selectivity” (8). Thus, proof for involvement of any one receptor subtype in a given system currently requires use of more than one antagonist. Additionally, if the selectivity of a novel muscarinic antagonist or putative agonist is to be assessed, it should be through use of recombinant mAChRs expressed in cell lines rather than with native receptors.

Cloning and sequencing of genes encoding mAChRs have led to major advances in understanding of their chemical nature and function (8,12–14). These experiments demonstrated that mAChRs are coupled to guanine nucleotide-binding proteins and are referred to as G protein-coupled receptors (GPCR) (8,15,16). The guanine nucleotide regulatory protein to which the receptors are coupled has three subunits (α , β , and γ) that link the receptor to effectors that produce second messenger molecules within the cell. Binding of muscarinic agonists to GPCRs leads to a variety of effector responses (see below). The ultimate observable response is a function of the tissue where the receptor is located.

Amino acid sequences of mAChR proteins expressed by cloned genes for the GPCRs have been deduced from the base sequence of the respective genes. These GPCRs

are components of the cell membrane and consist of seven hydrophobic transmembrane helical domains as well as hydrophilic extracellular and intracellular domains (17). The N-terminus of the GPCR protein is extracellular, and the C-terminus is intracellular. This proposed arrangement and the deduced amino acid sequence for the human type M_1 mAChR has been reported (12). Computer-assisted molecular modeling has made it possible to obtain three-dimensional models of the mAChR (17); a proposed top-view model of the M_1 mAChR is shown in Figure 9.4 (18). This model suggests that the quaternary nitrogen of acetylcholine participates in an ionic bond with the free carboxylate group of an aspartate residue (ASP105 in transmembrane domain 3)—one of the receptor functional groups that was originally hypothesized to be involved in receptor binding of acetylcholine.

A current model for mAChRs that better describes signal transduction is illustrated in Figure 9.5 (14). In this model, acetylcholine binds to the mAChR located in the cell membrane, and this ligand-receptor interaction is translated, presumably by a conformational perturbation, through the receptor protein to the receptor-coupled guanine nucleotide regulatory protein (G protein). A relationship between the guanine nucleotide regulatory protein and the effector is illustrated in Figure 9.6. In this scheme, the G protein is in the inactive state, with guanosine diphosphate (GDP) bound to its α subunit. Upon interaction of an agonist with the mAChR, the α subunit releases GDP and binds GTP. The α subunit-GTP complex then dissociates from the $\beta\gamma$ subunits. Both the α subunit-GTP complex and the $\beta\gamma$ subunits interact with membrane-bound effectors (phospholipase C or adenylate cyclase) or ion channels (K^+ and Ca^{2+}), either independently or in a parallel manner. The α subunit possesses GTPase activity and quickly hydrolyzes the GTP to GDP to terminate signal transmission, at which time the α , β , and γ subunits reassociate and migrate back to the receptor protein. Characteristics of the α subunit determine the classification of the particular G protein:

- G_s increases adenylyl cyclase activity and increases Ca^{2+} currents;
- G_i decreases adenylyl cyclase activity and increases K^+ currents;
- G_o decreases Ca^{2+} currents; and
- G_q increases phospholipase C activity.

The $\beta\gamma$ subunits are involved with receptor-operated K^+ currents and with activity of adenylyl cyclase and phospholipase C.

Signal transduction at the stimulatory “odd-numbered” mAChRs (i.e., M_1 , M_3 , and M_5) is via coupling with a $G_{q/11}$ protein that is involved with mobilization of intracellular calcium. Agonist binding to these receptors results in activation of phospholipase C, with subsequent production of the second messengers, diacylglycerol and inositol-1,4,5-triphosphate (IP_3). Stimulation of IP_3 ion channel receptors leads to release of intracellular calcium from the endoplasmic reticulum. The diacylglycerol

TABLE 9.1 Muscarinic Acetylcholine receptor Subtypes^a

Receptor	G Protein	Tissue Location	Cellular Response	Function
M ₁	G _{q/11}	CNS, gastric and salivary glands, autonomic ganglia, enteric nerves	PLC activation ($\uparrow$ IP ₃ & $\uparrow$ DAG $\rightarrow$ $\uparrow$ Ca ²⁺ & PKC); depolarization and excitation ($\uparrow$ sEPSP); PLA ₂ and PLD ₂ activation; $\uparrow$ AA	$\uparrow$ Cognitive function $\uparrow$ Seizure activity $\uparrow$ Secretions $\uparrow$ Autonomic ganglia depolarization $\downarrow$ DA release and locomotion
M ₂	G _i /G _o	Autonomic nerve terminals; CNS; heart; smooth muscle	Inhibition of adenylyl cyclase ($\downarrow$ cAMP) & voltage-gated Ca ²⁺ channels; activation of inwardly rectifying K ⁺ channels	$\downarrow$ Heart rate $\uparrow$ Smooth muscle contraction Neural inhibition in periphery via autoreceptors and heteroreceptor $\downarrow$ Ganglionic transmission Neural inhibition in CNS $\uparrow$ Tremors, hypothermia & analgesia
M ₃	G _{q/11}	CNS (less than other mAChRs), smooth muscle, glands, heart	Same as M ₁	$\uparrow$ Smooth muscle contraction (e.g., bladder) $\uparrow$ Salivary gland secretion $\uparrow$ Food intake, body fat deposits Inhibits dopamine release Synthesis of nitric oxide
M ₄	G _i /G _o	CNS	Same as M ₂	Inhibition of autoreceptor- and heteroreceptor-mediated transmitter release in CNS Analgesia Cataleptic activity Facilitates dopamine release
M ₅	G _{q/11}	Low levels in CNS & periphery; predominate mAChRs in dopaminergic neurons of substantia nigra & ventral tegmentum area	Same as M ₁	Mediates dilation of cerebral arteries Facilitates dopamine release Augments drug-seeking behavior and reward

^aCNS, central nervous system; PLC, Phospholipase C; IP₃, inositol-1, 4, 5-triphosphate; DAG, diacylglycerol; PLD₂, phospholipase D; AA, arachidonic acid; PKC, protein kinase C; sEPSP, slow excitatory postsynaptic potential; mAChRs, muscarinic acetylcholine receptor subtypes; PLA, phospholipase A; cAMP, cyclic adenosine monophosphate; VTA, ventral tegmentum area.

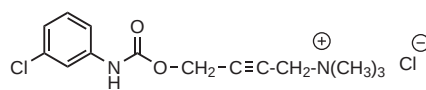
Adapted from Westfall TC, Westfall DP. Neurotransmission: the autonomic and somatic motor nervous systems. In: Brunton LL, Lazo JS, Parker KL, eds. Goodman and Gilman's The Pharmacological Basis of Therapeutics, 11th Ed. New York: McGraw-Hill, 2006, pp. 137-181; with permission.

produced, along with calcium, activates protein kinase C, which phosphorylates proteins to afford various physiologic responses. The M₁, M₃, and M₅ receptors also stimulate phospholipase A₂ and phospholipase D. Activation of phospholipase A₂ results in release of arachidonic acid, with subsequent synthesis of eicosanoids (C₂₀ fatty acids).

The “even-numbered” mAChR subtypes (i.e., M₂ and M₄) are coupled to G_i/G_o proteins, whose activation inhibits adenylyl cyclase. This results in a decrease in cyclic adenosine monophosphate, inhibition of voltage-gated calcium channels, and activation of inwardly rectifying potassium channels (19). The result is hyperpolarization and inhibition of these excitable membranes.

The M₁ receptors are sometimes described as “neural” due to their abundance in the cerebral cortex, hippocampus, and striatum. M₁ receptors have been implicated in Alzheimer disease and are thought to be involved with memory and learning. Early studies suggested that the

agonist McN-A-343 was selective for the M₁ receptor, but more recent evidence indicates otherwise. It can show moderate selectivity for M₄ receptors. Additionally, M₁ receptors are found at autonomic ganglia, enteric nerves, and salivary and gastric glands. Agonists at M₁ receptors show the greatest promise for treatment of the cholinergic deficit associated with Alzheimer disease.



McN-A-343

M₂ receptors are found in abundance in the heart, where activation exerts both negative chronotropic and inotropic actions, and stimulate contraction of smooth muscle. Activation of M₂ autoreceptors located on nerve

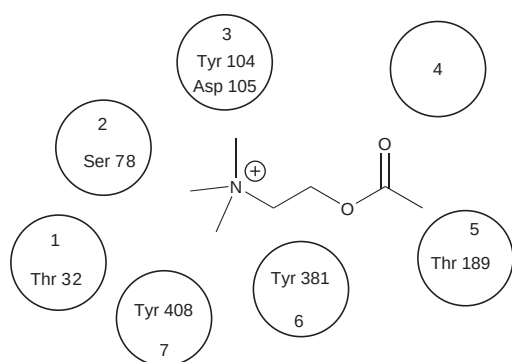


FIGURE 9.4 Model of acetylcholine interaction with muscarinic M_1 receptor. Circles represent seven transmembrane domains (18).

terminals affords neural inhibition by decreasing acetylcholine release.

M_3 receptors are found in abundance in smooth muscle and glands, where their stimulation leads to contraction and secretion, respectively. Knowledge of this effect on smooth muscle of the bladder has led to development and subsequent approval of several M_3 receptor antagonists for treatment of overactive bladder (see below). Although widely distributed in the CNS, their concentration in the CNS is lower than those of other mAChRs. M_3 mAChRs function to decrease neurotransmitter release.

M_4 receptors are found in the striatum and basal forebrain, where they decrease transmitter release in both the CNS and periphery. Their activation in smooth muscle and secretory glands leads to inhibition of potassium and calcium channels.

M_5 receptors are the least characterized of the mAChRs, and there is evidence for their existence in the CNS and periphery, whereby they can regulate dopamine release in the CNS.

Nicotinic Receptors

nAChRs are found at the skeletal neuromuscular junction, adrenal medulla, and autonomic ganglia. nAChRs have been the focus of intensive research interest, even though the majority of clinically effective cholinergic medicinal agents are either muscarinic agonists or

antagonists (20). The best known nicotinic agonist is nicotine, which is an alkaloid obtained from *Nicotiana tabacum* plant. (Fig. 9.2). Interest in nAChRs stems from the availability of the receptor protein from the electric organs of the electric eel (*Electrophorus electricus*) and the marine ray (*Torpedo californica*) and the important role they play in myasthenia gravis, an autoimmune disease.

The multiplicity of nAChRs is based on different structural requirements for agonists and antagonists acting at the autonomic ganglia and the skeletal neuromuscular junction and is supported by molecular biology research (20). Both ganglionic neuronal (NN) and neuromuscular (somatic muscle) (NM) nAChRs are classified as ligand-gated ion channel receptors that are structurally and functionally related to other ligand-gated ion channel receptors, such as those for γ -aminobutyric acid (GABA), 5-hydroxytryptamine (serotonin), and glycine (21). The nAChR creates a transmembrane ion channel (the gate), and acetylcholine (the ligand) serves as a gatekeeper by binding with the nAChR to modulate passage of ions, principally K^+ and Na^+ , through the channel.

The nAChR was the first neurotransmitter receptor to be isolated and purified in the active form using the same molecular biologic techniques described earlier for isolation and purification of mAChRs. The primary sequence of nAChRs has been deduced from cloning and sequencing of genes that encode their receptor proteins (22,23). nAChRs are pentameric transmembrane proteins made up of subunits designated as α , β , γ (or ϵ), and δ (24).

The nAChR of skeleton (somatic) muscle tissue is a transmembrane glycoprotein consisting of four types of subunits— α , β , γ (or ϵ), and δ . Only the α_1 subtype of the α subunit is present in muscle. In a mature muscle end plate, the γ subunit is replaced by an ϵ subunit. This change in gene expression encoding the γ and ϵ subunits affects ligand selectivity along with receptor turnover and/or tissue location.

One class of neuronal nAChRs exists as a heteromeric pentamer composed of α (α_2 to α_6) and β (β_2 to β_4) subunits (25–27)—for example, $\alpha_4\beta_3$ with a stoichiometry of two α_4 and three β_3 subunits. Another class of functional homomeric nAChRs is composed of α_7 through α_{10} subunits. The diversity of the subunits and the pentameric structure suggest that a large number of nAChR subtypes can exist.

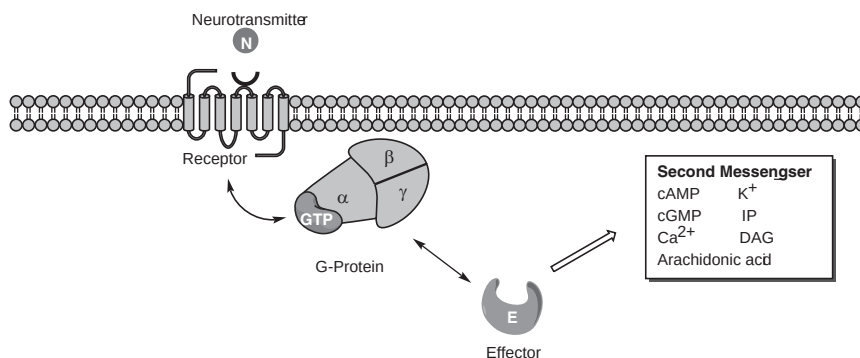


FIGURE 9.5 Model of signal transduction by a G protein-coupled receptor. This illustrates a proposed relationship between receptor, G protein, effector, and various second messengers.

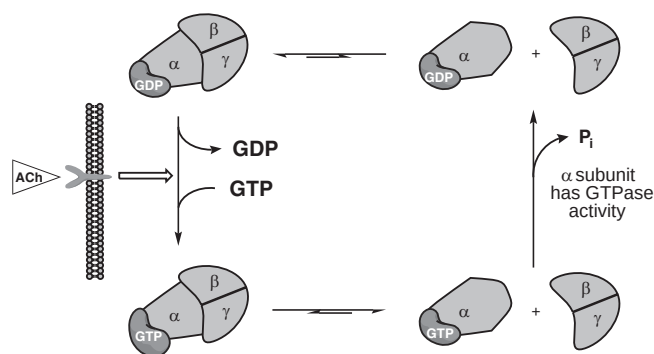


FIGURE 9.6 Diagram of a GTPase cycle and subunit association/dissociation proposed to control signal transduction between muscarinic G protein-coupled receptors and the effector. The acetylcholine (ACh)-receptor interaction facilitates GTP binding and activates the α subunit. The α subunit-GTP complex then dissociates from the $\beta\gamma$ subunit, and each is free to activate effector proteins. The duration of separation is determined by the rate of α subunit-mediated GTP hydrolysis.

The five subunits of each nAChR protein in muscle tissue are arranged around a central pore that serves as the ion channel. Based on molecular modeling of the deduced primary structure of the individual subunits, it has been proposed that each subunit [α , β , γ (or ϵ), and δ] possesses a hydrophilic extracellular N-terminus, a hydrophilic intracellular C-terminus, and four α helical hydrophobic transmembrane domains (M_1 to M_4) (Fig. 9.7) (26,28). (Note: Do not confuse these transmembrane domains of the nAChRs with the domains of the mAChRs.) A pentameric arrangement of these five amphipathic subunits makes up the walls of the ion channel. Two acetylcholine binding sites exist on the extracellular domain of each nAChR molecule. In Figure 9.7, one binding site is located on each α subunit at the $\alpha\gamma$ and $\alpha\delta$ interfaces (28). The binding sites show a positive cooperativity, even though the binding sites are not adjacent to each other in the pentameric receptor. Some central and peripheral nAChR subtypes are summarized in Table 9.2 (11).

Our knowledge and understanding of mAChR and nAChRs have advanced tremendously from the time

when these receptors were only ethereal concepts. This understanding provides the basis for the rational design of new selective therapeutic agents to treat diseases associated with cholinergic neurons.

DRUGS AFFECTING CHOLINERGIC NEUROTRANSMISSION

Acetylcholine Neurochemistry

The neurochemistry of acetylcholine includes its biosynthesis, storage, release, and metabolism. These are illustrated in Figure 9.8.

Biosynthesis

Acetylcholine is biosynthesized in cholinergic neurons by the enzyme-catalyzed transfer of the acetyl group from acetyl coenzyme A (acetyl-S-CoA) to choline, a quaternary ammonium alcohol (Fig. 9.9) (29). The enzyme catalyzing this reaction, choline acetyltransferase, is also biosynthesized in the cholinergic neuron. Choline used for biosynthesis of acetylcholine in the CNS comes from several sources. About 35% to 50% of the choline produced by AChE-catalyzed hydrolysis of acetylcholine in the synaptic space is transported into the neuron (see below). This accounts for about half of the choline required for acetylcholine synthesis (30). Two stores of choline are the lipid phosphatidylcholine, found in the plasma, and phosphorylcholine, stored in lipid. Although, choline will not cross the blood-brain barrier, phosphatidylcholine and phosphorylcholine will. Once in the CNS, these phosphatidylcholine esters can be hydrolyzed to afford choline (30). Choline can be biosynthesized from the amino acid serine (Fig. 9.9), but most of the choline used to form acetylcholine is recycled after AChE-catalyzed hydrolysis of acetylcholine in the synaptic space. The choline acetyltransferase-catalyzed reaction is not the rate-limiting step in acetylcholine biosynthesis. Active uptake of choline into the neuron is rate-determining. Choline is taken up into cells by both low-affinity and high-affinity transport sites. The low-affinity ($K_m = 10$ to $100 \mu M$) transporter is found

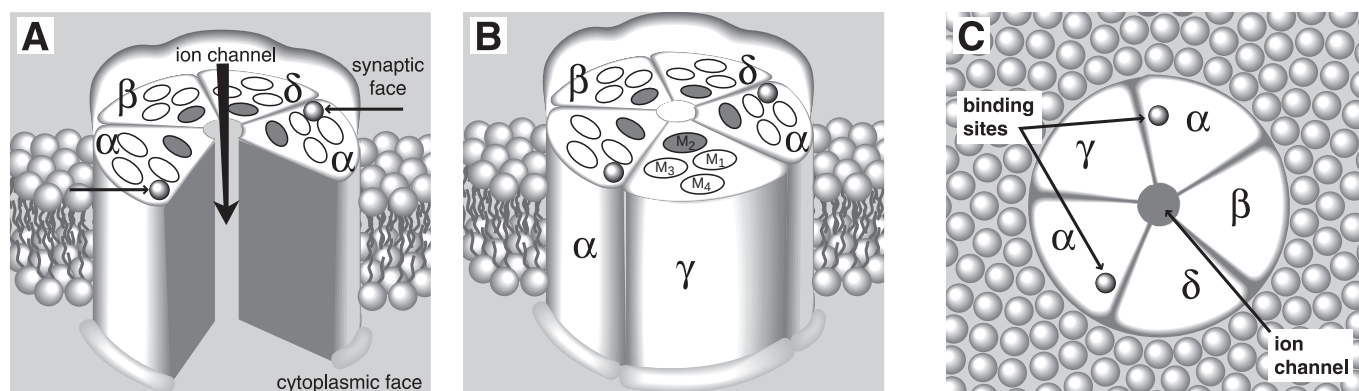


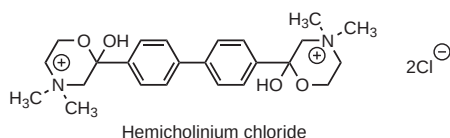
FIGURE 9.7 nAChR: (A) Longitudinal view (γ subunit removed) showing the internal ion channel. Acetylcholine binding sites on the α subunits are indicated by the arrows. These are located at the $\alpha\gamma$ and $\alpha\delta$ interfaces. (B) Each of the five transmembrane subunits (α , α , β , δ , and γ) are composed of four hydrophobic membrane spanning segments (M_1 to M_4). (C) Top view of the nAChR showing the subunits surrounding the ion channel.

TABLE 9.2 Nicotinic Acetylcholine Receptor Subtypes

Receptor	Location	Membrane Response	Molecular Mechanism	Agonists	Antagonists
Neuromuscular (N_M) (α_1) ₂ (β_1) ϵ (α_1) ₂ (β_1) $\gamma\delta$	Skeletal (somatic) neuro-muscular junction (postjunctional)	Excitatory; end plate depolarization; contraction (skeletal muscle)	Increased Na^+ & K^+ permeability	ACh; nicotine; succinylcholine	Atracurium; Vecuronium; <i>d</i> -tubocurarine; pancuronium; α -conotoxin; α -bungarotoxin
Ganglionic neuronal (N_N) (α_3) ₂ (β_4) ₃	Autonomic ganglia; adrenal medulla	Excitatory; depolarization firing of postganglionic neuron; depolarization & secretion of catecholamines	Increased Na^+ & K^+ permeability	ACh; nicotine; epibatidine; dimethylphenyl-piperazinium	Trimethaphan; mecamylamine; dihydro- β erythrodine; erysodine lophotoxin
Central neuronal (α_4) ₂ (β_4) ₃ (α -bungarotoxin insensitive)	CNS; pre- & postjunctional	Pre- & postsynaptic excitation; presynaptic control of transmitter release	Increased Na^+ & K^+ permeability	Cytisine; epibatidine; Anatoxin A	
(α_7) ₅ (α -bungarotoxin sensitive)	CNS; pre- and postsynaptic	Same as central neuronal	Increased Ca^{2+} permeability	Anatoxin A	Methyllycaconitine; α -conotoxin; α -bungarotoxin

Adapted from Westfall TC, Westfall DP. Neurotransmission: the autonomic and somatic motor nervous systems. In: Brunton LL, Lazo JS, Parker KL, eds. Goodman and Gilman's The Pharmacological Basis of Therapeutics, 11th Ed. New York: McGraw-Hill, 2006. pp. 137–181; with permission.

in cells that synthesize choline-containing phospholipids (e.g., corneal epithelium). The high-affinity ($K_m = 1$ to $5 \mu M$) transporter is a sodium-choline cotransporter and is found in cholinergic nerve endings, where it is responsible for uptake of most of the choline recycled from the synapse (30). Because this high-affinity uptake site is saturable at greater than $10 \mu M$, the rate of choline uptake by the neuron is limited. Thus, choline uptake is considered to be the rate-determining step in biosynthesis of acetylcholine. Hemicholinium-3 is a research tool that inhibits reuptake of choline at high-affinity uptake sites.



Storage

Most newly biosynthesized acetylcholine is actively transported into cytosolic storage vesicles located in presynaptic nerve endings, where it is maintained with adenosine triphosphate (ATP) (10:1 ratio) along with calcium and magnesium ions until it is released. Some acetylcholine remains in the cytosol and eventually is hydrolyzed. Only the stored form serves as the functional neurotransmitter.

Release

Release of acetylcholine from the storage vesicles is initiated by an action potential that has traveled down the axon to the presynaptic nerve membrane. This action potential leads to opening of voltage-dependent calcium channels, affording an influx of Ca^{2+} and exocytotic

EFFORTS TO MODULATE ACETYLCHOLINE BIOSYNTHESIS

Efforts to develop therapeutic agents based on regulation of acetylcholine biosynthesis have not been successful. Dexpanthenol, the dextrorotatory enantiomer of the alcohol derived from pantothenic acid (a vitamin), was once used as a cholinomimetic agent to help reestablish normal smooth muscle tone in the gastrointestinal tract following surgery. Pantothenic acid is essential for the biosynthesis of coenzyme A (CoA). The apparent rationale for the therapeutic use of dexpanthenol was that it would be biotransformed to pantothenic acid, which would be incorporated into CoA. This would lead to increased intracellular levels of acetyl CoA, which would facilitate increased biosynthesis of acetylcholine. The limited therapeutic success of dexpanthenol, difficulty with administration, and effectiveness of synthetic cholinergic agonists led to its discontinuation. The quaternary pyridinium salt, *trans*-*N*-methyl-4-(1-naphthylvinyl)pyridinium iodide, is an effective inhibitor of choline acetyltransferase in vitro, but it has proven to be a poor inhibitor in whole animal experiments.

Although efforts have been made to design cholinergic agents based on the mechanism of biosynthesis of acetylcholine, such agents would be expected to have nonselective effects, because it currently is thought that acetylcholine is biosynthesized by the same mechanism in all cholinergic neurons.

release of acetylcholine into the synapse. The increase in intracellular Ca^{2+} can induce fusion of acetylcholine storage vesicles with the presynaptic membrane before release of the neurotransmitter. Each synaptic vesicle contains a quantum of acetylcholine; one quantum represents between 12,000 and 60,000 molecules of

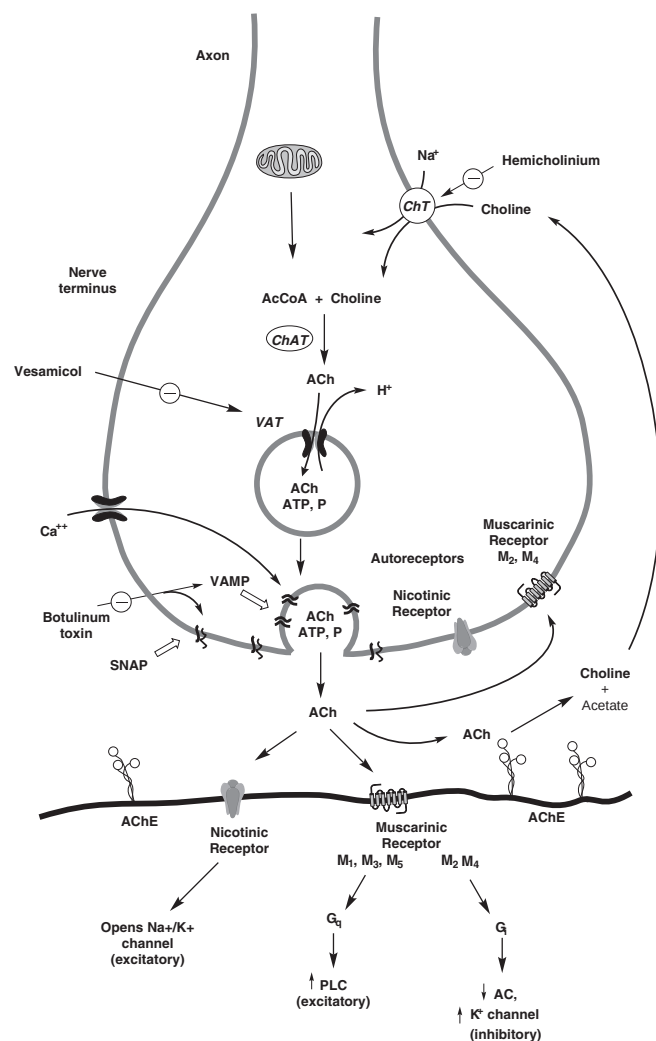


FIGURE 9.8 General cholinergic nerve junction showing location of receptor sites and biosynthesis, storage, release, and hydrolysis of acetylcholine.

acetylcholine. A single action potential causes release of several hundred quanta of acetylcholine into the synapse.

Metabolism

Acetylcholine in the synapse can bind with cholinergic receptors on postsynaptic or presynaptic membranes to produce a response. Free acetylcholine that is not bound to a receptor is hydrolyzed by AChE. Hydrolysis is the physiologic mechanism for terminating the action of acetylcholine. Enough AChE is present in the synapse to hydrolyze approximately 3×10^8 molecules of acetylcholine in 1 millisecond; thus, adequate enzyme activity exists to hydrolyze all the acetylcholine ($\sim 3 \times 10^6$ molecules) released by one action potential. A number of useful therapeutic cholinomimetic agents have been developed based on the ability of the compounds to inhibit AChE; these agents are addressed later in this chapter.

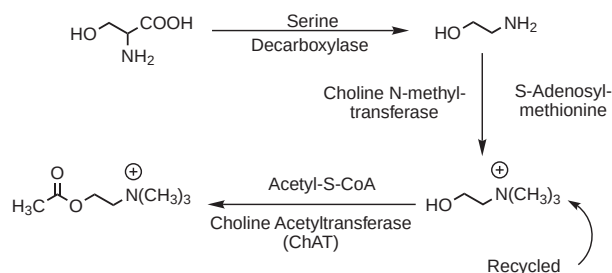


FIGURE 9.9 Biosynthesis of acetylcholine.

Stereochemistry

One shortcoming of early models for cholinergic receptors was that they did not account for the observed stereoselectivity of the receptors for agonist and antagonist ligands. Even though acetylcholine is achiral, many synthetic and naturally occurring agonists and antagonists possess chirality; usually, one enantiomer is many times more active than the other. It was apparent to early receptor investigators that the stereochemistry of cholinergic ligands was important for receptor binding. In this regard, the stereochemical-activity relationships of cholinergic ligands have been studied extensively to provide a rational basis for design of cholinergic drugs as well as to describe the properties and functions of cholinergic receptors.

The stereochemistry of acetylcholine resides in the different arrangements in space of its atoms by virtue of rotation about σ bonds (i.e., conformational isomerism). Because of relatively unrestricted rotation about these single covalent bonds, acetylcholine can exist in an infinite number of conformations. Most studies of the conformational isomerism of acetylcholine have focused on torsion angles between the ester oxygen atom and the quaternary nitrogen resulting from rotation about the α - β bond. Four of these conformations are illustrated by Newman projections in Figure 9.10.

Nuclear magnetic resonance (NMR) studies in aqueous solution revealed that the preferred conformation between the ester oxygen and the quaternary nitrogen of

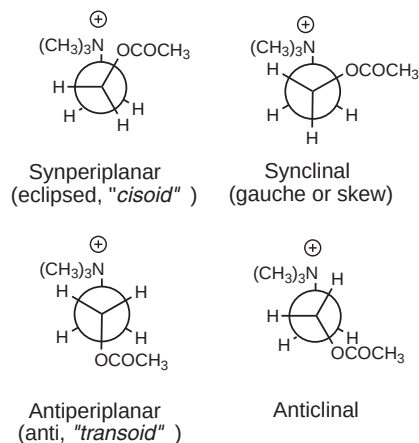
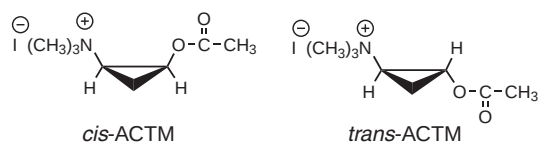


FIGURE 9.10 Conformational isomers of acetylcholine.

acetylcholine was synclinal (gauche or skew). Using x-ray data, the synclinal conformation was observed for acetylcholine in the solid crystalline state as well. The same conclusion was also obtained for the preferred conformation of acetylcholine using molecular orbital calculations. These experimental and theoretical determinations of the acetylcholine conformation differ from the antiperiplanar conformation that might be expected when using molecular models to visualize minimum steric interactions. The synclinal conformation would be stabilized by intramolecular electrostatic interactions between the quaternary nitrogen and the carbonyl oxygen.

It must be emphasized that the experimentally determined synclinal conformation of acetylcholine is only that measured in aqueous solution (NMR) or the crystalline state (x-ray). This might not be the conformation preferred by the receptors. Indeed, the conformation of receptor-bound acetylcholine could be much different and might not be a thermodynamically preferred conformation.

In recognition of this possibility, conformationally restricted acetylcholine analogs have been synthesized and pharmacologically evaluated in an effort to determine the conformation of acetylcholine when it binds to cholinergic receptors. The most significant study in this regard is that of Armstrong et al. (31), who synthesized and evaluated the muscarinic and nicotinic activity of *cis*- and *trans*-isomers of a conformationally rigid model of acetylcholine, *cis*- and *trans*-2-acetoxycyclopropyl-1-trimethylammonium iodide (ACTM). Because this model is based on the cyclopropane ring, the ester and quaternary ammonium functional groups cannot change their relative positions by bond rotation. The *cis*- and *trans*-isomers are rigidly constrained to the conformations shown. The *cis*-isomer is similar to the synperiplanar conformation of acetylcholine, and the *trans*-isomer approximates the anticlinal conformation. The (+)-*trans*-enantiomer was observed to be equally as or more potent, depending on the pharmacologic test used, than acetylcholine at mAChRs; it was much more potent than the (–)-*trans*-enantiomer. The racemic *cis*-compound had almost no activity in the same mAChR test system, and all compounds were very weak nicotinic agonists.



The important conclusion drawn from this study (31) was that acetylcholine would most probably interact with mAChRs in its less favored anticlinal conformation. The most active isomer, the (+)-*trans*-enantiomer, of these cyclopropane analogs was found to have a torsion angle of 137 degrees (anticlinal) between the ester oxygen and the quaternary nitrogen. This is significantly different from the 60-degree torsion angle in the synclinal conformation found by NMR and x-ray determinations to be the preferred conformation.

Stereochemistry of cholinergic ligands and stereoselectivity of receptors has played an important role in design of cholinergic ligands as therapeutic agents. This role becomes apparent in subsequent sections.

Acetylcholine Mimetics—Muscarinic Agonists

Interaction of cholinergic agonists with mAChRs leads to well-defined pharmacologic responses depending on the tissue or organ in which the receptor is located. These responses include contractions of smooth muscle, vasodilation, increased secretion from exocrine glands, miosis, and decreased heart rate and force of contraction.

Acetylcholine

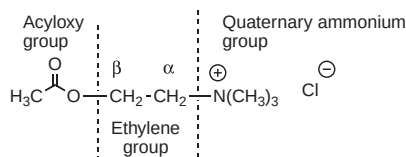
Acetylcholine is the prototypical muscarinic and nicotinic agonist, because it is the physiologic chemical neurotransmitter for the cholinergic nervous system. However, it is a poor therapeutic agent due to its lack of specificity for nAChRs or mAChRs and due to the chemical and physicochemical properties associated with its ester and quaternary ammonium salt functional groups. It is quite stable in the solid crystalline form but undergoes rapid hydrolysis in aqueous solution. This hydrolysis is accelerated in the presence of catalytic amounts of either acid or base. For this reason, acetylcholine cannot be administered orally due to rapid hydrolysis in the gastrointestinal tract. Even when administered parenterally, its pharmacologic action is fleeting as a result of hydrolysis by butyrylcholinesterase (also known as pseudocholinesterase or plasma cholinesterase) in serum. The quaternary ammonium functional group of acetylcholine imparts excellent water solubility, but quaternary ammonium salts are poorly absorbed across lipid membranes due to their high hydrophilic and ionic character. Thus, even if acetylcholine were stable enough to be administered orally, it would be poorly absorbed. When used during ocular surgery to produce complete miosis within seconds, acetylcholine must be directly instilled into the anterior chamber. It cannot be administered topically, because it is not lipophilic enough to penetrate the cornea, and it must be reconstitution immediately before instillation due to its hydrolytic lability.

Structure–Activity Relationship for Muscarinic Activity

The necessity to design compounds that would serve as therapeutic muscarinic alternatives to acetylcholine and as probes to study the role of acetylcholine in muscarinic neurotransmission led to an exhaustive study of the structural features required for the muscarinic action of acetylcholine. SARs that developed from these studies have provided the basis for the design of all muscarinic agonists currently used as therapeutic agents.

To review the SAR, it is logical to divide the structure of acetylcholine into the three components shown below

to examine the effects of chemical modification of each group.



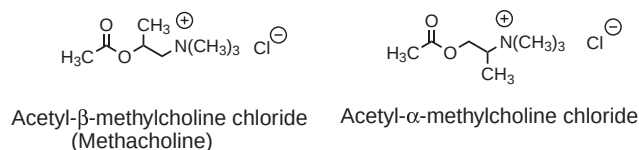
MODIFICATION OF THE QUATERNARY AMMONIUM GROUP Analogs of acetylcholine in which the nitrogen atom was replaced by arsenic, phosphorus, sulfur, or selenium have been synthesized (10,32,33). Although these analogs exhibited some of the activity of acetylcholine, these compounds are less active and are not used clinically. It was concluded that only compounds possessing a positive charge on the atom in the position of the nitrogen had appreciable muscarinic activity.

Compounds in which all three methyl groups on the nitrogen are replaced by larger alkyl groups are inactive as agonists. When the methyl groups are replaced by three ethyl groups, the resulting compound is a cholinergic antagonist. Replacement of only one methyl group by an ethyl or propyl group affords a compound that is active, but much less so than acetylcholine (32). Furthermore, successive replacement of one, two, or three of the methyl groups with hydrogen atoms to afford a tertiary, secondary, or primary amine, respectively, leads to successively diminishing muscarinic activity (34,35).

MODIFICATION OF THE ETHYLENE BRIDGE Synthesis of acetic acid esters of quaternary ammonium alcohols of greater length than choline led to a series of compounds with activity that was rapidly reduced as the chain length increased. This observation led Ing (10) to postulate his Rule of Five. This rule suggests that there should be no more than five atoms between the nitrogen and the terminal hydrogen atom for maximal muscarinic potency. Present concepts suggest that the mAChR cannot successfully accommodate molecules larger than acetylcholine and still produce its physiologic effect. Although larger analogs of acetylcholine can bind to the receptor, they lack efficacy and demonstrate antagonist properties.

Replacement of the hydrogen atoms of the ethylene bridge by alkyl groups larger than methyl affords compounds that are much less active than acetylcholine. Introduction of a methyl group on the carbon β to the quaternary nitrogen affords acetyl- β -methylcholine (methacholine), which has muscarinic potency almost equivalent to that of acetylcholine and much greater muscarinic than nicotinic selectivity.

A methyl group on the carbon α to the quaternary nitrogen affords acetyl- α -methylcholine. Although activity relative to acetylcholine is reduced at both mAChRs and nAChRs, it exhibits greater nicotinic than muscarinic potency. This compound is not currently used as a therapeutic agent.



Addition of methyl groups to either one or both of the ethylene carbons results in chiral molecules. mAChRs display stereoselectivity for the enantiomers of methacholine. Thus, *S*-(+)-methacholine (see Fig. 9.11) is equipotent with acetylcholine, whereas the *R*-(-)-enantiomer is approximately 20-fold less potent. Acetylcholinesterase hydrolyzes the *S*-(+)-isomer much slower (approximately half the rate) than acetylcholine. The *R*-(-)-isomer is not hydrolyzed by AChE and even acts as a weak competitive inhibitor of the enzyme. This stability toward AChE hydrolysis as well as the AChE inhibitory effect of the *R*-(-)-enantiomer can explain why racemic *R,S*-methacholine produces a longer duration of action than acetylcholine. The mAChR and AChE exhibit little stereoselectivity for the optical isomers of acetyl- α -methylcholine.

MODIFICATION OF THE ACYLOXY GROUP As would be predicted by the Rule of Five (10), when the acetyl group is replaced by higher homologs (i.e., the propionyl or butyryl groups), the resulting esters are less potent than acetylcholine. Choline esters of aromatic or higher molecular weight acids possess cholinergic antagonist activity.

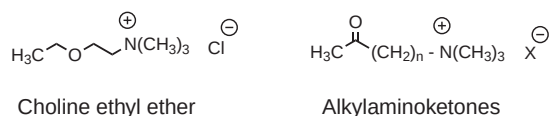


Because the fleeting pharmacologic action and chemical instability of acetylcholine result from its rapid hydrolysis, a logical approach to the development of better muscarinic therapeutic agents was to replace the acyloxy functional group with a functional group more resistant to hydrolysis. This led to synthesis of the carbamic acid ester of choline (carbachol), a potent cholinergic agonist possessing both muscarinic and nicotinic activity. Esters derived from carbamic acid are referred to as carbamates, and because their carbonyl carbon is less electrophilic, the carbamates are more stable than carboxylate esters to hydrolysis. Carbachol exhibits greater stability than acetylcholine toward hydrolysis by gastric acid, AChE, or butyrylcholinesterase and therefore is administered orally.

This same chemical logic was extended to methacholine and led to synthesis of its carbamate ester, bethanechol, an orally effective potent muscarinic agonist with almost no nicotinic activity at therapeutic doses. mAChRs exhibit stereoselectivity for the two optical isomers of bethanechol, and similar to methacholine, the *S*-(+)-enantiomer exhibits greater binding affinity at mAChRs than does the *R*-(-)-enantiomer in isolated receptor preparations.

The profound muscarinic activity of the alkaloid muscarine provided substantial rationale for synthesizing ethers of choline. Muscarine, which is obtained from the red variety of mushroom (*Amanita muscaria*) as well as other mushrooms, is one of the oldest known cholinergic agonists and is the compound for which mAChRs were named (see Fig. 9.2). Muscarine was used in many pharmacologic experiments during the latter 19th century and the early part of the 20th century, and its use preceded the discovery and chemical characterization of acetylcholine (2). The chemical structure of muscarine (see Fig. 9.11), however, was not completely characterized until 1957. Muscarine possesses three chiral centers (C2, C3, and C5). Thus, eight optical isomers (four enantiomeric pairs) are possible. Of these, only the naturally occurring alkaloid, 2*S*,3*R*,5*S*(+)-muscarine [also called *L*-(+)-muscarine], is correctly referred to as muscarine. The C5 carbon of (+)-muscarine has the same absolute configuration as the analogous chiral β carbon in *S*-(+)-methacholine. Thus, acetylcholine and methacholine can adopt three-dimensional structures through molecular flexibility, similar to *L*-(+)-muscarine (see Fig. 9.11).

Other choline ethers and alkylaminoketones have been synthesized and evaluated for muscarinic activity. Choline ethyl ether exhibits significant muscarinic activity and is chemically stable, but it has not been used clinically. The most potent ketone derivatives possess the carbonyl on the carbon δ to the quaternary nitrogen; this is the same relative position as the carbonyl in acetylcholine. This suggests that these carbonyl groups bind by either a hydrogen bond or other dipole-dipole interaction with an appropriate group on the mAChR. Furthermore, the activity of these ethers and ketones demonstrates that neither the ester functional group nor a carbonyl is required for muscarinic agonist activity.



The classic SAR for muscarinic agonist activity can be summarized as follows:

- The molecule must possess a nitrogen atom capable of bearing a positive charge, preferably a quaternary ammonium salt.

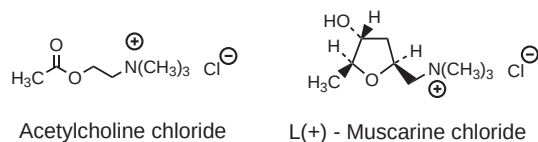


FIGURE 9.11 Structural relationships between acetylcholine, *L*-(+)-muscarine, and *S*-(+)-methacholine.

- For maximum potency, the size of the alkyl groups substituted on the nitrogen should not exceed the size of a methyl group.
- The molecule should have an oxygen atom, preferably an ester-like oxygen, capable of participating in a hydrogen bond.
- There should be a two-carbon unit between the oxygen atom and the nitrogen atom.

It is important to note that this SAR was based on *in vitro* and *in vivo* pharmacologic evaluations performed over a 60-year period without the luxury of modern, highly refined biologic testing systems (i.e., protein-binding assays, cell membrane-binding assays, and single-cell models) that are considered to be state-of-the-art today for pharmacologic evaluation of new medicinal agents. This is why some classic muscarinic agonists and many of the more modern agents do not adhere to this SAR. Indeed, SAR rules are not static; they should change as new experimental data refine the structural and stereochemical requirements for muscarinic agonist activity.

SPECIFIC MUSCARINIC AGONISTS

Methacholine Chloride (Provocholine) Methacholine, acetyl β -methylcholine, (see previous structure p. 12 and SAR discussion) is marketed as the racemic mixture. It is a selective muscarinic agonist with very little activity at nAChRs. Although methacholine chloride is marketed as the racemic mixture, the *S*-(+)-enantiomer is 240-fold more potent than the *R*-(-)-isomer at mAChRs. In addition, AChE hydrolyzes *S*-(+)-methacholine at approximately 54% the rate of acetylcholine, whereas the *R*-(-)-enantiomer is a weak inhibitor. Methacholine chloride is used via inhalation for the diagnosis of asthma. The resulting bronchospasm can be relieved with bronchodilators. Methacholine chloride is available as a powder that is reconstituted for inhalation.

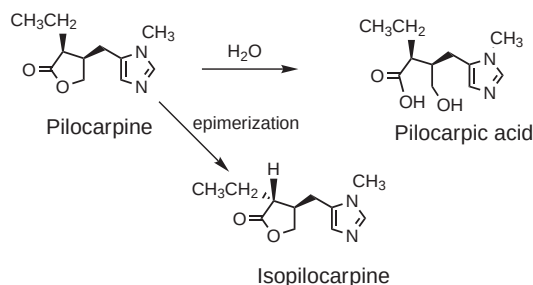
Carbachol Chloride (Isopto Carbachol) Carbachol (see structure p. 12), the carbamate analog of acetylcholine, exhibits affinity for both mAChRs and nAChRs. Because it is a carbamate ester, carbachol is more resistant toward acid-, base-, or enzyme (AChE)-catalyzed hydrolysis than acetylcholine. It is also reported to exhibit weak anticholinesterase activity. Both of these actions work to prolong the duration of action of carbachol. Because of erratic absorption and its actions at nAChRs, use of carbachol has been limited to the treatment of glaucoma and for the induction of miosis in ocular surgery. Carbachol is available as an intraocular solution and an ophthalmic solution.

Bethanechol Chloride (Urecholine) Bethanechol (see structure p. 12), the carbamate analog of methacholine, is selective for mAChRs and exhibits almost no affinity for nAChRs. It is used to treat postsurgical and postpartum urinary retention and abdominal distention. Bethanechol is administered orally, because there is

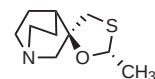
danger of a cholinergic crisis if it is given by intravenous or intramuscular injection.

Pilocarpine Hydrochloride (Isopto Carpine) Pilocarpine hydrochloride, the salt of an alkaloid obtained from *Pilocarpus jaborandi*, is an example of a muscarinic agonist with affinity for the M_3 mAChR that does not adhere to the traditional SAR of muscarinic agonists. In 1876, Langley reported that extracts containing the alkaloid stimulated the end organs of parasympathetic neurons. The structure of pilocarpine was reported in 1901. It possesses two chiral centers at C3 and C4. Thus, four optical isomers (two enantiomeric pairs) are possible, and of these, the naturally occurring alkaloid is 3*S*,4*R*-(+)-pilocarpine, with a pK_a of 6.8 and a logD at pH 7.4 of 1.03.

Pilocarpine is marketed as tablets (Salogen), an ophthalmic solution, and gel. It penetrates the eye well and is the miotic of choice for open-angle glaucoma and to terminate acute angle closure attacks. It is also used for the treatment of xerostomia (dryness of the mouth) caused by radiation therapy of the head and neck, Sjögren syndrome, and mucositis following chemotherapy. Because pilocarpine is a lactone, its solutions are subject to 1) hydrolysis to afford the pharmacologically inactive pilocarpic acid, and 2) base-catalyzed epimerization at C3 in the lactone to give isopilocarpine, an inactive stereoisomer of pilocarpine. Epimerization is not believed to be a serious problem if the drug is properly stored. Its solutions can be stored at room temperature, but the gel should be refrigerated and labeled with a 2-week expiration date when dispensed.

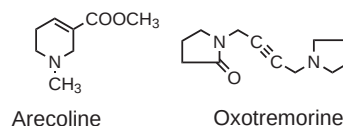


Cevimeline Hydrochloride (Evoxac) Cevimeline is a non-classical muscarinic agonist. It is a quinuclidine derivative that exhibits partial direct M_1 receptor agonist activity in the CNS and affinity for M_3 receptors in epithelial tissue of lacrimal and salivary glands. Its elimination half-life is 3 to 5 hours. It is metabolized by CYP2D6, CYP3A3, and CYP3A4 to inactive metabolites, the *cis*- and *trans*-sulfoxide, *N*-oxide, and glucuronide. Cevimeline hydrochloride is available as an oral capsule for the treatment of xerostomia (dry mouth) associated with Sjögren syndrome. Before its approval, pilocarpine was the only drug for this condition.

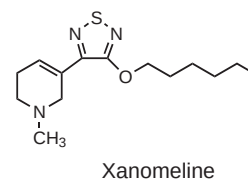


Cevimeline

FUTURE MUSCARINIC AGONISTS Current research interest in the design and synthesis of new muscarinic agonists is focused on discovering agents that might be effective in the treatment of Alzheimer disease and other cognitive disorders. Investigators are searching actively for muscarinic agonists that exhibit selectivity for mAChRs in the brain. Among these compounds are analogs of arecoline, oxotremorine, and McN-A-343 (see above) as well as many other novel chemical structures possessing muscarinic agonist activity.



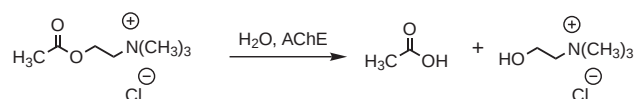
Arecoline is of historical interest because its structure, like those of many other early medicinal agents, was determined and confirmed by a 19th century German pharmacist, E. Jahns (2). Xanomeline can be viewed as a nonclassical bioisostere of the ester moiety of arecoline. It is a muscarinic M_1/M_4 agonist that is showing promise in clinical trials for the treatment of Alzheimer disease (36). Although it is not tolerated at orally effective doses, transdermal delivery systems are showing promise.



Xanomeline

Acetylcholinesterase Inhibitors

Another means of producing a cholinergic response is to interfere with the mechanism by which the action of acetylcholine is terminated. Thus, inhibition of its rapid hydrolysis by AChE increases the concentration of acetylcholine in the synapse and results in production of both muscarinic and nicotinic effects.



Therapeutic Application

Acetylcholinesterase inhibitors (AChEIs), sometimes referred to as anticholinesterases, are classified as indirect cholinomimetics, because their principle mechanism of action does not involve binding to cholinergic receptors, but indirectly increasing the synaptic concentration of acetylcholine. These agents are used therapeutically

to improve muscle strength in myasthenia gravis. They are also used in open-angle glaucoma to decrease intraocular pressure by stimulating contraction of the ciliary muscle and sphincter of the iris. This facilitates outflow of aqueous humor via the canal of Schlemm. Recently, AChEIs have found use in the treatment of symptoms of Alzheimer disease and similar cognitive disorders (37,38), which are conditions characterized by a cholinergic deficiency in the cortex and basal forebrain. They are used extensively as insecticides and are in military arsenals as chemical warfare agents.

Mechanism of Acetylcholinesterase-Catalyzed Hydrolysis of Acetylcholine

Extensive studies of AChE have resulted in purification and amino acid sequencing of the enzyme from several sources, as well as the description of its quaternary structure from x-ray crystallographic and molecular modeling studies (39). The active site of AChE consists of the esteratic site at which hydrolysis of the ester occurs and an “anionic-binding site” where the cationic portion of acetylcholine binds (Fig. 9.12). Using *Torpedo californica*, Sussman et al. (39) showed that the active site lies at the bottom of a deep, narrow gorge that is 18 to 20 Å deep and is lined with aromatic amino acids (39). The amino acids composing the gorge in AChEs from a number of species are highly conserved (40).

Describing AChE as possessing an “anionic site” is a misnomer. This site was originally proposed to be the free carboxylate group of a glutamate residue, but Sussman et al. (39), using AChE from *Torpedo californica*, showed that it is actually uncharged and lipophilic. Rather than an ionic bond being responsible for binding of the quaternary ammonium group, the quaternary nitrogen binds to the π -electrons of a tryptophan residue forming a cation- π bonding interaction. This is supported by mutagenesis studies using human

AChE (41). The aromatic residues found in the gorge can act as low-affinity sites that guide the substrate to the active site while the strong cation- π interaction is responsible for binding the choline portion of acetylcholine (40) (see Fig. 9.12).

Like other serine hydrolases, the functional catalytic unit of AChE is composed of a catalytic triad of glutamate, histidine, and serine at the esteratic site (42,43). The serine hydroxyl group serves as the nucleophile, while the histidine residue increases the nucleophilicity by serving as a general acid-base catalyst. The glutamic acid residue is believed to provide stabilization to the transition state (43,44). Also found at the esteratic site is an “oxyanion hole” formed by hydrogen bonding between *NH* groups of two glycine residues and an alanine (43,45). These hydrogen bonds with the carbonyl oxygen stabilize the transition state.

The proposed mechanism of hydrolysis is illustrated in Figure 9.13. Transition state B is unstable and collapses to form choline and acetylated AChE (C); this form of the enzyme is referred to as the acylated enzyme. As long as the enzyme is acylated, it cannot bind another molecule of acetylcholine; the enzyme is in an inactive state. The acylated enzyme then undergoes rapid hydrolysis (D) to regenerate the active form of AChE and a molecule of acetic acid (E) (43).

Hydrolysis of the acylated enzyme (deacylation) is important in the development of AChEIs. If the enzyme becomes acylated by a functional group (i.e., carbamyl or phosphate) that is more stable toward hydrolysis than a carboxylate ester, the enzyme remains inactive for a longer period of time. Application of this chemical principle regarding rates of hydrolysis led to discovery and design of two classes of AChEIs, the reversible inhibitors and the irreversible inhibitors.

Reversible Inhibitors of Acetylcholinesterase

MECHANISM OF ACTION Reversible AChEIs are 1) those compounds that are substrates for and react with AChE to form an acylated enzyme, which is more stable than the acetylated enzyme but still capable of undergoing hydrolytic regeneration or 2) those that bind to AChE with greater affinity than acetylcholine but do not react with the enzyme as a substrate. Both types of inhibitors have found clinical applications. Those that acylate AChE include the aryl carbamates, such as esters of carbamic acid and phenols (e.g., physostigmine). Alkyl carbamates (esters of carbamic acid and alcohols), such as carbachol and bethanechol, which are structurally related to acetylcholine, are also substrates for and competitively inhibit AChE because they are hydrolyzed very slowly by AChE. For reasons previously discussed, carbachol and bethanechol are more resistant than acetylcholine to AChE-catalyzed hydrolysis.

When aryl carbamate AChEIs, such as physostigmine and its analogs, bind to the catalytic site of AChE, hydrolysis of the carbamate occurs by transesterification by the carbamoyl group to the serine residue, forming what

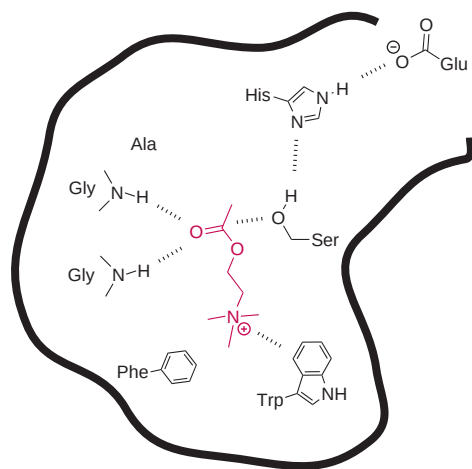


FIGURE 9.12 Binding of acetylcholine to catalytic site of acetylcholinesterase; role of serine and histidine residues is illustrated.

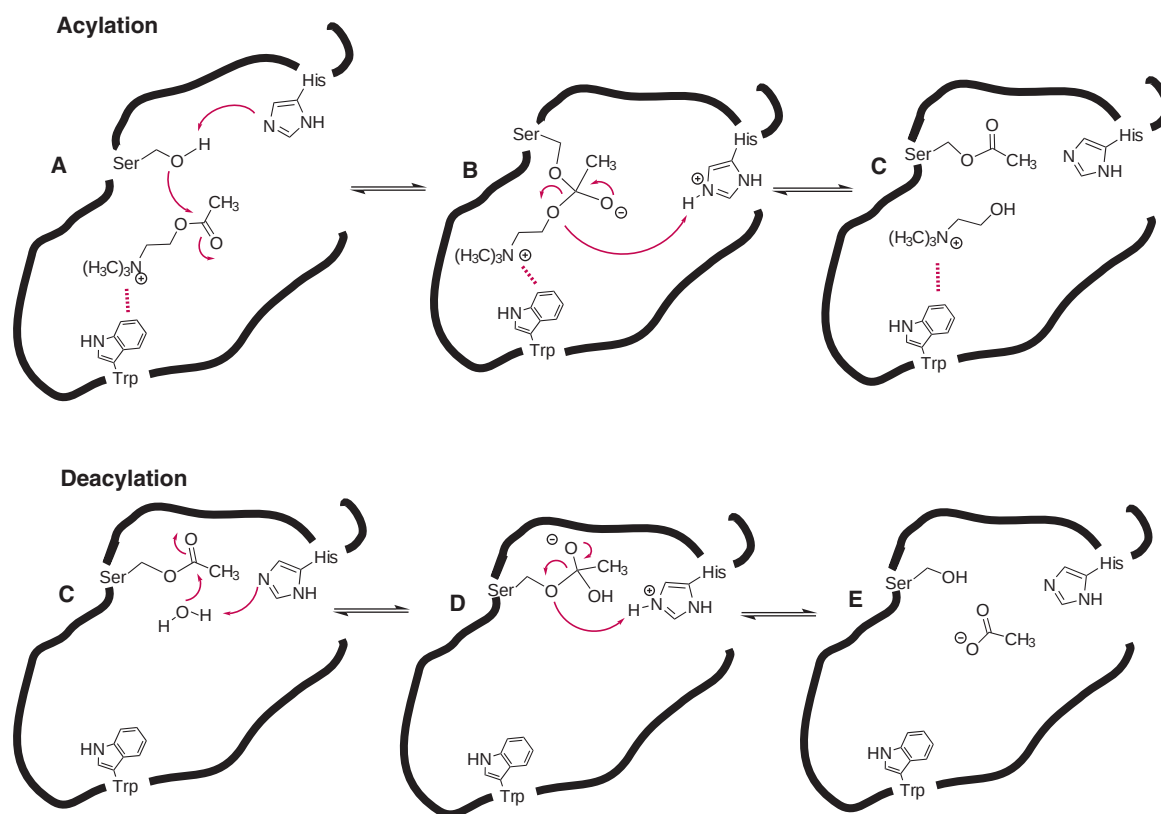
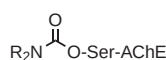


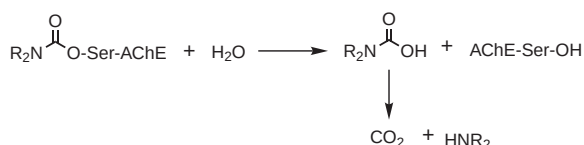
FIGURE 9.13 Mechanistic steps in the hydrolysis of acetylcholine (ACh).

is termed a “carbamylated enzyme.” The rate of carbamylation follows this order: unsubstituted carbamic acid esters > methylcarbamic acid esters > dimethylcarbamic acid esters (46).



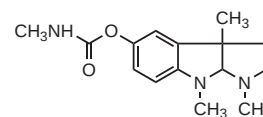
Carbamylated AChE

Regeneration of active AChE by hydrolysis of the carbamylated enzyme is much slower than hydrolysis of the acetylated enzyme. The rate for hydrolytic regeneration of the carbamylated AChE is measured in minutes (e.g., the half-life for methyl carbamylated enzyme is ~15 minutes); the rate of hydrolytic regeneration of acetylated AChE is measured in a fraction of a millisecond (e.g., the half-life for acetylated enzyme is ~0.2 milliseconds). Despite the longer time required to regenerate the carbamylated enzyme, the active form of AChE eventually is regenerated. Therefore, these inhibitors are considered to be reversible.



Aryl carbamates are superior to alkyl carbamates as AChEIs, because they have better affinity for AChE and, therefore, carbamate AChE more efficiently. Physostigmine and other aryl carbamates exhibit inhibition constants (K_i) on the order of 10^{-9} to 10^{-8} M and are three to four orders of magnitude more effective than alkyl carbamates, such as carbachol ($K_i \sim 10^{-5}$ M). This is to be expected because phenoxide anions are more stable and, hence, are better leaving groups than alkoxide anions. Phenoxide anions are stabilized through resonance with the aromatic ring. Thus, the therapeutically effective carbamate inhibitors of AChE are derived from phenols.

SPECIFIC AGENTS

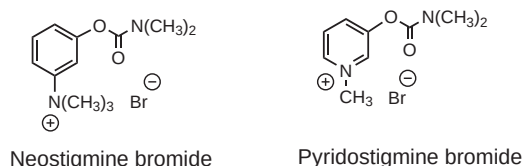


Physostigmine

Physostigmine The classic AChEI, physostigmine, is an alkaloid obtained from seeds of the Calabar bean (*Physostigma venenosum*) (38). Its parasympathomimetic effects were recognized long before its structure was elucidated in 1923. In 1929, Stedman found that

the mechanism of the parasympathomimetic effects of physostigmine was inhibition of AChE; it inhibits AChE by acting as a substrate and carbamylating the enzyme. Acetylcholinesterase is carbamylated at a slow rate, but physostigmine has exceptionally high affinity ($K_i \sim 10^{-9}$ M) for the catalytic site of the enzyme. By comparison, the K_s for acetylcholine is on the order of 10^{-4} M. Thus, physostigmine is classified as a reversible AChEI that carbamylates the enzyme at a slow rate; the carbamylated AChE is also regenerated quite slowly. Because physostigmine is a tertiary amine with a pK_a of 8.2 (BH^+) rather than a quaternary ammonium salt, it is more lipophilic than many other AChEIs and can diffuse across the blood–brain barrier. The tertiary amine also imparts pH dependence to its ability to inhibit AChE, because its affinity for AChE is greater when the amine is protonated. Physostigmine is metabolized in vivo by esterases to the phenol and has an elimination half-life of 1 to 2 hours. Its aqueous solutions are subject to hydrolytic decomposition to form eseroline, which undergoes light-catalyzed oxidation to form rubreserine, a red-colored compound (Fig. 9.14). Both degradation products are inactive as AChEIs.

Physostigmine has been used for many years in ophthalmology for the treatment of glaucoma. More recently, the salicylate salt has been used in hospital emergency rooms to treat overdoses of compounds possessing significant anticholinergic CNS effects, such as atropine and tricyclic antidepressants. Physostigmine's ability to cross the blood–brain barrier has led to renewed interest in this molecule, and it is also one of a number of centrally active AChEIs being investigated as indirect cholinomimetics for use in the treatment of Alzheimer disease and other cognitive disorders.

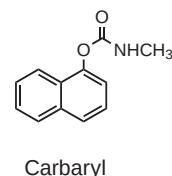


Neostigmine (Prostigmin) The discovery that physostigmine and other aryl carbamates inhibit AChE reversibly led to efforts to find other AChEIs possessing this activity. Most of this research involved incorporation of the required structural features of both physostigmine and acetylcholine into the new molecules. This led to synthesis of neostigmine, a compound resembling physostigmine but having a much simpler structure. Neostigmine retains the substituted carbamate group, the benzene ring, and

the nitrogen atom of the first heterocyclic ring of physostigmine. The distance between the ester and the quaternary ammonium group is approximately the same as that found in acetylcholine and physostigmine. Because of its quaternary ammonium group, it lacks central activity. Neostigmine is metabolized by plasma cholinesterases, in liver, and in skeletal muscle to 3-hydroxyphenyltrimethylammonium chloride, 3-hydroxyphenyldimethylamine, and their *O*-glucuronide conjugates (Fig. 9.15). It has an elimination half-life of 15 to 90 minutes. Neostigmine is indicated for prophylaxis of postoperative abdominal distension and urinary retention, myasthenia gravis, and reversal of neuromuscular blockade.

Pyridostigmine (Mestinon) Pyridostigmine, a closely related structure to neostigmine that incorporates the charged nitrogen into a pyridine ring, acts by the same mechanism as physostigmine, and, like neostigmine, it lacks CNS activity. It is orally effective and, compared to neostigmine, has a longer duration of action and a lower incidence of side effects. Thus, it is a better choice for oral therapy of myasthenia gravis. It is approved for U.S. military use as an adjunct for prophylaxis of soman nerve gas exposure. It is also administered parenterally to reverse the effects of nondepolarizing neuromuscular blocking agents. Its elimination half-life is 1 to 2 hours. Its metabolism pathway and metabolites are similar to those for neostigmine.

Carbaryl (Sevin) Carbaryl is a reversible, carbamate-derived AChEI that has tremendous economic impact as an insecticide for use on houseplants and vegetables as well as for control of fleas and ticks on pets. Its structural relationship to physostigmine and neostigmine is readily apparent. A number of other carbamate AChEIs are also commercially available for this use.



Edrophonium Chloride (Enlon, Reversol) Edrophonium is a quaternary ammonium-substituted phenol. Because it is a phenol rather than a carbamate ester of a phenol, it does not carbamylate AChE. It does, however, inhibit AChE in a reversible manner, and also exhibits a direct cholinomimetic effect at skeletal muscle. Edrophonium

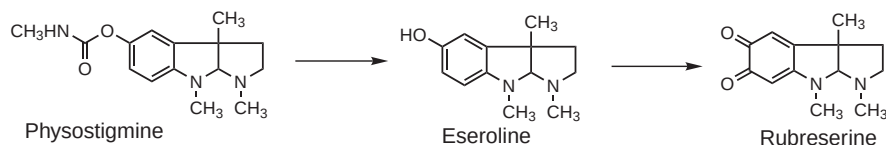


FIGURE 9.14 In vitro degradation of physostigmine.

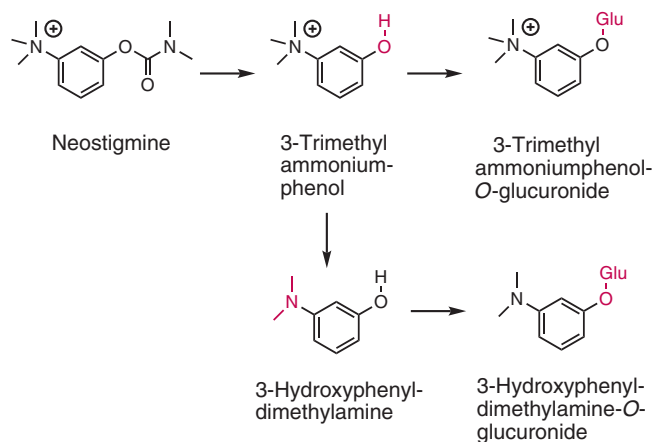
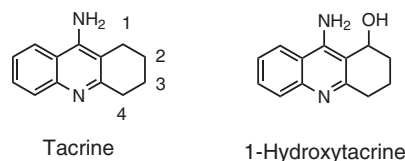


FIGURE 9.15 Metabolism of neostigmine bromide.

is used intravenously for the diagnosis of myasthenia gravis, where it acts rapidly to increase muscle strength. It is also administered intramuscularly to rapidly reverse the effects of nondepolarizing neuromuscular blocking agents like *d*-tubocurarine and gallamine. It is not effective, however, at reversing the effects of the depolarizing blockers such as succinylcholine and decamethonium. Its elimination half-life is 1.3 to 2.4 hours.



REVERSIBLE ACETYLCHOLINESTERASE INHIBITORS FOR TREATMENT OF ALZHEIMER DISEASE Of all the age-related disorders in which dementia is a component, Alzheimer disease (AD) is probably the best known. Much effort has been expended to discover the cause of AD. Autopsy examination of the brains of patients who had AD has revealed microscopic structural changes characteristic of the disease. In addition, neurotransmitter dysfunction involving reduction in acetylcholine, serotonin, norepinephrine, dopamine, and glutamate levels has been reported. For a review of AD and the search for therapies, see Rzeszutarski (47). It is known that in AD patients, there is widespread atrophy in the primary motor and sensory cortices and cerebellum. There is a disruption in cholinergic innervation in these areas of the brain, along with decreases in choline acetyltransferase, high-affinity nicotinic acetylcholine receptor binding, and choline transporter sites (48–51). Impairment of short-term memory is the first observable symptom of the disease, and progressive memory impairment, severe mood changes, and depression coupled with loss of judgment and reasoning ability follow. The U.S. Food and Drug Administration has approved four AChEIs for the treatment of AD: tacrine, donepezil, rivastigmine, and galantamine. Although these four AChEIs are not without problems, they do provide some benefit in early to mild AD. Their clinical effectiveness in advanced AD has yet to be shown.



Tacrine Hydrochloride (Cognex) Tacrine, an aminoacridine synthesized in the 1930s, is a nonclassical cholinesterase inhibitor that binds to both AChE and butyrylcholinesterase (52). It was approved in 1993 for treatment of AD. Approximately 20% of tacrine-treated patients show improvement. However, its use is limited due to hepatotoxicity and development of safer AChEIs. Tacrine is extensively metabolized by CYP1A2 to the 2-, 3-, and 4-hydroxy metabolites. All show reduced AChEI activity relative to tacrine, with the major metabolite, 1-hydroxytacrine, being the most active. Its elimination half-life is between 1.5 and 4 hours, with metabolites being excreted via the urine.

Donepezil (Aricept) Donepezil is another “nonclassical,” centrally acting, reversible, noncompetitive AChEI that was approved in 1997 for treatment of mild-to-moderate AD and dementia. Its selectivity for AChE is 570 to 1,250-fold than for butyrylcholinesterase, and it also exhibits greater affinity for brain AChE than for peripheral AChE (53). When compared to tacrine, donepezil exhibits greater CNS AChE selectivity, longer elimination half-life (70 to 104 hours in subjects older than 55 years), and little or no potential for hepatotoxicity. Donepezil is metabolized by CYP2D6 and CYP3A4 via O-demethylation, N-debenzylation, oxidation to the *cis*-N-oxide, and O-glucuronidation (Fig. 9.16). Donepezil is hepatically metabolized, and the predominant route for the elimination of both parent drug and its metabolites is renal. The major plasma metabolites of donepezil include O-dealkylation to M1 and M2 metabolites and

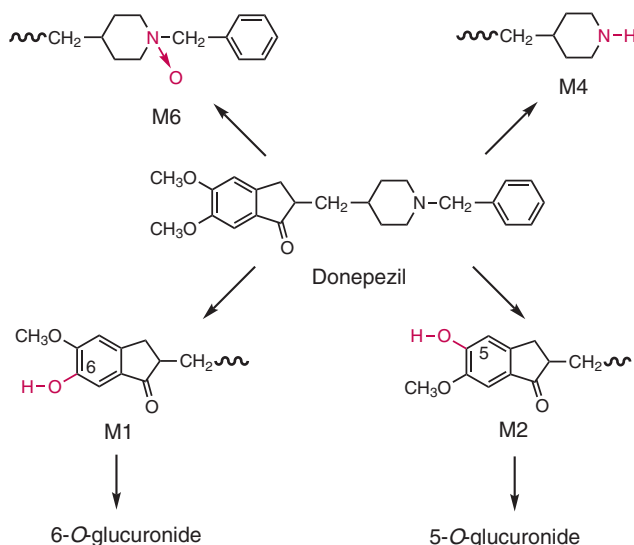
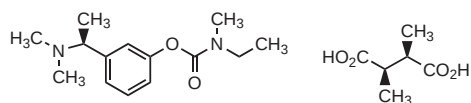


FIGURE 9.16 Metabolism of donepezil.

N-oxidation (M6) (Fig. 9.16). Unchanged donepezil accounted for approximately 25% of the dose and 17% in urine. Urinary metabolites included the *O*-glucuronides of M1 and M2, but also M4 the *N*-debenzylated metabolite. The 6-*O*-desmethyl metabolite (M1) accounted for 11% of the dose, and it exhibits AChE inhibitory activity comparable to that of the parent compound.

Rivastigmine Tartrate (Exelon) Rivastigmine is a centrally selective, arylcarbamate AChEI that was approved in 2000 for oral administration in the treatment of AD. It has an elimination half-life of 1.4 to 1.7 hours. Rivastigmine inhibits both plasma cholinesterase and acetylcholinesterase (unlike donepezil, which selectively inhibits acetylcholinesterase). It is thought that rivastigmine works by inhibiting these cholinesterase enzymes, which would otherwise break down acetylcholine. However, it inhibits AChE for up to 10 hours due to the slow hydrolysis of the carbamylated enzyme. Thus, it has been referred to as a pseudo-irreversible AChEI (53). When given orally, rivastigmine is well absorbed with a bioavailability of about 40%. Peak plasma concentrations are seen in about 1 hour, and peak spinal fluid is seen at 1.4 to 3.8 hours. When given by once-daily transdermal patch, the pharmacokinetic profile of rivastigmine is much smoother as compared with capsules, with lower peak plasma concentrations and reduced plasma concentration fluctuations. The 9.5 mg/24 hours rivastigmine patch provides comparable exposure to 12 mg/day capsules (the highest recommended oral dose). The drug crosses the blood–brain barrier. It is rapidly and extensively hydrolyzed by plasma and CNS cholinesterases with minimal involvement of hepatic CYP450 enzymes. The phenolic metabolite is excreted primarily via the kidneys. Like donepezil, rivastigmine exhibits a low potential of hepatotoxicity and drug–drug interactions due to lack of hepatic metabolism by CYP450 enzymes.



Rivastigmine tartrate

Galantamine Hydrobromide (Razadyne) Galantamine, introduced in 2001, is an alkaloid found in plants of the family Amaryllidaceae, which includes the daffodil (*Narcissus pseudonarcissus*) and snowflake (*Leucojum aestivum*). It is a reversible inhibitor of AChE, but it does not appear to inhibit butyrylcholinesterase. Because it is a tertiary amine and can cross the blood–brain barrier, it is indicated for treatment of mild-to-moderate AD and dementia. It has been used outside the United States for more than 30 years as an anticurare agent in anesthesia. Galantamine differs from other cholinesterase inhibitors, because it allosterically binds to nAChRs, giving it a dual cholinergic action. It is primarily metabolized (75%) by CYP2D6 to *O*-desmethyl and *O*-desmethylnormethyl

metabolites and by CYP3A4 to the normethyl metabolite, along with other minor metabolites (Fig. 9.17). Unlike tacrine, galantamine is not associated with hepatotoxicity. Its elimination half-life is 5.7 hours.

Irreversible Inhibitors of Acetylcholinesterase

MECHANISM OF ACTION The chemical logic involved in the development of effective AChEIs was to synthesize compounds that would be substrates for AChE and result in an acylated enzyme more stable to hydrolysis than a carboxylate ester. Phosphate esters are very stable to hydrolysis, being even more stable than many amides. Application of this chemical property to the design of AChEI compounds led to derivatives of phosphoric, pyrophosphoric, and phosphonic acids that are effective inhibitors of AChE. These act as inhibitors by the same mechanism as the carbamate inhibitors, except that they form phosphate esters with the enzyme. The rate of hydrolysis of the phosphorylated enzyme is much slower than that of the carbamylated enzyme, and its rate is measured in hours (e.g., the half-lives for diethyl phosphates are ~8 hours). Because the duration of action of these compounds is much longer than that of carbamate esters, they are referred to as irreversible inhibitors of AChE.

An important difference between irreversible phosphoester-derived AChEIs and reversible (carbamate-derived) AChEIs is that the phosphorylated AChE can undergo a process known as aging (Fig. 9.18). Aging plays an important role in the toxicity of these irreversible AChEIs. It is the result of hydrolysis of one or more of the phosphoester bonds while leaving the AChE phosphorylated. This reaction affords an anionic phosphate in which the phosphorus atom is much less electrophilic and, therefore, much less likely to undergo hydrolytic regeneration than the original phosphoester. Aging occurs over a period of time and depends on the rate of the P–O bond cleavage reaction. Prior to aging of the

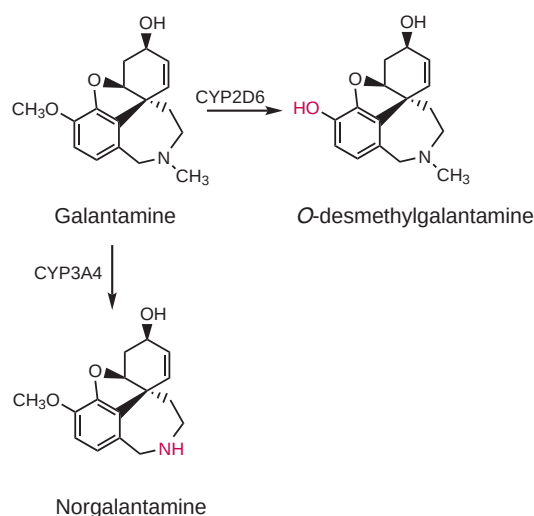


FIGURE 9.17 Metabolism of galantamine.

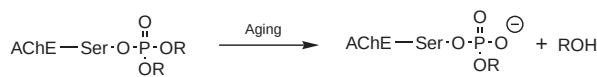
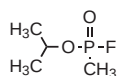


FIGURE 9.18 Aging of phosphorylated AChE.

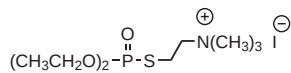
enzyme, the antidotes to phosphate ester poisoning can be effective.

Only those phosphorus-derived AChEIs that have at least one phosphoester group undergo the aging process. Knowledge of the chemical mechanisms associated with irreversible inhibition of AChE and the aging process led to development of deadly phosphorus-derived chemical warfare “nerve agents,” one of which is sarin (GB is the two-letter North Atlantic Treaty Organization designation for this nerve agent). When sarin phosphorylates AChE, only one aging reaction takes place, and then the enzyme becomes refractory to regeneration by the currently available antidotal agents. Other organophosphate nerve agents include tauban (GA), soman (GD), GF, and VX.



Sarin

SPECIFIC AGENTS



Echothiophate iodide

Echothiophate Iodide (Phospholine Iodide) Echothiophate iodide has found therapeutic application for the treatment of glaucoma and strabismus. Echothiophate is applied topically as a solution and is the only irreversible AChEI for the treatment of glaucoma. The decrease in intraocular pressure observed can last up to 4 weeks. Phosphoester AChEIs exhibit cataractogenic properties; thus, their use should be reserved for patients who are refractory to other forms of treatment (i.e., short-acting miotics, β -blockers, epinephrine, and possibly, carbonic anhydrase inhibitors). Because of its toxicity, echothiophate is not used for its systemic action. Selectivity of echothiophate for the AChE catalytic site was enhanced by incorporation of a quaternary ammonium salt functional group two carbons removed from the phosphoryl group.

Insecticidal AChEIs A number of lipophilic phosphoester AChEIs have been designed as insecticides; the structures of some of these are shown in Figure 9.19. This group of irreversible AChEI insecticides is beneficial to agricultural production throughout the world. In addition to being extremely lipophilic, another physicochemical property common to these compounds is a high vapor pressure. This combination of physicochemical

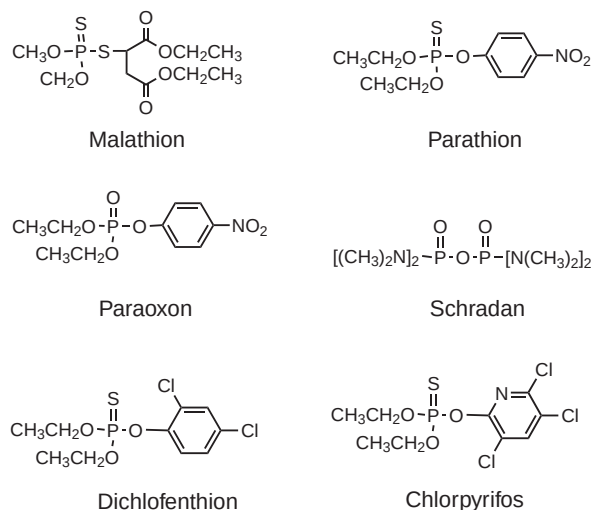


FIGURE 9.19 Irreversible acetylcholinesterase inhibitors used as insecticides.

properties makes it imperative that these compounds be used with extreme caution in the presence of humans and other mammals to prevent inhalation of the vapors and their absorption through the skin. Both routes of exposure cause a number of poisoning accidents every year, some of which are fatal.

Some of these irreversible AChEI insecticides have a sulfur atom bonded to the phosphorus atom with a coordinate-covalent bond. These compounds exhibit little AChEI activity, but they are rapidly bioactivated via desulfurization by microsomal oxidation in insects to afford the corresponding oxo-derivatives (phosphate esters), which are quite potent. A good example of this bioactivation phenomenon is illustrated by the commercially available insecticide parathion and its bioactivation to a toxic metabolite paraoxon.

Malathion (Oxide). Malathion (Fig. 9.19) is a dithio-phosphate ester that has found use both as an aerial insecticide and clinically as a mitocide for topical treatment of lice infestations of the hair and scalp. It will kill both hatched lice and their eggs (nits) within 3 seconds after application. Compared to other organophosphate AChEIs, malathion exhibits lower transdermal absorption. On intact skin, less than 10% of a topical dose is systemically absorbed. Similar to parathion, malathion is bioactivated in insects to its toxic phosphate ester metabolite. It is much less toxic in humans, mammals, and birds than in insects. This selectivity toward insects is achieved because, in humans, plasma esterases hydrolyze the carboxylate esters to less toxic carboxylic acid metabolites that are rapidly eliminated in urine as carboxylate anions. This does not occur in insects. Acute toxicity with malathion is rare and usually occurs only after oral ingestion. The lethal dose in mammals is approximately 1 g/kg.

ANTIDOTES FOR IRREVERSIBLE AChEIs

Background The marked toxicity of phosphate ester irreversible AChEIs, their widespread use as insecticides, and their proliferation as chemical warfare agents posed serious problems that stimulated research to develop antidotes for these agents. This required rational use of reaction kinetics, organic reaction mechanisms, and synthetic organic chemistry. Water is a nucleophile capable of rapidly hydrolyzing acetylated AChE and regenerating the active enzyme. Phosphorylated AChE (irreversibly inhibited), however, was known to involve a phosphate ester of serine. It is well established from reaction kinetic studies that the rate of hydrolysis is much slower for organic phosphate esters than for carboxylate esters and that a significantly stronger nucleophile than water would be required for efficient cleavage of phosphate esters. The problem required the design of reagents capable of efficiently catalyzing phosphate ester cleavage to regenerate active AChE while being safe enough for use as therapeutic agents. The resolution of this problem is an elegant example of application of chemical principles to the solution of a therapeutic problem (54–56).

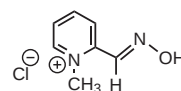
Hydroxylamine (NH_2OH) is a strong nucleophilic compound that efficiently cleaves phosphate esters and significantly increases the rate of reactivation of phosphorylated AChE, but only at toxic concentrations (57). The nucleophilicity of hydroxylamine prompted the development of a number of structurally related hydroxylamines in the hope of eliminating its toxicity. The toxicity inherent in hydroxylamine would most probably be present in any structurally related compound, but this toxicity might be minimized if sufficiently small doses could be used. It would be logical to design a compound that would have a high degree of selectivity and strong binding affinity for AChE and also carries a hydroxylamine-like nucleophile into close proximity to the phosphorylated serine residue. This design was achieved by the synthesis of hydroxylamine derivatives of organic compounds possessing a functional group bearing a positive charge.

Reaction of hydroxylamine with aldehydes or ketones affords oximes, which possess the desired nucleophilic oxygen atom. A pyridine ring was considered an attractive carrier for the oxime function, because such groups are common in a number of biochemical systems (e.g., NAD^+ and NADP^+), indicating a possible low order of toxicity. Furthermore, three readily available positional isomers of pyridine aldehyde can be easily converted to oximes. Finally, the nitrogen atom of the pyridine ring can be converted to a quaternary ammonium salt by treatment with methyl iodide. This cationic charge would be expected to increase affinity of the compound for the anionic-binding site of the phosphorylated AChE.



Oxime

The three isomeric pyridine aldoxime methiodides were synthesized and biologically evaluated. Of these, the most effective is the isomer derived from 2-pyridinylaldehyde. This compound, known as pralidoxime chloride (2-PAM, or 2-pyridine aldoxime methyl chloride) currently is the only available agent proven to be clinically effective as an antidote for poisoning by phosphate ester AChEIs. The proposed mechanism for regeneration of AChE by 2-PAM is illustrated in Figure 9.20. The initial step involves binding of the quaternary ammonium nitrogen of 2-PAM to the anionic-binding site of phosphorylated AChE. This places the nucleophilic oxygen of 2-PAM in close proximity to the electrophilic phosphorus atom. Nucleophilic attack of the oxime oxygen results in breaking of the ester bond between the serine oxygen atom and the phosphorus atom. The final products of the reaction are the regenerated active form of AChE and phosphorylated 2-PAM.



Pralidoxime chloride (2-PAM)

Pralidoxime is administered subcutaneously, intramuscularly, or intravenously, and it must be given within a short period of time after enzyme phosphorylation, generally a few hours after exposure, for it to be effective due to the aging process of the phosphorylated enzyme. Little reactivation is likely if given 36 hours after exposure. If the phosphorylated AChE has aged, 2-PAM will not regenerate the enzyme. For this reason, as well as because new phosphate ester AChEIs capable of aging rapidly are being developed as insecticides and chemical warfare agents, there is a continuing effort to discover new and better substitutes for 2-PAM. This research is focused on finding substitutes for 2-PAM that are better nucleophiles and, therefore, more effective generators of active AChE, as well as compounds that cross the blood-brain barrier to regenerate phosphorylated AChE in the brain.

Acetylcholine Antagonists—Muscarinic Antagonists

Muscarinic antagonists are compounds that have high binding affinity for mAChRs but exhibit no intrinsic activity. When the antagonist binds to the receptor, it is

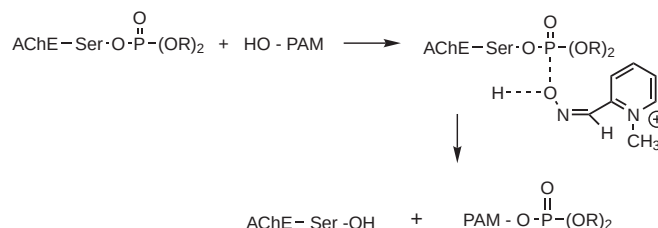


FIGURE 9.20 Reactivation of AChE with 2-PAM.

proposed that the receptor protein undergoes a conformational perturbation that is different from that produced by an agonist. Therefore, antagonist binding to the receptor produces no response. Muscarinic antagonists are often referred to as anticholinergics, antimuscarinics, cholinergic blockers, antispasmodics, or parasympatholytics. The term “anticholinergic” refers, in a pure sense, to medicinal agents that are antagonists at both mAChRs and nAChRs. Common usage of the term, however, has become synonymous with muscarinic antagonist, and it is used as such in this section.

Therapeutic Application

Muscarinic antagonists are employed as both prescription drugs and over-the-counter medications. Because they act as competitive (reversible) antagonists of acetylcholine, these compounds have pharmacologic effects that are opposite those of muscarinic agonists. Responses of muscarinic antagonists include decreased contractions of smooth muscle of the gastrointestinal and urinary tracts, dilation of the pupils, and reduced gastric, mucociliary, and salivary secretions. It follows that these compounds have therapeutic value in treating smooth muscle spasms associated with increased tone of the gastrointestinal tract or with overactive bladder, in ophthalmologic examinations, and in treatment of gastric ulcers. Compounds possessing muscarinic antagonist activity are common components of cold and flu remedies that act to reduce nasal and upper respiratory tract secretions.

In addition to reducing gastric motility, anticholinergic agents decrease gastric acid secretion and were once widely used to manage peptic ulcers. Histamine H_2 antagonists and, more recently, the proton pump inhibitors have largely replaced them for this use. When used systemically, they tend to produce undesirable side effects, such as blurred vision, photophobia, dry mouth, and difficulty in urination. These side effects tend to reduce patient compliance.

Anticholinergic agents exhibit a mydriatic action and, thus, must be used with caution due to their effect on intraocular pressure. Drainage of the canal of Schlemm is restricted by the iris when the pupil is dilated, and this can cause an increase in intraocular pressure. Hence, muscarinic antagonists are contraindicated in patients with glaucoma.

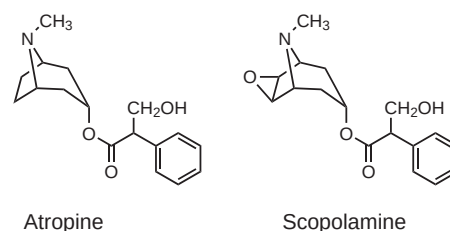
The aforementioned side effect of causing difficulty in urination has been used as an indication for several anticholinergic agents—darifenacin, trospium, solifenacin, tolterodine, and oxybutynin—for the treatment of overactive bladder.

Centrally acting belladonna alkaloids, such as scopolamine, have been used in transdermal delivery systems for prevention of motion sickness. They are most effective when used prophylactically; they have less effect when used after nausea and vomiting have begun. Several synthetic muscarinic antagonists have been used to treat parkinsonism and to block the extrapyramidal effects of antipsychotic agents. The anticholinergic alkaloid atropine is used for treatment

of central and peripheral symptoms associated with poisoning by organophosphorous anticholinesterase agents.

Specific Agents—Solanaceous Alkaloids

The earliest known anticholinergic agents were alkaloids found in the family Solanaceae, a large family of plants that includes potatoes. *Atropa belladonna* (deadly nightshade), *Hyoscyamus niger* (black henbane), and *Datura stramonium* (jimsonweed, thorn apple) are plants that have significant historical importance to our understanding of the parasympathetic nervous system. Pharmacologic effects of extracts from these plants have been recognized since the Middle Ages, although these effects were not associated with the autonomic nervous system until the latter part of the 19th century. (–)-Hyoscyamine, isolated as atropine, and scopolamine are the two alkaloids that have found the widespread clinical applications.

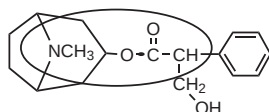


ATROPINE

Atropine is the tropic acid ester of tropine and is marketed as the sulfate salt. The naturally occurring alkaloid, (–)-hyoscyamine, undergoes base-catalyzed racemization during isolation from plants of the Solanaceae to give (±)-hyoscyamine or atropine. It was the first compound shown to block the effects of muscarine and electrical stimulation of the parasympathetic nervous system. Atropine sulfate has a number of clinical uses; two of the most common are treatment of bradycardia and as a preoperative agent to reduce secretions before surgery. Its use for management of parkinsonism has been supplanted by newer agents with fewer peripheral side effects. It has been used in ophthalmology as a cycloplegic agent to paralyze the iris and ciliary muscle in treatment of iritis and uveitis and as a cycloplegic/mydriatic agent. Atropine is contraindicated in glaucoma due to its ability to increase intraocular pressure during mydriasis. Its prolonged duration of mydriasis makes other drugs more attractive for this purpose. In poisoning by organophosphate nerve agents and insecticides, atropine decreases muscarinic cholinergic actions (e.g., lacrimation, salivation, sweating, bradycardia, and breathing problems) associated with this poisoning. It only treats the symptoms and does not reverse the underlying AChE inhibition. Approximately 50% of an oral dose of atropine is eliminated in the urine unchanged, 24% is eliminated as noratropine, 15% is eliminated as *N*-oxide, and less than 5% undergoes nonenzymatic ester hydrolysis. Its elimination half-life is 4 hours in adults and 6.5 hours in children.

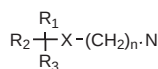
SCOPOLAMINE Scopolamine, another Solanaceous alkaloid, is chemically and pharmacologically similar to atropine. Scopolamine is the generic name given to (–)-hyoscyine, the naturally occurring alkaloid. The racemic compound, isolated during extraction of the alkaloid from plants, is atropine. Scopolamine is marketed as the hydrobromide salt, because it is less deliquescent than some of its other salts. Interestingly, scopolamine is a CNS depressant at usual therapeutic doses, whereas atropine and other antimuscarinic agents are CNS stimulants. It has been used for the treatment of uveitis, iritis, and parkinsonism, but its most widespread use is for the treatment of motion sickness. For this indication, scopolamine is used in a transdermal patch applied to the skin behind the ear and is well-absorbed percutaneously following application. Plasma levels are observed within 4 hours and peak levels within 24 hours. It crosses the placenta and the blood–brain barrier. Although not well characterized, scopolamine is extensively metabolized in the liver and conjugated with less than 5% of the total dose appearing unchanged in the urine. Following patch removal, plasma levels decline linearly with an observed half-life of 9.5 hours.

Structure–activity Relationship Atropine, the prototype anticholinergic agent, provided the structural model that guided the design of synthetic muscarinic antagonists for almost 70 years. The circled portion of the atropine molecule depicts the segment resembling acetylcholine.



Although the amine functional group is separated from the ester oxygen by more than two carbons, the conformation assumed by the tropane ring orients these two atoms such that the intervening distance is similar to that in acetylcholine. One important structural difference between atropine and acetylcholine, both of which are esters of amino alcohols, is the size of the acyl portion of the molecules. Based on the assumption that size was a major factor in blocking action, many substituted acetic acid esters of amino alcohols were prepared and evaluated for biologic activity.

It became apparent that the most potent antagonists were those that possessed two lipophilic ring substituents on the carbon α to the carbonyl of the ester moiety. This is the first of the classic SARs for muscarinic antagonist activity, and this SAR became defined more precisely as research on these antagonists continued. The SAR for muscarinic antagonists can be summarized as follows:



1. Substituents R_1 and R_2 should be carbocyclic or heterocyclic rings for maximal antagonist potency.

The rings can be identical, but the more potent compounds have different rings. Generally, one ring is aromatic and the other saturated or possessing only one olefinic bond. Substituents R_1 and R_2 , however, can be combined into a fused aromatic tricyclic ring system, such as that found in propanteline (Fig. 9.22). The size of these substituents is limited. For example, substitution of naphthalene rings for R_1 and R_2 affords compounds that are inactive, apparently due to steric hindrance of the binding of these compounds to the mAChR.

2. The R_3 substituent can be a hydrogen atom, a hydroxyl group, a hydroxymethyl group, or a carboxamide, or it can be a component of one of the R_1 and R_2 ring systems. When this substituent is either a hydroxyl group or a hydroxymethyl group, the antagonist is usually more potent than the same compound without this group. The hydroxyl group presumably increases binding strength by participating in a hydrogen bond interaction at the receptor.
3. The X substituent in the most potent anticholinergic agents is an ester, but an ester functional group is not an absolute necessity for muscarinic antagonist activity. This substituent can be an ether oxygen, or it can be absent completely.
4. The N substituent is a quaternary ammonium salt in the most potent anticholinergic agents. This

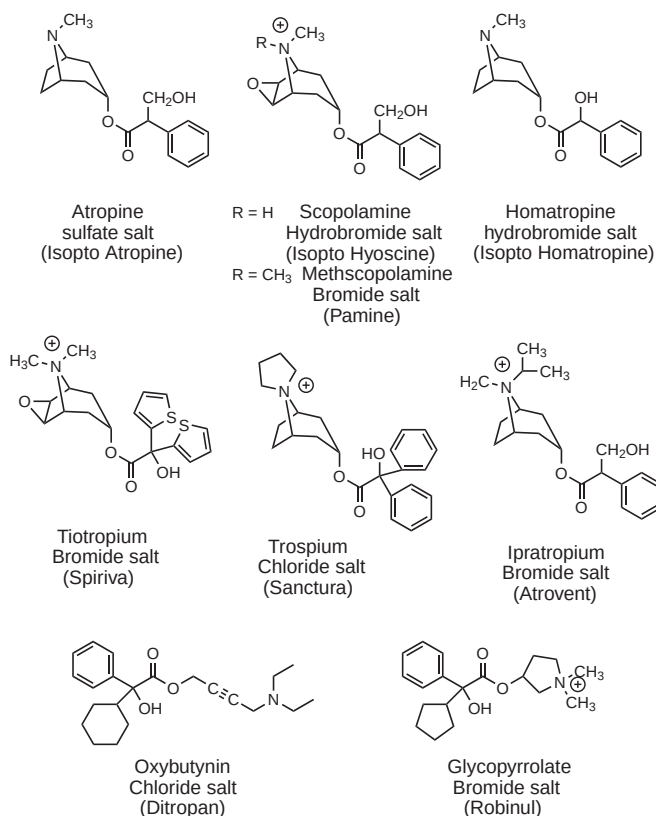


FIGURE 9.21 Anticholinergic aminoalcohol esters.

is not a requirement, however, because tertiary amines also possess antagonist activity, presumably by binding to the receptor in the cationic (conjugate acid) form. The alkyl substituents are usually methyl, ethyl, propyl, or isopropyl.

5. The distance between the ring-substituted carbon and the amine nitrogen apparently is not critical; the length of the alkyl chain connecting these can be from two to four carbons. The most potent anticholinergic agents have two methylene units in this chain.

Muscarinic antagonists must compete with agonists for a common receptor. Their ability to do this effectively is because the large groups R_1 and R_2 enhance binding to the receptor. Because antagonists are larger than agonists, this suggests that groups R_1 and R_2 bind outside the binding site of acetylcholine. It has been suggested that the area surrounding the binding site of acetylcholine is hydrophobic in nature (58). This accounts for the fact that in potent cholinergic antagonists, groups R_1 and R_2 must be hydrophobic (usually phenyl, cyclohexyl, or cyclopentyl). This concept is also supported by the current models for mAChRs.

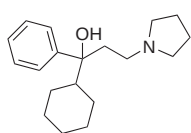
Figures 9.21 and 9.22 and Table 9.3 include structures and pharmacologic properties of some of the anticholinergic agents that have found clinical application. These compounds reflect the SAR features that have been described. All these compounds are effective when administered orally or parenterally. Anticholinergic agents possessing a quaternary ammonium functional group are generally not well absorbed from the gastrointestinal tract due to their ionic character. These drugs are useful primarily in the treatment of ulcers or other conditions for which a

reduction in gastric secretions and reduced motility of the gastrointestinal tract are desired. Those antagonists having a tertiary nitrogen are much better absorbed and distributed following all routes of administration and are especially useful when systemic distribution is desired. The tertiary amino-derived anticholinergic agents readily cross the blood-brain barrier. These have proven to be particularly beneficial in the treatment of Parkinson disease and other diseases requiring a central anticholinergic effect.

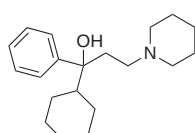
All these drugs display pronounced selectivity for mAChRs; however, some of those possessing the quaternary ammonium functional group exhibit nicotinic antagonist activity at high doses. With the exception of the M_3 antagonists, solifenacin and darifenacin, these agents display no marked selectivity for other mAChR subtypes.

RECENT MUSCARINIC ANTAGONISTS More recently discovered muscarinic antagonists display a higher affinity for the receptors compared with the older agents, as exemplified by quinuclidinylbenzilate, which has structural features common to the classic anticholinergic agents. Radiolabeled quinuclidinylbenzilate was instrumental in the development of mAChR labeling techniques as well as the discovery of subtypes of mAChRs. This latter research depended also on the M_1 -selective antagonist pirenzepine, a compound having a novel structure for muscarinic antagonist activity. A number of compounds structurally related to pirenzepine have demonstrated a similar M_1 selectivity; among these is telenzepine (59). Because of their selectivity for muscarinic M_1 receptors, pirenzepine and telenzepine have been evaluated in clinical trials for the treatment of duodenal ulcers. It is of interest to note that AFDX-116, structurally similar to pirenzepine, is a

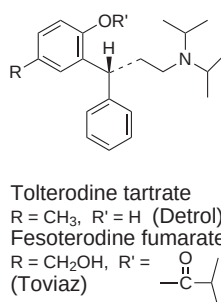
Anticholinergic aminoalcohols:



Procyclidine
hydrochloride salt
(Kemadrin)

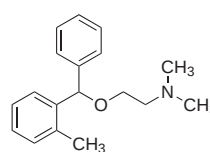


Trihexyphenidyl
hydrochloride salt
(Artane)

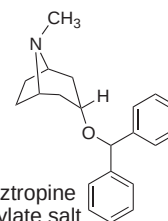


Tolterodine tartrate
 $R = \text{CH}_3$, $R' = \text{H}$ (Detrol)
Fesoterodine fumarate
 $R = \text{CH}_2\text{OH}$, $R' = -\text{C}(=\text{O})\text{CH}_3$
(Toviaz)

Anticholinergic aminoethers:

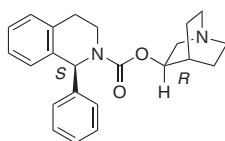


Orphenadrine
citrate salt
(Norflex)

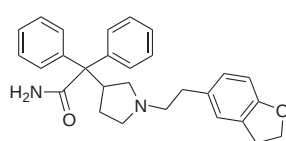


Benztropine
mesylate salt
(Cogentin)

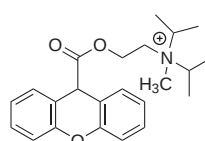
Miscellaneous anticholinergic agents:



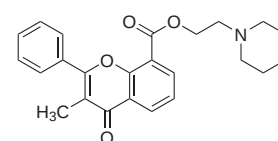
Solifenacin
succinate salt
(Vesicare)



Darifenacin
(Enablex)



Propantheline
bromide salt
(Pro-Banthine)



Flavoxate
hydrochloride salt
(Urispas)

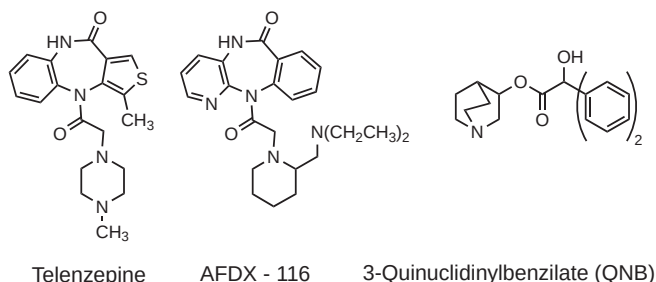
FIGURE 9.22 Aminoalcohol, aminoether, and miscellaneous anticholinergic agents.

TABLE 9.3 Anticholinergic Agents

Name	Calculated ^a LogP and LogD (pH 7)	Half-Life	Metabolism	Indications	Comments
Atropine	1.53 -1.21	3.5 ± 1.5 h	Hydrolysis; N-dealkylation; N-oxide	Bradycardia; parkinsonism; cycloplegic/mydriatic	Nonselective muscarinic antagonist; stimulates the CNS
Scopolamine	0.76 0.29	8 h	Almost completely metabolized (liver)	Uveitis; iritis; parkinsonism; motion sickness	Nonselective muscarinic antagonist; CNS depressant
Homatropine (Isopto Homatropine)	1.57 -1.17	—	—	Cycloplegic/mydriatic	Nonselective muscarinic antagonist; less potent and shorter duration than atropine
Ipratropium bromide (Atrovent)	—	2 h	Hydrolysis	Bronchodilator (oral inhalation); seasonal rhinitis (nasal spray)	Nonselective muscarinic antagonist; slow onset after inhalation
Tiotropium bromide (Spiriva)	—	5–6 d	CYP2D6 and CYP3A4; hydrolysis; N-dealkylation; glucuronide conjugation	Chronic obstructive pulmo- nary disease (oral inhalation)	Equal affinity for M ₁ , M ₂ , and M ₃ receptors
Trospium chloride (Sanctura)	—	20 h	Hydrolysis; conjugation	Urinary and gastrointestinal antispasmodic	High affinity for M ₁ and M ₃ recep- tors; lesser affinity for M ₂
Oxybutynin (oral: Ditropan and Ditropan XL; transdermal: Oxytrol)	5.19 3.93	2–5 h	CYP3A4; hydrolysis; N-dealkylation	Overactive bladder	Nonselective muscarinic antagonist
Solifenacin (Vesicare)	3.70 1.70	55 h	4-(R)-Hydroxy (active); N-glucuronide; N-oxide; 4R-hydroxy-N-oxide	Overactive bladder	Selective M ₃ antagonist
Tolterodine (Detrol)	5.77 2.79	2–4 h (extensive metabolizers); 9.6 h (poor metabolizers)	Primary pathway: CYP2D6 (pri- mary); 7% of Caucasians and 2% of African Americans lack CYP2D6; CYP3A4 is the primary pathway in the latter. Metabolites: 5-hydroxy- methyl (active), 5-carboxylic acid, N-dealkylated-5-carboxylic acid	Overactive bladder	Nonselective muscarinic antagonist
Fesoterodine fumarate (Toviaz)	5.08 ± 0.28 2.09	7	Hydrolysis to 5-hydroxymethyl tolterodine, followed by CYP2D6 and CYP3A4 to give carboxy, carboxy- N-desisopropyl, N-desisopropyl metabolites	Overactive bladder	Prodrug of 5-hydroxymethyl tol- terodine, a nonselective muscarinic agonist
Darifenacin (Enblex)	4.50 2.25	—	CYP2D6 (primary; see tolterodine above); hydroxylation of the dihydrobenzofuran; ring opening (dihydrobenzofuran); N-dealkylation	Overactive bladder	Selective M ₃ antagonist

^a Values calculated using ACD Lab Solarius, Chemical Abstracts Service, 2006, Columbus, OH (values for quaternary compounds are not listed).

muscarinic antagonist exhibiting selectivity for cardiac M_2 receptors.

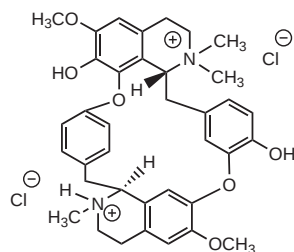


NICOTINIC ANTAGONISTS—NEUROMUSCULAR BLOCKING AGENTS

Nicotinic antagonists are chemical compounds that bind to nAChRs but have no efficacy. All therapeutically useful nicotinic antagonists are competitive antagonists; in other words, the effects are reversible with acetylcholine. There are two subclasses of nicotinic antagonists—skeletal neuromuscular blocking agents and ganglionic blocking agents—classified according to the two populations of nAChRs, NM and NN, respectively. This section emphasizes nicotinic antagonists used clinically as neuromuscular blocking agents. These medicinal agents should not be confused with those skeletal muscle relaxant compounds that produce their effects through the CNS.

History

In terms of the historical perspective, tubocurarine, the first known neuromuscular blocking drug, was as important to the understanding of nicotinic antagonists as atropine was to that of muscarinic antagonists. The neuromuscular blocking effects of extracts of curare were first reported as early as 1510, when explorers of the Amazon River region of South America found natives using these plant extracts as arrow poisons. Early research with these crude plant extracts indicated that the active components caused muscle paralysis by effects on either the nerve or the muscle (remember that the concept of neurochemical transmission was not introduced until the late 19th century). In 1856, however, Bernard described the results of his experiments, which demonstrated unequivocally that curare extracts prevented skeletal muscle contractions by an effect at the neuromuscular junction, rather than the nerve innervating the muscle or the muscle itself (60).



d-Tubocurarine chloride

Much of the early literature concerning the effects of curare is confusing and difficult to interpret. This is not at all surprising considering that this research was performed using crude extracts, many of which came from different plants. It was not until the late 1800s that scientists recognized that curare extracts contained quaternary ammonium salts. This knowledge prompted the use of other quaternary ammonium compounds to explore the neuromuscular junction. In the meantime, curare extracts continued to be used to block the effects of nicotine and acetylcholine at skeletal neuromuscular junctions and to explore the nAChRs.

In 1935, King (61) isolated a pure alkaloid, which he named *d*-tubocurarine, from a tube curare of unknown botanical origin. The word “tube” refers to the container in which the South American natives transported their plant extract. It was almost 10 years later that the botanical source for *d*-tubocurarine was clearly identified as *Chondodendron tomentosum*. The structure that King assigned to tubocurarine possessed two nitrogen atoms, both of which were quaternary ammonium salts (e.g., a *bis*-quaternary ammonium compound). It was not until 1970 that the correct structure was reported by Everett et al. (62). The correct structure, shown here, has only one quaternary ammonium nitrogen; the other nitrogen is a tertiary amine salt. Nevertheless, the incorrect structure of tubocurarine served as the model for the synthesis of all the neuromuscular blocking agents in use today. These compounds have been of immense therapeutic value for surgical and orthopedic procedures and have been essential to research that led to the isolation and purification of nAChRs.

The potential therapeutic benefits of the neuromuscular blocking effects of tubocurarine as well as the difficulty in obtaining pure samples of the alkaloid encouraged medicinal chemists to design structurally related compounds possessing nicotinic antagonist activity. Using the incorrectly assigned *bis*-quaternary ammonium structure of tubocurarine, as reported by King (61), as a guide, a large number of compounds were synthesized and evaluated. It became apparent that a *bis*-quaternary ammonium compound having two quaternary ammonium salts separated by 10 to 12 carbon atoms (similar to the distance between the nitrogen atoms in tubocurarine) was a requirement for neuromuscular blocking activity. The rationale for this structural requirement was that in contrast to mAChRs, nAChRs possessed two anionic-binding sites, both of which had to be occupied for a neuromuscular blocking effect. It is important to observe that the current transmembrane model for the nAChR protein has two anionic sites in the extracellular domain.

Some of the new *bis*-quaternary ammonium agents produced depolarization of the postjunctional membrane at the neuromuscular junction before causing blockade; other compounds, such as tubocurarine, did not produce this depolarization. Thus, the structural features of the remainder of the molecule determined whether the nicotinic antagonist was a depolarizing or a nondepolarizing neuromuscular blocker.

Therapeutic Application

Neuromuscular blocking agents are used primarily as an adjunct to general anesthesia. Their pharmacokinetics are shown in Table 9.4. They produce skeletal muscle relaxation that facilitates operative procedures such as abdominal surgery. Furthermore, they reduce the depth requirement for general anesthetics; this decreases the overall risk of a surgical procedure and shortens the postanesthetic recovery time. Muscles producing rapid movements are the first to be affected by neuromuscular blocking agents. These include muscles of the face, eyes, and neck. Muscles of the limbs, chest, and abdomen are affected next, with the diaphragm (respiration) being affected last. Recovery generally is in the reverse order.

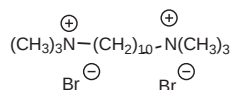
Neuromuscular blocking agents have also been used in the correction of dislocations and the realignment of fractures. Short-acting neuromuscular blocking agents, such as succinylcholine, are routinely used to assist in tracheal intubation. When choosing a neuromuscular blocking agent, four questions must be considered:

1. Will the compound produce the desired neuromuscular blockade?
2. What is its duration of action?
3. What are its adverse effects?
4. What is its relative cost?

Side Effects

Adverse reactions to most, but not all, of the neuromuscular blocking agents can include hypotension, bronchospasm, and cardiac disturbances. The depolarizing agents also cause an initial muscle fasciculation before relaxation. Many of these agents cause release of histamine and subsequent cutaneous (flushing, erythema, urticaria, and pruritus), pulmonary (bronchospasm and wheezing), and cardiovascular (hypotension) effects.

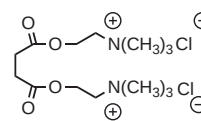
Specific Depolarizing Neuromuscular Blocking Agents



Decamethonium bromide

Decamethonium Bromide

Decamethonium was one of the first neuromuscular blocking agents synthesized. An SAR study on a series of *bis*-quaternary ammonium compounds with varying numbers of methylene groups separating the nitrogen atoms demonstrated that maximal neuromuscular blockade occurred with 10 to 12 unsubstituted methylene groups. Activity diminished as the number of carbons was either decreased or increased. The compound with six methylene groups, hexamethonium, is a nicotinic antagonist at autonomic ganglia (ganglionic blocking agent). All compounds in this series that possessed neuromuscular blocking activity also caused depolarization of the postjunctional membrane.



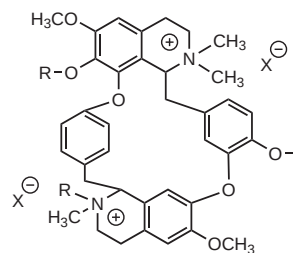
Succinylcholine chloride

Succinylcholine Chloride (*Anectine*)

Succinylcholine is a depolarizing neuromuscular blocking agent that represents a dimer of acetylcholine bonded through the α -carbon atom of each. The molecule can exist in an extended conformation (antiperiplanar). This would account for the appropriate separation of the quaternary nitrogens. Succinylcholine is rapidly hydrolyzed and rendered inactive both in aqueous solution and by plasma esterases; this chemical instability must be considered when preparing solutions for parenteral administration. This same chemical property, however, gives the compound a brief duration of action. As a result, succinylcholine is frequently used for rapid induction of neuromuscular blockade and when blockade of short duration is desired (Table 9.4). As such, it is used primarily to produce muscle relaxation during endotracheal intubation or endoscopic procedures. The depolarizing property is undesirable in neuromuscular blockers, so most research efforts have been directed toward the design of nondepolarizing agents.

Specific Nondepolarizing Neuromuscular Blocking Agents

Compounds in this class have one or two quaternary ammonium groups. Those with only one quaternary ammonium group, however, exist as *bis*-cations in vivo due to the second positive charge being on a protonated tertiary amine. The various structures of these compounds serve primarily as a “scaffold” to position two positive charges in the correct three-dimensional orientation for interaction with the transmembrane nAChRs.



d-Tubocurarine chloride (R = H, X = Cl)
Metocurine iodide (R = CH₃, X = I)

d-Tubocurarine and Metocurine (*Metubine Iodide*)

The prototype of this class is *d*-tubocurarine. It is administered intravenously and has a relatively long duration of action. Only about 1% of a dose is demethylated in the liver, and it is primarily excreted unchanged in the urine and bile. *d*-Tubocurarine preparations contain bisulfites and, thus, can potentiate allergic reactions in patients with bisulfate allergy. It is the most potent inducer of

TABLE 9.4 Properties of Clinically Useful Neuromuscular Blocking Agents

Agent	Time of Onset (min)	Duration of Action (min)	Half-life (min)	Mode of Elimination
Succinylcholine	1–1.5	6–8	<1	Hydrolysis by plasma cholinesterases
<i>d</i> -Tubocurarine	4–6	80–120	173	Renal elimination, liver clearance
Vecuronium	2–4	30–40	65–80	Liver metabolism and clearance renal elimination
Pancuronium	4–6	120–180	89–140	Renal elimination, liver metabolism and clearance
Pipecuronium	2–4	80–100	137–161	Renal elimination, liver metabolism and clearance
Rocuronium	1–2	30–40	84–131	Liver metabolism and clearance
Atracurium	2–4	30–40	16–20	Hofmann degradation, hydrolysis by plasma cholinesterases
Mivacurium	2–4	12–18	1.8–2.0	Hydrolysis by plasma cholinesterases
Doxacurium	4–6	90–120	72–96	Renal elimination, liver metabolism and clearance

Adapted from Taylor P. Agents acting at the neuromuscular junction and autonomic ganglia. In: Brunton LL, Lazo JS, Parker KL, eds. Goodman and Gilman's The pharmacological Basis of Therapeutics, 11th Ed. New York: McGraw-Hill, 2006:217–236; with permission.

histamine release of all the nondepolarizing neuromuscular blockers.

Reaction of *d*-tubocurarine with methyl iodide affords metocurine iodide (see above), in which the two phenolic hydroxyl groups of *d*-tubocurarine are changed to the methyl ethers and the tertiary amine becomes quaternary. This agent is approximately fourfold more potent than *d*-tubocurarine in neuromuscular blocking activity. Like *d*-tubocurarine, it has a long duration of action and is eliminated (predominantly unchanged) via the kidney.

Steroid-Based Neuromuscular Blocking Agents

An ideal neuromuscular blocking agent would be a nondepolarizing compound that is metabolically inactivated and rapidly eliminated. Efforts to design such a neuromuscular blocker have resulted in the development of several synthetic neuromuscular agents. Those that have found clinical use are either aminosteroids derived from (+)-malouetine (an aminosteroid found in the rain forest of central Africa) (Fig. 9.23) or tetrahydroisoquinoline derivatives (Fig. 9.24).

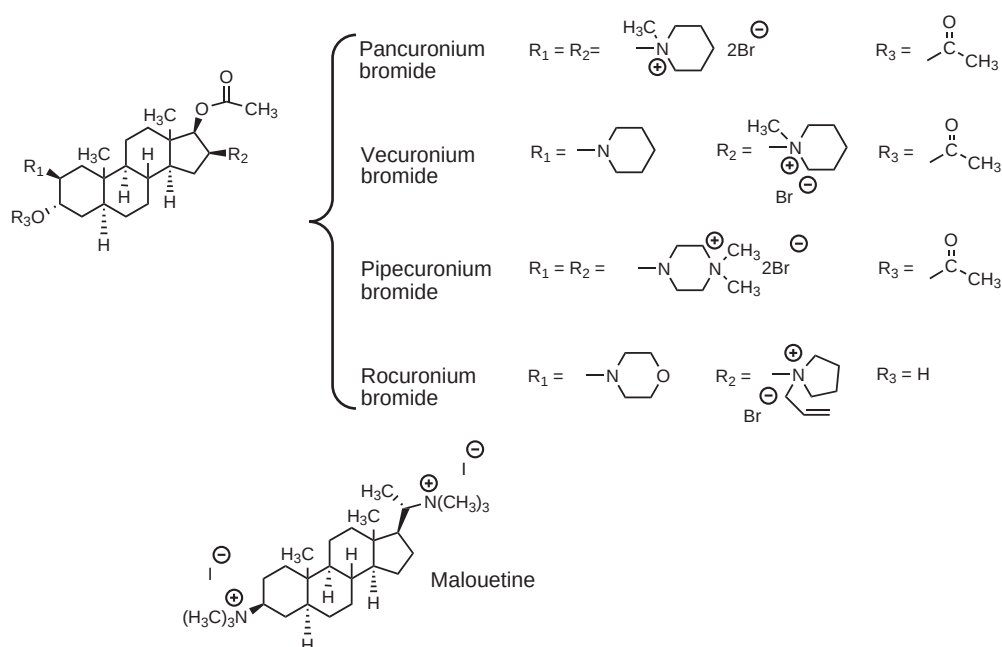


FIGURE 9.23 Aminosteroid-based neuromuscular blocking agents.

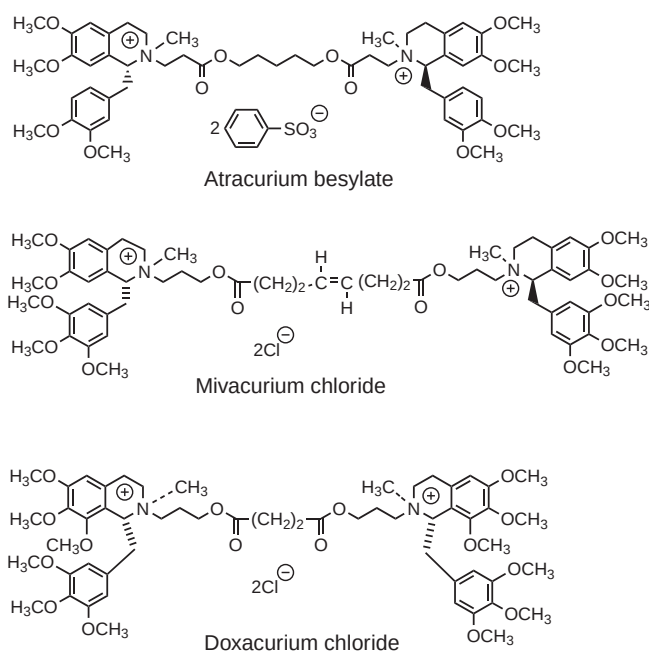


FIGURE 9.24 Tetrahydroisoquinoline-based neuromuscular blocking agents.

PANCURONIUM BROMIDE (PAVULON) Pancuronium, a long-acting agent, is more active than tubocurarine. It can cause increases in heart rate and blood pressure and should not be used in patients with coronary artery disease. Pancuronium undergoes hydrolysis in the liver to the active 3-hydroxy metabolite and the inactive 17-hydroxy and 3,17-dihydroxy metabolites; it is excreted primarily in the urine, with small amounts in the bile.

VECURIUM BROMIDE (NORCURON) Removal of the methyl group from the quaternary piperidinium group at position 3 of pancuronium affords vecuronium, an intermediate-acting agent. Vecuronium has the advantage of not inducing histamine release at normal doses and of not exhibiting significant cardiovascular effects. One-third of an administered dose of vecuronium is hydrolyzed to the 3-hydroxy, 17-hydroxy, and 3,17-dihydroxy metabolites, all of which are active. Accumulation of the 3,17-dihydroxy metabolite is responsible for prolonged neuromuscular blockade in patients receiving long-term therapy with vecuronium.

PIPECURONIUM BROMIDE (ARDUAN) Pipecuronium bromide, a long-acting neuromuscular blocking agent, exhibits minimal cardiovascular effects. Like pancuronium and vecuronium, pipecuronium undergoes some hydrolysis but is excreted primarily unchanged in the urine with very small amounts in the bile. Pipecuronium can be used in patients with coronary artery disease, but neuromuscular blockade is prolonged in patients with renal failure.

ROCURONIUM BROMIDE (ZEMURON) Rocuronium bromide is an intermediate-acting agent with a duration of action

similar to vecuronium and atracurium but with a more rapid onset. It does not appear to cause significant histamine release.

Tetrahydroisoquinoline-Based Neuromuscular Blocking Agents

ATRACURIUM BESYLATE (TRACRIUM) Atracurium besylate is a nondepolarizing neuromuscular blocker in which the quaternary ammonium groups are located in two substituted tetrahydroisoquinoline rings separated by an aliphatic diester. Its duration of action is slightly longer than that of succinylcholine. Atracurium is not metabolized in the liver; rather, it undergoes hydrolysis of the

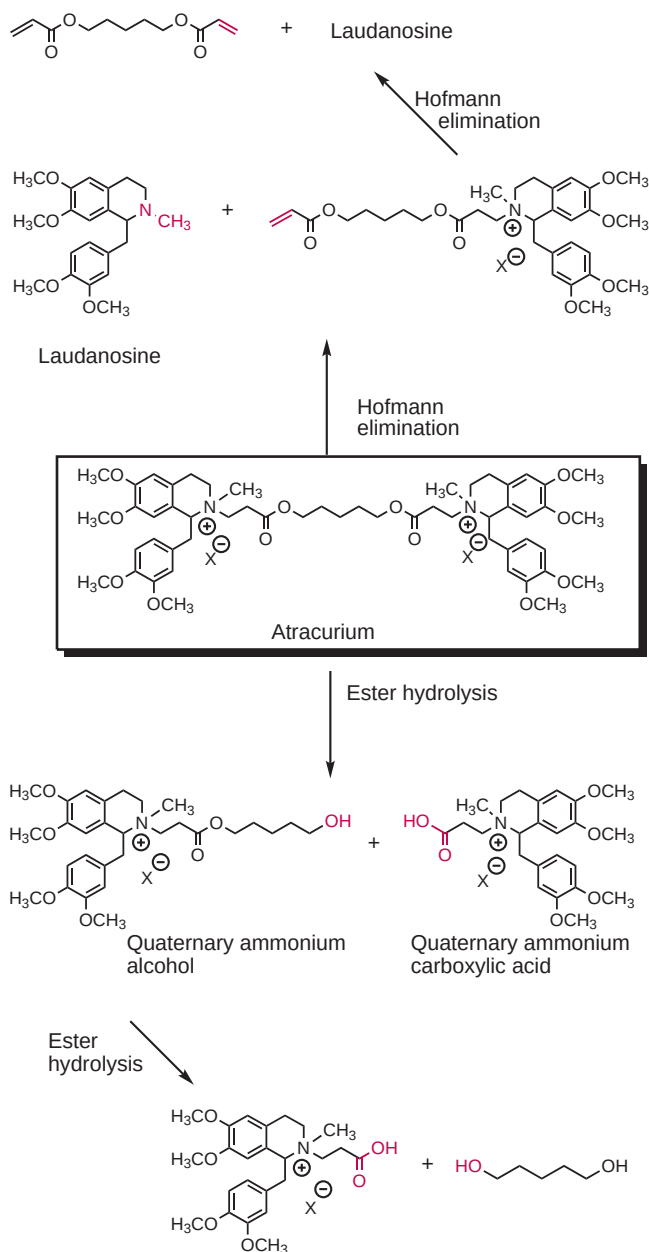


FIGURE 9.25 Inactivation of atracurium by Hofmann elimination and hydrolysis.

ester functional groups that connect the two quaternary nitrogens. It also undergoes Hofmann elimination, a nonenzymatic, base-catalyzed decomposition, to yield laudanosine, which is inactive (Fig. 9.25) (63,64). Thus, termination of the effects of atracurium is independent of renal elimination. Because of this unusual metabolic profile, it is useful in patients with hepatic or renal disease.

MIVACURIUM CHLORIDE (MIVACRON) Mivacurium chloride is a mixture of three stereoisomers, with the *trans-trans* (92% to 96%) and the *cis-trans* diesters being equipotent. The *cis-cis* diester produces only minimal (<5%) neuromuscular blockade. It is hydrophilic, has a small volume of distribution, and is distributed primarily to extracellular

fluids. Mivacurium is short acting (Table 9.4), with mean elimination half-lives for the *trans-trans* and *cis-trans* stereoisomers of 2.0 and 1.8 minutes, respectively, in adults receiving opioid/nitrous oxide/oxygen anesthesia. It is rapidly hydrolyzed and does not undergo Hofmann elimination like atracurium.

DOXACURIUM CHLORIDE (NUROMAX) Doxacurium is a mixture of three *trans-trans* stereoisomers, a *dl* pair [1*R*,1'*R*,2*S*,2'*S* and 1*S*,1'*S*,2*R*,2'*R*] and a *meso* form 1*R*,1'*S*,2*S*,2'*R*. Doxacurium is hydrophilic, has a small volume of distribution, and is distributed primarily to extracellular fluids. It is not metabolized by plasma cholinesterase or hepatic enzymes and does not undergo Hofmann elimination.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Kathryn Neill, PharmD

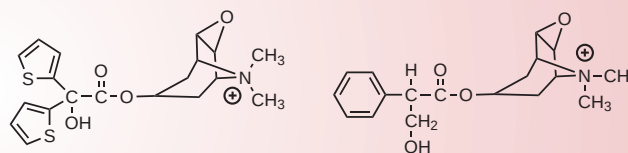
The pharmacist reviews the package insert and tells LT that he does not need to take ipecac or try to make himself expel the medication. There is little chance he will experience any symptoms, but the main symptom he may experience is dry mouth. The pharmacist also recommends LT take a regular daily dose of tiotropium using the inhalation device.

Chemical Analysis

Kathryn Neill, S. William Zito, and Victoria Roche

Tiotropium is a long-acting anticholinergic agent with an affinity for muscarinic receptors M₁–M₅. It causes bronchodilation through inhibition of M₃ receptors in bronchial smooth muscle; effects can last more than 24 hours. Tiotropium is

the dithienyl derivative of *N*-methyl scopolamine, a quaternary ammonium analog of naturally occurring scopolamine in *Atropa belladonna*.



Tiotropium bromide

N-Methyl scopolamine

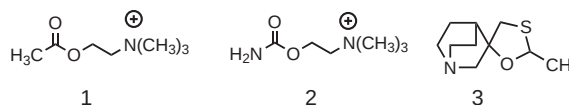
The bioavailability of inhaled tiotropium is approximately 19% in healthy volunteers, and therapeutic bronchodilation is predominantly a site-specific effect. Gastrointestinal absorption is minimal, with an oral solution resulting in bioavailability of only about 2%.

CASE STUDY

S. William Zito and Victoria Roche

NM is a 43-year-old man of East Indian heritage. He presents to the ophthalmology department of the hospital where you are on rotation. He explains to the ophthalmologist that he has been experiencing blurred vision in his left eye. He is a senior executive in a software company and his work involves a lot of reading and computer work. He has been having this problem for the last 3 to 4 months, and it is becoming increasingly difficult for him to work at the computer. On examination he had a visual acuity of 20/20 in the right eye and 20/40 in the left eye. The lens in the left eye showed a small central cataract whereas the right eye was normal. Anterior segment examination showed normal anterior chambers with normal pupils. Fundus examination showed no abnormality and intraocular pressure was 14 mm Hg in the left eye and 16 mm Hg in the right. Retinal examination also was normal in both the eyes. It was decided to conduct cataract surgery

in the left eye via phacoemulsification. The ophthalmologist wants to use a cholinergic agonist for the induction of meiosis. Your preceptor shows you the following structures and asks you which one you would recommend for this surgery.



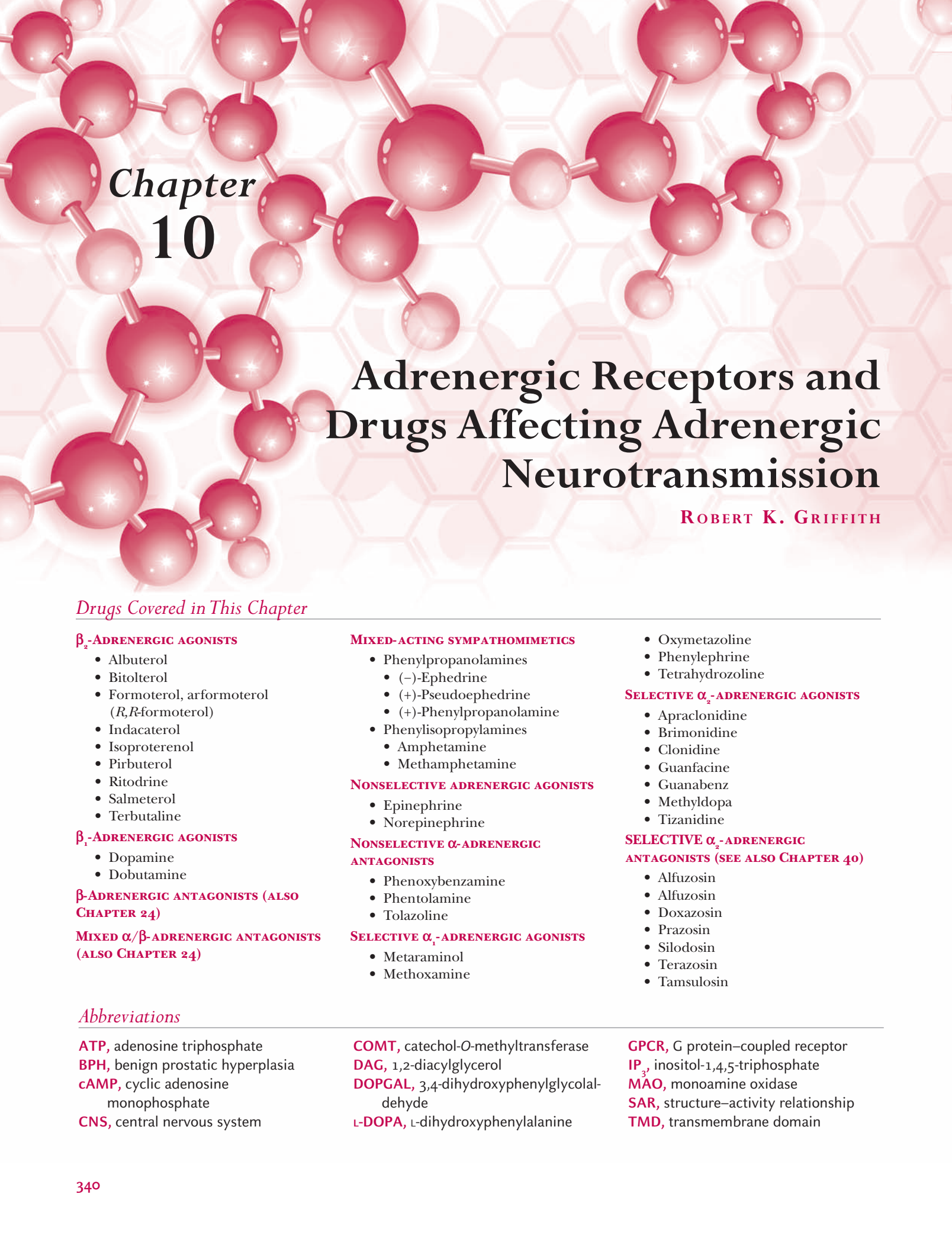
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 10

Adrenergic Receptors and Drugs Affecting Adrenergic Neurotransmission

ROBERT K. GRIFFITH

Drugs Covered in This Chapter

β_2 -ADRENERGIC AGONISTS

- Albuterol
- Bitolterol
- Formoterol, arformoterol (*R,R*-formoterol)
- Indacaterol
- Isoproterenol
- Pirbuterol
- Ritodrine
- Salmeterol
- Terbutaline

β_1 -ADRENERGIC AGONISTS

- Dopamine
- Dobutamine

β -ADRENERGIC ANTAGONISTS (ALSO CHAPTER 24)

MIXED α/β -ADRENERGIC ANTAGONISTS (ALSO CHAPTER 24)

MIXED-ACTING SYMPATHOMIMETICS

- Phenylpropanolamines
 - (–)-Ephedrine
 - (+)-Pseudoephedrine
 - (+)-Phenylpropanolamine
- Phenylisopropylamines
 - Amphetamine
 - Methamphetamine

NONSELECTIVE ADRENERGIC AGONISTS

- Epinephrine
- Norepinephrine

NONSELECTIVE α -ADRENERGIC ANTAGONISTS

- Phenoxybenzamine
- Phentolamine
- Tolazoline

SELECTIVE α_1 -ADRENERGIC AGONISTS

- Metaraminol
- Methoxamine

- Oxymetazoline
- Phenylephrine
- Tetrahydrozoline

SELECTIVE α_2 -ADRENERGIC AGONISTS

- Apraclonidine
- Brimonidine
- Clonidine
- Guanfacine
- Guanabenz
- Methyldopa
- Tizanidine

SELECTIVE α_2 -ADRENERGIC ANTAGONISTS (SEE ALSO CHAPTER 40)

- Alfuzosin
- Alfuzosin
- Doxazosin
- Prazosin
- Silodosin
- Terazosin
- Tamsulosin

Abbreviations

ATP, adenosine triphosphate

BPH, benign prostatic hyperplasia

cAMP, cyclic adenosine monophosphate

CNS, central nervous system

COMT, catechol-O-methyltransferase

DAG, 1,2-diacylglycerol

DOPGAL, 3,4-dihydroxyphenylglycolaldehyde

L-DOPA, L-dihydroxyphenylalanine

GPCR, G protein–coupled receptor

IP₃, inositol-1,4,5-triphosphate

MAO, monoamine oxidase

SAR, structure–activity relationship

TMD, transmembrane domain

SCENARIO

SJ is a 28-year-old white woman with pneumonia and acute psychosis. She has used illicit drugs extensively. Upon arrival to the emergency department, SJ was disoriented and combative and was sedated. She was started on intravenous antibiotics, fluids, and dopamine 800 mg in 250 mL of D5W at 5 mcg/kg/min. Hours later the patient remained combative and believed the hospital staff were trying to poison her. When the nurse

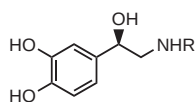
went to retrieve another dose of sedation, SJ disconnected the intravenous tubing and drank the remaining dopamine infusion. Upon finding the bag empty, the nurse called the pharmacy for recommendations.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

Adrenergic drugs are a broad class of agents employed in the treatment of disorders of widely varying severity. Adrenergic drugs include popular prescription drugs, such as albuterol for asthma and atenolol for hypertension, as well as many common over-the-counter cold remedies, such as the nasal decongestant pseudoephedrine.

Adrenergic drugs act on effector cells through adrenoceptors that normally are activated by the neurotransmitter norepinephrine (noradrenaline), or they can act on the neurons that release the neurotransmitter. The term “adrenergic” stems from the discovery early in the 20th century that administration of the hormone adrenaline (epinephrine) had specific effects on selected organs and tissues similar to the effects produced by stimulation of the sympathetic (adrenergic) nervous system. For a number of years, adrenaline was thought to be the neurotransmitter in the sympathetic nervous system, but it was also recognized that the effects of administered epinephrine were not quite identical to those of sympathetic stimulation. Finally, in the 1940s, norepinephrine was identified as the true neurotransmitter at the terminus of the sympathetic nervous system (1,2). Adrenoceptors are widely located in various organs and tissues as well as on neurons of both the peripheral nervous system and central nervous system (CNS).



Norepinephrine, R = H
Epinephrine, R = CH₃

Norepinephrine and epinephrine are members of a class of pharmacologically active substances known as catecholamines, because they contain within their structures both an amine and an *ortho*-dihydroxybenzene moiety, which is known by the common chemical name of catechol. Many adrenergic drugs are also catecholamines, and their structure–activity relationships (SARs) will be discussed.

Neurons at the terminus of an adrenergic neuron fiber release norepinephrine to influence the target

tissue through binding to receptors on cells of the tissue or organ. The cells bearing the receptors are called effector cells, because they produce the effect seen by adrenergic stimulation.

In general, stimulation of the sympathetic nervous system causes what is known as a “fight-or-flight” response. These effects include an increased rate and force of heart contraction, a rise in blood pressure, a shift of blood flow to skeletal muscles, dilation of bronchioles and pupils, and an increase in blood glucose levels through gluconeogenesis and glycogenolysis. These responses are what one might predict in a mammal that is enraged and preparing to fight or in one that is frightened and preparing to flee. Drugs can influence adrenergic responses through a variety of mechanisms, and an understanding of these mechanisms requires knowledge regarding the details of norepinephrine biosynthesis, storage, release, and fate after release.

BIOSYNTHESIS, STORAGE, AND RELEASE OF NOREPINEPHRINE

Biosynthesis of norepinephrine takes place within adrenergic neurons near the terminus of the axon and junction with the effector cell. The biosynthetic pathway (Fig. 10.1) begins with the active transport of the amino acid L-tyrosine into the adrenergic neuron cell (1). In the first step within the cytoplasm, the enzyme tyrosine hydroxylase (tyrosine-3-monooxygenase) oxidizes the 3' position of tyrosine to form the catechol amino acid L-dihydroxyphenylalanine (L-DOPA). This is the rate-limiting step in norepinephrine biosynthesis, and the activity of tyrosine hydroxylase is carefully controlled (3). The enzyme is under feedback inhibition control by product catecholamines and is controlled through a complex pattern of phosphorylation/dephosphorylation, in which phosphorylation by protein kinases activates the enzyme and dephosphorylation by phosphatases decreases activity.

In the second step, L-DOPA is decarboxylated to dopamine by aromatic L-amino acid decarboxylase, another cytoplasmic enzyme. Aromatic L-amino acid decarboxylase was discovered in the 1930s and originally

CLINICAL SIGNIFICANCE

Through the years, an increased understanding of adrenergic receptors and compounds that modulate their activity has led to decreased morbidity and mortality as well as increased quality of life for millions of patients worldwide; multiple drugs in this category are on the Top 200 list of prescriptions dispensed. Agents affecting adrenergic neurotransmission are used to treat a variety of clinical conditions, including hypertension, hypotension, angina, heart failure, arrhythmias, and asthma. In fact, our understanding of structure-activity relationships led to an ability to alter the pharmacodynamic effects of various agents even before the specific receptor types were identified.

A thorough understanding of the structure-activity relationships of these compounds and application of the principles of medicinal chemistry to alter their pharmacodynamic effects has enhanced the desired properties and diminished the incidence and/or severity of associated unwanted adverse effects. Furthermore, therapeutic utility and patient compliance has been improved with the advent of orally active agents with longer elimination times and decreased frequency of dosing. For clinicians, a thorough understanding of the resultant pharmacodynamic properties of these agents is necessary to substantiate therapeutic decision making for an individual patient. For example, a patient with hypotension resulting in shock is treated with an adrenergic agonist that will increase blood pressure. If the patient's hypotension results from distributive shock, an agent

with α_1 -stimulant properties is necessary. If hypotension results from cardiogenic shock, however, a β_1 -agonist drug would be most effective, and, in actuality, an α_1 stimulant agent may cause the patient to further decompensate by increasing cardiac workload without lending support to the failing heart. Manipulation of three sites on the basic phenylethanolamine structure alter these receptor specificities, which allows a clinician to tailor drug selection for an individual patient based on the specific desired pharmacological effects of each compound.

A second example would be selection of a β_1 -selective receptor blocking agent like metoprolol, as opposed to a nonspecific β -antagonist like propranolol, to treat a patient with asthma who has experienced a myocardial infarction. This allows the patient to benefit from the potential mortality reduction associated with the β -blocking agent while minimizing the risk of decreasing β_2 -agonist utility for bronchospasm. These are only a couple of simple illustrations of altering the structure-activity relationship of adrenergic compounds to increase their utility for specific disease processes, outlining the importance of understanding the pharmacodynamic impact of these alterations.

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named DOPA decarboxylase. Researchers discovered subsequently that DOPA decarboxylase is not specific for L-DOPA and decarboxylates other aromatic L-amino

acids having the L (or S) absolute configuration, such as 5-hydroxy-L-tryptophan, L-tryptophan, and L-tyrosine. Nevertheless, the enzyme is often referred to by the older name.

The dopamine formed in the cytoplasm by decarboxylation of L-DOPA is then taken up by active transport into storage vesicles or granules located near the terminus of the adrenergic neuron. Within these vesicles, the enzyme dopamine β -hydroxylase stereospecifically introduces a hydroxyl group in the (R) absolute configuration on the carbon atom β to the amino group to generate the neurotransmitter norepinephrine. Norepinephrine is stored in the vesicles in a 4:1 complex with adenosine triphosphate (ATP) in such quantities that each vesicle in a peripheral adrenergic neuron contains between 6,000 and 15,000 molecules of norepinephrine (4). The norepinephrine remains in the vesicles until released into the synapse during signal transduction. When a wave of depolarization reaches the terminus of an adrenergic neuron, it triggers the transient opening of voltage-dependent calcium channels, causing an influx of calcium ions. This influx of calcium ions triggers fusion of the storage vesicles with the neuronal cell

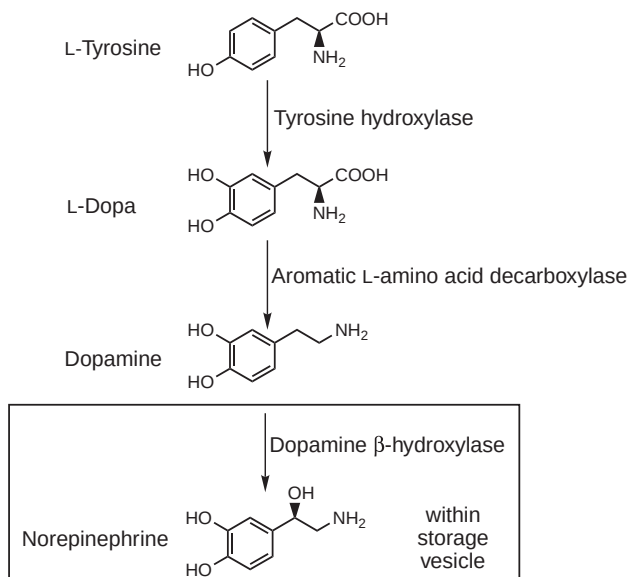


FIGURE 10.1 Biosynthesis of norepinephrine.

membrane, spilling the norepinephrine and other contents of the vesicles into the synapse through exocytosis (Fig. 10.2). The pathway for epinephrine biosynthesis in the adrenal medulla is the same as for norepinephrine with the additional step of conversion of norepinephrine to epinephrine by the enzyme phenylethanolamine-*N*-methyltransferase.

REUPTAKE AND METABOLISM OF NOREPINEPHRINE FOLLOWING RELEASE

After its release, norepinephrine diffuses through the intercellular space to bind reversibly to α - or β -adrenoceptors on the effector cell, inducing a conformational change in the receptor. This conformational change triggers a biochemical cascade that results in a physiologic response by the effector cell. As well as the receptors on effector cells, there are also α_2 -adrenoceptors that respond to norepinephrine on the presynaptic neuron, which, when stimulated by norepinephrine, inhibit the release of additional norepinephrine into the synapse. It must be noted that there are also postsynaptic α_2 -receptors, particularly in the CNS. Once it has been released and is stimulating its various receptors, there must be mechanisms for removing the norepinephrine from the synapse and terminating the adrenergic impulse. The most important of these mechanisms for removing the norepinephrine is transmitter recycling through active transport reuptake into the presynaptic neuron. This process, called uptake-1, is efficient, and in some tissues, up to 95% of released norepinephrine is likely removed from the synapse by this mechanism (5). Part of the norepinephrine taken into the presynaptic neuron by uptake-1 is metabolized to 3,4-dihydroxyphenylglycolaldehyde (DOPGAL) by mitochondrial monoamine oxidase (MAO) (Fig. 10.3), and part of it is sequestered in the storage vesicles to be used again

as a neurotransmitter. A less efficient uptake process, called uptake-2, operates in a variety of other cell types but only in the presence of high concentrations of norepinephrine. That portion of released norepinephrine that escapes uptake-1 diffuses out of the synapse and is metabolized in extraneuronal sites by catechol-*O*-methyltransferase (COMT), which methylates the *meta* hydroxyl group of norepinephrine. Norepinephrine is also metabolized to DOPGAL by MAO at extraneuronal sites, principally the liver and blood platelets. Both DOPGAL and normetanephrine are subject to further metabolism, as outlined in Figure 10.3 (6). These pathways are also important to drugs that are structural analogs of norepinephrine. In particular, drugs that are catechols (a 1,2-diphenolic moiety) are subject to metabolism by COMT, and drugs with unsubstituted aliphatic amino groups are often substrates for MAO.

As summarized in Figure 11.3, following biosynthesis and storage in a vesicle, norepinephrine release into the synapse is triggered by depolarization-induced calcium influx. The norepinephrine in the synapse interacts with postsynaptic G protein-linked α - or β -receptors on the effector cell, triggering effector cell response, or with presynaptic α_2 -receptors on the neuron, which inhibit release of more norepinephrine. Most of the synaptic neurotransmitter is transported back into the presynaptic neuron by uptake-1 active transport. Some of the norepinephrine is metabolized by MAO, and the remainder is stored in a vesicle to be used again. That portion of norepinephrine not captured by uptake-1 diffuses out of the synapse and is metabolized by COMT and MAO.

CHARACTERIZATION OF ADRENERGIC RECEPTOR SUBTYPES

The discovery of subclasses of adrenergic receptors and the ability of relatively small-molecule drugs to stimulate differentially or block these receptors represented

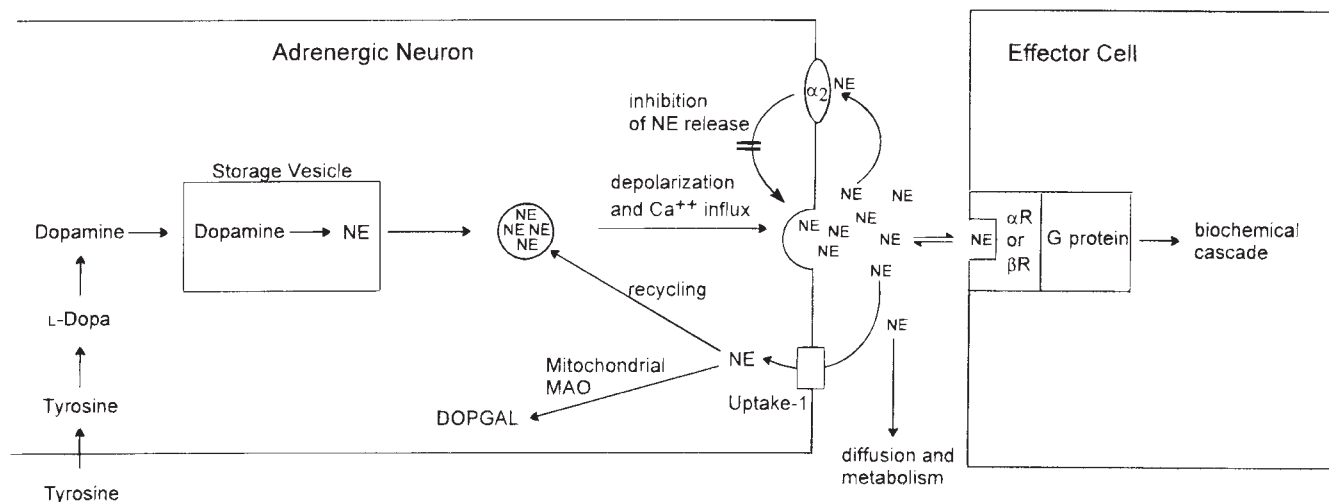


FIGURE 10.2 Neurotransmission events in an adrenergic neuron and effector cell. DOPGAL, 3,4-dihydroxyphenylglycolaldehyde; MAO, monoamine oxidase; NE, norepinephrine; α R, α -adrenoceptor; β R, β -adrenoceptor.

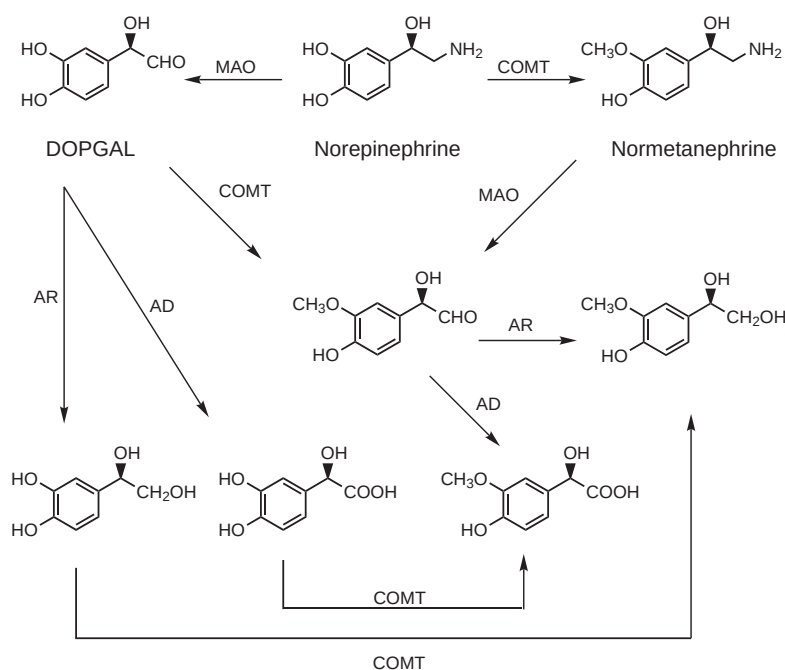
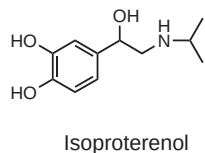


FIGURE 10.3 Metabolism of norepinephrine. AD, aldehyde dehydrogenase; AR, aldehyde reductase; COMT, catechol-O-methyltransferase; MAO, monoamine oxidase.

a major advance in several areas of pharmacotherapeutics. Adrenergic receptors were subclassified by Ahlquist (7) in 1948 into α - and β -adrenoreceptor classes according to their responses to different adrenergic receptor agonists, principally norepinephrine, epinephrine, and isoproterenol. These catecholamines are able to stimulate α -adrenoceptors in the following descending order of potency: epinephrine > norepinephrine > isoproterenol. In contrast, β -adrenoceptors are stimulated in the following descending order of potency: isoproterenol > epinephrine > norepinephrine.



In the years since Ahlquist's original classification, additional small-molecule agonists and antagonists have been used to allow further subclassification of α - and β -receptors into the α_1 and α_2 subtypes of α -receptors and the β_1 , β_2 , and β_3 subtypes of β -adrenoceptors. The powerful tools of molecular biology have been used to clone, sequence, and identify even more subtypes of α -receptors for a total of six. Currently, three types of α_1 -adrenoceptors, called α_{1A} , α_{1B} , and α_{1D} , are known. (There is no α_{1C} because identification of a supposed α_{1C} was found to be incorrect.) Currently, three subtypes of α_2 , known as α_{2A} , α_{2B} , and α_{2C} , are also known (8). The data derived from molecular biology provide a wealth of information on

the structures and biochemical properties of both α - and β -receptors. Intensive research continues in this area, and the coming years can provide evidence of additional subtypes of both α - and β -receptors. At this time, however, only the α_1 -, α_2 -, β_1 -, and β_2 -receptor subtypes are sufficiently well differentiated by their small-molecule binding characteristics to be clinically significant in pharmacotherapeutics, although therapeutic agents acting selectively on β_3 -adrenoceptors to induce fat catabolism could become available in the near future (9).

The adrenoceptors, both α and β , are members of a receptor superfamily of membrane-spanning proteins, including muscarinic, serotonin, and dopamine receptors, which are coupled to intracellular guanosine triphosphate-binding proteins (G proteins), which determine the cellular response to receptor activation (10). All these receptors exhibit a common motif of a single polypeptide chain that is looped back and forth through the cell membrane seven times with an extracellular N-terminus and intracellular C-terminus. The human β_2 -adrenoceptor was one of the first to be cloned and thoroughly studied (Fig. 10.4) (11). The seven transmembrane domains, TMD1 through TMD7, are composed primarily of lipophilic amino acids arranged in α -helices connected by regions of hydrophilic amino acids. The hydrophilic regions form loops on the intracellular and extracellular faces of the membrane. In all the adrenoceptors, the agonist/antagonist recognition site is located within the membrane-bound portion of the receptor. This binding site is within a pocket formed by the membrane-spanning regions of the peptide, as illustrated in Figure 10.5 for epinephrine bound to the

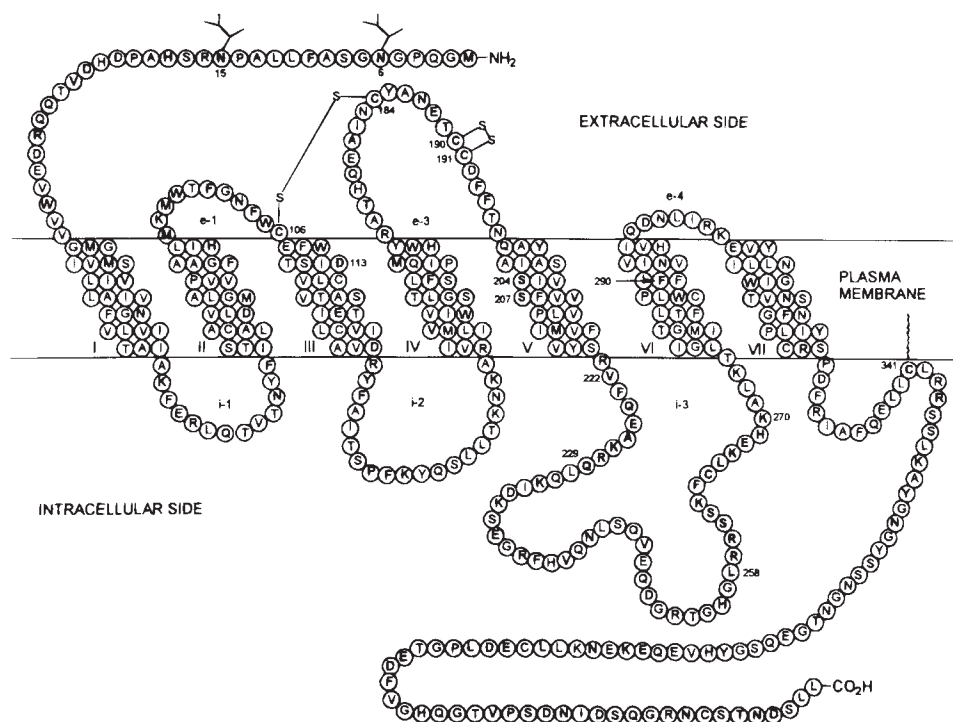


FIGURE 10.4 Human β_2 -adrenergic receptor: amino acid sequence of the human β_2 -receptor showing the seven transmembrane domains (I–VII), the connecting intracellular and extracellular loops, extracellular glycosylation sites at asparagines 6 and 15, and intrachain disulfide bonds between cysteines 106 to 184 and 190 to 191. Also indicated are the amino acids identified as participating in neurotransmitter binding—aspartate 113 in transmembrane domain III, which binds the positively charged amine of the neurotransmitter, and serines 204 and 207 of transmembrane domain V, which form hydrogen-bonds with the catechol hydroxyls. Phenylalanine 290 can participate in agonist binding as well. Amino acids 222 to 229 and 258 to 270 of the third intracellular loop are critical for G protein coupling, and palmitoylated cysteine 341 is critical for proper adenylyl cyclase activation. (Reprinted with permission from the Annual Review of Pharmacology and Toxicology, Volume 32 © 1992 by Annual Reviews, <http://www.annualreviews.org>.)

human β_2 -receptor. All the adrenoceptors are coupled to their effector systems through a G protein, which is linked through reversible binding interactions with the third intracellular loop of the receptor protein.

Salient features of the extensively studied β_2 -adrenoreceptor are indicated in Figure 10.4. Binding studies with selectively mutated β_2 -receptors have provided strong evidence for binding interactions between agonist functional groups and specific residues in the transmembrane domains of adrenoceptors. Such studies indicate that Asp113 in transmembrane domain 3 (TMD3) of the β_2 -receptor is the acidic residue that forms a bond, presumably ionic or a salt bridge, with the positively charged amino group of catecholamine agonists. An aspartic acid residue is also found in a comparable position in all the other adrenoceptors as well as other known G protein-coupled receptors (GPCRs) that bind substrates having positively charged nitrogens in their structures. Elegant studies with mutated receptors and analogs of isoproterenol demonstrated that serines at 204 and 207 of TMD5 are the residues that form hydrogen bonds with the catechol hydroxyls of β_2 -agonists (12). Furthermore, the evidence indicates that Ser204 interacts with the *meta* hydroxyl group of the ligand, whereas Ser207 interacts

specifically with the *para* hydroxyl group. Serine residues are found in corresponding positions in TMD5 of the other known adrenoceptors. Evidence indicates that the phenylalanine residue of TMD6 is also involved in ligand-receptor bonding with the catechol ring. Studies such as these and others that indicated the presence of specific disulfide bridges between cysteine residues of the β_2 -receptor led to the binding scheme shown in Figure 10.5.

Structural differences exist among the various adrenoceptors with regard to their primary structure, including the actual peptide sequence and length. Each of the adrenoceptors is encoded on a distinct gene, and this information was crucial to the proof that each adrenoceptor is, indeed, distinct although related. The amino acids that make up the seven transmembrane regions are highly conserved among the various adrenoceptors, but the hydrophilic portions are quite variable. The largest differences occur in the third intracellular loop connecting TMD5 and TMD6, which is the site of linkage between the receptor and its associated G protein. Compare the diagram of the β_2 -receptor in Figure 10.4 with that of the α_2 -receptor in Figure 10.6 (13).

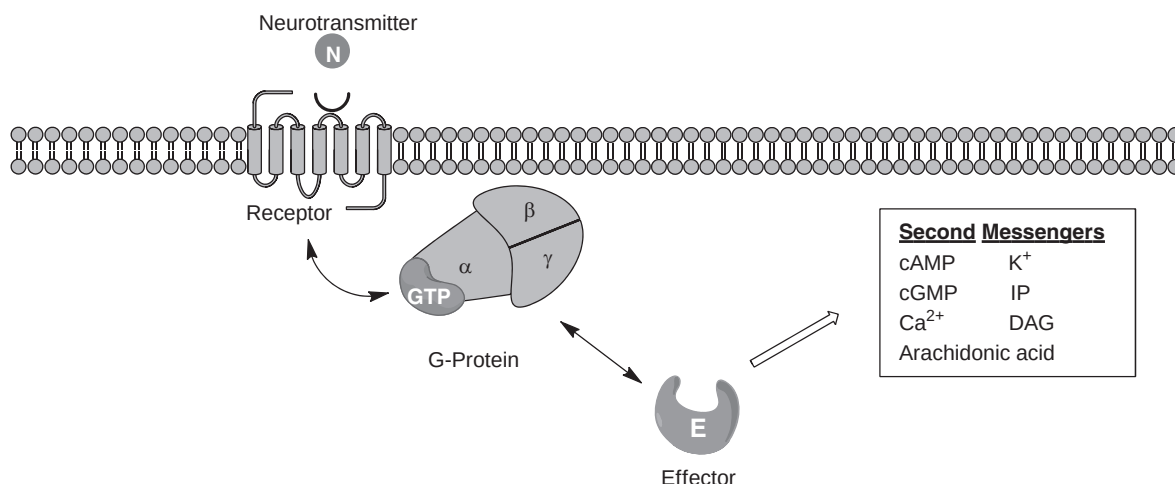
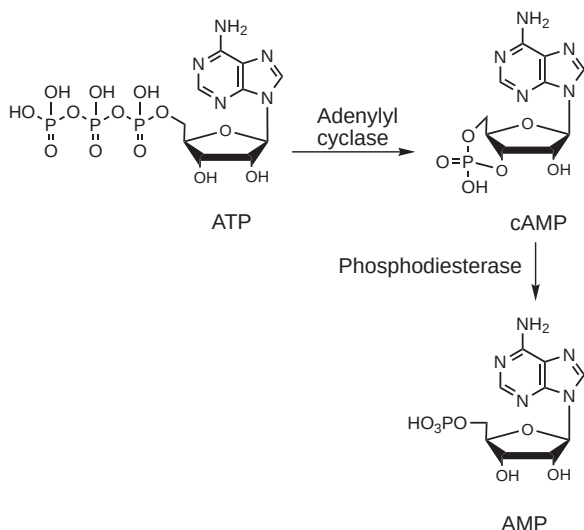


FIGURE 10.5 Proposed arrangement for the transmembrane helices of the β_2 -adrenergic receptor depicting the binding site for epinephrine as viewed from the extracellular side. (Reprinted with permission from the Annual Review of Pharmacology and Toxicology, Volume 32 © 1992 by Annual Reviews, <http://www.annualreviews.org>.)

Effector Mechanisms of Adrenergic Receptors



Each adrenoceptor is coupled through a G protein to an effector mechanism. Effector mechanisms are proteins that are able to translate the conformational change caused by activation of the receptor into a biochemical event within the cell. All the β -adrenoceptors are coupled via specific G proteins (G_s) to the activation of adenylyl cyclase (14). Thus, when the receptor is stimulated by an agonist, adenylyl cyclase is activated to catalyze the formation of cyclic adenosine monophosphate (cAMP) from ATP. Called a second messenger for the β -adrenoceptors, cAMP is known to function as a second messenger for a number of other receptor types. cAMP is considered to be a messenger because it can diffuse through the cell for at least short distances to modulate biochemical events remote from the synaptic cleft. Modulation of biochemical events by cAMP includes a

phosphorylation cascade of other proteins. cAMP is also rapidly deactivated by hydrolysis of the phosphodiester bond by the enzyme phosphodiesterase. The α_2 -receptor can use more than one effector system depending on the location of the receptor. To date, however, the best-understood effector system of the α_2 -receptor appears to be similar to that of the β -receptors, except that linkage via a G protein (G_i) leads to inhibition of adenylyl cyclase instead of activation.

α_1 -Adrenoreceptor is linked through yet another G protein to a complex series of events involving hydrolysis of polyphosphatidylinositol (15). The first event set in motion by activation of the α_1 -receptor is activation of the enzyme phospholipase C. Phospholipase C catalyzes the hydrolysis of phosphatidylinositol-4,5-bisphosphate. This hydrolysis yields two products, each of which has biologic activity as second messengers of the α_1 -receptor (see Chapter 7). These two products are 1,2-diacylglycerol (DAG) and inositol-1,4,5-trisphosphate (IP_3). The latter, IP_3 , causes the release of calcium ions from intracellular storage sites in the endoplasmic reticulum, resulting in an increase in free intracellular calcium levels. Increased free intracellular calcium is correlated with smooth muscle contraction. The other product, DAG, is thought to activate cytosolic protein kinase C, which can induce slowly developing contractions of vascular smooth muscle. The end result of a complex series of protein interactions triggered by agonist binding to the α_1 -receptor includes increased intracellular free calcium, which leads to smooth muscle contraction. When the smooth muscle innervated by α_1 -receptors is in vascular walls, stimulation leads to vascular constriction.

Receptor Localization

The generalization made in the past about synaptic locations of adrenoreceptor subtypes was that all

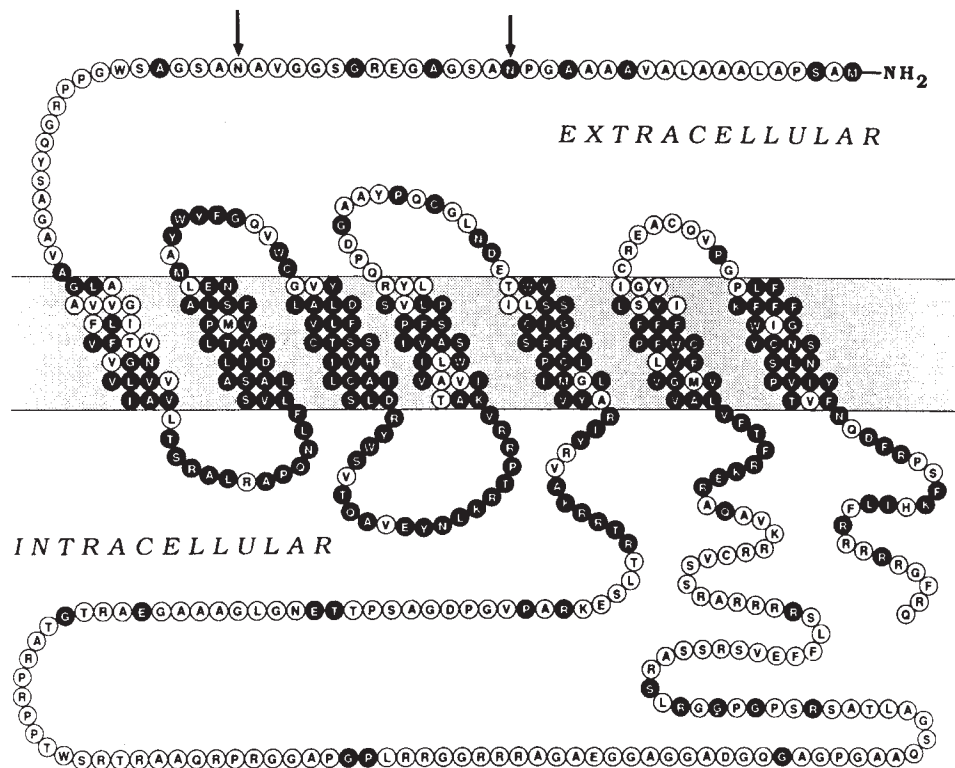


FIGURE 10.6 Human kidney α_2 -adrenergic receptor: amino acid sequence of the human kidney α_2 -receptor showing the seven transmembrane domains and the connecting intracellular and extracellular loops. Note particularly the large third intracellular loop, which is the G protein-binding site. The arrows point to the sites of glycosylation. Amino acids in black circles are those identical to the amino acids in the human platelet α_2 -receptor. (From Regan JW, Koblika TS, Yang-Feng TL, et al. Cloning and expression of a human kidney cDNA for an α_2 -adrenergic receptor subtype. *Proc Natl Acad Sci U S A* 1988;85:6301–6305, with permission.)

α_1 -, β_1 -, β_2 -, and β_3 -receptors are postsynaptic receptors that are linked to stimulation of biochemical processes in the postsynaptic cell. Presynaptic β -receptors, however, are known to occur, although their function is unclear. Traditionally, the α_2 -receptor has been viewed as a presynaptic receptor that resides on the outer membrane of the nerve terminus or presynaptic cell and reacts with released neurotransmitter. The α_2 -receptor serves as a sensor and modulator of the quantity of neurotransmitter present in the synapse at any given moment. Thus, during periods of rapid nerve firing and neurotransmitter release, the α_2 -receptor is stimulated and causes an inhibition of further release of neurotransmitter. This is a well-characterized mechanism for modulation of neurotransmission. Not all α_2 -receptors are presynaptic, but the physiologic significance of postsynaptic α_2 -receptors is less well understood (16).

Therapeutic Relevance of Adrenergic Receptor Subtypes

The clinical utility of receptor-selective drugs becomes obvious when one considers the adrenoceptor subtypes and effector responses of only a few organs and tissues innervated by the sympathetic nervous system. The major adrenoceptor subtypes are listed in Table 10.1. For

example, the predominant response to adrenergic stimulation of smooth muscle of the peripheral vasculature is constriction causing a rise in blood pressure. Because this response is mediated through α_1 -receptors, an α_1 -antagonist would be expected to cause relaxation of the blood vessels and a drop in blood pressure with clear implications for treating hypertension. The presence of α_1 -adrenoceptors in the prostate gland also leads to the use of α_1 -antagonists in treating benign prostatic hyperplasia. The principal therapeutic uses of adrenergic agonists and antagonists are shown in Table 10.2. A smaller number of β_2 -receptors on vascular smooth muscle mediate arterial dilation, particularly to skeletal muscle, and a few antihypertensives act through stimulation of these β_2 -receptors. (Adrenergic antihypertensives are discussed more thoroughly in Chapter 24.) Adrenergic stimulation of the lungs causes smooth muscle relaxation and bronchodilation mediated through β_2 -receptors. Drugs acting as β_2 -agonists are useful for alleviating respiratory distress in persons with asthma or other obstructive pulmonary diseases (see Chapter 39). Activation of β_2 -receptors in the uterus also causes muscle relaxation, and so some β_2 -agonists are used to inhibit uterine contractions in premature labor. Adrenergic stimulation of the heart causes an increase in rate and force of contraction,

TABLE 10.1 Selected Tissue Responses to Stimulation of Adrenoceptor Subtypes

Organ or Tissue	Major Receptor Type	Response
Arterioles, vascular bed	α_1, α_2	Constriction
Skeletal muscle	β_2	Dilation
Eye (radial muscle)	α_1	Contraction (papillary dilation)
Heart	β_1	Increased rate and force
Lungs	β_2	Relaxation (bronchodilation)
Liver	α_1, β_2	Increased gluconeogenesis and glycogenolysis
Fat cells	α_1, β_3	Lipolysis
Uterus (pregnant)	α_1	Contraction
	β_2	Relaxation
Intestine	α_1, β_2	Decreased motility

which is mediated primarily by β_1 -receptors. Drugs with β_1 -blocking activity slow the heart rate and decrease the force of contraction. These drugs have utility in treating hypertension, angina, and certain cardiac arrhythmias (see Chapters 21 and 24).

From the preceding discussions of the biosynthesis, storage, release, and fate of norepinephrine, one can readily conceive of a number of possible sites of drug action for adrenergic drugs. As mentioned, there are drugs that act directly on the receptors as agonists and antagonists, drugs that affect storage and release from vesicles, drugs that affect neurotransmitter biosynthesis, and drugs that affect uptake and catabolism of norepinephrine and epinephrine. These categories are discussed in turn. Most adrenergic drugs fit into well-defined classes with readily defined SARs, but a few adrenergic drugs do not permit such straightforward structural definition of their activity. We begin with a discussion of phenylethanolamine (or phenethanolamine) agonists, which do have reasonably clear SARs. Although many of these drugs directly stimulate adrenoceptors, others exhibit what is termed “indirect activity.” Indirect agonists do not directly bind to and activate adrenergic receptors; rather, they are taken up into the presynaptic neuron, where they cause the release of norepinephrine, which can diffuse into the receptor causing the observed response. Mixed-acting drugs have both a direct and an indirect component to their action, and the relative amount of direct versus indirect activity for a given drug varies considerably with its chemical structure, the tissue preparation examined, and the experimental animal species.

TABLE 10.2 Principal Therapeutic Uses of Adrenergic Agonists and Antagonists

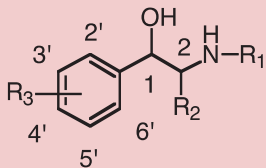
Adrenoceptor	Drug Action	Therapeutic Uses
α_1	Agonists	Shock, hypotension (to raise blood pressure)
		Nasal decongestants
	Antagonists	Antihypertensives
α_2	Agonists	Benign prostatic hyperplasia
		Antihypertensives
		Glaucoma
		Analgesia
β_1	Antagonists	Sedatives
		Antihypertensives
		Antiarrhythmics
β_2	Agonists	Bronchodilators (asthma and chronic obstructive pulmonary disorder)
		Glaucoma
β_3	Agonists	Weight loss (investigational drugs)

STRUCTURE–ACTIVITY RELATIONSHIPS OF ADRENERGIC AGONISTS

Phenylethanolamine Agonists

The structures of many clinically useful phenylethanolamine-type adrenergic agonists are summarized in Table 10.3. Agents of this type have been extensively studied over the years since discovery of the naturally occurring prototypes, epinephrine and norepinephrine, and the structural requirements and tolerances for substitutions at each of the indicated positions have been established (2). In general, a primary or secondary aliphatic amine separated by two carbons from a substituted benzene ring is minimally required for high agonist activity in this class. Because of the basic amino groups (pK_a range, ~ 8.5 to 10), all these agents are highly positively charged at physiologic pH. By definition, agents in this class have a hydroxyl group on C1 of the side chain, β to the amine, as in epinephrine and norepinephrine. This hydroxyl-substituted carbon must be in the (*R*) absolute configuration for maximal direct activity as in the natural neurotransmitter, although most drugs currently are sold as mixtures of both (*R*) and (*S*) stereoisomers at this position (racemates). Given these features in common, the nature of the other substituents determines receptor selectivity and duration of action. In the following discussions, keep in mind that saying a drug is selective for a given receptor does not mean it has no activity at other

TABLE 10.3 Phenylethanolamine Adrenergic Agonists



Drug	R ¹	R ²	R ³	Receptor Activity
Norepinephrine	H	H	3',4'-diOH	$\alpha + \beta$
Epinephrine	CH ₃	H	3',4'-diOH	$\beta \geq \alpha$
α -Methylnorepinephrine	H	CH ₃	3',4'-diOH	$\alpha + \beta$
Ethylnorepinephrine	H	CH ₂ CH ₃	3',4'-diOH	$\beta > \alpha$
Isoproterenol	CH(CH ₃) ₂	H	3',4'-diOH	General β
Isoetharine	CH(CH ₃) ₂	CH ₂ CH ₃	3',4'-diOH	selective β_2
Colterol	C(CH ₃) ₃	H	3',4'-diOH	Selective β_2
Metaproterenol	CH(CH ₃) ₂	H	3',5'-diOH	Selective β_2
Terbutaline	C(CH ₃) ₃	H	3',5'-diOH	Selective β_2
Albuterol	C(CH ₃) ₃	H	3'-CH ₂ OH, 4'-OH	Selective β_2
Phenylephrine	CH ₃	H	3'-OH	α
Metaraminol	H	CH ₃	3'-OH	α
Methoxamine	H	CH ₃	2',5'-diOCH ₃	α
Ephedrine, pseudoephedrine	CH ₃	CH ₃	H	$\alpha + \beta$
Phenylpropanolamine	H	CH ₃	H	$\alpha + \beta$
Salmeterol	-(CH ₂) ₆ -O-(CH ₂) ₄ -C ₆ H ₅	H	3'-CH ₂ OH, 4'-OH	$\beta_2 > \beta_1$
Formoterol	-CH(CH ₃)CH ₂ -C ₆ H ₄ -4-OCH ₃	H	3'-NH-COH, 4'-OH	$\beta_2 > \beta_1$
Arformoterol (<i>R,R</i> -formoterol)	-CH(CH ₃)CH ₂ -C ₆ H ₄ -4-OCH ₃	H	3'-NH-COH, 4'-OH	$\beta_2 > \beta_1$

receptors and that the clinically observed degree of selectivity is frequently dose-dependent.

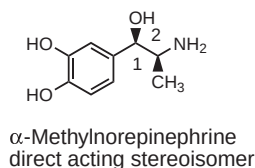
R¹, Substitution on the Amino Nitrogen

We have already seen that as R¹ is increased in size from hydrogen in norepinephrine to methyl in epinephrine to isopropyl in isoproterenol, that activity at α -receptors decreases, and that activity at β -receptors increases. These compounds were used to define α - and β -activity long before receptor proteins could be isolated and characterized. The activity at both α - and β -receptors is maximal when R¹ is methyl as in epinephrine, but α -agonist activity is dramatically decreased when R¹ is larger than methyl and is negligible when R¹ is isopropyl, as in isoproterenol, leaving only β -activity. In fact, the β -activity of isoproterenol actually is enhanced over norepinephrine and epinephrine. Presumably, the β -receptor has a large lipophilic binding pocket adjacent to the amine-binding

aspartic acid residue, which is absent in the α -receptor. As R¹ becomes larger than *t*-butyl into aryl- α -methylalkyl groups, affinity for α_1 -receptors, but not intrinsic activity, returns, which means large lipophilic groups can afford compounds with α_1 -blocking activity (e.g., labetalol, a mixed α/β -antagonist). The *N*-substituent can also provide selectivity for different β -receptors, with a *t*-butyl group affording selectivity for β_2 -receptors. For example, with all other features of the molecules being constant, colterol is a selective β_2 -agonist, whereas isoproterenol is a nonselective β -agonist. When considering use as a bronchodilator, a nonselective β -agonist, such as isoproterenol, has undesirable cardiac stimulatory properties because of its β_1 -activity, which is greatly diminished in a selective β_2 -agonist, such as albuterol. Also, an arylalkyl group (where the alkyl chain ranges from 2 to 11 carbon/oxygen atoms) can provide β -selectivity with increased lipophilicity and cell penetration for longer duration of action.

R², Substitution α to the Basic Nitrogen, Carbon-2

Small alkyl groups, methyl or ethyl, can be present on the carbon adjacent to the amino nitrogen, carbon-2 in Table 10.3. Such substitution slows metabolism by MAO but has little overall effect on duration of action in catecholamines, because they remain substrates for COMT. Resistance to MAO activity is more important in non-catechol, indirect-acting phenylethylamines. An ethyl group in this position diminishes α -activity far more than β -activity, affording compounds with β -selectivity, such as ethylnorepinephrine. Substitution on this carbon also introduces another asymmetric center into these molecules producing pairs of diastereomers, which can have significantly different biologic and chemical properties. For example, maximal direct activity in the stereoisomers of α -methylnorepinephrine resides with the *erythro* stereoisomer with the 1*R*,2*S* absolute configuration (17). The configuration of C2 has a great influence on receptor binding, because the 1*R*,2*R* diastereomer of α -methylnorepinephrine has primarily indirect activity, even though the absolute configuration of the hydroxyl-bearing C1 is the same as in norepinephrine. With respect to α -activity, this additional methyl group also makes the direct-acting 1*R*,2*S* stereoisomer of α -methylnorepinephrine more selective for α_2 -adrenoceptors than for α_1 -adrenoceptors. This has important consequences in the antihypertensive activity of α -methyldopa, which is discussed later and in Chapter 24. The same stereochemical relationships hold for metaraminol and other phenylethanolamines, in which stereochemical properties have been investigated.

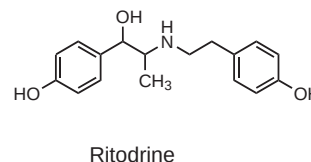


R³, Substitution on the Aromatic Ring

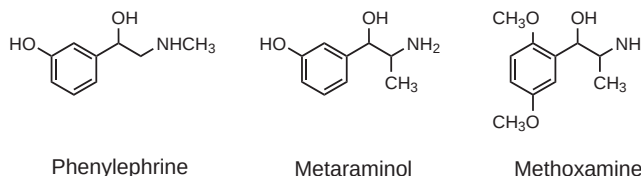
The natural 3',4'-dihydroxy-substituted benzene ring in norepinephrine provides excellent receptor activity for both α - and β -sites. Such catechol-containing compounds have poor oral activity, however, because they are rapidly metabolized by COMT. Alternative substitutions have been found that retain good activity but are more resistant to COMT metabolism. 3',5'-Dihydroxy compounds are not good substrates for COMT but provide selectivity for β_2 -receptors. Thus, because of its 3',5'-dihydroxy ring substitution pattern, metaproterenol is an orally active bronchodilator with little of the β_1 cardiac stimulatory properties possessed by 3',4'-dihydroxy ring substitution pattern for isoproterenol.

Other substitutions are possible that enhance oral activity and provide selective β_2 -activity, such as the 3'-hydroxymethyl and 4'-hydroxy substitution pattern of albuterol, the 3'-amino or 3'-formylamino, which are also resistant to COMT. At least one of the groups must be capable of forming hydrogen bonds, and if there is only one, it should be at the 4' position to retain β -activity.

For example, ritodrine has only a 4'-OH for R³ yet retains good β -activity, with the large substituent on the nitrogen making it β_2 -selective. Ritodrine has been administered to pregnant women to prevent premature labor, consistent with β_2 -adrenoceptor stimulation relaxing the uterus.

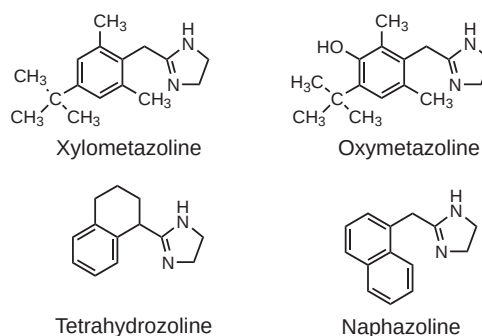
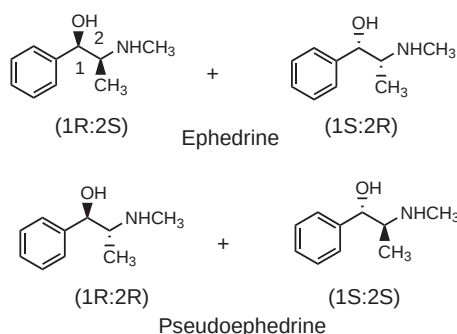


If R³ is only a 3'-OH or 3'-sulfonamide, however, activity is reduced at α_1 sites but almost eliminated at β sites, thus affording selective α_1 -agonists, such as phenylephrine and metaraminol. Further indication that α_1 sites have a wider range of substituent tolerance for agonist activity is shown by the 2',5'-dimethoxy substitution of methoxamine, which is a selective α_1 -agonist as a result of 5'-*O*-demethylation to 5'-hydroxy metabolite, which also has β -blocking activity at high concentrations. In keeping with the presence of α_1 -adrenoceptors in the peripheral vasculature, all three of these agents cause vasoconstriction.



When the phenyl ring has no phenolic substituents (i.e., R³ = H), these phenylethanolamines can have both direct and indirect activity. Direct activity (i.e., agonist) is the stimulation of an adrenoceptor by the drug itself; indirect activity is the result of displacement of norepinephrine from its storage granules or reuptake inhibition, resulting in nonselective stimulation of the adrenoceptors by the displaced norepinephrine. Because norepinephrine stimulates both α - and β -adrenoceptors, indirect activity cannot be selective. Stereochemistry of the various substituents can also have a role in determining the extent of direct/indirect activity.

For example, ephedrine and pseudoephedrine have the same substitution pattern, but substitution of both carbons 1 and 2 means four stereoisomers are possible. Racemic ($\pm$)-ephedrine is a mixture of the *erythro* enantiomers 1*R*,2*S* and 1*S*,2*R*, whereas the *threo* pair of enantiomers, 1*R*,2*R* and 1*S*,2*S*, are known as racemic pseudoephedrine (ψ -ephedrine). As discussed for α -methylnorepinephrine, (–)-ephedrine is the naturally occurring stereoisomer and has the 1*R*,2*S* absolute configuration with a mixed direct activity on both α - and β -receptors and some indirect activity. Its 1*S*,2*R*(+)-enantiomer exhibits primarily indirect activity. 1*S*,2*S*(+)-Pseudoephedrine has virtually no direct receptor activity and is mostly indirect acting.

FIGURE 10.7 Imidazoline α_1 -adrenergic agonists.

Norepinephrine and Epinephrine

Norepinephrine has limited clinical application because of the nonselective nature of its action, which causes both vasoconstriction and cardiac stimulation. In addition, it must be given intravenously, because it has no oral activity (poor oral bioavailability) as a result of its rapid metabolism by intestinal and liver COMT and MAO, 3'-O-glucuronidation/sulfation in the intestine, and low lipophilicity. Rapid metabolism by MAO and COMT limits its duration of action to only 1 or 2 minutes, even when given by infusion. The drug is used to counteract various hypotensive crises, because its α -activity raises blood pressure, and as an adjunct treatment in cardiac arrest, where its β -activity stimulates the heart.

Epinephrine is far more widely used clinically than norepinephrine, although it also lacks oral activity for the same reasons as norepinephrine. Epinephrine, similar to norepinephrine, is used to treat hypotensive crises and, because of its greater β -activity, to stimulate the heart in cardiac arrest. The β_2 -activity of epinephrine leads to its administration intravenously and in inhalers to relieve bronchoconstriction in asthma and to application in inhibiting uterine contractions. Because it has significant α -activity, epinephrine has been used in nasal decongestants. Constriction of dilated blood vessels in mucous membranes shrinks the membranes and reduces nasal congestion, although significant rebound congestion can limit its utility.

Selective α -Adrenergic Agonists

α_1 -Agonist Phenylethanalamines: Metaraminol, Methoxamine, and Phenylephrine

Metaraminol, methoxamine, and phenylephrine are selective for α_1 -receptors and have minimal cardiac stimulatory properties. Because they are not substrates for COMT, their duration of action is significantly longer than that of norepinephrine. Their α_1 -agonist activity makes them strong vasoconstrictors, however, and their primary systemic use is limited to treating hypotension during surgery or severe hypotension accompanying shock. Methoxamine is bioactivated by 5'-O-demethylation to an active 5'-phenolic metabolite. The β -blocking activity of methoxamine, which is seen at high concentrations, affords some use in treating tachycardia. Phenylephrine, which is also a selective α_1 -agonist, is used similarly to metaraminol and methoxamine for hypotension. It also has widespread use as a nonprescription nasal decongestant in both oral and

topical preparations. However, its oral bioavailability is less than 10% because of its hydrophilic properties and intestinal 3'-O-glucuronidation/sulfation. Phenylephrine preparations applied topically to the eye constrict the dilated blood vessels of bloodshot eyes and, in higher concentrations, are used to dilate the pupil during eye surgery.

2-Arylimidazoline α_1 -Agonists

In addition to phenylethanalamine derivatives, α -adrenoceptors accommodate a very diverse assortment of structures. The imidazoline derivatives in Figure 10.7 are also selective α_1 -agonists and, therefore, are called vasoconstrictors/vasopressors. They all contain a one-carbon bridge between C2 of the imidazoline ring (pK_a range, 10 to 11) and a phenyl substituent; therefore, the general skeleton of a phenylethylamine is contained within the structures. Lipophilic substitution on the phenyl ring *ortho* to the methylene bridge appears to be required for agonist activity at α_1 - and α_2 -receptors (15). Presumably, the bulky lipophilic groups attached to the phenyl ring at the *meta* or *para* positions provide selectivity for the α_1 -receptor by diminishing affinity for α_2 -receptors. These highly ionic compounds are widely used only in topical preparations as nasal decongestants and eye drops (Table 10.4). Systemically, they are potent vasoconstrictors.

α_2 -Adrenergic Agonists: 2-Aminoimidazoles and Other α_2 -Agonists

Three subtypes of α_2 -adrenoceptors, α_{2A} , α_{2B} , and α_{2C} , are recognized. Each has a role in the different clinical

TABLE 10.4 Imidazoline α_1 -Agonists in Over-the-Counter Vasoconstrictors

Drug	Nasal Decongestant	Eye Drops
Xylometazoline	Otrivin, Inspire	—
Oxymetazoline	Afrin, Duration, Neo-Synephrine, Vicks Sinex	Visine L.R. Ocu Clear
Tetrahydrozoline	—	Murine, Visine, Soothe
Naphazoline	4-Way Fast Acting, Privine	Naphcon, Clear Eyes

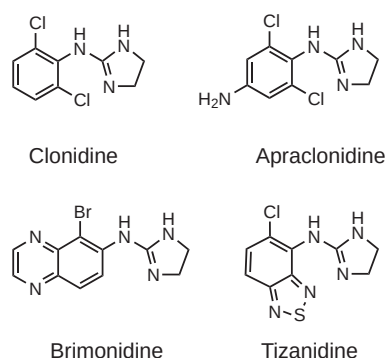
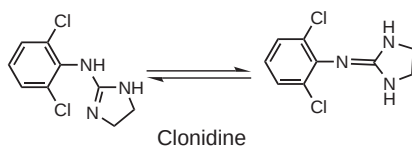


FIGURE 10.8 Imidazoline α_2 -adrenergic agonists.

applications of α_2 -agonists, which include use as antihypertensives (see Chapter 24), antiglaucoma drugs, and analgesics. The first of these drugs, clonidine, was introduced as an antihypertensive, an effect attributed to central α_{2A} -adrenoceptors in cardiovascular control areas of the brain (18).

CLONIDINE (CATAPRES) Closely related structurally to the imidazoline nasal decongestants is clonidine and other developed analogs (Fig. 10.8). Clonidine was originally synthesized as a vasoconstricting nasal decongestant but, in early clinical trials, was found to have dramatic hypotensive effects—in contrast to all expectations for a vasoconstrictor (19). Subsequent pharmacologic investigations showed that clonidine not only has some α_1 -agonist (vasoconstrictive) properties in the periphery but also is a powerful α_2 -adrenergic agonist and exhibits specific binding to nonadrenergic imidazoline binding sites in the CNS (mainly in the medulla oblongata), causing inhibition of sympathetic output (sympathoinhibition) (see Chapter 24). Because of its peripheral activity on extraneuronal vascular postsynaptic α_{2B} -receptors (18), initial doses of clonidine can produce a transient vasoconstriction and an increase in blood pressure that is soon overcome by vasodilation as clonidine penetrates the blood–brain barrier and interacts with CNS α_{2A} -receptors.



Similar to the imidazoline α_1 -agonists, clonidine has lipophilic *ortho*-dichloro substituents on the phenyl ring, but the most readily apparent difference between clonidine and the α_1 -agonists in Figure 10.7 is the replacement of the CH_2 bridge on C1 of the imidazolidine by an amine NH. This makes the imidazolidine ring part of a guanidino group, and the uncharged form of clonidine exists as a pair of tautomers as shown.

Clonidine has a $\text{p}K_a$ of 8.3 and is approximately 80% ionized at physiologic pH. Its experimental $\log D_{\text{pH } 7.4} = 1.03$. The positive charge is shared through resonance by all three nitrogens of the guanidino group. Steric crowding by the bulky *ortho*-chlorine groups does not permit a

coplanar conformation of the two rings, as illustrated in Figure 10.9.

In addition to its use as an antihypertensive, clonidine has been proven useful in a wide variety of conditions. Clonidine has sedative properties and has been used to treat attention-deficit hyperactivity disorder, nicotine and opiate withdrawal, and glaucoma, among other uses. Epidural anesthesia has been found to be enhanced by α_2 -agonists (20), and clonidine is available in an injectable form for epidural administration.

TIZANIDINE (ZANAFLEX) Tizanidine (Fig. 10.8) is a centrally active muscle relaxant analog of clonidine that is approved for use in reducing spasticity associated with cerebral or spinal cord injury. Tizanidine has a $\text{p}K_a$ of 7.4 and is approximately 50% ionized at physiologic pH. Its $\log D_{\text{pH } 7.4} = 1.8$. Its mechanism of action for reducing spasticity suggests presynaptic inhibition of motor neurons at the α_2 -adrenergic receptor sites, reducing the release of excitatory amino acids and inhibiting facilitatory ceruleo-spinal pathways, thus resulting in a reduction in spasticity. Tizanidine only has a small fraction of the antihypertensive action of clonidine, presumably because of action at a selective subgroup of α_{2C} -adrenoceptors, which appear to be responsible for the analgesic and antispasmodic activity of imidazoline α_2 -agonists (20).

APRACLONIDINE (IOPIDINE) AND BRIMONIDINE (ALPHAGAN) The other imidazoline α_2 -agonists in Figure 10.8 that are clinically used for treatment of glaucoma are apraclonidine ($\text{p}K_a = 9.22$, $\log D_{\text{pH } 7.4} = 0.01$) and brimonidine ($\text{p}K_a = 7.4$, $\log D_{\text{pH } 7.4} = 0.49$). Stimulation of α_2 -receptors in the eye reduces production of aqueous humor and enhances outflow of aqueous humor, thus reducing intraocular pressure, and also has a neuroprotective effect apparently through α_{2A} -receptors located in the retina (21,22). Animal and human studies suggest that apraclonidine's primary mechanism of action can be related to a reduction of aqueous humor formation, whereas brimonidine lowers intraocular pressure by reducing aqueous humor production and increasing uveoscleral outflow. Brimonidine is approximately 1,000-fold more selective for α_2 -receptors than are clonidine or apraclonidine and is a first-line agent for treating glaucoma. It exhibits minimal effect on blood pressure and heart rate. Although both are applied topically to the eye, measurable quantities of these drugs are detectable in plasma, so caution must be employed when cardiovascular agents are also being coadministered to the patient. Plasma brimonidine levels peaked within 1 to 4 hours and declined with a systemic half-life of approximately 3 hours. Brimonidine has been reported to enter the brain and can potentially cause fatigue and/or drowsiness in some patients. Brimonidine is primarily metabolized by aldehyde oxidase.

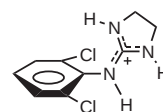
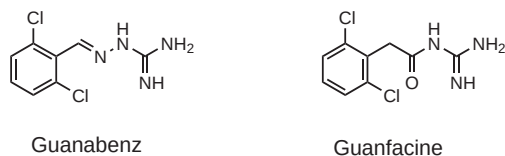


FIGURE 10.9 Protonated clonidine.

GUANFACINE (TENEX) AND GUANABENZ (WYTENSIN) After the discovery of clonidine, extensive research into the SAR of central α_2 -agonists showed that the imidazoline ring was not necessary for activity in this class but that the phenyl ring required at least one *ortho* chlorine or methyl group. Two clinically useful antihypertensive agents resulting from this effort are guanfacine and guanabenz (see Chapter 24). These are ring-opened analogs of clonidine, and their mechanism of action is the same as that of clonidine.



METHYLDOPA (ALDOMET) Although structurally unrelated to the aminoimidazolines or the guanidines, the pro-drug L- α -methyldopa (methyldopa) is an α_2 -agonist acting in the CNS via its active metabolite, α -methylnorepinephrine (Fig. 10.10). Methyldopa is transported across the blood–brain barrier, where it is decarboxylated by aromatic L-amino acid decarboxylase in the brain to α -methyldopamine, which is then stereospecifically hydroxylated to 1*R*,2*S*- α -methylnorepinephrine. This stereoisomer is a selective α_2 -agonist and acts as an antihypertensive agent much like clonidine to inhibit sympathetic neural output from the CNS, thus lowering blood pressure. α -Methylnorepinephrine and α -methyldopamine do not cross the blood–brain barrier because of their hydrophilicity. Originally synthesized as a norepinephrine biosynthesis inhibitor, methyldopa was thought to act through a combination of inhibition of norepinephrine biosynthesis through DOPA decarboxylase inhibition and metabolic decarboxylation to generate α -methylnorepinephrine. The latter was thought to replace norepinephrine in the nerve terminal and, when released, to have less intrinsic activity than the natural neurotransmitter. This latter mechanism is an example of the concept of a false neurotransmitter. (The antihypertensive properties for methyldopa are further described in Chapter 24.)

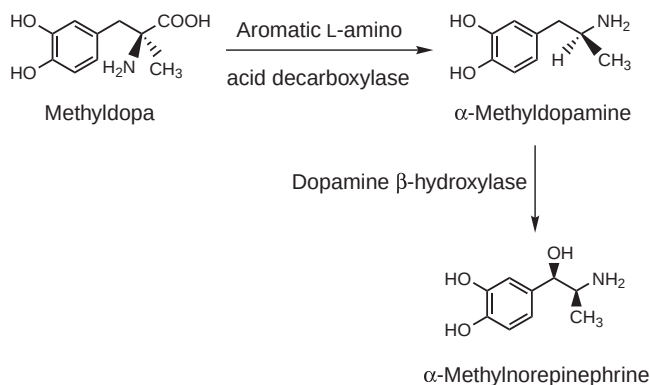


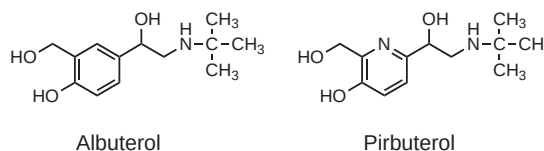
FIGURE 10.10 Methyldopa bioactivation.

β -Adrenergic Agonists

β_2 -Agonist Phenylethanolamines

Most of the β_2 -selective adrenergic agonists listed in Table 10.3 are used primarily as bronchodilators in asthma and other constrictive pulmonary conditions (see Chapter 39 for pharmacokinetic details about these drugs). Isoproterenol is a nonselective β -agonist ($\beta_2/\beta_1 = 1$), and the cardiac stimulation caused by its β_1 -activity and its lack of oral activity have led to its diminished use in favor of more selective β_2 -agonists.

SHORT-ACTING β_2 -ADRENERGIC AGONISTS

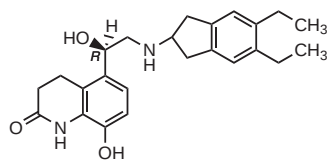
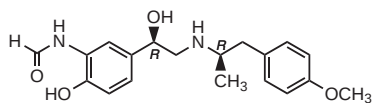
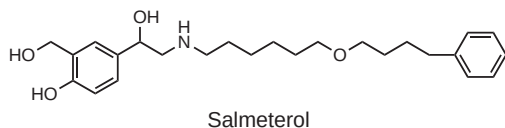


Albuterol (Ventolin, Proventil), Pirbuterol, (Maxair) Terbutaline (Bricanyl, Brethine) The noncatechol, selective β_2 -agonists, such as racemic albuterol ($\beta_2/\beta_1 = 60$), metaproterenol, and terbutaline, are available in oral dosage forms as well as in inhalers. All have similar activities and durations of action. Pirbuterol is an interesting analog of albuterol in which the benzene ring has been replaced by a pyridine ring, altering its pharmacokinetic properties. Similar to albuterol, pirbuterol is a selective β_2 -agonist currently available only for administration by inhalation.

Studies with the *R/S* enantiomers of albuterol have shown that the *S*-enantiomer can be proinflammatory, exacerbating airway reactivity to a variety of spasmogens and, thereby, enhancing bronchial muscle contraction, thus opposing the bronchodilation effects of the *R*-enantiomer levalbuterol. Moreover, racemic albuterol exhibits enantioselective presystemic metabolism. Levalbuterol undergoes more rapid metabolism (sulfation) than the *S*(+)-isomer, resulting in a lower oral bioavailability and rapid elimination. Because of its slower metabolism, *S*-albuterol has a higher and more prolonged tissue concentration than levalbuterol, increasing airway reactivity. These adverse effects of *S*-albuterol are completely avoided by using the *R*-enantiomer, levalbuterol (Xopenex). Thus, the removal of *S*-albuterol from racemic albuterol increases the clinical potency of levalbuterol, such that bronchodilator efficacy is achieved at one-fourth the dose of racemic albuterol along with a marked reduction in side effects.

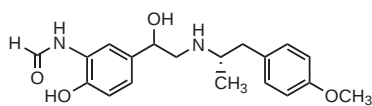
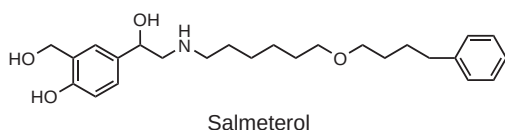
LONG-ACTING β_2 -ADRENERGIC AGONISTS The mechanisms behind the long duration of the bronchodilating effect of the β_2 -adrenoceptor agonists formoterol, arformoterol, salmeterol, and indicaterol are only partially understood. Their long duration is attributed to their higher lipophilicity and greater receptor affinity compared to those of the short-acting agonists at the β_2 -adrenoceptor. Lipophilicity determines the amount of drug entering into the cell membrane of the β_2 -adrenoceptor and how

much β_2 -adrenoceptor agonist must remain at the receptor for maximal sustained activity.



Salmeterol (Serevent) A β_2 -agonist with a slow onset and extended duration of action is salmeterol. Salmeterol has the same phenyl ring substitution R^3 as albuterol but also an unusually long and lipophilic group R^1 on the nitrogen. The octanol–water partition coefficient, $\log P$, for salmeterol is 3.88, compared with 0.66 for albuterol. Salmeterol is approximately 50-fold more selective than albuterol for the β_2 -receptor. Substantial evidence indicates that its long duration of action results from a specific binding interaction (“anchoring”) of the phenyl group at the end of the extended lipophilic side chain with a specific region of the β_2 -receptor, affording salmeterol a unique binding mechanism (Fig. 10.11).

Salmeterol is usually prescribed for severe persistent asthma following previous treatment with a short-acting β -agonist, such as albuterol. The noticeable differences between salmeterol and albuterol are the onset of action and their duration of action (see Chapter 39 for pharmacokinetic details). When used regularly every day as prescribed, inhaled salmeterol decreases the number and severity of asthma attacks. It is not used, however, for relieving an acute asthma attack that has already started.



Formoterol (Foradil) and Arformoterol (Brovana) Formoterol has 3'-formylamino and 4'-hydroxy ring R^3 substitution pattern but also an alkoxyphenylethyl lipophilic group R^1 on the nitrogen, similar to ritodrine. Although it is less lipophilic ($\log P = 1.6$) than salmeterol, it has a 12-hour duration of action similar to that for salmeterol. It is administered as an inhaled dry powder, because it is unstable to heat and moisture. Formoterol is a mixture of *R,R*(-)- and *S,S*(+)-stereoisomers, with the *R,R*-isomer having approximately 1,000-fold more affinity for the β_2 -receptor than the *S,S*-isomer. Arformoterol is the single *R,R* active stereoisomer. Formoterol and arformoterol have a faster onset of action than salmeterol as a result of lower lipophilicity.

Indacaterol (Arcapta, Neohaler) Indacaterol is the newest bronchodilator, which is U.S. Food and Drug Administration approved for use in chronic obstructive pulmonary disease. Although the calculated $\log P$ for indacaterol is the same as salmeterol ($\log P = 3.88$), the pK_a of the phenolic OH of indacaterol ($pK_a = 6.7$) is significantly lower than that of salmeterol ($pK_a = 10.2$), leading to a significant amount of indacaterol existing in solution at physiologic pH (7.4) as the zwitterion (protonated amine/phenoxide), whereas salmeterol is predominately the protonated amine cation (23). This difference in ionization has been shown to affect interactions with lipid membranes (24). As a result, indacaterol has a much longer duration of action of 24 hours, permitting once-daily dosing (25).

β_1 -Adrenergic Agonists

DOPAMINE Although not strictly an adrenergic drug, dopamine is a catecholamine with properties related to the cardiovascular activities of the other agents in this chapter. Dopamine acts on specific dopamine receptors

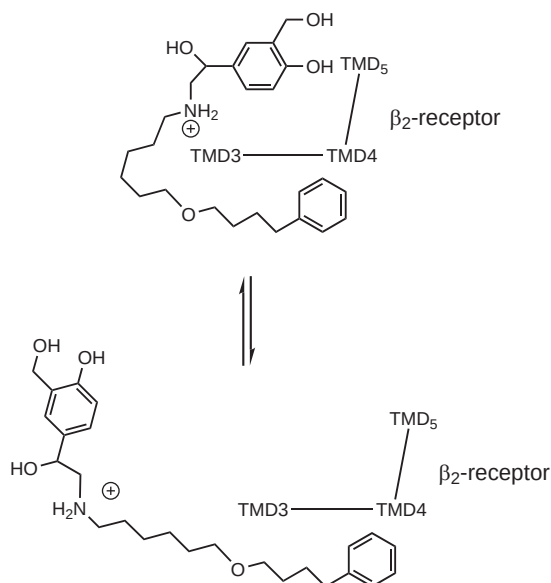


FIGURE 10.11 Mechanism of action for salmeterol.

to dilate renal vessels, increasing renal blood flow. Dopamine also stimulates cardiac β_1 -receptors through both direct and indirect mechanisms. It is used to correct hemodynamic imbalances induced by conditions such as shock, myocardial infarction, trauma, or congestive heart failure. As a catechol and primary amine, dopamine is rapidly metabolized by COMT and MAO and, similar to dobutamine, has a short duration of action with no oral activity. It is administered as an intravenous infusion.

DOBUTAMINE (DOBUTREX) Not all the adrenergic agonists with direct activity have an aliphatic β -hydroxyl group such as the agents discussed so far. One of these is the catechol dobutamine. Dobutamine is a dopamine analog with a bulky arylalkyl group on the nitrogen and one chiral (asymmetric) center. Racemic ($\pm$)-dobutamine has direct activity on both α_1 - and both β -receptors, but because of some unusual properties of its two enantiomers, the overall pharmacologic response looks similar to that of a selective β_1 -agonist. The *S*(-)-enantiomer of dobutamine exhibits β_1 -agonist activity and is also an α_1 -agonist and vasopressor. The *R*(+)-isomer is an α_1 -antagonist and general $\beta_1\beta_2$ -agonist; thus, when the racemate is used clinically, the α -effects of the enantiomers cancel each other, leaving primarily the β_1 -effects (26,27). The stereochemistry of the methyl substituent does not affect the ability of the drug to bind to the α_1 -receptor but does affect the ability of the molecule to activate the receptor. That is, the stereochemistry of the methyl group affects intrinsic activity but not affinity. Because both stereoisomers are β_1 -agonists, with the (-)-isomer having approximately one-tenth the potency of the (+)-isomer, the net effect is β_1 stimulation. Dobutamine is used as a cardiac stimulant after surgery or congestive heart failure. As a catechol, dobutamine is readily metabolized by COMT and has a short duration of action with no oral activity.

Mixed-Acting Sympathomimetics

PHENYLPROPANOLAMINES

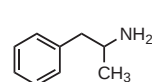
(-)-Ephedrine As previously discussed, ephedrine is a natural product isolated from several species of ephedra plants, which have been used for centuries in folk medicines in a variety of cultures worldwide (28). Its occurrence in ephedra is approximately 80% to 90% of the alkaloid content. Other substances also found include (+)-pseudoephedrine (10% to 15%) and *N*-methylephedrine (2% to 5%). Pure ephedrine was first isolated and crystallized in 1887 from a Chinese herbal medicine called Ma Huang. Its sympathomimetic activity was not recognized until 1917, and the pure drug was used clinically even before epinephrine and norepinephrine were isolated and characterized. Ephedrine does not have any phenolic substituents on the phenyl ring, giving it a mixed-acting response and good oral activity, because it is not a substrate for COMT. Lacking hydrogen-bonding phenolic substituents, ephedrine is less polar than the other compounds discussed thus far and crosses the blood-brain barrier far better than the catechols do.

Because of its ability to penetrate the CNS, ephedrine has been used as a stimulant and exhibits side effects related to its action in the brain. Ephedrine is widely used for many of the same indications as epinephrine, including use as a bronchodilator, vasopressor, cardiac stimulant, and nasal decongestant.

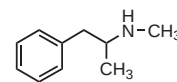
(+)-Pseudoephedrine Pseudoephedrine, as previously discussed, is the threo diastereomer of ephedrine, with virtually no direct activity and fewer CNS side effects than ephedrine. (+)-Pseudoephedrine is widely used as a nasal decongestant but is also used in the illicit manufacture of the widely abused drug methamphetamine. For this reason, sale of pseudoephedrine over-the-counter products is becoming increasingly restricted.

(+)-Phenylpropanolamine Phenylpropanolamine (Table 10.3) is the *N*-desmethyl analog of ephedrine, also known as stereoisomers of norephedrine and norpseudoephedrine, which is as a stimulant, nasal decongestant, and anorectic agent. Lacking the *N*-methyl group, however, phenylpropanolamine has none of the β_2 -agonist activity of ephedrine, is slightly less lipophilic, and therefore, does not enter the CNS as well as ephedrine. Phenylpropanolamine, similar to ephedrine, is a mixture of *erythro* enantiomers with mixed direct and mostly indirect activity. In the United States, phenylpropanolamine is no longer sold without a prescription due to a proposed increased risk of stroke in younger women. In Europe, however, it is still available both by prescription and over the counter. In Canada, it was withdrawn from the market, and in the United Kingdom, it is available in many “all-in-one” cough and cold medications, which also contain acetaminophen (paracetamol) or another analgesic and caffeine. It is still available for veterinary use in dogs as a treatment for urinary incontinence.

PHENYLISOPROPYLAMINES



Amphetamine



Methamphetamine

Amphetamine and Methamphetamine Other methyl-substituted phenylethylamines (phenylisopropylamines), such as *S*(+)-amphetamine and *S*(+)-methamphetamine, which lack both ring substituents and a side-chain hydroxyl, are sufficiently lipophilic to cross the blood-brain barrier readily and cause dramatic CNS stimulation, which gives them serious abuse potential. The clinical utility of *S*(+)-amphetamine and its derivatives is entirely based on CNS stimulant and central appetite suppressant effects. The only amphetamine product available that is used for the treatment of attention-deficit disorder combines the sulfate, saccharate, and aspartate salts of dextroamphetamine and amphetamine (total amphetamine base, 3.1 mg; Adderall). (These agents

and the methoxylated amphetamines are discussed in Chapter 19.)

STRUCTURE–ACTIVITY RELATIONSHIPS OF ADRENERGIC ANTAGONISTS

Nonselective α -Antagonists

Phenoxybenzamine

Because α -agonists cause vasoconstriction and raise blood pressure, one would expect α -antagonists to be therapeutically used as antihypertensive agents. An old but powerful drug in this class is phenoxybenzamine (dibenzylamine), a β -haloalkylamine that alkylates α -receptors. β -Haloalkylamines are present in nitrogen mustard anticancer agents and are highly reactive alkylating agents. The acid salt of phenoxybenzamine is stable, but at physiologic pH, equilibrium exists between the protonated drug and free base. The unshared electrons of the unprotonated amino group are nucleophilic and displace the β -chlorine atom in an intramolecular reaction to form a highly reactive aziridinium ion (Fig. 10.12). If this occurs in the vicinity of an α -receptor, a nucleophile group X on the receptor can open the aziridinium ion in a nucleophilic reaction to form a covalent bond between the receptor and the drug. The substituents attached to the haloalkylamine provide selectivity for binding to α -adrenoceptors so that the nucleophile generally is part of the target receptor. The nucleophile X is presumably part of an amino acid side chain, such as a cysteine thiol, serine hydroxyl, or lysine amino group, but the specific site of covalent attachment to the α -receptor has not been determined. Because the reaction in which phenoxybenzamine forms covalent bonds with the receptors is irreversible, new receptors must be synthesized before the effects can be overcome. Therefore, the α -blockade is long-lasting.

Unfortunately, other biomolecules besides the target α -receptor are also alkylated. Because of its receptor nonselectivity and toxicity, the use of phenoxybenzamine largely is limited to alleviating the sympathetic effects of pheochromocytoma. This tumor of chromaffin cells of the adrenal medulla produces large amounts of epinephrine and norepinephrine, which are released into the bloodstream, producing hypertension and generalized sympathetic stimulation.

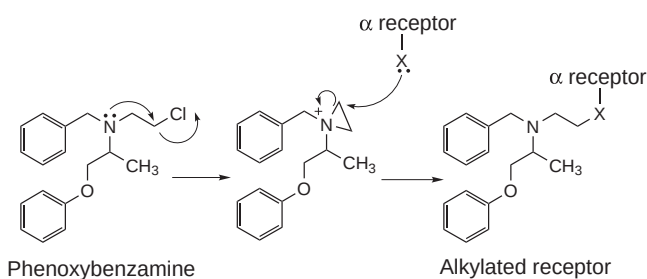
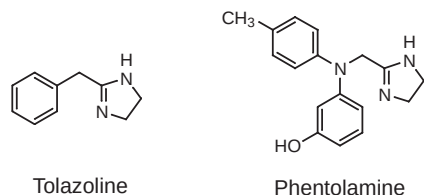


FIGURE 10.12 Phenoxybenzamine alkylation of α -adrenoceptors. X is a nucleophile, such as S, N, or O.

Tolazoline and Phentolamine



Tolazoline and phentolamine are two imidazoline nonselective α -antagonists that also have antihypertensive activity, although they have been replaced in general clinical use by far better agents. Tolazoline has clear structural similarities to the imidazoline α_1 -agonists, such as naphazoline and xylometazoline (Fig. 10.7), but does not have the lipophilic substituents required for agonist activity. The resemblance of phentolamine is not as readily apparent, but extensive molecular modeling studies have provided a topologic scheme for α_1 -antagonist SAR (19). This pattern, however, cannot be readily visualized without computer graphics and is beyond the scope of this chapter.

Both phentolamine and tolazoline are potent but rather nonspecific α -antagonists. Both drugs stimulate gastrointestinal smooth muscle, an action blocked by atropine, which would indicate cholinergic activity, and they both stimulate gastric secretion, possibly through release of histamine. Because of these and other side effects, the clinical applications of tolazoline and phentolamine are also limited to treating the symptoms of pheochromocytoma and hypertensive crises induced by certain illicit drugs.

Selective α_1 -Antagonists

Prazosin, Doxazosin, Terazosin, Tamsulosin, Alfuzosin, and Silodosin

Prazosin, the first known selective α_1 -blocker, was discovered in the late 1960s (29) and is now one of a small group of selective α_1 -antagonists that includes three other quinoxaline antihypertensives terazosin, doxazosin, and alfuzosin and the nonquinazoline benzensulfonamides tamsulosin and silodosin (Fig. 10.13). Prazosin, doxazosin, and terazosin contain a 4-amino-6,7-dimethoxyquinazoline ring system attached to a piperazine ring, whereas alfuzosin has a rotatable propylenediamine group (an open piperazine ring). The other structural differences are the heterocyclic acyl groups attached to the second nitrogen of the piperazine or the propyl chain. The differences in these groups afford dramatic differences in some of the pharmacokinetic properties of these agents (Table 10.5). For example, reduction of the furan ring for prazosin to the tetrahydrofuran ring of terazosin increases its duration of action by altering its rate of metabolism. Some of the important clinical parameters of the quinazolines are shown in Table 10.5. The long half-lives and durations of action for terazosin, doxazosin, tamsulosin, and silodosin permit once-a-day dosing and generally lead to increased patient compliance.

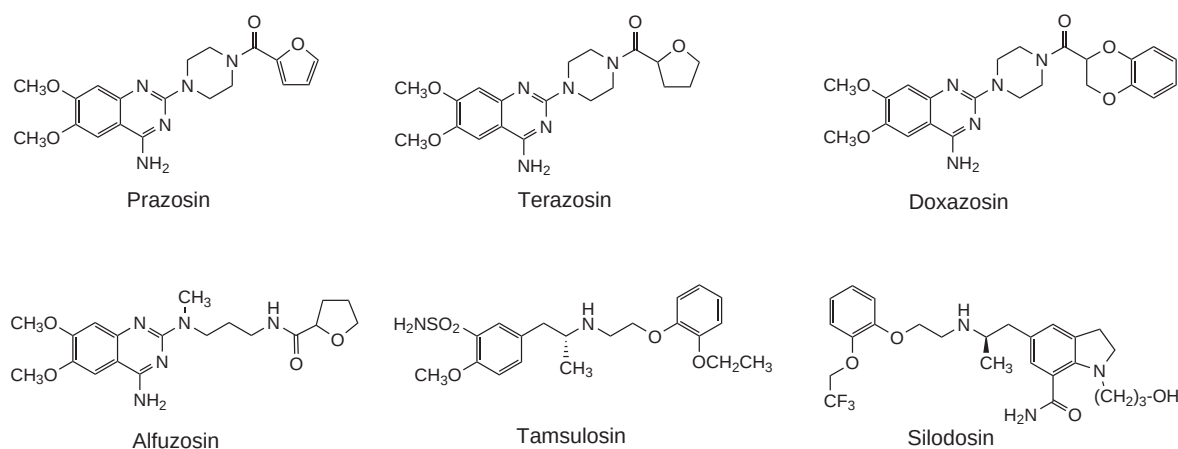


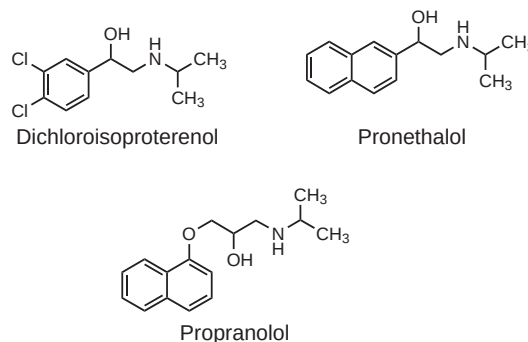
FIGURE 10.13 Selective α_1 -adrenergic antagonists.

Prazosin is an antihypertensive agent, as are terazosin and doxazosin. The latter two were subsequently discovered to block α_1 -receptors in the prostate gland and alleviate the symptoms of benign prostatic hyperplasia (BPH) (see Chapter 40). The subsequently developed tamsulosin and alfuzosin are more selective for the subtype of α_1 -adrenoceptor found in the prostate gland, α_{1A} , over those found in vascular tissue. The most recently developed drug, silodosin, is the most selective of the group for α_{1A} -receptors found in the prostate (30). Thus, tamsulosin, alfuzosin, and silodosin are first-line drugs for the treatment of BPH and have no utility in treating hypertension. They have fewer cardiovascular side effects than terazosin and doxazosin in treating BPH. (Details on the use and pharmacokinetics of α_1 -antagonists in treating hypertension and BPH are provided in Chapters 24 and 40, respectively.)

β -Adrenergic Antagonists

In the 1950s, dichloroisoproterenol, a derivative of isoproterenol in which the catechol hydroxyls had been replaced by chlorines, was discovered to be a β -antagonist that blocked the effects of sympathomimetic amines

on bronchodilation, uterine relaxation, and heart stimulation (31). Although dichloroisoproterenol had no clinical utility, replacement of the 3,4-dichloro substituents with a carbon bridge to form a naphthylethanolamine derivative did afford a clinical candidate, pronethalol, which was introduced in 1962 only to be withdrawn in 1963 because of tumor induction in animal tests.



Structure–Activity Relationships

Shortly thereafter, a major innovation in drug development for the β -adrenergic antagonists was introduced when it was discovered that an oxymethylene bridge,

TABLE 10.5 Selected Clinical Parameters of α_1 -Adrenergic Antagonists

Drug	Trade Name	cLogP/logD _{pH 7.0}	Half-Life (hours)	Duration of Action (hours)	Bioavailability (%)
Prazosin	Minipres	−1.1/−1.3	2–3	4–6	45–65
Terazosin	Hytrin	−1.0/−1.0	12	>18	90
Doxazosin	Cardura	0.7/0.5	22	18–36	65
Tamsulosin	Flomax	2.2/0.5	14–15	>24	<50 with food 50–90 fasting
Alfuzosin	Uroxatral	−1.0/−2.7	3–5	>48	65
Silodosin	Rapaflo	2.33/0.48	13	>24	18–43 with food

^aChemical Abstracts, American Chemical Society, calculated using Advanced Chemistry Development (ACD/Labs) Software V8.19 for Solaris (1994–2011 ACD/Labs).

OCH_2 , could be inserted into the aryethanolamine structure of pronethalol to afford propranolol, an aryloxypropanolamine and the first clinically successful β -blocker. Note that along with the introduction of the oxymethylene bridge, the side chain has been moved from C2 of the naphthyl group to the C1 position. In general, the aryloxypropanolamines are more potent β -blockers than the corresponding aryethanolamines, and most of the β -blockers currently being used clinically are aryloxypropanolamines. β -Blockers have found wide use in treating hypertension (Chapter 24) and certain types of glaucoma.

Initially, it might appear that lengthening the side chain would prevent appropriate binding of the required functional groups to the same receptor site. Molecular models, however, show that the side chains of aryloxypropanolamines can adopt a conformation that places the hydroxyl and amine groups into approximately the same position in space (Fig. 10.14). Although the simple two-dimensional drawing in Figure 10.14 exaggerates the true degree of overlap, elaborate molecular modeling studies confirm that the aryloxypropanolamine side chain can adopt a low-energy conformation that permits close overlap with the aryethanolamine side chain (32).

Stereochemistry of the β -Adrenergic Antagonists

A factor that sometimes causes confusion when comparing the structures of aryethanolamines with aryloxypropanolamines is the stereochemical nomenclature of the side-chain carbon bearing the hydroxyl group. For maximum effectiveness in receptor binding, the hydroxy group must occupy the same region in space as it does for the phenylethanolamine agonists in the *R* absolute configuration. Because of the insertion of an oxygen atom in the side chain of the aryloxypropanolamines, the Cahn-Ingold-Prelog priority of the substituents around the asymmetric carbon changes, and the isomer with the required special arrangement now has the *S* absolute configuration. This is

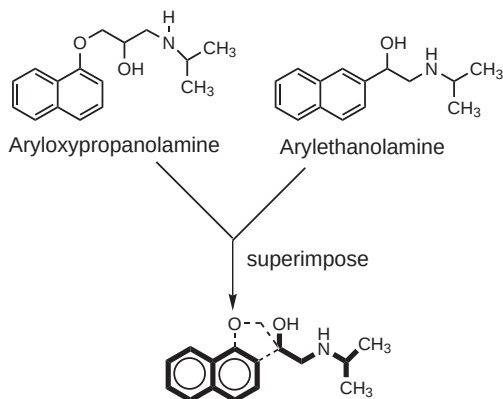


FIGURE 10.14 Overlap of aryloxypropanolamines and aryethanolamines. The structures of prototype β -antagonists propranolol and pronethalol can be superimposed so the critical functional groups occupy the same approximate regions in space, as indicated by the bold lines in the superimposed drawings. The dotted lines are those parts that do not overlap but are not necessary to receptor binding.

an effect of the nomenclature rules; the groups still have the same spatial arrangements (Fig. 10.15).

PROPRANOLOL AND OTHER β -BLOCKERS Propranolol was initially introduced for the treatment of angina pectoris and later underwent trials as an antiarrhythmic. During clinical trials as an antianginal, propranolol was discovered to have antihypertensive properties, and it has been widely employed for that purpose for decades (33). Propranolol rapidly became widely used for a variety of cardiac arrhythmias as well. In addition, because of its high lipophilicity and ability to penetrate the CNS, propranolol has found use in treating disorders of the CNS, such as anxiety.

At approximately this same time, a new series of 4-substituted phenyloxypropanolamines emerged that selectively inhibited sympathetic cardiac stimulation. These observations led to the recognition that not all β -receptors were the same, which in turn led to the introduction of β_1 and β_2 nomenclature to differentiate cardiac β_1 -receptors from others. Development of β -blockers proceeded rapidly, and today, a large number of additional drugs, both nonselective β -antagonists and selective β_1 -antagonists, are available on the world market. (Clinical use of β -blockers as antihypertensives is covered in Chapter 24.) Of those antagonists that are selective for the cardiac β_1 -receptor, most have also some β_2 -antagonist properties at the higher levels of therapeutic dosing (dose dependent). With the exception of sotalol, all the drugs shown in Figures 10.16 and 10.17 are aryloxypropanolamines. Metipranolol (Fig. 10.16) is an exception to the general rule that 4-substituted aryloxypropanolamines are selective β_1 -blockers.

Other than β_1 -selectivity of 4-substituted aryloxypropanolamines, little obvious structural pattern relates β -blockers to specific clinical applications, with the exception of esmolol. Esmolol is the methyl ester of a carboxylic acid, which makes it susceptible to hydrolysis by serum esterases. The acid metabolite generated by hydrolysis is essentially inactive and readily excreted as its zwitterion. For this reason, esmolol has a half-life of only approximately 8 minutes and is used to control supraventricular tachycardia during surgery when a short-acting β_1 -adrenergic antagonist is desirable. Nebivolol (Fig. 10.17) is also unique in this group of selective β_1 -blockers in that it has

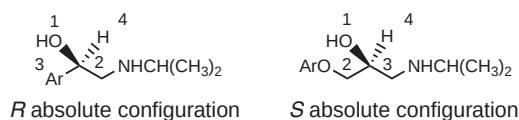


FIGURE 10.15 Stereochemical nomenclature for aryethanolamines versus aryloxypropanolamines. The relative positions in space of the four functional groups are the same in the two structures; however, one is designated *R* and the other *S*. This is because the introduction of an oxygen atom into the side chain of the aryloxypropanolamine changes the priority of two of the groups used in the nomenclature assignment.

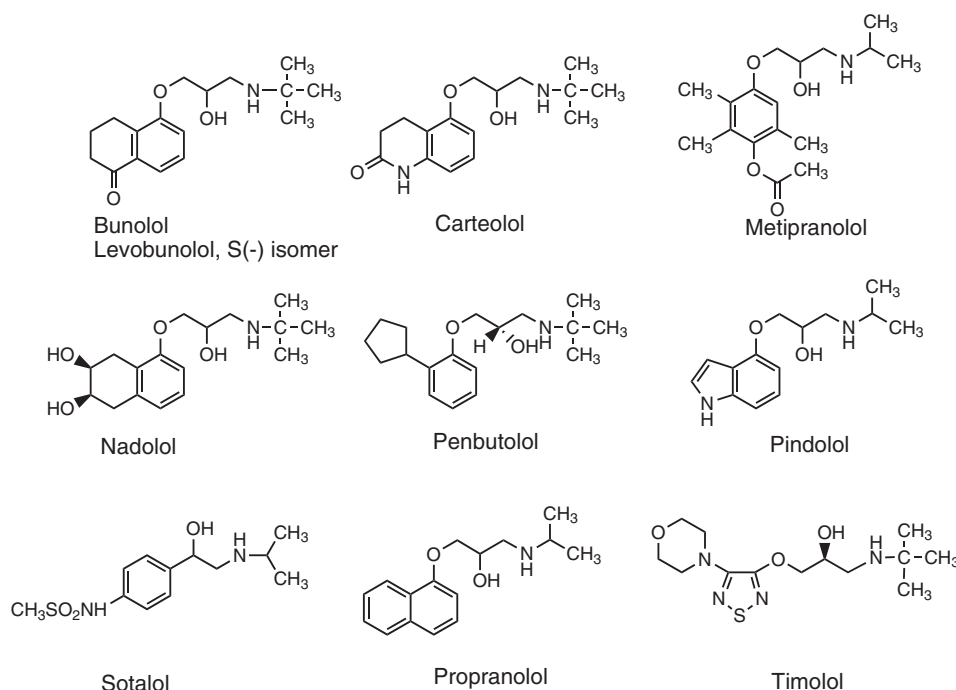


FIGURE 10.16 Nonselective β -adrenergic antagonists.

a nitric oxide potentiating effect (34). This gives the drug vasodilating effects in addition to its β -effects on the heart.

Another physicochemical parameter with some clinical correlation is the relative lipophilicity of different agents. Propranolol is by far the most lipophilic of the available β -blockers, and it enters the CNS far better than the less lipophilic agents, such as atenolol or nadolol. Lipophilicity as measured by octanol–water partitioning also correlates with the primary site of clearance, as seen

in Table 10.6. The more lipophilic drugs are primarily cleared by the liver, whereas the more hydrophilic agents are cleared by the kidney. This could influence the choice of agents in cases of renal failure or liver disease. Several of the β -blockers must be dose adjusted in patients with impaired renal function, as indicated in Table 10.6.

β -Blockers have also found extensive use in treating glaucoma, although the mechanism of action in treating glaucoma is more difficult to explain. The

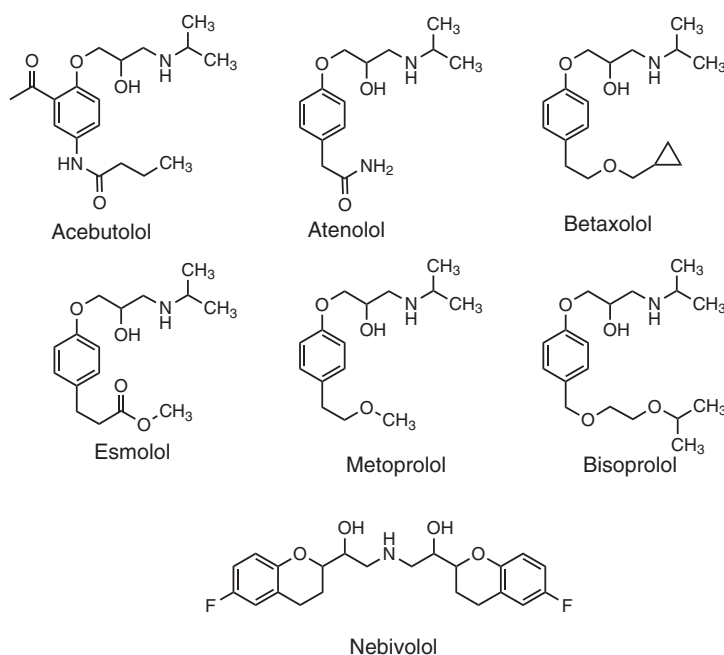


FIGURE 10.17 Selective β_1 -adrenergic antagonists.

TABLE 10.6 Pharmacologic/Pharmacokinetic Properties of β -Adrenergic Blocking Agents

Drug	Adrenergic Receptor Blocking Activity	Membrane Stabilizing Activity	Intrinsic Sympathomimetic Activity	$\text{cLogP}/\log D_{\text{pH } 7.4}^a$	Extent of Absorption (%)	Absolute Oral Bioavailability (%)	Half-Life (h)	Protein Binding (%)	Metabolism/Excretion
Acebutolol (Sectral)	β_1^b	+	+	2.67/0.52	90	20–60	3–4	26	Hepatic; renal excretion, 30%–40%; nonrenal excretion, 50%–60% (bile)
Atenolol (Tenormin)	β_1^b	0	0	0.1/–2.0	50	50–60	6–9	16–16	~50% excreted unchanged in feces
Betaxolol (Kerlone, Betoptic)	β_1^b	+	0	2.7/0.6	~100	89	14–22	~50	Hepatic; >80% recovered in urine, 15% unchanged
Bisoprolol (Zebeta)	β_1^b	0	0	2.2/0.1	≥90	80	9–12	~30	~50% excreted unchanged in urine, remainder as inactive metabolites; <2% excreted in feces
Esmolol (Brevibloc)	β_1^b	0	0	1.9/–0.22	na ^f	na ^f	0.15	55	Rapid metabolism by esterases in cytosol of red blood cells
Metoprolol (Lopressor) Metoprolol, LA	β_1^b	0 ^c	0	1.8/–0.34	95	40–50 77	3–7	12	Hepatic; renal excretion, <5% unchanged
Carteolol (Cartrol, Ocupress)	$\beta_1 \beta_2$	0	++	1.7/–0.4	80	85	6	23–30	50%–70% excreted unchanged in urine
Nadolol (Corgard)	$\beta_1 \beta_2$	0	0	1.3/0.84	30	30–50	20–24	30	Urine, unchanged
Nebivolol (Bystolic)	β_1^b	0	0	3.7/2.3	nd ^g	12% EM ^h 96% PM ^h	10 EM 30–50 PM	98	38%–67% in urine; 44%–13% in feces
Penbutolol (Levitol)	$\beta_1 \beta_2$	0	+	4.2/2.1	~100	~100	5	80–98	Hepatic (conjugation, oxidation); renal excretion of metabolites (17% as conjugate)
Pindolol (Visken)	$\beta_1 \beta_2$	+	+++	1.97/–0.2	95	~100	3–4 ^d	40	Urinary excretion of metabolites (60%–65%) and unchanged drug (35%–40%)
Propranolol (Inderal) Propranolol, LA	$\beta_1 \beta_2$	++	0	3.1/1	90	30 9–18	3–5 8–11	90	Hepatic; <1% excreted unchanged in urine
Sotalolol (Betapace)	$\beta_1 \beta_2$	0	0	0.3/–1.8	nd ^g	90–100	12	0	Not metabolized; excreted unchanged in urine
Timolol (Blocadren, Timoptic)	$\beta_1 \beta_2$	0	0	–4.3/–2	90	75	4	10	Hepatic; urinary excretion of metabolites and unchanged drug
Labetalol ^e (Normodyne)	$\beta_1 \beta_2 \alpha_1$	0	0	2.9/0.99	100	30–40	5.5–8	50	55%–60% excreted in urine as conjugates or unchanged drug
Carvedilol (Coreg)	$\beta_1 \beta_2 \alpha_1$	0	0	4.2/3.2	>90	25–35	7–10	98	

^a cLogP and LogD calculated using Advanced Chemistry Development (ACD/Labs) Software V8.19 for Solaris (1994–2011 Chemical Abstracts, American Chemical Society).^b Inhibits β_2 -receptors (bronchial and vascular) at higher doses.^c Detectable only at doses much greater than required for β -blockade.^d In elderly hypertensive patients with normal renal function, half-life is variable (7–15 hours).^e Not labelatol monograph.^f Not applicable (available intravenously only).^g No data.^h EM, extensive metabolizers in CYP2D6; PM, poor metabolizers in CYP2D6.

0 = none; + = low; ++ = moderate; +++ = high.

β -blockers lower intraocular pressure by decreasing the amount of aqueous humor fluid produced in the eye by the ciliary body, and β_2 -receptors have been found in that tissue. Observations have shown, however, that the ciliary body has no adrenergic innervation, that the effect is not stereoselective, and that a correlation of activity exists with decreased ciliary blood flow and decreased dopamine levels, indicating that the mechanism of action of β -blockers in treating glaucoma is unusual (35).

Mixed α/β -Adrenergic Antagonists

Labetalol and carvedilol are antihypertensives with α_1 -, β_1 -, and β_2 -blocking activity (Fig. 10.18) (see Chapter 24). In terms of SAR, recall from the earlier discussion of phenylethanolamine agonists that the type of *N*-substituents, such as *N*-isopropyl and *N*-*t*-butyl, eliminated α_1 -receptor activity; however, arylalkyl groups with α -methyl substituent returned α_1 -affinity but not intrinsic activity (c.f., dobutamine). Thus, these two drugs have structural features that permit binding to the α_1 -receptor and nonselectively to both β -receptors. The β -blocking activity of labetalol is approximately 1.5-fold that of its α_1 -blocking activity. Carvedilol has an estimated β -blocking activity 10- to 100-fold its α_1 -blocking activity. Carvedilol also has unique properties that make it particularly good for treating heart failure. In addition to both α_1 - and β -blocking activity, the carbazole portion of the molecule also has antioxidant properties that provide further protection to the heart by scavenging free radicals (36,37).

DRUGS AFFECTING NOREPINEPHRINE/EPINEPHRINE BIOSYNTHESIS

Hypothetically, inhibitors of any of the three enzymes involved in the conversion of tyrosine to norepinephrine could be used as drugs to moderate adrenergic transmission. Inhibitors of the rate-limiting enzyme tyrosine hydroxylase would be the most logical choice. One inhibitor of tyrosine hydroxylase, metyrosine or α -methyl-L-tyrosine, is in limited clinical use to help control hypertensive episodes and other symptoms of catecholamine overproduction in patients with the rare adrenal tumor pheochromocytoma (38). Metyrosine, a competitive inhibitor

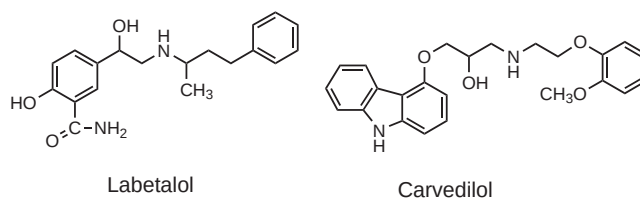
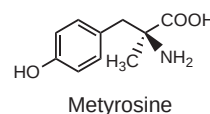
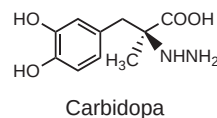


FIGURE 10.18 Mixed α/β -adrenergic antagonists.

of tyrosine hydroxylase, inhibits the production of catecholamines by the tumor. Although metyrosine is useful in treating hypertension caused by excess catecholamine biosynthesis in pheochromocytoma tumors, it is not useful for treating essential hypertension. The drug metyrosine is the (*S*)-stereoisomer of α -methyltyrosine. The enantiomer, (*R*)- α -methyltyrosine, does not bind to the active site of tyrosine hydroxylase and, thus, has no useful pharmacologic activity.



Powerful inhibitors of the next enzyme in the pathway, aromatic L-amino acid decarboxylase (e.g., carbidopa), have proven clinically useful, but not as modulators of peripheral adrenergic transmission. Rather, these agents are used to inhibit the metabolism of exogenous L-DOPA administered in the treatment of Parkinson disease (see Chapter 13).



The next enzyme in the biosynthetic pathway to norepinephrine and epinephrine, dopamine β -hydroxylase, has been the subject of extensive research into its chemical mechanism and the subject of many enzyme inhibition studies. The inhibitors known to date, however, are primarily of basic biochemical research interest and have no therapeutic relevance. The same is true of phenylethanolamine-*N*-methyltransferase, the last enzyme in the biosynthesis of epinephrine in the adrenal medulla.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Kathryn Neill, PharmD

Based on the infusion record, SJ ingested approximately 200 mL of dopamine solution. After reviewing the administration and solubility information for dopamine, the pharmacist recommended no intervention for the ingested dopamine. In addition, the pharmacist instructed the nurse that the dopamine infusion could be restarted if needed for continued treatment of hypotension with vigilant monitoring based on the patient's mental status.

Chemical Analysis

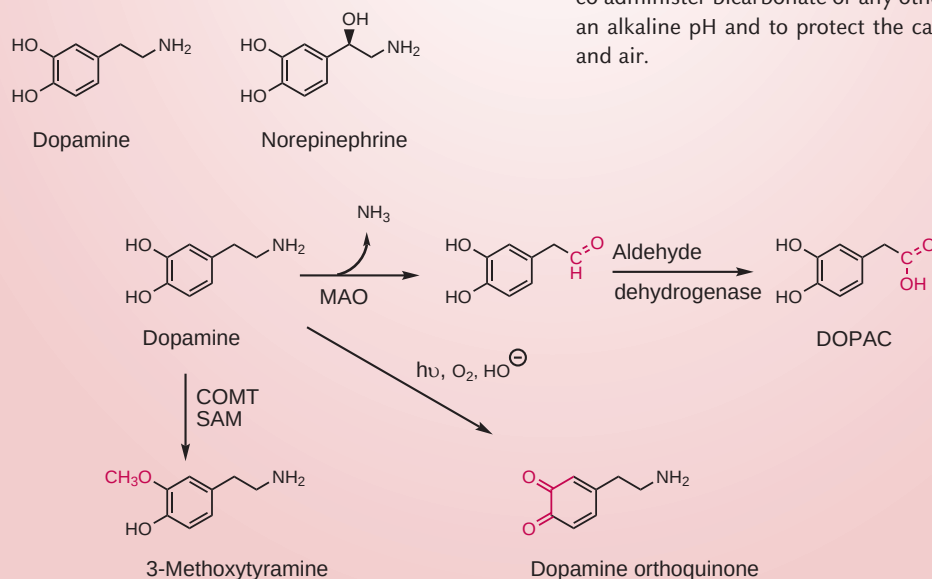
Victoria Roche, S. William Zito, and Kathryn Neill

Dopamine is a catechol-containing primary phenethylamine manufactured as a water-soluble hydrochloride salt. It is a precursor molecule in the biosynthesis of the adrenergic neurotransmitter norepinephrine and the neurohormone epinephrine. It lacks the β -OH group found on the true endogenous adrenergic agonists and, as a result, has a lower affinity for α and β adrenoceptors than norepinephrine or epinephrine. However, dopamine's close structural similarity with norepinephrine allows it to compete with its "sister" neurotransmitter for access to catecholaminergic reuptake sites, which leaves more norepinephrine in adrenergic synapses to act on postsynaptic receptors. Because it promotes the action of the true adrenergic neurotransmitter at its receptors, dopamine can be viewed as an indirect-acting adrenergic agonist. Another mechanism through which dopamine can act indirectly as an adrenergic agonist is by stimulating the release of the adrenergic neurotransmitter from vesicles in the presynaptic terminals of adrenergic neurons.

Dopamine increases renal blood flow through the stimulation of specific renal dopamine receptors and exerts dose-dependent pharmacodynamic effects through the direct and indirect stimulation of adrenergic receptors, as described above. By directly stimulating β receptors and causing release of norepinephrine from sympathetic storage sites, dopamine produces increases in myocardial inotropy and chronotropy. Stimulation of α receptors increases at higher doses and results in vasoconstriction.

Dopamine is administered intravenously because it is unstable in the gastrointestinal tract. The lack of substituents (e.g., CH_3) on the α -carbon makes dopamine exceptionally vulnerable to prehepatic (gut) and first pass (liver) deamination by monoamine oxidase, which would yield the inactive 3,4-dihydroxyphenylacetaldehyde. This short-lived aldehyde intermediately would be rapidly biotransformed to 3,4-dihydroxyphenylacetic acid by cytosolic aldehyde dehydrogenase. The catechol nucleus also would be vulnerable to rapid, inactivating COMT methylation to 3-methoxytyramine, using S-adenosylmethionine as the methyl-donating cofactor. As a result of these inactivating biotransformations, there would essentially be no active drug delivered to SJ's bloodstream after oral ingestion of an aqueous dopamine solution, so the patient is not at risk for dopamine toxicity as a result of her misadventure.

It should also be noted that dopamine, like all catechols, is unstable to base and can also auto-oxidize on exposure to light ($h\nu$) and oxygen in the air. The decomposition product generated by any of these conditions is the inactive orthoquinone. If the dopamine infusion is restarted, care should be taken to not co-administer bicarbonate or any other intravenous solution with an alkaline pH and to protect the catecholic solution from light and air.



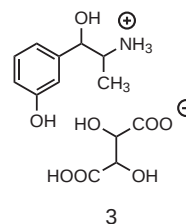
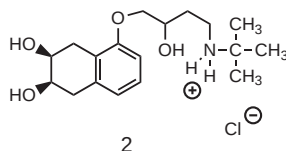
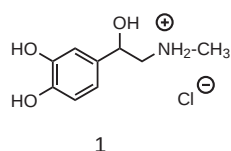
CASE STUDY

Victoria Roche and S. William Zito

It is a hot and humid Saturday afternoon, and you are chatting casually with a patient in the full-service neighborhood apothecary you've established in your hometown. Things were relatively quiet until BW, a 49-year-old woman who lives right around the corner from your pharmacy, phones in a panic. Her husband, LW, had been doing some yard work when he disturbed a nest of wasps with his weed whacker. He suffered several stings on his hands and arms before he was able to make it back to

the house. His limbs immediately began to swell and, as BW was putting ice packs on them, LW lost color in his face and began to breathe in a labored manner. You tell her to phone 911 ask your pharmacist coworker to call LW's physician for the needed prescription, grab the life-saving medication, and run to assist. You find LW clearly hypotensive and in the beginning stages of respiratory crisis.

One of the three structures drawn below needs to be administered stat. Which one did you bring with you to save the day?



1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 11

Serotonin Receptors and Drugs Affecting Serotonergic Neurotransmission

RICHARD A. GLENNON AND MALGORZATA DUKAT

*Drugs Covered in This Chapter**

ANTIEMETIC DRUGS (5-HT₃ RECEPTOR ANTAGONISTS)

- Alosetron
- Dolasetron
- Granisetron
- Ondansetron
- Palonosetron
- Tropisetron

DRUGS FOR THE TREATMENT OF MIGRAINE (5-HT_{1D/1F} RECEPTOR AGONISTS)

- Almotriptan
- Eletriptan
- Frovatriptan
- Naratriptan

- Rizatriptan
- Sumatriptan
- Zolmitriptan

DRUG FOR THE TREATMENT OF IRRITABLE BOWEL SYNDROME (5-HT₄ AGONISTS)

- Tegaserod

DRUGS FOR THE TREATMENT OF NEUROPSYCHIATRIC DISORDERS

- Buspirone
- Citalopram
- Clozapine
- Desipramine
- Fluoxetine

- Imipramine
- Olanzapine
- Propranolol
- Quetiapine
- Risperidone
- Tranylcypromine
- Trazodone
- Ziprasidone
- Zotepine

HALLUCINOGENIC AGENTS

- Lysergic acid diethylamide
- 2,5-dimethyl-4-bromoamphetamine
- 2,5-dimethoxy-4-iodoamphetamine

Abbreviations

cAMP, cyclic adenosine monophosphate

CNS, central nervous system

5-CT, 5-carboxamidotryptamine

DOB, 2,5-dimethyl-4-bromoamphetamine

DOI, 2,5-dimethoxy-4-iodoamphetamine

EMDT, 2-ethyl-5-methoxy-*N,N*-dimethyltryptamine

GABA, γ -aminobutyric acid

5-HT, serotonin

5-HTP, 5-hydroxytryptophan

IBS, irritable bowel syndrome

IBS-C, irritable bowel syndrome with constipation

IBS-D, irritable bowel syndrome with diarrhea

LCAP, long-chain arylpiperazine

L-DOPA, L-dihydroxyphenylalanine

LSD, lysergic acid diethylamide

MAO, monoamine oxidase

MAOI, monoamine oxidase inhibitor

mCPBG, *m*-chlorophenylbiguanide

mCPG, *m*-chlorophenylguanidine

mCPP, *m*-chlorophenylpiperazine

nM, nanomoles/L

MT, melatonin

MTR, melatonin receptor

NET, norepinephrine reuptake transporter

8-OH DPAT, 8-hydroxy-2-(di-*n*-propylamino)tetralin

PMDT, 2-phenyl-5-methoxy-*N,N*-dimethyltryptamine

SAFIR, structure–affinity relationship

SAR, structure–activity relationship

SERT, serotonin transporter

SSRI, selective serotonin reuptake inhibitor

*Drugs available outside the U.S. are shown in italics.

SCENARIO

Jill T. Johnson, PharmD, BCPS

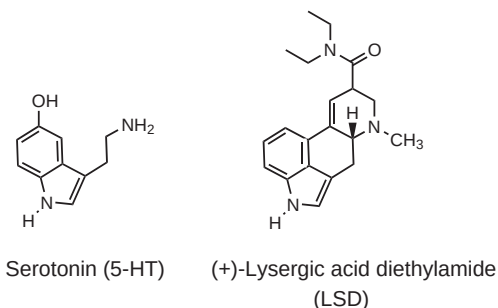
MB is a 34-year-old woman with migraines. She experiences photophobia and severe headaches with nausea and vision changes about twice per month. Recently she was prescribed sumatriptan to take as abortive therapy once she begins to feel the migraine

aura. After using sumatriptan for several months, taking it routinely up to 300 mg per day for 15 days of the month, she realized it was not working as well as it had been.

SEROTONIN

Serotonin could be considered the “baby boomer” of neurotransmitters: It was first identified in the late 1940s, its adolescent years were troubled, it made the drug scene in the 1960s, and it nearly died of an overdose in the early 1970s. It could be considered the original “sex, drugs, and rock-and-roll” receptor (as will be described below [see also Chapter 19], serotonin receptors have been implicated in sexual behavior, drug abuse [especially that involving classical hallucinogens], and the perception of sound)—but, it does much more.

At one time, it was remarked that “serotonin doesn’t do anything” (1). On reaching its middle years, serotonin matured and became an important topic of study, a household name, and more complicated than ever. Serotonin has been associated with, among other things, anxiety, depression, schizophrenia, drug abuse, sleep, dreaming, hallucinogenic activity, headache, cardiovascular disorders, sexual behavior, and appetite control. Television ads now routinely refer to serotonin and serotonin receptor antagonists. This, subsequently, prompted the comment that “it almost appears that serotonin is involved in everything” (1). A review of the current patent literature provides an indication of some of the claims being made for serotonergic agents (Table 11.1). Tens of thousands of papers have been published on serotonin. Much is known—but an incredible amount remains to be learned.



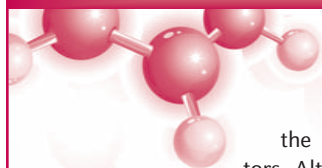
A hormonal substance was independently identified in the late 1940s by two groups of investigators, one in the United States and the other in Italy. In the

United States, the substance was called serotonin, whereas in Italy, it was termed enteramine. Its total synthesis in the early 1950s confirmed that both substances were the same structure: 5-hydroxytryptamine (5-HT). Serotonin (5-HT) was later detected in numerous plant and animal species, and in the mid-1950s, it was identified in the central nervous system (CNS) of animals. A neurotransmitter role was proposed. 5-HT was implicated in a variety of central and peripheral physiologic actions. It seemed to be involved in vasoconstriction and vasodilation, regulation of body temperature, sleep, and hormonal regulation, and early evidence suggested that it could be involved in depression. The structural similarity between 5-HT

TABLE 11.1 Some Indications and Treatment Claims for Novel Serotonergic Agents in the Patent Literature

Aggression disorders	Esophagitis	Obsessive-compulsive
Alcoholism	Gastric motility	Pain
Alzheimer’s disease	Head injury	Panic disorders
Amnesia	Headache	Parkinson’s disease
Anorexia	Hypertension	Psychosis
Bulimia	Impotence	Raynaud’s disease
Cardiac failure syndrome	Irritable bowel	Schizophrenia
Cardiovascular disorders	Ischemia	Sedation
Cerebrovascular disorders	Migraine	Sexual dysfunction
Cognition disorders	Movement	Sleep disorders
Depression	Nausea	Substance abuse
Drug abuse	Neurodegenerative disease	Substance dependence
Emesis	Obesity	Thromboembolism

CLINICAL SIGNIFICANCE



Discovery of the different types of serotonin receptors during the past few years has created the potential to target these receptors. Altering the chemical structures may improve tolerability, reduce the risk of side effects, improve efficacy, enhance compliance, or simply provide an alternative drug should another in the class fail to provide relief for a given patient.

The clinical effects of serotonin receptors are multifaceted. At this time, only a few of the identified receptors have drugs that are currently marketed for use in humans. Buspirone stimulates the 5-HT_{1A} receptor to cause antianxiety effects. The 5-HT_{1D} agonists, or the “triptans,” vary by side chains on essentially the same core structure to create compounds with different affinities for the 5-HT_{1D} receptors and, likely, for other serotonin receptors as well. The varying affinities of each triptan for receptors change the profiles of their effectiveness and adverse or complementary effects. As a rule, the triptans work to treat migraine headaches by causing vasoconstriction.

Unfortunately, they also may cause coronary vasoconstriction, making them contraindicated in patients with underlying coronary artery disease. Antagonists of the 5-HT₃ receptor, ondansetron, and granisetron, work to lessen emesis in chemotherapy-induced and radiation-associated emesis as well as postoperative nausea and vomiting. A variety of drugs work to inhibit selective serotonin reuptake without acting on any specific serotonin receptor. Drugs such as fluoxetine, paroxetine, sertraline, and fluvoxamine have been effective in the treatment of depression, obsessive-compulsive disorder, and panic disorder with varying degrees of side effects like weight gain, weight loss, and drowsiness.

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and the then recently discovered hallucinogenic agent (+)-lysergic acid diethylamide (LSD) intrigued investigators. The observation led to speculation that 5-HT could be involved in the mechanism of action of this psychoactive substances and could also have a role in certain mental disorders. LSD was shown to behave as a potent 5-HT receptor agonist in certain peripheral receptor assays and as a potent antagonist in others. The late 1960s and early 1970s, however, witnessed a decline in 5-HT research as the result of three factors: 1) sophisticated experimental techniques were still lacking for the investigation of the central actions of 5-HT; 2) apart from ergolines (LSD-related agents), only a few potent 5-HT agonists or antagonists had been developed; and 3) it was becoming increasingly difficult to understand how a single putative neurotransmitter substance could be involved in so many different central and peripheral actions. As a consequence, research interest in 5-HT entered the “doldrums.” Subsequent development of histochemical fluorescence techniques and 5-HT radioligand binding methodology led to the mapping of serotonergic pathways, identifying binding sites in the brain, and measuring the affinity (i.e., K_i values) of serotonergic agents for 5-HT receptors. This rekindled interest in 5-HT receptors—big time. Much of the early work on serotonin receptors and their ligands has been reviewed (2–4); as a result, a substantial amount of the older literature is not cited here, and the interested readers are urged to consult these reviews for references to the primary literature.

Serotonin Biosynthesis, Catabolism, and Function as Targets for Drug Manipulation

5-HT is biosynthesized from its dietary precursor L-tryptophan (5) (Fig. 11.1). Serotonergic neurons contain tryptophan hydroxylase (L-tryptophan-5-monooxygenase) that converts tryptophan to 5-hydroxytryptophan (5-HTP), in what is the rate-limiting step in 5-HT biosynthesis, and aromatic L-amino acid decarboxylase (a nonselective decarboxylase previously called 5-HTP decarboxylase) that decarboxylates 5-HTP to 5-HT. This latter enzyme is also responsible for the conversion of L-dihydroxyphenylalanine (L-DOPA) to dopamine (see Chapter 13). The major route of metabolism of 5-HT is oxidative deamination by monoamine oxidase (MAO; specifically, by MAO-A) to the unstable 5-hydroxyindole-3-acetaldehyde, which is either reduced to 5-hydroxytryptophol (~15%) or to the oxidized product 5-hydroxyindole-3-acetic acid (~85%) under normal physiologic conditions. In the pineal gland, 5-HT is acetylated by 5-HT N-acetyltransferase to N-acetylserotonin, which undergoes O-methylation by 5-hydroxyindole-O-methyltransferase to melatonin.

RADIOLIGANDS

Radioligand binding techniques measure the affinity of agonists and antagonists for their respective receptors (i.e., K_i values). Radioligands are receptor agonists or antagonists to which a radioactive atom (label) is covalently attached.

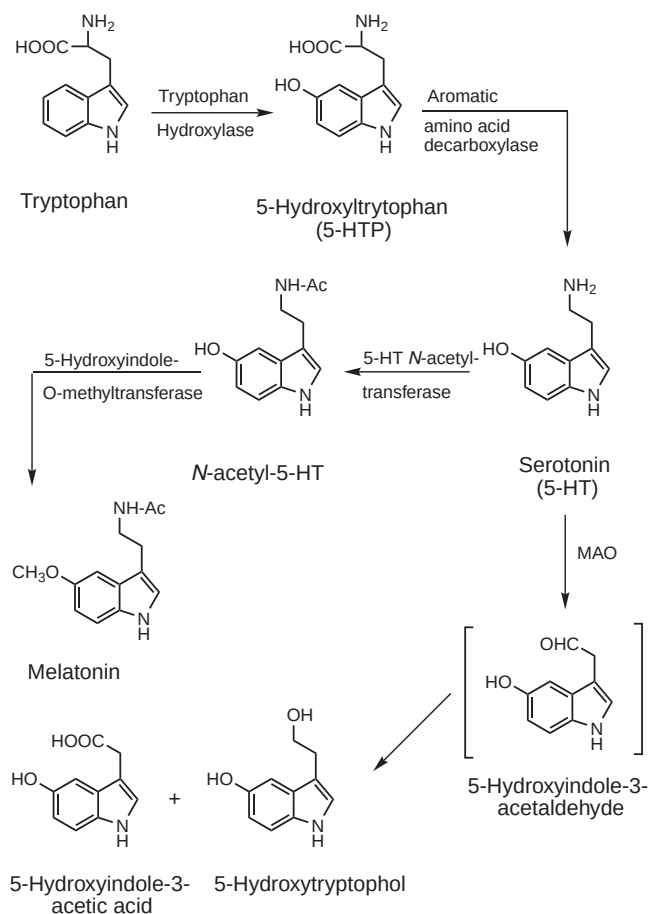


FIGURE 11.1 Biosynthesis and catabolism of serotonin. In the pineal, serotonin is converted to melatonin. Melatonin is an agonist at melatonin (MT), not 5-HT, receptors. Ac, acetyl.

Distinct types of melatonin receptors (MTRs; MTR₁/MTR₂) have been identified.

Each step in 5-HT biosynthesis, metabolism, and function is a hypothetical target for drug manipulation (Fig. 11.2). Tryptophan depletion, by reducing or restricting dietary tryptophan consumption, results in decreased 5-HT biosynthesis; conversely, tryptophan “loading,” by increasing dietary tryptophan, results in the overproduction of 5-HT. The latter effect also can occur in nonserotonergic neurons, such as in dopaminergic neurons, because of the nonselective nature of aromatic amino acid decarboxylase. Inhibitors of tryptophan hydroxylase, such as *para*-chlorophenylalanine, are used as pharmacologic tools; they are not used therapeutically.

Therapeutically exploited serotonergic targets include presynaptic receptors, postsynaptic receptors, the reuptake mechanism (i.e., the serotonin transporter [SERT]), second messenger systems, and 5-HT metabolism. MAO inhibitors (MAOI) effectively interfere with the oxidative deamination of 5-HT to increase synaptic concentrations of 5-HT. The MAOI tranylcypromine,

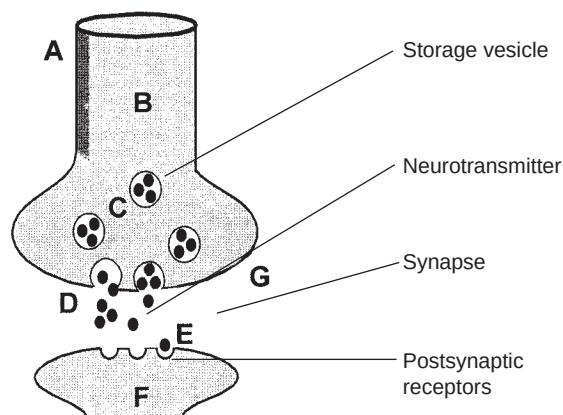
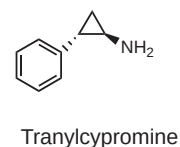


FIGURE 11.2 Steps involved in serotonergic neurotransmission. The serotonin precursor tryptophan is taken up into the neuron (A) and converted to 5-HT (B). Synthesized 5-HT is stored in synaptic vesicles (C). Under the appropriate conditions, the synaptic vesicles migrate to and fuse with the cell membrane, releasing their store of 5-HT (D). Released neurotransmitter interacts with postsynaptic receptors (E) and, in the case of G protein-coupled receptors, activates second messenger systems (F). The action of 5-HT is terminated (G) either by diffusion of 5-HT away from the synapse, with subsequent metabolism, or by the 5-HT being taken back up by 5-HT transporters into the presynaptic neuron (i.e., reuptake), where it can be re-stored in synaptic vesicles or is metabolized (by MAO).

for example, has been employed since the 1960s as an antidepressant agent.



A problem associated with many MAOIs is that they are notoriously nonselective and can interfere with the metabolism of other neurotransmitters, amines found in certain foods, and exogenously administered amine-containing therapeutic agents. Serotonin receptors and SERT are discussed below.

SEROTONIN RECEPTORS

Initially, 5-HT was thought to interact with what were termed 5-HT receptors. Today, seven distinct families or populations of serotonergic receptors have been identified, 5-HT₁ through 5-HT₇, and several are divided into subpopulations (Table 11.2). The discovery of the individual populations and subpopulations of 5-HT receptors follows the approximate order of their numbering and, as a consequence, more is known about 5-HT₁ and 5-HT₂ receptors than about 5-HT₆ and 5-HT₇ receptors. Factors contributing to our current lack of understanding about

TABLE 11.2 Classification and Nomenclature for the Various Populations of 5-HT Receptors

Populations and Subpopulations	Second Messenger System ^a	Currently Accepted Name ^b	Comments
5-HT₁			
5-HT _{1A}	AC(–)	5-HT _{1A}	Cloned and pharmacologic 5-HT _{1A} receptors
5-HT _{1B}	AC(–)	5-HT _{1B}	Rodent homolog of 5-HT _{1B} receptors
5-HT _{1Bβ}			A mouse homolog of h5-HT _{1B} receptors
5-HT _{1D}			Sites identified in binding studies using human and calf brain homogenates
5-HT _{1Dα}	AC(–)	h5-HT _{1D}	A cloned human 5-HT _{1D} subpopulation
5-HT _{1Dβ}	AC(–)	h5-HT _{1B}	A second cloned human 5-HT _{1D} subpopulation; human counterpart of rat 5-HT _{1B}
5-HT _{1E}	AC(–)	5-HT _{1E}	Sites identified in binding studies using brain homogenates and cloned receptors
5-HT _{1Eα}			An alternate name that has been used for cloned human 5-HT _{1E} receptors
5-HT _{1Eβ}	AC(–)	5-HT _{1F}	A cloned mouse homolog of 5-HT _{1F} receptors
5-HT _{1F}			A cloned human 5-HT ₁ receptor population
5-HT₂			
5-HT ₂	PI	5-HT _{2A}	Original “5-HT ₂ ” (sometimes called 5-HT _{2α}) receptors
5-HT _{2F}	PI	5-HT _{2B}	5-HT ₂ -like receptors originally found in rat fundus
5-HT _{1C}	PI	5-HT _{2C}	Originally described as 5-HT _{1C} (5-HT _{2β}) receptors
5-HT₃			
5-HT ₃	Ion channel	5-HT ₃	An ion channel receptor
5-HT₄			
5-HT ₄	AC(+)	5-HT ₄	5-HT ₄ population originally described in functional studies
5-HT _{4S}			Short form of cloned 5-HT ₄ receptors
5-HT _{4L}			Long form of cloned 5-HT ₄ receptors
5-HT _{4(b)–4(d)}			Recently identified human 5-HT ₄ receptor isoforms
5-HT₅			
5-HT _{5A}	?	5-HT _{5A}	Cloned mouse, rat, and human 5-HT ₅ receptors
5-HT _{5B}	?	5-HT _{5A}	Cloned mouse and rat 5-HT _{5A} -like receptor
5-HT₆			
5-HT ₆	AC(+)	5-HT ₆	Cloned rat and human 5-HT receptor
5-HT₇			
5-HT ₇	AC(+)	5-HT ₇	Cloned rat, mouse, guinea pig, and human 5-HT receptors

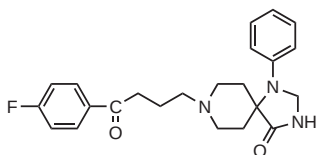
^a AC = adenylate cyclase; (–) = negatively coupled; (+) = positively coupled; PI = phospholipase coupled.

^b Currently accepted names are taken from Hoyer et al. (9).

the function of certain 5-HT receptor populations (e.g., 5-HT_{1E} or 5-HT₅ receptors) is the absence of agonists and/or antagonists with selectivity for these receptors.

History

Tritiated LSD (³H]LSD), the first radioligand used to identify a brain 5-HT binding site, suggested it could be a “hallucinogen receptor.” Tritiated 5-HT (³H]5-HT)–labeled serotonergic sites displayed high affinity for LSD. Thus, not only did 5-HT and LSD share structural similarity, there was now evidence that these agents could be acting via a common receptor type. According to the *interconvertible receptor conformation hypothesis* that was popular at the time, 5-HT (known to be an agonist) interacted with the agonist conformation of the receptor, whereas ³H]LSD (LSD being known to be a partial agonist) labeled both the agonist and antagonist conformations. A search was initiated for 5-HT receptor antagonists that could serve to label the antagonist conformation of 5-HT receptors. After the serendipitous discovery that a tritiated version of the dopamine antagonist spiperone not only labeled dopaminergic receptors but also labeled nondopaminergic receptors in other brain regions, it was shown that 5-HT displayed modest affinity for some of these sites, indicating they could represent 5-HT receptors.



Spiperone

Spiperone was also shown to antagonize some of the pharmacologic effects of 5-HT in functional assays. These data, coupled with the additional observation that 5-HT receptor agonists tended to display higher affinity for ³H]5-HT–labeled sites, whereas 5-HT antagonists displayed higher affinity for ³H]spiperone–labeled sites, led to the conclusion that ³H]5-HT and ³H]spiperone label two distinct populations (not conformations) of sites, termed 5-HT₁ and 5-HT₂ receptors, respectively (6). Soon thereafter, 5-HT₁ receptors were found to consist of 5-HT_{1A} and 5-HT_{1B} subpopulations. Earlier, during the 1950s, Gaddum and Picarelli had demonstrated the existence of two populations of serotonergic receptors in isolated guinea pig ileum and termed these receptors 5-HT-D (because phenoxybenzamine or dibenzylamine blocked the actions of 5-HT at this receptor) and 5-HT-M (because morphine and cocaine blocked the actions of 5-HT at the second population). Later, 5-HT-D receptors were found to be similar to 5-HT₂ receptors, and 5-HT-M receptors were eventually renamed 5-HT₃ receptors. By the early 1980s, 5-HT_{1A}, 5-HT_{1B}, 5-HT₂, and 5-HT₃ receptors had been identified, and interest in 5-HT research exploded. Molecular biology intervened in the late 1980s and early 1990s; new populations of serotonergic receptors were cloned and expressed. Perhaps the multitude of actions of 5-HT, previously thought impossible

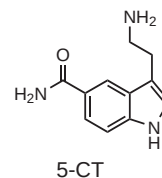
to understand, are mediated by multiple subtypes of 5-HT receptors. This led to attempts to develop selective agonists and antagonists for each subpopulation (2,7).

Table 11.2 lists the receptor classification and nomenclatures that have been employed for serotonergic receptors. Care should be used when reading the older primary literature because 5-HT receptor nomenclature has changed so dramatically and, often, can be confusing and very frustrating to comprehend.

All of the seven serotonergic receptor populations (and subpopulations) have been cloned and, together with the cloning of other neurotransmitter receptors, has led to generalizations regarding amino acid sequence homology (8). Any two receptors with amino acid sequences that are approximately 70% to 80% identical in their transmembrane-spanning segments are called the *intermediate-homology group*. This group of receptors could be members of the same subfamily and have highly similar to nearly indistinguishable pharmacologic profiles or second messenger systems. A *low-homology group* (~35% to 55% transmembrane homology) consists of distantly related receptor subtypes from the same neurotransmitter family, and a *high-homology group* (~95% to 99% transmembrane homology) consists of species homologs from the same gene in different species (8). Species homologs of the same gene reveal high sequence conservation in regions outside the transmembrane domains, whereas intraspecies receptor subtypes usually are quite different (8). Current 5-HT receptor classification and nomenclature require that several criteria be met before a receptor population can be adequately characterized. Receptor populations must be identified on the basis of drug binding characteristics (*operational* or *recognitory criteria*), receptor–effector coupling (*transductional criteria*), and gene and receptor structure sequences for the nucleotide and amino acid components, respectively (*structural criteria*) (7–9).

5-HT₁ Receptor Family

5-HT₁ receptors were one of the first two populations of 5-HT receptors to be identified (6), and 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, and 5-HT_{1F} receptor subpopulations have since been defined. 5-HT_{1C} receptors were initially described, but subsequent classification (employing the previously mentioned criteria) resulted in their being moved to the 5-HT₂ receptor family and being renamed 5-HT_{2C} receptors. With the exception of 5-HT_{1E} receptors, all 5-HT₁ receptors exhibit high affinity for 5-carboxamidotryptamine (5-CT).



5-CT

5-HT_{1A} Receptors and Agonists

5-HT_{1A} receptors are, as are all 5-HT receptors except for 5-HT₃ receptors, G protein–coupled receptors that consist of seven transmembrane-spanning helices connected

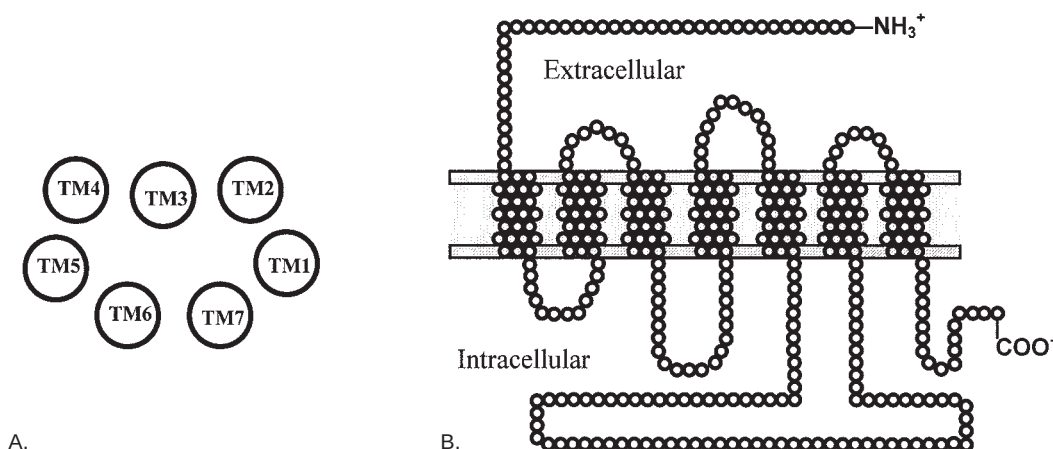
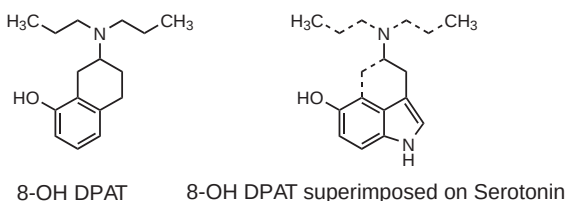


FIGURE 11.3 Top (A) and side (B) views of a schematic representation of a typical G protein-coupled receptor (GPCR). In B, the transmembrane-spanning helical portions are numbered, from left to right, as TM1 to TM7. The seven helices are connected by extracellular and intracellular loops. The large intracellular loop between TM5 and TM6 is believed to be associated with coupling to a second messenger system. The helices are arranged in such a manner that TM1 is adjacent to TM7, as shown in A. Molecular graphics studies suggest that agonists could bind in a manner that utilizes an aspartate residue in TM3 (common to all G protein-coupled 5-HT receptors) and residues in the TM4, TM5, and TM6 regions, whereas antagonists likely utilize the aspartate moiety but residues in the TM6, TM7, and TM1 regions.

by intracellular and extracellular loops (see Fig. 11.3 for a schematic representation of a generalized G protein receptor structure). The receptors are negatively coupled to an adenylate cyclase*¹ second messenger system, and the 5-HT_{1A} receptors located in the raphe nuclei correspond to somatodendritic autoreceptors (Fig. 11.4). 5-HT_{1A} receptors differ significantly in structure from most other 5-HT receptors and exhibit a substantial similarity to adrenergic receptors, which likely explains why a number of adrenoceptor agents bind at 5-HT_{1A} receptors with high affinity (see below). Cloned 5-HT_{1A} receptors and 5-HT_{1A} receptor ligands have been reviewed (9,10–18).

Structure–Activity Relationship of 5-HT_{1A} Receptor Agonists



Numerous 5-HT-related tryptamines bind with high affinity at 5-HT_{1A} receptors, but most are notoriously nonselective. One of the most selective 5-HT_{1A} receptor agonists is the aminotetralin derivative 8-hydroxy-2-(di-n-propylamino)tetralin (8-OH DPAT), and its early discovery was significant in advancing understanding of 5-HT_{1A} receptors. Furthermore, because the structure of 8-OH DPAT is similar to that of 5-HT (see 8-OH

DPAT/5-HT superimposition), its activity indicated that an intact indole nucleus was not required for 5-HT_{1A} receptor action. Although numerous 8-OH DPAT

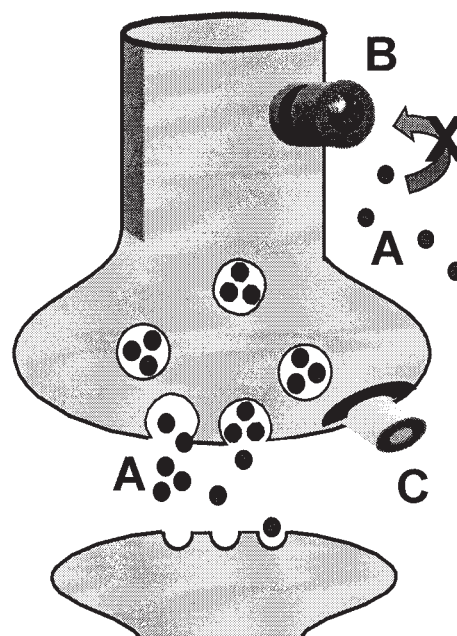


FIGURE 11.4 Typical nerve ending showing the cell body (i.e., somatodendritic) autoreceptors (B) and the terminal autoreceptors (C). Neurotransmitter molecules are also shown (A). 5-HT can interact with cell body autoreceptors (B) to regulate synthesis, and with terminal autoreceptors (C) to regulate release. Shown above is a drug molecule blocking the cell body autoreceptor and preventing an interaction with the neurotransmitter.

*This enzyme is known by two names, *adenylate cyclase* (EC 4.6.1.1), its official name from the International Union of Biochemistry and Molecular Biology Nomenclature Committee, or its alternative name, *adenylyl cyclase*.

derivatives have been reported, none is used therapeutically because of low oral bioavailability. This has led to efforts to develop novel aminotetralins with greater oral bioavailability.

LONG-CHAIN ARYLPYPERAZINES Simple arylpiperazines (i.e., those bearing no N_4 -substituent or only a small N_4 -substituent), such as 1-(phenyl)piperazine (Fig. 11.5), bind with modest to reasonably high affinity at multiple receptor types and are considered nonselective agents. Long-chain arylpiperazines (LCAPs) are piperazines possessing a long-chain N_4 substituent and represent the largest class of 5-HT_{1A} receptor ligands. Buspirone (Fig. 11.5), the first arylpiperazine approved for clinical use as an anxiolytic agent, and the structurally related gepirone and ipsapirone bind at 5-HT_{1A} receptors and behave as agonists or partial agonists. Structure–activity relationships (SARs) and structure–affinity relationships (SAFIRs) have been formulated, and this has led to LCAPs with enhanced 5-HT_{1A} receptor affinity and selectivity (12–15). With the LCAPs, there is substantial structural latitude for 5-HT_{1A} receptor binding (14,15).

The aryl portion of these agents (Fig. 11.6) typically is a phenyl, substituted phenyl, or heteroaryl group (such as 2-pyrimidinyl). The intact piperazine ring seems to be optimal for binding to 5-HT_{1A} receptors. A spacer or linker separates the N_4 -nitrogen atom of the piperazine and the terminus or terminal structural moiety. There has been controversy as to whether the spacer participates in binding to the receptor or whether it acts simply as a “connector”; in any event, a chain of two to five atoms is common. The terminus typically is an amide or imide, but it has been shown that neither is required for binding. Alternatively, the terminus can be a

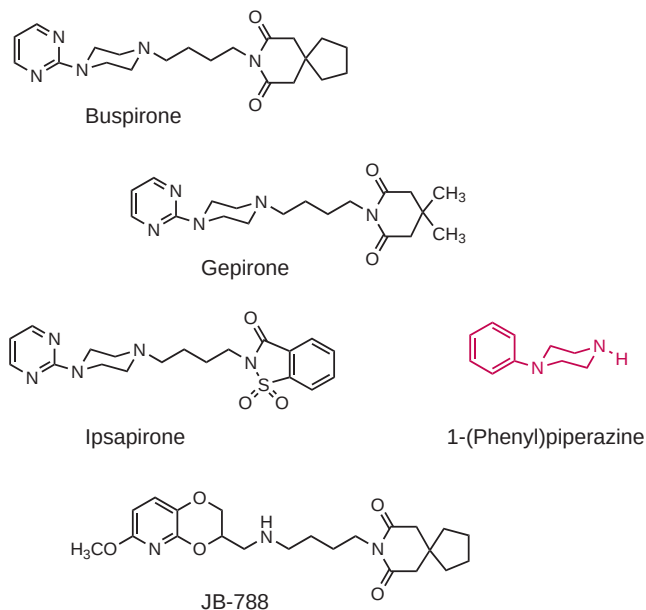


FIGURE 11.5 5-HT_{1A} receptor agonists (arylpiperazines).

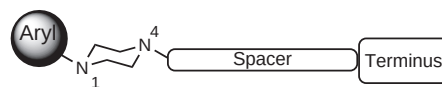


FIGURE 11.6 General structure of long-chain arylpiperazines (LCAPs).

phenyl or some other aryl or heteroaryl substituent (14). With respect to spacer length, when the spacer is $-(CH_2)_n$, 2 to 4 methylene groups appear optimal. Chain length (n) can influence affinity and selectivity. When the terminus contains a heteroarylamide, $n = 4$ seems to be optimal, whereas when the terminus is an alkylamide, optimal affinity is associated with $n = 2$. A region of bulk tolerance is associated with the terminus, or at least a portion thereof, and very bulky groups have been introduced into this part of the molecule (12–15). Some LCAPs are nonselective and variously bind at other populations of 5-HT receptors, dopamine receptors, or adrenoceptors.

Structure–activity Relationships of 5-HT_{1A} Receptor Antagonists

Many 5-HT_{1A} receptor antagonists possess a 2-methoxyphenyl group with structural similarity to buspirone. BMY 7378 and NAN-190 were among the first agents shown to be very low-efficacy partial agonists at 5-HT_{1A} receptors and were used as antagonists for many years (Fig. 11.7). Certain aminotetralins [e.g., S(–)UH-301] and arylpiperazines (e.g., WAY 100135 and WAY 100635) represent

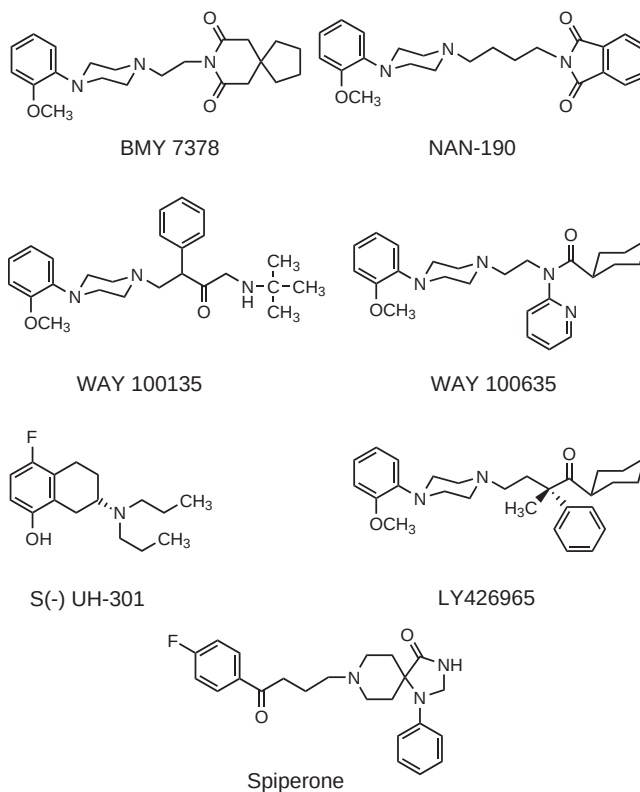


FIGURE 11.7 5-HT_{1A} receptor antagonists.

new classes of 5-HT_{1A} receptor antagonists, termed “silent antagonists,” because they are “seemingly” without any 5-HT_{1A} agonist action. The alkylpiperidine spiperone is a 5-HT_{1A} antagonist, but spiperone displays high affinity for D₂-dopamine receptors and 5-HT_{2A} receptors.

Molecular graphics studies suggest that 5-HT and 5-HT_{1A} receptor agonists interact with amino acid residues associated with helices 4, 5, and 6 (Site 1), whereas 5-HT_{1A} receptor antagonists likely interact with amino acid residues in helices 1, 2, 7, and, perhaps, 6 (Site 2) (16). The basic amine for both types of agents is thought to bind at a common aspartate residue found in TM helix 3 (Fig. 11.3). The 5-hydroxy group of 5-HT is thought to form a hydrogen bond with the threonine residue in TM5 (16).

5-HT_{1A} Receptor Agonists: Clinical Significance

In preclinical studies, 5-HT_{1A} receptor agonists have demonstrated anxiolytic, antidepressant, antiaggressive, and perhaps, anticraving, anticonvulsant, antiemetic, and neuroprotective properties (15,17). Evidence also exists indicating that 5-HT_{1A} receptors could be involved in sleep, impulsivity, alcoholism, sexual behavior, appetite control, thermoregulation, and cardiovascular function (17,19,20). The main focus of drug development for 5-HT_{1A} receptors is their therapeutic potential for the treatment of anxiety and depression (15,19). Buspirone (Buspar) was the first LCAP to become clinically available as an anxiolytic agent. A number of structurally related agents hold promise as novel anxiolytics (11,12,21); one of the newest is JB-788 (22). 5-HT_{1A} receptor agents could also be useful in the treatment of depression (15), and there seems to be a relationship between 5-HT metabolism, depression, and violent behavior. The anxiolytic actions of 5-HT_{1A} receptor (partial) agonists could involve, primarily, presynaptic somatodendritic 5-HT_{1A} receptors, whereas the antidepressant actions of 5-HT_{1A} receptor agents could primarily involve postsynaptic 5-HT_{1A} receptors (17). Gepirone produced marked improvement in depressed patients, and buspirone was effective in the treatment of mixed anxious-depressive patients. 5-HT₁ and, possibly, 5-HT_{1A} receptors have been implicated in obsessive-compulsive disorders.

5-HT_{1A} Receptor Antagonists: Clinical Significance

A new direction in 5-HT_{1A} receptor research targets the development of 5-HT_{1A} receptor antagonists (15,23). Agents such as the acknowledged dopaminergic antagonist spiperone and the β -adrenoceptor antagonist propranolol were among the first to see application as 5-HT_{1A} receptor antagonists. These agents are, obviously, nonselective; they bind at other populations of neurotransmitter receptors with comparable or higher affinities than they display at 5-HT_{1A} receptors. The next generation of 5-HT_{1A} receptor antagonists, the LCAPs BMY 7378 and NAN-190, possessed postsynaptic antagonist character but also behaved as low-efficacy

partial agonists (14,23) (Fig. 11.7). A third generation of agents—“silent” 5-HT_{1A} receptor antagonists—has been developed and includes WAY 100635, WAY 100135 (a structural relative of BMY 7378 and NAN-190), and S(-)UH-301 (a derivative of the 5-HT_{1A} agonist 8-OH DPAT); these are both presynaptic and postsynaptic 5-HT_{1A} receptor antagonists (23,24). Silent 5-HT_{1A} receptor antagonists, such as WAY 100135 and S(-)UH-301, are not intrinsically inactive and can indirectly produce non-5-HT_{1A} serotonin-mediated actions (25,26). These antagonists presumably block presynaptic 5-HT_{1A} autoreceptors, increasing the synaptic concentration of 5-HT, which results in the activation of other 5-HT receptor populations. Human evaluation of “so-called” silent and selective 5-HT_{1A} receptor antagonists should prove interesting and could open new vistas in 5-HT_{1A} research and therapeutics. For example, pretreatment of patients with 5-HT_{1A} receptor antagonists accelerates the effects of selective serotonin reuptake inhibitors (SSRIs) and enhance their clinical efficacy as antidepressants (27). The 5-HT_{1A} receptor antagonist WAY 100635 enhances the anorectic effect of citalopram in animals (28) and, thus, may be of benefit in weight reduction. Combination therapy using an SSRI plus a 5-HT_{1A} receptor antagonist, including the β -blocker pindolol, which binds at 5-HT_{1A} receptors, has been reported (29). A new LCAP, LY426965, is more metabolically stable than WAY 100635 and is orally bioavailable. In combination with fluoxetine, LY426965 increase extracellular levels of 5-HT beyond that achievable by fluoxetine alone, and it is being examined for the treatment of depression and as a smoking cessation agent (30). The therapeutic potential of 5-HT_{1A} receptor antagonists is quite intriguing.

5-HT_{1B} Receptors and Agents

Early studies identified 5-HT_{1B} receptors in rodent brain homogenates using radioligand binding techniques but failed to find them in human brain. 5-HT_{1B} receptors are located both presynaptically, where they regulate the release of 5-HT (Fig. 11.4), and postsynaptically (31). Like 5-HT_{1A} receptors, they are negatively coupled to adenylate cyclase. (See 5-HT_{1D} Receptors for further related discussion.)

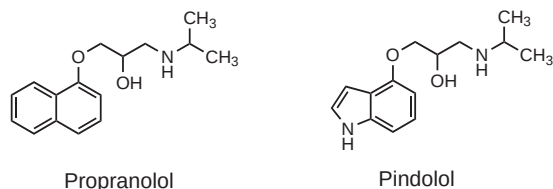
5-HT_{1B} Receptors: Clinical Significance

Rodent 5-HT_{1B} receptors have been implicated as having a role in thermoregulation, respiration, appetite control, sexual behavior, aggression, locomotor activity, sleep regulation, sensorimotor inhibition, and anxiety (32).

5-HT_{1D} Receptors

5-HT_{1D} receptors were first identified by radioligand binding techniques, and they are widely distributed throughout the CNS (33). They are G protein-linked and are coupled to inhibition of adenylate cyclase. Two

human subpopulations of 5-HT_{1D} receptors, 5-HT_{1D α} and 5-HT_{1D β} receptors, display approximately 77% sequence homology, and their pharmacologic properties are nearly indistinguishable. Because of the high degree of species homology with rat and mouse 5-HT_{1B} receptors, human 5-HT_{1D β} receptors have been renamed h5-HT_{1B} receptors. Human 5-HT_{1D α} receptors have been renamed h5-HT_{1D}. Most agents that bind at 5-HT_{1B} receptors bind at 5-HT_{1D} receptors.



Curious exceptions have been noted with certain aryloxyalkylamines, however, such as the β -blockers, propranolol and pindolol, which exhibit very low affinity ($K_i \sim 5,000$ nM) for human (h) 5-HT_{1D} receptors (34,35). The major functional difference between rat 5-HT_{1B} receptors and h5-HT_{1B} receptors has been attributed to both the presence of a threonine residue at position 355 (i.e., Thr³⁵⁵) in TM7 of the latter and the presence of an asparagine residue at the corresponding position in 5-HT_{1B} receptors; site-directed mutagenesis studies have demonstrated that conversion of Thr³⁵⁵ to an asparagine (i.e., a T355N mutant) accounts for the binding differences of certain ligands (e.g., aryloxyalkylamines such as propranolol). Combined ligand SAR, site-directed mutagenesis, and molecular modeling

studies have led to the conclusion that although most typical serotonergic agonists bind in the central cavity formed by TM3, TM4, TM5, and TM6 (Site 1) (Fig. 11.3), propranolol most likely occupies the region defined by TM1, TM2, TM3, and TM7 (Site 2). The higher affinity of propranolol for T355N mutant 5-HT_{1B} receptors relative to the wild-type receptors was specifically attributed to the formation of two hydrogen bonds between the receptor asparagine and the ether and hydroxyl oxygen atoms of propranolol (35).

5-HT_{1D} Agonists and Antagonists

There are few 5-HT_{1D}-selective agonists, but one agent commonly referred to as a prototypical 5-HT_{1D} receptor agonist is sumatriptan (Imitrex). Sumatriptan, however, exhibits only 2- to 20-fold greater selectivity for the 5-HT_{1D} receptors than for certain other populations of 5-HT₁ (especially 5-HT_{1A}) receptors, binds at h5-HT_{1D} and h5-HT_{1B} receptors with nearly identical affinity and also binds at 5-HT_{1F} receptors (36). SARs for 5-HT_{1D} receptor agonists have been reported for many indolealkylamines or tryptamine derivatives, which bind with high affinity but with little selectivity. Newer agents displaying high affinity and reasonable selectivity for h5-HT_{1D}/h5-HT_{1B} receptors over other populations of 5-HT receptors (37) include, for example, zolmitriptan (Zomig), naratriptan (Amerge), rizatriptan (Maxalt), and alniditan. Of these, all are tryptamine derivatives or sumatriptan-related structures except for the benzopyran alniditan. Many of these are commercially available or currently undergoing clinical trials. Other investigational agonists are shown in Figure 11.8.

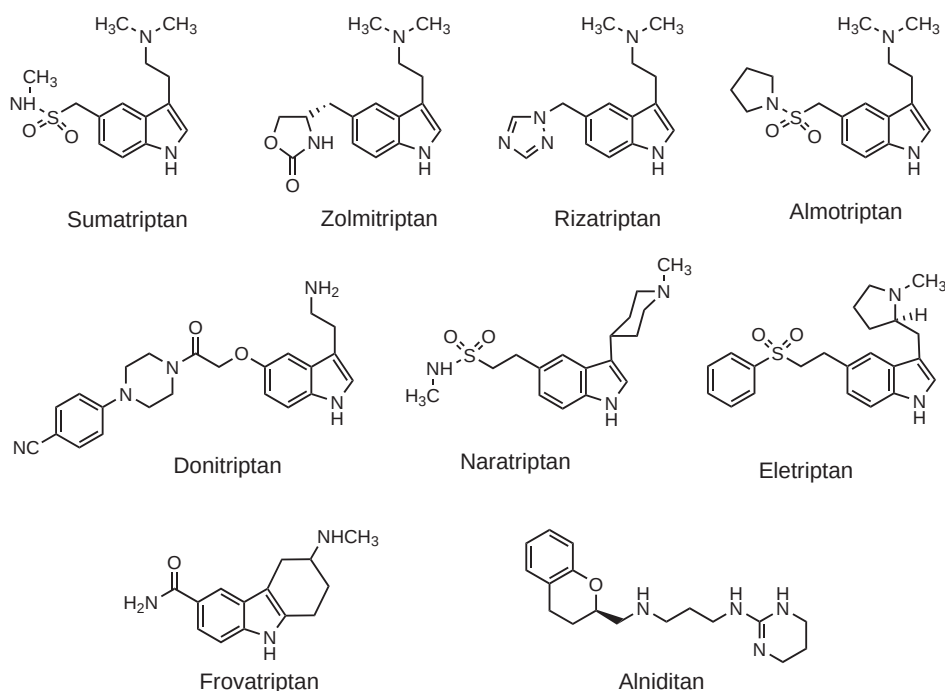
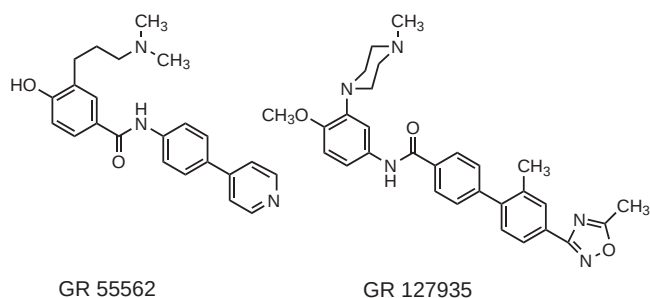


FIGURE 11.8 5-HT_{1D} receptor agonists.



Several 5-HT_{1D} receptor antagonists have been developed, including GR127935 (high affinity for h5-HT_{1D}/h5-HT_{1B} receptors but possibly a low-efficacy partial agonist) and GR55562. Both of these agents antagonize many of the effects of sumatriptan.

5-HT_{1D} Receptors: Clinical Significance

The clinical significance of 5-HT_{1D} receptors remains largely unknown. These receptors are speculated to be involved in anxiety, depression, and other neuropsychiatric disorders, but this remains to be substantiated. However, recent studies show that 5-HT_{1D} receptors are the dominant species in human cerebral blood vessels. Sumatriptan and several closely related agents are clinically effective in the treatment of migraine, and logical extrapolation implies a role for 5-HT_{1D} receptors in this disorder. Agents with 5-HT_{1D} receptor agonist activity that have found application in the treatment of migraine are, as a group, termed *triptans*, because the first agent introduced was *sumatriptan*. As efficacious as the triptans may be however, it is unknown if their activity involves action only in the periphery or in the CNS as well (38). Sumatriptan is an h5-HT_{1B} and h5-HT_{1D} receptor agonist. It is also an agonist at 5-HT_{1F} receptors. Most triptans share a similar binding profile. The vasoconstrictor properties of sumatriptan probably are mediated by its action on arterial smooth muscle. The triptans are also believed to inhibit the activation of peripheral nociceptors (38), and this could be related to the localization of 5-HT_{1D} receptors on peptide nociceptors.

Relatively little sumatriptan normally penetrates the blood–brain barrier. Although it has been speculated that transient changes in blood–brain barrier permeability could occur during migraine attacks (38), agents with greater lipophilicity (and, hence, enhanced ability to penetrate the blood–brain barrier) have been introduced, including zolmitriptan and rizatriptan (Table 11.3). Their greater lipophilicity, however, does not seem to correlate with significantly improved clinical efficacy over sumatriptan (38). Other triptans (Fig. 11.8) currently being examined include eletriptan, almotriptan, donatriptan, and frovatriptan (37).

In general, the newer triptans (e.g., zolmitriptan, rizatriptan, and naratriptan) have a higher oral bioavailability and a longer plasma half-life than sumatriptan (39,40) (Table 11.3). Most triptans also bind at 5-HT_{1F} receptors, and 5-HT_{1F} receptor agonists have demonstrated efficacy in

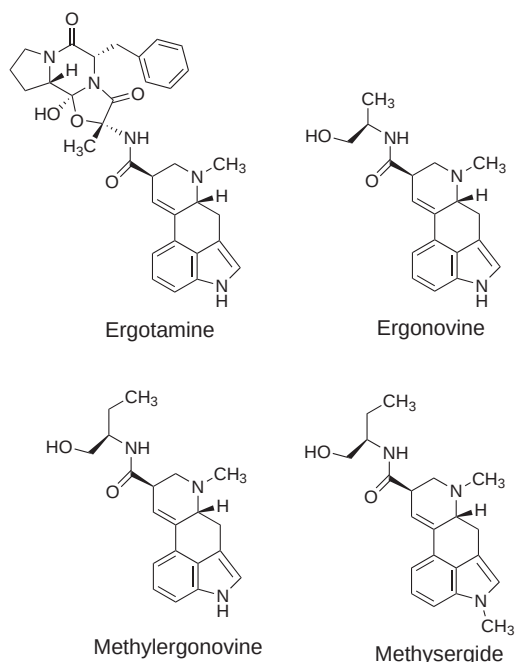
the treatment of migraine (41). The 5-HT receptor binding characteristics of various triptans have been compared (37).

The safety of the triptans has been established; more than 8 million patients have been treated for more than 340 million attacks with sumatriptan alone. All triptans narrow coronary arteries by 10% to 20% at clinical doses and should not be administered to patients with coronary or cerebrovascular disease. Triptans with potential for significant drug–drug interactions include sumatriptan, naratriptan, rizatriptan, almotriptan, and MAOIs; rizatriptan and propranolol; zolmitriptan and cimetidine; zolmitriptan, naratriptan, and eletriptan; CYP3A4-metabolized drugs; and P-glycoprotein pump inhibitors.

The rational employment of triptans should be governed by the use of these medications for patients with disability associated with migraine. Patients with greater than 10 days of at least 50% disability during 3 months have benefited from treatment with triptans as their first-line treatment for acute attacks. When the decision has been made to treat with a triptan, the patient should be instructed to treat early in the attack, when the pain is at a mild phase. This approach increases the likelihood of achieving a pain-free response, with fewer adverse events and with lower likelihood of the headache recurring.

5-HT_{1E} Receptors and Agents

In early binding experiments using [³H]5-HT as radioligand, masking of brain 5-HT_{1A} and 5-HT_{1B} receptors resulted in biphasic competition curves providing evidence for additional 5-HT₁-like receptor populations. One of these was the 5-HT_{1D} receptors; the other was termed 5-HT_{1E} receptors.



The low affinity of 5-CT and ergotamine for 5-HT_{1E} receptors allowed their differentiation from 5-HT_{1D}

TABLE 11.3 Pharmacokinetics of the 5-HT₁ Agonists (the Triptans)

Parameters	Sumatriptan	Zolmitriptan	Naratriptan	Rizatriptan	Almotriptan	Frovatriptan	Eletriptan
Trade name	Imitrex	Zomig	Amerge	Maxalt	Axert	Frova	Relpax
LogP (calc) ^a	0.7 ± 0.6	1.6 ± 0.4	1.4 ± 0.6	0.9 ± 0.6	1.9 ± 0.6	0.9 ± 0.4	3.1 ± 0.6
LogD (pH 7) (calc) ^a	-1.7	-0.8	-1.2	-1.4	-0.5	-2.1	0.18
Bioavailability (%)							
Oral (PO)	14–15 ^b	40–50 ^b	70 ^c	40–50 ^b	70–80	20–30 ^b	50 ^d
Nasal	17	102	—	—	—	—	—
Subcutaneous (SC)	97	—	—	—	—	—	—
Protein binding (%)	14–20	25	28–30	14	35	15	85
Volume of distribution (L/kg)	PO: 50	PO: 7	PO: 170	PO: 110–140	PO: 180–200	PO: 3–4	PO: 138
Elimination half-life (h)	PO: 2.5	PO: 2–3	PO: 5–6	PO: 2–3	PO: 3–4	PO: 25	PO: 4–5
Major metabolites (%)	SC: 2.5	Nasal: 3–4	Hepatic: 50%	Indolacetate	Indolacetate	N-Demethyl (act)	PO Elderly: 6
	Indoleacetic acid	N-Demethyl (act) ^e : 4	Hepatic: 50%	Indolacetate	Indolacetate	N-Demethyl (act)	N-Demethyl (act)
	Glucuronides	Indoleacetate: 31	Hepatic: 50%	N-Demethyl (act)	GABA	N-Ac demethyl	
	Hepatic: 60%			6-OH	N-demethyl		
				Hepatic: 60%			
Metabolizing enzymes	MAO-A	CYP3A4	CYP3A4	MAO-A	CYP3A4/CYP2D6: 12%	CYP1A2	CYP3A4
Excretion (%)							
	Urine metab: ~60	Urine metab: 60	Urine metab: 30	Urine metab: 80	Urine metab: 75	Urine metab: 10–30	Urine metab: ~90%
	Feces metab: ~40	Feces metab: 30	Feces metab: ~15	Feces metab: 12	Feces metab: 10	Feces metab: 60	Unchanged: <10
	Unchanged: 3–22	Unchanged: <10	Unchanged: 50	Unchanged: 14	Unchanged: 40–50		
Time to peak concentration (min)	SC: 12 (5–20)	PO: 120–240	PO: 60–180	PO: 60–90	PO: 60–240	PO: 120–240	PO: 60–90
	Nasal: 60–90	Nasal: 180–240			SC: <30		
	PO: 60–120						
Onset (min)	SC: <10	PO: 60	PO: 60–180	PO: 30–120	PO: 60–120	PO: 120	PO: <60
	Nasal: <15				SC: 60–120		
	PO: <30						
Dosage range (mg)	SC: 6	PO: 1.25–5.00	PO: 1.0–2.5	PO: 5–10	PO: 6.25–12.5	PO: 2.5–5.0	PO: 20–40
	Nasal: 5–20	Max PO: 10/24 h	Max PO: 5/24 h	Max PO: 25/24 h	Max PO: 25/24 h	Max PO: 7.5/24 h	Max PO: 80/24 h
	PO: 25–100	Duration PO: <24 h	Duration PO: <24 h	Duration PO: 14–16 h	Duration PO: <24 h	Duration PO: <24 h	Duration PO: 18 h
	Max PO: 200/24 h						
	Duration PO: 2–4 h						

^aChemical Abstracts, American Chemical Society, calculated using Advanced Chemistry Development (ACD/Labs) Software V8.14 for Solaris (1994–2006 ACD/Labs).

^bFirst-pass metabolism.

^cDelayed by food.

^dSlower onset during migraine attack.

^eact= active metabolite

receptors. No tryptamine analog binds at 5-HT_{1E} receptors with substantially higher affinity than 5-HT ($K_i \sim 10$ nM), and even simple *O*-methylation of 5-HT reduces its affinity for this receptor by approximately 100-fold (42). Ergolines, such as ergonovine (Ergotrate), methylergonovine (Methergine), and methysergide (Sansert), bind to 5-HT_{1E} receptors with K_i values in the 50 to 150 nM range (42). Studies indicate that these receptors are negatively coupled to adenylate cyclase. No 5-HT_{1E}-selective receptor agonists or antagonists have yet been reported (43); this has created a problem for investigating this receptor subpopulation. One problem stalling development of selective agents is the lack of 5-HT_{1E} receptors in rodent (i.e., mouse, rat) brain, the animal species commonly employed in preclinical drug development; however, the recent discovery of this receptor type in guinea pig brain bodes well for future studies.

5-HT_{1F} Receptors

The newest 5-HT₁ receptor subpopulation to be cloned is the human 5-HT_{1F} receptor (44), which exhibits intermediate (~50% to 70%) amino acid sequence homology with other 5-HT₁ receptor subpopulations. The receptors are coupled to inhibition of adenylate cyclase. Detection of these receptors in the uterus and mesentery suggests a possible role in vascular contraction. Although their distribution in the brain appears to be limited, distributional similarities with h5-HT_{1B} receptors have been observed. A 4-(3-indolyl)piperidine, LY-334370, and an aminocarbazole, LY-344864, were identified as the first

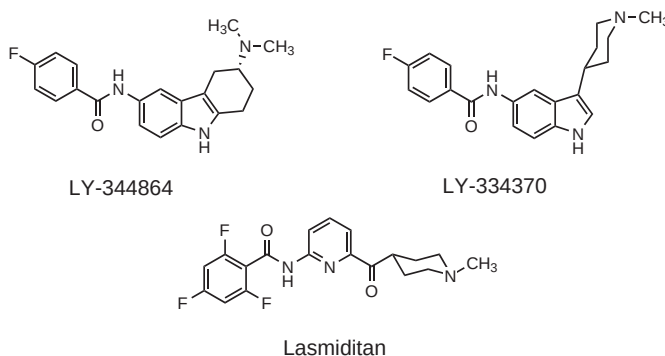
5-HT_{1F}-selective agonists (45) with potential for the treatment of migraine. A more selective (nearly 300-fold more selective over 5-HT_{1E}) agent, lasmiditan, has been recently identified (46). Preliminary evidence suggests that lasmiditan, unlike most of the triptans, will not constrict the coronary artery (46). The nonselective 5-HT₁ receptor antagonist methiothepin has been shown to act as a 5-HT_{1F} receptor antagonist. The SAFIR for the binding of tryptamines at 5-HT_{1F} receptors has been reported (43). Interestingly, there is a statistically significant correlation between the affinities of several dozen tryptamine derivatives at 5-HT_{1E} and 5-HT_{1F} receptors (43), indicating common or similar binding requirements; interestingly, 5-HT_{1F} but not 5-HT_{1E} (vide supra), receptors allow substitution at the tryptamine 5-position. This opens the door for the development of additional 5-HT_{1F}- versus 5-HT_{1E}-selective agents. Many agents that bind at 5-HT_{1E} receptors typically bind as well at 5-HT_{1F} receptors; however, not all 5-HT_{1F} receptor ligands bind at 5-HT_{1E} receptors (see below).

5-HT_{1F} Receptors: Clinical Significance

The clinical significance of 5-HT_{1F} receptors is unknown at this time. The binding of sumatriptan to this receptor population suggests a relationship between 5-HT_{1F} receptor binding and antimigraine activity. Other antimigraine agents, including naratriptan, rizatriptan, and zolmitriptan, also bind at 5-HT_{1F} receptors (37). Studies show that 5-HT_{1D} receptors are the dominant species in human cerebral blood vessels but that 5-HT_{1F} receptors are also expressed both in neural and vascular tissue; however, 5-HT_{1F} receptor agents could have a role in migraine as well (41). Indeed, lasmiditan could represent a prototype for a new generation of antimigraine agents that, because they do not bind at 5-HT_{1B/1D} receptors, are likely to display reduced coronary vasoconstrictor action associated with the triptans.

ERGOLINES

Ergolines, a large group of indole alkaloids with varied effects known for more than 2,000 years, are isolated from the ergot fungus, *Claviceps purpurea*, a plant parasite principally infecting rye. Eating rye grain contaminated with ergot caused a severely debilitating and painful disease during the Middle Ages called St. Anthony's Fire (ergotism), but in small doses, ergot was known to midwives for centuries for its ability to stimulate uterine contractions. Gangrene with burning pain in the extremities was one of two common presentations of ergot poisoning, which also could produce convulsions, hallucinations, severe psychosis, and death. St. Anthony was the patron saint of those who were stricken, and the Order of St. Anthony provided care for these patients. Outbreaks of "dancing mania," which occurred between the 13th and 16th centuries, sometimes have been attributed to ergotism, and one appealing—if unprovable—theory proposes that the women accused of witchcraft in the Salem trials of 1692 were suffering from ergot-induced psychosis and convulsions. The pharmacology of the various ergolines is complex, and they exhibit affinity for α -adrenergic, dopaminergic, and serotonergic receptor systems. The ergolines have been largely displaced by other more selective and effective drugs.



5-HT₂ Receptor Family

Serotonin receptors were first divided into 5-HT₁ and 5-HT₂ receptor families in 1979 (6), and the latter was subsequently divided into the subfamilies 5-HT_{2A}, 5-HT_{2B}, and 5-HT_{2C} (formerly 5-HT_{1C}) receptors. Now, the term "5-HT₂" refers to a receptor family, not to an individual population of receptors. Ketanserin (Fig. 11.9) was

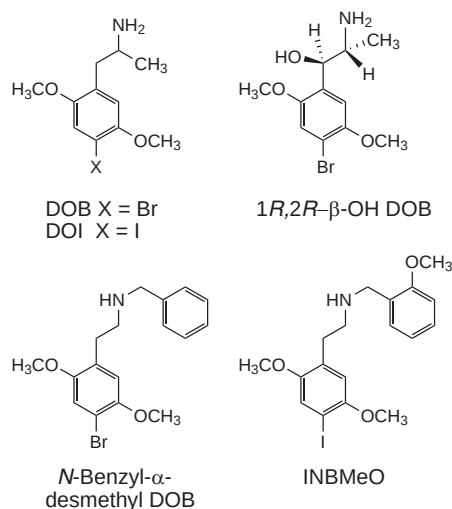
identified early on as a 5-HT₂ receptor antagonist with no affinity for 5-HT₁ receptors, and [³H]ketanserin was introduced as a radioligand to label 5-HT₂ receptors. 1-(2,5-Dimethoxy-4X-phenyl)-2-aminopropane, where X = -Br and -I (DOB and DOI, respectively), was introduced as a 5-HT₂ receptor agonist. A significant amount of pharmacology was published, and structure–activity studies led to the development of many novel agents. Many of the original agents thought to be 5-HT₂ selective, including standard antagonists such as ketanserin and the agonists DOB and DOI, were later shown to bind nonselectively both to 5-HT_{2A} and 5-HT_{2C} receptors. Consequently, pharmacologic actions originally thought to be 5-HT₂ mediated could actually involve 5-HT_{2A} receptors, 5-HT_{2C} receptors, or a combination of 5-HT_{2A} and 5-HT_{2C} receptors. The structures of the three 5-HT₂ receptor subpopulations were found to be consistent with those of transmembrane-spanning G protein–coupled receptors, and the receptors all use a phospholipase C second messenger system. Approximately 70% to 80% sequence homology is found among the three receptor subtypes (10). Only relatively recently have novel agents with subpopulation selectivity been reported.

5-HT_{2A} Receptors

5-HT_{2A} receptors, formerly termed 5-HT₂ receptors, have been extensively reviewed (47–51). 5-HT_{2A} receptors have been cloned from various species, including human, and exhibit a high degree (>90%) of species homology. Significant (78%) amino acid sequence homology is found between the transmembrane portions of cloned 5-HT_{2A} receptors and 5-HT_{2C} receptors; this could explain the observed similarities in the binding of various ligands at the two receptor subpopulations. Evidence was provided that 5-HT_{2A} receptors exist in a high-affinity state and a low-affinity state (sometimes referred to as 5-HT_{2H} and 5-HT_{2L} states, respectively); under normal conditions, the low-affinity state predominates. The tritiated antagonist, [³H]ketanserin, displays comparable affinity for both states, whereas agonists display higher affinity for the high-affinity state (e.g., when a tritiated agonist is employed as radioligand).

5-HT_{2A} AGONISTS The SAFIRs for 5-HT_{2A} receptor binding have been reviewed (47,51). Most indolealkylamines are nonselective 5-HT_{2A} receptor ligands, and typically bind with high affinity at the tritiated agonist-labeled high-affinity state. Investigations suggest that all indolealkylamines could not bind in the same manner at 5-HT_{2A} receptors (52). Phenylalkylamines, such as DOB and DOI, act as 5-HT_{2A} receptor agonists or high-efficacy partial agonists (see Chapter 19) and are significantly more selective than the indolealkylamines because of the low affinity of the former for non-5-HT_{2A} sites, but they do not differentiate between 5-HT₂ receptor subpopulations. [³H]DOB and [¹²⁵I]DOI have been introduced as agonist radioligands (53). Interestingly, although *N*-alkylation of DOB-type

compounds typically results in decreased affinity, it was found that *N*-benzyl- α -desmethyl DOB is a very high-affinity compound and, furthermore, that it behaves as a 5-HT_{2A/2C} agonist (54). The structurally related INBMeO has been introduced as a radioligand to label 5-HT_{2A/2C} receptors (55). Another 5-HT_{2A/2C} agonist with DOB-like effects is the 1*R*,2*R*-isomer of β -hydroxy DOB (β -OH DOB) (56,57).



5-HT_{2A} ANTAGONISTS One of the largest and more selective classes of 5-HT_{2A} receptor antagonists is the *N*-alkylpiperidines. The best-known examples are ketanserin and ritanserin. Although numerous ketanserin-related derivatives have been reported, their SAR still has not been completely defined. Nevertheless, far less than the entire structure of ketanserin is required for high affinity. Some 5-HT_{2A} receptor antagonists, although fairly selective for 5-HT_{2A/2C} receptors versus most other populations of 5-HT receptors, bind with modest to high affinity at dopaminergic, histaminergic, and/or adrenergic receptors. The tricyclic antipsychotics, atypical antipsychotics (risperidone, clozapine, and olanzapine) (Fig. 11.9), and tricyclic antidepressants also bind at 5-HT_{2A} receptors. Spiperone (Fig. 11.7) has been employed as a 5-HT_{2A} receptor antagonist with 1,000-fold selectivity for 5-HT_{2A} versus 5-HT_{2C} receptors, but spiperone is also a potent dopamine receptor antagonist, a 5-HT_{1A} receptor antagonist, and a 5-HT₇ receptor antagonist. Spiperone, volinanserin (MDL 100,907 or M100907), and AMI-193 were the first 5-HT_{2A}-versus 5-HT_{2C}-selective antagonists available (58,59) (Fig. 11.10). The binding selectivity of various antagonists (and agonists) at 5-HT_{2A}, 5-HT_{2B}, and 5-HT_{2C} receptors has been compared (60). Spiperone and AMI-193 bind at 5-HT_{2A} receptors with 1,000- to 3,000-fold selectivity relative to 5-HT_{2C} receptors but display high affinity for 5-HT_{1A} and D₂ dopamine receptors. A newer member of this series, KML-010, is a spiperone-related derivative that lacks affinity for 5-HT_{2C} and 5-HT_{1A} receptors and binds with low affinity at D₂-dopamine receptors (59).

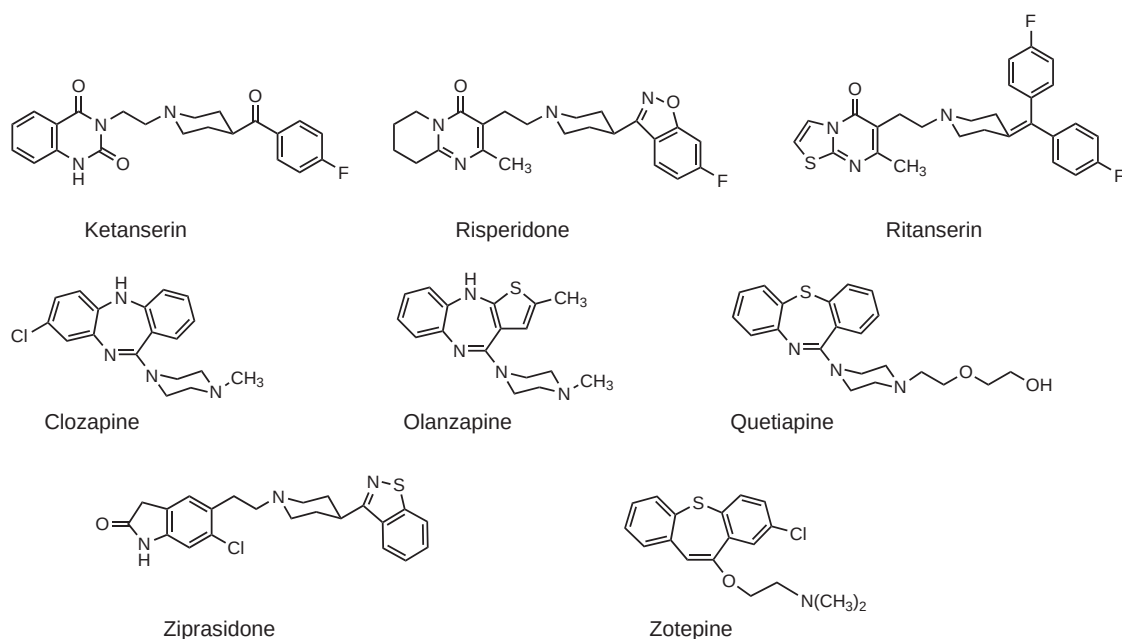


FIGURE 11.9 5-HT₂ receptor antagonists.

Volinanserin is a widely used pharmacologic tool with greater than 100-fold selectivity over most other receptor types (58). Pimavanserin and nelotanserin (ADP-125) display 10-fold and 250-fold selectivity, respectively, for 5-HT_{2A} versus 5-HT_{2C} receptors; one of the most selective antagonists is pruvanserin (EMD-281,014) with about 4,000-fold selectivity (Fig. 11.10).

5-HT_{2A} RECEPTORS: CLINICAL IMPLICATIONS The potential therapeutic roles of 5-HT_{2A} ligands and the possible involvement of 5-HT_{2A} receptors in modulating normal physiologic functions and various pathologic and psychopathologic conditions have been extensively reviewed (3,11,20). 5-HT_{2A} receptors appear to have a role in thermoregulation and sleep, and they could be involved in appetite control, learning, and, along with various

other serotonergic receptor populations, cardiovascular function and muscle contraction. Many of the clinical implications of 5-HT_{2A} receptors could actually involve 5-HT_{2C} receptors or a combination of 5-HT_{2A} and 5-HT_{2C} receptors, due to the high homology between the two receptor populations resulting in many antagonists that bind to both with relatively little selectivity. For example, 5-HT_{2A} (and/or 5-HT_{2C}) antagonists could be useful for the treatment of anxiety (particularly posttraumatic stress disorder) and sleep, cognitive, and mood disorders (61,62). With the recent development of subpopulation-selective agents, this is currently an important area of research. For example, nelotanserin, pimavanserin, pruvanserin, and volinanserin are being examined for their effectiveness in treating insomnia, schizophrenia, depression, and anxiety.

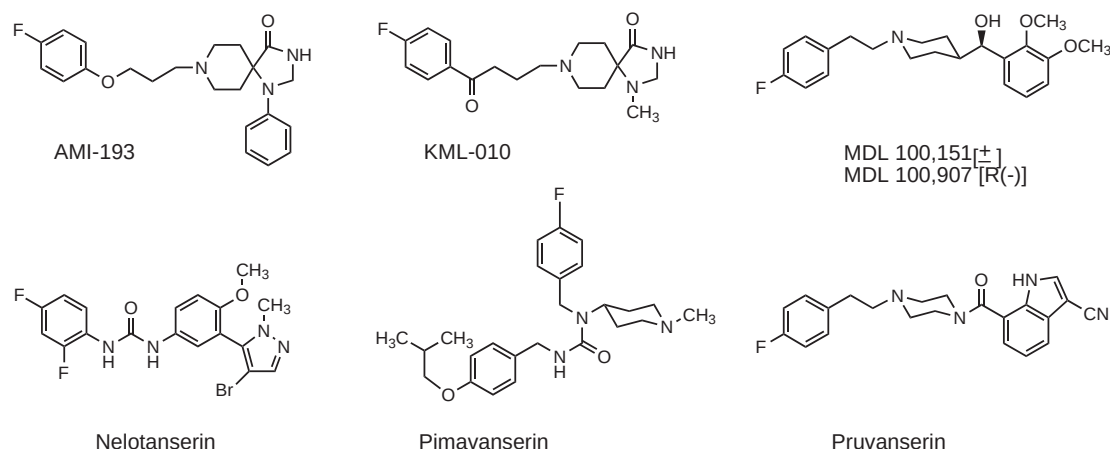


FIGURE 11.10 Examples of 5-HT_{2A} subpopulation selective agents.

Antipsychotic Agents and Antidepressants Various typical and atypical antipsychotic agents (see Chapter 14) and antidepressants (see Chapter 18) bind with relatively high affinity at 5-HT_{2A} receptors as antagonists (15,63). Although no direct correlation exists between their receptor affinities and clinically effective doses, evidence suggests that these disorders involve, at least to some extent, 5-HT_{2A} receptors. For example, chronic administration of 5-HT_{2A} antagonists results in a paradoxical downregulation of 5-HT_{2A} receptors. Such a downregulation would be of benefit in the treatment of depression. Several agents with 5-HT_{2A} antagonist action possess antipsychotic activity; an example is the atypical antipsychotic risperidone. Some 5-HT_{2A} antagonists also bind at dopamine receptors. Indeed, the atypical antipsychotics clozapine, olanzapine, quetiapine, risperidone, ziprasidone, and zotepine, bind both at 5-HT_{2A} and dopamine D₂ receptors (63,64) and often at other serotonergic and nonserotonergic receptors. Although this can obfuscate the role of 5-HT_{2A} antagonism as being important for (atypical) antipsychotic activity, it has been suggested that certain types of schizophrenia could actually be more responsive to the combined effect. That is, D₂-dopaminergic antagonist antipsychotics have been claimed to be more effective for treating the positive symptoms of schizophrenia, whereas the 5-HT_{2A} antagonists could be more effective in treating the negative symptoms; this has led to the development of the serotonin–dopamine antagonists. This theory also suggests that increasing the 5-HT_{2A} component of binding could be related to a decrease in extrapyramidal side effects associated with these types of agents. However, a recent study has found that whereas certain atypical antipsychotics (e.g., clozapine, olanzapine, risperidone) are more efficacious than typical antipsychotics agents against overall positive and negative symptoms of schizophrenia, this was not true for certain others (e.g., quetiapine, ziprasidone, zotepine) (65). Nevertheless, all agents produced decreased extrapyramidal stimulation as an undesirable side effect. From preclinical studies, there are indications that certain 5-HT_{2A} receptor antagonists also possess anxiolytic properties; for example, ritanserin (Fig. 11.9) has been demonstrated to produce both antipsychotic and antianxiety effects in humans.

Classical Hallucinogens 5-HT_{2A} receptors can be involved in the actions of the classical hallucinogens (66) (see Chapter 19). Although indolealkylamine (e.g., 5-methoxy-*N,N*-dimethyltryptamine) and ergoline-related (e.g., LSD) classical hallucinogens are fairly nonselective agents that bind to multiple populations of serotonergic receptors, the phenylalkylamine hallucinogens (e.g., DOB, and DOI) are much more 5-HT₂-selective agonists. Furthermore, a significant correlation exists between the human hallucinogenic potencies of classical hallucinogens and their 5-HT_{2A} receptor affinities (66). Interestingly, phenylalkylamine hallucinogens also bind at 5-HT_{2B} and 5-HT_{2C} receptors, and here, too, a significant correlation is found between human potency

and receptor affinity for 17 different agents (67). Recent studies suggest that 5-HT_{2A} receptors can have a more prominent role than 5-HT_{2B} or 5-HT_{2C} receptors for the behavioral actions of hallucinogens (67), and differences may exist in the manner in which hallucinogens activate the different receptor populations (68,69).

An interesting twist, with potential therapeutic ramifications, was the development of the 1*R*,2*R* isomer of β -OH DOB (56). 5-HT_{2A} receptors are found in the eye, and activation of these receptors can reduce intraocular pressure and could be of benefit for the treatment of glaucoma. However, 5-HT_{2A} agonists such as DOB are hallucinogenic. A DOB analog, β -OH DOB, was developed as a less lipid-soluble version of DOB. With its reduced lipophilicity, and because of its route of administration (ocular installation), the adverse effects of this agent should be minimized. Other 5-HT₂ agonists are now being examined in this regard (57).

5-HT_{2B} Receptors

The rat fundus preparation is a peripheral tissue assay that has been used as a functional assay for serotonergic action for more than 50 years. Long-standing questions concerning the pharmacologic similarity of serotonergic fundus receptors (now called 5-HT_{2B} receptors) to the 5-HT₂ family of receptors were answered once they were cloned (70). The 5-HT_{2B} receptors exhibit approximately 70% homology to 5-HT_{2A} and 5-HT_{2C} receptors, and like 5-HT_{2A} receptors, they appear to couple functionally to phosphoinositol hydrolysis. Nevertheless, rat and human 5-HT_{2B} receptors display more than 90% transmembrane sequence homology. Therefore, most agents that bind at rat 5-HT_{2B} receptors also bind with similar affinity at human 5-HT_{2B} receptors. There are, however, some exceptions (71). The standard 5-HT_{2A} receptor antagonist ketanserin and the 5-HT_{2A} receptor agonists DOI and DOB display higher affinity for 5-HT_{2A} and 5-HT_{2C} receptors than for 5-HT_{2B} receptors (67). Evidence suggests that human 5-HT_{2B} receptors, like human 5-HT_{2A} receptors, also exist in high-affinity and low-affinity states (71). 5-HT_{2B} receptors are found on cardiovascular tissue. Activation of such receptors by agents with a 5-HT_{2B} agonist character could result in cardiac valvulopathy; valvular heart disease associated with the anorectic agent fenfluramine could involve its metabolism to norfenfluramine—a high-affinity 5-HT_{2B} agonist (72). The designer drug 3,4-methylenedioxymethamphetamine (MDMA; Ecstasy) and its *N*-desmethyl analog MDA also show a 5-HT_{2B} agonist character (73). A series of 1-substituted β -carboline (e.g., LY-23728, LY-287375, and LY-266097) have been reported to be the first 5-HT_{2B}-selective antagonists (74).

In the periphery, 5-HT_{2B} receptors seem to be involved in muscle contraction; however, their function in the CNS (if any) is still a matter of speculation. On the basis of some preliminary studies, and considering their central distribution in brain, 5-HT_{2B} receptors could be involved, at least in

rodents, in anxiety, cognition, food intake, neuroendocrine regulation, locomotor coordination, and balance (75).

5-HT_{2C} Receptors

The 5-HT_{2C} receptors, formerly called 5-HT_{1C} receptors, originally were identified in various regions of the brain using autoradiographic and radioligand binding techniques. Cloned human 5-HT_{2C} receptors display a high amino acid sequence homology with 5-HT_{2A} receptors, and like 5-HT_{2A} receptors, they are coupled to phosphoinositol hydrolysis. As previously mentioned, some pharmacologic functions once attributed to “5-HT₂” receptors actually could involve a 5-HT_{2C} receptor mechanism. For example, the hyperthermic activity of a series of phenylisopropylamines is significantly correlated not only with 5-HT_{2A} but also with their 5-HT_{2C} receptor affinity. Numerous atypical antipsychotic agents bind at 5-HT_{2C} receptors as well as at 5-HT_{2A} receptors; however, no significant correlation exists between their atypical properties and binding affinity. 5-HT_{2C} receptors can have a greater role than 5-HT_{2A} receptors in migraine. Other studies suggest that 5-HT_{2C} receptor modulators could be useful in the treatment of obesity, schizophrenia, depression, anxiety, drug abuse, erectile dysfunction, urinary incontinence, and Parkinson disease (76). Several selective agents are now available.

A series of isotryptamine derivatives, including Ro 60-0175 (ORG-35030), has been shown to display high selectivity for 5-HT_{2C} versus 5-HT_{2A} receptors and to possess 5-HT agonist activity (77); however, some of the results could not be replicated (78). Structurally related tricyclic analogs, such as Ro 60-0332 (ORG-35035), also have been examined and display more than 100-fold selectivity (79). 10-Methoxy-9-methylpyrazino[1,2-a]indole, Ro 60-0175, and Ro 60-0332 all were active in animal models predictive of therapeutic utility for obsessive-compulsive disorders, panic disorders, and depression (79). WAY-163909 (Fig. 11.11), a full agonist at 5-HT_{2C} receptors with weak partial agonist action at 5-HT_{2B} receptors and inactive at 5-HT_{2A} receptors, was effective in animal models of obesity, psychotic-like behavior, and depression (80). Lorcaserin (APD-356) (Fig. 11.11) is a 5-HT_{2C} receptor agonist developed for the treatment of obesity (81); although it has completed phase III clinical trials, it has not yet been approved by the U.S. Food and Drug Administration. 1*R*,3*S*(-)-*trans*-1-Phenyl-3-dimethylamino-1,2,3,4-tetrahydronaphthalene, or 1*R*,3*S*(-)-*trans*-PAT (Fig. 11.11), a full agonist at 5-HT_{2C} receptors, is an inverse agonist and competitive antagonist at 5-HT_{2A} and 5-HT_{2B} receptors and produced anorexia in animals (82).

Interestingly, selective 5-HT_{2C} receptor antagonists appear to target some of the same actions as 5-HT_{2C} receptor agonists. Perhaps the first 5-HT_{2C/2B}-selective antagonist was SB-206553, which was identified in the 1990s; continued work with this molecule ultimately resulted in SB-243213—actually, an inverse agonist (83,84). The latter displays greater than 100-fold selectivity over the other two populations of 5-HT₂ receptors and is being

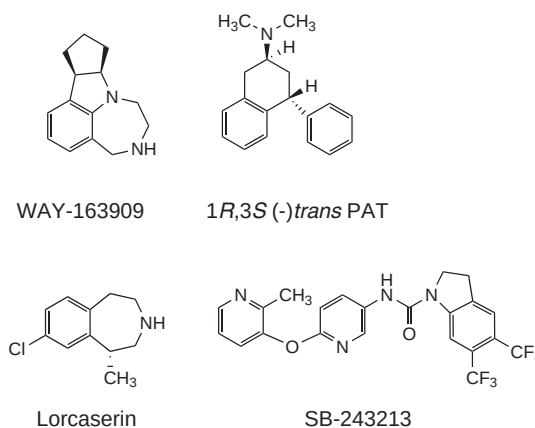
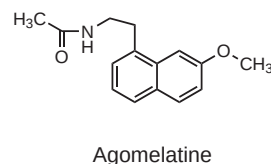


FIGURE 11.11 5-HT_{2C} receptor-selective agonists and antagonists.

examined for its potential use in the treatment of anxiety, depression, and schizophrenia. SB-243213 is claimed to possess an improved anxiolytic profile relative to the benzodiazepines and could have utility in the treatment of schizophrenia and motor disorders (83). It should be noted that agomelatine (Valdoxan), although not strictly a 5-HT_{2C}-selective receptor antagonist, has been found more effective than fluoxetine in a randomized double-blind study in patients with severe major depressive disorder (85), and is currently in clinical trials in the United States. Initially developed as a melatonin (MT) receptor agonist, agomelatine is a nonselective MT₁/MT₂ receptor agonist with 5-HT_{2C} receptor antagonist character (Chapter 18).



It is still not known with confidence specifically what pharmacologic effects are related to what 5-HT₂ receptor subpopulation. However, with the availability of subtype-selective agents, the problem comes closer to being solved.

5-HT₃ Receptor Family

Unlike with most 5-HT receptor populations, early 5-HT₃ pharmacologic studies relied almost exclusively on functional (i.e., peripheral tissue) assays. It was a number of years before radioligands were available to identify 5-HT₃ receptors in brain. 5-HT₃ receptors, ligand-gated ion channel receptors, are members of the Cys-loop family that includes nicotinic acetylcholine, γ -aminobutyric acid (GABA)_A and glycine receptors, and a Zn²⁺-activated cation channel (86,87). They consist of five subunits surrounding a cation-permeable (Na⁺, Ca²⁺, K⁺), water-filled pore. Each subunit is composed of four transmembrane-spanning helices (TM1 to TM4)

with the TM2 domains of each forming the channel pore (Fig. 11.12). Both, a large N-terminus with a Cys-loop and a short C-terminus are located extracellularly (86,87). Approximately 70% to 80% of 5-HT₃ receptors in brain are located presynaptically (87). Evidence indicates that 5-HT₃ receptors modulate release of other neurotransmitters, including dopamine, acetylcholine, GABA, and 5-HT (88).

Five 5-HT₃ subunits have been cloned. 5-HT_{3A}, 5-HT_{3B}, 5-HT_{3C}, and 5-HT_{3E} subunits are similar in their topology, whereas the 5-HT_{3D} subunit lacks most of the N-terminus including the Cys-loop. The 5-HT_{3A} subunit is the only one that forms functional homomeric receptors when expressed in *Xenopus* oocytes. The other subunits are unable to form functional homomeric receptors in vitro, but they can assemble to functional heteromeric receptors with the 5-HT_{3A} subunit (87). This could be explained by lack of a tryptophan residue in the N-terminus of all four subunits (5-HT_{3B}, 5-HT_{3C}, 5-HT_{3D}, and 5-HT_{3E}) shown to be essential for binding. Conversely, the latest reports indicate that subunits 5-HT_{3C}, 5-HT_{3D}, and 5-HT_{3E} could be present on the cell surface when expressed alone in CHO cells (89). 5-HT_{3A}- and 5-HT_{3AB} receptors are the most studied to date. 5-HT_{3AB} receptors differ from 5-HT_{3A} in that they have higher single-channel conductance, a lower Ca²⁺ permeability, faster activation and deactivation, and a lower potency for 5-HT. The subunit composition of recombinant 5-HT_{3AB} receptors in HEK293 cells has been shown to be B-B-A-B-A, but this could not be the

case for native 5-HT₃ receptors (87). The orthosteric ligand binding site is believed to be located at the interface of two adjacent subunits where it is formed by three loops (A to C) of the “principal” and three loops (D to F) of the “complementary” subunit as shown for acetylcholine binding protein and adapted for 5-HT₃ receptors. To fully activate the ion channel of homomeric 5-HT_{3A} receptors, three molecules of agonist are necessary, whereas in the case of heteromeric 5-HT_{3AB} receptors, with presumed stoichiometry of 5-HT_{3(A)2(B)3}, only two agonist molecules are necessary (87).

Structure–Activity Relationships of 5-HT₃ Agonists

Only a few 5-HT₃ receptor agonists have been identified (Fig. 12.13), and the topic has been comprehensively reviewed (90). Many tryptamine analogs bind at 5-HT₃ receptors in a nonselective manner. Simple *O*-methylation of 5-HT significantly decreases its affinity for 5-HT₃ receptors. Ergolines either do not bind or bind only with very low affinity. 5-HT is a nonselective 5-HT₃ receptor agonist that binds only with modest affinity ($K_i \sim 500$ to 1,000 nM). Its 2-methyl analog, 2-methyl 5-HT ($K_i = 1,200$ nM) (Fig. 11.13), is somewhat more selective but binds with slightly lower affinity than 5-HT. Although 2-methyl 5-HT may be only a partial agonist, it has found widespread application in 5-HT₃ research due to its greater selectivity over 5-HT. Recently, however, 2-methyl 5-HT was shown to bind with high affinity at 5-HT₆ receptors (see below). The *N,N,N*-trimethyl quaternary amine analog of 5-HT, 5-HTQ, binds with approximately 10-fold greater affinity and is much more selective than 5-HT; however, because of its quaternary nature, it could not readily penetrate the blood–brain barrier when administered systemically. Using cloned mouse 5-HT₃ receptors, 5-HT and 5-HTQ act as full agonists, suggesting that the quaternary nature of 5-HTQ has little effect on efficacy, whereas 2-methyl 5-HT and tryptamine act as partial agonists. Another example of a low-affinity ($K_i \sim 1,000$ nM) 5-HT₃ agonist is phenylbiguanide. *m*-Chlorophenylbiguanide (*m*CPBG), which binds in the low nanomolar range ($K_i \sim 20$ to 50 nM) and retains agonist character, has largely replaced phenylbiguanide. Because of its polar nature, *m*CPBG does not readily penetrate the blood–brain barrier. *m*-Chlorophenylguanidine (MD-354; *m*CPG) shows that the entire biguanide moiety is not required for serotonergic activity. Adding multiple chloro groups to *m*CPBG or *m*CPG increases their lipophilicity and affinity (90).

Simple arylpiperazines were among the first serotonergic agents investigated at 5-HT₃ receptors (Fig. 12.13). Many are nonselective 5-HT₃ ligands (see previous discussion of 5-HT_{1A} receptors). Depending on the particular substitution pattern, they can behave as 5-HT₃ agonists, partial agonists, or antagonists (90). This nonselectivity probably accounts for the initial lack of interest in arylpiperazines as 5-HT₃ ligands, but today, there is renewed interest in these

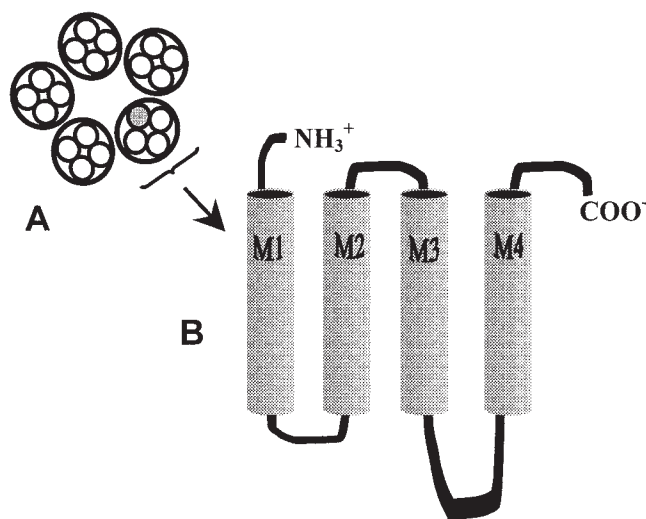


FIGURE 11.12 Top (A) and side (B) views of a schematic representation of an idealized ion channel receptor (such as the 5-HT₃ receptor). Ion channel receptors are pentameric units arranged to form a pore or ion channel. Each subunit consists of four transmembrane-spanning amino acid chains (M₁–M₄) constructed such that the M₂ chain faces the channel. The transmembrane portions are connected by extracellular and intracellular loops. In the serotonin family, only 5-HT₃ receptors have been identified as ion channel receptors.

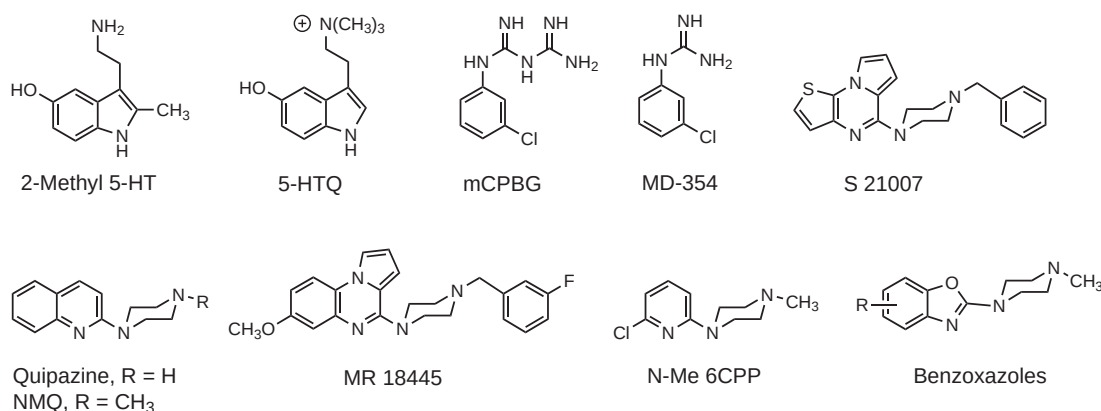


FIGURE 11.13 5-HT₃ receptor agonists or partial agonists.

types of agents. Quipazine was the first arylpiperazine shown to bind at 5-HT₃ receptors, even though it is also a 5-HT_{2A} agonist. It binds with much higher affinity than 5-HT at 5-HT₃ receptors ($K_i \sim 1$ nM) and was subsequently shown to act as an agonist in certain assays and as an antagonist in others. Interestingly, its structure was quite different from that of other 5-HT₃ antagonists known at that time. Early structure–affinity studies showed that its fused pyridine ring attached to *N*₄-piperazine nitrogen distance (~ 5.5 Å) was similar to that of 5-HT. Other findings indicated that 1) the *N*₄-piperazine nitrogen atom, but not the *N*₁-piperazine nitrogen atom, was important for binding; 2) the quinoline ring nitrogen atom was a major contributor to binding; 3) the benzene ring portion of the quinoline nucleus was not required for binding, but its presence was optimal for high affinity; and 4) *N*₄-methylation (*N*-methylquipazine) enhances 5-HT₃ receptor selectivity (Fig. 11.13) (90). With the availability of newer arylpiperazines, it has been possible to conduct more comprehensive structure–activity studies. A summary of quipazine SAR is shown in Figure 11.14; results of other SAR studies and several pharmacophoric models have been described (90). Appropriate structural modification of arylpiperazines can result in rather selective 5-HT₃ agonists (Fig. 11.13). For example, ring-fused quipazine-related analogs, such as the pyrrolo[1,2-*a*]quinoxalines, represent novel 5-HT₃ receptor agonists. Some are full agonists, whereas others (e.g., MR 18445) are partial agonists (90–93).

Structure–Activity Relationship of 5-HT₃ Antagonists

Bemesetron (MDL-72222) was the first selective 5-HT₃ receptor antagonist (Fig. 11.15). Its development stems from the structural modification of cocaine, an agent that had been previously shown to be a weak 5-HT₃ antagonist. Since then, many hundreds of 5-HT₃ antagonists have been identified as antiemetics (93,94). Many of these

agents belong to the structural class of compounds broadly referred to as keto compounds and contain an amide, reverse amide, ester, reverse ester, carbamoyl, or ketone function. Typical of these 5-HT₃ antagonists is retention of the bulky tropane or tropane-like amine group. Some of the more widely used or newer antiemetic agents include dolasetron (Anzemet), granisetron (Kytril), itasetron, renzapride, ricasetron, tropisetron, WAY-100289, zacopride, and zatosetron. It should be noted that some of these keto compounds also bind at 5-HT₄ receptors.

A related group of antagonists that possess an imidazole or related heterocyclic terminal amine include ondansetron (Zofran), alosetron (Lotronex), fabesetron, and ramosetron (Fig. 11.16). Many others have been described (93,94). The SARs of 5-HT₃ antagonists have been reviewed in detail (93–95).

Studies have identified pharmacophoric features (Fig. 11.17) that are common to many 5-HT₃ receptor antagonists.

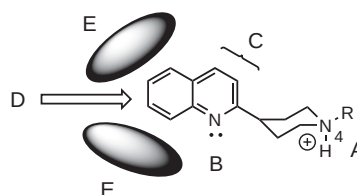


FIGURE 11.14 Structure–activity composite for quipazine analogs binding at 5-HT₃ receptors. (A) The *N*₄-piperazine nitrogen atom, but not the *N*₁-piperazine nitrogen atom, is important for binding; a 5-position-CH₃ is tolerated and results in somewhat greater 5-HT₃ selectivity. (B) The quinoline nitrogen atom is required for high affinity, and its replacement by an sp²-hybridized carbon atom results in a more than 100-fold decrease in affinity. (C) Substituents in this region are tolerated and can influence intrinsic activity. (D) An aromatic moiety (e.g., benzene ring or isosteric structure), although not required for binding, results in optimal affinity. (E) Regions of limited bulk tolerance.

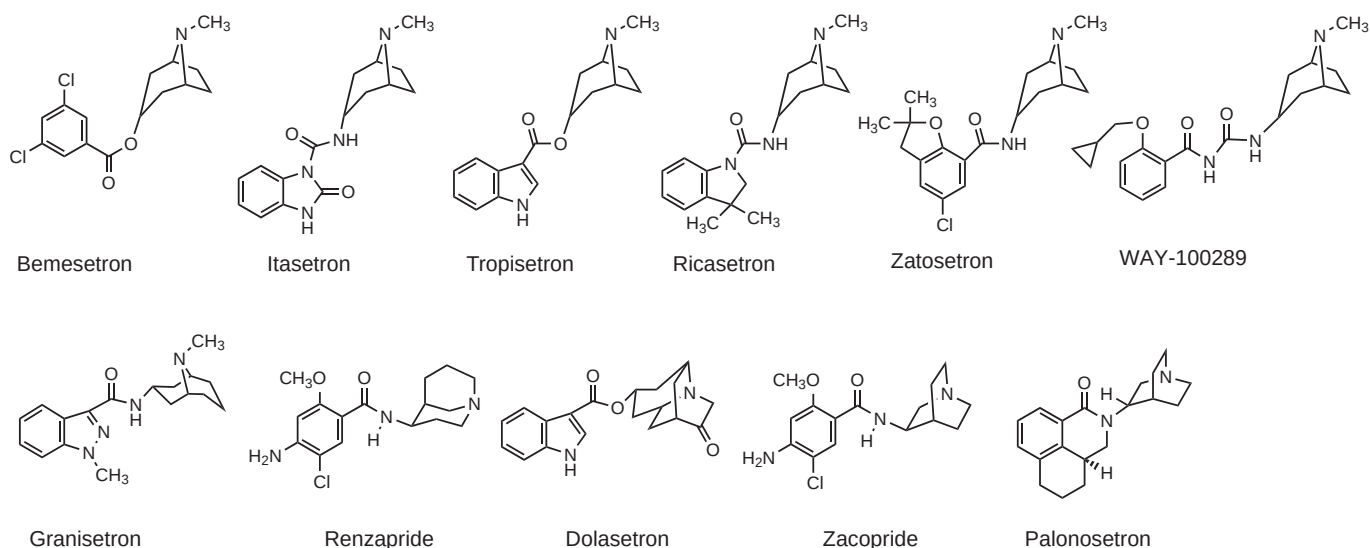


FIGURE 11.15 5-HT₃ receptor antagonists.

5-HT₃ Receptors: Clinical Implications

One of the most noteworthy clinical success stories in 5-HT research relates to the antiemetic properties of 5-HT₃ receptor antagonists. Ondansetron was introduced as an antiemetic in the 1990s, and 5-HT₃ receptor antagonists are now the “gold standard” for treatment of chemotherapy- and radiation-induced nausea and vomiting (96). Twenty or so years ago, nausea and vomiting were inevitable side effects that forced many patients to delay or avoid chemotherapy (97). With the current antiemetic therapy, nausea and vomiting can be prevented in nearly 80% of patients (97). The most commonly employed 5-HT₃ receptor antagonists are ondansetron, granisetron, dolasetron and, in Europe, tropisetron; a newer 5-HT₃ antagonist in clinical use in the United States is palonosetron (98). The various 5-HT₃ antagonists are commonly perceived as being of comparable efficacy and safety (99); however, they vary widely in their pharmacologic and

pharmacokinetic properties (96–99) (Table 11.4). For example, their duration of action and elimination half-lives differ considerably. Ondansetron displays the shortest half-life (~4 hours), whereas the half-life of palonosetron has been reported to be up to 128 hours (99). Another difference in their pharmacology is that ondansetron is a competitive 5-HT₃ antagonist, whereas granisetron and tropisetron (and, perhaps, palonosetron) produce an insurmountable antagonism (98,99). Selection of a particular 5-HT₃ antiemetic follows specific guidelines that are related, at least in part, to such factors as the emeticity of the chemotherapeutic regimen, side effect tolerability, patient history, and financial considerations. Patients who are refractory to the effect of a particular antiemetic may benefit by switching antiemetic agents—improvement could be related to different routes of metabolism (96) (Table 11.4).

Preclinical and limited clinical studies suggest that 5-HT₃ receptor antagonists could potentially be of benefit for the treatment of alcohol and substance abuse, anxiety, autism, bipolar disorder, cognitive impairment,

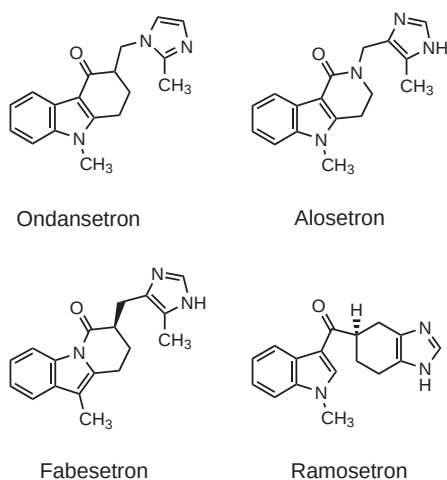


FIGURE 11.16 Imidazole-containing 5-HT₃ receptor antagonists.

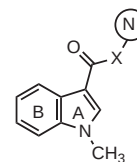


FIGURE 11.17 A composite pharmacophore model for 5-HT₃ receptor antagonists. An aromatic centroid (A) to oxygen (O) distance of 3.3 to 3.5 Å is thought to be optimal. Distances calculated from the terminal amine (N) to the oxygen atom (O) and from the terminal amine N to centroid A are from 5.1 to 5.2 and from 6.7 to 7.2 Å, respectively. It has been speculated that ring A is not required for binding and acts as a spacer. Ring B can be more important for binding, and associated hydrophobic binding regions have been proposed.

TABLE 11.4 Pharmacokinetics of the 5-HT₃ Antagonists (Setrons)

Parameters	Ondansetron	Dolasetron	Granisetron	Alosetron	Palonosetron	Tropisetron
Trade name	Zofran	Anzemet	Kytril	Lotronex	Aloxi	Navoban
CLogP (calc) ^a	2.1 ± 0.5	2.8 ± 0.5	1.5 ± 0.5	0.88 ± 0.8	2.6 ± 0.5	3.6 ± 0.3
LogD (pH 7) (calc) ^a	1.5	2.8	-1.5	0.4	0.01	0.8
Bioavailability (%)	56–70 ^d	Hydrodolasetron: 60–80	60 ^d	50–60 ^{bc}	IV	60 (60–100)
Protein binding (%)	70–76	Hydrodolasetron: 70–80	65	82	62	71
Volume of distribution (L/kg)	PO: 2.2–2.5	Hydrodolasetron: PO: 5.8–10	PO: 3.9	PO: 70 (65–95)	IV: 6.8–12.5	IV: 500
		PO: 6–7				
Elimination half-life (h)	PO: 3–6	PO: <10 min	IV: 4–5	PO: 1.5–2.0	PO: 30–40	EM: PO: 6–8
	Elderly: PO: 11	Hydrodolasetron: PO: 4–9	PO: ~6			PM: PO: 30
Major metabolites (%)	Hydroxylation	Hydrodolasetron	<i>N</i> -Demethyl	6-Hydroxylation	<i>N</i> -oxide	Hydrolyzation
	Glucuronidation	Hydroxylation	Hepatic	<i>N</i> -Demethyl	6S-Hydroxy	Glucuronides
	Hepatic	<i>N</i> -Demethyl			Hepatic: 50%	
Metabolizing enzyme (%)	CYP3A4	Carbonyl reductase	CYP3A4	CYP2C9: 30	CYP2D6	CYP2D6
	CYP2D6	CYP2D6		CYP3A4: 20	CYP3A4	
		CYP3A4 (<i>N</i> -oxide)		CYP1A2: 10	CYP1A2	
Time to peak plasma concentration (h)	PO: 1–2	Hydrodolasetron IV: <0.5	PO: 2–3	PO: 0.5–2	IV: 30 s	EM: PO: 3
		Hydrodolasetron PO: <1				PM: PO: 4
Excretion (%)	Urine metab: 40–60	Urine metab: 45	Urine metab: 48	Urine metab: 70	Urine metab: 80	Urine metab: ~70
	Feces metab: 25	Feces: 30	Feces metab: 38	Feces metab: 25		Feces metab: ~15
	Unchanged: <10	Unchanged hydrodolasetron: 60	Unchanged: <10	Unchanged: <10	Unchanged: 40	Unchanged: <10
Duration (h)	————	————	8–24	1–10	>24	————

IM, intramuscular; IV, intravenous; PO, oral.

^a Chemical Abstracts, American Chemical Society, calculated using Advanced Chemistry Development (ACD/Labs) Software V8.14 for Solaris (1994–2006 ACD/Labs).

^b First-pass metabolism.

^c Food delays absorption and peak plasma concentrations.

^d Food increase extent of absorption.

depression, eating disorders, gastrointestinal disorders, pain, and schizophrenia (87).

Very little is known about the potential therapeutic utility of 5-HT₃ receptor agonists (90).

5-HT₄ Receptors and Agents

A novel population of serotonergic receptors, originally identified in primary cell cultures of mouse embryo colliculi neurons and later termed 5-HT₄ receptors, have broad tissue distribution and are positively coupled to adenylate cyclase (100). In the brain, 5-HT₄ receptors appear to be localized

on neurons and can mediate the slow excitatory responses to 5-HT. Peripherally, these receptors facilitate acetylcholine release in guinea pig ileum and can have a role in peristalsis. The uniqueness of this receptor type and its potential therapeutic utility spurred initial interest in drug development. Human 5-HT₄ receptors have been cloned and display low transmembrane sequence homology (<50%) with other 5-HT receptors. In fact, two 5-HT₄ isoforms have been isolated, a long form (5-HT_{4L}) and a short form (5-HT_{4S}). These isoforms are splice variants and differ only in their C-terminus ends, with identical transmembrane regions

(101). In general, the potency of agonists to stimulate cyclic adenosine monophosphate (cAMP) was greater for the 5-HT_{4S} receptor than for the 5-HT_{4L} receptor. A mouse 5-HT_{4L} receptor has been cloned, and a human pseudogene has been identified that codes for a 5-HT₄-like receptor. Indeed, several new human 5-HT₄ receptor isoforms have been cloned and expressed (102). The new 5-HT₄ receptors have been termed 5-HT_{4(b)}, 5-HT_{4(c)}, and 5-HT_{4(d)}; the stimulatory pattern of cAMP formation in response to the 5-HT₄ agonist renzapride was found to be different for the various isoforms, suggesting that the splice variants could differ in the manner by which they trigger signal transduction following receptor activation (102). In the rat gastrointestinal tract, both 5-HT_{4L} and 5-HT_{4S} receptors are expressed, whereas only 5-HT_{4S} receptors are found in the heart, with localization almost exclusively in the atrium. The 5-HT_{4a} receptors have been cloned from human atrium and appear to correspond to the rodent 5-HT_{4S} isoform. It has been proposed that the cardiac effects of 5-HT are mediated by this short splice variant, whereas 5-HT_{4L} determines the neuronal effects of 5-HT (102).

Although 5-HT₃ receptors are ion channel receptors and 5-HT₄ receptors represent G protein-coupled receptors (Table 11.2), a number of 5-HT₃ receptor ligands are active at 5-HT₄ receptors. Even more interesting is that a number of 5-HT₃ antagonists, or what were considered at one time to be 5-HT₃-selective antagonists (e.g., renzapride and zacopride), actually exhibited 5-HT₄ agonist activity. Even today, there is considerable structural similarity among various 5-HT₃ and 5-HT₄ receptor ligands. In addition to their lack of selectivity for 5-HT₄ versus other 5-HT receptors, many early 5-HT₄ receptor ligands suffered from several other disadvantages, such as their affinity for other receptor types, inability or difficulty in penetrating the blood-brain barrier, and hydrolytic instability (103).

Structure-Activity Relationships of 5-HT₄ Agonists

In general terms, 5-HT₄ agonists can be divided into several different categories (Fig. 11.18) (90): tryptamines (e.g., 5-HT and 5-CT, with 2-methyl 5-HT and 5-methoxy-*N,N*-dimethyltryptamine being nearly inactive),

benzamides (particularly those bearing a 2-methoxy-4-amino-5-chloro substitution pattern, e.g., SC 53116, renzapride, zacopride, and cisapride), benzimidazolones (e.g., BIMU 8), quinolines (e.g., SDZ 216,908), naphthalimides (e.g., SRS 56532), benzoates (ML-10302), and ketones (e.g., RS 67333).

Structure-Activity Relationships of 5-HT₄ Antagonists

The 5-HT₃ antagonist tropisetron was the first agent to see application as a 5-HT₄ antagonist, and its low affinity for 5-HT₄ receptors prompted a search for higher affinity agents. Various agents have been identified (94,104,105), and 5-HT₄ antagonists are derived from structural classes similar to those from which the 5-HT₄ agonists are derived. These include indole esters and amides (e.g., GR 113,808), benzoates (e.g., SB 204070), benzimidazolones (e.g., DAU 6285), imidazoles (e.g., SC 53606), and ketones (e.g., RS 100235) (Fig. 11.19). These are just a few representative examples of the many agents that have been examined as 5-HT₄ antagonists. Structure-activity details for several different receptor preparations have been reviewed (94,104,105). It is worth noting that apart from 5-HT₃ receptors, 5-HT₄ receptors are the only other population of serotonergic sites that seem to accommodate quaternary amines.

5-HT₄ Receptors: Clinical Implications

Selective 5-HT₄ agents have been recently developed, and studies regarding their clinical potential are still in their infancy. Peripheral actions currently being examined include irritable bowel syndrome (IBS), gastrointestinal tract motility, bladder contraction, gastroesophageal reflux, corticosteroid secretion, and atrial contractility. Cisapride is available as a prokinetic drug that enhances gastrointestinal activity. With respect to their central effects, it has been suggested that 5-HT₄ agonists can restore deficits in cognitive function and that 5-HT₄ antagonists could be useful as anxiolytics or in the treatment of dopamine-related disorders. It is further speculated that 5-HT₄ receptors may be involved in memory and learning, and it has been noted that 5-HT₄ receptors

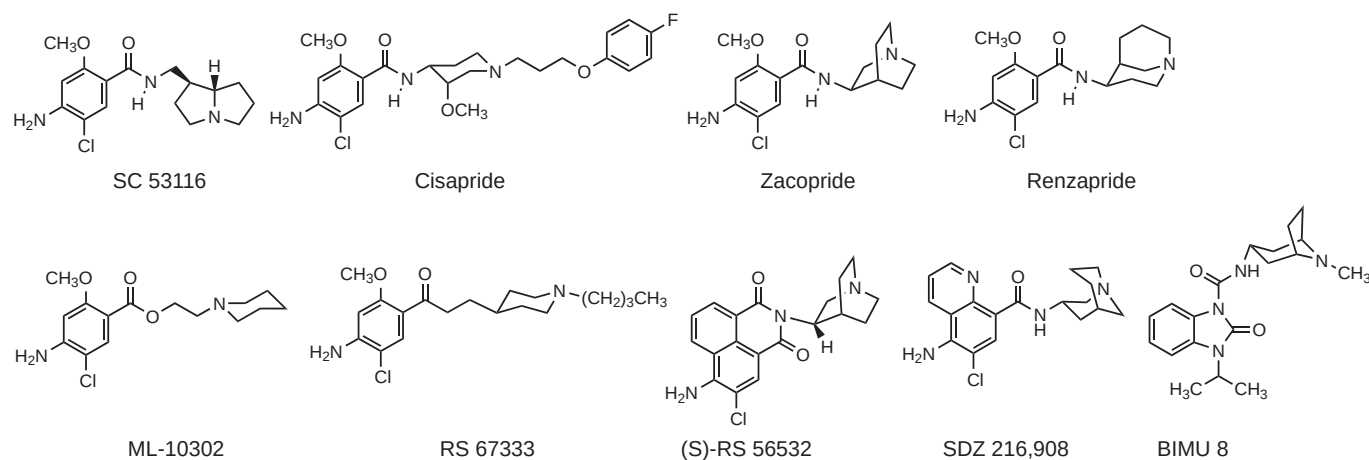


FIGURE 11.18 5-HT₄ receptor agonists.

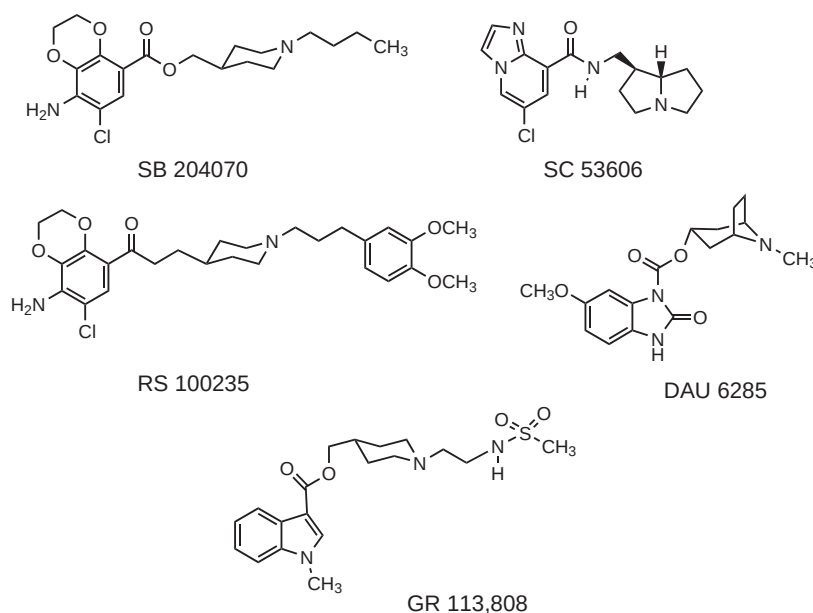


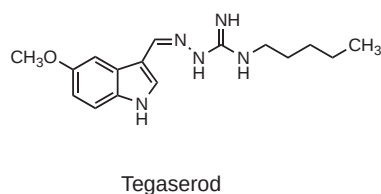
FIGURE 11.19 5-HT₄ receptor antagonists.

are markedly decreased in patients with Alzheimer disease (106,107). A high density of 5-HT₄ receptors in the nucleus accumbens has led some to speculate that these receptors may be involved in the reward system and that they could influence drug self-administration behavior. Other central roles are also beginning to emerge; for example, repeated administration of antidepressants decreases the responsiveness of central 5-HT₄ receptors to activation (108). It would appear that therapeutic roles exist for both 5-HT₄ antagonists and 5-HT₄ agonists. However, it has been cautioned that the use of highly potent and selective 5-HT₄ agonists could result in cardiovascular side effects (107). If different 5-HT₄ receptor isoforms can be shown to mediate the various effects for which 5-HT₄ receptors have been implicated, the potential exists for the development of selective agents. Another problem associated with 5-HT₄ agents is their lack of oral bioavailability (109).

IBS is one of the most common gastrointestinal disorders in the United States, accounting for more than 3.5 million doctor visits per year (110). IBS is characterized by abdominal discomfort or pain associated with altered bowel function (i.e., constipation [IBS-C]), diarrhea (IBS-D), or alternating constipation and diarrhea. Until recently, treatment has been limited by the poor efficacy or side effects of available agents. Agents commonly used to treat IBS include laxatives, antispasmodics and smooth muscle relaxants (e.g., dicyclomine and hyoscyamine), and tricyclic antidepressants, but only 40% of patients are satisfied with these medications (110). Because more than 95% of all 5-HT in the body is found in the gut, it would seem logical that serotonergic agents should be of benefit in the treatment of IBS.

In general, peristaltic and secretory reflexes are initiated by 5-HT acting at 5-HT_{1P} receptors (111)—that is, a population of 5-HT receptors found only in the gut. 5-HT₃

receptors are associated with excitation of the gastrointestinal tract, resulting in increased motility, secretion, and excitation (110) as well as signaling to the CNS (111); 5-HT₃ antagonists reduce colonic transit and improve fluid absorption (110). The 5-HT₃ antagonists tend to be constipating (111). The 5-HT₄ receptors mediate both excitatory and inhibitory effects on gut function (110).



Alosetron (Fig. 11.16), a 5-HT₃ antagonist, and tegaserod (Zelnorm), a 5-HT₄ agonist, are two of the most recent entries for the treatment of IBS. Recent clinical trials have found that both agents are more effective than placebo for the treatment of IBS-C and IBS-D (110). Tegaserod acts by accelerating small bowel and colonic transit in patients with IBS. It is rapidly absorbed following oral administration, with a bioavailability of approximately 10%, except that food reduces the bioavailability by 40% to 65%. Peak plasma concentrations are reached in approximately 1 hour. Tegaserod is approximately 98% bound to plasma proteins, primarily to α_1 -acid glycoprotein. Its volume of distribution is approximately 368 L/kg. Tegaserod undergoes presystemic acid-catalyzed hydrolysis in the stomach, followed by hepatic oxidation to its principal inactive metabolite (5-methoxyindole-3-carboxylic acid), its acyl glucuronide, and three isomeric *N*-glucuronides. The terminal half-life is approximately 11 hours following intravenous administration. Approximately two-thirds of the orally administered dose of tegaserod is excreted unchanged in feces,

with the remainder excreted in urine, primarily as glucuronide. Tegaserod exhibits dose-proportional kinetics when given twice daily at therapeutic doses for 5 days, with no relevant accumulation. No dosage adjustment is required in elderly patients or those with mild to moderate hepatic or renal impairment. No clinically relevant drug–drug interactions have been identified with tegaserod. However, in 2007, tegaserod was removed from the U.S. market due to increased risks of heart attack or stroke and was made available only through a restricted distribution program. As of 2008, tegaserod is available only in emergency life-threatening situations.

5-HT₅ Receptors and Agents

Two 5-HT₅ receptors, expressed primarily in the mouse CNS, have been identified as 5-HT_{5A} and 5-HT_{5B} receptors (112). The two 5-HT₅ receptors exhibit 77% amino acid sequence homology but less than 50% homology with other cloned serotonergic receptors. To some extent, 5-HT₅ receptors appear to resemble 5-HT₁ receptors (e.g., high affinity for 5-HT and 5-CT); however, their low homology with other 5-HT₁ receptors suggests that they represent a distinct family of receptors. Only 5-HT_{5A} receptors have been identified in humans (112). Human 5-HT_{5A} receptors are G protein–coupled receptors with a complex second messenger system (113).

Radiolabeled LSD binds to both 5-HT_{5A} and 5-HT_{5B} receptors, with 5-CT having 10-fold greater affinity for human 5-HT_{5A} receptors than 5-HT, which binds with modest affinity ($K_i = 100$ to 250 nM). The SAR for the binding of various ligands at 5-HT_{5A} receptors has been reviewed elsewhere (114). Ergotamine and methiothepin bind with high affinity at human 5-HT_{5A} receptors, whereas agents such as spiperone, sumatriptan, yohimbine, ketanserin, propranolol, zacopride, and clozapine bind with much lower affinity ($K_i > 1,000$ nM). To date, no 5-HT_{5A}–selective agonists or antagonists have been reported.

5-HT₅ Receptors: Clinical Implications

Pharmacologic functions of 5-HT₅ receptors are currently unknown. It has been speculated, on the basis of their localization, that they could be involved in motor control, feeding, anxiety, depression, learning, memory consolidation, adaptive behavior, and brain development (112). 5-HT_{5A} receptors also could be involved in a neuronally driven mechanism for regulating astrocyte physiology with relevance to gliosis; disruption of 5-HT neuronal–glial interactions can be involved in the development of certain CNS pathologies, including Alzheimer disease, Down syndrome, and some drug-induced developmental deficits. Recent evidence indicates that genes that encode for the human 5-HT_{5A} receptor could be involved in schizophrenia (115) and that spinal 5-HT_{5A} receptors could have a role in nociception and control of pelvic floor musculature (116).

5-HT₆ Receptors and Agents

A novel G protein–coupled serotonergic receptor that appears to be localized exclusively in the CNS was cloned from rat brain and named 5-HT₆. This receptor exhibits only 40% transmembrane homology with 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, 5-HT_{2A}, and 5-HT_{2C} receptors. Both LSD and 5-HT display modest affinity for 5-HT₆ receptors ($K_i \sim 50$ to 150 nM). Of interest is that a number of typical and atypical antipsychotic agents and tricyclic antidepressants bind with K_i values in the nanomolar range.

The human 5-HT₆ receptor was cloned, and its gene structure, distribution, and pharmacology were found to be similar to those of the rat receptor (117). Like the rat receptor, the human receptor is positively coupled to adenylate cyclase. 5-HT binds at human 5-HT₆ receptors with moderate affinity ($K_i = 65$ nM), and one of the highest affinity, albeit nonselective, agents is the antipsychotic methiothepin ($K_i = 0.4$ nM). Agents that bind at human 5-HT₆ receptors with K_i less than 50 nM include 5-methoxytryptamine, bromocriptine, octoclotheptin, the atypical antipsychotic agents clozapine and olanzapine, and the typical antipsychotics chlorpromazine, loxapine, and fluphenazine (118). Agents with K_i greater than 500 nM include 5-CT, sumatriptan, quipazine, ketanserin, 8-OH DPAT, haloperidol, risperidone, and mesulergine (118). A number of other antipsychotic agents, both typical and atypical, as well as antidepressants bind with low nanomolar affinity (114,117,118).

5-HT₆ Agonists and Antagonists

2-Ethyl-5-methoxy-*N,N*-dimethyltryptamine (EMDT) represented the first reasonably selective 5-HT₆ agonist (119), and Ro 04-6790 and Ro 63-0563 represented the first 5-HT₆–selective antagonists (120) (Fig. 11.20). These were soon followed by the antagonists SB-271046 (121), MS-245 (119,122), and PMDT (2-phenyl-5-methoxy-*N,N*-dimethyltryptamine, also known as BGC20-761) (119). Since then, work has continued on related structure types, leading to agents with greater metabolic stability and bioavailability (114). It should be noted that most of the early 5-HT₆ antagonists contained an arylsulfonamide moiety. Interestingly, the importance of this functionality was an independent discovery from several different laboratories, and this structural feature is now commonplace among many 5-HT₆ agents. In some cases, the sulfonamido moiety can be replaced by a sulfone (e.g., naphthylsulfones) (Fig. 11.20). Numerous structural modifications have now been reported, and one of the interesting findings is that the basic side chain of MS-245–type compounds can be moved to the indole 4-position (Fig. 11.20) (123,124) or removed altogether (e.g., amino-BSS) (114) with retention of antagonist action. 5-Sulfonamidotryptamines also bind with high affinity at 5-HT₆ receptors and, depending on the nature of their pendent substituents, act as 5-HT₆ agonists, partial agonists, or antagonists (125,126). A comprehensive review of 5-HT₆–related agents and their SARs has been published (127).

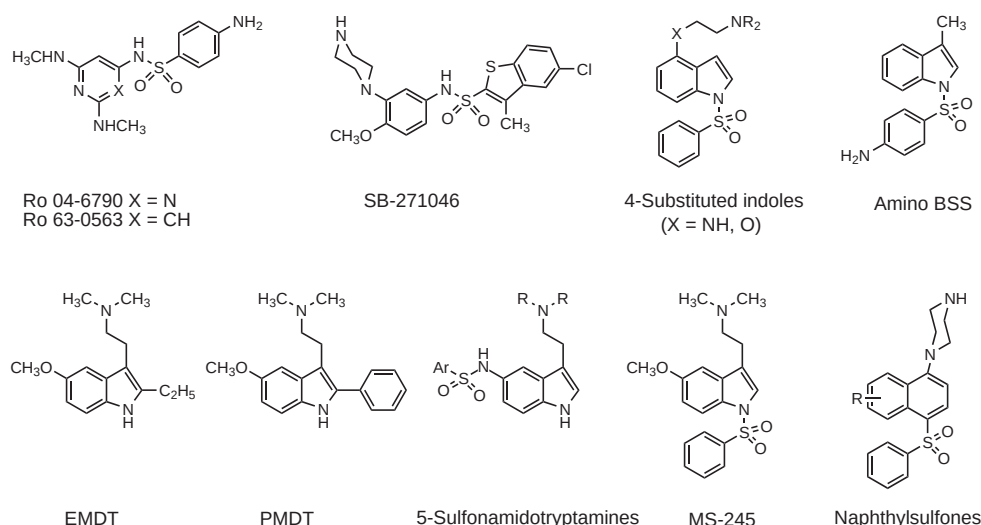


FIGURE 11.20 5-HT₆ receptor agonists and antagonists.

5-HT₆ Receptors: Clinical Implications

The exact clinical significance of 5-HT₆ receptors is unknown at this time. The high affinity of various antipsychotics, particularly atypical antipsychotics (see Chapter 14), and antidepressants suggests a possible connection between 5-HT₆ receptors and certain psychiatric disorders (128). The different binding profiles of atypical antipsychotics can be responsible for their atypical nature (e.g., D₂:5-HT_{2A} ratio); for example, certain agents, such as clozapine, can be classified as atypical on the basis of their binding with higher affinity at 5-HT_{2A} than at D₂ receptors. However, antipsychotics that produce the fewest extrapyramidal side effects in humans (e.g., clozapine, olanzapine, and fluperlapine) also possess high affinity for 5-HT₆ receptors (118). The atypical antipsychotic agent risperidone, which produces some extrapyramidal symptoms, binds with 1,000-fold higher affinity at 5-HT_{2A} than at 5-HT₆ receptors; thus, affinity of agents for 5-HT₆ receptors can contribute to the difference between typical and certain atypical antipsychotics (118). Furthermore, preclinical studies indicate that combinations of a 5-HT₆ antagonist and a 5-HT_{2A} antagonist were effective in models of psychosis and cognition (129). PMDT differs from most other 5-HT₆ antagonists in that it combines both types of antagonist action in the same molecule (114). In 5-HT₆ knockout mice, a behavioral syndrome is produced that seems to involve an increase in cholinergic function. Blocking the receptors in rats with 5-HT₆ antagonists produces a similar effect. This has led to speculation that one of the roles of 5-HT₆ receptors may be to control cholinergic neurotransmission and that 5-HT₆-selective antagonists could be useful in the treatment of anxiety and memory deficits. Other studies have shown that although 5-HT₆ antagonists could not influence basal levels of dopamine by themselves, they apparently increase amphetamine-induced increases in brain dopamine and can potentiate certain dopamine-mediated behavioral effects (130,131). The exact mechanisms

underlying this process are not understood, but 5-HT₆ receptors have a role in neuronal plasticity (132) and can influence the actions of dopaminergic agents. Evidence also suggests that 5-HT₆ receptors could be involved in motor function, mood-dependent behavior, anxiety disorders, appetite control, anticonvulsant activity, and early growth processes involving 5-HT (117,119,123). With the newly identified 5-HT₆ agonists and antagonists, interest in the therapeutic potential of such agents is on the upswing.

5-HT₇ Receptors and Agents

Like 5-HT₅ receptors, 5-HT₇ receptors were once considered to be orphan receptors. Rat, mouse, guinea pig, and human 5-HT₇ receptors have now been cloned and are expressed mainly in the CNS (114,133). Structural analysis of the 5-HT₇ receptor suggests a seven transmembrane-spanning G protein-coupled receptor. These receptors are positively coupled to adenylate cyclase, and several splice variants have been identified. Alternative splicing in rat and human receptors results in four 5-HT₇ receptor isoforms that vary with respect to the length of their C-terminus chains (114,134). In rat, the isoforms are named 5-HT_{7a}, 5-HT_{7b}, and 5-HT_{7c}. Two of the isoforms are homologous in rat and human (5-HT_{7a} and 5-HT_{7b}). The third human isoform is named 5-HT_{7d}. These different isoforms could have important functional consequences, such as different distribution or G protein-coupling efficiency or different susceptibility to desensitization (134,135). Apparently, the three human isoforms are pharmacologically indistinguishable and show similar affinity for various ligands. Evidence suggests that 5-HT₇ receptors are constitutively active and that the degree of constitutive activity could vary among the isoforms. Nonselective agents with K_i values at 5-HT₇ receptors of 10 nM or less include 5-HT, 5-methoxytryptamine, LSD, methiothepin, and mesulergine; those with K_i values in the range of 10 to 100 nM include 8-OH

DPAT (long considered a 5-HT_{1A}-selective agonist!), spiperone, ritanserin, metergoline, mianserin, and chlorpromazine; those with K_i values in the range of 100 to 1,000 nM include NAN-190, sumatriptan, and haloperidol; and those with K_i values of greater than 1,000 nM include 2-methyl 5-HT, tropisetron, pindolol, and ketanserin. Reportedly, 5-HT, 5-CT, and 8-OH DPAT act as agonists, whereas methiothepin, mianserin, mesulergine, ritanserin, spiperone (a 5-HT_{1A}, 2-HT_{2A}, and D₂ antagonist), NAN-190 (a 5-HT_{1A} antagonist), and clozapine act as antagonists. Numerous antidepressants and antipsychotic agents bind at 5-HT₇ receptors with nanomolar or subnanomolar affinity ($K_i \leq 10$ nM), including fluphenazine, acetophenazine, chlorprothixene, zotepine, clozapine, fluperlapine, pimozide, tiospirone, and risperidone (114,135).

5-HT₇ Antagonists

Several reasonably selective 5-HT₇ agents have been identified. The first reported 5-HT₇ antagonist was SB-258719 ($K_i \sim 30$ nM) (136), and attempts to optimize binding affinity and selectivity led to SB-269970 (*S*-isomer, $K_i = 1.3$ nM) (Fig. 11.21). Both compounds displayed some inverse agonist action. The high in vivo blood clearance of SB-269970 resulted in further structural modification, leading to compounds such as SB-656104 (*S*-isomer, $K_i = 2$ nM) (114). Another early series of 5-HT₇ antagonists was the DR compounds: DR4004 was the first of these to show activity as a competitive antagonist; structural modification resulted in others, including DR4365 ($K_i = 4$ nM) (137). Other antagonists Fig. 11 include phenylpyrrole-containing LCAPs (Fig. 11.21), which because of their structural similarity to other 5-HT_{1A} ligands could have been expected to—and do—bind at 5-HT_{1A} receptors (138); arylpiperazinosulfonamides; and tetrahydroisoquinolinesulfonamides (139). Actually, the latter two types of compounds have been demonstrated to act as inverse agonists (139).

5-HT₇ Agonists

N-Arylaminoimidazolines (Fig. 11.22) were identified as the first 5-HT₇ agonists (140); however, they have not been pursued because of their profound effects

on blood pressure and heart rate, which are probably a consequence of their affinity for α -adrenoceptors. Several new 5-HT₇ agonists have been recently reported, including the piperazinylhexanones (141) and 2-amino-tetralins (142) (Fig. 11.22); the latter can function either as agonists (e.g., R = *n*Pr) or antagonists (e.g., R = Me), depending on the nature of the R group. Pharmacophore models have been proposed for 5-HT₇ agonists (143), antagonists (144), and inverse agonists (139).

5-HT₇ Receptors: Clinical Implications

Because of the previous unavailability of 5-HT₇-selective agents, the pharmacology of the 5-HT₇ system is still relatively unexplored. Nevertheless, studies with nonselective agents, 5-HT₇ receptor knockout animals, and some of the first few selective agents that were identified have provided some tantalizing clues (114,135,145–147). The 5-HT₇ receptors could be involved in mood and learning as well as in neuroendocrine and vegetative behaviors. The 5-HT₂ ligand ritanserin, certain tricyclic antidepressants (e.g., amitriptyline), classical antipsychotic agents (e.g., chlorpromazine), and nonclassical antipsychotic agents (e.g., clozapine) bind with K_i values of less than 100 nM (128). On this basis, it has been speculated that 5-HT₇ receptors may have a role in certain neuropsychiatric disorders. Consistent with these suggestions, 5-HT₇ receptors are sensitive to antidepressant treatment (148). The 5-HT₇ receptors have been implicated in serotonergic regulation of circadian rhythm, leading to suggestions that 5-HT₇-selective agents could be effective in the treatment of jet lag or sleep disorders of a circadian nature (149). 5-HT₇ receptors could also be involved in sleep disorders, anxiety, memory and cognition, epilepsy, pain, migraine, and thermoregulation. In the periphery, 5-HT produces both contraction and relaxation of coronary artery from various species (150). It has been proposed that relaxation of coronary artery may be mediated by 5-HT₇ receptors. Agents active at 5-HT₇ receptors could thus be effective in the treatment of coronary heart disease. Now that newer, more selective agents are finally available, many of these hypotheses can be further tested.

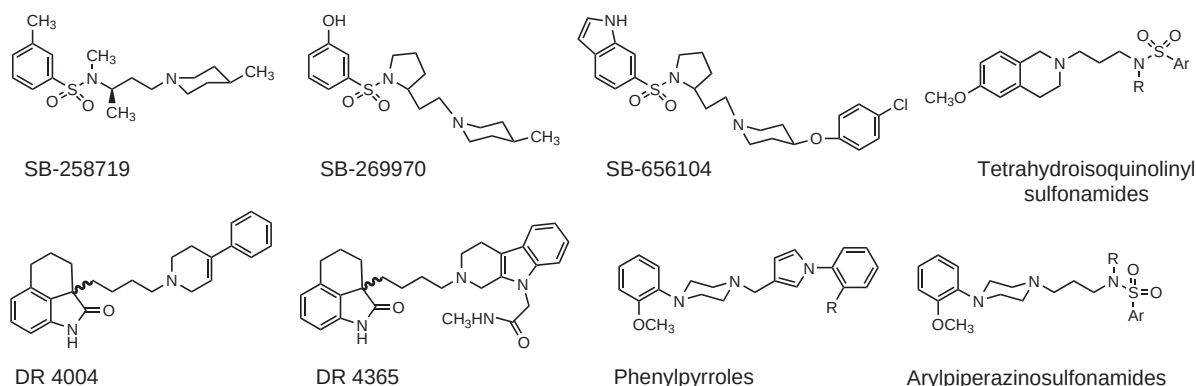


FIGURE 11.21 5-HT₇ receptor antagonists.

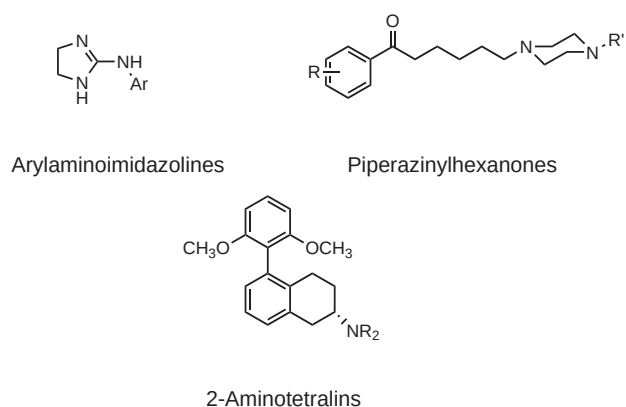


FIGURE 11.22 5-HT₇ receptor agonists.

THE SEROTONIN TRANSPORTER

The actions of 5-HT are terminated by its diffusion away from the synapse, by enzymatic degradation, and by reuptake into the presynaptic terminal (see Chapter 18 for further discussion). After reuptake, once 5-HT is inside the neuron, it can be re-stored in storage vesicles or metabolized. The 5-HT reuptake process involves a high-affinity transporter protein that is localized in the presynaptic terminal membrane. The 5-HT reuptake transporter (SERT) regulates the duration and magnitude of postsynaptic response to 5-HT. A different transporter is associated with different neurotransmitters (e.g., norepinephrine reuptake transporter [NET] transports norepinephrine). SERT has been cloned and expressed (138), and its putative structure is roughly similar to the general receptor structure shown in Figure 11.3 except that 1) it consists of 12 membrane-spanning helices, 2) both the amino

terminus and the carboxy terminus are located on the intracellular side, and 3) it has an exaggerated extracellular loop between TM3 and TM4 (Fig. 11.23). SERT possesses approximately 50% homology with the NET and the dopamine transporter. For 5-HT transport, a ternary complex of protonated 5-HT, Na⁺, and Cl⁻ binds to the transporter protein to form a quaternary complex; the transporter undergoes a conformational change to release 5-HT into the cytoplasm of the neuron (151).

The 5-HT transporter has been implicated as having a role in affective disorders (Chapter 18). Agents that block the transporter and, thereby, increase synaptic levels of 5-HT are useful for the treatment of depression, obsessive-compulsive behavior, and panic disorders. Tricyclic antidepressants (e.g., imipramine, desipramine) block the 5-HT transporter and the NET to varying degrees. Some display a preference for one transporter over the other, but most are nonselective (152). SSRIs display greater selectivity for SERT than for NET. The first SSRI to be used clinically was fluoxetine; several other agents have since become available. The SARs of SSRIs have been reviewed elsewhere (153); see Chapter 18 for further discussion of antidepressants and examples. Certain drugs of abuse (e.g., cocaine) also block the 5-HT transporter, although cocaine's primary mechanism of action likely involves the dopamine transporter.

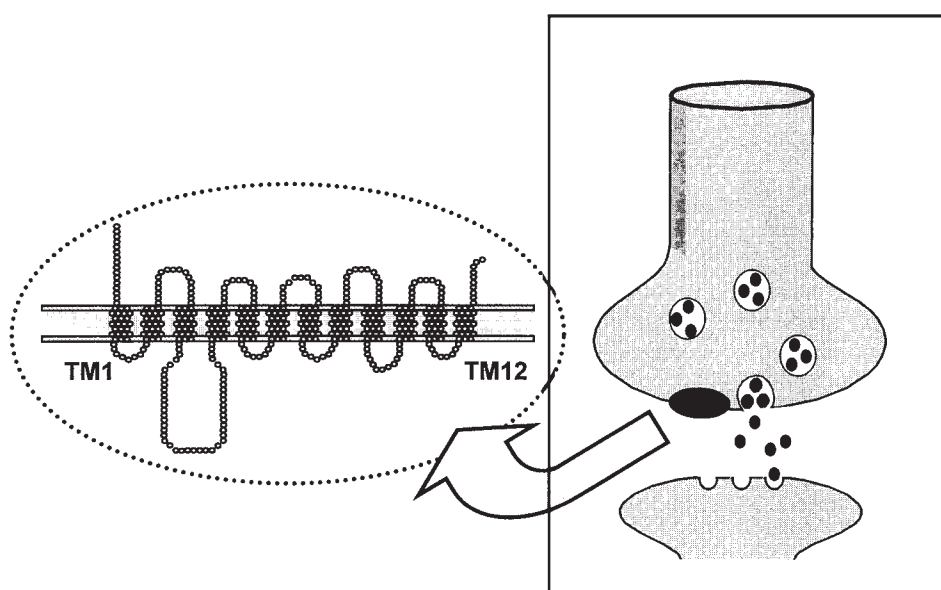
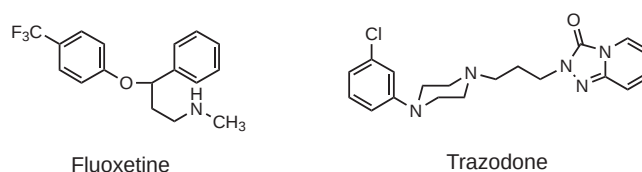


FIGURE 11.23 Schematic of a neuron showing the general location and basic structure (inset) of a serotonin transporter (SERT). Note that the transporter possesses 12 transmembrane-spanning helices (TM1–TM12). Both the amino terminus (attached to TM1) and the carboxy terminus (attached to TM12) are on the intracellular side.

Various tricyclic antidepressants and SSRIs, including fluoxetine, also bind at 5-HT_{2A} and 5-HT_{2C} receptors (154,155). The role, if any, derived from a direct interaction of these agents with 5-HT₂ receptors versus their interaction at SERT remains to be determined. 5-HT₂ antagonists typically downregulate 5-HT₂ receptors. The antidepressant trazodone, for example, is a weak SSRI but binds at 5-HT₂ receptors and is also a 5-HT₂ antagonist (155). The 5-HT_{2C} agonist *m*-chlorophenylpiperazine (*m*CPP) induces panic attacks in patients with panic disorder and increases obsessive compulsions in patients with obsessive-compulsive disorder (156), implicating a role for this specific 5-HT_{2C} subpopulation. The 5-HT_{2C} receptor antagonists could be useful targets for the development of novel agents to treat these disorders. This issue is complicated, however, by findings that trazodone is metabolized to *m*CPP and that, in some instances, trazodone possesses 5-HT_{2C} agonist properties (157). In any event, long-term treatment with tricyclic antidepressants (and MAOIs) leads to a downregulation in the number of 5-HT₂ receptors, the time course for which approximates the clinical response in depressed patients (152). Some SSRIs produce adaptive changes involving decreased responsiveness of 5-HT₂ receptors, whereas electroconvulsive therapy increases the number of 5-HT₂ receptors (152). Several 5-HT receptor populations have been implicated in the actions of antidepressants (e.g., 5-HT₁, 5-HT₂, 5-HT₆, and 5-HT₇), but SERT remains an attractive target for the development of novel psychotherapeutic agents.

SUMMARY

5-HT is a major neurotransmitter in the brain and is also involved in a number of peripheral actions. Seven families or populations of 5-HT receptors have been identified (5-HT₁ to 5-HT₇), and several are divided into distinct subpopulations (Table 11.2). Excluding splice variants, 14 different populations and subpopulations of 5-HT receptors have been cloned. Over the past 30 years, selective agonists and antagonists have been developed and identified for many of the subpopulations, but subpopulations remain for which selective agents have yet to be developed. The availability of such agents is important, because it aids functional investigations of the different 5-HT receptors. In addition to acting directly on 5-HT receptors, therapeutic agents with other mechanisms are available for influencing serotonergic transmission, including SSRIs and MAO inhibitors. Studies with 5-HT receptors have led to the introduction of agents useful for treating anxiety (e.g., buspirone), migraine (e.g., sumatriptan), irritable bowel syndrome (e.g. tegaserod), and chemotherapy-induced emesis (e.g., ondansetron); numerous other agents are currently in clinical trials for the treatment of depression, schizophrenia, and obsessive-compulsive and other disorders. Investigations also have led to a greater understanding of cardiovascular pharmacology, obesity, neurodegenerative disorders, aggression, sexual behavior, and drug abuse, just to mention a few examples. To reiterate a phrase from the introduction, “[I]t almost appears that 5-HT is involved in everything” (1).

SCENARIO: OUTCOME AND ANALYSIS

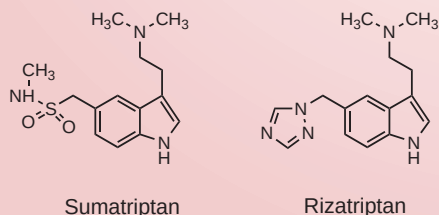
Outcome

Jill T. Johnson, PharmD, BCPS

The pharmacist referred her back to her physician to seek a different “triptan.” The pharmacist also counseled the patient about using the drug for aborting migraines, not as prophylaxis. The pharmacist described medication overuse headaches that one can experience from taking triptans too often. Because the maximum dose of sumatriptan for migraines is 200 mg per day, the patient likely was experiencing overuse headaches. She was prescribed rizatriptan, which has worked to abort her migraines for many months. She takes rizatriptan only for aborting migraines and limits its use to not more than 15 mg in 24 hours. She now has fewer than 3 migraines in a month.

Chemical Analysis

Victoria Roche and S. William Zito

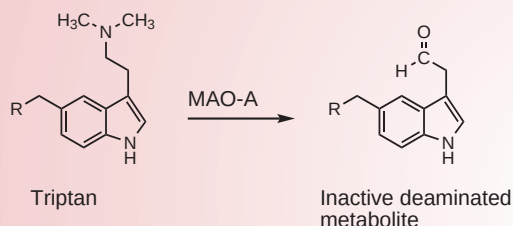


Both of the triptans associated with this scenario are indoleethylamines (tryptamines) that abort migraine headaches through their agonist action at 5-HT_{1D} receptors. Sumatriptan was the first molecule in this class to be developed, and its selectivity for this receptor subtype, while acceptable, is not absolute. Other serotonergic receptors stimulated by sumatriptan include the 1B (formerly classified as a 1D receptor), 1A, and 1F subtypes, the latter of which also may be beneficial in treating migraines. Triptans work by constricting central vessels that dilate during migraine headaches. Overuse of vasoconstrictors results in rebound vasodilation, which could explain why MB's inappropriate (daily) use of sumatriptan gave her less relief from headache pain than expected.

The sulfonamide sidechain of sumatriptan produces a more polar triptan compared to other analogs that followed, and penetration of the blood–brain barrier under normal physiological conditions is low. Though rizatriptan's triazole substituent is certainly not the most lipophilic of rings, the drug's log P is higher than that of sumatriptan (1.4 vs. 0.8). The more lipophilic structure confers a higher oral bioavailability on rizatriptan and allows a more facile penetration of the blood–brain barrier to reach central sites of action. The actual therapeutic impact of this enhanced distribution profile is, however, uncertain. The triazole ring of rizatriptan also confers a higher level of 5-HT_{1D} selectivity than the methanesulfonamide moiety of sumatriptan.

SCENARIO: OUTCOME AND ANALYSIS (*Continued*)

As ethylamines, both sumatriptan and rizatriptan are vulnerable to oxidative deamination by MAO-A, and this is the major biotransformation route for both drugs. The *N*-dealkylated metabolite of rizatriptan (which forms before deamination) is known to be equally active with the parent structure, although it is generated to a minor (14%) extent.



Either triptan should be effective to abort MB's headaches if taken as directed, which is during the very early phase of a migraine when the pain level is low. Likewise, either triptan, if taken on a chronic basis rather than as needed, will promote rebound vasodilation and undermine therapeutic efficacy when it's needed most.

CASE STUDY

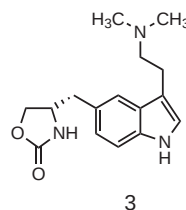
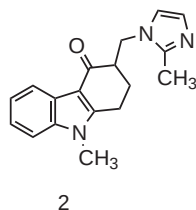
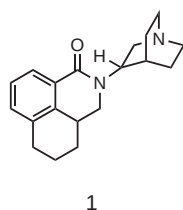
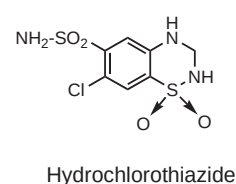
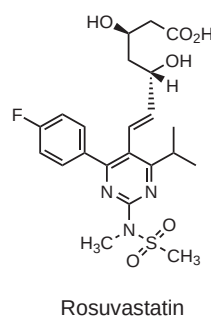
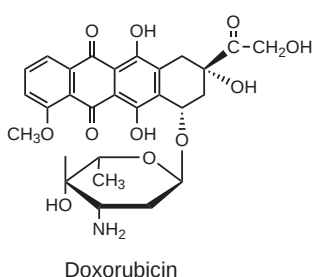
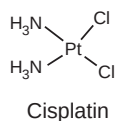
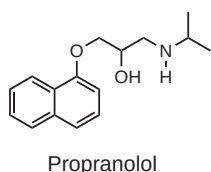
Victoria Roche and S. William Zito

Sr. MT is a 61-year-old woman with newly diagnosed ovarian cancer, and is a member of the Sisters of St. Agnes. The mission of this order of nuns includes community outreach related to social justice and health care, and Sr. MT is beloved by all those whom she has served over the years, including members of your family. Sr. MT will soon begin cisplatin/doxorubicin chemotherapy known as the AP regimen. She has been told that these two drugs induce severe nausea and vomiting that, in addition to occurring during or shortly after therapy, can also be delayed for several days after drug administration. Sr. MT has moderate coronary artery disease, and she is currently taking rosuvastatin to lower serum lipids and is taking propranolol (β -blocker)/hydrochlorothiazide (diuretic) antihypertensive therapy. Though she believes the propranolol in her blood pressure medication is helping her feel less anxious

about the discomforting side effects of her chemotherapy, she also wants to minimize the disabling nausea and vomiting so she can continue serving her community for as long as possible.

Recognizing the value of serotonergic receptor antagonists in the treatment of chemotherapy-induced nausea, you recall the structures of three serotonin-related agents you studied in Medicinal Chemistry class, and contemplate their value in easing your friend's way in the days ahead.

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

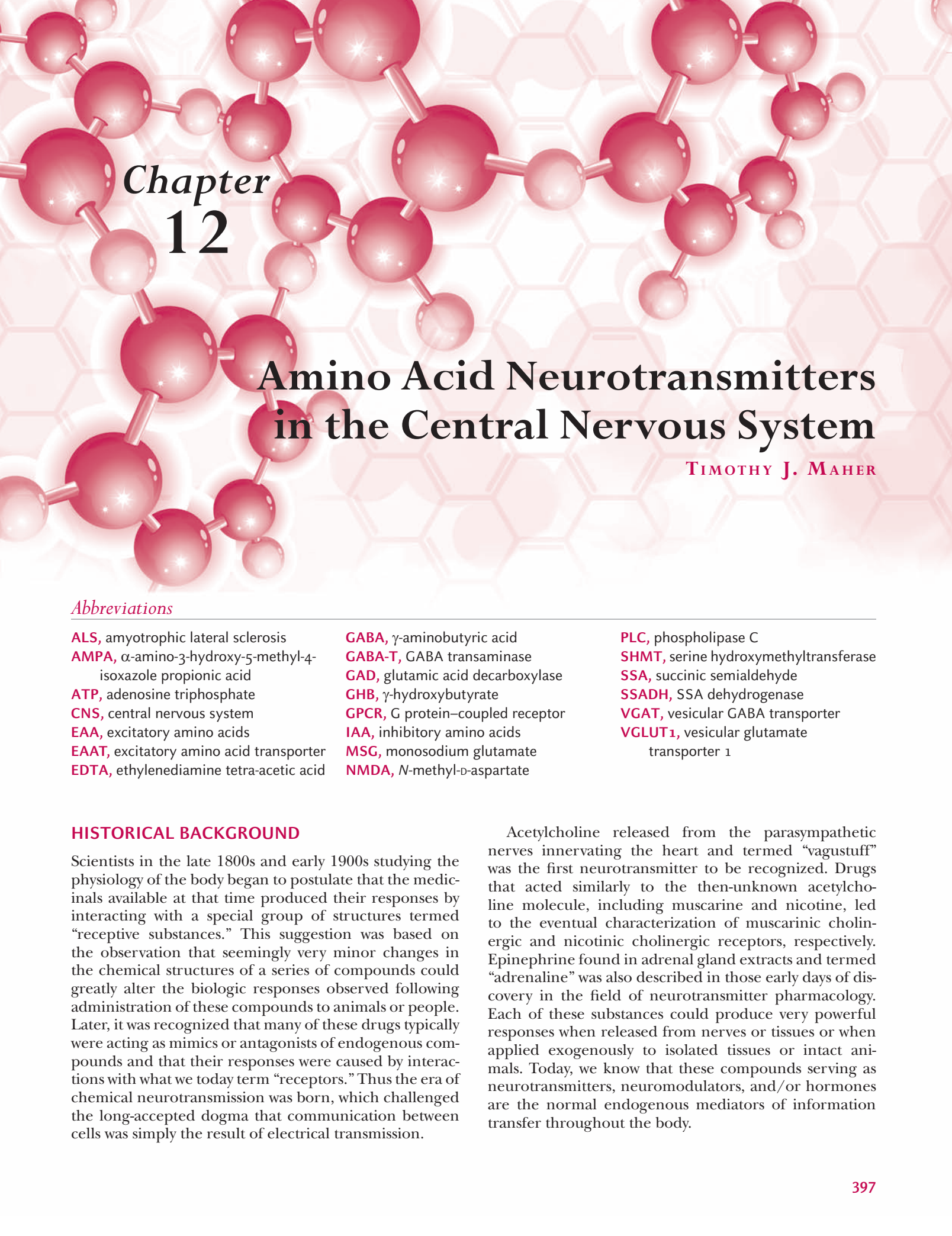


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Chapter 12

Amino Acid Neurotransmitters in the Central Nervous System

TIMOTHY J. MAHER

Abbreviations

ALS, amyotrophic lateral sclerosis
AMPA, α -amino-3-hydroxy-5-methyl-4-isoxazole propionic acid
ATP, adenosine triphosphate
CNS, central nervous system
EAA, excitatory amino acids
EAAT, excitatory amino acid transporter
EDTA, ethylenediamine tetra-acetic acid

GABA, γ -aminobutyric acid
GABA-T, GABA transaminase
GAD, glutamic acid decarboxylase
GHB, γ -hydroxybutyrate
GPCR, G protein-coupled receptor
IAA, inhibitory amino acids
MSG, monosodium glutamate
NMDA, *N*-methyl-D-aspartate

PLC, phospholipase C
SHMT, serine hydroxymethyltransferase
SSA, succinic semialdehyde
SSADH, SSA dehydrogenase
VGAT, vesicular GABA transporter
VGLUT1, vesicular glutamate transporter 1

HISTORICAL BACKGROUND

Scientists in the late 1800s and early 1900s studying the physiology of the body began to postulate that the medicinals available at that time produced their responses by interacting with a special group of structures termed “receptive substances.” This suggestion was based on the observation that seemingly very minor changes in the chemical structures of a series of compounds could greatly alter the biologic responses observed following administration of these compounds to animals or people. Later, it was recognized that many of these drugs typically were acting as mimics or antagonists of endogenous compounds and that their responses were caused by interactions with what we today term “receptors.” Thus the era of chemical neurotransmission was born, which challenged the long-accepted dogma that communication between cells was simply the result of electrical transmission.

Acetylcholine released from the parasympathetic nerves innervating the heart and termed “vagustuff” was the first neurotransmitter to be recognized. Drugs that acted similarly to the then-unknown acetylcholine molecule, including muscarine and nicotine, led to the eventual characterization of muscarinic cholinergic and nicotinic cholinergic receptors, respectively. Epinephrine found in adrenal gland extracts and termed “adrenaline” was also described in those early days of discovery in the field of neurotransmitter pharmacology. Each of these substances could produce very powerful responses when released from nerves or tissues or when applied exogenously to isolated tissues or intact animals. Today, we know that these compounds serving as neurotransmitters, neuromodulators, and/or hormones are the normal endogenous mediators of information transfer throughout the body.

The earliest characterizations of chemical compounds as neurotransmitters involved studies of physiologic systems in the periphery and used both chemical and physiologic techniques. Later, investigators started to apply some of the same techniques used in characterizing these peripheral neurotransmitters and their receptors to the central nervous system (CNS), which is comprised of the brain and spinal cord. These investigations were often much more difficult because of the greater complexity of structures and functions in the CNS. The seemingly less well defined and more ambiguous areas (e.g., limbic system, temporal lobe, and prefrontal cortex) in the CNS as well as the smaller and more diverse synaptic connections (e.g., axo-axonal and somato-dendritic) presented significant challenges to early investigators. In the periphery, stimulation of a nerve innervating a target tissue often led to an easily observable end point (e.g., changes in heart rate or blood pressure, hormone release), but stimulation of nerves in the brain or spinal cord often failed to produce easily observable and quantifiable endpoints (e.g., sensations of warmth, fear, memory, and hallucinations), making such studies very challenging, especially when performed in experimental animals.

Early studies by Falk and Hillarp used formaldehyde to visualize neurons containing the monoamines dopamine, norepinephrine, and epinephrine (fluoresces green) as well as serotonin (fluoresces yellow) in brain slices. With improvements in the chemical techniques for measuring these classical monoamines and acetylcholine and, eventually, the enzymes involved with their synthesis and degradation, investigators began to map the neurons in the CNS that used these neurotransmitters and began to appreciate the neurochemical basis of drug action in the brain and spinal cord. In the late 1950s and early 1960s, however, investigators began to observe the neuronal effects of simple amino acids in these brain tissues. The suggestion that simple amino acids could function as sophisticated neurotransmitters was met with considerable resistance and skepticism. Amino acids are typically reserved for protein synthesis and serve many metabolic functions, and they did not appear to have the same unique functional characteristics of the classical monoamine neurotransmitters and acetylcholine. As studies continued to support the neurotransmitter role of amino acids such as γ -aminobutyric acid (GABA), glutamic acid (glutamate), and glycine, however, overwhelming evidence has led to their nearly universal acceptance as the major neurotransmitters in the CNS. Quantitatively, the amino acids are the most prevalent neurotransmitters in the CNS. For example, it has been estimated that more than 40% of all neurons in the CNS use GABA as a neurotransmitter substance and that less than 10% use one of the other classical monoamines, such as norepinephrine and serotonin, or acetylcholine.

Although overall neuronal functioning is rather complex, a simplistic view of the two major effects of neurotransmitters involves changes in membrane potential toward states of either depolarization or hyperpolarization (Fig. 12.1). Such changes in membrane potential

usually result from alterations in membrane conductance to ions such as Na^+ , K^+ , Ca^{2+} , and Cl^- , which typically flow down their electrochemical gradients when a selective ion channel is opened. Ion channels need only open for 0.1 to 100 milliseconds to effect this change. Thus, when a neuron is to be depolarized, Na^+ or Ca^{2+} typically enters the cell, making the membrane potential more positive and more likely to reach the threshold potential and lead to an action potential (Fig. 12.1A). On the other hand, when a neuron is hyperpolarized, typically either K^+ moves out of the cell into the extracellular space or Cl^- moves into the cell (Fig. 12.1B). Both of these ion fluxes result in a hyperpolarization of the membrane, which decreases the likelihood that a neuronal cell will become excited and fire. Thus, these two opposing alterations in neuronal membrane potentials commonly are referred to as excitatory (depolarization) and inhibitory (hyperpolarization). When amino acid neurotransmitters produce their effects on neuronal function, they commonly are categorized as either excitatory amino acids (EAAs) or inhibitory amino acids (IAAs) (Fig. 12.2). The major established EAA neurotransmitter in the CNS is glutamate. Other endogenous EAAs include aspartate (aspartic acid), cysteine, and homocysteine. The major endogenous IAAs in the CNS include GABA and glycine, but much evidence also supports a role for taurine and β -alanine.

NEUROTRANSMITTER CRITERIA

For a compound to be classified as a neurotransmitter, a number of criteria must be satisfied. First, the neurotransmitter must be present. As a rule, the concentrations of the neurotransmitter in various parts of the CNS will vary such that a unique and unequal distribution can be demonstrated. For some amino acids, this unequal distribution is difficult to demonstrate, because most of these amino acids are also used to support protein synthesis and for intermediary metabolism, unlike the monoamine neurotransmitters, for which an unequal distribution is relatively easy to demonstrate. Although substances such as glucose and sodium are present in the CNS, their regional distribution is rather uniform, and few would consider either of these to be a classical neurotransmitter. Besides determining the concentration of the neurotransmitter itself, evidence that can also be used to support this criterion includes the distribution of specific receptors, reuptake systems, and enzymes involved with synthesis and/or degradation of the particular amino acid neurotransmitter. Today, with the aid of molecular biologic techniques (e.g., mRNA, cloning techniques, and in situ hybridization), the location and quantification of proteins that serve as receptors, transporters, or metabolizing enzyme are more easily achieved. Second, the neurotransmitter must be released when the nerve cell is stimulated. To date, the vast majority of amino acid neurotransmitter release, like that of the monoamines and acetylcholine, has been found to be Ca^{2+} dependent and originating from presynaptic storage vesicles. Release

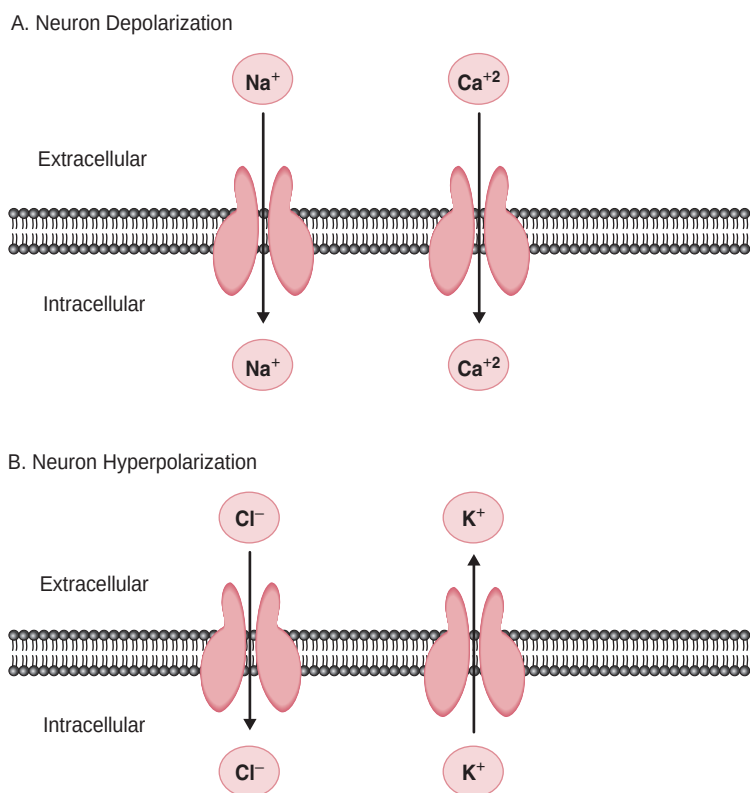
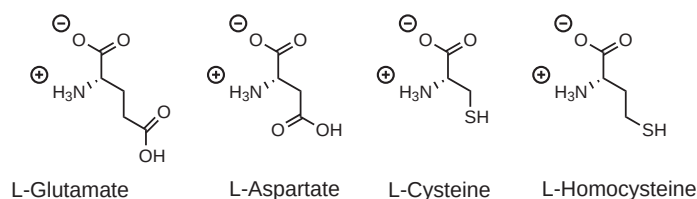


FIGURE 12.1 Electrophysiology of excitatory and inhibitory neurotransmitters. The tendency of an ion to move across the membrane depends on the difference in its electrochemical gradient on either side of the membrane. The electrochemical gradient depends on the difference in the concentration of the ion between the two sides of the membrane, the charge of the ion, and the transmembrane potential (the difference in voltage between the two sides of the membrane).

can be determined using brain slices prelabeled with radioactive neurotransmitter or with *in vivo* microdialysis, in which the extracellular fluid surrounding neurons is continuously sampled. Furthermore, there should be a mechanism in place that terminates the action of the

neurotransmitter (e.g., reuptake, removal into astroglia or other cells, metabolism, or some other form of inactivation). And finally, for a substance to be considered a neurotransmitter, the exogenous administration or application of the suspected neurotransmitter must mimic the

Excitatory amino acids:



Inhibitory amino acids:

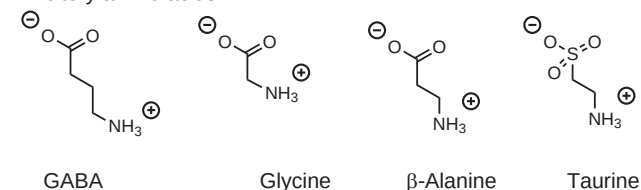


FIGURE 12.2 Endogenous excitatory and inhibitory amino acid neurotransmitters.

CLINICAL SIGNIFICANCE



Amino acids are the most abundant neurotransmitters in the CNS, and the majority of neurons in the mammalian brain utilize either glutamate or GABA as their primary neurotransmitters. These amino acids are responsible for almost all of the fast signaling between neurons, with a modulatory role for many of the other neurotransmitters. Glutamate is the most abundant excitatory neurotransmitter and GABA is the most abundant inhibitory neurotransmitter in the human CNS. Thus, these amino acids serve to regulate reciprocally the level of excitability of most neurons in the CNS and therefore have been implicated as important mediators of many critical physiological processes as well as pathophysiological conditions that underlie brain function and/or dysfunction. Pharmacological studies utilizing drugs that selectively inhibit/block or mimic the actions of GABA or glutamate support the belief that these two neurotransmitters, because of their often opposing excitatory and inhibitory actions, control, to a large degree, the overall excitability of the CNS. Glutamate is the major excitatory neurotransmitter and is distributed in all regions of the brain. Three major types of glutamate receptors have been identified: *N*-methyl- *D*-aspartate (NMDA), α -amino-3-hydroxy-5-methylisoxazole-4-propionic acid (AMPA), and kainate (KA) receptors. GABA, metabolically derived from glutamate, is the major inhibitory neurotransmitter in the brain. Glycine is an inhibitory neurotransmitter found mostly in the brain stem and spinal cord but which has also a role as a co-agonist on certain glutamate receptors.

Glutamate is involved in cognitive functions in the brain such as learning and memory, and it has been implicated in epileptic seizures. GABA and glycine are the most important

inhibitory amino acid neurotransmitters in the brain and brainstem/spinal cord, respectively. These inhibitory amino acids are of particular interest because many therapeutically useful drugs act by selectively affecting these two neurotransmitter systems. Specific inhibitors of GABA reuptake transporters have been shown to have antiseizure and/or antinociceptive properties in laboratory animals. Receptors for both inhibitory and excitatory amino acid neurotransmitters are either ionotropic (i.e., their activation results in enhanced membrane ion conductance) or metabotropic (i.e., their activation results in increased intracellular levels of a second messenger). GABA_A receptors are ionotropic receptors that lead to increased Cl⁻ ion conductance, whereas GABA_B receptors are metabotropic receptors that are coupled to G proteins and thereby alter neuronal excitability.

Subsequent chapters will describe the mechanisms of action of the sedative-hypnotic benzodiazepines, nonbenzodiazepines, barbiturates, and anesthetic agents at GABA_A receptors; baclofen and γ -hydroxybutyric acid at GABA_B receptors; GABA reuptake inhibitors as antiseizure agents, tiagabine; GABA-transaminase inhibitors, gabaculine, valproate, vigabatrin; and GABA analogs as antiseizure drugs and analgesics for neuropathic pain, pregabalin, gabapentin. The hallucinogenic drug phencyclidine (more commonly known as PCP) antagonizes glutamate noncompetitively at the NMDA receptor.

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effects (e.g., depolarization, hyperpolarization, activation of protein kinase C, inhibition of adenylyl cyclase, etc.) that are observed when the neurotransmitter is released endogenously following presynaptic nerve cell stimulation.

EXCITATORY AMINO ACID NEUROTRANSMITTERS

Glutamate

The major EAA neurotransmitter in the CNS is L-glutamate. (Note that all amino acids with the exception of glycine mentioned in this chapter are considered to be in the "L" configuration; thus, the "L" will be omitted, unless specifically indicated as "D"). Early studies in invertebrates demonstrated the potent actions of the acidic amino acid glutamate, as well as another acidic amino acid, aspartate. Acceptance of glutamate as a neurotransmitter in the CNS was delayed for many years as neuroscientists attempted to distinguish its role as a component of protein and peptides (e.g., glutathione), an important

intermediate in numerous metabolic processes, and a precursor to the inhibitory neurotransmitter GABA. Whereas glutamate is found in all cells within the CNS, an unequal distribution of this amino acid and of aspartate has been demonstrated.

Glutamate Synthesis, Storage, and Release

Glutamate is synthesized in the CNS via the transamination of α -ketoglutarate (Fig. 12.3), which is produced from glucose in the Krebs cycle. Glutamate can also be synthesized from glutamine via glutaminase. Once produced, glutamate can be stored in neuronal vesicles. With the vast number of compounds that can feed into the Krebs cycle and the various sources of glucose, it is not surprising that the control of glutamate synthesis in the CNS is still poorly understood.

The oral ingestion of glutamate in protein, as the individual amino acid in the form of a dietary supplement or in the form of the flavor enhancer monosodium glutamate (MSG), fails to significantly increase CNS levels of this amino acid neurotransmitter. Following

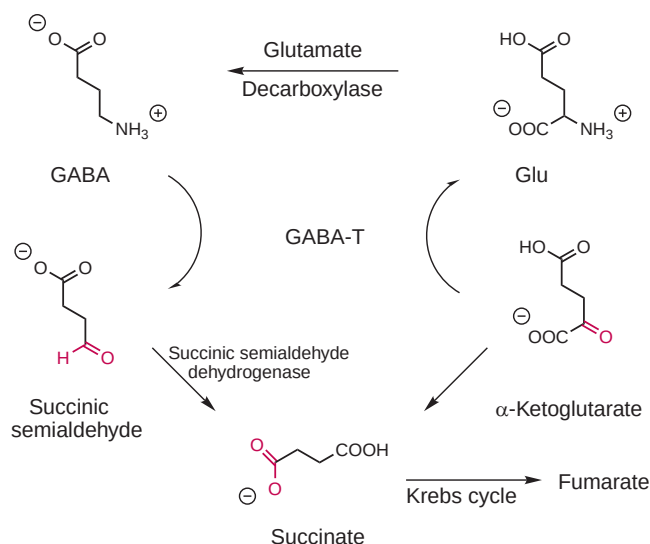
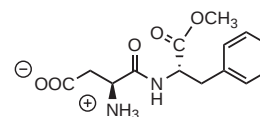


FIGURE 12.3 Glutamate and GABA biosynthesis.

its ingestion, blood glutamate levels do increase transiently, but largely because of the ability of the blood–brain barrier to regulate the entry of glutamate into the CNS, no significant changes in brain glutamate are observed. This ability to regulate synaptic glutamate levels in brain despite fluctuations in peripheral levels is very important, because uncontrolled variations in CNS glutamate would lead to serious consequences, including seizures and coma for increases and decreases of glutamate, respectively. Within the blood–brain barrier, there exist transport proteins that are responsible for controlling the influx of amino acids into the CNS. Of the three major types—acidic, basic, and neutral amino acid transporters—it is the acidic amino acid transporter that carries glutamate. These transporters function by facilitated diffusion and cannot operate against a concentration gradient (as seen with active transport). In actuality, the acidic amino acid transporter normally functions to move glutamate out of the CNS. There have been studies in which glutamate, often in the form of MSG, has been injected subcutaneously into neonatal mice or nonhuman primates and hypothalamic lesions noted. The lesions are mostly restricted to the circumventricular organs (those near the fourth ventricle in the brain), where the blood–brain barrier is significantly diminished. Moreover, when MSG is directly injected into the brains of laboratory animals, cellular necrosis is observed. However, when MSG is given orally in very high doses, the homeostatic processes in liver and other tissues help to regulate plasma glutamate concentrations, and large fluctuations are not observed. The blood–brain barrier also functions to keep CNS glutamate at appropriate levels. The studies in which glutamate in the form of MSG is parenterally injected do not reflect the chemical fluctuations that occur following oral ingestion and should not be interpreted as an indicator of toxicity of MSG (1).

Similar concerns have been voiced regarding the safety of the acidic amino acid aspartate in the artificial sweetener aspartame (L-aspartyl-L-phenylalanine methylester). As with MSG, however, the homeostatic mechanisms regulating plasma aspartate levels following aspartame consumption do not allow for large increases in plasma aspartate that could alter CNS function. In fact, the concern regarding the safety of aspartame results from the other amino acid present, phenylalanine, which much more readily crosses the blood–brain barrier and can alter CNS function. This is especially important for people who unknowingly are heterozygous for the genetic metabolic disorder of phenylketonuria (2).



Aspartame

The vesicular glutamate transporter 1 (VGLUT1) transports glutamate into storage vesicles in preparation for the Ca^{2+} -mediated release of this neurotransmitter that uses adenosine triphosphate (ATP) to concentrate glutamate against its concentration gradient (ATP-dependent transporter). Unrelated to any of the other families of neurotransmitter vesicle transporters, VGLUT1 originally was thought to be an inorganic phosphate membrane transporter. Today, it is known that VGLUT1 uses a vacuolar ATPase to cleave ATP to provide the energy required to generate and maintain an electrochemical proton gradient that allows the concentration of glutamate to remain extremely high in synaptic vesicles compared to that in the cytoplasm—and especially compared to that in the synapse. Although probably not the basis of their main pharmacologic actions, known inhibitors of VGLUT1 include the ergot bromocriptine (a dopamine agonist used in the treatment of Parkinson disease and hyperprolactinemia) and some of the azo-dyes, including Evans Blue (3). Recently, a family of endogenous inhibitory proteins has been identified that not only blocks glutamate vesicular transport but also GABA vesicular transport. The significance of the role that these proteins, known as IPF α , IPF β , and IPF γ , have in regulating glutamatergic neurotransmission is still poorly understood (4). As more is learned about such proteins and other yet unknown regulatory mechanism, a better appreciation regarding the pathology of various disease states can be realized, and therapeutic interventions can be designed based on this knowledge.

Like most other recognized neurotransmitter substances, vesicular release of glutamate occurs in a Ca^{2+} -dependent manner following depolarization of the presynaptic terminal. In cell culture or isolated tissue studies, agents that bind Ca^{2+} , such as ethylenediamine tetra-acetic acid (EDTA) or the use of Ca^{2+} -free perfusion solutions, the vesicular release of glutamate can be diminished or prevented. Thus, this is

one of the techniques that can be employed to distinguish, in part, between the neurotransmitter and metabolic roles of glutamate in a particular biologic system.

Glutamate Reuptake and Metabolism

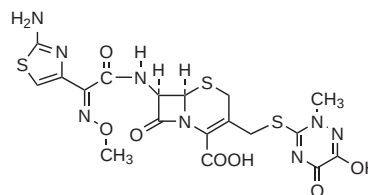
Transporter proteins that regulate the synaptic concentrations of glutamate are essential in keeping the basal levels of this EAA neurotransmitter low and in helping to terminate the responses to neuronally released glutamate. These high-affinity, sodium-dependent transporters are so efficient at sequestering glutamate from the synaptic cleft that the concentration of glutamate inside the presynaptic terminal is most often several thousand-fold greater than that found in the synapse (5).

These EAA transporters (EAATs) have been identified and categorized into five major subtypes: EAAT1, EAAT2, EAAT3, EAAT4, and EAAT5 (Table 12.1 and Fig. 12.4). Their distribution tends to be primarily cellular, and relative anatomic distribution of the subtypes in brain for each subtype is such that EAAT1 and EAAT2 are mostly found in astroglial cells in the cerebellum and forebrain, respectively. The astroglial reuptake of neuronally released glutamate is believed to be very important in terminating the glutamate-mediated synaptic signal. The EAAT3 and EAAT4 subtypes are highly concentrated in the cortical and cerebellar Purkinje neurons, respectively, whereas EAAT5 is most abundant in the retina. The reuptake of glutamate or aspartate via any of the EAAT subtypes is coupled to the cotransport of H^+ and $3Na^+$ into astroglia or neurons, with the extrusion of K^+ , as shown in Figure 12.4.

The EAATs are capable of transporting D-aspartate, aspartate, and glutamate but do not transport D-glutamate. The known inhibitors of glutamate reuptake transport are classified as either competitive (these, as a rule, act as substrates and are transported) or noncompetitive (these are nontransported), but no therapeutically useful agents have yet been discovered. The actions of glutamate are terminated via its reuptake into the presynaptic neuron, from where it originally was released, into the surrounding glia and into the postsynaptic neuron that houses the various target EAA receptors (6). Following its release, glutamate that is taken up via EAATs into neurons or astroglial cells can be recycled, in part, directly into vesicles for subsequent release or, more likely, is converted to glutamine via glutamine synthetase and is available

for eventual recycling to glutamate via the activity of the enzyme glutaminase, as shown in Figure 12.4.

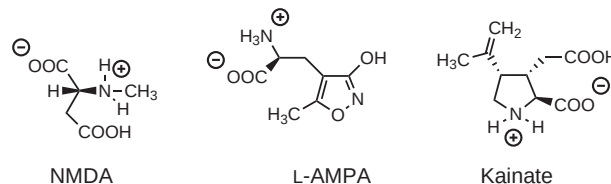
Ischemic brain injury has been shown to upregulate EAAT2, which can provide some neuroprotection by increasing the removal of glutamate from the extracellular space where it can be in contact with excitatory receptors and further the ischemic insult (7). Recently, two endogenous inflammatory mediators, tumor necrosis factor- α and nuclear factor- κB , have been shown to upregulate EAAT2, suggesting that selectively modulating this transporter might be a pharmacologic approach to reducing ischemic brain damage in the future. Similarly, the β -lactam antibiotic ceftriaxone, which enhances the activity of some EAATs, has been effective in attenuating neurodegeneration in an animal model of amyotrophic lateral sclerosis (ALS; also known as Lou Gehrig disease) (8). The EAATs continue to be a promising target for therapeutic interventions involving diseases of the CNS, which involve excessive excitatory amino acid neurotransmission.



Ceftriaxone

EAA Receptors

The receptors for glutamate and the other EAAs are categorized into major two groups: ionotropic and metabotropic (Fig. 12.4). The three ionotropic receptor types are classified based on their selective affinity for the EAA ligands: N-methyl-D-aspartate (NMDA), α -amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA), and kainite (Table 12.2). AMPA is a homologue of the neurotoxin ibotenic acid found in *Amanita* mushrooms, whereas kainate is a neurotoxin isolated from red algae.



These ligand-gated ion channel receptors are composed of homo- or heterotetramers of individual subunits that confer cation selectivity. Each subunit has an extracellular N-terminus with three transmembrane domains, an intramembrane reentrant "p-loop" between the first and third transmembrane domains, and an intracellular C-terminus (Fig. 12.5) (9). The metabotropic glutamate receptors belong to the larger family of G protein-coupled receptors (GPCRs). When activated, these metabotropic receptors can alter the activity of effector proteins, such as adenylyl cyclase and phospholipase C (PLC).

TABLE 12.1 Excitatory Amino Acid Transporters

Subtype	Predominant Brain Distribution
EAAT1	Cerebellar glia
EAAT2	Forebrain glia
EAAT3	Cortical neurons
EAAT4	Cerebellar Purkinje neurons
EAAT5	Retina

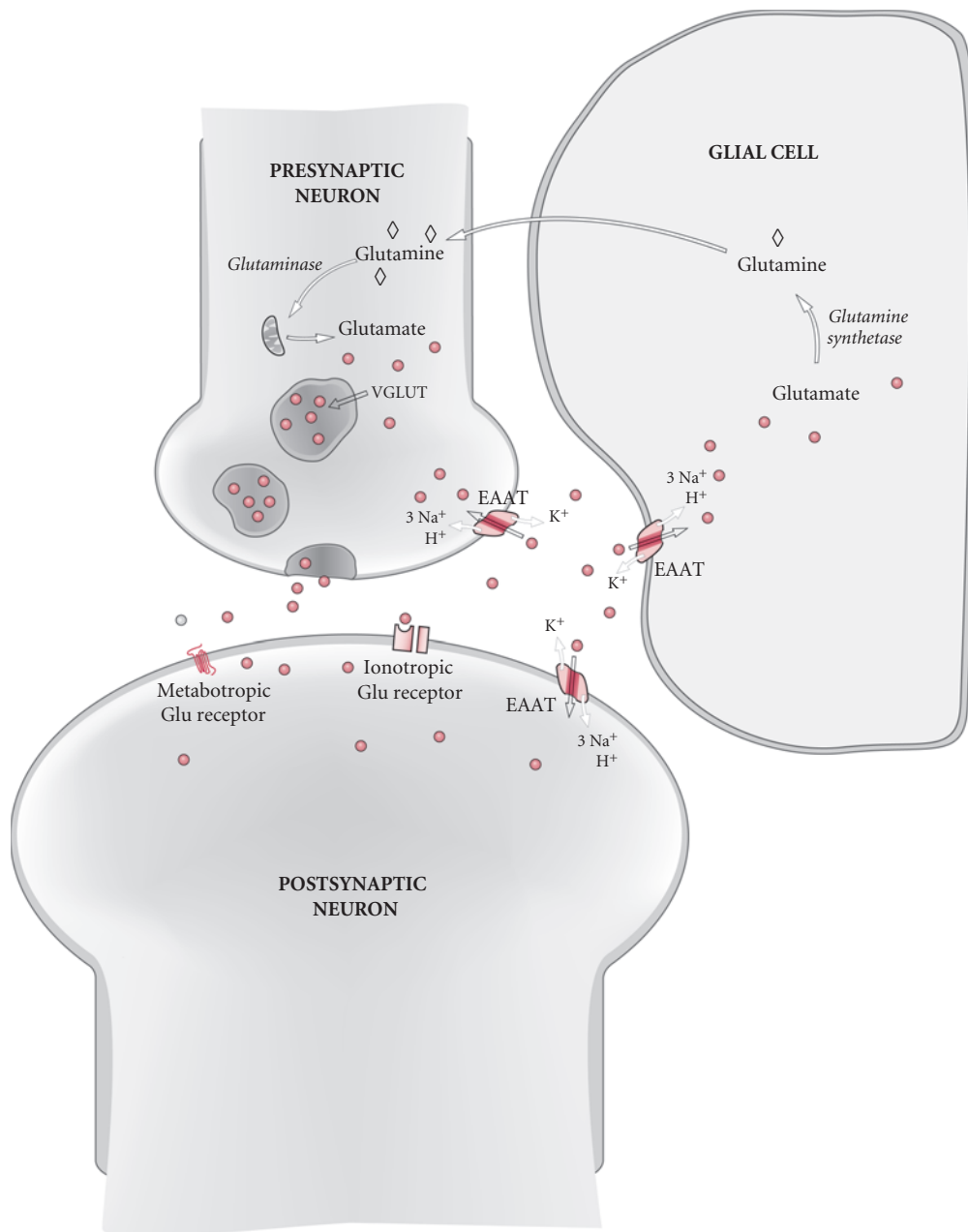


FIGURE 12.4 The glutamate neuron showing the formation of glutamate (o) from glutamine (◊), ionotropic and metabotropic receptors, storage vesicles, and reuptake transporters.

To date, at least eight distinct EAA metabotropic receptor subtypes appear to exist. The EAA receptors, in balance with the receptors for the IAAs, likely are crucial for the regulation of neuronal plasticity involving dynamic changes in neurons as a response to environmental stimuli. Neuronal plasticity accounts for the ability of an organism to learn and adapt, which includes both long-term potentiation and long-term depression.

Ionotropic Receptors

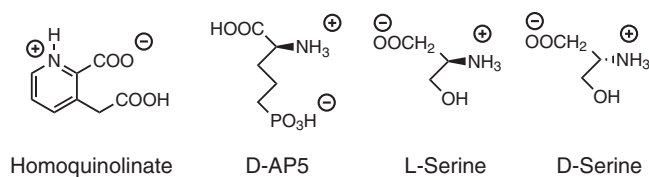
NMDA RECEPTOR The NMDA receptor is a heterotetramer comprised of a number of subunit forms—GLU_{N1},

GLU_{N2A}, GLU_{N2B}, GLU_{N2C}, GLU_{N2D}, GLU_{N3A}, and GLU_{N3B}—that can confer unique pharmacology to individual receptors (Table 12.2). Splice variants can also lead to a number of isoforms of the above subunits with the potential of changing the binding characteristics and functions on the receptor. Activation of the NMDA receptor requires the binding of two agonists: glutamate to the GLU_{N2} subunit, and glycine to a binding site on the GLU_{N1} subunit (10). Glycine appears to act as an important co-agonist positive modulator at a unique recognition site on the GLU_{N1} subunit, and unlike the actions of glycine as an IAA neurotransmitter, this recognition

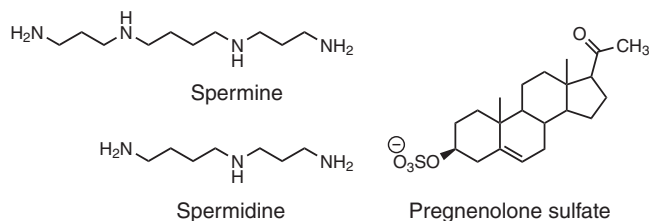
TABLE 12.2 Ionotropic Glutamate Receptor Subtypes

Subtype	Subunits	Agonists	Actions
AMPA	GluR1 GluR2 GluR4	Glutamate or AMPA	Increases Na ⁺ influx, increases K ⁺ efflux
Kainate	GluR5 GluR6 GluR7 KA1 KA2	Glutamate or kainate	Increases Na ⁺ influx, increases K ⁺ efflux
NMDA	NR1 NR2A NR2B NR2C NR2D	Glutamate or NMDA with glycine	Increases Na ²⁺ and Ca ²⁺ influx, increases K ⁺ efflux

site is not sensitive to blockade by strychnine (see the following section on glycine). Thus, this glycine binding site is often referred to as the strychnine-insensitive receptor site. Agonists at the glutamate binding site on the GLU_{N2} subunit include NMDA, glutamate, aspartate, and homoquinolinic acid. The best-characterized antagonist at this site is D-2-aminophosphonopentanoate. D-Serine is a well-known agonist at the GLU_{N1} glycine site. D-Serine, which normally is found only in glia and astrocytes, is synthesized from L-serine via serine racemase and is thought to be one of the important endogenous modulators of NMDA receptor activity.

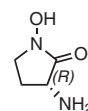


The NMDA receptor has also many sites for channel modulation by pharmacologic agents. Endogenous inhibitory channel modulators include Mg²⁺, Zn²⁺, and protons. Neurosteroids can either inhibit or potentiate NMDA receptor channel function depending on how the subunits comprising the tetrameric structure are assembled. For instance, the neurosteroid pregnenolone sulfate inhibits NMDA receptors that are assembled as GLU_{N1}/GLU_{N2C} but potentiates those assembled as GLU_{N1}/GLU_{N2A} and GLU_{N1}/GLU_{N2B} (11). Polyamines, such as spermine and spermidine, are known to be positive channel modulators.



A number of important NMDA channel antagonists (Fig. 12.6) also exist, and these include amantadine (an antiviral agent that also releases dopamine and is used in Parkinson disease), ketamine (a dissociative anesthetic agent that acts via the NMDA receptor), phencyclidine (a psychoactive drug of abuse also known as PCP), and memantine (recently approved for the treatment of Alzheimer disease). Whereas the antitussive agent dextromethorphan, via its metabolite dextrorphan, is known to block within the NMDA channel, its cough-suppressant activity likely is not the result of action at this site. The psychotomimetic effects observed with the abuse of this compound, however, are likely mediated by the NMDA receptor. An endogenous antagonist of the NMDA receptor is Mg²⁺, which normally prevents the flow of Ca²⁺ through the channel. When glutamate or another suitable agonist binds, along with glycine, the inhibition normally maintained by Mg²⁺ is relieved, Ca²⁺ can flow, and the cell can depolarize. Some studies in experimental animals have demonstrated the NMDA receptor-mediated neuroprotective effects of Mg²⁺ in models of stroke and other CNS insults.

One of the first NMDA receptor antagonists to be identified was dizocilpine, commonly known as MK-801 (Fig. 12.6). Initially, much excitement was generated with the discovery of MK-801, because the ability to block the excessive intracellular cation flow (Ca²⁺, Na⁺) that follows neuronal hypoxic insults, as are seen following cerebral vascular accidents (stroke) or head trauma, might lead to effective treatments for such pathologic conditions. Whereas MK-801 was very effective in decreasing infarct size in various rodent models of stroke, poor clinical efficacy was noted in humans. The generation of severe psychotic behaviors was also deemed to be unacceptable.



(R)-HA-966

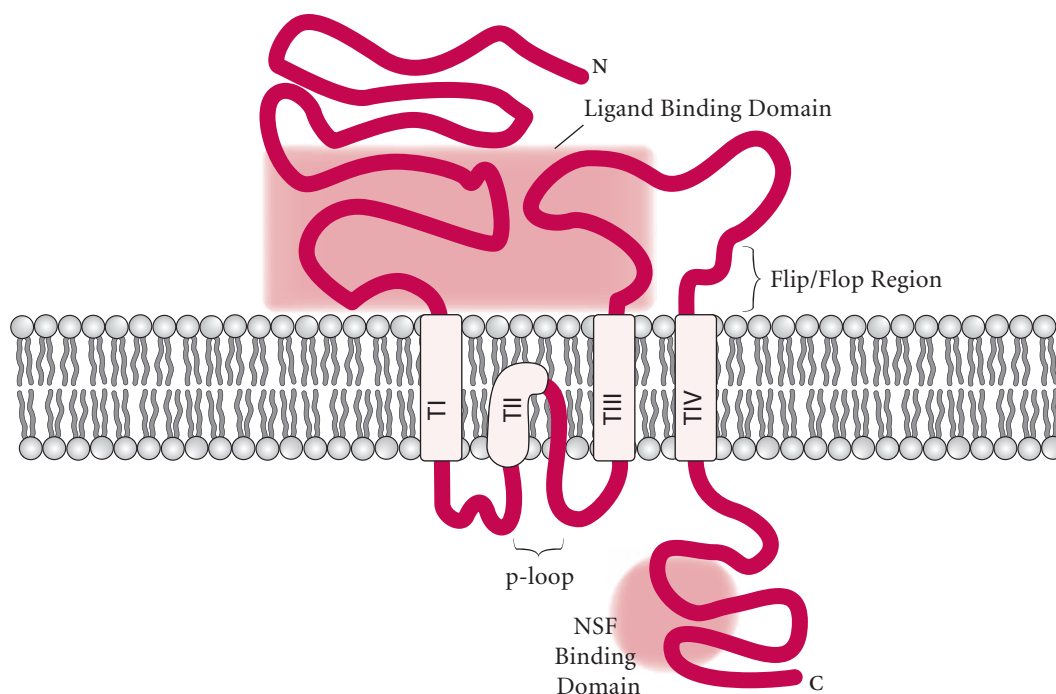
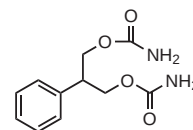


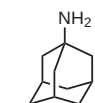
FIGURE 12.5 The ionotropic glutamate receptor showing the p-loop, ligand binding domain, *N*-ethylmaleimide-sensitive fusion protein (NSF) binding domain (shaded area), flip-flop region, and the N- and C-terminal regions.

The strychnine-insensitive binding site on the NMDA receptor has also been a target for researchers, because an agent that would antagonize this binding site might be useful in preventing the neuronal damage that occurs following hypoxic insults that lead to excessive glutamate release or in controlling electrical neuronal dysfunction associated with epilepsy. One agent, *R*(+)-3-amino-1-hydroxypyrrolidin-2-one [(*R*)-HA-966], is an antagonist at the strychnine-insensitive binding site on the NMDA receptor that, unfortunately, has not been used therapeutically because of its hepatotoxic

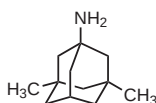
properties unrelated to the NMDA receptor. However, the dicarbamate anticonvulsant felbamate (see Chapter 17) produces part of its activity by inducing conformational (allosteric) changes in the receptor (allosterism) that alters the binding of glycine at the NMDA glycine binding site. Felbamate also has interactions at the AMPA and kainate receptors that contribute to its anticonvulsant activity (12,13).



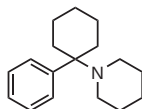
Felbamate



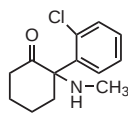
Amantadine



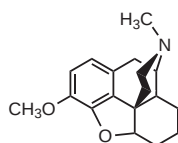
Memantine



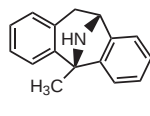
Phencyclidine



Ketamine



Dextromethorphan



Dizocilpine

FIGURE 12.6 NMDA channel antagonists.

AMPA RECEPTORS Another ionotropic EAA receptor is preferentially activated by AMPA. Although at one time, this receptor was referred to as the quisqualate receptor, the AMPA receptor mediates fast synaptic activity via the influx of Na^+ and, in some neurons, K^+ efflux (Table 12.2). The AMPA receptor, which is found in high abundance in the cerebral cortex and hippocampus, is comprised of four subunits: GluR1, GluR2, GluR3, and GluR4. One of these subunits, GluR2, when present prevents the formation of an ionophore that can efficiently conduct Ca^{2+} . However, when the

AMPA receptor does not contain a GluR2 subunit, Ca^{2+} is capable of being conducted through the ion channel. On the extracellular loop between transmembrane segments III and IV there exists a region termed the “flip-flop” that is sensitive to splice variants of the gene coding for each subunit. Such splice variants can lead to significant differences in the desensitization kinetics of the receptor (14). Moreover, intracellular sites on the C-terminus, where modulatory proteins such as *N*-ethylmaleimide-sensitive fusion protein and protein interacting with C kinase can bind, allow another important site where regulation of receptor trafficking can be influenced.

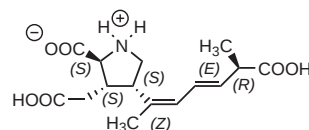
Recently, the discovery that some of the 2,3-benzodiazepines (e.g., GYK 53655) can selectively bind the AMPA receptor has aided in studies to understand its location and functions, especially where mixed populations of EAA receptors are present (15). Agents known to potentiate AMPA receptor activity include piracetam (a drug for improving mental performance), a cyclic derivative of GABA, and the benzothiazide diuretic, cyclothiazide (Fig. 12.7). Some evidence also suggests that certain barbiturates and volatile anesthetics have binding sites on the AMPA receptor. Riluzole (2-amino-6-fluoromethoxy benzothiazole), approved for the treatment of ALS and which acts partly by reducing glutamate release and altering AMPA receptor expression, has been found to have potential utility in treatment-resistant unipolar depression, bipolar depression, and generalized anxiety disorders (16).



Agents that act to cause conformational changes that positively modulate the AMPA receptor have been termed “ampakines” and have been suggested to improve memory, to enhance the activity of certain antipsychotic agents, to improve attention-deficit hyperactivity disorder, to improve Parkinson disease symptoms, and to provide neuroprotection following CNS ischemic insults (17).

KAINATE RECEPTORS The ionotropic kainate receptor is a heterotetramer comprised of subunits GluR5, GluR6,

GluR7, KA1, and KA2 (Table 12.2). The subunits GluR5, GluR6, and GluR7 can form functional homo- and heteromeric receptors; the KA1 and KA2 subunits require the presence of the GluR5, GluR6, and GluR7 subunits to properly assemble into functional receptors. Similarities exist between the binding characteristics of AMPA receptors and kainate receptors such that few pharmacologic agents are available that effectively differentiate between the two (18). Kainate receptors tend to be more sensitive to the phycotoxin domoic acid, an EAA found in red algae (*Pseudonitzschia* sp.), than are AMPA receptors. Domoic acid causes amnesic shellfish poisoning that in humans can cause nausea, vomiting, abdominal cramping, short-term memory loss, seizures, arrhythmias, and death in severe cases following ingestion. Domoic acid bioaccumulates in shellfish that feed on the red algae phytoplankton.



Domoic acid

Kainate receptors are located postsynaptically and mediate neuronal excitation, as do NMDA and AMPA receptors, but kainate receptors have also been found to be located presynaptically. Such presynaptic receptors appear to regulate the release of GABA in the hippocampus and glutamate in some other brain regions. Some kainate receptors have also been shown to be linked to a pertussis toxin-sensitive G protein, which via interaction with PLC can in turn act to influence nearby voltage-dependent Ca^{2+} channels (19). This dual signaling capability of kainate receptors appears to be unique for an EAA ionotropic receptor and can facilitate the role of this receptor subtype in influencing both short- and long-term synaptic plasticity (adaptive changes to the synapse affecting the efficacy of neurotransmission) in the CNS.

Metabotropic Glutamate Receptors

Another class of EAA receptors are the G protein-coupled metabotropic receptors (m), which are categorized into one of three groups based on their pharmacology, sequence similarity, and intracellular effector signaling systems. Group I, which includes mGlu1 and mGlu5, are positively coupled to PLC via $G_{q/11}$. Group II, which includes mGlu2 and mGlu3, and group III, which includes mGlu4, mGlu6, mGlu7, and mGlu8, are negatively coupled via $G_{i/o}$ to adenylyl cyclase (20). The term “metabotropic” is used to indicate that activation of these receptors results in alterations in metabolic processes within the cell. All of the metabotropic receptors are

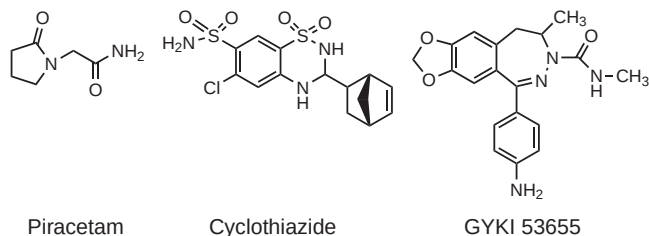
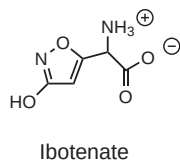


FIGURE 12.7 AMPA agonists.

activated by glutamate and ibotenate, but none is activated by NMDA, AMPA, or kainate.



Few useful selective agonists or antagonists are available for these receptors subtypes, but some researchers have used knockout mice to explore the role of a particular metabotropic receptor. For instance, mice lacking the mGlu5 receptor fail to display the reinforcing and locomotor stimulant effects of cocaine, suggesting that an agent that is selective at antagonizing the mGlu5 receptor might find utility in treating patients dependent on cocaine (21). Because the mGlu5 receptor is located in areas of the brain thought to be involved with anxiety, compounds that negatively modulate this receptor subtype have been shown to be anxiolytic in animal studies. The group II mGlu receptors are also highly concentrated in areas involved with anxiety, such that agents that are selective positive modulators for this receptor subtype are found to act as anxiolytics. Finally, positive modulators of the mGlu4 receptor subtype have been suggested as an approach for treating Parkinson disease (22). As more information is revealed about the distribution, function, binding characteristics, and trafficking of the metabotropic receptor subtypes in the CNS, advances in the therapy of disease states such as Parkinson disease, schizophrenia, chronic pain, epilepsy, depression, and drug dependence can be realized (23).

A steady accumulation of evidence supports a role for glutamate and its mGlu1 and mGlu5 receptors in depression and antidepressant activity. Furthermore, evidence also implicates glutamate release, which can result in activation of NMDA and mGlu1 and mGlu5 receptors, an underlying cause for depression and anxiety. Studies with NMDA receptor antagonists of mGlu1 and mGlu5 receptors as well as positive modulators of AMPA receptors have demonstrated antidepressant-like activity in a variety of preclinical models. The concept of NMDA antagonists as antidepressants has generated considerable interest in the NMDA receptor as a target for new antidepressant therapies (see Chapter 18). Several studies have shown that chronic antidepressant treatment can modulate NMDA receptor expression and function. Preclinical studies with NMDA receptor antagonists have demonstrated their potential antidepressant properties. Other studies, which have demonstrated that NMDA receptors have a role in the development and maintenance of alcohol dependence, suggest that NMDA receptors that are upregulated by alcohol could be a target for therapeutic interventions (24).

INHIBITORY AMINO ACID NEUROTRANSMITTERS

γ -Aminobutyric Acid

The major IAA neurotransmitter in the mammalian CNS is GABA (Fig. 12.2). Initially found to be involved with neuromuscular transmission in the lobster, GABA has since been well characterized as a neurotransmitter in the brain and spinal cord of mammals. In the periphery, in tissues such as the liver, spleen, sympathetic ganglia, and splenic nerve, GABA and its receptors are also found. The levels are very low, however, and their function in these non-CNS tissues is poorly understood. While levels of GABA in the CNS are very high (millimoles per gram of tissue) compared to those of the classical monoamine neurotransmitters (nanomoles per gram of tissue), the distribution of GABA in the CNS is unique, with an unequal distribution, thus supporting a neurotransmitter role. Unlike many of the other putative amino acid neurotransmitters that are also components of CNS proteins, GABA is not known to be incorporated into proteins, thus making it easier to support a role for GABA as a neurotransmitter.

The synthesis of GABA uses glutamate and the enzyme glutamic acid decarboxylase (GAD), the levels of which closely parallel those of GABA (Fig. 12.3). Although glutamate itself functions as an important neurotransmitter, the control of GABA synthesis appears to be dependent on the activity of GAD and its cofactor, pyridoxal phosphate. Glutamate can be synthesized from α -oxoglutaric acid, α -ketoglutarate, and glutamine (25) (Fig. 12.3). Glucose and pyruvate can also be converted to glutamate via metabolism involving the Krebs cycle. Two isoforms of GAD derived from different genes have been characterized: GAD-65 and GAD-67. The lower molecular weight GAD-65 has a much higher affinity for its pyridoxal cofactor than does the larger GAD-67. There also appears to be different cellular and subcellular distribution patterns between the two GAD isoforms, but the significance of their unique roles in GABAergic neurotransmission is not fully understood (26).

In a similar manner to most other classical neurotransmitters, GABA, once synthesized, is stored in vesicles located in the presynaptic nerve terminals, as shown in Figure 12.8. A specific vesicular GABA transporter protein (VGAT) has been identified that transports largely GABA into the storage vesicles (27). The VGAT lacks complete selectivity, because the other major IAA neurotransmitter glycine is also a substrate for VGAT, which has been shown to be present in glycinergic terminals. Unlike the well-defined vesicular transporters for the monoamine neurotransmitters, which are characterized by 12 transmembrane-spanning segments, VGAT only contains 10 such transmembrane-spanning segments. However, similar to the monoamine vesicular transporters, VGAT appears to be driven by an H^+ electrochemical gradient generated by an ATP-dependent H^+ pump in the vesicle plasma membrane. Whereas vigabatrin potently

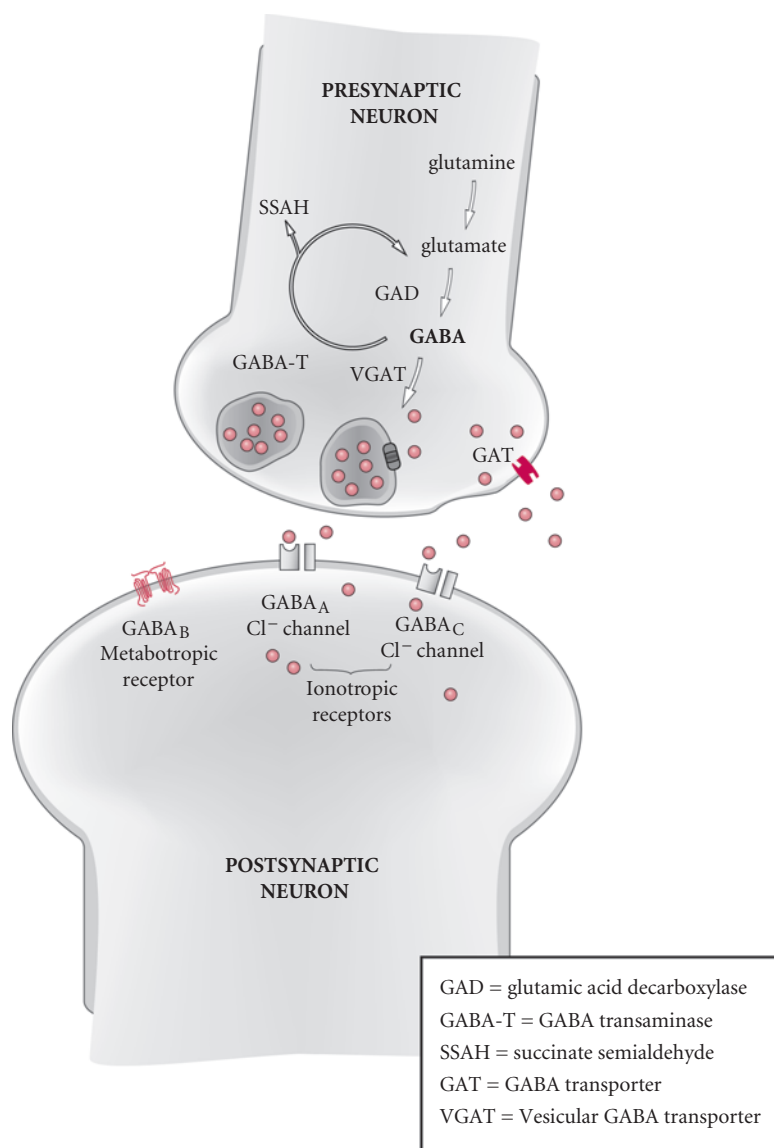
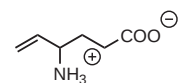
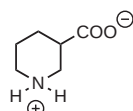


FIGURE 12.8 The GABA neuron, showing the formation of GABA, ionotropic and metabotropic receptors, and reuptake transporters.

inhibits VGAT and nipecotic acid weakly inhibits VGAT, the role of VGAT inhibition in the overall pharmacologic actions of these agents is not well understood, because both agents probably act largely via non-GABA-mediated mechanisms.



Vigabatrin

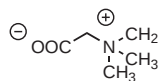


Nipecotate

Nerve stimulation leading to depolarization of GABAergic presynaptic terminals causes the Ca^{2+} -mediated exocytosis of vesicles containing GABA. Once released, GABA diffuses across the synapse to the

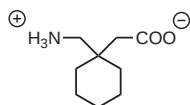
postsynaptic side, where interaction with GABA receptors takes place, leading to hyperpolarization of the postsynaptic neuron, as shown in Figure 12.8 (these receptor subtypes are described in detail below). Some regulation of GABA release occurs via presynaptic GABA_B autoreceptors that decrease subsequent neurotransmitter release. The termination of the GABA-mediated signal takes place as a result of reuptake transporter systems that transport GABA back into the presynaptic neuron or into surrounding glial cells and astrocytes. This dual neuronal-glial reuptake process is distinct from that observed with the classical monoamines, for which neuronal reuptake largely predominates. Four transporters with 12 transmembrane-spanning segments have been identified and cloned: GAT-1, GAT-2, GAT-3, and BGT-1 (28). All four are capable of transporting GABA, but

they can also transport other IAAs, particularly glycine, β -alanine, and taurine (Fig. 12.2). One of the transporters, BGT-1, is capable of transporting the dimethylated glycine derivative betaine and thus is also known as the betaine transporter.

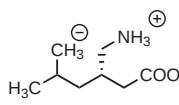


Betaine

A number of inhibitors of these transporters are known, but only one currently is available for therapeutic use. Tiagabine, by selectively inhibiting GAT-1, which can be the major GABA neuronal reuptake transporter, is an approved agent for the treatment of partial seizures (see Chapter 17). Clinical trials are under way for its use in the treatment of anxiety, neuropathic pain, and insomnia (29). Two GABA analogues that have been shown to interact with GAT-1 are gabapentin and pregabalin. The concentrations required to inhibit GAT, however, are very high, and a GABAergic mechanism of action most likely does not explain their anticonvulsant activity (30). Despite their close structural similarity to GABA, inhibition of a voltage-gated calcium channel containing the $\alpha_2\delta_1$ subunit is believed to be responsible for their anticonvulsant actions.

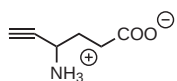


Gabapentin

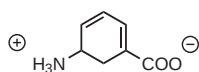


Pregabalin

Metabolism of GABA occurs via a series of enzymatic steps starting with GABA transaminase (GABA-T), as shown in Figure 12.3. This reversible mitochondrial enzyme uses pyridoxal phosphate as a cofactor and is widely distributed throughout the CNS. Transamination results in the formation of succinic semialdehyde (SSA), which can be further metabolized to succinate via SSA dehydrogenase (SSADH) or, alternatively, reduced to γ -hydroxybutyrate (GHB) (see Fig 12.12). Other metabolites of unknown significance include carnitine, homoanserine, homopantothenic acid, and a number of γ -butyryl derivatives, including γ -butyrylsine, γ -butyrylhistidine, and γ -butyrylcholine. Vigabatrin (γ -vinyl-GABA), an irreversible inhibitor of GABA-T, increases the levels of GABA by inhibiting its metabolism and is used for the treatment of seizure disorders in some countries. Other inhibitors of GABA-T include gabaculline and acetylenic GABA. These agents display anticonvulsant activity in experimental animal models of epilepsy.



Acetylenic GABA



Gabaculline

GABA_A Receptors

The initial observations that application of GABA was capable of hyperpolarizing neurons, as well as the lack of activity of structurally related amino acids and other compounds, suggested that a receptor likely mediated this response. At least three different families of GABA receptors have been identified to date: GABA_A, GABA_B, and GABA_C (Fig. 12.9). The GABA_A receptor was the first to be identified and was found to be a neurotransmitter-gated ion channel (ionotropic) that, when activated, allows the entry of chloride ion into the cell, thereby hyperpolarizing the neuron and making cell firing less likely to occur. The GABA_A receptor (Fig. 12.9) is a member of the Cys-loop (cysteine–cysteine loop located on the N-terminal region) family of receptors that also include the strychnine-sensitive glycine, nicotinic acetylcholine, and serotonin 5-HT₃ receptors. The pentameric GABA_A receptor is composed of various combinations of subunits (α , β , γ , δ , ϵ , π , ρ , and θ). Various subunit isoforms have been identified with 6α , 3β , 3γ , and 3ρ subunits (some have suggested that the ρ subunits are found only in the GABA_C receptor). Each subunit is a four

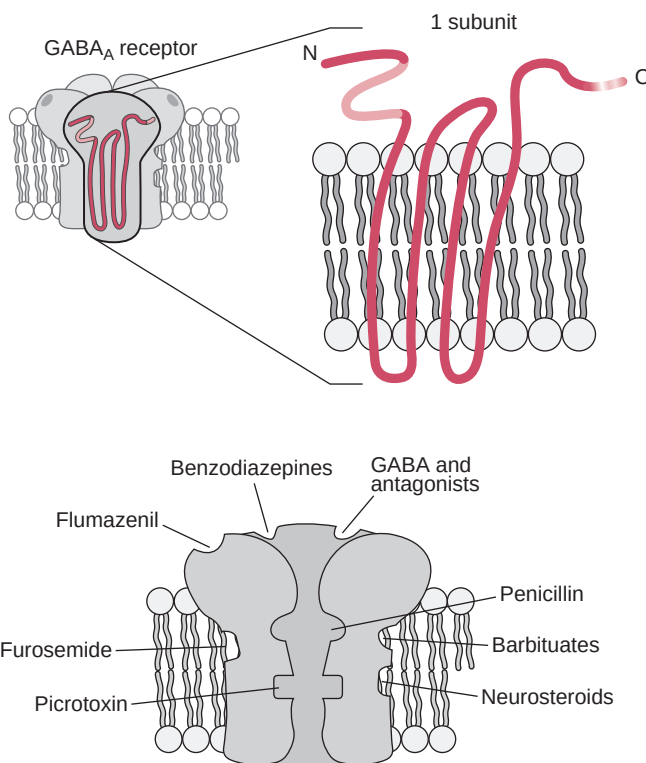


FIGURE 12.9 GABA_A receptor. (A) The structure of the GABA_A receptor, showing the N- and C-terminal regions of the four membrane-spanning regions for each of the five subunits. (B) The major binding sites on the GABA_A receptor. (From Chou J, Strichartz GR, Lo EH. Pharmacology of excitatory and inhibitory neurotransmitters, Chapter 10. In: Golan DE, Tashjian A, Tashjian A, et al., eds. Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy. Baltimore, MD: Lippincott Williams & Wilkins, 2004:139–160, with permission.)

transmembrane-spanning protein that, when arranged in a circle, forms the ion channel with a diameter of approximately 8 nm that remains closed until GABA binds to its ligand recognition site. One of the most common GABA_A receptor conformations in the mammalian CNS consists of a pair of α_1 subunits, a pair of β subunits, and a single γ_2 subunit, as shown in Figure 12.10. Other commonly identified receptors contain α_1 , α_2 , α_3 , and α_5 as well as forms of the β subunit and, typically, the γ_2 subunit, and always in a 2:2:1 stoichiometry (31). The binding of two molecules of GABA, as shown in Figure 12.10, probably to the individual β subunits near the α - β interface, is believed to be required for normal receptor activation. In fact, among GABA_A receptors, 17 different combinations of subunits have been identified. Having a family of receptors that all share the same basic structure but that differ in specific subunit composition allows greater levels of functional diversity. Each of the receptor varieties varies in binding affinity, channel activity, and the degree to which GABA binding is affected by different endogenous modulators. As a result, inhibitory neurotransmission can be more finely controlled. Such receptor heterogeneity also allows more control at the genomic level, in turn allowing post-synaptic cells to respond to changing environmental needs or variable activity at the synapse (plasticity). In contrast, altered subunit compositions could also be partly responsible for an increased susceptibility to pathophysiologic conditions in individuals.

A major class of compounds that modulate GABA_A receptor function is that of the benzodiazepines (e.g., diazepam). The binding site for benzodiazepines likely is the α subunit in proximity to the β subunit, as shown in Figure 12.10. The form of the γ subunit appears to help to determine the affinity of the individual benzodiazepine to the receptor. The benzodiazepines appear to bind more efficiently to receptors containing γ_2 than to the γ_1 subunit, as shown in Table 12.3. Similarly, very low-affinity binding is observed if a receptor contains the α_6 subunit. In fact, the α_6 subunit seems to confer binding preference to inverse agonists (agents that stabilize the inactive/resting receptor), such as Ro194603. The benzodiazepines do not bind to the GABA (ligand-recognition) site on the receptor and can only produce effects if presynaptic GABA has been released and is present at the receptors.

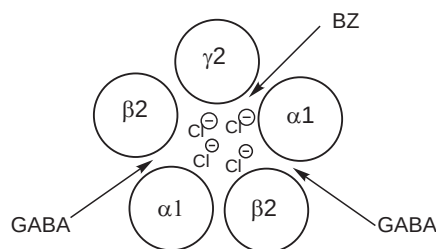
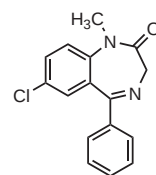
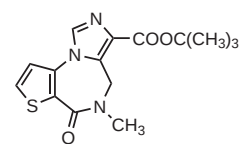


FIGURE 12.10 Extracellular view of the GABA_A receptor showing the location of benzodiazepine (BZ) and low-affinity GABA binding sites.

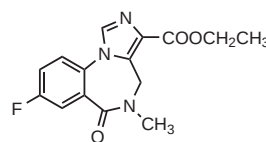


Diazepam

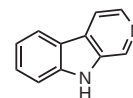


Ro-19-4603

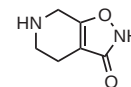
Benzodiazepines are allosteric modulators, or ligands that bind at a secondary binding site on a receptor that is distinct from the primary ligand binding site, resulting in conformational changes in the structure of the receptor that either activates or, sometimes, inhibits the receptor. Thus, the benzodiazepines allosterically modulate the GABA_A receptor, increasing the frequency of the chloride channel opening when GABA is bound, thus potentiating the response of exogenously released GABA. Clinically the benzodiazepines are very safe when used alone in the absence of CNS depressants, because they are not active on GABA receptors alone. This is in contrast to the barbiturates that can directly activate the GABA_A receptor when present at higher concentration and, thus, have a much lower therapeutic index. Flumazenil, a benzodiazepine receptor antagonist, is used for the treatment of severe overdoses of benzodiazepines. Flumazenil competitively antagonizes the binding and allosteric effects of benzodiazepine agonists as well as benzodiazepine inverse agonists, such as the β -carbolines.



Flumazenil

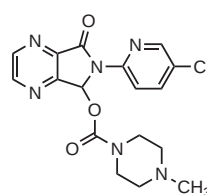


β -Carboline

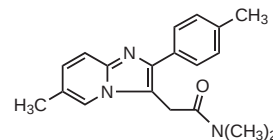


Gaboxadol

Using molecular biologic techniques, point mutations of the α subunits have revealed that the sedative effects of the benzodiazepines likely result from an interaction with the α_1 subunit, whereas the anxiolytic effects result from an interaction at the α_2 subunit (32,33), as shown in Figure 12.10 and Table 12.3. Nonbenzodiazepine receptor agonists, such as the sedative-hypnotics indiplon, zaleplon, zopiclone, and zolpidem (see Chapter 16), are α_1 subunit-preferring ligands, as shown in Table 12.3 (34).



Zopiclone



Zolpidem

An agent that acts on the GABA_A receptor that does not interact with the benzodiazepine binding site is gaboxadol (previously called THIP). This sedative-hypnotic

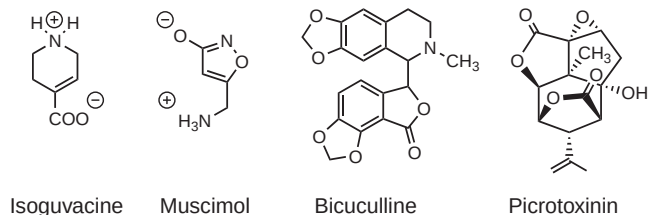
TABLE 12.3 A Comparison of the Binding Affinities (nm) of Benzodiazepine and Nonbenzodiazepine Ligands for GABA_A Receptor Subtypes (45)

Compound	GABA _A Receptor Subtypes					
	$\alpha_1\beta\gamma_2$	$\alpha_2\beta\gamma_2$	$\alpha_3\beta\gamma_2$	$\alpha_4\beta\gamma_2$	$\alpha_5\beta\gamma_2$	$\alpha_6\beta\gamma_2$
Benzodiazepines						
Diazepam	16	17	17	>10,000	15	>10,000
Clonazepam	1.3	1.7	2.0	—	—	>10,000
Triazolam	1.8	1.2	3.0	—	1.2	—
Flumazenil	1.0	1.1	1.5	107	2.4	90
Nonbenzodiazepines						
Zaleplon	130	1,820	1,530	>10,000	490	>10,000
Zolpidem	17	290	357	—	>15,000	—
Zopiclone	19	33	—	—	—	—

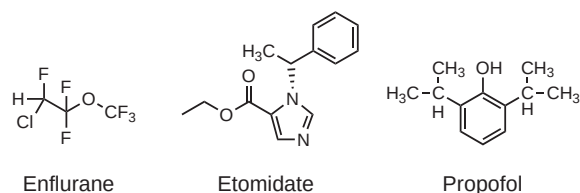
binds with high affinity to the extrasynaptic $\alpha_4\beta\delta$ GABA_A receptor and does not alter sleep onset or REM sleep as do the benzodiazepines. On the other hand, gaboxadol increases slow-wave sleep (35).

The barbiturates bind to a different portion of the GABA_A receptor and, similar to the benzodiazepines, enhance the activity of GABA. The binding of a barbiturate leads to an increase in the duration of ion channel opening. Higher concentrations of the barbiturates can directly open the chloride channel; thus, overdoses of barbiturates can lead to life-threatening, CNS-mediated respiratory and cardiovascular depression.

A number of compounds are known to directly activate the GABA_A receptor. Of these compounds, muscimol, the decarboxylated metabolite of ibotenate, and isoguvacine are the most well known. Unlike GABA, which does not cross the blood–brain barrier, these agents do cross the blood–brain barrier and display GABA-mimetic activity following peripheral administration. A number of compounds can also bind to the GABA_A receptor and antagonize the actions of GABA. Bicuculline, by binding to the GABA recognition site and preventing GABA from binding, produces convulsions in experimental animals. Picrotoxin, via its active metabolite picrotoxinin, binds in the chloride channel to prevent ion flow when the receptor is activated by GABA. Picrotoxin does not alter GABA binding but, instead, prevents ions from flowing and is a potent convulsant.



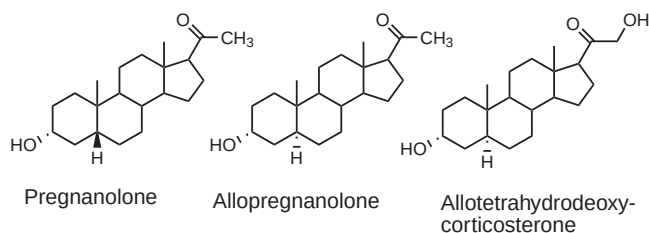
With the many possible subunit assembly sequences of GABA_A receptors, advances in medicinal chemistry can someday lead to the design of compounds that target specific pentameric subunit assemblies to preferentially produce specific effects (36). For instance, located on the base of the dendritic spines of hippocampal pyramidal cells are GABA_A receptors with an α_5 subunit composition. These receptors are thought to counteract the excitatory input because of glutamatergic NMDA receptor activation involved in learning and memory. Activation of GABA_A receptors in this area disrupts learning and memory, whereas NMDA receptor activation improves learning and memory. Administration of an inverse agonist of the GABA_A receptor with selectivity for the α_5 subunit improves memory performance in experimental animals (37). Furthermore, α_3 -selective agonists can be useful in treating schizophrenia, because this subtype of GABA_A receptor appears to have a role in decreasing the release of dopamine from overactive neurons in the mesolimbic system (38).



General anesthetics (see Chapter 14) also appear to have interactions with the GABA_A receptor in producing their various anesthetic effects, including immobilization, respiratory depression, and hypnosis via binding to hydrophobic pockets within the receptor. Using point-mutated knock-in mice, general anesthetic agents, such as enflurane, etomidate, and propofol, have been found to interact with the β_3 subunit of the GABA_A receptor to produce immobilization and hypnosis (38). The heart

rate and body temperature depressant effects of etomidate and propofol, however, do not appear to be mediated by the sub- β_3 subunit. Studies are under way to improve our understanding of the molecular mechanisms of this chemically varied group of agents used as general anesthetics.

Neurosteroids are also capable of modulating the activity of GABA at the GABA_A receptor by binding to a site distinct from that used by GABA, benzodiazepines, and barbiturates. These steroids, including the progesterone metabolites pregnenolone, allopregnenolone, and allotetrahydrodeoxycorticosterone, enhance GABA-mediated inhibitory activity. The presence of the δ subunit in the pentameric GABA_A receptor greatly increases the affinity of steroid binding and efficacy. High concentrations of these steroids can also result in direct activation of the receptor.



Thus, it can be possible in the future to selectively target the various modulatory sites on the GABA_A receptor to produce preferential pharmacologic effects—for example, benzodiazepines that are anxiolytic without sedative effects, antischizophrenic agents that lack sedation and

extrapyramidal side effects, and general anesthetics that do not alter respiratory and/or heart rates.

GABA_B Receptors

As studies continued, the application of GABA to cells occasionally was found to produce effects that were not blocked by the standard GABA receptor antagonist bicuculline. Standard GABA agonists, such as muscimol and isoguvacine, were also ineffective as agonists in such systems. This led to the suggestion that a different subtype of GABA receptor mediated these observed effects. In 1998, the structure of the GABA_B receptor was elucidated and found to be a GPCR heterodimer comprised of GABA_{B1} and GABA_{B2} subunits (Fig. 12.11). Each subunit had a seven membrane-spanning segment, which allowed their intracellular C-termini to connect in a 1:1 stoichiometric fashion (39). The GABA_{B1} subunit was also found to exist in two isoforms, GABA_{B1a} and GABA_{B1b}. Some have suggested that the GABA_{B1a} subunit is preferentially located presynaptically, whereas the GABA_{B1b} subunit is located postsynaptically. Within this metabotropic receptor, the GABA_{B1} subunits contain the GABA binding domain extracellularly, whereas the GABA_{B2} subunit couples to the G-protein mechanism intracellularly. Both subunits are required for normal receptor function, as shown in Figure 12.11. Knockout mice lacking either of these subunits typically display epileptiform seizures, hyperalgesia, hyperlocomotion, and impaired memory function (40). Besides the normal endogenous agonist GABA, a structural analogue, (–)-baclofen, acts stereoselectively as an agonist at the GABA_B receptor.

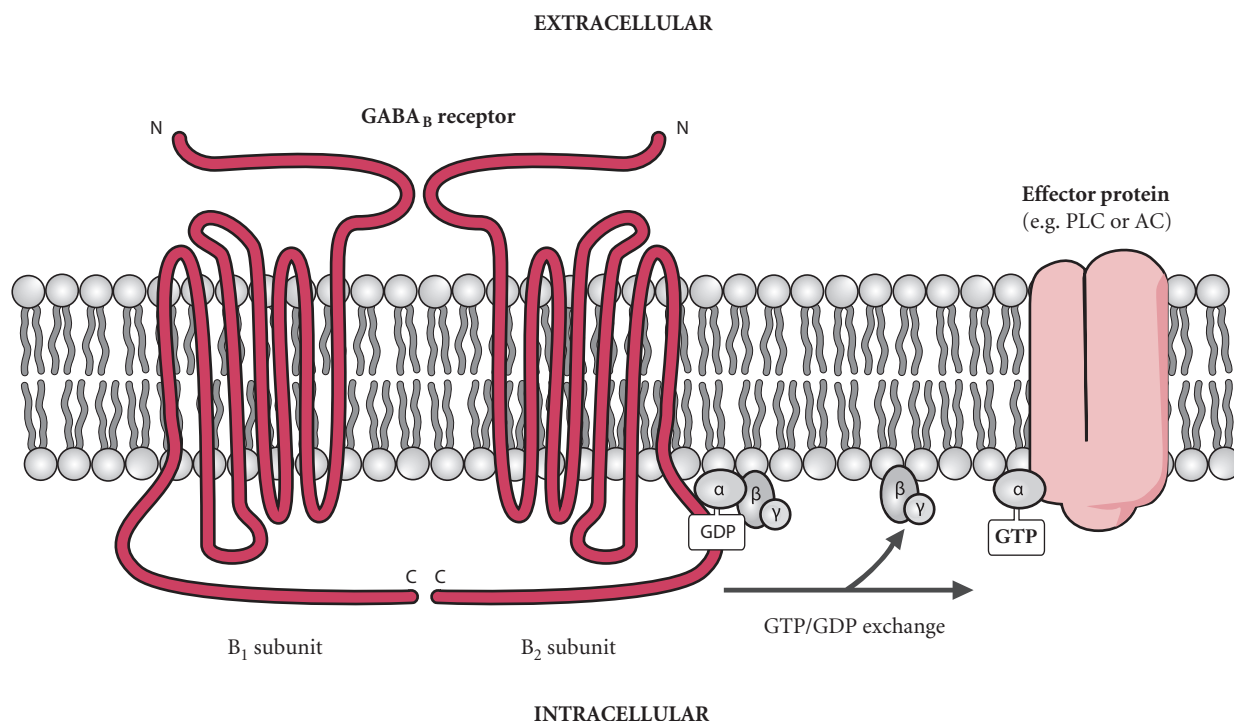
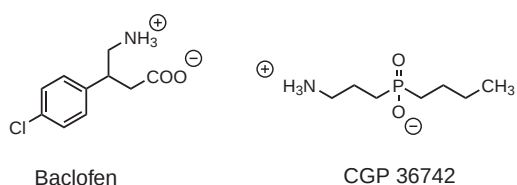


FIGURE 12.11 The GABA_B receptor, showing its B₁ and B₂ subunits and its mechanism of signal transduction.



(±)-Baclofen, which enters the brain via a selective transporter, has been approved for therapeutic use since 1972 for the treatment of spasticity as a muscle relaxant (41). Besides relaxing skeletal muscles, baclofen is capable of exacerbating absence seizures, decreasing cognitive function, increasing food intake, and decreasing drug-seeking behaviors. Drug development of selective GABA_B receptor antagonists can someday lead to therapeutically useful agents for the treatment of absence seizures, depression, anxiety, and cognitive impairments. One such compound currently undergoing clinical trials is the orally bioavailable GABA_B receptor antagonist CGP-36742, which is in clinical trials for the treatment of mild cognitive impairment and Alzheimer disease. The inhibition of GABA_B receptors is believed to cause disinhibition of somatostatin (a peptide neurotransmitter in the CNS) release, which then allows activation of NMDA receptors in the hippocampus that normally are required for memory and cognitive functions (42).

γ-Hydroxybutyrate

Another endogenous compound that binds to and weakly activates the GABA_B receptor is GHB (43). This compound, which can be formed as a result of the metabolism of GABA, has a unique distribution in the CNS, but it does not parallel closely that of GABA, as shown in Figure 12.12. Levels of GHB are increased significantly in patients and knockout mice lacking the enzyme SSADH, which is responsible for the conversion of SSA to succinic acid, with eventual entry into the citric acid cycle. A deficiency of this enzyme in patients is caused by a rare, autosomal recessive inheritance and is termed “GHB aciduria” (44).

Endogenous GHB likely interacts with a recently discovered GHB receptor, but the function of GHB in the CNS is not completely understood. A specific GHB receptor was reportedly cloned in 2003 and found to have no sequence homology with the GABA_B receptor—or with any other known GPCR (45). This putative receptor is a member of the GPCR family and is distributed in most brain regions known to bind GHB. The association is not perfect, however, because high levels of mRNA for the GHB receptor are found in the cerebellum even though little GHB normally binds here.

In the United States, GHB is approved for the treatment of cataplexy associated with the sleeping disorder narcolepsy (46). Narcolepsy is most often characterized by a lack of the CNS 11-amino acid polypeptide orexin (also known as hypocretin) or, possibly, its receptors. Patients with narcolepsy experience excessive bouts of sudden-onset daytime sleepiness and have been treated in the past with

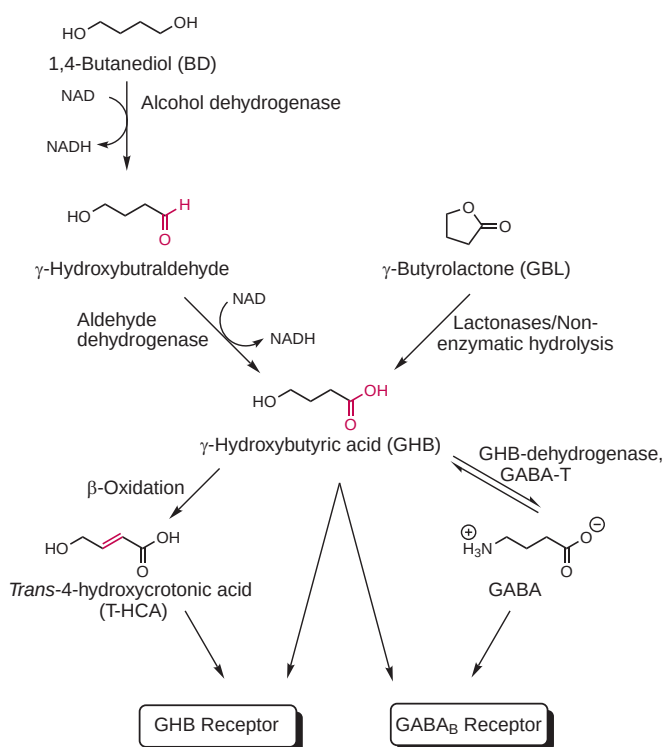
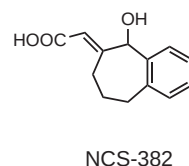


FIGURE 12.12 Biotransformation of 1,4-butanediol (BD) and γ-butyrolactone (GBL) and the receptor sites for their metabolites, γ-hydroxybutyrate (GHB), GABA, and trans-4-hydroxycrotonic acid (T-HCA).

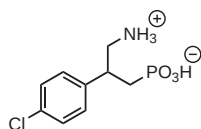
stimulants, such as methylphenidate and amphetamine, or with modafinil (see Chapter 22) or antidepressants. The use of GHB in narcolepsy takes a different approach in that dosing is at bedtime and, being a CNS depressant, it acts to enhance sleep quality. Patients then awake the following morning and experience a rebound insomnia, which counteracts any propensity for daytime sleepiness. Although not a U.S. Food and Drug Administration–approved use, GHB is being investigated for its efficacy in treating fibromyalgia. Exogenously administered GHB appears to produce its clinical and toxic effects via interaction with GABA_B receptors, because most studies have been able to block its effects with GABA_B receptor antagonists but not with the GHB receptor antagonist NCS-382 (although NCS-382 is not an ideal antagonist because it can function as a partial agonist).



In addition, GHB has been used for illicit purposes. Because GHB produces CNS sedation with amnesia, some have used this compound for drug-facilitated sexual assault (e.g., “date rape”). As governmental

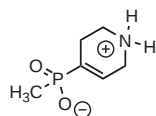
regulations on the availability of GHB became tighter, the use of GHB precursors, such as γ -butyrolactone and 1,4-butanediol, became popular (47). Figure 12.12 diagrams the biotransformation of γ -butyrolactone and 1,4-butanediol to GHB and GABA, explaining their relationship to GHB and GABA_B receptors. The popularity of these compounds as drugs of abuse is facilitated by their ready availability as industrial solvents. Furthermore, GHB can be further metabolized by β -oxidation to *trans*-4-hydroxycrotonic acid, which can also bind to the GHB receptor. Body builders use GHB and its precursor compounds as well in an attempt to enhance the release of growth hormone, which actually increases due to their ability to produce sleep. Most often, however, chronic abuse leads to tolerance, with a need to increase dosing and a propensity to produce physical dependence (48). A serious and difficult-to-treat withdrawal syndrome from GHB abuse has been documented.

GABA_C Receptor



Phaclofen

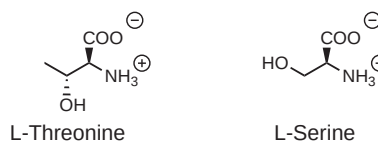
Another receptor that binds GABA but is not antagonized by bicuculline (GABA_A) or phaclofen (GABA_B) and is not influenced by either the benzodiazepines or barbiturates is known as the GABA_C receptor. The endogenous neurotransmitter GABA is an order of magnitude more potent on the GABA_C receptor as compared with the GABA_A receptor, and the responses to activation of the GABA_C receptor are much slower and sustained as compared with the rapid and brief responses following GABA_A receptor activation. The GABA_C receptor is most abundant in the retina, with significant levels in the spinal cord and pituitary gland. This ligand-gated ion chloride channel is a pentamer of subunits that form the ion channel. Some have suggested that the ρ subunit is unique to the GABA_C receptor (49). On the extracellular domain are binding sites for zinc, which is a potent modulator of receptor activity. The most well-described antagonist is TPMPA. Interestingly, isoguvacine, an agonist at the GABA_A receptors, acts as an antagonist at the GABA_C receptor. Much information regarding the location, function, and pharmacology of the GABA_C receptor is needed to begin to take advantage of this receptor for therapeutic purposes.



TPMPA

GLYCINE

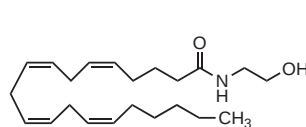
The simplest of all amino acids, glycine, which classically causes hyperpolarization of neurons in the CNS, satisfies all the required criteria for neurotransmitter candidacy (50). Glycine in the CNS is unevenly distributed, with the highest concentrations found in the ventral gray areas of the spinal cord and in the medulla oblongata in the brainstem (51). The retina is also a structure with high concentrations of glycine (52). The synthesis of glycine in glycinergic neurons appears to be from L-serine and to be regulated by the enzyme serine hydroxymethyltransferase (SHMT), the concentrations of which in the CNS parallel those of glycine. Another substrate of SHMT, however, is L-threonine, which can also serve as a precursor of glycine. Administration of large doses of L-serine to animals fails to alter CNS glycine levels, but administration of L-threonine, which passes the blood-brain barrier more readily than L-serine does, increases CNS glycine levels (53). L-Threonine has been used as an orphan drug to treat a rare form of spasticity, familial spastic paraparesis, in humans (54). The synthesis of glycine could also occur via a transamination reaction involving glutamate in which glycine and α -ketoglutarate are produced. The understanding of the synthesis of glycine in the CNS for its neurotransmitter functions has been a difficult hurdle to overcome, likely because of the varied roles of glycine in the synthesis of protein, porphyrin, bile salts, and nucleic acids in addition to intermediary metabolic processes involving one-carbon fragments.



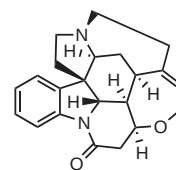
Glycine is stored in vesicles, and its uptake is mediated via VGAT, which is also capable of transporting GABA. Release of glycine occurs from presynaptic terminals in a Ca^{2+} -dependent fashion and, similarly, is taken back up into these terminals and, possibly, nearby glial cells (55,56). Endocannabinoids released from the postsynaptic membrane can act retrogradely at presynaptic glycinergic terminals via the cannabinoid-1 receptor to inhibit the release of glycine (57). Reuptake is via Na^+/Cl^- -dependent neurotransmitter transporters known as GLYT-1 and GLYT-2. These 12 transmembrane-spanning transporters are located in areas where glycine is believed to act as a neurotransmitter or coagonist at NMDA receptors. The GLYT-1 transporters are largely found on glial cells and at glutamatergic neurons, whereas the GLYT-2 transporter is found mostly on presynaptic glycinergic neurons (58). Few pharmacologic tools are available to examine the role of either form of the membrane transporters.

The glycine receptor mediating the inhibitory actions of glycine is similar to the GABA_A receptor and other members of the Cys-loop family in being a pentameric, circular ion channel that allows the conduction of chloride (59). Unlike the GABA receptor, however, the inhibitory glycine receptor can be a homopentamer of α subunits or a heteropentamer comprised of two α subunits and three β subunits. Four isoforms of the α subunits have been identified, whereas only one β subunit is known. The neurotransmitter glycine, in addition to taurine, D-alanine, L-alanine, β -alanine, hypotaurine, serine, and β -aminobutyric acid, can bind to any of three sites on the receptor to lead to activation. Positive modulators of the inhibitory glycine receptor include zinc, neurosteroids, propofol, ethanol, and volatile anesthetics, such as isoflurane (60,61).

The best-described antagonist of the inhibitory glycine receptor is the convulsant strychnine. Strychnine binds to a different site than that which recognizes glycine. Sometimes, the inhibitory glycine receptor is referred to as the “strychnine-sensitive” glycine receptor to distinguish it from the glycine modulatory site on the glutamatergic NMDA receptor. The GABA_A receptor antagonist picrotoxin can also inhibit the IAA glycine receptor. More recently, the endocannabinoids, such as anandamide and 2-arachidonylglycerol, have also been shown to antagonize the activity of IAA glycine at this receptor (57). Ginkgolide B has similarly been shown to act as an antagonist at the IAA glycine receptor. Agents that act as agonists at the IAA glycine receptor can find utility as anticonvulsants, muscle relaxants, sedatives, and general anesthetics.



Anandamide



Strychnine

As more information regarding the control of the synthesis of the neurotransmitter pool of glycine becomes available and more compounds with selectivity for the IAA glycine receptor are discovered or developed, useful therapeutic agents that take advantage of modulating glycinergic neurotransmission can be realized.

FUTURE DIRECTIONS

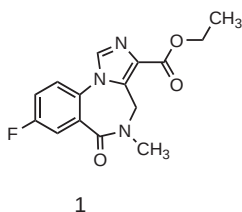
The amino acids have an essential role in the normal control of neurotransmission in the CNS, but during disease, these same amino acids can mediate a number of the pathologic consequences that are observed. Currently there are many therapeutic agents available that take advantage of one or more aspects of amino acidergic neurotransmission to produce their pharmacologic effects. As agents are developed that are more specific for a particular isoform configuration of an amino acid receptor or transporter, or other aspect of amino acid neurotransmitter function, the treatment of various diseases likely will become more effective. Associations between disease states and receptor isoform configurations, transporter density, and activity of enzymes involved in the synthesis and degradation of amino acid neurotransmitters can help to improve our understanding of the pathology of these diseases and lead to improved therapies.

CASE STUDY

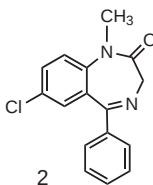
S. William Zito and Victoria Roche

SW is a 67-year-old white man who was recently in a car accident and sustained minor soft tissue bruising and shoulder and neck muscle spasms. SW is an academic and spends most of his day working at his computer. The neck and shoulder muscle spasms are severe enough to make it difficult to work for more than 20 minutes at a time before having to take a break. SW visits his primary care physician (PCP) seeking relief from

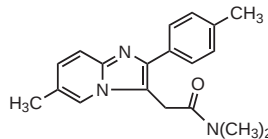
his muscle spasms so he can complete a writing commitment on time. A review of SW's current medication includes Lipitor for hyperlipidemia (10 mg daily) and Uloric for gout (40 mg daily). Clinical laboratory tests indicate that both conditions are under control. His PCP knows that SW is a health care educator who likes chemistry and presents him with the following three structure choices and asks him for his opinion.



1



2



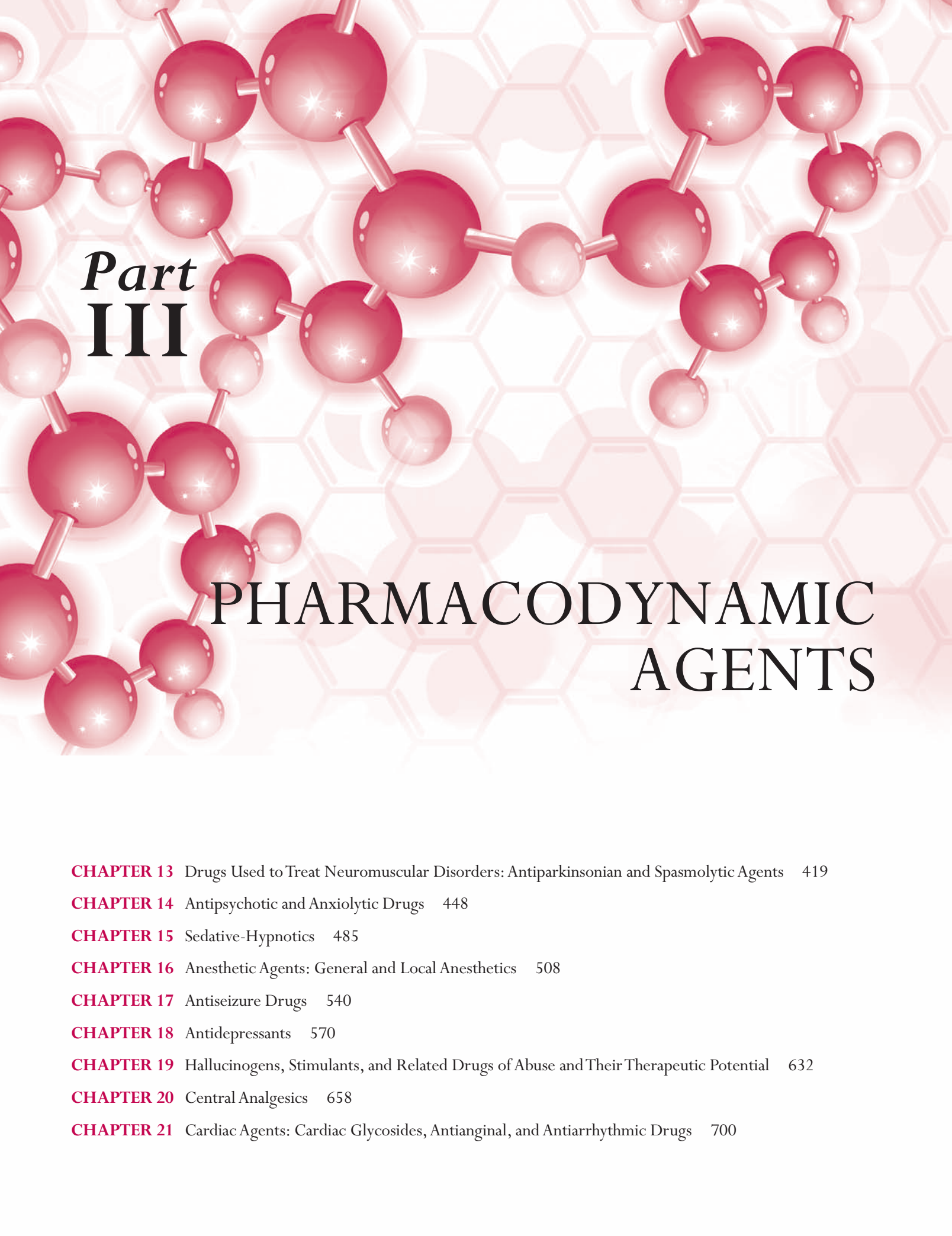
3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Part **III**

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Chapter 13

Drugs Used to Treat Neuromuscular Disorders: Antiparkinsonian and Spasmolytic Agents

RAYMOND G. BOOTH

Drugs Covered in This Chapter*

- | | | |
|--------------------------|-------------------------|--------------------|
| • Amantadine | • Entacapone | • Rotigotine |
| • Apomorphine | • <i>Istradefylline</i> | • Safinamide |
| • Baclofen | • L-DOPA | • <i>Sarizotan</i> |
| • Botulinum toxin-type A | • Memantine | • Selegiline |
| • Bromocriptine | • <i>Pergolide</i> | • Tizanidine |
| • Cabergoline | • Pramipexole | • Tolcapone |
| • Dantrolene | • Rasagiline | |
| • Diazepam | • Ropinirole | |

Abbreviations

- | | | |
|--|---|--|
| A-5,6-DTN , 2-amino-5,6-dihydroxy-1,2,3,4-tetrahydronaphthalene | GDP , guanosine diphosphate | MPTP , 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine |
| ATP , adenosine triphosphate | GPCR , G protein-coupled receptor | NMDA , <i>N</i> -methyl-D-aspartate |
| CNS , central nervous system | GTP , guanosine triphosphate | PD , Parkinson disease |
| COMT , catechol- <i>O</i> -methyltransferase | 5-HT , 5-hydroxytryptamine | PET , positron emission tomography |
| DA , dopamine | HVA , homovanillic acid | SAMe , <i>S</i> -adenosyl-L-methionine |
| DAT , DA transporter | L-DOPA/levodopa , L-dihydroxyphenylalanine | SPECT , single photon emission computed tomography |
| DOPAC , 3,4-dihydroxyphenylacetic acid | MAO , monoamine oxidase | UPS , ubiquitin-proteasome system |
| DOPAL , 3,4-dihydroxyphenylacetaldehyde | MPDP⁺ , 1-methyl-4-phenyl-2,3-dihydropyridinium | VMAT2 , vesicular monoamine transporter 2 |
| FDA , U.S. Food and Drug Administration | MPP⁺ , 1-methyl-4-phenylpyridinium | |
| Gα , G protein α -subunit | MPPP , 1-methyl-4-propionoxy-4-phenylpiperidine | |
| GABA , γ -aminobutyric acid | | |

*Drugs listed include those available in the United States; those available outside the United States are shown in italics.

SCENARIO

David Hayes, PharmD

A 56-year-old man is referred to a neurology clinic by his primary care physician secondary to a 6-month unilateral intermittent tremor in his left arm that is triggered by stress. His evaluation at the clinic reveals normal cognition, absence of oculomotor dysfunction, mild left arm rigidity and resting tremor, with normal gait and postural reflexes. The neurologist's impression is early stage of Parkinson disease. His medications include HCTZ 25 mg

daily, lovastatin 20 mg at bedtime, and a daily multivitamin. His medical history is significant for hypertension and hyperlipidemia. Should pharmacotherapy be started for this patient, and which would be best suited to his condition?

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

OVERVIEW OF NEUROMUSCULAR DISORDERS

Neuromuscular disorders covered in this chapter include the neurodegenerative movement disorder, Parkinson disease (PD), and various spasticity disorders. PD is characterized by debilitating tremor, rigidity, and bradykinesia. The neuropathology (but not the etiology) of PD, with resulting dopamine (DA) neurotransmitter deficit, has been well defined for decades; however, pharmacotherapy of the disorder remains far from satisfactory. The development of prophylactic and, perhaps, curative pharmacotherapy of PD requires advances in understanding of the causes and pathogenesis of the disease, and this chapter reviews some current research in these areas in addition to available pharmacotherapy. Considered are genetic, maturational, metabolic, and environmental contributors to risk for the disease.

Spasticity disorders covered here are characterized by a general increase in tonic stretch reflexes and flexor muscle spasms together with muscle weakness. Muscle spasticity may accompany a number of different disorders but mostly is associated with cerebral palsy, multiple sclerosis, spinal cord injury, and stroke. These disorders do not share a similar pathophysiology with neurodegenerative diseases, such as parkinsonism. Accordingly, drugs used to treat spastic neuromuscular disorders have mechanisms of action that differ from those used to treat PD. Nevertheless, most drugs described in this chapter to treat neurodegenerative or spastic neuromuscular disorders have in common the ability to reduce muscle tone by virtue of their action on the central nervous system (CNS).

PARKINSON DISEASE

Clinical Features and Neuropathology

PD, first described by James Parkinson (1) in 1817, typically presents after age 55 and affects approximately 1% to 2% of the population over age 65; after age 84, incidence increases to 3% to 5% per year (2). Clinically, PD presents as a classic triad of signs: 1) resting tremor that improves with voluntary activity; 2) bradykinesia or slow initiation and paucity of voluntary movements; and 3)

rigidity of muscle and joint motility that includes postural disturbances. Dementia is about six times more common among elderly patients with PD, and there can be other spontaneous or treatment-associated neuropsychiatric disturbances, including hallucinations and depression (3). Although L-dihydroxyphenylalanine (L-DOPA or levodopa) pharmacotherapy has decreased morbidity and mortality, mortality among PD patients still is approximately 60% greater than in age-matched controls (4).

Neuropathologically, PD is a slowly progressive, neurodegenerative disorder of the extrapyramidal dopaminergic pathway that has cell bodies in the A9 substantia nigra region and nerve terminals in the corpus striatum (Fig. 13.1). The disease is characterized by gradual destruction of DA-containing neurons in the pars compacta component of the pigmented midbrain substantia nigra, leading to a deficiency of the neurotransmitter in DA nerve terminals of the corpus striatum (3). Degenerative changes in the pigmented nuclei of the noradrenergic locus ceruleus region also are typical, as is the appearance of intraneuronal inclusions called Lewy bodies (5). Neurochemically, the striatal DA deficiency accounts for the major motor symptoms of the disease, but loss of noradrenergic nerve terminals with concomitant reductions in norepinephrine may account for several of the nonmotor features seen in PD, including fatigue and abnormalities of blood pressure regulation. The mainstay of pharmacologic treatment (3), developed in the mid-1960s, continues to be replacement therapy with the α -amino acid L-dihydroxyphenylalanine, the biochemical precursor to the catecholamine neurotransmitters DA and norepinephrine (Fig. 13.2).

Pathophysiology

PD primarily affects the nigrostriatal pathway in the part of the brain known as the basal ganglia, which consists of five interconnected, subcortical nuclei that span the telencephalon (forebrain), diencephalon, and mesencephalon (midbrain). These nuclei include the striatum (caudate and putamen), globus pallidus, subthalamic nucleus, substantia nigra pars compacta, and substantia nigra pars reticulata. Medium-sized spiny neurons that produce the inhibitory amino acid transmitter

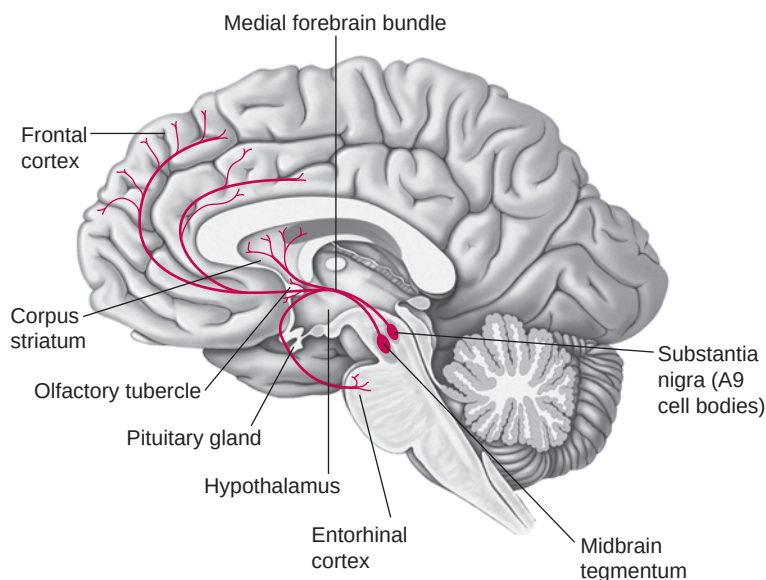


FIGURE 13.1 Some dopaminergic pathways in the brain. The nigrostriated pathway (A9 cell bodies in substantia nigra; nerve terminals in striatum) is degenerated in PD.

γ -aminobutyric acid (GABA) are the principal neurons in the striatum. They receive input from descending corticostriatal projections mediated by the excitatory amino acid neurotransmitter L-glutamic acid, as well as dopaminergic input from the substantia nigra. There are also acetylcholine-producing interneurons in the striatum. Both GABA-producing and acetylcholine-producing neurons in the striatum respond to DA through several DA receptor types.

Dopamine Receptors

DA receptors are grouped into excitatory D_1 types (D_1 and much less prevalent D_5 receptors), and inhibitory D_2 types (D_2 with long and short splice variants, and less abundant D_3 and D_4 types) (6). All are members of the G protein-coupled receptor (GPCR) superfamily and consist of 387 to 477 amino acids. GPCRs are thermodynamically flexible, which manifests as thermal instability when attempts are made to extract the protein from lipid membranes with a detergent, making crystal generation difficult (7). Currently, the only human GPCR x-ray crystal structures solved are for the β_2 -adrenergic (8) and adenosine A_{2A} (9) receptors. DA receptors, putatively, are structurally similar to these other GPCRs, consisting of seven relatively hydrophobic transmembrane regions, linked by four extracellular and four intracellular loop segments, with an extracellular amino terminus and cytoplasmic carboxy terminus; a schematic representation based on crystal structure of the human β_2 -adrenergic GPCR (8) is shown in Figure 13.3 (10).

The third intracellular loop segment varies most among the DA receptor subtypes, and this is likely where the specific α -subunit of the heterotrimeric (α , β , γ) guanine nucleotide-binding (G) protein binds. The

inherent structural flexibility of GPCRs translates to numerous thermodynamic conformations (7). Certain GPCR conformations are constitutively active; that is, the GPCR is able to activate the associated G protein α -subunit ($G\alpha$) in the absence of a ligand. In other cases, agonist ligands bind and stabilize GPCR conformations that lead to $G\alpha$ activation. Upon receptor activation, the $G\alpha$ subunit releases guanosine diphosphate (GDP) in exchange for binding guanosine triphosphate (GTP) and the β/γ -dimer dissociates. The $G\alpha$ and $G\beta/\gamma$ subunits then activate various intracellular effector systems, including adenylyl cyclase (stimulated by D_1 -like and inhibited by D_2 -like receptors) and phospholipase C (stimulated by D_2/D_4 and inhibited by D_3 receptors), to result in physiologic and/or psychological effects (and side effects, where drug therapy is concerned). It is now realized that the same GPCR can couple to different $G\alpha$ proteins to result in “multifunctional” signaling that provides a mechanistic basis for “functional selectivity” with regard to design of drugs targeting DA receptors (11–14); this concept is discussed in Chapter 14 relative to antipsychotic drugs; there are no known functionally selective drugs used to treat PD.

Neural Mechanisms in Parkinson Disease

The neuroanatomic connections of the basal ganglia, cerebral cortex, and motor neurons of the spinal cord are complex, and current understanding is incomplete. Figure 13.4 shows a schematic representation of basal ganglia anatomic connections (3,15–18) that can be used as a framework to understand pharmacologic management of the disease. In this model, DA released from nerve terminals of dopaminergic neuron cell bodies originating in the substantia nigra modulates activity of

CLINICAL SIGNIFICANCE



There is some degree of debate as to when to begin therapy, and many clinicians believe that early treatment of PD could lead to a decrease in the progression of disease. The approach to initial treatment should take into consideration the patient-specific factors (e.g., age, comorbid conditions); treatment specific factors (augmentation of the synthesis of brain DA, stimulation of dopamine release from presynaptic sites, direct stimulation of dopamine receptors, decreasing re-uptake of dopamine at presynaptic sites, or decreasing metabolism of DA or its precursor L-dopa); and disease-related factors (rate of progression, presenting phenomenology, related physical disability) to result in a decreased rate of progression and preservation of functionality. Current guidelines and consensus agree that levodopa/carbidopa, second-generation dopamine agonist, and monoamine oxidase-B (MAO-B) can be considered for initial therapy. Attention to the chemistry of these agents may provide an understanding of their benefit and potential detriment for specific patients. Though levodopa/carbidopa therapy is the most effective therapy to treat motor symptoms, there is the potential for dyskinesia and motor fluctuations, which may occur for some as early as within 5 years but for the majority within 15 years. There also is concern that this approach may result in increased oxidative pressure and disease progression. Therefore, it is generally accepted that levodopa/carbidopa

therapy should be reserved for those with disability. Second-generation dopamine agonists may hold value as initial therapy but may be limited by nausea and vomiting, psychiatric disturbances, and postural hypotension; interestingly, these agents may convey some neuroprotective qualities. MAO-B inhibitors can be considered, but the lack of evidence supporting selegiline as monotherapy, along with it having an amphetamine metabolite that holds potential for benefit (L-dopa potentiating) and detriment (vasoactive properties), may persuade some to consider rasagline, an agent for which sufficient clinical evidence exists for efficacy as initial monotherapy. Furthermore, some preliminary clinical data has demonstrated a neuroprotective effect at the lower dose range for this agent. Therefore, in a younger patient population the clinician should be mindful of the chemistry of dopamine agonists as they relate to improvement in motor function, but acknowledge that motor fluctuation may occur and recognize the need to reserve these agents for patients who exhibit disability. Clearly, the utility of rasagline or second-generation dopamine agonist potentially exhibiting neuroprotective effects to address early-stage PD without disability should be recognized.

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inhibitory GABA neurons in the striatum. In turn, striatal GABA neurons, through a series of “direct” and “indirect” neuronal pathways, modulate neuronal outflow to the thalamus that provides excitatory (glutamatergic) input to the motor cortex.

In the direct pathway are two sequential inhibitory GABAergic links that provide input directly to the thalamus. The first set of GABAergic neurons in striatum contains a predominance of excitatory DA D_1 -type receptors. Thus, the net effect of DA D_1 -mediated stimulation of striatal GABA neurons in the direct pathway is increased excitatory outflow from the thalamus to the motor cortex.

The indirect pathway uses two sequential GABAergic links, like the direct pathway, that are followed by an excitatory glutamatergic link and another inhibitory GABAergic input to the thalamus. The first set of striatal GABAergic neurons in the indirect pathway contains a predominance of inhibitory D_2 -type receptors. Thus, the net effect of DA D_2 -mediated modulation of striatal GABAergic neurons in the indirect pathway is to reduce excitatory outflow from the thalamus to the motor cortex.

In the normal condition, the direct pathway is dominant, but in PD, reduced levels of striatal DA negates this preference, and the indirect pathway becomes more

apparent, with a net effect of decreased excitatory input to the motor cortex. The clinical result is that the striatal DA deficiency accounts for the major motor symptoms of PD, particularly bradykinesia. The mainstay of pharmacologic treatment (3) continues to be replacement therapy with the immediate biochemical precursor of DA, L-DOPA, (see Fig. 13.2), in an effort to produce nearly physiologic agonism of both D_1 and D_2 DA receptors.

Etiology

Genetic Component

Although the neuropathology is well defined, the cause of PD is unknown. The development of effective pharmacotherapeutic and prophylactic therapy will require advances in understanding the etiology of the disease. Currently, there are several, convergent theories regarding the cause of PD: “proteolytic stress,” which recently has been characterized in connection with rare PD genetic mutations as well as environmental and/or endogenous neurotoxins; mitochondrial dysfunction; and oxidative metabolism—any and all of which may lead to “oxidative stress.” This section describes these and some alternative proposals regarding the etiology of PD.

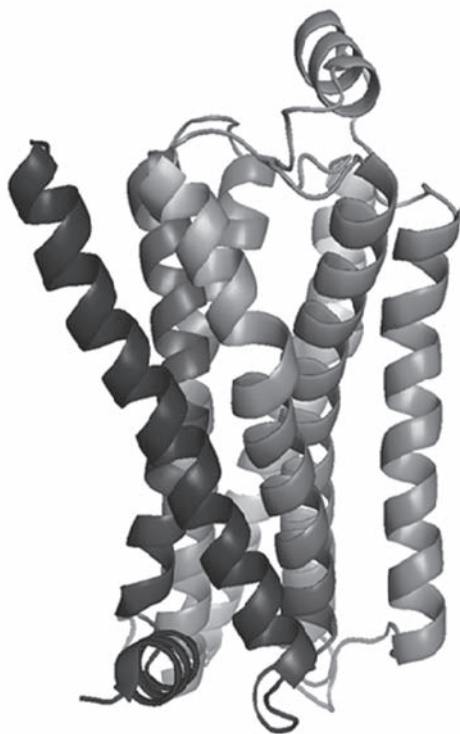


FIGURE 13.3 G protein-coupled receptor (GPCR) secondary structure model based on the crystal structure of the human β_2 -adrenergic GPCR.

is a highly conserved, abundant, 140-amino acid polypeptide expressed mainly in cerebral nerve terminals. Aggregation of α -synuclein molecules leads to pathologic inclusions (synucleinopathy) that characterize many neurodegenerative disorders, including PD (23). The function of α -synuclein is not well-characterized, but it appears to play a role in regulating DA homeostasis, including modulation of DA synthesis, release, and reuptake at nerve terminals. In vitro studies using neuronal cells (24) indicate that downregulation of α -synuclein enhances DA biosynthesis, suggesting that dysfunction

of α -synuclein may lead to increased intraneuronal levels of DA that become neurotoxic (see later discussion). Nevertheless, several studies failed to detect mutations in the α -synuclein gene in large family samples (25,26) or in studies of identical and heterozygous twins (27,28).

Mutations in four other genes—*DJ-1*, *PINK1*, *parkin*, and leucine-rich repeat kinase 2 (*LRRK-2*)—are unequivocally associated with development of familial PD (29), and several other genetic mutations also are implicated. It is proposed that such mutations may lead to excessive production of damaged proteins and/or dysfunction of protein clearance mechanisms in the brain (30,31). In normal physiologic conditions, damaged proteins usually are degraded and cleared by the ubiquitin-proteasome system (UPS). Increased production of damaged proteins or decreased UPS-mediated degradation and clearance may lead to protein aggregation and proteolytic stress, negatively affecting cellular structures and processes (32,33). For example, the UPS protein parkin mediates engulfment of dysfunctional mitochondria by autophagosomes (34), and dysfunctional parkin may fail to remove dysfunctional mitochondria. *PINK1* appears to function prior to parkin in the same pathway to maintain mitochondrial integrity and functioning in both muscles and dopaminergic neurons (35,36). In cases of familial PD associated with mutations of *LRRK-2* (a kinase encoding the protein dardarin), protein accumulation and Lewy body formation have been identified postmortem (37), but such data currently are unavailable for the parkin (a UPS enzyme), *DJ-1*, and *PINK-1* mutations.

In summary, although compelling evidence exists for apparently rare cases of genetically linked PD, primarily involving early onset of the disorder, the majority of cases are not associated with known genetic mutations and are considered to be sporadic (38). It should be noted, however, that synucleinopathy has been detected in sporadic cases of PD (39), suggesting that hypotheses involving protein aggregation may be relevant to the etiology of the more common forms of the disease. Moreover, although PD associated with genetic mutation is rare (<10% of cases), the study of these cases has facilitated

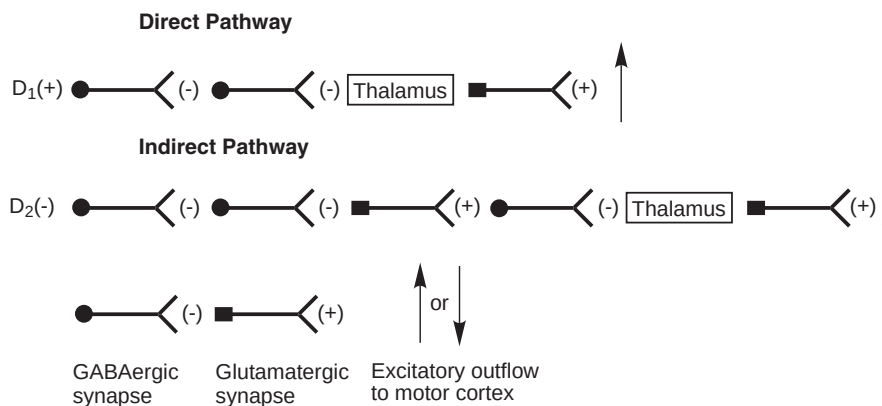


FIGURE 13.4 Striatal dopaminergic stimulation of the direct and indirect pathways modulates the thalamus excitatory outflow to the motor cortex.

understanding of the molecular pathways that lead to neurodegeneration, especially involving dopaminergic neurons.

In contrast to PD forms that are associated with genetic mutations, little evidence suggests that the disorder is autoimmune related (40). Likewise, although a form of parkinsonism associated with influenza virus-induced encephalitis did occur in a 1918 epidemic, recent studies indicate no evidence for a communicable infectious etiology of PD (41).

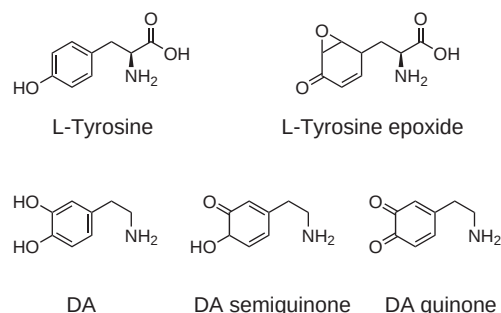
Oxidative Metabolism Component

Analysis of postmortem brain tissue from PD patients has revealed decreased levels of glutathione (42), increased lipid peroxidation (43), and increased oxidation of DNA (44) and proteins (45). Accordingly, oxidative metabolism involving the synthesis and catabolism of DA has been implicated in the PD disease process (46) through production of chemically reactive products, including epoxides (47,48), free radicals (49), and quinones (50,51).

Synthesis of DA, as well as the catecholamine neurotransmitters norepinephrine and epinephrine, proceeds by initial aromatic oxidation (*meta*-position) of the precursor amino acid L-tyrosine to L-DOPA, in a rate-limiting step catalyzed by tyrosine hydroxylase (L-tyrosine-3-mono-oxygenase; see Fig. 13.2). The mechanism of this hydroxylation may involve direct insertion of an oxygen atom into L-tyrosine to form L-DOPA, or through an intermediate arene epoxide (48) (Fig. 13.5), as is case for metabolic conversion of benzene to phenol (52) and of estradiol to catechol-estrogens (47,53). It is hypothesized that an analogous epoxide intermediate, formed during the conversion of tyrosine to L-DOPA, may alkylate nucleophilic functionalities on proteins, RNA, or DNA to produce neurotoxicity (47).

Once formed, DA can auto-oxidize to the corresponding electrophilic semiquinone and quinone (Fig. 13.5) species that can alkylate protein sulfhydryl groups, including glutathione (50,54). Manganese ion can catalyze this oxidation of DA, and the resulting semiquinone and quinone species have been implicated (51) in a PD-like syndrome among South American manganese miners (55). The auto-oxidation of DA also leads to the formation of the polymeric black pigment neuromelanin (54). The pigment is increasingly deposited in catecholaminergic neurons with advancing age, however, and it has been suggested that its accumulation in nigral neuronal cells eventually causes cell death (5).

In addition to neurotoxic metabolites derived from the auto-oxidation of DA, the monoamine oxidase (MAO)-catalyzed oxidation of DA may be relevant to PD. During MAO catalysis of DA and the other monoamine neurotransmitters (norepinephrine and serotonin), hydrogen peroxide is generated (see Equation 1 in Fig. 13.5) that can undergo a redox reaction with superoxide, according to the Haber-Weiss reaction (56), to form the extremely cytotoxic hydroxyl radical (see Equation 2 in Fig. 13.5).



Equation 1:



Equation 2:



FIGURE 13.5 Putative and known compounds involved in DA synthesis and catabolism, some of which may cause DA neuronal cell death.

Dysfunction of mitochondrial redox processes also is hypothesized to play a role in PD neuronal cell death (57,58). Postmortem samples of brain and peripheral tissue from PD patients indicate selective mitochondrial complex I deficiency, including in the substantia nigra and other sites (59,60). It is hypothesized that resulting decreased adenosine triphosphate (ATP) synthesis and altered homeostasis of reactive species such as hydrogen peroxide, superoxide, and free hydroxyl radicals might contribute to neuronal cell death in PD. In fact, mitochondrial dysfunction and altered oxidative metabolism of DA may converge such that decreased energy production by mitochondria may impede vesicular storage of DA, leading to increased cytoplasmic concentration of DA that can undergo MAO-catalyzed oxidation to hydroxyl radicals, as well as auto-oxidize to quinones and semiquinones (61).

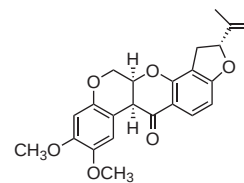
In summary, toxic products from normal oxidative metabolism and/or altered mitochondrial function may contribute to the progressive loss of DA neurons that occurs normally with maturation and aging at a rate of approximately of 13% per decade after age 45 (62–64). Clinical symptoms of PD emerge as losses of DA neurons exceed 65% (65). Conceivably, the symptoms of parkinsonism could be produced by two processes, a specific disease-related insult combined with pathologic changes resulting from normal aging. This two-pronged pathophysiology may explain why PD is a progressive disorder of late onset (66). Hypotheses concerning the initial pathologic event have centered on environmental neurotoxins.

Environmental Neurotoxins Component

There has been accumulated compelling evidence to suggest that environmental toxicants play an important

role in the etiology (and perhaps treatment) of PD. For example, as described earlier, the high prevalence of PD among South American miners probably is associated with manganese neurotoxicity. There is also evidence linking herbicides and pesticides to PD. For example, there is a remarkably high correlation between incidence of PD and use of pesticides in an agricultural region of Québec (67). In fact, *N,N*-dimethyl-4,4'-bipyridinium dichloride (Paraquat, Fig. 13.6) is one of the most widely used herbicides in the world (68) and is structurally similar to the selective DA neurotoxicant 1-methyl-4-phenylpyridinium (MPP⁺; Fig. 13.6), which was marketed as the herbicide Cyperquat (69,70). Paraquat, like MPP⁺ and the pesticide rotenone (see later discussion), inhibits mitochondrial complex I and cellular respiration, although only at concentrations (~10 mmol/L) considerably higher than are required for MPP⁺ (~10 μmol/L) or rotenone (~10 nmol/L) (69). Some investigators have reported Paraquat-induced postmortem losses of nigral DA neurons and degeneration of striatal DA nerve terminals, as well as decreased locomotor activity (71). Others, however, have found no evidence of neurotoxic changes after exposure to Paraquat (72), and Paraquat neurotoxicity, unlike MPP⁺-associated parkinsonism, probably is not highly selective for DA neurons (69). Finally, the ability of Paraquat itself to gain entry to the brain remains uncertain (70) because it is a (double) quaternary ammonium compound, unlike the much more lipophilic molecule rotenone, discussed in the following paragraph.

Rotenone is an isoflavone, many of which are found in roots and stems of several plants; this association allows rotenone to be labeled as “natural” and to be used in organic food farming (68). Rotenone is a lipophilic molecule that crosses the blood–brain barrier and neuronal cell membranes where it is proposed to inhibit mitochondrial complex I to cause degeneration of nigrostriatal DA neurons (69,70), producing hypokinesia and rigidity, as well as accumulation of fibrillar cytoplasmic inclusions that contain ubiquitin and α-synuclein (73).



Rotenone

Another environmental link to PD is the lower incidence of the disease in cigarette smokers than in non-smokers (74,75). It has been proposed that something in cigarette smoke may protect against a toxicant (environmental or endogenous) relevant to parkinsonian neuropathology. Several mechanisms might account for this inverse correlation, which might enhance understanding of neuroprotective or even curative pharmacotherapy in neurodegenerative disorders. For example, several constituents in tobacco smoke are MAO inhibitors, including farnesylacetone, a selective inhibitor of MAO-B (76,77), which is thought to be the predominant MAO isoform

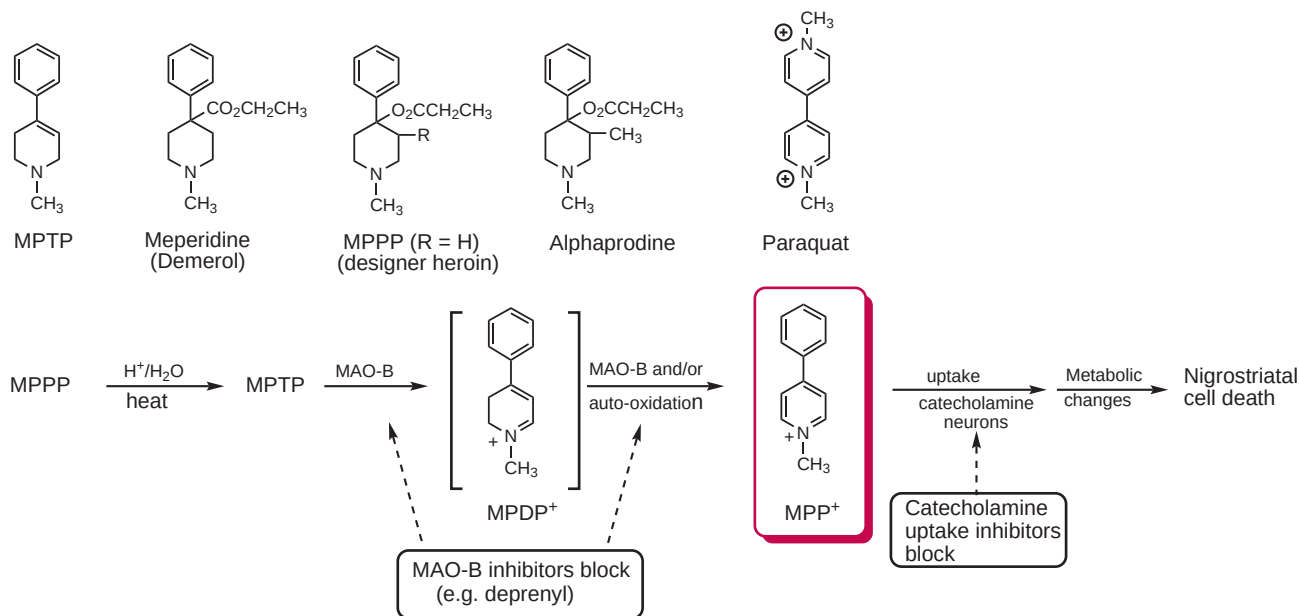


FIGURE 13.6 Phenylpiperidine analgesics and metabolic activation of MPTP. In the presence of acid and heat, MPPP forms MPTP that undergoes an MAO-B-catalyzed two-electron oxidation to MPDP⁺. MPDP⁺ undergoes a two-electron auto-oxidation (also can be catalyzed by MAO-B) to MPP⁺ that is accumulated into DA neurons by the DA transporter and subsequently into mitochondria where it disrupts cellular respiration, producing neuronal cell death.

responsible for metabolism of DA in brain (glial cells). Moreover, prolonged nicotine exposure can increase activity of the UPS system (78), discussed earlier.

There also is an inverse correlation between risk of developing PD and coffee and caffeine consumption (79). Caffeine is a nonselective adenosine A_1/A_2 receptor antagonist that reduces parkinsonism-like tremor in rodent models of PD (80). The adenosine A_{2A} receptor subtype is expressed abundantly on medium spiny neurons. Aberrant interneuronal signaling associated with A_{2A} -linked intracellular kinase activation is proposed to account for parkinsonism-like signs and the dyskinetic motor response produced by DA agonist therapy in rodent and primate models of PD (81). Accordingly, adenosine A_{2A} receptor antagonists, such as istradefylline (see later discussion, Fig. 13.13), have been proposed as pharmacotherapy for PD (82). Antagonism of adenosine A_1 receptors also may reduce parkinsonism-like tremor in rodents (80), but no clinical candidate anti- A_1 drugs have emerged.

In conclusion, there are several potential links between the environment and the etiology and pathophysiology of PD, including exposure to pesticides, herbicides, well water, wood pulp mills, and rural living (83,84). Pathologic mechanisms involving environmental neurotoxins may converge with dysfunctional oxidative metabolism. A particularly well-studied example of such convergence involves the discovery that 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP, Fig. 13.6) produces a severe parkinsonism syndrome in humans and some laboratory animals that is remarkably similar to the idiopathic disease. As such, MPTP has been a useful tool to study the etiology and pharmacotherapy of PD.

Parkinsonism Caused by MPTP

The cyclic tertiary amine MPTP (Fig. 13.6) induces a form of parkinsonism in humans and monkeys similar in neuropathology and motor abnormalities to idiopathic PD (85–87). The role of MPTP in parkinsonian disorders was revealed by a serendipitous series of events. In 1977, a 23-year-old college student suddenly developed parkinsonian symptoms, with severe rigidity, bradykinesia, and mutism. The abrupt and early onset of symptoms was so atypical that the patient initially was thought to have catatonic schizophrenia. The subsequent diagnosis of parkinsonism was substantiated by a therapeutic response to L-DOPA, whereupon the patient was referred to the National Institute of Mental Health in Bethesda, Maryland. The patient admitted having synthesized and used several illicit drugs, after which the psychiatrist who had elicited the patient's history visited his home and collected glassware that had been used for chemical syntheses. Chemical analysis revealed several pyridines, including MPTP, formed as by-products during synthesis of the target molecule, 1-methyl-4-propionoxy-4-phenylpiperidine (MPPP, Fig. 13.6). MPPP, known as “designer heroin” or “synthetic heroin,” is the reverse ester of the narcotic analgesic meperidine (Fig. 13.6)

and lacks only the 3-methyl moiety of the analgesic α -1,3-dimethyl-4-phenyl-4-propionoxypiperidine or alphaprodine (Nisentil, Fig. 13.6). Initially, it was unclear whether MPTP or other constituents of the injected mixture accounted for the neurotoxicity that produced parkinsonism.

After the patient returned home, he continued to abuse drugs and died of an overdose; autopsy revealed degeneration of the substantia nigra—the hallmark neuropathologic feature of PD. Subsequently, other patients were identified with virtually identical parkinsonian symptoms who had also been receiving intravenous injections of MPPP preparations containing varying amounts of MPTP. In several patients, MPTP was the principal or sole constituent injected, providing the first definitive evidence that MPTP is a parkinsonism-producing neurotoxicant. Both the clinical and neuropathologic features of MPTP-induced parkinsonism resemble idiopathic PD more closely than any previous animal or human disorder elicited by toxins, metals, viruses, or other means. Accordingly, understanding the molecular pathophysiology of MPTP neurotoxicity has shed light on the neurodegenerative mechanisms in idiopathic parkinsonism.

Mechanisms of Neuronal Cell Death in MPTP-Induced Parkinsonism

Consideration of the chemical structure of MPTP would suggest that the compound is relatively chemically inert, because no highly reactive functional group is present. Almost immediately, it was recognized that MPTP might undergo some type of metabolic activation to a more reactive metabolite. Researchers soon discovered that brain MAO-B catalyzes the two-electron oxidation of MPTP at the allylic α -carbon to give the unstable intermediate product 1-methyl-4-phenyl-2,3-dihydropyridinium (MPDP⁺), which subsequently undergoes a further two-electron oxidation to the stable 1-methyl-4-phenylpyridinium species (MPP⁺) via auto-oxidation, disproportionation, and enzyme-catalyzed mechanisms (Fig. 13.6) (88–90). Inhibitors of MAO-B subsequently were shown to prevent MPTP-induced parkinsonism in primates (91), and it currently is accepted that MPP⁺ is the major metabolite of MPTP responsible for the destruction of DA neurons, although a role for the unstable dihydropyridinium species MPDP⁺ has not been ruled out.

The relationship of MAO and MPTP has neurobiologic relevance beyond MPTP neurotoxicity. MAO catalyzes the α -carbon oxidation of the monoamine neurotransmitters (e.g., DA, norepinephrine, and serotonin) (Fig. 13.2). Oxidation of a heterocyclic tertiary amine (i.e., MPTP) by MAO suggests a novel physiologic role for this enzyme. For example, MAO could be important in regulating the oxidation state of pyridine systems, such as those involving NADH, which is relevant to mitochondrial function (see later discussion) (92). Interestingly, biochemical and epidemiologic evidence suggests that cigarette smokers have depressed MAO-B activity (93), which may be relevant to the lower incidence of PD seen in smokers (74,75). Nicotine is not a particularly potent

inhibitor of MAO, and in fact, nicotine increases the neurotoxicity of MPTP (94). As mentioned earlier, however, other components of cigarette smoke do inhibit MAO (76,77), and cigarette smoke protects against MPTP-induced depletion of striatal DA in mice (95).

Although extensive metabolic, biochemical, and toxicologic investigations have established that the nigrostriatal neurodegenerative properties of MPTP are mediated by the MAO-B–derived metabolite, MPP⁺, this bioactivation reaction must proceed outside of the target nigrostriatal DA neurons because they apparently do not contain MAO-B (96). It is thought that MPTP is oxidized to MPDP in MAO-B–rich glial cells near striatal nerve terminals and nigral cell bodies; the conjugate base MPDP presumably diffuses out of glial cells and is subsequently oxidized to the MPP⁺ metabolite. The MPP⁺ is sequestered into striatal dopaminergic nerve terminals via the DA neurotransmitter, which accepts MPP⁺ as a substrate (Fig. 13.6) (97). Intraneuronally, MPP⁺ is concentrated into mitochondria, where it selectively inhibits complex I of the electron-transport chain, inhibiting NADH oxidation and, eventually, depleting the nigrostriatal neuronal cell of ATP (98,99). Thus, it is proposed that the mechanism of nigrostriatal cell death induced by MPTP (via MPP⁺) is energy failure at the level of the mitochondrial respiratory chain (100,101).

Several sequential factors may account for the selective damage of nigrostriatal DA neurons by MPTP (Fig. 13.6). First, MPTP binds selectively to MAO-B, which is highly concentrated in glial cells in human substantia nigra and corpus striatum. Then, the MPP⁺ produced from MPTP is selectively accumulated by DA neurotransmitters into nigral DA cells and striatal DA nerve terminals. Finally, within nigral cell bodies, MPP⁺ binds to neuromelanin and may be gradually released in a depot-like fashion, maintaining a toxic intracellular concentration of MPP⁺ that inhibits mitochondrial respiration.

The serendipitous discovery and subsequent scientific investigation of the mechanism of parkinsonism produced by MPTP refocused study of the etiology and pathogenesis of idiopathic PD. For example, discovery of the selective ability of MPTP to induce nigral cell death has stimulated broad interest in identifying potential environmental or endogenous toxicants that may be causative agents in PD, as discussed earlier. Likewise, the mechanism of MPP⁺ to cause DA cell death via inhibition of mitochondrial respiration provides support for theories involving mitochondrial dysfunction and oxidative metabolism in general (oxidative stress) in the etiology of idiopathic PD. Specifically, it has been documented that there is a 30% to 40% reduction in mitochondrial complex I activity in the substantia nigra of patients with PD (102,103). In general, mitochondrial dysfunction and disorders of oxidative metabolism, which can include a genetic component (29), now are considered to be critical components of most theories of nigral cell degeneration in PD. Delineation of the neurobiochemical mechanism of MPTP-induced parkinsonism also has led to new pharmacotherapeutic

approaches aimed at slowing neurodegeneration in PD, focusing, notably, on MAO-B and oxidative stress. Clinical studies to evaluate the effectiveness of coadministration of an MAO-B inhibitor plus the antioxidant vitamin E to slow progression of neurodegeneration in PD, however, have not yielded encouraging results (104–106). In fact, as indicated in the next section, despite progress toward understanding the etiology and pathophysiology of PD summarized earlier, no treatment has conclusively been proven to stop or even reliably slow the progressive neurodegeneration in the disease (3).

Pharmacotherapy of PD

Due to the failure of various neuroprotection strategies to provide unequivocal disease-modifying benefit for PD patients, the mainstay of PD pharmacotherapy continues to be palliative or symptomatic, involving replacement of the DA deficiency in striatum. This is accomplished currently by one or more of the following means: 1) augmentation of the synthesis of brain DA; 2) stimulation of DA release from presynaptic sites; 3) direct stimulation of DA receptors; 4) decreasing reuptake of DA at presynaptic sites; or 5) decreasing metabolism of DA or its precursor L-DOPA. Some of these treatments, notably DA receptor agonists and MAO-B inhibitors (decrease DA metabolism), might also provide putative neuroprotective benefit (3).

L-DOPA Therapy

More than 40 years after its introduction, L-DOPA (Fig. 13.2) remains the most effective symptomatic pharmacotherapy for PD (3). Despite controversy regarding long-term efficacy, adverse effects, and even potential neurotoxicity, most PD patients derive a substantial benefit from L-DOPA over the entire course of their illness. Moreover, L-DOPA increases life expectancy among patients with PD, and survival is significantly reduced if the initiation of L-DOPA therapy is delayed (107).

In 1960, Ehringer and Hornykiewicz (108) assayed DA in the brains of patients dying with PD and found that tissue concentrations of DA in the corpora striata of many of these patients averaged only 20% of normal. Signs of PD in patients resembled behavioral changes in rats treated with reserpine or other amine-depleting agents. These findings led Birkmayer and Hornykiewicz (109) to administer high oral doses of racemic DOPA to PD patients in Vienna in 1960. Subsequent clinical trials led by Barbeau in Montréal in the early 1960s (110) and by Cotzias in New York in the late 1960s (112) confirmed this effect of racemic DOPA. Barbeau and Cotzias also demonstrated the greater potency and safety of the physiologic enantiomer L-DOPA (111,112).

Development of L-DOPA as a therapeutic agent in PD is a rare example of a rationally predicted and logically pursued clinical treatment in a neuropsychiatric disorder, based on neurochemical pathology and basic pharmacologic theory (113). The effectiveness of L-DOPA treatment requires its penetration into the CNS and local decarboxylation to

DA. DA does not cross the blood–brain diffusion barrier because its amino moiety is protonated under physiologic conditions ($pK_a = 10.6$ [NH_2]), making it relatively hydrophilic (114). However, the precursor amino acid to DA, L-DOPA, is less basic ($pK_a = 8.72$ [NH_2]) and polar at physiologic pH, and thus more able to penetrate the CNS, in part facilitated by transport into brain with other aromatic and neutral aliphatic amino acids (114–116).

BIOSYNTHESIS AND METABOLISM L-DOPA is normally a trace intermediary metabolite in the biosynthesis of catecholamines. L-DOPA is readily decarboxylated by the cytoplasmic enzyme L-aromatic amino acid decarboxylase (“DOPA decarboxylase”) to form DA (see Fig 13.2). The main by-products of DA are 3,4-dihydroxyphenylacetic acid (DOPAC) and homovanillic acid (HVA; 3-methoxy-4-hydroxyphenylacetic acid; Fig. 13.2). The effects observed after systemic administration of L-DOPA have been attributed to its peripheral and cerebral metabolites, mainly DA, with much less conversion to norepinephrine by β -hydroxylation, or epinephrine formed by *N*-methylation of norepinephrine by phenylethanolamine-*N*-methyltransferase (Fig. 13.2). A small amount of L-DOPA is *O*-methylated by catechol-*O*-methyltransferase (COMT) to L-3-*O*-methyldopa (L-3-methoxytyrosine or 3-methoxy-4-hydroxyphenylalanine, Fig 13.2), which accumulates in the CNS because of its long half-life. Most exogenous L-DOPA, however, is rapidly decarboxylated to DA in peripheral tissues, including liver, heart, lung, and kidney (Fig 13.7). In fact, only about

1% of an orally administered dose of L-DOPA reaches the brain; thus, L-DOPA, by itself, has very limited dose effectiveness (117). In humans, appreciable quantities of L-DOPA enter the brain only when administered alone in doses (3 to 6 g daily) high enough to compensate for losses caused by peripheral metabolism.

L-DOPA peripheral decarboxylation can be competitively inhibited (118) by coadministration of carbidopa (combined with L-DOPA in Sinemet) or benserazide (combined with L-DOPA in Prolopa in countries other than the United States) (Fig. 13.8). Such polar decarboxylase inhibitors do not appreciably penetrate the brain to inhibit cerebral decarboxylase, thus markedly increasing the proportion of L-DOPA that reaches the brain for conversion to DA (Fig 13.7), allowing for a 2.5- to 30-fold lower dose (0.2 to 1.2 g/d) of L-DOPA (17). Patients with PD are typically started on a combination product, either alone or with other adjunctive agents discussed later. Products containing extended-release preparations of decarboxylase inhibitors should provide more sustained benefits with less “wearing off” of benefit after several hours, but the bioavailability and performance of these products is variable. In general, tissue uptake of L-DOPA is highly dependent on competition with other aromatic and neutral aliphatic amino acids and can be decreased substantially by a protein meal.

Pyridoxine (vitamin B_6) is the cofactor for L-aromatic amino acid decarboxylase. In high doses, vitamin B_6 can reverse the therapeutic effects of L-DOPA by increasing

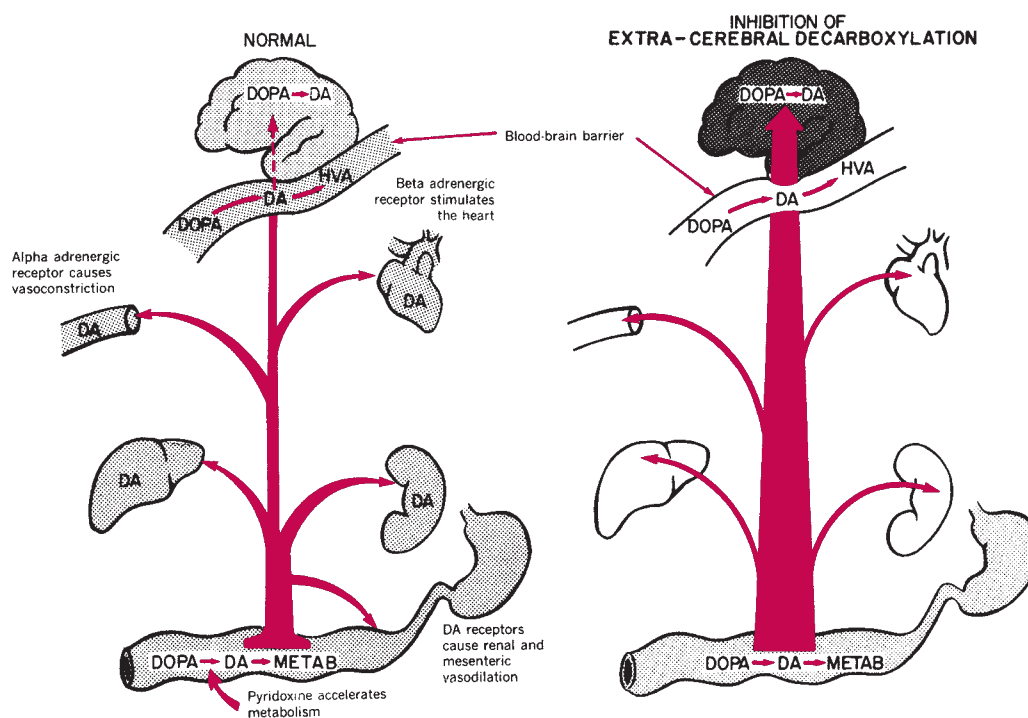


FIGURE 13.7 Diagrammatic representation of the peripheral decarboxylation of levodopa to form dopamine (DA) and the mode of action of extracerebral decarboxylase on levodopa metabolism and distribution in vivo. The concurrent administration of levodopa and a decarboxylase inhibitor decreases the amount of levodopa required to elicit a therapeutic response in parkinsonism. HVA, homovanillic acid.

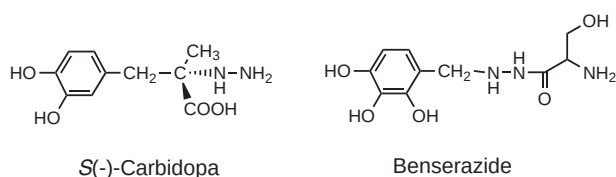


FIGURE 13.8 Drugs that inhibit peripheral L-aromatic amino acid decarboxylase

peripheral decarboxylase activity. Competitive blockade of peripheral decarboxylation with carbidopa or benserazide, however, minimizes this potential effect of pyridoxine.

SIDE EFFECTS Common adverse effects of L-DOPA therapy are nausea and vomiting, likely due to direct gastrointestinal irritation as well as stimulation by DA of the chemoreceptor trigger zone in the area postrema of the brainstem, an emesis-inducing center that is largely unprotected by the blood-brain diffusion barrier. An important advantage of combining L-DOPA with a peripheral decarboxylase inhibitor is a marked reduction of required L-DOPA dose; thus, there is less risk of emesis or other adverse effects associated with formation of excess DA. These can include activation of peripheral adrenergic and DA receptors, in part by releasing endogenous adrenergic catecholamines (119), as summarized in Figure 13.7. Theoretically, vasoconstriction and hypertension might occur by stimulation of peripheral α -adrenoceptors, tachycardia by stimulation of cardiac β -adrenoceptors, and direct renal and mesenteric vasodilatation by DA, although DA agonists and L-DOPA usually induce hypotension. However, such effects are rarely encountered clinically with the use of a peripheral decarboxylase inhibitor with L-DOPA (17).

After about 5 years of continuous treatment with L-DOPA, at least half of PD patients develop fluctuating motor responses, and nearly three-quarters do so by 15 years (120). These fluctuations (so-called “on-off” effects) include “off” periods of immobility and “on” periods with abnormal involuntary movements or dyskinesias. These phenomena may reflect progression of the disease with striatal nerve terminal degeneration and further lowering of DA levels, and possibly increased sensitivity of DA receptors.

Psychiatric disturbances such as hypersexuality, mania, visual hallucinations, and paranoid psychosis are common and sometimes severe adverse responses to treatment with L-DOPA or direct DA agonists (discussed later). Such behavioral disturbances probably reflect excessive stimulation of DA receptors in mesolimbic or mesocortical DA systems. These side effects complicate clinical management of PD patients, including those with clinical depression commonly associated with PD or dementia that sometimes arises in late stages of the disease. Serotonin-enhancing antidepressants usually are well tolerated with minor risk of worsening bradykinesia (120). Use of antipsychotic drugs, however, is limited to those with minimal risk of worsening bradykinesia and other aspects of extrapyramidal motor

dysfunction (121,122). Clozapine, although potentially toxic and relatively expensive, is best tolerated and may have particular efficacy for visual hallucinations in PD patients. Moderate doses of quetiapine and ziprasidone are tolerated, but their efficacy is not established, and other second-generation antipsychotic agents including olanzapine, risperidone, paliperidone, and even the DA partial agonist aripiprazole usually are poorly tolerated by PD patients due to increased bradykinesia (121–124).

Dopamine Receptor Agonists

MECHANISM OF ACTION The nigrostriatal neurodegeneration that proceeds over the course of PD limits the number of striatal nerve terminals that are available to decarboxylate L-DOPA to DA. Drugs that act directly to stimulate DA receptors, however, do not require functioning dopaminergic nerve terminals and can be useful in the management of late-stage disease problems during L-DOPA therapy. DA receptor agonists currently available are nonselective, and without balanced activity at D_1 -type and D_2 -type receptors, for example, clinically used agents are full or partial agonists primarily at D_2 -type receptors. Use of DA agonist monotherapy (i.e., without L-DOPA) has been suggested as initial therapy for PD based on the hypothesis that oxidative metabolites of DA (formed from exogenous and endogenous L-DOPA) may be neurotoxic. At present, however, no substantial evidence supports an indirect neuroprotective effect of DA receptor agonists (3). Meanwhile, DA receptor agonists have a longer duration of action (8 to 24 hours) compared to L-DOPA (6 to 8 hours) and may be less likely than L-DOPA to induce on/off effects and dyskinesias. In fact, the DA receptor agonists pramipexole and ropinirole (see later discussion) do produce reduced motor fluctuations compared to L-DOPA; however, they also produce increased incidence of other adverse effects (125,126), such as nausea and vomiting (presumably from activation of the chemoreceptor trigger zone), sedation, and hallucinations and other psychiatric disturbances that are particularly troublesome for elderly patients. Thus, the DA agonists usually are given in combination with a reduced dose of L-DOPA/carbidopa, but monotherapy may be used for younger patients better able to tolerate side effects (3).

STRUCTURE–ACTIVITY RELATIONSHIP Because there is currently no validated three-dimensional orientation of the amino acid residues at the ligand-binding site for D_1 - or D_2 -type DA receptors, rational design of selective DA receptor agonists (and antagonists) is guided by quantitative structure–activity relationship studies based on probe molecules and studies of ligand docking to DA molecular models built based on homology to solved GPCR crystal structures.

Little information can be gained concerning the conformational requirements for DA receptor activation using DA itself because its ethylamine side chain has unlimited flexibility, with unrestricted rotation about the β -carbon–phenyl bond (Fig. 13.9). Accordingly, compounds in which the catechol ring and the amino-ethyl

moiety of DA are held in rigid conformation have been synthesized to probe molecular determinants for receptor binding and activation. Such rigid analogs include aporphines, notably starting with the morphine acid-rearrangement product, *R*(-)-apomorphine (Fig. 13.9).

The alkaloid apomorphine has been employed in experimental neuropharmacology since the late nineteenth century (127,128). The *trans*- α -rotamer conformation of DA (Fig. 13.9) most closely aligns with apomorphine, as confirmed by the x-ray crystal structure of apomorphine (129). In contrast, *R*(-)-isoapomorphine (Fig. 13.9) mimics the structure of DA in the *trans*- β -rotameric conformation and has less activity than apomorphine as a DA receptor agonist. The analog *R*(-)-1,2-dihydroxyaporphine (Fig. 13.9) mimics the *cis*- α -rotamer conformation of DA and is inactive as a DA receptor agonist (130). In addition to the aporphines, the semi-rigid aminotetralin, 2-amino-5,6-dihydroxy-1,2,3,4-tetrahydronaphthalene (A-5,6-DTN, Fig. 13.9), that has its benzene ring and amino side chain in a *trans*- α -rotamer conformation, is a more potent DA agonist than its 6,7-dihydroxy congener (A-6,7-DTN, Fig. 13.9) with a *trans*- β -rotamer conformation (131). Studies of these rigid dopaminomimetic compounds suggest that the preferred conformation of DA is the extended *trans* conformation (α - or β -rotamer). Results from mutagenesis, molecular modeling, and computational chemistry

indicate that ligand activity at D_1 versus D_2 receptors is critically dependent on the position of protonated nitrogen moieties of candidate ligands (as in the *trans* conformation of DA) that can support high-affinity ionic bonding at a proposed anionic aspartic acid residue in the third transmembrane α -helix (D3.32), as a preferred docking site of DA receptors (132–134).

Aporphine-Type Dopamine Receptor Agonists

R(-)-Apomorphine (Fig. 13.9) was reintroduced as a useful adjunct in the therapy of PD (135), following years of neglect after promising early observations (136). Lack of oral bioavailability, short duration of action, and potent central emetic action discouraged its clinical use. Nevertheless, in 1993, *R*(-)-apomorphine received regulatory approval in the United Kingdom for control of refractory motor dysfunction and wide fluctuations in responses (“on-off” syndrome) to L-DOPA or DA agonists (later approved in the U.S. as Apokyn) (137,138). Improved motility in response to an acute challenge dose of apomorphine also can predict responsiveness to L-DOPA treatment (139,140). *R*(-)-Apomorphine is an agonist for both D_1 - and D_2 -type DA receptors. With a pK_a of about 9, it is mostly protonated at physiologic pH, but sufficiently lipophilic to cross the blood–brain barrier readily. Apomorphine can be administered subcutaneously by intermittent self-administration

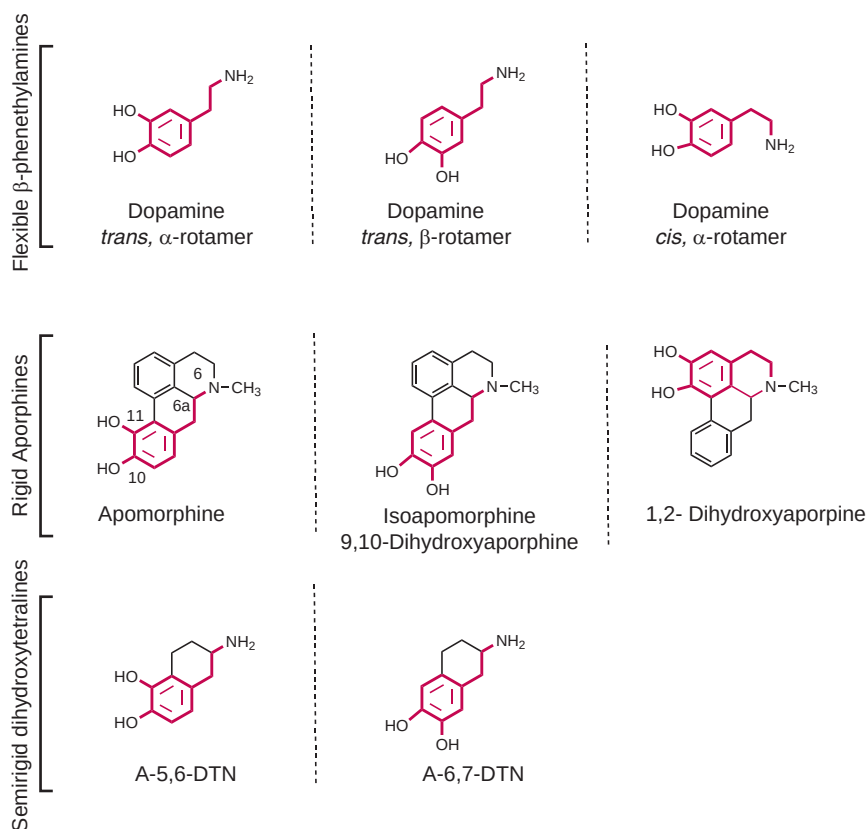


FIGURE 13.9 Conformations of dopamine in the *trans*- α -rotameric, *trans*- β -rotameric, and *cis*-rotameric forms, and their structural relationships to the rigid DA analogs apomorphine, isoapomorphine, and 1,2-dihydroxyaporphine. Also shown are the corresponding semirigid dihydroxyaminotetralin analogs of DA, A-5,6-DTN, and A-6,7-DTN.

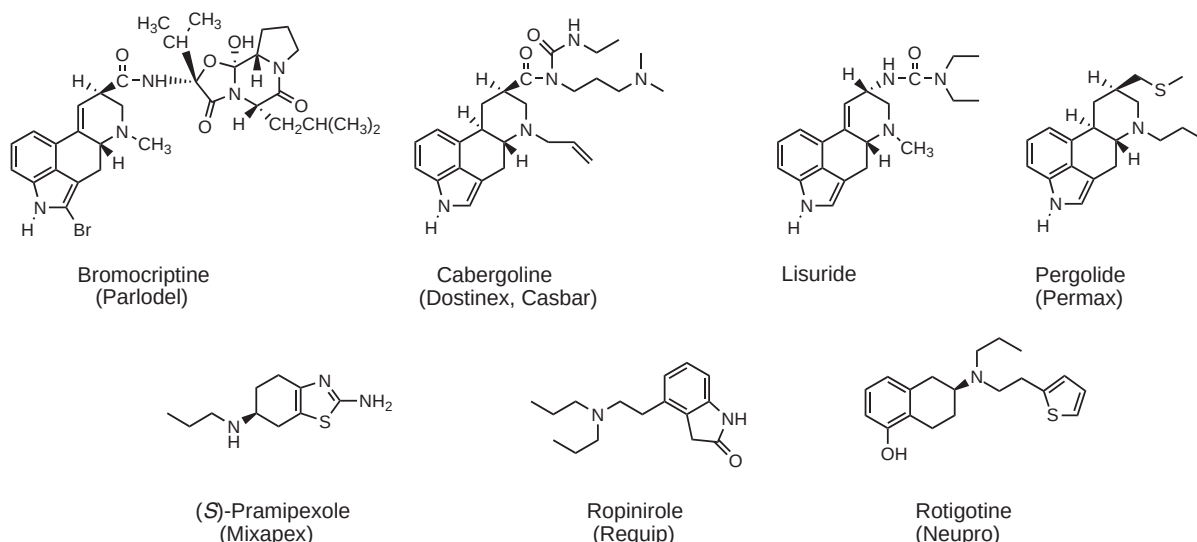


FIGURE 13.10 DA receptor agonist drugs that target primarily D_2 -type receptors.

with a small self-injector (Penject) or by continuous infusion with a portable minipump.

Ergot-Type Dopamine Receptor Agonists

Bromocriptine (Fig. 13.10) is an ergot alkaloid peptide that is a partial agonist at D_2 and D_3 DA receptors (141) that lacks appreciable activity at D_1 -type or D_4 receptors (142). It was the first direct DA agonist to be used in the treatment of PD, after its development for use at lower doses as a prolactin inhibitor (143). Bromocriptine inhibits prolactin release from anterior pituitary mammotrophic cells that selectively express D_2 DA receptors. These receptors respond to DA produced in the hypothalamus, released into the hypophysiportal blood vessels, and carried to the pituitary to act as a prolactin-inhibitory hormone. Bromocriptine is an effective prolactin inhibitor at daily doses of 1 to 5 mg, for which it is used to treat hyperprolactinemia associated with pituitary adenomas or to suppress prolactin output in prolactin-sensitive metastatic carcinoma of the breast. As a D_2 partial agonist, bromocriptine acts as an agonist at pituitary D_2 receptors that are normally in a high-sensitivity state. At daily doses of 10 to 20 mg, bromocriptine and other D_2 partial agonist ergolines appear to act as D_2 full agonists with antiparkinson effects; this full agonist activity probably reflects the supersensitized status of denervated DA receptors in PD (144,145). Bromocriptine is absorbed after oral administration, but approximately 90% undergoes extensive first-pass hepatic metabolism; the remainder is hydrolyzed in the liver to inactive metabolites eliminated mainly in bile, and the overall elimination half-life is approximately 3 hours.

Another ergoline, cabergoline (Fig. 13.10), is a full agonist at D_2 receptors and a partial agonist at D_3 and D_4 receptors, without appreciable activity at D_1 -type receptors (141,142), and with a relatively long half-life of approximately 48 hours (3). Cabergoline is superior to placebo for treating motor signs of PD, but its

comparative efficacy versus L-DOPA is poorly documented (146). Lisuride (Fig. 13.10) is an ergoline partial agonist at D_2 , D_3 , and D_4 receptors with little activity at D_1 -type receptors (141,142). This relatively short-acting agent is being evaluated in patients with advanced PD, using either patch or infusion delivery methods (3,147).

The peptide component of bromocriptine evidently is unnecessary for dopaminergic activity. Pergolide (formerly Permax, Fig. 13.10) was the first nonpeptide ergoline used successfully to treat PD, as well to inhibit release of prolactin (143,148). Pergolide shows greater agonist effects at both D_2 - and D_1 -type DA receptors than does bromocriptine, but it was recently withdrawn from clinical use due to association with valvular heart disease (149,150). This adverse cardiac effect is due to activation of serotonin 5-HT_{2B} receptors by pergolide, as has been hypothesized for other drugs with potent 5-HT_{2B} agonist activity, including cabergoline (Fig. 13.10) (151,152). Moderate cardiac tricuspid valve regurgitation was more frequent in patients taking cabergoline repeatedly in relatively high doses for PD than in newly exposed or untreated controls (153). However, doses of cabergoline used to treat hyperprolactinemia are much lower (154) and may avoid valvular heart disease (154,155). Ergolines such as bromocriptine and lisuride that lack 5-HT_{2B} agonist activity do not appear to induce cardiac valvular damage (150,156,157).

Other Small-Molecule Dopamine Receptor Agonists

Currently, pramipexole and ropinirole (Fig. 13.10) are among the most commonly prescribed direct DA agonists for PD in the United States (17). They were introduced primarily for advanced stages of PD to limit fluctuations in response to L-DOPA therapy and as a “rescue” therapy when L-DOPA became insufficiently effective. These direct D_2 DA agonists are relatively well tolerated. Moreover, in light of concern that L-DOPA might itself cause DA neuronal toxicity through formation of reactive

oxidized by-products, pramipexole and ropinirole are sometimes used as first-line treatments. In fact, a study in patients initially treated with pramipexole demonstrated a reduction in loss of striatal nerve terminals (104,105). An additional advantage of these agents is that they have relatively prolonged dopaminergic actions (long half-life), to provide more sustained clinical benefit with less risk of motor fluctuations than with L-DOPA (126).

Rotigotine (Fig. 13.10) is a relatively new, nonergoline, synthetic DA agonist that is administered by transdermal patch and used to treat both early and advanced PD (158). Rotigotine is an agonist at D_1 as well as D_2 and D_3 receptors (159), similar to apomorphine. It is highly lipophilic and undergoes extensive first-pass hepatic clearance when given orally, encouraging use of the long-acting transdermal patch formulation (159). Rotigotine provides benefits in PD both alone and combined with L-DOPA and is better tolerated than other DA agonists (160–162).

The direct DA agonists generally produce similar adverse effects, including initial nausea and vomiting, postural hypotension, and fatigue. These effects are especially likely with the ergolines. Additional risks include psychotic reactions when DA agonists are given alone or with L-DOPA. These reactions include hallucinations, delusions, and confusion, and are most likely in elderly PD patients with symptoms of dementia. Treatment is similar to psychotic reactions encountered during L-DOPA therapy.

Experimental Dopamine D_1 Receptor Agonists

The opposing actions of the direct and indirect pathways in the basal ganglia (Fig. 13.4) suggest that coordinated neurotransmission requires activation of the direct pathway and attenuation of the indirect pathway involving DA neurotransmission in the basal ganglia. Such neuromodulation may require a balance of stimulatory actions at D_1 -type receptors and inhibitory actions at D_2 -type receptors. Consistent with a role for D_1 receptors in the direct output pathway, their stimulation represents a plausible pharmacotherapeutic approach for PD.

Clinical trials in PD with early, selective D_1 partial agonists (Fig. 13.11) such as the benzergoline CY-208-243 and the *R*-(+)-phenylbenzazepine SKF-38393 found these drugs to be either short acting or ineffective, suggesting that full D_1 agonist activity may be required (163,164). Several analogs of SKF-38393 (Fig. 13.11) are full-efficacy D_1 agonists, including *R*-(+)-SKF-81297 and its 6-halo derivatives, 6-Br-APB and 6-Cl-APB (SKF-82958) (165,166). In MPTP-lesioned monkeys, 6-Cl-APB produced antiparkinson effects (167), but its duration of action was less than 1 hour (168) and it produced severe dyskinetic effects (169). Moreover, *R*-(+)-SKF-81297, as well as the benzo-phenanthridine dihydrexidine (Fig. 13.11) (168), showed beneficial results only in monkeys with severe parkinsonism. It has been suggested that long-acting D_1 agonists may be most useful in late stages of PD (170).

Dihydrexidine (Fig. 13.11) was the first full-efficacy D_1 agonist to be developed, although it also has some

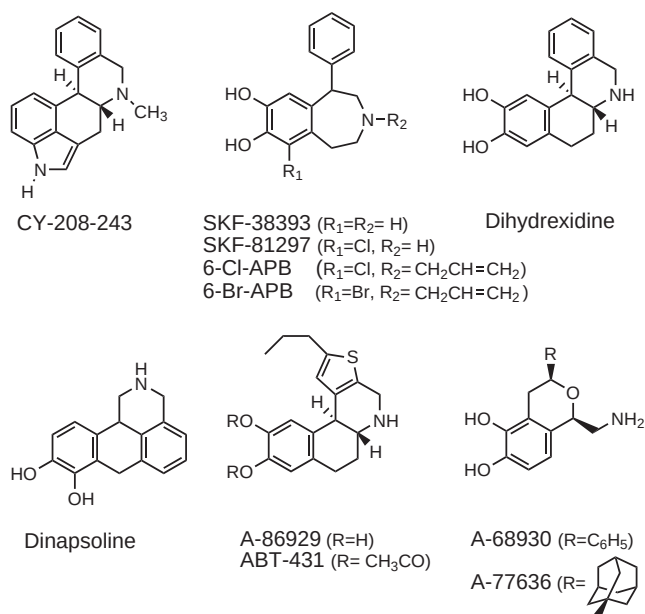


FIGURE 13.11 Experimental DA receptor agonist drugs that target primarily D_1 receptors.

D_2 -type activity (171,172). In MPTP-lesioned monkeys, dihydrexidine essentially eliminated all parkinsonian signs, and this effect was fully blocked by a selective D_1 antagonist but not by a selective D_2 antagonist (173). Continuous administration of dihydrexidine to rats for 2 weeks produced very little change in D_1 receptor density or D_1 receptor-mediated DA-stimulated adenylyl cyclase activity, suggesting that tolerance might not develop to its antiparkinson effects. In PD patients, however, dihydrexidine has a narrow therapeutic index, with dose-limiting adverse effects that include flushing, hypotension, and tachycardia after single intravenous doses (174).

A D_1 receptor pharmacophore model developed for dihydrexidine was used to design other novel molecular structures as full-efficacy D_1 agonists (175), including, dinapsoline (Fig. 13.11) (176), the dihydrexidine isostere A-86929, and its diacetyl prodrug ABT-431. These analogs are full-efficacy D_1 agonists with sustained antiparkinson effects in MPTP-lesioned monkeys (177). In PD patients, ABT-431 was highly effective against bradykinesia but produced dyskinesias (178).

Several isochromans also are full D_1 agonists. In primates pretreated with MPTP, A-68930 produced seizures, whereas the analog A-77636 showed antiparkinson effects without inducing seizures (Fig. 13.11) (179). However, A-77636 showed rapid desensitization to its beneficial effects (180), possibly related to its prolonged action that exceeds 20 hours (169). Other conformationally restricted analogs containing the β -phenyldopamine pharmacophore continue to be reported, including some with impressive D_1 -type over D_2 -type receptor agonist activity (181); however, in vivo studies with D_1 agonists have, so far, given disappointing results.

Monoamine Oxidase Inhibitors

Given the role of MAO in catabolism of DA (Fig. 13.2), leading to neurotoxic oxidation products of both endogenous (Fig. 13.5) and exogenous (Fig. 13.6) substrates, MAO inhibition has the potential to boost levels of brain DA and prevent formation of MAO-derived oxidation products that could cause neurotoxicity. Furthermore, by reducing the oxidation of DA, MAO inhibitors can extend the duration of response to L-DOPA and allow use of lower doses (106).

MAO is expressed as two isoforms, A and B. Histochemical and immunohistochemical studies reveal that in human brain MAO-A is found primarily in noradrenergic neurons, whereas MAO-B is found primarily in serotonergic and histaminergic neurons as well as in glial cells. There is no consensus on which form may predominate in brain DA neurons (182); thus, there is some uncertainty as to whether to target MAO-A or -B in treatment of PD. Less is known about the structural requirements for highly specific reversible MAO-A inhibitors (183) compared to selective MAO-B inhibitors (see Chapter 18, several of which produce clinical benefits in PD. A compelling link between PD etiology and MAO is that levels of MAO-B are increased in the brains of PD patients as a consequence of gliosis (184). Involvement of MAO-B in age-related neurologic disorders that include PD also is suggested by observations that human and rodent MAO-B levels increase with aging (185,186). Increased MAO-B levels would be expected to not only diminish DA levels, but also to increase levels of potentially neurotoxic oxidation products. Another link between MAO and PD is that 3,4-dihydroxyphenylacetaldehyde (DOPAL), the aldehyde that is the MAO-derived two-electron oxidation product of DA (Equation 1 in Fig 13.5), has been implicated in the aggregation of α -synuclein (187).

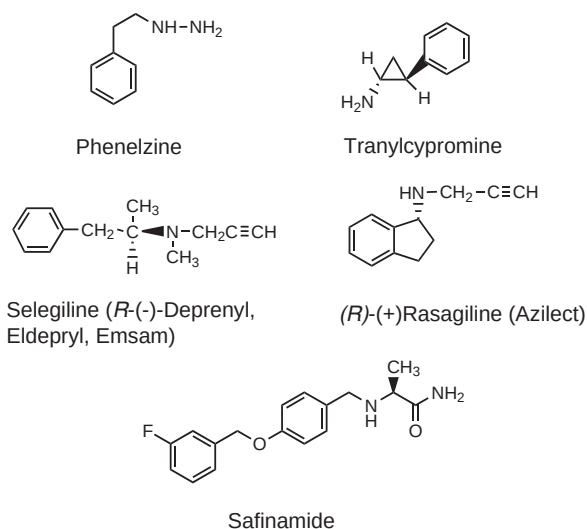
Selegiline and Rasagiline

Long-acting, irreversible, nonselective MAO-A/B inhibitors, such as phenelzine and tranylcypromine (Fig. 13.12; also, see Chapter 18) are contraindicated in combination with L-DOPA because of the risk of inducing hypertensive crises and delirium (17,188).

Selegiline (L-deprenyl; *R*-[$-$]-*N*,2-dimethyl-*N*-2-propynyl-phenethylamine) and rasagiline (Fig. 13.12) are propargylamine-type selective irreversible inhibitors of MAO-B. In addition to potentiating DA actions and allowing for reduction of L-DOPA dose, it has been proposed that MAO-B inhibitors may prevent formation of neurotoxic oxidation products of DA and slow neurodegeneration in PD; however, data from recent clinical studies do not support this attractive “neuroprotective” hypothesis (3,104–106). Nevertheless, MAO-B inhibitors have a beneficial effect on motor fluctuations because of their levodopa-sparing effect (106).

Selegiline and rasagiline undergo extensive hepatic metabolism. Selegiline is *N*-dealkylated via CYP2B6 and CYP2C19 to (–)-methamphetamine and, subsequently, to (–)-amphetamine, which has vasoactive activity similar to (+)-amphetamine (189). The amphetamine metabolites

MAO Inhibitors:



COMT Inhibitors:

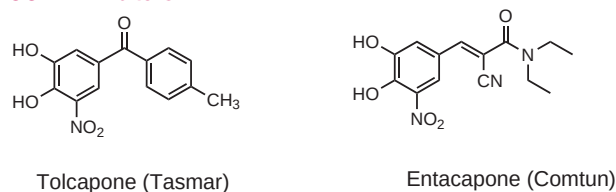


FIGURE 13.12 Drugs that inhibit monoamine oxidase (MAO) or catechol-*O*-methyltransferase (COMT).

of selegiline may contribute to its other pharmacologic property of DA and norepinephrine reuptake inhibition, thus potentiating the pharmacologic effects of L-DOPA. The amphetamine metabolites of selegiline have been associated with cardiovascular (orthostatic hypotension) and psychiatric (hallucinations) side effects.

Rasagiline is *N*-dealkylated primarily by CYP1A2 to (*R*)-1-aminoindan, which does not have vasoactive activity (189). Recently, rasagiline underwent clinical trials to assess its ability to provide neuroprotection (190). Rasagiline at a dose of 1.0 mg/d met endpoints consistent with a disease-modifying (neuroprotective) effect, but rasagiline at 2 mg/d failed; the authors of the study hypothesize that the stronger symptomatic effect of the higher dose masked an underlying disease-modifying effect. Rasagiline has recently been approved by United States FDA for treatment of PD.

Safinamide (Fig. 13.12) is an α -aminoamide derivative that is a reversible selective MAO-B inhibitor shown to provide benefits in early PD (191,192). In addition to its MAO-B inhibitor properties, safinamide also blocks voltage-dependent sodium and calcium channels and inhibits glutamate release (192). It is currently in phase III trials in the United States and in Europe (3).

Catechol-*O*-Methyltransferase Inhibitors

The peripheral metabolism of L-DOPA given alone leads to very limited access of the amino acid to the CNS. It is rapidly

decarboxylated by L-aromatic amino acid decarboxylase and 3-*O*-methylated by COMT (Fig. 13.2). Inactivation of L-DOPA and DA can be slowed by using a COMT inhibitor.

COMT is methyltransferase that uses as its methyl donor the cofactor S-adenosyl-L-methionine (SAME). COMT converts L-DOPA and catecholamines preferentially to their *m*-methoxy derivatives (Fig. 13.2). These include 3-*O*-methyl-DOPA and 3-*O*-methyl-DA (3-methoxytyramine, 3-methoxy-4-hydroxyphenethylamine), as well as the 3-*O*-methylated, deaminated compound HVA, the major final metabolite of DA in humans. Treatment with L-DOPA can reduce tissue concentrations of SAME (115), with uncertain consequences that should be limited by co-treatment with a COMT inhibitor. COMT acts in both peripheral and cerebral tissues, although the effect of COMT inhibitors to potentiate L-DOPA probably occurs mainly in peripheral tissues (193,194).

Tolcapone and Entacapone

Tolcapone and entacapone (Fig. 13.12) are reversible COMT inhibitors. Examination of their chemical structures reveals obvious similarities, and the molecular mechanisms by which these drugs interact with human COMT are proposed to be similar (195,196). Although the mechanisms of action and pharmacotherapeutic effects are similar for tolcapone and entacapone, they differ with respect to pharmacokinetic properties and adverse effects. Tolcapone has a relatively longer duration of action (8 to 12 hours) and acts both in the brain and periphery, whereas entacapone has a shorter duration of action (2 hours) and acts mostly in the periphery. Some common adverse effects of these agents are predictable and attributable to increased brain DA (e.g., nausea, vivid dreams, confusion, and hallucinations). A potentially fatal adverse effect, however, occurs only with tolcapone; after marketing, three fatal cases of fulminant hepatic failure were observed, leading to its market withdrawal in some countries and restriction in the United States to only those patients who have not responded to other therapies and who have appropriate monitoring for hepatic toxicity. The unforeseen hepatotoxicity associated with tolcapone has left entacapone as the only COMT inhibitor in wide clinical use; however, since the labeling restrictions in 1998, there have been no additional reports of hepatic fatality associated with entacapone (197). The mechanism by which liver damage is induced by tolcapone is believed to involve uncoupling of mitochondria oxidative phosphorylation, significantly reducing cellular generation of ATP (198,199). Additionally, it was shown that tolcapone (but not entacapone) induces cytotoxic pro-oxidant radical formation in hepatocytes (200). Finally, both COMT inhibitors may cause severe diarrhea and produce increased dyskinesias that may require a reduction in the dose of L-DOPA (201).

Approved and Experimental Drugs for PD That Target Nondopaminergic Proteins

Glutamate Antagonists

The dyskinesias associated with L-DOPA therapy may involve overactivity of thalamocortical excitatory glutamatergic

input to the motor cerebral cortex (Fig. 13.4). In addition, excessive release of glutamate due to synaptic overactivity is hypothesized to lead to “excitotoxicity,” resulting from excess neuronal Ca^{2+} influx due to opening of *N*-methyl-D-aspartate (NMDA) ion channel receptors (202), at which glutamate is a coagonist (along with glycine).

Amantadine and its dimethylated congener memantine (Fig. 13.13) are NMDA glutamate receptor antagonists that might provide neuroprotective effects (203). Both have moderately beneficial effects early in PD, can enhance the effects of L-DOPA, and can perhaps limit the severity of dyskinesias induced by L-DOPA therapy (204). Also, memantine has been used as a spasmolytic agent in the treatment of both PD and dementia (205). Amantadine was originally developed as an antiviral agent, and its use in PD patients with influenza revealed unexpected improvement in PD symptoms (206). The pK_a of this primary amine is 10.8; thus, it exists mainly in the protonated form at physiologic pH. The lipophilic cage-like structure of both amantadine and memantine likely facilitates their entry into brain and may provide resistance to metabolism by oxidative enzymes; both drugs are excreted mostly unchanged in the urine (207,208). In addition, amantadine causes release of DA and norepinephrine from intraneuronal storage sites and blocks reuptake of DA and was initially considered a DA-potentiating agent for use in mild PD (209).

Adenosine Receptor Antagonists

Adenosine is a nucleoside signaling molecule that acts at four GPCR subtypes, A_1 , A_{2A} , A_{2B} , and A_3 . The mRNA for A_{2A} receptor protein is highly concentrated in the striatum, nucleus accumbens, and olfactory tubercle, and co-localizes with D_2 receptor mRNA in these brain regions (210). Activation of A_{2A} receptors inhibits GABA release in striatum (211). Thus, antagonism of A_{2A} receptors is expected to increase GABA-mediated inhibition of the medium spiny output neurons to help compensate for loss of DA D_1 receptor-stimulated

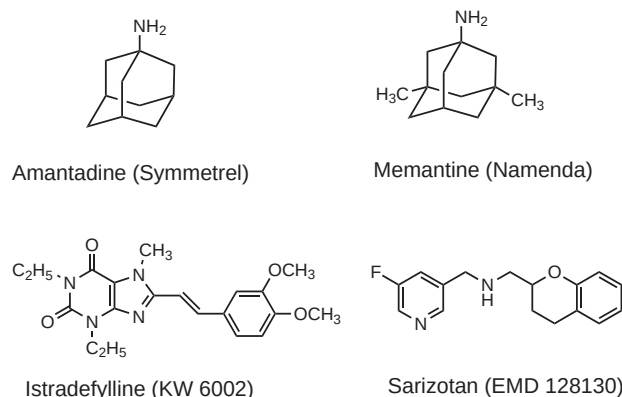


FIGURE 13.13 Approved and experimental drugs for PD that target nondopaminergic proteins.

GABA release and D_2 receptor-mediated inhibition of these neurons in PD (212) (see Fig. 13.4). Adenosine A_{2A} receptors also oppose the actions of D_1 and D_2 receptors on gene expression (213) and second-messenger systems and reduce the binding affinity of DA for D_2 receptors (214). An A_{2A} antagonist presumably would block these A_{2A} receptor-mediated inhibitory effects on DA neurotransmission. Activation of A_{2A} receptors also stimulates release of acetylcholine in striatum (215), and it has long been known that muscarinic receptor antagonists can ameliorate some signs of PD. As mentioned, there is an inverse relationship between the risk of developing PD and consumption of caffeine, a nonselective antagonist at adenosine A_1 and A_2 GPCRs (216,217).

Istradefylline (formerly KW 6002; Fig. 13.13) is a potent and selective antagonist at adenosine A_{2A} receptors that was shown to improve motor disability in primate models of PD (218). A recent double-blind, placebo-controlled clinical trial (219) showed that in patients stabilized on L-DOPA and other PD drug regimens, istradefylline-treated subjects had significant reductions in motor fluctuations and the drug was, in general, well tolerated. In a subsequent study assessing istradefylline as monotherapy for PD, the drug was well tolerated but did not demonstrate efficacy to improve motor symptoms (220). Studies with A_1 receptor antagonists lag behind; however, there is evidence that they reduce parkinsonian-type tremor in rodents (221).

Serotonin 5-HT_{1A} Agonists

Dysfunction of neurotransmission mediated by 5-hydroxytryptamine (5-HT; serotonin) occurs in the basal ganglia of patients with PD, and excessive serotonergic transmission may contribute to dyskinesias associated with dopaminergic treatments (222). The central and peripheral psychological and physiologic effects of 5-HT are mediated by 14 serotonin receptor subtypes grouped into the 5-HT₁ to 5-HT₇ families. The 5-HT₁ family consists of five GPCRs subtypes, 5-HT_{1A}, 1B, 1D, 1E, and 1F (223). The 5-HT_{1A} autoreceptor is expressed presynaptically on 5-HT terminals. When 5-HT_{1A} autoreceptors are activated, serotonin release is decreased (224), and diminished dyskinesias occur in rodent and primate models of PD (225). Activation of 5-HT_{1A} receptors also can reverse parkinsonism-like catalepsy induced by the DA D_2 antagonist haloperidol (221); thus, 5-HT_{1A} receptor activation might also counteract losses of nigrostriatal DA neurotransmission in PD. Moreover, in patients with advanced PD, intact striatal 5-HT terminals are an important site of decarboxylation of exogenous L-DOPA to DA. A 5-HT_{1A} agonist might act at striatal serotonergic terminals to limit release of DA produced by L-DOPA treatment and release from 5-HT terminals as a “false transmitter.”

Sarizotan (formerly EMD-128130; Fig. 13.13) is an aminomethylchroman derivative that has high affinity ($K_i \sim 0.05$ nM) and full-efficacy agonist activity at

human 5-HT_{1A} receptors (226). Sarizotan also has appreciable affinity ($K_i \sim 3$ nM/L) at human DA D_2 -type receptors. It is a partial agonist at D_2 but demonstrates no apparent activation of D_3 and D_4 receptors. Given by itself to MPTP-lesioned monkeys, sarizotan had no effect on the severity of motor deficits or on beneficial responses to L-DOPA, but it reduced L-DOPA-induced choreiform dyskinesias by more than 90% (224). Thus, there appears to be no compelling clinical evidence to support concern that sarizotan might limit release of DA from 5-HT terminals via 5-HT_{1A} activation or from DA nerve terminals via D_2 activation. The beneficial effects of sarizotan in parkinsonian monkeys may be specific to 5-HT_{1A} agonism because effects were reversed by a selective 5-HT_{1A} antagonist. In PD patients with dyskinesias resulting from L-DOPA therapy, adding sarizotan pharmacotherapy produced significant increases in periods of time without dyskinesia and significant reduction in periods of time with troublesome dyskinesias (227). A subsequent clinical study that assessed improvement of dyskinesias gave mixed results, depending on the dyskinesia rating scale that was used; however, the drug proved to be safe and relatively well tolerated (228). A recent study using the MPTP monkey PD model confirmed that sarizotan produces a sustained antidyskinetic effect while maintaining L-DOPA antiparkinsonian effects (229), suggesting the 5-HT_{1A} agonist field may prove fruitful in PD drug discovery.

Adjunct Therapy—Muscarinic and Histamine H₁ Antagonist Drugs

Cholinergic interneurons in the striatum exert mainly excitatory effects on GABAergic output from the striatum. Drugs that increase cholinergic neurotransmission (e.g., the cholinesterase inhibitor physostigmine and the direct agonist carbachol) have long been known to aggravate parkinsonism, whereas centrally active muscarinic antagonists (such as the belladonna alkaloids, including atropine) have moderately beneficial effects (17,230). Accordingly, before the discovery of L-DOPA, drug therapy for parkinsonism depended primarily on the limited efficacy of the natural belladonna alkaloids and newer synthetic antimuscarinic alkaloids, as well as histamine H₁ antagonists that also exert central antimuscarinic actions (Table 13.1). Synthetic centrally active muscarinic antagonists include benztropine mesylate, trihexylphenidyl, procyclidine, and biperiden; histamine H₁ antagonists with antimuscarinic activity include diphenhydramine and the phenothiazine ethopropazine.

The muscarinic and histamine H₁ antagonist drugs in Table 13.1 have limited therapeutic benefit in PD. Moreover, they exert a range of undesirable adverse effects associated with blockade of peripheral muscarinic receptors, including, dry mouth, impaired visual accommodation, constipation, urinary retention, and tachycardia. Adverse CNS effects include delirium,

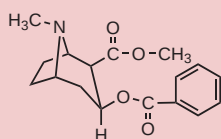
CLINICAL APPLICATIONS: RADIOPHARMACEUTICALS IN DIAGNOSIS OF PD

Even for an experienced clinical neurologist, diagnosis of PD can be difficult to confirm, especially in the early stages of this disease. Signs of PD vary markedly among patients and in the same person over time; disability can fluctuate dramatically, and progression of the disorder is unpredictable. In addition, a number of conditions mimic PD and vary in their responses to antiparkinson drugs. Given these difficulties, brain imaging techniques using positron emission tomography (PET) and single photon emission computed tomography (SPECT) are increasingly used for diagnosis of PD and to assess DA nerve terminal function.

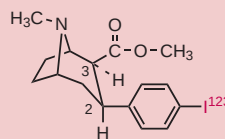
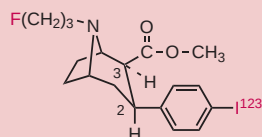
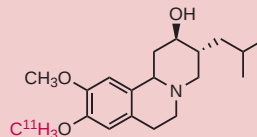
Spatial resolution is greater with PET, but SPECT technology is less expensive and more widely accessible in many clinical settings. In addition, positron-emitting nuclides used in PET imaging have very short half-lives (^{11}C , 20 minutes; ^{18}F , 109 minutes) and usually require an on-site cyclotron for

their production. SPECT nuclides have longer half-lives (^{123}I , 13 hours; $^{99\text{m}}\text{Tc}$, 6 hours) and often can be supplied commercially, and [$^{99\text{m}}\text{Tc}$]-labeled radioligands can be prepared locally as needed.

Quantitative assessment of nigrostriatal presynaptic DA nerve terminal function by PET using [^{18}F]-labeled 6-fluoro-L-DOPA ([^{18}F]DOPA) has proved useful for the early diagnosis of PD (231). Also, it has been known for some time that cocaine binds to the DA transporter (DAT) protein; however, cocaine also binds to other monoamine transporters and is rapidly hydrolyzed at its benzoyl ester function, making it impractical for use as an imaging agent (232). Linking the phenyl ring of cocaine directly to a tropane system, however, yields nonhydrolyzable, long-acting phenyltropanes that have been developed into clinically useful radiopharmaceuticals.



Cocaine

[^{123}I]-β-CIT
(DopaScan)[^{123}I]-FP-CIT
(DAT Scan)(+) - [^{11}C] - Dihydrotetrabenazine

The first tropane-based imaging to emerge was [^{123}I]-2-β-carbomethoxy-3-β-(4-iodophenyl)tropane, known as [^{123}I]β-CIT or RTI-55 (233). The radioactive iodine atom in the para position of the phenyl ring provides for the imaging properties of [^{123}I]β-CIT. A limiting factor with [^{123}I]β-CIT, however, is that it must be administered 8 hours before imaging to achieve peak uptake to DA nerve terminal regions. In an effort to improve the pharmacokinetic profile of [^{123}I]β-CIT, Neumeyer et al. (234) developed the *N*-3-fluoropropyl analog of β-CIT, known as [^{123}I]FP-CIT or [^{123}I]iof-lupane. This DA nerve terminal imaging agent rapidly reaches its target sites (DA reuptake proteins) so that patients can be imaged 1 to 2 hours after injection of [^{123}I]FP-CIT. A radioligand suitable

for PET imaging is obtained by replacing the fluorine atom with [^{18}F] in FP-CIT (235). DAT scan ([^{123}I]FP-CIT) is approved for clinical use, and both [^{123}I]β-CIT and [^{123}I]FP-CIT are used as research tools to assess DA nerve terminal function in neurodegenerative and neuropsychiatric disorders.

An alternative biomarker for assessing DA neuronal loss in PD is the vesicular monoamine transporter 2 (VMAT2) protein. VMAT2 is the transporter that moves nerve terminal cytoplasmic DA into synaptic vesicles for storage and subsequent exocytotic release. Recently, the PET radioligand (+)-[^{11}C]-dihydrotetrabenazine was reported as an investigational imaging agent for VMAT2 (236–238).

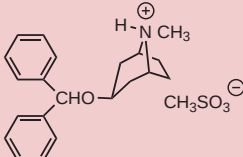
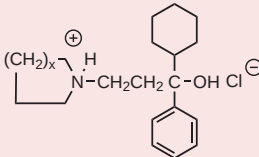
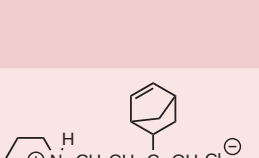
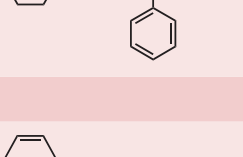
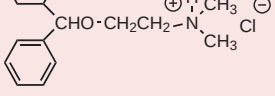
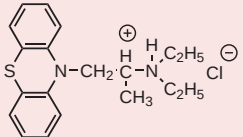
confusion, memory impairment, and psychotic symptoms. Despite their relatively unfavorable benefit/risk ratio, these drugs still are used in treatment of PD in combination with L-DOPA, particularly to help control tremor (230). They demonstrate better efficacy to control adverse extrapyramidal neurologic effects of potent D_2 -receptor antagonist antipsychotic drugs such as haloperidol (17).

SPASTICITY DISORDERS

Clinical Evaluation

Spasticity is characterized by skeletal muscle spasms and an increase in tonic stretch reflexes, sometimes with accompanying muscle weakness. Spasticity often is associated with cerebral palsy, multiple sclerosis, spinal cord injury, or stroke. The mechanisms that underlie clinical

TABLE 13.1 Muscarinic and Histamine H₁ Antagonist Drugs for Parkinson Disease

Class and Generic Name	Trade Name	Chemical Structure	Initial Single Oral Dose
Synthetic anticholinergic agents			
Benztrapine mesylate	Cogentin		0.5–1.0 mg
Trihexyphenidyl hydrochloride	Artane Pipanol (x = 2) Tremen		1.0–2.0 mg
Procyclidine hydrochloride	Kemadrin (x = 1)		2.0–5.0 mg
Biperiden hydrochloride	Akineton		1.0–2.0 mg
Antihistamine			
Diphenhydramine hydrochloride	Benadryl		25.0 mg
Phenothiazine (Antihistamine)			
Ethopropazine hydrochloride	Parsidol		50.0 mg

spasticity appear to involve damage to descending pathways in the spinal cord that results in hyperexcitability of motor neurons (239). In addition to lack of accurate experimental models, a limiting factor in characterizing the pathophysiology of spasticity and effectiveness of antispasmodic drugs continues to be lack of quantitative methodology for assessment of the spasmodic condition (240,241). Simple tests of muscle tone or reflex latency have not been fruitful; however, global clinical assessments, such as the number of painful spasms per day, have been more useful. Even the combined subjective impressions of improvement by the patient, family, and physician, however, may not establish that a particular drug is efficacious. Furthermore, spasm frequently coexists with pain, and spasmolytic drug efficacy may be related to both skeletal muscle relaxation

and analgesia (242,243). A straightforward strategy to establish whether a spasmolytic drug produces any benefit is gradual withdrawal of the drug. The diversity of neurologic disorders that culminate in spasticity and the subjectivity of many of the measurements make it difficult to establish efficacy of any one of the spasmolytic drugs. In summary, clinical evidence for efficacy of oral antispasmodic agents is scarce and weak.

Spasmolytic Drugs

Skeletal Muscle Relaxants

BACKGROUND The drugs used for spasmolytic conditions are diverse in their chemical structures and their sites and mechanisms of action. The first drug recognized to exhibit spasmolytic activity was antodyne or

3-phenoxy-1,2-propanediol. In guinea pigs and rabbits, antodyne produced prolonged paralysis without impairing consciousness. Antodyne was introduced into clinical medicine in 1910 as an analgesic and antipyretic. The duration of its skeletal muscle relaxant effect, however, was too short-lived to be clinically useful. In 1943, structure–activity relationship studies of a

series of simple glyceryl ethers related to antodyne led to the development and introduction of mephenesin (Table 13.2) in 1946 (244). Pharmacologic studies revealed that mephenesin selectively depressed polysynaptic, while sparing monosynaptic, spinal cord reflexes. The relative safety and selective action of mephenesin on the spinal cord led to its use as the first widely prescribed

TABLE 13.2 Skeletal Muscle Relaxants

Class and Generic Name	Trade Name	Chemical Structure	Initial Single Oral Dose
Glycerol monoethers and derivatives			
Mephenesin			1–2 g (oral)
Chlorphenesin carbamate	Maolate (R ₁ = H, R ₂ = Cl)		800 mg (oral)
Methocarbamol	Robaxin (R ₁ = OCH ₃ , R ₂ = H)		1–2 g (oral) 500 mg IM 1–3 g daily (IV)
Substituted alkanediols and derivatives			
Meprobamate	Equanil Miltown (R = H)		400 mg
Carisoprodol	Rela Soma (R = CH(CH ₃) ₂)		250–350 mg
Metaxalone	Skelaxin		800 mg
Benzazole			
Chlorzoxazone	Paraflex		250–750 mg
Miscellaneous			
Orphenadrine citrate	Norflex		100 mg (oral) 60 mg IM or IV
Cyclobenzaprine	Flexaril		10 mg

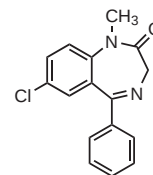
centrally acting skeletal muscle relaxant. Accordingly, mephenesin is the prototype of the interneuronal blocking type of muscle relaxant, albeit mephenesin itself no longer is used. In general, the pharmacology of the mephenesin-like muscle relaxants is remarkably similar to that of sedative-hypnotics. Indeed, the only apparent difference is that the spasmolytics have greater selectivity for modulating effects mediated by the spinal cord, thus producing less sedation than general sedative-hypnotics. Both classes produce a reversible, nonspecific depression of the CNS, and sedation, dizziness, and muscle weakness are common side effects.

Table 13.2 shows several compounds with mephenesin-like pharmacologic profiles that have been developed and marketed as antispasmodic muscle relaxants. Chlorphenesin carbamate and methocarbamol are carbamate analogs of mephenesin that are designed to have longer lasting effects; however, their duration of action has not been carefully evaluated and reported. The alcohol carbon of methocarbamol is chiral, and the *R*-(+)-enantiomer has greater muscle relaxant activity in mice (245); however, the drug is available only as the racemate. Meprobamate is the principal metabolite of carisoprodol; thus, the pharmacology of the two overlap. In clinical studies, carisoprodol has modest efficacy for treatment of low back pain associated with sprain or strain. Carisoprodol and meprobamate have sedative and anxiolytic effects similar to benzodiazepines, such as diazepam, and carry a similar liability for abuse and dependency (241). Metaxalone, on the other hand, is not associated with abuse, and although no data from high-quality clinical trials are available, it generally is reported to achieve muscle relaxation without excessive sedation. Chlorzoxazone is another agent marketed as a skeletal muscle relaxant, but with unclear efficacy in spasmodic conditions. A considerable drawback to empirical use of chlorzoxazone is that idiosyncratic hepatotoxicity and several sensitivity reactions (urticaria, erythema, and pruritus) have been reported (241). Orphenadrine is an ethanolamine ether that is related both chemically and pharmacologically to diphenhydramine-type histamine H_1 antagonists. Although its efficacy is not clearly established, any muscle relaxant effects of orphenadrine may be caused by CNS anticholinergic (muscarinic antagonist) activity. Orphenadrine also is an NMDA receptor antagonist; however, it is unclear if this activity may lead to antispasmodic effects. The anti-muscarinic activity of orphenadrine certainly leads to unpleasant peripheral side effects, such as dry mouth, blurred vision, and urinary retention. Cyclobenzaprine is another agent with prominent antimuscarinic activity that is used as a skeletal muscle relaxant, although its efficacy is modest. It is proposed that the muscle relaxant effects of cyclobenzaprine may result from antagonism of serotonin 5-HT₂ receptors in descending neurons of the spinal cord (246), but anticholinergic mechanisms cannot be ruled out. Although the presence of a double bond in the cycloheptyl ring of cyclobenzaprine is the only structural difference from the tricyclic antidepressant amitriptyline,

cyclobenzaprine is not an efficacious antidepressant. Cyclobenzaprine has a long plasma half-life (1 to 3 days), and accumulation on multiple dosing contributes to its high incidence of sedation.

In summary, it should be noted that there is no clear evidence that so-called skeletal muscle relaxants directly inhibit muscle contraction (as is the case for dantrolene, discussed later) or of whether their effects are related more to sedation and other nonspecific effects. In the United States, the drugs discussed above are approved for treatment of musculoskeletal conditions, whereas the drugs discussed below (baclofen, dantrolene, tizanidine) are approved specifically for muscle spasticity (243).

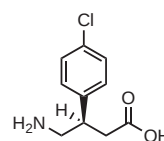
DIAZEPAM



Diazepam (Valium)

Although benzodiazepines, such as diazepam, often are used as skeletal muscle relaxants, they are not approved by the U.S. Food and Drug Administration (FDA) for this purpose (243). Diazepam and other benzodiazepines are proposed to exert skeletal muscle relaxant effects via their binding as agonists at the benzodiazepine receptor of the GABA_A receptor complex, which enhances GABA potency to increase chloride conductance (see Chapter 17). The muscle relaxant properties of classical benzodiazepines, such as diazepam, appear to be mediated mainly by the GABA_A α_2 and α_3 subunits (247,248). The result is neuronal hyperpolarization, probably at both supraspinal and spinal sites for spasmolytic activity. The actions of diazepam are sufficient to relieve spasticity in patients with lesions affecting the spinal cord and in some patients with cerebral palsy (242). Few clinical trials that include appropriate control groups have been conducted to assess diazepam efficacy as a muscle relaxant. It is currently generally accepted that diazepam is no more efficacious than, for example, carisoprodol, cyclobenzaprine, or tizanidine (i.e., efficacy is marginal) (240,241). Moreover, diazepam produces drowsiness and fatigue in most patients at doses required to significantly reduce muscle tone.

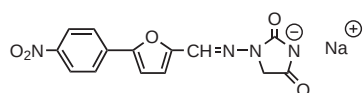
BACLOFEN



Baclofen
Racemic mixture (Lioresal)

Baclofen (β -*p*-chlorophenyl-GABA) is a GABA_B receptor agonist and is one of the most commonly used antispastic agents (243). The (*R*)-(-)-enantiomer is the active isomer at GABA_B receptors, but the racemate (Lioresal) is approved for use as a spasmolytic agent. The molecular mechanisms of GABA_B receptors in muscle spasticity are not understood any better than other putative receptor-based mechanisms discussed earlier. It is proposed that baclofen inhibits spinal cord monosynaptic and polysynaptic reflexes via GABA_B receptor-mediated opening of neuronal potassium channels that leads to hyperpolarization of primary afferent fiber terminals. In spinal motor neurons, baclofen suppresses excitability by reducing calcium persistent inward currents (239). Adverse effects of oral baclofen include sedation, excessive weakness, vertigo, and psychological disturbances. Baclofen is completely absorbed after oral administration, undergoes minimal hepatic metabolism, and is excreted mainly as the parent compound in urine and feces, with a half-life of approximately 3 to 4 hours. Intrathecal administration via an implanted infusion pump is used to control severe spasticity and pain that is not responsive to medication given by oral or other parenteral routes.

DANTROLENE

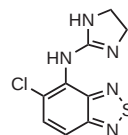


Dantrolene sodium (Dantrium)

Dantrolene is a hydantoin derivative that acts peripherally to reduce spasticity and is indicated for use in spinal cord injury, stroke, cerebral palsy, and multiple sclerosis (249). The site of action of dantrolene is believed to be at the sarcoplasmic reticulum in skeletal muscle cells. Dantrolene binds to a calcium channel protein (ryanodine receptor) on the sarcoplasmic reticulum to close the channel and inhibit the release of calcium; the alkaloid ryanodine activates the same receptor to open the channel. Dantrolene is believed to act directly on the contractile mechanism of skeletal muscle to decrease the force of contraction in the absence of any demonstrated effects on neural pathways, on the neuromuscular junction, or on the excitable properties of the muscle fiber membranes (249). Cardiac muscle and smooth muscle are minimally affected by dantrolene, likely because calcium release from sarcoplasmic reticulum of these muscle cell types occurs via a mechanism that differs from skeletal muscle. The muscle relaxant effect of dantrolene on skeletal muscle, however, is not specific, and generalized muscle weakness occurs as a major adverse side effect. Like other hydantoin, dantrolene is a weak base ($pK_a = 7.5$) that can cross the blood-brain barrier; thus, CNS depressant side effects (e.g., sedation) are common. Dantrolene sodium salt is slowly absorbed from the gastrointestinal tract. The mean half-life of the drug in

adults is approximately 9 hours after a 100-mg dose. It is slowly metabolized by the liver to give the 5-hydroxy and acetamido (nitro reduction and acetylation) metabolites, as well as unchanged drug, excreted in the urine. Interestingly, dantrolene also is valuable in alleviating the signs of malignant hyperthermia. This rare, genetically determined condition, which can be triggered by a variety of stimuli, including inhalation anesthetics and neuromuscular blocking drugs, involves an impaired ability of the sarcoplasmic reticulum to sequester calcium. For treating malignant hyperthermia, dantrolene is administered intravenously.

TIZANIDINE



Tizanidine (Zanaflex)

Tizanidine is a centrally acting adrenergic α_2 receptor agonist used to treat chronic muscle spasticity conditions. It is the only antispastic skeletal muscle relaxant to be studied for treatment of low back pain, in which its use is primarily as an adjunctive medication with analgesics (243). Tizanidine is postulated to act presynaptically to produce α_2 receptor-mediated decreased release of the excitatory neurotransmitters glutamate and aspartate from spinal interneurons (239). Tizanidine is structurally related to the α_2 -agonist clonidine that is used to treat hypertension; however, the blood pressure-lowering potency of tizanidine is approximately 10% to 20% that of clonidine. Nevertheless, patients may experience hypotension with tizanidine, together with muscle weakness, that may result in dizziness and falls in mobile patients. Tizanidine is rapidly and almost completely absorbed from the gastrointestinal tract; however, the estimated bioavailability is only 10% to 15% because of extensive first-pass metabolism, mainly by CYP1A2 (250), which results in oxidative degradation of the imidazoline ring and hydroxylation of the aromatic system (251). Elevated liver enzyme values are not frequent with tizanidine use. Hepatic injury and death because of liver failure have been reported, however, and this complication should be considered in view of its marginal antispasmodic efficacy (240,241). Other frequently reported side effects of tizanidine are drowsiness and dry mouth. Clonidine also has been used to treat spasticity, having been shown to suppress polysynaptic reflexes in vivo in lab animals (239); however, data from well-controlled clinical trials is not reported.

BOTULINUM TOXIN-TYPE A Botulinum toxin was identified in 1897 by the microbiologist Emile van Ermengem, who was the first to correlate botulism food poisoning with a bacterium found in raw, salted pork and the postmortem

tissue of victims who had consumed the contaminated meat; he later isolated the bacterium that was subsequently named *Clostridium botulinum* (252). There are seven immunologically distinct forms of botulinum neurotoxin, designated as serotypes A, B, C1, D, E, F, and G, all consisting of a 100-kda heavy chain and a 50-kda light chain (253). For all the serotypes, the C-terminal region of the toxin binds with high affinity to synaptic vesicle protein (SV2) receptors expressed on presynaptic acetylcholine motor nerve endings. After binding, botulinum toxin is taken up into the motor neuron via receptor-mediated endocytosis (254). Once in the cytoplasm, the light chain, a zinc-containing protease, targets synaptobrevin proteins (e.g., NSAP-25, VMAP, syntaxin) (255,256) that mediate fusion of acetylcholine-containing synaptic vesicles. The result is inhibition of acetylcholine release at the neuromuscular junction, thereby abolishing the motor end-plate potential (257) and, thus, inducing temporary muscle denervation and relaxation. The major clinical difference between the toxin serotypes is the duration of blockade of acetylcholine release. For type A botulinum toxin, clinical muscle relaxation persists for 4 to 6 months, longer than for the other serotypes. In 1989, the FDA approved type A botulinum toxin (Botox) for treatment of involuntary muscle contraction

disorders. There are now several different brands of type A botulinum toxin available, and it should be noted that dosage is measured in “units” that differ for each brand (258).

Administration of botulinum toxin is by injection into target muscles that are believed responsible for the clinical muscle spasticity (258). Despite the relatively high toxicity of botulinum toxin, it has proven safe because the protein does not diffuse beyond 2 cm from the injection site (256). The clinical effect of the toxin is observed about 4 days after injection, is optimal at about 1 month, and lasts about 3 to 4 months (258). The result is a weakening and relaxation of muscle spasticity that facilitates physical stretching and lengthening of the afflicted muscles, as well as allowing for an opportunity to strengthen antagonist muscles to restore the proper balance between the two (256,258). Clinical evidence continues to accumulate that supports the safe use of botulinum toxin to reduce muscle spasticity and improve voluntary muscle movement and active function in certain patients. Injection of botulinum toxin also is used cosmetically to relax facial muscles, the contraction of which results in alteration of the form and structure of the overlying connective tissue (i.e., wrinkles).

SCENARIO: OUTCOME AND ANALYSIS

Outcome

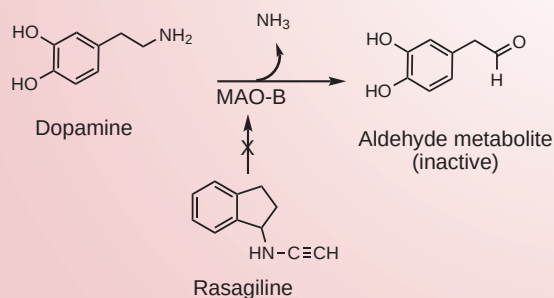
David Hayes, PharmD

After careful consideration, the patient is started on rasagiline secondary to his early-stage PD, lack of disability, and the potential for this agent to provide some neuroprotection. He is seen in the neurology clinic 3 months after starting this treatment and reports that he has not experienced any tremor and has tolerated the medication well.

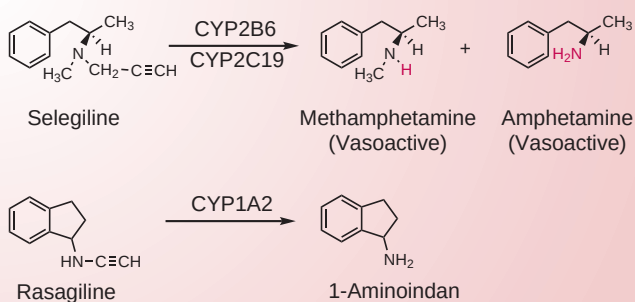
Chemical Analysis

Victoria Roche and S. William Zito

Rasagiline is an inhibitor of monoamine oxidase B (MAO-B), the enzyme responsible for the inactivating deamination of dopamine.



Rasagiline is vulnerable to CYP-mediated *N*-dealkylation but, as an indan-base structure, it does not produce the potent vasoconstricting amphetamine and methamphetamine metabolites that selegiline, a related phenethylamine-based MAO-B inhibitor, does. While these metabolites have some potential benefit in augmenting L-DOPA-mediated dopaminergic action in the CNS, their value may be overridden by the increase in blood pressure that patients will experience. Given that this patient was on a thiazide diuretic to lower elevated blood pressure, the amphetamine metabolites could have been highly problematic. Rasagiline would definitely be the safer MAO-B inhibitor for use in this hypertensive patient.



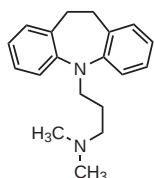
CASE STUDY

FB is a 64-year-old Cultural Studies faculty member at a prestigious university. A beloved teacher and well-respected scholar, he has spent his professional life totally engaged in his work and loving every moment of it. Nine months ago FB began to experience a slight unsteadiness when standing and walking, and these symptoms have now progressed to the point where he must use a power chair to negotiate the campus. Despite repeated attempts to pinpoint a diagnosis, his physicians are uncertain about what's causing his decline, although multiple sclerosis has not been ruled out. FB's job requires a great deal of reading, and he is currently conducting a thorough review of the literature for a new textbook he is writing. For as long as he can, he wants to

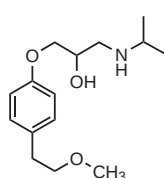
stand while teaching his courses, rather than deliver his lectures from his chair.

FB has begun to experience painful muscle spasms in his lower limbs that heat and massage do not adequately control, and you are being asked for an opinion related to an appropriate therapeutic agent to relieve them. FB had a myocardial infarction 5 years ago and is currently taking the β -blocker metoprolol to keep his cardiac function and blood pressure in check. Depressive episodes related to his future prompted his physician to start him on the tricyclic antidepressant imipramine. He is continuing this therapy despite occasional dizziness and some problematic drowsiness.

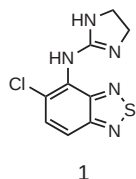
Consider the three potential therapeutic choices drawn below.



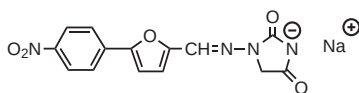
Imipramine



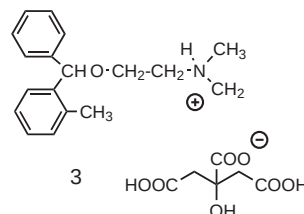
Metoprolol



1



2



3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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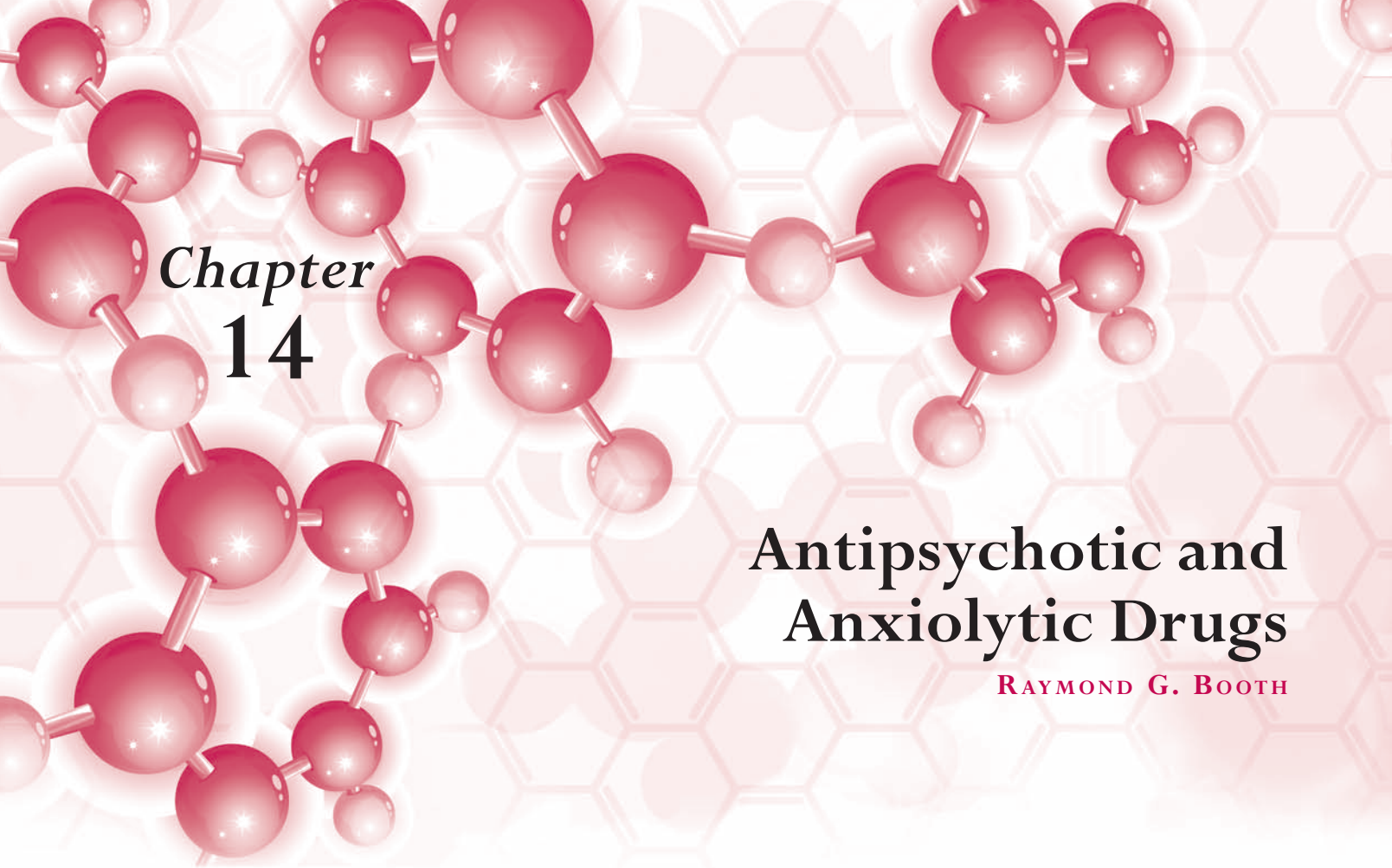
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Chapter 14

Antipsychotic and Anxiolytic Drugs

RAYMOND G. BOOTH

*Drugs Covered in This Chapter**

BENZAMIDE CLASS

BENZAZEPINES

- Asenapine
- Clozapine
- Loxapine
- Olanzapine
- Quetiapine

BENZISOXAZOLES AND BENZISOTHIAZOLES

- Iloperidone
- Lurasidone
- Paliperidone
- Risperidone
- Ziprasidone

BENZODIAZEPINES

- Chlordiazepoxide
- Diazepam
- Flurazepam
- Oxazepam

BUTYROPHENONE CLASS

- Droperidol
- Haloperidol
- Pimozide
- *Spiperone*
- *Trifluoperidol*

MISCELLANEOUS ANTIPSYCHOTIC AGENTS

- Aripiprazole

- Molindone
- Sertindole

MISCELLANEOUS ANXIOLYTIC AGENTS

- Buspirone

NONBENZODIAZEPINE AGONISTS

- Eszopiclone
- Zaleplon
- Zolpidem

PHENOTHIAZINE CLASS

THIOXANTHENES CLASS

- Thiothixene

Abbreviations

BZR, benzodiazepine receptor

cAMP, cyclic adenosine
monophosphate

βCCE, β-carboline-3-carboxylic acid
ethyl ester

CNS, central nervous system

DMCM, 6,7-dimethoxy-4-ethyl-β-
carboline-3-carboxylic acid methyl
ester

DSM-IV-TR, *Diagnostic and Statistical
Manual of Mental Disorders, Fourth
Edition, Text Revision*

FDA, U.S. Food and Drug
Administration

GABA, γ-aminobutyric acid

GPCRs, G protein-coupled receptors

HPTP, haloperidol
1,2,3,6-tetrahydropyridine

HPP⁺, haloperidol pyridinium

5-HT, serotonin

MPTP, 1-methyl-4-phenyl-1,2,3,6-
tetrahydropyridine

MPDP⁺, 1-methyl-4-phenyl-2,3-
dihydropyridinium

MPP⁺, 1-methyl-4-phenylpyridinium

NMDA, *N*-methyl-D-aspartate

7-OH-DPAT, (+)-7-hydroxy-*N,N*-di-*n*-
propyl-2-aminotetralin

SSRI, selective serotonin reuptake
inhibitor

*Drugs available outside the United States are shown in italics.

SCENARIO

David Hayes, PharmD

Ben is a 25 year old who was diagnosed with paranoid schizophrenia 1 year ago, after having a psychotic episode consisting of auditory hallucinations and delusions of people hiding in his attic. He was encouraged by his mother to see a psychiatrist, at which time he was initiated on olanzapine 5 mg daily. Six months ago he started feeling paranoid and believing that the FBI was following him, so his psychiatrist increased his dose to 10 mg daily; since then Ben has been stable. Ben has no medical history other than schizophrenia and takes no other medications. His family history is significant—his father had type 2 diabetes

and obsessive compulsive disorder. Ben recently went to see his primary care physician and it was noted that he had gained 25 lb since his visit 1 year ago. In addition, during this visit he was diagnosed with prediabetes, with a fasting blood glucose level of 119 mg/dL. The clinical question is whether to switch his antipsychotic in favor of one with fewer metabolic effects or to leave the antipsychotic as it is since Ben is stable.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

OVERVIEW OF MENTAL ILLNESSES

Mental illnesses that can be treated with psychotropic drugs are broadly categorized as psychoses, neuroses, and mood (depression, bipolar) disorders. Different classes of psychotropic agents differ in their ability to modify symptoms of these mental illnesses; thus, an appropriate diagnosis is critical to selecting an efficacious psychotropic drug. This chapter is focused on the medicinal chemistry of drugs that are used to treat psychoses (including the manic phase of bipolar illness) and anxiety disorders. The definitive diagnostic criteria for psychiatric disorders in the United States are well described in the *Diagnostic and Statistical Manual of Mental Disorders of the American Psychiatric Association* (DSM-IV-TR) (1). The psychoses (e.g., schizophrenia) are among the most severe mental illnesses and commonly include symptoms of delusions and sensory hallucinations. In anxiety disorders (neuroses), the ability to comprehend reality is retained, but mood changes (anxiety, panic, dysphoria) and thought (obsessions, irrational fears) and behavioral (rituals, compulsions, avoidance) dysfunction can be disabling. Mood and panic disorders often include dysfunction of the autonomic nervous system (e.g., altered patterns of sleep and appetite) in addition to psychic abnormalities. Depression (see Chapter 18) can lead to self-harm and suicide. In general, antipsychotic agents, which can have severe neurologic and metabolic (obesity, type 2 diabetes) side effects, should be used to treat only the most severe mental illnesses (i.e., psychoses such as schizophrenia).

SCHIZOPHRENIA

Definition

An historical definition of schizophrenia began approximately 100 years ago, with the German psychiatrist Emil Kraepelin's description of a type of dementia that was characterized as a severe, chronic mental disorder without known external causation wherein functional deterioration progresses with the symptoms of hallucinations, delusions, thought disorder, incoherence, blunted affect,

negativism, stereotyped behavior, and lack of insight (2). The deterioration progresses to catatonia and hebephrenia (illogical, incoherent, and senseless thought processes and actions, delusions, and hallucinations). Meanwhile, the Swiss psychiatrist Eugen Bleuler coined the term "schizophrenia" to take into account the perceived "schism" or splitting in mental functioning (3).

A modern definition of schizophrenia comes from DSM-IV (1). The diagnostic criteria for schizophrenia require two or more of the following characteristic symptoms to be present for a significant proportion of time during a 1-month period: delusions, hallucinations, disorganized speech, or grossly disorganized or catatonic behavior. There is, however, flexibility in the diagnostic criteria that leaves room for professional psychiatric judgment. For example, it is enough if hallucinations consist of a voice maintaining a running commentary on the patient's behavior or there are two or more voices that converse with each other. Also, for a significant proportion of time, one or more areas of social functioning, such as work, interpersonal relationships, or self-care, are markedly below the level achieved before the onset of symptoms. Continuous symptoms must persist for 6 months. Finally, before a diagnosis of schizophrenia is made, affective disorders as well as drug/alcohol abuse or other medical conditions must be ruled out. According to DSM-IV parameters, the common estimate for schizophrenia incidence is about 1% of the population.

Etiology of Schizophrenia

A neurobiologic basis for schizophrenia and related psychotic syndromes remains elusive. Compelling evidence linking genetic factors to the etiology of schizophrenia is not apparent despite enormous progress in the field of molecular genetics and numerous investigations of hereditary factors associated with psychotic illnesses. Current epidemiologic evidence suggests that individual variation in susceptibility to schizophrenia involves alleles of moderate to small effect in multiple genes (4). Investigations of environmental causative factors have

CLINICAL SIGNIFICANCE



As you review the chemistry of the antipsychotics and their receptor affinity and interaction (e.g., agonist, partial agonist, antagonism, and reverse agonist), think about potential for benefit and/or adverse effects. It is believed that those agents with lower affinity to D_2 receptor and higher affinity to $5-HT_{2A}$ may reduce the probability of extrapyramidal syndromes and provide effective therapy for the symptoms schizophrenia and/or psychosis. It has been determined that antipsychotics exhibit metabolic effects; however, the exact mechanism of metabolic effects such as weight gain has not been fully elucidated. There is some data to support that agents with affinity and antagonistic activity at the $5-HT_{2C}$ and/or histamine H_1 GPCR receptors have greater metabolic impact. Olanzapine and clozapine both share high to moderate affinity and antagonistic activity at these receptors and both have

been determined to have the significant impact on weight gain in their clinical evaluation. Conversely, agents with affinity but that act as a partial agonist, such as aripiprazole, have demonstrated lesser effects on weight gain. Therefore, an understanding of receptor affinity and interaction in relation to the potential for benefit and/or adverse effects tempered with clinic evidence may provide the clinician with support for the choice of an initial therapy or aid in the decision to switch secondary to the occurrence of an adverse effect such as weight gain. This consideration may be most important for those patients who are already overweight and/or at risk for metabolic diseases such as type 2 diabetes.

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focused on prenatal and perinatal risk factors for brain damage. For example, studies have examined the incidence of schizophrenics who were born under conditions of obstetrical complications, influenza epidemics, food shortages, and Rh factor incompatibility. Gene–environment interactions have been linked to the etiology and pathology of schizophrenia, with a role for inflammatory processes most clearly implicated (5). In fact, regulators of inflammatory processes such as cytokines are a common link between the genetic and environmental components of schizophrenia etiology, and it is noteworthy that inflammatory processes involved in maternal infection, obstetric complications, neonatal hypoxia, and brain injury involve cytokines. Gene–environment interactions involving chromosomal abnormalities and obstetric/perinatal complications may converge with precipitating factors such as stress or chemical agents to result in a neurodegenerative component of schizophrenia (6). Neuroanatomic hypotheses include increased ventricular volume; however, neuropathologic changes associated with schizophrenic brains are not obvious as in, for example, Parkinson disease. In contrast, neurochemical abnormalities are well documented. Alterations of brain dopaminergic neurotransmission in psychoses have been studied for more than 30 years, and this field of psychobiologic research generally revolves around the “dopamine hypothesis” of schizophrenia.

The Dopamine Hypothesis

The dopamine hypothesis of schizophrenia arose from observations that the first relatively safe and effective antipsychotic drugs, the phenothiazines, such as chlorpromazine, used in the early 1950s affected brain dopamine metabolism (7). Simply put, the dopamine hypothesis

of schizophrenia suggests that the disease results from increased dopaminergic neurotransmission and that approaches that decrease dopaminergic neurotransmission will alleviate psychotic symptoms. Most antipsychotic agents have activity to limit dopaminergic neurotransmission (8–10), providing some indirect evidence to support the dopamine hypothesis of schizophrenia. In a seminal study (11), the average daily dose of an antipsychotic was found to correlate well with affinity for dopamine D_2 -type receptors. Moreover, extrapyramidal side effects of antipsychotic drugs correlate with their dopamine D_2 affinity. It should be noted, however, that functional interaction of antipsychotic drugs with the D_2 receptor is complex, involving antagonism, inverse agonism, and partial agonism. For example, recent studies show that essentially all clinically used antipsychotic drugs are D_2 inverse agonists or partial agonists (12), suggesting that biochemical as well as clinical effects may not be explained by simple blockade of endogenous agonist (dopamine) access to the D_2 receptor.

The dopamine hypothesis of schizophrenia and pharmacotherapy involving targeting of dopamine receptors (especially the D_2 -type) has dominated research directions, but it should be noted that this approach is somewhat “pharmacocentric.” Consideration of potential new drugs as antipsychotic agents usually is limited to compounds that have demonstrated behavioral or biochemical evidence of antidopaminergic actions. In fact, this circular approach may be practical considering the lack of proven alternative neuropharmacologic explanations of antipsychotic drug activity; nevertheless, after about 50 years of research focusing on brain dopaminergic systems, uncontested evidence linking the etiology of psychotic illnesses to the neurobiology of dopaminergic systems remains elusive. Alternative

explanations, especially those involving adrenergic and serotonergic receptor systems, probably will gain popularity as more atypical antipsychotic drugs (e.g., clozapine) are introduced and also found to have actions at these receptor systems, perpetuating and expanding a pharmacocentric approach to antipsychotic drug design and development.

Dopamine Receptors in Schizophrenia

As discussed in Chapter 13, there are five different dopamine receptors, D_1 , D_2 (with short and long splice variants), D_3 , D_4 , and D_5 —all are G protein-coupled receptors (GPCRs), and as is the case for almost all GPCRs, three-dimensional x-ray crystal structural information is not available. Currently, there are no drugs or even medicinal chemical probes that are specific enough to distinguish between the five subtypes; thus, dopamine receptors often are classified as the D_1 -type (includes D_5), which stimulate adenylyl cyclase, or the D_2 -type (includes D_3 and D_4), which inhibit adenylyl cyclase.

Several chemical probes are available that can distinguish between the general D_1 -type and D_2 -type receptor families (Fig. 14.1). The *R*(+)-isomer of the benzazepine derivative, SKF 38393, is used for research as a selective D_1 -type partial agonist. Meanwhile, the structurally related benzazepine derivative, *R*(+)-SCH 23390, is used as a selective D_2 -type receptor antagonist. Although not very selective for D_1 -type over D_2 -type receptors, the rigid benzophenanthridine derivative (–)-dihydroxidine is a useful research tool, because it is a D_1 -type full-efficacy agonist (13,14). Selective D_2 -type full agonists, such as the pyrazole derivative (–)-quinpirole, and D_2 -type antagonists, such as (–)-sulpiride, also are available to researchers. The dopamine D_3 receptor subtype is of particular neuropharmacologic

interest because of its preferential distribution in certain limbic regions of mammalian brain, notably in the nucleus accumbens of the basal forebrain. It is proposed that highly D_3 -selective drugs might be developed as antipsychotic agents with preferential limbic antidopaminergic actions while sparing the extrapyramidal basal ganglia, presumably decreasing the neurologic movement disorder side effects associated with antipsychotic drug therapy. The tetrahydronaphthalene, (+)-7-hydroxy-*N,N*-di-*n*-propyl-2-aminotetralin (7-OH-DPAT), and some of its congeners are particularly promising D_3 -selective lead agents. The benzazepine clozapine, the prototype of second-generation atypical antipsychotics (see later discussion), shows relatively greater affinity for the D_4 dopamine receptor subtype in addition to its relatively high affinity for serotonin 5-HT₂, adrenergic α_1 and α_2 , muscarinic M_1 , and histamine H_1 receptors (15).

The dopamine D_1 -type and D_2 -type receptor families are differentially distributed in mammalian forebrain dopaminergic pathways. The extrapyramidal nigrostriatal pathway, which plays a key role in locomotor coordination, consists of neurons with cell bodies in the A9 pars compacta of the substantia nigra in the midbrain. These neurons project to the basal ganglia structures of the caudate nucleus and putamen (collectively referred to as striatum) in the forebrain (Fig. 14.2). Degeneration of neurons in the nigrostriatal pathway is the hallmark pathologic feature of Parkinson disease, clinically manifested as bradykinesia, muscular rigidity, resting tremor, and impairment of postural balance. Blockade of dopamine receptors on cholinergic neurons in striatum is associated with the sometimes severe extrapyramidal, parkinsonian-like side effects (muscular rigidity, bradykinesia, akathisia) that frequently occur with antipsychotic drug treatment.

The mesolimbic and mesocortical pathway, involved in integration of emotions, behaviors, and higher thought processes, consists of neurons with cell bodies in the A10 ventral tegmentum. These neurons project to limbic forebrain structures, including the nucleus accumbens and amygdala, and to higher levels of cerebral function, such as the frontal cortex (Fig. 14.2). According to the dopamine hypothesis, increased dopaminergic neurotransmission in limbic pathways contributes to the “positive” symptoms (e.g., hallucinations and excited delusional behavior that can be reduced with typical antipsychotic drugs) but not necessarily to the “negative” symptoms (e.g., catatonia) observed in the clinical manifestation of schizophrenia.

Typical antipsychotic drugs act in both extrapyramidal and limbic brain regions at D_2 -type dopamine receptors that can be located postsynaptically (on cell bodies, dendrites, and nerve terminals of other neurons) as well as presynaptically on dopamine neurons. Dopamine receptors located presynaptically on dopamine cell bodies and nerve terminals are called autoreceptors and act to negatively modulate neuronal firing and dopamine synthesis and release (Fig. 14.3). Low concentrations of certain dopamine agonists can stereospecifically activate dopamine D_2 -type autoreceptors to decrease dopamine synthesis (16) and release (17), thus reducing dopaminergic

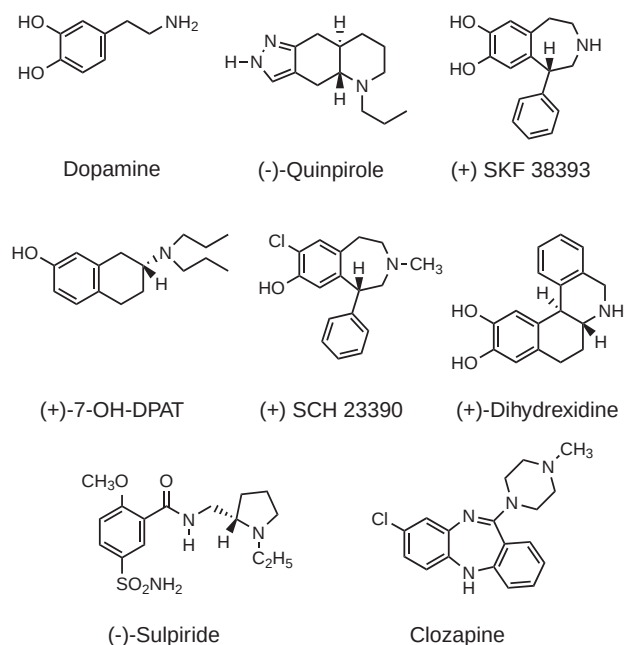


FIGURE 14.1 Structures of compounds useful for characterizing dopamine receptors.

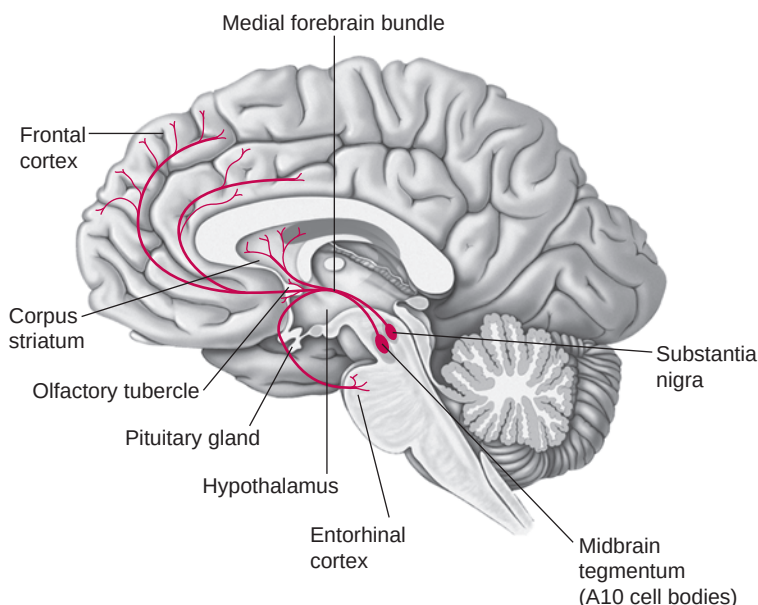


FIGURE 14.2 Some dopamine pathways in human brain.

neurotransmission. Consistent with the dopamine hypothesis of schizophrenia, selective dopamine autoreceptor agonists could, theoretically, be pharmacotherapeutic agents in schizophrenia and related mental illnesses.

GPCR Functional Selectivity

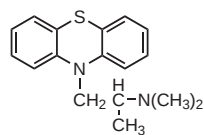
It is now realized that the same GPCR can couple to different $G\alpha$ proteins to result in “multifunctional” signaling that provides a mechanistic basis for “functional selectivity” with regard to design of drugs targeting DA receptors (18,19). Molecular mechanisms to account for GPCR multifunctional signaling involve the concept of “GPCR permissiveness,” which assumes a high degree of flexibility in the interactions between a ligand, receptor, and G protein (20). These interactions occur mainly between the G proteins and the second and third intracellular loops and carboxy-terminal tail of the receptor. Some factors that influence this interaction include receptor:G protein ratios and amounts, alternative GPCR splicing, and conformational changes in the G protein and/or receptor.

A critical assumption of GPCR multifunctional signaling theory is that a heterogeneity of active and inactive receptor conformations exists and that ligands differ in their ability to stabilize different receptor conformations, as described in the “stimulus trafficking” hypothesis (21). Of particular relevance to medicinal chemistry, it follows that upon binding, ligand chemical structural parameters are among the most important determinants of GPCR conformation stabilized. It follows that ligand stereochemistry and other structural parameters influence the type of $G\alpha$ protein and intracellular signaling pathway activated, resulting in ligand-specific functional outcomes (22–24). Ligands that show such functional selectivity can be exploited for drug design purposes. A clinically relevant example is the second-generation (atypical) antipsychotic drug aripiprazole (25), which interacts with dopamine D_2 receptors to produce agonist, partial agonist, inverse agonist, or neutral

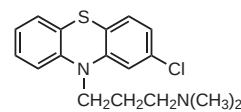
antagonist functional effects, depending on the D_2 receptor cellular milieu (e.g., G protein complement and concentration) and particular location (e.g., presynaptic vs. postsynaptic and extrapyramidal vs. limbic brain regions).

Other Brain Neurotransmitter Systems in Schizophrenia

In addition to postsynaptic dopamine receptors and presynaptic dopamine D_2 -type autoreceptors, heteroreceptors, such as adenosine (A_2) (26), histamine (H_1) (27), and serotonin ($5-HT_{1A}$) (28), located on or near presynaptic dopaminergic nerve terminals in the striatum (extrapyramidal) or nucleus accumbens (limbic) regions of brain can modulate dopamine synthesis (and release) by altering the activity of tyrosine hydroxylase, the rate-limiting step in catecholamine biosynthesis. Similarly, activation of adrenergic (α_2) autoreceptors in the limbic structure hippocampus negatively modulates the release of the neurotransmitter norepinephrine (29). It is proposed that atypical antipsychotic drugs, such as clozapine, may interact with these other neurotransmitter receptor systems (i.e., histamine, serotonin, and adrenergic) instead of (or in addition to) dopamine receptor systems. Preceding the introduction of the first clinically successful phenothiazine-type neuroleptic, chlorpromazine, the first phenothiazine to be used to treat psychiatric patients in the 1940s (unsuccessfully) was promethazine, an “antihistamine” H_1 antagonist.



Promethazine



Chlorpromazine

Serotonin $5-HT_{2A}$ and $5-HT_{2C}$ GPCRs that are expressed in the mesolimbic and mesocortical pathways also are

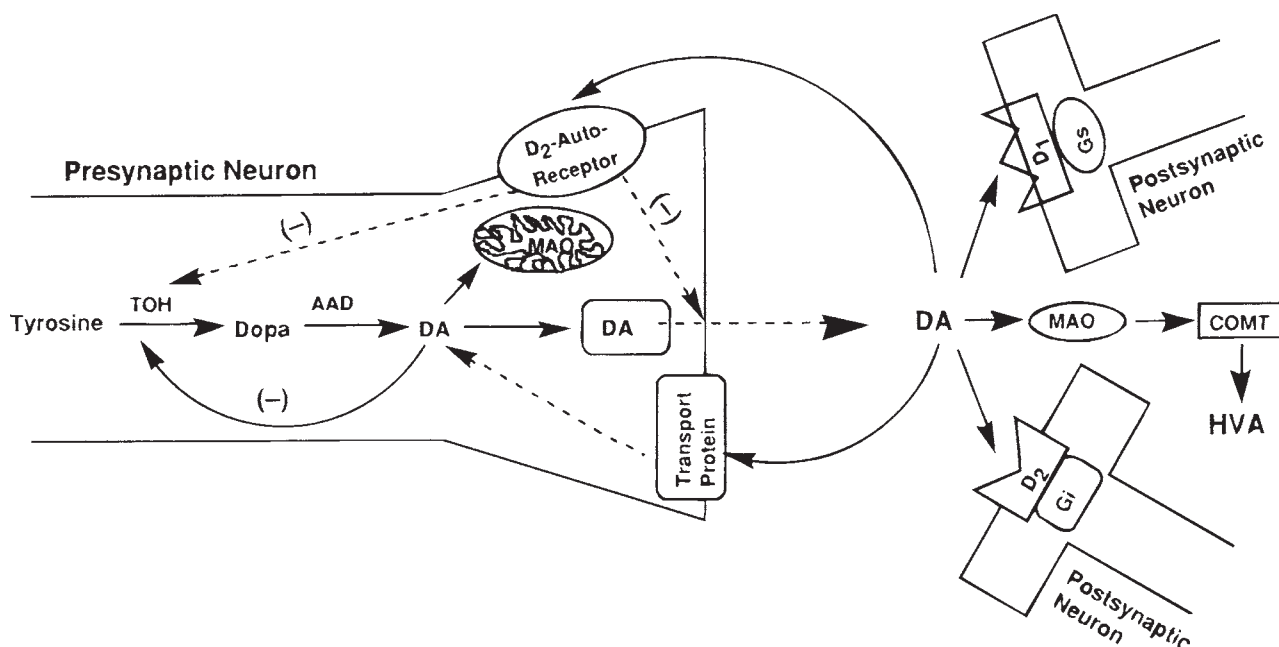


FIGURE 14.3 Tyrosine is hydroxylated in a rate-limiting step by tyrosine hydroxylase (TOH) to form dihydroxyphenylalanine (DOPA), which is decarboxylated by L-aromatic amino acid decarboxylase (AAD) to form dopamine (DA). Newly synthesized DA is stored in vesicles, from which release occurs into the synaptic cleft by depolarization of the presynaptic neuron in the presence of Ca^{2+} . The DA released into the synaptic cleft may go on to stimulate postsynaptic D_1 - and D_2 -type autoreceptors that negatively modulate DA synthesis (via inhibition of TOH) and release. The action of synaptic DA is inactivated largely via reaccumulation into the presynaptic neuron by high-affinity DA neurotransport proteins located on the nerve terminal membrane. Free cytoplasmic DA negatively modulates DA synthesis via end-product (feedback) inhibition of TOH by competition with bipterin cofactor. Cytoplasmic pools of DA may undergo metabolic deamination by monoamine oxidase (MAO), an enzyme bound to the outer membrane of mitochondria, to form dihydroxyphenylacetaldehyde, which oxidizes to dihydroxyphenylacetate (DOPAC). The DA or DOPAC may undergo methylation by catechol-O-methyltransferase (COMT), ultimately forming homovanillic acid (HVA), a metabolite excreted in urine.

linked to the pathophysiology and treatment of schizophrenia. For example, activation of $5\text{-HT}_{2\text{A}}$ receptors mediates psychotomimetic effects of some classes of hallucinogenic drugs (30), activation of $5\text{-HT}_{2\text{C}}$ receptors leads to antipsychotic effects in various rodent models of schizophrenia, and some second-generation (atypical) antipsychotic drugs such as clozapine have higher affinity for human serotonin 5-HT_2 -type receptors than dopamine D_2 -type receptors (15). Moreover, obesity associated with especially atypical antipsychotic drugs (e.g., clozapine) is thought to be due to their antagonist or inverse agonist actions at $5\text{-HT}_{2\text{C}}$ receptors (31), whereas selective activation of $5\text{-HT}_{2\text{C}}$ receptors reduces food intake (32). The other member of the 5-HT_2 family is the $5\text{-HT}_{2\text{B}}$ GPCR, which although also expressed in brain, is best known for its notorious role in the periphery where its activation has been linked to cardiac valvulopathy (33–35). Thus, the desired 5HT_2 -type pharmacology for treatment of schizophrenia without propensity for weight gain appears to be $5\text{-HT}_{2\text{A}}$ receptor antagonism or inverse agonism and/or $5\text{-HT}_{2\text{C}}$ receptor agonism, with no tolerance for activation of $5\text{-HT}_{2\text{B}}$ receptors (36). As for dopamine receptors, our understanding of the role of serotonin 5-HT_2 GPCRs (as well as other 5-HT receptor subtypes) in schizophrenia comes mainly from our understanding of drug pharmacology at these receptors; this pharmacocentric approach becomes more complex in light of recent evidence that

all three 5-HT_2 subtypes demonstrate multifunctional signaling (37–40); that is, the same drug may be an agonist, partial agonist, inverse agonist, or antagonist at the same receptor, depending on the intracellular milieu.

Pharmacotherapy of Schizophrenia and Related Psychoses

The most widely used class of drugs in the treatment of psychotic disorders is the so-called neuroleptics. This term suggests that such medicines “take hold” (*lepsis*) of the central nervous system (CNS) to suppress movement as well as behavior. Although the connotation has been stretched to include biochemical and clinical antagonism of dopamine D_2 receptors, debilitating extrapyramidal movement side effects are implicit in the clinical definition of neuroleptic antipsychotic drugs. Indeed, the term “neuroleptic” is so synonymous with neurologic side effects that newer antipsychotic drugs, without substantial risk of extrapyramidal effects, are referred to as atypical neuroleptic drugs. Also implied in the term “atypical” is a mechanism of antipsychotic action other than (or in addition to) postsynaptic D_2 receptor blockade. There is currently a trend by researchers and clinicians to refer to typical neuroleptics, with relatively potent dopamine D_2 receptor antagonist activity and high propensity for extrapyramidal side effects, as “first generation,” whereas, atypical antipsychotics, which have relatively less affinity

for D_2 receptors and less incidence of movement disorder side effects, are called “second generation.”

In general, pharmacotherapy with typical first-generation neuroleptics benefits patients with schizophrenia or other psychiatric illnesses marked by agitation, aggressive and impulsive behavior, and impaired reasoning. Positive symptoms respond better to treatment with typical neuroleptics, whereas negative symptoms are not appreciably affected. In general, the first-generation neuroleptics provide calming, mood-stabilizing, and antihallucinatory effects, and their beneficial impact on psychiatric medicine is unquestioned despite their sometimes severe extrapyramidal side effects. Chemical classes of typical neuroleptics include the phenothiazines, thioxanthenes, and butyrophenones. Second-generation antipsychotic drug classes include the dibenzodiazepines and benzisoxazoles, which have less potential for extrapyramidal side effects and have significant activity at brain serotonin 5-HT₂-type, adrenergic α_1/α_2 , muscarinic, and/or histamine H_1 receptors, in addition to some activity at dopamine receptors.

First-Generation (Typical) Antipsychotic Drugs

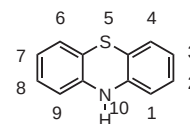
MECHANISM OF ACTION Given that the pathogenesis of schizophrenia and related psychiatric disorders is unknown, it is perhaps naïve to suggest how drugs act at the molecular level to relieve the symptoms of these disorders. Nevertheless, it generally is agreed that the antipsychotic mechanism of action of neuroleptics includes modulation of dopamine neurotransmission in the mesolimbic–mesocortical pathways. This is achieved via direct interaction with D_2 -type receptors, at which they have very high affinity ($K_i < 5$ nmol/L) and functional activity spanning the spectrum of inverse agonism, antagonism, and partial agonism. It is noted that phenothiazine neuroleptics (e.g., chlorpromazine) also have appreciable affinity ($K_i < 100$ nmol/L) at human cloned dopamine D_1 -type, serotonin 5-HT_{2A}, 5-HT_{2C}, 5-HT₆, 5-HT₇, adrenergic α_1 -type and α_2 -type, cholinergic M_{1-5} , and histamine H_1 GPCRs (15) that may account for clinical antipsychotic efficacy and certainly account for their numerous side effects (see later discussion). Likewise, the butyrophenone-type antipsychotic drug haloperidol binds with very high affinity ($K_i < 10$ nmol/L) at dopamine D_2 -type, high affinity ($K_i < 20$ nmol/L) at adrenergic α_1 -type, and moderate affinity ($K_i < 100$ nmol/L) at dopamine D_1 and serotonin 5-HT_{2A} (but not 5-HT_{2C}) GPCRs (15).

SIDE EFFECTS Many of the side effects associated with typical antipsychotic agents can be attributed to their interaction with the GPCRs mentioned earlier in both the CNS and periphery. For example, antipsychotic drug side effects such as sedation, hypotension, sexual dysfunction, and other autonomic effects reflect blockade of histamine H_1 and adrenergic α_1/α_2 receptors. Antimuscarinic (M_{1-5}) actions of neuroleptics account for cardiac, ophthalmic, gastrointestinal, and genitourinary side effects. Such untoward antimuscarinic actions also are characteristic of atypical antipsychotics such as clozapine. On the other hand, it has been proposed that anticholinergic activity may be beneficial in

controlling negative symptoms in patients with schizophrenia. The parkinsonian-like movement side effects of neuroleptics result from antagonism of dopamine D_2 receptors in the nigrostriatal pathway, and the severity of these extrapyramidal side effects increases with the ratio of their antidopaminergic to anticholinergic potency. Extrapyramidal side effects occur in 30% to 50% of patients receiving standard doses of typical neuroleptics and tend to occur during the first to eighth week of therapy. Extrapyramidal side effects include acute dystonias (e.g., facial grimacing, torticollis, and oculogyric crisis), akathisia (motor restlessness), and parkinsonian-type symptoms, such as bradykinesia, cogwheel rigidity, tremor, masked face, and shuffling gait. The higher the D_2 potency of the neuroleptic is, the worse the side effects, some of which can be reversed using anticholinergic drugs. Tardive dyskinesia occurs in 15% to 25% of patients after prolonged treatment with typical neuroleptics and is characterized by stereotyped, involuntary, repetitive, choreiform movements of the face, eyelids, mouth (grimaces), tongue, extremities, and trunk. There also are metabolic and endocrine side effects of neuroleptics, such as weight gain, hyperprolactinemia, and gynecomastia. Relatively common dermatologic reactions (e.g., urticaria and photosensitivity) also are observed especially with the phenothiazines. Interestingly, anticholinergic and dopaminergic agents worsen tardive dyskinesia, whereas antidopaminergic agents tend to suppress the symptoms. The pathophysiology of tardive dyskinesia is not known, and the disorder essentially is irreversible.

Meanwhile, antagonism of dopamine D_2 -type receptors in the chemoreceptor trigger zone in the brainstem is responsible for beneficial antiemetic effects produced by neuroleptics. Several phenothiazines (e.g., promethazine and prochlorperazine) are marketed to exploit this pharmacologic effect.

Development of Phenothiazine and Thioxanthene Neuroleptics



Phenothiazine

Although the phenothiazine nucleus was synthesized in 1883, and although it was used as an anthelmintic for many years, it has no antipsychotic activity. The basic structural type from which the phenothiazine antipsychotic drugs trace their origins is the antihistamines of the benzodioxane type I (Fig. 14.4). In 1937, Bovet (41) hypothesized that specific substances antagonizing histamine should exist, tried various compounds known to act on the autonomic nervous system, and was the first to recognize antihistaminic activity. With the benzodioxanes as a starting point, many molecular modifications were carried out in various laboratories in a search for other types of antihistamines. The benzodioxanes led to ethers of ethanolamine of type II, which after further modifications led to the benzhydryl

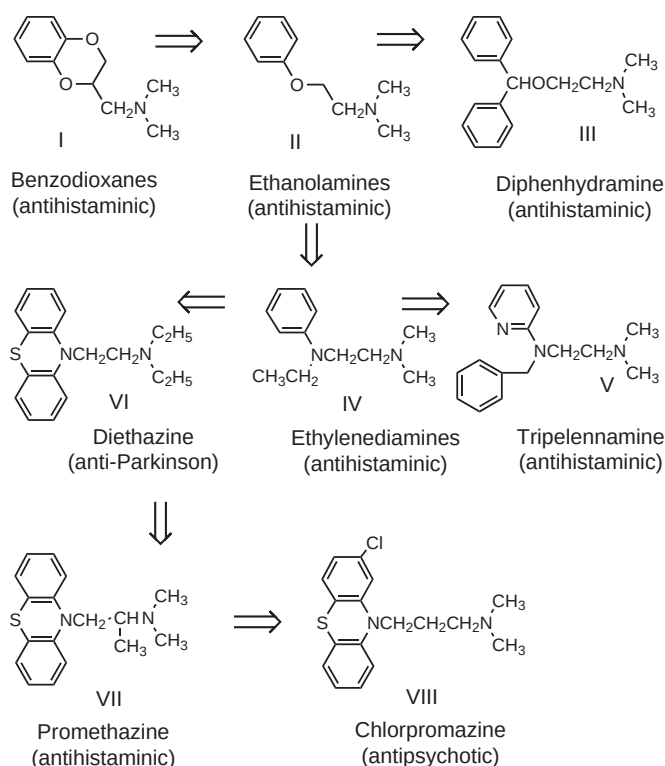


FIGURE 14.4 Development of phenothiazine-type antipsychotic drugs.

ethers (type III), which are characterized by the clinically useful antihistamine diphenhydramine, or to ethylenediamine (type IV), which led to antihistamine drugs, such as tripeleennamine (type V). Further modification of the ethylenediamine type of antihistamine resulted in the incorporation of one of the nitrogen atoms into a phenothiazine ring system, which produced phenothiazine (type VI), a compound that was found to have antihistaminic properties and, similar to many other antihistaminic drugs, a strong sedative effect. Diethazine (type VI) is more useful in the treatment of Parkinson disease (because of its potent antimuscarinic action) than in allergies, whereas promethazine (type VII) is clinically used as an antihistaminic. After the ability of promethazine to prolong barbiturate-induced sleep in rodents was discovered, the drug was introduced into clinical anesthesia as a potentiating agent.

To enhance the sedative effects of such phenothiazines, Charpentier and Courvoisier synthesized and evaluated many modifications of promethazine. This research effort eventually led to the synthesis of chlorpromazine (type VIII) in 1950 at the Rhône-Poulenc Laboratories (42). Soon thereafter, the French surgeon Laborit and his coworkers described the ability of this compound to potentiate anesthetics and produce artificial hibernation (43). They noted that chlorpromazine, by itself, did not cause a loss of consciousness but did produce only a tendency to sleep and a marked disinterest in the surroundings. The first attempts to treat mental illness with chlorpromazine alone were made in Paris in 1951 and early 1952 by Paire and Sigwald. In 1952, Delay and Deniker began their important work with chlorpromazine (44). They were convinced that

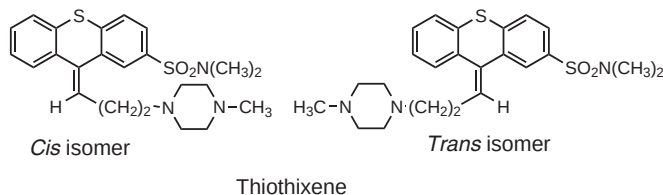
chlorpromazine achieved more than symptomatic relief of agitation or anxiety and that this drug had an ameliorative effect on psychosis. Thus, what initially involved minor molecular modifications of an antihistamine that produced sedative side effects resulted in the development of a major class of drugs that initiated a new era in the drug therapy for the mentally ill. More than anything else in the history of psychiatry, the phenothiazines and related drugs have positively influenced the lives of schizophrenic patients, enabling them to assume a greatly improved role in society.

More than 24 phenothiazine and the related thioxanthene derivatives are used in medicine, most of them for psychiatric conditions. The structures, generic and trade names, doses, and effects of phenothiazine-type and thioxanthene-type neuroleptics currently in use are listed in Table 14.1.

STRUCTURE-ACTIVITY RELATIONSHIPS It is presumed that phenothiazine and thioxanthene neuroleptics mediate their pharmacologic effects mainly through interactions at D_2 -type dopamine receptors. Examination of the x-ray structures of dopamine (in the preferred *trans*- α -rotamer conformation) and chlorpromazine shows that these two structures can be partly superimposed (Fig. 14.5) (45). In the preferred conformation of chlorpromazine, its side chain tilts away from the midline toward the chlorine-substituted ring.

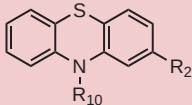
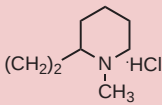
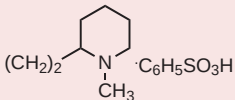
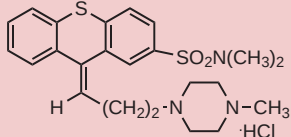
The electronegative chlorine atom on ring “a” is responsible for imparting asymmetry to this molecule, and the attraction of the amine side chain (protonated at physiologic pH) toward the ring containing the chlorine atom indicates an important structural feature of such molecules. Phenothiazine and related compounds lacking a chlorine atom in this position are, in most cases, inactive as neuroleptic drugs. In addition to the ring “a” substituent, another major requirement for therapeutic efficacy of phenothiazines is that the side-chain amine contain three carbons separating the two nitrogen atoms (Fig. 14.5). Phenothiazines with two carbon atoms separating the two nitrogen atoms lack antipsychotic efficacy. Compounds such as promethazine (Fig. 14.4, VII) are primarily antihistaminic and are less likely to assume the preferred conformation.

When thioxanthene derivatives that contain an olefinic double bond between the tricyclic ring and the side chain are examined, it can be seen that such structures can exist in either the *cis* or *trans* isomeric configuration. For example, the *cis* isomer of the neuroleptic thiothixene (marketing was recently discontinued) is several-fold more active than both the *trans* isomer and the compound obtained from saturation of the double bond.



Structure D in Figure 14.5 shows that the active structure of dopamine does not superimpose with a *trans*-like

TABLE 14.1 Phenothiazine and the Thioxanthene Derivatives Used as Neuroleptics*

Phenothiazine-Type Generic Name	Trade Name	 R ₁₀	R ₂	Adult Antipsychotic Oral Dose Range (mg/d)	Side Effects**			
					Sedative Effects	Extra- pyramidal Effects	Hypotensive Effects	Other Effects
Chlorpromazine hydrochloride	Thorazine	(CH ₂) ₃ N(CH ₃) ₂ ·HCl	Cl	300–800	+++	++	Oral ++ IM +++	Antiemetic dose 10–25 mg every 4–6 h
Thioridazine hydrochloride	Mellaril		SCH ₃	200–600	+++	+	++	
Mesoridazine mesylate	Serentil			75–300	+++	+	++	
Perphenazine	Trilafon	(CH ₂) ₃ –N–CH ₂ CH ₂ OH	Cl	8–32	++	+++	+	
Prochlorperazine edisyate, maleate	Compazine	(CH ₂) ₃ –N–CH ₃	Cl	75–100	++	+++	+	Antiemetic dose 5–10 mg every 4–6 h
Fluphenazine hydrochloride	Permitil, Prolixin	(CH ₂) ₃ –N–CH ₂ CH ₂ OH ·2HCl	CF ₃	1–20	+	+++	+	
Trifluoperazine hydrochloride	Stelazine	(CH ₂) ₃ –N–CH ₃ ·2HCl	CF ₃	6–20	+	+++	+	
Thiethylperazine maleate	Torecan	(CH ₂) ₃ –N–CH ₃	SCH ₂ CH ₃		+	+	+	Antiemetic dose 10–30 mg daily
Thioxanthene-Type								
Thiothixene hydrochloride	Navane			6–30	++	++	++	

*The phenothiazine derivatives that are effective in the treatment of nausea and vomiting are included in this listing. **+++ , high; ++ , medium; + , low. IM, intramuscular.

conformer of chlorpromazine that would be predicted to be inactive.

LONG-ACTING NEUROLEPTICS The duration of action of many of the neuroleptics with a free hydroxyl moiety (OH, as in perphenazine, fluphenazine, and acetophenazine; Table 14.1) can be considerably prolonged by the preparation of long-chain fatty acid esters (Table 14.2). Thus, fluphenazine decanoate and fluphenazine enanthate were the first of these esters to appear in clinical use and are longer acting, with fewer side effects, than the unesterified precursor. The ability to treat patients with a single intramuscular injection every 1 to 2 weeks with the enanthate or every 2

to 3 weeks with the decanoate ester means that problems associated with patient compliance to the drug regimens and with drug malabsorption can be reduced. Table 14.2 lists long-acting forms of phenothiazine-type and thioxanthene-type neuroleptics that are derivatives available in the United States and other countries. Long-acting forms of other typical (haloperidol) and atypical antipsychotics are given in the corresponding sections below.

METABOLISM Increasing evidence suggests that the metabolism of neuroleptic drugs is of major significance in the effects of these drugs. Although considerable information about the metabolism of the extensively studied

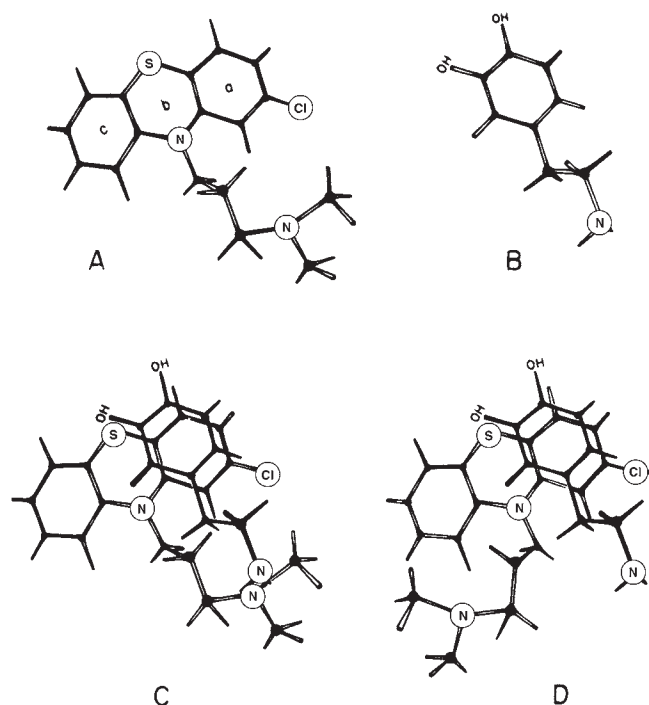


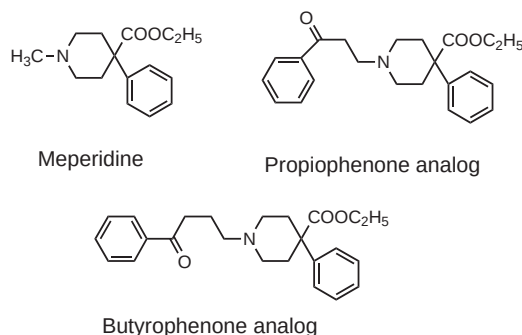
FIGURE 14.5 Conformations of chlorpromazine (A), dopamine (B), and their superposition (C) as determined by x-ray crystallographic analysis. The a, b, and c in (A) designate rings. Also shown (D) is another conformation in which the alkyl side chain of chlorpromazine is in the *trans* conformation (ring a and amino side chain), which is not superimposable onto dopamine. (Adapted from Horn AS, Snyder SH. Chlorpromazine and dopamine: conformational similarities that correlate with the antischizophrenic activity of phenothiazine drugs. *Proc Natl Acad Sci USA* 1971;68:2325–2328, with permission.)

chlorpromazine is available, information about many of the other drugs administered for prolonged periods is scant. Generally, however, the liver microsomal cytochrome P450–catalyzed metabolic pathways for neuroleptics are similar to those for many other drugs. Some metabolic pathways for chlorpromazine are shown in Figure 14.6. It should be kept in mind that during metabolism, several processes can and do occur for the same molecule. For example, chlorpromazine can be demethylated, sulfoxidized, hydroxylated, and glucuronidated to yield 7-*O*-glu-nor-CPZ-SO. The combination of such processes leads to more than 100 identified metabolites. Evidence indicates that the 7-hydroxylated derivatives and, possibly, other hydroxylated derivatives, as well as the mono- and disubstituted products (nor₁-CPZ, nor₂-CPZ), are active in vivo and at dopamine D₂ receptors, whereas the sulfide (CPZ-SO) is inactive. Although the thioxanthenes are closely related to the phenothiazines in their pharmacologic effects, there seems to be at least one major difference in metabolism: Most of the thioxanthenes do not form ring-hydroxylated derivatives. Metabolic pathways for phenothiazines and thioxanthenes are significantly altered, both quantitatively and qualitatively, by a number

of factors, including species, age, gender, interaction with other drugs, and route of administration.

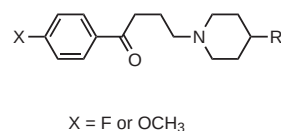
Development of Butyrophenone Neuroleptics

In the late 1950s, Janssen and coworkers synthesized the propiophenone and butyrophenone analogs of meperidine in an effort to increase its analgesic potency (46). Both the propiophenone and butyrophenone analogs had greater analgesic potency than meperidine, but, only the butyrophenone analog also displayed activity resembling that of chlorpromazine.



The structure–activity results of Janssen and coworkers showed that it was possible to eliminate the analgesic activity and, simultaneously, to enhance the chlorpromazine-like neuroleptic activity in the butyrophenone series.

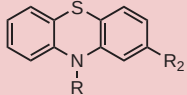
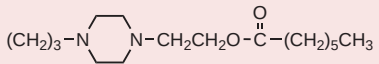
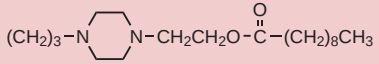
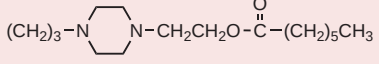
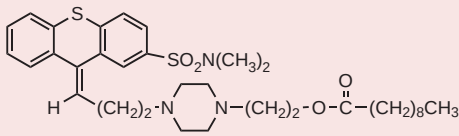
STRUCTURE–ACTIVITY RELATIONSHIPS All butyrophenone derivatives displaying high neuroleptic potency have the following general structure:



The attachment of a tertiary amino group to the fourth carbon of the butyrophenone skeleton is essential for neuroleptic activity; lengthening, shortening, or branching of the three-carbon propyl chain decreases neuroleptic potency. Replacement of the keto moiety (e.g., with the thioketone group as in the butyrothienones, with olefinic or phenoxy groups, or reduction of the carbonyl group) decreases neuroleptic potency. In addition, most potent butyrophenone compounds have a fluorine substituent in the para position of the benzene ring. Variations are possible in the tertiary amino group without loss of neuroleptic potency; for example, the basic nitrogen usually is incorporated into a six-membered ring (piperidine, tetrahydropyridine, or piperazine) that is substituted in the para position.

In most respects, the pharmacologic effects of butyrophenones differ in degree, but not in kind, from those of the piperazine phenothiazines. Consistent with its higher affinity for dopamine D₂ receptors (15), haloperidol produces a higher incidence of extrapyramidal reactions than chlorpromazine, but sedative effects in moderate

TABLE 14.2 Long-Acting Neuroleptics for Intramuscular Depot Injection

Phenothiazines Generic Name			Dosage Range (mg)	Typical Duration of Action (wk)
	R	R ₂		
Fluphenazine enanthate		CF ₃	25–100	112
Fluphenazine decanoate		CF ₃	25–200	2–3
Perphenazine enanthate		Cl	25–100	1–2
Thioxanthene				
Flupenthixol decanoate			100–200	1–2

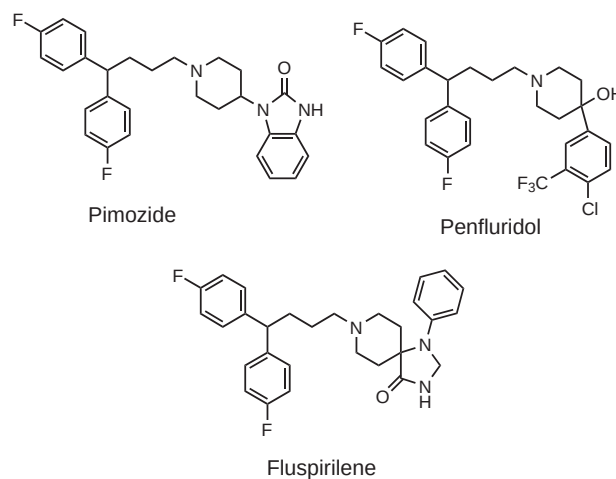
Adapted from Simpson and Lee (48) and Baldessarini and Tarsy (9,10).

doses are less than those observed with phenothiazines (47). Butyrophenones have less prominent autonomic effects than the phenothiazine-type antipsychotic drugs, and only mild hypotension occurs. Also, haloperidol has less propensity to induce weight gain than chlorpromazine and most of the second-generation antipsychotic drugs (see later discussion).

Haloperidol was introduced for the treatment of psychoses in Europe in 1958 and in the United States in 1967 (48). It is an effective alternative to more familiar antipsychotic phenothiazine drugs and also is used for the manic phase of bipolar (manic-depressive) disorder. Haloperidol decanoate has been introduced as depot maintenance therapy. When injected every 4 to 6 weeks, the drug appears to be as effective as daily orally administered haloperidol. Other currently available (mostly in Europe) butyrophenones include the very potent spiperone (spiroperidol) as well as trifluoperidol and droperidol (Fig. 14.7). Droperidol, a short-acting, sedating butyrophenone, is used in anesthesia for its sedating and antiemetic effects and, sometimes, in psychiatric emergencies as a sedative-neuroleptic. Droperidol often is administered in combination with the potent narcotic analgesic fentanyl for preanesthetic sedation and anesthesia.

Modification of the haloperidol butyrophenone side chain by replacement of the keto function with a di-4-fluorophenylmethane moiety results in diphenylbutyl piperidine neuroleptics, such as pimozide, penfluridol, and fluspirilene. The diphenylbutyl piperidine neuroleptics have a longer duration of action than the

butyrophenone analogs. All are effective in the control of schizophrenia, and pimozide in particular has been shown to be useful in treating acute exacerbation of schizophrenia and in reducing the rate of relapse in chronic schizophrenic patients (49).



Pimozide (Orap) also is used for treatment of Tourette syndrome (49), a movement disorder that is characterized by facial tics, grimaces, strange and uncontrollable sounds, and sometimes, involuntary shouting of obscenities. This disorder may be misdiagnosed by clinicians as schizophrenia. Typically, the onset of Tourette syndrome occurs at age 10, and standard treatment for Tourette syndrome in the past has been the neuroleptics, such as

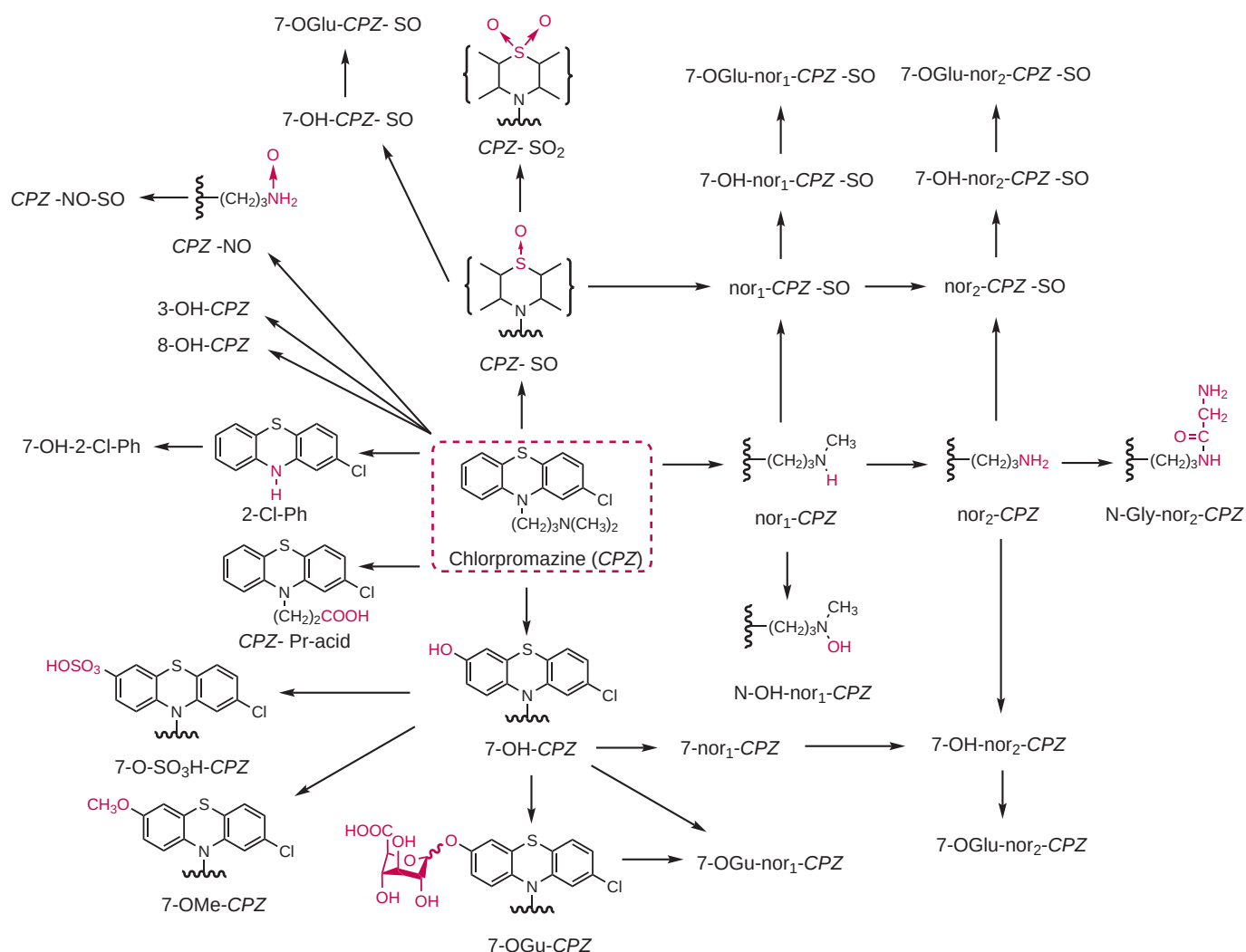


FIGURE 14.6 Metabolism of chlorpromazine. CPZ, chlorpromazine; NO, *N*-oxide; O-Glu, *O*-glucuronide; O-SO₃H, sulfate; Ph, phenothiazine; Pr-acid, propionic acid; SO, sulfoxide; SO₂, sulfone.

haloperidol. Chronic treatment of Tourette syndrome with haloperidol as well as with pimozide carries the risk of producing tardive dyskinesia. Penfluridol and fluspirilene, although not currently available in the United States, are other examples of long-acting neuroleptics in this structure class.

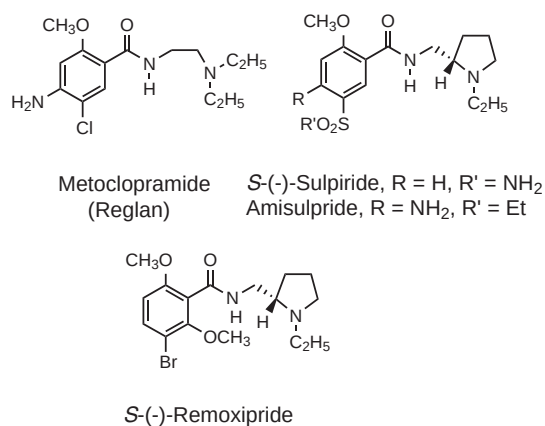
METABOLISM Haloperidol is readily absorbed from the gastrointestinal tract. Peak plasma levels occur 2 to 6 hours after ingestion. The drug is concentrated in the liver and CNS. Approximately 15% of a given dose is excreted in the bile, and approximately 40% is eliminated through the kidney. Figure 14.8 shows the typical oxidative metabolic pathway of butyrophenones as exemplified by haloperidol (50,51).

Benzamide Derivatives with Neuroleptic Activity

Similar to the phenothiazine antiemetics (e.g., promethazine), certain benzamide derivatives such as

metoclopramide antagonize dopamine D₂-type receptors in the chemoreceptor trigger zone of the brainstem and have neuroleptic activity (52,53). Metoclopramide is an efficacious antiemetic drug that modifies gastric motility, but it is not clinically effective for psychoses. Its antagonism of D₂ receptors in myenteric motor neurons increases acetylcholine release to result in a prokinetic effect on the gastrointestinal tract (54). Metoclopramide also binds to serotonin 5-HT₃ and 5-HT₄ receptors with low affinity (15), and these systems also are proposed to account for its effects to lower esophageal sphincter tone and stimulate antral and small intestine contractions, making it useful to treat gastroesophageal reflux disease and to promote gastric emptying (54).

Several analogs of metoclopramide in which the side chain is incorporated into a pyrrolidine ring include *S*(-)-sulpiride and *S*(-)-remoxipride. Both drugs display neuroleptic properties.



Sulpiride produces a relatively low incidence of extrapyramidal side effects, putatively because of a preferential effect on limbic versus extrapyramidal (striatum) tissue. The hydrophilic properties of sulpiride may account for its poor oral absorption, limited penetration into the CNS, and resulting low potency. The racemic para-amino congener of sulpiride, amisulpride, is used as an antipsychotic agent outside the United States. Remoxipride was a promising neuroleptic that is comparable to haloperidol in potency and efficacy and has less incidence of extrapyramidal and autonomic side effects. Life-threatening aplastic anemia, however, was reported with remoxipride use, which prompted its withdrawal from the market.

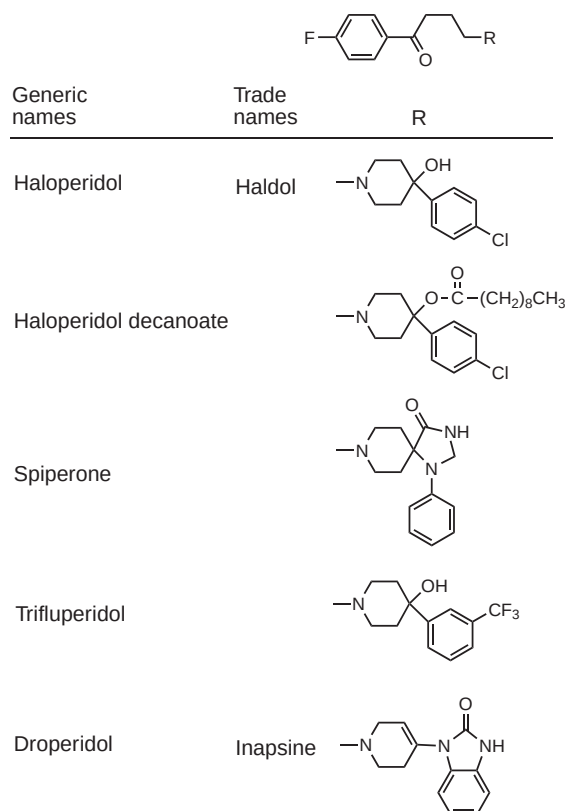


FIGURE 14.7 Haloperidol and its analogs.

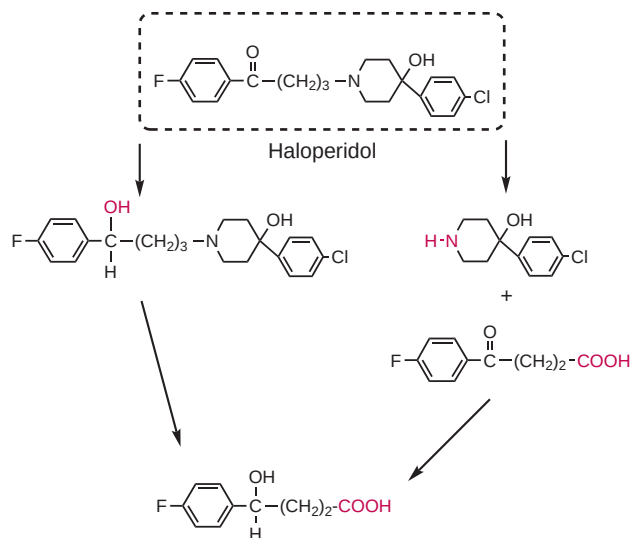


FIGURE 14.8 Metabolism of haloperidol.

Second-Generation (Atypical) Antipsychotic Drugs

The second generation of so-called atypical antipsychotic drugs emerged in 1990 with the introduction of the benzazepine derivative, clozapine, which was initially indicated for schizophrenia resistant to treatment by typical antipsychotics such as the phenothiazines and butyrophenones. This was followed by approval of the first benzisoxazole and benzisothiazole derivatives such as risperidone (1994) and ziprasidone (2001). The arylpiperazine quinolinone derivative, aripiprazole (introduced in 2002), occasionally is referred to as a “third-generation” antipsychotic drug because its mechanism of action is hypothesized to involve GPCR functional selectivity; however, its GPCR binding profile is characteristic of other second-generation antipsychotics, and it is classified as such here. It is reiterated that atypical second-generation antipsychotics interact with several aminergic neurotransmitter GPCRs, have a mechanism of antipsychotic action other than (or in addition to) postsynaptic dopamine D₂ receptor blockade, and produce less incidence of extrapyramidal movement disorder side effects.

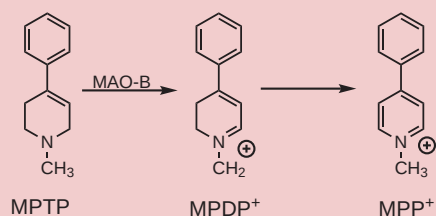
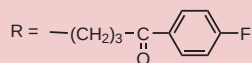
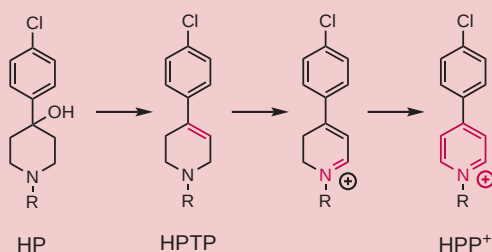
Mechanism of Action for Antipsychotic Efficacy and Side Effects

Currently, it is generally accepted that the efficacy of second-generation (atypical) antipsychotic drugs involves their actions at both dopamine D₂-type and serotonin 5-HT_{2A} receptors. It is naïve with current understanding to definitively classify such actions as inverse agonism, agonism, partial agonism, or neutral antagonism given that ligands interacting at dopamine and serotonin GPCRs are functionally selective (21–25) (i.e., drug functional activity depends on the neurophysiologic neuroanatomic circumstances). It is likely that activity of atypical antipsychotics at other serotonin GPCRs (5-HT_{1A}, 5-HT₆, and/or 5-HT₇), adrenergic (α₁ and/or α₂) GPCRs, acetylcholine

NEUROTOXICOLOGY OF HALOPERIDOL

As discussed in the section on side effects of neuroleptics, untoward extrapyramidal effects occur in 30% to 50% of patients receiving standard doses of typical neuroleptics. Extrapyramidal side effects include acute dystonias, akathisia, and parkinsonian-type symptoms, such as bradykinesia, cogwheel rigidity, tremor, masked face, and shuffling gait. Tardive dyskinesia, a severe extrapyramidal side effect that occurs in 15% to 25% of patients after prolonged neuroleptic treatment, is characterized by stereotyped, involuntary, repetitive, choreiform movements of the face, eyelids, mouth, tongue, extremities, and trunk. The pathophysiology of tardive dyskinesia is not known, and the disorder essentially is irreversible.

The severe and sometimes irreversible dyskinesias associated with haloperidol pharmacotherapy are hypothesized to result from neurotoxicity involving dopaminergic systems. In fact, the central and peripheral biochemical and resulting pathophysiologic features of haloperidol-induced movement disorders are similar to the potent dopaminergic neurotoxicant and parkinsonian-producing agent, 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) (see Chapter 13).



Microsomal-catalyzed dehydration of haloperidol yields the corresponding haloperidol 1,2,3,6-tetrahydropyridine derivative (HPTP) that is a close analog of the parkinsonian-inducing neurotoxicant MPTP. Long-term (58-week) administration of HPTP to nonhuman primates alters both presynaptic and postsynaptic dopaminergic neuronal function, which may contribute to the parkinsonian-type side effects and tardive dyskinesia that are relatively common with haloperidol pharmacotherapy (55). In baboons treated chronically with HPTP, animals developed orofacial dyskinesia, and histopathologic studies revealed volume loss in the basal forebrain and hypothalamus, along with other neuronal cell loss that may be relevant to the pathophysiology of tardive dyskinesia (56). In humans and baboons, HPTP is oxidized *in vivo* to the corresponding haloperidol pyridinium species (HPP⁺) similar to the oxidation of MPTP to its ultimate neurotoxic species MPP⁺. HPP⁺ is neurotoxic to dopaminergic and, especially, serotonergic neurons *in vivo* in rats (57), and HPP⁺ has been identified in the urine of humans treated with haloperidol (58). Furthermore, in a recent study involving psychiatric patients who were treated chronically with haloperidol, the severity of tardive dyskinesia and parkinsonism was associated with an increased serum concentration ratio of HPP⁺ to haloperidol (59), providing compelling clinical evidence for the neurotoxicity of HPP⁺. Investigations continue to determine if neuroleptic-induced pathology of the extrapyramidal motor system, such as that associated with tardive dyskinesia, may be related to production of MPP⁺/HPP⁺-type species in humans.

See Chapter 13 for a related discussion of Parkinson disease caused by MPTP.

muscarinic (M_1 , M_5) GPCRs, and/or histamine H_1 GPCRs also accounts for their psychotherapeutic effects, especially modulation of “negative” signs/symptoms (8,9,60). Certainly, antipsychotic drug action at these other neurotransmitter systems in the brain and periphery can contribute to untoward side effects (e.g., sedative, cardiovascular, gastrointestinal, ophthalmic, sexual, urinary).

The high 5-HT_{2A} receptor affinity of second-generation antipsychotics may account for their low propensity

to cause extrapyramidal side effects, but reduced affinity for D₂ receptors likely plays a role. Nevertheless, antagonism of presynaptic 5-HT_{2A} receptors that inhibit dopamine release from striatal dopaminergic nerve terminals could increase dopaminergic neurotransmission in the striatum to modulate postsynaptic D₂ blockade and reduce extrapyramidal symptoms.

Weight gain and general metabolic dysregulation that leads to type 2 diabetes and cardiovascular disease is problematic for both first-generation (typical) and

second-generation (atypical) antipsychotic drugs; however, this issue previously was obscured for the first-generation drugs due to concern about their much more serious extrapyramidal side effects. The mechanism for antipsychotic-induced weight gain and metabolic syndrome is not known for certain, but likely involves serotonin 5-HT_{2C} and/or histamine H₁ GPCR antagonism or inverse agonism (31,32,36,61,62). Meta-analyses consistently show that clozapine and olanzapine produce the most weight gain in the most patients, whereas quetiapine, risperidone, and chlorpromazine produce intermediate effects, and haloperidol, ziprasidone, and aripiprazole cause smaller changes in weight.

With regard to antipsychotic efficacy between the first-generation (typical) and the second-generation (atypical) drugs, a recent meta-analysis indicates there is no evidence for differences (63–65). Furthermore, there is no evidence for superiority of one second-generation versus another second-generation antipsychotic drug with regard to treatment of positive or negative symptoms of schizophrenia (66–70). Differences between first-generation and second-generation antipsychotic drugs, as well as differences among second-generation antipsychotic drugs, are only in regard to their side effect profile (8–10,60–71).

Benzazepine and Related Analogs

The structures of the benzazepine derivatives, clozapine, olanzapine, loxapine, amoxapine, and quetiapine, and the related dibenzo-oxepine asenapine, are shown in Figure 14.9.

SPECIFIC DRUGS

Clozapine The dibenzazepine clozapine is the prototype of second-generation atypical antipsychotic drugs that have minimal extrapyramidal side effects and do not produce tardive dyskinesia with long-term use. Clozapine also appears to effectively alleviate the negative symptoms of schizophrenia and has proven to be beneficial in treating patients who do not respond adequately to typical first-generation neuroleptic agents, such as the phenothiazines or butyrophenones. A serious drawback to the use of clozapine, however, is the potentially fatal agranulocytosis that is reported to occur in 1% to 2% of patients; decreased white blood cell count is not common with other second-generation antipsychotics, including drugs from the benzazepine (e.g., olanzapine, quetiapine) and benzisoxazole (e.g., risperidone) chemical classes (66–70,72).

The GPCR profile for clozapine is very well characterized with respect to binding (15) but not with respect to function. According to the literature, most researchers and clinicians seem to assume that clozapine is an antagonist at various GPCRs (8–10,66–70); however, the functional selectivity hypothesis (18–21) suggests this view may be too simplistic. Moreover, neurotransmitter receptor systems for which clozapine has no apparent affinity, such as glutamate *N*-methyl-D-aspartate (NMDA)

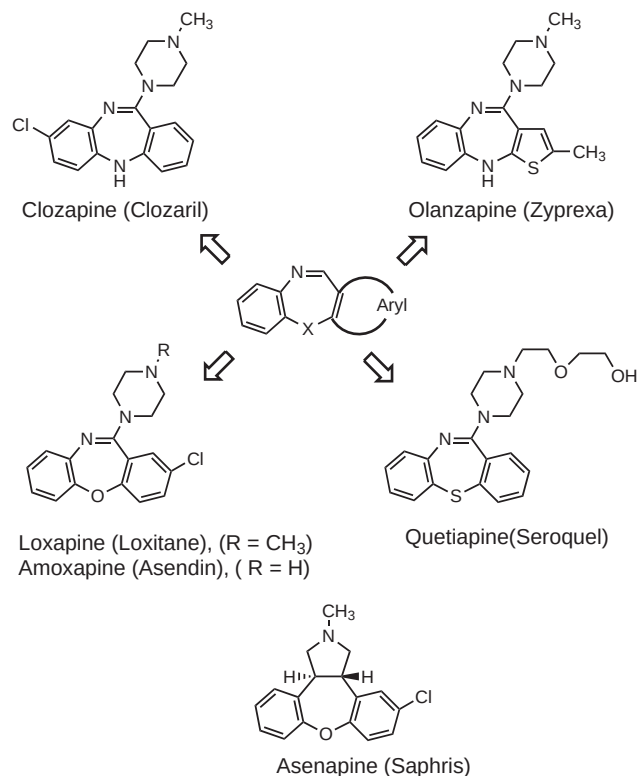


FIGURE 14.9 Benzazepine derivatives.

receptors (15,73), may be relevant to its antipsychotic mechanism of action. Thus, receptor binding profiles can offer only a clue to the mechanism of action for predicting efficacy, especially, in the absence of functional selectivity data. Although receptor binding profiles may be useful to help predict side effects, they fall short to advance understanding of pharmacologic and physiologic mechanisms involved. For example, myriad studies link the histamine H₁ (especially) and serotonin 5-HT_{2C} affinity of clozapine to its notorious weight gain side effect; however, functional outcomes of clozapine interaction with these sites relative to its ability to cause weight gain remain unknown (74,75).

Clozapine has moderate affinity (K_i ~250 to 500 nmol/L) at human dopamine D₁, D₂, and D₃, but moderate affinity (K_i ~50 nmol/L) at D₄ and D₅ GPCRs. It also has moderate affinity (K_i ~25 to 100 nmol/L) at cholinergic muscarinic M₃, M₄, and M₅ GPCRs but high affinity (K_i ~10 nmol/L) at M₁ and M₂ GPCRs. High affinity (K_i ≤10 nmol/L) also is apparent at serotonin 5-HT_{2A}, 5-HT_{2B}, 5-HT_{2C}, 5-HT₆, and 5-HT₇ GPCRs (moderate affinity at 5-HT_{1A}). Clozapine has very high affinity (K_i ≤5 nmol/L) at histamine H₁ and adrenergic α₁-type GPCRs (moderate affinity at α₂). The incidence of clozapine-induced weight gain is the highest among second-generation antipsychotic drugs except olanzapine, and clozapine also produces a higher incidence of sedation and seizures (61,66–70).

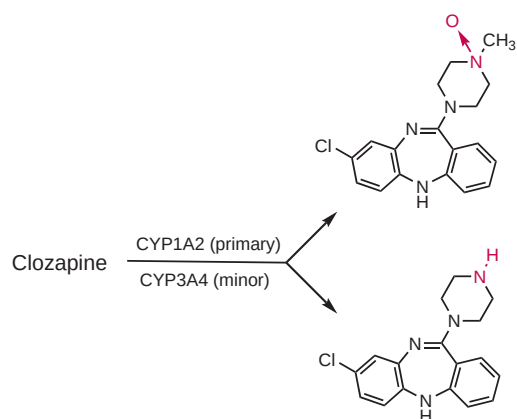


FIGURE 14.10 Metabolism of clozapine.

Clozapine is orally active and metabolized mainly by CYP1A2 with an important contribution by CYP3A4, although there is a high degree of between-subject and within-subject variability (Fig. 14.10) (76,77). The main products of metabolism are inactive hydroxyl and *N*-oxide derivatives, and the half-life is approximately 12 hours. *N*-Desmethylozapine is an active metabolite with a receptor binding profile generally similar to clozapine (15), although with higher affinity at 5-HT_{1A} GPCRs. It is noted that caffeine is metabolized primarily by CYP1A2, and it has been observed that some patients who consume caffeine-containing beverages while taking clozapine show signs of increased arousal and extrapyramidal symptoms, and removal of caffeine results in resolution of these problems (78), suggesting a clinically relevant drug interaction. Also, cigarette smoking induces activity of CYP1A2, and patients who smoke while taking clozapine may have significantly lower serum levels of clozapine (79,80).

Olanzapine The thienobenzodiazepine olanzapine (Fig. 14.9) shares close chemical structural resemblance to clozapine and has a similar receptor binding profile, except it has higher affinity at dopamine D₂ and serotonin 5-HT_{2A} receptors (15). The side effect profile is similar to clozapine, with the exception of agranulocytosis that is not usually seen with olanzapine use. Olanzapine is well absorbed after oral administration and is metabolized mainly by CYP1A2 to inactive metabolites similar to those seen with clozapine (*N*-oxide and *N*-demethylation) along with methyl oxidation and phase 2 glucuronidation (77,81) (Fig. 14.11). The drug had a variable half-life of approximately 20 to 50 hours. There is a long-acting injectable formulation of olanzapine pamoate complex available.

Loxapine The dibenzoxazepine loxapine (Fig. 14.9) has equally high affinity for dopamine D₂-type, serotonin 5-HT₂-type, and histamine H₁ GPCRs, but relatively low affinity at muscarinic GPCRs (15). It undergoes phase 1 aromatic hydroxylation to yield several phenolic

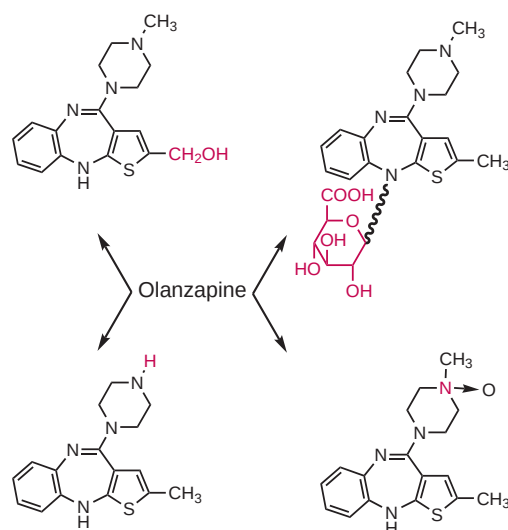


FIGURE 14.11 Metabolism of olanzapine.

metabolites that have even higher affinity for D₂-type receptors than the parent. Loxapine also undergoes *N*-desmethylation to form amoxapine (Fig. 14.9), which is used to treat depression, psychotic depression, and depression accompanied by anxiety or agitation. As with loxapine, aromatic hydroxylation of amoxapine produces metabolites that have D₂ antagonist activity similar to haloperidol. Unlike loxapine, amoxapine has moderate affinity (*K_i* ~25 nmol/L) for norepinephrine and serotonin neurotransmitters and blocks reuptake of these neurotransmitters, a correlate of antidepressant activity. Other benzazepine derivatives, as well as other antipsychotic drugs in general, have no appreciable affinity for norepinephrine, serotonin, or dopamine neurotransmitters (15).

Quetiapine Quetiapine (Fig. 14.9) is a dibenzothiazepine with a receptor binding profile similar to that of clozapine, binding with high affinity at histamine H₁ GPCRs and with moderate affinity at dopamine D₂/D₃ (but not D₄), serotonergic 5-HT_{2A} (but not 5-HT_{2C}), adrenergic α₁ and α₂, and muscarinic M₁ (but not M₂ to M₅) GPCRs. Thus, the quetiapine GPCR affinity profile differs from clozapine mainly regarding its selective affinity at M₁ versus other muscarinic receptors and its selective affinity for 5-HT_{2A} versus 5-HT_{2C} receptors. Quetiapine is 100% bioavailable, but first-pass metabolism yields at least 20 metabolites, with the major metabolites shown in Figure 14.12. The major products are the sulfoxide (catalyzed by CYP3A4) and the carboxylic acid (76,82). It does not appear that CYP2D6 or CYP1A2 is involved in quetiapine metabolism, and cigarette smoking does not affect the pharmacokinetics of this drug (77). The half-life of quetiapine is approximately 6 hours.

Asenapine The receptor binding profile of asenapine (Fig. 14.9), as reported by its manufacturers

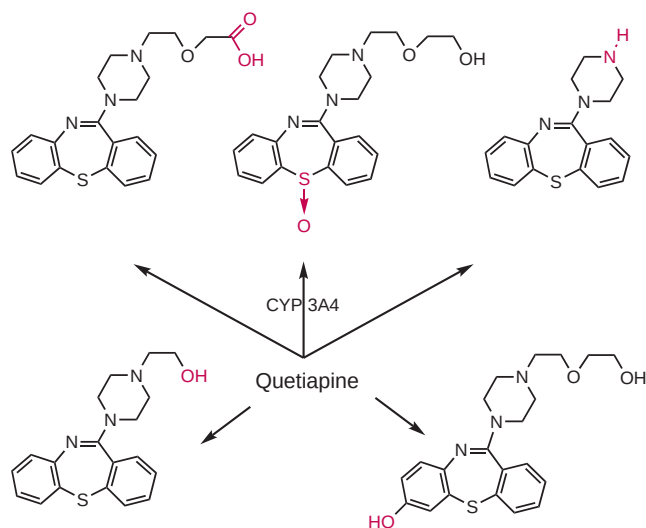


FIGURE 14.12 Major metabolic products of quetiapine.

(83), indicates this benzazepine analog significantly differs from clozapine and the other benzazepines in that it has very high affinity ($K_i < 1$ nmol/L) for all of the serotonin 5-HT GPCRs associated with antipsychotic activity (i.e., 5-HT_{1A}, 5-HT_{2A}, 5-HT_{2C}, 5-HT₆, and 5-HT₇). It is reported that asenapine antagonizes the ability of serotonin to activate these receptors (83), but functional selectivity is not reported. Asenapine also antagonizes the ability of dopamine, norepinephrine, and histamine to activate D₂-type, α_2 -type, and H₁ receptors, respectively, without regard to functional selectivity (83). Also in contrast to other benzazepines, asenapine has essentially no affinity ($K_i > 10$ μ mol/L) for cholinergic muscarinic (M₁ to M₅) GPCRs. Despite some uniqueness regarding its GPCR binding profile, there does not appear to be any efficacy advantages for asenapine over other antipsychotic drugs (first and second generation) for treatment of schizophrenia or bipolar disorder (84,85). Moreover, its side effect profile is similar to other first- and second-generation antipsychotics, except it (expectedly) produces a lower incidence of extrapyramidal effects compared to haloperidol and weight gain is less than with olanzapine (84,85). When given as sublingual tablet, bioavailability is about 35% (<2%, orally) with a half-life of 24 hours. The drug is extensively metabolized to compounds with little or no contributing activity. Direct conjugation by UGT1A4 appears to be a major route of metabolism, giving asenapine-*N*-glucuronide along with *N*-demethylation and aromatic oxidation (primarily by CYP1A2) followed by conjugation (83,84,86) (Fig. 14.13). Approximately 50% of the administered dose is lost via the kidney, and 40% is lost via the fecal route.

Benzisoxazole and Benzisothiazole Derivatives

Neuroanatomic and neurophysiologic interactions between dopaminergic and serotonergic systems,

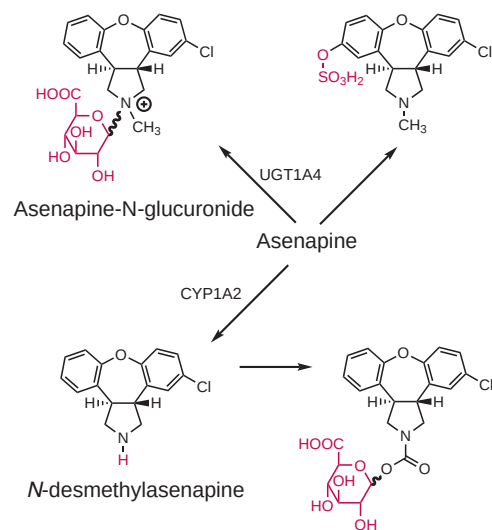


FIGURE 14.13 Major metabolic plasma products for asenapine.

together with evidence that several benzazepine-type antipsychotic agents have high affinity for 5-HT_{2A} receptors, led to the proposal that a combination of D₂/5-HT_{2A} receptor antagonism is required for efficacy in schizophrenia (87,88). Combining the chemical features present in potent benzamide D₂ antagonists (e.g., remoxipride) with those of the benzothiazolyl piperazine 5-HT_{2A} antagonists (e.g., tirosiprone) led to the development of the 3-(4-piperidinyl)-1,2-benzisoxazole nucleus present in the 5-HT_{2A}/D₂ antagonists ($K_i < 5$ nmol/L) risperidone and ziprasidone (Fig. 14.14), which also have high affinity ($K_i \sim 5$ nmol/L) at histamine H₁ and adrenergic α_1 receptors.

SPECIFIC DRUGS

Risperidone and Paliperidone Risperidone has very high affinity ($K_i \sim 0.5$ nmol/L) for serotonin 5-HT_{2A} and high affinity ($K_i \sim 5$ nmol/L) for dopamine D₂ GPCRs (15). It is proposed that the 5-HT_{2A} antagonist activity of risperidone uninhibits dopaminergic neurotransmission in the striatum and cortex, reducing the severity of D₂ antagonist-induced extrapyramidal side effects and alleviating negative symptoms of schizophrenia, while maintaining a blockade of limbic system D₂ receptors (89). In practice, risperidone and its active metabolite marketed as paliperidone (Fig. 14.14) demonstrate a higher incidence of extrapyramidal side effects compared to other second-generation antipsychotics, and they are associated with risk for development of tardive dyskinesia (90,91). Perhaps even more unusual for a second-generation antipsychotic than high D₂-type (including D₃ and D₄) affinity is the essentially zero affinity of risperidone for all five muscarinic GPCRs (15). Other aspects of its GPCR binding profile are unremarkable, including high affinity ($K_i < 5$ nmol/L) for histamine H₁ and adrenergic α_1 -type (moderate affinity at α_2 -type) receptors. Risperidone is one of the few antipsychotic drugs with very high affinity

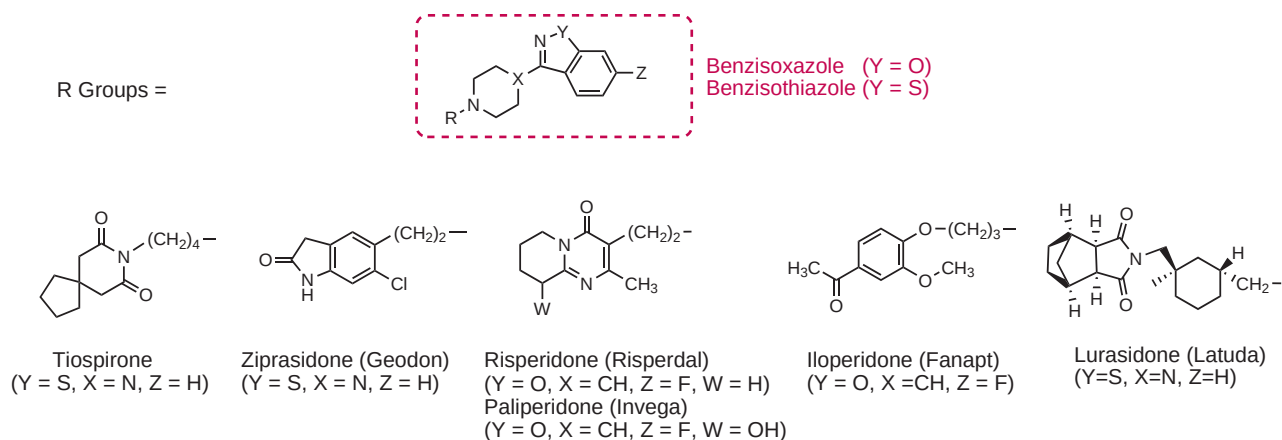


FIGURE 14.14 Benzisoxazole and benzisothiazole antipsychotic agents.

for 5-HT_{2A} receptors that does not also bind tightly to 5-HT_{2C} ($K_i \sim 100$ nmol/L) receptors; nevertheless, it produces significant weight gain on par with most other antipsychotics (61).

Risperidone is well absorbed orally and undergoes hepatic CYP2D6- and CYP3A4-catalyzed *N*-dealkylation and 9-hydroxylation; the racemic version of the 9-hydroxy metabolite is marketed separately for schizophrenia as paliperidone (Fig. 14.14). Paliperidone is not extensively metabolized by the liver and is excreted largely unchanged through the kidney. The half-life of risperidone and its 9-hydroxy metabolite (paliperidone) is approximately 22 hours. Patients differ regarding proportions of the (+)- and (–)-enantiomers of paliperidone produced in vivo, with the (+)-enantiomer arising from hydroxylation of risperidone by CYP2D6 and the (–)-enantiomer being formed mainly by action of CYP3A4 (92). There does not appear to be much difference between risperidone and paliperidone regarding pharmacotherapeutic activity or side effect profile (93). Most clinical trials for both drugs were conducted versus placebo, but the limited data available for comparison to other antipsychotics do not show superiority of these agents to other second-generation antipsychotics (94,95). Both drugs are available as long-acting injectable forms—risperidone in a physical complex with carbohydrate microspheres and paliperidone as the palmitate fatty acid ester.

Ziprasidone Ziprasidone is chemically similar to risperidone but with a substitution of piperziny and benzisothiazole for piperidiny and benzisoxazole and with minor aromatic modification (Fig. 14.14). Like risperidone, ziprasidone is a high-affinity antagonist at 5-HT_{2A}, 5-HT_{2C}, and D₂ receptors as well as at adrenergic α_1/α_2 and histamine H₁ receptors. Moreover, ziprasidone can activate 5-HT_{1A} receptors (96) that regulate dopaminergic neurotransmission in brain regions involved in critical cognitive functions, and this activity is thought to be important regarding antipsychotic efficacy (97). It also is suggested that ziprasidone 5-HT_{1A} partial agonist activity

contributes to its relatively low incidence of extrapyramidal side effects. Interestingly, ziprasidone has relatively low propensity for inducing weight gain (61) despite its significant affinity at histamine H₁ and serotonin 5-HT_{2C} receptors, which are most often associated with this side effect (31,32,36,61,62). Ziprasidone (half-life, 6 hours) has an oral bioavailability of approximately 60%, which can be enhanced in the presence of fatty foods. It is extensively metabolized (<5% excreted unchanged) by aldehyde oxidase, which results in reductive cleavage of the S–N bond, and then by *S*-methylation. Ziprasidone also can undergo CYP3A4-catalyzed *N*-dealkylation and *S*-oxidation (Fig. 14.15) (98).

Lurasidone Limited data from the manufacturer indicate that lurasidone (Fig. 14.14) is similar to risperidone regarding mainly high affinity ($K_i \sim 2$ nmol/L) for serotonin 5-HT_{2A} and dopamine D₂ GPCRs; under the reported assay conditions, lurasidone is an antagonist at these receptors and a partial agonist at 5-HT_{1A} receptors ($K_i \sim 7$ nmol/L) (99). Lurasidone also is a very high affinity ($K_i \sim 0.5$ nmol/L) antagonist at the 5-HT₇ GPCR. There is no significant activity ($K_i > 10$ μ mol/L) at muscarinic and histamine H₁ GPCRs.

Currently, there are no rigorous clinical trial data available comparing lurasidone to other antipsychotic drugs; it was approved for treating schizophrenia based on its superiority to placebo only. The most common adverse events reported by the manufacturer are somnolence, akathisia, nausea, parkinsonism, and agitation, along with weight gain (100). Under similar study conditions, lurasidone-induced weight gain was about one-fourth of the olanzapine-induced weight gain. Metabolism is reported (package insert) as oxidative *N*-dealkylation, hydroxylation of the norbornane ring, and *S*-oxidation, mainly by CYP3A4, with a half-life of 18 hours, suitable for once-daily dosing.

Iloperidone This benzisoxazole is very similar to risperidone regarding both its chemical structure and its

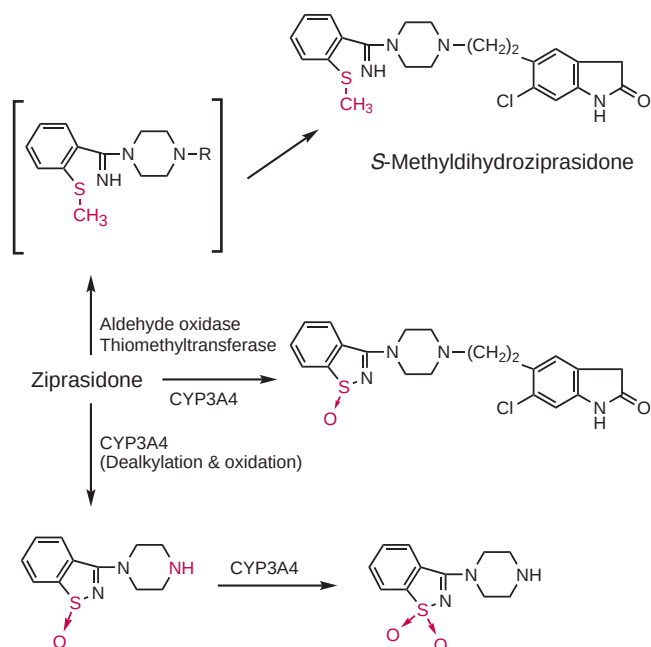


FIGURE 14.15 Metabolism of ziprasidone.

GPCR binding profile (15). Currently, it is not possible to determine with certainty and accuracy how iloperidone may differ from risperidone clinically because clinical trial data currently are available only from industry-sponsored supplemental reports that did not undergo the usual peer-review process (101). In any event, in the few studies where iloperidone was compared to other antipsychotic drugs (haloperidol, risperidone, ziprasidone), there was no difference in antipsychotic efficacy. Not surprisingly, iloperidone produces less severe extrapyramidal side effects than haloperidol, but no comparisons were made with atypical (second-generation) antipsychotics. Iloperidone produces significant weight

gain (>7% in acute studies), about the same as haloperidol and much more than risperidone (2.1 vs. 1.8 kg); interestingly, iloperidone and risperidone have about the same affinity at serotonin 5-HT_{2C} and histamine H₁ receptors, which are thought to be responsible for antipsychotic-induced weight gain. As is the case for risperidone, iloperidone is well absorbed orally and undergoes hepatic CYP2D6- and CYP3A4-catalyzed *N*-dealkylation. A majority of iloperidone is recovered unchanged in feces, indicating biliary excretion. Overall, there are currently no comparison data with other atypical antipsychotic drugs that indicate that iloperidone is more efficacious, produces less side effects, or is less expensive; thus, the clinical relevance of iloperidone is not substantiated.

MISCELLANEOUS CHEMICAL CLASSES OF ANTIPSYCHOTIC DRUGS

Aripiprazole (Abilify) Aripiprazole (Fig. 14.16) is an arylpiperazine quinolinone derivative that has received relatively extensive attention in the clinical and basic science literature, albeit its efficacy is not superior to other second-generation or even first-generation antipsychotic drugs (including for negative symptoms) (63–71), and its GPCR affinity profile is not very different from other second-generation antipsychotic drugs (15). For example, characteristic of other second-generation antipsychotic drugs, aripiprazole has significant affinity ($K_i \sim 50$ nmol/L) at adrenergic α_1 -type and α_2 -type and histamine H₁ GPCRs, although contribution of these molecular targets to antipsychotic efficacy is unknown. Aripiprazole is similar to risperidone in that it lacks appreciable affinity at cholinergic M₁–M₅ muscarinic receptors, which offers advantages regarding the side effect profile.

The affinity of aripiprazole for dopamine D₂/D₃ receptors is very high ($K_i \sim 1$ nmol/L); however, it has a low propensity to cause untoward extrapyramidal

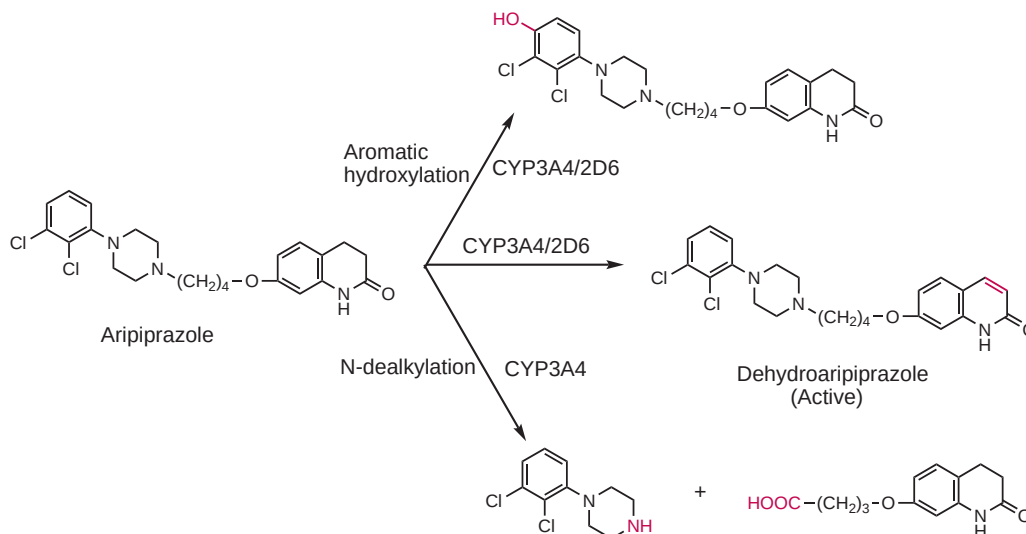


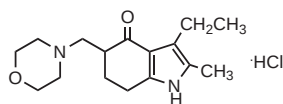
FIGURE 14.16 Structure and metabolism of aripiprazole.

symptoms and hyperprolactinemia (102). Partly, this may be explained by the GPCR functional profile of aripiprazole, which is better characterized than most other second-generation antipsychotic drugs. For example, it is a partial agonist at D_2 -type (including D_3/D_4) receptors (103), whereas D_2 antagonism or inverse agonism produces extrapyramidal and hyperprolactinemia side effects. In fact, the D_2 -type functional activity of aripiprazole depends on the anatomic location and type of cell type that expresses the D_2 -type receptor; that is, aripiprazole is a functionally selective drug (25,103,104). Thus, it is a full agonist that activates brain dopamine D_2 autoreceptors to negatively modulate dopamine synthesis and release (105); according to the dopamine hypothesis, this type of antidopaminergic activity would be beneficial to treat schizophrenia (16,17). Meanwhile, aripiprazole is a partial agonist regarding D_2 -mediated inhibition of cyclic adenosine monophosphate (cAMP) second messenger formation, and characteristic of a partial agonist, aripiprazole antagonizes the ability of dopamine to inhibit cAMP formation (106,107).

Aripiprazole also has significant affinity ($K_i < 10$ nmol/L) for serotonin GPCRs, notably 5-HT_{1A} , 5-HT_{2A} , 5-HT_{2C} , and 5-HT_7 (15), all thought to be important in the pathophysiology and pharmacotherapy of schizophrenia (8,9,60). Aripiprazole is a partial agonist at 5-HT_{2A} and 5-HT_{2C} receptors (103); the 5-HT_{2C} partial agonist properties of aripiprazole may reduce its potential for weight gain because 5-HT_{2C} activation is associated with satiety (32,108).

Aripiprazole has good oral bioavailability (90%) with a half-life of approximately 75 hours. It undergoes hepatic CYP3A4- and CYP2D6-catalyzed *N*-dealkylation and hydroxylation as well as dehydrogenation to dehydroaripiprazole (Fig. 14.16), which is an active metabolite with a half-life of about 90 hours (109).

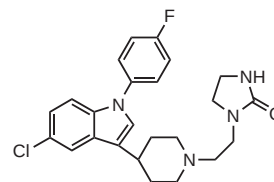
Molindone (Moban)



Molindone hydrochloride, a tetrahydro-indolone derivative, is a neuroleptic agent that is structurally unrelated to any of the other marketed neuroleptics. Molindone has no appreciable affinity at serotonin, cholinergic, adrenergic, or histaminergic GPCRs (15). Even at the dopamine D_2 GPCR, its affinity ($K_i \sim 10$ nmol/L) is less potent than haloperidol; however, it nonetheless can produce extrapyramidal side effects. Metabolism studies in humans show molindone to be rapidly absorbed and metabolized when given orally. There are 36 recognized metabolites, with less than 2% to 3% unmetabolized molindone being excreted in urine and feces. Clinical studies show that the antipsychotic effects of molindone last more than 24

hours, suggesting that one or more metabolites may contribute to its activity in vivo (110).

Sertindole (Serdolect)



Sertindole is an indole-containing compound with high affinity ($K_i < 5$ nmol/L) at serotonin $5\text{-HT}_{2A/2C}$, dopamine $D_{2/3/4}$, and adrenergic α_{1A} GPCRs. Meanwhile, sertindole has essentially no affinity at cholinergic M_1 to M_5 GPCRs, and this is apparent clinically in that it has low propensity to cause anticholinergic side effects (111), which, in part, gives some justification for its reintroduction in 2002 after being withdrawn in 1998. It is about as effective as haloperidol in the treatment of acute and chronic positive symptoms of schizophrenia, but with much lower incidence of extrapyramidal side effects, and it is more effective at treating negative symptoms (105). Also, compared to other antipsychotics, sertindole is relatively nonsedating (112), perhaps, reflecting its low affinity for histamine H_1 receptors (15) as well as muscarinic receptors. Sertindole has a long plasma half-life of approximately 3 days that offers advantages regarding patient compliance (112).

ANXIETY AND ANXIETY DISORDERS

Definitions

Anxiety can be defined as a sense of apprehensive expectation. In reasonable amounts and at appropriate times, anxiety is helpful (e.g., anxiety before an examination may cause a student to initiate an appropriate study plan). Too much anxiety, however, can be deleterious. Anxiety can be considered pathologic when it is either completely inappropriate to the situation or is in excess of what the situation normally should call for. An example of the former is nocturnal panic attacks—episodes of extreme anxiety that arise out of one of the most physiologically quiet times of the day, stage III/IV sleep (113). An example of the latter is specific phobias—for example, an irrational fear to venture outside of one's home.

According to the DSM-IV (1), abnormal anxiety is that level of anxiety that interferes with normal social or occupational functioning. This definition is helpful to distinguish between normal and pathologic levels of anxiety. To meet general DSM-IV criteria, anxiety symptoms must not be caused by an exogenous factor (e.g., caffeine) or a medical condition (e.g., hyperthyroidism). Examples of anxiety disorders include specific phobias, generalized anxiety disorder (chronic abnormally high level of anxiety), social phobia (e.g., fear of public speaking), obsessive-compulsive disorder, panic disorder with or without agoraphobia

(avoidance of situations believed by the patient to precipitate panic attacks), and posttraumatic stress disorder.

Etiology of Anxiety Disorders

Studies of patients with anxiety disorders have not revealed a general gross neuroanatomic lesion. In vivo functional imaging studies, however, show altered blood oxygen levels (as a manifestation of blood flow) or glucose utilization in specific brain areas in patients with anxiety conditions (114,115), including obsessive-compulsive disorder (116,117), panic disorder (118,119), specific phobia (120,121), and posttraumatic stress disorder (122,123), mostly implicating the prefrontal cortex and hippocampus (and other limbic areas) as being involved in the anatomy of pathologic anxiety. It is important to note that there is significant comorbidity for anxiety and major depressive disorders in both children and adults, and it is not clear if one illness has primacy or is part of the other (124,125). Likewise, although there may be genetic predisposition to general distress that can lead to anxiety and/or depression, no clear genetic evidence suggests specific symptoms of either disorder.

A variety of neurotransmitters, neuromodulators (e.g., adenosine), and neuropeptides (e.g., cholecystokinin, corticotropin-releasing factor, and neuropeptide Y) are suggested to be involved in the pathophysiology of anxiety. Currently, abundant evidence exists to document the involvement of the neurotransmitters γ -aminobutyric acid (GABA), norepinephrine, and serotonin in anxiety, and research increasingly is revealing that these neurotransmitter systems have complex anatomic and functional interrelationships.

GABA Receptors

The major inhibitory neurotransmitter in the mammalian CNS, GABA, is widespread, with approximately one-third of all synapses in the CNS using this neurochemical for intercellular communication. The two major classes of GABA receptors (126) are ionotropic GABA_A and metabotropic GABA_B receptors. It is currently recommended by the International Union of Pharmacology to not use the term “GABA_C” receptor to describe a gene product that was described, precloning, as a GABA receptor–like protein with pharmacology distinct from GABA_A and GABA_B receptors (126). It appears that so-called GABA_C receptors may be a GABA_A subtype that expresses a certain ρ subunit (see later discussion), although the overall subunit makeup of this putative GABA_A subtype is not established.

GABA_A Receptor

The GABA_A receptor is a member of the gene superfamily of ligand-gated ion channels that is known as the “cys-loop” family because of the presence of a cysteine loop in their N-terminal domain (126,127). These receptors exist as heteropentameric subunits arranged around a central ion channel (Fig. 14.17). The five polypeptide subunits are composed of an extracellular region, four membrane-spanning α -helical cylinders, and a large

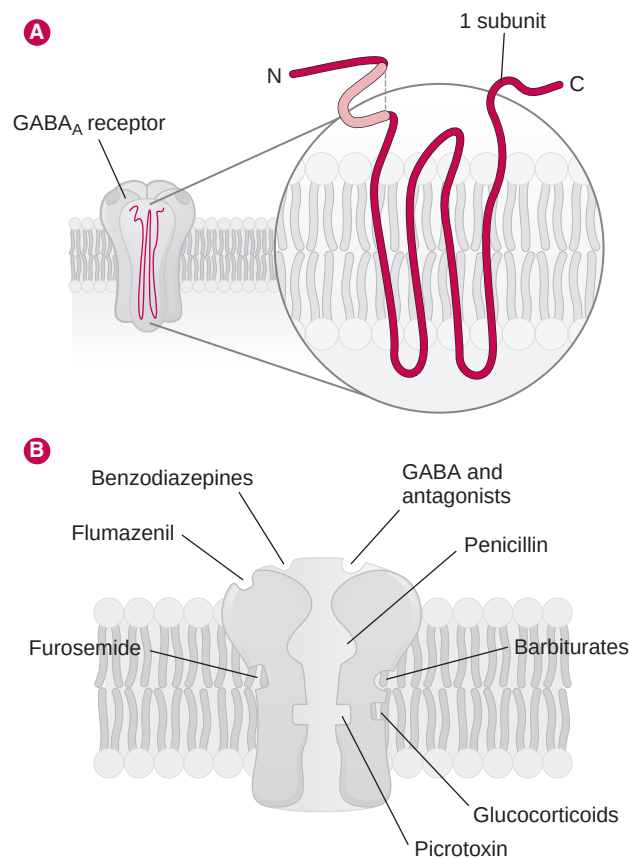


FIGURE 14.17 Schematic representation of the γ -aminobutyric acid (GABA_A) receptor. The GABA_A receptors have a pentameric structure composed predominantly of α , β , and γ subunits arranged, in various proportions, around a central ion channel that conducts chloride. Each subunit has four membrane-spanning regions and a cysteine loop in the extracellular N-terminal domain (dashed line). The type and proportion of α and γ subunit composition affects affinity, pharmacologic activity, and efficacy of ligands. (Chou J, Strichartz GR, Lo EA. Pharmacology of excitatory and inhibitory neurotransmission. In: Golan DE, ed. Principles of Pharmacology. Philadelphia: Lippincott Williams & Wilkins, 2005:142, with permission.)

intracellular cytoplasmic loop. The GABA_A ion channel conducts chloride and is defined by the second of the four membrane-spanning α -helical cylinders. The first GABA_A polypeptide subunit was sequenced in 1987 (128), and so far, 19 different subunits have been isolated. These polypeptides are denoted as α_{1-6} , β_{1-3} , γ_{1-3} , δ , ϵ , π , θ , and ρ_{1-3} . The subunits can combine in varied proportions, and alternatively spliced variants are common. Thus, many possible receptor subtypes may exist. The major (60%) GABA_A receptor isoform in the adult mammalian (rat) brain consists of α_1 , β_2 , and γ_2 subunits (GABA_{A1a}) (129).

The GABA_A extracellular N-terminal region contains a number of distinct binding sites for neuroactive drugs (e.g., barbiturates, benzodiazepines, β -carbolines, and neurosteroids). The benzodiazepines, commonly prescribed as anxiolytic (and sedative) agents, bind to the benzodiazepine receptor (BZR), which is defined mostly

by the α and γ subunits (126). The α and γ subunit composition can dramatically affect affinity and efficacy of BZR ligands (130,131). Early research on the BZR gave rise to the pharmacologic concept of inverse agonism in addition to the better known concepts of agonism and antagonism. Inverse agonist compounds bind to the BZR on the GABA_A receptor complex and negatively modulate GABA binding and neurophysiologic activity (i.e., agonists decrease chloride conductance), producing physiologic effects opposite those of GABA (e.g., anxiogenesis and proconvulsant action). The BZR agonist ligands potentiate GABA binding and activity to increase chloride conductance, enhancing physiologic effects of GABA (e.g., sedation and anticonvulsant activity). The BZR antagonists occupy the receptor but have no intrinsic activity to modulate GABA binding and function. A clinical example of a BZR antagonist is the compound flumazenil, which is used to reverse benzodiazepine-induced sedation in overdose. There also have been developed agents that are partial agonists and inverse partial agonists at the BZR/ GABA_A receptor complex. The existence of a GABA_A receptor complex that recognizes benzodiazepines has implications for our understanding of both normal and pathologic anxiety states and suggests the existence of endogenous GABA_A receptor ligands. Thus, anxiety could conceivably be either a lack of an endogenous GABA_A receptor agonist or a relative excess of a GABA_A receptor antagonist or inverse agonist.

GABA_B Receptors

The GABA_B receptors are GPCRs that exist as two major subtypes, $\text{GABA}_{B(1)}$ and $\text{GABA}_{B(2)}$. The $\text{GABA}_{B(1)}$ subtype can be expressed as $\text{GABA}_{B(1a)}$ and $\text{GABA}_{B(1b)}$ isoforms that differ in their extracellular NH_2 -terminal domains but are derived from the same gene (126). Interestingly, it was discovered early on that compared to native GABA_B receptors, recombinant $\text{GABA}_{B(1a)}$ and $\text{GABA}_{B(1b)}$ receptors expressed in heterologous cells display 100- to 150-fold lower affinity for agonist ligands. Likewise, recombinant $\text{GABA}_{B(1a)}$ and $\text{GABA}_{B(1b)}$ receptors were shown to couple inefficiently to

their effector systems (predominantly via $\text{G}\alpha_i$ and $\text{G}\alpha_o$). These surprising pharmacologic findings were explained by the discovery that recombinant $\text{GABA}_{B(1a)}$ and $\text{GABA}_{B(1b)}$ receptors expressed in heterologous cells are retained in the endoplasmic reticulum. In fact, it turned out that $\text{GABA}_{B(1)}$ receptors do not traffic to the cell membrane surface in the absence of $\text{GABA}_{B(2)}$ receptors. This remarkable discovery that the $\text{GABA}_{B(2)}$ receptor co-expresses on the cell surface with the $\text{GABA}_{B(1a)}$ or $\text{GABA}_{B(1b)}$ receptor to form a functional heterodimeric GPCR was reported simultaneously by three industry research groups in 1998 (130–135). The GABA_B receptors were the first GPCR shown to function not as a single protein but, rather, as two distinct subunits, neither of which is functional by itself (Fig. 14.18). Homodimerization and/or heterodimerization (and oligomerization) now are documented for many GPCRs and may account for the diverse signaling functionality for this protein family.

Drugs Used in the Treatment of Anxiety

Benzodiazepines

The benzodiazepines are efficacious drugs for treatment of anxiety disorders (140), putatively, via their interaction with GABA_A receptors (see Chapter 15 for a discussion of benzodiazepines and their sedative-hypnotic utility). Chlordiazepoxide was the first benzodiazepine to be marketed for clinical use in 1960. Its effectiveness and wide margin of safety were major advances over compounds, such as barbiturates, used previously. A variety of new benzodiazepines followed, each with some minor differences from the competition. The major factors considered when selecting an agent include rate and extent of absorption, presence or absence of active metabolites, and degree of lipophilicity. These factors help to determine how a benzodiazepine is marketed and used; for example, an agent that is rapidly absorbed, highly lipid soluble, and without active metabolites would be useful as a sedative but less useful for treatment of a chronic

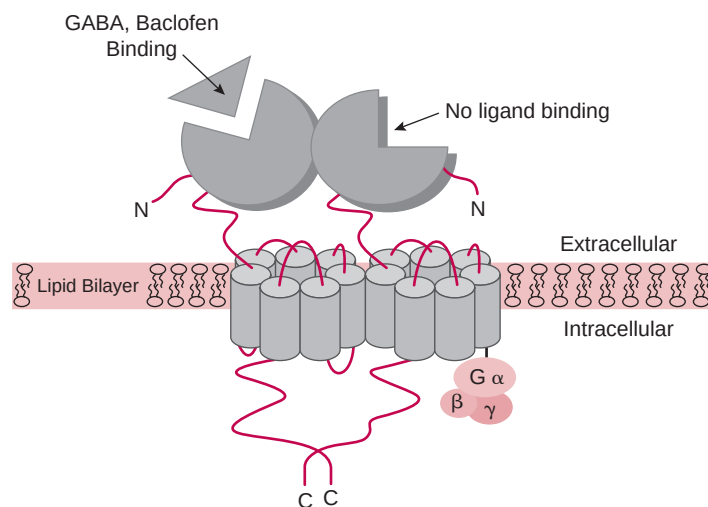
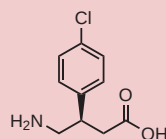


FIGURE 14.18 Schematic representation of the heterodimeric γ -aminobutyric acid (GABA_B) receptor.

GABA_B RECEPTORS

The structure of functional heterodimeric GABA_{B(1)}/GABA_{B(2)} GPCRs (Fig. 14.18) is assumed to be very different and considerably more complex in comparison to other aminergic GPCRs that are able to function as monomers. Functional GABA_B receptors are proposed to contain a binding pocket that consists of two globular lobes separated by a hinge region. Functional GABA_B receptors are proposed to contain a binding pocket that consists of two globular lobes separated by a hinge region. According to the Venus flytrap model, the two lobes close on ligand binding; however, it is thought that ligands bind in only one of the lobes (136–138). The individual GABA_{B(1)} and GABA_{B(2)} subunits can affect a number of other membrane and cytoplasmic proteins to result in complex signaling and a diverse array of pharmacologic and physiologic effects. For example, GABA_B receptors modulate activity of calcium channels, potassium channels, adenylyl cyclase, and phospholipase C via G_{αi}, G_{αo}, and G_{βγ} proteins. Meanwhile, preclinical data suggest that drugs that affect GABA_B receptor function may produce, for example, anxiolytic, anticonvulsant, and antidepressant effects as well as muscle relaxant and analgesic effects (137,138).



R(-)-Baclofen

The only drug currently in clinical use that selectively interacts with GABA_B receptors is baclofen (β-*p*-chlorophenyl-GABA). Baclofen was first synthesized in 1962 and was shown to have potent muscle relaxant and analgesic activity. In 1972, the racemate was marketed to treat spasticity disorders. In 1980, the *R*(-)-enantiomer of baclofen was shown to stereoselectively interact (as an agonist) with what is now known as the GABA_B receptor (139). The role of GABA_B receptors in anxiety, however, is not clear, and baclofen (racemate) is approved only for use as a spasmolytic agent (see Chapter 13).

anxiety state. On the other hand, a compound with slower absorption, active metabolites, and low lipophilicity would be a more effective antianxiety agent but less helpful as a soporific.

DEVELOPMENT OF BENZODIAZEPINE ANXIOLYTICS In the 1950s, the medicinal chemist Sternbach noted that “basic groups frequently impart biological activity,” and in accordance with this observation, he synthesized a series of compounds by treating various chloromethylquinazoline N-oxides with amines to produce what he hoped would be products with “tranquilizer” activity at the New Jersey

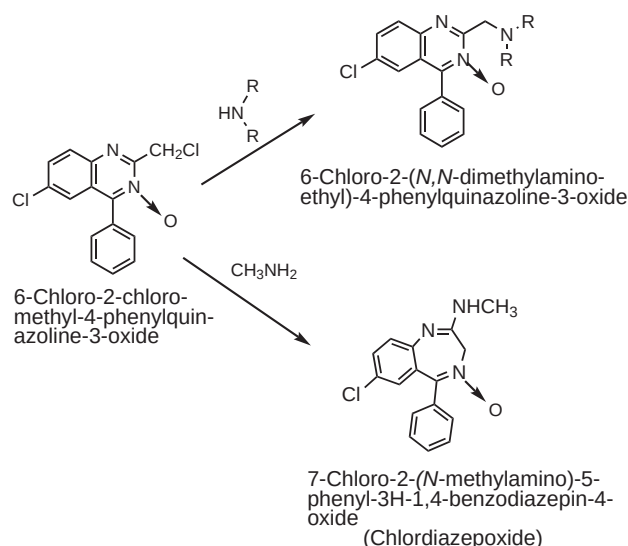
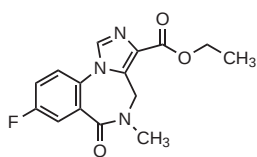


FIGURE 14.19 Synthesis of chlordiazepoxide.

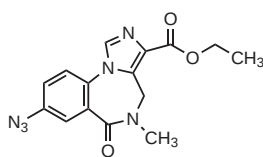
laboratories of Hoffman LaRoche (141,142). Sternbach's studies included the reaction of 6-chloro-2-chloromethyl-4-phenylquinazoline-3-oxide with methylamine, which yielded the unexpected rearrangement product 7-chloro-2-(*N*-methylamino)-5-phenyl-3H-1,4-benzodiazepin-4-oxide (Fig. 14.19). This product was given the code name RO 50690 and screened for pharmacologic activity in 1957. Subsequently, Randall et al. (143,144) reported that RO 50690 was hypnotic and sedative and had anti-strychnine properties similar to the propanediol meprobamate, a sedative that has tranquilizer (anxiolytic) properties only at intoxicating doses. Renamed chlordiazepoxide, RO 50690 was marketed in 1960 as Librium, a safe and effective anxiolytic agent.

Chlordiazepoxide turned out to have rather remarkable pharmacologic properties and tremendous potential as a pharmacotherapeutic product, but it possessed a number of unacceptable physical chemical properties. In an effort to enhance its “pharmaceutical elegance,” structural modifications of chlordiazepoxide were undertaken that eventually led to the synthesis of diazepam in 1959. In contrast to the maxim that basic groups impart biologic activity, diazepam contains no basic nitrogen moiety. Diazepam, however, was found to be 3- to 10-fold more potent than chlordiazepoxide and was marketed in 1963 as the anxiolytic drug Valium. Subsequently, thousands of benzodiazepine derivatives were synthesized, and more than two dozen benzodiazepines are in clinical use in the United States (Fig. 14.20).

A major advance in the BZR field was made in 1981 with the first report that the imidazobenzodiazepinone derivative, flumazenil, binds with high affinity to the BZR and blocks the pharmacologic effects of the classical benzodiazepines *in vitro* and *in vivo* (149).



Flumazenil



RO 15-4513

Unlike agonists, binding of [^3H]flumazenil to the BZR is not affected by modulators such as GABA and several ions that induce changes in receptors (150). The insensitivity of flumazenil to changes in BZR conformation suggests that the ligand does not induce a conformational change in the receptor to trigger a biologic response and is a pure antagonist (148). Such benzodiazepine antagonists have been used to characterize the pharmacologic nature of the BZR, and several of these agents, including

flumazenil, are used to treat benzodiazepine overdose. Other imidazobenzodiazepinone derivatives are not true BZR antagonists but, rather, have inverse agonist activity. For example, RO 15-4513 is reported to be a partial inverse agonist that produces anxiogenic-like effects in rats (146), a pharmacologic activity quite different from a true BZR competitive antagonist, such as flumazenil.

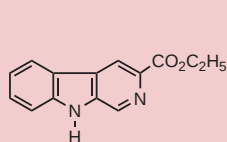
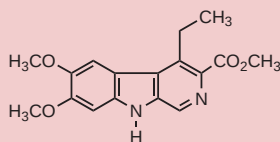
MECHANISM OF ACTION OF ANXIOLYTIC BENZODIAZEPINES The BZR ligands, regardless of intrinsic activity, do not directly alter transmembrane chloride conduction to produce their observed characteristic physiologic anxiolytic or anxiogenic effects. The BZR is an allosteric modulator of GABA binding to the GABA_A receptor complex that, in turn, modulates the transmembrane conductance of chloride. In the presence of BZR agonists or partial agonists, affinity and functional potency of GABA at GABA_A receptors are enhanced maximally or submaximally, respectively, and conductance of chloride is increased. Inverse agonists and partial inverse agonists reduce the effect of GABA, and GABA_A receptor-mediated conductance of chloride is accordingly decreased. The GABA_A receptor-chloride channels thus become either more or less sensitive to GABA in the presence of BZR agonists or inverse agonists, respectively. BZR competitive antagonists block access of agonists to the BZR but have no intrinsic activity to affect GABA-modulated conductance of chloride.

A representation of the relationship between ligand interaction with the BZR and intrinsic activity to modulate GABA_A receptor function is shown in Figure 14.21. The interaction of agonists, competitive antagonists, and inverse agonists with the BZR, as shown in the figure, is a simplistic rendering of the proposed three-state model of the BZR and GABA_A receptor interrelationship (151,152). This model is based on the hypothesis that the BZR and GABA_A receptor exist in three spontaneously oscillating conformational states, functionally described as “active” or agonist, “neutral” or “resting,” and “inactive” or inverse agonist. The BZR agonists and partial agonists bind to and stabilize the “active” state, inducing a conformational change in the GABA_A receptor complex that results in chloride channel opening, which may lead to an anticonvulsant or anxiolytic effect. The BZR inverse agonists and partial inverse agonists bind to and stabilize the “inactive” state, resulting in the chloride channel remaining closed, which may lead to a convulsant or anxiogenic effect. The BZR competitive antagonists presumably bind equally well to both states (hence, they bind to a “neutral” state) and affect no change in GABA_A receptor function or chloride conductance, but access of agonists to the BZR is blocked.

The classical BZR is located on the GABA_A receptor complex mainly at the interface of the α and γ subunits (126) that can be rendered benzodiazepine-insensitive by a point mutation in the α subunit, replacing a critical histidine residue for arginine (153). Different α and γ subunit compositions give rise to subtypes of the BZR receptor that are pharmacologically distinct with regard to ligand affinity and intrinsic activity (154–156), providing a

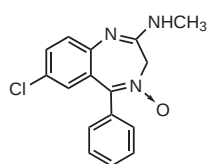
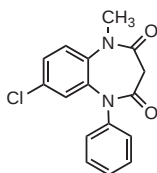
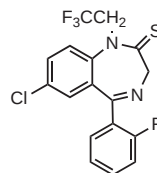
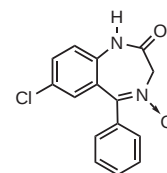
ENDOGENOUS BZR LIGANDS

An endogenous ligand with affinity for the BZR of the GABA_A receptor complex has not been conclusively identified. Several compounds of endogenous origin, however, that inhibit the binding of radiolabeled benzodiazepines to the BZR have been reported. In 1980, Braestrup et al. (145) reported the presence in normal human urine of β -carboline-3-carboxylic acid ethyl ester (β CCE), which has very high affinity for the BZR complex. It was subsequently shown, however, that β CCE formed as an artifact from Braestrup's extraction procedure, during which the urine extract was heated with ethanol at pH 1, a condition favoring formation of the ethyl ester from β -carboline-3-carboxylic acid, a tryptophan metabolite.

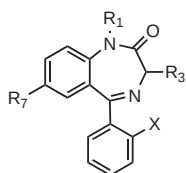
 β CCE

DMCM

Although β CCE actually was shown not to be of endogenous origin, its discovery as a high-affinity BZR ligand stimulated research that led to the synthesis of a series of β -carboline derivatives with a variety of intrinsic activities, presumably mediated through the BZR that is associated with the GABA_A receptor complex. For example, although β CCE is considered to be a partial inverse agonist at this site, 6,7-dimethoxy-4-ethyl- β -carboline-3-carboxylic acid methyl ester (DMCM) appears to be a full inverse agonist (146). In fact, β CCE blocks the convulsions produced by the very potent convulsant DMCM (147). These effects are even more complex and interesting in light of the approximately 10-fold higher affinity that β CCE shows for the BZR labeled by [^3H]diazepam when compared to DMCM (148). The β -carbolines currently are important research tools to probe the agonist, competitive antagonist, inverse agonist, and partial agonist/inverse agonist pharmacophores of the BZR/GABA receptor complex.

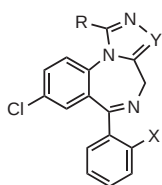
Chlordiazepoxide
(Librium)Clobazam
(Frisium)Quazepam
(Doral)

Demoxepam



Class A Benzodiazepines

Generic name	Trade name	R ₁	R ₃	R ₇	X
Clonazepam	Klonopin	H	H	NO ₂	Cl
Clorazepate	Tranxene	H	COOK	Cl	H
Diazepam	Valium	CH ₃	H	Cl	H
Flurazepam	Dalmane	(CH ₂) ₂ NC ₂ H ₅) ₂	H	Cl	F
Halazepam	Paxipam	CH ₂ CF ₃	H	Cl	H
Lorazepam	Ativan	H	OH	Cl	Cl
Oxazepam	Serax	H	OH	Cl	H
Temazepam	Restoril	CH ₃	OH	Cl	H



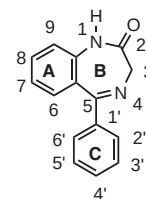
Class B Benzodiazepines

Generic name	Trade name	R	X	Y
Alprazolam	Xanax	CH ₃	H	N
Estazolam	ProSom	H	H	N
Midazolam	Versed	CH ₃	F	CH
Triazolam	Halcion	CH ₃	Cl	N

FIGURE 14.20 Some commercially available benzodiazepines.

mechanistic basis for development of ligands that are anxiolytic (i.e., anxiolysis in the absence of sedation, muscle relaxation, amnesia, and ataxia). Thus, current drug discovery approaches target specific α and γ molecular subunits of the GABA_A receptor complex in the quest for benzodiazepine and nonbenzodiazepine (see later discussion) drugs that demonstrate anxiolytic activity. As a group, currently used benzodiazepines are not α subtype-selective. In studies using nonhuman primates, it has been suggested that GABA_A α_2 , α_3 , and α_5 subunits mediate anxiolytic and muscle relaxant effects of benzodiazepines, whereas α_1 receptors mediate the sedative effects (157). Several putative anxiolytic compounds have reached the clinic; however, they have not exhibited the degree of anxiolytic activity predicted from preclinical testing and, usually, have lower efficacy than standard benzodiazepines (131). Of possible clinical importance, “uncoupling” of the BZR/GABA_A receptor complex has been observed in response to chronic benzodiazepine exposure both in vitro (158) and in vivo (159). In the absence of exogenous influences, however, coupling efficiency appears to be determined by the composition and stoichiometry of the α subunits (156), whereas benzodiazepine affinity, intrinsic activity, and efficacy are determined by the nature of both the α and γ subunits (131,155,156).

STRUCTURE–ACTIVITY RELATIONSHIPS



The structure–activity relationship for classical 5-phenyl-1,4-benzodiazepine-2-one anxiolytic agents has been described by Sternbach and other investigators (141,148,151,160). Thousands of benzodiazepine derivatives with a variety of substituents have been synthesized that interact with the BZR; however, classical quantitative structure–activity relationship and molecular modeling techniques have been used to reduce this myriad of structures to the minimal common molecular features necessary for binding (161–164). The pharmacologic activity continuum (agonist, antagonist, inverse agonist) displayed by BZR ligands would seem to suggest that such diverse functional activity be mediated by ligand interaction with different sites on the GABA_A receptor–chloride channel complex. This continuum of activity, however, is

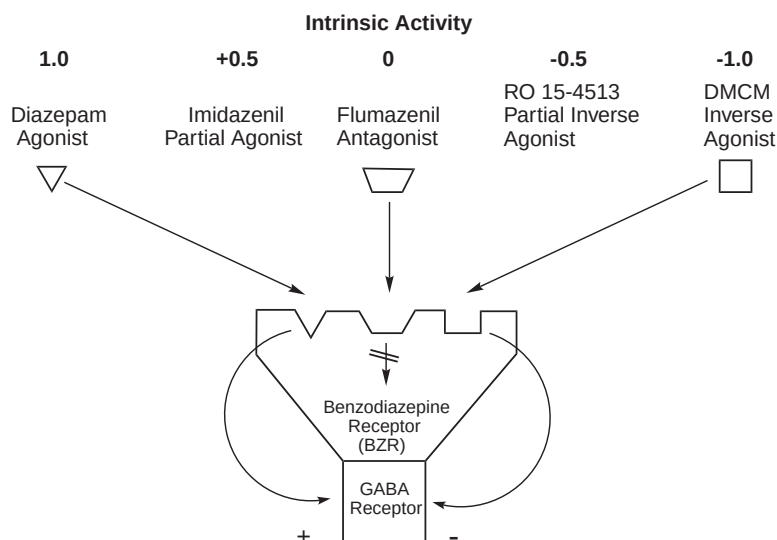


FIGURE 14.21 Ligand interaction with the γ -aminobutyric acid (GABA_A)/benzodiazepine receptor complex.

displayed by ligands within the same chemical class, and small modifications in the chemical structure of a ligand can shift the intrinsic activity from agonist to antagonist to inverse agonist. Moreover, each functional class of BZR ligands can competitively inhibit the binding of the other two classes as well as functionally antagonize each other. These observations suggest that the binding sites of functionally diverse BZR receptor ligands, at least, overlap. Nevertheless, most BZR pharmacophore models that describe ligand functional activity are based initially on the BZR pharmacophore for ligand binding activity at a single binding domain, and this approach is used here to summarize the structure–activity relationship for benzodiazepine derivatives at the BZR receptor.

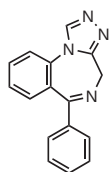
Ring A In general, the minimum requirement for binding of 5-phenyl-1,4-benzodiazepin-2-one derivatives to the BZR includes an aromatic or heteroaromatic ring (ring A), which is believed to participate in π - π stacking with aromatic amino acid residues of the receptor. Substituents on ring A have varied effects on binding of benzodiazepines to the BZR, but such effects are not predictable based on electronic or (within reasonable limits) steric properties. It is generally true, however, that an electronegative group (e.g., halogen or nitro) substituted at the 7-position markedly increases functional anxiolytic activity, although effects on binding affinity in vitro are not as dramatic. On the other hand, substituents at positions 6, 8, or 9 generally decrease anxiolytic activity. Other 1,4-diazepine derivatives in which ring A is replaced by a heterocycle generally show weak binding affinity in vitro and even less pharmacologic activity in vivo when compared to phenyl-substituted analogs.

Ring B A proton-accepting group is believed to be a structural requirement of both benzodiazepine and non-benzodiazepine ligand binding to the GABA_A receptor, putatively for interactions with a histidine residue that

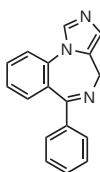
serves as a proton source in the $\text{GABA}_A \alpha_1$ subunit (156). For the benzodiazepines, optimal affinity occurs when the proton-accepting group in the 2-position of ring B (i.e., the carbonyl moiety) is in a coplanar spatial orientation with the aromatic ring A. Substitution of sulfur for oxygen at the 2-position may affect selectivity for binding to GABA_A BZR subpopulations, but anxiolytic activity is maintained. Substitution of the methylene 3-position or the imine nitrogen is sterically unfavorable for antagonist activity but has no effect on agonist (i.e., anxiolytic) activity. Derivatives substituted with a 3-hydroxy moiety have comparable potency to nonhydroxylated analogs and are excreted faster. Esterification of a 3-hydroxy moiety also is possible without loss of potency. Neither the 1-position amide nitrogen nor its substituent is required for in vitro binding to the BZR, and many clinically used analogs are not *N*-alkylated (Fig. 14.20). Although even relatively long *N*-alkyl side chains do not dramatically decrease BZR affinity, sterically large substituents like *tert*-butyl drastically reduce receptor affinity and in vivo activity. Neither the 4,5-double bond nor the 4-position nitrogen (the 4,5-[methyleneimino] group) in ring B is required for in vivo anxiolytic activity, although in vitro BZR affinity is decreased if the $\text{C}=\text{N}$ bond is reduced to $\text{C}-\text{N}$. It is proposed that in vivo activity of such derivatives results from oxidation back to $\text{C}=\text{N}$. It follows that the 4-oxide moiety of chlordiazepoxide can be removed without loss of anxiolytic activity.

Ring C The 5-phenyl ring C is not required for binding to the BZR in vitro. This accessory aromatic ring may contribute favorable hydrophobic or steric interactions to receptor binding, however, and its relationship to ring A planarity may be important. Substitution at the 4'-(*para*)-position of an appended 5-phenyl ring is unfavorable for agonist activity, but 2'-(*ortho*)-substituents are not detrimental to agonist activity, suggesting that limitations at the *para* position are steric, rather than electronic, in nature.

1,2-Annellation



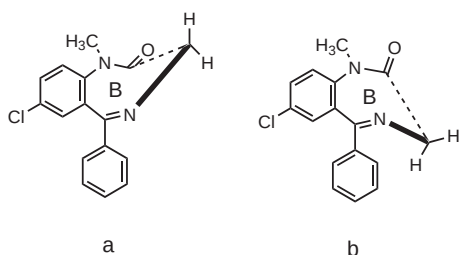
s-Triazolo[4,3a][1,4]benzodiazepine



Imidazo[1,5a][1,4]benzodiazepine

Annulating the 1,2-bond of ring B with an additional “electron-rich” (i.e., proton acceptor) ring, such as s-triazole or imidazole, also results in pharmacologically active benzodiazepine derivatives with high affinity for the BZR (Fig. 14.21). For example, the s-triazolo-benzodiazepines triazolam, alprazolam, and estazolam and the imidazo-benzodiazepine midazolam are clinically effective anxiolytic agents (Fig. 14.20).

STEREOCHEMISTRY



Most clinically useful benzodiazepines do not have a chiral center; however, the seven-membered ring B may adopt one of two possible boat conformations, *a* and *b*, that are “enantiomeric” (mirror images) to each other. Nuclear magnetic resonance studies indicate that the two conformations can easily interconvert at room temperature, making it impossible to predict which conformation is active at the BZR a priori. Evidence for stereospecificity for binding to the BZR was provided by introducing a 3-substituent into the benzodiazepine nucleus to provide a chiral center and enantiomeric pairs of derivatives (148). In vitro BZR binding affinity and in vivo anxiolytic activity of several 3-methylated enantiomers was found to reside in the *S*-isomer. Moreover, the *S*-enantiomer of 3-methyldiazepam was shown to stabilize conformation *a* for ring B, whereas the *R*-enantiomer stabilizes conformation *b*. Also, the 3-*S*-configuration and *a* conformation for ring B are present in both the crystalline state (165) and in solution (166) for 3-methyldiazepam. Despite the enantioselectivity demonstrated for benzodiazepines, the commonly used 3-hydroxylated derivatives (e.g., lorazepam and oxazepam) are commercially available only as racemic mixtures.

PHYSIOCHEMICAL AND PHARMACOKINETIC PROPERTIES The physiochemical and pharmacokinetic properties of the various benzodiazepines vary widely, and these properties

have clinical implications. For example, depending on the nature of substituents, particularly with regard to electronegative substituents, the lipophilicity of the benzodiazepines may vary by more than three orders of magnitude, which affects absorption, distribution, and metabolism. In general, most benzodiazepines have relatively high lipid:water partition coefficients (logP values) and are completely absorbed after oral administration and rapidly distributed to the brain and other highly perfused organs. A notable exception is clorazepate (Fig. 14.20), which is rapidly decarboxylated at the 3-position to *N*-desmethyldiazepam and, subsequently, quickly absorbed. Overall, the clinical and pharmacokinetic properties of clorazepate are similar to chlordiazepoxide and diazepam (see later discussion).

Most benzodiazepines and their metabolites bind to plasma proteins. The degree of protein binding is dependent on lipophilicity of the compound and varies from approximately 70% for more polar benzodiazepines, such as alprazolam, to 99% for very lipophilic derivatives, such as diazepam. Hepatic microsomal oxidation, including *N*-dealkylation and aliphatic hydroxylation by a wide variety of cytochrome P enzymes, accounts for the major metabolic disposition of most benzodiazepines. Subsequent conjugation of microsomal metabolites by glucuronyl transferases yields polar glucuronides that are excreted in urine. In general, the rate and product of benzodiazepine metabolism vary, depending on route of administration and the individual drug.

SPECIFIC DRUGS

Chlordiazepoxide Chlordiazepoxide is well absorbed after oral administration, and peak blood concentration usually is reached in approximately 4 hours. Intramuscular absorption of chlordiazepoxide, however, is slower and erratic. The half-life of chlordiazepoxide is variable but usually quite long (6 to 30 hours). Hepatic metabolism is mainly by CYP3A4 to give the initial *N*-desmethylation product, *N*-desmethylchlordiazepoxide, which undergoes deamination to form the demoxepam (Fig. 14.22), which is extensively metabolized, with less than 1% of a dose of chlordiazepoxide excreted as demoxepam. Demoxepam can undergo four different metabolic fates. Removal of the *N*-oxide moiety yields the active metabolite, *N*-desmethyldiazepam (desoxydemoxepam). This product is a metabolite of both chlordiazepoxide and diazepam and can be hydroxylated to yield oxazepam, another active metabolite that is rapidly glucuronidated and excreted in the urine. Another possibility for metabolism of demoxepam is hydrolysis to the “opened lactam,” which is inactive (Fig. 14.22). The two other metabolites of demoxepam are the products of ring A hydroxylation (9-hydroxydemoxepam) or ring C hydroxylation (4'-hydroxydemoxepam), both of which are inactive. The majority of a dose of chlordiazepoxide is excreted as glucuronide conjugates of oxazepam and other phenolic (9- or 4'-hydroxylated) metabolites. As with diazepam (see next section), repeated administration of

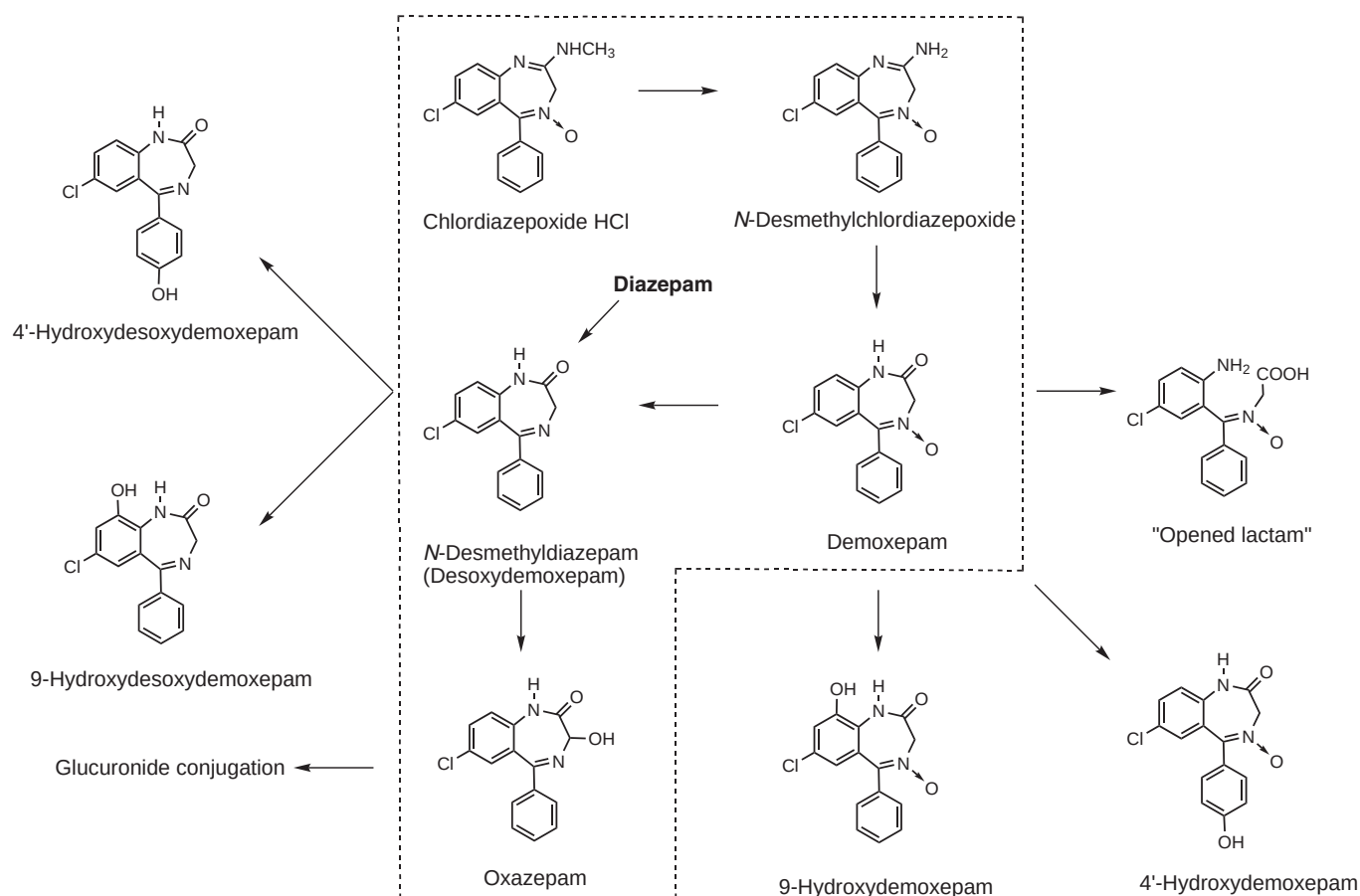


FIGURE 14.22 Metabolism of chlordiazepoxide and related benzodiazepines.

chlordiazepoxide can result in accumulation of parent drug and its active metabolites, which may have important clinical implications, including excessive sedation.

Diazepam Diazepam is rapidly and completely absorbed after oral administration. Maximum peak blood concentration occurs in 2 hours, and elimination is slow, with a half-life of approximately 20 to 50 hours. As with chlordiazepoxide, the major metabolic product of diazepam (by CYP3A4) is *N*-desmethyldiazepam, which is pharmacologically active and undergoes even slower metabolism than its parent compound. Repeated administration of diazepam or chlordiazepoxide leads to accumulation of *N*-desmethyldiazepam, which can be detected in the blood for more than 1 week after discontinuation of the drug. Hydroxylation of *N*-desmethyldiazepam at the 3-position gives the active metabolite oxazepam (Fig. 14.22).

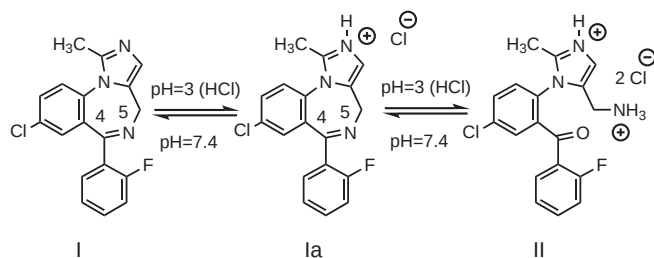
Oxazepam Oxazepam is an active metabolite of both chlordiazepoxide and diazepam and is marketed separately as a short-acting anxiolytic agent. Oxazepam is rapidly inactivated to glucuronidated metabolites that are excreted in the urine (Fig. 14.22). The half-life of oxazepam is approximately 4 to 8 hours, and cumulative effects with chronic therapy are much less than with

long-acting benzodiazepines, such as chlordiazepoxide and diazepam. Lorazepam is the 2'-chloro derivative of oxazepam and has a similarly short half-life (2 to 6 hours) and similar pharmacologic activity.

Flurazepam Flurazepam is administered orally as the dihydrochloride salt. It is rapidly ¹N-dealkylated (primarily by CYP3A4) to give the 2'-fluoro derivative of *N*-desmethyldiazepam, and it subsequently follows the same metabolic pathways as chlordiazepoxide and diazepam (Fig. 14.22) (also see Fig. 15.7 for metabolism). The half-life of flurazepam is fairly long (~7 hours); consequently, it has the same potential as chlordiazepoxide and diazepam to produce cumulative clinical effects and side effects (e.g., excessive sedation) and residual pharmacologic activity, even after discontinuation.

Midazolam The pK_a of midazolam is 6.2 (see Appendix), making it one of the few benzodiazepines that is highly water soluble ($pH < 4$), as well as highly lipid soluble ($pH > 4$). It is the most commonly used benzodiazepine as a premedication for anesthesia, with a quick onset (1 to 2 minutes) and recovery (20 minutes) after a bolus injection (167). The time to full alertness after midazolam premedication, however, is

about 40 minutes, which is much longer than for nonbenzodiazepine agents such as propofol (10 minutes). The intravenously administered preparation is a dihydrochloride salt preparation that is buffered to pH 3; at this pH, acid-catalyzed diazepine ring opening at the 4,5-double bond occurs that assists water solubility but renders the compound inactive. The dihydrochloride salt preparation consists of about 8% to 85% of the ring-opened structure II and 15% to 20% of the ring-closed form Ia (168). The diazepine ring completely reforms to the active midazolam compound (I) upon intravenous injection (i.e., at pH 7.4) (168). Midazolam undergoes hepatic metabolism mainly by CYP3A4 and CYP3A5 to yield primarily hydroxylated derivatives that likely do not contribute to pharmacologic activity at usual doses; these are subsequently excreted as glucuronide conjugates (167).

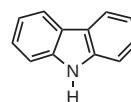


DETAILED PHARMACOKINETIC ANALYSIS Detailed pharmacokinetic analysis for most benzodiazepines is complex. Two-compartment models may be adequate to describe the disposition of most derivatives, but three-compartment models are necessary for highly lipophilic agents, such as diazepam. The distribution of such lipophilic drugs is further complicated by enterohepatic circulation. Thus, the usually stated elimination half-life of benzodiazepines may not adequately account for the pharmacodynamics of the distributive phase of the drug, which can be clinically important. For example, the distributive (α) half-life of diazepam is approximately 1 hour, whereas the elimination (β) half-life is approximately 1.5 days acutely and even longer after chronic dosing that results in accumulation of drug. Furthermore, plasma concentration and clinical effectiveness of benzodiazepines are difficult to correlate, and only a two-fold increase in clinically effective levels produces sedative side effects. Consequently, despite the long half-life of many benzodiazepines, they are not safe or effective when given in one daily dose and usually are divided into two to four doses per day for treatment of daytime anxiety. Both therapeutic and toxic effects may persist several days after discontinuation of chronically administered, long-acting benzodiazepines, such as chlordiazepoxide and diazepam. Thus, short-acting benzodiazepines, such as oxazepam, that are rapidly metabolized to inactive products should be considered in elderly or hepato-compromised patients.

Nonbenzodiazepine Agonists at the Benzodiazepine Receptor

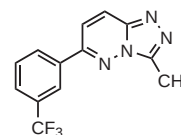
Relatively few structural classes of nonbenzodiazepine compounds have clinically relevant affinity for the BZR and show pharmacologic activity in vivo. Examples of these classes include the β -carbolines, triazolopyridazines, cyclopyrrolones, pyrazolopyrimidines, and imidazopyridines. Representative drugs of these classes in current clinical use are the pyrazolopyrimidine zaleplon, the cyclopyrrolone eszopiclone, and the imidazopyridazine zolpidem (also see Chapter 15). These nonbenzodiazepine BZR ligands show greater selectivity for GABA_A receptors containing the α_1 subunit; however, it should be noted that the α_2 , α_3 , α_5 , and γ subunits may be important in mediating anxiolytic effects of BRZ agonists (131).

β -CARBOLINE



Several β -carbolines have approximately 10-fold higher affinity for the BZR when compared to diazepam. The ethyl ester of β CCE, identified in human urine extracts as an artifact of the extraction procedure, has very high affinity for the BZR. Although β CCE and other β -carbolines are not endogenous BZR ligands (see previous discussion) and are not currently approved for clinical use, they are used to characterize different GABA_A/BZR subtypes (based on α and γ subunit composition) and function (e.g., the partial agonist abecarnil) toward the discovery of anxiolytic drugs. In this regard, the β -carboline ring system is planar in comparison to the boat conformation of the 1,4-benzodiazepines (see the discussion of stereochemistry); thus, β -carbolines have been useful to extend structure-activity relationship information for the agonist, antagonist, and inverse agonist pharmacophores of the various GABA_A/BZR subtypes.

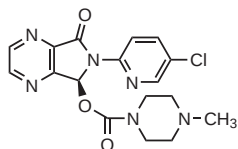
CL 218,872



The triazolopyridazine CL 218,872 is another research tool used to probe BZR heterogeneity, because it is known to have selective high affinity at BZR subtypes containing the α_1 subunit, hypothesized to mediate anxiolytic actions (131,169). CL 218,872 has lower efficacy than benzodiazepines in potentiating GABA-gated chloride currents, but it produces anxiolytic effects in animal models at substantially lower doses than those required

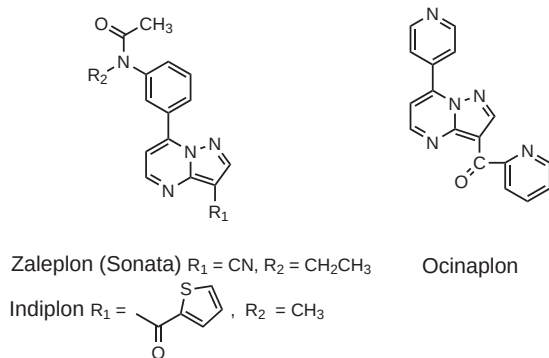
to produce untoward side effects (e.g., sedation, ataxia, and muscle relaxation) (127,169).

ESZOPICLONE (LUNESTA)



The cyclopyrrolone zopiclone is described as a “super-agonist” at BZRs with the subunit composition $\alpha_1 \beta_2 \gamma_2$ and $\alpha_1 \beta_2 \gamma_3$, because it potentiates the GABA-gated current more than the benzodiazepine (flunitrazepam) reference agonist (170). Racemic zopiclone has been available in Europe since 1992, and the higher affinity *S*-enantiomer (eszopiclone) was marketed in the United States in 2005, primarily to treat insomnia, because of its rapid onset and moderate duration (half-life, ~6 hours) of hypnotic-sedative effect (171). Less than 10% of orally administered eszopiclone is excreted unchanged, because it undergoes extensive CYP3A4- and CYP2E1-catalyzed oxidation and demethylation to metabolites excreted primarily in urine (see Fig. 15.12 and Chapter 15).

PYRAZOLOPYRIMIDINES



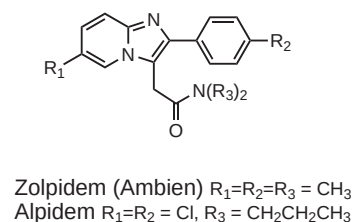
The pyrazolopyrimidines zaleplon, indiplon, and ocinalon have selective high affinity for α_1 -containing BZRs but also produce effects at other BZR/GABA_A subtypes. In patients with insomnia, zaleplon is effective to decrease sleep latency and does not appear to induce withdrawal symptoms or rebound insomnia on discontinuation. Indiplon is similar and currently under review by the U.S. Food and Drug Administration (FDA). Zaleplon is absorbed rapidly and reaches peak plasma concentrations in approximately 1 hour, with a half-life of approximately 1 hour as well. Less than 1% of a dose of zaleplon is excreted unchanged, because most is oxidized by aldehyde dehydrogenase and CYP3A4 to inactive metabolites, which are converted to glucuronides and eliminated in urine (see Fig. 15.13).

Indiplon has demonstrated efficacy to improve latency to sleep onset and total sleep time when compared to placebo, with limited rebound insomnia and tolerance (172). Currently, however, the drug is not yet approved by the U.S. FDA.

Ocinalon was studied primarily for its putative anxiolytic activity rather than for its sedative effects. In rats, ocinalon produced muscle relaxation, ataxia, and sedation only at doses 25-fold higher than the effective anxiolytic dose (169). Likewise, in patients with generalized anxiety disorder, ocinalon produced anxiolytic effects at doses that do not cause greater incidence of sedation or dizziness than placebo. Development of ocinalon, however, was discontinued due to liver toxicity.

As a group, the pyrazolopyrimidines may be useful compounds to study discrepancies observed using molecular (GABA_A subunit-selective) approaches versus transgenic animals and other *in vivo* models in the quest for anxiolytic drugs. For example, for some compounds, partial agonism at a particular α subunit may be sufficient to produce a full anxiolytic activity *in vivo*, whereas other compounds must be full agonists for clinically relevant anxiolysis. Currently, it appears that high relative potency and high relative efficacy at multiple receptor subtypes can account for anxiolytic properties of certain compounds, including for ocinalon.

IMIDAZOPYRIDINES



The imidazopyridines, zolpidem and alpidem, represent another example of α_1 subunit-selective BZR/GABA_A ligands that have clinical profiles different from those of typical benzodiazepines. For example, although the agonist effects of zolpidem on GABA_A receptors qualitatively resemble those of benzodiazepines, clinically it shows a weaker anticonvulsant effect and a stronger sedative effect, which may mask anxiolytic effects. Zolpidem was marketed as a sedative-hypnotic in the United States in 1993, and it appears to be effective in shortening sleep latency and prolonging total sleep time, without affecting sleep stages, in patients with insomnia (173). Zolpidem is readily absorbed from the gastrointestinal tract and is extensively metabolized by the liver to inactive oxidized products, with a half-life of approximately 2 hours (see Fig. 15.11). Alpidem is similar to zolpidem in that it apparently induces no significant changes in sleep parameters (174) and has no effect on memory or muscle tone (175). Alpidem was

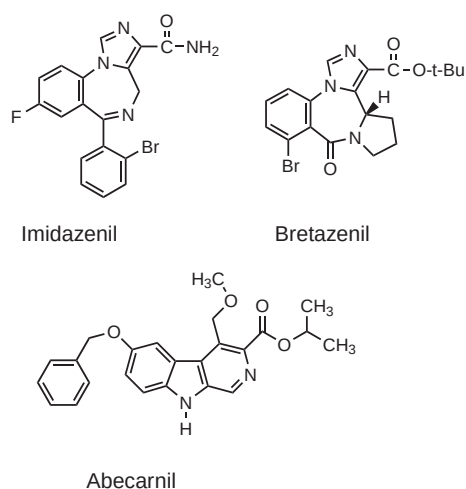


FIGURE 14.23 GABA_A receptor partial agonists.

found to be of at least equal efficacy to lorazepam in the treatment of patients with generalized anxiety disorder (176); however, it was withdrawn because of hepatotoxicity (177).

GABA_A Partial Allosteric Modulators

Partial agonists of the GABA_A receptor complex offer some theoretical and practical advantages over full agonists. For example, compared to the benzodiazepine-type full agonists, partial agonists seem to have lesser side effects, such as sedation, ataxia, and potentiation of alcohol. Also, there may be less abuse potential associated with partial agonists. Three partial agonists of GABA_A receptors were investigated: imidazenil, bretazenil, and abecarnil (Fig. 14.23). Abecarnil is reported by some investigators to have preferential affinity for BZR/GABA_A α_1 subunits, whereas imidazenil and bretazenil are not subtype selective. In any event, abecarnil and bretazenil exhibit anxiolytic activity in animal models, but data from clinical trials do not support the anxiolytic profile predicted from preclinical results (131).

Imidazenil is an imidazobenzodiazepine carboxamide that has higher BZR affinity than diazepam but is only about half as efficacious at modulating GABA effects on chloride currents. Consistent with the general pharmacologic principle that partial agonists may show antagonist functional effects in competition with a more efficacious agonist, imidazenil blocks the sedative and ataxic effects of diazepam (178). Interestingly, however, imidazenil does not block the anticonvulsant effects of diazepam; accordingly, it has been proposed as an alternative to flumazenil in the alleviation of benzodiazepine-induced withdrawal symptoms (178). Bretazenil has qualitatively similar binding and clinical characteristics as imidazenil (179). Its anxiolytic activity comes with significant sedation, however, and this led to discontinuation of its development. Abecarnil is a β -carboline with anxiolytic

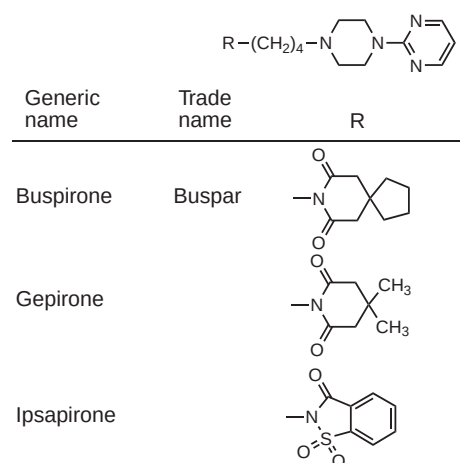


FIGURE 14.24 Azapirone-type 5-HT_{1A} partial agonists.

properties. Typical of other partial allosteric modulators, abecarnil demonstrates antianxiety and anticonvulsant activities, with little or no development of tolerance to these effects (180). Like bretazenil, however, doses of abecarnil required to produce anxiolysis also produce sedation, and it is unlikely that this drug lead will be developed (97).

Miscellaneous Anxiolytic Agents

SEROTONIN RECEPTOR-ACTIVE AGENTS

In the development of anxiolytic agents that do not act via the GABA_A receptor complex, serotonin receptors have been the focus of intensive research in recent years, because preclinical and clinical evidence supports the involvement of serotonin in anxiety (181). For example, serotonin 5-HT_{1A} receptors are found in relatively high density in the septohippocampal region of the brain, which is involved in the modulation of anxiety (182). In the structures of the limbic system, 5-HT_{1A} receptors are predominantly postsynaptic, whereas presynaptic 5-HT_{1A} receptors are found in the dorsal and median raphe nuclei. Presynaptic 5-HT_{1A} receptors function as autoreceptors to inhibit serotonergic neurotransmission, and postsynaptic receptor activation also results in decreased neuronal activity.

Azapirones The pyrimidinylbutylpiperazines (azapirones), buspirone, ipsaperone, and gepirone (Fig. 14.24), are partial agonists at brain 5-HT_{1A} receptors and have anxiolytic activity in humans (181,183). Their anxiolytic effects appear only after several days of treatment, and although it is well established that agonistic activity is required, the optimal level of intrinsic activity is still a matter of debate. Thus, it is unclear whether their mechanism of action is to acutely increase serotonergic activity or chronically decrease serotonergic activity (184). Buspirone is the only one of these agents currently marketed in the United States. It also has

antidopaminergic activity that complicates interpretation of its interaction with 5-HT_{1A} receptors regarding anxiolytic effects. In any event, buspirone is shown to be effective in the treatment of generalized anxiety disorders that are mild to moderate in severity, although it is not superior in efficacy to benzodiazepines and not as well tolerated (183).

SEROTONIN REUPTAKE INHIBITORS Several selective serotonin reuptake inhibitors (SSRIs; see Chapter 18), including escitalopram, fluoxetine, fluvoxamine, paroxetine, and sertraline (Fig. 14.25), demonstrate efficacy as first-line treatment of some anxiety disorders, with the purported advantage that they lack the addictive properties of benzodiazepines (181). Specifically, the SSRIs have been shown to be effective in obsessive-compulsive disorder (185), panic disorder (186), and social phobia (187). The mechanism of action of these agents in anxiety may differ with their role in the treatment of depression; however, current understanding centers on functional imaging studies that show SSRI treatment can dampen brain excitability (181). A recent review of the literature (188) that included data on the SSRIs escitalopram, fluoxetine, and paroxetine in comparison to several benzodiazepines, an azapirone (buspirone), a second-generation antipsychotic (olanzapine; see previous section), and other non-SSRI antidepressants (venlafaxine, duloxetine; see Chapter 18) concluded that efficacy of the drug classes and specific drugs is similar.

ATYPICAL ANTIPSYCHOTIC DRUGS The second-generation (atypical) antipsychotic drugs olanzapine, quetiapine, and risperidone have been investigated for efficacy as pharmacotherapy for generalized anxiety disorder and

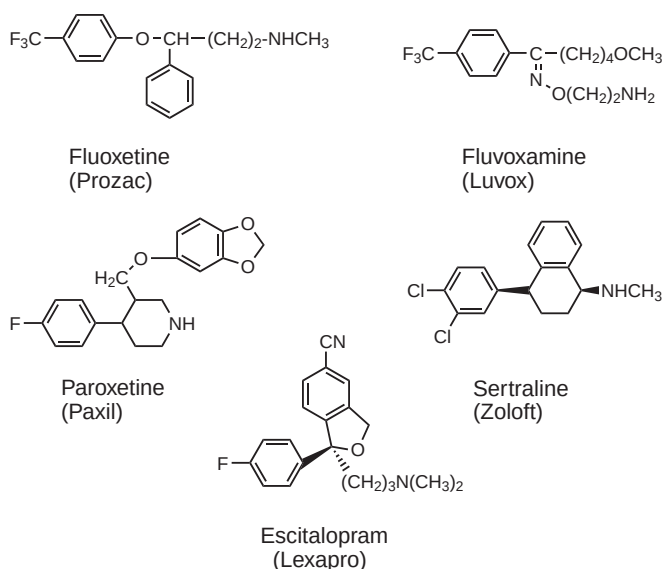


FIGURE 14.25 Structures of some selective serotonin reuptake inhibitors (SSRIs).

social phobia (189,190). Notably, quetiapine was more efficacious than placebo, but study patients were more likely to drop out due to adverse events, including weight gain, sedation, and extrapyramidal side effects. When quetiapine was compared with antidepressants, there was no significant difference in efficacy-related outcomes, but, again, more participants in the quetiapine groups dropped out due to adverse events. The scarce data available for olanzapine and risperidone are inconclusive. Overall, in contrast to antidepressants and benzodiazepines, the long-term risk and benefit of atypical antipsychotic drugs in the treatment of anxiety disorders is not currently established.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

David Hayes, PharmD

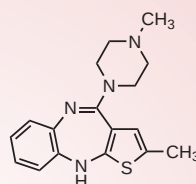
The decision is made to switch to aripiprazole because the metabolic effects are generally less often caused by its receptor affinity and interaction, and it has demonstrated good efficacy among patients with schizophrenia. In addition, Ben is encouraged to institute an exercise and diet regimen and an appointment with a dietician is made. Ben returns to his psychiatrist in 3 months after starting his new treatment and reports that he has not experienced any psychotic or delusional episodes. He sees his primary care physician a week later and has lost 21 lb since his last visit and has a fasting blood glucose of 99 mg/dL.

Chemical Analysis

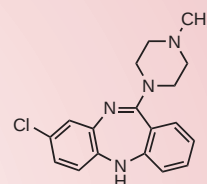
Victoria Roche and S. William Zito

Olanzapine is a thienobenzodiazepine antipsychotic agent that exhibits a high affinity for two main receptor subtypes: dopamine D_2 and serotonin 5-HT_{2A} . The presence of an aromatic moiety and a nitrogen atom capable of becoming cationic at a physiological pH contribute to high affinity at both receptor surfaces. Though 5-HT_{2C} is believed to be the serotonin receptor most closely linked to weight gain, it is known that 5HT_{2A} and 5HT_{2C} receptors share a strong structural similarity, and ligands of one subtype often bind with comparable affinity to the other. Some authors claim that some therapeutic activities ascribed to antagonism of the $2A$ subtype might be caused by, at least partially, antagonism of the $2C$ receptor. Along with its chlorinated

dibenzdiazepine analog clozapine, olanzapine prompts some of the most disturbing weight gain of the atypical antipsychotics.

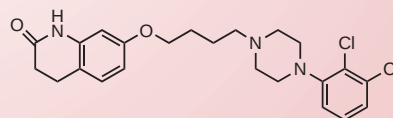


Olanzapine



Clozapine

Aripiprazole is a quinolinone-based atypical antipsychotic agent that acts at a wider variety of dopamine and serotonin receptor surfaces than the diazepine-containing molecules. Rather than antagonizing 5-HT_{2C} receptors like olanzapine does, it acts as a partial agonist. That pharmacodynamic action might help Ben feel less hungry and, therefore, consume fewer calories. Weight control and proper nutrition are essential in keeping prediabetic patients from progressing into full-blown disease. Like the other agents that bind to receptors expecting arylethylamine ligands, aripiprazole contains two aromatic nuclei and an ionizable piperazino nitrogen atom.



Aripiprazole

CASE STUDY

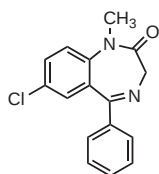
Victoria Roche and S. William Zito

As a Board-certified psychiatric pharmacist, you have enjoyed a professionally rewarding practice in Hollywood, where the demand for your services seems to be essentially nonstop. You have recently been brought in on a high-profile case involving SN, an actor whose public antics have always been fodder for the tabloids but whose behavior has become increasing aberrant both in public and in private. SN has a history of substance abuse, which has negatively impacted his mental health. His type A personality is exacerbated by the 1 to 2 pots of coffee he drinks daily, particularly after a night of serious partying. He has chain-smoked unfiltered cigarettes for years.

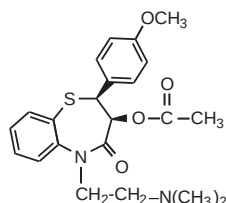
Upon a brief hospitalization after an even shorter stint in a celebrity detoxification center, a diagnosis of schizophrenia is made. SN's medication profile was thoroughly reviewed during

his hospitalization and every drug he had been taking was discontinued except for his low-dose diazepam (for anxiety) and diltiazem (for paroxysmal supraventricular tachycardia). Your advice about a suitable antipsychotic agent to manage his schizophrenia is now being sought. While interviewing SN you learn that he expects to return to his highly rated daytime soap opera and is already compulsively memorizing his lines. Though he appears physically fit to the casual observer, he is obsessing over the 10 lbs he gained during his days away from his in-home fitness center and personal trainer. He's convinced that, for every pound he's gained, the camera will add five and he'll soon be out of work.

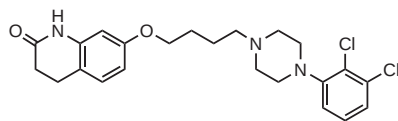
Consider the structure of the three antipsychotic agents drawn below and decide which might be the best initial therapeutic choice for this patient.



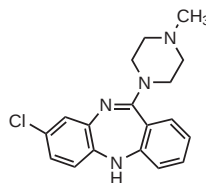
Diazepam



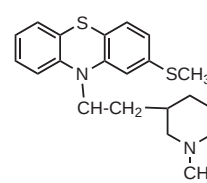
Diltiazem



1



2



3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

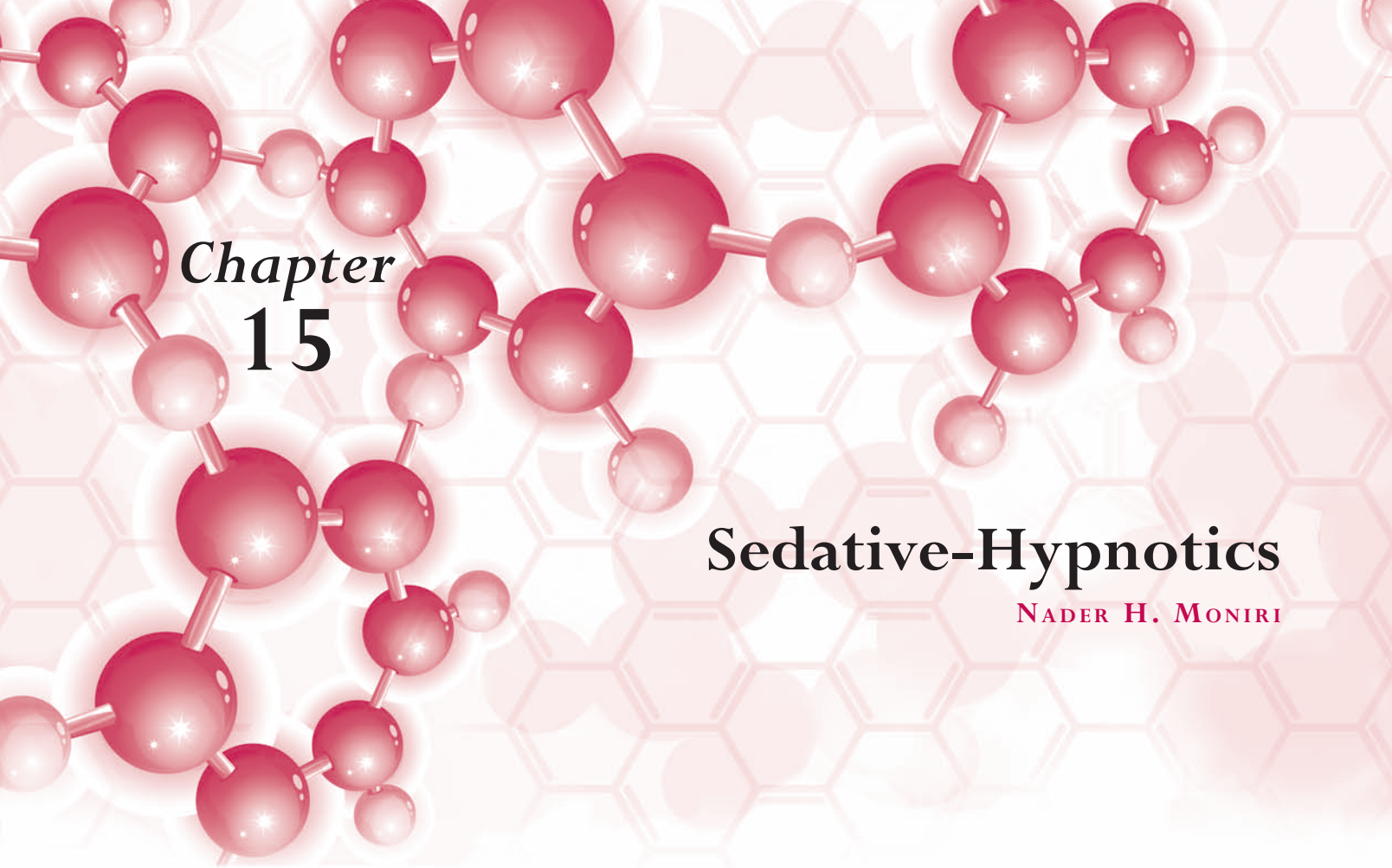
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Chapter 15

Sedative-Hypnotics

NADER H. MONIRI

Drugs Covered in This Chapter

BARBITURATES

- Amobarbital
- Butobarbital
- Pentobarbital
- Phenobarbital
- Secobarbital
- Thiamylal
- Thiobarbital

BENZODIAZEPINE SEDATIVE-HYPNOTICS

- Estazolam
- Flurazepam

- Quazepam
- Temazepam
- Triazolam

HISTAMINE H₁ RECEPTOR ANTAGONISTS

- Diphenhydramine
- Doxepin
- Doxylamine

MELATONIN RECEPTOR AGONISTS

- Ramelteon

NONBENZODIAZEPINE SEDATIVE-HYPNOTICS

- Eszopiclone
- Zaleplon
- Zolpidem

Abbreviations

AA-NAT, arylalkylamine
N-acetyltransferase

AMP, adenosine monophosphate

AMPA, α -amino-3-hydroxy-5-methyl-4-
isoxazole propionic acid

APR, antiplanar region

ARAS, ascending reticular activating
system

BZ, benzodiazepine

CNS, central nervous system

EEG, electroencephalography

ERR, electron-rich region

FDA, U.S. Food and Drug
Administration

FRAR, freely rotating aromatic ring
region

GABA_A, γ -aminobutyric acid A receptors

GPCR, G protein-coupled receptor

HIOMT, hydroxyindole-O-
methyltransferase

MT, melatonin

NPS, neuropeptide S

PAR, planar aromatic region

PKA, protein kinase A

REM, rapid eye movement

SAR, structure-activity relationship

SCN, suprachiasmatic nucleus

TMH, transmembrane helix

TMN, tuberomammillary nucleus

SCENARIO

Susan W. Miller, PharmD, CGP, FASCP

WK, a 78-year-old white woman, presents to the ambulatory care clinic for a medication therapy management (MTM) consult before her appointment for a routine examination with her primary care provider. WK is accompanied by her daughter, with whom she lives. WK's medical problems (and medications) are type 2 diabetes (glipizide 5 mg po QD), hypertension (verapamil SR 180 mg po QD), osteoporosis (alendronate 70 mg po once weekly, calcium carbonate 600 mg/vitamin D 200 iu po BID), and insomnia (zolpidem 5 mg po Q HS). WK's problems have been controlled, but at today's visit, an 8-lb weight gain in 6 months is noted. Her hemoglobin A_{1c} level is now 7.6% but it had been stable at 6.8%, and her blood pressure is 126/78 mm Hg. The daughter reports that her mom

has begun to wander through the house at night and has been found on several occasions eating warmed-up leftovers at 2 AM. The daughter also reports that WK has begun to experience memory loss. WK denies these behaviors and reports that she falls asleep just fine, but she is not sleeping well during the night. WK is also concerned about her recent weight gain and she confides that, as a result of her visit to the dermatologist last month, she was prescribed a new medicine, itraconazole capsules, for the newly diagnosed fungal infection in the fingernails on her right hand.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

The general objective of this chapter is to provide the reader with an in-depth understanding of the molecular and chemical basis of sedative-hypnotic drug action. The reader is first introduced to neuronal regulation of the sleep/wake cycle and the effects on regulating the stages of sleep. Emphasis is placed on introducing the targets of sedative-hypnotic agents, particularly γ -aminobutyric acid A receptors (GABA_A), melatonin (MT) MT₁/MT₂ receptors, and histamine H₁ receptors. Specific agents that modulate the functions of these receptors to promote sleep will be discussed from the perspectives of structure–activity relationships, as well as physiochemical and pharmacokinetic properties that affect their activity.

NEUROBIOLOGY OF SLEEP

Sleep is a reversible process that is typified by sensory and motor inactivity as well as reduced cortical responses to external stimuli. This process is distinct from complete states of unconsciousness (e.g., coma), in which decreased cortical activity is unresponsive to all external stimuli. In human physiology, sleep/wake cycles are regulated to a large degree by endogenous circadian rhythms as well as homeostatic mechanisms, which in turn are governed by a myriad of neuronal pathways. Critical experiments performed in felines in the late 1940s demonstrated that impairment of the reticular formation within the brainstem triggered behavioral and electroencephalographic (EEG) activity consistent with coma-like states, suggesting that this area of the brain was involved in regulating sleep and wakefulness (1). This was contrary to observations that showed that felines with lesions of ascending sensory neurons distal to the reticular formation displayed no impairment to wakefulness or sleep, indicating

that sleep/wake regulation by the reticular formation was independent of sensory neurons that *leave* it (2). Further experiments showed that the reticular formation contained a high density of afferent input and that direct stimulation of the reticular formation facilitated wakefulness (3). Taken together, these key observations suggested that sleep and wakefulness were likely controlled by factors within the reticular formation and that this structure behaves as a relay center that transmits afferent input to the cortex. These and other pioneering studies led to the ascending reticular activating system (ARAS) hypothesis of sleep/wake regulation and subsequent identification of ARAS function as being critical in modification of sleep/wake transitions. Many decades of ensuing research on sleep/wake responses of the ARAS have revealed that projections to and from various nuclei within the brainstem can modulate ARAS activity to facilitate wakefulness and sleep. These include serotonergic neurons of the raphe nucleus within the reticular formation itself; cholinergic neurons of the pedunculopontine tegmentum and noradrenergic neurons of the locus coeruleus, within the pons; as well as dopaminergic neurons of the substantia nigra and ventral tegmental areas of the midbrain. These structures themselves are regulated in a highly orchestrated manner primarily by the excitatory neurotransmitter glutamate and the inhibitory neurotransmitter GABA to regulate their respective activities.

In addition to the cell structures that make up the brainstem ARAS, there are additional populations of highly specialized cells that can modulate sleep/wakefulness either independently or in concert with cells of the ARAS. For example, stimulation of the tuberomammillary nucleus (TMN) within the posterior hypothalamus facilitates wakefulness and arousal, and these cells are predominantly active only during wakeful periods (4). TMN-mediated regulation of wakefulness

is dependent on histaminergic neurons that densely project into the cortex as well as other wake-promoting structures within the brainstem (5). The firing rate of these cortical-projecting histaminergic TMN neurons decreases upon sleep and increases upon arousal, whereas lesioning of histaminergic neurons within the TMN promotes sleep (6–8). Biochemically, the release of histamine from these neurons is increased upon and during wakefulness and is under the control of glutaminergic-stimulatory and GABAergic-inhibitory circuits (9,10). Studies with histamine receptor knock-out mice and in vivo pharmacologic studies have established that the TMN effects of histamine on wakefulness and sleep are mediated primarily by histamine H_1 receptors.

Whereas the posterior-hypothalamic TMN is involved in regulating arousal, the anterior-hypothalamus, specifically the suprachiasmatic nucleus (SCN), is involved in induction of sleep. This has been demonstrated by studies that show that lesions to the posterior hypothalamus (e.g., TMN) induce sleepfulness, whereas lesions of the anterior hypothalamus are associated with wakefulness (11). Decades of further work have established that the SCN is critical for the maintenance of circadian rhythms, in essence behaving as the brain's endogenous master clock, thereby regulating sleep/wake cycles and synchronizing circadian phase timings with other brain structures. Importantly, the SCN receives visual input from the retina via the retinohypothalamic tract, and this key connection serves to synchronize circadian rhythms to daylight (12). The firing rate of SCN neurons is strongly correlated with

day/night cycles, which serve to establish an approximate 24-hour circadian rhythm. Although the SCN is largely under the control of glutaminergic and GABAergic neurons, which originate in the preoptic nucleus, it also works in concert with the pineal gland to regulate circadian rhythms (13). As such, inhibitory GABAergic outflow from SCN neurons to the pineal gland is enhanced during the daylight hours and declines proportionally to decreases in daylight. Light-sensitive neurons from the SCN project to the hypothalamic paraventricular nuclei and through the lateral horn of the spinal cord primarily by way of vasopressin and GABAergic neurons. The signal is then relayed to the pineal gland through noradrenergic ganglionic sympathetic fibers, which directly innervate the pineal gland. Hence, during periods of light, GABA output is elevated, which inhibits the release of norepinephrine at the pineal gland. On the contrary, during periods of darkness, the inhibitory GABAergic outflow from the SCN is reduced, resulting in increased levels of norepinephrine released at the level of the pineal gland (12). Here, norepinephrine acts primarily through pineal β_1 - and α_{1B} -adrenergic receptors to stimulate a rapid and profound (~10-fold) synthesis and release of melatonin, another key regulator of circadian rhythms and the sleep/wake cycle, into the bloodstream (14).

The synthesis of melatonin occurs within the pineal gland, where the amino acid L-tryptophan is converted to serotonin, which is subsequently acetylated to yield N-acetylserotonin by the enzyme arylalkylamine N-acetyltransferase (AA-NAT). N-acetylserotonin is converted to melatonin by the

CLINICAL SIGNIFICANCE



Medications from several classes are useful pharmacotherapeutic agents for managing insomnia.

The benzodiazepines are safe and effective and are the most commonly prescribed sedative-hypnotic agents. Within the class of benzodiazepines, those with shorter elimination half-lives, such as the triazolobenzodiazepines triazolam and estazolam, provide improved sleep patterns and minimal daytime sedation. Relative duration of action, as well as sedative-hypnotic potency and risk of residual morning depressant effects, can be readily assessed from an evaluation of benzodiazepine structure. The nonbenzodiazepine GABA_A agonists of zolpidem, zaleplon, and eszopiclone have comparable efficacy to the benzodiazepines in improving sleep patterns and can be selectively used to reduce sleep latency, nighttime awakenings, and/or total sleep time. The melatonin receptor agonist ramelteon is a newer alternative as a sedative-hypnotic agent. The antihistamine H_1 -receptor antagonists diphenhydramine and doxylamine, members of the highly sedative aminoalkylether class of antihistamines, as

well as the antidepressants trazodone and doxepin (a member of the tertiary tricyclics well known for sedative side effects) are useful as sedative-hypnotic agents, but selection of these should be based on the side effect profile of the individual agent. Barbiturates are well established as effective sedative-hypnotic agents and their well-defined structure-activity relationships can help guide clinicians in drug product selection when they are truly indicated; however, their associated toxicities of central nervous system depression, physical dependence, and tolerance as well as their activity as potent hepatic enzyme inducers have limited their use as sedative-hypnotic agents. Identifying the chemical classes and their associated physical chemical properties and metabolic fates will greatly facilitate the understanding and utilization of these useful therapeutic agents.

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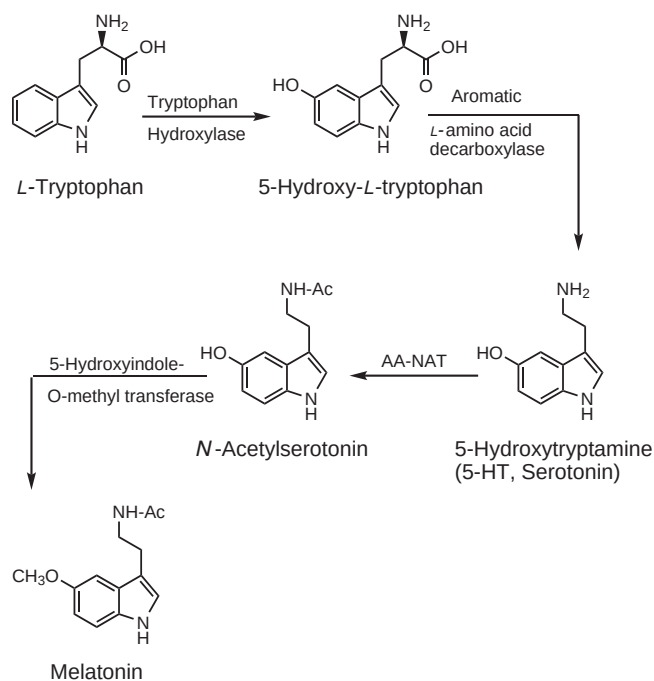


FIGURE 15.1 Biosynthesis of melatonin.

enzyme 5-hydroxyindole-*O*-methyltransferase (HIOMT) (Fig. 15.1). The activity of AA-NAT is greatly influenced by β_1 -adrenergic receptors, which are activated upon the release of norepinephrine via the SCN circuit described earlier. β_1 -receptor-mediated increases in intracellular cyclic adenosine monophosphate (AMP) stimulate AA-NAT expression and also increase protein kinase A (PKA)-mediated phosphorylation of AA-NAT, which stabilizes and activates the enzyme (15). The rhythm of AA-NAT activity is thus correspondingly coupled to light exposure because increases in light lead to decreases in AA-NAT activity. Activity of AA-NAT can also be effectively abolished by lesions of the SCN, further demonstrating that melatonin synthesis is dependent on retinohypothalamic input (16). Once synthesized, melatonin diffuses into capillary blood vessels and rapidly reaches all tissues of the body, where it can agonize its cognate G protein-coupled receptors (GPCRs), MT1R and MT2R, leading to various physiologic responses. With specific regard to sleep, melatonin has profound effects in the SCN, where it agonizes MT1R to mediate the acute inhibition of SCN neurons, thereby promoting sleep. Additionally, agonism of MT2R in the SCN is associated with the phase-shifting effects on circadian rhythms.

HUMAN SLEEP CYCLES

EEG has afforded researchers the ability to assess the electrical activity of the brain during the course of sleep, and these findings have revealed the presence of distinct EEG activity that correlates to five stages of sleep. These stages, which cycle throughout the night, include four non-rapid eye movement (REM) stages, often referred

to as “quiet sleep,” as well a single stage of REM sleep, referred to as “active sleep.” Non-REM sleep stages correlate to periods of low metabolic rate and low neuronal activity; hence, brain activity and EEG events are “quiet.” In non-REM sleep, heart rate and blood pressure are also reduced due to an increase in parasympathetic nervous system activity coupled with a decrease in sympathetic activity. As a result of these autonomic reflexes, non-REM sleep also causes pupillary constriction, which diminishes the amount of light that is allowed to enter the retina (17). The initial period of sleep is stage 1, which lasts only a brief period of time (5 to 15 minutes) and represents the period of transitioning from wakefulness to sleep. Although EEG activity in awake humans mainly consists of high-frequency (15 to 25 Hz) alpha-wave events, the progression to stage 1 sleep results in the emergence of slower theta waves with lower frequency (4 to 10 Hz) as the individual progresses to the light or drowsy sleep characterized by stage 1. In stage 2 sleep, which often lasts 15 to 20 minutes, background theta waves continue but are periodically interrupted by characteristic sleep spindles, which are short bursts of higher frequency (12 to 15 Hz) EEG events. Stage 2 of the sleep cycle is still considered a light sleep; however, skeletal muscle activity is decreased in this stage compared to the transitional stage 1. Stages 3 and 4, often referred to as slow-wave sleep, are characterized by appearance of higher amplitude, slower frequency (0.5 to 4 Hz) delta waves, which occur less often in stage 3 and more often in stage 4. Both of these stages represent deep sleep because the characteristic delta activity is least similar to arousal-state alpha activity. If awakened in this stage of sleep, an individual will likely be confused and disoriented (18).

The fifth stage of sleep is referred to as REM sleep and is characterized by increases in eye movement, respiration rate, and brain activity. Electroencephalographic changes in REM sleep include high-voltage firing spikes, which seem to originate in the pons, lateral geniculate nucleus, and occipital cortex, and these spikes are correlated with bursts of rapid eye movements associated with this stage of sleep (19). The increased brain activity in REM sleep is consistent with an increase in dreaming and an increase in metabolic rate, thus the naming convention “active sleep”; yet, it is inversely proportional to skeletal muscle tonicity, which effectively paralyzes skeletal movements.

The sleep stages are normally cycled every 90 to 100 minutes throughout the night, and it is important to note that sleep does not necessarily advance sequentially through the five stages. Healthy adults typically enter sleep through a progression through the stages, with the first REM stage occurring after 75 to 90 minutes of non-REM sleep and a typical return to stage 2 or 3 sleep thereafter. As sleep progresses over the course of a night, the amount of time spent in REM sleep increases such that over a typical 8-hour sleep cycle, approximately 55% to 60% of sleep time is spent in stages 1 and 2 (light sleep), 20% in stages 3 and 4 (deep sleep), and 20% to 25% in stage 5 (REM sleep).

PHARMACOLOGIC TARGETS OF SEDATIVE-HYPNOTIC AGENTS

As described earlier in the brief introduction, there are many physiologic and biochemical factors that can influence the various stages of sleep, and as such, numerous targets for pharmacologic intervention have arisen. In fact, nearly every neurotransmitter system in the mammalian brain, including the adenosinergic, serotonergic, dopaminergic, adrenergic, histaminergic, and cholinergic systems, have at one time or another been associated with induction of sleep or wakefulness. Despite this wide breadth of potential biochemical targets, the development of sedative-hypnotic agents has primarily focused on 1) agents that cause CNS depression via agonism of GABA_A receptors, and 2) agents that modulate hypothalamic histamine or melatonin circadian systems that, as described earlier, regulate sleep and arousal. This chapter will focus on structure–activity relationships (SARs) and pharmacodynamics of these agents, including barbiturates, benzodiazepines, nonbenzodiazepine GABA_A agonists, melatonin receptor agonists and histamine H₁ receptor antagonists. Development of agents that modulate the activity of novel protein targets such as orexin and neuropeptide S receptors will also be discussed towards the end of the chapter.

GABA_A Receptors

GABA is the major inhibitory neurotransmitter in the mammalian central nervous system (CNS) and is critical in balancing neuronal excitation. GABA is widely distributed throughout the CNS and can be found at concentrations up to 1,000-fold (high $\mu\text{mol/L}$ to low mmol/L) greater than that of monoaminergic neurotransmitters in various CNS nuclei. GABA-induced physiologic functions are mediated by at least two distinct classes of membrane-bound receptors, ionotropic GABA_A receptors and metabotropic GABA_B receptors. GABA_B receptors belong to the second messenger-linked GPCR superfamily and share homology with metabotropic glutamate receptors. Meanwhile, GABA_A receptors are ligand-gated ion channels that modulate conductance of chloride ions through the cell membrane upon binding of GABA. The activation of GABA_A receptors on excitable neurons leads to membrane hyperpolarization, which facilitates an increase in the firing threshold potential and consequently reduces the likelihood of generating an action potential. Hence, agonism of GABA receptors leads to neuronal inhibition and CNS depression, and not surprisingly, GABA_A receptors are important targets for treatment of a variety of CNS disorders in which CNS depression provides a therapeutic benefit, including anxiety (see Chapter 14), anesthesia (see Chapter 16), convulsions and seizures (see Chapter 17), and sleep disorders. In this regard, drugs that increase GABA_A-mediated chloride flux (e.g., GABA_A agonists) provide anxiolytic, anesthetic, anticonvulsant, and sedative-hypnotic activity, whereas agents that block the chloride channel (e.g., picrotoxin) can lead to convulsions and a heightened state of arousal.

The GABA_A receptor-channel complex is related to the nicotinic acetylcholine receptor superfamily, which also includes the strychnine-sensitive glycine receptors and serotonin 5-HT₃ receptors. In the human brain, GABA_A complexes are formed by oligomerization of individual subunits that produce heteropentameric complexes consisting of α , β , γ , δ , ϵ , or ρ subunits, which assemble a ligand-gated channel with a central pore that allows for the conductance of chloride (Fig. 15.2) (20). Moreover, the α , β , and γ subunits of GABA_A receptors are expressed as distinct alternatively spliced variants (e.g., $\alpha_{[1-6]}$, $\beta_{[1-3]}$, $\gamma_{[1-3]}$). Functional channels typically require multiple α and β subunits in combination with another subunit, allowing for tremendous diversity in the makeup of the channels (Fig. 15.2). The distribution of various GABA_A receptors varies according to the subtype combinations. The $\alpha_1\beta_2\gamma_2$, $\alpha_2\beta_3\gamma_2$, and $\alpha_3\beta_3\gamma_2$ subtypes are the most abundant receptors, accounting for nearly 80% of GABA_A receptors in the mammalian brain (20). The α_1 subunit is the most widely expressed, whereas α_4 , α_5 , and α_6 subunits are restricted to localized neuronal populations (20). Two distinct GABA binding sites are formed at the two α and β subunit interfaces, whereas the binding sites for GABA-modulating drugs such as barbiturates and benzodiazepines are at allosteric sites (Fig. 15.2). Upon binding of barbiturates or benzodiazepines to their respective allosteric sites on the GABA_A receptor, alterations in the conductance of Cl[−] occur, as described for each class of agent in the following sections.

Barbiturates

Barbiturates are potent CNS depressants with sedative-hypnotic, anesthetic, and anticonvulsant activity and were widely considered sedative-hypnotic agents of choice until the marketing of benzodiazepines in the late 1960s. Barbiturates played a crucial role in our understanding of GABA_A receptor function, and although there are special instances that call for barbiturate use, they have largely been replaced as sedative-hypnotic agents by other agents due to safety concerns, which include tolerance, dependence, potential for abuse, and a relatively low toxicity threshold that can lead to overdosage and poisoning. Currently, there are five barbiturates approved by

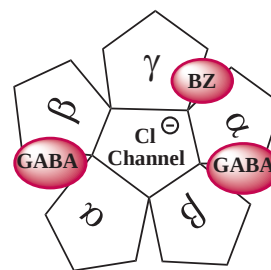


FIGURE 15.2 The pentameric assembly of GABA_A receptors, whose monomers assemble to form the chloride channel at the center. The distinct GABA and benzodiazepine binding sites of this $\alpha\beta\gamma$ -type receptor are noted.

the U.S. Food and Drug Administration (FDA) for use as sedative-hypnotics: amobarbital, butabarbital, pentobarbital, phenobarbital, and secobarbital.

BARBITURATE MECHANISM OF ACTION Upon binding of GABA to the receptor, the chloride ion (Cl^-) conductance channel opens in bursts of approximately 1 millisecond, 5 millisecond, and 10 millisecond of duration, followed by brief periods of closure. Channel opening increases in frequency and duration upon increases in GABA concentrations. Binding of barbiturates to their GABA_A binding site causes an increase in the binding of GABA to the receptor, facilitating an enhancement of the actions of GABA and leading to prolongation of the longest-duration open state (21). Although the exact site of barbiturate binding remains elusive, it is known that this site is distinct from both the GABA and benzodiazepine (BZ) sites and that the barbiturate site likely does not require specific subunits, unlike the BZ site (see later discussion). Higher therapeutic concentrations of barbiturates can also enhance binding of benzodiazepines, whereas supratherapeutic concentrations of barbiturates can cause opening of the channel independently of GABA, although the clinical importance of this response has not been widely studied (22). Notably, in addition to their augmentation of GABA_A responses, the physiologic effects of higher concentrations of barbiturates can also be mediated by kainite-sensitive α -amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA) glutamate receptors, as well as voltage-gated Na^+ and Ca^{2+} channels.

PHARMACOLOGIC EFFECTS OF BARBITURATES Barbiturates act as reversible inhibitors of virtually all excitable neurons and can produce dose-dependent CNS depression, which ranges in effect from weak sedation to general anesthesia. With regard to their effects on sleep, barbiturates significantly decrease the time it takes to fall asleep (sleep latency), increase the total time of sleep, and also decrease occurrences of nighttime awakenings. As a consequence of these effects, barbiturates significantly impair psychomotor abilities and also decrease memory and cognitive performance. Additionally, barbiturates can cause physical dependence, and sudden withdrawal of these agents can lead to serious neuropsychiatric symptoms and convulsions of excitable tissue, which can facilitate seizures and respiratory spasms. Long-term use of barbiturates has also been shown to lead to tolerance, in which higher doses of the agent are needed to achieve the same therapeutic endpoints (e.g., sleep). Because these agents have relatively narrow safety margins, higher doses that result as a consequence of tolerance can lead to toxicity. Due to these concerns, the use of barbiturates in treatment of insomnia has been virtually discontinued, and when these agents are used for treatment of insomnia, they are used in special cases and/or restricted to only short-term use. Other, older barbiturate-like drugs (e.g., chloral hydrate, glutethimide, and ethylchlorvynol) also act on the GABA_A receptor to produce sedative-hypnotic effects

but have fallen out of favor due to numerous untoward effects. These agents have been reviewed in previous editions of this text.

STRUCTURE–ACTIVITY RELATIONSHIPS OF BARBITURATES Barbiturates are derivatives of a cyclized ureide of malonic acid, commonly referred to as barbituric acid (2,4,6-trioxypyrimidine) (Fig. 15.3), which is itself devoid of sedative-hypnotic, anxiolytic, and anticonvulsant activity. As shown in Figure 15.3, barbituric acid can undergo pH-dependent keto-enol tautomerization through transfer of either imino hydrogen or methylene hydrogen to a keto oxygen. This tautomerization is based on a fairly acidic carbon ($\text{p}K_a \sim 4$) at the 2-position, which can be stabilized in either molecular form based on the acidity of the solution. Empirical and computational studies have demonstrated that the tri-keto form is the most stable in aqueous solutions, whereas the 4,6-dialcohol tautomeric forms are the least stable (23,24). Although barbituric acid itself lacks pharmacologic activity, addition of 5,5-disubstituents to the barbituric backbone yields compounds with potent sedative-hypnotic, anxiolytic, and anticonvulsant activity. Termed *barbiturates*, the 5,5-disubstituted barbituric acids all possess a high degree of lipophilicity and, as weak acids, can be easily converted to sodium salts by treatment with sodium hydroxide.

Although thousands of barbiturate-like compounds have been synthesized, the 5,5-disubstituted barbituric acid backbone is the primary pharmacophore required for sedative-hypnotic activity, and efforts to further derivatize this backbone lead to a general loss in activity. These efforts show that esterification of either of the 1,3-diazine nitrogens decreases hypnotic activity. Although substitution of these nitrogens with aliphatic carbons retained anticonvulsant effects, it led to only weak hypnotic activity for *N*-methylated substituents, and this activity was lost upon increases in chain length or bulk (25,26). Likewise, esterification of the 5-position substituents (e.g., 5-phenyl, 5-methylester) yielded agents with analgesic activity but only weak hypnotic effects (27). Although early work on barbiturate derivatization demonstrated the importance of the 5-position substitutions on CNS depressant activity, the inclusion of polar functional groups at the 5-position

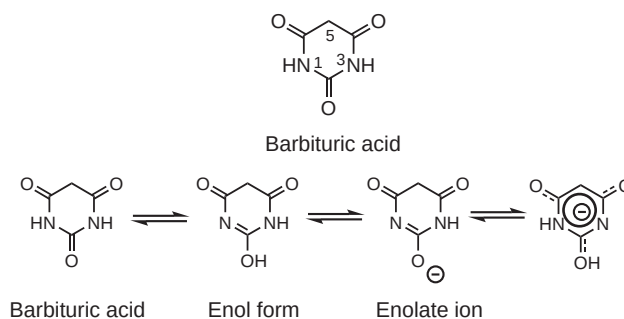


FIGURE 15.3 The structure of barbituric acid (top) and tautomerization of barbituric acid.

resulted in compounds that were fully devoid of sedative-hypnotic or anticonvulsant activity (28).

There are currently five barbiturates approved by the FDA for short-term sedative-hypnotic use and treatment of insomnia (Fig. 15.4). Structurally, pharmacologic activity is imparted to all five compounds due to the lipophilic di-substitutions at the 5-position (e.g., R_1 and R_2 ; Table 15.1). With the exception of secobarbital, which contains an allylic group at R_1 , the remaining agents contain an ethyl substitution at this position (Fig. 15.4). Differences in these sedative-hypnotic barbiturates lie predominately at the R_2 position, where substitutions can affect the potency, rate of onset, and duration of action of the various congeners. The activity of these agents is strongly influenced by the lipophilicity of both substituents at the 5-position. As the number of carbon atoms at the R_2 position increases, the lipophilicity of the barbiturate will also increase such that comparative predictions on activity and time to onset can be made. For example, pentobarbital is more potent and has faster onset compared to butobarbital, which contains one less methylene group and is thus less lipophilic (Fig. 15.4, Table 15.1). Although the lipophilicity of the molecule greatly influences its ability to cross the blood-brain barrier, leading to potency and affecting the time of onset (see later discussion), too high a degree of lipophilicity will offset the required hydrophilicity that is necessary for dissolution and solubility of the compound in aqueous fluids. Hence, a limit to lipophilicity is reached and pharmacologic activity will begin to decrease if this threshold is surpassed.

The first commercially marketed barbiturate, barbital, contained identical 5,5-diethyl substituents, and as such, a plane of symmetry. However, many clinically useful barbiturates contain distinct 5,5-disubstitutions and substituted nitrogens, which form an asymmetric chiral center at the C-5 carbon and/or within one of the 5-position alkyl chains. Although barbiturates are dispensed as racemic mixtures, *L*-stereoisomers typically have twice the potency of the respective *D*-stereoisomers, consistent with

observations that demonstrate stereoselectivity of GABA_A receptors.

A final note regarding the SAR of barbiturates is more critical in the role of this class as anesthetic agents and as such is only briefly discussed here. Specifically, modification of the 2-position oxygen of the barbiturate backbone with the larger sulfur atom yields thiobarbiturate derivatives with increased lipophilicity, faster time of onset, and shorter duration of action compared to the oxy-derivatives. For example, thiopentobarbital and thiamylal have much faster onsets and shorter durations than their respective oxy-congeners, pentobarbital and secobarbital, respectively (Fig. 15.4). Because thiobarbiturates have a specialized role as anesthetics, the reader is referred to Chapter 16 for further discussion of these agents.

PHARMACOKINETICS AND METABOLISM OF BARBITURATES Barbiturate salts that are approved for use as sedative-hypnotics are rapidly and completely absorbed following oral ingestion, in contrast to the free acids, which are absorbed at a much slower rate. The time of onset and duration of action are also significantly influenced by the lipophilicity of the individual agent and the route of administration. The five sedative-hypnotic barbiturates described here are all highly lipophilic, with corresponding logP values above 1.5, compared to barbital, which has a logP of 0.65 (Table 15.1) (29). Given the high proportion of cardiac output that is directed toward the brain, more lipophilic barbiturates are more rapidly distributed to the CNS and will have a faster time of onset. As with other CNS-acting agents, the time of onset is also dependent on the route of administration, with intravenous administration being the fastest and having near immediate effects, followed by intramuscular administration, which leads to faster brain distribution than the 10 to 60 minutes typical of orally administered sedative-hypnotics (Table 15.1).

Lipophilicity facilitates the relatively short duration of action of the barbiturates due to redistribution from the brain to other body compartments upon equilibration. This redistribution of barbiturates from the brain to other sites (e.g., adipose tissue) is the major mechanism that contributes to the loss of sedative-hypnotic activity of these agents, leading to durations of 3 to 8 hours for the short- and intermediate-acting agents and 10 to 16 hours for phenobarbital, which has the least lipophilicity and therefore the longest corresponding duration of action (Table 15.1). Because lipophilicity and route of administration can affect both time to onset and duration of action of barbiturates, these factors will influence a particular barbiturate's place in therapy. The anticonvulsant barbiturates are typically less lipophilic (e.g., phenobarbital) and correspondingly have slow onsets and long durations upon oral dosing, whereas the sedative-hypnotic agents described here are typically administered orally to achieve an optimum balance between achievement of sleep onset and duration. On the contrary, anesthesia-inducing barbiturates such as thiamylal are termed ultra-short acting due to their extremely high

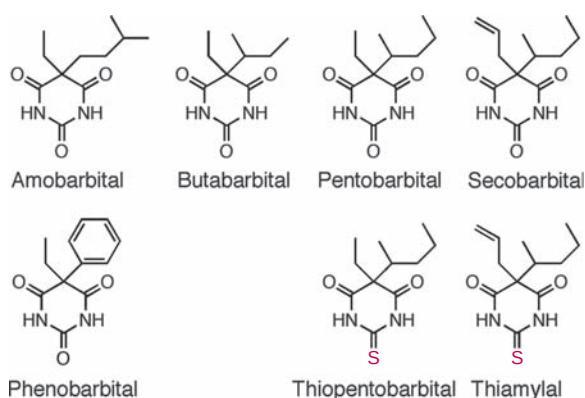
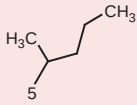
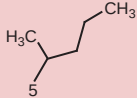
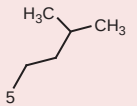
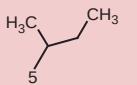
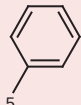


FIGURE 15.4 The barbiturates that are indicated for sedative-hypnotic use and the thiobarbiturates underneath their respective oxybarbiturate congeners.

TABLE 15.1 Pharmacokinetic Parameters of Barbiturates Approved for Sedative-Hypnotic Use

Barbiturate	R ₁	R ₂	LogP	Onset Time (min) ^d	Duration of action (hour)	Classification
Pentobarbital			2.10 ^a	10–15	3–4	Short-acting
Secobarbital			2.36 ^b	10–15	3–4	Short-acting
Amobarbital			2.07 ^d	45–60	6–8	Intermediate acting
Butobarbital			1.60 ^b	45–60	6–8	Intermediate acting
Phenobarbital			1.46 ^b	30–60	10–16	Long-acting

^a From Freese IE, Levin BC, Pearce R, et al. Correlation between the growth inhibitory effects, partition coefficients and teratogenic effects of lipophilic acids. *Teratology*. 1979;20:413–440.

^b Average determinations from Slater B, McCormack, A, Avdeef A, et al. pH-metric log P.4. Comparison of partition coefficients determined by HPLC and potentiometric methods to literature values. *J. Pharm. Sci* 1994;83:1280–1283

^c From Kakemi K, Arita T, Hori R, et al. Absorption and excretion of drugs. XXXI. On the relationship between partition coefficients and chemical structures of barbituric acid derivatives. *Chem Pharm Bull*. 1967;15:1705–17

^d Upon oral administration.

lipophilicity ($\log P > 3$), which confers immediate onset and very short durations upon intravenous injection.

Although loss of barbiturate sedative-hypnotic activity occurs primarily due to redistribution, the agents are also metabolically transformed. Metabolism of barbiturates is dependent on their individual degrees of lipid solubility—the more lipophilic compounds generally undergo greater metabolic transformations. It is important to understand that metabolism of barbiturates leads to metabolites that lack a high degree of lipophilicity and that, as a result, lack sedative-hypnotic activity. To achieve loss of lipophilicity, barbiturates are typically conjugated with glucuronide or sulfate conjugates that are hydrophilic and will allow for renal excretion of the agents. The most important pathway of barbiturate metabolism is oxidation or removal of the substituents at the 5-position carbon (Fig. 15.5). These phase 1 oxidation transformants are typically alcohols, or phenols in the case of phenobarbital, which can appear in the urine as either free or conjugated metabolites as a result of further phase 2 reactions leading to glucuronides or sulfates. The formed alcohols can also be further oxidized to yield ketones or carboxylic acid metabolites,

which can be excreted freely or also conjugated (Fig. 15.5). Although not as significant as the route described earlier, oxidative desulfuration of thiobarbiturates can also occur, leading to more hydrophilic oxybarbiturates. Oxybarbiturate metabolites of thiamylal and thiopentobarbital are subsequently oxidized by the phase 1 and phase 2 reactions described earlier (Fig. 15.5).

Benzodiazepines

As with the barbiturates, benzodiazepines modulate the function of GABA_A receptors, leading to neuronal hyperpolarization and CNS depressant effects. In contrast to the barbiturate binding site of GABA_A receptors, the benzodiazepine binding site has been well characterized and is known to be formed by the interface of α and γ subunits. Not surprisingly, GABA_A receptors that lack the γ subunit (e.g., $\alpha_6\beta\delta$) are completely insensitive to all benzodiazepines. Additionally, it has been shown that isoforms of the α subunit have differential benzodiazepine binding affinities, with the α_1 subunit-containing receptors (often termed BZ₁ receptors) having high benzodiazepine affinity and the α_2 , α_3 , and α_5 subunit-containing receptors (BZ₂ receptors) having

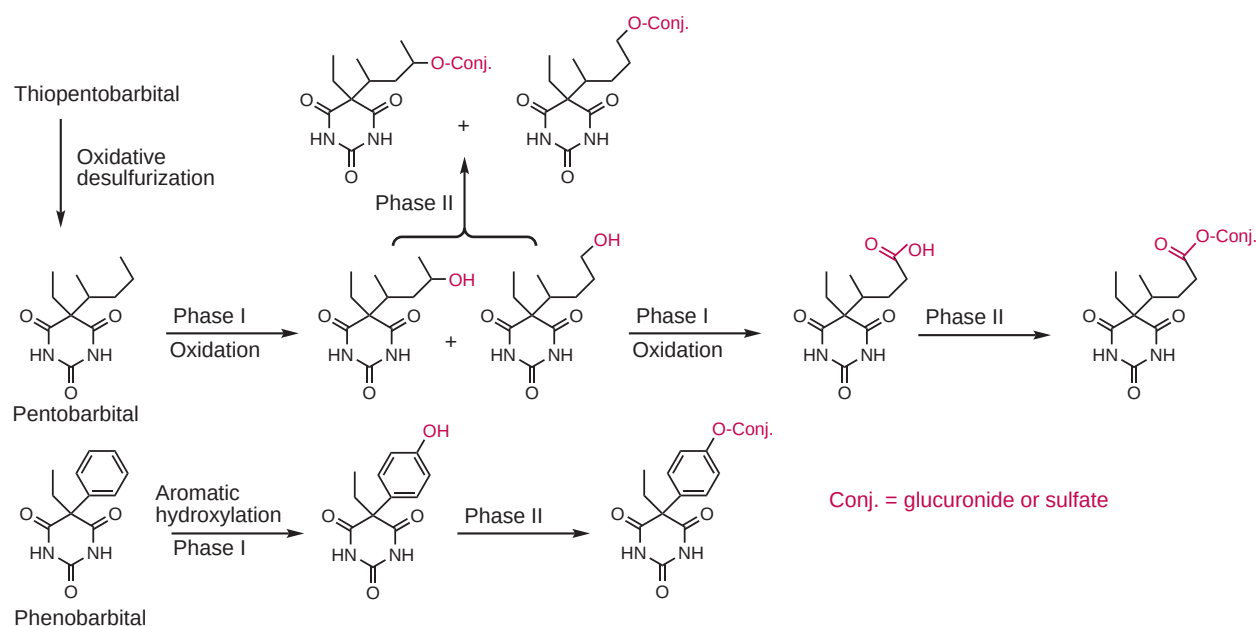


FIGURE 15.5 Major routes of metabolism of barbiturates. Thiopentobarbital and pentobarbital are used as examples to describe metabolism of alkyl thio- and oxybarbiturates, whereas phenobarbital exhibits the metabolism of an aromatic substituent. Coloration indicates site of metabolism.

moderate benzodiazepine affinity. Furthermore, some combinations that include α_4 or α_6 subunits are completely insensitive to benzodiazepine agonists, even in the presence of an appropriate γ subunit (20).

Binding of benzodiazepines to the allosteric BZ binding site can produce agonist, partial agonist, inverse agonist, or antagonist physiologic responses. Benzodiazepine agonists are used clinically as sedative-hypnotic agents as well as for the treatment of anxiety and convulsant disorders, in accordance with their ability to allosterically augment GABA-mediated inhibitory Cl^- conductance through the GABA_A channel. Specifically, binding of benzodiazepines to the BZ site leads to increases in the binding-on rate and affinity of one GABA molecule to the first GABA binding site of the receptor (30–32). The increased affinity and rate of binding of GABA correlate with an increase in the frequency of the open-channel bursts. It is important to distinguish that benzodiazepines increase the frequency of channel opening, whereas barbiturates increase the duration of the open channel, and as such, the two classes of agents have distinct molecular effects on the GABA_A receptor. Furthermore, benzodiazepines lack any effects in the absence of GABA, unlike the barbiturates. The reader is referred to Chapter 14 for more detailed descriptions of benzodiazepine binding and function.

PHARMACOLOGIC EFFECTS OF BENZODIAZEPINES All benzodiazepines are capable of producing sedative-hypnotic effects; however, currently only five shorter acting benzodiazepine agents are approved by the FDA for use as sedative-hypnotics: estazolam, flurazepam, quazepam, temazepam, and triazolam. Residual hypnosis and

psychomotor and cognitive inhibition are common side effects associated with their use. All of these agents are agonists of the BZ site of GABA_A receptors containing α and γ subunits. Physiologically, all of the agents significantly reduce sleep latency and the number and duration of nighttime awakenings, resulting in an increase in total sleep time. The clinical use of these agents as sedative-hypnotics is not free from untoward effects, which are primarily manifested as excessive residual next-day sleepiness, tolerance upon long-term use, and withdrawal upon discontinuation. Another effect that is clinically important is the ability of benzodiazepines to produce a high degree of anterograde amnesia, in which recent events are not transferred to long-term memory (33).

STRUCTURE–ACTIVITY RELATIONSHIPS OF BENZODIAZEPINES The SARs of benzodiazepines are described in detail in Chapter 14, and the reader is referred to that chapter for the particulars of benzodiazepine binding requirements. Regarding the SAR of sedative-hypnotic benzodiazepines, all five agents contain the 5-phenyl-1,4-benzodiazepine-2-one backbone required for GABA_A activity, and all include an electronegative 7-position chloro substitution as required on ring A (see Chapter 14 and Fig. 15.6). Similarly, all contain a pendant 5-phenyl ring (ring C) that is required for in vivo agonism and that can be substituted with ortho-electronegative halogens to increase lipophilicity, as is the case with flurazepam, quazepam, and triazolam. Interestingly, parasubstitution on ring C leads to inactive benzodiazepines, suggesting that steric restrictions are important at this site. Of note, several benzodiazepines, including

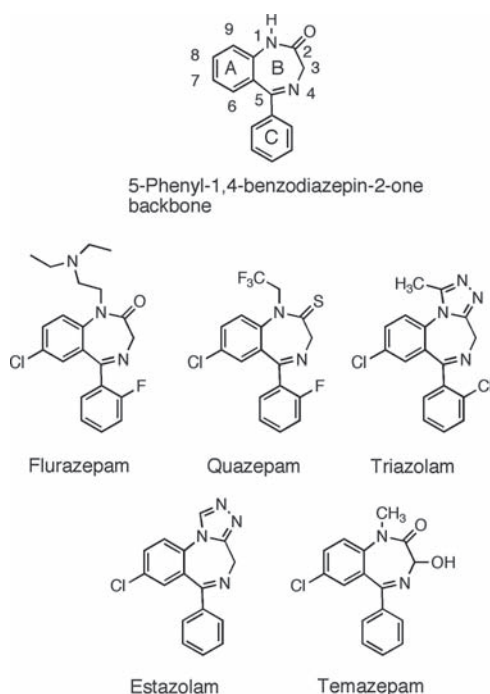


FIGURE 15.6 The benzodiazepine backbone and the benzodiazepines that are indicated for sedative-hypnotic use.

estazolam, are metabolized at the *para*- (4') position, leading to inactivation of the agent. The major structural differences that impart distinct pharmacokinetic properties to the sedative-hypnotic benzodiazepines are located within ring B, the major site of metabolic transformation of these agents. The structural features of ring B that contribute to metabolism are discussed in the following section.

PHARMACOKINETICS AND METABOLISM OF BENZODIAZEPINES Benzodiazepine agents, like the barbiturates, are lipophilic and can be easily absorbed upon oral ingestion and rapidly distributed to the brain. Among the oral benzodiazepine formulations specifically indicated for the treatment of insomnia, flurazepam is absorbed most rapidly, achieving mean peak plasma concentrations within 0.5 to 1.0 hour. Temazepam and triazolam reach mean peak plasma concentrations within 0.5 to 2.0 hours, whereas estazolam and quazepam take longer (Table 15.2).

The sedative-hypnotic duration and side effect potential of benzodiazepines is greatly influenced by metabolism of the parent agent and the activity of resulting metabolites. Flurazepam has an elimination half-life of approximately 2 hours and is primarily metabolized by CYP3A4-mediated *N*-dealkylation and hydroxylation yielding active *N*-desalkyl and *N*-hydroxyethyl metabolites with elimination half-lives of 47 to 100 and 2 to 4 hours, respectively (Fig. 15.7, Table 15.2) (34). After a single 30-mg dose, the *N*-hydroxyethyl is undetectable in the plasma after 24 hours. On the contrary, levels of *N*-desalkylflurazepam are five- to sixfold higher after 7 days of administration than they are after 24 hours,

suggesting buildup of this active metabolite. Conjugation of *N*-hydroxyethylflurazepam in phase 2 reactions allows for urinary excretion, and this conjugate is the major urinary metabolite accounting for up to 55% of a single dose (Fig. 15.7). The long half-life of these metabolites explains the clinical findings that flurazepam exhibits faster sleep latency and decreases in total wake time following several days of use and also why it is still efficacious for one to two nights following discontinuation of the parent. These observations are especially important in elderly populations, in whom the elimination half-life of *N*1-desethylflurazepam is significantly higher (up to 160 hours).

Quazepam is metabolized by microsomal CYP450 oxidases to yield 2-oxoquazepam, which is subsequently *N*-dealkylated to *N*-desalkyl-2-oxoquazepam (Fig. 15.7) (35). Both of these metabolites are pharmacologically active and have long durations, with mean plasma elimination half-lives of 40 and 73 hours, respectively (Table 15.2) (35). Both 2-oxoquazepam and *N*-desalkyl-2-oxoquazepam can be further hydroxylated at the 3-position, yielding 3-hydroxy-2-oxoquazepam and 3-hydroxy-*N*-desalkyl-2-oxoquazepam, respectively. These hydroxylated metabolites are inactive and can be *O*-glucuronidated and excreted readily in the urine (Fig. 15.7). As a consequence of the slow elimination of multiple active metabolites of both flurazepam and quazepam, residual hypnotic effects, including excessive daytime drowsiness, oversedation, and cognitive decline and confusion, are common and clinically relevant adverse effects.

These problems led to the development of newer triazolobenzodiazepines, such as estazolam and triazolam, with high GABA_A receptor affinity and relatively shorter durations. The 1,4-triazolo ring of these congeners prevents oxidative metabolism typical of the benzodiazepines, which as in the case of flurazepam and quazepam results in formation of active metabolites with long elimination half-lives. The presence of the fused fourth ring of these agents also changes the numbering convention, such that numbering priority proceeds around the triazole ring (Fig. 15.8). Estazolam has a longer elimination half-life (10 to 24 hours) compared to flurazepam, but this agent is primarily metabolized by CYP3A4 yielding 4'-hydroxyestazolam as a major metabolite and 1-oxoestazolam as a minor metabolite (Fig. 15.8, Table 15.2). Additionally, a 4-hydroxyestazolam metabolite has been identified in humans and is also detected in the urine at high levels (36). Although these metabolites have weak pharmacologic activity, three important factors prevent them from contributing to any significant sedative-hypnotic effects. The first is the low affinities of the metabolites for the GABA_A receptor. In particular, 4'-hydroxyestazolam is sterically hindered from optimal GABA_A binding, consistent with SAR findings that parasubstitutions of ring C lead to decreased potency. The second factor that impedes the activity of estazolam metabolites is their decreased lipophilicity compared to the parent drug. This decrease facilitates the final

TABLE 15.2 Pharmacokinetic Parameters of Benzodiazepines Approved for Sedative-Hypnotic Use

Drugs	Trade name	Log P ^a	Time to peak conc. (hrs)	Parent elimination half-life (hrs)	Major metabolites (t _{1/2} , hrs)	Predominant CYP isoform(s)
Flurazepam	Dalmane	2.35	0.5–1.0	2 ca.	N ¹ -desalkyl (47–100, active) N ¹ -hydroxyethyl (2–4, active)	3A4
Quazepam	Doral	4.03	2 ca.	39	2-Oxo (40, active) N-desalkyl (73, active)	3A4/2C9
Estazolam	Prosom	3.51	0.5–6.0	10–24	4'-Hydroxy; 4-Hydroxy; 1-Oxo (all inactive)	3A4
Triazolam	Halcion	2.42	<2	1.5–5.5	α-Hydroxy (50–100% active) 4-Hydroxy (inactive)	3A4
Temazepam	Restoril	2.19	1.2–1.6	0.4–0.6	O-glucuronide	–

^a From Sangster Research Laboratories LOGKOW database

factor, which is that the circulating metabolites are found in low concentrations due to direct excretion of the hydrophilic free metabolites or their rapid *O*-glucuronidation, which readily allows for urinary excretion of the agents (Fig. 15.8).

Triazolam has a short duration of approximately 4 hours and is metabolized in humans to six metabolites, one of which, α-hydroxytriazolam, has been shown to retain 50% to 100% of the potency of the parent compound. Although only small amounts of the parent drug

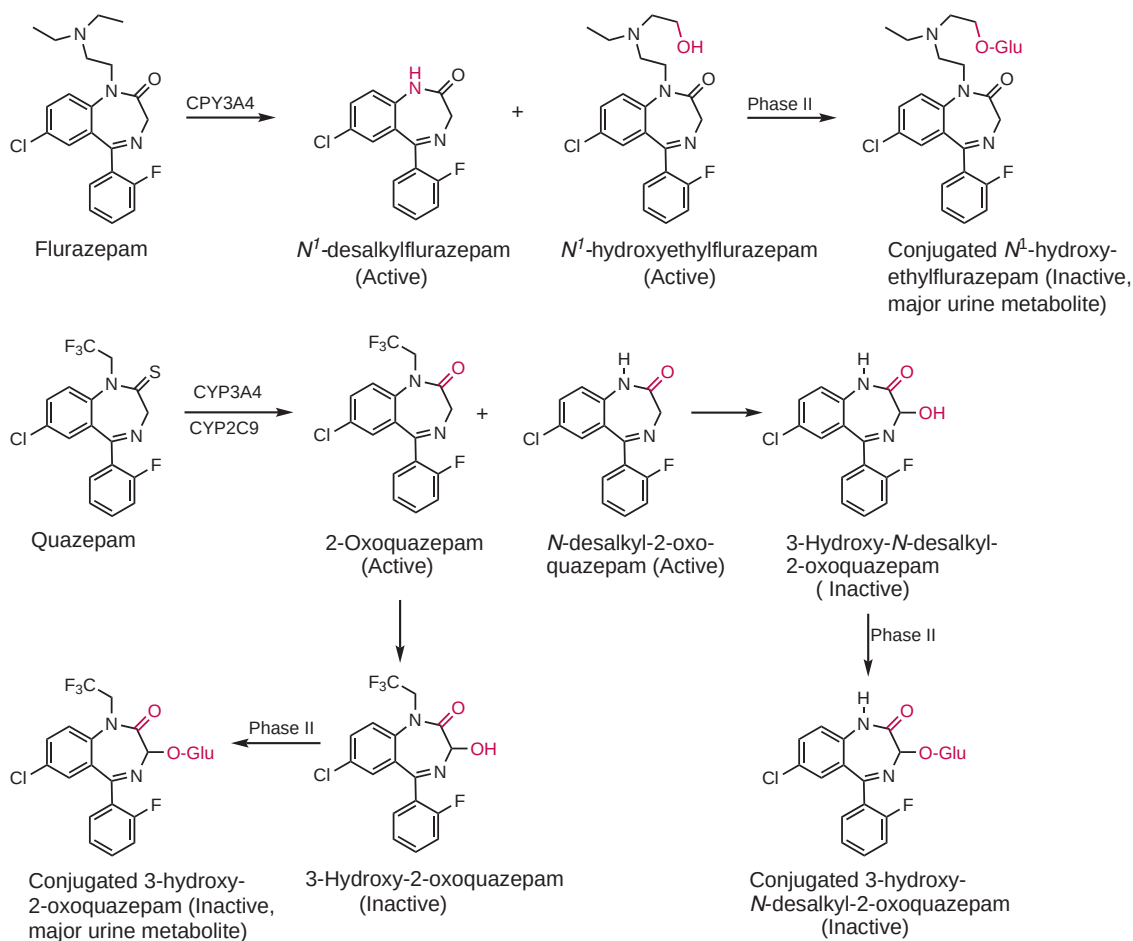


FIGURE 15.7 Metabolism of flurazepam and quazepam leads to formation of multiple long-acting active metabolites. Coloration indicates site of metabolism.

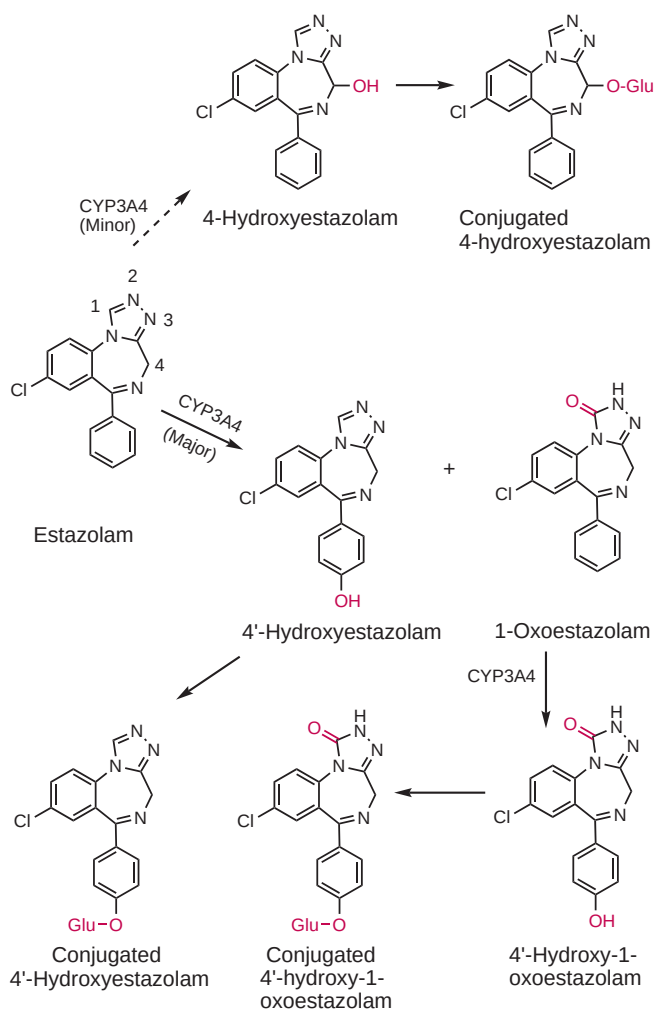


FIGURE 15.8 Metabolism of estazolam. Coloration indicates site of metabolism.

are found in the urine, α -hydroxytriazolam and 4-hydroxytriazolam are the principal glucuronidated metabolites found in the urine, and these account for approximately 80% of triazolam excretion (Fig. 15.9) (37). There is no evidence of accumulation of triazolam metabolites, and the clinical effects of the active α -hydroxytriazolam are unclear because this metabolite has been detected in the plasma primarily in its glucuronidated form. Taken together, the relatively short half-life of the parent coupled with the urinary excretion of glucuronidated metabolites afford triazolam a relatively short duration of action compared with other benzodiazepines.

Temazepam is unique among the sedative-hypnotic benzodiazepines in that it contains a 3-hydroxy group, and as such, the agent circumvents phase I oxidation reactions prior to conjugation and excretion. This feature affords a 0.4- to 0.6-hour elimination half-life and rapid excretion of the parent, while also providing the additional benefit of bypassing oxidative hepatic metabolism (Fig. 15.9). Based on this route of metabolism, temazepam has no active metabolites. In fact, the only significant metabolites that have been described are the *O*-glucuronide of

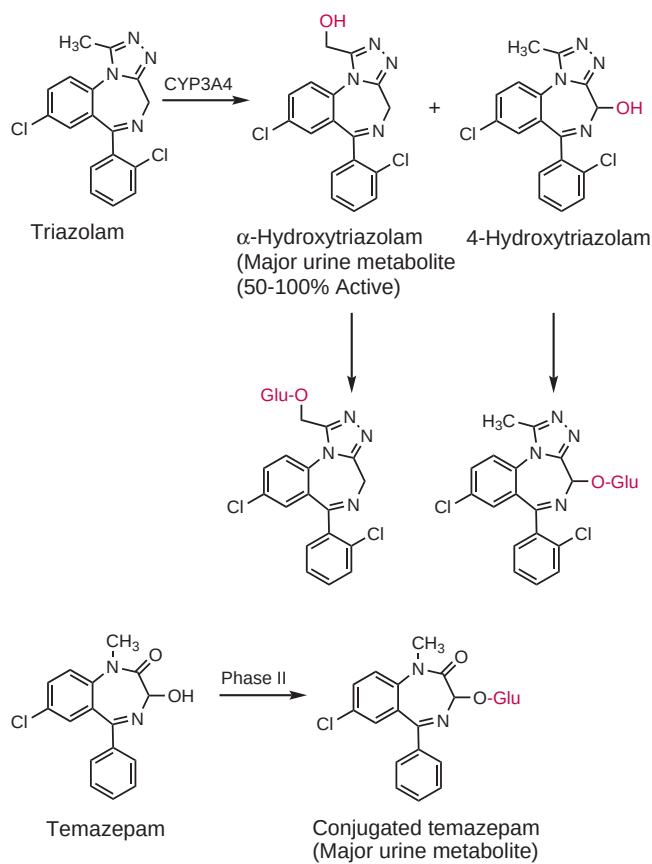


FIGURE 15.9 Metabolism of triazolam and temazepam. Coloration indicates site of metabolism.

the parent, which is the major metabolite (>90%), and the *O*-glucuronide of *N*-desmethyletemazepam (<10%). Importantly, these products are formed independently of CYP450 isoforms and are rapidly excreted in the urine with half-lives of approximately 2 hours.

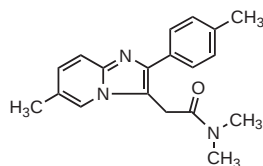
Benzodiazepines are used as sedative-hypnotic agents, as well as general muscle relaxants, anxiolytics, and anti-convulsants, yet as mentioned previously, their use can be confounded by presentation of undesirable side effects. Although death by benzodiazepine overdose is rare and typically only results as a consequence of concomitant use with other CNS depressants, these agents are not free from adverse effects and toxicities. Nearly all benzodiazepines have been reported to cause dose-dependent alterations in behavior, specifically bizarre, uninhibited, and confused behaviors. The propensity of benzodiazepines to cause tolerance and their potential for physical dependence and abuse also limit their use. Additionally, misuse of short-acting hypnotic benzodiazepines such as flunitrazepam (Rohypnol) has had profound social and medicolegal implications due to their use as “date rape” or “robbery” drugs. This illicit use is based on the ability to produce rapid hypnosis in combination with anterograde amnesia. In addition to cognitive effects, benzodiazepines may also exhibit respiratory depressant effects depending on the dose and the duration, and as such, they are contraindicated in patients with pulmonary conditions or sleep apnea. These

factors contributed to the development of nonbenzodiazepine sedative-hypnotics, which were devoid of many of the adverse effects induced by classical benzodiazepines.

Nonbenzodiazepine GABA_A Agonists

Due to advances in molecular biology, genetics, and pharmacology in the late 1980s to early 1990s, numerous lines of evidence demonstrated that different GABA_A receptor subtypes may bring about distinct functions depending on their localization in the brain. Future work revealed that α_2 , α_3 , and α_5 subtype-containing GABA_A receptors (BZ₂) play critical roles in the anxiolytic, anticonvulsant, muscle relaxant, and cognitive impairment properties of classical benzodiazepines, whereas modulation of the α_1 -containing GABA_A receptor subtypes (BZ₁) was shown to be the key to benzodiazepine-induced sedation and hypnosis (38–42). Importantly, these findings revealed that it could be feasible to design functionally selective drugs that act as selective agonists at specific GABA_A subtypes to yield the appropriate pharmacotherapeutic outcomes. For example, selective α_5 - or α_3 -acting agents could have anxiolytic properties while being devoid of sedative effects, whereas selective α_1 -acting agents would behave specifically as sedative-hypnotic agents. Although classical benzodiazepines described here and in Chapter 14 are nonselective and demonstrate sedative-hypnotic, anxiolytic, and other effects, the discovery of novel nonbenzodiazepine compounds with a high degree of selectivity for α_1 subtype-expressing GABA_A receptors allowed for the specific use of these agents as sedative-hypnotics. Currently, three structurally distinct nonbenzodiazepines—zolpidem, eszopiclone, and zaleplon, which are often referred to as the Z-drugs—are approved in the United States for treatment of insomnia, and several other investigational agents are in various stages of clinical trials.

ZOLPIDEM



Zolpidem is an imidazopyridine that is a highly selective agonist of the α_1 subunit-expressing GABA_A receptors (BZ₁), demonstrating 5- and 10-fold greater affinity for α_1 versus α_2 and α_3 subtypes, respectively (43). Similar to the benzodiazepines, zolpidem lacks appreciable affinity for α_4 and α_5 receptors (43). Of the three marketed nonbenzodiazepine drugs, zolpidem also has the greatest functional potency at potentiating GABA currents (44). The hypnotic effects of zolpidem parallel those seen for temazepam and triazolam, but based on its selective pharmacologic profile, zolpidem demonstrates weaker anxiolytic, anticonvulsant, and muscle relaxant effects compared to the classical benzodiazepines. Zolpidem has been shown to significantly improve sleep latency and prolong the duration of

sleep in healthy volunteers and in patients with insomnia. Studies that compare the sleep outcomes of zolpidem with those of benzodiazepines have generally revealed comparable onset and sleep durations, while also showing that zolpidem leads to significantly fewer nighttime awakenings and, at the same time, does not cause residual morning sedation, confusion, or memory impairment (44–48). Further studies demonstrated that zolpidem exhibits significant improvements with regard to sleep latency, and subjective accounts indicated improved sleep quality compared to benzodiazepines. Extensive review of trial data that compare sleep outcomes of zolpidem to various benzodiazepines shows that zolpidem is comparable or superior to various benzodiazepines (49–51).

Zolpidem is currently available in the United States in two different formulations, immediate or controlled release. Oral administration of both formulations leads to bioavailability of approximately 70% and peak maximal plasma concentrations within 1.6 hours (Table 15.3). Although plasma concentrations of zolpidem begin to decrease at approximately 2 hours following administration of the immediate-release formulation, the extended-release formulation produces a more sustained peak plasma concentration that results in higher plasma concentrations over an 8-hour period. Direct comparison of the two formulations shows that plasma concentration profiles are generally identical for 2 hours following administration, after which decreases in blood levels are seen with the immediate-release formulation. The controlled-release formulation allows for this peak blood level to be extended for approximately 1 to 2 additional hours, affording correspondingly sustained blood levels throughout the night. Administration of the agent with food doubles the time required to reach peak and lowers the peak concentration by 30%. Accumulation of zolpidem does not seem to occur upon repeated exposure, regardless of the formulation. Recently, the FDA has approved a zolpidem oral spray formulation, which is distinguished by the fact that it allows for significantly faster time of onset of approximately 15 minutes (Table 15.3) and, as such, is clinically useful for patients with difficulties in achieving sleep initiation.

Structure–Activity Relationships of Zolpidem There have been many medicinal–chemical efforts that have attempted to define the SARs of zolpidem and to characterize interaction of the agent with the BZ₁ receptor. Anzini et al. (52) have integrated these models using the structurally similar, but non- α_1 -selective, imidazopyridine anxiolytic agent alpidem. Based on these results, they have described a freely rotating aromatic ring region (FRAR), an electron-rich region (ERR), an antiplanar region (APR), and a planar aromatic region (PAR) (Fig. 15.10), which contribute to the binding of alpidem to BZ receptors. Studies on each of these regions have demonstrated that the replacement of alpidem's electronegative chloro groups at both the FRAR and PAR with methyl groups, as in zolpidem, does not affect receptor binding affinity, but significantly increases the selectivity for

TABLE 15.3 Pharmacokinetic Parameters of Nonbenzodiazepines Approved for Sedative-Hypnotic Use.

Drugs	Trade name	Log P ^a	Time to peak conc. (hrs)	Parent elimination half-life (mean hrs)	Major metabolites (t _{1/2} , hrs)	Predominant CYP isoform(s)
Zolpidem	Ambien	2.31	Immediate-release 1.6 ^a /1.6 ^b Control-release 1.5 ^c Oral spray 0.25	Immediate-release 2.5 ^a /2.6 ^b Control-release 2.6 ^c	None active	3A4 (major) 2C9, 1A2, and 2D6 (minor)
Eszopiclone	Lunesta	−0.34	1	6.5	<i>N</i> -oxide (inactive) <i>N</i> -desmethyl (active, lower affinity)	1A2, 3A4
Zaleplon	Sonata	1.23	1	1	None active	3A4 (minor)

^a 5 mg tablets.^b 10 mg tablets.^c 12.5 mg tablets.

α_1 -subtype receptors (53). Further studies on the ERR have shown that conversion of either of the imidazole nitrogens to hydrogen bond donors leads to complete loss of selectivity for α_1 subtype GABA_A receptors (52). These data suggest a role for the imidazole nitrogen's ability to behave as hydrogen bond acceptors within the GABA_A receptor binding site. (52). The importance of the ERR is also demonstrated by studies that convert the imidazole of zolpidem to its azaisostere congener. This simple change does not affect binding to α_1 -subtype receptors but decreases binding to α_2 and α_3 such that potency at these subtypes is negligible, similar to that seen for the α_5 subtype (54). The antiplanar group has also been shown to be critical in facilitating binding to GABA_A receptors by allowing for hydrogen bonding interactions. Molecular modeling studies have shown that the antiplanar carbonyl group of zolpidem can hydrogen bond to Ser²⁰⁴ or near Thr²⁰⁶/Gly²⁰⁷ within the backbone of loop C of α_1 subunits, as well as to Arg¹⁹⁴ within loop F of γ_2 subunits, which forms $\alpha_1\gamma_2$ complexes (55). Finally, the antiplanar group also contributes to binding selectivity and affinity, as sterically hindered bulky amide nitrogen substitutions show decreased binding to α_1 -subtype receptors. This study also identified three additional α_1 and three additional γ_2 residues that are within 5 Å of zolpidem on molecular docking simulations, suggesting that other hydrogen-bonding or salt-bridge interactions can occur within the ligand binding domain.

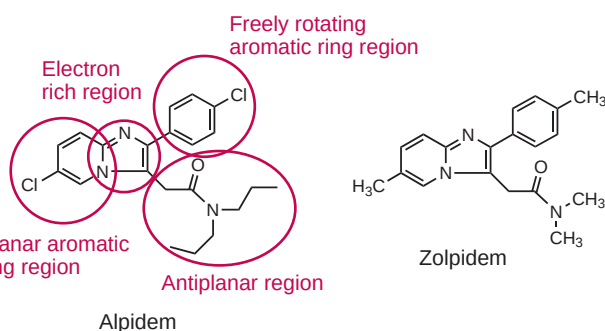


FIGURE 15.10 Structural relationship between of alpidem and zolpidem.

Metabolism of Zolpidem Zolpidem is rapidly eliminated, with a mean half-life of approximately 2.5 hours (Table 15.3). Repeated doses do not seem to accumulate, and elimination of the parent is achieved through extensive metabolism with only trace amounts of unchanged drug being found in urine or feces (56). In humans, zolpidem can be oxidatively metabolized to yield four primary metabolites, M II, M IV, M X, and M XI, all of which lack pharmacologic activity. As shown in Figure 15.11, the major metabolite is formed by CYP3A4-mediated hydroxylation of the *p*-tolyl methyl group yielding M III, which is subsequently oxidized to the corresponding carboxylic acid, M I (57). The M I metabolite is the principal metabolite found in human urine, accounting for approximately 70% to 85% of the administered dose. Microsomal enzyme-mediated hydroxylation of the *p*-methyl substituent on the imidazopyridine backbone can also occur, leading to formation of metabolite M IV, which can be further oxidized to the corresponding carboxylic acid M II, which accounts for approximately 10% of the administered dose. Importantly, M III was also shown to be formed by CYP2D6 and to a lesser degree by CYP1A2, whereas M I V formation by CYP1A2 and CYP3A4 was more modest (57). Because expression of CYP1A2 and CYP2D6 in human liver is generally significantly less than that of CYP3A4, the clinical contribution of these enzymes in transformation of zolpidem is expected to be minor. More recent studies have confirmed these suspicions and demonstrated that the contribution of CYP enzymes to zolpidem metabolism is greatest for CYP3A4 (61%), followed by CYP2C9 (22%), CYP1A2 (14%), and less than 3% for CYP2D6 (58). Because M III and M IV are not detectable in human urine or feces, but the respective carboxylic acids M I and M II are, it was proposed that rapid formation of these secondary metabolites occurs. Indeed, studies in rodents have demonstrated that the conversion of the alcohol metabolites to their corresponding acids occurs very rapidly and requires a nonmicrosomal enzyme, perhaps alcohol dehydrogenase (59). Finally, as shown in Figure 15.11, CYP3A4-mediated oxidation of zolpidem can also lead to formation of the M X and M XI minor metabolites, which are hydroxylated

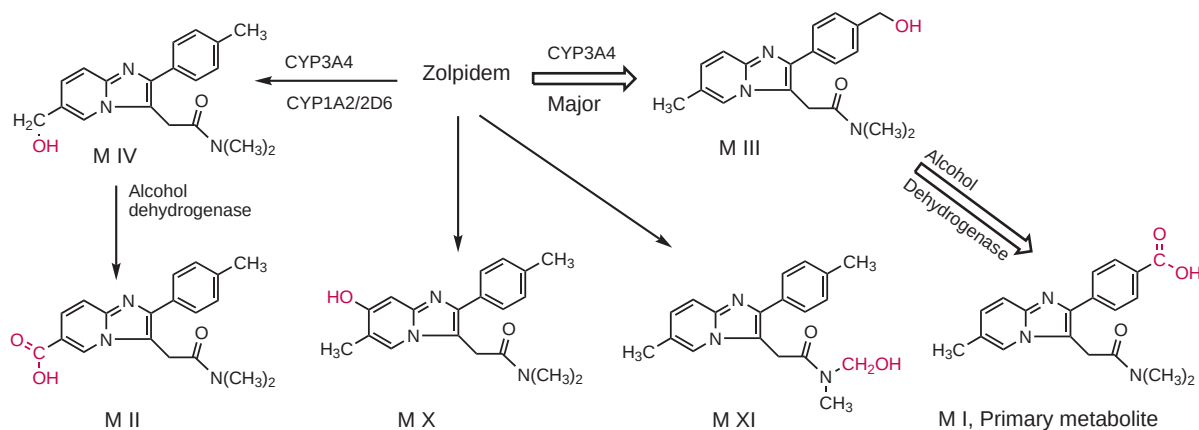
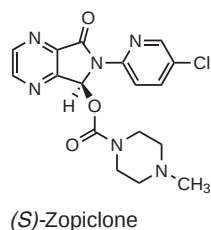


FIGURE 15.11 Metabolism of zolpidem. Coloration indicates site of metabolism.

on the imidazopyridine ring and the substituted amide nitrogen, respectively.

In elderly populations, dose adjustments must be made to account for the 50% increase in elimination half-life of the drug (60). Similarly, in patients with hepatic dysfunction, the plasma concentration of zolpidem doubles, with an increase in elimination half-life to a mean of 10 hours. Interestingly, there has been recent interest in the likelihood of differential gender-based metabolism of zolpidem based on the ability of testosterone to positively contribute to CYP3A4 activity such that higher circulating free testosterone increases CYP3A4 activity (60,61). Given this correlation, women are likely to have higher plasma concentrations of zolpidem and, as such, may be more likely to have higher degrees of zolpidem-mediated adverse effects (62).

ESZOPICLONE



Eszopiclone, a pyrrolopyrazine cyclopyrrolone, is the active (*S*)-enantiomer of zopiclone, a racemic mixture that is no longer marketed in the United States. Eszopiclone significantly improves sleep latency and sleep maintenance, increases the time spent in stage III and IV sleep, and is distinguished by the fact that its approval is not limited to short-term utilization and that it has no potential for development of tolerance or abuse. Several trials have demonstrated that hypnotic efficacy is maintained and eszopiclone is well tolerated upon chronic dosing, lasting up to 12 months (63). In contrast to the racemic zopiclone, eszopiclone is devoid of substantial residual next-day sedative and cognitive effects. Unlike zolpidem, eszopiclone is not subtype selective. Yet, despite this lack of selectivity, it does not behave as classical benzodiazepines with respects to its pharmacodynamics and pharmacologic activity (see next section on SAR). Eszopiclone binds with higher

affinity to α_1 receptors but also has appreciable affinity for α_3 subunits, and thus has hypnotic and other CNS activities. The affinity of the agent for α_1 -subtype receptors is less than the other marketed nonbenzodiazepine agonists.

Eszopiclone is rapidly absorbed following oral administration, and peak plasma concentrations are reached within 1 hour with an elimination half-life of 6.5 hours, the longest of the nonbenzodiazepine hypnotics (Table 15.3). Similar to zolpidem, eszopiclone does not accumulate following repeated administration, and administration after a high-fat meal delays the time to peak plasma concentration by 1 hour and decreases this concentration by more than 20% (64). Eszopiclone dosing in the elderly must also be adjusted to account for 41% greater drug levels and a significantly longer half-life of 9 hours in this population. Similar dose adjustments must be made in patients with liver dysfunction as total exposure to eszopiclone increases twofold at the 2-mg dose, although the time to peak concentration and the concentration itself were unaffected (64).

Structure–Activity Relationships of Eszopiclone Eszopiclone (Fig. 15.12) exhibits 50-fold greater binding affinity for GABAA receptors than does the (*R*)-enantiomer. This leads to significant improvement in GABA potency, but more importantly, enantiomeric separation of the (*S*)-enantiomer seems to reduce adverse effects such as residual sedative effects seen with racemic zopiclone, despite the fact that the racemic mixture has a shorter half-life (5.0 vs. 6.5 hours). Racemization of one enantiomer to the other does not occur in mammals *in vivo*. Although SAR studies for cyclopyrrolones have not been performed to a great degree, it is known that this subclass recognizes a distinct site of the GABAA receptor complex that is allosteric to the recognition site for classical benzodiazepines (65). The regional distribution and specificity of sites labeled by radiolabeled cyclopyrrolones are similar to those labeled by classical benzodiazepine ligands, suggesting that the cyclopyrrolone binding site resides on the BZ site of the GABAA complex (65). However, unlike classical benzodiazepine agonists, binding of cyclopyrrolones is not affected by GABA or barbiturates. Furthermore, radioreceptor displacement binding studies revealed a noncompetitive

interaction of cyclopyrrolones at sites labeled by classical benzodiazepines, whereas displacement of radiolabeled cyclopyrrolones by classical benzodiazepines occurred competitively, suggesting that cyclopyrrolones recognize a site on the BZ receptor complex that is allosteric to the recognition site of classical benzodiazepines (65). Others have suggested that perhaps cyclopyrrolones can interact with one of multiple BZ binding sites on the GABA_A receptor and, in doing so, cause allosteric changes in the ligand affinity at the remaining sites, potentiating further binding (66). With regard to binding within the ligand recognition site, molecular modeling and mutagenesis studies have shown that eszopiclone is within 4 Å of many amino acid residues within the GABA-BZ complex binding pocket, and such proximity allowed anchoring and stabilization by multiple hydrogen bond interactions with Arg144, Tyr209, Tyr159, and various other residues (55). These studies also revealed that the structural requirements for eszopiclone binding are different than those of zolpidem, and these differences may account for the lack of selectivity of eszopiclone (55).

Metabolism of Eszopiclone Eszopiclone is metabolized extensively by CYP3A4 and CYP1A2 isozymes, which yield the primary metabolites (*S*)-*N*-desmethylezopiclone and (*S*)-*N*-oxidezopiclone, respectively. The oxide metabolite is inactive, whereas *N*-desmethylezopiclone exhibits lower potency and affinity at GABA_A receptors and therefore has only very weak hypnotic activity compared to the parent (Fig. 15.12). Interestingly, due to its lower affinity and lack of subtype selectivity, together with its weak sedative effects, this metabolite has been investigated as a novel anxiolytic agent (67). Only small amounts (<7%) of unchanged eszopiclone are excreted in the urine, and greater than 75% of a dose is excreted via this route as inactive transformants of the two primary metabolites.

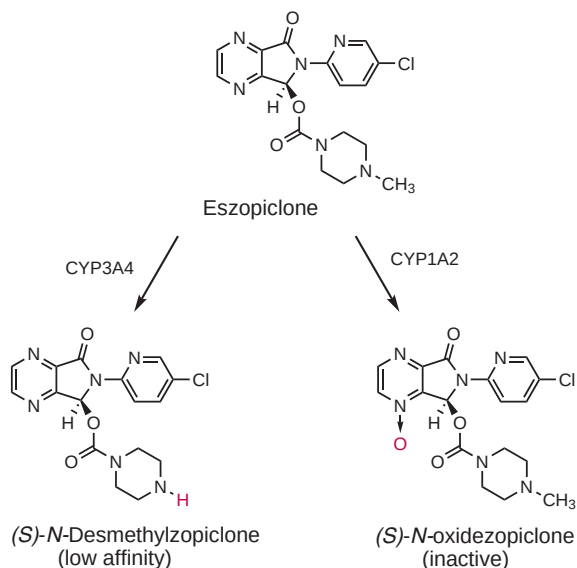
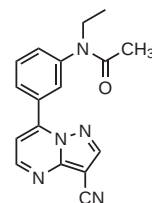


FIGURE 15.12 Metabolism of eszopiclone. Coloration indicates site of metabolism.

ZALEPLON



Zaleplon, a pyrazolopyrimidine, is a hypnotic agent that has a pharmacologic profile similar to zolpidem but is primarily distinguished by its extremely rapid onset and short half-life. Similar to zolpidem, zaleplon has greater affinity at the α_1 - versus α_2 - and α_3 -containing receptors, albeit with one-third to one-half the potency of zolpidem (68). Unlike zolpidem, which does not recognize α_5 -containing receptors with any appreciable affinity, zaleplon is able to potentiate the effects of GABA at α_5 -containing receptors; however, its affinity for α_5 -containing receptors is approximately 15-fold less than for α_1 -containing receptors (68). Hence, zaleplon is considered an α_1 -selective GABA modulator, which lacks the classical benzodiazepine effects on sleep.

The distinguishing features that afford zaleplon a unique place in the therapy of insomnia are its rapid rate of onset combined with its relatively faster rate of excretion. Oral administration of zaleplon leads to a mean peak plasma concentration in less than 1 hour, regardless of dose, the fastest of the nonbenzodiazepines (69). Although zaleplon is completely absorbed following oral administration, it is subject to first-pass metabolism, leading to an absolute bioavailability of 30% (Table 15.3) (69). As with zolpidem and eszopiclone, administration with heavy meals, particularly high-fat foods, delays the time to peak plasma concentration to 3 hours and reduces the peak plasma concentration by 35% (70,71). Following oral administration, zaleplon is rapidly eliminated with a mean elimination half-life of 1 hour, an effect that is delayed significantly in patients with hepatic dysfunction. Taken together, these pharmacokinetic properties facilitate the rapid onset and short duration hypnotic effects (70,71).

Clinically, zaleplon is distinguished by its ability to significantly reduce sleep latency for up to 5 weeks compared to placebo, an effect that can be contributed to its rapid onset (71,72). However, numerous clinical trials have failed to consistently show that a 50 or 10-mg nightly dose of zaleplon significantly improves sleep duration, decreases the number of awakenings, or improves overall sleep quality compared with placebo, likely due to the short half-life of the agent, which precludes hypnotic effects throughout the night. Higher nightly doses of zaleplon (20 mg) do significantly improve either sleep latency or duration, but effects on total sleep quality and number of awakenings compared to placebo have been inconsistent in clinical trials at this dose (71). Tolerance and withdrawal effects have not been reported during short-term treatment (up to 5 weeks), and the agent also seems to be free from rebound insomnia and residual next-day sedative effects, similar to other nonbenzodiazepine hypnotics (72). As a result of

these effects, zaleplon is approved only for patients with difficulties falling asleep; however, an extended-release formulation that could improve sleep maintenance and duration has been under development for some time.

Structure–Activity Relationships of Zaleplon Unlike for zolpidem and eszopiclone, extensive studies that examine the SARs of zaleplon have not been performed with regard to binding potency and interaction within the binding pocket. However, there have been numerous studies that compare the binding and function of zaleplon to the closely related compound indiplon (Fig. 15.13), allowing for deduction of SAR from pharmacologic results. These studies show that zaleplon exhibits approximately 7- and 10-fold greater selectivity for α_1 - over α_3 - and α_5 -containing GABA_A receptors, respectively, whereas indiplon demonstrates only 1.6- and 4-fold selectivity (73). Importantly, indiplon has 3-, 12-, and 7-fold higher affinity at α_1 -, α_3 -, and α_5 -containing receptors, respectively, than does zaleplon. Further work has demonstrated that indiplon is greater than 100-fold more potent than zaleplon at potentiating GABA currents through α_1 -containing receptors but is also 46- and 25-fold more potent at doing so through α_3 - and α_5 -containing receptors, respectively (44,73). This work suggests that the thiophene-2-carbonyl moiety of indiplon (Fig. 15.13) drives higher binding affinity and thus potency, but decreases overall selectivity for one subtype over another. Meanwhile, zaleplon contains only a cyano group at this position, suggesting that electronic rather than steric factors influence selectivity.

Metabolism of Zaleplon As described earlier, zaleplon is subjected to significant first-pass metabolism, which accounts for its low oral bioavailability. Following oral administration of the agent in humans, less than 0.1% of the drug is recovered unchanged in the urine. Animal studies have demonstrated distinct interspecies variability in metabolism of orally administered zaleplon due to interspecies differences in activity of hepatic aldehyde oxidase, which is the principle metabolic enzyme responsible for transformation of zaleplon (74). In rodents and canines, which have decreased expression and activity of aldehyde oxidase, zaleplon is predominantly metabolized by CYP3A4 to *N*-desethylzaleplon, with only minor oxidation of zaleplon by aldehyde oxidase to yield 5-oxozaleplon. In humans, these metabolic routes are interchanged due to greater activity of aldehyde oxidase, and as a result, the major metabolite is 5-oxozaleplon, whereas CYP3A4-mediated transformation, yielding *N*-desethylzaleplon, is a minor route (Fig. 15.13). Both primary metabolites are inactive and, once formed, can either be directly eliminated in the urine or oxidized further and excreted as a glucuronide conjugate. In addition, the *N*-desethylzaleplon metabolite can be rapidly converted, presumably also via aldehyde oxidase, to 5-oxo-*N*-desethylzaleplon (Fig. 15.13) (70). Following oral administration of radiolabeled zaleplon to healthy volunteers, 70% of the dose is recovered in the urine within 2 days, almost exclusively as the two primary metabolites or their glucuronide conjugates (70). The 5-oxozaleplon metabolite

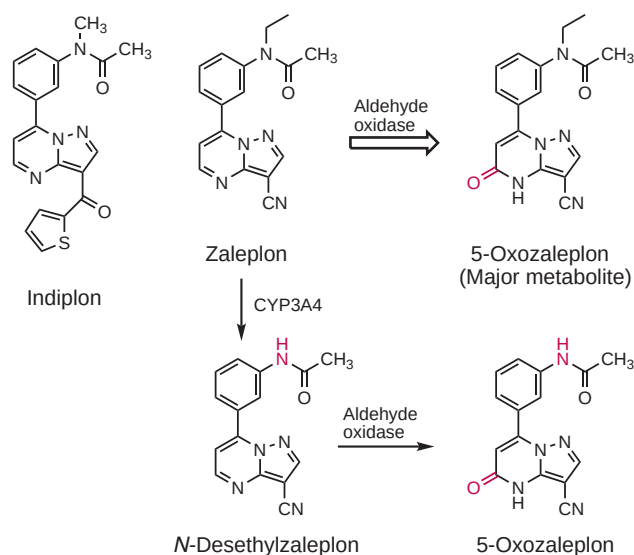


FIGURE 15.13 Structural relationship of zaleplon to indiplon and metabolism of zaleplon. Coloration indicates site of metabolism.

also predominates in the feces, where more than 15% of an administered dose can be found within 6 days.

The nonbenzodiazepines represent major advances in treatment of insomnia. They are highly effective hypnotic agents with negligible residual daytime drowsiness and are generally free from the psychomotor and cognitive side effects of classical benzodiazepines. Despite their primary use at nighttime and their relatively short half-lives, these agents are also not completely devoid of adverse effects. Recently, there has been an increase in reported occurrences of nonbenzodiazepine-induced sleepwalking and associated amnesic sleep-related complex behaviors such as sleep driving, sleep eating, sleep cooking, sleep talking, and sleep sex (75–83). These effects have been reported at therapeutic and supertherapeutic doses and are potentiated by alcohol consumption. Importantly, patients do not have any recall or memory of these events. Although these unintended effects are rare, they are potentially very serious because they can affect social, emotional, and physical health and pose risks for fatal incidents.

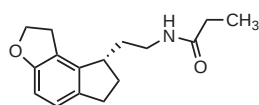
Melatonin Receptor Agonists

As described previously, melatonin is a neurohormone that is primarily synthesized in the pineal gland from its precursor serotonin (Fig. 15.1). Because melatonin synthesis is concurrent with sleep, the increase in endogenous nighttime melatonin levels correlates with the onset of sleepiness. The sleep-promoting and circadian effects of melatonin are due to agonism of both MT1 and MT2 receptors (MT1R, MT2R), both of which are present in very high density (~4 fmol/mg protein) within the SCN (84). Agonism of SCN MT1 receptors directly facilitates inhibition of SCN neuron firing promoting sleep, whereas activation of SCN MT2 receptors affects the circadian rhythm settings related to the central clock (85,86).

Melatonin itself is a poor chemotherapeutic agent due to its poor absorption and low oral bioavailability of less than

10%, as well as its ubiquitous effects on sleep and circadian rhythms. Moreover, because melatonin is a supplement that is not regulated by the FDA, preparations vary in their purity and concentration, and as a result, poor sleep outcomes are often encountered. The significant effects of MT receptor agonism on sleep coupled with the relatively poor nature of melatonin as a drug prompted intense medicinal-chemical efforts to develop novel small-molecule congeners of melatonin that behave as potent agonists of MT receptors. These efforts led to the successful launch of the first in-class melatonin receptor agonist (*S*)-ramelteon, which has an excellent sleep and safety profile. Other investigational MT agonists, such as tasimelteon, are also being developed for treatment of insomnia.

RAMELTEON (ROZEREM)



(*S*)-Ramelteon

(*S*)-Ramelteon (TAK-375) selectively binds to both MT1R and MT2R, recognizing the human receptors with extremely high affinity of 14 and 112 pmol/L, respectively, exhibiting 8- to 10-fold greater affinity for MT1R compared to MT2R and 6-fold greater affinity than melatonin for MT1R (87). The high potency and slight preference for MT1R correlate to the ability of the drug to primarily reduce sleep latency as opposed to regulation of phase-circadian rhythms. With regard to functional pharmacology, in Chinese hamster ovary cells ectopically expressing the human MT1R and MT2R, ramelteon was 4- and 17-fold more potent at inhibiting cyclic AMP production compared to melatonin, respectively, consistent with observations that demonstrate coupling of MT receptors to $G_{\alpha_{i/o}}$ proteins (87). Moreover, ramelteon does not have any appreciable affinity for a third physiologically relevant melatonin binding site known as MT3R, which is a melatonin-sensitive quinine reductase that is not involved in the sleep/wake functions (88,89). Taken together, the absence of binding to brain receptors other than MT1R/MT2R and the potent activity of ramelteon at these melatonin receptors make ramelteon an efficacious sleep-inducing agent that is free from many of the CNS side effects common to other sedative-hypnotics.

Animal studies demonstrate that ramelteon significantly decreases wakefulness and increases both short-wave and REM sleep compared with placebo (88,90). In clinical trials, no effects on short-wave sleep have been noted, but the agent leads to significant decreases in sleep latency and increases in total sleep duration over a 5-week period compared to placebo (91–93). Importantly, ramelteon, consistent with its melatonin receptor-related mechanism, which is free of endogenous GABAergic signaling, does not produce residual next-day sedation or decreases in psychomotor function (93). Likewise, due to its unique mechanism, ramelteon does not hinder learning,

cognition, and memory like the benzodiazepine agents do, and it also lacks the potential for abuse, does not lead to tolerance, and poses no risk of withdrawal or rebound insomnia upon discontinuation (92–94). One potential adverse effect that can be predicted for melatonin receptor agonists is based on the ability of melatonin to influence prolactin and testosterone levels via agonism of MT1 and MT2 receptors located within endocrine and reproductive tissue. Similarly, ramelteon has been reported to affect mean free and total testosterone levels (88). Further studies are necessary to gauge the clinical consequences, if any, that exogenous melatonin receptor agonists have on endocrine and reproductive function. In summary, from a sleep therapy standpoint, melatonin receptor agonists such as ramelteon have highly efficacious hypnotic effects while being free of many of the cognitive and residual effects common to GABA_A modulating agents.

Structure–Activity Relationships of Ramelteon SARs for melatonin and its congeners have been reviewed at length previously (95–98). Early studies demonstrated that the indole backbone of melatonin was not required for activity so long as it was replaced by an aromatic isostere as in ramelteon (which contains an indane) or as in the investigational MTR agonists tasimelteon and agomelatine, which are currently undergoing clinical trials for treatment of insomnia (Fig. 15.14) (99). It has been proposed that the importance of the aromatic ring system here is to offer optimum distance between the amide side chain at position 3 and the 5-methoxy substitution (100,101). The aromatic moiety has also been suggested to interact with aromatic receptor residues within the binding pocket through *pi*–*pi* stacking. Although the aromatic portion contributes to spacing and likely *pi*–*pi* interactions, the 3-position amide and the 5-methoxy side chains are responsible for binding and functional activation of the receptors by interacting with two proposed binding pockets. The amide group is critical for agonist activity at both MT receptors and is thought to interact with key serine (Ser¹¹⁰ and Ser¹¹⁴) and asparagine (Asn¹⁷⁵) residues in transmembrane helix (TMH) III of MT1 and TMH IV of MT2, respectively (102). Meanwhile, the 5-methoxy moiety is critical for functional effects. It has been proposed that the methoxy oxygen interacts with His¹⁹⁵ and His²⁰⁸ within TMH VI of MT1R and MT2R, respectively (101,103). Movement of the methoxy group to the 4-, 6-, or 7-positions significantly decreases functional activity, suggesting that distance to these conserved histidine residues is critical in maintaining binding affinity and functional activity (97). Additionally, Val¹⁹², which is located nearly one turn above His¹⁹⁵ within the ligand binding domain has been shown to be involved in binding the methyl component of the methoxy group (101). Finally, the binding pocket of MTRs is highly stereoselective, an important feature that must be considered for drug design. As a consequence of the three-dimensional requirements of binding to MT receptors, ramelteon is dispensed as the enantiomerically pure (*S*)-enantiomer, which has 500-fold greater affinity for MT1 than does (*R*)-ramelteon.

Metabolism of Ramelteon (S)-Ramelteon is rapidly absorbed after oral administration, and mean peak plasma concentrations are achieved at 0.75 hours. In a study in healthy volunteers who were administered radiolabeled ramelteon, 84% of recovered radioactivity was found in the urine and 4% in the feces, suggesting that at least 84% of the drug was absorbed. However, ramelteon is subject to extensive first-pass metabolism, leading to an absolute oral bioavailability of only approximately 2% (104). Ramelteon has a short elimination half-life of 1 to 2.5 hours, and once-nightly dosing does not lead to accumulation of the drug. The time to peak plasma concentration was delayed by 45 minutes upon administration with food, and the peak plasma concentrations were decreased by 22% in this case (104). Ramelteon is primarily metabolized by oxidation in phase I reactions, with secondary metabolites being excreted as glucuronide conjugates (105). Hepatic CYP1A2 is the major isoform responsible for transformation (49%), whereas CYP2C19 (42%) and CYP3A4 (8.6%) isoforms are also contributors to ramelteon metabolism in the liver. On the contrary, in the intestines, only CYP3A4 contributes to transformation (106). The major metabolite of ramelteon in humans is the hydroxylated propionamide M II metabolite (Fig. 15.15), which is active, has a 2- to 5-hour half-life, and has 20- to 100-fold greater systemic exposure than that of the parent, suggesting slower removal from the circulation (104,105). Moreover, this metabolite has one-fifth to one-tenth the binding affinity of the parent for human MT1 and MT2 receptors and shows potent hypnotic effects in animals, suggesting that it may contribute to the sedative-hypnotic effects of ramelteon (88). In addition to the M II metabolite, ramelteon can be oxidized to the ring-opened M I metabolite or transformed to a keto-metabolite, M III (Fig. 15.15), both of which are inactive. Biotransformation of the M II and M III metabolites by sequential oxidation steps can lead to formation of M IV. The rank order of prevalence of the four metabolites in human serum is M II, M IV, M I, and M III, and all three of the hydroxylated metabolites can be excreted as glucuronide conjugates (Fig. 15.15). Importantly, ramelteon is also subject to a significant drug interaction upon coadministration with the antidepressant fluvoxamine, a strong CYP1A2 inhibitor. Coadministration of these agents leads to a greater than 100-fold increase in total systemic exposure of ramelteon and should be avoided.

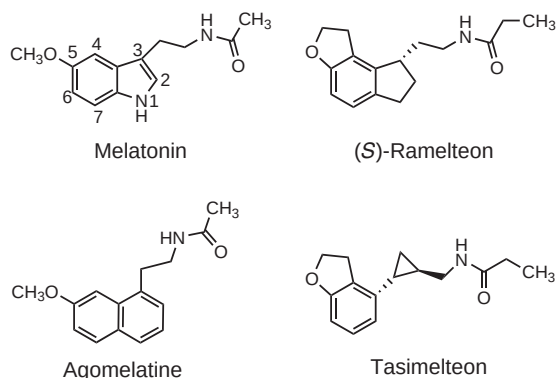


FIGURE 15.14 Structures of melatonin and the melatonin receptor agonists (S)-ramelteon, agomelatine, and tasimelteon.

Histamine H_1 Receptor Antagonists

The first generation of ethanolamine ether histamine H_1 receptor antagonists (i.e., antihistamines) that cross the blood-brain barrier are used for acute insomnia due to their sedation-promoting side effects. In particular, diphenhydramine and doxylamine, both high-affinity (low-nanomolar) H_1 receptor antagonists, are sold as over-the-counter sleep aids. Recently, the tricyclic antidepressant doxepin, which is a potent subnanomolar affinity H_1 antagonist, has also been approved for treatment of insomnia at lower doses than those used for depression (Silenor). The benefit of these agents for treatment of insomnia comes from H_1 receptor antagonism within the TMN of the posterior hypothalamus, where the normal release of histamine during the day causes arousal, and its decreased release at night reduces arousal responses. Antagonism of H_1 receptors within the TMN promotes sedation and drowsiness by inhibiting TMN outflow to other brain structures such as the dorsal raphe nucleus, paraventricular nucleus and locus coeruleus. Overall, antihistamines bring about increased drowsiness and sedation with marginal beneficial effects on sleep latency and total sleep time. There is very little in the way of rigorous clinical trial data that support their sedative-hypnotic efficacy, and to the contrary, some results reveal that these agents disrupt sleep architecture by delaying the onset to, and duration of, REM sleep. Regardless, these drugs are used heavily as sedative-hypnotics, and in the case of diphenhydramine and doxylamine, patients are able to readily self-medicate due to the ease in their availability. While diphenhydramine, doxylamine, and doxepin also exhibit antimuscarinic activity, leading to corresponding side effects such as dry mouth, urinary retention, and blurred vision, a major side effect of these agents is excessive daytime drowsiness and residual next-day sedation, which are not uncommon. These next-day effects can lead to impaired cognition and performance and are attributed to the relatively longer half-lives. Finally, nightly use of first-generation antihistamines as sleep aids has been associated with tolerance to the hypnotic effect over long-term use. These agents are discussed further with respects to their SARs and metabolic transformations in Chapter 32.

FUTURE SEDATIVE-HYPNOTICS

As demonstrated within this chapter, GABA $_A$ receptor-acting hypnotics continue to be the mainstay for the pharmacotherapeutic management of insomnia. Although the pharmaceutical industry has a number of newer nonbenzodiazepine GABA $_A$ agonists such as indiplon under development, these agents are likely to have similar pharmacologic profiles as the currently used Z-drugs. The arrival of ramelteon to the marketplace was a major advance in development of sedative-hypnotics because it represented the first new class to be approved in several decades. Despite this relative paucity of novel sedative-hypnotics, there are several promising new CNS targets that have been exploited for their effects on the sleep/wake cycle, including 5-HT $_{2A}$, orexin, and neuropeptide S (NPS) receptors. As a result, the pharmaceutical industry

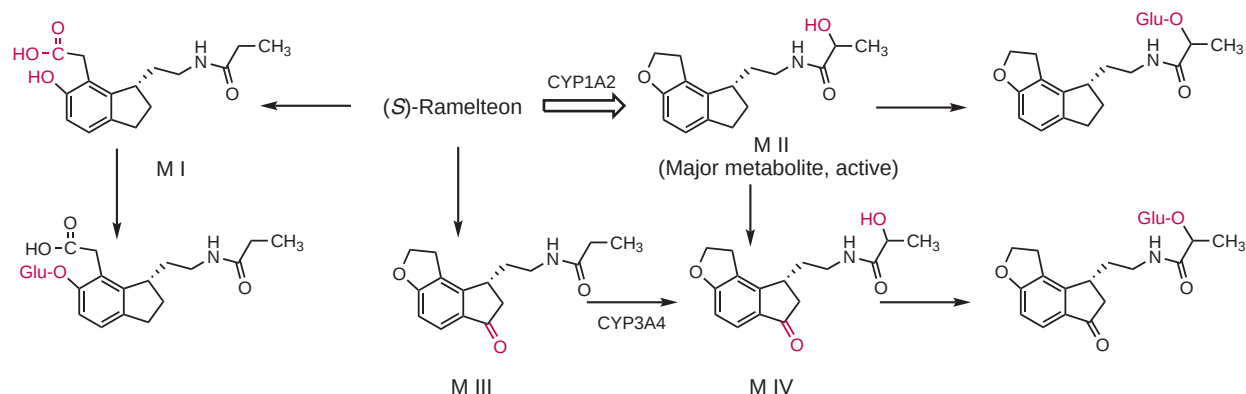


FIGURE 15.15 Metabolism of (S)-ramelteon. Coloration indicates site of metabolism.

has developed small-molecule antagonists of these targets with the hope of inducing and promoting sleep. Ritanserin, eplivanserin, and esmirtazapine are all 5-HT_{2A} receptor antagonists that are currently under investigation in various stages of clinical trials for treatment of insomnia. Because the effects of serotonin in the human brain are both broad and profound, further studies with these agents should shed light on the complexities of this system in regard to its regulation of the sleep/wake cycle.

Numerous lines of preclinical evidence support a key role of the orexin neuropeptides in promotion of wakefulness and alertness. Orexin neurons have been shown to extensively innervate key sleep/wake-regulating brain regions, in particular, the histaminergic neurons of the TMN. Binding of orexins to their cognate GPCRs, OX1 and OX2, has been demonstrated to promote wakefulness and alertness, suggesting that orexin receptor antagonism may have opposite effects. Indeed, almorexant is a first-in-class

orexin receptor antagonist that is currently in phase III clinical trials for treatment of insomnia. This agent exhibits potent hypnotic effects in animals and humans, including those with insomnia, with minimal adverse effects and tolerability issues, suggesting that orexin receptor antagonism may provide a unique approach to treat insomnia.

Similarly, NPS is another recently described neuropeptide that is expressed in distinct brain regions, most notably in the locus coeruleus. As with the orexins, injection of NPS has been found to significantly increase wakefulness and suppress all stages of sleep. These effects are mediated by the recently discovered NPS receptor, another GPCR, and suggest that antagonism of NPS receptors could lead to a hypnotic effect. Because GPCRs are relatively amicable drug targets—approximately 40% to 50% of all marketed drugs target a GPCR—these novel targets offer promising strategies for the development of new sedative-hypnotic agents.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

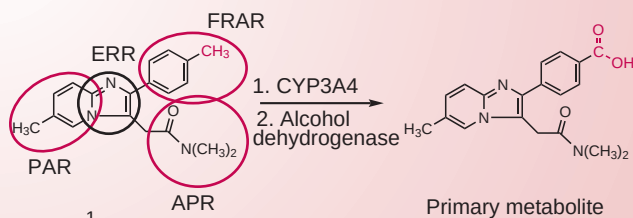
Susan W. Miller, PharmD, CGP, FASCP

The pharmacist's review concluded that one of the warnings associated with the use of zolpidem includes new onset of behavioral changes and complex behaviors. Sleep-eating is one of these complex behaviors and the consumption of additional food may account for the weight gain and the elevated hemoglobin A_{1c} level. The pharmacist's recommendations for the MTM were to taper the zolpidem over 4 days to discontinue its use and to initiate therapy with ramelteon dosed at 8 mg po 30 minutes before bedtime on day 5 of the taper. (Use of ramelteon is also associated with the development of complex behaviors.) The pharmacist also recommended discontinuation of the itraconazole because of the potential for drug interaction with the verapamil and zolpidem (now discontinued) and initiation of topical therapy with ciclopirox for treatment of the fingernail onychomycosis. WK should continue to monitor her diabetes and weight and return for a follow-up visit in 3 months. No changes were suggested for the diabetes or hypertension therapies at this time.

Chemical Analysis

S. William Zito and Victoria Roche

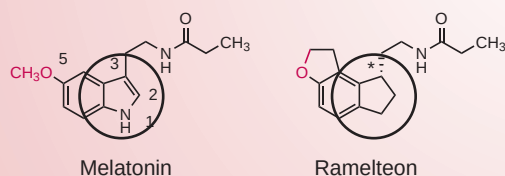
Zolpidem is an imidazopyridine, which is a highly selective agonist of the α_1 -subunit of the GABA_A receptors. SAR studies have revealed four regions responsible for zolpidem's binding to the GABA_A receptors: 1) the freely rotating aromatic ring (FRAR), 2) an electron-rich region (ERR), 3) an antiplanar region (APR), and 4) a planar aromatic region (PAR). The methyl substituents found on FRAR and PAR regions and the hydrogen-bond-accepting imidazole nitrogen are responsible for zolpidem's α_1 selective ability.



SCENARIO: OUTCOME AND ANALYSIS (Continued)

Zolpidem is a short-acting nonbenzodiazepine hypnotic that works like the BZs by binding to the γ -aminobutyric acid receptors. Zolpidem is used for the short-term treatment of insomnia. It works quickly (usually within 15 minutes) and has a short half-life (2 to 3 hours). The onset of zolpidem toxicity (behavioral changes and complex behaviors such as sleep-eating) is most likely caused by the addition of itraconazole, which is a CYP 3A4 inhibitor. Because zolpidem is metabolized by CYP 3A4, its metabolism will be inhibited by itraconazole and lead to behavioral toxicity.

Ramelteon is an analog of melatonin, the biological activity of which is related to circadian rhythm and induction of sleep. There are three melatonin receptors (MT₁, MT₂, and MT₃) found in the CNS. MT₁ is associated with induction of sleep and MT₂ with setting the body's "biological clock." The role for MT₃ has yet to be determined. Ramelteon changes the indole ring of melatonin to an indane ring and constrains the 5-methoxy group into a furan ring to form an indenofuran moiety.

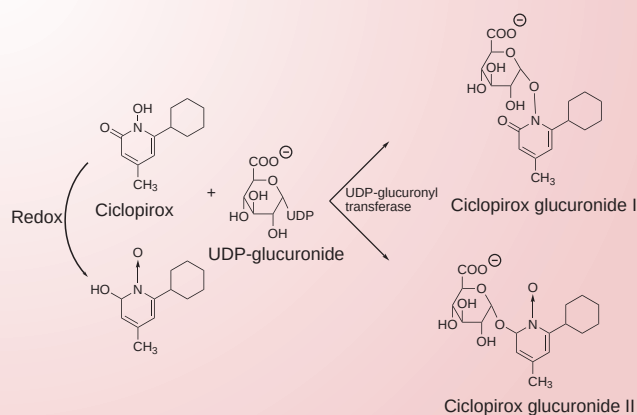


The amide side chain on position 3 is critical for agonist activity at both MT₁ and MT₂ receptors. Ramelteon contains a stereochemical center at position 3 and the (*S*)-enantiomer has 500-fold greater affinity for MT₁ than does (*R*)-ramelteon.

Ramelteon is primarily metabolized by phase I oxidation followed by excretion of the glucuronide conjugates. CYP1A2

is the major isoform responsible for transformation (49%), whereas CYP2C9 (42%) and CYP3A4 (8.6%) isoforms also contribute. The low percent of metabolism of ramelteon by CYP3A4 would not preclude the use of itraconazole; however, because WK is taking verapamil, which is metabolized by CYP3A4, the decision to discontinue itraconazole is justified.

Ciclopirox is a hydroxylated pyridinone that is used topically to treat superficial dermatophytic infections, primarily onychomycosis. Ciclopirox has a unique mechanism of action through chelation of polyvalent cations, such as Fe³⁺, which causes inhibition of a number of metal-dependent enzymes within the fungal cell. Ciclopirox is metabolized by phase II glucuronidation and does not inhibit any CYP450 isoforms and would not interfere with the metabolism of the other of WK's medications.



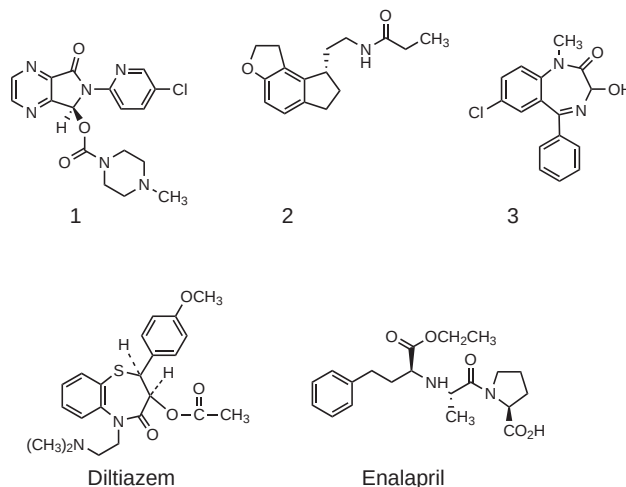
CASE STUDY

S. William Zito and Victoria Roche

BB is a 40-year-old school teacher who suffers from chronic insomnia. She experiences 3 to 5 awakenings every night and then finds it difficult to fall asleep again. She also experiences daytime fatigue and is unable to concentrate on her teaching responsibilities. Upon questioning, BB vaguely remembers being involved in a stressful family inheritance dispute years ago just before the onset of her sleep difficulty. BB also reports anxiety at bedtime because of the prospect of another sleepless night. BB has a history of hypertension, which is under control by treatment with diltiazem HCl (480 mg daily) and enalapril maleate (10 mg daily).

BB has been diagnosed with psychophysiological insomnia, sometimes referred to as "behavioral insomnia." The mainstay of treatment for this type of chronic insomnia is behavioral therapy, which can take 2 to 3 months to learn and provides clinical benefit. BB's physician wants to prescribe a sedative-hypnotic treatment until she experiences the benefits of the behavioral therapy. Evaluate the three choices for use in this case, given below.

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

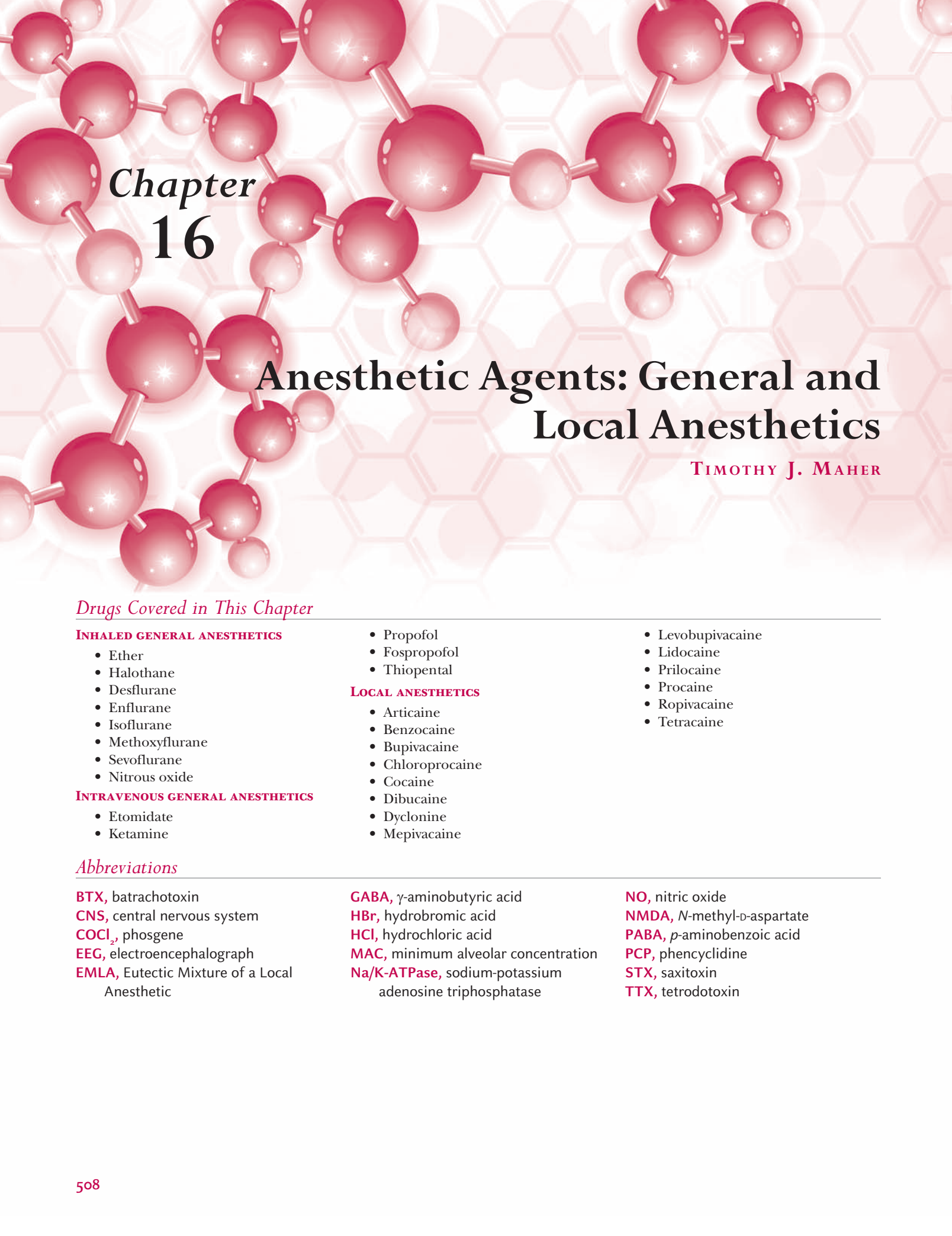


2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 16

Anesthetic Agents: General and Local Anesthetics

TIMOTHY J. MAHER

Drugs Covered in This Chapter

INHALED GENERAL ANESTHETICS

- Ether
- Halothane
- Desflurane
- Enflurane
- Isoflurane
- Methoxyflurane
- Sevoflurane
- Nitrous oxide

INTRAVENOUS GENERAL ANESTHETICS

- Etomidate
- Ketamine

- Propofol
- Fospropofol
- Thiopental

LOCAL ANESTHETICS

- Articaine
- Benzocaine
- Bupivacaine
- Chloroprocaine
- Cocaine
- Dibucaine
- Dyclonine
- Mepivacaine

- Levobupivacaine
- Lidocaine
- Prilocaine
- Procaine
- Ropivacaine
- Tetracaine

Abbreviations

BTX, batrachotoxin

CNS, central nervous system

COCl₂, phosgene

EEG, electroencephalograph

EMLA, Eutectic Mixture of a Local Anesthetic

GABA, γ -aminobutyric acid

HBr, hydrobromic acid

HCl, hydrochloric acid

MAC, minimum alveolar concentration

Na/K-ATPase, sodium-potassium adenosine triphosphatase

NO, nitric oxide

NMDA, *N*-methyl-D-aspartate

PABA, *p*-aminobenzoic acid

PCP, phencyclidine

STX, saxitoxin

TTX, tetrodotoxin

SCENARIO

Paul Arpino, R.Ph.

CDL is a 70-year-old obese man scheduled for carpal tunnel surgery. A review of his medical file indicates a history of obstructive sleep apnea and benign prostatic hypertrophy (BPH). CDL sleeps with a continuous positive pressure airway device and his BPH is treated with tamsulosin, 0.4 mg daily. Given that patients with sleep apnea are at high risk for respiratory depression, the clinical team decides that a peripheral nerve block would be a better alternative to both neuraxial and general anesthesia.

During the preoperative assessment before the scheduled day of surgery, the team discovers that CDL has an undefined allergy to procaine (Novocain) and that he experienced severe blistering after a dental procedure many years ago and was told he cannot receive “drugs like Novocain again.”

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

Anesthesia, defined as a loss of sensation with or without loss of consciousness, can be effectively achieved with a wide range of drugs with very diverse chemical structures. The list of such compounds includes not only the classic anesthetic agents, such as the general and local anesthetics, but also many central nervous system (CNS) depressants, such as analgesics, sedative/hypnotics (barbiturates and benzodiazepines), anticonvulsants, and skeletal muscle relaxants. Although various mechanisms of action are attributed to these agents, ultimately they all produce their anesthetic actions by interfering with conduction in sensory neurons and sometimes also motor neurons. Many of these agents are routinely used today in clinical practice to facilitate surgical and medical procedures. This chapter will focus on those agents typically classified as “general” and “local” anesthetics.

GENERAL ANESTHETICS

Prior to the mid-1800s, pain-producing surgical and dental procedures typically were undertaken without the aid of effective anesthetic agents. Chemical methods available at the time included intoxication with ethanol, hashish (cannabis), or opium, whereas physical methods included packing a limb in ice, creating ischemic conditions with tourniquets, inducing unconsciousness by a blow to the head, or the most common technique, employing strong-armed assistants to hold down the helpless patient during the entire painful surgical procedure. Additionally, at this time, many practicing physicians had been erroneously taught that pain was a requirement for effective healing; therefore, the observation of a patient in terrible pain was viewed as part of the normal healing process. These factors, along with the lack of knowledge regarding aseptic techniques or the availability of suitable infection-fighting agents, made surgical procedures a last resort approach to treating disease.

There have been many accounts of the first demonstration by the Hartford dentist Horace Wells of the use of nitrous oxide as a general anesthetic for surgery in 1844. Wells first observed the anesthetic actions of

nitrous oxide at a public demonstration of “laughing gas.” One of the volunteers, a pharmacy clerk named Samuel Cooley, injured his leg while under the influence of this gas and appeared to experience no pain. The next day, Wells inhaled the gas himself and, with the aid of a colleague, had one of his own teeth extracted without any sensation of pain. Wells then began routinely using nitrous oxide for dental procedures in his own practice. In 1845, he attempted to demonstrate the anesthetic effects of nitrous oxide at the Massachusetts General Hospital in Boston. This demonstration was considered to be a failure, however, because the patient cried out in the middle of the procedure. Following this unfortunate incident, the use of nitrous oxide was minimal until it resurfaced in dental practice during the mid-1860s, when it was combined with oxygen and made available in steel cylinders. This gas is still commonly used today, especially in combination with other anesthetic and analgesic agents.

The general anesthetic that gained greatest popularity shortly after the failed demonstration of Wells was diethyl ether. William Morton, a Boston dentist, was familiar at the time with the use of nitrous oxide by Wells. He also had heard of the interesting effects of diethyl ether and began to experiment on animals and himself with this volatile liquid. In 1846, he was allowed an opportunity to demonstrate the anesthetic actions of diethyl ether at, again, the Massachusetts General Hospital. In the famed “Ether Dome,” which still stands today, Morton administered diethyl ether with a specially designed delivery device to the nervous patient, and the surgical procedure was performed without apparent pain. Following this demonstration, word of its success spread quickly, and soon, dental and medical practices throughout the United States and Europe were employing diethyl ether as a general anesthetic agent. Today, diethyl ether is no longer used in procedures because of its toxicity and dangerous physical properties (e.g., it is flammable and explosive!).

Other general anesthetic agents that enjoyed early popularity were chloroform and cyclopropane. Chloroform vapor depresses the CNS of a patient, allowing a doctor to perform various otherwise painful surgical procedures.

CLINICAL SIGNIFICANCE



Anesthetics are a structurally diverse class of medications that enable clinicians to perform surgery and other noxious procedures.

Understanding the essential components of the anesthetic state (i.e., immobilization, analgesia, and amnesia) as well as the medicinal chemistry of the various anesthetic agents allows the clinician to optimize therapy to meet patient specific needs. The patient undergoing a minimally invasive ambulatory surgical procedure may only require a local anesthetic with adjunctive pain control. Alternatively, patients undergoing a major surgical procedure may require general anesthesia with several different classes of anesthetics as well as several adjunctive medications to counteract deleterious emergence reactions related to the anesthetics. In both cases, a thorough understanding of the basic chemical properties of the drugs and their respective mechanisms of action will prove invaluable to making appropriate clinical decisions.

The practice of anesthesia is typically not considered to be therapeutic; therefore, the practice as well as the development

of new agents is aimed at reducing adverse reactions, maintaining optimal physiologic conditions during procedures, and minimizing postoperative complications related to the procedure itself. The study of medicinal chemistry gives us hope for future treatment options, and knowledge of structure-activity relationships fosters the development of new medications and administration techniques. New generations of drugs are being created by modifying the structures of existing compounds to improve the side effect and pharmacokinetic profiles. Clinicians will have to stay up to date with new developments in anesthesia practices, the molecular actions of anesthetics, and the pharmacokinetic properties of the drugs to provide the best therapeutic outcomes for patients.

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In 1847, the Scottish obstetrician James Young Simpson first used chloroform for general anesthesia during childbirth. The use of chloroform during surgery expanded rapidly thereafter in Europe. In the United States, chloroform replaced diethyl ether as an anesthetic at the beginning of the 20th century; however, it was quickly abandoned due to its cardio and CNS toxicity. Cyclopropane is a hydrocarbon with anesthetic properties like diethyl ether, except it is also explosive and is no longer used. As described later in this chapter, the inhalational general anesthetic agents used today are typically hydrocarbons and halogenated ethers (Cl, Br, or F); nitrous oxide is the exception. Table 16.1 lists the characteristics of the “ideal” general anesthetic agent. Unfortunately, the agent that fulfills all these characteristics is currently unknown.

Stages of General Anesthesia

The ideal general anesthetic state is characterized by a loss of all sensations and includes analgesia and muscle relaxation. Neuronal depression in specific areas of the CNS is believed to be largely responsible for such an anesthetic state. The areas involved include many cortical regions that are represented by excitatory pyramidal cells and inhibitory/excitatory stellate cells. Excitation of the pyramidal cells helps to maintain consciousness, whereas the degree of inhibition or excitation of stellate cells determines the overall activity level of the pyramidal cells with which they synapse. As the concentration of the anesthetic agent increases in the brain, the degree of overall neuronal depression also increases, resulting in progressively deeper stages of anesthesia. Based on observations using diethyl ether, Guedel in 1920 originally described this progression as four distinct stages,

and Gillespie subsequently further subdivided these stages (Fig. 16.1), as described in the following sections.

Stage 1: Analgesia

Characterized by a mild depression of higher cortical neurons, this stage is suitable for minor surgical procedures that do not require significant neuromuscular relaxation. Depression of thalamic centers probably accounts for the observed analgesia, because many of the neuronal systems that mediate pain sensation traverse through this anatomic area. Some general anesthetic agents do not possess significant analgesic activity, but they all produce a loss of consciousness that, in turn, can produce some degree of insensitivity to painful stimuli.

TABLE 16.1 Characteristics of the Ideal General Anesthetic Agent

Rapid and pleasant induction of surgical anesthesia
Rapid and pleasant withdrawal from surgical anesthesia
Adequate relaxation of skeletal muscles
Potent enough to permit adequate oxygen supply in mixture
Wide margin of safety
Nontoxic
Absence of adverse effects
Nonflammable/nonexplosive
Chemically compatible with anesthetic devices
Nonreactive
Inexpensive

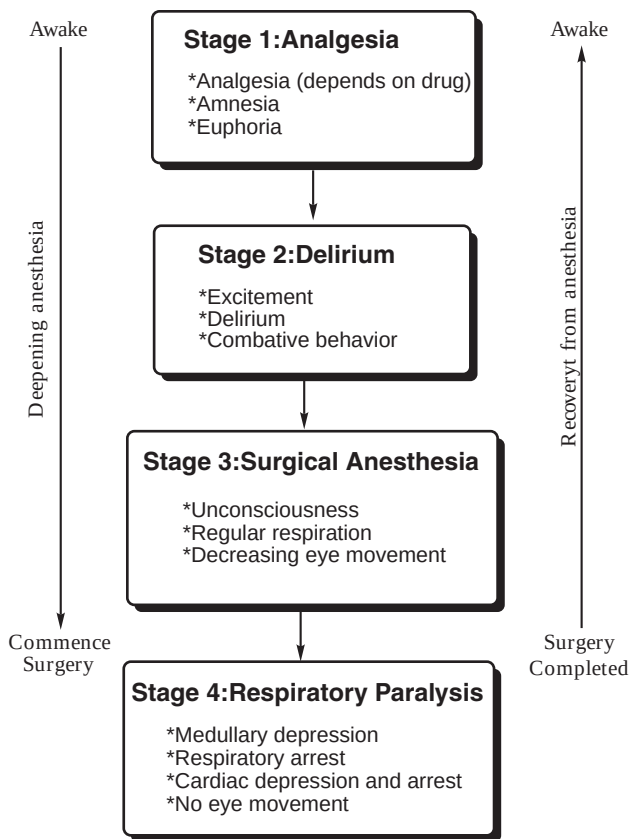


FIGURE 16.1 Stages of general anesthesia.

Stage 2: Delirium

As depression of inhibitory neurons in the CNS progresses, especially in the reticular formation (a network of neurons in the brainstem), a resultant excitation of cortical motor neurons leads to significant involuntary muscle activity, such as urination, delirium, uncontrolled skeletal muscular movements, and increased heart rate, blood pressure, and respiration. These paradoxical responses are caused by suppression of inhibitory neurons that normally function to closely regulate such neuronal activity. Ideally, an anesthetic agent should produce little or no excitatory phase. Together, stages 1 and 2 comprise the induction period, which ideally should be of short duration.

Stage 3: Surgical Anesthesia

This stage is divided into four planes characterized by increasing CNS depression: first, loss of spinal reflexes; second, decreased skeletal muscle reflexes; third, paralysis of intercostal muscles; and fourth, loss of most muscle tone. Stage 3 is also characterized by regular breathing, a loss of many reflexes, and roving eyeball movements.

Stage 4: Respiratory Paralysis

Characterized by respiratory and vasomotor paralysis, this stage represents an overdose or toxic level that should be avoided. Normally, this stage is never reached, because the anesthesiologist is careful to monitor abdominal

respiration to prevent apnea, blood pressure to prevent hypotension, and heart rate to prevent asystole (a state of no cardiac electrical activity).

Modern General Anesthetic Agents

Although these stages have been described for diethyl ether, an anesthetic agent no longer used today. Some of today's clinically useful general anesthetic agents fail to follow this described pattern of anesthetic progression. Some attempts have been made to correlate changes in the electroencephalograph (EEG) with the depth of anesthesia. Most of these studies, however, have failed to yield a reliable predictor for anesthesiologists to use. Additionally, concomitant drugs used as preanesthetic agents can alter the EEG while not altering the depth of anesthesia. Rather than describing specific stages or using EEG patterns, a number of useful signs that more accurately reflect the depth of anesthesia for most of the anesthetic agents are currently used. When during the initial period of anesthetic administration a patient has irregular respiratory depth and rate, is still swallowing, and blinks the eyes when the eyelashes are touched, the desired surgical stage of anesthesia likely has not been reached. However, when a loss of the eyelash reflex occurs along with rhythmic breathing, however, a level of adequate surgical anesthesia has generally begun. If a patient at this stage exhibits elevations in blood pressure, increased respiration rate, or increased jaw tension when a surgical incision is attempted, the subject is considered to be "light" and typically requires additional anesthesia to facilitate further surgical manipulations. These responses decrease further—until they are abolished—as the depth of anesthesia progresses. By monitoring reflexes, blood pressure, and respiration rate and depth, today's anesthesiologist is capable of effectively maintaining an appropriate depth of surgical anesthesia without producing unwanted medullary depression.

Pharmacokinetic Principles of Volatile Anesthetics

The production and maintenance of the anesthetic state is believed by most to be dependent on the concentration, or partial pressure, of the anesthetic agent in yet unknown areas of the brain. Obviously, the concentration of the anesthetic agent in the gas mixture administered, as well as the rate and depth of respiration of the patient, will influence the rate of anesthesia induction. The rate at which delivery of anesthetic agents to these sites occurs is dependent on their physicochemical properties, particularly their solubility in lipid and blood (Fig. 16.2).

Administration of Volatile Anesthetics

The administration of gaseous or volatile liquid anesthetics involves a number of sophisticated devices that have been refined over the years to aid the anesthesiologist in carefully controlling the amount of anesthetic delivered to the patient while minimizing the exposure of the



FIGURE 16.2 Dynamic equilibria existing during the state of anesthesia.

surgical personnel to these agents. Early systems used a gauze pad in a mask placed over the nose and mouth of the patient. The anesthesiologist would then put drops of the volatile anesthetic on the gauze pad, and as the patient breathed, the anesthetic was delivered to the lungs. This procedure was somewhat effective, but it allowed little or no control over the amount of anesthetic and oxygen delivered to the patient. Thus, the anesthetic agent not inhaled was allowed to evaporate into the surrounding area in the surgical suite and posed a significant risk to the surgical personnel. Today, flowmeters, vaporizers, and absorber devices are routinely available, allowing precise determination and control of the amount of volatile anesthetic, oxygen, and carbon dioxide administered while preventing significant exposure to workers. Typically, oxygen is bubbled through a volatile anesthetic liquid, and the resultant gas mixture is delivered to the patient for continual inhalation. Many of these devices are described in greater detail elsewhere (1).

The inhaled anesthetic concentration is controlled by the anesthesiologist, who can either increase or decrease this concentration depending on the observed depth of anesthesia. Eventually, with continued administration, the concentration of anesthetic in the bronchiolar alveoli reaches equilibrium with that in the inspired gas mixture (Fig. 16.2). Transfer from the alveolar space to the blood proceeds quickly, and depending on the concentrations of anesthetic used and its physiochemical characteristics, equilibrium with the arterial blood is achieved. However, before appreciable amounts of anesthetic agent dissolved in the blood will enter the brain, the blood must be saturated with the anesthetic. Therefore, anesthetics that are

more soluble in the blood (have a low blood/gas partition coefficient, Fig. 16.2) will require a longer time to achieve saturation of the blood–brain compartment. In such cases, the time for induction will be prolonged. On the other hand, an anesthetic that is poorly soluble in blood (has a high blood/gas partition coefficient, Fig. 16.2) will quickly saturate the blood compartment and then rapidly enter the tissues to produce a short induction period. Similarly, agents with high blood/gas partition coefficients will require a longer time for recovery from anesthesia. The solubility of an agent in the blood is usually expressed as the blood/gas partition coefficient, which is the ratio of the concentration of anesthetic in blood to that in the gas phase at equilibrium (Table 16.2). These values correspond well with the oil/gas partition coefficient, which is easier to determine experimentally. The blood/gas partition coefficient can be very high (e.g., 12) for soluble agents, such as methoxyflurane, and extremely low (e.g., 0.47) for poorly soluble agents, such as nitrous oxide.

The solubility of the anesthetic in tissue is expressed as the tissue/blood partition coefficient. Because the concentration of the anesthetic in the brain is probably of most interest, the brain/blood partition coefficient is more useful. Because the solubility of the anesthetic in lean tissues is essentially equal to that in blood, the tissue/blood or brain/blood partition coefficient is typically close to a value of 1. In fatty tissues, however, the partition coefficient can be much larger due to lipid solubility. The rate of blood flow to a particular organ will also influence the rate at which anesthetics reach their sites of action. The brain, liver, and kidneys have relatively high

TABLE 16.2 Partition Coefficients, MACs, and Metabolism of Some General Anesthetics

Anesthetics	Partition Coefficients at 37°C			MAC (vol %) ^a		% Metabolism
	Oil/Gas	Blood/Gas	Without N ₂ O	With N ₂ O (%)	MAC-Awake (Vol %)	
Methoxyflurane	970	12	0.16	0.07 (56)	—	50
Halothane	224	2.3	0.77	0.29 (66)	0.4	20
Enflurane	99	1.9	91.7	0.60 (70)	0.4	2.4
Isoflurane	97	1.4	1.15	0.50 (70)	0.4	0.17
Sevoflurane (2)	53	0.60	1.71	0.66 (64)	0.6	4–6
Desflurane (3)	19	0.42	6.0	2.83 (60)	2.4	0.02
Nitrous oxide	1.4	0.47	104	—	60	None

^aMAC = minimum alveolar concentration, expressed as volume %, that is required to produce immobility in respect to a standard surgical incision in 50% of middle-aged humans.

blood flows, whereas skeletal muscle at rest and fat tissues have relatively poor blood flows and would be expected to accumulate less of the anesthetic agent.

Reversal of the anesthetic state and recovery requires a reduction in the concentration of the anesthetic in the brain. This is achieved by stopping the delivery of the anesthetic through the lungs. As the patient continues to breathe, the anesthetic is continually removed, which favors diffusion from the brain to the blood, to the alveoli, and finally, to the expired air. The rate at which this occurs generally parallels that of induction, because the solubility of an agent in the brain and in the blood determines how quickly these compartments will return to a preanesthetic state. The main route of elimination is via the expired air, which can be mostly captured by gas-scavenging devices with absorbers, but some metabolism of these agents does take place (discussed below).

Minimum Alveolar Concentration

The minimum alveolar concentration (MAC) is defined as the concentration at 1 atmosphere of anesthetic in the alveoli that is required to produce immobility in 50% of adult patients subjected to a standard surgical incision. A further increase to 1.3 MAC will frequently cause immobility in 99% of patients. At equilibrium, the concentration (or partial pressure) of an anesthetic in the alveoli is equal to that in the brain, and it is this concentration in the brain that most closely reflects the concentration at the site responsible for the anesthetic actions. Thus, the MAC is often used as a measure of the potency of individual anesthetic agents. The MAC of many of the volatile and gaseous anesthetics in use today is shown in Table 16.2.

When anesthetic agents are used in combination, the MACs for inhaled anesthetics are simply additive. For instance, the anesthetic depth achieved with 0.5 MAC of enflurane plus 0.5 MAC of nitrous oxide is equivalent to that produced by 1.0 MAC of either agent alone. The combination of two anesthetics is a very common practice, because this technique allows for a reduction in the patient exposure to any one of the individual agents, thereby decreasing the likelihood of adverse reactions.

Many factors influence the MAC via a number of different mechanisms (Table 16.3). Factors that have been shown to increase the MAC for many volatile anesthetics include elevated catecholamines in the CNS following pharmacologic treatments, hyponatremia, and hyperthermia. Factors known to decrease MAC include ethanol ingestion, clonidine, lithium, lidocaine, centrally administered opioids, and drugs that decrease central catecholamine levels. Additionally, hyponatremia, hypotension, hypothermia, hypoxia, increasing age, and pregnancy have also been shown to decrease MAC. Plasma potassium, hypertension, gender, and the duration of anesthesia typically generally have minimal to no effect on the MAC (2).

Another term, the “MAC-Awake,” is used to describe the concentration of anesthetic at which appropriate

TABLE 16.3 Factors That May Alter MAC

Increase MAC	Increased catecholamine levels in CNS Hyponatremia Hyperthermia
Decrease MAC	Decreased catecholamine levels in CNS Alcohol Ingestion Hyponatremia Clonidine Hypotension Lidocaine Hypothermia Lithium Hypoxia Opioids Increased age Pregnancy
No effect on MAC	Plasma potassium Gender Hypertension Duration of anesthesia

responses to verbal commands are lost in 50% of the patients tested. At this concentration, amnesia and a loss of awareness are evident, and the patient is said to be in a state of hypnosis. The MAC-Awake occurs at concentrations significantly lower (e.g., 50% to 75% lower) than those required for surgical anesthesia.

Theories About the Mechanisms of Anesthesia

Meyer-Overton Theory

In the early 1900s, Hans Meyer and Charles Overton suggested that the potency of a substance as an anesthetic was directly related to its lipid solubility, or oil/gas partition coefficient (Table 16.2) (3–5). This has commonly been referred to as the “unitary theory of anesthesia.” They used olive oil, octanol, and other “membrane-like” lipids to determine the lipid solubility of the agents available at that time. Compounds with high lipid solubility required lower concentrations (i.e., lower MAC) to produce anesthesia. It was later postulated that the interaction of the anesthetic molecules with a hydrophobic portion of the nerve membrane caused a distortion of the nerve membrane near the channels that conducted Na^+ , those that mediated the fast action potentials and neuronal cell firing. The presence of this critical volume of anesthetic dissolved within the membrane caused the membrane to “bloat” and cause a “squeezing in” on the Na^+ channel to interfere with Na^+ conductance and normal neuronal depolarization. In support of this theory, it was found that at high pressures (40 to 100 atmospheres), the anesthetic actions of many of these agents could be partially reversed, presumably by compressing membranes back to their original conformation. Arguing against this theory, however, is the finding that not all highly lipid-soluble substances are capable of producing anesthesia. Additionally, more recent work involving protein–drug interactions has seriously challenged this theory. Today, more than 150 years after the first demonstration of the use of a volatile anesthetic agent, most theories about the mechanisms of anesthesia suggest that multiple selective

lipid–protein membrane interactions, involving numerous receptor types, are responsible for the anesthesia produced and that no reason exists to believe that all anesthetic agents need to produce their effects via identical actions. Thus, a single molecular target for anesthetic actions is no longer required (6).

Stereochemical Aspects

The volatile anesthetics isoflurane, desflurane, enflurane, and halothane each contain an asymmetric carbon and, thus, can exist as (+)- or (–)-enantiomers. Although all of the commercially available preparations are racemates, some researchers have been able to determine the anesthetic properties of individual enantiomers. The (+)-enantiomer of isoflurane is at least 50% more potent as an anesthetic in the rat than is the (–)-enantiomer (7). In that study, the MAC values were 1.06% and 1.62% for the (+)- and (–)-enantiomers, respectively. In another study, however, the potency of the individual isoflurane enantiomers to depress myocardial activity was not found to be different, suggesting possible involvement of mechanisms dissimilar from those responsible for producing anesthesia (8). These findings argue against the original and simple lipid-solubility theory of anesthesia, and they support a more complex mechanism, probably likely involvement of proteins in the form of receptor–anesthetic interactions (9).

Ion Channel and Protein Receptor Hypotheses

More recently, investigators have determined the effects of anesthetics on a number of protein receptors within the CNS. Features that support the likelihood of an interaction with a protein include 1) the steep dose–response

curves observed, 2) the stereochemical requirements of various anesthetics, 3) the finding that increasing the molecular weight and corresponding lipid solubility of an anesthetic can actually decrease or abolish anesthetic activity, and 4) the finding that specific ion channels and neurotransmitter receptor systems are required for most of the observed effects of the anesthetics. What appears to be emerging as a central theme for the mechanism of action of general anesthetics involves the interaction of the anesthetics with receptors that allosterically modulate the activity of ion channels (e.g., chloride and potassium) or with the ion channel directly (e.g., sodium). Many other mechanisms are also emerging to help explain the mechanisms of action of the general anesthetics.

CHLORIDE CHANNEL The ion channel that has received the most investigative attention is that for chloride (Fig. 16.3). Both the γ -aminobutyric acid_A (GABA_A) and the glycine_A (strychnine-sensitive) receptors are ligand-gated ion channels and linked to chloride channels that normally mediate inhibitory responses within the CNS. Halothane, isoflurane, and other volatile anesthetics are capable of inhibiting the synaptic destruction of GABA, thereby increasing the GABAergic neurotransmission, which typically is inhibitory in nature (10). Studies have also demonstrated the ability of these anesthetics to enhance the binding of GABA or other allosteric modulators within the GABA receptor complex (11). In one such study, (+)-isoflurane was significantly more potent than the (–)-enantiomer at enhancing GABAergic function (12). The volatile anesthetics, and many of the intravenous general anesthetic agents, bind to discrete cavities within the GABA_A

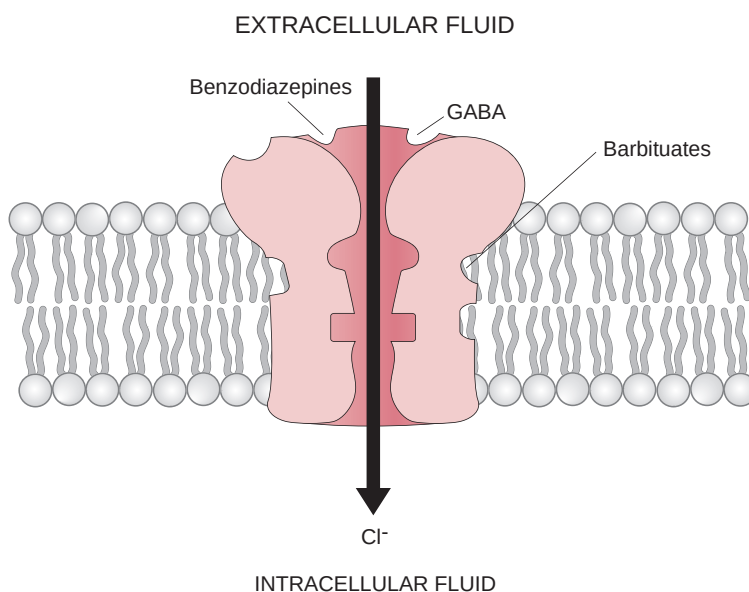


FIGURE 16.3 The GABA_A receptor controls the chloride ion channel and resulting in hyperpolarization of the neuron. Benzodiazepines and barbiturates can produce anesthesia by allosterically enhancing GABA opening of chloride channels, which are located at inhibitory synapses on pyramidal cells.

receptor complex to enhance GABA neurotransmission (13,14). Studies using mutant chimeric GABA_A receptors have identified a specific binding site for general anesthetics located between transmembrane segments 2 and 3 (15). At therapeutic concentrations, just about all of the inhalational general anesthetics are capable of enhancing GABAergic function, whereas at considerably higher concentrations, many also can act directly as GABA mimetics (16). Recent studies have demonstrated an effect of these agents not only on the synaptic GABA_A receptor function that mediates phasic neuronal responses but also on those extrasynaptic GABA_A receptors that mediate tonic neuronal activity (17). Other specific anesthetic agents can alter GABA_A receptor function via different mechanisms. For instance, propofol, an intravenous general anesthetic, appears to slow the desensitization of the GABA_A receptor during bouts of rapid, repetitive activation at inhibitory synapses (18). Most of these agents also potentiate the actions of glycine, the other important inhibitory amino acid neurotransmitter (16). The combination of GABAergic and glycinergic potentiation by the general anesthetics probably accounts for the vast majority of the observed activity of the inhalational agents as well as that of the barbiturates.

SODIUM CHANNELS One channel that has received much attention regarding the mediation of drug-induced anesthetic actions is the ligand-gated Na⁺ channel within the

N-methyl-D-aspartate (NMDA) receptor complex. When activated by the excitatory amino acid neurotransmitter, glutamate, an increase in the conductance to Na⁺ occurs that promotes neuronal depolarization (Fig. 16.4) (19). Compounds known to stimulate NMDA receptors are typically capable of increasing alertness and of acting as convulsants, whereas pharmacologic agents that act as antagonists at this site are usually sedatives, anticonvulsants, and dissociative anesthetics (e.g., ketamine). Halothane has been demonstrated to specifically antagonize the glutamate-stimulated depolarization of neurons (20), whereas isoflurane has been shown to decrease glutamate release and enhance its removal from the synaptic cleft (21). Glutamate acting at NMDA and other non-NMDA receptors within the CNS is probably one of the most important excitatory inputs that supports consciousness. It is not surprising that the general anesthetics would act by altering neurotransmission in this system (22). Others have reported an interaction of general anesthetics with the neuronal nicotinic acetylcholine receptor-linked Na⁺ channel (23). Voltage-gated Na⁺ channels in small, nonmyelinated hippocampal axons also appear to be inhibited by general anesthetics, such as isoflurane (24).

POTASSIUM CHANNELS Potassium ion channels have also been suggested as a site for general anesthetic agents. Increasing K⁺ conductance normally functions to maintain the polarized state of neurons and to assist in the

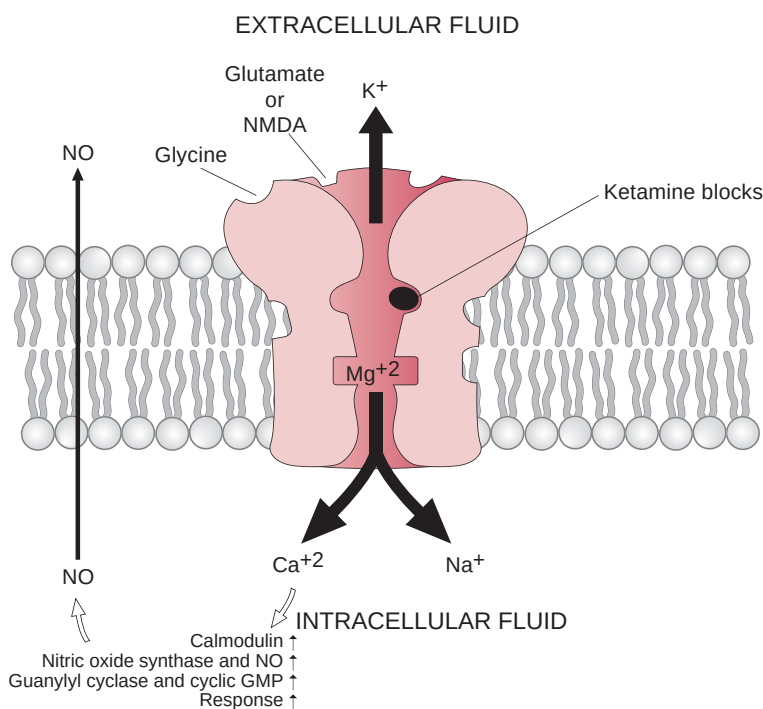
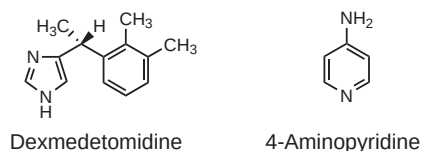


FIGURE 16.4 Glutamate or NMDA receptors in the CNS. Binding of agonists (glutamate or NMDA) opens the channel, allowing K⁺ to flow outward to extracellular fluid and sodium and calcium ions to flow into the nerve cells. Increased intracellular fluid calcium ion concentration triggers a cascade that produces a response and liberates the retrograde neuronal messenger nitric oxide (NO). Ketamine can produce anesthesia by blocking these NMDA-controlled channels, which are located at excitatory synapses on pyramidal cells (3). Glycine acts as a positive allosteric modulator at the NMDA receptor.

repolarization of neurons following their stimulation-induced depolarization (Fig. 16.5). Thus, enhancing the activity of certain K^+ channels would be expected to result in a decreased likelihood of neuronal excitation. A novel, anesthetic-sensitive K^+ current [$I_{K(an)}$] has been identified that is stereoselectively activated by isoflurane (25). Mice with a targeted deletion of the TREK-1 two-pore-domain K^+ channel show significantly reduced sensitivity to general anesthetics compared to wild-type controls (26). Additionally, certain α_2 -adrenoceptor agonists (e.g., dexmedetomidine) when injected produce an anesthetic state that is mediated by a G protein–coupled receptor that allosterically modulates K^+ channels. These responses can be antagonized by pertussis toxin and 4-aminopyridine, agents that inactivate G proteins and block K^+ channels, respectively, lending further support to the role of this ion channel (27). Similarly, G protein–mediated mechanisms appear to be involved with the action of morphine via the μ -opioid receptor (Fig. 16.5).



Halogenated Hydrocarbons and Ethers

Historical Aspects

ETHER The useful volatile anesthetics, with the exception of nitrous oxide, are halogenated hydrocarbons and ethers. Diethyl ether (Fig. 16.6), one of the first

agents to be introduced as a general anesthetic, has high potency with significant analgesic and neuromuscular relaxing effects. This agent is extremely flammable, and when mixed with air, oxygen, or nitrous oxide, is explosive. Induction with diethyl ether is very slow; significant time is spent progressing through the delirium stage. Irritation of the respiratory tract by diethyl ether can lead to excessive bronchial secretions, complicating adequate ventilation. In addition to its unpleasant induction and adverse effects, recovery is similarly prolonged and can be accompanied by vomiting. These pharmacologic and physical characteristics of diethyl ether have limited the utility of this anesthetic in humans.

SHORT-CHAIN HYDROCARBONS Many of the short-chain alkanes, alkenes, and alkynes are capable of producing an anesthetic state when administered to patients. Potency generally increases as chain length increases. However, because of their flammability and increased propensity to cause cardiovascular toxicity, these nonsubstituted hydrocarbons are not useful as anesthetic agents.

CHLOROFORM Another of the earlier anesthetic agents to be used was chloroform ($CHCl_3$). This halogenated hydrocarbon was first officially used in the United States in 1847; however, its toxicity seriously limited its utility. The addition of halogens to the hydrocarbon backbone increases potency and volatility, as well as decreases flammability. Similar effects are also observed with such substitutions on ethers. As an anesthetic agent, chloroform is very potent and possesses significant analgesic and neuromuscular

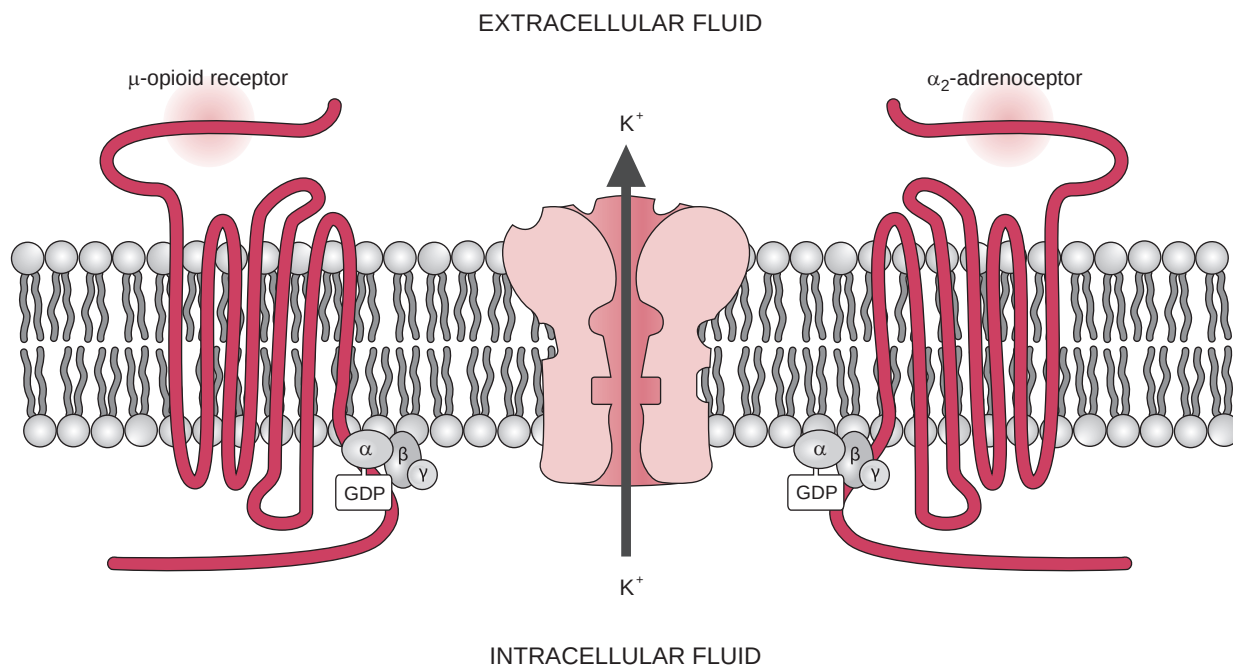


FIGURE 16.5 Morphine and α_2 -agonists activate their respective G proteins, which hyperpolarize neurons by lowering intracellular fluid K^+ ion concentration.

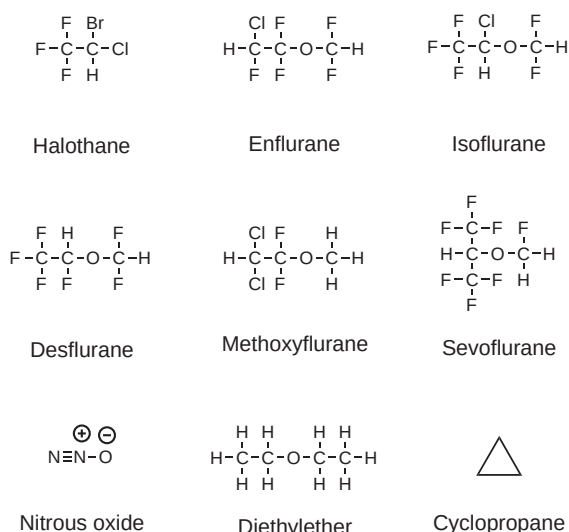


FIGURE 16.6 General anesthetics.

relaxing activity. Chloroform, a known carcinogen, has the disadvantage of being both hepatotoxic and nephrotoxic, in addition to producing adverse cardiovascular effects, such as arrhythmias and severe hypotension. As a result of these toxicities, chloroform has an unacceptable therapeutic index that prohibits its use in anesthesia. Two other halogenated compounds that found limited use in the mid-1900s included trichloroethylene and fluoro-xene. Neither of these agents is used today. Knowledge regarding the influence of the halogen substitutions on the potency and flammability of hydrocarbons and ethers, however, significantly contributed to our understanding of the structure–activity relationship of volatile anesthetics and the eventual design of substantially improved agents.

FLAMMABILITY The occurrence of fires in operating rooms is of great concern to all participants in the surgical procedure. Although the introduction of “nonflammable” agents, such as halothane, enflurane, and isoflurane, has substantially decreased this hazard, such fires still occur. Three essential ingredients are required for any combustion: 1) an ignition source (e.g., a laser), 2) a combustible material (e.g., gauze, drapes, or rubber tubes), and 3) an oxidizing agent (e.g., oxygen or nitrous oxide). Many substances are flammable in pure oxygen, nitrous oxide, or mixtures, but not air. Certain substances are flammable in nitrous oxide at concentrations that are too low to permit ignition in pure oxygen (28). The concentrations required for combustion, as indicated in Table 16.4, are higher than those generally encountered, except possibly during induction.

Clinically Useful Inhalation Agents

Fluorinated Hydrocarbons

The structure, physical properties and partition coefficients of the volatile anesthetics are given in Tables 16.2 and 16.5, respectively. Toxic degradation products

TABLE 16.4 Relative Flammability of “Nonflammable” Anesthetics

	Halothane (%)	Enflurane (%)	Isoflurane (%)
MFC of agent in 30% O ₂ with remaining atmosphere N ₂ O	4.75	5.75	7.0
MAC of agent given in above atmosphere	0.28	0.65	0.46
MAC in humans in the absence of N ₂ O	0.75	.68	1.15
MFC/MAC in N ₂ O	17	8.9	15.2

MFC, minimum flammable concentration; MAC, minimum effective alveolar concentration

are formed by reaction of the anesthetic agent with the basic substances such as soda lime, used as carbon dioxide absorbents during anesthesia. This reaction results in the conversion of halothane to 2-bromo-2-chloro-1,1-difluoroethylene, sevoflurane to 2-(fluoromethoxy)-1,1,3,3,3-pentafluoro-1-propene (Compound A), and desflurane, isoflurane, and enflurane to carbon monoxide. Compound A forms a glutathione S-conjugate, which undergoes hydrolysis to cysteine S-conjugates and bioactivation of the cysteine S-conjugates by renal cysteine conjugate β-lyase to give nephrotoxic metabolites.

HALOTHANE Halothane (Fig. 16.6) was introduced into medical practice in the United States in 1956 as a non-flammable, nonexplosive, halogenated volatile anesthetic that is usually mixed with air or oxygen. The presence of

TABLE 16.5 Physicochemical Properties of Clinically Useful Volatile Anesthetics

Generic Name (Trade Name) and Structure	Boiling Point (°C)	Chemically Stable ^a
Desflurane (Suprane) F ₂ HC–O–CHF–CF ₃	23.5	Yes
Enflurane (Ethrane) F ₂ HC–O–CF ₂ –CHFCl	56.5	Yes
Halothane (Fluothane) F ₂ C–CHBrCl	50.2	No
Isoflurane (Forane) F ₂ HC–O–CHCl–CF ₃	48.5	Yes
Methoxyflurane (Penthrane) H ₂ C–O–CF ₂ –CHCl ₂	104.7	No
Nitrous oxide N ₂ O	–8.0	Yes
Sevoflurane (Ultane) (CF ₃) ₂ CH–O–CH ₂ F	58.5	No

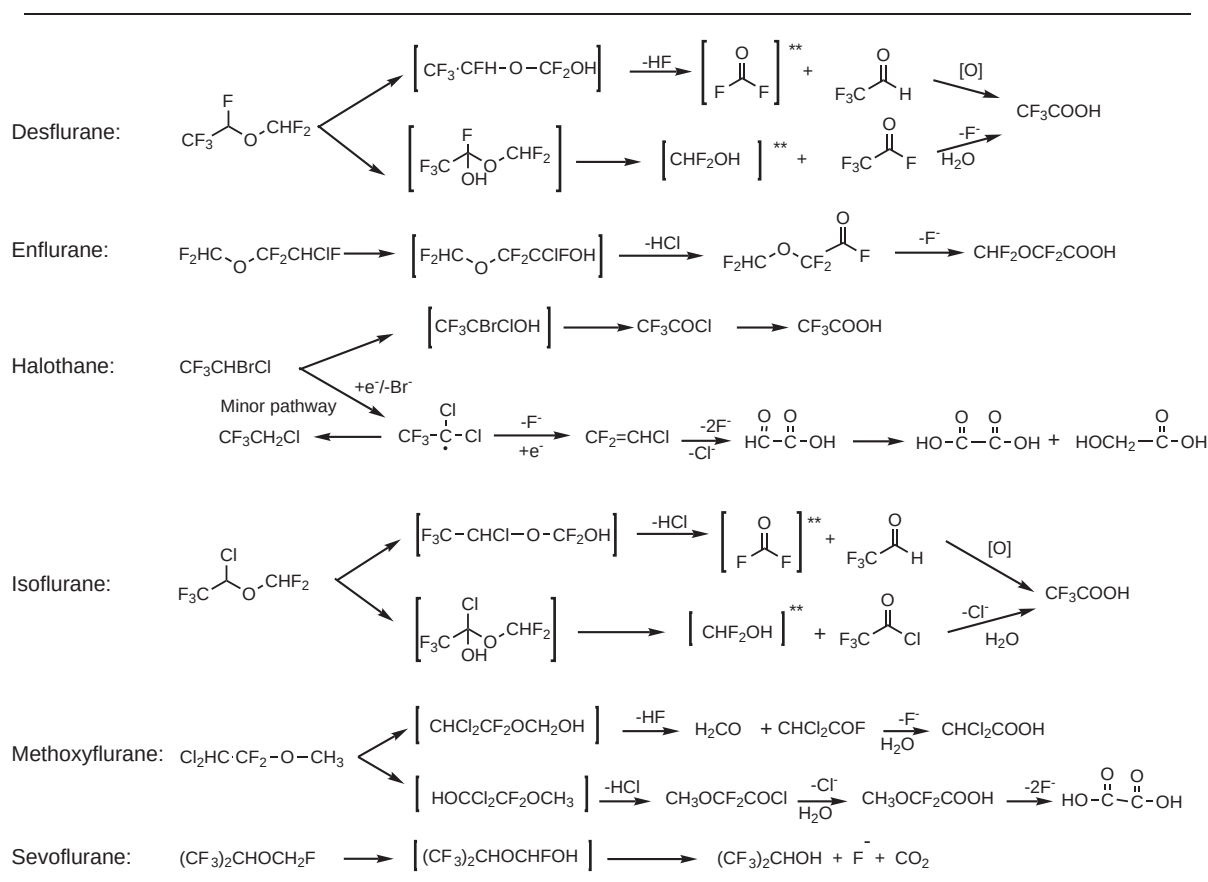
^aIndicates stability to soda lime, ultraviolet light, and common metals.

the carbon-halogen bonds contributes to its nonflammability, volatility, and high lipid solubility (blood/gas partition coefficient = 2.3). This clear liquid with a sweet odor was developed based on predictions that its halogenated structure would provide chemical stability, an intermediate blood solubility, and significant anesthetic potency. Halothane is the only useful volatile anesthetic possessing a bromine atom, which has been suggested to contribute to its potency. Similarly, the addition of fluorine atoms, of which halothane has three, contributes to its increased potency, volatility, and relative chemical stability of the hydrocarbon skeleton (Table 16.5).

Halothane produces rapid onset and recovery from anesthesia with high potency when used alone or in combination with nitrous oxide. Most metals, with the exception of chromium, nickel, and titanium, are easily tarnished by halothane. Although halothane is relatively stable, it is subject to spontaneous oxidative decomposition to hydrochloric acid (HCl), hydrobromic acid (HBr), and phosgene (COCl_2). For this reason, it comes in dark, amber glass containers with thymol added as a preservative to minimize decomposition. Halothane can permeate into the rubber components of the anesthetic

delivery devices, which might account for some slowing of the induction onset and recovery. Approximately 20% of an administered dose is metabolized, which accounts, in part, for the increased hepatotoxicity observed with this agent (Fig. 16.7).

ENFLURANE Enflurane (Fig. 16.6) was introduced into medical practice in the United States in 1973 and is a clear, colorless, nonflammable liquid with a mild, sweet odor. Although relatively stable chemically, enflurane does not attack aluminum, copper, iron, or brass and is soluble in rubber, which can prolong induction/recovery times, as seen with halothane (Table 16.5). Enflurane has an intermediate solubility in blood (blood/gas partition coefficient = 1.9) and significant potency. Most of its pharmacologic properties are similar to those of halothane, although there can be slightly less nausea, vomiting, arrhythmias, and postoperative shivering than observed with halothane. High concentrations of enflurane, however, are more likely to produce convulsions and circulatory depression. Enflurane also relaxes the uterus and, thus, should not be used as an anesthetic during labor. Metabolism via CYP2E1 accounts for 2% of an



*Data from references 28 and 29.

**Decomposition of unstable intermediates: $\text{CF}_2\text{O} \rightarrow 2\text{F}^- + \text{CO}_2$; $\text{CHF}_2\text{OH} \rightarrow 2\text{F}^- + \text{HCOOH}$;

FIGURE 16.7 Proposed metabolites of fluorinated anesthetics.

inhaled dose and includes metabolism to form a fluoride ion and fluoromethoxydifluoroacetic acid (Fig. 16.7) (30). During recovery, enflurane leaves the fatty tissues rapidly and, therefore, is not available for a prolonged period of time for significant metabolism to proceed.

ISOFLURANE Isoflurane (Fig. 16.6) was introduced in the United States in 1981 and is a potent anesthetic agent with many similarities to its isomer enflurane (potent, nonflammable, and intermediate blood solubility; with blood/gas partition coefficient = 1.4). However, it does produce significantly fewer cardiovascular effects than enflurane and can be used safely with epinephrine without a concern for arrhythmia production. Isoflurane has a more pungent odor than halothane and, thus, can cause irritation to the throat and respiratory tract, triggering coughing and laryngospasm. To overcome this problem, it is often supplemented with intravenous agents. Less than 0.2% of an administered dose is metabolized, mostly to fluoride and trifluoroacetic acid (Fig. 16.7). As discussed below, some minimal potential for hepatotoxicity is associated with a trifluoroacetyl halide metabolite.

A comparative assessment of the volatile anesthetic properties of enflurane, halothane, and isoflurane is shown in Table 16.6.

DESFLURANE Desflurane (Fig. 16.6) was introduced in the United States in 1992 and is a pungent, volatile agent that is nonflammable and noncorrosive to metals. With poor blood solubility (blood/gas partition coefficient = 0.42), similar to that of nitrous oxide, desflurane rapidly induces anesthesia. Because the boiling point

of desflurane is close to room temperature, a specially designed, heated vaporizer is used to deliver the anesthetic with appropriate concentrations of oxygen either alone or in combination with nitrous oxide. Recovery from the anesthetic state is also rapid, being approximately twice as rapid as that with isoflurane. Because of the rapid induction and recovery associated with desflurane, this anesthetic has gained popularity in outpatient surgical procedures. Desflurane is rather pungent, so patients often are induced with an intravenous anesthetic agent and then maintained with desflurane. Desflurane is not metabolized to any great extent and, therefore, has not been associated with hepatotoxicity or nephrotoxicity (31). Metabolites, mostly trifluoroacetate, account for less than 0.02% of the administered dose (Fig. 16.7). Although desflurane can react with soda lime or Baralyme to form carbon monoxide, no reports of adverse outcomes in patients have appeared.

SEVOFLURANE Sevoflurane (Fig. 16.6) is a nonflammable, nonirritating, pleasant-odored volatile anesthetic available for use in the United States. Similar to desflurane in many of its pharmacologic actions, except sevoflurane which has low blood solubility (blood/gas partition coefficient = 0.60), higher potency, and the advantage of not being irritating to the respiratory tract. Induction and recovery are rapid. Sevoflurane undergoes significantly more metabolism (CYP2E1) than desflurane, however, and as much as 3% of an administered dose can be recovered as hexafluoroisopropanol (Fig. 16.7). Some fluoride ion can also be produced, but the incidence of nephrotoxicity or hepatotoxicity appears low, especially when used infrequently for short periods of time. There have been concerns regarding the reactivity of sevoflurane with soda lime or Baralyme, in which a potentially toxic olefin byproduct termed “Compound A” (2-(fluoromethoxy)-1,1,3,3,3-pentafluoro-1-propene) can be formed. With appropriate precautions, however, sevoflurane can be used safely in both children and adults.

METHOXYFLURANE Methoxyflurane (Fig. 16.6) is seldom used because of its propensity to cause renal toxicity. It is the most potent of the agents discussed here, and it has high solubility in blood (blood/gas partition coefficient = 12). Induction and recovery would be expected to be slow. Chemically, it is rather unstable, and as much as 50% of an administered dose can be metabolized. Toxic metabolites significantly limit its utility as a general anesthetic (Fig. 16.7).

TOXICITY OF FLUORINATED GENERAL ANESTHETICS Although few signs of toxicity usually are observed during the short-term, infrequent administration of general anesthetics, a few well-defined toxic effects have been noted. For instance, halothane and methoxyflurane are known to produce hepatotoxicity and nephrotoxicity, respectively. Both of these toxic reactions are believed to result from highly reactive metabolites of the parent compound.

TABLE 16.6 Comparative Assessment of Enflurane (E), Halothane (H), and Isoflurane (I)

Property	Superior	Intermediate	Inferior
Stability	I = E	—	H
Blood solubility	I	E	H
Pungency	H	I	E
Respiratory depression	H	I	E
Circulatory depression	I	H	E
Induction of arrhythmias	I	E	H
Muscle relaxation	I = E	—	H
Increased intracranial pressure/cerebral blood flow	I	E	H
Seizure activity	H = E	—	E
Metabolism	I	E	H
Toxicity	I	E	H

Adapted from Wade JC, Stevens WC. Isoflurane: an anesthetic for the eighties? *Anesth Analg* 1981;60:666–682; with permission.

Overall, however, the therapeutic ratio for most of the general anesthetics approaches 4:1 (32).

Hepatotoxicity Hepatitis caused by halothane occurs in 1 in 20,000 patients exposed to this anesthetic and is thought to result from the binding of a reactive free radical metabolite to liver tissue (Fig. 16.7). The resultant abnormal molecular product in the liver is viewed by the immune system as a foreign substance (i.e., an antigen), which then sensitizes cells to produce antibodies. Some have suggested that the trifluoroacetyl halide metabolite is responsible for the initiation of halothane hepatitis. Interestingly, both enflurane and isoflurane can be metabolized to the acylated halides and produce a similar immune-mediated syndrome, although to a much lesser extent. Additionally, there appears to be cross-reactivity among these three agents, because the antigen formed is similar enough in structure to elicit the immune system response. Some investigations have suggested that a genetic susceptibility factor could be responsible, in part, for this serious form of hepatitis.

Halothane also can produce another form of hepatotoxicity. This is a self-limiting hepatic dysfunction characterized by elevated liver transaminase enzymes, which probably results from impaired oxygenation of the hepatocytes during exposure to this anesthetic. Isoflurane and enflurane have also been reported to produce a similar elevation of liver enzymes, although to a lesser extent than halothane.

Malignant Hyperthermia This rare (1 in 15,000 anesthetic uses) but potentially fatal complication associated with the use of certain anesthetics (e.g., halothane) is characterized by a rapid rise in core body temperature associated with hypermetabolic reactions in the skeletal muscle of genetically susceptible subjects. Such individuals appear to have an autosomal dominant-mediated defect in the Ca^{2+} -release channel commonly referred to as the ryanodine receptor. The large amounts of heat generated, massive increase in oxygen consumption, and production of carbon dioxide can quickly lead to death or permanent neurologic damage unless appropriate supportive treatment, including rapid cooling, 100% oxygen, and control of acidosis, is promptly initiated. The administration of the skeletal muscle relaxant dantrolene, which blocks release of Ca^{2+} from the sarcoplasmic reticulum, reduces muscle rigidity and heat production, which significantly improves the prognosis of the patient. Besides the fluorinated volatile anesthetics, some depolarizing neuromuscular blocking agents (e.g., succinylcholine) and some neuroleptics (e.g., haloperidol) are also reportedly associated with similar malignant hyperthermic syndromes, although the underlying mechanism mediating these can differ somewhat from those associated with the general anesthetics.

Nephrotoxicity Fluorinated anesthetics that undergo metabolism to form inorganic fluoride ion have the

potential to produce damage to the renal tubular cells. Of the fluorinated anesthetics, methoxyflurane is the only agent commonly associated with nephrotoxicity. Methoxyflurane is metabolized (Fig. 16.7) to produce plasma fluoride ion levels in excess of the threshold value for renal damage of 40 $\mu\text{mol/L}$. Others, such as sevoflurane, have only very rarely been associated with nephrotoxicity—and then usually in patients with severe renal compromise. Plasma levels of fluoride only reach 15 to 20 $\mu\text{mol/L}$ following 2.5 MAC-hour exposure to enflurane (33). The rates of metabolic defluorination of the useful anesthetic agents are as follows: methoxyflurane > enflurane = sevoflurane > isoflurane > desflurane = halothane.

Low-level Chronic Exposure Typically, patients are exposed to greater-than-MAC concentrations of the volatile anesthetics for limited periods of time, such as a number of hours during a surgical procedure, and not for extended periods of time (e.g., days or weeks). Because surgical and dental personnel, however, can be exposed to low levels of the general anesthetics for prolonged periods over many years or even decades, the ability of such agents to produce chronic toxicity is of paramount concern. Although the occupational exposure to these agents has been minimized with improved waste gas-scavenging devices, some epidemiologic studies have demonstrated increased levels of spontaneous abortions, congenital birth defects in offspring, and increased rates of certain cancers in chronically exposed medical personnel (34).

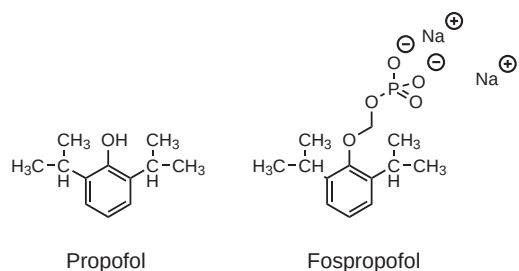
Nitrous Oxide

Commonly called “laughing gas,” nitrous oxide (N_2O) is a gas at room temperature, the least potent of the inhalation anesthetics used today and poor blood solubility (blood/gas partition coefficient = 0.47) (Table 16.5). With an MAC value in excess of 105%, this colorless, tasteless, and odorless to slightly sweet-smelling gas is not normally capable of producing surgical anesthesia when administered alone. The MAC for nitrous oxide has been demonstrated to be between 105% and 140% and, thus, cannot achieve surgical anesthesia under conditions at standard barometric pressure. To demonstrate that the MAC was greater than 100%, Bert in 1879 used a mixture of 85% nitrous oxide with oxygen at 1.2 atmospheres in a pressurized chamber. Only at this elevated pressure could an MAC adequate for surgical anesthesia be achieved. Decreasing the oxygen content of a nitrous oxide mixture to values less than 20% to allow an increase in the concentration of nitrous oxide to greater than 80% can be dangerous, because hypoxia would be expected to result. Thus, when administered alone, nitrous oxide finds utility as an anesthetic agent during certain procedures (e.g., dental) in which full surgical anesthesia is not required. Most commonly, however, nitrous oxide is used in combination with other general anesthetics, because it is capable of decreasing the concentration of the added anesthetic required to produce an adequate depth of anesthesia for surgical procedures.

Although no firm underlying mechanisms have been demonstrated, some authors have suggested that irreversible oxidation of the cobalt atom in vitamin B₁₂ by nitrous oxide can lead to inactivation of enzymes dependent on this vitamin, with resultant metabolic aberrations. Such examples have included methionine synthetase and thymidylate synthetase, which are essential in the synthetic pathways leading to the production of myelin and thymidine, respectively. Should these enzymes be impaired during the sensitive periods of in utero development, the potential for malformations can unfortunately be realized. To date, no studies have been able to demonstrate conclusively that low-level exposure to nitrous oxide is associated with a meaningful disruption of crucial metabolic functions to produce the above-described toxicity; however, measures including improved waste gas-scavenging systems should be taken to minimize exposure of personnel.

Clinically Useful Intravenous General Anesthetic Agents

Propofol



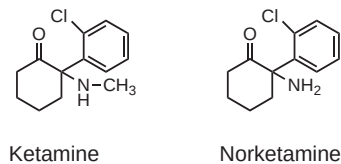
One of the most commonly used parenteral anesthetics used in the United States is propofol (Diprivan). Used intravenously, propofol is not chemically related to the barbiturates or other intravenous anesthetics. Propofol appears to act via enhancing GABAergic neurotransmission within the CNS. This occurs most likely at the GABA_A receptor complex, but at a site distinct from where the benzodiazepines bind. Because of its poor water solubility (partition coefficient ~6,200), propofol is formulated as a 1% or 2% emulsion with soybean oil, egg lecithin, and glycerol. Sodium metabisulfite (an antioxidant) or ethylenediaminetetraacetic acid (metal chelating agent) is also included in the parenteral dosage form for stability. Because of the likelihood of bacterial contamination of open containers, propofol should be either administered or discarded shortly after sterility seals are broken. Following intravenous administration of a dose of 2.0 to 2.5 mg/kg, a state of hypnosis is achieved within 30 to 60 seconds, which lasts for approximately 5 to 10 minutes. A longer anesthetic state can be achieved by additional propofol dosing or, as typically is the case, maintenance with a volatile anesthetic agent. Blood pressure and heart rate usually are decreased following propofol administration. Propofol is highly bound to plasma proteins (approximately 98%). Metabolism of propofol proceeds

rapidly via hepatic conversion to its glucuronide and sulfate conjugates, with less than 0.3% excreted unchanged. Because this agent produces a rapid induction and recovery and is infrequently associated with episodes of vomiting, propofol has found utility as an anesthetic agent in outpatient surgical environments.

Fospropofol

Due to its water solubility, the phosphate ester prodrug of propofol (Lusedra), fospropofol, avoids the emulsion formulation concerns described earlier for propofol. All of the pharmacodynamic effects of fospropofol are attributed to propofol, which is liberated following hydrolytic metabolism by serum alkaline phosphatases. Typical dosing is 6.5 mg/kg, with supplemental doses of 1.6 mg/kg as needed. While formaldehyde and phosphate are also released by this metabolic conversion, the levels of these compounds do not increase to levels beyond those normally found endogenously and thus do not pose any toxicity concerns, except perhaps in overdose situations. Due to its requirement for conversion to the active propofol, the onset of fospropofol is delayed (4 to 10 minutes) when compared to that for propofol (30 to 60 seconds) and has a prolonged duration of anesthetic action.

Ketamine

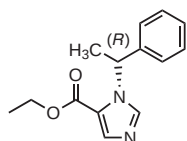


Ketamine hydrochloride is an injectable, very potent, rapidly acting anesthetic agent. As with propofol, its duration of anesthetic activity is also relatively short (10 to 25 minutes). Ketamine does not relax skeletal muscles and, therefore, can only be used alone in procedures of short duration that do not require muscle relaxation. Recovery from anesthesia can be accompanied by "emergence delirium," which is characterized by visual, auditory, and confusional illusions. Disturbing dreams and hallucinations can occur up to 24 hours after the administration of ketamine. Its elimination half-life is 2 to 3 hours, and its volume of distribution is 2 to 3 L/kg. Ketamine has an oral bioavailability of less than 16%. Termination of the acute action of ketamine is largely a result of its redistribution from the brain into other tissue; however, the formation of the glucuronide conjugate and metabolism in the liver to a number of metabolites does occur. One of these metabolites of interest, norketamine, is formed via the action of CYP2B6. This *N*-demethylated derivative retains significant activity at the NMDA receptor and can account for some of the longer-lasting effects of this anesthetic agent. Eventual conversion of norketamine to hydroxylated metabolites and subsequent conjugation leads to metabolites that can be renally eliminated. Less than 4% of a dose is excreted unchanged in the urine.

Ketamine is capable of producing a “dissociative” anesthesia, which is characterized by EEG changes indicating a dissociation between the thalamocortical and limbic systems (35). These neuronal systems, which normally are associated with one another, help to maintain the neuronal connections required for consciousness. When dissociated, the subject will appear to be cataleptic, with the eyes open in a slow, nystagmic gaze (oscillating movement of the eyeball) (1). A potent analgesic and amnesic effect is produced, as is an increase in muscle tone in some areas. Although patients can appear to be awake, they are incapable of communicating and do not remember the event or the people around them. Blood pressure and heart rate usually are increased following ketamine administration.

Ketamine appears to act similarly to phencyclidine (PCP; also known as Angel Dust), which acts as an antagonist within the cationic channel of the NMDA receptor complex (36). By preventing the flow of cations through this channel, ketamine prevents neuronal activation, which normally is required for the conscious state. The analgesic activity of ketamine, however, is more likely the result of an interaction with an opioid receptor or the less well-understood non-opioid sigma receptor. Other studies have suggested a possible involvement of serotonin receptors and muscarinic receptors (37). Ketamine, like PCP, has a significant potential for abuse.

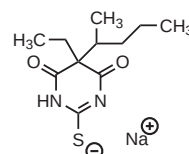
Etomidate



R-Etomidate

Etomidate is the ester of a carboxylated imidazole, with a partition coefficient of 2,000 and a weak base pK_a of 4.5, that is available as the *R*-(+)-isomer solubilized in 35% propylene glycol for intravenous injection in addition to being available for rectal administration. It is a potent, short-acting hypnotic agent (<3 minutes) without analgesic activity and with a rapid onset of action. This agent is useful for the induction of anesthesia in hemodynamically unstable patients prone to hypotension because of hypovolemia, coronary artery disease, or cardiomyopathies. Recovery is similarly rapid following discontinuance of the drug. Etomidate is hydrolyzed by hepatic esterases to the corresponding inactive carboxylic acid, with subsequent renal and biliary excretion terminating its action. Its apparent elimination half-life is approximately 5 to 6 hours, with a volume of distribution of 5 to 7 L/kg. Changes in hepatic blood flow or hepatic metabolism will have only moderate effects on etomidate disposition. Concerns regarding the ability of etomidate to precipitate myoclonic jerks and inhibit adrenal steroid synthesis have been reported.

Ultrashort-Acting Barbiturates



Thiopental

Thiopental, an ultrashort-acting barbiturate (partition coefficient ~390), is used intravenously to produce a rapid unconsciousness for surgical and basal anesthesia. This agent is used initially to induce anesthesia, which then can be maintained during the surgical procedure with a general anesthetic agent. The induction typically is very rapid and pleasant. (The ultrashort-acting barbiturates are discussed in Chapter 15.)

LOCAL ANESTHETICS

Local anesthetic agents are drugs that, when given either topically or administered directly into a localized area, produce a state of local anesthesia by reversibly blocking nerve conductances that transmit the sensations of pain from this localized area to the brain. Unlike the anesthesia produced by general anesthetics, the anesthesia produced by local anesthetics is without loss of consciousness or impairment of vital central cardiorespiratory functions. Local anesthetics block nerve conductance by binding to selective sites on the Na^+ channels in the excitable membranes, thereby reducing Na^+ passage (i.e., conductance) through the pores and, thus, interfere with the generation of action potentials. Although local anesthetics decrease the excitability of nerve membranes, they do not affect the neuron's resting potential. Local anesthetics, in contrast to analgesic compounds, do not interact with the pain receptors or inhibit the release or the biosynthesis of pain mediators.

The Discovery of Local Anesthetics

As with many modern drugs, the initial leads for the design of clinically useful local anesthetics originated from natural sources. As early as 1532, the anesthetic properties of coca leaves (*Erythroxylon coca* Lam) became known to Europeans from the natives of Peru, who chewed the leaves for a general feeling of well-being and to reduce hunger. Saliva from chewing the leaves was often used by the natives to relieve painful wounds. The active principle of the coca leaf, however, was not discovered until 1860 by Niemann, who obtained a crystalline alkaloid from the leaves, to which he gave the name cocaine, and who noted the anesthetic effect on the tongue (see Fig. 16.8 for structure of cocaine). Although Moréno y Maiz in 1868 first asked the question of whether cocaine could be used as a local anesthetic, Von Anrep in 1880, after many animal experiments, recommended that cocaine

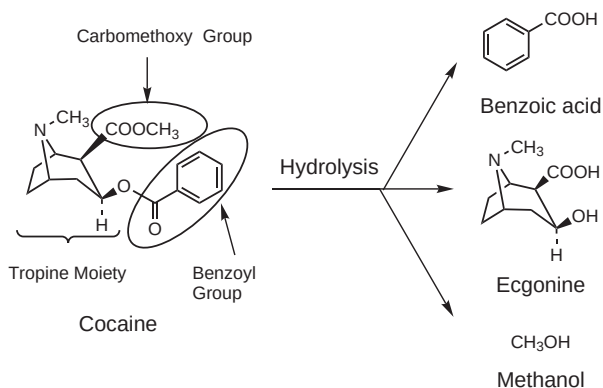
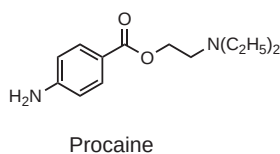
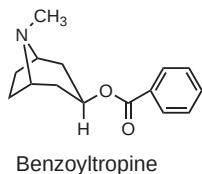


FIGURE 16.8 Structures of cocaine and its hydrolysis products.

be used clinically as a local anesthetic. The first report of successful surgical use of cocaine appeared in 1884 by Koller, an Austrian ophthalmologist. This discovery led to the rapid development of new local anesthetic agents and anesthetic techniques (38).

Cocaine dependence (or addiction) is psychological dependency on the regular use of cocaine. The use of cocaine, depending on the severity, can cause mood swings, paranoia, insomnia, psychosis, high blood pressure, tachycardia, panic attacks, cognitive impairments, and drastic changes in the personality that can lead to aggressive, compulsive, criminal, and/or erratic behaviors. The symptoms of cocaine withdrawal range from moderate to severe: dysphoria, depression, anxiety, psychological and physical weakness, pain, and compulsive craving.

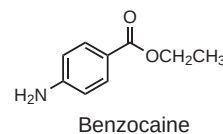
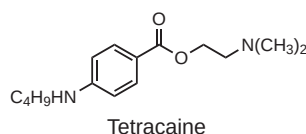
Although the structure of cocaine was not known until 1924, many attempts were made to prepare new analogs of cocaine that lacked its addicting liability and other therapeutic shortcomings, such as allergic reactions, tissue irritations, and poor stability in aqueous solution. Also, cocaine is easily decomposed to hydrolysis products, ecgonine and benzoic acid, when the solution is sterilized (Fig. 16.8).



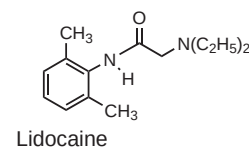
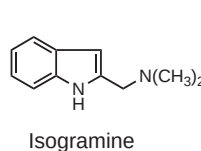
When the chemical structure of ecgonine became known, the preparation of active compounds containing the ecgonine nucleus accelerated. It was soon realized that a variety of benzoyl esters of amino alcohols, including benzoyltropine, exhibited strong local anesthetic properties without any of cocaine's addiction liability. Thus, removal of the 2-carbomethoxy group from cocaine also abolished its addiction liability. This discovery eventually led to the synthesis of procaine in 1905 (known as Novocain), which then became the prototype for local anesthetics for nearly half a century, largely because it lacked the severe local and systemic toxicities of cocaine.

Although the intrinsic potency of procaine was low and its duration of action relatively short compared with that of cocaine, it was found that these deficiencies could be remedied when procaine was combined with a vasoconstrictor, such as epinephrine. Vasoconstrictor agents reduce the local blood supply and, thereby, prolong the residence time of the local anesthetic at the injection site.

Following the introduction of procaine, hundreds of structurally related analogs were prepared and their local anesthetic properties examined in an attempt to identify agents with enhanced potency and duration of action compared to the weak and short-acting procaine. Among these compounds, tetracaine remains the most potent, long-acting, ester-type local anesthetic agent, which is used in spinal anesthesia.



The topical anesthetic agent benzocaine was synthesized by Ritsert in 1890 and found to have good anesthetic properties and low toxicity. However, due to its limited water solubility, except at low pH values as a result of the lack of a basic aliphatic amino group, the preparation of pharmaceutically acceptable parenteral solutions could not be achieved.



The serendipitous discovery of the local anesthetic activity of another natural alkaloidal product, isogramine, in 1935 by von Euler and Erdtman was the next major turning point in the development of clinically useful local anesthetic agents. This observation led to the synthesis of lidocaine (Xylocaine) by Löfgren in 1946; lidocaine was the first nonirritating, amide-type local anesthetic agent with good local anesthetic properties yet less prone to allergic reactions than procaine analogs, and was found to be stable in aqueous solution due to its more stable amide functionality. Structurally, lidocaine can be viewed as an open-chain analog of isogramine and, thus, is a bioisosteric analog of isogramine.

Since the discovery of lidocaine in the 1940s, much more progress has been achieved in the fields of neurophysiology and neuropharmacology than in the synthesis of local anesthetics by medicinal chemists. Most of this research has significantly increased our understanding of how nerve conduction occurs and how compounds interact with the neuronal membranes to produce local anesthesia. It should be noted, however, that although a number of current clinically useful local anesthetic

agents have been introduced into the market, the ideal local anesthetic drug has, unfortunately, not yet been realized.

Characteristics of an Ideal Local Anesthetic

The ideal local anesthetic should produce reversible blockade of sensory neurons with a minimal effect on the motor neurons. It also should possess a rapid onset, have a sufficient duration of action for the completion of major surgical procedures without any systemic toxicity, and be easily sterilized and not inordinately expensive (Table 16.7). Hopefully through further structure–activity relationship studies, particularly with regard to their selective actions on the voltage-gated Na^+ channels, the ideal local anesthetic agent can be realized. Additional leads for the design of ideal local anesthetics could also come from a more systematic metabolic and toxicity study of currently available agents. To understand the chemical aspects of local anesthetics and, thus, to provide a proper background for practical uses of these compounds, it is necessary to have a working knowledge of basic neuroanatomy and electrophysiology of the nervous system.

Neuroanatomy and Electrophysiology of the Nervous System

Neuroanatomy

Sensory neurons (afferent neurons) transmit sensory electrochemical impulses from sensory endings (receptors) in the skin and other sensory organs toward the CNS (i.e., brain and spinal cord) where the information is processed. On the other hand, motor neurons (efferent neurons) transmit electrochemical impulses from the CNS toward motor endings or other target (effector) cells that, when stimulated, produce a response such as contraction of muscle or stimulation/inhibition of sweat glands or exocrine glands. The transmission of a nerve impulse along an axon occurs

as a result of electrochemical changes in the Na^+ and K^+ potential across the neuronal membrane. Such neurons are bundled together into a cable-like structure and wrapped in a connective tissue sheath (the perineurium), called a nerve. Within a nerve, each axon is wrapped by a layer of connective tissue called the endoneurium. Finally, the entire nerve is wrapped in a layer of connective tissue called the epineurium (much like an electrical cable of wires wrapped with a plastic casing), as shown in Figure 16.9. (39). A nerve provides a common pathway for the transmission of electrochemical impulses. Thus, each nerve is a cord-like structure that contains groups of neurons in small bundles. The cell bodies of the sensory neurons are found at the point at which the nerve enters the vertebrae and can be seen as enlargements on the nerve bundles (spinal or dorsal root ganglion). The cell bodies (anterior horn cell) of the motor neurons are found within the gray matter of the spinal cord.

Figures 16.9 and 16.10 also illustrate that each axon has its own membranous covering, the endoneurium (often called the nerve membrane), which is tightly surrounded by a myelin sheath of the Schwann cell. The myelin sheath is not continuous along the fiber, with intermittent gaps or interruptions at the nodes of Ranvier, which serve to facilitate neuronal conduction.

Electrophysiology of Nerve Membrane

RESTING POTENTIAL Most nerves have a resting membrane potential (unstimulated or polarized state) of approximately -70 to -90 mV as a result of a slight imbalance of electrolyte ions (e.g., sodium, potassium, calcium, magnesium, and chloride) across the nerve membranes, between the intracellular cytoplasm and the extracellular fluid (40). In the polarized state, the nerve membrane is somewhat impermeable to Na^+ as seen by the low intracellular Na^+ concentration, whereas K^+ flows in and out of the cell with greater ease, indicating that the neuronal membrane is highly permeable to K^+ . A high K^+ concentration is retained intracellularly by the attractive forces

TABLE 16.7 Characteristics of the Ideal Local Anesthetic Agent

Produces a reversible blockade
Selective for sensory neurons with no effect on motor neurons
Rapid onset
Sufficient duration of action
Chemically stable when sterilized
No systemic toxicity
Wide margin of safety
Compatible with other coadministered drugs
Absence of adverse effects
Inexpensive

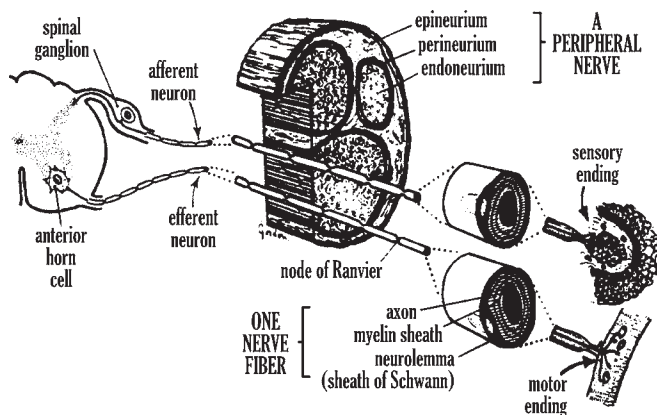


FIGURE 16.9 Diagram showing the various parts of a peripheral nerve. (Adapted from Ham AW. Histology, 6th Ed. Philadelphia: JB Lippincott, 1969:524, with permission.)

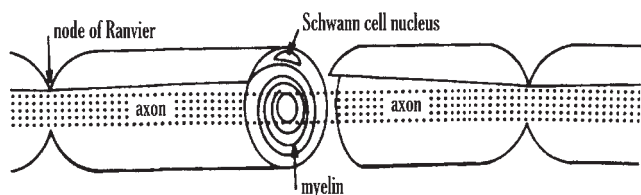


FIGURE 16.10 Single myelinated nerve fiber.

provided by the negatively charged intracellular protein molecules. Thus, the predominant intracellular cations are K^+ (~110 to 170 mmol/L), and the predominant extracellular cations are Na^+ (~140 mmol/L) and chloride (~110 mmol/L).

Because changes in the intracellular or extracellular concentration of K^+ would be expected to markedly alter the resting membrane potential, electrophysiologists modeled the excitable cell as if it were an electrochemical, or Nernst, cell. Thus, the resting potential for K^+ can be determined by the familiar Nernst equation:

$$E = -RT/zF \ln [K^+]_i / [K^+]_o$$

in which E = membrane potential, intracellular minus extracellular; R = gas constant; T = temperature; z = ion valence; F = Faraday's constant; $[K^+]_i$ = activity of intracellular K^+ ; and $[K^+]_o$ = activity of extracellular K^+ .

ACTION POTENTIAL In most cells, action potentials are transient transmembrane depolarizations that result from the influx of Na^+ through a brief opening of the voltage-gated Na^+ channels upon excitation of the cell (40). The transmembrane potential during an action potential changes from -70 to approximately $+40$ mV (a total net change of 110 mV) (depolarized state) and then promptly returns to the resting (polarized state) potential; the entire event typically lasts approximately 1 millisecond (Fig. 16.11). An action potential is capable of traveling long distances along the neuron because open Na^+ gates stimulate neighboring Na^+ gates to open.

The transmembrane potential for Na^+ at the peak of its action potential can be predicted from the Nernst equation by substituting appropriate Na^+ concentrations for those of K^+ . Thus, it appears that the excitable nerve membrane can be transformed from a potassium electrode to a sodium electrode during the active process (41).

As the cell approaches its peak action potential, the membrane permeability to Na^+ again decreases (Na^+ inactivation). If no other event occurred, this cell would slowly return to its resting potential, but the cell again becomes highly permeable to K^+ , allowing K^+ to flow out into extracellular fluid and quickly restore the membrane potential (repolarization). After an action potential, the cell would therefore be left with a small increase in Na^+ and a decrease in K^+ . To explain how the nerve is restored to its original electrolyte composition at the resting potential, it is necessary to postulate a mechanism

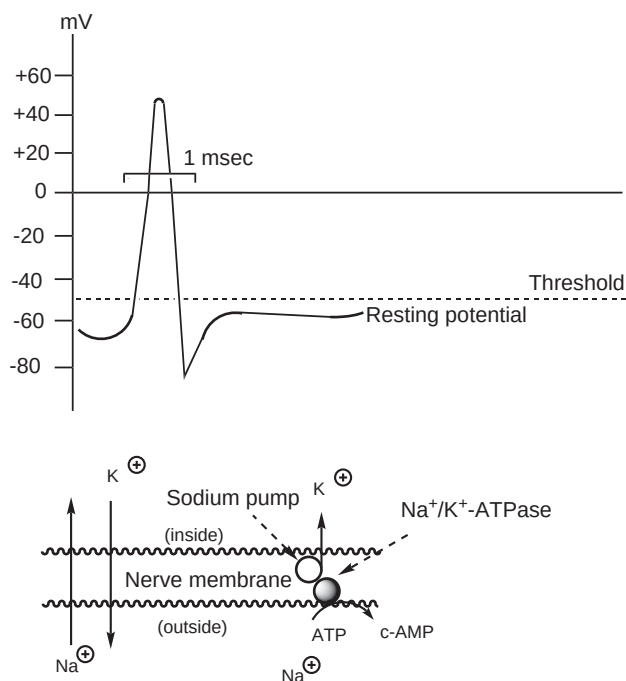


FIGURE 16.11 Relationship between membrane action potential and ionic flux across the nerve membrane.

by which intracellular Na^+ could be rapidly transported (effluxed) from inside the membrane to the outside extracellular fluid and extracellular K^+ could be transported (influxed) from extracellular fluid to inside the membrane. This is accomplished by the “sodium pump,” which requires energy from the splitting of adenosine triphosphate by sodium-potassium adenosine triphosphatase (Na/K -ATPase) to adenosine monophosphate. This pump transports three Na^+ to the outside of the membrane for every two K^+ that enter the inside of the membrane.

THRESHOLD The voltage necessary to change localized electrochemical differences into a propagated action potential is called the threshold voltage, which is closely related to the stimulus duration—the longer the stimulus, the lower the threshold voltage. An electrical stimulus of less than a certain voltage can only result in local electrochemical changes and cannot elicit a propagated action potential.

REFRACTORINESS A state of absolute refractoriness (i.e., complete inexcitability) occurs immediately after an impulse has been propagated, and no stimulus, no matter how strong or long, can produce an excited state. Shortly thereafter, the axon becomes relatively refractory; it responds with a propagated impulse only to stimulation that is greater than the normal threshold. The length of the refractory period is affected by the frequency of stimulation and by many drugs (Fig. 16.12).

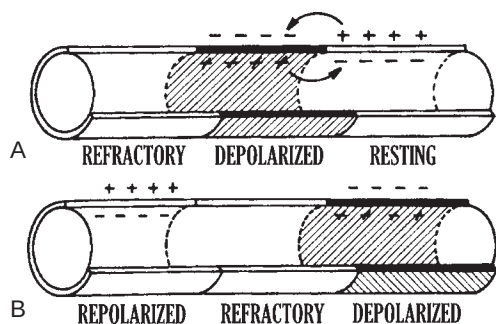


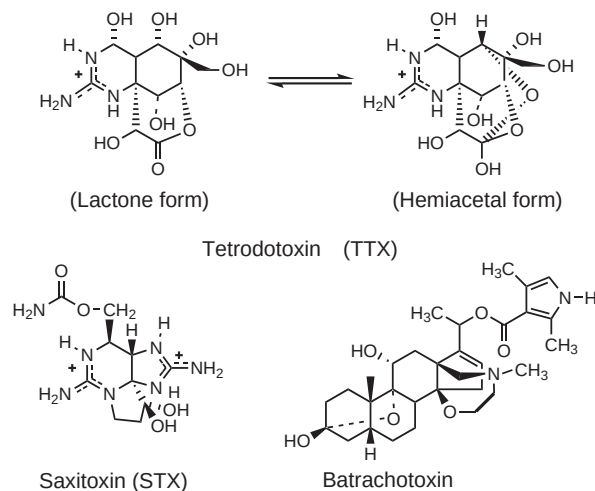
FIGURE 16.12 Impulse propagation. (A) The wave of depolarization passes down the nerve, followed by a wave of refractoriness. (B) The wave of refractoriness is followed by a wave of repolarization. (Adapted from De Jong RH, Freund FG. Physiology of peripheral nerve and local anesthesia. *Int Anesthesiol Clin* 1970;8:35–53, with permission.)

CONDUCTANCE VELOCITY AND NODAL CONDUCTION The conduction velocity is the velocity at which an impulse is conducted along the nerve and is proportional to the diameter of the axon. Because longitudinal resistance is inversely proportional to cross-sectional area, impulses are conducted faster in large-diameter axons. The squid giant axon, used in many neurophysiology investigations, is unmyelinated and exceptionally large (~500 to 1,000 μm); therefore, impulses are conducted rapidly along the axon. However, contraction of the mantle of a squid, which this axon controls, is an uncomplicated procedure that does not require a complex sensorimotor feedback system. Perhaps during evolution, vertebrates developed a complicated input–output system of many axons collected in bundles, as shown in Figure 16.9.

Conduction along these neurons would be slow if they were not insulated with a myelin coat of connective tissue, interrupted at intervals by the nodes of Ranvier, where electrical current enters and exits. These ionic fluxes occurring at the nodes allow the electrical impulse to jump along the axon from node to node much faster than could occur in an unmyelinated axon (42).

SODIUM CHANNEL The voltage-gated Na^+ channels are discrete, membrane-bound glycoproteins that mediate Na^+ permeability and, thus, are responsible for the generation of action potentials in skeletal muscle, nerve, and cardiac tissues (43–50). Our understanding of the structural domains and binding sites on voltage-gated Na^+ channels has evolved considerably since the first cloning and expression studies of the channel protein by Noda (43). The channel gating kinetics have been extensively studied with the use of selective blockers of Na^+ channels, such as tetrodotoxin (TTX) and saxitoxin (STX), and by site-directed mutagenesis (49). Both TTX and STX bind stoichiometrically to the outer opening of the channels and are detected with patch-clamp electrophysiologic techniques on the cut-open squid giant axon (50). Neher and Sakmann, two German scientists, were awarded the

Nobel Prize in 1991 for physiology and medicine for their work on ion channels with these neurotoxins (50).



Recent mapping of receptor binding sites within the channel protein for lipid-soluble neurotoxins, such as batrachotoxin (BTX), and for local anesthetics using site-directed mutagenesis has provided further insight regarding these channels (49). For example, mammalian voltage-gated Na^+ channels contain one large α -subunit and one or two smaller β -subunits (15). The primary structure of the α -subunit is composed of four homologous domains (D1 to D4), each with six transmembrane segments (S1 to S6) and a hydrophobic loop thought to dip into the membrane to align the aqueous pore into a pseudotetrameric arrangement (Fig. 16.13) (47–49).

Furthermore, the voltage sensor for activation gating and the structure for fast inactivation gating have been delineated to involve the positively charged S4 segments of each domain (also known as the ion selectivity filter, which contains positively charged amino acid residues).

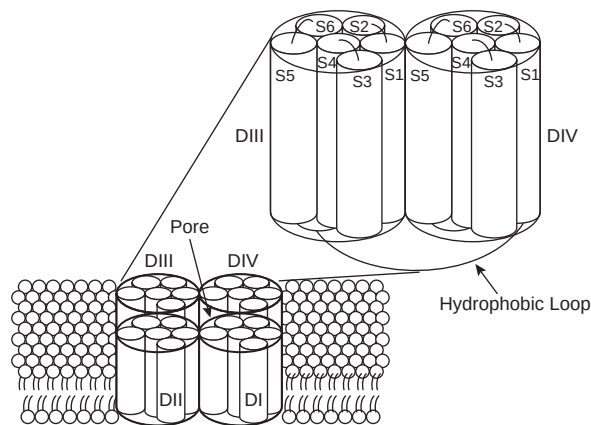


FIGURE 16.13 A schematic representation of the α subunit of the sodium channel and the pore-forming unit. (From Anger T, Madge DJ, Mulla M, et al. Medicinal chemistry of neuronal voltage-gated sodium channel blockers. *J Med Chem* 2001;44:115–137, with permission.)

The selective filter discriminates Na^+ from other ions (i.e., Na^+ passes through this pore approximately 12-fold faster than does K^+). Sodium channels open and close as they switch between several conformational states: the resting/closed form (polarized nonconducting state), the open channel (depolarized conducting state), and the inactivated form (polarized nonconducting state).

At resting potential, the Na^+ channels are in a resting/closed polarized state and are impermeable to the passage of Na^+ . On activation, the channels undergo conformational changes to an open depolarized state, allowing the rapid influx of Na^+ across the neuronal membrane.

Thus, when the threshold potential is exceeded, most of the Na^+ channels are in an open, or conducting, state. At the peak of the action potential, the open channels spontaneously convert to an inactivated polarized state by the “sodium pump” (i.e., nonconducting and non-activatable), leading to a decrease in Na^+ permeability. When a Na^+ channel is in the inactivated polarized state, it cannot be opened without first being transformed to the normal resting/closed form.

Therapeutic Considerations for Using Local Anesthetic Drugs

Since the discovery of cocaine in 1880 as a surgical local anesthetic, several thousand new compounds have been tested and found to produce anesthesia by blocking nerve conductance. Among these agents, approximately 20 are currently clinically available in the United States as local anesthetic preparations (Table 16.8). Table 16.9 contains chemical structures of the different types of agents in current or recent use.

Pharmaceutical Preparations

Local anesthetic agents generally are prepared in various dosage forms: aqueous solutions for parenteral injection, and creams and ointments for topical applications. Thus, chemical stability and aqueous solubility become primary factors in the preparations of suitable pharmaceutical dosage forms.

In general, compounds containing an amide linkage have greater chemical hydrolytic stability than do the ester types. In this regard, an aqueous solution of an amino ester-type local anesthetic is more likely to hydrolyze under normal conditions and cannot withstand heat sterilization as a result of base-catalyzed hydrolysis of the ester.

Local anesthetic activity usually increases with increasing lipid solubility. Unfortunately, this increase in lipid solubility is often inversely related to water solubility. For this reason, a suitable parenteral dosage form might not be available for these agents because of poor water solubility under acceptable conditions. For example, benzocaine, which lacks a sufficiently basic aliphatic amino group needed for salt formation, is insoluble in water at neutral pH. Protonation of the aromatic amino group in benzocaine results in a salt with a $\text{p}K_a$ of 2.78, which is too acidic and, therefore, unsuitable for use as a parenteral dosage form for injection. For this reason, benzocaine and its closely related analog, butamben, are used mostly in creams or ointments to provide topical anesthesia of accessible mucous membranes or skin for burns, cuts, or inflamed mucous surfaces.

Many attempts have been made to substitute oils, fats, or fluid polymers for the aqueous vehicle commonly used in injectable local anesthetics. Unfortunately, the pharmacologic results of these experiments have been quite disappointing, often as a result of the undesirable toxicity of the nonaqueous vehicle.

The only commonly accepted organic additives to local anesthetics are vasoconstrictors, such as epinephrine and levonordefrin (α -methylnorepinephrine). These compounds often increase the frequency of successful anesthesia and, to a limited degree, increase the duration of activity by reducing the rate of drug loss from the injection site, by constricting arterioles that supply blood to the area of the injection. The effect of these vasoconstrictors is less pronounced if the vasoconstrictors are added to a local anesthetic solution that is injected in an area that has profuse venous drainage but is remote from an arterial supply. By slowing the diffusion of the local anesthetic away from the targeted site of injection, the exposure of other tissues in the body is likely minimized such that the local anesthetic never reaches high enough concentrations to produce unwanted toxicities. However, there is a flip side to the benefit of the use of vasoconstrictors: A prolonged local anesthetic effect long after the surgical procedure is completed can lead to prolonged numbness and inadvertent soft tissue damage due to mechanical irritation (e.g., biting one's lip or tongue following dental procedures) as a result of the continued loss of pain sensations. Soft tissue injury to the lip in children 4 to 7 years of age following mandibular nerve block has been reported to be as high as 16%. Recently, an approach to reverse the vasoconstrictor-induced prolonged anesthetic state has utilized phentolamine, an α -adrenoceptor antagonist. Phentolamine mesylate in doses of 0.4 to 0.8 mg in adults and adolescents and 0.2

WHAT ARE THE SOURCES OF THESE NEUROTOXINS?

TTX is a potent neurotoxin isolated from the ovaries and liver of many marine species of Tetraodontidae, especially the Japanese fugu (or puffer fish). STX, produced by the marine dinoflagellates, *Gonyaulax catenella* or *Gonyaulax tamarensis*, is found concentrated in certain bivalve shellfish (e.g., mussels and clams). Consumption of contaminated mussels or clams causes paralytic shellfish poisoning, which is usually associated with toxic “red tide” environmental episodes in various coastal regions. BTX, a cardiotoxic and neurotoxic steroid isolated originally from the poisonous dart frog, *Phyllobates terribilis*, is a lipid-soluble neurotoxin that is at least 10-fold more toxic than TTX.

TABLE 16.8 Clinically Available Local Anesthetics

Generic Name	Trade Name	Recommended Application
Articaine	Septocaine, Septanest	Parenteral (dental)
Benoxinate	Oxybuprocaine	Mainly in ophthalmology
Benzocaine	Americaine, Anbesol, Benzodent, Orajel, Oratect, Rid-A-Pain, Hurricaine	Topical
Benzyl alcohol		Topical, mainly in combination with pramoxine
Bupivacaine	Marcaine, Sensorcaine	Parenteral
Butamben	Butesin	Topical
Chloroprocaine	Nesacaine	Parenteral
Dibucaine	Nupercainal, Cinchocaine	Topical
Dyclonine	Sucrets	Topical (mucosal only)
Etidocaine	Duranest	Parenteral
Ethyl chloride		Extracutaneous, temperature decreasing
Eugenol		Topical, especially in dentistry
Levobupivacaine	Chirocaine	Parenteral
Lidocaine	Xylocaine, L-Caine, DermaFlex, Dilocaine, Lidoject, Lignocaine, Octocaine,	Parenteral, topical
Mepivacaine	Carbocaine, Polocaine, Isocaine	Parenteral, topical
Menthol	Chloraseptic lozenges, Dermoplast, Pramegel, Pontacaine ointment	Topical, mainly in combination with benzocaine or pramoxine or tetracaine
Phenol	Anbesol	Topical, mainly in combination with benzocaine
Pramoxine	Prax, Tronothane	Topical
Prilocaine	Citanest	Parenteral, topical
Procaine	Novocain	Parenteral
Proparacaine	Alcaine, Ophthaine, Ak-Taine	Mainly in ophthalmology
Propoxycaine	Blockaine, Ravocaine	Parenteral
Ropivacaine	Naropin	Parenteral
Tetracaine	Pontocaine, Amethocaine, Prax	Parenteral, topical

to 0.4 mg in children reverses the vasoconstriction and allows for a more rapid diffusion of the local anesthetic from the injection site and a recovery of sensation (51).

Administration of a local anesthetic in a carbonic acid–carbon dioxide aqueous solution rather than the usual solution of a hydrochloride salt appreciably improves the time of onset and duration of action without causing increased local or systemic toxicity.

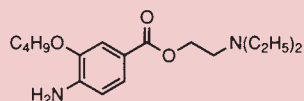
Carbon dioxide is believed to potentiate the action of local anesthetics by initial indirect depression of the axon, followed by diffusion trapping of the active form of the local anesthetic within the nerve. Use of the carbonate salt appears to be one pharmaceutical modification

of the classic local anesthetic agents that can result in significant clinical advantages.

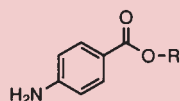
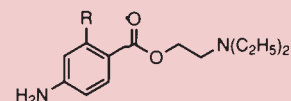
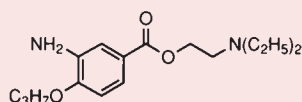
A Eutectic Mixture of a Local Anesthetic (EMLA) cream containing 2.5% lidocaine and 2.5% prilocaine (or etidocaine) is used for the topical application of local anesthetic through the keratinized layer of the intact skin to provide dermal or epidermal analgesia. This mode of administration allows the use of higher concentrations of local anesthetic with minimal local irritation and lower systemic toxicity. The use of EMLA creams, especially those containing prilocaine, on mucous membranes is not recommended, however, because of the faster absorption of the drugs and, therefore, the increasing

TABLE 16.9 Structures of Local Anesthetics

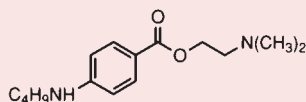
Amino Esters



Benoxinate (pKa = 9.0)

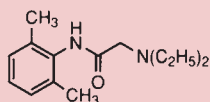
Benzocaine, R= C₂H₅ (pKa = 2.8)
Butamben, R= n-C₄H₉ (pKa = 2.5)Procaine, R= H (pKa = 8.8)
Chlorprocaine, R=Cl (pKa = 9.0)
Propoxycaine, R=OC₃H₇ (pKa = 9.1)

Proparacaine (pKa = 9.1)

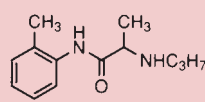


Tetracaine (pKa = 8.4)

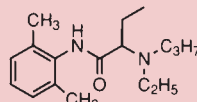
Amino amides



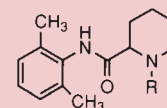
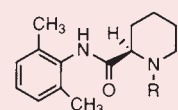
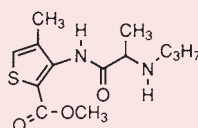
Lidocaine (pKa = 7.8)



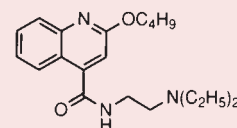
Prilocaine (pKa = 7.9)



Etidocaine (pKa = 7.7)

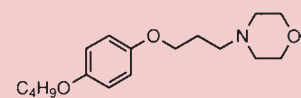
Mepivacaine, R= CH₃ (pKa = 7.6)
Bupivacaine, R= n-C₄H₉ (pKa = 8.1)Ropivacaine, R= n-C₃H₇ (pKa = 8.2)
Levobupivacaine, R= n-C₄H₉ (pKa = 8.1)

Articaine (pKa = 7.8)



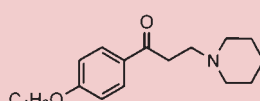
Dibucaine (pKa = 8.8)

Amino ethers



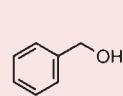
Pramoxine (pKa = 7.1)

Amino ketone

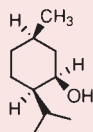


Dyclonine (pKa = 8.2)

Alcohols

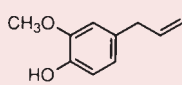


Benzyl alcohol



Menthol

Phenols



Eugenol



Phenol

risk of systemic toxicity, such as methemoglobinemia, specifically with prilocaine (52).

Toxicity and Side Effects

The side effects and toxicity of local anesthetics seem to be related to their actions on other excitable membrane proteins, such as in the Na⁺ and K⁺ channels in the heart, the nicotinic acetylcholine receptors in the neuromuscular junctions, and the nerve cells in the CNS. In general,

neuromuscular junctions and the CNS are more susceptible than the cardiovascular system to the toxic effects of local anesthetics.

The actions on skeletal muscles tend to be transient and reversible, whereas the CNS side effects can be more deleterious. The primary effect of the toxicity seems to be convulsions, followed by severe CNS depression, particularly of the respiratory and cardiovascular centers. This can be related to an initial depression of inhibitory

neurons, such as GABAergic systems, causing convulsions, followed by depression of other neurons, leading to general depression of the CNS.

The amino amide-type local anesthetics (i.e., lidocaine derivatives) are, in general, more likely to produce CNS side effects than the amino ester-type compounds (procaine analog). However, it should be noted that the toxic effects observed depend heavily on the route and site of administration as well as on the lipid solubility and metabolic stability of a given local anesthetic molecule. For example, most amide-type local anesthetics, such as lidocaine, are first degraded via *N*-dealkylation by hepatic enzymes (see Fig. 16.17). Unlike lidocaine, however, the initial metabolic degradation of prilocaine in humans is hydrolysis of the amide linkage to give *o*-toluidine and *N*-propylalanine. Formation of *o*-toluidine and its metabolites can cause methemoglobinemia in some patients (52). For this reason, prilocaine is much more likely than other local anesthetics to cause methemoglobinemia.

In contrast, allergic reactions to local anesthetics, although rare, are known to occur exclusively with *p*-aminobenzoic acid (PABA) ester-type local anesthetics (53). Whether the formation of PABA upon ester hydrolysis is solely responsible for this hypersensitivity remains to be determined. However, the preservative compounds, such as methylparaben, used in the preparation of amide-type local anesthetics are metabolized to the PABA-like substance, *p*-hydroxybenzoic acid. Thus, patients who are allergic to amino ester-type local anesthetics should be treated with a preservative-free amino amide-type local anesthetic.

Amide-type local anesthetics (e.g., procainamide and lidocaine) also possess antiarrhythmic activity when given parenterally and at a subanesthetic dosage. Although this action is likely an extension of their effects on Na⁺ channels in cardiac tissues, some evidence suggests a distinctly different mechanism of action with respect to the modulation of channel receptors and the location of binding sites for these compounds (54,55).

Chemical and Pharmacodynamic Aspects of Local Anesthetics

Mechanism of Action

Local anesthetics act by decreasing the excitability of nerve cells without affecting the resting potential. Because the action potential, or the ability of nerve cells to be excited, is associated with the movement of Na⁺ across the nerve membranes, anything that interferes with the movement of these ions will interfere with cell excitability. For this reason, many hypotheses have been suggested to explain how local anesthetics regulate the changes in Na⁺ permeability that underlie the nerve impulse. These hypotheses include direct action on ionic channels that interfere with ionic fluxes and interaction with phospholipids and calcium that reduces membrane

flexibility and responsiveness to changes in electrical fields.

The nonspecific membrane actions of local anesthetics can be easily ruled out, because most clinically useful agents, in contrast to general anesthetics, possess a defined set of structure–activity relationships. At much higher drug concentrations, local anesthetics also bind and block K⁺ channels.

INTERACTION WITH PHOSPHOLIPIDS AND CALCIUM Calcium exists in the membrane in a bound state. Many investigators believe that the release of the bound calcium is the first step in membrane depolarization and that this release leads to the changes in ionic permeability described previously. It has been suggested that local anesthetics displace the bound calcium from these sites and form more stable bonds, thereby inhibiting ionic fluxes. The following evidence has been offered in support of this theory: Both calcium and local anesthetics bind to phospholipids *in vitro*, reducing their flexibility and responsiveness to changes in electrical fields (56–58). Also, membrane excitability and instability increase in calcium-deficient solutions. Local anesthetics counteract this abnormal increase in excitability, and more local anesthetic is necessary to block excitation in calcium-poor solutions (59). Direct proof of this hypothesis, however, is lacking because of the difficulty in measuring temporal Ca²⁺ movements *in vivo*. It is also possible that the aforementioned cause-and-effect relationship between intracellular free Ca²⁺ and membrane excitability is the result of an Na⁺–Ca²⁺ exchange reaction; that is, the influx of Na⁺ displaces the membrane-bound calcium, which leads to an increase of intracellular free Ca²⁺ and, thereby, increases cellular excitability.

Local anesthetics interact differently, however, with neuronal phospholipids with or without the presence of cholesterol. Thus, the interactions of local anesthetics with the cellular membranes actually may help to explain some of the observed differences in toxicity of the individual local anesthetic agents (60).

ACTION ON VOLTAGE-SENSITIVE SODIUM CHANNELS As mentioned, the voltage-sensitive Na⁺ channels are membrane-bound glycoproteins that mediate Na⁺ permeability. On excitation, these channels undergo conformational changes from a closed to an open state, thus allowing a rapid influx of sodium. The movement of Na⁺ is blocked by the neurotoxins TTX and STX and by local anesthetics (61). Most electrophysiologists and neuropharmacologists now agree that the mechanism of action of local anesthetics results primarily from their binding to one or more sites within the Na⁺ channels, thus blocking Na⁺ conductance (62). However, the exact location of these binding sites and whether all local anesthetics interact with a common site remain matters of dispute.

ACTION ON SODIUM CONDUCTANCE Local anesthetics block Na^+ conductance by two possible modes of action: tonic inhibition and phasic inhibition (63,64). Tonic inhibition results from the binding of local anesthetics to non-activated closed Na^+ channels and, thus, is independent of channel activation. Phasic inhibition is accomplished when local anesthetics bind to activated, open states (conducting) or to inactivated states (nonconducting) of the Na^+ channels. Thus, it is not surprising that a greater phasic inhibition usually is obtained with repetitive depolarization and is referred to as use-dependent blockade.

Two reasons have been suggested to explain this observation. First, channel inactivation during depolarization increases the number of binding sites that normally are inaccessible to local anesthetics at resting potential. Second, both the open and the inactivated channels possess binding sites with a higher affinity; therefore, local anesthetics bind more tightly and result in a more stable nerve block.

Furthermore, it is generally agreed that most of the clinically useful local anesthetics exert their actions by binding to the inactivated forms of the channels and, thus, prevent their transition to the original resting state (64). Because most of these drugs exhibit both tonic and phasic inhibitions, whether tonic and phasic block results from drug interaction at the same or different sites remains unclear.

LOCAL ANESTHETICS BINDING TO SODIUM CHANNELS Most of the clinically useful local anesthetics are tertiary amines with a pK_a of 7.0 to 9.0. Thus, under physiologic conditions, both protonated forms (onium ions) and the un-ionized, molecular forms are available for binding to the channel proteins. In fact, the ratio between the onium ions $[\text{BH}^+]$ and the un-ionized molecules $[\text{B}]$ can be easily calculated based on the pH of the medium and the pK_a of the drug molecule by the Henderson-Hasselbalch equation:

$$\text{pH} = \text{pK}_a - \log [\text{BH}^+]/[\text{B}]$$

The effect of pH changes on the potency of local anesthetics has been extensively investigated (65). Based on these studies, it was concluded that local anesthetics block the action potential by first penetrating the nerve membrane in their un-ionized forms and then binding to a site within the channels in their onium forms. Perhaps the most direct support for this hypothesis comes from the experimental results of Narahashi et al. (66,67), who studied the effects of internal and external perfusion of local anesthetics (both tertiary amines and quaternary ammonium compounds), at different pH values, on the Na^+ conductance of the squid giant axon. The observation that both tertiary amines and quaternary ammonium compounds produce greater nerve blockage when applied internally indicates an axoplasmic site of action for these compounds.

Furthermore, only the tertiary amines exhibit a reduction in their local anesthetic activities when the internal pH is raised from 7.0 to 8.0. Because the increase of internal pH to 8.0 favors the existence of the un-ionized

forms, this result again suggests that the onium ions are required for binding to the channel receptors. Narahashi and Frazier (68) further estimated that approximately 90% of the blocking actions of lidocaine can be attributed to onium forms of the drug molecule, whereas only approximately 10% can result from un-ionized molecule and, perhaps, at a hydrophobic binding site other than the primary binding site. Benzocaine, because of its lack of a basic amine group ($\text{pK}_a = 2.78$), and other neutral anesthetics, such as benzyl alcohol, have been suggested to bind to this hydrophobic binding site.

In 1984, Hille (69) proposed a unified theory involving a single binding site in the Na^+ channels for both onium ions (protonated tertiary amines and quaternary ammonium compounds) and un-ionized forms of local anesthetics. As depicted in Figure 16.14, a number of pathways are available, depending on the size, pK_a , and lipid solubility of the drug molecules as well as the voltage and frequency-dependent modulation of the channel states, for a drug to reach its binding sites. Protonated anesthetic molecules $[\text{BH}^+]$ and quaternary ammonium compounds reach their target sites via the hydrophilic pathway externally (pathway b in Fig. 16.14), which is available only during channel activation.

The lipid-soluble anesthetic molecules, on the other hand, diffuse across the neuronal membrane in their un-ionized forms. They can interact with the same binding sites from either the hydrophilic pathway (pathway b' in Fig. 16.14) on reprotonation to their onium ions $[\text{BH}^+]$ or via the hydrophobic pathway (pathway a in Fig. 16.14) in their un-ionized forms $[\text{B}]$. Benzocaine and other nonbasic local anesthetic molecules use this hydrophobic pathway and, thus, bind in the hydrophobic domain to produce their actions. Site-directed mutagenesis studies (70–73) suggest that local anesthetics bind to the hydrophobic amino acid residues near the center and the intracellular end of the S6 segment in the domain

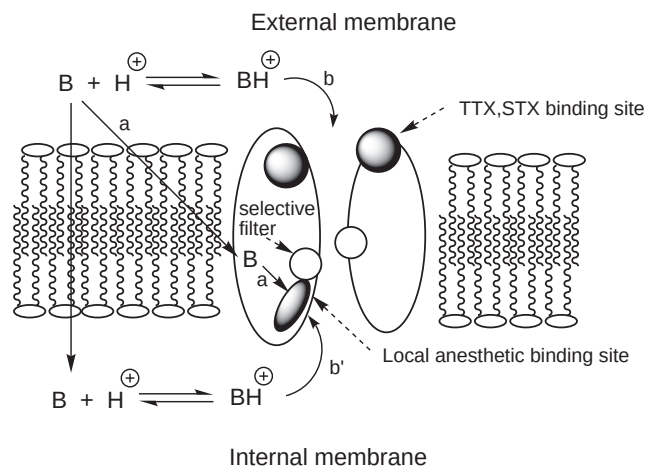


FIGURE 16.14 Model of a sodium channel, as suggested by Hille (69), depicting a hydrophilic pathway (denoted by b and b') and a hydrophobic pathway (denoted by a) by which local anesthetics can reach their binding sites.

D4, whereas the BTX receptor is within segment S6 in domain D1 of the α subunit of the Na^+ channels (Fig. 16.14) (49).

Structure–Activity Relationships

A quick perusal of Table 16.9 reveals that many diverse chemical structures possess local anesthetic properties: amino esters (procaine analogs), amino amides (lidocaine analogs), amino ethers (pramoxine), amino ketones (dyclonine), alcohols (benzyl alcohol and menthol), and phenols (eugenol and phenol). Although it would seem that there is no obvious structure–activity relationship among these agents, most of the clinically useful local anesthetics are tertiary amines with $\text{p}K_a$ values of 7.0 to 9.0. These compounds exhibit their local anesthetic properties by virtue of the interactions of the onium ions with a selective binding site within the Na^+ channels (Fig. 16.14). For this reason, any structural modifications that alter the lipid solubility, $\text{p}K_a$, and metabolic inactivation have a pronounced effect on the ability of a drug molecule to reach or interact with the hypothetical binding sites, thus modifying its local anesthetic properties.

LIPHILIC PORTION The lipophilic portion of the molecule is essential for local anesthetic activity. For most of the clinically useful local anesthetics, this portion of the molecule consists of either an aromatic group directly attached to a carbonyl function (the amino ester series) or a 2,6-dimethylphenyl group attached to a carbonyl function through an $-\text{NH}-$ group (the amino amide series) (Fig. 16.15). Both groups are highly lipophilic

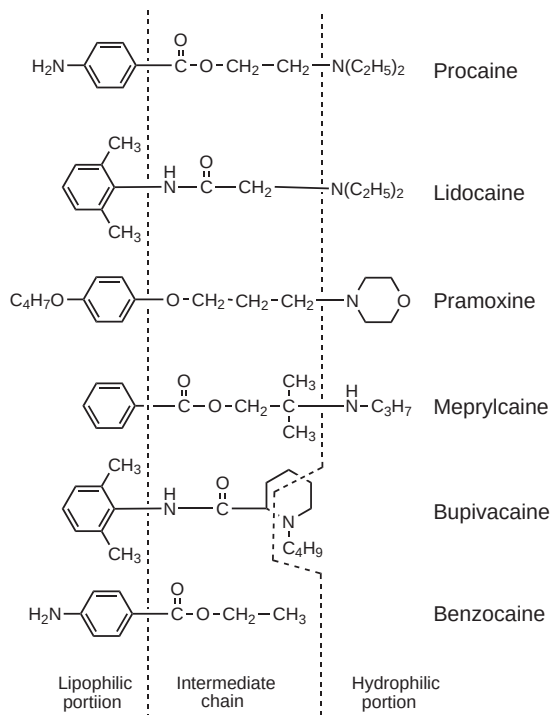


FIGURE 16.15 Structure–activity relationship comparison of local anesthetics.

and appear to play an important role in the binding of local anesthetics to the channel proteins. Structural modification of this portion of the molecule has a profound effect on its physical and chemical properties, which in turn alters its local anesthetic properties.

In the amino ester series, an electron-donating substituent in the *ortho* or *para* (or both) positions increases local anesthetic potency. Such groups as an aromatic amino (procaine, chlorprocaine, and propoxycaine), an alkylamino (tetracaine), or an alkoxy (proparacaine and propoxycaine) group contribute electron density to the aromatic ring by both resonance and inductive effects, thereby enhancing local anesthetic potency over nonsubstituted analogs (e.g., mepylcaine).

As illustrated in Figure 16.16, resonance is expected to give rise to a zwitterionic form (i.e., the electrons from the amino group can be resonance delocalized onto the carbonyl oxygen). Although neither drawn structure of procaine in Figure 16.16 can accurately represent the structure of procaine when it interacts with the local anesthetic binding site, it is reasonable to assume that the greater the resemblance to the zwitterionic form, the greater the affinity for the binding site (i.e., binding from both the hydrophilic pathway b' and hydrophobic pathway a in Fig. 16.14). This is particularly true for the affinity of benzocaine for its binding site, because it lacks a basic amine group. Therefore, it can only bind from the hydrophobic pathway a. Thus, addition of any aromatic substitution that can enhance the formation of the resonance form will produce more potent local anesthetic agents. Electron-withdrawing groups, such as nitro ($-\text{NO}_2$), reduce the local anesthetic activity.

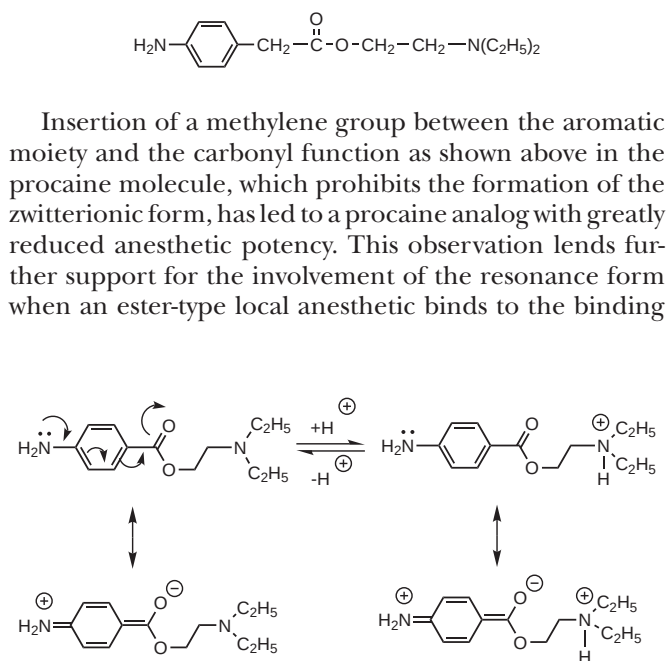


FIGURE 16.16 Possible zwitterionic forms for procaine.

site. When an amino or an alkoxy group is attached to the *meta* position of the aromatic ring, however, no resonance delocalization of their electrons is possible. The addition of this function only increases (alkoxy group) or decreases (amino group) the lipophilicity of the molecule (e.g., benoxinate, $\log D_{\text{pH } 7.4} = 3.19$; and proparacaine, $\log D_{\text{pH } 7.4} = 2.05$).

Furthermore, tetracaine is approximately 50-fold more potent than procaine. Experimentally, this increase in potency cannot be correlated solely with the 2,500-fold increase of lipid solubility by the *n*-butyl group ($\log D_{\text{pH } 7.4} = 2.73$ vs. procaine $\log D_{\text{pH } 7.4} = -0.32$). Perhaps part of this potentiation of local anesthetic activity can be attributed to the electron-releasing property of the *n*-butyl group via the inductive effect, which indirectly enhances the electron density of the *p*-amino group, which in turn increases the formation of the zwitterionic form available for interaction with the binding site proteins via both the hydrophobic and the hydrophilic pathways of the receptor.

Another important aspect of aromatic substitution has been observed from structure–activity relationship studies. In the amino amides (lidocaine analogs), the *o,o'*-dimethyl groups are required to provide suitable protection from amide hydrolysis to ensure a desirable duration of action. Similar conclusions can be made to rationalize the increase in the duration of action of propoxycaine by the *o*-propoxy group. The shorter duration of action, however, observed with chloroprocaine when compared with that of procaine can only be explained by the inductive effect of the *o*-chloro group, which pulls the electron density away from the carbonyl function, thus making it more susceptible to nucleophilic attack by the plasma cholinesterases.

INTERMEDIATE CHAIN The intermediate chain almost always contains a short alkyl chain of one to three carbons in length linked to the aromatic ring via several possible organic functional groups. The nature of this intermediate chain determines the chemical stability of the drug, which also influences the duration of action and relative toxicity. In general, amino amides are more resistant to metabolic hydrolysis than the amino esters and, thus, have a longer duration of action. The placement of small alkyl groups (i.e., branching), especially around the ester function (e.g., meprylcaine) or the amide function (e.g., bupivacaine, etidocaine, prilocaine, or ropivacaine), also hinders esterase- or amidase-catalyzed hydrolysis, prolonging the duration of action (Fig. 16.15 and Table 16.9). It should be mentioned, however, that prolonging the duration of action of a compound usually increases its systemic toxicities unless it is more selective toward the voltage-gated Na^+ channel, as in the case of levobupivacaine (74,75).

In the lidocaine series, lengthening of the alkyl chain from one to two or three increases the pK_a of the terminal tertiary amino group from 7.7 to 9.0 or 9.5, respectively. Thus, lengthening of the intermediate chain effectively

reduces local anesthetic potency as a result of a reduction of onium ions under physiologic conditions. As mentioned earlier, the onium ions are required for effective binding of the amino amide–type local anesthetics to the channel binding sites.

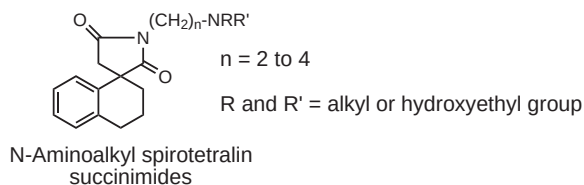
HYDROPHILIC PORTION Most clinically useful local anesthetics have a tertiary alkylamine, which readily forms water-soluble salts with mineral acids, and this portion is commonly considered to be the hydrophilic portion of the molecule (Fig. 16.9). The necessity of this portion of the molecule for amino ester–type local anesthetics remains a matter of debate. The strongest opposition for requiring a basic amino group for local anesthetic action comes from the observation that benzocaine, which lacks the basic aliphatic amine function, has potent local anesthetic activity. For this reason, it is often suggested that the tertiary amine function in procaine analogs is needed only for the formation of water-soluble salts suitable for pharmaceutical preparations. With the understanding of the voltage-activated Na^+ channel and the possible mechanism of action of local anesthetics previously discussed, however, it is quite conceivable that the onium ions produced by protonation of the tertiary amine group are also required for binding in the voltage-gated Na^+ channels (Fig. 16.14).

From Table 16.9, the hydrophilic group in most of the clinically useful drugs can be in the form of a secondary or tertiary alkyl amine or part of a nitrogen heterocycle (e.g., pyrrolidine, piperidine, or morpholine). As mentioned earlier, most of the clinically useful local anesthetics have pK_a values of 7.5 to 9.0. The effects of an alkyl substituent on the pK_a depend on the size, length, and hydrophobicity of the group; and thus, it is difficult to see a clear structure–activity relationship among these structures. It is generally accepted that local anesthetics with higher lipid solubility and lower pK_a values appear to exhibit more rapid onset and lower toxicity.

Stereochemistry

Are there any stereochemical requirements of local anesthetic compounds when they interact with the Na^+ channel binding sites? A number of clinically used local anesthetics do contain a chiral center (i.e., bupivacaine, etidocaine, mepivacaine, and prilocaine) (Table 16.9), but in contrast to other classes of drugs (e.g., cholinergics), the effect of optical isomerism on isolated nerve preparations revealed a lack of stereospecificity. In a few cases (e.g., prilocaine, bupivacaine, and etidocaine), however, small differences in the total pharmacologic profile of optical isomers have been noted when administered in vivo (76–78). Whether these differences result from differences in uptake, distribution, and metabolism or from direct binding to the Na^+ channel have not been determined. When structural rigidity has been imposed on the molecule, however, as in the case of some aminoalkyl spiro-*etralin* succinimides (79), differences in local anesthetic potency of the enantiomers have been observed (range, 1:2 to 1:10). Although these differences in enantiomers

clearly are not as pronounced as those with other pharmacologic agents, such as adrenergic antagonists or anticholinergic drugs, steric requirements are necessary for effective interaction between a local anesthetic agent and its proposed channel binding sites.



Stereochemistry of the local anesthetics, however, plays an important role in their observed toxicity and pharmacokinetic properties. For example, ropivacaine and levobupivacaine, the only optically active local anesthetics currently being marketed, have considerably lower cardiac toxicities than their close structural analog, bupivacaine (80). Furthermore, the degree of separation between motor and sensory blockade is more apparent with ropivacaine and levobupivacaine relative to bupivacaine at a lower end of the dosage scale (81). Thus, the observed cardiac toxicity of bupivacaine has been attributed to the *R*(+)-bupivacaine enantiomer (76–78). The exact mechanisms for this enantiomeric difference remain unknown. Longobardo and colleagues observed a stereoselective blockade on the cardiac hKv1.5 channels by the *R*(+)-enantiomers of bupivacaine, ropivacaine, and mepivacaine (82,83). It should be noted that *S*(-)-bupivacaine, which is approved by the U.S. Food and Drug Administration and marketed under the name of Chirocaine, has even less CNS toxicity than ropivacaine.

Metabolism of Local Anesthetics

An understanding of the metabolism of local anesthetics is important in clinical practice because the overall toxicity of a drug depends not only on its uptake and tissue distribution but also on how it is deactivated in vivo. The amino ester-type local anesthetics are rapidly hydrolyzed by plasma cholinesterase (also known as pseudocholinesterase), which is widely distributed in body tissues. These compounds can therefore be metabolized in the blood, kidneys, and liver and, to a lesser extent, at the site of administration. For example, both procaine and benzocaine are easily hydrolyzed by cholinesterase into PABA and the corresponding *N,N*-diethylaminoethyl alcohol.

It is not surprising that potential drug interactions exist between the amino ester-type local anesthetics and other clinically important drugs, such as cholinesterase inhibitors or atropine-like anticholinergic drugs (see Chapter 9). These compounds either inhibit or compete with local anesthetics for cholinesterases, therefore prolonging local anesthetic activity and/or toxicity. Another potential drug interaction with clinical significance can be envisioned between benzocaine and sulfonamides; that is, the hydrolysis of benzocaine to PABA can antagonize the antibacterial activity of sulfonamides.

The amino amide-type local anesthetics, however, are metabolized primarily in the liver, involving CYP1A2 isozymes (84). A general metabolic scheme for lidocaine is shown in Figure 16.17.

Marked species variations occur in the quantitative urinary excretion of these metabolites. For example, rats produce large quantities of the 3-hydroxy derivatives of both lidocaine and monoethylglycinexylidide, which are subsequently conjugated and recycled in the bile. Significant quantities of these two metabolites, however, are not produced by guinea pigs, dogs, or humans. Therefore, it is unlikely that biliary excretion is a major pathway for excretion in humans. Species variability is important primarily when the acute and chronic toxicity of nonester-type local anesthetic agents is being evaluated.

Although the exact mechanism for the CNS toxicity of lidocaine remains unclear, the metabolic studies of lidocaine provide some insight for future studies. Of all the metabolites of lidocaine, only monoethylglycinexylidide (and not glycinexylidide) contributes to some of the CNS side effects of lidocaine. This observation suggests that the toxicities of lidocaine are, perhaps, related to the

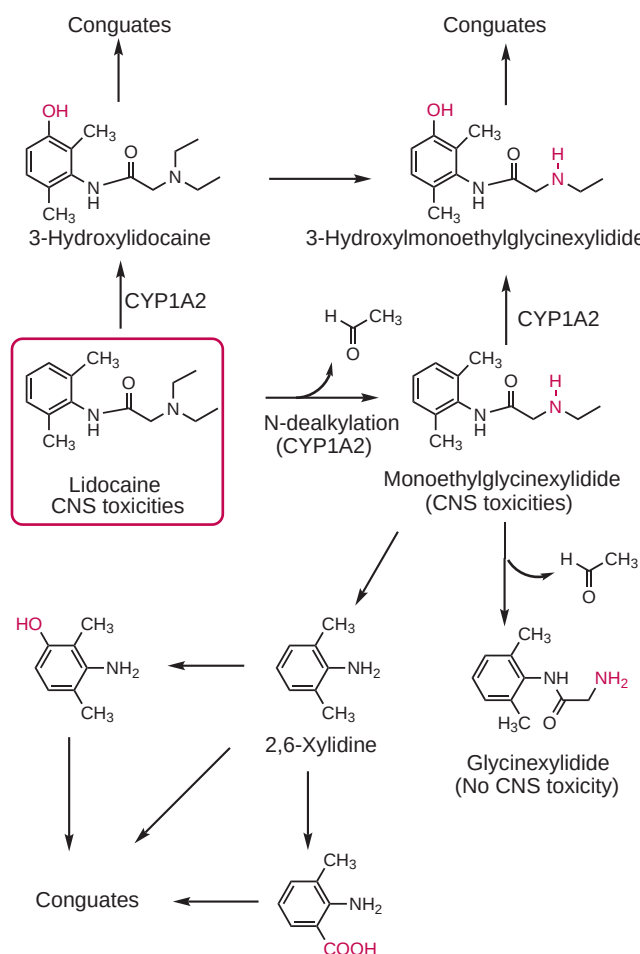
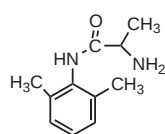
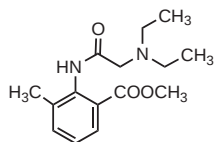


FIGURE 16.17 Metabolic scheme for lidocaine.

removal of the *N*-ethyl groups of lidocaine after crossing the blood–brain barrier. Support for this hypothesis can be obtained from the fact that reaction of a tryptophan derivative with formaldehyde under physiologic conditions gives rise to a β -carboline derivative, which is a CNS convulsant (Fig. 16.18). Advances in the GABA_A receptor–benzodiazepine receptor–chloride ion channels and their role in the mechanism of action of benzodiazepine anticonvulsants lends further support to this hypothesis (i.e., many β -carbolines are inverse agonists at the benzodiazepine binding site).



Tocainide



Tolycaine

To minimize these unwanted side effects of lidocaine, tocainide and tolycaine have been prepared and found to possess good local anesthetic activity without any appreciable CNS side effects. Tocainide, which lacks the vulnerable *N*-ethyl group but has an α -methyl group to prevent degradation of the primary amine group from amine oxidase, has desirable local anesthetic properties. Tolycaine has an *o*-carbomethoxy substituted for one of the *o*-methyl groups of lidocaine. The carbomethoxy group is fairly stable in tissues but is rapidly hydrolyzed in the blood to the polar carboxylate group and, thus, is unable to cross the blood–brain barrier.

For this reason, tolycaine lacks any CNS side effects, even though it still contains the *N*-ethyl group. It should be noted, however, that both tocainide and tolycaine are primarily used clinically as antiarrhythmic agents.

Furthermore, the metabolism of nonester-type drugs, especially lidocaine derivatives, is also known to be more

prone to being influenced by enzyme induction or inhibition due to other coadministered medications (e.g., cimetidine and barbiturates).

Common Agents Used for Local Anesthesia

Local anesthetics are widely used in many primary care settings. Techniques for their administration in these settings include topical application, local infiltration, field block, and peripheral nerve block. Their use can be maximized by an understanding of their potencies, durations of action, routes of administration, and pharmacokinetic and side effect profiles. The generic name, trade name, and recommended application are given in Table 16.8, and the chemical structures of these agents can be found in Table 16.9.

Articaine

Articaine (Table 16.9) has been widely used in dentistry since its U.S. Food and Drug Administration approval in 2000 due to its quick onset and short duration of action. The structure of articaine differs from the structures of all other amino amide-type local anesthetics in that it contains the bioisosteric thiophene ring instead of a benzene ring and a carbomethoxy group. This renders the molecule more lipophilic and, thus, makes it easier to cross lipid membranes.

Its local anesthetic potency is approximately 1.5-fold that of lidocaine, even though it has similar pK_a (7.8) and smaller $\log D_{pH\ 7.4}$ (1.65 vs. $\log D_{pH\ 7.4}$ of 2.26 for lidocaine) and plasma protein binding (76%) properties. Articaine is metabolized primarily by plasma cholinesterases because of the presence of an ester group and, therefore, has a much shorter duration of action than lidocaine (i.e., only approximately one-fourth that of lidocaine). Articaine undergoes rapid hydrolysis of the carbomethoxy group to give articainic acid, which is eliminated either unchanged (75%) or as its glucuronides (25%). Compared with other short-acting, amino amide-type local anesthetics, such as mepivacaine, lidocaine, or prilocaine, articaine is said to be a much safer drug for regional anesthesia and is the drug of choice for dental procedures.

Benzocaine

Benzocaine (Table 16.9) is used topically by itself or in combination with menthol or phenol in nonprescription dosage forms such as gels, creams, ointments, lotions, aerosols, and lozenges to relieve pain or irritation caused by such conditions as sunburn, insect bites, toothache, teething, cold sores or canker sores in or around the mouth, and fever blisters. Benzocaine is a lipophilic local anesthetic agent with a short duration of action.

Like most amino ester-type local anesthetics, it is easily hydrolyzed by plasma cholinesterase. However, because of its low pK_a , it is un-ionized under most physiologic conditions and, therefore, can only bind to the lipid site in the sodium channel ($\log D_{pH\ 7.4} = 1.91$) (Fig. 16.14). When administered topically to abraded skin, it

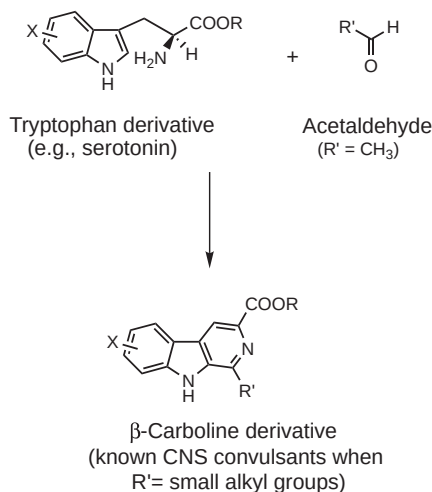


FIGURE 16.18 Reaction of a tryptophan derivative with acetaldehyde under physiologic conditions.

can easily cross membranes into the systemic circulation to cause systemic toxicities. Furthermore, being a PABA derivative, it has similar allergenic properties to procaine and is contraindicated with sulfonamide antibacterial agents.

Bupivacaine and Levobupivacaine

Bupivacaine hydrochloride (Table 16.8) is a racemic mixture of the *S*(-)- and *R*(+)-enantiomers. Bupivacaine has higher lipid solubility ($\log D_{\text{pH } 7.4} = 2.54$) and a much decreased rate of hepatic degradation compared with lidocaine. For this reason, bupivacaine has significantly greater tendency than lidocaine to produce cardiotoxicity. Because of its greater affinity for voltage-gated Na^+ channels, the *R*(+)-enantiomer confers greater cardiotoxicity than racemic bupivacaine.

It was not surprising to see the approval of levobupivacaine, the *S*(-)-enantiomer of ($\pm$)-bupivacaine, as the second optically active, amino amide-type local anesthetic for parenteral applications. Like ropivacaine, levobupivacaine has a lower cardiotoxicity than bupivacaine, but it also has a lower CNS toxicity than both ropivacaine and lidocaine.

Possible pathways for metabolism of bupivacaine include CYP1A2 aromatic 3-hydroxylation, CYP3A4 *N*-dealkylation, and to a minor extent, the amide hydrolysis. Only the *N*-dealkylated product, however, has been identified in urine after epidural or spinal anesthesia.

Chloroprocaine

Chloroprocaine (Fig. 16.8) is a very short-acting, amino ester-type local anesthetic used to provide regional anesthesia by infiltration as well as by peripheral and central nerve block, including lumbar and caudal epidural blocks. The presence of a chlorine atom *ortho* to the carbonyl of the ester function increases its lipophilicity ($\log D_{\text{pH } 7.4} = 0.95$) and its rate of hydrolysis by plasma cholinesterase at least threefold compared to procaine and benzocaine. Thus, chloroprocaine can be used in maternal and neonatal patients with minimal placental passage of chloroprocaine. The lower plasma cholinesterase activity in the maternal epidural space must still have sufficient activity for degrading chloroprocaine and, thus, not allowing it to cross the placental barrier. Like PABA, the hydrolysis product of chloroprocaine, 4-amino-2-chlorobenzoic acid, also inhibits the action of sulfonamides. Therefore, its use with sulfonamides should be avoided.

Lidocaine

Lidocaine (Fig. 16.8) is the most commonly used amino amide-type local anesthetic. Lidocaine is very lipid soluble ($\log D_{\text{pH } 7.4} = 2.26$) and, thus, has a more rapid onset and a longer duration of action than most amino ester-type local anesthetics, such as procaine and tetracaine. It can be administered parenterally (with or without epinephrine) or topically either by itself or in combination

with prilocaine or etidocaine as a eutectic mixture that is very popular with pediatric patients. The use of lidocaine–epinephrine mixtures should be avoided, however, in the areas with limited vascular supply to prevent tissue necrosis. Lidocaine is also frequently used as a class IB antiarrhythmic agent for the treatment of ventricular arrhythmias, both because it binds and inhibits Na^+ channels in the cardiac muscle and because of its longer duration of action than amino ester-type local anesthetics (Chapter 21).

CNS changes are the most frequently observed systemic toxicities of lidocaine. The initial manifestations are restlessness, vertigo, tinnitus, slurred speech, and eventually, seizures. Subsequent manifestations include CNS depression with a cessation of convulsions and the onset of unconsciousness and respiratory depression or cardiac arrest. This biphasic effect occurs because local anesthetics initially block the inhibitory GABAergic pathways, resulting in stimulation, and eventually block both inhibitory and excitatory pathways (i.e., block the Na^+ channels associated with the NMDA receptors, resulting in overall CNS inhibition) (85).

Lidocaine is extensively metabolized in the liver by CYP3A4 *N*-dealkylation and aromatic hydroxylation catalyzed by CYP1A2 (Fig. 16.17). Lidocaine also possesses a weak inhibitory activity toward the CYP1A2 isozymes and, therefore, can interfere with metabolism of other medications (86).

Mepivacaine

Mepivacaine hydrochloride (Fig. 16.8) is an amino amide-type local anesthetic agent widely used to provide regional analgesia and anesthesia by local infiltration, peripheral nerve block, and epidural and caudal blocks. The pharmacologic and toxicologic profile of mepivacaine is quite similar to that of lidocaine, except that mepivacaine is less lipophilic ($\log D_{\text{pH } 7.4} = 1.95$) and has a slightly longer duration of action but lacks the vasodilator activity of lidocaine. For this reason, it serves as an alternate choice for lidocaine when addition of epinephrine is not recommended in patients with hypertensive vascular disease.

Mepivacaine undergoes extensive hepatic metabolism catalyzed by CYP3A4 and CYP1A2, with only a small percentage of the administered dosage (<10%) being excreted unchanged in the urine. The major metabolic biotransformations of mepivacaine are *N*-dealkylation (to give the *N*-demethylated compound 2',6'-pipecoloxylidide) and aromatic hydroxylations. These metabolites are excreted as their corresponding glucuronides.

Ropivacaine

S-Ropivacaine hydrochloride (Fig. 16.8) is the first optically active, amino amide-type local anesthetic marketed in recent years. It combines the anesthetic potency and long duration of action of ($\pm$)-bupivacaine with a side effect profile intermediate between those of

SCENARIO: OUTCOME AND ANALYSIS

Outcome

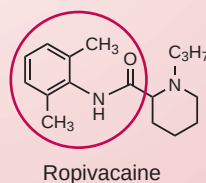
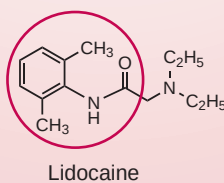
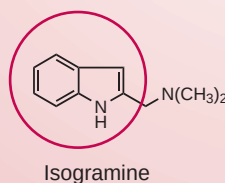
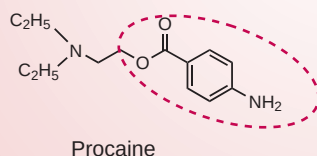
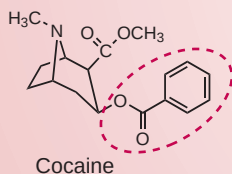
Paul Arpino

After consulting with the covering pharmacist and the hospital's allergy specialist, the team decides to proceed with the type of anesthesia and surgery as planned. On the day of the surgery, CDL received the peripheral nerve block with ropivacaine and required minimal adjunctive medications during the procedure. Postoperatively, CDL's pain was well controlled and there was no evidence of an allergic reaction.

Chemical Analysis

S. William Zito and Victoria Roche

Procaine is a local anesthetic developed from the investigation of cocaine analogs that were synthesized in an effort to enhance the local anesthetic properties and reduce cocaine's addiction and acute toxicity liability. Cocaine and procaine are benzoic acid derivatives and both are known to cause allergic reactions.



The aminoalkyl side chain is not necessary for local anesthetic activity but is useful for forming water-soluble salts for parenteral administration. The para-amino phenyl substituent enhances local anesthetic effect by increasing the electron density of the carbonyl oxygen through electron induction.

Ropivacaine is a local anesthetic developed from another natural alkaloid, isogramine. Structure activity studies of isogramine led to the development of the lidocaine class (anilides) of local anesthetics.

The lidocaine types of local anesthetics are bioisosteres of isogramine. The amino alkyl side chain serves to form water-soluble salts for parenteral administration. As anilides, these compounds have a longer duration of action compared with the benzoic acid ester class, and the 2,6-dimethyls of the aromatic ring also serve to increase duration of action.

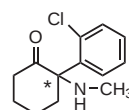
This class of local anesthetics is less irritating upon injection and, most importantly for this case, have no cross-allergic sensitivity with the benzoic acid class of local anesthetics. General anesthesia must be avoided in this case because it suppresses upper airway muscle activity, and it may impair breathing by allowing the airway to close. General anesthesia thus may increase the number of and duration of sleep apnea episodes of this patient.

CASE STUDY

S. William Zito and Victoria Roche

BB is a 19-year-old pregnant woman in her second trimester. Until now she has had an unremarkable pregnancy. However, today she presents to the emergency department complaining of 3 days of right lower quadrant pain, anorexia, and persistent nausea and vomiting. A physical examination of BB revealed vital signs of blood pressure, 127/68 mm Hg; heart rate, 86 beats per minute; respiratory rate, 18 breaths per minute, and a temperature of 97.5 °F. A diagnosis is made of acute appendicitis and surgery is recommended. Preoperative testing showed slight leucocytosis with a white blood cell count of 12,000/mm³; her hemoglobin was 12.1 gm/dL, with a hematocrit level of 34.9% and platelet count of 306,000/mm³. BB's medical history was unexceptional; she has had no previous health issues and she is not taking any medications except for prenatal multivitamins, which contain folic acid and iron. She reports no surgical history and that her parents have had operations for which, to her knowledge, there were no anesthetic sequelae. The surgical trauma team expects the operation to take 45 minutes and plans to

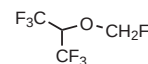
use general anesthesia using rapid induction and securing her airway with a small cuffed endotracheal tube. The following three general anesthetics are proposed. Which one would you recommend?



1



2



3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

bupivacaine and lidocaine. Although ropivacaine has a pK_a nearly identical to that of bupivacaine, it is two- to threefold less lipid soluble ($\log D_{pH\ 7.4} = 2.06$) and has a smaller volume of distribution, a greater clearance, and a shorter elimination half-life than bupivacaine in humans.

The metabolism of ropivacaine in humans is mediated by hepatic CYP1A2 and, to a minor extent, by CYP3A4 (87). The major metabolite is 3-hydroxyropivacaine, and the minor metabolite is *S*-2',6'-pipecoloxylidide (an *N*-dealkylated product).

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Chapter 17

Antiseizure Drugs

BARBARA LeDUC

Drugs Covered in This Chapter

PRIMARY DRUGS FOR TREATMENT OF SEIZURES

- Carbamazepine
- Clonazepam
- Clorazepate Dipotassium
- Diazepam
- Ethosuximide
- Fosphenytoin
- Lorazepam
- Midazolam
- Phenytoin

ALTERNATIVE DRUGS FOR TREATMENT OF SEIZURES

- Ethotoin
- Mephentyoin
- Mephobarbital
- Methsuximide
- Phenobarbital
- Phensuximide
- Primidone

ADJUNCT DRUGS FOR TREATMENT OF SEIZURES

- Dimethadione
- Ezogabine

- Felbamate
- Gabapentin
- Lacosamide
- Lamotrigine
- *S*(-)-Levetiracetam
- Oxcarbazepine
- Pregabalin
- Rufinamide
- Tiagabine
- Topiramate
- Trimethadione
- Valproic acid
- Vigabatrin
- Zonisamide

Abbreviations

ABC, adenosine triphosphate binding cassette
AED, antiepileptic drug
AMPA, L- α -amino-3-hydroxy-5-methyl-4-isoxazole propionate
AUC, area under the curve
CBMA, 3-carbamoyl-2-phenylpropionaldehyde
CBZ, carbamazepine
C_{max}, maximum plasma concentration
CNS, central nervous system

C_{ss}, steady-state plasma concentration
CRMP-2, collapsin-response mediator protein 2
D5W, 5% dextrose in water
EEG, electroencephalography
FDA, U.S. Food and Drug Administration
GABA, γ -aminobutyric acid
GAT1, GABA transporter 1
HPPH, 5-(4'-hydroxyphenyl)-5-phenylhydantoin
IM, intramuscular

IV, intravenous
KA, kainate
MES, maximal electroshock
MHD, 10-monohydroxycarbamazepine
NMDA, N-methyl-D-aspartate
2-OHIS, 2-hydroxyiminostilbene
PEMA, phenylethylmalonamide
UGT, uridine diphosphoglucuronosyltransferase
V_d, volume of distribution

SCENARIO

Michael Gonyeau, PharmD

BG is a 70-year-old white woman who has been admitted to your hospital after having a tonic clonic seizure at home, which was witnessed by her husband, who states that BG has been complaining of 5 days of nausea and vomiting, has seemed more confused, and has had problems walking (she almost fell down the stairs twice in the past 2 days). Her medical history is significant for seizure disorder, hypertension, end-stage renal disease on hemodialysis, and coronary artery disease. She has no known allergies and currently takes the following medications: phenytoin 300 mg XL po once daily, lisinopril 10 mg po daily, and atorvastatin 20 mg

po daily. Notable laboratory data include reduced renal function (creatinine clearance, 39 mL/min), albumin 1.5 mg/dL, and phenytoin level 9 mcg/mL (therapeutic goal, 10–20 mcg/mL). Based on the patient's recent seizure and the subtherapeutic phenytoin level, the admitting physician gives a 500 mg po bolus dose of phenytoin, increases the daily dose to 400 mg XL po daily, asks the pharmacist for any further recommendations.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

As early as 2000 BC, it was recognized that some people suffered from convulsive seizures. The term “epilepsy,” based on the Greek word *epilambanein* (meaning “to seize”), was first used by Hippocrates. In the world's first scientific monograph on epilepsy, entitled *On the Sacred Disease* (~400 BC), Hippocrates disputed the myth that the cause of epilepsy is supernatural and the cure magic. He described epilepsy as a disease of the brain, which should be treated by diet. At the same time, Hippocrates provided the first classification of epilepsy, which is still used. He distinguished true (idiopathic) epilepsy (i.e., a disorder for which the cause is unknown) from symptomatic (organic) epilepsy (i.e., a disorder resulting from a physiologic abnormality, e.g., brain injury, tumor, infection, intoxication, or metabolic disturbances).

Two opinions were put forward as to the causes of epilepsy. One was that epilepsy is a single disease entity and all forms of it have a common cause. On the other hand, it was proposed that different types of epilepsy result from different chemical, anatomic, or functional disorders. At the Symposium on Evaluation of Drug Therapy in Neurologic and Sensory Disease, the general opinion was that “epilepsy is a symptom complex characterized by recurrent paroxysmal aberrations of brain functions, usually brief and self-limited” (1).

All forms of epilepsy originate in the brain and appear to be the result of changes in neuronal activity. In turn, these changes, such as an excessive neuronal discharge, can be brought about by a disturbance of physicochemical function and electrical activity of the brain. However, the cause of this abnormality is not clearly understood.

The most important property of the nerve cell is its excitability. It responds to excitation by generating an action potential, which can lead to repeated discharges. All normal neurons can become epileptic if subjected to excessive excitation. DeRobertis et al. (2) list two possible mechanisms for convulsive disorders: a loss of the normal inhibitory control mechanism, and a chemical supersensitivity that increases excitability of neuronal elements.

The origin of the seizures was established as early as the 19th century by Jackson (3). According to him, an intense discharge of gray matter in various regions of the brain initiates the seizures. As a result, it is only a normal reaction of the brain to initiate convulsive seizures. The discharge of excessive electrical (nervous) energy has, indeed, been substantiated by brain-wave studies made possible by electroencephalography (EEG).

Attempts to classify epileptic seizures have been only partially successful, primarily due to limited knowledge regarding the pathologic processes of the brain. At the turn of the century, a classification of seizures had been published (4), and even more attempts appeared during the 1950s and 1960s (5–9). In 1981, the Commission on Classification and Terminology of the International League Against Epilepsy put forward a new proposal (10). The classification outlined in Table 17.1 is a short version of this proposal, which is based on clinical seizure type, ictal (seizure-induced) EEG expression, and interictal (occurring between attacks or paroxysms) EEG expression.

SEIZURE CLASSIFICATION

Seizures result from the sudden, excessive firing of neurons. They are broadly classified as either partial seizures, in which the abnormal firing initially occurs in a small number of neurons but can spread to adjacent areas, or generalized seizures, in which virtually the entire brain is affected simultaneously (11). Seizures can be characterized by clinical symptoms and by EEG patterns. In addition, computed tomography and magnetic resonance imaging of the head are used in virtually all patients with suspected epilepsy to aid in identifying the seizure type.

Partial (Local, Focal) Seizures

Partial seizures are divided into three categories: simple partial, complex partial, and partial progressing to generalized seizures. The key distinction between simple and complex partial seizures is the level of consciousness of the person undergoing the seizure. In partial seizures, the initial

TABLE 17.1 Classification of Epileptic Seizures

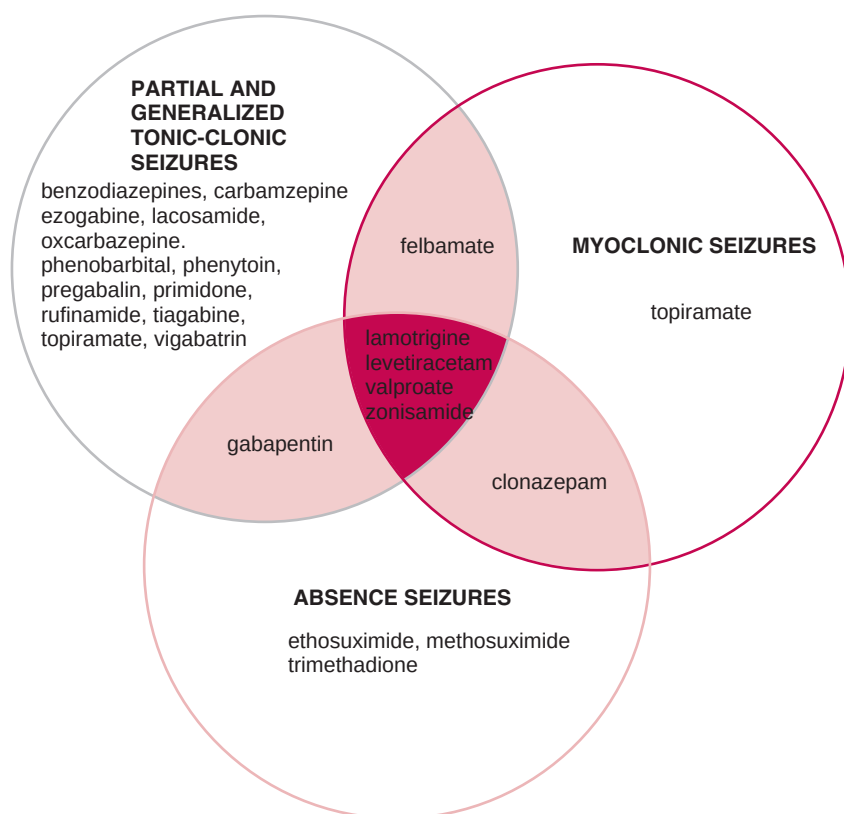
Classification	Subtypes
Partial (local, focal) seizures	Simple (consciousness not impaired)
	Complex partial seizures (psychomotor seizures) 1. Beginning as simple partial seizures, progressing to complex seizures 2. With impairment of consciousness at onset
	Partial seizures evolving to secondarily generalized tonic-clonic convulsions
Generalized seizures (convulsive or nonconvulsive)	Absence seizures: typical (petit mal) and atypical
	Myoclonic
	Clonic
	Tonic
	Tonic-clonic (grand mal)
	Atonic
Unclassified epileptic seizures (includes some neonatal seizures)	

neuronal discharge originates from a specific, limited cortical area, which is termed a “focus.” Development of the focus is thought to be caused by scarring after head trauma, infection, or oxygen deprivation. The abnormal EEG seizure patterns are restricted to one region of the brain, at least at the onset. These types of seizures are possible at all ages but are most frequent in the elderly. Partial seizures respond fairly well to antiseizure drugs (most commonly referred to as antiepileptic drugs [AEDs]) (Fig. 17.1) (12).

Medications to combat partial seizures are effective against secondary or generalized tonic-clonic seizures as well. These seizures respond well to carbamazepine, hydantoins, and barbiturates, although this latter group unfortunately displays substantial sedative effects. The newer AEDs (gabapentin, lamotrigine, levetiracetam, oxcarbazepine, tiagabine, topiramate, and zonisamide) are useful for either monotherapy or as adjunctive drugs. On the other hand, oxazolidinediones and succinimides are ineffective in the treatment of partial seizures. Valproate is appropriate when tonic-clonic seizures are combined with either myoclonic or absence seizures.

Simple Partial Seizures

The specific symptoms displayed during a simple partial seizure will depend on the area of the brain that is affected, and they will occur on the opposite side of

**FIGURE 17.1** The antiseizure drugs used in treatment of the various seizures.

CLINICAL SIGNIFICANCE



Seizure disorders can be devastating to a patient's quality of life.

Experiencing a seizure is associated

with a loss of control of both the body as well as one's control of life events. The

unpredictable nature of seizure occurrence and the resultant fear of recurrence can have a significantly negative effect on quality of life. Restrictions placed on patients with epilepsy include revocation of driver's licenses, potential physical limitations, work absenteeism, and various emotional and mental issues related to the disease and to side effects of many antiepileptic medications. Some of these factors may in turn result in poor adherence to antiepileptic medications, which is associated with increased mortality and morbidity; side effects of antiepileptic drugs are attributed to almost half of all recorded treatment failures.

Medicinal chemistry plays a vital role in the understanding of epilepsy, particularly in its treatment. Molecular agents used to treat seizures exert varying effects on neuronal function through their structure-activity relationships and chemical interactions with ion channels (carbamazepine, phenytoin, ethosuximide, zonisamide) and their similarities to naturally occurring neurotransmitters, such as GABAs (benzodiazepines, barbiturates, topiramate, gabapentin, tiagabine). They can directly affect ion channels or indirectly influence synthesis, metabolism, or function of neurotransmitters or receptors that control channel opening and closing.

Many patients cannot be controlled on monotherapy, and a significant number of patients will require drug

combinations. Freedom from seizures can be achieved by combining drugs with different or overlapping or similar mechanisms of action to attain synergy in an individual patient. There have been a number of newer antiepileptic drugs approved since 1993, and a solid understanding of basic chemical principles of antiseizure drugs is critical in making appropriate clinical recommendations. The pharmacokinetics of antiseizure medications are very complex and require a strong foundational knowledge base rooted in medicinal chemistry. Monitoring side effects—specifically the multitude of drug interactions associated with these agents—is extremely important to ensure patient safety and efficacy in the clinical setting, with the ultimate goal to keep patients seizure free and maintain their quality of life.

Improved understanding of the medicinal chemistry aspects of antiepileptic drugs, as well as mechanisms of action and pharmacokinetics, will help the clinician make a rational choice of agent based on knowledge about which is most effective against each type of seizure as well as minimize adverse effects and drug interactions and improve patients' quality of life.

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the body from the lesion. Combinations of symptoms are frequent, making accurate diagnosis a challenge. The temporal lobes are the most common origination site for partial seizures. Seizure foci located within the temporal lobe result in psychic symptoms, such as fear, panic, or hallucinations; autonomic signs, such as flushing and sweating; or unpleasant smells or tastes. Frontal lobe foci usually present with motor symptoms. Focal motor attacks most commonly start in one hand, one foot, or one side of the face. Often, however, the onset of focal seizures is not specific. Should the focal motor seizures spread to contiguous cortical areas, there can be an orderly sequence of repeated events (movement of hands, face, and legs) known as an epileptic march. This type of seizure is termed a "jacksonian seizure." If the unilateral movements characteristic of focal motor or jacksonian seizures steadily spread to the other half of the body, a generalized seizure can follow. In contrast, seizures beginning in the parietal lobes are termed "sensory seizures" and present with altered sensations, tingling, numbness, or pain. Foci in the occipital lobes produce nystagmus, blinking, or visual disturbances, such as flashing lights or the appearance of strange colors.

In simple partial seizures, the patient's consciousness is not impaired. Thus, the person remains able to respond to simple commands, perform simple deliberate movements, and recall events that occurred during the seizure.

Complex Partial Seizures

When consciousness is impaired, the seizure is classified as a complex partial seizure. Impaired consciousness will be manifested by the person's staring, inability to respond to simple commands, inaccurate recall, or amnesia of events occurring during the seizure. Clouding of consciousness can be apparent initially or can appear subsequent to the start of a simple partial seizure. With the exception of the mental status, the other symptoms of complex partial seizures are similar to those outlined earlier. Many complex seizures have bilateral hemispheric involvement and frequently are accompanied by automatisms (i.e., repetitive involuntary movements, including chewing, swallowing, or wringing the hands).

Partial Seizures Evolving to Secondarily Generalized Seizures

The abnormal electrical activity responsible for any type of partial seizure can generalize throughout the brain,

thus evolving into a secondary generalized tonic-clonic seizure. The diagnosis of partial seizures is therefore difficult to establish. The generalization can occur rapidly, or it can occur slowly enough that the symptoms of the partial seizure are experienced by the patient as an “aura” before the generalized tonic-clonic phase. Auras are comprised of symptoms such as seeing blinking lights or hearing unusual sounds, and they serve as an important warning for the patient. The symptoms of tonic-clonic seizures are described below.

Generalized Seizures (Convulsive or Nonconvulsive)

These disorders are generalized from the outset, and they show simultaneous involvement of both cerebral hemispheres and loss of consciousness. It is not possible to single out one anatomic or functional system in one hemisphere of the brain that is responsible for the clinical symptoms. The initial neuronal discharge spreads quickly into the entire gray matter or at least the greater part of the gray matter. The EEG pattern consists of bilateral, essentially synchronous and symmetric discharges from the start and indicates the widespread nature of neuronal discharge. The cause is rarely known, but it is usually attributed to diffuse lesions, to toxic and metabolic disturbances, or to constitutional genetic factors. People of all ages are affected by generalized convulsions. There are several classes of generalized seizures, and useful drugs are selected according to the seizure type that transpires (Fig. 17.1).

Absence (Petit Mal) Seizures

Both typical and atypical absence seizures bring about brief loss of consciousness. Typical absence seizures have a rapid onset and cessation, which can cause them to be misinterpreted as daydreaming; however, the person cannot be alerted or awakened during the seizures. Absence seizures often begin with a change in facial expression, which is followed by a period of motionless, blank staring. After the brief interruption in consciousness (typically ~10 seconds), the activity that was in progress before the seizure is resumed. The individuals have neither memory of events during the seizure nor postictal confusion. Typical absence seizures (petit mal) are more common in children than in adults. They are particularly disabling because they tend to occur many times daily. Most children will respond to drug treatment, but a small percentage will go on to develop generalized tonic-clonic seizures as adults. Complex typical absence seizures can include additional phenomena, such as clonic or myoclonic motions, automatisms, or more elaborate behaviors. Between 25% and 40% of affected children have a family history of absence seizures.

Atypical absence seizures have a slower onset and cessation, and they last longer (up to several minutes) than typical absence seizures. They can include clonic motions, automatisms, or autonomic symptoms. Differential diagnosis between these types is made on the basis of the EEG. Typical absence seizures display a 3-Hz spike-and-wave

EEG pattern, but the pattern of atypical absence seizures is slower, usually in the range of 1.5 to 2.5 Hz.

Both forms of absence seizure often occur as part of one of the recognized epilepsy syndromes. Typical absence seizures respond fairly well to AEDs; ethosuximide and valproate are first-choice drugs. Clonazepam is effective but sedating, and tolerance to the antiabsence effects can develop. Lamotrigine can be useful. Treatment of atypical absence seizures with AEDs is less successful.

Lennox-Gastaut syndrome is a mixed seizure disorder combining the atypical absence seizures with tonic, tonic-clonic, or myoclonic motor patterns. The syndrome begins in childhood and usually includes mental retardation. Although adequate control of the seizures is rarely achieved, valproate, phenytoin, felbamate, lamotrigine, topiramate, and clonazepam have been useful.

Myoclonic Seizures

Myoclonic seizures consist of sudden, very brief, jerking contractions that can involve the entire body or be confined to limited areas, such as the face and neck. The contractions can affect individual muscles or groups, with simultaneous contraction of both extensor and flexor muscles. These seizures occur in all age groups, with symptoms ranging from rapid tremors to falling down. No loss of consciousness is detectable due to the brief duration of the seizure. Myoclonic seizures often occur in combination with other seizure types. Valproate and clonazepam are used most often to treat myoclonic seizures; lamotrigine and topiramate also have shown some efficacy.

Tonic Seizures

Tonic seizures occur mostly in children and are characterized by increased tone in extensor muscles, resulting in falling to the ground. Although brief, the duration of contractions is somewhat longer than in myoclonic seizures. Vocalization can occur as a result of contraction of thoracic muscles, forcing air past the larynx. Brief periods of apnea and postictal tiredness can be associated.

Atonic Seizures

In atonic seizures, a very sudden decrease in muscle tone occurs, leading to a head drop, drooping of a limb, or loss of all muscle tone, resulting in falling. The risk of injury is high in these sudden “drop attacks.” Atonic seizures are more common in children.

Attaining good control of tonic or atonic seizures is difficult. Valproate, felbamate, lamotrigine, benzodiazepines, and topiramate have proven to be effective in some individuals.

Clonic Seizures

Clonic seizures nearly always occur in babies or young children. A loss or impairment of consciousness occurs simultaneously with a decrease in muscle tone or with a generalized tonic contraction, and it is followed by period of asymmetric jerking motions.

Tonic-Clonic (Grand Mal) Seizures

Generalized tonic-clonic seizures represent a maximal epileptic response of the brain. These seizures are characterized by the absence of an aura and by tonic stiffening of all muscle groups, causing the patient to fall. The initial contraction can be flexor and is rapidly followed by prolonged extension. Subsequently, there is a period of bilateral symmetric jerking of the extremities. The seizure can be associated with loss of bladder control and biting of the tongue or inside of the mouth. There is a pronounced postictal state (altered state of consciousness lasting from 5–30 minutes) following the seizure, and the person can pass directly into sleep before waking several hours later.

Status Epilepticus

Status epilepticus is a condition in which there is a single prolonged seizure lasting more than 5 minutes or in which there is insufficient time between multiple seizures to permit recovery. Several types exist, depending on the type of seizure involved (i.e., tonic-clonic, simple partial, complex partial, or absence). Tonic-clonic status epilepticus is both the most common and the most life-threatening. Pharmacologic treatment of most forms of status epilepticus can include intravenous (IV) administration of diazepam or lorazepam, fosphenytoin, and phenobarbital. Although lorazepam is not approved by the U.S. Food and Drug Administration (FDA) for this purpose, it sometimes is preferred due to its longer half-life.

Absence status epilepticus is a condition of impaired consciousness, perhaps including mild motor symptoms, that lasts from 30 minutes to 12 hours. It can be distinguished from ongoing seizures as a result of organic or toxic causes by the spike-and-wave EEG pattern that is characteristic of absence seizures. The usual pharmacologic treatment of absence status uses diazepam or lorazepam, followed by ethosuximide.

MECHANISMS OF ACTION FOR THE ANTISEIZURE DRUGS

Seizures result from bursts of abnormal synchronous discharging by a network of neurons. Although the mechanisms of seizure generation are still poorly understood, the causes of abnormal firing appear to involve neuronal ion channels and an imbalance between excitatory and inhibitory synaptic function. Various AEDs exhibit different mechanisms of action on neuronal function, causing them to show selective efficacy against different seizure types (13) (Fig. 17.2).

Ion Channels

Sodium and chloride ions are present at greater concentration outside the cell, whereas potassium, organic cations, and charged proteins are more numerous within the cell. Because the membrane is permeable only to small ions and not large ions or proteins, the neuronal membranes maintain a charge separation, resulting in a

“resting potential” in the range of -50 to -80 mV versus the outside of the cell.

An increase in interior negativity, termed “hyperpolarization,” decreases the resting potential (e.g., to -90 mV), thus making it more difficult for a neuron to reach threshold and subsequently fire. A reduction in interior negativity, termed “depolarization,” can result in generation of an action potential if the depolarization is sufficient to reach threshold (approximately -40 mV). Neuronal firing is initiated by an influx of sodium ions.

After each depolarization, voltage-dependent sodium channels adopt an inactive state and remain refractory to reopening for a period of time. While those channels are unable to open, rapid repetitive firing is diminished, and spread of electrical seizure activity to adjacent brain regions is suppressed (14). Stabilization and prolongation of this inactive state appears to be the primary mechanism of action of phenytoin, carbamazepine, and lamotrigine and can be instrumental in the antiseizure actions of phenobarbital, oxcarbazepine, valproate, topiramate, and zonisamide (Fig. 17.2).

Alterations in the structure or function of an ion channel caused by mutations in a gene encoding one of the channel’s subunits are termed “channelopathies.” Initially, these abnormalities were associated with cardiac and muscular disorders, but today, it is recognized that channelopathies are responsible for several forms of epilepsy (15,16). Presently, most of the discovered channel mutations appear to be associated with the development of idiopathic generalized epilepsy; most partial seizures are believed to be acquired. Minor alterations in gene structure or expression, however, can predispose a given individual to partial seizures. It is estimated that 40% of adult and childhood epilepsy can result from genetic factors.

Mutations in sodium and potassium channels are most common, because they give rise to hyperexcitability and burst firing. Mutations in the sodium channel subunit gene *SCN2A1* have been associated with benign familial neonatal epilepsy, mutations in *SCN1A* have been associated with severe myoclonic epilepsy of infancy, and mutations in *SCN1A* and *SCN1B* have been associated with generalized epilepsy with febrile seizures. Mutation of the *SCN1B* gene has been shown to reduce the channel’s response to phenytoin (17). The potassium channel genes *KCNQ2* and *KCNQ3* are implicated in some cases of benign familial neonatal epilepsy. Chloride channels have been implicated as well. Mutations of the *CLCN2* gene have been found to be altered in several cases of classical idiopathic generalized epilepsy subtypes: childhood and juvenile absence epilepsy, juvenile myoclonic epilepsy, and epilepsy with grand mal on awakening. Mutations of γ -aminobutyric acid (GABA) A receptor (GABA_A) subunits also have been detected. The gene encoding the α_1 subunit, *GABRG1*, has been linked to juvenile myoclonic epilepsy; mutated *GABRG2*, encoding an abnormal γ subunit, has been associated with generalized epilepsy with febrile seizures and childhood absence epilepsy. Lastly, mutations of calcium channel subunits have been

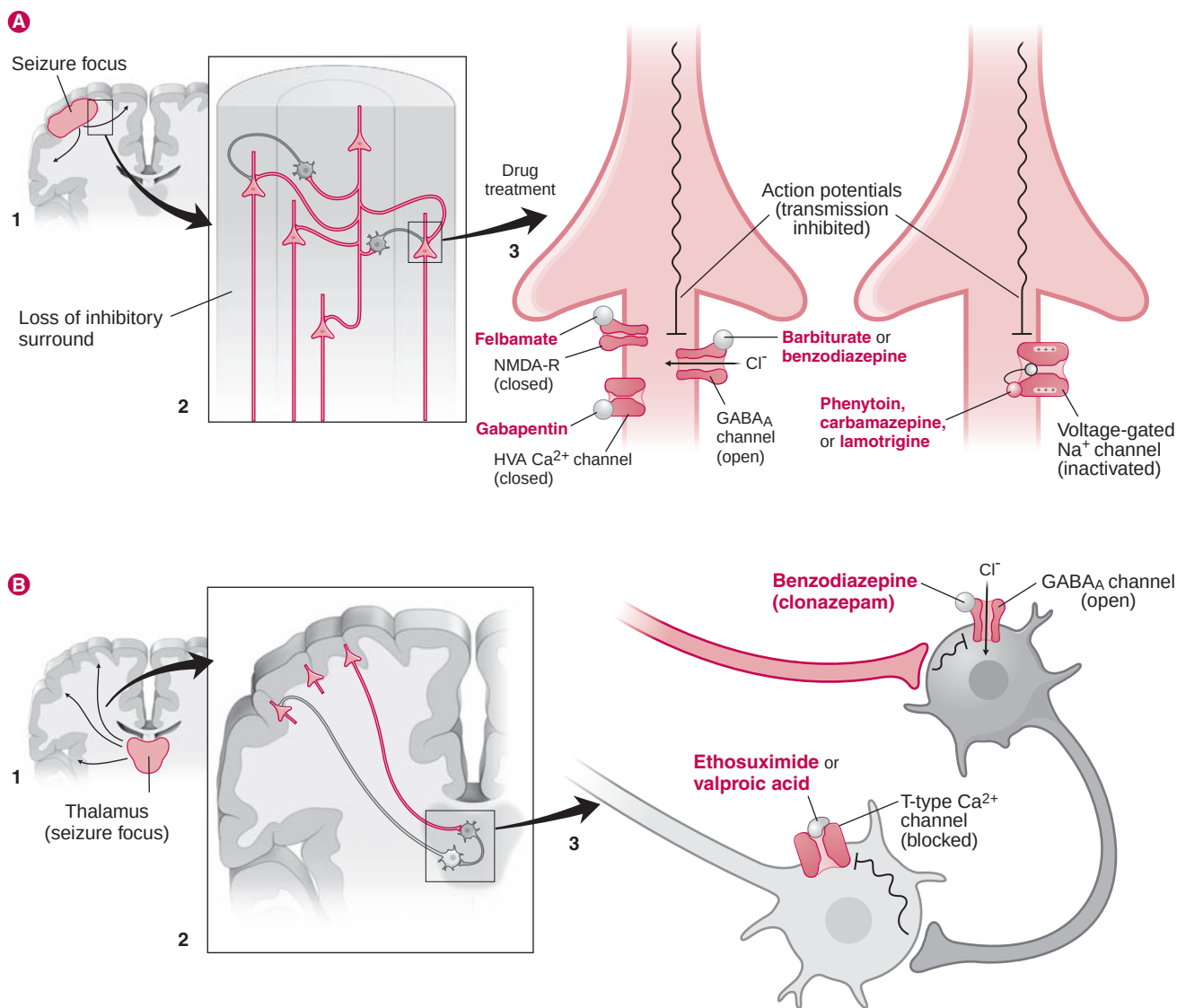


FIGURE 17.2 Mechanisms of antiseizure drugs. (A) Drugs for partial seizures: 1) Seizure activity spreads due to loss of surround inhibition; 2) drugs act in the surrounding region to 3) enhance γ -aminobutyric acid (GABA) inhibition or prolong Na^+ channel inactivation. (B) Drugs for absence seizures: 1) The seizure activity results from cyclic activity between the thalamus and the cortex; 2) drugs prevent synchronization by either 3) deactivating the inhibitory reticular neurons and decreasing hyperpolarization of the thalamic relay neurons or blocking T-type Ca^{2+} channels to prevent burst firing by the thalamic relay neurons. (From Griffin EA Jr, Lowenstein DH. Pharmacology of abnormal electrical neurotransmission in the central nervous system. In: Golan DE, Tashjian A, Armstrong E, et al., eds. Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy, 2nd Ed. Baltimore: Lippincott Williams & Wilkins, 2008, pp. 225–236, with permission.)

identified in juvenile absence epilepsy (mutation in *CACNB4*; the B4 subunit of the L-type calcium channel) and idiopathic generalized epilepsy (*CACNA1I*) (15,16).

Synaptic Inhibition and Excitation

For a neuron, whether an action potential is generated depends on the balance between excitatory and inhibitory stimulation. The predominant inhibitory neurotransmitter in the brain is GABA. It is synthesized from an amino acid, glutamic acid, by glutamic acid decarboxylase and is inactivated by GABA-transaminase. GABA

binds to two receptor types, GABA_A and GABA_B. GABA_A receptors occur on chloride ion channels, and the binding of GABA causes chloride influx and neuronal hyperpolarization. GABA_B receptors are linked via G proteins and second messengers to potassium and calcium channel activity, also mediating inhibition in the central nervous system (CNS). GABA_B receptors also can have a role in oscillatory rhythms in some forms of epilepsy (18) (Fig. 17.3).

A number of AEDs augment GABA-mediated inhibition or affect GABA concentration. Benzodiazepines,

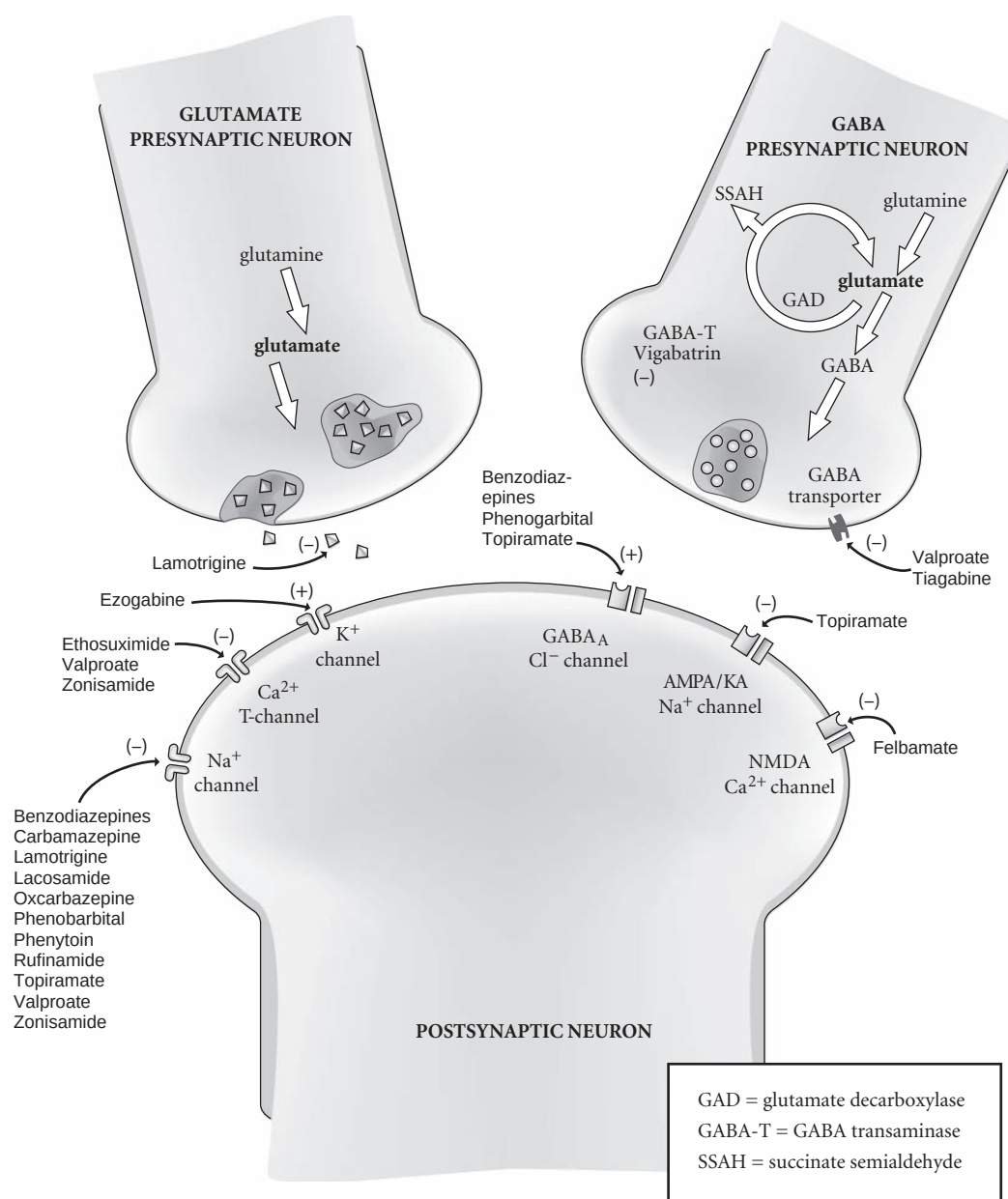
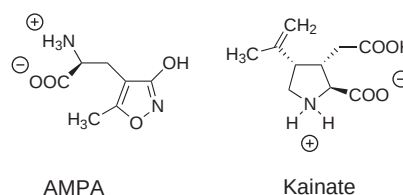


FIGURE 17.3 A summary of the sites of action for the antiepileptic drugs (Adapted from Taylor CP. Mechanisms of new antiepileptic drugs. In: Delgado-Escueta AV, Wilson WA, Olsen RW, et al., eds. Jasper's Basic Mechanisms of the Epilepsies, 3rd Ed. (Advances in Neurology, Vol 79.) Philadelphia: Lippincott Williams & Wilkins, 1999:1012–1027, with permission.)

barbiturates, and perhaps topiramate enhance the action of GABA on the GABA_A chloride channel. Tiagabine decreases the reuptake of GABA, and perhaps, gabapentin decreases its metabolism.

The ionotropic neurotransmitter glutamate provides excitatory neurotransmission via two types of receptors, the *N*-methyl-D-aspartate (NMDA) and L- α -amino-3-hydroxy-5-methyl-4-isoxazole propionate (AMPA) and kainate (KA) receptors (see Chapter 12). NMDA, AMPA, and KA are specific agonists for these receptors that are used in experiments to distinguish the glutamate ionotropic receptors.

Activation of these ligand-gated channels enables sodium and calcium influx and potassium efflux, facilitating depolarization. Blockade of the NMDA receptor by felbamate or of the AMPA/KA receptor by phenobarbital and topiramate inhibits depolarization (Fig. 17.3).



Low-threshold T-type calcium currents act as pacemakers for normal brain activity, particularly the thalamic oscillatory currents thought to be involved in the generation of absence seizures (19). Drugs such as ethosuximide, the oxazolidinediones, and zonisamide, which inhibit T-type currents, are effective against absence seizures but ineffective against partial or other seizure types. In addition, Kv7 potassium channels are suggested as the site of action of certain antiepileptic drugs (20).

ANTISEIZURE DRUGS

Introduction

The primary use of AEDs is in the prevention and control of epileptic seizures. Theoretically, the ideal AED should, among other things, completely suppress seizures in doses that do not cause sedation or other undesired CNS toxicity. The AED should be well tolerated and highly effective against various types of seizures and be devoid of undesirable side effects on vital organs and functions. Its onset of action should be rapid after parenteral injection for control of status epilepticus, and it should have a long duration of effect after oral administration for prevention of recurrent seizures.

The first effective remedy, potassium bromide, was introduced by Locock in 1857 (21). This drug was largely replaced by phenobarbital in 1912, when Hauptmann tried this sedative in epilepsy (22). Its great value was recognized at once, and it is still commonly prescribed.

The usefulness of both bromide and phenobarbital in convulsive disorders was discovered by chance, but phenytoin was developed in 1937 as the result of a study of potential AEDs in animals by Putnam and Merritt (23,24). Bromide is highly effective in humans and is relatively nonsedating. Treatment of convulsive disorders using bromide, phenobarbital, and phenytoin constitutes an important advance in clinical therapy.

Many of the standard AEDs that contain the ureide structure, as shown in Figure 17.4, have been used clinically for more than 30 years without much change in their ureide structures. Small changes in the X substituent of the ureide structure can cause significant changes in the type of seizures controlled, which will be discussed for each of the respective drugs. As a result of rapid developments in molecular biologic techniques for the study of the neurophysiology of epilepsy and in the interactions of AEDs with neurotransmitters at ion channels or brain receptors (AMPA/KA glutamate receptors), a new generation of clinically available AEDs has emerged. These AEDs include felbamate, gabapentin, lamotrigine, levetiracetam, oxcarbazepine, tiagabine, topiramate, and zonisamide. Their mechanisms of action are targeted toward ion channels and brain receptors either by enhancing brain GABA activity (e.g., tiagabine) or by inhibiting excitatory amino acids (L-Glu; e.g., lamotrigine and felbamate) (Fig. 17.2). These new-generation AEDs also exhibit

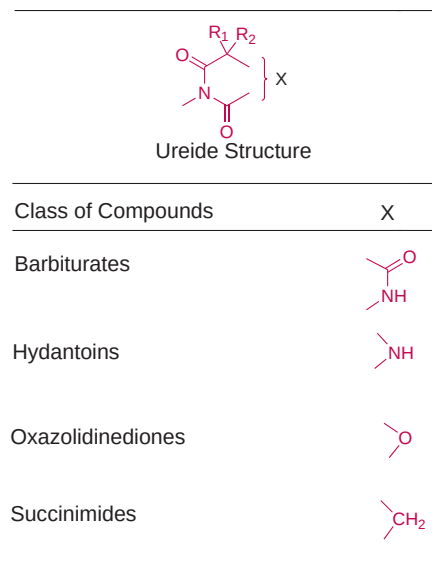


FIGURE 17.4 Structure of anticonvulsant drugs containing the ureide structure.

limited drug interactions with fewer adverse effects. A rational approach to the drug discovery process is necessary to develop new leads to novel effective therapy and to use structure–activity relationships to fine-tune the pharmacology of existing AEDs with the same or better efficacy and fewer adverse effects (25).

Approximately 60% of patients with epilepsy become seizure-free with monotherapy using frontline drugs, such as carbamazepine, benzodiazepine (clonazepam and diazepam), ethosuximide, or phenytoin. Alternative monotherapy drugs include phenobarbital and primidone. Another 20% have their epilepsy controlled by the addition of a second AED called adjunct drugs (e.g., felbamate, gabapentin, lamotrigine, levetiracetam, oxcarbazepine, tiagabine, topiramate, valproic acid, and zonisamide). Despite recent advances in neurobiology and significant insight regarding the molecular dysfunction of epilepsy, the remaining 20% do not completely respond to the current frontline therapeutic drugs and most often are prescribed more than two AEDs without any obvious benefit (11). Recently, much effort has been made to discover new AEDs effective in refractory seizures and partial complex seizures.

Seizure control requires continuous antiseizure action and is not achievable unless plasma concentrations remain relatively consistent at therapeutic levels throughout all 24 hours of the day. Therefore, a knowledge of the therapeutic ranges of plasma concentrations, time to peak serum concentrations, and elimination half-lives helps to guide AED administration to achieve consistent therapeutic serum concentrations that control seizures without causing intolerable toxicity (26). Enzymatic biotransformation is the principal determinant of the pharmacokinetic properties for most AEDs, although some drugs are excreted by the kidneys predominantly

as unchanged drug. Most AEDs exhibit linear enzyme kinetics, in which changes in daily dose lead to proportional changes in serum concentration if clearance remains constant. The traditional concept of administering a drug at intervals equal to one elimination half-life, however, does not apply to some drugs, in which the half-life of biologic activity can exceed its elimination half-life. The standard AEDs have the greatest potential to be involved in pharmacokinetic drug interactions when they are coadministered with other AEDs or other drugs (27). These interactions usually involve changes in the rate of biotransformation or in the protein binding of one or both coadministered drugs (26). Drug-induced changes in the pharmacokinetics for many of the AEDs are particularly pronounced in children, requiring a higher oral dose per kilogram body weight than in adults to obtain an effective plasma concentration (28).

This chapter surveys the structure–activity relationship, mechanism of action, metabolism, and pharmacokinetic parameters for the new generation of AEDs (felbamate, gabapentin, lamotrigine, oxcarbazepine, levetiracetam, tiagabine, topiramate, zonisamide, and vigabatrin) and the standard AEDs (phenytoin, carbamazepine, phenobarbital, primidone, valproate, ethosuximide, and the benzodiazepines) as well as for several of the older AEDs that are less commonly used today. The application of AEDs in the treatment of various kinds of epilepsies is shown in Figure 17.1; this illustration is based on AEDs used in clinical therapy. Table 17.2 lists the AEDs, their mechanisms of action, and some of the pharmacokinetic properties.

Drugs Effective against Partial and Generalized Tonic-Clonic Seizures

Hydantoin

The hydantoins have a five-membered ring structure containing two nitrogens in an ureide configuration (Fig. 17.4) and were tested as antiepileptics by Merritt and Putnam (23,24). These drugs suppressed electrically induced convulsions in animals but were ineffective against convulsions induced by pentylenetetrazole, picrotoxin, or bicuculline. The structures for the clinically available hydantoins are listed in Figure 17.5.

PHENYTOIN (DIPHENYLHYDANTOIN) Phenytoin is the prototype and most commonly prescribed member of the hydantoin family of drugs. Bioequivalency is a problem with the hydantoins as a result of their very poor water solubility (~32 mg/L) and low therapeutic ratio. Phenytoin exhibits nonlinear pharmacokinetics that exaggerate the effects of changes in the fraction of dose absorbed. Its apparent pK_a is in the range of 8.06 to 8.33 and, thus, can form a water-soluble sodium salt (~15 mg/mL at solution pH > 11). Aqueous solutions of phenytoin sodium (pH 11 to 12) gradually absorb carbon dioxide, neutralizing the alkalinity of the solution and causing partial hydrolysis and crystallization of free phenytoin resulting in turbid solutions. When phenytoin sodium is

administered intramuscularly (IM), its absorption can be erratic as a result of crystallization of insoluble phenytoin at the injection site due to the decrease in pH from 11.5. Phenytoin sodium injection is physically and chemically incompatible in 5% dextrose in water (D5W), normal saline, or with parenteral solutions of many drugs, especially salts of basic drugs. The nature of the incompatibility depends on several factors, including the type of salt, concentrations of the drugs, diluents used, resulting pH of the final admixture (must be pH > 11), and temperature. Phenytoin sodium capsules also will absorb carbon dioxide over time, resulting in the formation of free phenytoin with a different dissolution profile.

Mechanism of Action Phenytoin is indicated for initial monotherapy or adjunct treatment of complex partial or tonic-clonic seizures, convulsive status epilepticus, and prophylaxis. It often is selected for initial monotherapy due to its high efficacy and relatively low incidence of side effects (29). Phenytoin is not used in the treatment of absence seizures, because it can increase their frequency of occurrence (30). Phenytoin binds to and stabilizes the inactivated state of sodium channels, thus producing a use-dependent blockade of repetitive firing and inhibition of the spread of seizure activity to adjacent cortical areas (31).

Pharmacokinetics Phenytoin sodium from immediate-release capsules is rapidly absorbed and generally attains peak serum concentration in 1.5 to 3.0 hours; extended-release phenytoin sodium is absorbed more slowly, attaining peak serum concentration in 4 to 12 hours. The oral bioavailability for sodium phenytoin (70% to 100%) can vary enough among formulations from different manufacturers to result in a subtherapeutic serum concentration, and therefore, it can be ineffective in controlling seizures. Alternatively, switching formulation may result in a toxic blood concentration. Capsules of sodium phenytoin will absorb carbon dioxide, causing dissociation or neutralization to free phenytoin, thus altering its oral bioavailability (gastrointestinal dissolution), pharmacokinetics, and plasma concentrations, which can cause breakthrough seizures. Therapeutic plasma concentrations for phenytoin are usually 7.5 to 20.0 µg/mL, although in some patients, seizure control is not achieved at these plasma concentrations. Phenytoin is highly protein bound (Table 17.2).

Phenytoin is metabolized predominately by CYP2C9 to its primary metabolite, 5-(4'-hydroxyphenyl)-5-phenylhydantoin (HPPH) (Fig. 17.5) (32). Approximately 60% to 75% of an oral dose is excreted as HPPH glucuronide or sulfate metabolites. Approximately 1% is excreted unchanged in urine, and some undergoes enterohepatic circulation. Other minor metabolites also appear in urine. Up to 10% of the oral phenytoin can be excreted unchanged by the kidneys at toxic doses. Phenytoin is notorious for displaying nonlinear pharmacokinetics, because the route of metabolism is a saturable process. Therefore, small

TABLE 17.2 Mechanism(s) of Action and Pharmacokinetics for Antiseizure Drugs

Drug	Mechanism of Action ^a	Elimination Half-Life in Children (hours)	Elimination Half-Life in Adults (hours)	Time to Steady-State Plasma Concentration (hours)	Protein Binding (%)	LogD (pH 7.4)
Carbamazepine (Tegretol)	A	14–27 (children), 8–28 (neonates)	14–27	2–8	66–89	2.2
Clonazepam (Klonopin)	A, B	20–40	20–40	—	95–98	2.3
Diazepam (Diastat)	A, B	17	36	—	40	2.6
Ethosuximide (Zarontin)	C	20–60	20–60	7–10	0	0.38
Ezogabine (Potiga)	F	—	6–10	—	80	0.81 ^d
Felbamate (Felbatol)	E	15.9	20–23	—	20–25	1.2 ^d
Gabapentin (Neurontin)	D	4	5–7	2–3	<3%	–1.3 ^d
Lacosamide (Vimpat)	A	—	13	1–4 days	<15	0.9 ^d
Lamotrigine (Lamictal)	A, E	19	25–70	—	55	–0.2 ^d
Levetiracetam (Keppra)	I	—	6–8	2 days	<10	–0.67 ^d
Oxcarbazepine (Trileptal)	A	—	7–11 ^b	2–3	40	1.25 ^d
Phenobarbital	A, B	37–73	40–136	12–21	40–60	1.53
Phenytoin (Dilantin)	A	5–14 (children), 10–60 (neonates)	12–36	7–28	69–96	2.39
<i>p</i> -Hydroxyphenytoin (metabolite)						1.72
Pregabalin (Lyrica)	D	—	5–6.5	24–48	0	–1.38 ^d
Primidone (Mysoline)	A, B	23	3–7	4–7	20–30	0.91
Rufinamide (Banzel)	A	—	6–10	—	34	0.05 ^d
Tiagabine (Gabitril)	A	3–6	7–9	—	96	3.2 ^d
Topiramate (Topamax)	A, B	10–15, 6–8 ^c	20–30, 12–15 ^c	—	15	2.9 ^d
Valproic acid (Depakene, Depakote, Depacon)	A, C, G	8–15	6–15	1–4 3–5 (sustained release)	80–95	0.13
Vigabatrin (Sabril)	H	5.7	7.5	—	0	–2.6 ^d
Zonisamide (Zonegran)	A, C	—	27–46	10–12	40	–0.10 ^d

^aSee the *Mechanism of Action* sections of this chapter for discussion. A, sodium channels; B, γ -aminobutyric acid A receptor currents; C, T-calcium currents; D, P/Q calcium currents; E, glutamate receptor antagonist; F, potassium channel agonist; G, GAT1 antagonist; H, γ -aminobutyric acid–transaminase inhibitor; I, unknown.

^bMonohydroxy metabolite.

^cIn the presence of enzyme-inducing drugs, such as carbamazepine, phenobarbital, or primidone.

^dLogD calculated at pH 7.4 using Advanced Chemistry Development (ACD Labs) PhysChem Suite V12, 2010.

increases in dosage can produce substantial increases in plasma phenytoin concentrations; the steady-state plasma concentration (C_{ss}) can double or triple as a result of a 10% or more increase in dosage, possibly resulting in toxicity. Phenytoin also induces CYP3A4 and uridine diphosphoglucuronosyltransferases (UGTs; increased glucuronidation). Therefore, plasma concentrations for drugs metabolized by these isoforms will be affected (see also Chapter 4). Thus, the addition of phenytoin to an AED regimen can reduce their plasma levels by inducing their CYP3A4 metabolism for carbamazepine, felbamate,

lamotrigine, oxcarbazepine, tiagabine, valproate, and zonisamide. Other drugs, for which metabolism is induced by phenytoin, include methadone, theophylline, warfarin, and oral contraceptives. On the other hand, plasma phenytoin levels are increased by carbamazepine, felbamate, cimetidine, warfarin, chloramphenicol, isoniazid, and disulfiram. Plasma phenytoin levels are decreased by rifampin, antacids, and valproate (free phenytoin levels remain the same).

The pharmacokinetics of phenytoin are significantly affected by age. Its rate of elimination is strongly dose

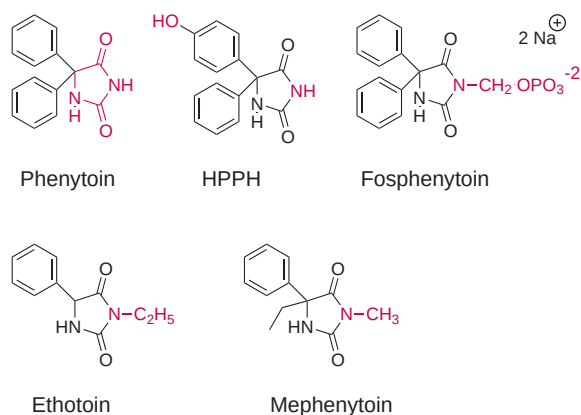


FIGURE 17.5 Hydantoin.

dependent (nonlinear) at all ages. The elimination half-life for phenytoin in children increases with age because of an age-dependent decrease in its rate of metabolism (28). The combination of these factors makes it difficult to predict the phenytoin plasma concentrations following the dose (dose per kilogram) adjustments in neonates and children, particularly when phenytoin is coadministered with other liver enzyme-inducing AEDs, such as phenobarbital and carbamazepine.

Adverse Effects Drug interactions, especially with other AEDs and CYP3A4 substrates, are extensive. Although toxic effects can begin in the upper normal plasma range ($>20 \mu\text{g/mL}$), serious toxicity is rare. CNS effects are most frequent and include nystagmus, ataxia, dysarthria, and sedation. Gingival hyperplasia, usually reversible, is common. Aromatic anticonvulsants, such as phenytoin, have been associated with a number of toxic effects, including a drug-induced hypersensitivity syndrome that manifests with a triad of reactions, such as rash, agranulocytosis, thrombocytopenia, lymphadenopathy, Stevens-Johnson syndrome, and hepatitis, that have occurred in affected individuals. Although anticonvulsant-induced hypersensitivity reactions are relatively rare events, occurring with an incidence of between 1:1,000 and 1:10,000 exposures, these idiosyncratic reactions can be potentially life-threatening. The hypersensitivity syndrome is consistent with an immune etiology, with symptoms typically appearing within 2 to 8 weeks after the initiation of therapy and generally abating on discontinuation of phenytoin. Dermal metabolism of phenytoin to reactive metabolites can be related to the occurrence of hypersensitivity rashes (33). Although the mechanism by which the aromatic anticonvulsants induce hypersensitivity reactions has not been well characterized, recent studies have suggested that the immune response elicited by phenytoin can be caused, at least in part, by its metabolism into chemically reactive metabolites and can be the critical step in the formation of protein adducts and subsequent immune responses. Identification of the reactive metabolite(s) for phenytoin has been difficult because

the rate at which human liver microsomes convert phenytoin to metabolites capable of binding to proteins is quite low. It also is important to note that phenytoin is typically administered in doses ranging from hundreds of milligrams to grams per day, suggesting that substantial amounts of reactive metabolites could be formed on a daily basis. Therefore, it is conceivable that individuals with a low capacity to detoxify reactive metabolites via glucuronidation or other secondary pathways can represent individuals at risk for idiosyncratic toxicity.

The incidence of phenytoin toxicity can be increased in the elderly or in those patients with hepatic or renal impairment, as a result of alterations in its pharmacokinetics. Plasma level determinations can be indicated in these cases. Although a role for P-glycoprotein transporter alleles in the development of phenytoin toxicity remains controversial, phenytoin is a robust substrate for the non-ABC efflux transporter RLIP76. Because RLIP76 has been found to be overexpressed in excised human epileptic foci, its action can account for treatment failures; conversely, inhibition of transport can cause toxicity (34). There is a 2% to 3% increase in the risk of fetal epilepsy syndrome if the mother is taking phenytoin. Phenytoin is contraindicated in cardiac patients with bradyarrhythmias. Induction of CYP2C19 by ginkgo biloba can increase phenytoin clearance and precipitate serious seizures (35).

FOSPHENYTOIN (CEREBYX) Fosphenytoin sodium (Fig. 17.5) is a soluble prodrug disodium phosphate ester of phenytoin (142 mg/mL) that was developed as a replacement for parenteral phenytoin sodium to circumvent the pH and solubility problems associated with parenteral phenytoin sodium formulations (36). Unlike phenytoin, fosphenytoin is freely soluble in aqueous solutions and is rapidly absorbed by the IM route (37). It is rapidly metabolized (conversion half-life, 8 to 15 minutes) to phenytoin by *in vivo* phosphatases. Therapeutic free (unbound) and total plasma phenytoin concentrations are consistently attained after IM or IV administration of fosphenytoin (26). It is administered IV after benzodiazepines for control of status epilepticus or whenever there is a need to rapidly achieve therapeutic plasma concentrations. Severe bradycardiac adverse events to fosphenytoin, including some fatalities, have been reported (38). A dose reduction in patients who are elderly or have renal or hepatic impairment has been suggested.

ETHOTOIN (PEGANONE) Ethotoin differs from phenytoin in that one phenyl substituent at position 5 has been replaced by hydrogen, and the N-H at position 3 is replaced by an ethyl group (Fig. 17.5). Ethotoin can be used for treatment of tonic-clonic and complex partial (psychomotor) seizures. Because it is considered to be less toxic but also less effective and more sedating than phenytoin, ethotoin usually is reserved for use as an add-on drug (39). Ethotoin does not share phenytoin's

profile of antiarrhythmic action. The metabolism of ethotoin, like phenytoin, is saturable and nonlinear. Its administration is contraindicated in patients with hepatic abnormalities and hematologic disorders.

MEPHENYTOIN Mephenytoin is *N*-methylated at position 3 with an ethyl group replacing one of the phenyl substituents at position 5 (Fig. 17.5). It is indicated for focal and jacksonian seizures in patients refractory to less toxic AEDs. Mephenytoin produces more sedation than phenytoin and should be used only when safer drugs have failed, because it is associated with an increased incidence of serious toxicities, such as severe rash, agranulocytosis, and hepatitis (40). Its *N*-desmethyl metabolite, 5-phenyl-5-ethylhydantoin, contributes to both efficacy and toxicity for mephenytoin. The drug is no longer commercially available inside the United States but is still available outside the United States

Iminostilbenes

CARBAMAZEPINE Carbamazepine (CBZ; Tegretol) (Fig. 17.6) was approved by the FDA in 1968 and is presently indicated as initial or adjunct therapy for complex partial, tonic-clonic, and mixed-type seizures. It is one of the two safest and most effective older AEDs for these seizure types (phenytoin is the other) and is chosen for monotherapy as a result of its high effectiveness and relatively low incidence of side effects (40). Its tricyclic structure resembles that of the psychoactive drugs imipramine, chlorpromazine, and maprotiline and also shares some structural features with the AEDs phenytoin, clonazepam, and phenobarbital. In addition, CBZ has been found to be effective for treatment of bipolar disorder and trigeminal neuralgia.

Mechanism of Action In animals, the profile of antiseizure properties for CBZ is similar to that of phenytoin. CBZ is effective in the maximal electroshock (MES) test (electrically induced seizure test) but is ineffective against pentylenetetrazole-induced seizures. CBZ is not

effective for absence or myoclonic seizures and, indeed, can exacerbate their onset (30,41). Like phenytoin, CBZ acts on voltage-dependent sodium channels to prevent the spread of seizures. CBZ depresses synaptic transmission in the reticular activating system, thalamus, and limbic structures. In a double-blind, crossover study in patients whose seizures were not controlled completely by combinations of AED, CBZ was equal in efficacy to phenobarbital and phenytoin in controlling seizure frequency, and side effects were minimal.

Pharmacokinetics After the administration of an oral dose, CBZ is slowly absorbed, with the attainment of peak concentration from immediate-release tablets in 4 to 5 hours and from extended-release tablets in 3 to 12 hours. The normal half-life averages between 12 and 17 hours; however, due to autoinduction, the half-life can range from 8 to 29 hours. The half-life for CBZ-10,11-epoxide is 5 to 8 hours. Therapeutic plasma concentrations range from 4 to 12 $\mu\text{g/mL}$ (in adults) and can require a month to achieve a stable therapeutic concentration for the desired antiseizure effect because of induction of hepatic metabolizing enzymes.

CBZ is principally metabolized by CYP3A4 to its 10,11-epoxide, with CYP2C8 and CYP1A2 having minor roles. CBZ epoxide is hydrolyzed by epoxide hydrazide to inactive 10,11-dihydroxy CBZ. CBZ epoxide is active and appears to be more toxic than CBZ (42,43). However, CBZ not only induces CYP3A4 activity but also its own metabolism (an autoinducer) as well as UGT and the increased formation of glucuronide metabolites. Like phenytoin, CBZ has been associated with a number of toxic effects, including a drug-induced hypersensitivity syndrome. Although phenytoin-induced hypersensitivity reactions are relatively rare events, they can be potentially life-threatening. Although the mechanism by which CBZ induces hypersensitivity reactions has not been well characterized, recent studies have suggested that the immune reaction can be caused, at least in part, by its metabolism into chemically reactive metabolites, which can be the critical step in the formation of protein adducts and subsequent immune responses. Identification of the reactive metabolite(s) has been difficult, because the rate at which human liver microsomes convert CBZ to metabolites capable of binding to proteins is quite low. Furthermore, it is also important to note that CBZ is typically administered in doses ranging from hundreds of milligrams to grams per day, suggesting that substantial amounts of reactive metabolites can be formed on a daily basis. Therefore, it is conceivable that individuals with a low capacity to detoxify reactive metabolites via glucuronidation or other pathways can be at risk for idiosyncratic toxicity. Although an arene oxide was originally proposed as the reactive species, either an iminoquinone metabolite derived from the CBZ metabolite (CBZ-IQ) or 2-hydroxyiminostilbene (2-OHIS) are potential candidates for the reactive metabolite, because quinone- and iminoquinone-type metabolites have been implicated

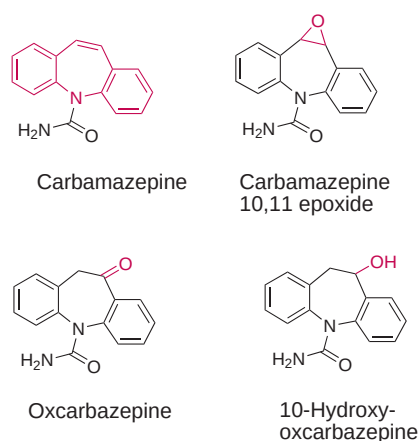


FIGURE 17.6 Iminostilbenes.

in drug-induced hepatotoxicity (see also Chapter 4). CYP3A4-catalyzed secondary metabolism of 2-hydroxycarbamazepine to 2-OHIS or to CBZ-IQ, followed by nonenzymatic reduction to 2-OHIS, can well underlie subsequent development of drug-induced hypersensitivity to CBZ (Fig. 17.7) (44). Based on the formation of glutathione and *N*-acetylcysteine conjugates, it has been suggested that 2-OHIS is the target for the formation of protein adducts, which could lead to localized idiosyncratic toxicities. Like phenytoin, CBZ is highly protein bound (Table 17.2) and is extensively transformed. Approximately 72% of an oral dose is excreted in the urine as metabolites and 3% as unchanged drug. The 28% found in the feces can be the result of incomplete absorption and enterohepatic cycling. As previously mentioned, interindividual variability in apparent plasma half-life and total-body clearance is related to the phenomenon of autoinduction.

Adverse Effects Gastric upset from CBZ can be diminished by taking the drug after meals. Common toxicities include blurred vision, dizziness, drowsiness, and ataxia. Tremor, depression, hyponatremia, and cardiac disturbances are seen at high serum concentrations. Idiosyncratic rashes are common; rarer severe idiosyncratic effects include aplastic anemia, agranulocytosis, thrombocytopenia, and jaundice. Therefore, patients receiving CBZ should have periodic blood count determinations and liver function tests. Both CBZ and oxcarbazepine can reduce plasma 25-hydroxyvitamin D levels (45). CBZ increases levels of phenytoin and decreases levels of felbamate, lamotrigine, oral contraceptives, theophylline, valproate, and zonisamide. CBZ levels are increased by propoxyphene, erythromycin, chloramphenicol, isoniazid, verapamil, and cimetidine. CBZ levels are decreased by phenobarbital, phenytoin, felbamate, and primidone. Lamotrigine and valproic acid can elevate CBZ epoxide levels. Administration of the extended-release form has been associated with fewer side effects and with improved seizure control (46).

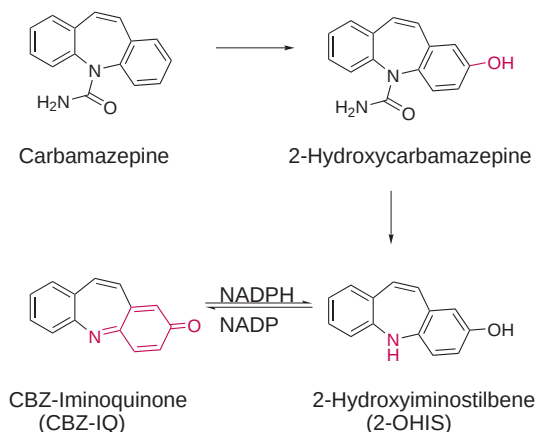


FIGURE 17.7 Bioactivation pathway for carbamazepine.

Macrolide antibiotics inhibit CBZ metabolism, thus increasing CBZ plasma levels and decreasing clearance with the potential for toxicity effects. Drug-induced changes in CBZ pharmacokinetics are particularly pronounced in children (28).

CBZ should be used with caution in patients with a history of congestive heart failure or cardiac arrhythmias (because it can aggravate them) and with a history of hematologic reactions to other drugs or hypersensitivity to tricyclic antidepressants. Blood levels should be monitored in patients with renal or hepatic impairment.

OXCARBAZEPINE Oxcarbazepine (Trileptal) is the 10-keto analog of CBZ (Fig. 17.6). It is indicated as monotherapy or adjunctive therapy for partial seizures in adults with epilepsy, as monotherapy for the treatment of partial seizures in children 4 years of age or older, and as adjunct therapy in children 2 to 4 years of age.

Mechanism of Action Although oxcarbazepine is less potent than CBZ, its mechanism of action is similar (47). The majority of the pharmacologic activity for oxcarbazepine is attributed to its primary metabolite, 10-monohydroxycarbamazepine (MHD) (Fig. 17.6), the plasma levels of which can be ninefold higher than those for CBZ. Both oxcarbazepine and MHD produce a blockade of voltage-dependent sodium channels, thus decreasing repetitive firing and spread of electrical activity. An additional action on calcium and potassium channels can contribute to the therapeutic effect. Like carbamazepine, oxcarbazepine can worsen juvenile myoclonic or absence seizures (41).

Pharmacokinetics Oxcarbazepine is completely absorbed, and food has no effect on its absorption. Unlike CBZ, it does not cause autoinduction of its own metabolism. The metabolism of oxcarbazepine is different from that of CBZ. Oxcarbazepine is reduced by cytosolic enzymes to MHD before its *O*-glucuronidation. More than 95% of its oral dose is excreted as conjugated metabolites, with approximately 4% of the drug converted to inactive 10,11-dihydroxy CBZ. Unlike CBZ, no epoxide or aromatic hydroxylation metabolites are formed. The half-life is 2 hours for oxcarbazepine and 9 hours for the active 10-monohydroxy metabolite. In patients with impaired renal function, the half-life for MHD is prolonged to 19 hours, with a doubling in its area under the plasma concentration curve. Peak plasma concentration following an oral dose occurs at approximately 4.5 hours.

Oxcarbazepine induces CYP3A4/5 and UGT and inhibits CYP2C19, producing significant alterations on the plasma concentration of other drugs. Therefore, oxcarbazepine decreases felodipine bioavailability and lowers plasma levels for lamotrigine, CBZ, CBZ epoxide, calcium channel blockers, and oral contraceptives (48). Oxcarbazepine increases plasma levels of phenobarbital and phenytoin. Unlike CBZ, oxcarbazepine has no effect on plasma levels of risperidone or olanzapine (49,50). The

plasma levels for oxcarbazepine or MHD are decreased by CBZ, phenobarbital, phenytoin, valproate, and verapamil. Serum MHD can decrease during pregnancy but increase following delivery (51). Oxcarbazepine clearance is decreased in renal impairment and the elderly. In children, a higher dose per kilogram for oxcarbazepine than in adults is required to obtain an effective plasma concentration.

Adverse Effects Patients with hypersensitivity reactions to carbamazepine can be expected to show cross-sensitivity (e.g., rash) or related problems to oxcarbazepine. The improved toxicity profile for oxcarbazepine when compared to CBZ can result from absence of the epoxide or CBZ-iminoquinone metabolites (47). The most common side effects are headache, dizziness, nystagmus, blurred vision, somnolence, nausea, ataxia, and fatigue. The incidence of adverse effects has been related to elevated serum MHD concentrations (52). Adverse effects on cognitive status, hyponatremia, and serious dermatologic reactions have been reported, as has hyponatremia (53).

Barbiturates

The barbiturates are substituted pyrimidine derivatives with an ureide configuration (Fig. 17.4). They are lipophilic weak acids ($pK_a = 7$ to 8) that are well distributed into brain (see Appendix A for the respective pK_a values). Although many barbiturates display sedative-hypnotic activity (see Chapter 15), only a few have antiseizure properties. Paradoxically, many barbiturates cause convulsions at larger doses. The barbiturates clinically useful as AEDs are phenobarbital, mephobarbital, and primidone (Fig. 17.8). In laboratory animals, phenobarbital is effective by several tests in nontoxic doses. It is active against MES, and it elevates the threshold for pentylenetetrazole stimulation. The mechanism of antiseizure action for the barbiturates is unknown but is thought to involve blockade of sodium channels and enhancement of GABA-mediated inhibitory transmission.

PHENOBARBITAL Phenobarbital is commonly used for convulsive disorders and is the drug of choice for seizures in infants up to 2 months of age. Phenobarbital is indicated for the treatment of partial and generalized tonic-clonic seizures in all age groups, although it is less effective than phenytoin or CBZ in adults (40). Although occasionally

used as monotherapy, it usually is combined with another AED. Phenobarbital can be administered parenterally, as its sodium salt, for emergency control of acute convulsive disorders associated with eclampsia (although magnesium sulfate is the standard treatment), meningitis, tetanus, and toxic reactions to strychnine or local anesthetics. Because of its slow onset of action, it is administered after benzodiazepines for the treatment of status epilepticus.

Pharmacokinetics Phenobarbital is a weak acid ($pK_a = 7.4$; $\log D_{pH\ 7.4} = 1.53$) that is approximately 50% ionized at physiologic pH and is well distributed into the CNS. Its oral absorption is slow but nearly complete, with an oral bioavailability of 80% to 100%, and it shows linear kinetics. Phenobarbital is 40% to 60% protein bound and exhibits a long plasma half-life of 2 to 6 days, which yields an extremely stable plasma concentration. Approximately 25% to 50% of a phenobarbital dose is excreted unchanged in the urine. The remainder is metabolized primarily by hydroxylation to its inactive metabolite, 5-*p*-hydroxyphenyl-5-ethyl-barbituric acid, which is then conjugated as its glucuronide or sulfate and is excreted in the urine. Some of the conjugated metabolites can appear in the feces from enterohepatic cycling. Alkalinizing the urine or increasing the urine flow substantially increases the rate of excretion of unchanged phenobarbital and its metabolites. Phenobarbital is a potent liver enzyme-inducing drug of CYP3A4 and increases the ability of the liver to metabolize many drugs, when taken concurrently, that are normally metabolized by CYP3A4. It also induces UGTs and increased formation of glucuronidation. No conclusive evidence, however, shows that phenobarbital induces its own metabolism (autoinduces), as does CBZ.

Because of its inducing effect on hepatic enzymes, phenobarbital has many drug interactions, decreasing plasma levels of CBZ, valproate, lamotrigine, tiagabine, zonisamide, warfarin, theophylline, cimetidine, and other CYP3A4 substrates. Serum concentrations of phenobarbital are increased by valproate.

Adverse Effects Serious toxicity is rare; however, drowsiness is the most common side effect reported for phenobarbital. Of the barbiturates, only phenobarbital, mephobarbital, and primidone are antiseizure at subhypnotic doses. The sedative effect of phenobarbital limits its use in older children and adults, although tolerance to the sedative effects often develops. When compared to phenytoin or CBZ, phenobarbital shows more sedation, irritability, paradoxical hyperactivity, and impaired intellectual function. This can prove to be particularly troublesome in children, especially those of school age, and in the elderly. Quite rare are idiosyncratic hypersensitivity reactions to phenobarbital that include rash, agranulocytosis, aplastic anemia, and hepatitis. Although the mechanism by which the phenobarbital and other aromatic anticonvulsants induce hypersensitivity reactions

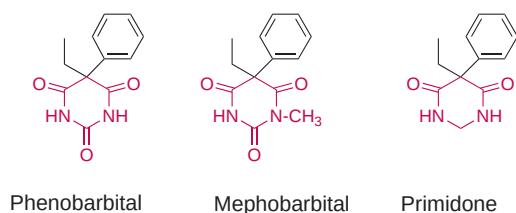


FIGURE 17.8 Barbiturates.

has not been well characterized, recent studies have suggested that the immune response elicited can be caused, at least in part, by the metabolism of phenobarbital into chemically reactive metabolites, which can be the critical step in the formation of protein adducts and subsequent immune responses (see discussion for CBZ). Long-term use of phenobarbital can precipitate folate, vitamin K, or vitamin D deficiencies.

Phenobarbital should be used with caution in patients with hepatic impairment; therefore, a dose reduction can be needed. It should be avoided in patients with renal impairment. Barbiturates are known to cause fetal abnormalities and a neonatal coagulation defect responsive to vitamin K.

MEPHOBARBITAL (MEBARAL) Mephobarbital is a barbiturate-derivative AED with a pK_a of 7.7 ($\log D_{pH\ 7.4} = 1.84$). Approximately 50% of an oral dose of mephobarbital is absorbed from the gastrointestinal tract. The plasma concentrations required for its therapeutic effects are unknown. The principal route of mephobarbital metabolism is *N*-demethylation by the liver to form phenobarbital, which can be excreted in the urine unchanged and as its *p*-hydroxy metabolite and glucuronide or sulfate conjugates. Conversion to the 4-hydroxy metabolite is stereoselective, being catalyzed by either CYP2C19 (*R*-enantiomer) or CYP2B6 (*S*-enantiomer); individuals who are CYP2C19-poor metabolizers show decreased clearance (54). Approximately 75% of a single oral dose of mephobarbital is converted to phenobarbital. It has not been determined whether mephobarbital contributes to the antiseizure effect or whether it results from its active metabolite, phenobarbital. Similarly, it is unclear whether mephobarbital, like phenobarbital, is a potent inducer of the enzymes involved in the metabolism of other drugs, but because the drug is chemically and pharmacologically similar to phenobarbital and is metabolized to phenobarbital, this possibility is likely.

Mephobarbital is less commonly used in the treatment of generalized and partial seizures. Like phenobarbital, it is classified as a long-acting barbiturate. No evidence exists that it is more effective than phenobarbital in equivalent doses; however, it can be less sedating in children.

PRIMIDONE (MYSOLINE) Primidone is the 2-deoxy derivative of phenobarbital (Fig. 17.8) and is approved by the FDA for initial or adjunctive treatment of simple partial, complex partial, and tonic-clonic seizures. It is less effective against these types of seizures than is phenytoin or CBZ, and it shares the antiseizure and sedative actions of phenobarbital. Although not approved for the purpose, it often is used to treat benign familial tremor (essential tremor).

Pharmacokinetics Approximately 60% to 80% of an oral dose of primidone is absorbed and slowly metabolized by the liver to phenobarbital and phenylethylmalonamide (PEMA) (55,56). All three molecules have antiseizure

effects, but PEMA appears to be weaker and to be the more toxic metabolite. During chronic therapy, approximately 15% to 25% of an oral dose of primidone is excreted in the urine unchanged, 15% to 25% is metabolized to phenobarbital, and 50% to 70% is excreted as PEMA (half-life, 24 to 48 hours). The phenobarbital metabolite can be excreted in the urine unchanged, as its *p*-hydroxy metabolite, and as glucuronide or sulfate conjugates. Following an oral dose, the peak plasma levels for primidone are reached in approximately 4 hours, with a reported half-life of 10 to 12 hours. Plasma concentrations in the range of 8 to 12 $\mu\text{g/mL}$ control seizures and minimize adverse effects. Primidone shows antiseizure activity before the phenobarbital levels reach therapeutic range. Only after chronic dosing of primidone are the levels of phenobarbital significant, suggesting autoinduction. Serum levels of chronically administered primidone exceed those of its metabolite, phenobarbital, thus demonstrating that it has antiseizure activity independent of phenobarbital. When primidone is coadministered with enzyme-inducing AEDs, the levels of its phenobarbital metabolite can be two- to threefold higher than those in the noninduced state. Protein binding of primidone and PEMA is negligible, and the phenobarbital metabolite is approximately 50% protein bound.

Primidone use is associated with decreases in CBZ, lamotrigine, valproate, tiagabine, and zonisamide serum levels. Primidone levels are increased by nicotinamide and isoniazid. Hydantoins increase the plasma concentrations of primidone, phenobarbital, and PEMA. CBZ increases levels of phenobarbital derived from primidone. Primidone levels are decreased by succinimides, CBZ, and acetazolamide.

Adverse Effects As with phenobarbital, serious toxicity for primidone is rare, although it can cause disabling sedation, irritability, and decreased mental functioning in a number of persons. Ataxia, dysphoria, idiosyncratic rash, leukopenia, agranulocytosis, lymphadenopathy, hepatitis, and a systemic lupus erythematosus-like syndrome have been reported adverse effects for primidone. Deficiencies of folic acid and of vitamins D and K are possible with long-term therapy of primidone, as is a folate-responsive megaloblastic anemia. Measurement of the complete blood cell count should be performed at 6-month intervals (40).

Benzodiazepines

This class of drugs has been widely used as sedative-hypnotics and antianxiety drugs (see Chapters 15 and 14, respectively). In laboratory animals, benzodiazepines display outstanding antiseizure properties against seizures induced by MES and pentylenetetrazole. The benzodiazepines diazepam, lorazepam, clonazepam, clorazepate dipotassium, and midazolam are effective for seizure control (Table 17.2 and Fig. 17.9). All benzodiazepines enter cerebral tissue rapidly. Although the duration of action is short for diazepam (2 hours) and midazolam

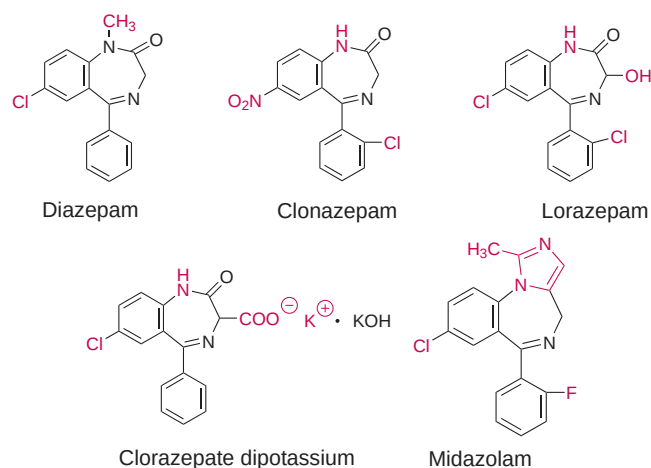


FIGURE 17.9 Benzodiazepines.

(3 to 4 hours), longer for clonazepam (24 hours), and much longer for lorazepam (up to 72 hours), no correlation exists with the plasma concentration–time profiles for these drugs (57,58). Diazepam and lorazepam can be administered either IV or IM for control of status epilepticus; however, absorption is slower from the IM site. Midazolam is particularly useful for treating status epilepticus, because its imidazole ring is open at low pH, allowing it to be dissolved in aqueous solution for IM injection, but is closed at physiologic pH, increasing lipophilicity with rapid IM absorption, brain penetration, and fast onset of action. When administered IM, midazolam has a faster onset of action than diazepam and lorazepam due to its rapid absorption from the injection site, with an efficacy at least equal to that of IV diazepam (57,58). Seizure arrest usually is attained within 5 to 10 minutes. In addition, the use of midazolam by the buccal route for initial treatment of acute seizure activity has been shown to be more effective and without increased respiratory depression as compared to the use of rectal diazepam (59). Midazolam has been used intranasally as well (60,61). Lorazepam, although not approved by the FDA for this purpose, is preferred by some clinicians for the treatment of status epilepticus as a result of its longer duration of action.

MECHANISM OF ACTION The benzodiazepines are thought to produce their antiseizure effects primarily by enhancing the effect of the inhibitory neurotransmitter GABA on the GABA_A chloride channel (Figs. 17.1 and 17.2). Additional evidence suggests that the benzodiazepines can diminish voltage-dependent sodium, potassium, and calcium currents in a manner independent of the GABA_A/benzodiazepine receptor complex or alter expression of GABA_A receptor subunit expression (62).

DIAZEPAM (VALIUM) Diazepam is given orally for adjunctive control of convulsive disorders, as a rectal gel (Diastat) for refractory patients with epilepsy on a stable regimen

of AEDs who require intermittent use of diazepam to control bouts of increased seizure activity, and parenterally as part of the regimen for the treatment of status epilepticus or other severe, recurrent seizures. Rectal diazepam gel is an effective and well-tolerated therapy for acute repetitive seizures (59).

Pharmacokinetics Orally administered diazepam is less effective as an AED, because tolerance to the antiseizure effects of diazepam develops within a short period. Diazepam gel is rapidly absorbed rectally, having greater than 90% bioavailability. In addition to cluster seizures, it has been proven to be useful to control prolonged febrile seizures in children.

Intravenously administered diazepam is the route of choice for rapid control of status epilepticus. Because of its high lipid solubility, IV diazepam enters the CNS rapidly. However, the initial high brain concentration is reduced quickly as a result of its redistribution; thus, status epilepticus can return. To prevent the return of status epilepticus, the initial dose of diazepam is followed sequentially by parenteral phenytoin (fosphenytoin) and phenobarbital as needed for control of tonic-clonic status epilepticus. For absence status epilepticus, diazepam usually is followed by ethosuximide.

The half-life for diazepam is 46 hours, and diazepam is metabolized by CYP2C19 and CYP3A4 to desmethyldiazepam, an active metabolite with a half-life of 71 hours. Diazepam is 95% protein bound. Cimetidine, by inhibiting CYP3A4, decreases the metabolism and clearance of diazepam. Drugs that affect the activity of CYP2C19 or CYP3A4 can alter diazepam kinetics, and vice versa.

Adverse Effects The most frequent side effect for diazepam is somnolence; dizziness, ataxia, headache, nervousness, euphoria, and rash occur less frequently. Excessive use of rectal diazepam can produce rebound seizures (63). Intravenous administration can produce infrequent respiratory depression and hypotension. Other sedative drugs, such as barbiturates, valproate, narcotics, phenothiazines, monoamine oxidase inhibitors, and antidepressants, can potentiate the effects of diazepam.

Because diazepam clearance is decreased in the elderly and in patients with hepatic insufficiency, a dosage reduction can be warranted. Intravenous diazepam should be used cautiously in patients who are elderly, very ill, or have limited pulmonary reserve, because respiratory depression has occurred. IV Diazepam is rarely given to patients for absence status (typical and atypical) seizures, because it may precipitate tonic status epileptic seizures.

CLONAZEPAM Clonazepam (Klonopin) was approved in 1975 for monotherapy or adjunctive treatment of akinetic (atonic), myoclonic, and absence variant seizures (64). Clonazepam also was found to be effective in controlling absence seizures, but because of its high incidence of side effects, it is rated second to ethosuximide. It can be useful, however, in absence seizures when succinimide

therapy has failed. It is considered to be a third-line drug after 1) ethosuximide or valproate and 2) lamotrigine or valproate for the treatment of absence seizures. It is ineffective for treatment of generalized clonic-tonic seizures.

Clonazepam is well absorbed, 95% to 98% protein bound, and extensively metabolized by CYP3A4. Clonazepam displays a wide spectrum of antiseizure activities and is one of the most potent AEDs. Side effects are common, however, and the development of tolerance is more frequent than with ethosuximide or valproate. Sedation is prominent, especially early in treatment. Drowsiness, ataxia, and behavioral changes can be disabling, but slowly increasing its dose over a 2-week period is recommended to minimize adverse effects. Diplopia (double vision), headaches, nystagmus (involuntary eye movement), and other neurologic effects have been reported with the use of clonazepam.

Serum levels of clonazepam are decreased by the enzyme-inducing properties of phenobarbital, phenytoin, and CBZ. Concurrent administration of amphetamines, methylphenidate, ethanol, antianxiety drugs, or antipsychotics can cause CNS depression or altered respiration. The combined administration of clonazepam and valproate can cause absence status, and in patients displaying a mixed seizure pattern, clonazepam can precipitate grand mal seizures.

CLORAZEPATE DIPOTASSIUM Clorazepate dipotassium (Tranxene) was approved for use as an AED by the FDA in 1981. It is less commonly used as adjunct therapy for the management of partial seizures in adults and children older than 9 years. Clorazepate is decarboxylated by the acidity of the stomach to desmethyldiazepam, which is also the major active metabolite of diazepam. Therefore, clorazepate exhibits the profile and properties of diazepam.

LORAZEPAM (ATIVAN) Lorazepam is a benzodiazepine (Fig. 17.6), with pK_a values of 1.3 and 11.5. Its parenteral solutions are formulated with polyethylene glycol, and dilution with diluents can cause its crystallization from solution. Lorazepam is also used IV or IM for the management of status epilepticus. Although IV diazepam has been used more extensively, some clinicians prefer IV lorazepam due to its more prolonged duration of effect. It has been recommended for initial treatment of generalized convulsive status epilepticus in the elderly (65). A long-acting AED, such as IV phenytoin or fosphenytoin, can be added to IV benzodiazepine therapy for the management of recurring seizures associated with status epilepticus.

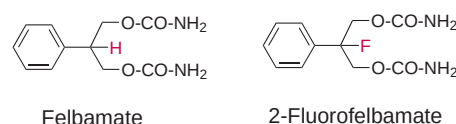
Intramuscular lorazepam is slowly absorbed with peak plasma concentrations (C_{max}) reached in approximately 60 to 90 minutes. The half-life for unconjugated lorazepam is approximately 16 hours when given IV or IM, and it is eliminated in the urine as its major metabolite, lorazepam 3-*O*-glucuronide. Lorazepam glucuronide has no demonstrable CNS activity in animals but has a

half-life of approximately 18 hours. Lorazepam is 85% bound to plasma proteins. Drugs that inhibit or induce the oxidative metabolism of benzodiazepines (CYP3A4) are less likely to affect lorazepam, because it undergoes only glucuronide conjugation.

OTHER BENZODIAZEPINES In addition to their anxiolytic and sedative-hypnotic properties, several other benzodiazepines also display antiseizure activity. They include clobazam and nitrazepam, which are used outside the United States; these drugs have not demonstrated any clinical advantage over clonazepam.

Biscarbamates

FELBAMATE (FELBATOL)



Felbamate is a dicarbamate that is structurally similar to the antianxiety drug meprobamate. It was approved by the FDA for antiseizure use in 1993. Following the occurrence of rare cases of aplastic anemia and of severe hepatotoxicity associated with the use of felbamate during early 1994, however, a black box warning was added to the drug's package insert (53). Despite this, felbamate continues to be used in many patients, although not as a first-line treatment. These toxicity effects can be attributed to the formation of toxic metabolites (66). Although felbamate use is now uncommon, it is used for severe refractory seizures, either partial, myoclonic, or atonic, or in Lennox-Gastaut syndrome.

Mechanism of Action Although its mechanism of action is unknown, felbamate alters the gating behavior of the NMDA receptor by binding to the receptor pore and lowering calcium currents (Fig. 17.2) (67).

Metabolism and Pharmacokinetics Although the metabolism of felbamate has not been fully characterized, felbamate is esterase hydrolyzed to its monocarbamate metabolite, 2-phenyl-1,3-propanediol monocarbamate, which is subsequently oxidized via aldehyde dehydrogenase to its major human metabolite 3-carbamoyl-2-phenylpropionic acid (Fig. 17.10). Other metabolites include the *p*-hydroxy and mercapturic acid metabolites of felbamate, which have been identified in human urine. Felbamate is a substrate for CYP2C19, with minor activity for CYP3A4 and CYP2E1. Thompson et al. (68) have provided evidence for the formation of the reactive metabolite, 3-carbamoyl-2-phenylpropionaldehyde (CBMA), from the alcohol oxidation of 2-phenyl-1,3-propanediol monocarbamate. CBMA then undergoes spontaneous elimination to another reactive intermediate, 2-phenylpropenal (more commonly known as atropaldehyde), which is proposed to have a role in the

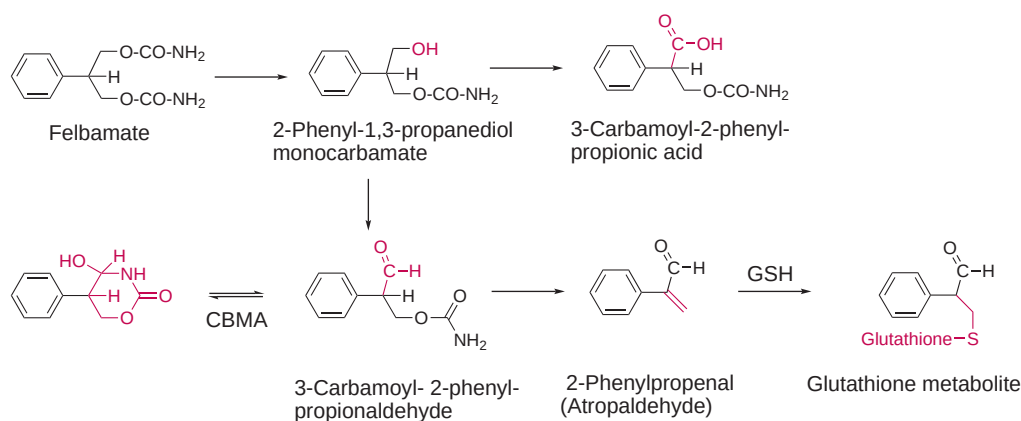


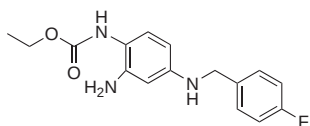
FIGURE 17.10 Metabolism of felbamate.

development of toxicity during felbamate therapy. CBMA or a further product has been shown to provoke an immune response in mice (69). Evidence for *in vivo* atropaldehyde formation was confirmed with the identification of its mercapturic acid conjugates in human urine after felbamate administration. This is consistent with the hypothesis that atropaldehyde reacts rapidly with thiol nucleophiles, such as glutathione, to form mercapturates (see Chapter 4). More recently, a fluorine analog of felbamate was synthesized in which the benzylic C₂ hydrogen of the propane chain was replaced with a fluorine atom, preventing the formation of atropaldehyde and confirming that the acidic benzylic hydrogen plays a pivotal role in its formation (70). This analog is presently undergoing drug development. Felbamate administration exhibited linear kinetics, with a half-life of 20 to 23 hours in the absence of enzyme-inducing AEDs. Approximately 50% of an oral dose of felbamate is excreted unchanged.

Adverse Effects Felbamate has exhibited the rare occurrence of aplastic anemia and of severe hepatotoxicity, which can be associated with the *in vivo* formation of reactive metabolites. Felbamate increases phenytoin and valproate serum levels but decreases the level of CBZ. The increase in valproate plasma concentrations by felbamate is through the inhibition of β -oxidation. No clinically relevant pharmacokinetic interactions have been noted between felbamate and lamotrigine, clonazepam, vigabatrin, or the active monohydroxy metabolite of oxcarbazepine. The enzyme inducers phenytoin, phenobarbital, and CBZ decrease felbamate plasma levels by increasing its clearance.

Miscellaneous Antiepileptic Drugs

EZOGBABINE (RETIGABINE; POTIGA)



Ezogabine (retigabine in Europe) is an anticonvulsant possessing a new mechanism of action: It affects certain voltage-gated potassium channels known to be expressed in brain. It is indicated for the adjunctive treatment of partial-onset seizures in adults.

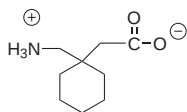
Mechanism of Action Ezogabine stabilizes the open state of Kv7.2 to Kv7.5 potassium channels (products of *KCNQ2* to *KCNQ5* genes). Kv7.2 and Kv7.3 are the predominant components of the slow voltage-gated M channel, which regulates neuronal excitability (71). Ezogabine produces a hyperpolarizing shift in their voltage gating, increases the channels' open time, and decreases substantially their closed time, reducing excitability. In addition, it produces faster opening of the channels during depolarization, thus suppressing repetitive firing (72). Kv7.2 and Kv7.3 are present on the axon initial segments of CA1 pyramidal neurons. The drug does not affect cardiac Kv7.1 channels because these lack a requisite tryptophan residue. It is noteworthy that mutations in *KCNQ2* and *KCNQ3* are known to be associated with a form of human epilepsy.

Pharmacokinetics Ezogabine is rapidly absorbed, having a bioavailability of up to 60%. It reaches C_{max} in approximately 1.5 hours, has a half-life of approximately 6.3 hours, and has a volume of distribution (V_d) of 6.2 L/kg. It is not metabolized by CYP450. Ezogabine's predominant metabolites are two *N*-glucuronides formed by UGT1A1, UGT1A4 and UGT1A9, plus lesser acetylated derivatives (73). Ezogabine has been shown to modestly increase the clearance of lamotrigine, whereas lamotrigine increased the AUC of ezogabine by 15%, probably as a result of competition for renal elimination. Phenytoin and CBZ can increase the clearance of ezogabine.

Adverse Effects In clinical studies, urinary retention and dizziness have caused withdrawal of the drug. Weight gain, somnolence, headache, fatigue, confusion, and slurred speech have been noted, although the drug is

generally well tolerated (74). Serious adverse events have included nephritis and urinary retention. Monitoring patients for urinary retention can be required.

GABAPENTIN (NEURONTIN)



Mechanism of Action Gabapentin is a water-soluble amino acid originally designed to be a GABA-mimetic analog capable of penetrating the CNS. Surprisingly, it has no direct GABA-mimetic activity. As with pregabalin, the mechanism of action involves drug binding to the α -2- δ subunit of the P/Q-type calcium channels, decreasing their activity. In addition, gabapentin raises brain GABA levels in patients with epilepsy (75). Recent studies have demonstrated gabapentin binding to calcium channels in a manner that can be allosterically modulated (76).

Gabapentin is indicated as an adjunct for use against partial seizures with or without secondary generalization, in patients older than 12 years, and as adjunct for the treatment of partial seizures in children 3 to 12 years of age. It also is approved for the treatment of postherpetic neuralgia.

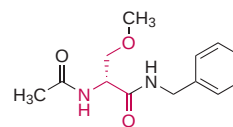
Pharmacokinetics The pharmacokinetic properties for gabapentin are generally favorable, with a bioavailability of 60% when given in low doses and somewhat less when given at higher doses as a result of saturable intestinal uptake by the L-amino acid transporter (77). The L-amino acid transporter is very susceptible to substrate saturation (low K_m value). Its absorption and distribution into the CNS appear to be dependent on this amino acid transporter. Following the administration of an oral dose, gabapentin reaches peak plasma concentration in 2 to 3 hours. Additionally, it exhibits linear pharmacokinetics. Moreover, it is not extensively metabolized and is not an inducer of hepatic metabolizing enzymes. The elimination of unmetabolized gabapentin occurs by the renal route. Although its therapeutic range is not well characterized, gabapentin has a broad therapeutic index. This implies that a wide range of doses can be used, based on individual patient needs, without significant limitation because of dose-dependent side effects. Protein binding is negligible. Its elimination half-life of 5 to 7 hours is not affected by the dose or by other drugs, and its short half-life necessitates multiple daily administration.

Adverse Effects Adverse effects of gabapentin are uncommon and not serious. The CNS effects include mild to moderate sedation, fatigue, ataxia, headache, dizziness, and diplopia. Gabapentin can exacerbate myoclonus, but the effect is mild and does not require discontinuance of the drug (53,78). It has been associated

with the development of neuropsychiatric adverse events in children.

Drug interactions are infrequent with gabapentin. It does not induce hepatic metabolizing enzymes, nor do other AEDs affect its metabolism and elimination. Antacids can decrease absorption. Gabapentin dosage can need to be decreased in patients with renal disease or in the elderly.

LACOSAMIDE (VIMPAT)



Lacosamide is an amino acid-related compound that was approved by the FDA in 2008 for adjunctive treatment of partial-onset seizures in patients age 17 years or older. Although the drug acts on voltage-gated sodium channels, it has a mechanism of action different from phenytoin and CBZ.

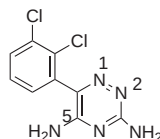
Mechanism of Action Lacosamide is a functionalized amino acid amide, the optical antipode of the natural amino acid L-serine (outlined in color), and enhances sodium channel slow inactivation with no effect on fast inactivation. Following opening, sodium channels adopt a “fast inactivated state” before returning to the resting state. This fast inactivated state is acted on by classical anticonvulsants such as phenytoin, CBZ, or lamotrigine. However, after repetitive firing, sodium channels can adopt a “slow inactivated state” over several seconds. It is this state that is enhanced by lacosamide, thus prolonging depolarization and diminishing rapid repetitive firing (79,80). In addition, lacosamide binds collapsin-response mediator protein 2 (CRMP-2), which is involved in polarization, regulation of gene expression, and growth of axons and dendrites. Whether this contributes to the drug’s anticonvulsant antiepileptic effects is presently unknown. (79)

Pharmacokinetics Lacosamide’s oral bioavailability is complete without any food effect. It displays linear pharmacokinetics, with a half-life of approximately 12 to 13 hours and reaching C_{max} in 1 to 4 hours. It is less than 15% bound by plasma proteins and shows a V_d of 0.6 L/kg. The drug is converted by hepatic CYP2C19 to an inactive O-desmethyl metabolite; this metabolite plus 40% of the unchanged drug are excreted renally (80). Although CYP2C19 is polymorphic, no significant differences in lacosamide pharmacokinetics have been reported in poor metabolizers (81). Dosage adjustments can be needed in patients with moderate hepatic or severe renal impairment. Drug interactions are minimal. Lacosamide has no effect on CYP450s and thus does not affect other anticonvulsant levels (79). Only small decreases in lacosamide

plasma concentrations are seen with CBZ, phenytoin, or phenobarbital coadministration. Lacosamide does not affect oral contraceptive efficacy (80).

Adverse Effects Because mild PR interval prolongation has been reported, there can be an increased risk of atrial fibrillation in patients with cardiac disease. Dizziness, headache, nausea, vomiting, diplopia (double vision), and blurred vision have occurred. Only one case of severe hypersensitivity reaction was noted in clinical trials.

LAMOTRIGINE (LAMICTAL)



Lamotrigine is a 5-phenyl-1,2,4-triazine derivative indicated as monotherapy or as an adjunct for partial seizures in adults, as adjunct in patients with Lennox-Gastaut syndrome, and as adjunct for partial seizures in children 2 years of age and older. Lamotrigine can have additional benefit in combating myoclonic and typical absence seizures. It is approved for use in the maintenance treatment of bipolar disorder.

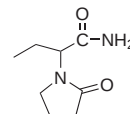
Mechanism of Action The most probable explanation for lamotrigine's efficacy is its ability to produce a blockade of sodium channel repetitive firing. In addition, lamotrigine appears to reduce glutaminergic excitatory transmission and inhibit neuronal nicotinic acetylcholine receptors (82,83).

Pharmacokinetics Following oral administration, lamotrigine is absorbed rapidly and completely, exhibiting linear pharmacokinetics and modest protein binding (55%). Lamotrigine is metabolized predominantly by *N*-glucuronidation and subsequent urinary elimination of its major metabolite, the quaternary 2-*N*-glucuronide (80% to 90%), the minor 5-amino-*N*-glucuronide (8% to 10%), and unchanged drug (8% to 10%) (84). Lamotrigine's usual elimination half-life of 24 to 35 hours is reduced to 13 to 15 hours in patients taking enzyme-inducing AEDs. The presence of valproate increases the lamotrigine half-life substantially by inhibiting *N*-glucuronidation, necessitating a reduction in dose to avoid toxicity. Hepatic disease patients can demonstrate a reduced capacity for lamotrigine glucuronidation, thus reducing its rate of clearance.

Adverse Effects The usefulness of lamotrigine is limited by the increased incidence of serious rashes, particularly in children or patients taking valproate (53). This increased incidence, however, can be attenuated by very slow dose escalation, because most rashes appear

within the first 8 weeks of treatment. The drug should be discontinued if a rash appears at any time. In 2010, the FDA warned that lamotrigine can cause aseptic meningitis. Furthermore, lamotrigine can be associated with development of myoclonus after 2 to 3 years of drug treatment (85). Additional common side effects associated with lamotrigine therapy include dizziness, diplopia, headache, ataxia, blurred vision, somnolence, and nausea.

S-(–)-LEVETIRACETAM (KEPPRA)



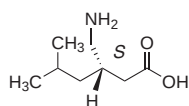
S-(–)-Levetiracetam is a pyrrolidone derivative unrelated to the structures of other AEDs. It is indicated as an adjunct in the treatment of partial-onset seizures in adults, and it has shown some benefit in clinical trials for generalized tonic-clonic seizures and myoclonic seizures in adults and children (86,87).

Mechanism of Action The mechanism of action for S-(–)-levetiracetam is unknown. It does not appear to interact with any of the recognized excitatory or inhibitory neural mechanisms. A CNS-specific binding site for S-(–)-levetiracetam has been identified as the synaptic vesicle protein (SV2A). Knockout animals without SV2A proteins accumulated presynaptic Ca²⁺ during consecutive action potentials that destabilized synaptic circuits and induced epilepsy. Thus, it appears that SV2A has a major role in the antiepileptic properties of S-(–)-levetiracetam, which acts by modulating the functions of SV2A, expression of the associated calcium sensor protein synaptotagmin, and the regulation of Ca²⁺-mediated synaptic transmission (88). These data support previous indications that S-(–)-levetiracetam possesses a mechanism of action distinct from that of other antiepileptic drugs. Three SV2 isoforms (SV2A, SV2B, and SV2C) have been identified, each of which has a unique distribution in brain, suggesting synapse-specific functions as well as antagonism of neuronal synchronization (78,89).

Pharmacokinetics S-(–)-Levetiracetam displays rapid and complete absorption, although food slows the rate but not the extent of absorption. It exhibits linear pharmacokinetics and is minimally protein bound (90). Approximately 60% of an oral dose is excreted into the urine unchanged, and 24% to 30% is excreted as its carboxylic acid metabolite, with an elimination half-life in adults of approximately 7 hours. Although S-(–)-levetiracetam is not metabolized by hepatic CYP450, UGT, or epoxide hydrolase, it is esterase hydrolyzed to its carboxylic acid metabolite (loss of the amido group), which is not affected by the hepatic metabolizing enzymes.

Adverse Effects The risk of clinically relevant drug interactions is minimal with *S*-(-)-levetiracetam because it does not alter the pharmacokinetics of coadministered drugs by inhibition or induction of hepatic enzymes (91). Toxic effects include mild to moderate somnolence, asthenia, ataxia, and dizziness; these effects seldom require discontinuance. An increase in the incidence of behavioral abnormalities in children and in adults having a previous history of neuropsychiatric problems has been noted (92). Its use in the elderly or in patients with renal impairment will require an individualization of dose, and an additional dose is needed after renal dialysis. Levetiracetam was associated with developmental toxicity in the offspring of pregnant animals.

PREGABALIN (LYRICA)



Pregabalin is a GABA analog with similarities in structure and effects to gabapentin; however, it is 3 to 10 times more potent as an AED than is gabapentin. It is the pharmacologically active *S*-enantiomer of 3-isobutyl GABA and is FDA approved for the adjunctive treatment of partial-onset seizures in adults.

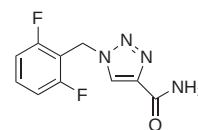
Mechanism of Action The drug acts by binding to the α -2 δ subunit of the P/Q-type voltage-sensitive calcium channels present on presynaptic neurons, modulating their activity and decreasing neuronal calcium currents (93).

Pharmacokinetics The absorption of pregabalin is rapid and nearly complete following oral administration. It displays linear pharmacokinetics, with a half-life of 5 to 6.5 hours and reaching C_{max} in approximately 1.5 hours. Although the rate of absorption is decreased by food, there is no clinically detectable decrease in the AUC or half-life; thus it can be administered with or without food. Pregabalin is not bound to plasma proteins and shows only minor metabolism to *N*-methylated (0.9%) and unidentified (0.4%) metabolites. Neither induction nor inhibition of the metabolism of other anticonvulsant agents has been demonstrated; no interaction with oral contraceptives has been reported. However, CBZ and other enzyme-inducing drugs can reduce pregabalin plasma concentrations. Dose adjustments are indicated in renal insufficiency, and supplemental doses are required following renal dialysis.

Adverse Effects Common adverse effects include ataxia, dizziness, somnolence, and confusion. In postmarketing studies, behavioral effects (agitation, irritability, and depression) and weight gain were the most frequent cause of drug discontinuance (93). Instances of severe

constipation requiring withdrawal of the drug have been cited. Precautions include withdrawal symptoms of insomnia, nausea, headache, and diarrhea, plus increased seizure occurrence on abrupt discontinuance. Therefore, it is recommended that the drug be tapered off over a minimum of 1 week. In studies, an increase in the cardiac PR interval was seen, but neither this nor heart block has been reported as an adverse effect. Because the medication has been associated with life-threatening rash with respiratory symptoms, caution is indicated for patients with a history of hives. Concomitant use of other angioedema-provoking agents such as the angiotensin-converting enzyme inhibitors can lead to a further increase in risk. The use of pregabalin in patients receiving thiazolidinedione antidiabetic drugs can further increase weight gain and peripheral edema. Infrequent increases in the creatine kinase plasma levels are known, and a few cases of rhabdomyolysis have been reported. Patients should be monitored for creatine kinase elevation; discontinuance is recommended for myopathy symptoms or significant creatine kinase elevation.

RUFINAMIDE (BANZEL)



Rufinamide is a triazole derivative that differs structurally from other anticonvulsants.

It is FDA approved as an orphan drug for adjunctive treatment of seizures associated with Lennox-Gastaut syndrome in children 4 years of age or greater and in adults. It was determined to be cost effective for treatment of Lennox-Gastaut syndrome. Additionally, the drug is being evaluated for adjunctive treatment of refractory partial-onset seizures; the European Union brand name is Inovelon.

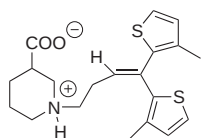
Mechanism of Action Rufinamide is thought to act by prolonging the inactive state of voltage-gated sodium channels, thereby limiting repetitive firing (94). High plasma concentrations inhibit the mGlu receptor (95).

Pharmacokinetics The bioavailability of rufinamide is approximately 85%. It should be given with food as this further increases absorption. The drug is 34% plasma protein bound, reaches maximum concentrations between 4 and 6 hours after administration, and has a half-life of 8 to 12 hours. It is extensively metabolized: Only 4% is excreted unchanged in urine and feces. The predominant metabolite (78%) is due to carboxylesterase hydrolysis of the carboxamide group, leading to an inactive metabolite; additionally, there is minor non-CYP450-mediated oxidative cleavage at the benzylic carbon. Rufinamide is a weak inhibitor of CYP2E1 and a

weak inducer of CYP3A4, but does not affect any other CYPs (94,95). Rufinamide plasma concentrations are affected by several drug interactions (81). Rufinamide concentrations are increased up to 70% by valproic acid, whereas they are decreased by CBZ, phenobarbital, primidone, vigabatrin, and phenytoin. Rufinamide increases plasma concentrations of phenobarbital and phenytoin and decreases concentrations of lamotrigine, CBZ, and triazolam. Additionally, rufinamide lowers the efficacy of oral contraceptives.

Adverse Effects QT interval shortening occurred in up to 65% of patients; thus familial short QT syndrome is an absolute contraindication. There can be an increased risk if combined with other agents known to shorten QT intervals. Multiorgan hypersensitivity syndrome was observed in one patient. Common adverse effects include fatigue, drowsiness, ataxia, somnolence, dizziness, headache, and nausea (95). Abrupt discontinuance can precipitate rebound seizures.

TIAGABINE (GABITRIL)



Mechanism of Action Tiagabine is a nipecotic acid derivative with an improved ability to cross the blood–brain barrier. It was rationally designed to be a GABA uptake inhibitor based on the fact that nipecotic acid (piperidine-3-carboxylic acid) inhibits GABA uptake by glial cells. Tiagabine binds to GABA transporter 1 (GAT1), blocking the uptake of GABA into both neurons and glia, thus enhancing GABA-mediated inhibition (Fig. 17.2) (53,96). Tiagabine is presently approved for adjunct use in patients with epilepsy who are older than 12 years and have partial seizures not controlled by first-line drugs.

Pharmacokinetics Tiagabine is well absorbed, with an oral bioavailability of 90% to 95%. It displays linear pharmacokinetics, with a plasma half-life of 5 to 8 hours, necessitating a multiple daily dosing regimen. It also is highly protein bound (96%). The major pathway of metabolism for tiagabine is oxidation by CYP3A4, followed by glucuronidation. Its pharmacokinetics are altered by the coadministration of enzyme-inducing AEDs, even though tiagabine does not appear to induce or inhibit hepatic microsomal metabolizing enzymes.

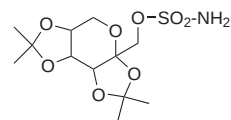
Adverse Effects Side effects are more common with tiagabine than with other adjunct drugs and most often involve the CNS. They include somnolence, headache, dizziness, tremor, abnormal thinking, depression, and psychosis. Furthermore, recent reports have implicated tiagabine in the development of nonconvulsive status

epilepticus (97). There is an increased risk of seizure in patients being treated for off-label psychiatric indications. Tiagabine can interfere with visual color perception (98).

Tiagabine does not affect the hepatic metabolism of other AEDs, but its half-life is decreased by enzyme-inducing AEDs, such as CBZ, phenytoin, and barbiturates. Other CYP3A4-inducing drugs can act similarly. Valproate decreases the protein binding of tiagabine, increasing its plasma concentration in these patients.

Hepatic disease causes decreased clearance of tiagabine, and a dose reduction can be required. Renal disease does not affect elimination.

TOPIRAMATE (TOPAMAX)



Topiramate is a sulfamate-substituted monosaccharide derived from fructose with a broad spectrum of AED activity. It is FDA approved for monotherapy or as an adjunct drug for partial or primary generalized tonic-clonic seizures in patients older than 10 years, as adjunct therapy in children age 2 to 10 years with partial-onset seizures, and in persons older than 2 years with Lennox-Gastaut syndrome (40,53). Topiramate also is approved for the prophylaxis of migraine headaches.

Mechanism of Action The mechanism of action for topiramate is unknown, but several actions are thought to contribute to its AED activity (99). It blocks repetitive firing by acting on sodium channels, can enhance GABA_A-mediated chloride flux, and appears to be an antagonist at the AMPA and KA receptors, thus blocking the effect of glutamate (Fig. 17.2) (100,101).

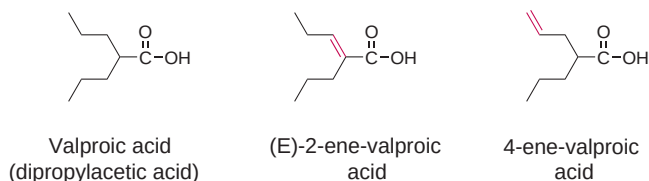
Pharmacokinetics Topiramate is rapidly absorbed, with at least an 80% to 95% oral bioavailability that is unaffected by food. Following an oral dose of topiramate, peak plasma concentration is reached in 1 to 4 hours exhibiting linear pharmacokinetics (102). Protein binding is minimal (<20%), and the usual elimination half-life is 20 to 30 hours, allowing a twice-daily dosing regimen. In the absence of enzyme-inducing drugs, approximately 70% to 80% of the drug is excreted unchanged in the urine, with the remainder as metabolites resulting from oxidation and hydrolysis. Enzyme-inducing AEDs alter the pharmacokinetics of topiramate by reducing its plasma levels and increasing its rate of elimination.

In children from 4 to 17 years of age, topiramate exhibits linear pharmacokinetics with a 50% increase in clearance rate compared to adults (102). Topiramate can require up to a 50% dose reduction in patients with renal insufficiency, and a replacement dose can be needed after renal dialysis. Topiramate has demonstrated teratogenicity in animal studies.

Adverse Effects Common CNS side effects associated with topiramate therapy include drowsiness, dizziness, impaired concentration and memory, speech and language difficulties, and confusion. These effects develop during the first weeks of therapy and can decline over time. Acute closed-angle glaucoma caused by topiramate requires immediate evaluation (103). Only rare hepatic or bone marrow effects have been noted thus far; however, an increased incidence of renal stones is troublesome and probably related to the drug's activity as a carbonic anhydrase inhibitor, reducing citrate excretion and increasing urinary pH. Use of additional carbonic anhydrase inhibitors, a ketogenic diet, or a family history of nephrolithiasis can be considered as contraindications for using topiramate.

Topiramate is not devoid of potential interaction properties: It induces CYP3A4 and inhibits CYP2C19, thus significantly increasing plasma phenytoin levels. Topiramate can also decrease the effectiveness of oral contraceptives.

VALPROIC ACID AND ITS DERIVATIVES



Valproate is available as valproic acid (Depakene), divalproex sodium (Depakote), and valproate sodium (Depacon) for IV use. Its AED properties were discovered serendipitously when it was used as a solvent for potential new AEDs undergoing clinical testing. It is effective against both MES test—and pentylenetetrazole-induced seizures in animals and possesses a satisfactory margin of safety. Because the pK_a of valproic acid is 4.7, the drug is completely ionized at physiologic pH; thus, the valproate ion is almost certainly the pharmacologically active species.

Mechanism of Action Although its mechanism of action is not clearly established, valproate appears to increase the inhibitory effect of GABA, possibly by activation of glutamic acid decarboxylase or inhibition of GABA-transaminase (Fig. 17.2). The high drug concentrations required to achieve treatment levels cast doubt on the clinical relevance of this effect. Furthermore, valproate has been shown to decrease the uptake of GABA into cultured astrocytes; this action can contribute to the AED efficacy (104). Valproate is known to produce a blockade of high-frequency repetitive firing by slowing the rate of Na^+ recovery from inactivation, a mechanism consistent with the actions of phenytoin and CBZ. Valproate blocks the low-threshold T-type Ca^{2+} channel. Consequently, the overall therapeutic utility of valproate is likely caused by multiple effects.

Valproate is indicated for initial or adjunct treatment of absence seizures or as an adjunct when absence seizures occur in combination with either tonic-clonic seizures, myoclonic seizures, or both. For patients with unambiguous idiopathic generalized epilepsy, valproate often is the drug of choice, because it controls absence, myoclonic, and generalized tonic-clonic seizures well (105). It also is approved by the FDA for use in complex partial seizures, occurring with or without other seizure types in adults or children 10 years of age or older. In new patients with typical absence seizures, ethosuximide is preferred to valproate due to valproate's risk of producing hepatotoxicity. In a comparative trial, sodium valproate and ethosuximide were equally effective when either drug was given alone or in combination with other AEDs in children with typical absence seizures. In atypical absence seizures (Lennox-Gastaut syndrome), sodium valproate is more effective, whereas in myoclonic seizures, it is less effective than clonazepam. Valproate is approved by the FDA for use in bipolar disorder and as a treatment for migraine headaches.

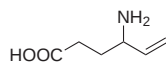
Pharmacokinetics Valproate undergoes rapid and complete absorption, which is only slightly slowed by food. It is 90% protein bound, and its clearance is dose dependent as a result of an increase in the free fraction of the drug at higher doses. It is metabolized almost entirely by the liver, with 30% to 50% of an orally administered dose being eliminated in the urine as its acyl glucuronide conjugate, 40% from mitochondrial β -oxidation, approximately 15% to 20% by ω -oxidation, and less than 3% is excreted unchanged in urine. Its major active metabolite is (E)-2-ene valproate (*trans*-2-ene valproate). Its 4-ene metabolite has been proposed to be a reactive metabolite responsible for the hepatotoxicity of valproate (see Chapter 4). Other metabolites found in the urine include 3-oxo- and 4-hydroxyvalproate. The elimination half-life for valproate ranged from 9 to 16 hours following oral dosing regimens of 250 to 1,000 mg. Patients who are not taking enzyme-inducing AEDs (CBZ, phenytoin, and phenobarbital) will clear valproate more rapidly; therefore, monitoring of AED plasma concentrations should be intensified whenever concurrent AEDs are introduced or withdrawn.

Adverse Effects The most commonly observed side effects for valproate are gastrointestinal (anorexia, nausea, and indigestion). These effects can be minimized by selecting divalproex sodium, which is enterically coated, and by initiating therapy at a low dose. More importantly, however, valproate is associated with the development of fatal hepatotoxicity, especially in children or when coadministered with other AEDs. Frequent monitoring of liver function tests is mandatory for determining the onset of toxicity, particularly during the first 6 months of treatment. Tremors, hematologic dyscrasias, pancreatitis, stupor, depression, behavioral anomalies, and coma have also been observed with valproate therapy.

Valproate, a substrate for hepatic CYP2C19 and CYP2C9, has an extensive pattern of drug interactions. It increases the plasma concentrations of lamotrigine, CBZ, and phenobarbital and the free fraction of phenytoin by either displacing these drugs from plasma proteins or inhibiting their metabolism. Phenytoin, phenobarbital, and CBZ cause decreased plasma concentrations of valproate, whereas felbamate increases valproate levels.

Because of its propensity for causing liver damage, valproate therapy should be avoided in persons with liver disease. It should be used with caution before surgery because it can produce thrombocytopenia and inhibition of platelet aggregation. During pregnancy, it has been associated with an increased risk of fetal epilepsy syndrome and spina bifida.

VIGABATRIN (SABRIL)



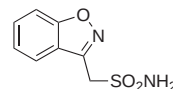
Vigabatrin is γ -vinyl GABA, a synthetic derivative of GABA that was FDA approved in 2009 as monotherapy in the treatment of infantile spasms (West syndrome) in children of 1 month to 2 years of age and as adjunctive treatment of adults with refractory complex partial seizures who are not adequately controlled by other anticonvulsants. Vigabatrin exists as two enantiomers, but only the *S*(+) form is pharmacologically active. The drug is supplied as the racemic mixture (106). Because of the risk of permanent visual field constriction, vigabatrin is available only through a restricted distribution program. The drug should be withdrawn if patients do not show substantial clinical benefit within 3 months.

Mechanism of Action While the exact mechanism for vigabatrin's anticonvulsant effect is unclear, the drug is known to be an irreversible inhibitor of GABA-transaminase, the enzyme responsible for GABA degradation. Although brain GABA concentrations are increased, there is no correlation between the plasma concentration and drug efficacy. Peak antiepileptic response is not seen until 2 to 4 weeks after initiating vigabatrin therapy. Furthermore, drug concentration has no direct relationship to seizure frequency.

Pharmacokinetics Oral bioavailability is up to 80%. The C_{max} is 2.5 hours in infants and 1 hour in children; the half-life is 5.7 hours in infants and 7.5 hours in adults. It does not bind to plasma proteins, is widely distributed ($V_d = 1.1$ L/kg), and does not undergo significant metabolism. Nevertheless, it has been shown to induce CYP2C9. The drug is excreted unchanged by the kidneys, warranting a dose reduction in renal impairment (106). Coadministration of vigabatrin increases CBZ plasma concentration by 24%, while decreasing phenytoin plasma levels by 16% to 20%.

Adverse Effects Vigabatrin carries a black box warning; up to 30% of patients suffer retinopathy manifesting as irreversible bilateral visual field constriction. Periodic vision testing is required for patients taking vigabatrin; it should not be used in patients at high risk for other types of irreversible vision loss. Gradual tapering of the dose is recommended when discontinuing the drug to avoid increased rebound in seizure frequency.

ZONISAMIDE (ZONEGRAN)



Mechanism of Action Zonisamide is a sulfonamide derivative that is indicated as an adjunct for partial seizures in patients older than 16 years whose seizures are not controlled by first-line drugs. In Japan, it is used for myoclonic seizures as well. Apparently, it has more than one mechanism of action—all of which are, as of yet, unidentified. It is known to produce blockade of both sodium and T-type calcium channels (Fig. 17.2) (107). Because it also affects dopaminergic transmission, bipolar or schizoaffective disorder patients can improve.

Pharmacokinetics The absorption for orally administered zonisamide is slow but nearly complete. Its pharmacokinetics are nonlinear, with a half-life of 50 to 70 hours when administered alone or 27 to 46 hours when administered concurrently with enzyme-inducing AEDs. Protein binding is moderate (<50%). An oral dose of zonisamide is completely absorbed, with peak plasma concentration occurring in 2 to 6 hours. Although the presence of food will delay the attainment of its peak plasma concentration, oral bioavailability does not appear to be altered. More than one-third of each oral dose is excreted in the urine in an unchanged form. The routes of metabolism for zonisamide include acetylation to form its *N*-acetyl metabolite, reduction by CYP3A4/CYP2D6, and the formation of an open-ring metabolite, 2-sulfamoylacetyl phenol. These metabolites subsequently are eliminated unconjugated or glucuronidated in the urine with an elimination half-life of 63 hours. Its coadministration with enzyme-inducing AEDs, such as phenytoin, CBZ, or phenobarbital, and with valproate will alter its pharmacokinetics by reducing its half-life and serum concentration. The half-life for zonisamide is decreased to 27 hours in the presence of phenytoin, to 38 hours in the presence of either CBZ or phenobarbital, and to 46 hours with valproate. Other drugs that inhibit or induce CYP3A4 could affect the metabolism of zonisamide. Zonisamide should be used with caution in patients with hepatic or renal disease. It also has shown to be teratogenic in animal studies.

Adverse Effects Zonisamide is contraindicated in patients with a history of allergy to sulfonamides. The most frequent

side effects include somnolence, anorexia, dizziness, agitation, confusion, headache, cognitive impairment, and memory loss. In addition, an incidence of drug-induced psychosis has been noted (108). Reports from both the United States and Europe have indicated that development of renal stones can occur with use of this drug. A family history of nephrolithiasis can be a contraindication, and urinary monitoring for hypercalciuria can be warranted in bedridden patients or those receiving multiple AEDs. Although the incidence of severe rashes attributable to zonisamide is low, sulfonamides are associated with Stevens-Johnson syndrome. Thus, it is recommended to discontinue the drug immediately should a rash occur.

Drugs Effective against Absence Seizures

Drugs that are effective against absence seizures include the five-membered ureides, the oxazolidinediones, and the succinimides (Fig. 17.11); clonazepam and lamotrigine, which have been previously discussed, are also effective. An examination of the structure–activity relationship for the five-membered heterocyclic ureides (Fig. 17.4) reveals that a small substructural difference between ring N (hydantoins), ring O (oxazolidinediones), and ring CH₂ (methylene and succinimides) results in switching from AEDs effective against partial and generalized tonic-clonic seizures to those effective against absence seizures.

Oxazolidinediones

These compounds are some of the oldest AEDs in use, having been introduced into antiseizure therapy between 1946 and 1948. At that time, no effective drugs were available to control absence seizures (*petit mal* disorders). Therefore, the FDA acceptance of trimethadione in 1946 and of paramethadione in 1948 for the control of absence seizures was rapid. At present, trimethadione (Tridione) (Fig. 17.11) is indicated only for control of absence seizures refractory to treatment with other AEDs. It is ineffective against other seizure types. Trimethadione is a prodrug and is metabolized by *N*-demethylation to dimethadione, which is effective in the pentylene-tetrazole test and which acts by decreasing T-type calcium currents. Trimethadione is rapidly absorbed, is not protein bound, and has a half-life of 16 to 24 hours. The half-life of dimethadione, however, is substantially longer (i.e., 6 to 13 days),

and dimethadione accumulates to concentrations greater than the parent drug. Because of its potentially fatal side effects, including aplastic anemia, nephrosis, idiosyncratic rashes, and exfoliative dermatitis, trimethadione is rarely used today. It causes malformations or fetal death in up to 87% of pregnancies. Paramethadione is no longer clinically available in the United States.

Succinimides

Because oxazolidinediones are toxic, an extensive search was undertaken to replace them with less toxic drugs. Substituting the ring O in the oxazolidinediones with a methylene group gave the antiseizure succinimides. The clinically used succinimides include ethosuximide, methsuximide, and phensuximide, which were introduced between 1951 and 1958 (Fig. 17.11) and widely accepted for the treatment of absence seizures.

MECHANISM OF ACTION Succinimides suppress the paroxysmal 3-Hz spike-and-wave activity associated with the lapses of consciousness associated with absence (*petite mal*) seizures, thus reducing the frequency of seizures and raising the threshold to seizures. The proposed mechanism of action involves a decrease in T-type calcium channel activity (Fig. 17.2).

Succinimides are indicated for the monotherapy of absence seizures or with concomitant therapy when additional forms of seizures occur in combination with absence seizures. These drugs are readily absorbed from the gastrointestinal tract and display very low protein binding. The drug interactions for the succinimides are less extensive than with the oxazolidinediones. They can increase plasma phenytoin levels, decrease plasma primidone levels, and either increase or decrease valproate levels, although the changes might not be clinically significant.

Renal or hepatic disease does not appear to enhance their toxicity. Extreme caution is advised, however, because succinimides can cause morphologic changes to kidneys and liver. Periodic monitoring of the blood count, hepatic function, and urinalysis is recommended with the use of succinimides.

ETHOSUXIMIDE (ZARONTIN) Ethosuximide is the drug of choice for treatment of simple absence seizures (109). It is not effective against partial complex or tonic-clonic seizures and can increase the frequency of grand mal attacks. Thus, it must be administered in combination with other AEDs when treating persons with mixed seizure types. Ethosuximide is a substrate for both CYP3A4 and CYP2E1. The major metabolite for ethosuximide is 3-(1-hydroxyethyl) succinimide, which is inactive and excreted unconjugated into the urine. Several additional metabolites have been characterized recently (110). Approximately 20% of an oral dose is excreted unchanged.

Although ethosuximide is thought to be the least toxic of the succinimides, it can cause gastrointestinal disturbances and dose-related CNS effects, such as drowsiness, dizziness, ataxia, sleep disturbances, and depression.

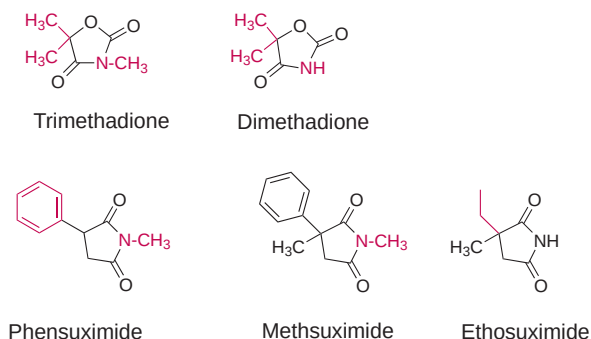


FIGURE 17.11 Drugs effective against absence seizures.

Idiosyncratic hypersensitivity reactions include severe rashes, leukopenia, agranulocytosis (some fatal), systemic lupus erythematosus, and parkinsonian-like symptoms. In addition to being less toxic than trimethadione, ethosuximide offers a wider range of protection against different kinds of absence seizures.

METHSUXIMIDE Although methsuximide is less commonly used, it can be indicated for the control of absence seizures refractory to other drugs. Although it does not precipitate tonic-clonic convulsions, it often is combined with phenytoin or phenobarbital when absence seizures coexist with tonic-clonic symptoms. Much of the efficacy of methsuximide is attributed to its desmethyl metabolite. The half-life of methsuximide is between 2.6 and 4.0 hours, but the half-life for *N*-desmethysuximide is 25 hours, causing it to accumulate substantially. Concentrations of greater than 40 mcg/mL can be associated with toxicity. Methsuximide is considered to be more toxic than ethosuximide.

PHENSUXIMIDE Phensuximide occasionally is used for the treatment of absence seizures refractory to other drugs, although it is considered to be less effective than ethosuximide. It is excreted in both urine and bile, and it can cause harmless pink to red discoloration of the urine. It should be used with caution in patients with acute intermittent porphyria.

Drugs Effective against Myoclonic Seizures

Clonazepam and valproate commonly are used to control myoclonic seizures. Studies suggest that lamotrigine and topiramate can be effective as well, although neither is approved by the FDA for this indication.

Drugs for Status Epilepticus

Diazepam, administered IV or IM, is a drug of choice for rapid control of status epilepticus. Lorazepam, although not approved by the FDA for the purpose, is preferred by some clinicians for its longer duration of action (40), and midazolam is preferred for its more rapid onset of action (58). Because of its high lipid solubility, IV diazepam enters the CNS rapidly. The initially high brain concentration is

quickly reduced, however, due to its redistribution, increasing the chance of recurring status epilepticus. Concomitant IV injection of diazepam and phenobarbital or fosphenytoin has been suggested to overcome this difficulty.

ANTIEPILEPTIC DRUGS IN PHASE III CLINICAL TRIALS

Antiepileptic drugs in phase III clinical trials are shown in Figure 17.12. Brivaracetam is the 4-*n*-propyl analog of levetiracetam (Keppra), which is an SV2A ligand, and thereby can regulate neurotransmitter release. Carisbamate (RWJ 333369) is a monocarbamate derivative of felbamate, which is an antagonist at KA receptors. Clobazam is a benzodiazepine that has long been used in Europe and has U.S. orphan drug status for treatment of Lennox-Gastaut syndrome.

Eslicarbazepine (Stedesa) is a prodrug converted to Slicarbazepine, an active metabolite of oxcarbazepine by hepatic esterases. Like carbamazepine, it is a sodium channel antagonist.

Perampanel (E2007) is the first-in-class highly selective noncompetitive AMPA-type glutamate receptor antagonist.

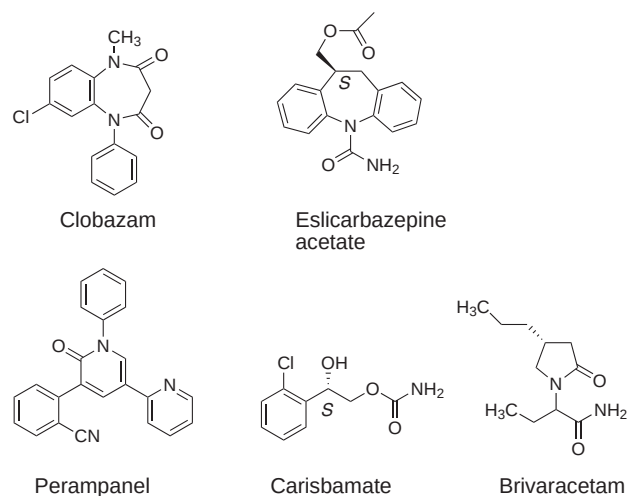


FIGURE 17.12 Antiepileptic drugs in phase III clinical trials.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Michael Gonyeau, PharmD

Though it may at first appear that the patient experienced a seizure caused by a subtherapeutic phenytoin level, the corrected phenytoin level reveals that the patient experienced the symptoms of nausea and vomiting, ataxia (problems walking), and the seizure that precipitated the hospitalization from phenytoin toxicity. Because of the nature of phenytoin kinetics, namely 90% protein binding, corrected phenytoin levels should be ascertained

in any patient with hypoalbuminemia or end-stage renal disease because the free fraction of the drug will be increased in such patients and the potential for toxicity exists even at therapeutic or subtherapeutic observed levels of total phenytoin. The admitting physician only looked at the observed level, which is subtherapeutic (therapeutic range, 10–20 mcg/mL), and increased the patient's dose because she had a recent seizure. However, because of the patient's corrected phenytoin level based on her medical history of end-stage renal disease and her hypoalbuminemia level (36 mcg/mL), increasing her phenytoin dose at this

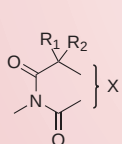
SCENARIO: OUTCOME AND ANALYSIS (Continued)

time will compound the current problem by increasing phenytoin levels even higher in a non-dose-dependent fashion because of its nonlinear kinetics. The end result could be more seizures, increased confusion, coma, or death. The pharmacist in this situation quickly identified the toxic phenytoin level and discussed with the physician, and the phenytoin was held and the patient's symptoms resolved over the next several days with close observation and monitoring in the hospital setting.

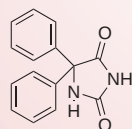
Chemical Analysis

S. William Zito and Victoria Roche

Phenytoin is a hydantoin AED. Structurally it is a five-member ring containing two nitrogens in a ureide configuration. Other AEDs containing the ureide moiety are barbiturates ($x = -\text{CONH}-$), oxazolidinediones ($x = -\text{O}-$) and the succinamides ($x = -\text{CH}_2-$).



Ureide Structure



Phenytoin

Seizures result from abnormal synchronous neuronal discharge in the CNS involving ion channels and an excitatory/

inhibitory synaptic imbalance. Various structural types of AEDs show selective efficacy versus different seizure types. Phenytoin is effective in the treatment of partial and generalized tonic-clonic seizures. Its mechanism of action is to stabilize the inactive state of the voltage-gated Na^+ channel. When phenytoin binds to the sodium channel it produces a use-dependent blockade of repetitive firing, thus inhibiting the spread of seizure activity.

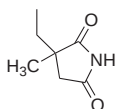
Phenytoin is metabolized predominately by CYP2C9. Approximately 60 to 75% of an oral dose is excreted as 5-(4'-hydroxyphenyl)-5-phenylhydantoin (HPPH) glucuronide or sulfate (HPPH). Minor amounts are excreted unchanged in the urine, and some undergo enterohepatic circulation. Phenytoin is notorious for displaying nonlinear pharmacokinetics because the route of metabolism is a saturable process, making the therapeutic index very narrow. The steady-state plasma concentration (C_{ss}) can double or triple as a result of a 10% or more increase in dosage, possibly resulting in toxicity. Phenytoin also induces CYP3A4 and uridine diphosphoglucuronosyltransferases (increases glucuronidation); therefore, co-administered drugs metabolized by these isoforms will be affected. In addition, carbamazepine, felbamate, cimetidine, warfarin, chloramphenicol, isoniazid and disulfiram can increase plasma phenytoin levels. The key problem in this case is the pharmacokinetic parameter of plasma binding. Phenytoin is highly protein bound (90%), and because this patient is in the end stage of renal disease and is hypoalbuminemic, the patient was actually suffering from phenytoin toxicity and not subtherapeutic dosing.

CASE STUDY

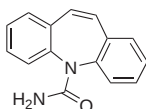
S. William Zito and Victoria Roche

BJ a 50-year-old woman who reports increased tonic clonic seizures to her neurologist during their most recent visit. She is currently taking oxcarbazepine and gabapentin, and previous trials with phenytoin, phenobarbital, lamotrigine and valproic acid

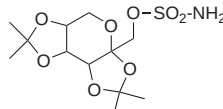
as either monotherapy or adjunct therapy were unsuccessful in seizure control. The physician wants to try adding one of the following three antiepileptic drugs.



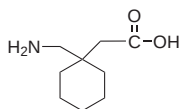
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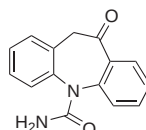
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3



Gabapentin



Oxcarbazepine

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 18

Antidepressants

DAVID A. WILLIAMS

Drugs Covered in This Chapter*

SELECTIVE NOREPINEPHRINE REUPTAKE INHIBITORS (SNRIs)

- Amoxapine
- Desipramine
- Maprotiline
- Nisoxetine
- Nortriptyline
- Protriptyline
- *Reboxetine*
- *R-Atomoxetine*

SELECTIVE SEROTONIN REUPTAKE INHIBITORS (SSRIs)

- Citalopram
- Escitalopram (*Scitalopram*)
- Fluoxetine
- Fluvoxamine
- Paroxetine
- Sertraline

NOREPINEPHRINE AND SEROTONIN REUPTAKE INHIBITORS (NSRIs)

- Amitriptyline
- Clomipramine
- Desvenlafaxine
(*Odesmethylvenlafaxine*)
- Doxepin
- *Dosulepin*
- Duloxetine
- Imipramine
- *Lofepramine*
- Milnacipran
- Trimipramine
- Venlafaxine

DOPAMINE AND NOREPINEPHRINE REUPTAKE INHIBITORS (DNRIs)

- Bupropion

SEROTONIN ANTAGONIST/REUPTAKE INHIBITORS (SARIs)

- Aripiprazole
- Trazodone
- *Vilazodone*

NORADRENERGIC AND SPECIFIC SEROTONERGIC ANTIDEPRESSANTS (NASSAs)

- Mirtazapine

MONOAMINE OXIDASE INHIBITORS (MAOIs)

- *Moclobemide*
- Phenelzine
- Tranylcypromine

MOOD STABILIZERS

- Lithium carbonate

MELATONERGIC AGONIST/SEROTONIN ANTAGONIST

- *Agomelatine*

Abbreviations

ACh, acetylcholine

ACTH, adrenocorticotrophic hormone

ADHD, attention-deficit hyperactivity disorder

ATP, adenosine triphosphate

AUC, area under the curve

BDNF, brain-derived neurotrophic factor

CNS, central nervous system

CRF, corticotropin-releasing factor

CYP, cytochrome P450

DA, dopamine

DAT, dopamine reuptake transporter

DCPP, 2,3-dichlorophenylpiperazine

DSM-IV, *Diagnostic and Statistical Manual of Mental Disorders*

ECT, electroconvulsive therapy

FDA, U.S. Food and Drug Administration

GABA, γ -aminobutyric acid

GI, gastrointestinal

GSK-3, glycogen synthase kinase-3

HPA, hypothalamus-pituitary-adrenal axis

*Drugs listed include those available in the United States. Drugs available outside the United States are shown in italics.

Abbreviations (Continued)

5-HT , serotonin	nAChRs , nicotinic receptors	RIMAs , reversible inhibitors of MAO-A
IC₅₀ , half maximal inhibitory concentration	NaSSAs , noradrenergic and specific serotonergic antidepressants	SARIs , serotonin antagonist/reuptake inhibitors
IP , intraperitoneal	NE , norepinephrine	SERT , serotonin reuptake transporter
MAO , monoamine oxidase	NET , norepinephrine reuptake transporter	SNRIs , selective norepinephrine reuptake inhibitors
MAOIs , monoamine oxidase inhibitors	NK₁ , neurokinin-1	SSRIs , selective serotonin reuptake inhibitors
m-CPP , <i>m</i> -chlorophenylpiperazine	NMDA , <i>N</i> -methyl-D-aspartate	TCAs , tricyclic antidepressants
MDD , major depression disorder	NSRIs , norepinephrine and serotonin reuptake inhibitors	TM , transmembrane domain
MIF-1 , melanocyte-stimulating hormone release-inhibiting factor-1	ODV , <i>O</i> -desmethylvenlafaxine	UV , ultraviolet
MT , melatonin receptor	P-gp , P-glycoprotein	
nM , nanomoles/L		

SCENARIO

Nancy Ordonez, PharmD

GP is a 45-year-old man who presents to the internal medicine clinic for a follow-up visit for his diabetes and lipids. His medical history is significant for uncontrolled hypertension, type 2 diabetes mellitus, coronary artery disease, and peripheral neuropathy. Six months before this visit he had complained of decreased energy, increased appetite, and loss of interest in his hobbies of golf and running. He was diagnosed with depression at that visit and was started on venlafaxine 37.5 mg twice daily, which was gradually increased to 150 mg twice daily. GP still is not seeing any improvement in his depressive symptoms. His current medications are simvastatin 40 mg, metoprolol 20 mg, amitriptyline

25 mg at bedtime, glyburide 20 mg, and lisinopril 10 mg once daily. He has gained 20 lb in 6 months. His blood pressure readings from today are 160/90 and 165/85, and from his blood pressure log his readings have ranged from 165 to 185 systolic and 75 to 90 diastolic. In addition, GP states that his peripheral neuropathy has worsened significantly even though he is maintaining good glucose control, and he complains that amitriptyline makes him quite drowsy in the morning.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

*Canst thou not minister to a mind diseas'd
Pluck from the memory a rooted sorrow,
Raze out the written troubles of the brain
And with some sweet oblivious antidote
Cleanse the stuff'd bosom of that perilous stuff
Which weighs upon the heart?*

—William Shakespeare

INTRODUCTION

Depression is an ancient and prevalent mental condition that has been referenced throughout history in song, poetry, and literature. In a depressed state, one feels hopeless and experiences an overwhelming sense of despair. Depression immobilizes a person, afflicting men and women, rich and poor, and young and old alike. Depression makes one feel exhausted, worthless, helpless, and hopeless. It is an illness that involves the body, mood, and thoughts and affects the way one eats and sleeps, the way one feels about oneself, and the way one thinks about things. It often interferes with normal functioning, causing pain and suffering not only to those who suffer from depression but also to those around them. Serious depression can destroy family life and the life of the person who is ill. A depressive disorder is not the

same as a passing blue mood. Without treatment, symptoms can last for weeks, months, or years. Appropriate treatment, however, can help most people who suffer from depression. Negative thinking fades as treatment begins to take effect. Unfortunately, many people do not recognize that depression is a treatable illness. Much of this suffering is unnecessary because *depression is one of the most treatable mental illnesses*.

“Depression is the flaw in love. The meaninglessness of every enterprise and every emotion, the meaninglessness of life itself, becomes self-evident. The only feeling left in this loveless state is insignificance.”

“My depression had grown on me as that vine had conquered the oak; it had been a sucking thing that had wrapped itself around me, ugly and more alive than I. It had a life of its own that bit by bit asphyxiated all of my life out of me. My moods belonged to the depression as surely as the leaves on that oak tree's high branches belonged to the vine.”

“Drug therapy hacks through the vine. You can feel it happening, how the medication seems to be poisoning the parasite so that bit by bit it withers away. You feel the weight going, feel the way that the branches can recover much of their natural bent. But even with the vine gone, you may still have a few leaves and shallow roots and the rebuilding of your self cannot be achieved with any drugs that now exist.

Rebuilding of the self in and after depression requires love, insight and most of all, time.”

—Andrew Solomon, *The Noonday Demon: An Atlas of Depression*

Depression affects approximately 10% to 12% of the Western population (United States, ~10%; Europe, 9% to 12%; Australia, ~18%), with women outnumbering men 2:1. One in 4 women and 1 in 10 men can expect to develop depression during their lifetime. Thus, depression ranks among the top five diseases in these countries. In the United States alone, approximately 19 million adults will be afflicted yearly with depression, and at least 50% of those with major depression will suffer one or more repeated episodes of depression during their adult lifetime (1,2). Depression affects at least 1 in 50 children under the age of 12 years and 1 in 20 teenagers, mostly girls. The increase in the rate of depression among adolescent girls is related more to physical changes that occur during puberty, suggestive of hormonal changes. Premenstrual syndrome and postpartum depression are additional conditions involving depression that specifically affect women and are suggestive of hormonal involvement in the pathogenesis of depression. About half of all cases of depression go unrecognized and untreated, and approximately 10% to 15% of those with depression will take their own lives yearly.

Depression in the elderly (17% to 35%) often is dismissed as a normal part of aging and may go undiagnosed and untreated, causing needless suffering for the family and for the individual who could otherwise live a fruitful life. Often, the symptoms described usually are physical, and the older person often is reluctant to discuss feelings of hopelessness, sadness, loss of interest in normally pleasurable activities, or extremely prolonged grief after a loss. Some symptoms may be the result of adverse (side) effects

of medication that the elderly person is taking for other physical problems, or they may be caused by a concurrent illness. Improved recognition and treatment of depression will make life more enjoyable and fulfilling for the depressed elderly person, the family, and the caretakers.

The economic cost for depressive illnesses in the United States is estimated to be \$53 billion per year (2005), but the cost in human suffering cannot be estimated. In 2009, antidepressant drugs (primarily norepinephrine and serotonin reuptake inhibitors [NSRIs] and selective serotonin reuptake inhibitors [SSRIs]) ranked in the top 50 drugs dispensed, third in total prescriptions written, and third in total dollar prescription sales, at approximately \$12.5 billion (~5% of total prescription drug sales).

TYPES OF DEPRESSIVE DISORDERS

Why are some people more susceptible to depression than others? Hippocrates, the father of medicine, theorized that we are born into one of four primary temperament styles and that each style has its own unique outlook on life: Choleric (aggressive), Sanguine (emotional), Phlegmatic (passive), and Melancholy (analytical). Of these four styles, the introverted Melancholy is the most perfection driven and depression prone. The analytical Melancholy influence gives one tremendous attention to detail, but it also can create stress, anxiety, and depression. In fact, the term “melancholy” has become synonymous with depression. People with the Melancholy temperament style are by their very nature sensitive, judgmental, and critical. People with this temperament style become depressed primarily because they fail to reach their own incredibly high standards. This depression often leads to suicide, violence against others, or both.

CLINICAL SIGNIFICANCE



The antidepressant class of drugs are well known and widely used today not only for the treatment of mood disorders, such as depression, and bipolar disorders, but also for the treatment of neuropathic pain, smoking cessation, and obsessive-compulsive disorders, among others. The application of medicinal chemistry in the development of newer antidepressants, such as selective serotonin uptake inhibitors (SSRIs), in the past 20 years has significantly affected the treatment of depression. The SSRIs have become first-line therapy in the treatment of major depressive disorder, and they are used as adjunct treatment for patients with other mood disorders. This new generation of antidepressants has more favorable side effect and pharmacokinetic profiles than the older generations of antidepressants, such as tricyclic antidepressants and monoamine oxidase inhibitors. This has resulted in improved patient compliance

and therapeutic outcomes for patients with major depressive disorder. Modifications of the structure–activity relationships of the antidepressant drugs have produced medications with multiple indications. For example, both duloxetine and bupropion are indicated for major depressive disorder, but duloxetine also is used for diabetic neuropathic pain and bupropion for smoking cessation. With a growing number of antidepressants on the market, it is more important than ever for the clinician to be knowledgeable about drugs’ structure–activity relationships, adverse effects, drug interactions, and pharmacokinetic properties to properly select the optimal drug regimen for the patient.

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Major Depression Disorder

Major depression disorder (MDD; also called unipolar depression) is the most serious type of depression; it is manifested through a combination of symptoms that interfere with the ability to work, study, sleep, eat, and enjoy once-pleasurable activities. MDD may reoccur several times during a lifetime (3,4). Many people with major depression cannot continue to function normally. Major depression seems to run in families, suggesting that depressive illnesses can be inherited. Early signs (prodromal symptoms) of major depression include changes in brain function among those individuals having low self-esteem, who consistently view themselves and the world with pessimism or who are readily overwhelmed by stress. The treatments for major depression are medication, psychotherapy, and, in extreme cases, electroconvulsive therapy.

Dysthymia

This is a mild, chronic depression that lasts for 2 years (1 year for children and adolescents) or longer and is characterized by chronic symptoms that do not disable but that keep one from functioning well or from feeling good about himself or herself (4). Many of those with dysthymia also experience major depressive episodes at some point in their lives. Most people may not realize that they are depressed and continue to function at work or school, but often with the feeling that they are “just going through the motions.” Antidepressants or psychotherapy can help.

Bipolar Disorder (Manic-Depressive Illness)

Bipolar disorders can be divided into bipolar I (episodes of severe mood swings from mania to depression), bipolar II (milder episodes of hypomania that alternate with depression), cyclothymia (milder episodes of bipolar II; low-grade bipolar II), or rapid-cycling bipolar disorder (more than four episodes within a year affecting women more than men) (4). Clinical studies over the years have provided evidence that monoamine signaling and hypothalamic-pituitary-adrenal axis disruption are integral to the pathophysiology of bipolar disorder (5). Bipolar disorders also seem to run in families and affect men and women equally. Not nearly as prevalent as the other forms of depressive disorders, bipolar disorders are characterized by cyclical periods of depression (lows) with periods of abnormal behavior (highs) known as hypomania or mania. Sometimes, the mood switches are dramatic and rapid, but most often, they are gradual. When in the depressed cycle, an individual can exhibit the symptoms of a depressive disorder. When in the hypomanic/manic cycle, the individual may be overactive and overtalkative and have a great deal of energy. The hypomanic/manic cycle often affects thinking, judgment, and social behavior in ways that cause serious problems and embarrassment. For example, the individual in the hypomanic/manic cycle may feel elated and full of grand

schemes that might range from unwise business decisions to romantic sprees. Lithium, carbamazepine, topiramate, and valproic acid are effective mood-stabilizing treatments for bipolar disorders. In addition, the atypical antipsychotics, olanzapine and aripiprazole (see Chapter 14), are used as mood stabilizers for acute bipolar mania or in conjunction with antidepressants (e.g., Symbyax, a combination of olanzapine and fluoxetine).

Other Types of Depressive Disorders

Other less common types of depression include seasonal affective disorder, a popular name that describes a type of depression that happens during particular seasons of the year (seasonal pattern) but that it is not a *Diagnostic and Statistical Manual of Mental Disorders* (DSM-IV) (4) diagnosis. This disorder involves symptoms of depression that occur during the fall and winter seasons, when the days are shorter and there is less exposure to natural sunlight. When the spring and summer seasons begin and there is greater exposure to longer hours of daylight, the symptoms of depression disappear. Adjustment disorder with depressed mood is a type of depression that results when a person has something bad happen that depresses them (e.g., loss of one's job can cause this type of depression). It generally fades as time passes and the person gets over whatever it was that happened (4). Additional factors involved in its onset include stresses at home, work, or school, and symptoms may persist for as long as 6 months.

BIOLOGIC BASIS OF DEPRESSION

Existing treatments for MDD usually take weeks to months to achieve their antidepressant effects, and about 25% of these patients do not have adequate improvement even after months of treatment to achieve response and remission, which commonly results in morbidity and disruption in personal, professional, family, and social life. In addition, increased risk of suicide attempts is a major public health concern during the first month of standard antidepressant therapy. Thus, new antidepressant treatment strategies presenting a rapid improvement of depressive symptoms—within hours or even a few days—and whose effects are sustained would have an enormous impact on public health. Thus, improved therapeutics that can exert their antidepressant effects within hours or a few days of their administration are urgently needed.

Current theories regarding the causes of depression support the role of the neurotransmitters serotonin (5-HT) and norepinephrine (NE) in depression and their interrelationships with each other and with dopamine (DA) (Fig. 18.1). Although the precise nature of the depression is not fully understood at the level of the chemistry in the brain, several theories have been proposed to explain the role of NE and 5-HT in the causes of depression (6).

However, evidence for cholinergic mechanisms in depression and adrenergic mechanisms in mania

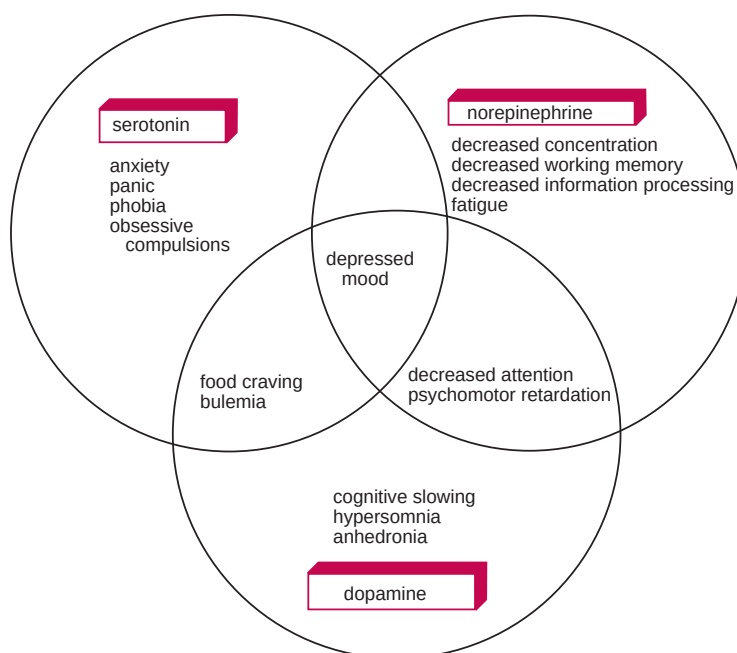


FIGURE 18.1 Neurotransmitter deficiency syndromes and their interactions.

has been building over the last several decades, especially the clinical relationships between tobacco use and depression, which suggest important effects of nicotine in MDD. About 120 years ago, Willoughby (*Lancet*, Pilocarpine in Threatening Mania, *Lancet* 1889;133:1030–1031.) first reported the use of pilocarpine (a cholinomimetic) to treat acute mania. During the 1970s, Janowsky and Davis (6) and others hypothesized that a cholinergic imbalance (i.e., hypercholinergic tone) was a primary factor in depressive illnesses. Since that time, a number of key findings from both animal and human studies have supported this hypothesis, with focus specifically on central nicotinic cholinergic rather than muscarinic pathways. Further support for this hypothesis is the loss of high-affinity nicotinic cholinergic receptors in Alzheimer patients. Most clinically prescribed antidepressants target NE and 5-HT neurotransmitter systems; however, many of the drugs that target these systems, such as SSRIs or selective norepinephrine reuptake inhibitors (SNRIs) and tricyclic antidepressants (TCAs), also act as potent noncompetitive antagonists of nicotinic acetylcholine receptors (nAChRs) at clinically effective doses for treatment of major depression (6). The rapid monoamine effects of the antidepressants and their delayed therapeutic onset of action have led to reconsideration of alternative hypotheses regarding the mechanism of action of antidepressants.

Animal studies with the *N*-methyl-D-aspartate (NMDA) antagonist ketamine (Chapters 12 and 16) have consistently shown antidepressant effects within a few hours of its administration, and thus, NMDA produces its antidepressant response in a much shorter period of time than existing antidepressant medications. Another alternative approach to improved treatment of depression

focuses on the dual-acting melatonin receptor agonists/serotonin receptor antagonists. Melatonin is a neurohormone secreted by the pineal gland in the brain that is an important regulator of the body's circadian (sleep/wake) rhythms, reproduction, and other hormones, which are thrown off balance in depression. Understanding the molecular basis of the NMDA and the dual-acting melatonin receptor agonists/serotonin receptor antagonists can lead to the development of improved antidepressant pharmacotherapy rather than simply furthering our knowledge of current standard antidepressants.

Monoamine Hypothesis

The monoamine hypothesis proposes that depression results from a deficiency in 5-HT and/or NE (7, 9–13) and that antidepressant therapy aims to correct these deficiencies. The role of DA in depression remains unclear. Schildkraut et al. (7) postulated that depression arises as a consequence of a deficiency of NE and that the effects caused by catecholamine depletion can be reversed by TCAs. Depression in animals following reserpine administration can be reversed by TCAs, suggesting that the stimulating effects for desipramine (a secondary amine TCA) and monoamine oxidase (MAO) inhibitors (MAOIs) were NE based.

During the late 1960s, Kielholz, a Swiss psychiatrist, argued that different antidepressants did quite different things and that it was important to select the right antidepressant for the right patient. He differentiated the TCAs based on whether they possessed the ability to sedate (e.g., trimipramine), to stimulate (e.g., desipramine or nortriptyline), or to improve mood (e.g., clomipramine).

By the end of the decade, the broad consensus was that secondary amine TCAs (like desipramine or nortriptyline) inhibited the reuptake of NE into noradrenergic neurons and that blocking 5-HT reuptake by tertiary amine TCAs (such as clomipramine) gave an explanation for the mood elevation of some antidepressants.

The catecholamine hypothesis was then modified to include 5-HT in the etiology of depression (9,10). It should be noted, however, that not all inhibitors of monoamine reuptake are antidepressants, because cocaine, a potent inhibitor of NE and DA reuptake, is not an antidepressant but, rather, an addictive stimulant. Subsequent studies with inhibitors of monoamine biosynthesis appear to confirm Kielholz's opinion and Schildkraut's modified theory that clinical depression is the result of a deficiency in both 5-HT and NE and that the antidepressive mechanism of action most likely affects levels of both.

Inhibition of 5-HT biosynthesis reverses the therapeutic effects of treatment with antidepressants that have predominantly 5-HT reuptake inhibitory activity (e.g., fluoxetine) but less so with those that have predominantly NE reuptake inhibitory activity (e.g., desipramine) (11,12). Also, inhibition of 5-HT biosynthesis produced a relapse in depressed patients who had their depression successfully treated with imipramine or tranylcypromine, whereas inhibitors of DA or NE synthesis had no effect on these depressed patients (13). A depletion of catecholamines reverses the therapeutic effects of desipramine more than that of fluoxetine or sertraline. These studies appear to confirm that the antidepressive action of antidepressants is, indeed, a function of their monoamine activity.

Kielholz further reasoned that because NE reuptake inhibitors (activating antidepressants) were likely to trigger suicide, the greatest hazard of an antidepressant, and 5-HT reuptake inhibitors would be less likely to lead to suicide that it would be worth developing drugs that selectively inhibited the reuptake of 5-HT, producing agents more useful for the treatment of depression.

Changes in NE and 5-HT levels do not affect mood in everyone. Some evidence also suggests that, for a subset of patients, DA plays a role in depression. Dopaminergic substances have been used as antidepressants when other measures have failed. The DA hypothesis of schizophrenia and the emphasis on other neurotransmitters, most notably NE and 5-HT, in the pathogenesis of depression have focused attention away from DA and its role in affective disorders. Recent clinical evidence suggests the involvement of DA in several subtypes of depression, psychomotor retardation, and diminished motivation and in seasonal mood disorder (11). The biochemical evidence in patients with depression indicates diminished DA turnover. In addition, a considerable amount of pharmacologic evidence exists regarding the efficacy of antidepressants with dopaminergic effects in the treatment of depression. However, the role of DA in depression must be understood in the context of existing theories involving other neurotransmitters that may act independently

and interact with DA and other neurochemicals to contribute to depression (12).

Thus, the clinical efficacy/toxicity produced by these antidepressants is due to a combination of inhibitory actions on different targets including reuptake transporters for the neurotransmitters 5-HT, NE, DA, muscarinic acetylcholine receptors, nAChRs, histamine H_1 receptors, and α -adrenergic receptors (10).

Limitations of the Monoamine Hypothesis

Although the monoamine deficiency hypothesis of depression has been commonly used to explain the mechanism of actions of the antidepressant drugs, a growing body of evidence has been accumulating over the past two decades supporting alternative nonmonoamine mechanisms of action: the cholinergic hypothesis of depression (6,8), NMDA antagonists, dual-acting melatonin receptor agonists/serotonin receptor antagonists (14), and neuropeptides.

Almost 40 years ago, the cholinergic theory of depression and mania hypothesized a balance between cholinergic and adrenergic systems, suggesting that hyperactivity of the cholinergic system over the adrenergic system would lead to symptoms of depression, mediated through excessive neuronal nicotinic receptor activation. Thus the therapeutic actions of many antidepressants may be, in part, mediated through inhibition of these nAChRs (6). On the other hand, hypochocholinergic activity would lead to the symptoms of mania (6). Consistent with this hypothesis, physostigmine (a cholinesterase inhibitor-hypercholinergic response; Chapter 9), when administered to normal subjects, produces symptoms of depression: anxiety, irritability, aggressiveness, and hostility (6). When physostigmine was administered to patients with MDD, these depressive symptoms were more pronounced and longer lasting in these patients (6); physostigmine also induced depression in patients with acute mania. The hypercholinergic effects of physostigmine are mediated primarily through acetylcholine activation of neuronal nAChRs, not muscarinic receptors. Over the past 20 years, various groups have reported on the nAChR inhibitory actions of classic TCAs, including imipramine, nortriptyline, amitriptyline, and desipramine, and the serotonin reuptake inhibitors, including fluoxetine, sertraline, paroxetine, and citalopram (8). Mecamylamine, a potent ganglionic blocker as a hypotensive agent that is devoid of monoamine reuptake, is a potent nAChR receptor antagonist that is undergoing clinical trials as an antidepressant to reduce symptoms of major depression and bipolar disorders (to be discussed later). nAChRs modulate not only the release of monoamines, but also γ -aminobutyric acid (GABA), glutamate, and various neuropeptides. Perhaps one of the most interesting antidepressants to have nAChR inhibitory activity is the atypical antidepressant, bupropion. Bupropion is unique because it is metabolized to hydroxybupropion, which is an SNRI and a reuptake inhibitor of DA, with

little or no direct action on serotonergic neurotransmission, and is used in nicotine cessation.

In addition, transdermal nicotine administration (patch) has been shown to substantially improve the depressive symptoms of patients with MDD, an effect believed to be due to nicotine's antagonist effects on nAChRs. These effects have been attributed primarily to inhibition of a high-affinity nAChR subtype in the brain, the $\alpha_4\beta_2$ subunit, representing the predominant mammalian brain nAChR subtype. Thus, cigarette smoking may exert antidepressant effects in patients with MDD who were refractory to SSRI treatment (7), supporting the view that depressed smokers self-medicate by smoking.

Thus, there is an increasing amount of evidence supporting a possible role of neuronal nAChRs in depression as well as in the clinical effectiveness of antidepressants and that the therapeutic action of many antidepressants may be mediated, at least partially, through inhibition of excessive neuronal nAChR activity.

Major depression is frequently accompanied by alterations in circadian (light/dark) rhythms of behavior, sleep, reproduction, and the secretion of hydrocortisone and other hormones (14). These circadian disturbances in depression are associated with an abnormal pineal release of melatonin, a key synchronizer of biologic circadian rhythms that is tightly coupled to sleep/wake and seasonal cycles. Melatonin production appears to be lower in depression even though the administration of melatonin itself is ineffective in the treatment of MDD. Aging is also associated with a weakened response of the circadian clock to environmental stimuli. These observations support the resynchronization of sleep/wake rhythms by melatonergic mechanisms as a new strategy for improving the pharmacotherapy of MDD.

Receptor Sensitivity Hypothesis

The receptor sensitivity hypothesis proposes that it is not simply the level of NE or 5-HT in the synapse that matters but, rather, the sensitivity of the postsynaptic receptors to these neurotransmitters. Those with depression, it is speculated, possess postsynaptic receptors that have grown hypersensitive to NE and 5-HT because of their depletion in the synaptic cleft. Thus, a low level of NE and 5-HT at their respective receptors can lead to changes in the receptors themselves, even if there are no clinical signs of depression. Increased receptor sensitivity (hypersensitivity) and an increase in receptor numbers on the neuronal cell membrane are events that may correlate with the start of depression. This theory is an important step toward understanding the long delay between administration of antidepressant drugs and their clinical response. According to this hypothesis, relief from the symptoms of depression following chronic administration of antidepressants comes from a normalization of receptor sensitivity (reducing receptor hypersensitivity) by increasing the concentration of NE and 5-HT in the synaptic cleft. Therefore, the use of reuptake inhibitors and the MAOIs as antidepressants increases the concentration of NE

and/or 5-HT in the synaptic cleft and, over time, causes the postsynaptic neuron to compensate by decreasing receptor sensitivity (desensitization) and the number of receptor sites (decrease in the expression of NE and 5-HT receptors: downregulation). Approximately one-third of depressed patients, however, fail to respond to antidepressant therapy. For some unknown reason, the increased concentrations of NE and/or 5-HT fail to desensitize the postsynaptic receptors.

Permissive Hypothesis

The permissive hypothesis (10) emphasizes the importance of the balance between 5-HT and NE in regulating mood, not the absolute levels of these neurotransmitters or their receptors. If 5-HT levels are too low, the balanced control of the NE system is lost, permitting abnormal levels of NE to cause mania, as seen in bipolar disorders. If the NE levels fall, the balanced control of the 5-HT system is lost, allowing abnormal levels of 5-HT to cause the person to exhibit the symptoms of depression.

Hormonal Hypothesis

The hormonal hypothesis suggests that changes in the hypothalamus-pituitary-adrenal axis (HPA) can influence the levels of 5-HT, NE, and acetylcholine (ACh) released by nerve cells in the brain and, subsequently, their function (15). In the event of stress, ACh stimulates the hypothalamus to produce a hormone locally in the brain called corticotrophin-releasing factor (CRF), which in turn stimulates the pituitary gland to secrete adrenocorticotrophic hormone (ACTH) into the blood, where it stimulates the adrenal glands to release hydrocortisone (cortisol), which prepares the body for dealing with stress. Stress also directly stimulates the adrenal gland to secrete epinephrine and NE. Hydrocortisone can cause depression, especially when released in higher-than-usual amounts. The release of hydrocortisone may push the individual over the edge into depression or contribute to the component of anxiety, which so often accompanies depressive illnesses. Approximately 50% of those with MDD have elevated hydrocortisone levels as a result of hyper-CRF activity.

Moreover, one of the most replicated findings in biologic psychiatry is that large numbers of unmedicated depressed patients exhibit HPA hyperactivity. The available evidence suggests that nAChRs play important roles in mediating stress-related and possibly depression-inducing neuroendocrine effects of ACh. ACh (hypercholinergic activity), in response to stress, can stimulate the HPA through activation of nAChRs. Thus, antidepressants may reduce symptoms of depression, in part, through blockade of nAChRs involved with stress-induced activation of the HPA (8).

Living organisms operate in a state of imbalance, and the neural (autonomic) and endocrine systems have evolved to modify the rates of biochemical pathways to maintain homeostasis. One of the hallmarks of these regulatory systems is the short-lived nature of the nerve signals produced. The half-life of neurotransmitters is measured in seconds, whereas those of the circulating hormones

may be minutes or hours. A rationale for the short-lived nature of the neurotransmitters is to permit these signaling pathways to quickly reset themselves to meet the next challenge. Readjustments (i.e., plasticity) in these systems include uncoupling of receptor responses from signaling events, degradation of receptors, and up- and downregulation of signaling molecules that affect the primary signaling pathway. These hypotheses are not mutually exclusive, and in each, it is assumed that the more extreme the event, the more severe the clinical outcome.

GENERAL APPROACHES TO TREATMENT OF DEPRESSION

Before 1950, there were no antidepressants—at least not as we know them today. The two treatments for depressive illness were either amphetamine stimulants, which often were ineffective and had the general effect of increasing energy and activity (16), or electroconvulsive therapy, which was effective but had the disadvantage of terrifying and often endangering the patient. Not until the late 1950s was the first generation of antidepressants discovered (the TCAs and MAOIs), not by design but by chance. While searching for “chlorpromazine-like” compounds to treat schizophrenia, imipramine was recognized by Kuhn (16) for its antidepressant properties, thus becoming the forerunner for the tricyclic class of monoamine reuptake inhibitor antidepressants (i.e., the TCAs). The second compound to be discovered was the antitubercular drug isoniazid (17), which proved to have powerful mood-enhancing properties, becoming the forerunner of the MAOIs. With the introduction of imipramine and isoniazid, the theory and treatment of depression changed. These early studies still summarize much of our current knowledge regarding the therapeutic effects of antidepressant treatments.

Between 1960 and 1980, TCAs were the major pharmacologic treatment for depression (18). The TCAs, however, have many other actions in addition to blocking monoamine reuptake and nicotinic receptors, including anticholinergic, antihistaminergic, and cardiotoxic side effects that are related to their affinity for muscarinic, histamine, and α_1 -adrenergic receptors as well as their action on cardiac and central nervous system (CNS) sodium channels in membranes. The improved safety, tolerability, and reuptake selectivity of the newer antidepressants (i.e., SSRIs, SNRIs, and NSRIs) have resulted in displacement of the TCAs as the first choice for prophylactic treatment of major depression. The TCAs occupy a narrower—but still important—role in the psychopharmacologic therapy.

The early MAOIs irreversibly inhibited the oxidative deamination of the neurotransmitter monoamines, the proposed mechanism of their antidepressant activity. The biggest liability for these MAOIs was their potential to cause life-threatening hypertensive reactions, resulting from the irreversible inhibition of both MAO-A and MAO-B, which decreases the intestinal and hepatic degradation of dietary sources of tyramine. This inhibition of MAO allows excessive amounts of dietary tyramine, a

weak sympathomimetic vasoconstrictor, to be absorbed from the food, resulting in increased blood pressure (refer to the section on MAOIs in this chapter for more details). Inhibition of MAOs also can alter the pharmacokinetics of monoamine over-the-counter and prescription drugs, allowing them to accumulate in the blood and, thus, increasing their potential for causing adverse drug effects and drug–drug interactions. Minimizing drug and food interactions of these early MAOIs inspired the development of a new generation of MAOIs that are both reversible and selective for MAO-A. The demise of the early MAOIs allowed the TCAs to become the gold standard for the treatment of depressive disorders.

The discovery that certain antihistaminic agents without the condensed aromatic ring systems are selective inhibitors of 5-HT reuptake with little affinity for the other neuroreceptors and almost devoid of cardiotoxicity questioned the need for the 10,11-ethylene bridge for the TCAs. Thus, the search for inhibitors that selectively blocked 5-HT reuptake without the seven-membered central ring of the TCAs resulted in the synthesis of the diarylpropylamine analogs of the TCAs (Fig. 18.2). Thus, during the late 1960s and early 1970s, antihistamine molecules were structurally manipulated in the search for compounds that selectively inhibited 5-HT reuptake with greater potency. The initial breakthrough came with the synthesis of *Z*-zimeldine (the *cis*-isomer; also known as zimelidine, Zemid, patented in 1971), the first SSRI that selectively inhibited the presynaptic reuptake of 5-HT without the adverse events associated with the multireceptor activities of the TCAs (19,20). Zimelidine was synthesized from the manipulation of the antihistamine pheniramine into a diaryl allylamine, the *cis*-isomer (rigid analog) of the propylamine group (Fig. 18.2).

Other structural changes that enhanced its potency and selectivity for blocking 5-HT reuptake was moving the regional position of the 2-pyridyl ring of pheniramine to the 3-pyridyl position and substitution of a halogen into the 4-position of the phenyl ring (2-substitutions selectively block NE reuptake). The secondary amine and primary metabolite, norzimelidine, was 15 times more potent than zimelidine for blocking 5-HT reuptake. On the other hand, *E*-zimelidine (the *trans*-isomer) is an inhibitor of both 5-HT and NE reuptake, whereas its corresponding secondary amine is a potent and selective inhibitor of NE reuptake. It is not unusual for geometric isomers to differ markedly from each other with regard to their receptor or transporter selectivity, affinity, and pharmacodynamic properties. Thus, zimelidine became the first SSRI to be marketed as an antidepressant, but unfortunately, several cases of Guillain-Barré syndrome (an autoimmune disorder attacking the peripheral nervous system) were associated with the use of this drug and led to its withdrawal from the market in 1983. During postmarketing clinical studies, zimelidine showed an increase in the number of suicide attempts over what had been expected—this adverse event was to become a major issue with the SSRIs 20 years later.

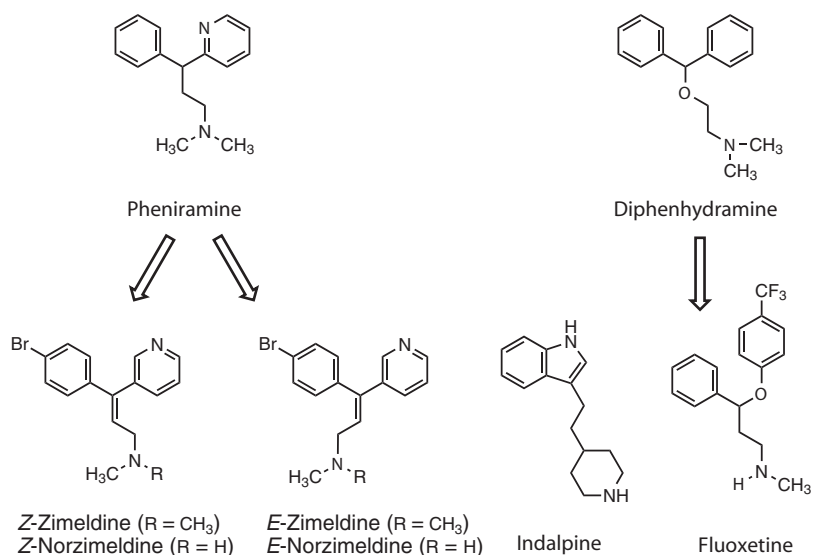


FIGURE 18.2 Structural relationship between antihistamines and antidepressants that block the reuptake of 5-HT.

The success of zimelidine as an SSRI, however, led to the discovery and marketing of several nontricyclic SSRIs from multiple pharmaceutical companies worldwide. Another manipulation of the antihistamines produced indalpine (patented 1977 by Rhône Poulenc) (Fig. 18.2). It produced responses in patients who had not responded to the TCAs or MAOIs but then ran into trouble, because clinical trials suggested that it might cause agranulocytopenia (a lowering of the white blood cell count). For the most part, this is not a serious problem, but in rare cases, if undetected, it can be fatal. It was removed from the European market in 1985 and was never marketed in the United States.

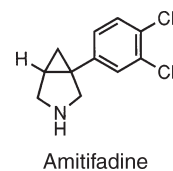
Other SSRIs developed during this period that have become household words include paroxetine (patented 1975 by Ferrosan to SmithKlineBeecham to GlaxoKline), citalopram (patented 1979 by Lindberg, licensed to Forest Labs), fluoxetine (patented 1982 by Lilly), and sertraline (patented 1985 by Pfizer).

Scientists at Lilly Research Laboratories synthesized more than 50 phenoxypropylamines derived from the antihistamine diphenhydramine (Benadryl) before discovering fluoxetine (21,22) (Fig. 18.2). The first of these compounds was nisoxetine, a potent SNRI (see Fig. 18.8) that was clinically developed but never marketed. Other derivatives that have since been marketed by Lilly include atomoxetine (an SNRI) and duloxetine (an NSRI).

Fluoxetine (Prozac) was heralded as the prototype for the next generation of SSRI antidepressants, possessing fewer adverse effects and with a greater margin of safety when overdoses are consumed compared with the TCAs and also lacking the food-interaction toxicity of MAOIs. Fluoxetine was marketed in 1987, and within a few years, it boasted worldwide sales of nearly \$1.2 billion a year. During the past 20 years, nontricyclic selective NE or 5-HT reuptake inhibitors, reuptake inhibitors of both NE and 5-HT, and reversible selective MAOIs have been approved for use in depression. These newer additions allow exploration

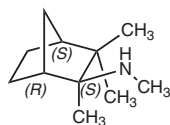
of the roles of NE versus 5-HT using treatments that are devoid of confounding receptor activities. Conscious targeting of more than one neurotransmitter activity (e.g., of serotonergic and noradrenergic mechanisms or of NE and DA), while retaining specificity, is the target for the development of the next generation of antidepressants.

The majority of antidepressants in current use selectively inhibit the reuptake of 5-HT and/or NE. Based on the previously neglected role proposed for DA in depression, it has been hypothesized that a “broad-spectrum” antidepressant will produce a more rapid onset and/or higher efficacy than agents inhibiting the reuptake of 5-HT and/or NE (23). Broad-spectrum antidepressants are compounds that inhibit the reuptake of NE, 5-HT, and DA, the three biogenic amines most closely linked to depression. The pharmacologic profile of one such compound, amitifadine (DOV 21947), an azabicyclo[3.1.0]hexane, has recently been described and is in phase III clinical trials (23,24). It is a potent inhibitor of NE, 5-HT, and DA reuptake by their corresponding membrane transporter proteins. The plasma concentrations of DOV 21947, following both single and multiple doses, appear to be sufficient to inhibit NE, 5-HT, and DA reuptake.



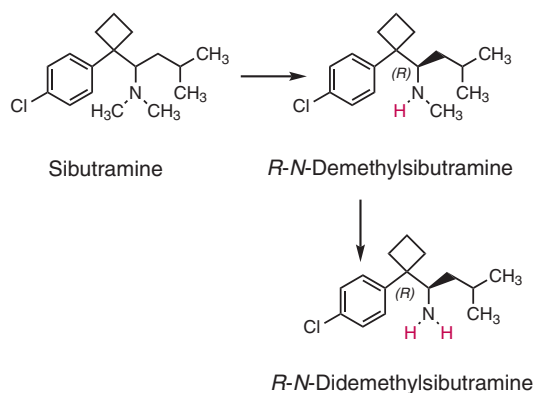
S-(+)-Mecamylamine is a potent noncompetitive antagonist of nAChRs. Racemic mecamylamine was originally developed as a ganglionic-blocking agent for the treatment of moderately severe to severe essential hypertension (Chapter 24). However, it soon became apparent that mecamylamine was capable of crossing the blood-brain

barrier and distributing itself into brain tissues, thus exerting CNS effects such as preventing nicotine-induced seizures in several animal species at doses lower than those necessary to reduce blood pressure or elicit other peripheral signs of postganglionic blocking activity. In contrast to its competitive blockade of preganglionic nAChRs in autonomic ganglia, mecamylamine is a non-competitive inhibitor of neuronal nAChRs and binds to the $\alpha_4\beta_2$ nAChR subunit in the open channel of the receptor. Mecamylamine is in phase III clinical trials as an augmentative antidepressant strategy for patients with MDD who are refractory to SSRI treatment. If clinical trials are successful and the U.S. Food and Drug Administration (FDA) approves mecamylamine as an antidepressant, this new drug paves the way for further investigations using nAChRs as targets for the action of safer antidepressants and novel antiaddictive nicotine compounds.



S(+)-Mecamylamine

The antiobesity agent, racemic *R,S*-sibutramine is a 5-HT and NE reuptake inhibitor and has an antidepressant profile similar to that of a “triple-acting” antidepressant in animals (25). It has two active metabolites, desmethylsibutramine and didesmethylsibutramine (the terms “desmethyl” and “demethyl” are used interchangeably). The half maximal inhibitory concentrations (IC_{50}) for the *R*-enantiomers of these metabolites were more potent as in vitro inhibitors of DA and NE reuptake than 5-HT (25). Both *R*-enantiomers had significantly greater anorectic effects than those of their respective *S*-enantiomers as well as that of *R,S*-sibutramine. The results suggest that these enantioselective metabolites of sibutramine could be safe and effective treatments for binge-eating disorder in obese patients.



Traditionally, antidepressants have been classified according to their structure (i.e., secondary or tertiary amine TCAs) or their principal mechanism of action (i.e.,

MAOIs and SSRIs). With the appearance of increasing numbers of second- and third-generation antidepressants, however, a better way of classifying and describing the antidepressants was necessary. For the purposes of this chapter, the antidepressants are organized into eight classes plus mood stabilizers (Table 18.1) and discussed according to their distinct and different mechanisms of action (Fig. 18.3). Considerable overlap exists in their mechanism of actions and uses, but these different classes of antidepressants work by distinct mechanisms, have different side effect profiles, and may be favored for different types of depressive illnesses.

A key step that determines the intensity and duration of monoamine signaling at synapses is the reuptake of the released neurotransmitter into nerve terminals through high-affinity plasma membrane transporters (26). Reuptake is the process of rapidly removing the monoamine neurotransmitters from the synaptic cleft and allowing most of the released neurotransmitter to be recycled for further use. The advantage of reuptake is that it is faster than passive diffusion through the membrane. Any monoamine neurotransmitter remaining in the synaptic cleft is then absorbed and metabolized into inactive metabolites. The monoamine reuptake transporter protein binds the released neurotransmitter in the extracellular fluid and transports the monoamine across the presynaptic plasma membrane back into the intracellular fluid of the presynaptic neuron (Fig. 18.4). Monoamine transporters (Fig. 18.5; see the sidebar for more detail) are embedded in the plasma membrane of the nerve terminals (perisynaptically) of dopaminergic, noradrenergic, and serotonergic neurons rather than intrasynaptically (along the portion of the nerve terminal forming the synapse). They are members of a larger sodium-dependent transporter family and represent a major mechanism terminating the action of released monoamine neurotransmitters in the synaptic cleft. These transporters are important targets for many antidepressive drugs and substances of abuse (e.g., cocaine). Transporter proteins are specific to their respective neurotransmitter: serotonin reuptake transporter (SERT), norepinephrine reuptake transporter (NET), and dopamine reuptake transporter (DAT). None of the reuptake antidepressants exhibit significant affinity for DAT, which may be related to their ineffectiveness in types of depression that are resistant to the SNRIs, SSRIs, and NSRIs. The TCAs and nontricyclic NSRIs block both NET and SERT, the SSRIs selectively block SERT, and SNRIs selectively block NET. The antidepressant reuptake inhibitors also may contribute to relief of depression by decreasing the expression of their respective transporter proteins.

Figure 18.6 graphically illustrates the selectivity of the reuptake inhibitors for their respective transporters. The selectivity ratio for inhibiting SERT (ratio >1) is obtained by dividing the affinity of the inhibitor (K_i) for inhibiting SERT with its affinity (K_i) for inhibiting NET, whereas the selectivity ratio for inhibiting NET (ratio <1) is obtained by dividing its affinity (K_i) for inhibiting NET with the affinity (K_i) for inhibiting SERT. For example,

TABLE 18.1 Antidepressant Classes with Examples of Generic and Trade Names

Drug Classes	Abbreviation	Drug Examples	Trade Names
Selective Norepinephrine Reuptake Inhibitor	SNRIs	Amoxapine Atomoxetine Desipramine Maprotiline Nortriptyline Roboxetine	Asendin Strattera Norpramin Aventyl, Pamelor Vestra
Selective Serotonin Reuptake Inhibitor	SSRIs	Citalopram Escitalopram Fluoxetine Fluvoxamine Paroxetine Sertraline	Celexa Lexapro Prozac Luvox Paxil Zoloft
Norepinephrine and Serotonin Reuptake Inhibitor	NSRIs	Amitriptyline Clomipramine Desvenlaxine Doxepine Duloxetine Imipramine Milnacipran Trimipramine Venlafaxine	Elavil Anafranol Pristiq Sinequan Cymbalta Tofranil Ixel (in EU) Surmontil Effexor
Dopamine and Norepinephrine Reuptake Inhibitor	DNRIs	Bupropion	Wellbutrin, Zyban
Serotonin Receptor Modulators	SRMs	Aripipazole	Abilify
Serotonin-2 Antagonists/ Serotonin Reuptake Inhibitor	SARI	Trazodone Vilazodone	Desyrel, Oleptro Viibryd
α_2 –Noradrenergic Antagonist/ Selective Serotonin Antagonist	NaSSA	Mirtazapine	Remeron
Monoamine Oxidase Inhibitors	MAOIs	Moclobemide Phenelzine Tranylcypamine	Nardil Parnate
Mood Stabilizers		Lithium Valproic acid Carbamazepine	Eskalith, Lithobid Depakote, Depakene, Depacon Tegretol

a value of approximately 1 for amitriptyline means that amitriptyline will inhibit both NET and SERT at the same concentration (i.e., no selectivity with regard to their mechanism of antidepressant activity). The value of -30 for desipramine means that desipramine is 30 times more potent at inhibiting the NET than the SERT, although the SSRIs with selectivity ratio values of greater than 100 are more than 100 times more potent at inhibiting the SERT than the NET. Furthermore, because the selectivity ratio for most SSRIs is more than 100, a plasma concentration of any SSRI that will produce inhibition of the SERT will produce no physiologically meaningful inhibition of NET (27–29). The converse will be true regarding selectivity for the NET. Clinically, such selectivity ratios of greater than 100 translate into being able to produce all the physiologic effects mediated by inhibiting one transporter without causing any effects that will be produced by inhibiting the other uptake transporter. When the selectivity ratio is less than 30, such as with fluoxetine, the

difference is small enough that inhibition of both reuptake transporters may occur under therapeutic doses and, thus, can contribute to the broad antidepressant activity of the drug.

Most scientists agree that MAOIs, SSRIs, SNRIs, and NSRIs improve depression by boosting the levels of NE and/or 5-HT in the brain, but what is not established is how increased concentrations of NE and 5-HT and their synergism translate into reducing depression. One problem with the original monoamine model was that whereas plasma concentrations of the antidepressant and binding to the monoamine transporter occur almost immediately, chronic administration of antidepressants is needed before clinical efficacy is attained (18). The therapeutic effect of an antidepressant is almost always observed after a period of 3 to 6 weeks of treatment. This suggests that certain adaptive changes are occurring with chronic administration of these drugs that may be important for their antidepressant

TRANSPORTER PROTEINS

The monoamine transporter protein (molecular weights, 60 to 80 kD) is a string of amino acids that weaves in and out of the presynaptic membrane 12 times (Fig. 18.4)—that is, 12 transmembrane domains (TMs) with a large extracellular loop between TM₃ and TM₄. Both the N- and C-termini of the transporters are located within the cytoplasm. There are six potential sites of phosphorylation by protein kinase A and protein kinase C, which regulate the transporters. The large extracellular loop and the cytoplasmic parts of the N- and C-termini do not appear to be the target sites for the transporter inhibitors (i.e., antidepressants). Rather, the areas important for selective monoamine affinity appear to be localized within TM₁ to TM₃ and TM₈ to TM₁₂ that project into the synapse, and these areas of the transporters have a common binding site for the monoamine and many of its inhibitors (Fig. 18.4). To transport protonated 5-HT (5-HT⁺), SERT cotransports one sodium ion (Na⁺) and one chloride ion (Cl⁻) while countertransporting a potassium ion (K⁺) (Fig. 18.5). The SERT then flips inside the cell, releasing the 5-HT⁺ and the Na⁺ and Cl⁻ into the cytoplasm of the neuron. On releasing the 5-HT⁺ and the Na⁺ and Cl⁻, SERT flips back out, with the unoccupied binding site exposed to the synaptic cleft, ready to receive and transport another 5-HT⁺ molecule. To transport protonated NE (NE⁺), NET also cotransports Na⁺ and Cl⁻ with intracellular K⁺ stimulation and no K⁺ efflux. The initial complex of the monoamine, Na⁺, and Cl⁻ with the transporter protein creates a conformational change in the transporter protein. The driving force (electrical potential) for the energetically unfavorable transport of the monoamine is the Na⁺ concentration gradient. The Na⁺, K⁺ transporter (Na⁺, K⁺-ATPase) maintains the extracellular Na⁺ concentration as well as the intracellular K⁺ concentration. The Na⁺, K⁺-ATPase transports three Na⁺ ions for each two K⁺ ions pumped into the cell. Unlike channels that stay open or closed, transporters undergo conformational changes (changes in their three-dimensional shape) and move one monoamine molecule in each cycle.

action (see previous Receptor Sensitivity Hypothesis section). This concept has been the motivating force for much of the development of alternative nonmonoamine theories. Over the years, mechanisms such as downregulation of β -adrenergic receptors, desensitization of presynaptic α_2 -adrenoceptors, increased postsynaptic 5-HT receptor sensitivity, downregulation of 5-HT₂ receptors, and desensitization of presynaptic 5-HT_{1A} receptor have been cited either as the final common pathway or as one of many possible final common pathways.

It should be remembered that the neurotransmitter and downregulation hypotheses are incomplete explanations for how antidepressants work. Antidepressants most likely set off an intricate chain of reactions that occur between the time the patient first takes them and the following few weeks, when they finally produce their

effect. What the neurotransmitter and downregulation hypotheses do provide are useful—if simplistic—models for comprehending at least some of the basic biochemical processes triggered by antidepressants.

ANTIDEPRESSANTS IN PSYCHOTHERAPY

The introduction of the second-generation classes of antidepressants in the 1980s and 1990s (SSRIs, SNRIs, NSRIs, and serotonin receptor modulators [serotonin antagonist/reuptake inhibitors {SARIs} and noradrenergic and specific serotonergic antidepressants {NaSSAs}]) (Table 18.1) has been regarded as the major pharmacologic advance in the treatment of depression since the appearance of the TCAs and MAOIs. The SSRIs, SNRIs, and NSRIs have proven to be effective for a broad range of depressive illnesses, dysthymia, several anxiety disorders, and bulimia. The SSRIs are the most widely prescribed antidepressant drugs and rank in the top 50 drugs in terms of total United States sales for 2004. In addition to being the usual first-line treatments for major depression, the SSRIs also are first-line treatments for panic disorder, obsessive-compulsive disorder, social phobia, posttraumatic stress disorder, and bulimia. They also may be the best medications for treatment of dysthymia and generalized anxiety disorder.

Differences in general tolerability of the different classes of antidepressants and in their side effect profiles are well known and generally accepted. Compared with the TCAs, the SSRIs cause significantly more nausea, diarrhea, agitation, sexual dysfunction, anorexia, insomnia, nervousness, and anxiety, whereas the TCAs cause more cardiotoxicity, dry mouth, constipation, dizziness, sweating, and blurred vision (18). Although the SSRIs, SNRIs, and NSRIs possess improved safety margins and lesser cardiovascular adverse effects than TCAs and utility for treating other nondepressive disorders, they offer no real gain in efficacy compared with the first-generation TCAs. Despite pharmacologic differences in their mechanisms of actions, the general view has been that all antidepressants are of equal efficacy. Only within the last 10 years has this general assumption come under serious challenge from comparisons of antidepressants with dual mechanisms of action that can be acting in a complementary and perhaps synergistic manner to improve depression versus those with a single mechanism of action (11). Often, a variety of antidepressants will be prescribed and the dosage adjusted before the most effective antidepressant or combination of antidepressants is found. Although some improvements may be seen during the first few weeks, the antidepressants must be taken regularly for 3 to 4 weeks (and, in some cases, for as many as 8 weeks) before the full therapeutic effect occurs. Patient compliance can become an issue, because patients often are tempted to stop medication too soon as a result of feeling better and, thus, thinking they no longer need the medication. They also may have problems with the adverse effects, or they may think

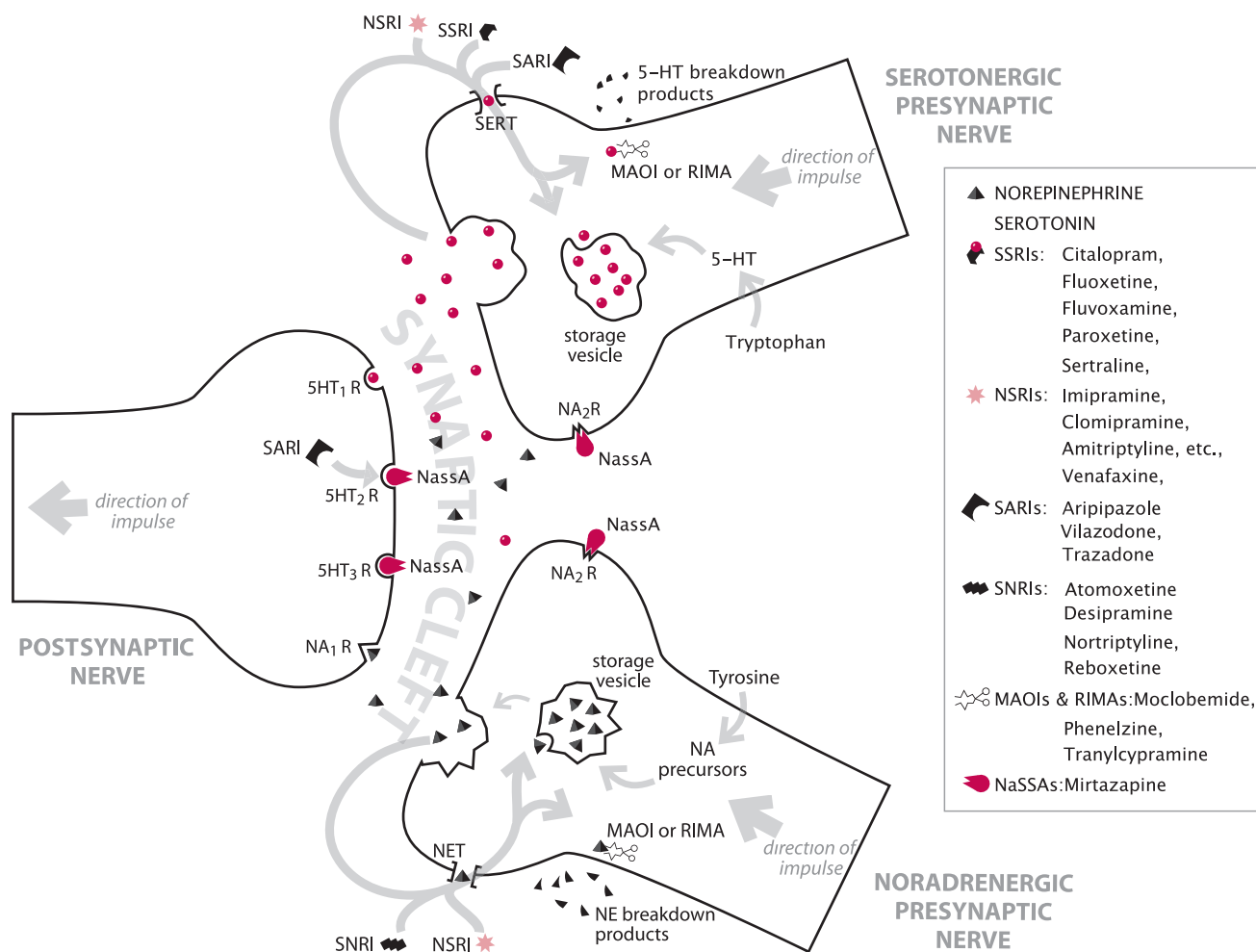


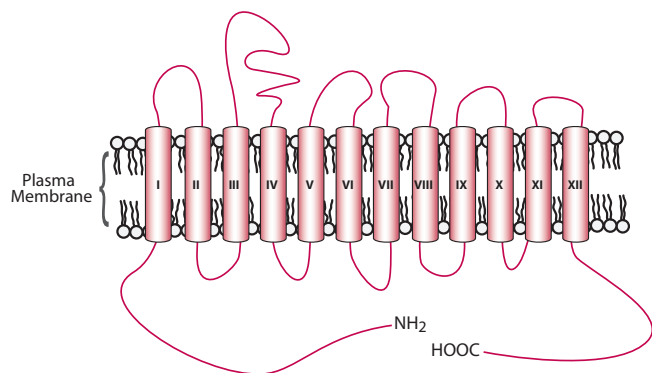
FIGURE 18.3 Sites of action of the antidepressants.

the medication is not helping at all. It is important for the patient to keep taking the antidepressant until it has a chance to work, although side effects often appear before antidepressant activity does. Once the individual is feeling better, it is important to continue the medication for at least 4 to 9 months (or longer) to prevent a recurrence of the depression. Antidepressants alter the brain chemistry; therefore, they must be stopped gradually to give the brain time to adjust. For some individuals with bipolar disorder or chronic major depression, antidepressant therapy may need to be maintained indefinitely (30). All patients who are prescribed antidepressants should be informed that although the drugs are not associated with tolerance and craving, discontinuation/withdrawal symptoms may occur on stopping, on missing doses, or occasionally, on reducing the dose of the drug. These symptoms usually are mild and self-limiting, but occasionally, they can be severe, particularly if the drug is stopped abruptly. Symptoms of antidepressant withdrawal include nausea, vomiting, anorexia, headache, restlessness, agitation, "chills," insomnia,

and, sometimes, hypomania, panic-anxiety, and extreme motor restlessness, especially if an antidepressant (particularly an MAOI) is stopped suddenly after regular administration for 8 weeks or more.

Clinical trials have concluded that 2 or 3 of every 100 children and teenagers treated with antidepressants might be at higher risk of suicidal behavior. Data regarding suicidal behavior vary among the antidepressants, which leads to the conclusion that no antidepressant is free from risk at this time. Most of the suicides have occurred with the SSRIs, especially paroxetine (Paxil) and fluoxetine (Prozac); only fluoxetine has been proven to be effective and is approved for treating pediatric depression. Approximately 7% of antidepressant prescriptions are written for children. Thus, in 2007, the FDA approved that "black box labeling" be included on antidepressant product packaging, alerting physicians to avoid prescribing antidepressants to children because of possible risks of suicidal behavior and to watch for signs of worsening depression or suicidal thoughts in children who are taking any antidepressant.

EXTRACELLULAR



INTRACELLULAR

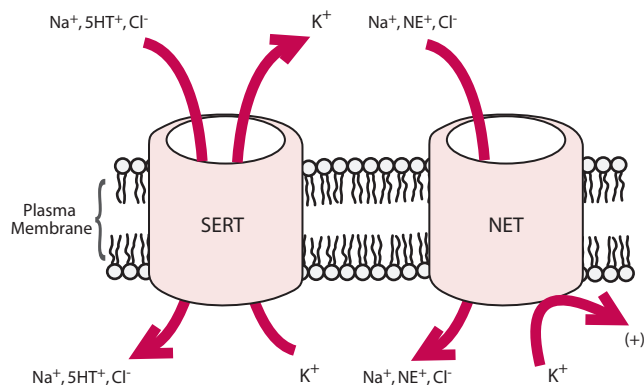
FIGURE 18.4 Monoamine reuptake transporter.

EFFECT OF PHYSICOCHEMICAL PROPERTIES AND STEREOCHEMISTRY ON ANTIDEPRESSANT EFFICACY

Small substituent changes in molecular structure can affect the pharmacokinetic and pharmacodynamic (clinical) properties of antidepressant drug molecules, resulting in profound differences between their transporter selectivity and their antidepressant effect—for example, a difference of a 2-chloro group between the structurally related antidepressants imipramine and clomipramine or the *o*- versus *p*-substituents between fluoxetine (SSRI) and atomoxetine (SNRI) (Fig. 18.7). Furthermore, a seemingly simple isosteric replacement of a sulfur atom in the central ring of chlorpromazine with an ethylene group to give a seven-membered azepine ring (clomipramine) or the replacement of the methylpiperidylidene at the 5-position for the antihistamine/5-HT antagonist cyproheptadine with a dimethylaminopropylidene for the antidepressant reuptake inhibitor amitriptyline has profound effects on the physicochemical properties, pharmacokinetics, mechanisms of action, and therapeutic activities (Fig. 18.7). On the other hand, the physicochemical differences can translate into differences in *in vitro* and *in vivo* pharmacologic and clinical properties, as exemplified between mianserin and mirtazapine (Fig. 18.7) (31). The isosteric replacement of a benzene ring (mianserin) with a pyridine (mirtazapine) resulted in significant changes in their dipole moments, lipophilicity ($\log P$), pK_a values, and electronegativity, resulting in different mechanisms of action and regioselectivity in formation of hydroxylated metabolites.

Many antidepressants are stereoisomers and contain either a chiral center or a center of unsaturation by which chiral metabolites could result (32,33). Often, such drugs are marketed as a mixture of the

EXTRACELLULAR



INTRACELLULAR

FIGURE 18.5 Model of the NET and SERT and the ion-coupled NE and 5-HT reuptake. Reuptake of 5-HT is dependent on the cotransport of Na^+ and Cl^- and countertransport of K^+ . Reuptake of NE is dependent on the cotransport of Na^+ and Cl^- with intracellular K^+ stimulation but without K^+ efflux.

resultant enantiomers (racemates) or of geometric isomers. These enantiomers or geometric isomers may differ markedly from each other with regard to their

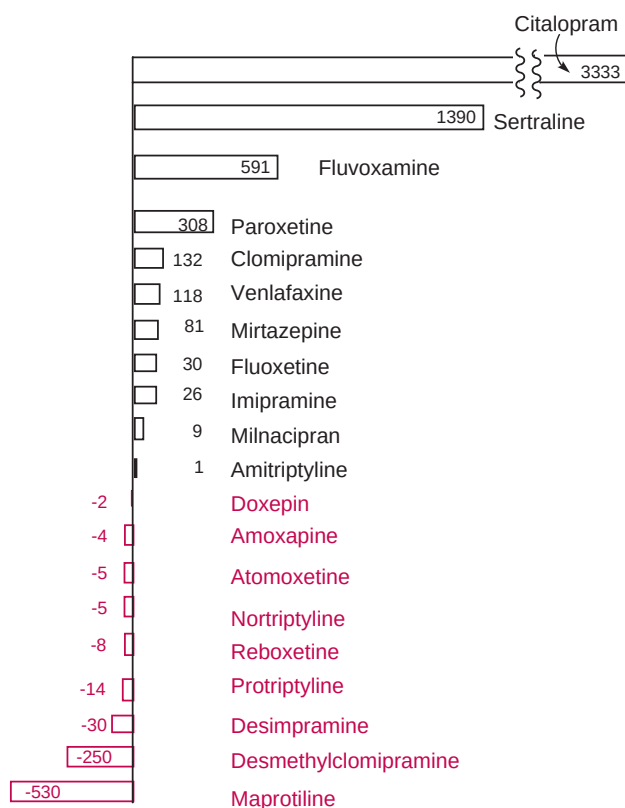


FIGURE 18.6 In vitro selectivity ratios for reuptake inhibitors.

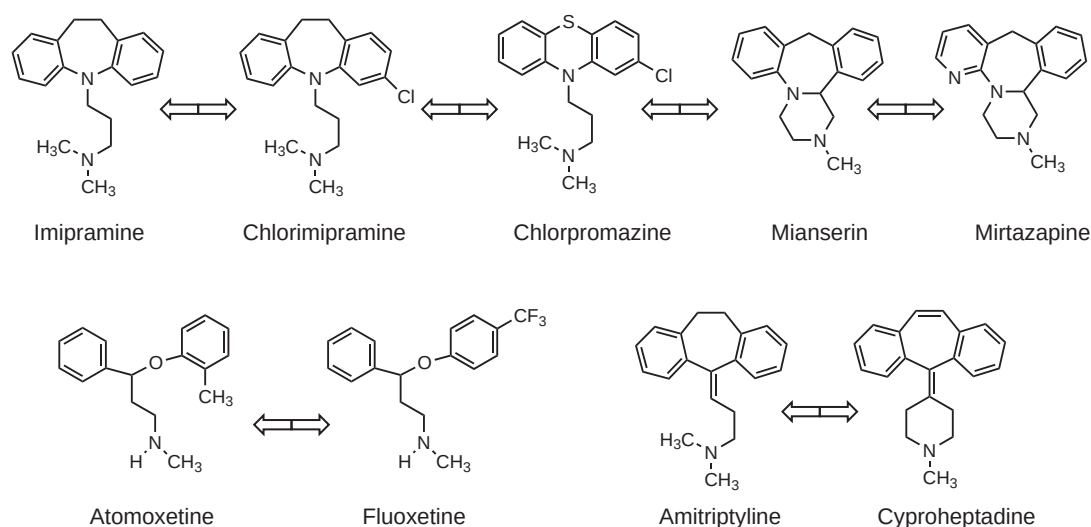


FIGURE 18.7 Structural relationship between selected molecular structures.

pharmacodynamic and/or pharmacokinetic properties. Increased knowledge about the molecular structure of specific drugs targets and an awareness of several possible advantages to using single enantiomers rather than racemic mixtures of drugs have led to an increased emphasis on understanding the role of chirality in drug development of antidepressant drugs. Several notable examples of antidepressants (the SSRIs) currently are available in which the individual enantiomers or geometric isomers differ considerably with regard to factors such as binding to SERTs and NETs, interactions with receptors and metabolizing enzymes, and clearance rates from the body. Examples of the effects of chiral centers or geometric centers on such properties include racemic mixtures (e.g., fluoxetine, citalopram, bupropion, and trimipramine), single enantiomers [e.g., (+)-sertraline, (–)-paroxetine, (–)-escitalopram, and (–)-atomoxetine]; or geometric isomers (e.g., *Z*-zimelidine, *Z*-doxepin, *E*-dosulepin, and *E*-fluvoxamine). Recent developments in analytical and preparative resolution of racemic and geometric drug mixtures as well as increased interest in developing new drugs that interact with specific targets, which have been described in detail at the molecular level, have resulted in increased emphasis on stereochemistry in antidepressant drug development.

DRUG METABOLISM AND DRUG-DRUG INTERACTIONS

Antidepressant treatment carries increased risks of adverse drug events because of age-related physiologic changes, polypharmacotherapy, and individual variability in drug metabolism (e.g., genetic factors, concurrent disease, diet, and eating habits). Pharmacokinetic drug-drug interactions occur when one medication

(the precipitant drug) significantly affects the plasma concentration, half-life, or both of another medication (the object drug) by altering its absorption, distribution, metabolism, or elimination. For object drugs with narrow therapeutic indices, even small elevations in plasma drug concentration can cause potentially serious adverse reactions. Pharmacodynamic drug-drug interactions occur when the precipitant drug affects the ability of the object drug to bind with its therapeutic target (e.g., transporter protein) or receptor. Some compounds compete directly for binding to a receptor; others indirectly affect the ability of an object drug to interact with its site of action. Many medications can bind to multiple receptor types (e.g., first-generation TCAs), causing diverse adverse reactions. Thus, before adding a new drug to an existing antidepressant regimen, it is wise to determine whether any medications can be eliminated. Reduction of total drug burden, adjustment of dose levels, and careful selection of an appropriate agent are important steps toward avoiding adverse drug interactions. In addition, the documented and potential drug interactions of the various classes of antidepressants—and of the specific drugs within each class—should be considered. Each patient should be treated individually and monitored carefully during the initiation and maintenance of antidepressant therapy.

The infamous drug-drug interactions between mibefradil (Propulsid) and terfenadine (Seldane) and other drugs metabolized by cytochrome P450 (CYP) 3A4 resulted in these drugs being withdrawn from the market, which placed a spotlight on hepatic drug metabolism as a significant participant in drug-drug interactions. Nearly all the antidepressants are metabolized by at least one CYP isoform, and these drugs and their metabolites may be substrates or inhibitors for the CYPs (Table 18.2).

TABLE 18.2 Cytochrome P450 (CYP) Metabolism^a

Antidepressant	Major CYP	Minor CYP	Metabolism Pathway	CYP Inhibitor
Amitriptyline	2D6	3A4, 2C19, 2C9, 1A2, and 2B6	2D6, 3A4, 2C19, and 1A2 <i>N</i> -demethylation 2D6 and 2B6 <i>E</i> -10-hydroxylation	2D6, 2C19, 1A2, and 2C9 ^b
Clomipramine	2D6	3A4, 2C19, 1A2, and 2C9	2D6, 3A4, 2C19, 2C9, and 1A2 <i>N</i> -demethylation 2D6 2- and 8-hydroxylation	2D6
Desipramine	2D6	1A2 and 2C19	1A2 and 2C19 <i>N</i> -demethylation 2D6 and 2C19 2-hydroxylation	2D6 and 2C19 ^b
<i>E</i> -Doxepin	2D6 and 2C9	3A4, 1A2, and 2C19	2D6 and 2C9 hydroxylation 3A4 and 1A2 <i>N</i> -demethylation	
<i>Z</i> -Doxepin	2C19	2C9, 3A4, and 1A2	2C19 and 2C9 <i>N</i> -demethylation 3A4 and 1A2 <i>N</i> -demethylation	
Imipramine	2D6	3A4, 2C19, 1A2, 2B6, and 2C9	1A2 and 2C19 <i>N</i> -demethylation 2D6, 1A2, 3A4, and 2C19 2- and 10-hydroxylation	1A2, 2D6, and 2C19
Maprotiline	2D6	1A2	2D6 and 1A2 <i>N</i> -demethylation	
Nortriptyline	2D6	3A4, 2C19, and 1A2	2D6, 3A4, 2C19, and 1A2 <i>N</i> -demethylation 2D6 <i>E</i> -10-hydroxylation	2D6 and 2C19 ^b
Trimipramine	2D6	2C9 and 2C19	2-hydroxyl, 10-hydroxyl, and <i>N</i> -demethylation	2D6
(±)-Citalopram	2D6	1A2 and 2C19	2D6, 2C19, and 3A4 <i>N</i> -demethylation 2D6 <i>N</i> -oxidation	1A2 ^b and 2D6 ^b
Escitalopram	2D6	1A2 and 2C19	2D6, 2C19, and 3A4 <i>N</i> -demethylation 2D6 <i>N</i> -oxidation	
(±)-Fluoxetine	2D6	3A4, 2C9, 2C19, and 1A2	2D6 and 2C19 <i>N</i> -dealkylation 1A2 and 2C9 <i>O</i> -dealkylation	2D6 (<i>S</i> -fluoxetine) 1A2, ^b 2C19, ^b 3A4, ^b 2C9, ^b and 2D6 ^b
Sertraline	3A4	2D6 and 2C19	3A4, 2C19, 2D6, and 2C9 <i>N</i> -demethylation	2C19, ^b 3A4, ^b and 2D6 ^b
Fluvoxamine	1A2	3A4, 2C19, and 2D6	2D6 and 1A2 <i>O</i> -dealkylation	2C19, 1A2, ^b and 3A4 ^b
Paroxetine	2D6	3A4, 1A2, and 2C9	2D6 cleavage methylene dioxy	2D6, ^c 2C19, ^b and 3A4 ^b
Venlafaxine	2D6	3A4, 2C19, and 2C9	2D6, 2C19, and 2C9 <i>O</i> -demethylation 2C19, 2C9, and 3A4 <i>N</i> -demethylation	2D6 ^b
Reboxetine	3A4		<i>O</i> -deethylation, aromatic ring hydroxylation	2D6 and 3A4
Atomoxetine	2D6	3A4, 2C19, and 1A2	2D6, 1A2, 2B6 and 2C19 4-hydroxylation 2C19, 3A4 and 2B6 <i>N</i> -demethylation	
Duloxetine	2D6		4- and 6-hydroxylation	2D6
Mirtazapine	2D6	3A4 and 1A2	2D6 and 1A2 8-hydroxylation 3A4 <i>N</i> -demethylation 3A4 and 1A2 <i>N</i> -oxidation	3A4 ^b and 2D6 ^b
Bupropion	2B6	2E1, 3A4, and 2D6	Hydroxylation <i>t</i> -butyl group	2D6 ^b
Trazodone	3A4	2D6	<i>N</i> -dealkylation to chlorophenylpiperazine Aromatic ring hydroxylation	
Tranlycypromine		3A4, 2A6, 2D6, 2C19, and 2C9	Aromatic ring hydroxylation	2A6 ^c
Moclobemide	2C19	2D6 and 1A2	Aromatic ring hydroxylation Morpholine ring oxidation	2D6 ^b

^aRendic S. Human P450 metabolism database. Available at: http://www.gentest.com/human_p450_database. Accessed July 11, 2003.^bWeak CYP inhibitor.^cMechanism-based inhibition.

When an antidepressant is metabolized by more than one CYP isoform in parallel, the antidepressant is unlikely to be affected by drug interactions or genetic polymorphisms and to cause clinically significant drug interactions via CYP isoform inhibition (34). However, if the drug is metabolized (depending on enzyme kinetics) primarily by CYP3A4 or the polymorphic CYP2C18 or CYP2D6 isoforms, the potential for drug–drug interactions increases. Therefore, knowledge regarding the metabolic pathways of antidepressants and knowledge about substrates and inhibitors of the CYP isoforms can assist in the selection of a proper drug and its dose, thus minimizing the risks of drug–drug interactions (35).

P-Glycoprotein Transporters

P-glycoprotein (P-gp) is a member of the adenosine triphosphate (ATP)-binding cassette superfamily of membrane transport proteins responsible for the efflux of many drugs (also see Chapter 5). It represents a major component of the blood–brain barrier and the intestinal barrier and contributes to renal and biliary elimination of drugs. At the blood–brain barrier, P-gp is localized in the apical membrane of brain capillary endothelial cells and transports substrates into the blood. Therefore, P-gp limits the penetration into and retention of numerous compounds, including the antidepressants, within the brain and thus modulates their effectiveness and CNS toxicity. Inhibition of P-gp could significantly increase drug concentrations in the CNS. Additionally, P-gp is highly expressed in the apical membrane of epithelial cells in the small and large intestine, where it transports drugs out of the cells into the intestinal lumen, thus limiting the bioavailability of compounds such as paclitaxel and HIV protease inhibitors.

The role of P-gp in affecting the pharmacokinetic parameters of antidepressants and contributing to numerous pharmacokinetic drug–drug interactions with coadministered drugs has not been thoroughly elucidated. Antidepressants, particularly TCAs, have a significant potential for inducing adverse drug reactions and, therefore, are subject to numerous drug–drug interactions. The role of P-gp in causing clinically relevant drug interactions is becoming more and more obvious (36). Concentrations of paroxetine and venlafaxine increased two to three times in the brains of mice without P-gp (i.e., knockout mice) after single-dose administration and after treatment for 11 days (34), suggesting that these antidepressants are P-gp substrates and that their pharmacokinetics might be influenced by coadministered P-gp inhibitors. In contrast, fluoxetine exhibited no P-gp substrate characteristics (34), indicating that not all antidepressants share these properties. For citalopram, the results are contradictory.

The potency of sertraline and its *N*-demethylated metabolite, desmethylsertraline, and paroxetine for inhibiting P-gp was comparable with that of quinidine (36). Fluoxetine, norfluoxetine, fluvoxamine, reboxetine, and *O*-demethylparoxetine showed intermediate inhibition,

and citalopram, desmethylcitalopram, venlafaxine, and *N*-desmethylvenlafaxine showed only weak inhibition. No inhibition was found for *O*-desmethylvenlafaxine.

Inhibition of P-gp by drugs may play an important role in drug safety by increasing plasma and brain concentrations of coadministered drugs and, thus, causing adverse drug reactions. No evidence for a potent drug–drug interaction was found with fluoxetine, although sertraline and paroxetine exhibited the greatest potential for affecting the pharmacokinetics of coadministered drugs at the level of P-gp (36,37). At usual therapeutic doses of paroxetine and sertraline, however, the IC_{50} for the inhibition of P-gp is approximately 250 times higher than the plasma concentration for paroxetine and approximately 500 times higher than that for sertraline (37), suggesting that even if the accumulation of sertraline within the cell (e.g., in the biliary or renal system) is taken into account, the P-gp inhibition observed in vitro might not be clinically relevant. This is substantiated by the fact that sertraline, fluvoxamine, and citalopram did not have a clinically relevant influence on the pharmacokinetic parameters of digoxin, a P-gp prototype substrate. However, in addition to being an inhibitor of CYP2D6, paroxetine is a substrate for this same isoform, the activity of which is regulated by a genetic polymorphism. In individuals who lack this active enzyme (i.e., poor metabolizers), plasma paroxetine concentrations are up to 25 times higher than those in individuals who are extensive metabolizers. Therefore, one cannot exclude that in patients who are poor metabolizers, administration of high doses of paroxetine may translate into clinically relevant changes in the pharmacokinetics of concomitantly administered P-gp substrates.

It remains to be studied whether the inhibition of P-gp by the newer antidepressants might lead to drug–drug interactions in patients. Such interactions might, for instance, be relevant when drugs with low oral bioavailability because of substantial transport back into the gut lumen are to be coadministered, as has been shown for loperamide when given in combination with quinidine (38).

SPECIFIC DRUGS

Selective Norepinephrine Reuptake Inhibitors (SNRIs)

Despite the current popularity of the SSRIs for the treatment of depression, the noradrenergic neurons should not be overlooked, because they also influence the depressed mood (39). The noradrenergic system appears to be associated with increased drive, whereas the serotonergic system relates more to changes in mood (Fig. 18.1). Thus, the different symptoms of depression may benefit from drugs acting mainly on one or other of the neurotransmitter systems (18). The SNRIs (Fig. 18.8) seem to be at least as effective as the SSRIs in the treatment of depressive illness (39) by acting specifically at noradrenergic sites. Thus, the SSRIs and SNRIs influence

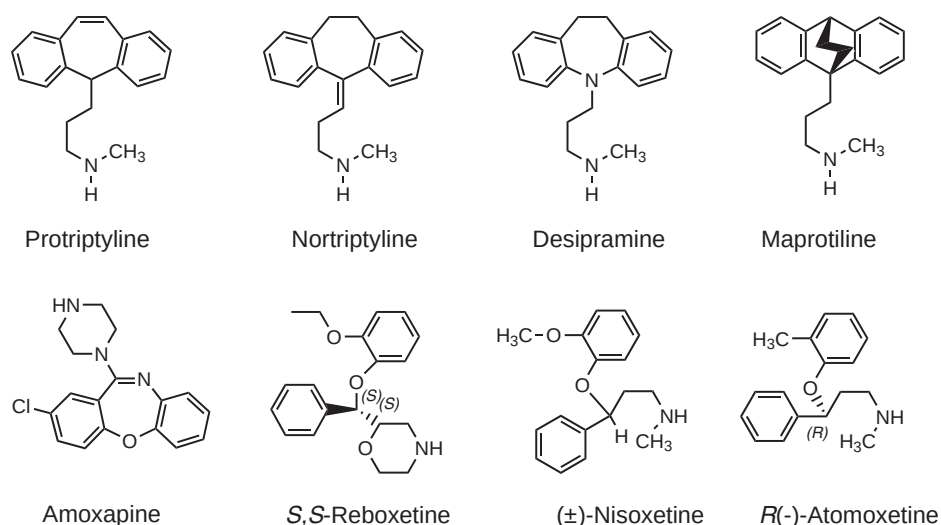


FIGURE 18.8 Selective norepinephrine reuptake inhibitors (SNRIs).

depression by parallel, independent pathways. The SNRIs have a role in the treatment of depression, either alone or as adjunct therapy. The SNRI antidepressants are well tolerated but possess different adverse event profiles.

The selectivity ratios (Fig. 18.6) show that the SNRIs, as a group, are potent selective inhibitors of the NET and that the secondary amine TCAs are substantially more potent with regard to their inhibition of NE reuptake compared with the SSRIs. Their *in vitro* affinity for inhibiting the NET essentially mirrors more or less their clinical efficacy as SNRIs (11): desipramine > protriptyline > amitriptyline = nortriptyline > reboxetine > maprotiline > amoxapine > imipramine > paroxetine. The level of affinity of the SNRIs for NET is not predictive for antidepressant activity.

Tricyclic Secondary Amine Antidepressants

STRUCTURE–ACTIVITY RELATIONSHIP OF THE TCAs The tricyclic ring structure can be found in a variety of different drugs and, for the most part, represents a method for medicinal chemists to restrict the conformational mobility of two phenyl rings attached to a common carbon or hetero atom. The tricyclic ring structure is formed by joining the two phenyl rings into 6-6-6 or 6-7-6 ring systems, in which the central ring is either a six-membered or seven-membered carbocyclic or heterocyclic ring, respectively. Small molecular changes, such as ring flexibility, substituents, or heteroatoms in the tricyclic ring structure, can bring about significant changes in physicochemical, electronegativity (dipole moments), and pharmacodynamic properties (e.g., anticholinergics [antimuscarinic], cholinesterase inhibitors, antihistamine, antipsychotics, and antidepressants). This suggests that the tricyclic structure is not associated with affinity for any particular receptor but, rather, contributes to a range of multiple CNS pharmacodynamic (adverse) effects because of increased lipophilicity. The most common tricyclic ring found in drugs is the near-planar phenothiazine ring common to most of the antipsychotic drugs (Fig. 18.9). The TCAs are classified as such

because they contain a 6-7-6 ring arrangement in which the central seven-membered ring is either carbocyclic or heterocyclic, saturated or unsaturated, which is fused to two phenyl rings (Fig. 18.8; also, see Fig. 18.17 for NSRIs). The side chain may be attached to any one of the atoms in the central seven-membered ring, but it must be three carbon atoms, either saturated (propyl) or unsaturated (propylidene), and have a terminal amine group (secondary or tertiary). The TCAs differ structurally from the antipsychotic phenothiazines in that the two phenyl (aromatic) rings are connected by a two-carbon link to form a central seven-membered ring instead of a sulfur bridge.

The TCAs are subdivided into a dihydrodibenzazepine ring (also known as iminodibenzyl, from which the name imipramine is derived); a dibenzocycloheptene ring, whereby the ring nitrogen of imipramine is replaced by an exocyclic olefinic group (e.g., amitriptyline); a dibenzoxepin ring as a bioisosteric modification of imipramine (e.g., doxepin); and a dibenzocycloheptatriene ring, which replaces the dihydroethylene group of imipramine

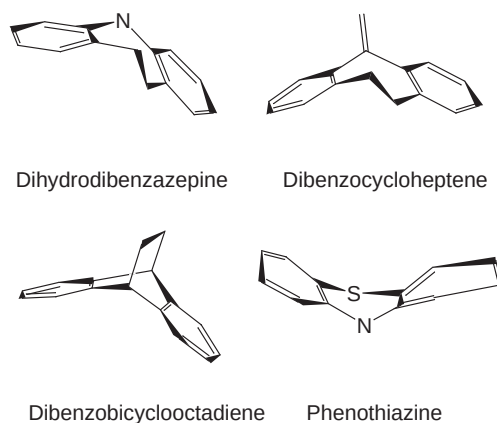


FIGURE 18.9 Three-dimensional models of the tricyclic and tetracyclic ring systems.

with an olefinic ethylene group (e.g., protriptyline) or a tetracyclic (bicyclic ring; e.g., maprotiline) derivative. The tricyclic ring system has little significance regarding selectivity for inhibiting the NET or SERT, but it appears to be important for DAT inhibition.

The secondary and tertiary amine TCAs differ markedly with regard to their selectivity ratios (Fig. 18.6) and their pharmacodynamic and/or pharmacokinetic properties (Tables 18.3, 18.8 and 18.9). Substituting a halogen (i.e., chlorine, clomipramine) or cyano group into the 3-position of the dihydrodibenzazepine ring enhances preferential affinity for SERT. 3-Cyanoimipramine was investigated as a potent SSRI but was never marketed as an antidepressant agent. It is used as a research probe for studying SERTs. Branching the propyl side chain with a 2-methyl group (as in trimipramine) significantly reduces the affinity (~100 times) of imipramine for both the SERT and NET. The *Z* (*cis*) geometry for the propylidene group in chiral TCAs appears to be important for transporter selectivity and affinity (e.g., doxepin).

Studies correlating the binding of the TCAs with the SSRIs found that the TCAs and SSRIs bind to different sites on the transporter and that the TCAs may act as a modulator of monoamine reuptake by producing conformational changes in the transporter, affecting affinity of the monoamine neurotransmitter (40).

Although similar in a two-dimensional plane to the antipsychotic (neuroleptic) phenothiazines, the ethylene bridge linking the two phenyl rings of the TCAs causes the two phenyl rings to be twisted out of the plane, leading to less rigid and more conformationally mobile molecular structures than the phenothiazines as shown in Fig 18.9. The conformational mobility for the TCAs, including ring inversion of the tricyclic ring system, flexing of the CH₂-X bridge (X = CH₂, O, N, or S) in the central seven-membered ring, and flexibility of the alkyl side chain, can result in substantial changes in the overall shape of the molecules. This, in turn, can affect transporter affinity and selectivity, diverse neuroreceptor affinity, and the drug's physicochemical properties. The rate of ring flexibility seems to be correlated with their differences in clinical potency. A dibenzazepine ring system exhibits a greater degree of conformational ring flexing, whereas the dibenzocycloheptene ring system and inserting heteroatoms into the benzylic position reduces the rate of ring flexing in TCAs and their potency (41). Thus, the differences in the pharmacologic activity between these TCAs allow selective binding to their respective transporter proteins. The angle between the two aromatic rings ranges from 106 to 110 degrees for the TCAs, and the large lipophilic ring enhances the affinity of TCA to block the CNS muscarinic, H₁-, and α_1 -adrenergic receptors and to block sodium channels, contributing to its multiple pharmacodynamic effects.

TABLE 18.3 Pharmacokinetics of the Selective Norepinephrine Reuptake Inhibitors (SNRIs)

Parameters	Desipramine	Nortriptyline	Amoxapine	Protriptyline	Maprotiline	Reboxetine	Atomoxetine
Oral bioavailability (%)	60–70	32–79	ND	77–93	66–75	95	>70%
Lipophilicity (logD _{7.4})	1.03	1.66	2.99	—	2.00	—	1.20
Protein binding (%)	91	92	92	92	88	>97	>97
Volume of distribution (L/kg)	17–42	14–22	ND	22	15–28	12–28	93–328 (250)
Elimination half-life (h)	30 (12–30)	30 (18–44)	8 (8–30)	67–89	43 (27–58)	12	4 (3–6) (EM) ^a 17–21 (PM)
Major active metabolites	None	10- <i>E</i> -hydroxy	8-OH (30 h) 7-OH (6.5 h)	None	(60–90 h)	None	4-Hydroxy
Peak plasma concentration (h)	4–6	7–9	90 min	24–30	8–24	12	1–2
Excretion (%)	Urine	Urine 40	Urine 60 (6 d)	Urine 50 (16 d)	Urine 65 (21 d)	Urine ~90%	Urine
	Feces	Feces minor	Feces 7–18	Feces minor	Feces 30	Feces minor	Feces
Plasma half-life (h)	14–62	18–93	8	54–198	21–52	2–4	4.3
Time to steady-state concentration	ND	ND	ND		7 d	ND	ND

ND, not determined.

^aEM = extensive metabolizer; PM = poor metabolizer.

PHARMACOKINETICS COMMON TO SECONDARY AMINE TCAs The secondary amine TCAs are rapidly and well absorbed following oral administration. Although the pharmacokinetics are approximately similar within the tertiary and secondary amine groups, the pharmacokinetics are different between the two groups (Table 18.3; also see Table 18.9 for NSRIs). The secondary amine TCAs have relatively high bioavailability. Their primary routes of hepatic metabolism are *N*-demethylation to inactive primary amine metabolites and aromatic ring hydroxylation (Table 18.2). Despite the fact that serum plasma levels are reached within 1 to 2 days, their onset of antidepressant action typically is at least 2 to 3 weeks or longer. Their volume of distribution is very high, suggesting distribution into the CNS and protein binding. Elimination is primarily as metabolites and their conjugates via renal elimination. Renal and liver function can affect the elimination and metabolism of the parent secondary amine TCA and its metabolites, leading to increased potential for adverse effects, especially in patients (i.e., the elderly) with renal disease.

MECHANISMS OF ACTION COMMON TO SECONDARY AMINE TCAs The exact mechanism of action for the secondary amine TCAs is unclear, but the secondary amine TCAs exhibit substantially more affinity than the SSRIs and the tertiary TCAs for inhibiting the NE transporter. None of the secondary TCAs has significant affinity for the DA transporter. Blocking the reuptake of NE increases its concentration in the synaptic cleft and its ability to interact with synaptic NE receptors. When drugs are selective for a transporter, differences in potency become clinically irrelevant, because the plasma concentration can be dose adjusted to achieve inhibition of the desired transporter without affecting the other transporters. During chronic therapy with the TCAs, adaptive changes at the noradrenergic receptor occur (i.e., downregulation) as a result of neurotransmitter hypersensitivity from low concentrations of NE at the postsynaptic receptor. These changes involve the α_1 -adrenergic receptor.

THERAPEUTIC USES COMMON TO ALL TCAs The efficacy of the secondary and tertiary amine TCAs in the clinical treatment of depressive illness is recommended for various conditions, including major depressive episodes, dysthymia, panic disorder, social phobia, bulimia, narcolepsy, attention-deficit disorder with or without hyperactivity, migraine headache and various other chronic pain syndromes, enuresis in children, and obsessive-compulsive disorder (clomipramine). The TCAs possibly are useful as well for a broader range of depressive conditions described as dysthymia or depressive neurosis and even for prolonged or pathologic mourning, agoraphobia without panic attacks, and some of the symptoms (e.g., nightmares) in posttraumatic stress disorder.

ADVERSE EFFECTS COMMON TO ALL TCAs The family of TCAs has many undesirable side effects and behaves like “five

drugs wrapped into one.” They not only block the reuptake of NE and 5-HT but also block muscarinic receptors (anticholinergic), α_1 -adrenergic receptors, H_1 -receptors (antihistamine), and sodium channels. The common side effects and appropriate responses are given in Table 18.4.

DRUG-DRUG INTERACTIONS COMMON TO THE SECONDARY AMINE TCAs The secondary amine TCAs were once first-line therapy for depression because of their efficacy in a broad range of depressive disorders. Today, however, these agents generally are reserved for second-line treatment because of their narrow therapeutic-to-toxicity ratios and troublesome adverse effect profiles. Even the better-tolerated nortriptyline is fatal in overdose and may have significant adverse effects at therapeutic dose levels. Most TCAs are metabolized by multiple CYP enzymes and, thus, are likely to be object drugs for many common medications. Because these TCAs have narrow therapeutic indices, any interference with their metabolism can lead to serious adverse reactions resulting from increased plasma concentrations (e.g., arrhythmias, seizures, and confusion). Such reactions are both more common and more likely to be life threatening in elderly patients because of age-related pharmacokinetic alterations. Therefore, although specific secondary amine TCAs are useful for some conditions (e.g., major depression), administration with other drugs should be done cautiously.

Concurrent administration of these TCAs and MAOIs is contraindicated, and at least 2 weeks should elapse between discontinuance of TCA therapy and initiation of MAOI therapy, and vice versa, to allow washout. Coadministration of SNRIs, TCAs, and MAOIs is potentially hazardous and may result in severe adverse effects associated with hypertension.

TABLE 18.4 Common Side Effects with Secondary Amine Tricyclic Antidepressants and Recommendations

Side Effects	Treatment Recommendations
Dry mouth	Drink sips of water; chew sugarless gum; clean teeth daily
Constipation	Diet rich in bran cereals, prunes, fruit, and vegetables
Bladder complaints (weak urine stream, emptying difficulty, painful urination)	Consult physician
Sexual problems	Consult physician
Blurred vision	Should pass with time
Dizziness	Rise slowly from the bed or chair
Daytime drowsiness	Do not drive; take medication at bedtime; commonly will pass with time

Because protein binding of secondary amine TCAs is high, displacement interactions with other highly protein-bound drugs with narrow therapeutic indices, although not yet fully evaluated, may be important. Concurrent use of the secondary amine TCAs with anticholinergic or sympathomimetic drugs requires close supervision and careful adjustment of the dosage because of potential additive anticholinergic effects (i.e., spastic colon) and increased blood pressure and heart rate. An additional disadvantage of these TCAs is their toxicity from overdosage, especially among those being treated for depression who may have suicidal thoughts.

In addition, secondary amine TCAs are inhibitors of sodium channels and, thus, can slow ventricular conduction at therapeutic doses. If the patient overdoses or drug interactions result in increased plasma concentration of the TCA, severe conduction block contributing to cardiotoxicity may result in ventricular arrhythmias. Also, changes in CNS conduction can result in seizures. Patients who are sensitive to one TCA may be sensitive to other TCAs.

The effect of smoking on the activity of CYP1A2 does not seem to have an effect on the plasma concentrations of the secondary TCAs because CYP1A is not involved with the *N*-dealkylation to their primary amine metabolites.

For common patient information and recommendations, see Table 18.5.

UNIQUE PROPERTIES FOR THE SPECIFIC SECONDARY AMINE TCAs

Desipramine Desipramine is a dihydrodibenzazepine secondary amine TCA that also is the active metabolite of imipramine (Fig. 18.8). Desipramine appears to have a bioavailability comparable to the other secondary TCAs (Table 18.3). Desipramine is distributed into milk in concentrations similar to those present at steady-state in maternal plasma. This drug is metabolized primarily by CYP2D6 to its 2-hydroxy metabolite and by CYP1A2

and CYP2C19 to its *N*-demethylated (primary amine) metabolite (Table 18.2).

Desipramine exhibits a greater potency and selectivity for the NET than do the other secondary TCAs (Fig. 18.6). Its antidepressant effect results from increases in the level of NE in CNS synapses, and long-term administration causes a downregulation of α_1 -adrenoceptors and desensitization of presynaptic α_2 -receptors, equilibrating the noradrenergic system and, thus, correcting the dysregulated output of depressed patients. The SSRIs do not produce this effect. Desipramine also downregulates NET, but not SERT. Substantial loss of NET-binding sites takes 15 days to occur and is accompanied by a marked reduction of NET function in vivo. Desipramine has weak effects on 5-HT reuptake.

Nortriptyline Nortriptyline is a secondary amine dibenzocycloheptene TCA (Fig. 18.8) as well as the major metabolite of amitriptyline. Similar to desipramine, nortriptyline appears in mother's milk and is metabolized by CYP2D6 to the primary amine and by ring hydroxylation to its *E*-10-hydroxy metabolite (Table 18.2). Approximately one-third of a dose of nortriptyline is excreted in urine as metabolites within 24 hours, and small amounts are excreted in feces via biliary elimination.

Amoxapine Amoxapine is a dibenzoxazepine TCA (Fig. 18.8) with antidepressant and antipsychotic effects that has shown therapeutic effectiveness in patients with delusional depression. Additionally, it is the *N*-desmethyl metabolite of the antipsychotic loxapine. Amoxapine differs structurally from the other secondary TCAs in that it has both a nitrogen and an oxygen atom in its seven-membered central ring and a piperazinyl ring rather than a propylamino side chain attached to the central ring.

Amoxapine is a less potent inhibitor of neuronal NE reuptake compared with the other secondary TCAs, with a mechanism of action similar to that of desipramine.

TABLE 18.5 Patient Information and Recommendations for Secondary Amines Tricyclic Antidepressants (TCAs)

Patient Information	Recommendations
Potential drug–drug and drug–health interactions	Share medical conditions, other medicines (including over-the-counter and herbal medicines), allergies to TCAs, fertility status, or breast-feeding with pharmacist
Seizures, breathing difficulties, fever and sweating, loss of bladder control, muscle stiffness, unusual weakness or tiredness	Discontinue therapy and consult physician (TCAs may increase risk of seizures)
Course of therapy	Complete full course of therapy
Discontinuation of therapy	Consult physician; abrupt discontinuation not recommended, because it may cause nausea, headache, and malaise
Alcohol use	Avoid alcohol
Central nervous system depressants	May exacerbate TCAs
Drowsiness, dizziness, blurred vision	Avoid driving or performing tasks requiring alertness and coordination
Sun/sunlamp exposure	Avoid prolonged exposure because of photosensitivity

Amoxapine shares the toxic potentials of the TCAs, and the usual precautions of TCA administration should be observed. Amoxapine resembles the atypical antipsychotic drugs in its intermediate affinity as an antagonist of DA-2 and of 5-HT₂ receptors.

Amoxapine is rapidly and almost completely absorbed from the gastrointestinal (GI) tract. Its pharmacokinetics are shown in Table 18.3. Amoxapine and its 8-hydroxyamoxapine metabolite have been detected in human milk at concentrations below steady-state therapeutic concentrations. Amoxapine has the shortest elimination time (~8 hours) of the secondary TCAs. It is metabolized in the liver principally to 8-hydroxyamoxapine and to 7-hydroxyamoxapine. Both of these metabolites are pharmacologically active and have half-lives of 30 and 6.5 hours, respectively. The hydroxylation of amoxapine is inhibited by ketoconazole, suggesting the involvement of CYP3A4.

Protriptyline Protriptyline is a dibenzocycloheptene TCA that differs from the other tricyclics by having an unsaturated ethylene bridge joining the two aromatic rings and a secondary aminopropyl side chain (Fig. 18.8). Protriptyline is completely absorbed from the GI tract and slowly eliminated. Its pharmacokinetic data are shown in Table 18.3. Metabolism data are limited for protriptyline, but it is most likely metabolized via the same pathways as the other TCAs (Table 18.2). Very little drug is excreted in the feces via the bile.

Protriptyline exhibits high selectivity for the NET, but with less potency than desipramine. Its mechanism of action is similar to that of desipramine. Minimal effect on 5-HT reuptake has been observed.

Maprotiline Maprotiline is a secondary amine dibenzobicyclooctadiene (a tetracyclic antidepressant) that differs structurally from the TCAs by having an ethylene bridge in its central ring, resulting in a rigid bicyclic molecular skeleton (Fig. 18.8).

Maprotiline exhibits the highest affinity and selectivity for the NET (Fig. 18.6). Its antidepressant mechanism of action is similar to that of desipramine, with an onset of action of up to 2 to 3 weeks.

Maprotiline is slowly but completely absorbed from the GI tract, and like the other TCAs, it is metabolized by the polymorphic CYP2D6 and CYP2C19 isoforms in the liver, primarily to pharmacologically active *N*-desmethylomaprotiline and to maprotiline-*N*-oxide. Its pharmacokinetics are shown in Table 18.3. Maprotiline is distributed into breast milk at concentrations similar to those found at steady-state in maternal blood. The elimination half-life of maprotiline averages 43 hours (60 to 90 hours for its *N*-desmethyl metabolite).

Maprotiline shares the toxic potentials of the TCAs, and the usual precautions of TCA administration should be observed. Although most of the TCAs have been reported to induce seizures, it is generally recognized that maprotiline may be associated with a higher incidence of

dose-dependent seizures compared with the other secondary TCAs. Maprotiline has been reported to produce sedation in depressed patients and to reduce aggressive behavior in animals. Maprotiline also shares the anticholinergic and cardiovascular effects of the secondary TCAs and may cause electrocardiographic changes, tachycardia, and postural hypotension.

NONTRICYCLIC SECONDARY AMINE ANTIDEPRESSANTS

Reboxetine Reboxetine is a nontricyclic SNRI in which the propylamine side chain of the TCAs is constrained into a morpholine ring (Fig. 18.8). It is a potent and selective ligand for the NET, with a mechanism of action similar to that of desipramine and with low affinity for β -adrenergic and muscarinic receptors. Reboxetine is used for the treatment of major depressive disorders. It is a chiral compound that is marketed as a racemic mixture of *R,R*- and *S,S*-reboxetine. The antidepressant activity for reboxetine appears to reside with the *S,S*(+)-enantiomer, which has approximately twofold the inhibition potency of the *R,R*-enantiomer (42). It is well tolerated, with different adverse event profiles, and it seems to be at least as effective as the SSRIs in the treatment of depressive illness. Currently, it is available only in Europe and is under FDA review. It preferentially inhibits the reuptake of NE (5-HT:NE ratio = 8). Reboxetine is not metabolized by the polymorphic isoforms CYP2D6 or CYP2C19 and may offer a valuable alternative to the secondary amine TCAs in the treatment of major depression. Reboxetine is likely to become a promising alternative for patients who have failed treatment with or do not tolerate serotonergic antidepressants. Reboxetine has been shown to be effective and well tolerated in the treatment of panic disorder.

Reboxetine's oral bioavailability is greater than 90% for both *R,R*(-)-reboxetine and *S,S*(+)-reboxetine, indicating that stereoisomerism has no significant effect on absorption and first-pass metabolism. The pharmacokinetics of reboxetine are linear after multiple oral doses of 12 mg/d (Table 18.3). Food affects the rate but not the extent of absorption. In human liver microsomes, each reboxetine enantiomer was metabolized to one primary metabolite, *O*-desethylreboxetine, and three minor metabolites, two arising via oxidation of the ethoxy aromatic ring and a third yet-unidentified metabolite (43). The metabolism of both reboxetine enantiomers in humans is mediated principally via CYP3A4, and they are competitive inhibitors of CYP2D6 and CYP3A4. Less than 10% of the unmetabolized drug is cleared renally, with the other 90% being excreted hepatically.

After a single, 4-mg dose, the plasma concentration in the elderly (mean age, 81 years) was twice that of younger subjects, although in both groups, the plasma concentration was similar after 2 hours. The area under the curve (AUC) was nearly four times greater in the elderly than in the younger subjects, and the elimination half-life was twice as long (24 ± 6 vs. 12 ± 3 hours). Renal clearance also was reduced. The increased plasma concentrations

of reboxetine observed in elderly subjects supports a reduction of the starting dose to 4 mg/d (in two divided doses) for elderly patients (44). Because of reduced metabolic clearance, reboxetine plasma concentrations also are increased in patients with hepatic or renal dysfunction. In these populations, reboxetine should be used with caution, and a dosage reduction is indicated. No ethnic differences have been observed with *R,R*(-)- or *S,S*(+)-reboxetine.

Drug Interactions. Reboxetine seems to be an antidepressant that has negligible interference with the pharmacokinetics of other drugs; thus, fewer drug–drug interactions are expected. It also may be possible to use reboxetine in combination with MAOIs, because it has no inhibitory effect on this enzyme, which would avoid tyramine-induced hypertensive reactions.

Adverse Effects. Reboxetine is relatively well tolerated, with insomnia, sweating, constipation, and dry mouth being commonly reported adverse events. Hypotension and urinary hesitancy occur at lower rates than with the TCAs. When compared with the SSRIs, reboxetine is associated with lower rates of nausea, somnolence, and diarrhea.

STRUCTURE–ACTIVITY RELATIONSHIPS FOR THE PHENOXYPHENYLPROPYLAMINES

(±)-Nisoxetine Nisoxetine (Fig. 18.8) was the initial phenoxyphenylpropylamine synthesized in the Lilly research laboratories during the early 1970s from the rearrangement of an oxygen atom in diphenylhydramine, a diphenylmethoxyethylamine, to a phenoxyphenylpropylamine (Fig. 18.2). Nisoxetine was discovered to be a potent and very selective SNRI, with little affinity for other receptors. It underwent clinical studies as an alternative to Lilly's best-selling antidepressant, nortriptyline, but without the adverse effects associated with the tricyclic secondary amines. It was never marketed, however, because of a greater interest in developing its 4-trifluoromethyl analog, fluoxetine, an SSRI.

The type and position of the ring substitution plays a critical role in the mechanism of action for these phenoxyphenylpropylamines (see Table 18.6 for structure–activity relationships of the phenoxypropylamines). The unsubstituted molecule is a weak SSRI. However, 2-substitutions into the phenoxy ring (except for the 2-trifluoromethyl) yields compounds with high potency and selectivity for blocking NE reuptake (an SNRI), whereas the 4-substitution results in compounds having potent SSRI activity, with the 4-trifluoromethyl group (fluoxetine) being the most potent and selective for SERT (28,29). The substantial changes in transporter selectivity for NET and SERT and the differences in affinity are more likely attributed to the bulky 2-(ortho)-substituted groups, which restrict the flexibility of the aromatic rings, thereby enhancing alignment

of the hydrogen-bond acceptor group (the methoxy) with a donor group on the binding site on the NET for NE that is not available for the 5-HT binding site. The *R*-isomer of nisoxetine has 20 times greater affinity than its *S*-isomer for NET. The NET K_i for nisoxetine is 0.8 nM and is 40 times more selective for NET than for SERT. Its tertiary amine is approximately 100 times less effective at inhibiting NET. Increasing the size of the methylamino with ethyl or larger alkyl eliminates all activity. The 2- and 4-analogs exhibited weak effects on neuronal uptake of DA and lack affinity for other neuroreceptors at therapeutic concentrations. Substituting the 2-methoxy with the isosteric 2-methylthio (thionisoxetine) produced a more potent SNRI ($K_i = 0.2$ nM for the *R*-enantiomer) and 600 times more selective for NET than for SERT. Thionisoxetine is approximately 10 times more potent than nisoxetine at inhibiting NET, and unlike nisoxetine, it reduces food consumption in rodents and has been studied for the treatment of obesity and eating disorders. Substitution of the phenoxy group with a naphthyl group and the phenyl ring with the isosteric thienyl (thiophene) group results in a drug with dual inhibition of NE and 5-HT reuptake (i.e., duloxetine) (see Fig. 18.20 and the discussion of reuptake inhibitors of both NE and 5-HT for a description of duloxetine).

***R*(-)-Atomoxetine** *R*(-)-Atomoxetine, 2-methylphenoxyphenylpropylamine, was marketed in 2003 as a “non-stimulant” treatment for attention-deficit hyperactivity disorder (ADHD) in both adults and children and for treatment of adult depression. The 2-methyl substitution (c.f., nisoxetine; Fig. 18.8) confers selectivity for inhibiting NE reuptake (Table 18.6) (21,22,45). The *R*-enantiomer is 10 times more potent than the *S*-enantiomer as a NET reuptake inhibitor. Atomoxetine has a low propensity for anticholinergic and adverse cardiovascular effects.

Pharmacokinetics. Atomoxetine is well absorbed from the GI tract and cleared primarily by metabolism, with the majority of the dose being excreted into the urine. Atomoxetine is metabolized primarily by CYP2D6 to its major active metabolite, 4-hydroxyatomoxetine, which is eliminated as its glucuronide (Fig. 18.10). Peak plasma concentrations of atomoxetine occur 1 to 2 hours after oral administration. Significant differences are seen in the elimination half-life between normal metabolizers, extensive metabolizers, and poor metabolizers (Table 18.3). Atomoxetine exhibited an elimination half-life of 3 to 6 hours for normal and extensive metabolizers and 17 to 21 hours for poor metabolizers (46). CYP2C19 is the other enzyme primarily responsible for the formation of its minor metabolite *N*-desmethyldatomoxetine (46).

Adverse Effects. At therapeutic doses, no serious drug-related adverse effects have been encountered. Adverse

TABLE 18.6 Structure–Activity Relationships for Phenoxyphenylpropylamines

R (drug)	Inhibition of Reuptake (K_i nM) (20)	
	5-HT	NE
H	102	200
2-OCH ₃ (nisoxetine)	1,371	2.4
2-SCH ₃ (thionisoxetine)	130	0.2
2-CH ₃ (atomoxetine)	390	3.4
2-F	898	5.3
2-I	25	0.4
2-CF ₃	1,489	4,467
3-CF ₃	16	1,328
4-CF ₃ (fluoxetine)	17	2,703
4-CF ₃ (norfluoxetine, NH ₂)	17	2,176
4-CH ₃	95	570
4-OCH ₃	71	1,207
4-Cl	142	568
4-F	638	1,276

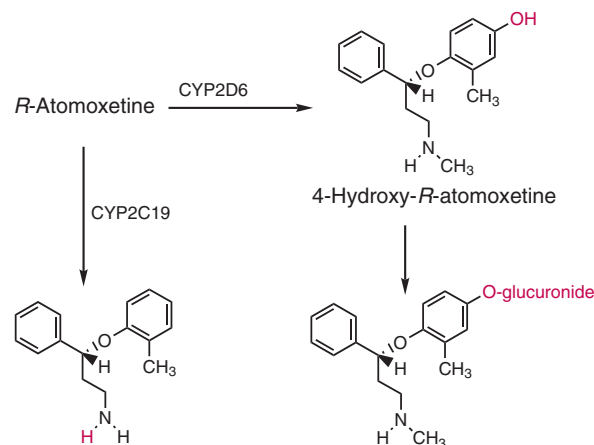
effects have included modest increases in diastolic blood pressure and heart rate, anorexia, weight loss, somnolence, dizziness, GI effects (nausea), dry mouth, and skin rash.

Therapeutic Uses. Atomoxetine is used as a safe and well-tolerated “nonstimulant” treatment of ADHD in both adults and children and of depression. Among children and adolescents age 8 to 18 years, atomoxetine was superior to placebo in reducing symptoms of ADHD and in improving social and family functioning symptoms. Oral atomoxetine is promoted as an alternative to conventional ADHD therapy with methylphenidate, dextroamphetamine, and pemoline. It also can be a replacement for bupropion or for TCAs. Onset of action is approximately 7 days.

Selective 5-HT Reuptake Inhibitors (SSRIs)

Serotonin Hypothesis of Depression

5-HT is a major player in depressive illness, and serotonergic pathways are closely related to mood disorders,

**FIGURE 18.10** Metabolism of *R*-atomoxetine.

especially depression (Fig. 18.1) (11,47). Thus, drugs affecting the 5-HT levels in the neural synapse and serotonergic pathways may lead to effective therapy of depression.

5-HT is synthesized from tryptophan, packaged into vesicles, and released into the synaptic cleft following an action potential (see Chapter 11). Once in the synaptic cleft, 5-HT interacts with both the pre- and postsynaptic serotonergic receptors. Evidence implicating multiple abnormalities in serotonergic pathways as a cause of depression includes:

1. Low urinary concentrations of 5-HT's major metabolite, 5-hydroxyindoleacetic acid.
2. A low density of brain and platelet SERT in depressed individuals.
3. A high density of brain and platelet 5-HT binding sites.
4. A low synaptic concentration of tryptophan, which is used in 5-HT synthesis.

Of these, the low level of SERTs in depressed patients has received the most attention in the development and synthesis of the SSRIs. The precise antidepressant mechanism of action for the SSRIs eludes neuroscientists, but the SSRIs have been shown to alleviate depression and are the most commonly used drugs in the therapy for depression. Claims of decreased adverse effects (adverse drug reactions) and less toxicity in overdose than both the MAOIs and the TCAs, together with increased safety, have led to their extensive use, and several are ranked in the top 50 prescription drugs dispensed in the United States during the year 2009.

The SSRIs are proven treatments for depression, obsessive-compulsive disorder, and panic disorder and are helpful in a variety of other conditions as well. The most substantial benefit to the SSRIs compared with the TCAs is their reduced adverse effect profile and the fact that they are better tolerated. Although the SSRIs

have become the most commonly prescribed drugs for depression, there are clinical situations in which TCAs may be more appropriate (e.g., melancholic depression). Meaningful differences between the individual SSRIs are largely related to their pharmacokinetics, metabolism to active metabolites, inhibition of CYP isoforms, effect of drug–drug interactions, and the half-life of the individual SSRI.

The SSRIs are expensive, and it has been common to have noncompliant patients (especially elderly) relapse because they cannot afford their medications. Persuading the patient to take his or her medication as prescribed is extremely important for potentially suicidal patients with depression. Because of this, it is exceedingly important that patients receive the lowest effective (and, thereby, the most cost-effective) dose of any drug they are prescribed. Also, the SSRIs have a history of increased risk of suicide for reasons that are not clearly understood.

The Discovery of SSRIs

Although the TCAs, as a group, are effective antidepressants, their adverse event profile and high potential for toxicity have limited their use. The early antidepressants indicated that 5-HT might play a significant role in depression. Therefore, medicinal chemists set out in search of the ideal SSRI with the goal for developing drugs with:

- High affinity and selectivity for the SERT.
- Ability to slow or inhibit the transporter when bound to it.
- Low affinity for the multiple neuroreceptors known to be responsible for many of the adverse effects of the TCAs (e.g., acetylcholine, histamine, and adrenergic receptors).
- No inhibition of the fast sodium channels that cause the cardiotoxicity problems associated with TCAs.

Initial success occurred with the synthesis of zimelidine, in which the central ring of amitriptyline was opened to form a diphenylpropylidene analog. Zimelidine displayed selective inhibition of 5-HT reuptake, with minimal inhibition of NE reuptake. Most importantly, zimelidine was without the adverse event profile exhibited by the TCAs. Thus, zimelidine became the template for the second-generation SSRIs as shown in Figure 18.11.

Mechanisms of Action Common to the SSRIs

The SSRIs preferentially act to inhibit SERT with minimal or no affinity for NET and DAT (11). These drugs have a high and selective affinity for SERT (Fig. 18.6) and, therefore, block 5-HT from binding to SERT and being absorbed into presynaptic cells. The excess 5-HT in the synaptic cleft means overactivation of the postsynaptic receptors. Over an extended period of time, this causes downregulation of pre- and postsynaptic receptors, a

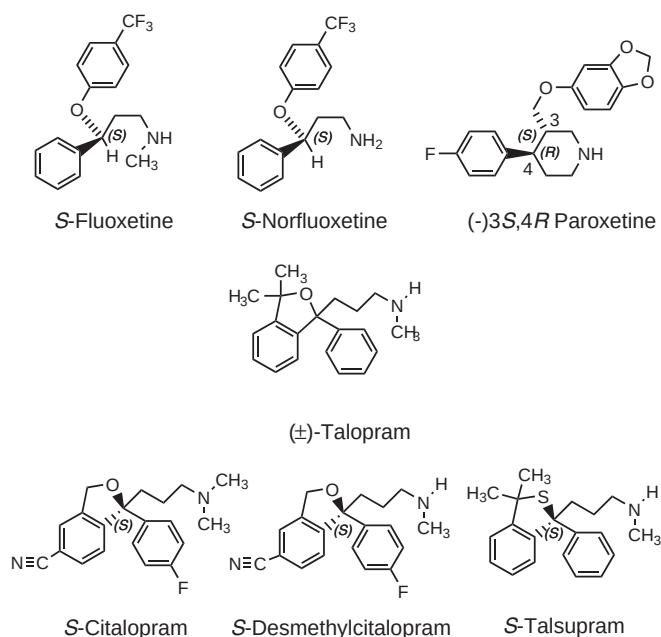


FIGURE 18.11 Phenoxyphenylalkylamine SSRIs.

reduction in the amount of 5-HT produced in the CNS, and a reduction in the number of SERTs expressed. Long-term administration of SSRIs causes downregulation of the SERT, but not the NET. Substantial loss of SERT binding sites takes 15 days to occur and is accompanied by a marked reduction of SERT function in vivo. These compensatory responses at receptors and transporters are thought to produce the antidepressant effects of SSRIs. This onset delay may, in part, explain the delayed onset of action of SSRIs in the treatment of depression (48). Similar to the binding of 5-HT, SSRIs likely bind to SERT at the same site as 5-HT does, although it has not been determined conclusively. Although not as selective as the SSRIs, drugs of abuse, such as cocaine, fenfluramine, and 3,4-methylenedioxymethamphetamine (“Ecstasy”), are inhibitors of SERT.

The affinity data for the SSRIs show that the SSRIs, as a group, are very potent and selective inhibitors for SERT compared with their affinity for NET and DAT (Fig. 18.6) and are more potent inhibitors of 5-HT reuptake than are the tertiary amine TCAs, with the exception of clomipramine. None of the SSRIs has substantial effect on NET or DAT. Of the SSRIs, sertraline exhibits the most potent inhibition of DAT, although it is still 100 times less potent in terms of inhibiting DAT versus SERT. Therefore, the plasma concentration of sertraline would have to be increased by as much as 100 times to inhibit DAT. When drugs are this selective for reuptake transporters, differences in potency become clinically irrelevant because the plasma concentration can be dose-adjusted to achieve inhibition of the desired transporter without affecting the other transporters. Clomipramine displays less affinity for SERT than citalopram, fluvoxamine, paroxetine,

or sertraline does and is more potent than fluoxetine. In terms of the ability to inhibit the NET, the SSRIs are two to three times less potent than the SNRI TCA, desipramine.

Their *in vitro* potency for selectively inhibiting the SERT more or less mirrors their clinical efficacy as SSRIs (11): paroxetine > sertraline > clomipramine > fluoxetine > citalopram > fluvoxamine > imipramine > amitriptyline > reboxetine > venlafaxine = milnacipran > desipramine. Clinically, however, all the SSRIs are equally effective over time, suggesting that these variations in potency do not affect efficacy or adverse effects. The SSRIs have less affinity for α_1 , α_2 , H_1 , and muscarinic receptors, which may explain the adverse effect profile differences between TCAs and SSRIs.

The results in Table 18.7 show the therapeutic doses that produce approximately 70% to 80% inhibition of the SERT (48). The inhibition of SERT is relevant to the antidepressant efficacy of the SSRIs and suggests that approximately 70% to 80% inhibition of this SERT usually is necessary to produce an antidepressant effect. Higher doses of these drugs do not produce a greater antidepressant response on average (i.e., a flat dose-response curve for antidepressant efficacy) but do increase the incidence and severity of adverse effects mediated by excessive 5-HT reuptake inhibition. Obviously, the results shown in Table 18.7 pertain to the average patient. A patient who has a rapid clearance of the SSRI may need a higher-than-average dose to achieve an effective concentration, whereas a patient who has a slow clearance may do better in terms of the ratio of efficacy to adverse effects on a minimum dose.

The β -adrenergic blocker pindolol blocks the presynaptic 5-HT_{1A} receptors, thereby increasing 5-HT neuronal transmission. The 5-HT_{1A} receptors do not

require prolonged exposure (several weeks) to excessive amounts of 5-HT to promote downregulation. This results in augmentation and acceleration of the antidepressant effect of the SSRIs when combined with a 5-HT_{1A} inhibitor. Bordet et al. (49) demonstrated the accelerated antidepressant response of pindolol with paroxetine.

Pharmacokinetics Common to the SSRIs

The SSRIs share a number of pharmacokinetic characteristics (48,50) (Table 18.8). They are well absorbed orally, although the presence of food in the stomach may alter the absorption of some SSRIs. Food, however, does not affect the AUC and does not appear to affect clinical efficacy. The SSRIs are highly lipophilic and are highly plasma protein bound.

Current SSRIs tend to be characterized by high volumes of distribution, which results in relatively long plasma concentration (typically 4 to 8 hours). The SSRIs display a range of elimination half-life values for the parent drugs, from half-life values of approximately 20 hours for paroxetine and fluvoxamine to 2 days for fluoxetine. Only sertraline and citalopram exhibit linear pharmacokinetics, whereas fluvoxamine, fluoxetine, and paroxetine exhibit nonlinear pharmacokinetics (i.e., changes in plasma concentration are not proportional to dose) as a result of their longer plasma half-lives within the usual therapeutic ranges (Table 18.8). Sertraline stands out as having the best effects on depression among all antidepressants. Fluoxetine and fluvoxamine are least likely to penetrate into breast milk. Thus, the SSRI antidepressants best suited for pharmacokinetic optimization of therapy are the following: sertraline, fluvoxamine, and citalopram. All the SSRIs are extensively metabolized by CYP isoforms to pharmacologically active *N*-demethylated metabolites, which are then excreted in urine and feces. Except for sertraline, the drugs fluoxetine, paroxetine, and fluvoxamine are metabolized by polymorphic CYP isoforms, a matter of concern for poor and extensive metabolizers who may need dose adjustments. Citalopram is metabolized almost equally by CYP2C19, CYP2D6, and CYP3A4 (Table 18.7). Peak plasma levels usually are reached in approximately 6 to 8 hours, and steady-state plasma levels are reached in approximately 7 to 10 days, except for fluoxetine (~4 weeks). The half-lives are variable depending on the specific SSRI and the presence and plasma concentration of an active metabolite, but the half-lives tend to be prolonged. No evidence indicates that serum drug monitoring of SSRIs is a useful strategy to predict response. The SSRIs in general exhibit a flat, dose-independent antidepressant response curve (i.e., the antidepressant activity does not improve with increasing dose, only side effects increase) (50).

Some of the key differences among the SSRIs are the result of differences in their pharmacokinetic properties and metabolism to active metabolites (Table 18.8). Fluoxetine is unique because of its long half-life and

TABLE 18.7 Relationship Between Dose, Plasma Level, Potency, and Serotonin Uptake

SSRI	Dose (mg/d)	Plasma Level	In Vitro Potency (IC ₅₀)	Inhibition of SERT
Citalopram	40	85 ng/mL (260 nM)	1.8	60%
Fluoxetine	20	200 ng/mL (300 nM/L) ^a	3.8	70%
Fluvoxamine	150	100 ng/mL (300 nM/L)	3.8	70%
Paroxetine	20	40 ng/mL (130 nM/L)	0.29	80%
Sertraline	50	25 ng/mL (65 nM/L)	0.19	80%

^aPlasma level for fluoxetine represents the total of fluoxetine plus norfluoxetine given comparable effects on SERT; parent SSRI alone shown for all others. Also, plasma levels are a total of both enantiomers for citalopram and fluoxetine. Values for the parent drug and for the respective major metabolite are in parentheses.

TABLE 18.8 Pharmacokinetics of the Selective Serotonin Reuptake Inhibitors (SSRIs)

Parameters	(±)-Fluoxetine	(-)-Sertraline	(-)-Paroxetine	<i>E</i> -Fluvoxamine	(±)-Citalopram [(+)-Escitalopram]
Oral bioavailability (%)	70	20–36	50	>50	80 (51–93)
Lipophilicity (logD _{7.4})	2.38	3.03	1.49	1.21	1.52
Protein binding (%)	95	96–98	95	77	~56 (70–80)
Volume of distribution (L/kg)	12–18	17 (9–25)	25 (12–28)	15–31	12–16
Elimination half-life (h)	50	24 (19–37)	22	15–20	36 (27–32); elderly ~48
Cytochrome P450 major isoform	2D6	3A4	2D6	2D6	2C19, 2D6, and 3A4
Major active metabolites	<i>O</i> -Desmethyl-fluoxetine (240 h) Norfluoxetine (96–364 h)	<i>N</i> -Desmethyl (62–104 h)	None	None	<i>N</i> -Desmethylcitalopram (30 h)
Peak plasma concentration (h)	6–8	4–8	2–8	3–8	4 (1–6)
Excretion (%)	Urine 25–50 Feces minor	Urine 51–60 Feces 24–32	Renal ~50 Feces minor	Urine ~40 (24 h) Feces minor	Feces 80–90 Urine <5
Plasma half-life (h)	1–4 d (norfluoxetine 7–15 d)	22–35	24	7–63	36 (23–75)
Time to steady-state concentration	~4 wk	7–10 d; elderly 2–3 wk	7–14 d	10 d	7 d

the long half-life of its active metabolite norfluoxetine (48,50). Although sertraline also has an active metabolite, it is 10 times less potent than sertraline and probably is not clinically relevant (51). Fluvoxamine and paroxetine have no active metabolites. Because of the differences in half-lives and activities of metabolites, a much longer washout period is necessary when switching from fluoxetine (a long-acting SSRI) to another SSRI or MAOI. These differences can cause considerable therapeutic delays in the treatment of refractory patients.

Adverse Effects Common to the SSRIs

The SSRIs are reported to have fewer side effects than the TCAs, which have strong anticholinergic and cardiotoxic properties (50). Among the SSRIs, there are few differences in adverse effects. The adverse effects observed for the SSRIs include nausea, diarrhea, anxiety, agitation, insomnia, and sexual dysfunction. Fewer patients have discontinued SSRIs than TCAs (amitriptyline and imipramine and not nortriptyline, desipramine, doxepin, and clomipramine).

Sexual dysfunction is reported in both men and women, such as of decreased libido, anorgasmia, ejaculatory incompetence, ejaculatory retardation, or inability to obtain or maintain an erection. The basic

pharmacologic similarities among the SSRIs suggest that the effects on sexual function should be similar for each drug and that no one SSRI was more likely to cause the reported sexual adverse effects than another. Moreover, evidence suggests that SSRI-induced sexual dysfunction may be dose related and may be treated by simply lowering its dose. In patients who cannot have their SSRI dosage reduced, another option is simply to wait and reassess sexual function after several months. If the above measures are ineffective in managing SSRI-induced sexual dysfunction, the next step is to consider an alternative antidepressant without serotonergic activity (e.g., bupropion). Orgasm difficulties and impotence occurred more frequently with paroxetine as compared with sertraline and fluoxetine. The addition of amantadine, cyproheptadine, yohimbine, or sildenafil has been reported to be effective in some patients with SSRI-induced sexual dysfunction.

Drug Interactions Common to the SSRIs

The most serious drug–drug interaction for the SSRIs is their potential to produce the “serotonin syndrome” (i.e., hyperserotonergic effect), which typically develops within hours or days following the addition of another serotonergic agent to a drug regimen that already

includes serotonergic-enhancing drugs. Symptoms of the 5-HT syndrome include agitation, diaphoresis, diarrhea, fever, hyperreflexia, incoordination, confusion, myoclonus, shivering, or tremor.

The 5-HT syndrome interaction between MAOIs and SSRIs is the most important drug interaction for the SSRIs, necessitating a washout ranging from 2 to 5 weeks depending on the plasma half-life of the SSRI. These differences in washout times for the SSRIs when switching to an MAOI are key differences between SSRIs and should be remembered if an MAOI is planned as a possible subsequent treatment in the event of SSRI failure. The differences among SSRIs are not important when a patient is switched from an MAOI to an SSRI. However, in this case, a 10- to 14-day washout for the MAOI is necessary, regardless of which SSRI is used, to allow regeneration of monoamine oxidase. The drug interaction between TCAs and SSRIs is of particular importance because of the potential for the development of toxic TCA concentrations, 5-HT syndrome, and subsequent adverse effects (48,50).

Coadministration of the antihistamine cyproheptadine or other 5-HT antagonists with SSRIs might be expected to result in a pharmacodynamic interaction (i.e., reduced effectiveness for the SSRI). Cyproheptadine acts to block postsynaptic 5-HT. Lack of antidepressant efficacy has been reported when cyproheptadine was given concurrently with fluoxetine and paroxetine.

Clinically, the potency of the SSRIs to inhibit CYP2D6 decreases from paroxetine to fluoxetine to norfluoxetine and then to fluvoxamine, with sertraline and citalopram being metabolized by CYP3A4 and CYP2C19, respectively, explaining the extent of differences in pharmacokinetic interactions between the SSRIs and other CYP2D6 substrates. Fluvoxamine is associated with drug interactions from its inhibition of CYP1A2, CYP2C9, CYP2C19, and CYP3A4 (see Table 4.10 for inhibition drug interactions). Because all the SSRIs are extensively metabolized in the liver, it is possible that other drugs that inhibit or induce hepatic CYP microsomal enzyme systems may alter SSRI plasma concentrations (AUCs) (Table 18.7). The SSRIs may inhibit or interfere with the metabolism of other frequently prescribed drugs that are CYP hepatically metabolized, increasing the potential for drug-drug interactions (Table 18.2; see also Tables 4.5 and 4.6). Although similar drug interactions are possible with other SSRIs, there is considerable variability among the drugs in the extent to which they inhibit CYP2D6. Fluoxetine and paroxetine appear to be more potent in this regard than sertraline. The extent to which this potential interaction may become clinically important depends on the extent of inhibition of CYP2D6 by the SSRI and the therapeutic index of the concurrently administered drug. The drugs for which this potential interaction is of greatest concern are those that are metabolized principally by CYP2D6 and

have a narrow therapeutic index. Caution should be exercised whenever concurrent therapy with fluoxetine and other drugs metabolized by CYP2D6 is considered. The clinical significance of these possible interactions with the CYP isoforms is questionable, however, because there is no known correlation between plasma concentration and therapeutic response for any of the SSRIs (50). If an interaction is suspected, the patient's SSRI dosage can be easily adjusted.

The SSRIs are highly protein bound and may affect the pharmacodynamic effect of other protein-bound drugs with narrow therapeutic indices (e.g., warfarin). The changes appear to be clinically significant, however, only for fluoxetine, fluvoxamine, and paroxetine (50). Close monitoring of prothrombin time and international normalized ratio is necessary if these drugs are used together.

The SSRIs have a high toxic to therapeutic ratio and, therefore, are safer than the TCAs or MAOIs in acute overdose. SSRI overdoses can result in drowsiness, tremor, nausea and vomiting, seizures, electrocardiographic changes, and coma. Fatalities are uncommon with pure SSRI overdoses.

Therapeutic Uses Common to the SSRIs

The primary uses for the SSRIs include MMD and bipolar depression (fluoxetine, paroxetine, sertraline, and citalopram), "atypical" depression (i.e., depressed patients with unusual symptoms such as hypersomnia, weight gain, and interpersonal rejection sensitivity; fluoxetine, paroxetine, sertraline, and citalopram), anxiety disorders, panic disorder (sertraline and paroxetine), dysthymia, premenstrual syndrome, postpartum depression, dysphoria, bulimia nervosa (fluoxetine), obesity, borderline personality disorder, obsessive-compulsive disorder (fluvoxamine, fluoxetine, paroxetine, and sertraline), alcoholism, rheumatic pain, and migraine headache. Among the SSRIs, there are more similarities than differences; however, the differences between the SSRIs could be clinically significant.

The SSRIs, such as paroxetine and fluoxetine, need stronger pediatric use warnings because of the possible risks of suicidal thoughts and behavior in some children and teenagers. Such risks may be unrelated to any specific SSRI. Recent clinical trials have concluded that 2 or 3 of every 100 young people treated with antidepressants might be at higher risk of suicidal behavior. Only fluoxetine has been proven to be effective and is approved for the treatment of pediatric depression.

Phenoxyphenylalkylamines

(±)-FLUOXETINE

Structure-Activity Relationship Fluoxetine is a 3-phenoxy-3-phenylpropylamine that exhibits selectivity and high affinity for human SERT and low affinity for NET (Fig. 18.11). It is marketed as a racemic mixture of *R*- and

S-fluoxetine. Its selectivity for SERT inhibition depends on the position of the substituent in the phenoxy ring (Table 18.6). Mono-substitution in the 4-(para) position of the phenoxy group (with an electron-withdrawing group, e.g., trifluoromethyl group, as in fluoxetine) results in selective inhibition of 5-HT reuptake. Disubstitution (2,4- or 3,4-substitution) results in loss of SERT selectivity. Constraining fluoxetine into semirigid analogs, such as MDL28618A or a phenylpiperidine (i.e., femoxetine), maintains selectivity for SERT, but both have approximately 10% of the affinity of fluoxetine for SERT (52) (Fig. 18.12). The *trans*-(1*S*,2*S*)-MDL28618A stereoisomer is approximately 10 times more potent than the *cis*-(-)-enantiomer (52,53). The *trans*-(3*R*,4*S*)-(+)-enantiomer of femoxetine has approximately 10% the affinity of fluoxetine for SERT (54). *N*-Demethylation of femoxetine to its secondary amine enhances affinity for SERT by 10 times (comparable to fluoxetine). Femoxetine is an analog not only of fluoxetine and paroxetine but also of the (3*R*,4*S*)-diastereomer of a paroxetine analog.

Mechanism of Action Fluoxetine is a potent and selective inhibitor of 5-HT reuptake, but not of NE or DA uptake in the CNS. Its mechanism of action is common to the SSRIs. Fluoxetine does not interact directly with postsynaptic 5-HT receptors and has weak affinity for the other neuroreceptors. Both enantiomers of fluoxetine display similar affinities for human SERT. The NE:5-HT selectivity ratio, however, indicates that the *S*-enantiomer is approximately 100 times more selective for SERT inhibition than the *R*-enantiomer. The *R*-(+)-stereoisomer is approximately eight times more potent an inhibitor of SERT and has a longer duration of action than the *S*-(-)-isomer. However, the *S*-(-)-norfluoxetine metabolite is seven times more potent as an inhibitor of the 5-HT transporter than the *R*-(+)-metabolite, with a selectivity ratio approximately equivalent to that of *S*-fluoxetine (54,55).

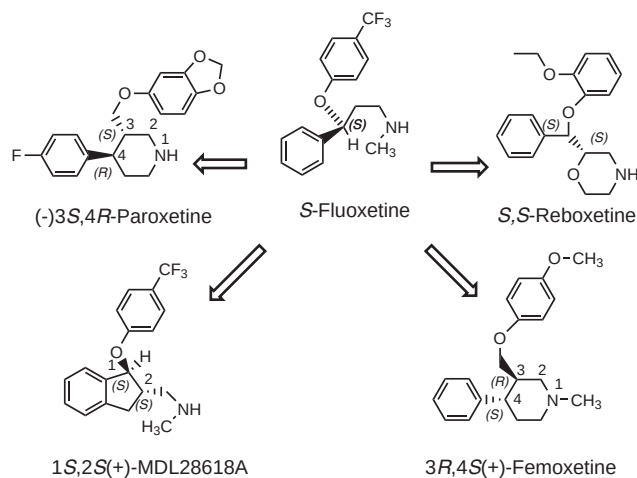


FIGURE 18.12 Derivatives of fluoxetine.

Pharmacokinetics The pharmacokinetics of fluoxetine fit the general characteristics of the SSRIs (Table 18.8). Of particular importance is its long half-life contributing to its nonlinear pharmacokinetics. In vitro studies show that fluoxetine and norfluoxetine are potent inhibitors of CYP2D6 and CYP3A4 and less potent inhibitors of CYP2C9, CYP2C19, and CYP1A2. Fluoxetine is metabolized primarily by CYP2D6 *N*-demethylation to its active metabolite norfluoxetine and, to a lesser extent, *O*-dealkylation to form the inactive metabolite *p*-trifluoromethylphenol. Following oral administration, fluoxetine and its metabolites are excreted principally in urine, with approximately 73% as unidentified metabolites, 10% as norfluoxetine, 10% as norfluoxetine glucuronide, 5% as fluoxetine *N*-glucuronide, and 2% as unmetabolized drug.

Both *R*- and *S*-norfluoxetine were less potent than the corresponding enantiomers of fluoxetine as inhibitors of NE uptake. Inhibition of 5-HT uptake in cerebral cortex persisted for more than 24 hours after administration of *S*-norfluoxetine similarly to fluoxetine. Thus, *S*-norfluoxetine is the active *N*-demethylated metabolite responsible for the persistently potent and selective inhibition of 5-HT uptake in vivo (54).

The pharmacokinetics of fluoxetine in healthy geriatric individuals do not differ substantially from those in younger adults. Because of its relatively long half-life and nonlinear pharmacokinetics, the possibility of altered pharmacokinetics in geriatric individuals could exist, particularly in those with systemic disease and/or in those receiving multiple medications concurrently. The elimination half-lives of fluoxetine and norfluoxetine do not appear to be altered substantially in patients with renal or hepatic impairment.

Drug Interactions Fluoxetine and its norfluoxetine metabolite, like many other drugs metabolized by CYP2D6, inhibit the activity of CYP2D6 and, potentially, may increase plasma concentrations of concurrently administered drugs that also are metabolized by this enzyme. Fluoxetine may make normal CYP2D6 metabolizers resemble poor metabolizers. Fluoxetine can inhibit its own CYP2D6 metabolism, resulting in higher-than-expected plasma concentrations during upward dose adjustments. Therefore, switching from fluoxetine to another SSRI or other serotonergic antidepressant requires a washout period of at least 5 weeks or a lower-than-recommended initial dose with monitoring for adverse events.

Fluoxetine is highly protein bound and may affect the free plasma concentration and, thus, the pharmacologic effect of other highly protein-bound drugs (e.g., warfarin sodium).

PAROXETINE

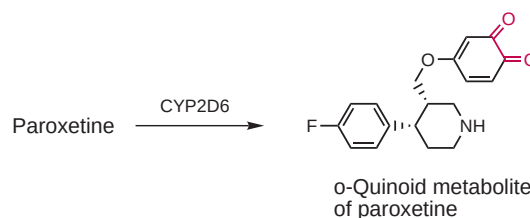
Structure-Activity Relationship Paroxetine is a constrained analog of fluoxetine in which the linear phenylpropylamine group has been folded into a piperidine ring (Fig. 18.11). Paroxetine contains two chiral centers, with

the possibility of four stereoisomers. One of these stereoisomers, the (3*S*,4*R*)-(-)-enantiomer, is marketed as paroxetine. Paroxetine is a potent and selective inhibitor of SERT and displays high affinity for human SERT and little affinity for NET and DAT (Fig. 18.6). Converting the secondary amine of the piperidine ring into a tertiary amine with a methyl group reduces affinity for SERT by 100 times (Fig. 18.13). Substituting the 4-fluoro with either a hydrogen or methyl reduces affinity for human SERT by approximately 10 times; replacing the 3,4-methylenedioxy group with a 4-methoxy group in the phenoxy ring also reduces affinity by a factor of 10. Stereochemical factors affect affinity of the paroxetine molecule for SERT. Therefore, substitution into the 2-(ortho) position of either aromatic ring decreases affinity for rat SERT by as much as 10 to 100 times, with the greatest loss occurring with substitution in the phenoxy ring. In vitro binding studies suggest that paroxetine is a more selective and potent inhibitor of 5-HT reuptake than fluoxetine. The drug essentially has no effect on NE or DA reuptake and does not show affinity for other neuroreceptors. Its onset of action is 1 to 4 weeks.

Pharmacokinetics Paroxetine appears to be slowly but well absorbed from the GI tract after oral administration, with an oral bioavailability of approximately 50%, suggesting first-pass metabolism (Table 18.8), and reaching peak plasma concentrations in 2 to 8 hours. Food does not substantially affect the absorption of paroxetine. Paroxetine is distributed into breast milk. Approximately 80% of an oral dose of paroxetine is oxidized by CYP2D6 to a catechol intermediate, which is then either O-methylated or O-glucuronidated. These conjugates are then eliminated in the urine.

Paroxetine exhibits a preincubation-dependent increase in inhibitory potency of CYP2D6 consistent with a mechanism-based inhibition of CYP2D6 (55). The

inactivation of CYP2D6 occurs via the formation of an o-quinonoid reactive metabolite.



The methylenedioxy has been associated with mechanism-based inactivation of other CYP isoforms (56,57). In contrast, fluoxetine, a potent inhibitor of CYP2D6 activity, did not exhibit a mechanism-based inhibition of CYP2D6. As a result of mechanism-based inhibition, saturation of CYP2D6 at clinical doses appears to account for its nonlinear pharmacokinetics observed with increasing dose and duration of paroxetine treatment, which results in increased plasma concentrations of paroxetine at low doses. The elderly may be more susceptible to changes in doses and, therefore, should be started off at lower doses. After oral administration, paroxetine and its metabolites are excreted in both urine and feces.

Oral administration of a single dose resulted in unmetabolized paroxetine accounting for 2% and metabolites accounting for 62% of the excretion products. The effect of age on the elimination of paroxetine suggests that hepatic clearance of paroxetine can be reduced, leading to an increase in elimination half-life (e.g., to ~36 hours) and increased plasma concentrations. The metabolites of paroxetine have been shown to possess no more than 2% of the potency of the parent compound as inhibitors of 5-HT reuptake; therefore, they are essentially inactive.

Because paroxetine is a potent mechanism-based inhibitor of CYP2D6, this type of inhibition yields nonlinear and long-term effects on drug pharmacokinetics, because the inactivated or complexed CYP2D6 must be replaced by newly synthesized CYP2D6 protein. Thus, coadministration of paroxetine with CYP2D6-metabolized medications should be closely monitored or, in certain cases, avoided, as should upward dose adjustment of paroxetine itself.

(±)-CITALOPRAM In trying to create a new antidepressant to inhibit NE reuptake, Lundbeck chemists accidentally synthesized two new compounds (talopram and tasulopram) having the phenylspiro-isobenzofuran nucleus (Fig. 18.11). These compounds were potent SNRIs, but considering that a number of suicide attempts were reported during clinical studies with these compounds, Lundbeck discontinued the studies. Undeterred, the chemists subsequently modified talopram by addition of a 5-cyano to the phthalan ring and a 4-fluoro to the benzene ring in formation of citalopram (Fig. 18.11). Therapeutic activity for

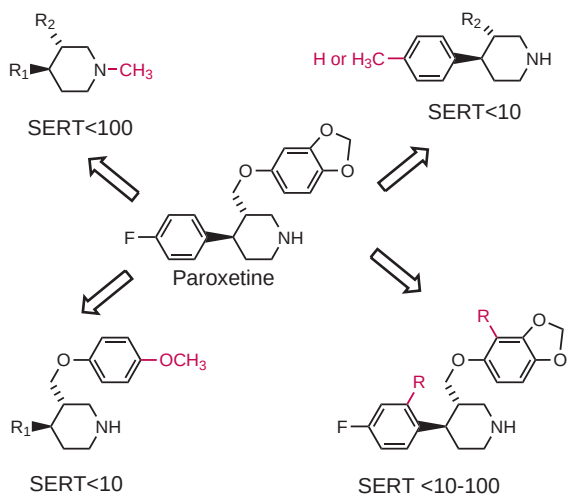


FIGURE 18.13 Structural changes to paroxetine and the resulting decrease in SERT affinity.

($\pm$)-citalopram resides in the *S*(+)-isomer. Isosteric substitution of the isobenzofuran ring in citalopram with an isobenzothiophene yields talsupram, which changes selectivity from an inhibitor of SERT to a potent inhibitor of NET (Fig. 18.11). Citalopram was marketed in the United States in 1996 as the most selective SSRI and, therefore, as the least likely to cause the adverse effects observed with most of the other antidepressants (Fig. 18.6).

Mechanism of Action Citalopram, primarily through its *S*-enantiomer, blocks 5-HT reuptake, leading to potentiation of serotonergic activity in the CNS. Citalopram exhibits the greatest in vitro selectivity for 5-HT reuptake inhibition compared with the other SSRIs (Fig. 18.6). The drug essentially has no effect on NE or DA reuptake, nor does it show affinity for other neuroreceptors.

Pharmacokinetics The pharmacokinetics of citalopram are shown in Table 18.8. Unlike several of the other SSRIs, citalopram does not undergo first-pass metabolism; it has an oral bioavailability of approximately 80%. The drug is metabolized via hepatic *N*-demethylation to its major metabolite, *N*-desmethylecitalopram, almost equally by CYP2C19, CYP2D6, and CYP3A4 (Table 18.7). The major metabolite exhibits approximately 50% of the potency of citalopram as an inhibitor of 5-HT reuptake. Because the metabolite concentration in the plasma is lower than that of citalopram, it should not add significantly to citalopram's antidepressant effects. Citalopram exhibits dose-proportional linear pharmacokinetics in a dosage range of 10 to 60 mg/d; plasma levels increase proportionately with each increasing dose. Approximately 10% to 12% of an oral dose was recovered in the urine as unmetabolized drug. The clearance of orally administered citalopram was reduced by 37% and 17% in patients with hepatic and renal function impairment, respectively.

Citalopram and its desmethyl metabolite are weak inhibitors of the CYP isoforms, suggesting a low potential for drug interactions. Although no relevant in vivo interactions between citalopram and CYP2D6-metabolized medications have been reported, caution is advised when coadministering citalopram with potential object drugs, especially those having narrow therapeutic indices, in elderly patients. Because citalopram is metabolized in parallel by CYP2C19, CYP2D6, and CYP3A4, it would have little inhibitory effect on the metabolism of other drugs metabolized by these isoenzymes. Citalopram is less highly protein bound than the other SSRIs, reducing the potential for drug interactions with protein-bound drugs having narrow therapeutic indices.

ESCITALOPRAM Escitalopram is the *S*-enantiomer of citalopram that binds with high affinity and selectivity to the human SERT equivalent to ($\pm$)-citalopram. It has

been reported that nearly all the activity resides in the *S*-enantiomer and that *R*-citalopram actually counteracts the action of the *S*-enantiomer (58,59). Studies show that escitalopram exhibits twice the activity of citalopram and is at least 27 times more potent than the *R*-enantiomer. The *R*-enantiomer inhibits the *S*-enantiomer at SERT (58,59). Escitalopram's mechanism of action is common to the SSRIs.

The pharmacokinetics for escitalopram do not exhibit stereoisomer selectivity and, therefore, are similar to those for citalopram (Table 18.7). Likewise, it exhibits linear pharmacokinetics so that plasma levels increase proportionately and predictably with increased doses, and its half-life of 27 to 32 hours is consistent with once-daily dosing. It also has been found that *R*-citalopram is cleared more slowly than the *S*-enantiomer. Therefore, when the drug is used as a racemic mixture (citalopram), the inactive isomer predominates at steady-state. This is an added incentive for use of the enantiomerically pure escitalopram. Escitalopram has negligible effects on CYP isoforms, suggesting a low potential for drug-drug interactions. Escitalopram is indicated for patients with major depressive disorder, generalized anxiety disorder, panic disorder, and social anxiety disorder.

Escitalopram is metabolized to *S*-desmethylecitalopram by CYP2C19 (37%), CYP2D6 (28%), and CYP3A4 (35%) and to *S*-didesmethylecitalopram (only by CYP2D6) in human liver microsomes and in expressed cytochromes. Escitalopram and its desmethyl metabolite are negligible inhibitors of CYP1A2, CYP2C9, CYP2C19, CYP2E1, and CYP3A and are weakly inhibited by CYP2D6. *R*-Citalopram and its metabolites have properties very similar to those of the corresponding *S*-enantiomers. Because escitalopram is biotransformed by three CYP isoforms in parallel, escitalopram is unlikely to be affected by drug interactions or genetic polymorphisms and is unlikely to cause clinically important drug interactions via CYP inhibition.

Phenylalkylamine

SERTRALINE Although sertraline appears to differ structurally from the other SSRIs, it is a phenylaminotetralin, in which the diphenylbutylamine nucleus is constrained into a rigid bicyclic ring system (Fig. 18.14). In the early work with the discovery of SSRIs at Pfizer, tetratraline was initially synthesized in 1978. Animal studies showed it to be a stimulant and to block NE and DA uptake, a use that Pfizer was not interested in pursuing. Subsequently, one or two chlorine atoms were introduced into tetratraline to produce new molecules that were potent inhibitors of 5-HT reuptake in the brain. One of the dichloro compounds was to become known as sertraline.

Sertraline contains two chiral centers and only the *S,S*(+)-diastereomer is marketed. The *R,R*, *R,S*, and *S,R*-diastereomers are significantly weaker as inhibitors of 5-HT reuptake. Sertraline was marketed in the United

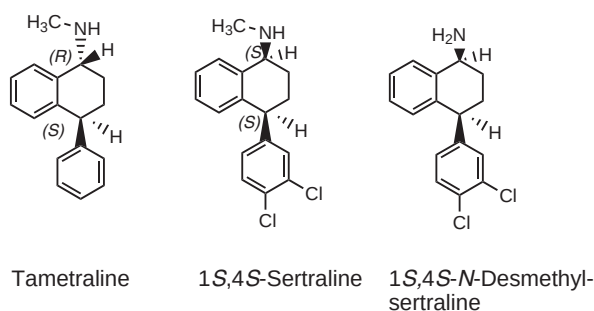


FIGURE 18.14 Phenylalkylamine SSRIs.

States in 1992, emphasizing its pharmacokinetic differences from the other SSRIs.

Mechanism of Action Sertraline is a potent and selective inhibitor of the neuronal reuptake SERT. In vitro binding studies suggest that sertraline has a substantially higher selectivity for inhibiting 5-HT reuptake than other SSRIs or TCAs, including clomipramine (Fig. 18.6). It has only weak effects on neuronal uptake of NE and DA. Its mechanism of action is common to the SSRIs. Sertraline is very selective, lacking affinity for other neuroreceptors at therapeutic concentrations.

Pharmacokinetics Sertraline appears to be slowly but well absorbed from the GI tract after oral administration. The oral bioavailability of sertraline in humans ranges from 20% to 36% (Table 18.8), suggesting extensive first-pass metabolism to its *N*-desmethylated metabolite. Food enhances its oral absorption, decreasing the time to achieve peak plasma concentrations from approximately 8 to 6 hours. After multiple doses, steady-state plasma sertraline concentrations are proportional and linearly related to dose (half-life: single dose, 24 hours; multiple dose, 24 hours). *N*-Desmethylsertraline, sertraline's principal metabolite, exhibits dose-dependent pharmacokinetics. Sertraline and *N*-desmethylsertraline are distributed into breast milk. Although in elderly patients the elimination half-life is increased to approximately 36 hours, this effect does not appear to be clinically important and does not warrant dosing alterations. Sertraline is primarily metabolized to *N*-desmethylsertraline through the action of CYP2B6, CYP2C9, CYP2C19, CYP2D6, and CYP3A4 (Fig. 18.15). Of these, CYP2B6 appears to be the major isoform involved (60). *N*-Desmethylsertraline is approximately 5 to 10 times less potent as an inhibitor of 5-HT reuptake than sertraline. The formation of sertraline ketone can occur either through the action of CYP or MAO, but studies have not been reported to indicate the role of either or both. The formation of sertraline *N*-carbamoyl glucuronide is a unique product that was shown to form in vitro through the action primarily of UGT2B7. The involvement of multiple enzymes in the metabolism of sertraline suggests that no single drug

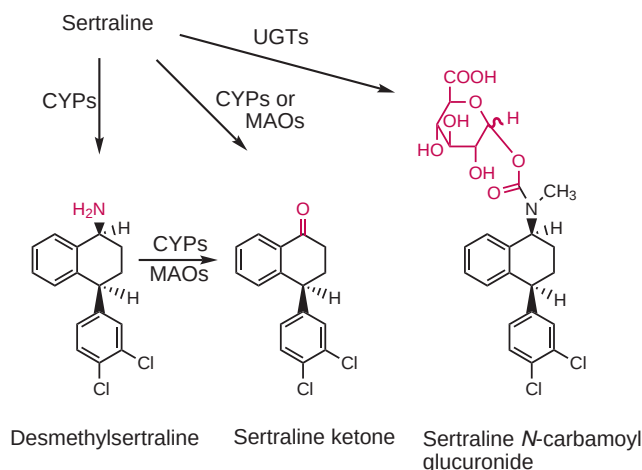


FIGURE 18.15 Metabolic products formed from sertraline metabolism.

could significantly alter sertraline pharmacokinetics and reduce any drug–drug interactions. Sertraline and *N*-desmethylsertraline undergo further metabolism via oxidative deamination and ring hydroxylation and glucuronide conjugation. *N*-Desmethylsertraline has an elimination half-life approximately 2.5 times that of sertraline. Following oral administration, sertraline and its conjugated metabolites are excreted in both urine and feces, and unmetabolized sertraline accounts for less than 5% of oral dose. Plasma clearance of sertraline was approximately 40% lower in geriatric patients. The elimination half-life of sertraline in patients with hepatic disease was prolonged to a mean of 52 hours, compared with 22 hours in individuals without hepatic disease.

Drug Interactions Sertraline is not a potent inhibitor of CYP3A4, and because CYP2D6 metabolism is a minor pathway for sertraline, drug–drug interactions with these isoforms are unlikely to be of clinical importance. Sertraline is metabolized by more than one CYP isoform in parallel; therefore, genetic polymorphisms are unlikely to cause clinically significant drug interactions via CYP isoform inhibition. Caution is advised, however, when coadministering sertraline with potential object drugs, especially those with narrow therapeutic indices in elderly patients. For example, sertraline has been shown to reduce the clearance of desipramine and imipramine as a result of CYP2D6 inhibition.

Because sertraline is highly protein bound, patients receiving it concurrently with any highly protein-bound drug should be observed for potential adverse effects associated with combined therapy.

Additional SSRIs

FLUVOXAMINE Fluvoxamine is a nontricyclic SERT inhibitor that is structurally unique among the SSRIs by being the (*E*)-isomer of a 2-aminoethyl oxime ether of

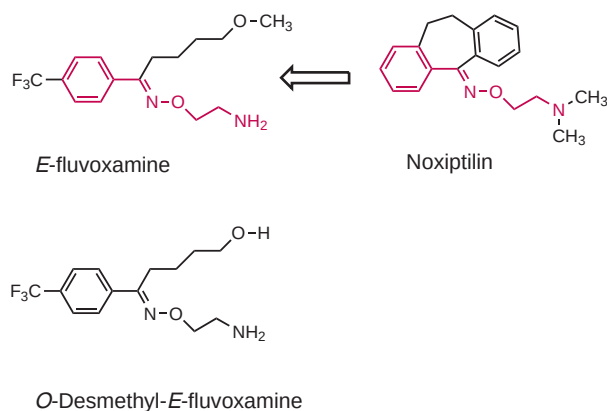


FIGURE 18.16 Miscellaneous SSRIs.

an aralkylketone (Fig. 18.16). The C=N double bond is isosteric with the propylidene group in amitriptyline and, thus, imparts geometric *E* or *Z* stereoisomerism to fluvoxamine. The oxime ether is found in a previously marketed analog of amitriptyline called noxiptilin (1966 by Bayer AG [Germany]). Thus, fluvoxamine may be considered to be an open-chain analog of the tricyclic noxipitilin. The 4-trifluoromethyl group or other electronegative group is essential for SERT affinity and selectivity. The C=N double bond also enhances the susceptibility of fluvoxamine to photoisomerization by ultraviolet (UV)-B light (290 to 320 nm). When fluvoxamine solutions were exposed to UV-B light, photoisomerization to the pharmacologically inactive *Z*-isomer occurred. Thus, fluvoxamine solutions should be protected from sunlight to prevent loss of antidepressant efficacy. No studies have been reported regarding its solid-state stability to UV-B light.

Mechanism of Action Fluvoxamine is a highly selective inhibitor of 5-HT reuptake at the presynaptic membrane. Potency data from in vitro affinity studies suggest that fluvoxamine is less potent than the other SSRIs (e.g., paroxetine, sertraline, and citalopram). Its mechanism of action is similar to that of the other SSRIs. Fluvoxamine appears to have little or no effect on the reuptake of NE or DA. In vitro studies have demonstrated that fluvoxamine possesses virtually no affinity for other neuroreceptors. Its onset of action is similar to the other SSRIs (2 to 4 weeks).

Pharmacokinetics Fluvoxamine is well absorbed, with a bioavailability of approximately 50%, probably because of first-pass metabolism (Table 18.8). At steady-state doses, fluvoxamine demonstrates nonlinear pharmacokinetics over a dosage range of 100 to 300 mg/d, which results in higher plasma concentrations at higher doses than would be predicted by lower dose kinetics (single dose, 15 hours; multiple dosing, 22 hours). Food does not significantly affect oral bioavailability. The mean apparent volume of distribution for fluvoxamine reflects its

lipophilic nature, extensive tissue distribution, and protein binding. Fluvoxamine is distributed into breast milk. Fluvoxamine is preferentially metabolized by CYP2D6 in the liver by *O*-demethylation to its alcohol metabolite, which subsequently is oxidized to a carboxylic acid. Oxidative deamination and nine other metabolites have been identified, none of which shows significant pharmacologic activity.

Adverse Effects The adverse effects for fluvoxamine include symptoms of drowsiness, nausea or vomiting, abdominal pain, tremors, sinus bradycardia, and mild anticholinergic symptoms. Toxic doses could produce seizures and severe bradycardia.

Drug Interactions In vitro studies have shown fluvoxamine to be a potent inhibitor of CYP1A2, to inhibit CYP3A4 and CYP2C19, and to weakly inhibit CYP2D6. The bioavailability of fluvoxamine is significantly decreased in smokers compared with nonsmokers, possibly because of induction of CYP1A metabolism of fluvoxamine. Therefore, interactions with drugs that inhibit CYP1A2 also should be considered (e.g., theophylline and caffeine).

Therapeutic Uses Fluvoxamine is approved for use in obsessive-compulsive disorders.

Norepinephrine and Serotonin Reuptake Inhibitors (NSRIs)

The NSRI antidepressant drugs in this class block both the NET and SERT (i.e., they combine the mechanisms of action for both the SSRIs and SNRIs), exhibiting dual affinity for NET and SERT (low NE:5-HT potency ratio). These dual inhibitors, as a group, do not show a significant separation in selectivity for the NET and SERT reuptake transporter. Historically, the tertiary amine TCAs displayed dual inhibition of 5-HT and NE presynaptic reuptake, but they also bind to other types of neuroreceptors, which is responsible for their narrow therapeutic window and adverse effects. Clinical studies suggest that dual-acting inhibitors of 5-HT and NE reuptake may be more beneficial than selective inhibitors in managing depression. This has given impetus to the search for nontricyclic NSRIs and has led to a second group of NSRIs, such as venlafaxine and duloxetine.

Tricyclic Tertiary Amine Antidepressants

The TCAs in this class belong to the tertiary amine TCAs (Fig. 18.17). The relatively low bioavailability for the tertiary amine TCAs suggests first-pass metabolism (*N*-demethylation) to their secondary amine active metabolites (nor or desmethyl metabolites) and aromatic ring hydroxylation (Table 18.2). Despite the fact that steady-state serum plasma levels are reached within 1 to 2 days, their onset of antidepressant action typically is at least 2 to 3 weeks or longer. Their volume of distribution

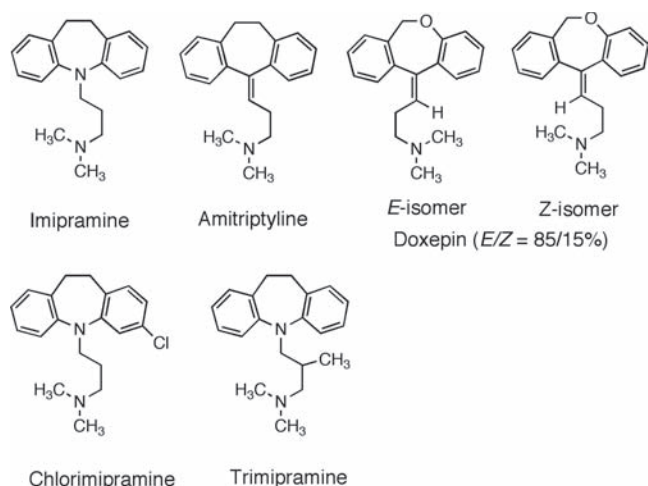


FIGURE 18.17 Tricyclic norepinephrine and serotonin reuptake inhibitors.

is very high, suggesting distribution into the CNS and protein binding. Excretion is primarily as metabolites via renal elimination. Renal and liver function can affect the elimination and metabolism of the parent TCA and its metabolites, leading to increased potential for adverse effects, especially in patients (i.e., elderly) with renal disease.

Meaningful differences between the tertiary TCAs are largely related to their pharmacokinetics, metabolism to active metabolites, inhibition of CYP isoforms, potential for drug–drug interactions, and half-life of the tertiary TCA.

MECHANISMS OF ACTION COMMON TO THE TERTIARY AMINE TCAs The exact mechanism of action for the tertiary TCAs is unclear, but it is known that the parent tertiary amine TCA exhibits dual inhibition of the NET and SERT. The result is an increase in both NE and 5-HT concentrations in the synaptic cleft. The *in vivo* antidepressant activity for these TCAs is more complex, however, because of the formation of secondary amine TCA metabolites, which in many cases annuls the 5-HT affinity of the parent TCA, leading to NET selectivity. The plasma concentrations for the secondary amine metabolites usually are higher than those of their parent tertiary amine TCA because of rapid *N*-demethylation metabolism. Note that none of the tertiary amine TCAs have any significant affinity for the DAT. During chronic therapy with the tertiary amine TCAs, downregulation of the noradrenergic and serotonergic receptors occurs, which is a result of neurotransmitter hypersensitivity caused by the continued high concentrations of NE and 5-HT at the postsynaptic receptor. The tertiary TCAs are less potent inhibitors than the SSRIs for SERT.

THERAPEUTIC USES COMMON TO TRICYCLIC TERTIARY ANTIDEPRESSANTS For the most part, the therapeutic uses for the tertiary amine

TCAs are very similar as a group, but these TCAs may be used in different cases of depression because of their variability in their dual mechanism of action as SNRIs and SSRIs. Their efficacy in treating depression suggests that mixed inhibition of NET and SERT influences depression by parallel and independent pathways. The tertiary TCAs may offer an option in the treatment of major depression for patients who have failed treatment with or who do not tolerate SNRIs or SSRIs.

ADVERSE EFFECTS COMMON TO TERTIARY TCAs Because of their potent and multiple pharmacodynamic effects at histamine H_1 , muscarinic, and α_1 -adrenergic receptors, the tertiary TCAs exhibit greater anticholinergic, antihistaminic, and α_1 -antiadrenergic adverse effects than the secondary TCAs do. Increased cardiotoxicity or frequency of seizures is higher for the tertiary TCAs than for the secondary TCAs, because they are potent inhibitors of sodium channels, leading to changes in nerve conduction. Cardiotoxicity can occur at plasma concentrations approximately 5 to 10 times higher than therapeutic blood levels. These concentrations can occur in individuals who take an overdose of the TCA or who are slow metabolizers and develop higher plasma concentrations on what usually are therapeutic doses.

DRUG–DRUG INTERACTIONS COMMON TO TERTIARY TCAs For the most part, drug–drug interactions for the tertiary amine TCAs are very similar to those for the secondary amine TCAs. These reactions, however, may be more pronounced.

The concurrent use of tertiary amine TCAs with SSRIs and other serotonergic drugs may result in 5-HT syndrome (see the discussion of drug interactions of SSRIs). Coadministration of a tertiary amine TCA with an MAOI is potentially hazardous and may result in severe adverse effects associated with 5-HT syndrome.

Because protein binding of the tertiary amine TCAs is high, displacement interactions with other highly protein-bound drugs and drugs with narrow therapeutic indices, although not fully evaluated, may be important. Concurrent use of tertiary amine TCAs with anticholinergic drugs requires close supervision and careful adjustment of the dosage because of potential additive anticholinergic effects (i.e., spastic colon). An additional disadvantage of TCAs is their toxicity from overdosage, especially in those who are being treated for depression who are having suicidal thoughts.

The plasma concentrations of tertiary amine TCAs usually are lower and plasma concentrations of their secondary amine metabolites usually are higher as a result of CYP1A induction.

PATIENT INFORMATION COMMON TO THE TERTIARY AMINE TCAs Common patient information and recommendations are similar to those given to patients who are prescribed secondary TCAs (Table 18.5).

UNIQUE PROPERTIES OF SPECIFIC TERTIARY AMINE TCAs

Amitriptyline Amitriptyline is a tertiary amine dibenzocycloheptadiene TCA with a propylidene side chain extending from the central carbocyclic ring (Fig. 18.17). The diarylpropylideneamine moiety for amitriptyline makes it sensitive to photo-oxidation; therefore, its hydrochloride solutions should be protected from light to avoid ketone formation and precipitation.

Pharmacokinetics. Amitriptyline is rapidly absorbed from the GI tract and from parenteral sites. Its pharmacokinetics are shown in Table 18.9. Amitriptyline and its active metabolite, nortriptyline, are distributed into breast milk. Amitriptyline is primarily (65%) metabolized by *N*-demethylation by CYP2D6 to nortriptyline and hydroxylation to its *E*-10-hydroxy metabolite. Nortriptyline is pharmacologically active as a secondary amine TCA. Amitriptyline shows approximately equal affinity for SERT and NET.

Imipramine Imipramine is a 10,11-dihydrodibenzazepine tertiary amine TCA (Fig. 18.17) that is marketed as hydrochloride and pamoate salts, both of which are administered orally. Although the hydrochloride salt may be administered in divided daily doses, imipramine's long duration of action suggests that the entire oral daily

dose may be administered at one time. On the other hand, imipramine pamoate usually is administered as a single daily oral dose. Imipramine preferentially inhibits 5-HT reuptake over NE; however, the formation of its *N*-desmethyl metabolite removes whatever 5-HT activity imipramine had, with the net result of enhanced noradrenergic activity from inhibition of NE reuptake at the presynaptic neuronal membrane. Imipramine shares the pharmacologic and adverse effect profiles of the other tertiary TCAs.

The pharmacokinetics for imipramine are shown in Table 18.9. Imipramine is completely absorbed from the GI tract. Imipramine is primarily metabolized by CYP2D6 to its 2- and 10-hydroxylated metabolites and *N*-demethylated via CYP2C19 and CYP1A2 to desipramine, its *N*-monodemethylated metabolite, an SNRI.

Therapeutic Uses. Besides being used in the clinical treatment of depression, imipramine also has been used for the treatment of functional enuresis in children who are at least 6 years of age (25 mg daily administered 1 hour before bedtime, not to exceed 2.5 mg/kg daily).

Doxepin Doxepin is a tertiary amine dibenzoxepine derivative with an oxygen replacing one of the ethylene carbons in the bridge. The oxygen introduces

TABLE 18.9 Pharmacokinetics of the Tricyclic Norepinephrine and Serotonin Reuptake Inhibitors (NSRIs)

Parameters	Amitriptyline	Clomipramine	Doxepin	Imipramine	Trimipramine
Oral bioavailability (%)	31–61	~50	13–45	29–77	44
Lipophilicity (logD _{7.4})	2.53	2.52	2.49	2.28	2.35
Protein binding (%)	95	96–98	ND	89–95	95
Volume of distribution (L/kg)	12–18	17 (9–25)	12–28	23 (15–31)	~31
Elimination half-life (h)	10–26	32 (19–37)	11–16	11–25	9–11
Cytochrome P450 major isoform	2D6	2D6	2D6 and 2C19	2D6 and 2C19	2D6 and 2C19
Major active metabolites	Nortriptyline	Desclomipramine (54–77 h)	Nordoxepin (~30 h)	Desipramine	Desmethyltrimipramine (30 h)
Peak plasma concentration (h)	2–12	2–6 (4.7)	<2	1–2	<2
Excretion (%)	Urine 25–50	Urine 51–60 (14 d)	Renal ~50	Urine ~40 (24 h)	Feces 80–90
	Feces minor	Feces 24–32	Feces minor	Feces minor	Urine <5
Plasma half-life (h)	21 (10–46)	34 (22–84)	8–24	18 (9–24)	9
Time to steady-state concentration	—	1–2 wk	ND	ND	ND

ND, not determined.

asymmetry into the tricyclic ring system, resulting in the formation of two geometric isomers: *E* (*trans*) and *Z* (*cis*) (Fig. 18.17). No commercial attempt was made to separate the isomers; thus, doxepin is administered as an 85:15 mixture of *E*- and *Z*-stereoisomers, with the *Z*-isomer being the more active stereoisomer for inhibiting the reuptake of 5-HT (61). The *E*-isomer, on the other hand, inhibits the reuptake of NE. Unless otherwise specified, the reported in vitro and in vivo studies with doxepin were done with the 85:15 geometric mixture.

Mechanism of Action. Because doxepin is administered as an 85:15 mixture of geometric isomers, its mechanism of action and antidepressant properties reflect this ratio. Therefore, doxepin's selectivity for inhibiting presynaptic NE reuptake is most likely caused by the 85% presence of the *E*-isomer in the geometric mixture. Its antidepressant activity is similar to amitriptyline. Data suggest NE reuptake inhibitory potency comparable to imipramine and clomipramine; the fact that doxepin is an 85:15 mixture of *E*- and *Z*-geometric isomers clouds its true efficacy for SERT or NET. The formation of *N*-desmethyldoxepin results in inhibition of NE reuptake with enhanced noradrenergic activity. As a result of these mixed effects SERT and NET, doxepin shares the pharmacologic and adverse effect profiles of the other TCAs.

The pharmacokinetics for oral doxepin are described in Table 18.9. After oral dosing, no significant difference was found between the bioavailability of the *E*- and *Z*-isomers. The plasma concentrations of the doxepin isomers remained roughly those of the administered drug, whereas the ratio for the metabolites, *E*-*N*-desmethyldoxepin and *Z*-*N*-desmethyldoxepin, were approximately 1:1 (61,62). This similarity in ratios of metabolites is attributed to *E*-doxepin being primarily metabolized in parallel by CYP2D6 and CYP2C19, whereas *Z*-doxepin is primarily metabolized only by CYP2C19 and not at all by CYP2D6 (Table 18.2) (61). Its *Z*-*N*-demethylated metabolite is pharmacologically more active than its *E*-metabolite as an inhibitor of 5-HT and NE reuptake (61). Both isomers of doxepin showed large volumes of distribution and relatively short half-lives in plasma, suggestive of extensive distribution and/or tissue binding. Renal clearances did not differ for the isomers.

Clomipramine Clomipramine is considered to be the most powerful antidepressant ever made. This dihydrodibenzazepine TCA, with actions on both SERT and NET, was the last of the major TCAs to come to market. Initially, the FDA regarded it as another "me-too" drug, and accordingly, they did not license it. Subsequently, however, it was licensed for the treatment of obsessive-compulsive disorders. Clomipramine differs from imipramine only by the addition of a 3-chloro group (Fig. 18.17).

Mechanism of Action. Clomipramine is different from the other TCAs, exhibiting preferential selectivity for inhibiting the reuptake of 5-HT at the presynaptic neuronal membrane. Its antidepressant mechanism of action as an inhibitor of the SERT is reduced in vivo, however, because of the formation of its active metabolite, *N*-desmethyldomipramine, which inhibits the reuptake of NE. As a result of its common structure with the other TCAs, clomipramine shares the pharmacologic and adverse effect profiles of the other TCAs.

The efficacy of clomipramine relative to the other TCAs in the treatment of obsessive-compulsive disorder may be related to its potency in blocking 5-HT reuptake at the presynaptic neuronal membrane, suggesting a dysregulation of 5-HT for the pathogenesis of obsessive-compulsive disorder. Clomipramine appears to decrease the turnover of 5-HT in the CNS, probably because of a decrease in the release and/or synthesis of 5-HT.

Although in vitro studies suggest that clomipramine is approximately four times more potent than fluoxetine as a 5-HT reuptake inhibitor, in vivo studies suggest the opposite. This difference has been attributed to the relatively long elimination half-lives for fluoxetine and its principal serotonergic metabolite norfluoxetine. In addition, metabolism of clomipramine to its *N*-desmethyl secondary amine metabolite decreases the potency and selectivity of 5-HT reuptake inhibition of clomipramine, but not fluoxetine.

Pharmacokinetics. Clomipramine appears to be well absorbed from the GI tract following oral administration, with an oral bioavailability of approximately 50%, suggesting some first-pass metabolism (Table 18.9). Food does not appear to substantially affect its bioavailability. Clomipramine and its active metabolite, *N*-desmethyldomipramine, exhibit nonlinear pharmacokinetics at 25 to 150 mg daily. At dosages exceeding 150 mg daily, their elimination half-lives may be considerably prolonged, allowing plasma concentrations of both to accumulate, which may increase the incidence of plasma concentration-dependent adverse effects, particularly seizures. Because of the relatively long elimination half-lives of clomipramine and *N*-desmethyldomipramine, their steady-state plasma concentrations generally are achieved within approximately 1 to 2 weeks. Plasma concentrations of *N*-desmethyldomipramine generally are greater than those for clomipramine at steady-state conditions. Clomipramine crosses the placenta and is distributed into breast milk.

Clomipramine is primarily metabolized by CYP2D6 *N*-dealkylation to its pharmacologically active metabolite, the 2- and 8-hydroxylated metabolites and their glucuronides, and clomipramine *N*-oxide (Fig. 18.18). *N*-Dealkylation also involves CYP3A4, CYP2C19, CYP2C9, and CYP1A2. Like all the other secondary amine TCAs, *N*-desmethyldomipramine is significantly more potent

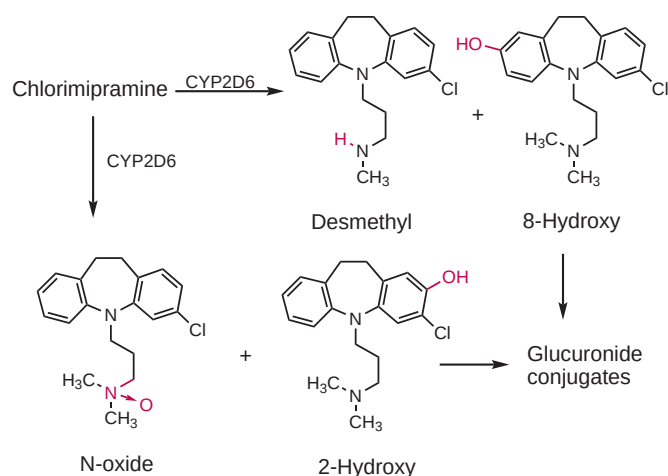


FIGURE 18.18 Metabolism of chlorimipramine.

as an inhibitor of NE reuptake than clomipramine. Although *N*-desmethylclomipramine is pharmacologically active, its efficacy in obsessive-compulsive disorder is not known. 8-Hydroxycloimipramine and 8-hydroxydesmethylclomipramine also are pharmacologically active, but their clinical importance remains unknown. The hydroxylation and *N*-demethylation of clomipramine highlight CYP2D6 polymorphism in healthy adults who were phenotyped as either extensive metabolizers or poor metabolizers of clomipramine. Interindividual variation in plasma concentrations may be caused by genetic differences in the metabolism of the drug. In addition, the CYP1A2 ring hydroxylates clomipramine. Less than 1% of an oral dose of clomipramine was excreted unmetabolized into the urine, with 8-hydroxycloimipramine glucuronide as the principal metabolite found in the urine. The effects of renal clearance suggest that clomipramine and desmethylclomipramine should be decreased in patients with renal impairment.

Pharmacogenetic differences in the metabolism of clomipramine after a single oral dose are apparent as increased plasma clomipramine concentrations in Indian and Pakistani patients compared with Caucasians. In Japanese patients, substantial interindividual variation in demethylation and hydroxylation of clomipramine was observed, although the prevalence of poor demethylators and poor hydroxylators of clomipramine has been estimated to be less than 1%.

If inhibition of SERT is critical to the desired clinical effect for clomipramine, then a patient may fail to respond because of higher levels of *N*-desmethylclomipramine as opposed to the parent drug. However, in a patient who has responded well and has stabilized to a dose of clomipramine, the drug might lose efficacy if exposed to an environmental agent that is capable of inducing CYP1A or CYP3A4.

Adverse Effects. Male patients taking clomipramine should be informed of sexual dysfunction as a side effect associated with antidepressants having

significant serotonergic activity. Sexual dysfunction in men appears as ejaculatory incompetence, ejaculatory retardation, decreased libido, or inability to obtain or maintain an erection. Sexual dysfunction is dose related and may be treated simply by lowering the drug dose.

Trimipramine Trimipramine also is a dihydrodibenzazepine TCA that differs structurally from imipramine in that the 5-propyl side chain is branched by a methyl group creating a chiral center (Fig. 18.17). Trimipramine is marketed as a racemic mixture. No data are available regarding the activity of the enantiomers. Apparently, branching the propyl side chain reduces affinity by 100 times for both SERT and NET, but the selectivity ratio favors the SERT. Although trimipramine has the weakest binding affinity for the monoamine transporters, it shares the pharmacologic and toxicity actions of the other TCAs and is used primarily in the treatment of depression.

The pharmacokinetics for trimipramine are shown in Table 18.9. Trimipramine is rapidly absorbed. Trimipramine demonstrates stereoselectivity in its metabolism to its three major metabolites. (–)-Trimipramine is primarily metabolized via CYP2D6 hydroxylation to 2-hydroxytrimipramine, whereas (+)-trimipramine is preferentially metabolized by CYP2C19 *N*-demethylation to desmethyltrimipramine. Desmethyltrimipramine is further hydroxylated to 2-hydroxydesmethyltrimipramine. (–)-Trimipramine is metabolized by CYP3A4/5 to an unknown metabolite (63,64). Most of the oral dose is excreted in urine in 72 hours, primarily as *N*-demethylated or hydroxylated and conjugated metabolites. The pharmacokinetics of trimipramine in geriatric individuals (≥65 years of age) do not differ substantially from those in younger adults.

Trimipramine is one of the antidepressants with the most pronounced differences in pharmacokinetics caused by the CYP2D6 genetic polymorphism (63,64). Its bioavailability and systemic clearance depend significantly on the CYP2D6 isoform with a linear dose relationship. Its mean bioavailability was 44% in individuals without CYP2D6 (poor metabolizers) but 16% and 12% in those individuals with two and three active genes of CYP2D6 (fast and ultrarapid metabolizers), respectively. Consequently, the mean total clearances of the oral dose were 27, 151, and 253 L/h in poor, extensive, and ultrarapid metabolizers, respectively. The 44% bioavailability combined with low systemic clearance of trimipramine in poor metabolizers of CYP2D6 substrates results in a very high exposure to trimipramine with the risk of adverse drug reactions. However, the presystemic elimination may result in subtherapeutic drug concentrations in carriers of CYP2D6 gene duplications with a high risk of poor therapeutic response (63,64).

OTHER TRICYCLIC NSRIs Figure 18.19 shows two tricyclic 5-HT/NE reuptake inhibitors. One is an analog of imipramine, and the second is an isostere of doxepin.

Dosulepin Dosulepin (formerly dothiepin) is the thio isostere of doxepin and is marketed in Europe as its single *E*-geometric isomer (Fig. 18.19), in contrast to the active *Z*-geometric isomer for doxepin. Its antidepressant activity is mediated by inhibition of both the NET and SERT, with preferential affinity for SERT (65). It exhibits greater overall in vitro affinity for NET and SERT than its oxygen isostere doxepin, consistent with its greater potency. Its overall therapeutic efficacy is similar to that of amitriptyline.

Although dosulepin is readily absorbed from the GI tract, it is extensively first-pass metabolized by *S*-oxidation in the liver to its primary active metabolite, dosulepin *S*-oxide, and by *N*-demethylation to its minor metabolite, desmethyldosulepin (northiaden) (65). Dosulepin is excreted in the urine, mainly in the form of its metabolites; small amounts of the drug also are excreted in the feces and distributed into breast milk.

The incidence of anticholinergic side effects is less among patients treated with dosulepin than with amitriptyline and without cardiotoxicity at therapeutic doses. The sedative/anxiolytic activity of dosulepin is similar to that of amitriptyline. Dosulepin is an effective treatment for patients with depressive symptoms of varying severity and coexisting anxiety (65).

Lofepramine Lofepramine differs from imipramine by the attachment of a *p*-chlorophenacyl moiety to the *N*-aminopropyl side chain (Fig. 18.19). This change confers enhanced lipophilicity and the potential of more rapid distribution into the CNS with greater in vitro affinity and selectivity for NET. Its mechanism of antidepressant action is attributed to its rapid metabolism to the secondary amine metabolite, desipramine, which selectively inhibits the neuronal uptake of NE (66).

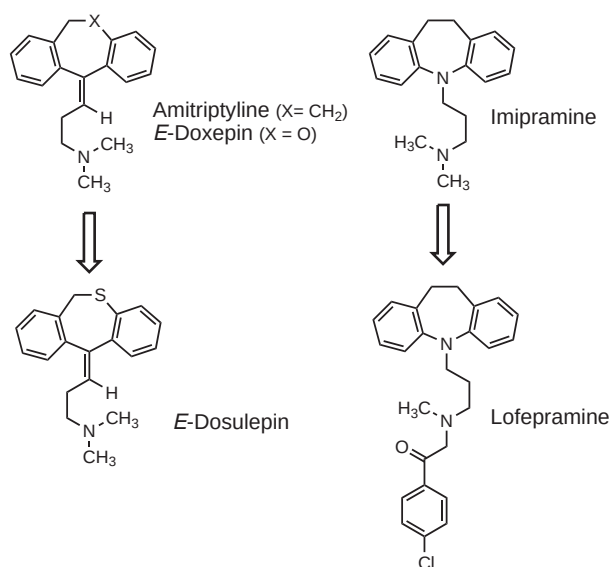


FIGURE 18.19 Future tricyclic NSRIs.

Lofepramine has an oral bioavailability of less than 10% as a result of the extensive first-pass metabolism. Most of a dose of lofepramine is excreted via the urine as metabolites, with a mean elimination half-life of approximately 2 hours. The adverse effects of lofepramine are similar to the secondary TCAs. Lofepramine displays reversible hepatotoxicity (jaundice and hepatitis), usually within 8 weeks of initiation of therapy.

Oral lofepramine is effective in the treatment of various types of depression and is similar in efficacy to imipramine and amitriptyline while inducing fewer adverse effects. This suggests its potential use in patients who are unable to tolerate other antidepressants.

NONTRICYCLIC NOREPINEPHRINE AND SEROTONIN REUPTAKE INHIBITORS

Clinical studies suggest that compounds that increase the synaptic availability of both NE and 5-HT have greater efficacy than single-acting drugs in the treatment of major depression. Thus began the efforts to design a drug that combined the properties of SSRIs and SNRIs that only blocked SERTs and NETs and without the unwanted adverse effects of TCAs (67). Currently, the nontricyclic dual inhibitors of 5-HT and NE uptake are venlafaxine, milnacipran, and duloxetine (Fig. 18.20). A structurally related nontricyclic SNRI is atomoxetine. Based on preclinical studies, venlafaxine, milnacipran, and duloxetine inhibit the reuptake of 5-HT and NE both in vitro and in vivo in the following order of decreasing potency: duloxetine > milnacipran > venlafaxine. All the dual reuptake inhibitors exhibit low affinity at neuronal receptors of the other neurotransmitters, suggesting a low side effect potential. Venlafaxine, milnacipran, and duloxetine have repeatedly been shown to be as efficacious as TCA drugs in treating major depressive disorders. These nontricyclic antidepressants have been reported to produce a faster and greater antidepressant response than an SSRI alone, suggesting that these two mechanisms of action can be

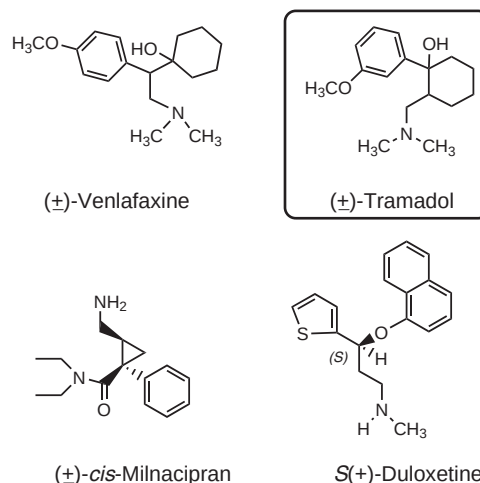


FIGURE 18.20 Nontricyclic NSRIs and related drug tramadol.

synergistic in terms of mediating antidepressant efficacy. Meaningful differences between the nontricyclic NSRIs are largely related to their pharmacokinetics, metabolism to active metabolites, inhibition of CYP isoforms, effect of drug–drug interactions, and the half-life of the nontricyclic NSRI.

Venlafaxine Venlafaxine is a methoxyphenylethylamine antidepressant that resembles an open TCA with one of the aromatic rings replaced by a cyclohexanol ring and a dimethylaminomethyl group rather than a dimethylaminopropyl chain (Fig. 18.20). It is structurally similar to the atypical opioid, tramadol (Ultram).

Venlafaxine and its active metabolite, *O*-desmethylvenlafaxine (ODV), have dual mechanisms of action, with preferential affinity for 5-HT reuptake and weak inhibition of NE and DA reuptake. Venlafaxine is approximately 30 times more potent as an inhibitor of SERT than of NET (68). Because of the 30-fold difference in transporter affinities, increasing the dose of venlafaxine from 75 to 375 mg/d can sequentially inhibit SERT and NET. Thus, venlafaxine displays an ascending dose-dependent antidepressant response in contrast to the flat dose–antidepressant response curve observed with the SSRIs. This sequential action for venlafaxine also is consistent with its dose-dependent adverse effect profile. Its mechanism of action is similar to imipramine.

Venlafaxine is rapidly and well absorbed (90%), but with a bioavailability of 15%, which has been attributed to first-pass metabolism (Table 18.10). Food delays its absorption but does not impair the extent of absorption. Venlafaxine is distributed into breast milk. Venlafaxine is primarily metabolized in the liver by CYP2D6 to its primary metabolite, ODV, which is approximately equivalent in pharmacologic activity and potency to venlafaxine. In vitro studies indicate that CYP3A4 also is involved in the metabolism of venlafaxine to its minor and less active metabolite, *N*-desmethylvenlafaxine (Fig. 18.21). Protein binding for venlafaxine and ODV is low and is not a problem for drug interactions. In patients with hepatic impairment, elimination half-lives were increased by approximately 30% for venlafaxine and approximately 60% for ODV (Table 18.10). In patients with renal function impairment, elimination half-lives were increased by approximately 40% to 50% for venlafaxine and for ODV. At steady-state doses, venlafaxine and ODV exhibit dose-proportional linear pharmacokinetics over the dose range of 75 to 450 mg/d. Steady-state concentrations of venlafaxine and ODV are attained within 3 days with regular oral dosing. Venlafaxine and its metabolites are excreted primarily in the urine (87%).

The potential for cardiotoxicity with venlafaxine during normal use and for various toxicities in overdose situations is a key concern. Venlafaxine displays minimal

TABLE 18.10 Pharmacokinetics of Nontricyclic Norepinephrine and Serotonin Reuptake Inhibitors (NSRIs)

Parameters	Venlafaxine	Desvenlafaxine (ODV)	Milnacipran	Duloxetine
Oral Bioavailability (%)	~15	80	90	>70
Lipophilicity (logD _{7.4})	0.70	0.74	−0.53	3.09
Protein binding (%)	25–30	30	15–30	>90
Volume of distribution (L/kg)	8 (5–19)	3.4	17 (9–25)	1,940
Elimination half-life (hours)	4 (2–7)	11	~8	11–16
Cytochrome P450 major isoform	2D6	3A4	None	2D6/1A2
Major active metabolites	ODV	None	None	4-hydroxy
Peak plasma concentration (h)	2	7.5	32–48	6–10
Excretion (%)	Urine 87	Urine 45	Urine 90	Urine >70
	Feces 2		Feces, minor	Feces 15
Plasma half-life (h)	2		NA	
Time to steady-state concentration	3 d	3 d	32–48 h	3–7d

NA, not available.

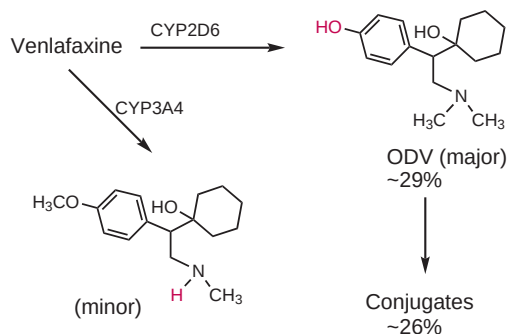


FIGURE 18.21 Metabolism of venlafaxine.

in vitro affinity for the other neural neurotransmitter receptors and, thus, a low probability for adverse effects. To minimize GI upset (e.g., nausea), venlafaxine can be taken with food without affecting its GI absorption. Venlafaxine should be administered as a single daily dose with food at approximately the same time each day. The extended-release capsules should be swallowed whole with fluid and should not be divided, crushed, chewed, or placed in water.

Whenever venlafaxine is being discontinued after more than 1 week of therapy, it generally is recommended that the patient be closely monitored and the dosage of the drug be tapered gradually to reduce the risk of withdrawal symptoms.

Although venlafaxine is a weak inhibitor of CYP2D6, variability has been observed in the pharmacokinetic parameters of venlafaxine in patients with hepatic or renal function impairment. As a precaution, elderly patients taking venlafaxine concurrently with a drug that has a narrow therapeutic index and also is metabolized by CYP2D6 should be carefully monitored. Concurrent use of CYP3A4 inhibitors with venlafaxine has been shown to interfere with its metabolism and clearance. Similar to the other antidepressants that block 5-HT reuptake, venlafaxine may interact pharmacodynamically to cause toxic levels of 5-HT to accumulate, leading to the 5-HT syndrome.

Desvenlafaxine Desvenlafaxine (ODV), as previously described, is the main active metabolite of venlafaxine, which is metabolized by CYP3A4 (*N*-demethylation) and *O*-glucuronidation (Fig. 18.21). Similar to venlafaxine, desvenlafaxine is an antidepressant of the SNRI reuptake inhibitor class and has been approved for use in the United States and Canada as an antidepressant, but not approved for use in the European market. Non-FDA-approved (off-label) use is as a non-hormonal-based vasomotor (hot flash) treatment for menopause. The European Union has not approved desvenlafaxine for any indication because, in relation to its parent, venlafaxine, desvenlafaxine seemed to be less effective with no advantages in terms of safety and tolerability. Since venlafaxine is already approved for the treatment of MDD and

is almost entirely transformed into desvenlafaxine, it would be expected that efficacy and safety of desvenlafaxine in the treatment of MDD would be very similar to those of venlafaxine. Desvenlafaxine is approximately 10-fold more potent at inhibiting 5-HT reuptake than NE reuptake. When most normal metabolizers take venlafaxine, approximately 70% of the parent drug is metabolized into desvenlafaxine, so the effects are very similar. Desvenlafaxine is structurally related to the atypical opioid tramadol (Ultram). The most common side effect of desvenlafaxine is nausea (30% to 50% vs. 9% to 11% for placebo), which is the most common reason for discontinuation. Other adverse reactions are dizziness, insomnia, sweating, constipation, somnolence, decreased appetite, priapism, night terrors, anxiety, and delayed ejaculation, which are consistent with the adverse reactions of other SNRIs. Suicidal risk was significant in some patients in each study.

Milnacipran ($\pm$)-Milnacipran is the cis-aminomethyl derivative of phenylcyclopropanecarboxamide (Fig. 18.20) that acts by inhibiting both NE and 5-HT reuptake. It is structurally different from the other SNRIs and currently is only available in Europe as a racemic mixture, with both enantiomers exhibiting antidepressant activity. Substituting the aminomethyl group of milnacipran with an aminopropyl gives a milnacipran homolog that exhibits antidepressant activity as a potent NMDA receptor antagonist. A glutamate hypothesis is being investigated as an alternative mechanism of depression (see the subsection on NMDA antagonists).

Milnacipran selectively inhibits the reuptake of 5-HT (selectivity ratio = 9) at the presynaptic membrane site, thus increasing the concentration of 5-HT in the synaptic cleft (69). Although milnacipran is not a TCA, its mechanism of action is similar to that of imipramine and its binding and reuptake inhibition profile more closely resembles that of the TCAs. Milnacipran has weak affinity for adrenergic, muscarinic, and H_1 receptors and, therefore, is expected to be devoid of the prominent side effects observed for the TCAs. In clinical studies, milnacipran showed antidepressant efficacy similar to that of TCAs and SSRIs.

In humans, milnacipran distinguishes itself from many other antidepressants by its simple pharmacokinetics. It is rapidly absorbed, with a high oral bioavailability, and it exhibits linear pharmacokinetics over a dose range of 25 to 200 mg/d (Table 18.10) (70). It circulates in the blood and distributes in the body principally as unmetabolized drug. Steady-state plasma levels are reached within 32 to 48 hours after twice-daily oral administration, and its metabolism does not involve the CYP enzyme system. Approximately 50% of the dose is excreted in urine as unmetabolized drug, and another 14% is excreted as its *N*-glucuronide conjugate. The remaining eliminated drug is composed of conjugated phase I inactive metabolites. Because the unmetabolized drug is the only compound responsible for the activity of milnacipran, no

dosage adjustment is needed in patients presenting with liver impairment.

Milnacipran has proven to be a very safe drug, with an adverse event profile clearly superior to that of TCAs and, to a certain extent, that of SSRIs. Only approximately 10% of patients experience side effects, and only dysuria occurred more frequently (2%) with milnacipran than with TCAs or SSRIs. Therefore, milnacipran seems to be an antidepressant with a very favorable benefit–risk ratio, although with a slower onset of action than the TCAs.

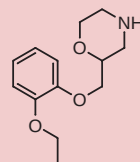
Duloxetine Duloxetine (Fig. 18.20) has been approved for the treatment of depression and diabetic peripheral neuropathic pain. It is another analog in the line of fluoxetine-based products from Lilly, in which the phenyl and phenoxy groups of fluoxetine have been respectively replaced with the benzene isostere, thiophene, and a naphthyloxy group (previously described in the section on fluoxetine). Duloxetine exhibits dual inhibition with high affinity for the SERTs and NETs, with a fivefold preferential inhibition of the SERT (68). Duloxetine appears to be a more potent *in vitro* blocker of SERTs and NETs than venlafaxine. In humans, duloxetine has a low affinity for the other neuroreceptors, suggesting low incidence of unwanted adverse effects.

Duloxetine appears to be fairly well absorbed after oral doses, with peak plasma levels in 6 to 10 hours and linear pharmacokinetics (Table 18.10) (71). The drug is extensively metabolized in the liver to active metabolites, with 72% of an oral dose primarily excreted in the urine as conjugated metabolites and up to 15% appearing in the feces. Its elimination half-life, time to steady-state blood levels, and mean volume of distribution are shown in Table 18.10.

N-Demethylation to an active metabolite (CYP2D6) and hydroxylation of the naphthyl ring (CYP1A2) at the 4-, 5-, or 6-position are the main metabolic pathways for duloxetine. Its *O*-metabolites are primarily excreted into the urine as glucuronide, sulfate, and *O*-methylated conjugation products (Fig. 18.22). The major metabolites found in plasma also were found in the urine (71). A minor metabolite is oxidation of the thiophene ring to a reactive intermediate which reportedly leads to a

VILOXAZINE

Viloxazine, which was originally developed for the treatment of depression and nocturnal enuresis, was withdrawn from consideration for business reasons. It deserves some discussion because of its early relationship with the development of the NE and 5-HT reuptake inhibitors.



Viloxazine

Research scientists at ICI in England (now AstraZeneca) recognized that some β -blockers exhibited CNS 5-HT antagonist activity at high doses. To improve the CNS distribution for the β -blockers, the medicinal chemists at ICI constrained the ethanolamine side chain of the β -blockers into a morpholine ring, resulting in the synthesis of viloxazine.

Viloxazine selectively inhibits the presynaptic reuptake of NE reuptake (approximately half as potent as imipramine) and is a weak inhibitor of mouse brain 5-HT reuptake. No appreciable *in vivo* inhibition of MAO activity was observed. It differed from the TCAs in not exhibiting the TCA adverse effect profile. Viloxazine produces antidepressant activities both similar to and different from the tricyclics desimipramine, imipramine, and amitriptyline. These actions seem to be of relevance with respect to the antidepressant action of this drug.

Presently, viloxazine is classified as an orphan drug, under the trade name of Catatrol, for the treatment of narcolepsy and cataplexy.

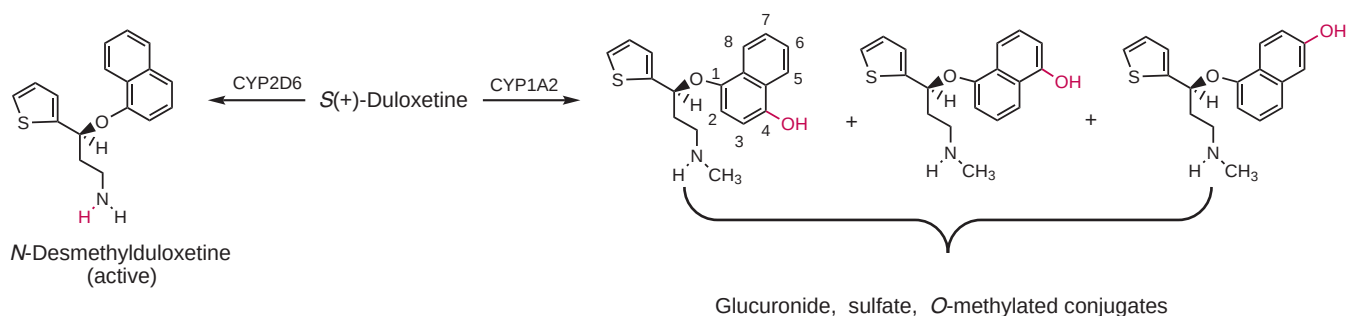


FIGURE 18.22 Metabolism of duloxetine.

jaundice-like condition. Preclinical data for 4-hydroxyduloxetine suggest it has a similar pharmacologic profile as duloxetine, with selective inhibition of SERT but less activity at NET.

Adverse effects have included insomnia, somnolence, headache, nausea, diarrhea, and dry mouth. Mild

OTHER USES FOR ANTIDEPRESSANT DRUGS

In addition to the wide use of antidepressants to treat depression, chronic neuropathic pain disorders, including fibromyalgia and diabetic and other peripheral neuropathic syndromes, have responded at least partly to treatment with tertiary TCAs and the NSRIs duloxetine, milnacipran, and venlafaxine. The NSRIs seem to be superior to the SSRIs (72).

withdrawal symptoms on abrupt discontinuation have been described in studies with healthy subjects.

Duloxetine is a moderately potent CYP2D6 inhibitor (intermediate between paroxetine and sertraline). Thus, duloxetine should be used with caution when CYP2D6 substrates and inhibitors are coadministered.

DA and Norepinephrine Reuptake Inhibitors

BUPROPION Bupropion (amfebutamone) is a phenylisopropylaminoketone that is structurally related to the phenylisopropylamine CNS stimulant, methamphetamine; the phenylisopropylaminoketone, cathinone (a constituent in khat); and the anorexiant, diethylpropion (Fig. 18.23). Although structurally similar to the CNS stimulants, bupropion exhibits distinctive different pharmacologic and therapeutic effects mostly because of its metabolites. The absence of the tricyclic ring system in bupropion results in a better adverse effect profile than with the TCAs. The tertiary butyl group in bupropion prevents its *N*-dealkylation to metabolites that could possess sympathomimetic and/or anorexiogenic properties.

Wellbutrin and Zyban (an aid in smoking cessation treatment) are trade name products for bupropion. Therefore, the potential exists for an overdose toxicity in a patient receiving multiple brand name and generic

prescriptions containing bupropion for the treatment of depression, smoking cessation, and other off-label uses.

Mechanism of Action The mechanism of antidepressant action for bupropion is more complex because of its metabolism to its three principal metabolites (Fig. 18.23), which contribute to its antidepressant mechanism of action because their accumulated plasma concentrations are higher than those of bupropion, with a longer duration of action (see the discussion of pharmacokinetics below). Bupropion appears to be a selective inhibitor of DA reuptake at the DA presynaptic neuronal membrane and an SNRI (Table 18.2), and also induces the release of DA and NE. An additional mechanism of action for bupropion is as a noncompetitive antagonist of several nAChRs. Thus, the dual antidepressant and antinicotinic activity of bupropion is mediated by its stimulatory action on the DA and NE systems and inhibition of the neuronal nAChRs. The neuronal nAChRs are ligand-gated ion channels of the CNS that regulate synaptic activity from both pre- and postsynaptic sites. In addition, bupropion blocks noncompetitively the activation of $\alpha_3\beta_2$, $\alpha_4\beta_2$, and α_7 neuronal nAChRs and is approximately 50 and 12 times more effective blocking $\alpha_3\beta_2$ and $\alpha_4\beta_2$ (the brain nicotine binding site) than α_7 receptors. Bupropion at high concentration failed to displace nicotine from the $\alpha_4\beta_2$ receptors. Bupropion inhibition of $\alpha_3\beta_2$ and $\alpha_4\beta_2$ ion channel receptors on neuronal nAChRs involves initial binding to the ion channels in the resting state, decreasing the probability of the ion channels opening and further interaction with a binding domain in the ion channel shared with the tricyclic antidepressants. Bupropion does not exhibit clinically significant anticholinergic, antihistaminic, or α_1 -adrenergic blocking activity or MAO inhibition.

Bupropion is also effective in nicotine dependence by blocking nicotine's pharmacologic effects,

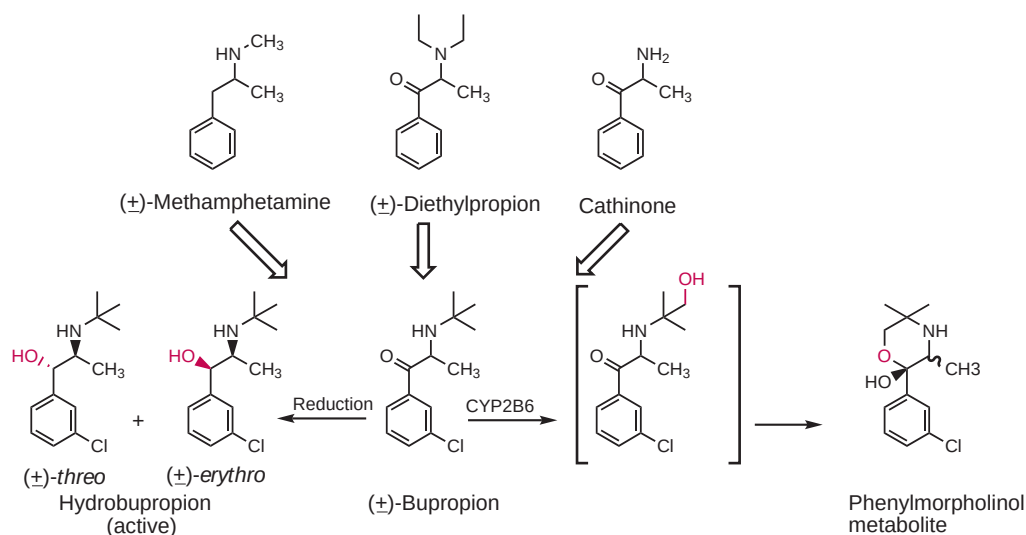


FIGURE 18.23 Bupropion, related derivatives, and metabolites.

suggesting that bupropion possesses some selectivity for neuronal nAChRs underlying these various nicotinic effects (73) and craving associated with smoking cessation, which suggests that the principal mode of action by bupropion as an aid in smoking cessation is on the withdrawal symptoms following smoking cessation. The efficacy of bupropion in smoking cessation does not appear to depend on the presence of depression. The current presumed mechanism of action of bupropion involves modulation of dopaminergic and noradrenergic systems that have been implicated by increasing extracellular CNS DA concentrations, most likely as a result of its inhibition of DAT and NET (Table 18.2). As nicotine CNS concentrations drop with smoking cessation, the firing rates of noradrenergic neurons increase, which may be the basis for the withdrawal symptoms. Thus, during withdrawal, bupropion and its active metabolite, hydroxybupropion, reduce the firing rates of these noradrenergic neurons in a dose-dependent manner, attenuating the symptoms of smoking cessation. Furthermore, its ability to block the nicotinic receptors also may prevent relapse by attenuating the reinforcing properties of nicotine but probably cannot acutely reduce smoking.

Bupropion is extensively metabolized in humans, with its three major hydroxylated metabolites reaching plasma levels higher than those of bupropion itself. These hydroxylated metabolites share many of the pharmacologic properties of bupropion, so they can play a greater role in attenuating the withdrawal and relapse by which bupropion exerts its activity in smoking cessation (73).

Pharmacokinetics Bupropion is absorbed from the GI tract, with a low oral bioavailability as a result of first-pass metabolism. The pharmacokinetic properties of bupropion are shown in Table 18.11. Food does not seem to substantially affect its peak plasma concentration or AUC. After oral administration, peak plasma concentrations usually are achieved within 2 hours for bupropion and 3 hours for sustained-released bupropion products, followed by a biphasic decline for bupropion. Plasma concentrations are dose proportional (linear pharmacokinetics) after single doses of 100 to 250 mg/d. The fraction of a dose excreted unmetabolized was less than 1%.

Bupropion hydroxylation of the tert-butyl group to hydroxybupropion intermediate, which cyclizes to a phenylmorpholinol metabolite, is mediated exclusively by CYP2B6 (74,75). Other metabolites include reduction of the aminoketone to amino-alcohol isomers, *threo*-hydrobupropion, and *erythro*-hydrobupropion (Fig. 18.23). Further oxidation of the bupropion side chain results in the formation of *m*-chlorobenzoic acid, which is eliminated in the urine as its glycine conjugate. Hydroxybupropion is approximately 50% as potent as bupropion, whereas *threo*-hydrobupropion and *erythro*-hydrobupropion have 20% of the potency of bupropion. Peak plasma concentrations for hydroxybupropion are approximately 20 times the peak level of the parent drug at steady-state, with an elimination half-life of approximately 20 hours. Thus, CYP2B6-catalyzed bupropion hydroxylation is a clinically important bioactivation and elimination pathway. The times to peak concentrations for the *erythro*-hydrobupropion

TABLE 18.11 Pharmacokinetics of Dopamine and Norepinephrine Inhibitors

Parameters	Bupropion	Trazodone	Mirtazapine
Oral bioavailability (%)	5–20	65 (60–70)	~50
Lipophilicity (logD _{7.4})	2.59	1.56	2.90
Protein binding (%)	80	90–95	85
Volume of distribution (L/kg)	19–21	0.84 (0.5–1.2)	107
Elimination half-life (h)	21 (14–16) ~20 (S,S)-hydroxy	7 (4–9)	16–40
Cytochrome P450 major isoform	2B6	3A4	3A4, 2D6
Major active metabolites	(S,S)-hydroxy	<i>m</i> -chlorophenylpiperazine	None
Peak plasma concentration (h)	2–3 8 (S,S)-hydroxy	1–2	2
Excretion (%)	Urine 87 Feces 10	Urine 70–75 Feces 25–30	Urine 75 Feces 15
Plasma half-life (h)	3–4 20 (S,S)-hydroxy	6 (4–8)	2
Time to steady-state concentration (d)	5 8 (S,S)-hydroxy	3–7	5

and *threo*-hydrobupropion metabolites are similar to that of the hydroxybupropion metabolite. The plasma levels of the *erythro*-hydrobupropion correlate with several side effects, such as insomnia and dry mouth. The elimination half-lives of *erythro*-hydrobupropion and *threo*-hydrobupropion, however, are longer (~33 and 37 hours, respectively), and the steady-state AUCs are 1.5 and 7.0 times that of bupropion, respectively. The hepatic clearance in patients with liver disease was increased from 19 to 29 hours. The median observed time to maximum plasma concentration was 19 hours for hydroxybupropion and 31 hours for *threo/erythro*-hydrobupropion. The mean half-lives for hydroxybupropion and *threo/erythro*-hydrobupropion were increased by 5 and 2 times, respectively, in patients with severe hepatic cirrhosis compared with healthy volunteers. Bupropion and its metabolites are distributed into breast milk.

In geriatric patients, the apparent half-life of hydroxybupropion averaged approximately 34 hours. Reduction in renal or hepatic function may affect elimination of the major metabolites, because these compounds are moderately polar and are likely to be metabolized further or conjugated in the liver before urinary excretion.

The pharmacokinetic parameters for bupropion and hydroxybupropion did not differ between smokers and nonsmokers, but adolescent girls exhibited increased AUCs and volume of distribution (normalized to body weight) and longer elimination half-life than did boys for bupropion and its major metabolite, hydroxybupropion. No differences in clearance between men and women, however, were observed.

Drug Interactions Inhibition studies with the SSRIs and bupropion suggest that bupropion is a potent CYP2D6 inhibitor (75). Bupropion hydroxylation was strongly inhibited by, in the following order: paroxetine > fluvoxamine > sertraline > desmethylsertraline > norfluoxetine > nefazodone > fluoxetine; it was only weakly inhibited by venlafaxine, ODV, citalopram, and desmethylcitalopram. The inhibition of bupropion hydroxylation in vitro by SSRIs suggests the potential for clinical drug interactions. Therefore, coadministration of drugs that inhibit CYP2D6 warrants careful monitoring. Because of its selective inhibition of DA reuptake, pharmacodynamic interactions with DA agonists (e.g., levodopa) and antagonists should be anticipated. Coadministration of bupropion with drugs that lower the seizure threshold should be avoided because of the risk of serious seizures. Drugs that affect metabolism by CYP2B6 (e.g., methadone, nicotine, and cyclophosphamide) also have the potential to interact with bupropion.

Therapeutic Uses Besides being used to treat depression, bupropion is a nonnicotine aid in the cessation of smoking. The efficacy of bupropion in smoking cessation is comparable to that of nicotine replacement therapy

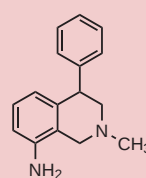
and should be considered as a second-line treatment in smoking cessation (72,73). It possesses a broad spectrum of infrequent adverse effects, however, with potential drug metabolism interactions with TCAs, β -adrenergic blocking drugs, and class Ic antiarrhythmics.

Serotonin Receptor Modulators

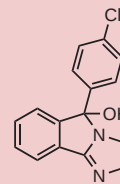
SEROTONIN ANTAGONIST/REUPTAKE INHIBITORS

General Mechanism of Action Serotonin receptor modulators are a class of antidepressants, the function of which is to modulate the concentration of 5-HT in the brain. A neuromodulator functions as a “volume control in the brain and nervous system,” regulating the other neurotransmitters through its receptors in the brain in response to external stimuli. As previously described, the serotonergic system modulates a large number of physiologic events, such as temperature regulation, sleep, learning and memory, behavior, sexual function, hormonal secretions, and immune activity, and is implicated in stress, anxiety, aggressiveness, and depression disorders. Of the various types of 5-HT receptors (see Chapter 11) mediating serotonergic activity, the 5-HT_{1B} receptors play an important role in modulating the serotonergic system. The 5-HT_{1B} receptors are autoreceptors localized on serotonergic neuron terminals, where they inhibit the release of 5-HT and its biosynthesis; they also are heteroreceptors located on nonserotonergic terminals, where they inhibit the release of other neurotransmitters, such as ACh, GABA, and NE. Excessive amounts of 5-HT in the

MISCELLANEOUS NOREPINEPHRINE AND DA REUPTAKE INHIBITORS



Nomifensine



Mazindol

Nomifensine is a substituted phenylpiperidine (an aminophenyltetrahydroisoquinoline) structurally related to sertraline that was marketed as a stimulatory antidepressant in the mid-1970s but later withdrawn because of a high incidence of hemolytic anemia. Nomifensine inhibits the NET and DAT. It displays high affinity for NET (human $pK_i = 7.8$), moderate affinity for DAT ($pK_i = 6.6$), and a low affinity for SERT (5-HT:NE ratio = 65).

Mazindol (Sanorex) is a phenyl-substituted imidazobenzisoxindole that inhibits both NET and DAT. It exhibits high affinity for the NET (rat $pK_i = 9.3$), good affinity for DAT (rat $pK_i = 7.8$), and a 5-HT:NE ratio of 224. DA reuptake inhibitors suppress appetite; thus, mazindol is approved to be marketed as an appetite suppressant.

brain may cause relaxation, sedation, and a decrease in sexual drive; inadequate amounts of 5-HT can lead to psychiatric disorders. Therefore, the 5-HT receptor modulator antidepressants exert their antidepressant effects by mechanisms that enhance noradrenergic or serotonergic transmission by acting as mixed 5-HT₂ antagonists/5-HT reuptake inhibitors (SARI; trazodone) and α_2 -adrenergic antagonists/5-HT₂ and 5-HT₃ antagonists (NaSSA; mirtazapine). Chronic antidepressant treatment with SARIs and NaSSAs modulate 5-HT receptor expression and, in turn, 5-HT function.

Arylpiperazines

Trazodone. Trazodone is a phenylpiperazine–triazolopyridine antidepressant (calculated logP = 2.8) that is structurally unrelated to most of the other antidepressant classes (Fig. 18.24). It acts as a dual 5-HT agonist and 5-HT reuptake inhibitor.

Mechanism of Action. Although the exact mechanism of action is unknown, trazodone acts as an antagonist at 5-HT_{2A} receptors and is a weak inhibitor of 5-HT reuptake at the presynaptic neuronal membrane, potentiating the synaptic effects of 5-HT. Its mechanism of action is complicated by the presence of its metabolite, *m*-chlorophenylpiperazine (Fig. 18.25), which is a 5-HT_{2C} agonist.

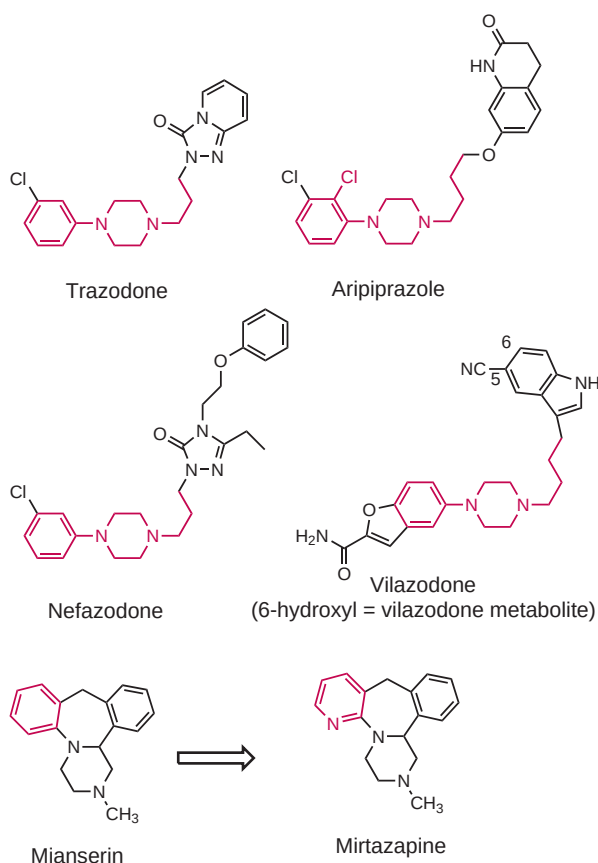


FIGURE 18.24 Serotonin receptor modulators.

At therapeutic dosages, trazodone does not appear to affect the reuptake of DA or NE within the CNS. It has little anticholinergic activity and is relatively devoid of toxic cardiovascular effects. The increase in serotonergic activity with long-term administration of trazodone decreases the number of postsynaptic serotonergic (i.e., 5-HT₂) and β -adrenergic binding sites in the brains of animals, decreasing the sensitivity of adenylate (or adenyl) cyclase to stimulation by β -adrenergic agonists. It has been suggested that postsynaptic serotonergic receptor modification is mainly responsible for the antidepressant action observed during long-term administration of trazodone. Trazodone does not inhibit MAO and, unlike amphetamine-like drugs, does not stimulate the CNS.

Trazodone is rapidly and almost completely absorbed from the GI tract following oral administration, with an oral bioavailability of approximately 65% (Table 18.11). Peak plasma concentrations of trazodone occur approximately 1 hour after oral administration when taken on an empty stomach or 2 hours when taken with food. At steady-state, its plasma concentrations exhibit wide interpatient variation.

Trazodone is extensively metabolized in the liver by *N*-dealkylation to its primary circulating active metabolite, *m*-chlorophenylpiperazine (*m*-CPP), which subsequently undergoes aromatic hydroxylation to *p*-hydroxy-*m*-CPP (Fig. 18.25) (76). In vitro studies indicate that CYP3A4 is the major isoform involved in the production of *m*-CPP from trazodone (and CYP2D6 to a lesser extent). 4'-Hydroxy-*m*-CPP and oxotriazolopyridinepropionic acid (the major metabolite excreted in urine) are conjugated with glucuronic acid. Less than 1% of a dose is excreted unmetabolized. *m*-CPP is of significant interest because of its 5-HT_{2C} agonist and 5-HT_{2A} antagonist activities that may contribute to trazodone antidepressant action.

Trazodone therapy has been associated with several cases of idiosyncratic hepatotoxicity (see Chapter 4). Although the mechanism of hepatotoxicity remains unknown, the generation of an iminoquinone, an epoxide reactive metabolite, or both may play a role in the initiation of trazodone-mediated hepatotoxicity (Fig. 18.25) (76). Studies have shown that the bioactivation of trazodone involves, first, aromatic hydroxylation of the 3-chlorophenyl ring, followed by its oxidation to a reactive iminoquinone intermediate, which then reacts with glutathione. Alternately, the aromatic hydroxylated derivative is oxidized in the triazolopyridinone ring to an electrophilic epoxide followed by hydrolytic ring opening to the stable diol metabolite or nucleophilic conjugates through attachment by various nucleophiles (:Nu). The pathway involving trazodone bioactivation to the iminoquinone also has been observed with many parahydroxyanilines (e.g., acetaminophen) (see Chapter 4), including the structurally related antidepressant nefazodone. The reactive intermediates consume the available glutathione, allowing the reactive intermediate to react with hepatic tissue, leading to liver damage.

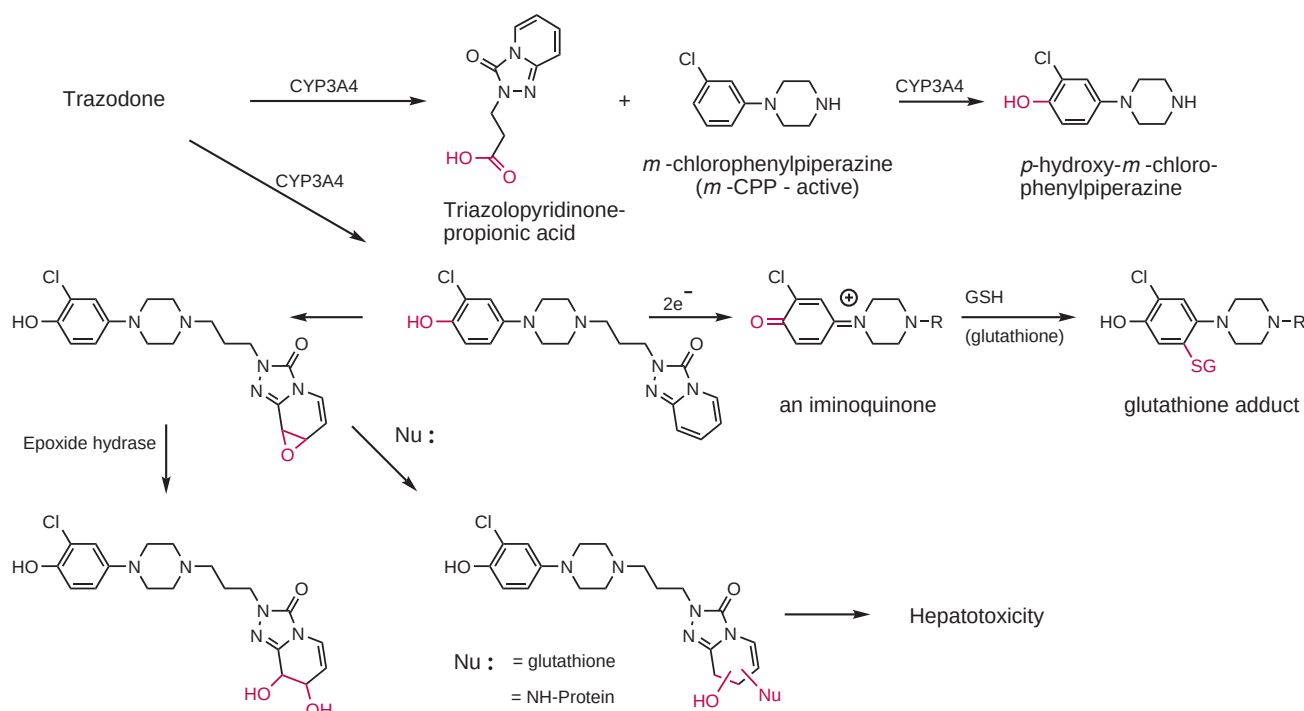


FIGURE 18.25 Metabolism of trazodone.

Unlike the TCAs, trazodone does not block the fast sodium channels and, thus, does not have significant arrhythmic activity. Compared with the SSRIs, it has a lesser tendency to cause drug-induced male sexual dysfunction as a side effect. Although trazodone displays α_1 -adrenergic blocking activity, hypotension is relatively uncommon. Signs of overdose toxicity include nausea, vomiting, and decreased level of consciousness. Trazodone produces a significant amount of sedation in normal and mentally depressed patients (principally from its central α_1 -adrenergic blocking activity and antihistaminic action).

Drug Interactions. Trazodone possesses serotonergic activity; therefore, the possibility of developing 5-HT syndrome should be considered in patients who are receiving trazodone and other SSRIs or serotonergic drugs concurrently. When trazodone is used concurrently with drugs metabolized by CYP3A4, caution should be used to avoid excessive sedation. Trazodone can cause hypotension, including orthostatic hypotension and syncope; concomitant administration of antihypertensive therapy may require a reduction in dosage of the antihypertensive agent.

The possibility of drug–drug interactions with trazodone and other substrates, inducers, and/or inhibitors of CYP3A4 exists (76).

Therapeutic Uses. Trazodone is used primarily in the treatment of insomnia, mental depression, or depression/anxiety disorders. The drug also has shown some efficacy in the treatment of benzodiazepine or alcohol dependence, diabetic neuropathy, and panic disorders.

Aripiprazole. Aripiprazole (Fig. 18.24) is an atypical antipsychotic but also an arylpiperazine antidepressant used for acute manic and mixed episodes associated with bipolar disorder in both pediatric patients age 10 to 17 years and adults and as an adjunct for major depression. In addition, it is often used as maintenance therapy in bipolar disorders in conjunction with a mood stabilizer such as lithium or valproate. Perhaps due to its dopaminergic mechanism of action, there is some evidence to suggest that aripiprazole blocks cocaine-seeking behavior in animal models without significantly affecting other rewarding behaviors (such as food self-administration).

Aripiprazole's mechanism of action is different from those of the other antidepressants but similar to trazodone as a partial agonist at the 5-HT_{1A} receptor and antagonist at the 5-HT_{2A} receptor. It is an antagonist at the 5-HT₇ receptor and acts as a partial agonist at the 5-HT_{2C} receptor, both with high affinity.

Aripiprazole undergoes extensive metabolism by CYP3A4 and CYP2D6 to produce its active circulating metabolites dehydroaripiprazole and 2,3-dichlorophenylpiperazine (DCPP), similar to *m*-CPP for trazodone and nefazodone (Fig. 18.26). It is unknown whether DCPP contributes to aripiprazole's antidepressant action, but the possibility cannot be excluded. Aripiprazole displays linear kinetics and has an elimination half-life of approximately 75 hours. Steady-state plasma concentrations are achieved in about 14 days. Peak concentration is achieved 3 to 5 hours after oral dosing. Bioavailability of the oral tablets is about 90%,

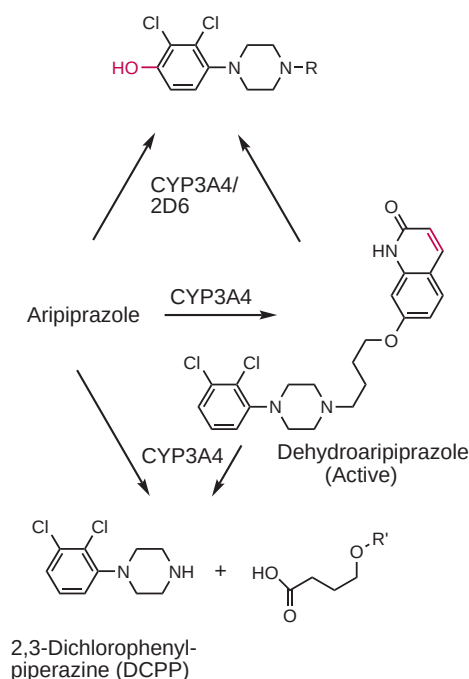


FIGURE 18.26 Major metabolites of aripiprazole.

with approximately 40% of the aripiprazole plasma concentration being present as the dehydroaripiprazole, which are both excreted via feces and urine. The long half-life for aripiprazole is responsible for many of its adverse side effects. Recent studies suggest that the long half-life, which results in lower doses, may actually prevent formation of clinically significant hepatotoxic metabolites such as those that are responsible for the toxicity of nefazodone (77). Aripiprazole and its active metabolite dehydroaripiprazole have been identified as substrates for the efflux transporter P-gp in the brain, which contributes to wanted or unwanted clinical effects in patients treated with aripiprazole.

Vilazodone. Vilazodone is a phenylpiperazine-2-benzofurancarboxamide antidepressant that is structurally related to trazodone and aripiprazole with a calculated logP at pH 7.4 is 4.44 (Fig. 18.24). It acts as a dual 5-HT receptor partial agonist and 5-HT reuptake inhibitor (79).

Mechanism of Action. The mechanism of the antidepressant effect of vilazodone is not fully understood but could be related to its enhancement of CNS serotonergic activity through selective inhibition of 5-HT reuptake. In addition, vilazodone is also a partial agonist at 5-HT_{1A} receptors. Vilazodone binds with high affinity to the 5-HT reuptake site ($K_i = 0.1$ nM), but not to the NE ($K_i = 56$ nM) or DA ($K_i = 37$ nM) reuptake sites. Vilazodone is a potent and selective inhibitor of 5-HT reuptake ($IC_{50} = 1.6$ nM). As a 5-HT_{1A} receptor partial agonist, it binds selectively with high affinity to 5-HT_{1A} receptors ($IC_{50} = 2.1$ nM).

NEFAZODONE

Nefazodone is a phenylpiperazine antidepressant structurally related to trazodone, but it differs pharmacologically from trazodone, the SSRIs, the MAOIs, and the TCAs (Fig. 18.24). When compared with trazodone, nefazodone displays approximately twice the affinity potency for SERT. Nefazodone therapy, however, was associated with life-threatening cases of idiosyncratic hepatotoxicity, and as a result, nefazodone was withdrawn from both the North American and European markets in 2003. The mechanism of hepatotoxicity remains unknown, but nefazodone, being structurally similar to trazodone (Fig. 18.24), is metabolized to *p*-hydroxynefazodone, *m*-CPP, and phenoxyethyltriazolodione. In turn, *p*-hydroxynefazodone is thought to be oxidized to an iminoquinone and/or an epoxide reactive metabolite, which may play a role in the initiation of nefazodone-mediated hepatotoxicity (78).

Vilazodone does not prolong the QTc interval, as occurs with the TCAs, and is below the threshold for clinical concern. Clinical trials of vilazodone showed significant antidepressant efficacy with an onset of effect in 1 week, unlike other antidepressants. The 1-week onset of its antidepressant action has been linked to its partial agonist at the 5-HT_{1A} receptor.

Pharmacokinetics. The antidepressant activity for vilazodone activity is due primarily to the parent drug. Vilazodone exhibits dose-proportional linear pharmacokinetics, and steady-state is achieved in about 3 days. Elimination of vilazodone is primarily by hepatic metabolism, with a terminal half-life of approximately 25 hours. Peak plasma concentrations occur 4 to 5 hours after oral administration. The absolute bioavailability of vilazodone is 72% with food. The peak concentration of vilazodone with food (high-fat or light meal) increases oral bioavailability approximately 147% to 160%, and AUC is increased by approximately 64% to 85%. Vilazodone is widely distributed and approximately 96% to 99% protein bound. Vilazodone is extensively metabolized by CYP3A4 to 6-hydroxyvilazodone (inactive metabolite) with minor contributions from CYP2C19 and CYP2D6. In vitro studies have shown that CYP1A2, CYP2A6, CYP2C9, and CYP2E1 have minimal contribution to the metabolism of vilazodone. Non-CYP pathways include carboxylesterase. Unlike trazodone, no phenylpiperazine metabolite was observed. Only 1% of the oral dose is recovered in the urine and 2% is recovered in the feces as unchanged vilazodone. In vitro studies indicate that vilazodone is unlikely to inhibit or induce the metabolism of other CYP substrates (except for CYP2C8) and did not alter the pharmacokinetics CYP2C19, CYP2D6, and CYP3A4 substrates. Renal impairment or mild or moderate hepatic impairment did not affect the clearance of vilazodone; thus no dose adjustment is required. Coadministration

of vilazodone with ethanol or with a proton pump inhibitor (e.g., pantoprazole) did not affect the rate or extent of vilazodone absorption. Vilazodone is excreted into the milk of lactating rats, and the effect on lactation and nursing in humans is unknown. Breast-feeding in women treated with vilazodone should be considered only if the potential benefit outweighs the potential risk to the child. A pharmacokinetic study in elderly (>65 years old) versus young (24 to 55 years old) subjects demonstrated that the pharmacokinetics were generally similar between the two age groups. No dose adjustment is recommended on the basis of age. Greater sensitivity of some older individuals to vilazodone cannot be ruled out. After adjustment for body weight, the systemic exposures between males and females are similar.

Adverse Effects. The most commonly observed adverse reactions in vilazodone-treated MDD patients with an incidence greater than 5% were diarrhea, nausea, vomiting, and insomnia. Sexual dysfunction was minimal with vilazodone. As with all antidepressants, use vilazodone cautiously in patients with a history or family history of bipolar disorder, mania, or hypomania. Vilazodone has safety risks associated with the induction of suicidal thoughts in young adults, adolescents, and children (a black box warning).

Drug Interactions. The risk of using vilazodone in combination with other CNS-active drugs has not been evaluated. Because of potential interactions with MAOIs, vilazodone should not be prescribed concomitantly with an MAOI or within 14 days of discontinuing or starting an MAOI. Based on the drug's mechanism of action and the potential for 5-HT toxicity (serotonin syndrome), caution is advised when coadministered with other drugs that may affect the serotonergic neurotransmitter systems (e.g., MAOIs, SSRIs, SNRIs, triptans, buspirone, tramadol, tryptophan products). Because 5-HT release by platelets plays an important role in hemostasis, concurrent use of a nonsteroidal anti-inflammatory drug or aspirin with vilazodone may potentiate the risk of abnormal bleeding. Altered anticoagulant effects, including increased bleeding, have been reported when SSRIs and SNRIs are coadministered with warfarin. Thus, patients receiving warfarin therapy should be carefully monitored when vilazodone is initiated or discontinued. Concomitant use of vilazodone and strong or moderate inhibitors of CYP3A4 (e.g., ketoconazole) can increase vilazodone plasma concentrations by approximately 50%, requiring a dosage adjustment. No dose adjustment is recommended when coadministered with mild inhibitors of CYP3A4 (e.g., cimetidine). Concomitant use of vilazodone with inducers of CYP3A4 has the potential to reduce vilazodone plasma levels. Concomitant administration of vilazodone with inhibitors of CYP2C19 and CYP2D6 is not expected to alter plasma concentrations of vilazodone. These isoforms are minor elimination pathways in the metabolism of vilazodone. Coadministration of vilazodone with substrates for CYP1A2, CYP2C9,

CYP3A4, or CYP2D6 is unlikely to result in clinically significant changes in the concentrations of the CYP substrates. Vilazodone coadministration with mephenytoin resulted in a small (11%) increase in mephenytoin biotransformation, suggestive of a minor induction of CYP2C19. In vitro studies have shown that vilazodone is a moderate inhibitor of CYP2C19 and CYP2D6. In vitro studies also suggest that vilazodone may inhibit the biotransformation of substrates of CYP2C8; thus coadministration of vilazodone with a CYP2C8 substrate may lead to an increase in the plasma concentration of the CYP2C8 substrate. Chronic administration of vilazodone is unlikely to induce the metabolism of drugs metabolized by CYP1A1, CYP1A2, CYP2A6, CYP2B6, CYP2C9, CYP2C19, CYP2D6, CYP2E1, and CYP3A4/5. Because vilazodone is highly bound to plasma protein, administration to a patient taking another drug that is highly protein bound may cause increased free concentrations of the other drug.

NORADRENERGIC AND SPECIFIC SEROTONERGIC ANTIDEPRESSANTS (NASSAs)

Mirtazapine Mirtazapine is a pyrazinopyridobenzazine antidepressant that is an isostere of the antidepressant mianserin (Fig. 18.24). A seemingly simple isosteric replacement of an aromatic methine group (CH) in mianserin with a nitrogen to give a pyridine ring (mirtazapine) has significant effects on the physicochemical properties, pharmacokinetics, mechanisms of action, and antidepressant activities (Table 18.12) (31). Noteworthy differences between receptor affinity and transporter affinity, pharmacokinetics, regioselectivity in the formation of metabolites, and toxicity are observed for mianserin and mirtazapine and their antidepressant mechanisms of action. The pyridine ring increases the polarity of the molecule and decreases the measured partition coefficient and the basicity. Mianserin is a potent inhibitor of NET, whereas mirtazapine has negligible effects on the inhibition of NET ($pK_i = 7.1$ vs. 5.8, respectively).

Mianserin is currently marketed in Europe as an antidepressant. Mianserin has not been approved for use in the United States because of its serious adverse effects

TABLE 18.12 Physicochemical Properties of Mirtazapine and Mianserin

Properties	Mirtazapine	Mianserin
pK_a	7.1	7.4
Lipophilicity ($\log D_{7.4}$)	2.72	3.17
Polarity	2.63 debye	0.82 debye
NET affinity (pK_i)	5.8	7.1
5-HT release	Yes	No

of agranulocytosis and leukopenia. Mirtazapine has not exhibited this adverse effect

Mechanism of Action. Animal studies indicate that the efficacy of mirtazapine as an atypical antidepressant results from enhancing central noradrenergic and serotonergic activity, possibly through blocking central presynaptic α_2 -adrenergic receptors. Blocking these receptors inhibits the negative feedback loop, which increases the release of NE into the synapse. Mirtazapine also is a potent antagonist at 5-HT₂ and 5-HT₃ receptors, and it shows no significant affinity for 5-HT_{1A} or 5-HT_{1B} receptors. Additionally, it displays some anticholinergic properties and produces sedative effects (because of potent histamine H₁ receptor antagonism) and orthostatic hypotension (because of moderate antagonism at peripheral α_1 -adrenergic receptors). Its antidepressant effect is comparable to the TCAs and may be better than some SSRIs, especially in patients with depression of the melancholic type, but at higher doses, it may cause drowsiness and weight gain. The drug generally is well tolerated, producing no more adverse events (including anticholinergic events) than the SSRIs and fewer adverse events than the TCAs.

The pharmacokinetics for mirtazapine are shown in Table 18.11. Mirtazapine absorption is rapid and complete, with a bioavailability of approximately 50% as a result of first-pass metabolism. The rate and extent of mirtazapine absorption are minimally affected by food. Dose and plasma levels are linearly related over a dose range of 15 to 80 mg. The elimination half-life of the (–)-enantiomer is approximately twice that of the (+)-enantiomer. In woman of all ages, the elimination half-life is significantly longer than in men (mean half-life, 37 vs. 26 hours).

After oral administration, mirtazapine undergoes first-pass metabolism by *N*-demethylation and ring hydroxylation to its 8-hydroxy metabolite, followed by *O*-glucuronide conjugation (Fig. 18.27) (80). In vitro

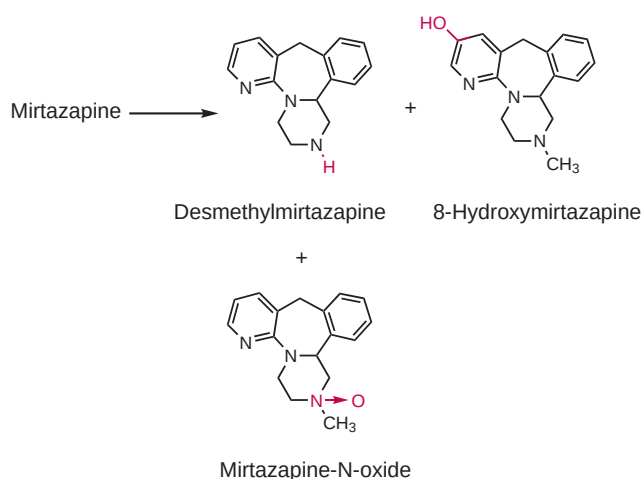


FIGURE 18.27 Metabolism of mirtazapine.

studies indicate that CYP2D6 and CYP1A2 are involved in the formation of the 8-hydroxy metabolite and that CYP3A4 is responsible for the formation of the *N*-desmethyl and *N*-oxide metabolites. The 8-hydroxy and *N*-desmethyl metabolites possess weak pharmacologic activity, but their plasma levels are very low and, thus, are unlikely to contribute to the antidepressant action of mirtazapine. Clearance for mirtazapine may decrease in patients with hepatic or renal impairment, increasing its plasma concentrations. Therefore, it should be used with caution in these patients. In vitro studies have shown mirtazapine to be a weak inhibitor of CYP1A2, CYP2D6, and CYP3A4.

Monoamine Oxidase Inhibitors

The discovery of MAOIs resulted from a search for derivatives of isoniazid (isonicotinic acid hydrazide) (Fig. 18.28) with antitubercular activity. During clinical trials with this hydrazine derivative, a rather consistent beneficial effect of mood elevation was noted in depressed patients with tuberculosis. Although no longer used clinically, iproniazid (Fig. 18.28), the first derivative to be synthesized, was found to be hepatotoxic at dosage levels required for antitubercular and antidepressant activity. The antidepressant activity of iproniazid, however, prompted a search for other MAOIs, which resulted in the synthesis of hydrazine and nonhydrazine MAOIs that were relatively less toxic than iproniazid.

The MAOIs can be classified as hydrazines (e.g., phenelzine) and nonhydrazines (e.g., tranylcypromine), which can block the oxidative deamination of naturally occurring monoamines. MAOIs can also be classified according to their ability to selectively inhibit MAO-A or MAO-B (nonselective inhibitors). The currently available MAOI antidepressants (phenelzine and tranylcypromine; Fig. 18.28) are considered to be irreversible inhibitors of both MAO-A and MAO-B. The mechanism of antidepressant action of the MAOIs suggests that an increase in free 5-HT and NE and/or alterations in other amine concentrations within the CNS are mainly responsible for their antidepressant effect.

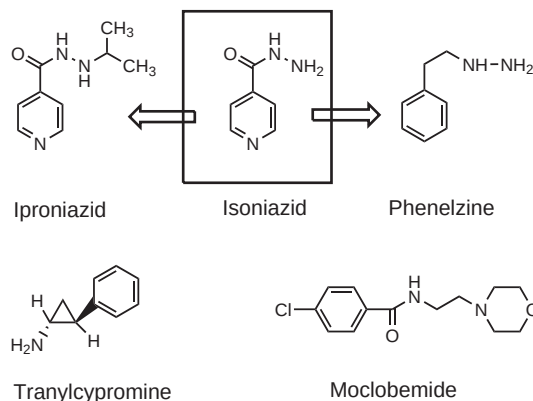


FIGURE 18.28 MAOI antidepressants.

Mechanisms of Action Common to MAOIs

An enzyme found mainly in nerve tissue and in the liver and lungs, MAO catalyzes the oxidative deamination of various amines, including epinephrine, NE, DA, and 5-HT. At least two isoforms of MAO exist, MAO-A and MAO-B, with differences in substrate preference, inhibitor specificity, and tissue distribution. The MAO-A substrates include 5-HT, and the MAO-B substrates include phenylethylamine. Tyramine, epinephrine, NE, and DA are substrates for both MAO-A and MAO-B. The cloning of MAO-A and MAO-B has demonstrated unequivocally that these enzymes consist of different amino acid sequences and also has provided insight regarding their structure, regulation, and function (51). Both MAO-A and MAO-B knockout mice exhibit distinct differences in neurotransmitter metabolism and behavior (51). The MAO-A knockout mice have elevated brain levels of 5-HT, NE, and DA, and they manifest aggressive behavior similar to human males with a deletion of MAO-A. In contrast, MAO-B knockout mice do not exhibit aggression, and only levels of phenylethylamine are increased. Both MAO-A and MAO-B knockout mice show increased reactivity to stress. These knockout mice are valuable models for investigating the role of monoamines in psychoses and in neurodegenerative and stress-related disorders.

The pharmacologic effects of MAOIs are cumulative. A latent period of a few days to several months may occur before the onset of the antidepressant action, and effects may persist for up to 3 weeks following discontinuance of therapy.

Adverse Effects Common to MAOIs

Common side effects for the nonselective MAOIs include difficulty getting to sleep and broken sleep, daytime insomnia, agitation, dizziness on standing that results in fainting (orthostatic hypotension), dry mouth, tremor (slight shake of muscles of arms and hands), syncope, palpitations, tachycardia, dizziness, headache, confusion, weakness, overstimulation including increased anxiety, constipation, GI disturbances, edema, dry mouth, weight gain, and sexual disturbances.

Drug Interactions Common to MAOIs

The most significant drug interaction limiting the efficacy of the nonselective MAOIs is with certain foods that have the potential to cause hypertensive crisis because of the release and potentiation of catecholamines. The severity and consequences of such interactions vary among individuals from only minor increases in blood pressure to substantial and rapid increases in blood pressure within 20 minutes. These patients may experience symptoms associated with brain hemorrhage or cardiac failure.

Hypertensive crises with MAOIs have occurred in some patients following ingestion of foods containing large amounts of tyramine or tryptophan. In general, patients taking MAOIs should avoid protein foods that have undergone protein breakdown by aging, fermentation, pickling, smoking, or bacterial contamination. Some of the common foods to avoid are shown in Table 18.13.

TABLE 18.13 Foods to Be Avoided Due to Potential Monoamine Oxidase Inhibitor–Food Interactions

Cheeses Cheddar Camembert Stilton Processed cheese Sour cream	Meats Chicken livers Genoa salami Hard salami Pepperoni Lebanon bologna
Spirits Chianti Champagne Alcohol-free/reduced wines	Fruit Figs (overripe/canned) Raisins Overripe bananas
Fish Pickled herring Anchovies Caviar	Dairy product Yogurt
Miscellaneous Shrimp paste Chocolate Meat tenderizers (papaya)	Vegetable products Yeast extract Pods-broad beans Bean curd Soy sauce Avocado

Patients should be warned against eating foods with a high tyramine content because hypertensive crisis may result. Excessive amounts of caffeine also reportedly may precipitate hypertensive crisis.

The MAOIs interfere with the hepatic metabolism of many prescription and nonprescription (over-the-counter) drugs and may potentiate the actions of their pharmacologic effects (i.e., cold decongestants, sympathomimetic amines, general anesthetics, barbiturates, and morphine).

Therapeutic Uses Common to MAOIs

The MAOIs are indicated in patients with atypical (exogenous) depression and in some patients who are unresponsive to other antidepressive therapy. They rarely are a drug of first choice. Unlabeled uses have included bulimia (having characteristics of atypical depression), treatment of cocaine addiction (phenelzine), night terrors, posttraumatic stress disorder, some migraines resistant to other therapies, seasonal affective disorder (30 mg/d), and some panic disorders. A list of information that should be transmitted to the patient concerning use of MAOIs is shown in Table 18.14.

Nonselective MAOI Antidepressants

PHENELZINE Phenelzine is a hydrazine MAOI (Fig. 18.28). Its mechanism of action is the prolonged, irreversible inhibition of MAO-A and MAO-B. Phenelzine has been used with some success in the management of bulimia nervosa. The MAOIs, however, are potentially dangerous in patients with binge eating and purging behaviors, and the American Psychiatric Association states that MAOIs should be used with caution in the management of bulimia nervosa.

Limited information is available regarding MAOI pharmacokinetics of phenelzine (Table 18.15). Phenelzine

TABLE 18.14 Common Information for Patients Taking Monoamine Oxidase Inhibitors (MAOIs)

Patient Information	Recommendation
Discontinuance of therapy or dose adjustment	Consult physician
Adding medication (prescription/over-the-counter)	Consult physician
Tyramine-containing foods and over-the-counter products	Avoid
Drowsiness, blurred vision	Avoid driving or performing tasks requiring alertness or coordination
Dizziness, weakness, fainting	Arise from sitting position slowly
Alcohol use	Avoid alcohol
Onset of action	Effects may be delayed for a few weeks
Severe headache, palpitation, tachycardia, sense of constriction in throat or chest, sweating, stiff neck, nausea or vomiting	Consult physician
New physician or dentist	Inform practitioner of MAOI use

seems to be well absorbed following oral administration; however, maximal inhibition of MAO occurs within 5 to 10 days. Acetylation of phenelzine to its inactive acetylated metabolite appears to be a minor metabolic pathway (79). Phenelzine is a substrate as well as an inhibitor

TABLE 18.15 Pharmacokinetics of the Monoamine Oxidase Inhibitors (MAOIs)

Parameters	Phenelzine (Nardil)	Tranylcypromine (Parnate)	Moclobemide
Oral bioavailability (%)	NA	~50	50–90
Lipophilicity ($\log D_{7.4}$)	1.10	0.54	1.47
Protein binding (%)	NA	NA	50
Volume of distribution (L/kg)	NA	1.1–5.7	1.2
Elimination half-life (h)	NA	2.5 (1.5–3.2)	1.5
Peak plasma concentration (h)	2–3	1.5 (0.7–3.5)	0.82
Excretion route	Urine	Urine Feces	Renal Feces

NA, not available.

of MAO, and major identified metabolites of phenelzine include phenylacetic acid and *p*-hydroxyphenylacetic acid. Phenelzine also elevates brain GABA levels, probably via its β -phenylethylamine metabolite. The clinical effects of phenelzine may continue for up to 2 weeks after discontinuation of therapy. Phenelzine is excreted in the urine mostly as its *N*-acetyl metabolite. Interindividual variability in plasma concentrations have been observed among patients who are either slow or fast acetylators. Slow acetylators of hydrazine MAOIs may yield exaggerated adverse effects after standard dosing. If adverse neurologic reactions occur during phenelzine therapy, phenelzine-induced pyridoxine deficiency should be considered. Pyridoxine supplementation can correct the deficiency while allowing continuance of phenelzine therapy.

TRANLYCYPROMINE Tranylcypromine is a nonhydrazine, irreversible MAO inhibitor antidepressant agent that was designed as the cyclopropyl analog of amphetamine (Fig. 18.28). Instead of exhibiting amphetamine-like stimulation, its mechanism of action is nonselective, irreversible inhibition of MAO. Its onset of antidepressant action is more rapid than for phenelzine. Tranylcypromine is well absorbed following oral administration (Table 18.15). Metabolism occurs via aromatic ring hydroxylation and *N*-acetylation (81). It is a competitive inhibitor of CYP2C19 and CYP2D6 and a noncompetitive inhibitor of CYP2C9. Most metabolism studies suggest that tranylcypromine is not metabolized to amphetamine, contrary to debate. Maximal MAO inhibition, however, occurs within 5 to 10 days. The GI absorption of the tranylcypromine shows interindividual variation and may be biphasic in some individuals, achieving an initial peak within approximately 1 hour and a secondary peak within 2 to 3 hours. It has been suggested that this apparent biphasic absorption in some individuals may represent different absorption rates. Following discontinuance of tranylcypromine, the drug is excreted within 24 hours. On withdrawal of tranylcypromine, MAO activity is recovered in 3 to 5 days (possibly in up to 10 days). However, concentrations of urinary tryptamine, an indicator of MAO-A inhibition, return to normal within 72 to 120 hours.

Reversible MAO-A Inhibitor Antidepressants

The major goal for developing new reversible inhibitors of MAO-A (RIMAs) is to avoid the severe, life-threatening hypertensive reactions that can occur with irreversible inhibitors. Irreversible inhibition of intestinal and hepatic MAO-A can lead to inhibition of tyramine degradation, thus allowing excessive amounts of naturally occurring tyramine to be absorbed from the food. Because these reversible compounds form unstable complexes with the MAO-A subtype, they can be easily displaced from MAO-A by tyramine. Thus, it becomes possible for ingested tyramine to be metabolized, diminishing the need for the dietary restrictions that plague the use of older irreversible nonselective MAOIs. This new class of selective and reversible inhibitors of MAO-A includes moclobemide (Fig. 18.28).

MOCLOBEMIDE Moclobemide is a benzamide derivative containing a morpholine ring with a pK_a of 6.2 and a partition coefficient of 40 in an octanol/pH 7.4 buffer solution. Moclobemide is not currently available commercially in the United States but is available in the United Kingdom and Australia.

Mechanism of Action Moclobemide is an RIMA that preferentially inhibits MAO-A (~80%) and, to a lesser extent, MAO-B (20% to 30% inhibition), thereby increasing the concentration of 5-HT, NE, and other catecholamines in the synaptic cleft and in storage sites. During chronic therapy with the MAOIs, adaptive changes at the noradrenergic and serotonergic receptors occur (“downregulation”) as a result of neurotransmitter hypersensitivity because of prolonged concentrations of NE and 5-HT at the postsynaptic receptor (see the discussion of the receptor sensitivity hypothesis for details). This mechanism is likely the basis for its antidepressant activity. Inhibition of MAO-A by moclobemide is short acting (maximum, 24 hours) and reversible. This is in contrast to phenelzine, which is long acting and irreversible in its binding to both MAO-A and MAO-B.

The pharmacokinetics (Table 18.15) for moclobemide are linear only up to 200 mg; at higher doses, nonlinear pharmacokinetics are observed (82). Although well absorbed from the GI tract, the presence of food reduces the rate but not the extent of absorption of moclobemide. Small quantities of moclobemide are distributed into human breast milk. Moclobemide undergoes a complex metabolism, initially involving morpholine carbon and nitrogen oxidation, deamination, and aromatic hydroxylation. The *N*-oxide and ring-opened metabolites retain some in vitro MAO-A inhibition. Moclobemide is a weak inhibitor of CYP2D6 in vitro. It is extensively metabolized in the liver by oxidation and is eliminated primarily into the urine as conjugates. Less than 1% of an administered dose of moclobemide is eliminated unmetabolized.

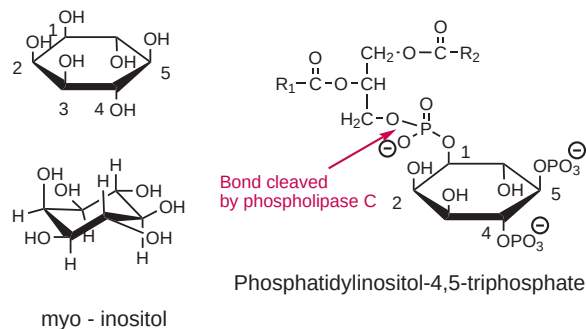
Because moclobemide is partially metabolized by the polymorphic isozymes CYP2C19 and CYP2D6, plasma concentrations of moclobemide may be affected in patients who are poor metabolizers. In patients who are slow metabolizers, the AUC for moclobemide is 1.5 times greater than the AUC in patients who are extensive metabolizers and receiving the same dose. This increase is within the normal range of variation (up to twofold) typically seen in patients.

Drug interactions for the RIMAs include interaction with SSRI antidepressants, which can cause the 5-HT syndrome (see the discussion of SSRIs). The effect of stimulant drugs, such as methylphenidate and dextroamphetamine (used to treat ADHD), may be increased. Some over-the-counter cold and hay fever decongestants (i.e., sympathomimetic amines) can have increased stimulant effects. Selegiline, a selective MAO-B used for Parkinson disease, should not be used concurrently with the RIMAs. Unlike the irreversible MAOIs, no significant interactions with foods occur because the selective inhibition of MAO-A does not stop the metabolism of tyramine. The RIMAs must not be taken concurrently with a nonreversible MAOI.

Mood Stabilizers

Manic depression, or bipolar affective disorder, is a prevalent mental disorder with a global impact. Mood stabilizers have acute and long-term effects and, at a minimum, are prophylactic for manic or depressive disorders. Lithium is the classic mood stabilizer and exhibits significant effects on mania and depression but may be augmented or substituted by some antiepileptic drugs. The biochemical basis for mood stabilizer therapies and the molecular origins of bipolar disorder are unknown. Lithium ion directly inhibits two signal transduction pathways. It suppresses inositol trisphosphate signaling through depletion of intracellular inositol and inhibits glycogen synthase kinase-3 (GSK-3), a multifunctional protein kinase. A number of GSK-3 substrates are involved in neuronal function and organization and, therefore, present plausible targets for manic depression. Despite these intriguing observations, it remains unclear how changes in inositol trisphosphate signaling underlie the origins of bipolar disorder (83).

Inositol (myo-inositol), a naturally occurring isomer of glucose, is a key intermediate of the phosphatidylinositol signaling pathway, a second messenger system used by noradrenergic, serotonergic, and cholinergic receptors. The suggestion that lithium might treat mania via its reduction of inositol levels led to experiments showing that oral doses of inositol reverse the behavioral effects of lithium in animals and the side effects of lithium in humans. Cerebrospinal fluid levels of inositol are low in depressed individuals (84). The effectiveness of inositol in treating manic depression was shown in a double-blind trial, which showed that large doses of inositol (12 g) increased inositol concentrations in human cerebrospinal fluid by 70% and led to improvement in depressed patients compared to placebo (84). Valproic acid and carbamazepine are antiepileptic drugs with mood-stabilizing properties that also inhibit inositol trisphosphate signaling through the inositol-depletion mechanism.



Inositol significantly reduced the number of panic attacks per week in patients as compared to fluvoxamine and without the nausea and tiredness that are common with fluvoxamine. Inositol has few known side effects, thus making it attractive for administration to patients with manic depression who are ambivalent about taking other antidepressant drugs (83).

Lithium

Lithium (from the Greek word *lithos*, meaning “stone”) is a monovalent cation that competes with sodium, potassium, calcium, and magnesium ions at intracellular binding sites; at sugar phosphatases; at protein surfaces; at carrier binding sites; and at transport sites. Lithium readily passes through sodium channels, and high concentrations can block potassium channels. In the 1870s, claims for the healthful effects of lithium fueled the markets for products such as Lithia Beer and Lithia Springs Mineral Water (in 1887, analysis of Lithia Springs Mineral Water proved the water to be rich not only in lithium but also in potassium, calcium, magnesium, fluoride, and other essential trace minerals). In 1890, the Lithia Springs Sanitarium (Georgia) was established using natural lithium water to treat alcoholism, opium addiction, and compulsive behavior, even though manic depression had not been identified as a form of mental illness until the early 1900s.

Lithium's mood-stabilizing properties were revitalized in the 1940s when an Australian physician, John Cade, hypothesized that a toxin in the blood was responsible for bipolar illness. Believing that uric acid would protect individuals from this toxin, he began studying the effects of a mixture of uric acid and lithium in rats. Lithium carbonate was used to dissolve the uric acid. He observed a calming effect of this combination on the rats and subsequently determined that the lithium, rather than the uric acid, was responsible for this calming effect. He then speculated that lithium might be useful in humans as a mood attenuator, subsequently administered lithium to a sample of patients with bipolar disorder, and discovered that lithium not only decreased the symptoms of mania but also prevented the recurrence of both depression and mania when taken regularly by these patients. After a decade of clinical trials, the FDA approved lithium for treatment of mania in 1970.

Lithium carbonate (Eskalith) is the most commonly used salt of lithium to treat manic depression. Lithium carbonate dosage forms are labeled in mg and mEq/dosage unit, and lithium citrate (Lithobid) is labeled as mg equivalent to lithium carbonate and mEq/dosage unit. Lithium is effectively used to control and prevent manic episodes in 70% to 80% of those with bipolar disorder as well as to treat other forms of depression. Those who respond to lithium for depression often are those who have not responded to TCAs after several weeks of treatment. When giving lithium in addition to their antidepressants, some of these people have shown significant improvement.

MECHANISM OF ACTION Lithium therapy for disorders is believed to be effective because of its ability to reduce signal transduction through the phosphatidylinositol signaling pathway (Fig. 18.29) (84,85). In this pathway, the second messengers diacylglycerol (DAG) and inositol 1,4,5-trisphosphate are produced from the enzymatic hydrolysis of phosphatidylinositol-4,5-bisphosphate (a membrane phospholipid) by the receptor-mediated activation of the membrane-bound, phosphatidylinositol-specific phospholipase C. The second messenger activity for inositol 1,4,5-trisphosphate is terminated by its

hydrolysis in three steps by inositol monophosphatases to inactive inositol, thus completing the signaling pathway. To recharge the signaling pathway, inositol must be recycled back to phosphatidylinositol bisphosphate by inositol phospholipid-synthesizing enzymes in the CNS, because inositol is unable to cross the blood-brain barrier into the CNS in sufficient concentrations to maintain the signaling pathway. By uncompetitive inhibition of inositol phosphatases in the signaling pathway, the therapeutic plasma concentrations of lithium ion deplete the pool of inositol available for the resynthesis of phosphatidylinositol-4,5-bisphosphate, ultimately decreasing its cellular levels and, thereby, reducing the enzymatic formation of the second messengers. Thus, lithium ion restores the balance among aberrant signaling pathways in critical regions of the brain.

The effects of lithium ion on disorders are surprisingly specific because of the inability of inositol to cross the blood-brain barrier and replenish depleted inositol levels. Lithium ion exerts its greatest influence on this signaling pathway when the lithium ion concentration is at saturation conditions.

The clinical efficacy of lithium in the prophylaxis of recurrent affective episodes in bipolar disorder is characterized by a lag in onset and remains for weeks to months after discontinuation. Thus, the long-term therapeutic effect of lithium likely requires reprogramming of gene expression. Protein kinase C and GSK-3 signal transduction pathways are perturbed by chronic lithium at therapeutically relevant concentrations and have been implicated in modulating synaptic function in nerve terminals (85).

PHARMACOKINETICS The absorption of lithium is rapid and complete within 6 to 8 hours. The absorption rate of slow-release capsules is slower, and the total amount of lithium absorbed is lower than with other dosage forms. Lithium is not protein bound. The elimination half-life for elderly patients (39 hours) is longer than that for adult patients (24 hours), which in turn is longer than that for adolescent patients (18 hours). The time to peak serum concentration for lithium carbonate is dependent on the dosage form (tablets, 1 to 3 hours; extended tab, 4 hours; slow release, 3 hours). Steady-state serum concentrations are reached in 4 days, with the desirable dose targeted to give a maintenance lithium ion plasma concentration range of 0.6 to 1.2 mEq/L, with a level of 0.5 mEq/L for elderly patients. The risk of bipolar recurrence was approximately threefold greater for patients with lithium dosages that gave plasma concentrations of 0.4 to 0.6 mEq/L. Adverse reactions are frequent at therapeutic doses, and adherence is a big problem. Toxic reactions are rare at serum lithium ion levels of less than 1.5 mEq/L. Mild to moderate toxic reactions may occur at levels from 1.5 to 2.5 mEq/L, and severe reactions may be seen at levels from 2.0 to 2.5 mEq/L, depending on individual response. The onset of therapeutic action for clinical improvement is 1 to 3 weeks. Renal elimination of lithium ion is 95%, with 80% actively reabsorbed in the proximal tubule. The rate of lithium ion urinary excretion decreases with age. Fecal elimination is less than 1%.

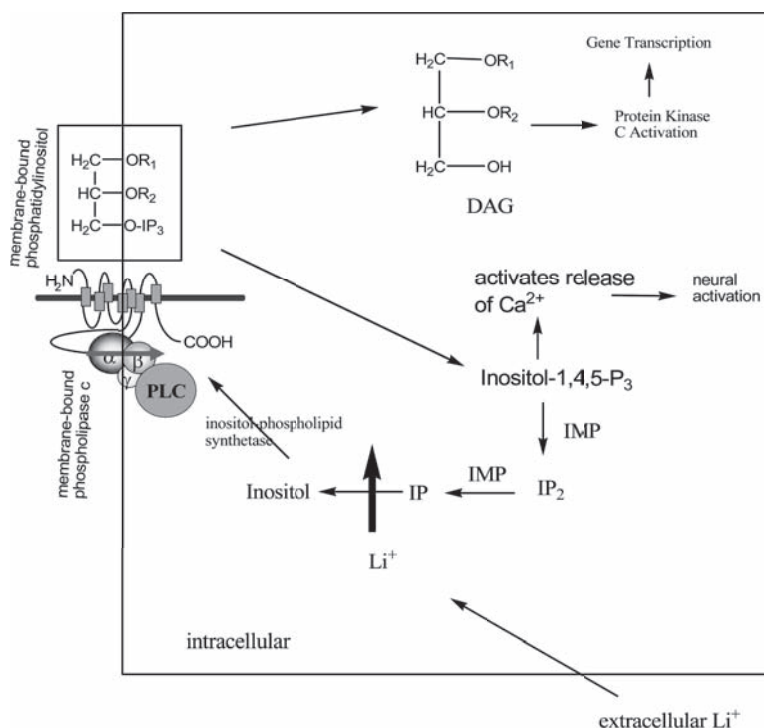


FIGURE 18.29 Intracellular phosphatidylinositol signaling pathway and site of action for lithium. Phospholipase C (PLC) is a membrane-bound enzyme. DAG, diacylglyceride; IMP, inositol monophosphate; IP, inositol monophosphate.

DRUG INTERACTIONS Lithium pharmacokinetics may be influenced by a number of factors, including age. Elderly patients require lower doses of lithium to achieve serum concentrations similar to those observed in younger adults as a result of reduced volume of distribution and reduced renal clearance. Lithium ion clearance decreases as the glomerular filtration rate decreases with increasing age. Reduced lithium ion clearance is expected in patients with hypertension, congestive heart failure, or renal dysfunction. Larger lithium ion maintenance doses are required in obese compared with nonobese patients. The most clinically significant pharmacokinetic drug interactions associated with lithium involve drugs that are commonly used in the elderly and that can increase serum lithium ion concentrations. People who are taking lithium should consult their physician before taking the following drugs: acetazolamide, antihypertensives, angiotensin-converting enzyme inhibitors, nonsteroidal anti-inflammatory drugs, calcium channel blockers, carbamazepine, thiazide diuretics, hydroxyzine, muscle relaxants, neuroleptics, table salt, baking powder, tetracycline, TCAs, MAOIs, and caffeine. The tolerability of lithium is lower in elderly patients. Lithium toxicity can occur in the elderly at concentrations considered to be “therapeutic” in the general adult populations. Serum concentrations of lithium ion need to be markedly reduced in the elderly population—and particularly so in the very old and frail.

ADVERSE EFFECTS Common side effects of lithium include nausea, loss of appetite, and mild diarrhea, which usually taper off within the first few weeks. Dizziness and hand tremors also have been reported. Increased production

of urine and excessive thirst are two common side effects that usually are not serious problems, but patients with kidney disease should not be given lithium. Taking the day’s dosage of lithium at bedtime also seems to help with the problem of increased urination. Other side effects of lithium include weight gain, hypothyroidism, increased white blood cell count, skin rashes, and birth defects.

While on lithium, a patient’s blood level must be closely monitored. If the blood level of lithium ion is too low, the patient’s symptoms will not be relieved. If the blood level of lithium ion is too high, there is a danger of a toxic reaction.

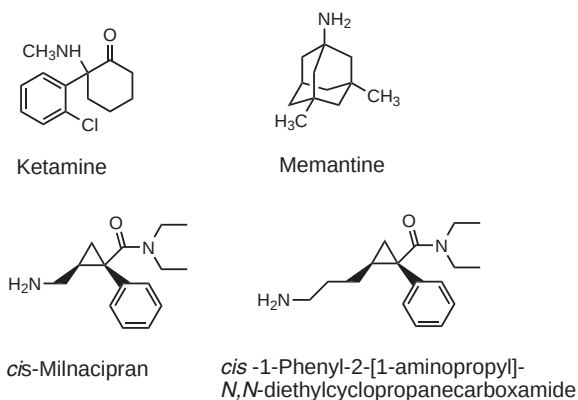
THERAPEUTIC USES For many years, lithium has been the treatment of choice for bipolar disorder, because it can be effective in smoothing out the mood swings common to this condition. Its use must be carefully monitored, however, because the range between an effective and a toxic dose is small.

Nonmonoaminergic Antidepressants

N-Methyl-D-Aspartate Antagonists

Despite intensive research, the problems of treating antidepressant-resistant patients and that existing treatments for MDD usually take weeks to months to achieve response and remission of depression have not yet been solved. The past decade has seen a steady accumulation of evidence supporting a role for the excitatory amino acid neurotransmitter glutamate and its mGluR1 and mGluR5 receptors in depression and antidepressant activity (86,87). Glutamate plays an essential role as a neurotransmitter in many physiologic functions, and an increase in glutamate release can

result in activation of NMDA receptors, an underlying cause for depression and anxiety. The NMDA receptor is a ligand-gated ion channel that mediates excitatory synaptic transmission in the CNS (Chapter 12). This channel opening and receptor activation are triggered by synaptically released glutamate and require the binding of glycine, which is a coagonist. Studies with NMDA receptor antagonists of mGluR1 and mGluR5 receptors, as well as positive modulators of AMPA receptors, have antidepressant-like activity in a variety of preclinical models. Furthermore, evidence implicates disturbances in glutamate metabolism, NMDA, and mGluR1/5 receptors in depression and suicidality.



Several preclinical studies in the 1990s revealed an important role for NMDA receptor antagonists in the mechanism of action of antidepressants and have generated considerable interest in the NMDA receptor as a target for new antidepressant therapies (85–88). Recent human studies have shown that the NMDA receptor antagonist ketamine leads to rapid and relatively sustained antidepressant effects in patients with treatment-resistant major depression (88,89). These findings have given rise to the hypothesis that the NMDA antagonist ketamine (an anesthetic) (Chapter 16) might have potential as an antidepressant, which has been validated in drug-free patients with MDD. A significant reduction in depression was observed 3 hours after a single infusion of ketamine, and this effect was sustained for at least 72 hours (88). This rapid antidepressant effect of ketamine has been replicated using larger sample sizes and treatment-resistant patients with depression. This rapid effect has a high therapeutic value in depressed patients who are suicidal, who might benefit from such a rapid and marked effect because their acute mortality risk is not considerably diminished with conventional antidepressants due to their long delay in onset of action (usually 2 to 3 weeks). Suicidal thoughts were reduced 24 hours after a single ketamine infusion. Ketamine therapy could be extended to other disorders in which NMDA receptors are implicated in bipolar disorders and addiction. The use of ketamine for the treatment of bipolar disorder is currently in phase III clinical trials.

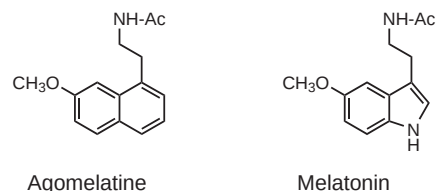
cis-1-Phenyl-2-[1-aminopropyl]-N,N-diethylcarboxamide, an NMDA receptor antagonist and homolog of milnacipran, produces sustained relief from depressive symptoms. Several studies have shown that chronic

antidepressant treatment can modulate NMDA receptor expression and function. Preclinical studies with this and other NMDA receptor antagonists have demonstrated their potential antidepressant properties.

Thus, the therapeutic effects of monoaminergic antidepressants, ketamine, and other NMDA antagonists may be mediated by increased AMPA-to-NMDA glutamate receptor throughput in critical neuronal circuits, and ketamine directly mediates this throughput. Because the monoaminergic antidepressants work indirectly and gradually, this may explain, in part, the lag of onset of several weeks to months that is observed with traditional antidepressants (88,89). These studies suggest that an intimate relationship exists between regulation of monoaminergic and excitatory amino acid neurotransmission and antidepressant effects.

The combination of traditional antidepressant drugs (e.g., imipramine) and uncompetitive NMDA receptor antagonists (e.g., memantine) may produce enhanced antidepressive effects as a result of synergism. This observation may be of particular importance for the treatment of antidepressant-resistant patients. Most interesting was the observation that fluoxetine, which was inactive in the forced swimming test in rats when given alone, showed a positive effect when combined with memantine (2.5 and 5 mg/kg).

Melatonergic Receptor Agonist/Serotonin Antagonist



The discovery and development of melatonin receptor agonists as antidepressants was motivated by the need for more potent drugs with better pharmacokinetics, a shorter onset of antidepressant action, and lesser adverse effects than the traditional monoamine reuptake inhibitors. Melatonin receptor agonists are analogs of melatonin. The chemical structure of agomelatine is very similar to that of melatonin, except that the indole ring of melatonin has been replaced with the bioisosteric naphthalene ring. Agomelatine, a dual-acting melatonin agonist/5-HT_{2C} receptor antagonist, is the first antidepressant with a non-monoamine reuptake mechanism of action (14). In 2009, agomelatine was approved in Europe for the treatment of MDD in adults and is currently under review by FDA for approval in 2011.

Melatonin is the primary neurohormone secreted by the pineal gland in the brain that regulates the body's circadian (light/dark) rhythm. The two-step synthesis of melatonin from 5-HT is regulated by the light/dark cycle. Circadian rhythm is the body's internal 24-hour "clock" that plays a critical role in when we fall asleep and when we wake up. When it is dark, your body produces more melatonin; when it is light, the production of melatonin

drops. Being exposed to bright lights in the evening or too little light during the day can disrupt the body's normal melatonin cycles. A core symptom of depression and a common problem in people with depression is a disruption in the normal melatonin sleep/wake rhythm.

MECHANISM OF ACTION Agomelatine is unique among FDA-approved melatonergic receptor agonists (e.g., ramelteon) because it exhibits affinity for both human melatonin receptor (MT) subtypes (MT₁ and MT₂) and has 5-HT_{2C} receptor antagonist properties (90). The antidepressant efficacy of agomelatine involves either complementary or synergistic actions on the MT and 5-HT_{2C} receptors. Binding studies indicate that it has no effect on monoamine reuptake and no affinity for α -adrenergic, β -adrenergic, histaminergic, cholinergic, dopaminergic, or benzodiazepine receptors. The MT₁ and MT₂ receptors are G protein-coupled, seven-membrane receptors responsible for the effects of melatonin on human circadian rhythms and on reproduction, which is also affected by the 24-hour sleep/wake cycle. These receptors are located in specific areas of the brain and in some peripheral organs. The MT₁ receptor is located in the hypothalamus, which is involved in circadian rhythm, and in the pituitary gland, which may be involved with melatonin's effect on reproduction. It can also be found in other areas of the brain. The MT₂ receptor is found mainly in the retina but can also be found in the lung, cardiovascular tissue, immune cells, duodenum, and adipocytes. When agomelatine binds to and activates both the MT₁ and MT₂ receptors, it triggers numerous physiologic processes, including the release of melatonin from the pineal gland, induction of sleep, and resynchronization of the altered sleep/wake circadian rhythm of depressed patients by delaying the sleep phase rhythm. It retains its antidepressant efficacy without the typical side effects of the other antidepressants such as sexual dysfunction and drug discontinuation symptoms commonly seen with the SSRIs and SNRIs (14). Hyperactivity of the 5-HT_{2C} receptors may contribute to the symptoms of MDD and may also be responsible for many of the negative side effects of SSRI and SNRI medications. Antagonism of 5-HT_{2C} receptors by agomelatine results in an increase in the release of DA and NE in the frontal cortex, the site most associated with depression in the brain, which may contribute to its overall antidepressant efficacy. Agomelatine does not affect the extracellular levels of 5-HT. The drug possesses a pharmacologic profile entirely distinct from SNRIs and SSRIs, making it a significant advance in the treatment of depression.

PHARMACOKINETICS Agomelatine is rapidly absorbed after oral administration, with an oral bioavailability of less than 5% at its therapeutic oral dose as a result of first-pass hepatic metabolism (90). Its bioavailability is increased in women who use oral contraceptives but is reduced by smoking, which induces CYP1A2. The peak plasma concentration is reached within 1 to 2 hours. In the therapeutic dose range, agomelatine exhibits linear pharmacokinetics, and at higher oral doses, a saturation of the first-pass effect occurs. Food intake does not modify its bioavailability or the absorption

rate but is increased with high-fat food. Its steady-state volume of distribution is about 35 L, plasma protein binding is 95%, and its pharmacokinetics are not modified with age or in patients with renal impairment, but the free fraction is doubled in patients with hepatic impairment. Following oral administration, agomelatine is rapidly metabolized mainly via hepatic CYP1A2 with a minor contribution from CYP2C9 and CYP2C19. The major metabolites are hydroxylated and *O*-demethylated agomelatine, which are inactive and rapidly glucuronide conjugated and eliminated in the urine. Elimination is rapid, with a mean plasma half-life of 2 to 3 hours. Excretion is mainly (80%) urinary in the form of metabolites, whereas unchanged parent drug recovery in urine is negligible. The pharmacokinetics of agomelatine are not modified after repeated administration.

THERAPEUTIC USES A reduction in sleep disruption, a core symptom of depression and a common problem in people with depression, without any daytime sedation is one of the clinical benefits of agomelatine treatment. Agomelatine is indicated for the treatment of MDD episodes in adults and is not recommended for use in children and adolescents younger than 18 years of age due to a lack of data on safety and efficacy. Because only limited clinical data are available on the use of agomelatine in patients ≥ 65 years old who have MDD episodes, caution should be exercised when prescribing it to these patients. From the first week of treatment, its onset of antidepressant efficacy, as measured by onset of sleep and the quality of sleep, was significantly improved without daytime drowsiness as assessed by patients. Agomelatine at 25 to 50 mg has been shown to be effective in more severely depressed patients whatever the intensity of depressive symptoms. It has comparable antidepressant activity to venlafaxine (75 to 150 mg/d) and was more effective than 50 to 100 mg/d of sertraline. Agomelatine was superior to placebo for relapse prevention at 10 months, and treatment adherence rates were higher than with venlafaxine and sertraline over 6 months. The commonly reported adverse effects in the clinical trials of agomelatine are headache, nausea, and diarrhea. It is found to increase the level of liver enzymes, and thus, monitoring of enzyme level is warranted before starting therapy and every 6 weeks thereafter. It is also contraindicated in patients with hepatic impairment.

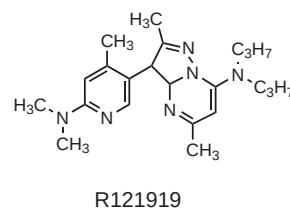
Neuropeptides

The pharmacologic treatment of depressive illness has been dominated by drugs that directly target monoamine neurotransmitter systems. Monoamine transport inhibitors are first-line treatments for depression. Current antidepressants exhibit a delayed onset of therapeutic action, and a significant number of patients are nonresponsive to this treatment regimen (54). Moreover, many patients discontinue treatment because of adverse side effects, including nausea, sexual dysfunction, anorexia, mouth dryness, and cardiotoxicity. A complementary strategy is to identify other treatments that target other neurotransmitter and neuromodulators in the brain. Neuropeptides have been shown to be attractive targets for depression (91,92). Neuropeptides

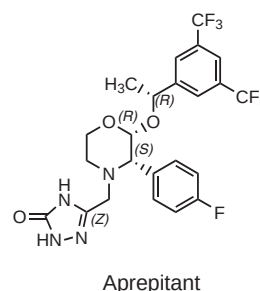
are short-chain amino acid neurotransmitters and neuro-modulators often localized in brain regions that mediate emotional behaviors and the response to stress (91,92). The neuropeptides that have been identified in stress include the tachykinins (substance P and neurokinin A), corticotropin-releasing factor, vasopressin, galanin, brain-derived neurotrophic factor (BDNF), and melanocyte-inhibiting factor. Thus, drugs that are antagonists at these neuropeptide receptors might exhibit a lower incidence of adverse effects, because such antagonists would not be expected to bind to the NE and 5-HT neurotransmitter receptors. The expression of BDNF in individuals with MDD is decreased, suggesting that BDNF plays a role in the pathophysiology of depression as a regulator of neuronal signaling pathways. Antidepressant drugs increase the BDNF brain levels and, therefore, enhance the mechanism of action of the antidepressant drugs. Drugs that boost the levels of BDNF may lead to the development of novel therapeutic agents for the treatment of MDD and bipolar disorders.

CORTICOTROPIN-RELEASING FACTOR Increasing evidence suggests that the neuroendocrine changes seen in patients suffering from affective disorders may be causally related to the course of depression. The most robustly confirmed neuroendocrine finding among patients with affective disorders is hyperactivity of the hypothalamic-pituitary-adrenocortical (HPA) system, resulting from hyperactive hypothalamic CRF neurons. Moreover, one of the most replicated findings in biologic psychiatry is that large numbers of unmedicated depressed patients exhibit HPA hyperactivity. The available evidence suggests that nAChRs play important roles in mediating stress-related and possibly depression-inducing neuroendocrine effects of ACh. ACh (hypercholinergic activity), in response to stress, can stimulate the HPA through activation of nAChRs. Abnormal HPA activity has been implicated in conditions related to stress, including HPA overactivation in depression, eating and substance abuse disorders, irritable bowel syndrome, inflammation, and cardiovascular dysfunction. Preclinical and clinical evidence suggests that both genetic and environmental factors contribute to the development of these HPA system abnormalities. CRF is a 41-amino acid neuropeptide that initiates and regulates the HPA axis response to stress, and it has been intensively studied in the pathophysiology and treatment of depression (93). In humans, the CRF system consists of CRF and two G protein-coupled CRF receptors (CRF₁ and CRF₂). The CRF₁ receptors play an important role in mediating the HPA response to stress. Additionally, CRF is capable of reproducing the hormonal changes that are characteristically seen in depressed patients. Postmortem and endocrine studies suggest that both hypothalamic and extrahypothalamic concentrations of CRF are elevated in proportion to antidepressant treatment. High CRF concentrations tend to reestablish the HPA imbalance. The careful manipulation of CRF concentrations with high-affinity CRF₁ antagonists, such as R121919, may hold therapeutic promise for sufferers of depression. Thus, antidepressants may reduce

symptoms of depression, in part, through blockade of AChRs involved with stress-induced activation of the HPA (8).



SUBSTANCE P Substance P is an 11-amino acid neuropeptide belonging to the tachykinin family, mediating its biologic actions through activation of G protein-coupled tachykinin (neurokinin-1 [NK₁]) receptors. Its proposed physiologic roles include inflammation, pain, GI and respiratory function, stress responses, and emesis. Substance P is uniquely associated with the monoamine neurotransmitters 5-HT and NE. The 5-HT neurons coexpress substance P, and the firing of NE neurons is modulated by substance P. Preclinical studies have supported a role of the substance P-NK₁ receptor system in stress-related disorders, which has guided the antidepressant development of centrally active NK₁ receptor antagonists, such as aprepitant (94). Aprepitant (Emend) has been approved for the prevention of nausea and vomiting due to cancer chemotherapy. The NK₁ antagonists are generally well tolerated and exhibit less nausea and sexual dysfunction than some currently used antidepressants.

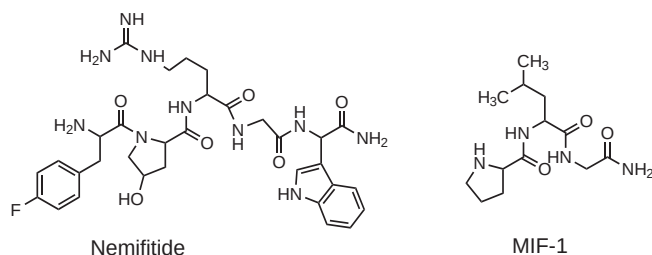


VASOPRESSIN The nonapeptide vasopressin is well known for its role on fluid metabolism (see Chapter 6), but it also is a key regulator of the HPA axis. Stress stimulates the release of vasopressin in the pituitary gland, where it strongly potentiates the effects of CRF on adrenocorticotrophic hormone release. These findings suggest that HPA axis dysregulation in depression might be associated with the development of centrally acting vasopressin receptor antagonists for the treatment of depression.

GALANIN Since its discovery in 1983, the neuropeptide galanin has been found to be involved in a wide range of functions, including pain sensation, sexual activity, feeding, and learning and memory (95). Galanin is widely distributed in the central and peripheral nervous systems and in the endocrine system, and it acts as an inhibitory neuromodulator of NE and 5-HT in the brain. The

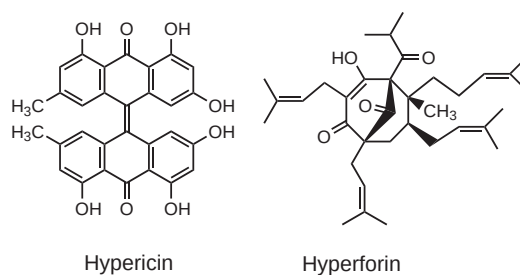
29- to 30-amino acid sequence of galanin is conserved (almost 90% among species), indicating the importance of the molecule among species. Galanin is colocalized with ACh, 5-HT, and NE in neurons or in brain regions implicated in cognitive and affective behavior, suggesting a possible role in the regulation of 5-HT and NA neurotransmission in depressive states and during the course of antidepressant therapy. Three galanin receptor subtypes have been cloned and studied, but little is known about their specific contributions to behavioral processes. In the CNS, galanin inhibits ACh release, suggesting a possible role for galanin in cholinergic dysfunction; inhibits neurotransmitter release and neuronal firing rate; and inhibits signal transduction by inhibition of phosphatidylinositol hydrolysis, leading to symptoms of depression. Thus, blocking the inhibitory effects of galanin on monoamine neurotransmitters with galanin receptor antagonists would be predicted to mimic or augment the action of the other monoamine classes of antidepressants.

MELANOCYTE-INHIBITING FACTOR



Nemifitide is a peptide analog of melanocyte-stimulating hormone release-inhibiting factor currently in clinical development for the potential treatment of moderate to severely depressed patients (96). It is rapidly absorbed, with a peak plasma concentration of 10 minutes and an elimination half-life of 15 to 30 minutes in most subjects. The pharmacokinetic results indicate that the dose is proportional in the dose range investigated. No evidence indicated systemic accumulation of drug following five daily doses. No serious adverse events or clinically significant systemic adverse events occurred at any of the doses investigated in the more than 100 subjects dosed in these studies. Drug-related adverse events were limited to local and transient skin reactions (pain and/or erythema) at the injection site, especially at the high doses administered. Melanocyte-stimulating hormone release-inhibiting factor-1 (MIF-1) has been shown to have antidepressant activity when administered subcutaneously (10 mg for 5 consecutive days) in a double-blind study of 20 depressed patients who all met the DSM-III-R criteria for major depression. After the 5-day treatment with MIF-1, these patients all exhibited substantial improvement in their symptoms of depression (97). Moreover, the potential clinical efficacy of combining MIF-1 (0.01 mg/kg intraperitoneal [IP]) with small doses of the TCAs amitriptyline (5 mg/kg IP) or desipramine (1.25 mg/kg IP) may be of benefit in the therapy of depressed patients.

Herbal Therapy



During the past few years, much interest has been generated regarding the use of herbs in the treatment of both depression and anxiety (98). Recent studies, however, have revealed potentially fatal interactions between herbal remedies and traditional drugs.

St. John's wort (*Hypericum perforatum*), an herb used extensively in the treatment of mild to moderate depression in Europe, has aroused interest in the United States. A bushy, low-growing plant covered with yellow flowers in summer, St. John's wort has been used for centuries in many folk and herbal remedies. In Germany, *Hypericum* is used in the treatment of depression more than any other antidepressant. The scientific studies that have been conducted regarding its use have been short term, however, and have used several different doses. St. John's wort works like the SSRIs, in that it not only increases the availability of 5-HT in synaptic clefts by blocking its reuptake but also increases the availability of NE, which increases energy and alertness, and DA, which increases the feeling of well-being.

Ingestion of St. John's wort increases the expression (i.e., upregulation) of intestinal P-gp and CYP3A4 in the liver and intestine, which impairs the absorption and stimulates the metabolism of other CYP3A4 substrates (e.g., the protease inhibitors indinavir and nevirapine, oral contraceptives, and TCAs [e.g., amitriptyline]), resulting in their subtherapeutic plasma levels. Hyperforin, the principal component in St. John's wort (2% to 4% in the fresh herb), contributes to the induction of CYP3A4. Furthermore, it not only inhibits the neuronal reuptake of 5-HT, NE, and DA, like many other antidepressants, but also inhibits GABA and L-glutamate uptake. This broad-spectrum effect is obtained by an elevation of the intracellular sodium ion concentration, probably resulting from activation of sodium conductive pathways not yet finally identified but most likely to be ionic channels. This makes hyperforin the first member of a new class of compounds with a pre-clinical antidepressant profile because of a completely novel mechanism of action (98,99). Hypericin, the other component in St. John's wort, also may exhibit inhibitor action on key neuroreceptors and may be responsible for the phototoxicity/photosensitivity of St. John's wort.

The National Institutes of Health conducted a double-blind, 3-year study of patients with major depression of moderate severity using St. John's wort and sertraline. This study did not support the use of St. John's wort in

the treatment of major depression, but a possible role for St. John's wort in the treatment of milder forms of depression was suggested. Health care providers should alert their patients about potential drug interactions with St. John's wort. Some other frequently used herbal supplements that have not been evaluated in large-scale clinical trials are ephedra, ginkgo biloba, echinacea, and ginseng. Any herbal supplement should be taken only after consultation with the physician or other health care provider.

ELECTROCONVULSIVE THERAPY

Electroconvulsive therapy (ECT) has been in use since the late 1930s to treat a variety of severe mental illnesses, most notably major depression. Use of ECT is beneficial particularly for individuals whose depression is severe or life threatening or who cannot take antidepressant medication. Often, ECT is effective in cases where antidepressant drugs do not provide sufficient relief of symptoms.

ECT remains the "gold standard" for the treatment of major depression and a variety of other psychiatric and neurologic disorders (100). Because of the effectiveness and resurgence of ECT, more patients are considered to be

good candidates for this treatment option. Overall, these patients are medication refractory and elderly and, thus, are more sensitive to polypharmacy. Additionally, these patients tend to have more coexisting medical problems.

In recent years, ECT has been much improved. A muscle relaxant is given before treatment, which is done under brief anesthesia. Electrodes are placed at precise locations on the head to deliver electrical impulses. The stimulation causes a brief (~30 seconds) seizure within the brain. The person receiving ECT does not consciously experience the electrical stimulus. For full therapeutic benefit, at least several sessions of ECT, typically given at the rate of three per week, are required. Electroconvulsive therapy appears to increase the sensitivity of postsynaptic 5-HT receptors and upregulation of 5-HT_{1A} postsynaptic receptors.

Side effects may result from the anesthesia, the ECT treatment, or both. Common side effects include temporary short-term memory loss, nausea, muscle aches, and headache. Some people may have longer-lasting problems with memory after ECT. Sometimes, a person's blood pressure or heart rhythm changes. If these changes occur, they are carefully watched during the ECT treatments and are immediately treated.

SCENARIO: OUTCOME AND ANALYSIS

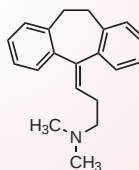
Outcome

Nancy Ordonez, PharmD

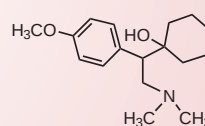
There are several issues that need to be addressed during GP's current visit. GP's depressive symptoms have not improved even though the dose of venlafaxine has been increased to a total of 300 mg/d. In addition, GP's uncontrolled hypertension needs to be addressed as well as his worsening peripheral neuropathy. Venlafaxine can cause hypertension, especially in higher doses, and with GP's uncontrolled hypertension, venlafaxine may not be the best therapeutic option for his depression. Amitriptyline is indicated for peripheral neuropathy; however, it can cause sedation and has cardiovascular side effects. GP's venlafaxine will be gradually tapered and discontinued, and amitriptyline will be discontinued. Duloxetine was started at 20 mg twice daily and will be increased gradually. Duloxetine is indicated for depression and diabetic neuropathy. Blood pressures, blood glucose, as well as his signs of improvement of his depressive symptoms and peripheral neuropathy will be monitored. GP scheduled a follow-up visit in 3 weeks.

Chemical Analysis

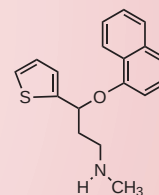
Victoria F. Roche and S. William Zito



Amitriptyline



Venlafaxine



Duloxetine

Amitriptyline is a tertiary tricyclic antidepressant that acts in parent drug form to inhibit the reuptake of central serotonin. The increased serotonin in central synapses can lead to sedation, which explains GP's morning drowsiness. The strong structural overlap with antimuscarinic and antihistaminic agents also adds to the sedative potential of this agent, and it explains some of its

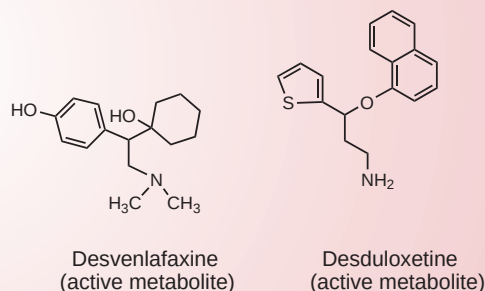
SCENARIO: OUTCOME AND ANALYSIS (Continued)

peripheral adverse effects including dry mouth, blurred vision, urinary retention, tachycardia, and orthostatic hypotension.

Venlafaxine and duloxetine both inhibit the reuptake of central norepinephrine as well as serotonin, with duloxetine being the more potent. Venlafaxine's tertiary amine structure may help explain its preference for blocking serotonin reuptake; however, selectivity will be lost at the high doses GP was taking. As norepinephrine concentrations in the CNS and periphery rise, so can blood pressure. Dependence can develop when venlafaxine is used chronically, which necessitates a step-down discontinuation regimen to avoid withdrawal symptoms. Venlafaxine is O-dealkylated by CYP2D6 to an equally active phenolic metabolite, desvenlafaxine. In addition to serving as a 2D6 substrate, venlafaxine inhibits this isoform. Both of these metabolic realities could lead to drug-drug interactions with other 2D6 substrates, such as metoprolol, which is the most highly dependent of all β -blockers on 2D6-mediated metabolism.

The secondary amine duloxetine is the more potent inhibitor of serotonin and norepinephrine reuptake and might be able to manage GP's depressive symptoms in lower doses. In therapeutic doses, it preferentially inhibits the serotonin reuptake transporter

and, therefore, would be less likely to raise blood pressure than high doses of venlafaxine. Like venlafaxine, duloxetine has shown efficacy in the treatment of diabetic neuropathic pain. However, also like venlafaxine, it is both a substrate (N-dealkylation) and an inhibitor of CYP2D6, and a more pronounced antihypertensive effect may be noted from GP's maintenance dose of metoprolol.



As noted above, GP's blood pressure should be closely monitored and any needed adjustment in antihypertensive therapy made as he is stabilized on this new antidepressant medication.

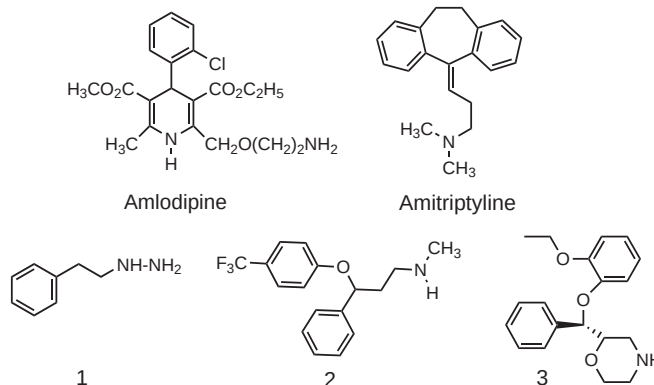
CASE STUDY

Victoria Roche and S. William Zito

MF is a 32-year-old computer programmer who has struggled with depression for years. As an adolescent, he experienced suicidal ideation and once made an unsuccessful attempt to end his life. MF married 8 months ago and he and his wife, a marketing executive for a national chain of high-end wine and cheese shops, have moved to Bordeaux, France, for a 3-year opportunity related to her work. His wife is very loving and supportive, and he felt better than he had in years during their courtship and after the wedding. But now, with the move complete, those old feelings of despair and panic are starting to close in again. Never highly compliant with taking his medication, his wife urged him to resume his therapy, but

he confides that his most currently prescribed antidepressant, amitriptyline, "just makes him feel bad all over." His current prescriptions also include low-dose amlodipine for modestly elevated blood pressure, another medication with which he is noncompliant.

With his wife at his side, MF has consulted with a psychiatrist in their new community about the return of his symptoms. This psychiatrist ordered a pharmacogenetic workup and it was found that MF is of the poor 2D6 metabolizer and slow acetylator phenotype. If you were lucky enough to be working as a psych pharmacist in beautiful French "wine country," which of the three drugs drawn below would you recommend?



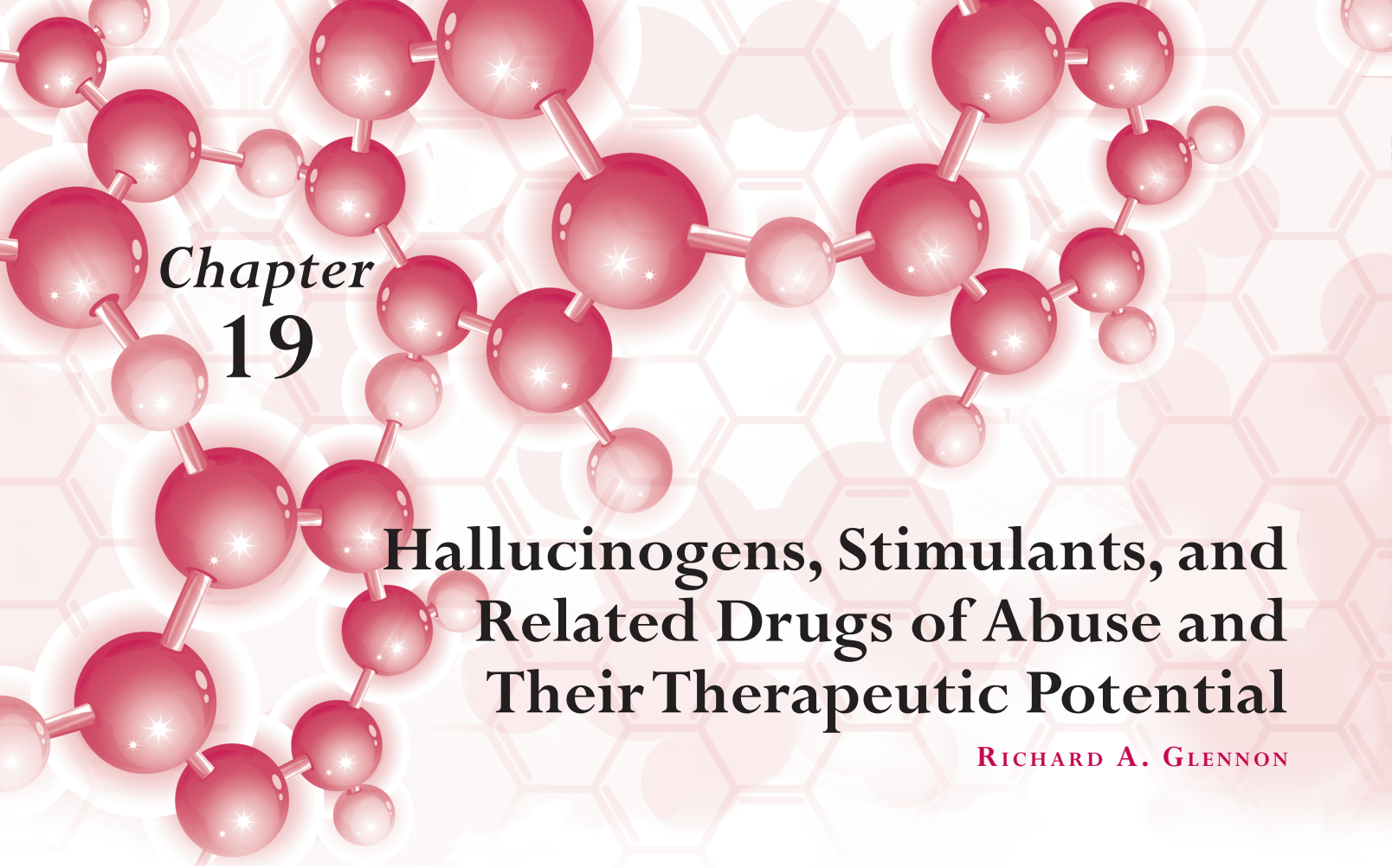
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 19

Hallucinogens, Stimulants, and Related Drugs of Abuse and Their Therapeutic Potential

RICHARD A. GLENNON

Abbreviations

A, adenosine
ADHD, attention-deficient hyperactivity disorder
2-AG, 2-arachidonylglycerol
cAMP, cyclic adenosine monophosphate
CB, cannabinoid
CBD, cannabidiol
DA, dopamine
DARPP-32, dopamine- and cAMP-regulated phosphoprotein, 32 kd
DET, *N,N*-diethyltryptamine
DMA, dimethoxyphenyl-2-aminopropane
DMH, 1,1-dimethylheptyl
DMT, *N,N*-dimethyltryptamine
DOB, 1-(4-bromo-2,5-dimethoxyphenyl)-2-aminopropane
DOI, 1-(2,5-dimethoxy-4-iodophenyl)-2-aminopropane
DOM, 1-(2,5-dimethoxy-4-methylphenyl)-2-aminopropane
DPT, *N,N*-di-*n*-propyltryptamine
d-TMP, *d*-threo-methylphenidate
ED₅₀, median effective dose

α-EtT, α-ethyltryptamine
FAAH, fatty acid amide hydrolase
GABA, γ-aminobutyric acid
GPCR, G protein-coupled receptor
5-HT, serotonin
IP₃, inositol triphosphate
LSD, lysergic acid diethylamide
MAGL, monoacylglycerol lipase
MAO, monoamine oxidase
MDA, 1-(3,4-methylenedioxyphenyl)-2-aminopropane
MDMA, *N*-methyl-1-(3,4-methylenedioxyphenyl)-2-aminopropane
MET, MethylEthylTryptamine or more correctly *N*-ethyl-*N*-methyltryptamine
α-MeT, α-methyltryptamine
NANM, *N*-allylnormetazocine
NE, norepinephrine
NMDA, *N*-methyl-D-aspartate
5-OMe α-Met, 5-methoxy-α-methyltryptamine
PCA, para-chloroamphetamine

PCP, phencyclidine
PDE, phosphodiesterase
PKA, phosphokinase A
PLC, phospholipase C
PMA, para-methoxyamphetamine
PMMA, *N*-methyl-1-(4-methoxyphenyl)-2-aminopropane
PP-1, protein phosphatase-1
PP-2A, protein phosphatase-2A
pTAP, 1-(para-tolyl)-2-aminopropane
REM, rapid eye movement
SAR, structure-activity relationship
Ser¹³⁷, serine¹³⁷
SSRI, selective serotonin reuptake inhibitor
Δ⁹-THC, Δ⁹-tetrahydrocannabinol
Thr³⁴, threonine³⁴
TMA, trimethoxyphenyl-2-aminopropane

SCENARIO

Mark D. Watanabe, PharmD, PhD, BCPP

JN, a 23-year-old male athlete from the United States, was offered some tablets of “flatliners” while at a dance club during a vacation in London. Having never tried them before, JN had heard that it was somehow related to “Ecstasy” and was willing to accept the offer. Shortly after ingestion, he felt his energy level increasing, his heart beating faster and stronger,

and some of his anxiety about being in a foreign country going away. Eventually, he danced for several hours without stopping.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

PSYCHOTOMIMETIC/HALLUCINOGENIC AGENTS

Why study psychotomimetic/hallucinogenic agents? It has been argued that investigations of such agents might shed light on mental illnesses and their potential treatment. Although studies with such agents have certainly contributed to our understanding of mental disorders, it is now recognized that there are many kinds of mental illnesses and that the actions—and putative mechanisms of action—of psychotomimetic/hallucinogenic agents are only tangentially related to their etiology or treatment. It has also been argued that investigations of psychotomimetic/hallucinogenic agents might contribute to a greater general understanding of basic neurochemical mechanisms and neurotransmitter function. This research approach has been much more rewarding. Studies with such agents have contributed *significantly* to what is currently known about G protein–coupled receptors (GPCRs) (e.g., cannabinoid receptors and serotonin [5-HT] receptors) and ion channel receptors (e.g., phencyclidine [PCP] receptors and excitatory amino acid receptors). Subsequent work with these receptors has identified new receptor subtypes that are being targeted for the development of novel therapeutic agents. Indeed, the past 10 to 20 years have witnessed an explosion of interest in the investigation of psychoactive substances because of their relevance to neurochemical mechanisms. Another important reason to study psychotomimetic agents is because they represent a large group of abused substances, and pharmacists are members of a health care team responsible for the dissemination of drug abuse prevention and treatment information. In addition, the past one to two decades have seen the popularization of controlled substance analogs (i.e., *designer drugs*), and the future will likely witness the introduction of yet more. So, yet another reason to study these agents is to prepare for the future: An understanding of presently available agents and their structure–activity relationships (SARs) will be instructive because many designer drugs emanate from the clandestine application of these same SAR principles at

the street level. Also, some agents currently in clinical use possess abuse liability that is underrecognized. An understanding of how these drugs work can lead to new treatment modalities. Finally, investigations of these agents presage the development of novel therapies for the treatment of various neuropsychiatric, neurologic, and other disorders (see later discussion).

DEFINITIONS AND CLASSIFICATION

“Psychotomimetics” and “hallucinogens” are commonly used terms, and they are frequently used interchangeably. However, little agreement exists regarding what constitutes such agents or exactly what they do. Because the actions of these agents are largely subjective, the best information should come from those who experience these agents; yet, by experiencing their effect, one might not be in the best position to accurately describe the effects they produce (1b). In contrast, an “outside” observer can never fully and accurately describe the effects of these agents. This has led to problems of definition. The most widely accepted definition of a psychotomimetic substance is that provided by Hollister (2): Psychotomimetic/hallucinogenic agents are those that upon administration of a single effective dose consistently produce changes in thought, mood, and perception with little memory impairment; produce little stupor, narcosis, or excessive stimulation; produce minimal autonomic side effects; and are nonaddicting.

Although certain opioid analgesics occasionally produce psychotomimetic effects, they are effectively eliminated from this category of agents because they do not meet the necessary criteria (e.g., they can be addicting). Likewise, chronic administration of high doses of stimulants, such as amphetamine and cocaine, sometimes produce hallucinogenic episodes (i.e., amphetamine psychosis and cocaine psychosis). These agents are not considered to be hallucinogens, however, because multiple doses typically are required to produce this effect. The Hollister criteria served a very useful function in narrowing the list of agents that

CLINICAL SIGNIFICANCE



The manipulation of structure-activity relationships to alter pharmacological activity for recreational benefit lies at the heart of the “designer drug” concept. The clinical impact is clearly evident in the class of drugs discussed in this chapter, i.e., hallucinogens and stimulants. Many of the illicit street drugs intended to produce euphoria and enhanced awareness were developed based on an understanding of how specific modifications in chemical structure can lead to desired changes in bioavailability and pharmacodynamics. Unfortunately, although these approaches provide greater insight regarding the scientific basis of drug action, they are not without risk.

In a recent editorial (1a), David Nichols, whose laboratory studied the MDMA (“ecstasy”) analogue 4-methylthioamphetamine

(4-MTA) in the 1990s, lamented the eventual consequences of his published research that resulted in motivating street chemists to synthesize 4-MTA for use as a club drug. Given the putative mechanism of action of 4-MTA, its inappropriate use has led to some potentially predictable fatalities due to elevated serotonin levels. Dr. Nichols’ concerns highlight a central challenge of medicinal chemistry in drug development: for every alteration in molecular structure intended for improved therapeutic efficacy and safety, there is often a parallel possibility of increasing abuse and toxicity.

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belong to this category of drugs, and Hollister was able to identify several classes of psychotomimetic agents: lysergic acid derivatives (e.g., lysergic acid diethylamide [LSD]), phenylethylamines (e.g., mescaline), indolealkylamines (e.g., *N,N*-dimethyltryptamine), other indolic derivatives (e.g., ibogaine and the harmala derivatives), piperidyl benzilate esters (e.g., JB-329), phenylcyclohexyl compounds (e.g., PCP), and miscellaneous agents (e.g., kawain, dimethylacetamide, and cannabinoids) (2).

Over time, it was demonstrated that *psychotomimetic agents* represent a *behaviorally heterogeneous* class of psychoactive agents. For example, human subjects can readily differentiate between the actions produced by certain agents in this category, and cross-tolerance develops among some of these agents but not between others. Likewise, it is possible to differentiate between the actions of certain of these agents in humans and using various animal procedures. Subcategorization was necessary. Today, it is well recognized that some psychotomimetics/hallucinogens act primarily via a serotonergic mechanism, that cannabinoids produce many of their behavioral effects via cannabinoid receptors, and that PCP-like agents produce their effects via PCP receptors. This is not to imply that there now exists a full understanding of how these agents work, but it does support the concept that these agents do not belong to a homogeneous mechanistic class.

Human Versus Animal Studies: Applicability of Animal Models

Human subjects should be best suited to provide the most reliable assessment of the actions and potency

of psychotomimetic agents, and considerable human data are available for some agents. However, much less information is available for most. Often, what human information is available comes from studies that were not well controlled, that included limited subject populations, or that investigated only a few drug doses. Some of what is known even comes from anecdotal reports. Very few clinical studies with psychotomimetic agents were sanctioned following the early 1960s. Although some human evaluations are once again being allowed, for a period of approximately 30 years, information concerning psychotomimetic substances relied—and continues to rely—heavily on the use of animal studies. This raises several questions: Do animal models exist that can accurately reflect human hallucinogenic activity? Indeed, do animals hallucinate? Many attempts have been made to develop animal models of psychotomimetic/hallucinogenic activity, but to date, no single animal model can account for the actions of these agents as a class (3).

Drug Discrimination Paradigm

One animal technique that has seen widespread application for the investigation of psychoactive agents is the drug discrimination paradigm (4). It must be emphasized at the outset that this method does not represent a model of psychotomimetic activity. Indeed, the technique has general applicability and has been employed to study a wide variety of centrally acting agents, including stimulants, antidepressants, barbiturates, anxiolytics, opiates, and many other drug classes. The technique may be viewed as a “*drug detection*” procedure. Specifically, animals (typically rats, pigeons, or

monkeys) are trained to recognize or *discriminate* the stimulus effects of a training drug from vehicle; humans have also been used as subjects in some drug discrimination studies. Many centrally acting agents produce an interoceptive cue, or stimulus, that subjects recognize. When animals are used, they are taught to make a particular response (e.g., to respond on one lever of a two-lever operant apparatus or Skinner box) when administered a training drug and to make a different response (e.g., to respond on the second of the two levers) when administered saline vehicle. After a period of time, the animals learn the stimulus cue and associate it with one of the two levers; that is, the animals make >80% of their responses on the training-drug lever (i.e., >80% drug-appropriate responding) when administered the training dose of the training drug, and <20% of their responses on the same lever, when administered vehicle. Doses of training drug less than those of the training dose result in a decreased percentage of drug-appropriate responding. The effect is dose related, and a dose–response curve can be constructed. A median effective dose (ED_{50} value) can be calculated as a measure of potency.

Once trained, these animals can be used in what is referred to as tests of substitution or stimulus transfer (i.e., tests of stimulus generalization). In such tests, other agents (i.e., challenge drugs) are administered to the animals to determine if they produce stimulus effects similar to those of the training drug. Stimulus generalization is said to have occurred when animals make >80% of their responses on the training drug-appropriate lever following administration of some dose of challenge drug. Stimulus generalization or substitution implies that the *challenge drug* and the *training drug* are producing similar stimulus effects in the animals. It should be noted that no claim has ever been made that the agents—the training drug and a challenge drug—are producing identical effects; rather, there is an implication that the agents are capable of producing a common stimulus effect or a behavioral cue common to the two agents (e.g., a drug that produces effects A and B may be recognized by animals trained to a drug that produces effects B and C; although this is an uncommon occurrence, it should be recognized that it is possible). Thus, not only is it possible to determine if two agents are producing similar stimulus effects, it is also possible to compare their relative potencies by calculating an ED_{50} value for the challenge drug. Hence, results are qualitative and quantitative.

Other studies that can be conducted are tests of stimulus antagonism. That is, a specific training drug can be administered together with another agent; if the combination results in <20% training drug-appropriate responding, stimulus antagonism is said to have occurred. Although this technique can be used in the development of novel antagonists for a series

of agents for which an antagonist is unknown, it is more common to use receptor-selective antagonists to investigate mechanisms of action. Drug discrimination, then, is a very powerful tool for investigating the actions and mechanisms of action of many different kinds of centrally acting agents. Specific examples of stimulus generalization and stimulus antagonists will be described later.

The drug discrimination procedure has seen broad application in the investigation of centrally acting agents, and a wide variety of different training drugs has been used. When a psychotomimetic/hallucinogenic agent is used as the training drug, it should be possible to identify other agents that produce similar stimulus effects (5). In this manner, it has been demonstrated that the psychotomimetics represent a behaviorally heterogeneous group of agents, much in the same way that humans have been able to differentiate the effects of these agents. Animals trained to discriminate LSD, for example, do not recognize PCP, and animals trained to discriminate PCP do not recognize LSD. Neither LSD- nor PCP-trained animals recognize tetrahydrocannabinol (Δ^9 -THC). However, LSD-trained animals recognize mescaline, 1-(2,5-dimethoxy-4-methylphenyl)-2-aminopropane (DOM), and certain other hallucinogens. Using this technique, then, it has been possible to identify what are termed “*classical hallucinogens*” (6). The classical hallucinogens are LSD-like agents that share common discriminative stimulus properties and that might act via a common mechanism of action. The remaining psychotomimetic agents will be referred to here as nonclassical agents; these groups of agents act by different mechanisms and produce distinct effects common to members within each group.

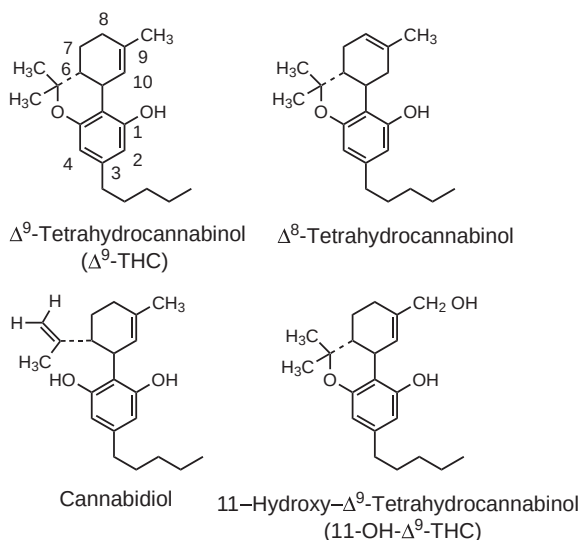
PSYCHOACTIVE DRUGS OF ABUSE: NONCLASSICAL AGENTS

The term “*nonclassical agents*” is used here to differentiate these psychoactive agents from the *classical hallucinogens* that will be discussed later in this chapter. Several categories of agents are described here, but there is no implication that these classes of agents produce similar effects or act via similar mechanisms.

Cannabinoids

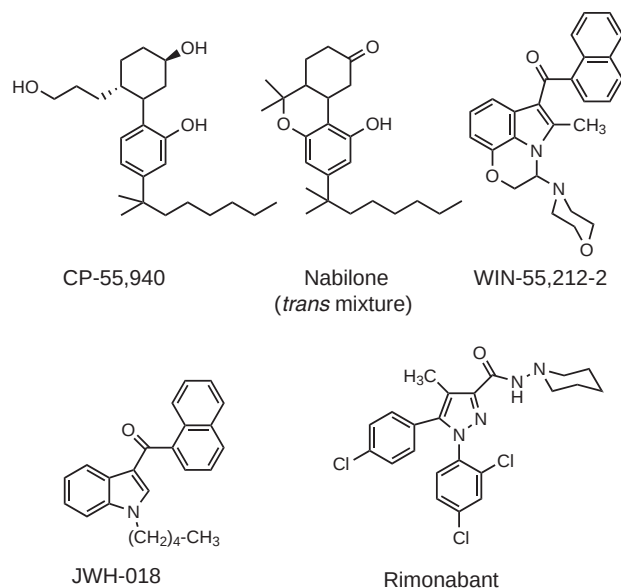
The marijuana or cannabis plant represents one of the oldest and most widely used psychoactive substances in the world and has been cultivated since approximately 6000 BC. Botanically, there are three major species of the plant—*Cannabis sativa*, *Cannabis indica*, and *Cannabis ruderalis*. Reference is made to three preparations, listed here in order of increasing potency: *Bhang* typically refers to the leaves and stems of the plant, *ganja* is prepared from the flowering tops of the

plant, and *hashish* is the pure resin. Although marijuana is orally active, inhalation by smoking is more frequently used as a route of administration. One of the major active constituents of the plant is Δ^9 -THC (often referred to simply as THC). Δ^9 -THC is rapidly and efficiently absorbed by inhalation; it is absorbed into body tissue and is slowly released back into circulation. Deuterium-labeled Δ^9 -THC has been detected in human plasma up to nearly 2 weeks postadministration. A major metabolite of Δ^9 -THC is 11-hydroxy- Δ^9 -THC. Evidence suggests that tolerance develops to Δ^9 -THC and that it does not usually lead to physical dependence. Marijuana can produce impairment of performance, memory, and learning; controversy exists over whether it produces an amotivational syndrome.



Over the years, many cannabinoids and related structures (i.e., synthetic cannabinoids), such as CP-55,940 and nabilone (a mixture of *trans* isomers: *RR* and *SS*), were prepared and evaluated (7–9). Noncannabinoids, such as WIN-55,212-2 and JWH-018, can also produce THC-like actions. An herbal product commonly known as Spice or K2 has been shown to contain various synthetic cannabinoids including JWH-018.

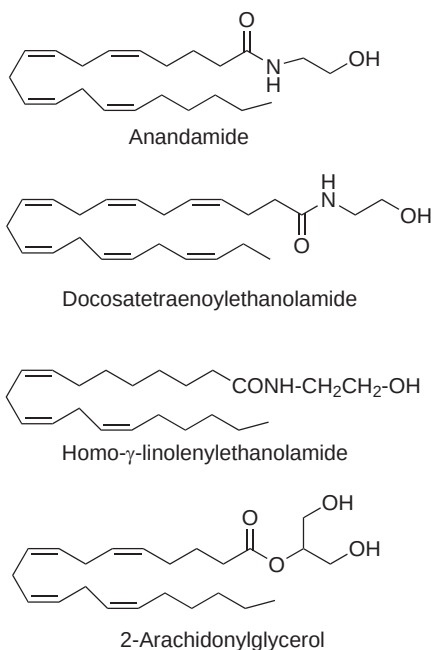
The pyrazole analog rimonabant (SR141716A) was found to be one of the first cannabinoid antagonists, and extensive studies were conducted to help unravel the actions of cannabinoids such as Δ^9 -THC (7,8). Rimonabant effectively attenuated the effects of Δ^9 -THC and WIN-55,212-2 (7) as well as the discriminative stimulus effects of Δ^9 -THC. Thus, in addition to cannabinoids and cannabinoid-related structures such as CP-55,940 and nabilone, there are other structural classes of cannabinoid ligands, including indolic derivatives such as WIN-55,212-2 and JWH-018, pyrazoles such as rimonabant, and fatty acid derivatives such as anandamide and 2-arachidonylglycerol (2-AG).



Mechanism of Action

For many years, it was thought that Δ^9 -THC acts in a nonspecific manner; however, in the early 1990s, two populations of cannabinoid (CB) receptors were identified: CB-1 and CB-2 receptors (8,9). Human forms of these receptors have been cloned, and both types are G protein-coupled, seven-helix transmembrane-spanning receptors. These receptors are differentially expressed: CB-1 receptors, which likely mediate the psychoactive effects of THC-related agents, are found primarily in the brain, whereas CB-2 receptors, which possibly are involved in immunomodulatory (and other) actions, are found almost exclusively in the periphery. The identification of such receptors suggested the possible existence of an endogenous ligand, and several now have been identified. The best investigated of these is the eicosanoid derivative arachidonylethanolamide or anandamide. Anandamide binds at CB-1 receptors with an affinity similar to that of Δ^9 -THC (10). Related structures have also been identified including docosatetraenylethanolamide and homo- γ -linolenylethanolamide (10). Pharmacologically, anandamide seems to be a THC-like agent. Although the actions of anandamide may not be identical to those of Δ^9 -THC, particularly regarding *in vivo* studies, differences may be related to the metabolic instability of anandamide. For example, in drug discrimination studies, a Δ^9 -THC stimulus failed to consistently or reliably generalize to anandamide; however, the more metabolically stable methanandamide, a chain-methylated analog of anandamide, produced THC-like effects. Furthermore, methanandamide has been used as a training drug, and the methanandamide stimulus generalizes to Δ^9 -THC (11). Another type of endocannabinoid is 2-AG (12). Despite their structural similarity, anandamide-type endocannabinoids are metabolized primarily by fatty acid amide hydrolase (FAAH), whereas

2-AG is metabolized mainly by monoacylglycerol lipase (MAGL) (8,9). It is now recognized that cannabinoids and synthetic cannabinoids influence the *endocannabinoid system*. The endocannabinoid system is composed of endogenous cannabinoids (i.e., endocannabinoids), enzymes that synthesize and degrade them, and cannabinoid receptors (12).



Structure–Activity Relationships

Structure–activity relationships (SARs) both for THC-like actions and for CB receptor binding have been formulated (7,9). Structure–activity studies can be discussed on the basis of several different types of behavioral assays in rodents, and it has been shown for 60 cannabinoids that behavioral potencies are highly correlated with receptor binding affinities (13). THC-like discriminative effects probably offer a more specific method of detecting and measuring cannabimimetic effects and are particularly useful for formulating SARs (4). Using this approach, it has been demonstrated that SARs for THC-like stimulus effects are not necessarily identical to those for the analgesic, antiemetic, or anticonvulsant actions of cannabinoids. An early study showed that animals trained to discriminate intraperitoneal dosing of Δ^9 -THC recognized hashish smoke and that animals trained to discriminate hashish smoke recognized Δ^9 -THC, supporting the concept that Δ^9 -THC likely accounts for the stimulus actions of hashish. A number of cannabinoids now have been evaluated. Cannabidiol (CBD), for example, does not produce THC-like stimulus effects. Relative to Δ^9 -THC (ED_{50} = 0.43 mg/kg intraperitoneal), some 11-hydroxy metabolites are quite potent, such as 11-OH Δ^9 -THC (ED_{50} = 0.10 mg/kg) and 11-OH Δ^8 -THC (ED_{50} = 0.38 mg/kg). One of the more potent cannabinoids is Δ^8 -THC-DMH (ED_{50} = 0.05 mg/kg), in which the 4-pentyl

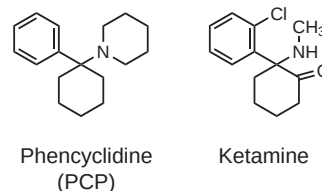
moiety of THC has been replaced with a 1,1-dimethylheptyl (DMH) group; its 11-hydroxyl analog, 11-OH Δ^8 -THC-DMH (ED_{50} = 0.002 mg/kg), is even more potent (4). One of the most extensively studied cannabinoid ligands is WIN-55,212-2. Molecular modeling and site-directed mutagenesis studies suggest that cannabinoids, CP-55,940, and anandamide bind in a similar fashion but in a manner that differs from the binding of WIN-55,212-2.

Therapeutic Potential

The discovery of CB-1 and CB-2 receptors, CB receptor antagonists, agents showing selectivity between the two CB receptor populations, endocannabinoids, and novel chemical tools with which to investigate these receptors (7,9,14) has generated renewed interest in the cannabinoids. Already approved for use are dronabinol (Marinol) and nabilone (Cesamet). Dronabinol is the synthetic *trans*(–)-isomer of Δ^9 -THC, whereas nabilone is a synthetic chromenone derivative (a mixture of *RR* and *SS* isomers). Both are approved for treatment of chemotherapy-induced nausea and vomiting, as well as for the anorexia and weight loss seen in patients with AIDS. Sativex, whose major active ingredients are natural Δ^9 -THC and CBD, is approved for use in several countries (not in the United States) for the treatment of neuropathic pain and spasticity associated with multiple sclerosis.

The CB-1 selective antagonist rimonabant was in clinical trials for the treatment of obesity, but its U.S. New Drug Application was withdrawn due to increased risk of suicidal ideation. Related analogs are still under investigation. CB-2 antagonists are being explored for their use in chronic pain, and several classes of FAAH inhibitors are being examined for their analgesic and anxiolytic actions (12). Various other compounds that influence the endocannabinoid system are being investigated for the treatment of glaucoma, Tourette syndrome, Parkinson disease, epilepsy, drug abuse, immune disorders, and several types of neuropsychiatric disorders (12,15).

PCP-Related Agents



PCP [1-(1-phenylcyclohexyl)piperidine] was introduced as a dissociative anesthetic during the late 1950s. Shortly after its introduction, clinical studies were terminated because of the occurrence of schizophrenic-like psychotomimetic effects (i.e., PCP can be considered a true *psychotomimetic* agent), particularly during emergence from anesthesia. This might have been the end of the story

except that additional attempts were made to exploit the anesthetic effects of PCP, leading to the development of novel agents such as ketamine; it was theorized that PCP-like states might provide a good model for investigating schizophrenia, leading to studies of PCP's mechanism of action; and PCP (e.g., "Angel Dust"), administered by inhalation, injection, ingestion, or smoking (as with PCP-laced parsley, tobacco, or marijuana), and ketamine (e.g., "Special K") emerged as drugs of abuse, leading to investigations of their abuse liability. Shortly thereafter, it was discovered that PCP behaves as an *N*-methyl-D-aspartate (NMDA) antagonist. Because NMDA receptors had been implicated in seizures and trauma, PCP and related arylcycloalkylamines were explored as potential antiepileptics and neuroprotective agents. Furthermore, because PCP is relatively easy to manufacture, it has become popular on the clandestine market, particularly being used among children and young adolescents (16,17).

Actions

In humans, PCP can produce disorientation, confusion, incoordination, delirium, impaired memory, and euphoria (17,18); higher doses produce additional adverse effects including death. PCP has a history of producing aggression and violent behavior. Because PCP often is consumed together with other substances, it sometimes has been difficult to establish exactly which effects are produced by PCP and which might be related to possible drug interactions. PCP has seen extensive investigation in animals, and it appears to produce effects similar to those of amphetamine-like stimulants and central depressants. PCP is self-administered by animals, and tolerance develops to its behavioral effects upon repeated exposure to the drug (18). PCP has both direct and indirect effects on dopaminergic systems; this might account, at least in part, for some of the amphetamine-like effects of PCP and might contribute to the production of its schizophrenic-like actions. The PCP model of schizophrenia was particularly attractive because PCP seemed to produce both the positive and negative symptoms associated with this disorder. PCP has also been widely investigated as a training drug in animals using drug discrimination studies (4).

Mechanism of Action

N-Allylnormetazocine (NANM; SKF-10047) produces some effects reminiscent of those produced by PCP. At one time, NANM was considered a prototypic σ opiate receptor ligand. It is now recognized that the σ receptors are *not* a class of opiate receptors and that the low-affinity NANM is only one of very few opiates that bind at these receptors. Subsequent structure-activity studies showed that NANM simply possesses certain minimal pharmacophoric features that are required for σ receptor binding (19). Nevertheless, the behavioral similarities between NANM and PCP led to early investigations regarding the binding of PCP at σ receptors, and because of its affinity (albeit low) for these receptors,

the σ receptors were renamed NANM-PCP receptors or σ /PCP receptors. This confusion continued for several years until it was demonstrated that agents with much higher affinity and selectivity than PCP for σ receptors failed to produce PCP-like actions in animals (18). Later, it was shown that PCP antagonizes the effects of the excitatory amino acid NMDA. [3 H]PCP has been used to label putative PCP binding sites, and PCP binding and NMDA binding display similar regional distribution in brain. It is established that PCP is a noncompetitive NMDA receptor antagonist.

The NMDA receptor (Fig. 19.1) is a ligand-gated ion channel receptor that regulates the flow of cations (Na^+ , Ca^{2+}) into certain neurons. The receptor complex possesses multiple binding sites, similar to the benzodiazepine/ γ -aminobutyric acid (GABA) receptor complex that allows the binding of glutamate, glycine, polyamines, and other ligands that can modulate the actions of NMDA. Like the NMDA antagonist dizocilpine (MK-801), PCP binds at a site (i.e., the PCP site) that is believed to be located within the ion channel. Drug discrimination studies have shown that PCP-trained animals recognize NMDA antagonists that bind at PCP receptors; for example, MK-801 is nearly 10-fold more potent than PCP. Furthermore, animals trained to discriminate MK-801 recognize PCP and other PCP-related agents. Consistent with early findings that PCP produces a psychotic state in humans, PCP has been shown to produce a pattern of metabolic, neurochemical, and behavioral changes in animals that reproduce almost exactly those seen in patients with schizophrenia. Consequently, this provides new insight regarding the mechanisms underlying such disorders and offers an animal model for the evaluation of novel antipsychotic agents (20).

Structure-Activity Relationships

SARs for PCP-like actions have not been particularly well defined, and what little is known stems primarily from drug discrimination studies. The PCP stimulus does not generalize to opioids, sympathomimetic stimulants, anticholinergic agents, or classical hallucinogens and only partially generalizes to depressants, such as barbiturates. In general, the stimulus properties of PCP are not shared by members

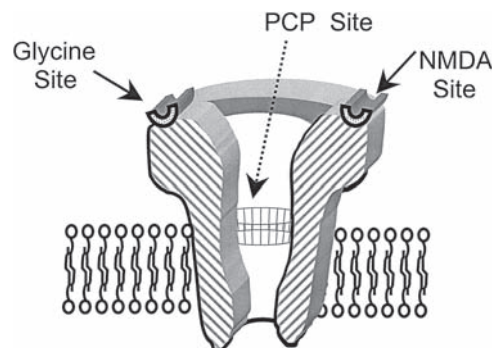
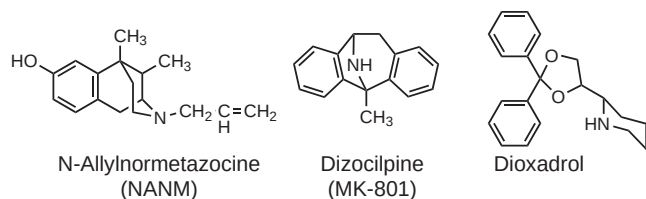


FIGURE 19.1 NMDA ion channel receptor showing binding sites for glycine, NMDA, and PCP.

of other drug classes (21). The PCP stimulus generalizes to ketamine and other structurally related derivatives of PCP, such as TCP, an analog of PCP in which the phenyl ring has been replaced by the isosteric 2-thienyl group.



PCP does not possess a chiral center. Several 1,3-dioxolanes possessing an asymmetric center produce PCP-like effects and have proven to be useful for investigating PCP-like actions. Dioxadrol, or 2-(2,2-diphenyl-1,3-dioxolan-4-yl)piperidine, and etoxadrol (i.e., dioxadrol in which one of the phenyl groups has been replaced by an ethyl group) are examples of such dioxolanes. The (+)-isomer of dioxadrol, dexoxodrol, but not the (–)-isomer levoxodrol, binds at PCP receptors and is recognized by PCP-trained animals (21).

PSYCHOACTIVE DRUGS OF ABUSE: CLASSICAL HALLUCINOGENS

Classical hallucinogens are agents that meet the Hollister definition (2) and that, in addition, bind at 5-HT₂ receptors and are recognized by DOM-trained animals in tests of stimulus generalization (5). The classical hallucinogens all possess the general structure Ar-C-C-N, where Ar is a substituted phenyl, 3-indolyl, or substituted 3-indolyl moiety; C-C is an ethyl or branched ethyl chain; and N is a primary, secondary, or tertiary amine. This will be further discussed. (See Chapter 10 for additional information on 5-HT receptors.)

Classification

There are two major structural categories of classical or arylalkylamine hallucinogens: the *indolealkylamines* and the *phenylalkylamines*. The indolealkylamines are further divided into the simple *N*-substituted tryptamines, the α -alkyltryptamines, the ergolines (or lysergamides), and, tentatively, the β -carbolines. The phenylalkylamines consist of phenylethylamines and phenylisopropylamines. In humans, examples from the different categories seem to produce similar effects. It should be noted, however, that relatively few agents have been examined in comprehensive and carefully controlled clinical situations. Furthermore, no claim is made that these agents produce identical effects in humans. Each category—and, indeed, even certain examples from within a given category—may produce effects that make them somewhat different from the others. As if to underscore the behavioral similarity among these agents, however, examples from each of the above categories produce common DOM-like stimulus effects in animals (Table 19.1).

TABLE 19.1 Results of Stimulus Generalization Studies with Examples from the Various Categories of Classical Hallucinogens Using Animals Trained to Discriminate DOM from Vehicle

Category	Example	ED ₅₀ Value for DOM-Stimulus Generalization (mg/kg) ^a
<i>N</i> -Alkyltryptamines	DMT	5.8
α -Alkyltryptamines	α -MeT	3.1
Lysergamides	(+)-LSD	0.05
β -Carbolines	Harmaline	6.2
Phenylethylamines	Mescaline	14.6
Phenylisopropylamines	DOB	0.2

^aData from Glennon and Young (4) and Glennon (5). DMT, *N,N*-dimethyltryptamine; DOB, 2-[4-bromo-2,5-dimethoxyphenyl]-1-aminopropane; LSD, lysergic acid diethylamide; α -MeT, α -methyltryptamine.

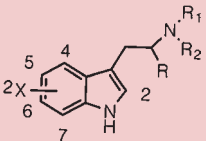
Indolealkylamines

N-Alkyltryptamines

One of the best-investigated hallucinogens is *N,N*-dimethyltryptamine (DMT) (Table 19.2), which is the prototype of this subclass of agents. Although readily synthesized in the laboratory, DMT is also a naturally occurring substance. Its actions are characterized by a rapid onset (typically <5 minutes) and short duration of action (~30 minutes). Like some other members of this family, DMT is not active via oral administration; it is generally administered by inhalation or by smoking. Although less common, DMT can also be injected. Some indolealkylamines are sensitive to the acidic conditions of the stomach. The corresponding secondary amine, *N*-monomethyltryptamine, and primary amine, tryptamine, are inactive as psychoactive substances, both because they are not sufficiently lipophilic to readily penetrate the blood–brain barrier and because what little does get into the brain is rapidly metabolized by monoamine oxidase (MAO). Other tertiary amine derivatives, such as the *N*-ethyl-*N*-methyltryptamine, *N,N*-diethyltryptamine (DET), *N,N*-di-*n*-propyltryptamine (DPT), and some secondary amines, are also hallucinogenic in humans. If the *N*-alkyl or *N,N*-dialkyl substituents are bulky and lipophilic enough, these tryptamines can be orally active (Table 19.2).

The effect of substitution in the pyrrole portion of DMT has not been extensively investigated in humans. In contrast, substitution in the benzenoid ring can enhance or diminish potency depending on the specific nature and location of the substituents. Table 19.2 shows some of the more frequently encountered derivatives of DMT, their common names, and their approximate human potency. 5-HT is not hallucinogenic and does not readily penetrate the blood–brain barrier

TABLE 19.2 Psychoactive Indolealkylamine and Related Agents

Agent	Common Name				Approximate Hallucinogenic Dose (mg)*	DOM Stimulus Generalization Potency (mg/kg)**
		R ₁ /R ₂	R	X		
Tryptamine		H/H	H	H	Likely inactive	Inactive
N-Methyltryptamine	NMT	CH ₃ /H	H	H	Likely inactive	Inactive
(±)α-Methyltryptamine	α-MeT	H/H	CH ₃	H	5–20 (smoked) 15–30 (po)	3.13
N,N-Dimethyltryptamine	DMT	CH ₃ /CH ₃	H	H	60–100 (smoked) 4–30 (iv)	5.80
N,N-Diethyltryptamine	DET	C ₂ H ₅ /C ₂ H ₅	H	H	50–100 (po)	2.45
N-Ethyl-N-methyltryptamine	MET	C ₂ H ₅ /CH ₃	H	H	Unknown	—
N,N-Di-n-propyltryptamine	DPT	n-C ₃ H ₇ /n-C ₃ H ₇	H	H	100–250 (po)	2.20
N,N-Di-isopropyltryptamine	DIPT	i-C ₃ H ₇ /i-C ₃ H ₇	H	H	25–100 (po)	2.60
(±)α-Ethyltryptamine	α-EtT	H/H	C ₂ H ₅	H	100–150 (po)	6.62
4-Hydroxy DMT	Psilocin	CH ₃ /CH ₃	H	4-OH	10–20 (po)	—
4-Methoxy DMT	4-OMe DMT	CH ₃ /CH ₃	H	4-OCH ₃	Unknown	3.53
5-Hydroxytryptamine	5-HT	H/H	H	5-OH	Inactive	Inactive
5-OH DMT	Bufotenine	CH ₃ /CH ₃	H	5-OH	Likely inactive	Inactive
5-Methoxy DMT	5-OMe DMT	CH ₃ /CH ₃	H	5-OCH ₃	6–20 (smoked) 2–3 (iv)	1.22
(±)5-Methoxy-α-MeT	5-OMe α-MeT	H/H	CH ₃	5-OCH ₃	2.5–4.5 (po)	0.50
6-Methoxy DMT	6-OMe DMT	CH ₃ /CH ₃	H	6-OCH ₃	Likely inactive	Inactive
7-Methoxy DMT	7-OMe DMT	CH ₃ /CH ₃	H	7-OCH ₃	Likely inactive	Inactive

*Data primarily from reference 22. Key: po = oral, iv = intravenous.

**Drugs were administered via the ip route reference 5.

when administered systemically. *N,N*-Dimethylserotonin [bufotenine (5-OH DMT)] has been reported to be a weak hallucinogen, but the results of human studies are controversial. It, too, likely does not readily penetrate the blood–brain barrier and produces considerable peripheral effects (e.g., facial flushing and cardiovascular actions) that prevent evaluation of an extended dose range. *O*-Methylation of bufotenine results in 5-OMe DMT, one of the more potent *N*-alkyltryptamines. A naturally occurring substance, 5-OMe DMT is a constituent of a number of plants used in various concoctions prepared by South American Indians for ceremonial and visionary purposes. Bufotenine and 5-OMe DMT are also found in the skin of certain frogs and may have given rise to the phenomenon of “toad licking.” Psilocin is 4-hydroxy DMT. Like bufotenine, with a polar

hydroxyl group, psilocin might not have been expected to enter the brain, yet it is hallucinogenic. Although this phenomenon has never been adequately explained, it has been speculated that the 4-hydroxyl group forms a hydrogen bond with the terminal amine and that this reduces polarity just enough that psilocin penetrates the blood–brain barrier. Psilocin and its phosphate ester, psilocybin, are widely found in certain species of mushrooms and have given rise to the terms “shrooms” and “shrooming.” There are no reports that 6-methoxy DMT or 7-methoxy DMT are hallucinogenic. It is quite difficult to make strict potency comparisons within this series because of the different routes of administration that have been used (Table 19.2).

In tests of stimulus generalization, the DOM stimulus has been shown to generalize to DMT, DET, DPT,

4-OMe DMT, 5-OMe DMT, and a number of other DMT analogs, but not to 5-OH DMT, 6-OMe DMT, or 7-OMe DMT.

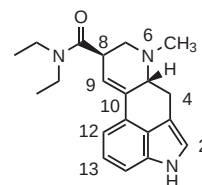
The metabolism of these agents has not been well investigated. The indolealkylamine 5-HT is a substrate for oxidative deamination by MAO. What evidence exists suggests that other indolealkylamines are also substrates for this enzyme system.

α -Alkyltryptamines

Tryptamine is not psychoactive. Introduction of an α -methyl group seemingly enhances lipophilicity and sufficiently protects against metabolism such that α -methyltryptamine (α -MeT) (Table 19.2) is approximately twice as potent as DMT. As a general rule of thumb, α -MeTs, where such agents have been investigated, typically are twice as potent as their corresponding DMT counterpart. Otherwise, their SAR is essentially the same as that of the DMT analogs. For example, 5-methoxy- α -methyltryptamine (5-OMe α -MeT) is approximately twice as potent as 5-OMe DMT. Introduction of the α -methyl group results in the creation of an asymmetric center, and the *S*(+)-isomers of α -MeTs are more potent than their *R*(-)-enantiomers. Homologation of the α -methyl group to an α -ethyl group affords α -ethyltryptamines. α -Ethyltryptamine (α -EtT) has been reported to be hallucinogenic, with effects somewhat distinguishable by human subjects from those of LSD and mescaline (23). Interestingly, α -EtT was clinically available during the early 1960s as an antidepressant because of its actions as an MAO inhibitor; however, it was removed from the market about a year after its introduction. It may be the MAO inhibitory effect that allowed the actions of α -EtT to be distinguished from those of LSD and mescaline (see also the later section on designer drugs). During the mid-1990s, α -EtT made an appearance on the clandestine market as a designer drug (i.e., “ET”). ($\pm$)- α -Methyltryptamine, ($\pm$)-5-methoxy- α -methyltryptamine and both of its optical isomers, and ($\pm$)- α -ethyltryptamine are recognized by DOM-trained animals in tests of stimulus generalization (4).

Ergolines or Lysergamides

(+)-LSD (i.e., LSD) is perhaps the best known—and, certainly, one of the most potent—of the classical hallucinogens. Although LSD itself is not naturally occurring, many related ergolines are found in nature. In terms of potency, LSD is at least 3,000-fold more potent than mescaline, with doses of 100 μ g showing activity. Certain structurally modified analogs of LSD retain hallucinogenic activity; although many derivatives are possible, relatively few have been investigated in humans. Structural changes often can reduce the activity of a pharmacologically active substance. Here is an instance in which a structural change resulting in even a 1,000-fold decrease in potency can afford a very active agent. Some work has been reported on the SAR of LSD (24,25).



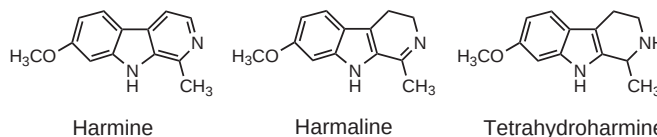
(+)-Lysergic acid diethylamide
(LSD)

In humans, LSD has been thoroughly investigated (24); no other hallucinogen has been as extensively studied as this agent. Its actions in humans can be divided into three major categories: perceptual (altered shapes and colors and heightened sense of hearing), psychic (alterations in mood, depersonalization, visual hallucinations, and an altered sense of time), and somatic (nausea, blurred vision, and dizziness). In terms of principal effects, there seems to be little difference between LSD, psilocybin, and mescaline.

Although LSD has been sold on the clandestine market in tablet form, it is not uncommon to find this material available on “blotter paper” because of its high potency. A sheet of porous paper is impregnated with a solution of LSD, and the sheet can later be cut to afford the appropriate dose.

β -Carbolines

The β -carbolines represent a very interesting class of agents generally referred to as the harmala alkaloids. Several are naturally occurring. In South America, β -carbolines are constituents of certain vines and lianas (e.g., *Banisteriopsis caapi*), and in the Old World, β -carbolines are constituents of Syrian Rue (*Peganum harmala*).



South American Indians prepare a variety of concoctions and snuffs, the most notable of which is ayahuasca, that are used for their hallucinogenic and visionary healing properties. In fact, the first written account of the use of these substances was made by a member of the Columbus expedition in 1493. There is little question that the concoctions are psychoactive; however, these plant preparations usually consist of admixtures in which certain tryptamines, such as DMT or 5-OMe DMT, sometimes have been identified. Some β -carbolines possess activity as MAO inhibitors; thus, the MAO inhibitory effect of the β -carbolines might be simply potentiating the effect of any other tryptaminergic hallucinogens present in the admixture by interfering with their metabolism. Studies with individual β -carbolines, especially under carefully controlled clinical settings, have been very limited. The three most commonly occurring β -carbolines are harmine, harmaline, and tetrahydroharmine, and evidence

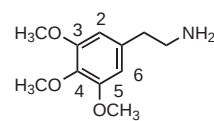
suggests that harmine and harmaline are hallucinogenic in humans (with potencies not greater than that of DMT) (26,27). Harmaline has seen some limited experimental application as an adjunct to psychotherapy (28). Like other classical hallucinogens, certain β -carbolines bind at 5-HT_{2A} receptors, and in animal studies, DOM stimulus generalization occurs to harmaline (28). Using harmaline-trained animals, harmaline stimulus generalization occurs to DOM. To date, however, few β -carbolines have been investigated, so they are only tentatively categorized as classical hallucinogens.

Although the scientific community has been aware of the psychoactive effects of the β -carbolines or β -carboline-containing natural substances for more than 100 years, only in the past decade or so have these agents become popular “on the street.” The use of β -carboline-containing plants has moved out of the jungle and has given rise to a variety of religious movements in some South American cities. Recent books and movies are also helping to popularize the use of these preparations, and they are now being encountered in North America. In fact, as of 2006, the use of “ayahuasca tea” by certain religious groups has been permitted in the United States under the Religious Freedom Restoration Act.

Phenylalkylamines

Phenylalkylamines, phenylethylamines and phenylisopropylamines, represent the largest group of classical hallucinogens (29,30). The phenylethylamines are the α -desmethyl counterparts of the phenylisopropylamines; as with the indolealkylamines, the presence of the α -methyl group increases the agent's lipophilicity and reduces its susceptibility to metabolism by MAO. As a consequence, the phenylethylamines typically produce effects that are qualitatively similar to those of their corresponding phenylisopropylamines but typically less potent. Phenylethylamine counterparts of weak phenylisopropylamines might be inactive. Literally hundreds of analogs have been examined in human and animal studies (29).

PHENYLETHYLAMINES Phenylethylamines are usually less potent analogs of the phenylisopropylamines. Some hallucinogenic phenylisopropylamines are claimed to possess some stimulant character that may be minimized or altogether absent in the corresponding phenylethylamines. The phenylisopropylamines also possess a chiral center that is absent in the phenylethylamines. Otherwise, the SARs of the two groups of agents are relatively similar; consequently, the phenylethylamines will not be discussed in detail here. The most common—and, indeed, one of the oldest known—phenylethylamine hallucinogen is mescaline. A constituent of peyote (and other) cactus, mescaline is a relatively weak hallucinogenic agent (total human dose ~350 mg). Like many of the hallucinogens, mescaline is listed as a Schedule I substance; however, the use of peyote in certain Native American Indian religious practices is sanctioned.



Mescaline

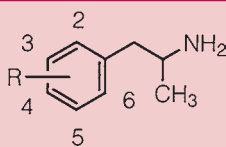
PHENYLISOPROPYLAMINES Structural modification of mescaline and related substances by introduction of an α -methyl group and by deletion or rearrangement of the position of its methoxy groups results in a series of agents known as the phenylisopropylamines. As might have been expected, introduction of an α -methyl group, to afford 3,4,5-trimethoxyphenyl-2-aminopropane (TMA) or α -methylnescaline, doubles the potency of mescaline. Although different nomenclatures exist for the dimethoxy- and trimethoxyphenylisopropylamines, the one used herein is a commonly used nomenclature: The position of methoxy groups is given by indicating its position, and the number of methoxy groups is indicated by a prefix. For example, α -methylnescaline is 3,4,5-TMA, indicating that it is a trimethoxy analog and that the methoxy groups are situated at the 3-, 4-, and 5-positions. Dimethoxy analogs are referred to as DMAs (dimethoxyphenyl-2-aminopropanes).

There are three possible monomethoxyphenylisopropylamines: the ortho-methoxy analog OMA, the meta-methoxy analog MMA, and the para-methoxy analog PMA (Table 19.3). Although PMA is specifically listed as a Schedule I substance, none of these three analogs is hallucinogenic. PMA possesses weak central stimulant actions and is an abused substance; several deaths have been attributed to PMA overdose within the past few years.

There are six isomeric DMA analogs. These have not been thoroughly investigated in humans, and few produce DOM-like stimulus effects in animals (Table 19.3). None is more potent than DOM. The most potent agent and one that has been evaluated in humans is 1-(2,5-dimethoxyphenyl)-2-aminopropane (2,5-DMA). There are also six different TMA analogs (Table 19.3). Here, most show some activity, but the 2,4,5-trimethoxy analog 2,4,5-TMA (sometimes referred to simply as TMA) is the most potent of the series. Most of the trimethoxy analogs are recognized by DOM-trained animals, but none is more potent than DOM itself (5). The presence of the 2,5-methoxy substitution pattern in 2,5-DMA and 2,4,5-TMA might be noted.

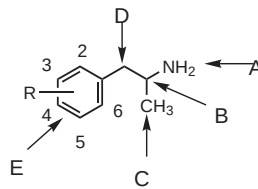
The DMAs and TMAs are methoxy-substituted derivatives of the parent phenylisopropylamine known as amphetamine (Fig. 19.2). Amphetamine undergoes several different routes of metabolism; one of these is para-hydroxylation (a route that seems more important in rodents than in humans). Initially, it was thought that the greater potency of 2,4,5-TMA over that of 2,5-DMA might be related to the 4-position of the former being blocked to metabolism by para-hydroxylation. Keeping the 2,5-dimethoxy substitution intact, different 4-position

TABLE 19.3 Psychoactive Phenylisopropylamines and Related Agents

Agent						Human Hallucinogenic Dose (mg)*	DOM-stimulus Gen. Potency (umol/kg) [†]
	R ₂	R ₃	R ₄	R ₅	R ₆		
Amphetamine	H	H	H	H	H	NH	NSG
OMA	OCH ₃	H	H	H	H	NH	NSG
MMA	H	OCH ₃	H	H	H	NH	NSG
PMA	H	H	OCH ₃	H	H	NH	NSG
2,3-DMA	OCH ₃	OCH ₃	H	H	H	(?)	NSG
2,4-DMA	OCH ₃	H	OCH ₃	H	H	>60(?)	21.0
2,5-DMA	OCH ₃	H	H	OCH ₃	H	120 (80–160)	23.8
2,5-DMA, R(-)	OCH ₃	H	H	OCH ₃	H	(?)	14.0
2,6-DMA	OCH ₃	H	H	H	OCH ₃	(?)	NSG
3,4-DMA	H	OCH ₃	OCH ₃	H	H	>500 (?)	NSG
3,5-DMA	H	OCH ₃	H	OCH ₃	H	(?)	NSG
2,3,4-TMA	OCH ₃	OCH ₃	OCH ₃	H	H	>100 (?)	29.8
2,3,5-TMA	OCH ₃	OCH ₃	H	OCH ₃	H	>80 (?)	63.0
2,3,6-TMA	OCH ₃	OCH ₃	H	H	OCH ₃	>30 (?)	—
2,4,5-TMA	OCH ₃	H	OCH ₃	OCH ₃	H	30 (20–40)	13.7
2,4,6-TMA	OCH ₃	H	OCH ₃	H	OCH ₃	38 (25–50)	13.9
3,4,5-TMA	H	OCH ₃	OCH ₃	OCH ₃	H	175 (100–250)	24.2
MEM	OCH ₃	H	OC ₂ H ₅	OCH ₃	H	35 (20–50)	22.9
DOM	OCH ₃	H	CH ₃	OCH ₃	H	7 (3–10)	1.8
DOM, R(-)	OCH ₃	H	CH ₃	OCH ₃	H	(?)	0.9
DOM, S(+)	OCH ₃	H	CH ₃	OCH ₃	H	(?)	6.9
DOET	OCH ₃	H	C ₂ H ₅	OCH ₃	H	4 (2.5–5)	0.9
DOPR	OCH ₃	H	n-C ₃ H ₇	OCH ₃	H	4 (3–4.5)	0.6
DOIP	OCH ₃	H	i-C ₃ H ₇	OCH ₃	H	(?)	2.9
DOBU	OCH ₃	H	n-C ₄ H ₉	OCH ₃	H	(?)	3.2
DOAM	OCH ₃	H	n-C ₅ H ₁₁	OCH ₃	H	(?)	NSG
DOT	OCH ₃	H	SCH ₃	OCH ₃	H	8 (5–10)	—
DON	OCH ₃	H	NO ₂	OCH ₃	H	4 (3–4.5)	2.7
DOF	OCH ₃	H	F	OCH ₃	H	(?)	5.8
DOC	OCH ₃	H	Cl	OCH ₃	H	2.5 (1.5–3)	1.2
DOB	OCH ₃	H	Br	OCH ₃	H	2 (1–3)	0.6
DOB, R(-)	OCH ₃	H	Br	OCH ₃	H	1.0–1.5 (?)	0.3
DOI	OCH ₃	H	I	OCH ₃	H	2.5 (1.5–3)	1.2
DOOC	OCH ₃	H	COOH	OCH ₃	H	(?)	NSG
DOOH	OCH ₃	H	OH	OCH ₃	H	(?)	NSG

*Data are primarily from reference 29. Where a dose range was reported in the literature, the arithmetic mean is also provided here for comparison (original range is given in parenthesis). The values should not be taken as a measure of precision. Doses are approximate and no implication is made that the different agents produce an identical effect. Key: NH = not hallucinogen, (?) = material has not been well studied or that its actions or potency are unknown.

[†]Drug discrimination data represent ED₅₀ values and are from references 5,30. NSG = no stimulus generalization.



The chemical structure shows a benzene ring with a substituent R at position 1. The ring carbons are numbered 1 through 6 starting from the R group and moving clockwise. Attached to carbon 6 is a chiral center (carbon 7) which is bonded to an amino group (NH₂), a methyl group (CH₃), and a hydrogen atom (H). Labels A, B, C, D, and E point to the following features: A points to the terminal amine (NH₂), B points to the chiral center (C7), C points to the α-methyl group (CH₃), D points to the β-position (C6), and E points to the aromatic ring (C1-C6).

Position	Amphetamine-like action	DOM-like actions
A: Terminal amine	N-Methyl > NH ₂ > NHR > NR ₂	NH ₂ > NHR > NR ₂
B: Chiral center	S(+) > (±) > R(-)	R(-) > (±) > S(+)
C: α-Methyl group	Homologation decreases potency Replacement by H decreases potency	Homologation decreases potency Replacement by H decreases potency
D: β-Position	β-OH: reduces potency β=O: retains activity and potency	β-OH: not well investigated β=O: not well investigated
E: Aromatic substitution	Unsubstituted aromatic ring preferred	2,5-Dimethyl-substitution preferred 4-Substitution further modulates activity

FIGURE 19.2 Comparative structure–activity relationships for the amphetamine-like stimulant actions and the DOM-like action of the phenylisopropylamines (4,30).

substituents were examined. This led to a series of agents, such as DOM and 1-(4-bromo-2,5-dimethoxyphenyl)-2-aminopropane (DOB) (Table 19.3). These 4-substituted 2,5-dimethoxy analogs represent some of the most potent members of the series.

DOM represents the prototype member of this family of agents. Increasing the length of the 4-methyl group to an ethyl or *n*-propyl group (i.e., DOET and DOPR, respectively) results in enhanced potency on a molar basis. Further extension of the alkyl chain results in a decrease in potency or loss of action. Substitution at the 4-position by electron-withdrawing groups, particularly those with hydrophobic character, also results in active agents, such as DOB (Table 19.3), which is quite a potent agent and has been misrepresented on the clandestine market as LSD both in tablet and “blotter” form.

When optical isomers have been examined, activity resides primarily with the *R*(-)-isomer; the *S*(+)-isomers typically are less active, are inactive, or have received little study. For example, although not well investigated, it appears that *R*(-)-DOM and *R*(-)-DOB show activity at total human doses of less than 4 mg and less than 1 mg, respectively. *N*-Monomethylation reduces potency or abolishes activity; for example, the *N*-monomethyl analogs of DOM and DOB are approximately 10% as potent as their primary amine counterparts. The SARs for the DOM-like actions of phenylisopropylamines are summarized in Table 19.3 and Figure 19.2.

Table 19.3 also provides a comparison of the approximate human doses of various phenylisopropylamines when administered via the oral route. These agents represent a mere sampling of the agents that have been examined; it can be imagined, using only those functional groups shown in the table, how many different analogs are possible on the basis of structural rearrangement.

There is no reason to suspect that each of these agents produces identical effects. In fact, the actions of some of these agents have been reported to be quite unique, ranging from hallucinations and closed-eye imagery to intellectual and sensory enhancement to erotic arousal (29).

Classical Hallucinogens: Mechanism of Action

Given that the arylalkylamines may not be producing identical effects, a common mechanism of action may not be expected. LSD was one of the first hallucinogens to be investigated mechanistically; another agent to see extensive investigation is mescaline. Interestingly, from a potency perspective, these two agents seem to represent opposite extremes. LSD has been proposed to produce its effects via numerous mechanisms, including those involving serotonergic, dopaminergic, histaminergic, adrenergic, and other receptors. LSD binds with high affinity at many different receptor populations and acts as an agonist at some, an antagonist at others, and a partial agonist at yet others. For many years, it was supposed that mescaline might be acting via a dopaminergic or adrenergic mechanism because of its structural similarity to dopamine (DA) and norepinephrine (NE). As early as the late 1950s, it was speculated, because of its structural similarity to 5-HT, that LSD might be working through a serotonergic mechanism. Significant experimental evidence supported this claim. Controversy existed, however, regarding whether LSD was a serotonergic agonist or antagonist. Furthermore, later studies revealed the existence of at least 14 populations of 5-HT receptors (see Chapter 10). With the subsequent availability of 5-HT₂ selective antagonists, it was demonstrated that several of these antagonists (e.g., ketanserin and

pirenperone) were particularly effective in blocking the stimulus effects of DOM and of DOM stimulus generalization to other hallucinogens, such as LSD, in tests of stimulus antagonism. It was later shown that the classical hallucinogens bind at 5-HT₂ receptors and that their receptor affinities were significantly correlated with both their DOM stimulus generalization potencies and their human hallucinogenic potencies (30). The classical hallucinogens are now thought to produce their effect by acting as agonists at 5-HT₂ receptors in the brain (i.e., the 5-HT₂ hypothesis of hallucinogen action). Radiolabeled analogs of DOB and DOI (e.g., [³H]DOB and [¹²⁵I]DOI, respectively) are now available for the investigation of 5-HT₂ pharmacology.

More recently, it has been demonstrated that 5-HT₂ receptors actually represent a family of 5-HT receptors that consist of 5-HT_{2A}, 5-HT_{2B}, and 5-HT_{2C} receptor subpopulations. Fewer than three dozen arylalkylamines have been compared, but it appears that they show little selectivity for one subpopulation versus another. Various pharmacologic studies with selective antagonists or using antagonist correlation analysis, however, suggest that it may be the 5-HT_{2A} subtype that has a predominant role in the behavioral actions of these agents (30,31). Although the 5-HT_{2A} receptors might be responsible for those actions that the classical hallucinogens have in common, other neurochemical mechanisms may account for their differences. For example, LSD is a very promiscuous agent that binds with high affinity at many receptor populations for which most other classical hallucinogens show little to no affinity. Many of the indolealkylamines bind with high affinity at multiple populations of 5-HT receptors, and some display comparable or higher affinity at these receptors (e.g., 5-HT_{1A}, h5-HT_{1D}, and 5-HT₆) than they do at 5-HT_{2A} receptors. The phenylalkylamines are quite selective for 5-HT₂ receptors but, as mentioned earlier, display little selectivity for the three 5-HT₂ subpopulations. Some β -carbolines, although they bind at 5-HT₂ receptors, also possess activity as MAO inhibitors. Thus, these differences might account for their somewhat different actions. The one feature that all the classical hallucinogens have in common (i.e., the *common component hypothesis*) is that they bind at 5-HT_{2A} receptors (5).

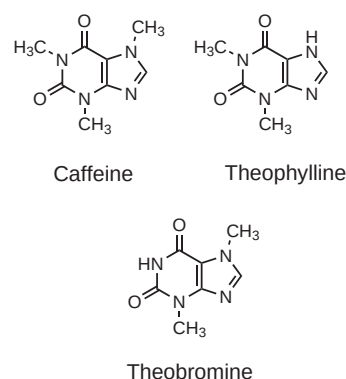
CENTRAL STIMULANTS

Introduction, Classification, and Definitions

The ubiquitous term “stimulant” typically conjures up agents such as the tropane analog cocaine or the phenylisopropylamine stimulant amphetamine. However, stimulants can be divided into several categories. Terms such as “psychostimulant,” “psychomotor stimulant,” “behavioral stimulant,” or, simply, “stimulant” typically refer to agents that produce a central stimulatory effect where actions are manifested mostly in motor activity, whereas the term “analeptic” refers to agents that have a stimulant effect primarily on autonomic centers such

as those involved in the regulation of respiration and circulation. Nicotine and related nicotinic agents also possess stimulant properties but are best discussed with other cholinergic agents. Analeptics include agents such as nikethamide, strychnine, and pentylenetetrazol. The boundary between analeptics and psychomotor stimulants is not sharply defined. Caffeine, for example, has been classified as an analeptic but is also considered a psychomotor stimulant. Caffeine is probably the best known of a series of xanthines; in fact, caffeine, which is found in coffee, tea, chocolate, guarana, mate, and other naturally occurring substances, is probably the most widely used psychoactive substance in the world. Although analeptics have never represented significant abuse problems, evidence exists for caffeine abuse (32).

Xanthines and Caffeine-Related Stimulants



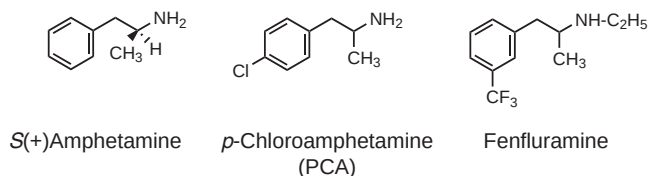
Caffeine (or 1,3,7-trimethylxanthine) is a naturally occurring methylated xanthine, as are its close relatives theophylline (1,3-dimethylxanthine) and theobromine (3,7-dimethylxanthine). A major metabolite of caffeine is paraxanthine, or 1,7-dimethylxanthine. All four compounds produce relatively similar, although not necessarily identical, effects (e.g., motor stimulation, relaxation of smooth muscle, stimulation of cardiac muscle). The different actions of the xanthines are thought to be mediated by (at least) one mechanism or, more commonly (depending on dose), a combination of mechanisms, at which some xanthines are more effective than others. In general, these xanthines act as nonselective competitive inhibitors at adenosine receptors (33) and of the enzyme(s) cyclic nucleotide phosphodiesterase (PDE; of which there are 14 human subtypes and >50 isoforms) (34); they can mobilize intracellular stores of calcium (e.g., from the endoplasmic reticulum) and bind with rather modest affinity at benzodiazepine binding sites on GABA_A receptors (33). Other mechanisms might be involved. For example, the mechanism of action of the xanthines in the treatment of airway diseases probably involves their ability to cause bronchodilation via inhibition of PDE in the lung, whereas the immunomodulatory and anti-inflammatory actions of theophylline seem to result from induction of histone deacetylase (35,36).

The psychostimulant effects of caffeine-related xanthines are probably a direct consequence of their actions at adenosine receptors and, to a substantially lesser extent, on their ability to inhibit cyclic nucleotide PDE. Indeed, the stimulant properties of a series of xanthines have been correlated with their blockade of adenosine receptors (33). Adenosine receptors are now recognized to consist of a family of four subtypes (i.e., A_1 – A_3): A_1 adenosine receptors, which are negatively coupled to adenylate cyclase; A_{2A} (with high affinity for agonists) and A_{2B} (with low affinity for agonists) receptors, which are positively coupled to adenylate cyclase; and A_3 receptors, which are coupled to phospholipase C (PLC) and also negatively coupled to adenylate cyclase; stimulation of PLC generates inositol triphosphate (IP_3), which, in turn, increases calcium levels (33). Adenosine binds with low affinity both at A_{2B} and A_3 receptors; in contrast, adenosine binds with higher affinity at A_1 and A_{2A} receptors, supporting a role for the latter in the stimulant actions of caffeine and related xanthines (33). The inhibitory control of neurotransmission exerted by A_1 receptor activation might account for the effects of caffeine on arousal and vigilance by its acting as a competitive antagonist (37). However, caffeine is believed to stimulate motor activity by countering the indirect tonic control exerted by activation of A_{2A} receptors on dopaminergic (primarily D_2) and glutamatergic transmission (37,38).

There are some similarities in the psychomotor stimulant actions of caffeine, amphetamine, and cocaine, and these involve both DA-dependent and DA-independent mechanisms; indeed, all ultimately seem to share a somewhat common component of action involving the modulation of DA- and cyclic adenosine monophosphate (cAMP)–regulated phosphoprotein (DARPP-32), a protein regulator of intracellular signaling (see Fig. 19.8). Direct activation of D_1 receptors by DA or DA agonists or indirect activation of these receptors by amphetamine and cocaine (e.g., by presynaptic release of DA or blockade of DA reuptake; see later sections) increases phosphorylation of DARPP-32 by protein kinase A (PKA) at a particular threonine moiety (i.e., Thr³⁴) to inhibit protein phosphatase-1 (PP-1); this results in enhanced signaling and increased psychomotor stimulation. Activation of these receptors also decreases phosphorylation of DARPP-32 at Thr⁷⁵ by activation of a specific protein phosphatase (PP-2A); phosphorylation of Thr⁷⁵ has an inhibitory effect on Thr³⁴ phosphorylation (see Fig. 19.8). Stimulation of A_{2A} receptors has a similar effect, but activation of these receptors results in decreased motor activity in rodents; this is believed to be related to the tonic inhibitory control of dopaminergic and glutamatergic systems by A_{2A} receptors (38). Caffeine, being a competitive A_{2A} receptor antagonist, reduces its (i.e., adenosine's A_{2A} receptor-mediated) tonic activation of the cAMP/PKA pathway and produces a dose-dependent increase in phosphorylation of DARPP-32 at Thr⁷⁵ (37). Caffeine also increases phosphorylation of Thr⁷⁵ by inhibition of PP-2A, which, by having an inhibitory effect

on phosphorylation of Thr³⁴, further amplifies its effect (37). The eventual outcome of blocking activation of A_{2A} adenosine receptors is disinhibition of its inhibitory control on dopaminergic and/or glutamatergic activity. Ultimately, then, amphetamine and cocaine indirectly regulate downstream effector mechanisms by having a particular effect on DARPP-32 via DA receptors, whereas caffeine regulates downstream effector mechanisms by having an effect on DARPP-32 via adenosine A_{2A} receptors. Both result in psychomotor stimulation. It might be noted that the psychomotor stimulant actions of caffeine are biphasic: Low doses produce A_{2A} receptor-mediated stimulation, whereas higher doses produce depression of activity. It has been speculated that higher caffeine doses result in blockade of A_1 adenosine receptors to produce this depressant effect (37).

Phenylisopropylamine Stimulants: Amphetamine-Related Agents



The simplest unsubstituted phenylisopropylamine, 1-phenyl-2-aminopropane, or amphetamine, serves as a common structural template for hallucinogens and psychostimulants. Amphetamine produces central stimulant, anorectic, and sympathomimetic actions, and it is the prototype member of this class (39). It is common to refer to amphetaminergic structures and amphetaminergic activity, but amphetamine may be more of an exception than a rule. Most substituted derivatives of amphetamine (i.e., phenylisopropylamine) lack central stimulant activity; in fact, pharmacologically, there are a greater number of “non-amphetamine-like” than “amphetamine-like” derivatives of amphetamine (e.g., see phenylisopropylamine hallucinogens). Relatively few derivatives of amphetamine retain the actions of amphetamine; still fewer retain the potency of amphetamine. This section will focus almost exclusively on the psychostimulant actions of amphetamine, and it should be recognized that these SARs are not necessarily identical to those for its anorectic or sympathomimetic actions.

Structure–Activity Relationships for Amphetamine-like Stimulant Action

In general, the SARs for amphetamine-like psychostimulant actions of the phenylisopropylamines are quite distinct from those for the DOM-like (or hallucinogenic) actions of the phenylisopropylamines, even though both share a common structural skeleton. The SARs for the two actions are summarized in Figure 19.2. The stimulus effects of amphetamine analogs have been reviewed elsewhere (4, 40).

ARYL-SUBSTITUTED DERIVATIVES In general, incorporation of substituents into the aromatic ring of amphetamine reduces or abolishes amphetamine-like psychostimulant activity. The sympathomimetic agent 4-hydroxyamphetamine lacks central stimulant action and is unlikely to penetrate the blood–brain barrier because of the presence of the polar aromatic hydroxyl group. Masking of the hydroxyl group in the form of its methyl ether affords the Schedule I substance *p*-methoxyamphetamine (PMA; also known as 4-methoxyamphetamine). PMA is a weak central stimulant with approximately 10% the potency of amphetamine. 4-Methylamphetamine (1-(*p*-tolyl)-2-aminopropane [pTAP]) has also been found on the clandestine market and is, at best, a weak central stimulant. Incorporation of electron-withdrawing substituents results in agents that generally lack central stimulant properties. For example, *p*-chloroamphetamine (PCA) is a 5-HT-releasing agent that saw evaluation as a potential antidepressant. Another related analog is the 5-HT-releasing agent fenfluramine, which was used for some time as an appetite suppressant. Both of these latter agents are still widely used as pharmacologic tools in basic neuroscience research.

AMINE SUBSTITUTION In general, the primary amines are more potent than the secondary amines, and the secondary amines are more potent than the tertiary amines, as central stimulants. With regard to secondary amines, as the length of the amine substituent increases, activity decreases; the *N*-monoethyl and *N*-mono-*n*-propyl amines retain stimulant character but are somewhat less potent than amphetamine itself. Larger *N*-substituents typically result in agents with little to no psychostimulant character. The one exception is the *N*-monomethyl derivative methamphetamine. Methamphetamine (e.g., “crystal,” “ice,” or “meth”) is at least as potent as amphetamine as a central stimulant; in most studies, it may be two- to threefold more potent than amphetamine. Methamphetamine is the most widely abused synthetic substance in the world. *N*-Hydroxylation of amphetamine has little effect on stimulant action. *N,N*-Dimethylamphetamine has been seized from clandestine laboratories, but it has never been certain whether this agent was being prepared for its possible stimulant actions or was an undesired byproduct of methamphetamine synthesis.

α-SUBSTITUENTS Amphetamine possesses an α-methyl group. As already mentioned at the beginning of this chapter, α-demethylation (to afford phenylethylamine or 2-phenyl-1-aminoethane in the case of amphetamine) results in agents with decreased lipophilicity and increased susceptibility to metabolism. Phenylethylamine lacks central stimulant activity. Homologation of the α-methyl group to, for example, an α-ethyl or α-*n*-propyl group results in a decrease or loss of central stimulant activity. The presence of the α-methyl group in amphetamine creates a chiral center; hence, amphetamine exists as a pair of optical isomers. With respect to central stimulant

actions, the *S*(+)-isomer (i.e., dextroamphetamine) is several-fold more potent than its *R*(–)-enantiomer (i.e., levoamphetamine); this is not necessarily the case with other actions produced by amphetamine, particularly those produced in the periphery, such as its cardiovascular actions.

β-SUBSTITUENTS The β-position has not been particularly well investigated. Perhaps the best-studied derivatives are ephedrine and norephedrine—and even these agents have not been especially well investigated. Ephedrine and norephedrine are phenylpropanolamines that may be viewed as the β-hydroxy analogs of methamphetamine and amphetamine, respectively. Actually, β-hydroxylation of amphetamine or methamphetamine results in the creation of a new chiral center; hence, a total of four optical isomers are possible in each case. These eight structures are shown in Figure 19.3. Relatively little comparative information is available regarding the central stimulant actions of these phenylpropanolamine isomers.

During the 1970s, there was a problem with what were termed “look-alike drugs.” Look-alikes available on the clandestine market were made to resemble amphetamine and methamphetamine, both in action and physical appearance, to circumvent the control of amphetamine. The major constituents of these agents were various combinations of ephedrine, norephedrine, and caffeine. Although the look-alikes are no longer the problem they once were, the 1990s witnessed the introduction of “herbal dietary supplements.” These supplements were—and still are—available in some health food and herbal shops, and even more so on the Internet; several dozen such preparations have appeared on the market. The major ingredients of many of these preparations are various combinations of ephedrine and caffeine (or of ephedrine-containing

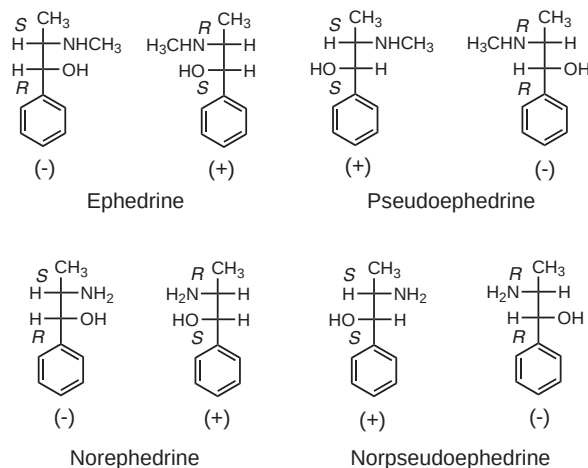
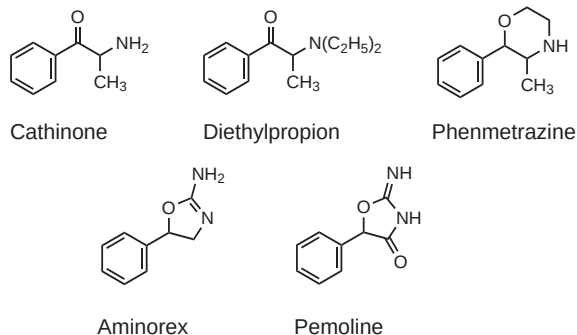


FIGURE 19.3 Structures of β-oxidized analogs of methamphetamine (ephedrine and pseudoephedrine) and amphetamine (norephedrine and norpseudoephedrine). Note that norpseudoephedrine is also known as cathine.

natural products [e.g., ma huang or ephedra] or caffeine-containing natural products [e.g., guarana or kola nut]). Interestingly, although ephedrine and caffeine possess stimulant character of their own, evidence suggests that these agents may potentiate one another's actions (41).

Although β -hydroxylation of amphetamine results in decreased central stimulant actions, this may be the result of the decreased ability of norephedrine to penetrate the blood-brain barrier, or it may be a clue that the presence of a β -oxygen substituent is inherently detrimental to activity. Support for the former is derived from the shrub *Catha edulis*. Commonly known as khat or kat, *C. edulis* is a plant indigenous to certain regions of the Middle East and eastern portions of Africa. The fresh shrub is sold openly in local markets and is used for its central stimulant character, much in the same way as the West uses coffee. Khat is used to prepare an infusion, or the fresh leaves are simply chewed. For more than 50 years, it was thought that the active constituent was the phenylpropanolamine cathine or (+)-norpseudoephedrine (Fig. 19.3). In the late 1970s, however, a more potent compound was isolated from fresh leaves and shown to be what is now called cathinone. Cathinone, which is simply β -ketoamphetamine or an oxidized analog of norephedrine, is at least as potent as amphetamine as a central stimulant.



Certain anorectic agents, such as diethylpropion, also possess a benzylic keto group. The anorectic agent phenmetrazine, 3-methyl-2-phenylmorpholine, and aminorex possess a benzylic oxygen atom in the form of an ether. All three of these agents possess stimulant character. A related stimulant is pemoline (available as a magnesium salt). Hence, it is specifically the hydroxyl analogs that seem to possess weak stimulant actions, and this is likely a result of their reduced lipophilicity and not because they simply possess an oxygen atom at the β or benzylic position.

Metabolism of Amphetamine

In humans, (+)-amphetamine has a half-life of approximately 7 hours. Some of the metabolic products of amphetamine metabolism are shown in Figure 19.4. Although a significant portion of amphetamine is excreted unchanged, it also undergoes both phase

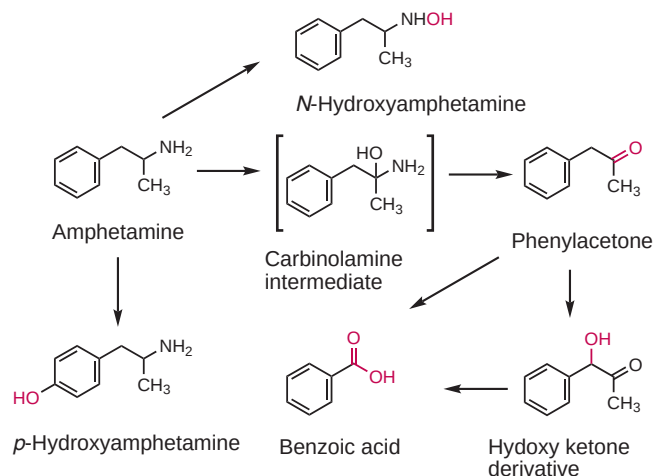


FIGURE 19.4 Some products of amphetamine metabolism.

1 (functionalization to more polar derivatives) and phase 2 (conjugation) metabolism (42). The phase 1 metabolism of amphetamine analogs is catalyzed by two enzyme systems: cytochrome P450 and flavin monooxygenase. The latter system oxidizes secondary and tertiary amine analogs of amphetamine. Amphetamine undergoes hydroxylation on the α -carbon, the β -carbon, the terminal amine, and the aromatic ring. These metabolites are subsequently oxidized, where possible, or conjugated.

Amphetamine is oxidized to phenylacetone (P2P) via a presumed carbinolamine intermediate. The phenylacetone is further oxidized directly to benzoic acid or, first, to a hydroxy keto analog that is subsequently converted to benzoic acid. Amphetamine can also undergo aromatic hydroxylation to *p*-hydroxyamphetamine. Initial work with rats indicated that *p*-hydroxylation is a major route of metabolism; however, subsequent studies showed that benzoic acid is the major metabolite in humans. Subsequent oxidation at the benzylic position by DA β -hydroxylase affords *p*-hydroxynorephedrine. Alternatively, direct oxidation of amphetamine by DA β -hydroxylase can afford norephedrine. Amphetamine and related derivatives also undergo *N*-hydroxylation, and the *N*-hydroxy derivatives can be further oxidized to nitroso, nitro, and oximino compounds. Some evidence suggests that the oximino derivative is hydrolyzed to phenylacetone. Additional metabolites are possible as well. In phase 2 reactions, ring-hydroxylated metabolites are conjugated to their corresponding glucuronides. Sulfation of the enol form of phenylacetone has been reported. Approximately 23% of methamphetamine is excreted unchanged, 18% is excreted as *p*-hydroxymethamphetamine, and 14% is excreted as the demethylated product (42).

Mechanism of Action of Amphetamine

Amphetamine is an indirect-acting dopaminergic and noradrenergic agonist; that is, amphetamine causes an

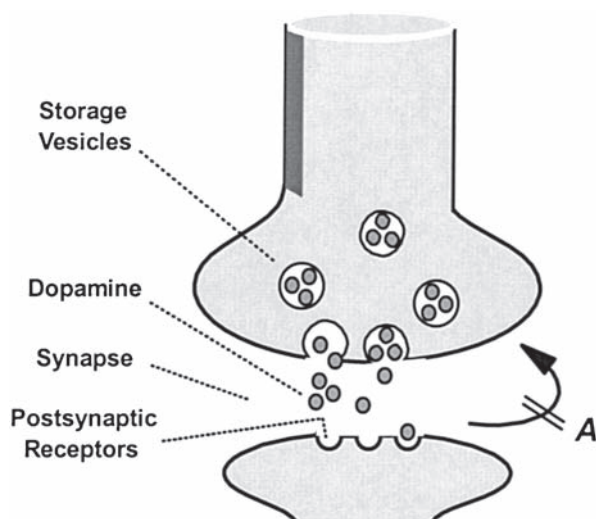


FIGURE 19.5 Schematic of a dopaminergic nerve terminal. Amphetamine increases synaptic concentration of DA primarily by causing its release from presynaptic terminals, whereas cocaine increases synaptic concentration by preventing its reuptake (A).

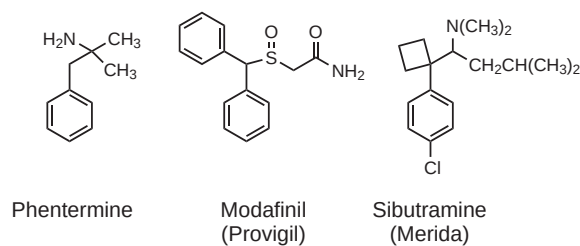
increase in the synaptic concentrations of the neurotransmitters DA and NE. Amphetamine enhances the release of DA and, to a lesser extent, prevents the reuptake of DA into presynaptic terminals (Fig. 19.5). The stimulant actions of amphetamine can be attenuated by the administration of DA antagonists, such as the antipsychotic phenothiazine chlorpromazine and the butyrophenone haloperidol. Chronic administration of high doses of amphetamine may result in “amphetamine psychosis,” which exhibits symptoms similar to those of acute paranoid psychosis. This is consistent with a role for DA in the central actions of amphetamine and, further, with the DA antagonist mechanisms proposed for most antipsychotic agents (see also Chapter 14). Similar psychotic episodes have been associated with khat (“khat psychosis” or “cathinone psychosis”) and cocaine (“cocaine psychosis”).

Clinical Applications

Although phenylisopropylamines generally are known for their abuse liability, several have gained clinical acceptance. Indeed, certain agents display reduced stimulant character and/or are infrequently abused. Ephedrine (and, later, phenylpropanolamine) and amphetamine were two of the first agents used to treat obesity. Caffeine has also been used to treat obesity and, when given in combination with ephedrine (or amphetamine), the combination has a supra-additive effect (43,44). Alternatives to synthetic products include herbal preparations that contain ephedrine (e.g., ma huang) or caffeine (e.g., as in guarana). Recent studies indicate that ephedrine–caffeine combinations are associated with an increased risk of psychiatric, autonomic, cardiovascular, and other side effects (44). The central stimulant actions of phenylisopropylamines led to their structural modification and subsequent introduction of

anorectic agents, such as diethylpropion, phenmetrazine, and phentermine. Despite a lack of widespread abuse, many of these agents retain central stimulant properties. (±)-Fenfluramine was developed in the 1960s and marketed as Pondimin. Although structurally related to amphetamine, fenfluramine is devoid of central stimulant character. Unlike amphetamine-related agents that act primarily via noradrenergic and dopaminergic mechanisms, fenfluramine is a 5-HT-releasing agent. The more potent isomer in reducing food intake, (+)-fenfluramine (i.e., dexfenfluramine), was introduced clinically during the 1980s. Fenfluramine was also available in combination with phentermine (i.e., phen-fen). Unfortunately, some patients treated with this combination displayed symptoms of valvular heart disease, resulting in the voluntary withdrawal from the market in 1997 of fenfluramine-containing anorectic agents. (±)-Fenfluramine is metabolized to its primary amine norfenfluramine, and evidence suggests that valvulopathy is the result of the agonist action of norfenfluramine on cardiac 5-HT_{2B} receptors (45,46). Today, it is not unusual for new drug candidates to be examined for 5-HT_{2B} agonist action during the early stages of their development in order to eliminate this side effect.

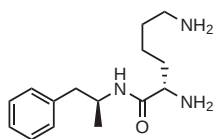
Other serotonergic agents have been evaluated for their antiobesity actions including selective serotonin reuptake inhibitors (SSRIs; e.g., fluoxetine; see Chapter 18) (43).



Sibutramine (Meridia), an agent with an amphetamine-like structural skeleton initially developed as an antidepressant, is an inhibitor of 5-HT and NE reuptake. Animal studies indicated that sibutramine reduces food intake by decreasing meal duration rather than feeding frequency, suggesting an effect on intrinsic satiation mechanisms (43). Sibutramine stimulates thermogenesis and effectively reduces amounts of visceral fat accumulation (47). Side effects of sibutramine include increased heart rate and increased blood pressure that have been attributed to its adrenergic action; in fact, because of its cardiovascular side effects, sibutramine was voluntarily withdrawn from the clinical market in 2010. These and other approaches and strategies to the treatment of obesity have been extensively reviewed (43,47).

Causes of excessive daytime sleepiness are numerous and include intrinsic sleep disorders, such as obstructive sleep apnea/hypopnea syndrome and narcolepsy;

circadian rhythm sleep disorders, such as jet lag; and sleep disorders associated with neuropsychiatric conditions, such as anxiety and depression (48–50). In many instances, excessive daytime sleepiness is commonly treated by addressing the underlying cause; however, the specific etiology of narcolepsy is unknown. Narcolepsy can also be characterized by brief periods of muscle paralysis (cataplexy). Hence, a need exists for agents that can effectively treat narcolepsy and cataplexy. Because of the increased behavioral activation (i.e., arousal, alertness, and motor activity) caused by psychostimulants, their use has been the mainstay for the treatment of narcolepsy. Agents such as methylphenidate (see later discussion) and amphetamine frequently are used, and methamphetamine and caffeine are used less commonly. Pemoline (Cyclert) has also seen some application but has been reported to produce liver toxicity (48–50) and was removed from the US market in 2005. Adderall, a combination of amphetamine salts [equal amounts of (+)-amphetamine saccharate, (+)-amphetamine sulfate, (±)-amphetamine aspartate, and (±)-amphetamine sulfate], was introduced in 1996; although used primarily for the treatment of attention-deficit hyperactivity disorder (ADHD), it is also used in the treatment of narcolepsy. Recently introduced for the treatment of ADHD is lisdexamfetamine (Vyvanse).



Lisdexamfetamine

Lisdexamfetamine is (+)-amphetamine acylated with an *N*-(L)-lysinyll function; in vivo, lisdexamfetamine is a prodrug that is slowly hydrolyzed to (+)-amphetamine. Newer agents for the treatment of narcolepsy include sodium oxybate ($\text{HO-CH}_2\text{CH}_2\text{CH}_2\text{COO}^- \text{Na}^+$; presumed to act through a GABAergic mechanism) and modafinil. Modafinil (Provigil) is a new agent that seems to promote wakefulness without producing the arousing effects associated with many other stimulants; it has also been approved for the treatment of obstructive sleep apnea/hypopnea syndrome (51). The exact mechanism of action of modafinil is unknown but has been shown by various investigators to involve DA, NE, histamine, 5-HT, and/or GABA receptors; modafinil also binds at hypocretin (orexin receptors) (52,53). Modafinil showed reduced stimulant character relative to methylphenidate (see later discussion), and its potential for abuse in patients with narcolepsy has been demonstrated to be quite low (48). Nevertheless, modafinil binds at the DA transporter and is only about eightfold less potent than cocaine as an inhibitor of DA reuptake (54). Furthermore, its more potent *R*-isomer

(armodafinil, Nuvigil), but not its *S*-isomer, is a potent D_2 receptor partial agonist (55). These dopaminergic actions might account for some of the stimulant actions associated with modafinil. Hepatic metabolism responsible for the clearance of modafinil includes amide hydrolysis to modafinilic acid, its primary inactive metabolite. *S*-Oxidation and aromatic ring hydroxylation occur via CYP2C9. Less than 10% is excreted as unchanged drug.

Motor suppression seen in patients with cataplexy is similar to the motor suppression seen in healthy individuals during rapid eye movement (REM) sleep. Consequently, agents previously found to decrease REM sleep have been evaluated for the treatment of cataplexy (50,52). Agents that suppress REM sleep include those that increase noradrenergic, serotonergic, and dopaminergic signaling. Tricyclic antidepressants, certain SSRIs, dual NE-serotonin reuptake inhibitors (e.g., venlafaxine), and MAO inhibitors have found application in the treatment of cataplexy, as has sodium oxybate (52,53). Psychostimulants also might be of some anticataplexic benefit, but modafinil generally produces little improvement (52).

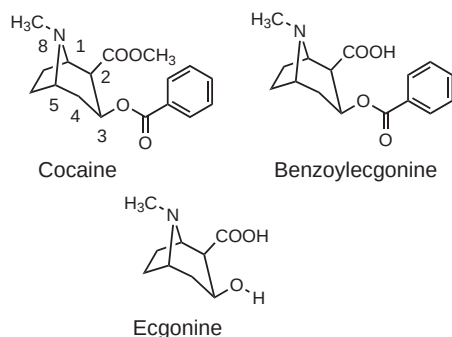
Cocaine-Related Agents

There are eight possible stereoisomeric forms of methyl 3-(benzoyloxy)-8-methylazabicyclo[3.2.1]octan-2-carboxylate, of which one is simply referred to as “cocaine.” Chemically, cocaine is 2*R*-carbomethoxy-3*S*-benzyloxy-1*R*-tropane. Cocaine is naturally occurring in a variety of plants, particularly the *Erythroxylon coca* species, which is indigenous to some countries in South America. In addition to its stimulant actions, cocaine possesses vasoconstrictor actions and is a local anesthetic; it has served as a template for the development of other therapeutically useful agents, including local anesthetics and 5-HT₃ receptor antagonists.

Cocaine has a very interesting history. The coca plant has been used by South American Indians for religious and mystical purposes and as a stimulant, both to increase endurance and to alleviate hunger, for millennia. The plant was introduced into Europe during the 1800s, and at the end of the 19th century, cocaine use was quite popular and very socially acceptable. Various cocaine-containing preparations were available, and it was also used to “fortify” wines (e.g., *Vin Coca*). For a period of approximately 20 years, until just after the turn of the last century, it was a constituent of the soft drink Coca-Cola. Additionally, cocaine was used for therapeutic reasons but was later supplanted by amphetamine.

Cocaine is active via nearly every possible route of administration; however, insufflation of “snow” or “coke” represents one of the most popular routes. Administered in this manner, peak effects and plasma levels are achieved within 30 minutes (56). Smoking the freebase form of cocaine (“crack”) results in an even more rapid effect. The freebase form, rather than the hydrochloride salt, is used for smoking, because the

temperatures required for vaporization of the salt results in considerable decomposition (56). Intravenously administered cocaine can achieve peak blood levels within a few minutes. Cocaine is metabolized to benzoylecgonine, the methyl ester of ecgonine, and to a lesser extent, to ecgonine, norcocaine, and hydroxylated derivatives.



Mechanism of Action of Cocaine

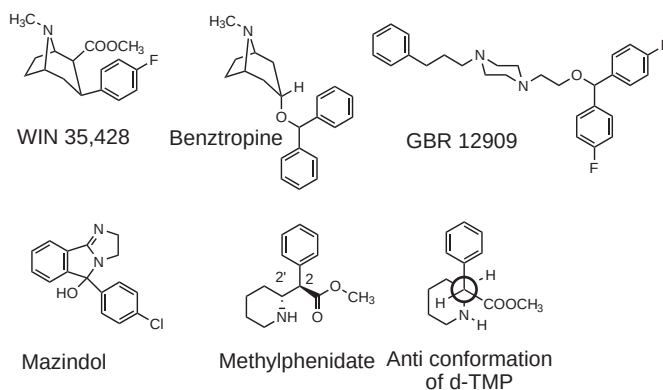
Cocaine has been shown to block the reuptake of NE, 5-HT, and DA; however, the reinforcing and stimulant nature of cocaine seems to be related primarily to blockade of DA reuptake, leading to the “dopamine hypothesis” of cocaine action (57). [^3H]Cocaine was used in an attempt to identify a “cocaine receptor,” and this target was later shown to be similar to the DA transporter. Currently, it is thought that cocaine produces its reinforcing effects by interfering with DA reuptake (Fig. 19.5) by blocking the DA transporter (58). Although the human DA transporter has been cloned, it is unknown whether the DA and cocaine binding domains are identical or how much they overlap (57). There are also indications that NE might have a larger role in the actions of cocaine and other psychostimulants than previously appreciated.

Cocaine-Like Structure–Activity Relationships and Cocaine-Like Agents

Because cocaine binds at the DA transporter, this provides a convenient method for the investigation and formulation of SAR; these have been reviewed elsewhere (56,57). Important features for the binding of cocaine analogs include configuration, substituent at C_2 , stereochemistry at C_2 , substituent at N_8 , and substituents at C_3 . With respect to cocaine analogs, inversion of configuration can decrease activity. The C_2 -position is quite important: Epimerization from β to α reduces activity by 30- to 200-fold, and hydrolysis of the ester to the acid (i.e., benzoylecgonine) reduces activity by more than 1,500-fold. Although an ester function seems to be important, the methyl group can be replaced by other substituents (e.g., phenyl or benzyl) with relatively little effect. A basic nitrogen atom appears to be optimal. Replacement of the N_8 -methyl group by other substituents, such as a

small alkyl or benzyl group, has only a small negative influence on activity, whereas quaternization or acylation (of norcocaine) reduces activity by 33- and 111-fold, respectively (57).

Other DA transport blockers are known, and their SARs have been investigated (58). One of the oldest and most widely investigated analogs is WIN-35,428, and [^3H]WIN-35,428 is commercially available as a radioligand. Others include methylphenidate (see next paragraph), bupropion, GBR 12909, and mazindol (58,59). These latter compounds produce varying degrees of cocaine-like actions and, thus, are being examined as structural leads for the development of therapies for the treatment of cocaine abuse (58). Because there are currently nearly 2 million cocaine users in the United States, various novel pharmacotherapies are being pursued, including GABAergic agents, dopaminergic agents, adrenoceptor antagonists, vasodilators, and cocaine vaccines (60).



Methylphenidate (Ritalin) is used for the treatment of ADHD and, to a lesser extent, for the treatment of narcolepsy. Methylphenidate possesses two chiral centers (i.e., one at the 2-position and one at the 2'-position) and, as such, exists as four possible isomers: a *threo* pair and an *erythro* pair (61). The SARs of methylphenidate isomers and analogs have been extensively investigated (62). X-ray crystallographic, conformational, computational, and other studies indicate the following stereochemistry for methylphenidate: (+)-*threo* = 2*R*,2'*R*, (-)-*threo* = 2*S*,2'*S*, (+)-*erythro* = 2*R*,2'*S*, and (-)-*erythro* = 2*S*,2'*R* (59–61). The structure of the (+)-*threo* isomer (i.e., d-*threo*-methylphenidate [d-TMP or (+)-TMP]) is shown; the *anti* conformation of d-TMP is particularly favored by a hydrogen bond between the piperidine nitrogen atom NH and the carbonyl oxygen atom of the ester group (61,63).

The psychostimulant action of methylphenidate (i.e., the actions desired for its use in ADHD) is associated exclusively with the *threo* isomeric pair; the *erythro* isomeric pair is essentially inactive in this regard (62). Ritalin, for example, is ($\pm$)-*threo*-methylphenidate. Following oral administration of ($\pm$)-*threo*-methylphenidate [($\pm$)-TMP],

it undergoes enantiomeric hepatic metabolism such that (-)-TMP is metabolized faster (to the corresponding ritalinic acid) than is the (+)-TMP (62,64). This is not necessarily the case following other routes of administration (e.g., transdermal) where first-pass hepatic metabolism is circumvented (64).

Studies have shown that (+)-TMP is the eutomer form of the drug and that it is about 10 times more potent than (-)-TMP (62). Recently, (+)-TMP was introduced as dexmethylphenidate (Focalin; a Schedule II substance).

Methylphenidate (and its various isomers) has been around for more than 50 years, and its mechanism of action has been extensively investigated. (Note: Some studies were influenced by evaluation of various mixtures of its diastereomers or of enantiomeric pairs.) Methylphenidate blocks the reuptake of NE and DA. Although its actions are attributed mainly to its ability to block DA reuptake, it [i.e., ($\pm$)-TMP and d-TMP] is actually more potent as an inhibitor of NE than DA reuptake. However, there is evidence for a ceiling effect on NE, but not DA, shifting the actions to a more dopaminergic profile at higher drug doses (62).

DESIGNER DRUGS

Designer drugs, or controlled substance analogs, are the end result of the application of known SARs at the clandestine level. That is, knowledge of the established SARs of a particular class of abused substances can be applied at the clandestine level for the development of novel agents of abuse. What is particularly frightening about this concept is that the novel agents are not necessarily—or even commonly—examined for action or toxicity before they are put on the illicit market. The term “designer drug” was first introduced in reference to novel opiate-related analogs that appeared on the clandestine market approximately three decades ago; today, the term is applied more generically to any class of abusable substance. Furthermore, the term is now commonly applied to nearly any substance, novel or not to the scientific literature, that is new to the street scene. Designer drugs have appeared that are structurally related to the hallucinogens and stimulants discussed earlier; the present discussion will focus on some of these agents.

Specific Examples

Because some designer drugs result from the clandestine application of known hallucinogen or psychostimulant SARs, it should be possible to legitimately forecast the actions and, perhaps, even the approximate potencies of novel street drugs on the basis of available SAR data. In fact, this sometimes is the case. For example, “Nexus” made an appearance on the East Coast of the United States in the early 1990s. Nexus is α -desmethyl DOB, or 2-(4-bromo-2,5-dimethoxyphenyl)-1-amino-

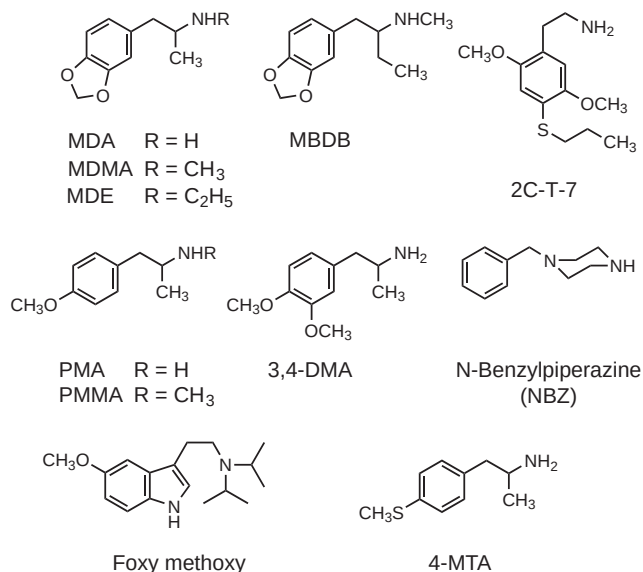
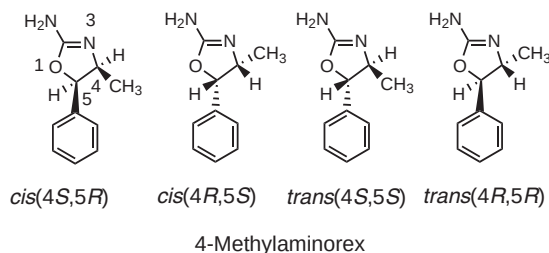


FIGURE 19.6 Examples of designer drugs.

ethane. Knowing that DOB is a potent phenylisopropylamine hallucinogen and that α -demethylation typically reduces the potency of phenylisopropylamines, it might have been suspected that Nexus would be a DOB-like hallucinogen with reduced potency. This has been supported by the results of drug discrimination studies in animals. Furthermore, this material, also known as 2C-B, has been shown to be active in humans at 12 to 24 mg relative to approximately 2 mg for DOB (29). In the last year or two, a number of related agents have been found on the clandestine market and, like 2C-B, are phenylethylamine analogs of their phenylisopropylamine counterparts; for example, 2C-C, 2C-I, 2C-N, 2C-E, and 2C-P are the phenylethylamine analogs of DOC, DOI, DON, DOET, and DOPR, respectively (Table 19.3 and Fig. 19.6). Another agent attracting recent attention is 2C-T-7 (e.g., “Blue Mystic” or “Triptasy”) (65), which has been recently controlled as a Schedule I substance. Recently controlled indolealkylamine analogs include α -MeT (AMT), α -EtT (“ET”), and 5-methoxy-*N,N*-diisopropyltryptamine (“Foxy Methoxy”) (Fig. 19.6); these agents had been previously shown to produce DOM-like stimulus effects in animals (66).



Stimulant designer drugs have also appeared. For example, CAT, or methcathinone, has been found on the illicit American market. Interestingly, it seems that methcathinone was a popular drug of abuse in the former Soviet Union (where it was known under a variety of names including ephedrone), but reports of this agent were never published in either the scientific or lay literature of that time. Methcathinone is the *N*-monomethyl analog of cathinone. Indeed, structurally, methcathinone is to cathinone what methamphetamine is to amphetamine. Methcathinone, which might be viewed as an oxidation product of ephedrine (hence the name ephedrone), is a potent central stimulant that is at least as potent as methamphetamine. Recently, a number of new cathinone analogs have appeared; the best known of these is 'bath salts'. Three common constituents of 'bath salts', mephedrone, methylenedioxypyrovalerone, and mephylone (or MDMC), were scheduled in 2011. Another example of a stimulant designer drug is 4-methylaminorex (U4Euh), which has been misrepresented on the illicit market as cocaine or methamphetamine. 4-Methylaminorex, an alkylated version of the anorectic/stimulant aminorex, contains two chiral centers and, hence, exists as four optical isomers. Typically, it is a mixture of the two *cis* isomers that has been confiscated by law enforcement officials, and *cis* 4-methylaminorex is now classified as a Schedule I substance. Interestingly, all four isomers behave as amphetamine-like agents, with the *trans*-(4*S*,5*S*) isomer being the most active, having a potency slightly greater than that of (+)-amphetamine itself (4). Yet other examples include piperazines. Several piperazine analogs have been reported to produce amphetamine-like effects in humans, and one in particular, *N*-benzylpiperazine (known either alone or in combination with other piperazines as "Rapture"), has been recently classified as a Schedule I substance (Fig. 19.6).

Not all designer drugs result in actions that are entirely predictable. One of the most popular of such agents is *N*-methyl-1-(3,4-methylenedioxyphe-nyl)-2-aminopropane (MDMA; e.g., "Ecstasy," "XTC," or "Adam") (Fig. 19.6). MDMA is the *N*-monomethyl analog of 1-(3,4-methylenedioxyphe-nyl)-2-aminopropane (MDA). MDA was popular during the 1960s, where it was known on the street as the "Love Drug." It was reported to produce effects in humans akin to a combination of cocaine and LSD. It has since been shown that MDA produces both amphetamine-like (i.e., central nervous system stimulant) and DOM-like (i.e., hallucinogen-like) stimulus effects in animals and, furthermore, that animals trained to discriminate MDA recognize central stimulants, such as amphetamine and cocaine, as well as classical hallucinogens, such as LSD, mescaline, and DOM. Interestingly, the stimulant actions of MDA appear to be associated with the *S*(+)-isomer, whereas the DOM-like actions are associated with the *R*(-)-isomer. Knowing that *N*-monomethylation of phenylisopropylamine

stimulants enhances their potency, whereas the corresponding change is detrimental to DOM-like actions, it would have been predicted that MDMA would probably behave as an amphetamine-like stimulant. Consistent with this prediction, amphetamine-trained (but not DOM-trained) animals recognized MDMA in tests of stimulus generalization. Furthermore, animals trained to discriminate MDMA recognized amphetamine but not DOM. However, MDMA was claimed to produce *empathogenic* effects in humans (i.e., increased empathy and sociability and enhanced feelings of well-being) and was used for several years as an adjunct to psychotherapy before emergency scheduling under the Controlled Substances Act as a Schedule I substance. It was argued that MDMA produces a unique, nonamphetamine-like effect (67). Although both optical isomers are active, the *S*(+)-isomer is the more active of the two. A closely related agent is its *N*-ethyl homolog MDE ("Eve"). The general consensus today is that MDMA is probably an empathogen with amphetamine-like stimulant side effects. Homologation of the α -methyl group of phenylisopropylamine stimulants and hallucinogens typically diminishes their potency or abolishes their activity; however, the α -ethyl analog of MDMA [*N*-methyl-1-(3,4-methylenedioxyphe-nyl)-2-aminobutane] retains MDMA-like actions (68) (Fig. 19.6). Another agent, sold as a substitute for MDMA, is 4-MTA (e.g., "Flatliners" or "Golden Eagles") (Fig. 19.6); this agent produces MDMA-like stimulus effects in animals but did not produce either DOM-like or cocaine-like effects (65).

A closely related agent is *N*-methyl-1-(4-methoxyphenyl)-2-aminopropane (PMMA). PMMA is a hybrid structure of two phenylisopropylamine stimulants: PMA and methamphetamine (Fig. 19.6). Interestingly, PMMA lacks significant central stimulant actions, and unlike PMA and methamphetamine, PMMA is not recognized by (+)-amphetamine-trained animals. Because PMMA is structurally related to metabolites of MDMA, it was examined in MDMA-trained animals and found to be several times more potent than MDMA. Animals have been trained to discriminate PMMA from vehicle, and PMMA stimulus generalization occurred to ($\pm$)-MDMA and *S*(+)-MDMA, but not to DOM, (+)-amphetamine, *R*(-)-MDMA, or *R*(-)-PMMA. Another psychoactive agent that has not been well investigated is 3,4-DMA (Table 19.3). 3,4-DMA may be viewed as an *O*-methyl ring-opened analog of MDA (Fig. 19.6). Although 3,4-DMA was not recognized by either DOM- or (+)-amphetamine-trained animals, it was recognized by MDA- and PMMA-trained animals. These results, coupled with the earlier discussion of MDMA, suggest that phenylisopropylamines might not best be described as merely central stimulants or hallucinogens; a third action needs to be accounted for. Although MDMA is widely abused, a contributing factor may be related to its amphetaminergic actions.

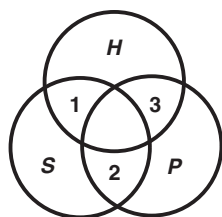


FIGURE 19.7 The behavioral effects of arylalkylamines may be described as falling into one or more of three different stimulus categories: classical hallucinogen (*H*), central stimulant (*S*), or PMMA-like (*P*). See text for further discussion.

It is not yet known if agents that fall into this third pharmacologic category possess abuse potential; consequently, they have been referred to simply as “other” agents. It has been proposed that the behavioral actions of the phenylisopropylamines can be described by the Venn diagram shown as Figure 19.7. As depicted in that figure, the three types of actions are classical hallucinogen (*H*), stimulant (*S*), and PMMA-like (*P*) (69). Because MDMA possesses both PMMA-like and (+)-amphetamine-like activity, it is perhaps best represented by Intersect 2. As mentioned earlier, *R*(-)-MDA is hallucinogenic, and *S*(+)-MDA is a stimulant. Both isomers possess PMMA-like activity. Thus, *R*(-)-MDA is best represented by Intersect 3, whereas *S*(+)-MDA is best represented by Intersect 2. The common intersect (shaded area) describes the actions of (±)-MDA. Using this classification system, it should be possible to classify the various phenylisopropylamines as falling into one or more categories. Furthermore, there is no reason to suspect that this classification system will be limited to the phenylisopropylamines; that is, there is evidence that the indolealkylamines might be classified in a similar manner. For example, *S*(+)-α-EtT produces both DOM- and PMMA-like effects, but not (+)-amphetamine-like effects, whereas *R*(-)-α-EtT produces (+)-amphetamine- and PMMA-like effects, but not DOM-like effects (70). The classification scheme suggests that there will be at least three different SARs and three different mechanisms of action. Certain agents, because they fall into more than one category, may represent mechanistic and structure-activity composites. The same may be said of arylalkylamine designer drugs; indeed, it may be the particular “mix” of actions that makes certain designer drugs so attractive as drugs of abuse.

Perhaps the most worrisome aspects about designer drugs are that almost none have been investigated under controlled clinical settings, that relatively little is known about their toxicity or long-term effects, and that medical professionals generally are unfamiliar with them (or with

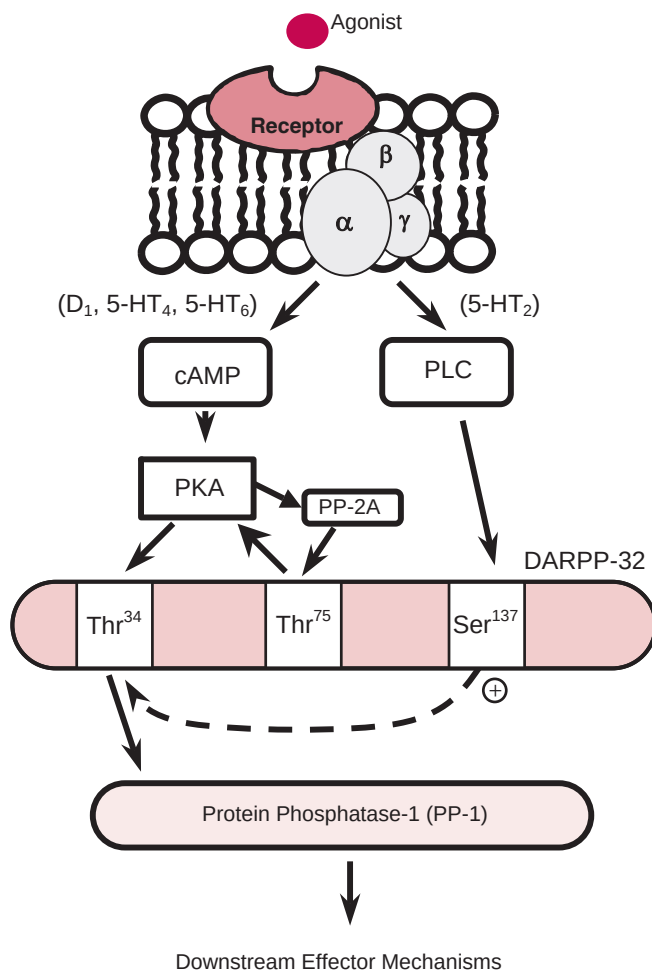


FIGURE 19.8 A simplistic schematic showing the proposed involvement of DARPP-32 in neurotransmission. Agonists (e.g., dopamine and serotonin) can interact with certain postsynaptic G protein-coupled receptors to activate second messenger systems. Agonist action at D_1 receptors increases cAMP levels and causes PKA to phosphorylate DARPP-32 at Thr³⁴; activation of D_1 receptors also decreases the phosphorylation state of DARPP-32 at Thr⁷⁵ by what appears to involve PKA-dependent activation of a protein phosphatase (PP-2A). Likewise, the action of 5-HT at 5-HT₄ and 5-HT₆ receptors increases phosphorylation of Thr³⁴ and decreases phosphorylation of Thr⁷⁵. Interaction of 5-HT at 5-HT₂ receptors activates PLC and promotes phosphorylation of DARPP-32 at Ser¹³⁷. Phosphorylation at Thr³⁴ inhibits protein phosphatase-1 (PP-1). Inhibition of PP-1 enhances signaling. Phosphorylation of Thr⁷⁵ has an inhibitory effect on Thr³⁴ phosphorylation (conversely, decreased Thr⁷⁵ phosphorylation disinhibits PKA to increase Thr³⁴ phosphorylation), whereas phosphorylation of Ser¹³⁷ prevents dephosphorylation of Thr³⁴. Other receptors can impinge on this mechanism (72–77).

the treatment of their overdose) in emergency room settings. The situation is further exacerbated by the broad availability of Web sites describing such agents to potential users (71).

NEURONAL PLASTICITY AND DRUGS OF ABUSE

Release of neurotransmitter from presynaptic terminals results in the activation of postsynaptic neurotransmitter receptors that are coupled to complex effector mechanisms. Through modulation of postsynaptic pathways, the state of the neuron can be altered such that neurons become more or less responsive to the neurotransmitter (72). This process is referred to as functional plasticity. One of the most exciting recent findings with implications for the treatment of drug abuse (as well as other neuropsychiatric disorders) involves the regulation of DARPP-32, an integrator of intracellular signaling (Fig. 19.8). Interaction of DA at D_1 -like receptors (D_1/D_5) activates adenylate cyclase, which increases cAMP levels; this, in turn, regulates phosphorylation of DARPP-32 by PKA. Interaction of DA at D_2 -like receptors ($D_2/D_3/D_4$, which are negatively coupled to cAMP) has an effect that is essentially opposite that of activation of D_1 receptors. Phosphorylation of a specific amino acid residue (i.e., threonine³⁴ [Thr³⁴]), induced by D_1 agonists, converts DARPP-32 to an inhibitor of PP-1; thus, when phosphorylated at Thr³⁴, DARPP-32 behaves as an amplifier of PKA-mediated signaling through its ability to inhibit PP-1. The actions of DARPP-32 can also be modulated by phosphorylation (or dephosphorylation) of Thr⁷⁵ (Fig. 19.8). Activation of D_1 receptors decreases the phosphorylation state of DARPP-32 at Thr⁷⁵ by a process that involves PKA-dependent activation of protein phosphatase-2A (PP-2A); this disinhibits phosphorylation of Thr³⁴ by PKA (i.e., results in enhanced phosphorylation of Thr³⁴). The result is potentiation of dopaminergic signaling. Together, PKA and PP-1 regulate the phosphorylation state of downstream neuronal effector proteins. Additionally, DARPP-32 can

be phosphorylated at serine¹³⁷ (Ser¹³⁷), and this phosphorylation decreases the rate of dephosphorylation of Thr³⁴.

5-HT causes an increase in phosphorylation of Thr³⁴ (via activation of 5-HT₄ and 5-HT₆ receptors, which are positively coupled to cAMP) and Ser¹³⁷ (via activation of 5-HT₂ receptors, which are coupled to phospholipase C) and a decrease in the phosphorylation of Thr⁷⁵ (via activation of 5-HT₄ and 5-HT₆ receptors). Hence, 5-HT inhibits PP-1 through what might be considered a synergistic mechanism (73). Other receptors that might modulate DARPP-32 include glutamate, GABA, adenosine, nitrous oxide, and opioid receptors. Hence, it has been speculated that various drugs of abuse, including amphetamine, methamphetamine, cocaine, opioids (e.g., morphine), nicotine, and ethanol involve a DARPP-32 mechanism; caffeine was discussed earlier. Furthermore, agents such as antidepressants, antipsychotics, and antiparkinsonian drugs have been shown to influence phosphorylation of DARPP-32. Classical hallucinogens, as described earlier, are thought to act by activation of 5-HT_{2A} receptors. It has been shown that LSD increases phosphorylation of Ser¹³⁷ (74). In theory, DARPP-32 should be modulated by various designer drugs that act via a dopaminergic or serotonergic mechanism. Only selected aspects of intracellular integration have been mentioned here, and others already have been implicated. The state of the art has been recently described (38,73–77). Nevertheless, the actions of many drugs of abuse might involve such postsynaptic events and certainly require further attention.

ACKNOWLEDGEMENT

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SCENARIO: OUTCOME AND ANALYSIS

Outcome

Mark D. Watanabe, PharmD, PhD, BCPP

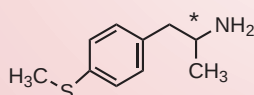
When JN took a break from the dance floor, he found that he was sweating profusely, short of breath, and quite dizzy. Given that he was in generally excellent physical shape from athletic training, he became alarmed about his weakened and disoriented state. Later, JN confided to a friend that although “flatliners” gave him a boost of energy, he was disappointed that he did not experience a more pleasant “buzz” as he had expected.

Chemical Analysis

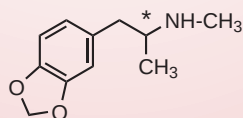
S. William Zito and Victoria Roche

The designer drug offered to JN was referred to as “flatliners.” It is structurally related to the arylisopropylamine classic hallucinogens. It is 4-MTA (4-methylthioamphetamine). JN was told it would act like Ecstasy. Ecstasy is *N*-Methyl-3,4-methylenedioxyamphetamine (MDMA). MDMA is the *N*-monomethyl analogue of MDA, or 1-(3,4-methylenedioxyphenyl)-2-aminopropane (also known as 3,4-methylenedioxyamphetamine).

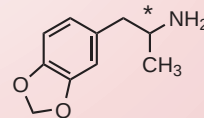
MDA produces both amphetamine-like (i.e., CNS stimulant) and DOM-like (i.e., hallucinogen-like) stimulus effects in animals. Interestingly, the stimulant actions of MDA appear to be associated with the *S*-(+)-isomer, whereas the DOM-like actions are associated with the *R*-(-)-isomer. SAR studies have shown that *N*-monomethylation of phenylisopropylamine stimulants enhances their CNS-stimulant potency and decreases or eliminates their DOM-like actions, it is therefore predicted that MDMA probably behaves as an amphetamine-like stimulant. In addition, MDMA seems to have unique non-amphetamine-like effects referred to as *empathogenic* (i.e., increased empathy and sociability and enhanced feelings of well being) and was used for several years as an adjunct to psychotherapy. The general consensus today is that MDMA is probably an empathogen with amphetamine-like stimulant side effects. In this case, JN experienced the amphetamine-like effects of Ecstasy (increased energy, increased stamina, tachycardia, decreased anxiety) from 4-MTA; however, he was disappointed with the lack of hallucinogenic effects (the “buzz”). This is not surprising because 4-MTA has been shown to produce MDMA-like stimulus effects in animals but did not produce either DOM-like or cocaine-like effects.



4-MTA



MDMA



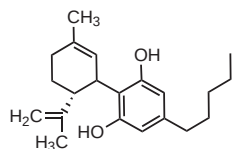
MDA

CASE STUDY

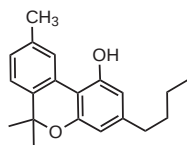
S. William Zito and Victoria Roche

You are one of the lucky few students to draw the police forensic laboratory as one of your elective rotations. On your second day there, your preceptor asks you to evaluate a recent case involving JB, a young man who was arrested and charged with possession of a drug of abuse. Upon arrest, JB was red eyed, very thirsty,

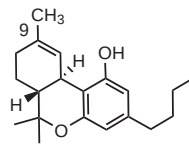
hungry, and bothered by bright light. Despite of the fix he was in, JB was laid back, laughing inappropriately and chuckling to himself. The forensic laboratory analyzed the confiscated substance and it was shown to contain the following three main compounds. Which one of them is responsible for JB's symptoms?



1



2



3

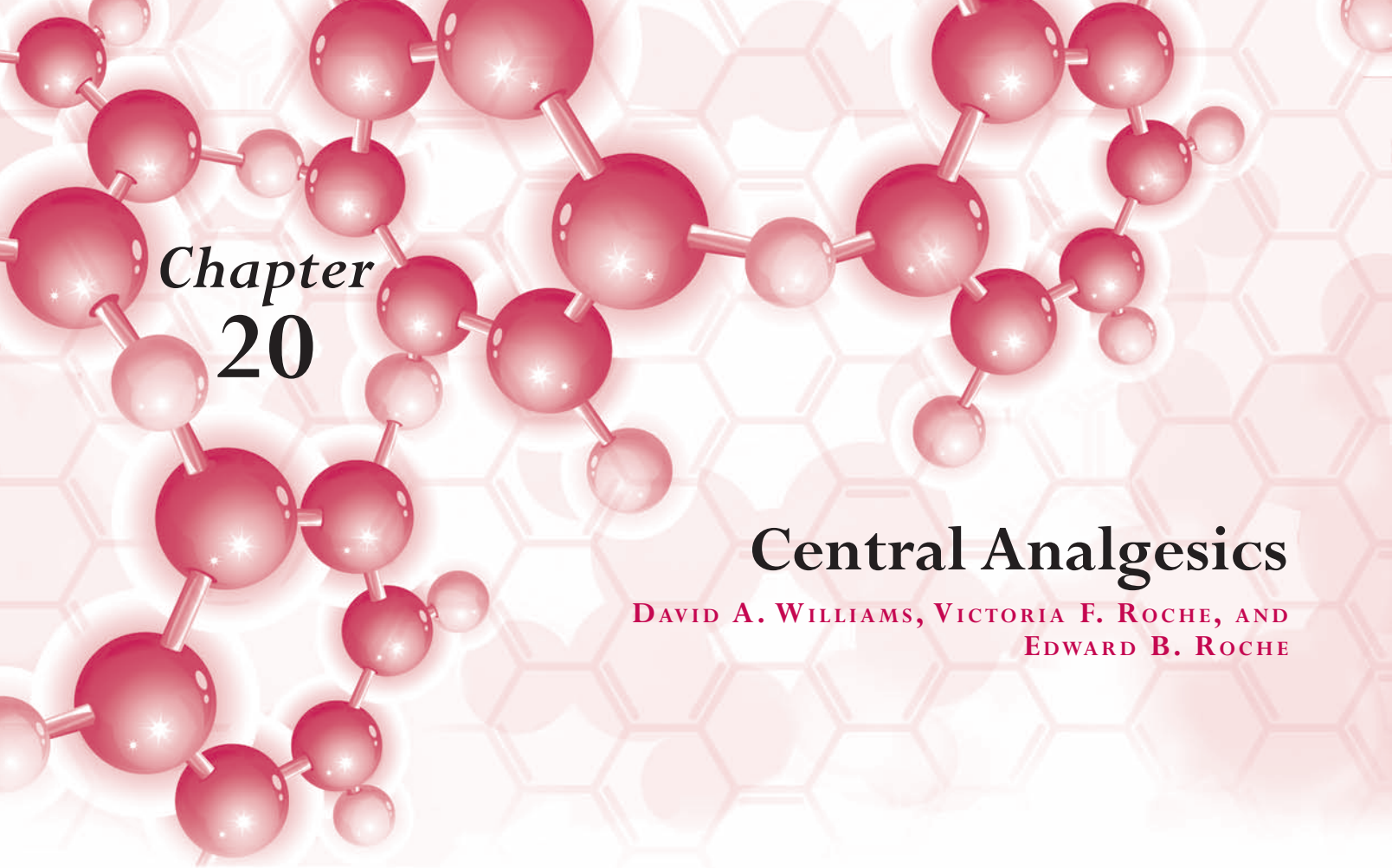
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 20

Central Analgesics

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Drugs Covered in This Chapter

NOCICEPTIVE ANALGESICS

MULTICYCLIC FULL μ AGONISTS

- Codeine
- Hydrocodone
- Hydromorphone
- Levorphanol
- Morphine
- Oxycodone
- Oxymorphone

MULTICYCLIC PARTIAL μ AGONISTS

- Buprenorphine

MULTICYCLIC κ AGONISTS/ μ ANTAGONISTS

- Butorphanol
- Nalbuphine
- Pentazocine

MULTICYCLIC OPIOID ANTAGONISTS

- Methylnaltrexone bromide
- Nalmefene
- Naloxone
- Naltrexone

FLEXIBLE FULL μ AGONISTS

- Alfentanil
- Fentanyl
- Meperidine
- Methadone
- Remifentanyl
- Sufentanil

DUAL-ACTION ANALGESICS

- Tapentadol
- Tramadol

ANTIDIARRHEALS

- Difenoxin
- Diphenoxylate
- Loperamide

NEUROPATHIC ANALGESICS

FIRST-LINE AGENTS

- Desipramine
- Duloxetine
- Gabapentin
- Lidocaine

- Nortriptyline
- Pregabalin
- Venlafaxine

SECOND-LINE AGENTS

- Methadone
- Morphine
- Oxycodone
- Tramadol

THIRD-LINE AGENTS

- Bupropion
- Carbamazepine
- Oxcarbazepine
- Paroxetine

FUTURE MEDICATIONS

- Cannabidiol
- Capsaicin
- Lacosamide
- Δ^9 -Tetrahydrocannabinol

MEDICAL FOOD

- Metanx

Abbreviations

cAMP, cyclic adenosine monophosphate

CNS, central nervous system

DADLE, [D-Ala², D-Leu⁵] enkephalin

DPDPE, [D-Pen², D-Pen⁵] enkephalin

DSLET, [D-Ser², Leu⁵] enkephalin-Thr

FDA, U.S. Food and Drug
Administration

GABA, γ -aminobutyric acid

GI, gastrointestinal

5-HT, serotonin

IUPHAR, International Union of
Pharmacology

Abbreviations (Continued)

IV, intravenous

MAOI, monoamine oxidase inhibitor

NE, norepinephrine

NMDA, *N*-methyl-D-aspartate

N/OFQ, nociceptin/orphanin FQ

P-gp, P-glycoprotein

ppN/OFQ, preproN/OFQ

SAR, structure–activity relationship

SSRI, selective serotonin reuptake inhibitor

TCA, tricyclic antidepressant

TRPV1, transient receptor potential vanilloid 1

SCENARIO

Jill T. Johnson, PharmD, BCPS

JB is a 25-year-old African-American man with sickle cell disease who was admitted to the hospital for a vaso-occlusive crisis. He says his pain started in his low back and now includes his chest and abdomen. The pain is severe and gnawing, and the patient scored the pain as 10 out of 10. At home, he does not take opioids routinely but does take folic acid and hydroxyurea. He says he is allergic to morphine and codeine and has a history of rash

when he takes either. He says intravenous pain medications work the best when he is in a crisis, and he prefers meperidine to hydromorphone.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter.)

INTRODUCTION

This chapter begins by describing types of pain and the mechanisms by which noxious stimuli result in the perception of pain. The pharmacologic treatment of pain should target the specific mechanism of pain rather than suppressing the symptoms of pain. Nociceptive (acute), chronic, and neuropathic pain are caused by different neurophysiologic processes and therefore respond to different pharmacologic treatment modalities. Neuropathic pain can be one of the most debilitating forms of chronic pain. Therefore, accurate diagnosis and treatment of pain should be sought early, and appropriate treatment should be rendered to increase the likelihood of a good outcome (relief of pain), without restricting a person's quality of life and general well-being.

A number of classes of drugs are used to relieve nociceptive pain. The opioid analgesics, also inappropriately known as narcotics, are the first-line treatment for acute or nociceptive pain; thus, the use of the term antinociceptive to describe the opioids. Agents that decrease acute and some types of chronic pain are referred to as analgesics, or analgetics. Although analgetic is grammatically correct, common use has made analgesic preferable to analgetic for the description of the pain-killing drugs. On the other hand, neuropathic or chronic pain relieving agents include antidepressants, antiseizure agents, and local anesthetics. Narcotic analgesic literally means that the agents cause sleep or loss of consciousness (narcosis) in conjunction with their analgesic effect, something the opioids do not do. The term “narcotic” has become associated with the addictive properties of opioids and other central nervous system (CNS) depressants. Because the great therapeutic value of the opioids lies in their ability to induce analgesia without causing narcosis, and because

not all opioids are addicting, the term “narcotic analgesic” is misleading and will not be used further in this chapter. The nonsteroidal anti-inflammatory agents have primarily a peripheral site of action as inhibitors of cyclooxygenase, are useful for mild to moderate pain, and often have an anti-inflammatory effect associated with their pain-killing action (Chapter 31). Local anesthetics inhibit pain transmission by inhibition of voltage-regulated sodium channels (Chapter 16). These agents are often highly toxic when used in concentrations sufficient to relieve chronic or acute pain in ambulatory patients. Dissociative anesthetics (ketamine) and other compounds that act as inhibitors of *N*-methyl-D-aspartate-activated glutamate receptors in the brain are effective antinociceptive agents when used alone or in combination with opioids (Chapter 16). Most CNS depressants (e.g., ethanol, barbiturates, and antipsychotics) will cause a decrease in pain perception. Current research into the antinociceptive effects of centrally acting α -adrenergic-, cannabinoid-, and nicotinic-receptor agonists may yield clinically useful analgesics working by nonopioid mechanisms.

The chapter concludes by describing the mechanism by which nerve damage results in neuropathic pain and the mechanisms of action of the major classes of drugs used for clinical neuropathic pain relief. Neuropathic pain can be very difficult to treat, with only 40% to 60% of patients achieving partial relief. Many of the pharmacologic treatments for chronic neuropathic pain decrease the sensitivity of nociceptors or desensitize C-fibers such that they transmit fewer signals. Agents for the treatment of neuropathic pain include the tricyclic and dual-acting reuptake inhibitor antidepressants (Chapter 18); these agents are useful either alone or in combination with opioids in treating certain cases of neuropathic pain, especially the anticonvulsants

pregabalin (Lyrica) and gabapentin (Neurontin), which inhibit voltage-regulated Ca^{2+} ion channels (Chapter 17), and topical lidocaine (Chapter 16). Opioid analgesics, including the dual-action agent tramadol, are recognized as useful agents but are not recommended as first-line treatments. Tricyclic antidepressants and anticonvulsants have long been the mainstay of treatment of neuropathic pain.

Research in one or more of the above areas may lead to new drugs, but at present, severe acute and some types of chronic pain are generally treated most effectively with opioid agonists, and neuropathic pain is most effectively treated with antidepressants and antiseizure agents. The cannabinoid receptors and the endogenous cannabinoids have become the focus of chronic pain regulation.

OVERVIEW OF PAIN

Pain is defined as an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage (1,2). Pain is part of the body's defense system producing a reflexive retraction from the painful stimulus and to protect the affected body part while it heals. Most pain resolves promptly once the painful stimulus is removed and the body has healed, but sometimes pain persists despite removal of the stimulus and apparent healing of the body (1). Pain sometimes arises in the absence of any detectable stimulus, damage, or disease. The experience of physical pain is familiar to everyone; however, the intensity, character, and tolerability of each person's pain are subjective. Differences in pain perception and tolerance thresholds are associated with, among other factors, ethnicity, genetics, and sex. For example, women have lower pain perception and tolerance thresholds than men, and this sex difference appears to apply to all ages, including newborn infants. Pain is essential for protection of our bodies from injury and recognition of the presence of injury (3). Pain is the most common reason for visiting the emergency department in more than 50% of cases and for 30% of family practice visits. According to the National Center for Health Statistics, 46 million Americans undergo inpatient surgical procedures each year and experience acute surgical pain; pain severity was reported as moderate to severe by 45% of patients in the emergency departments (4). Examples of clinical pain include pain following surgery and pain associated with injury. Pain management becomes more important as people approach death (5). Pain can significantly interfere with a person's quality of life and general well-being. Observation becomes critical when self-reporting of pain, and specific behaviors or types of pain (e.g., dull, throbbing, stabbing), location and intensity of pain, how and when it occurs, and the effects of pain are needed in order to recommend appropriate interventions (5). Because everyone feels pain differently, standardized pain scales are used in the clinic and in clinical trials to provide clues as to the intensity of a patient's

pain. To help compensate for this problem, one common type of pain scale shows a series of numbered cartoon faces moving from 0 (smiling and pain-free) to 10 (weeping in agony). The person in pain is asked which face matches up with the intensity of the pain the patient is experiencing. These scales can be used to assess the progression of pain and/or the effectiveness of pain therapy (3).

An aging adult may not respond to pain in the way that a younger person would. Their ability to recognize pain may be blunted by illness or the use of multiple prescription drugs (3,5). Depression may also keep the older adult from reporting they are in pain. The older adult may refrain from reporting pain and also stop doing activities because they hurt too much. Decline in self-care activities (dressing, grooming, walking, etc.) may also be indicators that the older adult is experiencing pain.

Acute Pain

Acute pain is defined as the normal, predicted physiologic response to an adverse chemical, thermal, or mechanical stimulus associated with surgery, trauma, or acute illness (1). Acute pain results from activation of the pain receptors (nociceptors) at the site of tissue damage. This type of pain generally accompanies surgery, traumatic injury, tissue damage, and inflammatory processes. Acute pain provides a warning signal that something is wrong and in need of further examination. Acute pain is typically self-limiting and resolves over days to weeks, but it can persist for months or longer as healing occurs (3). Acute pain can activate the sympathetic branch of the autonomic nervous system to produce such responses as hypertension, tachycardia, sweating, shallow respiration, restlessness, facial grimacing, pallor, and pupil dilation, as well as unnecessary physical and emotional distress (2).

Pain can be divided into nociceptive and neuropathic pain (2). Nociceptive pain and neuropathic pain are caused by different neurophysiologic processes and therefore respond to different treatment modalities. Nociceptive pain is mediated by stimulation of nociceptors, which are located in skin, bone, connective tissue, muscle, and viscera and respond only to noxious stimuli approaching or exceeding harmful intensity (2). Nociceptors serve a biologically useful role at localizing noxious chemical (alcohol in an open wound), thermal (heat or cold), and mechanical (crushing, tearing, etc.) stimuli. Nociceptive pain can be divided into visceral, deep somatic, or superficial somatic pain (1–3). Visceral pain stems from nociceptors in the viscera (organs), tends to be vague in distribution, and is usually described as deep, aching, squeezing, and spasmodic. Examples of nociceptive pain include postoperative pain, pain associated with trauma, and the chronic pain of arthritis. Nociceptive pain usually responds to opioids and nonsteroidal anti-inflammatory drugs. Deep somatic pain originates from stimulation of nociceptors in ligaments, tendons, bones, blood vessels, fasciae, and muscles and is a constant pain described as sharp, aching, throbbing, or gnawing (6). Examples include sprains and broken bones.

Superficial pain is initiated by activation of nociceptors in the skin or superficial tissues and is sharp, well-defined, and clearly localized (also described as sharp, aching, throbbing, or gnawing). Examples of injuries that produce superficial somatic pain include minor wounds and minor (first-degree) burns (6). On the other hand, neuropathic pain (chronic pain) involves the abnormal processing of stimuli from nerve damage in the peripheral or central nervous system; it is believed to serve no useful purpose.

The nociceptive pain mechanism is highly complex (2). Initially, tissue damage releases chemical mediators, such as prostaglandins, bradykinin, serotonin, substance P, and histamine, which subsequently stimulate nociceptors, generating an action potential (an electrical impulse) (Fig. 20.1). The action potential travels from the site of peripheral injury along afferent neurons or fibers, designated C-fibers and A-delta (Δ)-fibers, to receptors at the dorsal horn of the spinal cord, from where the action potential ascends the spinothalamic tract to the brainstem and onto the thalamus and the midbrain. A Δ -fibers are thinly myelinated and transmit the action potentials more rapidly (~20 m/s) than unmyelinated C-fibers (~2 m/s), but more slowly than more thickly myelinated A-fibers. A Δ -axons are associated with acute pain and therefore constitute the reflex process that results in “pulling away” from noxious stimuli (e.g., yanking hand away from hot stove). These fibers are

associated with thermal and mechanical sensations. By contrast, the slowly conducting, unmyelinated C-fibers are responsible for the slow, dull, longer-lasting, second pain (2). The lack of myelination is the cause for the slow conduction of C-fibers. Because of their higher rate of conduction, A Δ -neurons are responsible for the sensation of the sharp first pain and respond to a low stimulus threshold. On the other hand, C-fibers respond to stronger stimulus intensity and cannot be normally activated, but become responsive during inflammation or intense mechanical, chemical, or thermal stimuli. Finally, from the thalamus, the nociceptive message is transmitted to the somatosensory cortex, parietal lobe, frontal lobe, and limbic system, where the stimulus is processed, causing perception of pain (2).

Chronic Pain

While acute pain is usually a transitory response to a noxious stimulus to nociceptors lasting only until the noxious stimulus is removed or the underlying damage or pathology has healed, chronic pain is different (7). Tens of millions suffer from chronic pain that is unrelenting, not self-limiting, and can persist from months to years and even decades after the initial injury heals. The pain can be refractory, mild, or excruciating, episodic or continuous, or inconvenient or incapacitating; is not a symptom that exists alone; and usually extends beyond the expected period of healing (3,6,7). Pain signals keep firing in the nervous system for weeks, months, or even years. Some painful conditions, such as rheumatoid arthritis, osteoporosis, endometriosis, scleroderma, peripheral neuropathy, cancer, and idiopathic pain, may persist for years. Chronic pain serves no protective biologic function (7). Rather than being the symptom of a disease process, chronic pain is itself a disease process. Other problems associated with chronic pain, if not properly treated, include fatigue, sleeplessness, withdrawal from activity, increased need to rest, weakened immune system, and changes in mood including hopelessness, fear, depression, irritability, anxiety, and stress, and disability (8). Chronic, nonmalignant pain is predominately neuropathic and involves damage either to the peripheral or central nervous systems (7). The most common sources of chronic pain result from headaches, joint pain, pain from injury, and backaches (9). Other sources of chronic pain include tendinitis, sinus pain, carpal tunnel syndrome, and pain affecting specific parts of the body, such as the shoulders, pelvis, and neck. Generalized muscle or nerve pain can also develop into a chronic condition. The symptoms of chronic pain include mild to severe pain that does not go away and pain that can be described as shooting, burning, aching, feeling of discomfort, soreness, tightness, or stiffness (9).

Chronic pain may originate with an initial trauma or injury such as a sprained back, serious infection, or head injury, or there may be an ongoing cause of pain, such as arthritis or cancer, but some people suffer chronic pain in the absence of any past injury or

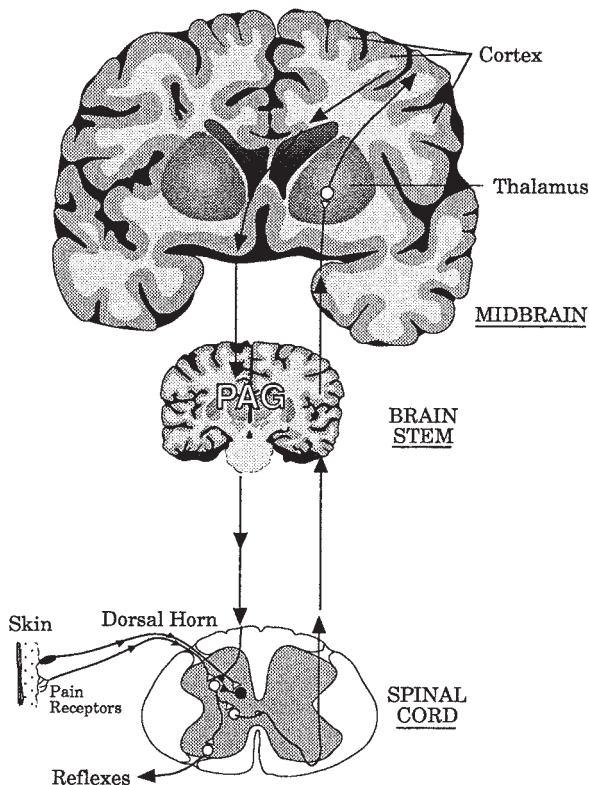


FIGURE 20.1 Location of endogenous opioid nerve tracts in the central nervous system. Endorphins and opioid receptors in the dorsal horn of the spinal cord, thalamus, and periaqueductal gray (PAG) areas are associated with the transmission of pain signals.

evidence of body damage (7). Common chronic pain complaints in older adults include headache, low back pain, cancer pain, arthritis pain, neurogenic pain (pain resulting from damage to the peripheral nerves or to the CNS itself), and psychogenic pain (pain not due to past disease or injury or any visible sign of damage inside or outside the nervous system) (10). A person may have two or more coexisting chronic pain conditions, such as chronic fatigue syndrome, endometriosis, fibromyalgia, inflammatory bowel disease, interstitial cystitis, or temporomandibular joint dysfunction. It is not known whether these disorders share a common cause. Medications and acupuncture are some treatments for chronic pain, which could include placebos, as well as using psychotherapy, relaxation, and behavioral modification.

Neuropathic pain is defined as pain arising as a direct consequence of a lesion or disease affecting the somatosensory system resulting in long-lasting pain (8) (to be discussed later in this chapter). Neuropathic pain is not one single disease but consists of a series of different diseases and conditions, such as nerve compression as a consequence of neoplasms, neuropathies due to metabolic disorders such as diabetes, and diseases of the CNS such as stroke and multiple sclerosis, that produce a common syndrome characterized by pain (10). An essential element in neuropathic pain is the combination of peripheral sensory loss and hypersensitivity in the painful area as consequence of an injury to the nervous system. As a result of damage to either C-fibers or A δ -fibers, these fibers, in response to the injury, become abnormally sensitive due to a combination of degeneration and

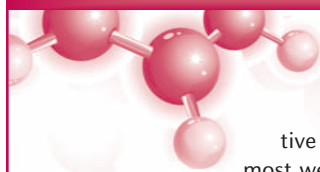
regeneration cellular changes with hypersensitization/hyperexcitability as the main factor causing the pain (8). Peripheral neuropathic pain is often described as a burning, tingling sensation, stabbing, or “pins and needles” (10). Phantom pain is a type of neuropathic pain from a part of the body that has been lost or from which the brain no longer receives signals (e.g., phantom limb pain is a common experience of amputees). Insensitivity to pain may also result from abnormalities in the nervous system as a result of damage to the nerves, such as spinal cord injury, or diabetes mellitus (diabetic neuropathy). These individuals are at risk of tissue damage due to undiscovered injury. People with diabetes-related nerve damage, for instance, sustain poorly healing foot ulcers as a result of decreased sensation (9).

OPIUM AND OPIOID ANALGESICS

History

The juice (*opium* in Greek) or latex from the unripe seed pods of the poppy *Papaver somniferum* is among the oldest recorded medications used by humans. The writings of Theophrastus around 200 BC describe the use of opium in medicine; however, evidence suggests that opium was used in the Sumerian culture as early as 3500 BC. The initial use of opium was as a tonic, or it was smoked. The pharmacist Surtürner first isolated an alkaloid from opium in 1803. He named the alkaloid morphine, after Morpheus, the Greek god of dreams. Codeine, thebaine, and papaverine are other medically important alkaloids that were later isolated from the latex of opium poppies.

CLINICAL SIGNIFICANCE



Central analgesics work to blunt painful stimuli operating through physiologically distinct nociceptive or neuropathic pathways. The most well-recognized centrally acting pain relievers are the opioid analgesics, which exert their agonist and/or antagonist action at defined receptors in the brain and spinal cord. Opioid agonists and partial agonist-antagonists generally act on μ and κ receptors to effect analgesia, and their structure–activity relationship (SAR) in the treatment of nociceptive pain is well defined. All of these receptors have subtypes that provide varying degrees of analgesia, euphoria or dysphoria, CNS depression, and perhaps the potential for tolerance. By modifying their structures, opioid agonist and antagonist properties can be changed to develop agents that require more or less hepatic metabolism, and thus, the duration of action, bioavailability, and risk of serious drug–drug interactions can be impacted. A simple change in nitrogen substituent can convert a potent μ agonist into a κ agonist/ μ antagonist,

an analgesic that would not be cross-tolerant with the original drug. Other changes in the chemical structures can yield agents with the same receptor selectivity profile but a much higher affinity, which corresponds to more potency on a milligram-to-milligram basis. Other alterations of the chemical structures can lead to improved central distribution profiles and/or positive changes related to respiratory depression liability, emesis, tolerance, and allergenicity. By altering the affinities for some receptors more than others, the addictive properties may also be manipulated.

Through an understanding of the relationship of chemical structures to biologic activity, the clinician can improve the selection of drug to the specific patient.

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Morphine was among the first compounds to undergo structure modification. Ethylmorphine (the 3-*O*-ethyl ether of morphine) was introduced as a medicine in 1898. Diacetylmorphine (heroin), which may be considered to be the first synthetic prodrug, was synthesized in 1874 and marketed as a nonaddicting analgesic, antidiarrheal, and antitussive agent in 1898.

OPIATE/OPIOID

The use of the terms “opiate” and “opioid” requires clarification. Until the 1980s, the term “opiate” was used extensively to describe any natural or synthetic agent that was derived from morphine. One could say an opiate was any compound that was structurally related to morphine. In the mid-1970s, the discovery of peptides in the brain with pharmacologic actions similar to morphine prompted a change in nomenclature. These peptides were not easily related to morphine structurally, yet their actions were like those produced by morphine. At this time, the term “opioid,” meaning opium- or morphine-like in terms of pharmacologic action, was introduced. The broad group of opium alkaloids, synthetic derivatives related to the opium alkaloids, and the many naturally occurring and synthetic peptides with morphine-like pharmacologic effects are called opioids. In addition to having pharmacologic effects similar to morphine, a compound must be antagonized by an opioid antagonist, such as naloxone, to be classed as an opioid. The neuronal-located proteins to which opioid agents bind and initiate biologic responses are called opioid receptors.

Endogenous Opioid Peptides and Their Physiologic Functions

Scientists had postulated for some time that based on the rigid structural and stereochemical structure–activity requirements essential for the analgesic actions of morphine and related opioids, they produce their effects by binding with specific receptors to cause their actions. It was also reasoned that morphine and the synthetic opioid derivatives are not the natural ligands for the opioid receptors and that some analgesic substance must exist within the brain. Techniques to prove these two points were not developed until the mid-1970s. Hughes et al. (11) used the electrically stimulated contractions of guinea pig ileum and the mouse vas deferens, which are very sensitive to inhibition by opioids, as bioassays to follow the purification of compounds with morphine-like activity from mammalian brain tissue. These researchers were able to isolate and determine the structures of two pentapeptides, Tyr-Gly-Gly-Phe-Met (Met-enkephalin) and Tyr-Gly-Gly-Phe-Leu (Leu-enkephalin), that caused the opioid activity (Fig. 20.2). The compounds were named enkephalins after the Greek word *Kaphale*, which translates as “from the head.”

At about the same time as Hughes and coworkers were making their discoveries, three other laboratories, using

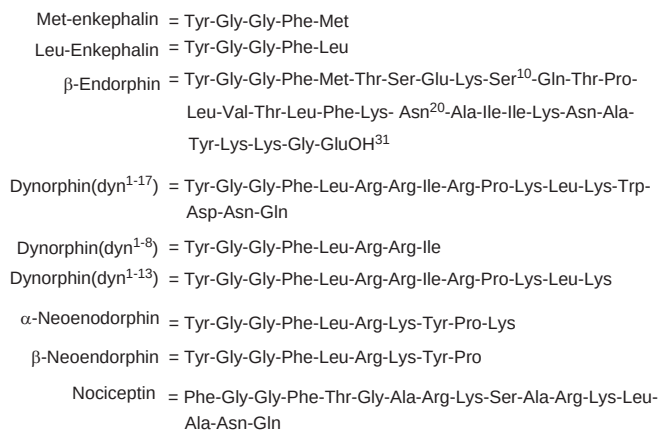


FIGURE 20.2 Precursor proteins to the endogenous opioid peptides.

a different assay technique, were able to identify endogenous opioids and opioid receptors in the brain (12–14). These scientists used radiolabeled opioid compounds (radioligands), with high specific activity, to bind to opioid receptors in brain homogenates (15). They demonstrated saturable binding (i.e., the tissue contains a finite number of binding sites that can all be occupied) of the radioligands and that the receptor-bound radioligands could be displaced stereoselectively by nonradiolabeled opioids. Discovery of the enkephalins was soon followed by the identification of other endogenous opioid peptides, including β -endorphin (16), the dynorphins (17), and the endomorphins (18).

The opioid peptides isolated from mammalian tissue are known collectively as endorphins, a word that is derived from a combination of endogenous and morphine. The opioid alkaloids and all of the synthetic opioid derivatives are exogenous opioids.

Opioid Peptides

The endogenous opioid peptides are synthesized as part of the structures of large precursor proteins (19). There is a different precursor protein for each of the major types of opioid peptides (Fig. 20.2). Pro-opiomelanocortin is the precursor for β -endorphin. Proenkephalin A is the precursor for Met- and Leu-enkephalin. Proenkephalin B (prodynorphin) is the precursor for dynorphin and α -neoenkephalin. The pronociceptin protein has been identified and contains only one copy of the active peptide, whereas the precursor protein for the endomorphins remains to be identified. All of the pro-opioid proteins are synthesized in the cell nucleus and transported to the terminals of the nerve cells from which they are released. The active peptides are hydrolyzed from the large proteins by proteases that recognize double basic amino acid sequences positioned just before and after the opioid peptide sequences.

Peptides with opioid activity have been isolated from sources other than mammalian brain. They include heptapeptide β -casomorphin (Tyr-Pro-Phe-Pro-Gly-Pro-Ile),

which is found in cow's milk and is a μ -opioid agonist (20), and dermorphin (Tyr-D-Ala-Phe-Gly-Tyr-Pro-Ser-NH₂), which is a μ -selective peptide isolated from the skin of South American frogs and is approximately 100-fold more potent than morphine in *in vitro* tests (21).

The endogenous opioids exert their analgesic action at spinal and supraspinal sites (Fig. 20.1). They also produce analgesia by a peripheral mechanism of action associated with the inflammatory process. In the CNS, the opioids exert an inhibitory neurotransmitter or neuromodulator action on afferent pain-signaling neurons in the dorsal horn of the spinal cord and on interconnecting neuronal pathways for pain signals within the brain. In the brain, the arcuate nucleus, periaqueductal gray, and thalamic areas are especially rich in opioid receptors and are sites at which opioids exert an analgesic action. In the spinal cord, concentrations of endogenous opioids are high in laminae 1, laminae 2, and trigeminal nucleus areas. All of the endogenous opioid peptides and the three major classes of opioid receptors appear to be at least partially involved in the modulation of pain. The actions of opioids at the synaptic level are described in Figure 20.3.

Analgesia that results from acupuncture or is self-induced by a placebo or biofeedback mechanisms is caused by release of endogenous endorphins. Analgesia produced by these procedures can be prevented by the previous dosage of a patient with an opioid antagonist. Electrical stimulation from electrodes properly placed in the brain causes endorphin release and analgesia. This procedure is used for the "self-stimulated" release of endorphins in patients with chronic pain who do not respond to

any other medical treatment. As with exogenously administered opioid drugs, tolerance develops to all procedures that work by release of endogenous opioids.

Structure–Activity Relationships of Opioid Peptides

Thousands of derivatives related to the endogenous opioid peptides have been prepared since the discovery of the enkephalins in 1975 (22) (Fig. 20.2). A thorough discussion of the SAR of these peptides would be a major task; however, some major trends have emerged and can easily be discussed. Some selected general SAR points for peptide opioids are as follows:

1. All of the endogenous opioid peptides, except for the endomorphins, have Leu- or Met-enkephalin as their first five amino acid residues.
2. The Tyr¹ at the first amino acid residue position of all the endogenous opioid peptides is essential for activity. Removal of the phenolic hydroxyl group or the basic nitrogen (amino terminus group) will abolish activity. The Tyr¹ free amino group may be alkylated (methyl or allyl groups to give agonists and antagonists), but it must retain its basic character. The structural resemblance between morphine and the Tyr¹ group of opioid peptides is especially obvious.
3. In addition to the phenol and amine groups of Tyr¹, the next most important moiety in the enkephalin structure is the phenyl group of Phe⁴. Removal of this group or changing its distance from Tyr¹ results in full or substantial loss in activity.

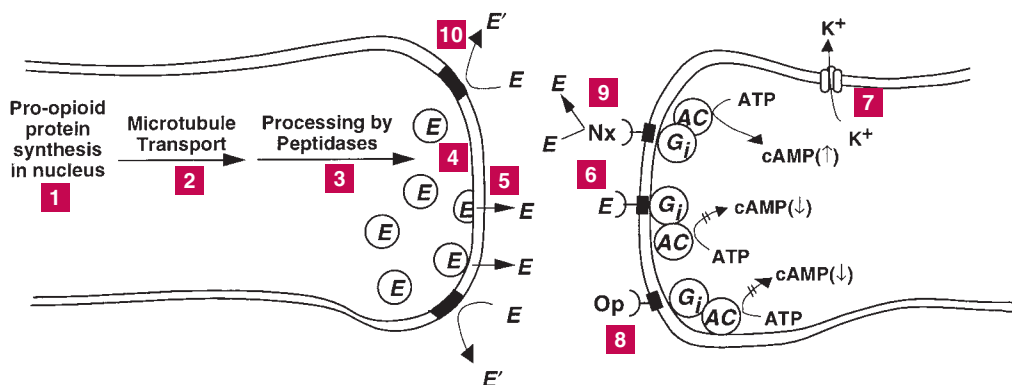


FIGURE 20.3 Schematic representation of a δ enkephalinergic nerve terminal. (1) Pro-opioid proteins (proenkephalin A) are synthesized in the cell nucleus. (2) Pro-opioid proteins undergo microtubular transport to the nerve terminal. (3) Active endogenous opioids (E) are cleaved from the pro-opioid proteins by the action of "processing" proteases. (4) The active peptides (E) are taken up and stored in presynaptic vesicles. (5) The peptides are released when the presynaptic neuron fires. (6) The endogenous opioid peptides bind to postsynaptic receptors and activate second messenger systems. (7) For all opioid receptors, the second messenger effect is primarily mediated by a G-inhibitory ($G_{i/o}$) protein complex, which promotes the inactivation of adenylyl cyclase (AC), a decrease in intracellular cyclic adenosine-3',5'-monophosphate (cAMP), and finally, an efflux of potassium ions (K^+) from the cell. The net effect is the hyperpolarization of the postsynaptic neuron and inhibition of cell firing. (8) Exogenous opioids (Op), such as morphine, combine with opioid receptors and mimic the actions of E. (9) Opioid antagonists, such as naloxone (Nx), combine with opioid receptors and competitively inhibit the actions of E or Op. (10) The action of E is terminated by a membrane-bound endopeptidase [EC3.4.24.11] (enkephalinase), which hydrolyzes the Gly³-Phe⁴ peptide bond of enkephalin. Other endopeptidases may be employed in the metabolism of different endogenous opioid peptides.

4. The enkephalins have several low-energy conformations, and different conformations likely are bound at different opioid receptor types and subtypes.
5. The replacement of the natural L-amino acids with unnatural D-amino acids can make the peptides resistant to the actions of several peptidases that generally rapidly degrade the natural endorphins. The use of a D-Ala in place of Gly² has been especially useful for protecting the peptides from the action of nonselective aminopeptidases. The placement of bulky groups into the structure (e.g., the addition of N-Me to Phe⁴) will also slow the action of peptidases. When evaluating new peptides for opioid activity, it often is difficult to tell if changes are caused by metabolic stability or receptor affinity.
6. Conversion of the terminal carboxyl group into an alcohol or an amide will protect the compound from carboxypeptidases.
7. Any introduction of unnatural D- or L-amino acids or bulky groups into the enkephalin structure will affect its conformational stability. The resultant peptides will have an increase or decrease in affinity for each of the opioid receptor types. The right combination of increases and/or decreases in receptor affinity will result in selectivity for a receptor type.

8. Structural changes that highly restrict the conformational mobility of the peptides (e.g., substitution of proline for Gly² or cyclization of the peptide) have been especially useful for the discovery of receptor-selective opioid peptides.

For examples of the above SARs, see the structures of the peptides given in Figures 20.4 to 20.6.

ENKEPHALIN PEPTIDES The effect of lengthening the amino acid chain of the enkephalin peptides deserves special consideration. As previously noted, the endogenous opioids found in mammals most often have Leu- or Met-enkephalin at their amino terminus end. Lengthening the carboxyl terminus can give the peptide greater affinity or selectivity for an opioid receptor type. This effect can be illustrated by the dynorphins, for which incorporation of the basic amino acids (especially Arg⁷) into the C-terminus chain results in a marked increase in affinity for κ receptors. The message-address analogy has been used to describe this effect. The first four amino acids (Tyr-Gly-Gly-Phe) are essential for peptide ligands to bind to and to activate all opioid receptor types. The N-terminus amino acids can then be referred to as carrying the “message” to the receptors. Adding additional amino acids to the C-terminus can “address” the message to a specific receptor type. The additional peptide chain may be affecting the address (selectivity) by providing new and favorable binding interactions to one of the

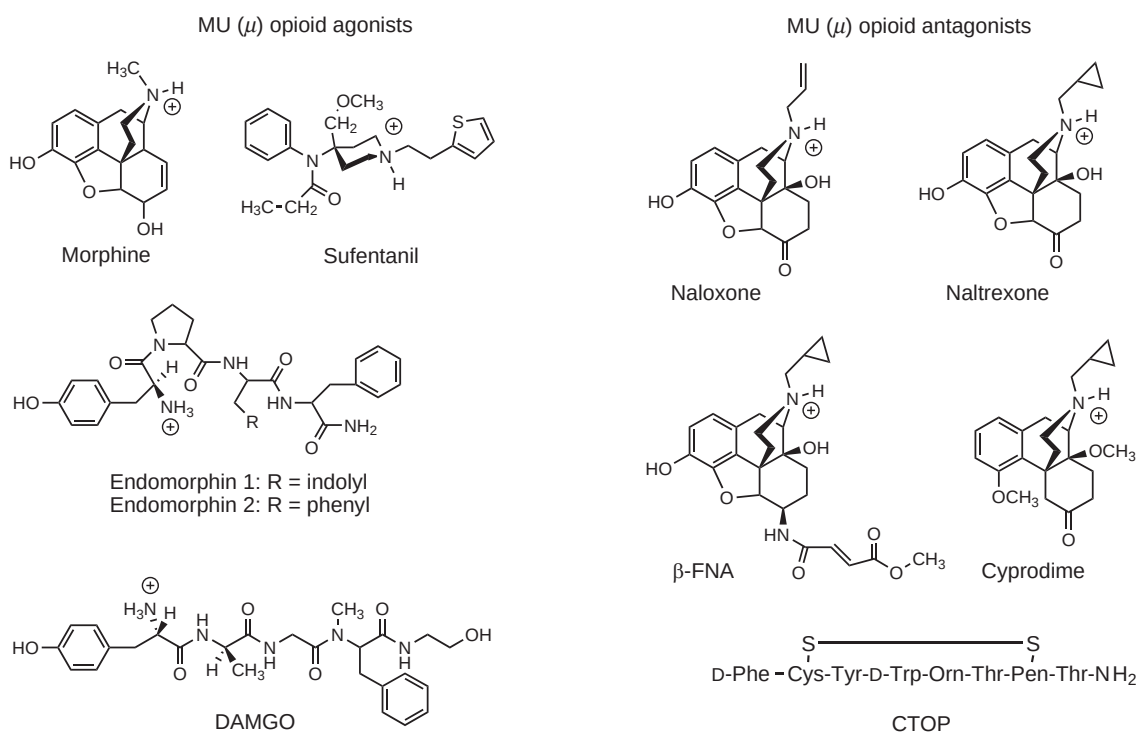


FIGURE 20.4 Structures of compounds selective for μ (OP_3) opioid receptors.

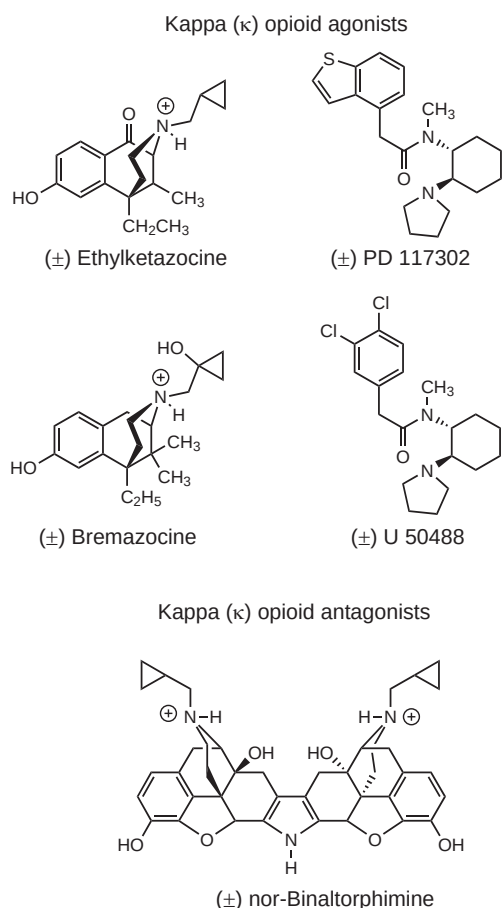


FIGURE 20.5 Structures of compounds selective for κ (OP_2) opioid receptors. (–)-Stereoisomers are the most active compounds.

receptor types. Alternatively, the additional peptide could be inducing a conformational change in the message portion of the peptide that favors interaction with one of the receptor types.

Opioid Receptors

Identification and Activation of Opioid Receptors

Identification of multiple opioid receptors has depended on the discovery of selective agonists and antagonists, the identification of sensitive assay techniques (23), and ultimately, the cloning of the receptor proteins (24). The techniques that have been especially useful are the radioligand binding assays on brain tissues and the electrically stimulated peripheral muscle preparations. Rodent brain tissue contains all three opioid receptor types, and special evaluation procedures (computer-assisted line fitting) or selective blocking (with reversible or irreversible binding agents) of some of the receptor types must be used to sort out the receptor selectivity of test compounds. The myenteric plexus-containing longitudinal strips of guinea pig ileum contain μ - and κ -opioid receptors. The contraction of these muscle strips is initiated by electrical stimulation and is inhibited by opioids. The vas deferens from mouse contains μ , δ , and κ receptors and reacts similarly to the guinea pig ileum to electrical stimulation and to opioids. Homogenous populations of opioid receptors are found in rat (μ), hamster (δ), and rabbit (κ) vas deferentia.

The signal transduction mechanism for μ , δ , and κ receptors is through $G_{i/o}$ proteins. Activation of opioid receptors is linked through the G protein to an inhibition of adenylate cyclase activity. The resultant decrease in cyclic adenosine monophosphate (cAMP) production, efflux of potassium ions, and closure of voltage-gated Ca^{2+} channels causes hyperpolarization of the nerve cell (25) and a strong inhibition of nerve firing.

It is now clear from work carried out in many laboratories over the last three decades that there are three major types of opioid receptors: μ , κ , and δ (26). All three of the receptor types have been well characterized and cloned (24). A nomenclature adopted by the International Union of Pharmacology (IUPHAR) in 1996 classifies the three opioid receptors by the order in which they were

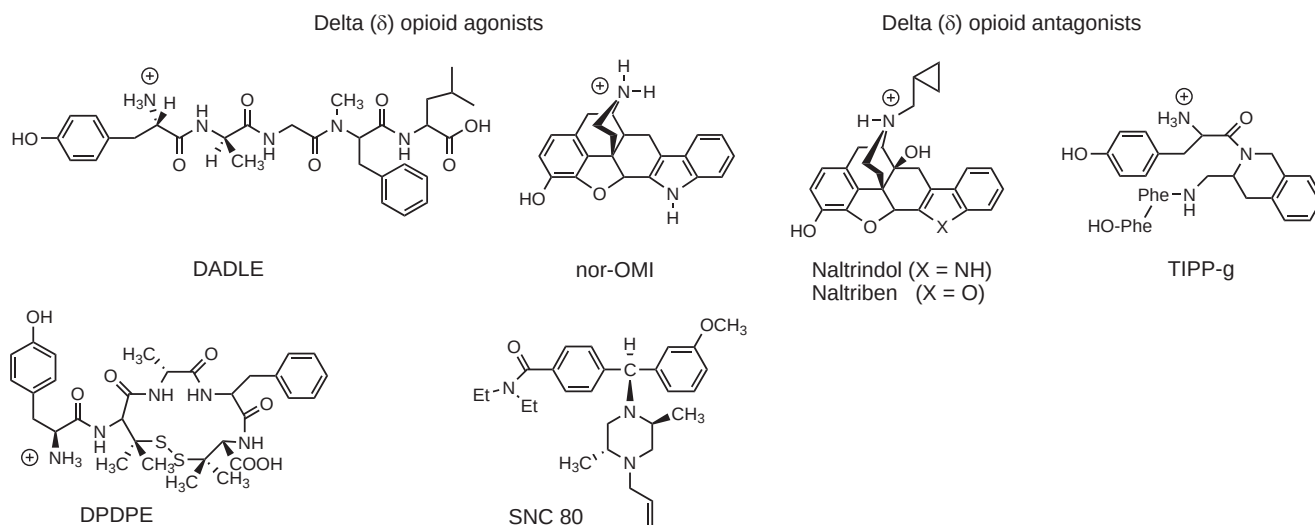


FIGURE 20.6 Structures of compounds selective for δ (OP_1) opioid receptors.

cloned (27). By this classification, δ -opioid receptors are OP_1 receptors, κ -opioid receptors are OP_2 receptors, and μ -opioid receptors are OP_3 receptors. The IUPHAR approved a new nomenclature in 2000, naming the receptors as MOP- μ , DOP- δ , and KOP- κ . In current literature, however, the opioid receptors often are referred to as DOR (δ), KOR (κ), and MOR (μ). There is evidence for subtypes of each of these receptors; however, the failure of researchers to find genomic evidence for additional receptors indicates that the receptor subtypes are posttranslational modifications (splice variants) of known receptor types (28). Receptor subtypes may also be known receptor types that are coupled to different signal transduction

systems. A fourth opioid receptor family (NOP, nociception receptor; formerly opioid orphan receptor) was identified from cDNA encoding, which has a high degree of homology to the classical opioid receptors, and named after its endogenous ligand, nociceptin/orphanin FQ (N/OFQ) peptide receptor (29,30). Table 20.1 lists the opioid receptor types, their known physiologic functions, and selective agonists and antagonists for each of the receptors. All three of the major opioid receptor types are located in human brain or spinal cord tissues, and each has a role in the mediation of pain. At this time, only μ and κ agonists are in clinical use as opioid analgesic drugs.

TABLE 20.1 Opioid Receptors

Receptor Subtype	δ (delta, DOP, OP_1)	K (kappa, KOP, OP_2)	μ (mu, MOP, OP_3)	NOP (N/OFQ peptide, OP_4 , ORL ₁)
Transduction Mechanism	Inhibition of adenylate cyclase, activation of inwardly rectifying K ⁺ channels, inhibition of Ca ²⁺ channels, phospholipase C stimulation			
G protein	G _{i/o}	G _{i/o}	G _{i/o}	G _{i/o}
Localization	Olfactory bulb Nucleus accumbens Caudate putamen neocortex	Cerebral cortex Nucleus accumbens hypothalamus	CNS (neocortex, thalamus, nucleus accumbens, hippocampus, amygdala) Vas deferens Myenteric gut neurons	Cortex, olfactory nucleus, hypothalamus, hippocampus, substantia nigra, locus coeruleus, spinal cord, amygdala,
Endogenous Ligand	Enkephalins β -endorphin	Dynorphins β -endorphin	Endomorphin 1 Endomorphin 2 β -endorphin)	N/OFQ peptide
Likely physiological roles	Analgesia GI motility Olfaction Immune stimulation Respiratory depression (rate) Cognitive function Motor integration	Regulation of nociception Analgesia Sedation Miosis Diuresis Dysphoria Neuroendocrine secretions	Analgesia (morphine-like) Euphoria Increased gastrointestinal transit time thermo regulation Immune suppression Respiratory depression (volume) Emetic effects Tolerance Physical dependence	Motor and balance control Reinforcement and reward Nociception Stress response Sexual behavior Aggression Autonomic control of physiological processes
Key selective agonists	DADLE (D-Ala ² -D-Leu ⁵ -enkephalin) DSLET (Tyr-D-Ser-Gly-Phe-Leu-Thr) DPDPE (D-Pen ² -D-Pen ⁵ -DADLE (δ_2) D-Ala ² -deltorphin II (δ_2)	Ethylketocyclazocine (EKC) Bremazocine Mr2034 dyn (1-17) Trifluadom U-50,488 (κ_1) Spiradolone (U-62,066)(κ_1) U-69,593 (κ_1) PD 117302 (κ_1) dyn 1-17 (κ_1) NalBzOH (κ_1)	Morphine Sufentanil DAMGO (Tyr-D-Ala-MePhe-NH-(CH ₂) ₂ -OH) PLO17 (Tyr-Pro-MePhe-D-Pro-NH ₂) BIT (affinity label) Meptazinol (μ_1 high affinity) Etonitazene	N/OFQ peptide Nociceptin (1-13)-NH ₂
Key Selective Antagonists	ICI 174864 FIT (affinity label) SUPERFIT (affinity label) Naltrindole (NTI) BNTX Naltriben (NTB) Naltrindol isothiocyanate (NTII)	TENA nor-BNI UPHIT	Naloxone Naltrexone CTOP Cyprodime β -FNA (affinity label) Naloxonazine	(N-phe ¹)-Nociceptin (1-13)-NH ₂ J-113397 UFP-101

Mu (μ)-Opioid Receptors

Endomorphin-1 (Tyr¹-Pro-Trp-Phe⁴-NH₂) and endomorphin-2 (Tyr¹-Pro-Phe-Phe⁴-NH₂) are endogenous opioid peptides with a high degree of selectivity for μ (MOP) receptors (18). A number of therapeutically useful compounds have been found that are selective for μ -opioid receptors (Fig. 20.4). All of the opioid alkaloids and most of their synthetic derivatives are μ -selective agonists. Morphine, normorphine, and dihydromorphine have 10- to 20-fold μ -receptor selectivity and were particularly important in early studies to differentiate the opioid receptors. Sufentanil and the peptides DAMGO (31) and dermorphin (32), all with 100-fold selectivity for μ over other opioid receptors, are frequently used in laboratory studies to demonstrate μ -receptor selectivity in cross-tolerance, receptor binding, and isolated smooth muscle assays. Studies with μ -receptor knockout mice have confirmed that all the major pharmacologic actions observed on injection of morphine (e.g., analgesia, respiratory depression, tolerance, withdrawal symptoms, decreased gastric motility, and emesis) occur by interactions with μ receptors (33).

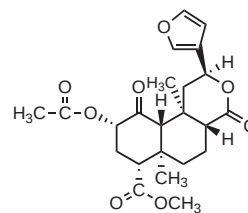
Naloxone and naltrexone are antagonists that have weak (5- to 10-fold) selectivity for μ receptors. Cyprodime is a selective nonpeptide μ antagonist (~30-fold selective for μ over κ and 100-fold selective for μ over δ) available for laboratory use (34). CTOP, a cyclic peptide analog of somatostatin, is a selective μ antagonist (35). There is evidence that μ_1 receptors are high-affinity binding sites that mediate pain neurotransmission, whereas μ_2 receptors control respiratory depression. Naloxonazine is a selective inhibitor of μ_1 -opioid receptors (36).

Kappa (κ)-Opioid Receptors

Ethylketazocine and bremazocine are 6,7-benzomorphan derivatives with κ -opioid receptor selectivity (Fig. 20.5). These two compounds were used in early studies to investigate κ (KOP) receptors. They are not highly selective, however, and their use in research has diminished. A number of arylacetamide derivatives, having a high selectivity for κ over μ or δ receptors, have been discovered. The first of these compounds, ($\pm$)-U50488, has a 50-fold selectivity for κ over μ receptors and has been extremely important in the characterization of κ -opioid activity (37). Other important agents in this class are ($\pm$) PD-117302 (38) and (–) CI-977 (39). Each of these agents has 1,000-fold selectivity for κ over μ or δ receptors. Evidence suggests that the arylacetamides bind to a subtype of κ receptors. In animals, including humans, κ agonists produce analgesia. Other prominent effects are diuresis, sedation, and dysphoria. Compared to μ agonists, κ agonists lack respiratory depressant, constipating, and strong addictive (euphoria and physical dependence) properties. It was hoped that κ agonists would become useful strong analgesics

that lacked addictive properties; however, clinical trials with several highly selective and potent κ agonists were aborted because of the occurrence of unacceptable sedative and dysphoric side effects. κ -Selective opioids with only a peripheral action have been shown to be effective in relieving inflammation and the pain associated with it (40). Scientific evidence suggests there are κ_1 , κ_2 , and κ_3 subtypes of κ receptors; however, the physiologic effects initiated by the κ receptor subtypes are not well defined (41).

The peptides related to dynorphin are the natural agonists for κ receptors. Their selectivity for κ over μ receptors is not very high. Synthetic peptide analogs have been reported that are more potent and more selective than dynorphin for κ receptors (42,43).



Salvinorin

The diterpene salvinorin A from *Salvia divinorum* has been reported to be a potent high-affinity and selective κ agonist (44,45). *Salvia divinorum* is a hallucinogenic plant in the mint family that has been used in traditional spiritual practices for its psychoactive properties by the Mazatecs of Oaxaca, Mexico, and its extracts have been used in the United States as legal hallucinogens. Salvinorin A thus represents the first known naturally occurring nonnitrogenous full κ agonist similar in efficacy to dynorphin, the naturally occurring peptide ligand for κ receptors. Results from structure–function studies suggested that the substituent at the 2-position of salvinorin A was critical for κ -opioid receptor binding and activation (45).

The major antagonist with good selectivity for κ receptors is norbinaltorphimine (46). This compound has approximately 100-fold selectivity for κ over δ receptors and an even greater selectivity for κ over μ receptors when tested during competitive binding studies in monkey brain homogenate. No medical use for a κ antagonist has been found.

Delta (δ)-Opioid Receptors

Enkephalins, the natural ligands at δ (DOP) receptors, are only slightly selective for δ over μ receptors. Changes in the amino acid composition of the enkephalins can give compounds with high potency and selectivity for δ receptors. The peptides most often used as selective δ receptor ligands (Fig. 20.6) are [D-Ala², D-Leu⁵] enkephalin (DADLE) (47), [D-Ser², Leu⁵] enkephalin-Thr (DSLET) (48), and the cyclic peptide [D-Pen²,

D-Pen⁵] enkephalin (DPDPE) (49). These and other δ -receptor selective peptides have been useful for in vitro studies, but their metabolic instability and poor distribution properties (i.e., penetration of the blood–brain barrier is limited by their hydrophilicity) have limited their usefulness for in vivo studies. Nonpeptide agonists that are selective for δ receptors have been reported. Derivatives of morphindoles were the first nonpeptide molecules to show δ selectivity in in vitro assays (50). SNC-80 is a newer and more selective δ -opioid receptor agonist (51). This compound produces analgesia after oral dose in several rodent models, and side effects appear minimal. Clinical trials with SCN-80 and other nonpeptide δ receptor agonists were attempted and aborted, primarily because of the convulsant action of δ receptor agonists. Radioligand binding studies in rodent brain tissue and in electrically stimulated vas deferentia have provided evidence of δ_1 and δ_2 receptors (52). The functional significance of this differentiation has not been determined.

Naltrindol and naltriben are highly selective nonpeptide antagonist for δ receptors (53,54). Naltrindol penetrates the CNS and displays antagonist activity that is selective for δ receptors in in vitro and in vivo systems. Peptidyl antagonists TIPP and TIPP- ψ are selective for δ receptors (55,56); however, their usefulness for in vivo studies and as clinical agents is limited by their poor pharmacokinetic properties. The δ -opioid receptor antagonists have shown clinical potential as immunosuppressants and in treatment of cocaine abuse.

NOP Receptor

A fourth opioid receptor named after its endogenous ligand, N/OFQ peptide (NOP; OP_4) has been identified and cloned based on homology with the cDNA sequence of the known (μ , δ , and κ) opioid receptors (29,30). Despite the homology in cDNA sequence with known opioid receptors, NOP did not bind the classical opioid peptide or nonpeptide agonists or antagonists with high affinity. Thus, the receptor was initially called the orphan opioid receptor or opioid-like receptor (OPLR-1). In subsequent studies, two research groups found a heptadecapeptide (Phe-Gly-Gly-Thr-Gly-Ala-Arg-Lys-Ser-Ala-Lys-Ala-Asn-Gln) to be the endogenous peptide for this receptor and named it nociceptin because it caused hyperalgesia (nociception) after intracerebral ventricular injection in mice (57). Another research group (58) named the heptapeptide orphanin FQ, after its affinity for the “orphan opioid receptor,” after the first and last amino acids in the peptide’s sequence (i.e., F = Phe and Q = Gln). Thus, the OPLR-1 receptor was renamed the NOP or N/OFQ peptide receptor, and the current name for the endogenous ligand is N/OFQ peptide. Human N/OFQ peptide is derived from a precursor protein,

preproN/OFQ (ppN/OFQ), which consists of 176 amino acids, and resembles dynorphin-A in structure, with the most notable difference being the replacement of the N-terminus Tyr¹ for dynorphin A with Phe¹ (58). Conflicting results have been published regarding the ability of N/OFQ peptide to produce hyperalgesia versus analgesia in rodent pain assay models. One study has established this compound to be a potent initiator of pain signals in the periphery, where it acts by releasing substance P from nerve terminals (59). N/OFQ peptide is thought to be an endogenous antagonist of dopamine transport that may act either directly on dopamine or by inhibiting γ -aminobutyric acid (GABA) to affect dopamine levels (60). Within the CNS, the actions of N/OFQ can be either similar or opposite to those of opioids depending on the location. It controls a wide range of biologic functions ranging from nociception to food intake, from memory processes to cardiovascular and renal functions, from spontaneous locomotor activity to gastrointestinal (GI) motility, and from anxiety to the control of neurotransmitter release at peripheral and central sites (57). Several commonly used opioid drugs, including etorphine and buprenorphine, have been demonstrated to bind to nociceptin receptors, but this binding is relatively insignificant compared to their binding at other opioid receptors. More recently, a range of selective ligands for NOP have been developed that show little or no affinity to other opioid receptors, which allows NOP-mediated responses to be studied in isolation (9). Injection of an N/OFQ peptide antagonist into the brains of laboratory animals results in an analgesic effect, raising hope for the use of these agents in the management of pain (57). The NOP receptors are widely distributed in the brain, and it is not surprising that many central actions of N/OFQ peptide have been suggested from animal studies, including supraspinal hyperalgesia, spinal analgesia, hyperphagia, depression, and inhibitions of anxiety, epilepsy, cough, motor activity, and learning and memory, as well as the regulation of cardiovascular, urogenital, GI, and immune systems. Many efforts have been endeavored in the development of agonists or antagonists of this novel member of the opioid receptor family. Table 20.1 summarizes the NOP receptor agonists and antagonists developed thus far.

Receptor Affinity Labeling Agents

A number of opioid receptor-selective affinity labeling agents (i.e., compounds that form an irreversible covalent bond with the receptor protein) have been developed (Fig. 20.7). These compounds have been important in the characterization and isolation of the opioid receptor types. Each of the affinity-labeling agents contains a pharmacophore that allows initial reversible binding to the receptor. Once reversibly bound to the receptor, an affinity labeling agent must have an electrophilic group positioned so that

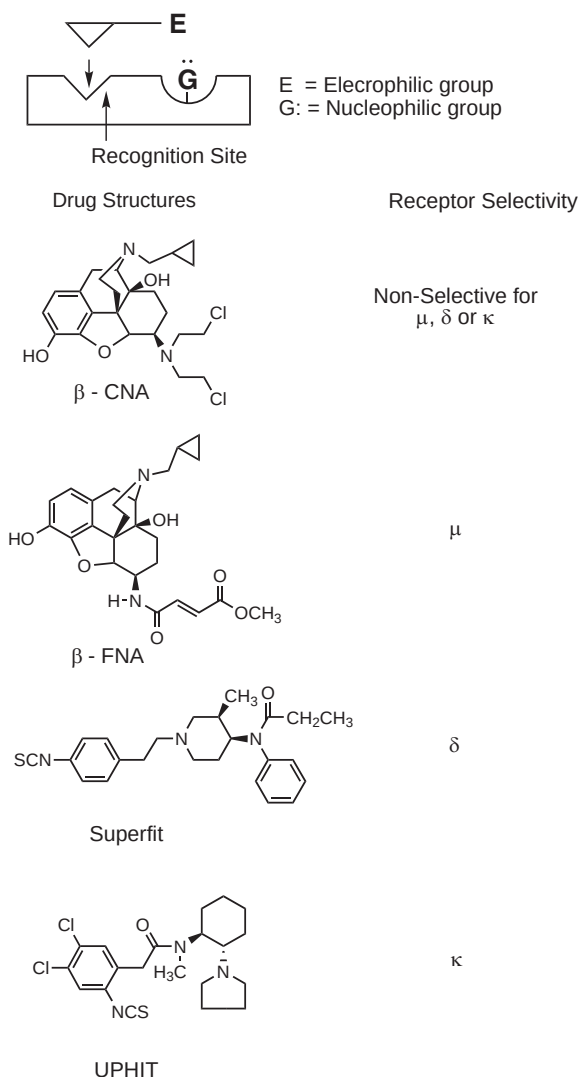


FIGURE 20.7 A representation of the concept of affinity labeling of receptors and affinity labeling agents for opioid receptors.

it can react with a nucleophilic group on the receptor protein. The receptor selectivity of these agents is dependent on 1) the receptor type selectivity of the pharmacophore, 2) the location of the electrophile within the pharmacophore structure so that when bound to the receptor it is positioned near a nucleophile, and 3) the relative reactivities of the electrophilic and nucleophilic groups.

An example of an important affinity labeling agent is β -CNA, which because of its highly reactive 2-chloroethylamine electrophilic group irreversibly binds to all three opioid receptor types (61). The structurally related compound β -FNA has a less reactive fumaramide electrophilic group and reacts irreversibly with only μ receptors (59). Derivatives of the fentanyl series, FIT and SUPERFIT, bind μ and δ receptors, but only the δ receptor is bound irreversibly (62,63).

Apparently, when these agents are bound to μ receptors, the electrophilic isothiocyanate group is not oriented in proper juxtaposition to a receptor nucleophile for covalent bond formation to occur. Incorporation of the electrophilic isothiocyanate into the structure of the highly κ receptor-selective arylacetamides has provided affinity labeling agents (UPHIT and DIPPA) for κ receptors (64,65).

OPIOID THERAPEUTIC CHALLENGES

Tolerance, Dependence, and Withdrawal

Patients who take opioids on a chronic basis will experience tolerance to many of their pharmacologic actions, including analgesia, and a dependence on their presence in the CNS.

Tolerance is the situation where an increasingly larger dose of drug is required to produce the same degree of biologic response that had previously been obtained with a lower dose. In other words, the body has adapted to the presence of drug and needs a bigger dose to be able to experience its effect. Tolerance to μ -opioid agonists is related to the attenuating impact of these agents on intracellular cAMP. As cAMP levels decrease in response to μ -receptor occupation, analgesia is realized. However, over the course of several weeks or longer, upregulation of adenylate cyclase enzymes occurs in an attempt to overcome μ agonist-induced inhibition. It then takes a larger dose of agonist to inhibit both the original population of adenylate cyclase enzymes and those generated in response to the original dose (66). In addition to analgesia, tolerance occurs to μ -opioid-induced euphoria and respiratory depression, but not to the constipating or papillary (miosis) effects of these drugs.

Dependence is the need for a drug to maintain "normal" physiologic and psychological functioning. If a μ -opioid agonist used on a chronic basis is suddenly withdrawn, the inhibition of adenylate cyclase abruptly stops, and the "super-sized" enzyme population begins to produce cAMP. The elevated cAMP concentrations are believed responsible for the physiologic symptoms of opioid withdrawal, which include tremor, alternating chills and sweating, increased heart rate and blood pressure, and abdominal cramping. Withdrawal symptoms last until enzymatic homeostasis is regained, which can take days (67). Psychologically, withdrawal leads to often uncontrollable drug craving and drug-seeking behavior. Drug craving, a learned response, may last a lifetime, and it is the main reason why addicts in recovery relapse.

Cross Tolerance

There are times when practitioners intentionally put patients tolerant to and dependent on opioids into a

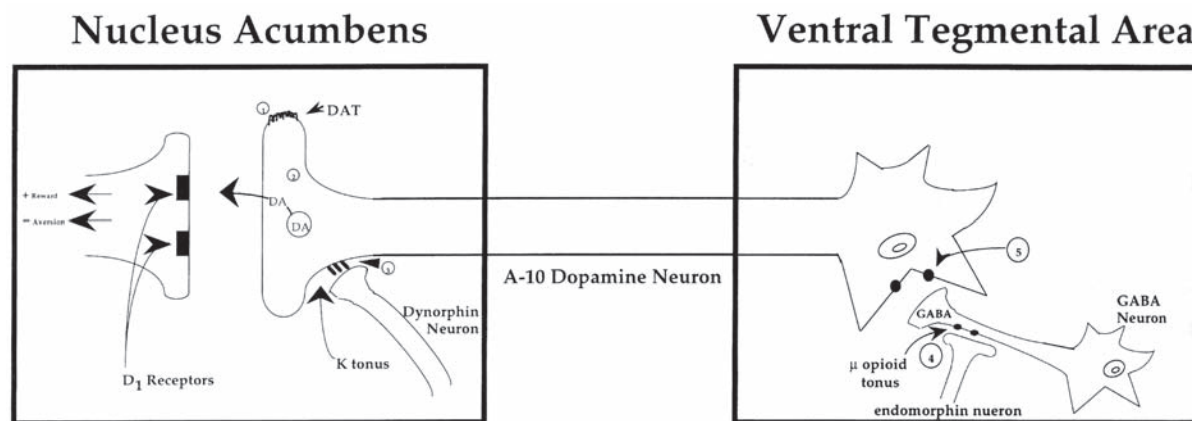


FIGURE 20.8 The neurochemical basis of drug abuse and addiction. The diagram is a representation of the brain's self-reward system. According to this theory, any agent that promotes stimulation of type 1 dopamine (D_1) receptors in the nucleus accumbens (NCA) potentiates self-reward and has the potential to be abused. Major drugs of abuse exert their actions at various sites within the self-reward system to increase dopamine (DA) in the NCA and stimulate D_1 receptors. Site 1: Cocaine blocks DA reuptake by the DA transporter (DAT) and greatly enhances DA action at D_1 . Site 2: Amphetamine, methamphetamine, and related drugs cause DA release with the resultant stimulation of D_1 . Site 3: Opioid κ agonists exert an inhibitory effect on DA neuronal firing, resulting in a decrease in DA release and aversion in animals. Site 4: Opioid μ agonists, such as morphine and heroin, exert an inhibitory action on γ -aminobutyric acid (GABA) interneurons in the DAT, thus removing the GABAergic inhibition on DA neuronal firing. Site 5: GABA agonists, such as gabapentin, enhance DA neuronal firing and DA release, and these agents may be useful in treating or preventing drug abuse and addiction. Other abused agents, such as nicotine and cannabis, also cause an increase in DA release in the NAC, but their exact neuronal connections to the self-reward system are not yet understood.

state of withdrawal, but these are few and far between. Unless you are attempting to rescue a patient from a life-threatening overdose or preparing to help an addicted patient on the road to recovery, withdrawal is an experience patients expect to avoid. Therefore, when abruptly switching a patient from one opioid agonist to another, practitioners must ensure that the two drugs are cross tolerant. Cross tolerance refers to the ability of one opioid agonist to substitute for another and produce the same biologic action without precipitating withdrawal symptoms. Cross-tolerant opioids exert their agonist effects through the same receptor subtype. In other words, one full μ agonist will be cross tolerant with another, but an analgesic that acts at the κ receptor will not be cross tolerant with another that works through the μ receptor. Similarly, a partial μ agonist would not be cross tolerant with a full μ agonist. Fortunately, opioid SARs will be explicit about receptor selectivity and intrinsic activity, so the chemically savvy practitioner will always be able to predict cross tolerance (or lack thereof) from a quick review of opioid structure.

Euphoria and Dysphoria

The ability of μ opioids to induce a profound euphoria is well known (68). Euphoria is a result of G_i -coupled μ -receptor stimulation in the thalamic, periaqueductal

gray, and arcuate nucleus areas of the brain. The opioid-induced decrease in cAMP inhibits the release of the inhibitory neurotransmitter GABA from the ventral tegmental area, which is upstream of the nucleus accumbens. The nucleus accumbens is the brain's self-reward or pleasure center and is activated by dopamine. GABA normally inhibits dopamine release from this cerebral "candy land," so the decrease in GABA levels induced by μ -receptor stimulation enhances dopamine release from these terminals. The subsequent elevation in dopamine receptor stimulation in the nucleus accumbens is perceived as a feeling of intense well-being or euphoria.

In contrast, κ agonists have the potential to induce dysphoria, a feeling of intense discomfort. Kappa agonists act directly on receptors in the nucleus accumbens. Stimulation of these receptors results in a direct inhibitory effect on presynaptic dopaminergic neurons, resulting in a decrease in dopamine and an attenuation of pleasure-related responses that translates as dysphoria (69).

Addiction

Because of the euphoric effects that μ -opioid agonists produce, these agents have the potential for abuse. The destructive impact that drug abuse can have on individuals and society as a whole is undeniable, and the use of opioids, both prescribed and otherwise

acquired, is on the rise. For example, despite representing only 4.6% of the global population, U.S. citizens are responsible for the consumption of 80% of opioids used around the world, including 99% of all available hydrocodone and two-thirds of all illicit drugs. In the decade spanning 1997 to 2007, the rise in the legal sale of euphoric opioids ranged from 222% for morphine to 1,293% for methadone, the latter of which is used in opioid addiction recovery as well as in the management of pain (70). Fortunately, patients using opioids for the treatment of pain are much less likely to experience euphoria or become addicted to these drugs than patients who do not need them for pain control (71). Nevertheless, addiction potential is viewed as a major use-limiting side effect by some health care providers, and patients, particularly those from underrepresented minority populations, can be undermedicated because of it (67,72–75). Fortunately, health care professionals are now recognizing the high importance of adequate pain control to quality of life and are taking steps to use opioids in more scientifically sound ways in patients who need them. The important role of the pharmacist in clinically competent pain management cannot be overestimated.

The above notwithstanding, opioid addiction is a serious social and health-related issue that merits additional discussion here. Addiction has been defined as “a primary, chronic neurobiologic disease with genetic, psychosocial and environmental factors influencing its development and manifestations,” and it involves a complex set of physiologic and psychological responses to a drug’s chemical message (76). The risk of addiction in chronic pain patients on opioids has been estimated at 3.3%, with genetic predisposition possibly accounting for 40% to 60% of that risk (77). Those interested in learning more about addiction as a disease state should consider viewing the program “Pleasure Unwoven,” which was recently produced by the Institute for Addiction Study (78).

Although tolerance and dependence are components of addiction, these two phenomena by themselves are not sufficient to define it. As alluded to earlier, true addiction has a strong psychosocial and genetic component, leading to drug craving and drug-seeking behavior in the absence of pain. Individuals must engage in destructive behavior patterns before they can be classified as addicted. Such destructive behaviors can include intentional overuse or compulsive use of opioids in the face of negative mental and/or social consequences, forging prescriptions, social isolation, and abuse of other psychoactive drugs. The “4Cs” of true addiction have been coined, and they include Compulsive use, inability to Control the quantity taken, drug Craving, and Continued use despite negative consequences (77).

Patients taking opioids for chronic pain will be tolerant to and dependent on them, but practitioners

should take care how they use the words “addiction” and “addict.” Fear of being looked upon or labeled as an addict may keep some patients who need pain relief from taking the drugs that could best provide it (67,74). Clinicians should also take care to distinguish true addiction from pseudoaddiction, where patients with undertreated pain seek additional doses or “clock watch” until the time for their next dose of opioid. Unlike true addiction, the drug-seeking behavior of pseudoaddicted patients disappears when their pain is relieved (67,74).

Clinicians should also recognize that opioids used on a chronic basis can sometimes prompt hyperalgesia, a situation characterized by new pain, exacerbation of existing pain, or experiencing pain from nonpainful events (e.g., a light touch) (67,79,80). Hyperalgesia has been noted in patients with cancer, chronic nonmalignant pain, and neuropathic pain. Patients with hyperalgesia experience pain relief when their dose of opioid is decreased, rather than increased. When decreasing the dose of opioid in a hyperalgesic patient, care must be taken to avoid precipitating withdrawal. Clinicians should consider rotating to a different opioid that is cross tolerant with the original drug to re-establish optimum pain control (81). *N*-methyl-D-aspartate (NMDA) receptor antagonists may also have a therapeutic role to play, as the hyperalgesic response is believed to be mediated, at least in part, by the NMDA neurotransmitter system (82).

Frequent office visits or telephone calls to the pharmacy or requests for interim prescriptions of pain medication could signal hyperalgesia, worsening of (or new) pathology leading to untreated pain, a disorganized lifestyle leading to missed doses, or physical/psychological dependence (79). Each of these possibilities would demand a different response from the health care professional, illustrating the importance of patient-specific assessment whenever a change in the degree of pain control is reported. Establishing specific goals for chronic opioid therapy, monitoring patients closely and continuously, and maintaining a positive and open patient–practitioner relationship throughout are all of paramount importance to achieving optimum patient care outcomes.

Several avenues of recovery are available for patients with a true opioid addiction (83). Opioid-related therapeutic approaches can include blunting the pleasure response to μ agonists with opioid antagonist therapy (introduced after an opioid-free period of at least 7 days), initiating therapy with a partial μ agonist (buprenorphine) at the first sign of intentional withdrawal, or switching the patient from a highly euphoric μ agonist like heroin to a safer but still cross-tolerant μ agonist (methadone) and stair-stepping down to full recovery over the course of months or years. These therapeutic uses for analgesic opioids will be mentioned within

the monographs of the drugs employed in addiction recovery. Common to all approaches is the need to work closely with the recovering patient so that he or she is ready from a physiologic, emotional, economic, and social support standpoint for a successful transition to each step in the recovery process.

Constipation

The GI tract has a high concentration of μ - and κ -opioid receptors. Stimulating intestinal μ receptors inhibits the release of acetylcholine from the myenteric plexus and halts propulsive peristalsis, resulting in constipation that can range from annoying to disabling. Because there is no tolerance to the effect of opioid agonists in the GI, the risk of GI complications with long-term opioid use is significant. Clinicians should recommend that all patients on chronic opioid therapy also use stool softeners and stimulant laxatives, such as Senna, bisacodyl, Milk of Magnesia or, when appropriate, magnesium citrate, to avoid fecal impaction (67).

PHYSICOCHEMICAL AND METABOLIC PROPERTIES OF OPIOIDS

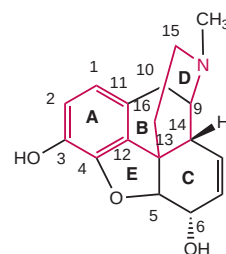
Lipophilicity, pK_a , and Pharmacokinetics

The centrally mediated therapeutic actions of opioid analgesics and antagonists demand transport across the blood–brain barrier to reach receptors in the brain and spinal cord. In addition to having all critical receptor binding functional groups in the proper orientation, the most potent opioids will also be highly lipophilic structures (Table 20.2). The SAR discussion below will highlight the impact on potency of eliminating polar functional groups not needed at the receptor and/or adding carbon-rich functional groups to both promote central distribution and augment hydrophobic interactions at the receptor surface. The pharmacokinetic data for the opioids are listed in Table 20.2.

All opioids require a cationic amine for anchoring to the conserved Asp residue of the G protein–coupled opioid receptors. Most commonly, the pK_a of this amine is between 8 and 10. Multicyclic opioids (morphines, morphinans, benazazocines, and oripavines) will either have or must generate a phenolic hydroxyl moiety, making them ultimately amphoteric structures. With a pK_a of approximately 11, this phenolic group is predominantly un-ionized at physiologic pH and participates as a hydrogen donor in an essential bond with a hydrogen-accepting His residue on μ - and κ -opioid receptors (84).

Stereochemistry

Morphine, the prototypical opioid agonist, has five asymmetric centers. Naturally occurring levorotatory (–) morphine has the 5*R*,6*S*,9*R*,13*S*,14*R* absolute configuration, which is the most active isomeric form at μ -opioid receptors.



Morphine with 4-phenylpiperidine pharmacophore highlighted

Ring C of pentacyclic (morphines) and tetracyclic (morphinans) opioids is forced into a pseudoboat conformation due to the presence of the 7,8-double bond and the restraining furan (E) ring. Activity will increase if this structural element is eliminated to allow the C ring to assume the more stable chair conformation. The B/C ring fusion in naturally occurring levorotatory morphine is *cis*, placing rings B and C at right angles to one another. This gives morphine a decidedly nonplanar T-shaped conformation. The 14*R* (B/C *cis*) configuration is indicated in two-dimensional drawings by a $C_{14}\beta$ substituent (in front of the plane of the paper). An α substituent at C_{14} (indicated by a dotted line) would denote 14*S* stereochemistry and the more planar B/C *trans* ring fusion. Because the absolute configuration at C_{14} influences molecular conformation and degree of fit at the receptor surface, it is not surprising that the degree of fit also impacts analgesic activity. In pentacyclic opioids, the B/C *cis* fused system is 10-fold more potent than the B/C *trans* system. In contrast, the B/C *cis* isomer of tetracyclic morphinans is twofold less active than the B/C *trans*. In all multicyclic opioids, only the B/C *cis* form is marketed.

The aromatic A ring can be viewed as a substituent on the 4-position of the piperidine (D) ring. The 4-phenylpiperidine structure is considered the pharmacophore for opioid analgesic action. In multicyclic opioids, the A ring is held rigidly in an axial orientation relative to the D ring, which forces a van der Waals interaction with the μ -opioid receptor Trp²⁹³ residue that is expecting to see the Tyr¹ residue of the enkephalins (84). For that reason, the aromatic A ring of analgesically active multicyclic opioids requires a phenolic hydroxyl group. If the orientation of the A ring can become equatorial to ring D (which occurs in flexible meperidine and anilidopiperidine-based opioids), then the phenyl ring will bind to the μ receptor Trp²⁸³ residue that normally binds with the enkephalin Phe⁴ residue (84). In that case, a phenolic OH is a binding liability.

Opioid receptors are highly stereoselective for the levorotatory isomers of morphine-like ligands. The dextrorotatory (+) isomers do not bind well to these receptors; however, both (+) and (–) isomers of many

TABLE 20.2 Pharmacokinetics of the Opioid Analgesics^{a,b}

Opioid	Log D _{pH 7.4} ^c	Oral Bioavailability (%)	Protein Binding (%)	V _d (L/kg)	T _{1/2} (h)	Duration (h) of Analgesia	Excretion (%)	T _{max} (h) for Analgesia	Onset (min) of Analgesia
Codeine	0.82	Well	0	~3	2–3	4–6	90 Renal	2–4	30–60
Hydrocodone	1.36	-	-	-	4–5	4	70 Renal <24 h	2	60
Hydromorphone	1.56	24	10–30	~3	11	4	-	1–2	30
Levorphanol	1.76	Rapid	40–50	10	11	4	-	-	10–60
Morphine	0.48	20–40	20–40	~4	2–5	4–5 sc	10 Fecal 90 Renal	-	10–30 sc
Oxycodone	0.38	60–90	45	~3	3–4	3–6	80 renal 20 unc	1–2	60
Oxymorphone	0.32	10	~10	3	7–9	4–6 iv	90 Renal	-	10 iv
Buprenorphine	2.08	15 TD 90 im	96	430 iv	2–6	4–6 im 7 d TD	30 Renal 7-Fecal	60 min TD	20 TD
Butorphanol	2.61	20 oral 70 nasal	80	10–13	5–6 nasal	5–6 oral 3–4 iv	80 renal 15 fecal	4–5 iv	15 nasal 60 iv
Nalbuphine	0.28	-	-	~4	3–5	3–6	10 renal 80 fecal	-	2–3 iv
Pentazocine	3.22	~20	370–415	-	2–3	2–3	70 unc	1–3 oral	15–30 2–3 iv
Naloxone	1.46	Rapidly inactivated orally	10	2–4	<2	45 min iv	50 renal <6h	-	2–3 iv
Naltrexone	1.51	5–40	20	1350 IV	4	24 iv 3 days oral	60–80 renal 10 fecal	-	-
Alfentanil	2.11	NA	92	~1	~2	10 min	80 renal	15 min im	2–3 iv
Fentanyl	2.94	50–65 buccal 92 TD	85	4–6	3–12 iv 20–27 TD	72 TD ~1 iv	75 renal 10 fecal 10 unc	8 min iv 30–50 min	12–24 TD < 1 iv
Meperidine	1.17	~50	60–85	3–5	3–4	2–4 sc, iv	10 unc 60 renal	30–60 min	10 sc
Methadone	3.23	85	70–90	3–4 iv	23 iv 12–24	2–10	21 unc	-	120 oral 30 iv
Remifentanil	1.70	NA	70	30–60 IV	3–10 m	3–10 min	90 renal	-	1–3 iv
Sufentanil	3.54	NA	90	2–3	2–3	30 min	80 renal 20 unc	-	1–3 iv
Tapentadol	1.38	32	20	540	4	4–6	70 renal	1–2	-
Tramadol	0.62	75	20	~3	5–7	4–6	30 renal 60 fecal	2–3	60

Abbreviations key: V_d, volume of distribution; T_{1/2}, elimination half-life; T_{max}, time to maximum analgesic effect; TD, transdermal; unc, unchanged; iv, intravenous; im, intramuscular; sc, subcutaneous; min, minutes

^aData for oral administration unless otherwise noted

^bData from Thomson Reuters Micromedex 2.0, accessed June 10, 2011

^cLog D_{pH 7.4} calculated using ACD/LogP v.12 Advanced Chemistry Development, Toronto Canada

μ agonists are potent antitussives. For example, almost everyone has purchased dextromethorphan (e.g., Robitussin DM) to control cough because it is available over the counter. Despite the fact that its μ -opioid receptor binding is exceptionally low, its ready availability has made it a focus of abuse, with approximately 10% of U.S. teenagers admitting to using it in large quantities in an attempt to attain euphoria (85,86). The drug antagonizes NMDA and σ -1 receptors, and some studies have documented a neuroprotective role for this compound related to its ability to combat glutamate-induced neurotoxicity (87–89). The levorotatory enantiomer of this over-the-counter product is called levomethorphan, and it is a potent, full μ -receptor agonist. It is not marketed, although levorphanol (Levo-Dromeran), its chemical cousin, is. Levorphanol is a potent prescription μ agonist analgesic with all of the anticipated risks and adverse effects common to this class of therapeutic agents.

Common Metabolic Reactions

The common metabolic pathways of the opioids are predictable from their structure. Many μ agonists have an *N*-methyl group that is readily *N*-dealkylated by CYP3A4 (90). From a therapeutic standpoint, nor-metabolites have limited clinical relevance due to a decrease in distribution-enhancing lipophilicity and the loss of an agonist-promoting hydrophobic interaction with the receptor.

In contrast, CYP2D6-mediated *O*-dealkylation of the 3-methoxy group of codeine analogs is required to generate the phenolic OH group essential for μ -receptor binding. Patients deficient in this activating enzyme show little or no response to codeine-based analgesics (90,91). While no clinically marketed opioid analgesic has a C_3 -ester, the need for metabolic liberation of the phenol would still hold true. For example, the 3-acetate ester of heroin is readily hydrolyzed in the CNS to generate the analgesically active monoacetylmorphine (Fig. 20.9). The 6-acetate ester is slower to cleave because its carbonyl carbon is less electrophilic due to a lack of resonance delocalization of electron density, but it will eventually succumb to hydrolysis to produce morphine.

Opioids are also subject to phase 2 metabolism prior to excretion. Phenols conjugate with either glucuronic acid or sulfate, and both conjugates are found in the urine of patients taking multicyclic opioids. These conjugates, like most, are inactive, and phase 2 conjugation in the GI tract impacts the oral bioavailability of many phenolic opioids (Table 20.2). Those administered by mouth must be given in doses higher than one would administer parenterally, and/or they are formulated to promote rapid absorption and limit exposure to intestinal transferase enzymes. Alcohols like those found at C_6 of morphine and codeine can also conjugate with

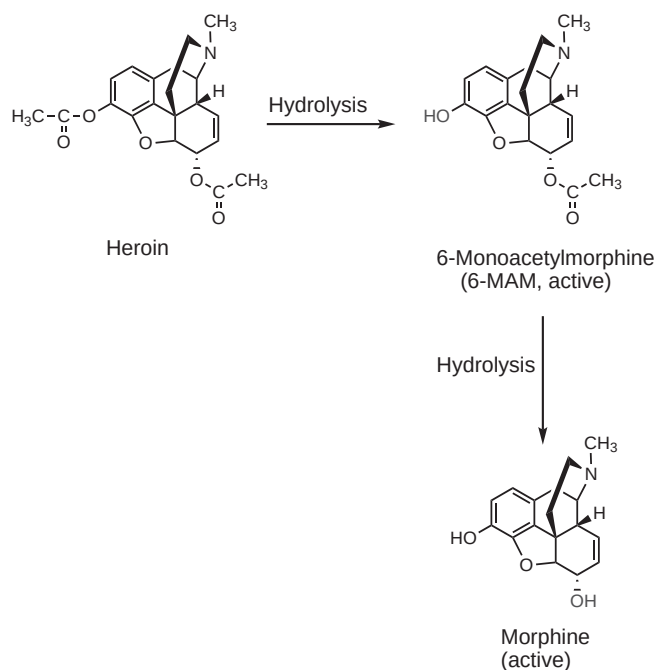


FIGURE 20.9 Heroin metabolism.

glucuronic acid. In a significant deviation from the norm, morphine's C_6 -glucuronide actually has a higher affinity for the μ receptors than the free alcohol and is believed to be responsible for a significant amount of morphine's and codeine's analgesic action (90,92). Glucuronidation at both C_3 and C_6 is catalyzed by UGT2B7 (90).

The metabolic pathway of codeine is illuminating in that it contains essentially all of the aforementioned biotransformations and includes reactions that are activating, inactivating, attenuating, and augmenting with regard to analgesic action (Fig. 20.10).

STRUCTURE-ACTIVITY RELATIONSHIPS OF OPIOID AGONISTS AND ANTAGONISTS

In the following paragraphs, SARs that can be broadly applied to various chemical classes of opioid agonists and antagonists are presented. Several opioid molecules are "one of a kind," and their unique SAR will be presented within the individual drug monograph.

Multicyclic Opioids

Multicyclic opioids include those with pentacyclic (morphine), tetracyclic (morphinan), tricyclic (benzazocine), or bicyclic (oripavine) fused ring systems (Figs. 20.11 to 20.13). Although not every structural prototype will contain all of the functional groups identified

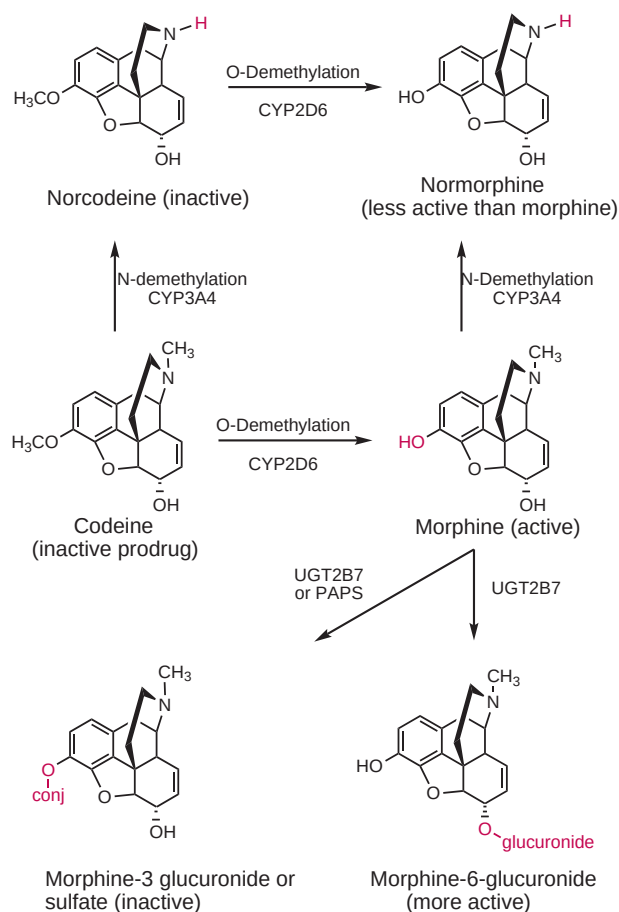
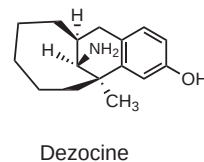


FIGURE 20.10 Codeine metabolism.

below, the following is a general discussion of key SARs that commonly apply to the multicyclic opioid analgesics and antagonists.

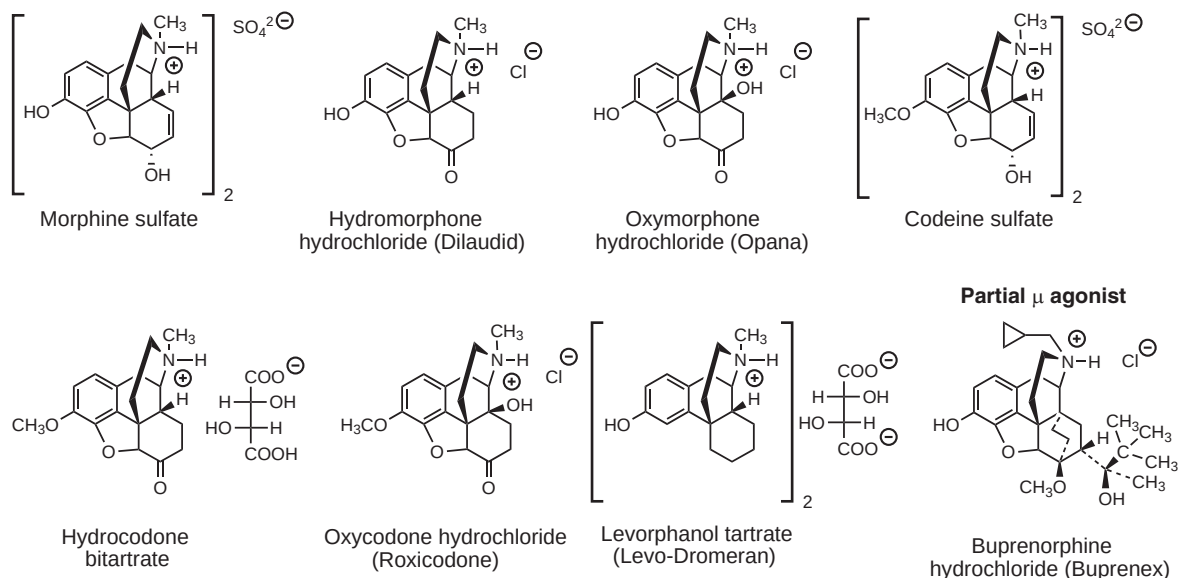
R₁

An amino nitrogen is an essential feature of all opioids because, in cationic conjugate acid form, it permits receptor anchoring through an essential electrostatic bond with the Asp residue conserved in all G protein-coupled receptors (84). With the exception of dezocine, an opioid agonist no longer available in the United States, only tertiary amines provide clinically significant opioid agonist or antagonist activity in the CNS.



The nitrogen substituent found on tertiary μ selective multicyclic opioid agonists is most commonly CH_3 . This group interacts hydrophobically with μ -receptor residues. An aralkyl substituent (e.g., phenethyl) would significantly enhance μ -receptor affinity and CNS distribution, but no such substituent is currently found on any multicyclic opioid.

Extending the length of the nitrogen substituent to three carbons or its equivalent and incorporating an area

FIGURE 20.11 Multicyclic μ agonists.

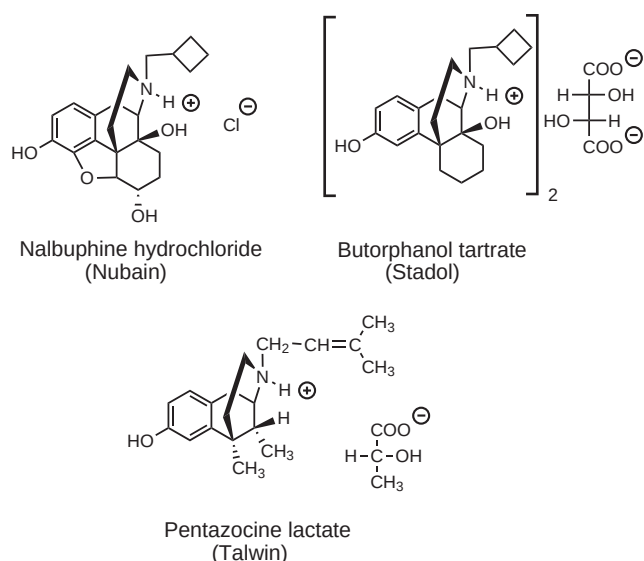


FIGURE 20.12 Multicyclic κ agonists/ μ antagonists.

of high electron density two atoms away from the nitrogen shifts the μ -receptor activity profile from agonist to antagonist. The antagonist site on the μ receptor is a generally hydrophobic narrow pocket formed between the side chains of Asp, Met, Tyr, Ile, and Gly residues (84). If the moiety providing the high electron density is a double bond (R_1 = allyl) or a highly strained cyclopropane ring (R_1 = cyclopropylmethyl), μ antagonist activity is predominant. If found on a pentacyclic opioid framework containing a 14β -OH group and a 7,8-dihydro-6-one (or equivalent) C ring, allyl and cyclopropylmethyl substituents provide a pure opioid antagonist. Pure antagonists have their highest affinity for the μ receptor but, in higher doses, will block all other opioid receptor subtypes too. They provide no analgesic action at any receptor at any

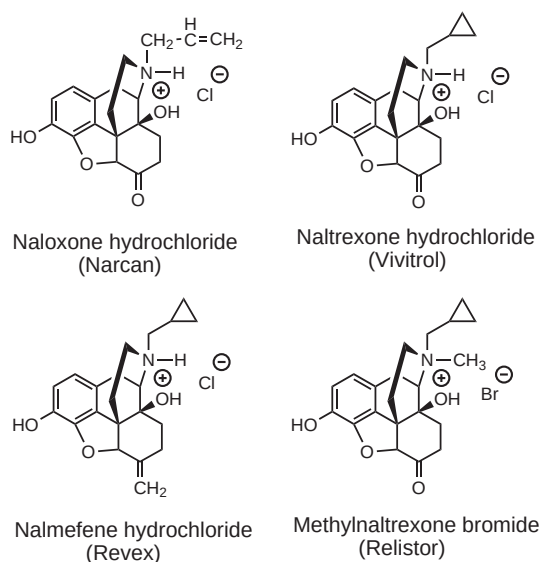


FIGURE 20.13 Multicyclic opioid antagonists.

dose. In other multicyclic systems, these two nitrogen substituents can provide κ agonism, but only in doses much higher than those needed to elicit μ antagonism. Strong dysphoria at these higher doses precludes their use as analgesics, and no such drugs are marketed.

If ring strain is eased as in cyclobutylmethyl, or if the allyl moiety is converted to dimethylallyl, significant μ antagonism remains but κ agonism (analgesia) becomes the therapeutically predominant action. While often potent, the level of μ antagonism exhibited by *N*-cyclobutylmethyl opioids will always be lower than that of their allyl or cyclopropylmethyl counterparts because of a less-than-optimal fit in (and affinity for) the narrow antagonist receptor cleft. However, because clinically significant μ antagonism will always accompany κ agonism, analgesics with cyclobutylmethyl or dimethylallyl nitrogen substituents will precipitate withdrawal in patients dependent on *N*-methyl- or *N*-aralkyl-substituted analgesics. In other words, these two chemically distinct classes of pain-relieving opioids are not cross tolerant.

R₂

A phenolic hydroxyl group or a methoxy ether will be found at C₃ of all marketed multicyclic opioids. As previously mentioned, the phenol is required for binding at μ and κ receptors, so the methoxy ethers (codeine analogs) are prodrugs. CYP2D6 *O*-dealkylation of these methoxy groups is activating, but it is a somewhat sluggish reaction. Only about 10% of a dose of codeine will be activated to morphine before activity-attenuating CYP3A4-mediated *N*-dealkylation occurs. Poor CYP2D6 metabolizers will not realize the full impact of codeine-based analgesics and require alternative analgesic therapy. Within the United States, the incidence of CYP2D6 deficiency has been estimated at 7.7% for Caucasians and between 1.9% and 7.3% for African Americans (93). These numbers are in line with estimates for these ethnic groups within the general population (94,95). Although the CYP2D6*3 and *4 alleles common in Caucasian poor metabolizers are present in only 1% or less of persons of Asian ancestry, the CYP2D6*10 allele associated with reduced CYP2D6 catalytic capability (intermediate metabolizer phenotype) is found in up to 50% of Asians (93–95).

The importance of CYP2D6 phenotype in the pharmacokinetics of 3-methoxy-substituted analgesics is powerfully illustrated by the tragic loss of life that occurred when a new mother, an ultrafast CYP2D6 metabolizer taking codeine for postpartum pain, nursed her newborn son who had the normal CYP2D6 metabolizer phenotype. Both mother and son were homozygous for the UGT2B7*2 (fast-metabolizing) allele of the C₆-glucuronidating transferase enzyme, meaning that both could rapidly convert morphine to the highly active and persistent 6-glucuronide metabolite. Over the course of 2 weeks, the infant was exposed to an excessive amount of morphine and its C₆-glucuronide metabolite, and he died of respiratory depression (96). One could envision the same tragic outcome if a CYP2D6 intermediate-metabolizer mother

given high doses of codeine in an attempt to relieve pain breastfed a normal (extensive metabolizer) or ultrarapid CYP2D6-metabolizing infant.

When present, the phenolic OH group will be vulnerable to inactivating prehepatic GI and first-pass (liver) phase 2 metabolism. Premature conjugation with glucuronic acid or with 3'-phosphoadenylyl sulfate significantly compromises the oral bioavailability of many phenolic opioids. Phenols also undergo in vitro oxidation to therapeutically inactive quinones. Phenolic drugs, including active opioids, should always be protected from light, base, and oxygen in the air or, over time, they will decompose.

R₃

Multicyclic opioids currently on the market have either a 14β-H or OH moiety, which provides a B/C *cis* or equivalent configuration. The polar 14β-OH slows penetration of the blood–brain barrier, but the negative impact this would have on analgesic potency is more than overridden by a significant increase in receptor affinity. At μ, the 14β-OH H-bonds very effectively with a Tyr residue, whereas at κ receptors, the bonding residue is Glu (84). Overall, a two- to threefold increase in opioid activity is achieved through the addition of a 14β-OH. Interestingly, there is a decrease in antitussive (cough suppression) action (97), and 14β-OH opioids are not used for this therapeutic purpose.

R₄

The impact of the C₆ substituent of multicyclic opioids on pharmacologic action is tied to the presence or absence of the 7,8-double bond. The 6α-OH found in naturally occurring morphine and codeine is believed to engage in a hydrogen bond with a μ-receptor Asn residue (84), but the bond formed is not strong. Despite the loss of this drug–receptor interaction, removal of the alcoholic OH group results in an approximate 10-fold increase in activity due to a significant increase in molecular lipophilicity (logD_{pH 7.4}) (98). If the 6α-OH is oxidized to provide an α,β-unsaturated ketone, the H-bonding role of the 6-substituent converts from H-donor to H-acceptor. In the rigid pseudoboat C ring that is ensured by the 7,8-olefin, a marked decrease in receptor affinity occurs, and activity decreases threefold. However, if the 7,8-double bond is reduced, the C ring becomes significantly more flexible and can properly position the H-accepting C₆ ketone for high-affinity receptor binding. A sixfold gain in analgesic potency is achieved by changing C ring structure from the natural 7,8-dehydro-6α-ol to 7,8-dihydro-6-one.

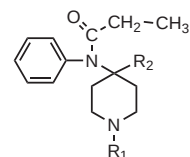


7,8-Dehydro-6α-ol 7,8-dihydro-6-one

The 6α-OH group of morphine and codeine (and the 6-glucuronide metabolite) is associated with mast cell

degranulation and histamine release, which can lead to an allergic response involving hypotension, intense pruritus, and rash (99). The mechanism involves activation of protein kinase A and inositol triphosphate kinase, which not only releases histamine but also stimulates the production of proinflammatory chemokines in mast cells (100). This adverse effect is particularly intense with codeine, and as a result, parenteral administration of this analgesic is generally not recommended. The delayed allergic reaction is not reversed by opioid antagonists, indicating that it is not mediated through the μ receptor. Pruritus and hypotension are not generally observed with 6-keto or 6-desoxy opioids, and the risk of nausea may also be lowered.

Flexible Opioids: Anilidopiperidines



Anilidopiperidine pharmacophore

The anilidopiperidine (also called anilinopiperidine) analgesics, commonly known as the fentanyls, are the only class of flexible opioids with more than two members (Fig. 20.14). As the aromatic phenyl ring of the fentanyls is believed to interact with the μ receptor Trp²⁸³ residue that normally binds Phe⁴ of enkephalin, no phenolic OH group is incorporated into their structures (84). These are highly lipophilic molecules that distribute rapidly across the blood–brain barrier, resulting in a fast and potent, albeit short-lived, analgesic response (101).

R₁

The SAR of the nitrogen substituent of all flexible opioids essentially parallels that of the multicyclics. *N*-Methyl or *N*-aralkyl substituents will always promote potent, selective μ agonism regardless of the class of opioids under discussion. The aralkyl substituents (e.g., R₁ = phenethyl, thienylethyl) of fentanyl and sufentanil promote a very fast penetration of the blood–brain barrier and a very high affinity at μ receptors through hydrophobic and van der Waals interactions. This results in an analgesic activity that runs between 80- and 800-fold that of morphine. The substantial lipophilicity of these aralkyl substituents contributes to the very high logD_{pH 7.4} values of fentanyl (2.94) and sufentanil (3.54) (102).

Replacing the lipophilic phenyl or thienyl ring found in fentanyl and sufentanil, respectively, with the polar tetrazolinone system found in alfentanil (logD_{pH 7.4} = 2.11) produces some complex but predictable changes in in vivo drug behavior. One might logically expect the decrease in molecular lipophilicity to translate to a more sluggish journey across the blood–brain barrier. However,

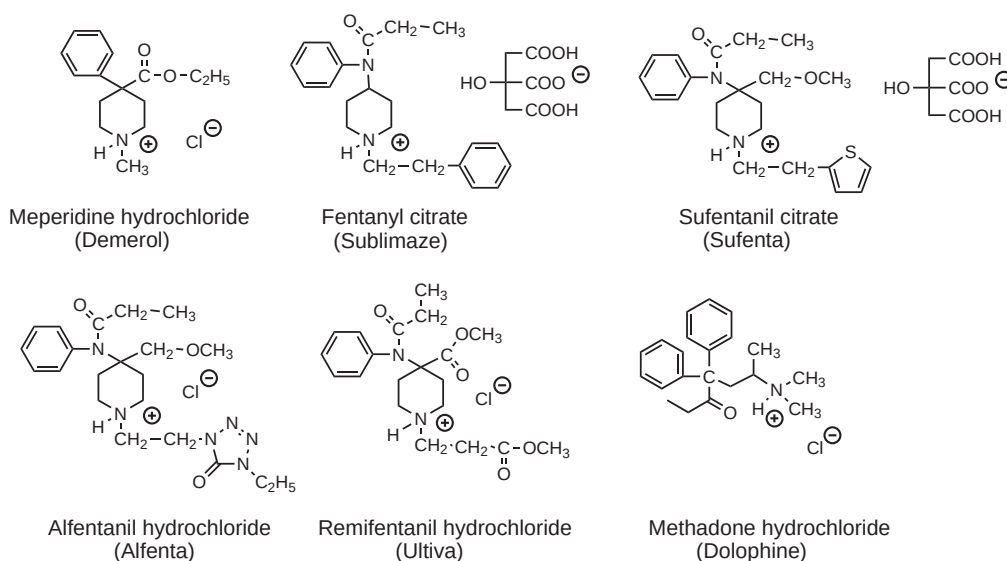


FIGURE 20.14 Flexible μ agonists.

this strongly electron-withdrawing ring system decreases the pK_a of the piperidino nitrogen atom to 6.5 compared to 8.4 for fentanyl's basic nitrogen atom, which significantly decreases the ratio of ionized to un-ionized conjugate forms in the bloodstream (103). Contrary to what would be expected from a simple comparison of $\log D_{pH\ 7.4}$ values, the lower pK_a allows for a faster penetration of the blood–brain barrier. However, this depressed ionized-to-un-ionized ratio is maintained at the receptor surface, which means less cationic drug is available for ion–ion anchoring with the μ receptor Asp. Therefore, despite a positive CNS distribution profile, the potency of alfentanil decreases to 25-fold that of morphine because of a lower μ -receptor affinity.

Mu antagonists are not found within the flexible opioid classes. The cyclopropylmethyl and allyl nitrogen substituents that provided such potent μ antagonism in the multicyclics simply attenuate agonist action in these less rigid structures, and they will not be encountered.

R₂

The methoxymethyl moiety at C₄ of the piperidine ring of sufentanil and alfentanil is lipophilic and promotes distribution to central sites of action. More importantly, its steric influence promotes high-affinity binding at the μ -receptor surface.

SPECIFIC DRUGS

Multicyclic Full μ Agonists (Fig. 20.11)

Morphine Sulfate

Morphine is the prototypical opioid agonist, and all synthetic multicyclic analgesics are based on the morphine nucleus to some degree. The presence of the N -CH₃ substituent, phenolic hydroxyl, and B/C *cis* configuration

assure potent and selective μ agonism. Morphine has very poor oral bioavailability because of extensive prehepatic (GI) and first-pass (liver) metabolism to the inactive 3-glucuronide and 3-sulfate conjugates (Table 20.2). The C₃ glucuronide undergoes extensive enterohepatic cycling, so if oral administration is required, large initial doses must be given followed by lower maintenance doses. Morphine's $\log D_{pH\ 7.4}$ is low (0.48) (102), and opioids of higher lipophilicity exhibit better oral bioavailability as well as a faster onset of analgesic action (Table 20.2) (67). Polar and metabolically vulnerable morphine is about sixfold more active by the intramuscular route than the oral route.

Morphine is metabolized by CYP3A4-mediated N -dealkylation to normorphine (90) which, at 1/20 the activity of the parent, is theoretically active but not clinically relevant. Prior to excretion, glucuronic acid conjugation may occur at the C₆-OH group. As previously noted, this glucuronide retains activity at the receptor and is under investigation as a therapeutic agent for the relief of postoperative pain (104). A potency ratio between morphine-6-glucuronide and the parent morphine in humans has been estimated at up to 50:1 (105). That ratio can rise to 100:1 when the drug is administered directly into the brain of animals (106). Although this polar metabolite should be readily excreted, it can actually accumulate in patients with renal failure or poor renal function (e.g., elderly), increasing the risk of inadvertent overdose (90,107).

Morphine is one of the safest opioids to use to treat the pain of myocardial infarction, unstable angina, and other ischemic disorders because of its positive hemodynamic effects (108,109). Its negative chronotropic effect and venous and arterial vasodilation result in a decrease in myocardial oxygen demand. Morphine decreases blood pressure secondary to histamine release,

an activity linked to the 6 α -OH group and/or the active 6 α -glucuronide metabolite, and must be used with caution in patients with hypovolemia. Extreme caution is advised with the use of all opioids in patients with head trauma because opioids can increase intracranial pressure, which in turn can exacerbate respiratory depression. Inhibition of cough reflex, however, is a valuable action in this situation because it helps keep intracranial pressure within defined limits. Because morphine lacks the 14 β -OH group that attenuates cough suppression, it is an opioid of choice in the treatment of patients with head injury (110). A relatively recent study cautions that inflammation associated with head injury may downregulate efflux proteins such as P-glycoprotein (P-gp) and multidrug-resistant protein, resulting in an accumulation of morphine and the active 6-glucuronide, respectively, in cerebral tissue (110).

Morphine is available in a wide variety of dosage forms, including solution for injection, oral solution, tablets, controlled- and extended-release tablets, and extended-release capsules (the latter marketed as Avinza and Kadian). Extended-release formulations of any opioid should only be administered to opioid-tolerant patients. They must be swallowed whole and not chewed, crushed, or dissolved in liquid. If patients have difficulty swallowing, the pellets in some extended-release capsules may be sprinkled on food (e.g., applesauce) and swallowed without chewing. To take an extended-release opioid in any way other than prescribed risks releasing and absorbing a very large dose of active drug, which can result in death from respiratory depression. The consumption of alcohol while on opioid agonists is never advised, but it could prove fatal when taking extended-release capsules, because alcohol can promote rapid capsule dissolution in the GI tract.

Morphine sulfate in combination with the opioid antagonist naltrexone hydrochloride is on the market as Embeda. This trade name refers to the fact that a core of naltrexone is embedded in a pellet of extended-release morphine. A variety of strengths is available, but the milligram-to-milligram ratio of agonist-to-antagonist is consistently held at 25:1. As with Avinza and Kadian, this combination product is for use only when continuous long-term analgesia requiring “around the clock” opioid agonist therapy is warranted. Patients taking this and all extended-release products must not be opioid-naïve, or potentially fatal respiratory depression could result.

Combination products such as Embeda have also an abuse-deterrent purpose. If the capsules or their contents were crushed and the active constituents extracted for intravenous (IV) injection (e.g., in an attempt to get high), the naltrexone hydrochloride would be extracted along with the morphine salt. If injected, the antagonist would antagonize opioid receptors, thereby blocking any euphoric responses an abuser might be seeking. Fortunately, morphine-induced hemodynamic responses would also be blocked, at least to some extent. If a patient tolerant to opioids injected this illicit formulation, a

naltrexone-induced withdrawal syndrome would most certainly be precipitated.

Again, opioid-tolerant patients are the only ones who should be prescribed this or any other extended-release opioid agonist. The risk of serious toxicity, specifically potentially fatal respiratory depression, prohibits the use of all extended-release products in opioid-naïve patients.

Hydromorphone Hydrochloride

Hydromorphone, known chemically as 7,8-dihydromorphone, differs from morphine only in the area of ring C. The 7,8-dihydro-6-one structural motif adds flexibility to the system and permits a more efficacious H-bond between the 6-ketone and the μ receptor. The increase in analgesic action arising from this single structural modification has been estimated at 6- to 10-fold. As μ receptor–elicited analgesia and side effects such as respiratory depression and constipation go hand-in-hand, the risk of toxicity with hydromorphone as compared to morphine also increases. Hydromorphone is available as a solution for injection, rectal suppositories, tablets, liquid for oral administration, and extended-release tablets. As with Avinza and Kadian, a black box warning has been issued for the extended-release formulation (Exalgo), because the potential for abuse and the risk of fatal respiratory depression are high. Like all extended-release opioids, Exalgo use is reserved for patients tolerant to and dependent on μ agonists who require high-potency pain management on a chronic basis.

Oxymorphone Hydrochloride

The addition of a hydroxyl group at 14 β increases analgesic action approximately two- to threefold over hydromorphone secondary to an increase in μ -receptor affinity. This makes oxymorphone at least 12-fold more potent than morphine. It is available as a solution for injection, tablets, and extended-release tablets. All precautions previously outlined for extended-release morphine and hydromorphone apply to Opana ER, the extended-release formulation of oxymorphone.

Codeine Sulfate, Hydrocodone Bitartrate, and Oxycodone Hydrochloride

Codeine, hydrocodone, and oxycodone are the 3-methoxy analogs of morphine, hydromorphone, and oxymorphone, respectively, and they have the same potency relationship to one another as their parent structures demonstrated. All are prodrugs that must undergo CYP2D6-mediated activation to the phenolic structure required for binding to μ receptors. Codeine and oxycodone are available as single-drug products. Although the potency of each is about 10- to 12-fold lower than their phenolic counterpart, oral bioavailability is greatly enhanced (Table 20.2). Both are available in tablet form and as a solution for injection. As previously mentioned, the risk of a severe allergic reaction to parenterally administered codeine phosphate is high due to the presence of the 6 α -OH group.

Oxycodone was one of the first μ agonist analgesics to be made available in controlled-release tablet form (Oxycontin), and there was much focus in the literature and lay press on the misuse of this drug by abusers when it was first released (111). Some abusers seek (and get) a very strong euphoric rush by dissolving a crushed Oxycontin tablet in alcohol. As can be imagined, this practice increases the likelihood of true addiction and/or death. It cannot be emphasized strongly enough that patients on opioids should avoid the consumption of alcohol due to the risk of additive CNS depressant actions, including coma and potentially fatal respiratory depression.

Hydrocodone is available only in analgesic combination with acetaminophen or aspirin and in some cough suppressant preparations (note the lack of the 14 β -OH, which permits its use as a cough suppressant). With a potency approximately half that of morphine, its combination with a nonsteroidal anti-inflammatory drug or *p*-aminophenol-based analgesic attacks pain on two mechanistic fronts, presumably allowing for a lower dose of each potentially toxic drug to be used.

Codeine use for longer than 6 months has recently been associated with an increased risk of serious or fatal cardiovascular events, and an increased risk of all-cause mortality has been noted when used for more than 30 days (112). Oxycodone, but not hydrocodone, is also associated with these serious adverse effects. The risk-to-benefit ratio of long-term use of codeine, oxycodone, and other potentially cardiotoxic opioids must always be carefully evaluated, particularly in the elderly and other vulnerable patient populations.

Levorphanol Tartrate

Levorphanol is the only μ agonist in the tetracyclic (morphinan) class of multicyclic opioids. Unlike the pentacyclic morphines, the B/C *cis* (14*R*) configuration is twofold less active than the B/C *trans* (14*S*) isomer (113,114), but the B/C *cis* structure is the only isomeric form commercially available. The analgesic potency of levorphanol is approximately four- to eightfold that of morphine due to enhanced molecular flexibility from the loss of both the restraining E ring and the 7,8-double bond, which leads to increased μ -receptor affinity despite the lack of a 14 β -OH. In addition, the loss of the E ring and 6-oxygen functions increases $\log D_{\text{pH } 7.4}$ (1.76) (102), which permits more facile transit across the blood-brain barrier. It is available as tablets and as a solution for injection.

Levorphanol's 6- to 8-hour duration of action is about twice that of morphine, which enhances utility in chronic pain. It has an 11- to 16-hour terminal half-life, however, and this allows the drug to accumulate over time. Patients should be monitored for unwanted or excessive sedation that might signal a build-up of drug in the blood and brain.

Multicyclic Partial μ Agonists

Buprenorphine Hydrochloride

This structurally unique opioid has an interesting and unique SAR (115–117). Despite the presence of the

N-cyclopropylmethyl group, which normally promotes therapeutically useful μ antagonist action, buprenorphine is a very potent, but partial, μ agonist with an analgesic potency between 25- and 40-fold that of morphine (113,118). A 0.3- to 0.4-mg dose of this oripavine-based analgesic is equipotent with 10 mg of morphine. This superior potency is attributed to high lipophilicity and a higher-than-normal affinity for μ receptors because of the additional hydrophobic and hydrogen bonding capabilities of the C₇ side chain. Typical of its partial agonist action, a bell-shaped dose-response curve has been described for many of the actions of this bicyclic opioid (115,119–121). Although not used clinically for this purpose, buprenorphine also antagonizes κ receptors (121).

The μ -receptor binding of buprenorphine is strong enough to be classified as pseudoirreversible (121,122), which provides an exceptionally long half-life of 37 hours and permits once-daily dosing. In addition, the slow dissociation from μ receptors minimizes withdrawal symptoms even if the drug is abruptly discontinued (123). Any discomfort a patient experiences after abrupt withdrawal of buprenorphine normally occurs 5 days to 2 weeks after the last dose was taken. The down side of pseudoirreversible binding is that it is difficult to reverse receptor-mediated side effects (124). The most problematic adverse reaction associated with this drug is respiratory depression. This unwanted effect is less severe than can be observed with full μ agonists, and fortunately, it is rarely life-threatening, but it can be clinically significant. If respiratory depression occurs in a patient taking buprenorphine, it can take up to 10-fold the normal dose of naloxone to compete for the μ receptors and reverse it. The respiratory depression induced by buprenorphine can outlast naloxone's duration of action, making the impact of a single high dose of antagonist inconsistent (121). A regimen of 2 to 3 mg of naloxone followed by a 4-mg/h continuous infusion has been shown to restore normal breathing in respiratory-depressed patients within an hour, and kinetic models to explain this clinical observation have been published (121).

Because buprenorphine has only partial μ agonist action, it will precipitate withdrawal in patients dependent on full μ agonists (85,125), but it also suppresses the symptoms of full-blown withdrawal. In other words, if μ receptors are occupied by a full agonist, they will not tolerate a stepping down of receptor stimulation, but if they are already in "crisis mode," some μ agonism is viewed as better than none.

Buprenorphine undergoes significant first-pass metabolism. The CYP3A4-generated *N*-dealkylated metabolite norbuprenorphine retains some analgesic activity (40-fold less than the parent drug), but the C₃ glucuronide is predictably inactive (126,127). This drug is given IV or sublingually to avoid inactivating conjugation in the GI and on first pass. The dose of buprenorphine may need to be adjusted if coadministered with CYP3A4 substrates or inhibitors or benzodiazepines, or if alcohol use is known or suspected.

A transdermal patch formulation of buprenorphine, marketed as Butrans, has recently been made available in 5-, 10-, and 20-mcg/h strengths. The previously described precaution on using extended-release opioids only in opioid-tolerant patients also applies to transdermal buprenorphine. In addition, patients must be advised to avoid allowing the applied patch to become hot, as the release of drug from the patch matrix is temperature dependent. Although buprenorphine can be hepatotoxic in high doses secondary to inhibition of mitochondrial respiration and subsequent adenosine triphosphate depletion, the nor metabolite is apparently nontoxic (128).

In addition to its use as an analgesic, buprenorphine is also employed in opioid addiction recovery. As previously noted, opioid addiction affects millions of people in the United States and has a huge financial impact on health care economics. In 2003, over 8% of U.S. emergency room visits were related to heroin use. It has been estimated that 6% to 15% of people in the United States abuse drugs (85), and the National Institute on Drug Abuse has estimated that illicit drug abuse (including opioid abuse) consumes over half a trillion dollars each year (129). The number of individuals over 50 years of age seeking treatment for drug and alcohol addiction has been increasing at a rate disproportionate to the general population, and in 2008, almost a quarter million “baby boomers” (or older) sought professional help for substance abuse (130). In 2008, heroin was second only to alcohol as the most abused substance in this older population.

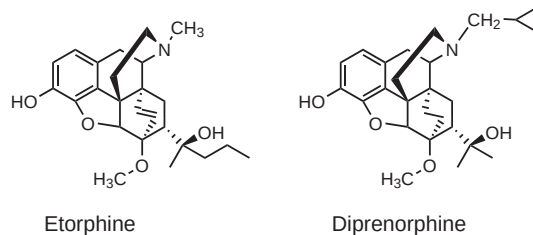
In 2-mg doses, buprenorphine blocks the euphoric effects of heroin and has been shown to be as active as 30 mg of methadone in addiction treatment programs. Doses between 2 and 32 mg daily are common, with an average daily dose of 16 mg. Treatment programs generally range from 6 months to over 2 years (83). Recovering addicts receiving their first dose of buprenorphine must be opioid-free for 7 to 10 days to avoid precipitation of a withdrawal episode. Alternatively, under the direct supervision of a physician, the patient can be allowed to enter the preliminary phase of opioid withdrawal and then be administered the first dose of buprenorphine, which will suppress the withdrawal symptoms (83). If a recovering addict relapses (common in the first years of abstinence) and increases the buprenorphine dose in an attempt to get euphoria, they will hit the partial agonist activity plateau, and their attempt to abuse will be thwarted (83). When the time comes, the patient can be withdrawn from buprenorphine with few, if any, withdrawal symptoms due to its slow dissociation from the receptors and slow disappearance from the body. The patient's receptors essentially wean themselves off of the drug.

Two sublingual formulations of buprenorphine are available for use in addiction recovery. Subutex contains only buprenorphine and is available in 2- and 8-mg strengths. Onset of action generally occurs within 30 to 60 minutes and peaks between 90 and 100 minutes. The same dose options are available for Suboxone, but these products contain 0.25 mg naloxone per mg of buprenorphine to block

euphoria if the sublingual tablets are crushed, extracted, and injected IV by would-be abusers (83). Naloxone has limited activity by the sublingual (and oral) route.

The direct drug cost of Suboxone has been estimated at \$250 per prescription, or about 10-fold that of methadone (85). However, the overall cost (which includes elements of adherence, regimen [every 2 to 3 days], and adverse events/toxicity), as well as the cost to patients, may actually be lower. The value of receiving therapy in one's own physician's office or pharmacy (greater accessibility, lower “stigma”) cannot be overestimated. The addition of these products to the U.S. marketplace is significant in that it allows recovering addicts to receive care from the hands of their own physicians and pharmacists (131). Physicians who engage in office-based treatment of addiction must have completed a minimum of 8 hours of specialized training. They are issued special Drug Enforcement Administration numbers indicating authorization to prescribe buprenorphine. Physicians may dispense a small supply of Subutex to patients in the office, while pharmacies primarily stock Suboxone for long-term therapy. The need for appropriate sensitivity to patient privacy when counseling should be obvious (132).

Predictably, replacing the *N*-cyclopropylmethyl substituent of buprenorphine with a methyl group provides a full μ agonist. The *N*-methylated analog that has also replaced the *t*-butyl group within the C_7 side chain with *N*-propyl and oxidized the 6,14-endoethano bridge to endoetheno is known as etorphine. Etorphine has an analgesic potency estimated at up to 200-fold that of morphine in humans for reasons related to superior receptor affinity and central distribution properties. Although never marketed for human use in the United States, etorphine has been widely used in veterinary medicine and as an immobilizer to permit the safe capture and/or handling of zoo animals and large game. Marketed alone or in combination with the sedative antiemetic acetylpromazine, etorphine has enjoyed the descriptive trade names of Captivon and Immobilon, respectively. Buprenorphine can also be converted into a potent and pure opioid antagonist (diprenorphine) simply by replacing the *t*-butyl moiety on the C_7 side chain with methyl.



Multicyclic κ Agonists/ μ Antagonists

Nalbuphine Hydrochloride

This pentacyclic *N*-cyclobutylmethyl opioid provides clinically useful analgesia elicited exclusively via the κ receptor. Potency is augmented by the lack of the 7,8-double bond and the presence of the 14 β -OH group, and overall, it is between

0.5- and 1-fold as active as morphine in the relief of pain. It is administered only by injection. The μ antagonist component of its activity is approximately one-fourth that of the prototypical μ opioid antagonist naloxone. The κ agonist/ μ antagonist activity profile assures a lack of cross tolerance with full or partial μ agonist analgesics, and patients stabilized on μ agonists cannot be switched to nalbuphine (or any other κ agonist) without precipitating withdrawal.

Kappa agonists produce less respiratory depression, constipation, euphoria, and potential for addiction than μ agonists (133). Sedation and diuresis are nalbuphine's major side effects. Like morphine, this analgesic decreases oxygen demand in the myocardium after infarct. Kappa receptor-related dysphoria is not as common with nalbuphine as with the other two marketed κ agonists, butorphanol and pentazocine.

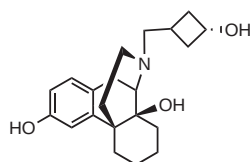
Women often respond better to κ opioids than men (134), and these compounds are often used in obstetrics. Gender differences in response may be due to differences in types of pain, pain perception, threshold, and/or pathways, steroid influence on drugs or pain relief, and/or opioid receptor density. Some men have shown a pronounced antianalgesic response to low-dose (5 mg) nalbuphine (134).

Butorphanol Tartrate

Butorphanol is another *N*-cyclobutylmethyl κ agonist analgesic with μ antagonist actions. The lack of the furan oxygen and 6 α -OH group provides a 5- to 10-fold higher analgesic potency compared to nalbuphine. This drug has enjoyed significant use in obstetrics and, while respiratory depression is possible, it is not usually a clinically significant problem. The ceiling effect for butorphanol-induced respiratory depression is lower than that observed with μ agonists, so the risk of fatality is slight, even when the drug is given in higher doses (135).

As with all κ agonists, receptor-mediated dysphoria and diuresis are possible with butorphanol, especially when higher doses are used (136,137). Although significant pulmonary hypertension and an increase in myocardial workload have been claimed (69), some indicate the hemodynamic impact of butorphanol is similar to morphine (138).

No oral dosage forms of butorphanol are available due to significant first-pass metabolism to an inactive *trans*-3'-hydroxy metabolite (139). However, a nasal spray is available for use when administration by injection is not warranted. The availability of this dosage form has increased the use of this analgesic in the outpatient setting.



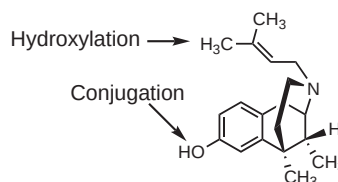
Trans-3'-hydroxybutorphanol metabolite

Pentazocine Lactate

The tricyclic (benzazocine) κ agonist pentazocine is missing carbons 6 and 7 of the C ring of multicyclic opioids, which compromises its ability to interact hydrophobically with opioid receptors. Although marketed as a racemic mixture, the analgesic action is found only in the levorotatory isomer, which is approximately half as active as morphine. Pentazocine is more likely than the other commercially available κ agonists to induce use-limiting dysphoria, especially at high doses (140). It also increases mean arterial and pulmonary arterial pressure and myocardial oxygen demand (138,141). This κ agonist should be avoided in myocardial infarction and congestive heart failure patients.

Pentazocine lactate is available as a solution for injection for the treatment of moderate to severe pain. Pentazocine hydrochloride is administered orally in combination with acetaminophen, but bioavailability is only 20% to 50% due to significant prehepatic and first-pass metabolism. Major inactivating reactions include phase 2 conjugation to the C₃-glucuronide and sulfate conjugates and hydroxylation of one of the terminal CH₃ groups of the dimethylallyl nitrogen substituent. The primary alcohol generated via hydroxylation can undergo further oxidation by alcohol and aldehyde dehydrogenase enzymes to provide a carboxylic acid metabolite.

An oral pentazocine hydrochloride/naloxone hydrochloride combination product designed to frustrate potential abusers is also commercially available. Marketed as Talwin NX, the philosophy behind its design is similar to buprenorphine/naloxone (Suboxone) and morphine/naltrexone (Embeda) combination products. If Talwin NX tablets are crushed in an attempt to extract pentazocine for IV injection, the antagonist would be extracted simultaneously. The injected naloxone would occupy the opioid receptors and block pentazocine-induced euphoria. If the drug is taken orally (as intended), the orally inactive naloxone antagonist will be destroyed on first pass so that the full analgesic effect of pentazocine can be realized.



Sites of pentazocine metabolism

Multicyclic Opioid Antagonists

Naloxone Hydrochloride, Naltrexone Hydrochloride, and Nalmefene Hydrochloride (Fig. 20.13)

Three pure opioid antagonists are marketed to reverse the central actions of opioid agonists. They all have a classic antagonist-directing nitrogen substituent on a pentacyclic scaffold and contain the requisite 14 β -OH and the

7,8-dihydro-6-one or 6-ene C ring. With a four-carbon *N*-cyclopropylmethyl substituent, naltrexone ($\log D_{\text{pH } 7.4} = 1.51$) is twice as potent as the allyl-substituted naloxone ($\log D_{\text{pH } 7.4} = 1.46$), presumably due to enhanced distribution to central sites of action. With the replacement of the polar C₆-keto of naltrexone with a lipophilic methylene moiety, nalmefene shows the anticipated rise in $\log P$ (2.66) and should distribute to the CNS at least as well as, if not better than, naltrexone.

Any of these opioid antagonists can be used IV to rescue patients from life-threatening opioid overdose. If administered in this situation, they will precipitate withdrawal symptoms but can save a life. Naloxone is orally inactive due to rapid inactivating allylic oxidation and phase 2 conjugation in the GI tract and liver, and it is only used as an emergency opioid agonist antidote. In contrast, naltrexone and nalmefene have sufficient bioavailability for oral administration, although only naltrexone is available in an oral (tablet) dosage form. This enhances its utility in opioid addiction and alcoholism recovery.

In combination with a sustained-release formulation of the atypical antidepressant bupropion, sustained-release naltrexone is finding use in the treatment of obesity in adult populations (142), and low-dose (4.5 mg/d) naltrexone has also shown efficacy in healing bowel ulcerations and enhancing immunity in active Crohn's disease (143). Other off-label uses for naltrexone hydrochloride include facilitating smoking cessation, posttraumatic stress disorder, pathologic gambling, bulimia, and cholestasis and uremia-induced pruritus.

Although currently only indicated to reverse the life-threatening events associated with opioid agonist overdose, the high oral bioavailability of nalmefene could make it potentially useful in recovery therapy for opioid and alcohol addiction. Clinical trials have shown that oral nalmefene was effective in preventing relapse in recovering alcoholics (144), and it blocked the μ agonist effects of fentanyl (145). Nalmefene has an extremely long half-life compared to naloxone (11 vs. 1 to 2 hours) because it dissociates very slowly from the μ receptor. For this reason, it also produces prolonged withdrawal symptoms when administered to opioid-dependent patients.

Methylnaltrexone Bromide

This quaternary analog of naltrexone is incapable of penetrating the blood-brain barrier, but it is highly effective in reversing the unwanted actions of opioid agonists in areas outside of the CNS, most importantly constipation (146,147). Dosed by patient weight and administered subcutaneously on an every-other-day basis, it restores bowel function in patients on chronic opioid analgesic therapy. Because it cannot distribute to the CNS, the risk of displacing opioids from central receptors, which would abolish analgesia and precipitate withdrawal, is essentially nonexistent. If needed, the dosing frequency can be increased to a maximum of once daily. Because

methylnaltrexone restores intestinal peristalsis, patients with bowel obstructions should not receive this drug until the blockage has been fully resolved.

Flexible μ Agonists (Fig. 20.14)

Meperidine Hydrochloride

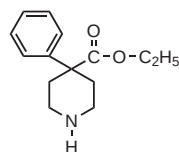
Meperidine can be viewed as the “stripped down” opioid agonist pharmacophore, 4-phenylpiperidine. In protonated form, meperidine's amino nitrogen will anchor the drug to the anionic μ receptor Asp residue, and the *N*-CH₃ will bind by hydrophobic forces to the μ receptor agonist site, albeit via a different binding mode (and potentially different hydrophobic residues) than the multicyclic opioids.

Because it is not immobilized in a fused polycyclic “body cast,” meperidine is structurally flexible and can adjust its conformation to fit the μ receptor in a way that is different from the multicyclics. Specifically, the phenyl ring at the piperidine C₄ can assume the more energetically stable equatorial conformation and bind to the μ receptor Trp²⁸³ residue that normally binds Phe⁴ of the enkephalin pentapeptides, rather than to the Trp²⁹³ that normally binds the enkephalin Tyr¹ residue. Therefore, a phenolic hydroxyl group in flexible opioids is neither required nor desired. Another major structural difference between meperidine and multicyclic opioids is that this flexible μ agonist is achiral.

The analgesic activity of meperidine is about one-tenth that of morphine. As a full μ agonist, it shows all typical μ agonist effects except inhibition of GI motility and cough (148). Meperidine has poor oral bioavailability and a variable absorption after intramuscular injection. With an elimination half-life of 3 to 4 hours, it is considered short-acting. It has found use in obstetrics because of its rapid onset and short duration. The view that meperidine has a low fetal respiratory depression liability has been challenged (149).

Once hailed as “the most widely used opioid analgesic in the United States” (150), meperidine has fallen out of favor, and many are advocating its restricted use (149,151). Meperidine undergoes CYP3A4-mediated *N*-dealkylation to a nor metabolite that has about half the analgesic potency of meperidine. An oral dose of meperidine generates more nor metabolite than an equivalent IV dose. The nor metabolite has an extended half-life of 14 to 21 hours in patients with normal renal function and up to 35 hours in patients with renal disease. The extended half-life is clinically important because normeperidine can induce neurologic side effects that are not consistently or reliably reversed by naloxone. The most serious of these adverse reactions is grand mal seizures secondary to an increase in central serotonin (5-HT) levels. Therefore, meperidine therapy must not be initiated and/or it should be promptly discontinued in patients exhibiting CNS excitation. The risk of seizure is considered high if the ratio of nor metabolite to parent drug is

greater than 1:1. Unfortunately, the dose of meperidine that can induce seizures in some patients overlaps with the dose commonly used in asymptomatic patients (260 to 540 mg/day), making it hard to predict when toxicity will manifest (151).



Normeperidine

The risk of serotonin syndrome–induced seizure, coma, and apnea (along with other potentially dangerous outcomes such as hypotension, cyanosis, and respiratory depression) is increased if meperidine is coadministered with monoamine oxidase inhibitors (MAOIs) (152). As might be expected, the risk of serious or fatal meperidine toxicity is significant if used in high doses, coadministered with cimetidine (which reduces meperidine clearance), or given to renally impaired patients (153,154). Urinary excretion of meperidine and normeperidine is pH dependent (154).

Like morphine and codeine, meperidine stimulates the release of histamine (155), although the mechanism does not appear to involve mast cell degranulation as is observed with the multicyclic analgesics (100). Meperidine can cause a precipitous drop in blood pressure, particularly in postsurgical patients or in those whose blood pressure is already compromised by disease or other hypotensive drugs.

The risk of potentially serious side effects and/or drug–drug interactions is prompting many hospitals to discourage and/or restrict meperidine use, especially in patient-controlled analgesia systems or when prolonged use is anticipated. Meperidine is still prescribed to treat postoperative or therapeutic hypothermia-related shivering (156), and it can be administered in low doses for short-term pain management. Treatment should be restricted to patients with normal renal function who have responded adversely to less toxic opioids, and the daily dose should be no greater than 600 mg. For more significant or prolonged pain, hydromorphone, morphine, or fentanyl is often recommended in place of meperidine.

Fentanyl Citrate

Fentanyl is a member of the anilidopiperidine class of opioid agonists and, as an *N*-phenethyl–substituted opioid structure, is a very sedative and euphoria-inducing analgesic with a potency approximately 50- to 100-fold that of morphine (157). The high activity of anilidopiperidines is due, in part, to their very high lipophilicity, which allows them to quickly penetrate the blood–brain barrier and concentrate in the CNS. Because they can leave the brain quickly too, their duration of analgesic action is short (30 to 60 minutes for fentanyl). They

are most commonly administered IV as adjuncts to anesthesia.

Transmucosal, transdermal, and nasal spray formulations of fentanyl are available to treat chronic pain, including the pain of cancer (158,159). The metered-dose nasal spray provides rapid delivery of dry fentanyl citrate powder that is readily absorbed in the highly vascularized nasal passages. The bioavailability of fentanyl by this route is 71%. The onset of analgesia is approximately 5 minutes, making it useful for as-needed use in breakthrough pain, and the duration of action is approximately 1 hour (159). The transdermal patch formulation (Duragesic) releases fentanyl free base at a rate that maintains therapeutic blood levels for 72 hours, providing a convenient and reliable mechanism for chronic pain patients to achieve analgesia at home (158).

The transmucosal formulations are used to treat breakthrough pain in opioid-tolerant cancer patients and in pediatric burn patients undergoing painful cleansings and dressing changes. Fentanyl citrate is incorporated into the buccal film (Onsolis) and lozenge on a stick (buccal lollipop, Actiq) dosage forms, which are placed or held between the cheek and gum. Approximately 25% of the dose will be absorbed into the bloodstream from the buccal cavity at the normal salivary pH of about 6.4 (158). The remaining drug, when swallowed, will be inactivated in the GI tract.

The effervescent buccal tablet formulation of fentanyl base (Fentora) contains an alkalinizing agent to raise the pH of the saliva and promote a faster and more complete absorption of the drug (~50% of the dose) (158). If there is a need to switch a patient from the lollipop or film to the effervescent tablet, the dose should be cut by 25% to 50% to avoid life-threatening toxicity (159). A sublingual tablet formulation of fentanyl base (Abstral) is also available which, when placed under the deepest sublingual area, disintegrates to release particles that cling to the mucosal surface, facilitating absorption (158). All transdermal and transmucosal dosage forms of fentanyl carry a black box warning related to unintentional overdose, toxicity, and intentional abuse.

Fentanyl is extensively biotransformed via CYP3A4-mediated *N*-dealkylation to an inactive metabolite and cannot be given orally (159). Because fentanyl is inactivated by this isoform, practitioners must be alert to the risk for potency-enhancing (CYP3A4 inhibitors or competitors) or attenuating (CYP3A4 inducers) interactions. Anilidopiperidines do not release histamine like meperidine and 6 α -hydroxylated multicyclics do, and there is no risk of histamine-mediated allergic responses when fentanyl is administered IV (155). The fentanyl adverse reaction of greatest concern is respiratory depression. This potentially fatal reaction is persistent and can outlast the action of antagonists used to reverse it (114,160). Between April 2005 and March 2007, the Centers for Disease Control and Prevention estimated that more than 1,000 people died from the inappropriate use of fentanyl (161).

Patients experiencing fentanyl-induced respiratory depression should be closely monitored until a sustained, normal pattern of breathing returns. The risk is significantly lower with sufentanil and alfentanil, as respiratory depression generally occurs only at doses higher than those needed to relieve pain or support anesthesia.

Sufentanil Citrate

Sufentanil has replaced fentanyl's phenethyl nitrogen substituent with 2-thienylethyl, another aralkyl moiety that has a comparable affinity for μ receptors. The 4-methoxymethyl substituent augments the lipophilicity provided by the thienyl ring, and together they enhance potency approximately 5- to 10-fold over fentanyl (e.g., 600- to 800-fold compared with morphine) (157). Despite the positive impact on central distribution of this enhanced lipophilicity, the stereochemistry and steric influence of the 4-substituent appear to be more important to sufentanil's potency than its other chemical properties (157). Sufentanil redistributes from brain to periphery within 17 minutes (much faster than fentanyl), resulting in shorter postanesthesia recovery times. A transdermal patch designed to release sufentanil over a week's time is in clinical trials and may soon be available for the treatment of chronic pain.

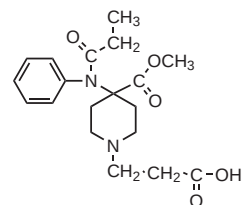
Alfentanil Hydrochloride

The potency-attenuating impact of alfentanil's tetrazolinone ring system has been described previously. While still 25-fold as active as morphine as an analgesic, at 0.25-fold fentanyl, it is the least potent of the currently marketed anilidopiperidine opioids (157). However, because the ratio of un-ionized to ionized conjugates in the bloodstream is significantly higher compared to fentanyl and sufentanil, alfentanil distributes and redistributes across the blood-brain barrier more rapidly than either of its predecessors (162). Its onset of action is approximately fourfold faster than fentanyl, and its use is reserved for very short surgical procedures requiring anesthesia.

Remifentanil Hydrochloride

Remifentanil has a unique and interesting chemistry (107,163,164). Instead of the traditional μ agonist-promoting *N*-methyl or aralkyl substituent, remifentanil's nitrogen substituent contains an electron-withdrawing ester that functions in a manner similar to alfentanil's tetrazolinone ring in promoting the lipophilic un-ionized conjugate at pH 7.4. This drug distributes rapidly across the blood-brain barrier, where it induces a highly sedative analgesia of very short duration. Unlike the hindered ester at C₄, the ester incorporated within the nitrogen substituent is very vulnerable to hydrolysis by serum esterases to the inactive carboxylic acid. This rapid inactivation in the bloodstream eliminates any concern of drug accumulation in patients with suboptimal liver or kidney function. Remifentanil is used by the IV route to induce and/or maintain anesthesia. Its biologic half-life

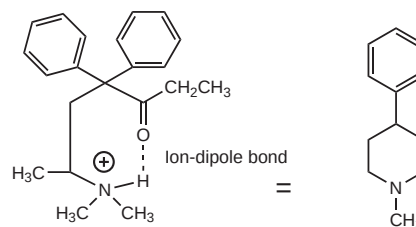
is a mere 3 to 10 minutes, which permits a rapid 3- to 5-minute onset of recovery from anesthesia.



Remifentanil carboxylic acid
(Inactive metabolite)

Methadone Hydrochloride

Methadone is one of only three marketed opioid analgesics that do not contain a piperidine ring. However, its highly flexible structure could approximate the 4-phenylpiperidine pharmacophore in vivo through the formation of an intramolecular ion-dipole bond. The *N*-methyl substituent accurately predicts its μ agonist activity profile, and the activity-attenuating impact of ketone reduction supports a receptor binding role for the C₃ carbonyl oxygen.



Methadone approximating the 4-phenylpiperidine pharmacophore

Methadone's lone chiral center, C₆, equates to the C₉ of morphine. While marketed as the racemic mixture, the *R*(-) isomer is responsible for methadone's analgesic action. The *S*(+) enantiomer antagonizes the NMDA receptor (165), which, although beneficial in neuropathic and opioid-resistant pain, may also be responsible for methadone's potentially serious cardiovascular toxicities such as a prolonged QT interval and torsades de pointes (83). The analgesic activity of methadone is shorter than its 15- to 60-hour terminal half-life would suggest, and it can induce respiratory depression that outlasts the analgesic phase. Methadone's naturally long half-life can be significantly increased in basified urine. Deaths from cardiac arrhythmia and respiratory depression have been noted, particularly as the drug is being introduced and/or the dose titrated. Dosing more frequently than once per day can allow drug accumulation, which sets the stage for serious and/or potentially fatal toxicity.

Methadone's prolonged duration is due to the generation of several active *N*-dealkylated and C₃-reduced (methadol) metabolites, some of which have long half-lives (Fig. 20.15). CYP2C19 catalyzes the stereoselective dealkylation of the active *R*(-) isomer of methadone and methadol, whereas CYP2B6 preferentially catalyzes

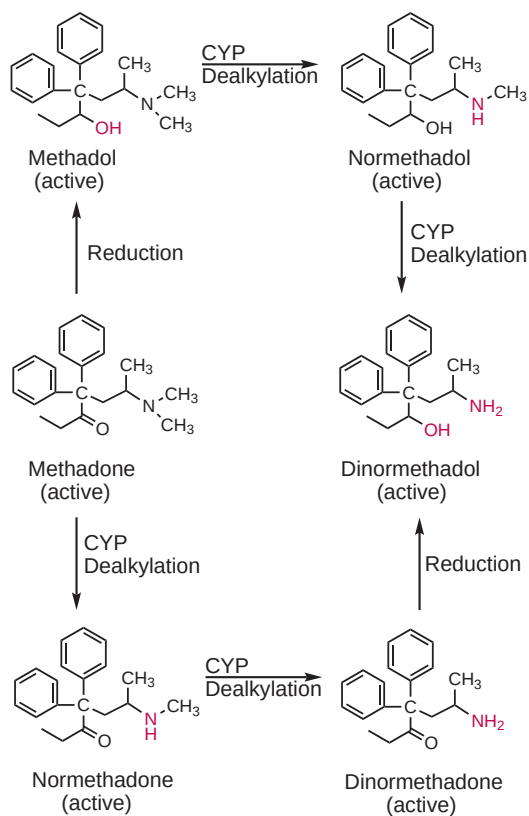


FIGURE 20.15 Methadone metabolism.

the dealkylation of the inactive *S*(+) isomers. CYP3A4 (and other isoforms) dealkylate both enantiomers with equal ease (166). While many believe that CYP3A4 is the major isoform involved in methadone dealkylation (83), others suggest that these other isoforms could be important players in the methadone biotransformation process (167). Because of the risk of serious/fatal cardiac and respiratory toxicity, pharmacists should be vigilant with regard to monitoring for metabolism-related interactions with CYP competitors, inhibitors, and inducers.

The *N*-dealkylated metabolites of methadone (but not methadol) can cyclize to form inactive pyrrolidine-based structures that are found in the urine of patients on methadone (Fig. 20.16) (168,169). These compounds are commonly known as EDDP and EDMP, which are acronyms for their chemical names. Methadone is incapable of generating a cyclic pyrrolidine metabolite because the *N,N*-dimethyl substitution pattern of the parent drug provides steric hindrance to nucleophilic attack by the un-ionized amine at the electrophilic carbonyl carbon. EDDP, and to a lesser extent, EDMP have recently been shown to be mechanism-based inhibitors of CYP2C19 (167). EDDP has an elimination half-life of 40 to 48 hours.

Methadone can be used orally (tablets, oral solution) or by injection to treat pain. Methadone is most commonly given by the oral route, but patients who cannot swallow oral dosage forms can “swish and spit” solutions of methadone that have been treated with an

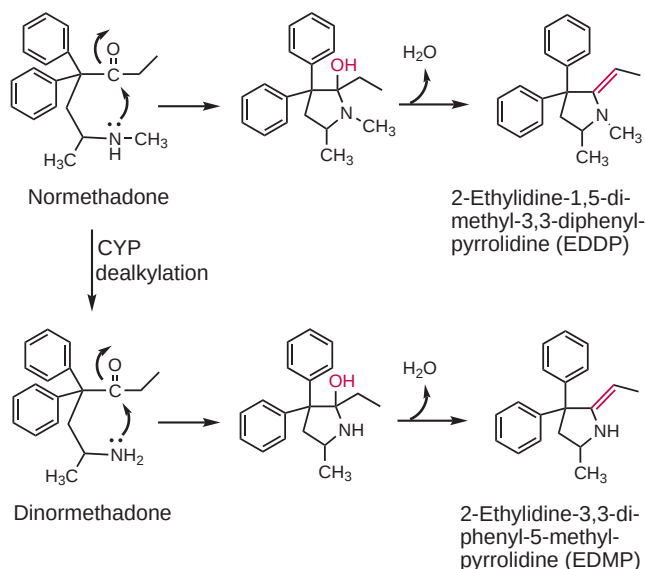


FIGURE 20.16 Formation of EDDP and EDMP.

alkalinizing agent to raise the pH of the saliva from 6.4 to 8.5. Holding the solution in the mouth for 2.5 minutes at a salivary pH of 8.5 results in the buccal absorption of 85% of the dose (99). First-pass metabolism would, of course, be avoided if methadone is given by the buccal route.

While certainly an effective analgesic, methadone has found its greatest use in addiction recovery (“methadone maintenance” or “methadone medication maintenance”) programs (170). Its high oral bioavailability, long duration of action, daily dosing regimen, slow tolerance development, and relative lack of physical dependence are definitely advantages in its use as a heroin (or other potent μ agonist) substitute (Table 20.2). Unlike buprenorphine, methadone used in addiction recovery is administered in strictly regulated, federal- and state-licensed methadone clinics. Patients are titrated upward from an initial 20- to 30-mg daily dose to a maintenance daily dose that is most commonly between 80 and 120 mg (83). Therapy commonly continues over 1 to 2 (or more) years. Compliance can be tracked by quantifying urinary levels of EDDP and calculating the EDDP:urine creatinine ratio. This analytical technique provides results related to the consumed quantity of methadone that are independent of patient hydration status and resistant to attempts to adulterate the sample with agents like soap, bleach, or methadone itself (171). When the time is right to discontinue methadone, any withdrawal symptoms experienced will generally be mild due to the prolonged duration of action of the drug and active metabolites. However, the patient will still be both tolerant to and dependent on methadone, which, unlike buprenorphine, is not a pseudoirreversible μ agonist. Therefore, the protocol for medically supervised drug discontinuation involves a “stair-step” dose-attenuation process, with reductions of less than 10% of the maintenance dose taking place no sooner than every 10 to 14 days.

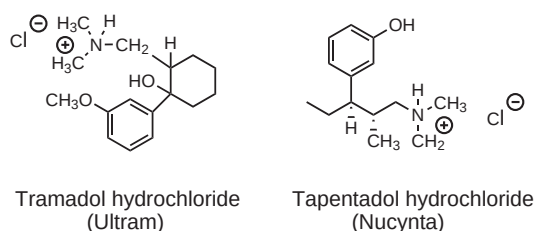


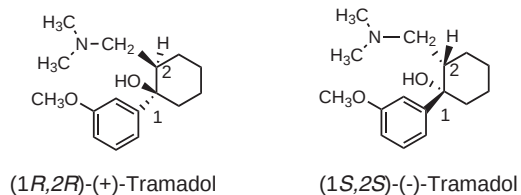
FIGURE 20.17 Dual-action opioids.

Dual-Action Opioids (Fig. 20.17)

Two novel, orally active analgesics on the U.S. market work by dual mechanisms of action: μ -receptor stimulation and inhibition of norepinephrine (NE) or 5-HT reuptake in the CNS (172). Inhibiting central monoamine reuptake in descending nociceptive pathways has been shown to alleviate chronic pain. NE reuptake inhibition appears to provide the stronger analgesic response. The rationale for the development of these dual-action analgesics is to lower dependence on μ -receptor stimulation for full analgesic efficacy, thus decreasing the risk of uncomfortable or dangerous μ agonist side effects. While the risk of respiratory depression, constipation, sedation, tolerance, dependence, and addiction liability are markedly low, all precautions associated with the use of μ -opioid analgesics are advised with the use of these dual-action pain-relieving agents.

Tramadol Hydrochloride

Tramadol has two asymmetric centers and is marketed as the racemic mixture of *cis* isomers. Each enantiomer, plus the *O*-dealkylated phenolic dextrorotatory metabolite, has a role to play in the activity profile of this unique analgesic. Specifically, the 1*S*,2*S*(-) isomer inhibits NE reuptake, the 1*R*,2*R*(+) isomer inhibits 5-HT reuptake, and the dextrorotatory isomer of the *O*-dealkylated metabolite stimulates μ -opioid receptors to produce analgesia sixfold greater than the parent drug (172–175). The μ agonist potency of *O*-dealkylated tramadol has been estimated at 10% of morphine after parenteral administration (173).



The mechanism of tramadol's analgesic action is complex and time-dependent (173,174). Over time, as the parent isomers are *O*-dealkylated to phenolic metabolites, the μ -receptor agonist component of activity increases while the monoamine reuptake inhibition mechanism decreases. As in codeine analogs, CYP2D6 catalyzes the *O*-dealkylation reaction, and patients deficient in this isoform will not experience the μ agonist component

of tramadol's action. Inactivating glucuronide or sulfate conjugation of the phenol will occur prior to urinary excretion. CYP3A4 and CYP2B6 catalyze inactivating *N*-dealkylation to secondary and primary amine metabolites (176).

Since the dextrorotatory (1*R*,2*R*) isomer of tramadol inhibits 5-HT reuptake, it has the potential to induce or exacerbate serotonin syndrome, a potentially life-threatening condition (172,177,178). The symptoms of serotonin syndrome can include agitation, coma, dangerous alterations in heart rate and blood pressure, hyperreflexia and loss of coordination, and GI distress. Tramadol also carries a risk of seizure, especially in patients taking drugs that increase central levels of monoamines (e.g., selective serotonin reuptake inhibitors [SSRIs], MAOIs, tricyclic antidepressants [TCAs]) or whose seizure risk threshold is otherwise reduced (179,180). Seizures have been documented in doses as low as 200 mg (which is close to the maintenance dose of 50 to 100 mg) and generally occur within 6 hours of drug ingestion (179). Withdrawal symptoms have been noted upon abrupt discontinuation of tramadol and, in addition to anticipated opioid withdrawal symptoms, can include paranoia, panic, sensory perception distortions, and hallucinations (172).

Tramadol is available in noninterchangeable immediate-release (every 4 to 6 hours) and extended-release (once daily) formulations. It is completely absorbed after oral administration and has a bioavailability of 75% (Table 20.2) (173). Drugs that increase central levels of 5-HT have been linked to suicide, and the tramadol manufacturer recently alerted the FDA and health care providers to the more stringent warnings related to suicide risk incorporated within the prescribing information for this dual-action analgesic (181,182).

Tapentadol Hydrochloride

Tapentadol was developed to couple time-independent μ -receptor agonism and selective NE reuptake inhibition and, in so doing, produce an analgesic with a consistently reliable therapeutic profile (183). Like tramadol, tapentadol has two chiral carbon atoms, but it is marketed as the pure 1*R*,2*R*(-) isomer (174). A phenolic drug, it is selective for the μ -opioid receptor, and its overall analgesic potency is estimated at one-half to one-third that of morphine despite having a 50-fold lower affinity for μ receptors. The potency discrepancy is explained by the synergistic mechanism of analgesic action (184). Tolerance to tapentadol develops slowly (174), and there is less physical dependence when compared to morphine and other μ agonists. When chronic therapy is to be concluded, tapering doses can minimize the risk of withdrawal symptoms (172,175).

Despite the apparent vulnerability of the *N,N*-dimethylamino substituent to CYP-mediated metabolism, only about 10% to 13% of a dose undergoes dealkylation via CYP2C9 and CYP2C19. Tapentadol is preferentially metabolized by glucuronic acid (55%) or sulfate (15%)

conjugation of the phenolic OH group. All tapentadol metabolites are inactive (175,183).

Like all μ agonists, tapentadol can induce respiratory depression, and the drug should be used with caution or avoided altogether in elderly or debilitated patients or in patients with respiratory disease (e.g., chronic obstructive pulmonary disease, asthma, sleep apnea). Inhibition of GI motility, while less frequent than is observed with oxycodone, is still common, and the drug should not be used in patients with compromised intestinal function (e.g., paralytic ileus). Nausea and vomiting can occur, but some claim the risk is less than with oxycodone (172,183).

Tapentadol should not be used in patients taking MAOI antidepressants, because the synergistic increase in circulating NE levels could precipitate a hypertensive crisis. There should be a minimum 14-day MAOI “wash-out period” before initiating tapentadol therapy (183).

While the risk of serotonin syndrome should be less than that observed with tramadol, NE reuptake inhibitors can also induce this potentially fatal reaction, especially if other drugs that increase central levels of 5-HT are being coadministered (183). Likewise, although this drug has not been shown to increase seizure risk, it was not evaluated in patients with seizure disorder. Practitioners should be mindful of the possibility for these serious adverse reactions and employ all appropriate precautions when using this drug.

Opioid-Based Antidiarrheals (Fig. 20.18)

Constipation is undoubtedly one of the most discomforting and, at times, serious complications of opioid analgesic use. However, medicinal chemists, following the old adage “When life hands you lemons, make lemonade,” have modified the structure of the flexible μ agonist

meperidine to limit blood–brain barrier penetration without compromising GI mobility inhibition (185). Three peripherally selective μ agonists, diphenoxylate, difenoxin, and loperamide, have enjoyed significant therapeutic use as antidiarrheals for years.

Diphenoxylate Hydrochloride and Difenoxin Hydrochloride

Of the three opioid-based antidiarrheals, diphenoxylate bears the closest resemblance to the parent meperidine structure. The only chemical difference between the two compounds is the nature of the nitrogen substituent; meperidine has the prototypical μ agonist-directing methyl group, while diphenoxylate has a more complex 3,3-diphenyl-3-cyanopropyl functional group. The ethylcarboxylate ester is readily cleaved by plasma esterases, yielding a carboxylic acid metabolite that is approximately fivefold as active as diphenoxylate as an antidiarrheal. The zwitterionic metabolite, which has been marketed as difenoxin, does not readily distribute into the CNS in therapeutic doses. In higher doses, the chemical reluctance to cross the blood–brain barrier is overcome, and central μ receptors can be stimulated.

For reasons both therapeutic (anticholinergic) and abuse related, diphenoxylate and difenoxin are only available in combination with atropine sulfate. If a potential abuser attempts to extract the antidiarrheal agent with the intent to inject and abuse, the atropine will be extracted and injected too, leading to nausea, weakness, sedation, and annoying peripheral effects such as dry mouth, blurred vision, and urinary retention. The diphenoxylate combination product is a Schedule V drug, indicating that there is a slight potential for abuse and/or addiction, whereas the difenoxin combination product is classified as Schedule IV (a medically useful drug with a low potential for abuse and/or addiction). If used alone, these antidiarrheals would be classified as Schedule II agents, indicating that they have a significant potential for abuse and/or addiction without the atropine “safety net.”

Loperamide Hydrochloride

Loperamide is a nonhydrolyzable amide analog of meperidine. Meperidine’s ethylcarboxylate ester has been replaced by a tertiary hydroxyl group, and the phenyl ring has been chlorinated at the *p* position. The nitrogen substituent is reminiscent of that incorporated into diphenoxylate and difenoxin, although a dimethylamide moiety has replaced the nitrile. While the highly lipophilic drug ($\log D_{\text{pH } 7.4} = 5.47$) acts as a full μ -receptor agonist, it lacks central μ agonist action because the efflux protein P-gp actively and extensively prevents drug uptake into the brain (186,187). This essentially abolishes its abuse potential and allows it to be available to patients without a prescription (188). Diphenoxylate, discussed earlier, is not a substrate for the P-gp efflux protein (189). In addition to and/or in

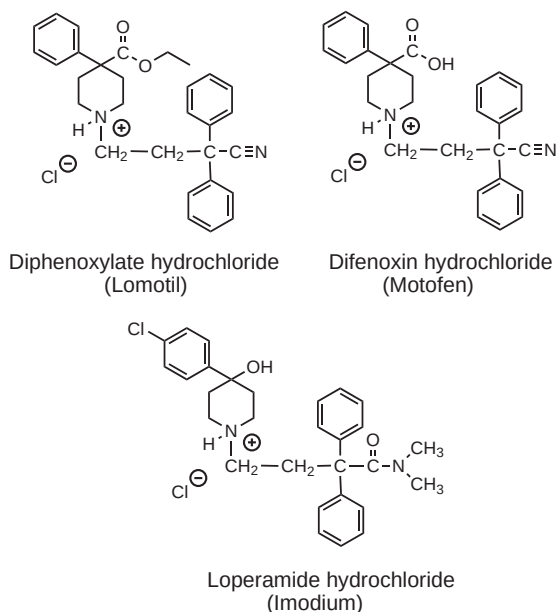


FIGURE 20.18 Opioid-based antidiarrheals.

concert with its μ agonist action, loperamide's GI motility inhibition has been linked to the μ agonist-mediated release of 5-HT (190) and calcium channel antagonism (191,192).

N-Demethylation of loperamide's tertiary amide group is catalyzed by CYP3A4 and CYP2C8 (193). Significant first-pass metabolism reduces oral bioavailability to approximately 40%. The desmethyl metabolite does not bind to μ receptors and is inactive as an antidiarrheal (194). Because loperamide also inhibits CYP3A4, there is risk of drug-drug interactions between this agent and other CYP3A4 substrates, although few have been reported in the literature. Loperamide has been compared favorably to diphenoxylate with regard to potency, duration and specificity of action, and safety margin (185).

NEUROPATHIC PAIN

Neuropathic pain generally arises from nerve damage due to physical injury or disease. A recent definition of neuropathic pain is "pain arising as a direct consequence of a lesion or disease affecting the somatosensory system" (8). It is estimated that 90% of neuropathic pain arises from peripheral injuries or diseases (e.g., postdiabetic neuralgia, postherpetic neuralgia, posttraumatic neuralgia, postpolio neuralgia, fibromyalgia, iatrogenic injuries). The remaining 10% is of central origin (e.g., stroke, Parkinson disease, multiple sclerosis, phantom limb pain) (195). This section will focus on peripheral neuropathic pain.

Nociceptive Versus Neuropathic Pain

Nociceptive pain is pain produced through disease or injury of nonneural tissues. Treatment of this kind of pain involves common analgesics (e.g., nonsteroidal anti-inflammatory drugs, opiates), and the pain subsides as the injury heals. As defined earlier, neuropathic pain emanates from lesions of the nervous system. There may be no observable injury, and the pain often increases in severity over time. However, regardless of the source of the lesion, the pain must involve the nociceptive pathways. The inverse of this is not true in that all lesions of the nociceptive pathways do not necessarily induce pain. Pain due to neuropathic processes generally has one or more of the following features: 1) widespread pain that is not explainable by nonneural injuries, 2) evidence of sensory deficit (e.g., numbness), 3) burning pain, 4) pain to light stroking of the skin, and 5) attacks of pain without seeming provocation (196).

Another descriptor of neuropathic pain is *allodynia* ("other power")—pain produced by a stimulus that does not normally produce pain.

Mechanisms of Neuropathic Pain

The mechanisms resulting in the characteristic persistence of neuropathic pain following nerve injury involve both functional and structural alterations in the nervous system. Figure 20.19 illustrates the involvement of both the primary afferent neurons and the CNS (2). Physiologic and transcriptional changes in the peripheral primary afferent sensory neurons that take place after the injury contribute to neuropathic pain. The alterations in

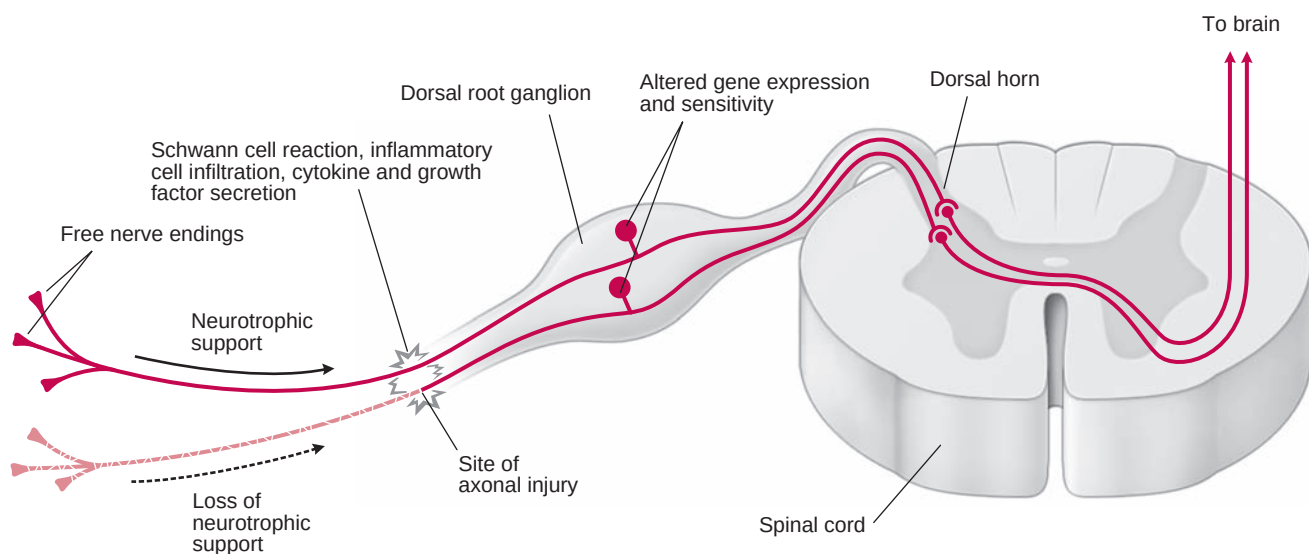


FIGURE 20.19 Schematization of neuropathic pain. Nerve injury results in a combination of negative signals and positive signals that alter the physiology of the nociceptive system. The loss of neurotrophic support alters gene expression in the injured nerve fiber, whereas the release of inflammatory cytokines alters gene expression in both the injured and adjacent uninjured nerve fibers. These changes in gene expression can lead to altered sensitivity and activity of nociceptive fibers and, thus, to the continued perception of injury that is characteristic of neuropathic pain. (Griffin RS, Woolf CJ. Pharmacology of analgesia. In: Golan DE, Tashjian AH Jr, Armstrong E, et al., eds. Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy, 2nd Ed. Philadelphia, PA: Lippincott Williams & Wilkins, 2008:272, with permission.)

neuronal function are induced by the release of inflammatory cytokines by macrophages. Alterations in the expression pattern of voltage-gated sodium channels are also seen in the injured sensory neurons. Specifically, $\text{Na}_v1.8$ and $\text{Na}_v1.9$ are downregulated, whereas $\text{Na}_v1.3$ is upregulated. The $\text{Na}_v1.3$ channels are normally not detectable in the primary sensory neurons, and their accelerated recovery from inactivation is thought to contribute to neuropathic pain. The efficacy of sodium channel blockers (e.g., local anesthetics and carbamazepine) in the treatment of trigeminal neuralgia is attributed to their role in the mechanism of neuropathic pain (2).

Structural changes caused by nerve injury also contribute to neuropathic pain. After peripheral nerve injury, there is an excitotoxic loss (via excessive stimulation by neurotransmitters) of inhibitory neurons in the dorsal horn. The resulting disinhibition contributes to the heightened pain sensitivity. In many respects, it is the loss of neuronal structures and supports that gives rise to the multiple targets and approaches to drug therapy (197). Although actual mechanisms in many cases are unknown, the preferred first-line medications either 1) support the inhibited nociceptive pathways modulated by NE and 5-HT (TCAs and dual-acting 5-HT-NE reuptake inhibitors), or 2) block the release of transmitters from the afferent peripheral neurons by binding to the $\alpha_2\delta$ subunit of the voltage-gated calcium channels or the voltage-gated sodium channels ($\text{Na}_v1.3$) (197,198).

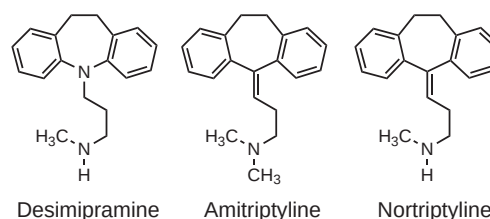
Treatment of Neuropathic Pain

The Neuropathic Pain Special Interest Group of the International Association for the Study of Pain has developed evidence-based guidelines for the treatment of neuropathic pain (199). These guidelines consider clinical efficacy, adverse effects, effects on the health-related quality of life, convenience of dosing, and cost.

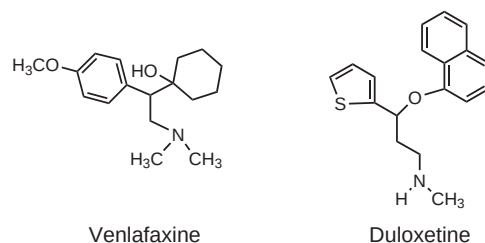
First-Line Medications

ANTIDEPRESSANTS (ALSO SEE CHAPTER 18) Numerous studies have shown the efficacy and beneficial effects of TCAs in the treatment of neuropathic pain from many causes, leading to off-label use in these conditions. Antidepressants most often used in neuropathic pain treatment include the TCAs and dual-acting 5-HT-NE reuptake inhibitors (198,200). The tertiary TCAs (amitriptyline, imipramine) are characterized by their ability to inhibit the reuptake of both 5-HT and NE from presynaptic terminals as a result of *N*-monodealkylation to secondary TCAs (Chapter 18), and the secondary TCAs/5-HT-NE reuptake inhibitors (desipramine, nortriptyline) selectively inhibit the reuptake of NE. The TCAs relieve pain independent of their antidepressant effect. Of the TCAs, the tertiary TCAs have been shown to be more effective for painful diabetic neuropathy, nerve injury pain, fibromyalgia, and spinal cord injury pain due to their dual action. Although the presence of depression is not necessary for their use or efficacy as neuropathic analgesics, these agents are frequently used in comorbid patients suffering

with neuropathic pain. The anticholinergic side effects associated with these TCA structures lead to a preference for the secondary amine derivatives (e.g., desipramine or nortriptyline). The cardiac toxicity associated with these compounds also limits their use. Studies have shown the SSRIs to have a weak analgesic effect.



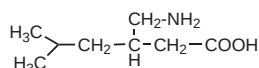
Duloxetine (Cymbalta) and venlafaxine (Effexor) are potent reuptake inhibitors of both 5-HT and NE and are effective not only for the treatment of major depressive disorder, but also as neuropathic analgesics for diabetic neuropathic pain and fibromyalgia, without the adverse effects of the TCAs. Duloxetine is generally well tolerated, and is approved by the FDA specifically for the treatment of diabetic peripheral neuropathic pain. The drug has been shown to be superior to placebo at doses of 60 and 120 mg/d in treating diabetic peripheral neuropathic pain (201). Its analgesic effect for reducing 24-hour average pain is seen early, typically within the first week, as contrasted with its antidepressant effect. Its efficacy is comparable to that of the tertiary TCAs commonly used for the treatment of diabetic peripheral neuropathic pain. The standard approach to treatment of diabetic peripheral neuropathy has been to use a TCA as a first-line trial of treatment. Duloxetine was more effective than pregabalin in producing at least a 50% reduction in pain, whereas treatment with pregabalin resulted in a greater proportion of patients with a 30% to 50% reduction in pain. Patients taking duloxetine would be less likely to have a partial pain response and require additional therapy with other costly medications. Therefore, duloxetine is an acceptable initial or alternative treatment for patients with diabetic neuropathic pain (201).



CALCIUM CHANNEL $\alpha_2\delta$ LIGANDS (ALSO SEE CHAPTER 21)

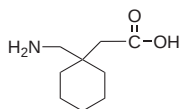
Pregabalin (Lyrica) Pregabalin binds to the $\alpha_2\delta$ subunit of the voltage-gated calcium channels in the CNS and inhibits release of several neurotransmitters. It has a use in epilepsy as an adjunct in the control of partial seizures. However, its primary use is in the treatment of neuropathic pain in fibromyalgia, diabetic

neuropathies, and postherpetic neuropathy. Pregabalin is well absorbed orally and follows linear pharmacokinetics. Its metabolism in humans is negligible, leading to 90% being recovered in urine as unchanged pregabalin. The *N*-methylated derivative is the major metabolite, and accounts for only 0.9% of the dose. The lack of interaction with metabolizing enzymes and the absence of protein binding leads to essentially no drug interactions. However, synergism with other CNS depressants may be seen.



Pregabalin

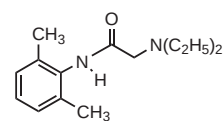
Gabapentin (Neurontin) Gabapentin is listed in many reviews as a calcium channel $\alpha_2\text{-}\delta$ subunit ligand. However, studies of binding to most major neurotransmitter receptor sites and voltage-gated ion channels have shown negative results (202). In addition, it does not affect GABA_A or GABA_B binding and is not converted to GABA or a GABA agonist or antagonist. Gabapentin's primary use is as an adjunct in the management of partial seizures in children and adults. Its use in neuropathic pain is off-label for diabetic neuropathic pain, fibromyalgia, and other chronic pain conditions. Gabapentin's bioavailability is not proportional to the dose, and it is not appreciably metabolized in humans.



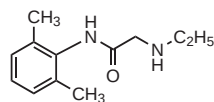
Gabapentin

LOCAL ANESTHETICS (ALSO SEE CHAPTER 16).

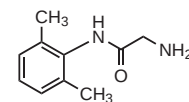
Lidocaine Patch (Lidoderm) The lidocaine patch is a transdermal dosage form containing 5% lidocaine and is effective in the treatment of postherpetic neuropathic pain. It is also useful in the treatment of allodynia from different types of neuropathic pain. It is a more chemically and biochemically stable amide type of local anesthetic, and like all other local anesthetics, acts as a blocker of voltage-gated sodium channels, shutting down the ion fluxes necessary for the propagation of neuronal signals. As might be expected, it is more useful in localized, as opposed to central, neuropathic pain. It is not known whether lidocaine is metabolized in the skin. Hepatic metabolism produces two dealkylated metabolites, monoethylglycinexylidide and glycinexylidide, that are active but less potent than lidocaine.



Lidocaine



MEGX



GX

Second-Line Medications

This group of medications specifically involves the dual-action analgesic tramadol and μ -opioid analgesics and is recommended for use in patients who have not responded to first-line medications. They may be recommended as first-line medications in cases of acute neuropathic pain, neuropathic pain due to cancer, and central neuropathic pain.

TRAMADOL (FIG. 20.17) As discussed earlier in this chapter, tramadol is a dual-action analgesic that acts through inhibition of neurotransmitter (5-HT and NE) reuptake and μ -receptor agonism. It has shown efficacy in some neuropathic pain conditions (199). To some extent, its adverse effects match other opioids, but its effect on the reuptake of 5-HT can interact with other selective 5-HT-NE reuptake inhibitors used in neuropathic pain treatment, leading to potential toxicity.

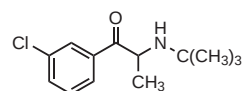
MORPHINE, OXYCODONE, AND METHADONE (FIGS. 20.11 AND 20.14)

A number of clinical trials have compared the μ agonists morphine, oxycodone, and methadone with the first-line agents tricyclic antidepressants gabapentin and pregabalin and found them to be as effective (199,203). However, concerns about their long-term use and potential for abuse have placed them in the second-line category.

Third-Line Medications

This group of compounds should be reserved for patients who do not respond well to first- and second-line medications. The criteria for placement in this category is that they have shown activity in only one randomized clinical trial or have presented inconsistent results in multiple trials (199).

BUPROPION (WELLBUTRIN, ZYBAN) (ALSO SEE CHAPTER 18)

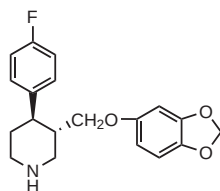


Bupropion

Bupropion is an antidepressant with weak reuptake inhibition of NE and dopamine. It showed neuropathic pain

efficacy in two clinical trials of the extended-release dosage form. More studies need to be done to further prove efficacy (204,205).

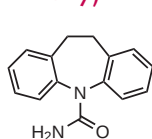
PAROXETINE (PAXIL) (ALSO SEE CHAPTER 18)



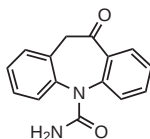
Paroxetine

Paroxetine is an SSRI antidepressant that may have analgesic properties. In a small number of clinical trials against diabetic neuropathic pain, it has shown efficacy greater than some first-line medications. However, more studies need to be done before it can be considered to be a first-line medication.

CARBAMAZEPINE (TEGRETOL) AND OXCARBAPINE (TRILEPTAL) (ALSO SEE CHAPTER 17)



Carbamazepine

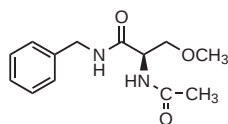


Oxcarbazepine

Carbamazepine and oxcarbazepine are both anticonvulsants. Oxcarbazepine is a derivative of carbamazepine with reduced toxicity. Carbamazepine has been studied in the treatment of postherpetic neuralgia, and oxcarbazepine has been studied in diabetic neuropathic pain. Inconsistent results in both cases lead to the need for further clinical study before any therapeutic recommendation on their use in neuropathic pain can be made.

Future Medications in the Treatment of Neuropathic Pain

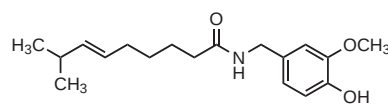
LACOSAMIDE (VIMPAT) (ALSO SEE CHAPTER 17)



Lacosamide

Lacosamide is a new anticonvulsant that acts as a slow inactivator of sodium channels. Chemically, it is *N*²-acetyl-*N*-benzyl-D-homoserinamide. Lacosamide has been studied in diabetic neuropathic pain and has presented evidence of efficacy. However, some later trials were inconclusive. Additional studies may provide the level of positive results needed for it to enter the mainstream of neuropathic pain treatments (198).

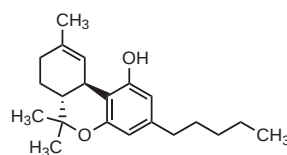
CAPSAICIN PATCH (8%) (QUTENZA)



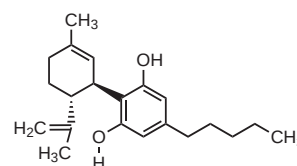
Capsaicin

Capsaicin is a natural compound isolated from Solanaceae family (chili peppers). The compound used in the patch is a synthetic equivalent of the natural substance. Capsaicin is an agonist for the transient receptor potential vanilloid 1 receptor (TRPV1), which is an ion channel-receptor complex expressed on nociceptive nerve fibers in the skin. Topical administration of capsaicin causes an initial enhanced stimulation of the TRPV1-expressing cutaneous nociceptors that may be associated with painful sensations. This is followed by pain relief thought to be mediated by a reduction in TRPV1-expressing nociceptive nerve endings. The efficacy of this patch in the treatment of postherpetic pain was shown in two multicenter studies. The activity in HIV neuropathic pain was inconsistent. The long-term effects of this treatment are not yet known (198).

CANNABINOIDS



Δ^9 -THC



Cannabidiol

The development of analogs of Δ^9 -tetrahydrocannabinol (Δ^9 -THC) and cannabidiol, the two major components from *Cannabis sativa*, has brought forth some interesting agonists of the cannabinoid type 1 (CB_1) and type 2 (CB_2) receptors (206). The CB_1 receptor has been shown to play a role in peripheral pain. Although many studies have established an analgesic activity of cannabinoids on neuropathic pain, there have not been enough controlled trials comparing them with other medications known to be effective.

Medical Food for Diabetic Neuropathy

METANX Metanx is a medical food under FDA Title 21. It is comprised of 6(*S*)-5-methyl-tetrahydrofolate calcium (3 mg), pyridoxal-5'-phosphate (35 mg), and methylcobalamin (methyl- B_{12}) (2 mg). All three of these ingredients are provided in their biologically active form and are indicated for the distinct nutritional requirements of patients with endothelial dysfunction with loss of protective sensation and neuropathic pain associated with diabetic peripheral neuropathy (207). Metanx relieves the numbness, tingling, and burning pain associated with diabetic neuropathic pain.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Jill T. Johnson, PharmD, BCPS

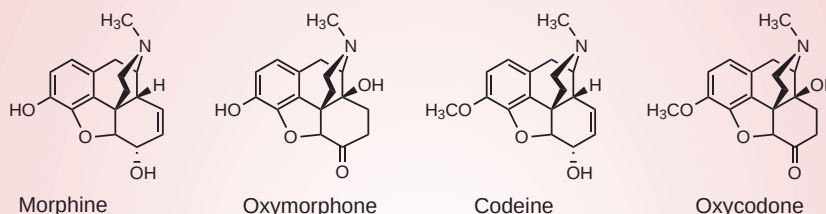
The pharmacist recommended IV hydration and IV hydromorphone 0.6 mg every 3 hours on a schedule with IV hydromorphone 0.2 mg every 3 hours as needed for breakthrough pain. The pharmacist avoided recommending meperidine due to the toxic metabolite, normeperidine, which has a longer duration but does not offer analgesia but instead puts the patient at risk for CNS stimulation including seizures. Although the pain medication was not enough at first, the dose was titrated upward to 1.2 mg every 3 hours. By the third day, the hydromorphone was changed to oral 2 mg every 4 hours. The patient was discharged with his pain almost completely resolved on day 5. The patient scored his pain severity as 2 out of 10. He was given a

prescription for oxycodone 5 mg/acetaminophen 500 mg, one tablet every 6 hours as needed for pain. He was given 5 days of medication. He was also counseled to continue taking folic acid and hydroxyurea.

Chemical Analysis

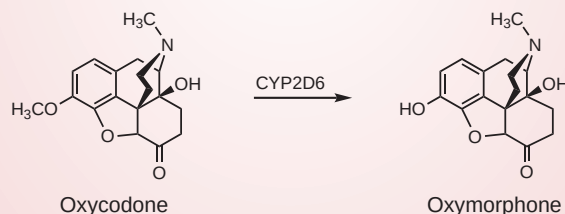
Victoria Roche and S. William Zito

Morphine and codeine are naturally occurring opium alkaloids. The 6 α -OH group found on each is associated with histamine release, and allergic responses to these analgesics can include rash, pruritus, and lowered blood pressure. Codeine is more problematic in this regard than morphine, especially if given parenterally. Opioids with a 6-keto group, such as oxymorphone and oxycodone, do not induce allergic reactions of this type.



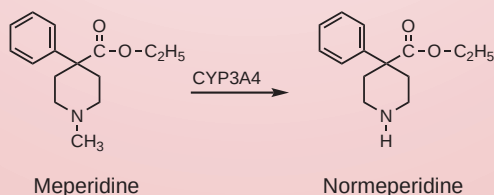
The N-CH₃ substituent on all four structures documents that they selectively exert their analgesic effect at the μ receptor. Oxymorphone is approximately 10 to 12 times more potent than morphine as an analgesic due to the 7,8-dihydro-6-one C ring and the 14 β -hydroxy substituent, both of which engage in high-affinity H-bonds with μ -receptor residues. Oxycodone is a prodrug of oxymorphone and requires CYP2D6-mediated activation to the essential

C₃ phenol. Unless JB is a poor CYP2D6 metabolizer, he should get the analgesic relief he requires from this agent, as it is approximately as potent as morphine. The oral bioavailability of oxycodone is significantly higher than oxymorphone because the methoxy ether blocks glucuronide conjugation in the gut and liver. This inactivating phase 2 reaction is the major reason why most phenolic opioids are given parenterally, sublingually, or via nasal spray.



Avoiding meperidine was a wise decision. This flexible μ -opioid has an analgesic potency about one-tenth that of morphine. It is N-dealkylated by CYP3A4 to a toxic metabolite (normeperidine) that has only about half the analgesic action of the parent structure and can induce grand mal seizures. The actions of normeperidine are long lasting and not reversed by naloxone, the "gold standard" opioid antagonist. The risk of seizure is high if the ratio of normeperidine to meperidine exceeds 1.0, and it should not be used in renally impaired patients. The dose of

meperidine that commonly induces seizures overlaps with that which elicits analgesia, making it very difficult to predict who will fall prey to this serious adverse reaction. In addition, like morphine and codeine, meperidine stimulates the release of histamine, and dangerous hypotensive episodes have occurred following its administration. The use of meperidine is generally restricted to patients with normal renal function whose pain cannot be controlled with less toxic agents or in the management of anesthesia-related shivering.



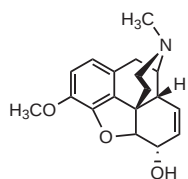
CASE STUDY

Victoria Roche and S. William Zito

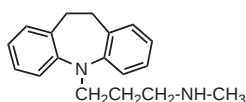
EJ is a 27-year-old woman who currently works as clerk in a local discount megastore. Although she's managing fairly well now, her life has not been an easy one. She has suffered from depression serious enough to warrant drug therapy from the time she turned 16. Initially placed on an SSRI, she was switched to desipramine, a norepinephrine selective reuptake inhibitor, after the suicidal ideation she experienced while taking the SSRI prompted an unsuccessful attempt to end her life. During her young adulthood, EJ's depression was complicated by bulimia, and she has experienced myocardial damage from her years of bingeing and purging. She had an emergency appendectomy at age 22, and she remembers suffering a bad reaction to the codeine she was given during the later part of her recovery,

which included an intensely itchy rash, low blood pressure, and visual hallucinations of dozens of multicolored parrots flying around her hospital room. She also remembers that the codeine seemed to do nothing for her pain.

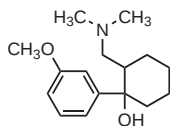
EJ is now facing yet another health challenge. After months of painful and hemorrhagic periods, she was diagnosed with a benign ovarian cyst and has been scheduled for surgery to remove it. If all goes as planned, she will stay in the hospital for 1 day before being discharged. As the pharmacist consulting with the surgical team, you are presented with EJ's medical history and asked to make a recommendation on postsurgical analgesia. You initiate a risk:benefit ratio of the following three options.



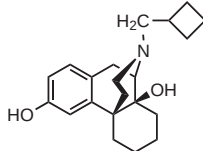
Codeine



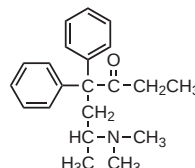
Desipramine



1



2



3

1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.

2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 21

Cardiac Agents: Cardiac Glycosides, Antianginal, and Antiarrhythmic Drugs

AHMED S. MEHANNA

Drugs Covered in This Chapter*

CARDIAC GLYCOSIDES POSITIVE INOTROPIC DRUGS

- Deslanatoside C (BP)
- Digitalis products
- Digitoxin (BP)
- Digoxin
- Lanatoside C (BP)
- Metildigoxin
- Ouabain (BP) (*G-strophanthin*)

NONGLYCOSIDIC POSITIVE INOTROPIC DRUGS

- Dobutamine
- Enoximone
- Inamrinone
- Milrinone
- Olprinone
- Pimobendan
- Vesnarinone

ANTIANGINAL NITRATES AND NITRITES

- Amyl nitrite (isoamyl nitrite)
- Erythryl tetranitrate
- Isosorbide dinitrate

- Nesiritide citrate
- Nicorandil
- Nitroglycerin
- Pentaerythritol tetranitrate

NITRIC OXIDE DONOR

- Molsidomine

CALCIUM CHANNEL BLOCKERS

- Amlodipine
- Bepridil
- Diltiazem
- Felodipine
- Nicardipine
- Nifedipine
- Nitrendipine
- Ranolazine
- Verapamil

ANTIARRHYTHMIC AGENTS

- Class IA
 - Disopyramide
 - Procainamide
 - Quinidine

- Class IB
 - Lidocaine
 - Mexiletine
 - Phenytoin
 - Tocainide
- Class IC
 - Encainide
 - Flecainide
 - Moricizine
 - Propafenone
- Class II
 - Propranolol
- Class III
 - Amiodarone
 - Bretylium
 - Dofetilide
 - Dronedarone
 - Ibutilide
 - Sotalol
- Class IV
 - Bepridil
 - Diltiazem
 - Verapamil

Abbreviations

ATP, adenosine triphosphate

ATPase, adenosine triphosphatase

AUC, area under the curve

AV, atrioventricular

cAMP, cyclic adenosine monophosphate

cGMP, cyclic guanosine monophosphate

CHF, congestive heart failure

COMT, catechol-*O*-methyltransferase

CYP, cytochrome P450

FDA, U.S. Food and Drug Administration

I_{Na}, sodium current

NO, nitric oxide

*Drugs listed include those available in the United States and outside the United States; drugs with the BP (British Pharmacopeia) designation are available only in Canada and countries other than the United States. Drugs in italics are not approved by the FDA.

Abbreviations (Continued)

PDE₃, phosphodiesterase 3
PDE₅, phosphodiesterase 5

P-gp, P-glycoprotein
SA, sinoatrial

SAR, structure–activity relationship

SCENARIO

Judy Cheng, PharmD

JS is a 58 year-old white man with a medical history of hypertension. He presented to the clinic today complaining of palpitation and dizziness for the past week. Electrocardiogram shows an irregular rhythm (i.e., atrial fibrillation) of 110 beats per minute. JS' blood pressure is 160/95 mmHg. He is diagnosed with new onset atrial

fibrillation. His current medications include lisinopril 5 mg daily, but JS has a significant history of medication nonadherence.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

Heart diseases are grouped into three major disorders: cardiac failure or contractile dysfunction (congestive heart failure), ischemic heart disease (with angina as its primary symptom), and cardiac arrhythmia.

DRUGS FOR THE TREATMENT OF HEART FAILURE

Congestive Heart Failure

Cardiac failure can be described as the inability of the heart to pump blood effectively at a rate that meets the needs of metabolizing tissues. This is the direct result of reduced contractility of the cardiac muscles, especially those of the ventricles, which causes a decrease in cardiac output, increasing the blood volume of the heart (hence the term “congested”). As a result, systemic blood pressure and renal blood flow are both reduced, which often leads to the development of edema in the lower extremities and the lung (pulmonary edema) as well as renal failure. A group of drugs known as the cardiac glycosides were found to reverse most of these symptoms and complications.

Drugs for the Treatment of Congestive Heart Failure

CARDIAC GLYCOSIDES: POSITIVE INOTROPIC DRUGS Cardiac glycosides are an important class of naturally occurring drugs, the actions of which include both beneficial and toxic effects on the heart. Their desirable cardiotonic action is of particular benefit in the treatment of congestive heart failure (CHF) and associated edema, and their preparations have been used as medicinal agents, as well as poisons, since 1500 BC. This dual application serves to highlight the toxic potential for this class of life-saving drugs. Despite the extended use and obvious therapeutic benefits of the cardiac glycosides, it was not until the famous monograph by William Withering in 1785, “*An Account of the Foxglove and Some of Its Medical Uses*,” that cardiac glycoside therapy started to become more standardized and rational (1–3). The therapeutic use of purified cardiac glycoside preparations has occurred only over the last century. Cardiac

glycosides represent one of the most important drug classes available to the physician for the treatment of CHF.

Chemistry of the Cardiac Glycosides Cardiac glycosides and other similar glycosides are composed of two portions: the sugar moiety and the nonsugar (aglycone) moiety.

Aglycones. The aglycone portion of the cardiac glycosides is a steroid nucleus with a unique set of fused rings, which makes these agents easily distinguished from other steroids. Rings A-B and C-D are cis fused, whereas rings B-C have a trans configuration. Such ring fusion gives the aglycone nucleus of cardiac glycosides the characteristic “U-shape,” as shown in Figure 21.1. The steroid nucleus also carries, in most cases, two angular methyl groups at C-10 and C-13. Hydroxyl groups are located at C-3, the site of the sugar attachment, and at C-14. The C-14 hydroxyl is normally unsubstituted. However, additional hydroxyl groups can be found at C-12 and C-16; the presence or absence of hydroxyl groups distinguishes the important differences in the genins: digitoxigenin, digoxigenin, and gitoxigenin (Fig. 21.2). These additional hydroxyl groups have significant impact on the partitioning and pharmacokinetics for each glycoside, to be discussed later. The lactone ring at C-17 is another major structural

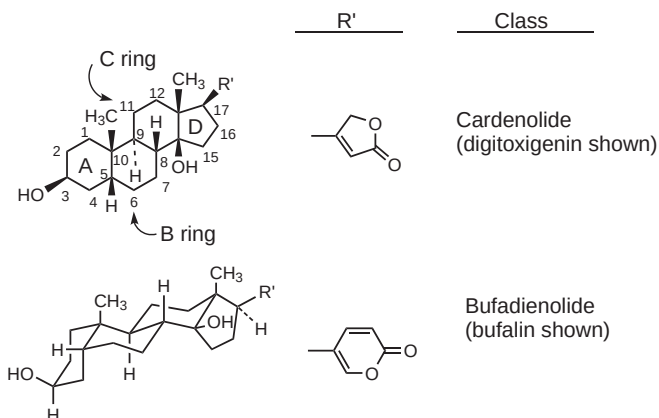


FIGURE 21.1 Cardenolide and bufadienolide aglycones.

CLINICAL SIGNIFICANCE

Digoxin, nitrates, and many antiarrhythmics are among the oldest cardiovascular agents around and still play an important role in managing different cardiovascular diseases. Cardiac glycosides such as digoxin have been used for decades for the management of heart failure as well as slowing down ventricular rate in patients with atrial fibrillation. Digoxin remains the only oral positive inotropic agent to date. Despite that, many questions still remain regarding the precise mechanism of action (beyond its positive inotropic effect), exactly what the most optimal therapeutic range should be, as well as what patient population will benefit the most from digoxin in terms of mortality and morbidity.

The role of nitrates in the management of angina is well established. However, the development of nitrate tolerance sometimes limits its long-term clinical benefits. Newer antianginals being developed have tried to overcome this issue by working as a direct nitric oxide donor (e.g., mosidomine). Others were developed to work through a completely different mechanism (e.g., ranolazine, a sodium channel blocker). Unlike other antianginals, ranolazine does not affect blood pressure and heart

rate. It can benefit patients whose blood pressure and heart rate may be too low or slow to tolerate other antianginals.

In recent years, many antiarrhythmic agents have given way to implantable defibrillation and surgical ablation for potentially life-threatening arrhythmias. However, they are still commonly used as adjunct therapy. Antiarrhythmic agents, although effective in controlling arrhythmia, also possess serious and potential lethal adverse events, and some have numerous significant clinical drug interactions. Newer antiarrhythmic agents continued to be developed in an attempt to minimize the risk of these severe toxicities. For example, the development of dronedarone, by eliminating the iodine moiety from amiodarone, significantly reduces clinical incidence of development of thyroid disorders. Understanding the medicinal chemistry of these agents will allow pharmacists to understand how the structures of these agents contribute to their efficacy and toxicity and to make the most optimal choices for individual patient scenarios.

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feature of the cardiac aglycones. The size and degree of unsaturation of the lactone ring vary with the source of the glycoside. In most cases, the cardiac glycosides of plant origin, also called cardenolides, possess a five-membered, α,β -unsaturated lactone ring, whereas those derived from animal origin, the bufadienolides, possess a

six-membered lactone ring with two conjugated double bonds (usually referred to as α -pyrone).

Sugars. The hydroxyl group at C-3 of the aglycone portion is conjugated to a monosaccharide or a polysaccharide with β -1,4-glucosidic linkages. The number and identity of sugars vary from one glycoside to another, as detailed subsequently. The most commonly found sugars in the cardiac glycosides are D-glucose, D-digitoxose, L-rhamnose, and D-cymarose (Fig. 21.3). These sugars exist predominately in the cardiac glycosides in the β -conformation. In some cases, the sugars exist in the acetylated form. The presence of an O-acetyl group on a sugar greatly affects the lipophilic character and pharmacokinetics of the entire glycoside, as subsequently discussed.

Sources and Common Names of Cardiac Glycosides. The cardiac glycosides occur mainly in plants and, in rare cases,

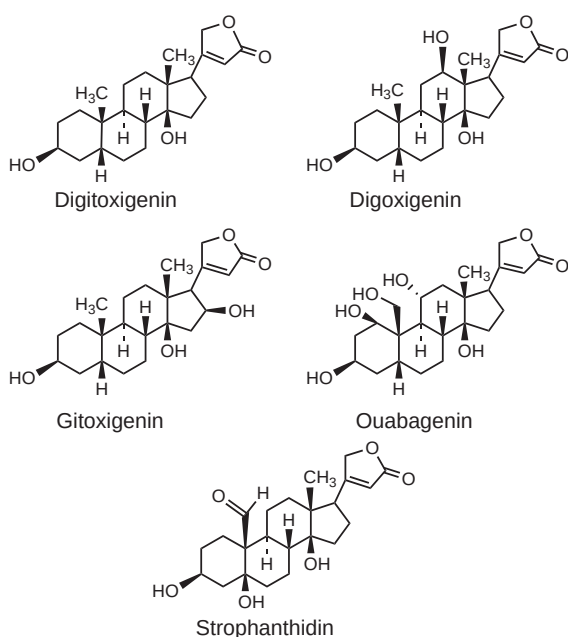


FIGURE 21.2 Major cardenolide aglycones.

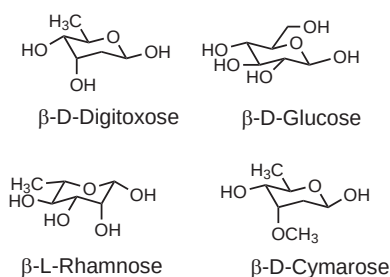
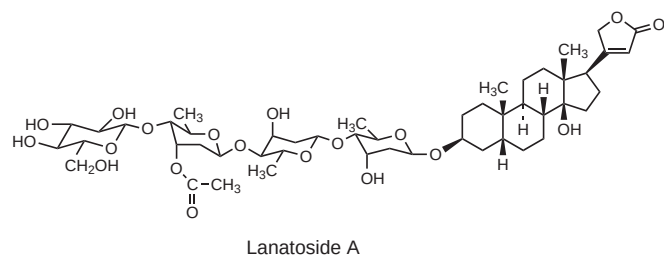


FIGURE 21.3 Selected sugars found in naturally occurring cardiac glycosides.

in animals, such as poisonous toads. *Digitalis purpurea* (foxglove plant), *Digitalis lanata*, *Strophanthus gratus*, and *Strophanthus kombe* are the major plant sources of the cardiac glycosides. Based on the nature and number of sugar molecules and the number of hydroxyl groups on the aglycone moiety, each combination of sugars and aglycones assumes different generic names. The site of the glycosides concentration in the plant, the types of glycosides, and the names of the structural components of these glycosides are summarized in Table 21.1.

Digitalis Lanata. Lanatoside A is composed of the aglycone digitoxigenin (genin indicates no sugar) connected to three digitoxose sugar molecules, the third of which carries a 3-acetyl group, and a terminal glucose molecule. In other words, the structure sequence is glucose₄-3-acetyldigitoxose₃-digitoxose₂-digitoxose₁-digitoxigenin.



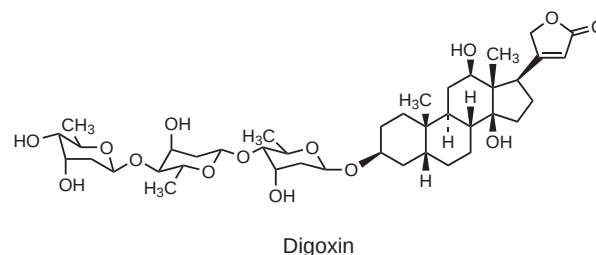
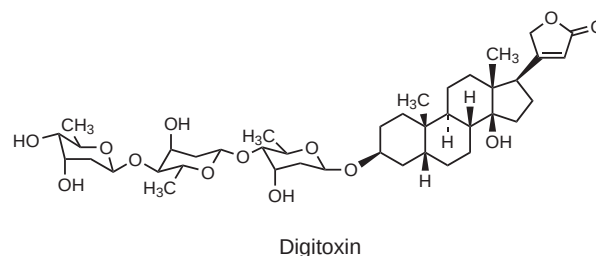
Lanatoside B has the identical sugar portion to lanatoside A, except that the aglycone has an extra hydroxyl group at C-16 and is given the name gitoxigenin. The structural sequence is glucose₄-3-acetyldigitoxose₃-digitoxose₂-digitoxose₁-gitoxigenin.

Lanatoside C has also the same sugars found in both lanatosides A and B; however, the aglycone has the steroid backbone of lanatoside A plus an additional hydroxyl group at C-12. This cardenolide is named digoxigenin. The structural sequence is

glucose₄-3-acetyldigitoxose₃-digitoxose₂-digitoxose₁-digoxigenin.

Partial hydrolysis of the glucose molecule and the acetate group from lanatoside A and C produces, respectively, two new and most important cardiac glycosides, digitoxin and digoxin, with the following sequences: digitoxin, (digitoxose)₃-digitoxigenin; and digoxin, (digitoxose)₃-digoxigenin.

Digitalis Purpurea.

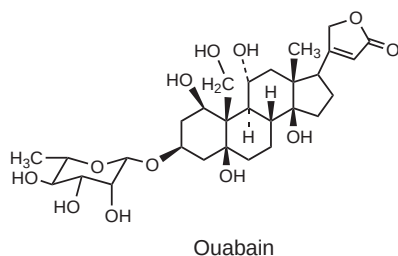


Purpurea glycosides A and B have structures identical to those of lanatosides A and B, but with no acetyl group on the third digitoxose. Therefore, the purpurea glycosides A and B are sometimes called desacetyl digitoxides A and B. Their sequences are as follows: purpurea glycoside A, glucose-(digitoxose)₃-digitoxigenin; and purpurea glycoside B, glucose-(digitoxose)₃-gitoxigenin. There is no purpurea glycoside C.

TABLE 21.1 Selected Natural Cardiac Glycosides and Their Sources

Source	Glycoside	Aglycone	Sugar ^a
<i>Digitalis lanata</i> (leaf)	Lanatoside A (digilanide A)	Digitoxigenin	Glucose-3-acetyldigitoxose-digitoxose-digitoxose
	Lanatoside B (digilanide B)	Gitoxigenin	
	Lanatoside C (digilanide C)	Digoxigenin	
<i>Digitalis purpurea</i> (leaf)	Purpurea glycoside A (desacetyl digilanide A)	Digitoxigenin	Glucose-digitoxose-digitoxose-digitoxose
	Purpurea glycoside B (desacetyl digilanide B)	Gitoxigenin	
<i>Strophanthus gratus</i> (seed)	G-Strophanthin	Oubagenin	Rhamnose
<i>Strophanthus kombe</i> (seed)	k-Strophanthoside	Strophanthidin	Glucose-glucose-cymarose

^aConjugated with the C-3 hydroxyl of the aglycone via the sugar to the far right. All sugars are conjugated via β-1,4-glucosidic bond.

Strophanthus Gratus and Strophanthus Kombe.

The glycosides extracted from the plants *Strophanthus gratus* and *Strophanthus kombe* are called G-strophanthin (or ouabain) and k-strophanthoside, respectively. The corresponding aglycone for ouabain is ouabagenin, and for k-strophanthoside, strophanthidin. Ouabagenin has a polyhydroxylated steroidal nucleus, and strophanthidin has an additional hydroxyl group at C-5 with an angular aldehyde group at C-10, replacing the traditional methyl group at that position (Fig. 21.2). Ouabagenin is conjugated only to a single molecule of L-rhamnose, whereas strophanthidin is conjugated to a molecule of cymarose, which is further linked to two molecules of glucose.

The medicinally used preparations are mainly obtained from *Digitalis purpurea* and *Digitalis lanata* plants. These glycosides are referred to as digitalis glycosides, cardiac glycosides, or simply, cardenolides. *Strophanthus* glycosides (e.g., ouabain) are no longer used therapeutically but were previously administered only intravenously because of poor oral absorption. Cardiac glycosides from animal sources (generally referred to as bufadienolides) are rare and of far less medicinal importance because of their high toxicity. Pharmaceutical preparations of whole plants and partially hydrolyzed glycosides of *Digitalis lanata* and *Digitalis purpurea* have been used clinically. Advancements in isolation and purification techniques, however, have made it possible to obtain highly purified digoxin preparations.

Pharmacology Cardiac glycosides affect the heart in a dual fashion, both directly (on the cardiac muscle and the specialized conduction system of sinoatrial [SA] node, atrioventricular [AV] node, and His-Purkinje system) and indirectly (on the cardiovascular system mediated by the autonomic nervous reflexes). The combined direct and indirect effects of the cardiac glycosides lead to changes in the electrophysiologic properties of the heart, including alteration of the contractility; heart rate; excitability; conductivity; refractory period; and automaticity of the atrium, ventricle, Purkinje fibers, AV node, and SA node. The heart response to the cardiac glycosides is a dose-dependent process and varies considerably between the normal heart and the heart with CHF. The cardio-effects observed after the administration of low doses of the cardiac glycosides (therapeutic doses) differ considerably from those observed at high doses (cardiotoxic doses).

The pharmacologic effects discussed subsequently relate mainly to therapeutic doses administered to patients with CHF. The effects of cardiac glycosides on the properties of the heart muscle and different sites of the conduction system are summarized in Table 21.2. The increased force and rate of myocardial contraction (positive inotropic effect) and the prolongation of the refractory period of the AV node are the effects most relevant to the CHF problem. Both of these effects result from the direct action of the cardiac glycosides on the heart. The indirect effects are manifested as increased vagal nerve activity, which probably results from the glycoside-induced sensitization of the baroreceptors of the carotid sinus to changes in the arterial pressure; in other words, any given increase in the arterial blood pressure results in an increase in the vagal activity (parasympathetic) coupled with a greater decrease in the sympathetic activity. The vagal effect with uncompensated sympathetic response results in decreased heart rate and decreased peripheral vascular resistance (afterload). Therefore, cardiac glycosides reverse most of the symptoms associated with CHF as a result of increased sympathetic system activity, including increased heart rate, vascular resistance, and afterload. The administration of cardiac glycosides to a patient with CHF increases cardiac muscle contraction, reduces heart rate, and decreases both edema and the heart size.

Biochemical Mechanism of Action The mechanism whereby cardiac glycosides cause a positive inotropic effect and electrophysiologic changes is still not completely known despite years of active investigation. Several mechanisms have been proposed, but the most widely accepted mechanism involves the ability of cardiac glycosides to inhibit the membrane-bound Na^+/K^+ -adenosine triphosphatase (Na^+/K^+ -ATPase) pump responsible for sodium/potassium exchange. To better understand the correlation between the pump and the mechanism of action of cardiac glycosides on the heart muscle contraction, one has to consider the sequence of

TABLE 21.2 Effects of Cardiac Glycosides on the Heart

	Atrium	Purkinje Ventricle	AV Fiber	SA Node	Node
Contractility	↑	↑	—	—	—
Excitability	o	Variable	↑	—	—
Conductivity	↑	↑	↓	↓	—
Refractory period	↓	↓	↑	↑	—
Automaticity	—	—	↑	—	↓

AV, atrioventricular; SA, sinoatrial; ↑ increased action; ↓ decreased action; o, no action; —, no data available.

events associated with cardiac action potential that leads ultimately to muscular contraction. The process of membrane depolarization/repolarization is controlled mainly by the movement of the three ions, Na^+ , K^+ , Ca^{2+} , in and out of the cell.

At the resting state (no contraction), the concentration of sodium is high outside the cell. On membrane depolarization, Na^+ fluxes in, leading to an immediate elevation of the action potential. Elevated intracellular sodium triggers the influx of Ca^{2+} , which occurs slowly and is represented by the plateau region of the cardiac action potential. The influx of calcium results in the efflux of potassium out of the myocardium. The Na^+/K^+ exchange occurs at a later stage of the action potential to restore the membrane potential to its normal level (for further detail, see the discussion of antiarrhythmic agents and their classification at the end of this chapter). The Na^+/K^+ exchange requires energy and is catalyzed by the enzyme Na^+/K^+ -ATPase. Cardiac glycosides are proposed to inhibit this enzyme, with a net result of reduced sodium exchange with potassium (i.e., increased intracellular sodium), which in turn results in increased intracellular calcium. Elevated intracellular calcium concentration triggers a series of intracellular biochemical events that ultimately result in an increase in the force of the myocardial contraction, or a positive inotropic effect. (The events that lead to muscle contraction are covered in further detail in the discussion of the mechanism of action of the calcium channel blockers later in this chapter.)

This mechanism of the cardiac glycosides via inhibiting the Na^+/K^+ -ATPase pump is in agreement with the fact that the action of the cardiac glycosides is enhanced by low extracellular potassium and inhibited by high extracellular potassium. The cardiac glycosides-induced changes in the electrophysiology of the heart can also be explained based on the inhibition of Na^+/K^+ -ATPase. It has been suggested that the intracellular loss of potassium because of inhibition of the pump causes a decrease in the cellular transmembrane potential approaching zero. This decrease in the membrane potential is sufficient to explain the increased excitability and other electrophysiologic effects observed after cardiac glycosides administration.

Structural Requirements for Intrinsic Activity Many hypotheses have been put forth to explain the cardiac glycoside structure–activity relationships (SARs). Some of the difficulty in arriving at a universally acceptable SAR model has been attributed to the early method of testing cardiac glycoside preparations and the lack of well-characterized cardiac glycoside “receptors.” Until the early 1970s, nearly all the cardiac glycosides were evaluated based on their cardiac toxicity rather than on more therapeutically relevant criteria. This was partly because of the belief that cardiac toxicity was, in fact, an extension of the desired cardiotonic action. Thus, comparisons of cardiac glycoside preparations were based

on the amount of drug required to cause cardiac arrest in test animals, usually anesthetized cats. More recently, most SAR studies have relied, at least initially, on results obtained with isolated cardiac tissue or whole-heart preparations. In these models, inotropic activity and contractility can be directly assessed. In addition, the recognition of cardiac Na^+/K^+ -ATPase as the probable receptor for the cardiac glycosides has made the inhibition of this enzyme system an important criterion for the cardiac glycosides activity.

Much of the interest in the effects of structural modification on cardiotonic activity results from the desire to develop agents with less potential for toxicity. Early studies based primarily on cardiac toxicity testing data suggested the importance of the steroid “backbone” shape, the 14- β -hydroxyl and the 17-unsaturated lactone for activity. Other studies have been directed toward characterizing the interaction of the cardiac glycosides with Na^+/K^+ -ATPase, the putative cardiac glycoside receptor. Using ATPase as a model for enzyme inhibition as the biologic end point, a number of hypotheses for cardiac glycoside receptor–binding interactions have been presented. Many of these hypotheses suggested that the 17-lactone has an important role in receptor binding. Using semisynthetic analogs, unsaturation in the lactone ring was discovered to be important, and the saturated lactone analog showed diminished activity (4,5). Further investigations of semisynthetic compounds in which the lactone was replaced with open-chain structures of varying electronic and steric resemblance to the lactone ring showed that the α,β -unsaturated lactone ring at C-17 was not an absolute requirement and that several α,β -unsaturated open-chain groups could be replaced with little or no loss in activity (4,5). For example, analogs possessing an α,β -unsaturated nitrile at the 17- β position had high activity. In light of this, most current theories point toward a key interaction of the carbonyl oxygen (or nitrile nitrogen) with the cardiac glycoside binding site on Na^+/K^+ -ATPase (6,7).

Some controversy, however, exists regarding this point. The importance of the remainder of the cardiac glycoside molecule cannot be ignored. Despite the apparently dominant role of the 17-lactone, the steroid (A-B-C-D) ring system provides the lead structure for cardiac glycoside activity. Lactones, when not attached to the steroid ring system, show no Na^+/K^+ -ATPase inhibitory activity. Some important steroid structural features have become apparent. The C-D *cis* ring juncture appears to be critical for activity in compounds possessing the unsaturated butyrolactone in the normal 17- β position. This requirement can be a reflection of changes in the spatial orientation of the 17-substituent (7). Moreover, the 14- β -OH is understood to be dispensable, and the contribution to activity previously attributed to this group is thought to be related to the need to retain the sp^3 and *cis* character of the C-D ring juncture. The earlier interpretation arose because the 14-deoxy analogs often had unsaturation in the D ring

in place of the 14- β -OH. This double bond markedly influenced the position of the C-17 substituent, thereby complicating the importance of the 14- β -OH group on cardiotonic activity. Finally, the A-B *cis* ring juncture also appears not to be mandatory for cardiac glycoside activity. This feature, however, is characteristic of all clinically useful cardiac glycosides and conversion to an A-B *trans* ring system as a rule leads to a marked drop in activity unless compensating structural modifications are made elsewhere in the molecule.

Pharmaceutical Preparations The cardiac glycoside preparations that have been used therapeutically range from powdered digitalis leaf to purified individual glycosides, including gitalin, lanatoside C, its partially hydrolyzed product deslanatoside C (desacetyl lanatoside C), digoxin, and ouabain (Table 21.3). Digoxin is the only cardiac glycoside commercially available for therapeutic use in the United States. To arrive at an effective plasma concentration, a large initial dose of digoxin (i.e., digitalizing or loading dose) is often given. The purpose of this large initial dose is to achieve a therapeutic blood and tissue level in the shortest possible time. Depending on the condition of the patient and the desired therapeutic goal, the loading dose can be much less than, or almost equal to, the dose that is likely to cause toxicity. Once the desired effect is obtained, the amount of drug eliminated from the body per day is replaced with a maintenance dose.

Absorption, Metabolism, and Excretion The therapeutic effects of all cardiac glycosides on the heart muscle are qualitatively similar; however, the glycosides differ largely in their pharmacokinetic properties, which are greatly influenced by the lipophilic character of each glycoside. In general, cardiac glycosides with more lipophilic character are absorbed faster and exhibit a longer duration of action as a result of a slower urinary rate of excretion. The lipophilicity of a cardiac glycoside is

measured by its partitioning between chloroform and water mixed with methanol: The higher the concentration of the cardiac glycoside in the chloroform phase, the higher its partition coefficient, and the more lipophilic it is. The partition coefficients for five cardiac glycosides are listed in Table 21.4. It is evident from a comparison of the coefficients that their lipophilicity is markedly influenced by the number of sugar molecules and the number of hydroxyl groups on the aglycone part of a given glycoside. Lanatoside C, with a partition coefficient of 16.2, is far less lipophilic than that of acetyldigoxin (partition coefficient, 98), which structurally differs only in lacking the terminal glucose molecule. Likewise, a comparison of digitoxin and digoxin structures reveals that they differ only by an extra hydroxyl in digoxin at C-12. This seemingly minor difference in their partition coefficients from 96.5 to 81.5 for digitoxin and digoxin, respectively, results in significant differences in their pharmacokinetic behavior (Table 21.5). Table 21.4 also illustrates that the presence of the 3-*O*-acetyl group on acetyldigoxin enhances its lipophilic character more than that of the desacetyl analog, digoxin (partition coefficients of 98 and 81.5, respectively). The glycoside G-strophanthin (ouabain) possesses a very low lipophilic character because of the presence of five free hydroxyl groups on the steroid nucleus of the aglycone ouabagenin.

Digoxin is the most frequently used cardiac glycoside. The absorption of digoxin from the gastrointestinal tract is a passive process that depends on the lipid solubility, dissolution, and membrane permeability of the drug. The oral bioavailability of digoxin following oral administration exhibits interindividual variability, ranging from 70% to 85% of an administered dose. This interindividual variability has been attributed to intestinal P-glycoprotein (P-gp) efflux and P-gp-dependent renal elimination. Although digoxin is not extensively metabolized, it is transported from intestinal enterocytes along its epithelium into the intestinal lumen (effluxed) by P-gp, which

TABLE 21.3 Cardiac Glycosides and Their Dosage Forms

Name ^a	Dosage Forms
Digoxin USP and BP	Tablets, elixir, pediatric
Digitalis powder (leaf) BP	Tablets, capsules
Digitoxin BP	Tablets, injection
Lanatoside C BP	Tablets
Ouabain BP	Injection (G-strophanthin)
Deslanatoside C BP	Injection (deacetyl lanatoside C)

^aThose cardiac glycosides with only the BP (British Pharmacopeia) designation are available only in Canada and other countries except the United States.

TABLE 21.4 Effect of Glycoside Structure on Partition Coefficient

Glycoside	Partition Coefficient (CHCl ₃ /16% aq.MeOH)
Lanatoside C (glucose-3-acetyldigitoxose–digitoxose ₂ –digoxigenin)	16.2
Digoxin (digitoxose ₃ –digoxigenin)	81.5
Digitoxin (digitoxose ₃ –digitoxigenin)	96.5
Acetyldigoxin (3-acetyldigitoxose–digitoxose ₂ –digoxigenin)	98.0
G-Strophanthin (rhamnose–ouabagenin)	Very low

TABLE 21.5 Comparison of the Pharmacokinetic Properties for Digoxin and Digitoxin

	Digoxin	Digitoxin
Gastrointestinal absorption	70%–85%	95%–100%
Average half-life	1–2 days	5–7 days
Protein binding	25%–30%	90%–95%
Enterohepatic cycling	5%	25%
Excretion	Kidneys; largely unchanged	Liver metabolism
Therapeutic plasma level	0.5–2.5 ng/mL	20–35 ng/mL
Digitalizing dose (mg)	Oral: 0.75–1.5 IV: 0.5–1.0	Oral: 0.8–1.2 IV: 0.8–1.2
Maintenance dose (oral mg)	0.125–0.5	0.05–0.2

IV, intravenous.

is also expressed in the kidney and liver. Alterations in P-gp transport can be the basis for several digoxin–drug interactions. For this reason, it is important to establish carefully the effective dose of digoxin for each patient to avoid digitalis toxicity.

Once the cardiac glycosides are absorbed, they bind to plasma proteins; digoxin has only 30% binding. The half-life of digoxin in patients with normal renal function is 1.5 to 2.0 days. Biliary excretion of digoxin is minimal. Digoxin is eliminated primarily unchanged by renal tubule excretion.

Contributing to the discontinuance of digitoxin as a therapeutic agent was a half-life range between 5 and 7 days because of its enterohepatic circulation. Approximately 25% of an absorbed dose of digitoxin is excreted in the bile unchanged, to be reabsorbed via enterohepatic circulation. Digitoxin, however, is extensively metabolized by the liver to a variety of metabolites, including (digitoxose)₂-digitoxigenin, (digitoxose)₁-digitoxigenin, and (digitoxose)₁-digitoxigenin. Trace amounts of digoxin have been discovered in the urine. The pharmacokinetic data for digoxin and digitoxin are summarized in Table 21.5.

Drug Interactions Digoxin–drug interactions are common causes of digitalis toxicity. Recently, the clinical significance of the P-gp–dependent renal tubular secretion of digoxin associated with the well-documented digoxin–quinidine interaction has been reported (8). The discovery that digoxin is actively secreted into the urine by the renal tubular cell via the P-gp efflux pump has led to the conclusion that the digoxin–quinidine interaction can be attributed to inhibition of renal tubular secretion

of digoxin by quinidine (a P-gp substrate). Quinidine competitively binds to P-gp in the renal tubule, reducing the renal secretion of digoxin by as much as 60% and raising digoxin’s plasma concentration to toxic levels. Other drugs that are substrates for renal P-gp are also likely to be associated with digoxin–drug interactions. Another documented digoxin–drug interaction associated with increased digoxin blood levels and toxicity is with verapamil. Unlike quinidine, verapamil inhibits intestinal P-gp efflux of digoxin, thereby blocking the intestinal secretion of digoxin into the lumen of the intestine and raising digoxin blood levels to toxic levels. On the other hand, the rifampin–digoxin interaction involves the rifampin induction of intestinal P-gp expression, thereby increasing the P-gp–mediated secretion of digoxin. This results in the lowering of digoxin blood levels to subtherapeutic concentrations. The P-gp transporters and their substrates, inhibitors, or inducers (see Table 4.16 for list) appear to play an important role in controlling the digoxin area under the curve (AUC) values through the renal tubular and intestinal secretion of digoxin and, subsequently, in digoxin–drug interactions and digitalis toxicity. Concurrent use of the cardiac glycosides with antiarrhythmics, sympathomimetics, β -adrenergic blockers, and calcium channel blockers, which are substrates for P-gp, can alter the control of arrhythmias.

The absorption of digoxin after oral administration can also be significantly altered by other drugs concurrently present in the gastrointestinal tract. For example, laxatives can interfere with the absorption of digoxin because of increased intestinal motility. The presence of the drug cholestyramine, an agent used to treat hyperlipoproteinemia, decreases the absorption of digoxin by binding to and retaining digoxin in the gastrointestinal tract. Antacids, especially magnesium trisilicate, and antidiarrheal adsorbent suspensions can also inhibit the absorption of the digoxin. Potassium-depleting diuretics, such as thiazides, can increase the possibility of digitalis toxicity because of the additive hypokalemia. Several other drugs that are known to bind to plasma proteins, such as thyroid hormones, have the potential to displace digoxin from its plasma-binding sites, thereby increasing its free drug concentration to a toxic level.

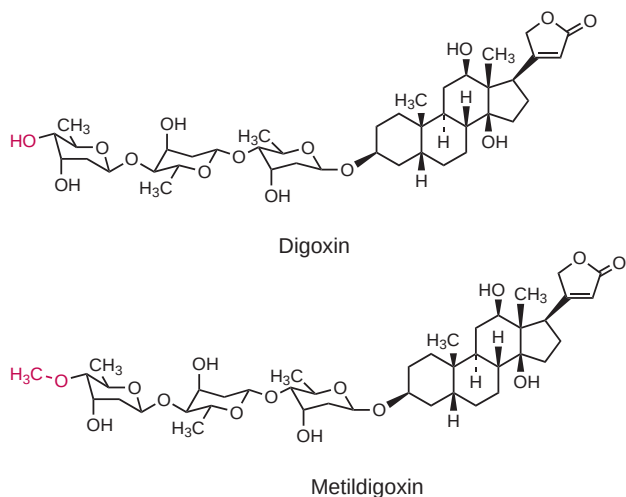
Therapeutic Uses Although the primary clinical use for digoxin is in the treatment of CHF, this agent is also used in cases of atrial flutter or fibrillation and paroxysmal atrial tachycardia.

Toxicity All cardiac glycosides preparations have the potential to cause toxicity. Because the minimal toxic dose of the glycosides is only two- to threefold the therapeutic dose, intoxication is quite common. In mild to moderate toxicity, the common symptoms are anorexia, nausea and vomiting, muscular weakness, bradycardia, and ventricular premature contractions. The nausea

is a result of excitation of the chemoreceptor trigger zone in the medulla. In severe toxicity, the common symptoms are blurred vision, disorientation, diarrhea, ventricular tachycardia, and AV block, which can progress into ventricular fibrillation. It is usually accepted that the toxicity of the cardiac glycosides results from inhibition of the $\text{Na}^+/\text{K}^+\text{-ATPase}$ pump, which results in increased intracellular levels of Ca^{2+} . Hypokalemia (decreased potassium), which can be induced by coadministration of thiazide diuretics, of glucocorticoids, or by other means, can be an important factor in initiating a toxic response. It has been shown that low levels of extracellular K^+ partially inhibit the $\text{Na}^+/\text{K}^+\text{-ATPase}$ pump. In a patient stabilized on a cardiac glycoside, the $\text{Na}^+/\text{K}^+\text{-ATPase}$ pump is already partially inhibited, and the hypokalemia only further inhibits the pump, causing an intracellular buildup of sodium, which leads to an increase in intracellular calcium levels. The high levels of calcium are responsible for the observed cardiac arrhythmias characteristic of the cardiac glycosides toxicity.

A common procedure used in treating cardiac glycoside toxicity is to administer potassium salts to increase extracellular potassium level, which stimulates the $\text{Na}^+/\text{K}^+\text{-ATPase}$ pump, resulting in decreased intracellular sodium levels and, thus, decreased intracellular calcium. In treating any cardiac glycoside-induced toxicity, it is important to discontinue administration of the drug in addition to administering a potassium salt, such as potassium chloride. Other drugs that can be useful in treating the tachyarrhythmias present during toxicity are lidocaine, phenytoin, and propranolol. Specific antibodies directed toward digoxin (Dig-Bind) have been used experimentally and proven to be very effective.

METILDIGOXIN



The drug is a semisynthetic derivative of digoxin, but with a methoxy group attached at position 4 of the terminal digitoxose molecule. The drug is marketed only

in Europe and has not been approved by the U.S. Food and Drug Administration (FDA). The major advantage of metildigoxin over digoxin is its rapid and complete absorption after oral administration. After oral administration, some demethylation to digoxin occurs, but 60% of the administered dose (oral or injection) is excreted in urine unchanged.

NONGLYCOSIDIC POSITIVE INOTROPIC AGENTS Nonglycosidic positive inotropic drugs can be divided into two main classes: those that act via stimulating the synthesis of cyclic adenosine monophosphate (cAMP), such as adrenergic and dopaminergic agonists; and those that inhibit the hydrolysis of cAMP, such as phosphodiesterase 3 (PDE3) inhibitors.

Phosphodiesterase 3 Inhibitors The mechanism of cardiac contraction involves a G protein signal transduction pathway, which regulates intracellular calcium concentrations. Activation of the G_s protein involves the formation of intracellular cAMP, which thereby increases intracellular calcium, stimulating cardiac muscle contraction. Relaxation occurs when the released cAMP is hydrolyzed by cytosolic cAMP-dependent PDE3, one of the phosphodiesterase isoforms. Therefore, inhibition of PDE3 increases intracellular cAMP, promoting cardiac muscle contraction but vasodilation of vascular smooth muscle.

The overall cardiostimulatory and vasodilatory actions of PDE3 inhibitors make them suitable for the treatment of heart failure, because vascular smooth muscle relaxation reduces ventricular wall stress and the oxygen demands placed on the failing heart. The cardiostimulatory effects of the PDE3 inhibitors increase inotropy, which further enhances stroke volume and ejection fraction. Clinical trials have shown that long-term therapy with PDE3 inhibitors increases mortality in heart failure patients. Therefore, these PDE3 inhibitors are not used for the long-term, chronic therapy of CHF. They are very useful, however, in treating acute, decompensated heart failure or temporary bouts of decompensated chronic failure. They are not used as a monotherapy. Instead, they are used in conjunction with other treatment modalities, such as diuretics, angiotensin-converting enzyme inhibitors, β -blockers, or cardiac glycosides. The PDE3 inhibitors contract cardiac muscle and are used for treating heart failure, whereas the phosphodiesterase 5 (PDE5) inhibitors are vasodilators and are also used for treating male erectile dysfunction (Chapter 40). Note that the generic names for PDE3 inhibitors end in “one,” and those for the PDE5 inhibitors end in “fil.”

Side Effects and Contraindications. The most common and severe side effects of PDE3 inhibitors are ventricular arrhythmias, some of which can be life threatening. Other side effects included headaches and hypotension, which are not uncommon for drugs that increase cAMP

in cardiac and vascular tissues, with other examples being β -agonists.

Milrinone (Primacor) and Inamrinone (Inocor). Although the digitalis glycosides can be the principal therapeutic agents for the treatment of CHF, they are not the only positive inotropic agents available. Among the “nonglycoside” inotropic agents are the bipyridines, inamrinone and milrinone, which are selective PDE3 inhibitors (Fig. 21.4). Inamrinone and milrinone are positive inotropes and vasodilators indicated for the short-term intravenous management of CHF in patients who have not responded adequately to digitalis, diuretics, and/or vasodilators (9,10). Milrinone is the drug of choice from among the currently available PDE3 inhibitors because of its greater selectivity for PDE3, shorter half-life (30 to 60 minutes), and fewer side effects. Inamrinone is associated with thrombocytopenia in 10% of patients. Inamrinone was introduced in 1978, and it produces both positive inotropic and concentration-dependent vasodilatory effects. Despite similar positive inotropic action to the cardiac glycosides, the inotropic action involves inhibition of PDE3. Inamrinone was approved for the short-term intravenous administration in patients with severe heart failure refractory to other measures. Although inamrinone is orally active, several adverse side effects have dampened enthusiasm for long-term oral inamrinone therapy. These effects include gastrointestinal disturbances, thrombocytopenia, and impairment of the liver function. For intravenous infusion, inamrinone lactate and milrinone lactate injection solutions can be diluted in sodium chloride injection. Inamrinone lactate for injection is preserved with sodium metabisulfite and needs protection from light. It should not be diluted with solutions

containing dextrose because a chemical reaction occurs in 24 hours. For milrinone, an immediate chemical interaction with furosemide with the formation of a precipitate is observed when furosemide is injected into an infusion of milrinone. Patients sensitive to bisulfites can also be sensitive to inamrinone lactate injection, which contains sodium metabisulfite.

The pharmacokinetics for inamrinone shows a half-life in healthy volunteers of approximately 3 to 4 hours, whereas in patients with CHF, the plasma half-life increases approximately 50% (5 to 8 hours). For infants younger than 4 weeks, the half-life is 12.7 to 22.2 hours, and for infants older than 4 weeks, the half-life is 3.8 to 6.8 hours. Time to peak effect is less than 10 minutes, with its duration of action ranging from 30 minutes to 2 hours depending on the dosage. Approximately 63% of inamrinone is eliminated via the urine as unchanged drug, and 18% is eliminated in the feces. Elderly patients are more likely to have age-related impairment of renal function, which can require adjustment of dosage in patients receiving inamrinone.

The limited success of inamrinone led to the development of structurally related newer agents, such as milrinone, with a mechanism of action similar to inamrinone. Milrinone, however, is 10-fold more potent than inamrinone. Furthermore, preliminary reports show milrinone to be better tolerated, with no apparent thrombocytopenia or gastrointestinal disturbances. Milrinone is excreted largely unchanged in the urine, and accordingly, patients with impaired renal function require reduced dosages.

The pharmacokinetics for milrinone following intravenous injections to patients with CHF showed an elimination half-life of 2 to 3 hours. Its primary route of excretion is via the urine as unchanged milrinone (83%) and its *O*-glucuronide metabolite (12%). In patients with renal function impairment, elimination of unchanged milrinone is reduced, suggesting that a dosage adjustment can be necessary.

Phosphodiesterase 3 Inhibitors not Approved by FDA

Enoximone. Enoximone is a selective PDE3 inhibitor with vasodilating and positive inotropic activity that does not cause changes in myocardial oxygen consumption. It is used in patients with CHF. US clinical trials were halted because of significant adverse effects on the nervous system. The drug is approved for use in the United Kingdom, with an oral bioavailability of 50% and 85% protein binding. The drug is not recommended for long-term use and is given only by injection for short-term use.

Olprinone. Olprinone has positive inotropic and vasodilator effects, which improve myocardial mechanical efficiency. Additionally, the drug augments cerebral blood flow by a direct vasodilatory effect on cerebral arteries, especially in patients with impaired cerebral circulation. Olprinone improves inadequate redistribution of

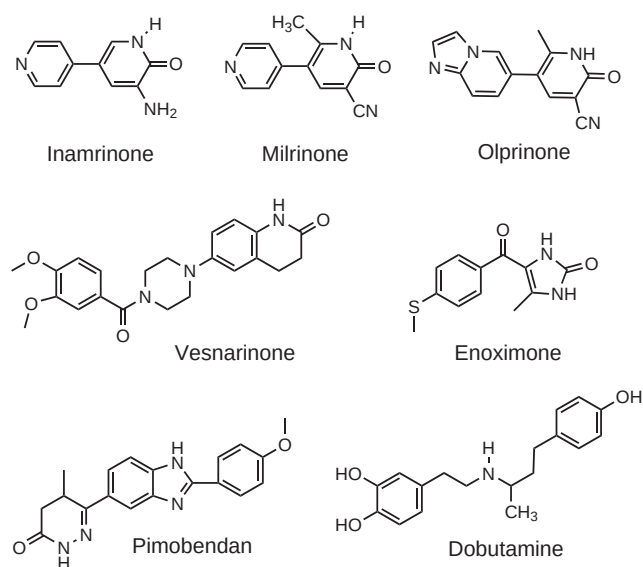


FIGURE 21.4 Miscellaneous inotropic agents.

brain perfusion and may prevent cerebral metabolic abnormalities in heart failure. The drug is given intravenously during acute heart failure. The drug is currently marketed in Japan.

Vesnarinone. This drug is a mixed PDE3 inhibitor and ion channel modifier with modest, dose-dependent, positive inotropic activity but minimal negative chronotropic activity. Vesnarinone improves ventricular performance most in patients with the worst degree of heart failure. The drug was withdrawn from US phase III clinical trials but is marketed in Japan.

Pimobendan. Pimobendan is a selective inhibitor of PDE3, as well as a calcium sensitizer with positive inotropic and vasodilator effects. It is marketed in Japan for human use and also approved for the management of heart failure in dogs, most commonly caused endocardiosis or dilated cardiomyopathy. As a calcium sensitizer, it increases the binding efficiency of cardiac myofibrils to the calcium ions that are already present without increasing the consumption of oxygen and energy. These effects result in decreased pressure and decreased cardiac preload and afterload (decreases the failing heart's workload). Pimobendan is absorbed rapidly, with an oral bioavailability of 60% to 65%, and metabolized by the liver to an active metabolite. The plasma half-life is 0.5 hours, and the half-life of its metabolite is 2 hours. Elimination is by excretion in the bile and then feces. Pimobendan is 90% to 95% bound to plasma proteins in circulation, which has implications in patients who are on concurrent drug therapies that are also highly protein bound. The drug is used as an adjunct therapy with other agents, and as is the case with other phosphodiesterase inhibitors, prolonged use is reported to cause an increase in mortality rate.

β -Adrenergic Receptor Agonists Another promising area for the development of new positive inotropic agents is β -adrenergic receptor agonists (11) (see Chapter 10). The myocardium has mostly β_1 -adrenergic receptors, and stimulation of these receptors by a variety of β -adrenergic agonists produces a potent positive inotropic response involving the G protein signal transduction process, increasing intracellular cAMP levels that lead to a cascade of events that ultimately produces an increase in intracellular Ca^{2+} , thereby increasing myocardial contractility (discussed in Chapters 7 and 10). Although many drugs possess β -adrenergic agonist activity, most have side effects that make them inappropriate for the treatment of CHF. For example, the well-known catecholamines, norepinephrine and epinephrine, are potent nonselective adrenergic receptor agonists. Because the actions of these agents are not limited to the myocardial β_1 -receptors, however, they produce undesirable positive chronotropic effects, exacerbate arrhythmias, and result in vasoconstriction. These effects limit their utility in the treatment of CHF.

Dobutamine (Dobutrex). Among the most promising β_1 -adrenergic agonists are those derived from dopamine, the endogenous precursor to norepinephrine. Dopamine is a potent stimulator of β_1 -receptors, but with many of the undesirable side effects described in Chapter 10. The new analogs of dopamine that have been developed retain the potent inotropic effect but possess fewer effects on heart rate, vascular tone, and arrhythmias. Dobutamine is a prime representative of this group of agents. Dobutamine is a potent β_1 -adrenergic agonist on the myocardium (as well as having α_1 -agonist and -antagonist activities; see Chapter 10 for details concerning its mechanism of action) with beneficial effects, the composite of a variety of actions on the heart and the peripheral vasculature. Dobutamine is active only by the intravenous route because of its rapid first-pass metabolism via catechol-O-methyltransferase (COMT). Therefore, its use is limited to critical care situations. Nonetheless, its parenteral success has led to the search and development of orally active drugs. One of the major limitations associated with β_1 -agonists is the phenomenon of myocardial β -receptor desensitization. This lowered responsiveness (desensitization) of the receptors appears to be due to a decrease in the number of β_1 -receptors and partial uncoupling of the receptors from adenylate cyclase.

DRUGS FOR THE TREATMENT OF ANGINA

Angina Pectoris

Angina pectoris is the chronic disease affecting the coronary arteries, which supply oxygenated blood from the left ventricle to all heart tissues, including the ventricles themselves. When the lumen of the coronary artery becomes restricted, it becomes less efficient in supplying blood and oxygen to the heart, and the heart is said to be "ischemic" (oxygen deficient). Angina is the primary symptom of ischemic heart disease and is characterized by a sudden, severe pain originating in the chest, often radiating to the left shoulder and down the left arm. Angina is further subclassified into typical or variant angina based on the precipitating factors and the electrophysiologic changes observed during the attack. Typical angina is the result of an advanced state of atherosclerosis and is provoked by food, exercise, and emotional factors. It is characterized by low ST segment of the electrocardiogram. Variant or acute angina results from sudden spasm in the coronary artery unrelated to atherosclerotic narrowing of the coronary circulation and can occur at rest. It is characterized by an increase in the ST segment of the electrocardiogram.

Antianginal Drugs

Therapy of angina is directed mainly toward alleviating and preventing anginal attacks by altering the oxygen supply/oxygen demand ratio to the cardiac muscle or dilating the coronary vessels. Three classes of drugs are found to be very efficient in this regard, although via

different mechanisms. These include organic nitrates, calcium channel blockers, and β -adrenergic blockers.

Organic Nitrates

Organic nitrates have dominated the treatment of acute angina over the last 100 years. Although the recent introduction of the calcium channel blockers and the β -blockers as antianginal agents has expanded the physician therapeutic arsenal, organic nitrates are still the class of choice in the treatment of acute anginal episodes.

OVERVIEW Organic nitrates are esters of simple organic alcohols or polyols with nitric acid. This class was developed after the antianginal effect of amyl nitrite (ester of isoamyl alcohol with nitrous acid) was first observed in 1857. Five members of this class are in clinical use today: amyl nitrite (amyl nitrite inhalant USP), nitroglycerin, isosorbide dinitrate, erythrityl tetranitrate, and pentaerythritol tetranitrate (Fig. 21.5). Two additional organic nitrates, tenitramine and propatyl nitrate, are currently available in Europe (Fig. 21.5). This class is, as a rule, referred to as organic nitrates, because all of these agents, except amyl nitrite, are nitrate esters. It should be noted that the generic names do not always precisely describe the chemical nature of the drug but, rather, are used for simplicity. For example, the drug nitroglycerin is not really a nitro compound, because a nitro compound means a nitro group attached to a carbon atom (i.e., $\text{NO}_2\text{-C}$); the correct chemical name of nitroglycerin is glyceryl trinitrate. Another example is amyl nitrite, the structure of which indicates that it is an ester of isoamyl alcohol with nitrous acid; the correct chemical name of this drug is isoamyl nitrite.

The chemical nature of these molecules as esters constitutes some problems in formulating these agents for clinical use. The small lipophilic ester character makes them volatile. Volatility is an important concern in drug formulation because of the potential loss of the active principle from the dosage form. In addition, moisture should

be avoided during storage to minimize the hydrolysis of the ester bond, which can lead to a decrease in the therapeutic effectiveness. Lastly, because these agents are nitrate esters, they possess explosive properties, especially in the pure concentrated form. Dilution in a variety of vehicles and excipients eliminates this potential hazard. The lipophilic nature of these esters, however, makes these agents very efficient in emergency treatment of anginal episodes as a result of their rapid absorption through biomembranes.

PHARMACOLOGIC ACTIONS The oxygen requirements of the myocardial tissues are related to the workload (oxygen demand) of the heart, which is, in part, a function of the heart rate, the systolic pressure, and the peripheral resistance of the blood flow (oxygen supply). Myocardial ischemia occurs when the oxygen supply is insufficient to meet the myocardial oxygen demand. This can occur, as explained previously, because of atherosclerotic narrowing of the coronary circulation (typical) or vasospasm of the coronary artery (variant). The nitrates have been shown to be effective in treating angina resulting from either cause. The vasodilating effect of organic nitrates on the veins leads to pooling of the blood in the veins and decreased venous return to the heart (decreased preload), whereas vasodilation of the coronary arterioles decreases the resistance of the peripheral tissues (decreased afterload). The decrease in both preload and afterload results in a generalized decrease in the myocardial workload, which translates into a reduced oxygen demand by the myocardium. Organic nitrates restore the balance between oxygen supply by venous dilation and oxygen demand by decreasing the myocardial workload.

BIOCHEMICAL MECHANISM OF ACTION The organic nitrates (Fig. 21.5) are pharmacologic sources of nitric oxide (NO) for the body. In the cardiovascular system, NO is naturally produced by vascular endothelial cells. This endothelial-derived NO has several important functions, including relaxation of vascular smooth muscle, inhibition of platelet aggregation (antithrombotic), and inhibition of leukocyte-endothelial interactions (anti-inflammatory). These actions involve NO-stimulated formation of cyclic guanosine monophosphate (cGMP) (see Chapter 24). Nitro dilators are drugs that mimic the actions of endogenous NO by releasing NO or forming NO within tissues. Free tissue sulfhydryl groups play a key role in the venodilation effect of nitroglycerin, which is supported by experimental evidence showing that prior administration of *N*-acetylcysteine, which should increase the availability of free sulfhydryl groups, resulted in an increase in the venodilating effect of organic nitrates. Similarly, pretreatment with reagents that react with free sulfhydryl groups, such as ethacrynic acid, blocked glyceryl trinitrate venodilation in vitro (12). A more complex mechanism for nitrate venodilation, however, was proposed by Ignarro et al. (14). They suggested that the nitrates act indirectly, by stimulating the enzyme

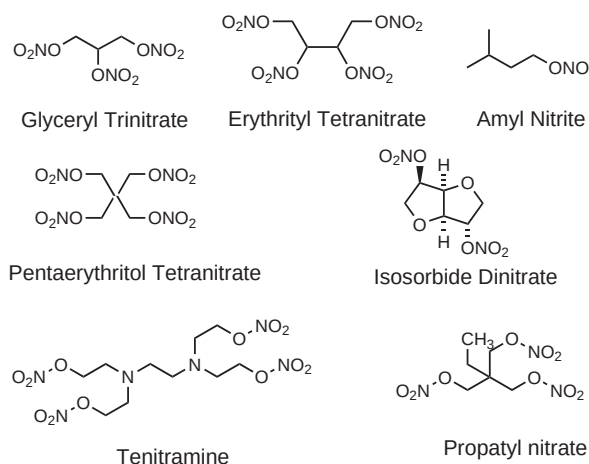


FIGURE 21.5 Organic nitrates and nitrites.

guanylate (also known as guanylyl) cyclase and, thereby, producing elevated levels of cGMP, which in turn leads to venodilation. The initial stimulation of soluble guanylate cyclase is believed to be mediated by a nitrate-derived nitrosothiol metabolite produced intracellularly. In support of this mechanism is the observation that a variety of synthetic nitrosothiols were found to increase markedly soluble guanylate cyclase activity and to produce venodilation *in vitro* (15–19). Such a mechanism is consistent with the requirement for free sulfhydryl groups previously described. A unifying mechanism suggests that the organic nitrates, through the formation of NO via a nitrosothiol-intermediate, activate soluble guanylate cyclase, increasing intracellular cGMP concentrations, which in turn blocks the Ca^{2+} -catalyzed vascular contractions (Fig. 24.3) (20–23). Depletion of sulfhydryl groups during this metabolic process can be a major factor in the development of nitrate tolerance, along with compensatory physiologic mechanisms (13). Data also exist suggesting that organic nitrates increase intraplatelet cGMP concentrations, thereby inhibiting platelet aggregation. These pharmacologic actions of organic nitrates appear to preferentially occur within portions of blood vessels containing damaged endothelium, thus making them extremely useful in the pharmacotherapy of acute ischemic events.

PHARMACEUTICAL PREPARATIONS AND DOSAGE FORMS Organic nitrates are administered by inhalation; by infusion; as sublingual, chewable, and sustained-release tablets; as capsules; as transdermal disks; and as ointments.

ABSORPTION, METABOLISM, AND THERAPEUTIC EFFECTS Organic nitrates are used for both treatment and prevention of painful anginal attacks. The therapeutic approaches to achieve these two goals, however, are distinctly different. For the treatment of acute anginal attacks (i.e., attacks that have already begun), a rapid-acting preparation is required. In contrast, preventative therapy requires a long-acting preparation with more emphasis on duration and less emphasis on onset. The onset of organic nitrate action is influenced not only by the specific agent chosen but also by the route of administration. Sublingual administration is used predominantly for a rapid onset of action. The duration of nitrate action is strongly influenced by rate of metabolism. All of the organic nitrates are subject to rapid first-pass metabolism not only by the action of glutathione-nitrate reductase in the liver, but also in extrahepatic tissues, such as the blood vessel walls themselves (24,25). In addition, rapid uptake into the vessel walls plays a significant role in the rapid disappearance of organic nitrates from the bloodstream. Sublingual, transdermal, and buccal administration routes have been used in an attempt to avoid at least some of the hepatic metabolism.

Acute angina is most frequently treated with sublingual glyceryl trinitrate. This sublingual preparation is rapidly absorbed from the sublingual, lingual, and buccal mucosa and most commonly provides relief within 2 minutes.

The duration of action is also short (~30 minutes). Other treatments include amyl nitrite by inhalation and sublingual isosorbide dinitrate. Amyl nitrite is by far the fastest-acting preparation, with an onset of action in approximately 15 to 30 seconds, but the duration of action is only approximately 1 minute. Isosorbide dinitrate is used as a long-acting agent and can be used to treat acute angina. Sublingually administered isosorbide dinitrate has a somewhat slower onset than glyceryl trinitrate (~3 minutes), but its action can last for 4 to 6 hours. Although the onset appears to be almost as rapid as that of glyceryl trinitrate, waiting an additional minute for relief can be deemed unacceptable by some patients.

To prevent recurring angina, long-acting organic nitrate preparations are used. Several agents fall into this category, such as orally administered isosorbide dinitrate, pentaerythritol tetranitrate, and erythrityl tetranitrate. In addition, a number of long-acting glyceryl trinitrate preparations are available. These include oral sustained-release forms, glyceryl trinitrate ointment, transdermal patches, and buccal tablets. Of these therapeutic options, isosorbide dinitrate and glyceryl trinitrate preparations are by far the most frequently used. At first, the whole concept of prophylactic nitrate use was met with skepticism by many physicians, both because early studies indicated that oral nitrates were almost completely broken down by first-pass metabolism (24,25) and because blood levels of the parent drug appeared to be virtually nil. These findings, in conjunction with several clinical studies showing equivocal efficacy, led Needleman et al. (25) to conclude, “There is no rational basis for the use of ‘long-acting’ nitrates (administered orally) in the prophylactic therapy of angina pectoris.” More recent studies, however, suggest that oral prophylactic nitrates can be effective if appropriate doses are used (26). Moreover, some metabolites of long-acting nitrates are active as venodilators, albeit less potent than the parent drug. An example of this is isosorbide dinitrate, which is metabolized primarily in the liver by glutathione-nitrate reductase, which also participates in the metabolism of other organic nitrates, catalyzing the denitration of the parent drug to yield two metabolites, 2- and 5-isosorbide mononitrate (27). Of these, the 5-isomer is still a potent vasodilator, and its plasma half-life of approximately 4.5 hours is much longer than that of isosorbide dinitrate itself. The extended half-life, because of the metabolite’s resistance to further metabolism, indicates that it can be contributing to the prolonged duration of action associated with use of isosorbide dinitrate (27).

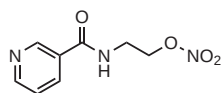
ADVERSE EFFECTS Most patients tolerate the nitrates fairly well. Headache and postural hypotension are the most common side effects of organic nitrates. Dizziness, nausea, vomiting, rapid pulse, and restlessness are among the additional side effects reported. These symptoms can be controlled by administering low doses initially and then gradually increasing the dose. Fortunately, tolerance to nitrate-induced headache develops after a few days

of therapy. Because postural hypotension can occur in some individuals, advise the patient to sit down when taking a rapid-acting nitrate preparation for the first time. An effective dose of nitrate as a rule produces a fall in upright systolic pressure of 10 mm Hg and a reflex rise in heart rate of 10 beats per minute.

Another concern associated with prophylactic nitrate use is the development of tolerance (26,28). Tolerance, usually in the form of a shortened duration of action, is commonly observed with chronic nitrate use. The clinical importance of this tolerance is, however, a matter of controversy. Because tolerance to nitrates has not been reported to lead to a total loss of activity, some physicians feel that it is not clinically relevant. In addition, an adjustment in dosage can compensate for the reduced response (26). It has also been reported that intermittent use of long-acting and sustained-release preparations can limit the extent of tolerance development.

DRUG INTERACTIONS The most significant interactions of organic nitrates are with those agents that cause hypotension, such as other vasodilators, alcohol, and tricyclic antidepressants, in which the potential for orthostatic hypotension can arise. On the other hand, concurrent administration with sympathomimetic amines, such as epinephrine and norepinephrine, can lead to a decrease in the antianginal efficacy of the organic nitrates.

Potassium Channel Opener



Nicorandil

Nicorandil is a nicotinamide-nitrate ester (Fig. 21.5) used for the treatment of angina pectoris and CHF in the United Kingdom and Europe, but is not approved by the FDA. It has a dual mechanism of action, and structurally, it is a hybrid between organic nitrates and potassium channel activators. Nicorandil, a nicotinamide derivative with a nitrate moiety, combines the smooth muscle-relaxing property of both nitrates and nicotinamide with its ability to increase potassium ion conductance. As a result, nicorandil has a direct (preload and afterload) vasodilating effect on normal and diseased coronary arteries and peripheral vessels. Although nicorandil contains a nitrate moiety, its pharmacologic properties differ from organic nitrates. The nicotinamide moiety is responsible for the effect on K^+ -adenosine triphosphate (ATP) channels, which produces vascular smooth muscle relaxation by increasing potassium flux through ATP-sensitive sarcolemmal potassium channels. This leads to hyperpolarization of the cell membrane and subsequent decreases in levels of cytoplasmic calcium (calcium channel blockade) and dilation of arterial resistance vessels. Other agents in this class are minoxidil and diazoxide. The nitrate

group explains its NO-like vasodilation on large coronary arteries, whereas its potassium channel-opening action is responsible for the dilatation of coronary resistance vessels, enabling it to decrease both preload and afterload and to increase coronary blood flow.

Nicorandil induces nitrate-like activation of soluble guanylate cyclase, increasing intracellular levels of cGMP with resultant dilation of venous capacitance vessels. Increases in cGMP are less than those observed with conventional nitrates, although the degree of vasodilation produced appears to be similar. Its oral bioavailability ranges from 75% to 80%. Food reduces the rate, but not the extent, of absorption. Nicorandil is extensively metabolized via denitration to inactive *N*-(2-hydroxyethyl)-nicotinamide, which undergoes further side-chain degradation to nicotinuric acid and, subsequently, nicotinamide and nicotinamide metabolites (e.g., nicotinic acid and *N*-methylnicotinamide). The nicotinamide derived from nicorandil merges into the endogenous pool of nicotinamide adenine dinucleoside coenzymes. Its elimination half-life is approximately 1 hour. Approximately 30% of nicorandil is excreted into the urine as metabolites, with less than 1% excreted unchanged.

Nitric Oxide Donor

Molsidomine (Corvaton) is an oral NO donor vasodilator known as a sydnone imine, a mesoionic compound that is soluble in both water and organic solvents.

Molsidomine is enzymatically metabolized by liver esterases to its active metabolite, linsidomine, which is spontaneously converted in the blood into its nitroso metabolite SIN-1A (Fig. 21.6). Molecular oxygen is required to release NO from SIN-1A. The beneficial effects of molsidomine in treatment of angina were recognized before the role of endogenous NO for causing vasodilation was identified. A possible explanation can revolve around activation of K^+ channels. Molsidomine has a slower onset and longer duration of action than conventional nitrates because of the relatively slow rate of conversion to linsidomine, which has a rapid onset and short duration of action. NO acts as a cellular messenger, leading to activation of soluble guanylate cyclase to

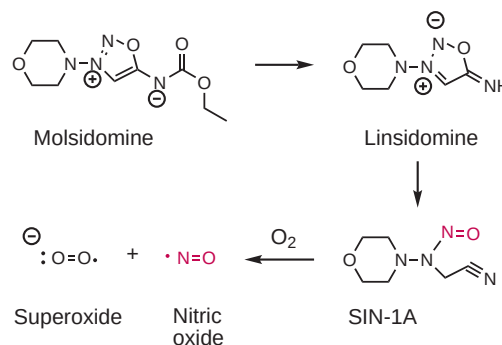


FIGURE 21.6 Molsidomine metabolism to nitric oxide.

release cGMP (see Chapter 24) and vasodilation. Because the metabolites of molsidomine generate NO without the need for enzymes or cofactors, other than the presence of oxygen, tolerance to prolonged exposure does not occur. This mechanism differs from organic nitrates in which cysteine or thiol donors are required for conversion into NO. It is believed that cysteine depletion is the major factor in the development of nitrate tolerance.

Molsidomine and lindosimine are therapeutic alternatives to traditional organic nitrates in treatment of stable angina, coronary vasospasm, and heart failure. Unlike organic nitrates, however, molsidomine and lindosimine have significant antiplatelet activity at therapeutic doses. The cogeneration of superoxide during the release of NO is a major concern limiting its therapeutic potential in the treatment of angina. Its short duration of action necessitates an increased dose frequency, which might be inconvenient in a clinical setting. The light sensitivity of molsidomine necessitates careful protection of infusion bags and tubing during administration.

The pharmacokinetics for molsidomine show 95% oral bioavailability, a plasma elimination half-life of approximately 5 hours (approximately 7 to 8 hours in the elderly), and volume of distribution of 98 L; 95% of the drug is renally eliminated. The pharmacokinetics and metabolism of molsidomine are impaired in the elderly, and thus, caution is warranted.

Calcium Channel Blockers

The second major therapeutic approach to the treatment of angina is the use of calcium channel blockers (29,30) (see also the corresponding subsection in Chapter 23). In the 1960s, it was recognized that inhibition of calcium ion (Ca^{2+}) influx into myocardial cells can be advantageous in preventing angina. Currently, the classes of calcium channel blockers approved for use in the prophylactic treatment of angina include the dihydropyridines nifedipine, nicardipine, and amlodipine; the benzothiazepine derivative diltiazem; the aralkyl amine derivative verapamil; the benzazepinone zatebradine, and the diaminopropanol ether bepridil (Fig. 21.7). These classes of drugs are reserved for treatment failures because serious arrhythmias can occur.

Zatebradine

The investigational drug zatebradine is structurally related to verapamil with a benzazepinone moiety replacing one of the dimethoxyphenyl groups seen in verapamil (Fig. 21.7). Despite its close structural similarity to verapamil, the drug has not been proven to have calcium channel-blocking activity or any effects on blood vessels. Zatebradine is reported to selectively block the SA node and to reduce the hyperpolarization phase of the action potential with a net result of strong bradycardiac effects, hence its value in treating patients with stable angina. Zatebradine is available as oral tablets and as an oral solution. After oral administration, the drug has high oral absorption, undergoes hepatic first-pass metabolism, and

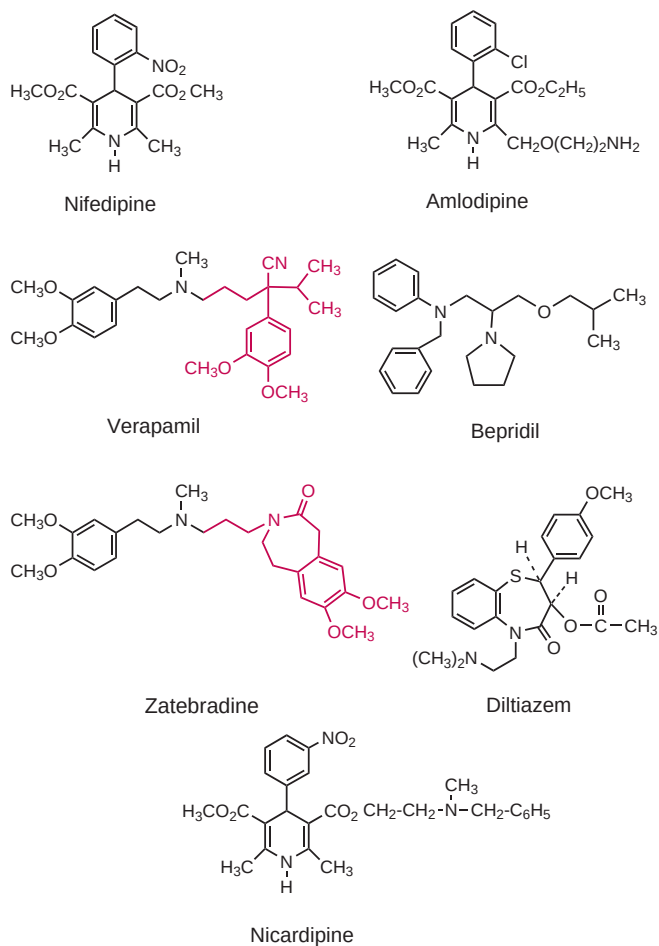


FIGURE 21.7 Calcium channel blockers.

is excreted in feces (40%). Several reports indicated that the drug causes severe visual adverse effects that may limit its clinical use for angina therapy.

Chemistry

The structural dissimilarity of these agents is apparent and serves to emphasize the fact that each is distinctly different from the others in its profile of effects. Although nifedipine and similar drugs belong to the dihydropyridine family, diltiazem belongs to the benzo[b-1,5]thiazepine family. Verapamil is structurally characterized by a central basic nitrogen to which alkyl and aralkyl groups are attached. It is noteworthy that diltiazem and verapamil are both chiral, possessing asymmetric centers. In each case, the dextro-rotatory [i.e., the (+)-enantiomer] is approximately one order of magnitude more potent as a calcium channel blocker than the levo-rotatory [i.e., the (–)-enantiomer]. Zatebradine substitutes one of the dimethoxyphenyl moieties of verapamil with a benzazepinone group.

Pharmacologic Effects

Calcium ions are known to play a critical role in many physiologic functions. Physiologic calcium is found in a

variety of locations, both intracellular and extracellular. Because calcium plays such a ubiquitous role in normal physiology, the overall therapeutic effect of the calcium channel blockers is often the composite of numerous pharmacologic actions in a variety of tissues. The most important of these tissues associated with angina are the myocardium and the arterial vascular bed. Because of the dependency of myocardial contraction on calcium, these drugs have a negative inotropic effect on the heart. Vascular smooth muscle depends also on calcium influx for contraction. Although the underlying mechanism is somewhat different, inhibition of calcium channel influx into the vascular smooth muscles by the calcium channel blockers leads to arteriolar vasodilation. The venous beds appear to be less affected by the calcium channel blockers. The negative inotropic effect and arterial vasodilation result in decreased heart workload and afterload, respectively. The preload is not affected because of a lesser sensitivity of the venous bed to the calcium channel blockers.

Mechanism of Action

The depolarization and contraction of the myocardial cells are mediated, in part, by calcium influx. As previously explained, the overall process consists of two distinct, inward ion currents: First, sodium ions flow rapidly into the cell through the “fast channels,” and subsequently, calcium enters more slowly through the “slow channels.” The calcium ions trigger contraction indirectly by binding and inhibiting troponin, a natural suppressor of the contractile process. Once the inhibitory effect of troponin is removed, actin and myosin can interact to produce the contractile response. The calcium channel blockers produce a negative inotropic effect by interrupting the contractile response. In vascular smooth muscles, calcium causes constriction by binding to a specific intracellular protein calmodulin to form a complex that initiates the process of vascular constriction. The calcium channel blockers inhibit vascular smooth muscle contraction by depriving the cell from the calcium ions.

The effects of the different classes of calcium channel blockers on the myocardium and the arteries vary from class to class. Although verapamil and diltiazem affect both the heart and the arteriolar bed, the dihydropyridines have much less effect on the cardiac tissues and higher specificity for the arteriolar vascular bed. Therefore, both verapamil and diltiazem are used clinically in the management of angina, hypertension, and cardiac arrhythmia, whereas the dihydropyridines are used more frequently as antianginal and antihypertensive agents. Because nicardipine has a less negative inotropic effect than nifedipine, it can be preferred over nifedipine for patients with angina pectoris or hypertension who have also CHF dysfunction.

The recognition of the pivotal role of calcium flux on biologic functions led to the reexamination of several therapeutic agents already in clinical use to see if their effects were also mediated through calcium-dependent

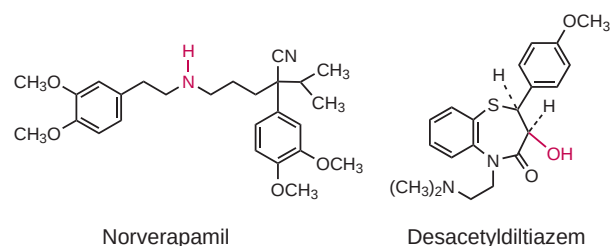
mechanisms. Interestingly, many drugs were found to influence calcium movement and availability. In many cases, however, this effect was not found to contribute significantly to the desirable pharmacologic activity, with other mechanisms playing more dominant roles.

Pharmaceutical Preparations

Calcium channel blockers are administered as oral tablets and capsules as regular or sustained-release forms. Verapamil and diltiazem are also administered by injection.

Absorption, Metabolism, and Excretion

The calcium channel blockers are rapidly and completely absorbed after oral administration (see Table 25.11 for summary of their pharmacokinetic parameters). Prehepatic first-pass metabolism by CYP3A4 enzymes occurs with some orally administered calcium channel blockers, especially verapamil, with its low bioavailability of 20% to 35%. The bioavailability of diltiazem is 40% to 67%, of nicardipine 35%, of nifedipine 45% to 70%, and of amlodipine 64% to 90%.



Verapamil is metabolized by CYP3A4 *N*-demethylation to its principal metabolite, norverapamil, which retains approximately 20% of the activity of verapamil, and by *O*-demethylation (CYP2D6) into inactive metabolites. Diltiazem is metabolized by enzyme hydrolysis to its primary metabolite, desacetyl derivative, which retains approximately 25% to 50% of the activity of diltiazem. The oral bioavailability of diltiazem and verapamil can be increased with chronic use and increasing dose (i.e., bioavailability is nonlinear). Diltiazem undergoes *N*-demethylation by CYP3A4 and *O*-demethylation by CYP2D6. The *N*-demethylated metabolism pathway results in mechanism-based inhibition of CYP3A4. The major metabolite, detected following oral and continuous intravenous administration but not following rapid intravenous administration, is desacetyl diltiazem, which has one-quarter to one-half the arteriolar vasodilation activity of the parent compound. Its elimination half-life ranges from 5 to 8 hours, depending on the dosage. Its onset of action following oral administration is 30 to 60 minutes; 2% to 4% is excreted unchanged. For extended-release capsules, the onset of action is 2 to 3 hours. CYP3A4 inhibition by diltiazem and a substrate for CYP2D6 provide a rational basis for pharmacokinetically significant interactions when they are coadministered with drugs that are cleared primarily by CYP3A4- or CYP2D6-mediated

pathways. Considerable variability for verapamil can be observed in the elimination half-life of 1.5 to 7 hours. In addition, the plasma half-life cannot always accurately predict the duration of action because of the presence of active metabolites. Less than 5% of an orally administered dose of verapamil is excreted unchanged into the urine. Its protein binding is approximately 90%.

A well-documented drug interaction between digoxin and verapamil that increases the AUCs for digoxin has been attributed to verapamil blocking the intestinal P-gp efflux of digoxin (previously described in the section on digoxin–drug interactions). The fact that verapamil is a substrate for P-gp transport of drugs can be a potential source of other drug interactions. The oral coadministration of verapamil with CYP3A4 inhibitors (see Table 4.10 for a list of inhibitors) has resulted in at least a 100- to 200-fold increase in the blood AUCs for verapamil and, thereby, a toxic dose.

The dihydropyridines are metabolized largely to a variety of inactive metabolites. Their binding to plasma proteins is high, ranging from 70% to 98% depending on the individual agent: For verapamil, protein binding is 90%; for diltiazem, 70% to 80%; for nifedipine, 92% to 98%; for nicardipine, greater than 90%; and for amlodipine, greater than 90%. Less than 4% of the dihydropyridine dose is excreted unchanged into the urine. The duration of action of the calcium channel blockers ranges from 4 to 8 hours (verapamil, 4 hours; diltiazem, 6 to 8 hours; nifedipine, 4 to 8 hours; and nicardipine, 6 to 8 hours). Amlodipine has a 24-hour duration of action. Thus, it is the only calcium channel blocker that can be given once daily as a non-sustained-release product.

Adverse Effects

The most common side effects of the calcium channel blockers include dizziness, hypotension, headache, and peripheral and pulmonary edema. These symptoms are related mainly to the excessive vasodilation, especially with the dihydropyridines. Verapamil was reported to cause constipation to some patients.

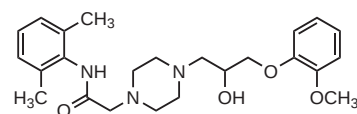
Drug Interactions

Clinically significant drug interactions between calcium channel blockers and coadministration of CYP3A4 inhibitors, such as 6 to 8 oz. of grapefruit juice, HIV protease inhibitors, and erythromycin, have resulted in a 100- to 200-fold increase in the AUC for some calcium channel blockers (31). On the other hand, the coadministration of CYP3A4 inducers, such as rifampin or phenobarbital, result in an approximately 50% decrease in the AUC of calcium channel blockers. With other vasodilators, antihypertensive drugs, and alcohol, excessive hypotension can arise because of an additive effect. The high protein-binding nature of these drugs precipitates a potential for mutual plasma displacement with other drugs known to possess the same property, such as oral anticoagulants, digitalis glycosides, oral hypoglycemic agents, sulfa drugs, and salicylates. Dose adjustment can be necessary in some cases.

Therapeutic Uses

Calcium channel blockers are clinically used as antianginal, antiarrhythmic, and antihypertensive agents (see corresponding subsection in Chapter 25).

Late Sodium Current Inhibitor



Ranolazine

Ranolazine (Ranexa) is used for chronic angina, and its mechanism of action is believed to alter the intracellular sodium level, which affects the sodium-dependent calcium channels during myocardial ischemia. Intracellular sodium and calcium overload play a key role in both electrical and contractility dysfunction of the heart in ischemia and heart failure. Inhibition of the late sodium current (I_{Na}) decreases intracellular sodium and calcium overload, thereby reducing their damaging effects on the heart. Ranolazine selectively inhibits late I_{Na} current and attenuates the abnormalities of ventricular repolarization and contractility associated with ischemia/reperfusion and heart failure. Thus, inhibition of late I_{Na} current may contribute to the cardioprotective effects of ranolazine. The drug indirectly prevents the calcium overload burden that causes cardiac ischemia without significantly changing blood pressure and heart rate. It is approved for the treatment of chronic angina in combination with calcium channel blockers, β -adrenoceptor antagonists, or nitrates in patients who have not achieved an adequate response with other antianginals. Its antianginal and anti-ischemic action is not dependent on heart rate or blood pressure reduction, and it does not increase myocardial workload.

The oral bioavailability of ranolazine from extended-release tablets is 76%, and plasma concentration is not affected by food. Metabolism is mainly by CYP3A4 and, to a lesser degree, by CYP2D6, with less than 5% being excreted unchanged in the urine and feces. After a single oral dose of ranolazine solution, approximately 75% of the dose is excreted in the urine and approximately 25% in feces. Its elimination half-life for extended-release tablets is 7 to 9 hours. Ranolazine is an inhibitor of P-gp transporter. Ranolazine plasma concentrations are increased by CYP3A4 inhibitors. The CYP2D6 inhibition has a negligible effect on ranolazine exposure.

β -Adrenergic Blocking Agents

The use of β -adrenergic blockers as antianginal agents is limited to the treatment of exertion-induced angina. Propranolol is the prototype drug in this class, but several newer agents have been approved for clinical use in the United States (see Chapter 10). Although these agents

can be used alone, they are often used in combination therapy with nitrates, calcium channel blockers, or both. In several instances, combination therapy was found to provide more improvement than with either agent alone. This, however, is not always the case.

Miscellaneous Coronary Vasodilators

Another approach to the treatment of myocardial insufficiency is the use of the coronary vasodilators dipyridamole and papaverine. Dipyridamole, a PDE3 inhibitor (Fig. 21.8), causes a long-acting and selective coronary vasodilation by increasing coronary blood flow via selective dilation of the coronary arteries. The state of the coronary arteries can determine the effect of dipyridamole on coronary blood flow and metabolic responses. Blood flow increased by 80% in unobstructed coronary vessels, whereas in stenotic coronary arteries, flow increased by approximately 40%. In patients with single-vessel coronary artery disease, intravenous dipyridamole increases flow to the ischemic area, probably by increasing collateral blood flow. Dipyridamole increases intracellular concentrations of the coronary vasodilator adenosine and cAMP and inhibits adenosine metabolism and uptake by vascular endothelial cells. The increased concentration of adenosine in vascular smooth muscle stimulates adenylate cyclase activity, leading to increased cAMP synthesis and, consequently, to relaxation of vascular smooth muscle (vasodilation). The effect of dipyridamole might not result only from its effect on adenosine but also from its ability to increase prostacyclin (vasodilator and platelet inhibitor) production by increasing cAMP concentration. Dipyridamole also increases intracellular cAMP by inhibiting PDE3, decreasing cAMP breakdown. Adenosine is a natural vasodilatory substance released by the myocardium during hypoxic episodes. Some structural similarity of adenosine to dipyridamole is apparent and substantiates this mechanism. Dipyridamole is used prophylactically, but its efficacy in reducing the incidence and severity of anginal attacks is not universally accepted.

Papaverine (Fig. 21.8) is a benzoisoquinoline vasodilator (an alkaloid found in opium) that produces generalized, nonspecific arteriolar dilatation and smooth muscle relaxation. Its oral bioavailability ranges from 30% to 50%, suggesting first-pass 6-*O*-demethylation metabolism by CYP3A4. Increased levels of intracellular cAMP secondary to inhibition of phosphodiesterase can contribute to its vasodilatation and relaxation effects without involving

nerve supply. Large doses of papaverine can cause hypotension and tachycardia. Other studies suggest that it also depresses cardiac conduction and prolongs the refractory period.

The natriuretic polypeptide nesiritide (Natreacor) is manufactured in the United States using recombinant DNA technology for intravenous use to treat cases of angina and CHF. Nesiritide has the same amino acid sequence of the natural, 32-amino acid natriuretic peptide that normally is released during cardiac ischemia and acts as vasodilator. Nesiritide acts as a coronary vasodilator by binding to the soluble guanylate cyclase receptors in vascular smooth muscles, leading to increased intracellular concentrations of the vasodilator cGMP.

DRUGS FOR THE TREATMENT OF CARDIAC ARRHYTHMIA

Arrhythmia

Arrhythmia is an alteration in the normal sequence of electrical impulse rhythm that leads to contraction of the myocardium. It is manifested as an abnormality in the rate, in the site from which the impulses originate, or in the conduction through the myocardium. The rhythm of the heart normally is determined by a pacemaker site called the SA node, which consists of specialized cells that undergo spontaneous generation of action potentials at a rate of 100 to 110 action potentials (“beats”) per minute. This intrinsic rhythm is strongly influenced by the vagus nerve, overcoming the sympathetic system at rest. This “vagal tone” brings the resting heart rate down to a normal sinus rhythm of 60 to 100 beats per minute. Sinus rates below this range are termed “sinus bradycardia,” and sinus rates above this range are termed “sinus tachycardia.” The sinus rhythm normally controls both atrial and ventricular rhythm. Action potentials generated by the SA node spread throughout the atria, depolarizing this tissue and causing atrial contraction. The impulse then travels into the ventricles via the AV node. Specialized conduction pathways within the ventricle rapidly conduct the wave of depolarization throughout the ventricles to elicit ventricular contraction. Therefore, normal cardiac rhythm is controlled by the pacemaker activity of the SA node. Abnormal or irregular cardiac rhythms (heartbeats) can occur when the SA node fails to function normally, when other pacemaker sites (e.g., ectopic pacemakers) trigger depolarization, or when a dysfunction occurs along the normal conduction pathways.

Causes of Arrhythmias

Many factors influence the normal rhythm of electrical activity in the heart. Arrhythmias can occur either because pacemaker cells fail to function properly or because of a blockage in transmission through the AV node. Underlying diseases, such as atherosclerosis, hyperthyroidism, or lung disease, can also be initiating

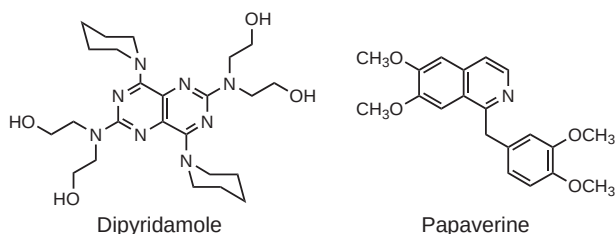


FIGURE 21.8 Miscellaneous coronary vasodilators.

factors. Some of the more common arrhythmias are those termed “ectopic,” which occur when electrical signals spontaneously arise in regions other than the pacemaker and then compete with the normal impulses. Myocardial ischemia, excessive myocardial catecholamine release, stretching of the myocardium, and cardiac glycoside toxicity have all been shown to stimulate ectopic foci. A second mechanism for the generation of arrhythmias is from a phenomenon called reentry. This occurs when the electrical impulse does not die out after firing but, rather, continues to circulate and reexcite resting heart cells into depolarizing. The result of this reexcitation can be a single, premature beat or runs of ventricular tachycardia. Reentrant rhythms are common in the presence of coronary atherosclerosis.

Drugs for the Treatment of Arrhythmias and Their Classification

It is widely accepted that most currently available antiarrhythmic drugs can be classified into four categories, which are grouped on the basis of their effects on the cardiac action potential and, consequently, on the electrophysiologic properties of the heart. To understand the basis of classification and the pharmacology of these agents, an understanding of normal cardiac electrophysiology is necessary.

Normal Physiologic Action

Normal cardiac contractions largely are a function of the action of a single atrial pacemaker, a fast and usually uniform conduction in predictable pathways, and a normal duration of the action potential and refractory period. Figure 21.9 depicts a normal cardiac action potential from a Purkinje fiber. The resting cell has a membrane potential of approximately -90 mV, with the inside of the cell being electronegative relative to the outside of the cell. This is termed the “transmembrane resting potential.” On excitation, the transmembrane potential reverses,

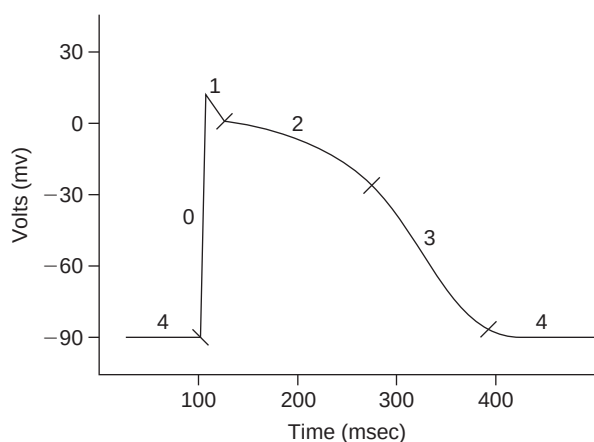


FIGURE 21.9 Cardiac action potential recorded from a Purkinje fiber.

and the inside of the membrane rapidly becomes positive with respect to the outside. On recovery from excitation, the resting potential is restored. These changes have been divided into five phases: Phase 0 represents depolarization and reversal of the transmembrane potential, phases 1 to 3 represent different stages of repolarization, and phase 4 represents the resting potential. During phase 0, which is also referred to as rapid depolarization, the permeability of the membrane for sodium ions increases, and sodium rapidly enters the cell, causing it to become depolarized. Phase 1 results from the ionic shift, which creates an electrochemical and concentration gradient that reduces the rate of sodium influx but favors the influx of chloride and efflux of potassium. Phase 2, the plateau phase, results from the slow inward movement of calcium, which is triggered by the rapid inward movement of sodium in phase 0. During this time, there is also an efflux of potassium that balances the influx of calcium, thus resulting in little or no change in membrane potential. Phase 3 is initiated by a slowing of the calcium influx coupled with a continued efflux of potassium. This continued efflux of potassium from the cell restores the membrane potential to normal resting potential levels. During phase 4, the Na^+ , K^+ -ATPase pump restores the ions to their proper local concentrations. The action potential is a coordinated sequence of ion movements in which sodium initially enters the cell, followed by a calcium influx, and finally, a potassium efflux returns the cell to its resting state. Several antiarrhythmic agents exert their effects by altering these ion fluxes.

Classification of Antiarrhythmic Drugs

CLASS I SODIUM CHANNEL BLOCKERS

Class IA Antiarrhythmic Drugs This class of antiarrhythmic drugs includes local anesthetics acting on nerve and myocardial membranes to slow conduction by blocking fast Na^+ channels, inhibiting phase 0 of the action potential (Fig. 21.9). Myocardial membranes show the greatest sensitivity. Class IA drugs decrease the maximal rate of depolarization without changing the resting potential. They also increase the threshold of excitability, increase the effective refractory period, decrease conduction velocity, and decrease spontaneous diastolic depolarization in pacemaker cells. The decrease in diastolic depolarization tends to suppress ectopic foci activity. Prolongation of the refractory period tends to abolish reentry arrhythmias. Table 21.6 summarizes these effects. Quinidine is considered to be the prototype drug for class IA.

Quinidine. Quinidine (Fig. 21.10) is widely used for acute and chronic treatment of ventricular and supraventricular arrhythmias, especially supraventricular tachycardia. It is a member of a family of alkaloids found in Cinchona bark (*Cinchona officinalis* L.) and is the diastereomer of quinine. Despite their structural similarity, quinidine and quinine differ markedly in their effects on the cardiac muscles, with the effects of quinidine being much more pronounced. Structurally, quinidine is composed

TABLE 21.6 Summary of the Cardiac Physiologic Effects of the Antiarrhythmic Drugs

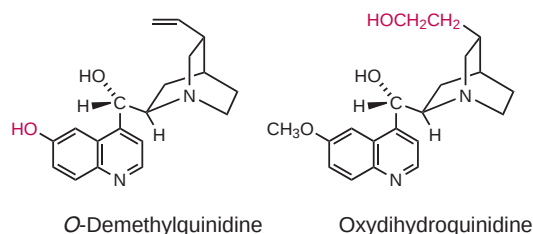
Classification	Mechanism of Action	Primary Sites of Action	Drug Examples
Class IA	Na ⁺ channel blockade	Atrial and ventricular tissue	Quinidine
	Intermediate rate of dissociation from sodium channels		Procainamide
	Slows phase 0 depolarization		Disopyramide
	Prolongs action potential duration		
	Slows conduction		
Class IB	Na ⁺ channel blockade	Ventricular tissue	Lidocaine
	Rapid rate of dissociation from sodium channels		Mexiletine
	Shortens phase 3 repolarization		Phenytoin
	Shortens action potential duration		Tocainide
Class IC	Na ⁺ channel blockade	Ventricular tissue	Flecainide
	Slows rate of dissociation from sodium channels		Encainide
	Markedly slows phase 0 depolarization		Propafenone
	Slows conduction		Moricizine
Class II	Blocks sympathetic stimulation of β_1 -adrenergic receptors	SA node	Propranolol
	Slows phase 4 depolarization	AV node	Sotalol
	Slows firing of SA node and conduction through AV node, prolonging repolarization		β_1 -Blockers
Class III	K ⁺ channel blockade (block delayed rectifier current)	Atrial and ventricular tissue	Amiodarone
	Prolongs phase 3 repolarization		Dronedarone Sotalol
	Prolongs duration of action potential, which prolongs refractory period		Bretylium
Class IV	Ca ²⁺ channel blockade	SA node	Verapamil
	Slows phase 4 depolarization	AV node	Diltiazem
	Slows firing of SA node and conduction through AV node, prolonging repolarization of AV node		

AV, atrioventricular; SA, sinoatrial.

of a quinoline ring and the bicyclic quinuclidine ring system, with a hydroxymethylene bridge connecting these two components. Examination of quinidine reveals two basic nitrogens, with the quinuclidine nitrogen ($pK_a = 11$) being the stronger of the two. Because of the basic character of quinidine, it is always used as water-soluble salt forms. These salts include quinidine sulfate, gluconate, and polygalacturonate. Good absorption (~95%) is observed with each of these forms after oral administration. In special situations, quinidine can be administered intravenously as the gluconate salt. The use of intravenous quinidine, however, is rare. The gluconate salt is particularly suited for parenteral use because of its high water solubility and lower irritant potential.

Quinidine's bioavailability appears to depend on a combination of metabolism and P-gp efflux. The

bioavailabilities of quinidine sulfate and gluconate are 80% to 85% and 70% to 75%, respectively. Once absorbed, quinidine is subject to hepatic first-pass CYP3A4 metabolism and is approximately 85% plasma protein bound, with an elimination half-life of approximately 6 hours. Quinidine is metabolized mainly in the liver, and renal excretion of unchanged drug is also significant (~10% to 50%).



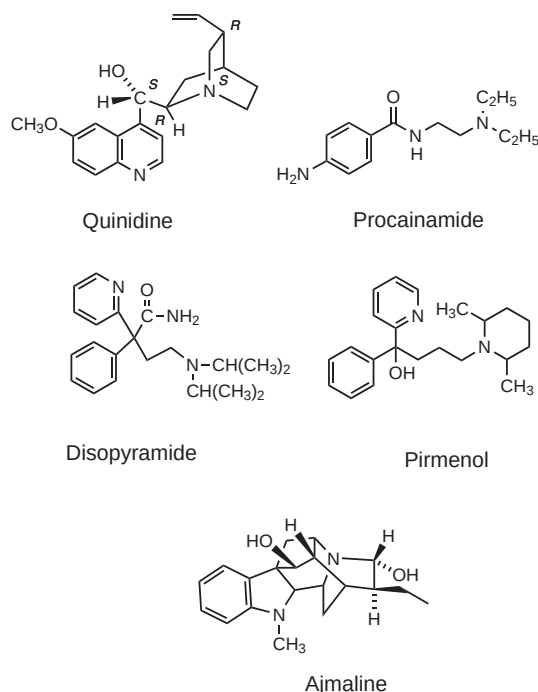


FIGURE 21.10 Class IA antiarrhythmics.

The metabolites are hydroxylated derivatives at either the quinoline ring through first-pass *O*-demethylation or at the quinuclidine ring through oxidation of the vinyl group. These metabolites possess only about one-third the activity of quinidine. Their contribution to the overall therapeutic effect of quinidine is unclear. The clinical significance of the well-documented digoxin–quinidine interaction was described previously in the section on digoxin–drug interactions. Apparently, quinidine (a P-gp substrate) inhibits the renal tubular secretion of digoxin via the P-gp efflux pump, resulting in increased plasma concentration for digoxin.

In addition, a common contaminant in quinidine preparations, dihydroquinidine, which is derived from reduction of the quinuclidine vinyl group at C-3 to an ethyl group, can also contribute to its activity (32). Although similar to quinidine in pharmacodynamic and pharmacokinetic behavior, this contaminant is both more potent as an antiarrhythmic and more toxic. Thus, levels of this contaminant can contribute to variability between commercial preparations. The most frequent adverse effects associated with quinidine therapy are gastrointestinal disturbances, such as nausea, diarrhea, and vomiting.

Procainamide. Procainamide (Fig. 21.10) is effective in the treatment of several types of cardiac arrhythmias. Its actions are similar to those of quinidine, yet procainamide can be effective in patients who are unresponsive to quinidine. The initial development of procainamide was stimulated by the observation that the local anesthetic procaine (the ester bio-isostere

of procainamide), when administered intravenously, produced significant, although short-lived, antiarrhythmic effects. Unfortunately, considerable central nervous system toxicity, in addition to the short duration, limited the usefulness of this agent. Moreover, procaine is not active orally because of its short duration of action caused by both chemical and plasma esterase hydrolysis. A logical modification of this molecule was the isosteric replacement of the ester with an amide group. This produced orally active procainamide, which is more resistant to both enzymatic and chemical hydrolysis. Peak plasma levels of procainamide are observed within 45 to 90 minutes after oral administration, and approximately 70% to 80% of the dose is bioavailable. Approximately half of this dose is excreted unchanged, and the remaining half undergoes acetylation metabolism in the liver. Metabolites of procainamide include *p*-aminobenzoic acid and *N*-acetylprocainamide. Interestingly, the acetylated metabolite is also active as an antiarrhythmic. Its formation accounts for up to one-third of the administered dose and is catalyzed by the liver enzyme *N*-acetyltransferase. Because acetylation is strongly influenced by an individual's genetic background, marked variability in the amounts of this active metabolite can be observed from patient to patient. Renal excretion dominates, with approximately 90% of a dose excreted as unchanged drug and metabolites. The elimination half-life is approximately 3.5 hours. A substantial percentage (60% to 70%) of patients on procainamide show elevated levels of antinuclear antibodies after a few months. Of these patients, between 20% and 30% develop a drug-induced lupus syndrome if therapy is continued. These adverse effects, which are attributed to the aromatic amino group, are observed more frequently and more rapidly in "slow acetylators." As a rule, the symptoms associated with procainamide-induced lupus syndrome subside fairly rapidly after the drug is discontinued. These problems, however, have discouraged long-term procainamide therapy.

Disopyramide. Disopyramide phosphate (Norpace) is used orally for the treatment of certain ventricular and atrial arrhythmias. Despite its structural dissimilarity to procainamide (Fig. 21.10), its cardiac effects are very similar. Disopyramide is rapidly and completely absorbed from the gastrointestinal tract. Peak plasma level is, as a rule, reached within 1 to 3 hours, and a plasma half-life of 5 to 7 hours is common. Approximately half of an oral dose is excreted unchanged in the urine. The remaining drug undergoes hepatic CYP3A4 metabolism, principally to the corresponding *N*-dealkylated metabolite. This metabolite retains approximately half the antiarrhythmic activity of disopyramide and is also subject to renal excretion. Adverse effects of disopyramide frequently are observed. These effects are primarily anticholinergic in nature and include dry mouth, blurred vision, constipation, and urinary retention.

Ajmaline. Ajmaline (Fig. 21.10) inhibits sodium influx and prolongs phase 3. The drug is a natural product alkaloid obtained from the roots of *Rauwolfia serpentina* and is used in other parts of the world to treat ventricular tachyarrhythmia and supraventricular arrhythmias. Ajmaline is given orally, intramuscularly, and as intravenous infusion over a 5-minute period. Being a *Rauwolfia* alkaloid with similar effects on the neurotransmitters, the drug has also been used in the differential diagnosis of Wolff-Parkinson-White syndrome.

Pirmenol. Pirmenol (Fig. 21.10) is a class IA antiarrhythmic drug marketed in Japan, but not in the United States, for the control of the ventricular arrhythmias. The drug is structurally similar to disopyramide, except that the amide functionality has been replaced by a secondary alcohol and the basic nitrogen is incorporated in a piperidine heterocyclic ring.

Class IB Antiarrhythmic Drugs

Lidocaine. Lidocaine, similar to procaine, is an effective, clinically used local anesthetic (Fig. 21.11) (see Chapter 16). Its cardiac effects, however, are distinctly different from those of procainamide or quinidine. Lidocaine normally is reserved for the treatment of ventricular arrhythmias and, in fact, is typically the drug of choice for emergency treatment of ventricular arrhythmias. Its utility in these situations results from the rapid onset of antiarrhythmic effects on intravenous infusion. In addition, these effects cease soon after the infusion is terminated. Thus, lidocaine therapy can be rapidly modified in response to changes in the patient's status. Lidocaine is effective as an antiarrhythmic only when given parenterally, and the intravenous route is the most common. Antiarrhythmic activity is not observed after oral administration because of the rapid and efficient first-pass CYP3A4 and 2D6 metabolism by the liver.

Parenterally administered lidocaine is approximately 60% to 70% plasma protein bound. Hepatic metabolism is rapid (plasma half-life, ~15 to 30 minutes) and primarily involves *N*-deethylation to yield monoethylglycinexylide, followed by amidase-catalyzed hydrolysis into *N*-ethylglycine and 2,6-dimethylaniline (2,6-xylydine) (Fig. 21.12).

Monoethylglycinexylide has good antiarrhythmic activity. It is not clinically useful, however, because it undergoes rapid enzymatic hydrolysis. The adverse effects of lidocaine include emetic and convulsant properties that predominantly involve the central nervous system and heart. The central nervous system effects can begin with dizziness and paresthesia and, in severe cases, ultimately lead to epileptic seizures.

Tocainide. Tocainide (Tonocard) (Fig. 21.11) is an α -methyl analog structurally related to monoethylglycinexylide, the active metabolite of lidocaine, which possesses very similar electrophysiologic effects to lidocaine. In contrast to lidocaine, tocainide is orally active, and its oral absorption is excellent. Like lidocaine, it is reserved for the treatment of ventricular arrhythmias. The α -methyl group is believed to slow the rate of metabolism and, thereby, to contribute to oral activity. The plasma half-life of tocainide is approximately 12 hours, and nearly 50% of the drug can be excreted unchanged in the urine. Adverse effects associated with tocainide are like those observed with lidocaine—specifically, gastrointestinal disturbances and central nervous system effects.

Mexiletine. Mexiletine (Mexitil) (Fig. 21.11) is similar to both lidocaine and tocainide in its effects and therapeutic application. It is used principally to treat and prevent ventricular arrhythmias. Like tocainide, mexiletine has very good oral activity and absorption properties. Clearance depends on CYP2D6 metabolism and renal excretion. A relatively long plasma half-life of approximately 12 to 16 hours is common. Adverse effects are similar to those experienced with tocainide and lidocaine.

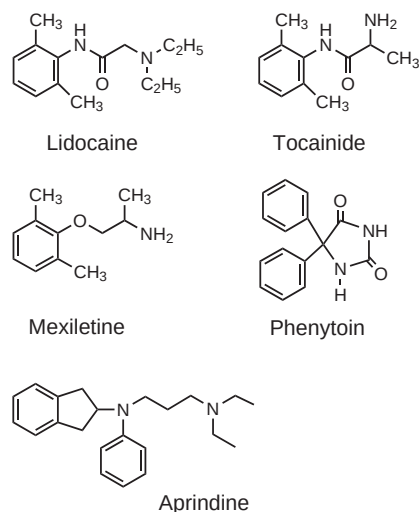


FIGURE 21.11 Class IB antiarrhythmics.

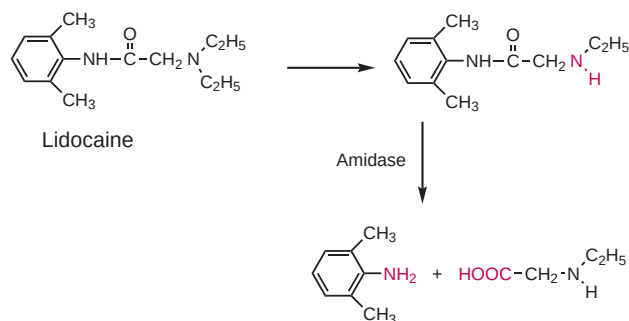


FIGURE 21.12 Metabolism of lidocaine.

Phenytoin. For 50 years, phenytoin (Fig. 21.11) has seen clinical use in the treatment of epileptic seizures (see Chapter 17). During this time, it was noticed that phenytoin also produced supposedly adverse cardiac effects. On closer examination, these adverse effects actually were found to be beneficial in the treatment of certain arrhythmias. Currently, phenytoin is used in the treatment of atrial and ventricular arrhythmias resulting from digitalis toxicity. It is not, however, officially approved for this use.

Phenytoin can be administered either orally or intravenously and is absorbed slowly after oral administration, with peak plasma levels achieved after 3 to 12 hours. It is extensively plasma protein bound (~90%), and the elimination half-life is between 15 and 30 hours. These large ranges reflect the considerable variability observed from patient to patient. Parenteral administration of phenytoin is limited to the intravenous route. Phenytoin for injection is dissolved in a highly alkaline vehicle (pH 12). This alkaline vehicle is required because phenytoin is weakly acidic and has very poor solubility in its un-ionized form. Reportedly, however, its phosphate ester fosphenytoin has water solubility advantages over phenytoin for injection. Intramuscular phenytoin is usually avoided, because it results in tissue necrosis at the site of injection and has erratic absorption as a result of high alkalinity. In addition, intermittent intravenous infusion is required to reduce the incidence of severe phlebitis.

Phenytoin metabolism is relatively slow and predominantly involves aromatic hydroxylation by the CYP2C family of enzymes to *p*-hydroxylated inactive metabolites (see Chapter 17). Phenytoin also induces its own metabolism and is subject to large interindividual variability. The major metabolite, 5-*p*-hydroxyphenyl-5-phenylhydantoin, accounts for approximately 75% of a dose. This metabolite is excreted through the kidney as the β -glucuronide conjugate. Phenytoin clearance is strongly influenced by its metabolism; therefore, agents that affect phenytoin metabolism can cause intoxication. In addition, because phenytoin is highly plasma protein bound, agents that displace phenytoin can also cause toxicity.

Aprindine. Aprindine (Fig. 21.11) is classified as class IB antiarrhythmic agent due its inhibitory effects on the sodium current without affecting the duration of the action potential. The drug is available in Europe, but has not been approved by the FDA. Aprindine is administered both orally and intravenously to treat ventricular and supraventricular arrhythmias. The drug is well absorbed after oral administration with extensive binding (85% to 90%) to plasma proteins and is largely metabolized by hepatic CYP2D6 through aromatic hydroxylation. Adverse effects reported for the drug included hypersensitivity, inducing agranulocytosis and visual disturbances.

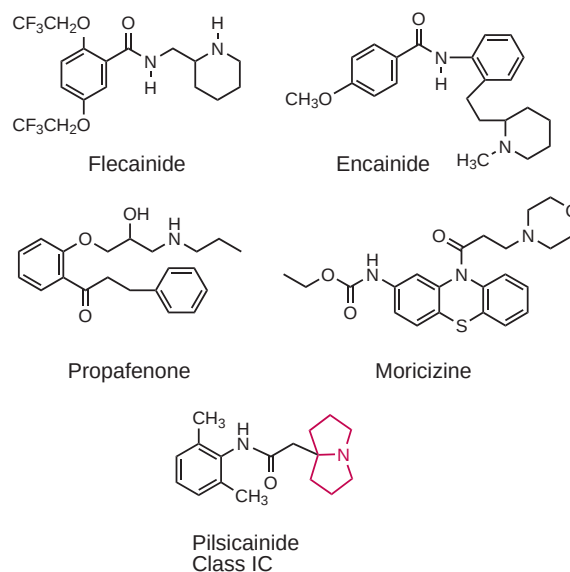


FIGURE 21.13 Class IC antiarrhythmics.

Class IC Antiarrhythmic Drugs

Flecainide. Flecainide (Tambacor) exhibits properties distinctly different from those of class IA or IB antiarrhythmic drugs. Flecainide is a fluorinated benzamide derivative (Fig. 21.13), available as the acetate salt. Flecainide has been approved by the FDA for the treatment of ventricular arrhythmias. Clinical studies suggest that this agent can be more effective than either quinidine or disopyramide in suppressing premature ventricular contractions. Oral flecainide is well absorbed, and the plasma half-life is approximately 14 hours. About half of an oral dose is CYP2D6 metabolized in the liver, and one-third is excreted unchanged in the urine. As with other antiarrhythmics, flecainide can produce adverse effects. The most severe is flecainide's occasional tendency to aggravate existing arrhythmias or to induce new ones. Although fewer than 10% of patients experience this effect, it can be life-threatening. Accordingly, it can be desirable to start therapy in the hospital. Other less serious side effects include blurred vision, headache, nausea, and abdominal pain.

Encainide. Encainide (Enkaid) (Fig. 21.13) represents another benzamide derivative, with similar pharmacologic properties to flecainide but with less negative inotropic effect.

Propafenone. Propafenone (Rythmol) (Fig. 21.13) is a class I, local anesthetic-type antiarrhythmic agent. Propafenone is structurally related to other class IC antiarrhythmic drugs and also to β -adrenergic receptor blockers. It is used primarily for ventricular and supraventricular arrhythmias. The drug is administered orally and intravenously; however, the parenteral dosage forms are not commercially available in the United States. After oral

administration, the drug is rapidly and almost completely absorbed from the gastrointestinal tract. Propafenone metabolism involves hepatic CYP2D6 enzymes. Its rate of metabolism is genetically determined by an individual's ability to metabolize the so-called phenotype compounds (fast or slow metabolizers) (see Chapter 9).

Moricizine. Moricizine (Ethmozine) (Fig. 21.13) is a phenothiazine analog that processes the same electrophysiologic effects on the heart as those of class IC antiarrhythmics. Despite its short half-life after oral administration due to CYP1A2 metabolism, its antiarrhythmic effects can persist for many hours, suggesting that some of its metabolites can be active.

Pilsicainide. Pilsicainide (Fig. 21.13) is an analog of lidocaine where the diethylamino group of lidocaine has been replaced with a pyrrolizidine bicyclic ring structure. Despite its structural similarity to lidocaine (Fig. 21.10), its electrophysiologic profile as a class IC antiarrhythmic is unlike that of lidocaine, which is a class IB antiarrhythmic drug. Apparently, class IC drugs prefer a bulky *N*-substituent. The drug is marketed in Japan and is not approved by the FDA. Pilsicainide is administered both orally and by injection to treat tachyarrhythmias and atrial fibrillation.

CLASS II ANTIARRHYTHMIC DRUGS Class II antiarrhythmic drugs (Fig. 21.14) are β -adrenergic receptor-blocking agents that block the role of the sympathetic nervous system in the genesis of certain cardiac arrhythmias. Their dominant electrophysiologic effect is to depress adrenergically enhanced calcium influx through β -receptor blockade. Drugs in this class decrease neurologically induced automaticity at normal therapeutic doses. At higher doses, these drugs can also exhibit anesthetic properties, which cause decreased excitability, decreased conduction velocity, and a prolonged effective refractory period. In normal therapeutic situations, the β -blocking effects are more important than any local anesthetic effects that these drugs can have. Propranolol is the prototype β -adrenergic blocker drug for class II (see Chapter 10).

Propranolol (+)-Propranolol, a nonselective β -adrenergic blocker, is the prototype for class II antiarrhythmics. Its pharmacology and pharmacokinetics are discussed in detail in Chapters 10 and 24. Its use as an antiarrhythmic is typically for the treatment of

supraventricular arrhythmias, including atrial flutter, paroxysmal supraventricular tachycardia, and atrial fibrillation. Propranolol is also reported to be effective in the treatment of digitalis-induced ventricular arrhythmias. Moreover, beneficial results can be obtained when propranolol is used in combination with other agents. For example, in certain cases, quinidine and propranolol together have proved to be more successful in alleviating atrial fibrillation than either agent has alone. Few serious adverse effects are associated with propranolol therapy.

Sotalol ($\pm$)-Sotalol (Betapace, Sotacor), a nonselective β -adrenergic blocker, is a methanesulfonanilide antiarrhythmic agent. As an antiarrhythmic, it is dually classified as class II and class III because of the similarity of its cardiac effects to both classes. Sotalol is used orally to suppress and prevent the recurrence of life-threatening ventricular arrhythmia.

CLASS III POTASSIUM CHANNEL BLOCKERS Class III drugs cause a homogeneous prolongation of the duration of the action potential. This results in a prolongation of the effective refractory period. It is believed that most of class III antiarrhythmic agents act through phase 3 of the action potential by blocking potassium channels. Figure 21.15 illustrates the chemical structures of members of class III. Bretylium is the prototype drug for this class.

Bretylium Tosylate Bretylium tosylate is a quaternary ammonium salt derivative (Fig. 21.15) originally developed for use as an antihypertensive. Its antiarrhythmic use is limited to emergency, life-threatening situations in which other agents, such as lidocaine and procainamide, have failed. Bretylium is used only in intensive care units and can be administered either intravenously or intramuscularly. The plasma elimination half-life is, as a rule, approximately 10 hours, and it is eliminated largely unchanged in the urine. The major adverse effect associated with bretylium tosylate is hypotension, including orthostatic hypotension, which can be very severe.

Amiodarone Initially developed as an antianginal (coronary vasodilator), amiodarone (Fig. 21.15) has antiarrhythmic effects that are somewhat similar to those of bretylium. It is approved by the FDA for the treatment of life-threatening ventricular arrhythmias that are refractory to other drugs. Its cardiac effects are not well characterized, but clinical studies indicate that it is primarily a class III agent but acts also as a class I, II, and IV antiarrhythmic. It has a unique mechanism of action that involves alteration of the lipid membrane in which ion channels and receptors are located. Its severe toxicity, however, makes it the drug of last choice. As with bretylium tosylate, use of this agent should be initiated in a hospital setting.

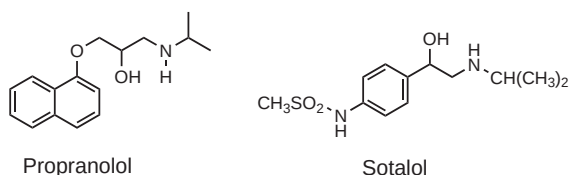


FIGURE 21.14 Class II inhibitors of depolarization.

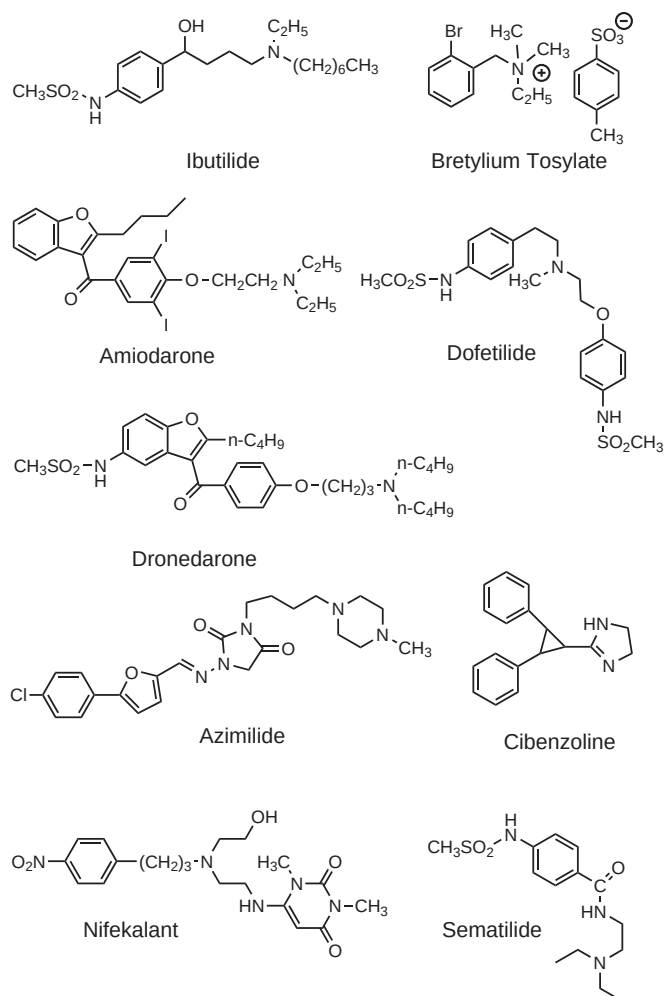


FIGURE 21.15 Class III antiarrhythmic agents.

Ibutilide Ibutilide (Corvert) is another methanesulfonanilide derivative (Fig. 21.15), but unlike sotalol, it lacks any β -adrenergic-blocking activity. Like sotalol, it exhibits electrophysiologic effects characteristic of class III. Ibutilide is used only by intravenous infusion as its fumarate salt.

Dofetilide Dofetilide (Tikosyn) is a bis-methanesulfonanilide derivative that is essentially the non- β -blocking moiety of the sotalol molecule (Fig. 21.15). It exhibits only class III electrophysiologic effects, but like ibutilide, it lacks any β -adrenergic-blocking activity. Dofetilide is used orally to suppress atrial fibrillation and flutter. It is more potent and selective than other class III methanesulfonanilides, including sotalol. Dofetilide is well absorbed from the gastrointestinal tract, with a bioavailability of 96% to 100%. The bioavailability of oral dofetilide is not affected by food or antacids. Protein binding is 60% to 70%. Dofetilide is metabolized by the hepatic CYP3A4 enzyme system via *N*-dealkylation and

N-oxidation to inactive or minimally active metabolites. Of the approximately 80% of a dose excreted in urine, approximately 80% is excreted unchanged, with the other 20% excreted as metabolites.

Dronedarone (Multaq) Dronedarone was approved by the FDA in 2009 for the treatment of atrial fibrillation and atrial flutter. Dronedarone is the benzofuran noniodinated derivative of amiodarone, with two additional functional group differences including a methanesulfonamido group on the benzofuran ring and dibutyl substitution on the basic nitrogen. The iodine groups of amiodarone were removed to reduce toxic effects of amiodarone on the thyroid and other organs. The methanesulfonamido group was added to dronedarone to reduce lipophilicity and thus reduce neurotoxic effects. Dronedarone exhibits amiodarone-like class III antiarrhythmic activity in vitro and in clinical trials. Dronedarone is less lipophilic than amiodarone, has a much smaller volume of distribution, and has an elimination half-life of 24 hours, in contrast to amiodarone's half-life of several weeks. As a result of these pharmacokinetic characteristics, dronedarone dosing may be less complicated than amiodarone. The drug's bioavailability after oral administration is very poor (4%). Concurrent administration with high-fat meal improves the bioavailability to 15%. Accordingly, the drug is recommended to be taken with food. After oral absorption, the drug has 98% plasma protein binding and is metabolized by the liver by CYP3A4 *N*-debutylation into an active metabolite. Renal excretion is minimal (6%), and 84% of the administered dose is excreted in feces as either the parent drug or its metabolite.

Therapeutic Precaution. Because heart failure is a major serious adverse effect of dronedarone, the FDA approved the addition of a black box warning to the package insert, a summary of which is as follows:

Warning: Heart failure: Multaq is contraindicated in patients with NYHA Class IV heart failure, or NYHA Class II–III heart failure with recent hospitalization or referral to a specialized heart failure clinic. In a placebo-controlled study in patients with severe heart failure requiring recent hospitalization or referral to a specialized heart failure clinic for worsening symptoms, patients given dronedarone had a greater than two-fold increase in mortality. Such patients should not be given dronedarone.

Azimilide Azimilide is a class III antiarrhythmic agent structurally unrelated to any of the other class III agents (Fig. 21.15). Azimilide is a potassium blocker similar to dofetilide or sotalol, but it blocks both I_{Kr} (rapid) and I_{Ks} (slow) inwardly rectifier potassium channels, which are responsible for repolarizing cardiac myocytes toward the end of the cardiac action potential. Blockages of both channels result in an increase of the QT interval and a prolongation of atrial and ventricular refractory

periods. The drug does not perform as a β -blocker like sotalol. Azimilide has not been approved by the FDA and is only available in Europe. Following oral administration, the drug is completely absorbed, with no effect of food. Protein binding is 94%. The drug is metabolized in the liver to an active carboxylate metabolite, but its concentration in plasma is less than 5% of the parent compound. Thus, it is considered to be therapeutically inactive. Renal excretion is approximately 10%. Its elimination half-life is 3 to 4 days; thus, azimilide can be dosed once daily, limiting major fluctuations in blood levels.

Cibenzoline Cibenzoline (Fig. 21.15) is marketed in Japan and has not been approved by the FDA. The drug is structurally unrelated to any of the well-known nuclei of the other families of antiarrhythmic agents. Cibenzoline is a diphenyl-cyclopropyl-2-imidazoline derivative with electrophysiologic properties that make it difficult to categorize under a specific class. Although based on its electrophysiologic effects on the potassium current the drug is classified as class III antiarrhythmic agent, it is also reported to have characteristics that fit classes IA, IC, and IV. Cibenzoline is given orally and by injection for the management of ventricular and supraventricular arrhythmias. The drug is well absorbed after administration, with 90% bioavailability and moderate plasma protein binding (50% to 60%). The drug is excreted mostly unchanged in urine, which requires attention if given to patients with renal insufficiency. Cibenzoline is also associated with several adverse effects including gastrointestinal tract disturbances, neurologic side effects, and most notably, hypoglycemia.

Nifekalant Nifekalant (Fig. 21.15) is another class III antiarrhythmic agent developed and marketed in Japan that is not available in the United States. Nifekalant is used intravenously as a lifesaving drug for the treatment of life-threatening ventricular arrhythmias. The drug is a derivative of uracil (pyrimidindione), which is

considered as a novel nucleus for class III antiarrhythmic agents. Very few clinical reports about the drug's adverse effects are available.

Sematilide Sematilide (Fig. 21.15) is an investigational class III antiarrhythmic drug that has not been approved by the FDA for clinical use. The drug is an analog of procainamide except that the aniline nitrogen has been substituted with a methylsulfone moiety. The methylsulfonamido procainamide derivative changed the molecule's selectivity from a sodium channel blocker (class IA antiarrhythmic) into a class III antiarrhythmic agent that selectively blocks the rapidly activating delayed rectifier K^+ current (I_{kr}) in atrial myocytes, and evidence supports the usefulness of the drug as a class III antiarrhythmic agent. Further studies are needed to assess sematilide's place in antiarrhythmic therapy, including comparisons with prototypical class III agents (e.g., amiodarone, sotalol) and other pure class III agents (e.g., dofetilide).

CLASS IV CALCIUM CHANNEL BLOCKERS Class IV calcium channel antiarrhythmic drugs (Fig. 21.7) comprise a group of agents that selectively block the slow inward current carried by calcium (i.e., calcium channel blockers). The slow inward current in cardiac cells has been shown to be of importance for the normal action potential in SA node cells. It has also been suggested that this inward current is involved in the genesis of certain types of cardiac arrhythmias. Administration of a class IV drug causes a prolongation of the refractory period in the AV node and the atria, a decrease in AV conduction, and a decrease in spontaneous diastolic depolarization. These effects block conduction of premature impulses at the AV node and, thus, are very effective in treating supraventricular arrhythmias. Verapamil and diltiazem are prototype drugs for this class (Fig. 21.7), but dihydropyridine drugs are less effective in cardiac tissues. Refer to the section on calcium channel blockers under antianginal drugs for pharmacokinetic information.

SCENARIO: OUTCOME AND ANALYSIS

Outcome

Judy Cheng, PharmD

JS was initiated on diltiazem sustained release 240 mg daily for ventricular rate control and warfarin 5 mg daily to prevent a thromboembolic event that could be caused by atrial fibrillation. The sustained-release formulation was chosen to improve compliance. The lisinopril therapy was kept for now. Blood pressure will be monitored closely, and lisinopril may be discontinued if the patient's blood pressure becomes too low after the addition of diltiazem. JS will also go to a warfarin clinic to have his INR (International Normalized Ratio: prothrombin time) monitored.

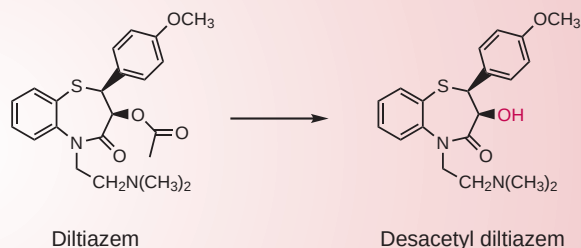
Chemical Analysis

S. William Zito and Victoria Roche

Diltiazem is the only clinically useful benzo[*b*-1,5]thiazepine chemical class of calcium channel blockers. Diltiazem possesses asymmetric centers and the dextrorotatory (i.e., the (+)-enantiomer) is approximately one order of magnitude more potent as a calcium channel blocker than the levorotatory (i.e., (–)-enantiomer).

Diltiazem affects both the heart and the arteriolar bed by inhibiting the influx of extracellular calcium across both the myocardial and

vascular smooth muscle cell membranes. This results in inhibition of the contractile processes and is used clinically in the management of angina, hypertension, and cardiac arrhythmias, often in combination with angiotensin-converting enzyme inhibitors such as lisinopril.



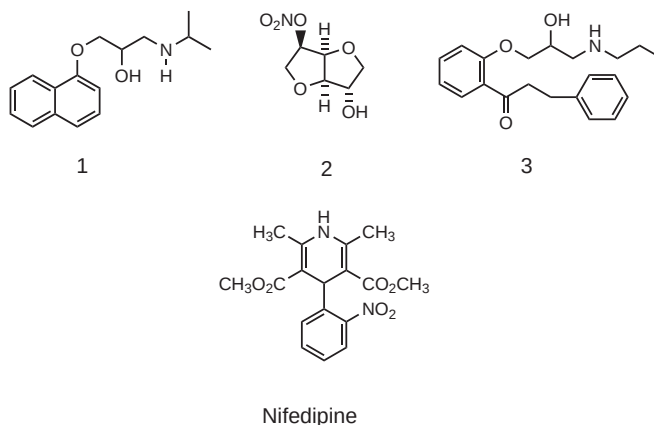
Diltiazem is primarily metabolized by enzyme hydrolysis to its desacetyl derivative, which retains approximately 25% to 50% of the activity and adds to its duration of action. Diltiazem also undergoes *N*-demethylation by CYP3A4 and *O*-demethylation by CYP2D6. The *N*-demethylated metabolism pathway results in mechanism-based inhibition of CYP3A4. Because R-warfarin is metabolized by CYP 3A4 to yield 6-, 8-, and 10-hydroxylated metabolites, it is important that JR's INR be monitored.

CASE STUDY

S. William Zito and Victoria Roche

SB is a fairly active 45-year-old white man. He works for a shipping company and plays third base in the company softball league on weekends. SB was diagnosed with type 2 diabetes 5 years ago, which he successfully controls with diet and metformin (850 mg daily). Knowing that diabetics are prone to silent ischemic heart disease, his endocrinologist orders an exercise stress test as a routine screening measure. The test implies that SB has silent myocardial ischemia at a peak heart rate of 171 bpm and peak blood pressure of 195/85 mm Hg. These results suggest

that SB may be at risk for sudden cardiac death, especially if he continues the high exertion associated with playing softball. The diagnosis was confirmed by the results of a Holter monitor that measured and recorded SB's electrocardiogram continuously for 48 hours and a thallium stress test. SB's physician suggests a reduction in physical activity and daily aspirin, and he wants to initiate combination drug therapy with a calcium channel blocker (nifedipine) and one of the three choices given below. Evaluate each choice in light of this case and make a recommendation.



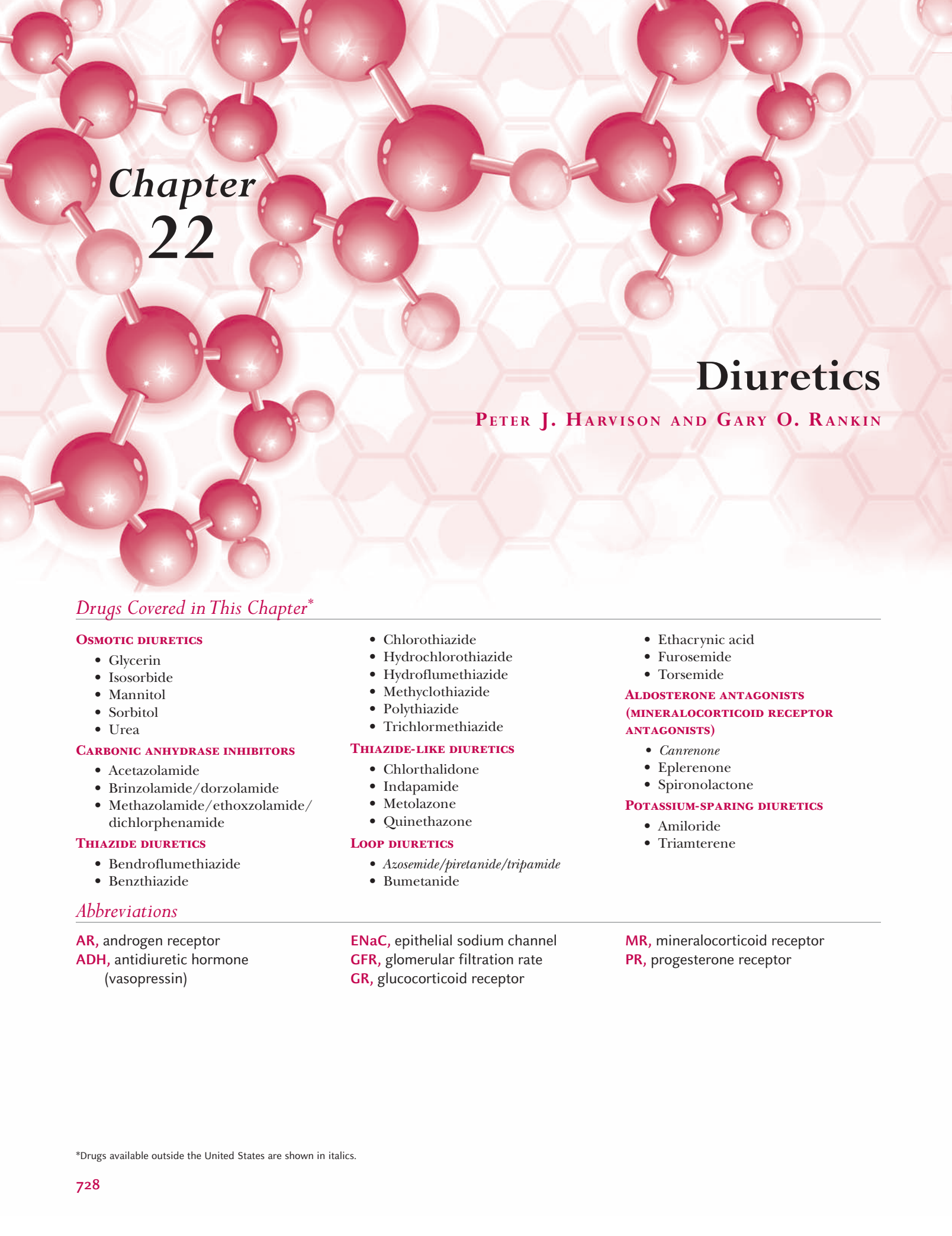
1. Conduct a thorough and mechanistic SAR analysis of the three therapeutic options in the case.
2. Apply the chemical understanding gained from the SAR analysis to this patient's specific needs to make a therapeutic recommendation.

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Chapter 22

Diuretics

PETER J. HARVISON AND GARY O. RANKIN

*Drugs Covered in This Chapter**

OSMOTIC DIURETICS

- Glycerin
- Isosorbide
- Mannitol
- Sorbitol
- Urea

CARBONIC ANHYDRASE INHIBITORS

- Acetazolamide
- Brinzolamide/dorzolamide
- Methazolamide/ethoxzolamide/dichlorphenamide

THIAZIDE DIURETICS

- Bendroflumethiazide
- Benzthiazide

- Chlorothiazide
- Hydrochlorothiazide
- Hydroflumethiazide
- Methyclothiazide
- Polythiazide
- Trichlormethiazide

THIAZIDE-LIKE DIURETICS

- Chlorthalidone
- Indapamide
- Metolazone
- Quinethazone

LOOP DIURETICS

- Azosemide/piretanide/tripamide
- Bumetanide

- Ethacrynic acid
- Furosemide
- Torsemide

ALDOSTERONE ANTAGONISTS (MINERALOCORTICOID RECEPTOR ANTAGONISTS)

- *Canrenone*
- Eplerenone
- Spironolactone

POTASSIUM-SPARING DIURETICS

- Amiloride
- Triamterene

Abbreviations

AR, androgen receptor

ADH, antidiuretic hormone
(vasopressin)

ENaC, epithelial sodium channel

GFR, glomerular filtration rate

GR, glucocorticoid receptor

MR, mineralocorticoid receptor

PR, progesterone receptor

*Drugs available outside the United States are shown in italics.

SCENARIO

Kim K. Birtcher, MS, PharmD, BCPS

TB, a 38-year-old white man, has been referred by his primary care physician (PCP) to the pharmacist-managed hypertension clinic. The patient states, "I was here for a physical. I haven't been to the doctor in awhile, and my wife encouraged me to come. I found out today that my blood pressure is a little high." The patient has no significant medical history or family history, and he is not taking any medications on a regular basis. His body mass index is 27, and his blood pressure is 162/92 mm Hg in the left arm and 164/88 mm Hg in the right arm. The baseline complete metabolic panel and fasting cholesterol panel from today are all within normal limits. The patient drinks wine occasionally with dinner, and he has never used tobacco. He works as an accountant. He is married and has two children living at home. The patient reports that he does not exercise on a regular basis,

and he eats fast food for lunch during the week. The patient receives prescriptions and counseling for lisinopril 10 mg PO daily and HCTZ 12.5 mg PO daily. The patient is given information about the DASH diet and how to make healthier choices at the fast food restaurants. He is encouraged to do regular daily exercise, and he agrees to walk for at least 30 minutes each day during his lunch break. He is encouraged to lose weight. The patient agrees to buy a blood pressure cuff and to monitor his blood pressure at home. He agrees to fax his blood pressure logs to the office weekly so his medications could be titrated per protocol by the pharmacist.

(The reader is directed to the clinical solution and chemical analysis of this case at the end of the chapter).

INTRODUCTION

Diuretics are chemicals that increase the rate of urine formation (1). By increasing the urine flow rate, diuretic usage leads to increased excretion of electrolytes (especially sodium and chloride ions) and water from the body without affecting protein, vitamin, glucose, or amino acid reabsorption. These pharmacologic properties have led to the use of diuretics in the treatment of edematous conditions resulting from a variety of causes (e.g., congestive heart failure, nephrotic syndrome, and chronic liver disease) and in the management of hypertension. Diuretic drugs also are useful as the sole agent or as adjunct therapy in the treatment of a wide range of clinical conditions, including hypercalcemia, diabetes insipidus, acute mountain sickness, primary hyperaldosteronism, and glaucoma.

The primary target organ for diuretics is the kidney, where these drugs interfere with the reabsorption of sodium and other ions from the lumina of the nephrons, which are the functional units of the kidney. The amount of ions and accompanying water that are excreted as urine following administration of a diuretic, however, is determined by many factors, including the chemical structure of the diuretic, the site or sites of action of the agent, the salt intake of the patient, and the amount of extracellular fluid present. In addition to the direct effect of diuretics to impair solute and water reabsorption from the nephron, diuretics also can trigger compensatory physiologic events that have an impact on either the magnitude or the duration of the diuretic response. Thus, it is important to be aware of the normal mechanisms of urine formation and renal control mechanisms to understand clearly the ability of chemicals to induce diuresis.

NORMAL PHYSIOLOGY OF URINE FORMATION

Two important functions of the kidney are 1) to maintain a homeostatic balance of electrolytes and water and 2) to excrete water-soluble end products of metabolism. The kidney accomplishes these functions through the formation of urine by the nephrons (Fig. 22.1). Each kidney contains approximately 1 million nephrons and is capable of forming urine independently. The nephrons are composed of a specialized capillary bed called the glomerulus and a long tubule divided anatomically and functionally into the proximal tubule, loop of Henle, and distal tubule. Each component of the nephron contributes to the normal functions of the kidney in a unique manner; thus, all are targets for different classes of diuretic agents.

Urine formation begins with the filtration of blood at the glomerulus. Approximately 1,200 mL of blood per minute flows through both kidneys and reaches the nephron by way of afferent arterioles. Approximately 20% of the blood entering the glomerulus is filtered into Bowman's capsule to form the glomerular filtrate. The glomerular filtrate is composed of blood components with a molecular weight less than that of albumin (~69,000 daltons) and not bound to plasma proteins. The glomerular filtration rate (GFR) averages 125 mL/min in humans but can vary widely even in normal functional states.

The glomerular filtrate leaves the Bowman's capsule and enters the proximal convoluted tubule (S1, S2 segments; Fig. 22.1), where the majority (50% to 60%) of filtered sodium is reabsorbed osmotically. Sodium reabsorption is coupled electrogenetically with the reabsorption of glucose, phosphate, and amino acids and nonelectrogenetically with bicarbonate reabsorption. Glucose and amino acids are completely reabsorbed

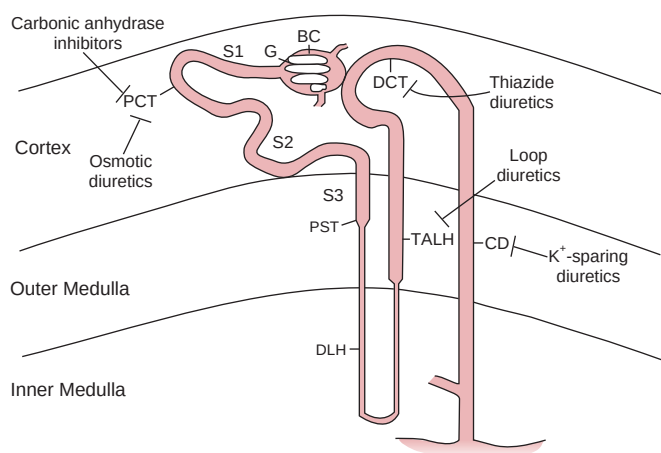


FIGURE 22.1 The nephron. BC, Bowman's capsule; CD, collecting duct; DCT, distal convoluted tubule; DLH, descending limb of the Loop of Henle; G, glomerulus; PCT, proximal convoluted tubule; PST, proximal straight tubule; TALH, thick ascending limb of the loop of Henle.

in this portion of the nephron, whereas phosphate reabsorption is between 80% and 90% complete. The early proximal convoluted tubule also is the primary site of bicarbonate reabsorption (80% to 90%), a process that is mainly sodium dependent and coupled to hydrogen ion secretion. The reabsorption of sodium and bicarbonate is facilitated by the enzyme carbonic anhydrase, which is present in proximal tubular cells and catalyzes the formation of carbonic acid from water and carbon dioxide. The carbonic acid provides the hydrogen ion, which drives the reabsorption of sodium bicarbonate.

Chloride ions are reabsorbed passively in the proximal tubule, where they follow actively transported sodium ions into tubular cells.

The reabsorption of electrolytes and water also occurs isosmotically in the proximal straight tubule or pars recta (S3 segment; Fig. 22.1). By the end of the straight segment, between 65% and 70% of water and sodium, chloride, and calcium ions; 80% to 90% of bicarbonate and phosphate; and essentially 100% of glucose, amino acids, vitamins, and protein have been reabsorbed from the glomerular filtrate. The proximal tubule also is the site for active secretion of weakly acidic and weakly basic organic compounds. Thus, many of the diuretics can enter luminal fluid not only by filtration at the glomerulus but also by active secretion.

The descending limb of the loop of Henle is impermeable to ions, but water can freely move from the luminal fluid into the surrounding medullary interstitium, where the higher osmolality draws water into the interstitial space and concentrates luminal fluid. Luminal fluid continues to concentrate as it descends to the deepest portion of the loop of Henle, where the fluid becomes the most concentrated. The hypertonic luminal fluid next enters the water-impermeable, thick ascending limb of the loop of Henle. In this segment of the nephron, approximately 20% to 25% of the filtered sodium and chloride ions are reabsorbed via a cotransport system ($\text{Na}^+/\text{K}^+/\text{2Cl}^-$) on the luminal membrane. Reabsorption of sodium and chloride in the medullary portion of the thick ascending limb is important for maintaining the medullary interstitial concentration gradient. Reabsorption of sodium chloride in the cortical component of the thick ascending limb of the loop of Henle and the early distal

CLINICAL SIGNIFICANCE



It is important for the clinician to understand the medicinal chemistry of diuretics to appropriately use them in individual patients. This diverse group of medications is classified

in many ways: mechanism of action, site of action, chemical class, and effect on urine contents. Knowledge of structure–activity relationships helps to predict indications, possible off-label uses, the magnitude of diuresis, potency, and the side effect profile.

Consequently, diuretics have a variety of uses. Thiazide diuretics may be used alone or in combination with other pharmacotherapy for the treatment of hypertension. Loop diuretics can provide immediate diuresis and are used for heart failure and in lieu of thiazides in patients with compromised renal function. In addition to more traditional uses, certain potassium-sparing diuretics provide added benefit to other pharmacotherapy in

patients with primary hyperaldosteronism, heart failure, or who have had an acute myocardial infarction. Carbonic anhydrase inhibitors have limited use for diuresis; however, they may be used to reduce intraocular pressure and treat acute mountain sickness.

A thorough understanding of the medicinal chemistry, mechanisms of action, and pharmacokinetics helps the clinician appropriately use available diuretics. As new medications are developed, the clinician will rely on these basic concepts to continue tailoring therapy to the individual patient with the goals to maximize outcomes, improve quality of life, and minimize adverse events.

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convoluted tubule contributes to urinary dilution, and as a result, these two nephron sections sometimes are called the cortical diluting segment of the nephron.

Luminal fluid leaving the early distal tubule next passes through the late distal tubule and cortical collecting tubule (collecting duct), where sodium is reabsorbed in exchange for hydrogen and potassium ions. This process is partially controlled by mineralocorticoids (e.g., aldosterone) and accounts for the reabsorption of between 2% and 3% of filtered sodium ions. Although the reabsorption of sodium ions from these segments of the nephron is not large, this sodium/potassium/hydrogen ion exchange system determines the final acidity and potassium content of urine. Several factors, however, can influence the activity of this exchange system, including the amount of sodium ions delivered to these segments, the status of the acid-base balance in the body, and the levels of circulating aldosterone.

The urine formed during this process represents only approximately 1% to 2% of the original glomerular filtrate, with more than 98% of electrolytes and water filtered at the glomerulus being reabsorbed during passage through the nephron. Thus, a change in urine output of only 1% to 2% could double urine volume. Urine leaves the kidney through the ureters and travels to the bladder, where it is stored until urination removes it from the body.

NORMAL REGULATION OF URINE FORMATION

The body contains several control mechanisms that regulate the volume and contents of urine. These systems are activated by changes in solute or water content of the body, by changes in systemic or renal blood pressure, and by a variety of other stimuli. Activation of one or more of these systems by diuretic drugs can modify the effectiveness of these drugs to produce their therapeutic response and may require additional therapeutic measures to ensure a maximal response.

The kidney has the ability to respond to changes in the GFR through the action of specialized distal tubular epithelial cells called the macula densa. These cells are in close contact with the glomerular apparatus of the same nephron and detect changes in the rate of urine flow and luminal sodium chloride concentration. An increase in the urine flow rate at this site (as can occur with the use of some diuretics) activates the macula densa cells to communicate with the granular cells and vascular segments of the juxtaglomerular apparatus. Stimulation of the juxtaglomerular apparatus causes renin to be released, which leads to the formation of angiotensin II and subsequent renal vasoconstriction. Renal vasoconstriction leads to a decrease in GFR and, possibly, a decrease in the effectiveness of the diuretic. Renin release also can be stimulated by factors other than diuretics, including decreased renal perfusion pressure, increased sympathetic tone, and decreased blood volume.

Another important regulatory mechanism for urine formation is antidiuretic hormone (ADH), also known as vasopressin, which is released from the posterior pituitary in response to reduced blood pressure and elevated plasma osmolality. In the kidney, ADH acts on the collecting tubule to increase water permeability and reabsorption. As a result, the urine becomes more concentrated, and water is conserved in the presence of ADH.

DISEASE STATES

The diuretic drugs are used primarily to treat two medically important conditions, edema and hypertension (2–5). Both conditions are common, although some patients exhibit refractory disease states that require additional modification of the drug regimen to include alternative diuretics or addition of nondiuretic drugs. Edema (excessive extracellular fluid) normally results from disease to the heart, kidney, or liver. Decreased cardiac function (e.g., congestive heart disease) can result in decreased perfusion of all organs (e.g., kidney) and limbs and an accumulation of edema fluid in the extremities, particularly around the ankles and in the hands. Left-sided heart failure can lead to the development of acute pulmonary edema, which is a medical emergency. Right-sided heart failure shifts extracellular fluid volume from the arterial circulation to the venous circulation, which leads to general edema formation.

Kidney dysfunction can lead to edema formation as a result of decreased formation of urine and the subsequent imbalance of water and electrolyte (e.g., sodium ion) homeostasis. Retention of salt and water results in an expansion of the extracellular fluid volume and edema formation. Thus, when salt intake exceeds salt excretion, edema can form. Edema formation also is associated with decreased protein levels in blood, as seen in nephrotic syndrome and liver disease. Cirrhosis of the liver leads to increased lymph in the space of Disse. Eventually, the increased lymph volume results in movement of fluid into the peritoneal cavity and ascites develops.

Hypertension develops from many causes and will be discussed in more detail elsewhere (see Chapter 24). In general, hypertension occurs when blood pressure is sustained at greater than 140/90 mm Hg. At this blood pressure level, patients are at increased risk for developing cardiovascular disease. One key element in controlling blood pressure is sodium ion, and early antihypertensive effects of diuretics are related to increased salt and water excretion. Additionally, however, diuretics have long-term effects resulting in decreased vascular resistance that contribute to blood pressure control. Although effects on vascular calcium-activated potassium channels have been proposed as contributing to the chronic antihypertensive effects of thiazide diuretics, the exact mechanisms of long-term effects remain to be determined.

Diuretics also are useful in treating a number of other conditions including increased cranial (trauma or surgery) or intraocular (glaucoma) pressure (i.e., osmotic

diuretics), diabetes insipidus (i.e., thiazides), hypercalcemia (i.e., loop diuretics), acute mountain sickness (i.e., carbonic anhydrase inhibitors), primary hyperaldosteronism (i.e., aldosterone antagonists), and osteoporosis (i.e., thiazides).

GENERAL THERAPEUTIC APPROACHES

Diuretic drugs may be administered acutely or chronically to treat edematous states. When immediate action to reduce edema (e.g., acute pulmonary edema) is needed, intravenous administration of a loop diuretic often is the approach of choice. Thiazide or loop diuretics normally are administered orally to treat nonemergency edematous states. The magnitude of the diuretic response is directly proportional to the amount of edema fluid that is present. As the volume of edema decreases, so does the magnitude of the diuretic response with each dose. If concern exists about diuretic-induced hypokalemia developing, then a potassium supplement or potassium-sparing diuretic may be added to the drug regimen. The development of hypokalemia is particularly important for patients with congestive heart failure who also are taking cardiac glycosides, such as digitalis. Digitalis has a narrow therapeutic index, and developing hypokalemia can potentiate digitalis-induced cardiac effects with potentially fatal results.

Diuretic drugs (thiazide and loop diuretics) are administered orally to help control blood pressure in the treatment of hypertension. Diuretics often are the first drugs used to treat hypertension, and they also may be added to other drug therapies used to control blood pressure with beneficial effects.

Diuretics have also been used illicitly by some athletes for “sport doping” (6). This is a consequence of the drugs’ ability to quickly produce weight loss (via increased excretion of water) and masking of urine contents (via dilution of other drugs or metabolites that might be present in the urine). Furosemide, triamterene, and hydrochlorothiazide are the most commonly used diuretics for sport doping since they are eliminated rapidly and are therefore more difficult to detect in urine samples taken at later time points after use. Another consideration is that some diuretics can degrade in urine samples; for example, thiazides may hydrolyze to aminobenzenedisulfonamide derivatives (7). Use of diuretics by athletes without a documented therapeutic need has been banned since 1988, and routine monitoring for these drugs and their degradation products is commonplace. A list of banned drugs, including diuretics, is maintained and updated by the International Olympic Committee and the World Anti-Doping Agency.

DIURETIC DRUG CLASSES

History

Compounds that increase the urine flow rate have been known for centuries. One of the earliest substances known

to induce diuresis is water, an inhibitor of ADH release. Calomel (mercurous chloride) was used as early as the 16th century as a diuretic, but because of poor absorption from the gastrointestinal tract and toxicity, calomel was replaced clinically by the organomercurials (e.g., chloromerodrin). The organomercurials represented the first group of highly efficacious diuretics available for clinical use. The need to administer these drugs parenterally, the possibility of tolerance, and their potential toxicity, however, soon led to the search for newer, less toxic diuretics. Today, the organomercurials are no longer used as diuretics, but their discovery began the search for many of the diuretics used today. Other compounds previously used as diuretics include the acid-forming salts (ammonium chloride) and methylxanthines (theophylline).

Structure Classification

The diuretics currently in use today (Table 22.1) are classified by their chemical class (thiazides), mechanism of action (carbonic anhydrase inhibitors and osmotics), site of action (loop diuretics), or effects on urine contents (potassium-sparing diuretics). These drugs vary widely in their efficacy (i.e., their ability to increase the rate of urine formation) and their site of action within the nephron. Efficacy often is measured as the ability of the diuretic to increase the excretion of sodium ions filtered at the glomerulus (i.e., the filtered load of sodium) and should not be confused with potency, which is the amount of the diuretic required to produce a specific diuretic response.

Efficacy is determined, in part, by the site of action of the diuretic. Drugs (e.g., carbonic anhydrase inhibitors) that act primarily on the proximal convoluted tubule to induce diuresis are weak diuretics because of the ability of the nephron to reabsorb a significant portion of the luminal contents in latter portions of the nephron. Likewise, drugs (potassium-sparing diuretics) that act at the more distal segments of the nephron are weak diuretics, because most of the glomerular filtrate has already been reabsorbed in the proximal tubule and ascending limb of the loop of Henle before reaching the distal tubule. Thus, the most efficacious diuretics discovered so far, the high-ceiling or loop diuretics, interfere with sodium chloride reabsorption at the ascending limb of the loop of Henle, which is situated after the proximal tubule but before the distal portions of the nephron and collecting tubule (Fig. 22.1).

Osmotic Diuretics

Mechanism of Action

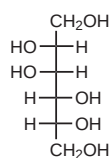
Osmotic diuretics are low molecular weight compounds that are freely filtered through the Bowman’s capsule into the renal tubules, are nonreabsorbable solutes, and are not extensively metabolized except for glycerin and urea (see Table 22.2 for their pharmacokinetic properties). Once in the renal tubule, osmotic diuretics have a limited reabsorption because of their high

TABLE 22.1 Diuretics: Sites and Mechanisms of Action

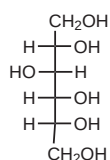
Class of Diuretic	Site of Action	Mechanism of Action
Osmotics	Proximal tubule	Osmotic effects decrease sodium and water reabsorption
	Loop of Henle	Increases medullary blood flow to decrease medullary hypertonicity and reduce sodium and water reabsorption
	Collecting tubule	Sodium and water reabsorption decreases because of reduced medullary hypertonicity and elevated urinary flow rate
Carbonic anhydrase inhibitors	Proximal convoluted tubule	Inhibition of renal carbonic anhydrase decreases sodium bicarbonate reabsorption
Thiazides and thiazide-like	Cortical portion of the thick ascending limb of loop of Henle and distal tubule	Inhibition of Na^+/Cl^- symporter
Loop or high-ceiling	Thick ascending limb of the loop of Henle	Inhibition of the luminal $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ transport system
Potassium-sparing	Distal tubule and collecting duct	Inhibition of sodium and water reabsorption by competitive inhibition of aldosterone (spironolactone) and blockade of sodium channel at the luminal membrane (triamterene and amiloride)

water solubility. When administered as a hypertonic (hyperosmolar) solution, these agents increase intraluminal osmotic pressure, causing water to pass from the body into the tubule. Since the osmotic agent and associated water are not reabsorbed from the nephron, a diuretic effect is observed. Osmotic diuretics increase the volume of urine and the excretion of water and almost all of the electrolytes.

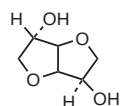
Polyols, such as mannitol, sorbitol, and isosorbide, provide this effect. Sugars, such as glucose and sucrose, also can have a diuretic effect by this mechanism. Although not a polyol, urea has a similar osmotic effect and has been used in the past as an osmotic diuretic.



Mannitol



Sorbitol



Isosorbide

Therapeutic Applications

Osmotic diuretics are not frequently used in medicine today except in the prophylaxis of acute renal failure, in which these drugs inhibit water reabsorption and maintain urine flow. They also may be helpful in maintaining

urine flow in cases where urinary output is diminished because of severe bleeding or traumatic surgical experiences. The osmotic diuretics also have been used to acutely reduce increased intracranial or intraocular pressure. They are not considered to be primary diuretic agents in treating ordinary edemas, because osmotic diuretics can expand extracellular fluid volume.

Adverse Effects

Osmotic diuretics induce few adverse effects, but expansion of the extracellular fluid volume can occur, as noted earlier. Alteration of blood sodium levels can be seen, and these drugs should not be used in anuric or unresponsive patients. If cranial bleeding is present, mannitol or urea should not be used.

Specific Drugs

MANNITOL Mannitol is the agent most commonly used as an osmotic diuretic. Sorbitol also can be used for similar reasons. These compounds can be prepared by the electrolytic reduction of glucose or sucrose.

Mannitol is administered intravenously in solutions of 5% to 50% at a rate of administration that is adjusted to maintain the urinary output at 30 to 50 mL/hour. Mannitol is filtered at the glomerulus and is poorly reabsorbed by the kidney tubule (8). The osmotic effect of mannitol in the tubule inhibits the reabsorption of water, and the rate of urine flow can be maintained. It also is used to reduce intracranial pressure by reducing cerebral intravascular volume.

ISOSORBIDE Isosorbide is basically a bicyclic form of sorbitol that is used orally to cause a reduction in intraocular

TABLE 22.2 Pharmacokinetic Properties of the Nonthiazide Diuretics

Drug	Trade Name	Relative Potency	Oral Absorption (%)	Peak Plasma	Half-Life	Duration of Effect	Elimination
Osmotic diuretics							
Glycerin			>80	1–1.5 h	0.5–0.75 h	4–6 h	Urine, as metabolites
Isosorbide			>80	1–1.5 h	5–9.5 h	5–6 h	Urine, as parent drug
Mannitol			<20	1–3 h IV	0.5–1.5 h	6–8 h	Urine, as parent drug
Urea			<10	1–2 h IV	NA	5–6 h	Urine and bacterial urease in gut
Loop diuretics: high-ceiling diuretics							
Furosemide	Lasix	1	11–90 ^a	4–5 h	0.5–4 h (>3 h) ^b	6–8 h	Urine/feces, as parent drug (60%–70%) and metabolites
Bumetanide	Bumex	40	80–100	<2 h	1–1.5 h (>3 h) ^b	5–6 h	Urine (major), as parent drug (~50%) and ~5 metabolites
Ethacrynic acid	Edecrin	0.7	>90	2 h	0.5–1 h	6–8 h	Urine/feces, as parent drug (30%–60%) and mercapturic acid
Torsemide	Demadex	3	80–100	1–2 h	0.8–4 h	6–8 h	Urine/feces (2:8), as parent drug
Carbonic anhydrase inhibitors							
Acetazolamide	Diamox		>90	1–3 h	6–9 h	8–12 h	Urine (major), as parent drug (70%–100%)
Methazolamide	Neptazane		>90	NA	~14 h	10–18 h	Urine, as parent drug (~25%) and metabolites
Ethoxzolamide	Cardrase		65	2.5–5.5 h			
Brinzolamide	Azopt		Topical use only (eye)	NA	~111 days ^c	8–12 h	Urine, as parent drug (major)
Dorzolamide	Trusopt		Topical use only (eye)	NA	4 months ^c	8–12 h	Urine, as parent drug (major)
Potassium-sparing diuretics (inhibitors of renal epithelial Na⁺ channels)							
Amiloride	Midamor	1	~50 ^a	3–4 h	6–9 h normal (21 h) ^b	24 h	Urine/feces (5:4), as parent drug
Triamterene	Dyrenium	0.1	>70	2–4 h	2–3 h	>24 h	Urine, as parent drug and metabolites
Mineralocorticoid receptor antagonists							
Spirolactone	Aldactone	20–40	>90 ^d	1–2 h	1–3 h (parent drug)	2–3 d	Urine/feces, as active metabolite(s)
Canrenone (<i>7</i> - α -thiospirolactone)			NA	NA	3–4 h	13–24 h	Urine
Eplerenone	Inspira	1	70	1.5 h	4–6 h	NA	Urine/feces (2:1), as metabolites (>95%)

^aFood affects bioavailability.^bIn patients with renal insufficiency.^cStrongly bound to red blood cells.^dFormulation affects bioavailability.

IV, intravenous; NA, data not available.

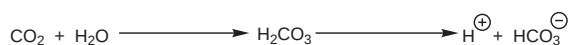
Data from AHFS Drug Information (2010), American Society of Health-System Pharmacists; Lexicomp Online (2011), Lexicomp, Inc.; Facts and Comparisons (2010), Wolters Kluwer Health.

pressure in glaucoma cases. Although a diuretic effect is noted, its ophthalmologic properties are its primary value.

Carbonic Anhydrase Inhibitors

Mechanism of Action

In 1937, it was proposed that the normal acidification of urine was caused by secretion of hydrogen ions by the tubular cells of the kidney. These ions were provided by the action of the enzyme carbonic anhydrase, which catalyzes the formation of carbonic acid (H_2CO_3) from carbon dioxide and water.



It also was observed that sulfanilamide rendered the urine of dogs alkaline because of the inhibition of carbonic anhydrase. This inhibition of carbonic anhydrase resulted in a lesser exchange of hydrogen ions for sodium ions in the kidney tubule. Sodium ions, along with bicarbonate ions, and associated water molecules were then excreted, and a diuretic effect was noted. The large doses required and the side effects of sulfanilamide prompted a search for more effective carbonic anhydrase inhibitors as diuretic drugs.

It was soon learned that the sulfonamide portion of an active diuretic molecule could not be monosubstituted or disubstituted (9,10). It was reasoned that a more acidic sulfonamide would bind more tightly to the carbonic anhydrase enzyme. Synthesis of more highly acidic sulfonamides produced compounds with activities greater than 2,500-fold that of sulfanilamide. Acetazolamide was introduced in 1953 as an orally effective diuretic drug. Before that time, the organic mercurials, which commonly required intramuscular injection, were the principal diuretics available.

Carbonic anhydrase inhibitors induce diuresis by inhibiting the formation of carbonic acid within proximal (proximal convoluted tubule; S2) and distal tubular cells to limit the number of hydrogen ions available to promote sodium reabsorption. For a diuretic response to be observed, more than 99% of the carbonic anhydrase must be inhibited. Although carbonic anhydrase activity in the proximal tubule regulates the reabsorption of approximately 20% to 25% of the filtered load of sodium, the carbonic anhydrase inhibitors are not highly efficacious diuretics. An increased excretion of only 2% to 5% of the filtered load of sodium is seen with carbonic anhydrase inhibitors because of increased reabsorption of sodium ions by the ascending limb of the loop of Henle and more distal nephron segments.

Therapeutic Applications

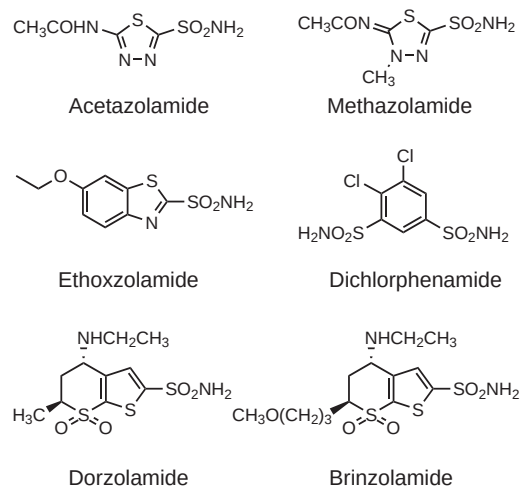
With prolonged use of the carbonic anhydrase inhibitor diuretics, the urine becomes more alkaline, and the blood becomes more acidic. When acidosis occurs, the carbonic anhydrase inhibitors lose their effectiveness as

diuretics. They remain ineffective until normal acid-base balance in the body has been regained. For this reason, this class of compounds is limited in its diuretic use. Today, they are most commonly used in the treatment of glaucoma, in which they inhibit carbonic anhydrase in the eye, reduce the rate of aqueous humor formation, and consequently, reduce intraocular pressure (11). These compounds also have found some limited use in the treatment of absence seizures, to alkalinize the urine, to treat familial periodic paralysis, to reduce metabolic alkalosis, and prophylactically, to reduce acute mountain sickness.

Adverse Effects

The carbonic anhydrase inhibitors generally do not produce serious side effects. However, monitoring for electrolyte disturbances may be necessary. Gastrointestinal (e.g., nausea, vomiting), nervous system (e.g., sedation, headache), and renal effects (kidney stones) have been reported. Although they contain sulfonamide groups, hypersensitivity reactions with other sulfonamide-containing drugs are not common. As noted earlier, these drugs can produce metabolic acidosis.

Specific Drugs



ACETAZOLAMIDE Acetazolamide, a thiadiazole derivative, was the first of the carbonic anhydrase inhibitors to be introduced as an orally effective diuretic, with a therapeutic effect that lasts approximately 8 to 12 hours (Table 22.2). As mentioned earlier, its diuretic action is limited because of the systemic acidosis it produces. The dose is 250 mg to 1 g per day.

METHAZOLAMIDE, ETHOXZOLAMIDE, AND DICHLORPHENAMIDE

Methazolamide is a close structural analog of acetazolamide in which one of the active hydrogens in the thiadiazole ring has been replaced by a methyl group. This decreases polarity of the compound and permits a greater penetration into the ocular fluid, where it acts as a carbonic anhydrase inhibitor, reducing intraocular

pressure. Its dose for glaucoma is 50 to 100 mg two to three times a day. Ethoxzolamide has a sulfonamide group attached to a fused 1,3-benzothiazole ring. Typical doses are 62.5 to 125 mg up to four times a day. Dichlorphenamide is a disulfonamide derivative that shares the same pharmacologic properties and clinical uses as the previously discussed compounds. The dose of dichlorphenamide is 25 to 100 mg one to three times a day. Distribution of these three drugs into the ocular compartment is not selective, and they can also inhibit multiple carbonic anhydrase isozymes in other tissues (11). Both of these properties can contribute to their side effects.

BRINZOLAMIDE AND DORZOLAMIDE Brinzolamide and dorzolamide contain ionizable amino groups and are the result of efforts to develop water-soluble carbonic anhydrase inhibitors that retain sufficient lipophilicity to penetrate the cornea (11). They are only indicated for topical eye administration in glaucoma patients. However, both drugs can be absorbed from ocular fluids into the blood. This results in prolonged half-lives (e.g., 111 days for brinzolamide in whole blood) because the drugs bind to carbonic anhydrase, especially in red blood cells.

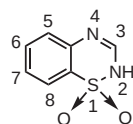
Benzothiadiazine or Thiazide Diuretics

Mechanism of Action

Further study of the benzene disulfonamide derivatives was undertaken to find more efficacious carbonic anhydrase inhibitors. These studies provided some compounds with a high degree of diuretic activity. Chloro and amino substitution gave compounds with increased activity, but these compounds were weak carbonic anhydrase inhibitors. When the amino group was acylated, an unexpected ring closure took place. These compounds possessed a diuretic activity independent of the carbonic anhydrase inhibitory activity, and a new series of diuretics called the benzothiadiazines was discovered (9,10,12).

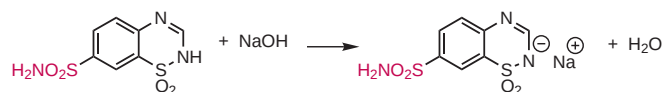
These diuretics are actively secreted in the proximal tubule and are carried to the loop of Henle and to the distal tubule. The major site of action of these compounds is in the distal convoluted tubule, where these drugs compete for the chloride binding site of the Na^+/Cl^- symporter and inhibit the reabsorption of sodium and chloride ions. For this reason, they are referred to as saluretics. They also inhibit the reabsorption of potassium and bicarbonate ions, but to a lesser degree.

Structure–Activity Relationship



The thiazide diuretics are weakly acidic (see Appendix A for their $\text{p}K_a$ values), with a benzothiadiazine

1,1-dioxide nucleus. The structure for the thiazide diuretics, relative activities, and pharmacokinetic properties for the thiazides are shown in Table 22.3. Chlorothiazide is the simplest member of this series, having a $\text{p}K_a$ of 6.7 and 9.5. The hydrogen atom at the 2-N is the most acidic because of the electron-withdrawing effects of the neighboring sulfone group. The sulfonamide group that is substituted at C-7 provides an additional point of acidity in the molecule but is less acidic than the 2-N proton. These acidic protons make possible the formation of a water-soluble sodium salt that can be used for intravenous administration of the diuretics.



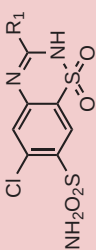
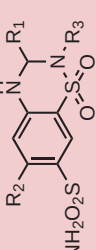
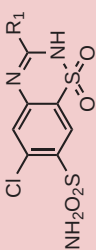
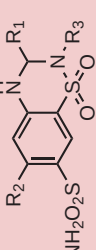
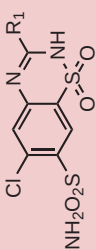
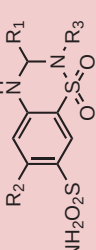
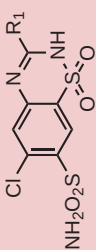
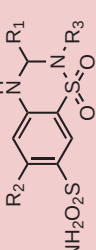
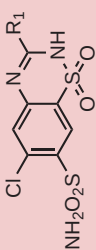
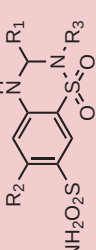
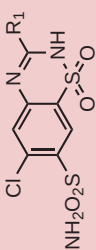
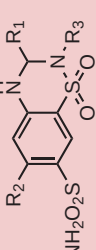
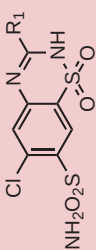
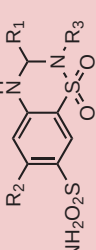
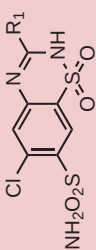
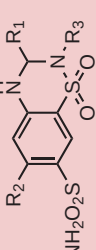
An electron-withdrawing group is necessary at position 6 for diuretic activity. Little diuretic activity is seen with a hydrogen atom at position 6, whereas compounds with a chloro or trifluoromethyl substitution are highly active (9,10,12). The trifluoromethyl-substituted diuretics are more lipid-soluble and have a longer duration of action than their chloro-substituted analogs. When electron-releasing groups, such as methyl or methoxyl, are placed at position 6, the diuretic activity is markedly reduced.

Replacement or removal of the sulfonamide group at position 7 yields compounds with little or no diuretic activity. Saturation of the double bond to give a 3,4-dihydro derivative produces a diuretic that is 10-fold more active than the unsaturated derivative. Substitution with a lipophilic group at position 3 gives a marked increase in the diuretic potency. Haloalkyl, aralkyl, or thioether substitution increases the lipid solubility of the molecule and yields compounds with a longer duration of action. Alkyl substitution on the 2-N position also decreases the polarity and increases the duration of diuretic action (Table 22.3). Although these compounds do have carbonic anhydrase activity, there is no correlation of this activity with their saluretic activity (excretion of sodium and chloride ions).

Therapeutic Applications

The thiazide diuretics are administered once a day or in divided daily doses. Some have a duration of action that permits administration of a dose every other day. Several of these compounds are rapidly absorbed orally and can show their diuretic effect in an hour (Table 22.3). These compounds are not extensively metabolized and are primarily excreted unchanged in the urine. Thiazide diuretics are used to treat edemas caused by cardiac decompensation as well as in hepatic or renal disease. They also commonly are used in the treatment of hypertension. Their effect may be attributed to a reduction in blood volume and a direct relaxation of vascular smooth muscle.

TABLE 22.3 Pharmacologic and Pharmacokinetic Properties for the Thiazide Diuretics

Generic Name	Trade Name	Structure	Structure II								
			Relative Potency ^a	Carbonic Anhydrase Inhibition ^b	Bioavailability	Peak Plasma (hours)	Half-life (hours)	Duration of Effect (hours)	Elimination		
Chlorothiazide	Diuril			0.8	2×10^{-6}	<25%	4	0.75–2	12–16	U	
Benzthiazide	Exna			1.3	$\sim 10^{-7}$	NA	4	3–3.9	12–16	NA	
Hydrochlorothiazide	HydroDiuril Esidrix			1.4	2×10^{-5}	>80%	4–6	6–15	12–16	U	
Trichlormethiazide	Diurese Metahydrin Naqua			1.7	6×10^{-5}	Var	6	2–7	24	U	
Methyclothiazide	Enduron Aquatensen			1.8		Var	6	NA	>24	U	
Polythiazide	Renese			2.0	5×10^{-7}	Var	6	NA	24–48	U + 30% M	
Hydroflumethiazide	Saluron Diucardin			1.3	2×10^{-4}	Inc	3–4	17	18–24	U + M	
Bendroflumethiazide	Naturetin			1.8	3×10^{-4}	>90%	4	8.5	6–12	U	

^aThe numerical values refer to potency ratios (in humans) with the natriuretic response to that of a standard dose of meralluride, which is given a value of one;

^b50% inhibition of carbonic anhydrase in vitro.

NA, data not available; Var, variable absorption; Inc, incomplete absorption.

Data from AHFS Drug Information (2010), American Society of Health-System Pharmacists; Lexicomp Online (2011), Lexicomp, Inc.; Facts and Comparisons (2010), Wolters Kluwer Health.

U, urine unchanged; M, metabolized.

Adverse Effects

Thiazide diuretics may induce a number of adverse effects, including hypersensitivity reactions, gastric irritation, nausea, and electrolyte imbalances, such as hyponatremia, hypokalemia, hypomagnesemia, hypochloremic alkalosis, hypercalcemia, and hyperuricemia. Individuals who exhibit hypersensitivity reactions to one thiazide are likely to have a hypersensitivity reaction to other thiazides and sulfamoyl-containing diuretics (e.g., thiazide-like and some high-ceiling diuretics). Potassium and magnesium supplements may be administered to treat hypokalemia or hypomagnesemia, but their use is not always indicated. These supplements usually are administered as potassium chloride, potassium gluconate, potassium citrate, magnesium oxide, or magnesium lactate. The salts are administered as solutions, tablets, or timed-release tablets. Generally, approximately 20 mEq of potassium is given daily. In cases of hypokalemia, 40 to 100 mEq/day may be administered. Potassium-sparing diuretics (e.g., triamterene or amiloride) also may be used to prevent hypokalemia. Combination preparations of hydrochlorothiazide or a potassium-sparing diuretic are available (e.g., Diazide and Moduretic).

Long-term use of thiazide diuretics also may result in decreased glucose tolerance and increased blood lipid (low-density lipoprotein cholesterol, total cholesterol, and total triglyceride) content.

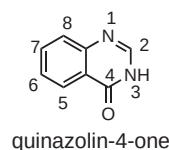
Thiazide-like Diuretics

Mechanism of Action

This is a structurally diverse group of sulfonamide derivatives that do not contain benzothiadiazine rings (Table 22.4). Nevertheless, they have the same mechanism of action and similar therapeutic activities and adverse effects as the thiazide diuretics.

Specific Drugs

QUINAZOLINONE DERIVATIVES: METOLAZONE AND QUINETHAZONE



Overview The quinazolin-4-one molecule has been structurally modified in a manner similar to the modification of the thiazide diuretics. Metolazone and quinethazone ($pK_a = 9.7$) are examples of this class (Table 22.4). The structural difference between the quinazolinone diuretics is the replacement of the 4-sulfone group ($-\text{SO}_2-$) with a 4-keto group ($-\text{CO}-$). Because of their similar structures, it is not surprising that the quinazolinones have a diuretic effect similar to that of the thiazides.

TABLE 22.4 Pharmacokinetic Properties for the Thiazide-Like Diuretics

Generic name	Trade name	Structure	Bioavailability	Peak Plasma	Half-life (hours)	Duration (hours)	Route of Elimination
Metolazone	Zaroxylon		<65%	8–12 h	14	12–24	Urine/feces (8:2), as parent drug (>70%)
Quinethazone	Hydromox		NA	6 h	6–15	18–24	Urine as parent drug
Chlorthalidone	Hygroton Thalitone [#]		Inc/var. >90%	4 h 2 h	35–50*	48–72	Urine, as parent drug (50–65%)
Indapamide	Lozol		>90%	2–3 h	14–18	8 wks	Urine/feces (6:2), as parent drug (<10%) and metabolites

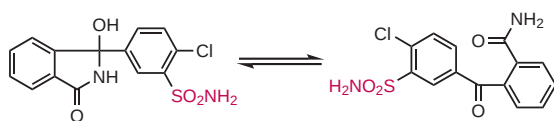
Data from AHFS Drug Information (2010). American Society of Health-System Pharmacists; Lexicomp Online (2011), Lexicomp, Inc.; Facts and Comparisons (2010), Wolters Kluwer Health.

*Strongly bound to red blood cells. [#]= not interchangeable with similar drug. NA = data not available. var. = variable absorption. inc. = incomplete absorption.

Therapeutic Applications The pharmacokinetic properties for the quinazolinone diuretics are listed in Table 22.4. They have a long duration of action, usually as a result of protein binding. Although chlorothiazide has a duration of action of 6 to 12 hours, quinethazone has a duration of 18 to 24 hours, and metolazone has a duration of 12 to 24 hours. Metolazone has a bioavailability of 65% (Zaroxolyn) and a prolonged onset to reach peak plasma concentrations of action ranging from 8 to 12 hours. Another formulation of metolazone (Mykrox) was more rapidly absorbed and had improved bioavailability (>90%) compared to Zaroxolyn, but it has been discontinued in the United States and other countries. Approximately 50% to 70% of metolazone is bound to carbonic anhydrase in the erythrocytes. Metolazone also has an increased potency, and the mode of action for both compounds is similar to that of the thiazide derivatives. In contrast to thiazide diuretics, metolazone may be effective as a diuretic when the GFR falls below 40 mL/min. The dose of quinethazone is 50 to 100 mg daily and that of metolazone is 2.5 to 20 mg given as a single oral dose. Side effects are similar to adverse effects induced by the thiazide diuretics.

PHthalimidine Derivatives: Chlorthalidone

Overview Chlorthalidone ($pK_a = 9.4$) is an example of a diuretic in this class of compounds that bears a structural analogy to the quinazolinones (Table 22.4). This compound may be named as a 1-oxo-isindoline or a phthalimidine. Although the molecule exists primarily in the phthalimidine form, the ring may be opened to form a benzophenone derivative. The benzophenone form illustrates the relationship to the quinazolinone series of diuretics.



Therapeutic Application Chlorthalidone has a long duration of action (48 to 72 hours) (see Table 22.4 for its other pharmacokinetic properties). Although quinethazone and metolazone are administered daily, chlorthalidone may be administered in doses of 25 to 100 mg three times a week. When chlorthalidone is formulated with the excipient povidone, the product, Thalitone, has greater bioavailability (>90%) and reaches peak plasma concentrations in a shorter time compared with its other products. Similar to the quinazolinones, it also is extensively bound to carbonic anhydrase in the erythrocytes. Chlorthalidone-induced effects on urine content and side effects are similar to those induced by thiazide diuretics.

Indolines: Indapamide

Overview The prototypic indoline diuretic is indapamide, which was reported as a diuretic in 1984. Indapamide contains a polar chlorobenzamide moiety

and a nonpolar lipophilic methylindoline group. In contrast to the thiazides, indapamide does not contain a thiazide ring, and only one sulfonamide group is present within the molecular structure ($pK_a = 8.8$). It is rapidly and completely absorbed from the gastrointestinal tract and reaches its peak plasma level in 2 to 3 hours, with a duration of action of up to 8 weeks. This prolonged duration of action is associated with its extensive binding to carbonic anhydrase in the erythrocytes. It exhibits biphasic kinetics, with a half-life of 14 to 18 hours and an elimination half-life of 24 hours. Indapamide is extensively metabolized (60% to 70%), with several of the metabolites being shown in Figure 22.2 (13,14). In vitro studies support aromatic hydroxylation as a metabolic route. Less than 10% of the drug is excreted unchanged, whereas the remaining 20% to 30% is eliminated via extrahepatic cycling.

Therapeutic Applications Uses of indapamide include the treatment of essential hypertension and edema resulting from congestive heart failure. Like metolazone, indapamide is an effective diuretic drug when GFR falls below 40 mL/min. The duration of action is approximately 24 hours, with the normal oral adult dosage starting at 2.5 mg given each morning. The dose may be increased to 5.0 mg/day, but doses beyond this level do not appear to provide additional results. Effects on urine content and side effects are similar to effects induced by thiazide diuretics.

High-Ceiling or Loop Diuretics

Mechanism of Action

This class of drugs is characterized more by its pharmacologic similarities than by its chemical similarities. Examples include furosemide, bumetanide, torsemide, and ethacrynic acid. These drugs produce a peak diuresis much greater than that observed with the other commonly used diuretics, hence the name high-ceiling diuretics. Their main

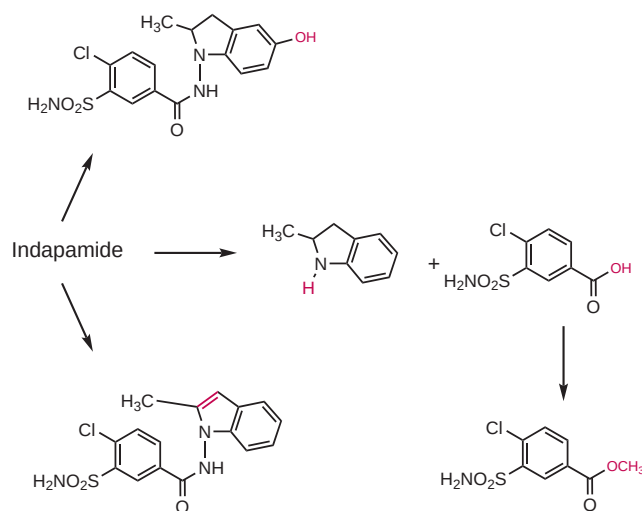
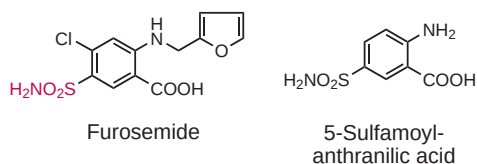


FIGURE 22.2 Metabolism of indapamide.

site of action is believed to be on the thick ascending limb of the loop of Henle, where they inhibit the luminal $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ symporter. These diuretics are commonly referred to as loop diuretics. Additional effects on the proximal and distal tubules also are possible. High-ceiling diuretics are characterized by a quick onset and short duration of activity (15). Their diuretic effect appears in approximately 30 minutes and lasts for approximately 6 hours. The pharmacokinetic properties for the loop diuretics are listed in Table 22.2.

Specific Drugs

FUROSEMIDE



Structure–Activity Relationships Furosemide is an example of a high-ceiling diuretic and may be regarded as a derivative of anthranilic acid or *o*-aminobenzoic acid. Research on 5-sulfamoylanthranilic acids at the Hoechst Laboratories in Germany showed them to be effective diuretics (10,12). The most active of a series of variously substituted derivatives was furosemide.

The chlorine and sulfonamide substitutions are features also seen in previously discussed diuretics. Because the molecule possesses a free carboxyl group, furosemide is a stronger acid than the thiazide diuretics ($\text{p}K_a = 3.9$). This drug is excreted primarily unchanged (16). A small amount of metabolism, however, can take place on the furan ring, which is substituted on the aromatic amino group (see Table 22.2 for its other pharmacokinetic properties).

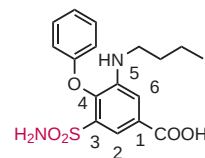
Therapeutic Applications Furosemide has a saluretic effect 8- to 10-fold that of the thiazide diuretics; however, it has a shorter duration of action (~6 to 8 hours). Furosemide causes a marked excretion of sodium, chloride, potassium, calcium, magnesium, and bicarbonate ions, with as much as 25% of the filtered load of sodium excreted in response to initial treatment. It is effective for the treatment of edemas connected with cardiac, hepatic, and renal sites. Because it lowers the blood pressure similar to the thiazide derivatives, one of its uses is in the treatment of hypertension.

Furosemide is orally effective but may be used parenterally when a more prompt diuretic effect is desired, such as in the treatment of acute pulmonary edema. The dosage of furosemide, 20 to 80 mg/day, may be given in divided doses because of the short duration of action of the drug and carefully increased up to a maximum of 600 mg/day.

Adverse Effects Clinical toxicity of furosemide and other loop diuretics primarily involves abnormalities of fluid

and electrolyte balance. As with the thiazide diuretics, hypokalemia is an important adverse effect that can be prevented or treated with potassium supplements or coadministration of potassium-sparing diuretics. Increased calcium ion excretion can be a problem for postmenopausal osteopenic women, and furosemide generally should not be used in these individuals. Hyperuricemia, glucose intolerance, increased serum lipid levels, ototoxicity, and gastrointestinal side effects might be observed as well. Hypersensitivity reactions also are possible with furosemide (a sulfonamide-based drug), and cross-reactivity with other sulfonamide-containing drugs is possible.

BUMETANIDE

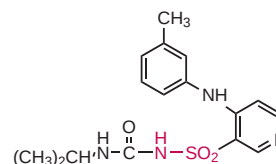


Bumetanide

Structure–Activity Relationships For bumetanide, a phenoxy group has replaced the customary chloro or trifluoromethyl substitutions seen in other diuretic molecules (17). The phenoxy group is an electron-withdrawing group similar to the chloro or trifluoromethyl substitutions. The amine group customarily seen at position 6 has been moved to position 5. These minor variations from furosemide produced a compound with a mode of action similar to that of furosemide, but with a marked increase in diuretic potency. The short duration of activity is similar, but the compound is approximately 50-fold more potent. Replacement of the phenoxy group at position 4 with a $\text{C}_6\text{H}_5\text{NH}$ - or $\text{C}_6\text{H}_5\text{S}$ - group also gives compounds with a favorable activity. However, when the butyl group on the C-5 amine is replaced with a furanylmethyl group, such as in furosemide, the results are not favorable.

Therapeutic Applications This compound also functions as a high-ceiling diuretic in the ascending limb of the loop of Henle. Its duration of action is approximately 4 hours. The uses of this compound are similar to those described for furosemide. The dose of bumetanide is 0.5 to 2 mg/day given as a single dose. Adverse effects are similar to those induced by furosemide.

TORSEMIDE



Torsemide